From First Courses to the Modular Degree

Book plan and chapter outlines

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Chapter 1

The book at a glance

This document collects the plan of the textbook *From First Courses to the Modular Degree* and the outline of every chapter, as they stand on 15 October 2026. The outlines are the specification from which each chapter is to be written; nothing in them is final text. Instructions for continuing the work are in the repository file `CONTINUE.md`.

Each outline lists the chapter's core results in teaching order (definitions, theorems, lemmas, examples), each with a precise statement, the planned level of rigour (proved here, proof sketched here, stated with reference), a reference, and the place where the paper uses it; then a section plan with target lengths, a coverage table, exercises, dependencies, notes for the writer, and points about the paper found while outlining.

1.1 Chapter list

Ch.	Title	Outline (lines)	Planned LaTeX (lines)
	Part I. Algebra		
1	Groups, group actions and matrix groups over finite fields	786	1910
2	Rings, fields, polynomials and characters	930	2080
3	Modules and commutative algebra	1110	2165
4	Representations of finite groups and the mod l world	1065	2120
	Part II. Number fields and local fields		
5	Galois theory, finite and infinite	990	2030
6	Algebraic number theory	1103	1820
7	Local fields	1111	2080
	Part III. Curves and elliptic curves		

Ch.	Title	Outline (lines)	Planned LaTeX (lines)
8	Algebraic curves, function fields and Riemann surfaces	1023	2150
9	Elliptic curves over fields	1088	2140
10	Elliptic curves over local fields: reduction, Kodaira types, Tate curves	1004	2150
11	Galois representations of elliptic curves	1078	2775
	Part IV. Modular forms and modular curves		
12	Modular forms and modular curves	974	1970
13	Hecke operators, newforms and twists	1005	1930
14	Modular curves over Q, Jacobians and the modular degree	1078	1985
15	Galois representations of modular forms, level lowering and congruences	1069	2780
	Part V. Cohomology, Selmer groups and L-functions		
16	Homological algebra and group cohomology	588	2000
17	Galois cohomology, local and global, and Selmer groups	1022	2110
18	Bloch-Kato Selmer groups, Tamagawa factors and the adjoint motive	1080	2630
19	L-functions: Euler products, local factors, the adjoint and the symmetric square	941	1975
	Part VI. Congruences, the modular degree and the paper		
20	Hecke algebras, congruence ideals and the theorem of Diamond, Flach and Guo	1065	2670

Ch.	Title	Outline (lines)	Planned LaTeX (lines)
21	Reading the paper: a guided tour	1884	2795
22	Computation, data and the statistical protocol	1380	1940
	Appendices		
A	Prerequisites: what we assume and where to look	written directly	about 1,000 (partly written)
B	Notation, conventions and index of symbols	1047	750
	Total	24,421	48,955 plus Appendix A

Planned lengths are guidelines, not caps (decision D2): the book is for online reading and reference, so a chapter may run longer where that makes an explanation clearer. At about 50 lines per page the book will run to roughly 1,100 to 1,400 pages.

1.2 Contents of this document

1. This overview.
2. The master plan (`PLAN.md`): audience, design principles, the scope of every chapter, the dependency graph, the paper-to-chapter map, writing conventions, the binding decisions of Section 7.5, and the production process.
3. The outline of each chapter, Chapters 1 to 22.
4. The outline of Appendix B, which fixes the book-wide conventions and indexes the symbols.

The memo for the paper's author (`important/paper_memo.md`), the full record of findings (`important/important_notes.md`) and the revised paper (`docs/note_v2.pdf`) are separate documents.

Part I

The master plan

Chapter 2

Master plan: a modular textbook leading to "Local factors of the adjoint L-function and the l-adic valuations of the modular degree"

This file is the single source of truth for everyone (human or agent) writing a chapter. Read all of it before writing anything.

2.1 1. Goal

The target text is the research note

> D. Soni, *Local factors of the adjoint L-function and the l-adic valuations of the modular degree* (version of 10 October 2026).

It proves, for an X_0(N)-optimal elliptic curve E/Q and an odd prime l not dividing 2N such that the mod l representation is absolutely irreducible over Q(sqrt(l*)), the inequality

```
v_l(deg phi_E)  >=  sum_{q | N} e_l(E,q) + sum_{q | N} t_l(E,q),
```

where e_l(E,q) is the l-adic valuation of the Euler factor at s = 1 of the adjoint L-function that the naive adjoint L-function omits (nonzero only at additive primes, read off the Kodaira type for tame q >= 5) and t_l(E,q) is the l-part of the valuation of the Tate parameter at a potentially multiplicative prime. The proof assembles: the theorem of Diamond, Flach and Guo (DFG04) on the length of the Sigma-imprimitive adjoint Selmer group and the congruence ideal; their formula for the variation of the congruence ideal with Sigma; the local Tamagawa formula of Fontaine and Perrin-Riou; a computation of the Tamagawa factor of the adjoint at a multiplicative prime; the comparison of the modular degree with the congruence number (Agashe, Ribet, Stein); the commutative algebra of Wiles's numerical criterion (DDT Lemma 4.17); and Ribet-type level lowering plus quadratic twisting. It also identifies the boundary of the hypothesis (mod 3 image the normaliser of a split Cartan subgroup), states conjectures beyond the hypotheses, and reports large computations on Cremona's tables with a pre-registered protocol.

The book must take a reader from the stated background to the point where every statement, every proof and every table of the paper can be read with understanding, and it must serve as a *refresher* for experts: a specialist who has forgotten the exact form of, say, the Fontaine-Perrin-Riou Tamagawa factor or the Atkin-Li twisting theorem opens the relevant chapter and finds the precise statement at the top.

2.2 2. Audience and assumed background

The reader has done:

- a first course in real analysis (sequences, series, continuity, differentiation, Riemann integration, uniform convergence, metric spaces at an elementary level);
- a first course in linear algebra (vector spaces over R and C, matrices, determinants, eigenvalues, diagonalisation, inner products, Jordan form at most as a statement);
- an introduction to complex analysis (holomorphic functions, Cauchy's theorem, residues, Laurent series, Mobius transformations at an elementary level);
- basic statistics (means, variances, distributions, hypothesis tests at an elementary level);
- Riley, Hobson and Bence, *Mathematical Methods for Physics and Engineering* (so: Fourier series and transforms, ordinary and partial differential equations, special functions, tensors, elementary group theory and representation theory of finite groups, elementary probability).

The reader has NOT done: abstract algebra beyond what RHB contains, Galois theory, algebraic number theory, topology, algebraic geometry, homological algebra, modular forms, p-adic numbers, or any Galois cohomology. Every one of these is built from scratch, with proofs where they are short enough to be instructive and precise statements with references where they are not.

Two reading modes must both work:

1. **Linear mode.** Chapters are in dependency order; a reader with the background above can read from Chapter 1 to Chapter 22.
2. **Refresher mode.** Each chapter opens with a *Core results at a glance* card: a numbered list of the definitions, theorems, lemmas and constructions of the chapter, each stated precisely (not summarised), each pointing to where in the chapter it is proved or discussed. An expert reads the card and stops.

2.3 3. Design principles

- **Modular.** One chapter = one LaTeX file = one topic that can be read on its own given the card of the earlier chapters. Chapters never `\input` other chapters. Cross-chapter references are by chapter label only (`\Cref{ch:slug}`), never by theorem number of another chapter, because chapters are written independently and in parallel.
- **Paper-driven.** Every chapter has a section *Where the paper uses this chapter* that names the exact lemmas, theorems, remarks, definitions or sections of the paper that depend on the chapter, and says which result of the chapter they use. Nothing is included that the paper does not need, directly or as a prerequisite of something it needs, except where a small amount of context makes a definition natural.
- **Three levels of rigour, always declared.** Each result in the card is marked *(proved here)*, *(proof sketched here)* or *(stated; reference)*. Elementary results (everything in Parts I and II, most of Part III, the commutative algebra of Part VI) are proved. Deep theorems (modularity, Serre's conjecture, level lowering, the DFG theorems, the Fontaine-Perrin-Riou formula, Tate local duality, Poitou-Tate) are stated precisely with their hypotheses, the shape of the proof is explained, and a standard reference with chapter or theorem number is given.
- **Precise statements over coverage.** A card entry that is wrong is worse than one that is missing. Standard references are cited with theorem numbers only when the writer is sure of the number; otherwise cite the chapter or section.
- **Worked examples from the paper.** The curves 11a1, 20a1, 121a1, 121c, 121d, 539c1, 980i1, 726a1 and 99A1 (Cremona labels) appear in the paper; use them as running examples where the chapter's topic allows.
- **Exercises.** Each chapter ends with 8 to 20 exercises, graded, some of them "check the paper's claim that ..." exercises.

- **The physics reader.** Where a construction has an RHB-style analogue (representations of finite groups, Fourier expansions, lattices, periods), say so in one sentence; do not belabour it.

2.4 4. Table of contents

Labels `ch:slug` are fixed and must be used exactly. File names are `chapters/chNN_slug.tex`.

2.4.1 Part I. Algebra

Chapter 1. Groups, group actions and matrix groups over finite fields (`ch:groups`, `ch01_groups.tex`) Scope: groups, subgroups, cosets, normal subgroups, quotients, homomorphism theorems, cyclic groups, direct and semidirect products, group actions, orbits and stabilisers, orbit-stabiliser, Sylow theorems (statement and the use for small groups), p-groups, the quaternion group Q_8, dihedral groups, finite abelian groups (structure theorem), profinite groups as inverse limits (definition, Z_p and the Galois groups to come). Matrix groups: GL_2(F), SL_2(F), PGL_2, PSL_2 over a field, orders over F_q, Borel subgroups, split and nonsplit Cartan subgroups and their normalisers, unipotent elements, the action on the projective line P^1(F_q) (lines in F_q^2), orbits of subgroups on lines. Full detail for GL_2(F_3) (order 48): its subgroups SL_2(F_3) (order 24), Q_8, the normalisers N_s(F_3) (order 8, dihedral) and N_ns(F_3) (order 16, semidihedral), the determinant-one subgroups of each (SL_2(F_3), C_4, Q_8), elements of determinant -1 and trace 0, orbits of N_s(F_3) on the four lines of F_3^2 (two orbits of size two) versus transitivity of N_ns(F_3) and GL_2(F_3). Also GL_2(F_5), GL_2(F_7) orders and the maximal subgroup types in general (Dickson's classification, statement). Paper hooks: Proposition prop:ns (images at l = 3, determinant-one subgroups, orbits on lines, the 3-division polynomial factorisation), Lemma lem:IV (unipotent image of inertia), Remark rem:types, Section sec:anomaly (image 3Ns), the Eisenstein case (Borel image), Lemma lem:h0.

Chapter 2. Rings, fields, polynomials and characters (`ch:rings`, `ch02_rings.tex`) Scope: rings, ideals, quotient rings, prime and maximal ideals, integral domains, fields of fractions, PIDs, UFDs, Euclidean domains, polynomial rings, Gauss's lemma, irreducibility criteria (Eisenstein, reduction mod p), Z/nZ and the Chinese remainder theorem, units (Z/nZ)^x, primitive roots, finite fields F_q (existence, uniqueness, Frobenius automorphism, subfields, multiplicative group cyclic), the l-th powers in F_q^x and the index [F_q^x : (F_q^x){l^k}] = l^{min(k, v_l(q-1))}, quadratic residues, the Legendre, Jacobi and Kronecker symbols, quadratic reciprocity (with proof via Gauss sums or Eisenstein's lemma), the characters chi_{-3}, chi_{-4}, chi_q (q odd) = (q/.) *with* $q = (-1)^{(q-1)/2}$ q, their conductors and the values at primes (chi_{-3}(p) = 1 iff p = 1 mod 3; chi_{-4}(p) = 1 iff p = 1 mod 4), Dirichlet characters, their conductors, primitive characters, orthogonality, Gauss sums (basic properties), the ring Z[zeta_3] and Z[i] as Euclidean domains, roots of unity, cyclotomic polynomials, valuations v_p on Q and the p-adic absolute value (definition only; completions in Chapter 7), l-adic valuation of integers and of rationals such as (q-1)/q, (q+1)^2 - a^2, 1 - q^{-2}. Paper hooks: Definition def:terms (valuations of rationals), the table of Section 1 (chi_{-3}, chi_{-4}), Lemma lem:euler, Lemma lem:tamfactor (l-th powers in F_q^x, the class c of the unit part), Lemma lem:twist (q* and chi_q), Lemma lem:crit (cyclic groups Z/r).

Chapter 3. Modules and commutative algebra (`ch:modules`, `ch03_modules.tex`) Scope: modules over commutative rings, submodules, quotients, homomorphisms, exact sequences, the snake lemma, Hom and tensor product (definitions, universal properties, right exactness, flatness of localisation), direct sums and products, free and projective modules, finitely generated modules over a PID (structure theorem; elementary divisors), torsion and torsion-free, the lattice/saturation notion (a submodule L of a free module M is saturated if M/L is torsion-free), Nakayama's lemma, localisation at a prime and at a maximal ideal, local rings, discrete valuation rings (definition, uniformiser, valuation, ideals are powers of the maximal ideal), completions and the rings Z_l, Q_l, Q_l/Z_l, the length of a module over a DVR and over a local ring (with the Jordan-Holder theorem, proved here: decision B2), Fitting ideals (definition via presentations, the case of a finitely generated module over a DVR: Fitt = l^{length}, Fitting ideal of a direct sum, of a quotient), the annihilator, Pontryagin duality for finite abelian groups and for the pair (Z_l-modules of finite type, discrete torsion Z_l-modules) via Hom(-, Q_l/Z_l), the duality

between T and T tensor Q_l/Z_l, perfect bilinear pairings and the Gram determinant (its valuation is independent of basis; for a saturated sublattice L of rank d, the cokernel of L -> Hom(L, O) has order #O/(Gram det)), alternating pairings, adjoints with respect to a pairing, finite free algebras over a DVR (finite flat), their maximal ideals and localisation (T tensor Z_l is a product of local rings, idempotents), generalised eigenspaces of commuting operators, semisimple versus non-semisimple operators, Gorenstein rings (definition: finite free O-algebra T with Hom_O(T, O) free of rank one over T; statement that Gorenstein plus a perfect self-adjoint pairing gives freeness results), the inequality half of Wiles's numerical criterion in its abstract form (the paper's Lemma lem:ddt417 is proved in full in Chapter 20; here prepare every ingredient: Ann, L[p], saturation, Gram determinant, the dual map). Paper hooks: Tamagawa factor as a Fitting ideal (Section 1.2, Lemma lem:tamfactor, Lemma lem:fpr), lengths of Selmer groups (Theorem thm:main), Lemma lem:ddt417, Lemma lem:hecke, Remark rem:free (freeness, Gorenstein), Lemma lem:crit (T/(I+J)), Lemma lem:h0 (Q_l/Z_l-modules, l^k-torsion), Pontryagin duals in Lemma lem:dfg21, the proof of Theorem thm:quant Step 1 (idempotents of T_l, generalised eigenspaces).

Chapter 4. Representations of finite groups and the mod l world (`ch:reps`, `ch04_reps.tex`) Scope: representations of a finite group over a field (any characteristic), subrepresentations, irreducibility, absolute irreducibility (irreducible after extending scalars to the algebraic closure), Schur's lemma in both forms, characters and the character table in characteristic zero (brief, with the RHB connection), Maschke's theorem and its failure in characteristic dividing the group order, semisimple versus non-semisimple modules, the Jordan-Holder theorem (proved in Chapter 3, cited here: decision B2) and composition factors, the Brauer-Nesbitt theorem (two semisimple representations with the same characteristic polynomials of all elements are isomorphic; in characteristic l, the char polys not just traces), restriction and induction, Frobenius reciprocity, Mackey's criterion for irreducibility of induced representations, Clifford theory for a subgroup of index 2 (the restriction of an irreducible is irreducible or splits into two conjugate characters; rho is isomorphic to rho tensor the sign character iff rho is induced from the index-2 subgroup), twisting by one-dimensional characters, the trace of an isomorphism rho -> rho tensor chi is zero when chi is nontrivial, the adjoint representation ad(rho) = End(V) with trace-zero part ad^0, the symmetric square and the identity ad^0 rho = Sym^2 rho tensor det^{-1} for two-dimensional rho, Hom_G(V, W) = (V^vee tensor W)^G, H^0 of ad^0 rho tensor chi as Hom(rho, rho tensor chi) with trace zero, representations of GL_2(F_3) and its subgroups on F_3^2 and on F_9^2 (the determinant-one subgroup C_4 of N_s(F_3) acts reducibly over F_9 as psi + psi^{-1} with psi of order 4, so N_s(F_3) acts as Ind psi), unipotent and "minus unipotent" elements and their fixed spaces, the representation theory needed for images of inertia (cyclic groups of order 2, 3, 4, 6 acting on a rank-two lattice over Z_l, eigenvalues roots of unity, the module structure over Z_l[zeta_3]), representations over Z_l and lattices (a G-stable lattice in a Q_l-representation, reduction mod l, homothetic lattices, Ribet's lemma stated: when the reduction is irreducible all stable lattices are homothetic), irreducible versus absolutely irreducible for mod l representations of Galois groups (foretaste), Dickson's list of subgroups of PGL_2(F_l) (statement). Paper hooks: Proposition prop:ns (induced representations, psi of order 4, Frobenius reciprocity, rho bar = rho bar tensor chi_{-3}), Lemma lem:h0 (trace-zero isomorphisms, ad rho bar tensor chi), Lemma lem:IV and Lemma lem:euler (eigenvalues of inertia, Z_3[zeta_3]-module structure, invariants on the adjoint), Lemma lem:old (Brauer-Nesbitt plus Chebotarev gives rho bar_h = rho bar), Section 4.2 (homothetic lattices: M_{f,lambda} and the dual of T_l E, important_notes.md point 26), the adjoint A_f = ad V and the trace form A_f(1) = Hom(A_f, Z_l(1)).

2.4.2 Part II. Number fields and local fields

Chapter 5. Galois theory, finite and infinite (`ch:galois`, `ch05_galois.tex`) Scope: field extensions, degree, algebraic and transcendental elements, minimal polynomials, splitting fields, separability, normal extensions, Galois extensions, the Galois group, the fundamental theorem of Galois theory, examples (quadratic fields, Q(zeta_n) with Galois group (Z/n)^x, Q(cube root of 2, zeta_3)), Kummer theory for cyclic extensions of degree n when zeta_n is in the base field (and the mod n version: $F^{x/(F}x)^n$ classifies Z/n-extensions), Artin-Schreier (statement), the Galois group of a polynomial and its action on roots, the discriminant and transitivity, cubic and quartic polynomials (the factorisation pattern of a quartic with Galois group a subgroup of GL_2(F_3) acting on lines, as used for the 3-division

polynomial), infinite Galois extensions and the Krull topology, profinite groups, the absolute Galois group G_Q = Gal(Qbar/Q) and G_K, continuous characters and representations of profinite groups, Frobenius elements (definition deferred to Chapter 6 but foreshadowed), the Chebotarev density theorem (statement, with the corollaries used: a Galois representation is determined up to semisimplification by the Frobenius characteristic polynomials at a density-one set of primes; two newforms with a_p congruent for almost all p have isomorphic residual semisimplifications), complex conjugation as an element of G_Q of order two, the quadratic characters of G_Q and Q(sqrt(d)), Q(sqrt(l*)) as the quadratic subfield of Q(zeta_l), restriction of representations to open subgroups. Paper hooks: the hypothesis "absolutely irreducible on Gal(Qbar/Q(sqrt(l)))*", Proposition prop:ns, Lemma lem:h0, Lemma lem:old (Chebotarev and Brauer-Nesbitt), Lemma lem:tamagawa-exp (Kummer theory: l^t-th roots of the Tate parameter), Lemma lem:tamfactor (Kummer classes).

Chapter 6. Algebraic number theory (`ch:ant`, `ch06_ant.tex`) Scope: number fields, rings of integers, integral closure, Dedekind domains and unique factorisation of ideals, norms of ideals, the splitting of primes in extensions (e, f, g; efg = n), ramified, inert, split primes, the discriminant and ramification, quadratic fields in full (splitting of p in Q(sqrt(d)) governed by the Kronecker symbol; the quadratic character attached to Q(sqrt(d)) as a Dirichlet character), cyclotomic fields (splitting of p in Q(zeta_n) governed by p mod n; Q(zeta_3) = Q(sqrt(-3)); Q(sqrt(l*)) inside Q(zeta_l)), Galois extensions: decomposition groups, inertia groups, the Frobenius element Frob_P, its conjugacy class, the Frobenius of an unramified prime in an abelian extension, the Artin map (statement of the main theorems of class field theory over Q: Kronecker-Weber, and the correspondence between Dirichlet characters and one-dimensional representations of G_Q), the class group and units (statements, examples), the rings Z[zeta_3] and Z[i], primes above 3 in Q(zeta_3) (3 = -zeta_3^2 (1 - zeta_3)^2, so Z_3[zeta_3] is a DVR with uniformiser pi = 1 - zeta_3, residue field F_3, pi^2 = -3 up to a unit), primes lambda above l in the coefficient field of a newform, the reduction mod lambda, the density of primes and the Chebotarev theorem in its number field form (Frobenius classes equidistribute), primes inert in Q(sqrt(l)) and complex multiplication by that field (a_p = 0 at inert primes). Paper hooks: Lemma lem:IV (Z_3[zeta_3] as DVR), Lemma lem:euler (Frobenius centralises or inverts sigma according to q mod 3; Z_l[zeta_3] = End_{I_q}(V)), Lemma lem:h0 (the paper's argument: a_p = 0 at primes inert in Q(sqrt(l*)) forces CM; this argument is flawed and the book's proof of lem:h0 in Chapter 11 does not use it, important_notes.md point 5), Lemma lem:crit and Lemma lem:old (rings of integers O of number fields, primes lambda | l), the twist by Q(sqrt(q)) (Section 1.4), the characters chi_{-3}, chi_{-4} of Q(sqrt(-3)) and Q(i).

Chapter 7. Local fields (`ch:local`, `ch07_local.tex`) Scope: absolute values, completions, Q_p and Z_p (as completions and as inverse limits), p-adic expansions, Hensel's lemma (two forms), the structure of Z_p^x and Q_p^x (Q_p^x = p^Z x mu_{p-1} x (1 + pZ_p); for odd p, 1 + pZ_p = Z_p via log/exp), roots of unity in Q_p, the Teichmuller lift, squares and l-th powers in Q_p^x (for l not p, x in Z_p^x is an l-th power iff its reduction is an l-th power in F_p^x), finite extensions of Q_p: ramification index, residue degree, unramified extensions (one of each degree, cyclic, generated by roots of unity of order prime to p, Galois group generated by Frobenius), totally ramified extensions, tamely and wildly ramified extensions, the maximal unramified extension Q_p^ur, the inertia group I_p, the wild inertia group P_p (pro-p), the tame quotient I_p/P_p = prod_{l not p} Z_l(1), the tame character t_l: I_p -> Z_l(1) (via l^k-th roots of p, or of any uniformiser), the relation Frob^{-1} t Frob = $t^{\{p\}\{-1\}}$ (equivalently Frob t Frob^{-1} = t^p) in Gal(Q_p^tame/Q_p), the absolute Galois group G_{Q_p} and its filtration, Kummer theory over Q_p: H^1(G_{Q_p}, mu_{l^k}) = $Q_p{}^{x/(Q_p x)^{\{l\}k}}$ (foreshadowing Chapter 17), the Kummer class of an element and its restriction to inertia (the restriction of the Kummer class of u in Q_p^x to I_p is v_p(u) times the tame character), the unramified quadratic extension of Q_p, the ramified quadratic extensions Q_p(sqrt(p*)) and their characters, the local behaviour of quadratic characters of G_Q (chi_q is unramified at p not q, ramified at q), the Weil group W_{Q_p} and Weil-Deligne representations (definition: (r, N) with r a representation of the Weil group and N nilpotent, Phi N Phi^{-1} = p^{-1} N for the geometric Frobenius Phi = Frob^{-1}, equivalently Frob N Frob^{-1} = p N for the arithmetic Frobenius Frob, which is the book's Frob: Appendix B, B-C2(f)), the inertia invariants and the local L-factor and conductor of a Weil-Deligne representation (definitions; the Artin conductor a(r) = codim V^{I} + Swan), special cases: unramified, tamely ramified through a finite cyclic quotient, the special (Steinberg) representation sp(2) and its unramified twists (conductor

*exponent 1), the twist of sp(2) by a ramified quadratic character (conductor exponent 2), representations with finite image of inertia of order 2, 3, 4, 6 (conductor exponent 2 for q >= 5; also for types IV and IV^* at q = 2, where the image of inertia is cyclic of order exactly 3 and tame: this case is settled, not open, see important_notes.md point 43). Paper hooks: Lemma lem:tamagawa-exp (Tate curve over the unramified quadratic extension, l^t-th roots of u), Lemma lem:tamfactor (tame character, Frob^{-1} t Frob = t^{q^{-1}}, Kummer class kappa, its restriction to I being n times the tame character, Z_l(1)/(q-1)), Lemma lem:IV and Remark rem:types (tame inertia is cyclic; wild at 2 and 3), Lemma lem:twist(iii) (unramified twist of Steinberg, ramified twist has conductor exponent 2), Lemma lem:euler (eigenvalues of Frobenius on inertia invariants), Remark rem:wild.

2.4.3 Part III. Curves and elliptic curves

Chapter 8. Algebraic curves, function fields and Riemann surfaces (`ch:curves`, `ch08_curves.tex`) Scope: affine and projective space, varieties over a field, the projective plane and homogeneous polynomials, smoothness and singular points, plane curves, the function field of a curve, points over extensions and the Galois action on points, the genus (via Riemann-Roch; for smooth plane curves (d-1)(d-2)/2), divisors, principal divisors, linear equivalence, the Picard group, Riemann-Roch (statement, with the standard consequences: a genus-one curve with a rational point is a plane cubic), morphisms of curves, the degree of a morphism (three equivalent descriptions: degree of the function field extension, number of preimages of a general point, the pullback on divisors), separable and inseparable, ramification of morphisms, the Riemann-Hurwitz formula, compact Riemann surfaces versus smooth projective curves over C (statement of the equivalence), differentials, holomorphic differentials and the genus, the Jacobian of a curve (definition as Pic^0, as a complex torus over C, as an abelian variety: statements), the Abel-Jacobi map, the Tate module and the first cohomology of a curve (H^1(X, Z) for the Riemann surface; its rank 2g; the intersection pairing / cup product; H^1(X, Z_l) as H^1(X, Z) tensor Z_l for the moment, with a remark on etale cohomology and on the Tate module of the Jacobian being H^1 dual), the cup product pairing is perfect and alternating on H^1(X, Z), correspondences on a curve and their action on H^1 and on the Jacobian, the adjoint of a correspondence under the cup product is the transpose correspondence. Paper hooks: the modular parametrisation X_0(N) -> E and its degree (Section 1), H^1(X_0(N), Z_l) with the cup product pairing (Section 4.1, Lemma lem:hecke), J_0(N) and its torsion (Remark rem:free), Tate modules of Jacobians.

Chapter 9. Elliptic curves over fields (`ch:ellcurves`, `ch09_ellcurves.tex`) Scope: Weierstrass equations (general and short), the quantities b_2, b_4, b_6, b_8, c_4, c_6, Delta, j, the discriminant and smoothness, changes of coordinates (u, r, s, t) and the transformation of c_4, c_6, Delta, the group law (chord-tangent; associativity via Riemann-Roch or stated), the group of points over any field, isogenies (definition, degree, separable isogenies and kernels, the dual isogeny, deg of multiplication by m is m^2), the structure of E[m] (Z/m x Z/m when char does not divide m), division polynomials (definition, degree (m^2 - 1)/2 for odd m, the 3-division polynomial of degree 4 and its roots as the x-coordinates of the four lines of E[3]), the Weil pairing (statement; nondegenerate, alternating, Galois-equivariant; so det of the Galois action on E[m] is the cyclotomic character), the Tate module T_l E = lim E[l^n] (free Z_l-module of rank 2) and V_l E, the Galois action (definition of rho_{E,l} and rho bar_{E,l} = action on E[l]), endomorphism rings (Z, an order in an imaginary quadratic field, or an order in a quaternion algebra), complex multiplication (definition; CM curves over Q have j among the 13 values; a_p = 0 at primes inert in the CM field), elliptic curves over C (lattices, the Weierstrass p-function, the uniformisation theorem, periods, the period lattice Lambda_E of a curve over Q and its area, isomorphism classes = lattices up to homothety = points of the upper half plane modulo SL_2(Z)), elliptic curves over finite fields (the Frobenius endomorphism, a_p = p + 1 - #E(F_p), the Hasse bound, the characteristic polynomial X^2 - a_p X + p of Frobenius on T_l E, the eigenvalues alpha, beta with alpha beta = p), the Mordell-Weil theorem (statement), rank and torsion, Mazur's theorem on torsion (statement), isogenies over Q and isogeny classes (finite; Cremona's class labels), rational l-isogenies and reducibility of E[l] (E[l] reducible iff E has a rational l-isogeny; rational l-torsion points; the structure Z/3 + mu_3), quadratic twists (definition E^(d): y^2 = x^3 + d a_2 x^2 + d^2 a_4 x + d^3 a_6 for a model y^2 = x^3 + ..., effect on c_4, c_6, Delta (d^2, d^3, d^6), same j, a_p(E^(d)) = chi_d(p) a_p(E) for p not dividing 2d, rho_{E^(d), l} =

rho_{E,l} tensor chi_d), the L-function L(E, s) (definition by Euler product, with the local factors at bad primes; analytic continuation as a statement, see Chapter 19), the Cremona and LMFDB labelling of curves (11a1, 121a1, 539c1, 980i1, 726a1, 99A1 explained). Paper hooks: everything; in particular Lemma lem:twist(ii) (models and invariants of twists), Lemma lem:h0 (CM and a_p = 0), Proposition prop:ns (3-division polynomial), Definition def:terms (v_q(j)), the twist of good reduction E' and a_q(E') in the table of Section 1, Conjecture conj:eis (torsion, isogeny class, Z/3 + mu_3), Section 1.1 (V, T, rho, rho bar).

Chapter 10. Elliptic curves over local fields: reduction, Kodaira types, Tate curves (`ch:reduction`, `ch10_reduction.tex`) Scope: minimal Weierstrass models over Z_p (existence, uniqueness up to change of coordinates, criterion v_p(Delta) < 12 or v_p(c_4) < 4 for p >= 5; Kraus's conditions at 2 and 3 as a remark, citing Kraus 1989, Acta Arith. 54, not Kraus90), reduction mod p, good, multiplicative (split and nonsplit) and additive reduction, the criterion via v_p(Delta) and v_p(c_4), the reduction map on points and E_0(Q_p), E_1(Q_p), the formal group (statement of what is needed: E_1 has no prime-to-p torsion), potentially good and potentially multiplicative reduction (criterion v_p(j) >= 0 or < 0), the semistable reduction theorem, the Neron model (definition by the universal property; existence as a statement; the special fibre, its identity component and the component group Phi_p, a finite etale group scheme over F_p), the geometric component group Phi_p(F_p bar) and the Tamagawa number c_p = #Phi_p(F_p), Kodaira symbols I_0, I_n, II, III, IV, I_0^, *I_n^*, IV^, *III^*, II^ *with the special fibres drawn, the orders of the geometric component groups (n; 1; 2; 3; 4; 4; 3; 2; 1) and the structure (Z/n; trivial; Z/2; Z/3; (Z/2)^2; Z/4 or (Z/2)^2 according to the parity of n; Z/3; Z/2; trivial), Tate's algorithm (full statement, with the steps; for p >= 5 the type is determined by v_p(Delta) of a minimal model for potentially good reduction: 0, 2, 3, 4, 6, 8, 9, 10 for I_0, II, III, IV, I_0^*, IV^, *III^*, II^, *and by v_p(Delta) = n or n + 6 for I_n and I_n^*), the cubic T^3 + a_2 T^2 + a_4 T + a_6 criterion for I_0^* at odd p, the conductor exponent f_p (0, 1, 2 for good, multiplicative, additive at p >= 5; Ogg's formula f_p = v_p(Delta) + 1 - m_p with m_p the number of components; the wild part at 2 and 3 and the bounds f_2 <= 8, f_3 <= 5), the conductor N = prod p^{f_p}, the Tate curve (E_q: the curve with parameter q in Q_p^x, |q| < 1; the uniformisation Q_p bar$^{x/q}$Z; j(E_q) with v_p(j) = -v_p(q); every curve with split multiplicative reduction is a Tate curve, nonsplit after the unramified quadratic extension; the Tate parameter of a curve with potentially multiplicative reduction, v_p(q_E) = -v_p(j_E) = n for type I_n and for type I_n^* at odd p; the unit part u = q_E p^{-n} and its class in F_p^x modulo l^k-th powers), the Galois module E_q[l^t] generated by mu_{l^t} and an l^t-th root of q (statement here, representation-theoretic consequences in Chapter 11), Kraus's theorem on the semistability defect: for p >= 5 and potentially good reduction the order of the image of inertia on T_l E (l not p) is 12/gcd(v_p(Delta), 12) (so 1, 6, 4, 3, 2, 3, 4, 6 for the types I_0, II, III, IV, I_0^*, *IV^*, III^, *II^*), the effect of quadratic twists on reduction (the paper's Lemma lem:twist (i) and (ii) proved in full: unramified twists change nothing; the twist by Q(sqrt(q*)) at an odd q exchanges I_n and I_n^*, I_0 and I_0^*, and for q >= 5 exchanges II and IV^*, III and III^*, IV and II^*; v_q(Delta) goes up by 6 modulo 12), the wild primes 2 and 3 (what changes: the Kodaira type no longer determines the inertia action; a type I_n^* at 2 may have potentially good reduction; the paper's Remark rem:wild), Cremona's tables: what is tabulated (Kodaira symbols, c_p, Delta, conductor), PARI's Kodaira codes (1 for I_0; n + 4 for I_n and -(n + 4) for I_n^*, *for n >= 1 only; -1 for I_0^*; +-2, +-3, +-4 for II/II^, *III/III^*, IV/IV^*). Paper hooks: Definition def:terms (t_l(E,q), v_q(j_E), potentially multiplicative primes), the table of Section 1, Section 1.4 (component groups, the exchange of types, the weight w_l), Lemma lem:twist, Lemma lem:tamagawa-exp (Tate curve over the unramified quadratic extension), Lemma lem:tamfactor (Tate parameter, unit part), Lemma lem:IV and Remark rem:types (Kraus's defect), Remark rem:wild, Proposition prop:forms, Section sec:wild, the appendix (Kodaira codes).

Chapter 11. Galois representations of elliptic curves (`ch:galrep`, `ch11_galrep.tex`) Scope: rho_{E,l}: G_Q -> Aut(T_l E) = GL_2(Z_l) and rho bar_{E,l}: G_Q -> GL_2(F_l), continuity, det rho_{E,l} = chi_l the l-adic cyclotomic character (via the Weil pairing), det rho bar_{E,3} = chi_{-3} cuts out Q(sqrt(-3)), complex conjugation has determinant -1 and trace 0, the criterion of Neron, Ogg and Shafarevich (rho_{E,l} unramified at p not l iff good reduction at p; potentially good iff the image of inertia is finite), the local representation at a prime of good reduction (unramified, Frobenius char poly X^2 - a_p X + p), at a prime of multiplicative reduction (the Tate curve: rho|$\{G\{$Q_p$\}\}$ is an

extension of an unramified character eps of order <= 2 by eps chi_l, the matrix form with eps kappa in the corner, kappa the Kummer class of the Tate parameter (not kappa alone: important_notes.md point 42), non-split; the paper's Lemma lem:tamagawa-exp proved in full: rho mod l^t unramified at p iff l^t | n_p, split or nonsplit; DDT Theorem 3.1(e)), at a prime of additive potentially good reduction with p >= 5 (image of inertia cyclic of order 2, 3, 4 or 6 by Kraus; eigenvalues roots of unity of that order; the paper's Lemma lem:IV in full: for types IV and IV^* the image of I_q in Aut(T_3 E) is cyclic of order 3, acting on E[3] as a nontrivial unipotent, E[3]^{I_q} a line, tame, Serre conductor exponent 1; and Remark rem:types: for II, II^* the image mod 3 is minus a unipotent, for I_n^* a ramified quadratic character times a unipotent, for I_0^, *III, III^* of order 2 or 4, with E[3]^{I_q} = 0 and Serre conductor exponent 2 in the II/II^/I_n^ cases), at a prime of potentially multiplicative additive reduction (rho = rho' tensor chi_q with rho' from the twist of multiplicative reduction), at l (the representation is crystalline/comes from a finite flat group scheme when E has good reduction at l; the Deligne-Fontaine description of rho bar|*{D_l} for weight 2 and level prime to l: either irreducible, "fundamental characters of level 2", and then a_l = 0 mod l, or reducible with an unramified quotient on which Frobenius acts by the unit root, so a_l mod l is determined by rho bar*/{D_l}; statements with reference to Edixhoven), the Artin conductor of rho bar and the Serre conductor N(rho bar) (definition: prod p^{dim rho bar - dim rho bar^{I_p} + Swan}; N(rho bar_{E,l}) divides the conductor of E (Carayol); N(rho bar) is prime to every prime where rho bar is unramified), the images of rho bar_{E,l}: Serre's open image theorem (statement), the classification of possible images for E/Q without CM (Zywina's lists for l = 3, 5, 7, 11, 13; for l = 3: GL_2(F_3), N_s(F_3), N_ns(F_3), Borel subgroups and their subgroups; the Sutherland labels 3B, 3B.1.1, 3Cs.1.1, 3Ns, 3Nn, 3Cs, 3Cn and what each means), reducible representations and rational isogenies (the Eisenstein case; Z/3 + mu_3 as the image 3Cs.1.1), irreducible versus absolutely irreducible for rho bar_{E,l} (with complex conjugation present, irreducible over F_l implies absolutely irreducible: because the image contains an element with distinct eigenvalues 1 and -1 for l odd; explain carefully), absolute irreducibility restricted to Gal(Qbar/Q(sqrt(l*))) and the dihedral/induced case: the paper's Proposition prop:ns proved in full (for l = 3: the image is N_s(F_3) iff rho bar is induced from Q(sqrt(-3)) iff rho bar = rho bar tensor chi_{-3} iff the 3-division polynomial factors as two irreducible quadratics), the paper's Lemma lem:h0 proved in full (H^0(Q, ad rho bar tensor chi_l bar) nonzero iff rho bar = rho bar tensor chi_l bar, which forces l = 3; the length h_l(E) of H^0(Q, ad T(1) tensor Q_l/Z_l) as the largest k with rho mod l^k = (rho tensor chi_l) mod l^k; the bound by the determinant argument: such an isomorphism gives chi_l^2 = 1 mod l^k on G_Q, which forces l = 3 and k = 1, so h_l(E) = 0 for l >= 5 and h_3(E) <= 1, with equality iff rho bar_{E,3} = rho bar_{E,3} tensor chi_{-3}, for every E with rho bar irreducible, CM or not; the paper's finiteness argument via a_p = 0 at inert primes confuses chi_l with chi_{l}* and is not used, important_notes.md point 5), the semisimplification and Brauer-Nesbitt applied to rho bar_{E,l} (the isomorphism class of an irreducible rho bar_{E,l} is determined by the a_p mod l for p in a density-one set), Serre's conjecture for rho bar_{E,l} (statement: it comes from a newform of weight 2 and level N(rho bar) when E has good reduction at l; this is the optimal-level theorem used in the paper; the general statement is in Chapter 15), the Galois representation of a quadratic twist, the computation of mod l images in practice (division polynomials, Sutherland's algorithm; statement). Paper hooks: Section 1.1 (V, T, rho, rho bar, A_f = ad V, delta_p = dim V^{I_p}), Theorem thm:main hypothesis, Lemma lem:tamagawa-exp, Lemma lem:IV, Remark rem:types, Lemma lem:euler (inertia images and invariants), Lemma lem:old (the local analysis at p = l, at p || N ramified or unramified), the optimal-level theorem paragraph of Section 3, Proposition prop:ns, Lemma lem:h0, Section sec:anomaly, the Eisenstein case (Section sec:eis), Conjecture conj:eis.

2.4.4 Part IV. Modular forms and modular curves

Chapter 12. Modular forms and modular curves (`ch:modforms`, `ch12_modforms.tex`) Scope: the upper half plane and the action of SL_2(R) and of SL_2(Z), the modular group, fundamental domains, the congruence subgroups Gamma_0(N), Gamma_1(N), Gamma(N) and their indices, cusps and their widths, elliptic points, the modular curves Y_0(N) and X_0(N) as Riemann surfaces (compactification by cusps, the genus formula for X_0(N), tables of small genera: X_0(11) has genus 1), modular forms and cusp forms of weight k for Gamma_0(N) (definitions: holomorphy at the cusps via q-expansions),

the spaces M_k(Gamma_0(N)) and S_k(Gamma_0(N)), q-expansions and the q-expansion principle, Eisenstein series (brief), the valence formula and dimension formulas (dim S_2(Gamma_0(N)) = genus of X_0(N)), weight 2 cusp forms as holomorphic differentials on X_0(N) (f(z) dz), Fourier coefficients and the Ramanujan-Petersson bound (statement), the Petersson inner product on S_k (definition, positive definite), the Sturm bound and integrality (forms with integer coefficients, S_2(Gamma_0(N); Z), S_2(Gamma_0(N); Z_l)), modular forms with character (nebentypus) briefly, the Atkin-Lehner involutions w_d and w_N (definition, action on q-expansions is not simple, action on cusps), the modular curve X_0(N) as a moduli space (pairs (E, C) with C cyclic of order N; the cusps as degenerate pairs; the Q-structure of X_0(N) as a statement, with reference), examples: S_2(Gamma_0(11)) is one-dimensional spanned by eta(z)^2 eta(11 z)^2, the form of 11a1; the first coefficients of f_{11a1}; physics aside: Fourier series on a quotient. Paper hooks: S_2(Gamma_0(N)), the newform f, q-expansions and congruences coefficientwise (Section 1, Lemma lem:crit), integral forms S_2(Gamma_0(N); Z) (Lemma lem:crit), the Petersson norm <f, f> (Section 1, Zagier, Watkins), the Atkin-Lehner involution w_N (Section 4.1, the pairing delta hat), X_0(N) as a curve (Section 1), the genus and dim S_2 (background to multiplicity one).

Chapter 13. Hecke operators, newforms and twists (`ch:hecke`, `ch13_hecke.tex`) Scope: the Hecke operators T_n on M_k(Gamma_0(N)) via double cosets and via the explicit formula on q-expansions (a_n(T_p f) = a_{pn}(f) + p^{k-1} a_{n/p}(f) for p not dividing N; U_p for p | N: a_n(U_p f) = a_{pn}(f)), commutativity, multiplicativity (T_{mn} = T_m T_n for coprime m, n; T_{p^{j+1}} = T_p T_{p^j} - p^{k-1} T_{p^{j-1}}, with k the weight), the Hecke algebra T = Z[T_n : n >= 1] inside End(S_2) (finite free over Z; of rank dim S_2 when acting faithfully, including the U_p), self-adjointness of T_n for p not dividing N with respect to the Petersson product and the non-self-adjointness of U_p, eigenforms and normalised eigenforms (a_1 = 1, then T_n f = a_n f), the eigencharacter phi_f: T -> Z or O, the coefficient field and its ring of integers, oldforms and newforms (the maps f(z) -> f(dz) for d | N/M; the old subspace; the new subspace as its orthogonal complement; Atkin-Lehner theory: the new subspace has a basis of newforms which are eigenforms for all T_n and U_p, with multiplicity one), the main theorem of Atkin-Lehner (strong multiplicity one: a newform is determined by a_p for p in any set of density one; two newforms of the same level with the same T_p eigenvalues for almost all p are equal), eigenvalues of U_p on oldforms (for h a newform of level M with p not dividing M, U_p on the span of h(z), h(pz) has characteristic polynomial X^2 - a_p(h) X + p, the p-stabilisations h - beta h(pz) and h - alpha h(pz); on the span of h(z), h(pz), h(p^2 z) the eigenvalue 0 appears with the depleted form h - a_p h(pz) + p h(p^2 z); for p || M, U_p acts by a_p(h) on h and the depleted form h - a_p h(pz) has eigenvalue 0; for p^2 | M, a_p(h) = 0 (Atkin-Lehner, Theorem 3) and U_p kills h), the non-semisimplicity of U_p when 0 is a repeated eigenvalue, the Sigma-depleted form f^Sigma = sum_{(n, Sigma) = 1} a_n q^n, its level M^Sigma = M prod_{p in Sigma} p^{delta_p} with delta_p = 2, 1, 0 for good, multiplicative, additive p (explain why), the structure of S_2(Gamma_0(N)) tensor Qbar as a direct sum of the spans of h(dz) over newforms h of level M | N and d | N/M, generalised eigenspaces of T and their relation to newforms, T tensor Q as a product of fields times the old part, the orthogonal complement f^perp of a newform f and its T-stability, twisting by Dirichlet characters (the Atkin-Li theorem (their Theorem 3.1, in the form used by the paper): for a newform g of level M and the quadratic character chi of conductor q with q^2 not dividing M, g tensor chi is a newform of level lcm(M, q^2)), the twist map iota: h -> h tensor chi on the whole space and its compatibility with Hecke operators T_p for p not dividing the conductor (iota intertwines T_p with chi(p) T_p), the integral structure: S_2(Gamma_0(N); Z) is a lattice, the pairing S_2(Gamma_0(N); Z) x T -> Z, (g, t) -> a_1(t g), is perfect (Ribet 1983 Theorem 2.2; Abbes-Ullmo Lemma 2.1; proved here via the q-expansion principle or stated with the proof idea), consequences (T is free of rank dim S_2 over Z; Hom(T, Z) = S_2(Z); a form is determined by its pairing with T), congruences between eigenforms modulo primes lambda (a_n(f) = a_n(g) mod lambda for all n versus for all n prime to a set), the recursion for a_{p^k} and multiplicativity, so that congruences at primes propagate to all n, the L-function of a newform L(f, s) = sum a_n n^{-s} = prod (1 - a_p p^{-s} + p^{1-2s})^{-1} prod_{p | N} (1 - a_p p^{-s})^{-1}, its analytic continuation and functional equation (statement), the Rankin-Selberg method and the formula <f, f> = (explicit factor) L(Sym^2 f, 2) (statement with the correct normalisation; the precise form in Chapter 19). Paper hooks: Lemma lem:crit (T, phi_f, I, J, f^perp, the perfect pairing, T/(I + J) cyclic of

order r), Lemma lem:old (oldforms, U_p eigenvalues, stabilisations, depleted forms, a_p = 0 for p^2 | M, multiplicativity and the recursion), Lemma lem:twist(iii) (Atkin-Li), Theorem thm:twist (the twist map iota, generalised eigenforms, g^perp -> f^perp), the proof of Theorem thm:quant Step 1 (strong multiplicity one, stabilisations, U_q eigenvalue 0 with multiplicity one, idempotents), Section 4.1 (f^Sigma, M^Sigma, delta_p), Section 1 (congruence number r_E, Petersson norm).

Chapter 14. Modular curves over Q, Jacobians and the modular degree (`ch:moddeg`, `ch14_moddeg.tex`) Scope: X_0(N) as a smooth projective curve over Q (statement; its cusps; its points over Q), the Jacobian J_0(N) (abelian variety over Q of dimension g; J_0(N)(C) = S_2(Gamma_0(N))^vee/H_1(X_0(N), Z)), the Hecke action on J_0(N) by correspondences (T_p as the correspondence (E, C) -> sum (E/D, (C + D)/D)), the Eichler-Shimura relation (T_p) * = *pi_p + hat pi_p on J_0(N) mod p for p not dividing N, with pi_p the p-power Frobenius endomorphism and hat pi_p = p pi_p^{-1} its dual, the Verschiebung (statement, with the consequence a_p(f) = trace of Frobenius; notation of Appendix B, B-C2(g)), the Hecke action on H^1(X_0(N), Z) and on H_1, the cup product pairing on H^1(X_0(N), Z) (perfect, alternating), adjoints: the adjoint of T_n under the cup product is w_N T_n w_N (equivalently T_n is self-adjoint for the pairing <x, w_N y>; proof or precise reference), so the modified pairing <x, w_N y> is perfect, alternating, and all T_n and U_p are self-adjoint for it (this is the pairing delta hat*! of DFG04 in weight 2 and trivial character), the Tate module T_l J_0(N) = H_1(X_0(N), Z) tensor Z_l and H^1(X_0(N), Z_l) (dual; etale cohomology as a statement), the modularity theorem (statement: every elliptic curve E/Q of conductor N has a newform f_E of weight 2 on Gamma_0(N) with a_p(f_E) = a_p(E) for all p; history: Wiles, Taylor-Wiles, Breuil-Conrad-Diamond-Taylor), the abelian variety J_f attached to a newform (quotient of J_0(N) by I_f J_0(N); Shimura's A_f, written J_f in the book because A_f is the paper's adjoint, Appendix B, B-C12(c)), for rational f the elliptic curve quotient, the modular parametrisation phi_E: X_0(N) -> E (sending the cusp infinity to O), the X_0(N)-optimal curve (unique in its isogeny class; equivalently the kernel of J_0(N) -> E is connected; also called the strong Weil curve; its relation to the Cremona label "1" is a convention that was checked only up to conductor 400000 and for some classes beyond; the paper's statement about optimality in the tables), the modular degree deg phi_E (definition as the degree of the morphism of curves; equivalently via the composition E^vee -> J_0(N) -> E being multiplication by deg phi_E; properties: multiplicative under nothing in particular, computability), the Manin constant c_E (definition, with the sign of omega_E chosen so that c_E > 0; Manin's conjecture c_E = 1; known results, each to be checked against the source: c_E is an integer (Edixhoven); if a prime p divides c_E then p = 2 or p^2 | N (Mazur78 for p not dividing 2N, Abbes-Ullmo for p^2 not dividing N); Cremona's verification; the role of the Manin constant in the formulas), the formula of Zagier and Watkins: deg phi_E = 4 pi^2 c_E^2 <f, f> / vol(Lambda_E), with phi_E^* omega_E = c_E 2 pi i f dz (for a general parametrisation phi, deg phi vol(Lambda_E) = 4 pi^2 c_phi^2 <f, f>), up to the precise normalisation (state the exact formula with the Manin constant; Zagier 1985, Watkins 2002), the congruence number r_E (definition as in the paper: the largest r such that f = g mod r for some integral g of level N orthogonal to f; the congruence ideal in T; Lemma lem:crit of the paper stated here and cited: it is proved in full in Chapter 15, decision B17 of Section 7.5, and restated in Chapter 20), the theorem of Agashe, Ribet and Stein (deg phi_E | r_E, and v_p(deg phi_E) = v_p(r_E) for p^2 not dividing N; the example 99A1 with deg = 4, r = 12; statement with proof idea: the modular degree is the congruence number computed in the Hecke algebra of the Jacobian's quotient, and the two differ by a factor controlled by the primes p with p^2 | N), multiplicity one and the Gorenstein property (the maximal ideal m of T tensor Z_l attached to (f, l); J_0(N)[m] is two-dimensional when l does not divide 2N and rho bar is irreducible: Mazur for prime level, Ribet-Stein Theorem 3.5 for J_1(N) with the exception of Serre weight l, the pull-back J_0(N)[m] -> J_1(N)[m'] with kernel in the Shimura subgroup which is Eisenstein; consequence (Tilouine): T_m is Gorenstein and T_l J_0(N)_m = H^1(X_0(N), Z_l)_m is free of rank two over T_m; the paper's Remark rem:free; multiplicity one is not needed for the equality in Lemma lem:hecke and in (eq:identity) of Theorem thm:main, nor for the first inequality of Theorem thm:quant to be an equality, when l is not in S_f: important_notes.md points 39, 56, 73, 74), the Eisenstein ideal, the cuspidal subgroup and the Shimura subgroup (definitions; Mazur's theorem for prime level; what "Eisenstein" means for a maximal ideal; why reducible rho bar is the Eisenstein case), the parametrisations by Shimura curves and the theorem of Ribet and Takahashi (statement: the ratio of the degrees of the

parametrisations by X_0(N) and by the Shimura curve of discriminant D is, up to primes where rho bar is reducible, the product of the c_q or n_q over q | D; the precise statement), the abc connection (deg phi_E « N^{2 + epsilon} is equivalent to a form of abc; statement with reference to ARS12 Section 2), Watkins's conjecture 2^{rank E(Q)} | deg phi_E and the theorem of Calegari and Emerton on curves of odd modular degree (statements), Mazur's theorem on J_0(N)(Q)*tors for prime N (statement), computing the modular degree (Watkins's method via the Petersson norm; Cremona's method via the period lattice; PARI's ellmoddegree). Paper hooks: the whole of Section 1 (optimal curve, modular degree, Petersson norm, Zagier, Watkins, Agashe-Ribet-Stein, congruence number), Lemma lem:crit (congruence number and congruence ideal), Lemma lem:hecke, Remark rem:free, Section 4.1 (H^1(X_0(N), Z_l), the pairing delta hat*!, Atkin-Lehner), the remark after Theorem thm:qual (99A1), Section 1.5 (Ribet-Takahashi), Section sec:two (Watkins, Calegari-Emerton), Conjecture conj:eis and Section sec:eis (Eisenstein ideal, cuspidal subgroup, Mazur), Open question 4.

Chapter 15. Galois representations of modular forms, level lowering and congruences (`ch:modgalrep`, `ch15_modgalrep.tex`) Scope: the Galois representation rho_{f, lambda}: G_Q -> GL_2(O_lambda) attached to a newform f of weight 2 (Eichler-Shimura construction via the Tate module of J_f, Shimura's abelian variety, Appendix B, B-C12(c); Deligne for higher weight as a statement), its properties: unramified outside N l, trace of Frob_p = a_p(f), det = chi_l, odd, irreducible; the residual representation rho bar_{f, lambda} and its semisimplification; the comparison with rho_{E, l} when f = f_E (they are isomorphic; the lattice M_{f, lambda} is homothetic to the dual of T_l E, which is T_l E(-1), when rho bar is irreducible, by Ribet's lemma on lattices, and equals H^1(E, Z_l) for the optimal curve with no hypothesis, as proved in Chapter 14: important_notes.md points 26, 40; ad is insensitive to the dual and the twist), local-global compatibility (Carayol; Langlands-Deligne): the restriction of rho_f to G_{Q_p} corresponds to the local component pi_p of f under the local Langlands correspondence; the local components of a weight 2 newform of trivial character: unramified principal series at p not dividing N (Frobenius eigenvalues alpha_p, beta_p), unramified twist of Steinberg at p || N (rho|*{G{Q_p}} is an extension of an unramified character eps by eps chi_l with eps(Frob_p) = a_p = +-1; DDT Theorem 3.1(e)), and at p^2 | N either a ramified twist of Steinberg (then the twist of f by the corresponding character has level with p exactly once, and a_p(f) = 0), or a ramified principal series, or supercuspidal; in all cases with p^2 | N one has a_p(f) = 0 and V^{I_p} = 0 (dim V^{I_p} = delta_p = 2, 1, 0), the conductor of rho_f equals N (Carayol), the Weil-Deligne representation at p and the local L-factor and epsilon factor (definitions recalled from Chapter 7), the non-split extension at a multiplicative prime has a unique stable line (used in Lemma lem:old), the local representation at l for l not dividing N (crystalline, finite flat, weight 2: Fontaine-Laffaille range; the description of rho bar|*{D_l} and the determination of a_l mod lambda from it (Deligne; Fontaine; Edixhoven Section 2): if rho bar|*{D_l} is irreducible then a_l = 0 mod lambda; if reducible, rho bar|*{D_l} has an unramified quotient on which Frobenius acts by a_l mod lambda), the Serre conductor and Serre weight of rho bar (the weight is 2 for rho bar*{E,l} with good reduction at l), Serre's conjecture (statement; the strong form: rho bar odd irreducible arises from a newform of weight k(rho bar) and level N(rho bar); the theorem of Khare and Wintenberger; for the paper only the "optimal level" direction for a representation already known to be modular is used), the optimal level theorem of Ribet, Carayol and Diamond in the form DDT Theorem 3.15: for l odd and rho bar irreducible and modular, absolutely irreducible on Gal(Qbar/Q(sqrt(-3))) if l = 3, rho bar arises from a newform of weight 2 and level N(rho bar) l^{delta(rho bar)} with nebentypus of order prime to l, delta = 0 when rho bar|*{D_l} is good; the specialisation to rho bar*{E', l} with l not dividing the conductor: it arises from a newform of weight 2 on Gamma_0(N(rho bar)) (the paper's argument that the nebentypus is trivial: det = chi_l bar forces a character trivial mod lambda of order prime to l), Ribet's level-lowering theorem (the case q || N with rho bar unramified at q: rho bar arises from level N/q; the original statement and proof idea via Shimura curves and the Ihara/Mazur principle), Mazur's principle, Carayol's lemma, the level-raising theorem of Ribet (statement: rho bar arises from level N q (q not dividing N l) iff a_q^2 = (q + 1)^2 mod lambda; the level-raising factor (1 - alpha/beta q^{-1})(1 - q^{-1}) (1 - beta/alpha q^{-1}) = (q - 1)((q + 1)^2 - a_q$^{2)/q}$3 ... give the exact form used in the paper, namely L_q^{nv}(A_f, 1)^{-1} = (1 - alpha/(beta q))(1 - 1/q)(1 - beta/(alpha q)) = ((q - 1)/q) ((q + 1)^2 - a_q$^{2)/q}$2), congruences between newforms: the congruence criterion (the paper's Lemma lem:crit; prove it here in full with the perfect pairing from Chapter 13 and the algebra from Chapter 3,

and cross-reference Chapter 14 and Chapter 20), the oldform lemma (the paper's Lemma lem:old proved in full: given a newform h of level M | N congruent to f at almost all primes, with p not dividing M for every p || N at which rho bar is unramified, the span of the h(dz) contains an eigenform g for the full Hecke algebra, orthogonal to f, congruent to f at ALL n; the prime-by-prime analysis: p not dividing l N, p = l, p^2 | N, p || N unramified, p || N ramified), the proof of the paper's Theorem thm:qual (cases (a), (b), (c)) as a worked application, including the examples 121a1, 539c1, 980i1 with the specific congruences, the results of Ribet-Takahashi, Pollack-Weston and Kim-Ota on the congruence ideal and the Tamagawa exponent t_q (statements only, with their hypotheses, as in the paper's Section 1.5 and Remark rem:ko), the Taylor-Wiles method and R = T theorems (one-page outline: deformation rings, the numerical criterion, patching; precise statement of what DFG04 Theorem 3.7 needs; details in Chapter 20). Paper hooks: Sections 3 (all of it: Lemma lem:crit, Lemma lem:old, the optimal-level theorem, cases (a), (b), (c), the example, the remark), Section 1.5 (history, Theorem thm:qual, Remark rem:ko), Section 4.2 (M_{f,lambda} homothetic to the dual of T_l E), Lemma lem:euler (local components: Steinberg and its twists), Lemma lem:twist(iii), Theorem thm:twist and Theorem thm:quant (twist map, eigenforms, levels), Theorem thm:dfg14 (the level-raising factor), Remark after Theorem thm:quant.

2.4.5 Part V. Cohomology, Selmer groups and L-functions

Chapter 16. Homological algebra and group cohomology (`ch:cohomology`, `ch16_cohomology.tex`) Scope: chain complexes, cohomology of a complex, exact sequences of complexes and the long exact sequence (with the connecting homomorphism constructed), the snake lemma, injective and projective resolutions and derived functors in outline (Ext and Tor as statements; Ext^1 classifies extensions), group cohomology H^n(G, M) for a group G and a G-module M (definition via the standard complex of inhomogeneous cochains; H^0 = M^G; H^1 = crossed homomorphisms modulo principal ones; H^1(G, M) = Hom(G, M) when G acts trivially; H^2 and extensions), functoriality in G and in M, the long exact sequence in M, restriction, corestriction, inflation, the inflation-restriction exact sequence 0 -> H^1(G/H, M^H) -> H^1(G, M) -> H^1(H, M)^{G/H} -> H^2(G/H, M^H), the conjugation action of G/H on H^1(H, M) (the formula (sigma . c)(h) = sigma c(sigma^{-1} h sigma), and its independence of choices), the cohomology of a finite cyclic group (H^1(, M) = ker(Norm)/ (t - 1)M, H^2 = M^G/Norm), the cohomology of Z and of Z_l (procyclic): H^1(, M) = M/(t - 1)M for a procyclic group acting continuously on a finitely generated Z_l-module (precise statement and proof), continuous cohomology of profinite groups with discrete, finite, and compact (Z_l-module of finite type) coefficients (definition as direct limits over open normal subgroups for discrete M; inverse limits for compact M; the pitfalls), the cohomology of a pro-l group versus a group of order prime to l (H^i(G, M) = 0 for i > 0 when G is profinite of order prime to l and M is an l-primary torsion module or Z_l-module of finite type; so cohomology of inertia factors through the maximal pro-l quotient when the coefficients are l-adic), H^1(G, M)_{tors} and its Fitting ideal (compute H^1(, M) = M/(t - 1)M for M = ad T with t acting unipotently as in the paper: full worked example reproducing the computation (t - 1)e = 0, (t - 1)h = -2n' e, (t - 1)f = n' h - n'^2 e, (t - 1)M = n'(Z_l e + Z_l h), H^1 = (Z_l/n')^2 + Z_l), cup products (definition on cochains; H^1 x H^1 -> H^2; graded commutativity), Pontryagin duality for cohomology (statement), Tate cohomology of finite groups (brief; H^1(G, M) is killed by #G, so H^1 of a group of order 2 is killed by 2, used for the real place), the Hochschild-Serre spectral sequence (statement; only the five-term exact sequence is used), the cohomology of Z/2 acting on a Z_l-module for l odd (vanishes), Galois cohomology as the special case G = G_K (notation H^i(K, M) = H^i(G_K, M); the next chapter). Paper hooks: Lemma lem:tamfactor (every step: inflation-restriction, H^1(, M) = M/(t - 1)M, the conjugation action of G/I, the explicit (t - 1)M computation, the Fitting ideal of the invariants), Lemma lem:dfg21 (Pontryagin duals, H^1(R, T^D) killed by 2), Lemma lem:h0 (H^0 of ad T(1) tensor Q_l/Z_l and its l^k-torsion), the definition of Tam_p (Section 1.2), Lemma lem:fpr.

Chapter 17. Galois cohomology, local and global, and Selmer groups (`ch:galcoh`, `ch17_galcoh.tex`) Scope: Galois cohomology of a field H^i(K, M) = H^i(G_K, M) with M a discrete G_K-module, Hilbert's theorem 90 (H^1(K, Kbar^x) = 0) and Kummer theory as H^1(K, mu_n) = $K^{x/(K}x)^n$ (proved), the Kummer map and the Kummer class of an element, H^1(K, Z_l(1)) = lim $K^{x/(K}x)^{\{l}k\}$ for a local field K (and the identification of Z_l(1) = T_l mu = lim mu_{l^k}), the Brauer

group and H^2(K, mu_n) (statement), local Galois cohomology for K = Q_p: finiteness of H^i(Q_p, M) for finite M, the local Euler-Poincare characteristic formula (statement), Tate local duality (H^i(Q_p, M) x H^{2 - i}(Q_p, M^(1)) -> H^2(Q_p, mu) = Q/Z perfect; statement with reference), unramified cohomology H^1_ur(Q_p, M) = H^1(G_{Q_p}/I_p, M^{I_p}) = ker(H^1(Q_p, M) -> H^1(I_p, M)) (via inflation-restriction), H^1_ur(Q_p, M) = M^{I_p}/(Frob_p - 1) M^{I_p} (for M a Z_l-module of finite type or a Q_l-vector space, by the procyclic computation), the annihilator of H^1_ur under Tate duality is H^1_ur of the dual (for p not l, finite M of order prime to p), the cohomology of the real place (H^1(R, M) is killed by 2; for l odd and M an l-adic module it vanishes), the cohomology of G_Q and of G_{Q,S} (the Galois group of the maximal extension unramified outside S), finiteness of H^1(G_{Q,S}, M) for finite M (statement), the Kummer sequence for an elliptic curve 0 -> E[n] -> E -> E -> 0 and the Kummer map E(K)/n E(K) -> H^1(K, E[n]), the classical n-Selmer group and the Shafarevich-Tate group (definitions, the exact sequence 0 -> E(Q)/n -> Sel_n(E/Q) -> Sha[n] -> 0, finiteness of Sel_n, the weak Mordell-Weil theorem as a consequence; this is the model for all Selmer groups), Selmer structures: a family of local conditions L_p in H^1(Q_p, M) for all p, unramified at almost all p, and the Selmer group H^1_L(Q, M); the dual Selmer structure L^ via local duality; the Greenberg-Wiles formula (the theorem of Wiles on the ratio #H^1_L(Q, M)/#H^{1-{L}}(Q, M^(1)) = #H^0(Q, M)/#H^0(Q, M^*(1)) times the product over p of #L_p/#H^0(Q_p, M); statement with reference; this is the origin of the exact sequence of DFG04 Lemma 2.1), the Poitou-Tate exact sequence (statement), comparing Selmer groups for two sets of local conditions: the exact sequence 0 -> H^1_L -> H^1_{L'} -> prod_p L'p/L_p -> ... with the cokernel controlled by the dual Selmer group (prove the elementary part; the paper's Lemma lem:dfg21 is this comparison with L = Bloch-Kato conditions and L' = no conditions at Sigma, for which the dual term is H^0(Q, W^D)* when H^1_f(Q, V^D) = 0), the cohomology of Q_l/Z_l-modules versus Z_l-modules: the exact sequences relating H^1(K, T), H^1(K, V), H^1(K, W) for T a lattice, V = T tensor Q_l, W = V/T (the divisible and torsion parts; H^1(K, T){tors} = image of H^0(K, W) ... prove the needed cases), H^1(K, T)_tors and H^1(K, W) for K = Q_p, p not l: H^1(Q_p, T) is finitely generated over Z_l with torsion H^0(Q_p, W)/(divisible) ... state exactly what is true and used (the paper: for p in Sigma, H^1_f(Q_p, T^D) is the torsion of H^1(Q_p, T^D), a finite group, because H^1_f(Q_p, V^D) = 0), the example of the Tate curve: H^1(Q_p, Z_l(1)) and the Kummer class of the Tate parameter, with its image in H^1(G/I, Z_l(1)^I) = Z_l(1)/(q - 1) = Z/l^{v_l(q - 1)} being the class of the unit part of the Tate parameter in F_p^x tensor Z_l (proved in full; this is the end of the paper's Lemma lem:tamfactor). Paper hooks: Lemma lem:tamagawa-exp and Lemma lem:tamfactor (Kummer theory, the class c' of the unit part, H^1(G/I, Z_l(1)^I)), Lemma lem:dfg21 and the paragraph after it (H^1_f(Q_p, V^D) = H^1_ur = (V^D){I_p}/ (Frob_p - 1) vanishes because det(1 - Phi_p | (V^D){I_p}) = L_p(A_f, 1)^{-1} is nonzero, with Phi_p the geometric Frobenius (the paper writes Frob_p there; Appendix B, B-C2(d)); the torsion of H^1(Q_p, T^D); H^1(R, T^D) killed by 2), Remark rem:ns (the exact sequence with H^0(Q, W^D) nonzero), the Selmer groups of Theorem thm:main.

Chapter 18. Bloch-Kato Selmer groups, Tamagawa factors and the adjoint motive (`ch:blochkato`, `ch18_blochkato.tex`) Scope: l-adic representations of G_Q (continuous, on finite-dimensional Q_l-vector spaces), lattices T, the divisible module W = V/T, Tate twists V(n) = V tensor Q_l(1)^{tensor n}, linear duals V^vee = Hom(V, Q_l) and the Cartier/Kummer dual V^D = V^vee(1) = Hom(V, Q_l(1)), T^D = Hom(T, Z_l(1)), W^D = $V^{D/T}D$ (linear dual ^vee, Pontryagin dual ^*, Kummer dual ^D: Appendix B, B-C4), the trace form on ad^0 V and the self-duality (ad^0 V)^vee = ad^0 V, so A(1) = Hom(A, Z_l(1)) = T^D for A = ad^0 T when l is odd (2 invertible), geometric representations and their local behaviour: at p not l, the Weil-Deligne representation, the inertia invariants and the local L-factor L_p(V, s) = det(1 - Phi_p p^{-s} | $V^{\{I_p\}}$){-1} with Phi_p = Frob_p^{-1} the geometric Frobenius (definition; convention fixed in Appendix B, B-C2(d); state the paper's convention that a good prime has eigenvalues alpha/beta, 1, beta/alpha on ad V, hence L_p(A_f, s) with s = 1 giving (1 - alpha/(beta p))(1 - 1/p)(1 - beta/(alpha p))), at l: de Rham, crystalline, semistable representations (Fontaine's rings B_dR, B_cris, B_st as black boxes; definitions of D_cris and the crystalline condition; the Tate module of an elliptic curve with good reduction at l is crystalline; the adjoint of a crystalline representation is crystalline; statements only, with references to Fontaine, Berger's survey, and DFG04 Section 2), the Bloch-Kato local conditions: H^1_f(Q_p, V) = H^1_ur(Q_p, V) = ker(H^1(Q_p, V) -> H^1(I_p, V)) for p not l, H^1_f(Q_l, V) = ker(H^1(Q_l, V) -> H^1(Q_l, V tensor B_cris)), H^1_f(R, V) = 0, H^1_g and H^1_e

(definitions; for the paper only H^1_f matters), the Bloch-Kato conditions on T and on W: H^1_f(Q_p, T) = preimage of H^1_f(Q_p, V) under H^1(Q_p, T) -> H^1(Q_p, V); H^1_f(Q_p, W) = image of H^1_f(Q_p, V) under H^1(Q_p, V) -> H^1(Q_p, W), the global Bloch-Kato Selmer groups H^1_f(Q, V), H^1_f(Q, T), H^1_f(Q, W), their relations, the Sigma-imprimitive Selmer group H^1_Sigma(Q, W) of DFG04 Section 2.1 (no condition at p in Sigma, the Bloch-Kato condition at l, unramified elsewhere), the duality between the Bloch-Kato conditions for V and V^D under Tate local duality (H^1_f(Q_p, V)^perp = H^1_f(Q_p, V^D) for p not l; at l the same by Bloch-Kato; statements), the Fontaine-Perrin-Riou Tamagawa factor at p not l: Tam_p(T) = Fitt_{Z_l}((H^1(I_p, T)*{tors}*)^{G{Q_p}}), the invariants inside the Fitting ideal (important_notes.md point 37) ... state the definition precisely as the paper does (the Fitting ideal of the G_{Q_p}-invariants of the torsion of H^1(I_p, T); FPR94 I.4.2.2; DFG04 p. 708), the alternative description as [H^1_f(Q_p, T) : H^1_ur(Q_p, T)] type index or as the order of H^1_f(Q_p, T)/(something) and the equivalence (Lemma lem:fpr of the paper: Fitt H^1_f(Q_p, T^D) = L_p(V, 1)^{-1} (or the appropriate twist) times Tam_p(T^D), and Tam_p(T^D) = Tam_p(T): state and prove what can be proved with the tools at hand, namely the finite group H^1_f(Q_p, T^D) = H^1(Q_p, T^D)_{tors} sits in an exact sequence with H^0(Q_p, W^D) and the unramified part, whose orders are the Euler factor and the Tamagawa factor; give the full argument for p not l, following FPR94 I.4 and the proof of DFG04 Theorem 2.15), the Tamagawa factor of an elliptic curve (Tam_p(T_l E) is the l-part of the Tamagawa number c_p = #Phi_p(F_p), as a sanity check: this is the classical fact and the paper's final remark in the proof of Lemma lem:tamfactor), the full computation of the paper's Lemma lem:tamfactor (Tam_q(ad T) at a prime of multiplicative reduction: v_l = v_l(n) + min(v_l(n), v_l(q - 1), v_l(c(u))), with c(u) the class of the reduction of the unit part u (the paper's c; Appendix B, B-C14), proved in full, every step, using Chapters 16 and 17; the consequence v_l(Tam_q(ad T)) >= t_l(E, q) with equality unless q = 1 mod l, l | n and the unit part of the Tate parameter is an l-th power), the Tamagawa factor at an additive prime (for q not l and l odd, v_l(Tam_q(ad T)) = 0 at every potentially good additive prime, by inflation-restriction through the finite image of inertia, important_notes.md point 64; at a potentially multiplicative additive prime the formula of Lemma lem:tamfactor applies to the Tate parameter, point 18; this answers the first half of the paper's Open question 3), the Tamagawa factor at a prime of good reduction is trivial, the Tamagawa number conjecture of Bloch and Kato (statement for a motive in outline: the l-part of the special value of the L-function equals the product of the order of the Selmer group, the Tamagawa factors, the periods and the orders of H^0 terms; the shape for the adjoint motive of a modular form as in DFG04 Theorem 0.1: L(ad f, 1)/(period) has l-adic valuation equal to the length of the Selmer group plus Tamagawa factors; precise statement of DFG04 Theorem 0.1 and Theorem 0.2 with the hypothesis l not in S_f), the exact sequence of DFG04 Lemma 2.1 (the paper's Lemma lem:dfg21: statement, hypotheses H^0(Q, V^D) = H^1_f(Q, V^D) = 0, the four terms, the Pontryagin duals; proof via the Greenberg-Wiles / Poitou-Tate comparison of Chapter 17), DFG04 Theorem 2.7 (H^0(Q, V^D) = H^1_f(Q, V^D) = 0 for l not in S_f; statement; it is the vanishing of the adjoint Selmer group with Q_l coefficients, which comes from Taylor-Wiles), H^0(Q, W^D) = H^0(Q, ad T(1) tensor Q_l/Z_l) and the paper's Lemma lem:h0 (restate and connect; proved in Chapter 11), the corrected exact sequence of Remark rem:ns when H^0(Q, W^D) is nonzero (length H^1_Sigma = length H^1_f + sum of local terms - h_l(E)), the adjoint motive of a modular form and of an elliptic curve (ad^0 of the Tate module; its L-function L(ad f, s) = L(Sym^2 f, s + 1) in the normalisation used; its Hodge numbers in one sentence; critical values; Flach's theorem that the Selmer group of Sym^2 is finite), the symmetric square and the Petersson norm (connection, details in Chapter 19). Paper hooks: Section 1.2 (the definition of Tam_p(M), the Bloch-Kato Selmer group), Theorem thm:main and its proof in Section 4.3 (every step of the assembly: Theorem thm:dfg37, Theorem thm:dfg27, Lemma lem:dfg21, Lemma lem:fpr, Lemma lem:h0), Lemma lem:tamfactor (in full), Remark rem:ko(1), Remark rem:ns, Section sec:tam, Open questions 2 and 3, Conjecture conj:euler (the term h_l(E)).

Chapter 19. L-functions: Euler products, local factors, the adjoint and the symmetric square (`ch:lfunctions`, `ch19_lfunctions.tex`) Scope: Dirichlet series and their convergence, Euler products, the Riemann zeta function (Euler product, analytic continuation and functional equation stated with the proof idea via theta functions, the physics-reader connection to Mellin transforms), Dirichlet L-functions (definition, Euler product, functional equation stated, L(1, chi) nonzero and Dirichlet's theorem stated), the Dedekind zeta function of a number field and its factorisation for abelian fields (statement),

the L-function of an elliptic curve (local factors: (1 - a_p p^{-s} + p^{1 - 2s})^{-1} at good p, (1 - a_p p^{-s})^{-1} at multiplicative p with a_p = +-1, 1 at additive p; the conductor; the completed L-function Lambda(E, s) and the functional equation with root number, as a consequence of modularity; L(E, s) = L(f_E, s)), L-functions of l-adic Galois representations (the local factor L_p(V, s) = det(1 - Phi_p p^{-s} | V^{I_p})^{-1} for p not l, with Phi_p = Frob_p^{-1} the geometric Frobenius, Appendix B, B-C2(d); at p = l via D_cris; the Weil-Deligne definition so that the factor is independent of l; the conjecture and theorem (for modular forms) that these are compatible), the L-function of a representation with finite image of inertia (examples: the three Kodaira cases), the adjoint and the symmetric square L-functions (definitions: L(ad f, s) attached to ad^0 rho_f, L(Sym^2 f, s) attached to Sym^2 rho_f; the identity ad^0 rho = Sym^2 rho tensor det^{-1} and L(Sym^2 f, s) = L(ad f, s - 1) in the paper's normalisation, with care about which normalisation each reference uses: Shimura, Gelbart-Jacquet, DFG04, PARI), the naive adjoint L-function (the Euler product of sum a_n^2 n^{-s} after removing the zeta factor, i.e. the Rankin-Selberg convolution L(f x f, s)/zeta^(N)(s - 1), where L(f x f, s) = zeta^(N)(2s - 2) sum a_n^2 n^{-s}: the naive local factor L_p^{nv}(A_f, s) equals the true one when delta_p > 0 and is 1 when delta_p = 0; explain why: the naive factor at p is built from a_p alone, and at an additive prime a_p = 0), the Rankin-Selberg method (unfolding; the theorem of Rankin and Shimura: for a newform f of weight 2 and level N, <f, f> = (explicit constant involving N, pi, and the local factors at bad primes) times L(Sym^2 f, 2); state the exact formula as in DFG04 Lemma 2.12 or Shimura 1976, with the correction factors at the bad primes, since the paper's Section 1 says the Petersson norm is the special value up to the local factors at bad primes), the true local factors at bad primes (the paper's table in Section 1 derived in full: this is Lemma lem:euler; the computation of (ad V)^{I_q} and the Frobenius eigenvalues for each Kodaira type at q >= 5: I_n (line on which the geometric Frobenius has eigenvalue q^{-1}, factor 1 - q^{-1-s}), I_n *(same, but the naive factor is 1, so e_l = v_l(q^2 - 1)), II/II^/IV/IV** (image of inertia of order 3 or 6, invariant line spanned by sqrt(-3) in Z_l[zeta_3] = End_{I_q}(T) (for V, read Q_l tensor Q(zeta_3) = End_{I_q}(V): important_notes.md point 12), Frobenius acts by chi_{-3}(q), factor 1 - chi_{-3}(q) q^{-s}, e_l = v_l(q - chi_{-3}(q))), III/III^* (order 4, eigenvalues +-i, invariant line spanned by the generator tau of the inertia image itself (not "the diagonal element", which need not exist over Q_l when l = 3 mod 4: important_notes.md point 34), Frobenius acts by chi_{-4}(q), e_l = v_l(q - chi_{-4}(q))), I_0^ *(V = V' tensor chi_q with V' unramified, ad V unramified with eigenvalues alpha/beta, 1, beta/alpha, L_q(A_f, 1)^{-1} = ((q - 1)/q)((q + 1)^2 - a_q(E')^2)/q2, e_l = v_l(q - 1) + v_l((q + 1)^2 - a_q(E')^2))), the wild primes 2 and 3 (the Kodaira type does not determine the factor; examples: type IV or IV^* at 2 has P_2(X) = 1 + 2X (a theorem, important_notes.md point 43); the paper's Remark rem:wild claims that type II at 2 has P_2 = 1 and that a type I_n^* at 2 with potentially good reduction has a cubic P_2, and both claims are false in general (336f1 has type II at 2 and P_2 = 1 + 2X; 1232l1 has type II^* at 2 and a cubic P_2; point 50); at every bad prime P_q is one of four shapes, 1 - X, the I_0^* cubic, 1 - cqX with c = +-1, or 1 (point 69)), how PARI computes these (lfunsympow(E, 2), lfuneuler; the polynomial P_q(X) with L_q(Sym f, s) = P_q(q^{-s})^{-1}, and e_l = v_l(P_q(q^{-2})/(1 - a_q^2 q^{-2}))), the definition of e_l(E, q) (the paper's Definition def:terms) and the failure of its invariance under quadratic twist (t_l(E, q) is twist-invariant, e_l(E, q) is not when the twist is ramified at q and changes the type there: 11a1 has e_3(E, 11) = 0, its twist 121d2 by chi_{-11} has type I_5^* at 11 and e_3 = 1; the paper's sentence "both terms are invariant under quadratic twist" is false for e_l, important_notes.md point 15), Flach's theorem on the finiteness of the Selmer group of Sym^2 (statement), special values and the Bloch-Kato conjecture for the adjoint (one page; the shape "L-value / period = Selmer times Tamagawa", the connection to the modular degree via Petersson norm), Deligne's conjecture on critical values in one paragraph, the functional equation and analytic continuation of L(Sym^2 f, s) (Shimura; Gelbart-Jacquet; statements). Paper hooks: Section 1 (Petersson norm, Zagier, Watkins, Rankin, Shimura, Flach, naive versus true adjoint L-function, DFG04 Lemma 2.12), Section 1.1 (delta_p, L_p(A_f, s), the naive factor, PARI's lfunsympow, Definition def:terms, the table), Lemma lem:euler (in full), Remark rem:wild, Theorem thm:dfg14 and the paragraph after it (the naive factors: level-raising factor at a good prime, 1 - p^{-2} at a multiplicative prime), Lemma lem:fpr (L_p(A_f, 1)^{-1} = L_p(A_f(1), 0)^{-1} is an l-adic integer of valuation e_l at an additive prime and v_l(1 - p^{-2}) at a multiplicative one), the appendix (rule(E, l)).

2.4.6 Part VI. Congruences, the modular degree and the paper

Chapter 20. Hecke algebras, congruence ideals and the theorem of Diamond, Flach and Guo (`ch:congruence`, `ch20_congruence.tex`) Scope: Hecke algebras as finite flat Z-algebras and their localisations T_m at maximal ideals m (the maximal ideal attached to (f, l): generated by l and the T_n - a_n(f); T tensor Z_l is the product of the T_m; the m-component of S_2(Gamma_0(N); Z_l) via idempotents; which newforms belong to m: those with eigencharacter congruent to phi_f mod m, i.e. with rho bar_g = rho bar_f), congruence modules and congruence ideals (for a surjection pi: T -> O with kernel wp and I = Ann(wp): the congruence ideal eta = pi(I); the congruence module O/eta = T/(wp + I); Hida's definition and the equality with the "largest modulus of a congruence f = g with g in f^perp" (the paper's Lemma lem:crit, in the integral and the Z_l form; Chapter 15 proves it in full (decision B17, Section 7.5), Chapter 20 restates it as a theorem of the theory of congruence modules and adds the commutative algebra viewpoint), the theorem of Hida on congruence modules and the adjoint L-value (statement: for weight 2 and level N, the congruence module of f is related to L(ad f, 1) / period up to primes dividing 2 N and the Euler factors; the precise statement in Hida 1981 and its refinement in DFG04), Wiles's numerical criterion (DDT Theorem 5.3 / Lemma 4.17; the number 5.3 is not verified, important_notes.md B26; in full: let O be a DVR, T a finite free O-algebra, pi: T -> O surjective, wp = ker pi, eta = pi(Ann wp); then T is a complete intersection iff ... ; the paper only needs the inequality half with a module L: the paper's Lemma lem:ddt417 stated and proved in full, in the paper's generality (L free of finite rank with a perfect O-bilinear pairing for which T is self-adjoint, L[wp] free of rank d with Gram determinant Delta: eta^d is contained in Delta O, with equality if L is free over T), with every ingredient from Chapter 3 (saturation, L[wp]^perp = L[I], the cokernel of L[wp] -> Hom(L[wp], O) has order #O/Delta)), the application to the modular degree (the paper's Lemma lem:hecke: as the paper states it, v_l(deg phi_E) = v_l(r_E) >= v_lambda(eta^empty_{f, lambda}) with equality if H^1(X_0(N), Z_l)*m is free over T_m; in fact v_l(deg phi_E) = v_lambda(eta^empty*{f, lambda}) for every l not dividing 2N with no freeness or multiplicity-one hypothesis, because the pairing on M_{f,lambda} = phi_E* ^{H}1(E, Z_l) has Gram determinant (deg phi_E)^2 (important_notes.md points 39, 56, 73, confirmed against DFG04; proved in Chapter 14), so (eq:identity) of Theorem thm:main is an equality whenever l is not in S_f; proof in full; the paper's Remark rem:free on when freeness holds, which the equality no longer needs), the lattices and ideals of Diamond, Flach and Guo (DFG04 Sections 1.5.3, 1.6.2, 1.7.3 as read for weight 2 and trivial character: the integral structure M(M^Sigma, 1)*{!, lambda} = H^1(X_0(M*Sigma), Z_l) with its Hecke action; the lattice M*{f^Sigma, lambda} = kernel of the kernel of the eigencharacter of f^Sigma (that is, the sublattice killed by ker phi_{f^Sigma}), of rank two; the pairing delta hat_! = cup product composed with the Atkin-Lehner involution w_{M^Sigma}, perfect and alternating, with all T_n and U_p self-adjoint; the naive Sigma-finite congruence ideal eta^Sigma_{f, lambda} = the ideal of values of delta hat_! on M_{f^Sigma, lambda}, equivalently the Gram determinant of the pairing on M_{f^Sigma, lambda} is (eta$^{\mathrm{Sigma})}$2 up to a unit (explain why a rank-two alternating pairing has Gram determinant a square)), DFG04 Proposition 1.4(c) (the paper's Theorem thm:dfg14: eta^Sigma = eta^empty prod_{p in Sigma} L_p^{nv}(A_f, 1)^{-1} when rho bar_f is irreducible; explain the three cases: additive primes change nothing, a good prime contributes Wiles's level-raising factor, a multiplicative prime contributes 1 - p^{-2}; proof idea via Ihara's lemma / the degeneracy maps, with reference), the exceptional set S_f (DFG04 (31): lambda | 2N or rho bar not absolutely irreducible on Gal(Qbar/Q(sqrt(l*)))), DFG04 Theorem 3.7 (the paper's Theorem thm:dfg37: for l not in S_f and Sigma a finite set of primes not containing l such that M_{f, lambda} is minimally ramified outside Sigma, H^1_Sigma(Q, W) is finite of length v_lambda(eta^Sigma); the meaning of "minimally ramified outside Sigma" and why the set of primes dividing N qualifies), DFG04 Theorem 2.7 (vanishing of H^0(Q, V^D) and H^1_f(Q, V^D)), DFG04 Theorem 0.1 and 0.2 (the Tamagawa number conjecture for the adjoint motive; statement), the Taylor-Wiles method in outline (deformation rings R_Sigma, the map R_Sigma -> T_Sigma, the numerical criterion, patching; the Selmer group H^1_Sigma(Q, W) as the tangent space / dual of the cotangent space of R_Sigma, which is why its length is the congruence ideal: the identity "length of Selmer group = valuation of congruence ideal" is the statement R = T plus complete intersection; DDT Chapters 2 to 5 as the reference), the assembly: the paper's proof of Theorem thm:main reproduced step by step with every input identified (Lemma lem:hecke,

Theorem thm:dfg14, Theorem thm:dfg37, Theorem thm:dfg27 plus Lemma lem:dfg21 plus Lemma lem:h0, Lemma lem:fpr, Definition def:terms, Lemma lem:tamfactor; the cancellation of the naive factors at multiplicative primes; the twist argument for I_n^ primes), the twist monotonicity theorem (the paper's Theorem thm:twist proved in full) and the proof of the paper's Theorem thm:quant in full (Steps 1 to 4, with the Hecke congruence ideal eta$^{\mathrm{Hecke(g}}$S) and its comparison with the cohomological one; the first inequality of thm:quant is an equality whenever l is not in S_f, with no multiplicity-one hypothesis, by applying the equality form of (eq:identity) to E and to E_S, important_notes.md point 74), the remarks after Theorem thm:quant (why the argument fails for potentially good types II, III, IV), the Kim-Ota formula and the Tamagawa exponent (statement, Remark rem:ko(2)), Open questions 1 and 2. Paper hooks: Section 4 in full (Sections 4.1, 4.2, 4.3, 4.4: Theorem thm:dfg14, Lemma lem:ddt417, Lemma lem:hecke, Remark rem:free, Theorem thm:dfg37, Theorem thm:dfg27, Lemma lem:dfg21, Lemma lem:fpr, Lemma lem:h0, the proof of Theorem thm:main, Remark rem:ko, Theorem thm:twist, the proof of Theorem thm:quant and the remark after it), Section 1 (Hida, DFG04, congruence ideal), Lemma lem:crit, Remark rem:ns.

Chapter 21. Reading the paper: a guided tour (`ch:paper`, `ch21_paper.tex`) Scope: a complete, self-contained commentary on the paper, in the paper's order, for a reader who has the cards of Chapters 1 to 20 at hand. For each definition, lemma, proposition, theorem, remark, conjecture and section of the paper: restate it precisely (in the paper's notation, which the book shares), say which chapters supply the tools, walk through the proof line by line expanding every step the paper compresses (the paper is terse; the tour is not), and point out the exact place where each hypothesis is used. Cover: the abstract and Section 1 (the local terms, Definition def:terms, the table; Theorem thm:main and the reading of its hypotheses; Conjecture conj:euler and the weights w_3; the component group inequalities (eq:main), (eq:twist) and Proposition prop:forms with its proof; the history and Theorem thm:qual, Theorem thm:quant, Proposition prop:ns, Conjecture conj:eis); Section 2 (Lemma lem:tamagawa-exp, Lemma lem:IV, Remark rem:types, Lemma lem:twist, Lemma lem:euler, Remark rem:wild, Lemma lem:tamfactor); Section 3 (Lemma lem:crit, Lemma lem:old, the optimal-level theorem, cases (a), (b), (c), the example with 121a1, 539c1, 980i1, the remark on l | N and 99A1); Section 4 (Theorem thm:dfg14, Lemma lem:ddt417, Lemma lem:hecke, Remark rem:free, Theorem thm:dfg37, Theorem thm:dfg27, Lemma lem:dfg21, Lemma lem:fpr, Lemma lem:h0, the proof of Theorem thm:main, Remark rem:ko, Theorem thm:twist, the proof of Theorem thm:quant, the remark after it); Section 5 (the proof of Proposition prop:ns, Remark rem:ns); Section 6 (the data and protocol; how to read each table; the calibration argument of Section sec:sharp; the Tamagawa factor test of Section sec:tam; the isolated terms of Section sec:iso; the two corollary forms; the 3Ns boundary; the wild primes and w_3; out-of-table curves; the Eisenstein case); Section 7 (the prime 2 and Watkins's conjecture); the open questions; the appendix (the PARI function rule(E, l) explained line by line, with the Kodaira codes). Also: a dependency diagram of the paper's results (which lemma feeds which theorem), a table "hypothesis -> where it is used -> what fails without it" (l odd; l not dividing 2N; absolute irreducibility over Q(sqrt(l*)); X_0(N)-optimality; multiplicity one, which the paper uses for the equality in thm:main and thm:quant but which is not needed there when l is not in S_f, important_notes.md points 39, 73, 74), a list of the exact statements imported from the literature with their numbering (DFG04 Theorem 0.1, 0.2, 2.7, 3.7, Proposition 1.4, Lemma 2.1, Theorem 2.15, (31), (57), p. 708; DDT Theorem 3.1(e), 3.15, Lemma 4.17; FPR94 I.4.2.2; ARS12 Theorem 2.2; Atkin-Li Theorem 3.1; Atkin-Lehner Theorem 3; Ribet83 Theorem 2.2; AU96 Lemma 2.1; Ribet-Stein Theorem 3.5; Kim-Ota Theorem 1.1, Definition 2.12; Kraus 1990; Zywina 2015; Sutherland 2016), and worked verifications of the numerical examples (121a1: e_3(E, 11) = v_3(12) = 1, deg = 6; 539c1: e_3(E, 7) = v_3(48) = 1, t_3(E, 7) = 1, deg = 2^5 3^2; 980i1: deg = 2^6 3^2 5; 726a1: 3Ns, the cancellation; 99A1: deg 4 versus r = 12). Paper hooks: all.

Chapter 22. Computation, data and the statistical protocol (`ch:computation`, `ch22_computation.tex`) Scope: PARI/GP for the reader who has never used it (installation, ellinit, ellglobalred and its output (conductor, minimal model change, the Kodaira symbols and Tamagawa numbers per prime, the local data structure with the Kodaira codes), ellap, ellan, elltors, ellisomat, ellmoddegree (and its memory needs), elldivpol, lfuncreate / lfunsympow / lfuneuler and the conventions for the symmetric square Euler factor, ellinit over Q_q and the Tate parameter (elltatepairing is not it: explain how to extract the Tate parameter / its valuation from v_q(j) and the unit part via the Tate curve

algorithm or via the q-adic curve ellinit(E, O(q^k)); note that ellinit(E, q) reduces E modulo q and is not the q-adic curve), valuation, factor), Cremona's tables and the ecdata repository (what the files contain: allcurves, allbsd, degphi (the modular degrees; not "alldegphi"), allgens, alllabels, allisog, the Tamagawa data as tabulated there; the files contain no per-prime Kodaira symbols, which PARI recomputes; the mod l images with Sutherland's labels (2Cs, 3B.1.1, 3Cs.1.1, 3Ns, 3Nn, 5B, ...), optimality and the Cremona label "1"; the LMFDB and its labels versus Cremona's), reading a Cremona label (11a1, 121a1, 121c, 121d, 539c1, 980i1, 726a1, 99A1 and the capitalisation convention), the paper's data protocol (working set N <= 300000, hold-out 300000 < N <= 400000, extension 400000 < N <= 500000, out-of-table curves with 500000 < N <= 4 million; pre-registration: the statement was written down before the hold-out was looked at; what "nothing was adjusted after the hold-out test" means; which statements were blind-tested and which were only confirmed; the role of the out-of-table set for Conjecture conj:euler), elementary statistics of the tests (what a "violation" is; the calibration argument of the paper's Section sec:sharp explained with the numbers: among bound-zero curves 53.2% have v_3 = 0, so under independence the 867,578 curves with a positive bound would give about 605,000 violations, there are none; what this does and does not show; equality rates and the mean excess as descriptive statistics; the base rates by Kodaira type; a short, honest discussion of multiple comparisons and of why a conjecture found on the working set must be tested on held-out data; the reader's statistics background is enough for this), the PARI function rule(E, l, iii) of the paper's appendix reproduced and explained line by line (ellglobalred's R[4] and R[5], the Kodaira code k (n + 4 for I_n and -(n + 4) for I_n^* for n >= 1 only; I_0^* is -1), the omitted factor r = P_q(q^{-2}) with the multiplicative correction, the exclusion of type III/III^* unless iii = 1, the weights w_3 at q = 3, the Tamagawa exponent from v_q(j), the correction ns3(E) via the factorisation of the 3-division polynomial), worked sessions: 121a1 returns [1, 1]; 726a1 returns [2, 2] with and without iii; 539c1; 980i1; 99A1 (where ellmoddegree gives 4 and the congruence number 12 is not given by PARI directly), computing e_l(E, q) by hand from the Kodaira type for q >= 5 and checking against lfunsympow, computing t_l(E, q), computing the mod 3 image class 3Ns via elldivpol, the twist lemma check (twist_lemma_check.gp: how to write it), reproducing a row of the paper's tables on a small range (a GP script that loops over curves of conductor up to 1000 from ellsearch (which needs PARI's elldata package) or from a downloaded ecdata file and counts violations and equalities), how the modular degree is computed (Watkins's algorithm in outline; Cremona's; why it is expensive), the repository named in the paper's appendix and what each script does (descriptive list only), a note on reproducibility practice (pinned data commits, re-running from a clean clone). Paper hooks: Section 1 (PARI's lfunsympow), Remark rem:wild, Lemma lem:euler (confirmed against PARI), Section 6 in full (sec:data and its subsections), the appendix, Section 1.4 (Cremona's tables and optimality), Proposition prop:ns (3-division polynomial), Section sec:eis (image labels 3Cs.1.1, 3B.1.1).

2.4.7 Back matter

Appendix A. Prerequisites: what we assume and where to look (`ch:prereq`, `chA_prereq.tex`) Short refresher cards for the assumed background: real analysis (completeness, uniform convergence, power series), linear algebra (eigenvalues, Jordan form, bilinear forms, Gram determinants, tensor products of vector spaces), complex analysis (Mobius transformations, the upper half plane, residues, Laurent expansions, uniform limits of holomorphic functions), elementary group theory and representations as in RHB, elementary probability and statistics (what a base rate is, what a hold-out set is). Each item with a pointer to where in the book it is first used.

Appendix B. Notation, conventions and index of symbols (`ch:notation`, `chB_notation.tex`) A table of every symbol used in the paper and the book, in the paper's notation (V, T, rho, rho bar, A_f, ad, Sym, delta_p, L_p, L_p^nv, P_q, e_l(E, q), t_l(E, q), n_q, q_E, u, c(u) (the paper's c in lem:tamfactor), c', c_E (the Manin constant; never a bare c), Tam_p, Fitt, H^1_f, H^1_Sigma, W, T^D, V^D, W^D, S_f, eta^Sigma_{f, lambda}, M_{f, lambda}, delta hat_!, phi_f, I, J, r_E, deg phi_E, Phi_q, Phig_q, c_q, w_l(E, q), w_3(E), h_l(E), chi_{-3}, chi_{-4}, chi_q, q, *l*, E^(q), E_S, N_s, N_ns, 3Ns, Kodaira symbols and PARI codes, ...) with the chapter where it is defined. Conventions: arithmetic versus geometric Frobenius; the normalisation of L(Sym^2 f, s) versus L(ad f, s); the sign conventions for Tate twists; "length" means Z_l-length; v_l is the l-adic valuation normalised by v_l(l) =

1.

Bibliography (`refs.bib`): every work cited in the paper, with full bibliographic data, plus the standard textbooks used as references (Silverman AEC and ATAEC, Diamond-Shurman, Serre's Local Fields and Abelian l-adic representations, Neukirch, Milne's notes on class field theory and on elliptic curves, Cassels-Frohlich, Washington, Atiyah-Macdonald, Eisenbud, Weibel, Brown, Neukirch-Schmidt-Wingberg, Rubin's Euler systems, Berger's survey on p-adic representations, Hida's books, Cremona's book, the LMFDB, the PARI manual, Lang's Algebra, Dummit-Foote, Serre's Linear representations of finite groups, Fulton's Algebraic curves, Hartshorne, Miranda's Riemann surfaces, Bosch-Lutkebohmert-Raynaud, Conrad's notes).

2.5 5. Dependency graph

```
1 groups -> 4 reps -> 11 galrep
2 rings  -> 3 modules -> 16 cohomology -> 17 galcoh -> 18 blochkato
2 rings  -> 5 galois -> 6 ant -> 7 local -> 10 reduction -> 11 galrep
8 curves -> 9 ellcurves -> 10 reduction
9 ellcurves -> 11 galrep
12 modforms -> 13 hecke -> 14 moddeg -> 15 modgalrep
11 galrep, 13 hecke, 7 local -> 15 modgalrep
3 modules, 13 hecke, 14 moddeg, 15 modgalrep, 18 blochkato, 19 lfunctions -> 20 congruence
7 local, 11 galrep, 18 blochkato -> 19 lfunctions
everything -> 21 paper
9, 10, 11, 14, 19, 21 -> 22 computation
```

2.6 6. Concept-to-chapter map for the paper

Notion in the paper	Chapter(s)
X_0(N)-optimal curve, modular parametrisation, modular degree, Manin constant	14
newform f_E, S_2(Gamma_0(N)), q-expansions, coefficientwise congruences	12, 13
l-adic valuations of integers and rationals	2
rho_{E,l}, rho bar, T_l E, V_l E, E[l], absolute irreducibility, restriction to Gal(Qbar/Q(sqrt(l*)))	4, 5, 9, 11
adjoint adˆ0 V, Symˆ2, trace form, A_f(1) = Hom(A_f, Z_l(1))	4, 18
Euler factors of the adjoint L-function, naive versus true, delta_p, P_q(X), e_l(E, q)	19
Kodaira types, component groups, Tamagawa numbers c_q, conductor exponents, Tate's algorithm	10
chi_{-3}, chi_{-4}, chi_q, q, *l*, Kronecker symbols	2, 6
the twist E' of good reduction, a_q(E'), quadratic twists and their invariants	9, 10
Tate parameter, unit part, l-th powers in F_qˆx, t_l(E, q), potentially multiplicative reduction	2, 7, 10
Sigma-imprimitive Selmer group, Bloch-Kato Selmer group, Hˆ1_f, Hˆ1_Sigma	17, 18
congruence ideal, congruence number r_E, etaˆSigma, M_{f, lambda}, the pairing delta hat_!	13, 14, 20

Notion in the paper	Chapter(s)
Fontaine-Perrin-Riou Tamagawa factor Tam_p(M) = Fitt((Hˆ1(I_p, M)_tors)ˆG)	3, 16, 18
the computation of Tam_q(ad T) at a multiplicative prime	16, 17, 18
Agashe-Ribet-Stein: deg phi_E divides r_E, equality away from pˆ2 | N	14
multiplicity one, freeness of Hˆ1(X_0(N), Z_l)_m over T_m, Gorenstein, J_0(N)[m]	3, 14, 20
Steinberg representation, unramified twists, ramified twist has conductor exponent 2	7, 15
normaliser of a split Cartan subgroup, 3Ns, GL_2(F_3) and its subgroups, orbits on lines	1, 4, 11
h_l(E) = length of Hˆ0(Q, ad T(1) tensor Q_l/Z_l), rho bar = rho bar tensor chi	4, 11, 18
induced representations, Frobenius reciprocity, characters of order 4	4
3-division polynomial, its factorisation pattern	5, 9, 11
PARI/GP, Cremona's tables, ecdata, LMFDB, Sutherland's labels	22
Eisenstein case, rational l-isogeny, Z/3 + mu_3, torsion, cuspidal subgroup, Eisenstein ideal	9, 11, 14
Petersson norm, Rankin-Selberg, L(Symˆ2 f, 2), Zagier, Watkins, Flach	12, 13, 14, 19
Hida's congruence modules, Wiles's numerical criterion, DDT Lemma 4.17	3, 20
Taylor-Wiles, R = T, DFG04 Theorems 0.1, 0.2, 2.7, 3.7, Proposition 1.4, Lemma 2.1, Theorem 2.15	18, 20
level lowering (Ribet, Carayol, Diamond; DDT Theorem 3.15), Serre conductor, Serre's conjecture	11, 15
level raising (Ribet), Wiles's level-raising factor	15, 19, 20
Kraus's semistability defect 12/gcd(v_q(Delta), 12)	10
image of inertia on T_l E at additive primes, unipotent elements, Z_3[zeta_3]	4, 6, 7, 11
Atkin-Li twisting theorem, Atkin-Lehner a_p = 0 for pˆ2 | level, oldforms, U_p eigenvalues, stabilisations, depleted forms	13
Chebotarev, Brauer-Nesbitt	4, 5
Deligne-Fontaine description of rho bar at l, a_l mod l	11, 15
DDT Theorem 3.1(e): local structure at multiplicative primes	11, 15
finite flat group schemes, "good at l"	11, 18
Kummer theory, Kummer classes, Hˆ1(G, Z_l(1)), tame character, Frobˆ{-1} t Frob = t^{qˆ{-1}}	5, 7, 16, 17
inflation-restriction, Hˆ1(, M) = M/(t - 1)M, conjugation action of G/I on Hˆ1(I, M)	16
Fitting ideals, lengths, Pontryagin duality, Gram determinants, saturated sublattices	3
perfect pairing S_2(Gamma_0(N); Z) x T -> Z (Ribet 1983, Abbes-Ullmo)	13

Notion in the paper	Chapter(s)
H^1(X_0(N), Z_l), cup product, Atkin-Lehner, Hecke self-adjointness	8, 14
Ribet-Takahashi, Pollack-Weston, Kim-Ota, Tamagawa exponent	14, 15, 20
Watkins's conjecture, Calegari-Emerton	14
the statistical protocol, calibration, hold-out sets	22
the function rule(E, l) and the Kodaira codes	10, 22
the worked examples 11a1, 20a1, 121a1, 539c1, 980i1, 726a1, 99A1	9, 14, 15, 21, 22

2.7 7. Conventions for writing

2.7.1 7.1 Files and structure

- One file per chapter: `chapters/chNN_slug.tex` (two-digit NN; appendices `chA_prereq.tex`, `chB_notation.tex`).
- A chapter file starts with `\chapter{Title}\label{ch:slug}` and contains, in this order:
  1. `\section*{Core results at a glance}`: a numbered `enumerate` list (use `\begin{enumerate}[label=\textbf{\thechapter.\arabic*}]` from `enumitem`), each item a precise statement of a definition, theorem, lemma, proposition or construction, with its rigour tag *(proved here) / (proof sketched here) / (stated; reference)*, and a `\Cref` to the numbered environment in the chapter where it lives. This card is the refresher. It must be complete: every numbered theorem, proposition and lemma of the chapter, and every definition the paper uses, appears on the card.
  2. `\section*{Where the paper uses this chapter}`: a short list naming the paper's lemmas, theorems, remarks, definitions and sections (by the paper's own labels and names, e.g. "Lemma lem:tamfactor (the Tamagawa factor at a multiplicative prime)") and which results of this chapter they use.
  3. The numbered sections of the exposition.
  4. `\section{Exercises}`.
  5. `\section{Notes and references}`: where to read more, which references were followed, which theorem numbers were cited.
- Chapters never contain `\input`, `\include`, `\newcommand`, `\renewcommand`, `\def`, `\DeclareMathOperator`, `\usepackage`, `\newtheorem`, `\begin{document}` or `\bibliography`. Everything needed is in `preamble.tex`. If a macro is missing, write `\operatorname{...}` inline.
- Labels: every label in a chapter is prefixed with the chapter slug: `\label{thm:groups:sylow}`, `\label{def:modules:fitting}`, `\label{eq:local:tame}`, `\label{sec:hecke:newforms}`, `\label{ex:reps:q8}`. Cross-chapter references use only `\Cref{ch:slug}` (chapter labels are listed in Section 4).
- Citations: `\cite{key}` with keys from `refs.bib`. Keys for the paper's own references are exactly the paper's keys: `Cremona97, Watkins02, ARS12, Zagier85, DFG04, Flach92, Hida81, FPR94, SilvermanATAEC, Sutherland16, RibetTakahashi97, Takahashi01, Pasten, PollackWeston11, KimOta, Kraus90, AtkinLi78, AtkinLehner70, Ribet83, AU96, Edixhoven92, DDT, Ribet90, Carayol89, Diamond95, Tilouine97, RibetStein, Zywina15, Mazur77, Dummigan06, CalegariEmerton09, Yazdani11, CaroPasten22, HatleyKundu, BKP, PARI`. Standard textbook keys: `SilvermanAEC, DiamondShurman, SerreLocalFields, SerreAbelian, SerreLinearReps, SerreCourse, Neukirch, NSW, MilneCFT, MilneEC, MilneANT, CasselsFrohlich, Washington, AtiyahMacdonald, Eisenbud, Weibel, Brown, Rubin, Berger, HidaMFG, HidaBook, CremonaBook, LMFDB, PARIusers, Lang, DummitFoote, Fulton, Hartshorne, Miranda,`

`BLR, Conrad, Shimura71, Shimura76, GelbartJacquet, Miyake, Koblitz, Knapp, Husemoller, Tate75 (Tate's algorithm; there is no key TateAlgorithm), Kodaira, NeronModels, Ogg, BlochKato, Fontaine, Deligne, Wiles95, TaylorWiles, BCDT, KhareWintenberger, Khare, Ribet76, Ribet91, Mazur78, MazurRibet, Zagier85, Cremona97, Stein, Loeffler, Sutherland16, Zywina15, Serre72`. If a chapter needs a reference not in `refs.bib`, add a complete entry to `bib/chNN.bib` (the chapter's own bib file, which main.tex includes), unless the reference is needed by more than one chapter: such shared references go into `refs.bib` (decision B14, Section 7.5).

- Environments (defined in `preamble.tex`, all numbered `chapter.counter` in one sequence): `theorem`, `proposition`, `lemma`, `corollary`, `conjecture`, `definition`, `example`, `exercise`, `construction`, `convention`, `remark`, `warning`, `paperlink` (a remark-style environment titled "In the paper", used to point at the exact place in the paper where a result is used). Proofs in `proof`; for sketches use `\begin{proof}[Sketch of proof]`.
- The paper's macros are the book's macros: `\Q \Z \F \R \C \N \T \degphi \ord \Phig \Gal \ad \Sym \nv \Frob \Fitt \Tam \len \Ns` plus the operators listed in `preamble.tex`. Use `\ad V` for the trace-zero adjoint exactly as the paper does.
- Kodaira symbols: `\mathrm{I}_n`, `\mathrm{I}_n^*`, `\mathrm{II}`, `\mathrm{II}^*`, `\mathrm{III}`, `\mathrm{III}^*`, `\mathrm{IV}`, `\mathrm{IV}^*`, `\mathrm{I}_0^*`.
- Code (PARI/GP, Python) goes in `\begin{lstlisting}[language=GP] ... \end{lstlisting}` or `verbatim`.

2.7.2 7.2 Typography rules (strict)

- **Never use an em dash or an en dash, in any form**: not the characters "—" or "–", not the LaTeX ligatures `---` or `--`. Use a colon, a semicolon, a comma, or parentheses. Double names are hyphenated with a plain hyphen: Bloch-Kato, Atkin-Lehner, Diamond-Flach-Guo, Taylor-Wiles, Fontaine-Perrin-Riou. Number ranges are written "pages 663 to 727" or "663-727" with a plain hyphen. The lint script rejects any dash of either kind.
- No Unicode mathematics in LaTeX source (write `\ell`, `\alpha`, not the characters).
- Sentences are short. Each section starts with one paragraph saying what it is for and where in the paper it is used.
- Every theorem has either a proof, a sketch clearly labelled as such, or the sentence "We do not prove this; see ..." with a precise reference. Never present a sketch as a proof.
- Name results by their authors on first appearance (the theorem of Agashe, Ribet and Stein), then by the book's label.

2.7.3 7.3 Length and depth

- Guideline 1,000 to 2,200 lines of LaTeX per chapter, the card excluded (Chapters 11, 15, 18, 20 and 21 up to 2,800). This is a guideline, not a cap (decision D2 in Section 7.5): the book is meant to be read online and used as a reference, so a chapter may run longer wherever the extra length makes an explanation clearer or a proof complete. Length must never come from padding or repetition. Appendices 400 to 800 lines.
- Depth: full proofs for everything elementary; for the deep theorems, the exact statement, the hypotheses, the idea of proof in a paragraph or a page, and the reference.
- The paper's own lemmas are proved in full in the book, with every compressed step expanded (the paper's proofs are written for experts). The paper's imported theorems (DFG04, DDT, FPR94, ARS12, Ribet, Carayol, Diamond, Atkin-Li, Atkin-Lehner, Kraus, Zywina, Tilouine, Ribet-Stein) are stated precisely and sketched.

2.7.4 7.4 Building and checking

- `tools/build_chapter.sh NN` compiles chapter NN alone (with the shared preamble and bibliography) into `build/chNN/` and prints the errors. A chapter must compile with zero errors before it is considered written. Warnings about undefined references to other chapters' labels are not allowed (use only `\Cref{ch:slug}` across chapters); undefined citations are errors to fix by adding bib entries.
- `tools/lint.py chapters/chNN_slug.tex` rejects dashes, forbidden commands, unprefixed labels, Unicode mathematics and stray `\input`. It must pass clean.
- `tools/build.sh` builds the whole book (`main.pdf`).

2.7.5 7.5 Decisions recorded after the outlines (binding)

These decisions are recorded in `important/important_notes.md` under the number given; they override anything above that disagrees with them.

- **B14 (bibliography).** main.tex reads all chapter bib files, so a key defined in two of them is a BibTeX error. Any reference needed by more than one chapter goes into `refs.bib`, not into a chapter bib file. Before adding a key, check `refs.bib` and the other chapters' bib files.

- **B17 (Lemma lem:crit).** Chapter 15 proves the paper's Lemma lem:crit in full. Chapter 14 states the congruence number and cites Chapter 15; Chapter 20 restates the lemma in the language of congruence modules and cites Chapter 15 for the proof.

- **B2 (Jordan-Holder).** The Jordan-Holder theorem is proved in Chapter 3 (for modules, over any ring); Chapter 4 cites it for the composition factors of representations and does not prove it again.

- **B24 (conventions).** The book-wide conventions B-C1 to B-C16 of Appendix B are binding on every chapter. Their specification, the index of symbols and the list of places where an outline disagrees with them are in `outlines/chB_notation.md` (Parts 1 to 3). In brief: Frob is the arithmetic Frobenius and Phi = Frobˆ{-1} the geometric one, L-factors use Phi; ˆvee is the linear dual, ˆ* the Pontryagin dual (and a pullback when written after a map), ˆD the Kummer dual, ˆdagger the adjoint of an operator; A_f is the paper's adjoint and Shimura's abelian variety is J_f; depletion is fˆSigma or fˆ{p}, never fˆ(p); the Manin constant is c_E.

- **D1 (strengthened results; author's decision, 15 October 2026).** Where the book proves more than the paper states, the book states its own stronger version as the theorem, and a Commentary gives the paper's statement and the difference. This covers: equality in (eq:identity) of Theorem thm:main and in the first inequality of Theorem thm:quant without multiplicity one (points 39, 73, 74); Theorem thm:qual (a) at q = 2 (point 43); Theorem thm:quant for every odd q (point 60); Proposition prop:forms with no exception (point 43); and the other strengthenings of `important/paper_memo.md`, Section A. The author intends to revise the paper accordingly.

- **D2 (length; author's decision).** No hard cap on chapter length; see Section 7.3. The medium is online (PDF and possibly HTML), not print.

- **D3 (quadratic characters; author's decision: "keep what is generally better").** The book writes the quadratic character of Q(sqrt(d)) as chi_d with d the discriminant, so the character of conductor q (q odd) is chi_{q*}, never chi_q; chi_l with a letter subscript is reserved for the l-adic cyclotomic character, and chi_3ˆcyc where a number is needed. This replaces the allowance for the paper's chi_q in Appendix B, convention B-C11; the remaining chi_q occurrences in the outlines are to be converted (about 40). The paper is to be rewritten with the same convention (memo, point 32).

- **D4 (Manin constant).** The sign convention c_E > 0 (so Manin's conjecture reads c_E = 1) is adopted, as already applied in the outlines.

2.8 8. Production process

1. The plan (this file), the preamble, the build tools and the bibliography skeleton are committed first.
2. One agent per chapter, in two stages, run as a pipeline per Part:
   - **Stage 1 (line-up).** The agent reads this plan, the paper and the preamble, then lines up the core theorems, concepts, lemmas, definitions, constructions and examples of its chapter in teaching order, with precise statements, rigour levels, references and paper hooks, and a section plan with line targets. Output: `outlines/chNN_slug.md`.
   - **Stage 2 (writing).** A fresh agent takes the line-up and writes the chapter independently to `chapters/chNN_slug.tex`, compiles it, lints it, and fixes every error.
3. Each written chapter is then reviewed by two independent agents (a sceptical mathematician who tries to find false statements, wrong hypotheses, wrong theorem numbers, gaps presented as proofs; and an editor who checks coverage against the line-up and this plan, accessibility for the stated audience, the completeness of the card, LaTeX hygiene and the typography rules). A fourth agent applies the fixes and recompiles. The mathematical review is repeated until no blocking findings remain (at most three rounds).
4. After all chapters exist: a completeness critic reads the paper and all the cards and lists any notion of the paper not covered; a cross-reference pass upgrades chapter-level references in Chapters 21 and 22 to precise theorem labels where they exist; the whole book is built; the bibliography is verified.
5. Everything is committed on the branch `claude/textbook-paper-prerequisites-clhh5a`.

2.9 9. Progress and measured cost (updated 16 October 2026)

Item	State
Outlines (22 chapters, Appendix B), bibliography, memo, revised paper	done (see CONTINUE.md)
D3 applied to the outlines; Chapter 18 hedge settled by point 73	done
Appendix A	written, 994 lines, compiles
Chapter 10 (pilot)	**written**: 2,344 lines, 48 pages, compiles, lint clean; full pipeline, three review rounds
Chapters 1 to 9, 11 to 22, Appendix B, preface update	not yet written

What the pilot taught. * Length: 2,344 lines against a plan average of about 2,200 per chapter; the outline is a good predictor. * Quality: the writer covered every outline item and strengthened several proofs; the reviewers found real errors in every round (mathematical findings per round: 5, 7, 7), so a review is worth its cost, but the number of findings did not fall with more rounds. The third round's fixes were applied without a check. * New results and paper findings: important_notes.md points 81 to 83, memo Section F. * **Cost: about USD 31 to 35 for the chapter on Opus 5.5** (95M cache-read tokens, 2.3M cache-write tokens, a small output). Almost all of it is context re-read on every tool call: each agent was told to read PLAN.md, both versions of the paper, the notation outline and the memo in full (about 170k tokens), then made 100 to 200 tool calls. The writer alone was about a third of the cost. Output tokens are a small part. * Time: about 70 minutes of agent time per chapter (the agents run one after another).

Budget mode (`args.lean: true` in tools/workflows/write_chapters.js). For a fixed budget (for example USD 100 of Console API credit), the workflow has a lean mode: each agent reads only PLAN.md 7.1 to 7.5, its outline, Part 1 of Appendix B, memo Sections B and C, and the paper only at its hooks, and is told to work in few, large steps; and the review is one combined (mathematical and

editorial) review, one fix, and one check of the fixes, with a second fix only if the check finds a blocking or major problem. Expected cost about USD 12 to 18 per chapter (an estimate; measure the first lean chapter). At that rate USD 100 writes about 5 to 7 chapters; the remaining 21 chapters and Appendix B need about USD 250 to 400 in lean mode, or about USD 700 to 800 with the full pipeline.

Priorities under a fixed budget. (1) Part I (Chapters 1 to 4, the easiest and shortest) in lean mode, measuring the first chapter; (2) the chapters the paper leans on most (14, 15, 20, 21); (3) the rest. Chapters 21 and 22 still come last, since they cite every other chapter.

Part II

Chapter outlines

Chapter 3

Line-up for Chapter 1: Groups, group actions and matrix groups over finite fields

Label `ch:groups`, file `chapters/ch01_groups.tex`, Part I (Algebra). Target length 1910 lines (card excluded; see Part 2).

This chapter is the first of the book: it relies on no earlier chapter, only on the assumed background (linear algebra over R and C, elementary group theory as in RHB Chapters 28 and 29). Every result is elementary; everything is proved here except Tychonoff's theorem, the topological characterisation of profinite groups, the isomorphisms PGL_2(F_5) = S_5 and the simplicity of PSL_2(F_7), the classification of groups of order 8, and Dickson's classification, which are stated with references.

Conventions fixed for the whole chapter (and inherited by the book):

- C1. Matrices act on column vectors on the left. For $v \in F^2 \setminus \{0\}$ the line through v is $[v] = Fv$; $\mathbb{P}^1(F)$ is the set of lines of F^2, with coordinates $[x : y]$; we write $\infty = [1 : 0]$ and identify $[x : 1]$ with $x \in F$, so that $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ acts by the Moebius formula $g \cdot x = (ax + b)/(cx + d)$ (check: $g[x : 1] = [ax + b : cx + d]$).
- C2. Conjugation is $g \cdot h = ghg^{-1}$. In a semidirect product $N \rtimes H$ the group H acts on N and $hnh^{-1} = \theta(h)(n)$.
- C3. C_n or $\mathbb{Z}/n$ is the cyclic group of order n; D_n is the dihedral group of order $2n$ (so D_4 has order 8; some authors write D_8); $V_4 = C_2 \times C_2$; Q_8 the quaternion group; SD_{16} the semidihedral group of order 16.
- C4. $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$ is a field (an integer prime to p is invertible modulo p by Bezout). $\mathbb{F}_q$ is a finite field with q elements, constructed in Chapter 2; in this chapter q is odd whenever Cartan subgroups are discussed, and the reader may take $q = p$ prime throughout. Warning: in this chapter q is a field size; in the paper q is a bad prime of E (the two agree in the paper's $\mathrm{GL}_2(\mathbb{F}_3)$, where the field size is $\ell = 3$).
- C5. "The image is $N_s(\mathbb{F}_3)$" always means "conjugate in $\mathrm{GL}_2(\mathbb{F}_3)$ to $N_s(\mathbb{F}_3)$"; the standard models of the subgroups (Section 6) are fixed once and for all.

Rigour tags: **PH** = proved here, **SK** = proof sketched here, **ST** = stated; reference.

Each line-up item gives: number and label-ready name, kind, rigour tag, the precise statement, a reference (DF = Dummit-Foote, 3rd edition; section numbers only where sure), and the paper's use (by the paper's `\label` names, or "background"). All group orders and subgroup claims for $\mathrm{GL}_2(\mathbb{F}_3)$ below were checked by hand and by a complete enumeration of the 48 elements and the 55 subgroups (16 conjugacy classes) of $\mathrm{GL}_2(\mathbb{F}_3)$, and the orders for $q = 5, 7$ by enumeration.

3.1 Section 1. Groups, subgroups, cosets, quotients and cyclic groups

1.1 `def:groups:group` (definition, background). A group, abelian group, subgroup, the subgroup $\langle S\rangle$ generated by a subset, the order $\#G$ of a group and the order of an element, cyclic groups $\langle g\rangle$; examples: $(\mathbb{Z},+)$, $\mathbb{Z}/n$, S_n, $\mathrm{GL}_2(F)$, the group of units $F^\times$ of a field. Reference: Dummit-Foote (DF) Chapter 1. Paper use: background.

1.2 `prop:groups:lagrange` (proposition, PH). Let $H \le G$ be groups with G finite. The left cosets gH partition G, all have $\#H$ elements, and $\#G = [G:H]\cdot\#H$. Consequently the order of every element divides $\#G$, and $g^{\#G} = 1$ for all $g \in G$. Reference: DF Section 3.2. Paper use: background (orders of the subgroups of $\mathrm{GL}_2(\mathbb{F}_3)$).

1.3 `thm:groups:cyclic` (theorem, PH). Let $C = \langle t\rangle$ be cyclic of order n. (i) Every subgroup of C is cyclic; for each $d \mid n$ there is exactly one subgroup of order d, namely $C[d] = \{x : x^d = 1\} = \langle t^{n/d}\rangle = C^{n/d}$ (the (n/d)-th powers). (ii) For $m \ge 1$ the m-th power map $x \mapsto x^m$ is an endomorphism of C with kernel $C[\gcd(m,n)]$ of order $\gcd(m,n)$ and image $C^m = C^{\gcd(m,n)}$ of index $\gcd(m,n)$; in particular for a prime ℓ and $k \ge 0$, $[C : C^{\ell^k}] = \ell^{\min(k, v_\ell(n))}$, and $x \in C$ is an ℓ^k-th power iff its class in $C/C^{\ell^{v_\ell(n)}} \simeq \mathbb{Z}/\ell^{v_\ell(n)}$ is divisible by ℓ^k (for $k \le v_\ell(n)$). (iii) The generators of C are the $\varphi(n)$ elements of order n; $\mathrm{Aut}(C) = (\mathbb{Z}/n)^\times$ acting by $t \mapsto t^a$; $\mathrm{Aut}(\mathbb{Z}) = \{\pm 1\}$. (iv) A homomorphism $\chi\colon G \to C$ from any group onto C satisfies $\chi^2 = 1$ (i.e. $\chi(g)^2 = 1$ for all g) iff $n \mid 2$. Reference: DF Section 2.3. Paper use: Lemma lem:tamfactor (the class c of the unit part of the Tate parameter in $\mathbb{F}_q^\times/(\mathbb{F}_q^\times)^{\ell^{v_\ell(q-1)}}$, restated for $\mathbb{F}_q^\times$ in Chapter 2), Lemma lem:crit ($\mathbb{T}/(I+J)$ cyclic of order r), Lemma lem:h0 ($\bar\chi_\ell$ surjects onto $\mathbb{F}_\ell^\times$, cyclic of order $\ell-1$, so $\bar\chi_\ell^2 = 1$ forces $\ell - 1 \mid 2$, i.e. $\ell = 3$).

1.4 `def:groups:normal` (definition, background). Normal subgroup, quotient group G/N, homomorphism, kernel, image, index, the canonical projection; simple group; the centre $Z(G)$, the centraliser $C_G(S)$ and the normaliser $N_G(S)$ of a subset; conjugacy classes; the commutator subgroup $[G,G]$ and the abelianisation. Reference: DF Sections 2.2, 3.1. Paper use: background (N_s, N_{ns} are normalisers; $\{\pm I\}$ is the centre of $\mathrm{GL}_2(\mathbb{F}_3)$).

1.5 `thm:groups:isothms` (theorem, PH). (First isomorphism theorem) a homomorphism $f\colon G \to H$ induces $G/\ker f \simeq f(G)$. (Second) for $H \le G$ and $N \trianglelefteq G$, $HN/N \simeq H/(H\cap N)$. (Third) for $N \le M$ both normal in G, $(G/N)/(M/N) \simeq G/M$. (Correspondence) subgroups of G/N correspond bijectively and inclusion-preservingly to subgroups of G containing N, normal ones to normal ones. Reference: DF Section 3.3. Paper use: background.

1.6 `lem:groups:index2` (lemma, PH). (i) A subgroup of index 2 is normal, and is the kernel of a surjection onto C_2. (ii) (Image of a kernel.) Let $\rho\colon G \to \Gamma$ and $\delta\colon \Gamma \to A$ be homomorphisms and $G_0 = \ker(\delta\circ\rho)$. Then $\rho(G_0) = \rho(G)\cap\ker\delta$. In particular, if $\det\rho\colon G \to \{\pm1\}$ has kernel G_0, the image of G_0 is the determinant-one subgroup $\rho(G)\cap\mathrm{SL}_2$ of the image of G. Reference: elementary. Paper use: Proposition prop:ns, first sentence of the proof (the image of $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{-3}))$ is the determinant-one subgroup of the image H of $\bar\rho_{E,3}$, because $\det\bar\rho_{E,3} = \chi_{-3}$ has kernel $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{-3}))$).

1.7 `prop:groups:conjclasses` (proposition, PH). Conjugacy is an equivalence relation; the class of g has $[G : C_G(g)]$ elements; a subgroup is normal iff it is a union of conjugacy classes; the number of conjugates of a subgroup H is $[G : N_G(H)]$. Reference: DF Section 4.3. Paper use: the class table of $\mathrm{GL}_2(\mathbb{F}_3)$ (6.1).

1.8 `prop:groups:lprimary` (proposition, PH). Let A be a finite abelian group and ℓ a prime. The ℓ-primary part $A[\ell^\infty] = \{a : \ell^k a = 0 \text{ for some } k\}$ is a subgroup of order $\ell^{v_\ell(\#A)}$, $A = \bigoplus_\ell A[\ell^\infty]$ (sum over the primes dividing $\#A$), and $A[\ell^\infty]$ is the unique Sylow ℓ-subgroup of A and the image of the idempotent projector; $v_\ell(\#A) = v_\ell(\#A[\ell^\infty])$. (Foretaste: $A[\ell^\infty] = A\otimes\mathbb{Z}_\ell$, Chapter 3.) The exponent of A divides $\#A$ and has the same prime divisors. Proof note (correction to the order of proofs): the splitting $A = A[m]\oplus A[n]$ for $\#A = mn$, $\gcd(m,n) = 1$, is Bezout; the order statement $\#A[\ell^\infty] = \ell^{v_\ell(\#A)}$ needs Cauchy's theorem for abelian groups ($\ell \mid \#A$ implies an element of order ℓ), which is proved here by induction on $\#A$ in three lines, ahead of the general Cauchy theorem 3.6. Reference: DF Section 5.2. Paper use: Definition def:terms and Section 1.4 of the paper ("the ℓ-part of the Tamagawa number", $v_\ell(\#\Phi_q^{\mathrm{geom}})$), Lemma lem:tamfactor ($H^1(I_q,M)_{\mathrm{tor}}$ as a finite abelian ℓ-group, "the ℓ-part of $c_q(E)$").

3.2 Section 2. Direct and semidirect products; dihedral, quaternion and semidihedral groups

2.1 `prop:groups:directproduct` (definition and proposition, PH). The direct product $G \times H$. (Recognition) if $H, K \trianglelefteq G$, $H \cap K = 1$ and $HK = G$, then $(h,k) \mapsto hk$ is an isomorphism $H \times K \to G$; in particular elements of H and K commute. Reference: DF Sections 5.1, 5.4. Paper use: background ($\mathrm{B}(\mathbb{F}_3) = \{\pm I\} \times S_3$ in 5.5; $\hat{\mathbb{Z}} = \prod_p \mathbb{Z}_p$ in 7.2).

2.2 `prop:groups:crtgroups` (proposition, PH). $C_m \times C_n \simeq C_{mn}$ if and only if $\gcd(m,n) = 1$. Consequently $\mathbb{Z}/n \simeq \prod_{p|n} \mathbb{Z}/p^{v_p(n)}$ as groups (the ring version, the Chinese remainder theorem, is in Chapter 2), and a cyclic group is the direct product of its ℓ-primary parts. Reference: DF Section 5.1. Paper use: background (Lemma lem:tamfactor, $\mathbb{Z}_\ell(1)/(q-1) \simeq \mathbb{Z}/\ell^{v_\ell(q-1)}$ as the ℓ-part of a cyclic group of order $q-1$).

2.3 `def:groups:semidirect` (construction and proposition, PH). Given groups N, H and a homomorphism $\theta\colon H \to \mathrm{Aut}(N)$, the semidirect product $N \rtimes_\theta H$ is the set $N \times H$ with $(n,h)(n',h') = (n\,\theta(h)(n'),\, hh')$; N is normal, H is a complement, and $hnh^{-1} = \theta(h)(n)$ (convention C2). (Recognition) if $N \trianglelefteq G$, $H \leq G$, $N \cap H = 1$ and $NH = G$, then $G \simeq N \rtimes_\theta H$ with $\theta =$ conjugation. Equivalently, a short exact sequence $1 \to N \to G \to Q \to 1$ splits (has a homomorphic section $Q \to G$) iff $G \simeq N \rtimes Q$. The semidirect product is direct iff θ is trivial. Examples to be given: $D_n = C_n \rtimes C_2$; $\mathrm{B} = \mathrm{U} \rtimes \mathrm{T}$ (5.5); $\mathrm{GL}_2(F) = \mathrm{SL}_2(F) \rtimes F^\times$ (5.1); $N_s = C_s \rtimes C_2$, $N_{ns} = C_{ns} \rtimes C_2$ (5.8); $\mathrm{SL}_2(\mathbb{F}_3) = Q_8 \rtimes C_3$ (6.4); $S_4 = V_4 \rtimes S_3$; foretaste: the tame Galois group $\left(\prod_{\ell \neq p} \mathbb{Z}_\ell(1)\right) \rtimes \hat{\mathbb{Z}}$ of $\mathbb{Q}_p$, with $\mathrm{Frob}\, t\, \mathrm{Frob}^{-1} = t^p$ (Chapter 7). Reference: DF Section 5.5. Paper use: background (Lemma lem:tamfactor uses $\mathrm{Frob}^{-1} t \mathrm{Frob} = t^{q^{-1}}$, made precise in Chapter 7).

2.4 `def:groups:dihedral` (definition and proposition, PH). For $n \geq 2$, $D_n = \langle r, s \mid r^n = s^2 = 1, srs^{-1} = r^{-1} \rangle$, of order $2n$, $= C_n \rtimes C_2$ with C_2 acting by inversion; the symmetry group of the regular n-gon for $n \geq 3$ (RHB analogue, one sentence). Facts: every element outside $\langle r \rangle$ has order 2; for $n \geq 3$ the centre is trivial if n is odd and $\{1, r^{n/2}\}$ if n is even; $D_2 = V_4$; $D_3 \simeq S_3$; $D_6 \simeq C_2 \times S_3$. (Recognition) if G is generated by r of order n and s of order 2 with $srs^{-1} = r^{-1}$ and $\#G = 2n$, then $G \simeq D_n$. D_4 has five elements of order 2 and two of order 4. Reference: DF Section 1.2. Paper use: Proposition prop:ns ($N_s(\mathbb{F}_3) \simeq D_4$, "dihedral"), Section sec:anomaly (image 3Ns); Chapter 4 ("the dihedral case").

2.5 `def:groups:quaternion` (definition and proposition, PH). $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$ with $i^2 = j^2 = k^2 = ijk = -1$; equivalently $\langle i, j \mid i^4 = 1, i^2 = j^2, jij^{-1} = i^{-1} \rangle$. Facts: -1 is the unique element of order 2; the six elements $\pm i, \pm j, \pm k$ have order 4; $Z(Q_8) = [Q_8, Q_8] = \{\pm 1\}$ and $Q_8/\{\pm 1\} \simeq V_4$; every subgroup is normal; every nontrivial subgroup contains -1, so Q_8 is not a semidirect product of two proper subgroups; $Q_8 \not\simeq D_4$ (count elements of order 2). (Recognition) a non-abelian group of order 8 with a unique element of order 2 is isomorphic to Q_8. Reference: DF Section 1.5. Paper use: Proposition prop:ns (the determinant-one subgroup of $N_{ns}(\mathbb{F}_3)$ is Q_8).

2.6 `def:groups:semidihedral` (definition and proposition, PH). $SD_{16} = \langle a, x \mid a^8 = x^2 = 1, xax^{-1} = a^3 \rangle$, of order 16, $= C_8 \rtimes C_2$ with C_2 acting by $a \mapsto a^3$. Facts: the elements $a^i x$ satisfy $(a^i x)^2 = a^{4i}$, so they have order 2 for i even and order 4 for i odd; the numbers of elements of orders $1, 2, 4, 8$ are $1, 5, 6, 4$; $Z(SD_{16}) = \{1, a^4\}$; the three subgroups of index 2 are $\langle a \rangle \simeq C_8$, $\langle a^2, x \rangle \simeq D_4$ and $\langle a^2, ax \rangle \simeq Q_8$. Comparison table of the four non-abelian groups of order 16 with a cyclic subgroup of index 2, by numbers of elements of orders $1, 2, 4, 8$: D_8 (dihedral of order 16): $1, 9, 2, 4$; SD_{16}: $1, 5, 6, 4$; Q_{16} (generalised quaternion): $1, 1, 10, 4$; $M_{16} = C_8 \rtimes C_2$ with $a \mapsto a^5$: $1, 3, 4, 8$. (Recognition) a group of order 16 generated by a of order 8 and x of order 2 with $xax^{-1} = a^3$ is SD_{16} (it is a quotient of the presented group, which has order at most 16). Reference: DF Section 5.5 (semidirect products of C_8 by C_2); the comparison table is a direct count. Paper use: Proposition prop:ns ($N_{ns}(\mathbb{F}_3)$ of order 16).

2.7 `thm:groups:order8` (theorem, SK). Every group of order 8 is isomorphic to exactly one of C_8, $C_4 \times C_2$, C_2^3, D_4, Q_8. Sketch: abelian case by 4.1; non-abelian case: there is an element r of order 4 (exponent 2 forces abelian), $\langle r \rangle$ is normal of index 2, any $s \notin \langle r \rangle$ has $s^2 \in \{1, r^2\}$ and $srs^{-1} = r^{-1}$, giving

D_4 or Q_8. Reference: DF Chapter 5 (table of groups of small order). Paper use: background (the names "dihedral" and "quaternion" in Proposition prop:ns are isomorphism types; the chapter identifies them by the recognition criteria 2.4, 2.5, so this theorem is context only).

3.3 Section 3. Group actions, orbit-stabiliser, p-groups and the Sylow theorems

3.1 `def:groups:action` (definition, background). A (left) action of G on a set X; the associated permutation representation $G \to \mathrm{Sym}(X)$; kernel, faithful, transitive; orbit Gx, stabiliser G_x, fixed set X^G; restriction of an action to a subgroup; examples: left multiplication on G/H; conjugation on G and on the set of subgroups; $\mathrm{GL}_2(F)$ on F^2, on $F^2 \setminus \{0\}$, on $\mathbb{P}^1(F)$. Reference: DF Sections 1.7, 4.1. Paper use: Proposition prop:ns ("orbits on the four lines").

3.2 `thm:groups:orbitstab` (theorem, PH). (Orbit-stabiliser) Let G act on X and $x \in X$. Then $gG_x \mapsto gx$ is a well-defined bijection $G/G_x \to Gx$, so $\#Gx = [G : G_x]$ when G is finite; $G_{gx} = gG_xg^{-1}$; the orbits partition X, and for finite X, $\#X = \sum_i [G : G_{x_i}]$ over orbit representatives. Reference: DF Section 4.1. Paper use: Proposition prop:ns.

3.3 `prop:groups:suborbits` (proposition, PH). Let G act transitively on a finite set X, $x \in X$, and $H \leq G$. (i) The H-orbits on X correspond bijectively to the double cosets $H\backslash G/G_x$; H is transitive iff $G = HG_x$. (ii) If $H \trianglelefteq G$, then G permutes the H-orbits transitively, so all H-orbits have the same size, which divides $\#X$. (iii) In particular if $[G : H] = 2$, either H is transitive or X is the union of exactly two H-orbits of equal size $\#X/2$, interchanged by every element of $G \setminus H$. Reference: elementary (DF Section 4.1, exercises). Paper use: Proposition prop:ns (the determinant-one subgroup C_4 of $N_s(\mathbb{F}_3)$ has the same two orbits of size two; the Galois-theoretic reading of orbits as factor degrees is Chapter 5).

3.4 `lem:groups:burnside` (lemma, PH). (Orbit-counting lemma) For a finite group acting on a finite set, the number of orbits is $\frac{1}{\#G}\sum_{g\in G} \#X^g$. Reference: DF Section 4.1 (exercises). Paper use: background (exercise: recount the orbits of 6.5 and 6.6 from the class table 6.1; foretaste of the Chebotarev densities of Chapter 22).

3.5 `thm:groups:classeq` (theorem, PH). (Class equation) For finite G, $\#G = \#Z(G) + \sum [G : C_G(g_i)]$ over representatives of the non-central conjugacy classes. Let p be a prime. (i) If a p-group P acts on a finite set X then $\#X \equiv \#X^P \pmod p$. (ii) A nontrivial finite p-group has nontrivial centre. (iii) A group of order p^2 is abelian. (iv) A group of order p^n has a normal subgroup of order p^k for every $0 \leq k \leq n$, and every subgroup of index p in a p-group is normal. Reference: DF Sections 4.3, 6.1. Paper use: background (used in 3.7; pro-p groups 7.3, the wild inertia of Chapter 7).

3.6 `thm:groups:cauchy` (theorem, PH). (Cauchy) If a prime p divides $\#G$, then G has an element of order p. (Proof: C_p acts on $\{(g_1, \ldots, g_p) : g_1 \cdots g_p = 1\}$ by rotation; apply 3.5(i).) Reference: DF Section 3.4 (abelian case) and Section 4.5. Paper use: background.

3.7 `thm:groups:sylow` (theorem, PH). (Sylow) Let $\#G = p^a m$ with $p \nmid m$. (i) G has a subgroup of order p^a (a Sylow p-subgroup). (ii) Every p-subgroup of G is contained in a Sylow p-subgroup, and any two Sylow p-subgroups are conjugate. (iii) The number n_p of Sylow p-subgroups equals $[G : N_G(P)]$ for any Sylow P, divides m, and satisfies $n_p \equiv 1 \pmod p$. (iv) $P \trianglelefteq G$ iff $n_p = 1$; a normal Sylow subgroup contains every p-element of G. Reference: DF Section 4.5. Paper use: background (6.3, 6.6, 6.7: $N_{ns}(\mathbb{F}_3)$ is a Sylow 2-subgroup of $\mathrm{GL}_2(\mathbb{F}_3)$ and $N_s(\mathbb{F}_3)$ lies in one; the subgroups of order 3 in Lemma lem:IV are the Sylow 3-subgroups).

3.8 `ex:groups:sylowsmall` (worked example, PH). (a) A group of order 15 is cyclic. (b) In S_4: $n_2 = 3$, $n_3 = 4$. (c) In $\mathrm{GL}_2(\mathbb{F}_3)$: $n_3 = 4$ (the four groups U_L of 5.6, one for each line L) and $n_2 = 3$ (the three conjugates of $N_{ns}(\mathbb{F}_3)$). (d) In $\mathrm{SL}_2(\mathbb{F}_3)$: $n_2 = 1$ (the Sylow 2-subgroup is Q_8) and $n_3 = 4$. Paper use: background for 6.4 to 6.7.

3.4 Section 4. Finite abelian groups

4.1 `thm:groups:finiteabelian` (theorem, PH). (Structure theorem) Every finite abelian group A is isomorphic to $\bigoplus_i \mathbb{Z}/\ell_i^{e_i}$ for primes ℓ_i and $e_i \geq 1$, and the multiset of prime powers $\ell_i^{e_i}$ (the elementary divisors) is determined by A. Equivalently $A \simeq \mathbb{Z}/d_1 \oplus \cdots \oplus \mathbb{Z}/d_r$ with $1 < d_1 \mid d_2 \mid \cdots \mid d_r$ (invariant factors), uniquely. For each prime ℓ the numbers $\#A[\ell^k]$, $k \geq 1$, determine the ℓ-primary part. Proof: reduce to ℓ-groups by 1.8; for an ℓ-group split off a cyclic subgroup generated by an element of maximal order; uniqueness by counting $\#A[\ell^k]$. The general version for finitely generated modules over a PID is Chapter 3. Reference: DF Section 5.2. Paper use: background (component groups $\mathbb{Z}/n$, $\mathbb{Z}/2$, $\mathbb{Z}/3$, $(\mathbb{Z}/2)^2$, $\mathbb{Z}/4$ of the paper's Section 1.4; $H^1(I_q, M)_{\mathrm{tor}} \simeq (\mathbb{Z}/\ell^{v_\ell(n)})^2$ in Lemma lem:tamfactor).

4.2 `cor:groups:cyclic` (corollary, PH). (i) A finite abelian group A is cyclic iff $\#A[\ell] \leq \ell$ for every prime ℓ, iff for every d the equation $da = 0$ has at most d solutions. (ii) Every finite subgroup of the multiplicative group $K^\times$ of a field K is cyclic (a polynomial of degree d has at most d roots). In particular $\mathbb{F}_p^\times$ is cyclic of order $p-1$, the group $\mu_n(K)$ of n-th roots of unity in K is cyclic, and the multiplicative group of the field $\mathbb{F}_q[\sqrt{\varepsilon}]$ of 5.8 is cyclic of order q^2-1. Reference: DF Sections 5.2, 9.5. Paper use: Lemma lem:tamfactor and Lemma lem:h0 (cyclicity of $\mathbb{F}_q^\times$ and of $\mathbb{F}_\ell^\times$; restated with field theory in Chapter 2); Proposition prop:ns ($C_{ns}(\mathbb{F}_3) \simeq C_8$).

4.3 `prop:groups:torsioncount` (proposition, PH). For a finite abelian group A and $n \geq 1$, the sequence $0 \to A[n] \to A \xrightarrow{n} A \to A/nA \to 0$ is exact, so $\#A[n] = \#A/nA$. For $A \simeq \bigoplus_i \mathbb{Z}/\ell^{e_i}$ (an ℓ-group), $\#A[\ell^k] = \#A/\ell^k A = \ell^{\sum_i \min(k,e_i)}$. Reference: elementary. Paper use: Lemma lem:tamfactor (counting the Frobenius-fixed points of $(\mathbb{Z}/\ell^{v_\ell(n)})^2$: the solutions y of $(q-1)y \equiv 0$ and $c'y \equiv 0 \pmod{\ell^{v_\ell(n)}}$ number $\ell^{\min(v_\ell(n), v_\ell(q-1), v_\ell(c'))}$), Lemma lem:h0 (length of a finite ℓ-group).

4.4 `ex:groups:componentgroups` (worked example, PH). The groups $\mathbb{Z}/n$, $\mathbb{Z}/2$, $\mathbb{Z}/3$, $(\mathbb{Z}/2)^2$, $\mathbb{Z}/4$ that occur as geometric component groups (Chapter 10) and their ℓ-parts: $v_\ell(\#\mathbb{Z}/n) = v_\ell(n)$; $(\mathbb{Z}/2)^2$ and $\mathbb{Z}/4$ have the same order but are not isomorphic (count $\#A[2]$); for odd ℓ only $\mathbb{Z}/n$ and $\mathbb{Z}/3$ contribute. Paper use: the paper's Section 1.4 ($v_\ell(\#\Phi_q^{\mathrm{geom}})$), Definition def:terms (background).

3.5 Section 5. The groups GL_2, SL_2, PGL_2, PSL_2; Borel, unipotent and Cartan subgroups

Throughout, F is a field and $\mathbb{F}_q$ a finite field with q elements (convention C4).

5.1 `def:groups:gl2` (definition and proposition, PH). $\mathrm{GL}_2(F)$, $\mathrm{SL}_2(F) = \ker(\det)$, $\mathrm{PGL}_2(F) = \mathrm{GL}_2(F)/F^\times I$, $\mathrm{PSL}_2(F) = \mathrm{SL}_2(F)/\{\pm I\}$. Facts: (i) $Z(\mathrm{GL}_2(F)) = F^\times I$ (a matrix commuting with all elementary matrices is scalar) and $Z(\mathrm{SL}_2(F)) = \{\pm I\}$; (ii) det is surjective and $d \mapsto \mathrm{diag}(1,d)$ is a section, so $\mathrm{GL}_2(F) \simeq \mathrm{SL}_2(F) \rtimes F^\times$; (iii) $\mathrm{SL}_2(F)$ is generated by the elementary matrices $\begin{pmatrix}1 & b\\ 0 & 1\end{pmatrix}$, $\begin{pmatrix}1 & 0\\ c & 1\end{pmatrix}$; (iv) if $\mathrm{char} F \neq 2$, the only element of order 2 in $\mathrm{SL}_2(F)$ is $-I$ (an element with $g^2 = 1$ is diagonalisable with eigenvalues ± 1 of product 1); (v) $\mathrm{PSL}_2(F)$ is the image of $\mathrm{SL}_2(F)$ in $\mathrm{PGL}_2(F)$, of index $[F^\times : (F^\times)^2]$. Reference: DF Section 1.4; Lang Chapter XIII. Paper use: background (Proposition prop:ns, Lemma lem:h0: $\det \bar\rho = \bar\chi_\ell$).

5.2 `prop:groups:orders` (proposition, PH). $\#\mathrm{GL}_2(\mathbb{F}_q) = (q^2-1)(q^2-q) = q(q-1)^2(q+1)$ (count ordered bases); $\#\mathrm{SL}_2(\mathbb{F}_q) = \#\mathrm{PGL}_2(\mathbb{F}_q) = q(q^2-1)$; $\#\mathrm{PSL}_2(\mathbb{F}_q) = q(q^2-1)/\gcd(2, q-1)$. More generally $\#\mathrm{GL}_n(\mathbb{F}_q) = \prod_{i=0}^{n-1}(q^n - q^i)$ (remark). Table (verified by enumeration):

q	GL_2	SL_2	PGL_2	PSL_2
3	48	24	24	12
5	480	120	120	60
7	2016	336	336	168

Reference: DF Section 1.4 (exercises), Lang Chapter XIII. Paper use: Proposition prop:ns (order 48), background for $\ell = 5, 7$ (Section sec:data).

5.3 `prop:groups:projline` (proposition, PH). Let $\mathbb{P}^1(F)$ be the set of lines of F^2 (convention C1); $\#\mathbb{P}^1(\mathbb{F}_q) = q+1$. (i) $\mathrm{GL}_2(F)$ acts on $\mathbb{P}^1(F)$ by $g[v] = [gv]$, through Moebius transformations in the coordinate $x = [x:1]$, $\infty = [1:0]$. (ii) The kernel of the action is $F^\times I$, so $\mathrm{PGL}_2(F)$ acts faithfully. (iii) $\mathrm{PGL}_2(F)$ acts sharply 3-transitively: for two triples of distinct lines there is exactly one element of $\mathrm{PGL}_2(F)$ carrying one to the other. (iv) The stabiliser of $\infty = [e_1]$ is the upper triangular group $\mathrm{B}(F)$, so $\mathbb{P}^1(F) \simeq \mathrm{GL}_2(F)/\mathrm{B}(F)$ and $[\mathrm{GL}_2(\mathbb{F}_q) : \mathrm{B}(\mathbb{F}_q)] = q+1$. (v) $\mathrm{SL}_2(F)$ is transitive, indeed 2-transitive, on $\mathbb{P}^1(F)$. Reference: Lang Chapter XIII; DF Section 4.1 (exercises). Paper use: Proposition prop:ns ("the four lines of $\mathbb{F}_3^2$"), Chapter 9 (the roots of the 3-division polynomial correspond to the lines of $E[3]$).

5.4 `thm:groups:smallpgl` (theorem; (i), (ii) PH, (iii) SK, (iv) ST). (i) The action on the four lines of $\mathbb{F}_3^2$ gives isomorphisms $\mathrm{PGL}_2(\mathbb{F}_3) \simeq S_4$ and $\mathrm{PSL}_2(\mathbb{F}_3) \simeq A_4$ (faithful by 5.3(ii), orders 24 and 12; $\mathrm{SL}_2(\mathbb{F}_3)$ is generated by elements of order 3, which act as 3-cycles). (ii) For $g \in \mathrm{GL}_2(\mathbb{F}_3)$, $\det g \in \{\pm 1\} = \mathbb{F}_3^\times$ equals the sign of the permutation of the four lines induced by g. (Proof: both are homomorphisms to $\{\pm 1\}$; $\mathrm{SL}_2(\mathbb{F}_3)$ lies in both kernels by (i), and both kernels have index 2.) (iii) $\mathrm{PGL}_2(\mathbb{F}_5) \simeq S_5$ and $\mathrm{PSL}_2(\mathbb{F}_5) \simeq A_5$ (sketch via the action on the five Sylow 2-subgroups of $\mathrm{PSL}_2(\mathbb{F}_5)$, or reference). (iv) $\mathrm{PSL}_2(\mathbb{F}_q)$ is simple for $q \geq 4$; $\mathrm{PSL}_2(\mathbb{F}_7)$ is simple of order 168 and isomorphic to $\mathrm{GL}_3(\mathbb{F}_2)$. Reference: (iii), (iv) Lang Chapter XIII; DF Section 4.6 and exercises. Paper use: Proposition prop:ns via (ii) (the sign of the permutation of the roots of the 3-division polynomial is $\det \bar\rho_{E,3} = \chi_{-3}$; the discriminant consequence is Chapter 5 and 11); (iii), (iv) background.

5.5 `def:groups:borel` (definition and proposition, PH). A Borel subgroup of $\mathrm{GL}_2(F)$ is the stabiliser B_L of a line $L \in \mathbb{P}^1(F)$; $\mathrm{B} = \mathrm{B}_{[e_1]}$ is the upper triangular group. Its unipotent radical $\mathrm{U} = \{\begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix}\} \simeq (F, +)$ is normal, the diagonal torus $\mathrm{T} = C_s(F)$ is a complement, so $\mathrm{B} = \mathrm{U} \rtimes \mathrm{T}$ with $\mathrm{diag}(a,d)$ acting on U by $b \mapsto (a/d)b$. Facts: $\#\mathrm{B}(\mathbb{F}_q) = q(q-1)^2$; the Borel subgroups are all conjugate and correspond bijectively to the $q+1$ lines; $N_{\mathrm{GL}_2(F)}(\mathrm{B}) = \mathrm{B}$; a subgroup of $\mathrm{GL}_2(F)$ stabilises a line iff it is contained in a Borel subgroup. For $q = 3$: $\#\mathrm{B}(\mathbb{F}_3) = 12$ and $\mathrm{B}(\mathbb{F}_3) = \{\pm I\} \times \langle \mathrm{U}, \mathrm{diag}(1,-1)\rangle \simeq C_2 \times S_3 \simeq D_6$; its image in $\mathrm{PGL}_2(\mathbb{F}_3) = S_4$ is the point stabiliser S_3. Reference: Lang Chapter XIII; Serre72 Section 2. Paper use: the Eisenstein case (Section sec:eis and Conjecture conj:eis: a rational ℓ-isogeny means image in a Borel subgroup, Chapter 11); Sutherland's labels 3B, 3B.1.1, 3B.1.2 (6.10).

5.6 `def:groups:unipotent` (definition and proposition, PH). $u \in \mathrm{GL}_2(F)$ is unipotent if $(u-1)^2 = 0$, equivalently its characteristic polynomial is $(X-1)^2$. (i) A unipotent $u \neq 1$ is conjugate to $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, and its fixed vectors form exactly one line, $\ker(u-1) = \mathrm{im}(u-1)$; it fixes no other line of F^2. (ii) If $\mathrm{char}F = p > 0$, a unipotent $u \neq 1$ has order p, and conversely every element of order p (indeed of p-power order) of $\mathrm{GL}_2(F)$ is unipotent ($x^p = 1$ gives $(x-1)^p = 0$). (iii) The unipotent elements of B are exactly U; for each line L the group U_L of unipotents fixing L pointwise is the unipotent radical of B_L. (iv) "Minus a unipotent": $-u$ with $u \neq 1$ unipotent has characteristic polynomial $(X+1)^2$, has no nonzero fixed vector when $\mathrm{char}F \neq 2$, still stabilises exactly one line, and has order $2p$ in characteristic p odd. (v) In $\mathrm{GL}_2(\mathbb{F}_3)$: the 8 elements of order 3 are the nontrivial unipotents and form one conjugacy class; the 4 subgroups of order 3 are the Sylow 3-subgroups U_L, one for each line L, and U_L fixes exactly the line L; the 8 elements of order 6 are the $-u$; the 8 elements of order 3 generate $\mathrm{SL}_2(\mathbb{F}_3)$. Reference: elementary linear algebra; Serre72 Section 2. Paper use: Lemma lem:IV (the image of I_q in $\mathrm{Aut}(E[3])$ is a nontrivial unipotent subgroup, so $E[3]^{I_q}$ is a line), Remark rem:types ("minus a unipotent": $E[3]^{I_q} = 0$; "a ramified quadratic character times a unipotent").

5.7 `lem:groups:borelcommutator` (lemma, PH). Let K be a field and $H \leq \mathrm{B}(K)$. Then $[H,H] \leq \mathrm{U}(K)$. If $\mathrm{char}K = p > 0$ then $\mathrm{U}(K)$ is an elementary abelian p-group, so: (i) a subgroup of $\mathrm{B}(K)$ of finite order prime to p meets $\mathrm{U}(K)$ trivially and is abelian; (ii) a finite subgroup H of $\mathrm{GL}_2(K)$ whose commutator subgroup is not a p-group stabilises no line of K^2. Applied with $K = \overline{\mathbb{F}}_3$ (or $\mathbb{F}_9$): $N_s(\mathbb{F}_3)$, $N_{ns}(\mathbb{F}_3)$, Q_8 (non-abelian, order prime to 3) and $\mathrm{SL}_2(\mathbb{F}_3)$, $\mathrm{GL}_2(\mathbb{F}_3)$ (commutator subgroup containing Q_8) stabilise no line of $\overline{\mathbb{F}}_3^2$, whereas the abelian group $C_4 = \langle J \rangle$ does (6.5). Reference: elementary. Paper use: Proposition prop:ns (the paper's "the first two act absolutely irreducibly on $\mathbb{F}_3^2$ (they are non-abelian and

act faithfully)"; non-abelian plus faithful is not enough in general (the Borel subgroup is a counterexample), and this lemma supplies the missing step; absolute irreducibility as a notion is Chapter 4).

5.8 `def:groups:cartan` (definition and proposition, PH). Let q be odd and fix a non-square $\varepsilon \in \mathbb{F}_q^\times$ (for $q = 3$, $\varepsilon = -1$). (i) The split Cartan subgroup is $C_s(\mathbb{F}_q) = \{\mathrm{diag}(a,d)\} \simeq (\mathbb{F}_q^\times)^2$, the subgroup stabilising each of the two lines $[e_1]$, $[e_2]$ (conjugates: the stabiliser of each of two distinct lines). Its normaliser is $N_s(\mathbb{F}_q) = C_s(\mathbb{F}_q) \cup wC_s(\mathbb{F}_q)$, $w = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, the stabiliser of the set $\{[e_1],[e_2]\}$ (proof that the normaliser is no bigger: C_s contains a non-scalar $\mathrm{diag}(1,d)$, $d \neq 1$, whose eigenlines are $[e_1]$, $[e_2]$; this needs $q \geq 3$). (ii) The nonsplit Cartan subgroup is $C_{ns}(\mathbb{F}_q) = \{\begin{pmatrix} a & \varepsilon b \\ b & a \end{pmatrix} : (a,b) \neq (0,0)\}$, the image of $\mathbb{F}_q[\sqrt{\varepsilon}]^\times$ (a field with q^2 elements, constructed here as $\{a + b\sqrt{\varepsilon}\}$; it is $\mathbb{F}_{q^2}$ of Chapter 2) acting on itself by multiplication in the basis $1, \sqrt{\varepsilon}$; \$\det = \$ norm $a^2 - \varepsilon b^2$, trace $= 2a$; it is cyclic of order $q^2 - 1$ by 4.2. Its normaliser is $N_{ns}(\mathbb{F}_q) = C_{ns}(\mathbb{F}_q) \cup C_{ns}(\mathbb{F}_q)\sigma$, $\sigma = \mathrm{diag}(1,-1)$, the matrix of the conjugation $a + b\sqrt{\varepsilon} \mapsto a - b\sqrt{\varepsilon}$; $\sigma c \sigma^{-1} = c^q$ for $c \in C_{ns}$. (iii) Intrinsically: a Cartan subgroup is the unit group $A^\times$ of a commutative semisimple subalgebra $A \subset M_2(\mathbb{F}_q)$ of dimension 2, split if $A \simeq \mathbb{F}_q \times \mathbb{F}_q$, nonsplit if A is a field; all split ones are conjugate, and all nonsplit ones are conjugate (a field $A \simeq \mathbb{F}_{q^2}$ makes $\mathbb{F}_q^2$ a one-dimensional A-vector space). Each Cartan subgroup has index 2 in its normaliser. Reference: Serre72 Section 2; Lang Chapter XIII. Paper use: Proposition prop:ns, Section sec:anomaly, Remark rem:ns, the abstract ("the normaliser of a split Cartan subgroup").

5.9 `prop:groups:cartanfacts` (proposition, PH). Let q be odd. (i) $\#C_s = (q-1)^2$, $\#N_s = 2(q-1)^2$, $\#C_{ns} = q^2 - 1$, $\#N_{ns} = 2(q^2-1)$. For $q = 3, 5, 7$: C_s: 4, 16, 36; N_s: 8, 32, 72; C_{ns}: 8, 24, 48; N_{ns}: 16, 48, 96; Borel: 12, 80, 252 (verified by enumeration). (ii) Every element of $N_s \setminus C_s$ and of $N_{ns} \setminus C_{ns}$ has trace 0. In N_s: $\begin{pmatrix} 0 & b \\ c & 0 \end{pmatrix}^2 = bc\,I$ and its determinant is $-bc$. In N_{ns}: $(c\sigma)^2 = N(c)\,I$ and $\det(c\sigma) = -N(c)$, where $N(c) = \det c$. (iii) $\det(C_{ns}) = \mathbb{F}_q^\times$ (the norm is surjective; count: the kernel has order $q+1$). (iv) Determinant-one subgroups: $C_s \cap \mathrm{SL}_2 \simeq C_{q-1}$; $\#(N_s \cap \mathrm{SL}_2) = 2(q-1)$; $C_{ns} \cap \mathrm{SL}_2 \simeq C_{q+1}$; $\#(N_{ns} \cap \mathrm{SL}_2) = 2(q+1)$. (v) Orbits on $\mathbb{P}^1(\mathbb{F}_q)$ (coordinate x of C1; $\mathrm{diag}(a,d)$ acts by $x \mapsto ax/d$): C_s has orbits $\{\infty\}$, $\{0\}$, $\mathbb{F}_q^\times$ (sizes $1, 1, q-1$); N_s has orbits $\{0,\infty\}$, $\mathbb{F}_q^\times$ (sizes 2, $q-1$); B has orbits $\{\infty\}$, $\mathbb{F}_q$ (sizes 1, q); C_{ns} acts transitively, indeed $C_{ns}/\mathbb{F}_q^\times I \simeq C_{q+1}$ acts simply transitively, because C_{ns} acts simply transitively on $\mathbb{F}_q^2 \setminus \{0\}$; hence N_{ns} is transitive. In particular C_{ns} and N_{ns} stabilise no line of $\mathbb{F}_q^2$. Reference: Serre72 Section 2. Paper use: Proposition prop:ns (orders 8 and 16, orbits on the four lines); Chapter 11 (Zywina's lists).

5.10 `thm:groups:dickson` (theorem, ST; the case $\ell = 3$ PH in 6.7). (Dickson) Let ℓ be a prime, $G \leq \mathrm{GL}_2(\mathbb{F}_\ell)$ a subgroup and $\bar{G}$ its image in $\mathrm{PGL}_2(\mathbb{F}_\ell)$. (a) If $\ell \mid \#G$, then either $G \supseteq \mathrm{SL}_2(\mathbb{F}_\ell)$ or G is contained in a Borel subgroup. (b) If $\ell \nmid \#G$, then exactly one of the following holds: $\bar{G}$ is cyclic and G is contained in a Cartan subgroup (split or nonsplit); $\bar{G}$ is dihedral (of order $2n$, $n \geq 2$) and G is contained in the normaliser of a Cartan subgroup but not in the Cartan subgroup; $\bar{G}$ is isomorphic to A_4, S_4 or A_5 (the exceptional case). Consequently, a subgroup of $\mathrm{GL}_2(\mathbb{F}_\ell)$ not containing $\mathrm{SL}_2(\mathbb{F}_\ell)$ is contained in a Borel subgroup, in the normaliser of a split or a nonsplit Cartan subgroup, or in the preimage of an exceptional subgroup of $\mathrm{PGL}_2(\mathbb{F}_\ell)$. Do not state this as a list of maximal subgroups without care: for $\ell = 3$, $N_s(\mathbb{F}_3) \subset N_{ns}(\mathbb{F}_3)$ in the standard models (6.6), and for $\ell = 5$, $N_s(\mathbb{F}_5)$ (order 32, a Sylow 2-subgroup) lies in a subgroup of order 96 with projective image S_4 (checked by enumeration). Reference: Serre72 Section 2; Dickson's 1901 book (add to `bib/ch01.bib` if cited). Paper use: background for Proposition prop:ns (the list of possible images at $\ell = 3$ that the paper attributes to Zywina15) and for Chapter 11; Chapter 4 restates it for PGL_2.

3.6 Section 6. The group GL_2(F_3) in full, and GL_2(F_5), GL_2(F_7)

Notation for this section: $J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ (order 4, $J^2 = -I$); $\sigma = \mathrm{diag}(1, -1)$; $w = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = J\sigma$; $u = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$. The paper's generator $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ of C_4 is $J^{-1} = -J$. The four lines of $\mathbb{F}_3^2$ are $\infty = [1 : 0]$, $0 = [0 : 1]$, $1 = [1 : 1]$, $-1 = [-1 : 1]$ (convention C1).

6.1 `prop:groups:gl23classes` (proposition, PH). $\mathrm{GL}_2(\mathbb{F}_3)$ has order 48, centre $\{\pm I\}$, commutator subgroup $\mathrm{SL}_2(\mathbb{F}_3)$, and exactly 8 conjugacy classes:

rep.	size	order	det	trace	char. poly	lines fixed	image in S_4
I	1	1	1	-1	$(X-1)^2$	4	identity
$-I$	1	2	1	1	$(X+1)^2$	4	identity
σ	12	2	-1	0	X^2-1	2	transposition
u	8	3	1	-1	$(X-1)^2$	1	3-cycle
J	6	4	1	0	X^2+1	0	double transposition
$-u$	8	6	1	1	$(X+1)^2$	1	3-cycle
$\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$	6	8	-1	-1	X^2+X-1	0	4-cycle
$\begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$	6	8	-1	1	X^2-X-1	0	4-cycle

The representative $\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$ of the first order-8 class lies in $C_{ns}(\mathbb{F}_3)$ (it is $1 + \sqrt{-1}$; trace $2 = -1$, determinant $2 = -1$); both order-8 classes meet $C_{ns}(\mathbb{F}_3)$: with a as in 6.6, a and a^3 have trace -1, while $a^5 = -a$ and $a^7 = -a^3$ have trace 1. Sizes sum to 48. Classes are determined by the characteristic polynomial except that $(X \mp 1)^2$ splits into the scalar and the non-scalar class. Element-order statistics: orders $1, 2, 3, 4, 6, 8$ occur $1, 13, 8, 6, 8, 12$ times. Reference: direct computation (verified by enumeration). Paper use: Proposition prop:ns, Lemma lem:IV, Remark rem:types (orders and fixed lines of the possible images of inertia mod 3), Lemma lem:h0.

6.2 `conv:groups:models` (convention). Standard models, fixed for the whole book (with $\varepsilon = -1$ in 5.8): $C_s(\mathbb{F}_3) = \{\mathrm{diag}(\pm 1, \pm 1)\} \simeq V_4$; $N_s(\mathbb{F}_3) = C_s(\mathbb{F}_3) \cup wC_s(\mathbb{F}_3) = \langle J, \sigma \rangle$; $C_{ns}(\mathbb{F}_3) = \{\begin{pmatrix} a & -b \\ b & a \end{pmatrix} : (a, b) \neq (0, 0)\}$ (the image of $\mathbb{F}_3[i]^\times = \mathbb{F}_9^\times$, $i^2 = -1$, $J \leftrightarrow i$); $N_{ns}(\mathbb{F}_3) = C_{ns}(\mathbb{F}_3) \cup C_{ns}(\mathbb{F}_3)\sigma$; $\mathrm{B}(\mathbb{F}_3)$ upper triangular; $\mathrm{U} = \langle u \rangle$; $Q_8 = \{\pm I, \pm J, \pm \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \pm \begin{pmatrix} 1 & -1 \\ -1 & -1 \end{pmatrix}\}$; $C_4 = \langle J \rangle = \{\pm I, \pm J\}$. "Image $N_s(\mathbb{F}_3)$" always means "conjugate to this model" (C5). With these models $N_s(\mathbb{F}_3) \subset N_{ns}(\mathbb{F}_3)$ and $C_4 = N_s(\mathbb{F}_3) \cap \mathrm{SL}_2(\mathbb{F}_3) = C_{ns}(\mathbb{F}_3) \cap \mathrm{SL}_2(\mathbb{F}_3)$ (verified).

6.3 `prop:groups:gl23subgroups` (proposition, PH; the writer may present the table with a remark that it was checked by computer, and prove only the rows used). $\mathrm{GL}_2(\mathbb{F}_3)$ has 55 subgroups in 16 conjugacy classes:

order	type	conjugates	stable lines	orbit sizes on $\mathbb{P}^1$	det onto $\mathbb{F}_3^\times$	has det -1, trace 0	name
1	1	1	4	1,1,1,1	no	no	
2	C_2	1	4	1,1,1,1	no	no	$\{\pm I\}$

order	type	conjugates	stable lines	orbit sizes on $\mathbb{P}^1$	det onto $\mathbb{F}_3^\times$	has det -1, trace 0	name
2	C_2	12	2	1,1,2	yes	yes	$\langle\sigma\rangle$, 3Cs.1.1
3	C_3	4	1	1,3	no	no	U
4	V_4	6	2	1,1,2	yes	yes	C_s, 3Cs
4	C_4	3	0	2,2	no	no	$\langle J\rangle$
6	S_3	4	1	1,3	yes	yes	3B.1.1
6	S_3	4	1	1,3	yes	yes	3B.1.2
6	C_6	4	1	1,3	no	no	$\{\pm I\}\times$ U
8	C_8	3	0	4	yes	no	C_{ns}, 3Cn
8	D_4	3	0	2,2	yes	yes	N_s, 3Ns
8	Q_8	1	0	4	no	no	Q_8
12	D_6	4	1	1,3	yes	yes	B, 3B
16	SD_{16}	3	0	4	yes	yes	N_{ns}, 3Nn
24	$\mathrm{SL}_2(\mathbb{F}_3)$	1	0	4	no	no	
48	$\mathrm{GL}_2(\mathbb{F}_3)$	1	0	4	yes	yes	

Consequences: the subgroups with surjective determinant form 9 conjugacy classes, and all of them except $C_{ns}(\mathbb{F}_3)$ contain an element of determinant -1 and trace 0; every subgroup of order divisible by 3 either contains $\mathrm{SL}_2(\mathbb{F}_3)$ or stabilises a line (Dickson 5.10(a) for $\ell = 3$). Reference: computation. Paper use: Proposition prop:ns, Section sec:eis (3Cs.1.1, 3B.1.1), Section sec:anomaly (3Ns).

6.4 `prop:groups:sl23` (proposition, PH). (i) $\mathrm{SL}_2(\mathbb{F}_3)$ has order 24 and element-order statistics $1, 1, 8, 6, 8$ for orders $1, 2, 3, 4, 6$; $-I$ is its unique element of order 2 (5.1(iv)). (ii) The 8 elements of order dividing 4 form the subgroup Q_8 of 6.2, which is the unique Sylow 2-subgroup of $\mathrm{SL}_2(\mathbb{F}_3)$, hence characteristic in $\mathrm{SL}_2(\mathbb{F}_3)$ and normal in $\mathrm{GL}_2(\mathbb{F}_3)$; it is isomorphic to Q_8 by 2.5 (non-abelian, unique involution), with $i = J$, $j = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$. (iii) $\mathrm{SL}_2(\mathbb{F}_3) = Q_8 \rtimes \mathrm{U} \simeq Q_8 \rtimes C_3$, with u permuting i, j, k cyclically up to sign. (iv) $[\mathrm{SL}_2(\mathbb{F}_3), \mathrm{SL}_2(\mathbb{F}_3)] = Q_8$, and $\mathrm{SL}_2(\mathbb{F}_3)$ has no subgroup of order 12. (v) $\mathrm{SL}_2(\mathbb{F}_3)/\{\pm I\} \simeq A_4$; $\mathrm{SL}_2(\mathbb{F}_3) \not\simeq S_4$ (one involution versus nine); $\mathrm{GL}_2(\mathbb{F}_3)/Q_8 \simeq S_3$. (vi) Q_8 acts transitively on the four lines; its image in S_4 is the normal V_4. Reference: direct computation; DF Chapter 4 (exercises). Paper use: Proposition prop:ns (determinant-one subgroups $\mathrm{SL}_2(\mathbb{F}_3)$ and Q_8).

6.5 `prop:groups:ns3` (proposition, PH). (i) $N_s(\mathbb{F}_3) = \langle J, \sigma\rangle$ has order 8 and is dihedral, $N_s(\mathbb{F}_3) \simeq D_4$, by 2.4 with $r = J$, $s = \sigma$, $\sigma J \sigma^{-1} = J^{-1}$. Its elements: $\pm I$, $\pm J$ and the four elements $\pm\sigma$, $\pm w$, which are exactly its elements of determinant -1, all of trace 0 and order 2. (ii) It is the normaliser of $C_s(\mathbb{F}_3)$ and the stabiliser of the pair of lines $\{\infty, 0\}$; it has 3 conjugates, one for each partition of the four lines into two pairs. (iii) Its determinant-one subgroup is $C_4 = \langle J\rangle \simeq C_4$, cyclic, abelian, generated by (the paper's) $-J$. (iv) Orbits on the four lines: $\{\infty, 0\}$ (the two lines of the Cartan subgroup) and $\{1, -1\}$ (the other two), each of size 2; C_4 has the same two orbits (J swaps ∞ and 0, and 1 and -1). (v) $N_s(\mathbb{F}_3)$ stabilises no line of $\mathbb{F}_3^2$, and no line of $\overline{\mathbb{F}}_3^2$ (5.7: non-abelian of order prime to 3). C_4 stabilises no line of $\mathbb{F}_3^2$ ($X^2 + 1$ is irreducible over $\mathbb{F}_3$), but stabilises the two eigenlines of J in $\mathbb{F}_9^2$, with eigenvalues $\pm i \in \mathbb{F}_9$, interchanged by σ (the representation-theoretic reading, $\psi \oplus \psi^{-1}$ with ψ of order 4 and $\mathrm{Ind}\,\psi$, is Chapter 4). (vi) Image in $\mathrm{PGL}_2(\mathbb{F}_3) = S_4$: the non-normal Klein group generated by the transpositions $(\infty\ 0)$ and $(1\ -1)$. (vii) $N_s(\mathbb{F}_3)$ is contained, with index 2, in the Sylow 2-subgroup $N_{ns}(\mathbb{F}_3)$ (6.6). Reference: direct computation. Paper use: Proposition prop:ns ("$N_s(\mathbb{F}_3)$ (order 8)", "the cyclic group of order 4 generated by $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$", "two orbits of size two (the two lines of the Cartan subgroup and the two others)"), Section sec:anomaly and Remark rem:ns (image 3Ns), Lemma lem:h0 (the case $\bar\rho \simeq \bar\rho \otimes \chi_{-3}$).

6.6 `prop:groups:nns3` (proposition, PH). (i) $C_{ns}(\mathbb{F}_3) \simeq C_8$, generated by $a = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$ $(= 1 + i$; $a^2 = 2i = -J$, $a^4 = -I)$. (ii) $N_{ns}(\mathbb{F}_3) = \langle a, \sigma\rangle$ has order 16 and $\sigma a \sigma^{-1} = a^3$ (conjugation is the Frobenius

$c \mapsto c^3$ of $\mathbb{F}_9$), so $N_{ns}(\mathbb{F}_3) \simeq SD_{16}$ by 2.6; element-order statistics $1, 5, 6, 4$ for orders $1, 2, 4, 8$. (iii) It is a Sylow 2-subgroup of $\mathrm{GL}_2(\mathbb{F}_3)$ ($48 = 16 \cdot 3$), with 3 conjugates. (iv) Its determinant-one subgroup is Q_8 (the $c\sigma$ with $N(c) = -1$, of order 4, together with $C_{ns} \cap \mathrm{SL}_2 = \langle J \rangle$); its three index-2 subgroups are $C_{ns}(\mathbb{F}_3) \simeq C_8$, $N_s(\mathbb{F}_3) = \langle a^2, \sigma \rangle \simeq D_4$ and $Q_8 = \langle a^2, a\sigma \rangle$. (v) Its elements of determinant -1 and trace 0 are the four involutions $c\sigma$ with $N(c) = 1$, namely $\pm\sigma, \pm w$ (the same four as in $N_s(\mathbb{F}_3)$); $C_{ns}(\mathbb{F}_3)$ itself contains none: its elements of determinant -1 are the four elements of order 8, of trace ± 1. (vi) $C_{ns}(\mathbb{F}_3)$, hence $N_{ns}(\mathbb{F}_3)$, acts transitively on the four lines ($C_{ns}/\{\pm I\} \simeq C_4$ acts simply transitively); image in S_4: a Sylow 2-subgroup D_4; Q_8 is also transitive. Reference: direct computation. Paper use: Proposition prop:ns ("$N_{ns}(\mathbb{F}_3)$ (order 16)", its determinant-one subgroup Q_8, transitivity; "the nonsplit Cartan subgroup itself being excluded by the last condition"), Section sec:anomaly (no curve with image $N_{ns}(\mathbb{F}_3)$ violates the inequality).

6.7 `thm:groups:threeimages` (theorem, PH). (Group-theoretic core of Proposition prop:ns.) Let $H \leq \mathrm{GL}_2(\mathbb{F}_3)$ satisfy: (a) $\det(H) = \mathbb{F}_3^\times$; (b) H stabilises no line of $\mathbb{F}_3^2$; (c) H contains an element of determinant -1 and trace 0. Then H is conjugate to exactly one of $\mathrm{GL}_2(\mathbb{F}_3)$, $N_{ns}(\mathbb{F}_3)$, $N_s(\mathbb{F}_3)$ (orders 48, 16, 8). Their determinant-one subgroups $H \cap \mathrm{SL}_2(\mathbb{F}_3)$, of index 2 in H, are $\mathrm{SL}_2(\mathbb{F}_3)$, Q_8 and $C_4 = \langle J \rangle$ respectively; the first two stabilise no line of $\overline{\mathbb{F}}_3^2$ (5.7), the third is abelian and stabilises two lines of $\mathbb{F}_9^2$. The subgroup $C_{ns}(\mathbb{F}_3)$ satisfies (a) and (b) but not (c). Proof (to be given in full): if $3 \mid \#H$, a Sylow 3-subgroup is some U_L (5.6); it is not normal in H, otherwise $H \leq N(\mathrm{U}_L) = \mathrm{B}_L$ would stabilise L; so H contains all four U_L, hence all 8 elements of order 3, which generate $\mathrm{SL}_2(\mathbb{F}_3)$ (they generate a subgroup of $\mathrm{SL}_2(\mathbb{F}_3)$ with at least 9 elements, and there is no subgroup of order 12 by 6.4(iv)); with (a), $H = \mathrm{GL}_2(\mathbb{F}_3)$. If $3 \nmid \#H$, H is a 2-group, so by Sylow it is conjugate into $N_{ns}(\mathbb{F}_3)$; run through the subgroups of SD_{16} using 6.6: of the index-2 subgroups only N_s satisfies (a), (b), (c); subgroups of order at most 4 with (a) are conjugate to $\langle \sigma \rangle$ or C_s and stabilise a line. Reference: proved here; for the list of images of $\bar{\rho}_{E,3}$ see Zywina15 and Chapter 11. Paper use: Proposition prop:ns (the list of three images, the determinant-one subgroups), Lemma lem:h0 (for $\ell = 3$, "absolutely irreducible on $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{-3}))$" fails exactly in the N_s case), Theorem thm:main hypothesis at $\ell = 3$.

6.8 `cor:groups:orbitdichotomy` (corollary, PH). For H as in 6.7, the following are equivalent: (i) H is conjugate to $N_s(\mathbb{F}_3)$; (ii) H has two orbits (each of size 2) on the four lines of $\mathbb{F}_3^2$; (iii) H is not transitive on the four lines; (iv) $H \cap \mathrm{SL}_2(\mathbb{F}_3)$ is abelian; (v) $H \cap \mathrm{SL}_2(\mathbb{F}_3)$ stabilises a line of $\overline{\mathbb{F}}_3^2$. Otherwise H is transitive on the four lines. Reference: 6.5, 6.6, 6.7. Paper use: Proposition prop:ns (the 3-division polynomial factors as two irreducible quadratics exactly in the N_s case and is irreducible otherwise; the translation from orbits to factors is Chapters 5 and 9), Section sec:data and the appendix (the PARI test `ns3`), Section sec:anomaly.

6.9 `prop:groups:cc` (proposition, PH). Let q be odd. An element $g \in \mathrm{GL}_2(\mathbb{F}_q)$ has $\det g = -1$ and $\mathrm{tr}\, g = 0$ iff its characteristic polynomial is $X^2 - 1$ iff g is conjugate to $\sigma = \mathrm{diag}(1, -1)$; such g has order 2, fixes exactly two lines (its eigenlines) and, for $q = 3$, swaps the other two. They form one class of $q(q+1)$ elements (12 for $q = 3$). Counts in the standard subgroups for $q = 3$: GL_2: 12; B: 6; N_s: 4; N_{ns}: 4; C_s: 2; C_{ns}: 0; SL_2: 0. Reference: elementary. Paper use: Proposition prop:ns ("contain an element of determinant -1 and trace 0 (complex conjugation)"); Chapter 11 (complex conjugation has determinant -1 and trace 0, and for ℓ odd its distinct eigenvalues ± 1 make irreducibility over $\mathbb{F}_\ell$ equivalent to absolute irreducibility).

6.10 `ex:groups:sutherland3` (worked example and convention; group facts PH, labels ST). Sutherland's labels at $\ell = 3$ (Sutherland16) attached to the standard models, defined intrinsically so that the row/column convention does not matter: 3Cs.1.1: image $\{\mathrm{diag}(1, \pm 1)\}$, i.e. $E[3] \simeq \mathbb{Z}/3 \oplus \mu_3$ (order 2, index 24); 3Cs: C_s (order 4); 3B.1.1: $\{\begin{pmatrix} 1 & * \\ 0 & * \end{pmatrix}\}$, a fixed nonzero vector, i.e. a rational point of order 3 (order 6); 3B.1.2: $\{\begin{pmatrix} * & * \\ 0 & 1 \end{pmatrix}\}$, a stable line on which the group acts by the determinant, with trivial quotient (order 6); 3Ns: $N_s(\mathbb{F}_3)$ (order 8); 3B: $\mathrm{B}(\mathbb{F}_3)$ (order 12); 3Nn: $N_{ns}(\mathbb{F}_3)$ (order 16); 3Cn: $C_{ns}(\mathbb{F}_3)$ (order 8; never the image for $E/\mathbb{Q}$, by 6.9 and the presence of complex conjugation, Chapter 11). Surjective image: $\mathrm{GL}_2(\mathbb{F}_3)$, no proper label. These are the 9 classes with surjective determinant in 6.3. Reference: Sutherland16 (tables of subgroup labels); Zywina15 (which occur for $E/\mathbb{Q}$). Paper use:

Section sec:anomaly and the abstract (3Ns), Section sec:eis and Conjecture conj:eis (3Cs.1.1, 3B.1.1, 3B), Section sec:data (images from the tables).

6.11 `prop:groups:imagesS4` (proposition, PH). Under $\mathrm{PGL}_2(\mathbb{F}_3) \simeq S_4$ (5.4): $\mathrm{GL}_2 \mapsto S_4$; $\mathrm{SL}_2 \mapsto A_4$; $N_{ns} \mapsto D_4$ (a Sylow 2-subgroup); $Q_8 \mapsto$ the normal V_4; $N_s \mapsto$ a non-normal V_4; $C_{ns} \mapsto C_4$; $C_s \mapsto C_2$ (a transposition); $\langle J \rangle \mapsto C_2$ (a double transposition); $\mathrm{B} \mapsto S_3$ (a point stabiliser); $\mathrm{U} \mapsto A_3$. Image orders 24, 12, 8, 4, 4, 4, 2, 2, 6, 3 (verified). Reference: computation. Paper use: background for Proposition prop:ns (the Galois group of the 3-division polynomial is the image in S_4, Chapter 5).

6.12 `ex:groups:gl2f5f7` (worked example, PH for orders, ST for the isomorphisms). For $q = 5$ and $q = 7$ ($\varepsilon = 2$ and $\varepsilon = 3$ respectively): $\#\mathrm{GL}_2 = 480$, 2016; $\#\mathrm{SL}_2 = 120$, 336; $\#\mathrm{PGL}_2 = 120$, 336; $\#\mathrm{PSL}_2 = 60$, 168; $\#\mathrm{B} = 80$, 252; $\#C_s = 16$, 36; $\#N_s = 32$, 72; $\#C_{ns} = 24$, 48; $\#N_{ns} = 48$, 96; $\#\mathbb{P}^1 = 6$, 8. $\mathrm{PGL}_2(\mathbb{F}_5) \simeq S_5$, $\mathrm{PSL}_2(\mathbb{F}_5) \simeq A_5$, $\mathrm{PSL}_2(\mathbb{F}_7) \simeq \mathrm{GL}_3(\mathbb{F}_2)$ simple. At $\ell = 5$ the exceptional case $\bar{G} \simeq S_4$ of 5.10 occurs inside $\mathrm{PGL}_2(\mathbb{F}_5) \simeq S_5$ and contains the image of $N_s(\mathbb{F}_5)$. For $\ell \geq 5$ the character χ_{ℓ^*} differs from $\bar{\chi}_\ell$, so the $\ell = 3$ coincidence of 6.7 has no analogue (Lemma lem:h0; Chapter 4). Reference: as 5.2, 5.4. Paper use: background (the computations at $\ell = 5, 7$, Section sec:data).

3.7 Section 7. Inverse limits and profinite groups

7.1 `def:groups:inverselimit` (definition and proposition; PH except Tychonoff, ST). An inverse system of groups $(G_n, \pi_n \colon G_{n+1} \to G_n)_{n \geq 1}$ (index set $\mathbb{N}$; the general directed index set in a remark, needed for Galois groups in Chapter 5); its inverse limit $\varprojlim G_n = \{(g_n) \in \prod G_n : \pi_n(g_{n+1}) = g_n\}$, with the universal property. Give each finite G_n the discrete topology and $\prod G_n$ the product topology; then $\varprojlim G_n$ is a closed subgroup, compact (Tychonoff) and Hausdorff. A profinite group is a topological group isomorphic to an inverse limit of finite groups. Reference: Neukirch Chapter IV, Sections 1 and 2; NSW Chapter I, Section 1. Paper use: background ($G_{\mathbb{Q}}$, $G_{\mathbb{Q}_p}$, I_p and continuity throughout; Section 1.1 of the paper).

7.2 `ex:groups:zp` (worked example, PH for the algebra). $\mathbb{Z}_p = \varprojlim \mathbb{Z}/p^n$ (a ring; its analysis and its description as a completion are Chapter 7); $\mathbb{Z}_p^\times = \varprojlim (\mathbb{Z}/p^n)^\times$, and an element of $\mathbb{Z}_p$ is a unit iff its image in $\mathbb{Z}/p$ is nonzero; $\mathbb{Z}_p$ is torsion-free and its open subgroups are the $p^n\mathbb{Z}_p$; $\hat{\mathbb{Z}} = \varprojlim \mathbb{Z}/n \simeq \prod_p \mathbb{Z}_p$ (by 2.2); foretaste: $\mathbb{Z}_\ell(1) = \varprojlim \mu_{\ell^n}$, isomorphic to $\mathbb{Z}_\ell$ as a group after choosing a compatible system of roots of unity (Chapters 5 and 7). Reference: Neukirch Chapter II, Section 1 (for $\mathbb{Z}_p$), Chapter IV, Section 2. Paper use: background (coefficients $\mathbb{Z}_\ell$, $T_\ell E$, $\mathbb{Z}_\ell(1)$ in Lemma lem:tamfactor, $\mathbb{Q}_\ell/\mathbb{Z}_\ell$ in Lemma lem:h0).

7.3 `prop:groups:profinite` (proposition; PH except (v), ST). Let $G = \varprojlim G_n$ be profinite and U_n the kernel of $G \to G_n$. (i) The U_n form a basis of open neighbourhoods of 1. (ii) A subgroup of G is open iff it is closed of finite index. (iii) A homomorphism from G to a finite group, continuous for the discrete topology, has open kernel; if the maps $G \to G_n$ are surjective, it factors through some G_n. (iv) Closed subgroups and quotients by closed normal subgroups of profinite groups are profinite; the image of a continuous homomorphism from a profinite group to a Hausdorff topological group (such as $\mathrm{GL}_2(\mathbb{Z}_\ell)$) is closed. (v) A topological group is profinite iff it is compact, Hausdorff and totally disconnected. (vi) A pro-p group is an inverse limit of finite p-groups; closed subgroups and quotients of pro-p groups are pro-p; every continuous homomorphism from a pro-p group to a finite group of order prime to p is trivial. Reference: Neukirch Chapter IV, Section 2; NSW Chapter I, Section 1. Paper use: background (Lemma lem:tamfactor: "the kernel of $I \to \langle t \rangle$ has trivial pro-ℓ quotient", made precise in Chapter 7; Lemma lem:h0: $\rho \bmod \ell^k$ factors through a finite quotient of $G_{\mathbb{Q}}$).

7.4 `prop:groups:gl2zl` (proposition, PH). Let ℓ be a prime. (i) $\mathrm{GL}_2(\mathbb{Z}_\ell) = \varprojlim \mathrm{GL}_2(\mathbb{Z}/\ell^n)$, and each reduction map $\mathrm{GL}_2(\mathbb{Z}/\ell^{n+1}) \to \mathrm{GL}_2(\mathbb{Z}/\ell^n)$ is surjective with kernel $\{1 + \ell^n X\} \simeq (M_2(\mathbb{F}_\ell), +)$ of order ℓ^4; so $\#\mathrm{GL}_2(\mathbb{Z}/\ell^n) = \ell^{4(n-1)}(\ell^2 - 1)(\ell^2 - \ell)$ (for $\ell = 3$, $n = 2$: $81 \cdot 48 = 3888$). (ii) The kernel $\Gamma(\ell) = 1 + \ell M_2(\mathbb{Z}_\ell)$ of reduction mod ℓ is a pro-ℓ group. (iii) If ℓ is odd, $\Gamma(\ell)$ is torsion-free (if $g = 1 + \ell^k A$ with $A \not\equiv 0 \bmod \ell$, then $g^m \equiv 1 + m\ell^k A \bmod \ell^{k+1}$ for $\ell \nmid m$ and $g^\ell \equiv 1 + \ell^{k+1} A \bmod \ell^{k+2}$); hence every finite subgroup of $\mathrm{GL}_2(\mathbb{Z}_\ell)$ maps isomorphically onto its image in $\mathrm{GL}_2(\mathbb{F}_\ell)$. For $\ell = 2$ this fails ($-I \in \Gamma(2)$). Reference: elementary; Serre72 Section 4 for context. Paper use: Lemma lem:IV (the image of I_q in

$\mathrm{Aut}(T_3E) = \mathrm{GL}_2(\mathbb{Z}_3)$ is cyclic of order 3 and its image in $\mathrm{Aut}(E[3])$ is nontrivial, consistently with (iii)), Lemma lem:h0 (the representations $\rho \bmod \ell^k$ and the largest k with $\rho \bmod \ell^k \simeq (\rho \otimes \chi_\ell) \bmod \ell^k$), Lemma lem:tamagawa-exp ($\rho \bmod \ell^t$).

7.5 `ex:groups:galoisprofinite` (worked example, ST; forward pointers). Statements, proved in Chapters 5 and 7: $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) = \varprojlim \mathrm{Gal}(K/\mathbb{Q})$ over finite Galois $K/\mathbb{Q}$; $\mathrm{Gal}(\mathbb{Q}(\mu_{\ell^\infty})/\mathbb{Q}) \simeq \mathbb{Z}_\ell^\times$ (the cyclotomic character); $\mathrm{Gal}(\overline{\mathbb{F}}_q/\mathbb{F}_q) \simeq \hat{\mathbb{Z}}$, topologically generated by Frobenius; a procyclic group (topologically generated by one element) is a quotient of $\hat{\mathbb{Z}}$; the tame inertia group $\prod_{\ell \neq p} \mathbb{Z}_\ell(1)$ is procyclic and the wild inertia group is pro-p. Reference: Neukirch Chapter IV, Section 1; Chapter II, Section 9; Chapter 5 and Chapter 7 of this book. Paper use: background (Section 1.1; Lemma lem:tamfactor; Lemma lem:IV "the image is tame since $q \neq 3$, hence cyclic").

3.8 Coverage check: PLAN.md scope and paper hooks to line-up items

PLAN.md scope item or paper hook	line-up items
groups, subgroups, cosets	1.1, 1.2
normal subgroups, quotients	1.4, 1.6
homomorphism theorems	1.5, 1.6
cyclic groups	1.3, 2.2
direct and semidirect products	2.1, 2.2, 2.3
group actions, orbits and stabilisers, orbit-stabiliser	3.1, 3.2, 3.3, 3.4
Sylow theorems (statement and use for small groups)	3.6, 3.7, 3.8
p-groups	3.5, 7.3(vi)
quaternion group Q_8	2.5, 6.4
dihedral groups	2.4, 6.5
finite abelian groups (structure theorem)	1.8, 4.1, 4.2, 4.3, 4.4
profinite groups as inverse limits, $\mathbb{Z}_p$, Galois groups to come	7.1, 7.2, 7.3, 7.4, 7.5
$\mathrm{GL}_2(F)$, $\mathrm{SL}_2(F)$, PGL_2, PSL_2 over a field	5.1, 5.4
orders over $\mathbb{F}_q$	5.2
Borel subgroups	5.5
split and nonsplit Cartan subgroups and their normalisers	5.8, 5.9
unipotent elements	5.6
action on $\mathbb{P}^1(\mathbb{F}_q)$, orbits of subgroups on lines	5.3, 5.9(v), 6.5(iv), 6.6(vi), 6.8
$\mathrm{GL}_2(\mathbb{F}_3)$ of order 48	6.1, 6.3
$\mathrm{SL}_2(\mathbb{F}_3)$ of order 24, Q_8	6.4
$N_s(\mathbb{F}_3)$ of order 8, dihedral	6.5
$N_{ns}(\mathbb{F}_3)$ of order 16, semidihedral	2.6, 6.6
determinant-one subgroups $\mathrm{SL}_2(\mathbb{F}_3)$, Q_8, C_4	6.4, 6.5(iii), 6.6(iv), 6.7
elements of determinant -1 and trace 0	6.9, 6.5(i), 6.6(v)
two orbits of size two of $N_s(\mathbb{F}_3)$ versus transitivity of $N_{ns}(\mathbb{F}_3)$, $\mathrm{GL}_2(\mathbb{F}_3)$	6.5(iv), 6.6(vi), 6.8
$\mathrm{GL}_2(\mathbb{F}_5)$, $\mathrm{GL}_2(\mathbb{F}_7)$ orders	5.2, 6.12
Dickson's classification (statement), maximal subgroup types	5.10, 6.3 (case $\ell = 3$)
Proposition prop:ns	1.6(ii), 3.3, 5.4(ii), 5.7, 5.9, 6.2, 6.4 to 6.9, 6.11

PLAN.md scope item or paper hook	line-up items
Lemma lem:IV	5.6, 6.1, 7.4
Remark rem:types	5.6(iv), 6.1
Section sec:anomaly (image 3Ns)	6.5, 6.8, 6.10
the Eisenstein case (Borel image), Section sec:eis, Conjecture conj:eis	5.5, 6.3, 6.10
Lemma lem:h0	1.3(iv), 6.7, 6.9, 7.3, 7.4
also: Lemma lem:tamfactor, Lemma lem:crit, Lemma lem:tamagawa-exp (background)	1.3, 2.2, 4.1, 4.3, 7.4

3.9 Part 2. Section-by-section plan with target line counts

Line counts are lines of LaTeX, card excluded (PLAN.md Section 7.3: 1,000 to 2,200).

section	title	contents	target lines
(starred)	Core results at a glance	the card: items 1.1 to 7.5, each stated precisely with tag and `\Cref`	(excluded)
(starred)	Where the paper uses this chapter	prop:ns, lem:IV, rem:types, sec:anomaly, sec:eis and conj:eis, lem:h0; lem:tamfactor, lem:crit as background	40
1	Groups, subgroups, cosets, quotients and cyclic groups	1.1 to 1.8; conventions C1 to C5 as a `convention` environment at the start	230
2	Products and the small groups	2.1 to 2.7; the comparison table of groups of order 16	200
3	Group actions and the Sylow theorems	3.1 to 3.8	260
4	Finite abelian groups	4.1 to 4.4	150
5	Matrix groups over a field	5.1 to 5.10; the orders table; one figure-free description of $\mathbb{P}^1(\mathbb{F}_3)$	330
6	The group $\mathrm{GL}_2(\mathbb{F}_3)$ in full	6.1 to 6.12; class table, subgroup table, the theorem 6.7 with full proof, `paperlink` boxes for prop:ns, lem:IV, rem:types, sec:anomaly	380
7	Inverse limits and profinite groups	7.1 to 7.5	170
8	Exercises	14 exercises, graded (list below)	110

section	title	contents	target lines
9	Notes and references	DF, Lang, Serre72, Sutherland16, Zywina15, Neukirch, NSW; which numbers were cited	40
	total		**1910**

Exercises (Section 8), graded easy to harder: 1. Show $\#\mathrm{GL}_2(\mathbb{Z}/9) = 3888$ and $\#\mathrm{SL}_2(\mathbb{Z}/9) = 648$. 2. Compute the conjugacy classes of D_4 and of Q_8; deduce that they have the same character degrees (RHB link). 3. Prove that $C_8 \times C_2$, D_8, SD_{16}, Q_{16}, M_{16} are pairwise non-isomorphic (element orders). 4. Find all subgroups of SD_{16} and identify each. 5. Verify the class table 6.1 and use the orbit-counting lemma 3.4 to count the orbits of $N_s(\mathbb{F}_3)$ on $\mathbb{P}^1(\mathbb{F}_3)$. 6. Check the paper's claim that the determinant-one subgroup of $N_s(\mathbb{F}_3)$ is cyclic of order 4 generated by $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$, and that of $N_{ns}(\mathbb{F}_3)$ is Q_8. 7. Check the paper's claim that $N_s(\mathbb{F}_3)$ has two orbits of size two on the four lines and $N_{ns}(\mathbb{F}_3)$ is transitive. 8. Show that $C_{ns}(\mathbb{F}_3)$ contains no element of determinant -1 and trace 0, and conclude that 3Cn cannot contain the image of a complex conjugation. 9. Show that a subgroup of $\mathrm{GL}_2(\mathbb{F}_3)$ of order 6 stabilises a line; classify the subgroups of order 6 up to conjugacy (two copies of S_3, one C_6). 10. Prove that $\det g$ is the sign of the permutation of the four lines for $g \in \mathrm{GL}_2(\mathbb{F}_3)$ (5.4(ii)). 11. For q odd, show that $N_{ns}(\mathbb{F}_q) \cap \mathrm{SL}_2(\mathbb{F}_q)$ has a unique element of order 2, namely $-I$, and that its other elements outside C_{ns} have order 4. 12. Show that $N_s(\mathbb{F}_5)$ is a Sylow 2-subgroup of $\mathrm{GL}_2(\mathbb{F}_5)$ and that its image in $\mathrm{PGL}_2(\mathbb{F}_5) \simeq S_5$ lies in a copy of S_4 (hard). 13. Show that $1 + 3M_2(\mathbb{Z}_3)$ is torsion-free and that $1 + 2M_2(\mathbb{Z}_2)$ is not. 14. Show that the image of a continuous homomorphism $\mathbb{Z}_p \to \mathrm{GL}_2(\mathbb{F}_\ell)$ is cyclic of p-power order; deduce that it is trivial when $p \nmid \#\mathrm{GL}_2(\mathbb{F}_\ell)$ (foretaste of tame inertia, Chapter 7).

3.10 Part 3. Dependencies and forward pointers

Results from other chapters used here: none. The chapter uses only PLAN.md Section 2 (linear algebra over a field: determinants, eigenvalues, change of basis; elementary group theory as in RHB). Two points need a word: (a) the finite fields: the chapter needs only $\mathbb{F}_p = \mathbb{Z}/p$ (Bezout) and the field $\mathbb{F}_q[\sqrt{\varepsilon}]$ built by hand in 5.8; general $\mathbb{F}_q$ is Chapter 2, and every statement for $\mathbb{F}_q$ may be read with $q = p$; (b) Tychonoff's theorem (7.1) and the topological characterisation of profinite groups (7.3(v)) are stated with references.

Forward pointers (chapters that use this one): * Chapter 2 (`ch:rings`): cyclicity of $\mathbb{F}_q^\times$ (4.2, restated with field theory), the index $[\mathbb{F}_q^\times : (\mathbb{F}_q^\times)^{\ell^k}]$ (1.3(ii)), $(\mathbb{Z}/n)^\times = \mathrm{Aut}(C_n)$ (1.3(iii)), Chinese remainder (2.2). * Chapter 3 (`ch:modules`): structure theorem over a PID generalises 4.1; $\mathbb{Z}_\ell$ as a DVR (7.2); lengths of finite ℓ-groups (4.3); Pontryagin duality of finite abelian groups. * Chapter 4 (`ch:reps`): representations of $N_s(\mathbb{F}_3)$, C_4, Q_8, $\mathrm{SL}_2(\mathbb{F}_3)$ on $\mathbb{F}_3^2$ and $\mathbb{F}_9^2$; absolute irreducibility (5.7 is the group-theoretic input); ψ of order 4 and $\mathrm{Ind}\,\psi$; Clifford theory for index 2 (3.3(iii) is the permutation analogue); Dickson for PGL_2 (5.10); unipotent and minus-unipotent fixed spaces (5.6). * Chapter 5 (`ch:galois`): profinite Galois groups and the Krull topology (7.1 to 7.5); the factorisation pattern of the 3-division polynomial as the orbit pattern on lines (6.8) and its discriminant via det = sign (5.4(ii)). * Chapter 6 (`ch:ant`) and Chapter 7 (`ch:local`): decomposition and inertia groups as stabilisers (3.2), the semidirect structure of the tame Galois group (2.3), pro-p wild inertia (7.3(vi)), $\mathbb{Z}_p$ (7.2). * Chapter 9 (`ch:ellcurves`): $E[m] \simeq (\mathbb{Z}/m)^2$ and $\mathrm{Aut}(E[m]) = \mathrm{GL}_2(\mathbb{Z}/m)$, $T_\ell E$ and $\mathrm{GL}_2(\mathbb{Z}_\ell)$ (7.4), the four lines of $E[3]$ (5.3). * Chapter 10 (`ch:reduction`): component groups as finite abelian groups (4.1, 4.4). * Chapter 11 (`ch:galrep`): Proposition prop:ns (6.5 to 6.10), Lemma lem:IV (5.6, 7.4), Remark rem:types (5.6(iv), 6.1), Zywina's lists and Sutherland's labels (6.10), the Eisenstein case (5.5), complex conjugation (6.9). * Chapter 15 (`ch:modgalrep`): Serre conductor exponents through fixed spaces of inertia images (5.6).

* Chapters 16 and 17 (`ch:cohomology`, `ch:galcoh`): cohomology of finite and profinite groups (7.3), inflation and restriction through quotients. * Chapter 21 (`ch:paper`) and Chapter 22 (`ch:computation`): the labels 3Ns, 3Nn, 3B, 3Cs.1.1 and the `ns3` test (6.8, 6.10).

3.11 Part 4. Notes for the writer

Conventions to fix (state them once, in Section 1, as `convention` environments): * C1 to C5 above. Column vectors, left action; lines $[v]$; Moebius coordinate with $\infty = [1:0] = [e_1]$. * D_n has order $2n$ (so $N_s(\mathbb{F}_3) \simeq D_4$; Dummit-Foote writes D_8). Say so in a footnote-free sentence. * Standard models 6.2, with $\varepsilon = -1$ at $q = 3$; $J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$; the paper's generator $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = -J = J^{-1}$ generates the same C_4. * C_s, N_s, C_{ns}, N_{ns}, B, U always mean the standard models; "image X" means "conjugate to X". Use the paper's macro `\Ns` only for the label 3Ns, and write $N_s(\mathbb{F}_3)$, $N_{ns}(\mathbb{F}_3)$ for the groups (the paper writes `N_{ns}`, `N_s` in subscripts). * Sutherland's labels: define each by an intrinsic property (6.10), not by matrices alone. Transposition (row versus column convention) swaps 3B.1.1 and 3B.1.2; the intrinsic definitions (fixed vector versus trivial quotient) are convention-free. The writer should double-check the label list and the index of each group against the tables of Sutherland16 before finalising.

Pitfalls: * The paper justifies the absolute irreducibility of $\mathrm{SL}_2(\mathbb{F}_3)$ and Q_8 by "non-abelian and faithful". That is not sufficient in general (a Borel subgroup is non-abelian and faithful); use 5.7 (commutator subgroup inside the unipotent radical). Do not "correct" the paper; supply the step. * $C_4 = \langle J \rangle$ is irreducible over $\mathbb{F}_3$ (no stable $\mathbb{F}_3$-line) but not absolutely irreducible. The paper's hypothesis is about absolute irreducibility; the distinction is exactly the N_s case. Keep "stabilises no line of K^2" (group language) here and leave the word "absolutely irreducible" to Chapter 4, with a pointer. * $N_s(\mathbb{F}_3)$ is contained in $N_{ns}(\mathbb{F}_3)$ (index 2, standard models); the two cases of Proposition prop:ns are distinguished by orbits and by the determinant-one subgroup, not by maximality. Never call $N_s(\mathbb{F}_\ell)$ maximal for $\ell = 3$ or $\ell = 5$ (6.12, 5.10). * $N_s(\mathbb{F}_3)$ itself acts absolutely irreducibly on $\mathbb{F}_3^2$; only its determinant-one subgroup does not. Do not write "the image $N_s(\mathbb{F}_3)$ is reducible". * Every element of $N \setminus C$ has trace 0, for both normalisers; the determinant -1, trace 0 elements of $\mathrm{GL}_2(\mathbb{F}_3)$ are 12 in number and all conjugate to $\mathrm{diag}(1,-1)$; $C_{ns}(\mathbb{F}_3)$ has none, which is the reason 3Cn does not occur. * Elements of order 3 of $\mathrm{GL}_2(\mathbb{F}_3)$ are unipotent (characteristic 3), whereas in $\mathrm{GL}_2(\mathbb{Z}_3)$ an element of order 3 has eigenvalues the primitive cube roots of unity (Lemma lem:IV); the two are compatible through 7.4(iii) and the $\mathbb{Z}_3[\zeta_3]$ computation of Chapter 4 and 11. Do not attempt the $\mathbb{Z}_3[\zeta_3]$ argument here. * Cartan subgroups need q odd here (ε a non-square; for $q = 2$, C_s is trivial and its normaliser is everything). The normaliser claim for C_s needs $q \geq 3$. * Do not confuse the field size q with the paper's bad prime q (C4). * Typography (PLAN.md 7.2): no dash ligatures; write "Diamond-Flach-Guo" with a plain hyphen; number ranges with "to"; no Unicode. Tables 6.1 and 6.3 should be `tabular` with `booktabs`.

Leave to other chapters: * Chapter 2: construction and uniqueness of $\mathbb{F}_q$, Frobenius automorphism, quadratic residues; Chapter 1 builds only $\mathbb{F}_q[\sqrt{\varepsilon}]$ by hand. * Chapter 4: characters, Schur, Maschke, absolute irreducibility, induced representations, Frobenius reciprocity, $\bar{\rho} \simeq \bar{\rho} \otimes \chi$, the trace-zero argument of Lemma lem:h0, Brauer-Nesbitt. * Chapter 5: the Galois group of a polynomial as a permutation group, the dictionary orbits/factors, the discriminant, the Krull topology, Chebotarev. * Chapter 11: which subgroups occur as images of $\bar{\rho}_{E,3}$ (Zywina15), the proof of Proposition prop:ns as a whole, Lemma lem:IV and Remark rem:types as statements about elliptic curves, the Serre conductor.

Chapter 4

Chapter 02: Rings, fields, polynomials and characters (`ch:rings`, `chapters/ch02_rings.tex`)

Stage 1 line-up. Part I (Algebra). Target length 2,080 lines of LaTeX (card excluded; see Part 3), inside the 1,000 to 2,200 range of PLAN.md Section 7.3.

Rigour tags: **P** = proved here, **S** = proof sketched here, **R** = stated; reference. Item format, as in the Chapter 5 and 6 outlines: number, label-ready name, kind, tag; the precise statement with all hypotheses; reference (book and section or chapter; a theorem number only where sure); the paper's use by its `\label` names, or "background". Every computation quoted below was checked by hand and by short Python/sympy scripts in `/tmp/ch02` (verification log, Part 7).

4.1 0. What this chapter is for

The paper uses this chapter in seven places, and the chapter is built around them.

1. **Definition def:terms and the table of Section 1.** The local terms $e_\ell(E,q)$ and $t_\ell(E,q)$ are ℓ-adic valuations of rational numbers: $v_\ell\big(P_q(q^{-2})/(1-a_q(E)^2q^{-2})\big)$, $v_\ell(-v_q(j_E))$, $v_\ell(q-\chi_{-3}(q))$, $v_\ell(q-\chi_{-4}(q))$, $v_\ell(q^2-1)=v_\ell(1-q^{-2})$, $v_\ell(q-1)+v_\ell((q+1)^2-a_q(E')^2)$. The chapter supplies v_p on $\mathbb{Q}$, the ring $\mathbb{Z}_{(p)}$, the fact that a rational whose denominator is a power of $q\neq\ell$ is ℓ-integral (Lemma lem:fpr: "a nonzero rational number which is an ℓ-adic integer"), the algebraic identity behind the I_0^* row, and the arithmetic of every entry (when it is positive, for $\ell=3$ and for $\ell\geq 5$).
2. **The characters χ_{-3}, χ_{-4}, χ_{q^*}, χ_S as Dirichlet characters** (table of Section 1, Lemma lem:euler, Proposition prop:forms, Lemma lem:twist, Theorems thm:twist, thm:quant, thm:qual and its Example): Kronecker symbols, conductors 3, 4, q, $\prod_{q\in S}q$, values at primes, the congruences $q\equiv\chi_{-3}(q) \pmod 3$ and $q\equiv\chi_{-4}(q) \pmod 4$, and the sign in $q^*=(-1)^{(q-1)/2}q$.
3. **Quadratic reciprocity**, which is what makes $(q^*/\cdot)$ "the quadratic character of conductor q" (Lemma lem:twist), and which Chapters 5 and 6 cite.
4. **Lemma lem:tamfactor and Section sec:tam**: ℓ-th powers in $\mathbb{F}_q^\times$, the index $[\mathbb{F}_q^\times:(\mathbb{F}_q^\times)^{\ell^k}]=\ell^{\min(k,v_\ell(q-1))}$, and the class c of the (reduced) unit part of the Tate parameter in $\mathbb{F}_q^\times/(\mathbb{F}_q^\times)^{\ell^{v_\ell(q-1)}}\simeq\mathbb{Z}/\ell^{v_\ell(q-1)}$, with "$v_\ell(c)\geq k$ iff u is an ℓ^k-th power".
5. **Lemma lem:crit** (and Lemma lem:hecke, Theorem thm:quant Step 1): $\mathbb{T}/(I+J)\simeq\mathbb{Z}/\phi_f(J)$ is cyclic, the exact sequence $0\to R/(I\cap J)\to R/I\times R/J\to R/(I+J)\to 0$, products of rings and idempotents ($\mathbb{T}\otimes\mathbb{Q}=\mathbb{Q}\times\mathbb{T}'$), and "a maximal ideal of residue characteristic ℓ contains $I+J$, so ℓ divides $\#\mathbb{T}/(I+J)=r$".

6. **Lemma lem:h0 in its corrected form** (important_notes.md points 5 and 6) and Proposition prop:ns: $a^2 \equiv 1 \pmod{\ell^k}$ for every unit a iff $\ell^k = 3$; on $(\mathbb{Z}/\ell)^\times$ the Legendre symbol is $\omega_\ell^{(\ell-1)/2}$, where $\omega_\ell(a) = a \bmod \ell$, and the two agree iff $\ell = 3$ (the Dirichlet side of "$\bar{\chi}_3 = \chi_{-3}$").
7. **Foundations for Chapters 3, 5, 6, 7**: ideals, PIDs and UFDs, Gauss's lemma, Eisenstein, finite fields (existence, uniqueness, Frobenius, subfields, cyclic multiplicative group), roots of unity and cyclotomic polynomials, Gauss sums ($g^2 = \ell^*$), $\mathbb{Z}[i]$ and $\mathbb{Z}[\zeta_3]$ Euclidean, $(\mathbb{Z}/n)^\times$ and primitive roots, v_p and $|\cdot|_p$ (definition only).

Split with the neighbouring chapters (decided here, consistent with what their outlines assume): * **Chapter 1** proves the group theory: cyclic groups and $[C : C^{\ell^k}] = \ell^{\min(k, v_\ell(n))}$ (its 1.3), finite subgroups of $K^\times$ cyclic (its 4.2), CRT for groups (its 2.2), $\mathrm{Aut}(\mathbb{Z}/n) = (\mathbb{Z}/n)^\times$. Chapter 2 *applies* these to $\mathbb{F}_q^\times$ and $(\mathbb{Z}/n)^\times$ and does not reprove them. * **Chapter 5** does the Galois theory: $\mathrm{Gal}(\mathbb{F}_{q^n}/\mathbb{F}_q)$ cyclic generated by Frobenius, the factorisation of $\Phi_n \bmod p$, irreducibility of all Φ_n, $\mathrm{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q}) \simeq (\mathbb{Z}/n)^\times$, $\mathbb{Q}(\sqrt{\ell^*}) \subset \mathbb{Q}(\zeta_\ell)$, Kummer theory over $\mathbb{F}_q$ (the Kummer character $\chi_u(\phi_q) = u^{(q-1)/\ell^k}$), quadratic characters of $G_\mathbb{Q}$. Chapter 2 gives existence, uniqueness, Frobenius as an automorphism of order r, subfields, cyclicity, the power criteria in $\mathbb{F}_q^\times$ as pure group theory, $\Phi_n \in \mathbb{Z}[x]$, Eisenstein for Φ_{p^k}, and $g^2 = q^*$. * **Chapter 6** does the number fields: splitting of p in $\mathbb{Q}(\sqrt{d})$ by $\left(\frac{D}{p}\right)$, $\chi_d(\mathrm{Frob}_p) = \chi_D(p)$, Dirichlet characters as characters of $G_\mathbb{Q}$, the primes of $\mathbb{Z}[i]$ and $\mathbb{Z}[\zeta_3]$, $\pi = 1 - \zeta_3$, $\mathbb{Z}_3[\zeta_3]$ a DVR, $\mathbb{Z}_\ell[\zeta_3]$, fundamental discriminants as field discriminants, a second proof of reciprocity. Chapter 2 gives the Kronecker and Dirichlet characters as functions on $\mathbb{Z}$, their conductors and values, reciprocity by Gauss sums, and $\mathbb{Z}[i]$, $\mathbb{Z}[\zeta_3]$ Euclidean with their unit groups (Chapter 6, item 4.5, cites this). * **Chapter 7** does $\mathbb{Q}_p$, $\mathbb{Z}_p$, Hensel, the structure of $\mathbb{Z}_p^\times$, the reduction of $u \in \mathbb{Z}_q^\times$; Chapter 2 gives v_p, $|\cdot|_p$, $\mathbb{Z}_{(p)}$ and its reduction map, and $(\mathbb{Z}/p^k)^\times$. * **Chapter 3** does localisation, DVRs, $\mathbb{Z}_\ell$, lengths, Fitting ideals, finite algebras over a DVR and their idempotents; Chapter 2 gives $\mathbb{Z}_{(p)}$ as the first example and idempotents of a product of rings.

4.2 1. Global conventions fixed in this chapter

- **C1 (rings).** Rings are commutative with 1, homomorphisms send 1 to 1, subrings contain 1. The zero ring is allowed; fields and integral domains are nonzero. $R^\times$ is the unit group. (The paper's $\mathbb{T}$ is such a ring; matrix rings appear only as examples and are flagged as non-commutative.)
- **C2 (primes and q).** p, q, ℓ denote primes unless stated. In Sections 6 and 7 of the chapter $q = p^r$ is a prime power and $\mathbb{F}_q$ a finite field with q elements. **In the paper q is a prime** (a bad prime of E), so $\mathbb{F}_q = \mathbb{Z}/q\mathbb{Z}$ there and "primitive root mod q" applies; say this in the paperlink of Section 7. ℓ is the paper's coefficient prime, odd in all uses. Chapter 1's warning (its C4: field size versus bad prime) is repeated once.
- **C3 (valuations).** $v_p \colon \mathbb{Q} \to \mathbb{Z} \cup \{+\infty\}$ with $v_p(0) = +\infty$; $|x|_p = p^{-v_p(x)}$, $|0|_p = 0$. For an ideal $r\mathbb{Z}$, $v_p(r\mathbb{Z}) := v_p(r)$. For $c \in \mathbb{Z}/\ell^m\mathbb{Z}$ put $v_\ell(c) := \min(v_\ell(\tilde{c}), m)$ for any lift $\tilde{c} \in \mathbb{Z}$ (well defined; so $v_\ell(0) = m$); then $v_\ell(c) \geq k$ iff $c \in \ell^k\mathbb{Z}/\ell^m\mathbb{Z}$, for $0 \leq k \leq m$. With this convention the paper's $\min(v_\ell(n), v_\ell(q-1), v_\ell(c))$ is meaningful also when $c = 0$, and when $v_\ell(q-1) = 0$ (then $c = 0$ in the zero group and the minimum is 0).
- **C4 (symbols).** $\left(\frac{a}{p}\right)$ Legendre (p odd prime), $\left(\frac{a}{n}\right)$ Jacobi (n odd positive), and the Kronecker symbol $\left(\frac{a}{n}\right)$ for all $a \in \mathbb{Z}$, $n \in \mathbb{Z}$, with $\left(\frac{a}{2}\right) = 0$ for a even, 1 for $a \equiv \pm 1 \pmod 8$, -1 for $a \equiv \pm 3 \pmod 8$; $\left(\frac{a}{-1}\right) = 1$ for $a \geq 0$ and -1 for $a < 0$; $\left(\frac{a}{1}\right) = 1$; $\left(\frac{a}{0}\right) = 1$ if $a = \pm 1$, else 0.
- **C5 (Dirichlet characters and their names).** A Dirichlet character mod n is a function $\chi \colon \mathbb{Z} \to \mathbb{C}$ coming from a homomorphism $(\mathbb{Z}/n)^\times \to \mathbb{C}^\times$, with $\chi(a) = 0$ when $\gcd(a, n) > 1$. Quadratic characters are named by a discriminant: $\chi_D(a) = \left(\frac{D}{a}\right)$ for $D \equiv 0, 1 \pmod 4$. So $\chi_{-3} = \left(\frac{-3}{\cdot}\right)$, $\chi_{-4} = \left(\frac{-4}{\cdot}\right)$. For an odd prime q, $q^* = (-1)^{(q-1)/2}q$ and χ_{q^*} is "the quadratic character of conductor q"; the paper writes χ_q for it, the book never does (PLAN.md 7.5, decision D3), so the paper's χ_3, χ_7, χ_{11} are the book's χ_{-3}, χ_{-7}, χ_{-11}, while χ_5, χ_{13} coincide. **Warning:** the paper also writes χ_ℓ for the ℓ-adic cyclotomic character (Section 4, Lemmas lem:tamfactor, lem:old, lem:h0) and $\bar{\chi}_\ell$ for its reduction. The book writes χ_{q^*} (or $\left(\frac{\cdot}{q}\right)$), never χ_q; the paper's χ_q appears only when quoting the

paper (inside `paperlink` environments), χ_ℓ with a letter subscript is the cyclotomic character, and χ_3^{cyc} is written where a number is needed (D3); and write $\omega_\ell\colon a \mapsto a \bmod \ell \in \mathbb{F}_\ell^\times$ for the Dirichlet character that corresponds to $\bar{\chi}_\ell$ (Chapters 5, 6 make the correspondence). These conventions agree with Chapter 5 (C3, C4) and Chapter 6 (4.2, 5.8).

- **C6 (roots of unity).** $\mu_n(K) = \{x \in K : x^n = 1\}$; $\zeta_n = e^{2\pi i/n} \in \mathbb{C}$; $\zeta_3 = (-1+\sqrt{-3})/2$ (never written ω, which the book reserves for differentials and fundamental characters: Appendix B, B-C15(b)) with $\sqrt{-3} := 1 + 2\zeta_3 = i\sqrt{3}$ (the sign fixed as in Chapter 6). Do not use π for $1-\zeta_3$ in this chapter (that is Chapter 6's notation); π is $3.14159\ldots$ here.
- **C7 (typography).** PLAN.md Section 7.2: no dash ligatures, ASCII source, labels `...:rings:...`.

4.3 2. Line-up of the core results, in teaching order

4.3.1 Section 1. Rings, ideals, quotients, products

1.1 `def:rings:ring` (definition and proposition; P). A ring (C1), its unit group $R^\times$, zero divisors, nilpotent elements, fields, integral domains, subrings, homomorphisms, kernel and image. Examples: $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$, $\mathbb{Z}/n\mathbb{Z}$, $R[x]$, $R_1 \times R_2$, the subring $\mathbb{Z}[i] \subset \mathbb{C}$, a commutative subring of $\mathrm{End}_{\mathbb{Z}}(A)$ for an abelian group A (the shape of the Hecke algebra $\mathbb{T}$). For every ring R there is a unique homomorphism $\mathbb{Z} \to R$; its kernel is $n\mathbb{Z}$ for a unique $n \geq 0$, the characteristic of R. A domain has characteristic 0 or a prime. A finite domain is a field ($x \mapsto ax$ is injective, hence bijective). Reference: Dummit-Foote (DF) 7.1 to 7.3. Paper: background ($\mathbb{T}$ is "the subring of $\mathrm{End}\, S_2(\Gamma_0(N))$ generated over $\mathbb{Z}$ by all T_n", Section 3).

1.2 `def:rings:ideal` (definition and proposition; P). An ideal $I \subset R$; the ideal $(a_1, \ldots, a_n)$ generated by elements; sum $I+J$, product IJ, intersection $I \cap J$, with $IJ \subset I \cap J$; the annihilator $\mathrm{Ann}_R(S) = \{r : rS = 0\}$ of a subset of an R-module (definition only; modules are Chapter 3); I and J are comaximal if $I + J = R$. The quotient ring R/I and the projection $R \to R/I$; kernels of homomorphisms are exactly the ideals. Every ideal of $\mathbb{Z}$ is $n\mathbb{Z}$ for a unique $n \geq 0$ (division with remainder), and $m\mathbb{Z} + n\mathbb{Z} = \gcd(m,n)\mathbb{Z}$, $m\mathbb{Z} \cap n\mathbb{Z} = \mathrm{lcm}(m,n)\mathbb{Z}$. Reference: DF 7.3, 7.4. Paper: Lemma lem:crit ($I = \ker \phi_f$, $J = \mathrm{Ann}_{\mathbb{T}}(f^\perp)$, $I+J$), Lemma lem:ddt417 ($\wp = \ker \pi$, $I = \mathrm{Ann}_{\mathbb{T}}(\wp)$, the congruence ideal $\eta = \pi(I)$).

1.3 `thm:rings:isomorphism` (theorem; P). (i) A homomorphism $\phi\colon R \to S$ induces an isomorphism $R/\ker\phi \simeq \phi(R)$. (ii) (Correspondence.) For an ideal I, $J \mapsto J/I$ is an inclusion-preserving bijection from ideals of R containing I to ideals of R/I, with $(R/I)/(J/I) \simeq R/J$; prime and maximal ideals correspond (after 2.1). (iii) If $\phi\colon R \to S$ is **surjective** with kernel I and $J \subset R$ is an ideal, then $\phi(J)$ is an ideal of S and ϕ induces $R/(I+J) \simeq S/\phi(J)$. (iv) In particular every quotient ring of $\mathbb{Z}$ is $\mathbb{Z}/r\mathbb{Z}$ for a unique $r \geq 0$, cyclic as an additive group. Reference: DF 7.3. Paper: Lemma lem:crit ("$\mathbb{T}/(I+J) \simeq \mathbb{Z}/\phi_f(J)$ is cyclic", with $\phi_f\colon \mathbb{T} \to \mathbb{Z}$ surjective because $\phi_f(T_1) = 1$; "equivalently, r generates the congruence ideal $\phi_f(J)$"), Lemma lem:hecke ($r_E\mathbb{Z}_\ell$ is the congruence ideal).

1.4 `prop:rings:two-ideals` (proposition; P). Let I, J be ideals of a ring R. The sequence of additive groups (indeed of R-modules)

$$0 \to R/(I \cap J) \xrightarrow{x \mapsto (x,x)} R/I \times R/J \xrightarrow{(y,z) \mapsto y - z} R/(I+J) \to 0$$

is exact. Consequently: (i) if $I \cap J = 0$, then R embeds in $R/I \times R/J$ with cokernel $R/(I+J)$; (ii) (Chinese remainder theorem for two ideals) if $I + J = R$ then $I \cap J = IJ$ and $R/IJ \simeq R/I \times R/J$ as rings; (iii) (general CRT) if $I_1, \ldots, I_k$ are pairwise comaximal then $R/(I_1 \cdots I_k) \simeq \prod_i R/I_i$. Reference: DF 7.6. Paper: Lemma lem:crit ("Dualising the inclusion $\mathbb{T} \subset \mathbb{T}/I \times \mathbb{T}/J$, whose cokernel is $\mathbb{T}/(I+J)$", with $I \cap J = 0$); the module analogue $L[\wp] \oplus L[I] \subset L$ in Lemma lem:ddt417 is Chapter 3; Chapter 6, item 2.4 (CRT for Dedekind domains) cites (iii).

1.5 `prop:rings:idempotents` (definition and proposition; P). An idempotent is $e \in R$ with $e^2 = e$; then $1-e$ is idempotent and $R \to Re \times R(1-e)$, $x \mapsto (xe, x(1-e))$, is a ring isomorphism (Re is a ring with identity e). R is isomorphic to a product of two nonzero rings iff it has an idempotent $e \neq 0, 1$. The ideals of $R_1 \times R_2$ are the $I_1 \times I_2$; its prime ideals are $P_1 \times R_2$ and $R_1 \times P_2$. Example: $\mathbb{Q}[x]/(x^2 - x) \simeq \mathbb{Q} \times \mathbb{Q}$; more generally $K[x]/(f) \simeq \prod_i K[x]/(f_i)$ for $f = \prod f_i$ with pairwise coprime f_i (by 1.4(iii)). Reference: DF 7.6; Atiyah-Macdonald Chapter 1 (exercises). Paper: Lemma lem:crit

("$\mathbb{T}\otimes\mathbb{Q}=\mathbb{Q}\times\mathbb{T}'$", $I=\mathbb{T}\cap(0\times\mathbb{T}')$, $J=\mathbb{T}\cap(\mathbb{Q}\times 0)$), Theorem thm:quant Step 1 (the idempotent of $\mathbb{T}_\ell$ projecting to the $\mathfrak{m}$-component; the general statement for finite $\mathbb{Z}_\ell$-algebras is Chapter 3).

1.6 `prop:rings:freshman` (proposition; P). If R has prime characteristic p then $\binom{p}{i}\equiv 0 \pmod p$ for $0<i<p$, so $(x+y)^p=x^p+y^p$ and $F\colon x\mapsto x^p$ is a ring endomorphism of R (the absolute Frobenius); also $(x+y)^{p^k}=x^{p^k}+y^{p^k}$. F is injective if R is a domain and bijective if R is a finite domain (a finite field). Corollary (Fermat): $a^p\equiv a \pmod p$ for all $a\in\mathbb{Z}$. Reference: DF 13.5. Paper: background (finite fields, Section 6; the Frobenius of Chapters 5 and 6).

4.3.2 Section 2. Prime and maximal ideals, domains, fractions, $\mathbb{Z}_{(p)}$

2.1 `thm:rings:prime-maximal` (definition and theorem; P). An ideal $P\neq R$ is prime if $ab\in P$ implies $a\in P$ or $b\in P$; an ideal $M\neq R$ is maximal if no ideal lies strictly between M and R. (i) R/P is a domain iff P is prime; R/M is a field iff M is maximal; maximal ideals are prime. (ii) The preimage of a prime ideal under a homomorphism is prime; the kernel of a homomorphism $R\to K$, K a field, is prime, and maximal if the map is surjective (or if its image is finite, by 1.1). (iii) If R/P is finite then P is prime iff maximal. (iv) The prime ideals of $\mathbb{Z}$ are 0 and $p\mathbb{Z}$; the maximal ones are the $p\mathbb{Z}$. (v) Every nonzero ring has a maximal ideal (Zorn's lemma; stated, with the elementary proof for Noetherian rings, Chapter 3); every non-unit lies in a maximal ideal. *P except (v), R.* Reference: DF 7.4. Paper: Lemma lem:crit ("$I+J$ lies in the kernel of $\mathbb{T}\to\mathcal{O}/\lambda$, a maximal ideal of residue characteristic ℓ"), Lemma lem:hecke and Theorem thm:quant ($\mathfrak{m}$, the maximal ideal of $\mathbb{T}\otimes\mathbb{Z}_\ell$ containing ℓ and $\ker\phi_f$).

2.2 `prop:rings:residue-char` (proposition; P). Let $\mathfrak{m}$ be a maximal ideal of a ring R such that $R/\mathfrak{m}$ has characteristic $\ell>0$ ("residue characteristic ℓ"). Then $\ell\cdot 1\in\mathfrak{m}$. If $\mathfrak{a}\subset\mathfrak{m}$ is an ideal with $R/\mathfrak{a}$ finite, then $\ell\mid\#(R/\mathfrak{a})$ (because $R/\mathfrak{a}$ surjects onto the field $R/\mathfrak{m}$, an $\mathbb{F}_\ell$-vector space of dimension ≥ 1, and additive orders multiply along the quotient). In particular a ring homomorphism $\mathbb{Z}/r\mathbb{Z}\to k$ into a field of characteristic ℓ exists iff $\ell\mid r$ (for $r\geq 1$). Reference: elementary. Paper: Lemma lem:crit, last step ("and ℓ divides $\#\mathbb{T}/(I+J)=r$"); Lemma lem:old ("after enlarging $\mathcal{O}$ and λ"); Chapter 6, item 7.3, cites this.

2.3 `thm:rings:fractions` (definition and theorem; P). For a domain R the field of fractions $\mathrm{Frac}(R)$ (pairs a/b, $b\neq 0$, modulo $ad=bc$) is a field containing R, with the universal property: every injective homomorphism $R\to K$ into a field extends uniquely to $\mathrm{Frac}(R)\to K$. Examples: $\mathrm{Frac}(\mathbb{Z})=\mathbb{Q}$, $\mathrm{Frac}(F[x])=F(x)$, $\mathrm{Frac}(\mathbb{Z}[i])=\mathbb{Q}(i)$, $\mathrm{Frac}(\mathbb{Z}[\zeta_3])=\mathbb{Q}(\zeta_3)=\mathbb{Q}(\sqrt{-3})$. Reference: DF 7.5. Paper: background ($\mathbb{Q}$; number fields in Chapter 6).

2.4 `prop:rings:zp-local` (definition and proposition; P). For a prime p let $\mathbb{Z}_{(p)}=\{a/b\in\mathbb{Q}: p\nmid b\}$. (i) $\mathbb{Z}_{(p)}$ is a subring of $\mathbb{Q}$ with $\mathbb{Z}_{(p)}^\times=\{a/b: p\nmid ab\}$; an integer prime to p is a unit of $\mathbb{Z}_{(p)}$. (ii) Its nonzero ideals are the $p^k\mathbb{Z}_{(p)}$, $k\geq 0$; it has exactly one maximal ideal $p\mathbb{Z}_{(p)}$ (it is a local ring; the first example of a discrete valuation ring, Chapter 3). (iii) The reduction map $\mathbb{Z}_{(p)}\to\mathbb{F}_p$, $a/b\mapsto\bar{a}\,\bar{b}^{-1}$, is a well-defined surjective ring homomorphism with kernel $p\mathbb{Z}_{(p)}$; it maps $\mathbb{Z}_{(p)}^\times$ onto $\mathbb{F}_p^\times$. (iv) $\mathbb{Z}_{(p)}=\{x\in\mathbb{Q}: v_p(x)\geq 0\}$ and $\mathbb{Z}_{(p)}^\times=\{x: v_p(x)=0\}$ (11.1). (v) $\mathbb{Z}=\bigcap_p\mathbb{Z}_{(p)}$. Reference: Atiyah-Macdonald Chapter 3 (localisation); Serre, Course in Arithmetic, Chapter II. Paper: Lemma lem:fpr ("$L_p(A_f,1)^{-1}$ is a nonzero rational number which is an ℓ-adic integer"), Definition def:terms (q is an ℓ-adic unit), Lemma lem:tamfactor (the class in $\mathbb{F}_q^\times$ of the unit part u of the Tate parameter is its reduction; Chapter 7 replaces $\mathbb{Z}_{(q)}$ by $\mathbb{Z}_q$). Forward: Chapter 3 (localisation, DVRs, $\mathbb{Z}_\ell$), Chapter 7 (completion), Chapter 16 ("an integer prime to ℓ is a unit of $\mathbb{Z}_\ell$").

4.3.3 Section 3. Euclidean domains, PIDs, UFDs; $\mathbb{Z}[i]$ and $\mathbb{Z}[\zeta_3]$

3.1 `def:rings:factorisation` (definitions and proposition; P). In a domain R: divisibility, units, associates, irreducible elements, prime elements; a prime element is irreducible; (a) is a nonzero prime ideal iff a is a prime element. R is a *Euclidean domain* if there is $N\colon R\setminus\{0\}\to\mathbb{Z}_{\geq 0}$ such that for a, $b\neq 0$ there are κ,ρ with $a=\kappa b+\rho$ and $\rho=0$ or $N(\rho)<N(b)$; a *principal ideal domain* (PID) if every ideal is principal; a *unique factorisation domain* (UFD) if every nonzero non-unit is a product of irreducibles, uniquely up to order and units. A greatest common divisor. Reference: DF 8.1 to 8.3. Paper: background.

3.2 thm:rings:euclid-pid-ufd (theorem; P). (i) A Euclidean domain is a PID. (ii) A PID is Noetherian (ascending chains of ideals stabilise) and a UFD; in a PID irreducible elements are prime, nonzero prime ideals are maximal, and $\gcd(a,b)$ exists with Bezout's identity $(a,b) = (\gcd(a,b))$. (iii) In a UFD irreducible elements are prime and gcds exist. (iv) Examples: $\mathbb{Z}$ (absolute value), $F[x]$ for a field F (degree; 4.1); $\mathbb{Z}[x]$ and $F[x,y]$ are UFDs (4.3) but not PIDs ($(2,x)$, (x,y)). Reference: DF 8.1 to 8.3. Paper: background ($\mathbb{Z}$, $\mathbb{F}_\ell[x]$; $\mathbb{Z}_\ell$ is a PID, Chapter 3).

3.3 prop:rings:integers (proposition; P). In $\mathbb{Z}$: Bezout, Euclid's lemma ($p \mid ab \Rightarrow p \mid a$ or $p \mid b$), the fundamental theorem of arithmetic $n = \pm\prod_p p^{v_p(n)}$, and $\gcd(m,n) = \prod p^{\min(v_p(m),v_p(n))}$, $\operatorname{lcm}(m,n) = \prod p^{\max(v_p(m),v_p(n))}$; the p-part of $n \neq 0$ is $p^{v_p(n)}$. Reference: DF 8.1, 8.3. Paper: background ($v_\ell(\deg \varphi_E)$, "the ℓ-part of r", Agashe-Ribet-Stein's $v_p(\deg \varphi_E) = v_p(r_E)$).

3.4 thm:rings:gauss-eisenstein-integers (theorem; P). In $\mathbb{Z}[i] = \{a+bi\}$ and $\mathbb{Z}[\zeta_3] = \{a+b\zeta_3\}$ (subrings of $\mathbb{C}$; $\zeta_3^2 = -1-\zeta_3$) put $N(z) = z\bar{z} = |z|^2$, so $N(a+bi) = a^2+b^2$ and $N(a+b\zeta_3) = a^2-ab+b^2$, nonnegative integers, multiplicative. (i) Both rings are Euclidean for N: for α, $\beta \neq 0$ there is κ with $N(\alpha - \kappa\beta) \leq \frac{1}{2}N(\beta)$ in $\mathbb{Z}[i]$ and $\leq \frac{1}{3}N(\beta)$ in $\mathbb{Z}[\zeta_3]$ (round $\alpha/\beta \in \mathbb{Q}(i)$, resp. $\mathbb{Q}(\zeta_3)$, to a nearest lattice point; the worst cases are the centre of a square of side 1, resp. the centroid of an equilateral triangle of side 1). Hence both are PIDs and UFDs. (ii) z is a unit iff $N(z) = 1$; $\mathbb{Z}[i]^\times = \{\pm 1, \pm i\} = \mu_4$ and $\mathbb{Z}[\zeta_3]^\times = \{\pm 1, \pm\zeta_3, \pm\zeta_3^2\} = \mu_6$. (iii) An element of prime norm is irreducible, hence prime. (iv) Examples: $2 = -i(1+i)^2$, $5 = (2+i)(2-i)$, $7 = (3+\zeta_3)(3+\zeta_3^2)$, $13 = (4+\zeta_3)(4+\zeta_3^2)$, $3 = -\zeta_3^2(1-\zeta_3)^2$, and a worked Euclidean algorithm in $\mathbb{Z}[i]$: $11+3i = (1-i)(1+8i)+(2-4i)$, $1+8i = (-2+i)(2-4i)+(1-2i)$, $2-4i = 2(1-2i)$, so $\gcd(11+3i, 1+8i) = 1-2i$, of norm 5 (checked). The split: the list of all primes of $\mathbb{Z}[i]$ and $\mathbb{Z}[\zeta_3]$ (p splits iff $p \equiv 1 \pmod 4$, resp. $\pmod 3$), the ramified prime $1-\zeta_3$, the DVR $\mathbb{Z}_3[\zeta_3]$ and $\mathbb{Z}_\ell[\zeta_3]$ are Chapter 6 (items 4.5 to 4.9), which cites (i) and (ii) from here. Reference: DF 8.1 ($\mathbb{Z}[i]$); Ireland-Rosen, Chapters 1 and 9. Paper: background for Lemma lem:IV ($\mathbb{Z}_3[\zeta_3]$), Lemma lem:euler ($\mathbb{Z}_\ell[\zeta_3]$, $\mathbb{Z}_\ell[i]$), and the table of Section 1 (χ_{-3}, χ_{-4} are the characters of $\mathbb{Q}(\zeta_3)$, $\mathbb{Q}(i)$).

3.5 ex:rings:non-ufd (example; P). In $\mathbb{Z}[\sqrt{-5}]$, $6 = 2\cdot 3 = (1+\sqrt{-5})(1-\sqrt{-5})$ are two factorisations into irreducibles (norms 4, 9, 6, 6; there is no element of norm 2 or 3), none of the four factors associate; so $\mathbb{Z}[\sqrt{-5}]$ is neither a UFD nor a PID ($(2, 1+\sqrt{-5})$ is not principal). The ideal-theoretic repair is Chapter 6 (items 2.2, 2.6). Similarly $4 = 2\cdot 2 = (1+\sqrt{-3})(1-\sqrt{-3})$ in $\mathbb{Z}[\sqrt{-3}] \subsetneq \mathbb{Z}[\zeta_3]$ (Chapter 6, Exercise 2). Paper: background (why the rings $\mathcal{O}$ of Lemmas lem:crit and lem:old are handled through ideals).

4.3.4 Section 4. Polynomial rings and irreducibility

4.1 prop:rings:poly-division (proposition; P). $R[x]$; degree; over a domain $\deg(fg) = \deg f + \deg g$, $R[x]$ is a domain and $R[x]^\times = R^\times$. (i) Division with remainder by a **monic** g over any ring: $f = hg + r$, $\deg r < \deg g$, uniquely. (ii) For a field F, $F[x]$ is Euclidean (degree), hence a PID and a UFD; its maximal ideals are the (f) with f irreducible. (iii) Factor theorem: $f(a) = 0$ iff $(x-a) \mid f$. (iv) A nonzero polynomial of degree d over a domain has at most d roots in the domain. (v) Polynomials versus functions: $x^p - x \in \mathbb{F}_p[x]$ is nonzero but vanishes on $\mathbb{F}_p$; over an infinite domain a polynomial is determined by its values. Reference: DF 9.1 to 9.5. Paper: background ((iv) is the input of Chapter 1, item 4.2: finite subgroups of $K^\times$ are cyclic).

4.2 prop:rings:derivative (proposition; P). The formal derivative f', with $(fg)' = f'g + fg'$. For $f \in F[x]$ and an extension field $K \supseteq F$: $a \in K$ is a multiple root of f iff $f(a) = f'(a) = 0$; f has no repeated root in any extension iff $\gcd(f, f') = 1$ in $F[x]$ (the gcd does not change under extension of the field). Consequences: in characteristic p, $x^{p^r} - x$ has derivative -1, so it has p^r distinct roots in any field where it splits; $x^n - 1$ is separable over F iff $\operatorname{char} F \nmid n$. Reference: DF 13.5. Paper: background (finite fields, 6.2; roots of unity, 8.1; separability in Chapter 5).

4.3 thm:rings:gauss-lemma (theorem, Gauss's lemma; P for (i), (ii), (iv); S for (iii)). Let R be a UFD with fraction field K. The content $c(f)$ of $0 \neq f \in R[x]$ is a gcd of its coefficients (defined up to units); f is primitive if $c(f)$ is a unit. (i) $c(fg) = c(f)c(g)$ up to units; a product of primitive polynomials is primitive. (ii) If $f \in R[x]$ factors as $f = gh$ in $K[x]$, then $f = g_1 h_1$ in $R[x]$ with $g_1 \in K^\times g$, $h_1 \in K^\times h$. Hence a primitive f of positive degree is irreducible in $R[x]$ iff it is irreducible in $K[x]$. (iii) $R[x]$ is a UFD (so $\mathbb{Z}[x_1,\ldots,x_n]$ and $F[x_1,\ldots,x_n]$ are UFDs). (iv) Corollaries: a monic $f \in \mathbb{Z}[x]$ that factors into monic polynomials in $\mathbb{Q}[x]$ has those factors in $\mathbb{Z}[x]$; the rational root test (if a/b in lowest terms is a root of

$a_n x^n + \cdots + a_0 \in \mathbb{Z}[x]$ then $b \mid a_n$ and $a \mid a_0$); a rational root of a monic integer polynomial is an integer. Reference: DF 9.3; Lang, Algebra IV.2. Paper: background (Chapter 6, item 1.2: the minimal polynomial of an algebraic integer lies in $\mathbb{Z}[x]$; Chapter 5: $\Phi_n \in \mathbb{Z}[x]$ irreducible).

4.4 thm:rings:eisenstein (theorem, Eisenstein's criterion; P). Let R be a domain, P a prime ideal of R, and $f = a_n x^n + \cdots + a_0 \in R[x]$, $n \geq 1$, with $a_n \notin P$, $a_i \in P$ for $i < n$, and $a_0 \notin P^2$. Then f is not a product of two polynomials of positive degree in $R[x]$; if R is a UFD and f is primitive, f is irreducible in $R[x]$ and in $K[x]$ (4.3). Variants: apply it to $f(x+a)$. Examples: $x^n - p$; $\Phi_p(x+1) = \big((x+1)^p - 1\big)/x$; $\Phi_{p^k}(x+1)$ (8.2); $\Phi_3(x+1) = x^2+3x+3$. Reference: DF 9.4. Paper: background (Chapter 5: irreducibility of Φ_p; Chapter 6, items 4.7 and 5.1: $\mathbb{Z}_3[\zeta_3]$ a DVR, $\mathbb{Z}[\zeta_{\ell^a}]$ is ℓ-maximal).

4.5 prop:rings:reduction-mod-p (proposition; P). Let $f \in \mathbb{Z}[x]$ of degree n and p a prime not dividing the leading coefficient. If $f = gh$ in $\mathbb{Z}[x]$ then $\bar{f} = \bar{g}\bar{h}$ in $\mathbb{F}_p[x]$ with $\deg \bar{g} = \deg g$, $\deg \bar{h} = \deg h$. Hence: (i) if f is primitive and $\bar{f}$ is irreducible in $\mathbb{F}_p[x]$, f is irreducible in $\mathbb{Q}[x]$; (ii) more generally the degrees of any factorisation of f over $\mathbb{Q}$ are sums of degrees of the irreducible factors of $\bar{f}$, for every such p, and two primes with incompatible patterns prove irreducibility. Warning: $x^4 + 1$ is irreducible over $\mathbb{Q}$ (Eisenstein after $x \mapsto x+1$) but reducible modulo every prime (checked for $p < 500$; proof: $\mathbb{F}_{p^2}$ contains a primitive 8-th root of unity, so every root generates a field of degree at most 2). Example: the 3-division polynomial of 11a1, $\psi_3 = 3x^4 - 4x^3 - 60x^2 - 237x - 21$, is irreducible modulo 5, hence over $\mathbb{Q}$. Reference: DF 9.4. Paper: Proposition prop:ns ("otherwise it [the 3-division polynomial] is irreducible": a test for a given curve), Section sec:data ("checked against the factorisation of the 3-division polynomial"); the Galois-theoretic meaning of the pattern is Chapter 5.

4.6 thm:rings:adjoin-root (theorem, Kronecker's construction; P). Let F be a field and $f \in F[x]$ irreducible of degree d. (i) $F[x]/(f)$ is a field containing F, of dimension d over F with basis $1, \bar{x}, \ldots, \bar{x}^{d-1}$, in which $\bar{x}$ is a root of f. (ii) If α is a root of f in a field $K \supseteq F$, evaluation induces an isomorphism $F[x]/(f) \simeq F[\alpha]$, and $F[\alpha]$ is a field. (iii) For every nonconstant $g \in F[x]$ there is a field $K \supseteq F$ of finite dimension over F in which g is a product of linear factors (iterate (i)). Examples: $\mathbb{F}_3[x]/(x^2+1) = \mathbb{F}_9$; $\mathbb{F}_q[x]/(x^2 - \varepsilon)$ for a non-square ε (Chapter 1, item 5.8). Splitting fields and their uniqueness in general are Chapter 5. Reference: DF 13.1, 13.4. Paper: background ($\mathbb{F}_9$ in Proposition prop:ns: "$\bar{\rho}| \otimes \mathbb{F}_9 = \psi \oplus \psi^{-1}$").

4.3.5 Section 5. $\mathbb{Z}/n\mathbb{Z}$, the Chinese remainder theorem, units and primitive roots

5.1 prop:rings:zn (proposition; P). (i) a mod n is a unit of $\mathbb{Z}/n\mathbb{Z}$ iff $\gcd(a,n) = 1$ (Bezout); $\mathbb{Z}/n\mathbb{Z}$ is a field iff it is a domain iff n is prime; $\mathbb{F}_p := \mathbb{Z}/p\mathbb{Z}$. (ii) $\#(\mathbb{Z}/n\mathbb{Z})^\times = \varphi(n)$; Euler's theorem $a^{\varphi(n)} \equiv 1 \pmod{n}$ for $\gcd(a,n) = 1$ (Lagrange, Chapter 1); $\varphi(p^k) = p^{k-1}(p-1)$; $\sum_{d|n} \varphi(d) = n$. (iii) $(\mathbb{Z}/n\mathbb{Z})^\times = \mathrm{Aut}(\mathbb{Z}/n)$, acting by multiplication (Chapter 1, item 1.3(iii)). Note $\mathbb{Z}/p^k\mathbb{Z}$ ($k \geq 2$) is not a field and is not $\mathbb{F}_{p^k}$. Reference: DF 0.3, 7.3. Paper: background ($\mathbb{F}_q = \mathbb{Z}/q\mathbb{Z}$ for the paper's prime q; the groups on which Dirichlet characters live).

5.2 thm:rings:crt (theorem, Chinese remainder theorem; P). For $n = \prod_i p_i^{k_i}$ the reduction maps give a ring isomorphism $\mathbb{Z}/n\mathbb{Z} \simeq \prod_i \mathbb{Z}/p_i^{k_i}\mathbb{Z}$; more generally $\mathbb{Z}/mn\mathbb{Z} \simeq \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}$ iff $\gcd(m,n) = 1$. Hence $(\mathbb{Z}/n\mathbb{Z})^\times \simeq \prod_i (\mathbb{Z}/p_i^{k_i}\mathbb{Z})^\times$ and φ is multiplicative. Proof from 1.4(iii); the group version is Chapter 1, item 2.2. Reference: DF 7.6. Paper: background (Dirichlet characters split over prime powers, 10.3(iv); the conductor $\prod_{q \in S} q$ of χ_S; the decomposition of a cyclic group into ℓ-parts in Lemma lem:tamfactor).

5.3 thm:rings:primitive-roots (theorem; P). (i) $(\mathbb{Z}/p\mathbb{Z})^\times = \mathbb{F}_p^\times$ is cyclic of order $p-1$ (Chapter 1, item 4.2); a generator is a primitive root mod p. (ii) For p odd and $k \geq 1$, $(\mathbb{Z}/p^k\mathbb{Z})^\times$ is cyclic of order $p^{k-1}(p-1)$: the class of $1+p$ has order p^{k-1} (5.4(iii)), and if g is a primitive root mod p with $g^{p-1} \not\equiv 1 \pmod{p^2}$ (true for g or $g+p$) then g is a primitive root mod p^k for every k. (iii) $(\mathbb{Z}/2\mathbb{Z})^\times$ and $(\mathbb{Z}/4\mathbb{Z})^\times$ are cyclic; for $k \geq 3$, $(\mathbb{Z}/2^k\mathbb{Z})^\times = \{\pm 1\} \times \langle 5 \rangle \simeq \mathbb{Z}/2 \times \mathbb{Z}/2^{k-2}$ is not cyclic (5 has order 2^{k-2}, and $-1 \notin \langle 5 \rangle$). (iv) $(\mathbb{Z}/n\mathbb{Z})^\times$ is cyclic iff $n \in \{1, 2, 4, p^k, 2p^k\}$ with p an odd prime (checked for $n < 400$). Reference: Ireland-Rosen, Chapter 4; Serre, Course in Arithmetic, Chapter II (in p-adic form). Paper: background (Chapter 7: the structure of $\mathbb{Z}_p^\times$; Dirichlet characters mod p^k; Chapter 5's $\chi_\ell(G_\mathbb{Q}) = \mathbb{Z}_\ell^\times$ maps onto the cyclic $(\mathbb{Z}/\ell^k)^\times$).

5.4 lem:rings:lte (lemma, lifting the exponent; P). (i) Let ℓ be an odd prime and $a, b \in \mathbb{Z}$ with

$\ell \mid a-b$ and $\ell \nmid ab$. Then $v_\ell(a^n-b^n) = v_\ell(a-b)+v_\ell(n)$ for every $n \geq 1$. (ii) For a odd and n even, $v_2(a^n-1) = v_2(a-1)+v_2(a+1)+v_2(n)-1$. (iii) Consequently, for ℓ odd, $(1+\ell)^{\ell^{k-1}} \equiv 1+\ell^k \pmod{\ell^{k+1}}$ and $1+\ell$ has order ℓ^{k-1} in $(\mathbb{Z}/\ell^k\mathbb{Z})^\times$. (iv) For primes $q \neq \ell$, ℓ odd, and d the order of q mod ℓ: $v_\ell(q^f-1)=0$ if $d \nmid f$, and $v_\ell(q^f-1) = v_\ell(q^d-1)+v_\ell(f/d)$ if $d \mid f$. (v) $v_\ell(q^2-1) = v_\ell(q-1)+v_\ell(q+1)$, and for odd $\ell \neq q$ at most one of the two terms is positive. Reference: elementary (Ireland-Rosen, Chapter 4, exercises). Paper: the I_n^* entry $v_\ell(q^2-1)$ of the table; background for Chapter 7 ($v_\ell(\#\mathbb{F}_{q^f}^\times)$, the ℓ-power roots of unity in unramified extensions) and Chapter 5 (smallest f with $\mu_{\ell^k} \subset \mathbb{F}_{q^f}$).

5.5 `cor:rings:squares-one` (corollary; P). Let ℓ be an odd prime and $k \geq 1$. The solutions of $a^2 \equiv 1 \pmod{\ell^k}$ are $a \equiv \pm 1$ (as $(\mathbb{Z}/\ell^k)^\times$ is cyclic of even order). Hence $a^2 \equiv 1 \pmod{\ell^k}$ for **every** a prime to ℓ iff $\varphi(\ell^k) = 2$ iff $\ell^k = 3$. In particular, if a homomorphism χ from a group onto $\mathbb{Z}_\ell^\times$ (or onto $(\mathbb{Z}/\ell^k)^\times$) satisfies $\chi^2 \equiv 1 \pmod{\ell^k}$, then $\ell^k = 3$. Example: $2^2 = 4 \not\equiv 1 \pmod 9$. Reference: elementary. Paper: Lemma lem:h0 in the corrected form of important_notes.md point 5 ("comparing determinants it forces $\bar{\chi}_\ell^2 = 1$, so $\ell = 3$"; and $h_3(E) \leq 1$ because $4 \not\equiv 1 \bmod 9$); used by Chapter 5, item galois:twist (c).

4.3.6 Section 6. Finite fields

6.1 `prop:rings:finite-field-basic` (proposition; P). Let F be a finite field. (i) $\operatorname{char} F = p$ is prime, $F \supseteq \mathbb{F}_p$ (the image of $\mathbb{Z}$), and $\#F = p^r$ with $r = \dim_{\mathbb{F}_p} F$. (ii) $F^\times$ is cyclic of order p^r-1 (Chapter 1, item 4.2, which rests on 4.1(iv)). (iii) Every $a \in F$ satisfies $a^{p^r} = a$, and $x^{p^r}-x = \prod_{a \in F}(x-a)$ in $F[x]$. (iv) $F = \mathbb{F}_p(\alpha) = \mathbb{F}_p[\alpha]$ for any generator α of $F^\times$. Reference: DF 14.3; Serre, Course in Arithmetic, I.1. Paper: Lemma lem:tamfactor ($\mathbb{F}_q^\times$ cyclic), background.

6.2 `thm:rings:finite-field-exists` (theorem, existence and uniqueness; P). Let $q = p^r$. (i) There is a field with q elements: in a field $K \supseteq \mathbb{F}_p$ where $x^q - x$ splits (4.6(iii)) its roots are q distinct elements (4.2) forming a subfield (closed under sums by 1.6, under products and inverses clearly). (ii) Any two fields K, K' with q elements are isomorphic: write $K = \mathbb{F}_p[\alpha]$ (6.1(iv)) and let f be the monic irreducible polynomial of α over $\mathbb{F}_p$ (the monic generator of the kernel of $\mathbb{F}_p[x] \to K$); f divides x^q-x, which is $\prod_{b \in K'}(x-b)$, so f has a root $\beta \in K'$, and $K \simeq \mathbb{F}_p[x]/(f) \simeq \mathbb{F}_p[\beta] = K'$ by 4.6(ii) and counting. Write $\mathbb{F}_q$. (iii) $\mathbb{F}_q \simeq \mathbb{F}_p[x]/(f)$ for every irreducible $f \in \mathbb{F}_p[x]$ of degree r. (iv) The isomorphism in (ii) is not unique (compose with powers of the Frobenius 6.3); inside a fixed algebraic closure the subfield with q elements is unique (Chapter 5). Reference: DF 14.3; Lang, Algebra V.5; Serre, Course in Arithmetic, I.1. Paper: background ($\mathbb{F}_9 = \mathbb{F}_3[i]$ in Proposition prop:ns; $\mathbb{F}_{q^2}$ of Chapter 1, item 5.8; residue fields in Chapters 6 and 7).

6.3 `prop:rings:frobenius` (proposition; P). Let $q = p^r$. (i) $\mathrm{Frob}_p \colon x \mapsto x^p$ is a field automorphism of $\mathbb{F}_q$ (1.6), its fixed points are exactly $\mathbb{F}_p$ (the p roots of x^p-x), and Frob_p has order exactly r (if $\mathrm{Frob}_p^s = 1$ with $s < r$, all q elements would be roots of $x^{p^s}-x$). (ii) For $d \mid r$, $\{x \in \mathbb{F}_q : x^{p^d} = x\}$ is a subfield with p^d elements and the only one; $\mathbb{F}_q$ contains a subfield with p^d elements iff $d \mid r$ ($\mathbb{F}_{p^d}^\times \subseteq \mathbb{F}_q^\times$ forces $p^d-1 \mid p^r-1$, i.e. $d \mid r$). (iii) The fixed field of Frob_p^s is $\mathbb{F}_{p^{\gcd(s,r)}}$. (iv) (Remark with forward pointer, not a numbered result.) Every automorphism of $\mathbb{F}_q$ is a power of Frob_p, and $\mathrm{Gal}(\mathbb{F}_{q^n}/\mathbb{F}_q)$ is cyclic of order n generated by $x \mapsto x^q$: Chapter 5, item galois:finite-field-galois. Reference: DF 14.3; Lang, Algebra V.5. Paper: background (the arithmetic Frobenius of Chapters 5 and 6 induces $x \mapsto x^q$ on residue fields; the "Frob" of Lemmas lem:euler and lem:tamfactor).

6.4 `prop:rings:irreducible-count` (proposition; P). In $\mathbb{F}_p[x]$ (more generally $\mathbb{F}_q[x]$), $x^{p^r}-x = \prod f$ over the monic irreducible f with $\deg f \mid r$, each once. Hence $p^r = \sum_{d \mid r} d N_d$, where N_d is the number of monic irreducibles of degree d, and $N_r = \frac{1}{r}\sum_{d \mid r} \mu(d) p^{r/d} > 0$ (Moebius inversion; positivity gives a second existence proof). Examples: over $\mathbb{F}_2$, $N_1, \ldots, N_4 = 2, 1, 2, 3$; over $\mathbb{F}_3$, $N_2 = 3$ (x^2+1, x^2+x+2, x^2+2x+2). Proof uses 6.1, 6.3(ii) and 4.2. Reference: Ireland-Rosen, Chapter 7; Lang, Algebra V.5. Paper: background.

6.5 `prop:rings:mu-in-fq` (proposition; P). Let $q = p^r$ and $n \geq 1$. (i) $\mu_n(\mathbb{F}_q)$ is cyclic of order $\gcd(n, q-1)$; it has n elements iff $n \mid q-1$. (ii) If $p \nmid n$, the least $f \geq 1$ with $\mu_n \subset \mathbb{F}_{q^f}$ (i.e. with n distinct n-th roots of unity) is the order of q in $(\mathbb{Z}/n)^\times$. (iii) In characteristic p, $x^{p^k m}-1 = (x^m-1)^{p^k}$: there are no nontrivial p-power roots of unity. (iv) For q odd: -1 is a square in $\mathbb{F}_q$ iff $q \equiv 1 \pmod 4$; if moreover

$p \neq 3$, -3 is a square in $\mathbb{F}_q$ iff $\mu_3 \subset \mathbb{F}_q$ iff $q \equiv 1 \pmod 3$ (because $(1+2\zeta_3)^2 = -3$ for a primitive cube root of unity ζ_3, and conversely $\zeta_3 = (-1+s)/2$ if $s^2 = -3$). (v) Let K be any field with $\operatorname{char} K \neq 3$ containing μ_3, and m an integer prime to 3; then $\zeta \mapsto \zeta^m$ is the identity of μ_3 if $m \equiv 1 \pmod 3$ and inversion if $m \equiv 2 \pmod 3$; likewise on μ_4 ($\operatorname{char} K \neq 2$, m odd) with $m \bmod 4$. By 10.5 this says $\zeta^m = \zeta^{\chi_{-3}(m)}$ on μ_3 and $\zeta^m = \zeta^{\chi_{-4}(m)}$ on μ_4. Reference: DF 14.3; Ireland-Rosen, Chapter 7. Paper: Lemma lem:euler ("Frobenius centralises σ if $q \equiv 1 \pmod 3$ and inverts it if $q \equiv 2$, so it acts on the line by $\chi_{-3}(q)$"; the same with μ_4 and $\chi_{-4}(q)$: the arithmetic input, with the Galois-theoretic input in Chapter 6, items 3.11, 3.12); Chapter 5 (Kummer theory needs $\mu_{\ell^k} \subset \mathbb{F}_q$ iff $\ell^k \mid q-1$); Chapter 7.

4.3.7 Section 7. Powers in $\mathbb{F}_q^\times$

7.1 `thm:rings:power-index` (theorem; P). Let q be a prime power, $m \geq 1$ and $d = \gcd(m, q-1)$. (i) The m-th power map on $\mathbb{F}_q^\times$ has kernel $\mu_d(\mathbb{F}_q)$, of order d, and image $(\mathbb{F}_q^\times)^m = (\mathbb{F}_q^\times)^d$, the unique subgroup of index d. (ii) (Generalised Euler criterion.) $u \in \mathbb{F}_q^\times$ is an m-th power iff $u^{(q-1)/d} = 1$. (iii) For a prime ℓ and $k \geq 0$:

$$[\mathbb{F}_q^\times : (\mathbb{F}_q^\times)^{\ell^k}] = \ell^{\min(k, v_\ell(q-1))}.$$

In particular, if $q \not\equiv 1 \pmod \ell$ the ℓ-th power map is bijective and every element is an ℓ^k-th power; if $q \equiv 1 \pmod \ell$, exactly a fraction $1/\ell$ of $\mathbb{F}_q^\times$ are ℓ-th powers. (iv) ($m = 2$, q odd.) The squares form the subgroup of index 2; $u^{(q-1)/2} = \pm 1$, with $+1$ iff u is a square. Proof: Chapter 1, item 1.3(ii), applied to the cyclic group $\mathbb{F}_q^\times$ (6.1). (Checked numerically for all primes $q < 400$, $\ell \leq 13$, $k \leq 4$, and for $q = 4, 8, 9, 16, 25, 27, 49, 64, 81$.) Reference: Ireland-Rosen, Chapters 4 and 7; Serre, Course in Arithmetic, I.3. Paper: Lemma lem:tamfactor ("u is an ℓ^k-th power in $\mathbb{F}_q^\times$, for $k \leq v_\ell(q-1)$"), Section sec:tam, Remark rem:ko(1) ("$q \equiv 1 \pmod \ell$").

7.2 `prop:rings:class-c` (proposition, the class of a unit modulo ℓ-power powers; P). Let q be a prime power, $\ell \nmid q$ a prime, $v = v_\ell(q-1)$, and $A = \mathbb{F}_q^\times/(\mathbb{F}_q^\times)^{\ell^v}$. (i) A is cyclic of order ℓ^v; the projection restricted to the ℓ-primary part $\mu_{\ell^v}(\mathbb{F}_q)$ of $\mathbb{F}_q^\times$ is an isomorphism onto A, and $u \mapsto u^{(q-1)/\ell^v}$ induces an isomorphism $A \simeq \mu_{\ell^v}(\mathbb{F}_q)$. (ii) A generator g of $\mathbb{F}_q^\times$ identifies A with $\mathbb{Z}/\ell^v$: $u = g^a \mapsto c(u) = a \bmod \ell^v$. Changing g multiplies $c(u)$ by a unit of $\mathbb{Z}/\ell^v$, so $v_\ell(c(u))$ (convention C3) does not depend on g. (iii) For $0 \leq k \leq v$: $u \in (\mathbb{F}_q^\times)^{\ell^k}$ iff $v_\ell(c(u)) \geq k$; for $k \geq v$, $(\mathbb{F}_q^\times)^{\ell^k} = (\mathbb{F}_q^\times)^{\ell^v}$. (iv) For $n \in \mathbb{Z} \setminus \{0\}$: $\min(v_\ell(n), v, v_\ell(c(u))) = \max\{k \leq \min(v_\ell(n), v) : u \in (\mathbb{F}_q^\times)^{\ell^k}\}$; it is ≥ 1 iff $\ell \mid n$, $q \equiv 1 \pmod \ell$ and u is an ℓ-th power in $\mathbb{F}_q^\times$. (v) For the paper's prime q and a unit $u \in \mathbb{Z}_{(q)}^\times$ (or $u \in \mathbb{Z}_q^\times$, Chapter 7) the class is that of the reduction $\bar{u} \in \mathbb{F}_q^\times$ (2.4(iii)). (The identification $A \simeq \mathbb{F}_q^\times \otimes \mathbb{Z}_\ell \simeq \mathbb{Z}_\ell(1)/(q-1)$ is Chapters 3, 5, 7; the Kummer interpretation $u^{(q-1)/\ell^k} = \chi_u(\phi_q)$ is Chapter 5, item galois:kummer-finite-field.) Reference: elementary. Paper: Lemma lem:tamfactor (statement: "$c \in \mathbb{Z}/\ell^{v_\ell(q-1)}$ the class of u in $\mathbb{F}_q^\times/(\mathbb{F}_q^\times)^{\ell^{v_\ell(q-1)}}$, so that $v_\ell(c) \geq k$ iff u is an ℓ^k-th power", and the "In particular" clause), Section sec:tam, Remark rem:ko(1); Chapter 21, item A2.7.

7.3 `ex:rings:powers` (worked example; P). (a) $q = 7$, $\ell = 3$: $v = 1$, the cubes are $\{1, 6\}$; $\bar{u} = -1$ is a cube, $\bar{u} = 2$ is not. So a multiplicative prime $q = 7$ with $3 \mid n_7$ and unit part $u \equiv -1 \pmod 7$ contributes $v_3(n_7) + 1$ to $v_3(\mathrm{Tam}_7(\mathrm{ad}^0\, T))$ by Lemma lem:tamfactor. (b) $q = 19$, $\ell = 3$: $v = 2$; 2 is a primitive root mod 19; the cubes are $\{1, 7, 8, 11, 12, 18\}$ and the ninth powers $\{1, 18\}$; $c(2) = 1$, $c(8) = 3$, $c(-1) = c(2^9) = 0$, so $v_3(c) = 0, 1, 2$. (c) $q = 13$, $\ell = 3$: u is a cube iff $u^4 = 1$ (compare Chapter 5, Exercise 8). (d) $q = 5$, $\ell = 3$: every element is a cube; $q = 11$, $\ell = 5$: $v = 1$, the fifth powers are $\{\pm 1\}$. Paper: Lemma lem:tamfactor, Section sec:tam.

4.3.8 Section 8. Roots of unity and cyclotomic polynomials

8.1 `def:rings:roots-of-unity` (definition and proposition; P). $\mu_n(K)$, $\mu(K) = \bigcup_n \mu_n(K)$; a primitive n-th root of unity is an element of order exactly n. (i) $\mu_n(K)$ is cyclic of order dividing n (Chapter 1, item 4.2), of order n iff K contains a primitive n-th root of unity, which requires $\operatorname{char} K \nmid n$ (4.2). (ii) $\mu_n(\mathbb{C}) = \langle \zeta_n \rangle$; its primitive elements are the ζ_n^a, $\gcd(a, n) = 1$, $\varphi(n)$ in number. (iii) $\mu(\mathbb{Q}) = \mu(\mathbb{R}) = \{\pm 1\}$,

$\mu(\mathbb{Q}(i)) = \mu_4$, $\mu(\mathbb{Q}(\zeta_3)) = \mu_6$ (3.4(ii)). Reference: DF 13.6. Paper: background (ζ_3, i, μ_{ℓ^k}, μ_3 and μ_4 in Lemma lem:euler).

8.2 `thm:rings:cyclotomic` (theorem; P). Let $\Phi_n(x) = \prod_{1\le a\le n,\ \gcd(a,n)=1}(x-\zeta_n^a) \in \mathbb{C}[x]$. (i) $x^n - 1 = \prod_{d|n}\Phi_d(x)$. (ii) $\Phi_n \in \mathbb{Z}[x]$, monic of degree $\varphi(n)$ (induction on n and 4.1(i)). (iii) $\Phi_n(x) = \prod_{d|n}(x^d-1)^{\mu(n/d)}$. (iv) $\Phi_p = x^{p-1}+\cdots+x+1$; $\Phi_{p^k}(x) = \Phi_p(x^{p^{k-1}})$; $\Phi_{2m}(x) = \Phi_m(-x)$ for odd $m>1$; $\Phi_3 = x^2+x+1$, $\Phi_4 = x^2+1$, $\Phi_6 = x^2-x+1$, $\Phi_8 = x^4+1$, $\Phi_{12} = x^4-x^2+1$. (v) $\Phi_n(1) = p$ if $n = p^k$ with $k\ge 1$, $\Phi_n(1) = 1$ if n has at least two distinct prime factors, $\Phi_1(1) = 0$; $\Phi_n(0) = 1$ for $n \ge 2$. (vi) Φ_{p^k} is irreducible over $\mathbb{Q}$ (Eisenstein at p for $\Phi_{p^k}(x+1)$, whose reduction mod p is $x^{\varphi(p^k)}$). The irreducibility of every Φ_n and $\mathrm{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q}) \simeq (\mathbb{Z}/n)^\times$ are Chapter 5 (item galois:cyclotomic). Reference: DF 13.6; Washington, Chapter 2. Paper: background (Chapter 5; Chapter 6, item 5.1 uses (v), (vi)).

8.3 `prop:rings:cyclotomic-mod-p` (proposition; P). Let p be a prime with $p \nmid n$. (i) $\Phi_n \bmod p$ divides x^n-1, so it is separable over $\mathbb{F}_p$ (4.2), and its roots in any field of characteristic p are exactly the elements of order n. (ii) For $a\in\mathbb{Z}$: $p \mid \Phi_n(a)$ iff $a \bmod p$ has order exactly n in $\mathbb{F}_p^\times$; hence every prime divisor $p \nmid n$ of $\Phi_n(a)$ satisfies $p \equiv 1 \pmod n$. (iii) There are infinitely many primes $p \equiv 1 \pmod n$ (apply (ii) to $\Phi_n(Nn)$, N large and divisible by any given finite set of primes). (iv) Φ_n splits into linear factors mod p iff $p \equiv 1 \pmod n$. The complete factorisation pattern ($\varphi(n)/f$ factors of degree f, f the order of p mod n) is Chapter 5 (item galois:cyclotomic-mod-p). Reference: Ireland-Rosen, Chapter 13 (or 7); Washington, Chapter 2. Paper: background (the primes $q \equiv 1 \pmod\ell$ of Lemma lem:tamfactor are those with $\mu_\ell \subset \mathbb{F}_q$; Chapter 6, item 5.2).

4.3.9 Section 9. Quadratic residues, the symbols of Legendre, Jacobi and Kronecker, reciprocity

9.1 `def:rings:legendre` (definition and proposition; P). For an odd prime p and $a\in\mathbb{Z}$: $\left(\frac{a}{p}\right) = 0$ if $p \mid a$, 1 if a is a nonzero square mod p, -1 otherwise. (i) Euler's criterion: $\left(\frac{a}{p}\right) \equiv a^{(p-1)/2} \pmod p$ (7.1(iv)). (ii) $\left(\frac{ab}{p}\right) = \left(\frac{a}{p}\right)\left(\frac{b}{p}\right)$; $a \mapsto \left(\frac{a}{p}\right)$ is the unique homomorphism $(\mathbb{Z}/p)^\times \to \{\pm1\}$ of order 2; there are $(p-1)/2$ residues and $(p-1)/2$ non-residues. (iii) $\left(\frac{-1}{p}\right) = (-1)^{(p-1)/2}$. (iv) For $p=3$: $\left(\frac{a}{3}\right) \equiv a \pmod 3$ for all a. Reference: Ireland-Rosen, Chapter 5; Serre, Course in Arithmetic, I.3. Paper: background; (iv) is the congruence "$q \equiv \chi_{-3}(q) \pmod 3$" of Proposition prop:forms (via 9.6, 10.5).

9.2 `prop:rings:quadratic-gauss-sum` (proposition; P, except the sign in (iii), R). Let q be an odd prime, K a field of characteristic $\ne q$ containing a primitive q-th root of unity ζ, and $g = \sum_{a=1}^{q-1}\left(\frac{a}{q}\right)\zeta^a \in K$. (i) $g^2 = q^* = (-1)^{(q-1)/2}q$ (as an element of K); in particular $g \ne 0$, since $q \ne 0$ in K. (ii) For b prime to q, $\sum_a\left(\frac{a}{q}\right)\zeta^{ab} = \left(\frac{b}{q}\right)g$. (iii) Over $\mathbb{C}$ with $\zeta = \zeta_q$: $g = \sqrt{q}$ if $q \equiv 1 \pmod 4$ and $g = i\sqrt{q}$ if $q \equiv 3 \pmod 4$ (Gauss; stated, not needed). (iv) Hence $\sqrt{q^*} \in \mathbb{Q}(\zeta_q)$ (used by Chapter 5, item galois:sqrt-lstar (b)). Proof of (i): $g^2 = \sum_{a,b}\left(\frac{ab}{q}\right)\zeta^{a+b}$, substitute $b = ac$ and use $\sum_{a=1}^{q-1}\zeta^{a(1+c)} = q-1$ or -1 according as $c \equiv -1$ or not. Reference: Ireland-Rosen, Chapter 6; Washington, Chapter 4. Paper: background (Theorem thm:main: $\mathbb{Q}(\sqrt{\ell^*}) \subset \mathbb{Q}(\zeta_\ell)$, via Chapter 5).

9.3 `thm:rings:qr` (theorem, quadratic reciprocity; P). Let $p \ne q$ be odd primes. (i) $\left(\frac{p}{q}\right)\left(\frac{q}{p}\right) = (-1)^{\frac{p-1}{2}\cdot\frac{q-1}{2}}$; equivalently $\left(\frac{q^*}{p}\right) = \left(\frac{p}{q}\right)$. (ii) (First supplement) $\left(\frac{-1}{p}\right) = (-1)^{(p-1)/2}$. (iii) (Second supplement) $\left(\frac{2}{p}\right) = (-1)^{(p^2-1)/8}$, i.e. 1 iff $p \equiv \pm1 \pmod 8$. Proof of (i) by Gauss sums in characteristic p: let f be the order of p mod q, $\zeta \in \mathbb{F}_{p^f}$ a primitive q-th root of unity (6.5(ii)), and g as in 9.2 with $K = \mathbb{F}_{p^f}$; then $g^2 = q^*$ and, by 1.6 and 9.2(ii), $g^p = \sum\left(\frac{a}{q}\right)\zeta^{ap} = \left(\frac{p}{q}\right)g$; since $g \ne 0$, $(q^*)^{(p-1)/2} = g^{p-1} = \left(\frac{p}{q}\right)$ in $\mathbb{F}_p$, and Euler's criterion gives $\left(\frac{q^*}{p}\right) = \left(\frac{p}{q}\right)$. Proof of (iii): with ζ a primitive 8-th root of unity in $\mathbb{F}_{p^2}$, $\tau = \zeta + \zeta^{-1}$ satisfies $\tau^2 = 2$ and $\tau^p = \zeta^p + \zeta^{-p}$, which is τ if $p \equiv \pm1 \pmod 8$ and $-\tau$ if $p \equiv \pm3 \pmod 8$. Alternatives: Eisenstein's lemma (Exercise 13); the splitting of p in $\mathbb{Q}(\zeta_q)$ (Chapter 6, item 5.4). Checked for all odd primes below 400. Reference: Ireland-Rosen, Chapters 5 and 6; Serre, Course in Arithmetic, I.3. Paper: Lemma lem:twist (the character of $\mathbb{Q}(\sqrt{q^*})$ is "the quadratic character of conductor q", 9.6), Theorem thm:main and the set S_f (ℓ^*), Chapters 5 (galois:quadratic-characters (c)) and 6 (4.2(iii)).

9.4 `def:rings:jacobi` (definition and proposition; P). For odd $n = \prod p_i^{e_i} > 0$ and $a\in\mathbb{Z}$, $\left(\frac{a}{n}\right) =$

$\prod\left(\frac{a}{p_i}\right)^{e_i}$. (i) It is multiplicative in a and in n, depends only on a mod n, and is 0 iff $\gcd(a,n)>1$. (ii) For odd coprime $m,n>0$: $\left(\frac{m}{n}\right)\left(\frac{n}{m}\right)=(-1)^{\frac{m-1}{2}\cdot\frac{n-1}{2}}$; $\left(\frac{-1}{n}\right)=(-1)^{(n-1)/2}$; $\left(\frac{2}{n}\right)=(-1)^{(n^2-1)/8}$ (from 9.3, because $n\mapsto(-1)^{(n-1)/2}$ and $n\mapsto(-1)^{(n^2-1)/8}$ are multiplicative on odd n). (iii) Warning: $\left(\frac{a}{n}\right)=1$ does not imply that a is a square mod n: $\left(\frac{2}{15}\right)=1$ but 2 is not a square mod 3. (iv) The algorithm computing $\left(\frac{a}{n}\right)$ by reciprocity without factoring. Checked for odd $m,n<200$. Reference: Ireland-Rosen, Chapter 5. Paper: background (computing Kronecker symbols; PARI's `kronecker`).

9.5 `prop:rings:kronecker` (definition and proposition; P for (i) to (iii), (v); S for (iv)). The Kronecker symbol $\left(\frac{a}{n}\right)$ for $a,n\in\mathbb{Z}$ (C4): for $n=\varepsilon 2^e\prod p_i^{e_i}$, $\varepsilon=\pm1$, $\left(\frac{a}{n}\right)=\left(\frac{a}{\varepsilon}\right)\left(\frac{a}{2}\right)^e\prod\left(\frac{a}{p_i}\right)^{e_i}$. Let $D\neq0$ with $D\equiv0,1 \pmod 4$ (a *discriminant*). (i) $n\mapsto\left(\frac{D}{n}\right)$ is completely multiplicative on $\mathbb{Z}\setminus\{0\}$ and vanishes exactly when $\gcd(D,n)>1$. (ii) For $n,n'>0$ with $n\equiv n' \pmod{|D|}$, $\left(\frac{D}{n}\right)=\left(\frac{D}{n'}\right)$; and $\left(\frac{D}{-n}\right)=\operatorname{sign}(D)\left(\frac{D}{n}\right)$. Hence $\chi_D(a):=\left(\frac{D}{a}\right)$ (for $a\neq0$, and $\chi_D(0)=0$ unless $D=1$) is a Dirichlet character modulo $|D|$ (Section 10), real (values in $\{0,\pm1\}$), with $\chi_D(-1)=\operatorname{sign}D$. Proof via Jacobi reciprocity 9.4(ii), writing $D=2^sD'$ with D' odd. (iii) Values at primes: $\chi_D(p)=\left(\frac{D}{p}\right)$ (Legendre) for odd p; $\chi_D(2)=0$ if $4\mid D$, 1 if $D\equiv1 \pmod 8$, -1 if $D\equiv5 \pmod 8$. (iv) A *fundamental discriminant* is $D\equiv1 \pmod 4$ squarefree, or $D=4m$ with $m\equiv2,3 \pmod 4$ squarefree. Every fundamental discriminant $D\neq1$ is uniquely a product of pairwise coprime *prime discriminants* -4, 8, -8, q^* (q odd prime), and χ_D is the product of the corresponding characters; each of these is primitive of conductor 4, 8, 8, q, so χ_D is primitive of conductor $|D|$ (10.3(iv)). (Checked: for the 39 fundamental discriminants with $|D|\le60$ the conductor of χ_D is $|D|$.) The identification of fundamental discriminants with discriminants of quadratic fields, and $\chi_D(p)$ as the splitting law, are Chapter 6 (items 1.5, 4.1 to 4.3). (v) $\chi_{DD'}=\chi_D\chi_{D'}$ on integers prime to DD', and $\chi_{Df^2}(n)=\chi_D(n)$ for $\gcd(n,f)=1$ (so a non-fundamental discriminant gives an imprimitive character). Checked: periodicity mod $|D|$ and the sign rule for all discriminants $|D|\le200$, $n<400$. Reference: Washington, Chapter 3; Cohen, A Course in Computational Algebraic Number Theory, Section 1.4 (add to `bib/ch02.bib` if cited). Paper: the table of Section 1 (χ_{-3}, χ_{-4}), Section 3 Example (χ_{-11}, χ_{-7}), Lemma lem:twist, Theorem thm:twist (χ_S); PLAN.md Section 6 row "$\chi_{-3},\chi_{-4},\chi_{q^*},q^*,\ell^*$, Kronecker symbols".

9.6 `prop:rings:qstar` (proposition; P). Let q be an odd prime and $q^*=(-1)^{(q-1)/2}q$. (i) $q^*\equiv1 \pmod 4$; q^* is the prime discriminant attached to q. (ii) For **every** integer n, $\left(\frac{q^*}{n}\right)=\left(\frac{n}{q}\right)$ (Legendre symbol of n mod q): at odd primes $p\neq q$ by 9.3(i); at $n=2$ because $q^*\equiv1 \pmod 8$ iff $q\equiv\pm1 \pmod 8$ iff $\left(\frac{2}{q}\right)=1$ (9.3(iii)); at $n=-1$ because $\operatorname{sign}(q^*)=(-1)^{(q-1)/2}=\left(\frac{-1}{q}\right)$; at q both vanish. So $\chi_{q^*}=\left(\frac{\cdot}{q}\right)$ as functions on $\mathbb{Z}$. (iii) χ_{q^*} is the unique Dirichlet character of order 2 and conductor q ($(\mathbb{Z}/q)^\times$ is cyclic, 5.3), which is why the paper calls it "the quadratic character of conductor q" and writes χ_q. By contrast, for $q\equiv3 \pmod 4$ the function $n\mapsto\left(\frac{q}{n}\right)$ is a character of conductor $4q$ ($\left(\frac{3}{\cdot}\right)$ has conductor 12). (iv) The same for the coefficient prime ℓ: $\ell^*\equiv1 \pmod 4$, and for primes $p\neq\ell$, ℓ^* is a square mod p (for odd p), resp. $\ell^*\equiv1 \pmod 8$ (for $p=2$), iff p is a square mod ℓ. (v) Table: $3^*=-3$, $5^*=5$, $7^*=-7$, $11^*=-11$, $13^*=13$; so in the paper's notation $\chi_3=\chi_{-3}$, $\chi_7=\chi_{-7}$, $\chi_{11}=\chi_{-11}$. Checked: (ii) for all odd primes $q<200$ and all $|n|\le300$. Reference: Washington, Chapter 3; Ireland-Rosen, Chapter 5. Paper: Section 1 (q^*, the twist E' by $\mathbb{Q}(\sqrt{q^*})$), Section 1.4 and Lemma lem:twist ("$\chi=\chi_q$ the quadratic character of conductor $q\ldots$ that is, by $\mathbb{Q}(\sqrt{q^*})$"), Theorem thm:main (ℓ^*), Section 3 Example ($\chi_{-11}=\chi_{11}$, $\chi_{-7}=\chi_7$), Theorems thm:twist, thm:quant (χ_S).

4.3.10 Section 10. Dirichlet characters, conductors, orthogonality, Gauss sums; the paper's characters

10.1 `def:rings:dirichlet` (definition and proposition; P). A Dirichlet character mod n (C5); the principal character χ_0 mod n; complete multiplicativity, n-periodicity, $\chi(a)$ a root of unity of order dividing the exponent of $(\mathbb{Z}/n)^\times$ for $\gcd(a,n)=1$; $\bar\chi=\chi^{-1}$; parity: $\chi(-1)=\pm1$ (even/odd). The characters mod n form a group $X(n)$ isomorphic (non-canonically) to $(\mathbb{Z}/n)^\times$ (via 5.2, 5.3), of order $\varphi(n)$. Characters with values in another field k containing enough roots of unity are defined the same way; the example used later is the $\mathbb{F}_\ell^\times$-valued character $\omega_\ell(a)=a$ mod ℓ of conductor ℓ and order $\ell-1$, which is the Dirichlet character of the reduced cyclotomic character $\bar\chi_\ell$ (Chapters 5, 6). Reference: Washington, Chapter

3; Serre, Course in Arithmetic, VI.1; Ireland-Rosen, Chapter 16. Paper: Lemma lem:twist(iii), Theorems thm:twist and thm:quant ($a_n(g\otimes\chi)=\chi(n)a_n(g)$), Section 3 Example ($a_p(121a1)\equiv\chi_{-11}(p)a_p(11a1)\pmod 3$).

10.2 `thm:rings:orthogonality` (theorem; P). For Dirichlet characters mod n: (i) $\sum_{a \bmod n}\chi(a)=\varphi(n)$ if $\chi=\chi_0$ and 0 otherwise; (ii) $\sum_{\chi \bmod n}\chi(a)=\varphi(n)$ if $a\equiv 1 \pmod n$ and 0 otherwise; (iii) for $\gcd(a,n)=1$, $\frac{1}{\varphi(n)}\sum_\chi \bar\chi(a)\chi(b)=1$ if $b\equiv a \pmod n$, else 0; (iv) the characters mod n form an orthonormal basis of the functions on $(\mathbb{Z}/n)^\times$ for $\langle f,h\rangle=\varphi(n)^{-1}\sum f\bar h$ (RHB analogue: the character table of a finite abelian group; finite Fourier analysis). Checked numerically for all characters mod $n<40$. Reference: Serre, Course in Arithmetic, VI.1; Washington, Chapter 3. Paper: background (Chapter 19, Dirichlet L-functions; projections onto character eigenspaces).

10.3 `prop:rings:conductor` (definition and proposition; P). Let χ be a character mod n and $d\mid n$. χ is *defined mod* d (induced from a character χ' mod d) if $\chi(a)=\chi'(a)$ for all a with $\gcd(a,n)=1$; equivalently $\chi(a)=1$ for all a with $\gcd(a,n)=1$ and $a\equiv 1 \pmod d$. (i) If χ is defined mod d_1 and mod d_2 then it is defined mod $\gcd(d_1,d_2)$. (ii) Hence there is a least such d, the *conductor* f_χ; it divides every d modulo which χ is defined; χ is induced by a unique character χ^{prim} mod f_χ, which is *primitive* (its own conductor is f_χ); χ is primitive iff $f_\chi=n$. (iii) χ and χ^{prim} agree on integers prime to n, but may differ at a with $\gcd(a,n)>1=\gcd(a,f_\chi)$: χ_{-3} regarded mod 6 vanishes at 2, while $\chi_{-3}(2)=-1$. (iv) Writing $\chi=\prod_p\chi_p$ with χ_p a character mod $p^{v_p(n)}$ (5.2), $f_\chi=\prod_p f_{\chi_p}$; a product of primitive characters with pairwise coprime conductors is primitive, with conductor the product. (v) Examples: the trivial character has conductor 1; the nontrivial character mod 4 has conductor 4 (not 2); $\left(\frac{\cdot}{q}\right)$ has conductor q; χ_8 ($=1$ iff $n\equiv\pm1 \pmod 8$) and χ_{-8} ($=1$ iff $n\equiv 1,3 \pmod 8$) have conductor 8; $\chi_S=\prod_{q\in S}\chi_{q^*}$ (S a finite set of odd primes) is primitive of conductor $\prod_{q\in S}q$. Reference: Washington, Chapter 3; Ireland-Rosen, Chapter 16 (or Miyake, Modular Forms, 3.1, which is in `refs.bib`). Paper: Lemma lem:twist ("the quadratic character of conductor q"), Lemma lem:twist(iii) (Atkin-Li: the level of $g\otimes\chi$ involves the conductor of χ), Theorems thm:twist and thm:quant ($\chi_S(n)=0$ when $\gcd(n,D)>1$, which is why "ι kills the coefficients at multiples of D"); Chapter 6, item 5.7.

10.4 `thm:rings:gauss-sums` (definition and theorem; P, except (iv), S). For a character χ mod n and $b\in\mathbb{Z}$ put $\tau(\chi,b)=\sum_{a \bmod n}\chi(a)\zeta_n^{ab}$ and $\tau(\chi)=\tau(\chi,1)$. (i) If $\gcd(b,n)=1$, $\tau(\chi,b)=\bar\chi(b)\tau(\chi)$. (ii) If χ is primitive, $\tau(\chi,b)=\bar\chi(b)\tau(\chi)$ for **all** b (both sides vanish when $\gcd(b,n)>1$), $|\tau(\chi)|^2=n$, $\tau(\chi)\tau(\bar\chi)=\chi(-1)n$ and $\overline{\tau(\chi)}=\chi(-1)\tau(\bar\chi)$. (iii) $\tau\left(\left(\frac{\cdot}{q}\right)\right)^2=q^*$ (9.2); $\tau(\chi_{-4})=i-(-i)=2i$; $\tau(\chi_{-3})=\zeta_3-\zeta_3^2=1+2\zeta_3=\sqrt{-3}$ (C6). (iv) For χ mod n induced from primitive χ^{prim} mod f with $\gcd(n/f,f)=1$: $\tau(\chi)=\mu(n/f)\chi^{\mathrm{prim}}(n/f)\tau(\chi^{\mathrm{prim}})$; e.g. χ_{-3} mod 6 has $\tau=(-1)(-1)\sqrt{-3}=\sqrt{-3}$; $\tau(\chi_0)=\mu(n)$ (a Ramanujan sum). Checked numerically: (i), (ii) for all 278 primitive characters of modulus $3\le n<40$; (iii), (iv) as stated. Reference: Washington, Chapter 4; Ireland-Rosen, Chapter 8. Paper: background (Chapter 5, galois:sqrt-lstar; Chapter 19, root numbers and functional equations of twisted L-functions).

10.5 `prop:rings:paper-characters` (proposition and worked table; P). The characters of the paper, as Dirichlet characters:

paper's name	Kronecker symbol	conductor	$\chi(-1)$	value at a prime p
χ_{-4}	$\left(\frac{-4}{\cdot}\right)$	4	-1	0 at $p=2$; $(-1)^{(p-1)/2}$, i.e. 1 iff $p\equiv 1 \bmod 4$
χ_{-3}	$\left(\frac{-3}{\cdot}\right)=\left(\frac{\cdot}{3}\right)$	3	-1	0 at 3; 1 iff $p\equiv 1 \bmod 3$ (so $\chi_{-3}(2)=-1$)
χ_{q^*}, q odd prime	$\left(\frac{q^*}{\cdot}\right)=\left(\frac{\cdot}{q}\right)$	q	$(-1)^{(q-1)/2}$	$\left(\frac{p}{q}\right)$ (0 at q; at $p=2$ also $\left(\frac{2}{q}\right)$)
$\chi_S=\prod_{q\in S}\chi_{q^*}$	$\left(\frac{\prod_{q\in S}q^*}{\cdot}\right)$	$\prod_{q\in S}q$	$\prod(-1)^{(q-1)/2}$	$\prod_{q\in S}\left(\frac{p}{q}\right)$
χ_{ℓ^*}	$\left(\frac{\ell^*}{\cdot}\right)=\left(\frac{\cdot}{\ell}\right)$	ℓ	$(-1)^{(\ell-1)/2}$	$\left(\frac{p}{\ell}\right)$

paper's name	Kronecker symbol	conductor	$\chi(-1)$	value at a prime p
ω_ℓ ($\mathbb{F}_\ell^\times$-valued)	none	ℓ	-1	$p \bmod \ell$

Facts proved: (i) **for every prime $p \neq 3$, $p \equiv \chi_{-3}(p) \pmod 3$; for every odd prime p, $p \equiv \chi_{-4}(p) \pmod 4$** (9.1(iv); $p \bmod 4$ directly). (ii) Hence (6.5(v)) for m prime to 3 the map $\zeta \mapsto \zeta^m$ of μ_3 is $\zeta \mapsto \zeta^{\chi_{-3}(m)}$, and for odd m on μ_4 it is $\zeta \mapsto \zeta^{\chi_{-4}(m)}$. (iii) $\left(\frac{\cdot}{\ell}\right) = \omega_\ell^{(\ell-1)/2}$ (Euler's criterion, with $\{\pm 1\} \subset \mathbb{F}_\ell^\times$), and $\left(\frac{\cdot}{\ell}\right) = \omega_\ell$ iff $\ell = 3$ (ω_ℓ has order $\ell - 1$); so $\chi_{-3} = \omega_3$ as $\{\pm 1\}$-valued characters mod 3. (iv) The type-III residue classes: $v_3(q - \chi_{-4}(q)) \geq 1$ iff $q \equiv \pm 1 \pmod{12}$ (for primes $q \geq 5$). The Galois-side statements ($\chi_D(\mathrm{Frob}_p) = \chi_D(p)$; χ_{-3} and χ_{-4} cut out $\mathbb{Q}(\sqrt{-3})$ and $\mathbb{Q}(i)$; $\bar{\chi}_3 = \chi_{-3}$ on $G_\mathbb{Q}$) are Chapter 5 (galois:quadratic-characters, galois:cyclotomic-character) and Chapter 6 (4.2, 5.7, 5.8). All congruences checked for $p < 5000$ (and (iv) for $q < 10^5$). Reference: as 9.5, 9.6. Paper: the table of Section 1, Lemma lem:euler, Proposition prop:forms ("since $q \equiv \chi_{-3}(q) \pmod 3$"), Lemma lem:twist, Theorems thm:twist and thm:quant, Section 3 Example, Section sec:anomaly ("the residue classes of the type III primes of the violating curves are $q \equiv 1$ and $q \equiv 11 \pmod{12}$": exactly the classes with $e_3 > 0$), Lemma lem:h0 and Proposition prop:ns ($\det \bar{\rho}_{E,3} = \chi_{-3}$ is $\bar{\chi}_3$; important_notes.md point 6).

4.3.11 Section 11. Valuations, the p-adic absolute value, and the paper's local terms

11.1 `def:rings:valuation` (definition and proposition; P). For a prime p, $v_p \colon \mathbb{Q} \to \mathbb{Z} \cup \{+\infty\}$, $v_p(0) = +\infty$, $v_p(p^k a/b) = k$ for $a, b \in \mathbb{Z}$ prime to p (C3). (i) $v_p(xy) = v_p(x) + v_p(y)$; (ii) $v_p(x + y) \geq \min(v_p(x), v_p(y))$, with equality if $v_p(x) \neq v_p(y)$; (iii) $v_p(x) \geq 0$ iff $x \in \mathbb{Z}_{(p)}$, $v_p(x) = 0$ iff $x \in \mathbb{Z}_{(p)}^\times$ (2.4); (iv) on $\mathbb{Z}$: $p^k \mid n$ iff $v_p(n) \geq k$, $v_p(\gcd) = \min$, $v_p(\mathrm{lcm}) = \max$ (3.3); (v) $v_p(r\mathbb{Z}) = v_p(r)$ for ideals, and "the p-part" $p^{v_p(n)}$; (vi) Legendre's formula $v_p(n!) = \sum_{k \geq 1} \lfloor n/p^k \rfloor$ (exercise). Normalisation $v_\ell(\ell) = 1$, as in Appendix B. Reference: Serre, Course in Arithmetic, Chapter II; Koblitz, Chapter I. Paper: Definition def:terms, Theorem thm:main ($v_\ell(\deg \varphi_E)$), the abstract, PLAN.md Section 6 row "ℓ-adic valuations of integers and rationals".

11.2 `def:rings:padic-abs` (definition and proposition; P for (i) to (iii), R for (iv)). $|x|_p = p^{-v_p(x)}$, $|0|_p = 0$; $|x|_\infty$ the usual absolute value. (i) $|\cdot|_p$ is an absolute value satisfying the ultrametric inequality $|x + y|_p \leq \max(|x|_p, |y|_p)$; $d_p(x, y) = |x - y|_p$ is a metric on $\mathbb{Q}$ in which $p^k \to 0$. (ii) Product formula: $\prod_{p \leq \infty} |x|_p = 1$ for $x \in \mathbb{Q}^\times$. (iii) $\mathbb{Q}$ is not complete for d_p (definition of the completion $\mathbb{Q}_p$ and of $\mathbb{Z}_p$: Chapter 7; $\mathbb{Z}_\ell$ as a profinite group: Chapter 1, item 7.2; as a DVR: Chapter 3). (iv) Ostrowski: every nontrivial absolute value on $\mathbb{Q}$ is equivalent to $|\cdot|_\infty$ or to one $|\cdot|_p$ (stated). Reference: Koblitz, Chapter I; Neukirch, Chapter II. Paper: background (the field $\mathbb{Q}_q$ and the ring $\mathbb{Z}_\ell$ throughout Sections 2 and 4).

11.3 `lem:rings:ell-integral` (lemma; P). Let $q \neq \ell$ be primes. (i) $v_\ell(q^k) = 0$ for all $k \in \mathbb{Z}$: q is an ℓ-adic unit. (ii) For $P \in \mathbb{Z}[X]$ of degree at most d and $k \geq 0$: $P(q^{-k}) \in \mathbb{Z}_{(\ell)}$ and $v_\ell(P(q^{-k})) = v_\ell\big(q^{kd} P(q^{-k})\big)$, the valuation of an integer; it is finite iff $P(q^{-k}) \neq 0$. (iii) Hence, for $\varepsilon = \pm 1$ and $a \in \mathbb{Z}$: $v_\ell\big(\frac{q-1}{q}\big) = v_\ell(q - 1)$; $v_\ell(1 - \varepsilon q^{-1}) = v_\ell(q - \varepsilon)$; $v_\ell(1 - q^{-2}) = v_\ell(q^2 - 1) = v_\ell(q - 1) + v_\ell(q + 1)$ (5.4(v)); $v_\ell\big(\frac{q-1}{q} \cdot \frac{(q+1)^2 - a^2}{q^2}\big) = v_\ell(q - 1) + v_\ell\big((q + 1)^2 - a^2\big)$ when $(q + 1)^2 \neq a^2$. (iv) For $q = 2$, $\ell = 3$: $v_3(1 + 2 \cdot 2^{-2}) = v_3(3/2) = 1$. All these valuations are ≥ 0. Reference: elementary. Paper: Definition def:terms ($e_\ell = v_\ell(P_q(q^{-2})/(1 - a_q^2 q^{-2}))$; at a multiplicative prime $a_q^2 = 1$ and the quotient is 1; at an additive prime $a_q = 0$), the table of Section 1, Lemma lem:euler (proof: "$e_\ell = v_\ell(1 - q^{-2}) = v_\ell(q^2 - 1)$", "$e_\ell = v_\ell(1 - \chi_{-3}(q) q^{-1}) = v_\ell(q - \chi_{-3}(q))$"), Lemma lem:fpr ("a nonzero rational number which is an ℓ-adic integer"), Remark rem:wild ("$P_2(X) = 1 + 2X$ and $e_3 = v_3(1 + \frac{1}{2}) = 1$").

11.4 `prop:rings:i0star-identity` (proposition; P). Let $q > 0$ and a be real, and $\alpha, \beta \in \mathbb{C}$ the roots of $X^2 - aX + q$. Then

$$\Big(1 - \frac{\alpha}{\beta q}\Big)\Big(1 - \frac{1}{q}\Big)\Big(1 - \frac{\beta}{\alpha q}\Big) = \frac{q - 1}{q} \cdot \frac{(q + 1)^2 - a^2}{q^2},$$

(expand, using $\alpha\beta = q$ and $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{a^2 - 2q}{q}$). Moreover $(q + 1)^2 - a^2 = (q + 1 - a)(q + 1 + a)$, and if

$|a| \le 2\sqrt{q}$ (the Hasse bound, Chapter 9) then $(q+1)^2 - a^2 \ge (q-1)^2 > 0$, so for $q \ge 2$ the valuation in 11.3(iii) is finite. Remark (forward pointer): for an elliptic curve $E'/\mathbb{F}_q$ with $a = q+1-\#E'(\mathbb{F}_q)$, the factors are $\#E'(\mathbb{F}_q)$ and the order of the group of its quadratic twist over $\mathbb{F}_q$ (Chapter 9). Checked symbolically. Reference: elementary. Paper: Lemma lem:euler (the displayed identity for type I_0^*), the table of Section 1, the abstract.

11.5 `prop:rings:local-terms` (proposition; P, given the table of Section 1 as stated in Chapters 19 and 21). Let $q \ge 5$ be a prime, $\ell \ne q$ a prime, and let e be the value of $e_\ell(E,q)$ prescribed by the table. Then e is a finite nonnegative integer, and: (i) types $\mathrm{II}, \mathrm{II}^*, \mathrm{IV}, \mathrm{IV}^*$: $e = v_\ell(q - \chi_{-3}(q))$, which is $v_\ell(q-1)$ if $q \equiv 1 \pmod 3$ and $v_\ell(q+1)$ if $q \equiv 2 \pmod 3$; for $\ell = 3$, $e \ge 1$ always (10.5(i)), and e is unbounded (e.g. $q = 163$: $e = v_3(162) = 4$); for $\ell \ge 5$, $e \ge 1$ iff $q \equiv \chi_{-3}(q) \pmod \ell$; (ii) types $\mathrm{III}, \mathrm{III}^*$: $e = v_\ell(q - \chi_{-4}(q))$; for $\ell = 3$, $e \ge 1$ iff $q \equiv \pm 1 \pmod{12}$; (iii) type I_n^*: $e = v_\ell(q^2-1)$; for $\ell = 3$, $e \ge 1$ always; for $\ell \ge 5$, $e \ge 1$ iff $q \equiv \pm 1 \pmod \ell$; (iv) type I_0^*, with $a = a_q(E')$: $e = v_\ell(q-1) + v_\ell((q+1)^2 - a^2)$; for $\ell = 3$, $e \ge 1$ iff $q \equiv 1 \pmod 3$ or $3 \mid a$; for $\ell \ge 5$, $e \ge 1$ iff $q \equiv 1 \pmod \ell$ or $a \equiv \pm(q+1) \pmod \ell$; (v) type I_n: $e = 0$; the term $t_\ell(E,q) = v_\ell(-v_q(j_E)) = v_\ell(n)$ and the Tamagawa refinement $v_\ell(n) + \min(v_\ell(n), v_\ell(q-1), v_\ell(c))$ of Lemma lem:tamfactor exceeds t_ℓ iff $\ell \mid n$, $q \equiv 1 \pmod \ell$ and $\bar{u} \in (\mathbb{F}_q^\times)^\ell$ (7.2(iv)). (Checked: (i) to (iii) for all primes $5 \le q < 10^5$; (iii) at $\ell = 5, 7, 11, 13$ for $q < 2 \cdot 10^4$.) Reference: elementary given the table. Paper: Definition def:terms, the table, Lemma lem:euler, Proposition prop:forms (proof: "$e_3(E,q) = v_3(q - \chi_{-3}(q)) \ge 1$ since $q \equiv \chi_{-3}(q) \pmod 3$"), Remark after Theorem thm:quant, (1) ("a single prime of type I_n^* ... forces $3 \mid \deg \varphi_E$, and ... a prime of type I_0^* with $q \equiv 1 \pmod 3$ does too"), Section sec:anomaly (residue classes mod 12), Section sec:iso, Section sec:tam.

11.6 `ex:rings:paper-values` (worked example; P for the arithmetic, the curve data quoted from the paper). (a) 121a1, type II at 11: $\chi_{-3}(11) = -1$, $e_3 = v_3(11+1) = v_3(12) = 1$, and $\deg \varphi = 6$, $v_3(6) = 1$: the bound is attained (paper, Section 3 Example; also Appendix: `rule(...)` returns `[1, 1]`). (b) 539c1, type I_3^* at 7: $e_3 = v_3(7^2-1) = v_3(48) = 1$, $t_3 = v_3(3) = 1$, $\deg \varphi = 2^5 \cdot 3^2$, $v_3 = 2 = e_3 + t_3$. (c) 980i1, type I_5^* at 7, $\ell = 5$: $e_5 = v_5(48) = 0$, $t_5 = v_5(5) = 1$, $\deg \varphi = 2^6 \cdot 3^2 \cdot 5$. (d) 726a1, type III at 11: $\chi_{-4}(11) = -1$, $e_3 = v_3(12) = 1$. (e) Type IV at 2 with PARI's $P_2(X) = 1 + 2X$: $e_3 = v_3(3/2) = 1$; type II at 2 with $P_2 = 1$: $e_3 = 0$ (Remark rem:wild, Proposition prop:forms); but type II at 2 does not force $P_2 = 1$ (336f1 has $P_2 = 1 + 2X$ and $e_3 = 1$; important_notes.md point 50). (f) Type I_0^* at 7 with $a_7(E') = -2$ (for instance \$E'=\$11a3, in the class 11a, whose $a_7 = -2$ was recomputed by point counting): $e_3 = v_3(6) + v_3(64-4) = 1 + 1 = 2$. The curve is 539d1: type I_0^* at 7 with \$E'=\$11a3, $e_3 = 2$ and modular degree 72 (important_notes.md B23). (g) A curve of type I_n at 11 has $e_3 = 0$ there, while its twist by χ_{-11} (the paper's χ_{11}) has type I_n^* and $e_3 = v_3(120) = 1$: e_ℓ is not invariant under a twist ramified at q (important_notes.md point 15). Paper: Section 3 Example, Remark rem:wild, the Appendix, Definition def:terms; Exercise 4.

4.4 3. Section-by-section plan with target line counts

Line counts are lines of LaTeX, card excluded (PLAN.md Section 7.3: 1,000 to 2,200).

section	title	contents	target lines
(starred)	Core results at a glance	the card: every item of Part 2, stated as above, with tag and `\Cref`	(excluded)

section	title	contents	target lines
(starred)	Where the paper uses this chapter	def:terms and the table; lem:euler; prop:forms; lem:twist, thm:twist, thm:quant, thm:qual Example; lem:tamfactor, sec:tam, rem:ko(1); lem:crit, lem:hecke, thm:quant Step 1; lem:fpr; rem:wild; sec:anomaly; lem:h0 (corrected) and prop:ns	40
1	Rings, ideals, quotients and products	conventions C1 to C7 as `convention` environments; 1.1 to 1.6; `paperlink` for lem:crit (1.3, 1.4, 1.5)	190
2	Prime and maximal ideals, fractions, $\mathbb{Z}_{(p)}$	2.1 to 2.4; `paperlink` for the last step of lem:crit (2.2) and for lem:fpr (2.4)	130
3	Euclidean domains, PIDs, UFDs; $\mathbb{Z}[i]$ and $\mathbb{Z}[\zeta_3]$	3.1 to 3.5; the two lattice pictures in words (square, triangular lattice)	190
4	Polynomial rings and irreducibility	4.1 to 4.6; `warning` on x^4+1; the ψ_3 example with a `paperlink` for prop:ns	200
5	$\mathbb{Z}/n\mathbb{Z}$, the Chinese remainder theorem, units, primitive roots	5.1 to 5.5; `paperlink` for lem:h0 (5.5)	170
6	Finite fields	6.1 to 6.5; `paperlink` for lem:euler (6.5(v))	180
7	Powers in $\mathbb{F}_q^\times$	7.1 to 7.3; `paperlink` for lem:tamfactor and sec:tam, with the reading of c and $v_\ell(c)$ (C3)	110
8	Roots of unity and cyclotomic polynomials	8.1 to 8.3; a table of Φ_n for $n \le 12$	110
9	Quadratic residues and reciprocity	9.1 to 9.6; `paperlink` for lem:twist (q^*)	230

section	title	contents	target lines
10	Dirichlet characters	10.1 to 10.5; the table of 10.5 (`booktabs`); `paperlink` for the table of Section 1, prop:forms, thm:twist; `warning` on χ_ℓ (cyclotomic) versus χ_{q^*} (quadratic; never χ_q, which appears only when quoting the paper, decision D3) and on ω_ℓ versus $\left(\frac{\cdot}{\ell}\right)$	200
11	Valuations and the paper's local terms	11.1 to 11.6; `paperlink` for def:terms, lem:euler, rem:wild, the Section 3 Example	170
12	Exercises	the 17 exercises of Part 4	120
13	Notes and references	Dummit-Foote, Ireland-Rosen, Serre (Course in Arithmetic), Lang, Washington, Koblitz, Atiyah-Macdonald, Miyake; which section numbers were cited; the alternative proofs of reciprocity	40
	total		**2,080**

4.5 4. Coverage table and exercises

4.5.1 4.1 Coverage: PLAN.md scope, Section 6 map rows and paper hooks to line-up items

PLAN.md scope item, Section 6 row, or paper hook	line-up items
rings	1.1, 1.6
ideals, quotient rings	1.2, 1.3, 1.4, 1.5
prime and maximal ideals	2.1, 2.2
integral domains	1.1, 2.1
fields of fractions	2.3, 2.4
PIDs, UFDs, Euclidean domains	3.1, 3.2, 3.3, 3.5
polynomial rings	4.1, 4.2, 4.6
Gauss's lemma	4.3
irreducibility criteria: Eisenstein, reduction mod p	4.4, 4.5, 8.2(vi)
$\mathbb{Z}/n\mathbb{Z}$ and the Chinese remainder theorem	1.4, 5.1, 5.2
units $(\mathbb{Z}/n\mathbb{Z})^\times$, primitive roots	5.1, 5.3, 5.4, 5.5
finite fields: existence, uniqueness	6.1, 6.2, 6.4
finite fields: Frobenius automorphism, subfields	1.6, 6.3
finite fields: multiplicative group cyclic	6.1(ii) (proof in Chapter 1, item 4.2)

PLAN.md scope item, Section 6 row, or paper hook	line-up items
ℓ-th powers in $\mathbb{F}_q^\times$; $[\mathbb{F}_q^\times : (\mathbb{F}_q^\times)^{\ell^k}] = \ell^{\min(k, v_\ell(q-1))}$	7.1, 7.2, 7.3
quadratic residues; Legendre, Jacobi and Kronecker symbols	9.1, 9.4, 9.5
quadratic reciprocity with proof (Gauss sums)	9.2, 9.3 (Eisenstein's lemma: Exercise 13)
χ_{-3}, χ_{-4}, $\chi_{q^*} = (q^*/\cdot)$, $q^* = (-1)^{(q-1)/2}q$; conductors; values at primes	9.6, 10.3(v), 10.5
Dirichlet characters, conductors, primitive characters, orthogonality	10.1, 10.2, 10.3
Gauss sums (basic properties)	9.2, 10.4
$\mathbb{Z}[\zeta_3]$ and $\mathbb{Z}[i]$ as Euclidean domains	3.4 (primes, $1-\zeta_3$, $\mathbb{Z}_3[\zeta_3]$: Chapter 6, items 4.5 to 4.9)
roots of unity, cyclotomic polynomials	6.5, 8.1, 8.2, 8.3
valuations v_p on $\mathbb{Q}$, the p-adic absolute value (definition only)	2.4, 11.1, 11.2
ℓ-adic valuations of integers and of rationals	5.4, 11.3, 11.4
$(q-1)/q$, $(q+1)^2 - a^2$, $1 - q^{-2}$	
the valuations $e_\ell(E, q)$ of the paper's table	11.5, 11.6
Section 6 row: ℓ-adic valuations of integers and rationals	11.1, 11.3, 11.5
Section 6 row: $\chi_{-3}, \chi_{-4}, \chi_{q^*}, q^*, \ell^*$, Kronecker symbols (with Chapter 6)	9.5, 9.6, 10.3, 10.5
Section 6 row: Tate parameter, unit part, ℓ-th powers in $\mathbb{F}_q^\times$, $t_\ell(E, q)$ (with Chapters 7, 10)	2.4(iii), 7.1, 7.2, 7.3, 11.5(v)
hook: Definition def:terms (valuations of rationals)	11.1, 11.3, 11.5, 11.6
hook: the table of Section 1 (χ_{-3}, χ_{-4})	10.5, 11.3, 11.4, 11.5
hook: Lemma lem:euler	6.5(v), 10.5(i), (ii), 11.3, 11.4
hook: Lemma lem:tamfactor (ℓ-th powers in $\mathbb{F}_q^\times$, the class c of the unit part)	2.4(iii), 7.1, 7.2, 7.3, 11.5(v)
hook: Lemma lem:twist (q^* and χ_{q^*})	9.3, 9.6, 10.3, 10.5
hook: Lemma lem:crit (cyclic groups $\mathbb{Z}/r$)	1.2, 1.3, 1.4, 1.5, 2.1, 2.2
hook: Proposition prop:forms ($q \equiv \chi_{-3}(q) \pmod 3$)	9.1(iv), 10.5(i), 11.5(i)
also: Lemma lem:fpr (ℓ-integrality), Remark rem:wild ($P_2(1/4) = 3/2$)	2.4, 11.3, 11.6(e)
also: Theorems thm:twist, thm:quant (χ_S, conductor $\prod q$, vanishing at multiples of D); thm:quant Step 1 (idempotents)	1.5, 10.3, 10.5
also: Section 3 Example (χ_{-11}, χ_{-7}; $e_3 = v_3(12)$, $v_3(48)$)	9.6(v), 11.6
also: Section sec:anomaly (type III classes mod 12), sec:tam, rem:ko(1)	7.2, 10.5(iv), 11.5(ii), (v)
also: Lemma lem:h0 corrected (important_notes.md points 5, 6), Proposition prop:ns ($\mathbb{F}_9$, ψ_3, $\chi_{-3} = \omega_3$)	4.5, 4.6, 5.5, 10.5(iii)
also: what Chapters 5, 6, 16 assume from Chapter 2 (Part 5)	4.3, 4.4, 6.1 to 6.5, 8.2, 9.2, 9.3, 9.5, 10.1, 10.3, 3.4, 2.4, 11.1

4.5.2 4.2 Exercises (Section 12 of the chapter), graded from easy to harder

1. (The toy Hecke algebra; check the paper's claim in Lemma lem:crit.) Let $r \geq 1$ and $R = \{(a, b) \in \mathbb{Z}^2 : a \equiv b \pmod r\}$, a subring of $\mathbb{Z} \times \mathbb{Z}$, with $\phi(a, b) = a$, $I = \ker \phi$ and $J = \{(a, 0) \in R\}$. Show

$I \cap J = 0$, $R/(I+J) \simeq \mathbb{Z}/\phi(J) = \mathbb{Z}/r$, and that the cokernel of $R \to R/I \times R/J$ is $\mathbb{Z}/r$ (1.3, 1.4). Interpret r as a congruence between two "eigenforms".

2. Show that a ring homomorphism $\mathbb{Z}/r\mathbb{Z} \to k$, k a field of characteristic ℓ, exists iff $\ell \mid r$, and deduce the last step of Lemma lem:crit: if $I+J$ lies in a maximal ideal of residue characteristic ℓ, then ℓ divides $\#\mathbb{T}/(I+J)$ (2.2).
3. In $\mathbb{Z}[\zeta_3]$ show $N(\alpha - \kappa\beta) \le \frac{1}{3}N(\beta)$ can always be achieved; factor $5 + \zeta_3 = (2-\zeta_3)(2+\zeta_3)$ and $7 = (3+\zeta_3)(2-\zeta_3)$ (note $3 + \zeta_3^2 = 2 - \zeta_3$), and compute $\gcd(7, 5+\zeta_3)$ by the Euclidean algorithm. List the units and check $\mathbb{Z}[\zeta_3]^\times = \mu_6$.
4. Check the paper's values in the Example of Section 3: $e_3(121\text{a}1, 11) = v_3(12) = 1$, $e_3(539\text{c}1, 7) = v_3(48) = 1$ and $t_3(539\text{c}1, 7) = 1$, $e_5(980\text{i}1, 7) = 0$ and $t_5(980\text{i}1, 7) = 1$; compare with $v_3(6)$, $v_3(2^5 \cdot 3^2)$, $v_5(2^6 \cdot 3^2 \cdot 5)$.
5. Check the paper's claim (proof of Proposition prop:forms) that $q \equiv \chi_{-3}(q) \pmod 3$ for every prime $q \neq 3$, so that $e_3(E, q) \ge 1$ at every prime $q \ge 5$ of type II, II*, IV, IV*. Find the least prime q with $v_3(q - \chi_{-3}(q)) \ge 3$ (answer 53) and with ≥ 4 (answer 163).
6. Check the paper's statement in Section sec:anomaly that the type III primes with a positive term e_3 are those with $q \equiv 1$ or $11 \pmod{12}$. For $\ell = 5$, show that the type III primes with $e_5 > 0$ are those with $q \equiv \pm 1 \pmod{20}$.
7. Check the paper's Remark after Theorem thm:quant, (1): every prime $q \ge 5$ of type I_n^* has $e_3 \ge 1$, and a prime of type I_0^* has $e_3 \ge 1$ iff $q \equiv 1 \pmod 3$ or $3 \mid a_q(E')$. Prove 11.4 and compute e_3 for type I_0^* at $q = 7$ with $a_7(E') = -2$ (answer 2), at $q = 5$ and at $q = 11$ with $a_q(E') = 0$ (answers 2, 2).
8. Check Remark rem:wild: with $P_2(X) = 1 + 2X$, $e_3 = v_3(P_2(1/4)) = 1$; with $P_2 = 1$, $e_3 = 0$. Why is the denominator $1 - a_q^2 q^{-2}$ of Definition def:terms equal to 1 at an additive prime and to $1 - q^{-2}$ at a multiplicative one, so that $e_\ell = 0$ there?
9. (Lemma lem:tamfactor.) For $q = 7, 13, 19, 37$ and $\ell = 3$ list the cubes and the ninth powers in $\mathbb{F}_q^\times$ and compute $v_3(c(u))$ for $u = 2, 3, -1$. Check the paper's claim that $\min(v_\ell(n), v_\ell(q-1), v_\ell(c)) > 0$ iff $q \equiv 1 \pmod \ell$, $\ell \mid n$ and u is an ℓ-th power.
10. Prove 7.1(iii) without quoting Chapter 1. Show that the ℓ-th power map of $\mathbb{F}_q^\times$ is bijective iff $q \not\equiv 1 \pmod \ell$. Find the least f with $\mu_9 \subset \mathbb{F}_{2^f}$ (answer 6) and compute $v_3(2^{6m} - 1)$ by 5.4.
11. Show that the only solutions of $a^2 \equiv 1 \pmod{\ell^k}$ (ℓ odd) are ± 1, and deduce that a homomorphism χ onto $(\mathbb{Z}/\ell^k)^\times$ with $\chi^2 = 1$ forces $\ell^k = 3$. Explain how this gives $h_\ell(E) = 0$ for $\ell \ge 5$ and $h_3(E) \le 1$ (important_notes.md point 5; the Galois side is Chapter 5, galois:twist (c)).
12. Construct $\mathbb{F}_{16} = \mathbb{F}_2[x]/(x^4 + x + 1)$; show that $\bar{x}$ generates $\mathbb{F}_{16}^\times$; list the subfields and the orbits of the Frobenius on $\mathbb{F}_{16}$; check $N_4 = 3$ over $\mathbb{F}_2$.
13. Prove quadratic reciprocity by Eisenstein's lemma: for odd primes $p \neq q$, $\left(\frac{q}{p}\right) = (-1)^{\sum_{a=1}^{(p-1)/2} \lfloor 2qa/p \rfloor}$, and count lattice points in a rectangle.
14. Verify that $\left(\frac{D}{\cdot}\right)$ is periodic mod $|D|$ for $D = -3, -4, -7, 5, 8, 12$. Tabulate $\chi_{-3}, \chi_{-4}, \chi_{-7}, \chi_{-11}, \chi_5$ at the primes below 30. Show that $\left(\frac{3}{\cdot}\right)$ has conductor 12 while $\left(\frac{-3}{\cdot}\right)$ has conductor 3 (the sign in q^* matters).
15. Compute $\tau(\chi_{-4}) = 2i$, $\tau(\chi_{-3}) = \sqrt{-3}$, $\tau\left(\left(\frac{\cdot}{5}\right)\right) = \sqrt{5}$; show that χ_{-3} regarded mod 6 has Gauss sum $\sqrt{-3}$ and that $\tau(\chi_0) = \mu(n)$ for the principal character mod n.
16. Show that $x^4 + 1$ is irreducible over $\mathbb{Q}$ but reducible modulo every prime. Check the paper's claim (Proposition prop:ns) for 11a1, whose mod 3 image is $\mathrm{GL}_2(\mathbb{F}_3)$: its 3-division polynomial $3x^4 - 4x^3 - 60x^2 - 237x - 21$ is irreducible (reduce mod 5).
17. Show that if $p \nmid n$ and $p \mid \Phi_n(a)$ then $p \equiv 1 \pmod n$; deduce that there are infinitely many primes $\equiv 1 \pmod n$. Check with $\Phi_{12}(2) = 13$ and $\Phi_9(2) = 73$.

4.6 5. Dependencies and forward pointers

Results from other chapters used here: * PLAN.md Section 2 only, plus Chapter 1. From the background: vector spaces over a field and dimension (used for $\#F = p^r$, $[F[x]/(f) : F] = d$), complex numbers and $e^{2\pi i/n}$, elementary group theory (RHB), the floor function. * Chapter 1 (`ch:groups`): Lagrange (1.2);

cyclic groups, their subgroups and ℓ^k-th powers, $[C : C^{\ell^k}] = \ell^{\min(k, v_\ell(n))}$, $\mathrm{Aut}(C_n) = (\mathbb{Z}/n)^\times$ (1.3); the ℓ-primary decomposition (1.8); CRT for groups (2.2); finite abelian groups (4.1); finite subgroups of $K^\times$ are cyclic (4.2, whose proof uses 4.1(iv) here: a polynomial of degree d over a field has at most d roots, which Chapter 1 takes from the background); the field $\mathbb{F}_q[\sqrt{\varepsilon}]$ of 5.8 (here identified with $\mathbb{F}_{q^2}$). Chapter 2 must use, not redo, these. * Chapter 1 uses $v_\ell(n)$ for integers informally before 11.1; 11.1 is the formal definition (say so in a footnote-free sentence).

Black boxes imported from later chapters (stated with a forward `\Cref`, used only in examples and `paperlinks`, never in a proof of a numbered result): the table of Section 1 and the definition of P_q (Chapters 19, 21), the Hasse bound $|a_q| \le 2\sqrt{q}$ (Chapter 9), the Kodaira types and the paper's curve data (Chapters 10, 21, 22), the Galois meaning of the characters (Chapters 5, 6), the Tate parameter and its unit part (Chapters 7, 10).

Forward pointers (chapters that use this one): * Chapter 3 (`ch:modules`): ideals, quotients, Noetherian PIDs (3.2), $\mathbb{Z}_{(p)}$ as the first DVR and localisation (2.4), products of rings and idempotents (1.5, for finite $\mathbb{Z}_\ell$-algebras), annihilators (1.2), the module version of 1.4 in Lemma lem:ddt417, v_ℓ of ideals of $\mathbb{Z}_\ell$ (C3, 11.1). * Chapter 4 (`ch:reps`): $\mathbb{F}_9 = \mathbb{F}_3[i]$ (4.6, 6.2), characters of finite abelian groups and orthogonality (10.2). * Chapter 5 (`ch:galois`): Gauss's lemma and Eisenstein (4.3, 4.4), $\mathbb{Z}/n$ and $(\mathbb{Z}/n)^\times$ (5.1 to 5.3), finite fields: existence, uniqueness, Frobenius, cyclic multiplicative group (6.1 to 6.3), roots of unity in $\mathbb{F}_q$ (6.5), $\Phi_n \in \mathbb{Z}[x]$ and $x^n - 1 = \prod_{d|n} \Phi_d$ (8.2), Gauss sums $g^2 = \ell^*$ (9.2), Legendre and Kronecker symbols, q^*, quadratic reciprocity (9.1 to 9.6), the power criteria in $\mathbb{F}_q^\times$ (7.1, 7.2) which its galois:kummer-finite-field reinterprets, $a^2 \equiv 1 \bmod \ell^k$ (5.5) for galois:twist (c). * Chapter 6 (`ch:ant`): PIDs and UFDs (3.2), $\mathbb{Z}[i]$ and $\mathbb{Z}[\zeta_3]$ Euclidean with units μ_4, μ_6 (3.4, cited by its 4.5), Gauss's lemma (its 1.2), Eisenstein (its 4.7, 5.1), CRT (its 2.4), finite fields (its 2.3, 3.6), Kronecker symbols and reciprocity (its 4.1, 4.2, 5.4), Dirichlet characters, conductors, primitivity (its 5.7, 5.8), cyclotomic polynomials (its 5.1, 5.2), v_p (its 2.2). * Chapter 7 (`ch:local`): $|\cdot|_p$ and v_p (11.1, 11.2), $\mathbb{Z}_{(p)}$ and reduction (2.4), $(\mathbb{Z}/p^k)^\times$ and lifting the exponent (5.3, 5.4) for the structure of $\mathbb{Z}_p^\times$, $\mu_{\ell^k} \subset \mathbb{F}_q$ iff $\ell^k \mid q-1$ (6.5), ℓ-th powers in $\mathbb{F}_q^\times$ lifted by Hensel to $\mathbb{Z}_q^\times$ (7.1, 7.2). * Chapter 9 (`ch:ellcurves`): a_q and $\#E(\mathbb{F}_q) = q + 1 - a_q$ (11.4's remark), quadratic twists by χ_D with $a_p(E^{(d)}) = \chi_D(p) a_p(E)$ (9.5, 10.5). * Chapters 10, 11 (`ch:reduction`, `ch:galrep`): $t_\ell = v_\ell(n)$, the class of the unit part (7.2), $\mathbb{F}_9$ and χ_{-3} (10.5(iii)), ψ_3 (4.5). * Chapters 12, 13 (`ch:modforms`, `ch:hecke`): twisting newforms by Dirichlet characters, $a_n(g \otimes \chi) = \chi(n) a_n(g)$, conductors in Atkin-Li (10.1, 10.3); the Hecke algebra as a ring (1.1 to 1.5). * Chapter 14 (`ch:moddeg`) and Chapter 20 (`ch:congruence`): the congruence number r, $\mathbb{T}/(I+J) \simeq \mathbb{Z}/r$ (1.3, 1.4, 2.2), v_ℓ of integers (11.1). * Chapters 16, 17 (`ch:cohomology`, `ch:galcoh`): $\mathbb{Z}_{(\ell)}^\times$ and "q prime to ℓ is a unit" (2.4, 11.3), $v_\ell(q-1)$, the cyclic group $\mathbb{F}_q^\times/(\mathbb{F}_q^\times)^{\ell^v} \simeq \mathbb{Z}/\ell^v$ (7.2), which they identify with $\mathbb{Z}_\ell(1)/(q-1)$. * Chapter 19 (`ch:lfunctions`): Euler factors evaluated as rationals and their valuations (11.3 to 11.5), Dirichlet characters, orthogonality and Gauss sums (10.1 to 10.4). * Chapters 21, 22 (`ch:paper`, `ch:computation`): every hook of Part 0; PARI's `valuation`, `kronecker`, `znprimroot`, `ffgen`, `polisirreducible` as the computational counterparts (Chapter 22).

4.7 6. Notes for the writer

Conventions to fix (Section 1 of the chapter, as `convention` environments): C1 to C7 of Part 1. In addition: * Name every quadratic character by its discriminant inside the chapter ($\chi_D = \left(\frac{D}{\cdot}\right)$); give the dictionary to the paper's names once, in 9.6(v) and the table of 10.5. Never write χ_3 for χ_{-3} in the chapter's own text: in the paper χ_ℓ is the cyclotomic character, and with $\ell = 3$ the two readings collide (Part 8, point 4). * The Kronecker symbol at 2 and at -1 (C4): $\left(\frac{a}{2}\right)$ depends on a mod 8 and vanishes for even a; $\left(\frac{a}{-1}\right)$ is the sign of a, so $\chi_D(-1) = \mathrm{sign}\, D$ and χ_{-3}, χ_{-4} are odd. Do not confuse $\left(\frac{D}{2}\right)$ (Kronecker, top entry D) with the Jacobi symbol $\left(\frac{2}{D}\right)$; e.g. $\chi_{-3}(2) = \left(\frac{-3}{2}\right) = -1$ because $-3 \equiv 5 \pmod 8$, consistent with $2 \equiv 2 \pmod 3$. * The sign in q^*: $\left(\frac{q^*}{\cdot}\right)$ has conductor q, while $\left(\frac{q}{\cdot}\right)$ has conductor $4q$ when $q \equiv 3 \pmod 4$ (e.g. $\left(\frac{3}{\cdot}\right)$, conductor 12). The conductor of χ_{-4} is 4, not 2; of χ_{-3} is 3, not 6 or 12; $\chi_{\pm 8}$ have conductor 8. A quadratic field $\mathbb{Q}(\sqrt{d})$ has character χ_D with D the discriminant, not d ($\chi_{-1} = \chi_{-4}$, $\chi_2 = \chi_8$): that is Chapter 6's statement, but the naming convention starts here. * $\mathbb{F}_q$: in Sections 6 and 7

q is a prime power; in the paper q is a prime, so $\mathbb{F}_q = \mathbb{Z}/q\mathbb{Z}$; the paper's "class of u in $\mathbb{F}_q^\times$" means the class of the reduction $\bar{u}$ of the unit $u \in \mathbb{Z}_q^\times$ (7.2(v); Part 8, point 3). * $v_\ell(c)$ for $c \in \mathbb{Z}/\ell^v$ (C3) is independent of the generator chosen in 7.2(ii); make that explicit, since the paper writes "$c \in \mathbb{Z}/\ell^{v_\ell(q-1)}$" without naming the identification. * State $v_p(0) = +\infty$ and say when each rational in 11.3 is nonzero ($q \pm 1 \neq 0$; Hasse for $(q+1)^2 - a^2$). * Rigour: everything in this chapter is proved, except Zorn's lemma for maximal ideals (2.1(v)), R UFD implies $R[x]$ UFD (sketch, 4.3(iii)), the primitivity of χ_D for fundamental D (sketch via prime discriminants, 9.5(iv)), the sign of the quadratic Gauss sum (stated, 9.2(iii)), the formula 10.4(iv) (sketch), Ostrowski (stated). Never present these as proofs. * Keep the RHB remarks to one sentence each: characters of $(\mathbb{Z}/n)^\times$ as the character table of a finite abelian group (10.2); Gauss sums as finite Fourier transforms (10.4); lattices $\mathbb{Z}[i]$, $\mathbb{Z}[\zeta_3]$ in $\mathbb{C}$ (3.4).

Pitfalls: * Do not prove $\mathrm{Aut}(\mathbb{F}_q) = \langle \mathrm{Frob}_p \rangle$ or $\mathrm{Gal}(\mathbb{F}_{q^n}/\mathbb{F}_q)$ cyclic as a numbered result: it is Chapter 5's item galois:finite-field-galois. A one-line remark with a forward pointer is enough (6.3(iv)). * Do not prove the irreducibility of every Φ_n (Chapter 5); only Φ_{p^k} via Eisenstein. * Do not classify the primes of $\mathbb{Z}[i]$ and $\mathbb{Z}[\zeta_3]$, nor introduce $1 - \zeta_3$ as a uniformiser or $\mathbb{Z}_3[\zeta_3]$ (Chapter 6, items 4.5 to 4.9); 3.4 gives the Euclidean property, the units and a few factorisations. * Do not identify Dirichlet characters with characters of $G_\mathbb{Q}$, and do not write $\chi_D(\mathrm{Frob}_p)$: that is Chapter 6 (items 4.2, 5.7). 10.5 states only the values $\chi_D(p)$ of the functions on $\mathbb{Z}$. * The generalised Euler criterion needs $d = \gcd(m, q-1)$, not m: u is a cube in $\mathbb{F}_{11}^\times$ for every u, although $u^{10/3}$ is meaningless. * The Jacobi symbol $\left(\frac{a}{n}\right) = 1$ does not mean a is a square mod n (9.4(iii)). * $x^4 + 1$: reducible mod every prime, irreducible over $\mathbb{Q}$; the reduction test proves irreducibility only in one direction. * $\mathbb{Z}/p^k\mathbb{Z}$ is not $\mathbb{F}_{p^k}$. * $e_\ell(E, q)$ is not invariant under a quadratic twist ramified at q (important_notes.md point 15; 11.6(g)); the chapter must not repeat the paper's sentence "Both terms are invariant under quadratic twist" without that restriction. * At $q = 2, 3$ the Kodaira type does not determine the factor; the chapter only evaluates PARI's P_2 as a rational (11.6(e)). * Arithmetic versus geometric Frobenius (important_notes.md point 11) is not this chapter's business; 6.5(v) speaks of the map $\zeta \mapsto \zeta^q$, which is the arithmetic Frobenius of $\mathbb{F}_q$ acting on μ_3, and that is all Lemma lem:euler needs from here. * The paper's introduction describes the naive adjoint L-function as "the Euler product of $\sum a_n(f)^2 n^{-s}$"; this is not literally right at the good primes (Part 8, point 2). The chapter does not need the introduction's description; quote Definition def:terms and Section 1.1's definition of L_q^{nv} instead. * Typography (PLAN.md 7.2): no dash ligatures ("Dummit-Foote", "Ireland-Rosen", "Atkin-Li" with plain hyphens; ranges as "1 to 8"); ASCII source; the tables of 8.2 and 10.5 as `tabular` with `booktabs`.

References: Ireland-Rosen (A Classical Introduction to Modern Number Theory, 2nd ed.) and Cohen (A Course in Computational Algebraic Number Theory) are not in `refs.bib`: add them to `bib/ch02.bib` if cited. Dummit-Foote section numbers used above (7.1 to 7.6, 8.1 to 8.3, 9.1 to 9.5, 13.1, 13.4 to 13.6, 14.3) and the chapter numbers for Ireland-Rosen (4: $(\mathbb{Z}/n)^\times$; 5: reciprocity; 6: quadratic Gauss sums; 7: finite fields; 8: Gauss and Jacobi sums; 13: cyclotomic fields; 16: Dirichlet characters), Serre's Course in Arithmetic (I.1 finite fields, I.3 reciprocity, II p-adic fields, VI.1 characters), Washington (2 cyclotomic fields, 3 Dirichlet characters, 4 Gauss sums) and Koblitz (I) were recalled, not re-verified against the books: check them, or cite the chapter only. No theorem numbers are cited in this outline.

4.8 7. Verification log (computations done for this outline)

Python 3 with sympy 1.14, scripts `check1.py` to `check10.py` in `/tmp/ch02`. * Index formula and generalised Euler criterion: for every prime $q < 400$, $\ell \in \{2, 3, 5, 7, 11, 13\}$, $\ell \neq q$, $k \leq 4$: the index of the ℓ^k-th powers is $\ell^{\min(k, v_\ell(q-1))}$ and u is an ℓ^k-th power iff $u^{(q-1)/\gcd(\ell^k, q-1)} = 1$. No mismatch. The same index formula, cyclicity of $\mathbb{F}_q^\times$ and $\#\mu_n(\mathbb{F}_q) = \gcd(n, q-1)$ ($n < 30$) in $\mathbb{F}_q$ for $q = 4, 8, 9, 16, 25, 27, 49, 64, 81$ built as $\mathbb{F}_p[x]/(f)$. * The class c: for primes $q < 300$ and $\ell \in \{3, 5, 7\}$ with $\ell \mid q - 1$, with c the discrete logarithm mod ℓ^v for a primitive root: $v_\ell(c) \geq k$ iff u is an ℓ^k-th power, for all u and $0 \leq k \leq v$. No mismatch. Examples of 7.3 recomputed. * Characters: $\chi_{-3}(p) = 1$ iff $p \equiv 1 \pmod 3$, $\chi_{-4}(p) = 1$ iff $p \equiv 1 \pmod 4$, $\chi_{-3}(2) = -1$, $\chi_{-4}(2) = 0$, $\chi_{-3}(3) = 0$, $p \equiv \chi_{-3}(p) \pmod 3$ and $p \equiv \chi_{-4}(p) \pmod 4$: all primes below 5000. $\left(\frac{q^*}{n}\right) = \left(\frac{n}{q}\right)$ for all odd primes $q < 200$ and all $0 < |n| \leq 300$. Kronecker periodicity mod $|D|$ and $\left(\frac{D}{-n}\right) = \mathrm{sign}(D)\left(\frac{D}{n}\right)$ for every discriminant $|D| \leq 200$, $n < 400$. Conductor of χ_D equals $|D|$ for the 39 fundamental discriminants with $|D| \leq 60$ (brute force). * Reciprocity and supplements for odd

primes below 400; Jacobi reciprocity and $\left(\frac{2}{n}\right)$ for odd $m, n < 200$; Eisenstein's lemma (Exercise 13) for primes below 200; $\left(\frac{2}{15}\right) = 1$ with 2 not a square mod 15. * Gauss sums: $g^2 = q^*$ and the sign ($\sqrt{q}$ or $i\sqrt{q}$) for primes $q < 60$; $\tau(\chi_{-4}) = 2i$, $\tau(\chi_{-3}) = i\sqrt{3} = \sqrt{-3}$, $\tau(\chi_{-3} \bmod 6) = i\sqrt{3}$; for every character mod n, $3 \le n < 40$: orthogonality (i), and for the 278 primitive ones $|\tau|^2 = n$ and $\tau(\chi, b) = \bar{\chi}(b)\tau(\chi)$ for all b. * Cyclotomic: $\deg \Phi_n = \varphi(n)$ and $\Phi_n(1)$ as in 8.2(v) for $n < 60$; $p \mid \Phi_n(a)$ iff a has order n mod p ($2 \le n < 40$, $p < 200$, $p \nmid n$); $\Phi_{12}(2) = 13$, $\Phi_9(2) = 73$. $x^4 + 1$ reducible mod every prime below 500. ψ_3 of 11a1 irreducible over $\mathbb{Q}$ and irreducible mod 5. * Primitive roots: $(\mathbb{Z}/n)^\times$ cyclic iff $n \in \{1, 2, 4, p^k, 2p^k\}$ for $n < 400$; 5 has order 2^{k-2} mod 2^k and $-1 \notin \langle 5 \rangle$ for $3 \le k < 16$; $a^2 \equiv 1 \pmod{\ell^k}$ only for $a \equiv \pm 1$, and all units square to 1 only for $\ell^k = 3$ ($\ell \le 11$, $k \le 4$). * Lifting the exponent (i) for $\ell \in \{3, 5, 7\}$, $a, b < 60$, $n < 40$; (ii) for odd $a < 200$, even $n < 40$; 5.4(iv) for $\ell \le 13$, $q < 300$, $f < 30$. * Irreducible counts N_r over $\mathbb{F}_2$ $(2, 1, 2, 3, 6, 9, 18, 30)$ and $\mathbb{F}_3$ $(3, 3, 8, 18, 48)$, and $p^r = \sum_{d \mid r} d N_d$. * Euclidean constants: the maximum over a fundamental domain of the squared distance to the nearest lattice point is $1/2$ for $\mathbb{Z}[i]$ and $1/3$ for $\mathbb{Z}[\zeta_3]$ (grid search). Factorisations $5 + \zeta_3 = (2 - \zeta_3)(2 + \zeta_3)$, $3 + \zeta_3^2 = 2 - \zeta_3$; the gcd example in $\mathbb{Z}[i]$. * Local terms: $\min_q v_3(q - \chi_{-3}(q)) = 1$ and $\min_q v_3(q^2 - 1) = 1$ over primes $5 \le q < 10^5$; type III: $v_3(q - \chi_{-4}(q)) > 0$ iff $q \equiv \pm 1 \pmod{12}$ ($q < 10^5$); $v_\ell(q^2 - 1) > 0$ iff $q \equiv \pm 1 \pmod{\ell}$ for $\ell = 5, 7, 11, 13$, $q < 2 \cdot 10^4$; least q with $v_3(q - \chi_{-3}(q)) \ge 3$ is 53, with ≥ 4 is 163. Values: $11 - \chi_{-3}(11) = 12$, $v_3(48) = 1$, $v_5(48) = 0$, $11 - \chi_{-4}(11) = 12$, $v_3(P_2(1/4)) = v_3(3/2) = 1$. The I_0^* identity 11.4 symbolically. a_p of 11a1 by point counting: $a_2, a_3, a_5, a_7, a_{13}, a_{17}, a_{19} = -2, -1, 1, -2, 4, -2, 0$, so $e_3 = v_3(6) + v_3(60) = 2$ for 11.6(f). * Twists: for $y^2 = x^3 + a_2x^2 + a_4x + a_6$ and its twist $y^2 = x^3 + da_2x^2 + d^2a_4x + d^3a_6$ the invariants are d^2c_4, d^3c_6, $d^6\Delta$ (symbolic), used in Part 8, point 1. * Rankin's identity: for $(p, a_p) = (5, 1), (7, -2), (13, 4)$, $\sum_k a_{p^k}^2 X^k = (1 + pX) \big/ \big((1 - \alpha^2 X)(1 - pX)(1 - \beta^2 X)\big)$ (exact power series to X^9), used in Part 8, point 2.

4.9 8. Points for important_notes.md

New points in the area of this chapter (valuations of rationals, q^* and the quadratic characters, ℓ-th powers in $\mathbb{F}_q^\times$), not among the 28 already recorded. Points 5, 6 (on χ_ℓ versus χ_{ℓ^*} in Lemma lem:h0 and Section 1.3), 15 (e_ℓ not twist invariant) and 23 (c_q used for two things) are used above and not repeated. None of the four points changes a result of the paper.

1. **Lemma lem:twist, proof of (ii): the invariants of the twist by $\mathbb{Q}(\sqrt{q^*})$.** Location: "the twist has a model with invariants q^2c_4, q^3c_6, $q^6\Delta$ and the same j-invariant". Problem: the quadratic twist by d of $y^2 = x^3 + a_2x^2 + a_4x + a_6$ is $y^2 = x^3 + da_2x^2 + d^2a_4x + d^3a_6$, with invariants d^2c_4, d^3c_6, $d^6\Delta$ (checked symbolically). For the twist by $\mathbb{Q}(\sqrt{q^*})$, $d = q^*$, this gives q^2c_4, $(q^*)^3c_6 = (-1)^{(q-1)/2}q^3c_6$, $q^6\Delta$. For $q \equiv 3 \pmod 4$ the triple $(q^2c_4, q^3c_6, q^6\Delta)$ belongs to the twist by $q = -q^*$, that is, to $E^{(q)} \otimes \chi_{-4}$, which is also ramified at 2. The next sentence of the proof writes the model with q^* correctly. Fix: write "$(q^*)^2c_4 = q^2c_4$, $(q^*)^3c_6$, $(q^*)^6\Delta = q^6\Delta$", or "invariants whose q-adic valuations are those of c_4, c_6, Δ increased by 2, 3, 6". Effect: none, since only q-adic valuations are used there. Confidence: high.

2. **Introduction (paragraph "The classical route to the degree . . . "): the description of the naive adjoint L-function.** Location: "The L-function that appears there is the *naive* one: the Euler product of $\sum a_n(f)^2 n^{-s}$, which has the true local factor at the primes of good or multiplicative reduction and the trivial factor at the additive primes." Problem: at a good prime p, $\sum_k a_{p^k}^2 p^{-ks} = (1 + p^{1-s})L_p(\mathrm{Sym}^2 f, s)$, not $L_p(\mathrm{Sym}^2 f, s)$ (Rankin's identity; checked as an exact power series identity for several (p, a_p)). At multiplicative primes ($a_{p^k}^2 = 1$) and additive primes ($a_{p^k} = 0$, $k \ge 1$) the description is right. Globally $\sum a_n(f)^2 n^{-s} = L^{\mathrm{nv}}(\mathrm{Sym}^2 f, s)\, \zeta^{(N)}(s-1)/\zeta^{(N)}(2s-2)$, where $\zeta^{(N)}$ omits the Euler factors at $p \mid N$. Fix: "the naive one, $\prod_p L_p^{\mathrm{nv}}(\mathrm{Sym}^2 f, s)$, which has the true local factor at the primes of good or multiplicative reduction and the trivial factor at the additive primes; it differs from $\sum a_n(f)^2 n^{-s}$ by the factor $\zeta^{(N)}(s-1)/\zeta^{(N)}(2s-2)$". Effect: none on the results; Definition def:terms and Section 1.1 define the naive factor prime by prime, which is what is used. (This belongs to Chapter 19's area; it was found while checking Definition def:terms.) Confidence: high.

3. **Lemma lem:tamfactor, statement: "$c \in \mathbb{Z}/\ell^{v_\ell(q-1)}$ the class of u in $\mathbb{F}_q^\times/(\mathbb{F}_q^\times)^{\ell^{v_\ell(q-1)}}$".** Problem (expository): $u = q_E q^{-n}$ lies in $\mathbb{Z}_q^\times$, not in $\mathbb{F}_q^\times$; what is meant is the reduction $\bar{u} \in \mathbb{F}_q^\times$ (and likewise "u is an ℓ^k-th power in $\mathbb{F}_q^\times$" means $\bar{u}$; by Hensel this is the same as $u \in (\mathbb{Z}_q^\times)^{\ell^k}$, Chapter 7). The identification of the cyclic group $\mathbb{F}_q^\times/(\mathbb{F}_q^\times)^{\ell^v}$ with $\mathbb{Z}/\ell^v$ needs a choice of generator; $v_\ell(c)$ does not depend on it, but this should be said, together with the reading $v_\ell(c) = v$ when $c = 0$ (so that the formula is right when $\bar{u}$ is an ℓ^v-th power, and when $v = 0$). Fix: "let $\bar{u} \in \mathbb{F}_q^\times$ be the reduction of u and c its class in $\mathbb{F}_q^\times/(\mathbb{F}_q^\times)^{\ell^v} \simeq \mathbb{Z}/\ell^v$, $v = v_\ell(q-1)$ (the identification depending on a generator; $v_\ell(c) := \min(v_\ell(\tilde{c}), v)$ for a lift $\tilde{c}$ does not)". Effect: none. Confidence: high. (The proof's identification of c' is important_notes.md point 7, a different issue.)

4. **Notation: the paper's χ_q (quadratic, conductor q; the book's χ_{q^*}, decision D3) versus χ_ℓ (cyclotomic), and prime-indexed versus discriminant-indexed names.** Locations: Lemma lem:twist ("$\chi = \chi_q$ the quadratic character of conductor q", for every odd prime q, so $q = 3$ is allowed); the proofs of Lemmas lem:tamfactor ("$\chi_\ell\varepsilon$"), lem:old ("$\varepsilon\chi_\ell$"), lem:h0 and Section 1.3 ($\bar{\chi}_\ell$, the cyclotomic character); Section 3 Example (χ_{-11}, χ_{-7}, which Lemma lem:twist would call χ_{11}, χ_7); the table of Section 1 (χ_{-3}, χ_{-4}, by discriminant). Problem (expository): with $q = \ell = 3$ the symbol χ_3 means two different characters (the quadratic χ_{-3}, and the 3-adic cyclotomic character, which agrees with it only mod 3); and the same quadratic character is named both by its conductor (χ_{11}) and by its discriminant (χ_{-11}). Fix: name the quadratic characters by their discriminant throughout (χ_{q^*} in Lemma lem:twist, $\chi_S = \chi_{\prod_{q\in S} q^*}$), or write ψ_q for the quadratic character of conductor q; one sentence in Section 1.1 suffices. Effect: none. Confidence: high that both usages occur; the importance is low.

Chapter 5

Chapter 03: Modules and commutative algebra (`ch:modules`, `chapters/ch03_modules.tex`)

Stage 1 line-up. Part I (Algebra). Target length 2,165 lines of LaTeX (card excluded; see Part 3), inside the 1,000 to 2,200 range of PLAN.md Section 7.3.

Rigour tags: **P** = proved here, **S** = proof sketched here, **R** = stated; reference. Item format, as in the Chapter 2 and 5 outlines: number, label-ready name, kind, tag; the precise statement with all hypotheses; reference (book and chapter or section; a theorem number only where sure, otherwise marked "check"); the paper's use by its `\label` names, or "background". Every computation quoted below was checked by hand and by short Python/sympy scripts in `/tmp/ch03` (verification log, Part 7).

5.1 0. What this chapter is for

The paper uses this chapter in nine places, and the chapter is built around them.

1. **Lengths and Fitting ideals over $\mathbb{Z}_\ell$** (Section 1.2: $\mathrm{Tam}_p(M) = \mathrm{Fitt}_{\mathbb{Z}_\ell}\big(H^1(I_p, M)^{G_{\mathbb{Q}_p}}_{\mathrm{tors}}\big)$; Lemma lem:tamfactor: the Fitting ideal of $(\mathbb{Z}/\ell^v)\oplus(\mathbb{Z}/\ell^m)$ is $\ell^{v+m}\mathbb{Z}_\ell$; Lemma lem:fpr: $\mathrm{Fitt}_{\mathbb{Z}_\ell} H^1_f(\mathbb{Q}_p, T^D) = L_p(A_f,1)^{-1}\,\mathrm{Tam}_p(T^D)$; Theorem thm:main, Theorem thm:dfg37, Remark rem:ns, Conjecture conj:euler: $\mathrm{length}_{\mathbb{Z}_\ell}$ of Selmer groups and of H^0). The chapter proves that over a DVR the Fitting ideal of a finite-length module is ϖ^{length}, that length is additive along exact sequences, and that Pontryagin duality preserves length.
2. **Pontryagin duality** (Lemma lem:dfg21: the duals $H^1_f(\mathbb{Q}_p, T^D)^*$ and $H^0(\mathbb{Q}, W^D)^*$; Lemma lem:h0 and Theorem thm:main: $W = \mathrm{ad}^0 T \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$, $W^D = V^D/T^D$): duality for finite abelian groups and between finitely generated $\mathbb{Z}_\ell$-modules and cofinitely generated discrete ℓ-primary modules; the duality between T and $T \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$; $W[\ell^k] = T/\ell^k T$.
3. **Saturated sublattices, perfect pairings, Gram determinants, alternating pairings, adjoints** (Lemma lem:ddt417: "the pairing identifies $L/L[\wp]^\perp$ with $\mathrm{Hom}(L[\wp], \mathcal{O})$, because $L[\wp]$ is saturated", "Q is the cokernel of $L[\wp] \to \mathrm{Hom}(L[\wp], \mathcal{O})$, of order $\#\mathcal{O}/\Delta$"; Section 4.1: $\hat{\delta}_!$ perfect and alternating with all Hecke operators self-adjoint, "the Gram determinant of the pairing on $M_{f^\Sigma,\lambda}$ is $(\eta^\Sigma_{f,\lambda})^2$ up to a unit"; Lemma lem:hecke: $d = 2$, $\Delta = (\eta^\emptyset)^2$).
4. **Congruence ideals as commutative algebra** (Lemma lem:crit: $I = \ker \phi_f$, $J = \mathrm{Ann}_{\mathbb{T}}(f^\perp) = \mathrm{Ann}_{\mathbb{T}}(I)$, "dualising the inclusion $\mathbb{T} \subset \mathbb{T}/I \times \mathbb{T}/J$, whose cokernel is $\mathbb{T}/(I+J)$", "embeds in $\mathbb{Q}f/\mathbb{Z}f$, so it is cyclic"; Lemma lem:ddt417: $\wp$, $I = \mathrm{Ann}_{\mathbb{T}}(\wp)$, $\eta = \pi(I)$, $L[\wp]$, $L[I]$, $\wp \cap I = 0$, $L[\wp]^\perp = L[I]$, the equality case $L \simeq \mathbb{T}^d$). The chapter supplies every ingredient of Lemma lem:ddt417 and stops short of the lemma itself, which Chapter 20 proves.

5. **Finite free algebras over $\mathbb{Z}_\ell$, their maximal ideals and idempotents** (Lemma lem:hecke: "$\mathfrak{m}$ the maximal ideal of $\mathbb{T}\otimes\mathbb{Z}_\ell$ containing ℓ and $\ker\phi_f$", "the congruence ideal is computed in $\mathbb{T}_\mathfrak{m}$"; Theorem thm:quant Step 1: "projecting h to the $\mathfrak{m}$-component (an idempotent of $\mathbb{T}_\ell$, so the projection stays in the complement and the congruence survives)"; Remark rem:free: $\mathbb{T}_\mathfrak{m}$, $H^1(X_0(N),\mathbb{Z}_\ell)_\mathfrak{m}$).
6. **Generalised eigenspaces, semisimple and non-semisimple operators, common eigenvectors** (Lemma lem:crit: $\mathbb{T}\otimes\mathbb{Q}=\mathbb{Q}\times\mathbb{T}'$; Lemma lem:old: "$U_p$ is not semisimple on this span when 0 is a repeated eigenvalue, but each listed eigenvalue has an eigenvector", "for every choice of one such eigenvalue μ_p at each p there is a common eigenform"; Theorem thm:twist: generalised eigenforms, $g^\perp$ as the sum of the other generalised eigenspaces; Theorem thm:quant Step 1: "the generalised eigenspace of $\pi\ \ldots$ is the line of g^S").
7. **Gorenstein algebras and freeness** (Remark rem:free: "the localised Hecke algebra is then Gorenstein and the Tate module is free of rank two"; the equality clauses of Lemma lem:ddt417, Lemma lem:hecke and Theorem thm:main).
8. **Modules over DVRs and $\mathbb{Z}_\ell$** (Lemma lem:IV: "being torsion-free of $\mathbb{Z}_3$-rank 2, T is free of rank one over $\mathbb{Z}_3[\zeta_3]$"; Lemma lem:tamfactor: $H^1(I,M)=(\mathbb{Z}_\ell/n')^2\oplus\mathbb{Z}_\ell$ and its torsion; Lemma lem:crit: "the same statements hold with $\mathbb{Z}$ replaced by $\mathbb{Z}_\ell$"; Lemma lem:dfg21: "$H^1(\mathbb{R},T^D)$ is killed by 2, so $H^1_f(\mathbb{R},T^D)=0$ for ℓ odd").
9. **Foundations for Chapters 6, 7, 13, 14, 16, 17, 18, 20**: exact sequences and the snake lemma, Hom and $\otimes$, flatness, Noetherian rings, Nakayama, localisation, the DVR criterion, $\mathbb{Z}_\ell$ as a complete DVR, the topology of finitely generated $\mathbb{Z}_\ell$-modules, $\mathbb{Q}_\ell/\mathbb{Z}_\ell$ injective.

Split with the neighbouring chapters (decided here, consistent with what their outlines assume): * **Chapter 1** proves the structure theorem for finite abelian groups (its 4.1), $\#A[n]=\#A/nA$ (its 4.3), the ℓ-primary decomposition (its 1.8) and constructs $\mathbb{Z}_\ell=\varprojlim\mathbb{Z}/\ell^n$ as a profinite group (its 7.2). Chapter 3 generalises the structure theorem to finitely generated modules over a PID (5.3), makes $\mathbb{Z}_\ell$ a DVR (6.4), and proves Pontryagin duality for finite abelian groups (8.1), which Chapter 1's forward pointer assigns here. * **Chapter 2** proves ideals, quotient rings, the two-ideal exact sequence and CRT (its 1.4), idempotents of a product of rings (its 1.5), prime and maximal ideals (its 2.1, 2.2), $\mathbb{Z}_{(p)}$ (its 2.4), PIDs and UFDs (its 3.2), v_p (its 11.1), and defines $\mathrm{Ann}_R(S)$ (its 1.2). Chapter 3 uses these and does not reprove them; it gives the module version of the two-ideal sequence (2.2), localisation in general (4.5), the existence of maximal ideals in Noetherian rings (4.1, which Chapter 2's item 2.1(v) defers here), and the lifting of idempotents for finite algebras over a complete DVR (10.3, which Chapter 2's item 1.5 defers here). * **Chapter 4** (outlined in parallel) does representations of groups: subrepresentations, irreducibility, Schur, Maschke, semisimple *modules over group algebras*, Brauer-Nesbitt, G-stable lattices in $\mathbb{Q}_\ell$-representations, reduction mod ℓ, homothety of lattices and Ribet's lemma. Chapter 3 does modules over *commutative* rings, lattices as finitely generated free $\mathcal{O}$-modules (saturation, index, elementary divisors, Gram determinants) with no group action, and semisimplicity of a single operator or a commuting family (10.7), not of representations. The Jordan-Holder theorem is needed here first (for length over a local ring, 7.1); Chapter 3 proves it for modules over a ring (the proof does not use commutativity), and Chapter 4 should cite `\Cref{ch:modules}` for it and add only the reading for $k[G]$-modules (composition factors of representations). If Chapter 4's outline also proves it, the writers keep one proof (Part 6, coordination note). * **Chapter 6** uses the DVR criterion (6.1), torsion-free modules over a DVR are free (6.3), localisation (4.5), Noetherian rings (4.1), $\mathbb{Z}_\ell$ (6.4) and "$R\otimes\mathbb{Z}_\ell$ is the product of its localisations at the maximal ideals above ℓ" (10.4). Chapter 3 does not treat $\mathbb{Z}_3[\zeta_3]$, $\mathbb{Z}_\ell[\zeta_3]$ or $\mathcal{O}_\lambda$ beyond naming them as examples. * **Chapter 7** does $\mathbb{Q}_p$ as an analytic completion (absolute values, Cauchy sequences, Hensel's lemma, the structure of $\mathbb{Z}_p^\times$ and $\mathbb{Q}_p^\times$, finite extensions). Chapter 3 does $\mathbb{Z}_\ell$ algebraically: as $\varprojlim\mathbb{Z}/\ell^n$ and as the ℓ-adic completion of $\mathbb{Z}$, a complete DVR; $\mathbb{Q}_\ell=\mathbb{Z}_\ell[1/\ell]$; $\mathbb{Q}_\ell/\mathbb{Z}_\ell$; the ℓ-adic topology of finitely generated $\mathbb{Z}_\ell$-modules. Idempotent lifting (10.3) is proved by a Newton iteration that needs only completeness, not Hensel. * **Chapter 16** proves complexes, long exact sequences, Ext and Tor, group cohomology, and the length identity of Lemma lem:dfg21 in its own item lengths-in-exact-sequences. Chapter 3 proves the snake lemma (Chapter 16 recalls it), additivity of length (7.1), Fitting ideals (7.4 to 7.7), Pontryagin duality of modules (8.1 to 8.3), closedness of submodules and topological Nakayama

(6.6), $\mathbb{Q}_\ell/\mathbb{Z}_\ell$ injective (6.5). * **Chapters 13, 14** prove the perfect pairing $S_2(\Gamma_0(N);\mathbb{Z}) \times \mathbb{T} \to \mathbb{Z}$, the freeness of $\mathbb{T}$ over $\mathbb{Z}$, the Hecke-equivariance of the cup product and of w_N, multiplicity one and the Gorenstein property of $\mathbb{T}_\mathfrak{m}$ (Mazur, Tilouine, Ribet-Stein). Chapter 3 supplies the abstract algebra those statements are about: 9.6 (an algebra paired perfectly with a module), 9.5 (adjoints, twisted pairings), 11.4 (Gorenstein), 11.5 (Mazur's freeness criterion). * **Chapter 18** defines Tam_p (its item 7.1) and proves Lemma lem:fpr (its items 7.4, 7.5) and Lemma lem:tamfactor (its item 8.2); **Chapter 20** proves Lemma lem:ddt417, Lemma lem:hecke and the assembly. Chapter 3 gives the Fitting ideal and every ingredient of Lemma lem:ddt417 (11.1, 11.3, 9.3, 9.4) and checks the statement on toy algebras (11.6), but does not prove the lemma.

5.2 1. Global conventions fixed in this chapter

- **C1 (rings and modules).** As Chapter 2, C1: rings are commutative with 1, homomorphisms preserve 1, modules are unital. R is a ring, M, N are R-modules. An R-algebra is a ring A with a ring homomorphism $R \to A$. The paper's $\mathbb{T}$ is a commutative ring; its modules $S_2(\Gamma_0(N);\mathbb{Z})$, $H^1(X_0(N),\mathbb{Z}_\ell)$, L are modules over a commutative ring.
- **C2 (DVR notation).** $\mathcal{O}$ is a discrete valuation ring with uniformiser ϖ, residue field $k = \mathcal{O}/\varpi\mathcal{O}$, fraction field K, and valuation $v\colon K^\times \to \mathbb{Z}$ normalised by $v(\varpi) = 1$, $v(0) = +\infty$. For an ideal $\mathfrak{a} = a\mathcal{O}$ put $v(\mathfrak{a}) = v(a)$. For $\mathcal{O} = \mathbb{Z}_\ell$: $\varpi = \ell$, $k = \mathbb{F}_\ell$, $v = v_\ell$ (Chapter 2, C3). The paper writes $\mathcal{O}$ and λ, v_λ; for rational newforms $\mathcal{O} = \mathbb{Z}_\ell$ and $\lambda = \ell$.
- **C3 (torsion and duals; three different duals).** For an ideal $\mathfrak{a} \subset R$, $M[\mathfrak{a}] = \{m \in M : \mathfrak{a}m = 0\}$ (the paper's $L[\wp]$, $L[I]$); $M[x] = M[xR]$; $M[\ell^k]$, $M[\ell^\infty] = \bigcup_k M[\ell^k]$; M_{tors} the torsion submodule over a domain. **Three duals, never confused:** $M^\vee = \mathrm{Hom}_R(M,R)$ (the linear dual; the paper's $\mathrm{Hom}(L[\wp],\mathcal{O})$); $M^* = \mathrm{Hom}(M,\mathbb{Q}/\mathbb{Z})$, or $\mathrm{Hom}_{\mathbb{Z}_\ell}(M,\mathbb{Q}_\ell/\mathbb{Z}_\ell)$ for ℓ-primary or $\mathbb{Z}_\ell$-modules (the Pontryagin dual; the paper's * in Lemma lem:dfg21); $T^D = \mathrm{Hom}(T,\mathbb{Z}_\ell(1))$ (the Cartier dual, Chapters 17 and 18; not used here).
- **C4 (lattices).** Over a PID R with fraction field K: a *lattice* is a finitely generated free R-module; a lattice in a finite-dimensional K-vector space V is a finitely generated R-submodule spanning V (free of rank $\dim V$). $L' \subset L$ is *saturated* if L/L' is torsion-free (5.5). G-stable lattices and homothety are Chapter 4.
- **C5 (pairings).** A pairing is an R-bilinear map $\langle\ ,\ \rangle\colon L \times M \to R$. Gram matrix $G = (\langle e_i, f_j\rangle)$ for bases (e_i), (f_j). For $S \subset M$: $S^\perp = \{x \in L : \langle x, S\rangle = 0\}$; for $S \subset L$: ${}^\perp S$ or, when $L = M$ and the pairing is symmetric or antisymmetric, $S^\perp$. A pairing $L \times L \to R$ is *alternating* if $\langle x, x\rangle = 0$ for all x. "Self-adjoint" for an endomorphism t: $\langle tx, y\rangle = \langle x, ty\rangle$.
- **C6 (length and orders).** $\mathrm{length}_R(M)$ is the length (7.1); "length" without subscript means $\mathrm{length}_{\mathbb{Z}_\ell}$, as in PLAN.md Appendix B. "$\#\mathcal{O}/\Delta$" (the paper's notation) means $\#k^{v(\Delta)}$ and needs k finite; for a general DVR the book writes $\mathrm{length}_\mathcal{O}(\mathcal{O}/\Delta\mathcal{O}) = v(\Delta)$.
- **C7 (localisation versus factor; the paper's $\mathbb{T}_\mathfrak{m}$).** For a finite free $\mathbb{Z}$-algebra $\mathbb{T}$ and a maximal ideal $\mathfrak{m} \ni \ell$, $\mathbb{T}_\mathfrak{m}$ **always** means $(\mathbb{T} \otimes \mathbb{Z}_\ell)_\mathfrak{m}$, the direct factor of $\mathbb{T} \otimes \mathbb{Z}_\ell$ cut out by an idempotent (10.3, 10.4); it is also the $\mathfrak{m}$-adic completion of $\mathbb{T}$. The algebraic localisation of $\mathbb{T}$ itself is written $\mathbb{T}_{(\mathfrak{m})}$ and is not complete. Likewise $L_\mathfrak{m} = e_\mathfrak{m}(L \otimes \mathbb{Z}_\ell)$ for a $\mathbb{T}$-module L. This is the paper's usage (Lemma lem:hecke, Remark rem:free, Theorem thm:quant).
- **C8 (matrices).** Relations are rows: a presentation $R^m \xrightarrow{A} R^n \to M \to 0$ means $M = R^n/(\text{row space of } A)$ with $A \in \mathrm{M}_{m\times n}(R)$; Fitting ideals use the $n \times n$ minors of A (7.4).
- **C9 (Frobenius).** This chapter contains no Frobenius; its examples quoting Lemma lem:tamfactor use the arithmetic convention of Chapter 16 and say so. (important_notes.md point 11 is not this chapter's business.)
- **C10 (typography and labels).** PLAN.md Section 7.2: no dash ligatures, ASCII source, labels `...:modules:...`; the paper's macros `\Fitt`, `\len`, `\Tam`, `\ad`, and the preamble's `\Hom`, `\End`, `\Ann`, `\coker`, `\rank`, `\tors`, `\Zl`, `\Ql`, `\Qlmod`, `\Ext` (only in remarks). The paper's $\wp$ is `\wp`.

5.3 2. Line-up of the core results, in teaching order

5.3.1 Section 1. Modules, submodules, quotients, homomorphisms

1.1 def:modules:module (definition and proposition; P). An R-module; submodules; the submodule $\sum Rm_i$ generated by a family; finitely generated (f.g.) and cyclic modules; quotient modules M/N; homomorphisms, kernel, image, cokernel $\operatorname{coker} f = N/f(M)$; $\operatorname{Hom}_R(M,N)$ is an R-module; $\operatorname{End}_R(M)$ is a (non-commutative in general) ring. A cyclic module Rm is isomorphic to $R/\operatorname{Ann}_R(m)$. Examples: $\mathbb{Z}$-modules are abelian groups; K-modules are vector spaces; a $K[x]$-module is a K-vector space with one endomorphism (the dictionary used in 10.5 to 10.8); an $R/\mathfrak{a}$-module is an R-module killed by $\mathfrak{a}$; an ideal is a submodule of R. Reference: Dummit-Foote (DF) 10.1 to 10.3; Atiyah-Macdonald (AM) Chapter 2. Paper: background.

1.2 def:modules:algebra-modules (definition and examples; P). For a ring homomorphism $\phi\colon R \to S$ every S-module is an R-module (restriction of scalars); an R-algebra A acting on an R-module M is a homomorphism of R-algebras $A \to \operatorname{End}_R(M)$; the action is *faithful* if this map is injective. A module over $R_1 \times R_2$ is the same as a pair (M_1, M_2), $M = e_1M \oplus e_2M$ for the idempotents $e_1 = (1,0)$, $e_2 = (0,1)$ (Chapter 2, item 1.5). Running examples, stated with forward pointers: (a) $\mathbb{T} \subset \operatorname{End}_{\mathbb{Z}} S_2(\Gamma_0(N);\mathbb{Z})$ acting faithfully (Chapters 12, 13); (b) $\mathbb{T}$ acting on $H^1(X_0(N),\mathbb{Z}_\ell)$ (Chapter 14); (c) $T = T_3E$ with σ of order 3, $\det\sigma = 1$: a module over $\mathbb{Z}_3[X]/(X^2+X+1) = \mathbb{Z}_3[\zeta_3]$ (Lemma lem:IV; Chapter 6, item 4.8); (d) the toy algebras of 11.2. Reference: DF 10.1; AM Chapter 2. Paper: Lemma lem:IV, Lemma lem:crit, Lemma lem:ddt417 ("a commutative $\mathcal{O}$-algebra ... acting on an $\mathcal{O}$-module L").

1.3 thm:modules:isomorphism (theorem; P). (i) $M/\ker f \simeq f(M)$. (ii) For submodules $A, B \subset M$: $(A+B)/B \simeq A/(A\cap B)$. (iii) For $B \subset A \subset M$: $(M/B)/(A/B) \simeq M/A$, and submodules of M/B correspond to submodules of M containing B. (iv) If $f\colon M \to N$ is surjective with kernel K_0 and $A \subset M$, then f induces $M/(K_0+A) \simeq N/f(A)$ (the module form of Chapter 2, item 1.3(iii)). Reference: DF 10.2. Paper: Lemma lem:crit ("$\mathbb{T}/(I+J) \simeq \mathbb{Z}/\phi_f(J)$"), the congruence module $\mathbb{T}/(\wp+I) \simeq \mathcal{O}/\eta$ of 11.1.

5.3.2 Section 2. Exact sequences, the snake lemma, direct sums and products, free and projective modules

2.1 def:modules:exact (definition and proposition; P). A sequence $\cdots \to M_{i-1} \xrightarrow{f_{i-1}} M_i \xrightarrow{f_i} M_{i+1} \to \cdots$ is exact at M_i if $\ker f_i = \operatorname{im} f_{i-1}$; short exact sequences $0 \to M' \to M \to M'' \to 0$. *Splitting lemma:* for a short exact sequence the following are equivalent: (a) there is $s\colon M'' \to M$ with $ps = \mathrm{id}$; (b) there is $r\colon M \to M'$ with $ri = \mathrm{id}$; (c) there is an isomorphism $M \simeq M' \oplus M''$ compatible with i and p. Examples: $0 \to \mathbb{Z} \xrightarrow{n} \mathbb{Z} \to \mathbb{Z}/n \to 0$ does not split for $n \geq 2$; $0 \to \mathbb{Z}_\ell \xrightarrow{\ell^k} \mathbb{Z}_\ell \to \mathbb{Z}/\ell^k \to 0$; $0 \to T \to V \to W \to 0$ for $V = T \otimes \mathbb{Q}_\ell$, $W = T \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$ does not split when $T \neq 0$ (V has no nonzero torsion). Reference: DF 10.5; AM Chapter 2. Paper: Lemma lem:dfg21 (a four-term exact sequence), background.

2.2 prop:modules:two-submodules (proposition; P). For submodules A, B of M the sequence

$$0 \to M/(A\cap B) \xrightarrow{x\mapsto(x,x)} M/A \oplus M/B \xrightarrow{(y,z)\mapsto y-z} M/(A+B) \to 0$$

is exact. If $A \cap B = 0$ then $M \hookrightarrow M/A \oplus M/B$ with cokernel $M/(A+B)$; the sum $A+B$ is direct iff $A \cap B = 0$. (Module form of Chapter 2, item 1.4.) Reference: elementary. Paper: Lemma lem:crit ("the inclusion $\mathbb{T} \subset \mathbb{T}/I \times \mathbb{T}/J$, whose cokernel is $\mathbb{T}/(I+J)$", with $I \cap J = 0$), Lemma lem:ddt417 ("the sum $L[\wp] + L[I]$ is direct").

2.3 lem:modules:snake (lemma, the snake lemma; P). Given a commutative diagram with exact rows $M' \xrightarrow{i} M \xrightarrow{p} M'' \to 0$ and $0 \to N' \xrightarrow{j} N \xrightarrow{q} N''$ and vertical maps f', f, f'', there is an R-linear $\delta\colon \ker f'' \to \operatorname{coker} f'$, defined by: lift $x'' \in \ker f''$ to $x \in M$; then $q(f(x)) = 0$, so $f(x) = j(y')$ for a unique $y' \in N'$; put $\delta(x'') = [y']$. It is well defined, and $\ker f' \to \ker f \to \ker f'' \xrightarrow{\delta} \operatorname{coker} f' \to \operatorname{coker} f \to \operatorname{coker} f''$ is exact; it begins with $0 \to$ if i is injective and ends with $\to 0$ if q is surjective. Corollaries (P): (a) for a short exact sequence $0 \to A \to B \to C \to 0$ and $n \in R$, the multiplication-by-n maps give an exact sequence $0 \to A[n] \to B[n] \to C[n] \to A/nA \to B/nB \to C/nC \to 0$; (b) for an endomorphism ϕ of a

finite abelian group, $\#\ker\phi = \#\operatorname{coker}\phi$ (compare Chapter 1, item 4.3). Reference: AM Chapter 2 (check the proposition number); Weibel, Lemma 1.3.2 (as Chapter 16 cites it). Paper: background (Chapter 16 recalls it and builds every connecting map on it; Lemma lem:h0 and the sequence $0 \to T \to V \to W \to 0$ in Chapters 16, 17).

2.4 `lem:modules:five` (lemma, the five lemma; P). In a commutative ladder of two exact rows of five modules, if the four outer vertical maps are isomorphisms, so is the middle one (the two halves: injectivity from the second and fourth being injective and the first surjective; surjectivity dually). Reference: Weibel 1.3; DF 10.5 (exercise). Paper: background (comparisons of long exact sequences in Chapters 16, 17).

2.5 `def:modules:sums-products` (definition and proposition; P). External and internal direct sums $\bigoplus_i M_i$ and direct products $\prod_i M_i$, with their universal properties; finite sums equal finite products; $\operatorname{Hom}_R(\bigoplus_i M_i, N) = \prod_i \operatorname{Hom}_R(M_i, N)$ and $\operatorname{Hom}_R(N, \prod_i M_i) = \prod_i \operatorname{Hom}_R(N, M_i)$. $M = A \oplus B$ internally iff $A + B = M$ and $A \cap B = 0$ iff there is an idempotent $e \in \operatorname{End}_R(M)$ with $eM = A$, $(1-e)M = B$. Reference: DF 10.3; AM Chapter 2. Paper: Lemma lem:ddt417 ($L[\wp] \oplus L[I]$), Lemma lem:hecke ($H^1 \otimes \mathbb{Z}_\ell = \bigoplus_{\mathfrak{m}'} H^1_{\mathfrak{m}'}$).

2.6 `def:modules:free` (definition and proposition; P). A basis; free modules $R^{(I)}$ and their universal property; every module is a quotient of a free module, so has a presentation $F_1 \to F_0 \to M \to 0$ with F_0, F_1 free; M is *finitely presented* if F_0, F_1 can be taken of finite rank. Rank is well defined: if $R \neq 0$ and $R^m \simeq R^n$ then $m = n$ (tensor with $R/\mathfrak{m}$ for a maximal ideal $\mathfrak{m}$, which exists by 4.1 for Noetherian R and by Zorn's lemma in general, Chapter 2, item 2.1(v)). A matrix $A \in \mathrm{M}_n(R)$ is invertible iff $\det A \in R^\times$ (adjugate). Reference: DF 10.3; AM Chapter 2. Paper: background (presentations for Fitting ideals, 7.4).

2.7 `def:modules:projective` (definition and proposition; P). For an R-module P the following are equivalent: (a) every surjection $M \to P$ splits; (b) P is a direct summand of a free module; (c) $\operatorname{Hom}_R(P, -)$ is exact; (d) every map from P lifts along every surjection. Such P is *projective*. Examples: free modules; eR for an idempotent e (so each factor R_i of $R = \prod R_i$ is a projective R-module); $\mathbb{Z}/2$ over $\mathbb{Z}/6 \simeq \mathbb{Z}/2 \times \mathbb{Z}/3$ is projective and not free. Over a local ring a finitely generated projective module is free (4.3(v)); over a PID every submodule of a free module is free (5.1). Consequence used later: if $\mathbb{T}$ is finite free over $\mathbb{Z}_\ell$ and $e \in \mathbb{T}$ is idempotent, $e\mathbb{T}$ is projective, hence free, over $\mathbb{Z}_\ell$. Reference: DF 10.5; Weibel 2.2. Paper: Lemma lem:hecke ("Apply Lemma lem:ddt417 to $\mathcal{O} = \mathbb{Z}_\ell$, $\mathbb{T}_{\mathfrak{m}}$": $\mathbb{T}_{\mathfrak{m}}$ is a finite free $\mathbb{Z}_\ell$-algebra), Remark rem:free.

5.3.3 Section 3. Hom and tensor product; flatness

3.1 `prop:modules:hom` (proposition; P). (i) $\operatorname{Hom}_R(R, M) = M$; $\operatorname{Hom}_R(R^n, M) = M^n$; $\operatorname{Hom}_R(R/\mathfrak{a}, M) = M[\mathfrak{a}]$. (ii) $\operatorname{Hom}_R(-, N)$ and $\operatorname{Hom}_R(N, -)$ are left exact: $0 \to M' \to M \to M''$ exact gives $0 \to \operatorname{Hom}(N, M') \to \operatorname{Hom}(N, M) \to \operatorname{Hom}(N, M'')$ exact, and $M' \to M \to M'' \to 0$ exact gives $0 \to \operatorname{Hom}(M'', N) \to \operatorname{Hom}(M, N) \to \operatorname{Hom}(M', N)$ exact. (iii) They are not right exact: $\operatorname{Hom}_{\mathbb{Z}}(\mathbb{Z}/2, -)$ applied to $0 \to \mathbb{Z} \xrightarrow{2} \mathbb{Z} \to \mathbb{Z}/2 \to 0$, and $\operatorname{Hom}_{\mathbb{Z}}(-, \mathbb{Z})$ applied to the same sequence. (iv) The linear dual $M^\vee = \operatorname{Hom}_R(M, R)$; for M free with basis $(e_i)_{i \le n}$ the dual basis $(e_i^\vee)$; $M \to M^{\vee\vee}$ is an isomorphism for f.g. free M; over a domain $M^\vee$ is torsion-free for every M. Reference: DF 10.5; AM Chapter 2. Paper: Lemma lem:ddt417 ($\operatorname{Hom}(L[\wp], \mathcal{O})$), Lemma lem:crit (dualising), Lemma lem:dfg21 (duals; Pontryagin version in Section 8).

3.2 `thm:modules:tensor` (theorem; P). (i) The tensor product $M \otimes_R N$ exists and is unique up to unique isomorphism, with the universal property for R-bilinear maps; construction as a quotient of the free module on $M \times N$. (ii) $R \otimes M = M$; $(\bigoplus M_i) \otimes N = \bigoplus (M_i \otimes N)$; $R^m \otimes R^n = R^{mn}$; $(R/\mathfrak{a}) \otimes M = M/\mathfrak{a}M$; commutativity and associativity. (iii) Adjunction: $\operatorname{Hom}_R(M \otimes N, P) \simeq \operatorname{Hom}_R(M, \operatorname{Hom}_R(N, P))$. (iv) Base change: for an R-algebra S, $S \otimes_R M$ is an S-module, $S \otimes_R R^n = S^n$, and $S \otimes_R (R/\mathfrak{a}) = S/\mathfrak{a}S$; the tensor product of two R-algebras is an R-algebra. (v) Examples (checked): $\mathbb{Z}/m \otimes \mathbb{Z}/n \simeq \mathbb{Z}/\gcd(m, n)$; $\mathbb{Q} \otimes_{\mathbb{Z}} \mathbb{Z}/n = 0$; $\mathbb{Z}_\ell \otimes_{\mathbb{Z}} \mathbb{Z}/n \simeq \mathbb{Z}/\ell^{v_\ell(n)}$; $\mathbb{Q}_\ell/\mathbb{Z}_\ell \otimes_{\mathbb{Z}} A = 0$ for every torsion abelian group A (a divisible group tensored with a torsion group); $\mathbb{Q}_\ell/\mathbb{Z}_\ell \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell = 0$. Reference: DF 10.4; AM Chapter 2. Paper: $T \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$ (Theorem thm:main, Lemma lem:h0), $\mathbb{T} \otimes \mathbb{Z}_\ell$, $\mathbb{T} \otimes \mathbb{Q}$ (Lemmas lem:crit, lem:hecke, Theorem thm:quant).

3.3 `prop:modules:right-exact` (proposition; P). $- \otimes_R N$ is right exact: $M' \to M \to M'' \to 0$ exact gives $M' \otimes N \to M \otimes N \to M'' \otimes N \to 0$ exact (via 3.1(ii) and the adjunction 3.2(iii)). It is not left

exact: $0 \to \mathbb{Z} \xrightarrow{2} \mathbb{Z}$ tensored with $\mathbb{Z}/2$. N is *flat* if $- \otimes_R N$ is exact. Free and projective modules are flat. Reference: DF 10.5; AM Chapter 2. Paper: background.

3.4 `prop:modules:flat` (proposition; P, except (ii) in full generality, S). (i) For a multiplicative set S, $S^{-1}R$ is a flat R-module (proof after 4.5: $S^{-1}R \otimes M \simeq S^{-1}M$ and localisation is exact). (ii) Over a PID a module is flat iff it is torsion-free (P for finitely generated modules, by 5.3; the general case S, since tensor products commute with directed unions and every f.g. submodule of a torsion-free module is free). Hence $\mathbb{Q}$, $\mathbb{Z}_{(\ell)}$, $\mathbb{Z}_\ell$, $\mathbb{Q}_\ell$ are flat over $\mathbb{Z}$, and $\mathbb{Q}_\ell$ is flat over $\mathbb{Z}_\ell$. (iii) Consequences, for $R \to S$ flat (P): kernels, images, cokernels, finite intersections of submodules commute with $S \otimes_R -$; for an ideal $\mathfrak{a}$ generated by $a_1, \dots, a_k$, $(M[\mathfrak{a}]) \otimes_R S = (M \otimes_R S)[\mathfrak{a}S]$ (kernel of $M \to M^k$); for a f.g. module M with generators $m_1, \dots, m_n$, $\mathrm{Ann}_S(M \otimes_R S) = \mathrm{Ann}_R(M)S$ (kernel of $R \to M^n$); for $L' \subset L$ free over a PID and S a flat PID over it, the saturation commutes with base change. Reference: AM Chapters 2, 3; DF 10.5. Paper: Lemma lem:crit ("the same statements hold with $\mathbb{Z}$ replaced by $\mathbb{Z}_\ell$"; "the pairing being perfect after tensoring with $\mathbb{Z}_\ell$"), Lemma lem:hecke ("$r_E\mathbb{Z}_\ell$ is the congruence ideal $\phi_f(\mathrm{Ann}_{\mathbb{T}} \ker \phi_f)\mathbb{Z}_\ell$": Ann commutes with $\otimes\mathbb{Z}_\ell$), Theorem thm:quant Step 1 ($\mathbb{Z}_\ell$-version of Lemma lem:crit).

5.3.4 Section 4. Noetherian rings, Nakayama's lemma, localisation, local rings

4.1 `prop:modules:noetherian` (definition and proposition; P, except (iv), R). (i) For an R-module M the following are equivalent: every ascending chain of submodules stabilises; every nonempty set of submodules has a maximal element; every submodule is f.g. Such M is *Noetherian*; R is Noetherian if it is as an R-module. (ii) Submodules, quotients and extensions of Noetherian modules are Noetherian; over a Noetherian ring every f.g. module is Noetherian, so submodules of f.g. modules are f.g. and f.g. modules are finitely presented. (iii) A nonzero Noetherian ring has a maximal ideal, and every proper ideal lies in one (maximal element of the set of proper ideals containing it); this is the proof promised in Chapter 2, item 2.1(v). (iv) Hilbert's basis theorem: R Noetherian implies $R[x]$ Noetherian (stated). (v) Examples: fields, PIDs (Chapter 2, item 3.2), $\mathbb{Z}_\ell$, every ring that is f.g. as a module over a Noetherian ring (the Hecke algebra $\mathbb{T}$ over $\mathbb{Z}$, $\mathbb{T} \otimes \mathbb{Z}_\ell$ over $\mathbb{Z}_\ell$). Reference: AM Chapters 6, 7 (Hilbert: Theorem 7.5, check); DF 12.1, 15.1. Paper: background ($\mathbb{T}$, $\mathbb{T}_\mathfrak{m}$, $H^1(X_0(N), \mathbb{Z}_\ell)$ are f.g. over Noetherian rings).

4.2 `lem:modules:determinant-trick` (lemma; P). Let M be generated by n elements, $\mathfrak{a} \subset R$ an ideal and $\phi \in \mathrm{End}_R(M)$ with $\phi(M) \subset \mathfrak{a}M$. Then $\phi^n + a_1\phi^{n-1} + \cdots + a_n = 0$ with $a_i \in \mathfrak{a}^i$ (Cayley-Hamilton applied to a matrix of ϕ with entries in $\mathfrak{a}$; adjugate argument over the commutative ring $R[\phi]$). Corollaries: (a) if M is f.g. and $\mathfrak{a}M = M$, there is $x \equiv 1 \pmod{\mathfrak{a}}$ with $xM = 0$; (b) a surjective endomorphism of a f.g. module is an isomorphism. Reference: AM Proposition 2.4 (check the number) and Chapter 2. Paper: background (proof of 4.3; 11.5).

4.3 `thm:modules:nakayama` (theorem, Nakayama's lemma; P). Let $(R, \mathfrak{m}, k)$ be a local ring (4.4) and M a f.g. R-module. (i) If $\mathfrak{m}M = M$ then $M = 0$ (more generally for any ideal inside the Jacobson radical). (ii) If $N \subset M$ and $N + \mathfrak{m}M = M$, then $N = M$. (iii) $x_1, \dots, x_r$ generate M iff their images span the k-vector space $M/\mathfrak{m}M$; the minimal number of generators of M is $\dim_k M/\mathfrak{m}M$. (iv) A homomorphism $f\colon M \to N$ of f.g. modules with N f.g. is surjective iff $f \otimes k$ is; a matrix in $\mathrm{M}_n(R)$ is invertible iff its reduction mod $\mathfrak{m}$ is; an endomorphism of R^n is an automorphism iff it is so mod $\mathfrak{m}$. (v) A f.g. projective R-module is free. Example: over $\mathbb{Z}_\ell$, a f.g. module M with $M/\ell M \simeq \mathbb{F}_\ell^2$ is generated by two elements; an endomorphism of $(\mathbb{Z}/\ell^k)^n$ that is invertible mod ℓ is invertible. Reference: AM Proposition 2.6, Corollaries 2.7, 2.8 (check); Eisenbud Chapter 4. Paper: Remark rem:free ("$J_0(N)[\mathfrak{m}]$ is two-dimensional" gives two generators, through 11.5; Chapter 14), Lemma lem:h0 in the corrected form of important_notes.md point 5 (a lift of an element of exact order ℓ^k is an isomorphism $\rho \bmod \ell^k \to (\rho \otimes \chi_\ell) \bmod \ell^k$ because it is one mod ℓ), Chapter 16 (topological Nakayama, 6.6).

4.4 `def:modules:local-ring` (definition and proposition; P). A ring is *local* if it has exactly one maximal ideal. R is local iff its non-units form an ideal; then $R^\times = R \setminus \mathfrak{m}$. The Jacobson radical (intersection of the maximal ideals); $1 + x$ is a unit for x in it. Examples: fields; $\mathbb{Z}_{(p)}$ (Chapter 2, item 2.4); DVRs (6.1); $\mathbb{Z}/p^n$; $k[\varepsilon]/(\varepsilon^2)$; $\mathbb{Z}_\ell[x]/(x^2)$; each factor $\mathbb{T}_\mathfrak{m}$ of a finite free $\mathbb{Z}_\ell$-algebra (10.3). Non-examples: $\mathbb{Z}$; $\mathbb{Z}_\ell \times \mathbb{Z}_\ell$; $\mathbb{Z}_{(11)}[x]/(x^2 + x - 1)$ (two maximal ideals, 10.3). Reference: AM Chapter 1. Paper: Lemma lem:hecke, Remark rem:free, Theorem thm:quant ($\mathbb{T}_\mathfrak{m}$ local).

4.5 thm:modules:localisation (definition and theorem; P). For a multiplicative set $S \subset R$: the ring $S^{-1}R$ and the module $S^{-1}M$ (fractions m/s, with $m/s = m'/s'$ iff $u(s'm - sm') = 0$ for some $u \in S$). (i) Universal property: $R \to S^{-1}R$ is initial among homomorphisms inverting S. (ii) $S^{-1}M \simeq S^{-1}R \otimes_R M$, and $M \mapsto S^{-1}M$ is exact (hence 3.4(i)). (iii) Ideals: $\mathfrak{b} \mapsto \mathfrak{b} \cap R$ and $\mathfrak{a} \mapsto S^{-1}\mathfrak{a}$ give a bijection between prime ideals of $S^{-1}R$ and prime ideals of R not meeting S. (iv) For a prime $\mathfrak{p}$, $R_{\mathfrak{p}} = (R \setminus \mathfrak{p})^{-1}R$ is local with maximal ideal $\mathfrak{p}R_{\mathfrak{p}}$; $M_{\mathfrak{p}}$. (v) Local nature: $M = 0$ iff $M_{\mathfrak{m}} = 0$ for every maximal $\mathfrak{m}$; a map is injective (surjective) iff it is so after localising at every maximal ideal. (vi) Localisation commutes with finite direct sums, quotients, Ann of f.g. modules (3.4(iii)) and Hom out of finitely presented modules (stated). (vii) Examples: $\mathbb{Z}_{(p)}$; $\mathbb{Q} = \mathbb{Z}_{(0)}$; $\mathbb{Z}[1/\ell]$; $(\mathbb{Z}/n)_{(p)} \simeq \mathbb{Z}/p^{v_p(n)}$; for $R = R_1 \times R_2$ and $\mathfrak{p} = \mathfrak{p}_1 \times R_2$, $R_{\mathfrak{p}} \simeq (R_1)_{\mathfrak{p}_1}$ (localisation at a maximal ideal of a product picks out a factor; this is why, for a finite algebra over a complete DVR, localisation at $\mathfrak{m}$ is a direct factor, 10.3). Reference: AM Chapter 3; DF 15.4. Paper: Lemma lem:hecke, Remark rem:free, Theorem thm:quant ($\mathbb{T}_{\mathfrak{m}}$, $H^1(X_0(N), \mathbb{Z}_\ell)_{\mathfrak{m}}$; with convention C7).

5.3.5 Section 5. Finitely generated modules over a PID; torsion; lattices and saturation

5.1 thm:modules:submodule-free (theorem; P). Let R be a PID. A submodule of a free R-module of rank n is free of rank at most n (induction on n, projecting to the last coordinate, whose image is a principal ideal). Corollaries: over a PID, f.g. modules are finitely presented; f.g. projective modules are free. Reference: DF 12.1. Paper: background (5.3, 5.5).

5.2 thm:modules:smith (theorem, Smith normal form; P for Euclidean domains, which covers $\mathbb{Z}$, $F[x]$ and every DVR; S for a general PID). For $A \in \mathrm{M}_{m\times n}(R)$ there are $U \in \mathrm{GL}_m(R)$, $V \in \mathrm{GL}_n(R)$ with $UAV = \mathrm{diag}(d_1, \dots, d_s, 0, \dots)$, $d_i \neq 0$, $d_1 \mid d_2 \mid \cdots \mid d_s$. For each k, $d_1 \cdots d_k$ generates the ideal of $k \times k$ minors of A (the k-th determinantal divisor, unchanged by invertible row and column operations), so the d_i are unique up to units. Examples (checked): $\left(\begin{smallmatrix} 2 & 4 \\ 6 & 8 \end{smallmatrix}\right) \sim \mathrm{diag}(2, 4)$; the relation matrix with rows $(2, 4, 4)$, $(-6, 6, 12)$, $(10, -4, -16)$ (DF 12.1) $\sim \mathrm{diag}(2, 6, 12)$; $\left(\begin{smallmatrix} 3 & 6 \\ 9 & 3 \end{smallmatrix}\right) \sim \mathrm{diag}(3, 15)$ over $\mathbb{Z}$, $\sim \mathrm{diag}(3, 3)$ over $\mathbb{Z}_3$. Reference: DF 12.1 (exercises), Lang III.7; Jacobson, Basic Algebra I, 3.7 (not in refs.bib; add to `bib/ch03.bib` if cited). Paper: background (5.3, 5.6, 7.6).

5.3 thm:modules:structure-pid (theorem, structure of f.g. modules over a PID; P). Let R be a PID and M a f.g. R-module. (i) (Invariant factors.) $M \simeq R^r \oplus R/(d_1) \oplus \cdots \oplus R/(d_s)$ with d_i nonzero non-units and $d_1 \mid \cdots \mid d_s$; r and the ideals (d_i) are uniquely determined. (ii) (Elementary divisors.) $M \simeq R^r \oplus \bigoplus_j R/(p_j^{a_j})$ with p_j primes, $a_j \geq 1$; the multiset of the ideals $(p_j^{a_j})$ is uniquely determined. (iii) Proof: existence from a presentation and 5.2; uniqueness: $r = \dim_K(M \otimes K)$, and for each prime p the numbers $\dim_{R/p}(p^k M_{\mathrm{tors}}/p^{k+1}M_{\mathrm{tors}})$ determine the p-primary part. (iv) Special cases: $R = \mathbb{Z}$ gives f.g. abelian groups and recovers Chapter 1, item 4.1; $R = \mathcal{O}$ a DVR gives $M \simeq \mathcal{O}^r \oplus \bigoplus_i \mathcal{O}/\varpi^{a_i}$ (6.3); $R = K[x]$ gives the rational canonical form and, over an algebraically closed field, the Jordan form (used in 10.7). (v) Example: the module of the second example in 5.2 is $\mathbb{Z}/2 \oplus \mathbb{Z}/6 \oplus \mathbb{Z}/12$, with elementary divisors $2, 2, 3, 4, 3$, that is $\mathbb{Z}/2 \oplus \mathbb{Z}/2 \oplus \mathbb{Z}/4 \oplus \mathbb{Z}/3 \oplus \mathbb{Z}/3$. Reference: DF 12.1, Theorems 5 and 6 (check); Lang III.7. Paper: Lemma lem:tamfactor ($H^1(I, M) = (\mathbb{Z}_\ell/n')\bar{e} \oplus (\mathbb{Z}_\ell/n')\bar{h} \oplus \mathbb{Z}_\ell \bar{f}$, torsion $(\mathbb{Z}/\ell^{v_\ell(n)})^2$), Lemma lem:IV.

5.4 def:modules:torsion (definition and proposition; P). Over a domain R with fraction field K: torsion elements, the torsion submodule M_{tors}, torsion-free modules, $\mathrm{rank}\, M = \dim_K(M \otimes_R K)$. (i) M/M_{tors} is torsion-free; M_{tors} is the kernel of $M \to M \otimes K$. (ii) Over a PID, a f.g. torsion-free module is free, and $M \simeq M_{\mathrm{tors}} \oplus M/M_{\mathrm{tors}}$ (the splitting is not canonical). (iii) If $R \subset R'$ is an extension of domains with R' a PID finite free of rank e over R (example $\mathbb{Z}_3 \subset \mathbb{Z}_3[\zeta_3]$, $e = 2$), a f.g. R'-module that is torsion-free over R of R-rank n is free over R' of rank n/e. Reference: DF 12.1. Paper: Lemma lem:IV ("Being torsion-free of $\mathbb{Z}_3$-rank 2, T is free of rank one over $\mathbb{Z}_3[\zeta_3]$"; Chapter 6, item 4.8, cites (iii)), Lemma lem:tamfactor ("its torsion is $(\mathbb{Z}/\ell^{v_\ell(n)})^2$"), the paragraph after Lemma lem:dfg21 ("the torsion of $H^1(\mathbb{Q}_p, T^D)$, a finite group").

5.5 def:modules:saturated (definition and proposition; P). Let R be a PID with fraction field K, M a lattice (C4) and $L \subset M$ a submodule. L is *saturated* if M/L is torsion-free. Equivalent: (a) L saturated; (b) $L = M \cap (KL)$ inside $M \otimes K$; (c) L is a direct summand of M; (d) (for $R = \mathcal{O}$ a DVR) $L \cap \varpi M = \varpi L$, i.e. $L/\varpi L \to M/\varpi M$ is injective. The *saturation* $\mathrm{sat}(L) = M \cap KL$ is the smallest

saturated submodule containing L, of the same rank, and $\mathrm{sat}(L)/L = (M/L)_{\mathrm{tors}}$. Saturated submodules: kernels of maps $M \to N$ with N torsion-free; $M \cap W$ for a K-subspace $W \subset M \otimes K$; $M[\mathfrak{a}]$ for any action of a ring $\mathbb{T}$ on M and any f.g. ideal $\mathfrak{a} \subset \mathbb{T}$ ($M/M[\mathfrak{a}] \hookrightarrow M^k$); orthogonal complements for a pairing (9.2). Non-examples: $3\mathbb{Z} \subset \mathbb{Z}$; $\mathbb{Z}(3,3) \subset \mathbb{Z}^2$ (while $\mathbb{Z}(1,3)$ is saturated). Reference: elementary (Bourbaki, Algebra VII, for "pure submodules"; cite DF 12.1 exercises). Paper: Lemma lem:ddt417 ("because $L[\wp]$ is saturated"), Lemma lem:crit ($L = f^\perp \cap S_2(\Gamma_0(N);\mathbb{Z})$ and $\mathbb{Z}f = S_2(\Gamma_0(N);\mathbb{Z}) \cap \mathbb{Q}f$ are saturated), Section 4.1 ($M_{f^\Sigma,\lambda} = H^1[\ker\phi]$ is saturated), PLAN.md scope ("the lattice/saturation notion").

5.6 `thm:modules:stacked-bases` (theorem, stacked bases; P). Let R be a PID, M free of rank n and $L \subset M$ a submodule of rank d. There are a basis $e_1,\dots,e_n$ of M and nonzero $d_1 \mid \cdots \mid d_d$ such that $d_1e_1,\dots,d_de_d$ is a basis of L. The ideals (d_i) (the *elementary divisors of L in M*) are unique. Consequences: $M/L \simeq R^{n-d} \oplus \bigoplus_i R/(d_i)$; L is saturated iff every d_i is a unit; if $d = n$ and B is the matrix of a basis of L in a basis of M, then M/L is torsion of length $v(\det B)$ over a DVR and of order $|\det B|$ over $\mathbb{Z}$ (the *index* $[M : L]$). Reference: DF 12.1 (Theorem 4, check); Lang III.7. Paper: Lemma lem:crit ("$\mathbb{Z}f \oplus L$ has index r'"), 9.3, 9.4, 11.6.

5.7 `prop:modules:dual-inclusion` (proposition; P). Let R be a PID, $L \subset M$ lattices of the same rank n with M/L torsion (finite when $R = \mathbb{Z}$ or $\mathbb{Z}_\ell$). Then restriction $M^\vee \to L^\vee$ is injective and $L^\vee/M^\vee \simeq \bigoplus_i R/(d_i)$, with the same invariant factors as M/L (dual stacked bases $e_i^\vee$ and $e_i^\vee/d_i$). For $R = \mathbb{Z}$ or $\mathbb{Z}_\ell$: $L^\vee/M^\vee \simeq (M/L)^*$, the Pontryagin dual (8.1), so $[L^\vee : M^\vee] = [M : L]$. Variant used by the paper: if $A \hookrightarrow B$ is an injection of lattices of the same rank with finite cokernel C, then $B^\vee \hookrightarrow A^\vee$ with cokernel $\simeq C^*$. Reference: elementary. Paper: Lemma lem:crit ("Dualising the inclusion $\mathbb{T} \subset \mathbb{T}/I \times \mathbb{T}/J$, whose cokernel is $\mathbb{T}/(I+J)$, shows that $\mathbb{Z}f \oplus L$ has index r' in $S_2(\Gamma_0(N);\mathbb{Z})$": with 9.6, $\mathrm{Hom}(\mathbb{T}/I,\mathbb{Z}) \oplus \mathrm{Hom}(\mathbb{T}/J,\mathbb{Z}) = S_2[I] \oplus S_2[J] = \mathbb{Z}f \oplus L$ and $\mathrm{Hom}(\mathbb{T},\mathbb{Z}) = S_2(\Gamma_0(N);\mathbb{Z})$).

5.8 `prop:modules:congruence-decomposition` (proposition, the congruence module of a decomposition; P). Let R be a PID with fraction field K, M a lattice, $M \otimes K = V_1 \oplus V_2$ a decomposition into K-subspaces with projections p_1, p_2, and $L_i = M \cap V_i$ (saturated). (i) $L_1 \oplus L_2 \subset M$ has finite colength (torsion quotient), and $x \mapsto p_i(x)$ induces isomorphisms $M/(L_1 \oplus L_2) \simeq p_1(M)/L_1 \simeq p_2(M)/L_2$. (ii) If $V_1 = Kf$ is a line with $f \in M$ and $L_1 = Rf$ (i.e. f is primitive), the quotient embeds in $Kf/Rf \simeq K/R$, so it is cyclic, $\simeq R/(r)$, generated by the class of any $h \in M$ with $p_1(h) = f/r$; then $rh = f + g_0$ with $g_0 \in L_2$, so $f \equiv -g_0 \pmod{rM}$; and (r) is the smallest ideal (r') for which $f \equiv g \pmod{r'M}$ with some $g \in L_2$ (the *congruence modulus* of f against L_2). (iii) Example (checked): $M = \mathbb{Z}^2$, $f = (1,a)$, $g = (1,b)$, $V_1 = \mathbb{Q}f$, $V_2 = \mathbb{Q}g$: $M/(\mathbb{Z}f \oplus \mathbb{Z}g) \simeq \mathbb{Z}/(a-b)$ and $f \equiv g \pmod{(a-b)}$ coordinatewise. Reference: Hida81 (congruence modules); Ribet83; DDT Section 4 (cite the section). Paper: Lemma lem:crit (the whole second paragraph of its proof: "embeds in $\mathbb{Q}f/\mathbb{Z}f$ … so it is cyclic, generated by the class of a form h with projection f/r'; then $r'h = f + g_0$"), Lemma lem:ddt417 ($Q = L/(L[\wp] \oplus L[I])$ is the congruence module of the decomposition $L \otimes K = L[\wp]_K \oplus L[I]_K$), Theorem thm:quant Step 1 (the $\mathbb{Z}_\ell$-version).

5.3.6 Section 6. Discrete valuation rings, completions, $\mathbb{Z}_\ell$, $\mathbb{Q}_\ell$, $\mathbb{Q}_\ell/\mathbb{Z}_\ell$

6.1 `thm:modules:dvr` (definition and theorem; P for (a) to (d), R for (e)). Let $\mathcal{O}$ be a Noetherian local domain which is not a field, with maximal ideal $\mathfrak{m}$ and fraction field K. The following are equivalent: (a) $\mathfrak{m}$ is principal; (b) $\mathcal{O}$ is a PID; (c) every nonzero ideal is a power of $\mathfrak{m}$; (d) there is a surjective homomorphism $v\colon K^\times \to \mathbb{Z}$ with $v(x+y) \geq \min(v(x),v(y))$ and $\mathcal{O} = \{x : v(x) \geq 0\} \cup \{0\}$; (e) $\mathcal{O}$ is integrally closed (with $\dim \mathcal{O} = 1$). Such $\mathcal{O}$ is a *discrete valuation ring* (DVR). A generator ϖ of $\mathfrak{m}$ is a *uniformiser*; every $x \in K^\times$ is uniquely $u\varpi^{v(x)}$ with $u \in \mathcal{O}^\times$; $\mathcal{O}^\times = \{v = 0\}$; the nonzero ideals are $\varpi^k\mathcal{O}$. Key step (a)⇒(c),(d): Krull's intersection theorem in the principal case, $\bigcap_k \mathfrak{m}^k = 0$, proved directly (if $x = \varpi y_1 = \varpi^2 y_2 = \cdots$, the chain $(y_1) \subset (y_2) \subset \cdots$ stabilises, $y_{k+1} = uy_k$, so $(1-\varpi u)y_k = 0$ and $y_k = 0$). Reference: AM Proposition 9.2 (check); Serre, Local Fields, Chapter I, Section 1. Paper: background ($\mathcal{O}$ in Lemma lem:ddt417 and Section 4.1; Chapter 6, item 4.7, applies (a) to $\mathbb{Z}_3[\zeta_3]$ in Lemma lem:IV).

6.2 `ex:modules:dvr-examples` (examples; P where stated). DVRs: $\mathbb{Z}_{(p)}$ (Chapter 2, item 2.4); $K[[x]]$; $\mathbb{Z}_\ell$ (6.4); and, with proofs in other chapters, $\mathbb{Z}_3[\zeta_3]$ and $\mathbb{Z}_\ell[\zeta_3]$ for $\ell \equiv 2 \pmod 3$ (Chapter 6, items 4.7, 4.9), the completions $\mathcal{O}_\lambda$ of rings of integers (Chapters 6, 7). Non-DVRs: $\mathbb{Z}$ (not local), $\mathbb{Z}/p^2$ and $\mathcal{O}[\varepsilon]/(\varepsilon^2)$ (not domains), $\mathbb{Z}_\ell \times \mathbb{Z}_\ell$ (not local), $\mathbb{Z}_\ell[x]$ (not local, dimension two), $\mathbb{Z}_{(11)}[x]/(x^2+x-1)$ (a domain with two maximal ideals). Paper: background.

6.3 `prop:modules:dvr-modules` (proposition; P). Let $\mathcal{O}$ be a DVR and M a f.g. $\mathcal{O}$-module. (i) $M \simeq \mathcal{O}^r \oplus \bigoplus_{i=1}^{s} \mathcal{O}/\varpi^{a_i}$, $a_i \geq 1$, with r and the multiset $\{a_i\}$ unique (5.3). (ii) M torsion-free implies M free. (iii) $\dim_k M/\varpi M = r + s$ (the minimal number of generators) and $\dim_k M[\varpi] = s$. (iv) $M[\varpi^k] \simeq \bigoplus_i \mathcal{O}/\varpi^{\min(k,a_i)}$ if $r = 0$, and $M/\varpi^k M \simeq (\mathcal{O}/\varpi^k)^r \oplus \bigoplus_i \mathcal{O}/\varpi^{\min(k,a_i)}$. (v) M is cyclic iff $r + s \leq 1$; a torsion module with $\dim_k M[\varpi] \leq 1$ is cyclic. (vi) A submodule L of a free M is saturated iff $L \cap \varpi M = \varpi L$ (5.5(d)). Reference: DF 12.1; Serre, Local Fields, Chapter I. Paper: Lemma lem:IV, Lemma lem:tamfactor, Lemma lem:h0 (a finite H^0 whose ℓ-torsion has order ℓ is cyclic, so its length is the largest k with an element of exact order ℓ^k; Chapter 16, item h0-and-torsion).

6.4 `thm:modules:zl` (definition and theorem; P for (i) to (vi), R for (vii)). $\mathbb{Z}_\ell = \varprojlim_n \mathbb{Z}/\ell^n$, a ring (Chapter 1, item 7.2, constructs it as a profinite group). (i) $x = (x_n) \in \mathbb{Z}_\ell$ is a unit iff $x_1 \neq 0$ in $\mathbb{F}_\ell$. (ii) Every nonzero x is uniquely $\ell^k u$ with $k \geq 0$, $u \in \mathbb{Z}_\ell^\times$; $\mathbb{Z}_\ell$ is a domain and a DVR with uniformiser ℓ, residue field $\mathbb{F}_\ell$, ideals $\ell^k\mathbb{Z}_\ell$, and valuation extending v_ℓ on $\mathbb{Z}$. (iii) $\mathbb{Z} \to \mathbb{Z}_\ell$ is injective, $\mathbb{Z}_{(\ell)} \subset \mathbb{Z}_\ell$, and $\mathbb{Z}_\ell/\ell^n\mathbb{Z}_\ell \simeq \mathbb{Z}/\ell^n$; an integer prime to ℓ is a unit of $\mathbb{Z}_\ell$, so a $\mathbb{Z}_\ell$-module killed by an integer prime to ℓ is zero. (iv) $\mathbb{Q}_\ell := \mathrm{Frac}(\mathbb{Z}_\ell) = \mathbb{Z}_\ell[1/\ell]$. (v) $\mathbb{Z}_\ell$ is flat over $\mathbb{Z}$ (3.4(ii)); for a f.g. abelian group $A \simeq \mathbb{Z}^r \oplus A_{\mathrm{tors}}$, $A \otimes_{\mathbb{Z}} \mathbb{Z}_\ell \simeq \mathbb{Z}_\ell^r \oplus A_{\mathrm{tors}}[\ell^\infty]$; for finite A, $\#(A \otimes \mathbb{Z}_\ell) = \ell^{v_\ell(\#A)}$ ("the ℓ-part"; Chapter 1, item 1.8). (vi) Completeness: every f.g. $\mathbb{Z}_\ell$-module M satisfies $M \simeq \varprojlim M/\ell^n M$ (by 6.3(i) and the case $M = \mathbb{Z}_\ell$, $\mathbb{Z}/\ell^a$). (vii) More generally, the I-adic completion $\hat{R} = \varprojlim R/I^n$ of a Noetherian ring is Noetherian and flat over R, and $\hat{M} = M \otimes_R \hat{R}$ for f.g. M (Artin-Rees; stated). The analytic description of $\mathbb{Z}_\ell$ and $\mathbb{Q}_\ell$ (absolute value, Cauchy sequences, Hensel's lemma, $\mathbb{Z}_p^\times$) is Chapter 7. Reference: Serre, Local Fields, Chapter II, Section 1; AM Chapter 10. Paper: everywhere ($\mathbb{Z}_\ell$-coefficients); Lemma lem:crit ("$\mathbb{Z}$ replaced by $\mathbb{Z}_\ell$ and r by its ℓ-part"), Lemma lem:dfg21 and the paragraph after it ("$H^1(\mathbb{R}, T^D)$ is killed by 2, so $H^1_f(\mathbb{R}, T^D) = 0$ for ℓ odd"), Lemma lem:fpr ("an ℓ-adic integer").

6.5 `prop:modules:qlzl` (proposition; P). (i) $\mathbb{Q}_\ell/\mathbb{Z}_\ell \simeq \mathbb{Z}[1/\ell]/\mathbb{Z} \simeq (\mathbb{Q}/\mathbb{Z})[\ell^\infty] \simeq \varinjlim_k \mathbb{Z}/\ell^k$ (transition maps multiplication by ℓ); it is isomorphic, as an abstract group, to $\mu_{\ell^\infty}(\overline{\mathbb{Q}})$ (no Galois action here). (ii) It is divisible and ℓ-primary torsion; $(\mathbb{Q}_\ell/\mathbb{Z}_\ell)[\ell^k] = \ell^{-k}\mathbb{Z}_\ell/\mathbb{Z}_\ell \simeq \mathbb{Z}/\ell^k$; its proper $\mathbb{Z}_\ell$-submodules are exactly these. (iii) $\mathrm{End}_{\mathbb{Z}_\ell}(\mathbb{Q}_\ell/\mathbb{Z}_\ell) = \mathbb{Z}_\ell$, $\mathrm{Hom}_{\mathbb{Z}_\ell}(\mathbb{Z}/\ell^k, \mathbb{Q}_\ell/\mathbb{Z}_\ell) \simeq \mathbb{Z}/\ell^k$, $\mathrm{Hom}_{\mathbb{Z}_\ell}(\mathbb{Z}_\ell, \mathbb{Q}_\ell/\mathbb{Z}_\ell) = \mathbb{Q}_\ell/\mathbb{Z}_\ell$. (iv) $\mathbb{Q}_\ell/\mathbb{Z}_\ell$ is an injective $\mathbb{Z}_\ell$-module, and $\mathbb{Q}/\mathbb{Z}$ an injective $\mathbb{Z}$-module (divisible modules over a PID are injective: Baer's criterion, proved for a PID); hence $\mathrm{Hom}(-, \mathbb{Q}_\ell/\mathbb{Z}_\ell)$ is exact. Reference: Weibel 2.3; DF 10.5. Paper: Theorem thm:main and Lemma lem:h0 ($W = \mathrm{ad}^0 T \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$), Lemma lem:dfg21 (Pontryagin duals), Chapter 16 ("$\mathbb{Q}_\ell/\mathbb{Z}_\ell$ injective").

6.6 `prop:modules:zl-topology` (proposition; P for (i) to (iii), S for (iv)). Give f.g. $\mathbb{Z}_\ell$-modules the ℓ-adic topology (basis of neighbourhoods $\ell^n M$). (i) A f.g. $\mathbb{Z}_\ell$-module is compact, Hausdorff and complete. (ii) Every submodule of a f.g. $\mathbb{Z}_\ell$-module is f.g. (4.1) and closed; every $\mathbb{Z}_\ell$-linear map between f.g. modules is continuous with closed image. (iii) $\mathrm{GL}_n(\mathbb{Z}_\ell) = \mathrm{Aut}_{\mathbb{Z}_\ell}(\mathbb{Z}_\ell^n)$ is profinite (Chapter 1, item 7.4). (iv) *Topological Nakayama:* a compact Hausdorff $\mathbb{Z}_\ell$-module A on which $\mathbb{Z}_\ell$ acts continuously and with $A/\ell A$ finite is f.g. (sketch: by Pontryagin duality 8.2, A^* is discrete ℓ-primary with $A^*[\ell] = (A/\ell A)^*$ finite, hence cofinitely generated, so $A = A^{**}$ is f.g.; NSW, Chapter V, Section 2, or Rubin, Appendix B, for the duality of compact modules). Reference: NSW Chapter I and V; Serre, Galois Cohomology, Chapter I (not yet in refs.bib; `SerreGaloisCoh` is to be added to `refs.bib`, decision B14, important_notes.md B21). Paper: Chapter 16 ($(t-1)M$ closed; continuous cohomology with $\mathbb{Z}_\ell$-modules of finite type), Lemma lem:tamfactor ("every $\mathbb{Z}_\ell$-module M' of finite type").

5.3.7 Section 7. Length, annihilators, Fitting ideals

7.1 `thm:modules:jordan-holder` (definition and theorem; P). A composition series of M is a chain $0 = M_0 \subsetneq M_1 \subsetneq \cdots \subsetneq M_n = M$ with simple quotients. (i) Jordan-Holder: any two composition series have the same length and the same factors up to order and isomorphism (proof by induction on the minimal length, comparing M_{n-1} and M'_{n-1} through $M_{n-1} \cap M'_{n-1}$; commutativity of R is not used). The common length is $\mathrm{length}_R(M)$; $\mathrm{length}_R(M) = \infty$ if there is no composition series. (ii) M has finite length iff it is Noetherian and Artinian (stated). (iii) Additivity: for $0 \to M' \to M \to M'' \to 0$, $\mathrm{length}(M) = \mathrm{length}(M') + \mathrm{length}(M'')$; for an exact sequence $0 \to M_1 \to \cdots \to M_k \to 0$ of modules of finite length, $\sum_i (-1)^i \mathrm{length}(M_i) = 0$. (iv) Over a local ring $(R, \mathfrak{m}, k)$ every simple module is $\simeq k$;

if $\mathfrak{m}^N M = 0$ and M is f.g. then $\text{length}(M) = \sum_{i<N} \dim_k \mathfrak{m}^i M/\mathfrak{m}^{i+1}M$; over a Noetherian local ring a f.g. module has finite length iff it is killed by a power of $\mathfrak{m}$. (v) If $R \to R'$ is a local homomorphism of local rings with R' finite over R and residue extension of degree f, then $\text{length}_R(M) = f\,\text{length}_{R'}(M)$ for every R'-module of finite length. Reference: AM Chapter 6; Lang X.3; Eisenbud 2.4. Paper: Theorem thm:main, Theorem thm:dfg37, Remark rem:ns (length H^1_Σ, length H^1_f), Lemma lem:h0 ($h_\ell(E)$), Chapter 16 (lengths-in-exact-sequences uses (iii)); Chapter 4 cites (i) for representations.

7.2 `prop:modules:length-dvr` (proposition; P). Let $\mathcal{O}$ be a DVR. (i) $\text{length}(\mathcal{O}/\varpi^a) = a$; a f.g. $\mathcal{O}$-module has finite length iff it is torsion, and then $\text{length}(\bigoplus_i \mathcal{O}/\varpi^{a_i}) = \sum_i a_i$. (ii) If k is finite, $\#M = (\#k)^{\text{length}\,M}$; over $\mathbb{Z}_\ell$, $\text{length}_{\mathbb{Z}_\ell}(M) = v_\ell(\#M)$ for finite M; over $\mathcal{O}_\lambda$ with residue field $\mathbb{F}_{\ell^f}$, $v_\ell(\#M) = f\,\text{length}(M)$. (iii) $\text{length}(M[\varpi^k])$ and $\text{length}(M/\varpi^k M)$ from 6.3(iv); for a finite $\mathcal{O}$-module, $\text{length}\,M[\varpi^k] = \text{length}\,M/\varpi^k M$ (snake lemma, 2.3(b)). Examples: $\text{length}_{\mathbb{Z}_\ell}(\mathbb{Z}/\ell^v \oplus \mathbb{Z}/\ell^m) = v+m$; $\text{length}_{\mathbb{Z}_\ell}(\mathbb{Z}_\ell/n') = v_\ell(n')$. Reference: as 7.1. Paper: Theorem thm:main ("$\text{length}_{\mathbb{Z}_\ell}$"), Lemma lem:tamfactor (orders $\ell^{v_\ell(n)} \cdot \ell^{\min(\dots)}$), PLAN.md Appendix B ("length means $\mathbb{Z}_\ell$-length").

7.3 `def:modules:annihilator` (definition and proposition; P). $\text{Ann}_R(M) = \{r : rM = 0\}$, $\text{Ann}_R(m)$, and $M[\mathfrak{a}]$ (C3). (i) $\text{Ann}(R/\mathfrak{a}) = \mathfrak{a}$; $\text{Ann}(M \oplus N) = \text{Ann}\,M \cap \text{Ann}\,N$; $\text{Ann}(M)$ is the kernel of $R \to M^n$, $r \mapsto (rm_i)$, for generators $m_1, \dots, m_n$. (ii) $M[\mathfrak{a}] = \text{Hom}_R(R/\mathfrak{a}, M)$; $M[\mathfrak{a}] \cap M[\mathfrak{b}] = M[\mathfrak{a}+\mathfrak{b}]$; $M[\mathfrak{a}] + M[\mathfrak{b}] \subset M[\mathfrak{a}\cap\mathfrak{b}]$. (iii) For a ring $\mathbb{T}$ and an ideal $\mathfrak{a}$, $\mathbb{T}[\mathfrak{a}] = \text{Ann}_{\mathbb{T}}(\mathfrak{a})$; if $L \simeq \mathbb{T}^r$ then $L[\mathfrak{a}] = \text{Ann}_{\mathbb{T}}(\mathfrak{a})^r$. (iv) Flat base change (3.4(iii)) for f.g. M and f.g. $\mathfrak{a}$. (v) Over a DVR: $\text{Ann}(\bigoplus \mathcal{O}/\varpi^{a_i}) = \varpi^{\max a_i}$. "$M$ is killed by η" means $\eta \subset \text{Ann}(M)$. Reference: AM Chapters 1, 2. Paper: Lemma lem:crit ($J = \text{Ann}_{\mathbb{T}}(f^\perp)$, "$\phi_f(\text{Ann}_{\mathbb{T}}\, I)$"), Lemma lem:ddt417 ($I = \text{Ann}_{\mathbb{T}}(\wp)$, $L[\wp]$, $L[I]$, "Q is killed by η", "$L \simeq \mathbb{T}^d$ then $Q \simeq (\mathbb{T}/(\wp + I))^d$"), Lemma lem:hecke, Theorem thm:quant Step 1 ($\eta^{\text{Hecke}} = \pi(\text{Ann}_{\mathbb{T}} \ker \pi)$), PLAN.md scope ("Ann").

7.4 `def:modules:fitting` (definition and theorem; P for the definition and for presentations over a PID, S for the independence in general). Let M be a finitely presented R-module with a presentation $R^m \xrightarrow{A} R^n \to M \to 0$ (C8). The (zeroth) *Fitting ideal* $\text{Fitt}_R(M)$ is the ideal generated by the $n \times n$ minors of A (0 if $m < n$); more generally $\text{Fitt}_i(M)$ uses the $(n-i)\times(n-i)$ minors. It does not depend on the presentation (sketch: two presentations are related by invertible row and column operations, by adding redundant relations, and by adding a generator together with a relation expressing it, none of which changes the ideal of maximal minors; Eisenbud, Section 20.2, for the full proof). Reference: Eisenbud Chapter 20; Northcott, Finite free resolutions (not in refs.bib; add to `bib/ch03.bib` if cited); Mazur-Wiles 1984, Appendix (add to `bib/ch03.bib` if cited). Paper: Section 1.2 (Tam_p), Lemma lem:fpr, PLAN.md scope ("definition via presentations").

7.5 `prop:modules:fitting-properties` (proposition; P). Let M, N be finitely presented. (i) $\text{Fitt}(R/\mathfrak{a}) = \mathfrak{a}$; $\text{Fitt}(0) = R$; $\text{Fitt}(R^n) = 0$ for $n \geq 1$. (ii) $\text{Fitt}(M \oplus N) = \text{Fitt}(M)\,\text{Fitt}(N)$ (block-diagonal presentation; the maximal minors of a block-diagonal matrix are sums of products of maximal minors of the blocks). (iii) If $M \to N$ is surjective, $\text{Fitt}(M) \subset \text{Fitt}(N)$ (a presentation of N is obtained from one of M by adding relations). (iv) $\text{Fitt}(M) \subset \text{Ann}(M)$ (adjugate), and $\text{Ann}(M)^n \subset \text{Fitt}(M)$ if M has n generators. (v) $\text{Fitt}_S(S \otimes_R M) = \text{Fitt}_R(M)S$ for every ring homomorphism $R \to S$ (no flatness needed). (vi) For $0 \to M' \to M \to M'' \to 0$: $\text{Fitt}(M')\,\text{Fitt}(M'') \subset \text{Fitt}(M)$ (block-triangular relations define a module surjecting onto M; then (iii)); equality fails in general: over $R = k[\varepsilon]/(\varepsilon^2)$ with $M = R$, $M' = \varepsilon R \simeq k$, $M'' = R/\varepsilon R \simeq k$ one has $\text{Fitt}(M) = R$ but $\text{Fitt}(M')\,\text{Fitt}(M'') = (\varepsilon)^2 = 0$. (vii) Over a domain, $\text{Fitt}(M) \neq 0$ iff M is torsion. Reference: Eisenbud 20.2; Mazur-Wiles 1984, Appendix. Paper: PLAN.md scope ("Fitting ideal of a direct sum, of a quotient"), Lemma lem:tamfactor (the Fitting ideal of the invariants, a submodule, compared with that of the whole torsion: by (vi) and 7.6 it is smaller in length).

7.6 `thm:modules:fitting-dvr` (theorem; P). (i) Over a PID, $\text{Fitt}(R^r \oplus \bigoplus_i R/(d_i)) = 0$ if $r > 0$ and $= (\prod_i d_i)$ if $r = 0$ (5.2: the product of the invariant factors generates the ideal of maximal minors). (ii) Over a DVR $\mathcal{O}$, for a f.g. torsion module, $\text{Fitt}_{\mathcal{O}}(M) = \varpi^{\text{length}(M)}\mathcal{O}$; hence $v(\text{Fitt}(M)) = \text{length}(M)$, and over $\mathbb{Z}_\ell$, $\text{Fitt}_{\mathbb{Z}_\ell}(M) = \#M \cdot \mathbb{Z}_\ell = \ell^{v_\ell(\#M)}\mathbb{Z}_\ell$. (iii) Consequently, over a DVR the Fitting ideal of finite-length modules is multiplicative in short exact sequences (7.1(iii)), and $\text{Fitt}(N) \supset \text{Fitt}(M)$ for every submodule or quotient N of M. (iv) Fitt $\neq$ Ann: $(\mathbb{Z}/\ell)^2$ and $\mathbb{Z}/\ell^2$ both have Fitting ideal $\ell^2\mathbb{Z}_\ell$, with annihilators $\ell\mathbb{Z}_\ell$ and $\ell^2\mathbb{Z}_\ell$. Reference: Eisenbud 20.2; Mazur-Wiles 1984, Appendix. Paper:

Section 1.2 ($\mathrm{Tam}_p(M)$ is ℓ^{length} of the invariants of the torsion of $H^1(I_p, M)$), Lemma lem:tamfactor ($v_\ell(\mathrm{Tam}_q(\mathrm{ad}^0 T)) = v_\ell(n) + \min(\dots)$ is the length of $\mathbb{Z}/\ell^v \oplus \mathbb{Z}/\ell^m$), Lemma lem:fpr ($\mathrm{Fitt}_{\mathbb{Z}_\ell} H^1_f(\mathbb{Q}_p, T^D)$ is ℓ to the length), PLAN.md scope ("over a DVR Fitt = ℓ^{length}"), Section 6 map row (Tamagawa factor).

7.7 `ex:modules:fitting-examples` (worked examples; P, all checked). Over $\mathbb{Z}$ (relations as rows): $\left(\begin{smallmatrix}2&4\\6&8\end{smallmatrix}\right)$: Fitt = (8), module $\mathbb{Z}/2 \oplus \mathbb{Z}/4$, Ann = (4); the DF matrix of 5.2: Fitt = (144), Ann = (12); $\left(\begin{smallmatrix}3&6\\9&3\end{smallmatrix}\right)$: Fitt = (45), and over $\mathbb{Z}_3$ Fitt $= 9\mathbb{Z}_3$ (length 2); a presentation with more relations than generators, rows $(6,0), (0,4), (2,2)$: module $\mathbb{Z}/2 \oplus \mathbb{Z}/2$, Fitt = (4) (gcd of the three 2×2 minors $24, 12, -8$), Ann = (2); rows $(2), (4)$ on one generator: Fitt = (2). Over a non-PID: $R = \mathbb{Z}[x]$, $M = R/(2, x)$ (one generator, relation rows (2) and (x)): Fitt $= (2, x) =$ Ann, while $\mathrm{Fitt}(M \oplus M) = (2, x)^2 = (4, 2x, x^2) \subsetneq \mathrm{Ann} = (2, x)$. The paper's case: $\mathrm{Fitt}_{\mathbb{Z}_\ell}(\mathbb{Z}/\ell^v \oplus \mathbb{Z}/\ell^m) = \ell^{v+m}\mathbb{Z}_\ell$; for $\ell = 3$, $v = 2$, $m = 1$: $27\mathbb{Z}_3$ with annihilator $9\mathbb{Z}_3$. Paper: Lemma lem:tamfactor, Section 1.2.

5.3.8 Section 8. Pontryagin duality

8.1 `thm:modules:pontryagin-finite` (theorem; P). For a finite abelian group A let $A^* = \mathrm{Hom}(A, \mathbb{Q}/\mathbb{Z})$. (i) $(\mathbb{Z}/n)^* \simeq \mathbb{Z}/n$, so $A^* \simeq A$ (non-canonically, by Chapter 1, item 4.1). (ii) The evaluation map $A \to A^{**}$ is an isomorphism. (iii) $A \mapsto A^*$ is an exact contravariant functor (6.5(iv)), and $\#A^* = \#A$. (iv) For $B \subset A$, $B^\perp = \{\varphi : \varphi(B) = 0\} \simeq (A/B)^*$ and $B^* \simeq A^*/B^\perp$; $B \mapsto B^\perp$ is an inclusion-reversing bijection between subgroups of A and of A^*. (v) If A is ℓ-primary, $A^* = \mathrm{Hom}_{\mathbb{Z}_\ell}(A, \mathbb{Q}_\ell/\mathbb{Z}_\ell)$ and $\mathrm{length}_{\mathbb{Z}_\ell} A^* = \mathrm{length}_{\mathbb{Z}_\ell} A$; dualising an exact sequence of finite $\mathbb{Z}_\ell$-modules gives an exact sequence with arrows reversed. (vi) The physics reader's dictionary: $x \mapsto e^{2\pi i x}$ identifies A^* with the group of characters $A \to \mathbb{C}^\times$ of Chapter 2, item 10.2 (one sentence). Reference: Serre, Course in Arithmetic, VI.1; NSW Chapter I, Section 1. Paper: Lemma lem:dfg21 ("where * is the Pontryagin dual": the finite groups $H^1_f(\mathbb{Q}_p, T^D)$ and $H^0(\mathbb{Q}, W^D)$), the proof of Theorem thm:main (lengths of the four terms), Remark rem:ns.

8.2 `thm:modules:pontryagin-zl` (theorem; P for the f.g. and cofinitely generated modules, R for general compact and discrete modules). Let $\mathcal{C}$ be the f.g. $\mathbb{Z}_\ell$-modules and $\mathcal{D}$ the discrete ℓ-primary torsion $\mathbb{Z}_\ell$-modules W with $W[\ell]$ finite (*cofinitely generated*). For $M \in \mathcal{C}$ put $M^* = \mathrm{Hom}_{\mathbb{Z}_\ell}(M, \mathbb{Q}_\ell/\mathbb{Z}_\ell)$ (every such homomorphism is continuous for the ℓ-adic topology on M and the discrete one on $\mathbb{Q}_\ell/\mathbb{Z}_\ell$); for $W \in \mathcal{D}$ put $W^* = \mathrm{Hom}_{\mathbb{Z}_\ell}(W, \mathbb{Q}_\ell/\mathbb{Z}_\ell) = \varprojlim_k \mathrm{Hom}(W[\ell^k], \mathbb{Q}_\ell/\mathbb{Z}_\ell)$ with the inverse-limit topology. (i) $\mathbb{Z}_\ell^* = \mathbb{Q}_\ell/\mathbb{Z}_\ell$, $(\mathbb{Q}_\ell/\mathbb{Z}_\ell)^* = \mathbb{Z}_\ell$, $(\mathbb{Z}/\ell^a)^* \simeq \mathbb{Z}/\ell^a$. (ii) $M \mapsto M^*$ and $W \mapsto W^*$ are exact, map $\mathcal{C}$ to $\mathcal{D}$ and $\mathcal{D}$ to $\mathcal{C}$, and $M \to M^{**}$, $W \to W^{**}$ are isomorphisms. (iii) Every $W \in \mathcal{D}$ is $\simeq (\mathbb{Q}_\ell/\mathbb{Z}_\ell)^r \oplus F$ with F finite; r is the *corank*, equal to $\mathrm{rank}_{\mathbb{Z}_\ell} W^*$; W is finite iff its corank is 0; W is divisible iff $F = 0$. (iv) Lengths of finite modules are preserved. (v) General Pontryagin duality between compact and discrete ℓ-primary $\mathbb{Z}_\ell$-modules (stated). Reference: NSW Chapter I, Section 1; Rubin, Appendix B; Serre, Local Fields, Chapter XIII (check). Paper: Lemma lem:dfg21 (duals), Lemma lem:h0 ("$H^0(\mathbb{Q}, W^D)$ is finite"), Theorem thm:dfg37 ("$H^1_\Sigma(\mathbb{Q}, W)$ is finite"), Chapter 16 (pontryagin-duality-cohomology recalls (ii) to (iv)).

8.3 `prop:modules:t-and-w` (proposition, the duality between T and $T \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$; P). Let T be a free $\mathbb{Z}_\ell$-module of rank r, $V = T \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$, $W = T \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell/\mathbb{Z}_\ell$. (i) $0 \to T \to V \to W \to 0$ is exact (tensor $0 \to \mathbb{Z}_\ell \to \mathbb{Q}_\ell \to \mathbb{Q}_\ell/\mathbb{Z}_\ell \to 0$ with the flat T), and $W \simeq (\mathbb{Q}_\ell/\mathbb{Z}_\ell)^r$ is divisible of corank r. (ii) $W[\ell^k] \simeq T/\ell^k T$ via $t \otimes \ell^{-k} \mapsto t \bmod \ell^k$; under it the inclusion $W[\ell^k] \subset W[\ell^{k+1}]$ becomes multiplication by ℓ: $T/\ell^k \to T/\ell^{k+1}$, and multiplication by ℓ: $W[\ell^{k+1}] \to W[\ell^k]$ becomes reduction. (iii) $T \simeq \varprojlim_k W[\ell^k]$ (transition maps multiplication by ℓ): T is the Tate module of W. (iv) $W^* = \mathrm{Hom}(T \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell, \mathbb{Q}_\ell/\mathbb{Z}_\ell) \simeq \mathrm{Hom}_{\mathbb{Z}_\ell}(T, \mathrm{End}(\mathbb{Q}_\ell/\mathbb{Z}_\ell)) = \mathrm{Hom}_{\mathbb{Z}_\ell}(T, \mathbb{Z}_\ell) = T^\vee$, and $T^* \simeq T^\vee \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$. (v) Everything is functorial in T: a group acting $\mathbb{Z}_\ell$-linearly on T acts on V, W, $W[\ell^k] = T/\ell^k T$, and on the duals by $(g\varphi)(x) = \varphi(g^{-1}x)$. (vi) The same holds over any complete DVR $\mathcal{O}$ with $K/\mathcal{O}$ in place of $\mathbb{Q}_\ell/\mathbb{Z}_\ell$. Twists by $\mathbb{Z}_\ell(1)$, the Cartier dual $T^D = \mathrm{Hom}(T, \mathbb{Z}_\ell(1))$ and $W^D = V^D/T^D$ are Chapters 17 and 18 (Chapter 17 needs them first, for local duality); the Galois action is Chapters 16, 17. Reference: NSW Chapter I; Rubin, Section 1.1 (check). Paper: Theorem thm:main ($W = \mathrm{ad}^0 T \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$), Lemma lem:h0 ("its ℓ-torsion"; "the description of the length follows from the ℓ^k-torsion": $W^D[\ell^k] = T^D/\ell^k$), Lemma lem:dfg21 (W, T^D, W^D), PLAN.md scope ("the duality between T and $T \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$").

8.4 `cor:modules:cyclic-l-primary` (corollary; P). An ℓ-primary torsion $\mathbb{Z}_\ell$-module A with $\#A[\ell] \le \ell$ is 0, $\mathbb{Z}/\ell^h$ or $\mathbb{Q}_\ell/\mathbb{Z}_\ell$; if finite, its length is the largest k such that A has an element of exact order ℓ^k. (From

8.2(iii) and 6.3(v).) Reference: elementary. Paper: Lemma lem:h0 (with important_notes.md point 5: the ℓ-torsion of $H^0(\mathbb{Q}, W^D)$ is at most a line, so $h_\ell(E)$ is the largest k with $\rho \bmod \ell^k \simeq (\rho \otimes \chi_\ell) \bmod \ell^k$); Chapter 16, item h0-and-torsion (c) recalls it.

5.3.9 Section 9. Pairings: perfect pairings, Gram determinants, saturated sublattices, alternating pairings, adjoints

9.1 `def:modules:pairing` (definition and proposition; P). For a pairing $\langle\ ,\ \rangle\colon L \times M \to R$ (C5) let $\alpha\colon L \to M^\vee$, $x \mapsto \langle x, -\rangle$, and $\beta\colon M \to L^\vee$, $y \mapsto \langle -, y\rangle$. The pairing is *perfect* if α and β are isomorphisms, *nondegenerate* if they are injective. (i) For L, M free of rank n with bases (e_i), (f_j) and Gram matrix $G = (\langle e_i, f_j\rangle)$: the pairing is perfect iff $\det G \in R^\times$ (iff one of α, β is an isomorphism); over a domain it is nondegenerate iff $\det G \neq 0$. Example: $\langle x, y\rangle = 2xy$ on $\mathbb{Z}$ is nondegenerate, not perfect; it is perfect over $\mathbb{Z}_\ell$ for ℓ odd. (ii) Change of bases $e' = eP$, $f' = fQ$ gives $G' = P^{\mathsf{T}} G Q$ and $\det G' = \det P \det Q \det G$; for $L = M$ with the same basis change, $\det G' = (\det P)^2 \det G$. So the ideal ΔR generated by the *Gram determinant* $\Delta = \det G$ does not depend on the bases, and over a DVR its valuation $v(\Delta)$ is basis-independent (PLAN.md scope). (iii) Perfectness is preserved by base change $R \to S$ (the Gram determinant maps to a unit). Reference: Lang XV.1; Serre, Course in Arithmetic, V.1 (for $\mathbb{Z}$-lattices). Paper: Lemma lem:ddt417 ("a perfect $\mathcal{O}$-bilinear pairing", "Gram determinant $\Delta \neq 0$"), Lemma lem:crit ("the pairing being perfect after tensoring with $\mathbb{Z}_\ell$"), Section 4.1.

9.2 `prop:modules:orthogonal` (proposition; P). Let $\mathcal{O}$ be a DVR (or a PID) and $\langle\ ,\ \rangle\colon L \times L \to \mathcal{O}$ a perfect pairing on a lattice of rank n, not necessarily symmetric; write $S^\perp = \{y : \langle S, y\rangle = 0\}$ (right) and ${}^\perp S$ (left). (i) $S^\perp$ and ${}^\perp S$ are saturated for every subset S. (ii) For a submodule N of rank d: $\operatorname{rank} N^\perp = n - d$ and ${}^\perp(N^\perp) = \operatorname{sat}(N)$; for N saturated, ${}^\perp(N^\perp) = N$. (iii) If the pairing is symmetric or antisymmetric, left and right orthogonals agree. Reference: elementary. Paper: Lemma lem:ddt417 ($L[\wp]^\perp$), 9.3.

9.3 `thm:modules:gram-cokernel` (theorem; P). Let $\mathcal{O}$ be a DVR, L a lattice of rank n with a perfect pairing, $L' \subset L$ a **saturated** sublattice of rank d, and Δ the Gram determinant of the restriction of the pairing to L' (any basis). (i) The map $\rho\colon L \to \operatorname{Hom}_{\mathcal{O}}(L', \mathcal{O})$, $y \mapsto \langle -, y\rangle|_{L'}$, is surjective with kernel $L'^\perp$ (because L' is a direct summand and $L \simeq L^\vee$); so $L/L'^\perp \simeq \operatorname{Hom}(L', \mathcal{O})$. (ii) If $\Delta \neq 0$: $L' \cap L'^\perp = 0$, the restriction $\rho|_{L'}\colon L' \to \operatorname{Hom}(L', \mathcal{O})$ is injective, and

$$L/(L' \oplus L'^\perp) \ \simeq\ \operatorname{coker}\big(L' \to \operatorname{Hom}(L', \mathcal{O})\big) \ \simeq\ \textstyle\bigoplus_i \mathcal{O}/(\delta_i),$$

where the δ_i are the invariant factors of the Gram matrix; its length is $v(\Delta)$, and its order is $\#\mathcal{O}/\Delta\mathcal{O} = (\#k)^{v(\Delta)}$ when k is finite. (iii) $L'^\perp$ is saturated of rank $n - d$, and the Gram determinant of the restriction to $L'^\perp$ has the same valuation $v(\Delta)$. (iv) Saturation is necessary: in $L = \mathbb{Z}_\ell^2$ with the standard pairing, $L' = \mathbb{Z}_\ell \cdot \ell e_1$ has $\Delta = \ell^2$ while $L/(L' \oplus L'^\perp) \simeq \mathbb{Z}/\ell$ has length 1. Example (checked): on $\mathbb{Z}^2$ with Gram matrix $\left(\begin{smallmatrix} 9 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ (perfect, determinant -1), $L' = \mathbb{Z}e_1$ has $\Delta = 9$, $L'^\perp = \mathbb{Z}(1, -9)$, and $[L : L' \oplus L'^\perp] = 9$; (ii) was also checked on 500 random perfect pairings and saturated sublattices of rank ≤ 5 over $\mathbb{Z}$, and (iii) on 300 of them (non-symmetric pairings). Reference: elementary; Serre, Course in Arithmetic, V.1 (discriminants of sublattices); Eichler-style arguments in DDT Section 4 (cite the section). Paper: Lemma lem:ddt417 ("The pairing identifies $L/L[\wp]^\perp$ with $\operatorname{Hom}(L[\wp], \mathcal{O})$, because $L[\wp]$ is saturated"; "Hence Q is the cokernel of $L[\wp] \to \operatorname{Hom}(L[\wp], \mathcal{O})$, of order $\#\mathcal{O}/\Delta$"), PLAN.md scope ("for a saturated sublattice L of rank d, the cokernel of $L \to \operatorname{Hom}(L, \mathcal{O})$ has order $\#\mathcal{O}/$(Gram det)").

9.4 `prop:modules:alternating` (proposition; P, except the Pfaffian identity and the normal form in general rank, S). A pairing $L \times L \to R$ is alternating if $\langle x, x\rangle = 0$ for all x; then it is antisymmetric; conversely antisymmetric implies alternating if 2 is a nonzerodivisor in R (e.g. R a domain of characteristic 0, or ℓ odd). (i) The Gram matrix of an alternating pairing is antisymmetric with zero diagonal; its determinant is $\operatorname{Pf}(G)^2$ (Pfaffian; stated with the rank 4 case checked), and it vanishes in odd rank; so a perfect alternating pairing exists only in even rank. (ii) Rank two: $G = \left(\begin{smallmatrix} 0 & a \\ -a & 0 \end{smallmatrix}\right)$ with $a = \langle e_1, e_2\rangle$, so $\Delta = a^2$, and the ideal of values $\langle L, L\rangle = \{\langle x, y\rangle\}$ is $a\mathcal{O}$ (since $\langle x, y\rangle = a(x_1y_2 - x_2y_1)$). Hence, basis-free: $\Delta\mathcal{O} = (\text{ideal of values})^2$. (iii) Over a PID an alternating pairing has a basis in which G is block diagonal with blocks $\left(\begin{smallmatrix} 0 & d_i \\ -d_i & 0 \end{smallmatrix}\right)$, $d_1 \mid d_2 \mid \cdots$, and zeros (symplectic normal form; S); it is perfect iff all d_i are units

and there are no zero blocks. Reference: Lang XV.8 (check); Bourbaki, Algebra IX. Paper: Section 4.1 ("$\hat{\delta}_!$ ... perfect and alternating"; "the Gram determinant of the pairing on $M_{f^\Sigma,\lambda}$ is $(\eta^\Sigma_{f,\lambda})^2$ up to a unit", with $\eta^\Sigma_{f,\lambda}$ defined as the ideal of values), Lemma lem:hecke ($d=2$, Gram determinant $(\eta^\emptyset_{f,\lambda})^2$), Theorem thm:quant Step 2, PLAN.md scope ("rank-two Gram determinant is a square").

9.5 `prop:modules:adjoint` (proposition; P). Let $\langle\ ,\ \rangle$ be a perfect pairing on a lattice L over R. (i) Every $A \in \mathrm{End}_R(L)$ has a unique adjoint $A^\dagger$ with $\langle Ax, y\rangle = \langle x, A^\dagger y\rangle$; in matrices $A^\dagger = G^{-1}A^{\mathsf{T}}G$ (C5, C8: write the convention once); $(AB)^\dagger = B^\dagger A^\dagger$, $(A^\dagger)^\dagger = A$ for symmetric or antisymmetric pairings. (ii) Let $\mathbb{T} \subset \mathrm{End}_R(L)$ be a commutative subring of self-adjoint endomorphisms. For every ideal $\mathfrak{a} \subset \mathbb{T}$: $(\mathfrak{a}L)^\perp = L[\mathfrak{a}]$ (since $\langle ax, y\rangle = \langle x, ay\rangle$ and the pairing is perfect). (iii) Idempotents of $\mathbb{T}$ (or of $\mathbb{T}\otimes S$ for an R-algebra S) are self-adjoint; for an idempotent e, $L = eL \oplus (1-e)L$ is an orthogonal decomposition, and the restriction of the pairing to eL is perfect. (iv) Twisted pairings: let $w \in \mathrm{Aut}_R(L)$ with $A^\dagger = wAw^{-1}$ for all $A \in \mathbb{T}$, and put $[x,y] = \langle x, wy\rangle$. Then $[\ ,\ ]$ is perfect and every $A \in \mathbb{T}$ is self-adjoint for it. If $\langle\ ,\ \rangle$ is antisymmetric and $w^\dagger = w$ (for instance w an involution preserving the pairing, so $w^\dagger = w^{-1} = w$), then $[\ ,\ ]$ is antisymmetric, hence alternating when 2 is a nonzerodivisor. Reference: Lang XV.1 (check). Paper: Section 4.1 ("the Hecke operators, U_p included, are self-adjoint for it, because $w_{M^\Sigma} T_n w_{M^\Sigma}^{-1}$ is the adjoint of T_n for the cup product (this is what the composition with w is for)": the geometric inputs are Chapters 8 and 14), Lemma lem:ddt417 ("for which $\mathbb{T}$ is self-adjoint"), Lemma lem:hecke (the restriction of $\hat{\delta}_!$ to the $\mathfrak{m}$-component is perfect, by (iii)), PLAN.md scope ("adjoints").

9.6 `prop:modules:algebra-duality` (proposition; P). Let R be a PID (in practice $\mathbb{Z}$ or $\mathbb{Z}_\ell$), $\mathbb{T}$ a commutative R-algebra that is free of finite rank as an R-module, S a $\mathbb{T}$-module free of finite rank over R, and $\langle\ ,\ \rangle\colon S\times\mathbb{T}\to R$ a perfect R-bilinear pairing with $\langle ts, u\rangle = \langle s, tu\rangle$ (example: $\langle g, t\rangle = a_1(tg)$ on $S_2(\Gamma_0(N);\mathbb{Z})\times\mathbb{T}$, Chapter 13). (i) $s \mapsto \langle s, -\rangle$ is an isomorphism of $\mathbb{T}$-modules $S \simeq \mathbb{T}^\vee = \mathrm{Hom}_R(\mathbb{T}, R)$, where $(t\varphi)(u) = \varphi(tu)$. (ii) For every ideal $\mathfrak{a} \subset \mathbb{T}$: $S[\mathfrak{a}] \simeq \mathrm{Hom}_R(\mathbb{T}/\mathfrak{a}, R)$ (the functionals vanishing on $\mathfrak{a}$); it is saturated in S and of rank $\mathrm{rank}_R(\mathbb{T}/\mathfrak{a})$. (iii) If $I, J \subset \mathbb{T}$ are ideals with $I\cap J = 0$ and $\mathbb{T}/I$, $\mathbb{T}/J$ torsion-free, the inclusion $\mathbb{T} \hookrightarrow \mathbb{T}/I \times \mathbb{T}/J$ (2.2) has finite cokernel $\mathbb{T}/(I+J)$, and dually $S[I]\oplus S[J] \subset S$ with $S/(S[I]\oplus S[J]) \simeq \big(\mathbb{T}/(I+J)\big)^*$ for $R = \mathbb{Z}$ or $\mathbb{Z}_\ell$ (5.7), of the same order. (iv) $\mathbb{T}$ acts faithfully on S. Reference: Ribet83 (Theorem 2.2 as the paper cites it); DDT Section 4 (cite the section). Paper: Lemma lem:crit ("under it the forms killed by I are $\mathbb{Z}f$ and the forms killed by J are L"; "Dualising the inclusion ... shows that $\mathbb{Z}f \oplus L$ has index r'"), Theorem thm:quant Step 1 (the $\mathbb{Z}_\ell$-version).

5.3.10 Section 10. Finite free algebras over a DVR; idempotents; generalised eigenspaces; semisimplicity

10.1 `thm:modules:artinian-algebras` (theorem; P). Let A be a commutative algebra of finite dimension over a field F. (i) A has finitely many maximal ideals $\mathfrak{m}_1, \dots, \mathfrak{m}_s$ ($s \le \dim A$, by CRT), every prime ideal is maximal, and the Jacobson radical $\mathfrak{J} = \bigcap \mathfrak{m}_i$ is nilpotent (the chain $\mathfrak{J}^k$ stabilises; Nakayama). (ii) $A \simeq \prod_i A_{\mathfrak{m}_i}$, each factor local with nilpotent maximal ideal (CRT for the comaximal $\mathfrak{m}_i^N$, Chapter 2, item 1.4(iii)); the factors are cut out by idempotents $e_1, \dots, e_s$ with $\sum e_i = 1$, $e_i e_j = 0$. (iii) A is reduced iff $\mathfrak{J} = 0$ iff A is a product of fields. (iv) Every A-module V is $\bigoplus_i e_i V$ with $e_i V = V_{\mathfrak{m}_i}$. Reference: AM Theorem 8.7 (structure theorem for Artin rings; check) and Chapter 8. Paper: Lemma lem:crit ("$\mathbb{T}\otimes\mathbb{Q} = \mathbb{Q}\times\mathbb{T}'$"), Theorem thm:quant Step 1, Section 11 below.

10.2 `thm:modules:finite-algebra-dvr` (theorem; P). Let $\mathcal{O}$ be a DVR and $\mathbb{T}$ a commutative $\mathcal{O}$-algebra that is finite and free as an $\mathcal{O}$-module (a *finite flat* or *finite free* $\mathcal{O}$-algebra). (i) $\mathbb{T}$ is Noetherian; every maximal ideal of $\mathbb{T}$ contains ϖ (if $\mathfrak{M}$ is maximal, $\mathbb{T}/\mathfrak{M}$ is a field finite over the image of $\mathcal{O}$, which is then a field, so $\mathfrak{M}\cap\mathcal{O} = \varpi\mathcal{O}$); so the maximal ideals of $\mathbb{T}$ correspond to those of the finite k-algebra $\mathbb{T}/\varpi\mathbb{T}$, are finite in number, and have residue fields finite over k. (ii) $\mathbb{T}$ is torsion-free and $\mathbb{T} \subset \mathbb{T}_K := \mathbb{T}\otimes_{\mathcal{O}} K$, a finite-dimensional K-algebra (10.1); $\mathbb{T}$ is reduced iff $\mathbb{T}_K$ is reduced iff $\mathbb{T}$ embeds in a product of fields. (iii) A surjective $\mathcal{O}$-algebra homomorphism $\pi\colon \mathbb{T}\to\mathcal{O}$ has kernel $\wp$ with $\mathbb{T}/\wp \simeq \mathcal{O}$ ($\wp$ is a prime ideal, $\mathbb{T} = \mathcal{O}\cdot 1 \oplus \wp$ as $\mathcal{O}$-modules), and determines the maximal ideal $\mathfrak{m}_\pi = \pi^{-1}(\varpi\mathcal{O}) = \wp + \varpi\mathbb{T}$; it extends to $\pi_K\colon \mathbb{T}_K \to K$, which factors through exactly one local factor of $\mathbb{T}_K$ (10.1). Reference: AM Chapters 5, 8; DDT Section 4 (cite the section). Paper: Lemma lem:ddt417 ("$\mathbb{T}$ a commutative $\mathcal{O}$-algebra, finite and free as an $\mathcal{O}$-module", "$\pi\colon\mathbb{T}\to\mathcal{O}$ surjective"), Lemma lem:hecke ("$\mathfrak{m}$ the maximal ideal of $\mathbb{T}\otimes\mathbb{Z}_\ell$

containing ℓ and $\ker \phi_f$" is $\mathfrak{m}_{\phi_f}$), Lemma lem:crit ("a maximal ideal of residue characteristic ℓ").

10.3 `thm:modules:idempotent-lifting` (theorem; P). Let $\mathcal{O}$ be a **complete** DVR (e.g. $\mathbb{Z}_\ell$, or $\mathcal{O}_\lambda$ of Chapter 7) and $\mathbb{T}$ as in 10.2. (i) Every idempotent $\bar{e}$ of $\mathbb{T}/\varpi\mathbb{T}$ lifts to a unique idempotent of $\mathbb{T}$ (Newton iteration $e \mapsto 3e^2 - 2e^3$, which squares the error $e^2 - e$ up to a unit factor and converges ϖ-adically; uniqueness: two idempotents congruent mod ϖ are equal). (ii) $\mathbb{T} \simeq \prod_{\mathfrak{m}} \mathbb{T}_{\mathfrak{m}}$ over the maximal ideals $\mathfrak{m}$, where $\mathbb{T}_{\mathfrak{m}} = e_{\mathfrak{m}}\mathbb{T}$ is the localisation of $\mathbb{T}$ at $\mathfrak{m}$, a local ring, finite and free over $\mathcal{O}$ (2.7). (iii) For every $\mathbb{T}$-module L: $L = \bigoplus_{\mathfrak{m}} e_{\mathfrak{m}} L$ and $L_{\mathfrak{m}} = e_{\mathfrak{m}} L$; if L is f.g. free over $\mathcal{O}$, so is each $L_{\mathfrak{m}}$. (iv) A ring homomorphism from $\mathbb{T}$ to a local ring (for instance $\pi\colon \mathbb{T} \to \mathcal{O}$, or $\mathbb{T} \to k'$ a field) factors through exactly one $\mathbb{T}_{\mathfrak{m}}$: $\pi(e_{\mathfrak{m}_\pi}) = 1$ and $\pi(e_{\mathfrak{m}'}) = 0$ for $\mathfrak{m}' \neq \mathfrak{m}_\pi$. (v) Completeness is necessary: $\mathbb{Z}_{(11)}[x]/(x^2 + x - 1)$ has two maximal ideals above 11 (the polynomial has the roots 3 and 7 mod 11) but no idempotent other than $0, 1$ (it is a domain, the polynomial being irreducible over $\mathbb{Q}$). Reference: Eisenbud 7.5 (Hensel and idempotents; check); AM Chapter 10; DDT Section 4 (cite the section). Paper: Lemma lem:hecke ("the other local factors of $\mathbb{T} \otimes \mathbb{Z}_\ell$ are killed by ϕ_f"), Theorem thm:quant Step 1 ("the $\mathfrak{m}$-component (an idempotent of $\mathbb{T}_\ell$ ...)"), Remark rem:free, PLAN.md scope ("$\mathbb{T} \otimes \mathbb{Z}_\ell$ is a product of local rings, idempotents").

10.4 `cor:modules:hecke-type` (corollary and examples; P). Let $\mathbb{T}$ be a commutative ring that is free of finite rank as a $\mathbb{Z}$-module (the Hecke algebra, Chapter 13). Then $\mathbb{T} \otimes \mathbb{Z}_\ell$ is finite free over $\mathbb{Z}_\ell$, its maximal ideals are the $\mathfrak{m} \otimes \mathbb{Z}_\ell$ for the maximal ideals $\mathfrak{m} \ni \ell$ of $\mathbb{T}$, and $\mathbb{T} \otimes \mathbb{Z}_\ell = \prod_{\mathfrak{m} \ni \ell} \mathbb{T}_{\mathfrak{m}}$ with $\mathbb{T}_{\mathfrak{m}} = (\mathbb{T} \otimes \mathbb{Z}_\ell)_{\mathfrak{m}}$, which is also the $\mathfrak{m}$-adic completion of $\mathbb{T}$ (convention C7). Examples (checked): (a) $\mathbb{T} = \mathbb{Z}[x]/(x^2 + x - 1) = \mathbb{Z}[\frac{1+\sqrt{5}}{2}]$ (the Hecke algebra of level 23, Chapter 6, item 7.5): $x^2 + x - 1 \equiv (x - 3)(x - 7) \pmod{11}$, so $\mathbb{T} \otimes \mathbb{Z}_{11} \simeq \mathbb{Z}_{11} \times \mathbb{Z}_{11}$ with idempotent $e = (x - r_2)/(r_1 - r_2)$, $r_1 \equiv 7$, $r_2 \equiv 3$ the Hensel lifts (computed mod 11^6, and equal to the Newton limit of $(x-3)/4$); $x^2 + x - 1 \equiv (x - 2)^2 \pmod 5$, so $\mathbb{T} \otimes \mathbb{Z}_5$ is local (a ramified DVR); mod 2 and mod 3 the polynomial is irreducible, so $\mathbb{T} \otimes \mathbb{Z}_2$, $\mathbb{T} \otimes \mathbb{Z}_3$ are local (unramified DVRs with residue fields $\mathbb{F}_4$, $\mathbb{F}_9$). (b) $\mathbb{T} = \mathbb{Z}[x]/(x^2 - 4) \simeq \{(a, b) \in \mathbb{Z}^2 : a \equiv b \pmod 4\}$: $\mathbb{T} \otimes \mathbb{Z}_3 = \mathbb{Z}_3 \times \mathbb{Z}_3$ with idempotent $(x + 2)/4$; $\mathbb{T} \otimes \mathbb{Z}_2$ is local and not a domain. Reference: DDT Section 4 (cite the section); Chapter 6, item 7.3(iv), which cites this. Paper: Lemma lem:hecke, Remark rem:free, Theorem thm:quant (all their $\mathbb{T}_{\mathfrak{m}}$).

10.5 `thm:modules:generalised-eigenspaces` (theorem; P). (i) Let V be a finite-dimensional vector space over a field F and $\mathbb{T} \subset \mathrm{End}_F(V)$ a commutative subalgebra such that the characteristic polynomial of every $t \in \mathbb{T}$ splits over F. Then $V = \bigoplus_\varphi V(\varphi)$, the sum over the F-algebra homomorphisms $\varphi\colon \mathbb{T} \to F$ ("systems of eigenvalues"), with $V(\varphi) = \{v : (t - \varphi(t))^{\dim V} v = 0 \ \forall t \in \mathbb{T}\} \neq 0$; the projection onto $V(\varphi)$ is the idempotent $e_\varphi \in \mathbb{T}$ of 10.1; every $V(\varphi)$ contains a nonzero common eigenvector ($tv = \varphi(t)v$ for all t); every $\mathbb{T}$-stable subspace W satisfies $W = \bigoplus_\varphi (W \cap V(\varphi))$. (ii) (Integral version.) Let $\mathcal{O}$ be a complete DVR with fraction field K, L a lattice, $\mathbb{T} \subset \mathrm{End}_{\mathcal{O}}(L)$ a commutative subalgebra (finite free over $\mathcal{O}$), and suppose the eigenvalues of $\mathbb{T}$ on $V = L \otimes K$ lie in K. Then every $\varphi\colon \mathbb{T} \to K$ takes values in $\mathcal{O}$ (eigenvalues are integral over $\mathcal{O}$), $\mathfrak{m}_\varphi = \varphi^{-1}(\varpi\mathcal{O})$ is a maximal ideal of $\mathbb{T}$, $\mathfrak{m}_\varphi = \mathfrak{m}_\psi$ iff $\varphi \equiv \psi \pmod{\varpi}$ on $\mathbb{T}$, and $e_{\mathfrak{m}} L \otimes K = \bigoplus_{\varphi : \mathfrak{m}_\varphi = \mathfrak{m}} V(\varphi)$. (If the eigenvalues are not in K, replace $\mathcal{O}$ by the ring of integers of a finite extension, Chapter 7.) Reference: Lang XIV.2 (check); DF 12.3; DDT Section 4. Paper: Lemma lem:crit ("$\mathbb{T} \otimes \mathbb{Q} = \mathbb{Q} \times \mathbb{T}'$ with $\mathbb{T}'$ acting faithfully on $f^\perp$"), Theorem thm:twist ("$g^\perp$, the $\mathbb{T}$-stable complement of the g-line (the sum of the generalised eigenspaces of the other systems of eigenvalues)"), Theorem thm:quant Step 1 ("the generalised eigenspace of π", "the $\mathfrak{m}$-component of $S_2(\Gamma_0(N); \mathbb{Z}_\ell)$ is the sum of the line of g^S and of generalised eigenspaces ..."), PLAN.md scope ("generalised eigenspaces of commuting operators").

10.6 `prop:modules:idempotent-congruence` (proposition; P). In 10.5(ii) let $\mathfrak{m}$ be a maximal ideal of $\mathbb{T}$ and $e = e_{\mathfrak{m}}$. (i) e maps L into L and every $\mathbb{T}$-stable subspace $W \subset V$ into itself, and fixes $V(\varphi)$ pointwise for $\mathfrak{m}_\varphi = \mathfrak{m}$. (ii) If $x \in V(\varphi) \cap L$ with $\mathfrak{m}_\varphi = \mathfrak{m}$, $y \in L \cap W$, and $x \equiv y \pmod{\varpi^t L}$, then $x \equiv ey \pmod{\varpi^t L}$ with $ey \in L \cap W \cap eV$. Reference: elementary. Paper: Theorem thm:quant Step 1 ("projecting h to the $\mathfrak{m}$-component (an idempotent of $\mathbb{T}_\ell$, so the projection stays in the complement and the congruence survives) we may take h in the sum of the generalised eigenspaces above").

10.7 `prop:modules:semisimple-operators` (definition and proposition; P, with (iv) R). Let $A \in \mathrm{End}_F(V)$, $\dim V < \infty$. (i) A is *semisimple* if every A-stable subspace has an A-stable complement; for F perfect (characteristic 0 or finite) this holds iff the minimal polynomial of A is squarefree iff A

is diagonalisable over $\overline{F}$. (ii) If V has a cyclic vector for A (some v with $v, Av, \dots, A^{n-1}v$ a basis), the minimal polynomial equals the characteristic polynomial, so A is not semisimple as soon as the characteristic polynomial has a repeated root. (iii) A commutative subalgebra $\mathbb{T} \subset \mathrm{End}_F(V)$ is reduced iff each of its elements is semisimple (by 10.1(iii) and (i)); if one $t \in \mathbb{T}$ is not semisimple, its nilpotent part is a nonzero nilpotent element of $\mathbb{T}$. (iv) Jordan-Chevalley: $A = A_s + A_n$ with A_s semisimple, A_n nilpotent, commuting, both polynomials in A (stated). (v) Example (the linear algebra of U_p on oldforms; checked symbolically): on V with basis $v_0, \dots, v_e$ let $Uv_j = v_{j-1}$ for $j \geq 1$ and (a) $Uv_0 = av_0 - pv_1$ (with $p \neq 0$): the characteristic polynomial is $X^{e-1}(X^2 - aX + p)$, v_e is a cyclic vector, so for $e \geq 3$ the eigenvalue 0 has a single Jordan block of size $e - 1 \geq 2$ and U is not semisimple; nevertheless every eigenvalue has an eigenvector, e.g. $\ker U \ni v_0 - av_1 + pv_2$ for $e \geq 2$; (b) $Uv_0 = av_0$: characteristic polynomial $X^e(X - a)$, not semisimple for $e \geq 2$, with $U(v_0 - av_1) = 0$ for $e \geq 1$. The modular meaning ($v_j = h(p^j z)$; the depleted forms) is Chapter 13. Reference: DF 12.3; Lang XIV.2. Paper: Lemma lem:old ("The operator U_p is not semisimple on this span when 0 is a repeated eigenvalue, but each listed eigenvalue has an eigenvector"), Theorem thm:quant Step 1 ("the operators U_q ... have the eigenvalue 0 with multiplicity one"), PLAN.md scope ("semisimple versus non-semisimple operators"); 11.1 and Part 8, point 1 (the full Hecke algebra need not be reduced).

10.8 `lem:modules:common-eigenvector` (lemma; P). (i) Commuting operators $A_1, \dots, A_k$ on a nonzero finite-dimensional space over an algebraically closed field have a common eigenvector; more precisely, for every tuple (λ_i) whose joint generalised eigenspace $V(\lambda) = \bigcap_i \ker(A_i - \lambda_i)^{\dim V}$ is nonzero, $V(\lambda)$ contains a common eigenvector, and it can be chosen to be an eigenvector of every element of any commutative algebra $\mathbb{T}$ containing the A_i (apply 10.5(i) to $\mathbb{T}$ acting on the $\mathbb{T}$-stable $V(\lambda)$). (ii) If $V = V_1 \otimes \cdots \otimes V_k$ and A_i acts on the i-th factor, the joint generalised eigenspace for (λ_i) is $V_1(\lambda_1) \otimes \cdots \otimes V_k(\lambda_k)$, nonzero iff each λ_i is an eigenvalue of A_i on V_i. Reference: elementary; DF 12.3. Paper: Lemma lem:old ("Since V is the tensor product of these local spaces and $\mathbb{T}$ is commutative, for every choice of one such eigenvalue μ_p at each $p \mid N/M$ there is a common eigenform $g \in V \otimes \overline{\mathbb{Q}}$ of all of $\mathbb{T}$").

5.3.11 Section 11. Congruence ideals, Gorenstein algebras, and the ingredients of Lemma lem:ddt417

11.1 `prop:modules:congruence-ideal` (definition and proposition; P). Let $\mathcal{O}$ be a DVR with fraction field K, $\mathbb{T}$ a commutative $\mathcal{O}$-algebra finite and free as an $\mathcal{O}$-module, $\pi\colon \mathbb{T} \to \mathcal{O}$ a surjective $\mathcal{O}$-algebra homomorphism, $\wp = \ker \pi$, $I = \mathrm{Ann}_{\mathbb{T}}(\wp)$, and $\eta = \pi(I)$, the *congruence ideal* of π; the *congruence module* is $\mathbb{T}/(\wp + I) \simeq \mathcal{O}/\eta$ (1.3(iv)). Then: (i) $I \neq 0$ (in $\mathbb{T}_K$ the maximal ideal $\wp_K$ has a nonzero annihilator, 10.1, and $I = \mathbb{T} \cap \mathrm{Ann}_{\mathbb{T}_K}(\wp_K)$). (ii) The following are equivalent: (a) $\eta \neq 0$; (b) $\wp \cap I = 0$; (c) the local factor of $\mathbb{T}_K$ through which π_K factors is K itself, i.e. $\mathbb{T}_K = K \times \mathbb{T}'_K$ with π_K the first projection. *Proof of (a)⇒(b) in two lines:* choose $t \in I$ with $\pi(t)$ generating η; for $x \in \wp \cap I$, $\pi(t)x = tx + (\pi(t) - t)x$, where $tx \in I\wp = 0$ and $(\pi(t) - t)x \in \wp I = 0$ (as $\pi(t) - t \in \wp$); so $\pi(t)x = 0$ and $x = 0$, $\mathbb{T}$ being torsion-free. (b)⇒(a): if $\eta = 0$ then $I \subset \wp$, so $\wp \cap I = I \neq 0$ by (i). (a)⇔(c): if the factor A of $\mathbb{T}_K$ is not a field, $\mathrm{Ann}_A(\mathfrak{m}_A) \subset \mathfrak{m}_A$, so $\pi(I) = 0$; if it is K, the idempotent $e_\pi = (1, 0)$ has a nonzero multiple $ae_\pi \in \mathbb{T}$, which lies in I and has $\pi(ae_\pi) = a$. (iii) If $\eta \neq 0$: $\pi|_I\colon I \to \eta$ is an isomorphism (so I is free of rank one over $\mathcal{O}$); $\mathrm{Ann}_{\mathbb{T}}(I) = \wp$ (if $xI = 0$ then $\pi(x)\pi(t) = 0$); $I = \mathbb{T} \cap (K \times 0)$ and $\wp = \mathbb{T} \cap (0 \times \mathbb{T}'_K)$; and $\eta = \{a \in \mathcal{O} : ae_\pi \in \mathbb{T}\}$, the "denominator" of the idempotent e_π. (iv) If $\mathbb{T}_K$ is reduced, (c) holds automatically; it can fail to be reduced (10.7(v) and Part 8, point 1), and then (c) must be checked directly (for a newform it is multiplicity one, Chapter 13). (v) Compatibility: if $\mathcal{O}$ is complete and $\mathfrak{m} = \mathfrak{m}_\pi$, then $\ker \pi = \wp_{\mathfrak{m}} \times \prod_{\mathfrak{m}' \neq \mathfrak{m}} \mathbb{T}_{\mathfrak{m}'}$ and $\mathrm{Ann}_{\mathbb{T}}(\ker \pi) = \mathrm{Ann}_{\mathbb{T}_{\mathfrak{m}}}(\wp_{\mathfrak{m}}) \times 0$, so $\eta(\mathbb{T}, \pi) = \eta(\mathbb{T}_{\mathfrak{m}}, \pi|_{\mathbb{T}_{\mathfrak{m}}})$: **the congruence ideal is computed in** $\mathbb{T}_{\mathfrak{m}}$. The other factors $\mathbb{T}_{\mathfrak{m}'}$ lie in $\ker \pi$, not in the annihilator (Part 8, point 2). For a flat extension $\mathcal{O} \to \mathcal{O}'$ of DVRs (e.g. $\mathbb{Z}_{(\ell)} \to \mathbb{Z}_\ell$), $\eta(\mathbb{T} \otimes \mathcal{O}', \pi \otimes \mathcal{O}') = \eta\mathcal{O}'$ (3.4(iii)). Reference: DDT Section 4 (cite the section; Lemma 4.17 for the context); Hida81; Ribet83. Paper: Lemma lem:crit ("r generates the congruence ideal $\phi_f(J) = \phi_f(\mathrm{Ann}_{\mathbb{T}} I)$"; "$I \cap J = 0$"), Lemma lem:ddt417 ("$\wp \cap I = 0$"; important_notes.md point 19), Lemma lem:hecke ("the congruence ideal is computed in $\mathbb{T}_{\mathfrak{m}}$"), Theorem thm:quant Step 1 ($\eta^{\mathrm{Hecke}}(g^S)$ computed in $\mathbb{T}_{\mathfrak{m}}$), Section 6 map row (Hida's congruence modules).

11.2 `ex:modules:toy-congruence` (examples; P, all checked). (a) $\mathbb{T}_r = \{(a, b) \in \mathcal{O}^2 : a \equiv b$

$(\mathrm{mod}\ \varpi^r)\} \simeq \mathcal{O}[x]/(x(x-\varpi^r))$ (with $x=(\varpi^r,0)$), $\pi(a,b)=a$: $\wp = 0\times\varpi^r\mathcal{O}$, $I=\varpi^r\mathcal{O}\times 0$, $\eta=\varpi^r\mathcal{O}$, congruence module $\mathcal{O}/\varpi^r$, $e_\pi=(1,0)=x/\varpi^r$: "two eigenforms congruent modulo ϖ^r". (b) $\mathbb{Z}[x]/(x^2-4)$, $\pi(x)=2$: $I=\mathbb{Z}(x+2)$, $\eta=4\mathbb{Z}$, $e_\pi=(x+2)/4$, congruence module $\mathbb{Z}/4$. (c) $\mathbb{T}_3^\flat=\{(a,b,c)\in\mathcal{O}^3 : a\equiv b\equiv c \pmod{\varpi}\}$, π the first coordinate: $\eta=\varpi\mathcal{O}$ (three pairwise congruent eigenforms). (d) $\mathcal{O}[\varepsilon]/(\varepsilon^2)$, $\pi(\varepsilon)=0$: $\wp=I=\mathcal{O}\varepsilon$, $\eta=0$, $\wp\cap I\neq 0$. (e) $\mathbb{Z}[x]/(x^2(x-1))$: for $\pi(x)=0$ the factor of $\mathbb{T}_\mathbb{Q}$ is $\mathbb{Q}[x]/(x^2)$, $I=\mathbb{Z}(x-x^2)$ and $\eta=0$; for $\pi(x)=1$, $I=\mathbb{Z}x^2$, $\eta=\mathbb{Z}$, congruence module 0. Paper: Lemma lem:crit, Lemma lem:ddt417 (models), Exercises 8, 9.

11.3 `prop:modules:ddt-ingredients` (proposition, the module ingredients of Lemma lem:ddt417; P). In the setting of 11.1 with $\eta\neq 0$, let L be a $\mathbb{T}$-module that is free of finite rank over $\mathcal{O}$, with a perfect $\mathcal{O}$-bilinear pairing for which every element of $\mathbb{T}$ is self-adjoint. Then: (i) $L[\wp]$ and $L[I]$ are saturated in L (5.5); (ii) $L[\wp]\cap L[I]=0$ (an element killed by $\wp+I\ni\pi(t)$ is killed by $\pi(t)\neq 0$); (iii) $L[\wp]^\perp=L[I]$ and $L[I]^\perp=L[\wp]$ (for $y\in L[I]$, $x\in L[\wp]$: $\pi(t)\langle x,y\rangle=\langle tx,y\rangle=\langle x,ty\rangle=0$, using $(\pi(t)-t)x\in\wp x=0$; conversely if $y\perp L[\wp]$ then for every $s\in I$ and $z\in L$, $\langle sy,z\rangle=\langle y,sz\rangle=0$ since $sz\in L[\wp]$, so $sy=0$; the second equality by 9.2(ii)); (iv) more generally $(\mathfrak{a}L)^\perp=L[\mathfrak{a}]$ for every ideal (9.5(ii)); (v) if $L\simeq\mathbb{T}^r$ as $\mathbb{T}$-modules, then $L[\wp]=I^r$ (free of rank r over $\mathcal{O}$, by 11.1(iii)), $L[I]=\mathrm{Ann}_\mathbb{T}(I)^r=\wp^r$, and $L/(L[\wp]\oplus L[I])\simeq(\mathbb{T}/(\wp+I))^r\simeq(\mathcal{O}/\eta)^r$. The assembly (the quotient $Q=L/(L[\wp]\oplus L[I])$ is killed by η, is generated by $d=\mathrm{rank}\,L[\wp]$ elements, and is $\mathrm{coker}(L[\wp]\to\mathrm{Hom}(L[\wp],\mathcal{O}))$ of length $v(\Delta)$ by 9.3; hence $d\,v(\eta)\geq v(\Delta)$, with equality when L is free) is Lemma lem:ddt417, proved in Chapter 20. Reference: DDT, Lemma 4.17 and Section 4 (cite the section). Paper: Lemma lem:ddt417 (every sentence of its proof except the counting), Lemma lem:hecke, Theorem thm:quant Step 2, PLAN.md scope ("every ingredient: Ann, $L[\wp]$, saturation, $L[\wp]^\perp=L[I]$, the dual map").

11.4 `def:modules:gorenstein` (definition and proposition; P for (i) to (iii), R for (iv)). Let $\mathbb{T}$ be a commutative $\mathcal{O}$-algebra, finite and free as an $\mathcal{O}$-module, and $\mathbb{T}^\vee=\mathrm{Hom}_\mathcal{O}(\mathbb{T},\mathcal{O})$ with $(t\varphi)(u)=\varphi(tu)$. $\mathbb{T}$ is *Gorenstein* if $\mathbb{T}^\vee$ is free of rank one as a $\mathbb{T}$-module. (i) Equivalently there is a perfect $\mathcal{O}$-bilinear pairing $\mathbb{T}\times\mathbb{T}\to\mathcal{O}$ with $\langle st,u\rangle=\langle s,tu\rangle$ (take $\langle s,t\rangle=\lambda(st)$ for a generator λ of $\mathbb{T}^\vee$). (ii) For $\mathcal{O}$ complete, $\mathbb{T}$ is Gorenstein iff each local factor $\mathbb{T}_\mathfrak{m}$ (10.3) is; for $\mathbb{T}$ local with maximal ideal $\mathfrak{m}$ and residue field $k_\mathbb{T}$, $\mathbb{T}$ is Gorenstein iff $\dim_{k_\mathbb{T}}(\mathbb{T}/\varpi\mathbb{T})[\mathfrak{m}]=1$ (the socle of $\mathbb{T}/\varpi\mathbb{T}$ is a line) iff $\dim_{k_\mathbb{T}}\mathbb{T}^\vee/\mathfrak{m}\mathbb{T}^\vee=1$ (proof: $\mathbb{T}^\vee/\varpi\mathbb{T}^\vee=\mathrm{Hom}_k(\mathbb{T}/\varpi\mathbb{T},k)$ and $\mathbb{T}^\vee/\mathfrak{m}\mathbb{T}^\vee$ is dual to the socle; a cyclic $\mathbb{T}^\vee$ is free since $\mathbb{T}^\vee$ and $\mathbb{T}$ have the same $\mathcal{O}$-rank). (iii) Examples (checked): $\mathcal{O}[x]/(f)$ for monic f is Gorenstein (a generator of $\mathbb{T}^\vee$ is the coefficient of x^{n-1}); $\mathbb{T}_r$ of 11.2(a) is Gorenstein (socle dimension 1; with $\lambda(a,b)=(a-b)/\varpi^r$ the Gram matrix of $\lambda(st)$ in the basis $(1,1),(\varpi^r,0)$ is $\left(\begin{smallmatrix}0 & 1\\ 1 & \varpi^{-r}\end{smallmatrix}\right)$, determinant -1); $\mathbb{T}_3^\flat$ of 11.2(c) is not (socle of dimension 2), although $\mathbb{T}_3^\flat\otimes K=K^3$ is reduced; $\mathcal{O}[\varepsilon]/(\varepsilon^2)$ is Gorenstein and not reduced. (iv) Complete intersections are Gorenstein; for $\mathbb{T}$ local Gorenstein and π as in 11.1, Wiles's numerical criterion relates η to $\wp/\wp^2$ (stated with pointer to Chapter 20). Reference: Mazur77, Chapter II, Section 15 (check); Eisenbud Chapter 21; Tilouine97; DDT Section 4. Paper: Remark rem:free ("the localised Hecke algebra is then Gorenstein"), PLAN.md scope ("Gorenstein rings (definition: finite free $\mathcal{O}$-algebra $\mathbb{T}$ with $\mathrm{Hom}_\mathcal{O}(\mathbb{T},\mathcal{O})$ free of rank one over $\mathbb{T}$)"), Section 6 map row (multiplicity one, freeness, Gorenstein).

11.5 `thm:modules:mazur-freeness` (theorem, a freeness criterion in the style of Mazur; P). Let $\mathbb{T}$ be a local commutative $\mathcal{O}$-algebra, finite and free over $\mathcal{O}$, with maximal ideal $\mathfrak{m}$ and residue field $k_\mathbb{T}$, and let L be a $\mathbb{T}$-module, free of finite rank over $\mathcal{O}$, with a perfect $\mathcal{O}$-bilinear pairing for which $\mathbb{T}$ is self-adjoint (so $L\simeq L^\vee$ as $\mathbb{T}$-modules). Suppose (a) $\dim_{k_\mathbb{T}}L/\mathfrak{m}L=d$ and (b) $\mathrm{rank}_\mathcal{O}L=d\cdot\mathrm{rank}_\mathcal{O}\mathbb{T}$. Then $L\simeq\mathbb{T}^d$ and $\mathbb{T}$ is Gorenstein. (Proof: by Nakayama there is a surjection $\mathbb{T}^d\to L$, an isomorphism by (b) since both are free over $\mathcal{O}$ of the same rank; then $(\mathbb{T}^\vee)^d\simeq L^\vee\simeq L\simeq\mathbb{T}^d$, so $\dim\mathbb{T}^\vee/\mathfrak{m}\mathbb{T}^\vee=1$ and 11.4(ii) applies.) In the Hecke application (Chapter 14): $d=2$; (a) is "$J_0(N)[\mathfrak{m}]$ is two-dimensional over $\mathbb{T}/\mathfrak{m}$" read through Pontryagin duality (8.1, 8.2), and (b) holds because $H^1(X_0(N),\mathbb{Q})\otimes\mathbb{C}\simeq S_2\oplus\overline{S_2}$ and $S_2(\Gamma_0(N))\simeq\mathrm{Hom}(\mathbb{T},\mathbb{C})$ by 9.6, compatibly with the idempotent decomposition 10.3. Reference: Mazur77, Chapter II, Section 15 (check); Tilouine97; DDT Section 4 (cite the section). Paper: Remark rem:free ("since the localised Hecke algebra is then Gorenstein and the Tate module is free of rank two"), the equality clauses of Lemma lem:ddt417, Lemma lem:hecke and Theorem thm:main, Theorem thm:quant Step 4.

11.6 `rem:modules:ddt417-preview` (remark with checked examples; R for the lemma, P for the examples). Statement of the paper's Lemma lem:ddt417 in the notation of 11.1 and 11.3 (no proof; Chapter 20), with the table: which item supplies which step (saturation 5.5 and 11.3(i); directness

11.3(ii); $L[\wp]^\perp = L[I]$ 11.3(iii); the dual map and its cokernel 9.3; the Gram determinant 9.1, 9.4; equality 11.3(v)). Checks of the statement on the toy algebra $\mathbb{T}_r$ (all computed): $L = \mathbb{T}_r$ with $\langle s, t\rangle = \lambda(st)$: $L[\wp] = \mathcal{O} \cdot (\varpi^r, 0)$, $d = 1$, $\Delta = \varpi^r$, so $d\, v(\eta) = r = v(\Delta)$ (free case, equality); $L = \mathcal{O}^2$ with $\mathbb{T}_r$ acting diagonally and the standard pairing: $L[\wp] = \mathcal{O}e_1$, $\Delta = 1$, $v(\Delta) = 0 < r$ (strict inequality; L is not free over $\mathbb{T}_r$); $L = \mathbb{T}_r^2$ with the alternating pairing $\lambda(x_1y_2 - x_2y_1)$: $d = 2$, $\Delta = \varpi^{2r} = \eta^2$ (the shape of Lemma lem:hecke). Also: when $\eta = 0$ the inequality $\eta^d \subset \Delta\mathcal{O}$ is trivial, and if moreover L is free over $\mathbb{T}$ then $\Delta = 0$ ($I \subset \wp$ gives $I \cdot I = 0$, so the pairing vanishes on $L[\wp] = I^r$), so the hypothesis $\Delta \neq 0$ of the lemma already forces $\eta \neq 0$ in the free case. Paper: Lemma lem:ddt417, Lemma lem:hecke, Section 6 map row (DDT Lemma 4.17, Wiles's numerical criterion).

5.4 3. Section-by-section plan with target line counts

Line counts are lines of LaTeX, card excluded (PLAN.md Section 7.3: 1,000 to 2,200). The card lists all 64 items of Part 2 (numbered 3.1 to 3.64 in the book), about 190 lines, not counted.

section	title	contents	target lines
(starred)	Core results at a glance	the card: every item of Part 2, stated as above, with tag and `\Cref`	(excluded)
(starred)	Where the paper uses this chapter	Section 1.2 (Tam_p as a Fitting ideal); lem:tamfactor; lem:fpr; thm:main, thm:dfg37, rem:ns (lengths); lem:dfg21 (Pontryagin duals); lem:h0 (W, ℓ^k-torsion, cyclicity); lem:crit (dualising, congruence module, $\mathbb{T} \otimes \mathbb{Q} = \mathbb{Q} \times \mathbb{T}'$); lem:ddt417 (every ingredient); lem:hecke ($\mathbb{T}_{\mathfrak{m}}$, idempotents, restricted pairing); rem:free (Gorenstein, freeness); thm:quant Steps 1, 2 (generalised eigenspaces, idempotents); lem:old (non-semisimple U_p, common eigenforms); lem:IV (modules over $\mathbb{Z}_3[\zeta_3]$); Section 4.1 (alternating Gram determinant)	45
1	Modules, submodules, quotients, homomorphisms	conventions C1 to C10 as `convention` environments; 1.1 to 1.3; the running examples announced with forward `\Cref`s	110

section	title	contents	target lines
2	Exact sequences, the snake lemma, sums and products, free and projective modules	2.1 to 2.7; the snake lemma with the diagram (`tikz-cd` if the preamble has it, else an `array`); `paperlink` for lem:crit and lem:ddt417 (2.2)	170
3	Hom and tensor product; flatness	3.1 to 3.4; the examples of 3.2(v) in a small table; `paperlink` for lem:crit over $\mathbb{Z}_\ell$ and lem:hecke (3.4(iii))	170
4	Noetherian rings, Nakayama, localisation, local rings	4.1 to 4.5; `paperlink` for rem:free and lem:h0 (4.3)	180
5	Modules over a PID; lattices and saturation	5.1 to 5.8; worked Smith forms (5.2 examples); `paperlink` for lem:IV (5.4(iii)), lem:crit (5.7, 5.8), lem:ddt417 (5.5)	240
6	DVRs, completions, $\mathbb{Z}_\ell$, $\mathbb{Q}_\ell$, $\mathbb{Q}_\ell/\mathbb{Z}_\ell$	6.1 to 6.6; `warning` that the analytic theory of $\mathbb{Q}_p$ is Chapter 7; `paperlink` for lem:dfg21 ("killed by 2")	170
7	Length, annihilators, Fitting ideals	7.1 to 7.7; the examples of 7.7 as a `booktabs` table (presentation, module, Fitt, Ann, length); `paperlink` for Section 1.2 and lem:tamfactor; `warning` Fitt $\neq$ Ann	190
8	Pontryagin duality	8.1 to 8.4; `paperlink` for lem:dfg21, lem:h0	130
9	Pairings	9.1 to 9.6; the $\left(\begin{smallmatrix} 9 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ example and the non-saturated counterexample; `paperlink` for lem:ddt417 (9.3), Section 4.1 (9.4, 9.5), lem:crit (9.6)	200
10	Finite free algebras over a DVR; idempotents; generalised eigenspaces	10.1 to 10.8; the level-23 example worked mod 11, 5, 2, 3; the shift-operator matrix of 10.7(v); `paperlink` for lem:hecke, thm:quant Step 1, lem:old	210

section	title	contents	target lines
11	Congruence ideals and Gorenstein algebras	11.1 to 11.6; the toy algebras of 11.2 in one table ($\wp$, I, η, e_π, Gorenstein or not); the item-to-step table of 11.6; `paperlink` for lem:ddt417, rem:free	180
12	Exercises	the 17 exercises of Part 4	130
13	Notes and references	Atiyah-Macdonald, Dummit-Foote, Lang, Eisenbud, Serre (Local Fields; Course in Arithmetic), NSW, Weibel, DDT Section 4, Hida81, Mazur77, Tilouine97; which numbers were cited and which were checked; historical remark on Fitting (1936) and on Mazur's multiplicity one argument	40
	total		**2,165**

5.5 4. Coverage table and exercises

5.5.1 4.1 Coverage: PLAN.md scope, Section 6 map rows and paper hooks to line-up items

PLAN.md scope item, Section 6 row, or paper hook	line-up items
modules over commutative rings, submodules, quotients, homomorphisms	1.1, 1.2, 1.3
exact sequences	2.1, 2.2
the snake lemma	2.3 (and 2.4)
Hom and tensor product: definitions, universal properties	3.1, 3.2
right exactness of the tensor product; flatness of localisation	3.3, 3.4, 4.5(ii)
direct sums and products	2.5
free and projective modules	2.6, 2.7, 4.3(v), 5.1
f.g. modules over a PID: structure theorem, elementary divisors	5.2, 5.3, 5.6
torsion and torsion-free	5.4, 6.3(ii)
saturated sublattices (L saturated if M/L torsion-free)	5.5, 5.6, 9.2, 11.3(i)
Nakayama's lemma	4.2, 4.3, 6.6(iv)
localisation at a prime and at a maximal ideal; local rings	4.4, 4.5, 10.3(ii)

PLAN.md scope item, Section 6 row, or paper hook	line-up items
DVRs: definition, uniformiser, valuation, ideals are powers of the maximal ideal	6.1, 6.2, 6.3
completions and $\mathbb{Z}_\ell$, $\mathbb{Q}_\ell$, $\mathbb{Q}_\ell/\mathbb{Z}_\ell$ (coordinated with Chapter 7)	6.4, 6.5, 6.6
length over a DVR and over a local ring	7.1, 7.2
Fitting ideals: presentations; over a DVR Fitt $= \varpi^{\text{length}}$; direct sums; quotients	7.4, 7.5, 7.6, 7.7
annihilators	7.3, 11.1
Pontryagin duality for finite abelian groups	8.1
Pontryagin duality for (f.g. $\mathbb{Z}_\ell$-modules, discrete torsion $\mathbb{Z}_\ell$-modules) via $\mathrm{Hom}(-, \mathbb{Q}_\ell/\mathbb{Z}_\ell)$	8.2, 8.4, 6.5
the duality between T and $T \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$	8.3
perfect pairings, the Gram determinant, basis independence of its valuation	9.1
saturated L of rank d: cokernel of $L \to \mathrm{Hom}(L, \mathcal{O})$ has order $\#\mathcal{O}/(\text{Gram det})$	9.3
alternating pairings; rank-two Gram determinant is a square	9.4
adjoints with respect to a pairing	9.5, 9.6
finite free algebras over a DVR, maximal ideals, localisation; $\mathbb{T} \otimes \mathbb{Z}_\ell$ a product of local rings; idempotents	10.1, 10.2, 10.3, 10.4
generalised eigenspaces of commuting operators	10.5, 10.6, 10.8
semisimple versus non-semisimple operators	10.7
Gorenstein rings: definition; Gorenstein plus a perfect self-adjoint pairing gives freeness	11.4, 11.5
ingredients of Lemma lem:ddt417: Ann, $L[\wp]$, saturation, $L[\wp]^\perp = L[I]$, the dual map	7.3, 11.1, 11.3, 5.5, 9.3, 9.4, 11.6
Section 6 row: Tamagawa factor $\mathrm{Tam}_p(M) = \mathrm{Fitt}(H^1(I_p, M)_{\text{tors}})^G$ (with Chapters 16, 18)	7.4, 7.5, 7.6, 7.7
Section 6 row: multiplicity one, freeness of $H^1(X_0(N), \mathbb{Z}_\ell)_{\mathfrak{m}}$ over $\mathbb{T}_{\mathfrak{m}}$, Gorenstein, $J_0(N)[\mathfrak{m}]$ (with Chapters 14, 20)	4.3, 10.3, 11.4, 11.5
Section 6 row: Hida's congruence modules, Wiles's numerical criterion, DDT Lemma 4.17 (with Chapter 20)	5.8, 11.1, 11.2, 11.3, 11.6
Section 6 row: Fitting ideals, lengths, Pontryagin duality, Gram determinants, saturated sublattices	7.1 to 7.7, 8.1 to 8.4, 9.1 to 9.4, 5.5
hook: Tamagawa factor as a Fitting ideal (Section 1.2, Lemma lem:tamfactor, Lemma lem:fpr)	7.6, 7.7, 7.2
hook: lengths of Selmer groups (Theorem thm:main; also thm:dfg37, rem:ns, conj:euler)	7.1(iii), 7.2, 8.1(v), 8.2
hook: Lemma lem:ddt417	5.5, 9.1, 9.3, 9.4, 9.5, 11.1, 11.3, 11.6
hook: Lemma lem:hecke	3.4(iii), 9.5(iii), 10.3, 10.4, 11.1(v)
hook: Remark rem:free (freeness, Gorenstein)	4.3, 11.4, 11.5
hook: Lemma lem:crit ($\mathbb{T}/(I+J)$)	1.3(iv), 2.2, 3.4, 5.7, 5.8, 9.6, 10.1, 11.1
hook: Lemma lem:h0 ($\mathbb{Q}_\ell/\mathbb{Z}_\ell$-modules, ℓ^k-torsion)	4.3(iv), 6.3(v), 8.3, 8.4
hook: Pontryagin duals in Lemma lem:dfg21	8.1, 8.2, 6.4(iii) ($H^1(\mathbb{R}, T^D)$ killed by 2)
hook: proof of Theorem thm:quant Step 1 (idempotents of $\mathbb{T}_\ell$, generalised eigenspaces)	10.3, 10.5, 10.6, 11.1(v)

PLAN.md scope item, Section 6 row, or paper hook	line-up items
also: Lemma lem:IV (T free of rank one over $\mathbb{Z}_3[\zeta_3]$)	5.4(iii), 6.3, 1.2(c)
also: Lemma lem:old (non-semisimple U_p; common eigenform for chosen eigenvalues)	10.7(v), 10.8
also: Theorem thm:twist ($g^\perp$ as a sum of generalised eigenspaces)	10.5
also: Section 4.1 ($\hat{\delta}_!$ perfect, alternating, self-adjoint Hecke operators; Gram determinant $(\eta^\Sigma)^2$)	9.4, 9.5(iv)
also: Theorem thm:quant Step 2 and Step 4 (Lemma lem:ddt417 without freeness; free case)	11.3, 11.5
what Chapters 6, 14, 16, 20, 21 assume from Chapter 3 (Part 5)	4.1, 4.5, 5.4, 6.1, 6.3, 6.4, 6.5, 6.6, 7.1, 7.6, 8.2, 9.3, 10.3, 10.4, 11.3, 11.5

5.5.2 4.2 Exercises (Section 12 of the chapter), graded from easy to harder

1. Compute $\mathbb{Z}/m \otimes \mathbb{Z}/n$, $\mathbb{Q} \otimes \mathbb{Z}/n$, $\mathbb{Z}_\ell \otimes \mathbb{Z}/n$, $\mathbb{Q}_\ell/\mathbb{Z}_\ell \otimes \mathbb{Z}/n$, $\mathrm{Hom}(\mathbb{Z}/m, \mathbb{Z}/n)$ and $\mathrm{Hom}(\mathbb{Q}_\ell/\mathbb{Z}_\ell, \mathbb{Z}/\ell^k)$. Show that $\mathrm{Hom}(\mathbb{Z}/2, -)$ is not right exact and $- \otimes \mathbb{Z}/2$ is not left exact on $0 \to \mathbb{Z} \xrightarrow{2} \mathbb{Z} \to \mathbb{Z}/2 \to 0$ (3.1, 3.2, 3.3).
2. Prove corollary (a) of the snake lemma: for $0 \to A \to B \to C \to 0$ and $n \in R$ there is an exact sequence $0 \to A[n] \to B[n] \to C[n] \to A/n \to B/n \to C/n \to 0$. Apply it to $0 \to T \to V \to W \to 0$ with $n = \ell^k$ to get $W[\ell^k] \simeq T/\ell^k T$ (8.3(ii)).
3. Find the Smith normal form, the invariant factors, the elementary divisors, the Fitting ideal and the annihilator of the abelian group with relation rows $(2, 4, 4)$, $(-6, 6, 12)$, $(10, -4, -16)$; then of its tensor product with $\mathbb{Z}_2$ and with $\mathbb{Z}_3$ (answers: $\mathbb{Z}/2 \oplus \mathbb{Z}/6 \oplus \mathbb{Z}/12$; Fitt $= (144)$, Ann $= (12)$; over $\mathbb{Z}_2$: $\mathbb{Z}/2 \oplus \mathbb{Z}/2 \oplus \mathbb{Z}/4$, Fitt $= 16\mathbb{Z}_2$; over $\mathbb{Z}_3$: $\mathbb{Z}/3 \oplus \mathbb{Z}/3$, Fitt $= 9\mathbb{Z}_3$).
4. (Check the paper's claim in Lemma lem:IV.) Let T be free of rank 2 over $\mathbb{Z}_3$ with an automorphism σ satisfying $\sigma^2 + \sigma + 1 = 0$. Show that T is a torsion-free, hence free, module of rank one over $\mathbb{Z}_3[X]/(X^2 + X + 1)$ (5.4(iii), with the DVR property from Chapter 6), that $T/3T \simeq \mathbb{Z}_3[\zeta_3]/(\pi^2)$, and that the fixed vectors of σ on $T/3T$ form a line.
5. (Check the paper's claim in Lemma lem:tamfactor.) Show that $\mathrm{Fitt}_{\mathbb{Z}_\ell}(\mathbb{Z}/\ell^v \oplus \mathbb{Z}/\ell^m) = \ell^{v+m}\mathbb{Z}_\ell$ while its annihilator is $\ell^{\max(v,m)}\mathbb{Z}_\ell$, and that v_ℓ of the Fitting ideal of the module $\{x\bar{e} + y\bar{h}\}$ of the lemma's proof is $v_\ell(n) + \min(v_\ell(n), v_\ell(q-1), v_\ell(c'))$ once one knows its order (7.6). Why would the formula be wrong with Ann in place of Fitt?
6. Show that Fitting ideals are multiplicative in short exact sequences of finite $\mathbb{Z}_\ell$-modules, but not over $k[\varepsilon]/(\varepsilon^2)$ (7.5(vi)). Compute Fitt and Ann of $\mathbb{Z}[x]/(2, x)$ and of its square.
7. (Saturation is necessary.) On $\mathbb{Z}^2$ with Gram matrix $\left(\begin{smallmatrix} 9 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ compute $L'^\perp$ for $L' = \mathbb{Z}e_1$ and the index of $L' \oplus L'^\perp$; then show that for the standard pairing and $L' = \mathbb{Z} \cdot 3e_1$ the index is 3 while the Gram determinant is 9. Check the paper's claim in Lemma lem:ddt417 that the cokernel of $L[\wp] \to \mathrm{Hom}(L[\wp], \mathcal{O})$ "has order $\#\mathcal{O}/\Delta$", and say exactly where saturation enters.
8. (Check the paper's claim in Section 4.1.) Show that an alternating pairing on a free module of rank 2 has Gram determinant a^2 with $a\mathcal{O}$ the ideal of values, and that no perfect alternating pairing exists in odd rank. Compute the Pfaffian identity $\det G = \mathrm{Pf}(G)^2$ for rank 4.
9. (The toy congruence.) For $\mathbb{T}_r = \{(a, b) \in \mathbb{Z}_\ell^2 : a \equiv b \bmod \ell^r\}$ and $\pi(a, b) = a$ compute $\wp$, I, η, $\mathbb{T}_r/(\wp + I)$ and the denominator of the idempotent $(1, 0)$; verify the inequality of Lemma lem:ddt417 for $L = \mathbb{T}_r$ (pairing $\lambda(st)$, equality), for $L = \mathbb{Z}_\ell^2$ with the diagonal action (strict) and for $L = \mathbb{T}_r^2$ with an alternating pairing ($\Delta = \eta^2$).
10. (Check the paper's claim in Lemma lem:ddt417 that $\wp \cap I = 0$.) Show that $\wp \cap I = 0$ iff $\eta \neq 0$ (11.1(ii)); give the example $\mathcal{O}[\varepsilon]/(\varepsilon^2)$; show that when $\eta = 0$ the inequality of the lemma is trivial and that a free L then has $\Delta = 0$.
11. (Check the paper's claim in Lemma lem:crit.) Let $\mathbb{T}$ act on S with a perfect pairing $S \times \mathbb{T} \to \mathbb{Z}$ as

in 9.6, and I, J ideals with $I \cap J = 0$ and $\mathbb{T}/I$, $\mathbb{T}/J$ torsion-free. Show that $S/(S[I] \oplus S[J])$ has the same order as $\mathbb{T}/(I+J)$. Work it out for $\mathbb{T} = \mathbb{Z}[x]/(x^2-4)$, $I = (x-2)$, $J = (x+2)$ (answer 4), and interpret it as a congruence modulo 4 between two "eigenforms".

12. (Check the paper's claim in Lemma lem:hecke that "the congruence ideal is computed in $\mathbb{T}_{\mathfrak{m}}$".) For $\mathbb{T} \otimes \mathbb{Z}_\ell = \prod_{\mathfrak{m}'} \mathbb{T}_{\mathfrak{m}'}$ and π factoring through $\mathbb{T}_{\mathfrak{m}}$, show $\mathrm{Ann}(\ker \pi) = \mathrm{Ann}_{\mathbb{T}_{\mathfrak{m}}}(\wp_{\mathfrak{m}}) \times 0$. Do the other local factors lie in the annihilator, as the paper's sentence says, or in the kernel?
13. Idempotents: for $\mathbb{T} = \mathbb{Z}[x]/(x^2+x-1)$ show $\mathbb{T} \otimes \mathbb{Z}_{11} \simeq \mathbb{Z}_{11} \times \mathbb{Z}_{11}$ and compute the idempotent modulo 11^3 by the iteration $e \mapsto 3e^2 - 2e^3$ from $(x-3)/4$; show $\mathbb{T} \otimes \mathbb{Z}_5$, $\mathbb{T} \otimes \mathbb{Z}_2$, $\mathbb{T} \otimes \mathbb{Z}_3$ are local and $\mathbb{Z}_{(11)}[x]/(x^2+x-1)$ has no nontrivial idempotent although it has two maximal ideals above 11.
14. (Non-semisimple U_p; check the paper's claim in Lemma lem:old.) For the operator U of 10.7(v)(a) compute the characteristic polynomial, show that v_e is a cyclic vector, and find an eigenvector for each eigenvalue; deduce that for $e \geq 3$ the commutative algebra generated by U is not reduced. (With Chapter 13: the full Hecke algebra of level 88 acting on the span of $f_{11}(z), f_{11}(2z), f_{11}(4z), f_{11}(8z)$, where $a_2(f_{11}) = -2$.)
15. Gorenstein: show that $\mathcal{O}[x]/(f)$ (f monic) and $\mathbb{T}_r$ are Gorenstein and that $\{(a,b,c) \in \mathbb{Z}_\ell^3 : a \equiv b \equiv c \bmod \ell\}$ is not, by computing socles of the reductions mod ℓ. Which of these "Hecke algebras" admit a free module L with a perfect pairing for which they are self-adjoint and with $\dim L/\mathfrak{m}L = 1$?
16. (Check the paper's use of Pontryagin duals in Lemma lem:dfg21 and the proof of Theorem thm:main.) For an exact sequence $0 \to A \to B \to \bigoplus_i C_i^* \to D^* \to 0$ of finite $\mathbb{Z}_\ell$-modules show $\mathrm{length}\, B = \mathrm{length}\, A + \sum_i \mathrm{length}\, C_i - \mathrm{length}\, D$; deduce the paper's length identity when $H^0(\mathbb{Q}, W^D) = 0$, and the corrected one of Remark rem:ns when it has length $h_\ell(E)$.
17. Show that $\mathbb{Q}_\ell/\mathbb{Z}_\ell$ is an injective $\mathbb{Z}_\ell$-module, that $\mathrm{End}_{\mathbb{Z}_\ell}(\mathbb{Q}_\ell/\mathbb{Z}_\ell) = \mathbb{Z}_\ell$, that $(T \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell)^* \simeq \mathrm{Hom}(T, \mathbb{Z}_\ell)$ for T free of finite rank, and that a $\mathbb{Z}_\ell$-module killed by 2 is zero when ℓ is odd (the paper: "$H^1(\mathbb{R}, T^D)$ is killed by 2, so $H^1_f(\mathbb{R}, T^D) = 0$ for ℓ odd").

5.6 5. Dependencies and forward pointers

Results from other chapters used here: * PLAN.md Section 2 background: linear algebra (vector spaces, dimension, determinants, Cayley-Hamilton, Jordan form as a statement, bilinear forms and Gram matrices); elementary group theory. * Chapter 1 (`ch:groups`): the structure theorem for finite abelian groups (its 4.1, generalised in 5.3), $\#A[n] = \#A/nA$ (its 4.3), the ℓ-primary decomposition (its 1.8, used in 6.4(v)), $\mathbb{Z}_\ell$ as an inverse limit and $\mathrm{GL}_n(\mathbb{Z}_\ell)$ as a profinite group (its 7.2, 7.4, used in 6.4, 6.6). * Chapter 2 (`ch:rings`): rings, ideals, quotients (its 1.1 to 1.3), the two-ideal exact sequence and CRT (its 1.4, used in 2.2 and 10.1), idempotents and products of rings (its 1.5, used in 1.2, 10.1, 10.3), prime and maximal ideals and residue characteristic (its 2.1, 2.2, used in 10.2), $\mathbb{Z}_{(p)}$ (its 2.4, the first DVR, 6.2), PIDs, UFDs, Euclidean domains (its 3.1, 3.2, used in Section 5), v_p (its 11.1, C3), $\mathrm{Ann}_R(S)$ as a definition (its 1.2). * Black boxes imported from later chapters (stated with a forward `\Cref`, used only in examples and `paperlink`s, never in a proof of a numbered result): $\mathbb{Z}_3[\zeta_3]$ is a DVR (Chapter 6, used in Exercise 4 and 1.2(c)); the Hecke algebra, the perfect pairing $S_2(\Gamma_0(N); \mathbb{Z}) \times \mathbb{T} \to \mathbb{Z}$ and the oldform spaces (Chapters 12, 13, used in 9.6, 10.7(v), Exercises 11, 14); the cup product, w_N and $J_0(N)[\mathfrak{m}]$ (Chapters 8, 14, used in 9.5(iv), 11.5); the extensions $\mathcal{O}_\lambda$ (Chapters 6, 7, used in 10.5(ii)); Tam_p, H^1_f, W^D (Chapters 16 to 18, used in `paperlink`s only).

Forward pointers (chapters that use this one): * Chapter 4 (`ch:reps`): Jordan-Holder (7.1(i)) for composition factors of representations; modules over $\mathbb{Z}_\ell$ and $\mathbb{F}_\ell$, lattices, saturation (5.5) and free modules over a DVR (6.3) for G-stable lattices; Hom and $\otimes$ (3.1, 3.2) for $\mathrm{Hom}_G(V,W) = (V^\vee \otimes W)^G$ and ad^0; Nakayama (4.3) for reduction mod ℓ of lattices. * Chapter 6 (`ch:ant`): the DVR criterion (6.1, used at its 2.1, 4.7, 4.9), torsion-free modules over a DVR are free and the rank count (5.4(iii), 6.3, used at its 4.8), localisation (4.5, its 2.1), Noetherian rings (4.1), $\mathbb{Z}_\ell$ (6.4), $R \otimes \mathbb{Z}_\ell = \prod R_{\mathfrak{m}}$ (10.4, its 7.3(iv)), f.g. modules over a PID (5.3, its 1.4, 3.1). * Chapter 7 (`ch:local`): $\mathbb{Z}_\ell$ and $\mathbb{Q}_\ell$ as algebraic objects (6.4), complete DVRs and idempotent lifting (6.1, 10.3), $\mathbb{Q}_\ell/\mathbb{Z}_\ell$ (6.5). * Chapters 8, 14 (`ch:curves`, `ch:moddeg`): perfect and alternating pairings, adjoints and twisted pairings (9.1, 9.4, 9.5) for the cup product on $H^1(X_0(N), \mathbb{Z})$ and $\langle x, w_N y\rangle$; the idempotent decomposition $H^1 \otimes \mathbb{Z}_\ell = \bigoplus H^1_{\mathfrak{m}}$ orthogonal for the pairing (9.5(iii), 10.3).

Chapter 14 only (not Chapter 8, which does not use Gorenstein rings; the users are Chapters 14 and 20, important_notes.md B5): Gorenstein and the freeness criterion (11.4, 11.5) for Remark rem:free. * Chapter 13 (`ch:hecke`): the Hecke algebra as a finite free $\mathbb{Z}$-algebra (10.2, 10.4), the algebra-module duality (9.6), generalised eigenspaces (10.5), the shift operator and non-semisimplicity of U_p (10.7(v)), common eigenvectors (10.8). * Chapter 15 (`ch:modgalrep`): Lemma lem:crit and Lemma lem:old (5.7, 5.8, 9.6, 10.1, 10.8, 11.1). * Chapter 16 (`ch:cohomology`): the snake lemma (2.3), Hom and $\otimes$ (Section 3), the structure of f.g. $\mathbb{Z}_\ell$-modules (5.3, 6.3), closedness of submodules and topological Nakayama (6.6), length and its additivity (7.1, 7.2), Fitting ideals of finite $\mathbb{Z}_\ell$-modules and of direct sums (7.5, 7.6), Pontryagin duality, exactness, $A^{**} = A$, lengths preserved (8.1, 8.2), $W[\ell^k] = T/\ell^k T$ (8.3), $\mathbb{Q}_\ell/\mathbb{Z}_\ell$ injective (6.5(iv)), cyclic ℓ-primary modules (8.4). * Chapter 17 (`ch:galcoh`): $0 \to T \to V \to W \to 0$ and its torsion (8.3), Pontryagin duality (8.2), lengths (7.1). * Chapter 18 (`ch:blochkato`): Fitting ideals over $\mathbb{Z}_\ell$ (7.6) in the definition of Tam_p (Chapter 18, item 7.1) and in Lemma lem:fpr (Chapter 18, items 7.4, 7.5) and Lemma lem:tamfactor (Chapter 18, item 8.2); T, V, W and their duals (8.3); lengths of Selmer groups (7.2, 8.2). * Chapter 20 (`ch:congruence`): everything in Sections 9 to 11: congruence ideals (11.1, 11.2), the ingredients of Lemma lem:ddt417 (11.3, 9.3, 9.4), the computation of the congruence ideal in $\mathbb{T}_\mathfrak{m}$ (11.1(v)), Gorenstein and freeness (11.4, 11.5), the congruence module of a decomposition (5.8), idempotents and generalised eigenspaces for Theorem thm:quant (10.3, 10.5, 10.6). * Chapters 21, 22 (`ch:paper`, `ch:computation`): the card entries cited in Chapter 21's A2.7, A3.2, A4.3, A4.4, A4.10; PARI's `matsnf` (Smith normal form) and `mathnf` as computational counterparts of 5.2, 5.6 (Chapter 22, one sentence).

5.7 6. Notes for the writer

Conventions to fix (Section 1 of the chapter, as `convention` environments): C1 to C10 of Part 1. In addition: * **Three duals.** Use $M^\vee$ for $\mathrm{Hom}_R(M, R)$, M^* for the Pontryagin dual and never use * for the linear dual; the paper's $\mathrm{Hom}(L[\wp], \mathcal{O})$ is $L[\wp]^\vee$, its $H^1_f(\mathbb{Q}_p, T^D)^*$ is Pontryagin. The adjoint of an operator is $A^\dagger$ (9.5; * is the Pontryagin dual only, and a pullback when written after a map: Appendix B, B-C4). * **Orthogonals of non-symmetric pairings.** The paper's pairings are alternating, so left and right orthogonals agree; state 9.2, 9.3 for general perfect pairings with right orthogonals and say this once. * **"Order $\#\mathcal{O}/\Delta$".** The paper's Lemma lem:ddt417 is stated for a general DVR but counts orders; write lengths ($v(\Delta)$, $d\,v(\eta)$) and remark that for finite k this is the paper's count (Part 8, point 4). * $\mathbb{T}_\mathfrak{m}$ always means the factor of $\mathbb{T} \otimes \mathbb{Z}_\ell$ (C7). Do not write $\mathbb{T}_\mathfrak{m}$ for the algebraic localisation of $\mathbb{T}$ at $\mathfrak{m}$, which is not complete and is not a direct factor of anything. * **$\wp \cap I = 0$.** State and prove 11.1(ii) in the two-line form: it holds iff $\eta \neq 0$; do not justify it by "$\mathbb{T}$ is reduced", because the full Hecke algebra (with the U_p) need not be reduced (10.7(v); Part 8, point 1). * **The congruence ideal in $\mathbb{T}_\mathfrak{m}$.** In 11.1(v) say explicitly that the other local factors lie in $\ker \pi$ and that their annihilator component is 0 (the paper's sentence in Lemma lem:hecke says "lie in the annihilator"; Part 8, point 2). * **Fitting versus annihilator versus order.** Over $\mathbb{Z}_\ell$ the Fitting ideal of a finite module is $\#M \cdot \mathbb{Z}_\ell$, not its annihilator; the Tamagawa factor is a Fitting ideal, so it sees the whole order (7.6(iv), Exercise 5). * **Notation $\mathrm{Tam}_p(M) = \mathrm{Fitt}(\dots)^{G_{\mathbb{Q}_p}}$.** In the card and the `paperlink`, write $\mathrm{Fitt}_{\mathbb{Z}_\ell}\big(H^1(I_p, M)^{G_{\mathbb{Q}_p}}_{\mathrm{tors}}\big)$, the Fitting ideal of the invariants, as Chapters 16 and 21 do; the paper's display puts the superscript outside the parentheses (Part 8, point 3). * **Saturation.** Every time a Gram determinant is converted into a cokernel order, say "saturated" and point to 9.3(iv). * **Completeness.** Idempotent lifting and $\mathbb{T} \otimes \mathbb{Z}_\ell = \prod \mathbb{T}_\mathfrak{m}$ need $\mathbb{Z}_\ell$ complete; give the example of 10.3(v). * **Proof levels.** Everything is proved except: Zorn's lemma for maximal ideals in general rings (2.6, referred to Chapter 2), Hilbert's basis theorem (4.1(iv)), flatness of torsion-free modules over a PID in the non-f.g. case (3.4(ii), sketch), the Smith form over a non-Euclidean PID (5.2, sketch), DVR criterion (e) (6.1, stated), Artin-Rees and completions in general (6.4(vii), stated), topological Nakayama (6.6(iv), sketch), finite length iff Noetherian and Artinian (7.1(ii), stated), independence of the Fitting ideal of the presentation in general (7.4, sketch), Pontryagin duality for general compact modules (8.2(v), stated), the Pfaffian identity and the symplectic normal form in general rank (9.4, sketch), Jordan-Chevalley (10.7(iv), stated), complete intersections are Gorenstein and Wiles's criterion (11.4(iv), stated). Never present these as proofs. * **Do not duplicate.** Do not prove Lemma lem:ddt417, Lemma

lem:hecke, Lemma lem:crit or Remark rem:free (Chapters 15, 20, 14); do not prove the analytic theory of $\mathbb{Q}_p$, Hensel's lemma or $\mathbb{Z}_p^\times$ (Chapter 7); do not do G-stable lattices, homothety, Ribet's lemma, Maschke or Brauer-Nesbitt (Chapter 4); do not do Ext, Tor, long exact sequences of complexes or group cohomology (Chapter 16; mention Ext^1 at most in one remark after 5.7); do not do $\mathbb{Z}_3[\zeta_3]$ as a DVR (Chapter 6); do not define Tam_p, H^1_f, T^D (Chapter 18) beyond quoting them in `paperlinks`. * **Coordination with Chapter 4 on Jordan-Holder.** Chapter 3 comes first and proves 7.1(i) for modules over a ring without using commutativity; Chapter 4 should cite it. If the Chapter 4 outline (written in parallel) also proves Jordan-Holder, the orchestrator should keep this proof and turn Chapter 4's into a citation, or keep Chapter 4's and make 7.1(i) here "stated; proved in Chapter 4" with a forward pointer; one proof only. * **Examples from the paper.** Use the paper's curves only where the topic allows: T_3E for a type IV curve as a $\mathbb{Z}_3[\zeta_3]$-module (1.2(c), Exercise 4); the Tamagawa module $\mathbb{Z}/\ell^v \oplus \mathbb{Z}/\ell^m$ (7.7); the level-11 oldforms at level 88 (Exercise 14). Most examples are toy algebras (11.2), which is deliberate: they are the smallest "Hecke algebras with a congruence". * **RHB remarks** (one sentence each): Pontryagin duality as the finite Fourier transform (8.1(vi)); generalised eigenspaces and the Jordan form of a matrix (10.5, 10.7); Gram matrices as metric tensors (9.1). * **Typography** (PLAN.md 7.2): no dash ligatures ("Atiyah-Macdonald", "Jordan-Holder", "Cayley-Hamilton", "Jordan-Chevalley" with plain hyphens; ranges as "1 to 8"); ASCII source (write "Jordan-Holder" without the umlaut, or `H\"older`); labels with the `modules:` infix; the tables of 7.7 and 11.2 with `booktabs`.

References: Atiyah-Macdonald (`AtiyahMacdonald`), Dummit-Foote (`DummitFoote`), Lang (`Lang`), Eisenbud (`Eisenbud`), Serre (`SerreLocalFields`, `SerreCourse`), NSW (`NSW`), Weibel (`Weibel`), Rubin (`Rubin`), DDT (`DDT`), Hida81, Ribet83, Mazur77, Tilouine97 are in `refs.bib`. Not in `refs.bib` (add to `bib/ch03.bib` if cited): Northcott, *Finite free resolutions* (1976); Mazur and Wiles, *Class fields of abelian extensions of* $\mathbb{Q}$, Invent. Math. 76 (1984), Appendix (Fitting ideals); Matsumura, *Commutative ring theory* (Gorenstein rings, Chapter 18); Jacobson, *Basic Algebra I* (Smith form); Bourbaki, *Algebra*, Chapters VII, IX. Theorem numbers recalled but not re-verified against the books (cite them only after checking, otherwise cite the chapter): AM Propositions 2.4, 2.6, Corollaries 2.7, 2.8, Theorem 7.5, Theorem 8.7, Proposition 9.2; DF Section 12.1 Theorems 4 to 6; Weibel Lemma 1.3.2; Eisenbud Section 20.2 and Corollary 7.5. No theorem number is cited in this outline without "check".

5.8 7. Verification log (computations done for this outline)

Python 3.13 with sympy 1.14, scripts `c1.py` to `c5.py` in `/tmp/ch03`. * Smith normal forms and Fitting ideals (`c1.py`): rows $(2,4),(6,8)$: $\mathrm{diag}(2,4)$, Fitt $=(8)$; rows $(2,4,4),(-6,6,12),(10,-4,-16)$: $\mathrm{diag}(2,6,12)$, Fitt $=(144)$; rows $(3,6),(9,3)$: $\mathrm{diag}(3,15)$, Fitt $=(45)$; rows $(4,0),(0,2)$: $\mathrm{diag}(2,4)$; one generator with rows $(2),(4)$: Fitt $=(2)$; rows $(6,0),(0,4),(2,2)$: $\mathrm{diag}(2,2)$, Fitt $=(4)$ (minors $24,12,-8$). Annihilators and the $\mathbb{Z}_2$, $\mathbb{Z}_3$ parts by hand. * Gram determinants (`c2.py`, `c5.py`): the pairing $\left(\begin{smallmatrix}9&1\\1&0\end{smallmatrix}\right)$, $L'=\mathbb{Z}e_1$: $L'^{\perp}=\mathbb{Z}(1,-9)$, index 9; standard pairing and $L'=\mathbb{Z}\cdot 3e_1$: Gram 9, index 3. Random tests: 200 symmetric perfect pairings on $\mathbb{Z}^4$ and 300 non-symmetric perfect pairings on $\mathbb{Z}^n$, $n\le 5$, with random saturated sublattices: $[L : L'\oplus L'^{\perp}] = |\Delta(L')| = |\Delta(L'^{\perp})|$ in every case. * Toy algebras (`c3.py`): $\mathbb{T}_r$ for $\ell=3$, $r=1,2$: Gram of $\lambda(st)$ on $(1,1),(3^r,0)$ is $\left(\begin{smallmatrix}0&1\\1&3^r\end{smallmatrix}\right)$ (determinant -1), $L[\wp]=\mathbb{Z}_3(3^r,0)$ with Gram $3^r=\eta$; the alternating pairing on $\mathbb{T}_2^2$: $\Delta=81=\eta^2$. Socle dimensions of the reductions mod 3: $\mathbb{T}_2$: 1; $\{(a,b,c) : a\equiv b\equiv c \bmod 3\}$: 2, with $\eta=3$ (brute force over small coefficients). $\mathbb{Z}[x]/(x^2-4)$: $I=\mathbb{Z}(x+2)$, $\eta=4$, idempotent $(x+2)/4$. $\mathbb{Z}[x]/(x^2(x-1))$: $\eta=0$ at $x\mapsto 0$, $\eta=1$ at $x\mapsto 1$ (by hand). * Shift operator (`c3.py`): characteristic polynomials X^2-aX+p, $X(X^2-aX+p)$, $X^2(X^2-aX+p)$ for $e=1,2,3$, with v_e cyclic (rank of the Krylov matrix $e+1$); $U(v_0-av_1+pv_2)=0$; the variant $Uv_0=av_0$: characteristic polynomial $X^e(X-a)$, $U(v_0-av_1)=0$ (by hand). * Idempotents (`c4.py`): x^2+x-1 factors mod ℓ as: irreducible ($\ell=2,3$), $(x-2)^2$ ($\ell=5$), $(x-3)(x-7)$ ($\ell=11$), split for $\ell=19,29,31$; Hensel roots mod 11^6: 728020, 1043540; the idempotent $(x-r_2)/(r_1-r_2)$ satisfies $e^2=e \bmod 11^6$ and equals the limit of the Newton iteration $e\mapsto 3e^2-2e^3$ from $(x-3)/4$. * By hand: the two-line proof of 11.1(ii) and of 11.3(ii), (iii) (no use of reducedness); $\mathrm{Ann}(\ker\pi)=\mathrm{Ann}_{\mathbb{T}_{\mathfrak{m}}}(\wp_{\mathfrak{m}})\times 0$ for a product; the $k[\varepsilon]$ counterexample to multiplicativity of Fitt; $\mathrm{Fitt}(\mathbb{Z}[x]/(2,x))^2=(4,2x,x^2)$; the Gorenstein criterion via socles; the proof of 11.5; the claim of 11.6 that a free L with $\eta=0$ has $\Delta=0$. * The paper's proofs of

Lemma lem:crit, Lemma lem:ddt417, Lemma lem:hecke, Remark rem:free and Theorem thm:quant Step 1 were reread sentence by sentence against the items of Sections 5 and 9 to 11; the findings are Part 8.

5.9 8. Points for important_notes.md

New points in the area of this chapter (commutative algebra of Lemmas lem:crit, lem:ddt417, lem:hecke, Remark rem:free, Section 1.2), not among the 32 already recorded. Point 1 refines point 19; the others are new. None changes a result of the paper. Each was checked as stated (Part 7).

1. **Lemma lem:ddt417, proof: the justification of $\wp \cap I = 0$ (refines important_notes.md point 19).** Location: "Since L is torsion-free and $\wp \cap I = 0$, the sum $L[\wp] + L[I]$ is direct." Point 19 proposes the justification "$\mathbb{T}$ is reduced ($\mathbb{T} \otimes \mathbb{Q}$ is a product of fields)". Problem: that justification is not available in general. The full Hecke algebra, which contains the U_p, need not be reduced: on the span of $h(z), h(pz), \ldots, h(p^e z)$ (h a newform of level M, $p \nmid M$) the operator U_p has characteristic polynomial $X^{e-1}(X^2 - a_p X + p)$ and the cyclic vector $h(p^e z)$, so for $e \geq 3$ the eigenvalue 0 has a Jordan block of size $e - 1 \geq 2$ (10.7(v); the paper itself says "U_p is not semisimple on this span when 0 is a repeated eigenvalue"). Example: level 88, $h = f_{11}$, $p = 2$, $e = 3$. So $\mathbb{T} \otimes \mathbb{Q}$ can contain nonzero nilpotents, and Lemma lem:ddt417 is in any case stated for an arbitrary finite free $\mathbb{T}$. Fix: the correct and elementary statement is "$\wp \cap I = 0$ if and only if $\eta \neq 0$" (11.1(ii)): if $\eta = 0$ there is nothing to prove; otherwise pick $t \in I$ with $\pi(t)$ generating η, and for $x \in \wp \cap I$ write $\pi(t)x = tx + (\pi(t) - t)x$, where $tx \in I\wp = 0$ and $(\pi(t) - t)x \in \wp I = 0$; so $\pi(t)x = 0$ and $x = 0$. Two further remarks for the same proof: (a) the step "$(\pi(t) - t)x \in L[I]$ (as $s(\pi(t) - t) \in \wp \cap I = 0$ for $s \in I$)" needs only $I\wp = 0$, since $\pi(t) - t \in \wp$; (b) the directness of $L[\wp] + L[I]$ needs only $\pi(t) \neq 0$: an element killed by $\wp + I \ni \pi(t)$ is killed by $\pi(t)$. The equality clause uses $\mathrm{Ann}_{\mathbb{T}}(I) = \wp$ (for $\mathbb{T}[I] = \wp$ in "$Q \simeq (\mathbb{T}/(\wp + I))^d$"), which also holds exactly when $\eta \neq 0$ (if $xI = 0$ then $\pi(x)\pi(t) = 0$). Also, if $\eta = 0$ and L is free over $\mathbb{T}$, then $\Delta = 0$, so the hypothesis $\Delta \neq 0$ excludes that case. In the applications $\eta \neq 0$ because the f-factor of $\mathbb{T} \otimes K$ is K (multiplicity one for the newform, or Step 1 of Theorem thm:quant for g^S). Confidence: high (the algebra is two lines; the Jordan block was checked symbolically).

2. **Lemma lem:hecke, proof: "the other local factors of $\mathbb{T} \otimes \mathbb{Z}_\ell$ are killed by ϕ_f and lie in the annihilator".** Problem: as written this is false. With $\mathbb{T} \otimes \mathbb{Z}_\ell = \mathbb{T}_\mathfrak{m} \times \mathbb{T}''$ and ϕ_f factoring through $\mathbb{T}_\mathfrak{m}$, one has $\ker \phi_f = \wp_\mathfrak{m} \times \mathbb{T}''$ and $\mathrm{Ann}(\ker \phi_f) = \mathrm{Ann}_{\mathbb{T}_\mathfrak{m}}(\wp_\mathfrak{m}) \times 0$: the factor $\mathbb{T}''$ lies in the kernel, and its component of the annihilator is 0 (an element $(0, b)$ with $b \neq 0$ does not kill $(0, 1_{\mathbb{T}''}) \in \ker \phi_f$). The conclusion, that the congruence ideal is computed in $\mathbb{T}_\mathfrak{m}$, is right: $\phi_f(\mathrm{Ann}(\ker \phi_f)) = \phi_f(\mathrm{Ann}_{\mathbb{T}_\mathfrak{m}}(\wp_\mathfrak{m}))$. Fix: "since ϕ_f kills the other local factors, $\ker \phi_f = \wp_\mathfrak{m} \times \mathbb{T}''$ and $\mathrm{Ann}(\ker \phi_f) = \mathrm{Ann}_{\mathbb{T}_\mathfrak{m}}(\wp_\mathfrak{m}) \times 0$, so the congruence ideal is computed in $\mathbb{T}_\mathfrak{m}$" (11.1(v)). Confidence: high (expository slip; no effect).

3. **Lemma lem:hecke, proof: the pairing on the $\mathfrak{m}$-component.** Location: "Apply Lemma lem:ddt417 to ... $L = H^1(X_0(N), \mathbb{Z}_\ell)_\mathfrak{m}$ with the pairing $\hat{\delta}_!$ of Section sec:dfg, perfect and alternating with $\mathbb{T}$ self-adjoint". Problem (small gap): Section 4.1 says $\hat{\delta}_!$ is perfect on $H^1(X_0(N), \mathbb{Z}_\ell)$, not on its $\mathfrak{m}$-component. The restriction is perfect because the idempotent $e_\mathfrak{m} \in \mathbb{T} \otimes \mathbb{Z}_\ell$ is self-adjoint (it is a limit of polynomials in self-adjoint operators, or: idempotents of a commutative algebra of self-adjoint operators are self-adjoint), so $H^1 \otimes \mathbb{Z}_\ell = e_\mathfrak{m} H^1 \oplus (1 - e_\mathfrak{m}) H^1$ is an orthogonal decomposition and a perfect pairing restricts to a perfect pairing on each summand (9.5(iii)). Also $L[\wp]$, computed in L, is the lattice $M_{f,\lambda}$ of the whole H^1, since $M_{f,\lambda}$ is killed by $\ker \phi_f \ni 1 - e_\mathfrak{m}$. The same remark applies to Step 2 of the proof of Theorem thm:quant. Fix: add one sentence. Confidence: high (the argument is standard; the omission is minor).

4. **Section 1.2 (and the proof of Lemma lem:tamfactor): the display $\mathrm{Tam}_p(M) = \mathrm{Fitt}_{\mathbb{Z}_\ell}\big(H^1(I_p, M)_{\mathrm{tor}}\big)^{G_{\mathbb{Q}_p}}$.** Problem (notation): the superscript $G_{\mathbb{Q}_p}$ is outside the parentheses, so the display reads as "the $G_{\mathbb{Q}_p}$-invariants of an ideal of $\mathbb{Z}_\ell$". What is meant, and what the proof of

Lemma lem:tamfactor computes ("the Fitting ideal of the G-invariants of the torsion of $H^1(I, M)$"), is $\mathrm{Fitt}_{\mathbb{Z}_\ell}\big(H^1(I_p, M)_{\mathrm{tor}}^{G_{\mathbb{Q}_p}}\big)$. Fix: move the superscript inside the parentheses. Confidence: high that the display is misbracketed (the LaTeX source has `\bigr)^{G_{\Q_p}}`); importance low. Chapters 16 and 21 already write the corrected form.

5. **Lemma lem:ddt417, statement and proof: orders for a general DVR.** Location: "Let $\mathcal{O}$ be a discrete valuation ring ..."; "of order $\#\mathcal{O}/\Delta$", "$\#\mathcal{O}/\Delta \le \#(\mathcal{O}/\eta)^d$". Problem (expository): for a DVR with infinite residue field these orders are infinite and the inequality is meaningless; the argument is right with lengths: $\mathrm{length}_{\mathcal{O}} Q = v(\Delta)$ (9.3), and Q, being generated by d elements and killed by η, is a quotient of $(\mathcal{O}/\eta)^d$, so $v(\Delta) \le d\, v(\eta)$. In the paper's applications $\mathcal{O} = \mathbb{Z}_\ell$, where the two formulations agree. Fix: replace orders by lengths, or assume the residue field finite. Confidence: high; importance low.

Checked and found correct (recorded so that later chapters need not recheck): the paper's Lemma lem:crit argument "$\mathbb{T} \otimes \mathbb{Q} = \mathbb{Q} \times \mathbb{T}'$, $I = \mathbb{T} \cap (0 \times \mathbb{T}')$, $J = \mathbb{T} \cap (\mathbb{Q} \times 0)$, $J = \mathrm{Ann}_{\mathbb{T}} I$" (11.1(iii)); "dualising the inclusion ... index r'" (5.7, 9.6); "embeds in $\mathbb{Q}f/\mathbb{Z}f$, so it is cyclic ... $r'h = f + g_0$" (5.8); the forms killed by I are $\mathbb{Z}f$ because $a_1(f) = 1$; "Every step holds verbatim over $\mathbb{Z}_\ell$" (3.4(iii)); Lemma lem:ddt417's "$L[\wp]$ is saturated" and "$L[\wp]^\perp = L[I]$" (11.3); Section 4.1's "Gram determinant $(\eta^\Sigma)^2$ up to a unit" for an alternating rank-two lattice whose ideal of values is η^Σ (9.4(ii)); Lemma lem:IV's "free of rank one over $\mathbb{Z}_3[\zeta_3]$" (5.4(iii)); Theorem thm:quant Step 1's projection by the idempotent (10.6); the length identity in the proof of Theorem thm:main (7.1(iii), 8.1(v)); "$H^1(\mathbb{R}, T^D)$ is killed by 2, so $H^1_f(\mathbb{R}, T^D) = 0$ for ℓ odd" (6.4(iii)).

Chapter 6

Chapter 04: Representations of finite groups and the mod l world (`ch:reps`, `chapters/ch04_reps.tex`)

Stage 1 line-up. Part I (Algebra). Target length 2,120 lines of LaTeX (card excluded; see Part 3), inside the 1,000 to 2,200 range of PLAN.md Section 7.3.

Rigour tags: **P** = proved here, **S** = proof sketched here, **R** = stated; reference. Item format, as in the Chapter 1, 2 and 5 outlines: number, label-ready name, kind, tag; the precise statement with all hypotheses; reference (book and chapter or section; a theorem number only where sure); the paper's use by its `\label` names, or "background". Every computation quoted below was checked by hand and by short Python/sympy scripts in `/tmp/ch04` (verification log, Part 7).

6.1 0. What this chapter is for

The paper uses representation theory in eight places, and the chapter is built around them.

1. **Proposition prop:ns.** For the image $N_s(\mathbb{F}_3)$: the determinant-one subgroup C_4 acts on $\mathbb{F}_9^2$ as $\psi \oplus \psi^{-1}$ with ψ of order 4; Frobenius reciprocity gives $\bar\rho \otimes \mathbb{F}_9 \simeq \operatorname{Ind}\psi$; an induced representation is invariant under the twist by the character of the index-two subgroup, so $\bar\rho \simeq \bar\rho \otimes \chi_{-3}$; the other two images ($\mathrm{GL}_2(\mathbb{F}_3)$, $N_{ns}(\mathbb{F}_3)$) have absolutely irreducible determinant-one subgroups ($\mathrm{SL}_2(\mathbb{F}_3)$, Q_8). The chapter proves all of this as statements about the standard representations of the subgroups of $\mathrm{GL}_2(\mathbb{F}_3)$ (Section 8), using Chapter 1 for the group theory.
2. **Lemma lem:h0.** "Trace-zero $G_{\mathbb{Q}}$-maps $\bar\rho \to \bar\rho \otimes \bar\chi_\ell$" (that is, $H^0(\mathrm{ad}^0\,\bar\rho \otimes \bar\chi_\ell)$ as $\mathrm{Hom}_G(V, V\otimes\chi)$ with trace zero, and $\mathrm{Hom}_G(V,W) = (V^\vee \otimes W)^G$); Schur ("a nonzero such map is an isomorphism"); the determinant comparison; the trace-zero lemma ($\operatorname{tr}\phi = \chi(g)\operatorname{tr}\phi$); Clifford theory for an index-two subgroup ("$\bar\rho \simeq \bar\rho\otimes\chi_{-3}$ iff induced iff not absolutely irreducible on the subgroup"); the length $h_\ell(E)$ as "the largest k with $\rho \bmod \ell^k \simeq (\rho\otimes\chi_\ell) \bmod \ell^k$", which is a statement about lattices (Section 10).
3. **Lemma lem:IV, Remark rem:types, Lemma lem:euler.** Cyclic groups of order $2, 3, 4, 6$ and (minus) unipotent elements acting on a rank-two space or lattice: eigenvalues, fixed spaces on V and on $\mathrm{ad}^0 V$, the commutant $k[\sigma]$ and its trace-zero line, reduction modulo ℓ (an element of order 3 in characteristic 0 becomes a nontrivial unipotent modulo 3), the failure of Maschke's theorem behind "$E[3]^{I_q}$ is a line while $T^{I_q} = 0$".
4. **Lemma lem:old.** Brauer-Nesbitt in characteristic ℓ (characteristic polynomials, not traces), the absolute irreducibility needed to pass from equal semisimplifications to an isomorphism over a common finite field (important_notes.md point 10), and "a non-split extension of two characters has a unique stable line".

5. **Section 4.2 and Lemma lem:fpr.** $A = \mathrm{ad}^0 T$; the trace form and the self-duality of $\mathrm{ad}^0 T$ for ℓ odd, so $T^D = \mathrm{Hom}(A, \mathbb{Z}_\ell(1)) \simeq A(1)$; homothetic lattices (and dual lattices, point 26) have the same adjoint; the homothety lemma ("all stable lattices homothetic when the reduction is irreducible").
6. **Lemma lem:tamfactor.** The basis $e = e_1 \otimes e_2^\vee$, h, f of $\mathrm{ad}^0 T$ and the formulas for the adjoint action of a unipotent and of an upper triangular matrix.
7. **Section 1.1 and the Introduction.** $A_f = \mathrm{ad}^0 V$, the eigenvalues $\alpha/\beta, 1, \beta/\alpha$ of ad^0 of a matrix with eigenvalues α, β, and $\mathrm{ad}^0 \rho \simeq \mathrm{Sym}^2 \rho \otimes \det^{-1}$, which underlies $L_p(\mathrm{Sym}^2 f, s) = L_p(A_f, s-1)$.
8. **The hypothesis of Theorem thm:main** ("absolutely irreducible on $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{\ell^*}))$"), S_f, and DDT Theorem 3.15: absolute irreducibility, the distinct-eigenvalue criterion (complex conjugation), restriction to an index-two subgroup, the dihedral case of Dickson's classification read as "induced".

Split with the neighbouring chapters (decided here; each decision is repeated in Part 5 and Part 6): * **Chapter 1** proves the group theory of $\mathrm{GL}_2(\mathbb{F}_3)$: the models J, σ, w, u, a, the 55 subgroups, orbits on lines, the three images of its 6.7, unipotent and minus-unipotent elements as group elements (its 5.6), the commutator argument of its 5.7, $1 + \ell M_2(\mathbb{Z}_\ell)$ torsion-free (its 7.4), Dickson for $\mathrm{GL}_2(\mathbb{F}_\ell)$ (its 5.10). Chapter 4 cites these and adds only the representation-theoretic reading (Section 8 and 9). The alternative proof of absolute irreducibility for Q_8 by Maschke (item 3.7) is new here and complements Chapter 1's 5.7. * **Chapter 2** supplies $\mathbb{F}_9 = \mathbb{F}_3[i]$, finite fields, orthogonality for characters of finite abelian groups and Dirichlet characters, and $a^2 \equiv 1 \pmod{\ell^k}$ for all units iff $\ell^k = 3$ (its 5.5). * **Chapter 3** (being outlined in parallel; PLAN.md split) supplies modules over rings, Hom and $\otimes$, Nakayama, DVRs and $\mathbb{Z}_\ell$, finitely generated modules over a DVR (torsion-free implies free; cyclic iff the λ-torsion has length at most one), saturation, perfect pairings and Gram determinants, Pontryagin duality, generalised eigenspaces of a single operator. Chapter 4 uses these for $\mathcal{O}[G]$-lattices and does not redo them. **Jordan-Holder** is proved in Chapter 3 (for modules of finite length over any ring, where it is first needed for lengths over a local ring) and is cited here (item 3.4), which adds only the reading for $k[G]$- and $\mathcal{O}[G]$-modules (composition factors of representations): decision B2 of important_notes.md, PLAN.md 7.5. * **Chapter 5** proves the profinite and Galois forms: continuity, existence of stable lattices for compact images (its galois:continuous-reps (c)), Chebotarev plus Brauer-Nesbitt (its galois:chebotarev-brauer-nesbitt, galois:congruent-ap), the cyclotomic determinant argument (its galois:twist (c)), complex conjugation (its galois:complex-conjugation (d)), restriction to $G_{\mathbb{Q}(\sqrt{\ell^*})}$ (its galois:index-two-restriction). **Decision:** Chapter 4 proves, for an arbitrary group, Brauer-Nesbitt (5.2), the distinct-eigenvalue criterion (2.4), Clifford theory for index two (6.6), the trace-zero lemma (7.2), independence of $\bar{\rho}^{ss}$ from the lattice and the homothety lemma (10.2, 10.3); Chapter 5 cites them and adds only continuity and the Galois input. In particular Chapter 5's galois:reduction-ss and galois:complex-conjugation (d) become one-line applications of 10.2, 10.3 and 2.4 (the Chapter 5 writer should cite, not reprove). * **Chapter 6** proves the arithmetic of $\mathbb{Z}_\ell[\zeta_3]$ and $\mathbb{Z}_\ell[i]$ (its 4.7, 4.8, 4.9), the freeness of a lattice over $\mathbb{Z}_\ell[\sigma]$ and the Frobenius action $\chi_{-3}(q)$, $\chi_{-4}(q)$ on the invariant line (its 4.10 (iii) to (v)), and the twist detector by traces at inert primes (its 6.4). **Decision:** Chapter 4 proves the commutant lemma and the trace-zero line (9.2), which Chapter 6's 4.10 (i), (ii) cites ("Chapter 4 or linear algebra"), and states the $\mathbb{Z}_\ell[\zeta_3]$-module structure with a forward pointer to Chapter 6 (9.4 (v)); it gives an elementary proof of the mod 3 unipotence of an order-3 element (9.4 (ii)) that does not need the DVR. * **Chapter 11** proves the paper's Lemma lem:IV, Remark rem:types, Proposition prop:ns and Lemma lem:h0 as statements about elliptic curves; Chapter 4 supplies their representation-theoretic cores (8.5, 9.4, 10.6) and never mentions Kodaira types except in `paperlink` environments and in the shape table of 9.3, where the assignment type to shape is a black box from Chapters 10 and 11. * **Chapter 18** uses the trace form and the self-duality of $\mathrm{ad}^0 T$ (7.6) for $T^D = A(1)$ and adds the Tate twist and Pontryagin-dual statements; **Chapter 19** uses $\mathrm{ad}^0 \simeq \mathrm{Sym}^2 \otimes \det^{-1}$ (7.5) for the L-function normalisations.

6.2 1. Global conventions fixed in this chapter

- **C1 (representations).** G is a group (finite unless said otherwise; many statements hold for any group and are marked "any G"), k a field, R a commutative ring. A representation is a

homomorphism $\rho\colon G \to \mathrm{GL}(V)$ with V a finite-dimensional k-vector space (or a free R-module of finite rank), i.e. a left $k[G]$-module of finite dimension. Matrices act on column vectors on the left (Chapter 1, C1). "G-map" = "intertwiner" = $k[G]$-linear map; $\mathrm{Hom}_G(V,W)$; $V \simeq W$ means isomorphic as representations.

- **C2 (duals).** The linear dual is $V^\vee = \mathrm{Hom}_k(V,k)$ with $(g\lambda)(v) = \lambda(g^{-1}v)$, i.e. the matrix $\rho(g^{-1})^t$ in the dual basis $e_i^\vee$. The symbol * is reserved for Pontryagin duals (Chapter 3, the paper's Lemma lem:dfg21) and is not used for linear duals; PLAN.md's "$V^* \otimes W$" is written $V^\vee \otimes W$.
- **C3 (adjoints).** $\mathrm{End}_k(V) = V \otimes V^\vee$ with $g \cdot x = \rho(g)x\rho(g)^{-1}$. The full adjoint is written $\mathrm{End}(V)$ (or $\mathrm{Ad}\,V$, macro `\Ad`); the trace-zero adjoint is the paper's $\mathrm{ad}^0 V$ (macro `\ad`, which already contains the superscript 0: never write `\ad`). $\mathrm{Sym}^2 V$ (macro `\Sym`, which already contains the exponent 2) is the symmetric square. Matrix units $E_{ij} = e_i \otimes e_j^\vee$; for $\dim V = 2$ the basis of $\mathrm{ad}^0 V$ is the paper's $e = E_{12} = e_1 \otimes e_2^\vee$, $h = E_{11} - E_{22}$, $f = E_{21} = e_2 \otimes e_1^\vee$ (as in Chapter 16).
- **C4 (characters and twists).** A *character* of G with values in R is a homomorphism $\chi\colon G \to R^\times$; $R(\chi)$ is R with G acting through χ, and $M(\chi) = M \otimes_R R(\chi)$, $\rho \otimes \chi$. In Section 4 only, "the character of V" is the trace function, written tr_V (never χ_V, to avoid a clash with the paper's χ_{-3}, χ_ℓ). Galois Tate twists $M(1) = M(\chi_\ell)$ are Chapter 5's and 18's notation.
- **C5 (fields).** $\bar{k}$ is an algebraic closure; $V_K = K \otimes_k V$. $\mathbb{F}_9 = \mathbb{F}_3[i]$ with $i^2 = -1$ (Chapter 2). "Absolutely irreducible" means $V_{\bar{k}}$ irreducible (2.3).
- **C6 (standard models).** $J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, $\sigma = \mathrm{diag}(1,-1)$, $w = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $u = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, $a = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$ and the subgroups $C_s, N_s, C_{ns}, N_{ns}, Q_8, C_4 = \langle J \rangle, \mathrm{B}, \mathrm{U}$ of $\mathrm{GL}_2(\mathbb{F}_3)$ are those of Chapter 1, 6.2. The paper's generator $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ is J^{-1}. $\psi\colon C_4 \to \mathbb{F}_9^\times$ is $\psi(J) = i$. **Clash:** the matrix $\sigma = \mathrm{diag}(1,-1)$ of Chapter 1 and the generator σ of inertia of order 3 in the paper (Lemmas lem:IV, lem:euler). In Sections 8 the matrix keeps Chapter 1's name; in Section 9 the order-3 element is written σ as in the paper, and the matrix $\mathrm{diag}(1,-1)$ is written h (it is the basis vector h of $\mathrm{ad}^0 V$ viewed as a matrix). Say this in a `convention` environment.
- **C7 (lattices).** $\mathcal{O}$ is a discrete valuation ring with uniformiser λ, fraction field K, residue field $k = \mathcal{O}/\lambda$; in the paper $\mathcal{O} = \mathbb{Z}_\ell$ (or the ring of integers of a finite extension of $\mathbb{Q}_\ell$), $\lambda = \ell$. A lattice in a K-vector space V of dimension n is a finitely generated $\mathcal{O}$-submodule spanning V (free of rank n, Chapter 3). Two lattices L, L' are homothetic if $L' = cL$ for some $c \in K^\times$ (equivalently $L' = \lambda^m L$, $m \in \mathbb{Z}$).
- **C8 (no Frobenius here).** The chapter never uses Frobenius elements. Where the paper's Frobenius appears (9.1 (iv), 9.3) it is an abstract $F \in \mathrm{GL}(V)$ with $FgF^{-1} = g^q$ (the arithmetic convention of Chapter 5, C2, and Chapter 7); the geometric Frobenius F^{-1} gives the inverse eigenvalues (important_notes.md point 11).
- **C9 (typography).** PLAN.md Section 7.2: no dash ligatures, ASCII source, labels `...:reps:...`; "Jordan-Holder", "Brauer-Nesbitt", "Frobenius reciprocity" with plain hyphens; tables with `booktabs`.

6.3 2. Line-up of the core results, in teaching order

6.3.1 Section 1. Representations over a field

1.1 `def:reps:representation` (definition and examples; P). Let G be a group and k a field. A representation of G over k is a homomorphism $\rho\colon G \to \mathrm{GL}(V)$, V a finite-dimensional k-vector space; equivalently a $k[G]$-module structure on V. After a choice of basis, $\rho\colon G \to \mathrm{GL}_n(k)$; two matrix representations are isomorphic iff they are conjugate by one $P \in \mathrm{GL}_n(k)$. Degree $n = \dim V$, kernel, faithful. Examples: the trivial representation $\mathbf{1}$; one-dimensional representations = characters $\chi\colon G \to k^\times$ (they factor through G^{ab}); the permutation representation $k[X]$ of a G-set X, the regular representation

$k[G]$; the standard representation of a subgroup $H \leq \mathrm{GL}_2(k)$ on k^2; inflation along a homomorphism $G \to G'$ (so a representation of a finite quotient of a profinite group is one of the group, Chapter 5). RHB link (one sentence): RHB's representations are the case $k = \mathbb{C}$, G finite. Reference: Serre, Linear representations, Chapter 1; Dummit-Foote Section 18.1. Paper: background (V, T, ρ, $\bar{\rho}$ of Section 1.1 are representations of $G_{\mathbb{Q}}$).

1.2 `def:reps:constructions` (definition and proposition; P). Subrepresentation (a G-stable subspace), quotient, direct sum, tensor product ($g(v \otimes w) = gv \otimes gw$), dual $V^\vee$ (C2), $\mathrm{Hom}_k(V, W)$ with $(g\phi)(v) = g\,\phi(g^{-1}v)$, invariants V^G, determinant $\det \rho = \wedge^n \rho$. Facts: $\mathrm{tr}(g \mid V^\vee) = \mathrm{tr}(g^{-1} \mid V)$; $\mathrm{tr}(g \mid V \otimes W) = \mathrm{tr}(g \mid V)\,\mathrm{tr}(g \mid W)$; $\det(V \otimes \chi) = \chi^n \det V$; $V^{\vee\vee} \simeq V$; $(V \otimes W)^\vee \simeq V^\vee \otimes W^\vee$. For $\dim V = 2$: the alternating form $\omega(x, y) = \det[x \,|\, y]$ satisfies $\omega(gx, gy) = \det(g)\omega(x, y)$, so $v \mapsto \omega(v, -)$ is an isomorphism $V \otimes \det^{-1} \simeq V^\vee$, i.e. $V^\vee \simeq V \otimes \det^{-1}$. Reference: Serre, Linear representations, Sections 1.4 to 1.6. Paper: Section 4.2 (dual lattices, important_notes.md point 26; via 7.7), lem:tamfactor ($e_2^\vee$).

1.3 `prop:reps:homG` (proposition; P). For finite-dimensional V, W (or free R-modules of finite rank) the map $V^\vee \otimes W \to \mathrm{Hom}(V, W)$, $\lambda \otimes w \mapsto (v \mapsto \lambda(v)w)$, is a G-isomorphism, and $\mathrm{Hom}_G(V, W) = \mathrm{Hom}(V, W)^G = (V^\vee \otimes W)^G$. In particular $\mathrm{End}(V)^G = \mathrm{End}_G(V)$ and, for a character χ, $(\mathrm{End}(V) \otimes \chi)^G = \mathrm{Hom}_G(V, V \otimes \chi)$. Reference: Serre, Section 2.2 (in characteristic 0; the proof is characteristic-free). Paper: Lemma lem:h0 ("the ℓ-torsion of $H^0(\mathbb{Q}, W^D)$ is $H^0(\mathbb{Q}, \mathrm{ad}^0\,\bar{\rho} \otimes \bar{\chi}_\ell)$, the space of trace-zero $G_{\mathbb{Q}}$-maps"), Proposition prop:ns ($H^0(\mathbb{Q}, \mathrm{ad}^0\,\bar{\rho}(1)) \neq 0$), Lemma lem:tamfactor (the basis $e_1 \otimes e_2^\vee$).

1.4 `prop:reps:scalars` (proposition; P for (i) to (iii), R for (iv)). Let K/k be a field extension. (i) $\mathrm{Hom}_{K[G]}(V_K, W_K) = K \otimes_k \mathrm{Hom}_{k[G]}(V, W)$; in particular $\dim_K \mathrm{Hom}_{K[G]}(V_K, W_K) = \dim_k \mathrm{Hom}_{k[G]}(V, W)$ and $(V_K)^G = K \otimes_k V^G$ (the G-maps are the solutions of a homogeneous linear system with coefficients in k). (ii) If V, W are irreducible over k and $V_K \simeq W_K$, then $V \simeq W$ (a nonzero K-map gives a nonzero k-map, which is an isomorphism by Schur 2.2(i)). (iii) If V is absolutely irreducible (2.3) and W is any representation with $W_K \simeq V_K$, then $W \simeq V$ (same argument: W_K is irreducible, so is W). (iv) (Noether-Deuring) For arbitrary finite-dimensional V, W, $V_K \simeq W_K$ implies $V \simeq W$. Reference: Curtis-Reiner, Section 29 (add `CurtisReiner` to `bib/ch04.bib`). Paper: Proposition prop:ns (the isomorphism $\bar{\rho} \otimes \mathbb{F}_9 \simeq \mathrm{Ind}\,\psi$ lives over $\mathbb{F}_9$; $\bar{\rho} \simeq \bar{\rho} \otimes \chi_{-3}$ descends to $\mathbb{F}_3$ by (ii); see Part 8, point N1), Lemma lem:old ("over a common finite field", important_notes.md point 10).

6.3.2 Section 2. Irreducibility, Schur's lemma, absolute irreducibility

2.1 `def:reps:irreducible` (definition and examples; P). $V \neq 0$ is irreducible (simple) if its only subrepresentations are 0 and V; indecomposable if it is not a direct sum of two nonzero subrepresentations; reducible otherwise. Irreducible implies indecomposable, not conversely (3.3). For $\dim V = 2$: V is irreducible iff no line of k^2 is G-stable iff $\rho(G)$ lies in no Borel subgroup of $\mathrm{GL}_2(k)$ (Chapter 1, 5.5). Examples: every one-dimensional representation; the standard representation of $\mathrm{GL}_2(\mathbb{F}_q)$, of $N_s(\mathbb{F}_3)$, of $C_4 = \langle J \rangle$ on $\mathbb{F}_3^2$ (irreducible: $X^2 + 1$ has no root in $\mathbb{F}_3$); U $= \langle u \rangle$ and B on $\mathbb{F}_3^2$ are reducible. Reference: Serre, Section 1.4. Paper: Proposition prop:ns ("acting irreducibly on $\mathbb{F}_3^2$"), Theorem thm:qual and Lemma lem:old (hypothesis "$\bar{\rho}$ irreducible").

2.2 `thm:reps:schur` (theorem, Schur's lemma in both forms; P). Let V, W be irreducible representations of any group G over any field k. (i) Every G-map $V \to W$ is 0 or an isomorphism; $D = \mathrm{End}_G(V)$ is a division algebra of finite dimension over k. (ii) If k is algebraically closed, $\mathrm{End}_G(V) = k$, and $\dim \mathrm{Hom}_G(V, W) = 1$ if $V \simeq W$, 0 otherwise. (iii) The conclusions of (ii) hold over any k when V and W are absolutely irreducible (by 1.4(i)). (iv) Consequences: if V is absolutely irreducible, the centre $Z(G)$ acts through scalars (a central character); every absolutely irreducible representation of an abelian group is one-dimensional; for an irreducible V of an abelian group, $\mathrm{End}_G(V)$ is a field. Reference: Serre, Section 2.2 (Proposition 4, characteristic 0); Lang, Chapter XVII, Section 1. Paper: Lemma lem:h0 ("a nonzero such map is an isomorphism, $\bar{\rho}$ being irreducible"; and the one-dimensionality of $\mathrm{Hom}_G(\bar{\rho}, \bar{\rho} \otimes \bar{\chi}_\ell)$ behind "H^0 is cyclic"), Proposition prop:ns ("the third is abelian, hence absolutely reducible"), Lemma lem:old.

2.3 `def:reps:absirr` (definition and proposition; P for (i) to (iii), S for (iv)). V is absolutely irreducible if $V_{\bar{k}}$ is irreducible. (i) V is absolutely irreducible iff the map $k[G] \to \mathrm{End}_k(V)$ is surjective (the image of $\bar{k}[G]$ is $\bar{k} \otimes_k$ (image of $k[G]$)), and over $\bar{k}$ irreducibility is equivalent to surjectivity by Burnside's theorem 5.1(b); the writer may move 5.1 into this section); hence V_K is irreducible for every extension K/k.

(ii) Absolutely irreducible implies irreducible and $\mathrm{End}_G(V) = k$. (iii) In dimension 2: V is absolutely irreducible iff $\rho(G)$ fixes no line of $\bar{k}^2$, iff $\rho(G)$ is not conjugate in $\mathrm{GL}_2(\bar{k})$ into the upper triangular group. (iv) If k is finite or of characteristic 0 (perfect), an irreducible V is absolutely irreducible iff $\mathrm{End}_G(V) = k$. (Sketch: over a perfect field $V_{\bar{k}}$ is semisimple (stated, Curtis-Reiner Section 29); a semisimple module with endomorphism ring $\bar{k}$ is irreducible; and $\mathrm{End}_G(V_{\bar{k}}) = \bar{k} \otimes \mathrm{End}_G(V)$ by 1.4(i).) Reference: Curtis-Reiner, Section 29; Serre, Section 12.1; Lang, Chapter XVII, Section 3. Paper: the hypothesis of Theorem thm:main, Theorem thm:qual, DDT Theorem 3.15 and S_f ("absolutely irreducible when restricted to $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{\ell^*}))$"), Proposition prop:ns, Lemma lem:h0.

2.4 `prop:reps:distinct-eigenvalues` (proposition, the distinct-eigenvalue criterion; P). Let G be any group, V a representation over k of dimension n, and suppose some $g \in G$ (or some element of the image of $k[G]$) acts on V with n distinct eigenvalues in k. Then every G-stable subspace of $V_{\bar{k}}$ is a sum of eigenlines of g, all of which are defined over k; hence it is $U_{\bar{k}}$ for a G-stable subspace $U \subset V$. In particular V is irreducible iff it is absolutely irreducible. Special case ($n = 2$, $\mathrm{char}\, k \neq 2$): if some $c \in G$ has $\rho(c)^2 = 1$ and $\det \rho(c) = -1$ (eigenvalues $1 \neq -1$), irreducible and absolutely irreducible coincide. Proof: a subspace stable under a diagonalisable operator with distinct eigenvalues is the sum of the eigenlines it meets (Vandermonde). Reference: elementary. Paper: Lemmas lem:h0 and lem:old ("irreducible" there must mean absolutely irreducible; for $\bar{\rho}_{E,\ell}$, ℓ odd, complex conjugation supplies c: important_notes.md point 10), Proposition prop:ns ("contain an element of determinant -1 and trace 0"); applied to $G_{\mathbb{Q}}$ in Chapter 5 (galois:complex-conjugation (d)) and Chapter 11, which cite this item.

2.5 `ex:reps:c4` (worked example; P). $C_4 = \langle J \rangle$ acting on $\mathbb{F}_3^2$: irreducible (2.1), not absolutely irreducible. $\mathrm{End}_{C_4}(\mathbb{F}_3^2) = \mathbb{F}_3[J] \simeq \mathbb{F}_3[X]/(X^2+1) \simeq \mathbb{F}_9$, a field of dimension 2 (consistent with 2.3(iv) and 2.2(iv)). Over $\mathbb{F}_9 = \mathbb{F}_3[i]$: $J(1,-i)^t = i(1,-i)^t$ and $J(1,i)^t = -i(1,i)^t$, so $\mathbb{F}_9^2 = \psi \oplus \psi^{-1}$ with $\psi(J) = i$, a character of order 4; $\psi^{-1} = \psi^3$ is the Frobenius conjugate of ψ (the two summands are exchanged by $x \mapsto x^3$ on coordinates). Contrast: $\mathrm{U} = \langle u \rangle$ on $\mathbb{F}_3^2$ fixes exactly one line and is not a direct sum (3.3); $C_{ns}(\mathbb{F}_3) \simeq \mathbb{F}_9^\times$ is irreducible, not absolutely irreducible, with $\mathrm{End} = \mathbb{F}_9$. Reference: computation (Part 7). Paper: Proposition prop:ns ("$\bar{\rho}|_{\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{-3}))} \otimes \mathbb{F}_9 = \psi \oplus \psi^{-1}$ for a character ψ of order 4").

6.3.3 Section 3. Semisimplicity, Maschke, Jordan-Holder

3.1 `prop:reps:semisimple` (definition and proposition; P). For a $k[G]$-module V of finite dimension (any G) the following are equivalent: (a) V is a sum of irreducible subrepresentations; (b) V is a direct sum of irreducible subrepresentations; (c) every subrepresentation has a G-stable complement. Such V is *semisimple*; subrepresentations and quotients of semisimple representations are semisimple. Isotypic decomposition $V = \bigoplus_S V_S$ over the isomorphism classes S of irreducibles, $V_S \simeq S^{m_S}$, m_S the multiplicity; if S is absolutely irreducible, $m_S = \dim \mathrm{Hom}_G(S, V)$. Reference: Lang, Chapter XVII, Section 2; Serre, Section 2.6. Paper: background (Brauer-Nesbitt, semisimplifications).

3.2 `thm:reps:maschke` (theorem, Maschke; P). (i) Let G be finite and $\mathrm{char}\, k \nmid \#G$. Then every finite-dimensional representation of G over k is semisimple. Proof: if π is a k-linear projection onto a stable U, then $\pi' = \frac{1}{\#G} \sum_g g\pi g^{-1}$ is a G-equivariant projection onto U. (ii) (Integral form.) Let R be a commutative ring with $\#G \in R^\times$ and M an $R[G]$-module. Then $e_G = \frac{1}{\#G} \sum_g g$ is an idempotent projector onto M^G, $M = M^G \oplus (1 - e_G)M$ as $R[G]$-modules, and for every ideal I, $(M/IM)^G = M^G/IM^G$. In particular for $R = \mathbb{Z}_\ell$, $\ell \nmid \#G$ and T a $\mathbb{Z}_\ell[G]$-lattice: $(T/\ell T)^G = T^G/\ell T^G$, so invariants commute with reduction. Reference: Serre, Section 1.3 (Theorem 1). Paper: Lemma lem:euler and Remark rem:types (for $\ell \geq 5$ the finite inertia images of order $2, 3, 4, 6$ have order prime to ℓ, so their invariants on T and on $\mathrm{ad}^0 T$ reduce well; at $\ell = 3$ this fails, 9.4), Proposition prop:ns (Q_8 and C_4 have order prime to 3).

3.3 `ex:reps:maschke-fails` (worked examples; P). (a) $\mathrm{U} = \langle u \rangle \simeq C_3$ on $\mathbb{F}_3^2$: the line $\mathbb{F}_3 e_1$ is stable and has no stable complement (u fixes exactly one line, Chapter 1, 5.6); $\mathbb{F}_3^2$ is indecomposable, not semisimple, with composition factors $\mathbf{1}, \mathbf{1}$. (b) The regular representation $\mathbb{F}_p[C_p] \simeq \mathbb{F}_p[x]/(x-1)^p$ is uniserial with the single composition factor $\mathbf{1}$ of multiplicity p. (c) For an infinite group Maschke fails even in characteristic 0: $\mathbb{Z}$ acting on $\mathbb{Q}_\ell^2$ through $\begin{pmatrix} 1 & n' \\ 0 & 1 \end{pmatrix}$, $n' \neq 0$, is not semisimple (the shape of tame inertia at a multiplicative prime). (d) $\mathrm{B}(\mathbb{F}_3)$ on $\mathbb{F}_3^2$ is indecomposable with $\mathrm{End}_{\mathrm{B}} = \mathbb{F}_3$: "$\mathrm{End}_G(V) = k$" does not

imply irreducibility without semisimplicity. Paper: Lemma lem:IV (the image of I_q mod 3 is example (a)), Lemma lem:tamfactor and Lemma lem:euler type I_n (example (c)), Conjecture conj:eis (Borel images).

3.4 thm:reps:jordan-holder (theorem, Jordan-Holder; R: proved in Chapter 3, cited with \Cref{ch:modules}, decision B2). Let A be any ring (here $k[G]$ or $\mathcal{O}[G]$) and M an A-module of finite length (for instance a finite-dimensional representation, or a finite $\mathcal{O}[G]$-module). Then M has a composition series $0 = M_0 \subset M_1 \subset \cdots \subset M_r = M$ with simple quotients, and the multiset of isomorphism classes of the quotients M_i/M_{i-1} (the composition factors, with multiplicity) and the length r are independent of the series. Not proved here: Chapter 3 proves it (by induction on the minimal length of a composition series, comparing two series through $M_{r-1} \cap M'_{s-1}$); this item only restates it for $k[G]$- and $\mathcal{O}[G]$-modules. Reference: Lang, Chapter III (composition series); Dummit-Foote Section 3.4 (for groups; the module case is identical). Paper: background (semisimplification in Lemma lem:old; lengths of finite $\mathbb{Z}_\ell$-modules in Theorem thm:main, through Chapter 3).

3.5 def:reps:semisimplification (definition and proposition; P for (i), (ii); R for (iii)). (i) The semisimplification $V^{ss} = \bigoplus_i M_i/M_{i-1}$ of a finite-dimensional representation is well defined up to isomorphism (3.4); $V^{ss} \simeq V$ iff V is semisimple; $(\cdot)^{ss}$ is additive in short exact sequences. (ii) $\det(X - g \mid V) = \det(X - g \mid V^{ss})$ and $\mathrm{tr}(g \mid V) = \mathrm{tr}(g \mid V^{ss})$ for all g (block triangular form). (iii) If k is perfect, $(V^{ss})_K \simeq (V_K)^{ss}$ for every algebraic extension K/k (irreducible representations stay semisimple under extension of scalars; Curtis-Reiner, Section 29). Paper: Lemma lem:old ("$\bar\rho_h \simeq \bar\rho$" via semisimplifications), Chapter 5 galois:reduction-ss (cites 10.2).

3.6 prop:reps:nonsplit-line (proposition; P). Let V be 2-dimensional with a stable line L, G acting on L by χ_1 and on V/L by χ_2; in a basis $e_1 \in L$, e_2: $\rho = \begin{pmatrix} \chi_1 & b \\ 0 & \chi_2 \end{pmatrix}$ with $b(gh) = \chi_1(g)b(h) + b(g)\chi_2(h)$. (i) V is semisimple iff $V \simeq \chi_1 \oplus \chi_2$ iff V has a second stable line iff there is $c \in k$ with $b(g) = c(\chi_1(g) - \chi_2(g))$ for all g (a coboundary; Chapter 16 for the cohomological reading). (ii) If V is not semisimple (a non-split extension), L is its unique stable line; hence for two non-split extensions V, V' (with sub-characters χ_1, χ'_1 and quotient characters χ_2, χ'_2), every isomorphism $V \simeq V'$ maps L onto L', so $\chi_1 = \chi'_1$ and $\chi_2 = \chi'_2$. (iii) If $\chi_1 \neq \chi_2$ and V is semisimple, V has exactly two stable lines. Reference: elementary. Paper: Lemma lem:old (case $p \,\|\, N$, $\bar\rho$ ramified: "a non-split extension of two characters has a unique stable line, so the isomorphism ... matches the sublines and hence the quotients"), Lemma lem:tamagawa-exp and Lemma lem:tamfactor (the matrix shape $\left(\begin{smallmatrix} \chi_\ell\varepsilon & \kappa \\ 0 & \varepsilon \end{smallmatrix}\right)$).

3.7 prop:reps:dim2-absirr (proposition; P). Let $\rho\colon G \to \mathrm{GL}_2(k)$. (i) If $\rho(G)$ is abelian, ρ is absolutely reducible (commuting operators on $\bar{k}^2$ have a common eigenvector). (ii) If $\rho(G)$ is finite of order prime to $\mathrm{char}\, k$ and ρ is absolutely reducible, then $\rho_{\bar k} \simeq \chi_1 \oplus \chi_2$ (Maschke) and $\rho(G)$ is abelian. Hence: a finite non-abelian subgroup of $\mathrm{GL}_2(k)$ of order prime to $\mathrm{char}\, k$ acts absolutely irreducibly on k^2. (iii) If $H \leq G$ and $\rho|_H$ is absolutely irreducible, so is ρ. (iv) Applied to $\mathrm{GL}_2(\mathbb{F}_3)$: Q_8 (non-abelian, order 8 prime to 3) acts absolutely irreducibly on $\mathbb{F}_3^2$; so do $\mathrm{SL}_2(\mathbb{F}_3)$, $N_{ns}(\mathbb{F}_3)$, $\mathrm{GL}_2(\mathbb{F}_3)$ (they contain Q_8) and $N_s(\mathbb{F}_3) \simeq D_4$ (non-abelian of order 8); C_4 and $C_{ns}(\mathbb{F}_3)$ (abelian) do not. The Borel subgroup $\mathrm{B}(\mathbb{F}_3)$ is non-abelian and faithful but reducible: its order 12 is divisible by 3, so (ii) does not apply. Reference: elementary. Paper: Proposition prop:ns ("The first two act absolutely irreducibly on $\mathbb{F}_3^2$ (they are non-abelian and act faithfully)"): (ii) supplies the missing step of important_notes.md point 1 in the form "non-abelian, of order prime to 3" for Q_8, and (iii) for $\mathrm{SL}_2(\mathbb{F}_3) \supset Q_8$; it complements the commutator argument of Chapter 1, 5.7.

6.3.4 Section 4. Characters in characteristic zero (brief)

4.1 def:reps:trace-character (definition and proposition; P). The trace function $\mathrm{tr}_V(g) = \mathrm{tr}\,\rho(g)$ is a class function; $\mathrm{tr}_V(1) = \dim V$; $\mathrm{tr}_{V\oplus W} = \mathrm{tr}_V + \mathrm{tr}_W$; $\mathrm{tr}_{V\otimes W} = \mathrm{tr}_V\,\mathrm{tr}_W$; $\mathrm{tr}_{V^\vee}(g) = \mathrm{tr}_V(g^{-1})$, which for $k = \mathbb{C}$ and G finite is $\overline{\mathrm{tr}_V(g)}$ (eigenvalues are roots of unity). Reference: Serre, Section 2.1. RHB link: this is RHB's "character". Paper: background.

4.2 thm:reps:orthogonality (theorem; P for (i), (ii); S for (iii); R for (iv)). G finite, $k = \mathbb{C}$, $\langle \phi, \psi \rangle = \frac{1}{\#G}\sum_g \phi(g)\overline{\psi(g)}$. (i) For irreducible V, W: $\langle \mathrm{tr}_V, \mathrm{tr}_W \rangle = 1$ if $V \simeq W$ and 0 otherwise. (ii) $\dim \mathrm{Hom}_G(V, W) = \langle \mathrm{tr}_V, \mathrm{tr}_W \rangle$; V is irreducible iff $\langle \mathrm{tr}_V, \mathrm{tr}_V \rangle = 1$; two representations with the same character are isomorphic. (iii) The irreducible characters form an orthonormal basis of the class functions;

their number is the number of conjugacy classes; $\sum(\dim V_i)^2 = \#G$. (iv) Each $\dim V_i$ divides $\#G$. Proofs: (i), (ii) by averaging (Schur 2.2 applied to $\frac{1}{\#G}\sum g\phi g^{-1}$); (iii) by the regular representation and the centre of $\mathbb{C}[G]$. The abelian case is Chapter 2's orthogonality of Dirichlet characters (its 10.2). Reference: Serre, Chapter 2 (Theorems 3 to 7, Corollaries). RHB link: RHB's orthogonality theorems and character tables. Paper: background.

4.3 `cor:reps:traces-char0` (corollary; P). If $\operatorname{char} k = 0$, G any group, two semisimple finite-dimensional representations with the same trace functions are isomorphic. (Proof in 5.2: in characteristic 0 the multiplicities are traces of suitable elements of $k[G]$.) Paper: background (ℓ-adic representations, Chapter 5's galois:chebotarev-brauer-nesbitt in characteristic 0).

4.4 `ex:reps:tables` (worked example; P). Character tables of C_n (Chapter 2), D_4 and Q_8. The last two tables coincide (classes $\{1\},\{r^2\},\{r^{\pm1}\},\{s,sr^2\},\{sr,sr^3\}$ against $\{1\},\{-1\},\{\pm i\},\{\pm j\},\{\pm k\}$; four characters of degree 1 and one of degree 2) although $D_4 \not\simeq Q_8$ (Chapter 1, 2.5). The degree-2 representation of D_4 is defined over $\mathbb{Q}$; that of Q_8 is not defined over $\mathbb{R}$ (Frobenius-Schur indicator -1, stated), but its reduction modulo 3 is the representation of $Q_8 \subset \mathrm{SL}_2(\mathbb{F}_3)$ on $\mathbb{F}_3^2$ of Chapter 1, 6.2 (over a finite field every absolutely irreducible representation is defined over the field of its traces, stated; Serre, Section 12.1 for the characteristic-zero obstruction). Paper: background ($N_s(\mathbb{F}_3) \simeq D_4$ and Q_8 in Proposition prop:ns).

4.5 `ex:reps:traces-fail` (worked example and remark; P). In characteristic p traces do not determine semisimple representations: for $G = C_2 = \{1, s\}$ over $\mathbb{F}_3$, $\mathbf{1}^{\oplus 3}$ and $\mathrm{sgn}^{\oplus 3}$ both have trace $3 \equiv 0$ at 1 and $\pm 3 \equiv 0$ at s, but $\det(X - s)$ is $(X-1)^3$, respectively $(X+1)^3$. Remark: if $n < p$ (or $p = 0$), the traces of $g, g^2, \ldots, g^n$ determine $\det(X - g)$ by Newton's identities, so traces suffice for representations of dimension less than p; for $n = 2$ and $p \neq 2$ the trace and the determinant determine the characteristic polynomial. Paper: Lemma lem:old (Brauer-Nesbitt is applied with characteristic polynomials, which for $\dim 2$ and equal determinants $\bar{\chi}_\ell$ amounts to $a_p(h) \equiv a_p(f) \pmod{\lambda}$).

6.3.5 Section 5. Brauer-Nesbitt

5.1 `thm:reps:burnside` (theorem, the density theorem of Jacobson and Burnside; R, with the proof of (b) sketched). Let G be any group, $S_1, \ldots, S_r$ pairwise non-isomorphic irreducible finite-dimensional representations over k, $D_i = \mathrm{End}_G(S_i)$. (a) The image of $k[G] \to \prod_i \mathrm{End}_k(S_i)$ is $\prod_i \mathrm{End}_{D_i}(S_i)$. (b) If k is algebraically closed, the image is $\prod_i \mathrm{End}_k(S_i)$; in particular for every i there is $x \in k[G]$ acting on S_i as a projector of rank one and on every S_j, $j \neq i$, as 0. Reference: Lang, Chapter XVII, Section 3 (density theorem and Burnside's theorem); Curtis-Reiner, Section 27. Paper: background (used in 2.3 and 5.2).

5.2 `thm:reps:brauer-nesbitt` (theorem, Brauer-Nesbitt; P given 5.1). Let G be any group, k any field, and V_1, V_2 finite-dimensional representations of G over k with $\det(X - \rho_1(g)) = \det(X - \rho_2(g))$ in $k[X]$ for every $g \in G$. Then V_1 and V_2 have the same composition factors with multiplicities, i.e. $V_1^{ss} \simeq V_2^{ss}$. If $\operatorname{char} k = 0$, or if $\dim V_1, \dim V_2 < \operatorname{char} k$, equality of the traces $\operatorname{tr}\rho_1(g) = \operatorname{tr}\rho_2(g)$ for all g suffices. Proof (to be given in full): (0) replace V_i by V_i^{ss} (3.5(ii)). (1) Reduction to $k = \bar{k}$: for non-isomorphic irreducible S, S' over k, 5.1(a) gives $x \in k[G]$ acting as 1 on S and 0 on S', so $S_{\bar{k}}$ and $S'_{\bar{k}}$ have no common composition factor; hence the composition factors of $V_{\bar{k}}$ determine those of V. (2) Over $\bar{k}$ write $V_1 = A \oplus B$, $V_2 = A \oplus C$ with B, C semisimple without common irreducible constituent; cancelling in the factorial ring $\bar{k}[X]$ gives $\det(X - g \mid B) = \det(X - g \mid C)$ for all g. (3) Traces agree on G, hence on $\bar{k}[G]$ by linearity; with the rank-one elements of 5.1(b), the multiplicity of each irreducible S satisfies $m_S(B) \equiv m_S(C)$ in $\bar{k}$; as one of them is 0, both are divisible by $p = \operatorname{char} k$ (and are 0 if $p = 0$, which ends the proof). (4) So $B \simeq B'^{\oplus p}$, $C \simeq C'^{\oplus p}$, $\det(X - g \mid B) = \det(X - g \mid B')^p$, and $f \mapsto f^p$ is injective on $\bar{k}[X]$; so B', C' satisfy the hypothesis with smaller dimension, and induction gives $B' = C' = 0$. Reference: Curtis-Reiner, Section 30 (the theorem of Brauer and Nesbitt); the proof above follows the standard one (also in DDT, Section 1, for Galois representations, cited without number). Paper: Lemma lem:old ("$\bar{\rho}_h \simeq \bar{\rho}$ by Chebotarev and Brauer-Nesbitt"); Chapter 5's galois:chebotarev-brauer-nesbitt and galois:congruent-ap, Chapter 15.

5.3 `cor:reps:bn-consequences` (corollary; P). (i) If $\dim V_1 = \dim V_2 = 2$, $\operatorname{char} k \neq 2$ and $\det\rho_1 = \det\rho_2$, then $\operatorname{tr}\rho_1 = \operatorname{tr}\rho_2$ on G implies $V_1^{ss} \simeq V_2^{ss}$. (ii) Let $k \subset k'$ be fields, V_1 over k and V_2 over k' with $\det(X - \rho_1(g)) = \det(X - \rho_2(g))$ for all g. If $V_1 \otimes_k k'$ is irreducible (for example if V_1 is absolutely irreducible), then $V_2 \simeq V_1 \otimes_k k'$ (no semisimplicity of V_2 is assumed: V_2^{ss} is irreducible of full dimension,

so V_2 is irreducible). If V_1 is only irreducible over k, the conclusion can fail: C_4 on $\mathbb{F}_3^2$ (2.5) and $\psi \oplus \psi^{-1}$ over $\mathbb{F}_9$ have the same characteristic polynomials. Paper: Lemma lem:old (the residual representation $\bar{\rho}_h$ lives over the residue field of the coefficient ring of h, $\bar{\rho}$ over $\mathbb{F}_\ell$; (ii) with $\bar{\rho}$ absolutely irreducible gives $\bar{\rho}_h \simeq \bar{\rho} \otimes k'$: important_notes.md point 10), Section 3 of the paper ("that is, $a_p(g) \equiv a_p(f) \pmod{\lambda}$").

6.3.6 Section 6. Restriction, induction, Frobenius reciprocity, Mackey, Clifford

6.1 `def:reps:induction` (definition and proposition; P). Let $H \leq G$ have finite index m, W a representation of H over k (or an $R[H]$-module). $\mathrm{Res}_H^G V$ is V viewed as an H-module. $\mathrm{Ind}_H^G W = k[G] \otimes_{k[H]} W$; with coset representatives $G = \bigsqcup_{i=1}^m s_i H$ it is $\bigoplus_i s_i \otimes W$, and if $gs_i = s_j h$ then $g(s_i \otimes w) = s_j \otimes hw$; $\dim \mathrm{Ind}\, W = m \dim W$. Equivalently (finite index) the module of maps $f\colon G \to W$ with $f(hx) = hf(x)$, G acting by right translation. Trace formula (any field): $\mathrm{tr}(g \mid \mathrm{Ind}\, W) = \sum_{i:\, s_i^{-1} g s_i \in H} \mathrm{tr}(s_i^{-1} g s_i \mid W)$. Index two: if H has index 2 and $s \notin H$, then $\mathrm{tr}(h \mid \mathrm{Ind}\, W) = \mathrm{tr}(h \mid W) + \mathrm{tr}(s^{-1}hs \mid W)$ for $h \in H$ and $\mathrm{tr}(g \mid \mathrm{Ind}\, W) = 0$ for $g \notin H$; for a character ψ of H, in the basis $1 \otimes 1$, $s \otimes 1$: $h \mapsto \mathrm{diag}(\psi(h), \psi^s(h))$ with $\psi^s(h) = \psi(s^{-1}hs)$, and $s \mapsto \begin{pmatrix} 0 & \psi(s^2) \\ 1 & 0 \end{pmatrix}$. Reference: Serre, Sections 3.3, 7.1, 7.2. Paper: Proposition prop:ns (Ind ψ), Lemma lem:h0 ("induced from $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{-3}))$"); Chapter 6's 6.4 and 6.5 use the trace formula.

6.2 `thm:reps:frobenius-reciprocity` (theorem, Frobenius reciprocity; P). For $H \leq G$ of finite index, W a representation of H and V of G (over a field or a commutative ring), there are natural isomorphisms $\mathrm{Hom}_G(\mathrm{Ind}_H^G W, V) \simeq \mathrm{Hom}_H(W, \mathrm{Res}_H V)$ and $\mathrm{Hom}_G(V, \mathrm{Ind}_H^G W) \simeq \mathrm{Hom}_H(\mathrm{Res}_H V, W)$. In characteristic 0 for finite G: $\langle \mathrm{tr}_{\mathrm{Ind}\, W}, \mathrm{tr}_V \rangle_G = \langle \mathrm{tr}_W, \mathrm{tr}_{\mathrm{Res}\, V} \rangle_H$. Reference: Serre, Section 7.2 (Theorem 13). Paper: Proposition prop:ns ("$\bar{\rho} \simeq \mathrm{Ind}\, \psi$ by Frobenius reciprocity": a nonzero H-map $\psi \to \mathrm{Res}\, \bar{\rho}_{\mathbb{F}_9}$ gives a nonzero G-map $\mathrm{Ind}\, \psi \to \bar{\rho}_{\mathbb{F}_9}$, surjective as $\bar{\rho}_{\mathbb{F}_9}$ is irreducible, hence an isomorphism by dimensions).

6.3 `prop:reps:projection-formula` (proposition; P). For $H \leq G$ of finite index: (i) $\mathrm{Ind}_H^G(W \otimes \mathrm{Res}_H U) \simeq \mathrm{Ind}_H^G W \otimes U$; in particular, for a character χ of G trivial on H, $\mathrm{Ind}\, W \otimes \chi \simeq \mathrm{Ind}\, W$. (ii) $\mathrm{Ind}_H^G \mathbf{1} = k[G/H]$; for H of index 2 and $\mathrm{char}\, k \neq 2$, $\mathrm{Ind}_H^G \mathbf{1} \simeq \mathbf{1} \oplus \chi_H$, χ_H the character with kernel H. (iii) Transitivity, additivity, compatibility with duals and with extension of scalars. Reference: Serre, Section 7.2. Paper: Proposition prop:ns ("so $\bar{\rho} \simeq \bar{\rho} \otimes \chi_{-3}$"), Lemma lem:h0.

6.4 `thm:reps:mackey` (theorem; R for (a), S for (b), P for (c)). (a) (Mackey's formula) For $H, K \leq G$ of finite index, $\mathrm{Res}_K \mathrm{Ind}_H^G W \simeq \bigoplus_{s \in K \backslash G / H} \mathrm{Ind}_{K \cap sHs^{-1}}^K W^s$, $W^s(x) = W(s^{-1}xs)$. (b) (Mackey's irreducibility criterion, characteristic 0, G finite) $\mathrm{Ind}_H^G W$ is irreducible iff W is irreducible and for every $s \in G \setminus H$ the representations W^s and W of $H_s = sHs^{-1} \cap H$ have no common irreducible constituent. (c) (Index two, any characteristic.) Let $H \leq G$ have index 2, $s \notin H$, W an absolutely irreducible representation of H. Then $\mathrm{Res}_H \mathrm{Ind}\, W \simeq W \oplus W^s$, and $\mathrm{Ind}\, W$ is absolutely irreducible iff $W_{\bar{k}} \not\simeq W^s_{\bar{k}}$. Proof of (c): if $W \not\simeq W^s$, the only H-submodules of $W \oplus W^s$ over $\bar{k}$ are 0, W, W^s and the sum, and s swaps the two summands, so no proper nonzero G-submodule exists; if $W \simeq W^s$, $\mathrm{End}_G(\mathrm{Ind}\, W_{\bar{k}}) = \mathrm{Hom}_H(W, W \oplus W^s)$ has dimension 2 over $\bar{k}$ (6.2), so $\mathrm{Ind}\, W_{\bar{k}}$ is not irreducible (Schur 2.2(ii)). Reference: Serre, Sections 7.3, 7.4. Paper: Proposition prop:ns (Ind ψ is irreducible since $\psi^s = \psi^{-1} \neq \psi$), Lemma lem:h0.

6.5 `thm:reps:clifford` (theorem, Clifford; P). Let $N \trianglelefteq G$ have finite index and V an irreducible representation of G over any field. Then $\mathrm{Res}_N V$ is semisimple, its irreducible constituents form one G-orbit under $W \mapsto W^g$, and they all occur with the same multiplicity. Proof: for an irreducible N-submodule W, $\sum_g gW$ is a nonzero G-submodule, hence V, and a sum of irreducibles $gW \simeq W^{g^{-1}}$ is semisimple (3.1). Reference: Curtis-Reiner, Section 49; Serre, Section 8.1. Paper: background for 6.6.

6.6 `thm:reps:clifford-index-two` (theorem, Clifford theory for an index-two subgroup; P). Let G be any group, $H \leq G$ of index 2, $s \in G \setminus H$, $\chi_H\colon G \to \{\pm 1\}$ the character with kernel H, k a field with $\mathrm{char}\, k \neq 2$, and V an absolutely irreducible representation of G of dimension n. Exactly one of the following holds. (A) $\mathrm{Res}_H V$ is absolutely irreducible; then $V \not\simeq V \otimes \chi_H$, and $\mathrm{Ind}_H^G \mathrm{Res}_H V \simeq V \oplus (V \otimes \chi_H)$. (B) $\mathrm{Res}_H V$ is not absolutely irreducible; then $n = 2d$, there is an absolutely irreducible representation W of H over $\bar{k}$ of dimension d with $W \not\simeq W^s$, $\mathrm{Res}_H V_{\bar{k}} \simeq W \oplus W^s$, $V_{\bar{k}} \simeq \mathrm{Ind}_H^G W$, and $V \simeq V \otimes \chi_H$ over k. Moreover $V \simeq V \otimes \chi_H$ iff (B) holds; in that case $\mathrm{Hom}_G(V, V \otimes \chi_H)$ is a line over k, spanned by an

isomorphism ϕ with $\phi^2 \in k^\times$ and $\operatorname{tr}\phi = 0$, whose eigenspaces over $\bar{k}$ are W and W^s. For $n = 2$: (B) iff $V_{\bar{k}} \simeq \operatorname{Ind}_H^G \psi$ for a character ψ of H with $\psi^s \neq \psi$. Proof: (A) an isomorphism $V \to V \otimes \chi_H$ commutes with H, so is a scalar (Schur), forcing $\chi_H = 1$; the induced statement is 6.3(i), (ii). (B) by Clifford 6.5 over $\bar{k}$ the constituents of Res_H are conjugate, so at least two of dimension d, whence $n \geq 2d$; Frobenius reciprocity 6.2 gives a surjection $\operatorname{Ind} W \to V_{\bar{k}}$, so $n \leq 2d$; hence equality, $V_{\bar{k}} \simeq \operatorname{Ind} W$, and $W \not\simeq W^s$ by 6.4(c); $V \simeq V \otimes \chi_H$ by 6.3(i) over $\bar{k}$ and descent 1.4(iii). Finally $\phi^2 \in \operatorname{End}_G(V) = k$, and $\operatorname{tr}\phi = 0$ by 7.2. Reference: Serre, Section 8.2 (characteristic 0); the argument is characteristic-free once $\operatorname{char} k \neq 2$. Paper: Lemma lem:h0 ("for $\ell = 3$ and $\bar{\rho}$ irreducible, $\bar{\rho} \simeq \bar{\rho} \otimes \chi_{-3}$ holds if and only if $\bar{\rho}$ is induced from $\operatorname{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{-3}))$, that is, if and only if its restriction there is not absolutely irreducible": correct for $\bar{\rho}$ absolutely irreducible, point 10, with "induced" read over $\bar{\mathbb{F}}_3$, point N1 of Part 8), Proposition prop:ns, the hypothesis of Theorem thm:main and S_f; Chapter 5's galois:index-two-restriction is this theorem plus continuity.

6.7 `prop:reps:dihedral` (proposition; P). Let $\operatorname{char} k \neq 2$ and $\rho\colon G \to \mathrm{GL}_2(k)$ absolutely irreducible with finite projective image $P(\rho) \subset \mathrm{PGL}_2(\bar{k})$. (i) $P(\rho)$ is not cyclic (a group whose image modulo scalars is cyclic is abelian; 3.7(i)). (ii) The following are equivalent: $P(\rho)$ is dihedral of order $2m$, $m \geq 2$ (with $D_2 = V_4$); $\rho \simeq \rho \otimes \chi$ for some character χ of order 2; $\rho_{\bar{k}} \simeq \operatorname{Ind}_H^G \psi$ for some subgroup H of index 2. (iii) The characters $\chi \neq 1$ with $\rho \simeq \rho \otimes \chi$ correspond bijectively to the cyclic subgroups of index 2 of $P(\rho)$ (via $H = \ker\chi =$ the preimage): exactly one if $m \geq 3$, exactly three if $P(\rho) \simeq V_4$. Proof: if $\rho_{\bar{k}} = \operatorname{Ind}\psi$, in the basis of 6.1 $\rho(H)$ is diagonal and $\rho(G \setminus H)$ antidiagonal, so $P(\rho)$ is a cyclic group (image of H in the diagonal torus modulo scalars) extended by an involution acting by inversion; conversely the preimage H of a cyclic subgroup of index 2 has abelian image, so $\operatorname{Res}_H \rho$ is absolutely reducible (3.7(i)) and 6.6(B) applies. Reference: elementary; Serre72, Section 2 for the dihedral case of Dickson's list. Paper: Proposition prop:ns and Section sec:anomaly (the projective image of $N_s(\mathbb{F}_3)$ is V_4, so its representation is induced in three ways, one of them from the determinant-one subgroup, 8.2), the hypothesis of Theorem thm:main (for $N_{ns}(\mathbb{F}_3)$, with projective image D_4, the representation is induced from $C_{ns}(\mathbb{F}_3)$, not from the determinant-one subgroup, 8.3; so "dihedral" alone is not the excluded case: the hypothesis singles out the field $\mathbb{Q}(\sqrt{\ell^*})$), Dickson 11.2.

6.3.7 Section 7. Twisting, the adjoint, the trace-zero lemma, the symmetric square, the trace form

7.1 `prop:reps:twist` (proposition; P). Let $\chi\colon G \to k^\times$ (or $R^\times$) be a character. (i) $\operatorname{tr}(\rho \otimes \chi)(g) = \chi(g)\operatorname{tr}\rho(g)$; $\det(X - (\rho\otimes\chi)(g)) = \det(X - \chi(g)\rho(g))$; $\det(\rho\otimes\chi) = \chi^n \det\rho$. (ii) V is (absolutely) irreducible iff $V\otimes\chi$ is; $\operatorname{Hom}_G(V\otimes\chi, W\otimes\chi) = \operatorname{Hom}_G(V,W)$; $(V\otimes\chi)^\vee = V^\vee\otimes\chi^{-1}$; $\operatorname{Res}_{\ker\chi}(V\otimes\chi) = \operatorname{Res}_{\ker\chi} V$. (iii) $\operatorname{End}(V\otimes\chi) = \operatorname{End}(V)$ as representations (the same space with the same action), hence $\operatorname{ad}^0(V\otimes\chi) = \operatorname{ad}^0 V$. (iv) If $V \simeq V \otimes \chi$ then $\chi^n = 1$ on G (determinants); for $n = 2$, $\chi^2 = 1$. Reference: elementary. Paper: Lemma lem:twist and Theorem thm:twist (twisting; the Galois statement $\rho_{E^{(d)},\ell} = \rho_{E,\ell} \otimes \chi_d$ is Chapter 9), Lemma lem:h0 ("comparing determinants it forces $\bar{\chi}_\ell^2 = 1$"), proof of Theorem thm:main ("$\operatorname{ad}^0 T$ is unchanged by the twist"), Lemma lem:euler (types I_n^*, I_0^*: $\operatorname{ad}^0 V = \operatorname{ad}^0 V'$).

7.2 `lem:reps:trace-zero` (lemma, the trace-zero lemma; P). Let $\phi\colon V \to V \otimes \chi$ be a G-map, i.e. $\phi\rho(g) = \chi(g)\rho(g)\phi$ for all g. Then $(1 - \chi(g))\operatorname{tr}\phi = 0$ for every $g \in G$, since $\operatorname{tr}\phi = \operatorname{tr}(\rho(g)^{-1}\phi\rho(g)) = \chi(g)\operatorname{tr}\phi$. Hence $\operatorname{tr}\phi = 0$ if $\chi \neq 1$ (over a field), or, over a ring R, if $1 - \chi(g) \in R^\times$ for some g. If moreover V is absolutely irreducible, $\operatorname{Hom}_G(V, V\otimes\chi)$ has dimension at most one and a nonzero element is an isomorphism (Schur 2.2). Reference: elementary. Paper: Lemma lem:h0 ("an isomorphism $\phi\colon \bar{\rho} \to \bar{\rho}\otimes\chi_{-3}$ has trace 0, since $\operatorname{tr}\phi = \operatorname{tr}(\bar{\rho}(g)^{-1}\phi\bar{\rho}(g)) = \chi_{-3}(g)\operatorname{tr}\phi$"), Proposition prop:ns; Chapter 5's galois:twist (b) and Chapter 6's 6.5 cite it.

7.3 `def:reps:adjoint` (definition and proposition; P). $\operatorname{End}(V) = V \otimes V^\vee$ with $g \cdot x = \rho(g)x\rho(g)^{-1}$; $\operatorname{ad}^0 V$ its trace-zero part (C3). (i) If $n = \dim V$ is invertible in k (or in R), $\operatorname{End}(V) = k \cdot 1 \oplus \operatorname{ad}^0 V$ as representations, $x = \frac{\operatorname{tr} x}{n}1 + (x - \frac{\operatorname{tr} x}{n}1)$; for $n = 2$ this needs $\operatorname{char} k \neq 2$ (ℓ odd). (ii) If $\rho(g)$ has eigenvalues $\lambda_1, \ldots, \lambda_n$ in $\bar{k}$, then $\operatorname{Ad}\rho(g)$ has eigenvalues λ_i/λ_j on $\operatorname{End}(V)$; for $n = 2$ it has λ_1/λ_2, 1, λ_2/λ_1 on $\operatorname{ad}^0 V$, and $\det(X - \operatorname{Ad}\rho(g) \mid \operatorname{ad}^0 V) = (X-1)\big(X^2 - (\operatorname{tr}(g)^2/\det g - 2)X + 1\big)$. (iii) Formulas in the basis e, h, f

(C3; checked symbolically, Part 7): $\mathrm{Ad}\,\mathrm{diag}(x,y) = \mathrm{diag}(x/y, 1, y/x)$; for $t = \begin{pmatrix} 1 & n' \\ 0 & 1 \end{pmatrix}$: $(t-1)e = 0$, $(t-1)h = -2n'e$, $(t-1)f = n'h - n'^2 e$; for $F = \begin{pmatrix} \varepsilon q & c' \\ 0 & \varepsilon \end{pmatrix}$: $FeF^{-1} = qe$, $FhF^{-1} = h - 2c'\varepsilon^{-1}e$, $FfF^{-1} = q^{-1}f + c'\varepsilon^{-1}q^{-1}h - c'^2\varepsilon^{-2}q^{-1}e$. Reference: elementary. Paper: Section 1.1 ($A_f = \mathrm{ad}^0 V$), the eigenvalues $\alpha/\beta, 1, \beta/\alpha$ in Lemma lem:euler and Chapter 19, Lemma lem:tamfactor (the three formulas for t and the action of F on e and h; with $\varepsilon = \pm 1$, $\varepsilon^{-1} = \varepsilon$ as the paper writes); Chapter 16 cites these.

7.4 `prop:reps:H0-twisted-adjoint` (proposition; P). Let V be a representation over k (or a free R-module of rank n with R local) and χ a character. (i) $(\mathrm{End}\, V \otimes \chi)^G = \mathrm{Hom}_G(V, V \otimes \chi)$ (1.3). (ii) If n is invertible, $(\mathrm{End}\, V \otimes \chi)^G = (\mathrm{ad}^0 V \otimes \chi)^G \oplus R(\chi)^G$ and $(\mathrm{ad}^0 V \otimes \chi)^G = \{\phi \in \mathrm{Hom}_G(V, V \otimes \chi) : \mathrm{tr}\,\phi = 0\}$. (iii) If $\chi \neq 1$ (over R local: if the reduction $\bar\chi$ of χ to the residue field is nontrivial), then $R(\chi)^G = 0$, so every G-map $V \to V \otimes \chi$ has trace 0 and $(\mathrm{ad}^0 V \otimes \chi)^G = \mathrm{Hom}_G(V, V \otimes \chi)$ (agreeing with 7.2). (iv) If V is absolutely irreducible and n is invertible: $(\mathrm{ad}^0 V)^G = 0$; and for $\chi \neq 1$, $(\mathrm{ad}^0 V \otimes \chi)^G \neq 0$ iff $V \simeq V \otimes \chi$, in which case it is a line. Reference: elementary. Paper: Lemma lem:h0 ("the ℓ-torsion of $H^0(\mathbb{Q}, W^D)$ is $H^0(\mathbb{Q}, \mathrm{ad}^0\,\bar\rho \otimes \bar\chi_\ell)$, the space of trace-zero $G_{\mathbb{Q}}$-maps $\bar\rho \to \bar\rho \otimes \bar\chi_\ell$"; by (iii) the trace condition is automatic, as Chapter 21's A4.11 step (2) observes with $\mu_{\ell^k}(\mathbb{Q}) = 0$), Proposition prop:ns ("$H^0(\mathbb{Q}, \mathrm{ad}^0\,\bar\rho_{E,3}(1)) \neq 0$").

7.5 `prop:reps:sym2` (proposition; P). Let $\dim V = 2$ with basis e_1, e_2, $\omega(x,y) = \det[x\,|\,y]$, and $\mathrm{Sym}^2 V$ the symmetric square (quotient of $V \otimes V$ by $x \otimes y - y \otimes x$), with basis e_1^2, e_1e_2, e_2^2; if $\rho(g)$ has eigenvalues λ, μ, then $\mathrm{Sym}^2\rho(g)$ has $\lambda^2, \lambda\mu, \mu^2$. (i) The map $\Xi\colon \mathrm{Sym}^2 V \to \mathrm{ad}^0 V$, $\Xi(xy)(v) = \omega(v,x)y + \omega(v,y)x$, is well defined, lands in trace zero, and satisfies $\Xi(g \cdot m) = \det(g)\,\rho(g)\Xi(m)\rho(g)^{-1}$, so it is a G-map $\mathrm{Sym}^2 V \otimes \det^{-1} \to \mathrm{ad}^0 V$. In coordinates $\Xi(e_1^2) = -2e$, $\Xi(e_1e_2) = h$, $\Xi(e_2^2) = 2f$; so Ξ is an isomorphism iff 2 is invertible (over $\mathbb{Z}$ its image has index 4; over $\mathbb{Z}_\ell$ it is an isomorphism of lattices iff ℓ is odd). (ii) Hence $\mathrm{ad}^0\rho \simeq \mathrm{Sym}^2\rho \otimes \det^{-1}$ and $\mathrm{Sym}^2\rho \simeq \mathrm{ad}^0\rho \otimes \det$; and, for $\mathrm{char}\,k \neq 2$, $V \otimes V \simeq \mathrm{Sym}^2 V \oplus \det$ and $V \otimes V^\vee \simeq \mathrm{ad}^0 V \oplus \mathbf{1}$. (iii) (Normalisations; Galois input from Chapters 9, 18, 19.) With $V = V_\ell E$, $\det = \chi_\ell$: $\mathrm{Sym}^2 V \simeq \mathrm{ad}^0 V(1)$ and $\mathrm{Sym}^2(V^\vee) \simeq \mathrm{ad}^0 V(-1)$; the classical $L(\mathrm{Sym}^2 f, s)$ is the L-function of $\mathrm{Sym}^2(V^\vee) = \mathrm{Sym}^2 H^1$, whence $L_p(\mathrm{Sym}^2 f, s) = L_p(A_f, s-1)$ (Chapter 19 for the L-factors). Reference: elementary; Fulton-Harris style computation. Paper: Section 1.1 ("In the normalisation of the symmetric square, $L_p(\mathrm{Sym}^2 f, s) = L_p(A_f, s-1)$"), the Introduction (Flach's theorem on $\mathrm{Sym}^2 f$; Rankin-Shimura), Chapters 18, 19.

7.6 `prop:reps:trace-form` (proposition; P). Let T be free of rank n over a commutative ring R with a G-action. (i) $B(x,y) = \mathrm{tr}(xy)$ on $\mathrm{End}_R(T)$ is G-invariant ($B(gxg^{-1}, gyg^{-1}) = B(x,y)$) and perfect (the matrix units satisfy $B(E_{ij}, E_{kl}) = \delta_{jk}\delta_{il}$, a permutation Gram matrix of determinant ± 1). (ii) For $n = 2$ the Gram matrix of B on $\mathrm{ad}^0 T$ in the basis e, h, f is $\begin{pmatrix} 0 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 0 \end{pmatrix}$, of determinant -2: B is perfect on $\mathrm{ad}^0 T$ iff $2 \in R^\times$. (iii) Hence for $R = \mathbb{Z}_\ell$, ℓ odd, and $A = \mathrm{ad}^0 T$: $x \mapsto B(x, -)$ is a G-isomorphism $A \simeq A^\vee = \mathrm{Hom}_{\mathbb{Z}_\ell}(A, \mathbb{Z}_\ell)$, and for every character $\chi\colon G \to \mathbb{Z}_\ell^\times$, $\mathrm{Hom}_{\mathbb{Z}_\ell}(A, \mathbb{Z}_\ell(\chi)) \simeq A(\chi)$; likewise with $\mathbb{Q}_\ell$ and $\mathbb{Q}_\ell/\mathbb{Z}_\ell$ coefficients (via Chapter 3). For $\ell = 2$ the image of A in $A^\vee$ has index 2. Reference: elementary (Gram determinants: Chapter 3). Paper: Section 4.2 ("$T^D = \mathrm{Hom}(A, \mathbb{Z}_\ell(1)) \simeq A(1)$ (via the trace form)", with $\chi = \chi_\ell$; this is where ℓ odd enters), Lemma lem:h0 ($W^D = \mathrm{ad}^0 T(1) \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$), Lemma lem:fpr; Chapter 18 cites this item.

7.7 `prop:reps:adjoint-invariance` (proposition; P). (i) $\mathrm{ad}^0(V \otimes \chi) = \mathrm{ad}^0 V$ (7.1(iii)). (ii) $x \mapsto -x^t$ (the negative transpose, $x^t \in \mathrm{End}(V^\vee)$) is a G-isomorphism $\mathrm{ad}^0 V \simeq \mathrm{ad}^0(V^\vee)$, and also $\mathrm{End}\, V \simeq \mathrm{End}(V^\vee)$ (either sign works for modules; $-x^t$ also respects brackets). (iii) Let $L \subset V$ be an $\mathcal{O}$-lattice and $L' = cL$, $c \in K^\times$. Then $\mathrm{End}_{\mathcal{O}}(L') = \mathrm{End}_{\mathcal{O}}(L)$ and $\mathrm{ad}^0 L' = \mathrm{ad}^0 L$ inside $\mathrm{End}_K(V)$, where $\mathrm{ad}^0 L = \mathrm{End}_{\mathcal{O}}(L) \cap \mathrm{ad}^0 V$; and under (ii) $\mathrm{ad}^0(L^\vee)$ corresponds to $\mathrm{ad}^0 L$, $L^\vee = \mathrm{Hom}_{\mathcal{O}}(L, \mathcal{O})$. (iv) Consequently a lattice homothetic to L, to $L^\vee$, or to a twist $L \otimes \chi$ has the same adjoint lattice as L, up to these canonical identifications. Reference: elementary. Paper: Section 4.2 ("the lattice $M_{f,\lambda}$ of DFG04 and $T_\ell E$ are homothetic when $\bar\rho$ is irreducible, so their adjoints coincide": by 10.3 and (iv), with $M_{f,\lambda}$ homothetic to the dual of $T_\ell E$, important_notes.md point 26), proof of Theorem thm:main (type I_n^*), Theorem thm:quant Step 3 ("$A_g = A_f$").

6.3.8 Section 8. $\mathrm{GL}_2(\mathbb{F}_3)$ and its subgroups on $\mathbb{F}_3^2$ and $\mathbb{F}_9^2$

8.1 `prop:reps:gl23-table` (proposition and table; P, the table checked by computer, Part 7). For the standard models (C6) acting on $\mathbb{F}_3^2$ by the standard representation ρ (and on $\mathbb{F}_9^2$ after extension of scalars):

H	order	det onto $\mathbb{F}_3^\times$	irreducible over $\mathbb{F}_3$	absolutely irreducible	$\dim \mathrm{End}_H$	$\dim(\mathrm{ad}^0\,\rho)^H$	$\dim \mathrm{Hom}_H(\rho, \rho\otimes\det)$ (trace-zero part)	$\rho \simeq \rho\otimes\det$	projective image
$\mathrm{GL}_2(\mathbb{F}_3)$	48	yes	yes	yes	1	0	0 (0)	no	S_4
$\mathrm{SL}_2(\mathbb{F}_3)$	24	no	yes	yes	1	0	1 (0)	(det = 1)	A_4
$N_{ns}(\mathbb{F}_3)$	16	yes	yes	yes	1	0	0 (0)	no	D_4
Q_8	8	no	yes	yes	1	0	1 (0)	(det = 1)	V_4 (normal)
$N_s(\mathbb{F}_3)$	8	yes	yes	yes	1	0	1 (1)	**yes**	V_4 (not normal)
$C_{ns}(\mathbb{F}_3)$	8	yes	yes	no	2	1	0 (0)	no	C_4
$C_4 = \langle J\rangle$	4	no	yes	no	2	1	2 (1)	(det = 1)	C_2
$C_s(\mathbb{F}_3)$	4	yes	no	no	2	1	2 (2)	yes	C_2
$\mathrm{B}(\mathbb{F}_3)$	12	yes	no	no	1	0	1 (1)	no	S_3
U	3	no	no	no	2	1	2 (1)	(det = 1)	C_3

Remarks to be made in the text: for the subgroups of $\mathrm{SL}_2(\mathbb{F}_3)$ the twist by det is trivial; $\mathrm{End}_{\mathrm{B}} = \mathbb{F}_3$ although B is reducible (3.3(d)); the trace-zero intertwiner for B is e (not invertible). Reference: Chapter 1, 6.3 to 6.6 for the groups; computation. Paper: Proposition prop:ns, Lemma lem:h0, Section sec:anomaly, Remark rem:ns.

8.2 `prop:reps:ns3-induced` (proposition; P). Let ρ be the standard representation of $N_s(\mathbb{F}_3) = \langle J, \sigma\rangle$ (Chapter 1, 6.5) and $C_4 = \langle J\rangle = N_s(\mathbb{F}_3) \cap \mathrm{SL}_2(\mathbb{F}_3)$. (i) ρ is absolutely irreducible (3.7(iv)), but $\mathrm{Res}_{C_4}\rho\otimes\mathbb{F}_9 \simeq \psi\oplus\psi^{-1}$, $\psi(J) = i$, with eigenvectors $v = (1,-i)^t$ and $\sigma v = (1,i)^t$ (2.5); $\psi^\sigma = \psi^{-1} \neq \psi$ since $\sigma J\sigma^{-1} = J^{-1}$. (ii) In the basis $(v, \sigma v)$ of $\mathbb{F}_9^2$, $\rho(J) = \mathrm{diag}(i,-i)$ and $\rho(\sigma) = w$, which is the matrix form of $\mathrm{Ind}_{C_4}^{N_s}\psi$ of 6.1 (with $s = \sigma$, $\sigma^2 = 1$); so $\rho\otimes\mathbb{F}_9 \simeq \mathrm{Ind}_{C_4}^{N_s}\psi$, as Frobenius reciprocity 6.2 also shows. Over $\mathbb{F}_3$, ρ is *not* induced from a character of C_4 ($\mathrm{Res}_{C_4}\rho$ is irreducible over $\mathbb{F}_3$). (iii) $\rho \simeq \rho\otimes\det$ over $\mathbb{F}_3$, with intertwiner $\phi = J$: $J\rho(g) = \det(g)\rho(g)J$ for all g (check on J and σ: $J\sigma = -\sigma J$); $\mathrm{tr}\,J = 0$, $J^2 = -1$, and the eigenlines of J over $\mathbb{F}_9$ are the two summands of (i), as 6.6(B) predicts. (iv) The projective image is V_4 (Chapter 1, 6.11), so by 6.7(iii) $\rho \simeq \rho\otimes\chi$ for each of the three characters χ of $N_s(\mathbb{F}_3)/\{\pm1\}$: their kernels are C_4 ($\chi = \det$, intertwiner J), $C_s(\mathbb{F}_3)$ (intertwiner σ) and $\{\pm1, \pm w\}$ (intertwiner w). (v) $(\mathrm{ad}^0\,\rho\otimes\det)^{N_s} = \mathbb{F}_3 J$, a line; $(\mathrm{ad}^0\,\rho)^{N_s} = 0$. Reference: computation (Part 7). Paper: Proposition prop:ns ("In the third case $\bar\rho|_{\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{-3}))}\otimes\mathbb{F}_9 = \psi\oplus\psi^{-1}$ for a character ψ of order 4, and $\bar\rho \simeq \mathrm{Ind}\,\psi$ by Frobenius reciprocity, so $\bar\rho \simeq \bar\rho\otimes\chi_{-3}$ and $H^0(\mathbb{Q}, \mathrm{ad}^0\,\bar\rho(1)) \neq 0$"; the isomorphism with $\mathrm{Ind}\,\psi$ is over $\mathbb{F}_9$, point N1), Lemma lem:h0, Section sec:anomaly, Remark rem:ns; Chapter 5's Exercise 12 (the three quadratic subfields correspond to (iv)).

8.3 `prop:reps:nns3` (proposition; P). Let ρ be the standard representation of $N_{ns}(\mathbb{F}_3) = \langle a, \sigma\rangle$ (Chapter 1, 6.6). (i) ρ and $\mathrm{Res}_{Q_8}\rho$ are absolutely irreducible (3.7(iv)); $Q_8 = N_{ns}(\mathbb{F}_3) \cap \mathrm{SL}_2(\mathbb{F}_3)$. (ii) $\rho \not\simeq \rho\otimes\det$: an isomorphism ϕ gives $\phi\rho(g)\phi^{-1} = \det(g)\rho(g)$, so taking traces $\mathrm{tr}\,\rho(g) = 0$ whenever $\det g = -1$ (characteristic $\neq 2$), but the four elements of order 8 and determinant -1 have trace ±1; $(\mathrm{ad}^0\,\rho\otimes\det)^{N_{ns}} = 0$. (iii) Nevertheless the projective image is D_4, so ρ is dihedral (6.7): over $\mathbb{F}_9$, $C_{ns}(\mathbb{F}_3) \simeq \mathbb{F}_9^\times$ acts with the eigencharacters θ, θ^3, $\theta(a) = 1 + i$ of order 8, $\theta^\sigma = \theta^3 \neq \theta$, and $\rho\otimes\mathbb{F}_9 \simeq \mathrm{Ind}_{C_{ns}}^{N_{ns}}\theta$; $\rho \simeq \rho\otimes\chi_{ns}$ for the character χ_{ns} with kernel $C_{ns}(\mathbb{F}_3)$, and for no other nontrivial

character. Reference: computation. Paper: Proposition prop:ns (the image $N_{ns}(\mathbb{F}_3)$: "the first two act absolutely irreducibly"), Section sec:anomaly ("no curve with image ... $N_{ns}(\mathbb{F}_3)$ violates"), and the reading of the hypothesis of Theorem thm:main: "dihedral" is not the excluded case, "induced from $\mathbb{Q}(\sqrt{-3})$" is.

8.4 `prop:reps:gl23-sl23` (proposition; P). The standard representations of $\mathrm{SL}_2(\mathbb{F}_3)$ and $\mathrm{GL}_2(\mathbb{F}_3)$ are absolutely irreducible (they contain Q_8; 3.7(iii), (iv)); $\mathrm{GL}_2(\mathbb{F}_3)$: $\rho \not\simeq \rho \otimes \det$ (elements of order 8 as in 8.3(ii)), $(\mathrm{ad}^0\,\rho \otimes \det)^{\mathrm{GL}_2} = 0$; projective images A_4, S_4 (exceptional in Dickson's list, 11.2). Paper: Proposition prop:ns (the image $\mathrm{GL}_2(\mathbb{F}_3)$, the "generic" case).

8.5 `thm:reps:three-images` (theorem, the representation-theoretic core of Proposition prop:ns; P). Let $H \leq \mathrm{GL}_2(\mathbb{F}_3)$ be conjugate to $\mathrm{GL}_2(\mathbb{F}_3)$, $N_{ns}(\mathbb{F}_3)$ or $N_s(\mathbb{F}_3)$ (by Chapter 1, 6.7, these are exactly the subgroups with surjective determinant, no stable line, and an element of determinant -1 and trace 0), $H_1 = H \cap \mathrm{SL}_2(\mathbb{F}_3)$ (of index 2, the kernel of $\det|_H$), and ρ the standard representation. The following are equivalent: (a) H is conjugate to $N_s(\mathbb{F}_3)$; (b) $\mathrm{Res}_{H_1}\rho$ is not absolutely irreducible; (c) $\rho \simeq \rho \otimes \det$; (d) $(\mathrm{ad}^0\,\rho \otimes \det)^H \neq 0$; (e) $\rho \otimes \mathbb{F}_9 \simeq \mathrm{Ind}_{H_1}^H \psi$ for a character ψ of H_1 of order 4; (f) H_1 is abelian. When they fail, $(\mathrm{ad}^0\,\rho \otimes \det)^H = 0$; when they hold, $(\mathrm{ad}^0\,\rho \otimes \det)^H$ is the line spanned by a trace-zero isomorphism. Proof: (b), (c), (e) are equivalent by 6.6 with $\chi_H = \det$ (det has kernel H_1 and values ± 1); (c) iff (d) by 7.4(iv); (a) iff (b) iff (f) by 8.2 to 8.4 (or by 3.7). Reference: proved here from Chapter 1. Paper: Proposition prop:ns (with Chapter 5: $\det\bar{\rho}_{E,3} = \chi_{-3} = \bar{\chi}_3$ cuts out $\mathbb{Q}(\sqrt{-3})$, so $\bar{\rho}(G_{\mathbb{Q}(\sqrt{-3})}) = H_1$), Lemma lem:h0 for $\ell = 3$, the hypothesis of Theorem thm:main at $\ell = 3$, Section sec:anomaly; Chapter 11 assembles the elliptic-curve statement.

6.3.9 Section 9. Unipotent elements and cyclic groups of order 2, 3, 4, 6 on rank-two spaces and lattices

9.1 `prop:reps:unipotent-invariants` (proposition; P). Let k be a field, $V = k^2$, $u \in \mathrm{GL}_2(k)$ unipotent with $u \neq 1$ (so $u = 1 + n'e$ in a suitable basis, $n' \neq 0$; Chapter 1, 5.6), and $\varepsilon \in k^\times$. (i) $V^u = \ker(u-1) = \mathrm{im}(u-1)$ is a line; if $\varepsilon \neq 1$, $V^{\varepsilon u} = 0$ (in particular for "minus a unipotent", $\varepsilon = -1$, $\mathrm{char}\,k \neq 2$). (ii) If $\mathrm{char}\,k \neq 2$: $(\mathrm{ad}^0\,V)^u = (\mathrm{ad}^0\,V)^{\varepsilon u} = k\,(u-1) = k\,e$, a line (by the formulas of 7.3(iii): $(t-1)h = -2n'e$, $(t-1)f = n'h - n'^2 e$); in characteristic 2 it is the plane $ke + kh$. The same holds for the group generated by u (or by εu), and for a group acting through $\langle\varepsilon\rangle \times \langle u\rangle$ with ε scalar. (iii) If $F \in \mathrm{GL}_2(k)$ satisfies $FuF^{-1} = u^q$ ($q \in \mathbb{Z}$; or $q \in \mathbb{Z}_\ell$ when u lies in a pro-ℓ group, Chapter 7), then F preserves $(\mathrm{ad}^0\,V)^u$ and acts on it by q: $F(u-1)F^{-1} = u^q - 1 = q(u-1)$ since $(u-1)^2 = 0$. Then F^{-1} acts by q^{-1}. Reference: elementary; Chapter 1, 5.6 for the group facts. Paper: Lemma lem:IV ("its image in $\mathrm{Aut}(E[3])$ is a nontrivial unipotent subgroup ... $E[3]^{I_q}$ is a line"), Remark rem:types ("minus a unipotent", "a ramified quadratic character times a unipotent": $E[3]^{I_q} = 0$), Lemma lem:euler type I_n ("$(\mathrm{ad}^0\,V)^{I_q}$ is a line with eigenvalue q^{-1}": by (iii), the eigenvalue is q for the arithmetic Frobenius and q^{-1} for the geometric one, important_notes.md point 11), Lemma lem:tamfactor ($(t-1)M = n'(\mathbb{Z}_\ell e \oplus \mathbb{Z}_\ell h)$ for ℓ odd, worked over $\mathbb{Z}_\ell$ in Chapter 16).

9.2 `lem:reps:commutant` (lemma; P). Let k be a field and $x \in \mathrm{End}_k(V)$, $\dim V = n$, whose characteristic polynomial has n distinct roots in $\bar{k}$ (more generally, whose minimal polynomial equals its characteristic polynomial). Then the commutant $\{y \in \mathrm{End}_k(V) : xy = yx\}$ is $k[x] \simeq k[X]/(\det(X - x))$, of dimension n. For $n = 2$ and x not scalar: the commutant is $k \cdot 1 \oplus k \cdot x$, and (if $\mathrm{char}\,k \neq 2$) its trace-zero part is the line spanned by $2x - \mathrm{tr}\,x$. Examples: for σ with $\sigma^2 + \sigma + 1 = 0$ the line is $k(2\sigma + 1)$, with $(2\sigma+1)^2 = -3$ ("the line spanned by $\sqrt{-3}$"); for τ with $\tau^2 = -1$ it is $k\tau$. The commutant is a field iff $\det(X - x)$ is irreducible over k: for $k = \mathbb{Q}_\ell$ and σ of order 3, iff $\ell \not\equiv 1 \pmod 3$. Reference: elementary linear algebra (cyclic vectors). Paper: Lemma lem:euler ("the invariants form the line spanned by $\sqrt{-3} \in \mathbb{Z}_\ell[\zeta_3] \simeq \mathrm{End}_{I_q}(V)$": the correct statement is $\mathrm{End}_{I_q}(V) = \mathbb{Q}_\ell[\sigma] \simeq \mathbb{Q}_\ell \otimes \mathbb{Q}(\zeta_3)$, $\mathrm{End}_{I_q}(T) = \mathbb{Z}_\ell[\sigma]$, important_notes.md point 12; types $\mathrm{III}, \mathrm{III}^*$: the line $k\tau$); Chapter 6's 4.10 (i), (ii) cites this item and adds the Frobenius action and the integral form.

9.3 `prop:reps:cyclic-shapes` (proposition and table; P for the linear algebra; the column "Kodaira types" is a black box from Chapters 10 and 11). Let K be a field and $g \in \mathrm{SL}_2(K)$ of finite order $m \in \{2, 3, 4, 6\}$ with $\mathrm{char}\,K \nmid m$. Then $g = -1$ if $m = 2$; otherwise g has distinct eigenvalues ζ, ζ^{-1}, ζ a primitive m-th root of unity, and characteristic polynomial Φ_m: $X^2 + X + 1$, $X^2 + 1$, $X^2 - X + 1$ for

$m = 3, 4, 6$; for $m = 6$, $g = -\sigma$ with $\sigma = -g$ of order 3. (Proof: $g^m = 1$ with $X^m - 1$ separable, so g is diagonalisable over $\bar{K}$; $\det g = 1$ pairs the eigenvalues; $g \neq \pm 1$ for $m \geq 3$.) Fixed spaces, with 9.1 and 9.2 (checked over $\mathbb{Q}$ and over $\mathbb{F}_3$, Part 7):

shape of the image of inertia	Kodaira types at $q \geq 5$ (Ch. 10, 11)	$\dim V^I$	$\dim(\mathrm{ad}^0 V)^I$	generator of $(\mathrm{ad}^0 V)^I$	action of F ($FgF^{-1} = g^q$) on it	mod 3: $\dim(T/3T)^I$
trivial	I_0	2	3	all of $\mathrm{ad}^0 V$	eigenvalues $\alpha/\beta, 1, \beta/\alpha$ (7.3(ii))	2
$\{\pm 1\}$	I_0^*	0	3	all of $\mathrm{ad}^0 V$	as for the twist E' (7.1(iii))	0
unipotent $u \neq 1$ (infinite)	I_n, $n \geq 1$	1	1	e	q (9.1(iii))	1 if $3 \nmid n$, 2 if $3 \mid n$
εu, ε ramified quadratic	I_n^*	0	1	e	q	0
order 3	IV, IV*	0	1	$2\sigma + 1$	$\chi_{-3}(q)$ (Chapter 6, 4.10)	1 (9.4(ii))
order 4	III, III*	0	1	τ	$\chi_{-4}(q)$ (Chapter 6, 4.10)	0
order 6 ($-\sigma$)	II, II*	0	1	$2\sigma + 1$	$\chi_{-3}(q)$ (Chapter 6, 4.10)	0

The degree of $L_q(A_f, s)^{-1}$ is $\dim(\mathrm{ad}^0 V)^I$ $(3, 3, 1, 1, 1, 1, 1)$ and $\delta_q = \dim V^I$ $(2, 0, 1, 0, 0, 0, 0)$. The eigenvalue q in rows 3 and 4 is for the arithmetic normalisation; the geometric Frobenius gives q^{-1}, which is the paper's $1 - q^{-1-s}$. Reference: Serre-Tate and Kraus90 for the shapes (Chapters 10, 11). Paper: Lemma lem:euler (every row), the table of Section 1, Definition def:terms (δ_p), Lemma lem:IV, Remark rem:types (last column; the complete version is important_notes.md point 22), Theorem thm:qual (a), (b), (c).

9.4 `prop:reps:reduction-cyclic` (proposition; P for (i) to (iv), R (forward) for (v)). Let ℓ be a prime, $T = \mathbb{Z}_\ell^2$, $g \in \mathrm{SL}_2(\mathbb{Z}_\ell)$ of finite order $m \in \{2, 3, 4, 6\}$, $\bar{g}$ its reduction modulo ℓ. (i) If $\ell \nmid m$: $\bar{g}$ has order m (the kernel $1 + \ell M_2(\mathbb{Z}_\ell)$ of reduction is pro-ℓ, Chapter 1, 7.4), Φ_m mod ℓ is separable, and by 3.2(ii) $(T/\ell T)^g = T^g/\ell T^g$ and $(\mathrm{ad}^0 T/\ell)^g = (\mathrm{ad}^0 T)^g/\ell$: invariants commute with reduction. (ii) $\ell = 3$, $m = 3$: $\bar{g}$ is a nontrivial unipotent (reduction is injective on finite subgroups for ℓ odd, Chapter 1, 7.4(iii); an element of order 3 of $\mathrm{GL}_2(\mathbb{F}_3)$ is unipotent, Chapter 1, 5.6(ii)); so $(T/3T)^g$ is a line although $T^g = 0$, and $\dim T/3T - \dim(T/3T)^g = 1$. Invariants do not commute with reduction (Maschke fails, 3.3(a)); on $\mathrm{ad}^0 T$ they still do here: $(\mathrm{ad}^0 T)^g = \mathbb{Z}_3(2g + 1)$ and $2g + 1 \not\equiv 0 \pmod 3$ spans $(\mathrm{ad}^0 T/3)^g$. (iii) $\ell = 3$, $m = 6$: $\bar{g} = -\bar{u}$ with $\bar{u}$ a nontrivial unipotent, $(T/3T)^g = 0$. (iv) $\ell = 3$, $m = 4$: $\bar{g}$ has characteristic polynomial $X^2 + 1$, irreducible over $\mathbb{F}_3$, eigenvalues $\pm i \in \mathbb{F}_9$, $(T/3T)^g = 0$; $m = 2$: $\bar{g} = -1$, $(T/3T)^g = 0$. (v) (Chapter 6, 4.8 to 4.10(v).) T is free of rank one over $\mathbb{Z}_\ell[g] \simeq \mathbb{Z}_\ell[X]/(\Phi_m)$, i.e. over $\mathbb{Z}_\ell[\zeta_3]$ ($m = 3, 6$) or $\mathbb{Z}_\ell[i]$ ($m = 4$), ℓ odd; for $\ell = 3$, $m = 3$: $T/3T \simeq \mathbb{Z}_3[\zeta_3]/(\pi^2)$, $\pi = 1 - \zeta_3$, which is the paper's route to (ii). Reference: Chapter 1 and Chapter 6. Paper: Lemma lem:IV (the algebraic content: the image of I_q in $\mathrm{Aut}(T_3 E)$ cyclic of order 3 with determinant 1 gives a nontrivial unipotent mod 3 and a fixed line; the paper's notation slip is important_notes.md point 13), Remark rem:types (iii), (iv), Theorem thm:qual (a), (b) (Serre conductor exponent 1 for IV, IV*), Lemma lem:euler ($\ell \geq 5$ versus $\ell = 3$).

6.3.10 Section 10. Representations over $\mathbb{Z}_\ell$: lattices, reduction, homothety

10.1 `def:reps:lattice` (definition and proposition; P). Conventions C7. A G-stable lattice $L \subset V$ gives $\rho_L \colon G \to \mathrm{GL}(L) \simeq \mathrm{GL}_n(\mathcal{O})$, conjugate in $\mathrm{GL}_n(K)$ to ρ. (i) Any two lattices satisfy $\lambda^a L \subset L' \subset \lambda^{-b} L$ for some $a, b \geq 0$. (ii) If G is finite, $L = \sum_{g \in G} \rho(g) L_0$ is a G-stable lattice for any lattice L_0; more generally whenever the stabiliser of L_0 has finite index in G (the compact and continuous case is Chapter 5, galois:continuous-reps (c)). (iii) If L is stable, so are $\lambda^m L$, $L^\vee = \mathrm{Hom}_{\mathcal{O}}(L, \mathcal{O}) \subset V^\vee$, $L \otimes \mathcal{O}(\chi)$ for $\chi \colon G \to \mathcal{O}^\times$, $\mathrm{End}_{\mathcal{O}}(L)$ and $\mathrm{ad}^0 L = \mathrm{End}_{\mathcal{O}}(L) \cap \mathrm{ad}^0 V$; $\mathrm{ad}^0 L$ is a direct summand of $\mathrm{End}_{\mathcal{O}}(L)$ (kernel of the split surjection tr), so $\mathrm{ad}^0 L / \lambda^m \mathrm{ad}^0 L = \mathrm{ad}^0(L/\lambda^m L)$. Reference: Serre, Abelian l-adic representations, Chapter I; Chapter 3 for lattices. Paper: Section 1.1 ($T = T_\ell E \subset V = V_\ell E$), Section 4.2 ($M_{f,\lambda}$, $A = \mathrm{ad}^0 T$), Lemma lem:h0 ($W^D[\ell^k] = \mathrm{ad}^0(T/\ell^k)(1)$).

10.2 `prop:reps:reduction-ss` (proposition; P). For a stable lattice L, $\bar\rho_L$ is the representation on $L/\lambda L$ over k, and ρ mod λ^m the one on $L/\lambda^m L$. (i) $\det(X - \bar\rho_L(g))$ is the reduction of $\det(X - \rho(g)) \in \mathcal{O}[X]$, independent of L. (ii) Hence (Brauer-Nesbitt 5.2) $\bar\rho_L^{ss}$ does not depend on L; it is written $\bar\rho^{ss}$. (iii) If $\bar\rho_L$ is irreducible for one L then $\bar\rho_{L'} \simeq \bar\rho_L$ for every stable L' (5.3(ii) with $k' = k$). Reference: Serre, Abelian l-adic representations, Chapter I; DDT, Section 1. Paper: Section 1.1 ($\bar\rho$), Lemma lem:old; Chapter 5's galois:reduction-ss is this item plus continuity.

10.3 `thm:reps:homothety` (theorem, the homothety lemma, which PLAN.md calls Ribet's lemma; P). If $\bar\rho_L$ is irreducible, every G-stable lattice $L' \subset V$ is homothetic to L: $L' = \lambda^m L$ for a unique $m \in \mathbb{Z}$. Proof: by 10.1(i) replace L' by $\lambda^{-m} L'$ with m maximal such that $\lambda^{-m} L' \subset L$; then the image of L' in $L/\lambda L$ is a nonzero stable subspace, hence everything, so $L' + \lambda L = L$ and $L' = L$ by Nakayama (Chapter 3). Reference: standard (Serre, Abelian l-adic representations, Chapter I; the converse direction is Ribet76, Section 2, see 10.4). Paper: Section 4.2 ("$M_{f,\lambda}$ ... and $T_\ell E$ are homothetic when $\bar\rho$ is irreducible": applied to $M_{f,\lambda}$ and $T_\ell E^\vee$ inside $V_\ell E^\vee$, important_notes.md point 26, then 7.7); Chapter 15 (comparison of $\rho_{f,\lambda}$ with $\rho_{E,\ell}$).

10.4 `ex:reps:nonhomothetic` (worked example and theorem; P for (i), R for (ii)). (i) Let $G \subset \mathrm{GL}_2(\mathbb{Z}_\ell)$ be upper triangular, containing $t = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$. $L = \mathbb{Z}_\ell e_1 \oplus \mathbb{Z}_\ell e_2$ and $L' = \mathbb{Z}_\ell e_1 \oplus \ell \mathbb{Z}_\ell e_2$ are both stable and not homothetic; $\bar\rho_L$ is a non-split extension (3.6: $\bar t$ is a nontrivial unipotent), while in the basis $e_1, \ell e_2$ of L' the matrix of $g = \begin{pmatrix} a & b \\ 0 & d \end{pmatrix}$ is $\begin{pmatrix} a & \ell b \\ 0 & d \end{pmatrix}$, so $\bar\rho_{L'}$ is the direct sum of the two characters: same semisimplification (10.2), different reductions. (Elliptic-curve analogue: two ℓ-isogenous curves, Chapters 9 and 11.) (ii) (Ribet's lemma.) If V is irreducible of dimension 2 and $\bar\rho^{ss} \simeq \chi_1 \oplus \chi_2$, then there is a stable lattice whose reduction is a non-split extension with sub-character χ_1 (and one with sub-character χ_2). Reference: Ribet76, Section 2. Paper: background (the Eisenstein case: Section sec:eis and Conjecture conj:eis concern reducible $\bar\rho$, where lattices are not unique up to homothety).

10.5 `prop:reps:twist-length` (proposition; P). Let G be any group, T free of rank n over $\mathcal{O}$ with a G-action such that $\bar T = T/\lambda T$ is absolutely irreducible, and $\chi \colon G \to \mathcal{O}^\times$ a character. For $m \geq 1$ put $M_m = \mathrm{Hom}_G(T/\lambda^m, (T/\lambda^m)(\chi))$. (i) M_m is a cyclic $\mathcal{O}$-module, isomorphic to $\mathcal{O}/\lambda^j$ for some $j \leq m$: its λ-torsion is $\mathrm{Hom}_G(\bar T, \bar T(\bar\chi))$ (maps with image in $\lambda^{m-1} T/\lambda^m T \simeq \bar T$), of dimension at most one by Schur 2.2, and an $\mathcal{O}$-module of finite length with λ-torsion of length at most one is cyclic (Chapter 3). (ii) An element $\phi \in M_m$ with $\lambda^{m-1}\phi \neq 0$ is an isomorphism $T/\lambda^m \simeq (T/\lambda^m)(\chi)$: $\lambda^{m-1}\phi \neq 0$ means ϕ mod $\lambda \neq 0$, a nonzero map of irreducible representations of the same dimension, hence surjective, and ϕ is surjective by Nakayama, hence bijective (finite sets of the same size, or ranks). (iii) Let $X = \mathrm{End}_{\mathcal{O}}(T)(\chi) \otimes_{\mathcal{O}} K/\mathcal{O}$. Then $X[\lambda^m] = \mathrm{End}(T/\lambda^m)(\chi)$, $X^G[\lambda^m] = M_m$, and X^G is either $\mathcal{O}/\lambda^h$ with $h = \max\{m : T/\lambda^m \simeq (T/\lambda^m)(\chi)\}$, or $K/\mathcal{O}$ when such isomorphisms exist for every m, in which case $T \simeq T(\chi)$ (an inverse limit of nonempty finite sets is nonempty). (iv) If n is invertible in $\mathcal{O}$ and $\bar\chi \neq 1$, the same holds for $\mathrm{ad}^0 T(\chi) \otimes K/\mathcal{O}$ (7.4(iii)). (v) If $T/\lambda^m \simeq (T/\lambda^m)(\chi)$ then $\chi^n \equiv 1 \pmod{\lambda^m}$ on G (determinants, 7.1(iv)). For $n = 2$, $\mathcal{O} = \mathbb{Z}_\ell$, ℓ odd, and χ with $\chi(G) = \mathbb{Z}_\ell^\times$ (the cyclotomic character, Chapter 5): $x^2 \equiv 1 \pmod{\ell^m}$ for all $x \in \mathbb{Z}_\ell^\times$ forces $\ell^m = 3$ (Chapter 2, 5.5), so $h = 0$ for $\ell \geq 5$ and $h \leq 1$ for $\ell = 3$. Reference: elementary. Paper: Lemma lem:h0 (the length $h_\ell(E)$ "is the largest k such that ρ mod $\ell^k \simeq (\rho \otimes \chi_\ell)$ mod ℓ^k" and its finiteness: (iii) and (v) give the corrected statement of important_notes.md point 5, with absolute irreducibility as in point 10), Conjecture conj:euler, Remark rem:ns, Section sec:anomaly ($h_3(E) \leq 1$ without the numerical

check); Chapter 11 applies it with $G = G_{\mathbb{Q}}$, $\chi = \chi_\ell$, Chapter 18 with the Pontryagin-dual language.

6.3.11 Section 11. The mod ℓ world: a foretaste, and Dickson's list

11.1 `rem:reps:mod-l-galois` (remark with a dictionary; P for the translations, the Galois inputs cited). For $\bar\rho\colon G_{\mathbb{Q}} \to \mathrm{GL}_2(\mathbb{F}_\ell)$ continuous (finite image, Chapter 5), ℓ odd: (i) complex conjugation c has $\bar\rho(c)^2 = 1$, $\det\bar\rho(c) = -1$ (Chapters 5, 9), so irreducible iff absolutely irreducible (2.4); (ii) the hypothesis of Theorem thm:main concerns $\mathrm{Res}_H\,\bar\rho$ for the index-two subgroup $H = G_{\mathbb{Q}(\sqrt{\ell^*})}$, and by 6.6 it fails iff $\bar\rho \simeq \bar\rho \otimes \chi_{\ell^*}$ iff $\bar\rho \otimes \bar{\mathbb{F}}_\ell$ is induced from H; (iii) Lemma lem:h0 concerns $\bar\rho \simeq \bar\rho \otimes \bar\chi_\ell$ (7.4, 10.5), which needs $\bar\chi_\ell^2 = 1$, i.e. $\ell = 3$, where $\bar\chi_3 = \chi_{-3} = \det\bar\rho$ and 8.5 applies; for $\ell \geq 5$ the two conditions differ (important_notes.md points 5 and 6). (iv) A dictionary table: V, T, ρ, $\bar\rho$, $\rho \bmod \ell^k$, $A_f = \mathrm{ad}^0\,V$, $A = \mathrm{ad}^0\,T$, $T^D = \mathrm{Hom}(A, \mathbb{Z}_\ell(1)) \simeq A(1)$, $W = A \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$, W^D, each with the item of this chapter that constructs it. Paper: Theorem thm:main hypothesis, S_f, Lemma lem:h0, Proposition prop:ns, Section 1.3 (point 6).

11.2 `thm:reps:dickson` (theorem; R; restated from Chapter 1, 5.10 in projective form). Let ℓ be a prime and $P \leq \mathrm{PGL}_2(\mathbb{F}_\ell)$. If $\ell \mid \#P$, then P contains $\mathrm{PSL}_2(\mathbb{F}_\ell)$ or fixes a point of $\mathbb{P}^1(\mathbb{F}_\ell)$. If $\ell \nmid \#P$, then P is cyclic, dihedral (including V_4), or isomorphic to A_4, S_4 or A_5. Representation-theoretic reading for an absolutely irreducible $\bar\rho$ with projective image P (6.7): cyclic is impossible; dihedral means induced from an index-two subgroup; exceptional and "large" ($P \supseteq \mathrm{PSL}_2$) images are not induced. At $\ell = 3$: $\mathrm{GL}_2(\mathbb{F}_3) \mapsto S_4$, $N_{ns}(\mathbb{F}_3) \mapsto D_4$, $N_s(\mathbb{F}_3) \mapsto V_4$ (8.1); $N_s(\mathbb{F}_3)$ is not maximal (important_notes.md point 2). Reference: Serre72, Section 2; Chapter 1, 5.10. Paper: background for Proposition prop:ns (Zywina's list), S_f.

6.4 3. Section-by-section plan with target line counts

Line counts are lines of LaTeX, card excluded (PLAN.md Section 7.3: 1,000 to 2,200). Items per section: 4, 5, 7, 5, 3, 7, 7, 5, 4, 5, 2 (54 items).

section	title	contents	target lines
(starred)	Core results at a glance	the card: every item of Part 2, stated as above, with tag and `\Cref`	(excluded)
(starred)	Where the paper uses this chapter	prop:ns; lem:h0 (with the corrected reading); lem:IV, rem:types, lem:euler; lem:old; Section 4.2, lem:fpr; lem:tamfactor; Section 1.1 and Sym^2; hypothesis of thm:main, S_f, DDT 3.15; conj:euler, rem:ns, sec:anomaly	40
1	Representations over a field	conventions C1 to C9 as `convention` environments; 1.1 to 1.4; one sentence on RHB	150

section	title	contents	target lines
2	Irreducibility, Schur's lemma, absolute irreducibility	2.1 to 2.5; `paperlink` for the hypothesis of thm:main and for lem:h0 / lem:old (point 10)	190
3	Semisimplicity, Maschke, Jordan-Holder	3.1 to 3.7; `paperlink` for lem:IV (3.3(a)), lem:old (3.6), prop:ns (3.7, point 1)	200
4	Characters in characteristic zero	4.1 to 4.5; the tables of D_4 and Q_8 (`booktabs`); one paragraph on RHB; `warning` on traces in characteristic p	130
5	Brauer-Nesbitt	5.1 to 5.3, with the full proof of 5.2; `paperlink` for lem:old	140
6	Restriction, induction, Frobenius reciprocity, Mackey, Clifford	6.1 to 6.7; `paperlink` for prop:ns and lem:h0 (induced over $\mathbb{F}_9$, point N1)	240
7	Twisting, the adjoint, the symmetric square, the trace form	7.1 to 7.7; the e, h, f formulas in a displayed array; `paperlink` for lem:h0, lem:tamfactor, Section 4.2 (point 26), Section 1.1 ($L_p(\mathrm{Sym}^2 f, s) = L_p(A_f, s-1)$)	210
8	$\mathrm{GL}_2(\mathbb{F}_3)$ and its subgroups on $\mathbb{F}_3^2$ and $\mathbb{F}_9^2$	8.1 to 8.5; the table of 8.1 (`booktabs`, landscape-free: split into two tables if needed); `paperlink` for prop:ns, sec:anomaly	200
9	Unipotent elements and cyclic groups of order $2, 3, 4, 6$	9.1 to 9.4; the shape table of 9.3; `paperlink` for lem:IV, rem:types, lem:euler	180
10	Representations over $\mathbb{Z}_\ell$: lattices, reduction, homothety	10.1 to 10.5; `paperlink` for Section 4.2 and lem:h0 ($h_\ell(E)$, point 5)	200
11	The mod ℓ world and Dickson's list	11.1, 11.2; the dictionary table	70
12	Exercises	the 17 exercises of Part 4.2	130

section	title	contents	target lines
13	Notes and references	Serre (Linear representations; Abelian l-adic representations), Curtis-Reiner, Lang, Dummit-Foote, Serre72, Ribet76, DDT; which numbers were cited; RHB Chapter 29	40
	total		**2,120**

6.5 4. Coverage table and exercises

6.5.1 4.1 Coverage: PLAN.md scope, Section 6 map rows and paper hooks to line-up items

PLAN.md scope item, Section 6 row, or paper hook	line-up items
representations of a finite group over a field (any characteristic)	1.1, 1.2, 1.3, 1.4
subrepresentations	1.2, 2.1
irreducibility	2.1, 2.2
absolute irreducibility (irreducible after extending scalars to the algebraic closure)	2.3, 2.4, 2.5, 3.7
Schur's lemma in both forms	2.2
characters and the character table in characteristic zero (brief, RHB connection)	4.1, 4.2, 4.3, 4.4
Maschke's theorem and its failure in characteristic dividing the group order	3.2, 3.3, 9.4(ii)
semisimple versus non-semisimple modules	3.1, 3.3, 3.6
Jordan-Holder theorem and composition factors	3.4, 3.5
Brauer-Nesbitt (characteristic polynomials, not traces, in characteristic ℓ)	4.5, 5.1, 5.2, 5.3, 10.2
restriction and induction	6.1, 6.3
Frobenius reciprocity	6.2
Mackey's criterion for irreducibility of induced representations	6.4
Clifford theory for index two (restriction irreducible or two conjugate characters; $\rho \simeq \rho \otimes \mathrm{sgn}$ iff induced)	6.5, 6.6, 6.7
twisting by one-dimensional characters	7.1
the trace of an isomorphism $\rho \to \rho \otimes \chi$ is zero when χ is nontrivial	7.2, 7.4(iii)
the adjoint $\mathrm{End}\, V$ and $\mathrm{ad}^0 V$	7.3, 7.7
Sym^2 and $\mathrm{ad}^0 \rho = \mathrm{Sym}^2 \rho \otimes \det^{-1}$ for $\dim 2$	7.5
$\mathrm{Hom}_G(V, W) = (V^\vee \otimes W)^G$; $H^0(\mathrm{ad}^0 \rho \otimes \chi)$ as trace-zero $\mathrm{Hom}_G(\rho, \rho \otimes \chi)$	1.3, 7.4
the trace form and the self-duality of ad^0 for ℓ odd	7.6

PLAN.md scope item, Section 6 row, or paper hook	line-up items
representations of $\mathrm{GL}_2(\mathbb{F}_3)$ and its subgroups on $\mathbb{F}_3^2$ and $\mathbb{F}_9^2$; $C_4 \subset N_s(\mathbb{F}_3)$ acts as $\psi \oplus \psi^{-1}$, $N_s(\mathbb{F}_3)$ as $\mathrm{Ind}\,\psi$	2.5, 3.7, 8.1, 8.2, 8.3, 8.4, 8.5
unipotent and minus-unipotent elements and their fixed spaces	9.1, 3.3(a), 9.4(ii), (iii)
cyclic groups of order $2, 3, 4, 6$ on rank-two $\mathbb{Z}_\ell$-lattices: eigenvalues, the $\mathbb{Z}_\ell[\zeta_3]$-module structure, invariants on the adjoint	9.2, 9.3, 9.4
G-stable lattices, reduction mod ℓ, homothety	10.1, 10.2, 10.3, 10.4
Ribet's lemma (irreducible reduction forces all stable lattices homothetic)	10.3 (and the converse lemma of Ribet, 10.4(ii))
irreducible versus absolutely irreducible for mod ℓ	2.4, 11.1
Galois representations (foretaste); the criterion via an element with distinct eigenvalues	
Dickson's list (stated, from Chapter 1)	11.2, 6.7
Section 6 row: $\rho_{E,\ell}$, $\bar\rho$, T, V, absolute irreducibility, restriction to $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{\ell^*}))$	1.1, 2.3, 2.4, 6.6, 10.1, 10.2, 11.1
Section 6 row: adjoint $\mathrm{ad}^0 V$, Sym^2, trace form, $A_f(1) = \mathrm{Hom}(A_f, \mathbb{Z}_\ell(1))$	7.3, 7.5, 7.6, 7.7, 10.1(iii)
Section 6 row: N_s, 3Ns, $\mathrm{GL}_2(\mathbb{F}_3)$ and subgroups, orbits on lines	8.1 to 8.5 (orbits: Chapter 1)
Section 6 row: $h_\ell(E)$, $\bar\rho \simeq \bar\rho \otimes \chi$	7.2, 7.4, 10.5, 6.6
Section 6 row: induced representations, Frobenius reciprocity, characters of order 4	6.1, 6.2, 6.4(c), 8.2
Section 6 row: image of inertia on $T_\ell E$ at additive primes, unipotent elements, $\mathbb{Z}_3[\zeta_3]$	9.1, 9.3, 9.4
Section 6 row: Chebotarev, Brauer-Nesbitt	5.2, 5.3, 10.2 (Chebotarev: Chapter 5)
hook: Proposition prop:ns (induced representations, ψ of order 4, Frobenius reciprocity, $\bar\rho \simeq \bar\rho \otimes \chi_{-3}$)	1.4(ii), 2.5, 3.7, 6.2, 6.3, 6.4(c), 6.6, 6.7, 7.4, 8.1 to 8.5
hook: Lemma lem:h0 (trace-zero isomorphisms, $\mathrm{ad}^0 \bar\rho \otimes \chi$, the length h_ℓ)	1.3, 2.2, 2.4, 6.6, 7.1(iv), 7.2, 7.4, 10.5
hook: Lemma lem:IV (eigenvalues of inertia, $\mathbb{Z}_3[\zeta_3]$, unipotent mod 3)	3.3(a), 9.1, 9.3, 9.4
hook: Lemma lem:euler (invariants on the adjoint, the trace-zero line, End_{I_q})	7.3(ii), 9.1(iii), 9.2, 9.3, 3.2(ii)
hook: Remark rem:types (minus unipotent, order 2 and 4 mod 3)	9.1(i), 9.3, 9.4(iii), (iv)
hook: Lemma lem:old (Brauer-Nesbitt plus Chebotarev gives $\bar\rho_h \simeq \bar\rho$; unique stable line)	1.4, 3.6, 5.2, 5.3(ii), 10.2
hook: Section 4.2 (homothetic lattices $M_{f,\lambda}$ and $T_\ell E$; $A = \mathrm{ad}^0 T$; $T^D = \mathrm{Hom}(A, \mathbb{Z}_\ell(1)) \simeq A(1)$)	7.6, 7.7, 10.1, 10.3
hook: $A_f = \mathrm{ad}^0 V$ and the trace form $A_f(1) = \mathrm{Hom}(A_f, \mathbb{Z}_\ell(1))$	7.3, 7.6
also: Lemma lem:tamfactor (basis e, h, f, adjoint action of t and of $\rho(\mathrm{Frob})$)	7.3(iii), 9.1, 3.6
also: Section 1.1 ($L_p(\mathrm{Sym}^2 f, s) = L_p(A_f, s-1)$; eigenvalues $\alpha/\beta, 1, \beta/\alpha$)	7.3(ii), 7.5
also: proof of Theorem thm:main and Theorem thm:quant ($\mathrm{ad}^0 T$ unchanged by twists)	7.1(iii), 7.7
also: Conjecture conj:euler, Remark rem:ns, Section sec:anomaly ($h_\ell(E)$, image 3Ns)	8.2, 8.5, 10.5

PLAN.md scope item, Section 6 row, or paper hook	line-up items
also: the hypothesis of Theorem thm:main, Theorem thm:qual, DDT 3.15, S_f	2.3, 2.4, 6.6, 6.7, 8.5, 11.1

6.5.2 4.2 Exercises (Section 12 of the chapter), graded from easy to harder

1. Show that $C_4 = \langle J \rangle$ acting on $\mathbb{F}_3^2$ is irreducible but not absolutely irreducible, that its commutant is a field with 9 elements, and decompose $\mathbb{F}_9^2$ into eigenlines.
2. Prove that every absolutely irreducible representation of a finite abelian group is one-dimensional, and give an irreducible two-dimensional representation of C_8 over $\mathbb{F}_3$.
3. Show that the standard representation of $\mathrm{U} = \langle u \rangle$ on $\mathbb{F}_3^2$ is indecomposable but not irreducible, and that $\mathbb{F}_p[C_p]$ is uniserial. Why does this not contradict Maschke's theorem?
4. Give two non-isomorphic semisimple representations of C_2 over $\mathbb{F}_3$ with the same trace function. Show that for two-dimensional representations with the same determinant in characteristic $\neq 2$, traces suffice.
5. Check the paper's claim (Proposition prop:ns) that the determinant-one subgroup C_4 of $N_s(\mathbb{F}_3)$ acts on $\mathbb{F}_9^2$ as $\psi \oplus \psi^{-1}$ with ψ of order 4, and that the standard representation of $N_s(\mathbb{F}_3)$ over $\mathbb{F}_9$ is isomorphic to $\mathrm{Ind}_{C_4}^{N_s} \psi$: find the basis $(v, \sigma v)$ explicitly. Show that over $\mathbb{F}_3$ it is not induced from a character of C_4.
6. Check the paper's claim (Proposition prop:ns) that Q_8 and $\mathrm{SL}_2(\mathbb{F}_3)$ act absolutely irreducibly on $\mathbb{F}_3^2$, by Maschke's theorem (3.7) rather than by commutators; explain why "non-abelian and faithful" is not enough for the Borel subgroup $\mathrm{B}(\mathbb{F}_3)$.
7. Show that the standard representation of $N_s(\mathbb{F}_3)$ is isomorphic to its twist by each of the three non-trivial characters of $N_s(\mathbb{F}_3)/\{\pm 1\}$, with intertwiners J, σ, w. Show that the standard representation of $N_{ns}(\mathbb{F}_3)$ is induced from $C_{ns}(\mathbb{F}_3)$ but not isomorphic to its twist by det.
8. Check the paper's claim (Lemma lem:h0) that an isomorphism $\phi\colon \bar{\rho} \to \bar{\rho} \otimes \chi_{-3}$ has trace 0; show more generally that $(\mathrm{ad}^0 V \otimes \chi)^G = \mathrm{Hom}_G(V, V \otimes \chi)$ for every nontrivial χ when $\dim V$ is invertible.
9. Let $H \leq G$ have index 2 and V be absolutely irreducible of dimension 2, $\mathrm{char}\, k \neq 2$. Prove directly that $V \simeq V \otimes \chi_H$ iff $\mathrm{tr}\, \rho(g) = 0$ for all $g \notin H$, and deduce that the standard representation of $\mathrm{GL}_2(\mathbb{F}_3)$ is not isomorphic to its twist by det.
10. Prove $\mathrm{ad}^0 V \simeq \mathrm{Sym}^2 V \otimes \det^{-1}$ for $\dim V = 2$ by the explicit map Ξ of 7.5, and show that over $\mathbb{Z}$ the image of $\mathrm{Sym}^2 \mathbb{Z}^2$ has index 4 in $\mathrm{ad}^0 \mathbb{Z}^2$. Deduce the eigenvalues α/β, 1, β/α from $\alpha^2, \alpha\beta, \beta^2$.
11. Compute the Gram matrix of the trace form on $\mathrm{ad}^0 \mathbb{Z}_\ell^2$ in the basis e, h, f, and check the paper's identification $T^D = \mathrm{Hom}(A, \mathbb{Z}_\ell(1)) \simeq A(1)$ (Section 4.2) for ℓ odd. What goes wrong for $\ell = 2$?
12. Check the formulas of Lemma lem:tamfactor: for $t = 1 + n'e$, $(t-1)e = 0$, $(t-1)h = -2n'e$, $(t-1)f = n'h - n'^2 e$; and for $F = \begin{pmatrix} \varepsilon q & c' \\ 0 & \varepsilon \end{pmatrix}$ with $\varepsilon = \pm 1$, $FeF^{-1} = qe$, $FhF^{-1} = h - 2c'\varepsilon e$.
13. For each of $g = -1$, g of order 3, 4, 6 in $\mathrm{SL}_2(\mathbb{Z})$, $g = u$ and $g = -u$, compute $\dim V^g$ and $\dim(\mathrm{ad}^0 V)^g$ over $\mathbb{Q}$ and over $\mathbb{F}_3$, and check the paper's claims in Lemma lem:IV and Remark rem:types about $E[3]^{I_q}$.
14. Let $FuF^{-1} = u^q$ with $u \neq 1$ unipotent. Show that F acts on $(\mathrm{ad}^0 V)^u$ by q, and explain why the paper's factor $1 - q^{-1-s}$ at a multiplicative prime requires the geometric Frobenius (important_notes.md point 11).
15. Show that for a finite group of order prime to ℓ invariants commute with reduction modulo ℓ, and that this fails for $g \in \mathrm{SL}_2(\mathbb{Z}_3)$ of order 3 (Lemma lem:IV), although it holds for the invariants of ad^0.
16. In $\mathbb{Q}_\ell^2$ with the upper triangular group, exhibit two stable lattices that are not homothetic and whose reductions are non-isomorphic but have the same semisimplification. Prove the homothety lemma 10.3.
17. Let T be a free $\mathbb{Z}_\ell$-module of rank 2 with an action of G such that $T/\ell T$ is absolutely irreducible, and $\chi\colon G \to \mathbb{Z}_\ell^\times$ surjective. Show that $\mathrm{Hom}_G(T/\ell^m, (T/\ell^m)(\chi))$ is cyclic and that the length of $H^0(G, \mathrm{ad}^0 T(\chi) \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell)$ is at most 1, and is 0 unless $\ell = 3$ (the corrected Lemma lem:h0,

important_notes.md point 5).

6.6 5. Dependencies and forward pointers

Results from other chapters used here (by chapter label only): * Chapter 1 (ch:groups): the standard models and the class and subgroup tables of $\mathrm{GL}_2(\mathbb{F}_3)$ (its 6.1 to 6.6), the three images (its 6.7), projective images (its 6.11), unipotent and minus-unipotent elements (its 5.6), Borel subgroups and the commutator lemma (its 5.5, 5.7), Cartan subgroups and normalisers (its 5.8, 5.9), $1+\ell M_2(\mathbb{Z}_\ell)$ pro-ℓ and torsion-free for ℓ odd (its 7.4), index-two subgroups (its 1.6), D_4, Q_8, SD_{16} (its 2.4 to 2.6), Dickson (its 5.10). Chapter 4 must use, not redo, these. * Chapter 2 (ch:rings): $\mathbb{F}_9 = \mathbb{F}_3[i]$, finite fields and their Frobenius, characters of finite abelian groups and orthogonality (its 10.2), cyclotomic polynomials Φ_m, $a^2 \equiv 1 \pmod{\ell^k}$ (its 5.5). * Chapter 3 (ch:modules): modules, Hom, $\otimes$, exact sequences; Nakayama; DVRs, $\mathbb{Z}_\ell$, lattices (finitely generated torsion-free modules over a DVR are free); cyclicity criterion for finite-length modules over a DVR; saturated submodules and direct summands; Gram determinants and perfect pairings; $\mathbb{Q}_\ell/\mathbb{Z}_\ell$ and $T \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$; Pontryagin duality (only in remarks). * Background (PLAN.md Section 2): linear algebra (eigenvalues, diagonalisation, cyclic vectors, Vandermonde), RHB's representation theory of finite groups (Chapter 29 of RHB, 3rd edition).

Black boxes imported from later chapters (stated with a forward \Cref, never used in the proof of a numbered result of this chapter): Galois representations, continuity, complex conjugation, $\det \bar{\rho}_{E,\ell} = \bar{\chi}_\ell$ (Chapters 5, 9, 11), the assignment of inertia shapes to Kodaira types (Kraus; Chapters 10, 11), the Frobenius action $\chi_{-3}(q)$, $\chi_{-4}(q)$ on the invariant lines and the $\mathbb{Z}_\ell[\zeta_3]$-module structure (Chapter 6), Tate twists and L-factors (Chapters 18, 19). They appear only in paperlink environments, in the "Kodaira types" and "action of F" columns of the table of 9.3, in 9.4(v), 7.5(iii) and 11.1.

Forward pointers (chapters that use this one): * Chapter 5 (ch:galois): 2.4 (complex conjugation criterion), 5.2 and 5.3 (Chebotarev plus Brauer-Nesbitt, congruent a_p), 6.6 (galois:index-two-restriction), 7.1(iv) and 7.2 (galois:twist), 10.1 to 10.3 (galois:reduction-ss). * Chapter 6 (ch:ant): 9.2 (its 4.10 (i), (ii)), 6.1 trace formula and 7.2 (its 6.4, 6.5), 5.2 (via Chapter 5). * Chapter 7 (ch:local): 3.3(c) and 9.1 (unipotent tame inertia, Weil-Deligne shapes), 9.3 (finite inertia of order $2, 3, 4, 6$). * Chapter 9 (ch:ellcurves): twists $\rho \otimes \chi_d$ (7.1); $V^\vee \simeq V \otimes \det^{-1}$ and the Weil pairing (1.2). * Chapter 11 (ch:galrep): 2.4, 3.6, 6.6, 7.4, 8.5 (Proposition prop:ns), 9.1, 9.3, 9.4 (Lemma lem:IV, Remark rem:types), 10.5 (Lemma lem:h0), 11.1. * Chapter 13 (ch:hecke): twisting (7.1) on the Galois side of the Atkin-Li theorem (motivation only). * Chapter 15 (ch:modgalrep): 3.6 (unique stable line in Lemma lem:old), 5.3 (Lemma lem:old), 10.2, 10.3, 7.7 ($M_{f,\lambda}$ versus $T_\ell E$). * Chapter 16 (ch:cohomology): 1.3, 7.3 (E_{ij}, the basis e, h, f, the conjugation action), 9.1, 3.6 (extensions and cocycles), 3.2(ii) (cohomology of groups of order prime to ℓ is the same averaging). * Chapter 17 (ch:galcoh): 7.6 (duals), 3.2(ii). * Chapter 18 (ch:blochkato): 7.6 (self-duality of ad^0, $T^D \simeq A(1)$), 7.7, 10.1(iii), 10.5 ($h_\ell(E)$). * Chapter 19 (ch:lfunctions): 7.3(ii) (eigenvalues of ad^0), 7.5 ($\mathrm{ad}^0 \simeq \mathrm{Sym}^2 \otimes \det^{-1}$ and the shift $L(\mathrm{Sym}^2 f, s) = L(A_f, s-1)$), 9.3 (degrees of the local factors). * Chapter 20 (ch:congruence): 7.7 and 10.3 (lattices $M_{f,\lambda}$). * Chapters 21, 22 (ch:paper, ch:computation): 8.1 to 8.5 (the ns3 test, image 3Ns), 9.3 (Lemma lem:euler), 10.5.

6.7 6. Notes for the writer

Conventions to fix (Section 1 of the chapter, as convention environments): C1 to C9 of Part 1. In addition: * Macros: \ad is ad^0 and \Sym is Sym^2 (both already carry the superscript). Write the full adjoint as \End(V) or \Ad V. There is no \O macro: write \mathcal{O}. \Ind, \Res, \tr, \Hom, \End, \GL, \SL, \PGL exist. \Res is restriction here, never the resultant. * Use the paper's basis e, h, f of $\mathrm{ad}^0 V$ (C3) and the matrices of C6; keep the paper's σ for the order-3 element of Section 9 and rename Chapter 1's $\mathrm{diag}(1, -1)$ there (C6). * Characters: χ, ψ, θ for one-dimensional representations; tr_V for

trace functions. The paper's χ_ℓ (cyclotomic) and χ_q (quadratic; the book's χ_{q^*}, decision D3) appear only in `paperlink`s (Chapter 2, C5, and important_notes.md point 32). * Rigour tags on the card exactly as in Part 2. Theorem numbers are cited only where marked as sure: Serre, Linear representations, Theorem 1 (Maschke), Proposition 4 (Schur), Theorem 13 (Frobenius reciprocity) were recalled from memory and should be checked, otherwise cite sections (1.3, 2.2, 7.2). Curtis-Reiner and Lang are cited by section only. `CurtisReiner` is not in `refs.bib`: add it to `bib/ch04.bib` (Curtis and Reiner, Representation theory of finite groups and associative algebras, Interscience 1962). RHB's chapter number (29 in the 3rd edition) should be checked.

Pitfalls: * **Traces versus characteristic polynomials.** In characteristic ℓ, Brauer-Nesbitt needs characteristic polynomials (4.5); for $\dim 2$ with equal determinants (the paper's situation, $\det = \bar{\chi}_\ell$) traces suffice (5.3(i)). Say both. * **Irreducible versus absolutely irreducible.** Every place where the paper says "irreducible" and then uses Schur in the form $\mathrm{End} = k$, Clifford theory, or Brauer-Nesbitt over a larger field, absolute irreducibility is what is needed (important_notes.md point 10). For $\bar{\rho}_{E,\ell}$ the two coincide by 2.4 (complex conjugation). The example to show: C_4 on $\mathbb{F}_3^2$ (2.5), and 5.3(ii)'s counterexample. * **"Induced" over which field.** $\bar{\rho}_{E,3}$ with image $N_s(\mathbb{F}_3)$ is induced from $G_{\mathbb{Q}(\sqrt{-3})}$ only after extending scalars to $\mathbb{F}_9$; over $\mathbb{F}_3$ its restriction to $G_{\mathbb{Q}(\sqrt{-3})}$ is irreducible (8.2(ii); Part 8, point N1). The twist isomorphism $\bar{\rho} \simeq \bar{\rho} \otimes \chi_{-3}$ does hold over $\mathbb{F}_3$ (1.4(ii)). * **Absolute irreducibility of Q_8 and $\mathrm{SL}_2(\mathbb{F}_3)$** (important_notes.md point 1). Give both proofs: Chapter 1's commutator lemma and 3.7 (non-abelian of order prime to 3, then containment). Do not "correct" the paper in the text; supply the step in a `paperlink`. * $N_s(\mathbb{F}_3)$ **itself is absolutely irreducible**; only its determinant-one subgroup is not. Never write "the image $N_s(\mathbb{F}_3)$ is reducible". Its representation is induced in three ways (projective image V_4), and the representation of $N_{ns}(\mathbb{F}_3)$ is also induced (from C_{ns}); the hypothesis of Theorem thm:main excludes only "induced from $\mathbb{Q}(\sqrt{\ell^*})$" (6.7, 8.3). * **Maschke over $\mathbb{Z}_\ell$.** Invariants commute with reduction only when the group order is prime to ℓ (3.2(ii)); the paper's Lemma lem:IV is exactly the failure at $\ell = 3$ (9.4(ii)). The elementary proof of 9.4(ii) (Chapter 1, 7.4(iii) and 5.6(ii)) does not need $\mathbb{Z}_3[\zeta_3]$; give it, and point to Chapter 6 for the paper's route. * **Mackey in characteristic p.** The general criterion 6.4(b) is a characteristic-0 statement ($\mathrm{End} = k$ does not imply irreducible without semisimplicity, 3.3(d)); only the index-two case 6.4(c) is needed and it holds in every characteristic for absolutely irreducible W. * **Clifford 6.6 needs** $\mathrm{char}\, k \neq 2$ (so that $\chi_H \neq 1$ and $\mathrm{Ind}\, \mathbf{1} = \mathbf{1} \oplus \chi_H$), and absolute irreducibility of V. * **The splitting** $\mathrm{End}\, V = k \oplus \mathrm{ad}^0 V$**, the isomorphism** $\mathrm{ad}^0 \simeq \mathrm{Sym}^2 \otimes \det^{-1}$ **and the perfectness of the trace form on** ad^0 **all need 2 invertible** (ℓ odd); give the index-4 and index-2 computations over $\mathbb{Z}$ and $\mathbb{Z}_2$ (7.5, 7.6). * **Twist direction for** Sym^2**.** $\mathrm{Sym}^2 V_\ell E \simeq \mathrm{ad}^0 V(1)$ but the classical $L(\mathrm{Sym}^2 f, s)$ is that of $\mathrm{Sym}^2(V^\vee) \simeq \mathrm{ad}^0 V(-1)$, whence $L_p(\mathrm{Sym}^2 f, s) = L_p(A_f, s-1)$. Keep 7.5(iii) as a remark with pointers; the L-factor bookkeeping is Chapter 19's. (Cross-outline note: Chapter 21's item C2 says that on $V_\ell E$ the geometric Frobenius has eigenvalues α, β; on $V_\ell E$ it is the arithmetic Frobenius that has eigenvalues α, β, and the geometric one has α^{-1}, β^{-1}; the statement is right for $V_\ell E^\vee = H^1$. This does not affect any number in the paper.) * **Frobenius conventions** (important_notes.md point 11). This chapter never uses Frobenius; 9.1(iii) and 9.3 state the eigenvalue for an abstract F with $FgF^{-1} = g^q$ (arithmetic) and note that F^{-1} (geometric) gives the inverse. * **Name of 10.3.** PLAN.md calls the homothety statement "Ribet's lemma"; in the literature Ribet's lemma usually means the converse construction 10.4(ii). Call 10.3 "the homothety lemma" and mention both names once. * **Dual lattices** (important_notes.md point 26). $M_{f,\lambda}$ of DFG04 sits in H^1, dual to the Tate module; 10.3 gives homothety with $T_\ell E^\vee$, and 7.7 identifies the adjoints. State 7.7(iv) in exactly this form. * $h_\ell(E)$ (important_notes.md point 5). Prove 10.5 for an abstract group and character; do not reproduce the paper's CM sentence. The application to $G_\mathbb{Q}$ and χ_ℓ ($h_\ell = 0$ for $\ell \geq 5$, $h_3 \leq 1$) is one line with Chapter 5's $\chi_\ell(G_\mathbb{Q}) = \mathbb{Z}_\ell^\times$ and Chapter 2's 5.5. * End_{I_q} (important_notes.md point 12): state 9.2 for V with field coefficients and point to Chapter 6 for T and $\mathbb{Z}_\ell[\zeta_3]$; $\mathbb{Z}_\ell[\zeta_3]$ is not a domain for $\ell \equiv 1 \pmod 3$. * Typography (PLAN.md 7.2): no dash ligatures; "Jordan-Holder" without diacritics in the source is acceptable, or `H\"older`; tables with `booktabs`; the wide table of 8.1 may be split into two.

What to leave to other chapters: * Chapter 1: all group theory of $\mathrm{GL}_2(\mathbb{F}_3)$ (subgroups, orders, orbits on lines, the three images, commutators, maximality), Dickson's proof; this chapter only cites. * Chapter 3: modules, Nakayama, DVRs, lattices, Gram determinants, Pontryagin duality, lengths over a DVR. * Chapter 5: continuity, profinite groups, existence of lattices for compact images, Chebotarev, complex

conjugation, the cyclotomic character and the determinant argument for χ_ℓ, the field $\mathbb{Q}(\sqrt{\ell^*})$. * Chapter 6: $\mathbb{Z}_\ell[\zeta_3]$, $\mathbb{Z}_\ell[i]$ as rings, freeness of T over $\mathbb{Z}_\ell[\sigma]$, the Frobenius action $\chi_{-3}(q)$, $\chi_{-4}(q)$ on the invariant lines, the twist detector at inert primes, CM. * Chapter 11: Lemma lem:IV, Remark rem:types, Proposition prop:ns, Lemma lem:h0 as statements about elliptic curves, the Kodaira-type to inertia-shape dictionary (with Chapter 10, Kraus), the Serre conductor. * Chapters 16 to 19: cohomology (H^1 of cyclic groups, extensions as H^1), Tate twists, Bloch-Kato, L-factors.

How the important_notes.md points in this chapter's area are handled: * **Point 1** (prop:ns, "non-abelian and faithful"): 3.7 gives the Maschke proof for Q_8 (order prime to 3) and containment for $\mathrm{SL}_2(\mathbb{F}_3)$; the Borel counterexample is 3.7(iv); Exercise 6. Chapter 1's 5.7 is cited as the alternative. * **Point 5** (lem:h0, finiteness and $h_\ell \le 1$): 10.5 (cyclicity, the length as the largest m with $T/\ell^m \simeq (T/\ell^m)(\chi)$, the determinant constraint); Exercise 17; the Galois specialisation is one sentence pointing to Chapters 5 and 11. * **Point 10** (lem:old and lem:h0, "irreducible" should be "absolutely irreducible"): 2.4 (distinct eigenvalues), 5.3(ii) (with the C_4 counterexample showing why), 6.6 (Clifford needs absolute irreducibility), 1.4 (common finite field). * **Point 12** (lem:euler, $\mathrm{End}_{I_q}(V)$ versus $\mathrm{End}_{I_q}(T)$): 9.2 states the commutant over a field, with the remark that it is a field iff $\ell \not\equiv 1 \pmod 3$; 9.4(v) points to Chapter 6 for the integral statement. * **Point 26** (Section 4.2, $M_{f,\lambda}$ homothetic to the dual of $T_\ell E$): 7.7 and 10.3; the `paperlink` of Section 10 says "homothetic to the dual of $T_\ell E$; the adjoints are canonically identified". * Also touched: point 11 (9.1(iii), 9.3: arithmetic versus geometric eigenvalue), point 13 (9.4: write $\mathbb{Z}_3[X]/(X^2+X+1)$), point 22 (9.3 last column and 9.4: the complete mod-3 picture), point 2 (11.2: $N_s(\mathbb{F}_3)$ not maximal), point 6 (11.1(iii)), and the new points N1, N2 of Part 8.

6.8 7. Verification log (computations done for this outline)

All scripts were short Python/sympy programs in `/tmp/ch04` (`gl23.py`, `sym.py`). * $\mathrm{GL}_2(\mathbb{F}_3)$ enumerated (48 elements) with $\mathbb{F}_9 = \mathbb{F}_3[i]$ arithmetic; the subgroups GL_2, SL_2, N_{ns}, Q_8, N_s, C_4, C_{ns}, C_s, B, U, $\{\pm1\} \times$ U generated from the models C6 (orders $48, 24, 16, 8, 8, 4, 8, 4, 12, 3, 6$). For each: the number of stable lines over $\mathbb{F}_3$ and over $\mathbb{F}_9$ ($0,0$ for $\mathrm{GL}_2, \mathrm{SL}_2, N_{ns}, Q_8, N_s$; $0,2$ for C_4, C_{ns}; $2,2$ for C_s; $1,1$ for B, U, $\{\pm1\} \times$ U); $\dim \mathrm{End}_H$, $\dim(\mathrm{ad}^0\,\rho)^H$, $\dim \mathrm{Hom}_H(\rho, \rho \otimes \det)$ and its trace-zero part, the existence of an invertible intertwiner $\rho \to \rho \otimes \det$, and the order of the projective image: the table of 8.1. * $N_s(\mathbb{F}_3)$: $\rho \simeq \rho \otimes \chi$ for the three characters with kernels C_4, C_s, $\{\pm1, \pm w\}$ (invertible intertwiners found); with $X = [v \,|\, \sigma v]$, $v = (1, -i)^t$: $X^{-1} J X = \mathrm{diag}(i, -i)$, $X^{-1}\sigma X = w$ (8.2(ii)); the elements of determinant -1 of $N_s(\mathbb{F}_3)$ all have trace 0, those of $N_{ns}(\mathbb{F}_3)$ and $\mathrm{GL}_2(\mathbb{F}_3)$ have traces $0, \pm1$, the nonzero ones exactly the elements of order 8. $N_{ns}(\mathbb{F}_3) \simeq N_{ns} \otimes \chi_{ns}$ for the character with kernel C_{ns} (invertible intertwiner found). * Symbolic (sympy): $\mathrm{Ad}(g)\Xi = \Xi\, \mathrm{Sym}^2(g) \det(g)^{-1}$ for general g with $\Xi(e_1^2) = -2e$, $\Xi(e_1 e_2) = h$, $\Xi(e_2^2) = 2f$; the Gram matrix of $\mathrm{tr}(xy)$ on e, h, f and its determinant -2; $(t-1)e = 0$, $(t-1)h = -2n'e$, $(t-1)f = n'h - n'^2 e$ for $t = \begin{pmatrix} 1 & n' \\ 0 & 1 \end{pmatrix}$; $\mathrm{Ad} \begin{pmatrix} \varepsilon q & c' \\ 0 & \varepsilon \end{pmatrix}$ on e, h, f (7.3(iii)); $\mathrm{Ad}\,\mathrm{diag}(x, y) = \mathrm{diag}(x/y, 1, y/x)$. * For $g \in \{-1, \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}, J, \begin{pmatrix} 1 & -1 \\ 1 & 0 \end{pmatrix}, u, -u\}$: orders $2, 3, 4, 6, \infty, \infty$, characteristic polynomials $(X+1)^2$, X^2+X+1, X^2+1, X^2-X+1, $(X-1)^2$, $(X+1)^2$; $\dim V^g = 0, 0, 0, 0, 1, 0$ and $\dim(\mathrm{ad}^0 V)^g = 3, 1, 1, 1, 1, 1$ over $\mathbb{Q}$, with generators $2\sigma + 1$ (orders 3 and 6, up to a factor 2), J (order 4), e ($\pm u$); over $\mathbb{F}_3$: $\dim V^g = 0, 1, 0, 0, 1, 0$ and $\dim(\mathrm{ad}^0 V)^g = 3, 1, 1, 1, 1, 1$ (the table of 9.3 and 9.4). * By hand: the Brauer-Nesbitt example 4.5; $\Xi(e_1^2)$, $\Xi(e_1 e_2)$, $\Xi(e_2^2)$ from the formula of 7.5; the matrix of $\mathrm{Ind}\,\psi$ in 6.1; the intertwiners J, σ, w of 8.2 (via $J\sigma = -\sigma J$, $wJ = -Jw$, $w\sigma = -\sigma w$); $\theta(a) = 1 + i$, $\theta(a)^3 = 1 - i$ (8.3); the determinant obstruction $x^2 \equiv 1 \pmod{\ell^m}$ for all units only for $\ell^m = 3$ (ℓ odd).

6.9 8. Points for important_notes.md

New points in this chapter's area, not among the 32 already recorded. Both are expository; neither changes a result.

N1. Proposition prop:ns, proof ("$\bar\rho \simeq \operatorname{Ind}\psi$"), and Lemma lem:h0, proof ("induced from $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{-3}))$"): the induction lives over $\mathbb{F}_9$, not over $\mathbb{F}_3$. Location: Section 5, proof of Proposition prop:ns, third case; Section 4.2, proof of Lemma lem:h0, the sentence "For $\ell = 3$ and $\bar\rho$ irreducible, $\bar\rho \simeq \bar\rho \otimes \chi_{-3}$ holds if and only if $\bar\rho$ is induced from $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{-3}))$". Problem: ψ takes values in $\mathbb{F}_9^\times$ (order 4), and in the $N_s(\mathbb{F}_3)$ case the restriction of $\bar\rho$ to $G_{\mathbb{Q}(\sqrt{-3})}$ is irreducible over $\mathbb{F}_3$ (its image $C_4 = \langle J\rangle$ has no stable $\mathbb{F}_3$-line, X^2+1 being irreducible over $\mathbb{F}_3$). So $\bar\rho$ is not induced from a character of $G_{\mathbb{Q}(\sqrt{-3})}$ over $\mathbb{F}_3$; only $\bar\rho \otimes \mathbb{F}_9 \simeq \operatorname{Ind}\psi$ holds. Fix: write "$\bar\rho \otimes \mathbb{F}_9 \simeq \operatorname{Ind}\psi$ by Frobenius reciprocity, so $\bar\rho \otimes \mathbb{F}_9 \simeq (\bar\rho \otimes \chi_{-3}) \otimes \mathbb{F}_9$, hence $\bar\rho \simeq \bar\rho \otimes \chi_{-3}$ (a $G_{\mathbb{Q}}$-map between absolutely irreducible representations defined over $\mathbb{F}_3$ exists over $\mathbb{F}_3$ as soon as it exists over $\mathbb{F}_9$)", and in Lemma lem:h0 "induced from $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{-3}))$ after extension of scalars to $\bar{\mathbb{F}}_3$". Confidence: high (the $\mathbb{F}_3$-irreducibility of C_4 and the explicit isomorphism over $\mathbb{F}_9$ were checked by computer, Part 7). Importance: low (expository; the conclusion $\bar\rho \simeq \bar\rho \otimes \chi_{-3}$ over $\mathbb{F}_3$ is correct).

N2. Lemma lem:euler, proof, types $\mathrm{III}, \mathrm{III}^*$: "the invariant line of $\mathrm{ad}^0 V$ is spanned by the diagonal element". Location: Section 2, proof of Lemma lem:euler. Problem: for $\ell \equiv 3 \pmod 4$ the generator τ of the image of inertia (eigenvalues $\pm i$) has no eigenvectors over $\mathbb{Q}_\ell$, so there is no "diagonal element" in $\mathrm{ad}^0 V$ over $\mathbb{Q}_\ell$; the invariant line of $\mathrm{ad}^0 V$ is $\mathbb{Q}_\ell\tau$ (9.2), which is diagonal, equal to $i\,\mathrm{diag}(1,-1)$, only in an eigenbasis over $\mathbb{Q}_\ell(i)$. The Frobenius eigenvalue on the line is $\chi_{-4}(q)$ either way ($F\tau F^{-1} = \tau^q = \chi_{-4}(q)\tau$), so nothing changes. This is the order-4 companion of important_notes.md point 12. Fix: "the invariant line of $\mathrm{ad}^0 V$ is spanned by τ itself (diagonal in an eigenbasis over $\mathbb{Q}_\ell(i)$), on which Frobenius acts by $\chi_{-4}(q)$, since $\tau^q = \chi_{-4}(q)\tau$". Confidence: high. Importance: low.

No other new problem was found in this chapter's area. Checked and found correct: the trace-zero computation in Lemma lem:h0; "a nonzero such map is an isomorphism" (given absolute irreducibility, point 10); the list of determinant-one subgroups and the orbit statement in Proposition prop:ns (Chapter 1); ψ of order 4 and $\psi^{-1} = \psi^\sigma$; "the third is abelian, hence absolutely reducible"; the e, h, f formulas and the action of $\rho(\mathrm{Frob})$ on e, h in Lemma lem:tamfactor (with $\varepsilon^{-1} = \varepsilon$); "$T^D = \mathrm{Hom}(A, \mathbb{Z}_\ell(1)) \simeq A(1)$ via the trace form" (needs ℓ odd, which the paper assumes); "$\mathrm{ad}^0 T$ is unchanged by the twist"; "a non-split extension of two characters has a unique stable line" in Lemma lem:old; the eigenvalues $\zeta_3^{\pm 1}$ and the $\mathbb{Z}_3[\zeta_3]$ argument of Lemma lem:IV (modulo point 13); "$L_p(\mathrm{Sym}^2 f, s) = L_p(A_f, s-1)$" against 7.5(iii).

Chapter 7

Chapter 05: Galois theory, finite and infinite (ch:galois, chapters/ch05_galois.tex)

Stage 1 line-up. Part II (Number fields and local fields). Target length 2,030 lines of LaTeX (card excluded; see Part 4), inside the 1,000 to 2,200 range of PLAN.md Section 7.3.

7.1 0. What this chapter is for

The paper uses Galois theory in five places, and the chapter is built around them.

1. The hypothesis of Theorem thm:main, Theorem thm:dfg37, Theorem thm:dfg27 and the set S_f of Section 4.2: "$\bar\rho_{E,\ell}$ is absolutely irreducible when restricted to $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{\ell^*}))$". The chapter must define $G_{\mathbb{Q}}$, its open subgroup $G_{\mathbb{Q}(\sqrt{\ell^*})}$ of index two, the field $\mathbb{Q}(\sqrt{\ell^*}) \subset \mathbb{Q}(\zeta_\ell)$ and its quadratic character χ_{ℓ^*}, the restriction of a representation to an open subgroup, and the equivalence (for $\bar\rho$ absolutely irreducible) "not absolutely irreducible on $G_{\mathbb{Q}(\sqrt{\ell^*})}$" iff "$\bar\rho \simeq \bar\rho \otimes \chi_{\ell^*}$" iff "$\bar\rho$ is induced from $G_{\mathbb{Q}(\sqrt{\ell^*})}$" (used in Lemma lem:h0 and Proposition prop:ns).
2. Proposition prop:ns: the image of $G_{\mathbb{Q}(\sqrt{-3})}$ under $\bar\rho_{E,3}$ is the determinant-one subgroup of the image (because $\det \bar\rho_{E,3} = \chi_{-3}$ cuts out $\mathbb{Q}(\sqrt{-3})$), and the factorisation pattern of the 3-division polynomial is the orbit pattern of the image on the four lines.
3. Lemma lem:old: "$\bar\rho_h \simeq \bar\rho$ by Chebotarev and Brauer-Nesbitt": a semisimple mod ℓ representation of $G_{\mathbb{Q}}$ is determined by the characteristic polynomials of Frobenius on a density-one set.
4. Lemma lem:tamagawa-exp: Kummer theory, "$F(u^{1/\ell^t}) = F$ iff u is an ℓ^t-th power in F" when $\mu_{\ell^t} \subset F$.
5. Lemma lem:tamfactor: the Kummer class κ of the Tate parameter, as an explicit 1-cocycle with values in $\mathbb{Z}_\ell(1)$, and the conjugation rule $\kappa(\sigma\tau\sigma^{-1}) = \chi_\ell(\sigma)\kappa(\tau)$, which is the abstract form of $\mathrm{Frob}\ t\ \mathrm{Frob}^{-1} = t^q$, i.e. $\mathrm{Frob}^{-1}\, t\, \mathrm{Frob} = t^{q^{-1}}$, of Chapter 7.
6. Lemma lem:h0, Conjecture conj:euler and Section 1.3 of the paper: the difference between the reduced cyclotomic character $\bar\chi_\ell$ and the quadratic character $\chi_{\ell^*} = \bar\chi_\ell^{(\ell-1)/2}$ (they coincide only for $\ell = 3$), and the determinant argument "$\det(\rho\otimes\chi) = \chi^2 \det\rho$" which shows that $\rho \bmod \ell^k \simeq (\rho\otimes\chi_\ell) \bmod \ell^k$ forces $\ell = 3$ and $k = 1$ (item galois:twist below; see also Section 8 of this outline, "Points for important_notes.md").

Everything else in the chapter (field extensions, splitting fields, the fundamental theorem, cyclotomic fields, Galois groups of polynomials, profinite groups, continuous representations, the cyclotomic character,

complex conjugation, Chebotarev) is the minimal path from the reader's background (RHB, linear algebra) to these five points, plus what Chapters 6, 7, 11, 15, 17 need from Galois theory.

7.2 1. Global conventions fixed in this chapter

- **C1 (one algebraic closure).** $\overline{\mathbb{Q}} \subset \mathbb{C}$ is the field of algebraic numbers; every algebraic extension of $\mathbb{Q}$ is taken inside $\overline{\mathbb{Q}}$; $G_{\mathbb{Q}} = \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$; for $K \subset \overline{\mathbb{Q}}$, $G_K = \mathrm{Gal}(\overline{\mathbb{Q}}/K) \subset G_{\mathbb{Q}}$. For a general field F, $F^{\mathrm{sep}} \subset \overline{\mathbb{F}}$ is a separable closure and $G_F = \mathrm{Gal}(F^{\mathrm{sep}}/F)$.
- **C2 (Frobenius is arithmetic).** Wherever a Frobenius element appears (the black box of Section 9, made precise in Chapter 6), Frob_p is the *arithmetic* Frobenius: it induces $x \mapsto x^p$ on residue fields, acts on μ_n ($p \nmid n$) by $\zeta \mapsto \zeta^p$, and $\chi_\ell(\mathrm{Frob}_p) = p$. This is the paper's convention (proof of Lemma lem:tamfactor: "the arithmetic Frobenius acts ... as $q^{-1}\rho(\mathrm{Frob})$").
- **C3 (quadratic characters).** For $d \in \mathbb{Q}^\times \setminus (\mathbb{Q}^\times)^2$, $\chi_d \colon G_{\mathbb{Q}} \to \{\pm 1\}$ is $\sigma \mapsto \sigma(\sqrt{d})/\sqrt{d}$; it depends only on d modulo squares and has kernel $G_{\mathbb{Q}(\sqrt{d})}$. For an odd prime q put $q^* = (-1)^{(q-1)/2}q$; χ_{q^*} is the character of $\mathbb{Q}(\sqrt{q^*})$, "the quadratic character of conductor q", which the paper writes χ_q (the book never does: PLAN.md 7.5, decision D3); χ_{-3} and χ_{-4} are the characters of $\mathbb{Q}(\sqrt{-3})$ and $\mathbb{Q}(i)$. Chapter 2 defines the same symbols as Dirichlet characters (Kronecker symbols); the two agree through $\chi_d(\mathrm{Frob}_p) = \left(\frac{D}{p}\right)$ (Kronecker symbol) for every prime $p \nmid D$, D the discriminant of $\mathbb{Q}(\sqrt{d})$, proved in Chapter 6.
- **C4 (the cyclotomic character and its reduction).** $\chi_\ell \colon G_{\mathbb{Q}} \to \mathbb{Z}_\ell^\times$ is defined by $\sigma(\zeta) = \zeta^{\chi_\ell(\sigma)}$ for all $\zeta \in \mu_{\ell^\infty}$; $\bar{\chi}_\ell \colon G_{\mathbb{Q}} \to \mathbb{F}_\ell^\times$ is its reduction. $\chi_{\ell^*} = \bar{\chi}_\ell^{(\ell-1)/2}$ is the quadratic character of $\mathbb{Q}(\sqrt{\ell^*})$; $\chi_{\ell^*} = \bar{\chi}_\ell$ if and only if $\ell = 3$.
- **C5 (representations).** A representation of a profinite group G is a continuous homomorphism $\rho \colon G \to \mathrm{GL}_n(k)$ where k is a finite field or $\overline{\mathbb{F}}_\ell$ with the discrete topology, or a finite extension E of $\mathbb{Q}_\ell$ (or $\mathbb{Z}_\ell$-lattice inside E^n) with the ℓ-adic topology. "ρ continuous into $\mathrm{GL}_n(\overline{\mathbb{Q}}_\ell)$" means: into $\mathrm{GL}_n(E)$ for some finite $E/\mathbb{Q}_\ell$, continuously. For T a free $\mathbb{Z}_\ell$-module with continuous action, $\rho \bmod \ell^k$ is the action on $T/\ell^k T$ and $\bar{\rho} = \rho \bmod \ell$. Irreducible, absolutely irreducible, semisimplification ρ^{ss}, $\rho \otimes \chi$, Ind, Res are as in Chapter 4. "ρ is absolutely irreducible on G_K" means: $(\rho \otimes \bar{k})|_{G_K}$ is irreducible.
- **C6 (Kummer characters live in μ_n).** $\chi_a \colon G_F \to \mu_n$ is a homomorphism to μ_n; it is identified with a homomorphism to $\mathbb{Z}/n\mathbb{Z}$ only after a primitive ζ_n is chosen. $\mathbb{Z}_\ell(1) = \varprojlim_k \mu_{\ell^k}$ (transition maps $\zeta \mapsto \zeta^\ell$), written additively when convenient, with G_F acting through χ_ℓ.
- **C7 (typography).** No em dash, no en dash, no double or triple hyphen, ASCII source, `\ell` not a Unicode letter, labels prefixed `...:galois:`.

7.3 2. Line-up of the core results, in teaching order

Format of each item: **name** (kind). Statement. *Rigour.* Reference. Paper use.

7.3.1 Section 1. Field extensions and algebraic elements

galois:field-extension-degree (definition + proposition). A field extension L/F is a field L containing F as a subfield; its degree is $[L : F] = \dim_F L \in \{1, 2, \dots\} \cup \{\infty\}$. *Tower law:* for $F \subset L \subset M$, $[M : F] = [M : L]\,[L : F]$ (with the convention $\infty \cdot n = \infty$): if (x_i) is an F-basis of L and (y_j) an L-basis of M, then $(x_i y_j)$ is an F-basis of M. *Proved here.* Lang, Algebra, V.1 (Proposition 1.2); Dummit-Foote 13.2. Background.

galois:algebraic-element (definition + proposition). $\alpha \in L$ is algebraic over F if $f(\alpha) = 0$ for some nonzero $f \in F[x]$, transcendental otherwise. If α is algebraic, the kernel of $F[x] \to L$, $x \mapsto \alpha$, is generated by a unique monic irreducible polynomial $m_{\alpha,F}$, the minimal polynomial; $F(\alpha) = F[\alpha] \simeq F[x]/(m_{\alpha,F})$ and $[F(\alpha) : F] = \deg m_{\alpha,F}$; $g(\alpha) = 0$ for $g \in F[x]$ iff $m_{\alpha,F} \mid g$. Conversely if $[F(\alpha) : F] < \infty$ then α is algebraic. Examples: $\sqrt{2}$, ζ_n, $2^{1/3}$; π is transcendental (stated). *Proved here.* Lang V.1; Dummit-Foote 13.1, 13.2. Background.

galois:algebraic-extensions (proposition). (a) A finite extension is algebraic. (b) If $\alpha_1, \dots, \alpha_r$ are algebraic over F then $F(\alpha_1, \dots, \alpha_r)$ is finite over F, of degree at most $\prod \deg m_{\alpha_i, F}$. (c) The elements of L algebraic over F form a subfield of L. (d) If M/L and L/F are algebraic, so is M/F. (e) A finite extension is generated by finitely many algebraic elements. *Proved here.* Lang V.1. Background.

galois:algebraic-closure (theorem). (a) Every field F has an algebraic closure $\overline{\mathbb{F}}$: an algebraic extension of F which is algebraically closed; it is unique up to F-isomorphism (not up to unique isomorphism). (b) If L/F is algebraic and $\sigma\colon F \to \Omega$ is an embedding into an algebraically closed field Ω, then σ extends to an embedding $L \to \Omega$. (c) $\overline{\mathbb{Q}} = \{z \in \mathbb{C} : z \text{ algebraic over } \mathbb{Q}\}$ is an algebraic closure of $\mathbb{Q}$; it is countable, and $\mathbb{C}$ is not algebraic over $\mathbb{Q}$. *Stated; reference* for (a) and (b) (Zorn's lemma argument sketched in one paragraph); (c) proved here from the fundamental theorem of algebra (assumed from complex analysis). Lang V.2; Dummit-Foote 13.4. Background: $\overline{\mathbb{Q}}$ and $G_{\mathbb{Q}}$ throughout the paper.

7.3.2 Section 2. Splitting fields, normality, separability

galois:splitting-field (theorem). Let $f \in F[x]$ be nonconstant. A splitting field of f over F is an extension $L = F(r_1, \dots, r_n)$ with $f = c\prod(x - r_i)$ in $L[x]$. (a) A splitting field exists, of degree at most $n!$ over F (in fact dividing $n!$; Exercise). (b) *Extension of isomorphisms:* if $\phi\colon F \to F'$ is a field isomorphism, L a splitting field of f over F and L' a splitting field of $\phi(f)$ over F', then ϕ extends to an isomorphism $L \to L'$; the number of such extensions is at most $[L : F]$, with equality if f is separable. (c) In particular any two splitting fields of f over F are F-isomorphic, and the splitting field of f inside a fixed $\overline{\mathbb{F}}$ is unique. The same holds for a family of polynomials (splitting field of a set $S \subset F[x]$). *Proved here* ((a), (b), (c) for one polynomial; the family case sketched). Lang V.3; Dummit-Foote 13.4. Background.

galois:normal-extension (definition + proposition). An algebraic extension L/F inside $\overline{\mathbb{F}}$ is normal if the following equivalent conditions hold: (i) every irreducible polynomial in $F[x]$ with one root in L splits into linear factors in $L[x]$; (ii) L is the splitting field over F of a family of polynomials; (iii) every F-embedding $L \to \overline{\mathbb{F}}$ has image L (so $\mathrm{Emb}_F(L, \overline{\mathbb{F}}) = \mathrm{Aut}(L/F)$). A finite extension L/F has a normal closure (the smallest normal extension of F inside $\overline{\mathbb{F}}$ containing L: the splitting field of the minimal polynomials of generators), unique inside $\overline{\mathbb{F}}$, finite over F. Normality is not transitive: $\mathbb{Q} \subset \mathbb{Q}(\sqrt{2}) \subset \mathbb{Q}(2^{1/4})$. *Proved here.* Lang V.3 (Theorem 3.3); Dummit-Foote 14.1. Background.

galois:separability (definition + proposition). A polynomial $f \in F[x]$ is separable if it has no repeated root in $\overline{\mathbb{F}}$, equivalently $\gcd(f, f') = 1$. (a) An irreducible f is inseparable iff $f' = 0$ iff $\mathrm{char}\, F = p > 0$ and $f \in F[x^p]$. (b) F is perfect (every irreducible polynomial is separable) iff $\mathrm{char}\, F = 0$ or $\mathrm{char}\, F = p$ and $F = F^p$; fields of characteristic 0, finite fields and algebraically closed fields are perfect. (c) An algebraic element is separable if its minimal polynomial is; an algebraic extension is separable if all its elements are; over a perfect field every algebraic extension is separable. (d) For L/F finite, the separable degree $[L : F]_s = \#\mathrm{Emb}_F(L, \overline{\mathbb{F}})$ divides $[L : F]$, with equality iff L/F is separable; separable degree is multiplicative in towers. (e) *Primitive element theorem:* a finite separable extension is simple, $L = F(\theta)$; more generally a finite extension is simple iff it has finitely many intermediate fields. *Proved here* ((a) to (d) and (e) for infinite F; finite F by cyclicity of $\mathbb{F}_q^\times$ from Chapter 2). Lang V.4 (Theorem 4.6); Dummit-Foote 13.5, 14.4. Background.

galois:separable-closure (definition). $F^{\mathrm{sep}} = \{\alpha \in \overline{\mathbb{F}} : \alpha \text{ separable over } F\}$ is a field, Galois over F (Section 3), and equals $\overline{\mathbb{F}}$ when F is perfect; in particular $\mathbb{Q}^{\mathrm{sep}} = \overline{\mathbb{Q}}$, $\mathbb{F}_q^{\mathrm{sep}} = \overline{\mathbb{F}}_q$, $\mathbb{Q}_p^{\mathrm{sep}} = \overline{\mathbb{Q}}_p$. *Proved here.* Lang V.4. Background (definition of G_F).

7.3.3 Section 3. Galois extensions and the fundamental theorem

galois:dedekind-independence (theorem, Dedekind-Artin). Let G be a group (or monoid) and L a field. Distinct homomorphisms $\chi_1, \dots, \chi_n\colon G \to L^\times$ are linearly independent over L: if $\sum a_i\chi_i(g) = 0$ for all $g \in G$ with $a_i \in L$, then all $a_i = 0$. In particular distinct field embeddings $L \to \Omega$ are linearly independent over Ω. *Proved here.* Lang VI.4 (Theorem 4.1); Dummit-Foote 14.2 (Theorem 7). Used for Artin's theorem and Hilbert 90.

galois:artin-fixed-field (theorem, Artin). Let L be a field, $G \subset \mathrm{Aut}(L)$ a finite subgroup, and $F = L^G = \{x \in L : \sigma x = x \text{ for all } \sigma \in G\}$. Then L/F is finite, normal and separable, $[L : F] = \#G$, and

$\mathrm{Aut}(L/F) = G$. Conversely, for any finite extension L/F, $\#\,\mathrm{Aut}(L/F) \le [L:F]$, with equality iff L/F is normal and separable. *Proved here* (via Dedekind independence and the primitive element theorem). Lang VI.1 (Theorem 1.8); Dummit-Foote 14.2. Background.

galois:galois-extension (definition + theorem). An algebraic extension L/F is Galois if it is normal and separable; then $\mathrm{Gal}(L/F) := \mathrm{Aut}(L/F)$. For L/F finite the following are equivalent: (i) L/F is Galois; (ii) L is the splitting field of a separable polynomial over F; (iii) $L^{\mathrm{Aut}(L/F)} = F$; (iv) $\#\,\mathrm{Aut}(L/F) = [L:F]$. If L/F is Galois and $F \subset M \subset L$, then L/M is Galois. *Proved here.* Lang VI.1; Dummit-Foote 14.2. Background.

galois:fundamental-theorem (theorem, fundamental theorem of Galois theory, finite case). Let L/F be a finite Galois extension with group $G = \mathrm{Gal}(L/F)$. The maps $H \mapsto L^H$ and $M \mapsto \mathrm{Gal}(L/M)$ are mutually inverse, inclusion-reversing bijections between the subgroups $H \le G$ and the intermediate fields $F \subset M \subset L$. Under them: $[L:M] = \#\,\mathrm{Gal}(L/M)$ and $[M:F] = [G:\mathrm{Gal}(L/M)]$; for $\sigma \in G$, $\mathrm{Gal}(L/\sigma M) = \sigma\,\mathrm{Gal}(L/M)\sigma^{-1}$; M/F is Galois iff $\mathrm{Gal}(L/M)$ is normal in G, and then restriction $\sigma \mapsto \sigma|_M$ induces an isomorphism $G/\,\mathrm{Gal}(L/M) \simeq \mathrm{Gal}(M/F)$; intersections of fields correspond to the subgroup generated, composita to intersections of subgroups. *Proved here.* Lang VI.1 (Theorem 1.1); Dummit-Foote 14.2 (Theorem 14). Background: every "cuts out", "fixed field", "index two" statement in the paper.

galois:translation-theorem (proposition, "shifting"). Let L/F be Galois (finite or infinite) and M/F an arbitrary extension, L and M inside a common field. Then LM/M is Galois and restriction $\mathrm{Gal}(LM/M) \to \mathrm{Gal}(L/L\cap M)$, $\sigma \mapsto \sigma|_L$, is an isomorphism (of topological groups in the infinite case, Section 7). Corollaries: (a) if L_1, L_2 are Galois over F then so is L_1L_2, and $\mathrm{Gal}(L_1L_2/F) \to \mathrm{Gal}(L_1/F)\times\mathrm{Gal}(L_2/F)$ is injective, bijective iff $L_1\cap L_2 = F$; (b) for K/F finite, $[LK:K] = [L:L\cap K]$. *Proved here* (finite case; infinite case follows from Section 7). Lang VI.1; Dummit-Foote 14.4. Background: restriction of representations of G_F to G_K; "unramified base change does not change inertia" in Chapter 7.

7.3.4 Section 4. Finite fields, quadratic and cyclotomic extensions; worked examples

galois:finite-field-galois (theorem). Let $q = p^r$ and $n \ge 1$; work inside a fixed $\overline{\mathbb{F}}_p$. (a) $\mathbb{F}_{q^n}$ is the splitting field of $x^{q^n} - x$ over $\mathbb{F}_q$; every extension of $\mathbb{F}_q$ of degree n is $\mathbb{F}_q$-isomorphic to it, and inside $\overline{\mathbb{F}}_p$ it is unique. (b) $\mathbb{F}_{q^n}/\mathbb{F}_q$ is Galois, and $\mathrm{Gal}(\mathbb{F}_{q^n}/\mathbb{F}_q)$ is cyclic of order n, generated by the q-power Frobenius $\phi_q(x) = x^q$. (c) The intermediate fields are the $\mathbb{F}_{q^d}$, $d \mid n$, with $\mathbb{F}_{q^d}$ the fixed field of $\langle \phi_q^d \rangle$. (d) An irreducible $g \in \mathbb{F}_q[x]$ of degree d with a root α has the d distinct roots $\alpha, \alpha^q, \dots, \alpha^{q^{d-1}}$ and splits in $\mathbb{F}_{q^d}$; so every finite extension of a finite field is Galois with cyclic group. *Proved here* (existence and uniqueness of $\mathbb{F}_q$ and cyclicity of $\mathbb{F}_q^\times$ recalled from Chapter 2). Lang V.5; Dummit-Foote 14.3. Background: the Frobenius of Section 9 and Chapter 6, unramified extensions in Chapter 7.

galois:cyclotomic-mod-p (proposition). Let $n \ge 1$, p a prime with $p \nmid n$, and f the order of p in $(\mathbb{Z}/n)^\times$. Then $\Phi_n \bmod p$ is separable and is the product of $\varphi(n)/f$ distinct monic irreducible polynomials of degree f in $\mathbb{F}_p[x]$; $\mathbb{F}_p(\mu_n) = \mathbb{F}_{p^f}$. (Proof: $\mu_n(\overline{\mathbb{F}}_p)$ is cyclic of order n, the Frobenius acts on it by $\zeta \mapsto \zeta^p$, and the orbits on the primitive n-th roots have size f.) *Proved here.* Lang VI.3. Background: foreshadows "p splits in $\mathbb{Q}(\zeta_n)$ into $\varphi(n)/f$ primes of residue degree f" (Chapter 6) and $\mathrm{Frob}_p \mapsto p \bmod n$ (item galois:frobenius-preview).

galois:quadratic-extensions (proposition). Let $\mathrm{char}\, F \ne 2$. (a) Every extension L/F of degree 2 is Galois and equals $F(\sqrt{d})$ for some $d \in F^\times \setminus F^{\times 2}$; $\mathrm{Gal}(L/F) = \{1, \sigma\}$ with $\sigma(\sqrt{d}) = -\sqrt{d}$. (b) Inside $\overline{\mathbb{F}}$, $F(\sqrt{d}) = F(\sqrt{d'})$ iff $d/d' \in F^{\times 2}$; so the quadratic extensions of F correspond bijectively to the nontrivial elements of $F^\times/F^{\times 2}$. (c) For $F = \mathbb{Q}$ the quadratic fields are the $\mathbb{Q}(\sqrt{d})$, $d \ne 0, 1$ a squarefree integer, pairwise distinct. (d) If $a, b \in F^\times$ are independent in $F^\times/F^{\times 2}$, then $F(\sqrt{a}, \sqrt{b})/F$ is Galois with group $\simeq V_4$ and exactly three quadratic subfields $F(\sqrt{a})$, $F(\sqrt{b})$, $F(\sqrt{ab})$. *Proved here.* Dummit-Foote 13.2 and 14.2 (examples). Background; convention C3; Lemma lem:twist and Theorem thm:twist (the fields $\mathbb{Q}(\sqrt{q^*})$ and $\mathbb{Q}(\sqrt{\prod_{q\in S} q^*})$ of the characters χ_{q^*}, χ_S).

galois:cyclotomic (theorem). Let $n \ge 1$, $\zeta_n = e^{2\pi i/n} \in \overline{\mathbb{Q}}$, $\mu_n = \langle \zeta_n \rangle$. (a) $\Phi_n = \prod_{(a,n)=1,\, 1\le a\le n} (x - \zeta_n^a)$ lies in $\mathbb{Z}[x]$, and $x^n - 1 = \prod_{d\mid n} \Phi_d$ (Chapter 2). (b) Φ_n is irreducible over $\mathbb{Q}$. (Proof: if ζ is a root of

a monic irreducible factor $g \in \mathbb{Z}[x]$ of Φ_n and $p \nmid n$ is prime, then ζ^p is a root of g; otherwise $g(x)$ and $h(x^p)$, with h the minimal polynomial of ζ^p, share a factor mod p, so $x^n - 1$ would have a repeated root mod p.) (c) $\mathbb{Q}(\zeta_n)/\mathbb{Q}$ is Galois of degree $\varphi(n)$, and $a \mapsto \sigma_a$, $\sigma_a(\zeta) = \zeta^a$ for all $\zeta \in \mu_n$, is an isomorphism $(\mathbb{Z}/n)^\times \to \mathrm{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q})$, independent of the choice of a primitive n-th root of unity. (d) For $m \mid n$, restriction $\mathrm{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q}) \to \mathrm{Gal}(\mathbb{Q}(\zeta_m)/\mathbb{Q})$ is reduction $(\mathbb{Z}/n)^\times \to (\mathbb{Z}/m)^\times$; $\mathbb{Q}(\zeta_m) \cap \mathbb{Q}(\zeta_n) = \mathbb{Q}(\zeta_{\gcd(m,n)})$ and $\mathbb{Q}(\zeta_m)\mathbb{Q}(\zeta_n) = \mathbb{Q}(\zeta_{\mathrm{lcm}(m,n)})$; $\mathbb{Q}(\zeta_n) = \mathbb{Q}(\zeta_{2n})$ for n odd; every subfield of $\mathbb{Q}(\zeta_n)$ is Galois over $\mathbb{Q}$ with abelian group. (e) $\mathbb{Q}(\zeta_3) = \mathbb{Q}(\sqrt{-3})$ ($2\zeta_3 + 1 = \sqrt{-3}$), $\mathbb{Q}(\zeta_4) = \mathbb{Q}(i)$, $\mathbb{Q}(\zeta_8) = \mathbb{Q}(i, \sqrt{2})$ with quadratic subfields $\mathbb{Q}(i)$, $\mathbb{Q}(\sqrt{2})$, $\mathbb{Q}(\sqrt{-2})$. *Proved here.* Lang VI.3; Dummit-Foote 13.6 and 14.5; Washington, Chapter 2. Paper use: $\mathbb{Q}(\zeta_3) = \mathbb{Q}(\sqrt{-3})$ (Proposition prop:ns, Theorem thm:qual, Lemma lem:h0, the optimal-level theorem quoted in Section 3); Chapter 6.

galois:sqrt-lstar (proposition). Let ℓ be an odd prime and $\ell^* = (-1)^{(\ell-1)/2}\ell$. (a) $\mathrm{disc}(\Phi_\ell) = (-1)^{(\ell-1)/2}\ell^{\ell-2} = \ell^* \cdot \ell^{\ell-3}$, a square times ℓ^*, so $\sqrt{\ell^*} \in \mathbb{Q}(\zeta_\ell)$ (item galois:discriminant). (b) Equivalently, the Gauss sum $g = \sum_{a=1}^{\ell-1} \left(\frac{a}{\ell}\right)\zeta_\ell^a$ satisfies $g^2 = \ell^*$ (Chapter 2) and $\sigma_a(g) = \left(\frac{a}{\ell}\right)g$. (c) $\mathbb{Q}(\sqrt{\ell^*})$ is the unique quadratic subfield of $\mathbb{Q}(\zeta_\ell)$ (the cyclic group $(\mathbb{Z}/\ell)^\times$ of even order has a unique subgroup of index 2, the squares), and σ_a acts on $\sqrt{\ell^*}$ by the Legendre symbol $\left(\frac{a}{\ell}\right)$. In terms of characters of $G_\mathbb{Q}$ (Section 7): $\chi_{\ell^*}(\sigma) = \left(\frac{\bar\chi_\ell(\sigma)}{\ell}\right)$, that is $\chi_{\ell^*} = \bar\chi_\ell^{(\ell-1)/2}$. (d) For $\ell = 3$: $\mathbb{Q}(\sqrt{-3}) = \mathbb{Q}(\zeta_3)$ and $\chi_{-3} = \bar\chi_3$; for $\ell \geq 5$, $\chi_{\ell^*} \neq \bar\chi_\ell$ ($\bar\chi_\ell$ has order $\ell - 1 > 2$). (e) Every quadratic field $\mathbb{Q}(\sqrt{d})$, d squarefree, lies in $\mathbb{Q}(\zeta_{|D|})$, D its discriminant ($D = d$ if $d \equiv 1 \bmod 4$, $D = 4d$ otherwise), using (c), $i \in \mathbb{Q}(\zeta_4)$ and $\sqrt{2} = \zeta_8 + \zeta_8^{-1}$. *Proved here* ((a) via $\mathrm{disc}(\Phi_\ell) = \pm\,\mathrm{N}(\Phi_\ell'(\zeta_\ell))$, sketched; checked by computer for $\ell \leq 13$). Washington, Chapter 2; Lang VI.3; Ireland-Rosen, Chapter 6 (Gauss sums). Paper use: the hypothesis of Theorem thm:main and the set S_f of Section 4.2 ($\mathbb{Q}(\sqrt{\ell^*})$), Lemma lem:h0 ($\bar\chi_3 = \chi_{-3}$), Proposition prop:ns, convention C4.

galois:x3minus2 (worked example). $L = \mathbb{Q}(2^{1/3}, \zeta_3)$ is the splitting field of $x^3 - 2$ over $\mathbb{Q}$, of degree 6 (it contains $\mathbb{Q}(2^{1/3})$ of degree 3 and $\mathbb{Q}(\zeta_3)$ of degree 2). With composition $(\sigma\tau)(x) = \sigma(\tau(x))$, let σ fix ζ_3 and send $2^{1/3} \mapsto \zeta_3 2^{1/3}$, and let τ be complex conjugation ($2^{1/3}$ fixed, $\zeta_3 \mapsto \zeta_3^2$). Then $\sigma^3 = \tau^2 = 1$, $\tau\sigma\tau = \sigma^{-1}$, and $\mathrm{Gal}(L/\mathbb{Q}) \simeq S_3$, acting faithfully on the three roots. Correspondence: $\{1\} \leftrightarrow L$; $\langle\tau\rangle \leftrightarrow \mathbb{Q}(2^{1/3})$; $\langle\sigma\tau\rangle \leftrightarrow \mathbb{Q}(\zeta_3^2 2^{1/3})$; $\langle\sigma^2\tau\rangle \leftrightarrow \mathbb{Q}(\zeta_3 2^{1/3})$ (each order-two subgroup is the stabiliser of the root it fixes); $\langle\sigma\rangle = A_3 \leftrightarrow \mathbb{Q}(\zeta_3) = \mathbb{Q}(\sqrt{-3}) = \mathbb{Q}(\sqrt{\mathrm{disc}})$, $\mathrm{disc}(x^3 - 2) = -108 = -3 \cdot 6^2$; $S_3 \leftrightarrow \mathbb{Q}$. The three cubic fields are conjugate and not normal; $\mathbb{Q}(\sqrt{-3})$ is the only normal proper subfield. *Proved here* (the correspondence was checked by hand). Dummit-Foote 14.2 (example). Paper use: background; the model of "the quadratic subfield cut out by the sign/determinant" (Proposition prop:ns, item galois:quartic-on-lines), and of the affine group $\mu_3 \rtimes (\mathbb{Z}/3)^\times$ (item galois:kummer-affine).

7.3.5 Section 5. Cyclic extensions: Hilbert 90, Kummer theory, Artin-Schreier

galois:norm-trace (definition and proposition). For L/F finite separable of degree n with F-embeddings $\sigma_1, \dots, \sigma_n \colon L \to \overline{\mathbb{F}}$, put $\mathrm{N}_{L/F}(a) = \prod \sigma_i(a)$ and $\mathrm{Tr}_{L/F}(a) = \sum \sigma_i(a)$. They lie in F, equal the determinant and the trace of the F-linear map $x \mapsto ax$ of L, N is multiplicative and Tr additive, both are transitive in towers, and for L/F Galois $\mathrm{N}_{L/F}(a) = \prod_{\sigma \in \mathrm{Gal}(L/F)} \sigma(a)$. *Proved here* (the determinant description sketched). Lang VI.5. Background: Hilbert 90; norms of ideals in Chapter 6.

galois:hilbert90 (theorem, Hilbert's Theorem 90). Let L/F be a finite cyclic extension of degree n with $\mathrm{Gal}(L/F) = \langle\sigma\rangle$. (a) For $b \in L^\times$: $\mathrm{N}_{L/F}(b) = 1$ iff $b = c/\sigma(c)$ for some $c \in L^\times$. (b) For $b \in L$: $\mathrm{Tr}_{L/F}(b) = 0$ iff $b = c - \sigma(c)$ for some $c \in L$. (c) (General form.) For any finite Galois L/F with group G, every family $(a_\sigma)_{\sigma \in G}$ in $L^\times$ with $a_{\sigma\tau} = a_\sigma\, \sigma(a_\tau)$ is of the form $a_\sigma = \sigma(c)/c$ for some $c \in L^\times$ (in the language of Chapter 16: $H^1(G, L^\times) = 1$). *Proved here* ((a), (b), (c) by Dedekind independence: take $c = \sum_\tau a_\tau \tau(\theta)$ with θ chosen so that $c \neq 0$; the profinite form $H^1(G_F, (F^{\mathrm{sep}})^\times) = 1$ is stated and left to Chapter 17). Lang VI.6. Paper use: through Kummer theory (Lemmas lem:tamagawa-exp and lem:tamfactor); Chapter 17.

galois:kummer-character (definition and proposition). Let $n \geq 1$ with $\mathrm{char}\, F \nmid n$ and $\mu_n \subset F$. For $a \in F^\times$ choose $\alpha \in F^{\mathrm{sep}}$ with $\alpha^n = a$, and let L/F be any Galois extension containing α. (a) The map $\chi_a \colon \mathrm{Gal}(L/F) \to \mu_n$, $\sigma \mapsto \sigma(\alpha)/\alpha$, is a homomorphism independent of the choice of α (the other roots are $\zeta\alpha$, $\zeta \in \mu_n \subset F$); its kernel is $\mathrm{Gal}(L/F(\alpha))$, so $F(\alpha)/F$ is Galois and cyclic of degree $d = \#\chi_a(\mathrm{Gal}(L/F))$, and d is the order of the class of a in $F^\times/F^{\times n}$; in particular $d \mid n$. (b) $\chi_{ab} = \chi_a\chi_b$, and $\chi_a = 1$ iff $a \in F^{\times n}$;

so $a \mapsto \chi_a$ is an injective homomorphism $F^\times/F^{\times n} \to \mathrm{Hom}(G_F, \mu_n)$ (continuous homomorphisms, Section 7). (c) The minimal polynomial of α is $x^d - \alpha^d$, and $x^n - a$ is irreducible over F iff $a \notin F^{\times p}$ for every prime $p \mid n$. (Without $\mu_n \subset F$ the correct criterion is Capelli's: $x^n - a$ is irreducible iff $a \notin F^{\times p}$ for all primes $p \mid n$ and $a \notin -4F^{\times 4}$ when $4 \mid n$; stated, Lang VI.9.) *Proved here* (Capelli stated). Lang VI.6 and VI.8. Paper use: Lemma lem:tamagawa-exp (with $F = \mathbb{Q}_q^{\mathrm{ur}}$, which contains μ_{ℓ^t} because $q \neq \ell$, Chapter 7: $F(u^{1/\ell^t}) = F$ iff $u \in F^{\times \ell^t}$), Lemma lem:tamfactor.

galois:kummer-cyclic (theorem). Let $\mathrm{char}\, F \nmid n$ and $\mu_n \subset F$. Every cyclic extension L/F of degree n is $L = F(\alpha)$ with $\alpha^n \in F^\times$. (Proof: with $\mathrm{Gal}(L/F) = \langle\sigma\rangle$, $\mathrm{N}_{L/F}(\zeta_n^{-1}) = \zeta_n^{-n} = 1$, so Hilbert 90 gives $c \in L^\times$ with $\sigma(c) = \zeta_n c$; then $c^n \in F$ and c has n distinct conjugates.) *Proved here.* Lang VI.6. Background (completes the classification).

galois:kummer-classification (theorem, Kummer theory). Let $\mathrm{char}\, F \nmid n$ and $\mu_n \subset F$. (a) $\Delta \mapsto F_\Delta = F(\{\alpha \in F^{\mathrm{sep}} : \alpha^n \in \Delta\})$ is an inclusion-preserving bijection from the subgroups $F^{\times n} \subset \Delta \subset F^\times$ to the abelian extensions of F inside F^{sep} of exponent dividing n; the inverse is $L \mapsto L^{\times n} \cap F^\times$. (b) The pairing $\mathrm{Gal}(F_\Delta/F) \times \Delta/F^{\times n} \to \mu_n$, $(\sigma, a) \mapsto \sigma(\alpha)/\alpha$ ($\alpha^n = a$), is bilinear and perfect: $\mathrm{Gal}(F_\Delta/F) \simeq \mathrm{Hom}(\Delta/F^{\times n}, \mu_n)$ and $[F_\Delta : F] = [\Delta : F^{\times n}]$ when this is finite (for infinite Δ the isomorphism is one of profinite groups, $\Delta/F^{\times n}$ discrete). (c) (The mod n form.) $a \mapsto \chi_a$ is an isomorphism $F^\times/F^{\times n} \simeq \mathrm{Hom}_{\mathrm{cont}}(G_F, \mu_n)$; the cyclic extensions of degree $d \mid n$ correspond to the cyclic subgroups of order d of $F^\times/F^{\times n}$; after fixing a primitive ζ_n, so that $\mu_n \simeq \mathbb{Z}/n$, the surjections $G_F \to \mathbb{Z}/n$ (a $\mathbb{Z}/n$-extension with a chosen identification of its group with $\mathbb{Z}/n$) correspond to the elements of exact order n of $F^\times/F^{\times n}$. (d) For $n = 2$, $\mathrm{char}\, F \neq 2$: $\mathrm{Hom}_{\mathrm{cont}}(G_F, \{\pm 1\}) \simeq F^\times/F^{\times 2}$, recovering galois:quadratic-extensions. *Proved here* for finite $\Delta/F^{\times n}$; the infinite case after Section 7 (sketched). Lang VI.8; Neukirch IV.3. Paper use: Lemma lem:tamfactor (Kummer classes of the Tate parameter and of its unit part), Lemma lem:tamagawa-exp; Chapter 7 ($\mathbb{Q}_p^\times/(\mathbb{Q}_p^\times)^{\ell^k}$), Chapter 17 ($H^1(K, \mu_n) = K^\times/K^{\times n}$, which removes the hypothesis $\mu_n \subset F$).

galois:kummer-affine (proposition). Let $\mathrm{char}\, F \nmid n$, without assuming $\mu_n \subset F$; let $a \in F^\times$, $\alpha^n = a$, $L = F(\mu_n, \alpha)$, and fix a primitive ζ_n. For $\sigma \in \mathrm{Gal}(L/F)$ write $\sigma(\zeta_n) = \zeta_n^{c(\sigma)}$ and $\sigma(\alpha) = \zeta_n^{k(\sigma)}\alpha$. Then $\sigma \mapsto \begin{pmatrix} c(\sigma) & k(\sigma) \\ 0 & 1 \end{pmatrix}$ is an injective homomorphism $\mathrm{Gal}(L/F) \to \left\{\begin{pmatrix} c & k \\ 0 & 1 \end{pmatrix} : c \in (\mathbb{Z}/n)^\times, k \in \mathbb{Z}/n\right\} \simeq \mathbb{Z}/n \rtimes (\mathbb{Z}/n)^\times$; its restriction to $\mathrm{Gal}(L/F(\mu_n))$ is the Kummer character. Example: $x^3 - 2$ over $\mathbb{Q}$ gives $\mathrm{Gal} \simeq \mathbb{Z}/3 \rtimes (\mathbb{Z}/3)^\times \simeq S_3$. *Proved here.* Lang VI.9 (examples). Paper use: the shape $\left(\begin{smallmatrix} \chi_{\ell^\varepsilon} & \kappa \\ 0 & \varepsilon \end{smallmatrix}\right)$ in the proof of Lemma lem:tamfactor (here with $\varepsilon = 1$ and finite level; the ℓ-adic form is galois:l-adic-kummer).

galois:kummer-finite-field (proposition). Let q be a prime power, $\ell \nmid q$ a prime, $v = v_\ell(q-1)$, and $u \in \mathbb{F}_q^\times$. (a) For $1 \le k \le v$ (so $\mu_{\ell^k} \subset \mathbb{F}_q$), the Kummer character $\chi_u \colon G_{\mathbb{F}_q} \to \mu_{\ell^k}$ satisfies $\chi_u(\phi_q) = u^{(q-1)/\ell^k}$ (because $\phi_q(\alpha)/\alpha = \alpha^{q-1}$); hence $u \in (\mathbb{F}_q^\times)^{\ell^k}$ iff $u^{(q-1)/\ell^k} = 1$, and $u \mapsto u^{(q-1)/\ell^v}$ is an isomorphism $\mathbb{F}_q^\times/(\mathbb{F}_q^\times)^{\ell^v} \simeq \mu_{\ell^v}(\mathbb{F}_q)$ of cyclic groups of order ℓ^v (this group is $\mathbb{F}_q^\times \otimes \mathbb{Z}_\ell$). (b) ($\ell$-adic form.) For any compatible system $(\alpha_k)_{k \ge 0}$ in $\overline{\mathbb{F}}_q$ with $\alpha_0 = u$, $\alpha_{k+1}^\ell = \alpha_k$, the element $\kappa_u(\phi_q) = (\phi_q(\alpha_k)/\alpha_k)_k = (\alpha_k^{q-1})_k$ of $\mathbb{Z}_\ell(1) = \varprojlim \mu_{\ell^k}(\overline{\mathbb{F}}_q)$ changes by an element of $(q-1)\mathbb{Z}_\ell(1)$ when the system is changed, and its class in $\mathbb{Z}_\ell(1)/(q-1)\mathbb{Z}_\ell(1) \simeq \mu_{\ell^v}(\mathbb{F}_q)$ is $u^{(q-1)/\ell^v}$. For $k \le v$, the class has ℓ-adic valuation $\ge k$ (as an element of $\mathbb{Z}/\ell^v$ after choosing a generator) iff $u \in (\mathbb{F}_q^\times)^{\ell^k}$. *Proved here.* Elementary; compare Serre, Local Fields, Chapter IV. Paper use: Lemma lem:tamfactor ("c the class of u in $\mathbb{F}_q^\times/(\mathbb{F}_q^\times)^{\ell^{v_\ell(q-1)}}$", "the class of the unit part of q_E in $\mathbb{F}_q^\times \otimes \mathbb{Z}_\ell$ by Kummer theory"); Chapter 7 and Chapter 17 complete the passage from $\mathbb{F}_q$ to $\mathbb{Q}_q$ (Hensel, inflation from G/I).

galois:artin-schreier (theorem). Let $\mathrm{char}\, F = p > 0$ and $\wp(b) = b^p - b$. (a) For $a \in F$ the polynomial $x^p - x - a$ is separable; if α is a root, its roots are $\alpha + i$, $i \in \mathbb{F}_p$; it splits over F if $a \in \wp(F)$ and is irreducible otherwise, and then $F(\alpha)/F$ is cyclic of degree p with group generated by $\alpha \mapsto \alpha + 1$. (b) Every cyclic extension of F of degree p is of this form. (c) $\mathrm{Hom}_{\mathrm{cont}}(G_F, \mathbb{Z}/p) \simeq F/\wp(F)$, the additive analogue of Kummer theory. Example: $x^p - x - 1$ is irreducible over $\mathbb{F}_p$, with splitting field $\mathbb{F}_{p^p}$. *Stated; reference* ((a) proved in a few lines; (b) from the additive Hilbert 90, sketched). Lang VI.6. Background (the p-extensions that Kummer theory cannot reach in characteristic p; context for wild ramification in Chapter 7 and Remark rem:wild).

7.3.6 Section 6. The Galois group of a polynomial; discriminants, cubics, quartics and the four lines

galois:galois-group-polynomial (definition and proposition). Let $f \in F[x]$ be separable of degree $n \geq 1$, R its set of n roots in F^{sep}, $L = F(R)$ its splitting field and $G_f = \mathrm{Gal}(L/F)$, the Galois group of f. (a) G_f acts faithfully on R; numbering the roots gives an injective homomorphism $G_f \to S_n$ whose image is well defined up to conjugation in S_n; $\#G_f = [L : F]$. (b) The G_f-orbits on R correspond bijectively to the monic irreducible factors of f in $F[x]$: the orbit of r is the set of roots of the minimal polynomial of r; its size is $[F(r) : F]$ and the stabiliser of r is $\mathrm{Gal}(L/F(r))$. In particular f is irreducible iff G_f is transitive on R, and then $n \mid \#G_f$. (c) For any Galois extension Ω/F containing L (for instance $\Omega = F^{\mathrm{sep}}$), $\mathrm{Gal}(\Omega/F)$ acts on R through its quotient G_f, with the same orbits; and for an intermediate field $F \subset M \subset \Omega$ the irreducible factors of f in $M[x]$ correspond to the orbits of $\mathrm{Gal}(\Omega/M)$ on R. *Proved here.* Dummit-Foote 14.6; Lang VI.2. Paper use: Proposition prop:ns ("the four roots of the 3-division polynomial correspond to the four lines", "factors as two quadratics", "irreducible otherwise"), Section sec:data and the appendix (the test `ns3`); Chapter 9 (division polynomials).

galois:discriminant (definition and proposition). For $f = a\prod_{i=1}^{n}(x - r_i)$ separable, put $\delta_f = \prod_{i<j}(r_i - r_j)$ (depends on the numbering up to sign) and $\mathrm{disc}(f) = a^{2n-2}\delta_f^2 = (-1)^{n(n-1)/2}a^{-1}\,\mathrm{Res}(f, f')$, an element of F given by a universal integer polynomial in the coefficients. (a) $\sigma(\delta_f) = \mathrm{sgn}(\sigma)\delta_f$ for $\sigma \in G_f \subset S_n$. (b) If $\mathrm{char}\, F \neq 2$: $G_f \subset A_n$ iff $\mathrm{disc}(f) \in F^{\times 2}$; the fixed field of $G_f \cap A_n$ is $F(\sqrt{\mathrm{disc}\, f})$, and the character of G_F attached to it by galois:quadratic-extensions is $\mathrm{sgn} \circ$(action on R). (c) For monic g, h: $\mathrm{disc}(gh) = \mathrm{disc}(g)\,\mathrm{disc}(h)\,\mathrm{Res}(g, h)^2$. (d) Formulas (all checked by computer algebra): $\mathrm{disc}(x^2 + bx + c) = b^2 - 4c$; $\mathrm{disc}(x^3 + bx^2 + cx + d) = b^2c^2 - 4c^3 - 4b^3d - 27d^2 + 18bcd$, so $\mathrm{disc}(x^3 + px + q) = -4p^3 - 27q^2$; $\mathrm{disc}(x^n - a) = (-1)^{(n-1)(n-2)/2}n^n a^{n-1}$; $\mathrm{disc}(\Phi_\ell) = (-1)^{(\ell-1)/2}\ell^{\ell-2}$ for ℓ an odd prime. *Proved here* ((a) to (c), the quadratic and cubic formulas; $\mathrm{disc}(x^n - a)$ and $\mathrm{disc}(\Phi_\ell)$ via $\mathrm{disc}(f) = \pm\prod_i f'(r_i)$, sketched). Dummit-Foote 14.6. Paper use: Proposition prop:ns through galois:quartic-on-lines (c): for $E/\mathbb{Q}$ the sign of the Galois action on the roots of ψ_3 is $\det \bar\rho_{E,3} = \chi_{-3}$, so $\mathrm{disc}(\psi_3) \in -3 \cdot (\mathbb{Q}^\times)^2$; checked: $\mathrm{disc}(\psi_3) = -3^3 \cdot 11^{10}$ for 11a1 and $-3^3 \cdot 11^4$ for 121a1.

galois:cubic (proposition). Let $\mathrm{char}\, F \neq 2$ and $f \in F[x]$ a separable cubic. If f is irreducible, $G_f = A_3 \simeq C_3$ when $\mathrm{disc}(f)$ is a square and $G_f = S_3$ otherwise. If f is a linear factor times an irreducible quadratic g, $G_f = C_2$ with splitting field $F(\sqrt{\mathrm{disc}\, g})$. Examples over $\mathbb{Q}$: $x^3 - 2$ ($\mathrm{disc} = -108$, $G_f = S_3$, galois:x3minus2); $x^3 - 3x + 1$ ($\mathrm{disc} = 81$, $G_f = C_3$, roots $2\cos(2\pi k/9)$ for $k = 1, 2, 4$, splitting field the real cubic subfield $\mathbb{Q}(\zeta_9 + \zeta_9^{-1})$ of $\mathbb{Q}(\zeta_9)$). *Proved here.* Dummit-Foote 14.6. Background (the resolvent cubic below; the cubic factor of ψ_3 for a Borel image, galois:quartic-on-lines).

galois:quartic (proposition). Let $\mathrm{char}\, F \neq 2$ and $f = x^4 + ax^3 + bx^2 + cx + d \in F[x]$ separable with roots $r_1, \ldots, r_4$. Put $\theta_1 = r_1r_2 + r_3r_4$, $\theta_2 = r_1r_3 + r_2r_4$, $\theta_3 = r_1r_4 + r_2r_3$ (one for each partition of the roots into two pairs). The resolvent cubic $R_f(y) = \prod(y - \theta_i)$ equals $y^3 - by^2 + (ac - 4d)y - (a^2d - 4bd + c^2)$, and $\mathrm{disc}(R_f) = \mathrm{disc}(f)$ (both identities checked symbolically), so the θ_i are distinct. S_4 permutes the three partitions through a surjection $S_4 \to S_3$ with kernel the normal $V_4 = \{1, (12)(34), (13)(24), (14)(23)\}$. (a) If f is irreducible, G_f is conjugate to one of S_4, A_4, D_4, C_4, V_4, and: R_f irreducible and $\mathrm{disc}\, f \notin F^{\times 2}$ gives S_4; R_f irreducible and $\mathrm{disc}\, f \in F^{\times 2}$ gives A_4; R_f split into linear factors gives V_4; R_f with exactly one root in F gives D_4 or C_4, and C_4 iff f is reducible over $F(\sqrt{\mathrm{disc}\, f})$ (Kappe-Warren). (b) For any separable quartic, the degrees of the irreducible factors are the orbit sizes of G_f (galois:galois-group-polynomial), and $\theta_i \in F$ iff G_f stabilises the i-th partition. *Proved here* ((a) except the D_4 versus C_4 criterion, which is sketched). Dummit-Foote 14.6. Paper use: background for Proposition prop:ns (the Galois group of ψ_3 is the image of $\bar\rho_{E,3}$ in $\mathrm{PGL}_2(\mathbb{F}_3) \simeq S_4$); Chapter 22.

galois:quartic-on-lines (proposition; the Galois-theoretic form of the last sentence of Proposition prop:ns). Let F be a field, V a two-dimensional $\mathbb{F}_3$-vector space, $\rho\colon G_F \to \mathrm{GL}(V) \simeq \mathrm{GL}_2(\mathbb{F}_3)$ a continuous homomorphism with image H, and $f \in F[x]$ a separable quartic whose root set R is in a G_F-equivariant bijection with the set $\mathbb{P}(V)$ of the four lines of V. Then: (a) G_F acts on R through $H \to \mathrm{PGL}_2(\mathbb{F}_3) \simeq S_4$ (the action on lines is faithful modulo the scalars $\{\pm 1\}$, Chapter 1), so G_f is isomorphic to the image of H in $\mathrm{PGL}_2(\mathbb{F}_3)$ and the splitting field of f is the fixed field of $\rho^{-1}(\{\pm 1\})$; (b) the irreducible factors of f over F correspond to the orbits of H on the four lines, with degrees the orbit sizes; (c) the sign of the permutation of the four lines induced by $g \in \mathrm{GL}_2(\mathbb{F}_3)$ is $\det g \in \{\pm 1\} = \mathbb{F}_3^\times$ (Chapter 1), so, if $\mathrm{char}\, F \neq 2$,

$F(\sqrt{\operatorname{disc} f})$ is the fixed field of $\ker(\det\circ\rho)$; (d) for the subgroups H with $\det(H)=\mathbb{F}_3^\times$, up to conjugacy (orders and patterns checked by enumerating all 16 conjugacy classes of subgroups of $\mathrm{GL}_2(\mathbb{F}_3)$):

H (Sutherland label)	$\#H$	orbits on the four lines	G_f	factorisation of f over F
$\mathrm{GL}_2(\mathbb{F}_3)$ (surjective)	48	4	S_4	irreducible
$N_{ns}(\mathbb{F}_3)$ (3Nn)	16	4	D_4	irreducible
$C_{ns}(\mathbb{F}_3)$ (3Cn)	8	4	C_4	irreducible
$N_s(\mathbb{F}_3)$ (3Ns)	8	2+2	V_4, not normal in S_4	two irreducible quadratics
$\mathrm{B}(\mathbb{F}_3)$ (3B)	12	1+3	S_3	linear times irreducible cubic
the two of order 6 (3B.1.1, 3B.1.2)	6	1+3	S_3	linear times irreducible cubic
$C_s(\mathbb{F}_3)$ (3Cs)	4	1+1+2	C_2	two linear times irreducible quadratic
$\langle\mathrm{diag}(1,-1)\rangle$ (3Cs.1.1)	2	1+1+2	C_2	two linear times irreducible quadratic

(e) In the $N_s(\mathbb{F}_3)$ case the two quadratic factors have distinct splitting fields $K_1\neq K_2$, and the third quadratic subfield of K_1K_2 is $F(\sqrt{\operatorname{disc} f})$, the fixed field of $\det\circ\rho$; in the N_s and N_{ns} cases the resolvent cubic has exactly one root in F (the image in S_4 stabilises one partition of the lines into pairs), in the GL_2 case none. (f) Consequently, if H fixes no line of V and contains an element of determinant -1 and trace 0, then H is conjugate to $\mathrm{GL}_2(\mathbb{F}_3)$, $N_{ns}(\mathbb{F}_3)$ or $N_s(\mathbb{F}_3)$ (Chapter 1), and f is irreducible in the first two cases and a product of two irreducible quadratics in the third. *Proved here.* Chapter 1 for the group theory (the orbit table, det = sign, the three images). Paper use: Proposition prop:ns (last two sentences of the statement and of the proof), Section sec:data ("the mod 3 images of the tables were checked against the factorisation of the 3-division polynomial"), Section sec:anomaly, the appendix (`ns3`). The input "the roots of ψ_3 are the x-coordinates of the four lines of $E[3]$, Galois-equivariantly" is Chapter 9; "$\det\bar\rho_{E,3}=\chi_{-3}$ and complex conjugation has determinant -1, trace 0" is Chapters 9 and 11 (and galois:complex-conjugation).

galois:ns-example (worked example). The curve $E\colon y^2=x^3-6x-8$ ($j=-1728$, not a CM j-invariant; $\Delta=-2^9\cdot3^3$) has $\psi_3=3x^4-36x^2-96x-36=3(x^2-4x-2)(x^2+4x+6)$, a product of two irreducible quadratics with discriminants 24 and -8, so $K_1=\mathbb{Q}(\sqrt{6})$, $K_2=\mathbb{Q}(\sqrt{-2})$ and the third quadratic subfield is $\mathbb{Q}(\sqrt{-3})$, as galois:quartic-on-lines (e) predicts; the image of $\bar\rho_{E,3}$ is therefore $N_s(\mathbb{F}_3)$ (by (d): two orbits of size two and surjective determinant), and indeed $a_p\equiv0 \pmod 3$ for every prime $p\equiv2 \pmod 3$, $5\le p\le71$ ($a_5=3$, $a_{11}=3$, $a_{17}=-6$, $a_{23}=6$, $a_{29}=6$, $a_{41}=6$, $a_{47}=12$, $a_{53}=-3$, $a_{59}=12$, $a_{71}=-6$), as $\bar\rho\simeq\bar\rho\otimes\chi_{-3}$ requires (galois:index-two-restriction); note $9\nmid a_5$. By contrast 11a1 has $\psi_3=3x^4-4x^3-60x^2-237x-21$, irreducible with irreducible resolvent cubic and $\mathrm{disc}=-3^3\cdot11^{10}$: $G_{\psi_3}=S_4$, image $\mathrm{GL}_2(\mathbb{F}_3)$. *Proved here* (all values computed by the outline author; the writer should recompute with PARI, `elldivpol(E,3)`, `factor`, `ellap`). Paper use: Proposition prop:ns, Section sec:anomaly; the paper's own example 726a1 (image 3Ns, type III at 11) may be added after checking it in PARI.

7.3.7 Section 7. Infinite Galois extensions, the Krull topology, $G_{\mathbb{Q}}$, the cyclotomic character, complex conjugation

galois:infinite-galois-limit (proposition and definition). Let L/F be Galois of arbitrary degree. (a) L is the union of its finite Galois subextensions K/F, these form a directed set under inclusion, and restriction

gives a group isomorphism $\mathrm{Gal}(L/F) \to \varprojlim_K \mathrm{Gal}(K/F)$. (b) The Krull topology on $\mathrm{Gal}(L/F)$ is the one in which the subgroups $\mathrm{Gal}(L/K)$, $K \subset L$ finite over F (equivalently finite Galois over F), form a basis of open neighbourhoods of 1; it coincides with the inverse-limit topology of Chapter 1 (directed index set), so $\mathrm{Gal}(L/F)$ is a profinite group: compact, Hausdorff and totally disconnected. (c) For L/F finite the topology is discrete. *Proved here* (Tychonoff and the topological facts about profinite groups from Chapter 1). Neukirch IV.1; NSW I.1. Background: every Galois group in the paper ($G_{\mathbb{Q}}$, $G_{\mathbb{Q}_p}$, I_p, Section 1.1).

galois:infinite-fundamental-theorem (theorem, Krull). Let L/F be Galois with group G. (a) $H \mapsto L^H$ and $M \mapsto \mathrm{Gal}(L/M)$ are mutually inverse inclusion-reversing bijections between the closed subgroups of G and the intermediate fields $F \subset M \subset L$. (b) For an arbitrary subgroup $H \leq G$, $\mathrm{Gal}(L/L^H) = \overline{H}$ (closure). (c) Open subgroups correspond to the intermediate fields finite over F, with $[G : \mathrm{Gal}(L/M)] = [M : F]$. (d) M/F is Galois iff $\mathrm{Gal}(L/M)$ is normal in G, and then restriction induces an isomorphism of topological groups $G/\mathrm{Gal}(L/M) \simeq \mathrm{Gal}(M/F)$ (quotient topology on the left). (e) $\mathrm{Gal}(L/\sigma M) = \sigma\,\mathrm{Gal}(L/M)\sigma^{-1}$. (f) Warnings: in $\mathrm{Gal}(\overline{\mathbb{F}}_p/\mathbb{F}_p) \simeq \hat{\mathbb{Z}}$ the subgroup $\phi_p^{\mathbb{Z}}$ is dense and not closed (its fixed field is $\mathbb{F}_p$); and a subgroup of finite index need not be open: $\mathrm{Gal}(\mathbb{Q}(\sqrt{-1}, \sqrt{2}, \sqrt{3}, \sqrt{5}, \dots)/\mathbb{Q}) \simeq \prod_{\mathbb{N}} \mathbb{Z}/2$ (Kummer theory with $\Delta = \mathbb{Q}^\times$, $n = 2$) has non-open subgroups of index 2 (kernels of non-continuous functionals; uses the axiom of choice). *Proved here* ((a) to (e) and the first warning; the second warning stated). Neukirch IV.1; Lang VI.14; Milne, Fields and Galois Theory, Chapter 7. Paper use: background ("open subgroups", "the field cut out by", "Gal(Qbar/Q(sqrt(l*)))").

galois:absolute-galois-group (definition and proposition). $G_F = \mathrm{Gal}(F^{\mathrm{sep}}/F)$ with the Krull topology (convention C1). (a) For K/F finite separable inside F^{sep}, G_K is an open subgroup of G_F of index $[K : F]$, every open subgroup is of this form, $\sigma G_K \sigma^{-1} = G_{\sigma K}$, and G_K is normal iff K/F is Galois, in which case $G_F/G_K \simeq \mathrm{Gal}(K/F)$. (b) $G_{\mathbb{F}_q} \simeq \hat{\mathbb{Z}}$, with $1 \mapsto \phi_q$; its open subgroups are the $n\hat{\mathbb{Z}} = G_{\mathbb{F}_{q^n}}$. (c) $G_{\mathbb{Q}} = \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$; for a number field $K \subset \overline{\mathbb{Q}}$, G_K is open of index $[K : \mathbb{Q}]$; so $G_{\mathbb{Q}(\sqrt{d})}$ has index 2 (d not a square) and $G_{\mathbb{Q}(\zeta_n)}$ is open and normal of index $\varphi(n)$. (d) $G_{\mathbb{Q}}$ is uncountable and non-abelian; its maximal abelian quotient is $\mathrm{Gal}(\mathbb{Q}^{\mathrm{ab}}/\mathbb{Q}) \simeq \hat{\mathbb{Z}}^\times$ with $\mathbb{Q}^{\mathrm{ab}} = \mathbb{Q}(\mu_\infty)$ (Kronecker-Weber, Chapter 6). *Proved here* except (d), stated. Neukirch IV.1 and VI.1. Paper use: Section 1.1 ($\rho\colon G_{\mathbb{Q}} \to \mathrm{GL}_2(\mathbb{Z}_\ell)$), the hypothesis of Theorem thm:main, Section 4.2.

galois:cyclotomic-character (definition and proposition). Let ℓ be a prime and $\mathbb{Q}(\mu_{\ell^\infty}) = \bigcup_k \mathbb{Q}(\mu_{\ell^k})$. (a) There is a unique continuous homomorphism $\chi_\ell\colon G_{\mathbb{Q}} \to \mathbb{Z}_\ell^\times$ with $\sigma(\zeta) = \zeta^{\chi_\ell(\sigma)}$ for all $\zeta \in \mu_{\ell^\infty}$ (so $\chi_\ell(\sigma) \bmod \ell^k$ is the a with $\sigma|_{\mathbb{Q}(\zeta_{\ell^k})} = \sigma_a$). (b) χ_ℓ is surjective with kernel $G_{\mathbb{Q}(\mu_{\ell^\infty})}$, so $\mathrm{Gal}(\mathbb{Q}(\mu_{\ell^\infty})/\mathbb{Q}) \simeq \mathbb{Z}_\ell^\times$ as topological groups. (c) $\bar{\chi}_\ell = \chi_\ell \bmod \ell\colon G_{\mathbb{Q}} \to \mathbb{F}_\ell^\times$ is surjective with kernel $G_{\mathbb{Q}(\zeta_\ell)}$, and $\chi_{\ell^*} = \left(\frac{\bar{\chi}_\ell}{\ell}\right)$ (galois:sqrt-lstar). (d) For any field F with $\mathrm{char}\, F \neq \ell$, the same definition gives $\chi_\ell\colon G_F \to \mathbb{Z}_\ell^\times$ (no longer surjective in general; trivial modulo ℓ^k iff $\mu_{\ell^k} \subset F$; $\chi_\ell(\phi_q) = q$ for $F = \mathbb{F}_q$), and $\mathbb{Z}_\ell(1) = \varprojlim_k \mu_{\ell^k}$ (transition $\zeta \mapsto \zeta^\ell$) is a free $\mathbb{Z}_\ell$-module of rank one on which G_F acts through χ_ℓ; $\mathbb{Z}_\ell(1)/\ell^k = \mu_{\ell^k}$; $M(1) = M \otimes \mathbb{Z}_\ell(1)$. *Proved here.* Washington, Chapter 2; Neukirch IV.1; Serre, Abelian l-adic representations, Chapter I. Paper use: Section 4 (χ_ℓ, $T(1)$, $\mathrm{Hom}(A, \mathbb{Z}_\ell(1))$), Lemma lem:h0 ($\bar{\chi}_\ell$, $\bar{\chi}_3 = \chi_{-3}$), Lemma lem:tamfactor ($\mathbb{Z}_\ell(1)$, $\mathbb{Z}_\ell(1)^I$, Frob acts on $\mathbb{Z}_\ell(1)$ by q), Lemma lem:old ($\bar{\chi}_\ell$ in the local analysis); Chapter 9 ($\det \rho_{E,\ell} = \chi_\ell$ via the Weil pairing).

galois:complex-conjugation (proposition). Fix the inclusion $\overline{\mathbb{Q}} \subset \mathbb{C}$ (C1) and let $c \in G_{\mathbb{Q}}$ be the restriction of complex conjugation. (a) c has order 2 ($i \in \overline{\mathbb{Q}}$); for $\sigma \in G_{\mathbb{Q}}$, the complex conjugation attached to the embedding $\overline{\mathbb{Q}} \xrightarrow{\sigma} \overline{\mathbb{Q}} \subset \mathbb{C}$ is $\sigma^{-1} c \sigma$; "a complex conjugation" means an element of this conjugacy class. (b) $\chi_\ell(c) = -1$ for every ℓ; $\chi_d(c) = -1$ iff $d < 0$; so $\chi_{\ell^*}(c) = (-1)^{(\ell-1)/2}$ and $\chi_{-3}(c) = \chi_{-4}(c) = -1$. (c) (Artin-Schreier, stated.) Every element of $G_{\mathbb{Q}}$ of finite order other than 1 is a complex conjugation; in particular it has order 2. (d) Let k be a field with $\mathrm{char}\, k \neq 2$ and $\rho\colon G_{\mathbb{Q}} \to \mathrm{GL}_2(k)$ a homomorphism with $\det \rho(c) = -1$ (for instance $\det \rho = \bar{\chi}_\ell$, ℓ odd, or $\det \rho = \chi_\ell$). Then $\rho(c)$ is conjugate to $\mathrm{diag}(1, -1)$, has determinant -1 and trace 0, and ρ is irreducible over k iff it is absolutely irreducible (a line of $\bar{k}^2$ stable under $\rho(G_{\mathbb{Q}})$ is an eigenline of $\rho(c)$, and both eigenlines are k-rational). *(a), (b), (d) proved here; (c) stated; reference.* Lang XI (real fields); Serre, Course in Arithmetic, Chapter I (for the signs). Paper use: Proposition prop:ns ("contain an element of determinant -1 and trace 0 (complex conjugation)"), Lemmas lem:h0 and lem:old (irreducible implies absolutely irreducible for $\bar{\rho}_{E,\ell}$, used silently), Lemma lem:dfg21 ($G_{\mathbb{R}} = \{1, c\}$, so $H^1(\mathbb{R}, -)$ is killed by 2; Chapter 17).

7.3.8 Section 8. Continuous characters and representations of profinite groups

galois:continuous-reps (definition and proposition). Let G be profinite and $\rho\colon G \to \mathrm{GL}_n(k)$ a homomorphism, k topologised as in C5. (a) If k is discrete (a finite field, $\overline{\mathbb{F}}_\ell$, any field with the discrete topology), ρ is continuous iff $\ker\rho$ is open iff ρ factors through a finite quotient G/U; then $\rho(G)$ is finite. For $G = G_F$ the field cut out by ρ is $F_\rho = (F^{\mathrm{sep}})^{\ker\rho}$, finite Galois over F with $\mathrm{Gal}(F_\rho/F) \simeq \rho(G_F)$. (b) If $k = E$ is a finite extension of $\mathbb{Q}_\ell$ with ring of integers $\mathcal{O}$ and uniformiser λ, a homomorphism $\rho\colon G \to \mathrm{GL}_n(\mathcal{O})$ is continuous iff every $\rho \bmod \lambda^m$ is; then $\rho(G)$ is compact, hence closed, $F_\rho = (F^{\mathrm{sep}})^{\ker\rho}$ is Galois (in general infinite) over F, and $\mathrm{Gal}(F_\rho/F) \simeq \rho(G_F)$ as topological groups. (c) Every continuous $\rho\colon G \to \mathrm{GL}_n(E)$ stabilises an $\mathcal{O}$-lattice $\Lambda \subset E^n$ (the stabiliser U of $\mathcal{O}^n$ is open, of finite index; take $\Lambda = \sum_{g\in G/U} \rho(g)\mathcal{O}^n$), so it is conjugate to a continuous $\rho\colon G \to \mathrm{GL}_n(\mathcal{O})$; its reduction $\bar\rho_\Lambda$ on $\Lambda/\lambda\Lambda$ is continuous for the discrete topology. (d) Every continuous $\rho\colon G \to \mathrm{GL}_n(\mathbb{C})$ has finite image ($\mathrm{GL}_n(\mathbb{C})$ has a neighbourhood of 1 containing no nontrivial subgroup); in particular continuous characters $G_{\mathbb{Q}} \to \mathbb{C}^\times$ have finite order. (e) Subrepresentations, quotients, the semisimplification ρ^{ss} (Jordan-Holder, Chapter 4), $\rho\otimes\rho'$, $\rho^\vee$, $\mathrm{ad}^0\,\rho$ of continuous representations are continuous. *Proved here* ((d) sketched). Serre, Abelian l-adic representations, Chapter I, Section 1; DDT, Section 1. Paper use: Section 1.1 (ρ, $\bar\rho$, T, V), Lemma lem:h0 ($\rho \bmod \ell^k$), Lemma lem:old, Lemma lem:tamagawa-exp ($\rho \bmod \ell^t$), Section 4.2 (lattices).

galois:quadratic-characters (proposition). (a) $d \mapsto \chi_d$ (C3) is an isomorphism $\mathbb{Q}^\times/\mathbb{Q}^{\times 2} \simeq \mathrm{Hom}_{\mathrm{cont}}(G_{\mathbb{Q}}, \{\pm1\})$, with $\ker\chi_d = G_{\mathbb{Q}(\sqrt{d})}$ (Kummer theory with $n = 2$); the nontrivial continuous quadratic characters of $G_{\mathbb{Q}}$ correspond to the quadratic fields. (b) $\chi_{dd'} = \chi_d\chi_{d'}$; for example $\chi_{-15} = \chi_{-3}\chi_5$ (no numerical χ_3: in the book a numerical subscript is always a quadratic χ_d with its sign, Appendix B, B-C6(b)), and for a set S of odd primes $\chi_S = \prod_{q\in S}\chi_{q^*}$ is the character of $\mathbb{Q}(\sqrt{\prod_{q\in S} q^*})$. (c) (Values at Frobenius; proved in Chapter 6, used here as a black box, item galois:frobenius-preview.) For $p \nmid D$: $\chi_d(\mathrm{Frob}_p) = \left(\frac{D}{p}\right)$, which is 1 iff p splits in $\mathbb{Q}(\sqrt{d})$; $\chi_{-3}(\mathrm{Frob}_p) = 1$ iff $p \equiv 1 \pmod 3$; $\chi_{-4}(\mathrm{Frob}_p) = 1$ iff $p \equiv 1 \pmod 4$; $\chi_{\ell^*}(\mathrm{Frob}_p) = \left(\frac{p}{\ell}\right)$ for $p \neq \ell$, directly from $\chi_{\ell^*} = \left(\frac{\bar\chi_\ell}{\ell}\right)$ and $\bar\chi_\ell(\mathrm{Frob}_p) = p \bmod \ell$. (Comparing with $\chi_{\ell^*}(\mathrm{Frob}_p) = \left(\frac{\ell^*}{p}\right)$ is quadratic reciprocity: a remark, with the proof left to Chapter 2.) (d) As Dirichlet characters (Chapters 2 and 6) χ_d has conductor $|D|$: χ_{q^*} has conductor q, χ_{-3} conductor 3, χ_{-4} conductor 4. *(a), (b) proved here; (c), (d) stated with Chapter 6 reference; the congruence conditions in (c) were checked numerically for $p < 2000$ and $\ell \le 13$.* Paper use: the table of Section 1 and Lemma lem:euler ($\chi_{-3}(q)$, $\chi_{-4}(q)$), Lemma lem:twist (χ_{q^*}, $\mathbb{Q}(\sqrt{q^*})$), Theorem thm:twist (χ_S), Lemma lem:h0, Proposition prop:ns (χ_{-3}).

galois:twist (proposition). Let $\rho\colon G \to \mathrm{GL}_n(k)$ and $\chi\colon G \to k^\times$ be continuous. (a) $\rho\otimes\chi$ is continuous, $\mathrm{tr}(\rho\otimes\chi)(g) = \chi(g)\,\mathrm{tr}\,\rho(g)$, $\det(\rho\otimes\chi) = \chi^n \det\rho$, and for $G = G_{\mathbb{Q}}$ the field cut out by $\rho\otimes\chi$ lies in $F_\rho F_\chi$. (b) If $\rho \simeq \rho\otimes\chi$ and $n = 2$ then $\chi^2 = 1$; if moreover $\chi \neq 1$ and ρ is absolutely irreducible, every isomorphism $\phi\colon \rho \to \rho\otimes\chi$ has trace 0 (Chapter 4). (c) Let $\rho\colon G_{\mathbb{Q}} \to \mathrm{GL}_2(\mathbb{Z}_\ell)$ be continuous, ℓ odd, $k \ge 1$. If $\rho \bmod \ell^k \simeq (\rho\otimes\chi_\ell) \bmod \ell^k$ then $\chi_\ell^2 \equiv 1 \pmod{\ell^k}$ on $G_{\mathbb{Q}}$, which (as $\chi_\ell(G_{\mathbb{Q}}) = \mathbb{Z}_\ell^\times$) forces $\ell = 3$ and $k = 1$: for $\ell \ge 5$, $\bar\chi_\ell^2 \neq 1$; for $\ell = 3$, $\chi_3^{\mathrm{cyc}}(\sigma) = 2$ for some σ and $4 \not\equiv 1 \pmod 9$. *Proved here.* Elementary. Paper use: Lemma lem:h0 ("comparing determinants it forces $\bar\chi_\ell^2 = 1$, so $\ell = 3$"; with (c), the length $h_\ell(E)$ is at most 1, and is 0 for $\ell \ge 5$, see Section 8 of this outline), Conjecture conj:euler, Section sec:anomaly ($h_3(E) = 1$), Lemma lem:twist and Theorem thm:twist ($a_p(f\otimes\chi) = \chi(p)a_p(f)$), Theorem thm:qual cases (b) and (c).

galois:restriction (definition and proposition). Let K/F be finite separable inside F^{sep} and $\rho\colon G_F \to \mathrm{GL}_n(k)$ continuous. (a) $\mathrm{Res}\,\rho = \rho|_{G_K}$ is continuous; $[\rho(G_F) : \rho(G_K)] \le [K : F]$, and it divides $[K : F]$ when K/F is Galois ($\rho(G_K)$ is then normal in $\rho(G_F)$ with quotient a quotient of $\mathrm{Gal}(K/F)$). (b) (Image of a kernel.) If K is the field cut out by a character $\delta\circ\rho$, with $\delta\colon \mathrm{GL}_n(k) \to A$ a homomorphism, then $\rho(G_K) = \rho(G_F) \cap \ker\delta$ (Chapter 1). (c) For an elliptic curve $E/\mathbb{Q}$ and ℓ odd, with $\det\bar\rho_{E,\ell} = \bar\chi_\ell$ (Chapter 9): $\bar\rho_{E,\ell}(G_{\mathbb{Q}(\zeta_\ell)}) = \bar\rho_{E,\ell}(G_{\mathbb{Q}}) \cap \mathrm{SL}_2(\mathbb{F}_\ell)$ and $\bar\rho_{E,\ell}(G_{\mathbb{Q}(\sqrt{\ell^*})}) = \{g \in \bar\rho_{E,\ell}(G_{\mathbb{Q}}) : \det g \in (\mathbb{F}_\ell^\times)^2\}$ (take $\delta = \det$, resp. $\delta = (\frac{\det}{\ell})$); for $\ell = 3$ both are the determinant-one subgroup of the image. (d) The field cut out by $\mathrm{Res}\,\rho$ is KF_ρ, and $\mathrm{Gal}(KF_\rho/K) \simeq \mathrm{Gal}(F_\rho/F_\rho\cap K)$ (galois:translation-theorem). (e) "ρ is absolutely irreducible on G_K" means that $\rho(G_K)$ fixes no proper nonzero subspace of $\bar k^n$; it depends only on the subgroup $\rho(G_K)$. *Proved here.* Paper use: the hypothesis of Theorem thm:main, Theorem thm:qual and the quoted DDT Theorem 3.15 ("absolutely irreducible when restricted to $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{\ell^*}))$"), S_f (Section

4.2), Proposition prop:ns (first sentence of the proof), Lemma lem:h0.

galois:index-two-restriction (proposition; the Galois form of Clifford theory for index two, Chapter 4). Let $H \subset G$ be an open subgroup of index 2 of a profinite group (for $G = G_{\mathbb{Q}}$: $H = G_K$, K quadratic), $\chi\colon G \to G/H \simeq \{\pm 1\}$, $s \in G \setminus H$, k a field with $\operatorname{char} k \neq 2$, and $\rho\colon G \to \mathrm{GL}_2(k)$ continuous and absolutely irreducible. The following are equivalent: (i) $\rho|_H$ is not absolutely irreducible; (ii) $\rho \simeq \rho \otimes \chi$ (over k, equivalently over $\bar{k}$, since $\mathrm{Hom}_G(\rho, \rho\otimes\chi)$ has dimension at most one); (iii) $\rho \otimes \bar{k} \simeq \mathrm{Ind}_H^G \psi$ for a continuous character $\psi\colon H \to \bar{k}^\times$. When they hold, $\rho|_H \otimes \bar{k} \simeq \psi \oplus \psi^s$ with $\psi^s(h) = \psi(s^{-1}hs) \neq \psi(h)$ for some h, and every isomorphism $\rho \to \rho \otimes \chi$ has trace 0. For $\bar{\rho} = \bar{\rho}_{E,\ell}$ and $K = \mathbb{Q}(\sqrt{\ell^*})$: (i) is the failure of the hypothesis of Theorem thm:main; for $\ell = 3$, $\chi_{-3} = \bar{\chi}_3$ and (ii) is $H^0(\mathbb{Q}, \mathrm{ad}^0\,\bar{\rho} \otimes \bar{\chi}_3) \neq 0$ (Lemma lem:h0); for $\ell \geq 5$, (ii) for χ_{ℓ^*} can hold (the "dihedral" case excluded by S_f) but is not the condition $\bar{\rho} \simeq \bar{\rho} \otimes \bar{\chi}_\ell$, which never holds (galois:twist (c)). *Proved here* from Chapter 4 (Clifford theory, Frobenius reciprocity) plus continuity (an induced representation from an open subgroup is continuous). Paper use: Lemma lem:h0 ("$\bar{\rho} \simeq \bar{\rho} \otimes \chi_{-3}$ iff $\bar{\rho}$ is induced from $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{-3}))$, that is, iff its restriction there is not absolutely irreducible"), Proposition prop:ns ("$\bar{\rho} \simeq \mathrm{Ind}\,\psi$ by Frobenius reciprocity, so $\bar{\rho} \simeq \bar{\rho} \otimes \chi_{-3}$"), Theorem thm:main hypothesis, Section 1.3 of the paper (see Section 8 of this outline).

galois:reduction-ss (proposition). Let $E/\mathbb{Q}_\ell$ be finite with ring of integers $\mathcal{O}$, uniformiser λ, residue field k, and $\rho\colon G \to \mathrm{GL}_n(E)$ continuous, G profinite. For two G-stable $\mathcal{O}$-lattices Λ, Λ' the reductions $\bar{\rho}_\Lambda$, $\bar{\rho}_{\Lambda'}$ have the same characteristic polynomials (the reductions of those of ρ, which lie in $\mathcal{O}[X]$), hence isomorphic semisimplifications (Brauer-Nesbitt, Chapter 4, applied to a common finite quotient of G); so $\bar{\rho}^{ss}$ is well defined. If $\bar{\rho}_\Lambda$ is irreducible, Λ' is homothetic to Λ (Ribet's lemma, Chapter 4). *Proved here* (Ribet's lemma quoted). Serre, Abelian l-adic representations, Chapter I; DDT, Section 1. Paper use: Lemma lem:old ("$\bar{\rho}_h$"), Section 4.2 ($M_{f,\lambda}$ and $T_\ell E$ homothetic, so their adjoints coincide).

galois:l-adic-kummer (construction and proposition). Let $\ell \neq \operatorname{char} F$ and $a \in F^\times$. Choose a compatible system $(\alpha_k)_{k\geq 0}$ in F^{sep} with $\alpha_0 = a$ and $\alpha_{k+1}^\ell = \alpha_k$. (a) $\kappa_a(\sigma) = (\sigma(\alpha_k)/\alpha_k)_k \in \mathbb{Z}_\ell(1)$ defines a continuous map $\kappa_a\colon G_F \to \mathbb{Z}_\ell(1)$ with $\kappa_a(\sigma\tau) = \kappa_a(\sigma) + \chi_\ell(\sigma)\kappa_a(\tau)$ ($\mathbb{Z}_\ell(1)$ written additively): a continuous 1-cocycle (Chapter 16). Equivalently, after choosing a generator of $\mathbb{Z}_\ell(1)$, $\sigma \mapsto \begin{pmatrix} \chi_\ell(\sigma) & \kappa_a(\sigma) \\ 0 & 1 \end{pmatrix}$ is a continuous homomorphism $G_F \to \mathrm{GL}_2(\mathbb{Z}_\ell)$; it is the action on the $\mathbb{Z}_\ell$-span of a basis $e_1 \in \mathbb{Z}_\ell(1)$, e_2 with $\sigma e_2 = e_2 + \kappa_a(\sigma)e_1$. (b) Changing the system changes κ_a by the coboundary $\sigma \mapsto (\chi_\ell(\sigma) - 1)z$, $z \in \mathbb{Z}_\ell(1)$; $\kappa_{ab} = \kappa_a + \kappa_b$ for the product systems; and $\kappa_a \bmod \ell^k$ restricted to $G_{F(\mu_{\ell^k})}$ is the Kummer character χ_a with $n = \ell^k$. (c) On $\ker\chi_\ell = G_{F(\mu_{\ell^\infty})}$ the map κ_a is a homomorphism, and for $\sigma \in G_F$, $\tau \in \ker\chi_\ell$: $\kappa_a(\sigma\tau\sigma^{-1}) = \chi_\ell(\sigma)\kappa_a(\tau)$. (d) If $K \supset F$ contains μ_{ℓ^k}, then $\kappa_a \bmod \ell^k$ vanishes on G_K iff $a \in (K^\times)^{\ell^k}$. *Proved here.* Serre, Abelian l-adic representations, Chapter IV (Tate curves; for the shape); NSW, Chapter VI. Paper use: Lemma lem:tamfactor ("κ the Kummer class of q_E", the matrix $\left(\begin{smallmatrix} \chi_\ell\varepsilon & \kappa \\ 0 & \varepsilon \end{smallmatrix}\right)$, "the restriction of κ to I is n times the tame character", and, via (c) with $\sigma = \mathrm{Frob}$, $\tau = t$ in inertia, $\mathrm{Frob}\; t\; \mathrm{Frob}^{-1} = t^q$), Lemma lem:tamagawa-exp (by (d) with $F = \mathbb{Q}_q$, $K = \mathbb{Q}_q^{\mathrm{ur}}$, $G_K = I_q$: inertia acts trivially on $E[\ell^t]$ iff the Tate parameter is an ℓ^t-th power in $\mathbb{Q}_q^{\mathrm{ur}}$; Chapter 7 turns this into $\ell^t \mid n_q$); Chapters 7, 10, 11, 17.

7.3.9 Section 9. Frobenius elements (foreshadowed) and the Chebotarev density theorem

galois:frobenius-preview (statement; the definitions and proofs are in Chapter 6, the local form in Chapter 7). Let $K/\mathbb{Q}$ be finite Galois and p a prime unramified in K. (a) For each prime $\mathfrak{P}$ of $\mathcal{O}_K$ above p there is a unique $\mathrm{Frob}_{\mathfrak{P}} \in \mathrm{Gal}(K/\mathbb{Q})$ with $\mathrm{Frob}_{\mathfrak{P}}(x) \equiv x^p \pmod{\mathfrak{P}}$ for all $x \in \mathcal{O}_K$ (arithmetic Frobenius, C2); the $\mathrm{Frob}_{\mathfrak{P}}$, $\mathfrak{P} \mid p$, form one conjugacy class Frob_p; the order of $\mathrm{Frob}_{\mathfrak{P}}$ is the residue degree, and p splits completely in K iff $\mathrm{Frob}_p = 1$. (b) For $K' \subset K$ Galois over $\mathbb{Q}$, $\mathrm{Frob}_p(K/\mathbb{Q})$ restricts to $\mathrm{Frob}_p(K'/\mathbb{Q})$. (c) In $\mathbb{Q}(\zeta_n)$, $p \nmid n$: $\mathrm{Frob}_p = \sigma_p$, so $\bar{\chi}_\ell(\mathrm{Frob}_p) = p \bmod \ell$ and $\chi_\ell(\mathrm{Frob}_p) = p$; in $\mathbb{Q}(\sqrt{d})$, $p \nmid D$: $\chi_d(\mathrm{Frob}_p) = \left(\frac{D}{p}\right)$. (d) In $G_{\mathbb{Q}}$: a Frobenius element at p is any $\sigma \in G_{\mathbb{Q}}$ whose restriction to each finite Galois K unramified at p lies in $\mathrm{Frob}_p(K/\mathbb{Q})$ and which comes from a decomposition group at a prime above p; it is well defined up to conjugation and up to the inertia group. Hence for a continuous ρ of $G_{\mathbb{Q}}$ unramified at p (trivial on the inertia groups at p) the conjugacy class of $\rho(\mathrm{Frob}_p)$, in particular its

characteristic polynomial, is well defined. (e) (Dedekind.) If $f \in \mathbb{Z}[x]$ is separable with leading coefficient a and $p \nmid a\,\mathrm{disc}(f)$, and $f \bmod p$ has irreducible factors of degrees $d_1, \dots, d_r$, then Frob_p acts on the roots of f with cycle type $(d_1, \dots, d_r)$. *Stated; reference* (Chapter 6; Neukirch I.9 and I.8). Paper use: Lemma lem:old ("$X^2 - a_p(h)X + p$" as characteristic polynomial of $\bar\rho_h(\mathrm{Frob}_p)$, the eigenvalues of $\bar\rho(\mathrm{Frob}_p)$), Lemma lem:euler, Lemma lem:tamfactor (Frob), Lemma lem:h0 (a_p at primes inert in $\mathbb{Q}(\sqrt{\ell^*})$).

galois:frobenius-on-lines (worked example, using the black box). For $E/\mathbb{Q}$ with good reduction at $p \neq 3$ and $p \nmid \mathrm{disc}(\psi_3)$, $\bar\rho_{E,3}(\mathrm{Frob}_p)$ has characteristic polynomial $X^2 - a_p X + p \bmod 3$ (Chapters 9, 11), and by galois:frobenius-preview (e) and galois:quartic-on-lines the degrees of the factors of $\psi_3 \bmod p$ are the orbit sizes of $\bar\rho(\mathrm{Frob}_p)$ on the four lines: $X^2 \pm X - 1$ (order 8) gives (4); $X^2 + 1$ (order 4) gives $(2,2)$; $X^2 - 1$ (determinant -1, trace 0) gives $(1,1,2)$; $(X \mp 1)^2$ gives $(1,3)$ when $\bar\rho(\mathrm{Frob}_p) \neq \pm 1$ and $(1,1,1,1)$ otherwise. Checked for 11a1 and 121a1 at all primes $p < 40$, $p \neq 3, 11$ (for 11a1: $p = 2, 5, 17, 23$ give (4); $p = 7, 13, 31$ give $(1,3)$; $p = 19, 37$ give $(2,2)$; $p = 29$ gives $(1,1,2)$). *Proved here* (given the black boxes). Paper use: Proposition prop:ns, Section sec:data (consistency of the image labels with ψ_3), Chapter 22.

galois:density (definition). For a set X of primes: its Dirichlet density is $\lim_{s\to 1^+} \sum_{p\in X} p^{-s} / \log \frac{1}{s-1}$ and its natural density is $\lim_{x\to\infty} \#\{p \le x : p \in X\}/\pi(x)$, when these exist. A set with a natural density has the same Dirichlet density; finite sets have density 0; "density one" means that the complement has density 0; "almost all p" means all but finitely many. *Definition; the comparison stated.* Serre, Course in Arithmetic, Chapter VI, Section 4. Paper use: Lemma lem:old ("for all primes $p \nmid \ell N$"), Section 3 ("for all $p \nmid 3N$").

galois:chebotarev (theorem, Chebotarev's density theorem). Let $K/\mathbb{Q}$ be finite Galois with group G and $C \subset G$ a conjugacy class. The set of primes p, unramified in K, with $\mathrm{Frob}_p = C$ has Dirichlet density $\#C/\#G$; it also has natural density $\#C/\#G$. The same holds for K/F finite Galois of number fields, with primes of F. Special cases: $K = \mathbb{Q}(\zeta_n)$ gives Dirichlet's theorem (density $1/\varphi(n)$ for each class $a \in (\mathbb{Z}/n)^\times$); $K = \mathbb{Q}(\sqrt{d})$: half the primes split; K the splitting field of $x^3 - 2$: $x^3 - 2$ has exactly one root mod p for density 1/2 of the primes (transpositions; precisely the $p \equiv 2 \bmod 3$), three roots for density 1/6 and none for density 1/3 (checked numerically: 0.516, 0.159, 0.325 for $5 \le p < 3000$). *Stated; reference.* Neukirch VII.13 (Dirichlet density); Serre, Abelian l-adic representations, Chapter I, Section 2.2 (with the natural-density form). Paper use: Lemma lem:old ("by Chebotarev and Brauer-Nesbitt").

galois:chebotarev-dense (corollary). Let $L/\mathbb{Q}$ be Galois (possibly infinite) and unramified outside a finite set S of primes (every finite subextension is), and let X be a set of primes of Dirichlet density one. Then the Frobenius elements $\mathrm{Frob}_{\mathfrak{P}} \in \mathrm{Gal}(L/\mathbb{Q})$ at primes $\mathfrak{P}$ of L above $p \in X \setminus S$ are dense in $\mathrm{Gal}(L/\mathbb{Q})$. (Proof: a basic open set is a coset of $\mathrm{Gal}(L/K)$ with $K/\mathbb{Q}$ finite Galois; the primes whose Frobenius in $\mathrm{Gal}(K/\mathbb{Q})$ is the class of the coset have density at least $1/\#\,\mathrm{Gal}(K/\mathbb{Q}) > 0$ and so meet $X \setminus S$.) *Proved here* from galois:chebotarev. Serre, Abelian l-adic representations, Chapter I, Section 2.2. Paper use: Lemma lem:old.

galois:chebotarev-brauer-nesbitt (corollary; the form the paper uses). Let k be a finite field, $\overline{\mathbb{F}}_\ell$, or a finite extension of $\mathbb{Q}_\ell$, and $\rho_1, \rho_2 \colon G_\mathbb{Q} \to \mathrm{GL}_n(k)$ continuous, semisimple, and unramified outside a finite set S. If $\det(X - \rho_1(\mathrm{Frob}_p)) = \det(X - \rho_2(\mathrm{Frob}_p))$ for all p in a set of density one (with $p \notin S$), then $\rho_1 \simeq \rho_2$. (Proof: let L be the field cut out by $\rho_1 \oplus \rho_2$, unramified outside S; $g \mapsto \det(X - \rho_1(g)) - \det(X - \rho_2(g))$ is continuous on $\mathrm{Gal}(L/\mathbb{Q})$ and vanishes on the dense set of Frobenius elements, hence everywhere; then Brauer-Nesbitt, Chapter 4: for finite or discrete k the image of $\mathrm{Gal}(L/\mathbb{Q})$ is finite and the finite-group form suffices; for ℓ-adic k use the form for an arbitrary group, or in characteristic 0 the fact that traces determine a semisimple representation.) *Proved here* from galois:chebotarev-dense and Chapter 4. DDT, Section 1; Serre, Abelian l-adic representations, Chapter I. Paper use: Lemma lem:old (the step $p \nmid \ell N$), Section 3 ("$\bar\rho_g \simeq \bar\rho$, that is, $a_p(g) \equiv a_p(f) \pmod\lambda$ for all $p \nmid 3N$"); Chapters 11 and 15.

galois:congruent-ap (corollary). Let $\mathcal{O}$ be the ring of integers of a finite extension of $\mathbb{Q}_\ell$, λ its maximal ideal, $k = \mathcal{O}/\lambda$, and $\rho_1, \rho_2 \colon G_\mathbb{Q} \to \mathrm{GL}_2(\mathcal{O})$ continuous and unramified outside a finite set S, with $\mathrm{tr}\,\rho_1(\mathrm{Frob}_p) \equiv \mathrm{tr}\,\rho_2(\mathrm{Frob}_p)$ and $\det\rho_1(\mathrm{Frob}_p) \equiv \det\rho_2(\mathrm{Frob}_p) \pmod\lambda$ for all p in a set of density one. Then $\bar\rho_1^{ss} \simeq \bar\rho_2^{ss}$ over k; if moreover $\bar\rho_1$ is absolutely irreducible, $\bar\rho_2 \simeq \bar\rho_1$ (a two-dimensional representation whose semisimplification is irreducible is irreducible). Conversely $\bar\rho_1^{ss} \simeq \bar\rho_2^{ss}$ gives the congruences at every $p \notin S$. Application (with the Chapter 15 black box: a newform h of weight 2 and level M with coefficients in $\mathcal{O}$ has $\rho_{h,\lambda} \colon G_\mathbb{Q} \to \mathrm{GL}_2(\mathcal{O})$, unramified outside $M\ell$, $\det = \chi_\ell$, $\mathrm{tr}\,\rho_{h,\lambda}(\mathrm{Frob}_p) = a_p(h)$ for $p \nmid M\ell$): if $a_p(h) \equiv a_p(f) \pmod\lambda$ for almost all p, then $\bar\rho_{h,\lambda}^{ss} \simeq \bar\rho_{f,\lambda}^{ss}$, and for $f = f_E$ with $\bar\rho_{E,\ell}$ irreducible

(hence absolutely irreducible, galois:complex-conjugation) $\bar{\rho}_{h,\lambda} \simeq \bar{\rho}_{E,\ell} \otimes_{\mathbb{F}_\ell} k$. Note that the two residual representations may be defined over different finite fields; compare them over a common finite extension. *Proved here* (given the Chapter 15 black box). Paper use: Lemma lem:old ("Consequently $\bar{\rho}_h \simeq \bar{\rho}$ by Chebotarev and Brauer-Nesbitt, $\bar{\rho}$ being irreducible"), Section 3 cases (a) to (c) and the Example (121a1 with 11a1, 539c1 with 11a1, 980i1 with 20a1).

7.4 3. Coverage check: PLAN.md scope, Section 6 map rows and paper hooks to line-up items

PLAN.md scope item, Section 6 row or paper hook	line-up items
field extensions, degree	galois:field-extension-degree
algebraic and transcendental elements, minimal polynomials	galois:algebraic-element, galois:algebraic-extensions, galois:algebraic-closure
splitting fields	galois:splitting-field
separability	galois:separability, galois:separable-closure
normal extensions	galois:normal-extension
Galois extensions, the Galois group	galois:dedekind-independence, galois:artin-fixed-field, galois:galois-extension
fundamental theorem of Galois theory	galois:fundamental-theorem, galois:translation-theorem
examples: quadratic fields	galois:quadratic-extensions, galois:quadratic-characters
examples: $\mathbb{Q}(\zeta_n)$ with group $(\mathbb{Z}/n)^\times$	galois:cyclotomic, galois:cyclotomic-mod-p, galois:finite-field-galois
examples: $\mathbb{Q}(2^{1/3}, \zeta_3)$	galois:x3minus2, galois:kummer-affine, galois:cubic
Kummer theory for cyclic extensions of degree n with $\zeta_n \in F$	galois:hilbert90, galois:kummer-character, galois:kummer-cyclic
mod n version: $F^\times/(F^\times)^n$ classifies $\mathbb{Z}/n$-extensions	galois:kummer-classification (c), galois:kummer-finite-field
Artin-Schreier (statement)	galois:artin-schreier
Galois group of a polynomial, action on roots	galois:galois-group-polynomial
discriminant and transitivity, irreducibility	galois:galois-group-polynomial (b), galois:discriminant
cubic and quartic polynomials	galois:cubic, galois:quartic
quartic with group in $\mathrm{GL}_2(\mathbb{F}_3)$ acting on lines; 3-division polynomial (two quadratics for $N_s(\mathbb{F}_3)$, irreducible for $N_{ns}(\mathbb{F}_3)$, $\mathrm{GL}_2(\mathbb{F}_3)$)	galois:quartic-on-lines, galois:ns-example, galois:frobenius-on-lines
infinite Galois extensions and the Krull topology	galois:infinite-galois-limit
infinite fundamental theorem	galois:infinite-fundamental-theorem
profinite groups	galois:infinite-galois-limit (with Chapter 1, items 7.1 to 7.5)
$G_{\mathbb{Q}}$ and G_K	galois:absolute-galois-group
continuous characters and representations of profinite groups	galois:continuous-reps, galois:quadratic-characters, galois:twist, galois:reduction-ss
Frobenius elements (foreshadowed)	galois:finite-field-galois, galois:cyclotomic-mod-p, galois:frobenius-preview
Chebotarev density theorem (statement)	galois:density, galois:chebotarev, galois:chebotarev-dense

PLAN.md scope item, Section 6 row or paper hook	line-up items
corollary: semisimple representation determined by Frobenius characteristic polynomials on a density-one set	galois:chebotarev-brauer-nesbitt
corollary: congruent a_p for almost all p give isomorphic residual semisimplifications (with Brauer-Nesbitt)	galois:congruent-ap
complex conjugation as an element of order two of $G_{\mathbb{Q}}$	galois:complex-conjugation
quadratic characters of $G_{\mathbb{Q}}$ and $\mathbb{Q}(\sqrt{d})$	galois:quadratic-characters, galois:quadratic-extensions
$\mathbb{Q}(\sqrt{\ell^*})$ as the quadratic subfield of $\mathbb{Q}(\zeta_\ell)$	galois:sqrt-lstar, galois:cyclotomic-character (c)
restriction of representations to open subgroups such as $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{\ell^*}))$	galois:restriction, galois:index-two-restriction, galois:translation-theorem
cyclotomic character, $\mathbb{Z}_\ell(1)$ (needed by Section 4 of the paper)	galois:cyclotomic-character
hook: hypothesis "absolutely irreducible on $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{\ell^*}))$" (Theorem thm:main, Theorem thm:qual, S_f, DDT 3.15)	galois:sqrt-lstar, galois:absolute-galois-group, galois:restriction, galois:index-two-restriction, galois:complex-conjugation (d)
hook: Proposition prop:ns	galois:restriction (b), (c); galois:index-two-restriction; galois:galois-group-polynomial; galois:discriminant; galois:quartic-on-lines; galois:ns-example; galois:complex-conjugation
hook: Lemma lem:h0	galois:sqrt-lstar (d), galois:cyclotomic-character, galois:twist, galois:index-two-restriction, galois:complex-conjugation (d)
hook: Lemma lem:old (Chebotarev and Brauer-Nesbitt)	galois:frobenius-preview, galois:chebotarev, galois:chebotarev-dense, galois:chebotarev-brauer-nesbitt, galois:congruent-ap, galois:reduction-ss
hook: Lemma lem:tamagawa-exp (Kummer theory, ℓ^t-th roots of the Tate parameter)	galois:kummer-character, galois:kummer-classification, galois:l-adic-kummer (d)
hook: Lemma lem:tamfactor (Kummer classes)	galois:l-adic-kummer, galois:kummer-affine, galois:kummer-finite-field, galois:cyclotomic-character (d)
Section 6 row: $\rho_{E,\ell}$, restriction to $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{\ell^*}))$	galois:continuous-reps, galois:restriction
Section 6 row: 3-division polynomial, its factorisation pattern	galois:quartic-on-lines, galois:frobenius-on-lines
Section 6 row: Chebotarev, Brauer-Nesbitt	galois:chebotarev-brauer-nesbitt, galois:congruent-ap
Section 6 row: Kummer theory, Kummer classes also used: χ_{-3}, χ_{-4}, χ_{q^*}, χ_S (table of Section 1, Lemma lem:euler, Lemma lem:twist, Theorem thm:twist)	galois:kummer-classification, galois:l-adic-kummer galois:quadratic-characters, galois:twist

7.5 4. Section-by-section plan with target line counts

Line counts are lines of LaTeX, card excluded (PLAN.md Section 7.3: 1,000 to 2,200).

section	title	contents	target lines
(starred)	Core results at a glance	the card: every item of Part 2, stated as above, with tag and `\Cref`	(excluded)
(starred)	Where the paper uses this chapter	thm:main hypothesis and S_f; prop:ns; lem:h0; lem:old; lem:tamagawa-exp; lem:tamfactor; $\chi_{-3}, \chi_{-4}, \chi_{q^*}, \chi_S$ in Section 1, lem:euler, lem:twist, thm:twist	40
1	Field extensions and algebraic elements	conventions C1 to C7 as `convention` environments; galois:field-extension-degree to galois:algebraic-closure	150
2	Splitting fields, normality, separability	galois:splitting-field to galois:separable-closure	190
3	Galois extensions and the fundamental theorem	galois:dedekind-independence to galois:translation-theorem; one picture-free subgroup/subfield diagram in `tabular`	200
4	Finite fields, quadratic and cyclotomic extensions	galois:finite-field-galois to galois:x3minus2	220
5	Cyclic extensions: Hilbert 90, Kummer, Artin-Schreier	galois:norm-trace to galois:artin-schreier; `paperlink` for lem:tamagawa-exp and lem:tamfactor	230
6	The Galois group of a polynomial	galois:galois-group-polynomial to galois:ns-example; the orbit table; `paperlink` for prop:ns	250
7	Infinite Galois theory, $G_{\mathbb{Q}}$, χ_ℓ, complex conjugation	galois:infinite-galois-limit to galois:complex-conjugation	200
8	Continuous representations of profinite groups	galois:continuous-reps to galois:l-adic-kummer; `paperlink` for the hypothesis of thm:main and lem:h0	230
9	Frobenius elements and Chebotarev	galois:frobenius-preview to galois:congruent-ap; `paperlink` for lem:old	170

section	title	contents	target lines
10	Exercises	15 exercises, graded (list below)	110
11	Notes and references	Lang, Dummit-Foote, Milne (Fields and Galois Theory), Neukirch, NSW, Washington, Serre (Abelian l-adic representations, Course in Arithmetic), DDT; which numbers were cited	40
	total		**2030**

Exercises (Section 10), graded from easy to harder: 1. Show $\mathbb{Q}(\sqrt{2},\sqrt{3}) = \mathbb{Q}(\sqrt{2}+\sqrt{3})$ (minimal polynomial x^4-10x^2+1) and list its intermediate fields. 2. Compute the Galois group of x^4-2 over $\mathbb{Q}$ (D_4, order 8) and over $\mathbb{Q}(i)$ (C_4, by Kummer theory). 3. Show that x^3-3x+1 has discriminant 81 and splitting field $\mathbb{Q}(\zeta_9+\zeta_9^{-1})$. 4. Prove $\operatorname{disc}(\Phi_\ell) = (-1)^{(\ell-1)/2}\ell^{\ell-2}$ and deduce $\sqrt{\ell^*} \in \mathbb{Q}(\zeta_\ell)$ without Gauss sums. 5. Show that $\mathbb{Q}(\sqrt{-5}) \subset \mathbb{Q}(\zeta_{20})$ and $\mathbb{Q}(\sqrt{6}) \subset \mathbb{Q}(\zeta_{24})$, and that neither lies in a smaller cyclotomic field. 6. Prove that the degree of a splitting field of a polynomial of degree n divides $n!$. 7. Prove that $x^p - x - 1$ is irreducible over $\mathbb{F}_p$, with splitting field $\mathbb{F}_{p^p}$. 8. Let $u \in \mathbb{F}_{13}^\times$. Show that u is a cube iff $u^4 = 1$, and compute the class of $u = 2$ in $\mathbb{F}_{13}^\times/(\mathbb{F}_{13}^\times)^3$ (illustrating galois:kummer-finite-field with $\ell = 3$, $v = 1$). 9. Show that $\phi_p^{\mathbb{Z}}$ is dense and not closed in $\operatorname{Gal}(\overline{\mathbb{F}}_p/\mathbb{F}_p)$, and that $\operatorname{Gal}(\mathbb{Q}(\mu_{\ell^\infty})/\mathbb{Q}) \simeq \mathbb{Z}_\ell^\times$. 10. Show that every continuous homomorphism from a profinite group to $\mathbb{C}^\times$ has finite image. 11. Let $\bar\rho\colon G_{\mathbb{Q}} \to \mathrm{GL}_2(\mathbb{F}_\ell)$, ℓ odd, with $\det\bar\rho = \bar\chi_\ell$. Prove that $\bar\rho$ is irreducible iff it is absolutely irreducible, and that $\bar\rho(G_{\mathbb{Q}(\sqrt{\ell^*})})$ is the set of elements of the image with square determinant. 12. Check the paper's claim (Proposition prop:ns) on $y^2 = x^3 - 6x - 8$: factor ψ_3, find the three quadratic subfields of its splitting field, deduce that the image of $\bar\rho_{E,3}$ is $N_s(\mathbb{F}_3)$, and verify $a_p \equiv 0 \pmod 3$ for the primes $p \equiv 2 \pmod 3$ below 50. 13. For 11a1, check that $\psi_3 = 3x^4 - 4x^3 - 60x^2 - 237x - 21$ has discriminant $-3^3 \cdot 11^{10}$, so that $\mathbb{Q}(\sqrt{\operatorname{disc}\psi_3}) = \mathbb{Q}(\sqrt{-3})$, and compare the factorisations of ψ_3 modulo 19 and 29 with $a_{19} = a_{29} = 0$. 14. Let $\rho\colon G_{\mathbb{Q}} \to \mathrm{GL}_2(\mathbb{Z}_3)$ be continuous. Show that $\rho \bmod 9 \not\simeq (\rho \otimes \chi_3^{\mathrm{cyc}}) \bmod 9$, with χ_3^{cyc} the 3-adic cyclotomic character,, and deduce, with the description of $h_\ell(E)$ in Lemma lem:h0 (proved in Chapter 11), that $h_3(E) \le 1$ and $h_\ell(E) = 0$ for $\ell \ge 5$. 15. Using Chebotarev, compute the densities of the primes p for which $x^3 - 2$ has 0, 1, 3 roots modulo p, and compare with a count of the primes below 3000.

7.6 5. Dependencies and forward pointers

Results from other chapters used here: * Chapter 1 (`ch:groups`): groups, cyclic groups and their ℓ^k-th powers (1.3), index-two subgroups and the image of a kernel (1.6), group actions and orbits (3.1 to 3.3), $\mathrm{GL}_2(\mathbb{F}_3)$: the four lines, $\mathrm{PGL}_2(\mathbb{F}_3) \simeq S_4$, det = sign (5.3, 5.4), the subgroup and orbit tables and the three images (6.3 to 6.9), profinite groups, inverse limits, $\mathbb{Z}_\ell$, $\hat{\mathbb{Z}}$, $\mathrm{GL}_2(\mathbb{Z}_\ell)$ (7.1 to 7.5). Chapter 5 must use, not redo, these. * Chapter 2 (`ch:rings`): polynomial rings, Gauss's lemma, Eisenstein, $\mathbb{Z}/n$ and $(\mathbb{Z}/n)^\times$, finite fields (existence, uniqueness, cyclic multiplicative group), cyclotomic polynomials and $x^n - 1 = \prod_{d|n}\Phi_d$, Gauss sums ($g^2 = \ell^*$), Legendre and Kronecker symbols, q^*, quadratic reciprocity. * Chapter 4 (`ch:reps`): irreducibility and absolute irreducibility, Schur, Jordan-Holder and semisimplification, Brauer-Nesbitt (needed for an arbitrary group, or at least for finite groups plus the characteristic-zero trace criterion), Clifford theory for index two, induction and Frobenius reciprocity, the trace-zero lemma, Ribet's lemma on lattices (statement). * Background (PLAN Section 2): linear algebra, the fundamental theorem of algebra (complex analysis), elementary point-set topology (product topology; Tychonoff is quoted in Chapter 1).

Black boxes imported from later chapters (stated here with a forward \Cref, never used in a proof of an earlier item): Frobenius elements and their properties (Chapter 6), Kronecker-Weber (Chapter 6), $\det \rho_{E,\ell} = \chi_\ell$ and the four lines of $E[3]$ as the roots of ψ_3 (Chapter 9), the Galois representations of newforms (Chapter 15).

Forward pointers (chapters that use this one): * Chapter 6 (ch:ant): primitive element theorem (number fields), Galois theory of $\mathbb{Q}(\zeta_n)$ and $\mathbb{Q}(\sqrt{d})$, $\mathbb{Q}(\sqrt{\ell^*}) \subset \mathbb{Q}(\zeta_\ell)$, decomposition and inertia groups inside $\mathrm{Gal}(K/\mathbb{Q})$, Frobenius (made precise there), Chebotarev in number-field form (restated there with the Frobenius made precise), quadratic characters as Dirichlet characters. * Chapter 7 (ch:local): finite fields and $G_{\mathbb{F}_q} \simeq \hat{\mathbb{Z}}$, unramified extensions, Kummer theory over $\mathbb{Q}_p$ and $\mathbb{Q}_p^{\mathrm{ur}}$, the ℓ-adic Kummer cocycle and the tame character, Frob t Frob$^{-1} = t^q$, the translation theorem ("unramified base change does not change inertia"). * Chapter 9 (ch:ellcurves): Galois action on $E[m]$ and $T_\ell E$, the 3-division polynomial and its roots, quadratic twists by χ_d. * Chapter 10 (ch:reduction): the Tate curve, $E_{\mathfrak{q}}[\ell^t]$ generated by μ_{ℓ^t} and an ℓ^t-th root of $\mathfrak{q}$. * Chapter 11 (ch:galrep): continuity of $\rho_{E,\ell}$, irreducible versus absolutely irreducible via complex conjugation, restriction to $G_{\mathbb{Q}(\sqrt{\ell^*})}$, the proofs of Proposition prop:ns and Lemma lem:h0, the isomorphism class of $\bar{\rho}_{E,\ell}$ determined by the a_p mod ℓ. * Chapter 15 (ch:modgalrep): Lemma lem:old (Chebotarev and Brauer-Nesbitt), congruences between newforms, the optimal-level theorem's hypothesis at $\ell = 3$. * Chapters 16 and 17 (ch:cohomology, ch:galcoh): Hilbert 90 in cohomological form, $H^1(K, \mu_n) = K^\times / K^{\times n}$, the Kummer class in $H^1(K, \mathbb{Z}_\ell(1))$, continuous cohomology of G_K, $G_{\mathbb{R}} = \{1, c\}$. * Chapter 18 (ch:blochkato): ℓ-adic representations of $G_{\mathbb{Q}}$, Tate twists $V(n)$, $\mathbb{Z}_\ell(1)$. * Chapters 20, 21, 22: the hypothesis $\ell \notin S_f$, the ns3 test, the image labels.

7.7 6. Notes for the writer

Conventions to fix (Section 1, as convention environments): C1 to C7 of Part 1. In addition: * Composition of automorphisms is $(\sigma\tau)(x) = \sigma(\tau(x))$; groups act on the left; $\sigma(\zeta) = \zeta^{\chi(\sigma)}$. * $\mathbb{Z}_\ell(1)$ is written multiplicatively as $\varprojlim \mu_{\ell^k}$ and additively when it carries cocycles; say once which. * The quadratic character of conductor q is always χ_{q^*}, never χ_q (the paper's χ_q appears only when quoting the paper; decision D3), and never "the Legendre symbol mod q as a character of $\mathbb{Z}$" without the identification through Frobenius; $\chi_{\ell^*} = \bar{\chi}_\ell^{(\ell-1)/2}$ is not $\bar{\chi}_\ell$ unless $\ell = 3$. The paper itself slips on this once (Section 8 below, point 2). * "Density" means Dirichlet density unless stated; "almost all" means all but finitely many. * Use the paper's macros and the preamble's (\Gal, \Frob, \Qbar, \Fbar, \Zl, \GL, \SL, \PGL, \Ind, \Res, \Aut, \Hom, \tr, \Tr, \Nm, \disc, \sgn); Res for the resultant must not clash with the restriction \Res: write the resultant as $\mathrm{Res}_x(f, g)$ or $\mathrm{res}(f, g)$ and say so. * Labels: def:galois:..., thm:galois:... etc.; the line-up names above are ready to be used as the label stems.

Pitfalls: * Do not present the infinite fundamental theorem without the word "closed"; give the $\phi_p^{\mathbb{Z}}$ example. * Kummer theory needs $\mu_n \subset F$ and $\mathrm{char}\, F \nmid n$; $x^3 - 2$ over $\mathbb{Q}$ is the example of what fails without roots of unity (not Galois, group S_3 of the affine shape). In the paper's uses the base field is $\mathbb{Q}_q^{\mathrm{ur}}$ (or $\mathbb{F}_q$ for $k \le v_\ell(q-1)$), which contains the relevant roots of unity because $q \neq \ell$; say so in the paperlink. * Frobenius: everything in Section 9 that mentions primes of number fields is a black box from Chapter 6; label the items "stated" and make sure no proof in Sections 1 to 8 depends on them. The finite-field Frobenius (galois:finite-field-galois, galois:cyclotomic-mod-p) is proved here and is the honest preview. * The quartic-on-lines proposition needs an equivariant bijection between roots and lines; for ψ_3 this is the map $\pm P \mapsto x(P)$ from Chapter 9 (distinct lines have distinct x-coordinates because $x(P) = x(Q)$ iff $Q = \pm P$). ψ_3 has leading coefficient 3; irreducibility over $\mathbb{Q}$ is up to this constant, and Dedekind's theorem needs $p \neq 3$. * $N_s(\mathbb{F}_3)$ acts absolutely irreducibly on $\mathbb{F}_3^2$; only its determinant-one subgroup C_4 does not. The N_s image has two orbits on lines although $E[3]$ is irreducible: the factorisation pattern detects the image, not reducibility. Reducible $E[3]$ always gives a linear factor of ψ_3 (a rational line), patterns $1+3$ or $1+1+2$. * Brauer-Nesbitt: the paper's use is over finite fields and finite images, so the finite-group form from Chapter 4 suffices; the residual representations of f and h live over possibly different finite fields, compare over a common one. Absolute irreducibility of $\bar{\rho}_{E,\ell}$ (from complex conjugation) is what upgrades "isomorphic semisimplifications" to "isomorphic"; make this explicit (the paper says only "irreducible"). * Natural versus Dirichlet density: Chebotarev with natural density is true but deeper; only positivity

of the Dirichlet density is needed for galois:chebotarev-dense. Do not claim more than is cited. * The Artin-Schreier theorem on real closed fields (complex conjugations are the only torsion in $G_{\mathbb{Q}}$) and the Artin-Schreier theory of degree-p extensions share a name; say so. * The tower $\mathbb{Q} \subset \mathbb{Q}(\sqrt{2}) \subset \mathbb{Q}(2^{1/4})$ (normality not transitive) and $\mathbb{Q}(2^{1/3})$ (not normal) should both appear, since "the field cut out by" is always a Galois extension and readers confuse the two. * Typography (PLAN.md 7.2): no dash ligatures anywhere ("Artin-Schreier", "Kappe-Warren", "Brauer-Nesbitt" with plain hyphens; ranges as "1 to 8"); ASCII source; the orbit table as `tabular` with `booktabs`. * The theorem numbers for Lang and Dummit-Foote kept in Sections 1 to 3 of Part 2 (Lang V.1 Proposition 1.2, Lang VI.1 Theorems 1.1 and 1.8, Lang V.4 Theorem 4.6, Lang VI.4 Theorem 4.1, Lang V.3 Theorem 3.3, Dummit-Foote 14.2 Theorems 7 and 14) come from the first session and were not re-verified here; check them against the books or cite the section only. Every other reference in this outline is by chapter or section. Ireland-Rosen and Milne's "Fields and Galois Theory" are not in `refs.bib`: add them to `bib/ch05.bib` if cited.

Leave to other chapters: * Chapter 2: construction of $\mathbb{F}_q$, quadratic reciprocity and its proofs, Gauss sums, Dirichlet characters. * Chapter 4: all representation theory proper (Schur, Maschke, Brauer-Nesbitt, Clifford, induction); this chapter only translates to profinite groups and continuity. * Chapter 6: rings of integers, primes, decomposition and inertia groups, the definition and existence of Frobenius, Dedekind's factorisation theorem, Kronecker-Weber, the Chebotarev theorem for number fields restated with the Frobenius made precise. * Chapter 7: $\mathbb{Q}_p$, $\mathbb{Q}_p^{\mathrm{ur}}$, inertia, the tame character, Kummer theory over local fields, Hensel. * Chapter 9 and 11: $\det \rho_{E,\ell} = \chi_\ell$ (Weil pairing), the division polynomials, the proofs of Proposition prop:ns and Lemma lem:h0 as statements about elliptic curves, Zywina's list. * Chapters 16 and 17: group cohomology; Hilbert 90 and Kummer theory in the form $H^1(K, \mu_n) = K^\times / K^{\times n}$ (this chapter proves the field-theoretic forms only, so that Chapter 17 can cite them); the identification of the image of the Kummer class of the Tate parameter with the class of its unit part (end of Lemma lem:tamfactor). * Chapter 15: the Galois representations of newforms (used here only as a black box in galois:congruent-ap).

7.8 7. Verification log (computations done for this outline)

All scripts were short Python/sympy programs run in `/tmp/ch05`. * $\mathrm{GL}_2(\mathbb{F}_3)$: all 48 elements, all 55 subgroups in 16 conjugacy classes; for each class: order, surjectivity of det, presence of an element of determinant -1 and trace 0, fixed lines, orbits on the four lines, image in S_4 and whether it lies in A_4. Result: the table in galois:quartic-on-lines; $\det g$ equals the sign of the permutation of the lines for all 48 elements. * $\mathrm{disc}(x^3 - 2) = -108$; $\mathrm{disc}(x^3 - 3x + 1) = 81$ (irreducible); $\mathrm{disc}(\Phi_\ell) = (-1)^{(\ell-1)/2}\ell^{\ell-2}$ for $\ell = 3, 5, 7, 11, 13$; $\mathrm{disc}(x^n - a)$ for $n \le 6$; the resolvent cubic formula and $\mathrm{disc}(R_f) = \mathrm{disc}(f)$ symbolically; the general cubic discriminant on x^3+x^2+x+1 (-16). * $\Phi_n \bmod p$ factors into $\varphi(n)/f$ distinct irreducibles of degree $f = \mathrm{ord}_n(p)$ for all $3 \le n < 40$, $p < 60$, $p \nmid n$. * For $5 \le p < 2000$: $x^2 + 3$ has a root mod p iff $p \equiv 1 \pmod 3$; $x^2 + 1$ iff $p \equiv 1 \pmod 4$; for $\ell \in \{3, 5, 7, 11, 13\}$, ℓ^* is a square mod p iff p is a square mod ℓ. * ψ_3 of 11a1 ($[0, -1, 1, -10, -20]$) and 121a1 ($[1, 1, 1, -30, -76]$): irreducible, disc $= -3^3 \cdot 11^{10}$ and $-3^3 \cdot 11^4$; factorisation degrees mod $p < 40$ match the orbit sizes predicted from $X^2 - a_p X + p \bmod 3$ (with a_p by point counting). Resolvent cubic of ψ_3(11a1) irreducible. * Search over $y^2 = x^3 + Ax + B$, $|A|, |B| \le 30$, $AB \ne 0$: ψ_3 is a product of two quadratics only for $(A, B) = (-6, \mp 8)$ ($j = -1728$); for $(-6, -8)$ the factors have discriminants 24 and -8, the product is -3 times a square, the resolvent cubic has exactly one rational root, and $a_p \equiv 0 \pmod 3$ for all $p \equiv 2 \pmod 3$, $p \le 71$, with $a_5 = 3$. * $x^3 - 2$: proportions of primes $5 \le p < 3000$ with 0, 1, 3 roots: 0.325, 0.516, 0.159.

7.9 8. Points for important_notes.md

Points found while preparing this outline, in the area of Chapter 5 (Galois theory, Kummer theory, quadratic and cyclotomic characters, restriction to $G_{\mathbb{Q}(\sqrt{\ell^*})}$, the 3-division polynomial). The gap in the proof of Proposition prop:ns ("non-abelian and faithful") is already recorded from the Chapter 1 outline and is not repeated.

1. **Lemma lem:h0, last sentence of the statement and of the proof; Section sec:anomaly.** Problem: the paper proves finiteness of $H^0(\mathbb{Q}, \mathrm{ad}^0\, T(1) \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell)$ "for E without complex multiplication" by saying that $\rho \simeq \rho \otimes \chi_\ell$ would force $a_p = 0$ at the primes inert in $\mathbb{Q}(\sqrt{\ell^*})$. That argument confuses the cyclotomic character χ_ℓ with the quadratic character χ_{ℓ^*} ($\rho \simeq \rho \otimes \chi_\ell$ is impossible for every E, since it would give $p\,a_p = a_p$ for all good $p \neq \ell$). The true statement is much stronger and elementary: by the paper's own description of the length, $h_\ell(E) \geq k$ iff $\rho \bmod \ell^k \simeq (\rho \otimes \chi_\ell) \bmod \ell^k$; taking determinants, this forces $\chi_\ell^2 \equiv 1 \pmod{\ell^k}$ on $G_\mathbb{Q}$, and since $\chi_\ell(G_\mathbb{Q}) = \mathbb{Z}_\ell^\times$ this happens only for $\ell = 3$, $k = 1$ ($4 \not\equiv 1 \bmod 9$). Hence $h_\ell(E) = 0$ for $\ell \geq 5$ and $h_3(E) \leq 1$, with equality iff $\bar\rho_{E,3} \simeq \bar\rho_{E,3} \otimes \chi_{-3}$, for every E with $\bar\rho_{E,\ell}$ irreducible, CM or not. Fix: replace the finiteness sentence of the proof by this determinant argument (one line), drop "for E without complex multiplication" there, and in Section sec:anomaly replace "$h_3(E) \leq 1$ because some prime $p < 1223$ inert in $\mathbb{Q}(\sqrt{-3})$ has $9 \nmid a_p$" by "$h_3(E) \leq 1$ by Lemma lem:h0" (the numerical check is valid but no longer needed). The paper's identification of the length with "the largest k with $\rho \bmod \ell^k \simeq (\rho \otimes \chi_\ell) \bmod \ell^k$" is correct when $\bar\rho$ is absolutely irreducible (then the ℓ-torsion of H^0 is at most one-dimensional, H^0 is cyclic, and an element of exact order ℓ^k is an isomorphism mod ℓ^k by Nakayama). Sure: high (the determinant computation is elementary; the outline's galois:twist (c) and Exercise 14 carry it).

2. **Section 1.3 of the paper ("Beyond the hypotheses"), paragraph after Conjecture conj:euler.** Problem: "The term $h_\ell(E)$ ... is positive exactly when $\bar\rho \simeq \bar\rho \otimes \chi_{\ell^*}$". For $\ell \geq 5$ this is wrong: $\bar\rho \simeq \bar\rho \otimes \chi_{\ell^*}$ (the case where $\bar\rho$ is induced from $\mathbb{Q}(\sqrt{\ell^*})$, excluded by S_f) can occur, but $h_\ell(E) = 0$ always for $\ell \geq 5$ (Lemma lem:h0 itself says so). The criterion of Lemma lem:h0 is $\bar\rho \simeq \bar\rho \otimes \bar\chi_\ell$, and $\bar\chi_\ell = \chi_{\ell^*}$ only for $\ell = 3$. Likewise the opening sentence of that subsection ("a correction term when the representation is induced from $\mathbb{Q}(\sqrt{\ell^*})$") is right only for $\ell = 3$. Fix: "is positive exactly when $\bar\rho \simeq \bar\rho \otimes \bar\chi_\ell$, which forces $\ell = 3$ (where $\bar\chi_3 = \chi_{-3}$); for $\ell = 3$ this happens precisely when ..." and "a correction term when $\ell = 3$ and the representation is induced from $\mathbb{Q}(\sqrt{-3})$". Sure: high.

3. **Lemma lem:tamfactor, proof, "Finally c' is the image of κ in $H^1(G/I, \mathbb{Z}_\ell(1)^I)$".** Problem: there is no natural map $H^1(G, \mathbb{Z}_\ell(1)) \to H^1(G/I, \mathbb{Z}_\ell(1)^I)$ (inflation goes the other way, and κ is not inflated: its restriction to I is n times the tame character). The value $c' = \kappa(\mathrm{Frob})$ depends on the lift of Frob (by multiples of n') and on the cocycle (by multiples of $q-1$). Fix: write $q_E = q^n u$, so $\kappa = n\kappa_q + \kappa_u$ (galois:l-adic-kummer (b)); κ_u is unramified, i.e. inflated from the class of u in $H^1(G/I, \mathbb{Z}_\ell(1)) = \mathbb{Z}_\ell(1)/(q-1) \simeq \mathbb{F}_q^\times \otimes \mathbb{Z}_\ell$ (galois:kummer-finite-field (b), plus Hensel, Chapter 7); choosing the lift of Frob that fixes a compatible system of ℓ-power roots of q (possible because $\mathbb{Q}_q(q^{1/\ell^\infty})$ is totally ramified) gives $c' = \kappa_u(\mathrm{Frob})$, and in general c' is well defined modulo $(n', q-1)\mathbb{Z}_\ell$, which does not change $\min(v_\ell(n), v_\ell(q-1), v_\ell(c'))$. The conclusion of the lemma is unaffected. Sure: high that the phrase is imprecise; high that the conclusion stands. (The Chapter 16 outline, writer note 10, records the indeterminacy; the Kummer side is in this chapter and Chapter 17.)

4. **Lemma lem:tamagawa-exp, statement and proof.** Problem (expository): the proof calls the Tate parameter u, while Lemma lem:tamfactor uses u for its unit part $q_E q^{-n}$; and "inertia acts trivially on $E[\ell^t]$ exactly when u is an ℓ^t-th power in $\mathbb{Q}_q^{\mathrm{ur}}$ up to units" mixes two statements. Fix: call the Tate parameter q_E; say "by Kummer theory over $\mathbb{Q}_q^{\mathrm{ur}}$, which contains μ_{ℓ^t} as $q \neq \ell$, inertia acts trivially on $E[\ell^t]$ iff q_E is an ℓ^t-th power in $\mathbb{Q}_q^{\mathrm{ur}}$; as the units of its valuation ring are ℓ^t-th powers (Hensel), this holds iff $\ell^t \mid v_q(q_E) = n_q$". Sure: high (no mathematical error, only precision).

5. **Proposition prop:ns, hypothesis "without complex multiplication".** Problem (harmless): the proof does not use it. The list $\mathrm{GL}_2(\mathbb{F}_3)$, $N_{ns}(\mathbb{F}_3)$, $N_s(\mathbb{F}_3)$ follows from surjectivity of $\det \bar\rho_{E,3} = \chi_{-3}$, the irreducibility of $E[3]$ and the presence of complex conjugation (an element of determinant -1 and trace 0), by pure group theory (Chapter 1, item 6.7), with no appeal to Zywina's classification, and the rest of the proof (Frobenius reciprocity, orbits on lines) is also CM-free. Fix: optional; either drop the hypothesis or note that it is not needed. Sure: high for the mathematics; whether the author wants the extra generality is a choice.

6. **Lemma lem:old (case $p \nmid \ell N$) and the proof of Lemma lem:h0: "$\bar{\rho}$ being irreducible".** Problem (minor gap): Chebotarev plus Brauer-Nesbitt give only $\bar{\rho}_h^{ss} \simeq \bar{\rho}^{ss}$ over a common finite field; to conclude $\bar{\rho}_h \simeq \bar{\rho}$ one needs $\bar{\rho}$ absolutely irreducible, and the equivalence "$\bar{\rho} \simeq \bar{\rho} \otimes \chi_{-3}$ iff induced from $\mathbb{Q}(\sqrt{-3})$ iff not absolutely irreducible there" is Clifford theory for an absolutely irreducible $\bar{\rho}$. Both are fine for $\bar{\rho}_{E,\ell}$ because irreducible implies absolutely irreducible in the presence of complex conjugation (ℓ odd). Fix: add "(hence absolutely irreducible, complex conjugation having distinct eigenvalues ± 1)" after "irreducible" in both places, and "over a common finite extension of $\mathbb{F}_\ell$" after "$\bar{\rho}_h \simeq \bar{\rho}$". Sure: high.

Chapter 8

Line-up for Chapter 06: Algebraic number theory (`ch:ant`, `chapters/ch06_ant.tex`)

Stage 1 output (core results in teaching order). Target length 1,820 lines of LaTeX excluding the card (Part 8 below). Rigour tags: **P** = proved here, **S** = proof sketched here, **R** = stated; reference. Frobenius is *arithmetic* throughout (Convention 3.7 below, the same as Chapter 5's convention C2): $\mathrm{Frob}_P(x) \equiv x^{N\mathfrak{p}} \pmod P$; the *geometric* Frobenius is its inverse, and local L-factors are written with the geometric Frobenius (3.7). Chapter 5's conventions C1 (one $\overline{\mathbb{Q}} \subset \mathbb{C}$), C3 (quadratic characters χ_d, χ_{q^*}, never χ_q: decision D3) and C4 (χ_ℓ, $\bar\chi_\ell$, $\chi_{\ell^*} = \bar\chi_\ell^{(\ell-1)/2}$) are inherited without change.

Item format: number, label-ready name, kind, rigour tag; the precise statement; reference (book and theorem number only where sure, otherwise book and chapter); the paper's use, by its `\label` names, or "background". Small computations (splitting of specific primes, discriminants, factorisations in $\mathbb{Z}[i]$ and $\mathbb{Z}[\zeta_3]$, Dedekind-Kummer examples, a_p of CM curves) were checked by Python scripts in `/tmp/ch06` (verification log, Part 12).

8.1 0. What the chapter must deliver (checked against PLAN.md Section 4 entry and Section 6 map)

Scope items of the PLAN entry, each mapped to an item below (full table in Part 10): number fields, rings of integers, integral closure (1.1-1.4); Dedekind domains and unique factorisation of ideals (2.1-2.2); norms of ideals (2.3); splitting of primes, e, f, g, $\sum e_i f_i = n$, ramified/inert/split (3.1-3.3, 3.5); discriminant and ramification (1.4, 3.4); quadratic fields in full, Kronecker symbol, the quadratic character of $\mathbb{Q}(\sqrt{d})$ as a Dirichlet character (4.1-4.4, 5.7, 5.8); cyclotomic fields, splitting governed by $p \bmod n$, $\mathbb{Q}(\zeta_3) = \mathbb{Q}(\sqrt{-3})$, $\mathbb{Q}(\sqrt{\ell^*}) \subset \mathbb{Q}(\zeta_\ell)$ (5.1-5.4); decomposition and inertia groups, Frob_P, its conjugacy class, arithmetic versus geometric Frobenius, Frobenius in abelian extensions (3.5-3.12, 5.6); Artin map, Kronecker-Weber, Dirichlet characters versus one-dimensional representations of $G_{\mathbb{Q}}$ (5.5-5.8); class group and units (1.6, 2.5, 2.6, 4.5); $\mathbb{Z}[\zeta_3]$, $\mathbb{Z}[i]$ (Euclidean), primes above 3, $\mathbb{Z}_3[\zeta_3]$ a DVR with uniformiser $\pi = 1 - \zeta_3$ (4.5-4.9); primes $\lambda \mid \ell$ in the coefficient field of a newform, reduction mod λ (7.1-7.5); densities and Chebotarev in number-field form (6.1-6.4); primes inert in $\mathbb{Q}(\sqrt{\ell^*})$ and CM (5.3, 6.4, 6.5).

Paper hooks of the PLAN entry: Lemma lem:IV ($\mathbb{Z}_3[\zeta_3]$ a DVR, T free of rank one over it, $E[3] = \mathbb{Z}_3[\zeta_3]/(\pi^2)$ with σ unipotent: 4.6, 4.7, 4.8); Lemma lem:euler (Frobenius centralises or inverts σ according to $q \bmod 3$: 3.11, 3.12; "$\mathbb{Z}_\ell[\zeta_3] = \mathrm{End}_{I_q}(V)$", the line spanned by $\sqrt{-3}$, the action of Frobenius on it by $\chi_{-3}(q)$, and the $\mathbb{Q}_\ell(i)$ analogue for the types III, III* with $\chi_{-4}(q)$: 4.9, 4.10); Lemma lem:h0 ($a_p = 0$ at primes inert in $\mathbb{Q}(\sqrt{\ell^*})$ forces CM, in the corrected reading of important_notes.md point 5: 5.3, 6.4, 6.5);

Lemma lem:crit and Lemma lem:old (rings of integers $\mathcal{O}$, primes $\lambda \mid \ell$, "after enlarging $\mathcal{O}$ and λ", "a prime above λ in $\mathcal{O}[\alpha_p]$", "a maximal ideal of residue characteristic ℓ": 7.1-7.4); the twist by $\mathbb{Q}(\sqrt{q^*})$ of Section 1.4, Lemma lem:twist and the character χ_S of Theorems thm:twist and thm:quant (4.3, 5.7, 5.8); χ_{-3}, χ_{-4} of $\mathbb{Q}(\sqrt{-3})$ and $\mathbb{Q}(i)$ (4.2, 5.8). Section 6 map rows pointing at Chapter 6: "$\chi_{-3}, \chi_{-4}, \chi_{q^*}, q^*, \ell^*$, Kronecker symbols" (4.1-4.3, 5.3, 5.7, 5.8); "image of inertia on $T_\ell E$ at additive primes, unipotent elements, $\mathbb{Z}_3[\zeta_3]$" (3.11, 3.12, 4.6-4.10).

What Chapter 6 supplies to Chapter 5 (which states these as black boxes): the definition and existence of Frob_P and its properties (3.8, 3.10), Dedekind's cycle-type theorem (3.9), $\chi_d(\mathrm{Frob}_p) = (D/p)$ (4.2), Kronecker-Weber (5.5), the Chebotarev theorem with the Frobenius made precise (6.2). Chapter 6 does not redo Chapter 5's Galois theory of $\mathbb{Q}(\zeta_n)$ and $\mathbb{Q}(\sqrt{d})$, its Chebotarev-Brauer-Nesbitt corollaries, or its index-two restriction criterion; it cites them.

8.2 1. Line-up, Section 1: number fields, integers, discriminant, units

8.2.1 1.1 def:ant:numberfield (definition) P

A *number field* is a field K with $[K : \mathbb{Q}] = n < \infty$. By the primitive element theorem (ch:galois) $K = \mathbb{Q}(\theta)$, and there are exactly n field embeddings $\sigma_1, \dots, \sigma_n \colon K \to \mathbb{C}$, namely $\theta \mapsto$ each root of the minimal polynomial of θ. Exactly r_1 of them have image in $\mathbb{R}$ and the remaining $2r_2$ come in complex-conjugate pairs; $n = r_1 + 2r_2$. K is *totally real* if $r_2 = 0$, *imaginary quadratic* if $n = 2$, $r_1 = 0$. Reference: Neukirch I.5; Marcus Ch. 2. Paper: background.

8.2.2 1.2 prop:ant:integral (definition and proposition) P

$\alpha \in K$ is an *algebraic integer* (integral over $\mathbb{Z}$) if it is a root of a monic polynomial in $\mathbb{Z}[X]$. Equivalent: (i) the minimal polynomial of α over $\mathbb{Q}$ lies in $\mathbb{Z}[X]$; (ii) $\mathbb{Z}[\alpha]$ is a finitely generated $\mathbb{Z}$-module; (iii) α lies in a subring of K that is finitely generated as a $\mathbb{Z}$-module. The set $\mathcal{O}_K$ of algebraic integers of K is a subring (the *ring of integers*), $\mathcal{O}_K \cap \mathbb{Q} = \mathbb{Z}$, $\mathrm{Frac}(\mathcal{O}_K) = K$, and for every $\alpha \in K$ there is $m \in \mathbb{Z}_{>0}$ with $m\alpha \in \mathcal{O}_K$. $\mathcal{O}_K$ is integrally closed in K. (Gauss's lemma from ch:rings gives (i).) Reference: Neukirch I.2; Marcus Ch. 2. Paper: Lemma lem:crit, Lemma lem:old (the ring $\mathcal{O}$ of a number field), background.

8.2.3 1.3 def:ant:tracenorm (definition and proposition) P

For $\alpha \in K$, $\mathrm{Tr}_{K/\mathbb{Q}}(\alpha) = \sum_i \sigma_i(\alpha)$ and $N_{K/\mathbb{Q}}(\alpha) = \prod_i \sigma_i(\alpha)$ are the trace and determinant of multiplication by α on the $\mathbb{Q}$-vector space K; so they lie in $\mathbb{Q}$, Tr is $\mathbb{Q}$-linear, N is multiplicative, and $N(\alpha) = 0$ iff $\alpha = 0$. If $K = \mathbb{Q}(\alpha)$ with minimal polynomial $X^n + c_{n-1}X^{n-1} + \cdots + c_0$ then $\mathrm{Tr}(\alpha) = -c_{n-1}$, $N(\alpha) = (-1)^n c_0$. For $\alpha \in \mathcal{O}_K$: $\mathrm{Tr}(\alpha), N(\alpha) \in \mathbb{Z}$, and $\alpha \in \mathcal{O}_K^\times$ iff $N(\alpha) = \pm 1$. In a tower $L \supseteq K \supseteq \mathbb{Q}$: $\mathrm{Tr}_{L/\mathbb{Q}} = \mathrm{Tr}_{K/\mathbb{Q}} \circ \mathrm{Tr}_{L/K}$, $N_{L/\mathbb{Q}} = N_{K/\mathbb{Q}} \circ N_{L/K}$. (RHB aside: trace and determinant of a linear map.) Reference: Neukirch I.2.5-2.6 (cite as I.2); Lang, Algebra VI.5. Paper: background (norms of π in $\mathbb{Z}[\zeta_3]$, $N(\sqrt{-3}) = 3$).

8.2.4 1.4 thm:ant:integralbasis (theorem) P

The trace form $(x, y) \mapsto \mathrm{Tr}_{K/\mathbb{Q}}(xy)$ is a nondegenerate symmetric bilinear form on K. $\mathcal{O}_K$ is a free $\mathbb{Z}$-module of rank n (an *integral basis* $\omega_1, \dots, \omega_n$ exists), and for any $\mathbb{Z}$-basis the integer $\mathrm{disc}(\omega_1, \dots, \omega_n) = \det(\mathrm{Tr}(\omega_i\omega_j)) = \det(\sigma_i(\omega_j))^2$ is independent of the basis: it is the *discriminant* d_K of K. If $\alpha_1, \dots, \alpha_n \in \mathcal{O}_K$ form a $\mathbb{Q}$-basis of K and $M = \mathbb{Z}\alpha_1 + \cdots + \mathbb{Z}\alpha_n$, then $\mathrm{disc}(\alpha_1, \dots, \alpha_n) = [\mathcal{O}_K : M]^2 \, d_K$; in particular if $\mathrm{disc}(\alpha)$ is squarefree then $M = \mathcal{O}_K$. For $K = \mathbb{Q}(\theta)$ with $\theta \in \mathcal{O}_K$ of minimal polynomial g: $\mathrm{disc}(1, \theta, \dots, \theta^{n-1}) = \mathrm{disc}(g) = \prod_{i<j}(\theta_i - \theta_j)^2$. (RHB aside: a lattice with a Gram matrix; the trace form is the Gram matrix.) Reference: Neukirch I.2.9-2.11 (cite as I.2); Marcus Ch. 2, Thm 9 and Thm 14 (check numbers; else cite the chapter). Paper: background (discriminants of $\mathbb{Q}(\sqrt{-3})$, $\mathbb{Q}(i)$, $\mathbb{Q}(\sqrt{q^*})$).

8.2.5 1.5 prop:ant:quadraticintegers (proposition) P

Let $d \in \mathbb{Z}$ be squarefree, $d \neq 0, 1$, $K = \mathbb{Q}(\sqrt{d})$. If $d \equiv 2, 3 \pmod 4$ then $\mathcal{O}_K = \mathbb{Z}[\sqrt{d}]$ and $d_K = 4d$; if $d \equiv 1 \pmod 4$ then $\mathcal{O}_K = \mathbb{Z}[(1+\sqrt{d})/2]$ and $d_K = d$. In both cases d_K is the *fundamental discriminant* D of K and $K = \mathbb{Q}(\sqrt{D})$; a fundamental discriminant is an integer $D \equiv 1 \pmod 4$ squarefree, or $D = 4m$ with $m \equiv 2, 3 \pmod 4$ squarefree. Examples: $\mathbb{Q}(i)$: $\mathcal{O} = \mathbb{Z}[i]$, $D = -4$. $\mathbb{Q}(\sqrt{-3})$: $\mathcal{O} = \mathbb{Z}[\zeta_3]$ with $\zeta_3 = (-1+\sqrt{-3})/2$, $D = -3$; note $\mathbb{Z}[\sqrt{-3}] \subsetneq \mathbb{Z}[\zeta_3]$ has index 2. $\mathbb{Q}(\sqrt{5})$: $\mathcal{O} = \mathbb{Z}[(1+\sqrt{5})/2]$, $D = 5$. $\mathbb{Q}(\sqrt{-7})$, $\mathbb{Q}(\sqrt{-11})$: $D = -7, -11$. The norm form: $N(a+b\zeta_3) = a^2 - ab + b^2$, $N(a+bi) = a^2 + b^2$. Reference: Marcus Ch. 2, Cor. to Thm 1 (cite chapter); Neukirch I.2 (Example). Paper: χ_{-3}, χ_{-4}, $\mathbb{Q}(\sqrt{q^*})$ (Section 1, 1.4), Lemma lem:IV.

8.2.6 1.6 thm:ant:units (theorem) R, imaginary quadratic case P

(Dirichlet's unit theorem.) $\mathcal{O}_K^\times \simeq \mu(K) \times \mathbb{Z}^{r_1+r_2-1}$, where $\mu(K)$, the roots of unity in K, is finite cyclic. For K imaginary quadratic, $\mathcal{O}_K^\times = \mu(K)$, which is $\{\pm 1\}$ except $\mu(\mathbb{Q}(i)) = \mu_4 = \{\pm 1, \pm i\}$ and $\mu(\mathbb{Q}(\sqrt{-3})) = \mu_6 = \{\pm 1, \pm\zeta_3, \pm\zeta_3^2\}$ (proved via $N(\alpha) = 1$ and the norm forms of 1.5). Example: $\mathbb{Z}[\sqrt{2}]^\times = \pm(1+\sqrt{2})^{\mathbb{Z}}$. Reference: Neukirch I.7.4; Marcus Ch. 5, Thm 38 (check; else chapter). Paper: background (units of $\mathbb{Z}[\zeta_3]$ in $3 = -\zeta_3^2\pi^2$, "$\pi^2 = -3$ up to a unit").

8.3 2. Line-up, Section 2: Dedekind domains, ideals, norms, class groups

8.3.1 2.1 thm:ant:dedekind (definition and theorem) P

A *Dedekind domain* is a Noetherian integrally closed integral domain, not a field, in which every nonzero prime ideal is maximal (some books allow fields; we exclude them so that the criterion below is exact). Theorem: $\mathcal{O}_K$ is a Dedekind domain (Noetherian since free of finite rank over $\mathbb{Z}$; integrally closed by 1.2; $\mathcal{O}_K/\mathfrak{p}$ is a finite integral domain, hence a field, because $\mathfrak{p} \cap \mathbb{Z} = p\mathbb{Z} \neq 0$). A Noetherian domain R is Dedekind iff R is not a field and every localisation $R_\mathfrak{p}$ at a nonzero prime is a discrete valuation ring (ch:modules supplies DVRs and localisation); PIDs that are not fields are Dedekind. Reference: Neukirch I.3.1, I.11.5 (cite I.3 and I.11); Atiyah-Macdonald Ch. 9. Paper: background for every "$\mathcal{O}$" and every λ.

8.3.2 2.2 thm:ant:idealfactorisation (theorem) P

Let R be a Dedekind domain with fraction field K. (i) Every nonzero proper ideal $\mathfrak{a}$ is a product $\mathfrak{a} = \mathfrak{p}_1^{a_1} \cdots \mathfrak{p}_r^{a_r}$ of distinct nonzero prime ideals with $a_i \geq 1$, uniquely up to order. (ii) The nonzero *fractional ideals* (finitely generated R-submodules $\mathfrak{a} \neq 0$ of K; equivalently $x\mathfrak{b}$ with $x \in K^\times$, $\mathfrak{b}$ an ideal) form an abelian group J_K under multiplication, free on the nonzero primes, with $\mathfrak{a}^{-1} = \{x \in K : x\mathfrak{a} \subseteq R\}$. (iii) For ideals, $\mathfrak{a} \mid \mathfrak{b}$ (i.e. $\mathfrak{b} = \mathfrak{a}\mathfrak{c}$) iff $\mathfrak{a} \supseteq \mathfrak{b}$; gcd is the sum, lcm the intersection, and $v_\mathfrak{p}(\mathfrak{a})$ denotes the exponent of $\mathfrak{p}$. (iv) For $\alpha \in K^\times$, $v_\mathfrak{p}(\alpha) := v_\mathfrak{p}(\alpha R)$ is the discrete valuation of $R_\mathfrak{p}$. Proof via "every nonzero ideal contains a product of primes" and the invertibility $\mathfrak{p}\mathfrak{p}^{-1} = R$ (Neukirch's proof), tying to ch:modules DVRs. Reference: Neukirch I.3.3 and I.3.8 (cite I.3); Marcus Ch. 3, Thm 16; Atiyah-Macdonald 9.3. Paper: background (every ideal statement: $\lambda \mid \ell$, v_λ, the ideal η in $\mathbb{Z}_\ell$).

8.3.3 2.3 prop:ant:idealnorm (definition and proposition) P

For a nonzero ideal $\mathfrak{a} \subseteq \mathcal{O}_K$ the *norm* is $N(\mathfrak{a}) = \#(\mathcal{O}_K/\mathfrak{a})$, finite. (i) $N(\mathfrak{a}\mathfrak{b}) = N(\mathfrak{a})N(\mathfrak{b})$. (ii) $N(\alpha\mathcal{O}_K) = |N_{K/\mathbb{Q}}(\alpha)|$. (iii) If $\mathfrak{p} \cap \mathbb{Z} = p\mathbb{Z}$ then $\mathcal{O}_K/\mathfrak{p} \simeq \mathbb{F}_{p^f}$ for $f = f(\mathfrak{p}|p)$ and $N(\mathfrak{p}) = p^f$. (iv) An ideal of prime norm is prime; if $N(\mathfrak{a})$ is prime to m then so is $\mathfrak{a}$ to $m\mathcal{O}_K$. (v) disc of a $\mathbb{Z}$-basis of $\mathfrak{a}$ equals $N(\mathfrak{a})^2 d_K$. Reference: Neukirch I.6.1-6.2 (cite I.6); Marcus Ch. 3, Thm 22. Paper: background (norms of π, of $\sqrt{-3}$, residue fields $\mathcal{O}/\lambda$ finite of characteristic ℓ in Lemma lem:crit).

8.3.4 2.4 prop:ant:crt (proposition) P

(i) (Chinese remainder theorem.) If $\mathfrak{a}_1, \dots, \mathfrak{a}_r$ are pairwise coprime ideals ($\mathfrak{a}_i + \mathfrak{a}_j = R$) of a commutative ring then $R/\mathfrak{a}_1 \cdots \mathfrak{a}_r \simeq \prod R/\mathfrak{a}_i$; in a Dedekind domain distinct primes are coprime, so $R/\mathfrak{a} \simeq \prod_i R/\mathfrak{p}_i^{a_i}$, and $R/\mathfrak{p}^a$ has length a with $\mathfrak{p}^k/\mathfrak{p}^{k+1} \simeq R/\mathfrak{p}$. (ii) In a Dedekind domain every nonzero ideal is generated by two elements, one of which may be chosen arbitrarily (nonzero) in it. (iii) A Dedekind domain with finitely many primes is a PID; a Dedekind domain is a UFD iff it is a PID. (iv) For a nonzero prime $\mathfrak{p}$ of $\mathcal{O}_K$ the localisation $\mathcal{O}_{K,\mathfrak{p}}$ is a DVR with maximal ideal $\mathfrak{p}\mathcal{O}_{K,\mathfrak{p}}$, residue field $\mathcal{O}_K/\mathfrak{p}$ and valuation $v_{\mathfrak{p}}$. Reference: Neukirch I.3.6, I.3.9, I.11 (cite I.3); Marcus Ch. 3 (Thm 17, Cor. 2 to Thm 16: check). Paper: background (the product decomposition $\mathcal{O} \otimes \mathbb{Z}_\ell$ in 7.4; $\mathbb{Z}_3[\zeta_3]/(\pi^2)$).

8.3.5 2.5 thm:ant:classgroup (definition and theorem) R

$\mathrm{Cl}(K) = J_K/P_K$, P_K the principal fractional ideals; $h_K = \#\,\mathrm{Cl}(K)$ is the *class number*; $h_K = 1$ iff $\mathcal{O}_K$ is a PID iff it is a UFD. Theorem (Minkowski): $\mathrm{Cl}(K)$ is finite; every ideal class contains an integral ideal of norm at most $M_K = (4/\pi)^{r_2}(n!/n^n)|d_K|^{1/2}$; and $|d_K| > 1$ for $K \neq \mathbb{Q}$, so every number field other than $\mathbb{Q}$ has a ramified prime (with 3.4). Examples: $h = 1$ for $\mathbb{Q}(i)$, $\mathbb{Q}(\sqrt{-3})$, $\mathbb{Q}(\sqrt{-7})$, $\mathbb{Q}(\sqrt{2})$, $\mathbb{Q}(\sqrt{5})$ (Minkowski bound < 2: 1.27, 1.10, 1.68, 1.41, 1.12; also Euclidean for $\mathbb{Z}[i]$, $\mathbb{Z}[\zeta_3]$ by ch:rings) and for $\mathbb{Q}(\sqrt{-11})$ (bound $2.11 > 2$, but 2 is inert since $-11 \equiv 5 \pmod 8$, so there is no ideal of norm 2); $h(\mathbb{Q}(\sqrt{-5})) = 2$; $h(\mathbb{Q}(\sqrt{-23})) = 3$ (stated). (RHB aside: $\mathcal{O}_K$ embedded by (σ_i) is a lattice in $\mathbb{R}^{r_1} \times \mathbb{C}^{r_2}$ of covolume $2^{-r_2}|d_K|^{1/2}$; Minkowski's theorem on lattice points is the tool.) Reference: Neukirch I.6.3 and I.5.3 (cite I.5-I.6), III.2.17 for $|d_K| > 1$ (cite III.2); Marcus Ch. 5. Paper: background (why $\mathcal{O}$ need not be a PID; nothing in the paper uses class groups).

8.3.6 2.6 ex:ant:sqrtminus5 (example) P

In $\mathbb{Z}[\sqrt{-5}]$ (the ring of integers of $\mathbb{Q}(\sqrt{-5})$, $D = -20$): $6 = 2 \cdot 3 = (1 + \sqrt{-5})(1 - \sqrt{-5})$ are two factorisations into irreducibles, none of which is prime; the ideal factorisation is $(2) = \mathfrak{p}_2^2$, $(3) = \mathfrak{p}_3\bar{\mathfrak{p}}_3$, $(1 + \sqrt{-5}) = \mathfrak{p}_2\mathfrak{p}_3$, $(1 - \sqrt{-5}) = \mathfrak{p}_2\bar{\mathfrak{p}}_3$ with $\mathfrak{p}_2 = (2, 1 + \sqrt{-5})$, $\mathfrak{p}_3 = (3, 1 + \sqrt{-5})$, $\bar{\mathfrak{p}}_3 = (3, 1 - \sqrt{-5})$, all of them non-principal, of norms $2, 3, 3$. Computed by Dedekind-Kummer (3.2) from $X^2 + 5$ modulo 2 and 3. Paper: background.

8.4 3. Line-up, Section 3: splitting of primes, ramification, decomposition and inertia groups, Frobenius

8.4.1 3.1 thm:ant:efg (definition and theorem) P

Let L/K be an extension of number fields of degree n and $\mathfrak{p}$ a nonzero prime of $\mathcal{O}_K$. In $\mathcal{O}_L$ write $\mathfrak{p}\mathcal{O}_L = \mathfrak{P}_1^{e_1} \cdots \mathfrak{P}_g^{e_g}$ with distinct primes $\mathfrak{P}_i$ (2.2). The $\mathfrak{P}_i$ are exactly the primes of $\mathcal{O}_L$ with $\mathfrak{P}_i \cap \mathcal{O}_K = \mathfrak{p}$ ("$\mathfrak{P}_i$ lies above $\mathfrak{p}$", $\mathfrak{P}_i \mid \mathfrak{p}$); $e_i = e(\mathfrak{P}_i|\mathfrak{p})$ is the *ramification index* and $f_i = f(\mathfrak{P}_i|\mathfrak{p}) = [\mathcal{O}_L/\mathfrak{P}_i : \mathcal{O}_K/\mathfrak{p}]$ the *residue degree*. Then

$$\sum_{i=1}^{g} e_i f_i = n.$$

Terminology: $\mathfrak{p}$ is *ramified* in L if some $e_i > 1$, *unramified* otherwise; *totally ramified* if $g = 1$, $e_1 = n$; *inert* if $g = 1$, $e_1 = 1$, $f_1 = n$; *split completely* if $g = n$ (so all $e_i = f_i = 1$); for L/K quadratic every $\mathfrak{p}$ is exactly one of ramified, inert, split. Multiplicativity in towers $K \subset M \subset L$: $e(\mathfrak{P}|\mathfrak{p}) = e(\mathfrak{P}|\mathfrak{P} \cap M)\, e(\mathfrak{P} \cap M|\mathfrak{p})$ and the same for f. Proof: $\mathcal{O}_L/\mathfrak{p}\mathcal{O}_L \simeq \prod_i \mathcal{O}_L/\mathfrak{P}_i^{e_i}$ (2.4) has dimension $\sum e_i f_i$ over $\mathcal{O}_K/\mathfrak{p}$ (filtration by $\mathfrak{P}_i^k/\mathfrak{P}_i^{k+1} \simeq \mathcal{O}_L/\mathfrak{P}_i$), and dimension n because $\mathcal{O}_L \otimes_{\mathcal{O}_K} \mathcal{O}_{K,\mathfrak{p}}$ is free of rank n over the DVR (hence PID) $\mathcal{O}_{K,\mathfrak{p}}$; for $K = \mathbb{Q}$ this is just "$\mathcal{O}_L \simeq \mathbb{Z}^n$" (1.4). Reference: Neukirch I.8 (Proposition 8.2); Marcus Ch. 3. Paper: background (every "prime $\lambda \mid \ell$", "prime inert in $\mathbb{Q}(\sqrt{\ell^*})$").

8.4.2 3.2 thm:ant:dedekindkummer (theorem, Dedekind-Kummer) P

Let K be a number field, $\theta \in \mathcal{O}_K$ with $K = \mathbb{Q}(\theta)$, $g \in \mathbb{Z}[X]$ the (monic) minimal polynomial of θ, and p a prime with $p \nmid [\mathcal{O}_K : \mathbb{Z}[\theta]]$ (for instance $p^2 \nmid \operatorname{disc}(g)$, by 1.4). Let $\bar{g} = \bar{g}_1^{e_1} \cdots \bar{g}_r^{e_r}$ be the factorisation of $\bar{g} = g \bmod p$ into distinct monic irreducibles in $\mathbb{F}_p[X]$, and let $g_i \in \mathbb{Z}[X]$ be monic lifts of $\bar{g}_i$. Then $\mathfrak{P}_i = (p, g_i(\theta)) = p\mathcal{O}_K + g_i(\theta)\mathcal{O}_K$ are distinct primes, $f(\mathfrak{P}_i|p) = \deg \bar{g}_i$, and $p\mathcal{O}_K = \mathfrak{P}_1^{e_1} \cdots \mathfrak{P}_r^{e_r}$. The same holds verbatim for a relative extension $L = K(\theta)$, $\theta \in \mathcal{O}_L$, and a prime $\mathfrak{p}$ of $\mathcal{O}_K$ coprime to the conductor of $\mathcal{O}_K[\theta]$ (stated). Proof (for $K/\mathbb{Q}$): $\mathbb{Z}[\theta]/p\mathbb{Z}[\theta] \simeq \mathbb{F}_p[X]/(\bar{g})$, and when $p \nmid [\mathcal{O}_K : \mathbb{Z}[\theta]]$ the inclusion induces $\mathbb{Z}[\theta]/p\mathbb{Z}[\theta] \simeq \mathcal{O}_K/p\mathcal{O}_K$; then the Chinese remainder theorem in $\mathbb{F}_p[X]$ and the bijection between ideals of $\mathcal{O}_K$ containing p and ideals of $\mathcal{O}_K/p$. Warning: the hypothesis on the index cannot be dropped (3.3(d)). Reference: Neukirch I.8 (Proposition 8.3); Marcus Ch. 3 (Theorem 27). Paper: background (all splitting computations; proof of 4.1 and 5.2).

8.4.3 3.3 ex:ant:dkexamples (worked examples) P

(a) $\mathbb{Q}(i)$, $g = X^2 + 1$, index 1: mod 2, $(X+1)^2$, so $2\mathbb{Z}[i] = (1+i)^2$ and indeed $2 = -i(1+i)^2$; mod 5, $(X-2)(X+2)$, so $5 = (5, i-2)(5, i+2) = (2+i)(2-i)$; mod 3 irreducible, so 3 is inert.

(b) $\mathbb{Q}(\zeta_3)$, $g = X^2 + X + 1$, index 1: mod 3, $(X-1)^2$, so $3\mathbb{Z}[\zeta_3] = (3, \zeta_3 - 1)^2 = (1-\zeta_3)^2$; mod 7, $(X-2)(X-4)$, so $7 = (7, \zeta_3 - 2)(7, \zeta_3 - 4)$, with generators $2 - \zeta_3$ and $2 - \zeta_3^2$ ($(2-\zeta_3)(2-\zeta_3^2) = 7$); mod 2 and mod 5, irreducible, so 2 and 5 are inert (residue fields $\mathbb{F}_4$, $\mathbb{F}_{25}$). (c) $\mathbb{Q}(\sqrt{-5})$: see 2.6 ($X^2 + 5 \equiv (X+1)^2 \bmod 2$, $\equiv (X-1)(X+1) \bmod 3$).

(c) The index matters: $\mathbb{Q}(\sqrt{-7})$ with $\theta = \sqrt{-7}$ ($[\mathcal{O}_K : \mathbb{Z}[\sqrt{-7}]] = 2$): $X^2 + 7 \equiv (X+1)^2 \bmod 2$ would wrongly suggest that 2 ramifies; with $\theta = (1+\sqrt{-7})/2$, $g = X^2 - X + 2 \equiv X(X-1) \bmod 2$, so 2 splits: $2 = \frac{1+\sqrt{-7}}{2} \cdot \frac{1-\sqrt{-7}}{2}$; and $2 \nmid d_K = -7$.

(d) A non-Galois cubic: $K = \mathbb{Q}(2^{1/3})$, $\mathcal{O}_K = \mathbb{Z}[2^{1/3}]$ (stated; $d_K = \operatorname{disc}(X^3 - 2) = -108 = -2^2 \cdot 3^3$): $X^3 - 2 \equiv X^3 \bmod 2$ and $\equiv (X+1)^3 \bmod 3$, so 2 and 3 are totally ramified; mod 5, $X^3 - 2 \equiv (X-3)(X^2+3X+4)$ with the quadratic irreducible, so $5\mathcal{O}_K = \mathfrak{P}_1\mathfrak{P}_2$ with $f = 1, 2$ (unequal residue degrees: impossible in a Galois extension, 3.5); mod 7 irreducible (inert); mod 31 three linear factors (split completely). All factorisations checked by computer (Part 12). Paper: background.

8.4.4 3.4 thm:ant:discram (theorem, Dedekind's discriminant theorem) P

A prime p ramifies in the number field K if and only if $p \mid d_K$. In particular only finitely many primes ramify in K, and (with Minkowski, 2.5) at least one prime ramifies in every $K \neq \mathbb{Q}$. Proof: $p \mid d_K$ iff the trace form of the $\mathbb{F}_p$-algebra $A = \mathcal{O}_K/p\mathcal{O}_K$ is degenerate (reduce an integral basis); $A \simeq \prod_i \mathcal{O}_K/\mathfrak{P}_i^{e_i}$ (2.4); the trace form of a finite separable field extension of $\mathbb{F}_p$ is nondegenerate (finite fields are perfect), and a nonzero nilpotent element lies in the kernel of the trace form; so the trace form of A is nondegenerate iff A is reduced iff all $e_i = 1$. Relative version (stated): for L/K, a prime $\mathfrak{p}$ of K ramifies in L iff $\mathfrak{p}$ divides the relative discriminant ideal $\mathfrak{d}_{L/K}$. Examples: $d = -4, -3, -7, -11, 5, -20$ for $\mathbb{Q}(i), \mathbb{Q}(\sqrt{-3}), \mathbb{Q}(\sqrt{-7}), \mathbb{Q}(\sqrt{-11}), \mathbb{Q}(\sqrt{5}), \mathbb{Q}(\sqrt{-5})$ (1.5); $d(\mathbb{Q}(\zeta_\ell)) = (-1)^{(\ell-1)/2}\ell^{\ell-2}$ (5.1); $d(\mathbb{Q}(2^{1/3})) = -108$. Reference: Neukirch III.2 (Corollary 2.12), and I.2 for the trace form; Marcus Ch. 3 (Theorem 24). Paper: background ($\mathbb{Q}(\sqrt{q^*})$ is ramified only at q, 4.3).

8.4.5 3.5 prop:ant:galoissplitting (proposition and definitions) P

Let L/K be Galois with group G, $\mathfrak{p}$ a prime of K. (i) G acts transitively on the set of primes $\mathfrak{P}$ of L above $\mathfrak{p}$. Hence all $e(\mathfrak{P}|\mathfrak{p}) = e$ and all $f(\mathfrak{P}|\mathfrak{p}) = f$ are equal, and $efg = n = \#G$. (ii) The *decomposition group* is $D_\mathfrak{P} = \{\sigma \in G : \sigma\mathfrak{P} = \mathfrak{P}\}$ (the stabiliser, Chapter 1), and the *inertia group* is $I_\mathfrak{P} = \{\sigma \in D_\mathfrak{P} : \sigma(x) \equiv x \pmod{\mathfrak{P}} \text{ for all } x \in \mathcal{O}_L\}$. Then $[G : D_\mathfrak{P}] = g$, $\#D_\mathfrak{P} = ef$, $I_\mathfrak{P} \trianglelefteq D_\mathfrak{P}$, and $D_{\sigma\mathfrak{P}} = \sigma D_\mathfrak{P} \sigma^{-1}$, $I_{\sigma\mathfrak{P}} = \sigma I_\mathfrak{P} \sigma^{-1}$; for G abelian $D_\mathfrak{P}$ and $I_\mathfrak{P}$ depend only on $\mathfrak{p}$. (iii) For $K \subset M \subset L$ with $H = \operatorname{Gal}(L/M)$: $D_\mathfrak{P}(L/M) = D_\mathfrak{P} \cap H$, $I_\mathfrak{P}(L/M) = I_\mathfrak{P} \cap H$; if M/K is Galois, the images of $D_\mathfrak{P}$ and $I_\mathfrak{P}$ in $G/H = \operatorname{Gal}(M/K)$ are $D_{\mathfrak{P}\cap M}$ and $I_{\mathfrak{P}\cap M}$. (iv) $\mathfrak{p}$ splits completely iff $D_\mathfrak{P} = 1$; $\mathfrak{p}$ is unramified iff $I_\mathfrak{P} = 1$. Proof of (i): if $\mathfrak{P}' \neq \sigma\mathfrak{P}$ for all σ, CRT gives $x \in \mathfrak{P}'$ with $x \equiv 1 \bmod \sigma\mathfrak{P}$ for all σ; then

$N_{L/K}(x) \in \mathfrak{P}' \cap \mathcal{O}_K = \mathfrak{p} \subset \mathfrak{P}$ but no factor $\sigma(x)$ lies in $\mathfrak{P}$. (RHB aside: an orbit and a stabiliser.) Reference: Neukirch I.9 (Proposition 9.1, Definition 9.2); Marcus Ch. 4 (Theorems 23, 28; check numbers, else cite the chapter). Paper: background for Lemma lem:euler and Lemma lem:tamfactor ("G/I", "I_q", "Frobenius normalises inertia"), Section 1.1 (I_p, $\delta_p = \dim V^{I_p}$).

8.4.6 3.6 thm:ant:residuegalois (theorem) P

In the situation of 3.5 write $\kappa(\mathfrak{P}) = \mathcal{O}_L/\mathfrak{P}$, $\kappa(\mathfrak{p}) = \mathcal{O}_K/\mathfrak{p}$ (finite fields, $\#\kappa(\mathfrak{p}) = N\mathfrak{p}$). Every $\sigma \in D_{\mathfrak{P}}$ induces an automorphism $\bar\sigma$ of $\kappa(\mathfrak{P})/\kappa(\mathfrak{p})$, and $\sigma \mapsto \bar\sigma$ is a surjective homomorphism $D_{\mathfrak{P}} \to \mathrm{Gal}(\kappa(\mathfrak{P})/\kappa(\mathfrak{p}))$ with kernel $I_{\mathfrak{P}}$. Hence $\#I_{\mathfrak{P}} = e$, $D_{\mathfrak{P}}/I_{\mathfrak{P}}$ is cyclic of order f, generated by the class of any σ with $\bar\sigma = (x \mapsto x^{N\mathfrak{p}})$ (Chapter 5, galois:finite-field-galois). Proof of surjectivity: choose $\bar\theta$ generating $\kappa(\mathfrak{P})^\times$, lift it to $\theta \in \mathcal{O}_L$ lying in every $\sigma\mathfrak{P} \neq \mathfrak{P}$ (CRT); the polynomial $h = \prod_{\sigma\in G}(X - \sigma\theta) \in \mathcal{O}_K[X]$ reduces to a polynomial with root $\bar\theta$, whose roots are the $\overline{\sigma\theta}$; the Frobenius image of $\bar\theta$ is a root $\overline{\sigma\theta} \neq 0$, which forces $\sigma \in D_{\mathfrak{P}}$. Reference: Neukirch I.9 (Proposition 9.4); Marcus Ch. 4 (Theorem 28); Serre, Local Fields, Ch. I. Paper: background (the exact sequence $1 \to I \to D \to \mathrm{Gal}(\bar\kappa/\kappa) \to 1$, used in profinite form in Lemma lem:tamfactor: "the group G/I").

8.4.7 3.7 conv:ant:frobenius (convention) P (a definition and a dictionary)

For $\mathfrak{P}$ unramified over $\mathfrak{p}$ the *arithmetic Frobenius* $\mathrm{Frob}_{\mathfrak{P}}$ is the element of 3.8; the *geometric Frobenius* is $\mathrm{Frob}_{\mathfrak{P}}^{-1}$. The book writes Frob_p for an arithmetic Frobenius at p (Chapter 5, C2, and the paper's proof of Lemma lem:tamfactor, "the arithmetic Frobenius"), so that Frob_p acts on μ_n ($p \nmid n$) by $\zeta \mapsto \zeta^p$, $\chi_\ell(\mathrm{Frob}_p) = p$, $\bar\chi_\ell(\mathrm{Frob}_p) = p \bmod \ell$, and for $E/\mathbb{Q}$ with good reduction at $p \neq \ell$ the characteristic polynomial of $\rho_{E,\ell}(\mathrm{Frob}_p)$ on $V_\ell E$ is $X^2 - a_p X + p$ (Chapters 9, 11). Local L-factors of a representation V of $G_{\mathbb{Q}}$ are written with the geometric Frobenius:

$$L_p(V,s) = \det\bigl(1 - \mathrm{Frob}_p^{-1}\, p^{-s} \;\big|\; V^{I_p}\bigr)^{-1},$$

so that $L_p(V(1),s) = L_p(V,s+1)$ (as $\chi_\ell(\mathrm{Frob}_p^{-1}) = p^{-1}$), $L_p(\mathbb{Q}_\ell(1),s) = (1-p^{-1-s})^{-1}$, and $L_p(V_\ell E^\vee, s) = (1 - a_p p^{-s} + p^{1-2s})^{-1}$ is the Euler factor of $L(E,s)$ at a good prime (on $V_\ell E$ itself the geometric Frobenius has eigenvalues α^{-1}, β^{-1}, on the dual $V_\ell E^\vee = H^1$ it has α, β). Dictionary: when the multiset of Frobenius eigenvalues on V^{I_p} is stable under inversion (e.g. $\mathrm{ad}^0 V$ at a good prime: $\alpha/\beta, 1, \beta/\alpha$; the lines on which Frobenius acts by ±1) the two conventions give the same factor; when it is not, they differ. The example that matters: for E with multiplicative reduction at p, $(\mathrm{ad}^0 V)^{I_p}$ is a line on which the arithmetic Frobenius acts by p and the geometric one by p^{-1}, and the true factor $L_p(\mathrm{ad}^0 V, s) = (1 - p^{-1-s})^{-1}$ of the paper's table needs the geometric one (Part 13, point 1). Reference: Deligne, "Les constantes des equations fonctionnelles des fonctions L" (conventions), DDT Section 1 (cite the section); Chapter 19 of this book for L-functions. Paper: Section 1.1 (definition of $L_p(A_f,s)$), the table of Section 1, Lemma lem:euler (type I_n: "a line with eigenvalue q^{-1}"), Lemma lem:tamfactor (arithmetic Frobenius, $\mathrm{Frob}^{-1}\, t\, \mathrm{Frob} = t^{q^{-1}}$), Lemma lem:fpr ($L_p(A_f,1) = L_p(A_f(1),0)$).

8.4.8 3.8 prop:ant:frobenius (definition and proposition) P

Let L/K be Galois with group G, $\mathfrak{p}$ a prime of K unramified in L, $\mathfrak{P} \mid \mathfrak{p}$. (i) There is a unique $\mathrm{Frob}_{\mathfrak{P}} \in G$ such that $\mathrm{Frob}_{\mathfrak{P}}(x) \equiv x^{N\mathfrak{p}} \pmod{\mathfrak{P}}$ for all $x \in \mathcal{O}_L$; it lies in $D_{\mathfrak{P}}$ and generates it, and its order is f. (ii) $\mathrm{Frob}_{\sigma\mathfrak{P}} = \sigma\, \mathrm{Frob}_{\mathfrak{P}}\, \sigma^{-1}$, so $\{\mathrm{Frob}_{\mathfrak{P}} : \mathfrak{P} \mid \mathfrak{p}\}$ is a conjugacy class $\mathrm{Frob}_{\mathfrak{p}}$ of G (the *Frobenius class*). (iii) $\mathfrak{p}$ splits completely in L iff $\mathrm{Frob}_{\mathfrak{p}} = \{1\}$. (iv) If G is abelian, $\mathrm{Frob}_{\mathfrak{P}}$ depends only on $\mathfrak{p}$ and is written $\mathrm{Frob}_{\mathfrak{p}} \in G$ (the "Frobenius of an unramified prime in an abelian extension"; over $\mathbb{Q}$ it is the Artin symbol of 5.6). (v) Functoriality: if $K \subset M \subset L$ with M/K Galois then $\mathrm{Frob}_{\mathfrak{P}}\,|_M = \mathrm{Frob}_{\mathfrak{P}\cap M}$; if $K \subset K' \subset L$ and $\mathfrak{p}' = \mathfrak{P} \cap K'$ has residue degree f' over $\mathfrak{p}$ then $\mathrm{Frob}_{\mathfrak{P}}(L/K') = \mathrm{Frob}_{\mathfrak{P}}(L/K)^{f'}$. (vi) For a ramified $\mathfrak{p}$ there is no well-defined Frobenius element, only a coset $\sigma I_{\mathfrak{P}}$ of elements of $D_{\mathfrak{P}}$ inducing $x \mapsto x^{N\mathfrak{p}}$; any such σ is called *a Frobenius at* $\mathfrak{P}$. Examples (proved in 4.2 and 5.2): in $\mathbb{Q}(\zeta_n)$, $p \nmid n$: $\mathrm{Frob}_p = \sigma_p$; in $\mathbb{Q}(\sqrt{d})$, $p \nmid D$: $\mathrm{Frob}_p(\sqrt{d}) = \left(\frac{D}{p}\right)\sqrt{d}$. Reference: Neukirch I.9; Marcus Ch. 4. Paper:

Lemma lem:old ("$\bar\rho(\mathrm{Frob}_p)$", "$X^2 - a_p(h)X + p$"), Lemma lem:euler, Lemma lem:tamfactor; supplies the black box galois:frobenius-preview of Chapter 5.

8.4.9 3.9 `prop:ant:frobsubfield` (proposition) P

Let $L/\mathbb{Q}$ be finite Galois with group G, $H \le G$, $M = L^H$, and p unramified in L; let $\mathfrak{P} \mid p$ and $F = \mathrm{Frob}_{\mathfrak{P}}$. Then the primes of M above p correspond bijectively to the orbits of $\langle F\rangle$ acting on G/H by left multiplication (the orbit of σH corresponds to $\sigma^{-1}\mathfrak{P}\cap M$), and the residue degree of the corresponding prime is the length of the orbit; all ramification indices are 1. Corollary (Dedekind's theorem, Chapter 5 galois:frobenius-preview (e)): if $f \in \mathbb{Z}[X]$ is separable with leading coefficient a, $p \nmid a\,\mathrm{disc}(f)$, L is the splitting field of f, and $\bar f$ has irreducible factors of degrees $d_1, \dots, d_r$ in $\mathbb{F}_p[X]$, then $\mathrm{Frob}_{\mathfrak{P}}$ permutes the roots of f with cycle type $(d_1, \dots, d_r)$. (Proof of the corollary: reduction mod $\mathfrak{P}$ is a $D_{\mathfrak{P}}$-equivariant bijection from the roots of f to the roots of $\bar f$ in $\kappa(\mathfrak{P})$, since $p \nmid a\,\mathrm{disc}(f)$, and $\mathrm{Frob}_{\mathfrak{P}}$ acts on the latter as $x \mapsto x^p$, whose cycles are the irreducible factors.) Example: 3.3(e) ($p = 5$: one fixed root and a transposition of the other two roots of $X^3 - 2$). Reference: Neukirch I.9 (Exercises) and I.8; Marcus Ch. 4. Paper: Proposition prop:ns via Chapter 5 (galois:frobenius-on-lines: the factorisation of the 3-division polynomial mod p), background.

8.4.10 3.10 `prop:ant:frobGQ` (proposition and definition) P

Let $\bar{\mathbb{Z}}$ be the integral closure of $\mathbb{Z}$ in $\overline{\mathbb{Q}}$ and p a prime. (i) The primes $\bar{\mathfrak{P}}$ of $\bar{\mathbb{Z}}$ above p are the compatible families $(\mathfrak{P}_L)_L$ of primes above p in the finite Galois $L/\mathbb{Q}$ ($\mathfrak{P}_L = \bar{\mathfrak{P}} \cap \mathcal{O}_L$); $G_{\mathbb{Q}}$ acts transitively on them. (ii) $D_{\bar{\mathfrak{P}}} = \varprojlim D_{\mathfrak{P}_L}$ and $I_{\bar{\mathfrak{P}}} = \varprojlim I_{\mathfrak{P}_L}$ are closed subgroups of $G_{\mathbb{Q}}$ (*a decomposition group* and *an inertia group at* p, written D_p, I_p once $\bar{\mathfrak{P}}$ is fixed), conjugate under $G_{\mathbb{Q}}$ for different choices, and $D_{\bar{\mathfrak{P}}}/I_{\bar{\mathfrak{P}}} \simeq \mathrm{Gal}(\overline{\mathbb{F}}_p/\mathbb{F}_p) \simeq \hat{\mathbb{Z}}$, topologically generated by the class of $x \mapsto x^p$. (iii) *A Frobenius element at* p is any $\mathrm{Frob}_p \in D_{\bar{\mathfrak{P}}}$ mapping to $x \mapsto x^p$; it is well defined up to $I_{\bar{\mathfrak{P}}}$ and up to conjugation in $G_{\mathbb{Q}}$, and its image in $\mathrm{Gal}(L/\mathbb{Q})$ lies in the class $\mathrm{Frob}_p(L/\mathbb{Q})$ of 3.8 for every finite Galois L unramified at p. (iv) A continuous representation ρ of $G_{\mathbb{Q}}$ is *unramified at* p if $\rho(I_{\bar{\mathfrak{P}}}) = 1$ for one (equivalently every) $\bar{\mathfrak{P}} \mid p$; then the conjugacy class of $\rho(\mathrm{Frob}_p)$, in particular $\det(X - \rho(\mathrm{Frob}_p))$, is well defined; for ρ with finite image this is the Frobenius class of 3.8 in the field cut out by ρ. (v) (Stated; Chapter 7.) Choosing an embedding $\overline{\mathbb{Q}} \to \overline{\mathbb{Q}}_p$ identifies $D_{\bar{\mathfrak{P}}}$ with $G_{\mathbb{Q}_p}$ and $I_{\bar{\mathfrak{P}}}$ with the inertia subgroup of $G_{\mathbb{Q}_p}$. Reference: Neukirch I.9 and IV.1 (cite chapters); Serre, Abelian l-adic representations, Ch. I. Paper: Section 1.1 (I_p, Frob_p, V^{I_p}), Lemma lem:old, Lemma lem:tamagawa-exp, Lemma lem:tamfactor, Lemma lem:euler; supplies Chapter 5's galois:frobenius-preview (d).

8.4.11 3.11 `prop:ant:tameconj` (proposition) S here, P in Chapter 7

Let L/K be finite Galois, $\mathfrak{P} \mid \mathfrak{p}$, $q = N\mathfrak{p}$ (a power of the residue characteristic p), and suppose $\#I_{\mathfrak{P}}$ is prime to p (*tamely ramified*). Then $I_{\mathfrak{P}}$ is cyclic, and for every $F \in D_{\mathfrak{P}}$ inducing $x \mapsto x^q$ on $\kappa(\mathfrak{P})$ and every $\tau \in I_{\mathfrak{P}}$: $F\tau F^{-1} = \tau^q$. Sketch: with ϖ a uniformiser of the DVR $\mathcal{O}_{L,\mathfrak{P}}$, the map $\theta\colon I_{\mathfrak{P}} \to \kappa(\mathfrak{P})^\times$, $\tau \mapsto \tau(\varpi)/\varpi \bmod \mathfrak{P}$, is an injective homomorphism independent of ϖ when $I_{\mathfrak{P}}$ has order prime to p (its kernel is the wild inertia group, a p-group), and $\theta(F\tau F^{-1}) = F(\theta(\tau)) = \theta(\tau)^q$. Reference: Serre, Local Fields, Ch. IV, Section 2 (Propositions 7 and 9; check numbers, else cite the section); Neukirch II.9 (local form). The profinite form $\mathrm{Frob}\; t\; \mathrm{Frob}^{-1} = t^q$ on the tame inertia quotient of $G_{\mathbb{Q}_q}$ is Chapter 7. Paper: Lemma lem:euler ("Frobenius centralises σ if $q \equiv 1 \pmod 3$ and inverts it if $q \equiv 2$"), Lemma lem:tamfactor ($\mathrm{Frob}^{-1}\, t\, \mathrm{Frob} = t^{q^{-1}}$, Chapter 7).

8.4.12 3.12 `cor:ant:frobsigma` (corollary) P (given 3.11)

Let q be a prime, $\ell \neq q$ an odd prime, and $\rho\colon G_{\mathbb{Q}} \to \mathrm{GL}_2(\mathbb{Z}_\ell)$ continuous with $\rho(I_q)$ finite of order m prime to q (for $\rho = \rho_{E,\ell}$ at a prime $q \ge 5$ of additive potentially good reduction, $m \in \{2, 3, 4, 6\}$ by Kraus, Chapters 10, 11). Then $\rho(I_q)$ is cyclic, and for a Frobenius $F \in D_q$ (3.10) and $\tau \in I_q$: $\rho(F)\rho(\tau)\rho(F)^{-1} = \rho(\tau)^q$. In particular, if $\rho(I_q) = \langle\sigma\rangle$ with σ of order 3 (or $\langle -\sigma\rangle$ of order 6), then $\rho(F)$

commutes with σ if $q \equiv 1 \pmod 3$ and conjugates it to σ^{-1} if $q \equiv 2 \pmod 3$; if $\rho(I_q) = \langle\tau\rangle$ of order 4, $\rho(F)$ commutes with τ if $q \equiv 1 \pmod 4$ and conjugates it to τ^{-1} if $q \equiv 3 \pmod 4$. The result does not depend on the choice of F (two choices differ by I_q, and $\rho(I_q)$ is abelian). Proof from 3.11 alone (no local fields): reduction mod ℓ is injective on the finite subgroup $\rho(I_q)$ of $\mathrm{GL}_2(\mathbb{Z}_\ell)$ ($1+\ell M_2(\mathbb{Z}_\ell)$ is torsion-free for ℓ odd, Chapter 1 item 7.4); apply 3.11 to the finite Galois extension L cut out by $\bar\rho$ (whose inertia group at q is $\bar\rho(I_q) \simeq \rho(I_q)$, of order prime to q, and F restricts to an element of $D_{\mathfrak{P}}$ inducing $x \mapsto x^q$), and lift: $\rho(F)\rho(\tau)\rho(F)^{-1}$ and $\rho(\tau)^q$ lie in the normal subgroup $\rho(I_q)$ of $\rho(D_q)$ and have the same reduction. Reference: as 3.11. Paper: Lemma lem:euler (types II, II*, IV, IV* and III, III*), Lemma lem:IV ("the image is tame since $q \neq 3$, hence cyclic").

8.5 4. Line-up, Section 4: quadratic fields in full; $\mathbb{Z}[i]$, $\mathbb{Z}[\zeta_3]$ and the rings $\mathbb{Z}_\ell[\zeta_3]$

8.5.1 4.1 thm:ant:quadsplit (theorem) P

Let $K = \mathbb{Q}(\sqrt{d})$, $d \neq 0, 1$ squarefree, with discriminant D (1.5), and p a prime. Then, with $\left(\frac{D}{p}\right)$ the Kronecker symbol (Chapter 2: the Legendre symbol for odd p; for $p = 2$ it is 0 if $2 \mid D$, $+1$ if $D \equiv \pm 1 \pmod 8$, -1 if $D \equiv \pm 3 \pmod 8$): p ramifies in K iff $\left(\frac{D}{p}\right) = 0$ (iff $p \mid D$); p splits iff $\left(\frac{D}{p}\right) = 1$; p is inert iff $\left(\frac{D}{p}\right) = -1$. Explicitly: for odd $p \mid D$, $p\mathcal{O}_K = (p, \sqrt{d})^2$; for odd $p \nmid D$ with $a^2 \equiv d \pmod p$, $p\mathcal{O}_K = (p, \sqrt{d} - a)(p, \sqrt{d} + a)$, the two factors distinct; for $p = 2$: $d \equiv 3 \pmod 4$ gives $(2) = (2, 1 + \sqrt{d})^2$, $d \equiv 2 \pmod 4$ gives $(2) = (2, \sqrt{d})^2$, $d \equiv 1 \pmod 8$ gives $(2) = (2, \frac{1+\sqrt{d}}{2})(2, \frac{1-\sqrt{d}}{2})$, and $d \equiv 5 \pmod 8$ gives 2 inert. Proof: 3.2 with $\theta = \sqrt{d}$ (p odd, or $d \not\equiv 1 \bmod 4$) or $\theta = (1 + \sqrt{d})/2$, $g = X^2 - X + \frac{1-d}{4}$. Checked by computer for 23 discriminants and all $p < 400$ (Part 12). Reference: Neukirch I.8 (Proposition 8.5); Marcus Ch. 3 (Theorem 25). Paper: χ_{-3}, χ_{-4} (table of Section 1), $\mathbb{Q}(\sqrt{\ell^*})$ (Theorem thm:main), $\mathbb{Q}(\sqrt{q^*})$ (Section 1.4, Lemma lem:twist).

8.5.2 4.2 prop:ant:quadchar (proposition) P

Let $K = \mathbb{Q}(\sqrt{d})$ with discriminant D and $\chi_d \colon G_{\mathbb{Q}} \to \{\pm 1\}$ its character (Chapter 5, C3). (i) For every prime $p \nmid D$: $\chi_d(\mathrm{Frob}_p) = \left(\frac{D}{p}\right)$; the right side is the value at p of the primitive real Dirichlet character $\chi_D = \left(\frac{D}{\cdot}\right)$ of conductor $|D|$ (Chapter 2). (ii) χ_d is unramified exactly at the primes $p \nmid D$ ($\chi_d(I_p) = \{\pm 1\}$ for $p \mid D$). (iii) The characters the paper uses: $\chi_{-4}(\mathrm{Frob}_p) = (-1)^{(p-1)/2}$ (p odd; conductor 4); $\chi_{-3}(\mathrm{Frob}_p) = \left(\frac{p}{3}\right)$, which is 1 iff $p \equiv 1 \pmod 3$ ($p \neq 3$, including $\chi_{-3}(\mathrm{Frob}_2) = -1$; conductor 3); for an odd prime q, χ_{q^*} (the paper's χ_q) has $\chi_{q^*}(\mathrm{Frob}_p) = \left(\frac{q^*}{p}\right) = \left(\frac{p}{q}\right)$ for all $p \neq q$ including $p = 2$ (quadratic reciprocity with its supplement, Chapter 2, or 5.4; conductor q); $\chi_{\pm 2}$ (discriminants $8, -8$) have conductor 8. Proof of (i): Frob_p fixes $\sqrt{d}$ iff $D_{\mathfrak{P}} = 1$ iff p splits (3.8(iii)) iff $\left(\frac{D}{p}\right) = 1$ (4.1). (ii) is 3.4 with 3.5(iv). (iii) is (i) for $D = -4, -3, q^*, \pm 8$. All congruence conditions checked for $p < 3000$ (Part 12). Reference: Neukirch I.8 and VI.1 (cite chapters); Washington Ch. 3 (quadratic characters as Dirichlet characters). Paper: the table of Section 1 and Lemma lem:euler ($\chi_{-3}(q)$, $\chi_{-4}(q)$), Definition def:terms, Proposition prop:forms ("$q \equiv \chi_{-3}(q) \pmod 3$"), Lemma lem:twist ($\chi_{q^*}$), Proposition prop:ns ($\chi_{-3}$); supplies Chapter 5's galois:quadratic-characters (c), (d).

8.5.3 4.3 prop:ant:ramonlyat (proposition) P

(i) For an odd prime q, $\mathbb{Q}(\sqrt{q^*})$, $q^* = (-1)^{(q-1)/2}q$, is the unique quadratic field ramified at q only; its discriminant is q^* (because $q^* \equiv 1 \pmod 4$). (ii) More generally, a quadratic field is unramified outside a finite set S of primes iff its discriminant is $D = D_0 \prod_{q \in T} q^*$ with T a set of odd primes in S and $D_0 \in \{1, -4, 8, -8\}$, $D_0 = 1$ unless $2 \in S$ (and $D \neq 1$). Every fundamental discriminant factors uniquely in this way ("prime discriminants"). (iii) For a finite set S of odd primes, $\chi_S := \prod_{q \in S} \chi_{q^*}$ is the character of $\mathbb{Q}(\sqrt{\prod_{q \in S} q^*})$, of discriminant $\prod_{q \in S} q^*$, conductor $D_S = \prod_{q \in S} q$, ramified exactly

at the primes of S. (iv) Hence a quadratic twist by χ_{q^*} (or χ_S) is unramified at every $p \notin S$: the Galois representation does not change on I_p for $p \notin S$, which is the arithmetic input of Lemma lem:twist (i). Proof: 3.4 and 4.1; a fundamental discriminant that is odd is $\equiv 1 \pmod 4$, which fixes the sign $(-1)^{(q-1)/2}$. Reference: Washington Ch. 3; Cox, Primes of the form x^2+ny^2, Section 5 (cite the section). Paper: Section 1.4 ($E^{(q)}$, the twist by $\mathbb{Q}(\sqrt{q^*})$), Lemma lem:twist (i), Theorem thm:twist (χ_S, D), Theorem thm:quant (E_S), Theorem thm:qual (b), (c).

8.5.4 4.4 ex:ant:quadexamples (worked examples) P

(a) $\mathbb{Q}(i)$, $D=-4$: 2 ramified, $p\equiv 1 \pmod 4$ split, $p\equiv 3 \pmod 4$ inert. (b) $\mathbb{Q}(\sqrt{-3})=\mathbb{Q}(\zeta_3)$, $D=-3$: 3 ramified, $p\equiv 1 \pmod 3$ split, $p\equiv 2 \pmod 3$ inert (including 2). (c) $\mathbb{Q}(\sqrt{-7})=\mathbb{Q}(\sqrt{7^*})$, $D=-7\equiv 1 \pmod 8$: 2 splits; $p\neq 7$ splits iff $p\equiv 1,2,4 \pmod 7$. (d) $\mathbb{Q}(\sqrt{-11})=\mathbb{Q}(\sqrt{11^*})$, $D=-11\equiv 5 \pmod 8$: 2 inert, 3 and 5 split, 7 inert; $p\neq 11$ splits iff p is a square mod 11. (e) $\mathbb{Q}(\sqrt{5})=\mathbb{Q}(\sqrt{5^*})$: $p\neq 5$ splits iff $p\equiv \pm 1 \pmod 5$. (f) $\mathbb{Q}(\sqrt{-5})$, $D=-20$: 2, 5 ramified, p splits iff $p\equiv 1,3,7,9 \pmod{20}$ (stated; Exercise). The paper's twists: 121a1 twisted by -11 (field (d)), 539c1 and 980i1 twisted by -7 (field (c)): Example after Theorem thm:qual. Paper: Section 3 Example, Lemma lem:twist.

8.5.5 4.5 prop:ant:gausseisenstein (proposition) P

(i) $\mathbb{Z}[i]$ and $\mathbb{Z}[\zeta_3]$ are Euclidean for the norms a^2+b^2 and a^2-ab+b^2 (Chapter 2, cited), hence PIDs and UFDs, with unit groups μ_4 and μ_6 (1.6) and class number 1 (2.5). (ii) Primes of $\mathbb{Z}[i]$, up to units: $1+i$ (with $2=-i(1+i)^2$); for $p\equiv 1 \pmod 4$ the two non-associate factors $a\pm bi$ of $p=a^2+b^2$; for $p\equiv 3 \pmod 4$ the rational prime p. Examples $5=(2+i)(2-i)$, $13=(3+2i)(3-2i)$. (iii) Primes of $\mathbb{Z}[\zeta_3]$, up to units: $\pi=1-\zeta_3$ (4.6); for $p\equiv 1 \pmod 3$ the two non-associate factors $a+b\zeta_3$, $a+b\zeta_3^2$ of $p=a^2-ab+b^2$; for $p\equiv 2 \pmod 3$ the rational prime p. Examples: $7=(3+\zeta_3)(3+\zeta_3^2)$, $13=(4+\zeta_3)(4+\zeta_3^2)$. (iv) Corollaries: a prime p is a sum of two squares iff $p=2$ or $p\equiv 1 \pmod 4$ (Fermat); $p=a^2-ab+b^2$ iff $p=3$ or $p\equiv 1 \pmod 3$, and then also $p=x^2+3y^2$ (all checked for $p<300$). Proof: 4.1 for the splitting pattern, plus "principal ideal of norm p" from (i). Reference: Ireland-Rosen, Chapters 1 and 9 (add to bib/ch06.bib); Marcus Ch. 3. Paper: background for Lemma lem:IV ($\mathbb{Z}_3[\zeta_3]$), Lemma lem:euler ($\mathbb{Z}_\ell[\zeta_3]$, $\mathbb{Z}_\ell[i]$), the table of Section 1 (χ_{-3}, χ_{-4}).

8.5.6 4.6 prop:ant:above3 (proposition) P

In $\mathbb{Z}[\zeta_3]$ put $\pi=1-\zeta_3$. (i) $\pi^2=-3\zeta_3$, $3=-\zeta_3^2\pi^2$, and $\sqrt{-3}:=1+2\zeta_3=\zeta_3-\zeta_3^2=\zeta_3\pi$ satisfies $(\sqrt{-3})^2=-3$; so π and $\sqrt{-3}$ are associates, $N(\pi)=3$, $(3)=(\pi)^2$ (3 is totally ramified) and $\pi^2=-3$ up to the unit ζ_3. (ii) $\mathbb{Z}[\zeta_3]/(\pi)=\mathbb{F}_3$, and $\zeta_3\equiv 1 \pmod\pi$. (iii) $\mathbb{Z}[\zeta_3]/(3)=\mathbb{Z}[\zeta_3]/(\pi^2)$ is a local ring of length 2 with $\mathbb{F}_3$-basis $1,\pi$, and multiplication by $\zeta_3=1-\pi$ on it has matrix $\left(\begin{smallmatrix}1&0\\-1&1\end{smallmatrix}\right)$ in this basis ($\zeta_3\cdot 1=1-\pi$, $\zeta_3\cdot\pi=\pi-\pi^2=\pi$): a nontrivial unipotent map whose fixed points form the line $\pi\mathbb{Z}[\zeta_3]/(\pi^2)$. (iv) The analogue at 2 in $\mathbb{Z}[i]$: $(2)=(1+i)^2$, $\mathbb{Z}[i]/(2)=\mathbb{F}_2[\varepsilon]/(\varepsilon^2)$ with $\varepsilon=1+i$, and multiplication by $i=1+\varepsilon$ is a nontrivial unipotent map (the reason the prime 2 is wild for χ_{-4}; background for Remark rem:wild). All identities checked symbolically (Part 12). Reference: Ireland-Rosen Ch. 9 (Section 1); Washington Ch. 1 and Lemma 1.4 (for $(1-\zeta_\ell)^{\ell-1}=\ell\cdot$unit; check the number, else cite Ch. 1). Paper: Lemma lem:IV ("$E[3]=T/3T\simeq\mathbb{Z}_3[\zeta_3]/(\pi^2)$ with σ acting as multiplication by $\zeta_3=1-\pi$ … a nontrivial unipotent automorphism … its fixed space is the line $\pi\mathbb{Z}_3[\zeta_3]/(\pi^2)$").

8.5.7 4.7 thm:ant:z3zeta3dvr (theorem) P (local-field interpretation in Chapter 7)

Let $\mathbb{Z}_3[\zeta_3]:=\mathbb{Z}_3[X]/(X^2+X+1)$ (Chapter 3 supplies $\mathbb{Z}_3$; X^2+X+1 is irreducible over $\mathbb{Q}_3$, being Eisenstein after $X\mapsto X+1$: X^2+3X+3), write ζ_3 for the class of X and $\pi=1-\zeta_3$. Then: (i) $\mathbb{Z}_3[\zeta_3]$ is a domain, free of rank 2 over $\mathbb{Z}_3$ with basis $1,\zeta_3$ (or $1,\pi$); (ii) it is a discrete valuation ring with maximal ideal (π), uniformiser π and residue field $\mathbb{F}_3$; every nonzero element is $u\pi^k$ with u a unit and

$k \geq 0$; (iii) $\pi^2 = -3\zeta_3$, so $3\mathbb{Z}_3[\zeta_3] = \pi^2\mathbb{Z}_3[\zeta_3]$ and $v_\pi(3) = 2$; (iv) $\mathbb{Z}_3[\zeta_3] \simeq \mathbb{Z}_3 \otimes_{\mathbb{Z}} \mathbb{Z}[\zeta_3]$, and it is the π-adic completion of $\mathbb{Z}[\zeta_3]$ (stated; Chapter 7); its fraction field $\mathbb{Q}_3(\zeta_3) = \mathbb{Q}_3(\sqrt{-3})$ is a totally ramified quadratic extension of $\mathbb{Q}_3$ (stated; Chapter 7). Proof of (ii): $\mathbb{Z}_3[\zeta_3]/(\pi) = \mathbb{Z}_3[X]/(X^2 + X + 1, 1 - X) = \mathbb{Z}_3/3 = \mathbb{F}_3$; if $a + b\zeta_3 \notin (\pi)$, i.e. $3 \nmid a + b$, then its norm $a^2 - ab + b^2 \equiv (a + b)^2 \not\equiv 0 \pmod 3$ is a unit of $\mathbb{Z}_3$ and $(a + b\zeta_3)^{-1} = (a + b\zeta_3^2)/(a^2 - ab + b^2)$; so the ring is local with principal maximal ideal (π), and $\bigcap_k \pi^k\mathbb{Z}_3[\zeta_3] = \bigcap_k 3^k\mathbb{Z}_3[\zeta_3] = 0$; a Noetherian local domain with nonzero principal maximal ideal and this property is a DVR (Chapter 3). Reference: Serre, Local Fields, Ch. I (Proposition 17 and Section I.6 for Eisenstein extensions; check numbers); Neukirch II.5. Paper: Lemma lem:IV ("$\mathbb{Z}_3[\zeta_3]$, a discrete valuation ring with uniformiser $\pi = 1 - \zeta_3$ and residue field $\mathbb{F}_3$").

8.5.8 4.8 cor:ant:z3zeta3modules (corollary) P

Let T be a free $\mathbb{Z}_3$-module of rank 2 and $\sigma \in \mathrm{Aut}_{\mathbb{Z}_3}(T)$ with $\sigma^2 + \sigma + 1 = 0$ (equivalently: σ of order 3 with $\det\sigma = 1$, Chapter 4). Then, with ζ_3 acting as σ: (i) T is a torsion-free $\mathbb{Z}_3[\zeta_3]$-module (if $xt = 0$ with $0 \neq x$, then $N(x)t = x\bar{x}t = 0$ with $0 \neq N(x) \in \mathbb{Z}_3$), finitely generated, hence free (DVR, Chapter 3), of rank $2/2 = 1$: $T \simeq \mathbb{Z}_3[\zeta_3]$; (ii) $T/3T \simeq \mathbb{Z}_3[\zeta_3]/(\pi^2)$, and σ acts on $T/3T$ as the nontrivial unipotent of 4.6(iii), with fixed line $\pi T/3T$; (iii) more generally the σ-fixed vectors of $T/\pi^k T$ form $\pi^{k-1}T/\pi^k T$, a line over $\mathbb{F}_3$. (Parts (i) and (ii) are the algebra of the paper's proof; Chapter 11 supplies σ from Kraus's theorem.) Reference: as 4.7. Paper: Lemma lem:IV, Theorem thm:qual (a) (Serre conductor exponent 1), Remark rem:types (contrast).

8.5.9 4.9 prop:ant:zlzeta3 (proposition) P, with Hensel's lemma cited from Chapter 7

For a prime ℓ put $\mathbb{Z}_\ell[\zeta_3] := \mathbb{Z}_\ell[X]/(X^2 + X + 1) \simeq \mathbb{Z}_\ell \otimes_{\mathbb{Z}} \mathbb{Z}[\zeta_3]$ and $\mathbb{Z}_\ell[i] := \mathbb{Z}_\ell[X]/(X^2 + 1)$. (i) $\ell = 3$: the ramified DVR of 4.7. (ii) $\ell \equiv 1 \pmod 3$: $X^2 + X + 1$ has two distinct roots in $\mathbb{Z}_\ell$ (distinct simple roots in $\mathbb{F}_\ell$, Hensel), and $\mathbb{Z}_\ell[\zeta_3] \simeq \mathbb{Z}_\ell \times \mathbb{Z}_\ell$. (iii) $\ell \equiv 2 \pmod 3$: $X^2 + X + 1$ is irreducible mod ℓ, and $\mathbb{Z}_\ell[\zeta_3]$ is an unramified DVR with uniformiser ℓ and residue field $\mathbb{F}_{\ell^2}$. The trichotomy is the splitting of ℓ in $\mathbb{Q}(\zeta_3)$ (ramified, split, inert; 4.4(b)) and is the case $K = \mathbb{Q}(\zeta_3)$ of 7.4. Likewise $\mathbb{Z}_\ell[i] \simeq \mathbb{Z}_\ell \times \mathbb{Z}_\ell$ for $\ell \equiv 1 \pmod 4$, an unramified DVR with residue field $\mathbb{F}_{\ell^2}$ for $\ell \equiv 3 \pmod 4$, and a ramified DVR with uniformiser $1 + i$ for $\ell = 2$. (iv) In all cases $\mathbb{Q}_\ell[\zeta_3] := \mathbb{Z}_\ell[\zeta_3][1/\ell] = \mathbb{Q}_\ell \otimes_{\mathbb{Q}} \mathbb{Q}(\zeta_3)$ carries the $\mathbb{Q}_\ell$-linear involution $\zeta_3 \mapsto \zeta_3^2$ (the nontrivial element of $\mathrm{Gal}(\mathbb{Q}(\zeta_3)/\mathbb{Q})$ extended by scalars), which negates $\sqrt{-3} = 1 + 2\zeta_3$; and $\sqrt{-3}$ is a unit of $\mathbb{Z}_\ell[\zeta_3]$ iff $\ell \neq 3$. The analogous statements hold for i in $\mathbb{Q}_\ell[i]$. Proof: Chapter 3 for the DVR criterion as in 4.7, CRT for (ii). Reference: Neukirch II.4 (Hensel), II.8 (extensions of valuations; cite the chapter). Paper: Lemma lem:euler ("$\mathbb{Z}_\ell[\zeta_3] \simeq \mathrm{End}_{I_q}(V)$", "the line spanned by $\sqrt{-3}$"; see 4.10 and Part 13, point 2), Lemma lem:IV.

8.5.10 4.10 prop:ant:commutant (proposition) P (the number theory behind the types II, IV, III in Lemma lem:euler)

Let F be a field of characteristic 0 (in the paper $F = \mathbb{Q}_\ell$, $\ell \neq q$), V a 2-dimensional F-vector space, and $\sigma \in \mathrm{GL}(V)$ with $\sigma^2 + \sigma + 1 = 0$ (order 3, determinant 1). (i) The commutant $\{x \in \mathrm{End}_F(V) : x\sigma = \sigma x\}$ is $F[\sigma] \simeq F[X]/(X^2 + X + 1) = F \otimes_{\mathbb{Q}} \mathbb{Q}(\zeta_3)$, of dimension 2 (a field iff F contains no primitive cube root of unity; for $F = \mathbb{Q}_\ell$ iff $\ell \not\equiv 1 \pmod 3$); the same holds for $-\sigma$ (order 6). (ii) The trace-zero part of $F[\sigma]$ is the line $F \cdot (2\sigma + 1)$, and $(2\sigma + 1)^2 = -3$: "the line spanned by $\sqrt{-3}$". (iii) If $\Phi \in \mathrm{GL}(V)$ satisfies $\Phi\sigma\Phi^{-1} = \sigma^q$ with q a prime $\neq 3$, then conjugation by Φ preserves $F[\sigma]$ and acts on it as $\mathrm{id}_F \otimes \mathrm{Frob}_q$, where $\mathrm{Frob}_q = \sigma_q \in \mathrm{Gal}(\mathbb{Q}(\zeta_3)/\mathbb{Q})$ (5.2); hence it acts on the line $F(2\sigma + 1)$ by $\chi_{-3}(\mathrm{Frob}_q) = \chi_{-3}(q)$ (4.2). (iv) The order-4 analogue: if $\tau \in \mathrm{GL}(V)$, $\tau^2 = -1$, the commutant is $F[\tau] \simeq F \otimes_{\mathbb{Q}} \mathbb{Q}(i)$, its trace-zero part is $F\tau$, and $\Phi\tau\Phi^{-1} = \tau^q$ (q odd) gives the action $\chi_{-4}(q)$ on $F\tau$. (v) Integral form: if $F = \mathbb{Q}_\ell$ and $T \subset V$ is a σ-stable $\mathbb{Z}_\ell$-lattice, then T is free of rank one over $\mathbb{Z}_\ell[\sigma] \simeq \mathbb{Z}_\ell[\zeta_3]$ (4.8 for $\ell = 3$; a DVR for $\ell \equiv 2 \pmod 3$; for $\ell \equiv 1 \pmod 3$ via the idempotents $(\sigma - \zeta^{-1})/(\zeta - \zeta^{-1})$, $\zeta \in \mathbb{Z}_\ell$ a primitive cube root of unity, whose denominator $\pm\sqrt{-3}$ is a unit), and the commutant of σ in $\mathrm{End}_{\mathbb{Z}_\ell}(T)$ is $\mathbb{Z}_\ell[\sigma] \simeq \mathbb{Z}_\ell[\zeta_3]$. So the correct reading of the paper's "$\mathbb{Z}_\ell[\zeta_3] \simeq \mathrm{End}_{I_q}(V)$" is $\mathrm{End}_{I_q}(T) = \mathbb{Z}_\ell[\sigma] \simeq \mathbb{Z}_\ell \otimes \mathbb{Z}[\zeta_3]$ and $\mathrm{End}_{I_q}(V) = \mathbb{Q}_\ell \otimes \mathbb{Q}(\zeta_3)$,

which is not a field (nor $\mathbb{Z}_\ell[\zeta_3]$ a domain) when $\ell \equiv 1 \pmod 3$; the argument of Lemma lem:euler needs only (ii) and (iii). Proof: σ has distinct eigenvalues over $\bar{F}$, so it is cyclic and its commutant is $F[\sigma]$ (Chapter 4 or linear algebra); $\operatorname{tr}\sigma = -1$ gives $\operatorname{tr}(2\sigma+1) = 0$; (iii) is $\Phi(2\sigma+1)\Phi^{-1} = 2\sigma^q + 1$, which is $2\sigma+1$ if $q \equiv 1$ and $2\sigma^2 + 1 = -(2\sigma+1)$ if $q \equiv 2 \pmod 3$; (iv) is the same with $\tau^q = \pm\tau$. Reference: elementary. Paper: Lemma lem:euler (types II, II*, IV, IV*: "$\mathbb{Z}_\ell[\zeta_3] \simeq \operatorname{End}_{I_q}(V)$", "Frobenius ... acts on the line by $\chi_{-3}(q)$"; types III, III*: $\chi_{-4}(q)$), with 3.12 supplying $\Phi = \rho(F)$; the table of Section 1.

8.6 5. Line-up, Section 5: cyclotomic fields and class field theory over $\mathbb{Q}$

8.6.1 5.1 thm:ant:cyclotomicintegers (theorem) P for prime powers, R in general

Let $n \geq 1$ and $K = \mathbb{Q}(\zeta_n)$, of degree $\varphi(n)$ with $\operatorname{Gal}(K/\mathbb{Q}) \simeq (\mathbb{Z}/n)^\times$, $a \mapsto \sigma_a$ (Chapter 5, galois:cyclotomic; not redone). (i) $\mathcal{O}_K = \mathbb{Z}[\zeta_n]$. (ii) For $n = \ell^a$, ℓ prime: $\prod_{b\in(\mathbb{Z}/\ell^a)^\times}(1-\zeta^b) = \Phi_{\ell^a}(1) = \ell$, each $1-\zeta^b$ is a unit times $1-\zeta$, so $\ell\mathcal{O}_K = (1-\zeta_{\ell^a})^{\varphi(\ell^a)}$: ℓ is totally ramified, with residue field $\mathbb{F}_\ell$. (iii) $d(\mathbb{Q}(\zeta_n)) = (-1)^{\varphi(n)/2}\, n^{\varphi(n)} / \prod_{p|n} p^{\varphi(n)/(p-1)}$; in particular $d(\mathbb{Q}(\zeta_\ell)) = (-1)^{(\ell-1)/2}\ell^{\ell-2}$, $d(\mathbb{Q}(\zeta_3)) = -3$, $d(\mathbb{Q}(\zeta_4)) = -4$, $d(\mathbb{Q}(\zeta_8)) = 256$. (iv) p ramifies in $\mathbb{Q}(\zeta_n)$ iff $p \mid n$, for $n \not\equiv 2 \pmod 4$ (recall $\mathbb{Q}(\zeta_n) = \mathbb{Q}(\zeta_{2n})$ for n odd). Proof of (i) for $n = \ell^a$: $\operatorname{disc}(1, \zeta, \dots)$ is $\pm$ a power of ℓ (3.4 bookkeeping), so the index is a power of ℓ, and $\mathbb{Z}[\zeta]$ is ℓ-maximal because $1-\zeta$ generates a totally ramified prime (Eisenstein at ℓ for $\Phi_{\ell^a}(X+1)$). General n: compositum argument (stated). Formula (iii) checked against $\operatorname{disc}(\Phi_n)$ for $n \leq 20$ (Part 12). Reference: Washington Ch. 2 (Theorem 2.6, Proposition 2.7; check numbers, else cite the chapter); Neukirch I.10. Paper: background ($\mathbb{Z}[\zeta_3]$, $\mathbb{Q}(\zeta_\ell)$).

8.6.2 5.2 thm:ant:cyclotomicsplitting (theorem) P

Let $n = p^a m$ with $p \nmid m$, and let f be the order of p in $(\mathbb{Z}/m)^\times$. In $\mathbb{Q}(\zeta_n)$: $e = \varphi(p^a)$, residue degree f, and $g = \varphi(m)/f$ primes above p. The inertia group of p is $\operatorname{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q}(\zeta_m)) \simeq (\mathbb{Z}/p^a)^\times \times \{1\}$ and the decomposition group is $(\mathbb{Z}/p^a)^\times \times \langle p\rangle$ inside $(\mathbb{Z}/n)^\times = (\mathbb{Z}/p^a)^\times \times (\mathbb{Z}/m)^\times$. For $p \nmid n$: $\operatorname{Frob}_p = \sigma_p$, p splits completely iff $p \equiv 1 \pmod n$, and is inert iff p generates $(\mathbb{Z}/n)^\times$ (possible only when this group is cyclic). Examples (checked, Part 12): $n = 3$: $p \equiv 1$ split, $p \equiv 2$ inert, 3 ramified; $n = 5$: 11 splits completely, 19 and 29 give $g = f = 2$, 2 and 3 are inert; $n = 7$: 2 gives $f = 3$, $g = 2$; 13 gives $f = 2$, $g = 3$; 29 splits completely. Proof: for $p \nmid n$, $\Phi_n \bmod p$ factors into $\varphi(n)/f$ irreducibles of degree f (Chapter 5, galois:cyclotomic-mod-p) and 3.2 applies ($p \nmid \operatorname{disc}\Phi_n$); $\sigma_p(\zeta) = \zeta^p \equiv \zeta^p$ identifies the Frobenius (3.8); for $p \mid n$ combine 5.1(ii) in $\mathbb{Q}(\zeta_{p^a})$ (totally ramified) with the unramified $\mathbb{Q}(\zeta_m)$ and multiplicativity of e, f (3.1). Reference: Neukirch I.10 (Proposition 10.3); Washington Ch. 2 (Theorem 2.13; check). Paper: background for the Frobenius identities $\chi_\ell(\operatorname{Frob}_p) = p$, $\bar{\chi}_\ell(\operatorname{Frob}_p) = p \bmod \ell$ (Chapter 5, C2), Lemma lem:euler (Frobenius on μ_3 and μ_4).

8.6.3 5.3 prop:ant:sqrtlstar (proposition) P

Let ℓ be an odd prime. (i) $\mathbb{Q}(\sqrt{\ell^*})$ is the unique quadratic subfield of $\mathbb{Q}(\zeta_\ell)$ (Chapter 5, galois:sqrt-lstar; not redone); it is ramified only at ℓ (4.3), with discriminant ℓ^*. (ii) For $p \neq \ell$ (including $p = 2$): $\chi_{\ell^*}(\operatorname{Frob}_p) = \left(\frac{p}{\ell}\right)$; so p splits in $\mathbb{Q}(\sqrt{\ell^*})$ iff p is a square mod ℓ, and p is *inert* in $\mathbb{Q}(\sqrt{\ell^*})$ iff p is a non-square mod ℓ. The inert primes have density 1/2 (6.3). (iii) For $\ell = 3$: $\mathbb{Q}(\sqrt{-3}) = \mathbb{Q}(\zeta_3)$, and the inert primes are the $p \equiv 2 \pmod 3$. (iv) $\mathbb{Q}(\sqrt{\ell^*})$ is imaginary iff $\ell \equiv 3 \pmod 4$. Proof of (ii): $\operatorname{Frob}_p = \sigma_p$ in $\operatorname{Gal}(\mathbb{Q}(\zeta_\ell)/\mathbb{Q}) \simeq \mathbb{F}_\ell^\times$ (5.2), restriction to the quadratic subfield is the quotient by the squares (3.8(v)). Checked for $\ell \leq 23$, $p < 2000$ (Part 12). Reference: Washington Ch. 2; Ireland-Rosen Ch. 6. Paper: Theorem thm:main (the hypothesis on $\operatorname{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{\ell^*}))$), the set S_f (Section 4.2), Lemma lem:h0 ("primes inert in $\mathbb{Q}(\sqrt{\ell^*})$"), Section sec:anomaly ("some prime $p < 1223$ inert in $\mathbb{Q}(\sqrt{-3})$").

8.6.4 5.4 cor:ant:qr (corollary; a second proof of quadratic reciprocity) P

For distinct odd primes p, q: $\left(\frac{q^*}{p}\right) = \left(\frac{p}{q}\right)$, i.e. $\left(\frac{p}{q}\right)\left(\frac{q}{p}\right) = (-1)^{\frac{p-1}{2}\frac{q-1}{2}}$; and $\left(\frac{-1}{p}\right) = (-1)^{(p-1)/2}$, $\left(\frac{2}{p}\right) = (-1)^{(p^2-1)/8}$ from $\mathbb{Q}(i) = \mathbb{Q}(\zeta_4)$ and $\mathbb{Q}(\sqrt{2}) \subset \mathbb{Q}(\zeta_8)$. Proof: compute $\chi_{q^*}(\mathrm{Frob}_p)$ by 4.2(i) and by 5.3(ii). (Chapter 2 proves reciprocity by Gauss sums; this proof shows why the character of $\mathbb{Q}(\sqrt{q^*})$ is the Legendre symbol mod q, and may be given as a remark.) Reference: Neukirch I.10 (Exercise) or Washington Ch. 2 (cite chapter). Paper: Lemma lem:twist (χ_{q^*} "of conductor q" is the character of $\mathbb{Q}(\sqrt{q^*})$).

8.6.5 5.5 thm:ant:kroneckerweber (theorem, Kronecker-Weber) R

Every finite abelian extension $K/\mathbb{Q}$ is contained in $\mathbb{Q}(\zeta_n)$ for some n; the least such n (taken $\not\equiv 2 \bmod 4$) is the *conductor* f_K of K, and the primes dividing f_K are exactly the primes ramified in K. Equivalently $\mathbb{Q}^{\mathrm{ab}} = \mathbb{Q}(\mu_\infty)$, and the cyclotomic characters induce $G_{\mathbb{Q}}^{\mathrm{ab}} = \mathrm{Gal}(\mathbb{Q}^{\mathrm{ab}}/\mathbb{Q}) \simeq \varprojlim_n (\mathbb{Z}/n)^\times = \hat{\mathbb{Z}}^\times = \prod_\ell \mathbb{Z}_\ell^\times$. For a quadratic field $f_K = |D|$ (Chapter 5, galois:sqrt-lstar (e) gives the inclusion directly). Reference: Washington Ch. 14; Neukirch V.1 (local form) and VI (global class field theory; cite chapters); Cox Section 8 (cite section). Paper: background (completes Chapter 5's galois:absolute-galois-group (d)); used through 5.7.

8.6.6 5.6 thm:ant:artinQ (theorem; Artin reciprocity over $\mathbb{Q}$, explicit form) P given 5.5

Let $K/\mathbb{Q}$ be abelian with $K \subset \mathbb{Q}(\zeta_n)$. (i) The *Artin map* $\mathrm{Art}_K \colon (\mathbb{Z}/n)^\times \to \mathrm{Gal}(K/\mathbb{Q})$, $a \mapsto \sigma_a|_K$, is surjective, with kernel H_K, and $K \mapsto H_K$ is an inclusion-reversing bijection between subfields of $\mathbb{Q}(\zeta_n)$ and subgroups of $(\mathbb{Z}/n)^\times$ (Galois theory). (ii) For every prime $p \nmid n$, $\mathrm{Frob}_p(K/\mathbb{Q}) = \mathrm{Art}_K(p \bmod n)$: the Frobenius of an unramified prime in an abelian extension depends only on $p \bmod n$ (3.8(iv), 3.8(v), 5.2); so $p \nmid n$ splits completely in K iff $p \bmod n \in H_K$, and the residue degree of p in K is the order of p in $(\mathbb{Z}/n)^\times/H_K$. (iii) (Stated.) Idelic form: the global Artin map $\mathbb{Q}^\times\backslash\mathbb{A}_{\mathbb{Q}}^\times \to G_{\mathbb{Q}}^{\mathrm{ab}}$ is surjective with kernel the connected component, and $\mathbb{A}_{\mathbb{Q}}^\times = \mathbb{Q}^\times \cdot (\mathbb{R}_{>0} \times \hat{\mathbb{Z}}^\times)$; under the normalisation in which a uniformiser at p maps to an arithmetic Frobenius, the idele $(1, \dots, 1, u_p, 1, \dots)$ with $u_p \in \mathbb{Z}_p^\times$ maps to an element of the inertia group at p acting on μ_{p^∞} by u_p^{-1} (sign pitfall: Deligne's normalisation sends uniformisers to geometric Frobenius; say which one is used, and avoid it in the chapter if possible). Reference: (i), (ii) elementary from 5.2, 5.5; (iii) Neukirch VI.5 and VI.6, Milne CFT Ch. V (cite chapters). Paper: background for "the quadratic character of conductor q" (Lemma lem:twist), the table of Section 1.

8.6.7 5.7 thm:ant:dirichletgalois (theorem) P given 5.5

(i) There is a bijection $\chi \leftrightarrow \psi_\chi$ between primitive Dirichlet characters χ (Chapter 2) and continuous homomorphisms $\psi \colon G_{\mathbb{Q}} \to \mathbb{C}^\times$ (these have finite image, Chapter 5, galois:continuous-reps (d)), characterised by $\psi_\chi(\mathrm{Frob}_p) = \chi(p)$ for all $p \nmid f_\chi$; concretely, ψ_χ is χ composed with $G_{\mathbb{Q}} \to \mathrm{Gal}(\mathbb{Q}(\zeta_{f_\chi})/\mathbb{Q}) \simeq (\mathbb{Z}/f_\chi)^\times$. It is multiplicative ($\psi_{\chi\chi'} = \psi_\chi\psi_{\chi'}$ after passing to the primitive character inducing $\chi\chi'$), and the field cut out by ψ_χ has conductor f_χ. (ii) ψ_χ is unramified at p iff $p \nmid f_\chi$; more precisely, with $f_\chi = p^a m$, $p \nmid m$, the image of I_p in $\mathrm{Gal}(\mathbb{Q}(\zeta_{f_\chi})/\mathbb{Q}) \simeq (\mathbb{Z}/p^a)^\times \times (\mathbb{Z}/m)^\times$ is $(\mathbb{Z}/p^a)^\times \times \{1\}$ (5.2), so $\psi_\chi(I_p) = \chi((\mathbb{Z}/p^a)^\times \times \{1\})$; and the Artin conductor of ψ_χ (Chapter 7) equals f_χ (stated). (iii) The same holds for characters with values in $\overline{\mathbb{Q}}_\ell^\times$ or $\overline{\mathbb{F}}_\ell^\times$ of finite order, via a fixed embedding of the roots of unity; and every continuous $G_{\mathbb{Q}} \to \mathbb{Z}_\ell^\times$ factors through $\hat{\mathbb{Z}}^\times$ (5.5), e.g. χ_ℓ is the projection $\hat{\mathbb{Z}}^\times \to \mathbb{Z}_\ell^\times$. (iv) One-dimensional representations of $G_{\mathbb{Q}}$ over $\mathbb{C}$ are thus "the same as" primitive Dirichlet characters; under this dictionary, twisting a newform's coefficients, $a_n \mapsto \chi(n)a_n$, corresponds to tensoring its Galois representation by ψ_χ (Chapters 13, 15). Proof: (i) and (ii) from 5.2 (inertia groups in $\mathbb{Q}(\zeta_n)$) and 5.5. Reference: Washington Ch. 3; Neukirch VII.2 (cite chapter). Paper: the characters χ_{-3}, χ_{-4}, χ_{q^*}, χ_S used both as Galois characters ($\rho \otimes \chi$, Lemma lem:h0, Proposition prop:ns) and as Dirichlet characters ($a_n(g \otimes \chi) = \chi(n)a_n(g)$, Lemma lem:twist (iii), Theorem thm:twist, Theorem thm:qual (b), (c)).

8.6.8 5.8 ex:ant:characters (worked example; the paper's characters in both languages) P

Galois character	field cut out	Dirichlet character	conductor	value at Frob_p, p unramified
χ_{-4}	$\mathbb{Q}(i)=\mathbb{Q}(\zeta_4)$	the nontrivial character mod 4	4	$(-1)^{(p-1)/2}$
$\chi_{-3}=\bar{\chi}_3$	$\mathbb{Q}(\sqrt{-3})=\mathbb{Q}(\zeta_3)$	$\left(\frac{\cdot}{3}\right)$	3	1 iff $p\equiv 1 \bmod 3$
χ_{q^*}, q odd	$\mathbb{Q}(\sqrt{q^*})\subset\mathbb{Q}(\zeta_q)$	$\left(\frac{\cdot}{q}\right)$	q	$\left(\frac{p}{q}\right)$
$\chi_S=\prod_{q\in S}\chi_{q^*}$	$\mathbb{Q}(\sqrt{\prod_{q\in S}q^*})$	$\prod_{q\in S}\left(\frac{\cdot}{q}\right)$	$\prod_{q\in S}q$	$\prod_{q\in S}\left(\frac{p}{q}\right)$
$\chi_{\ell^*}=\bar{\chi}_\ell^{(\ell-1)/2}$	$\mathbb{Q}(\sqrt{\ell^*})$	$\left(\frac{\cdot}{\ell}\right)$	ℓ	$\left(\frac{p}{\ell}\right)$
$\bar{\chi}_\ell$ (values in $\mathbb{F}_\ell^\times$)	$\mathbb{Q}(\zeta_\ell)$	$a\mapsto a \bmod \ell\in\mathbb{F}_\ell^\times$	ℓ	$p \bmod \ell$

Warning (important_notes.md point 6): $\chi_{\ell^*}=\bar{\chi}_\ell$ only for $\ell=3$. Reference: as 4.2, 5.7. Paper: table of Section 1, Lemma lem:euler, Lemma lem:twist, Theorem thm:twist, Lemma lem:h0, Proposition prop:ns, Conjecture conj:euler.

8.7 6. Line-up, Section 6: densities, Chebotarev in number-field form, complex multiplication

8.7.1 6.1 prop:ant:dedekindzeta (definition, theorem R, corollary P)

For a number field K, $\zeta_K(s)=\sum_{\mathfrak{a}} N\mathfrak{a}^{-s}=\prod_{\mathfrak{p}}(1-N\mathfrak{p}^{-s})^{-1}$ (sum over nonzero ideals, product over nonzero primes) converges absolutely for $\mathrm{Re}\, s>1$ (P, from 2.2, 2.3 and $\#\{\mathfrak{p}\mid p\}\le[K:\mathbb{Q}]$). Theorem (R): ζ_K extends meromorphically to $\mathrm{Re}\, s>1-1/[K:\mathbb{Q}]$ with a simple pole at $s=1$ (residue given by the analytic class number formula). Consequences (P given the pole): $\sum_{\mathfrak{p}} N\mathfrak{p}^{-s}\sim\log\frac{1}{s-1}$ as $s\to 1^+$, so the *Dirichlet density* $\delta(X)=\lim_{s\to1^+}\sum_{\mathfrak{p}\in X}N\mathfrak{p}^{-s}/\log\frac{1}{s-1}$ of a set X of primes of K is defined as in Chapter 5 (galois:density) with $N\mathfrak{p}$ in place of p; the primes of degree $f(\mathfrak{p}|p)\ge 2$ form a set of density 0, so the primes of degree one have density one. Reference: Neukirch VII.5 and VII.13 (cite chapters); Lang, Algebraic Number Theory, Ch. VIII (cite chapter). Paper: background ("density one", "almost all p" in Lemma lem:old and Section 3 via Chapter 5).

8.7.2 6.2 thm:ant:chebotarev (theorem, Chebotarev density theorem, number-field form) R

Let L/K be a finite Galois extension of number fields with group G and $C\subset G$ a conjugacy class. The set of primes $\mathfrak{p}$ of K, unramified in L, with $\mathrm{Frob}_{\mathfrak{p}}=C$ (3.8(ii)) has Dirichlet density $\#C/\#G$; it also has natural density $\#C/\#G$ (counting $N\mathfrak{p}\le x$). For G abelian the classes are elements and the statement is the generalised Dirichlet theorem. Chapter 5 (galois:chebotarev) states the case $K=\mathbb{Q}$ with the Frobenius as a black box; here it is restated with the Frobenius of 3.8, and for a general base. Reference: Neukirch VII.13 (Theorem 13.4); Serre, Abelian l-adic representations, Ch. I, Section 2.2. Paper: Lemma lem:old ("by Chebotarev and Brauer-Nesbitt"), Section 3 (the optimal-level theorem applied to "$a_p(g)\equiv a_p(f)$ for all $p\nmid 3N$").

8.7.3 6.3 cor:ant:chebotarev (corollary) P from 6.2

In the setting of 6.2: (i) every $\sigma\in G$ is $\mathrm{Frob}_{\mathfrak{P}}$ for infinitely many $\mathfrak{P}$ (a set of primes $\mathfrak{p}$ of positive density); (ii) the primes of K splitting completely in L have density $1/[L:K]$; for L/K not Galois, the primes of K having a degree-one prime above them in L have positive density (apply (i) in the Galois closure with

3.9); (iii) (Bauer) for L, L' Galois over K: $L \subset L'$ iff almost every prime splitting completely in L' splits completely in L; so a Galois extension is determined by its set of split primes; (iv) $K = \mathbb{Q}$, $L = \mathbb{Q}(\zeta_n)$: Dirichlet's theorem, density $1/\varphi(n)$ for each class in $(\mathbb{Z}/n)^\times$ (by 5.2); (v) L quadratic over $\mathbb{Q}$: the split and the inert primes each have density 1/2; in particular the primes inert in $\mathbb{Q}(\sqrt{\ell^*})$ (the p with $\left(\frac{p}{\ell}\right) = -1$) have density 1/2, and for $\ell = 3$ the primes $p \equiv 2 \pmod 3$ have density 1/2; (vi) if $L/\mathbb{Q}$ is Galois and unramified outside a finite set, the Frobenius elements at primes outside any density-zero set are dense in $\mathrm{Gal}(L/\mathbb{Q})$ (Chapter 5, galois:chebotarev-dense; not redone). Reference: Neukirch VII.13 (Corollaries); Serre, Abelian l-adic representations, Ch. I. Paper: Lemma lem:old, Lemma lem:h0 (the inert primes form a set of density 1/2), Section sec:anomaly.

8.7.4 6.4 `prop:ant:twistdetect` (proposition) P given 6.2 and Chapter 5

Let k be a field of characteristic $\neq 2$ (a finite field, $\overline{\mathbb{F}}_\ell$, or a finite extension of $\mathbb{Q}_\ell$), $\rho\colon G_\mathbb{Q} \to \mathrm{GL}_2(k)$ continuous, semisimple and unramified outside a finite set S, and K a quadratic field with character χ_K. Then $\rho \simeq \rho \otimes \chi_K$ if and only if $\operatorname{tr}\rho(\mathrm{Frob}_p) = 0$ for all (equivalently, for all but finitely many, or for a density-one subset of the) primes $p \notin S$ inert in K. Proof: $\det(X - (\rho \otimes \chi_K)(\mathrm{Frob}_p)) = X^2 - \chi_K(\mathrm{Frob}_p)\operatorname{tr}\rho(\mathrm{Frob}_p)X + \det\rho(\mathrm{Frob}_p)$, which equals $\det(X - \rho(\mathrm{Frob}_p))$ at split p always and at inert p iff $2\operatorname{tr}\rho(\mathrm{Frob}_p) = 0$; conclude with Chapter 5's galois:chebotarev-brauer-nesbitt ($\rho \otimes \chi_K$ is semisimple). Applications: (a) $\bar\rho_{E,3}$ irreducible satisfies $\bar\rho_{E,3} \simeq \bar\rho_{E,3} \otimes \chi_{-3}$ iff $3 \mid a_p$ for almost all $p \equiv 2 \pmod 3$ (Chapter 5, galois:ns-example); (b) $\rho_{E,\ell} \simeq \rho_{E,\ell} \otimes \chi_K$ iff $a_p = 0$ for almost all p inert in K ($\rho_{E,\ell}$ is semisimple: Faltings, stated; or irreducible, e.g. when $\bar\rho_{E,\ell}$ is). Note: this criterion concerns twisting by a *quadratic* character; it says nothing about $\rho \otimes \chi_\ell$, which is never isomorphic to ρ (Chapter 5, galois:twist (c); important_notes.md point 5). Reference: elementary given the cited results. Paper: Lemma lem:h0 (the finiteness sentence, corrected reading), Proposition prop:ns ("$\bar\rho \simeq \bar\rho \otimes \chi_{-3}$"), the hypothesis of Theorem thm:main.

8.7.5 6.5 `thm:ant:cm` (theorem) S for (ii), R for the rest; forward pointer to Chapter 9

Let $E/\mathbb{Q}$ be an elliptic curve with complex multiplication: $\mathrm{End}_{\overline{\mathbb{Q}}}(E) \neq \mathbb{Z}$; then $\mathrm{End}_{\overline{\mathbb{Q}}}(E)$ is an order $\mathcal{O}$ in an imaginary quadratic field K, the endomorphisms are defined over K, and $\sigma \in G_\mathbb{Q}$ acts on them through $\mathrm{Gal}(K/\mathbb{Q})$ (Chapter 9). (i) (R) $\mathcal{O}$ has class number one, so $K = \mathbb{Q}(\sqrt{-d})$ with $d \in \{1, 2, 3, 7, 11, 19, 43, 67, 163\}$, and $j(E)$ is one of 13 rational values. (ii) (S) $\rho_{E,\ell} \simeq \rho_{E,\ell} \otimes \chi_K$ for every ℓ, and $a_p(E) = 0$ for every prime p of good reduction that is inert in K. Sketch: take $\alpha \in \mathcal{O}$ with $\bar\alpha = -\alpha$ (e.g. $\alpha = \sqrt{-d}$ or $2\sqrt{-d}$); it acts on $V_\ell E$ invertibly, and $\rho(\sigma)\alpha\rho(\sigma)^{-1} = \sigma(\alpha) = \chi_K(\sigma)\alpha$, so $\alpha\colon \rho \to \rho \otimes \chi_K$ is an isomorphism; then $\operatorname{tr}\rho(\sigma) = \operatorname{tr}(\alpha^{-1}\rho(\sigma)\alpha) = \chi_K(\sigma)\operatorname{tr}\rho(\sigma)$, so $\operatorname{tr}\rho(\mathrm{Frob}_p) = a_p = 0$ when $\chi_K(\mathrm{Frob}_p) = -1$ (take $\ell \neq p$). (iii) (R) Conversely, if $a_p(E) = 0$ for almost all p inert in a quadratic field K, then $\rho_{E,\ell} \simeq \rho_{E,\ell} \otimes \chi_K$ (6.4(b)) and E has CM by K (Serre's open image theorem, Serre72: for E without CM the image of $\rho_{E,\ell}$ is open in $\mathrm{GL}_2(\mathbb{Z}_\ell)$, so the trace cannot vanish on the coset $\rho(G_\mathbb{Q} \setminus G_K)$); in particular K is imaginary. (iv) (R) For E without CM the set of p with $a_p(E) = 0$ has density 0 (Serre, 1981, "Quelques applications du theoreme de densite de Chebotarev"; add to `bib/ch06.bib`). (v) (R; conductor-discriminant formula, Chapters 7 and 19) Every prime ramified in K divides the conductor of E. (vi) Consequences for $\mathbb{Q}(\sqrt{\ell^*})$: "$a_p(E) = 0$ for almost all p inert in $\mathbb{Q}(\sqrt{\ell^*})$" holds iff E has CM by $\mathbb{Q}(\sqrt{\ell^*})$, which needs $\ell \equiv 3 \pmod 4$ and $\ell \in \{3, 7, 11, 19, 43, 67, 163\}$, and then $\ell \mid N$ by (v); so it never happens in the range $\ell \nmid N$ of Theorem thm:main. Examples (checked: $a_p = 0$ at every good inert $p < 300$, nonzero at many split p; Part 12): $y^2 = x^3 - x$ (conductor 32, CM by $\mathbb{Z}[i]$); $y^2 = x^3 + 1$ (36a1) and $y^2 + y = x^3 - 7$ (27a1), CM by $\mathbb{Q}(\sqrt{-3})$; 121b1 $= [0, -1, 1, -7, 10]$, CM by $\mathbb{Q}(\sqrt{-11}) = \mathbb{Q}(\sqrt{11^*})$. Warning for the writer: the paper's Lemma lem:h0 invokes this fact for $\rho \simeq \rho \otimes \chi_\ell$, which is the wrong character (important_notes.md point 5); the statement to present is (ii), (iii) for χ_{ℓ^*}, and the finiteness of $h_\ell(E)$ needs none of it ($h_\ell(E) \leq 1$ by the determinant argument, Chapter 5). Reference: Silverman ATAEC Ch. II (cite chapter) and Appendix A (the 13 j-invariants); Cox Section 14 (Deuring's reduction theorem); Serre72 Section 4.5 (cite section). Paper: Lemma lem:h0 (last sentence of the statement and the proof), Conjecture conj:euler ("without complex multiplication"), the data sections ("not CM").

8.8 7. Line-up, Section 7: coefficient fields of newforms, primes $\lambda \mid \ell$, reduction mod λ

8.8.1 7.1 prop:ant:coefffield (proposition) P

Let R be a commutative ring which is finitely generated as a $\mathbb{Z}$-module (in the paper the Hecke algebra $\mathbb{T}$, Chapter 13), and $\phi\colon R \to \mathbb{C}$ a ring homomorphism. Then every $\phi(t)$ is an algebraic integer (it is a root of the characteristic polynomial of multiplication by t on R/tors, Cayley-Hamilton; or $\mathbb{Z}[\phi(t)]$ is a finitely generated $\mathbb{Z}$-module, 1.2(ii)); $K_\phi = \mathbb{Q}(\phi(R))$ is a number field and $\phi(R)$ is a subring of $\mathcal{O}_{K_\phi}$ of finite index (an *order*). For $\sigma \in G_\mathbb{Q}$, $\sigma \circ \phi$ is another such homomorphism (the *Galois conjugates*). Applied to the eigencharacter $\phi_f\colon \mathbb{T} \to \mathbb{C}$, $T_n \mapsto a_n(f)$, of a newform f: the *coefficient field* $K_f = \mathbb{Q}(a_n(f) : n \geq 1)$ is a number field, every $a_n(f) \in \mathcal{O}_{K_f}$, and the conjugates $f^\sigma = \sum \sigma(a_n) q^n$ are newforms (stated; Chapter 13); for weight 2 and trivial character K_f is totally real (stated; Chapter 13). For $f = f_E$, $K_f = \mathbb{Q}$, $\mathcal{O} = \mathbb{Z}$ and the only prime above ℓ is $\lambda = \ell$ (as in the paper's Section 4: "$\lambda = \ell$"). Reference: Diamond-Shurman Ch. 6 (Section 6.5; check), Atiyah-Macdonald Ch. 5 (Proposition 5.1). Paper: Lemma lem:crit ("coefficients in the ring of integers $\mathcal{O}$ of a number field"), Lemma lem:old ("with coefficients in $\mathcal{O}$").

8.8.2 7.2 prop:ant:primesabove (proposition) P

Let K be a number field, ℓ a prime. (i) The primes $\lambda \mid \ell$ of $\mathcal{O}_K$ are the maximal ideals containing ℓ; there are at most $[K : \mathbb{Q}]$ of them (3.1); $k_\lambda = \mathcal{O}_K/\lambda \simeq \mathbb{F}_{\ell^f}$; reduction mod λ is the ring homomorphism $\mathcal{O}_K \to k_\lambda$, $a \mapsto \bar{a}$. (ii) For $a, b \in \mathbb{Z}$: $a \equiv b \pmod{\lambda}$ iff $a \equiv b \pmod{\ell}$ ($\lambda \cap \mathbb{Z} = \ell\mathbb{Z}$); for $a \in \mathcal{O}_K$, $\lambda \mid a$ for every $\lambda \mid \ell$ iff $a \in \mathrm{rad}(\ell\mathcal{O}_K)$, and $\ell \mid N_{K/\mathbb{Q}}(a)$ iff some $\lambda \mid \ell$ divides a. (iii) *Enlarging*: if $K \subset K'$ then $\lambda\mathcal{O}_{K'} \neq \mathcal{O}_{K'}$ and every prime λ' of K' dividing $\lambda\mathcal{O}_{K'}$ satisfies $\lambda' \cap \mathcal{O}_K = \lambda$; reduction mod λ' extends reduction mod λ along $k_\lambda \subset k_{\lambda'}$. (iv) *Lying over* (R/P): if $A \subset B$ is an integral extension of rings (e.g. an order $\mathcal{O}[\alpha]$ with α integral over $\mathcal{O}$, or $B = \mathcal{O}_{K'}$) then every prime of A is $\mathfrak{q} \cap A$ for some prime $\mathfrak{q}$ of B, and $\mathfrak{q}$ is maximal iff $\mathfrak{q} \cap A$ is. (v) Example of the step in Lemma lem:old: with α_p a root of $X^2 - a_p(h)X + p$ ($p \neq \ell$), $\mathcal{O}[\alpha_p]$ is integral over $\mathcal{O}$, so there is a prime λ' of $\mathcal{O}[\alpha_p]$ (or of the ring of integers of $K(\alpha_p)$) above λ, and if $X^2 - a_p(h)X + p \equiv (X-u)(X-w) \pmod{\lambda}$ then $\alpha_p \equiv u$ or $\alpha_p \equiv w \pmod{\lambda'}$ (λ' is prime); if $\alpha_p \equiv a_p(f)p$ with $a_p(f) = \pm 1$, then the other root $\beta_p = p/\alpha_p \equiv a_p(f)$ (since p is a unit mod λ'). Reference: Atiyah-Macdonald Ch. 5 (Theorem 5.10, Corollary 5.8); Neukirch I.8. Paper: Lemma lem:crit ("a prime $\lambda \mid \ell$ of $\mathcal{O}$"), Lemma lem:old ("after enlarging $\mathcal{O}$ and λ", "modulo a prime above λ in $\mathcal{O}[\alpha_p]$", the case $p \,\|\, N$ with $\bar{\rho}$ unramified).

8.8.3 7.3 prop:ant:congruencemaximal (proposition) P

Let R be a commutative ring finitely generated as a $\mathbb{Z}$-module, $\mathcal{O} = \mathcal{O}_K$, $\lambda \mid \ell$, and $\phi\colon R \to \mathcal{O}$ a ring homomorphism. (i) $\mathfrak{m} = \ker(R \xrightarrow{\phi} \mathcal{O} \to k_\lambda)$ is a maximal ideal of R with residue field a subfield of k_λ, of characteristic ℓ (a finite subring of a field is a field). (ii) If $\phi'\colon R \to \mathcal{O}$ is another homomorphism with $\phi'(t) \equiv \phi(t) \pmod{\lambda}$ for all $t \in R$ (for $R = \mathbb{T}$: $a_n(g) \equiv a_n(f)$ for all n, i.e. on generators), then ϕ and ϕ' define the same $\mathfrak{m}$. (iii) If $J \subset R$ is an ideal with R/J finite cyclic of order r and $J \subset \mathfrak{m}$, then $\ell \mid r$ (R/J surjects onto $R/\mathfrak{m} \neq 0$, of characteristic ℓ). (iv) For R finite over $\mathbb{Z}$, $R \otimes \mathbb{Z}_\ell$ is the product of its localisations at the maximal ideals containing ℓ (stated; Chapter 3), and ϕ extends to $R \otimes \mathbb{Z}_\ell \to \mathcal{O}_\lambda$ (7.4) factoring through the factor $R_\mathfrak{m}$. Reference: Atiyah-Macdonald Ch. 1, 5, 8 (cite chapters); DDT Section 4.1 (cite section). Paper: Lemma lem:crit ("$I + J$ lies in the kernel of $\mathbb{T} \to \mathcal{O}/\lambda$, a maximal ideal of residue characteristic ℓ, and ℓ divides $\#\mathbb{T}/(I+J) = r$"; "the same statements hold with $\mathbb{Z}$ replaced by $\mathbb{Z}_\ell$"), Lemma lem:hecke and the proof of Theorem thm:quant ("the maximal ideal $\mathfrak{m}$ of $\mathbb{T} \otimes \mathbb{Z}_\ell$ containing ℓ and $\ker \phi_f$", "the other local factors ... are killed by ϕ_f").

8.8.4 7.4 prop:ant:otimesZl (proposition) R here, P in Chapter 7

For a number field K and a prime ℓ: $\mathcal{O}_K \otimes_\mathbb{Z} \mathbb{Z}_\ell \simeq \prod_{\lambda|\ell} \mathcal{O}_{K,\lambda}$ and $K \otimes_\mathbb{Q} \mathbb{Q}_\ell \simeq \prod_{\lambda|\ell} K_\lambda$, where $\mathcal{O}_{K,\lambda}$ is the λ-adic completion, a complete DVR, free over $\mathbb{Z}_\ell$ of rank $e_\lambda f_\lambda$ ($\sum_\lambda e_\lambda f_\lambda = [K : \mathbb{Q}]$, 3.1), with uniformiser

any $\varpi \in \lambda \setminus \lambda^2$ and residue field k_λ; K_λ is a finite extension of $\mathbb{Q}_\ell$ of degree $e_\lambda f_\lambda$. The cases $K = \mathbb{Q}(\zeta_3)$, $\mathbb{Q}(i)$ are 4.9. Reference: Neukirch II.8 (Proposition 8.3; check), Serre, Local Fields, Ch. II. Paper: Section 4 ("λ", "$\mathcal{O}$ a discrete valuation ring" in Lemma lem:ddt417; DFG's $\mathcal{O}$, E_λ), Lemma lem:crit (over $\mathbb{Z}_\ell$), Lemma lem:IV ($\mathbb{Z}_3[\zeta_3]$), Lemma lem:euler ($\mathbb{Z}_\ell[\zeta_3]$).

8.8.5 7.5 ex:ant:level23 (worked example) P for the algebra, R for the modular input

$S_2(\Gamma_0(23))$ is two-dimensional, spanned by a newform f and its Galois conjugate, with coefficient field $K_f = \mathbb{Q}(\sqrt{5})$ (stated; LMFDB 23.2.a.a; the writer must check the q-expansion there, in particular that $a_2(f)$ is a root of $X^2 + X - 1$, before quoting coefficients). Primes of $\mathcal{O}_{K_f} = \mathbb{Z}[\frac{1+\sqrt{5}}{2}]$: $5 = (\sqrt{5})^2$ ramified; 11 splits (5 is a square mod 11), into two primes λ, λ' with residue field $\mathbb{F}_{11}$, giving two different reductions of f mod 11 that are exchanged by $\mathrm{Gal}(K_f/\mathbb{Q})$; 2 is inert ($5 \equiv 5 \pmod 8$), residue field $\mathbb{F}_4$, and a root of $X^2 + X - 1$ reduces to a root of $X^2 + X + 1$, a primitive cube root of unity in $\mathbb{F}_4$. So "$a_n(g) \equiv a_n(f) \pmod \lambda$" in Lemma lem:crit depends on λ, not only on ℓ. Paper: Lemma lem:crit, Lemma lem:old (the forms g, h may have non-rational coefficients), Section 3 Example (all forms there have rational coefficients, so $\lambda = \ell$).

8.9 8. Section-by-section plan with target line counts

Line counts are lines of LaTeX, card excluded (PLAN.md Section 7.3: 1,000 to 2,200).

section	title	contents	target lines
(starred)	Core results at a glance	the card: items 1.1 to 7.5, each stated as above, with tag and \Cref	(excluded)
(starred)	Where the paper uses this chapter	lem:IV; lem:euler; lem:h0 (corrected reading); lem:crit, lem:old; Section 1.4, lem:twist, thm:twist, thm:quant; table of Section 1 (χ_{-3}, χ_{-4}); the hypothesis of thm:main ($\mathbb{Q}(\sqrt{\ell^*})$); Section 1.1 ($I_p$, Frob_p, L_p)	40
1	Number fields and their rings of integers	1.1 to 1.6; convention: $\overline{\mathbb{Q}} \subset \mathbb{C}$ as in Chapter 5	230
2	Dedekind domains, ideals, norms and class groups	2.1 to 2.6	230
3	Splitting of primes, ramification and Frobenius	3.1 to 3.12; Convention 3.7 (arithmetic versus geometric) as a convention environment; paperlink for lem:euler (3.12)	330

section	title	contents	target lines
4	Quadratic fields, $\mathbb{Z}[i]$, $\mathbb{Z}[\zeta_3]$ and $\mathbb{Z}_\ell[\zeta_3]$	4.1 to 4.10; a table of splitting in $\mathbb{Q}(i)$, $\mathbb{Q}(\sqrt{-3})$, $\mathbb{Q}(\sqrt{-7})$, $\mathbb{Q}(\sqrt{-11})$, $\mathbb{Q}(\sqrt{5})$; `paperlink` for lem:IV (4.7, 4.8) and lem:euler (4.10)	300
5	Cyclotomic fields and class field theory over $\mathbb{Q}$	5.1 to 5.8; the table of 5.8; `paperlink` for lem:twist, thm:twist (χ_{q^*}, χ_S)	230
6	Densities, Chebotarev, complex multiplication	6.1 to 6.5; `paperlink` for lem:h0 with a `warning` on the χ_ℓ versus χ_{ℓ^*} confusion	160
7	Coefficient fields and primes above ℓ	7.1 to 7.5; `paperlink` for lem:crit and lem:old	140
8	Exercises	18 exercises (Part 10)	120
9	Notes and references	Neukirch, Marcus, Ireland-Rosen, Washington, Serre (Local Fields; Abelian l-adic representations; Serre72), Cox, Atiyah-Macdonald, Lang (Algebra; Algebraic Number Theory), Silverman ATAEC, Milne CFT, DDT; which theorem numbers were cited and which were left at chapter level	40
	total		**1820**

8.10 9. Dependencies and forward pointers

Results from other chapters used here (by chapter label only): * Chapter 1 (`ch:groups`): group actions, orbits and stabilisers (decomposition groups as stabilisers, 3.5, 3.9); cyclic groups and $(\mathbb{Z}/n)^\times$; profinite groups, $\hat{\mathbb{Z}}$, $\mathbb{Z}_\ell$; $1+\ell M_2(\mathbb{Z}_\ell)$ torsion-free for ℓ odd (item 7.4 there; used in 3.12). * Chapter 2 (`ch:rings`): rings, ideals, PIDs and UFDs, Euclidean domains and the proofs that $\mathbb{Z}[i]$ and $\mathbb{Z}[\zeta_3]$ are Euclidean (4.5), Gauss's lemma (1.2), Eisenstein's criterion (4.7, 5.1), CRT (2.4), finite fields, Legendre/Jacobi/Kronecker symbols and quadratic reciprocity (4.1, 4.2), Dirichlet characters, conductors, primitivity (5.7), cyclotomic polynomials. * Chapter 3 (`ch:modules`): finitely generated modules over a PID (1.4, 3.1), localisation, Noetherian rings, DVRs and the DVR criterion (2.1, 2.4, 4.7), $\mathbb{Z}_\ell$ and its completion properties (4.7, 4.9), torsion-free modules over a DVR are free (4.8), products of local rings for finite $\mathbb{Z}_\ell$-algebras (7.3). * Chapter 4 (`ch:reps`): Brauer-Nesbitt (6.4, via Chapter 5), traces of induced representations and the trace-zero lemma (6.5), commutants of a cyclic endomorphism (4.10), the module side of $\mathbb{Z}_\ell[\zeta_3]$ acting on lattices (4.8, 4.10). * Chapter 5 (`ch:galois`): all of finite Galois theory, primitive elements, $\mathrm{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q}) \simeq (\mathbb{Z}/n)^\times$, $\mathbb{Q}(\sqrt{\ell^*}) \subset \mathbb{Q}(\zeta_\ell)$, quadratic characters χ_d (C3), cyclotomic character (C4), Krull

topology and $G_{\mathbb{Q}}$, continuous representations, galois:chebotarev-brauer-nesbitt and galois:chebotarev-dense, galois:twist, galois:index-two-restriction. Chapter 6 supplies the black boxes Chapter 5 states (see Part 0). * Background: linear algebra (trace forms, Gram determinants, Cayley-Hamilton), complex analysis (only for the statement of the pole of ζ_K).

Black boxes imported from later chapters (stated with a forward `\Cref`, never used in a proof of an earlier item): Hensel's lemma and completions (Chapter 7: 4.9, 7.4); the tame-inertia relation in profinite form (Chapter 7: after 3.11); Kraus's theorem on the image of inertia (Chapters 10, 11: hypothesis of 3.12); the Galois action on endomorphisms of a CM curve and Deuring's theorem (Chapter 9: 6.5); Hecke algebras and newforms (Chapters 12, 13: 7.1, 7.5); Faltings's semisimplicity and Serre's open image theorem (Chapter 11: 6.4, 6.5).

Forward pointers (chapters that use this one): * Chapter 7 (`ch:local`): completions $\mathcal{O}_{K,\lambda}$, K_λ, $\mathcal{O}\otimes\mathbb{Z}_\ell = \prod\mathcal{O}_\lambda$ (7.4); $D_p \simeq G_{\mathbb{Q}_p}$ (3.10(v)); $\mathbb{Z}_3[\zeta_3]$ as the ring of integers of $\mathbb{Q}_3(\sqrt{-3})$ (4.7); ramified quadratic extensions $\mathbb{Q}_p(\sqrt{p^*})$ and the local behaviour of χ_{q^*} (4.2(ii), 4.3); tame inertia and $\mathrm{Frob}\ t\ \mathrm{Frob}^{-1} = t^q$ (3.11); geometric Frobenius in local L-factors (3.7); Artin conductors of characters (5.7(ii)). * Chapter 9 (`ch:ellcurves`): CM curves, $a_p = 0$ at inert primes (6.5), quadratic twists by χ_d with $a_p(E^{(d)}) = \chi_d(p)a_p(E)$ (4.2, 5.7), Frobenius characteristic polynomial $X^2 - a_pX + p$ (3.7). * Chapter 10 (`ch:reduction`): the twist by $\mathbb{Q}(\sqrt{q^*})$ is unramified outside q (4.3), χ_{q^*} at q. * Chapter 11 (`ch:galrep`): Lemma lem:IV in full (4.6 to 4.8), Remark rem:types, Lemma lem:euler's inertia analysis (3.12, 4.10), Proposition prop:ns (χ_{-3}, 6.4(a)), Lemma lem:h0 (5.3, 6.4, 6.5), unramified representations and $\rho(\mathrm{Frob}_p)$ (3.10). * Chapters 12, 13 (`ch:modforms`, `ch:hecke`): coefficient fields, orders, primes λ (7.1, 7.2), twisting newforms by Dirichlet characters (5.7(iv)), Atkin-Li (conductor q of χ_{q^*}, 5.8). * Chapter 15 (`ch:modgalrep`): λ-adic representations of newforms over $\mathcal{O}_\lambda$ (7.4), Lemma lem:old (7.2(v), 7.3), Chebotarev (6.2). * Chapter 19 (`ch:lfunctions`): Euler factors with geometric Frobenius (3.7), the values $\chi_{-3}(q)$, $\chi_{-4}(q)$ in the table of Section 1 (4.2, 4.10), Dedekind zeta and Dirichlet L-functions (6.1, 5.7). * Chapter 20 (`ch:congruence`): congruence ideals over $\mathcal{O}$, maximal ideals of $\mathbb{T}\otimes\mathbb{Z}_\ell$ (7.3). * Chapters 21, 22 (`ch:paper`, `ch:computation`): every hook in Part 0; the PARI checks of splitting and of CM.

8.11 10. Coverage table and exercises

PLAN.md scope item, Section 6 map row, or paper hook	line-up items
number fields	1.1
rings of integers, integral closure	1.2, 1.4, 1.5
Dedekind domains and unique factorisation of ideals	2.1, 2.2, 2.4
norms of ideals	1.3, 2.3
splitting of primes in extensions (e, f, g; $efg = n$)	3.1, 3.5
ramified, inert, split primes	3.1, 3.3, 4.1, 5.2
Dedekind-Kummer	3.2, 3.3, 2.6
the discriminant and ramification	1.4, 3.4, 5.1
quadratic fields in full; splitting governed by the Kronecker symbol	1.5, 4.1, 4.4
the quadratic character attached to $\mathbb{Q}(\sqrt{d})$ as a Dirichlet character	4.2, 5.7, 5.8
cyclotomic fields; splitting governed by $p \bmod n$	5.1, 5.2
$\mathbb{Q}(\zeta_3) = \mathbb{Q}(\sqrt{-3})$	3.3(b), 4.4(b), 4.6, 5.3(iii)
$\mathbb{Q}(\sqrt{\ell^*})$ inside $\mathbb{Q}(\zeta_\ell)$ (coordinated with Chapter 5)	5.3, 5.4
decomposition groups, inertia groups	3.5, 3.6, 3.10
the Frobenius element Frob_P and its conjugacy class	3.8, 3.10

PLAN.md scope item, Section 6 map row, or paper hook	line-up items
arithmetic versus geometric Frobenius	3.7
the Frobenius of an unramified prime in an abelian extension	3.8(iv), 5.6(ii)
class field theory over $\mathbb{Q}$: Artin map, Kronecker-Weber	5.5, 5.6
Dirichlet characters as one-dimensional representations of $G_{\mathbb{Q}}$	5.7, 5.8
class group and units (statements, examples)	1.6, 2.5, 2.6, 4.5
$\mathbb{Z}[\zeta_3]$ and $\mathbb{Z}[i]$ as Euclidean domains	4.5 (proof in Chapter 2)
$3 = -\zeta_3^2(1-\zeta_3)^2$; $\mathbb{Z}_3[\zeta_3]$ a DVR, $\pi = 1-\zeta_3$, residue field $\mathbb{F}_3$, $\pi^2 = -3$ up to a unit	4.6, 4.7 (local interpretation in Chapter 7)
primes λ above ℓ in the coefficient field of a newform; reduction mod λ	7.1, 7.2, 7.3, 7.4, 7.5
density of primes; Chebotarev in number-field form (coordinated with Chapter 5)	6.1, 6.2, 6.3
primes inert in $\mathbb{Q}(\sqrt{\ell^*})$ and CM by that field ($a_p = 0$ at inert primes)	5.3, 6.3(v), 6.4, 6.5
hook: Lemma lem:IV ($\mathbb{Z}_3[\zeta_3]$ as DVR)	4.6, 4.7, 4.8, 3.12
hook: Lemma lem:euler (Frobenius centralises or inverts σ by q mod 3)	3.11, 3.12
hook: Lemma lem:euler ($\mathbb{Z}_\ell[\zeta_3] = \mathrm{End}_{I_q}(V)$; $\chi_{-3}(q)$, $\chi_{-4}(q)$ on the invariant line; L-factor convention)	4.9, 4.10, 3.7
hook: Lemma lem:h0 ($a_p = 0$ at primes inert in $\mathbb{Q}(\sqrt{\ell^*})$ forces CM; important_notes.md point 5)	5.3, 6.4, 6.5
hook: Lemma lem:crit (rings of integers $\mathcal{O}$, $\lambda \mid \ell$, maximal ideal of residue characteristic ℓ)	7.1, 7.2, 7.3
hook: Lemma lem:old (enlarging $\mathcal{O}$ and λ; a prime above λ in $\mathcal{O}[\alpha_p]$)	7.2, 7.3, 6.2
hook: the twist by $\mathbb{Q}(\sqrt{q^*})$ (Section 1.4, Lemma lem:twist, Theorems thm:twist, thm:quant)	4.3, 4.4, 5.7, 5.8
hook: χ_{-3}, χ_{-4} of $\mathbb{Q}(\sqrt{-3})$ and $\mathbb{Q}(i)$ (table of Section 1, Proposition prop:forms)	4.2, 5.8, 4.10
Section 6 row: $\chi_{-3}, \chi_{-4}, \chi_{q^*}, q^*, \ell^*$, Kronecker symbols (Chapters 2, 6)	4.1, 4.2, 4.3, 5.3, 5.7, 5.8
Section 6 row: image of inertia on $T_\ell E$ at additive primes, unipotent elements, $\mathbb{Z}_3[\zeta_3]$ (Chapters 4, 6, 7, 11)	3.11, 3.12, 4.6, 4.7, 4.8, 4.10
also: Section 1.1 (I_p, Frob_p, V^{I_p}, L_p), Lemma lem:fpr ($L_p(A_f,1) = L_p(A_f(1),0)$)	3.7, 3.10
also: Chapter 5's black boxes (Frobenius, Dedekind's theorem, $\chi_d(\mathrm{Frob}_p)$, Kronecker-Weber, Chebotarev)	3.8, 3.9, 3.10, 4.2, 5.5, 6.2

Exercises (Section 8 of the chapter), graded from easy to harder: 1. Compute $\mathcal{O}_K$ and d_K for $\mathbb{Q}(\sqrt{-7})$ and $\mathbb{Q}(\sqrt{13})$, and factor 2, 3, 7, 13 in each. Show that Dedekind-Kummer with $\theta = \sqrt{-7}$ gives the wrong answer at 2, and explain why (3.3(d)). 2. Show that $\mathbb{Z}[\sqrt{-3}]$ is not integrally closed and that $4 = 2\cdot 2 = (1+\sqrt{-3})(1-\sqrt{-3})$ are two factorisations into irreducibles there; what happens in $\mathbb{Z}[\zeta_3]$? 3. Factor 2, 3, 5, 7, 13 in $\mathbb{Z}[i]$ and in $\mathbb{Z}[\zeta_3]$. Verify $3 = -\zeta_3^2(1-\zeta_3)^2$, $\sqrt{-3} = 1+2\zeta_3 = \zeta_3(1-\zeta_3)$ and $(1-\zeta_3)^2 = -3\zeta_3$. 4. Show that $p \neq 2, 5$ splits in $\mathbb{Q}(\sqrt{-5})$ iff $p \equiv 1, 3, 7, 9 \pmod{20}$, and that the primes

above 3 and 7 are not principal; deduce $h(\mathbb{Q}(\sqrt{-5})) \geq 2$. 5. In $K = \mathbb{Q}(2^{1/3})$ factor 5, 7 and 31 and check $\sum e_i f_i = 3$. Using 3.9 in the splitting field $\mathbb{Q}(2^{1/3}, \zeta_3)$, determine Frob_p (up to conjugacy) for $p = 5, 7, 31$ and explain why the residue degrees above 5 differ. 6. Prove directly that an odd prime p ramifies in $\mathbb{Q}(\sqrt{d})$ iff $p \mid d$, without 3.4. 7. Factor 2, 13, 29 in $\mathbb{Q}(\zeta_7)$ and check against 5.2. Identify the quadratic subfield and check that its splitting of each p agrees with $\left(\frac{p}{7}\right)$. 8. Prove quadratic reciprocity from 4.2 and 5.3 (5.4), including $\left(\frac{2}{p}\right)$ via $\mathbb{Q}(\zeta_8)$. 9. Show that $\mathbb{Q}(\sqrt{q^*})$ is the only quadratic field ramified only at the odd prime q, and list the quadratic fields unramified outside $\{3, 7\}$ (answer: $\mathbb{Q}(\sqrt{-3})$, $\mathbb{Q}(\sqrt{-7})$, $\mathbb{Q}(\sqrt{21})$). Which character is χ_S for $S = \{3, 7\}$, and what is its conductor? 10. Prove that $\mathbb{Z}_3[\zeta_3]$ is a DVR (4.7) and compute v_π of $1 + \zeta_3$, $2 + \zeta_3$, 3 and $\sqrt{-3}$ (answers $0, 1, 2, 1$). 11. Check the paper's Lemma lem:IV algebra: if $\sigma \in \mathrm{GL}_2(\mathbb{Z}_3)$ has order 3 and determinant 1, then $\mathbb{Z}_3^2$ is free of rank one over $\mathbb{Z}_3[\sigma] \simeq \mathbb{Z}_3[\zeta_3]$, and $\sigma \bmod 3$ is a nontrivial unipotent with a fixed line. Give an explicit such σ (for example the companion matrix of $X^2 + X + 1$). 12. Determine $\mathbb{Z}_\ell[\zeta_3]$ for $\ell = 2, 5, 7$. For $\ell = 7$ find the roots of $X^2 + X + 1 \bmod 7$ and mod 49 (answers $2, 4$ and $18, 30$) and write down the idempotents of $\mathbb{Z}_7[\zeta_3] \simeq \mathbb{Z}_7 \times \mathbb{Z}_7$ modulo 49. 13. Check the paper's claim in Lemma lem:euler: if $\Phi\sigma\Phi^{-1} = \sigma^q$ with σ of order 3, then $\Phi(2\sigma + 1)\Phi^{-1} = \chi_{-3}(q)(2\sigma + 1)$; state and prove the analogue with τ of order 4 and $\chi_{-4}(q)$, and deduce the entries $1 - \chi_{-3}(q)q^{-s}$ and $1 - \chi_{-4}(q)q^{-s}$ of the table of Section 1. 14. Check the paper's sign conventions: for E with split multiplicative reduction at q and $\rho(\mathrm{Frob}) = \left(\begin{smallmatrix} q & c' \\ 0 & 1 \end{smallmatrix}\right)$, show that the arithmetic Frobenius acts on $(\mathrm{ad}^0 V)^{I_q}$ by q, and deduce that the factor $1 - q^{-1-s}$ of the table requires the geometric Frobenius (3.7). 15. For $y^2 = x^3 - x$, show $a_p = 0$ for every $p \equiv 3 \pmod 4$ by the substitution $x \mapsto -x$, and compute $a_5 = -2$, $a_{13} = 6$, $a_{17} = 2$ by counting points. Relate to 6.5(ii). 16. For 11a1 show, using 6.4 and $a_2 = -2$, that $\bar{\rho}_{E,3} \not\simeq \bar{\rho}_{E,3} \otimes \chi_{-3}$. For $y^2 = x^3 - 6x - 8$ (Chapter 5, galois:ns-example) explain why $3 \mid a_p$ for the primes $p \equiv 2 \pmod 3$. 17. Let $K_f = \mathbb{Q}(\sqrt{5})$ (level 23, 7.5). Factor 2, 5, 11 in $\mathbb{Z}[\frac{1+\sqrt{5}}{2}]$, find the roots of $X^2 + X - 1 \bmod 11$ (answers 3, 7), and describe the two reductions mod the primes above 11. 18. Prove 7.3: if $\phi\colon \mathbb{T} \to \mathcal{O}$ and $\lambda \mid \ell$, then $\ker(\mathbb{T} \to \mathcal{O}/\lambda)$ is maximal of residue characteristic ℓ; deduce the last step of the paper's Lemma lem:crit ($\ell \mid \#\mathbb{T}/(I + J)$).

8.12 11. Notes for the writer

Conventions to fix (Section 1 and Section 3 of the chapter, as `convention` environments): * $\overline{\mathbb{Q}} \subset \mathbb{C}$ fixed (Chapter 5, C1); $\bar{\mathbb{Z}}$ its ring of algebraic integers; primes of $\bar{\mathbb{Z}}$ written $\bar{\mathfrak{P}}$. * Frobenius (3.7): $\mathrm{Frob}_{\bar{\mathfrak{P}}}$, Frob_p are *arithmetic*; Frob_p^{-1} is geometric; L-factors are $\det(1 - \mathrm{Frob}_p^{-1} p^{-s} \mid V^{I_p})^{-1}$. State the dictionary once, with the multiplicative example, and use the same words as Chapter 5 (C2). Appendix B (`ch:notation`) must record the same convention. * χ_d, χ_{q^*} (never χ_q, decision D3), χ_{-3}, χ_{-4}, χ_S, χ_{ℓ^*} exactly as in Chapter 5 (C3, C4); the Dirichlet character attached to χ_d is $\left(\frac{D}{\cdot}\right)$ with D the *discriminant*, not d (e.g. $\chi_{-1} = \chi_{-4}$ has conductor 4; χ_2 has conductor 8). * $\mathbb{Z}_\ell[\zeta_3]$ means $\mathbb{Z}_\ell[X]/(X^2 + X + 1) = \mathbb{Z}_\ell \otimes \mathbb{Z}[\zeta_3]$, not a subring of $\overline{\mathbb{Q}}_\ell$ (for $\ell \equiv 1 \pmod 3$ it is $\mathbb{Z}_\ell \times \mathbb{Z}_\ell$). Say so explicitly where it first appears (4.9). * $\sqrt{-3}$ always denotes $1 + 2\zeta_3 = \zeta_3 - \zeta_3^{-1}$ (fixing the sign); $\pi = 1 - \zeta_3$; $\zeta_3 = e^{2\pi i/3}$ under $\overline{\mathbb{Q}} \subset \mathbb{C}$. Do not use π for 3.14159… in the same section (write ϖ for a general uniformiser). * Labels prefixed `...:ant:...`; the names in the line-up are ready to use as label stems.

Pitfalls: * Dedekind-Kummer needs $p \nmid [\mathcal{O}_K : \mathbb{Z}[\theta]]$; give the $\mathbb{Q}(\sqrt{-7})$ counterexample (3.3(d)). Equal e and equal f hold only in Galois extensions; give the $\mathbb{Q}(2^{1/3})$ example. * "Unramified" for a representation means trivial on *every* inertia group above p (equivalently one, by conjugacy); $\rho(\mathrm{Frob}_p)$ is defined only up to conjugacy and only for unramified ρ. At a ramified prime, "a Frobenius" is a coset of inertia; the paper's "Frob" in Lemmas lem:euler and lem:tamfactor is such a lift, and every statement made there is independent of the lift (3.12; Chapter 7). * Arithmetic versus geometric Frobenius: the paper's $L_p(A_f, s)$ at a multiplicative prime and its $L_p(A_f, 1) = L_p(A_f(1), 0)$ are correct only with the geometric Frobenius (Part 13, point 1); never "fix" the table, fix the convention. For the other types the eigenvalue sets are inversion-stable and nothing changes. * The paper's "$\mathbb{Z}_\ell[\zeta_3] \simeq \mathrm{End}_{I_q}(V)$" is to be read as in 4.10(v) (Part 13, point 2); the argument only uses the trace-zero line and Frobenius's action on it. * Lemma lem:h0 (important_notes.md point 5): present the CM fact (6.5) for the *quadratic* character χ_{ℓ^*} and say explicitly that the finiteness of $h_\ell(E)$ follows from the determinant argument of Chapter 5, not from CM. Do not reproduce the paper's sentence "$\rho \simeq \rho \otimes \chi_\ell$ would force $a_p = 0 \ldots$". * $\chi_{\ell^*} \neq \bar{\chi}_\ell$ for $\ell \geq 5$

(important_notes.md point 6); in 5.8 put both in the table. * The Kronecker symbol at $p = 2$ ($D \equiv \pm 1$ versus ± 3 mod 8) and the sign of q^* are where errors creep in; the verification log checks them; keep the explicit $p = 2$ rows. * Idelic class field theory has two sign normalisations (5.6(iii)); if the chapter mentions ideles at all, state which, and prefer the explicit cyclotomic form 5.6(i), (ii), which is all the paper needs. * Chebotarev: only Dirichlet density positivity is needed downstream; do not present the natural-density version as proved. Brauer-Nesbitt is in Chapter 4 and its Galois form in Chapter 5; cite, do not redo. * 6.5: Deuring's theorem ($a_p = 0$ at inert good primes) has a two-line proof from "$\alpha\colon \rho \to \rho \otimes \chi_K$"; give it as a sketch with the Galois action on endomorphisms quoted from Chapter 9. The converse and Serre's density zero result are stated only. * Minkowski bound for $\mathbb{Q}(\sqrt{-11})$ is $2.11 > 2$: class number one there needs "2 is inert" (corrected in 2.5). * 7.5 depends on LMFDB data (23.2.a.a) that the outline author recalled but could not verify offline: check the Hecke polynomial $X^2 + X - 1$ of T_2 before quoting it, or replace the example by the algebra alone.

Leave to other chapters: * Chapter 2: Euclidean algorithms in $\mathbb{Z}[i]$ and $\mathbb{Z}[\zeta_3]$, Legendre/Jacobi/Kronecker symbols, quadratic reciprocity by Gauss sums, Dirichlet characters and conductors, cyclotomic polynomials, v_p on $\mathbb{Q}$. * Chapter 5: the Galois groups of $\mathbb{Q}(\zeta_n)$ and $\mathbb{Q}(\sqrt{d})$, $\mathbb{Q}(\sqrt{\ell^*}) \subset \mathbb{Q}(\zeta_\ell)$ via the discriminant or Gauss sums, Kummer theory, Chebotarev over $\mathbb{Q}$ and its Brauer-Nesbitt corollaries, the index-two criterion, complex conjugation and absolute irreducibility, the determinant argument for $h_\ell(E) \le 1$. * Chapter 7: completions, Hensel, $\mathbb{Q}_p$, local fields, $D_p \simeq G_{\mathbb{Q}_p}$, the higher ramification groups, tame inertia $\prod_{\ell \neq p} \mathbb{Z}_\ell(1)$ and Frob t $\mathrm{Frob}^{-1} = t^p$ in profinite form, the local Kronecker-Weber and local class field theory, Artin conductors, Weil-Deligne representations, the local interpretation of $\mathbb{Z}_3[\zeta_3]$. * Chapters 9 to 11: CM theory proper, Kraus's theorem, the inertia images of $\rho_{E,\ell}$, the elliptic-curve content of Lemmas lem:IV, lem:euler, lem:h0 and Proposition prop:ns; this chapter supplies only the arithmetic of the rings and fields involved. * Chapters 12, 13, 15: Hecke algebras, newforms, their Galois representations; here only the algebraic integrality and the primes λ.

8.13 12. Verification log (computations done for this outline)

Short Python/sympy scripts in `/tmp/ch06` (`check1.py`, `check2.py`, `check3.py`, and one-liners). * Splitting in quadratic fields versus the Kronecker symbol $\left(\frac{D}{p}\right)$ (including $p = 2$), by factoring the minimal polynomial of $\sqrt{d}$ or $(1 + \sqrt{d})/2$ mod p: 23 fundamental discriminants $(-3, -4, -7, -8, -11, -15, -19, -20, -23, 5, 8, 12, 13, -163, 17, 21, 24, -24, 28, -52, -115, 85, -67)$, all $p < 400$: no mismatch. * $p \neq \ell$ inert in $\mathbb{Q}(\sqrt{\ell^*})$ iff $\left(\frac{p}{\ell}\right) = -1$, including $p = 2$: $\ell \le 23$, $p < 2000$: no mismatch. $\chi_{-3}(p) = 1$ iff $p \equiv 1 \pmod 3$, $\chi_{-4}(p) = 1$ iff $p \equiv 1 \pmod 4$: $p < 3000$. Splitting in $\mathbb{Q}(\sqrt{-5})$ iff $p \equiv 1, 3, 7, 9 \pmod{20}$: $p < 3000$. Split primes of $\mathbb{Q}(\sqrt{-7})$ are $\equiv 1, 2, 4 \pmod 7$. * Discriminants of the quadratic generators equal D; $\mathrm{disc}(\Phi_n)$ equals the formula of 5.1(iii) for $n \in \{3, 4, 5, 7, 8, 9, 11, 12, 13, 15, 16, 20\}$ (e.g. -3, -4, 125, -16807, 256). * Splitting in $\mathbb{Q}(\zeta_n)$: $\Phi_n \bmod p$ has $\varphi(m)/f$ factors of degree f with multiplicity $\varphi(p^a)$, for all $3 \le n \le 40$, $n \not\equiv 2 \bmod 4$, $p < 80$: no mismatch. Examples of 5.2 confirmed. * Dedekind-Kummer examples: $X^3 - 2$ mod $2, 3, 5, 7, 11, 31$ (patterns 1^3, 1^3, (1)(2), (3), (1)(2), (1)(1)(1); mod 5: $(X + 2)(X^2 - 2X - 1)$); $X^2 + X + 1$ mod $2, 3, 5, 7, 13$ (mod 7 roots 2, 4); $X^2 + 7 \equiv (X + 1)^2$ and $X^2 - X + 2 \equiv X(X + 1)$ mod 2; $X^2 + 5$ mod $2, 3$; $X^2 + 1$ mod $2, 3, 5, 13$. * Identities in $\mathbb{Z}[\zeta_3]$ (reduction mod $z^2 + z + 1$): $\pi^2 = -3\zeta_3$, $-\zeta_3^2\pi^2 = 3$, $(1 + 2\zeta_3)^2 = -3$, $\zeta_3\pi = 1 + 2\zeta_3$, $(3 + \zeta_3)(3 + \zeta_3^2) = 7$, $(2 - \zeta_3)(2 - \zeta_3^2) = 7$, $(4 + \zeta_3)(4 + \zeta_3^2) = 13$, $1 + \zeta_3 = -\zeta_3^2$; in $\mathbb{Z}[i]$: $-i(1 + i)^2 = 2$, $(2 + i)(2 - i) = 5$, $(3 + 2i)(3 - 2i) = 13$. * Representations: $p = a^2 - ab + b^2$ iff $p = 3$ or $p \equiv 1 \pmod 3$ ($p < 200$); $p = x^2 + 3y^2$ iff $p \equiv 1 \pmod 3$ ($5 \le p < 300$); $p = x^2 + y^2$ iff $p = 2$ or $p \equiv 1 \pmod 4$ ($p < 300$). * Minkowski bounds $(2/\pi)\sqrt{|d|}$: 1.10 (-3), 1.27 (-4), 1.68 (-7), 1.80 (-8), 2.11 (-11), 2.85 (-20), 3.05 (-23); real: 1.12 (5), 1.41 (8). * CM curves: $a_p = 0$ at every good $p < 300$ inert in the CM field, and $a_p \neq 0$ at 26 to 29 split primes, for $y^2 = x^3 - x$ ($\Delta = 64$), $y^2 = x^3 + 1$ ($\Delta = -432$), 121b1 $[0, -1, 1, -7, 10]$ ($\Delta = -11^3$), 27a1 $[0, 0, 1, 0, -7]$ ($\Delta = -3^9$). For $y^2 = x^3 - x$: $a_5 = -2$, $a_{13} = 6$, $a_{17} = 2$, $a_{29} = -10$, $a_{37} = -2$. * 11a1: $a_2 = -2$, $a_3 = -1$, $a_5 = 1$, $a_7 = -2$, $a_{13} = 4$, $a_{17} = -2$, $a_{19} = 0$, $a_{23} = -1$, $a_{29} = 0$ (point counts). * Roots of $X^2 + X + 1$ mod 49: 18, 30; of $X^2 + X - 1$ mod 11: 3, 7. * Not verified (recalled): the q-expansion of 23.2.a.a (7.5); the theorem numbers flagged "check".

8.14 13. Points for important_notes.md

Points in the area of Chapter 6 not already in important/important_notes.md (points 5 and 6 there, on Lemma lem:h0 and on χ_{ℓ^*} versus $\bar{\chi}_\ell$, are used above and not repeated).

1. **Section 1.1 (definition of $L_p(A_f,s)$), Lemma lem:euler (type I_n), Lemma lem:fpr: the Frobenius in the local L-factor must be the geometric one.** Problem: Section 1.1 defines $L_p(A_f,s)=\det(1-\mathrm{Frob}_p\,p^{-s}\mid A_f^{I_p})^{-1}$ with $A_f=\mathrm{ad}^0\,V$, $V=V_\ell E$, without saying which Frobenius; the proof of Lemma lem:tamfactor uses the arithmetic Frobenius on $T=T_\ell E$, with $\rho(\mathrm{Frob})=\left(\begin{smallmatrix}\varepsilon q & c'\\ 0 & \varepsilon\end{smallmatrix}\right)$ and "the adjoint action sends e to qe". With that (arithmetic) Frobenius, the invariant line $(\mathrm{ad}^0\,V)^{I_q}=\mathbb{Q}_\ell e$ at a multiplicative prime has eigenvalue q, so the formula of Section 1.1 would give $1-q^{1-s}$, not the tabulated $1-q^{-1-s}$; and the proof of Lemma lem:euler says "a line with eigenvalue q^{-1}". The table, the value e_ℓ, PARI's factors, Lemma lem:fpr's "$L_p(A_f,1)^{-1}=L_p(A_f(1),0)^{-1}$" and the statement "$\det(1-\mathrm{Frob}_p\mid(V^D)^{I_p})=L_p(A_f,1)^{-1}$" are all correct with the *geometric* Frobenius Frob_p^{-1} (with the arithmetic one the last determinant at a multiplicative prime is $1-q^2$, which has the same ℓ-adic valuation but is a different number). Fix: in Section 1.1 write "$\det(1-\mathrm{Frob}_p^{-1}\,p^{-s}\mid A_f^{I_p})^{-1}$, with Frob_p an arithmetic Frobenius" (or "with Frob_p a geometric Frobenius"), and in the proof of Lemma lem:euler "eigenvalue q^{-1} for the geometric Frobenius". Only the multiplicative and I_n^* cases are affected; for the other types the eigenvalues (±1, $\alpha/\beta,1,\beta/\alpha$) are stable under inversion. No numerical statement changes. Sure: high (direct computation with the paper's own matrices; consistent with $L(V(1),s)=L(V,s+1)$ only for the geometric convention).

2. **Lemma lem:euler, proof (types II, II*, IV, IV*): "$\sqrt{-3}\in\mathbb{Z}_\ell[\zeta_3]\simeq\mathrm{End}_{I_q}(V)$".** Problem (imprecise, no effect on the result): V is a $\mathbb{Q}_\ell$-vector space, so $\mathrm{End}_{I_q}(V)$ is a $\mathbb{Q}_\ell$-algebra; it is $\mathbb{Q}_\ell[\sigma]\simeq\mathbb{Q}_\ell\otimes_{\mathbb{Q}}\mathbb{Q}(\zeta_3)$, which is a field only for $\ell\not\equiv1\pmod3$ and is $\mathbb{Q}_\ell\times\mathbb{Q}_\ell$ for $\ell\equiv1\pmod3$. The integral statement is $\mathrm{End}_{I_q}(T)=\mathbb{Z}_\ell[\sigma]\simeq\mathbb{Z}_\ell\otimes_{\mathbb{Z}}\mathbb{Z}[\zeta_3]$ (T is free of rank one over $\mathbb{Z}_\ell[\sigma]$ in all cases; 4.10(v)). Fix: write "$(\mathrm{ad}^0\,V)^{I_q}$ is the trace-zero part of $\mathrm{End}_{I_q}(V)=\mathbb{Q}_\ell[\sigma]\simeq\mathbb{Q}_\ell\otimes\mathbb{Q}(\zeta_3)$, the line spanned by $2\sigma+1$, whose square is -3". The argument (Frobenius acts on this line through $\mathrm{Gal}(\mathbb{Q}(\zeta_3)/\mathbb{Q})$, i.e. by $\chi_{-3}(q)$) is unchanged; the same remark applies to the types III, III* with $\mathbb{Q}_\ell\otimes\mathbb{Q}(i)$ and the element τ ("the diagonal element"). Sure: high.

3. **Lemma lem:IV, proof: "$\mathbb{Z}_3[\sigma]/(\sigma^2+\sigma+1)\simeq\mathbb{Z}_3[\zeta_3]$".** Problem (notation only): $\mathbb{Z}_3[\sigma]$ already denotes the subring of $\mathrm{End}(T)$ generated by σ, in which $\sigma^2+\sigma+1=0$; the intended object is $\mathbb{Z}_3[X]/(X^2+X+1)$, and the claim is $\mathbb{Z}_3[\sigma]\simeq\mathbb{Z}_3[X]/(X^2+X+1)$ (true because σ is not a scalar, so X^2+X+1 is its minimal polynomial). Fix: "T is a module over $\mathbb{Z}_3[\sigma]\simeq\mathbb{Z}_3[X]/(X^2+X+1)=\mathbb{Z}_3[\zeta_3]$". Sure: high; purely expository.

4. **Supplement to important_notes.md point 5 (Lemma lem:h0, CM).** Even in the corrected reading (with the quadratic character χ_{ℓ^*} in place of χ_ℓ), the CM alternative cannot arise in the range of Theorem thm:main: CM by $\mathbb{Q}(\sqrt{\ell^*})$ requires $\mathbb{Q}(\sqrt{\ell^*})$ imaginary of class number one, i.e. $\ell\in\{3,7,11,19,43,67,163\}$, and a prime ramified in the CM field divides the conductor (conductor-discriminant formula for $\rho_{E,\ell}\simeq\mathrm{Ind}\,\psi$), so $\ell\mid N$ (examples 27a1, 36a1 at $\ell=3$, 121b1 at $\ell=11$). So the phrase "for E without complex multiplication" in Lemma lem:h0 can be dropped altogether, as point 5 already suggests for other reasons. Sure: high for the class number one list and the examples; medium-high for the conductor statement (standard, but the outline author did not re-derive the conductor-discriminant formula for induced representations here).

Chapter 9

Line-up for Chapter 07: Local fields (ch:local, chapters/ch07_local.tex)

Stage 1 output (core results in teaching order). Part II (Number fields and local fields). Target length 2,080 lines of LaTeX excluding the card (Part 10 below), inside the 1,000 to 2,200 range of PLAN.md Section 7.3. Rigour tags: **P** = proved here, **S** = proof sketched here, **R** = stated; reference.

Item format (as in the Chapter 2 to 6 outlines): number, label-ready name, kind, rigour tag; the precise statement with all hypotheses; reference (book and theorem number only where sure, otherwise book and chapter or section); the paper's use by its `\label` names, or "background". Small computations (Hensel lifts, Teichmuller lifts, unit indices, the Tate parameter of 11a1, the tame relation at finite level, the adjoint of sp(2), quadratic extensions of $\mathbb{Q}_2$) were checked by Python scripts in `/tmp/ch07` (verification log, Part 14).

9.1 0. Purpose of the chapter, and what it must deliver

9.1.1 0.1 Purpose

The paper works at a bad prime $q \neq \ell$ with the restriction of $\rho = \rho_{E,\ell}$ to a decomposition group $G_{\mathbb{Q}_q}$. Every local statement it makes rests on a small amount of local field theory, which this chapter supplies in full:

1. **Lemma lem:tamagawa-exp.** Unramified base change does not change inertia; $\mu_{\ell^\infty} \subset \mathbb{Q}_q^{\mathrm{ur}}$; the units of $\mathbb{Q}_q^{\mathrm{ur}}$ are ℓ^t-th powers (Hensel); hence $x \in \mathbb{Q}_q^\times$ has an ℓ^t-th root in $\mathbb{Q}_q^{\mathrm{ur}}$ iff $\ell^t \mid v_q(x)$ (items 4.2, 4.4, 6.2).
2. **Lemma lem:tamfactor.** The tame character $t_\ell \colon I_q \to \mathbb{Z}_\ell(1)$, the identification of the maximal pro-ℓ quotient of I_q with $\mathbb{Z}_\ell(1)$ (kernel of supernatural order prime to ℓ), the relation Frob t Frob$^{-1} = t^q$, i.e. Frob$^{-1}\, t\, \mathrm{Frob} = t^{q^{-1}}$ (arithmetic Frob), the restriction of a Kummer class κ_x to I_q being $v_q(x)\, t_\ell$, the decomposition $\kappa_x = v_q(x)\kappa_q + \kappa_u$, the existence of a Frobenius lift fixing the ℓ-power roots of q, and the explicit group $\varprojlim \mathbb{Q}_q^\times/(\mathbb{Q}_q^\times)^{\ell^k} \simeq \mathbb{Z}_\ell \times \mathbb{Z}/\ell^{v_\ell(q-1)}$ (items 2.7, 5.3 to 5.5, 6.1, 6.2). The group cohomology is Chapter 16, the cohomological identification of c' is Chapter 17.
3. **Lemma lem:IV, Remark rem:types, Lemma lem:euler.** A finite image of inertia of order prime to q is cyclic and Frobenius acts on it by the q-th power; the Weil-Deligne description of the local representations (unramified, finite cyclic tame inertia, sp(2) and its twists); inertia invariants; the local L-factor with an explicit Frobenius convention; the adjoint of each shape and its L-factor (items 5.6, 7.1 to 7.6, 8.1, 8.2, 9.1 to 9.4).
4. **Lemma lem:twist (iii) and the conductor exponents in Lemma lem:twist (i), (ii), Lemma lem:IV, Remark rem:types.** The Artin conductor $a(V) = \mathrm{codim}\, V^I + \mathrm{sw}(V)$ and the Weil-Deligne conductor; tame quadratic characters at odd q; the unramified twist of Steinberg has exponent 1, its

twist by a ramified quadratic character has exponent 2; finite inertia of order $2, 3, 4, 6$ without fixed vectors has exponent 2 when tame (items 6.3, 6.4, 8.3, 8.4, 9.2, 9.3).

5. **Remark rem:wild and Section sec:wild.** Wild inertia P_q is pro-q; its image in $\mathrm{GL}_2(\mathbb{Z}_\ell)$ is a finite q-group; at $q = 2, 3$ the image of inertia need not be cyclic and the Swan conductor can be positive; an inertia image of order 3 is wild at $q = 3$ and tame at $q = 2$ (items 5.1, 5.7, 9.5).
6. **Section 1.1 (the definition of $L_p(A_f, s)$), Lemma lem:fpr and the paragraph after Lemma lem:dfg21.** The geometric Frobenius in L-factors, so that $L_p(M(1), s) = L_p(M, s + 1)$ (item 8.2; important_notes.md point 11).

It also discharges what earlier chapters deferred to it (by `\Cref{ch:local}`): Chapter 2 (the completion $\mathbb{Q}_p$, Hensel, the structure of $\mathbb{Z}_p^\times$, $u \in \mathbb{Z}_q^\times$ versus its reduction), Chapter 3 ($\mathbb{Q}_p$ analytically; Chapter 3 has $\mathbb{Z}_\ell$ algebraically), Chapter 5 (Kummer theory over $\mathbb{Q}_p$ and $\mathbb{Q}_p^{\mathrm{ur}}$, unramified base change, Frob t $\mathrm{Frob}^{-1} = t^q$), Chapter 6 (item 3.10(v): $D_p \simeq G_{\mathbb{Q}_p}$; item 3.11 in profinite form; item 4.7(iv): the local meaning of $\mathbb{Z}_3[\zeta_3]$; item 4.9: Hensel; item 5.7(ii): Artin conductors of characters; item 7.4: $\mathcal{O}_K \otimes \mathbb{Z}_\ell = \prod_\lambda \mathcal{O}_{K,\lambda}$), Chapter 16 (the imported facts of its tame-inertia-example and model-setting).

9.1.2 0.2 Global conventions fixed in this chapter (to be typeset as `convention` environments)

- **L1 (fields).** p is a prime; K denotes $\mathbb{Q}_p$ or a finite extension of it, with valuation ring $\mathcal{O}_K$, maximal ideal $\mathfrak{m}_K$, uniformiser ϖ_K, residue field $k_K = \mathbb{F}_q$ ($q = p^f$), normalised valuation v_K ($v_K(\varpi_K) = 1$) and normalised absolute value $|x|_K = q^{-v_K(x)}$. On $\mathbb{Q}_p$: v_p, $|\cdot|_p$, as in Chapter 2. When the paper's prime is meant, the local field is $\mathbb{Q}_q$ and $\ell \neq q$; in the general theory the residue characteristic is p and $\ell \neq p$. Say once that the letter q is used both for $\#k_K$ and for the paper's bad prime, which agree when $K = \mathbb{Q}_q$.
- **L2 (closures).** $\overline{\mathbb{Q}}_p$ is a fixed algebraic closure; $G_K = \mathrm{Gal}(\overline{\mathbb{Q}}_p/K)$; $K^{\mathrm{ur}} \subset K^{\mathrm{tame}} \subset \overline{\mathbb{Q}}_p$ the maximal unramified and maximal tamely ramified extensions; $I_K = \mathrm{Gal}(\overline{\mathbb{Q}}_p/K^{\mathrm{ur}})$, $P_K = \mathrm{Gal}(\overline{\mathbb{Q}}_p/K^{\mathrm{tame}})$; for $K = \mathbb{Q}_p$: I_p, P_p. Embeddings $\overline{\mathbb{Q}} \to \overline{\mathbb{Q}}_p$ identify $G_{\mathbb{Q}_p}$ with decomposition groups in $G_{\mathbb{Q}}$ (3.5).
- **L3 (Frobenius: both, named).** $\mathrm{Frob} = \mathrm{Frob}_K$ denotes an **arithmetic** Frobenius: any element of G_K inducing $x \mapsto x^q$ on $\bar{k}_K$, well defined modulo I_K; it acts on μ_m ($p \nmid m$) by $\zeta \mapsto \zeta^q$, and $\chi_\ell(\mathrm{Frob}) = q$ for $\ell \neq p$. The **geometric** Frobenius is $\Phi = \mathrm{Frob}^{-1}$. An eigenvalue λ of Frob is an eigenvalue λ^{-1} of Φ. Group-theoretic statements (the tame relation, conjugation actions, the matrices of Lemma lem:tamfactor) use Frob; local L-factors use Φ (8.2). This is the convention of Chapter 5 (C2), Chapter 6 (3.7), Chapter 16 and Chapter 21 (C2), and of DFG04 1.7.2 for L-factors.
- **L4 (the norm on the Weil group and the monodromy relation).** For w in the Weil group W_K, $d(w) \in \mathbb{Z}$ is defined by $w|_{K^{\mathrm{ur}}} = \mathrm{Frob}^{d(w)}$ and $\|w\| = q^{d(w)}$; so $\|\mathrm{Frob}\| = q$, $\|\Phi\| = q^{-1}$, and $\|w\| = \chi_\ell(w)$ for $\ell \neq p$. A Weil-Deligne representation satisfies $r(w)Nr(w)^{-1} = \|w\|N$, that is

$$r(\mathrm{Frob})\, N\, r(\mathrm{Frob})^{-1} = q\, N, \qquad r(\Phi)\, N\, r(\Phi)^{-1} = q^{-1} N.$$

  (PLAN.md writes "Frob N $\mathrm{Frob}^{-1} = p^{-1}N$": that is the geometric form; see Part 15, B-a.) This $\|\cdot\|$ agrees with Tate's normalisation in "Number theoretic background" ($\|\Phi\| = q^{-1}$).
- **L5 (Tate twists and the tame character).** $\mathbb{Z}_\ell(1) = \varprojlim_k \mu_{\ell^k}$ (Chapter 5, C6), written additively when it carries cocycles; G_K acts through χ_ℓ. The tame character $t_\ell \colon I_K \to \mathbb{Z}_\ell(1)$ is canonical (5.3). Wherever a $\mathbb{Z}_\ell$-valued version is needed, a generator $\zeta = (\zeta_{\ell^k})_k$ of $\mathbb{Z}_\ell(1)$ is fixed and $t_\zeta \colon I_K \to \mathbb{Z}_\ell$ is t_ℓ followed by $\mathbb{Z}_\ell(1) \simeq \mathbb{Z}_\ell$, $\zeta \mapsto 1$. Changing ζ multiplies t_ζ by a unit.
- **L6 (L-factors).** For a continuous representation V of G_K over a finite extension of $\mathbb{Q}_\ell$, $\ell \neq p$: $L(V, s) = \det(1 - \Phi\, q^{-s} \mid V^{I_K})^{-1}$, Φ geometric. For $K = \mathbb{Q}_p$ and a $G_{\mathbb{Q}}$-representation this is the paper's $L_p(V, s)$ (Section 1.1) read with the geometric Frobenius (important_notes.md point 11).
- **L7 (labels and typography).** Labels `...:local:...`; the names in the line-up are the label stems. No em or en dashes, no double hyphens anywhere (write "Grothendieck's monodromy theorem", "Weil-Deligne", "Hasse-Arf" with plain hyphens; ranges "1 to 8"); ASCII source; `\ell`; Kodaira symbols as in PLAN.md 7.1. Use `\operatorname{sp}` or `\mathrm{sp}` for the special representation and `\operatorname{sw}` for the Swan conductor (no new macros).

9.1.3 0.3 Scope check (PLAN.md Section 4 entry for Chapter 7)

Each scope item is mapped to line-up items in Part 11 (coverage table). In order of the PLAN entry: absolute values, completions (1.1 to 1.3); $\mathbb{Q}_p$ and $\mathbb{Z}_p$ as completions and as inverse limits, p-adic expansions (1.4, 1.5); Hensel in two forms (2.1, 2.2, 2.3); $\mathbb{Z}_p^\times$ and $\mathbb{Q}_p^\times$, $1+p\mathbb{Z}_p \simeq \mathbb{Z}_p$ via log / exp (2.5); roots of unity, Teichmuller (2.4, 2.6); squares and ℓ-th powers in $\mathbb{Q}_p^\times$ (2.7, 2.8); finite extensions, e, f (3.1 to 3.4); unramified extensions (4.1, 4.2); totally ramified, tamely and wildly ramified extensions (3.2, 3.3, 3.6, 5.1); $\mathbb{Q}_p^{\mathrm{ur}}$ (4.2); I_p, P_p, the tame quotient and the tame character (5.2, 5.3); the tame relation (5.4, 5.8); $G_{\mathbb{Q}_p}$ and its filtration (5.1, 5.2); Kummer theory over $\mathbb{Q}_p$, restriction of a Kummer class to inertia (6.1, 6.2); quadratic extensions and characters locally (6.3, 6.4); the Weil group and Weil-Deligne representations (7.1 to 7.6); inertia invariants, L-factor, Artin conductor (8.1 to 8.4); the special cases (9.1 to 9.5). Additions not in the PLAN entry but needed by deferrals or by the paper: local class field theory as a statement (6.5, for conductors of characters), ϵ-factors as a statement (8.5, which Chapter 15 "recalls from Chapter 7"), the decomposition-group comparison (3.5).

9.2 1. Line-up, Section 1: absolute values, completions, $\mathbb{Q}_p$ and $\mathbb{Z}_p$

9.2.1 1.1 def:local:absval (definition and proposition) P

An *absolute value* on a field F is a map $|\cdot|\colon F \to \mathbb{R}_{\geq 0}$ with $|x| = 0$ iff $x = 0$, $|xy| = |x||y|$ and $|x+y| \leq |x| + |y|$; it is *nonarchimedean* if $|x+y| \leq \max(|x|, |y|)$ (ultrametric). (i) $|\cdot|$ is nonarchimedean iff $|n \cdot 1| \leq 1$ for all $n \in \mathbb{Z}$. (ii) If nonarchimedean: $|x+y| = \max(|x|, |y|)$ whenever $|x| \neq |y|$ ("all triangles are isosceles"); $\mathcal{O} = \{|x| \leq 1\}$ is a subring (the valuation ring), $\mathfrak{m} = \{|x| < 1\}$ its unique maximal ideal, $\mathcal{O}^\times = \{|x| = 1\}$, $k = \mathcal{O}/\mathfrak{m}$ the residue field; open balls are closed, every point of a ball is a centre; a series $\sum a_n$ in a complete F converges iff $a_n \to 0$. (iii) $|\cdot|$ is *discrete* if $|F^\times|$ is a discrete subgroup of $\mathbb{R}_{>0}$; then $|F^\times| = c^{\mathbb{Z}}$, $\mathcal{O}$ is a DVR (Chapter 3) and $|x| = c^{v(x)}$ for the normalised valuation v. (iv) Two absolute values define the same topology iff $|\cdot|_1 = |\cdot|_2^s$ for some $s > 0$ (equivalent). (v) (Recalled from Chapter 2, item 11.2; R) Ostrowski: every nontrivial absolute value on $\mathbb{Q}$ is equivalent to $|\cdot|_\infty$ or to exactly one $|\cdot|_p$. Examples: $|\cdot|_p$ on $\mathbb{Q}$ with $\mathcal{O} = \mathbb{Z}_{(p)}$, $k = \mathbb{F}_p$; the $\mathfrak{p}$-adic absolute value $|x|_{\mathfrak{p}} = N\mathfrak{p}^{-v_{\mathfrak{p}}(x)}$ on a number field (Chapter 6, 2.2, 2.4(iv)). Reference: Neukirch II.3; Koblitz Ch. I; Cassels-Frohlich Ch. II (Cassels, "Global fields"). Paper: background (the field $\mathbb{Q}_q$ and the ring $\mathbb{Z}_\ell$ in Sections 2 and 4).

9.2.2 1.2 rem:local:ostrowski-product (remark) R

Product formula $\prod_{v \leq \infty} |x|_v = 1$ for $x \in \mathbb{Q}^\times$ (Chapter 2, 11.2(ii); cite, do not redo). Weak approximation for finitely many inequivalent absolute values (stated). Reference: Neukirch II.3. Paper: background.

9.2.3 1.3 thm:local:completion (construction and theorem) P

Let $(F, |\cdot|)$ be a field with an absolute value. There is a field $\hat{F}$ with an absolute value extending $|\cdot|$, complete (every Cauchy sequence converges), in which F is dense; it is unique up to a unique isometric F-isomorphism (the *completion*). Construction: Cauchy sequences modulo null sequences. If $|\cdot|$ is nonarchimedean: (i) $|\hat{F}^\times| = |F^\times|$ (the value group does not grow); (ii) $\hat{\mathcal{O}}$ is the closure of $\mathcal{O}$ and $\mathcal{O}/\mathfrak{m}^n \simeq \hat{\mathcal{O}}/\hat{\mathfrak{m}}^n$ for discrete $|\cdot|$ (in particular the residue field does not change); (iii) for discrete $|\cdot|$ with uniformiser ϖ, $\hat{\mathcal{O}} \simeq \varprojlim_n \mathcal{O}/\varpi^n\mathcal{O}$ as topological rings (this is the bridge to Chapter 3's algebraic completion). Reference: Neukirch II.4; Serre, Local Fields, Ch. II, Section 1. Paper: background.

9.2.4 1.4 def:local:qp (definition and theorem) P

$\mathbb{Q}_p$ is the completion of $\mathbb{Q}$ for $|\cdot|_p$, and $\mathbb{Z}_p = \{x \in \mathbb{Q}_p : |x|_p \leq 1\}$. Then: (i) $\mathbb{Z}_p$ is the closure of $\mathbb{Z}$ (and of $\mathbb{Z}_{(p)}$) in $\mathbb{Q}_p$; it is a complete DVR with uniformiser p, residue field $\mathbb{F}_p$, ideals $p^n\mathbb{Z}_p$, and v_p extends to $\mathbb{Q}_p$; (ii) $\mathbb{Z}_p \to \varprojlim_n \mathbb{Z}/p^n$, $x \mapsto (x \bmod p^n)_n$, is an isomorphism of topological rings (product

of discrete topologies on the right), so the analytic $\mathbb{Z}_p$ of this chapter is the algebraic $\mathbb{Z}_\ell$ of Chapter 3 (item 6.4) and the profinite group of Chapter 1 (item 7.2) at $\ell = p$; (iii) $\mathbb{Z}_p^\times = \{|x|_p = 1\}$, and $x \in \mathbb{Z}_p$ is a unit iff $x \not\equiv 0 \pmod p$; every $x \in \mathbb{Q}_p^\times$ is uniquely $x = p^{v_p(x)}u$ with $u \in \mathbb{Z}_p^\times$, so $\mathbb{Q}_p^\times = p^{\mathbb{Z}} \times \mathbb{Z}_p^\times$ and $\mathbb{Q}_p = \mathbb{Z}_p[1/p] = \operatorname{Frac}\mathbb{Z}_p$; (iv) $\mathbb{Z}_p$ is compact, $\mathbb{Q}_p$ is locally compact and totally disconnected; (v) for $q \neq p$ prime, $q \in \mathbb{Z}_p^\times$, so $q^{-1} \in \mathbb{Z}_p$ (the exponent q^{-1} of the tame relation). Reference: Serre, Course in Arithmetic, Ch. II, Section 1; Koblitz Ch. I; Neukirch II.1-II.2 (cite II.1, II.2). Paper: background (every $\mathbb{Z}_\ell$, $\mathbb{Q}_\ell$, $\mathbb{Q}_q$); the exponent $q^{-1} \in \mathbb{Z}_\ell$ in Lemma lem:tamfactor.

9.2.5 1.5 prop:local:expansions (proposition) P

Every $x \in \mathbb{Q}_p$ has a unique expansion $x = \sum_{i \geq v} a_i p^i$, $v = v_p(x)$, $a_i \in \{0, 1, \ldots, p-1\}$, $a_v \neq 0$ (for $x \neq 0$), the series converging in $\mathbb{Q}_p$; $x \in \mathbb{Z}_p$ iff $v \geq 0$; the digits of $x \in \mathbb{Z}_p$ are those of the residues $x \bmod p^n$. $x \in \mathbb{Q}$ iff the digit sequence is eventually periodic. Examples: $-1 = \sum_{i \geq 0}(p-1)p^i$; $1/(1-p) = \sum p^i$; in $\mathbb{Z}_5$, $\sqrt{-1} \equiv 2 + 1 \cdot 5 + 2 \cdot 5^2 + 1 \cdot 5^3 \pmod{5^4}$ (that is 182, item 2.3). (RHB aside: a "Laurent series in p", but with carries.) Reference: Koblitz Ch. I; Serre, Course in Arithmetic, Ch. II. Paper: background.

9.3 2. Line-up, Section 2: Hensel's lemma, units, roots of unity, powers

9.3.1 2.1 thm:local:hensel (theorem, Hensel's lemma in two forms) P

Let K be a field complete for a nonarchimedean absolute value with valuation ring $\mathcal{O}$, maximal ideal $\mathfrak{m}$ and residue field k (in this chapter K is $\mathbb{Q}_p$ or a finite extension; also the ℓ-adic fields). Let $f \in \mathcal{O}[X]$. (a) (Simple-root form.) If $a_0 \in \mathcal{O}$ satisfies $f(a_0) \equiv 0 \pmod{\mathfrak{m}}$ and $f'(a_0) \not\equiv 0 \pmod{\mathfrak{m}}$, there is a unique $a \in \mathcal{O}$ with $f(a) = 0$ and $a \equiv a_0 \pmod{\mathfrak{m}}$. (b) (Newton form.) If $a_0 \in \mathcal{O}$ satisfies $|f(a_0)| < |f'(a_0)|^2$, then the Newton sequence $a_{n+1} = a_n - f(a_n)/f'(a_n)$ lies in $\mathcal{O}$ and converges to a root a of f, which is the unique root with $|a - a_0| < |f'(a_0)|$; moreover $|a - a_0| = |f(a_0)/f'(a_0)|$ and $|f'(a)| = |f'(a_0)|$. (a) is (b) with $|f'(a_0)| = 1$. Proof: the Taylor expansion $f(x+h) = f(x) + f'(x)h + h^2 g(x,h)$ with $g \in \mathcal{O}[x,h]$ gives $|f(a_{n+1})| \leq |f(a_n)|^2/|f'(a_0)|^2$ and $|f'(a_n)| = |f'(a_0)|$ inductively; uniqueness from the same expansion. Warning: (a) is false without the hypothesis on $f'(a_0)$ ($X^2 - p$ has a root mod p, none in $\mathbb{Q}_p$; $X^2 + 7$ has no simple root mod 2, since $f' = 2X$, yet -7 is a square in $\mathbb{Z}_2$ by (b)). Reference: Neukirch II.4 (Hensel's lemma and its Newton form; cite the section); Serre, Course in Arithmetic, Ch. II, Section 2; K. Conrad, "Hensel's lemma" (expository notes; add to bib/ch07.bib only if cited). Paper: Lemma lem:tamagawa-exp (units of $\mathbb{Q}_q^{\mathrm{ur}}$ are ℓ^t-th powers), Lemma lem:tamfactor (an ℓ^k-th power in $\mathbb{F}_q^\times$ lifts), via 2.7 and 6.2; Chapter 6 (4.9: $\mathbb{Z}_\ell[\zeta_3]$).

9.3.2 2.2 thm:local:hensel-factor (theorem, factorisation form, with a corollary) S

Let $K, \mathcal{O}, k$ be as in 2.1. If $f \in \mathcal{O}[X]$ is primitive ($\max|a_i| = 1$) and $\bar{f} = \bar{g}\bar{h}$ in $k[X]$ with $\bar{g}, \bar{h}$ coprime, then $f = gh$ in $\mathcal{O}[X]$ with $\bar{g}, \bar{h}$ the reductions of g, h and $\deg g = \deg \bar{g}$. Corollary: if $f = a_n X^n + \cdots + a_0 \in K[X]$ is irreducible with $a_0 a_n \neq 0$, then $\max_i |a_i| = \max(|a_0|, |a_n|)$; in particular a monic irreducible f with $f(0) \in \mathcal{O}$ lies in $\mathcal{O}[X]$. (Sketch: successive approximation modulo $\mathfrak{m}^n$, using a Bezout relation for $\bar{g}, \bar{h}$.) Reference: Neukirch II.4 (cite the section). Paper: background (used in 3.1).

9.3.3 2.3 ex:local:hensel-examples (worked examples) P

(a) $\sqrt{-1} \in \mathbb{Z}_5$: the two roots are $\equiv 182$ and $\equiv 443 \pmod{5^4}$; 182 is the Teichmuller lift of 2 (2.4) and $182^2 + 1 = 53 \cdot 5^4$. (b) $\sqrt{-3} \in \mathbb{Z}_7$: roots $\equiv 37, 306 \pmod{7^3}$. (c) $X^2 + X + 1$ in $\mathbb{Z}_7$: roots $\equiv 18, 30 \pmod{49}$, so $\mathbb{Z}_7[\zeta_3] \simeq \mathbb{Z}_7 \times \mathbb{Z}_7$ (Chapter 6, 4.9(ii)). (d) $\sqrt{2} \in \mathbb{Z}_7$: roots $\equiv 108, 235 \pmod{7^3}$. (e) Squares in $\mathbb{Z}_2$: $u \in \mathbb{Z}_2^\times$ is a square iff $u \equiv 1 \pmod 8$ (2.1(b) with $f = X^2 - u$, $a_0 = 1$, $|f'(1)|^2 = 1/4$; conversely odd squares are $\equiv 1 \bmod 8$); so -7 is a square in $\mathbb{Z}_2$, -1, 3, 5 are not. (f) Squares in

$\mathbb{Z}_p$, p odd: $u \in \mathbb{Z}_p^\times$ is a square iff $\bar{u}$ is a square in $\mathbb{F}_p$. (g) The trichotomy of Chapter 6, 4.9, for $\mathbb{Z}_\ell[\zeta_3]$ and $\mathbb{Z}_\ell[i]$ is now proved: split iff X^2+X+1 (resp. X^2+1) has a simple root mod ℓ. Paper: Lemma lem:euler (the rings $\mathbb{Z}_\ell[\zeta_3]$, $\mathbb{Z}_\ell[i]$; important_notes.md point 12), background.

9.3.4 2.4 thm:local:teichmuller (theorem) P

Let K be complete and discretely valued with finite residue field $k = \mathbb{F}_q$. (i) For $a \in k$ and any lift $\tilde{a} \in \mathcal{O}$, the limit $\omega(a) = \lim_{n\to\infty} \tilde{a}^{q^n}$ exists and does not depend on $\tilde{a}$; $\omega(a)$ is the unique lift of a with $\omega(a)^q = \omega(a)$. (ii) $\omega\colon k^\times \to \mathcal{O}^\times$ is an injective homomorphism with image $\mu_{q-1}(K)$, and reduction $\mu_{q-1}(K) \to k^\times$ is an isomorphism; $\omega(0) = 0$; ω is multiplicative but not additive. (iii) $\mathcal{O}^\times = \mu_{q-1}(K) \times U^{(1)}$, $U^{(1)} = 1+\mathfrak{m}$, the projection to the first factor being $u \mapsto \omega(\bar{u})$. Examples: in $\mathbb{Z}_5$, $\omega(2) \equiv 182$, $\omega(3) \equiv 443$, $\omega(4) \equiv 624 \equiv -1 \pmod{5^4}$. Reference: Serre, Local Fields, Ch. II, Section 4; Serre, Course in Arithmetic, Ch. II, Section 3. Paper: Lemma lem:tamfactor (the class of the unit part u depends only on $\bar{u}$; important_notes.md point 31), background.

9.3.5 2.5 thm:local:units-structure (theorem) P

(i) For p odd: $\mathbb{Z}_p^\times = \mu_{p-1} \times (1+p\mathbb{Z}_p)$; the series $\log(1+x) = \sum_{n\geq 1}(-1)^{n+1}x^n/n$ and $\exp(x) = \sum x^n/n!$ converge on $p\mathbb{Z}_p$ and give mutually inverse isomorphisms of topological groups $1+p\mathbb{Z}_p \simeq p\mathbb{Z}_p$, carrying $1+p^n\mathbb{Z}_p$ onto $p^n\mathbb{Z}_p$ for $n \geq 1$; hence $1+p\mathbb{Z}_p \simeq \mathbb{Z}_p$, topologically generated by $1+p$, and

$$\mathbb{Q}_p^\times \simeq p^{\mathbb{Z}} \times \mu_{p-1} \times (1+p\mathbb{Z}_p) \simeq \mathbb{Z} \times \mathbb{Z}/(p-1) \times \mathbb{Z}_p.$$

(ii) For $p = 2$: $\mathbb{Z}_2^\times = \{\pm 1\} \times (1+4\mathbb{Z}_2)$, log and exp give $1+4\mathbb{Z}_2 \simeq 4\mathbb{Z}_2$, $1+4\mathbb{Z}_2 \simeq \mathbb{Z}_2$ generated by 5, and $\mathbb{Q}_2^\times \simeq \mathbb{Z} \times \mathbb{Z}/2 \times \mathbb{Z}_2$. (iii) These are the limits of the finite statements of Chapter 2 (5.3, 5.4: $(\mathbb{Z}/p^k)^\times$ cyclic for p odd, $\{\pm 1\} \times \langle 5\rangle$ for $p = 2$). (iv) (For a finite extension K with residue field $\mathbb{F}_q$; stated in the form used.) $K^\times = \varpi_K^{\mathbb{Z}} \times \mu_{q-1}(K) \times U_K^{(1)}$ and $U_K^{(1)}$ is a $\mathbb{Z}_p$-module, finitely generated; in particular it is uniquely m-divisible for every m prime to p. Reference: Serre, Course in Arithmetic, Ch. II, Section 3 (cite the section); Neukirch II.5. Paper: background (Lemma lem:tamfactor uses only (iv) for $m = \ell^k$, through 2.7).

9.3.6 2.6 cor:local:roots-of-unity (corollary) P

$\mu(\mathbb{Q}_p) = \mu_{p-1}$ for p odd and $\mu(\mathbb{Q}_2) = \{\pm 1\}$. For m prime to p: $\mu_m(\mathbb{Q}_p) = \mu_{\gcd(m,p-1)}$; in particular for a prime $\ell \neq p$, $\mu_{\ell^k} \subset \mathbb{Q}_p$ iff $\ell^k \mid p-1$, and $\#\mu_{\ell^\infty}(\mathbb{Q}_p) = \ell^{v_\ell(p-1)}$. More generally $\mu_m(K) = \mu_{\gcd(m,q-1)}(K)$ for $p \nmid m$ ($q = \#k_K$). $\mu_{p^n} \subset \mathbb{Q}_p$ only for $p^n = 2$ (and $p^n = 1$). Reference: as 2.5. Paper: Lemma lem:tamfactor ($\mathbb{Z}_\ell(1)^{G/I}$, "$\mathbb{Z}_\ell(1)/(q-1) \simeq \mathbb{Z}/\ell^{v_\ell(q-1)}$"), background for Chapter 17.

9.3.7 2.7 thm:local:powers (theorem; ℓ-th powers in $\mathbb{Q}_p^\times$ for $\ell \neq p$) P

Let $\ell \neq p$ be primes and $k \geq 1$; K finite over $\mathbb{Q}_p$ with residue field $\mathbb{F}_q$ and uniformiser ϖ (for $K = \mathbb{Q}_p$: $q = p$, $\varpi = p$). (i) $U_K^{(1)} = 1+\mathfrak{m}_K$ is uniquely ℓ^k-divisible (2.1(b) with $f = X^{\ell^k} - u$, $a_0 = 1$, $f'(1) = \ell^k$ a unit; or 2.5). (ii) $u \in \mathcal{O}_K^\times$ is an ℓ^k-th power in K iff its reduction $\bar{u}$ is an ℓ^k-th power in $\mathbb{F}_q^\times$ (2.4(iii) and (i)); by Chapter 2, 7.1, iff $\bar{u}^{(q-1)/\gcd(\ell^k, q-1)} = 1$. (iii) $x = \varpi^a u \in K^\times$ is an ℓ^k-th power iff $\ell^k \mid a$ and $\bar{u} \in (\mathbb{F}_q^\times)^{\ell^k}$. (iv) Hence

$$K^\times/(K^\times)^{\ell^k} \;\simeq\; \mathbb{Z}/\ell^k \times \mathbb{F}_q^\times/(\mathbb{F}_q^\times)^{\ell^k} \;\simeq\; \mathbb{Z}/\ell^k \times \mathbb{Z}/\ell^{\min(k, v_\ell(q-1))},$$

the first factor by v_K (canonical), the splitting by the choice of ϖ, the second isomorphism by a choice of generator of $\mathbb{F}_q^\times$. In particular $\#K^\times/(K^\times)^{\ell^k} = \ell^k \cdot \#\mu_{\ell^k}(K)$ (a first instance of the local Euler characteristic formula, Chapter 17). (v) Passing to the limit: $\widehat{K^\times} := \varprojlim_k K^\times/(K^\times)^{\ell^k}$ sits in a canonical exact sequence $0 \to \mathbb{F}_q^\times \otimes \mathbb{Z}_\ell \to \widehat{K^\times} \xrightarrow{v_K} \mathbb{Z}_\ell \to 0$, split by ϖ, so $\widehat{K^\times} \simeq \mathbb{Z}_\ell \times \mathbb{Z}/\ell^{v_\ell(q-1)}$; and

$\mathbb{F}_q^\times \otimes \mathbb{Z}_\ell \simeq \mathbb{F}_q^\times/(\mathbb{F}_q^\times)^{\ell^{v_\ell(q-1)}} \simeq \mu_{\ell^{v_\ell(q-1)}}(\mathbb{F}_q)$ (Chapter 2, 7.2; Chapter 5, galois:kummer-finite-field). (vi) (The case $\ell = 2$, p odd, written out.) $\mathbb{Q}_p^\times/(\mathbb{Q}_p^\times)^2$ has order 4, with representatives $1, u_0, p, u_0 p$, u_0 a non-square unit; $\mathbb{Q}_2^\times/(\mathbb{Q}_2^\times)^2$ has order 8, representatives $\pm 1, \pm 5, \pm 2, \pm 10$ (by 2.3(e)). Reference: Serre, Course in Arithmetic, Ch. II, Section 3 (squares); elementary from 2.1 to 2.5 otherwise. Paper: Lemma lem:tamfactor ("$H^1(G, \mathbb{Z}_\ell(1)) = \varprojlim \mathbb{Q}_q^\times/(\mathbb{Q}_q^\times)^{\ell^k}$", "$c$ the class of u", "u is an ℓ^k-th power in $\mathbb{F}_q^\times$" read as the reduction, important_notes.md point 31), Lemma lem:tamagawa-exp (via 6.2), Section sec:tam.

9.3.8 2.8 ex:local:powers-examples (worked examples) P

(a) $\mathbb{Q}_7^\times/(\mathbb{Q}_7^\times)^3 \simeq \mathbb{Z}/3 \times \mathbb{Z}/3$; the unit cubes reduce to $\{1, 6\}$ mod 7: -1 and 6 are cubes in $\mathbb{Q}_7$; 2 is not; 14 is not ($v_7 = 1$); $7^3 \cdot 6$ is. (b) $\mathbb{Q}_{19}^\times/(\mathbb{Q}_{19}^\times)^9 \simeq \mathbb{Z}/9 \times \mathbb{Z}/9$, $v_3(18) = 2$; the classes of $2, 8, -1$ have $v_3(c) = 0, 1, 2$ (Chapter 2, 7.3(b), now read in $\mathbb{Z}_{19}^\times$). (c) $\mathbb{Q}_{11}^\times/(\mathbb{Q}_{11}^\times)^5 \simeq \mathbb{Z}/5 \times \mathbb{Z}/5$; the unit fifth powers reduce to $\{\pm 1\}$ mod 11. For 11a1 the Tate parameter at 11 has $v_{11}(q_E) = 5$ and unit part $u \equiv -1 \pmod{11}$ (from $j = -2^{12}31^3/11^5$ and $q_E = j^{-1} + 744j^{-2} + \ldots$; Chapter 10 for the Tate parameter), so $q_E \in (\mathbb{Q}_{11}^\times)^5$: both the valuation and the unit part are fifth powers. (d) $\mathbb{Q}_5^\times/(\mathbb{Q}_5^\times)^3 \simeq \mathbb{Z}/3$: every unit of $\mathbb{Z}_5$ is a cube. Paper: Lemma lem:tamfactor and Section sec:tam (the extra term $\min(v_\ell(n), v_\ell(q-1), v_\ell(c))$), Chapter 16, Exercise 10 (11a1 at $\ell = 5$).

9.4 3. Line-up, Section 3: finite extensions of $\mathbb{Q}_p$

9.4.1 3.1 thm:local:extension-absval (theorem) P (existence via 2.2; equivalence of norms S)

Let K be complete for a nonarchimedean absolute value (in practice $\mathbb{Q}_p$ or a finite extension) and L/K an algebraic extension. Then $|\cdot|$ extends uniquely to an absolute value on L; for $[L : K] = n < \infty$ it is

$$|x|_L = |N_{L/K}(x)|^{1/n},$$

L is complete, and $\mathcal{O}_L = \{|x| \le 1\}$ is the integral closure of $\mathcal{O}_K$ in L. Uniqueness uses that all norms on a finite-dimensional vector space over a complete field are equivalent (S). Consequences: $\sigma \in \mathrm{Aut}(L/K)$ is an isometry; conjugate elements have the same absolute value; $\overline{\mathbb{Q}}_p$ carries a unique absolute value extending $|\cdot|_p$ (not complete; its completion $\mathbb{C}_p$ is mentioned only). Reference: Neukirch II.4 and II.6 (cite the sections); Serre, Local Fields, Ch. II, Section 2. Paper: background.

9.4.2 3.2 thm:local:ef (definition and theorem) P

For L/K finite of degree n (both complete, discretely valued, finite residue fields): the *ramification index* $e = e(L/K) = [|L^\times| : |K^\times|] = v_L(\varpi_K)$ and the *residue degree* $f = f(L/K) = [k_L : k_K]$ satisfy $ef = n$; $\mathcal{O}_L$ is free of rank n over $\mathcal{O}_K$ with basis $\{\alpha_i \varpi_L^j : 1 \le i \le f,\ 0 \le j < e\}$ (α_i lifting a basis of k_L/k_K); e and f are multiplicative in towers. Terminology: L/K is *unramified* if $e = 1$, *totally ramified* if $f = 1$, *tamely ramified* if $p \nmid e$, *wildly ramified* if $p \mid e$ (p the residue characteristic). (Residue fields are finite, hence perfect, so no separability condition is needed.) Comparison with Chapter 6, 3.1: one prime above, $g = 1$. Reference: Neukirch II.6; Serre, Local Fields, Ch. I, Section 4 and Ch. II, Section 2. Paper: background.

9.4.3 3.3 thm:local:eisenstein (theorem and examples) P

L/K is totally ramified of degree e iff $L = K(\pi)$ with π a root of an Eisenstein polynomial of degree e over $\mathcal{O}_K$; then π is a uniformiser of L and $\mathcal{O}_L = \mathcal{O}_K[\pi]$. Conversely the minimal polynomial of any uniformiser of a totally ramified L is Eisenstein. Examples (all proved): (a) $\mathbb{Q}_p(\zeta_{p^n})$: $\Phi_{p^n}(X+1)$ is Eisenstein, so the extension is totally ramified of degree $\varphi(p^n)$, uniformiser $1 - \zeta_{p^n}$, ring of integers $\mathbb{Z}_p[\zeta_{p^n}]$, and $\mathrm{Gal}(\mathbb{Q}_p(\zeta_{p^n})/\mathbb{Q}_p) \simeq (\mathbb{Z}/p^n)^\times$ is all inertia. (b) $\mathbb{Q}_3(\zeta_3) = \mathbb{Q}_3(\sqrt{-3})$, totally ramified quadratic, tame ($3 \nmid 2$), with ring of integers $\mathbb{Z}_3[\zeta_3]$ and uniformiser $\pi = 1 - \zeta_3$, $\pi^2 = -3\zeta_3$: this is the local meaning of Chapter 6, 4.7, and $\mathbb{Z}_3[\zeta_3]$ is the π-adic completion of $\mathbb{Z}[\zeta_3]$ (by 3.4). (c) $\mathbb{Q}_p(p^{1/m})$, $X^m - p$ Eisenstein:

totally ramified of degree m. (d) $\mathbb{Q}_p(\sqrt{p^*})$, p odd, $p^* = (-1)^{(p-1)/2}p$: totally ramified quadratic (6.3). Reference: Serre, Local Fields, Ch. I, Section 6; Neukirch II.5. Paper: Lemma lem:IV ("$\mathbb{Z}_3[\zeta_3]$, a discrete valuation ring with uniformiser $\pi = 1-\zeta_3$ and residue field $\mathbb{F}_3$", now as the ring of integers of $\mathbb{Q}_3(\sqrt{-3})$), Lemma lem:twist ($\mathbb{Q}_q(\sqrt{q^*})$).

9.4.4 3.4 thm:local:completion-numberfield (theorem) P

Let F be a number field and p a prime with $p\mathcal{O}_F = \prod_{\mathfrak{p}|p} \mathfrak{p}^{e_\mathfrak{p}}$, residue degrees $f_\mathfrak{p}$ (Chapter 6, 3.1). (i) The absolute values of F extending $|\cdot|_p$ are exactly the maps $x \mapsto p^{-v_\mathfrak{p}(x)/e_\mathfrak{p}}$, one for each $\mathfrak{p} \mid p$ (distinct primes give inequivalent absolute values). (ii) The completion $F_\mathfrak{p}$ is a finite extension of $\mathbb{Q}_p$ with $e(F_\mathfrak{p}/\mathbb{Q}_p) = e_\mathfrak{p}$, $f(F_\mathfrak{p}/\mathbb{Q}_p) = f_\mathfrak{p}$, and $\mathcal{O}_{F_\mathfrak{p}} = \varprojlim_n \mathcal{O}_F/\mathfrak{p}^n$. (iii) $F \otimes_\mathbb{Q} \mathbb{Q}_p \simeq \prod_{\mathfrak{p}|p} F_\mathfrak{p}$ and $\mathcal{O}_F \otimes_\mathbb{Z} \mathbb{Z}_p \simeq \prod_{\mathfrak{p}|p} \mathcal{O}_{F_\mathfrak{p}}$, hence $\sum_\mathfrak{p} e_\mathfrak{p} f_\mathfrak{p} = [F:\mathbb{Q}]$. (Proof of (iii): write $F = \mathbb{Q}(\theta)$, factor the minimal polynomial over $\mathbb{Q}_p$, CRT; or: $\mathcal{O}_F \otimes \mathbb{Z}_p = \varprojlim \mathcal{O}_F/p^n$ and CRT for $p^n\mathcal{O}_F = \prod \mathfrak{p}^{ne_\mathfrak{p}}$.) This is the result Chapter 6, 7.4 states and defers here. Examples: $\mathbb{Z}[\zeta_3] \otimes \mathbb{Z}_3 = \mathbb{Z}_3[\zeta_3]$ (ramified), $\mathbb{Z}[\zeta_3] \otimes \mathbb{Z}_7 \simeq \mathbb{Z}_7 \times \mathbb{Z}_7$ (split), $\mathbb{Z}[\zeta_3] \otimes \mathbb{Z}_2$ the unramified quadratic extension's ring of integers (inert); the coefficient ring $\mathcal{O}_\lambda$ of a newform (Chapters 13, 15). Reference: Neukirch II.8 (cite the section); Serre, Local Fields, Ch. II, Section 3. Paper: Lemma lem:crit and Lemma lem:old ("the same statements hold with $\mathbb{Z}$ replaced by $\mathbb{Z}_\ell$", primes $\lambda \mid \ell$), Lemma lem:ddt417 ($\mathcal{O}$ a DVR), Lemma lem:IV, Lemma lem:euler.

9.4.5 3.5 prop:local:decomposition (proposition) S

(i) Let L/F be a finite Galois extension of number fields, $\mathfrak{P} \mid \mathfrak{p}$. Restriction identifies $\mathrm{Gal}(L_\mathfrak{P}/F_\mathfrak{p})$ with the decomposition group $D_\mathfrak{P}$, and its inertia subgroup (3.2 and 5.1) with $I_\mathfrak{P}$ (Chapter 6, 3.5). (ii) An embedding $\iota\colon \overline{\mathbb{Q}} \to \overline{\mathbb{Q}}_p$ induces $G_{\mathbb{Q}_p} \to G_\mathbb{Q}$, $\sigma \mapsto \iota^{-1}\sigma\iota$; it is injective (because $\iota(\overline{\mathbb{Q}})$ is dense in $\overline{\mathbb{Q}}_p$, by Krasner's lemma, stated) with image the decomposition group $D_{\bar{\mathfrak{P}}}$ of the prime $\bar{\mathfrak{P}}$ of $\bar{\mathbb{Z}}$ determined by ι, and it carries I_p onto $I_{\bar{\mathfrak{P}}}$ and arithmetic Frobenius lifts to Frobenius elements (Chapter 6, 3.10(v), which states this and defers it here). Changing ι conjugates the image. (iii) Hence for a representation ρ of $G_\mathbb{Q}$, "$\rho|_{G_{\mathbb{Q}_p}}$", "$\rho(I_p)$", "V^{I_p}" and "$\rho(\mathrm{Frob}_p)$ for ρ unramified at p" are well defined up to conjugation. Reference: Neukirch II.8 and II.9 (cite the sections); Serre, Local Fields, Ch. II, Section 3. Paper: Section 1.1 (I_p, $\delta_p = \dim V^{I_p}$), Lemmas lem:tamagawa-exp, lem:tamfactor, lem:euler (all local arguments at q).

9.4.6 3.6 prop:local:tame-totally-ramified (proposition) P

Let L/K be totally ramified of degree e with $p \nmid e$. Then $L = K(\alpha)$ with $\alpha^e = \varpi$ for some uniformiser ϖ of K. L/K is Galois iff $\mu_e \subset K$ (iff $e \mid q-1$), and then $\mathrm{Gal}(L/K) \simeq \mu_e$ via $\sigma \mapsto \sigma(\alpha)/\alpha$, cyclic. Proof: $\pi_L^e = \varpi_K w$ with $w \in \mathcal{O}_L^\times$; as $k_L = k_K$, $w = w_0 w_1$ with $w_0 \in \mathcal{O}_K^\times$ and $w_1 \in 1 + \mathfrak{m}_L$, and w_1 is an e-th power by 2.7(i); so $(\pi_L w_1^{-1/e})^e = \varpi_K w_0$ is a uniformiser of K. Galois criterion: the conjugates of α are $\zeta\alpha$, $\zeta \in \mu_e$, and $\mu_e \subset L$ forces $\mu_e \subset K$ by 2.4 since $k_L = k_K$. Reference: Serre, Local Fields, Ch. IV, Section 2 (cite the section); Neukirch II.7. Paper: background for 5.2 (the tame extensions of $\mathbb{Q}_p^{\mathrm{ur}}$).

9.5 4. Line-up, Section 4: unramified extensions, $\mathbb{Q}_p^{\mathrm{ur}}$ and Frobenius

9.5.1 4.1 thm:local:unramified (theorem) P

Let K be finite over $\mathbb{Q}_p$ with residue field $\mathbb{F}_q$. (i) For each $f \geq 1$ there is, inside $\overline{\mathbb{Q}}_p$, exactly one unramified extension K_f/K of degree f, namely $K_f = K(\mu_{q^f-1})$. (ii) K_f/K is Galois and reduction gives an isomorphism $\mathrm{Gal}(K_f/K) \to \mathrm{Gal}(\mathbb{F}_{q^f}/\mathbb{F}_q)$; so $\mathrm{Gal}(K_f/K)$ is cyclic of order f, generated by the unique $\mathrm{Frob}_{K_f/K}$ with $\mathrm{Frob}(x) \equiv x^q \pmod{\mathfrak{m}}$ for $x \in \mathcal{O}_{K_f}$ (arithmetic Frobenius; on μ_{q^f-1} it is $\zeta \mapsto \zeta^q$). (iii) A finite L/K is unramified iff $L \subseteq K_f$ for some f; composita and subextensions of unramified extensions are unramified; every finite L/K has a maximal unramified subextension L_0, with $[L_0 : K] = f(L/K)$

and L/L_0 totally ramified. (iv) If L/K is unramified and $\alpha \in \mathcal{O}_L$ reduces to a generator of k_L/k_K, then $L = K(\alpha)$ (Hensel). Proof: lift the irreducible polynomial of a generator of $\mathbb{F}_{q^f}^\times$ (Hensel for the simple root), Teichmuller (2.4). Reference: Serre, Local Fields, Ch. III, Section 5; Neukirch II.7. Paper: Lemma lem:tamagawa-exp and Lemma lem:tamfactor ("the unramified quadratic extension of $\mathbb{Q}_q$").

9.5.2 4.2 def:local:qpur (definition and proposition) P

$K^{\mathrm{ur}} = \bigcup_f K_f = K(\mu_m : p \nmid m)$. (i) Its residue field is $\bar{\mathbb{F}}_q$; restriction gives $\mathrm{Gal}(K^{\mathrm{ur}}/K) \simeq \mathrm{Gal}(\bar{\mathbb{F}}_q/\mathbb{F}_q) \simeq \hat{\mathbb{Z}}$, topologically generated by the arithmetic Frobenius (Chapter 5, galois:finite-field-galois and the profinite theory). (ii) The valuation of K^{ur} is discrete with the same value group as K (each K_f is unramified), ϖ_K is a uniformiser, but K^{ur} is not complete. (iii) Hensel's lemma holds for $\mathcal{O}_{K^{\mathrm{ur}}}$ (a polynomial and an approximate root live in some complete $\mathcal{O}_{K_f}$, and the root found there lies in K^{ur}). (iv) For every m prime to p: $\mu_m \subset K^{\mathrm{ur}}$, every unit of $\mathcal{O}_{K^{\mathrm{ur}}}$ is an m-th power in K^{ur} (because $\bar{\mathbb{F}}_q^\times$ is m-divisible, then (iii)), and $x \in (K^{\mathrm{ur}})^\times$ is an m-th power iff $m \mid v_K(x)$. In particular $\mu_{\ell^\infty} \subset \mathbb{Q}_q^{\mathrm{ur}}$ for $\ell \neq q$. Reference: Serre, Local Fields, Ch. III, Section 5; Neukirch II.7 and II.9. Paper: Lemma lem:tamagawa-exp ("ℓ^t-th power in $\mathbb{Q}_q^{\mathrm{ur}}$"; the units of its valuation ring are ℓ^t-th powers: important_notes.md point 8), Lemma lem:tamfactor (G/I, Kummer theory over $\mathbb{Q}_q^{\mathrm{ur}}$); Chapter 5's paperlink for galois:kummer-character.

9.5.3 4.3 conv:local:frobenius (convention, with a dictionary) P

Convention L3 in a convention environment, with the dictionary: (a) Frob acts on μ_m ($p \nmid m$) by $\zeta \mapsto \zeta^q$, $\chi_\ell(\mathrm{Frob}) = q$, $\chi_\ell(\Phi) = q^{-1}$; (b) Frob is defined modulo I_K and up to conjugation in G_K; any statement about $\rho(\mathrm{Frob})$ for ρ ramified must be checked to be independent of the lift (as in 5.6 and 6.2(iii)); (c) the arithmetic eigenvalues λ correspond to geometric eigenvalues λ^{-1}; when the multiset of eigenvalues on the relevant space is stable under inversion ($\mathrm{ad}^0 V$ at a good prime, the lines with eigenvalue ± 1) the two conventions give the same L-factor, otherwise they do not (the Steinberg case, 9.3); (d) the paper uses the arithmetic Frob in Lemma lem:tamfactor ("$\mathrm{Frob}^{-1}\, t\, \mathrm{Frob} = t^{q^{-1}}$", $\rho(\mathrm{Frob}) = \left(\begin{smallmatrix} \varepsilon q & c' \\ 0 & \varepsilon \end{smallmatrix}\right)$) and the geometric one, without saying so, in Section 1.1, Lemma lem:euler and Lemma lem:fpr (important_notes.md point 11). Reference: Deligne73 Section 2; Tate79 Section 4 (add both to bib/ch07.bib). Paper: as in (d).

9.5.4 4.4 prop:local:unram-basechange (proposition) P

Let $K/\mathbb{Q}_p$ be a finite unramified extension of degree f. Then $K^{\mathrm{ur}} = \mathbb{Q}_p^{\mathrm{ur}}$, hence $I_K = I_{\mathbb{Q}_p}$ and $P_K = P_{\mathbb{Q}_p}$ as subgroups of $G_K \subseteq G_{\mathbb{Q}_p}$; the tame characters agree (5.3: p is a uniformiser of both); $\mathrm{Frob}_K = \mathrm{Frob}_{\mathbb{Q}_p}^f$ modulo I. Consequently, for a representation ρ of $G_{\mathbb{Q}_p}$: $\rho(I_p)$, V^{I_p}, tame or wild ramification, the Swan and Artin conductors (8.3, 8.4) and the monodromy operator N (7.2) are unchanged by restriction to G_K. More generally (stated) for any finite $K/\mathbb{Q}_p$, $I_K = I_{\mathbb{Q}_p} \cap G_K$ is open of index $e(K/\mathbb{Q}_p)$ in $I_{\mathbb{Q}_p}$. (Translation theorem, Chapter 5, galois:translation-theorem.) Paper: Lemma lem:tamagawa-exp and Lemma lem:tamfactor ("over the unramified quadratic extension of $\mathbb{Q}_q$ (which does not change the inertia group)"), Lemma lem:twist (i) ("invariant under unramified base change"; the field-theoretic half).

9.5.5 4.5 prop:local:cyclotomic-local (proposition) P

(i) For $\ell \neq p$: $\mathbb{Q}_p(\mu_{\ell^k})/\mathbb{Q}_p$ is unramified of degree the order of p in $(\mathbb{Z}/\ell^k)^\times$ (Chapter 2, 5.4(iv) for the ℓ-adic valuations), and Frob acts on μ_{ℓ^∞} by $\zeta \mapsto \zeta^p$: $\chi_\ell|_{G_{\mathbb{Q}_p}}$ is unramified with $\chi_\ell(\mathrm{Frob}) = p$. (ii) For $\ell = p$: $\mathbb{Q}_p(\mu_{p^n})/\mathbb{Q}_p$ is totally ramified with group $(\mathbb{Z}/p^n)^\times$ (3.3(a)), so $\chi_p(I_p) = \mathbb{Z}_p^\times$, and $\mathbb{Q}_p(\mu_{p^\infty}) \cap \mathbb{Q}_p^{\mathrm{ur}} = \mathbb{Q}_p$. (iii) Global consequence (with 3.5): for $p \neq \ell$, $\chi_\ell(\mathrm{Frob}_p) = p$ (Chapter 5, C4; Chapter 6, 5.2). Reference: Serre, Local Fields, Ch. IV, Section 4; Washington Ch. 2. Paper: Lemma lem:tamfactor ($\chi_\ell\varepsilon$ on the diagonal, $\chi_\ell(\mathrm{Frob}) = q$), Lemma lem:old ($\bar{\chi}_\ell$), background.

9.6 5. Line-up, Section 5: ramification groups, inertia, wild inertia, the tame quotient

9.6.1 5.1 def:local:ramgroups (definition and proposition) S (P for (i), (ii), (iii))

Let L/K be finite Galois with group G, π_L a uniformiser of L. For $i \geq -1$ the *lower ramification groups* are $G_i = \{\sigma \in G : v_L(\sigma(x) - x) \geq i+1 \text{ for all } x \in \mathcal{O}_L\}$. (i) $G_{-1} = G$, $G_0 = I(L/K)$ is the kernel of $G \to \mathrm{Gal}(k_L/k_K)$ (so $\#G_0 = e$), the G_i are normal in G and decreasing, and $G_i = 1$ for $i \gg 0$; if $\mathcal{O}_L = \mathcal{O}_K[\pi_L]$ (e.g. L/K totally ramified), $\sigma \in G_i$ iff $v_L(\sigma(\pi_L) - \pi_L) \geq i+1$. (ii) $\sigma \mapsto \sigma(\pi_L)/\pi_L \bmod \mathfrak{m}_L$ is an injective homomorphism $G_0/G_1 \to k_L^\times$, independent of π_L; so G_0/G_1 is cyclic of order prime to p. (iii) For $i \geq 1$, $\sigma \mapsto \sigma(\pi_L)/\pi_L - 1$ induces an injective homomorphism $G_i/G_{i+1} \to \mathfrak{m}_L^i/\mathfrak{m}_L^{i+1} \simeq (k_L, +)$; so G_1 is a p-group, the *wild inertia group* $P(L/K)$, the unique Sylow p-subgroup of G_0; L/K is tamely ramified iff $G_1 = 1$. (iv) (Stated.) Herbrand's function $\varphi_{L/K}(u) = \int_0^u dt/[G_0 : G_t]$ defines the *upper numbering* $G^{\varphi(u)} = G_u$; upper numbering is compatible with quotients ($(G/H)^v = G^v H/H$, Herbrand); for G abelian the jumps of the upper filtration are integers (Hasse-Arf). (v) Examples (proved): $\mathbb{Q}_2(i)/\mathbb{Q}_2$, $\pi = 1+i$: $\sigma(\pi) - \pi = -2i$, $v_L = 2$, so $G_0 = G_1 = G$, $G_2 = 1$; $\mathbb{Q}_2(\sqrt{2})/\mathbb{Q}_2$, $\pi = \sqrt{2}$: $v_L(-2\sqrt{2}) = 3$, so $G_0 = G_1 = G_2 = G$, $G_3 = 1$; $\mathbb{Q}_3(\sqrt{-3})/\mathbb{Q}_3$ and $\mathbb{Q}_p(\sqrt{p})/\mathbb{Q}_p$ (p odd), $\mathbb{Q}_p(\zeta_p)/\mathbb{Q}_p$: tame, $G_1 = 1$. Reference: Serre, Local Fields, Ch. IV, Sections 1 to 3 (cite the sections; the injections are its Propositions 7 and 9 of Section 2 only if the writer checks the numbers); Neukirch II.10. Paper: Remark rem:wild, Section sec:wild (wild primes 2, 3), background for 8.3.

9.6.2 5.2 thm:local:inertia-structure (theorem) S (P for (i), (iii))

Let $G = G_K$. (i) $I_K = \mathrm{Gal}(\overline{\mathbb{Q}}_p/K^{\mathrm{ur}})$ is a closed normal subgroup with $G/I_K \simeq \mathrm{Gal}(\bar{k}_K/k_K) \simeq \hat{\mathbb{Z}}$ (4.2), and for every finite Galois L/K the image of I_K in $\mathrm{Gal}(L/K)$ is $I(L/K) = G_0$. (ii) P_K, the closure of the union of the images of the wild inertia groups (equivalently $\mathrm{Gal}(\overline{\mathbb{Q}}_p/K^{\mathrm{tame}})$, K^{tame} the union of the finite tamely ramified extensions of K), is a normal pro-p subgroup of G, and it is the unique pro-p Sylow subgroup of I_K. (iii) $K^{\mathrm{tame}} = \bigcup_{p\nmid m} K^{\mathrm{ur}}(\varpi^{1/m})$ for any uniformiser ϖ of K: every finite tamely ramified extension of K^{ur} is $K^{\mathrm{ur}}(\varpi^{1/m})$ for some m prime to p (by 3.6 at a finite level, and ϖu with $u \in \mathcal{O}_{K^{\mathrm{ur}}}^\times$ an m-th power by 4.2(iv)), and these are Galois over K^{ur} because $\mu_m \subset K^{\mathrm{ur}}$. (iv) (The filtration of $G_{\mathbb{Q}_p}$, stated.) $G_K \supset I_K \supset P_K \supset \{1\}$ is the start of the upper ramification filtration G_K^u ($u \geq -1$): $G_K^{-1} = G_K$, $G_K^0 = I_K$, $\overline{\bigcup_{u>0} G_K^u} = P_K$; upper numbering is defined on G_K by passing to the limit (possible by (iv) of 5.1). Reference: Serre, Local Fields, Ch. IV; Neukirch II.9; Serre, Abelian l-adic representations, Ch. I (cite chapters). Paper: Lemma lem:tamfactor ("the kernel of $I \to \langle t\rangle$ has trivial pro-ℓ quotient"), Remark rem:wild, Section 1.1 (I_p).

9.6.3 5.3 thm:local:tame-character (theorem and definition) P

Let m be prime to p and ϖ a uniformiser of K. (i) $\theta_m \colon I_K \to \mu_m$, $\sigma \mapsto \sigma(\varpi^{1/m})/\varpi^{1/m}$, is a continuous surjective homomorphism, independent of the choice of the root $\varpi^{1/m}$ and of the uniformiser ϖ (for $\varpi' = u\varpi$, $u^{1/m} \in K^{\mathrm{ur}}$ is fixed by I_K), and of replacing K by an unramified extension (4.4). (ii) $\theta_{mm'}^{m'} = \theta_m$; the limit $\theta \colon I_K \to \varprojlim_{p\nmid m} \mu_m = \prod_{\ell\neq p} \mathbb{Z}_\ell(1)$ is surjective with kernel P_K:

$$I_K/P_K \;\simeq\; \prod_{\ell\neq p} \mathbb{Z}_\ell(1),$$

procyclic, abelian, of supernatural order prime to p (it has no quotient of order p). (iii) For $\ell \neq p$ the ℓ-component $t_\ell \colon I_K \to \mathbb{Z}_\ell(1)$, $\sigma \mapsto \left(\sigma(\varpi^{1/\ell^k})/\varpi^{1/\ell^k}\right)_k$ (for a compatible system of roots), is the **tame character**; it is surjective, its kernel is an extension of $\prod_{\ell'\neq\ell,p} \mathbb{Z}_{\ell'}(1)$ by P_K, of supernatural order prime to ℓ, and t_ℓ identifies the maximal pro-ℓ quotient $I_K^{(\ell)}$ of I_K with $\mathbb{Z}_\ell(1)$; a topological generator t of $I_K^{(\ell)}$ is "a generator of tame inertia" in the paper's sense. (iv) Consequently every continuous homomorphism from

I_K to a pro-ℓ group (for example to $1+\ell M_n(\mathbb{Z}_\ell)$, or to a group of unipotent matrices over $\mathbb{Z}_\ell$) factors through t_ℓ. Reference: Serre, Local Fields, Ch. IV, Section 2 (cite the section); Neukirch II.9; Serre-Tate68, Section 1 (add to `bib/ch07.bib`). Paper: Lemma lem:tamfactor ("I acts through its maximal pro-ℓ tame quotient, generated by an element t", "the kernel of $I \to \langle t\rangle$ has trivial pro-ℓ quotient", "the tame character"); Chapter 16 (tame-inertia-example, model-setting), which imports exactly (ii) to (iv).

9.6.4 5.4 `thm:local:tame-relation` (theorem) P

(i) For $\tau \in G_K$ and $\sigma \in I_K$: $t_\ell(\tau\sigma\tau^{-1}) = \tau \cdot t_\ell(\sigma) = \chi_\ell(\tau)\, t_\ell(\sigma)$ (the Galois action on $\mathbb{Z}_\ell(1)$), and likewise $\theta(\tau\sigma\tau^{-1}) = \tau(\theta(\sigma))$. (Proof: $\tau^{-1}(\varpi^{1/m}) = \zeta_\tau \varpi^{1/m}$ with $\zeta_\tau \in \mu_m$; $\sigma \in I_K$ fixes μ_m; apply τ.) (ii) Hence for every arithmetic Frobenius lift $\mathrm{Frob} \in G_K$ and every $t \in I_K/P_K$:

$$\mathrm{Frob}\; t\; \mathrm{Frob}^{-1} = t^q, \qquad \text{equivalently} \qquad \mathrm{Frob}^{-1}\; t\; \mathrm{Frob} = t^{q^{-1}},$$

where q^{-1} is the inverse of q in $\prod_{\ell \neq p} \mathbb{Z}_\ell$, the ring of exponents of the procyclic group I_K/P_K; for the geometric Frobenius, $\Phi\, t\, \Phi^{-1} = t^{q^{-1}}$. The same holds in the maximal pro-ℓ quotient $I_K^{(\ell)} = \mathbb{Z}_\ell(1)$ with $q^{-1} \in \mathbb{Z}_\ell$, and modulo $\ker t_\ell$ in $G_K/\ker t_\ell$. (iii) (Structure, S.) $\mathrm{Gal}(K^{\mathrm{tame}}/K) \simeq \big(\prod_{\ell\neq p}\mathbb{Z}_\ell(1)\big) \rtimes \hat{\mathbb{Z}}$, the generator $1 \in \hat{\mathbb{Z}}$ acting by multiplication by q; the extension splits (choose Frob fixing all $\varpi^{1/m}$, 5.5); it is topologically generated by t and Frob subject to $\mathrm{Frob}\; t\; \mathrm{Frob}^{-1} = t^q$ (stated; Iwasawa). This is Chapter 6, 3.11 in profinite form and the foretaste in Chapter 1 (its semidirect-product item). Reference: Serre, Local Fields, Ch. IV, Section 2; Neukirch II.9 (cite the section); NSW, Ch. VII, Section 5 (structure, stated). Paper: Lemma lem:tamfactor ("$\mathrm{Frob}^{-1}\, t\, \mathrm{Frob} = t^{q^{-1}}$" with the arithmetic Frobenius), Lemma lem:euler ("Frobenius centralises σ if $q \equiv 1 \pmod 3$ and inverts it if $q \equiv 2$"), Chapter 16 (frobenius-on-h1-inertia).

9.6.5 5.5 `cor:local:frob-fixing-roots` (corollary) P

Let ϖ be a uniformiser of K and $(\varpi^{1/m})_{p\nmid m}$ a compatible system of roots. There is an arithmetic Frobenius lift $\mathrm{Frob} \in G_K$ fixing every $\varpi^{1/m}$, hence fixing the field $K_\varpi = K(\varpi^{1/m} : p \nmid m)$; two such lifts differ by an element of $G_{K_\varpi} \cap I_K$, which lies in $\ker t_\ell$ for every ℓ. Proof: K_ϖ/K is totally ramified (each $K(\varpi^{1/m})$ is, by Eisenstein), so $K_\varpi \cap K^{\mathrm{ur}} = K$, and by the translation theorem (Chapter 5) restriction $G_{K_\varpi} \to \mathrm{Gal}(K^{\mathrm{ur}}/K)$ is surjective. For $K = \mathbb{Q}_q$, $\varpi = q$: a Frobenius fixing all ℓ-power roots of q, so that the Kummer cocycle κ_q of 6.2 vanishes on it. Paper: Lemma lem:tamfactor (the correct reading of "c' is the image of κ", important_notes.md point 7).

9.6.6 5.6 `cor:local:finite-tame-image` (corollary) P

Let H be a group and $\rho\colon G_K \to H$ a homomorphism with $\rho(I_K)$ finite of order prime to p (for example $\rho\colon G_K \to \mathrm{GL}_n(\mathbb{Q}_\ell)$ continuous with finite image of inertia of order prime to p). Then $\rho(P_K) = 1$, $\rho|_{I_K}$ factors through the procyclic I_K/P_K, so $\rho(I_K)$ is cyclic, generated by $\rho(t)$ for a topological generator t of I_K/P_K, and $\rho(\mathrm{Frob})\, h\, \rho(\mathrm{Frob})^{-1} = h^q$ for all $h \in \rho(I_K)$, for every Frobenius lift; the statement does not depend on the lift ($\rho(I_K)$ is abelian). In particular if $\rho(I_K) = \langle g\rangle$ has order m: $\rho(\mathrm{Frob})$ commutes with g iff $q \equiv 1 \pmod m$, and inverts it iff $q \equiv -1 \pmod m$. Proof: $\rho(P_K)$ is a finite quotient of a pro-p group inside a group of order prime to p; then 5.3 and 5.4. Compare Chapter 6, 3.11 and 3.12 (which prove the same for a finite Galois extension directly). Reference: Serre, Local Fields, Ch. IV, Section 2. Paper: Lemma lem:IV ("the image is tame since $q \neq 3$, hence cyclic"), Lemma lem:euler (types II, II*, IV, IV*: Frobenius centralises or inverts σ by q mod 3; types III, III*: by q mod 4), Remark rem:types.

9.6.7 5.7 `prop:local:wild-image` (proposition) P

Let $\ell \neq p$, $E/\mathbb{Q}_\ell$ finite with ring of integers $\mathcal{O}$, and $\rho\colon G_K \to \mathrm{GL}_n(\mathcal{O})$ continuous. Then $\rho(P_K)$ is a finite p-group, mapping injectively to $\mathrm{GL}_n(\mathcal{O}/\ell)$ (the kernel $1+\ell M_n(\mathcal{O})$ of reduction is pro-ℓ). Hence: (i) there is an open subgroup $I_1 \subseteq I_K$, normal in G_K, on which ρ factors through t_ℓ (namely $I_1 = I_K \cap \rho^{-1}(1+\ell^2 M_n(\mathcal{O}))$,

or $1+\ell M_n$ for ℓ odd); (ii) ρ is tamely ramified ($\rho(P_K)=1$) iff the image of inertia has no element of order p; automatically so if $\rho(I_K)$ is finite of order prime to p. Warning: for a general ρ nothing forbids wild ramification at $p\geq 5$ ($\mathrm{GL}_2(\mathbb{Z}_\ell)$ contains elements of order 5 when $\ell\equiv 1 \pmod 5$). For $\rho=\rho_{E,\ell}$ with potentially good reduction the characteristic polynomials of $\rho(\sigma)$, $\sigma\in I_K$, have integer coefficients independent of ℓ, so $\rho(I_K)$ has order in $\{1,2,3,4,6\}$ for $p\geq 5$ and is tame (Serre-Tate68, Kraus90; proved or stated in Chapters 10, 11, not here). Examples of wild images at $p=2,3$ (stated; Chapters 10, 11): Q_8 and $\mathrm{SL}_2(\mathbb{F}_3)$ at $p=2$; cyclic of order 3 or 6 at $p=3$ (wild there, since I/P has no quotient of order 3). Reference: Serre-Tate68, Section 2; Serre, Local Fields, Ch. IV. Paper: Remark rem:wild ("At $q=2,3$ the Kodaira type does not determine $L_q(A_f,s)$"), Lemma lem:IV ("tame since $q\neq 3$"), Lemma lem:twist (ii) at $q=3$ (important_notes.md point 28).

9.6.8 5.8 `ex:local:tame-explicit` (worked example) P

Let $\ell\neq q$ be primes, $m=\ell^k$, $L=\mathbb{Q}_q(\mu_m, q^{1/m})$. With a primitive $\zeta\in\mu_m$ fixed, $\sigma_{a,b}\colon \zeta\mapsto\zeta^a$, $q^{1/m}\mapsto \zeta^b q^{1/m}$ identifies $\mathrm{Gal}(L/\mathbb{Q}_q)$ with the subgroup $\{(a,b): a\in\langle q\rangle\subseteq(\mathbb{Z}/m)^\times,\ b\in\mathbb{Z}/m\}$ of the affine group (Chapter 5, galois:kummer-affine), with $(a,b)(a',b')=(aa',ab'+b)$. The inertia subgroup is $\{a=1\}\simeq\mathbb{Z}/m$, generated by $t=\sigma_{1,1}$, and $t_\ell \bmod m$ is $\sigma_{1,b}\mapsto\zeta^b$; $\mathrm{Frob}=\sigma_{q,0}$ is a Frobenius fixing $q^{1/m}$ (5.5); $\mathrm{Frob}\, t\,\mathrm{Frob}^{-1}=\sigma_{1,q}=t^q$ and $\mathrm{Frob}^{-1}\, t\,\mathrm{Frob}=\sigma_{1,q'}=t^{q'}$ with $qq'\equiv 1 \pmod m$. (Checked: $(q,\ell,k)=(7,3,2)$ gives t^7 and t^4; $(13,3,2)$ gives t^4 and t^7; $(2,3,2)$ gives t^2 and t^5.) Paper: Lemma lem:tamfactor (the relation, at every finite level), Exercise 8.

9.7 6. Line-up, Section 6: Kummer theory over $\mathbb{Q}_p$, quadratic characters, local class field theory

9.7.1 6.1 `prop:local:kummer-qp` (proposition) P for (i), (iii); R for (ii)

Let K be finite over $\mathbb{Q}_p$, $\ell\neq p$ (the statements (i), (ii) hold for any n in characteristic 0). (i) (Chapter 5, galois:kummer-character and galois:l-adic-kummer, recalled.) For $x\in K^\times$ and a compatible system $(x^{1/\ell^k})_k$ in $\overline{\mathbb{Q}}_p$, the Kummer cocycle $\kappa_x\colon G_K\to\mathbb{Z}_\ell(1)$, $\kappa_x(\sigma)=(\sigma(x^{1/\ell^k})/x^{1/\ell^k})_k$, satisfies $\kappa_x(\sigma\tau)=\kappa_x(\sigma)+\chi_\ell(\sigma)\kappa_x(\tau)$; changing the system changes it by $\sigma\mapsto(\chi_\ell(\sigma)-1)z$; $\kappa_{xy}=\kappa_x+\kappa_y$ for the product systems; $\kappa_x \bmod \ell^k$ vanishes on $G_{K'}$ iff $x\in(K'^\times)^{\ell^k}$, for any $K'\supseteq K(\mu_{\ell^k})$. (ii) (Stated here, proved in Chapter 17 via Hilbert 90.) The Kummer maps $K^\times/(K^\times)^{\ell^k}\to H^1(K,\mu_{\ell^k})$ and $\widehat{K^\times}=\varprojlim K^\times/(K^\times)^{\ell^k}\to H^1(K,\mathbb{Z}_\ell(1))$ are isomorphisms. (iii) With 2.7: $H^1(K,\mathbb{Z}_\ell(1))\simeq\mathbb{Z}_\ell\times\mathbb{Z}/\ell^{v_\ell(q-1)}$, the projection to $\mathbb{Z}_\ell$ being v_K and the torsion being the image of the units, $\mathbb{F}_q^\times\otimes\mathbb{Z}_\ell\simeq\mathbb{Z}_\ell(1)/(q-1)\mathbb{Z}_\ell(1)$; $H^1(K,\mu_{\ell^k})$ has order $\ell^k\cdot\#\mu_{\ell^k}(K)$. Reference: Serre, Local Fields, Ch. XIV (cite chapter); NSW Ch. VII (cite chapter); Chapter 17. Paper: Lemma lem:tamfactor ("κ the Kummer class of q_E in $H^1(G,\mathbb{Z}_\ell(1))=\varprojlim\mathbb{Q}_q^\times/(\mathbb{Q}_q^\times)^{\ell^k}$"), Lemma lem:tamagawa-exp.

9.7.2 6.2 `thm:local:kummer-inertia` (theorem; the Kummer class on inertia) P

Let K be finite over $\mathbb{Q}_p$ with uniformiser ϖ, residue field $\mathbb{F}_q$, $\ell\neq p$, $x\in K^\times$, $a=v_K(x)$, $x=\varpi^a u$ with $u\in\mathcal{O}_K^\times$. Choose compatible systems of ℓ-power roots of ϖ and of u, and give x the product system. (i) $\kappa_x=a\,\kappa_\varpi+\kappa_u$. (ii) $\kappa_\varpi|_{I_K}=t_\ell$ (5.3, computed with ϖ), and $\kappa_u|_{I_K}=0$ because $u^{1/\ell^k}\in K^{\mathrm{ur}}$ (4.2(iv)); hence

$$\kappa_x|_{I_K}=v_K(x)\cdot t_\ell,$$

independently of all choices (a cocycle restricted to the normal subgroup I_K on which $\chi_\ell=1$ is a homomorphism, and coboundaries $(\chi_\ell-1)z$ vanish there). (iii) κ_u is unramified: it factors through $G_K/I_K=\langle\mathrm{Frob}\rangle$, and $\kappa_u(\mathrm{Frob})\in\mathbb{Z}_\ell(1)$ is well defined modulo $(q-1)\mathbb{Z}_\ell(1)$; its class in $\mathbb{Z}_\ell(1)/(q-1)\mathbb{Z}_\ell(1)\simeq\mu_{\ell^{v_\ell(q-1)}}(\mathbb{F}_q)$ is $\bar u^{(q-1)/\ell^{v_\ell(q-1)}}$, i.e. the class of $\bar u$ in $\mathbb{F}_q^\times\otimes\mathbb{Z}_\ell$ (Chapter 5, galois:kummer-finite-field (b), applied

to the Teichmuller decomposition $u = \omega(\bar{u})u_1$ with u_1 uniquely ℓ-divisible, 2.4, 2.7(i)). In particular $\kappa_u \equiv 0$ modulo coboundaries iff $\bar{u} \in (\mathbb{F}_q^\times)^{\ell^{v_\ell(q-1)}}$. (iv) If Frob fixes the chosen roots of ϖ (5.5), then $\kappa_\varpi(\mathrm{Frob}) = 0$ and $\kappa_x(\mathrm{Frob}) = \kappa_u(\mathrm{Frob})$. For any other lift $\mathrm{Frob}\cdot\sigma$, $\sigma \in I_K$: $\kappa_x(\mathrm{Frob}\,\sigma) = \kappa_x(\mathrm{Frob}) + q\,a\,t_\ell(\sigma)$. So $c' = \kappa_x(\mathrm{Frob})$ is well defined modulo the ideal $(a, q-1)\mathbb{Z}_\ell$ (after $\mathbb{Z}_\ell(1) \simeq \mathbb{Z}_\ell$), and $\min(v_\ell(a), v_\ell(q-1), v_\ell(c'))$ does not depend on any choice. (v) (Inertia on ℓ^k-th roots.) I_K fixes the ℓ^k-th roots of x iff $\ell^k \mid a$; the image of I_K in $\mathrm{Gal}(K^{\mathrm{ur}}(x^{1/\ell^k})/K^{\mathrm{ur}}) \simeq \mu_{\ell^k}$ has order $\ell^{k-\min(k,v_\ell(a))}$. Equivalently, x is an ℓ^k-th power in K^{ur} iff $\ell^k \mid v_K(x)$. Reference: Serre, Abelian l-adic representations, Ch. IV, A.1 (Tate curves; cite the chapter); Silverman ATAEC Ch. V (cite the chapter); elementary from 5.3, 4.2. Paper: Lemma lem:tamagawa-exp (proof: "inertia acts trivially on $E[\ell^t]$ exactly when … $\ell^t \mid n_q$"; (v) is the correct form, important_notes.md point 8), Lemma lem:tamfactor ("the restriction of κ to I is n times the tame character", "$v_\ell(n') = v_\ell(n)$", and the identification of c': (iii), (iv) are the field-theoretic half of the fix in important_notes.md point 7; the cohomological half is Chapter 17), Definition def:terms and Section sec:tam ($t_\ell(E, q)$ and the class of the unit part).

9.7.3 6.3 `prop:local:quadratic-ext` (proposition) P

(i) p odd: $\mathbb{Q}_p$ has exactly three quadratic extensions (2.7(vi) and Kummer theory for $n = 2$, Chapter 5): the unramified one $\mathbb{Q}_p(\sqrt{u_0}) = \mathbb{Q}_p(\mu_{p^2-1})$ (u_0 a non-square unit), and the ramified $\mathbb{Q}_p(\sqrt{p})$, $\mathbb{Q}_p(\sqrt{u_0 p})$. The field $\mathbb{Q}_p(\sqrt{p^*})$ is $\mathbb{Q}_p(\sqrt{p})$ if $p \equiv 1 \pmod 4$ and $\mathbb{Q}_p(\sqrt{-p}) = \mathbb{Q}_p(\sqrt{u_0 p})$ if $p \equiv 3 \pmod 4$ (-1 is a non-square then). Both ramified ones are tame (3.6). (ii) p odd: the quadratic characters of $G_{\mathbb{Q}_p}$ are 1, the unramified η_{ur} ($\eta_{\mathrm{ur}}(\mathrm{Frob}) = -1$), and two ramified ones $\chi, \chi\eta_{\mathrm{ur}}$; every ramified one restricts on I_p to the unique character of order 2 of I_p, namely $\theta_2 \colon \sigma \mapsto \sigma(\sqrt{p})/\sqrt{p}$ (unique because I_p/P_p is procyclic and P_p is pro-odd); all have Artin conductor exponent $a \le 1$ (8.4). (iii) $p = 2$: $\mathbb{Q}_2$ has seven quadratic extensions, $\mathbb{Q}_2(\sqrt{d})$ for $d \in \{-1, \pm 5, \pm 2, \pm 10\}$; the unramified one is $\mathbb{Q}_2(\sqrt{5}) = \mathbb{Q}_2(\sqrt{-3})$ ($-15 \equiv 1 \bmod 8$); $\mathbb{Q}_2(\sqrt{-1})$, $\mathbb{Q}_2(\sqrt{-5}) = \mathbb{Q}_2(\sqrt{3})$ are wildly ramified with conductor exponent 2; $\mathbb{Q}_2(\sqrt{\pm 2})$, $\mathbb{Q}_2(\sqrt{\pm 10})$ with conductor exponent 3 (5.1(v) and 8.3). Reference: Serre, Course in Arithmetic, Ch. II, Section 3; Serre, Local Fields, Ch. IV. Paper: Lemma lem:twist (the twist by $\mathbb{Q}(\sqrt{q^*})$ is ramified only at q, tamely for q odd), Lemma lem:euler (types I_0^*, I_n^*: $V = V' \otimes \chi_{q^*}$), Remark rem:types (a ramified quadratic character times a unipotent), Theorem thm:qual (b), (c).

9.7.4 6.4 `prop:local:global-quadratic-local` (proposition) P

Let d be a nonsquare integer, D the discriminant of $\mathbb{Q}(\sqrt{d})$ and χ_d its character (Chapter 5, C3). Via 3.5, $\chi_d|_{G_{\mathbb{Q}_p}}$ is the character of $\mathbb{Q}_p(\sqrt{d})$. (i) It is unramified iff $p \nmid D$, and then $\chi_d(\mathrm{Frob}_p) = \left(\frac{D}{p}\right)$ (Chapter 6, 4.2). (ii) For an odd prime q and χ_{q^*} (the paper's χ_q): at $p \neq q$, including $p = 2$, χ_{q^*} is unramified with $\chi_{q^*}(\mathrm{Frob}_p) = \left(\frac{p}{q}\right)$ ($q^* \equiv 1 \pmod 4$, so $\mathbb{Q}_2(\sqrt{q^*})$ is unramified); at $p = q$ it is ramified, tame, $\chi_{q^*}|_{I_q} = \theta_2$, and its local conductor exponent is 1. (iii) χ_{-3} at 3: tame, exponent 1; χ_{-4} at 2: wild, exponent 2; $\chi_{\pm 2}$ at 2: exponent 3; in each case the product of $p^{a_p(\chi)}$ is the conductor of the Dirichlet character (Chapter 2, 10.3; the general statement is 6.5(iii)). (iv) For a finite set S of odd primes, $\chi_S = \prod_{q \in S} \chi_{q^*}$ is unramified outside S and at each $q \in S$ restricts to θ_2 on I_q. (v) Consequently, for any representation ρ of $G_{\mathbb{Q}}$: $(\rho \otimes \chi_{q^*})|_{I_p} = \rho|_{I_p}$ for $p \neq q$, and $(\rho \otimes \chi_{q^*})|_{I_q} = \rho|_{I_q} \otimes \theta_2$; so twisting by χ_{q^*} changes nothing at $p \neq q$ (inertia, V^{I_p}, conductor exponent; Frobenius changes by the sign $\chi_{q^*}(\mathrm{Frob}_p)$), and at q multiplies the inertia action by ± 1. Reference: as 6.3; Chapter 6, 4.2, 4.3. Paper: Lemma lem:twist (i) ("χ is unramified at p, so E and $E^{(q)}$ become isomorphic over an unramified extension of $\mathbb{Q}_p$") and (ii) at q, Theorem thm:twist (χ_S), Theorem thm:quant, Lemma lem:euler ($V = V' \otimes \chi_{q^*}$), Theorem thm:qual (b), (c).

9.7.5 6.5 `thm:local:lcft` (theorem; local class field theory for $\mathbb{Q}_p$, explicit form) R

(i) (Local Kronecker-Weber.) $\mathbb{Q}_p^{\mathrm{ab}} = \mathbb{Q}_p^{\mathrm{ur}} \cdot \mathbb{Q}_p(\mu_{p^\infty})$. (ii) There is a unique continuous homomorphism $\mathrm{Art} \colon \mathbb{Q}_p^\times \to G_{\mathbb{Q}_p}^{\mathrm{ab}}$ with: $\mathrm{Art}(p)$ is the arithmetic Frobenius on $\mathbb{Q}_p^{\mathrm{ur}}$ and trivial on $\mathbb{Q}_p(\mu_{p^\infty})$; for

$u \in \mathbb{Z}_p^\times$, $\mathrm{Art}(u)$ is trivial on $\mathbb{Q}_p^{\mathrm{ur}}$ and acts on μ_{p^∞} by $\zeta \mapsto \zeta^{u^{-1}}$. It is injective with dense image, $\mathrm{Art}(\mathbb{Z}_p^\times) = I_p^{\mathrm{ab}}$ (image of I_p), $\mathrm{Art}(1+p^n\mathbb{Z}_p)$ is the image of the upper ramification group $G_{\mathbb{Q}_p}^n$ ($n \geq 1$), and it extends to an isomorphism $\mathbb{Q}_p^\times \simeq W_{\mathbb{Q}_p}^{\mathrm{ab}}$ (7.1). (iii) For a character χ of $G_{\mathbb{Q}_p}$ of finite order, the Artin conductor exponent (8.4) equals the least $n \geq 0$ with $\chi \circ \mathrm{Art}(1+p^n\mathbb{Z}_p) = 1$ (with $1+p^0\mathbb{Z}_p := \mathbb{Z}_p^\times$); with Chapter 6, 5.7 this gives "the Artin conductor of ψ_χ equals the conductor of the Dirichlet character χ" (Chapter 6, 5.7(ii), stated there with a pointer here). Warning: Deligne's normalisation sends uniformisers to geometric Frobenius elements; the book uses the arithmetic one (as Chapter 6, 5.6(iii)) and never needs the other. Reference: Serre, "Local class field theory" in Cassels-Frohlich (cite the chapter); Milne CFT, Ch. I (cite the chapter); Neukirch V. Paper: background (conductors of the quadratic characters in Lemma lem:twist; Section 1 table: χ_{-3}, χ_{-4}).

9.8 7. Line-up, Section 7: the Weil group and Weil-Deligne representations

9.8.1 7.1 `def:local:weil-group` (definition and proposition) P

$W_K = \{w \in G_K : \ w|_{K^{\mathrm{ur}}} \in \mathrm{Frob}^{\mathbb{Z}}\}$, with $d\colon W_K \to \mathbb{Z}$ defined by $w|_{K^{\mathrm{ur}}} = \mathrm{Frob}^{d(w)}$ and $\|w\| = q^{d(w)}$ (L4). (i) W_K is a dense subgroup of G_K containing I_K; $W_K/I_K \simeq \mathbb{Z}$; it is a topological group when I_K (with its profinite topology) is declared open, and this topology is finer than the one induced from G_K. (ii) $\|\mathrm{Frob}\| = q$, $\|\Phi\| = q^{-1}$, $\|\cdot\|$ is trivial on I_K, and $\|w\| = \chi_\ell(w)$ for every $\ell \neq p$ and $w \in W_K$ (4.5). (iii) Unramified characters $\mu\colon W_K \to \Omega^\times$ (trivial on I_K) are determined by $\mu(\Phi) \in \Omega^\times$, which may be any element (not only units, unlike characters of G_K); $\|\cdot\|$ is one, with $\|\Phi\| = q^{-1}$. (iv) (With 6.5.) $W_{\mathbb{Q}_p}^{\mathrm{ab}} \simeq \mathbb{Q}_p^\times$, with $\|\mathrm{Art}(x)\| = |x|_p^{-1}$ in the book's normalisation ($\mathrm{Art}(p)$ arithmetic, $\|\mathrm{Art}(p)\| = p$). Reference: Tate79, Section 1; Deligne73, Section 2 (add both to `bib/ch07.bib`); Rohrlich94, Sections 1 to 3 (add). Paper: background (Lemma lem:euler and Lemma lem:twist (iii) speak of "local components" and "Steinberg"; the Galois side is the Weil-Deligne representation).

9.8.2 7.2 `thm:local:grothendieck` (theorem, Grothendieck's ℓ-adic monodromy theorem) P

Let $\ell \neq p$, $E/\mathbb{Q}_\ell$ finite, V a finite-dimensional E-vector space and $\rho\colon G_K \to \mathrm{GL}(V)$ continuous (or only $\rho\colon W_K \to \mathrm{GL}(V)$ continuous). Fix a generator ζ of $\mathbb{Z}_\ell(1)$ and let $t_\zeta\colon I_K \to \mathbb{Z}_\ell$ be as in L5. Then there are an open subgroup $I_1 \subseteq I_K$, normal in G_K, and a unique nilpotent $N \in \mathrm{End}_E(V)$ with

$$\rho(\sigma) = \exp\bigl(t_\zeta(\sigma)N\bigr) \quad \text{for all } \sigma \in I_1,$$

and $\rho(w)N\rho(w)^{-1} = \|w\|N$ for all $w \in W_K$ (so $\rho(\mathrm{Frob})N\rho(\mathrm{Frob})^{-1} = qN$). $N = 0$ iff $\rho(I_K)$ is finite. Replacing ζ by ζ^a ($a \in \mathbb{Z}_\ell^\times$) replaces N by $a^{-1}N$. Proof: choose a stable lattice; take $I_1 = I_K \cap \rho^{-1}(1+\ell^2 M_n(\mathcal{O}))$; by 5.7 and 5.3(iv), $\rho|_{I_1}$ factors through $t_\zeta(I_1) = \ell^m\mathbb{Z}_\ell$; put $N = \ell^{-m}\log\rho(\sigma_1)$ for $\sigma_1 \in I_1$ with $t_\zeta(\sigma_1) = \ell^m$ (log and exp converge on $1+\ell^2M_n(\mathcal{O})$ and $\ell^2M_n(\mathcal{O})$); the relation follows from 5.4(i), and $N \sim qN$ forces all eigenvalues of N to vanish. Reference: Serre-Tate68, Appendix (cite the section); Deligne73, Section 8; Tate79, Section 4.2; Rohrlich94, Section 4. Paper: background for Lemma lem:euler ("through a quadratic character times a unipotent when it is potentially multiplicative"), Lemma lem:tamfactor (the unipotent action of t); Chapters 11, 15.

9.8.3 7.3 `def:local:wd` (definition and proposition) P

Let Ω be a field of characteristic 0. A *Weil-Deligne representation* of W_K over Ω is a triple (r, N, V): V a finite-dimensional Ω-vector space, $r\colon W_K \to \mathrm{GL}(V)$ a homomorphism whose restriction to I_K has open kernel (so $r(I_K)$ is finite), and $N \in \mathrm{End}(V)$ nilpotent with

$$r(w)\,N\,r(w)^{-1} = \|w\|\,N \qquad (w \in W_K),$$

that is $r(\mathrm{Frob})Nr(\mathrm{Frob})^{-1} = qN$ and $r(\Phi)Nr(\Phi)^{-1} = q^{-1}N$ (L4). Morphisms commute with r and N. Constructions: $V \oplus V'$; $V \otimes V'$ with $N \otimes 1 + 1 \otimes N'$; the dual with $r^\vee(w) = r(w^{-1})^t$ and $-N^t$; $\mathrm{Hom}(V, V')$; $\mathrm{End}\, V$ and $\mathrm{ad}^0 V$ (trace zero) with $N_{\mathrm{ad}^0} = [N, \cdot]$; twisting by a character μ of W_K. (r, N) is *Frobenius-semisimple* if $r(\Phi)$ is semisimple; every (r, N) has a Frobenius-semisimplification (replace $r(\Phi)$ by its semisimple part; stated). (Check of the relation for ad^0: $r(w)[N, x]r(w)^{-1} = \|w\|[N, r(w)xr(w)^{-1}]$.) Reference: Deligne73, Section 8; Tate79, Section 4.1. Paper: background.

9.8.4 7.4 thm:local:wd-equivalence (theorem) S (construction P, equivalence R)

In the situation of 7.2, fix Φ (a geometric Frobenius lift) and ζ. Put $r(\Phi^n\sigma) = \rho(\Phi^n\sigma)\exp(-t_\zeta(\sigma)N)$ for $n \in \mathbb{Z}$, $\sigma \in I_K$. (i) (r, N) is a Weil-Deligne representation over E (P: r is a homomorphism by 5.4(i) and the relation for N; r is trivial on I_1). (ii) Its isomorphism class does not depend on Φ or ζ (S). (iii) (R) $\rho \mapsto \mathrm{WD}(\rho)$ is an equivalence between continuous representations of W_K on finite-dimensional E-spaces and Weil-Deligne representations over E; those that extend to G_K are the ones for which the eigenvalues of $r(\Phi)$ are ℓ-adic units. It is compatible with $\oplus$, $\otimes$, duals, ad^0, and unramified twists; $\mathbb{Q}_\ell(1)$ goes to $(\|\cdot\|, 0)$ (the unramified character with $\Phi \mapsto q^{-1}$). (iv) The Weil-Deligne representation of ρ (Frobenius-semisimplified) does not depend on ℓ for the Tate modules of elliptic curves (stated; Chapters 11, 15), which is why L-factors and conductors are independent of ℓ. Reference: Deligne73, Section 8 (cite the section); Tate79, Section 4.2; Rohrlich94. Paper: Section 1.1 ("$L_p(A_f, s)$" independent of ℓ), Lemma lem:euler, Lemma lem:twist (iii) ("local component ... Steinberg").

9.8.5 7.5 def:local:sp (definition and proposition) P (classification (iii) S)

(i) For $n \geq 1$, $\mathrm{sp}(n)$ is $V = \bigoplus_{i=0}^{n-1} \Omega e_i$ with $r(w)e_i = \|w\|^i e_i$ and $Ne_i = e_{i+1}$ ($e_n := 0$). The relation holds: $r(w)Nr(w)^{-1}e_i = \|w\|e_{i+1}$. On $\mathrm{sp}(2)$: Φ acts on e_0, e_1 by $1, q^{-1}$ (geometric), Frob by $1, q$; $\ker N = \Omega e_1$. For a character μ of W_K, $\mathrm{sp}(n) \otimes \mu$ is the *special (Steinberg) representation twisted by* μ; it is indecomposable; $\mathrm{sp}(2) \otimes \mu$ for μ unramified is the *unramified twist of Steinberg*, for μ ramified a *ramified twist*. (ii) (Every two-dimensional (r, N) with $N \neq 0$ is a twist of $\mathrm{sp}(2)$.) If $\dim V = 2$, $N \neq 0$ and (r, N) is Frobenius-semisimple, then $(r, N) \simeq \mathrm{sp}(2) \otimes \mu$ for a unique character μ of W_K. Proof: $\ker N = \mathrm{im}\, N$ is an r-stable line with character μ'; r on $V/\ker N$ is a character μ, and $N\colon V/\ker N \to \ker N$ is an isomorphism with $\mu' = \|\cdot\|\mu$; a Frobenius eigenvector e_0 outside $\ker N$ exists because $r(\Phi)$ is semisimple with the distinct eigenvalues $\mu(\Phi)$, $q^{-1}\mu(\Phi)$; $r(I_K)$ is finite, hence acts semisimply, through the same character $\mu|_{I_K} = \mu'|_{I_K}$ on $\ker N$ and on $V/\ker N$, hence by scalars; put $e_1 = Ne_0$. (iii) (S/R.) A Frobenius-semisimple $(r, 0)$ of dimension 2 is either a sum of two characters or irreducible; an irreducible one is induced from a character of $W_{K'}$, K'/K quadratic, except possibly when $p = 2$ ("primitive" representations). Reference: Tate79, Section 4.1.4; Rohrlich94, Sections 5 to 8. Paper: Lemma lem:twist (iii) ("unramified twist of the Steinberg representation"), Lemma lem:euler (type I_n: "unramified twist of the Steinberg representation"); Chapter 15 (local components: principal series, Steinberg, supercuspidal).

9.8.6 7.6 ex:local:tate-shape-wd (worked example) P

Let $\ell \neq p$, $x \in K^\times$ with $a = v_K(x)$, $\varepsilon\colon G_K \to \{\pm 1\}$ unramified, and ρ on $\mathbb{Q}_\ell e_1 \oplus \mathbb{Q}_\ell e_2$ given by

$$\rho(g) = \begin{pmatrix} \chi_\ell(g)\varepsilon(g) & \varepsilon(g)\kappa_x(g) \\ 0 & \varepsilon(g) \end{pmatrix}$$

(the twist by ε of the Kummer extension $\left(\begin{smallmatrix} \chi_\ell & \kappa_x \\ 0 & 1 \end{smallmatrix}\right)$ of Chapter 5, galois:l-adic-kummer; the corner of the twisted representation is $\varepsilon\kappa_x$, see Part 15, N1). (i) On I_K: $\rho(\sigma) = \left(\begin{smallmatrix} 1 & a\, t_\ell(\sigma) \\ 0 & 1 \end{smallmatrix}\right)$ (6.2(ii)); so $N = a\, u_\zeta E_{12}$ ($u_\zeta \in \mathbb{Z}_\ell^\times$ from identifying the e_1-line with $\mathbb{Z}_\ell(1)$), and $N \neq 0$ iff $a \neq 0$. (ii) If $a \neq 0$: $\mathrm{WD}(\rho) \simeq \mathrm{sp}(2) \otimes \mu_\varepsilon$ with μ_ε unramified, $\mu_\varepsilon(\Phi) = \varepsilon(\mathrm{Frob})$ (e_0 a Φ-eigenvector over e_2, $e_1 = Ne_0$); $V^{I_K} = \mathbb{Q}_\ell e_1$ with Φ acting by εq^{-1} and Frob by εq. (iii) If $a = 0$: ρ is unramified with Frob-eigenvalues εq, ε (distinct), $N = 0$. (iv)

Integrally ($\mathbb{Z}_\ell e_1 \oplus \mathbb{Z}_\ell e_2$): I_K acts on the lattice mod ℓ^k trivially iff $\ell^k \mid a$ (6.2(v)). Chapters 10 and 11 show that $T_\ell E$ at a prime of multiplicative reduction has exactly this shape with $x = q_E$ the Tate parameter and ε trivial or the unramified quadratic character (split or nonsplit). Paper: Lemma lem:tamagawa-exp, Lemma lem:tamfactor (the matrices, $te_2 = e_2 + n'e_1$ with $v_\ell(n') = v_\ell(n)$), Lemma lem:euler (type I_n), Definition def:terms ($\delta_q = 1$).

9.9 8. Line-up, Section 8: inertia invariants, local L-factors, conductors, ϵ-factors

9.9.1 8.1 `prop:local:invariants` (proposition) P

Let $\rho\colon G_K \to \mathrm{GL}(V)$ be continuous ($\ell \neq p$) with $\mathrm{WD}(\rho) = (r, N)$. Then

$$V^{\rho(I_K)} = (\ker N)^{r(I_K)},$$

and $\rho(\Phi) = r(\Phi)$ on it (for the Φ used in 7.4). In particular $\dim V^{I_K}$ and the characteristic polynomial of Frobenius on V^{I_K} depend only on the Weil-Deligne representation. Proof: if $v \in \ker N$ is $r(I_K)$-fixed then $\rho(\sigma)v = r(\sigma)\exp(t_\zeta(\sigma)N)v = v$; conversely, $\exp(tN)v = v$ for t in an open subgroup of $\mathbb{Z}_\ell$ gives $Nv = 0$ (take logarithms), and then $r(\sigma)v = v$. Examples: $\mathrm{sp}(2) \otimes \mu$ with μ unramified: Ωe_1; with μ ramified: 0; $(r, 0)$ with $r(I_K)$ finite: $V^{r(I_K)}$. Notation: the paper's $\delta_p = \dim V^{I_p}$ for $V = V_\ell E$ (Section 1.1). Reference: Tate79, Section 4.2; Deligne73, Section 8. Paper: Section 1.1 (δ_p, $A_f^{I_p}$), Lemma lem:euler ("$(\mathrm{ad}^0 V)^{I_q}$ is a line"), Lemma lem:dfg21 and the paragraph after it ($(V^D)^{I_p}$).

9.9.2 8.2 `def:local:L-factor` (definition, convention and proposition) P

For ρ as in 8.1 (or a Weil-Deligne representation (r, N)):

$$L(V, s) = \det\big(1 - \Phi\, q^{-s} \;\big|\; V^{I_K}\big)^{-1} = \det\big(1 - r(\Phi)q^{-s} \;\big|\; (\ker N)^{r(I_K)}\big)^{-1}, \qquad \Phi = \mathrm{Frob}^{-1} \text{ geometric.}$$

(i) It is well defined (any two geometric Frobenius lifts agree on V^{I_K}) and depends only on the Weil-Deligne class. (ii) $L(V \oplus V', s) = L(V, s)L(V', s)$. (iii) For μ unramified with $\mu(\Phi) = \lambda$: $L(\mu, s) = (1 - \lambda q^{-s})^{-1}$; $L(\mathbb{Q}_\ell(1), s) = (1 - q^{-1-s})^{-1}$; $L(V(1), s) = L(V, s+1)$, and more generally $L(V \otimes \|\cdot\|^k, s) = L(V, s+k)$. (iv) With the arithmetic Frob in place of Φ one would get $L(V(1), s) = L(V, s-1)$ and the factor $(1 - q^{1-s})^{-1}$ for $\mathbb{Q}_\ell(1)$; this is why the geometric convention is forced by the paper's Lemma lem:fpr ("$L_p(A_f, 1)^{-1} = L_p(A_f(1), 0)^{-1}$") and table (important_notes.md point 11). (v) Examples: $L(\mathrm{sp}(2) \otimes \mu, s) = (1 - \mu(\Phi)q^{-1-s})^{-1}$ for μ unramified, $= 1$ for μ ramified; for $E/\mathbb{Q}$ with good reduction at $p \neq \ell$, the arithmetic Frobenius on $V_\ell E$ has eigenvalues α, β ($\alpha + \beta = a_p$, $\alpha\beta = p$; Chapter 9), the geometric one α^{-1}, β^{-1}, so $L(V_\ell E, s) = (1 - a_p p^{-1-s} + p^{-1-2s})^{-1} = L_p(E, s+1)$ and $L(V_\ell E^\vee, s) = (1 - a_p p^{-s} + p^{1-2s})^{-1} = L_p(E, s)$ (as Chapter 6, 3.7). Reference: Deligne73, Section 8 (cite the section); Tate79, Section 4.1; DFG04 1.7.2 for the convention in the paper's source. Paper: Section 1.1 (definition of $L_p(A_f, s)$, L_p^{nv}), the table of Section 1, Definition def:terms, Lemma lem:euler, Lemma lem:fpr, the paragraph after Lemma lem:dfg21 ("$\det(1 - \mathrm{Frob}_p \mid (V^D)^{I_p}) = L_p(A_f, 1)^{-1}$", true with Frob_p read as geometric).

9.9.3 8.3 `def:local:swan` (definition and theorem) S (definition P, integrality R)

Let V be a finite-dimensional representation over a field Ω of G_K, W_K or just I_K, such that I_K acts through a finite quotient: there is a finite Galois L/K, $G = \mathrm{Gal}(L/K)$, with I_K acting through its image $G_0 = I(L/K)$ (5.2(i)). For an ℓ-adic ρ, apply this to $r|_{I_K}$ of 7.4, or to $\rho|_{P_K}$, which has finite image by 5.7. With $g_i = \#G_i$:

$$\mathrm{sw}(V) = \sum_{i \geq 1} \frac{g_i}{g_0} \dim\big(V/V^{G_i}\big), \qquad a(V) = \sum_{i \geq 0} \frac{g_i}{g_0} \dim\big(V/V^{G_i}\big) = \dim V/V^{I_K} + \mathrm{sw}(V).$$

(i) These do not depend on L (Herbrand); in upper numbering $\mathrm{sw}(V) = \int_0^\infty \dim(V/V^{G_K^u})\,du$ (S). (ii) (R; Artin, Serre) $\mathrm{sw}(V)$ and $a(V)$ are nonnegative integers. (iii) $\mathrm{sw}(V) = 0$ iff P_K acts trivially (tame); sw depends only on $V|_{P_K}$; both are additive in V and unchanged by unramified base change (4.4) and by twisting by unramified characters. (iv) For a character χ: $a(\chi) = 0$ if unramified, 1 if tamely ramified, and $1 + c$ if wildly ramified with largest upper jump c; this agrees with 6.5(iii). Examples: χ_{-4} at 2: $\mathrm{sw} = 1$, $a = 2$; $\chi_{\pm 2}$ at 2: $\mathrm{sw} = 2$, $a = 3$ (5.1(v)). (v) The same definitions apply verbatim to representations over $\mathbb{F}_\ell$ (where the image of inertia is finite); this is the local factor of the Serre conductor used in Chapter 11 and in the paper (Lemma lem:IV: "the Serre conductor exponent at q is $\dim E[3] - \dim E[3]^{I_q} = 1$, there being no wild part"). Reference: Serre, Local Fields, Ch. VI, Section 2 (cite the section); Serre, Linear representations of finite groups, Ch. 19 (Artin representation; cite the chapter). Paper: Lemma lem:IV, Remark rem:types, Remark rem:wild (wild primes), Lemma lem:twist (ii) ("the conductor exponents are the standard ones for q odd (tame)", important_notes.md point 28).

9.9.4 8.4 `def:local:conductor` (definition and proposition) P

The conductor exponent of a Weil-Deligne representation (r, N, V) is

$$a(r, N) = a(r) + \dim V^{r(I_K)} - \dim(\ker N)^{r(I_K)} = \dim V - \dim(\ker N)^{r(I_K)} + \mathrm{sw}(r),$$

and for ℓ-adic ρ, $a(\rho) := a(\mathrm{WD}(\rho)) = \dim V - \dim V^{\rho(I_K)} + \mathrm{sw}(\rho)$, where $\mathrm{sw}(\rho) = \mathrm{sw}(\rho|_{P_K})$ (8.1; $\rho(P_K) = r(P_K)$ since $P_K \subseteq \ker t_\zeta$). Properties: additive; invariant under unramified twists and unramified base change; $a = 0$ iff unramified; for ρ tamely ramified, $a(\rho) = \dim V - \dim V^{I_K}$. For $E/\mathbb{Q}$ and $\ell \neq p$, $a(V_\ell E|_{G_{\mathbb{Q}_p}})$ is the exponent of p in the conductor of E (Ogg-Saito, stated; Chapter 10). Reference: Serre-Tate68, Section 2 (conductor of an ℓ-adic representation); Tate79, Section 4.1; Rohrlich94. Paper: Lemma lem:twist (ii), (iii) ("conductor exponent 2"), Lemma lem:IV, Remark rem:types, Theorem thm:qual (Serre conductors), Section 1.4.

9.9.5 8.5 `def:local:epsilon` (theorem and definition) R

(Langlands, Deligne.) For a nontrivial additive character ψ of K and a Haar measure dx there is a unique way to attach $\epsilon(r, \psi, dx) \in \Omega^\times$ to representations r of W_K (with $r(I_K)$ finite), multiplicative in exact sequences, inductive in degree 0, and equal to Tate's local constant for characters; for Weil-Deligne representations $\epsilon(r, N, \psi, dx) = \epsilon(r, \psi, dx)\det\big(-\Phi \;\big|\; V^{r(I_K)}/(\ker N)^{r(I_K)}\big)$. The local root number is $w(V) = \epsilon(V, \psi, dx)/|\epsilon(V, \psi, dx)|$ (for V with real structure; stated). The paper does not use ϵ-factors; Chapter 15 recalls this item. Reference: Deligne73, Sections 4 to 5 and 8; Tate79, Sections 3 and 4. Paper: none (background for Chapter 15 and the functional equations of Chapter 19).

9.10 9. Line-up, Section 9: the local shapes the paper meets, and their adjoints

Standing hypotheses of this section: $K = \mathbb{Q}_q$ (q the paper's prime, so $\#k_K = q$), $\ell \neq q$, V a two-dimensional $\mathbb{Q}_\ell$-representation (or over a finite extension) of $G_{\mathbb{Q}_q}$ with $\det V|_{I_q} = 1$ (true for $V_\ell E$, whose determinant is χ_ℓ, unramified at q), $A = \mathrm{ad}^0 V$ the trace-zero adjoint (Chapter 4), Φ geometric. The linear algebra of the inertia actions is Chapter 4, items 9.1 to 9.4, and Chapter 6, 4.10; this section adds the Weil-Deligne description, the conductors and the L-factors. The dictionary to Kodaira types is a black box from Chapters 10 and 11 and is quoted only in the last column of 9.4.

9.10.1 9.1 `prop:local:unramified-shape` (proposition) P

If V is unramified with arithmetic Frobenius eigenvalues α, β: $\mathrm{WD}(V) = (r, 0)$, $a(V) = 0$, $L(V, s) = (1 - \alpha^{-1}q^{-s})^{-1}(1 - \beta^{-1}q^{-s})^{-1}$, A is unramified with eigenvalues $\alpha/\beta, 1, \beta/\alpha$ for both Frobenii (inversion-

stable set), and

$$L(A,s)^{-1}=\Big(1-\frac{\alpha}{\beta}q^{-s}\Big)(1-q^{-s})\Big(1-\frac{\beta}{\alpha}q^{-s}\Big),\qquad L(A,1)^{-1}=\frac{q-1}{q}\cdot\frac{(q+1)^2-a^2}{q^2}\ \text{ if } \alpha+\beta=a,\ \alpha\beta=q$$

(the algebraic identity is Chapter 2, 11.4). Paper: the I_0^* row of the table ($V=V'\otimes\chi_{q^*}$, $A=\mathrm{ad}^0\,V'$ unramified), Theorem thm:dfg14 (Wiles's level-raising factor at a good prime), Lemma lem:euler.

9.10.2 9.2 `prop:local:tame-cyclic-shape` (proposition) P

Suppose $\rho(I_q)$ is finite cyclic of order $m\in\{2,3,4,6\}$ with $q\nmid m$ (tame; 5.6), generated by g. Then $N=0$ and: (i) $m=2$: $g=-1$; for q odd and any ramified quadratic character χ of $G_{\mathbb{Q}_q}$ (6.3), $V\otimes\chi$ is unramified, $V^{I_q}=0$, $a(V)=2$, $L(V,s)=1$, and $A=\mathrm{ad}^0(V\otimes\chi)$ is unramified, so $L(A,s)$ is as in 9.1 for $V\otimes\chi$. (ii) $m\in\{3,4,6\}$: g has eigenvalues ζ,ζ^{-1} (ζ primitive of order m, Chapter 4, 9.3), so $V^{I_q}=0$, $a(V)=2$ (tame) and $L(V,s)=1$. Put $m'=3$ if $m\in\{3,6\}$ and $m'=4$ if $m=4$, and $\chi=\chi_{-3}$, resp. χ_{-4}. Then Frob g Frob$^{-1}=g^q$ (5.6), so ρ(Frob) commutes with g if $q\equiv1\pmod{m'}$ and conjugates g to g^{-1} if $q\equiv-1\pmod{m'}$ (for $m=6$, $g=-\sigma$ and $\sigma\mapsto\sigma^q$); in the first case $V\otimes\Omega(\zeta)$ is a sum of two characters of $W_{\mathbb{Q}_q}$ restricting to $\zeta^{\pm1}$-characters of I_q; in the second $V\otimes\Omega(\zeta)$ is induced from a character of W_{K_2}, K_2 the unramified quadratic extension (the two eigenlines are swapped by Frob). (iii) A^{I_q} is the line spanned by $2g-\operatorname{tr}g$ (that is $2\sigma+1$, with square -3, for $m=3,6$; $\tau=g$ for $m=4$; Chapter 4, 9.2), and both Frob and Φ act on it by $\chi(q)\in\{\pm1\}$ (for Φ: $\Phi g\Phi^{-1}=g^{q'}$ with $qq'\equiv1$, and $q'\equiv q\pmod{m'}$). Hence

$$L(A,s)^{-1}=1-\chi(q)\,q^{-s},\qquad a(A)=2.$$

Reference: elementary from 5.6, 8.1 to 8.4 and Chapters 4 and 6. Paper: Lemma lem:euler (types $\mathrm{II},\mathrm{II}^*,\mathrm{IV},\mathrm{IV}^*$ with χ_{-3}, types $\mathrm{III},\mathrm{III}^*$ with χ_{-4}, type I_0^* via (i)), the table of Section 1, Lemma lem:IV (cyclic tame image; the mod 3 computation is Chapter 4, 9.4 and Chapter 6, 4.8), Remark rem:types, Section 1.4 (conductor exponent 2 at additive primes).

9.10.3 9.3 `prop:local:steinberg-shape` (proposition) P

Let μ be an unramified character and χ a ramified quadratic character of $G_{\mathbb{Q}_q}$. (i) $V=\mathrm{sp}(2)\otimes\mu$: $V^{I_q}=\ker N$ is a line on which Φ acts by $\mu(\Phi)q^{-1}$ (Frob by μ(Frob)q); $a(V)=1$; $L(V,s)=(1-\mu(\Phi)q^{-1-s})^{-1}$. (ii) $V=\mathrm{sp}(2)\otimes\mu\chi$: $V^{I_q}=0$, $L(V,s)=1$, $a(V)=2a(\chi)$, which is 2 for q odd (tame χ) and 4 or 6 for $q=2$ (6.3(iii)). This is the content of "the twist of an unramified twist of Steinberg by a ramified quadratic character has conductor exponent 2" (Lemma lem:twist (iii), q odd). (iii) The adjoint: $\mathrm{ad}^0(\mathrm{sp}(2)\otimes\mu\chi^j)=\mathrm{ad}^0\,\mathrm{sp}(2)\simeq\mathrm{sp}(3)\otimes\|\cdot\|^{-1}$: in the basis E_{01}, $h=E_{00}-E_{11}$, $E_{10}=N$ (with $E_{ij}=e_i\otimes e_j^\vee$), Φ acts by $q,1,q^{-1}$ and $[N,E_{01}]=-h$, $[N,h]=2N$, $[N,N]=0$. So $A^{I_q}=\ker[N,\cdot]=\Omega N$, Φ acts on it by q^{-1} (Frob by q), $a(A)=2$ and

$$L(\mathrm{ad}^0\,V,s)=(1-q^{-1-s})^{-1}\qquad\text{for } V=\mathrm{sp}(2)\otimes\mu \text{ and } V=\mathrm{sp}(2)\otimes\mu\chi.$$

With the arithmetic Frobenius one would get $(1-q^{1-s})^{-1}$; the paper's table has $1-q^{-1-s}$ (important_notes.md point 11). (iv) The line ΩN is the line $\mathbb{Z}_\ell e$, $e=e_1\otimes e_2^\vee$, of Lemma lem:tamfactor (in the paper's basis, 7.6). Reference: Tate79, Section 4.1; Rohrlich94, Sections 9 to 11; elementary (verified symbolically). Paper: Lemma lem:twist (iii), Lemma lem:euler (types I_n and I_n^*: "$(\mathrm{ad}^0\,V)^{I_q}$ is a line with eigenvalue q^{-1}", "$V=V'\otimes\chi_{q^*}$ with V' Steinberg … $\mathrm{ad}^0\,V=\mathrm{ad}^0\,V'$"), Definition def:terms ($e_\ell=0$ at multiplicative primes: naive and true factors agree), Lemma lem:fpr ($L_p(A_f,1)^{-1}=1-p^{-2}$ at a multiplicative p), Theorem thm:dfg14 (the factor $1-p^{-2}$), Lemma lem:tamfactor.

9.10.4 9.4 `thm:local:shapes-table` (summary table) P (the last column R, forward)

For V two-dimensional, $\det V|_{I_q}=1$, $q\geq5$ (so every finite inertia image below is tame), the shapes and their invariants are:

shape of WD(V)	$\rho(I_q)$	$\dim V^{I_q}$	$a(V)$	$L(V,s)^{-1}$	$\dim A^{I_q}$	$L(A,s)^{-1}$ (Φ geometric)	Kodaira types (Ch. 10, 11)
unramified, Frob-eigenvalues α, β	1	2	0	$(1-\alpha^{-1}q^{-s})(1-\beta^{-1}q^{-s})$	3	$(1-\frac{\alpha}{\beta}q^{-s})(1-q^{-s})(1-\frac{\beta}{\alpha}q^{-s})$	I_0
(unramified)$\otimes\{\pm1\}$ χ, χ ramified quadratic		0	2	1	3	as for the untwisted row	I_0^*
$\mathrm{sp}(2)\otimes\mu$, μ unramified	unipotent, infinite	1	1	$1-\mu(\Phi)q^{-1-s}$	1	$1-q^{-1-s}$	I_n, $n\geq 1$
$\mathrm{sp}(2)\otimes\mu\chi$	$\pm$ unipotent	0	2	1	1	$1-q^{-1-s}$	I_n^*, $n\geq 1$
$N=0$, $\rho(I_q)$ cyclic of order 3 or 6	$\langle\sigma\rangle$ or $\langle-\sigma\rangle$	0	2	1	1	$1-\chi_{-3}(q)q^{-s}$	IV, IV*; II, II*
$N=0$, $\rho(I_q)$ cyclic of order 4	$\langle\tau\rangle$	0	2	1	1	$1-\chi_{-4}(q)q^{-s}$	III, III*

$\delta_q = \dim V^{I_q}$ is $2,0,1,0,0,0$ down the rows, and the naive factor of the paper is $L(A,s)$ when $\delta_q>0$ and 1 when $\delta_q=0$ (Section 1.1); so the omitted factor is $L(A,s)^{-1}$ in rows 2, 4, 5, 6 and nothing in rows 1, 3. The ℓ-adic valuations at $s=1$ ($e_\ell(E,q)$) are Chapter 2 (11.3 to 11.5) and Chapter 19. Writer's note: Chapter 4's table 9.3 already lists $\rho(I_q)$, $\dim V^{I_q}$, $\dim A^{I_q}$ and the generator of A^{I_q}; here cite it for those columns and add the columns $a(V)$, $L(V,s)$, $L(A,s)$. Reference: as 9.1 to 9.3; Kraus90 and Serre-Tate68 for the Kodaira column (Chapters 10, 11). Paper: Lemma lem:euler and the table of Section 1 (every row), Definition def:terms, Lemma lem:twist (ii), (iii), Remark rem:types, Section 1.4.

9.10.5 9.5 ex:local:wild-shapes (worked example and remarks) S

(i) At $q=3$ an inertia image of order 3 (or 6) is wild, since I_3/P_3 has no quotient of order 3 (5.3(ii)); the computation of 9.2(iii) still gives A^{I_3} spanned by $2\sigma+1$ when $\rho(I_3)$ is cyclic, but nothing forces cyclicity and $a(V)=2+\mathrm{sw}(V)$ with $\mathrm{sw}(V)$ possibly positive; this is why the paper's Lemmas lem:IV and lem:euler assume $q\geq 5$ and why the Kodaira type does not determine the factor at $q=2,3$ (Remark rem:wild). (ii) At $q=2$ an inertia image of order 3 is tame ($2\nmid 3$) and 9.2 applies verbatim: $\chi_{-3}(2)=-1$, so $L(A,s)^{-1}=1+2^{-s}$, i.e. $L(\mathrm{Sym}^2 f,s)^{-1}=L(A,s-1)^{-1}=1+2\cdot 2^{-s}$, $P_2(X)=1+2X$, and $e_3(E,2)=v_3(1+2\cdot 2^{-2})=v_3(3/2)=1$; this agrees with the paper's Remark rem:wild ("a type IV at 2 has $P_2(X)=1+2X$ and $e_3=1$"), since the image of inertia at 2 for types IV, IV* is cyclic of order exactly 3 (a theorem proved in Chapters 10 and 11, not here; important_notes.md point 43). (iii) At $q=2$ a non-abelian inertia image acting absolutely irreducibly (e.g. Q_8, $\mathrm{SL}_2(\mathbb{F}_3)$) has $A^{I_2}=\mathrm{End}_{I_2}(V)_{\mathrm{tr}=0}=0$ (Schur), so $L(A,s)=1$: consistent with "a type II at 2 has $P_2=1$" when the image is of that kind (again a black box from Chapters 10, 11); the paper's sentence is false in general, since the Kodaira type at 2 does not determine the image (336f1 has type II at 2 and $P_2=1+2X$; important_notes.md point 50). (iv) For $\mathrm{sp}(2)\otimes\mu\chi$ at $q=2$ with χ wild, $a=4$ or 6 (9.3(ii)); and a potentially good ρ at $q=2$ of Kodaira type I_n^* is possible (Definition def:terms: "$t_\ell=0$" then), because at $q=2$ the Kodaira type does not determine

the Weil-Deligne shape. Reference: Serre-Tate68, Section 2; Kraus90; Rohrlich94. Paper: Remark rem:wild, Section sec:wild, Definition def:terms (the I_n^* at 2 remark), Proposition prop:forms (IV, IV* at 2 with $e_3(E,2) \geq 1$).

9.11 10. Section-by-section plan with target line counts

Line counts are lines of LaTeX, card excluded (PLAN.md Section 7.3: 1,000 to 2,200). The card lists all 53 items (about 190 lines, not counted).

section	title	contents	target lines
(starred)	Core results at a glance	the card: items 1.1 to 9.5, each stated as above, with tag and `\Cref`	(excluded)
(starred)	Where the paper uses this chapter	lem:tamagawa-exp (4.2, 4.4, 6.2(v)); lem:tamfactor (5.3 to 5.5, 6.1, 6.2, 2.7); lem:IV and rem:types (5.6, 5.7, 8.3, 9.2); lem:twist (i) to (iii) (4.4, 6.3, 6.4, 8.4, 9.3); lem:euler and the table (8.1, 8.2, 9.1 to 9.4); rem:wild and sec:wild (5.1, 5.7, 9.5); Section 1.1, def:terms, lem:fpr, the paragraph after lem:dfg21 (8.2, convention L3)	40
1	Absolute values, completions, $\mathbb{Q}_p$ and $\mathbb{Z}_p$	1.1 to 1.5; RHB aside (digits and carries); `warning`: the analytic $\mathbb{Z}_p$ equals Chapter 3's algebraic $\mathbb{Z}_\ell$ (1.4(ii))	190
2	Hensel's lemma; units, roots of unity and powers	2.1 to 2.8; the conventions box L1; `paperlink` for lem:tamfactor (2.7: ℓ-th powers of units read on the reduction)	240
3	Finite extensions of $\mathbb{Q}_p$	3.1 to 3.6; `paperlink` for lem:IV ($\mathbb{Z}_3[\zeta_3]$ as ring of integers of $\mathbb{Q}_3(\sqrt{-3})$) and for Chapter 6's deferred items (3.4, 3.5)	190

section	title	contents	target lines
4	Unramified extensions, $\mathbb{Q}_p^{\mathrm{ur}}$ and Frobenius	4.1 to 4.5; convention L3 as `convention` 4.3 with the dictionary; `paperlink` for lem:tamagawa-exp (4.2(iv), 4.4)	150
5	Ramification groups, inertia and the tame quotient	5.1 to 5.8; the diagram $G_K \supset I_K \supset P_K$ with quotients $\hat{\mathbb{Z}}$, $\prod_{\ell \neq p} \mathbb{Z}_\ell(1)$; `paperlink` for lem:tamfactor (5.3, 5.4, 5.5) and lem:IV, lem:euler (5.6); `warning` on the sign: Frob t Frob$^{-1} = t^q$ versus $\Phi\, t\, \Phi^{-1} = t^{q^{-1}}$	270
6	Kummer theory over local fields; quadratic characters	6.1 to 6.5; `paperlink` for lem:tamagawa-exp (6.2(v), important_notes.md point 8) and lem:tamfactor (6.2(ii) to (iv), point 7); `warning` on the two normalisations of the Artin map	210
7	The Weil group and Weil-Deligne representations	7.1 to 7.6; convention L4; RHB aside: N as the "logarithm of the monodromy", like log of a unipotent matrix; `paperlink` for lem:twist (iii) and lem:euler ("Steinberg")	220
8	Inertia invariants, L-factors and conductors	8.1 to 8.5; convention L6 restated; `warning` (important_notes.md point 11): the paper's L_p needs Φ; `paperlink` for Section 1.1, lem:fpr, the paragraph after lem:dfg21	170
9	The local shapes and their adjoints	9.1 to 9.5, the table 9.4; `paperlink` for lem:euler (every row) and rem:wild (9.5)	220
10	Exercises	18 exercises (Part 11)	140

section	title	contents	target lines
11	Notes and references	Serre (Local Fields; Course in Arithmetic; Abelian l-adic representations; Linear representations), Neukirch, Koblitz, Cassels-Frohlich, Milne CFT, NSW, Silverman ATAEC, Serre-Tate68, Deligne73, Tate79, Rohrlich94; which numbers were cited (none of the theorem numbers flagged "check")	40
	total		**2080**

9.12 11. Coverage table and exercises

9.12.1 11.1 Coverage

PLAN.md scope item, Section 6 map row, or paper hook	line-up items
absolute values	1.1, 1.2
completions	1.3, 3.4
$\mathbb{Q}_p$ and $\mathbb{Z}_p$ as completions and as inverse limits	1.4
p-adic expansions	1.5, 2.3(a)
Hensel's lemma (two forms)	2.1 (simple root, Newton), 2.2 (factorisation), 2.3, 4.2(iii)
structure of $\mathbb{Z}_p^\times$ and $\mathbb{Q}_p^\times$; $1+p\mathbb{Z}_p \simeq \mathbb{Z}_p$ via log / exp (p odd)	2.5
roots of unity in $\mathbb{Q}_p$; Teichmuller lift	2.4, 2.6
squares and ℓ-th powers in $\mathbb{Q}_p^\times$; $u \in \mathbb{Z}_p^\times$ an ℓ-th power iff its reduction is	2.3(e), (f), 2.7, 2.8
finite extensions; ramification index, residue degree	3.1, 3.2, 3.4
unramified extensions (one of each degree, cyclic, roots of unity of order prime to p, Frobenius)	4.1, 4.3
totally ramified extensions	3.3, 3.6
tamely and wildly ramified extensions	3.2, 3.6, 5.1, 5.2
maximal unramified extension $\mathbb{Q}_p^{\mathrm{ur}}$	4.2, 4.4
inertia group I_p; wild inertia P_p (pro-p)	5.1, 5.2, 5.7
tame quotient $I_p/P_p = \prod_{\ell \neq p} \mathbb{Z}_\ell(1)$; tame character t_ℓ via roots of p or of any uniformiser	5.3
$\mathrm{Frob}^{-1}\, t\, \mathrm{Frob} = t^{p^{-1}}$, equivalently $\mathrm{Frob}\, t\, \mathrm{Frob}^{-1} = t^p$	5.4, 5.8
$G_{\mathbb{Q}_p}$ and its filtration	5.1(iv), 5.2(iv)
Kummer theory over $\mathbb{Q}_p$: $H^1(G_{\mathbb{Q}_p}, \mu_{\ell^k}) = \mathbb{Q}_p^\times/(\mathbb{Q}_p^\times)^{\ell^k}$ (foreshadowing Chapter 17)	6.1

PLAN.md scope item, Section 6 map row, or paper hook	line-up items
restriction of the Kummer class of $u \in \mathbb{Q}_p^\times$ to I_p is $v_p(u)$ times the tame character	6.2(ii)
unramified quadratic extension; ramified $\mathbb{Q}_p(\sqrt{p^*})$ and their characters	6.3
local behaviour of quadratic characters of $G_\mathbb{Q}$ (χ_{q^*} unramified at $p \neq q$, ramified at q)	6.4
Weil group $W_{\mathbb{Q}_p}$	7.1, 6.5(ii)
Weil-Deligne representations, (r, N), the relation (explicit convention)	7.2, 7.3, 7.4, convention L4
inertia invariants	8.1
local L-factor (explicit Frobenius convention)	8.2, 4.3, convention L6
Artin conductor $a(r) = \operatorname{codim} V^I + \text{Swan}$	8.3, 8.4
special cases: unramified; tamely ramified through a finite cyclic quotient	9.1, 9.2, 5.6
sp(2) and its unramified twists (conductor exponent 1)	7.5, 9.3(i)
twist of sp(2) by a ramified quadratic character (conductor exponent 2)	9.3(ii)
finite image of inertia of order $2, 3, 4, 6$ (conductor exponent 2 for $q \geq 5$)	9.2, 9.4
Section 6 row: Tate parameter, unit part, ℓ-th powers in $\mathbb{F}_q^\times$, $t_\ell(E, q)$, potentially multiplicative reduction (Chapters 2, 7, 10)	2.7, 2.8, 6.2, 7.6, 9.3
Section 6 row: Steinberg representation, unramified twists, ramified twist has conductor exponent 2 (Chapters 7, 15)	7.5, 9.3, 9.4
Section 6 row: image of inertia on $T_\ell E$ at additive primes, unipotent elements, $\mathbb{Z}_3[\zeta_3]$ (Chapters 4, 6, 7, 11)	3.3(b), 5.6, 5.7, 7.2, 9.2, 9.5
Section 6 row: Kummer theory, Kummer classes, $H^1(G, \mathbb{Z}_\ell(1))$, tame character, $\mathrm{Frob}^{-1} t \,\mathrm{Frob} = t^{q^{-1}}$ (Chapters 5, 7, 16, 17)	5.3, 5.4, 5.5, 6.1, 6.2
hook: Lemma lem:tamagawa-exp (Tate curve over the unramified quadratic extension; ℓ^t-th roots)	4.1, 4.2(iv), 4.4, 6.2(v), 7.6(iv)
hook: Lemma lem:tamfactor (tame character; $\mathrm{Frob}^{-1} t \,\mathrm{Frob} = t^{q^{-1}}$; Kummer class κ; restriction to I is n times t_ℓ; $\mathbb{Z}_\ell(1)/(q-1)$)	2.6, 2.7(v), 5.3, 5.4, 5.5, 6.1, 6.2, 7.6, 9.3(iv)
hook: Lemma lem:IV and Remark rem:types (tame inertia cyclic; wild at 2 and 3)	3.3(b), 5.6, 5.7, 8.3(v), 9.2, 9.5(i)
hook: Lemma lem:twist (iii) (unramified twist of Steinberg; ramified twist has exponent 2)	7.5, 9.3(ii), 8.4
hook: Lemma lem:twist (i), (ii) (unramified base change; tame quadratic characters at odd q; point 28)	4.4, 6.3, 6.4(v), 8.3(iii)
hook: Lemma lem:euler (eigenvalues of Frobenius on inertia invariants)	5.6, 8.1, 8.2, 9.1 to 9.4
hook: Remark rem:wild (and Section sec:wild, Definition def:terms at $q = 2$)	5.1(v), 5.7, 8.3, 9.5

PLAN.md scope item, Section 6 map row, or paper hook	line-up items
also: Section 1.1 (δ_p, $L_p(A_f,s)$), Lemma lem:fpr ($L_p(A_f(1),0)=L_p(A_f,1)$), paragraph after Lemma lem:dfg21	8.1, 8.2(iii), (iv)
also: Chapter 6's deferred items (3.10(v), 3.11 profinite, 4.7(iv), 4.9 Hensel, 5.7(ii) conductors, 7.4)	3.5, 5.4, 3.3(b), 2.3(g), 6.5(iii), 3.4
also: Chapter 16's imported facts (I/P, $I^{(\ell)}=\mathbb{Z}_\ell(1)$, kernel prime to ℓ, the relation, $\chi_\ell(\mathrm{Frob})=p$, Kummer on inertia)	5.3, 5.4, 4.5, 6.2

9.12.2 11.2 Exercises (Section 10 of the chapter), graded from easy to harder

1. Find the 5-adic expansions of $-1/3$ and $1/3$ to five digits, and prove that $x\in\mathbb{Q}_p$ is rational iff its expansion is eventually periodic.
2. Prove that $\mathbb{Z}_p\to\varprojlim\mathbb{Z}/p^n$ is an isomorphism of topological rings and that $\mathbb{Z}_{(p)}$ is dense in $\mathbb{Z}_p$ (so the $\mathbb{Z}_\ell$ of Chapter 3 is the $\mathbb{Z}_p$ of this chapter at $p=\ell$).
3. By Newton's iteration find $\sqrt{-1}\in\mathbb{Z}_5$ mod 5^4 (answers 182, 443), the roots of X^2+X+1 in $\mathbb{Z}_7$ mod 49 (18, 30) and $\sqrt{-3}\in\mathbb{Z}_7$ mod 7^3 (37, 306). Show that 182 is the Teichmuller lift of 2 mod 5^4 and deduce $\mathbb{Z}_7[\zeta_3]\simeq\mathbb{Z}_7\times\mathbb{Z}_7$.
4. Prove that $u\in\mathbb{Z}_2^\times$ is a square iff $u\equiv1\pmod 8$. List the seven quadratic extensions of $\mathbb{Q}_2$, show that $\mathbb{Q}_2(\sqrt{-3})=\mathbb{Q}_2(\sqrt{5})$ is the unramified one, and compute the ramification groups of $\mathbb{Q}_2(i)/\mathbb{Q}_2$ and $\mathbb{Q}_2(\sqrt{2})/\mathbb{Q}_2$ and their conductor exponents 2 and 3; compare with the conductors 4 and 8 of χ_{-4} and χ_2.
5. Show that log and exp give $1+p\mathbb{Z}_p\simeq p\mathbb{Z}_p$ for p odd and $1+4\mathbb{Z}_2\simeq4\mathbb{Z}_2$, but not $1+2\mathbb{Z}_2\simeq2\mathbb{Z}_2$. Deduce the structure of $\mathbb{Q}_p^\times$ and of $\mathbb{Q}_p^\times/(\mathbb{Q}_p^\times)^2$.
6. Compute $\mathbb{Q}_7^\times/(\mathbb{Q}_7^\times)^3$ and decide which of $2,-1,6,14,7^3\cdot6$ are cubes in $\mathbb{Q}_7$. Do the same for ninth powers in $\mathbb{Q}_{19}$ with $2,8,-1$ (valuations of the class c: $0,1,2$).
7. Check the paper's claim in Lemma lem:tamagawa-exp: for $x\in\mathbb{Q}_q^\times$ and $\ell\neq q$, x is an ℓ^t-th power in $\mathbb{Q}_q^{\mathrm{ur}}$ iff $\ell^t\mid v_q(x)$. Apply it to 11a1 at $q=11$ ($n_{11}=5$): ρ mod 5 is unramified at 11 and ρ mod 25 is not. Compare with $E(\mathbb{Q})\simeq\mathbb{Z}/5$.
8. Check the paper's claim $\mathrm{Frob}^{-1}\,t\,\mathrm{Frob}=t^{q^{-1}}$ (proof of Lemma lem:tamfactor) at finite level: in $\mathrm{Gal}(\mathbb{Q}_q(\mu_{\ell^k},q^{1/\ell^k})/\mathbb{Q}_q)$ (item 5.8) compute $\mathrm{Frob}\;t\;\mathrm{Frob}^{-1}$ and $\mathrm{Frob}^{-1}\,t\,\mathrm{Frob}$ for $(q,\ell,k)=(7,3,2)$ and $(2,3,2)$. Then prove $\Phi\,t\,\Phi^{-1}=t^{q^{-1}}$ for the geometric Frobenius.
9. Check the paper's claim that "the restriction of κ to I is n times the tame character" (Lemma lem:tamfactor), with κ the Kummer cocycle of q_E, $v_q(q_E)=n$. Show that $c'=\kappa(\mathrm{Frob})$ changes by $qn\,t_\ell(\sigma)$ when Frob is replaced by $\mathrm{Frob}\,\sigma$ ($\sigma\in I$) and by $(q-1)z$ when the roots are changed, and that $\min(v_\ell(n),v_\ell(q-1),v_\ell(c'))$ is independent of all choices.
10. For 11a1 ($j=-2^{12}31^3/11^5$) and $q_E=j^{-1}+744j^{-2}+750420j^{-3}+\dots$ show $v_{11}(q_E)=5$ and $q_E11^{-5}\equiv-1\pmod{11}$; deduce that q_E is a fifth power in $\mathbb{Q}_{11}$ and that $v_5(c)=1=v_5(11-1)$ (the data entering Chapter 16, Exercise 10).
11. Show that $I_K=I_{\mathbb{Q}_p}$ for $K/\mathbb{Q}_p$ finite unramified and that this fails for $K=\mathbb{Q}_p(\sqrt{p})$ (index 2). Check the paper's claim "the unramified quadratic extension of $\mathbb{Q}_q$ … does not change the inertia group".
12. For $\rho=\left(\begin{smallmatrix}\chi_\ell&\kappa_x\\0&1\end{smallmatrix}\right)$ compute N (7.2) and $\mathrm{WD}(\rho)$; show $\mathrm{WD}(\rho)\simeq\mathrm{sp}(2)$ if $v_p(x)\neq0$ and $\simeq1\oplus\|\cdot\|$ (with $N=0$) if x is a unit. Show that the twist by an unramified quadratic ε has corner $\varepsilon\kappa_x$ (Part 15, N1).
13. Compute $a(\mathrm{sp}(2)\otimes\mu)=1$ and $a(\mathrm{sp}(2)\otimes\mu\chi)=2$ for χ tamely ramified quadratic, and $a=4$ for $\chi=\chi_{-4}$ at 2. Check the paper's claim in the proof of Lemma lem:twist (iii) that the twist of an unramified twist of Steinberg by a ramified quadratic character has conductor exponent 2 (q odd).
14. Show $\mathrm{ad}^0\,\mathrm{sp}(2)\simeq\mathrm{sp}(3)\otimes\|\cdot\|^{-1}$ and $L(\mathrm{ad}^0\,\mathrm{sp}(2),s)=(1-q^{-1-s})^{-1}$. Show that with the arithmetic

Frobenius one gets $(1-q^{1-s})^{-1}$, and check the paper's claims that at a multiplicative prime $e_\ell(E,q)=0$ and $L_q(A_f,1)^{-1}=1-q^{-2}$ (Definition def:terms, Lemma lem:fpr).

15. Check the paper's claim in Lemma lem:euler for types IV, IV*, II, II*: if $\rho(I_q)$ is cyclic of order 3 or 6 ($q\neq 3$, $\det=1$) then $L(\mathrm{ad}^0\rho,s)^{-1}=1-\chi_{-3}(q)q^{-s}$; and the analogue with χ_{-4} for order 4. For $q=2$ and order 3 deduce $P_2(X)=1+2X$ and $e_3(E,2)=1$ (Remark rem:wild).
16. Check the paper's claim in Lemma lem:IV that an inertia image of order 3 at $q\geq 5$ is tame and cyclic; show that at $q=3$ an inertia image of order 3 is necessarily wild.
17. For $\rho(I_q)$ cyclic of order 4 with $\det=1$: show that $V\otimes\mathbb{Q}_\ell(i)$ is a sum of two characters of $W_{\mathbb{Q}_q}$ if $q\equiv 1 \pmod 4$ and is induced from the unramified quadratic extension if $q\equiv 3 \pmod 4$.
18. (Harder) Prove Grothendieck's monodromy theorem (7.2) for $\rho\colon G_{\mathbb{Q}_p}\to \mathrm{GL}_2(\mathbb{Z}_\ell)$, ℓ odd: show that $\rho(P_p)$ embeds in $\mathrm{GL}_2(\mathbb{F}_\ell)$, that ρ restricted to $I_p\cap\rho^{-1}(1+\ell M_2(\mathbb{Z}_\ell))$ factors through t_ℓ, and that N is nilpotent. Deduce: $\rho(I_p)$ finite iff $N=0$.

9.13 12. Dependencies and forward pointers

Results from earlier chapters used here (by chapter label only): * Chapter 1 (`ch:groups`): profinite groups, $\mathbb{Z}_p$ and $\hat{\mathbb{Z}}$ as inverse limits, pro-p groups and their maps to groups of order prime to p (5.2, 5.6, 5.7), semidirect products (5.4(iii)), $1+\ell M_n(\mathbb{Z}_\ell)$ pro-ℓ and torsion-free for ℓ odd (5.7, 7.2). * Chapter 2 (`ch:rings`): v_p, $|\cdot|_p$, Ostrowski and the product formula (1.1, 1.2), $\mathbb{Z}_{(p)}$ and reduction, $(\mathbb{Z}/p^k)^\times$ and lifting the exponent (2.5), finite fields, ℓ-th powers in $\mathbb{F}_q^\times$ and the class c (2.7, 6.2), the identity of 9.1 (11.4), the values $v_\ell(\cdot)$ of the omitted factors (9.4). * Chapter 3 (`ch:modules`): DVRs, $\mathbb{Z}_\ell$ algebraically, completions of Noetherian rings (1.3(iii), 1.4(ii)), finitely generated modules over a DVR (3.2), $\mathbb{Q}_\ell/\mathbb{Z}_\ell$. * Chapter 4 (`ch:reps`): unipotent elements and their invariants (9.1 there), commutants (9.2 there), the table of cyclic inertia shapes (9.3 there), reduction mod 3 of order-3 elements (9.4 there), $\mathrm{ad}^0 V$, $\mathrm{End}\,V=V\otimes V^\vee$, Schur's lemma (used in 9.5(iii) here), induction (9.2(ii)). * Chapter 5 (`ch:galois`): finite and infinite Galois theory, the translation theorem (4.4, 5.5), finite fields and $G_{\mathbb{F}_q}\simeq\hat{\mathbb{Z}}$, Kummer theory for fields containing μ_n, the affine group of x^n-a (5.8), the ℓ-adic Kummer cocycle and galois:kummer-finite-field (6.1, 6.2), quadratic extensions (6.3), the cyclotomic character, conventions C2 to C6. * Chapter 6 (`ch:ant`): e,f for number fields, Dedekind-Kummer, decomposition and inertia groups and Frobenius (3.4, 3.5), convention 3.7, 3.11 and 3.12 (finite-level tame relation; 5.4, 5.6 here are the profinite forms), $\mathbb{Z}_3[\zeta_3]$ and $\mathbb{Z}_\ell[\zeta_3]$ (2.3(g), 3.3(b)), quadratic characters and their conductors (6.4), Dirichlet characters as Galois characters (6.5(iii)).

Black boxes imported from later chapters (stated with a forward `\Cref`, never used in a proof of an earlier item): Hilbert 90 and $H^1(K,\mu_n)\simeq K^\times/(K^\times)^n$ (Chapter 17: 6.1(ii)); the Tate curve and the shape of $T_\ell E$ at multiplicative primes (Chapters 10, 11: 7.6's last sentence); Kraus's theorem and the Kodaira-type dictionary (Chapters 10, 11: 5.7, 9.4 last column, 9.5); the conductor of an elliptic curve equals the Artin conductor of $V_\ell E$ (Chapter 10: 8.4); independence of ℓ of the Weil-Deligne representation of $V_\ell E$ (Chapters 11, 15: 7.4(iv)); the symmetric square normalisation $L(\mathrm{Sym}^2 f,s)=L(A,s-1)$ and PARI's P_q (Chapter 19: 9.5(ii)).

Forward pointers (chapters that use this one): * Chapter 8 and Chapter 9 (`ch:curves`, `ch:ellcurves`, outlined in parallel): only $\mathbb{Q}_p$, $\mathbb{Z}_p$, Hensel (points over $\mathbb{Z}_p$ and reduction) and the arithmetic Frobenius of 4.3 (the characteristic polynomial X^2-a_pX+p is for Frob on $V_\ell E$); nothing about elliptic curves is done here. * Chapter 10 (`ch:reduction`): minimal models over $\mathbb{Z}_p$, Hensel (2.1), $E_1(\mathbb{Q}_p)$ (2.5 for formal-group style filtrations), the Tate curve over $\mathbb{Q}_p$ and over the unramified quadratic extension (4.1, 4.4), $v_q(q_E)$ and the unit part (2.7, 2.8, 6.2), tame versus wild at 2, 3 (5.7, 8.3), the conductor exponent (8.4), quadratic twists locally (6.3, 6.4). * Chapter 11 (`ch:galrep`): Lemma lem:tamagawa-exp in full (6.2(v), 7.6), Lemma lem:IV and Remark rem:types (5.6, 9.2, 8.3(v)), the Serre conductor (8.3(v), 8.4), Grothendieck's theorem and Neron-Ogg-Shafarevich (7.2), unramified base change (4.4). * Chapter 15 (`ch:modgalrep`): Weil-Deligne representations, local L- and ϵ-factors, sp(2) and its twists, the classification 7.5(ii), (iii) (principal series, Steinberg, supercuspidal on the Galois side), conductors 8.4, 9.3. * Chapter 16 (`ch:cohomology`): exactly the imported facts of its tame-inertia-example and model-setting: 4.2(i) ($G/I\simeq\hat{\mathbb{Z}}$), 4.5 ($\chi_\ell(\mathrm{Frob})=q$), 5.3(ii) to (iv), 5.4(ii), 6.2(ii). * Chapter 17 (`ch:galcoh`): Kummer theory

over $\mathbb{Q}_p$ (6.1, 6.2), $\widehat{\mathbb{Q}_p^\times}$ (2.7(v)), the unramified class κ_u (6.2(iii)), the Frobenius lift of 5.5, H^1_{ur} via $G/I \simeq \hat{\mathbb{Z}}$ (4.2), the local Euler characteristic count (2.7(iv)). * Chapter 18 (ch:blochkato): V^{I_p}, $L_p(V,s)$ with Φ (8.1, 8.2), $L_p(V(1),s) = L_p(V,s+1)$. * Chapter 19 (ch:lfunctions): the convention 8.2, the shapes table 9.4, the adjoint computations 9.1 to 9.3, the wild primes 9.5. * Chapter 20 (ch:congruence): the factor $1-p^{-2}$ (9.3(iii)) and Wiles's level-raising factor (9.1) in Theorem thm:dfg14. * Chapters 21, 22 (ch:paper, ch:computation): convention L3/L6 (Chapter 21's C2), 5.4, 6.2, 9.4; the Tate parameter and its unit part computed with PARI (2.8(c)).

9.14 13. Notes for the writer

9.14.1 13.1 Conventions (to be set as convention environments at first use)

- **Both Frobenii, named every time.** Convention L3 in Section 4 (item 4.3). In every display write Frob (arithmetic) or Φ (geometric), never an unqualified "Frobenius" in a formula. Group theory (t, conjugation, the matrices of Lemma lem:tamfactor, $\chi_\ell(\mathrm{Frob}) = q$) uses Frob; L-factors use Φ. State the one-line dictionary: arithmetic eigenvalue λ is geometric eigenvalue λ^{-1}. The book's Frob_p macro means the arithmetic one, as in Chapters 5, 6, 16 and 21.
- **The tame relation, in exactly these forms:** $\mathrm{Frob}\, t\, \mathrm{Frob}^{-1} = t^q$; equivalently $\mathrm{Frob}^{-1}\, t\, \mathrm{Frob} = t^{q^{-1}}$; for the geometric one $\Phi\, t\, \Phi^{-1} = t^{q^{-1}}$. Exponents live in $\prod_{\ell \neq p} \mathbb{Z}_\ell$ (on I/P) or in $\mathbb{Z}_\ell$ (on $I^{(\ell)}$). Give the finite-level check 5.8.
- **The sign in the Weil-Deligne relation:** $r(w) N r(w)^{-1} = \|w\| N$ with $\|\mathrm{Frob}\| = q$, $\|\Phi\| = q^{-1}$; so $\mathrm{Frob}\, N\, \mathrm{Frob}^{-1} = qN$ and $\Phi N \Phi^{-1} = q^{-1}N$. PLAN.md's "Frob N Frob$^{-1} = p^{-1}N$" is the geometric form (Part 15, B-a); write both and say which is which. Check it on sp(2) and on the Tate shape (7.6): $\rho(\mathrm{Frob})$ multiplies e_1 by εq and e_2 by ε, and N maps e_2 to e_1.
- **Normalisation of local factors:** $L(V,s) = \det(1 - \Phi q^{-s} \mid V^I)^{-1}$. Consequences to display: $L(\mathbb{Q}_\ell(1),s) = (1-q^{-1-s})^{-1}$, $L(V(1),s) = L(V,s+1)$, $L(V_\ell E,s) = L_p(E,s+1)$, $L(V_\ell E^\vee,s) = L_p(E,s)$, $L(\mathrm{ad}^0\,\mathrm{sp}(2),s) = (1-q^{-1-s})^{-1}$. The paper's $L_p(\mathrm{Sym}^2 f,s) = L_p(A_f,s-1)$ and PARI's P_q are Chapter 19's; mention them only in 9.5(ii) with a \Cref{ch:lfunctions}.
- **The choice of** ζ (generator of $\mathbb{Z}_\ell(1)$) enters t_ζ and N; N is defined up to $\mathbb{Z}_\ell^\times$; none of V^I, a, L depends on it.
- **Uniformiser-dependence:** t_ℓ on I_K is independent of ϖ; the cocycle κ_ϖ on G_K is not (it changes by κ_u); the Frobenius lift of 5.5 depends on ϖ and on the roots.
- **Notation clash:** q is both $\#k_K$ and the paper's prime; the quadratic character of conductor q is χ_{q^*} (Chapter 5, C3; the paper's χ_q, never written in the book, decision D3) and χ_ℓ the cyclotomic character (important_notes.md point 32); write θ_2 for the order-two character of inertia and η_{ur} for the unramified quadratic character; ε is reserved for the unramified character of the Tate shape (paper's notation); ϵ for ϵ-factors; e is both a ramification index and the basis vector $e = e_1 \otimes e_2^\vee$ (paper): say so.

9.14.2 13.2 Pitfalls

- Hensel (simple-root form) needs $f'(a_0)$ a unit; give $X^2 - p$ and $X^2 + 7$ over $\mathbb{Z}_2$ (the Newton form succeeds where the simple form fails). $\mathbb{Q}_p^{\mathrm{ur}}$ is not complete; Hensel holds there by passing to a finite level (4.2(iii)); say this, because Lemma lem:tamagawa-exp uses it.
- "u is an ℓ^k-th power in $\mathbb{F}_q^\times$" for $u \in \mathbb{Z}_q^\times$ means its reduction (important_notes.md point 31); by 2.7(ii) this is equivalent to $u \in (\mathbb{Z}_q^\times)^{\ell^k}$ for $\ell \neq q$. False for $\ell = q$ (e.g. $1+q$ is not a q-th power in $\mathbb{Z}_q$ for q odd).
- $\mathbb{Z}_\ell[\zeta_3]$ is $\mathbb{Z}_\ell[X]/(X^2+X+1)$, not a domain for $\ell \equiv 1 \pmod 3$ (2.3(c); important_notes.md point 12).
- I/P has no quotient of order p: an inertia image of order p is wild. At $q=3$ an image of order 3 is wild; at $q=2$ an image of order 3 is tame. Do not write "tame since $q \geq 5$" when "$q \nmid \#\rho(I)$" is meant.

- Grothendieck's theorem gives N only on an open subgroup of I; $\rho(I)$ itself can be $\{\pm 1\} \times$ (unipotent) (the I_n^* shape), and V^I must be computed as $(\ker N)^{r(I)}$ (8.1), not as $\ker N$.
- The Artin conductor of a twist: $a(\mathrm{sp}(2) \otimes \chi) = 2a(\chi)$, not $a(\chi) + 1$; and $a(V \otimes \chi) \neq a(V)$ in general for ramified χ. Lemma lem:twist (ii) at $q = 3$: twisting a good or multiplicative curve by the tame χ_{-3} gives exponent 2, but a general additive curve at 3 can have exponent up to 5 (point 28).
- Local class field theory normalisations: say "uniformiser to arithmetic Frobenius" and give the sign $u \mapsto u^{-1}$ on μ_{p^∞} (6.5(ii)); avoid using LCFT in any proof the paper needs (only 6.5(iii) for characters, as a statement).
- The Swan conductor is an integer only by a theorem (Artin); do not present 8.3(ii) as obvious.
- Do not claim Krasner's lemma or the equivalence of norms as proved (3.1, 3.5: S).
- Do not present the Kodaira-type column of 9.4 or the inertia images of elliptic curves as proved here: they are black boxes from Chapters 10 and 11.

9.14.3 13.3 What to leave to neighbouring chapters

- Chapter 2: v_p, $|\cdot|_p$ on $\mathbb{Q}$, Ostrowski, $(\mathbb{Z}/p^k)^\times$, powers in $\mathbb{F}_q^\times$, the class $c(u)$, the numerical identities of the omitted factors. Cite; do not redo.
- Chapter 3: $\mathbb{Z}_\ell$ as an algebraic completion and complete DVR, lengths, Fitting ideals, $\mathbb{Q}_\ell/\mathbb{Z}_\ell$. Chapter 7 adds only the analytic description and the equality 1.4(ii).
- Chapter 4: the linear algebra of unipotent and finite cyclic actions (9.1 to 9.4 there), ad^0, Schur. Cite its table 9.3.
- Chapter 5: Kummer theory for fields containing μ_n, the ℓ-adic Kummer cocycle (its definition and cocycle identity), the translation theorem, finite fields. Chapter 7 adds the local field input (Hensel, inertia, valuations).
- Chapter 6: global e, f, g, Dedekind-Kummer, global decomposition groups, the finite-level 3.11, 3.12, $\mathbb{Z}_3[\zeta_3]$ as a DVR.
- Chapters 9, 10, 11: everything about elliptic curves: minimal models, reduction types, Tate's algorithm, the Tate curve itself, $E_{\mathfrak{q}}[\ell^t]$, Kraus's theorem, Neron-Ogg-Shafarevich, the proofs of Lemmas lem:tamagawa-exp and lem:IV and of Remark rem:types, the Serre conductor of $\bar{\rho}_{E,\ell}$. Chapter 7 provides the local field statements they quote.
- Chapter 15: local Langlands, local components π_p of newforms, Carayol's theorem; Chapter 7 gives only the Galois side.
- Chapters 16, 17: group cohomology, $H^1(I, M) = M/(t-1)M$, inflation-restriction, Hilbert 90 in cohomological form, the identification of $H^1(K, \mathbb{Z}_\ell(1))$ with $\widehat{K^\times}$ and of c' with the class of the unit part (Chapter 17 proves the end of Lemma lem:tamfactor). Chapter 7 proves only the field-level statements 6.2.
- Chapter 19: the symmetric-square normalisation, PARI's P_q, $e_\ell(E, q)$, the naive factor, Lemma lem:euler in the language of Kodaira types.

9.14.4 13.4 How each relevant point of important_notes.md is handled

- **Point 7 (Lemma lem:tamfactor, "image of κ in $H^1(G/I, \mathbb{Z}_\ell(1)^I)$").** Items 5.5 (a Frobenius lift fixing the ℓ-power roots of q) and 6.2(i) to (iv) (the decomposition $\kappa = n\kappa_q + \kappa_u$, κ_u unramified with value the class of $\bar{u}$, the indeterminacy $(n, q-1)\mathbb{Z}_\ell$) give the field-theoretic half of the fix; Exercise 9 makes the reader check it. The cohomological statement is Chapter 17's. A `paperlink` at 6.2 quotes the paper's sentence and the corrected reading.
- **Point 8 (Lemma lem:tamagawa-exp, notation and "up to units").** Item 6.2(v) is the precise statement ("x is an ℓ^t-th power in $\mathbb{Q}_q^{\mathrm{ur}}$ iff $\ell^t \mid v_q(x)$"), with 4.2(iv) for the units; the chapter calls the Tate parameter q_E and its unit part u, as Chapters 10, 11, 21 do. Exercise 7.
- **Point 11 (geometric versus arithmetic Frobenius).** Conventions L3, L6, item 4.3 and 8.2(iv): both Frobenii are defined, the L-factor uses Φ, the group theory uses Frob; 9.3(iii) shows that the paper's $1 - q^{-1-s}$ requires Φ; 8.2(iii) gives $L(V(1), s) = L(V, s+1)$ (Lemma lem:fpr). Exercises 8, 14. A `warning` environment in Section 8 repeats the point.

- **Point 28 (Lemma lem:twist, "the conductor exponents are the standard ones for q odd (tame)").** Items 6.3, 6.4(ii), (v), 8.3(iii), 9.3(ii): quadratic characters of $\mathbb{Q}_q$, q odd, are tame of exponent 1; twisting a tame representation by them keeps it tame; at $q = 3$ general additive reduction can be wild (5.7, 9.5(i)). The `paperlink` at 9.3 says exactly which exchanges are tame.
- **B1 (which Frobenius has eigenvalues α, β on $V_\ell E$).** Item 8.2(v) states: the arithmetic Frobenius on $V_\ell E$ has eigenvalues α, β; the geometric one α^{-1}, β^{-1}; the geometric one on $V_\ell E^\vee$ has α, β; on $\mathrm{ad}^0 V$ the set $\alpha/\beta, 1, \beta/\alpha$ is the same for both. Part 15, B-b, records that the slip is still present in the Chapter 21 outline's C2(b).
- **Points 12, 13, 22, 29, 31, 32 (adjacent).** 12: 2.3(c), (g) and the pitfall above; 13: not this chapter's (Chapter 4, 6 notation); 22: 9.2, 9.4 give the complete list of V^I and conductors for every tame additive shape, which is the corrected Remark rem:types on the Galois side; 29: not this chapter's (Weierstrass models); 31: 2.7(ii) and the pitfall above; 32: the notation note in 13.1.

9.14.5 13.5 References and bibliography

`SerreLocalFields`, `SerreCourse`, `SerreAbelian`, `SerreLinearReps`, `Neukirch`, `Koblitz`, `CasselsFrohlich`, `MilneCFT`, `NSW`, `SilvermanATAEC`, `Kraus90`, `Washington`, `DFG04` are in `refs.bib`. Add to `refs.bib`, not to `bib/ch07.bib`, since each is needed by more than one chapter (decision B14; important_notes.md B16, B18, B23), after checking that no other chapter has added it (complete entries; check against the sources): `Deligne73` (P. Deligne, "Les constantes des equations fonctionnelles des fonctions L", Modular functions of one variable II, Lecture Notes in Math. 349, Springer, 1973, 501-597), `Tate79` (J. Tate, "Number theoretic background", Automorphic forms, representations and L-functions, Proc. Sympos. Pure Math. 33, Part 2, AMS, 1979, 3-26), `SerreTate68` (J.-P. Serre and J. Tate, "Good reduction of abelian varieties", Ann. of Math. 88 (1968), 492-517), `Rohrlich94` (D. E. Rohrlich, "Elliptic curves and the Weil-Deligne group", Elliptic curves and related topics, CRM Proc. Lecture Notes 4, AMS, 1994, 125-157). The `Deligne` key of `refs.bib` is the Bourbaki talk on modular forms and ℓ-adic representations, not the ϵ-factor paper; do not confuse them. No theorem numbers are cited from these works in this outline; the writer cites sections only unless checked against the source.

9.15 14. Verification log (computations done for this outline)

Python scripts `/tmp/ch07/check1.py`, `/tmp/ch07/check2.py` and hand computations. * Hensel lifts by Newton iteration: $\sqrt{-1} \in \mathbb{Z}_5$ is 182 or 443 mod 5^4 ($182^2 + 1 = 53 \cdot 625$); roots of $X^2 + X + 1$ mod 49: 18, 30; $\sqrt{-3}$ mod 7^3: 37, 306; $\sqrt{2}$ mod 7^3: 108, 235. * Teichmuller lifts in $\mathbb{Z}_5$ mod 5^4: $\omega(1, 2, 3, 4) = 1, 182, 443, 624$, each with fourth power 1; so $\omega(2) = \sqrt{-1}$. * Unit indices: for $p \in \{7, 13, 19, 31\}$, $\ell \in \{3, 5\}$, $k \in \{1, 2\}$, the index of ℓ^k-th powers in $(\mathbb{Z}/p^3)^\times$ equals $\ell^{\min(k, v_\ell(p-1))}$ (consistent with 2.7(ii), (iv)); $\ell^k \cdot \ell^{\min(k, v_\ell(p-1))} = \ell^k \gcd(\ell^k, p-1)$ for $p \le 37$, $k \le 3$ (2.7(iv), Euler characteristic form). * 11a1: $j = -2^{12}31^3/11^5$; with $q_E = j^{-1} + 744j^{-2} + 750420j^{-3}$, $v_{11}(q_E) = 5$ and $q_E 11^{-5} \equiv 10 \equiv -1 \pmod{11}$ (by hand: $2^{12} \equiv 4$, $31^3 \equiv 3$, product $\equiv 1$). The fifth powers in $\mathbb{F}_{11}^\times$ are $\{\pm 1\}$. * Cubes mod 7: $\{1, 6\}$. * Tame relation in the affine model of $\mathrm{Gal}(\mathbb{Q}_q(\mu_{\ell^k}, q^{1/\ell^k})/\mathbb{Q}_q)$: $\mathrm{Frob}\, t\, \mathrm{Frob}^{-1} = t^q$ and $\mathrm{Frob}^{-1} t\, \mathrm{Frob} = t^{q'}$, $qq' \equiv 1$, for $(q, \ell, k) = (7, 3, 2), (11, 5, 1), (13, 3, 2), (2, 3, 2)$ (outputs t^7, t^4; t, t; t^4, t^7; t^2, t^5). * $\mathrm{sp}(2)$ with $\Phi = \mathrm{diag}(1, q^{-1})$, $N = E_{10}$: $\Phi N \Phi^{-1} = q^{-1}N$; $[N, E_{01}] = -h$, $[N, h] = 2N$, $[N, N] = 0$; Φ-eigenvalues on E_{01}, h, E_{10}: $q, 1, q^{-1}$ (symbolic, sympy). Hence $\mathrm{ad}^0 \mathrm{sp}(2) \simeq \mathrm{sp}(3) \otimes \|\cdot\|^{-1}$ and $L(\mathrm{ad}^0 \mathrm{sp}(2), s) = (1 - q^{-1-s})^{-1}$. * Tate shape twisted by ε: $\left(\begin{smallmatrix} \chi\varepsilon & \varepsilon\kappa \\ 0 & \varepsilon \end{smallmatrix}\right)$ is a homomorphism when $\kappa(gh) = \kappa(g) + \chi(g)\kappa(h)$, while $\left(\begin{smallmatrix} \chi\varepsilon & \kappa \\ 0 & \varepsilon \end{smallmatrix}\right)$ would need $\kappa(gh) = \varepsilon(h)\kappa(g) + \chi(g)\varepsilon(g)\kappa(h)$ (hand computation; Part 15, N1). * Quadratic extensions of $\mathbb{Q}_2$: square classes $\pm 1, \pm 5, \pm 2, \pm 10$; $(-3) \cdot 5 = -15 \equiv 1 \pmod 8$ so $\mathbb{Q}_2(\sqrt{-3}) = \mathbb{Q}_2(\sqrt{5})$, unramified; $3 \cdot (-5) \equiv 1 \pmod 8$ so $\mathbb{Q}_2(\sqrt{3}) = \mathbb{Q}_2(\sqrt{-5})$. Ramification groups: $\mathbb{Q}_2(i)$, $\pi = 1 + i$, $\sigma(\pi) - \pi = -2i$ of $v_L = 2$: lower jump 1, $a = 2$; $\mathbb{Q}_2(\sqrt{2})$: $v_L(-2\sqrt{2}) = 3$, jump 2, $a = 3$; these match the conductors 4 and 8. * Conductor exponents: $a(\mathrm{sp}(2) \otimes \mu) = 0 + 2 - 1 = 1$; $a(\mathrm{sp}(2) \otimes \chi) = 2a(\chi) + 0 - 0$; finite cyclic tame inertia without fixed vectors in dimension 2: $a = 2$. Omitted factor at 2 for order-3 inertia: $L(A, s)^{-1} = 1 + 2^{-s}$, $P_2(X) = 1 + 2X$, $e_3 = v_3(3/2) = 1$. * Not verified (recalled): the theorem and section numbers of Serre's Local Fields, Neukirch, Deligne73, Tate79

and Rohrlich94; the outline cites chapters and sections only, and the writer must check them before citing anything finer.

9.16 15. Points for important_notes.md

9.16.1 15.1 New points about the paper

N1. Lemma lem:tamfactor, proof, first sentence: in the nonsplit case the corner of the matrix is $\varepsilon\kappa$, not the Kummer class κ. Location: "T has a basis e_1, e_2 in which $G = G_{\mathbb{Q}_q}$ acts through matrices $\left(\begin{smallmatrix}\chi_\ell\varepsilon & \kappa\\ 0 & \varepsilon\end{smallmatrix}\right)$, with ε unramified of order at most 2 and κ the Kummer class of q_E". Problem: for nonsplit reduction $T_\ell E \simeq T_\ell E_{q_E} \otimes \varepsilon$, and $T_\ell E_{q_E}$ has the matrices $\left(\begin{smallmatrix}\chi_\ell & \kappa\\ 0 & 1\end{smallmatrix}\right)$ with κ the Kummer cocycle of q_E; tensoring with ε gives the corner $\varepsilon\kappa$. A map $g \mapsto \left(\begin{smallmatrix}\chi_\ell\varepsilon & \kappa\\ 0 & \varepsilon\end{smallmatrix}\right)$ with κ a Kummer cocycle ($\kappa(gh) = \kappa(g) + \chi_\ell(g)\kappa(h)$) is not a homomorphism unless $\varepsilon = 1$. Fix: write the corner as $\varepsilon\kappa$ ("with ε unramified of order at most 2 and κ a Kummer cocycle of q_E"), or say "$\varepsilon\kappa$ is the cocycle"; then $c' = \varepsilon(\mathrm{Frob})\kappa(\mathrm{Frob}) = \pm\kappa(\mathrm{Frob})$. Effect: none. On I_q, $\varepsilon = 1$, so n' is unchanged; $v_\ell(c')$ is unchanged by the sign; the formula for Tam_q stands. (The Chapter 16 outline's item cocycle-in-the-corner already reads the paper's matrix this way: "the cocycle is $\varepsilon\kappa$".) Confidence: high (two-line computation; also in Part 14).

N2 (settled: important_notes.md point 43). Lemma lem:euler and Remark rem:wild: the $q \geq 5$ hypothesis is used only for the shape of the inertia image; the type IV value at 2 is the same formula. A remark for Chapters 19 and 21. The proof for the types $\mathrm{II}, \mathrm{II}^*, \mathrm{IV}, \mathrm{IV}^*$, $\mathrm{III}, \mathrm{III}^*$ uses only that $\rho(I_q)$ is cyclic of order m with $q \nmid m$ (item 9.2). For types IV, IV* at $q = 2$ the image of inertia is cyclic of order exactly 3 (never 6), hence tame: this is now a theorem (Chapter 11 proves it from $E[3]^{I_2} \simeq \Phi_2^{\mathrm{geom}}[3] = \mathbb{Z}/3$; Chapter 10 gives a second proof via the conductor exponent 2; important_notes.md point 43). So 9.2 applies verbatim and gives $P_2(X) = 1 + 2X$ and $e_3(E, 2) = v_3(2 - \chi_{-3}(2)) = 1$, which is what Remark rem:wild reports from PARI for type IV at 2 and what Proposition prop:forms needs ("$e_3(E, 2) \geq 1$ for all 92,812 curves of type IV or IV* at 2"). The exception in Proposition prop:forms disappears and "PARI's factor at 2 is $1 + 2X$ for every one of them" is a theorem. Fix (for the paper, recorded in point 43): replace the exception and the appeal to the 92,812 curves by this argument.

No other new problem was found in the paper in this chapter's area. The local statements of Lemmas lem:tamagawa-exp, lem:IV, lem:twist (iii), lem:euler, lem:tamfactor and Remark rem:wild were rechecked here; the known issues are points 7, 8, 11, 12, 13, 22, 28, 29, 31, 32 of important_notes.md.

9.16.2 15.2 Book-internal inconsistencies

- **B-a. PLAN.md, Chapter 7 entry: "Frob N Frob$^{-1} = p^{-1}N$".** With the book's convention (Frob arithmetic: Chapter 5 C2, Chapter 6 3.7, Chapter 16, Chapter 21 C2) the relation is Frob N Frob$^{-1} = pN$; the PLAN formula is the geometric form $\Phi N \Phi^{-1} = p^{-1}N$. This outline states both (convention L4). Action: none needed in the chapters; if PLAN.md is edited, write "$\Phi N \Phi^{-1} = p^{-1}N$ for the geometric Frobenius Φ".
- **B-b. Chapter 21 outline, convention C2(b), still carries the slip recorded as B1.** It says that on $V_\ell E$ the geometric Frobenius has eigenvalues α, β and the arithmetic one $p/\alpha = \beta$, $p/\beta = \alpha$, "the same set", and that $L_p(V_\ell E, s) = (1 - a_p p^{-s} + p^{1-2s})^{-1}$ with the geometric Frobenius. In fact the arithmetic Frobenius on $V_\ell E$ has eigenvalues α, β, the geometric one $\alpha^{-1} = \beta/p$, $\beta^{-1} = \alpha/p$ (a different set), so $\det(1 - \Phi p^{-s} \mid V_\ell E)^{-1} = (1 - a_p p^{-1-s} + p^{-1-2s})^{-1} = L_p(E, s+1)$; the factor $(1 - a_p p^{-s} + p^{1-2s})^{-1}$ is that of $V_\ell E^\vee$ (Chapter 6, 3.7; this outline, 8.2(v)). Nothing about $\mathrm{ad}^0 V$ changes. Action: the Chapter 21 writer must use 8.2(v).
- **B-c. Chapter 21 outline, A2.7 step (1).** It repeats the paper's matrix with corner κ "the Kummer class of q_E"; with N1 the corner is $\varepsilon\kappa$ (Chapter 16's cocycle-in-the-corner already says so). Step (7)'s indeterminacy argument is unaffected. Action: the Chapter 21 writer adds the factor ε.
- **B-d. Chapter 6 outline, 3.10(v) and 4.7(iv), 7.4, 5.7(ii); Chapter 2 outline, 11.2(iii); Chapter 3 outline, 6.4; Chapter 16 outline, imported facts.** All deferrals to Chapter 7 are honoured by items 3.5, 3.3(b), 3.4, 6.5(iii), 1.3, 1.4(ii), 5.3, 5.4, 6.2 respectively; no conflict found.

The normalisation of the Artin map in Chapter 6, 5.6(iii) (uniformiser to arithmetic Frobenius, units acting by u^{-1} on μ_{p^∞}) agrees with 6.5(ii) here.

- **B-e. Chapter 4 outline, table 9.3, row "unipotent".** It gives the mod 3 dimension of $(T/3T)^I$ for type I_n as "1 if $3 \nmid n$, 2 if $3 \mid n$"; this agrees with 6.2(v) and 7.6(iv) here (I acts trivially on $T/3T$ iff $3 \mid n$). No conflict; recorded because the same fact is stated in three chapters (4, 7, 11) and must stay consistent.

Chapter 10

Chapter 08: Algebraic curves, function fields and Riemann surfaces (`ch:curves`, `chapters/ch08_curves.tex`)

Stage 1 line-up. Part III (Curves and elliptic curves). Target length 2,150 lines of LaTeX (card excluded; see Part 3), inside the 1,000 to 2,200 range of PLAN.md Section 7.3.

Rigour tags: **P** = proved here, **S** = proof sketched here, **R** = stated; reference. Item format, as in the Chapter 2 to 6 outlines: number, label-ready name, kind, tag; the precise statement with all hypotheses; reference (book and chapter or section; a theorem number only where sure); the paper's use by its `\label` names, or "background". The small computations quoted below (genera, smoothness of the paper's Weierstrass models, Riemann-Hurwitz counts, the torus computations) were checked by hand and by Python/sympy scripts in `/tmp/ch08` (verification log, Part 6.6).

10.1 1. Purpose and global conventions

10.1.1 1.1 What this chapter is for

The paper uses curves in four places, and the chapter is built around them.

1. **The modular parametrisation and its degree** (Section 1, the abstract, Theorem thm:main, Lemma lem:hecke): "$\varphi_E\colon X_0(N)\to E$ its modular parametrisation", "the $X_0(N)$-optimal curve, for which the modular parametrisation has minimal degree", "$\deg\varphi_E$, its modular degree". The chapter supplies: smooth projective curves and morphisms between them, the degree of a nonconstant morphism in its equivalent forms (field degree, number of preimages, pullback of divisors, topological degree on H^2), $\varphi_*\varphi^* = \deg\varphi$ on divisor classes and on H^1, and the formula $\langle\varphi^*x,\varphi^*y\rangle_X = \deg\varphi\cdot\langle x,y\rangle_Y$ for the cup product, which is the cohomological meaning of the modular degree (item 11.5).
2. $H^1(X_0(M),\mathbb{Z}_\ell)$ **with the cup product, and the pairing** $\hat\delta_!$ (Section 4.1, Lemma lem:hecke, Theorem thm:quant Step 2, Lemma lem:ddt417 as applied): "it is perfect and alternating on $H^1(X_0(M^\Sigma),\mathbb{Z}_\ell)$, and the Hecke operators, U_p included, are self-adjoint for it, because $w_{M^\Sigma}T_nw_{M^\Sigma}^{-1}$ is the adjoint of T_n for the cup product". The chapter supplies H_1 and H^1 of a compact Riemann surface, rank $2g$, the cup product (perfect, alternating; Poincare duality), the base change to $\mathbb{Z}_\ell$, correspondences and their actions, and the general theorem that **the adjoint of a correspondence for the cup product is the transpose correspondence** (12.3), with the abstract lemma that

turns "$C^t = wCw$ for all C" into "every C is self-adjoint for $\langle x, w^*y\rangle$, which is perfect and alternating" (12.3(e)). The specific identity $T_n^t = w_N T_n w_N$ on $X_0(N)$ is Chapter 14's.

3. $J_0(N)$**, its torsion and its Tate module** (Remark rem:free, Section 4.2, important_notes.md point 26): "$J_0(N)[\mathfrak{m}]$ is two-dimensional", "the Tate module is free of rank two", "$M_{f,\lambda}$ and $T_\ell E$ are homothetic". The chapter supplies the Jacobian as Pic^0, as a complex torus $H^0(X,\Omega^1)^\vee/H_1(X,\mathbb{Z})$ and as an abelian variety (statements), Abel-Jacobi, $J[n] \simeq H_1(X,\mathbb{Z}/n) \simeq (\mathbb{Z}/n)^{2g}$, $T_\ell J \simeq H_1(X,\mathbb{Z}) \otimes \mathbb{Z}_\ell$, and **which is dual to which**: $H^1(X,\mathbb{Z}_\ell) = \mathrm{Hom}(T_\ell J, \mathbb{Z}_\ell)$, and Galois-equivariantly $H^1_{\text{et}} \simeq T_\ell J(-1)$ via the Weil pairing.
4. **Foundations for Chapters 9, 10, 12, 14, 15**: varieties and points over extensions with the Galois action (every $E(K)$, $E[m]$, $T_\ell E$ of the paper), Riemann-Roch and its consequences (a genus-one curve with a rational point is a Weierstrass cubic; $P \mapsto [(P)-(O)]$ is a bijection onto Pic^0), separability and the Frobenius morphism, ramification and Riemann-Hurwitz (isogenies are unramified; genus formula of $X_0(N)$ in Chapter 12), compact Riemann surfaces versus complex curves (so that $X_0(N)$, built analytically in Chapter 12, is an algebraic curve), holomorphic differentials (weight-two cusp forms are holomorphic differentials, Chapter 12).

This is the hardest audience gap of the book: a reader who knows complex analysis and linear algebra meets algebraic geometry and algebraic topology at once. The chapter therefore follows three rules.

- **Two languages, one dictionary.** The algebraic side is done in the classical language of Silverman's AEC, Chapters I and II: varieties are sets of $\bar{k}$-points cut out by polynomials, with G_k acting on coordinates; no schemes, no sheaves, no sheaf cohomology. The analytic side is done on compact Riemann surfaces with charts, exactly as in a first complex analysis course. The dictionary between them (Riemann existence, GAGA) is stated (R) once, in 8.3, and used freely afterwards; every other comparison is proved from it.
- **What is proved, sketched, stated.** Proved (P): everything that is a computation with polynomials, valuations, local normal forms of holomorphic maps, change of variables in integrals, or linear algebra of lattices: smooth points have DVR local rings (2.2), rational maps from smooth curves extend (3.2), the degree formulas (6.4), the consequences of Riemann-Roch (5.3, 5.6, 5.7), Riemann-Hurwitz from Riemann-Roch (7.2), the residue theorem (9.2), positivity in the bilinear relations (9.4), the cup product on the torus (10.4), the torsion and Tate module of a complex torus (11.6, 11.7), the pullback formula and the saturation criterion (11.5), the adjointness of correspondences (12.3). Sketched (S): the degree formula $\sum e_P = \deg\varphi$ (6.3, via Chapter 6's $\sum e_i f_i = n$), the genus of smooth plane curves (5.4), the Weierstrass embedding being an isomorphism (5.6), Riemann's bilinear relations in period form (9.4), Hodge decomposition (10.2). Stated (R): the Nullstellensatz, Bezout, the equivalence of categories of curves and function fields, Riemann-Roch, Riemann existence and GAGA, the classification of surfaces and the existence of a symplectic basis, Poincare duality over $\mathbb{Z}$, Abel's theorem and Jacobi inversion, the algebraicity of the Jacobian and its universal property, the Weil pairing and etale cohomology, the Weil conjectures for curves.
- **The physics reader's intuition is used, in one sentence each, where it is literally right**: contour integrals and residues (9.2: "total charge on a closed surface is zero"); periods of dz on a torus as the primitive vectors of a crystal lattice, and H^1 as the reciprocal lattice (10.4); the torus $\mathbb{R}^2/\mathbb{Z}^2$ as a periodic box with area form $dx \wedge dy$ (10.4); intersection number as a signed crossing count, like a winding number (10.1); Riemann-Hurwitz as Euler's $V - E + F$ for a branched cover (7.2, 8.4); positivity $\frac{i}{2}\int \omega \wedge \bar{\omega} = \int |f|^2 dx\, dy$ as a norm (9.4; the Petersson norm of Chapter 12 is this); lattices up to homothety (Chapter 9). Never belaboured.

10.1.2 1.2 Global conventions fixed in this chapter

- **C1 (fields).** k is a **perfect** field (characteristic 0 or a finite field are all the paper needs: $\mathbb{Q}$, $\mathbb{Q}_p$, $\mathbb{F}_p$, number fields); $\bar{k}$ an algebraic closure, $G_k = \mathrm{Gal}(\bar{k}/k)$ (Chapter 5, C1). Over $\mathbb{C}$ we write X for a compact Riemann surface and identify it with the smooth projective curve it defines (8.3).
- **C2 (curves).** A *curve over* k is a projective variety of dimension one defined over k (1.3); "variety" includes "irreducible over $\bar{k}$" (geometrically irreducible). Unless said otherwise curves are **smooth**

and projective; a singular or affine model is called a *model*. Morphisms are defined over k when k is named, over $\bar{k}$ otherwise. The genus is g; for X a compact Riemann surface it is also the topological genus (8.4).

- **C3 (points and Galois).** $C(L)$ is the set of points with coordinates in L (for $k \subseteq L \subseteq \bar{k}$); $\sigma \in G_k$ acts on $C(\bar{k})$ coordinatewise, on the left; $C(L) = C(\bar{k})^{G_L}$. This is the action behind $\rho_{E,\ell}$ (Chapters 9, 11).
- **C4 (pullback, pushforward, transpose, adjoint).** For a nonconstant morphism $\varphi\colon X \to Y$: φ^* is pullback (of functions, divisors, divisor classes, differentials, H^1 classes) and φ_* is pushforward (of divisors, divisor classes, H_1 classes) and, on H^1, the transpose of the transfer (12.2). A correspondence is written $\mathcal{C} = (C, \alpha, \beta)$ and its transpose $\mathcal{C}^t = (C, \beta, \alpha)$. **The adjoint of an endomorphism A for a pairing is written $A^\dagger$ in this chapter**, because * is taken by pullbacks; this is the book-wide convention of Appendix B (B-C4), and the paper and Chapter 4 write * for Pontryagin duals (Part 7, book-internal point B3).
- **C5 (duals).** $M^\vee = \mathrm{Hom}(M, \mathbb{Z})$ or $\mathrm{Hom}_{\mathbb{Z}_\ell}(M, \mathbb{Z}_\ell)$ for lattices; $V^\vee$ for linear duals and contragredient representations, as in Chapter 4, C2. Pontryagin duals (* in the paper) do not occur in this chapter.
- **C6 (orientation and the cup product).** A Riemann surface is oriented by its complex structure: in a chart $z = x+iy$, $dx \wedge dy = \frac{i}{2} dz \wedge d\bar{z}$ is positive. The cup product of H^1 classes is $\langle \alpha, \beta \rangle = \int_X \alpha \wedge \beta$ (de Rham representatives), an integer on integral classes. The intersection number $a \cdot b$ of transverse oriented loops counts crossings with sign $+1$ when (tangent of a, tangent of b) is a positive basis. Poincare duality PD: $H^1 \to H_1$ is fixed by $\int_{\mathrm{PD}(\alpha)} \beta = \int_X \alpha \wedge \beta$; with these choices PD is an isometry, $\langle \alpha, \beta \rangle = \mathrm{PD}(\alpha) \cdot \mathrm{PD}(\beta)$ (checked on the torus, 10.4). Other books use other signs; the paper needs only "perfect and alternating", the adjointness of 12.3 and ideals of values, none of which depends on the sign.
- **C7 (Frobenius).** As in Chapter 5, C2 and Chapter 6, 3.7: the arithmetic Frobenius of $G_{\mathbb{F}_q}$ acts on $\bar{\mathbb{F}}_q$-points by raising coordinates to the q-th power; on $T_\ell J$ it acts as the q-power Frobenius endomorphism; the geometric Frobenius is its inverse, and on $H^1 = (T_\ell J)^\vee$ the geometric Frobenius has the same eigenvalues as the arithmetic one on $T_\ell J$ (11.8(d)).
- **C8 (typography).** PLAN.md Section 7.2: no dash ligatures, ASCII source, labels `...:curves:...`; "Riemann-Roch", "Riemann-Hurwitz", "Abel-Jacobi" with plain hyphens; tables with `booktabs`.

10.2 2. Line-up of the core results, in teaching order

10.2.1 Section 1. Affine and projective space, varieties

1.1 `def:curves:affine-variety` (definitions and theorem; P except the Nullstellensatz, R). $\mathbb{A}^n = \mathbb{A}^n(\bar{k}) = \bar{k}^n$, with G_k acting coordinatewise and $\mathbb{A}^n(L) = \mathbb{A}^n(\bar{k})^{G_L}$. For $S \subseteq \bar{k}[x_1, \ldots, x_n]$, $V(S)$ is its zero set (an *algebraic set*); for $Y \subseteq \mathbb{A}^n$, $I(Y)$ is the ideal of polynomials vanishing on Y. The algebraic sets are the closed sets of the *Zariski topology*. (i) Hilbert's Nullstellensatz: $I(V(\mathfrak{a})) = \sqrt{\mathfrak{a}}$ for every ideal $\mathfrak{a}$ of $\bar{k}[x]$ (R). (ii) An *affine variety* is an algebraic set V with $I(V)$ prime; it is *defined over k* if $I(V)$ is generated by polynomials in $k[x]$ (for k perfect, equivalently $\sigma(V) = V$ for all $\sigma \in G_k$; R). (iii) The coordinate ring $\bar{k}[V] = \bar{k}[x]/I(V)$ is a domain; $k[V] = k[x]/(I(V) \cap k[x])$; the function field $\bar{k}(V) = \mathrm{Frac}\, \bar{k}[V]$ and $k(V) = \mathrm{Frac}\, k[V]$. (iv) $\dim V$ is the transcendence degree of $\bar{k}(V)$ over $\bar{k}$ (transcendence degree recalled from Lang VIII.1, R; Chapter 5 defines only algebraic and transcendental elements). Examples: $\mathbb{A}^1$, the parabola $y = x^2$ ($\bar{k}[V] \simeq \bar{k}[x]$), the affine Weierstrass cubics of 2.1. Reference: SilvermanAEC I.1; Hartshorne I.1; Fulton, Chapter 1. Paper: background.

1.2 `def:curves:projective-space` (definitions and proposition; P). $\mathbb{P}^n = \mathbb{P}^n(\bar{k}) = (\bar{k}^{n+1} \setminus 0)/\bar{k}^\times$ with homogeneous coordinates $[x_0 : \cdots : x_n]$ (for $n = 1$ this is Chapter 1's set of lines, C1 there). (i) G_k acts coordinatewise, and $\mathbb{P}^n(\bar{k})^{G_L} = \mathbb{P}^n(L)$: a G_L-fixed point has a representative with coordinates in L (scale a nonzero coordinate to 1; then each σ fixes every coordinate). (ii) Homogeneous ideals, projective algebraic sets $V(S)$ for homogeneous S, projective varieties (homogeneous prime $I(V)$), defined over k as in 1.1. (iii) The charts $U_i = \{x_i \neq 0\} \simeq \mathbb{A}^n$; the projective closure $\bar{V}$ of an affine variety (homogenise

the generators of $I(V)$) and dehomogenisation; $\mathbb{P}^n = \mathbb{A}^n \sqcup \mathbb{P}^{n-1}$, $\mathbb{P}^1 = \mathbb{A}^1 \sqcup \{\infty\}$, with PGL_2 acting by Moebius transformations (Chapter 1, C1). (iv) The function field of a projective variety is the field of quotients F/G of homogeneous polynomials of equal degree with $G \notin I(V)$; it equals that of any nonempty affine chart. Reference: SilvermanAEC I.2; Hartshorne I.2. Paper: background ($X_0(N)$ and E are projective curves).

1.3 `def:curves:curve-morphism` (definitions; P). A *curve over k* is a projective variety of dimension 1 defined over k (C2). For projective varieties V_1, $V_2 \subseteq \mathbb{P}^n$, a *rational map* $\varphi = [f_0 : \cdots : f_n] \colon V_1 \dashrightarrow V_2$ with $f_i \in \bar{k}(V_1)$, not all zero, such that $\varphi(P) \in V_2$ wherever defined; φ is *regular* at P if there is $g \in \bar{k}(V_1)^\times$ with all gf_i regular at P and some $(gf_i)(P) \neq 0$; a *morphism* is a rational map regular everywhere; φ is *defined over k* if it can be written with $f_i \in k(V_1)$ (for k perfect: iff $\varphi(\sigma P) = \sigma\varphi(P)$ for all σ, P; R). Isomorphisms. Example: $[x^2 : xy : y^2] \colon \mathbb{P}^1 \to \mathbb{P}^2$ onto the conic $XZ = Y^2$. Reference: SilvermanAEC I.3. Paper: background (φ_E is a morphism defined over $\mathbb{Q}$, Chapter 14).

10.2.2 Section 2. Plane curves and smoothness

2.1 `def:curves:plane-curve` (definition and proposition; P). A *plane curve of degree d* is $C = V(F) \subset \mathbb{P}^2$ for a homogeneous $F \in k[X, Y, Z]$ of degree d, irreducible over $\bar{k}$; its affine part is $f(x, y) = F(x, y, 1) = 0$. A point $P \in C$ is *singular* if $F_X(P) = F_Y(P) = F_Z(P) = 0$, *smooth* (nonsingular) otherwise; C is smooth if it has no singular point in $C(\bar{k})$. (i) Euler's identity $XF_X + YF_Y + ZF_Z = dF$; hence if $\mathrm{char}\, k \nmid d$, a point of $\mathbb{P}^2$ where the three partials vanish lies on C; on the affine part, P is singular iff $f(P) = f_x(P) = f_y(P) = 0$. (ii) At a smooth point the tangent line is $F_X(P)X + F_Y(P)Y + F_Z(P)Z = 0$. (iii) Examples (checked by Groebner bases over $\mathbb{Q}$, Part 6.6): the nodal cubic $y^2 = x^3 + x^2$ and the cuspidal cubic $y^2 = x^3$ have the single singular point $(0, 0)$; the Fermat curve $X^d + Y^d + Z^d = 0$ is smooth iff $\mathrm{char}\, k \nmid d$ (in characteristic $p = d$ it is the triple line $(X + Y + Z)^p$, not even irreducible); the cubics $y^2 + y = x^3 - x^2 - 10x - 20$ (11a1), $y^2 + xy + y = x^3 + x^2 - 30x - 76$ (121a1) and $y^2 = x^3 + x^2 + 4x + 4$ (20a1) are smooth over $\bar{\mathbb{Q}}$, including the point $[0 : 1 : 0]$ at infinity. The general smoothness criterion for Weierstrass cubics ($\Delta \neq 0$) is Chapter 9's. Reference: SilvermanAEC I.1 (Example 1.6 and the exercises; cite the section); Fulton, Chapters 3 and 5. Paper: background (the curves of the paper are given by such equations; Section 1, the Appendix `ellinit`).

2.2 `thm:curves:smooth-dvr` (theorem; P). Let C be a curve (affine plane model $f = 0$ suffices) and $P \in C(\bar{k})$ with local ring $\bar{k}[C]_P = \{g/h : h(P) \neq 0\} \subset \bar{k}(C)$ and maximal ideal $\mathfrak{m}_P$. Then P is smooth iff $\dim_{\bar{k}} \mathfrak{m}_P/\mathfrak{m}_P^2 = 1$, iff $\bar{k}[C]_P$ is a discrete valuation ring. At a smooth point of $f = 0$, any line through P other than the tangent line gives a uniformiser t_P (for instance $x - x(P)$ if $f_y(P) \neq 0$). The valuation $\mathrm{ord}_P \colon \bar{k}(C)^\times \to \mathbb{Z}$ is the order of vanishing; $f(P)$ is defined iff $\mathrm{ord}_P f \geq 0$. Proof: $\mathfrak{m}_P/\mathfrak{m}_P^2$ is spanned by $x - x(P)$, $y - y(P)$ subject to the one relation $f_x(P)(x - x(P)) + f_y(P)(y - y(P)) \equiv 0$; then Nakayama (Chapter 3, 4.3) and the DVR criterion (Chapter 3, 6.1(a)). Reference: SilvermanAEC II.1 (Proposition 1.1); Fulton 3.2; Hartshorne I.6. Paper: background (orders of zeros and poles everywhere below).

2.3 `thm:curves:bezout` (theorem; R). Two plane curves of degrees m and n without common component meet in exactly mn points of $\mathbb{P}^2(\bar{k})$ counted with intersection multiplicity. Special cases used: a line meets a curve of degree d (not containing it) in d points with multiplicity, so the divisor of a linear form on a smooth plane curve of degree d (4.1) has degree d; a line meets a cubic in three points (the chord-tangent construction of Chapter 9). Reference: Fulton 5.3; Hartshorne I.7. Paper: background.

2.4 `thm:curves:smooth-model` (theorem; R). Every curve, possibly singular or affine, is birational over k to a smooth projective curve, unique up to isomorphism (follows from 3.3). Example: the node $y^2 = x^3 + x^2$ is birational to $\mathbb{P}^1$ via $t = y/x$, $x = t^2 - 1$, $y = t(t^2 - 1)$ (P). Reference: Hartshorne I.6; Fulton, Chapter 7. Paper: background ("the curve given by an equation" always means its smooth projective model; Chapter 14's models of $X_0(N)$).

10.2.3 Section 3. Function fields, points over extensions, the Galois action

3.1 `def:curves:function-field` (definition and proposition; P for (ii), (iii); R for (i), (iv)). (i) A *function field of one variable over k* is a finitely generated field extension K/k of transcendence degree 1 in which k is algebraically closed; $k(C)$ is one (R). (ii) For C smooth, each $P \in C(\bar{k})$ gives the valuation

ord_P on $\bar{k}(C)$ (2.2), with $\operatorname{ord}_P(c) = 0$ for $c \in \bar{k}^\times$. (iii) If $f \in \bar{k}(C)^\times$, then $\operatorname{ord}_P f = 0$ for all but finitely many P (P on a plane model: zeros and poles lie on $V(f)$, $V(g)$ for $f = g/h$, finite sets by Bezout or by dimension). (iv) A function with no poles is constant (R here; P after 6.4). Reference: SilvermanAEC II.1 (Proposition 1.2); Hartshorne I.6. Paper: background.

3.2 `thm:curves:extend-morphism` (theorem; P). Let C be a smooth curve, $V \subseteq \mathbb{P}^n$ a projective variety and $\varphi = [f_0 : \cdots : f_n] \colon C \dashrightarrow V$ a rational map. Then φ is a morphism. Proof: at P put $m = \min_i \operatorname{ord}_P f_i$; then $t_P^{-m} f_i$ are regular at P and one is nonzero there. Corollaries: a rational map between smooth projective curves is a morphism; every $f \in \bar{k}(C)$ is a morphism $f = [f : 1] \colon C \to \mathbb{P}^1$ (constant or not). Reference: SilvermanAEC II.2 (Proposition 2.1). Paper: background.

3.3 `thm:curves:equivalence` (theorem; R). The functor $C \mapsto k(C)$, $(\varphi \colon C_1 \to C_2) \mapsto (\varphi^* \colon k(C_2) \to k(C_1))$ is an equivalence (contravariant) between the category of smooth projective curves over k with nonconstant morphisms defined over k, and function fields of one variable over k with k-embeddings. Consequences: two smooth projective curves with isomorphic function fields are isomorphic; a morphism of degree 1 (6.2) is an isomorphism. Reference: Hartshorne I.6 (Corollary 6.12); SilvermanAEC II.2 (Theorem 2.4). Paper: background (Section 1: φ_E corresponds to $\mathbb{Q}(E) \hookrightarrow \mathbb{Q}(X_0(N))$, of degree $\deg \varphi_E$).

3.4 `prop:curves:galois-action` (proposition; P). Let C be a curve over k. (i) G_k acts on $C(\bar{k})$, and $C(\bar{k})^{G_L} = C(L)$ for $k \subseteq L \subseteq \bar{k}$ (from 1.2(i)). (ii) G_k acts on $\bar{k}(C)$ through the coefficients, with fixed field $k(C)$ (R for the fixed field: Hilbert 90 for $\bar{k}(C)$, SilvermanAEC II.5); for $f \in \bar{k}(C)$ regular at P, $(\sigma f)(\sigma P) = \sigma(f(P))$, and $\operatorname{ord}_{\sigma P}(\sigma f) = \operatorname{ord}_P f$. (iii) A morphism φ defined over k satisfies $\varphi(\sigma P) = \sigma\varphi(P)$; so $\varphi(C_1(L)) \subseteq C_2(L)$, and the fibres $\varphi^{-1}(Q)$ over $Q \in C_2(k)$ are G_k-stable. (iv) Examples: the conic $X^2 + Y^2 + Z^2 = 0$ over $\mathbb{Q}$ has $C(\mathbb{Q}) = \emptyset$ (no real points) but $C(\mathbb{Q}(i)) \ni [1 : i : 0]$, so it is a genus-0 curve not isomorphic over $\mathbb{Q}$ to $\mathbb{P}^1$; Selmer's cubic $3X^3 + 4Y^3 + 5Z^3 = 0$ is smooth of genus 1 with no rational point (stated). Reference: SilvermanAEC I.1 to I.3 and II.2 (cite the sections). Paper: Section 1.1 ($\rho_{E,\ell}$ is the action of 3.4(i) on $E[\ell^n] \subset E(\bar{\mathbb{Q}})$; Chapters 9, 11), Proposition prop:ns (the action on $E[3]$ and on the roots of the 3-division polynomial), Section 4.2 ($G_{\mathbb{Q}}$ acting on H^1; 11.8).

10.2.4 Section 4. Divisors, differentials, the Picard group

4.1 `def:curves:divisor` (definitions and proposition; P given 6.4 for (iii)). For a smooth curve C: $\operatorname{Div}(C)$ is the free abelian group on $C(\bar{k})$; $\deg \sum n_P(P) = \sum n_P$; $D \geq 0$ (effective); $\operatorname{supp} D$; G_k acts, and $\operatorname{Div}_k(C) = \operatorname{Div}(C)^{G_k}$ (divisors *defined over* k; a G_k-orbit of points is one example). (i) The divisor of $f \in \bar{k}(C)^\times$ is $\operatorname{div}(f) = \sum_P \operatorname{ord}_P(f)(P)$ (a finite sum by 3.1(iii)); principal divisors form a subgroup, $\operatorname{div}(fg) = \operatorname{div} f + \operatorname{div} g$. (ii) $\operatorname{div}(f) = 0$ iff $f \in \bar{k}^\times$ (3.1(iv)). (iii) $\deg \operatorname{div}(f) = 0$: for f nonconstant, $\operatorname{div}(f) = f^*((0)) - f^*((\infty))$ for $f \colon C \to \mathbb{P}^1$, and both have degree $\deg f$ by 6.4(iii). Over $\mathbb{C}$ a second proof by the residue theorem (9.2). Example: on $\mathbb{P}^1$, $\operatorname{div}(x - a) = (a) - (\infty)$. Reference: SilvermanAEC II.3 (Proposition 3.1). Paper: background (Abel-Jacobi, 11.2; Chapter 9's group law).

4.2 `def:curves:picard` (definitions and proposition; P except (iii), R). $D \sim D'$ (linearly equivalent) iff $D - D'$ is principal. $\operatorname{Pic}(C) = \operatorname{Div}(C)/\operatorname{div}(\bar{k}(C)^\times)$; deg induces $\operatorname{Pic}(C) \to \mathbb{Z}$ with kernel $\operatorname{Pic}^0(C) = \operatorname{Div}^0(C)/\operatorname{Prin}(C)$. (i) $\operatorname{Pic}^0(\mathbb{P}^1) = 0$: a degree-0 divisor $\sum n_i(a_i) + n_\infty(\infty)$ is $\operatorname{div} \prod (x - a_i)^{n_i}$. (ii) G_k acts on $\operatorname{Pic}^0(C)$. (iii) The natural map $\operatorname{Div}_k^0(C)/\operatorname{div}(k(C)^\times) \to \operatorname{Pic}^0(C)^{G_k}$ is injective (Hilbert 90 for $\bar{k}(C)$) and bijective when $C(k) \neq \emptyset$ (R; Hilbert 90, Chapter 17). Reference: SilvermanAEC II.3; Milne, Jacobian varieties, Section 1 (cite the section). Paper: background (the definition of J in 11.1).

4.3 `def:curves:differentials` (definition and proposition; R for (i), (ii); P for (iii), (iv)). The space Ω_C of (meromorphic) differentials of C: the $\bar{k}(C)$-vector space generated by symbols df, $f \in \bar{k}(C)$, modulo $d(f + g) = df + dg$, $d(fg) = f\,dg + g\,df$, $da = 0$ for $a \in \bar{k}$. (i) Ω_C is one-dimensional over $\bar{k}(C)$, and df is a basis iff $\bar{k}(C)/\bar{k}(f)$ is a finite separable extension (in characteristic p: $d(f^p) = 0$). (ii) For a uniformiser t at P, every ω is $g\,dt$ and $\operatorname{ord}_P \omega := \operatorname{ord}_P g$ does not depend on t; $\operatorname{ord}_P \omega = 0$ for almost all P; $\operatorname{div}(\omega) = \sum \operatorname{ord}_P(\omega)(P)$; $\operatorname{div}(f\omega) = \operatorname{div} f + \operatorname{div} \omega$. (iii) Hence all $\operatorname{div}(\omega)$, $\omega \neq 0$, lie in one linear equivalence class, the *canonical class* K_C; a *canonical divisor* is any $\operatorname{div} \omega$. (iv) ω is *regular* (holomorphic) if $\operatorname{div} \omega \geq 0$; they form a $\bar{k}$-vector space. Examples (P): on $\mathbb{P}^1$, $\operatorname{div}(dx) = -2(\infty)$, so no nonzero regular differential; on a smooth cubic $y^2 = h(x)$, h squarefree of degree 3, $\operatorname{char} k \neq 2$, $\operatorname{div}(dx/y) = 0$ (at a point with $y = 0$, y is a uniformiser and $dx = 2y\,dy/h'(x)$; at ∞, $t = x/y$ and $\operatorname{ord}_\infty(dx/y) = 0$ by the computation $\operatorname{ord} x = -2$, $\operatorname{ord} y = -3$, $\operatorname{ord} dx = -3$). The "invariant differential" and its translation

invariance are Chapter 9's. Reference: SilvermanAEC II.4 (Propositions 4.2 and 4.3); Hartshorne II.8 and IV.1. Paper: background (weight-two cusp forms as holomorphic differentials $f(z)\,dz$ on $X_0(N)$: Chapter 12; $\varphi_E^*\omega_E = 2\pi i c f(z)dz$ with the Manin constant c: Chapter 14).

10.2.5 Section 5. Riemann-Roch and its consequences

5.1 `def:curves:riemann-roch-space` (definition and proposition; P except (iv), R). For $D \in \mathrm{Div}(C)$, $\mathcal{L}(D) = \{f \in \bar{k}(C)^\times : \mathrm{div}\, f + D \geq 0\} \cup \{0\}$, a $\bar{k}$-vector space, $\ell(D) = \dim \mathcal{L}(D)$. (i) $\mathcal{L}(0) = \bar{k}$; $\mathcal{L}(D) = 0$ if $\deg D < 0$ (by 4.1(iii)); $D \sim D'$ implies $\mathcal{L}(D) \simeq \mathcal{L}(D')$ (multiply by h with $D' = D + \mathrm{div}\, h$). (ii) $\ell(D+P) \leq \ell(D) + 1$ (the map $f \mapsto (t_P^{n+1} f)(P)$, n the coefficient of P in D, has kernel $\mathcal{L}(D)$); hence $\ell(D) \leq \deg D + 1$ for $D \geq 0$, and $\mathcal{L}(D)$ is finite-dimensional. (iii) Hence $\ell(D)$ and $\ell(K-D)$ depend only on the classes of D and $K - D$ in $\mathrm{Pic}(C)$. (iv) If D is defined over k, $\mathcal{L}(D)$ has a basis in $k(C)$ (R; Hilbert 90 for vector spaces, SilvermanAEC II.5 (Proposition 5.8)). Reference: SilvermanAEC II.5. Paper: background.

5.2 `thm:curves:riemann-roch` (theorem, Riemann-Roch; R, with the shape of the proof in one paragraph, S). Let C be a smooth projective curve over k and K a canonical divisor. There is an integer $g \geq 0$, the *genus* of C, such that for every $D \in \mathrm{Div}(C)$:

$$\ell(D) - \ell(K - D) = \deg D - g + 1.$$

Shape of the proof: Riemann's inequality $\ell(D) \geq \deg D + 1 - g$ (with g defined as the maximum defect) by comparing $\mathcal{L}(D)$ with $\mathcal{L}(D + nP)$; the correction term $\ell(K - D)$ is the dimension of the space of obstructions to prescribing principal parts ("repartitions", the algebraic Mittag-Leffler problem), which is dual to $\mathcal{L}(K - D)$ via residues (Serre duality). Over $\mathbb{C}$ the obstruction space is $H^1(X, \mathcal{O}(D))$. Reference: SilvermanAEC II.5 (Theorem 5.4, proof referred to Hartshorne and Lang); Hartshorne IV.1 (Theorem 1.3); Fulton, Chapter 8; Miranda, Chapter VI. Paper: background (everything below).

5.3 `cor:curves:rr-consequences` (corollary; P). (a) $\ell(K) = g$: the genus is the dimension of the space of regular differentials ($D = 0$, and $f \mapsto f\omega$ identifies $\mathcal{L}(\mathrm{div}\,\omega)$ with the regular differentials). (b) $\deg K = 2g - 2$ ($D = K$). (c) If $\deg D > 2g - 2$ then $\ell(D) = \deg D - g + 1$. (d) $g(\mathbb{P}^1) = 0$ ($\mathrm{div}\, dx = -2(\infty)$); conversely a curve C of genus 0 with $P \in C(k)$ is isomorphic over k to $\mathbb{P}^1$ ($\ell(P) = 2$ gives $f \in k(C)$ of degree 1, an isomorphism by 3.3, 6.4); in particular a curve with two points $P \neq Q$ and $(P) \sim (Q)$ has genus 0. (e) If $g = 1$: $\ell(D) = \deg D$ for $\deg D \geq 1$, $K \sim 0$, and a nonzero regular differential has no zeros. Reference: SilvermanAEC II.5 (Corollary 5.5). Paper: background.

5.4 `thm:curves:plane-genus` (theorem; S). A smooth plane curve of degree d over k with $\mathrm{char}\, k \nmid d$ (or any smooth plane curve with $\mathrm{char}\, k = 0$) has genus $(d-1)(d-2)/2$. Sketch: with affine equation $f(x, y) = 0$ and the line at infinity in general position, $\omega_0 = dx/f_y = -dy/f_x$ is regular and nowhere zero on the affine part, and $\mathrm{div}(\omega_0) = (d-3) \cdot (\text{line at infinity})$, of degree $d(d-3)$ by Bezout (2.3); so $2g - 2 = d(d-3)$ by 5.3(b). The regular differentials are exactly $h(x, y)\,\omega_0$ with $\deg h \leq d - 3$ ("adjoint differentials"), a space of dimension $\binom{d-1}{2}$. Reference: Fulton, Chapter 8 (cite the chapter); Hartshorne IV.1 and I.7 (cite the sections); Miranda, Chapter VII. Paper: background (lines and conics have genus 0; smooth cubics genus 1, Chapter 9).

5.5 `ex:curves:genus-examples` (examples; P given 5.4 and 7.2). (a) Lines and smooth conics: $g = 0$. (b) Smooth plane cubics, in particular every smooth Weierstrass cubic (11a1, 20a1, 121a1 of 2.1): $g = 1$, with $\omega = dx/(2y + a_1 x + a_3)$ regular and nowhere zero. (c) The Fermat quartic $X^4 + Y^4 + Z^4 = 0$: $g = 3$. (d) Hyperelliptic curves: the smooth projective model of $y^2 = h(x)$, h squarefree of degree $2g + 1$ or $2g + 2$, $\mathrm{char}\, k \neq 2$, has genus g (7.3(c)); $y^2 = x^5 - x$ has genus 2. (e) Singular plane models have smaller genus: the nodal and cuspidal cubics are rational (2.4); the general formula $g = \frac{(d-1)(d-2)}{2} - \sum_P \delta_P$ is stated (R; Fulton, Chapter 7; Hartshorne IV.1 (cite the section)). (f) The modular curves are not given as smooth plane curves; their genera come from Riemann-Hurwitz (7.3(f), Chapter 12): $g(X_0(11)) = 1$, $g(X_0(20)) = 1$, $g(X_0(121)) = 6$, $g(X_0(539)) = 49$ (computed from Chapter 12's genus formula, Part 6.6). Paper: background; Section 1 (the curves 11a1, 20a1, 121a1, 539c1).

5.6 `thm:curves:genus-one-weierstrass` (theorem; P for the equation, S for the isomorphism). Let C be a smooth projective curve of genus 1 over k and $O \in C(k)$. (i) There are $x \in \mathcal{L}(2(O))$ and

$y \in \mathcal{L}(3(O))$, defined over k, with $\mathcal{L}(2(O)) = \langle 1, x\rangle$ and $\mathcal{L}(3(O)) = \langle 1, x, y\rangle$ ($\operatorname{ord}_O x = -2$, $\operatorname{ord}_O y = -3$; by 5.3(e) and 5.1(iv)). (ii) The seven functions $1, x, y, x^2, xy, y^2, x^3$ lie in the six-dimensional $\mathcal{L}(6(O))$, so they satisfy a linear relation over k, in which y^2 and x^3 occur with nonzero coefficients (they are the only ones with a pole of order 6 at O); rescaling x, y gives

$$y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6, \qquad a_i \in k.$$

(iii) The morphism $\phi = [x : y : 1]\colon C \to \mathbb{P}^2$ is an isomorphism onto the projective cubic E defined by this equation, with $\phi(O) = [0 : 1 : 0]$; E is smooth. Sketch of (iii): $[k(C) : k(x)] = 2$ and $[k(C) : k(y)] = 3$ (6.4: degrees of pole divisors), so $k(C) = k(x, y)$ and ϕ is birational onto its image; a singular cubic is rational (2.4), which would contradict $g = 1$; so E is smooth and ϕ is an isomorphism by 3.3. (iv) Conversely a smooth Weierstrass cubic has genus 1 (5.4) and the point $O = [0 : 1 : 0]$ is defined over k. Reference: SilvermanAEC III.3 (Proposition 3.1). Paper: background (an elliptic curve over $\mathbb{Q}$ "is" a Weierstrass equation: Section 1, the Appendix (`ellinit([1,1,1,-30,-76])`); Chapter 9 starts from here).

5.7 `prop:curves:genus-one-pic` (proposition; P). Let C have genus 1 and $O \in C(k)$. The map $\kappa\colon C(\bar{k}) \to \operatorname{Pic}^0(C)$, $P \mapsto$ the class of $(P) - (O)$, is a bijection, commuting with G_k. Proof: surjective, because for D of degree 0, $\ell(D + (O)) = 1$ (5.3(e)), so $D + (O) \sim (P)$ for the unique zero P of a generator; injective, because $(P) \sim (Q)$ with $P \neq Q$ would force genus 0 (5.3(d)). Consequence: $C(\bar{k})$ inherits a group structure from $\operatorname{Pic}^0(C)$, with identity O; Chapter 9 identifies it with the chord-tangent law (via $(P) + (Q) + (R) \sim 3(O)$ for collinear P, Q, R on a Weierstrass cubic) and so gets associativity for free. Reference: SilvermanAEC III.3 (Proposition 3.4). Paper: background ($E = \operatorname{Pic}^0(E) = J(E)$, so E is its own Jacobian: 11.4, 11.5 and Chapter 14's $E^\vee \to J_0(N) \to E$).

10.2.6 Section 6. Morphisms and their degree; separability; the Frobenius morphism

6.1 `thm:curves:constant-or-surjective` (theorem; R for surjectivity, P for the rest). A morphism $\varphi\colon C_1 \to C_2$ of curves (C2) is constant or surjective. If it is nonconstant, $\varphi^*\colon \bar{k}(C_2) \to \bar{k}(C_1)$, $f \mapsto f \circ \varphi$, is an injection of fields and $\bar{k}(C_1)$ is a finite extension of $\varphi^*\bar{k}(C_2)$ (both have transcendence degree 1; finitely generated). Reference: SilvermanAEC II.2 (Theorem 2.3); Hartshorne II.6. Paper: background.

6.2 `def:curves:degree` (definitions; P). For nonconstant $\varphi\colon C_1 \to C_2$: $\deg\varphi = [\bar{k}(C_1) : \varphi^*\bar{k}(C_2)]$; the separable and inseparable degrees $\deg_s\varphi$, $\deg_i\varphi$ of this extension (Chapter 5, galois:separability), $\deg\varphi = \deg_s\varphi \deg_i\varphi$; φ is *separable* (purely inseparable) if the extension is. Constant maps have degree 0. The *ramification index* at P is $e_\varphi(P) = \operatorname{ord}_P(\varphi^* t_{\varphi(P)}) \geq 1$ (t_Q a uniformiser at Q). Pullback $\varphi^*\colon \operatorname{Div}(C_2) \to \operatorname{Div}(C_1)$, $(Q) \mapsto \sum_{\varphi(P)=Q} e_\varphi(P)(P)$; pushforward $\varphi_*\colon \operatorname{Div}(C_1) \to \operatorname{Div}(C_2)$, $(P) \mapsto (\varphi(P))$. Reference: SilvermanAEC II.2 and II.3. Paper: Section 1 ("$\deg\varphi_E$", the modular degree is the degree of the morphism $\varphi_E\colon X_0(N) \to E$ in this sense).

6.3 `thm:curves:degree-formula` (theorem; S). For nonconstant $\varphi\colon C_1 \to C_2$ and every $Q \in C_2(\bar{k})$:

$$\sum_{P \in \varphi^{-1}(Q)} e_\varphi(P) = \deg\varphi.$$

Sketch: the integral closure B of the DVR $A = \bar{k}[C_2]_Q$ in $\bar{k}(C_1)$ is a semilocal Dedekind domain, finite and free of rank $\deg\varphi$ over A (finiteness: R), whose maximal ideals are the $\mathfrak{m}_P$, $P \in \varphi^{-1}(Q)$, with $t_Q B = \prod \mathfrak{m}_P^{e_\varphi(P)}$; then the argument of Chapter 6, item 3.1 ($\sum e_i f_i = n$) applies with all residue degrees $f_i = 1$ because the residue fields are $\bar{k}$. Physics aside: the number of sheets of a branched cover, counted with multiplicity, does not jump. Reference: SilvermanAEC II.2 (Proposition 2.6); Hartshorne II.6 (Proposition 6.9). Paper: background (Section 1).

6.4 `cor:curves:degree-three-ways` (corollary; P given 6.3, except the "all but finitely many" in (ii), S). Let $\varphi\colon C_1 \to C_2$ be nonconstant. (i) $\deg\varphi = [\bar{k}(C_1) : \varphi^*\bar{k}(C_2)]$ (definition). (ii) $\#\varphi^{-1}(Q) = \deg_s\varphi$ for all but finitely many $Q \in C_2(\bar{k})$; in characteristic 0, or for separable φ, $\#\varphi^{-1}(Q) = \deg\varphi$ except over the finitely many branch points (7.1). (iii) $\deg\varphi^*D = \deg\varphi \cdot \deg D$; $\varphi_*\varphi^*D = (\deg\varphi)D$; $\varphi^*\operatorname{div}(f) = \operatorname{div}(\varphi^*f)$; $\varphi_*\operatorname{div}(h) = \operatorname{div}(N_{\bar{k}(C_1)/\varphi^*\bar{k}(C_2)}h)$ (R for the last); hence φ^* and φ_* induce homomorphisms of Pic and Pic^0

with $\varphi_*\varphi^* = [\deg\varphi]$ on $\mathrm{Pic}^0(C_2)$. (iv) (Over $\mathbb{C}$, proved in 8.2 and 10.3.) $\deg\varphi$ is the number of preimages of a general point, and φ^* on $H^2(\cdot,\mathbb{Z}) = \mathbb{Z}$ is multiplication by $\deg\varphi$. (v) $\deg(\psi\circ\varphi) = \deg\psi\deg\varphi$; $\deg\varphi = 1$ iff φ is an isomorphism (3.3). (vi) For nonconstant $f \in \bar{k}(C)$, the degree of $f\colon C \to \mathbb{P}^1$ is $[\bar{k}(C) : \bar{k}(f)]$, the degree of its divisor of zeros, and the degree of its divisor of poles; so $\deg\operatorname{div} f = 0$ (4.1(iii)) and a function without poles is constant (3.1(iv)). Reference: SilvermanAEC II.2, II.3 (cite the sections). Paper: Section 1 and Lemma lem:hecke via Chapter 14: "$\deg\varphi_E$" in all three senses, and the description "$E^\vee \to J_0(N) \to E$ is multiplication by $\deg\varphi_E$" of PLAN Chapter 14, which is (iii) on Pic^0 with $E = \mathrm{Pic}^0(E)$ (5.7).

6.5 `prop:curves:separability-frobenius` (proposition; P for (c) identities, S for (c) degree, R for (b)). (a) In characteristic 0 every nonconstant morphism of curves is separable (Chapter 5: characteristic-0 fields are perfect). (b) In characteristic p, φ is separable iff $\varphi^*\omega \neq 0$ for one (equivalently every) nonzero differential ω on C_2; and every nonconstant φ factors as $C_1 \xrightarrow{\pi_q} C_1^{(q)} \xrightarrow{\lambda} C_2$ with $q = \deg_i\varphi$ and λ separable. (c) For $k = \mathbb{F}_q$ and $C \subseteq \mathbb{P}^n$ defined over $\mathbb{F}_q$, the q-power Frobenius morphism $\pi_q\colon [x_0 : \cdots : x_n] \mapsto [x_0^q : \cdots : x_n^q]$ maps C to itself; $\pi_q^*\bar{k}(C) = \bar{k}(C)^q$, so π_q is purely inseparable of degree q (S) and $e_{\pi_q}(P) = q$ at every point; its fixed points are $C(\mathbb{F}_q)$; on $C(\bar{\mathbb{F}}_q)$ it coincides with the action of the arithmetic Frobenius of $G_{\mathbb{F}_q}$ (C7). Reference: SilvermanAEC II.2 (Proposition 2.11, Corollary 2.12) and II.4 (Proposition 4.2). Paper: background (Chapter 9: the Frobenius endomorphism and $a_p = p + 1 - \#E(\mathbb{F}_p)$; Chapter 14: Eichler-Shimura).

10.2.7 Section 7. Ramification and the Riemann-Hurwitz formula

7.1 `def:curves:ramification` (definition and proposition; P). φ is *unramified* at P if $e_\varphi(P) = 1$, *ramified* otherwise; $\varphi(P)$ is then a *branch point*. Ramification is *tame* at P if $\operatorname{char} k \nmid e_\varphi(P)$, *wild* otherwise. For separable φ only finitely many points are ramified (from the differential formula in 7.2: $\operatorname{ord}_P\varphi^*\omega > e_P \operatorname{ord}_{\varphi P}\omega$ at ramified P, and $\varphi^*\omega$ has finitely many zeros). Over $\mathbb{C}$ the local normal form is $z \mapsto z^e$ (8.2). Reference: SilvermanAEC II.2 (cite the section). Paper: background.

7.2 `thm:curves:riemann-hurwitz` (theorem; P given 5.3(b) and 4.3). Let $\varphi\colon C_1 \to C_2$ be a nonconstant separable morphism of smooth projective curves of genera g_1, g_2. Then

$$2g_1 - 2 \;\geq\; \deg\varphi\,(2g_2 - 2) + \sum_{P\in C_1} (e_\varphi(P) - 1),$$

with equality iff every ramification is tame (always in characteristic 0). Proof: for a nonzero ω on C_2 and $Q = \varphi(P)$, writing $t_Q = ut_P^e$, one gets $\operatorname{ord}_P(\varphi^*\omega) = e\operatorname{ord}_Q(\omega) + \operatorname{ord}_P(dt_Q/dt_P)$ and $\operatorname{ord}_P(dt_Q/dt_P) \geq e-1$, with equality iff $\operatorname{char} k \nmid e$; sum over P using 6.3 and take degrees with 5.3(b). Topological proof over $\mathbb{C}$ (S, in 8.4): triangulate C_2 with the branch points as vertices, lift, and count $V - E + F$. Reference: SilvermanAEC II.5 (Theorem 5.9); Hartshorne IV.2 (Corollary 2.4). Paper: background (Section 1; Chapter 12's genus formula for $X_0(N)$).

7.3 `cor:curves:rh-consequences` (corollary and examples; P). Let $\varphi\colon C_1 \to C_2$ be nonconstant and separable. (a) If $g_2 \geq 1$ then $g_1 \geq g_2$; if $g_1 = 0$ then $g_2 = 0$ (Lueroth's theorem, curve form). (b) If $g_1 = g_2 = 1$ then φ is unramified ($0 \geq \deg\varphi\cdot 0 + \sum(e_P - 1)$). In particular, in characteristic 0 every nonconstant morphism between genus-one curves (every isogeny, Chapter 9) is unramified, and $\#\varphi^{-1}(Q) = \deg\varphi$ for every Q. (c) The double cover $x\colon C \to \mathbb{P}^1$ of the hyperelliptic curve $y^2 = h(x)$ ($\deg h = 2g+1$ or $2g+2$, squarefree, $\operatorname{char} k \neq 2$) is branched at the $2g+2$ points (roots of h, and ∞ when $\deg h$ is odd), so $2g_C - 2 = 2(-2) + (2g+2)$ and $g_C = g$. (d) Fermat: the projection $[X : Y : Z] \mapsto [X : Y]$ of $X^d + Y^d + Z^d = 0$ has degree d, total ramification ($e = d$) over the d roots of $X^d + Y^d = 0$ and none elsewhere, so $2g - 2 = -2d + d(d-1)$, $g = (d-1)(d-2)/2$, agreeing with 5.4. (e) For a nonconstant $\varphi\colon X \to E$ over $\mathbb{C}$ with $g(E) = 1$: $\sum_P(e_\varphi(P)-1) = 2g(X)-2$, whatever $\deg\varphi$. Examples (with $g(X_0(N))$ from Chapter 12 and the degrees from Chapter 14 and the paper): $\varphi_{11a1}\colon X_0(11) \to 11a1$ has degree 1, so it is an isomorphism and $X_0(11)$ is an elliptic curve; $X_0(20) \to 20a1$ likewise; φ_{121a1} has degree 6 and total ramification $2\cdot 6 - 2 = 10$; φ_{539c1} has degree $2^5\cdot 3^2 = 288$ and total ramification 96. (f) Forward pointer: the genus formula of $X_0(N)$ (Chapter 12) is 7.2 for $X_0(N) \to X_0(1) \simeq \mathbb{P}^1$, of degree $[\mathrm{SL}_2(\mathbb{Z}) : \Gamma_0(N)]$, ramified only over $j = 0$, 1728, ∞. Reference: SilvermanAEC II.5 (exercises); Miranda,

Chapter II. Paper: Section 1 (the parametrisations of 11a1, 121a1, 539c1 quoted in the Example of Section 3 and the Appendix).

10.2.8 Section 8. Compact Riemann surfaces and smooth projective complex curves

8.1 def:curves:riemann-surface (definitions and examples; P for the examples). A *Riemann surface* is a connected Hausdorff second-countable topological surface with a maximal atlas whose transition maps are holomorphic; holomorphic and meromorphic functions, holomorphic maps between Riemann surfaces. Examples: (a) $\mathbb{C}$; (b) the Riemann sphere $\mathbb{P}^1(\mathbb{C})$ with charts z and $1/z$; (c) $\mathbb{C}/\Lambda$ for a lattice $\Lambda = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2$ (charts: local inverses of the projection; compact, genus 1; Chapter 9 does the Weierstrass $\wp$-function and the uniformisation of elliptic curves); (d) the smooth points of an affine plane curve $f(x, y) = 0$ (implicit function theorem: near a point with $f_y \neq 0$, x is a chart); (e) hence $C(\mathbb{C})$ for a smooth projective curve C over $\mathbb{C}$ is a compact Riemann surface ($\mathbb{P}^n(\mathbb{C})$ is compact). (f) (Forward pointer, Chapter 12.) $\Gamma_0(N)\backslash\mathbb{H}$ compactified by the cusps, with special charts at the cusps and elliptic points, is the compact Riemann surface $X_0(N)(\mathbb{C})$. Reference: Miranda, Chapters I and II; Forster, Section 1. Paper: background (Section 1: $X_0(N)$).

8.2 thm:curves:holomorphic-maps (theorem; P). Let $F\colon X \to Y$ be a nonconstant holomorphic map of compact Riemann surfaces. (a) Local normal form: for each P there are charts z at P and w at $F(P)$, centred at 0, with $w = z^e$; the integer $e = \mathrm{mult}_P(F) \geq 1$ is independent of the charts (if $w \circ F$ has a zero of order e, it is h^e with h holomorphic, $h'(0) \neq 0$, and h is a chart). (b) F is open and surjective, and its fibres are finite. (c) $d = \sum_{P \in F^{-1}(Q)} \mathrm{mult}_P(F)$ does not depend on $Q \in Y$ (locally constant by (a) and properness; Y connected): the *degree* of F. (d) $\mathrm{mult}_P(F) = 1$ except at finitely many points, so $\#F^{-1}(Q) = d$ except over finitely many Q. (e) F preserves orientation: in local coordinates the real Jacobian determinant is $|F'(z)|^2 \geq 0$. Reference: Miranda, Chapter II (Section 4; cite the section); Forster, Section 2. Paper: background.

8.3 thm:curves:riemann-existence (theorem; R). (a) (Riemann.) On a compact Riemann surface X the meromorphic functions separate points and give local coordinates; the field $\mathcal{M}(X)$ is a function field of one variable over $\mathbb{C}$. (b) (Equivalence; GAGA for curves.) $X \mapsto \mathcal{M}(X)$ and $C \mapsto C(\mathbb{C})$ give an equivalence between compact connected Riemann surfaces with nonconstant holomorphic maps and smooth projective curves over $\mathbb{C}$ with nonconstant morphisms, under which: meromorphic functions are rational functions; the order of a zero is ord_P; $\mathrm{mult}_P(F) = e_F(P)$; the analytic degree of 8.2(c) is $\deg F$ of 6.2; holomorphic differentials are regular algebraic differentials; divisors, linear equivalence and Pic agree. (c) The algebraic genus $g = \ell(K)$ equals the topological genus (8.4). Reference: Miranda, Chapters VI and VII (Riemann existence and the algebraic curve of a Riemann surface); Hartshorne, Appendix B (GAGA); Forster, Sections 14 and 16 (cite the sections). Paper: Section 1 ($X_0(N)$ is constructed analytically in Chapter 12 and is an algebraic curve over $\mathbb{Q}$ in Chapter 14; φ_E is at once a holomorphic map and a morphism, and $\deg \varphi_E$ means the same in both senses).

8.4 thm:curves:topology (theorem; R for (a), (b); S for (c)). (a) A compact Riemann surface is an orientable compact surface, homeomorphic to a sphere with g handles (classification of surfaces); $\chi(X) = 2 - 2g$. (b) This g is the genus of 5.2. (c) Topological Riemann-Hurwitz: for $F\colon X \to Y$ of degree d, $\chi(X) = d\,\chi(Y) - \sum_P (\mathrm{mult}_P F - 1)$ (lift a triangulation of Y having the branch points among its vertices; each face and edge has d lifts, a vertex Q has $\#F^{-1}(Q)$ lifts). Physics aside: Euler's $V - E + F$. Reference: Miranda, Chapters I and II (cite the chapters); Forster, Section 17. Paper: background.

10.2.9 Section 9. Holomorphic differentials, residues, periods

9.1 def:curves:holomorphic-differential (definitions; P). A meromorphic (holomorphic) 1-form on X: $f(z)\,dz$ in each chart, with f meromorphic (holomorphic), transforming by $f(z)\,dz = f(z(w))z'(w)\,dw$; $\mathrm{ord}_P\,\omega$, $\mathrm{div}\,\omega$; the residue $\mathrm{res}_P\,\omega$ (coefficient of z^{-1}, chart-independent); $H^0(X, \Omega^1)$, the space of holomorphic 1-forms. Smooth real and complex 1- and 2-forms, d, $\wedge$, $\int_X$ of 2-forms; Stokes' theorem on a compact surface with boundary (from the background real analysis, R). Line integrals $\int_\gamma \omega$; a holomorphic 1-form is

closed ($d(f\,dz) = \bar\partial f \wedge dz = 0$). Examples: dz on $\mathbb{C}/\Lambda$; $\mathbb{P}^1$ has none ($dz = -w^{-2}dw$). Reference: Miranda, Chapter IV. Paper: background (Chapter 12: $f \mapsto f(z)\,dz$ identifies $S_2(\Gamma_0(N))$ with $H^0(X_0(N), \Omega^1)$).

9.2 `thm:curves:residue` (theorem; P). For a meromorphic 1-form ω on a compact Riemann surface, $\sum_{P\in X} \operatorname{res}_P \omega = 0$ (Stokes on X minus small discs around the poles). Corollary: for $f \in \mathcal{M}(X)^\times$, $\deg \operatorname{div} f = \sum_P \operatorname{res}_P(df/f) = 0$ (argument principle); a second proof of 4.1(iii) over $\mathbb{C}$. Physics aside: the total charge on a closed surface is zero. Reference: Miranda, Chapter IV (cite the chapter). Paper: background.

9.3 `thm:curves:genus-dimension` (theorem; R via 8.3 and 5.3). $\dim_{\mathbb{C}} H^0(X, \Omega^1) = g$ and $\deg \operatorname{div} \omega = 2g-2$ for every meromorphic $\omega \neq 0$. Examples: $H^0(\mathbb{C}/\Lambda, \Omega^1) = \mathbb{C}\,dz$; $H^0(\mathbb{P}^1, \Omega^1) = 0$. Forward pointer: $\dim S_2(\Gamma_0(N)) = g(X_0(N))$ (Chapter 12), e.g. $\dim S_2(\Gamma_0(121)) = 6$. Reference: Miranda, Chapter VII. Paper: background (multiplicity one and the genus, Chapter 12).

9.4 `thm:curves:bilinear-relations` (theorem; P for (a), (b); S for (c), (d)). Let X have genus $g \geq 1$ and let $a_1, \dots, a_g, b_1, \dots, b_g$ be a symplectic basis of $H_1(X, \mathbb{Z})$ (10.1). (a) For $\omega, \eta \in H^0(X, \Omega^1)$: $\int_X \omega \wedge \eta = 0$ ($dz \wedge dz = 0$). (b) For $\omega \neq 0$ holomorphic: $\frac{i}{2}\int_X \omega \wedge \bar\omega > 0$ (locally $\frac{i}{2}|f|^2 dz \wedge d\bar z = |f|^2 dx \wedge dy$). (c) For closed 1-forms α, β:

$$\int_X \alpha \wedge \beta = \sum_{j=1}^{g} \Big(\int_{a_j} \alpha \int_{b_j} \beta - \int_{b_j} \alpha \int_{a_j} \beta \Big)$$

(S: cut X along the basis to a $4g$-gon and apply Stokes; checked on the torus, 10.4, where it gives $\int dx \wedge dy = 1$). (d) Riemann's bilinear relations, with periods $A_j(\omega) = \int_{a_j} \omega$, $B_j(\omega) = \int_{b_j} \omega$: $\sum_j (A_j(\omega)B_j(\eta) - B_j(\omega)A_j(\eta)) = 0$ and $\sum_j \operatorname{Im}\big(\overline{A_j(\omega)} B_j(\omega)\big) > 0$ for $\omega \neq 0$ (from (a) to (c); signs checked on $\mathbb{C}/(\mathbb{Z} + \tau\mathbb{Z})$ with $\omega = dz$, $A = 1$, $B = \tau$, where the inequality is $\operatorname{Im}\tau > 0$). (e) Consequence: a holomorphic form with all A-periods zero is zero, so the period map $H_1(X, \mathbb{Z}) \to H^0(X, \Omega^1)^\vee$ is injective with image a lattice (11.2). Physics aside: (b) is a norm $\int |f|^2$; the Petersson product of Chapter 12 is $\frac{i}{2}\int \omega \wedge \bar\eta$ up to normalisation, and the area of a period parallelogram is (b) for dz. Reference: Miranda, Chapter VIII (cite the chapter); Griffiths-Harris, Chapter 2, Section 2 (cite the section). Paper: background (Section 1: the Petersson norm and the area of the period lattice in the Zagier-Watkins formula, Chapter 14).

10.2.10 Section 10. H_1, H^1 and the cup product

10.1 `def:curves:homology` (definitions and theorem; R, with the torus and the picture of the g-holed surface). $H_1(X, \mathbb{Z})$: the abelianisation of $\pi_1(X)$, equivalently singular homology (closed oriented loops modulo boundaries, for the reader). (i) $H_1(X, \mathbb{Z})$ is free of rank $2g$, with a *symplectic basis* $a_1, \dots, a_g, b_1, \dots, b_g$ (the a_j around and the b_j through the j-th handle). (ii) The intersection pairing $H_1 \times H_1 \to \mathbb{Z}$ (signed count of transverse crossings, C6) is bilinear, alternating and perfect, with $a_i \cdot b_j = \delta_{ij}$, $a_i \cdot a_j = b_i \cdot b_j = 0$. (iii) $H_1(X, A) := H_1(X, \mathbb{Z}) \otimes A$ for $A = \mathbb{Z}/n, \mathbb{Z}_\ell, \mathbb{R}, \mathbb{C}$. Physics aside: the intersection number is a signed crossing count, like a winding or linking number. Reference: Miranda, Chapter VIII (Section 1); Forster, Section 21; Farkas-Kra III.1 (cite the sections). Paper: background (Chapter 14: $H_1(X_0(N), \mathbb{Z})$ and $J_0(N)(\mathbb{C}) = S_2^\vee / H_1$).

10.2 `def:curves:cohomology` (definitions and proposition; P for (i), R for (ii), S for (iii), R for (iv)). (i) $H^1(X, \mathbb{Z}) := \operatorname{Hom}(H_1(X, \mathbb{Z}), \mathbb{Z})$, free of rank $2g$; $H^1(X, A) := \operatorname{Hom}(H_1(X, \mathbb{Z}), A) = H^1(X, \mathbb{Z}) \otimes A$ for $A = \mathbb{Z}/n, \mathbb{Z}_\ell, \mathbb{R}, \mathbb{C}$ (since H_1 is free of finite rank). (This agrees with singular cohomology by the universal coefficient theorem, H_0 being free; R.) (ii) (de Rham.) $H^1_{\mathrm{dR}}(X) = \{\text{closed smooth 1-forms}\}/d\{\text{smooth functions}\} \simeq H^1(X, \mathbb{R})$ via $\alpha \mapsto (\gamma \mapsto \int_\gamma \alpha)$; under it $H^1(X, \mathbb{Z})$ is the lattice of classes with integral periods. $H^2(X, \mathbb{Z}) \simeq \mathbb{Z}$ via $\int_X$ (orientation, C6). (iii) (Hodge.) $H^1(X, \mathbb{C}) = H^0(X, \Omega^1) \oplus \overline{H^0(X, \Omega^1)}$ (S: the map is injective by 9.4(b) and both sides have dimension $2g$ by 9.3). (iv) For $g \geq 1$, X is a $K(\pi, 1)$ (its universal cover is $\mathbb{C}$ or the disc), so $H^1(X, M) = H^1(\pi_1(X), M)$ in the sense of Chapter 16, and the cup products agree (R); this is the bridge to the group-cohomological H^1 of $\Gamma_0(N)$ used by DFG04 (Chapter 14). Reference: Miranda, Chapter VIII; Forster, Sections 12 and 19 (de Rham, Dolbeault); Griffiths-Harris, Chapter 0 (cite the sections). Paper: Section 4.1 ("$H^1(X_0(M^\Sigma), \mathbb{Z}_\ell)$"), Lemma lem:hecke ($L = H^1(X_0(N), \mathbb{Z}_\ell)_{\mathfrak{m}}$), Theorem thm:quant Step 2.

10.3 thm:curves:cup-product (theorem; R for perfectness in general (Poincare duality), P for the rest given 10.1 and 9.4(c)). The cup product $\langle\ ,\ \rangle\colon H^1(X,\mathbb{Z})\times H^1(X,\mathbb{Z})\to H^2(X,\mathbb{Z})=\mathbb{Z}$, $\langle\alpha,\beta\rangle=\int_X\alpha\wedge\beta$, is bilinear, alternating ($\langle\alpha,\alpha\rangle=0$, so $\langle\alpha,\beta\rangle=-\langle\beta,\alpha\rangle$) and perfect: if (α_j,β_j) is the basis dual to a symplectic basis (a_j,b_j) ($\alpha_i(a_j)=\delta_{ij}$, $\alpha_i(b_j)=0$, $\beta_i(b_j)=\delta_{ij}$, $\beta_i(a_j)=0$), then by 9.4(c) $\langle\alpha_i,\beta_j\rangle=\delta_{ij}$ and $\langle\alpha_i,\alpha_j\rangle=\langle\beta_i,\beta_j\rangle=0$: the Gram matrix is $\left(\begin{smallmatrix}0 & I_g\\ -I_g & 0\end{smallmatrix}\right)$, of determinant 1. The map $\mathrm{PD}\colon H^1(X,\mathbb{Z})\to H_1(X,\mathbb{Z})$ defined by $\int_{\mathrm{PD}(\alpha)}\beta=\langle\alpha,\beta\rangle$ is an isomorphism (Poincare duality) and an isometry: $\langle\alpha,\beta\rangle=\mathrm{PD}(\alpha)\cdot\mathrm{PD}(\beta)$ ($\mathrm{PD}(\alpha_j)=b_j$, $\mathrm{PD}(\beta_j)=-a_j$). Base change: for any commutative ring R (in particular $\mathbb{Z}_\ell$, $\mathbb{F}_\ell$, $\mathbb{Q}_\ell$, $\mathbb{C}$) the extended pairing on $H^1(X,R)$ is perfect and alternating (Chapter 3, 9.1(iii)). (Over $\mathbb{R}$ nondegeneracy also follows from 9.4(b) and 10.2(iii).) Reference: Miranda, Chapter VIII; Griffiths-Harris, Chapter 0, Section 4 (cite the section); Hatcher, Section 3.3 (for Poincare duality in general; not in refs.bib). Paper: Section 4.1 ("perfect and alternating on $H^1(X_0(M^\Sigma),\mathbb{Z}_\ell)$"), Lemma lem:hecke and Lemma lem:ddt417 (the module L "with a perfect $\mathcal{O}$-bilinear pairing"), Theorem thm:quant Step 2, PLAN Section 6 row "$H^1(X_0(N),\mathbb{Z}_\ell)$, cup product".

10.4 ex:curves:torus (example; P). $X=\mathbb{C}/\Lambda$, $\Lambda=\mathbb{Z}\omega_1+\mathbb{Z}\omega_2$, $\mathrm{Im}(\omega_2/\omega_1)>0$; real coordinates $z=s\omega_1+t\omega_2$, $s,t\in\mathbb{R}/\mathbb{Z}$. $H_1=\Lambda$ with $a=[0,\omega_1]$, $b=[0,\omega_2]$, $a\cdot b=+1$. $H^1(X,\mathbb{Z})$ has basis ds, dt (dual to a, b), $\langle ds,dt\rangle=\int ds\wedge dt=1$, and $\mathrm{PD}(ds)=b$, $\mathrm{PD}(dt)=-a$. The holomorphic form $dz=\omega_1 ds+\omega_2 dt$ has periods (ω_1,ω_2), and $\frac{i}{2}\int dz\wedge d\bar z=\mathrm{Im}(\bar\omega_1\omega_2)$ is the area of the period parallelogram (9.4(b), (d)). $[-1]\colon z\mapsto -z$ acts by -1 on H_1 and H^1; multiplication by an integer m acts by m on H_1 and H^1 and by m^2 on H^2, so has degree m^2 (6.4(iv)). Physics aside: Λ is a crystal lattice, $H^1(X,\mathbb{Z})=\Lambda^\vee$ its reciprocal lattice, $ds\wedge dt$ the normalised volume form of the periodic box. Paper: background (genus one: $E(\mathbb{C})$, Chapter 9; Section 1, the period lattice Λ_E).

10.2.11 Section 11. The Jacobian: Pic^0, complex torus, abelian variety; torsion and Tate modules

11.1 def:curves:jacobian (definition; P). For a smooth projective curve C over k, $J(C)(\bar k):=\mathrm{Pic}^0(C)$, with its G_k-action (4.2). Examples: $J(\mathbb{P}^1)=0$; for $g=1$ and $O\in C(k)$, $J(C)\simeq C$ via 5.7. Reference: SilvermanAEC II.3; Milne, Jacobian varieties, Section 1. Paper: Remark rem:free ($J_0(N)$), Chapter 14.

11.2 thm:curves:abel-jacobi (theorem; P for (i), (ii), (iv); R for (iii)). Let X be a compact Riemann surface of genus $g\geq 1$. (i) The period map $H_1(X,\mathbb{Z})\to H^0(X,\Omega^1)^\vee$, $\gamma\mapsto(\omega\mapsto\int_\gamma\omega)$, is injective and its image Λ_X is a lattice (9.4(e) and 10.2(iii): a class pairing to zero with $H^0(\Omega^1)$ and its conjugate is zero in $H_1(X,\mathbb{R})$). So $J_{\mathrm{an}}(X)=H^0(X,\Omega^1)^\vee/\Lambda_X$ is a complex torus of dimension g. (ii) The *Abel-Jacobi map* $u\colon \mathrm{Div}^0(X)\to J_{\mathrm{an}}(X)$, $\sum_i((P_i)-(Q_i))\mapsto\left(\omega\mapsto\sum_i\int_{Q_i}^{P_i}\omega\right)$ mod Λ_X, is a well-defined homomorphism (changing paths changes the integral by a period). (iii) Abel's theorem: $\ker u=\mathrm{Prin}(X)$. Jacobi inversion: u is surjective. Hence $u\colon \mathrm{Pic}^0(X)\xrightarrow{\sim}J_{\mathrm{an}}(X)$ (R; the "only if" half of Abel is a residue computation, S optional). (iv) For a base point P_0, $\iota\colon X\to J_{\mathrm{an}}(X)$, $P\mapsto u((P)-(P_0))$, is holomorphic and injective (if $\iota(P)=\iota(Q)$, $P\neq Q$, then $(P)\sim(Q)$ and $g=0$ by 5.3(d)), and $\iota_*\colon H_1(X,\mathbb{Z})\to H_1(J_{\mathrm{an}},\mathbb{Z})=\Lambda_X$ is the period isomorphism of (i) (S). For $g=1$: $X\simeq\mathbb{C}/\Lambda_X$, the analytic form of 5.7 (Chapter 9 makes it explicit with $\wp$). Physics aside: Λ_X is the lattice of periods, as in a crystal. Reference: Miranda, Chapter VIII (Abel's theorem, Jacobi inversion); Forster, Sections 20 and 21; Griffiths-Harris, Chapter 2, Section 2 (cite the sections). Paper: Remark rem:free, Chapter 14 ($J_0(N)(\mathbb{C})=S_2(\Gamma_0(N))^\vee/H_1(X_0(N),\mathbb{Z})$ via $f\mapsto f(z)dz$), Section 1 (period lattice Λ_E).

11.3 thm:curves:jacobian-variety (theorem; R). Let C be a smooth projective curve of genus $g\geq 1$ over k. (i) There is an abelian variety $J=\mathrm{Jac}(C)$ over k of dimension g (a smooth projective connected group variety) with $J(\bar k)=\mathrm{Pic}^0(C)$ compatibly with G_k; $J(L)\supseteq\mathrm{Pic}^0_L(C)$ with equality when $C(L)\neq\emptyset$ (4.2(iii)). (ii) Over $\mathbb{C}$, $J(\mathbb{C})\simeq J_{\mathrm{an}}(C(\mathbb{C}))$ as complex Lie groups; the torus is algebraic because the cup product gives a Riemann form (positivity 9.4) (Lefschetz, Riemann). (iii) Universal property (Albanese): for $P_0\in C(k)$, $\iota\colon C\to J$, $P\mapsto[(P)-(P_0)]$, is a closed immersion for $g\geq 1$, and every morphism $C\to A$ to an abelian variety over k with $P_0\mapsto 0$ factors uniquely as $h\circ\iota$ with $h\colon J\to A$ a homomorphism. (iv) Mordell-Weil: $J(K)$ is finitely generated for a number field K; in particular $J(K)_{\mathrm{tors}}$ is finite. Reference: Milne, Jacobian varieties (Sections 1 to 6; cite the sections); Milne, Abelian varieties; Mumford, Abelian varieties (Section 24 for the Weil pairing); BLR, Chapter 9 (the Picard functor). Paper:

Section 1 (φ_E factors as $X_0(N) \xrightarrow{\iota} J_0(N) \to E$, Chapter 14), Remark rem:free, Conjecture conj:eis and Open question 4 (torsion; cuspidal subgroup, Chapter 14).

11.4 `prop:curves:jacobian-functoriality` (proposition; P for (i), (ii), (iv); S for (iii)). Let $\varphi\colon X \to Y$ be a nonconstant morphism of smooth projective curves. (i) $\varphi_*\colon J(X) \to J(Y)$ ("Albanese", pushforward of divisor classes) and $\varphi^*\colon J(Y) \to J(X)$ ("Picard", pullback) are homomorphisms, defined over k if φ is, with $\varphi_*\varphi^* = [\deg\varphi]$ on $J(Y)$ (6.4(iii)). (ii) Over $\mathbb{C}$, φ_* is induced on $H^0(X,\Omega^1)^\vee/\Lambda_X \to H^0(Y,\Omega^1)^\vee/\Lambda_Y$ by the transpose of $\varphi^*\colon H^0(Y,\Omega^1) \to H^0(X,\Omega^1)$ and on lattices by $\varphi_*\colon H_1(X,\mathbb{Z}) \to H_1(Y,\mathbb{Z})$ (change of variables $\int_\gamma \varphi^*\omega = \int_{\varphi_*\gamma}\omega$). (iii) φ^* on J is induced by the trace of differentials and by the transfer $\varphi^!\colon H_1(Y) \to H_1(X)$ (lift a loop avoiding the branch points to its $\deg\varphi$ lifts; 12.2). (iv) For $Y = E$ of genus 1 with origin $O = \varphi(P_0)$ and $J(E) = E$ (5.7), $\varphi_* \circ \iota_X = \varphi$: the morphism φ factors through $J(X)$ (the universal property 11.3(iii), made explicit), and $\varphi_* \circ \varphi^*\colon E \to J(X) \to E$ is multiplication by $\deg\varphi$. Reference: Milne, Jacobian varieties, Section 6 (cite the section); Miranda, Chapter VIII. Paper: Section 1 (the modular parametrisation and its degree; PLAN Chapter 14 "the composition $E^\vee \to J_0(N) \to E$ being multiplication by $\deg\varphi_E$" is (iv) with $E^\vee = E$).

11.5 `prop:curves:degree-cup-product` (proposition; P, with the identification $H_1(X) = H_1(J(X))$ of 11.2(iv) used as S). Let $\varphi\colon X \to Y$ be a nonconstant holomorphic map of compact Riemann surfaces, $d = \deg\varphi$. (i) $\langle\varphi^*x, \varphi^*y\rangle_X = d\,\langle x,y\rangle_Y$ for $x,y \in H^1(Y,\mathbb{Z})$ ($\int_X \varphi^*(\alpha\wedge\beta) = d\int_Y \alpha\wedge\beta$: away from the finitely many branch points φ is a d-sheeted orientation-preserving covering, 8.2). (ii) $\varphi_*\varphi^* = d$ on $H^1(Y,\mathbb{Z})$ and $\varphi_*\varphi^! = d$ on $H_1(Y,\mathbb{Z})$ (12.2 for φ_* on H^1 and $\varphi^!$). (iii) The following are equivalent: (a) $\varphi^*H^1(Y,\mathbb{Z})$ is saturated in $H^1(X,\mathbb{Z})$; (b) $\varphi_*\colon H_1(X,\mathbb{Z}) \to H_1(Y,\mathbb{Z})$ is surjective; (c) the homomorphism $\varphi_*\colon J(X) \to J(Y)$ has connected kernel; (d) $\varphi^*\colon J(Y) \to J(X)$ is injective. Proof of (b)⇔(c): for a surjective homomorphism of complex tori $V/\Lambda \to W/M$ induced by $A\colon V \to W$, the group of components of the kernel is $A^{-1}(M)/(\ker A + \Lambda) \simeq M/A(\Lambda)$, the cokernel on H_1. (a)⇔(b): φ^* on H^1 is the transpose of φ_* on H_1; φ_* has finite cokernel (by (ii)), so its transpose is injective with cokernel whose torsion is $\mathrm{Hom}(\mathrm{coker}\,\varphi_*, \mathbb{Q}/\mathbb{Z})$; it is torsion-free iff φ_* is surjective (elementary divisors, Chapter 3, 5.3). (c)⇔(d): the torus-map form of (a) (S). (iv) If $Y = E$ has genus 1 and (iii) holds, then $M = \varphi^*H^1(E,\mathbb{Z})$ is a saturated sublattice of rank 2 on which the cup product has Gram determinant d^2 and ideal of values $d\mathbb{Z}$ (by (i) and 10.3); by Chapter 3, 9.3, $[H^1(X,\mathbb{Z}) : M \oplus M^\perp] = d^2$; the same holds over $\mathbb{Z}_\ell$ with d replaced by $\ell^{v_\ell(d)}$. If (iii) fails, the index of M in its saturation is the order of the component group of $\ker(J(X) \to E)$. (v) Example: for an isogeny $\psi\colon E \to E'$ of degree n and $\varphi' = \psi\circ\varphi$, $(\varphi')^*H^1(E',\mathbb{Z})$ has index n in $\varphi^*H^1(E,\mathbb{Z})$ (on H_1, ψ_* is the inclusion of Λ in Λ', of index $n = \#\ker\psi$), so φ' never satisfies (iii) when $n > 1$ and (iii) holds for φ. Reference: elementary given 10.3 and 11.2; compare ARS12, Section 3 (cite the section) and Chapter 14. Paper: Section 1 (the modular degree, "minimal degree" of the optimal parametrisation), Section 4.1 and Lemma lem:hecke (for E optimal, Chapter 14 identifies $M_{f,\lambda} = \varphi_E^*H^1(E,\mathbb{Z}_\ell)$, and then (iv) computes the ideal $\eta^{\emptyset}_{f,\lambda}$: Part 7, point 1), Remark rem:free, important_notes.md point 26 (Part 7, point 2).

11.6 `prop:curves:torsion` (proposition; P over $\mathbb{C}$, R over k). (i) For a complex torus V/Λ of dimension g and $n \geq 1$: $(V/\Lambda)[n] = \frac{1}{n}\Lambda/\Lambda \simeq \Lambda/n\Lambda \simeq (\mathbb{Z}/n)^{2g}$ (via $x \mapsto nx$). Hence $J(X)[n] \simeq H_1(X,\mathbb{Z}/n)$ for a compact Riemann surface X, and $[n]\colon J \to J$ has degree n^{2g}. (ii) For C over k with $\mathrm{char}\,k \nmid n$: $J[n](\bar{k}) \simeq (\mathbb{Z}/n)^{2g}$ as groups, with a continuous action of G_k (R; Milne, Abelian varieties, Section 8). (iii) For a commutative ring $\mathbb{T}$ of endomorphisms of J and an ideal $\mathfrak{a} \subseteq \mathbb{T}$, $J[\mathfrak{a}] = \{P : tP = 0\ \forall t \in \mathfrak{a}\}$ (definition); $J[\ell]$ is a $\mathbb{T}/\ell\mathbb{T}$-module, and for a maximal ideal $\mathfrak{m} \ni \ell$, $J[\mathfrak{m}]$ is a $\mathbb{T}/\mathfrak{m}$-vector space (P). Paper: Remark rem:free ("$J_0(N)[\mathfrak{m}]$ is two-dimensional", "injects into $J_1(N)[\mathfrak{m}']$"), Conjecture conj:eis (torsion), Chapter 14.

11.7 `def:curves:tate-module` (definition and proposition; P over $\mathbb{C}$, R over k). For a prime ℓ: $T_\ell J = \varprojlim_n J[\ell^n]$ (transition maps multiplication by ℓ), $V_\ell J = T_\ell J \otimes \mathbb{Q}_\ell$. (i) Over $\mathbb{C}$, the isomorphisms $J[\ell^n] = \ell^{-n}\Lambda_X/\Lambda_X \to \Lambda_X/\ell^n\Lambda_X$, $x \mapsto \ell^n x$, are compatible with the transition maps and give a canonical isomorphism $T_\ell J \simeq \Lambda_X \otimes \mathbb{Z}_\ell = H_1(X,\mathbb{Z}) \otimes \mathbb{Z}_\ell = H_1(X,\mathbb{Z}_\ell)$; so $T_\ell J$ is free of rank $2g$ and $T_\ell J/\ell^n T_\ell J = J[\ell^n]$. (ii) **Duality**: $H^1(X,\mathbb{Z}_\ell) = \mathrm{Hom}(H_1(X,\mathbb{Z}),\mathbb{Z}_\ell) \simeq \mathrm{Hom}_{\mathbb{Z}_\ell}(T_\ell J,\mathbb{Z}_\ell)$ canonically: H^1 is the $\mathbb{Z}_\ell$-dual of the Tate module. (iii) Composing with Poincare duality (10.3), $H^1(X,\mathbb{Z}_\ell) \simeq H_1(X,\mathbb{Z}_\ell) \simeq T_\ell J$, an isomorphism of $\mathbb{Z}_\ell$-modules which depends on the orientation convention and is **not** Galois-equivariant when X is defined over a number field (11.8(c)), and which intertwines a correspondence acting by $\mathcal{C}^*$ on H^1 with

its transpose acting on $T_\ell J$ (12.3(c)). (iv) Over k with $\ell \neq \operatorname{char} k$: $T_\ell J$ is free of rank $2g$ over $\mathbb{Z}_\ell$ with a continuous G_k-action (R). For $g = 1$: $T_\ell J(E) = T_\ell E$ (Chapter 9). Reference: Milne, Abelian varieties, Section 15 (cite the section); Mumford, Abelian varieties, Section 18 (cite the section). Paper: Remark rem:free ("the Tate module is free of rank two"; "$H^1(X_0(N), \mathbb{Z}_\ell)_{\mathfrak{m}}$"), Section 4.2 ($M_{f,\lambda}$ versus $T_\ell E$, important_notes.md point 26), PLAN hook "Tate modules of Jacobians".

11.8 `thm:curves:weil-pairing-etale` (theorem; R). Let C be a smooth projective curve of genus g over k, J its Jacobian, $\ell \neq \operatorname{char} k$. (a) (Weil pairing.) There is a perfect, alternating, G_k-equivariant $\mathbb{Z}_\ell$-bilinear pairing $e_\ell \colon T_\ell J \times T_\ell J \to \mathbb{Z}_\ell(1)$ (from the canonical principal polarisation); over $\mathbb{C}$, after identifying $\mathbb{Z}_\ell(1)$ with $\mathbb{Z}_\ell$ by $\zeta_{\ell^n} = e^{2\pi i/\ell^n}$, it is the intersection pairing of $H_1(X, \mathbb{Z}_\ell)$ up to a universal sign. For $g = 1$ it is the Weil pairing of Chapter 9, and $\det \rho_{E,\ell} = \chi_\ell$ follows. (b) (Etale cohomology.) $H^1_{\mathrm{et}}(C_{\bar{k}}, \mathbb{Z}_\ell)$ is free of rank $2g$ with a continuous G_k-action, and $H^1_{\mathrm{et}}(C_{\bar{k}}, \mathbb{Z}_\ell) \simeq \operatorname{Hom}_{\mathbb{Z}_\ell}(T_\ell J, \mathbb{Z}_\ell)$ as G_k-modules (contragredient action, Chapter 4, C2); for $k \subseteq \mathbb{C}$ there is a comparison isomorphism $H^1_{\mathrm{et}}(C_{\bar{k}}, \mathbb{Z}_\ell) \simeq H^1(C(\mathbb{C}), \mathbb{Z}) \otimes \mathbb{Z}_\ell$ compatible with 11.7(ii) and with cup products (Artin). (c) Combining: $H^1_{\mathrm{et}}(C_{\bar{k}}, \mathbb{Z}_\ell) \simeq T_\ell J(-1) = T_\ell J \otimes \mathbb{Z}_\ell(-1)$ as G_k-modules, and the cup product $H^1 \times H^1 \to H^2_{\mathrm{et}} = \mathbb{Z}_\ell(-1)$ is perfect, alternating and G_k-equivariant. For $g = 1$: $H^1(E, \mathbb{Z}_\ell) \simeq (T_\ell E)^\vee \simeq T_\ell E(-1)$, with determinant χ_ℓ^{-1}. (d) (Weil.) For $k = \mathbb{F}_q$: the arithmetic Frobenius acts on $T_\ell J$ as the q-power Frobenius endomorphism π of J; $P(T) = \det(T - \pi \mid T_\ell J)$ has integer coefficients independent of ℓ, degree $2g$, roots α_i with $|\alpha_i| = \sqrt{q}$, and $\#C(\mathbb{F}_{q^n}) = q^n + 1 - \sum_i \alpha_i^n$; on $H^1 = (T_\ell J)^\vee$ the geometric Frobenius has the same eigenvalues α_i. For $g = 1$: $a_q = q + 1 - \#E(\mathbb{F}_q) = \operatorname{tr}(\text{arithmetic Frobenius} \mid T_\ell E) = \operatorname{tr}(\text{geometric Frobenius} \mid H^1(E))$. Reference: Milne, Jacobian varieties and Abelian varieties (Sections 16 and 19, cite the sections); Mumford, Abelian varieties, Section 20 (Weil pairing; cite the section); Milne, Etale cohomology (lecture notes; not in refs.bib), the comparison theorem; SilvermanAEC III.8 and V.2 for $g = 1$. Paper: Section 4.2 and important_notes.md point 26 ($M_{f,\lambda} \subset H^1$ is homothetic to the dual of $T_\ell E$, that is to $T_\ell E(-1)$, not to $T_\ell E$; the adjoints agree), Section 1.1 ($V = V_\ell E$), Chapter 6 C2/3.7 and important_notes.md point 11 and B1 (which Frobenius has which eigenvalues), Chapter 14 (Eichler-Shimura), Chapter 15 ($\rho_{f,\lambda}$ via T_ℓ of J_f, Shimura's abelian variety, or via H^1; A_f is the paper's adjoint, Appendix B, B-C12(c)).

10.2.12 Section 12. Correspondences, their actions, and adjoints

12.1 `def:curves:correspondence` (definitions and examples; P). A *correspondence* $\mathcal{C}$ from X to Y (smooth projective curves over k, or compact Riemann surfaces) is a triple (C, α, β) with C a finite disjoint union of smooth projective curves and $\alpha \colon C \to X$, $\beta \colon C \to Y$ morphisms nonconstant on every component; its *transpose* is $\mathcal{C}^t = (C, \beta, \alpha)$, from Y to X; sums are disjoint unions. Examples: (a) the graph $\Gamma_\varphi = (X, \mathrm{id}, \varphi)$ of a nonconstant $\varphi \colon X \to Y$, and its transpose $\Gamma_\varphi^t = (X, \varphi, \mathrm{id})$; (b) an automorphism w of X gives Γ_w, with Γ_w^t acting as $\Gamma_{w^{-1}}$ (12.2); (c) (forward pointer, Chapter 14) the Hecke correspondence $T_p = (X_0(Np), \alpha_1, \alpha_p)$ on $X_0(N)$ for a prime $p \nmid N$, with $\alpha_1(z) = z$, $\alpha_p(z) = pz$ on $\mathbb{H}$, and U_p similarly for $p \mid N$. Remark (R): the image of (α, β) is a divisor on $X \times Y$; correspondences up to the obvious equivalence form a ring under composition (fibre products); this chapter does not need the composition, only the actions below, because the Hecke algebra is defined as the ring of operators they generate (Chapter 13). Reference: Shimura71, Chapter 7 (cite the chapter); DiamondShurman, Section 6.3 (cite the section); Milne, Abelian varieties (correspondences); Fulton, Intersection theory (not in refs.bib). Paper: Section 4.1 (the Hecke operators T_n, U_p and w_{M^Σ} acting on $H^1(X_0(M^\Sigma), \mathbb{Z}_\ell)$), Lemma lem:hecke, Theorem thm:quant Steps 1 and 2.

12.2 `def:curves:correspondence-actions` (definitions and proposition; P for the definitions and (a), (d); S for (b), (c)). Let $\mathcal{C} = (C, \alpha, \beta)$ be a correspondence from X to Y. (i) On divisors: $\mathcal{C}_* = \beta_* \alpha^* \colon \operatorname{Div}(X) \to \operatorname{Div}(Y)$; it multiplies degrees by $\deg \alpha$, maps principal divisors to principal divisors (6.4(iii)), hence induces $\mathcal{C}_* \colon J(X) \to J(Y)$. (ii) On homology: the transfer $\alpha^! \colon H_1(X, \mathbb{Z}) \to H_1(C, \mathbb{Z})$ (a loop avoiding the branch points of α is replaced by the sum of its $\deg \alpha$ lifts; well defined, S) and $\mathcal{C}_* = \beta_* \alpha^! \colon H_1(X, \mathbb{Z}) \to H_1(Y, \mathbb{Z})$. (iii) On cohomology: $\alpha_* \colon H^1(C, \mathbb{Z}) \to H^1(X, \mathbb{Z})$ is the transpose of $\alpha^!$ (on de Rham classes, the trace: $(\alpha_* \eta)(x) = \sum_{c \in \alpha^{-1}(x)} \eta_c$ away from branch points; S), and $\mathcal{C}^* = \alpha_* \beta^* \colon H^1(Y, \mathbb{Z}) \to H^1(X, \mathbb{Z})$. (iv) On holomorphic differentials: $\mathcal{C}^* \omega = \alpha_* \beta^* \omega$ (the trace of a holomorphic form extends holomorphically over the branch points: P by Riemann's removable singularity theorem). Compatibilities: (a) $\langle \mathcal{C}^* \xi, \gamma \rangle = \langle \xi, \mathcal{C}_* \gamma \rangle$ for $\xi \in H^1(Y)$, $\gamma \in H_1(X)$ (the evaluation pairing): $\mathcal{C}^*$

on H^1 is the transpose of $\mathcal{C}_*$ on H_1 (P from the definitions). (b) Under Abel-Jacobi (11.2), $\mathcal{C}_*$ on $J(X)$ is induced by the transpose of $\mathcal{C}^*$ on holomorphic differentials and by $\mathcal{C}_*$ on H_1 (S, from 11.4(ii), (iii)); hence on $T_\ell J(X) \simeq H_1(X,\mathbb{Z}_\ell)$ (11.7(i)) a correspondence acts by $\mathcal{C}_*$, and on $H^1(X,\mathbb{Z}_\ell) = \mathrm{Hom}(T_\ell J,\mathbb{Z}_\ell)$ by the transpose. (c) For a graph: $\Gamma_\varphi^* = \varphi^*$ on H^1 and on differentials, $(\Gamma_\varphi)_* = \varphi_*$ on H_1 and on J (Albanese), $(\Gamma_\varphi^t)_* = \varphi^*$ on J (Picard) and $(\Gamma_\varphi^t)^* = \varphi_*$ on H^1. (d) Linearity in $\mathcal{C}$, and functoriality for composition with graphs: $(\Gamma_u \circ \mathcal{C} \circ \Gamma_v)$ acts as $u_*\mathcal{C}_*v_*$ on H_1 for automorphisms u, v (P). Reference: as in 12.1; Milne, Abelian varieties (cite the section on correspondences). Paper: Section 4.1, Remark rem:free (the Hecke action on $J_0(N)$ and on $T_\ell J_0(N)$ versus on $H^1(X_0(N),\mathbb{Z}_\ell)$), Lemma lem:hecke, Chapter 14.

12.3 `thm:curves:adjoint-transpose` (theorem; P over $\mathbb{R}$ by de Rham, given the trace description of α_* in 12.2(iii) (S); hence over $\mathbb{Z}$ and $\mathbb{Z}_\ell$). Let $\mathcal{C} = (C,\alpha,\beta)$ be a correspondence from X to Y. For all $\xi \in H^1(Y,\mathbb{Z})$ and $\eta \in H^1(X,\mathbb{Z})$:

$$\langle \mathcal{C}^*\xi, \eta\rangle_X = \langle \xi, (\mathcal{C}^t)^*\eta\rangle_Y,$$

that is, **the adjoint of a correspondence for the cup product is its transpose**: $(\mathcal{C}^*)^\dagger = (\mathcal{C}^t)^*$. Proof: both sides equal $\int_C \beta^*\xi \wedge \alpha^*\eta$, because $\int_X (\alpha_*\theta)\wedge\eta = \int_C \theta \wedge \alpha^*\eta$ for every closed 1-form θ on C (on the complement of the finitely many branch points α is a $\deg\alpha$-sheeted covering, and on each sheet this is the change of variables formula; the branch points have measure zero), and likewise for β with the order of the factors kept. Both sides are integers on integral classes, so the identity holds in $H^1(\cdot,\mathbb{Z})$ and after $\otimes\mathbb{Z}_\ell$. Corollaries (P): (a) for an automorphism w of X, $\langle w^*x, w^*y\rangle = \langle x,y\rangle$ (take $\mathcal{C} = \Gamma_w$; $(\Gamma_w^t)^* = (w^{-1})^*$), so w^* is an isometry; (b) $\langle \varphi^*\xi,\eta\rangle_X = \langle \xi, \varphi_*\eta\rangle_Y$ for a morphism φ, and 11.5(i), (ii) again; (c) $\mathrm{PD}\circ\mathcal{C}^* = (\mathcal{C}^t)_*\circ\mathrm{PD}$: under Poincare duality the action $\mathcal{C}^*$ on H^1 corresponds to the action of the transpose on H_1, hence on $T_\ell J$ (11.7(iii)); (d) the same with $\mathbb{Z}_\ell$, $\mathbb{Q}_\ell$ or $\mathbb{C}$ coefficients. (e) (Abstract self-adjointness lemma.) Let $\mathbb{T}$ be a commutative ring of endomorphisms of $H^1(X,\mathbb{Z}_\ell)$, each of the form $\sum c_i\mathcal{C}_i^*$ with $c_i \in \mathbb{Z}_\ell$, and w an involutive automorphism of X such that for every $t = \sum c_i\mathcal{C}_i^* \in \mathbb{T}$ the transposed operator $t^t = \sum c_i(\mathcal{C}_i^t)^*$ equals w^*tw^*. Put $[x,y] = \langle x, w^*y\rangle$. Then $[\ ,\ ]$ is perfect and alternating on $H^1(X,\mathbb{Z}_\ell)$ and every $t \in \mathbb{T}$ is self-adjoint for it: $[tx,y] = \langle x, t^tw^*y\rangle = \langle x, w^*ty\rangle = [x,ty]$; antisymmetric because $[y,x] = \langle y, w^*x\rangle = \langle w^*y, x\rangle = -[x,y]$ ((a) and $w^{*2} = 1$), hence alternating since $H^1(X,\mathbb{Z}_\ell)$ is torsion-free (Chapter 3, 9.4); perfect because w^* is an automorphism (Chapter 3, 9.5(iv)). Consequently every idempotent of $\mathbb{T}$ is self-adjoint, $H^1(X,\mathbb{Z}_\ell) = \bigoplus_{\mathfrak{m}} H^1(X,\mathbb{Z}_\ell)_{\mathfrak{m}}$ is an orthogonal decomposition, and $[\ ,\ ]$ is perfect on each summand (Chapter 3, 9.5(iii)): this is the sentence important_notes.md point 36 asks for. Reference: Shimura71, Chapter 7 (cite the chapter); DDT, Section 4.4 (cite the section); DFG04, Sections 1.6.1 and 1.8.2 (as the paper cites them). Paper: Section 4.1 ("the Hecke operators, U_p included, are self-adjoint for it, because $w_{M^\Sigma}T_n w_{M^\Sigma}^{-1}$ is the adjoint of T_n for the cup product (this is what the composition with w is for)"; "perfect and alternating"), Lemma lem:hecke and Theorem thm:quant Step 2 (Lemma lem:ddt417 applied to $L = H^1(X_0(N),\mathbb{Z}_\ell)_{\mathfrak{m}}$ "with a perfect $\mathcal{O}$-bilinear pairing for which $\mathbb{T}$ is self-adjoint"), important_notes.md point 36. The input "$T_n^t = w_N T_n w_N$ for all n, including U_p" on $X_0(N)$ is Chapter 14's.

12.4 `ex:curves:correspondence-examples` (examples; P). (a) On $X = \mathbb{C}/\Lambda$, the automorphism $[-1]$ acts by -1 on H^1; it is an isometry ($\langle -x,-y\rangle = \langle x,y\rangle$). (b) For $\varphi\colon X \to E$ of degree d, the correspondence $\Gamma_\varphi^t \circ \Gamma_\varphi$ on E (that is, $\varphi_*\varphi^*$) acts on $H^1(E)$ and on $E = J(E)$ by d (11.4(iv), 11.5(ii)); $\Gamma_\varphi \circ \Gamma_\varphi^t$ on X acts on $H^1(X)$ by $\varphi^*\varphi_*$, whose image is $\varphi^*H^1(E)$, and on $\varphi^*H^1(E)$ by d: so $\varphi^*\varphi_*/d$ is the orthogonal projector onto $\varphi^*H^1(E,\mathbb{Q})$ for the cup product (P by 12.3(b)), integral at $\ell \nmid d$. (c) On a genus-one curve E, for a nonconstant morphism $\psi\colon E \to E$ fixing O (an endomorphism, Chapter 9) the adjoint of ψ^* on $H^1(E,\mathbb{Z})$ is $\psi_* = (\deg\psi)(\psi^*)^{-1}$ (12.3(b) and 11.5(ii)); for $\psi = [m]$, $\psi^* = \psi_* = m$ is self-adjoint, but for $E = \mathbb{C}/\mathbb{Z}[i]$ and $\psi(z) = iz$, ψ^* is a rotation of order 4 and its adjoint is the inverse rotation: a correspondence need not be self-adjoint for the cup product (Exercise 13). (d) (Forward pointer, Chapter 14.) For $p \nmid N$, $T_p^t = w_N T_p w_N$ on $X_0(N)$ (in fact $= T_p$ on $\Gamma_0(N)$ as double cosets), while for $p \mid N$, $U_p^t = w_N U_p w_N \neq U_p$ in general: this is why the paper composes the cup product with w. Paper: Section 4.1, Lemma lem:hecke.

10.3 3. Section-by-section plan with target line counts

Line counts are lines of LaTeX, card excluded (PLAN.md Section 7.3: 1,000 to 2,200).

section	title	contents	target lines
(starred)	Core results at a glance	the card: every item of Part 2, stated as above, with tag and `\Cref`	(excluded)
(starred)	Where the paper uses this chapter	Section 1 (φ_E, $\deg \varphi_E$, optimal curve); Section 4.1 ($H^1(X_0(M^\Sigma), \mathbb{Z}_\ell)$, cup product, $\hat{\delta}_!$, adjoints); Lemma lem:hecke; Lemma lem:ddt417 as applied; Theorem thm:quant Steps 1, 2, 4; Remark rem:free ($J_0(N)[\mathfrak{m}]$, Tate module, freeness); Section 4.2 ($M_{f,\lambda}$ versus $T_\ell E$, point 26); Section 1.1 and Proposition prop:ns (Galois action on points)	40
1	Affine and projective varieties	a one-page roadmap ("two languages, one dictionary", what is proved, sketched, stated) as the opening paragraph; conventions C1 to C8 as `convention` environments; 1.1 to 1.3	150
2	Plane curves and smoothness	2.1 to 2.4; the pictures of the node and the cusp described in words; the Weierstrass models of 11a1, 20a1, 121a1 as smooth examples	140
3	Function fields, points over extensions, the Galois action	3.1 to 3.4; `paperlink`: $\rho_{E,\ell}$ is the Galois action on points (Section 1.1), the action on $E[3]$ in Proposition prop:ns; `warning`: a genus-zero curve need not have a rational point	140
4	Divisors, differentials, the Picard group	4.1 to 4.3; the examples on $\mathbb{P}^1$ and on $y^2 = h(x)$	150

section	title	contents	target lines
5	Riemann-Roch and its consequences	5.1 to 5.7; the shape of the proof of Riemann-Roch in one paragraph; the Weierstrass derivation written out; paperlink: the curves of the paper as Weierstrass equations	210
6	Morphisms and degree; separability; Frobenius	6.1 to 6.5; a table "degree in three (four) ways"; paperlink: the modular degree is $\deg \varphi_E$ in every sense	170
7	Ramification and Riemann-Hurwitz	7.1 to 7.3; the worked examples (hyperelliptic, Fermat, $X_0(121) \to$ 121a1, $X_0(539) \to$ 539c1)	120
8	Compact Riemann surfaces and complex curves	8.1 to 8.4; the dictionary table (algebraic versus analytic: function field, ord_P, $e_\varphi(P)$, deg, differentials, genus)	150
9	Holomorphic differentials, residues, periods	9.1 to 9.4; physics asides (charge, norm, area)	130
10	H_1, H^1 and the cup product	10.1 to 10.4; the torus computation in full; a convention box on signs (C6) and on "which is dual to which"; paperlink: Section 4.1, "perfect and alternating"	170
11	The Jacobian, torsion and Tate modules	11.1 to 11.8; paperlinks: Remark rem:free, Section 4.2 (dual of $T_\ell E$, point 26), Lemma lem:hecke (11.5(iv)); warning: $H^1(X, \mathbb{Z}_\ell) \simeq T_\ell J$ is not Galois-equivariant	240
12	Correspondences and adjoints	12.1 to 12.4; paperlink: Section 4.1 ($\hat{\delta}_!$, self-adjointness of T_n and U_p), Lemma lem:hecke and important_notes.md point 36 (12.3(e))	170
13	Exercises	the 18 exercises of Part 4.2	130

section	title	contents	target lines
14	Notes and references	SilvermanAEC I to III, Hartshorne I, II, IV and Appendix B, Fulton, Miranda, Forster, Griffiths-Harris, Farkas-Kra, Milne (Jacobian varieties, Abelian varieties, Etale cohomology), Mumford, Shimura71, DiamondShurman; which section numbers were cited	40
	total		**2,150**

10.4 4. Coverage table and exercises

10.4.1 4.1 Coverage: PLAN.md scope, Section 6 map row and paper hooks to line-up items

PLAN.md scope item, Section 6 row, or paper hook	line-up items
affine and projective space	1.1, 1.2
varieties over a field	1.1, 1.2, 1.3
the projective plane and homogeneous polynomials	1.2, 2.1
smoothness and singular points	2.1, 2.2, 2.4
plane curves	2.1, 2.3, 5.4, 5.5
the function field of a curve	1.1(iii), 1.2(iv), 3.1, 3.3
points over extensions and the Galois action on points	1.2(i), 3.4, 6.5(c)
the genus via Riemann-Roch; $(d-1)(d-2)/2$ for smooth plane curves	5.2, 5.3(a), 5.4, 5.5, 7.3(d)
divisors, principal divisors, linear equivalence	4.1, 4.2
the Picard group	4.2, 5.7, 11.1
Riemann-Roch (statement) with standard consequences	5.1, 5.2, 5.3
a genus-one curve with a rational point is a plane cubic	5.6 (and 5.7)
morphisms of curves	1.3, 3.2, 3.3, 6.1
degree: field degree, number of preimages of a general point, pullback on divisors	6.2, 6.3, 6.4 (and 8.2(c), 11.5(i) for the analytic and cohomological forms)
separable and inseparable morphisms	6.2, 6.5
ramification of morphisms	6.2, 7.1, 8.2(a)
the Riemann-Hurwitz formula	7.2, 7.3, 8.4(c)
compact Riemann surfaces versus smooth projective curves over $\mathbb{C}$ (equivalence)	8.1, 8.2, 8.3, 8.4
differentials, holomorphic differentials and the genus	4.3, 5.3(a), 9.1, 9.3
the Jacobian as Pic^0, as a complex torus, as an abelian variety (statements)	11.1, 11.2, 11.3

PLAN.md scope item, Section 6 row, or paper hook	line-up items
the Abel-Jacobi map	11.2
$H^1(X,\mathbb{Z})$ of rank $2g$; intersection pairing and cup product	10.1, 10.2, 10.3, 10.4
$H^1(X,\mathbb{Z}_\ell) = H^1(X,\mathbb{Z}) \otimes \mathbb{Z}_\ell$; remark on etale cohomology	10.2(i), 10.3, 11.8(b), (c)
the Tate module of the Jacobian is dual to H^1	11.6, 11.7, 11.8
the cup product is perfect and alternating on $H^1(X,\mathbb{Z})$	10.3
correspondences and their action on H^1 and on the Jacobian	12.1, 12.2, 12.4
the adjoint of a correspondence under the cup product is the transpose	12.3
Section 6 row: $H^1(X_0(N),\mathbb{Z}_\ell)$, cup product, Atkin-Lehner, Hecke self-adjointness (with Chapter 14)	10.3, 12.3, 12.3(e), 12.4(d)
hook: the modular parametrisation $X_0(N) \to E$ and its degree (Section 1)	6.2, 6.4, 7.3(e), 8.3, 11.4, 11.5
hook: $H^1(X_0(N),\mathbb{Z}_\ell)$ with the cup product (Section 4.1, Lemma lem:hecke)	10.2, 10.3, 11.5(iv), 12.2, 12.3
hook: $J_0(N)$ and its torsion (Remark rem:free)	11.1 to 11.3, 11.6, 11.7
hook: Tate modules of Jacobians	11.7, 11.8
also: Theorem thm:quant Step 2 (same pairing at level N for g^S)	10.3, 12.3(e)
also: Section 4.2 ($M_{f,\lambda}$ versus $T_\ell E$; important_notes.md point 26)	11.5(iv), 11.7(ii), 11.8(c)
also: important_notes.md point 36 (orthogonality of the $\mathfrak{m}$-decomposition)	12.3(e)
also: Section 1.1, Proposition prop:ns (Galois action on points, $E[3]$)	3.4
also: Section 1 (Petersson norm, area of Λ_E in the Zagier-Watkins formula: background)	9.4, 10.4
also: what Chapters 9, 10, 12, 14, 15 assume from Chapter 8 (Part 5)	2.1, 3.4, 4.3, 5.6, 5.7, 6.4, 6.5, 7.3(b), (f), 8.1, 8.3, 9.1, 9.3, 11.2 to 11.8, 12.1 to 12.3

10.4.2 4.2 Exercises (Section 13 of the chapter), graded from easy to harder

1. Find the singular points of $y^2 = x^3 + x^2$ and $y^2 = x^3$ (projective closures included); parametrise the node by $t = y/x$ and show that the cusp is the image of $t \mapsto (t^2, t^3)$. Show that $X^d + Y^d + Z^d = 0$ is smooth iff $\operatorname{char} k \nmid d$.
2. Check the paper's curves are smooth: for 11a1 ($y^2+y = x^3-x^2-10x-20$) and 121a1 ($y^2+xy+y = x^3+x^2-30x-76$) show that the three partial derivatives of the homogenised equation have no common zero on the curve over $\bar{\mathbb{Q}}$ (compute F_Z at $[0:1:0]$; on the affine part eliminate y). Which primes make the reduction singular? (Answer for 11a1: only 11; compare with Chapter 9's $\Delta = -11^5$.)
3. On $\mathbb{P}^1$ show $\operatorname{div}(dx) = -2(\infty)$, $\operatorname{Pic}^0(\mathbb{P}^1) = 0$, and $\ell(n(\infty)) = n+1$; check Riemann-Roch with $g = 0$.
4. On the smooth cubic $y^2 = h(x)$ (h squarefree of degree 3, $\operatorname{char} k \neq 2$) compute $\operatorname{ord}_\infty x = -2$, $\operatorname{ord}_\infty y = -3$, and show $\operatorname{div}(dx/y) = 0$. Deduce $g = 1$ from 5.3(b).
5. Let C have genus 1 and $O \in C(k)$. Show $\ell(n(O)) = n$ for $n \geq 1$, find bases of $\mathcal{L}(n(O))$ for $n \leq 6$ in terms of x, y, and derive the Weierstrass equation (5.6(ii)). Why does the argument fail for the conic $X^2 + Y^2 + Z^2 = 0$ over $\mathbb{Q}$, and for Selmer's cubic?
6. Prove 5.7: for genus one, $P \mapsto [(P) - (O)]$ is a bijection $C(\bar{k}) \to \operatorname{Pic}^0(C)$. Check G_k-equivariance.

7. For the Weierstrass cubic, show that $x\colon C \to \mathbb{P}^1$ has degree 2 and is ramified exactly at O and at the three points with $2y + a_1x + a_3 = 0$; verify Riemann-Hurwitz. Do the same for y (degree 3).
8. Compute the genus of the Fermat curve of degree d in two ways (5.4 and 7.3(d)), and of $y^2 = x^5 - x$ (answer 2).
9. Show that a nonconstant morphism between curves of genus one in characteristic 0 is unramified and that $\#\varphi^{-1}(Q) = \deg\varphi$ for every Q. Where does the argument use characteristic 0? (Frobenius, 6.5(c), is a counterexample to $\#\varphi^{-1}(Q) = \deg\varphi$.)
10. Check the paper's degrees against Riemann-Hurwitz: given $g(X_0(121)) = 6$, $g(X_0(539)) = 49$ (Chapter 12) and $\deg\varphi_{121a1} = 6$, $\deg\varphi_{539c1} = 288$ (the paper, Section 3 Example), compute the total ramification of the two parametrisations. Given $g(X_0(11)) = 1$ and $\deg\varphi_{11a1} = 1$, conclude $X_0(11) \simeq$ 11a1.
11. (The torus.) For $X = \mathbb{C}/(\mathbb{Z}+\tau\mathbb{Z})$ compute the cup product on $H^1(X,\mathbb{Z})$ in the basis ds, dt; show it is perfect and alternating; compute PD and check $\langle\alpha,\beta\rangle = \mathrm{PD}(\alpha)\cdot\mathrm{PD}(\beta)$; compute $\frac{i}{2}\int dz\wedge d\bar z = \operatorname{Im}\tau$ and check 9.4(d).
12. Show $J(X)[n] \simeq H_1(X,\mathbb{Z}/n)$ and $T_\ell J \simeq H_1(X,\mathbb{Z})\otimes\mathbb{Z}_\ell$ for a compact Riemann surface; deduce $H^1(X,\mathbb{Z}_\ell) \simeq \mathrm{Hom}_{\mathbb{Z}_\ell}(T_\ell J,\mathbb{Z}_\ell)$. Using the Weil pairing of an elliptic curve (Chapter 9), deduce $H^1(E,\mathbb{Z}_\ell) \simeq T_\ell E(-1)$ as Galois modules and compare determinants (χ_ℓ^{-1} versus χ_ℓ). Explain why the paper's "$M_{f,\lambda}$ and $T_\ell E$ are homothetic" should read "homothetic to the dual of $T_\ell E$" (important_notes.md point 26), and why the adjoints agree anyway.
13. Show that an automorphism w of X acts on $H^1(X,\mathbb{Z})$ as an isometry of the cup product. For $E = \mathbb{C}/\mathbb{Z}[i]$ and $\psi(z) = iz$, write the matrix of ψ^* in the basis of Exercise 11 and of its adjoint; conclude that the adjoint of a correspondence is its transpose and not the correspondence itself.
14. (Check the paper's claim in Section 4.1, abstractly.) Let $\langle\ ,\ \rangle$ be a perfect alternating pairing on a free $\mathbb{Z}_\ell$-module L, w an isometric involution of L, and $\mathbb{T}$ a commutative ring of endomorphisms with $t^\dagger = wtw$ for all $t\in\mathbb{T}$. Show that $[x,y] = \langle x, wy\rangle$ is perfect and alternating and that every $t\in\mathbb{T}$ is self-adjoint for it; deduce that the idempotents of $\mathbb{T}$ split L orthogonally (important_notes.md point 36).
15. (Pullback and degree.) For $\varphi\colon X\to Y$ of degree d prove $\langle\varphi^*x,\varphi^*y\rangle = d\langle x,y\rangle$ and $\varphi_*\varphi^* = d$. For $Y = E$ of genus one, show that the cup product on $\varphi^*H^1(E,\mathbb{Z})$ has Gram determinant d^2 and that, if $\varphi^*H^1(E,\mathbb{Z})$ is saturated, $[H^1(X,\mathbb{Z}) : \varphi^*H^1(E,\mathbb{Z})\oplus(\varphi^*H^1(E,\mathbb{Z}))^\perp] = d^2$ (with Chapter 3, 9.3).
16. (Optimality.) Let $\pi\colon V/\Lambda \to W/M$ be a surjective homomorphism of complex tori. Show that the component group of its kernel is $M/\pi(\Lambda)$. For an isogeny $\psi\colon E\to E'$ of degree n and $\varphi\colon X\to E$, show that $(\psi\circ\varphi)^*H^1(E',\mathbb{Z})$ has index n in $\varphi^*H^1(E,\mathbb{Z})$; deduce that if $\varphi^*H^1(E,\mathbb{Z})$ is saturated then $(\psi\circ\varphi)^*H^1(E',\mathbb{Z})$ is not (for $n>1$), and relate this to "minimal degree" in the paper's definition of the optimal curve (Section 1; Chapter 14 shows that every parametrisation of a curve isogenous to E factors through the optimal one).
17. (Frobenius.) For C over $\mathbb{F}_p$, show that π_p is a morphism with $\pi_p^*\bar{\mathbb{F}}_p(C) = \bar{\mathbb{F}}_p(C)^p$, that $d(\pi_p^*f) = 0$, that its fixed points are $C(\mathbb{F}_p)$, and that $e_{\pi_p}(P) = p$ everywhere. Explain why Riemann-Hurwitz does not apply to π_p.
18. (Residues.) Prove $\sum_P \operatorname{res}_P\omega = 0$ on a compact Riemann surface and deduce $\deg\operatorname{div} f = 0$. On $\mathbb{P}^1$ verify it for $\omega = dz/(z^2-1)$.

10.5 5. Dependencies and forward pointers

Results from other chapters used here: * Background (PLAN.md Section 2): complex analysis (holomorphic functions, zeros of finite order and local e-th roots, the open mapping theorem, Laurent series and residues, Riemann's removable singularity theorem, Moebius maps), real analysis (implicit function theorem, change of variables in double integrals, Stokes/Green's theorem in the plane, used on surfaces via partitions of unity), linear algebra (dual bases, Gram matrices, real versus complex dimensions). Elementary point-set topology (compactness, connectedness, quotient topology) as in Chapter 1's profinite section. * Chapter 1 (`ch:groups`): $\mathbb{P}^1(F)$ as lines and the Moebius action (its C1); $\mathbb{Z}_\ell$ as an inverse limit (its 7.2), used for $T_\ell J$. * Chapter 2 (`ch:rings`): polynomial rings, irreducibility, UFDs (its 3.2, 4.3: $k[x,y]$ is a UFD,

used for plane curves), the freshman's dream in characteristic p (its 1.6, for the Frobenius morphism), finite fields (its 6.1 to 6.3). * Chapter 3 (ch:modules): Nakayama (its 4.3) and the DVR criterion (its 6.1) for 2.2; localisation (its 4.5); finitely generated modules over a PID and saturation (its 5.3, 5.5) for 11.5(iii); perfect pairings, base change, Gram determinants, saturated sublattices (its 9.1, 9.3), alternating pairings (its 9.4), adjoints, twisted pairings and orthogonal idempotent decompositions (its 9.5) for 10.3, 11.5(iv), 12.3(e). Chapter 8 must use, not redo, these. * Chapter 5 (ch:galois): algebraic and transcendental elements, separability, separable and inseparable degree (its galois:separability), perfect fields, G_k and its Krull topology (its galois:absolute-galois-group), the cyclotomic character and $\mathbb{Z}_\ell(1)$ (its galois:cyclotomic-character), Hilbert 90 in field form (its Section 5). * Chapter 6 (ch:ant): Dedekind domains and $\sum e_i f_i = n$ (its 2.1, 3.1), whose proof 6.3 transports to curves; the Frobenius conventions (its 3.7). * Chapter 4 (ch:reps): contragredient (its C2) for 11.8(b).

Black boxes imported from later chapters (stated with a forward \Cref, used only in examples, exercises and paperlinks, never in a proof of a numbered result): the genus formula of $X_0(N)$ and $S_2(\Gamma_0(N)) \simeq H^0(X_0(N), \Omega^1)$ (Chapter 12; used in 5.5(f), 7.3(e), 9.3, Exercise 10); the modular degrees of 11a1, 121a1, 539c1 (the paper; Chapter 14); the Hecke correspondences on $X_0(N)$ and $T_n^t = w_N T_n w_N$ (Chapter 14; used in 12.1(c), 12.4(d)); the Weil pairing of an elliptic curve and Δ (Chapter 9; Exercises 2, 12); Hilbert 90 in cohomological form (Chapter 17; used in 4.2(iii), stated); group cohomology and its cup product (Chapter 16; 10.2(iv), a remark).

Forward pointers (chapters that use this one): * Chapter 9 (ch:ellcurves): curves, points over extensions and the Galois action (3.4), smoothness of plane cubics (2.1), regular differentials and $\operatorname{div}(dx/y) = 0$ (4.3), Riemann-Roch consequences: the Weierstrass equation (5.6) and the bijection $E \to \operatorname{Pic}^0(E)$ (5.7) for the group law and its associativity, the degree of a morphism and $\varphi_*\varphi^* = \deg$ (6.4) for isogenies and the dual isogeny, separability and the Frobenius morphism (6.5), "isogenies are unramified" (7.3(b)), complex tori $\mathbb{C}/\Lambda$ (8.1(c), 10.4, 11.2 for $g = 1$), $J[n]$ and $T_\ell J$ for $g = 1$ (11.6, 11.7), the general Weil pairing (11.8(a)) and the point count $a_q = \operatorname{tr}\operatorname{Frob}$ (11.8(d)) as the $g = 1$ cases. Chapter 9 does not redo Riemann-Roch. * Chapter 10 (ch:reduction): smoothness and singular points of reductions mod p (2.1, 2.2), function fields (3.1). * Chapter 12 (ch:modforms): Riemann surfaces and charts (8.1), holomorphic maps and degree (8.2), Riemann-Hurwitz (7.2, 8.4(c)) for the genus formula of $X_0(N)$, holomorphic differentials and $\dim H^0(\Omega^1) = g$ (9.1, 9.3) for $\dim S_2(\Gamma_0(N)) = g$, the positivity 9.4(b) behind the Petersson product. * Chapter 13 (ch:hecke): the Hecke algebra as operators generated by correspondences (12.1, 12.2). * Chapter 14 (ch:moddeg): everything in Sections 10 to 12: the Jacobian $J_0(N)$ (11.1 to 11.3), its torsion and Tate module (11.6, 11.7, 11.8), the Hecke correspondences and their actions (12.1, 12.2), the adjoint theorem and the pairing $\langle x, w_N y\rangle$ (12.3, 12.3(e)), the modular parametrisation and its degree (6.4, 11.4, 11.5), the cup product on $H^1(X_0(N), \mathbb{Z}_\ell)$ (10.3), the identification of $M_{f,\lambda}$ with $\varphi_E^* H^1(E, \mathbb{Z}_\ell)$ for optimal E (11.5(iii), (iv); Part 7, points 1 and 2), Eichler-Shimura (with 6.5(c), 11.8(d)). * Chapter 15 (ch:modgalrep): Tate modules of abelian varieties and H^1 (11.7, 11.8) for $\rho_{f,\lambda}$; "homothetic to the dual" (point 26). * Chapter 18 (ch:blochkato): H^1_{et} and Tate twists $T_\ell J(-1)$ (11.8(c)). * Chapter 20 (ch:congruence): $H^1(X_0(N), \mathbb{Z}_\ell)_{\mathfrak{m}}$ with a perfect alternating pairing for which $\mathbb{T}$ is self-adjoint (10.3, 12.3(e)) as the module L of Lemma lem:ddt417; 11.5(iv) (Part 7, point 1). * Chapters 21, 22 (ch:paper, ch:computation): Section 4.1 and Remark rem:free hooks; ellmoddegree computes $\deg \varphi_E$ of 6.2.

10.6 6. Notes for the writer

10.6.1 6.1 Conventions to state (Section 1 of the chapter, as convention environments)

C1 to C8 of Part 1.2. In addition, one convention box in Section 10, "Homology, cohomology, Tate module: which is dual to which", with exactly these lines: * covariant objects: $H_1(X, \mathbb{Z})$, the lattice Λ_X, $J(X) = H^0(X, \Omega^1)^\vee/\Lambda_X$, $J[n] = H_1(X, \mathbb{Z}/n)$, $T_\ell J = H_1(X, \mathbb{Z}_\ell)$; a morphism φ acts by φ_* and a correspondence $\mathcal{C}$ by $\mathcal{C}_* = \beta_*\alpha^!$; * contravariant objects: $H^1(X, \mathbb{Z}) = \operatorname{Hom}(H_1, \mathbb{Z})$, $H^1(X, \mathbb{Z}_\ell) = \operatorname{Hom}(T_\ell J, \mathbb{Z}_\ell)$, $H^0(X, \Omega^1)$; φ acts by φ^*, $\mathcal{C}$ by $\mathcal{C}^* = \alpha_*\beta^*$; $\mathcal{C}^*$ is the transpose of $\mathcal{C}_*$ (12.2(a)); * the two sides are identified by Poincare duality (10.3), which is an isometry for the cup product and the intersection

pairing and turns $\mathcal{C}^*$ into $(\mathcal{C}^t)_*$ (12.3(c)); it is canonical up to the global orientation sign; * with Galois action (11.8): $H^1_{\text{et}} = (T_\ell J)^\vee \simeq T_\ell J(-1)$; for an elliptic curve, $V_\ell E$ (the paper's V) has determinant χ_ℓ and $H^1(E, \mathbb{Q}_\ell) = V^\vee \simeq V(-1)$ has determinant χ_ℓ^{-1}; $\mathrm{ad}^0 V$ is the same for V, $V^\vee$, $V(n)$ (Chapter 4); * Frobenius (C7): the arithmetic Frobenius on $T_\ell J$ and the geometric Frobenius on H^1 have the same characteristic polynomial $P(T)$, with $\#C(\mathbb{F}_q) = q + 1 - \mathrm{tr}$.

10.6.2 6.2 Pitfalls

- **Never define H^1 by sheaf cohomology** and never use sheaf cohomology in a proof: $H^1(X, \mathbb{Z}) := \mathrm{Hom}(H_1(X, \mathbb{Z}), \mathbb{Z})$, with de Rham forms for computing cup products. The one place sheaves appear is the remark "$\ell(K - D) = \dim H^1(X, \mathcal{O}(D))$" in the sketch of 5.2, flagged as such.
- **Sign of the cup product.** State C6 once, check it on the torus (10.4), and say that the paper needs only properties invariant under a global sign: perfectness, alternation, the adjointness 12.3, ideals of values, Gram determinants (in even rank $\det(-G) = \det G$). Do not try to reconcile with every textbook's convention.
- **Adjoint versus transpose versus inverse.** The cup-product adjoint of $\mathcal{C}^*$ is $(\mathcal{C}^t)^*$ (12.3); for an automorphism the transpose acts as the inverse (12.3(a)); a correspondence is self-adjoint only if it equals its transpose (as an operator). On $X_0(N)$, T_p ($p \nmid N$) is its own transpose but U_p ($p \mid N$) is not, and this is why the paper composes with w (12.4(d), Chapter 14). The adjoint is written $A^\dagger$ (C4), never A^*.
- **Degrees.** $\varphi_*\varphi^* = [\deg \varphi]$ on E, and the endomorphism $[\deg \varphi]$ of E has degree $(\deg \varphi)^2$ (10.4): never write "$\deg(\varphi_*\varphi^*) = \deg \varphi$". The cup product on $\varphi^* H^1(E)$ is $\deg \varphi$ times that of E, so the Gram determinant is $(\deg \varphi)^2$ (11.5(iv)): rank-two alternating pairings have square Gram determinants (Chapter 3, 9.4(ii)).
- **Separability.** "Number of preimages = deg" holds only for separable maps and away from branch points (6.4(ii)); in characteristic p the Frobenius is a bijection on points of degree p. The paper works over $\mathbb{Q}$ and $\mathbb{C}$, but Chapters 9, 10 and 14 reduce mod p.
- **Rational points.** A genus-zero curve over $\mathbb{Q}$ need not be $\mathbb{P}^1$ over $\mathbb{Q}$ (3.4(iv)); a genus-one curve need not have a rational point (Selmer's cubic). 5.6 and 5.7 assume $O \in C(k)$.
- **Optimality and saturation** (11.5(iii)): saturation of $\varphi^* H^1(E, \mathbb{Z})$ is equivalent to connectedness of $\ker(J(X) \to E)$, which is Chapter 14's definition of an optimal quotient; over $\mathbb{Z}_\ell$ only the ℓ-part of the component group matters (important_notes.md point 27: when $E[\ell]$ is irreducible all curves of the class give the same v_ℓ).
- **Jacobians over non-closed fields.** $J(k) = \mathrm{Pic}^0_k(C)$ needs $C(k) \neq \emptyset$ (4.2(iii), 11.3(i)); $X_0(N)$ has the rational cusp ∞, so this is harmless in the paper.
- **Do not prove Riemann-Roch, Abel-Jacobi, Riemann existence or Poincare duality.** State them precisely with references and give the shape of the proof in a paragraph (5.2, 11.2(iii), 8.3, 10.3). Never present a sketch as a proof (PLAN.md 7.2).
- Physics asides: at most one sentence each, at the places listed in Part 1.1.
- Typography (PLAN.md 7.2): no dash ligatures anywhere ("Riemann-Roch", "Riemann-Hurwitz", "Abel-Jacobi", "Eichler-Shimura", "Griffiths-Harris" with plain hyphens; ranges as "1 to 8"); ASCII source; the dictionary table of Section 8 and the degree table of Section 6 with `booktabs`.

10.6.3 6.3 What to leave to Chapters 9, 12, 14 (and others)

- **Chapter 9** (`ch:ellcurves`): Weierstrass invariants b_i, c_4, c_6, Δ, j, the criterion $\Delta \neq 0$, changes of coordinates; the chord-tangent group law (Chapter 8 gives only 5.7, from which associativity follows); isogenies, the dual isogeny, $\deg[m] = m^2$ for elliptic curves over any field, separable isogenies and kernels; $E[m]$ and the Weil pairing for E (proofs or statements there; 11.8(a) is only the general statement); the $\wp$-function and the uniformisation $E(\mathbb{C}) \simeq \mathbb{C}/\Lambda_E$, the period lattice of a curve over $\mathbb{Q}$; elliptic curves over finite fields and Hasse. Chapter 8 must not do Weierstrass invariants or the group law.

- **Chapter 12** (ch:modforms): $\mathbb{H}$, $\Gamma_0(N)$, cusps and elliptic points, the charts making $X_0(N)(\mathbb{C})$ a compact Riemann surface, the genus formula (as an application of 7.2), $S_2(\Gamma_0(N)) \simeq H^0(X_0(N), \Omega^1)$ via $f \mapsto f(z)dz$ (with the check at cusps and elliptic points), the Petersson product. Chapter 8 gives the general theory and the pointers 5.5(f), 7.3(f), 8.1(f), 9.3.
- **Chapter 14** (ch:moddeg): $X_0(N)$ over $\mathbb{Q}$; $J_0(N)$; the Hecke correspondences T_p, U_p, w_N and the identities $T_n^t = w_N T_n w_N$; the Hecke action on H_1, H^1, $J_0(N)$, $T_\ell J_0(N)$; Eichler-Shimura; the pairing $\langle x, w_N y\rangle$ on $H^1(X_0(N), \mathbb{Z}_\ell)$ (an application of 12.3(e)); the modular parametrisation, optimal curves, the modular degree and its equivalent descriptions; the identification $M_{f,\lambda} = \varphi_E^* H^1(E, \mathbb{Z}_\ell)$ for optimal E (11.5 applied; Part 7). Chapter 8 must state everything for a general curve and a general correspondence, and use $X_0(N)$ only in examples and paperlinks.
- **Chapter 13** (ch:hecke): the Hecke algebra on S_2; **Chapter 15**: $\rho_{f,\lambda}$; **Chapter 16**: group cohomology and its cup product (10.2(iv) is only a bridge remark); **Chapter 3**: all pairing algebra.

10.6.4 6.4 How the relevant important_notes.md points are handled

- **Point 26** (Section 4.2: $M_{f,\lambda}$ is homothetic to the dual of $T_\ell E$). Chapter 8 supplies the facts: 11.7(ii) ($H^1 = \mathrm{Hom}(T_\ell J, \mathbb{Z}_\ell)$), 11.8(c) ($H^1_{\mathrm{et}} \simeq T_\ell J(-1)$, and for E: $H^1(E, \mathbb{Z}_\ell) \simeq T_\ell E(-1)$), and 11.5(iii), (iv) (for an optimal quotient the pullback of $H^1(E)$ is saturated). The paperlink of Section 11 says: "$M_{f,\lambda} \subset H^1(X_0(N), \mathbb{Z}_\ell)$ is homothetic to the dual of $T_\ell E$, that is to $T_\ell E(-1)$; for the optimal curve it is exactly $\varphi_E^* H^1(E, \mathbb{Z}_\ell) \simeq (T_\ell E)^\vee$ (Chapter 14); the adjoints agree because $\mathrm{ad}^0(T^\vee) \simeq \mathrm{ad}^0 T$ and $\mathrm{ad}^0(T(-1)) = \mathrm{ad}^0 T$ (Chapter 4)". Exercise 12. Part 7, point 2 sharpens the note.
- **Point 36** (Lemma lem:hecke and Theorem thm:quant Step 2: the pairing stays perfect on the $\mathfrak{m}$-component). 12.3(e) states and proves the needed sentence in general: every idempotent of a ring of operators all self-adjoint for $[\ ,\ ]$ is self-adjoint, the decomposition is orthogonal, and the restriction of a perfect pairing to each summand is perfect (with Chapter 3, 9.5(iii)). The paperlink of Section 12 quotes Lemma lem:hecke and adds this sentence. Exercise 14.
- **Point 27** (optimality plays no role at ℓ when $E[\ell]$ is irreducible) is consistent with 11.5(iii), (v): an isogeny of degree prime to ℓ changes $\varphi^* H^1$ by an index prime to ℓ. Mentioned in the pitfall above.
- **Points 11 and B1** (Frobenius conventions): 11.8(d) and C7 state which Frobenius has which eigenvalues on $T_\ell J$ and on H^1, consistently with Chapter 6, 3.7.
- **Point 35** (Lemma lem:hecke: other local factors lie in the kernel, not the annihilator) is Chapter 3's and Chapter 20's; Chapter 8 does not touch it, but Part 7, B1 records that the Chapter 21 outline still has the old wording.

10.6.5 6.5 References

SilvermanAEC (I.1 to I.3, II.1 to II.5, III.3, III.8; the theorem numbers II.1.1, II.2.1, II.2.3, II.2.4, II.2.6, II.2.11, II.5.4, II.5.9, III.3.1 are cited with confidence, the others by section), Hartshorne (I.6 Corollary 6.12, II.6 Proposition 6.9, IV.1 Theorem 1.3, IV.2 Corollary 2.4; Appendix B), Fulton (Algebraic curves: Chapters 3, 5, 7, 8), Miranda (Chapters I to VIII, by chapter only), Shimura71 (Chapter 7), DiamondShurman (Section 6.3), BLR (Chapter 9) are in refs.bib. Not in refs.bib, to be added to bib/ch08.bib if cited: Forster, *Lectures on Riemann Surfaces* (GTM 81); Griffiths-Harris, *Principles of Algebraic Geometry*; Farkas-Kra, *Riemann Surfaces* (GTM 71); Hatcher, *Algebraic Topology*; Milne, *Jacobian varieties* (in Cornell-Silverman, *Arithmetic Geometry*, 1986) and *Abelian varieties* (same volume, and his online notes); Mumford, *Abelian Varieties*; Milne, *Lectures on Etale Cohomology* (online notes); Lang, *Algebra* (already a standard key Lang). All section and chapter numbers for Miranda, Forster, Griffiths-Harris, Farkas-Kra, Milne and Mumford were recalled, not re-verified: cite them by chapter or section only, as marked above, and check before giving any theorem number.

10.6.6 6.6 Verification log (computations done for this outline)

Python 3 with sympy 1.14, scripts check1.py, check2.py in /tmp/ch08. * Genera of $X_0(N)$ from the formula $g = 1 + \mu/12 - \nu_2/4 - \nu_3/3 - \nu_\infty/2$ (Chapter 12): $N =$ 1: 0; 11: 1; 20: 1; 37: 2; 77: 7; 99: 9;

121: 6 ($\mu = 132$, $\nu_2 = \nu_3 = 0$, $\nu_\infty = 12$); 140: 19; 539: 49 ($\mu = 672$, $\nu_\infty = 16$); 726: 109; 980: 145. The known values $g(X_0(11)) = g(X_0(20)) = 1$, $g(X_0(37)) = 2$ are reproduced. * Riemann-Hurwitz totals for the paper's parametrisations (degrees from the paper and the Chapter 21 outline's ecdata check): 11a1 and 20a1: degree 1, total ramification 0; 121a1: degree 6, total 10; 539c1: degree 288, total 96; 980i1: degree 2880, total 288; 726a1: degree 144, total 216; 99a1: degree 4, total 16. * Smoothness by Groebner bases over $\mathbb{Q}$ of $\{F, F_X, F_Y, F_Z\}$ in each standard chart: the homogenised Weierstrass equations of 11a1 $[0, -1, 1, -10, -20]$, 121a1 $[1, 1, 1, -30, -76]$, 20a1 $[0, 1, 0, 4, 4]$ and the Fermat curves of degree $3, 4, 5$ have no singular point over $\bar{\mathbb{Q}}$; the nodal and cuspidal cubics have exactly the singular point $[0 : 0 : 1]$. * By hand: $\operatorname{div}(dx/y) = 0$ on $y^2 = h(x)$; the bilinear identity 9.4(c) and positivity 9.4(d) on $\mathbb{C}/(\mathbb{Z} + \tau\mathbb{Z})$ ($\int ds \wedge dt = 1$; $\frac{i}{2}\int dz \wedge d\bar{z} = \operatorname{Im}(\bar{\omega}_1\omega_2)$); $\mathrm{PD}(ds) = b$, $\mathrm{PD}(dt) = -a$ and the isometry property under C6 (and under the opposite convention, where PD changes sign on both factors); the adjointness 12.3 by the projection-formula computation in both orders of the factors; the component-group formula $M/A(\Lambda)$ in 11.5(iii); the self-adjointness and alternation computation of 12.3(e) (it needs w^* to be an isometry with $w^{*2} = 1$, which holds for any involutive automorphism); the rank-two Gram determinant d^2 in 11.5(iv); Riemann-Hurwitz for the hyperelliptic and Fermat examples; the $\psi(z) = iz$ example of 12.4(c).

10.7 7. Points for important_notes.md

New points in the area of this chapter (the modular parametrisation and its degree, $H^1(X_0(N), \mathbb{Z}_\ell)$ with the cup product and $\hat{\delta}_!$, $J_0(N)$ and its Tate module), not among the 38 points and B1, B2 already recorded. Points 26, 27, 35, 36 are used above and not repeated. None of the three points below changes a result of the paper; point 1 removes a hypothesis from one step.

1. **Lemma lem:hecke (and Remark rem:free, Theorem thm:quant Step 4): the inequality $v_\ell(\deg\varphi_E) \geq v_\lambda(\eta^{\emptyset}_{f,\lambda})$ is always an equality, without the freeness of $H^1(X_0(N), \mathbb{Z}_\ell)_{\mathfrak{m}}$.** Location: Lemma lem:hecke ("with equality if $H^1(X_0(N), \mathbb{Z}_\ell)_{\mathfrak{m}}$ is free over $\mathbb{T}_{\mathfrak{m}}$"), Remark rem:free ("So the equality ... holds under the hypotheses of Lemma lem:hecke"), Theorem thm:quant Step 4 ("By Lemma lem:hecke applied to E_S in the free case (Remark rem:free ...)"), Theorem thm:main ("with equality when ... free"). Argument (with the paper's own definition of η as the ideal of values of $\hat{\delta}_!$ on $M_{f,\lambda} = H^1(X_0(N), \mathbb{Z}_\ell)[\ker\phi_f]$):

   (a) $\varphi_E^*\omega_E$ is a nonzero multiple of $f(z)\,dz$, so $\varphi_E^* H^1(E, \mathbb{Q})$ is the two-dimensional f-part of $H^1(X_0(N), \mathbb{Q})$, and $\varphi_E^* H^1(E, \mathbb{Z}_\ell) \subseteq M_{f,\lambda}$ with finite index. (b) E optimal means $\ker(J_0(N) \to E)$ connected, so $\varphi_{E*}\colon H_1(X_0(N), \mathbb{Z}) \to H_1(E, \mathbb{Z})$ is surjective and $\varphi_E^* H^1(E, \mathbb{Z})$ is saturated (item 11.5(iii)); hence $M_{f,\lambda} = \varphi_E^* H^1(E, \mathbb{Z}_\ell)$. (c) $\langle \varphi_E^* x, \varphi_E^* y\rangle = \deg\varphi_E \langle x, y\rangle_E$ with $\langle\ ,\ \rangle_E$ unimodular (items 11.5(i), 10.3). (d) w_N preserves the f-part and acts on it by the Atkin-Lehner sign ± 1 (on $f\,dz$ and on its conjugate), so $\hat{\delta}_!(x, y) = \langle x, w_N y\rangle = \pm\langle x, y\rangle$ on $M_{f,\lambda}$. Hence the ideal of values is $\deg\varphi_E \cdot \mathbb{Z}_\ell$, that is, $v_\lambda(\eta^{\emptyset}_{f,\lambda}) = v_\ell(\deg\varphi_E)$, for every odd $\ell \nmid N$ (indeed for every ℓ for which η is defined). With Lemma lem:ddt417 this gives back $v_\ell(r_E) \geq v_\ell(\deg\varphi_E)$ (the divisibility of Agashe, Ribet and Stein) and, in the free case, their equality; with $v_\ell(\deg\varphi_E) = v_\ell(r_E)$ for $\ell^2 \nmid N$ (ARS12 Theorem 2.2, as the paper uses it) it gives $v_\ell(r_E) = v_\lambda(\eta^{\emptyset}_{f,\lambda})$ directly. In Step 4 of Theorem thm:quant the same argument for the optimal curve in the class of E_S gives $v_\lambda(\eta^{\emptyset}_{g,\lambda}) = v_\ell(r_{E_S})$ with no appeal to Remark rem:free (which partly answers important_notes.md point 24: the middle term of Theorem thm:quant needs ARS12 Theorem 2.2 but not multiplicity one beyond what that theorem uses). Fix: replace "with equality if ... free" by "with equality" in Lemma lem:hecke, adding the four lines (a) to (d); Remark rem:free then becomes an alternative route (or can be kept as context for the Gorenstein property). Effect: the equality clause of Theorem thm:main (eq:identity) holds under the hypotheses of the theorem without citing Tilouine97 and RibetStein; no number changes. Confidence: high for (a) to (d) as mathematics; medium on whether DFG04's normalisation of $\eta^{\emptyset}_{f,\lambda}$ agrees exactly with the paper's description "ideal of values of $\hat{\delta}_!$ on $M_{f,\lambda}$"

(if DFG04 normalise the pairing differently by a unit, nothing changes; by a non-unit, the paper's description itself would need correcting).

2. **Section 4.2 (supplement to important_notes.md point 26): for the optimal curve, $M_{f,\lambda}$ is exactly the dual of $T_\ell E$, with no irreducibility needed.** Location: "(the lattice $M_{f,\lambda}$ of DFG04 and $T_\ell E$ are homothetic when $\bar\rho$ is irreducible, so their adjoints coincide)". By point 1(a), (b), $M_{f,\lambda} = \varphi_E^* H^1(E,\mathbb{Z}_\ell) \simeq H^1(E,\mathbb{Z}_\ell)$, and φ_E is defined over $\mathbb{Q}$, so in etale cohomology this is an isomorphism of $G_\mathbb{Q}$-modules $M_{f,\lambda} \simeq H^1_{\text{et}}(E_{\bar\mathbb{Q}},\mathbb{Z}_\ell) \simeq \mathrm{Hom}(T_\ell E,\mathbb{Z}_\ell) \simeq T_\ell E(-1)$ (items 11.7(ii), 11.8(c)). Fix: "the lattice $M_{f,\lambda}$ is $\varphi_E^* H^1(E,\mathbb{Z}_\ell) \simeq \mathrm{Hom}_{\mathbb{Z}_\ell}(T_\ell E,\mathbb{Z}_\ell) \simeq T_\ell E(-1)$ for the optimal curve E, so $\mathrm{ad}^0 M_{f,\lambda} \simeq \mathrm{ad}^0 T_\ell E$"; the irreducibility of $\bar\rho$ is then not needed for this identification (it is needed elsewhere, for the homothety class of lattices of other curves in the class). Confidence: high (given point 1(a), (b), which use only the definition of optimality and the standard fact that $\varphi_E^*\omega_E$ is a multiple of $f(z)dz$).

3. **Remark rem:free: "$T_\ell J_0(N)$" versus "$H^1(X_0(N),\mathbb{Z}_\ell)$" as Hecke modules (expository).** Location: "the localised Hecke algebra is then Gorenstein and the Tate module is free of rank two \cite{Tilouine97}" followed by the conclusion for $H^1(X_0(N),\mathbb{Z}_\ell)_\mathfrak{m}$. The Tate module carries the Albanese action $\mathcal{C}_*$ of the Hecke correspondences and H^1 the action $\mathcal{C}^*$; Poincare duality turns $\mathcal{C}^*$ into $(\mathcal{C}^t)_*$ (item 12.3(c)), so the Hecke-equivariant identification is $x \mapsto w_{N*}\,\mathrm{PD}(x)$, using $T^t = w_N T w_N$ (Chapter 14), not PD alone; and neither identification is Galois-equivariant (twist by -1, point 26). Freeness transfers in either case ($t \mapsto w_N t w_N$ is an automorphism of $\mathbb{T}$ fixing $\mathfrak{m}$, because w_N acts on the f-part by a scalar; alternatively $\mathrm{Hom}_{\mathbb{Z}_\ell}(\cdot,\mathbb{Z}_\ell)$ preserves freeness over the Gorenstein ring $\mathbb{T}_\mathfrak{m}$). Fix: add "(via Poincare duality composed with w_N)" after "the Tate module". Effect: none. Confidence: high; importance low.

Book-internal inconsistencies found while preparing this outline:

- **B1. Chapter 21 outline, A4.4** (expanded proof of Lemma lem:hecke) still says the local factors $\mathfrak{m}' \neq \mathfrak{m}$ "are killed by ϕ_f and lie in $\mathrm{Ann}(\ker\phi_f)$", which important_notes.md point 35 (from the Chapter 3 outline) corrects to "lie in the kernel". The Chapter 21 writer must use the corrected wording. (A4.4 does already contain the point 36 sentence.)
- **B2. Chapter 21 outline, A4.6** says $M_{f,\lambda}$ "is homothetic to $T_\ell E$ for $\bar\rho$ irreducible", while its own A4.1 Commentary and important_notes.md point 26 say "homothetic to the dual of $T_\ell E$". A4.6 should be brought in line (and can use point 2 above).
- **B3. Notation for adjoints (settled as in Appendix B, B-C4: $A^\dagger$ for adjoints, $^\vee$ linear, * Pontryagin and pullback; Chapter 3 now writes $A^\dagger$).** Chapter 3, 9.5 wrote the adjoint of A as A^*; Chapter 4, C2 reserves * for Pontryagin duals (as the paper does); Chapters 8 and 14 need * for pullbacks (φ^*, T^*, $\mathcal{C}^*$). This outline writes $A^\dagger$ for adjoints (C4). Appendix B should fix one book-wide convention (suggestion: $A^\dagger$ for adjoints, $^\vee$ for linear duals, * for pullbacks and for the paper's Pontryagin duals, which never meet pullbacks in the same formula).
- **B4. Chapter 3 outline, Part 5** lists "Chapters 8, 14" as users of "Gorenstein and the freeness criterion (11.4, 11.5)"; Chapter 8 does not use them (Remark rem:free is Chapter 14's and Chapter 20's). Harmless; noted so that no writer looks for Gorenstein rings in Chapter 8.

Chapter 11

Chapter 09: Elliptic curves over fields (ch:ellcurves, chapters/ch09_ellcurves.tex)

Stage 1 line-up. Part III (Curves and elliptic curves). Target length 2,140 lines of LaTeX (card excluded; see Part 3), inside the 1,000 to 2,200 range of PLAN.md Section 7.3.

Rigour tags: **P** = proved here, **S** = proof sketched here, **R** = stated; reference. Item format, as in the Chapter 2, 5 and 6 outlines: number, label-ready name, kind, tag; the precise statement with all hypotheses; reference (book and section or chapter; a theorem number only where the outline author is sure, otherwise the section, and numbers marked "(verify)" must be checked against the book before they are printed); the paper's use by its `\label` names, or "background". Every computation quoted below (invariants, division polynomials, twists, point counts, periods, the data of the running examples) was checked by Python/sympy/mpmath scripts in `/tmp/ch09` against Cremona's `ecdata` files for conductors below 10,000 (verification log, Part 7).

11.1 0. What this chapter is for

PLAN.md says the paper hooks of this chapter are "everything". Concretely the chapter must deliver the following, and it is organised around them.

1. **The objects of Section 1.1 of the paper.** $E/\mathbb{Q}$ by a Weierstrass equation; $E[\ell]$, $T = T_\ell E$, $V = V_\ell E$, $\rho = \rho_{E,\ell}$, $\bar\rho = \bar\rho_{E,\ell}$, $\rho \bmod \ell^k$ as the action on $E[\ell^k]$; $\det\rho = \chi_\ell$ by the Weil pairing; complex conjugation with determinant -1; the Weil pairing as $T \simeq \mathrm{Hom}(T, \mathbb{Z}_\ell(1))$ (used for $A(1) = T^D$ in Section 4.2 and for important_notes.md point 26).
2. **Definition def:terms and the table of Section 1.** The invariant j_E and $v_q(j_E)$ (independent of the model, invariant under every quadratic twist, so $t_\ell(E,q)$ is twist-invariant); $a_q(E)$ and the characteristic polynomial $X^2 - a_q X + q$ of Frobenius with roots α, β, $\alpha\beta = q$, $\alpha + \beta = a_q$; the twist E' of good reduction at an I_0^* prime and the fact that only $a_q(E')^2$ enters; the statement "both terms are invariant under quadratic twist" in its correct form (important_notes.md point 15).
3. **Lemma lem:twist (ii) and Section 1.4** (models and invariants of twists): $E^{(d)}$ for a general Weierstrass model, c_4, c_6, Δ scaled by d^2, d^3, d^6, the same j, the difference between the twist by q and by q^* (important_notes.md point 29), $\rho_{E^{(d)},\ell} = \rho_{E,\ell} \otimes \chi_d$, $a_p(E^{(d)}) = \chi_d(p) a_p(E)$, composite twists E_S (Theorems thm:twist, thm:quant), and $V = V' \otimes \chi_{q^*}$ in Lemma lem:euler. (The effect on Kodaira types, Lemma lem:twist (i), (ii), is proved in Chapter 10.)
4. **Lemma lem:h0** (the CM facts): endomorphism rings, the thirteen CM j-invariants, $\rho \simeq \rho \otimes \chi_K$ and $a_p = 0$ at good primes inert in the CM field K, the fact that a prime ramified in K divides N

(so CM by $\mathbb{Q}(\sqrt{\ell^*})$ forces $\ell \mid N$: important_notes.md point 14), with the warning that the paper's sentence uses the wrong character (point 5).

5. **Proposition prop:ns** (the 3-division polynomial): $\psi_3 = 3x^4 + b_2x^3 + 3b_4x^2 + 3b_6x + b_8$, its four roots are the x-coordinates of the four lines of $E[3]$, Galois-equivariantly; a rational root iff a rational 3-isogeny; the factorisation pattern is the orbit pattern of the image on $\mathbb{P}^1(\mathbb{F}_3)$ (Chapters 1 and 5); $\operatorname{disc}\psi_3 \in -3\cdot(\mathbb{Q}^\times)^2$. Worked factorisations for 11a1, 121a1, 726a1 (the paper's 3Ns example), 20a1, 14a1, 27a1.
6. **Conjecture conj:eis and Section sec:eis** (torsion and isogenies): Mordell-Weil, Mazur's torsion theorem, rational ℓ-isogenies and reducibility of $E[\ell]$, isogeny classes, the structure $\mathbb{Z}/\ell \oplus \mu_\ell$, and the proof (important_notes.md point 17, decided here) that a non-CM curve with $E[3] \simeq \mathbb{Z}/3 \oplus \mu_3$ has no rational point of order 9, which turns the paper's bound $D \le 2$ from conjecture into a consequence of the first bound.
7. **Optimality and** ℓ (important_notes.md point 27; Section sec:oot): if $E[\ell]$ is irreducible, all isogenies in the class are prime to ℓ up to $[\ell^k]$, so T_ℓ, ρ, $h_\ell(E)$ and every ℓ-adic invariant of the adjoint are the same for all curves of the class; a trivial isogeny class gives $E[\ell]$ irreducible for every ℓ.
8. **Curves over** $\mathbb{C}$: lattices, $\wp$, uniformisation, $\mathrm{SL}_2(\mathbb{Z})\backslash\mathfrak{H}$, the period lattice Λ_E of a curve over $\mathbb{Q}$ and its area $\mathrm{vol}(\Lambda_E)$ (which Chapter 14 needs for the Zagier-Watkins formula), computed for 11a1.
9. **Curves over finite fields**: Frobenius, a_p, Hasse, the characteristic polynomial, supersingularity, the quadratic twist over $\mathbb{F}_q$ (the factorisation $(q+1)^2 - a^2 = \#E(\mathbb{F}_q)\#E^{\mathrm{tw}}(\mathbb{F}_q)$ behind the I_0^* row).
10. $L(E,s)$ **by its Euler product**, with the bad factors, and the **Cremona and LMFDB labels** of the running examples 11a1, 20a1, 121a1, 121c, 121d, 539c1, 980i1, 726a1, 99A1, with their basic data checked.

Split with the neighbouring chapters (decided here, consistent with PLAN.md Section 4 and with what the written outlines assume): * **Chapter 8** (outlined in parallel) supplies: projective plane, smooth plane curves, function fields, divisors, principal divisors, Pic^0, Riemann-Roch (statement) and its consequence "a genus-one curve with a rational point is a plane cubic", degree of a morphism (three descriptions), separable and inseparable morphisms, differentials, compact Riemann surfaces versus curves over $\mathbb{C}$, Jacobians as complex tori, Abel-Jacobi. Its finished outline (`outlines/ch08_curves.md`, split section 6.3) also proves, for a general genus-one curve, that a genus-one curve with a rational point has a Weierstrass equation (its 5.6), the bijection $P \mapsto$ class of $(P)-(O)$ onto Pic^0 (its 5.7, which gives associativity of the group law), and that every nonconstant morphism between genus-one curves in characteristic 0 is unramified (its 7.3(b)). Chapter 9 cites these from `\Cref{ch:curves}` and does not reprove them; it proves the identification of the Pic^0 structure with the chord-tangent law (item 2.3) and everything else in its PLAN.md scope. * **Chapter 10** supplies: minimal models (existence, uniqueness, the criterion), reduction types, the reduction map and its injectivity on prime-to-p torsion (formal group), the Neron model and Neron differential, Kodaira types, Tate's algorithm, conductor exponents, the Tate curve, and the *proofs* of Lemma lem:twist (i), (ii) on types. Chapter 9 uses the conductor N and the types of the running examples as data, with a forward pointer, and states the injectivity of reduction on torsion with a forward reference. * **Chapter 11** supplies: the images of $\bar\rho_{E,\ell}$ (Serre, Zywina, Sutherland labels), the full Neron-Ogg-Shafarevich criterion, local representations at bad primes and at ℓ, irreducible versus absolutely irreducible, Serre conductor, and the *proofs* of Proposition prop:ns and Lemma lem:h0. Chapter 9 gives the objects (ρ, $\bar\rho$, det, complex conjugation), the division polynomial, the CM facts, the isogeny facts and the easy direction "good reduction at $p \neq \ell$ implies unramified with $\rho(\mathrm{Frob}_p)$ of characteristic polynomial $X^2 - a_pX + p$". * **Chapter 14** supplies: modularity, $X_0(N)$, the optimal curve and its Cremona label, the modular degree, the Manin constant c_E and the formula $\deg\varphi_E = 4\pi^2 c_E^2\langle f,f\rangle/\mathrm{vol}(\Lambda_E)$. Chapter 9 defines Λ_E and its area and records the convention "Cremona's curve number 1 is the optimal curve" as a convention with the exception 990h (item 12.1). * **Chapter 12** supplies the fundamental domain for $\mathrm{SL}_2(\mathbb{Z})$, modular functions, the q-expansion of j, the surjectivity of j on $\mathfrak{H}$ (valence formula). Chapter 9 proves "lattices up to homothety = $\mathrm{SL}_2(\mathbb{Z})\backslash\mathfrak{H}$" and states the rest. * **Chapter 19** supplies analytic continuation and the functional equation of $L(E,s)$ and the local factors via Weil-Deligne representations; Chapter 9 defines $L(E,s)$ by its Euler product. * **Chapters 5 and 6** already contain the quartic-on-lines proposition (Chapter 5, galois:quartic-on-lines,

which needs "the roots of ψ_3 are the x-coordinates of the four lines, Galois-equivariantly" from here) and a preview of CM (Chapter 6, item 6.5, which quotes "the Galois action on endomorphisms" from here). Chapter 9 is the home of the elliptic-curve side of both; it cites the field-theoretic and number-theoretic sides. * **Chapter 2** has the identity $(1-\frac{\alpha}{\beta q})(1-\frac{1}{q})(1-\frac{\beta}{\alpha q}) = \frac{q-1}{q}\cdot\frac{(q+1)^2-a^2}{q^2}$ and quotes the Hasse bound and $\#E(\mathbb{F}_q) = q+1-a_q$ from here (its 11.4). * **Chapter 22** supplies PARI/GP (`ellinit`, `elldivpol`, `ellap`, `elltors`, `ellisomat`, `ellperiods`); Chapter 9 may show one-line GP commands in `lstlisting` but explains only the mathematics.

11.2 1. Global conventions fixed in this chapter

- **C1 (fields and curves).** K is a field, $\bar{K}$ a fixed algebraic closure, $G_K = \mathrm{Gal}(K^{\mathrm{sep}}/K)$. An elliptic curve over K is a pair (E, O), E a smooth projective curve of genus one over K and $O \in E(K)$; by Riemann-Roch (Chapter 8) it has a Weierstrass model (item 1.2). "$E/\mathbb{Q}$" always means a curve given by a Weierstrass equation with rational coefficients; from Section 9 on, by a global minimal model (Chapter 10), as in Cremona's tables. Whenever ℓ appears, $\ell \neq \mathrm{char}\, K$.
- **C2 (letters).** p, q, ℓ are primes. As in the paper, q is usually a bad prime of $E/\mathbb{Q}$, p any prime and ℓ the coefficient prime. In Section 8 the size of a finite field is written $\mathfrak{q} = p^r$ when it need not be prime (Chapter 1, C4 and Chapter 2, C2 make the same warning); for $\mathbb{F}_p$ with p prime the notation is a_p.
- **C3 (Weierstrass data).** a_1, a_2, a_3, a_4, a_6; b_2, b_4, b_6, b_8; c_4, c_6; Δ; $j = c_4^3/\Delta$; the invariant differential $\omega = dx/(2y + a_1x + a_3)$. These are Silverman's (AEC III.1) and PARI's conventions; PARI's `ellinit` returns them in this order (Chapter 22).
- **C4 (twists and characters).** For $d \in K^\times$ (char $K \neq 2$), $\chi_d\colon G_K \to \{\pm 1\}$ is the character of $K(\sqrt{d})/K$, $\sigma(\sqrt{d}) = \chi_d(\sigma)\sqrt{d}$. For $K = \mathbb{Q}$, $\chi_d(\mathrm{Frob}_p) = \left(\frac{D}{p}\right)$ (Kronecker symbol) for $p \nmid D$, with D the discriminant of $\mathbb{Q}(\sqrt{d})$ (Chapters 2 and 6). For an odd prime q, $q^* = (-1)^{(q-1)/2}q$ and χ_{q^*} is the quadratic character of conductor q (Chapter 2, C5); the paper writes χ_q, the book never does (decision D3). The paper also writes χ_ℓ for the cyclotomic character; in Chapter 9's own text the cyclotomic character is χ_ℓ^{cyc} or simply χ_ℓ only where no quadratic character of conductor ℓ is in sight, and $\bar{\chi}_\ell$ is its reduction; the `paperlink` environments use the paper's notation and say so.
- **C5 (Frobenius).** For E over a finite field $\mathbb{F}_\mathfrak{q}$ the Frobenius endomorphism $\pi\colon (x,y) \mapsto (x^\mathfrak{q}, y^\mathfrak{q})$ acts on $E(\bar{\mathbb{F}}_p)$ exactly as the arithmetic Frobenius $\mathrm{Frob}_\mathfrak{q} \in G_{\mathbb{F}_\mathfrak{q}}$ $(t \mapsto t^\mathfrak{q})$. For $E/\mathbb{Q}$ with good reduction at $p \neq \ell$, Frob_p in a group-theoretic statement is an **arithmetic** Frobenius (Chapter 6, 3.7; Chapter 21, C2(a)); $\rho_{E,\ell}(\mathrm{Frob}_p)$ has characteristic polynomial $X^2 - a_pX + p$. The Euler factor of E is $\det(1 - \mathrm{Frob}_p\, p^{-s} \mid V_\ell E)^{-1}$ with the arithmetic Frobenius on $V_\ell E$, equivalently with the geometric Frobenius on the dual $V_\ell E^\vee = H^1$; with the geometric Frobenius on $V_\ell E$ itself one gets $L_p(E, s+1)$, because $V_\ell E \simeq V_\ell E^\vee(1)$ (item 5.5). In the book's notation (Appendix B, B-C2, B-C3), with $\Phi_p = \mathrm{Frob}_p^{-1}$ the geometric Frobenius: $L_p(E,s) = L_p(V_\ell E^\vee, s) = \det(1 - \Phi_p\, p^{-s} \mid V_\ell E^\vee)^{-1}$ and $L_p(V_\ell E, s) = L_p(E, s+1)$. The chapter states this in a `convention` and Chapter 19 and Chapter 21 must match it (see Part 8, B3).
- **C6 (labels).** Cremona labels are written as the paper writes them (11a1, 121c, 99A1); "class 121c" means the isogeny class, "121c1" the curve. The capital letters of Cremona's book and of ARS12 (99A1) denote the same classes as the lower-case letters of the `ecdata` files. LMFDB labels are written 11.a2 (item 12.1).
- **C7 (Weil pairing sign).** The Weil pairing is Silverman's e_m (AEC III.8). Some books use the inverse; every statement in the book that depends on the sign (the comparison with $e^{2\pi i/m}$ over $\mathbb{C}$, item 7.4) says which. $\det\rho = \chi_\ell$ does not depend on the sign.
- **C8 (typography).** PLAN.md Section 7.2: no dash ligatures (write "Mordell-Weil", "Neron-Ogg-Shafarevich", "pages 33 to 186"), ASCII source, labels `...:ellcurves:...`; Kodaira symbols as in PLAN.md 7.1.

11.3 2. Line-up of the core results, in teaching order

11.3.1 Section 1. Weierstrass equations and their invariants

1.1 def:ellcurves:weierstrass (definition; P). A Weierstrass equation over K is

$$y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6, \qquad a_i \in K,$$

with projective closure $Y^2Z+a_1XYZ+a_3YZ^2 = X^3+a_2X^2Z+a_4XZ^2+a_6Z^3$ and the single point $O = [0 : 1 : 0]$ at infinity. Put $b_2 = a_1^2+4a_2$, $b_4 = 2a_4+a_1a_3$, $b_6 = a_3^2+4a_6$, $b_8 = a_1^2a_6+4a_2a_6-a_1a_3a_4+a_2a_3^2-a_4^2$, $c_4 = b_2^2 - 24b_4$, $c_6 = -b_2^3 + 36b_2b_4 - 216b_6$, $\Delta = -b_2^2b_8 - 8b_4^3 - 27b_6^2 + 9b_2b_4b_6$, and $j = c_4^3/\Delta$ when $\Delta \neq 0$. Identities: $4b_8 = b_2b_6 - b_4^2$ and $1728\Delta = c_4^3 - c_6^2$. If $\operatorname{char} K \neq 2$, $\eta = 2y + a_1x + a_3$ gives $\eta^2 = 4x^3 + b_2x^2 + 2b_4x + b_6$; if $\operatorname{char} K \neq 2, 3$, $(x, y) \mapsto ((36x + 3b_2), 108(2y + a_1x + a_3))$ gives the short form $y^2 = x^3 - 27c_4x - 54c_6$. Invariant differential $\omega = dx/(2y + a_1x + a_3) = dy/(3x^2 + 2a_2x + a_4 - a_1y)$, holomorphic and nowhere vanishing (Chapter 8 for the words). Both identities and the completed square were checked symbolically. Reference: Silverman AEC III.1. Paper: Section 1.1 and everywhere (E by a model; PARI's `ellinit([a1,a2,a3,a4,a6])` in the appendix, e.g. 121a1 $= [1, 1, 1, -30, -76]$), Lemma lem:twist (ii) (models $y^2 = x^3 + a_2x^2 + a_4x + a_6$).

1.2 prop:ellcurves:smooth (proposition; P). The curve of 1.1 is smooth iff $\Delta \neq 0$. If $\Delta = 0$ it has exactly one singular point, a node if $c_4 \neq 0$ and a cusp if $c_4 = 0$. A smooth Weierstrass curve with O is an elliptic curve; conversely every elliptic curve over K is isomorphic to a smooth Weierstrass curve with $O \mapsto [0 : 1 : 0]$ (via $x \in L(2O)$, $y \in L(3O)$; proved in Chapter 8, item 5.6, cited here, not reproved). Reference: AEC III.1.4(a) (verify) and III.3.1. Paper: background (the singular reductions of Chapter 10: node = multiplicative, cusp = additive).

1.3 prop:ellcurves:coordchange (proposition; P). The admissible changes of coordinates are $x = u^2x' + r$, $y = u^3y' + su^2x' + t$ with $u \in K^\times$, $r, s, t \in K$; they form a group. They transform $ua_1' = a_1 + 2s$, $u^2a_2' = a_2 - sa_1 + 3r - s^2$, $u^3a_3' = a_3 + ra_1 + 2t$, $u^4a_4' = a_4 - sa_3 + 2ra_2 - (t + rs)a_1 + 3r^2 - 2st$, $u^6a_6' = a_6+ra_4+r^2a_2+r^3-ta_3-t^2-rta_1$; $u^2b_2' = b_2+12r$, $u^4b_4' = b_4+rb_2+6r^2$, $u^6b_6' = b_6+2rb_4+r^2b_2+4r^3$, $u^8b_8' = b_8+3rb_6+3r^2b_4+r^3b_2+3r^4$; $u^4c_4' = c_4$, $u^6c_6' = c_6$, $u^{12}\Delta' = \Delta$, $j' = j$, $\omega' = u\omega$. Every isomorphism of Weierstrass curves sending O to O is of this form (S; AEC III.3.1(b)). All formulas checked symbolically by substitution. Consequences used later: for $K = \mathbb{Q}$ and a prime q, $v_q(j) = 3v_q(c_4) - v_q(\Delta)$ does not depend on the model, and $v_q(\Delta)$ changes by multiples of 12 (the minimality criterion is Chapter 10). Reference: AEC III.1, Table 3.1. Paper: Definition def:terms ($v_q(j_E)$, $t_\ell(E, q)$), Lemma lem:twist (ii) ("$v_q(\Delta)$ ... otherwise 12 is subtracted"), the appendix (`valuation(E.j, q)`).

1.4 thm:ellcurves:jinv (theorem; P). (i) Two elliptic curves over K are isomorphic over $\bar{K}$ iff they have the same j. (ii) Every $j_0 \in K$ is the j-invariant of a curve over K: $y^2 + y = x^3$ ($j = 0$), $y^2 = x^3 + x$ ($j = 1728$), and $y^2 + xy = x^3 - \frac{36}{j_0-1728}x - \frac{1}{j_0-1728}$ for $j_0 \neq 0, 1728$ (checked symbolically). Proof of (i) for $\operatorname{char} K \neq 2, 3$ with short models: $j = j'$ gives u with $c_4' = u^{-4}c_4$, $c_6' = u^{-6}c_6$ over $\bar{K}$; the general characteristic is stated (AEC Appendix A). Reference: AEC III.1.4(b), (c) (verify). Paper: Definition def:terms (j_E), Lemma lem:twist ("the same j-invariant"), Proposition prop:ns and Section sec:anomaly (CM j-values excluded).

1.5 prop:ellcurves:aut (proposition; P, $\operatorname{char} K \neq 2, 3$). $\operatorname{Aut}_{\bar{K}}(E)$ (automorphisms fixing O) is $\{\pm 1\}$ if $j \neq 0, 1728$, μ_4 if $j = 1728$ and μ_6 if $j = 0$, acting by u with $u^4c_4 = c_4$, $u^6c_6 = c_6$, $r = s = t = 0$ on a short model; $[-1](x, y) = (x, -y - a_1x - a_3)$. Reference: AEC III.10.1 (verify). Paper: background for quadratic twists (10.2: for $j \neq 0, 1728$ every twist is quadratic).

1.6 ex:ellcurves:running-invariants (worked example; P). The invariants of the running examples (all recomputed from the a-invariants; models are the global minimal ones of `ecdata`):

curve	$[a_1, a_2, a_3, a_4, a_6]$	c_4	c_6	Δ	j
11a1	$[0, -1, 1, -10, -20]$	$496 = 2^4 \cdot 31$	20008	-11^5	$-2^{12}31^3/11^5$
11a3	$[0, -1, 1, 0, 0]$	16	-152	-11	$-2^{12}/11$
20a1	$[0, 1, 0, 4, 4]$	-176	-2368	$-2^8 \cdot 5^2$	$2^4 \cdot 11^3/5^2$
99a1 (99A1)	$[1, -1, 1, -2, 0]$	81	135	$3^3 \cdot 11$	$3^9/11$

curve	$[a_1,a_2,a_3,a_4,a_6]$	c_4	c_6	Δ	j
121a1	$[1,1,1,-30,-76]$	$1441 = 11\cdot 131$	54703	-11^2	$-11\cdot 131^3$
121c1	$[1,1,0,-2,-7]$	121	5203	-11^4	-11^2
121c2	$[1,1,0,-3632,82775]$	14641	-72809693	-11^8	$-11\cdot 131^3$
121d1	$[0,-1,1,-40,-221]$	$1936 = 2^4 11^2$	202312	-11^7	$-2^{12}/11$
539c1	$[1,0,1,170,-3237]$	-8183	2808827	$-7^9\cdot 11^2$	$167^3/(7^3\cdot 11^2)$
726a1	$[1,1,0,-35,-51]$	1705	31339	$2^6\cdot 3^3\cdot 11^3$	$5^3\cdot 31^3/(2^6\cdot 3^3)$
980i1	$[0,0,0,1568,-72716]$	75264	62826624	$-2^8\cdot 5\cdot 7^{11}$	$2^{19}3^3/(5\cdot 7^5)$

The writer may add 77c1 $[1,1,0,4,11]$ ($\Delta = -7^3\cdot 11^2$, same j as 539c1), 140b1 $[0,0,0,32,212]$ ($\Delta = -2^8\cdot 5\cdot 7^5$, same j as 980i1) and 539d1 $[0,1,1,-16,-66]$ ($\Delta = -7^6\cdot 11$, same j as 11a3). Reference: Cremona97, `ecdata`. Paper: the Example of Section 3, the remark on 99A1, the appendix (121a1, 726a1); the valuations $v_q(j)$ feed t_ℓ (e.g. $v_7(j(539c1)) = -3$, $t_3 = 1$; $v_7(j(980i1)) = -5$, $t_5 = 1$).

11.3.2 Section 2. The group law

2.1 `def:ellcurves:chordtangent` (construction and proposition; P). For $P, Q \in E(\bar{K})$ the line through P and Q (tangent if $P = Q$) meets E in a third point R counted with multiplicity (a cubic in one variable with two known roots); $P + Q$ is the third intersection of the line through R and O, i.e. $-R$, where $-(x,y) = (x, -y - a_1x - a_3)$. Explicit formulas: if $x_1 \neq x_2$, $\lambda = \frac{y_2-y_1}{x_2-x_1}$, $\nu = \frac{y_1x_2-y_2x_1}{x_2-x_1}$; if $P = Q$ and $2y_1 + a_1x_1 + a_3 \neq 0$, $\lambda = \frac{3x_1^2+2a_2x_1+a_4-a_1y_1}{2y_1+a_1x_1+a_3}$, $\nu = \frac{-x_1^3+a_4x_1+2a_6-a_3y_1}{2y_1+a_1x_1+a_3}$; then $x_3 = \lambda^2 + a_1\lambda - a_2 - x_1 - x_2$, $y_3 = -(\lambda + a_1)x_3 - \nu - a_3$. Duplication: $x(2P) = \dfrac{x^4 - b_4x^2 - 2b_6x - b_8}{4x^3 + b_2x^2 + 2b_4x + b_6}$. The 2-torsion points $\neq O$ are the points with $2y + a_1x + a_3 = 0$, whose x-coordinates are the roots of $4x^3 + b_2x^2 + 2b_4x + b_6$. Reference: AEC III.2, Algorithm 2.3. Paper: background.

2.2 `thm:ellcurves:grouplaw` (theorem; P, with associativity from 2.3). $(E(\bar{K}), +)$ is an abelian group with identity O and inverse as in 2.1; for every field $K \subset L \subset \bar{K}$, $E(L)$ is a subgroup; $\sigma(P+Q) = \sigma P + \sigma Q$ for $\sigma \in G_K$; addition and negation are morphisms. Commutativity, identity and inverses are immediate from the construction; associativity is 2.3 (a direct verification is possible but long; mention it). Reference: AEC III.2.2, III.3.6 (verify). Paper: background (every group of points: $E(\mathbb{Q})_{\mathrm{tors}}$, $E[\ell]$, $E(\mathbb{F}_q)$).

2.3 `prop:ellcurves:pic0` (proposition; P given Chapter 8, item 5.7). The bijection $\kappa\colon E(\bar{K}) \to \mathrm{Pic}^0(E)$, $P \mapsto [(P) - (O)]$ of Chapter 8 (cited, not reproved) satisfies $\kappa(P+Q) = \kappa(P) + \kappa(Q)$ for the chord-tangent law; hence $+$ is associative. Corollary: a divisor $D = \sum n_P(P)$ is principal iff $\deg D = 0$ and $\sum [n_P]P = O$ in E. Proof (the part done here): the line through P, Q and the vertical line through R of 2.1 have divisors $(P) + (Q) + (R) - 3(O)$ and $(R) + (-R) - 2(O)$, so $(P) + (Q) \sim (P+Q) + (O)$. Reference: AEC III.3.4, III.3.5 (verify). Paper: background (and the model for Jacobians, Chapters 8 and 14).

2.4 `ex:ellcurves:torsion-points` (worked example; P). On 11a1, $P = (5,5)$ has order 5: $2P = (16,-61)$, $3P = (16,60)$, $4P = (5,-6)$, $5P = O$. On 20a1, $(-1,0)$ has order 2, $(0,2)$ order 3 and $(4,10)$ order 6. On 14a1 $[1,0,1,4,-6]$, $(2,-5)$ has order 3 and $(9,23)$ order 6. On 27a1 $[0,0,1,0,-7]$, $(3,4)$ has order 3. On 99a1, $(0,0)$ has infinite order (its multiples have growing denominators; rank 1 by `ecdata`). All checked with the formulas of 2.1. Paper: background for torsion (Section sec:eis), and the data of 12.2.

11.3.3 Section 3. Isogenies

3.1 `def:ellcurves:isogeny` (definition and proposition; P given Chapter 8). An isogeny $\varphi\colon E_1 \to E_2$ is a morphism with $\varphi(O) = O$; it is constant or surjective (Chapter 8). Its degree is $[\bar{K}(E_1) : \varphi^*\bar{K}(E_2)]$ ($\deg 0 := 0$), with separable and inseparable degrees; φ is defined over K if it is G_K-equivariant on points. $\mathrm{Hom}(E_1, E_2)$ is an abelian group, $\mathrm{End}(E)$ a ring, $[m] \in \mathrm{End}(E)$; $\deg(\psi \circ \varphi) = \deg\psi \deg\varphi$. Reference: AEC III.4. Paper: background (isogeny classes, Section 1; Conjecture conj:eis; Section sec:oot).

3.2 `thm:ellcurves:isog-hom` (theorem; S). (i) An isogeny is a group homomorphism (through 2.3: φ_* on Pic^0). (ii) A nonconstant isogeny has finite kernel of order $\deg_s\varphi$; if φ is separable, $\#\ker\varphi = \deg\varphi$,

φ is unramified (Chapter 8, item 7.3(b), cited), and $\bar{K}(E_1)/\varphi^*\bar{K}(E_2)$ is Galois with group $\ker\varphi$ acting by translations. Reference: AEC III.4.8, III.4.10 (verify). Paper: background.

3.3 `thm:ellcurves:quotient` (theorem; R). Let $\Phi \subset E(\bar{K})$ be a finite subgroup. There are an elliptic curve E' and a separable isogeny $\varphi\colon E \to E'$ with $\ker\varphi = \Phi$, unique up to isomorphism of E'; if Φ is G_K-stable (as a set), E' and φ can be taken over K. Write $E' = E/\Phi$. Explicit equations: Velu's formulas (Velu 1971, add to `bib/ch09.bib`). Reference: AEC III.4.12 and Exercise 3.13 (verify). Paper: Conjecture conj:eis ("rational ℓ-isogeny"), 9.5, 9.8.

3.4 `thm:ellcurves:dual` (theorem; S). For a nonconstant isogeny $\varphi\colon E_1 \to E_2$ of degree m there is a unique isogeny $\hat{\varphi}\colon E_2 \to E_1$ with $\hat{\varphi}\circ\varphi = [m]_{E_1}$; then $\varphi\circ\hat{\varphi} = [m]_{E_2}$, $\widehat{\psi\circ\varphi} = \hat{\varphi}\circ\hat{\psi}$, $\widehat{\varphi+\psi} = \hat{\varphi}+\hat{\psi}$, $\widehat{[m]} = [m]$, $\deg\hat{\varphi} = \deg\varphi$, $\hat{\hat{\varphi}} = \varphi$; $\hat{\varphi}$ is defined over K if φ is. The degree is a positive definite quadratic form on $\mathrm{Hom}(E_1,E_2)$ ($\deg(\varphi+\psi) - \deg\varphi - \deg\psi$ is bilinear because $\widehat{\varphi+\psi} = \hat{\varphi}+\hat{\psi}$). Sketch: $\hat{\varphi}$ is $E_2 \to \mathrm{Pic}^0(E_2) \xrightarrow{\varphi^*} \mathrm{Pic}^0(E_1) \to E_1$. Reference: AEC III.6.1, III.6.2 (verify). Paper: background for 9.8 and for Chapter 14 ($E^\vee \to J_0(N) \to E$ is $[\deg\varphi_E]$).

3.5 `cor:ellcurves:degm` (corollary; P given 3.4). (i) $\deg[m] = m^2$ for $m \neq 0$. (ii) If $\mathrm{char}\,K \nmid m$ then $[m]$ is separable and $E[m] \simeq (\mathbb{Z}/m)^2$. (iii) If $\mathrm{char}\,K = p > 0$ then $E[p^e]$ is 0 for all e or $\mathbb{Z}/p^e$ for all e. Proof of (i): $\widehat{[m]} = [m]$ and $\widehat{[m]}\circ[m] = [\deg[m]]$ give $[m^2] = [\deg[m]]$; $\mathrm{End}(E)$ is torsion-free. (ii) from 3.2 and the structure theorem (Chapter 1): $\#E[d] = d^2$ for every $d \mid m$ forces $(\mathbb{Z}/m)^2$. (iii) R. Over $\mathbb{C}$ there is a second proof by lattices (7.4). Reference: AEC III.6.2(d), III.6.4 (verify). Paper: Section 1.1 ($E[\ell]$ two-dimensional over $\mathbb{F}_\ell$), Lemma lem:IV ($E[3] = T/3T$).

3.6 `prop:ellcurves:cyclicfactor` (proposition; P given 3.3, 3.5; $\mathrm{char}\,K = 0$). Let $\varphi\colon E \to E'$ be a nonconstant isogeny and $a \geq 1$ the largest integer with $E[a] \subset \ker\varphi$. Then $\ker\varphi \simeq \mathbb{Z}/a \times \mathbb{Z}/ab$ for some $b \geq 1$, and $\varphi = \psi\circ[a]$ where $\psi\colon E \to E'$ is the isogeny with $\ker\psi = a\cdot\ker\varphi \simeq \mathbb{Z}/b$ (cyclic). If φ is defined over K, so is ψ ($a\cdot\ker\varphi$ is G_K-stable because $[a]$ commutes with G_K). An isogeny over K with cyclic kernel of order n is a *cyclic n-isogeny*; its kernel contains exactly one subgroup of each order $d \mid n$, and that subgroup is G_K-stable, so a cyclic n-isogeny over K factors into cyclic isogenies over K of prime degrees. Proof: finite subgroups of $E(\bar{K})_{\mathrm{tors}} \simeq (\mathbb{Q}/\mathbb{Z})^2$ have at most two generators (Chapter 1). Reference: elementary; AEC III.4. Paper: 9.6, 9.8 (cyclic 27-isogeny), 9.9.

11.3.4 Section 4. Division polynomials

4.1 `def:ellcurves:divpol` (definition; P). $\psi_0 = 0$, $\psi_1 = 1$, $\psi_2 = 2y + a_1x + a_3$, $\psi_3 = 3x^4 + b_2x^3 + 3b_4x^2 + 3b_6x + b_8$, $\psi_4 = \psi_2\big(2x^6 + b_2x^5 + 5b_4x^4 + 10b_6x^3 + 10b_8x^2 + (b_2b_8 - b_4b_6)x + b_4b_8 - b_6^2\big)$, and for $m \geq 2$, resp. $m \geq 3$,

$$\psi_{2m+1} = \psi_{m+2}\psi_m^3 - \psi_{m-1}\psi_{m+1}^3, \qquad \psi_2\psi_{2m} = \psi_m\big(\psi_{m+2}\psi_{m-1}^2 - \psi_{m-2}\psi_{m+1}^2\big).$$

These lie in $\mathbb{Z}[a_1,\dots,a_6,x,y]$; reduce with $\psi_2^2 = 4x^3 + b_2x^2 + 2b_4x + b_6$. Reference: AEC Exercise 3.7; Washington, *Elliptic Curves: Number Theory and Cryptography*, Section 3.2 (add to `bib/ch09.bib`). Paper: Proposition prop:ns, the appendix (`elldivpol(E, 3)` in `ns3`).

4.2 `prop:ellcurves:divpol-deg` (proposition; P by induction). For odd m, $\psi_m \in \mathbb{Z}[a_i][x]$ has degree $(m^2-1)/2$ and leading coefficient m; for even m, $\psi_m/\psi_2 \in \mathbb{Z}[a_i][x]$ has degree $(m^2-4)/2$ and leading coefficient $m/2$. Checked for $m \leq 9$ on 11a1 and 99a1 with the recursion above. Reference: AEC Exercise 3.7. Paper: background (PLAN.md scope: "degree $(m^2-1)/2$ for odd m").

4.3 `thm:ellcurves:multm` (theorem; S). For $P = (x,y) \in E$ with $mP \neq O$: $x(mP) = x - \dfrac{\psi_{m-1}\psi_{m+1}}{\psi_m^2}(P)$, and a similar formula for $y(mP)$ with ψ_m^3 in the denominator; $mP = O$ iff $\psi_m(P) = 0$ (for $P \neq O$, $m \geq 2$). For $m = 2$ this is the duplication formula of 2.1. Sketch: induction with the addition formulas, or comparison of divisors: $\mathrm{div}(\psi_m) = \sum_{T\in E[m]}(T) - m^2(O)$ when $\mathrm{char}\,K \nmid m$. Reference: AEC Exercise 3.7; Washington EC Theorem 3.6 (verify). Paper: background.

4.4 `cor:ellcurves:divpol-roots` (corollary; P given 3.5 and 4.3). Let m be odd with $\mathrm{char}\,K \nmid m$. The roots of $\psi_m(x)$ in $\bar{K}$ are exactly the x-coordinates of the points of $E[m] \smallsetminus \{O\}$; they are simple and there are $(m^2-1)/2$ of them. Indeed $x(P) = x(Q)$ iff $Q = \pm P$, no nonzero point of $E[m]$ has order 2, so

$P \mapsto x(P)$ is a G_K-equivariant bijection $(E[m] \smallsetminus \{O\})/\{\pm 1\} \to \{\text{roots}\}$, and the count $(m^2-1)/2$ equals the degree. Paper: Proposition prop:ns.

4.5 `prop:ellcurves:psi3` (proposition; P). (i) $x(2P) - x(P) = -\psi_3(x)/\psi_2(P)^2$ (polynomial identity, checked symbolically), so for $P \neq O$ not of order 2: $3P = O$ iff $\psi_3(x(P)) = 0$. (ii) Let $\operatorname{char} K \neq 3$. The four roots of ψ_3 correspond bijectively and G_K-equivariantly to the four lines of the $\mathbb{F}_3$-plane $E[3]$ (a line $\{O, P, -P\}$ goes to $x(P)$). Hence the factorisation pattern of ψ_3 over K is the orbit pattern of $\bar\rho_{E,3}(G_K)$ on $\mathbb{P}^1(\mathbb{F}_3)$ (Chapter 5, galois:quartic-on-lines; Chapter 1, 6.8): ψ_3 has a root in K iff $E[3]$ has a G_K-stable line iff E has a K-rational 3-isogeny (9.5). (iii) For $K = \mathbb{Q}$, $\operatorname{disc}(\psi_3) \in -3 \cdot (\mathbb{Q}^\times)^2$: the sign of the permutation of the four lines is $\det \bar\rho_{E,3} = \chi_{-3}$ (5.2 and Chapter 1, 5.4(ii)), so $\mathbb{Q}(\sqrt{\operatorname{disc}\psi_3}) = \mathbb{Q}(\sqrt{-3})$. Reference: elementary given the cited chapters. Paper: Proposition prop:ns (last two sentences of statement and proof: "the four roots of the 3-division polynomial correspond to the four lines of $\mathbb{F}_3^2$", "factors as two quadratics ... irreducible otherwise", "no rational root when $E[3]$ is irreducible"), Section sec:data, Section sec:anomaly, the appendix (`ns3`).

4.6 `ex:ellcurves:psi3-examples` (worked example; P, all factorisations and discriminants recomputed with sympy; images from `ecdata`):

curve	ψ_3 over $\mathbb{Q}$	$\operatorname{disc}\psi_3$	image at 3
11a1	$3x^4 - 4x^3 - 60x^2 - 237x - 21$, irreducible	$-3^3 \cdot 11^{10}$	surjective
121a1	$3x^4 + 5x^3 - 177x^2 - 909x - 1249$, irreducible	$-3^3 \cdot 11^4$	surjective
726a1	$(x^2 - x - 74)(3x^2 + 8x + 20)$	$-2^{12}3^9 11^6$	3Ns
338d1 $[1, 1, 0, 504, -13112]$	$(x^2 + 29x + 1858)(3x^2 - 82x - 172)$	$-2^6 3^3 13^{18}$	3Ns
20a1	$x(3x^3 + 4x^2 + 24x + 48)$	$-2^{16}3^3 5^4$	3B.1.1
14a1 $[1, 0, 1, 4, -6]$	$(x-2)(3x+1)(x^2 + 2x + 13)$	$-2^{12}3^3 7^6$	3Cs.1.1
27a1 (CM)	$3x(x-3)(x^2 + 3x + 9)$	-3^{21}	3Cs.1.1
6912d1 $y^2 = x^3 - 6x - 8$	$3(x^2-4x-2)(x^2+4x+6)$	$-2^{18}3^9$	3Ns (Chapter 5 example)

For 726a1 the quadratic factors have discriminants $297 = 3^3 \cdot 11$ and $-176 = -2^4 \cdot 11$, so they split over $\mathbb{Q}(\sqrt{33})$ and $\mathbb{Q}(\sqrt{-11})$, and the third quadratic subfield of their compositum is $\mathbb{Q}(\sqrt{-3})$, as Chapter 5 (e) predicts. For 20a1 the root $x = 0$ is the rational 3-torsion point $(0, 2)$; for 14a1 the root 2 is the rational point $(2, -5)$ and the root $-1/3$ gives the points over $\mathbb{Q}(\sqrt{-3})$ spanning the line μ_3 (the y-quadratic has discriminant $-784/27 = -3 \cdot (28/9)^2$). Every discriminant is -3 times a square, as 4.5(iii) says. Paper: Proposition prop:ns, Section sec:anomaly (726a1), Section sec:eis (3B.1.1, 3Cs.1.1).

11.3.5 Section 5. The Weil pairing, the Tate module and the Galois representations

5.1 `thm:ellcurves:weil` (theorem; S). Let $m \geq 1$ with $\operatorname{char} K \nmid m$. There is a pairing $e_m \colon E[m] \times E[m] \to \mu_m$ which is (i) bilinear, (ii) alternating ($e_m(T,T) = 1$, hence $e_m(S,T) = e_m(T,S)^{-1}$), (iii) nondegenerate, (iv) Galois-equivariant: $e_m(\sigma S, \sigma T) = \sigma(e_m(S,T))$ for $\sigma \in G_K$, (v) compatible: $e_{mm'}(S,T) = e_m(m'S, T)$ for $S \in E[mm']$, $T \in E[m]$, (vi) adjoint: $e_m(S, \hat\varphi T) = e_m(\varphi S, T)$ for an isogeny $\varphi \colon E \to E'$, $S \in E[m]$, $T \in E'[m]$. Sketch: for $T \in E[m]$ choose g with $\operatorname{div}(g) = [m]^*(T) - [m]^*(O)$ (2.3); $e_m(S,T) = g(X+S)/g(X)$. Reference: AEC III.8.1 (verify). Over $\mathbb{C}$, 7.4 gives a direct proof of (i) to (iii) by a determinant. Paper: Section 4.2 ($\det\rho = \chi_\ell$, $A(1)$), Lemma lem:h0 ("comparing determinants").

5.2 `cor:ellcurves:det` (corollary; P from 5.1). (i) For $\sigma \in G_K$ acting on $E[m]$ in an $\mathbb{Z}/m$-basis, $\det = \chi_m^{\mathrm{cyc}}(\sigma)$ where $\sigma(\zeta) = \zeta^{\chi_m^{\mathrm{cyc}}(\sigma)}$ on μ_m. (ii) If $E[m] \subset E(K)$ then $\mu_m \subset K$. (iii) If $L \subset E[\ell]$ is a G_K-stable line, then $E[\ell]/L \simeq \operatorname{Hom}(L, \mu_\ell)$ as G_K-modules; in particular a K-rational point of order ℓ

gives an exact sequence $0 \to \mathbb{Z}/\ell \to E[\ell] \to \mu_\ell \to 0$, and the two characters on a reducible $E[\ell]$ multiply to $\bar\chi_\ell$. Proof: $e_m(\sigma S, \sigma T) = e_m(S,T)^{\det}$ for σ acting by a matrix. Reference: AEC III.8.1.1 (verify). Paper: Proposition prop:ns ("$\det \bar\rho_{E,3} = \chi_{-3}$ cuts out $\mathbb{Q}(\sqrt{-3})$"), Conjecture conj:eis and Section sec:eis ($\mathbb{Z}/3 \oplus \mu_3$), Lemma lem:h0.

5.3 `def:ellcurves:tate` (definition and proposition; P given 3.5 and Chapter 1's inverse limits). $T_\ell E = \varprojlim_n E[\ell^n]$ (transition maps $[\ell]$) is a free $\mathbb{Z}_\ell$-module of rank 2; $V_\ell E = T_\ell E \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$; $T_\ell E/\ell^n T_\ell E \simeq E[\ell^n]$; $V_\ell E/T_\ell E \simeq E[\ell^\infty] \simeq (\mathbb{Q}_\ell/\mathbb{Z}_\ell)^2$. For $\ell = p = \operatorname{char} K > 0$, $T_p E$ is 0 or $\mathbb{Z}_p$ (R). The Weil pairings assemble to a perfect alternating G_K-equivariant pairing $e\colon T_\ell E \times T_\ell E \to \mathbb{Z}_\ell(1)$ (P from 5.1(v)). Reference: AEC III.7, III.8. Paper: Section 1.1 (T, V), Lemma lem:IV (T_3E, $E[3] = T/3T$), Lemma lem:tamfactor (basis e_1, e_2 of T).

5.4 `def:ellcurves:rho` (definition and proposition; P). $\rho_{E,\ell}\colon G_K \to \operatorname{Aut}_{\mathbb{Z}_\ell}(T_\ell E) \simeq \mathrm{GL}_2(\mathbb{Z}_\ell)$ (after a choice of basis), $\bar\rho_{E,\ell}\colon G_K \to \operatorname{Aut}(E[\ell]) \simeq \mathrm{GL}_2(\mathbb{F}_\ell)$, and $\rho \bmod \ell^k$ is the action on $E[\ell^k] = T/\ell^k T$. (i) $\rho_{E,\ell}$ is continuous: $E[\ell^n]$ is pointwise fixed by $G_{K(E[\ell^n])}$, where $K(E[\ell^n])$ (coordinates of ℓ^n-torsion points) is a finite Galois extension. (ii) $\det \rho_{E,\ell} = \chi_\ell^{\mathrm{cyc}}$ (5.3). (iii) An isogeny $\varphi\colon E \to E'$ over K induces an injective G_K-map $\varphi_\ell\colon T_\ell E \to T_\ell E'$ with $\operatorname{coker}\varphi_\ell \simeq (\ker\varphi)[\ell^\infty]$; so $V_\ell E \simeq V_\ell E'$, and $T_\ell E \simeq T_\ell E'$ if $\ell \nmid \deg\varphi$. Reference: AEC III.7, Serre's *Abelian ℓ-adic representations* I.1 (SerreAbelian). Paper: Section 1.1 (V, T, ρ, $\bar\rho$), Theorem thm:main hypothesis (the restriction of $\bar\rho$ to $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{\ell^*}))$, Chapter 11), Lemma lem:h0 ($\rho \bmod \ell^k$), Section 4.2 ($M_{f,\lambda}$ and $T_\ell E$ homothetic, Chapter 15).

5.5 `cor:ellcurves:dual-T` (corollary; P from 5.3). The Weil pairing gives G_K-isomorphisms $T_\ell E \simeq \operatorname{Hom}_{\mathbb{Z}_\ell}(T_\ell E, \mathbb{Z}_\ell(1)) = T_\ell E^\vee(1)$ and $E[\ell^k] \simeq \operatorname{Hom}(E[\ell^k], \mu_{\ell^k})$; equivalently $T_\ell E^\vee \simeq T_\ell E(-1)$ and $\rho^\vee \simeq \rho \otimes (\chi_\ell^{\mathrm{cyc}})^{-1}$ (Chapter 4's $V^\vee \simeq V \otimes \det^{-1}$ in dimension two). Consequently $\mathrm{ad}^0(T_\ell E^\vee) = \mathrm{ad}^0(T_\ell E)$ as G_K-modules (Chapter 4), and a lattice homothetic to $T_\ell E^\vee$ has the same adjoint as $T_\ell E$. Paper: Section 4.2 ("the lattice $M_{f,\lambda}$ of DFG04 and $T_\ell E$ are homothetic"), which is right up to dualising (important_notes.md point 26: the cohomology H^1 gives the dual; point 40: for the optimal curve $M_{f,\lambda} = H^1(E, \mathbb{Z}_\ell) = \operatorname{Hom}(T_\ell E, \mathbb{Z}_\ell)$, which by this item is exactly $T_\ell E(-1)$); Chapter 18 (T^D).

5.6 `cor:ellcurves:complex-conj` (corollary; P). Let $E/\mathbb{Q}$, ℓ odd, c a complex conjugation. Then $\bar\rho_{E,\ell}(c)$ has order dividing 2 and determinant $\chi_\ell(c) = -1$, hence eigenvalues $1, -1$, trace 0, and is conjugate to $\mathrm{diag}(1,-1)$; the same holds for $\rho_{E,\ell}(c)$ in $\mathrm{GL}_2(\mathbb{Z}_\ell)$. (Directly: $E(\mathbb{R})[\ell] \simeq \mathbb{Z}/\ell$ because $E(\mathbb{R}) \simeq S^1$ or $S^1 \times \mathbb{Z}/2$.) Paper: Proposition prop:ns ("contain an element of determinant -1 and trace 0 (complex conjugation)"); Chapter 11 uses it for "irreducible implies absolutely irreducible" (important_notes.md point 10).

5.7 `conv:ellcurves:frobenius` (convention and warning). C5 of Part 1 typeset as a `convention`: arithmetic Frobenius in group-theoretic statements, $\rho_{E,\ell}(\mathrm{Frob}_p)$ with characteristic polynomial $X^2 - a_p X + p$; the Euler factor of E is $\det(1 - \mathrm{Frob}_p\, p^{-s} \mid V_\ell E)^{-1} = \det(1 - \Phi_p p^{-s} \mid V_\ell E^\vee)^{-1}$, while $\det(1 - \Phi_p p^{-s} \mid V_\ell E)^{-1} = L_p(E, s+1)$ (by 5.5). On $\mathrm{ad}^0 V$ the two conventions differ only where an eigenvalue $q^{\pm 1}$ occurs (important_notes.md point 11; the L-factors of the adjoint are Chapters 18, 19).

11.3.6 Section 6. Endomorphisms and complex multiplication

6.1 `thm:ellcurves:end` (theorem; R, with P for the parts over $\mathbb{C}$ in 7.4). $\operatorname{End}_{\bar K}(E)$ has no zero divisors and deg is a positive definite quadratic form on it; it is isomorphic to $\mathbb{Z}$, to an order in an imaginary quadratic field, or (only if $\operatorname{char} K = p > 0$) to a maximal order in the quaternion algebra over $\mathbb{Q}$ ramified exactly at p and ∞. Reference: AEC III.9.4 and V.3.1 (verify). Paper: background for Lemma lem:h0.

6.2 `prop:ellcurves:normalised` (proposition; S). Let $\operatorname{char} K = 0$. The map $\alpha \mapsto [\alpha]^*\omega/\omega \in \bar K$ is an injective ring homomorphism $\operatorname{End}_{\bar K}(E) \to \bar K$; with it, $\operatorname{End}_{\bar K}(E) \simeq \mathcal{O} \subset \mathbb{C}$ "normalised". (i) Endomorphisms defined over K map into K; hence $\operatorname{End}_{\mathbb{Q}}(E) = \mathbb{Z}$ for every $E/\mathbb{Q}$ (an imaginary quadratic integer in $\mathbb{Q}$ is in $\mathbb{Z}$). (ii) If $\operatorname{End}_{\bar{\mathbb{Q}}}(E) = \mathcal{O} \subset F$ (F imaginary quadratic) for $E/\mathbb{Q}$, every endomorphism is defined over F, and for $\sigma \in G_{\mathbb{Q}}$ one has $\sigma \circ [\alpha] \circ \sigma^{-1} = [\sigma(\alpha)]$ on points, i.e. $[\alpha]^\sigma = [\alpha]$ if $\sigma|_F = 1$ and $[\bar\alpha]$ otherwise. Sketch: $([\alpha]^\sigma)^*\omega = \sigma(\alpha)\omega$ because ω is defined over $\mathbb{Q}$. Reference: AEC III.5.6 (verify); Silverman ATAEC II.2 (SilvermanATAEC). Paper: Lemma lem:h0 (via 6.4); Chapter 6, item 6.5 quotes this.

6.3 `def:ellcurves:cm` (definition). An elliptic curve E over a field of characteristic 0 has *complex multiplication* (CM) if $\operatorname{End}_{\bar K}(E) \neq \mathbb{Z}$; then $\operatorname{End}_{\bar K}(E) \simeq \mathcal{O}$, an order in an imaginary quadratic field F,

and E has CM by $\mathcal{O}$ (or by F). CM is an isogeny invariant (End $\otimes\mathbb{Q}$ is: $\alpha \mapsto \varphi\alpha\hat{\varphi}/\deg\varphi$; P). Paper: Conjecture conj:euler, Conjecture conj:eis, Proposition prop:ns, Lemma lem:h0, Section sec:data ("not CM").

6.4 `thm:ellcurves:13j` (theorem; R, numerical check P). If $E/\mathbb{Q}$ has CM by $\mathcal{O}$, then $\mathcal{O}$ has class number one, so $\operatorname{disc}\mathcal{O} \in \{-3,-4,-7,-8,-11,-12,-16,-19,-27,-28,-43,-67,-163\}$, and $j(E)$ is respectively 0, 1728, -3375, 8000, -32768, 54000, 287496, -884736, -12288000, 16581375, -884736000, -147197952000, -262537412640768000; conversely each value is the j-invariant of a curve over $\mathbb{Q}$ with CM by that order. These are $j(\tau)$ for $\tau = \frac{-1+\sqrt{D}}{2}$ or $\frac{\sqrt{D}}{2}$ (checked to 40 digits with `mpmath.kleinj`). Examples: 27a1, 36a1 ($j=0$, $F=\mathbb{Q}(\sqrt{-3})$), 32a1 ($j=1728$, $F=\mathbb{Q}(i)$), 121b1 ($j=-32768=-2^{15}$, $F=\mathbb{Q}(\sqrt{-11})$), 27a2 and 27a4 ($j=-12288000=-2^{15}\cdot 3\cdot 5^3$, order of discriminant -27). Reference: AEC Appendix C, Section 11 (verify the location); Silverman ATAEC II (Theorem II.4.3 and the class number one list, verify); Cox, *Primes of the form x^2+ny^2* (add to `bib/ch09.bib`). Paper: Lemma lem:h0 ("only for curves with complex multiplication by that field"), the data sections ("not CM"); important_notes.md point 14.

6.5 `thm:ellcurves:cm-ap` (theorem; (i), (ii) S, rest R). Let $E/\mathbb{Q}$ have CM by an order in F, χ_F the character of F. (i) $\rho_{E,\ell} \simeq \rho_{E,\ell}\otimes\chi_F$ for every ℓ. (ii) For every prime p of good reduction inert in F, $a_p(E)=0$ (Deuring: the reduction is supersingular). (iii) For p good and split in F, $a_p(E)\neq 0$ (ordinary). (iv) Every prime ramified in F divides the conductor N (for instance via $\rho_{E,\ell}\simeq \operatorname{Ind}_{G_F}^{G_{\mathbb{Q}}}\psi$ and the conductor of an induced representation, or Deuring's identification of $L(E,s)$ with a Hecke L-function); hence a curve with CM by $\mathbb{Q}(\sqrt{\ell^*})$ has $\ell \mid N$. (v) Conversely, if $a_p(E)=0$ for a set of primes inert in a quadratic field F of density $1/2$ (all but finitely many), then E has CM by F (Serre's open image theorem; Chapter 11). Sketch of (i), (ii): take $\alpha\in\mathcal{O}$ with $\bar{\alpha}=-\alpha$; by 6.2(ii), $\alpha\colon V_\ell E\to V_\ell E$ satisfies $\rho(\sigma)\alpha\rho(\sigma)^{-1}=\chi_F(\sigma)\alpha$, an isomorphism $\rho\to\rho\otimes\chi_F$; then $\operatorname{tr}\rho(\sigma)=\chi_F(\sigma)\operatorname{tr}\rho(\sigma)$ and $a_p=\operatorname{tr}\rho(\mathrm{Frob}_p)=0$ when $\chi_F(\mathrm{Frob}_p)=-1$ (8.6). Checked: $a_p=0$ at every good inert $p<500$ and $a_p\neq 0$ at every good split $p<500$ for 27a1, 36a1, 32a1, 121b1; for all 462 CM curves of conductor below 10,000 in `ecdata`, every prime dividing $\operatorname{disc}F$ divides N. Reference: AEC Appendix C.11; ATAEC II.10 (verify); Deuring 1941 (add to bib); Serre-Tate 1968 for (iv) (add to bib) or Serre's Local Fields VI for conductors of induced representations. Paper: Lemma lem:h0 (last sentence of statement and proof; the correct reading uses the quadratic character χ_{ℓ^*}, not the cyclotomic χ_ℓ: important_notes.md point 5; by (iv) the CM alternative cannot occur when $\ell\nmid N$: point 14), Conjecture conj:euler ("without complex multiplication"). Chapter 6, item 6.5 is the number-theoretic preview of this item and cites it.

11.3.7 Section 7. Elliptic curves over $\mathbb{C}$

7.1 `def:ellcurves:lattice` (definition and proposition; P). A lattice $\Lambda=\mathbb{Z}\omega_1+\mathbb{Z}\omega_2\subset\mathbb{C}$ (ω_1,ω_2 linearly independent over $\mathbb{R}$), oriented by $\operatorname{Im}(\omega_2/\omega_1)>0$; homothety $\Lambda'=c\Lambda$; covolume $\operatorname{vol}(\Lambda)=|\operatorname{Im}(\bar{\omega}_1\omega_2)|$ (the area of a fundamental parallelogram). $\sum_{\omega\in\Lambda\smallsetminus 0}|\omega|^{-s}$ converges for $s>2$ (count lattice points in annuli; RHB-style estimate). Eisenstein series $G_{2k}(\Lambda)=\sum_{\omega\neq 0}\omega^{-2k}$ ($k\geq 2$), $g_2=60G_4$, $g_3=140G_6$, $\Delta(\Lambda)=g_2^3-27g_3^2$, $j(\Lambda)=1728g_2^3/\Delta$; $G_{2k}(c\Lambda)=c^{-2k}G_{2k}(\Lambda)$, so j depends only on the homothety class. Reference: AEC VI.3; DiamondShurman 1.4. Paper: background (Section 1: "the area of the period lattice", Chapter 14). Physics aside (one sentence): a doubly periodic function is a Fourier series on a torus.

7.2 `thm:ellcurves:wp` (theorem; P). $\wp(z)=z^{-2}+\sum_{\omega\neq 0}\big((z-\omega)^{-2}-\omega^{-2}\big)$ converges normally on compact sets away from Λ, is even, meromorphic, Λ-periodic with double poles exactly at Λ; $\wp(z)=z^{-2}+\sum_{k\geq 1}(2k+1)G_{2k+2}z^{2k}$; $\wp'(z)^2=4\wp(z)^3-g_2\wp(z)-g_3$ (compare principal parts; Liouville); $\Delta(\Lambda)\neq 0$ (the values $\wp(\omega_1/2)$, $\wp(\omega_2/2)$, $\wp((\omega_1+\omega_2)/2)$ are the distinct roots of $4x^3-g_2x-g_3$); the field of Λ-periodic meromorphic functions is $\mathbb{C}(\wp,\wp')$. Reference: AEC VI.2, VI.3 (VI.3.1, VI.3.2, VI.3.5, verify). Paper: background.

7.3 `thm:ellcurves:cplxuniform` (theorem; (i) P, (ii) R with two proof routes named). (i) $z\mapsto[\wp(z):\wp'(z):1]$ ($z\notin\Lambda$), $0\mapsto O$, is a complex-analytic group isomorphism $\mathbb{C}/\Lambda\to E_\Lambda(\mathbb{C})$, $E_\Lambda\colon y^2=4x^3-g_2x-g_3$, under which dx/y pulls back to dz (the addition theorem of $\wp$ matches the chord-tangent law of 2.1). (ii) (Uniformisation.) For $A,B\in\mathbb{C}$ with $4A^3+27B^2\neq 0$ there is a unique lattice Λ with $g_2(\Lambda)=-4A$,

$g_3(\Lambda) = -4B$; so every $E/\mathbb{C}$ is $\mathbb{C}/\Lambda$ for a lattice unique up to homothety. Routes: surjectivity of $j\colon \mathfrak{H} \to \mathbb{C}$ (Chapter 12, valence formula), or $\Lambda = \{\int_\gamma \omega : \gamma \in H_1(E(\mathbb{C}), \mathbb{Z})\}$ and Abel-Jacobi (Chapter 8). Reference: AEC VI.3.6, VI.5.1 (verify). Paper: background; Section 1 (period lattice), Chapter 14.

7.4 `prop:ellcurves:cplxisog` (proposition; P). (i) Holomorphic maps $\mathbb{C}/\Lambda_1 \to \mathbb{C}/\Lambda_2$ sending 0 to 0 are $z \mapsto \alpha z$ with $\alpha\Lambda_1 \subset \Lambda_2$; they are group homomorphisms; isomorphisms iff $\alpha\Lambda_1 = \Lambda_2$; so $E_{\Lambda_1} \simeq E_{\Lambda_2}$ iff the lattices are homothetic. (ii) For $\alpha \neq 0$ the degree is $[\Lambda_2 : \alpha\Lambda_1]$, the kernel $\alpha^{-1}\Lambda_2/\Lambda_1$; $[m]$ has degree m^2 and $E[m] = \frac{1}{m}\Lambda/\Lambda \simeq (\mathbb{Z}/m)^2$ (second proof of 3.5). (iii) $\mathrm{End}(\mathbb{C}/\Lambda) = \{\alpha : \alpha\Lambda \subset \Lambda\}$ is $\mathbb{Z}$ or an order in an imaginary quadratic field, the latter iff $\tau = \omega_2/\omega_1$ is imaginary quadratic (P: α satisfies a monic integral quadratic, and $\alpha \notin \mathbb{R}$ if $\alpha \notin \mathbb{Z}$). (iv) For an oriented basis, $e_m(\omega_1/m, \omega_2/m) = e^{2\pi i/m}$ up to the sign convention C7 (AEC Chapter VI, exercises; verify the statement and the sign before printing), and e_m is $\exp(2\pi i \det(\cdot,\cdot)/m)$ on $\frac{1}{m}\Lambda/\Lambda$. Reference: AEC VI.4 (verify). Paper: background.

7.5 `thm:ellcurves:moduli` (theorem; P for (i), (ii); R for (iii) with forward pointer to Chapter 12). (i) Every lattice is homothetic to $\Lambda_\tau = \mathbb{Z} + \mathbb{Z}\tau$ with $\tau = \omega_2/\omega_1 \in \mathfrak{H}$ (oriented basis), and Λ_τ, $\Lambda_{\tau'}$ are homothetic iff $\tau' = \frac{a\tau+b}{c\tau+d}$ with $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$ (change of oriented basis). So homothety classes of lattices, isomorphism classes of $E/\mathbb{C}$, and $\mathrm{SL}_2(\mathbb{Z})\backslash\mathfrak{H}$ are in bijection. (ii) $j(\tau) := j(\Lambda_\tau)$ is holomorphic on $\mathfrak{H}$ and $\mathrm{SL}_2(\mathbb{Z})$-invariant; $j(i) = 1728$, $j(e^{2\pi i/3}) = 0$. (iii) $j\colon \mathrm{SL}_2(\mathbb{Z})\backslash\mathfrak{H} \to \mathbb{C}$ is a bijection; the standard fundamental domain $\{|\operatorname{Re}\tau| \le \frac{1}{2}, |\tau| \ge 1\}$; $j(\tau) = q^{-1} + 744 + 196884q + \cdots$, $q = e^{2\pi i\tau}$ (Chapter 12). Reference: AEC VI.4, VI.5 and Appendix C; DiamondShurman 1.1, 1.4. Paper: background (PLAN.md scope: "isomorphism classes = lattices up to homothety = points of the upper half plane modulo $\mathrm{SL}_2(\mathbb{Z})$"); Chapter 12 builds $X_0(N)$ on this.

7.6 `def:ellcurves:periodlattice` (definition and proposition; S). Let $E/\mathbb{Q}$ be given by a global minimal model (Chapter 10) with Neron differential $\omega = dx/(2y + a_1x + a_3)$. Its period lattice is $\Lambda_E = \{\int_\gamma \omega : \gamma \in H_1(E(\mathbb{C}), \mathbb{Z})\}$, and $E(\mathbb{C}) \simeq \mathbb{C}/\Lambda_E$ with $\omega \leftrightarrow dz$. (i) Λ_E is stable under complex conjugation; if $\Delta > 0$, $\Lambda_E = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2$ with $\omega_1 > 0$ and $\omega_2 \in i\mathbb{R}$ (rectangular; $E(\mathbb{R})$ has two components); if $\Delta < 0$, $\Lambda_E = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2$ with $\omega_1 > 0$ and $\operatorname{Re}\omega_2 = \pm\omega_1/2$ ($E(\mathbb{R})$ connected). ω_1 is the least positive real period. (ii) $\mathrm{vol}(\Lambda_E) = \omega_1 \operatorname{Im}\omega_2 = \frac{i}{2}\int_{E(\mathbb{C})} \omega \wedge \bar{\omega}$. (iii) Computation by the arithmetic-geometric mean (Cremona97 Section 3.7; Cohen's *A course in computational algebraic number theory*, Algorithm 7.4.7, add to `bib/ch09.bib`; PARI `E.omega`, `ellperiods`). (iv) Isogenous curves have commensurable period lattices ($\varphi^*\omega' = c\,\omega$, $c \in \mathbb{Q}^\times$, the subject of the Manin constant, Chapter 14). Reference: Cremona97 Chapter III; AEC VI. Paper: Section 1 ("$\deg\varphi_E$ is $\langle f, f\rangle$ divided by the area of the period lattice of E", Zagier85, Watkins02); Chapter 14 states the exact formula.

7.7 `ex:ellcurves:11a1-periods` (worked example; P numerically). For 11a1 ($\Delta = -11^5 < 0$), by the AGM: $\omega_1 = 1.2692093042795534\ldots$ (agrees with `ecdata`'s $\Omega = 1.26920930427955$), $\omega_2 = -\omega_1/2 + 1.4588166169384952\ldots\, i$, $\mathrm{vol}(\Lambda_E) = 1.8515436234559593\ldots$; $\tau = \omega_2/\omega_1$ reduced to the fundamental domain is $-\frac{1}{2} + 1.1493901061232524\ldots\, i$, and $1728\,J(\tau) = -757.67263786\ldots = -122023936/161051 = j(11a1)$ to 40 digits. Paper: background for Section 1 and Chapter 14 (11a1 has $\deg\varphi_E = 1$).

11.3.8 Section 8. Elliptic curves over finite fields

8.1 `def:ellcurves:frobendo` (definition and proposition; S). Let $E/\mathbb{F}_\mathfrak{q}$, $\mathfrak{q} = p^r$. The $\mathfrak{q}$-th power map $\pi\colon (x, y) \mapsto (x^\mathfrak{q}, y^\mathfrak{q})$ is an endomorphism of E of degree $\mathfrak{q}$, purely inseparable, and acts on $E(\overline{\mathbb{F}}_p)$ as the arithmetic Frobenius $\mathrm{Frob}_\mathfrak{q}$ (C5); π is written $\pi_\mathfrak{q}$ when the field must be shown (Appendix B, B-C2(g)). $E(\mathbb{F}_{\mathfrak{q}^n}) = \ker(1 - \pi^n)$. $1 - \pi$ is separable ($(1-\pi)^*\omega = \omega \neq 0$; AEC III.5.5 and II.4.2(c), verify), so $\#E(\mathbb{F}_\mathfrak{q}) = \deg(1 - \pi)$. Put $a_\mathfrak{q} = \mathfrak{q} + 1 - \#E(\mathbb{F}_\mathfrak{q})$. Reference: AEC II.2, III.5, V.1. Paper: the table of Section 1 ($a_q(E')$), Lemma lem:old ($X^2 - a_p(h)X + p$), Definition def:terms ($a_q(E)$).

8.2 `thm:ellcurves:hasse` (theorem, Hasse; P given 3.4 and 8.1). $|a_\mathfrak{q}| \le 2\sqrt{\mathfrak{q}}$. Proof: $\deg(m - n\pi) = m^2 - a_\mathfrak{q} mn + \mathfrak{q}n^2 \ge 0$ for all $m, n \in \mathbb{Z}$ (positive definite quadratic form, 3.4), so the discriminant $a_\mathfrak{q}^2 - 4\mathfrak{q} \le 0$. Checked for the good primes $p < 2000$ of seven running examples. Reference: AEC V.1.1. Paper: background; Chapter 2, 11.4 needs $(q+1)^2 - a^2 > 0$; Lemma lem:euler and the table (α, β complex conjugate of absolute value $\sqrt{q}$).

8.3 `thm:ellcurves:charpoly` (theorem; P given 5.3 and "deg = det on T_ℓ", which is S from the Weil pairing: AEC III.8.6, verify). For $\ell \neq p$, the endomorphism π acts on $T_\ell E$ with characteristic polynomial

$X^2-a_{\mathfrak{q}}X+\mathfrak{q}$ ($\det\pi_\ell=\deg\pi=\mathfrak{q}$ and $\det(1-\pi_\ell)=\deg(1-\pi)=\#E(\mathbb{F}_{\mathfrak{q}})$); $\pi^2-a_{\mathfrak{q}}\pi+\mathfrak{q}=0$ in $\mathrm{End}(E)$. Its roots $\alpha,\beta\in\mathbb{C}$ satisfy $\alpha\beta=\mathfrak{q}$, $\alpha+\beta=a_{\mathfrak{q}}$, $|\alpha|=|\beta|=\sqrt{\mathfrak{q}}$, and $\#E(\mathbb{F}_{\mathfrak{q}^n})=\mathfrak{q}^n+1-\alpha^n-\beta^n$; in particular $\#E(\mathbb{F}_{\mathfrak{q}^2})=\mathfrak{q}^2+1-(a_{\mathfrak{q}}^2-2\mathfrak{q})$; $Z(E/\mathbb{F}_{\mathfrak{q}},T)=\frac{1-a_{\mathfrak{q}}T+\mathfrak{q}T^2}{(1-T)(1-\mathfrak{q}T)}$. Checked on 11a1: $\#E(\mathbb{F}_{p^2})=15,35,60,180$ for $p=3,5,7,13$, by direct count in $\mathbb{F}_{p^2}$. Reference: AEC V.2 (Theorem V.2.3.1, verify). Paper: the table of Section 1 and Lemma lem:euler (α,β roots of $X^2-a_q(E')X+q$, "using $\alpha\beta=q$ and $\alpha+\beta=a_q(E')$"), Lemma lem:old, Section 4.1 (stabilisations, Chapter 13).

8.4 def:ellcurves:supersingular (definition and theorem; R, with (ii) P). (i) $E/\mathbb{F}_{\mathfrak{q}}$ is supersingular iff $E[p]=0$ iff $p\mid a_{\mathfrak{q}}$ iff $\mathrm{End}_{\overline{\mathbb{F}}_p}(E)$ is a quaternion order; otherwise ordinary. (ii) For $\mathfrak{q}=p\geq 5$: supersingular iff $a_p=0$ (Hasse: $|a_p|\leq 2\sqrt{p}<p$). Reference: AEC V.3.1 (verify). Paper: Lemma lem:h0 via 6.5 ($a_p=0$ at inert primes).

8.5 prop:ellcurves:twistFq (proposition; P). Let $\mathfrak{q}$ be odd, $E\colon y^2=f(x)$ over $\mathbb{F}_{\mathfrak{q}}$ (f a cubic) and $E^{\mathrm{tw}}\colon \varepsilon y^2=f(x)$ with ε a non-square. Then $\#E(\mathbb{F}_{\mathfrak{q}})=\mathfrak{q}+1+\sum_x\eta(f(x))$ with η the quadratic character of $\mathbb{F}_{\mathfrak{q}}^\times$, $a_{\mathfrak{q}}(E^{\mathrm{tw}})=-a_{\mathfrak{q}}(E)$, $\#E(\mathbb{F}_{\mathfrak{q}})+\#E^{\mathrm{tw}}(\mathbb{F}_{\mathfrak{q}})=2\mathfrak{q}+2$, and $(\mathfrak{q}+1)^2-a_{\mathfrak{q}}^2=\#E(\mathbb{F}_{\mathfrak{q}})\cdot\#E^{\mathrm{tw}}(\mathbb{F}_{\mathfrak{q}})$. Reference: elementary (Chapter 2, 9 and 11.4). Paper: the I_0^* row of the table ($v_\ell((q+1)^2-a_q(E')^2)=v_\ell(\#E'(\mathbb{F}_q))+v_\ell(\#E'^{\mathrm{tw}}(\mathbb{F}_q))$: a curious reading of e_ℓ the writer may give in one sentence), Lemma lem:euler.

8.6 thm:ellcurves:frobQ (theorem; S, using Chapter 10). Let $E/\mathbb{Q}$ have good reduction at p (minimal model, $p\nmid\Delta$) and $\ell\neq p$. Then $\rho_{E,\ell}$ is unramified at p and $\rho_{E,\ell}(\mathrm{Frob}_p)$ (arithmetic) has characteristic polynomial X^2-a_pX+p with $a_p=a_p(E)=p+1-\#\tilde{E}(\mathbb{F}_p)$, independently of the choice of Frob_p and of ℓ. Sketch: reduction $E[\ell^n]\to\tilde{E}[\ell^n]$ is injective (Chapter 10; AEC VII.3.1, verify) hence bijective by counting, and it is equivariant for $D_p\to G_{\mathbb{F}_p}$; so I_p acts trivially and Frob_p acts as π (8.3). This is the easy half of Neron-Ogg-Shafarevich (Chapter 11 states the criterion). Write α_p,β_p for the roots. Reference: AEC VII.7 (verify). Paper: Section 1.1, the table (α,β for E'), Lemma lem:old ("characteristic polynomial of $\bar{\rho}_h(\mathrm{Frob}_p)$ is $X^2-a_p(h)X+p$"), Lemma lem:h0 ($a_p=\mathrm{tr}\,\rho(\mathrm{Frob}_p)$).

8.7 ex:ellcurves:aptable (worked example; P). a_p for $p<40$ (point counts on the minimal model; at bad primes $a_p=p+1-\#\tilde{E}(\mathbb{F}_p)$ counting the singular point, which gives 1, -1, 0 for split multiplicative, nonsplit multiplicative and additive reduction; Chapter 10):

curve	a_2	a_3	a_5	a_7	a_{11}	a_{13}	a_{17}	a_{19}	a_{23}	a_{29}	a_{31}	a_{37}
11a1	-2	-1	1	-2	1	4	-2	0	-1	0	7	3
20a1	0	-2	-1	2	0	2	-6	-4	6	6	-4	2
99a1	-1	0	-4	-2	-1	-2	2	-6	4	-6	4	-6
121a1	-1	2	1	2	0	-1	5	-6	2	-9	-2	-3
539c1	1	-2	2	0	1	-4	-4	0	-4	-6	-10	-6
726a1	-1	-1	0	0	0	-6	6	-6	6	-6	4	-2
980i1	0	-3	1	0	-5	3	1	-6	6	-9	4	2

(726a1, image 3Ns: $3\mid a_p$ at the good $p\equiv 2\pmod 3$, as $\bar{\rho}\simeq\bar{\rho}\otimes\chi_{-3}$ requires; 11a1: $a_p\equiv 1+p\pmod 5$, item 9.7.) The coefficients of 11a1 agree with $q\prod_{n\geq 1}(1-q^n)^2(1-q^{11n})^2$ up to q^{20} (Chapter 12). Paper: the Example of Section 3 (congruences between these a_p, Chapter 15), Lemma lem:euler.

11.3.9 Section 9. Elliptic curves over $\mathbb{Q}$: Mordell-Weil, torsion, isogenies

9.1 thm:ellcurves:mordellweil (theorem, Mordell; R, with the outline of proof S). $E(\mathbb{Q})$ is finitely generated: $E(\mathbb{Q})\simeq\mathbb{Z}^r\oplus E(\mathbb{Q})_{\mathrm{tors}}$, r the rank, $E(\mathbb{Q})_{\mathrm{tors}}$ finite. Outline: weak Mordell-Weil ($E(\mathbb{Q})/2E(\mathbb{Q})$ finite, Chapter 17 via the Kummer map and Selmer groups) plus the descent with the canonical height. Reference: AEC VIII.6.7 (verify), VIII.4 to VIII.9. Paper: background (ranks in Section sec:eis "all of rank 0", Section sec:two and Watkins's conjecture $2^{\mathrm{rank}}\mid\deg\varphi_E$).

9.2 prop:ellcurves:torsred (proposition; R from Chapter 10, with the application P). For p of good reduction, reduction is injective on $E(\mathbb{Q})[m]$ for $p\nmid m$, and for $p\geq 3$ it is injective on all of $E(\mathbb{Q})_{\mathrm{tors}}$; so $\#E(\mathbb{Q})_{\mathrm{tors}}$ divides $\#\tilde{E}(\mathbb{F}_p)$ for every good $p\geq 3$. Nagell-Lutz (R): torsion points of a model with

integral coefficients have integral coordinates (with the usual caveat at 2). Application (P): for 11a1, $\#\tilde{E}(\mathbb{F}_3) = 5$ and $\#\tilde{E}(\mathbb{F}_7) = 10$, so $\#E(\mathbb{Q})_{\mathrm{tors}} \mid 5$, and $(5,5)$ has order 5: $E(\mathbb{Q})_{\mathrm{tors}} \simeq \mathbb{Z}/5$. Reference: AEC VII.3.1, VII.3.5, VIII.7.2 (verify). Paper: Conjecture conj:eis ($v_\ell(\#E(\mathbb{Q})_{\mathrm{tors}})$), Section sec:eis.

9.3 `thm:ellcurves:mazur` (theorem, Mazur; R). $E(\mathbb{Q})_{\mathrm{tors}}$ is isomorphic to $\mathbb{Z}/n$ with $1 \le n \le 10$ or $n = 12$, or to $\mathbb{Z}/2 \times \mathbb{Z}/2n$ with $1 \le n \le 4$; each occurs infinitely often. Data check: the torsion orders in `ecdata` below 10,000 are $1, \dots, 10, 12, 16$ (with 16 only from $\mathbb{Z}/2 \times \mathbb{Z}/8$). Reference: Mazur77; AEC VIII.7.5 (verify). Paper: Conjecture conj:eis ("$D \le 1$ for $\ell \ge 5$": v_5, $v_7 \le 1$ and no ℓ-torsion for $\ell \ge 11$; "a point of order 9"), Section sec:eis.

9.4 `def:ellcurves:isogclass` (definition and theorem; P for (i), (ii); R for (iii), (iv)). (i) Being isogenous over $\mathbb{Q}$ is an equivalence relation (3.4); the classes are the isogeny classes. (ii) Isogenous curves have $V_\ell E \simeq V_\ell E'$ (5.4), hence the same a_p at every prime p good for both (8.6), the same L-function (Section 11), and the same rank ($E(\mathbb{Q}) \otimes \mathbb{Q} \simeq E'(\mathbb{Q}) \otimes \mathbb{Q}$ via φ and $\hat{\varphi}$); they have the same conductor (R; Chapters 10, 11) but in general different torsion, Δ, j, Tamagawa numbers and periods. (iii) (Faltings) Conversely, E and E' over $\mathbb{Q}$ are isogenous iff $a_p(E) = a_p(E')$ for almost all p, iff $V_\ell E \simeq V_\ell E'$ for one ℓ. (iv) An isogeny class over $\mathbb{Q}$ is finite, with at most 8 curves (Kenku); `ecdata` below 10,000 has classes of sizes $1, 2, 3, 4, 6, 8$. Cremona's isogeny matrix lists the degrees of the cyclic isogenies between the curves of a class: 11a: $\left(\begin{smallmatrix} 1 & 5 & 5 \\ 5 & 1 & 25 \\ 5 & 25 & 1 \end{smallmatrix}\right)$; 20a: $\left(\begin{smallmatrix} 1 & 2 & 3 & 6 \\ 2 & 1 & 6 & 3 \\ 3 & 6 & 1 & 2 \\ 6 & 3 & 2 & 1 \end{smallmatrix}\right)$; 121a, 121c: one 11-isogeny; 121d: degrees $5, 5, 25$; 99a, 539c, 726a: one 2-isogeny; 980i: trivial. Reference: Faltings 1983 (add to `bib/ch09.bib`); Kenku 1982 (add); Cremona97 Chapter III. Paper: Section 1 ("the isogeny class of E", the optimal curve, Chapter 14), Conjecture conj:eis ("the number of curves in the isogeny class of E with a rational point of order ℓ"), Section sec:oot ("trivial isogeny class").

9.5 `prop:ellcurves:isogline` (proposition; P given 3.3). Let ℓ be a prime, E/K, $\mathrm{char}\, K \ne \ell$. The K-rational ℓ-isogenies from E (up to isomorphism of the target) correspond to the G_K-stable lines of $E[\ell]$. Hence $E[\ell]$ is reducible as an $\mathbb{F}_\ell[G_K]$-module iff E has a K-rational ℓ-isogeny. A K-rational point of order ℓ gives one (its span), but not conversely: 11a2 and 539a1 have rational 5- resp. 3-isogenies and no rational 5- resp. 3-torsion. If $E[\ell]$ is reducible, $\bar{\rho}^{\mathrm{ss}}_{E,\ell} = \psi_1 \oplus \psi_2$ with $\psi_1\psi_2 = \bar{\chi}_\ell$ (5.2(iii)). Data check (`ecdata`, conductor below 10,000, $\ell = 3, 5, 7$): for curve number 1 of every non-CM class, a Borel or split-Cartan label at ℓ (Chapter 11) occurs iff its row of the isogeny matrix has a degree divisible by ℓ (114,126 pairs; the 59 mismatches are all CM curves, for which `ecdata` lists no label at ℓ). Reference: AEC III.4.12, Exercise 3.13. Paper: Conjecture conj:eis ("with a rational ℓ-isogeny"), Section sec:eis, Conjecture conj:euler and Theorem thm:quant ("$E[\ell]$ irreducible"), Proposition prop:ns ("$E[3]$ irreducible", "no rational root").

9.6 `thm:ellcurves:mazurkenku` (theorem, Mazur and Kenku; R). (i) If $E/\mathbb{Q}$ has a rational ℓ-isogeny with ℓ prime, then $\ell \in \{2, 3, 5, 7, 11, 13, 17, 19, 37, 43, 67, 163\}$ (Mazur78). (ii) If $E/\mathbb{Q}$ has a rational cyclic n-isogeny, then $n \le 19$ or $n \in \{21, 25, 27, 37, 43, 67, 163\}$ (Mazur, Kenku). (iii) The values $n = 19, 27, 43, 67, 163$ occur only for CM curves; for $n = 27$ the curves have $j = -2^{15} \cdot 3 \cdot 5^3 = -12288000$ (CM by the order of discriminant -27; in `ecdata` the cyclic 27-isogeny joins 27a2 and 27a4, both with this j). Data check: the isogeny degrees occurring below conductor 10,000 are 1 to 16, 18, 19, 21, 25, 27, 37, 43, 67. Reference: Mazur78; Kenku's papers on $X_0(N)$ for $N = 39, 65, 91, 125, 169$ and the summary in Kenku 1982 (add to `bib/ch09.bib`; the writer should cite a survey with exact statement, e.g. the table in Lozano-Robledo, Math. Ann. 357 (2013), after checking it). Paper: Conjecture conj:eis, Section sec:eis, 9.8.

9.7 `prop:ellcurves:zlmul` (proposition; P). Let ℓ be odd and $E/\mathbb{Q}$ with a rational point P_0 of order ℓ. (i) In a basis (P_0, Q) of $E[\ell]$, $\bar{\rho}_{E,\ell} = \left(\begin{smallmatrix} 1 & * \\ 0 & \bar{\chi}_\ell \end{smallmatrix}\right)$ (5.2(iii)). (ii) The following are equivalent: the sequence $0 \to \mathbb{Z}/\ell \to E[\ell] \to \mu_\ell \to 0$ splits; $E[\ell] \simeq \mathbb{Z}/\ell \oplus \mu_\ell$ as $G_{\mathbb{Q}}$-modules; the image of $\bar{\rho}_{E,\ell}$ is conjugate to $\{\mathrm{diag}(1, d) : d \in \mathbb{F}_\ell^\times\}$ (label ℓCs.1.1: 3Cs.1.1 of order 2, 5Cs.1.1 of order 4; Chapter 1, 6.10, Chapter 11); E has two rational ℓ-isogenies, with kernels $\mathbb{Z}/\ell$ and μ_ℓ. (iii) In that case $\#\tilde{E}(\mathbb{F}_p) \equiv 0$ and $a_p \equiv 1 + p \pmod{\ell}$ for every good $p \ne \ell$ (trace of $\mathrm{diag}(1, p)$); the congruence $a_p \equiv 1 + p$ already follows from (i). Examples (all checked): 14a1, 19a1, 26a1 and 27a1 (CM) have $E[3] \simeq \mathbb{Z}/3 \oplus \mu_3$; 11a1 has $E[5] \simeq \mathbb{Z}/5 \oplus \mu_5$ ($a_p \equiv 1 + p \pmod 5$ for all good $p < 500$; its two 5-isogenies have kernels $\mathbb{Z}/5$ and μ_5; 11a1/μ_5 contains the rational group $E[5]/\mu_5 \simeq \mathbb{Z}/5$, so it is 11a3 (torsion $\mathbb{Z}/5$), and 11a1/$(\mathbb{Z}/5) \simeq$ 11a2 (torsion 1)); 20a1 has a rational 3-torsion point with non-split $E[3]$ (3B.1.1). Reference: elementary. Paper: Conjecture conj:eis ("$[E[3] \simeq \mathbb{Z}/3 \oplus \mu_3]$"), Section sec:eis ("the split Cartan image 3Cs.1.1": see Part 8, N1).

9.8 prop:ellcurves:no27 (proposition; P given 9.6(iii); this is important_notes.md point 17, decided to live here). Let $E/\mathbb{Q}$ be without CM and $E[3] \simeq \mathbb{Z}/3 \oplus \mu_3$. Then $E(\mathbb{Q})$ has no point of order 9; so $E(\mathbb{Q})[3^\infty] \simeq \mathbb{Z}/3$ and $v_3(\#E(\mathbb{Q})_{\mathrm{tors}}) = 1$. Proof: suppose $P \in E(\mathbb{Q})$ has order 9. Then $3P$ is a rational point of order 3, so $\langle 3P\rangle$ is the line $\mathbb{Z}/3$ (μ_3 has no nonzero rational point since $\chi_{-3} \neq 1$), and $\langle P\rangle \cap E[3] = \langle 3P\rangle$ meets μ_3 trivially. Let $\varphi\colon E \to E' = E/\mu_3$ (rational, 3.3) and $\hat\varphi\colon E' \to E$ its dual; $\varphi\circ\hat\varphi = [3]$ gives $\hat\varphi(E'[3]) \subset \ker\varphi = \mu_3$, with equality since $\#\hat\varphi(E'[3]) = 9/\#\ker\hat\varphi = 3$. Put $C = \hat\varphi^{-1}(\langle P\rangle) \subset E'$: it is $G_\mathbb{Q}$-stable of order $\#\ker\hat\varphi\cdot 9 = 27$. If $C \supset E'[3]$ then $\mu_3 = \hat\varphi(E'[3]) \subset \langle P\rangle$, a contradiction; so C has cyclic 3-part of order 27, i.e. C is cyclic (3.6), and $E' \to E'/C$ is a rational cyclic 27-isogeny. By 9.6(iii) E' has CM, hence so has E (6.3), a contradiction. Corollary (with Mazur, 9.3): for a non-CM E with a rational 3-isogeny, $v_3(\#E(\mathbb{Q})_{\mathrm{tors}}) + [E[3] \simeq \mathbb{Z}/3\oplus\mu_3] \le 2$, so the bound "$D \le 2$ for $\ell = 3$" of Conjecture conj:eis follows from its first bound and is not an additional conjecture; also $v_\ell(\#E(\mathbb{Q})_{\mathrm{tors}}) \le 1$ for $\ell = 5, 7$ and $= 0$ for $\ell \ge 11$. Data check: the 264 curves of conductor below 10,000 with image 3Cs.1.1 have torsion of order 3 or 6; the six curves there with a point of order 9 (54b3, 714i1, 1482l1, 1554n1, 1890r3, 6942n1) all have image 3B.1.1, as Section sec:eis reports for its range. Reference: the argument of important_notes.md point 17 (and Chapter 21 outline, A1.13); Mazur77, 9.6. Paper: Conjecture conj:eis, Section sec:eis ("We have not proved that the two cannot coexist, so the bound $D \le 2$ is part of the conjecture"), Open question 4.

9.9 lem:ellcurves:primetol (lemma; P given 3.6, 5.4, 9.5; important_notes.md point 27). Let $E/\mathbb{Q}$ and ℓ a prime with $E[\ell]$ irreducible. (i) Every rational isogeny $\varphi\colon E \to E'$ factors as $\varphi = \psi\circ[a]$ with $\ker\psi$ cyclic of order prime to ℓ (by 3.6, a cyclic kernel of order divisible by ℓ contains a rational subgroup of order ℓ, contradicting 9.5). (ii) Hence ψ induces $G_\mathbb{Q}$-isomorphisms $E[\ell^n] \simeq E'[\ell^n]$ and $T_\ell E \simeq T_\ell E'$; all curves of the isogeny class have the same $\rho_{E,\ell}$ up to conjugation in $\mathrm{GL}_2(\mathbb{Z}_\ell)$, the same $\bar\rho$, the same $\mathrm{ad}^0\,T$, the same $h_\ell(E)$ and the same ℓ-adic Tamagawa factors and Selmer groups of $\mathrm{ad}^0\,T$; and $E'[\ell]$ is irreducible. (iii) In particular the hypotheses and all ℓ-adic quantities of Theorem thm:main and Conjecture conj:euler except $\deg\varphi_E$ itself do not depend on which curve of the class is the optimal one; $t_\ell(E,q) = v_\ell(-v_q(j))$ is also unchanged, because ℓ-power-free isogenies change $v_q(j)$ at a potentially multiplicative prime by a factor prime to ℓ (Tate curve, Chapter 10; state with forward pointer). Reference: elementary. Paper: Section 1 (optimality), Theorem thm:qual ("the $X_0(N)$-optimal curve in the isogeny class of E"), Section 3 (E_0), Section sec:oot, important_notes.md point 27.

9.10 cor:ellcurves:trivialclass (corollary; P). If the isogeny class of $E/\mathbb{Q}$ has only one curve, then E has no rational ℓ-isogeny for any prime ℓ, so $E[\ell]$ is irreducible for every ℓ (including $\ell = 2$), and E is the $X_0(N)$-optimal curve of its class (Chapter 14). Example: 980i1. Paper: Section sec:oot ("kept those with trivial isogeny class (so that the curve is optimal and $E[\ell]$ is irreducible for every ℓ)").

11.3.10 Section 10. Quadratic twists

10.1 def:ellcurves:twist (definition and proposition; P). Let $\mathrm{char}\,K \neq 2$ and $d \in K^\times$. For a model $y^2 = x^3 + a_2x^2 + a_4x + a_6$ put $E^{(d)}\colon y^2 = x^3 + da_2x^2 + d^2a_4x + d^3a_6$ (equivalently $dy^2 = x^3 + a_2x^2 + a_4x + a_6$ via $(x,y) \mapsto (dx, d^2y)$). For a general model complete the square first: $\eta^2 = 4x^3 + b_2x^2 + 2b_4x + b_6$ becomes $\eta^2 = 4x^3 + db_2x^2 + 2d^2b_4x + d^3b_6$, and $y^2 = x^3 + db_2x^2 + 8d^2b_4x + 16d^3b_6$ is an integral model when the a_i and d are integral. $E^{(d)}$ depends only on d mod $(K^\times)^2$ up to K-isomorphism; $E^{(1)} \simeq E$, $(E^{(d)})^{(d')} \simeq E^{(dd')}$; $\psi\colon E^{(d)} \to E$, $(x,y) \mapsto (x/d, y/d^{3/2})$ is an isomorphism over $K(\sqrt{d})$ with $\psi^\sigma = [\chi_d(\sigma)]\circ\psi$. Twists in general are classified by $H^1(G_K, \mathrm{Aut}\,E)$ (R; AEC X.2, X.5). Reference: AEC X.2, X.5 (Proposition X.5.4, verify). Paper: Lemma lem:twist (ii) (the twist $y^2 = x^3 + q^*a_2x^2 + q^{*2}a_4x + q^{*3}a_6$), Section 1.4 ($E^{(q)}$), Theorem thm:twist and thm:quant (E_S).

10.2 prop:ellcurves:twistinv (proposition; P, checked symbolically and on examples). (i) For the first model of 10.1, $c_4(E^{(d)}) = d^2c_4$, $c_6(E^{(d)}) = d^3c_6$, $\Delta(E^{(d)}) = d^6\Delta$, $j(E^{(d)}) = j(E)$; for the integral model of a general curve the same holds after the change $u = 1/2$ (factors 2^4, 2^6, 2^{12}). (ii) Conversely, if $j(E) = j(E') \neq 0, 1728$ then $E' \simeq E^{(d)}$ over K with $d = c_6(E')c_4(E)/(c_6(E)c_4(E'))$. (iii) **The twist by q and the twist by q^*** (q an odd prime) are different curves when $q \equiv 3 \pmod 4$: they differ by the twist by -1, which changes the sign of c_6 and is ramified at 2. The invariants of the twist by q^* are $q^{*2}c_4$, $q^{*3}c_6$, $q^{*6}\Delta$; the q-adic valuations are the same as for the twist by q, which is all Lemma lem:twist uses (important_notes.md point 29). Example (checked): $121\mathrm{a}1^{(-11)} \simeq 121\mathrm{c}2$ (conductor 121),

while $121\text{a}1^{(11)} \simeq 1936\text{k}2$ (conductor $1936 = 2^4 \cdot 11^2$). Further checked identifications: $11\text{a}1^{(-11)} \simeq 121\text{d}2$, $11\text{a}3^{(-11)} \simeq 121\text{d}1$, $11\text{a}2^{(-11)} \simeq 121\text{d}3$, $539\text{c}1^{(-7)} \simeq 77\text{c}1$, $980\text{i}1^{(-7)} \simeq 140\text{b}1$, $11\text{a}1^{(-7)} \simeq 539\text{d}2$, $11\text{a}3^{(-7)} \simeq 539\text{d}1$, $20\text{a}1^{(-1)} \simeq 80\text{b}1$. Paper: Lemma lem:twist (ii) ("the twist has a model with invariants q^2c_4, q^3c_6, $q^6\Delta$ and the same j-invariant"), the Example of Section 3, Definition def:terms ("Both terms are invariant under quadratic twist", see 10.7).

10.3 `thm:ellcurves:twistgal` (theorem; P). For E/K, $d \in K^\times$ and $\ell \neq \operatorname{char} K$: $\rho_{E^{(d)},\ell} \simeq \rho_{E,\ell} \otimes \chi_d$, via $\psi_\ell \colon T_\ell E^{(d)} \to T_\ell E$ ($\psi(\sigma P) = \chi_d(\sigma)\sigma(\psi(P))$ from $\psi^\sigma = [\chi_d(\sigma)] \circ \psi$). Consequences: $\bar\rho_{E^{(d)},\ell} = \bar\rho_{E,\ell} \otimes \chi_d$, $E^{(d)}[2] \simeq E[2]$, $\operatorname{ad}^0 \rho$ unchanged (Chapter 4, 7.1(iii)), det unchanged; $E[\ell]$ is irreducible iff $E^{(d)}[\ell]$ is. Paper: Lemma lem:euler ("$V = V' \otimes \chi_{q^*}$", "$\operatorname{ad}^0 V = \operatorname{ad}^0 V'$"), proof of Theorem thm:main ("$\operatorname{ad}^0 T$ is unchanged by the twist"), Theorem thm:qual (b), (c) ("$\bar\rho_{E',3} = \bar\rho \otimes \chi$"), Theorem thm:quant ("$\bar\rho \otimes \chi_S$ is irreducible"), Remark after thm:quant ("adjoint lattices … coincide").

10.4 `cor:ellcurves:twistap` (corollary; P). Let $E/\mathbb{Q}$, $d \in \mathbb{Q}^\times$ with $\mathbb{Q}(\sqrt{d})$ of discriminant D, and p a prime of good reduction for E with $p \nmid D$. Then $E^{(d)}$ has good reduction at p (Chapter 10) and $a_p(E^{(d)}) = \chi_d(p)a_p(E)$, $\chi_d(p) = \left(\frac{D}{p}\right)$. Elementary proof for odd $p \nmid d\Delta$: $\#E(\mathbb{F}_p) = p + 1 + \sum_x \eta(f(x))$ and the substitution $x = dx'$ gives $a_p(E^{(d)}) = \eta(d)a_p(E)$ (8.5). General proof: 10.3 and 8.6. Checked for all good $p < 400$ (including $p = 2$ where allowed): 121d2 versus 11a1 (χ_{-11}), 121c2 versus 121a1, 77c1 versus 539c1 (χ_{-7}), 140b1 versus 980i1, 539d1 versus 11a3, 121d1 versus 11a3. Reference: AEC X.2; elementary. Paper: Lemma lem:twist (iii) and Theorem thm:twist ("$a_p(f) = \chi_S(p)a_p(g)$ for $p \nmid D$"), the Example of Section 3 ("$f_{11a} \otimes \chi_{-11}$ is the form of 121d"), Theorem thm:qual (b), (c), PLAN.md scope ("$a_p(E^{(d)}) = \chi_d(p)a_p(E)$ for p not dividing $2d$").

10.5 `prop:ellcurves:twistgood` (proposition; P given 10.4). Let q be an odd prime at which $E/\mathbb{Q}$ has type I_0^*, so that $E' = E^{(q^*)}$ has good reduction at q (Chapter 10; this is the paper's "twist of good reduction"). (i) $V_\ell E \simeq V_\ell E' \otimes \chi_{q^*}$ with $V_\ell E'$ unramified at q (10.3, 8.6). (ii) Any other twist $E^{(q^*d)}$ with $q \nmid D$ has good reduction at q and $a_q(E^{(q^*d)}) = \chi_d(q)a_q(E') = \pm a_q(E')$; so $a_q(E')^2$, the pair $\{\alpha/\beta, \beta/\alpha\}$ and $(q+1)^2 - a_q(E')^2$ depend only on E and q, not on the choice of the twist; this is why the table may say "the twist of good reduction". (iii) Example: 539d1 has type I_0^* at 7 and $E' = 539\text{d}1^{(-7)} \simeq 11\text{a}3$, $a_7(11\text{a}) = -2$, $(7+1)^2 - 4 = 60$, so $e_3(539\text{d}1, 7) = v_3(6) + v_3(60) = 2 = v_3(\deg \varphi_E)$ ($\deg \varphi_E = 72$). Paper: the table of Section 1 and the abstract (type I_0^*), Lemma lem:euler, Theorem thm:quant (S contains the I_0^* primes; E_S good at them).

10.6 `def:ellcurves:compositetwist` (definition and proposition; P). For a finite set S of odd primes, $\chi_S = \prod_{q \in S} \chi_{q^*}$ is the character of $\mathbb{Q}(\sqrt{d_S})$ with $d_S = \prod_{q \in S} q^*$ (a fundamental discriminant, $\equiv 1 \pmod 4$), of conductor $\prod_{q \in S} q$, unramified outside S (Chapters 2, 6); $E_S := E^{(d_S)} = E \otimes \chi_S$ is obtained by successive twists in any order. $a_p(E_S) = \chi_S(p)a_p(E)$ for $p \notin S$ good. Paper: Theorem thm:twist and Theorem thm:quant (E_S, χ_S, $D = \prod q$), the remark after thm:twist.

10.7 `rem:ellcurves:twistinvariance` (remark; P). Under every quadratic twist: j, hence $v_q(j)$ and $t_\ell(E, q) = v_\ell(-v_q(j))$ (when $v_q(j) < 0$), and $\operatorname{ad}^0 \rho_{E,\ell}$ are unchanged; a_q, Δ, the Kodaira type at a prime where the twist ramifies, and the naive Euler factor change. So $e_\ell(E, q)$, the ratio of the true to the naive adjoint factor, is not invariant under a twist ramified at q (important_notes.md point 15): 11a1 has type I_5 at 11 and $e_3 = 0$, its twist 121d2 has type I_5^* and $e_3 = v_3(11^2 - 1) = 1$ (and likewise 11a3, I_1, versus 121d1, I_1^*; types from Chapter 10). It is invariant under twists unramified at q. Paper: Definition def:terms (last sentence), Section 1.4.

11.3.11 Section 11. The L-function

11.1 `def:ellcurves:Lfunction` (definition and proposition; P for (i) to (iii), R for (iv)). Let $E/\mathbb{Q}$ have conductor N (Chapter 10) and minimal model; for every prime p put $a_p = p + 1 - \#\tilde E(\mathbb{F}_p)$ ($\tilde E$ the reduction, singular point included), so $a_p \in \{1, -1, 0\}$ for $p \mid N$ according as the reduction is split multiplicative, nonsplit multiplicative or additive. (i) $L(E, s) = \prod_{p \nmid N}(1 - a_p p^{-s} + p^{1-2s})^{-1} \prod_{p \mid N}(1 - a_p p^{-s})^{-1} = \sum_{n \geq 1} a_n n^{-s}$ converges absolutely for $\operatorname{Re} s > 3/2$ (Hasse, 8.2: $|a_n| \leq d(n)\sqrt{n}$). (ii) $a_{mn} = a_m a_n$ for coprime m, n; $a_{p^{k+1}} = a_p a_{p^k} - p a_{p^{k-1}}$ for $p \nmid N$; $a_{p^k} = a_p^k$ for $p \mid N$. (iii) For $p \neq \ell$ the local factor is $\det(1 - \operatorname{Frob}_p p^{-s} \mid V_\ell E^{I_p})^{-1}$ (C5): at good p by 8.6; at bad p the statement is Chapter 11 (dimension of V^{I_p} is 1 or 0, the paper's δ_p). Isogenous curves have the same L-function (9.4);

$L(E^{(d)}, s)$ has the same local factors as the twist $\sum \chi_d(n) a_n n^{-s}$ at the primes $p \nmid ND$ (10.4). (iv) (Modularity, Chapter 14; Chapter 19) $L(E, s) = L(f_E, s)$ for the newform f_E of level N, it has an analytic continuation to $\mathbb{C}$ and a functional equation under $s \mapsto 2 - s$. Example: 11a1, $L(E, s) = \sum a_n n^{-s}$ with $a_1, \ldots, a_{13} = 1, -2, -1, 2, 1, 2, -2, 0, -2, -2, 1, -2, 4$. Reference: AEC C.16 (verify), Cremona97 Chapter II, DiamondShurman 8.8 (verify). Paper: Section 1 ("$f = f_E$ ... newform attached to it"), Section 1.1 (δ_p; the naive versus true adjoint factors are Chapter 19), Lemma lem:old (multiplicativity, the recursion).

11.3.12 Section 12. Labels and the running examples

12.1 `conv:ellcurves:labels` (convention and facts; R for the data). (i) A Cremona label $N\,c\,k$ consists of the conductor N, an isogeny-class code c (a, b, ..., z, ba, bb, ...) and the number k of the curve in its class. In Cremona's tables curve number 1 of each class is the $X_0(N)$-optimal curve, with the single exception of the class 990h, whose optimal curve is 990h3 = $[1, -1, 1, -1568, -4669]$ (`ecdata`'s `degphi` file lists the modular degree 1728 of 990h with this equation); optimality is established in the tables for $N \le 400000$ and for most classes beyond (the paper, Section 6; Chapter 14). Cremona's book (Cremona97) and ARS12 print the class code in capitals: 99A1 is 99a1. (ii) LMFDB labels $N.c\,k$ use a different ordering both of the classes and of the curves in a class, so the same curve can have different letters and numbers; correspondence (from `ecdata`'s `alllabels` file): 11a1 = 11.a2, 11a2 = 11.a1, 11a3 = 11.a3, 14a1 = 14.a6, 20a1 = 20.a4, 27a1 = 27.a3, 77c1 = 77.c2, 99a1 = 99.a2, 121a1 = 121.a2, 121c1 = 121.c2, 121c2 = 121.c1, 121d1 = 121.d3, 140b1 = 140.b1, 539c1 = 539.d2, 539d1 = 539.a3, 726a1 = 726.b2, 980i1 = 980.b1. The LMFDB number 1 is not the optimal curve in general. The writer must not describe the LMFDB's ordering rules beyond "different from Cremona's" unless checked against the LMFDB documentation. Reference: Cremona97, CremonaBook (same work), LMFDB. Paper: every label (Section 3 Example, the remark on 99A1, the appendix, Section sec:data).

12.2 `ex:ellcurves:running` (worked example table; invariants P (1.6), data R from `ecdata`; Kodaira types from Chapter 10 and the Chapter 21 outline's Tate algorithm). The paper's curves:

curve	N	rank	$E(\mathbb{Q})_{\mathrm{tors}}$	class (degrees)	$\deg \varphi_E$	non-surjective $\bar{\rho}_\ell$ (`ecdata`)	types (Ch. 10)	role in the paper
11a1	11	0	$\mathbb{Z}/5$	11a: $5, 5, 25$	1	5Cs.1.1	I_5 at 11	congruent form in thm:qual (b), (c) at $\ell = 3$
20a1	$2^2 \cdot 5$	0	$\mathbb{Z}/6$	20a: $2, 3, 6$	1	2B, 3B.1.1	IV^* at 2, I_2 at 5	congruent form at $\ell = 5$ for 980i1
121a1	11^2	0	0	121a: 11	6	11B.1.6	II at 11	thm:qual (b), appendix `rule`
121c1, 121c2	11^2	0	0	121c: 11	6	11B.1.7, 11B.1.5	IV, IV^* at 11	121c2 = $121\mathrm{a}1^{(-11)}$
121d1	11^2	0	0	121d: $5, 5, 25$	24	5B.4.1	I_1^* at 11	$f_{11a} \otimes \chi_{-11}$; 121d1 = $11\mathrm{a}3^{(-11)}$
539c1	$7^2 \cdot 11$	1	$\mathbb{Z}/2$	539c: 2	$288 = 2^5 3^2$	2B	I_3^* at 7, I_2 at 11	thm:qual (c), $\ell = 3$

curve	N	rank	$E(\mathbb{Q})_{\mathrm{tors}}$	class (degrees)	$\deg \varphi_E$	non-surjective $\bar{\rho}_\ell$ (ecdata)	types (Ch. 10)	role in the paper
980i1	$2^2 \cdot 5 \cdot 7^2$	0	0	trivial	$2880 = 2^6 3^2 5$	none	IV* at 2, I_1 at 5, I_5^* at 7	thm:qual (c), $\ell = 5$
726a1	$2 \cdot 3 \cdot 11^2$	1	$\mathbb{Z}/2$	726a: 2	$144 = 2^4 3^2$	2B, 3Ns	I_6, I_3, III at $2, 3, 11$	3Ns example, appendix
99a1 (99A1)	$3^2 \cdot 11$	1	$\mathbb{Z}/2$	99a: 2	4	2B	III at 3, I_1 at 11	$\deg \varphi_E = 4$ versus $r_E = 12$

Supporting curves: 77c1 ($N = 77$, $\deg \varphi_E = 6$, $= 539\mathrm{c}1^{(-7)}$), 140b1 ($N = 140$, $\deg \varphi_E = 60$, $= 980\mathrm{i}1^{(-7)}$), 539d1 (I_0^* at 7, $\deg \varphi_E = 72$, $= 11\mathrm{a}3^{(-7)}$), 14a1 ($E[3] \simeq \mathbb{Z}/3 \oplus \mu_3$), 338d1 and 6912d1 (3Ns), 27a1, 32a1, 36a1, 121b1 (CM). Note that 539c1, 726a1 and 99a1 have rank 1. Reference: Cremona97 and ecdata (allcurves, allisog, degphi, galrep, alllabels). Paper: the Example of Section 3, the remark on 99A1 (ARS12 Table 1: $r_E = 12$; recomputed by the Chapter 14 outline, its 8.5 and 9.4, which the writer cites; important_notes.md B19), the appendix (121a1, 726a1), Section sec:anomaly (726a1), Section sec:eis.

11.4 3. Section-by-section plan with target line counts

Line counts are lines of LaTeX, card excluded (PLAN.md Section 7.3: 1,000 to 2,200).

section	title	contents	target lines
(starred)	Core results at a glance	the card: every item of Part 2, stated as above, with tag and \Cref	(excluded)

section	title	contents	target lines
(starred)	Where the paper uses this chapter	Section 1.1 ($V, T, \rho, \bar{\rho}$); def:terms (j, $v_q(j)$, a_q, twist invariance); the table (E', $a_q(E')$, α, β); Section 1.4 and lem:twist (ii); lem:euler ($V = V' \otimes \chi_{q^*}$); thm:twist, thm:quant (E_S); thm:qual and the Section 3 Example (twists, labels, a_p); lem:h0 (CM); prop:ns (ψ_3, det, complex conjugation); conj:eis and sec:eis (torsion, isogenies, $\mathbb{Z}/3 \oplus \mu_3$, item 9.8); Section 1 (isogeny class, period lattice); sec:oot (trivial class); the appendix (`ellinit`, `elldivpol`)	40
1	Weierstrass equations and their invariants	conventions C1 to C8 as `convention` environments; 1.1 to 1.6; the invariants table; `paperlink` for def:terms ($v_q(j)$) and lem:twist (ii)	190
2	The group law	2.1 to 2.4; a figure in words (chord and tangent); `paperlink` for torsion in sec:eis	150
3	Isogenies	3.1 to 3.6; `warning` that "isogenous" means over $\mathbb{Q}$ unless stated	180
4	Division polynomials	4.1 to 4.6; the factorisation table (`booktabs`); `paperlink` for prop:ns, sec:data, the appendix `ns3`	160
5	The Weil pairing, the Tate module and the Galois representations	5.1 to 5.7; `convention` on Frobenius (5.7); `paperlink` for Section 1.1, Section 4.2 (point 26), prop:ns	170
6	Endomorphisms and complex multiplication	6.1 to 6.5; the table of the 13 j-invariants with discriminants; `warning` on χ_ℓ versus χ_{ℓ^*} in lem:h0 (points 5, 14)	150

section	title	contents	target lines
7	Elliptic curves over $\mathbb{C}$	7.1 to 7.7; the 11a1 numbers; RHB aside (Fourier series on a torus); `paperlink` for Section 1 and Chapter 14	210
8	Elliptic curves over finite fields	8.1 to 8.7; the a_p table; `paperlink` for the table of Section 1 and lem:euler (α, β)	160
9	Elliptic curves over $\mathbb{Q}$: Mordell-Weil, torsion, isogenies	9.1 to 9.10; 9.8 with full proof and a `paperlink` to conj:eis and sec:eis (point 17); 9.9 with a `paperlink` to optimality (point 27)	200
10	Quadratic twists	10.1 to 10.7; `warning` on q versus q^* (point 29) and on curve-level versus class-level twist identifications; `paperlink` for lem:twist, lem:euler, thm:twist, thm:quant, the Section 3 Example	170
11	The L-function	11.1; forward pointers to Chapters 14 and 19	70
12	Labels and the running examples	12.1, 12.2; the big table; GP one-liners in `lstlisting` (`ellinit`, `ellglobalred`, `elltors`, `ellisomat`, `elldivpol`) with a pointer to Chapter 22	120
13	Exercises	the 17 exercises of Part 4	130
14	Notes and references	Silverman AEC and ATAEC, Cremona97, Washington EC, Cox, DiamondShurman, Husemoller, Knapp, MilneEC, Mazur77, Mazur78, Kenku, Faltings, Velu, Serre (SerreAbelian), Cohen (AGM); which theorem numbers were cited and checked	40
	total		**2,140**

11.5 4. Coverage table and exercises

11.5.1 4.1 Coverage: PLAN.md scope, Section 6 map rows and paper hooks to line-up items

PLAN.md scope item, Section 6 row, or paper hook	line-up items
Weierstrass equations (general and short)	1.1, 1.2
$b_2, b_4, b_6, b_8, c_4, c_6, \Delta, j$; discriminant and smoothness	1.1, 1.2, 1.6
changes of coordinates (u, r, s, t) and the transformation of c_4, c_6, Δ	1.3, 1.4
the group law (chord-tangent; associativity via Riemann-Roch)	2.1, 2.2, 2.3
the group of points over any field	2.2, 2.4
isogenies: definition, degree, separable isogenies and kernels	3.1, 3.2, 3.3, 3.6
the dual isogeny; $\deg[m] = m^2$	3.4, 3.5, 7.4
structure of $E[m]$ ($\mathbb{Z}/m \times \mathbb{Z}/m$ when char $\nmid m$)	3.5, 7.4
division polynomials; degree $(m^2-1)/2$ for odd m	4.1, 4.2, 4.3, 4.4
the 3-division polynomial, degree 4, roots = x-coordinates of the four lines of $E[3]$	4.5, 4.6
Weil pairing: nondegenerate, alternating, Galois-equivariant; det = cyclotomic	5.1, 5.2, 5.3
Tate module $T_\ell E$ (free of rank 2) and $V_\ell E$	5.3, 5.5
the Galois action: $\rho_{E,\ell}$ and $\bar{\rho}_{E,\ell}$	5.4, 5.6, 5.7
endomorphism rings ($\mathbb{Z}$, imaginary quadratic order, quaternion order)	6.1, 6.2, 7.4(iii)
complex multiplication; the 13 j-invariants; $a_p = 0$ at inert primes	6.3, 6.4, 6.5, 8.4
curves over $\mathbb{C}$: lattices, $\wp$, uniformisation, periods	7.1, 7.2, 7.3, 7.4
the period lattice Λ_E of a curve over $\mathbb{Q}$ and its area	7.6, 7.7
isomorphism classes = lattices up to homothety = $\mathrm{SL}_2(\mathbb{Z})\backslash\mathfrak{H}$	7.4, 7.5
curves over finite fields: Frobenius endomorphism, $a_p = p + 1 - \#E(\mathbb{F}_p)$	8.1, 8.7
the Hasse bound	8.2
characteristic polynomial $X^2 - a_p X + p$ on $T_\ell E$; $\alpha\beta = p$	8.3, 8.6
Mordell-Weil, rank and torsion	9.1, 9.2
Mazur's torsion theorem	9.3
isogenies over $\mathbb{Q}$ and isogeny classes (finite; Cremona's class labels)	9.4, 12.1
rational ℓ-isogenies and reducibility of $E[\ell]$; rational ℓ-torsion	9.5, 9.6, 5.2(iii)
the structure $\mathbb{Z}/3 + \mu_3$	9.7, 9.8
quadratic twists: definition, model $y^2 = x^3 + da_2x^2 + d^2a_4x + d^3a_6$	10.1
effect on c_4, c_6, Δ (d^2, d^3, d^6), same j	10.2
$a_p(E^{(d)}) = \chi_d(p)a_p(E)$; $\rho_{E^{(d)},\ell} = \rho_{E,\ell} \otimes \chi_d$	10.3, 10.4, 10.6
$L(E, s)$ by its Euler product with the bad factors; continuation stated	11.1

PLAN.md scope item, Section 6 row, or paper hook	line-up items
Cremona and LMFDB labels (11a1, 121a1, 539c1, 980i1, 726a1, 99A1 explained)	12.1, 12.2
Section 6 row: $\rho_{E,\ell}$, $\bar{\rho}$, $T_\ell E$, $V_\ell E$, $E[\ell]$ (with Chapters 4, 5, 11)	5.3, 5.4, 5.5, 5.6
Section 6 row: the twist E' of good reduction, $a_q(E')$, quadratic twists and their invariants (with Chapter 10)	8.3, 10.1, 10.2, 10.5
Section 6 row: 3-division polynomial, its factorisation pattern (with Chapters 5, 11)	4.4, 4.5, 4.6
Section 6 row: Eisenstein case, rational ℓ-isogeny, $\mathbb{Z}/3+\mu_3$, torsion (with Chapters 11, 14)	9.3, 9.5, 9.6, 9.7, 9.8
Section 6 row: the worked examples 11a1, 20a1, 121a1, 539c1, 980i1, 726a1, 99A1	1.6, 4.6, 8.7, 10.2, 12.2
hook: Lemma lem:twist (ii) (models and invariants of twists; q versus q^*)	10.1, 10.2
hook: Lemma lem:h0 (CM and $a_p = 0$)	6.2, 6.4, 6.5, 8.4
hook: Proposition prop:ns (3-division polynomial; $\det\bar{\rho}_3 = \chi_{-3}$; complex conjugation)	4.5, 4.6, 5.2, 5.6
hook: Definition def:terms ($v_q(j)$; twist invariance)	1.3, 1.6, 10.7
hook: the twist E' of good reduction and $a_q(E')$ in the table of Section 1	8.3, 8.5, 10.5
hook: Conjecture conj:eis (torsion, isogeny class, $\mathbb{Z}/3+\mu_3$)	9.3, 9.4, 9.7, 9.8
hook: Section 1.1 $(V, T, \rho, \bar{\rho})$	5.3, 5.4
also: Lemma lem:euler ($V = V' \otimes \chi_{q^*}$, $\alpha\beta = q$)	8.3, 10.3, 10.5
also: Theorems thm:twist, thm:quant (E_S, χ_S)	10.4, 10.6
also: Theorem thm:qual and the Section 3 Example (twists by -11, -7; labels; a_p)	8.7, 10.2, 10.4, 12.2
also: Lemma lem:old ($X^2 - a_pX + p$, multiplicativity of a_n)	8.3, 8.6, 11.1
also: Section 4.2 (T^D, $M_{f,\lambda}$ versus $T_\ell E$, point 26)	5.3, 5.5
also: Section 1 (isogeny class, optimal curve, area of the period lattice)	7.6, 7.7, 9.4, 12.1
also: Section sec:oot (trivial isogeny class) and point 27	9.9, 9.10
also: the appendix (`ellinit`, `elldivpol`, `ns3`)	1.1, 4.5, 12.1
also: what Chapters 1, 2, 4, 5, 6, 21 assume from Chapter 9 (Part 5)	3.5, 4.4, 4.5, 5.2, 5.4, 6.2, 6.5, 8.2, 8.5, 10.3, 10.4, 9.8

11.5.2 4.2 Exercises (Section 13 of the chapter), graded from easy to harder

1. Compute $b_2, b_4, b_6, b_8, c_4, c_6, \Delta, j$ for 11a1 $[0,-1,1,-10,-20]$; verify $4b_8 = b_2b_6 - b_4^2$, $1728\Delta = c_4^3 - c_6^2$ and $j = -2^{12}\cdot 31^3/11^5$ (1.1, 1.6).
2. From 1.3 show that $v_q(j) = 3v_q(c_4) - v_q(\Delta)$ is independent of the model, and that $v_q(\Delta)$ changes by multiples of 12. Deduce that a model with $v_q(\Delta) < 12$ is minimal at q (the converse direction is Chapter 10).
3. Using 2.1, check that $(5,5)$ on 11a1 has order 5 and list its multiples. With $\#\tilde{E}(\mathbb{F}_3) = 5$ and $\#\tilde{E}(\mathbb{F}_7) = 10$ and 9.2, show $E(\mathbb{Q})_{\mathrm{tors}} \simeq \mathbb{Z}/5$. Do the same for 20a1 ($\mathbb{Z}/6$).
4. Compute ψ_3 for 20a1 and 14a1, find their rational roots and the corresponding 3-torsion points, and explain the factorisation patterns $1+3$ and $1+1+2$ by the orbits on the four lines (4.5, 9.7).
5. Check the paper's claim (Proposition prop:ns, Section sec:anomaly) for 726a1: $\psi_3 = (x^2 - x -$

74)$(3x^2+8x+20)$; the two quadratic fields are $\mathbb{Q}(\sqrt{33})$ and $\mathbb{Q}(\sqrt{-11})$ and the third quadratic subfield of their compositum is $\mathbb{Q}(\sqrt{-3})$; and $3 \mid a_p$ for the good primes $p \equiv 2 \pmod 3$, $p < 40$ (use 8.7).

6. Check the paper's claim (Example of Section 3) that the twist of 121a1 by -11 lies in the class 121c with $\Delta = -11^8$: compute the twisted model by 10.1, its c_4, c_6, and identify it with 121c2 by 10.2(ii). Then twist by $+11$ and show the result is not isomorphic over $\mathbb{Q}$ to any curve of conductor 121 (it is 1936k2): the point of important_notes.md point 29.
7. Check the paper's claim that $f_{11a} \otimes \chi_{-11}$ is the form of 121d: verify $a_p(121\text{d}1) = \chi_{-11}(p)a_p(11\text{a}1)$ for $p \le 50$ by point counting. Show from j-invariants that 121d1 is the twist of 11a3, not of 11a1, and explain why the a_p cannot see the difference (9.4).
8. Check the paper's claim that the twist of 539c1 by -7 has conductor 77 and type I_3 at 7: show that it is 77c1 (same $j = 167^3/(7^3 \cdot 11^2)$), with $v_7(\Delta) = 3$ against 9 for 539c1 (Chapter 10 for the types).
9. For 11a1 and $p = 3, 5, 7, 13$ compute $\#E(\mathbb{F}_{p^2})$ from a_p by 8.3 and compare with a direct count $(15, 35, 60, 180)$. Show $|\alpha| = \sqrt{p}$.
10. Prove 8.5. Using it, check the paper's I_0^* term on 539d1: $E' = 11\text{a}3$, $a_7 = -2$, $e_3(E,7) = v_3(6) + v_3(60) = 2 = v_3(72)$, and show that the value does not depend on which twist of good reduction is used (10.5).
11. Show that $a_p(27\text{a}1) = 0$ for the primes $p \equiv 2 \pmod 3$, $p \le 50$, and $a_p(121\text{b}1) = 0$ for p inert in $\mathbb{Q}(\sqrt{-11})$, $p \le 50$, and explain both by 6.5. Compute $j\big(\frac{1+\sqrt{-163}}{2}\big)$ numerically and compare with 6.4.
12. Use the Weil pairing to show that complex conjugation acts on $E[\ell]$ (ℓ odd) with eigenvalues $1, -1$, and that a rational point of order ℓ forces $E[\ell]/(\mathbb{Z}/\ell) \simeq \mu_\ell$ (5.2, 5.6).
13. Show $a_p(11\text{a}1) \equiv 1 + p \pmod 5$ for every good p and check it for $p < 100$; explain by $E[5] \simeq \mathbb{Z}/5 \oplus \mu_5$ (9.7). Which of the two 5-isogenous curves has a rational 5-torsion point, and why?
14. Check the claim of 9.8 (important_notes.md point 17) in detail: prove that a non-CM curve with $E[3] \simeq \mathbb{Z}/3 \oplus \mu_3$ has no rational point of order 9, and deduce that the bound "$D \le 2$ for $\ell = 3$" of Conjecture conj:eis follows from its first bound and Mazur's theorem.
15. Check the paper's statement (Section sec:oot) that a curve with trivial isogeny class is optimal and has $E[\ell]$ irreducible for every ℓ (9.10). Show more generally (9.9) that if $E[\ell]$ is irreducible then $T_\ell E \simeq T_\ell E'$ for all E' in the class.
16. For $\Lambda = \mathbb{Z} + \mathbb{Z}i$ and $\Lambda = \mathbb{Z} + \mathbb{Z}e^{2\pi i/3}$ show $g_3 = 0$, resp. $g_2 = 0$, hence $j = 1728$, resp. 0, and compute End. For 11a1, compute ω_1 by the AGM (formula of 7.6(iii)) and check that the reduced τ gives $j(\tau) = j(E)$ (7.7).
17. Using the `alllabels` file of `ecdata`, translate 11a1, 539c1, 726a1, 980i1 into LMFDB labels. Explain why Cremona's curve number 1 is the optimal curve while the LMFDB's number 1 need not be, and look up the exception 990h.

11.6 5. Dependencies and forward pointers

Results from other chapters used here (by chapter label only): * Chapter 1 (`ch:groups`): finite abelian groups (structure; $\#E[d] = d^2$ for all $d \mid m$ forces $(\mathbb{Z}/m)^2$), $\mathrm{GL}_2(\mathbb{Z}/m)$, $\mathrm{GL}_2(\mathbb{Z}_\ell)$ and inverse limits (its 7.4), the four lines of $\mathbb{F}_3^2$ (5.3), det = sign on lines (5.4(ii)), Borel subgroups (5.5), the orbit table and Sutherland labels (6.8, 6.10), complex-conjugation elements (6.9). * Chapter 2 (`ch:rings`): finite fields, the quadratic character η of $\mathbb{F}_\mathfrak{q}^\times$, Kronecker symbols, q^* and χ_{q^*} (C5), v_p. * Chapter 3 (`ch:modules`): $\mathbb{Z}_\ell$, free modules of finite rank, inverse limits of $\mathbb{Z}/\ell^n$-modules, Pontryagin duality ($E[\ell^\infty]$ versus $T_\ell E$). * Chapter 4 (`ch:reps`): twisting by characters, $\mathrm{ad}^0(V \otimes \chi) = \mathrm{ad}^0 V$, $V^\vee \simeq V \otimes \det^{-1}$ in dimension two, semisimplification. * Chapter 5 (`ch:galois`): Galois theory of a polynomial (galois:galois-group-polynomial, galois:quartic-on-lines, galois:discriminant), profinite groups and continuity, the cyclotomic character, complex conjugation, quadratic characters of G_K. * Chapter 6 (`ch:ant`): imaginary quadratic orders and class number one (statements), splitting of p in $\mathbb{Q}(\sqrt{d})$, the Frobenius element Frob_p, D_p and I_p, the arithmetic/geometric convention (its 3.7), its CM preview 6.5. * Chapter 7 (`ch:local`, outlined in parallel): only the words "unramified at p", $I_p \subset D_p \simeq G_{\mathbb{Q}_p}$ (also in Chapter 6). * Chapter 8 (`ch:curves`):

projective plane and smooth plane curves, function fields, divisors and Pic^0, Riemann-Roch, Weierstrass model of a genus-one curve with a point (its 5.6), Pic^0 bijection (its 5.7), degree and separability of morphisms, unramified morphisms between genus-one curves (its 7.3(b)), differentials, the Frobenius morphism, compact Riemann surfaces, H_1 and Abel-Jacobi. * Background (PLAN.md Section 2): complex analysis (Liouville, Laurent series, meromorphic functions, the residue theorem for the $\wp$-function), linear algebra (quadratic forms, Cauchy-Schwarz for 8.2), real analysis (convergence of lattice sums).

Black boxes imported from later chapters (stated with a forward `\Cref`, never used in the proof of an earlier item of the chapter except 8.6, 9.2 and 9.9(iii), which say so): minimal models, conductor, Kodaira types, injectivity of reduction on prime-to-p torsion, the Tate curve (Chapter 10); Neron-Ogg-Shafarevich, mod ℓ images and their labels (Chapter 11); the fundamental domain and the bijectivity of j (Chapter 12); modularity, optimal curves, modular degree, Manin constant (Chapter 14); analytic continuation of $L(E,s)$ (Chapter 19).

Forward pointers (chapters that use this one): * Chapter 10 (`ch:reduction`): Weierstrass data and coordinate changes (1.1 to 1.3) for minimal models and Tate's algorithm; twists and their invariants (10.1, 10.2) for Lemma lem:twist (i), (ii); $\#\tilde{E}(\mathbb{F}_p)$ and a_p at bad primes (8.7, 11.1); the Tate curve's $E_{\mathfrak{q}}[\ell^t]$ uses $E[\ell^t]$ (3.5, 5.3); 9.9(iii). * Chapter 11 (`ch:galrep`): ρ, $\bar{\rho}$, $\det = \chi_\ell$, complex conjugation (5.2 to 5.6); the division polynomial (4.5, 4.6) for Proposition prop:ns; CM (6.5) and twists (10.3) for Lemma lem:h0; rational isogenies and $\mathbb{Z}/\ell \oplus \mu_\ell$ (9.5 to 9.8) for the Eisenstein case and the labels 3B.1.1, 3Cs.1.1; 8.6 (easy half of Neron-Ogg-Shafarevich). * Chapter 12 (`ch:modforms`): lattices, $\mathrm{SL}_2(\mathbb{Z})\backslash\mathfrak{H}$, $j(\tau)$ (7.1 to 7.5); 11a1's coefficients (8.7). * Chapter 13 (`ch:hecke`): $X^2 - a_pX + p$ and the recursion (8.3, 11.1); twisting newforms matches 10.4. * Chapter 14 (`ch:moddeg`): isogeny classes, labels and optimality convention (9.4, 9.9, 12.1), the dual isogeny (3.4), the period lattice and its area (7.6, 7.7), $L(E,s)$ (11.1), torsion (9.3) for Mazur's Eisenstein-ideal results. * Chapter 15 (`ch:modgalrep`): $\rho_{E,\ell}$ versus $\rho_{f,\lambda}$, $T_\ell E$ versus $M_{f,\lambda}$ (5.5, important_notes.md points 26, 40), twisting (10.3, 10.4) in Theorem thm:qual (b), (c). * Chapters 17, 18 (`ch:galcoh`, `ch:blochkato`): $E[n]$ and the Kummer sequence, $T(1)$, T^D (5.3, 5.5). * Chapter 19 (`ch:lfunctions`): $L(E,s)$, the Frobenius convention (5.7), α, β (8.3), the I_0^* factor via 8.5 and 10.5. * Chapters 21, 22 (`ch:paper`, `ch:computation`): every hook of Part 0, the running examples table (12.2), 9.8 for Conjecture conj:eis, the PARI commands. * Chapters 1, 2, 4, 5, 6 already point here for: $E[m] \simeq (\mathbb{Z}/m)^2$ and $T_\ell E$ (Chapter 1), $\#E(\mathbb{F}_q) = q + 1 - a_q$, Hasse and twists (Chapter 2), twists $\rho \otimes \chi_d$ and the Weil pairing (Chapter 4), $\det \rho_{E,\ell} = \chi_\ell$ and the roots of ψ_3 (Chapter 5), CM and $a_p = 0$ (Chapter 6). All are delivered: 3.5, 5.3, 8.1, 8.2, 10.1 to 10.4, 5.2, 4.4, 4.5, 6.2, 6.5.

11.7 6. Notes for the writer

Pitfalls and conventions: * **Characteristic.** Short Weierstrass forms and the Aut computation need $\mathrm{char}\, K \neq 2, 3$; the general formulas of 1.1 work in all characteristics. Over $\mathbb{Q}$ say once that "a model" means one with the general a_i, because Cremona's minimal models have $a_1, a_3 \neq 0$ in general (121a1, 726a1, 99a1). * **Twists of general models.** Never twist $[a_1, a_2, a_3, a_4, a_6]$ by multiplying a_2, a_4, a_6 directly unless $a_1 = a_3 = 0$; complete the square first (10.1). The scaling $u = 2$ in the integral model multiplies c_4, c_6, Δ by $2^4, 2^6, 2^{12}$; the minimal model of the twist must then be recomputed (Chapter 10). $E^{(d)}$ depends on d mod squares; χ_d has conductor $|D|$, not $|d|$. * q **versus** q^* **(point 29).** The paper's twist is by $\mathbb{Q}(\sqrt{q^*})$; its invariants are $q^{*2}c_4$, $q^{*3}c_6$, $q^{*6}\Delta$. When $q \equiv 3 \pmod 4$ the twist by q is a different curve, ramified at 2 (121a1: -11 gives 121c2, $+11$ gives 1936k2). * **Curve versus class.** A twist of a curve is a curve, a twist of a class is a class. The paper and the Chapter 21 outline sometimes say "the twist of 11a1 is 121d1"; the twist of 11a1 is 121d2 (I_5^* at 11), that of 11a3 is 121d1 (I_1^*); the twist of 121a1 is 121c2, not 121c1; the twist of 11a1 by -7 is 539d2. Statements about a_p, L, V_ℓ and (for ℓ prime to the isogeny degrees) T_ℓ are class-level; statements about j, Δ, types, torsion and $v_q(j)$ are curve-level (Part 8, B2). * **Frobenius (C5, 5.7).** π on $\tilde{E}$ is the arithmetic Frobenius. $\det(1 - \mathrm{Frob}\, X \mid V_\ell E) = 1 - a_pX + pX^2$, while $\det(1 - \Phi X \mid V_\ell E) = 1 - \frac{a_p}{p}X + \frac{1}{p}X^2$. Do not write "the geometric Frobenius on $V_\ell E$ has eigenvalues α, β" (that is the slip recorded in important_notes.md B1). * **Weil pairing sign (C7).** State Silverman's definition; the comparison with $e^{2\pi i/m}$ over $\mathbb{C}$ has a sign that differs between sources: check before printing. * **Division polynomials.** Normalisations differ ($\psi_2 = 2y + a_1x + a_3$ here and in PARI; some books

use $2y$ for short models). PARI's `elldivpol(E,3)` is ψ_3 with leading coefficient 3; "irreducible" means irreducible in $\mathbb{Q}[x]$. A $2+2$ factorisation detects the image $N_s(\mathbb{F}_3)$ only for $E[3]$ irreducible and non-CM; CM curves can give $2+2$ (121b1, Chapter 21 outline A1.12) and reducible $E[3]$ gives a linear factor. * **Mazur's theorems.** Keep the torsion theorem (Mazur77), the prime-degree isogeny theorem (Mazur78) and the cyclic n-isogeny list (Mazur and Kenku) apart, and say which n occur only for CM curves; 9.8 uses only "$n = 27$ forces CM". * **Uniformisation.** Do not prove 7.3(ii) via the surjectivity of j if Chapter 12 proves that surjectivity by the valence formula and cites Chapter 9: one of them must be a reference. Recommended: Chapter 9 states 7.3(ii) with reference (R) and names both routes. * **Period lattice.** Use the Neron differential of the global minimal model (Chapter 10); `ecdata`'s `allbsd` column Ω equals ω_1 for 11a1 ($\Delta < 0$); for $\Delta > 0$ check whether the column is ω_1 or $2\omega_1$ before quoting it. The area $\mathrm{vol}(\Lambda_E)$ is what Chapter 14's formula uses; do not confuse it with Ω. * **Optimality convention.** "Curve number 1 is optimal" is a property of Cremona's labelling (exception 990h), not of the LMFDB's; the LMFDB class letters also differ (539c = 539.d, 726a = 726.b, 980i = 980.b). * **Labels at ℓ.** The image labels (3Ns, 3Cs.1.1, 3B.1.1, 5Cs.1.1, ...) are Chapter 11's; use them here only in tables, with the intrinsic description (Chapter 1, 6.10; important_notes.md point 3). * **Theorem numbers.** Every AEC/ATAEC number in Part 2 marked "(verify)" was recalled, not checked against the book; only AEC V.1.1 (Hasse) is quoted without that mark. Print section numbers if in doubt. * Bibliography to add to `bib/ch09.bib` (complete entries, checked): Velu 1971 (C. R. Acad. Sci. Paris 273), Faltings 1983 (Invent. Math. 73), Kenku 1982 (J. Number Theory 15), Deuring 1941 (Abh. Math. Sem. Hamburg 14), Serre-Tate 1968 (Ann. of Math. 88), Washington's *Elliptic Curves: Number Theory and Cryptography* (2nd ed., 2008), Cox's *Primes of the form $x^2 + ny^2$*, Cohen's *A course in computational algebraic number theory*, and optionally Lozano-Robledo 2013. Check every entry before adding. Exception (decision B14, important_notes.md B16, B19): Serre-Tate 1968 (`SerreTate68`) and Kenku (`Kenku`) are needed by more than one chapter and go into `refs.bib`, not `bib/ch09.bib`.

What to leave to other chapters (and not do here): * Chapter 8: Riemann-Roch, existence of the Weierstrass model, the Pic^0 bijection, unramifiedness of isogenies, Jacobians. * Chapter 10: minimality, Tate's algorithm, Kodaira types, conductor, the formal group, the Tate curve, Lemma lem:twist (i), (ii) on types (Chapter 9 only supplies the models and invariants). * Chapter 11: images, Neron-Ogg-Shafarevich in full, Proposition prop:ns and Lemma lem:h0 in full (with points 1, 5, 6, 9, 10, 33), the Ind ψ over $\mathbb{F}_9$ issue (point 33), Serre's open image theorem. * Chapter 12: modular forms, the fundamental domain, j's q-expansion; Chapter 14: modularity, $X_0(N)$, optimal curves, the modular degree, the Manin constant, points 39 to 41; Chapter 19: the analytic theory of $L(E, s)$ and of the adjoint.

How the relevant important_notes.md points are handled: * **Point 14** (CM by $\mathbb{Q}(\sqrt{\ell^*})$ forces $\ell \mid N$): 6.5(iv), stated (R) with the data check (462 CM curves below conductor 10,000, no exception) and a `paperlink` to Lemma lem:h0. * **Point 15** (e_ℓ not twist-invariant): 10.7, with the corrected example pairs 11a1/121d2 and 11a3/121d1. * **Point 17** (the rational cyclic 27-isogeny): **decided: Chapter 9**, item 9.8, with full proof. Reason: every ingredient (isogenies and duals, quotients by finite subgroups, cyclic factorisation, the Weil pairing for μ_3, the CM list for $X_0(27)$, Mazur's torsion theorem) is in this chapter, and the statement is about torsion and isogenies, not about images. Chapter 11 restates it in the language of the label 3Cs.1.1 with a `\Cref{ch:ellcurves}`; Chapters 14 and 21 cite it. * **Point 26 and point 40** ($M_{f,\lambda}$ versus $T_\ell E$): 5.5 supplies $H^1(E, \mathbb{Z}_\ell) = \mathrm{Hom}(T_\ell E, \mathbb{Z}_\ell) \simeq T_\ell E(-1)$ exactly. * **Point 27** (optimality irrelevant at ℓ when $E[\ell]$ is irreducible): 9.9, proved. * **Point 29** (q versus q^*): 10.2(iii), with the 121a1 example. * **Point 33** ($\bar\rho = \mathrm{Ind}\,\psi$ only over $\mathbb{F}_9$): not in this chapter's scope; 4.6 only reads lines over $\mathbb{F}_3$. * Points 5, 6 (the character in Lemma lem:h0): 6.5 states the CM facts for the quadratic character and a `warning` says the cyclotomic χ_ℓ is not a quadratic character; the proof of Lemma lem:h0 is Chapter 11's.

11.8 7. Verification log (computations done for this outline)

Scripts in `/tmp/ch09/scripts` (`ec.py` helper; `check1.py` to `check9.py`); data in `/tmp/ch09/data` downloaded from the `ecdata` GitHub repository (`allcurves`, `allisog`, `degphi`, `galrep`, `allbsd`, `alllabels`, all for conductors 0 to 9,999). The LMFDB API was not reachable (captcha), so LMFDB labels come from `alllabels`. * Symbolic (sympy, generic a_i): $4b_8 = b_2b_6 - b_4^2$; $1728\Delta = c_4^3 - c_6^2$; the completed square;

the short form $108^2\eta^2 = X^3 - 27c_4X - 54c_6$ with $X = 36x + 3b_2$; every transformation formula of 1.3 (by substitution); the twist scalings d^2, d^3, d^6; $x(2P) - x(P) = -\psi_3/\psi_2^2$; the curve with prescribed j_0 (1.4). * Division polynomials: degrees and leading coefficients for $m = 2, \ldots, 9$ on 11a1 and 99a1 (4.2); factorisations and discriminants of ψ_3 for the eight curves of 4.6. * Invariants of all curves in 1.6, 4.6, 12.2 from their a-invariants. * Group law: orders of the points in 2.4. * Point counts: a_p of 8.7 (and at bad primes), Hasse for good $p < 2000$ on seven curves, $\#E(\mathbb{F}_{p^2})$ for 11a1, CM vanishing for 27a1, 36a1, 32a1, 121b1 ($p < 500$), $a_p \equiv 1 + p$ for 11a1 mod 5 and 14a1 mod 3, the twist relations of 10.4 ($p < 400$, also $p = 2$ where allowed), the Section 3 congruences (121a1 versus $\chi_{-11}\cdot$11a1 mod 3, 539c1 versus $\chi_{-7}\cdot$11a1 mod 3, $p < 1000$; 980i1 versus $\chi_{-7}\cdot$20a1 mod 5, $3 \le p < 1000$), the η-product for 11a1 up to q^{20}. * Twist identifications of 10.2 by comparing j and $c_6c_4'/(c_6'c_4)$ against all curves of the target conductors. * mpmath: the thirteen CM j-values as $1728J(\tau)$; the periods, area and reduced τ of 11a1 (AGM, Cohen 7.4.7 shape), $j(\tau) = j(E)$ to 40 digits; ω_1 agrees with `allbsd`. * `ecdata`: torsion orders, class sizes, isogeny degrees, the 27a isogeny matrix and j-values, label/isogeny consistency at $\ell = 3, 5, 7$, 3Cs.1.1 versus 9-torsion, the six curves with a point of order 9 (all 3B.1.1), CM curves and conductors, the 990h exception, modular degrees and images of the running examples, LMFDB labels. * Not recomputed here: Kodaira types at 2 and 3 (taken from the Chapter 21 outline's Tate algorithm), anything beyond conductor 10,000. ($r_E = 12$ for 99A1 is recomputed in the Chapter 14 outline, 8.5, and in the Chapter 13 outline; cite Chapter 14.)

11.9 8. Points for important_notes.md

New points about the paper in this chapter's area (not already recorded; each checked as stated):

- **N1. Section sec:eis: "the split Cartan image 3Cs.1.1".** Problem: 3Cs.1.1 is not the split Cartan subgroup but its subgroup $\{\mathrm{diag}(1, \pm 1)\}$ of order 2 (the image when $E[3] \simeq \mathbb{Z}/3 \oplus \mu_3$); the full split Cartan image is the label 3Cs (order 4), whose curves have no rational 3-torsion (e.g. 539a2, torsion trivial). Fix: "the image 3Cs.1.1 (inside a split Cartan subgroup, fixing a nonzero vector), i.e. $E[3] \simeq \mathbb{Z}/3 \oplus \mu_3$". Confidence: high (`ecdata` below 10,000: all 264 curves labelled 3Cs.1.1 have a rational point of order 3; Chapter 1, 6.10). Expository.
- **N2. Lemma lem:twist (ii), proof, the $\mathrm{I}_0 \leftrightarrow \mathrm{I}_0^*$ sentence (addendum to point 29).** Problem: for the twisted model $y^2 = x^3 + q^*a_2x^2 + q^{*2}a_4x + q^{*3}a_6$ the cubic of Tate's algorithm is $T^3 + \varepsilon a_2T^2 + a_4T + \varepsilon a_6$ with $\varepsilon = q^*/q$, not $T^3 + a_2T^2 + a_4T + a_6$ as written. Its roots are ε times those of the latter, so "distinct roots mod q" is unaffected. Fix: "whose cubic $T^3 + \varepsilon a_2T^2 + a_4T + \varepsilon a_6$, $\varepsilon = q^*/q$, has distinct roots modulo q because $T^3 + a_2T^2 + a_4T + a_6$ does". Confidence: high (one-line substitution $T = \varepsilon S$). Expository.
- No other new problem was found in the paper's statements about Weierstrass data, twists, torsion, isogenies, the 3-division polynomial or CM; the Example of Section 3 (twists, labels, congruences, modular degrees), the appendix's 121a1 model, and Section sec:eis's claims on 3Cs.1.1 and 9-torsion were all consistent with `ecdata` in the range checked.

Book-internal inconsistencies found: * **B1 extension (Frobenius on $V_\ell E$).** The Chapter 21 outline, C2(b), says that with the geometric Frobenius $L_p(V_\ell E, s) = (1 - a_pp^{-s} + p^{1-2s})^{-1}$ and $\det(1 - \Phi X \mid V) = (1 - \alpha X)(1 - \beta X)$. Both are wrong: with the geometric Frobenius on $V_\ell E$ the factor is $(1 - \frac{a_p}{p}p^{-s} + \frac{1}{p}p^{-2s})^{-1} = L_p(E, s + 1)$; the stated formula holds for $V_\ell E^\vee = H^1$ (or with the arithmetic Frobenius on $V_\ell E$). Nothing on $\mathrm{ad}^0 V$ changes. Chapter 9 fixes the convention in C5 and 5.7. * **B2. Curve-level twist identifications.** important_notes.md point 15 says the twist of 11a1 by χ_{-11} has type I_1^*; in fact $11a1^{(-11)} = 121d2$ has type I_5^* ($v_{11}(\Delta) = 11$); I_1^* is $121d1 = 11a3^{(-11)}$. The value $e_3 = v_3(120) = 1$ is the same, so the point stands. Likewise the Chapter 21 outline (A1.1, B4 item 6, Section 9 "11a1 twisted by -7 is 539d1", the verification line "exact twist relations 121a1/121c1, 11a1/121d1") pairs curves that are twists only up to isogeny: $11a1^{(-7)} = 539d2$, $11a3^{(-7)} = 539d1$, $121a1^{(-11)} = 121c2$. All quantities it uses at $\ell = 3$ are class-level (the isogenies have degree 5 or 11), so no conclusion changes; the wording should be class-level. * **Chapter 6 outline, 6.5 versus this chapter's 6.5.** Consistent; Chapter 6's item is the preview and should cite Chapter 9 for the Galois action on

endomorphisms and for (iv), as it already plans. No conflict. * **Chapter 5 outline, galois:ns-example.** The curve $y^2 = x^3 - 6x - 8$ is 6912d1 in Cremona's tables (conductor 6912); its data agree with this outline (4.6). Not an inconsistency; the label may be added there.

Chapter 12

Chapter 10: Elliptic curves over local fields: reduction, Kodaira types, Tate curves (`ch:reduction`, `chapters/ch10_reduction.tex`)

Stage 1 line-up. Part III (Curves and elliptic curves). Target length 2,150 lines of LaTeX (card excluded; see Part 3), inside the 1,000 to 2,200 range of PLAN.md Section 7.3.

Rigour tags: **P** = proved here, **S** = proof sketched here, **R** = stated; reference. Item format, as in the Chapter 2 to 9 outlines: number, label-ready name, kind, tag; the precise statement with all hypotheses; reference (book and section or chapter; a theorem number only where the outline author is sure, otherwise the section, and numbers marked "(verify)" must be checked against the source before they are printed); the paper's use by its `\label` names, or "background". Every computation quoted below (minimal models, Tate's algorithm on the running examples, Kodaira types and $v(\Delta)$, conductor exponents, Tamagawa numbers, Tate parameters and their unit parts, Kraus defects, the twist exchange of types, the inertia images at 2, PARI's Euler factors at 2 and 3) was checked with PARI/GP 2.17.2 (through `cypari2`) and Python against Cremona's `ecdata` files `allcurves` for conductors below 30,000 (verification log, Part 6.6). The special task (important_notes.md point 43) is settled in Part 7: **proved**.

12.1 0. Purpose of the chapter, and the split with its neighbours

12.1.1 0.1 Purpose

The paper works prime by prime. At each bad prime q it reads off a handful of local invariants of E: the Kodaira type, the geometric component group Φ_q^{geom} and the Tamagawa number c_q, $v_q(\Delta)$ and $v_q(j)$, the conductor exponent, the Tate parameter q_E and its unit part u at a potentially multiplicative prime, the order of the image of inertia at a potentially good prime (Kraus), and the effect on all of these of the quadratic twist by $\mathbb{Q}(\sqrt{q^*})$. This chapter defines every one of these objects, proves what can be proved at the book's level, states the rest precisely, and checks all of it on the paper's curves. Concretely it must deliver:

1. **Definition def:terms.** $v_q(j_E)$ as an invariant of the curve (not the model), the trichotomy good, multiplicative, additive, and the dichotomy potentially good ($v_q(j) \geq 0$) versus potentially multiplicative ($v_q(j) < 0$); the Tate parameter q_E with $v_q(q_E) = -v_q(j)$; the statement that $t_\ell(E, q) = v_\ell(-v_q(j))$ is $v_\ell(n)$ for I_n at every q and for I_n^* at odd q, while at $q = 2$ a symbol I_m^* is either potentially good or has $m = \nu + 4$ or $m = \nu + 8$ with $\nu = -v_2(j)$ (items 7.4, 7.7, 9.4).

2. **The table of Section 1 and Lemma lem:euler (the local half).** For $q \geq 5$ the Kodaira type is read off $v_q(\Delta_{\min})$ and $v_q(c_4)$ (item 5.3); the image of inertia at a potentially good prime has order $12/\gcd(v_q(\Delta), 12)$ (Kraus, item 6.6). The Euler factors themselves are Chapter 19's.
3. **Section 1.4 of the paper.** Neron models, $\Phi_q^{\mathrm{geom}} = \Phi_q(\overline{\mathbb{F}}_q)$ and its order and structure for each type, $c_q = \#\Phi_q(\mathbb{F}_q)$ and why $c_q \neq \#\Phi_q^{\mathrm{geom}}$ in general; the exchange of types under the twist by $\mathbb{Q}(\sqrt{q^*})$; the weight $w_\ell(E,q)$ and the formula $w_\ell(E,q) = \max(v_\ell \#\Phi_q^{\mathrm{geom}}(E), v_\ell \#\Phi_q^{\mathrm{geom}}(E^{(q)}))$ (items 4.1 to 4.4, 5.1, 8.4).
4. **Lemma lem:twist (i), (ii), proved in full** (Chapter 9 supplied the models and invariants of twists; here the types, component groups and conductor exponents), with important_notes.md points 28, 29, 45 built in (items 8.1, 8.2).
5. **Lemma lem:tamagawa-exp and Lemma lem:tamfactor (the curve-theoretic input).** The Tate curve, Tate's uniformisation theorem, the fact that a split multiplicative curve is a Tate curve and a nonsplit one becomes one over the unramified quadratic extension, $E_{\mathfrak{q}}[\ell^t]$ as an extension of $\mathbb{Z}/\ell^t$ by μ_{ℓ^t} generated by an ℓ^t-th root of $\mathfrak{q}$, and the unit part u (items 7.1 to 7.5). The Galois-representation consequences are Chapter 11's.
6. **Lemma lem:IV and Remark rem:types (the curve-theoretic input).** Kraus's theorem on the semistability defect (item 6.6), and the Serre-Tate description of the minimal extension of good reduction (item 6.5). Chapter 11 turns them into statements about $\rho(I_q)$ and $E[3]^{I_q}$.
7. **Remark rem:wild, Section sec:wild and Proposition prop:forms.** What changes at 2 and 3: wild conductor exponents ($f_2 \leq 8$, $f_3 \leq 5$), the classification of inertia images at 2, the theorem that types IV, IV* at 2 are exactly the tame additive potentially good curves at 2 and have inertia image cyclic of order 3 (item 6.8: this settles important_notes.md point 43 and removes the exception in Proposition prop:forms), and the facts at 3 (types II, II*, IV, IV* are always wild there) (items 6.7, 6.8, 9.1 to 9.4).
8. **The appendix of the paper.** PARI's Kodaira codes and the outputs of `ellglobalred` and `elllocalred` (item 5.5, 10.2), Cremona's tables (10.1).
9. **Discharging deferrals.** Chapter 9 sends here: minimality and its criterion, reduction types, $a_p \in \{1, -1, 0\}$ at bad primes, the reduction map and its injectivity on prime-to-p torsion (needed for Chapter 9's items 8.6 and 9.2), the Neron differential, Kodaira types, the conductor, the formal group, the Tate curve, Lemma lem:twist (i), (ii), and the invariance of $v_\ell(-v_q(j))$ under isogenies of degree prime to ℓ (its 9.9(iii)). Chapter 7 sends: the Tate curve itself, Kraus's theorem, the Kodaira column of its table 9.4 and the hypothesis of its 9.5(ii) (cyclic inertia of order 3 for type IV at 2). Chapter 1 sends: component groups as finite abelian groups. Chapter 2 sends: the unit part of the Tate parameter. Chapter 5 sends: $E_{\mathfrak{q}}[\ell^t]$ generated by μ_{ℓ^t} and an ℓ^t-th root of $\mathfrak{q}$. Chapter 6 sends: the twist by $\mathbb{Q}(\sqrt{q^*})$ is unramified outside q (used in 8.1). Chapter 8 sends: smoothness and singular points of reductions. Chapter 16 sends: $c_q = n$ (split), 1 or 2 (nonsplit), and the unit part of the Tate parameter of 11a1 at 11. All are delivered (Part 5).

12.1.2 0.2 Split with the neighbouring chapters (decided here; consistent with PLAN.md Section 4 and the written outlines)

- **Chapter 7** (written outline) supplies: $\mathbb{Q}_p$, Hensel, $\mathbb{Q}_p^{\mathrm{ur}}$, inertia and wild inertia, the tame character and the tame relation, Kummer theory over $\mathbb{Q}_p$, the restriction of a Kummer class to inertia, Weil-Deligne representations, the Artin and Swan conductors, local L-factors, and the local shapes table (its 9.4). Chapter 10 uses the Artin conductor to define f_p and never redoes local field theory.
- **Chapter 9** (written outline) supplies: Weierstrass equations, b_i, c_4, c_6, Δ, j, the transformation formulas, the group law, $E[m]$, the Weil pairing, $T_\ell E$, $\rho_{E,\ell}$, curves over finite fields, a_p at good primes, quadratic twists $E^{(d)}$ with their invariants (d^2, d^3, d^6), q versus q^* (point 29), $\rho_{E^{(d)}} = \rho_E \otimes \chi_d$, $L(E,s)$ as an Euler product, labels. Chapter 10 cites these and proves only the local statements.
- **Chapter 11** (outlined in parallel) supplies: Neron-Ogg-Shafarevich in full, the local Galois representation at each kind of bad prime (in particular the paper's Lemma lem:tamagawa-exp and Lemma lem:IV proved in full, Remark rem:types in its corrected form, point 22), the Serre conductor. Chapter 10 states the two inputs those proofs need (the Tate curve with its torsion, item 7.5; Kraus

and Serre-Tate, items 6.5 to 6.8) and the easy direction "good reduction implies unramified" only through the injectivity of reduction on torsion (item 2.6).

- **Chapter 12** (outlined in parallel) supplies: $j(\tau)$ and its q-expansion $q^{-1} + 744 + 196884q + \cdots$ over $\mathbb{C}$. Chapter 10 uses the same series as a formal power series with integer coefficients (item 7.2) and says so.
- **Chapter 16 and 17** supply the cohomology in Lemma lem:tamfactor; Chapter 10 gives only q_E, u, n and c_q.
- **Chapter 19** supplies the Euler factors $L_q(A_f, s)$, $P_q(X)$, $e_\ell(E, q)$, Lemma lem:euler in full and the Euler-factor half of Remark rem:wild. Chapter 10 supplies only the inertia images on which those computations rest, and gives the $P_2(X) = 1 + 2X$ consequence of item 6.8 as a `paperlink` with a forward pointer (Part 7).
- **Chapter 22** supplies PARI/GP in practice and the `ecdata` file formats. Chapter 10 explains the mathematics of `ellglobalred`, `elllocalred`, the Kodaira codes and the Tate parameter, with one-line GP examples.

12.2 1. Global conventions fixed in this chapter

- **R1 (local setting).** K is a field complete for a discrete valuation v, of characteristic 0, with valuation ring $\mathcal{O}$, maximal ideal $\mathfrak{m} = \varpi\mathcal{O}$, perfect residue field k of characteristic p, and $v(\varpi) = 1$. In every application K is $\mathbb{Q}_p$, a finite extension of $\mathbb{Q}_p$, or the completion of the maximal unramified extension $\mathbb{Q}_p^{\mathrm{ur}}$ (residue field $\bar{\mathbb{F}}_p$); Chapter 7, conventions L1, L2. When the paper's prime is meant, $K = \mathbb{Q}_q$ and $p = q$. The letter q is also the Tate parameter's variable; the chapter writes the Tate curve as E_{q} or E_t with a different letter where both occur in one display (writer's choice, fixed once), and calls the Tate parameter of E always q_E (important_notes.md point 8).
- **R2 (models).** "Model" means a Weierstrass equation with the general $a_1, \ldots, a_6$ (Chapter 9, C3), never only a short one: at $p = 2, 3$ short models are never minimal in general. Coordinate changes (u, r, s, t) and their effect on c_4, c_6, Δ are Chapter 9's item 1.3.
- **R3 (types and codes).** Kodaira symbols as in PLAN.md 7.1. PARI's codes (item 5.5) appear only in tables, in `lstlisting` and in `paperlink` environments.
- **R4 (two component groups).** $\Phi_p^{\mathrm{geom}} = \Phi_p(\bar{k})$ is the geometric component group (the paper's Φ_q^{geom}); $c_p = \#\Phi_p(k)$ is the Tamagawa number. Never write c_p for $\#\Phi_p^{\mathrm{geom}}$; never use c for the class of the unit part of the Tate parameter (write $c(u)$; important_notes.md point 23).
- **R5 (conductor exponent).** $f_p = f_p(E)$; the letter δ_p is reserved for the paper's $\dim V^{I_p}$ (Section 1.1), so the wild part is written $\mathrm{sw}_p(E)$ (Swan conductor), as in Chapter 7.
- **R6 (semistability defect).** For potentially good reduction Φ_K (or Φ_p over $\mathbb{Q}_p$) denotes $\mathrm{Gal}(L/K^{\mathrm{ur}})$ for the minimal extension L of good reduction (item 6.5). To avoid a clash with the component group Φ_p, the writer must name it differently in the chapter: G_{ss} **(semistability group)** is recommended, with "defect" $e = \#G_{\mathrm{ss}}$; this outline writes G_{ss}.
- **R7 (Frobenius).** As in Chapter 7 (L3): Frob is arithmetic in group-theoretic statements. Chapter 10 needs Frobenius only in item 2.2 (split versus nonsplit) and in the `paperlink` of item 6.8.
- **R8 (characters).** $q^* = (-1)^{(q-1)/2}q$ and χ_{q^*}, the quadratic character of conductor q (Chapter 2, C5; Chapter 9, C4; the paper's χ_q, never written in the book, decision D3). The paper's $E^{(q)}$ is the twist by q^*, not by q (point 29).
- **R9 (typography).** PLAN.md Section 7.2: no dash ligatures (write "Kodaira-Neron", "Neron-Ogg-Shafarevich", "Ogg-Saito", "Dokchitser-Dokchitser", ranges "1 to 11"); ASCII source; labels `...:reduction:...`. `preamble.tex` does not load TikZ (checked), and chapters may not add packages: the special fibres are therefore given as a `booktabs` table (components, multiplicities, intersection pattern in words) and, if the writer wishes, as small LaTeX `picture` environments (no package needed). Bibliography keys: the paper's and `refs.bib`'s (`Tate75` is Tate's algorithm; PLAN.md's list also names `TateAlgorithm`, which is not in `refs.bib`: use `Tate75`).

12.3 2. Line-up of the core results, in teaching order

12.3.1 Section 1. Integral and minimal Weierstrass models

1.1 `def:reduction:minimal` (definition and proposition; P). In the setting R1, a Weierstrass model of E/K is *integral* if all $a_i \in \mathcal{O}$; then $b_i, c_4, c_6, \Delta \in \mathcal{O}$. It is *minimal* if $v(\Delta)$ is least among integral models of E. (i) Minimal models exist (scale any model by $u = \varpi^{-k}$, $k \gg 0$, to make it integral; $v(\Delta) \in \mathbb{Z}_{\geq 0}$). (ii) Two minimal models differ by a change (u, r, s, t) with $u \in \mathcal{O}^\times$ and $r, s, t \in \mathcal{O}$ (S: AEC VII.1.3(d) (verify); the proof uses that $u \in \mathcal{O}^\times$ and then the integrality of r, s, t from the transformation formulas of Chapter 9, item 1.3, with care at $p = 2, 3$). (iii) The invariant differential $\omega = dx/(2y + a_1x + a_3)$ of a minimal model is well defined up to $\mathcal{O}^\times$ (the *Neron differential*); $v(\Delta_{\min})$, and $v(c_4)$, $v(c_6)$ of a minimal model, are invariants of E/K, and so is $v(j) = 3v(c_4) - v(\Delta)$ for any model. Reference: AEC VII.1. Paper: Section 2 ("Tate's algorithm ... for the Kodaira type"), Definition def:terms ($v_q(j_E)$), Lemma lem:twist (ii) ("a minimal model of E at q"); Chapter 9's period lattice uses the Neron differential.

1.2 `prop:reduction:min-criterion` (proposition; P for (i), (ii); R for (iii)). Let a model be integral. (i) (any p) If $v(\Delta) < 12$, or $v(c_4) < 4$, or $v(c_6) < 6$, the model is minimal. (ii) ($p \geq 5$) The model is minimal iff $v(\Delta) < 12$ or $v(c_4) < 4$; equivalently, a non-minimal integral model becomes minimal after $(x, y) \mapsto (\varpi^2 x, \varpi^3 y)$ repeated until one of the two holds. (iii) (Kraus's conditions, for $K = \mathbb{Q}_p$ with $p = 2, 3$; R) Integers c_4, c_6 with $\Delta = (c_4^3 - c_6^2)/1728 \in \mathbb{Z} \setminus \{0\}$ are the invariants of an integral Weierstrass model over $\mathbb{Z}$ iff $v_3(c_6) \neq 2$ and either $c_6 \equiv -1 \pmod 4$, or $v_2(c_4) \geq 4$ and $c_6 \equiv 0$ or $8 \pmod{32}$. Consequently a model over $\mathbb{Z}$ with $p^4 \mid c_4$, $p^6 \mid c_6$, $p^{12} \mid \Delta$ ($p \in \{2, 3\}$) is non-minimal at p iff $(c_4/p^4, c_6/p^6)$ satisfies the condition at p (the condition at the other prime being inherited). Proof of (i): (u, r, s, t) with $u \in \mathcal{O}$, $v(u) \geq 1$ would make $\Delta' = u^{-12}\Delta$ etc. non-integral. (ii): if $v(c_4) \geq 4$, $v(\Delta) \geq 12$, then $v(c_6) \geq 6$ ($c_6^2 = c_4^3 - 1728\Delta$), and for $p \geq 5$ the model $y^2 = x^3 - 27c_4\varpi^{-4}x - 54c_6\varpi^{-6}$ is integral (the factors 27, 54, 2, 3 are units). Reference: AEC VII.1.1 and Exercise 7.1 (verify); for (iii) A. Kraus, "Quelques remarques a propos des invariants c_4, c_6 et Δ d'une courbe elliptique", Acta Arith. 54 (1989), 75 to 80 (key `Kraus89`, add to `bib/ch10.bib`; not the paper's `Kraus90`); Cremona's book (CremonaBook) Section 3.2 for the algorithmic use. Data check: 20,000 random integral models all satisfy (iii); on the 8,248 pairs (curve, $p \in \{2, 3\}$) of `ecdata` below conductor 10,000 whose minimal model has $p^4 \mid c_4$, $p^6 \mid c_6$, $p^{12} \mid \Delta$, (iii) correctly says "minimal" every time. Paper: Lemma lem:twist (ii) ("is minimal exactly when the result is less than 12, otherwise 12 is subtracted": that is (ii), plus $v(c_4) \geq 4$ for potentially good reduction with $v(\Delta) \geq 12$).

1.3 `thm:reduction:global-minimal` (theorem; S). Every $E/\mathbb{Q}$ has a *global minimal model*: a model over $\mathbb{Z}$ minimal at every prime; it is unique up to (u, r, s, t) with $u = \pm 1$, $r, s, t \in \mathbb{Z}$, and its discriminant Δ_E is the minimal discriminant. Sketch: minimal models at the finitely many primes dividing Δ of a given integral model are glued because $\mathbb{Z}$ has class number one (AEC VIII.8.2 (verify)). Cremona's tables list the global minimal model, normalised by $a_1, a_3 \in \{0, 1\}$, $a_2 \in \{-1, 0, 1\}$ (CremonaBook 3.2, verify the normalisation before printing). Paper: Section 1 ("E"), the appendix (`ellinit([1,1,1,-30,-76])`), Section sec:data ("recomputed with PARI ... from the a-invariants").

1.4 `ex:reduction:minimal-examples` (worked example; P). (a) The twist of 121a1 $[1, 1, 1, -30, -76]$ ($\Delta = -11^2$) by -11, written with b-invariants as in Chapter 9 (item 10.1), has $v_{11}(\Delta) = 2 + 6 = 8 < 12$, hence is minimal: it is 121c2 ($\Delta = -11^8$). Twisting 121c2 back by -11 gives $v_{11}(\Delta) = 14$ and $v_{11}(c_4) = 3 + 2 = 5 \geq 4$, so it is not minimal at 11; one step of 1.2(ii) gives $v_{11}(\Delta) = 2$: 121a1 again. (b) 140b1 $[0, 0, 0, 32, 212]$ has $v_2(\Delta) = 8$, $v_2(c_4) = 9$, $v_2(j) = 19$: minimal by 1.2(i), although c_4 is very divisible by 2. (c) The short model $y^2 = x^3 - 27c_4x - 54c_6$ of 11a1 has discriminant $6^{12}\Delta$ and is not minimal at 2 and 3. All checked. Paper: background.

12.3.2 Section 2. Reduction modulo ϖ

2.1 `def:reduction:types` (definition and proposition; P). Let the model be minimal and $\tilde{E}$ its reduction over k (Chapter 8 for singular points of plane cubics; Chapter 9, item 1.2: one singular point, a node if $\tilde{c}_4 \neq 0$, a cusp if $\tilde{c}_4 = 0$). E has *good* reduction if $\tilde{E}$ is smooth, i.e. $v(\Delta) = 0$; *multiplicative* reduction if $\tilde{E}$ has a node, i.e. $v(\Delta) > 0 = v(c_4)$; *additive* reduction if $\tilde{E}$ has a cusp, i.e. $v(\Delta) > 0$, $v(c_4) > 0$.

Multiplicative reduction is *split* if the two tangent directions at the node are defined over k and *nonsplit* otherwise. Over $\mathbb{Q}$ with p bad, $a_p := p+1-\#\tilde{E}(\mathbb{F}_p)$ (singular point counted) is 1, -1, 0 in the three cases split, nonsplit, additive (P: count $\tilde{E}_{\mathrm{ns}}(\mathbb{F}_p)$ by 2.3). Reference: AEC VII.5; VII.2. Paper: Section 1.1 ($\delta_p = 2, 1, 0$), the table and Definition def:terms ($a_q(E)$, $a_q^2 = 1$ at multiplicative q), Lemma lem:old ($a_p(f) = \pm 1$ for $p \,\|\, N$), the appendix (`k >= 5` detects multiplicative primes).

2.2 `prop:reduction:split` (proposition; P). Let E have multiplicative reduction. (i) After moving the node to $(0,0)$ the tangent cone is $y^2 + a_1xy - a_2x^2$; reduction is split iff $T^2 + a_1T - a_2$ has a root in k. (ii) For p odd, reduction is split iff $-c_6$ is a square in $k^\times$ (equivalently $-c_4/c_6$, since $\tilde{c}_4^3 = \tilde{c}_6^2$ makes $\tilde{c}_4$ a square). (iii) Split and nonsplit are exchanged by the unramified quadratic twist, and become equal over the unramified quadratic extension of K. Data check: for all 127,592 pairs (curve, odd multiplicative p) below conductor 10,000, $a_p = \left(\frac{-c_6}{p}\right)$. Reference: ATAEC V.5 (verify); Cremona's book 3.2. Paper: Lemma lem:tamagawa-exp ("This holds for split and for nonsplit reduction"), Lemma lem:tamfactor (ε of order at most 2), Lemma lem:old ($\varepsilon(\mathrm{Frob}_p) = a_p(f)$).

2.3 `prop:reduction:Ens` (proposition; P). The nonsingular points $\tilde{E}_{\mathrm{ns}}(k)$ form a group under the chord-tangent law, and $\tilde{E}_{\mathrm{ns}}(k) \simeq k^+$ (cusp), $k^\times$ (split node), $\ker(N\colon l^\times \to k^\times)$ (nonsplit node, l/k the quadratic extension). For $k = \mathbb{F}_p$ the orders are p, $p-1$, $p+1$. Reference: AEC III.2.5 (verify) and Exercise 3.5. Paper: background (via 2.1: $a_p \in \{0, 1, -1\}$; and via 4.2: the identity component of the special fibre of the Neron model).

2.4 `prop:reduction:filtration` (proposition; P). Define $E_0(K) = \{P : \tilde{P} \in \tilde{E}_{\mathrm{ns}}(k)\}$ and $E_1(K) = \{P : \tilde{P} = \tilde{O}\}$. Then $E_0(K)$ is a subgroup, reduction $E_0(K) \to \tilde{E}_{\mathrm{ns}}(k)$ is a surjective homomorphism (surjectivity by Hensel, Chapter 7) with kernel $E_1(K)$:

$$0 \to E_1(K) \to E_0(K) \to \tilde{E}_{\mathrm{ns}}(k) \to 0.$$

Reference: AEC VII.2.1 (verify). Paper: Section 1.4 ($c_q = [E(\mathbb{Q}_q) : E_0(\mathbb{Q}_q)]$ via 4.2), background.

2.5 `thm:reduction:formal-group` (theorem; S for (i), (ii); P for (iii) given (ii)). (i) The power series $w(z) = z^3 + a_1zw + a_2z^2w + \cdots$ ($z = -x/y$, $w = -1/y$) and the addition law define a one-parameter formal group $\hat{E}$ over $\mathbb{Z}[a_1, \ldots, a_6]$, and z gives an isomorphism $E_1(K) \simeq \hat{E}(\mathfrak{m})$. (ii) For $r > v(p)/(p-1)$ the formal logarithm is an isomorphism $\hat{E}(\mathfrak{m}^r) \simeq \mathfrak{m}^r$ (additive group). (iii) Consequently: $E_1(K)$ has no nonzero torsion of order prime to p (for m prime to p, $[m]$ is invertible on $\hat{E}(\mathfrak{m})$ since $[m](z) = mz + \cdots$ with $m \in \mathcal{O}^\times$). Only (iii) is used later; (i), (ii) are stated so that the reader knows where (iii) comes from. Reference: AEC IV.2 to IV.6, VII.2.2, VII.3.1 (verify all). Paper: background (it is the input of 2.6).

2.6 `cor:reduction:torsion-injects` (corollary; P given 2.4, 2.5). Let m be prime to p. (i) Reduction $E_0(K)[m] \to \tilde{E}_{\mathrm{ns}}(k)$ is injective. (ii) If E has good reduction, $E(K)[m] \hookrightarrow \tilde{E}(k)$, and over K^{ur} (or its completion) reduction is an isomorphism $E[m] \to \tilde{E}[m]$ (injective by (i) applied over the finite unramified extensions, bijective by counting: both have m^2 elements, Chapter 9, item 3.5), compatible with $G_K \to G_k$; hence inertia acts trivially on $E[m]$ and on $T_\ell E$ for $\ell \neq p$. Reference: AEC VII.3.1, VII.7 (verify). Paper: Section 1.1 (good primes are unramified, $\delta_p = 2$); Chapter 9's items 8.6 and 9.2 and Chapter 11's Neron-Ogg-Shafarevich criterion use (ii).

2.7 `prop:reduction:index-finite` (proposition; S). $E(K)/E_0(K)$ is finite; it is trivial when the reduction is good, and its order is computed in 4.2 and 7.6. Reference: AEC VII.6.1 (Kodaira-Neron; verify). Paper: Section 1.4 (c_q).

12.3.3 Section 3. Potentially good and potentially multiplicative reduction

3.1 `def:reduction:potential` (definition and theorem; S). E/K has *potentially good* (resp. *potentially multiplicative*) reduction if it has good (resp. multiplicative) reduction over some finite extension of K. (i) E has potentially good reduction iff $v(j) \geq 0$; (ii) potentially multiplicative iff $v(j) < 0$ (and then the reduction is multiplicative over every finite extension over which it is semistable). Proof sketch: good reduction over L forces j integral ($j = c_4^3/\Delta$ with Δ a unit); conversely for $p \geq 3$ use a Legendre model $y^2 = x(x-1)(x-\lambda)$ over $K(E[2])$ and $j = 2^8(\lambda^2-\lambda+1)^3/(\lambda^2(\lambda-1)^2)$, $p = 2$ by a similar cubic model (cite). Multiplicative reduction has $v(j) = -v(\Delta) < 0$. Reference: AEC VII.5.5 (verify). Paper: Definition def:terms (the case split $v_q(j_E) < 0$), Remark rem:wild ("a type I_n^* at 2 has potentially good reduction in about half of the cases").

3.2 `thm:reduction:semistable` (theorem, semistable reduction; S). For every E/K there is a finite extension L/K over which E has good or multiplicative (semistable) reduction; one may take $L = K(E[m])$ for any $m \geq 3$ prime to p (R, Serre-Tate). Sketch: (i) of 3.1 for $v(j) \geq 0$; the Tate curve (7.3) for $v(j) < 0$. Reference: AEC VII.5.4 (verify); SerreTate68 Section 2 (key in `refs.bib`, shared by Chapters 7, 9, 10, 11: decision B14). Paper: Lemma lem:IV, Lemma lem:euler ("the inertia group acts on V through a finite cyclic quotient when the reduction is potentially good, and through a quadratic character times a unipotent when it is potentially multiplicative").

3.3 `rem:reduction:types-of-the-four-cases` (remark; P given 2.1, 3.1). The four classes: good (I_0); multiplicative (I_n, $n \geq 1$); additive potentially multiplicative (I_n^*, $n \geq 1$, item 7.7); additive potentially good (all other symbols, and at $p = 2$ also I_n^*). This is the first appearance of the symbols, used as names until Section 5 defines them. Paper: Definition def:terms, Section 1.1.

12.3.4 Section 4. Neron models and component groups

4.1 `def:reduction:neron` (definition and theorem; R). Let $R = \mathcal{O}$ (or $\mathbb{Z}_p$). A *Neron model* of E/K is a smooth separated group scheme $\mathcal{E}$ of finite type over R with generic fibre E such that for every smooth R-scheme S every K-morphism $S_K \to E$ extends uniquely to an R-morphism $S \to \mathcal{E}$ (Neron mapping property). It exists and is unique up to unique isomorphism (Neron); for elliptic curves it is the smooth locus of the minimal regular proper model. The writer gives one paragraph of intuition (the best smooth integral model whose R-points are all of $E(K)$) and no scheme theory beyond the words used. Reference: NeronModels; BLR 1.2 and 1.4 (verify); ATAEC IV.5, IV.6 (verify). Paper: Section 1.4 ("the special fibre of the Neron model of E at q").

4.2 `prop:reduction:component-group` (proposition; R for (i); P for (ii) given (i)). (i) $\mathcal{E}(R) = E(K)$; the identity component $\mathcal{E}^0$ of the special fibre $\mathcal{E}_k$ is $\tilde{E}_{\mathrm{ns}}$ (for the minimal model) and $\mathcal{E}^0(R) = E_0(K)$; the component group $\Phi = \mathcal{E}_k/\mathcal{E}_k^0$ is a finite etale group scheme over k, i.e. a finite abelian group $\Phi(\bar{k})$ with a continuous action of $\mathrm{Gal}(\bar{k}/k)$. (ii) $E(K)/E_0(K) \simeq \Phi(k)$ (Hensel on the smooth $\mathcal{E}$). Definitions: $\Phi_p^{\mathrm{geom}} = \Phi_p(\bar{\mathbb{F}}_p)$ (geometric component group), $c_p = \#\Phi_p(\mathbb{F}_p) = [E(\mathbb{Q}_p) : E_0(\mathbb{Q}_p)]$ (Tamagawa number). As $\Phi_p(\mathbb{F}_p)$ is the subgroup of Frobenius invariants of Φ_p^{geom}, $c_p \mid \#\Phi_p^{\mathrm{geom}}$, with equality iff Frobenius acts trivially on Φ_p^{geom}. Reference: ATAEC IV.9 Corollary 9.2 (verify), BLR 9.5 (verify). Paper: Section 1.4 (definitions of Φ_q^{geom} and c_q, "the statements below are about Φ_q^{geom}, not c_q"), Lemma lem:tamfactor (the check "the ℓ-part of the Tamagawa number $c_q(E)$"), (eq:main).

4.3 `prop:reduction:etale-base-change` (proposition; R). The formation of the Neron model commutes with unramified (more generally etale) base change, e.g. from K to a finite unramified extension K' or to $\widehat{K^{\mathrm{ur}}}$. Consequently the reduction type, the Kodaira symbol, Φ^{geom}, $v(\Delta_{\min})$ and f are unchanged under unramified extension, while c can change (only $\mathrm{Gal}(\bar{k}/k)$- invariants are counted). Reference: BLR 7.2, Theorem 1 (verify); ATAEC IV.6 (verify). Paper: Lemma lem:twist (i) (its proof: "the Kodaira type, the geometric component group and the conductor exponent are invariant under unramified base change"), Lemma lem:tamagawa-exp ("which does not change the inertia group").

4.4 `def:reduction:tamagawa` (definition and remark; P). The Tamagawa number c_p of $E/\mathbb{Q}$ at p (with $c_p = 1$ at good p); the global product $\prod_p c_p$ (one sentence: the factor in the Birch and Swinnerton-Dyer formula; Chapter 22 for `ecdata`'s columns). The ℓ-part of c_p at a multiplicative prime of type I_n is $\ell^{v_\ell(n)}$ when the reduction is split and 1 when it is nonsplit and ℓ is odd (item 7.6). Paper: Section 1.4, the history paragraph (Ribet-Takahashi's product of the n_q), Lemma lem:tamfactor ("as it should").

4.5 `prop:reduction:component-invariants` (proposition; P given 2.3 to 2.5 and 4.2). Let $q \neq \ell$ be primes, $E/\mathbb{Q}_q$ with bad reduction, $\Phi^{\mathrm{geom}} = \Phi(\bar{\mathbb{F}}_q)$, $k \geq 1$. (i) If the reduction is additive, there is a $\mathrm{Gal}(\bar{\mathbb{F}}_q/\mathbb{F}_q)$-equivariant isomorphism $E[\ell^k]^{I_q} = E[\ell^k](\mathbb{Q}_q^{\mathrm{ur}}) \simeq \Phi^{\mathrm{geom}}[\ell^k]$; in particular $E(\mathbb{Q}_q)[\ell^\infty] \simeq \Phi(\mathbb{F}_q)[\ell^\infty]$. (ii) If the reduction is multiplicative there is an exact sequence $0 \to \mu_{\ell^k} \otimes \varepsilon \to E[\ell^k]^{I_q} \to \Phi^{\mathrm{geom}}[\ell^k] \to 0$ with $\Phi^{\mathrm{geom}} \simeq \mathbb{Z}/n_q$ and ε the unramified character of 7.3 (trivial for split reduction). (iii) Hence, for ℓ odd, $\dim_{\mathbb{F}_\ell} E[\ell]^{I_q}$ is $1 + [\ell \mid n_q]$ for I_n ($n \geq 1$), 1 for $\mathrm{IV}, \mathrm{IV}^*$ when $\ell = 3$, and 0 for every other additive type (the additive $\#\Phi^{\mathrm{geom}}$ are $1, 2, 3, 4$ by 5.1). Proof: over each finite unramified $K'/\mathbb{Q}_q$, $E(K')/E_0(K') \simeq \Phi(k')$ (4.2), $E_1(K')$ is a $\mathbb{Z}_q$-module (2.5), hence uniquely ℓ-divisible, and $E_0(K')/E_1(K') \simeq \tilde{E}_{\mathrm{ns}}(k')$ (2.4), which is k'^+ (additive, uniquely ℓ-divisible) or a torus (multiplicative); apply the snake lemma for $[\ell^k]$ and take the union over K'. This is the decisive input of the Chapter 11

proof of important_notes.md point 43 (type IV, IV* at $q = 2$: $\Phi^{\mathrm{geom}} \simeq \mathbb{Z}/3$, so inertia fixes a line of $E[3]$) and of the second proofs of Lemmas lem:IV and lem:tamagawa-exp there. **Decision (orchestrator):** this lemma lives here; Chapter 11 cites it (its outline's item 2.2 becomes a citation, Part 8, B-items). Reference: SerreTate68 Section 1 (the argument); AEC VII.6, VII.7 (verify); ATAEC IV.9 for $E(K)/E_0(K) \simeq \Phi(k)$. Paper: Lemma lem:IV ("$E[3]^{I_q}$ is a line"), Remark rem:types ("$E[3]^{I_q} = 0$"), Lemma lem:tamagawa-exp, Lemma lem:tamfactor (the check with $M = T$: "the ℓ-part of the Tamagawa number $c_q(E)$"), Proposition prop:forms (via 6.8 and Part 7).

12.3.5 Section 5. Kodaira symbols and Tate's algorithm

5.1 `thm:reduction:kodaira-neron` (theorem with table; R, with the pictures described). The special fibre of the minimal regular model of E over R (with k algebraically closed, or over $\bar{k}$) is one of the following configurations; the Neron model's special fibre is the union of its multiplicity-one components minus their intersection points. With m = number of irreducible components of the special fibre over $\bar{k}$:

symbol	m	configuration (over $\bar{k}$)	Φ^{geom}	$\#\Phi^{\mathrm{geom}}$	possible c (over finite k)
I_0	1	smooth elliptic curve	0	1	1
I_1	1	nodal rational curve	0	1	1
I_n, $n \geq 2$	n	n lines of multiplicity 1 in a cycle	$\mathbb{Z}/n$	n	n (split); 1 (n odd) or 2 (n even) (nonsplit)
II	1	cuspidal rational curve	0	1	1
III	2	two lines tangent at one point	$\mathbb{Z}/2$	2	2
IV	3	three concurrent lines	$\mathbb{Z}/3$	3	1 or 3
I_0^*	5	one line of multiplicity 2 meeting four of multiplicity 1 ($\tilde{D}_4$)	$(\mathbb{Z}/2)^2$	4	1, 2 or 4
I_n^*, $n \geq 1$	$n+5$	a chain of $n+1$ lines of multiplicity 2 with two of multiplicity 1 at each end ($\tilde{D}_{n+4}$)	$\mathbb{Z}/4$ (n odd), $(\mathbb{Z}/2)^2$ (n even)	4	2 or 4
IV*	7	$\tilde{E}_6$: multiplicities $1,2,3,2,1,2,1$	$\mathbb{Z}/3$	3	1 or 3
III*	8	$\tilde{E}_7$: multiplicities $1,2,3,4,3,2,1,2$	$\mathbb{Z}/2$	2	2

symbol	m	configuration (over $\bar{k}$)	Φ^{geom}	$\#\Phi^{\mathrm{geom}}$	possible c (over finite k)
II^*	9	$\tilde{E}_8$: multiplicities $1,2,3,4,5,6,4,2,3$	0	1	1

Φ^{geom} is the group of multiplicity-one components (the dual group of the extended Dynkin diagram modulo its root lattice: the RHB aside "the centre of the simply connected compact Lie group of type $A_{n-1}, D_{n+4}, E_6, E_7, E_8$" is one sentence, optional). Reference: Kodaira; NeronModels; ATAEC IV.8, IV.9 Table 4.1 (verify); Tate75. Data check: every value of c in the last column, and only those, occurs among the 211,455 pairs (curve, bad prime) below conductor 10,000 (for $\mathrm{IV}, \mathrm{IV}^*$: $c \in \{1,3\}$; I_0^*: $\{1,2,4\}$; I_n^*: $\{2,4\}$). Paper: Section 1.4 (the list of orders $n; 1; 2; 3; 4$ and "\cite[IV.9]{SilvermanATAEC}"), (eq:main), (eq:twist), Section sec:forms, Conjecture conj:eis (the deficit uses $\#\Phi_q^{\mathrm{geom}}$), Section sec:two ($v_2(\#\Phi_q^{\mathrm{geom}}) = 2$ for I_0^*).

5.2 `thm:reduction:tate-algorithm` (theorem, Tate's algorithm; S: statement in full, proof by reference, each step made plausible by the change of coordinates it performs). Input: an integral model over R, ϖ, k (perfect). Output: the Kodaira symbol, m, f (via Ogg's formula 6.2), c, and a minimal model. Write $a_{i,j} = a_i/\varpi^j$ and $P(T)$, $Q(Y)$ for the auxiliary polynomials over k: 1. If $\varpi \nmid \Delta$: I_0. 2. Change coordinates so that $\varpi \mid a_3, a_4, a_6$ (singular point at $(0,0)$). If $\varpi \nmid b_2$: I_n with $n = v(\Delta)$; split iff $T^2 + a_1 T - a_2$ splits over k; $c = n$ (split), 1 or 2 by the parity of n (nonsplit). 3. If $\varpi^2 \nmid a_6$: II, $c = 1$. 4. If $\varpi^3 \nmid b_8$: III, $c = 2$. 5. If $\varpi^3 \nmid b_6$: IV, $c = 3$ if $Y^2 + a_{3,1}Y - a_{6,2}$ has a root in k, else $c = 1$. 6. Change coordinates so that $\varpi \mid a_1, a_2$, $\varpi^2 \mid a_3, a_4$, $\varpi^3 \mid a_6$. Let $P(T) = T^3 + a_{2,1}T^2 + a_{4,2}T + a_{6,3}$. If P has distinct roots: I_0^*, $c = 1 + \#\{\text{roots of } P \text{ in } k\}$. 7. If P has one simple and one double root: I_n^* for some $n \geq 1$, found by a subprocedure (translate the double root to 0, then alternately test the quadratics $Y^2 + a_{3,j}Y - a_{6,2j}$ and $a_{2,1}X^2 + a_{4,j+1}X + a_{6,2j+1}$ for distinct roots, translating when they have a double root; each translation raises n by one); $c \in \{2,4\}$ (4 iff the final quadratic splits). 8. If P has a triple root, translate it to 0 (so $\varpi^2 \mid a_2$, $\varpi^3 \mid a_4$, $\varpi^4 \mid a_6$). If $Y^2 + a_{3,2}Y - a_{6,4}$ has distinct roots: IV^*, $c = 3$ or 1 according as the roots are in k or not. 9. Otherwise translate so that $\varpi^3 \mid a_3$, $\varpi^5 \mid a_6$. If $\varpi^4 \nmid a_4$: III^*, $c = 2$. 10. If $\varpi^6 \nmid a_6$: II^*, $c = 1$. 11. Otherwise the model is not minimal: replace a_i by a_i/ϖ^i and go to step 1. The writer prints the explicit translations of steps 2 and 6 for $p \geq 3$ and for $p = 2$ (they differ only in how one divides by 2; Cremona's book gives both). Reference: Tate75; ATAEC IV.9, Algorithm 9.4 (verify the number); CremonaBook 3.2 (the version with explicit formulas). Data check: an independent Python implementation of steps 1 to 11 (structure of Cremona's book) agrees with PARI's `elllocalred` on the Kodaira symbol, f and c for all pairs (curve, bad prime) tested (Part 6.6). Paper: Section 2 ("We use Tate's algorithm ... for the Kodaira type and c_q"), Section sec:data (types recomputed with PARI), the appendix.

5.3 `cor:reduction:tame-table` (corollary; P from 5.2). Let $p \geq 5$ and the model be minimal. Then (i) $v(\Delta) = 0$: I_0; (ii) $v(c_4) = 0 < v(\Delta) = n$: I_n; (iii) if $v(c_4) > 0$ and $v(j) \geq 0$, the type is determined by $v(\Delta) \in \{2,3,4,6,8,9,10\}$: $\mathrm{II}, \mathrm{III}, \mathrm{IV}, \mathrm{I}_0^*, \mathrm{IV}^*, \mathrm{III}^*, \mathrm{II}^*$ respectively; (iv) if $v(c_4) > 0$ and $v(j) < 0$: $v(c_4) = 2$, $v(\Delta) = n + 6$, $v(j) = -n$, type I_n^*. In particular $f = 2$ for every additive type and $f = v(\Delta) + 1 - m$ can be read off. Proof: for $p \geq 5$ use the short model $y^2 = x^3 + Ax + B$ ($A = -27c_4$, $B = -54c_6$ up to units) and run steps 2 to 10 with $(v(A), v(B), v(\Delta))$. The resulting table, for $p \geq 5$ and a minimal model, is $(v(c_4), v(c_6), v(\Delta)) = (0,0,n)$ for I_n; $(\geq 1, 1, 2)$ for II; $(1, \geq 2, 3)$ for III; $(\geq 2, 2, 4)$ for IV; $(\geq 2, \geq 3, 6)$ for I_0^*; $(2, 3, n+6)$ for I_n^*; $(\geq 3, 4, 8)$ for IV^*; $(3, \geq 5, 9)$ for III^*; $(\geq 4, 5, 10)$ for II^* (checked on every such pair below conductor 10,000; compare ATAEC IV.9 Table 4.1 and Papadopoulos 1993, key `Papadopoulos93`, J. Number Theory 44 (1993), 119 to 152, add to `refs.bib` (shared key, decision B14), which also gives the tables at $p = 2, 3$). Data check (`ecdata`, conductor below 10,000, potentially good additive pairs with $p \geq 5$): $v_p(\Delta)$ is $2,3,4,6,8,9,10$ for $\mathrm{II}, \mathrm{III}, \mathrm{IV}, \mathrm{I}_0^*, \mathrm{IV}^*, \mathrm{III}^*, \mathrm{II}^*$ in all 9,699 cases. Paper: Lemma lem:twist (ii) (proof: "the type is determined by $v_q(\Delta)$ of a minimal model, which lies in $\{0,2,3,4,6,8,9,10\}$"), Lemma lem:IV ("$v_q(\Delta_E) \in \{4,8\}$"), the table of Section 1, Definition def:terms.

5.4 `prop:reduction:I0star-cubic` (proposition; P). Let p be odd and E have a model $y^2 = x^3 + a_2x^2 + a_4x + a_6$ with $p \mid a_2$, $p^2 \mid a_4$, $p^3 \mid a_6$. Then E has type I_0^* iff $P(T) = T^3 + (a_2/p)T^2 + (a_4/p^2)T + a_6/p^3$

has distinct roots mod p; then $c = 1 + \#\{\text{roots in } \mathbb{F}_p\}$. Conversely every curve of type I_0^* at odd p has such a model. Corollary (the paper's sentence, corrected): if $y^2 = x^3 + a_2x^2 + a_4x + a_6$ has good reduction at p, its twist $y^2 = x^3 + p^*a_2x^2 + p^{*2}a_4x + p^{*3}a_6$ has cubic $T^3 + \varepsilon a_2T^2 + a_4T + \varepsilon a_6$ with $\varepsilon = p^*/p = \pm 1$, whose roots are ε times those of $T^3 + a_2T^2 + a_4T + a_6$; so the twist has type I_0^* (important_notes.md point 45). Proof: step 6 of 5.2; the converse by step 6's coordinate change and completing the square (p odd). Paper: Lemma lem:twist (ii) ("the cubic $T^3 + a_2T^2 + a_4T + a_6$ of Tate's algorithm has distinct roots modulo q"), the table (E' for type I_0^*), Theorem thm:quant (E_S good at the I_0^* primes).

5.5 `conv:reduction:pari-codes` (convention and table; P as a dictionary). PARI's `elllocalred(E,p)` returns $[f_p, k_p, [u, r, s, t], c_p]$ and `ellglobalred(E)` returns $[N, [u, r, s, t], \prod c_p$, factorisation of N, the list of local data], so that the paper's `R[4]` is the factorised conductor and `R[5][i]` is $[f, k, [u, r, s, t], c]$ at the i-th bad prime (`K[i][2]` is k). The Kodaira code k_p:

code	symbol	m	$\#\Phi^{\mathrm{geom}}$	f at $p \geq 5$	$v(\Delta_{\min})$ at $p \geq 5$
1	I_0	1	1	0	0
$n+4$ $(n \geq 1)$	I_n	n	n	1	n
2	II	1	1	2	2
3	III	2	2	2	3
4	IV	3	3	2	4
-1	I_0^*	5	4	2	6
$-(n+4)$ $(n \geq 1)$	I_n^*	$n+5$	4	2	$n+6$
-4	IV^*	7	3	2	8
-3	III^*	8	2	2	9
-2	II^*	9	1	2	10

The formulas "$n+4$ for I_n, $-(n+4)$ for I_n^*" hold for $n \geq 1$ only: $-(0+4) = -4$ is IV^*, and I_0^* is -1 (Part 8, N4). Hence `k >= 5` detects multiplicative reduction and `abs(k) == 3` the types $\mathrm{III}, \mathrm{III}^*$, as the appendix's `rule` uses. Reference: PARI users' guide, `elllocalred` (PARIusers; verify the order of the output vector against the installed version: checked on PARI 2.17.2). Paper: the appendix (`K[i][2]`, the comments `III, III* (codes 3,-3)`, `k == 4 || k == 2`, `k == -4 || k == -2`, `k == -3`).

5.6 `ex:reduction:tate-running` (worked example; P, every number recomputed). Tate's algorithm by hand on the paper's curves (models are `ecdata`'s minimal ones): * 121a1 $[1, 1, 1, -30, -76]$ at 11: $v(\Delta) = 2$, $v(c_4) = 1$; step 2 with $(r, t) = (6, 2)$ gives $[1, 19, 11, 88, -22]$, and $v(a_6) = 1 < 2$: type II, $c = 1$, $f = 2$. * 121c2 $[1, 1, 0, -3632, 82757]$ at 11: $v(\Delta) = 8$; step 2 $(r, t) = (6, 8)$, step 6 $(s, t) = (5, 110)$ gives $[11, -11, 242, -4840, 46585]$ with $P(T) \equiv (T-4)^3 \pmod{11}$: triple root, step 8: IV^*, $c = 1$, $f = 2$. * 539d1 $[0, 1, 1, -16, -66]$ at 7: step 6 gives $[0, 7, 49, 0, -686]$, $P(T) = T^3 + T^2 - 2 = (T-1)(T^2 + 2T + 2)$ with the quadratic irreducible mod 7: I_0^*, $c = 2$. * 539c1 $[1, 0, 1, 170, -3237]$ at 7: step 6 gives $[7, 0, 49, 49, -3087]$, $P(T) = T^3 + T - 9 \equiv (T-1)(T^2 + T + 2)$, and $T^2 + T + 2 \equiv (T-3)^2 \pmod 7$: double root, step 7: I_3^*, $c = 4$, $f = 2$, $v(\Delta) = 9 = 3 + 6$. * 726a1 $[1, 1, 0, -35, -51]$: at 11 step 2 gives $[1, 19, 22, 77, -121]$, $v(a_6) = 2$, $v(b_8) = 2 < 3$: III, $c = 2$; at 2: I_6 nonsplit ($a_2 = -1$), $c = 2$; at 3: I_3 nonsplit, $c = 1$. * 20a1 $[0, 1, 0, 4, 4]$ at 2: $v(\Delta) = 8$, $v(c_4) = 4$; step 6 $(s, t) = (1, 2)$ gives $[2, 0, 4, 0, 0]$, $P(T) = T^3$; step 8: $Y^2 + Y$ has distinct roots in $\mathbb{F}_2$: IV^*, $c = 3$, $f = 8 + 1 - 7 = 2$. * 99a1 $[1, -1, 1, -2, 0]$ at 3: step 2 gives $[1, 5, 3, 6, 0]$, $v(b_8) = 2$: III, $c = 2$, $f = 3 + 1 - 2 = 2$. * 980i1 at 2: IV^*, $c = 1$; at 5: I_1; at 7: I_5^*, $c = 2$, $v_7(\Delta) = 11$. Paper: the Example of Section 3, the remark on 99A1, the appendix (121a1, 726a1), Section sec:anomaly (726a1, type III at 11).

12.3.6 Section 6. The conductor and the semistability defect

6.1 `def:reduction:conductor-exponent` (definition and proposition; R for independence of ℓ, P for the values). For $\ell \neq p$ let $V = V_\ell E$ as a G_K-representation. The conductor exponent is the Artin conductor

of Chapter 7: $f_p(E) = a(V) = \dim V - \dim V^{I_K} + \mathrm{sw}(V)$, where $\mathrm{sw}(V) \geq 0$ is the Swan conductor; it is independent of ℓ (Serre-Tate; Ogg) and vanishes iff wild inertia acts trivially. Values: $f = 0$ (good), $f = 1$ (multiplicative), $f = 2 + \mathrm{sw}$ (additive), with $\mathrm{sw} = 0$ for $p \geq 5$. Proof of the values given Chapter 11's Neron-Ogg-Shafarevich criterion and the Tate curve (7.5): good means V unramified; multiplicative means V^I a line with unipotent (tame) action; additive means $V^I = 0$; for $p \geq 5$ every finite subgroup of $\mathrm{GL}_2(\mathbb{Z}_\ell)$ arising (orders dividing 24) has order prime to p by 6.5(iii) (it embeds in $\mathrm{SL}_2(\mathbb{F}_3)$, of order 24), and in the potentially multiplicative case the ramified quadratic character is tame for p odd (7.3, 7.7); so wild inertia acts trivially. Reference: SerreTate68 Section 3 (verify), Ogg, ATAEC IV.10 (verify). Paper: Lemma lem:twist (i), (ii) (conductor exponents), Lemma lem:IV and Remark rem:types (the conductor of E versus the Serre conductor of $\bar{\rho}$), Theorem thm:qual ($q^2 \parallel N$ at additive $q \geq 5$), Section 1 ("Σ the set of bad primes"), Remark rem:wild.

6.2 `thm:reduction:ogg` (theorem, Ogg's formula; R). $f_p = v_p(\Delta_{\min}) + 1 - m_p$, with m_p the number of irreducible components of the special fibre of the minimal regular model over $\bar{k}$ (the column m of 5.1). Due to Ogg (Ogg) for $p \neq 2$ and in general to Saito (T. Saito, Duke Math. J. 57 (1988), 151 to 173, key `Saito88`, add to `bib/ch10.bib`, verify). Reference: ATAEC IV.11.1 (verify). Data check: no failure on the 211,455 pairs (curve, bad prime) below conductor 10,000. Paper: background (the proof of Lemma lem:twist (ii) can be phrased through it; the writer uses it in 8.2 (d) and in 8.3).

6.3 `thm:reduction:conductor-bounds` (theorem; R). $f_2 \leq 8$, $f_3 \leq 5$, $f_p \leq 2$ for $p \geq 5$; all values $2, \ldots, 8$ at 2 and $2, \ldots, 5$ at 3 occur for additive reduction over $\mathbb{Q}$. Reference: Brumer-Kramer, "The conductor of an abelian variety", Compositio Math. 92 (1994) (key `BrumerKramer94`, verify), or Lockhart-Rosen-Silverman, J. Algebraic Geom. 2 (1993) (verify); ATAEC IV.10 exercises (verify). Data check: the maxima below conductor 10,000 are 8 at 2 and 5 at 3. Paper: background (PLAN.md scope; Remark rem:wild).

6.4 `def:reduction:conductor` (definition and proposition; P for (i), R for (ii)). (i) The conductor of $E/\mathbb{Q}$ is $N = \prod_p p^{f_p}$; $p \mid N$ iff E has bad reduction at p; $p \parallel N$ iff multiplicative; $p^2 \mid N$ iff additive; $p^3 \nmid N$ for $p \geq 5$. (ii) Isogenous curves have the same conductor ($V_\ell E \simeq V_\ell E'$, Chapter 9, item 9.4, and 6.1). Examples (all checked): 11a1: 11; 20a1: $2^2 \cdot 5$; 99a1: $3^2 \cdot 11$; 121a1, 121c2, 121d1: 11^2; 539c1: $7^2 \cdot 11$; 726a1: $2 \cdot 3 \cdot 11^2$; 980i1: $2^2 \cdot 5 \cdot 7^2$. Paper: Section 1 (level N), Theorem thm:main ("$\ell \nmid 2N$"), Theorem thm:qual ("$q^2 \parallel N$"), Remark after Theorem thm:qual (99A1, $9 \cdot 11$).

6.5 `thm:reduction:serre-tate` (theorem; R, with the consequence P). Let E/K have potentially good reduction. (i) There is a smallest extension L of K^{ur} over which E has good reduction; $L = K^{\mathrm{ur}}(E[m])$ for every $m \geq 3$ prime to p, and L/K^{ur} is Galois. (ii) Put $G_{\mathrm{ss}} = \mathrm{Gal}(L/K^{\mathrm{ur}})$, the *semistability group*, and $e = \#G_{\mathrm{ss}}$ (the *semistability defect*). For every $\ell \neq p$ the image of I_K in $\mathrm{Aut}(T_\ell E)$ is isomorphic to G_{ss} (by Neron-Ogg-Shafarevich over L, Chapter 11, the kernel of $I_K \to \mathrm{Aut}\, T_\ell E$ is I_L). (iii) G_{ss} embeds in $\mathrm{SL}_2(\mathbb{Z}/m)$ for every $m \geq 3$ prime to p (the determinant is the cyclotomic character, trivial on I_K mod m), and in $\mathrm{Aut}(\tilde{E}_L)$ (it acts faithfully on the reduction over L); so $e \in \{1, 2, 3, 4, 6\}$ for $p \geq 5$, $e \mid 24$ for $p = 2$ and $e \mid 12$ for $p = 3$, and the reduction over L has $j(\tilde{E}_L) \in \{0, 1728\}$ when $e > 2$. Reference: SerreTate68 Section 2, Corollaries 2 and 3 (verify); Serre72 Section 5.6 (verify). Paper: Lemma lem:IV ("the order of the image of I_q in $\mathrm{Aut}(T_\ell E)$ for any $\ell \neq q$"), Lemma lem:euler, Remark rem:types.

6.6 `thm:reduction:kraus` (theorem, Kraus; S for $p \geq 5$, R for $p = 2, 3$). Let E/K have potentially good reduction, $K/\mathbb{Q}_p$ finite. (i) If $p \geq 5$, G_{ss} is cyclic of order $e = 12/\gcd(v(\Delta_{\min}), 12)$; by type: $\mathrm{I}_0, \mathrm{II}, \mathrm{III}, \mathrm{IV}, \mathrm{I}_0^*, \mathrm{IV}^*, \mathrm{III}^*, \mathrm{II}^*$ give $e = 1, 6, 4, 3, 2, 3, 4, 6$. (ii) If $p = 3$, G_{ss} is cyclic of order $1, 2, 3, 4, 6$ or the non-abelian group $\mathbb{Z}/3 \rtimes \mathbb{Z}/4$ of order 12. (iii) If $p = 2$, G_{ss} is cyclic of order $1, 2, 3, 4, 6$, or quaternion of order 8, or $\mathrm{SL}_2(\mathbb{F}_3)$ of order 24; for $K = \mathbb{Q}_2$ Kraus determines which in terms of $v(c_4), v(c_6), v(\Delta)$ and congruences on the unit parts. Sketch of (i): over $K' = K^{\mathrm{ur}}(\varpi^{1/e})$ (tame, cyclic) the change $u = \varpi^{v(\Delta)/12}$ makes Δ' a unit and c_4', c_6' integral (they have valuations ≥ 0 because $v(j) \geq 0$ and by 5.3's table), so E has good reduction over K'; conversely $e\, v(\Delta) \equiv 0 \pmod{12}$ is necessary (the discriminant of a good model over L is a unit and $v_L = e\, v$), so $L = K'$; tame implies cyclic (Chapter 7, the tame quotient is procyclic). Reference: Kraus90 (the paper's citation; the theorem numbers of Kraus90 are not certain: Dokchitser-Dokchitser cite its Theorem 1 for $p = 3$, Pineda-Martin cites its Theorem 2 for $p = 2$, Billerey quotes a Theorem 3 for the 2-adic criterion; cite Kraus90 without a number unless checked); Serre72 Section 5.6. Paper: Lemma lem:IV ("equals $12/\gcd(v_q(\Delta_E), 12)$ \cite{Kraus90}; for $v_q(\Delta_E) \in \{4, 8\}$ this is 3"), Remark rem:types ("defects 2 and 4", "defect 6"), Lemma lem:euler ("cyclic of order 3 or 6

\cite{Kraus90}"), PLAN.md Section 6 row "Kraus's semistability defect".

6.7 `thm:reduction:tame-additive` (theorem; R). (Dokchitser-Dokchitser, "Local invariants of isogenous elliptic curves", Trans. Amer. Math. Soc. 367 (2015), Theorem 3.1 (number checked in arXiv:1208.5519v4; journal pages to verify), resting on Kramer-Tunnell 1982, Proposition 8.20, for $l = 2$ and on Kraus90 for $l = 3$.) Let $K/\mathbb{Q}_l$ be finite and E/K have additive potentially good reduction. Then E has tame reduction, equivalently $f = 2$, iff (a) $l \geq 5$; or (b) $l = 3$ and the type is III, III* or I$_0^*$; or (c) $l = 2$ and the type is IV or IV*. Data check (`ecdata`): at 2, among the potentially good additive pairs with conductor below 30,000, $f_2 = 2$ occurs for exactly the 14,431 pairs of type IV or IV* and for no other type; at 3, below 10,000, types III, III*, I$_0^*$ always have $f_3 = 2$ and types II, II*, IV, IV* always have $f_3 \in \{3, 4, 5\}$. Keys to add: `DokDok15`, `KramerTunnell82` (Compositio Math. 46 (1982), 307 to 352). Paper: Remark rem:wild, Section sec:wild, the remark after Theorem thm:qual ("at $q = 3$ in cases (a) and (b), where Lemma lem:IV does not apply because the ramification is wild": this is (b)), Proposition prop:forms (via 6.8).

6.8 `cor:reduction:IV-at-2` (corollary; P given 6.5, 6.7, 7.7). Let $E/\mathbb{Q}_2$ have Kodaira type IV or IV*. Then: (i) E has potentially good reduction; (ii) $f_2 = 2$; (iii) G_{ss} is cyclic of order 3, so for every odd ℓ the image of I_2 in $\mathrm{Aut}(T_\ell E)$ is cyclic of order 3, generated by σ with $\det\sigma = 1$ and eigenvalues $\zeta_3^{\pm 1}$; (iv) every Frobenius lift Frob satisfies $\mathrm{Frob}\,\sigma\,\mathrm{Frob}^{-1} = \sigma^{-1}$ in the image. Proof: (i) by 7.7 an additive potentially multiplicative curve has type I$_m^*$; (ii) by 6.7(c); (iii) $f = 2 = \dim V - \dim V^I + \mathrm{sw}$ with $V^I = 0$ forces $\mathrm{sw}(V) = 0$, so the wild inertia group P_2 acts trivially on V and on $T_\ell E$ (Chapter 7: $\mathrm{sw} = 0$ iff P acts trivially), the image of I_2 is a finite quotient of the tame quotient $I_2/P_2 \simeq \prod_{l\neq 2}\mathbb{Z}_l(1)$, hence cyclic of odd order; it divides 24 (6.5(iii)), so it is 1 or 3, and not 1 because the reduction is additive; (iv) the tame relation of Chapter 7 ($\mathrm{Frob}\ t\ \mathrm{Frob}^{-1} = t^2$, and $\sigma^2 = \sigma^{-1}$). This is the curve-theoretic half of the answer to important_notes.md point 43 (Part 7); the Euler-factor half is the `paperlink` below. Data check: for all 14,431 pairs of type IV, IV* at 2 with conductor below 30,000, the four roots of the 3-division polynomial generate over $\mathbb{Q}_2$ fields with ramification indices $1, 3, 3, 3$ (so the projective image of inertia is cyclic of order 3, consistent with $G_{\mathrm{ss}} \simeq \mathbb{Z}/3$), and $f_2 = 2$; PARI's $P_2(X)$ is $1 + 2X$ on a sample of 400 of them. `paperlink`: with (iii), (iv) the proof of Lemma lem:euler for II, II*, IV, IV* applies verbatim at $q = 2$ (it uses only that the image of inertia is cyclic of order prime to q and the tame relation): the invariant line of $\mathrm{ad}^0 V$ is spanned by $2\sigma + 1$, the geometric Frobenius acts on it by $\chi_{-3}(2) = -1$, so $L_2(A_f, s)^{-1} = 1 + 2^{-s}$, $P_2(X) = 1 + 2X$ and $e_\ell(E, 2) = v_\ell(3/2)$, which is 1 for $\ell = 3$ and 0 for $\ell \geq 5$ (Chapter 7, item 9.5(ii), which assumed exactly (iii); Chapter 19 for the factor). Hence the exception in Proposition prop:forms is empty. Paper: Proposition prop:forms, Remark rem:wild ("a type IV at 2 has $P_2(X) = 1 + 2X$"), Section sec:wild ("a type IV or IV* at 2 has $e_3 = 1$"), important_notes.md point 43.

6.9 `ex:reduction:defects` (worked example; P). Semistability defects of the paper's additive primes: 121a1 at 11 (II, $v = 2$): $e = 6$; 121c1 (IV, $v = 4$) and 121c2 (IV*, $v = 8$) at 11: $e = 3$; 539d1 at 7 (I$_0^*$): $e = 2$; 726a1 at 11 (III, $v = 3$): $e = 4$; 99a1 at 3 (III, $f_3 = 2$, tame by 6.7): $e = 4$; 20a1, 140b1, 980i1 at 2 (IV*): $e = 3$ by 6.8; 80b1 = 20a1$^{(-1)}$ at 2 (I$_0^*$, $f_2 = 4$): G_{ss} cyclic of order 6 (the twist of the order-3 group by the wildly ramified character of $\mathbb{Q}_2(i)$). The potentially multiplicative primes (539c1 and 980i1 at 7, 121d1 at 11) have no defect (infinite image of inertia, item 7.5). Paper: the Example of Section 3, Section sec:anomaly (726a1).

12.3.7 Section 7. The Tate curve

7.1 `thm:reduction:tate-curve` (theorem, Tate; S). Let $\mathfrak{q} \in K^\times$ with $v(\mathfrak{q}) > 0$. Put $s_k(\mathfrak{q}) = \sum_{n\geq 1} n^k\mathfrak{q}^n/(1-\mathfrak{q}^n)$, $a_4(\mathfrak{q}) = -5s_3(\mathfrak{q})$, $a_6(\mathfrak{q}) = -(5s_3(\mathfrak{q}) + 7s_5(\mathfrak{q}))/12$. (i) These series have coefficients in $\mathbb{Z}$ (for a_6: $5n^3 + 7n^5 \equiv 0 \pmod{12}$), converge, and $E_\mathfrak{q}\colon y^2 + xy = x^3 + a_4(\mathfrak{q})x + a_6(\mathfrak{q})$ is an elliptic curve over K with $\Delta = \mathfrak{q}\prod_{n\geq 1}(1-\mathfrak{q}^n)^{24}$, $c_4 = 1 + 240s_3(\mathfrak{q})$ and $j(E_\mathfrak{q}) = 1/\mathfrak{q} + 744 + 196884\,\mathfrak{q} + \cdots \in \mathfrak{q}^{-1}\mathbb{Z}[[\mathfrak{q}]]$. (ii) The model is minimal with split multiplicative reduction of type I$_n$, $n = v(\mathfrak{q}) = -v(j)$. (iii) There is an isomorphism of groups $\phi\colon \bar{K}^\times/\mathfrak{q}^{\mathbb{Z}} \to E_\mathfrak{q}(\bar{K})$, given by explicit power series, which is G_K-equivariant; in particular $E_\mathfrak{q}(L) \simeq L^\times/\mathfrak{q}^{\mathbb{Z}}$ for every algebraic L/K. Proof sketch: the series are the q-expansions of the Weierstrass functions of the lattice $\mathbb{Z} + \mathbb{Z}\tau$ with $\mathfrak{q} = e^{2\pi i\tau}$ (Chapter 9, Section 7; Chapter 12), rewritten so that they make sense p-adically; the identities hold formally because they hold over $\mathbb{C}$. RHB aside (one sentence): the map $\mathbb{C}^\times \to \mathbb{C}/(\mathbb{Z} + \mathbb{Z}\tau)$, $u = e^{2\pi iz}$, is the complex picture; Tate's insight is that $\mathbb{C}^\times/\mathfrak{q}^{\mathbb{Z}}$

makes sense over K. Reference: ATAEC V.3, Theorem 3.1 (verify); Tate's 1959 manuscript (not needed). Paper: Lemma lem:tamagawa-exp ("E is a Tate curve with parameter ..."), Lemma lem:tamfactor.

7.2 `prop:reduction:j-inverse` (proposition; P). The series $j(\mathfrak{q})$ of 7.1 has an inverse $\mathfrak{q}(j) = j^{-1} + 744j^{-2} + 750420j^{-3} + 872769632j^{-4} + \cdots \in j^{-1}\mathbb{Z}[[j^{-1}]]$ (series reversion; coefficients checked with PARI's `serreverse`). For $j_0 \in K$ with $v(j_0) < 0$ the series converges and gives the unique $\mathfrak{q} \in K^\times$, $v(\mathfrak{q}) > 0$, with $j(E_\mathfrak{q}) = j_0$; $v(\mathfrak{q}) = -v(j_0)$. Proof: $\mathbb{Z}$-coefficients by induction; convergence because $v(j_0^{-1}) > 0$; uniqueness because j is injective on $\{v(\mathfrak{q}) > 0\}$ ($j(\mathfrak{q})\mathfrak{q} = 1 + 744\mathfrak{q} + \cdots$ is a unit power series). Paper: Definition def:terms ("the valuation of the Tate parameter"), the appendix (`valuation(j, q)`), Section sec:data ("the Tate parameters with `ellinit` over $\mathbb{Q}_q$").

7.3 `thm:reduction:tate-uniformisation` (theorem; S). Let E/K have $v(j) < 0$, let $\mathfrak{q}_E = \mathfrak{q}(j_E)$ (7.2) and $\gamma = -c_4/c_6 \in K^\times/(K^\times)^2$ (from any model). Then $E \simeq E_{\mathfrak{q}_E}$ over $K(\sqrt{\gamma})$, and over K iff γ is a square; i.e. E is the quadratic twist of $E_{\mathfrak{q}_E}$ by χ_γ. Moreover: E has split multiplicative reduction iff $\gamma \in (K^\times)^2$; nonsplit multiplicative iff $K(\sqrt{\gamma})/K$ is the unramified quadratic extension; additive (type I_m^*, 7.7) iff $K(\sqrt{\gamma})/K$ is ramified. In the multiplicative case $n := v(\Delta_{\min}) = -v(j) = v(\mathfrak{q}_E)$. Proof sketch: two curves with the same $j \neq 0, 1728$ are quadratic twists of each other (Chapter 9, item 10.2(ii)) and the twisting class is read off c_4, c_6; the reduction type of a twist by an unramified character is multiplicative (4.3), by a ramified one additive. Reference: ATAEC V.5, Theorem 5.3 and Lemma 5.2 (verify). Paper: Lemma lem:tamagawa-exp ("Over the unramified quadratic extension of $\mathbb{Q}_q$ (which does not change the inertia group) E is a Tate curve"), Lemma lem:tamfactor (the same), Lemma lem:euler ("through a quadratic character times a unipotent when it is potentially multiplicative"), Lemma lem:twist (ii) ("a ramified quadratic twist exchanges the two"), Section 1.1 (the paper's ε).

7.4 `def:reduction:tate-parameter` (definition and proposition; P). For $E/\mathbb{Q}_q$ with $v_q(j) < 0$, the *Tate parameter* is $q_E = \mathfrak{q}(j_E) \in \mathbb{Q}_q^\times$, with $v_q(q_E) = -v_q(j_E) =: \nu$ (for multiplicative reduction $\nu = n_q = v_q(\Delta_{\min})$, the paper's n); its *unit part* is $u = q_E q^{-\nu} \in \mathbb{Z}_q^\times$, and $\bar{u} \in \mathbb{F}_q^\times$ its reduction, whose class in $\mathbb{F}_q^\times/(\mathbb{F}_q^\times)^{\ell^k}$ ($k \le v_\ell(q-1)$) is the class $c(u)$ of Lemma lem:tamfactor (Chapter 2 does the finite-field arithmetic; important_notes.md point 31). (i) q_E depends only on j_E, so it is the same for E and all its quadratic twists (e.g. 539c1 at 7 and its twist 77c1). (ii) (isogenies; P) If $\varphi\colon E \to E'$ is an isogeny over $\mathbb{Q}_q$ of prime degree d and $v_q(j) < 0$, then $\nu(E') \in \{d\,\nu(E), \nu(E)/d\}$ (over a field where both are Tate curves, the kernel is μ_d, giving $\mathfrak{q} \mapsto \mathfrak{q}^d$, or generated by the image of $\mathfrak{q}^{1/d}$, giving $\mathfrak{q} \mapsto \mathfrak{q}^{1/d}$ up to roots of unity); so $v_\ell(\nu)$ and $t_\ell(E,q)$ are unchanged by isogenies of degree prime to ℓ (this proves Chapter 9's item 9.9(iii)). Paper: Definition def:terms ($t_\ell(E,q) = v_\ell(-v_q(j_E))$, "the ℓ-part of the valuation of the Tate parameter"), Lemma lem:tamfactor (q_E, $u = q_E q^{-n}$, c), Section sec:tam, the history paragraph ("$t_q = v_\ell(n_q) = t_\ell(E,q)$").

7.5 `prop:reduction:tate-torsion` (proposition; P for (i) given 7.1(iii) and Chapter 5's Kummer theory; (ii), (iii) stated here and proved in Chapter 11, whose outline proves the matrix shape as its item 3.2). Let $\ell \neq p$ (or any ℓ), $t \ge 1$. (i) In $E_\mathfrak{q}(\bar{K}) = \bar{K}^\times/\mathfrak{q}^\mathbb{Z}$, $E_\mathfrak{q}[\ell^t] = \{\zeta^a(\mathfrak{q}^{1/\ell^t})^b\}$ for a fixed ℓ^t-th root $\mathfrak{q}^{1/\ell^t}$ and a primitive ℓ^t-th root of unity ζ; there is an exact sequence of G_K-modules $0 \to \mu_{\ell^t} \to E_\mathfrak{q}[\ell^t] \to \mathbb{Z}/\ell^t \to 0$, the quotient generated by the class of $\mathfrak{q}^{1/\ell^t}$. (ii) In the basis $e_1 = (\zeta_{\ell^k})_k$, $e_2 = (\mathfrak{q}^{1/\ell^k})_k$ of $T_\ell E_\mathfrak{q}$, $\sigma \in G_K$ acts by $\left(\begin{smallmatrix}\chi_\ell(\sigma) & \kappa(\sigma)\\ 0 & 1\end{smallmatrix}\right)$ with κ the Kummer cocycle of $\mathfrak{q}$ (Chapter 5: $\sigma(\mathfrak{q}^{1/\ell^k}) = \zeta_{\ell^k}^{\kappa(\sigma)}\mathfrak{q}^{1/\ell^k}$). (iii) If $E \simeq E_{\mathfrak{q}_E} \otimes \varepsilon$ for a quadratic character ε (7.3), then $T_\ell E$ has the matrices $\left(\begin{smallmatrix}\chi_\ell\varepsilon & \varepsilon\kappa\\ 0 & \varepsilon\end{smallmatrix}\right)$: the corner is $\varepsilon\kappa$, not κ (important_notes.md point 42). The consequences for ramification ($\rho \bmod \ell^t$ unramified iff $\ell^t \mid n$, the paper's Lemma lem:tamagawa-exp) are proved in Chapter 11 from (ii) and Chapter 7's Kummer theory. Reference: ATAEC V.5 (verify); SerreAbelian IV.A.1.2 (verify). Paper: Lemma lem:tamagawa-exp ("The ℓ^t-torsion is generated by μ_{ℓ^t} and an ℓ^t-th root"), Lemma lem:tamfactor (the matrices), PLAN.md scope.

7.6 `prop:reduction:tate-tamagawa` (proposition; P given 7.1, 7.3). Let E have multiplicative reduction of type I_n. Split case: $E(K) \simeq K^\times/q_E^\mathbb{Z}$, $E_0(K)$ corresponds to $\mathcal{O}^\times q_E^\mathbb{Z}/q_E^\mathbb{Z}$, so $E(K)/E_0(K) \simeq \mathbb{Z}/n$ via v and $c = n$. Nonsplit case: $E(K)$ is the subgroup of $E_{q_E}(K') = K'^\times/q_E^\mathbb{Z}$ (K' unramified quadratic) on which Frobenius acts by $x \mapsto$ (conjugate of x)$^{-1}$; the component group is $\mathbb{Z}/n$ with Frobenius acting by -1, whose invariants have order 1 (n odd) or 2 (n even). Data check: c_p agrees with this rule for all 86,856 multiplicative pairs (curve number 1, conductor below 10,000). Reference: ATAEC IV.9 Table 4.1, V.4 (verify). Paper: Section 1.4 ($\#\Phi_q^{\mathrm{geom}} = n$ for I_n; c_q "counts only the components defined over $\mathbb{F}_q$"), Lemma lem:tamfactor (the check "the ℓ-part of the Tamagawa number $c_q(E)$, as it should"), important_notes.md

(Chapter 16's deferral).

7.7 `thm:reduction:potmult-additive` (theorem; S for p odd, R for $p = 2$). Let E/K have additive potentially multiplicative reduction, $\nu = -v(j) > 0$, and let $L = K(\sqrt{\gamma})$ be the ramified quadratic extension of 7.3, with different exponent $s_{L/K}$. Then E has type I_m^* with $m = \nu + 4(s_{L/K} - 1)$. For p odd $s_{L/K} = 1$, so $m = \nu$ (and $f = 2$, $v(\Delta_{\min}) = \nu + 6$); for $K = \mathbb{Q}_2$, $s_{L/K} \in \{2, 3\}$ (the conductor exponent of the quadratic character), so $m = \nu + 4$ (then $f_2 = 4$) or $m = \nu + 8$ (then $f_2 = 6$). Proof for p odd: twist the Tate curve by ϖ (5.4's model), run step 7 of 5.2. Reference: D. Lorenzini, "Models of curves and wild ramification", Pure Appl. Math. Q. 6 (2010), Theorem 2.8 (as cited by Praveen, arXiv:2607.02678; key `Lorenzini10`, verify statement and number before printing). Data check: below conductor 10,000 every potentially multiplicative additive pair has type I_m^* (at $p = 2$: 3,732 pairs; at 3: 7,805; at $p \geq 5$: all), and at $p = 2$ always $m - \nu \in \{4, 8\}$, with $m - \nu = 4$ exactly when $f_2 = 4$ and 8 when $f_2 = 6$; examples 208a1 (I_5^*, $\nu = 1$), 304b1 (I_7^*, $\nu = 3$), 832c1 (I_9^*, $\nu = 1$), 336d1 (I_{12}^*, $\nu = 8$). Paper: Definition def:terms ("$v_\ell(n)$ for a Kodaira symbol I_n at any q and I_n^* at odd q, while a symbol I_n^* at $q = 2$ can have potentially good reduction"), Section sec:wild ("the index n of the symbol is not the valuation of the Tate parameter at 2"), important_notes.md point 18 (the I_n^* primes at 2 in the proof of Theorem thm:main).

7.8 `ex:reduction:tate-parameters` (worked example; P, all recomputed by the series of 7.2 and by PARI's `ellinit(E, O(q^k)).tate`). Tate parameters $q_E = q^\nu u$ and $\bar{u} \in \mathbb{F}_q^\times$:

curve, prime	type	a_q	ν	$\bar{u}$ mod q	remark
11a1, 11	I_5	1	5	$10 \equiv -1$	$(\mathbb{F}_{11}^\times)^5 = \{\pm 1\}$, so $\bar{u}$ is a 5th power: the extra term of Lemma lem:tamfactor at $\ell = 5$ (11a1 is Eisenstein at 5, so this illustrates the lemma, not the theorem)
11a3, 11	I_1	1	1	8	
20a1, 5	I_2	-1	2	1	nonsplit
77c1, 7	I_3	-1	3	5	same q_E as 539c1 at 7 (I_3^*)
77c1 and 539c1, 11	I_2	1	2	3	
140b1, 7	I_5	-1	5	1	same q_E as 980i1 at 7 (I_5^*)
140b1 and 980i1, 5	I_1	-1	1	2	
726a1, 2	I_6	-1	6	1	
726a1, 3	I_3	-1	3	2	

Values of t_ℓ: $t_5(11a1, 11) = 1$; $t_3(539c1, 7) = t_3(77c1, 7) = 1$; $t_5(980i1, 7) = 1$; $t_3(726a1, 2) = v_3(6) = 1$; and $v_3(\nu) = 1$ at $q = 3$ for 726a1, which the appendix's `rule` counts as $t_3(E, 3)$ although Definition def:terms is stated for $q \neq \ell$ (Part 8, N5). Paper: Definition def:terms, the Example of Section 3 ("$t_3(E, 7) = v_3(3) = 1$"), Lemma lem:tamfactor, Section sec:tam, Section sec:anomaly ("a prime $p \equiv -1 \pmod 3$ with $t_3(E, p) \geq 1$": for 726a1 that is $p = 2$).

12.3.8 Section 8. Quadratic twists and reduction (Lemma lem:twist (i), (ii))

8.1 `lem:reduction:twist-unramified` (lemma; P given 4.3; the paper's Lemma lem:twist (i)). Let χ be a quadratic character of $G_{\mathbb{Q}}$ unramified at p (e.g. $\chi = \chi_{q^*}$ with $q \neq p$ odd, Chapter 6: $\mathbb{Q}(\sqrt{q^*})$ is unramified outside q, also at 2 since $q^* \equiv 1 \pmod 4$). Then E and $E \otimes \chi$ are isomorphic over the unramified extension $\mathbb{Q}_p(\sqrt{d})$ (or over $\mathbb{Q}_p$), so they have the same Kodaira type, the same Φ_p^{geom}, the same $v_p(\Delta_{\min})$ and the same f_p (4.3); c_p may differ (split versus nonsplit, 2.2(iii); e.g. 11a1 has $c_{11} = 5$, while its twist by a character inert at 11 has $c_{11} = 1$). Also $v_p(j)$ and the Tate parameter are equal. Paper: Lemma lem:twist (i), Section 1.4 ("it has the same reduction type as E at every prime other than q"), Theorem thm:twist, Theorem thm:quant.

8.2 `lem:reduction:twist-ramified` (lemma; P; the paper's Lemma lem:twist (ii) with points 28, 29, 45 built in). Let q be an odd prime and $E^{(q)} = E \otimes \chi_{q^*}$ the twist by $\mathbb{Q}(\sqrt{q^*})$. (a) $\mathrm{I}_n \leftrightarrow \mathrm{I}_n^*$ with the same $n \geq 1$ (every odd q). (b) $\mathrm{I}_0 \leftrightarrow \mathrm{I}_0^*$ (every odd q). (c) For $q \geq 5$: $\mathrm{II} \leftrightarrow \mathrm{IV}^*$, $\mathrm{III} \leftrightarrow \mathrm{III}^*$, $\mathrm{IV} \leftrightarrow \mathrm{II}^*$, and $v_q(\Delta_{\min})$ changes by $+6$ modulo 12. (d) $f_q(E^{(q)}) = 2$ when $E^{(q)}$ is additive, 1 when it is I_n with $n \geq 1$, 0 when good. Proof: model $y^2 = x^3 + a_2x^2 + a_4x + a_6$ (q odd: complete the square, Chapter 9, 10.1); the twist $y^2 = x^3 + q^*a_2x^2 + q^{*2}a_4x + q^{*3}a_6$ has invariants $q^{*2}c_4$, $q^{*3}c_6$, $q^{*6}\Delta$ and the same j (Chapter 9, 10.2(iii): q^*, not q; point 29). (a): $v_q(j) < 0$; by 7.3 the twist by a ramified character turns multiplicative into additive and back, and by 7.7 with q odd the additive type is I_n^* with $n = -v_q(j)$, the same n as the multiplicative side. (b): 5.4, including the sign $\varepsilon = q^*/q$ (point 45); conversely an I_0^* curve has such a model and its twist has good reduction. (c): 5.3 and 1.2(ii): the twisted model has $v_q(\Delta)$ raised by 6; if the result is ≥ 12 then (potentially good, so $3v(c_4) \geq v(\Delta) \geq 12$) $v(c_4) \geq 4$ and one step of 1.2(ii) subtracts 12; read off the new type from $\{2, 3, 4, 6, 8, 9, 10\}$. (d): quadratic characters of $\mathbb{Q}_q$ for odd q are tame (Chapter 7); a twist by a tame character keeps wild inertia trivial when it was trivial, and in the cases (a) to (c) the curve on the other side is tame (multiplicative, good, or $q \geq 5$), so $f = 2$ for the additive side (6.1). The paper's phrase "the conductor exponents are the standard ones for q odd (tame)" is right exactly for these exchanges and wrong for general additive reduction at $q = 3$ (point 28; 8.3 below). Data check: all 90,970 pairs (curve number 1, odd bad prime) below conductor 10,000 behave as stated, with the same n in (a) in all 78,351 cases, $v_q(\Delta)$ shifted by 6 mod 12 in all 67,745 cases with $q \geq 5$, and the types at the other primes unchanged. Paper: Lemma lem:twist (ii), Section 1.4 (the list of exchanges), Lemma lem:euler ("$V = V' \otimes \chi_{q^*}$ with V' Steinberg at q"), Theorem thm:qual (b), (c), Theorem thm:twist ((a), (b), (c) of the paragraph after it), Theorem thm:quant, the proof of Theorem thm:main ("$E_{\{p\}}$").

8.3 `rem:reduction:twist-at-3` (remark; S, with data; not claimed by the paper). At $q = 3$ the exchanges of (c) also hold, with equal (now possibly wild) conductor exponents: in all 5,122 pairs of types $\mathrm{II}, \mathrm{III}, \mathrm{IV}, \mathrm{IV}^*, \mathrm{III}^*, \mathrm{II}^*$ at 3 below conductor 10,000 (curve number 1), $\mathrm{II} \leftrightarrow \mathrm{IV}^*$, $\mathrm{IV} \leftrightarrow \mathrm{II}^*$ with $f_3 \in \{3, 4, 5\}$ unchanged, and $\mathrm{III} \leftrightarrow \mathrm{III}^*$ with $f_3 = 2$ (examples: 54b1 (II, $f_3 = 3$) and its twist; 27a1 (IV^*) to a type II curve; 36a1 (III) to III^*). Sketch: χ_{-3} is tame at 3, so sw and $V^I = 0$ are unchanged and f is unchanged; the twisted model has $v_3(\Delta)$ raised by 6, so $v_3(\Delta_{\min})$ changes by ± 6 and by Ogg's formula (6.2) m changes by ± 6; the only potentially good types with $m = 7, 8, 9, 1, 2, 3$ are $\mathrm{IV}^*, \mathrm{III}^*, \mathrm{II}^*, \mathrm{II}, \mathrm{III}, \mathrm{IV}$ once one knows that I_n^* ($n \geq 1$) at an odd prime is potentially multiplicative (7.7 gives the converse; the direction needed is checked in the data, 7,805 of 7,805, and follows from step 7 of 5.2 for p odd: writer to verify before promoting this to a proposition). Consequence: the formula $w_\ell(E, q) = \max(v_\ell \# \Phi_q^{\mathrm{geom}}(E), v_\ell \# \Phi_q^{\mathrm{geom}}(E^{(q)}))$ of Section 1.4 also holds at $q = 3$. Paper: Section 1.4 ("the three others for $q \geq 5$"), important_notes.md point 28, Part 8, N3.

8.4 `prop:reduction:twist-weight` (proposition; P given 5.1, 8.2). For $q \geq 5$ and ℓ odd, $w_\ell(E, q) = \max\big(v_\ell(\#\Phi_q^{\mathrm{geom}}(E)), v_\ell(\#\Phi_q^{\mathrm{geom}}(E^{(q)}))\big)$, and $w_\ell(E, 2) = v_\ell(\#\Phi_2^{\mathrm{geom}}(E))$, where w_ℓ is the paper's twist-symmetrised weight. Proof by the table: $\mathrm{I}_n/\mathrm{I}_n^*$ give $\max(v_\ell(n), v_\ell(4)) = v_\ell(n)$; $\mathrm{II}/\mathrm{IV}^*$ and $\mathrm{IV}/\mathrm{II}^*$ give $\max(0, v_\ell(3)) = [\ell = 3]$; $\mathrm{III}/\mathrm{III}^*$ give $v_\ell(2) = 0$; $\mathrm{I}_0/\mathrm{I}_0^*$ give 0; at 2, I_n gives $v_\ell(n)$, I_n^* gives $v_\ell(4) = 0$, $\mathrm{IV}, \mathrm{IV}^*$ give $[\ell = 3]$, the others 0, which is the paper's definition at $q = 2$. Paper: Section 1.4 (definition of w_ℓ and the displayed formula), (eq:twist), Proposition prop:forms, Section sec:wild ("with the weights of (eq:twist) at 2 counted as if 2 were odd").

8.5 `ex:reduction:twist-pairs` (worked example; P, every identification checked by j and by $c_6c_4'/(c_6'c_4)$, Chapter 9, 10.2). Curve-level twists (important_notes.md B8) and their local data:

E	type at q	d	$E^{(d)}$	type at q	other primes
11a1	I_5 at 11	-11	121d2	I_5^* ($v=11$)	none
11a3	I_1 at 11	-11	121d1	I_1^* ($v=7$)	none
121a1	II ($v=2$) at 11	-11	121c2	IV^* ($v=8$)	none
121c1	IV ($v=4$) at 11	-11	121a2	II^*	none
539c1	I_3^* at 7	-7	77c1	I_3	I_2 at 11 on both
539d1	I_0^* at 7	-7	11a3	I_0	I_1 at 11 on both
980i1	I_5^* at 7	-7	140b1	I_5	IV^* at 2 ($-7\equiv 1 \pmod 8$, so χ_{-7} is trivial on $G_{\mathbb{Q}_2}$) and I_1 at 5 on both
20a1	IV^* at 2	-4	80b1	I_0^*, $f_2=4$	a wild twist at 2: no rule of 8.2 applies

Paper: the Example of Section 3, Theorem thm:qual (b), (c), Theorem thm:quant, Definition def:terms (point 15: $e_3(11a1,11)=0$ but $e_3(121d2,11)=1$).

12.3.9 Section 9. The wild primes 2 and 3

9.1 `rem:reduction:wild-overview` (remark; P for the statements proved earlier, data for the rest). At $p=2,3$: (i) the Kodaira type does not determine $v(\Delta_{\min})$ or f (Ogg's formula still links them); (ii) f can exceed 2 (6.3); (iii) G_{ss} can be non-abelian or wild (6.6); (iv) the type does not determine G_{ss}, except that $\mathrm{IV},\mathrm{IV}^*$ at 2 force $G_{\mathrm{ss}}\simeq\mathbb{Z}/3$ (6.8) and $\mathrm{III},\mathrm{III}^*,\mathrm{I}_0^*$ at 3 are tame (6.7); (v) I_m^* at 2 may be potentially good, and when potentially multiplicative $m\neq -v_2(j)$ (7.7). Hence the Euler factor of the adjoint is not a function of the type at 2, 3 (Chapter 19; Remark rem:wild). Paper: Remark rem:wild ("At $q=2,3$ the Kodaira type does not determine $L_q(A_f,s)$"), Section sec:wild, Definition def:terms.

9.2 `prop:reduction:inertia-at-2` (proposition with table; P for the dictionary, data for the frequencies). Let $E/\mathbb{Q}_2$ have additive potentially good reduction, ℓ odd. The projective image of I_2 on $E[3]$ is read off the ramification indices of the fields generated over $\mathbb{Q}_2$ by the four roots of the 3-division polynomial (the four lines of $E[3]$, Chapter 9, 4.5): pattern $(1,1,1,1)$: $G_{\mathrm{ss}}\simeq\mathbb{Z}/2$ (E is the twist of a curve with good reduction by a ramified quadratic character); $(1,3,3,3)$: G_{ss} cyclic of order 3 ($f_2=2$) or 6 ($f_2\in\{4,6\}$, the twist of the order-3 case by a ramified quadratic character); $(2,2,2,2)$: cyclic of order 4; $(4,4,4,4)$: Q_8 or $\mathrm{SL}_2(\mathbb{F}_3)$. Observed below conductor 30,000 (all curves, $4\mid N$, $v_2(j)\geq 0$), by type and f_2:

type at 2	f_2 and pattern (number of curves)
IV	$f=2$, $(1,3,3,3)$: 6,421
IV^*	$f=2$, $(1,3,3,3)$: 8,010
II	$f=4$: $(1,3,3,3)$ 1,598, $(4,4,4,4)$ 1,617; $f=6$: $(1,1,1,1)$ 464, $(4,4,4,4)$ 1,308; $f=7$: $(4,4,4,4)$ 786
II^*	$f=3$: $(4,4,4,4)$ 3,626; $f=4$: $(1,1,1,1)$ 951; $f=6$: $(1,3,3,3)$ 358, $(4,4,4,4)$ 344
III	$f=3,5,7$: $(4,4,4,4)$ 3,148, 2,106, 686; $f=8$: $(2,2,2,2)$ 310, $(4,4,4,4)$ 330

type at 2	f_2 and pattern (number of curves)
III^*	$f=3,5,7$: $(4,4,4,4)$ 4,637, 968, 686; $f=8$: $(2,2,2,2)$ 310, $(4,4,4,4)$ 330
I_0^*	$f=4$: $(1,3,3,3)$ 1,924, $(4,4,4,4)$ 2,333; $f=5$: $(4,4,4,4)$ 2,544; $f=6$: $(1,3,3,3)$ 688, $(4,4,4,4)$ 684
I_m^* (pot. good)	$f=3$: $(4,4,4,4)$ 4,717; $f=4$: $(1,1,1,1)$ 1,527, $(4,4,4,4)$ 4,132; $f=5$: $(4,4,4,4)$ 1,714; $f=6$: $(1,1,1,1)$ 690, $(1,3,3,3)$ 534, $(4,4,4,4)$ 4,964; $f=7$: $(4,4,4,4)$ 786

Note: $f_2=2$ occurs only for $\mathrm{IV},\mathrm{IV}^*$ (6.7). The `paperlink` (Chapter 19 does the computation, Chapter 7 9.5 sets it up): G_{ss} cyclic of order 3 or 6 gives $P_2(X)=1+2X$ and $e_3(E,2)=1$; $G_{\mathrm{ss}}\simeq\mathbb{Z}/2$ gives the cubic P_2 of the twist of good reduction ($e_3(E,2)=v_3(9-a_2(E')^2)$, which is 2 when $a_2(E')=0$); Q_8 and $\mathrm{SL}_2(\mathbb{F}_3)$ act absolutely irreducibly, so $(\mathrm{ad}^0 V)^{I_2}=0$ and $P_2=1$; cyclic of order 4 gives $1\pm 2X$. In particular a type II or II^* at 2 can have $e_3(E,2)\in\{0,1,2\}$ and a potentially good I_m^* at 2 usually has $P_2=1$, contrary to the wording of Remark rem:wild and Section sec:wild (Part 8, N1). Paper: Remark rem:wild, Section sec:wild, Proposition prop:forms.

9.3 `rem:reduction:wild-at-3` (remark; R via 6.7, data). At 3: types $\mathrm{III},\mathrm{III}^*,\mathrm{I}_0^*$ are tame ($f_3=2$; G_{ss} cyclic of order 4, 4, 2); types $\mathrm{II},\mathrm{II}^*,\mathrm{IV},\mathrm{IV}^*$ are always wild ($f_3\in\{3,4,5\}$; $3\mid e$); I_n^* is potentially multiplicative with $f_3=2$. Below conductor 10,000 the distributions of f_3 for II and IV^* coincide ($879,280,107$ for $f=3,4,5$), as do those of IV and II^* ($628,177,71$), as 8.3 predicts. This is why Lemma lem:IV needs $q\geq 5$ (the conclusion "tame, Serre conductor exponent 1" fails at $q=3$) and why the weights w_3 at $q=\ell=3$ of Conjecture conj:euler are not explained by Lemma lem:euler. Paper: the remark after Theorem thm:qual ("so do they at $q=3$ in cases (a) and (b), where Lemma lem:IV does not apply because the ramification is wild"), Conjecture conj:euler ($w_3(E)$), Section sec:wild.

9.4 `rem:reduction:Instar-at-2` (remark; P given 7.7, data). A symbol I_m^* at 2 is potentially good in 61% of the cases below conductor 10,000 (5,935 of 9,667 pairs); otherwise $t_\ell(E,2)=v_\ell(\nu)$ with $\nu=m-4$ or $m-8$, so $v_\ell(m)$ is the wrong invariant: 304b1 (I_7^*, $\nu=3$: $t_3=1$, $v_3(7)=0$), 832c1 (I_9^*, $\nu=1$: $t_3=0$, $v_3(9)=2$). Paper: Definition def:terms, Remark rem:wild ("about half"), Section sec:wild ("the index n of the symbol is not the valuation of the Tate parameter at 2, which is why n was not the right invariant"), important_notes.md point 18.

12.3.10 Section 10. Cremona's tables and PARI

10.1 `rem:reduction:cremona-tables` (remark; R). Cremona's tables (Cremona97; `ecdata`) give for each curve the global minimal model, the conductor, rank, torsion, and (in the local data files) the Kodaira symbols and Tamagawa numbers per bad prime, the minimal discriminant and j; the modular degree and the images are in other files (Chapter 22 for file names and formats; the writer must not name files not checked there). Paper: Section sec:data ("with its Kodaira symbols, Tamagawa numbers, ..."), Section 1.4.

10.2 `ex:reduction:pari-session` (worked example; P, outputs checked on PARI 2.17.2). `E = ellinit([1,1,0,-35,-51])` (726a1); `ellglobalred(E)` returns $N=726$, the factorisation $[2,1;3,1;11,2]$ and local data $[1,10,*,2]$, $[1,7,*,1]$, $[2,3,*,2]$ (types I_6, I_3, III; $c=2,1,2$); `elllocalred(ellinit([0,1,0,4,4]),2)` gives $[2,-4,*,3]$ (20a1: IV^*, $f_2=2$, $c_2=3$); `ellinit(E, O(11^12)).tate` for 11a1 gives $q_E=10\cdot 11^5+2\cdot 11^6+\cdots$. Paper: the appendix (`ellglobalred`, `R[4]`, `R[5]`, `K[i][2]`), Section sec:data.

10.3 `ex:reduction:running-table` (worked example; P). Local data of the running examples (PARI and the independent Tate's algorithm agree):

curve	N	$\Delta_{\min}$	q: type, f_q, c_q, $\#\Phi_q^{\mathrm{geom}}$, $v_q(j)$
11a1	11	-11^5	11: I_5, 1, 5, 5, −5
20a1	$2^2 \cdot 5$	$-2^8 5^2$	2: IV^*, 2, 3, 3, 4; 5: I_2, 1, 2, 2, −2
99a1	$3^2 \cdot 11$	$3^3 \cdot 11$	3: III, 2, 2, 2, 9; 11: I_1, 1, 1, 1, −1
121a1	11^2	-11^2	11: II, 2, 1, 1, 1
121c1	11^2	-11^4	11: IV, 2, 1, 3, 2
121c2	11^2	-11^8	11: IV^*, 2, 1, 3, 1
121d1	11^2	-11^7	11: I_1^*, 2, 2, 4, −1
539c1	$7^2 \cdot 11$	$-7^9 11^2$	7: I_3^*, 2, 4, 4, −3; 11: I_2, 1, 2, 2, −2
539d1	$7^2 \cdot 11$	$-7^6 \cdot 11$	7: I_0^*, 2, 2, 4, 0; 11: I_1, 1, 1, 1, −1
726a1	$2 \cdot 3 \cdot 11^2$	$2^6 3^3 11^3$	2: I_6, 1, 2, 6, −6; 3: I_3, 1, 1, 3, −3; 11: III, 2, 2, 2, 0
980i1	$2^2 \cdot 5 \cdot 7^2$	$-2^8 \cdot 5 \cdot 7^{11}$	2: IV^*, 2, 1, 3, 19; 5: I_1, 1, 1, 1, −1; 7: I_5^*, 2, 2, 4, −5

Paper: Section 1.4 ($\#\Phi_q^{\mathrm{geom}}$ versus c_q: 121c1, 121c2 and 980i1 at 2 have $c = 1 < 3 = \#\Phi^{\mathrm{geom}}$), the Example of Section 3, the appendix.

12.4 3. Section plan with target line counts

Lines of LaTeX, card excluded (PLAN.md 7.3: 1,000 to 2,200). The card lists all 54 items of Part 2 (about 200 lines, not counted).

section	title	contents	target lines
(starred)	Core results at a glance	the card: every item of Part 2, stated as above, with tag and `\Cref`	(excluded)
(starred)	Where the paper uses this chapter	def:terms (1.1, 3.1, 7.2, 7.4, 7.7, 9.4); the table and lem:euler (5.3, 6.6); Section 1.4 (4.2, 5.1, 8.2, 8.4); lem:twist (8.1, 8.2); lem:tamagawa-exp and lem:tamfactor (2.2, 4.5, 7.3 to 7.6); lem:IV and rem:types (4.5, 6.5, 6.6); rem:wild, sec:wild, prop:forms (6.7, 6.8, 9.1 to 9.4); the appendix (5.5, 10.2)	40

section	title	contents	target lines
1	Integral and minimal models	conventions R1, R2 as `convention` environments; 1.1 to 1.4; `paperlink` for lem:twist (ii) (minimality after twisting)	170
2	Reduction modulo ϖ	2.1 to 2.7; the formal group only as far as 2.5(iii); `paperlink` for Section 1.1 (δ_p) and lem:tamagawa-exp (split/nonsplit)	210
3	Potentially good and potentially multiplicative reduction	3.1 to 3.3; `warning`: "potentially multiplicative" includes additive curves of type I_n^*	100
4	Neron models and component groups	4.1 to 4.5 (4.5 in full); convention R4 (Φ^{geom} versus c); `paperlink` for Section 1.4 and for Chapter 11's use of 4.5	190
5	Kodaira symbols and Tate's algorithm	5.1 to 5.6; the table of 5.1 (`booktabs`), the eleven steps as an `enumerate`, the $p \geq 5$ table, the PARI code table; worked examples 5.6; `paperlink` for the appendix	320
6	Conductor and semistability defect	6.1 to 6.9; convention R5, R6; full proof of 6.8; `paperlink` for lem:IV, rem:types, prop:forms (point 43)	260
7	The Tate curve	7.1 to 7.8; RHB aside ($\mathbb{C}^\times/\mathsf{q}^{\mathbb{Z}}$); `warning` on $\varepsilon\kappa$ (point 42); `paperlink` for def:terms, lem:tamagawa-exp, lem:tamfactor (no proof of lem:tamagawa-exp: Chapter 11)	280

section	title	contents	target lines
8	Quadratic twists and reduction	8.1 to 8.5 (8.1, 8.2 proved in full); `warning` on q versus q^* (point 29) and on the sign ε in the cubic (point 45); `paperlink` for lem:twist, Section 1.4, thm:quant	200
9	The wild primes 2 and 3	9.1 to 9.4 with the table of 9.2; `paperlink` for rem:wild, sec:wild (Part 8, N1)	140
10	Cremona's tables and PARI	10.1 to 10.3; GP one-liners in `lstlisting`, pointer to Chapter 22	80
11	Exercises	the 17 exercises of Part 4.2	120
12	Notes and references	AEC, ATAEC, Tate75, Kodaira, Neron, BLR, Ogg, Saito, SerreTate68, Serre72, Kraus90, Kraus89, Papadopoulos93, Dokchitser-Dokchitser, Kramer-Tunnell, Lorenzini, CremonaBook; which numbers were checked	40
	total		**2,150**

12.5 4. Coverage table and exercises

12.5.1 4.1 Coverage: PLAN.md scope, Section 6 map rows and paper hooks

PLAN.md scope item, Section 6 row, or paper hook	line-up items
minimal Weierstrass models over $\mathbb{Z}_p$: existence, uniqueness up to change of coordinates	1.1, 1.3
criterion $v_p(\Delta) < 12$ or $v_p(c_4) < 4$ for $p \geq 5$; Kraus's conditions at 2 and 3 as a remark	1.2
reduction mod p; good, multiplicative (split, nonsplit), additive; criterion via $v_p(\Delta)$, $v_p(c_4)$	2.1, 2.2, 2.3
the reduction map, $E_0(\mathbb{Q}_p)$, $E_1(\mathbb{Q}_p)$	2.4, 2.7
the formal group: E_1 has no prime-to-p torsion	2.5, 2.6
potentially good / potentially multiplicative; criterion $v_p(j) \geq 0$ or < 0	3.1, 3.3, 7.3
the semistable reduction theorem	3.2, 6.5
Neron model: universal property, existence, special fibre, identity component, Φ_p finite etale	4.1, 4.2, 4.3

PLAN.md scope item, Section 6 row, or paper hook	line-up items
$\Phi_p^{\mathrm{geom}} = \Phi_p(\bar{\mathbb{F}}_p)$ and $c_p = \#\Phi_p(\mathbb{F}_p)$	4.2, 4.4, 5.1, 10.3
Kodaira symbols with the special fibres; orders and structures of Φ^{geom}	5.1
Tate's algorithm in full; for $p \geq 5$ type from $v_p(\Delta)$ $(0, 2, 3, 4, 6, 8, 9, 10;\ n;\ n+6)$	5.2, 5.3, 5.6
the cubic $T^3 + a_2T^2 + a_4T + a_6$ criterion for I_0^* at odd p	5.4
conductor exponent f_p ($0, 1, 2$ for $p \geq 5$); Ogg's formula; wild part; $f_2 \leq 8$, $f_3 \leq 5$	6.1, 6.2, 6.3, 6.7
the conductor $N = \prod p^{f_p}$	6.4
the Tate curve $E_{\mathfrak{q}}$, $\bar{\mathbb{Q}}_p^\times/\mathfrak{q}^{\mathbb{Z}}$, $j(E_{\mathfrak{q}})$, $v_p(j) = -v_p(\mathfrak{q})$	7.1, 7.2
every split multiplicative curve is a Tate curve; nonsplit after the unramified quadratic extension	7.3
Tate parameter of a potentially multiplicative curve; $v_p(q_E) = n$ for I_n and I_n^* at odd p; unit part and its class	7.4, 7.7, 7.8
$E_{\mathfrak{q}}[\ell^t]$ generated by μ_{ℓ^t} and an ℓ^t-th root of $\mathfrak{q}$ (statement)	7.5
Kraus's theorem: $12/\gcd(v_p(\Delta), 12)$, the list $1, 6, 4, 3, 2, 3, 4, 6$	6.6, 6.9
twists: Lemma lem:twist (i), (ii) proved in full; exchanges; $v_q(\Delta)$ up by 6 mod 12	8.1, 8.2, 8.5
the wild primes: type does not determine the inertia action; I_n^* at 2 potentially good; Remark rem:wild	6.7, 6.8, 9.1 to 9.4
Cremona's tables; PARI's Kodaira codes	5.5, 10.1, 10.2, 10.3
Section 6 row: Kodaira types, component groups, Tamagawa numbers, conductor exponents, Tate's algorithm	4.2, 5.1 to 5.6, 6.1, 6.2
Section 6 row: the twist E' of good reduction, quadratic twists (with Chapter 9)	5.4, 8.2(b), 8.5
Section 6 row: Tate parameter, unit part, $t_\ell(E, q)$, potentially multiplicative reduction (with Chapters 2, 7)	7.2, 7.3, 7.4, 7.7, 7.8
Section 6 row: Kraus's semistability defect	6.5, 6.6
Section 6 row: the function `rule(E, l)` and the Kodaira codes (with Chapter 22)	5.5, 10.2
hook: Definition def:terms (t_ℓ, $v_q(j_E)$, potentially multiplicative primes)	1.1, 3.1, 7.4, 7.7, 9.4
hook: the table of Section 1	5.3, 6.6, 8.2
hook: Section 1.4 (component groups, exchange of types, w_ℓ)	4.2, 5.1, 8.2, 8.3, 8.4
hook: Lemma lem:twist	8.1, 8.2 (and 5.4, 1.2)
hook: Lemma lem:tamagawa-exp (Tate curve over the unramified quadratic extension)	2.2, 4.3, 7.3, 7.5 (proof in Chapter 11)
hook: Lemma lem:tamfactor (Tate parameter, unit part)	7.4, 7.5(iii), 7.6, 7.8, 4.5
hook: Lemma lem:IV and Remark rem:types (Kraus)	4.5, 6.5, 6.6
hook: Remark rem:wild, Section sec:wild	6.7, 6.8, 7.7, 9.1 to 9.4
hook: Proposition prop:forms	6.8, 8.4, Part 7

PLAN.md scope item, Section 6 row, or paper hook	line-up items
hook: the appendix (Kodaira codes)	5.5, 10.2
deferrals from Chapters 1, 2, 5, 6, 7, 8, 9, 16 (Part 0.1, item 9)	2.1, 2.6, 4.2, 5.1, 6.6, 6.8, 7.4, 7.5, 7.6, 8.1

12.5.2 4.2 Exercises (Section 11 of the chapter), graded from easy to harder

1. Show that a model with $v(\Delta) < 12$ is minimal, and that for $p \geq 5$ a model with $v(c_4) \geq 4$, $v(\Delta) \geq 12$ is not (1.2). Decide minimality at 2 and 3 of $y^2 = x^3 - 27c_4x - 54c_6$ for 11a1.
2. For 11a1 at 11 and 20a1 at 5 decide split or nonsplit both from $T^2 + a_1T - a_2$ and from $\left(\frac{-c_6}{p}\right)$, and compare with $a_{11} = 1$, $a_5 = -1$ (2.2).
3. Prove 2.3 for the cusp $y^2 = x^3$ and the split node $y^2 = x^3 + x^2$ by writing down the isomorphisms with k^+ and $k^\times$.
4. Using 2.6, show that $E(\mathbb{Q})_{\mathrm{tors}}$ of 11a1 injects into $\tilde{E}(\mathbb{F}_3)$ and $\tilde{E}(\mathbb{F}_7)$; deduce $E(\mathbb{Q})_{\mathrm{tors}} \simeq \mathbb{Z}/5$ (Chapter 9's application).
5. Run Tate's algorithm by hand for 121a1 and 121c2 at 11, and for 539d1 and 539c1 at 7, and check the paper's types in the Example of Section 3 (5.6).
6. Check the paper's claim (Lemma lem:twist (ii)) that the twist of 121a1 by -11 has type IV^* at 11, and show that the twist by $+11$ (1936k2) has additive reduction at 2 (point 29).
7. Check the paper's sentence "the cubic $T^3 + a_2T^2 + a_4T + a_6$ of Tate's algorithm has distinct roots modulo q" for the twist of 11a3 by -7 (it is 539d1): compute the cubic with the sign $\varepsilon = q^*/q$ and its roots mod 7 (5.4, point 45).
8. Prove $w_\ell(E, q) = \max(v_\ell \# \Phi_q^{\mathrm{geom}}(E), v_\ell \# \Phi_q^{\mathrm{geom}}(E^{(q)}))$ for $q \geq 5$ from 5.1 and 8.2 (8.4). Which curves of 10.3 have $c_q \neq \# \Phi_q^{\mathrm{geom}}$?
9. Verify Ogg's formula for every bad prime of the curves in 10.3 (6.2).
10. Using 6.6, compute the semistability defect at each additive prime $q \geq 5$ of the running examples, and show that type III at 11 (726a1) gives $e = 4$.
11. Prove 6.8 from 6.5, 6.7, 7.7 and Chapter 7's description of tame inertia. Then check the paper's claim (Remark rem:wild, Proposition prop:forms) that a type IV or IV^* at 2 has $P_2(X) = 1 + 2X$, using Chapter 7, item 9.5(ii).
12. Show that $a_6(\mathsf{q}) = -(5s_3 + 7s_5)/12$ has integer coefficients and compute $j(\mathsf{q})$ to the term $196884\mathsf{q}$ (7.1). Invert to get $\mathsf{q}(j) = j^{-1} + 744j^{-2} + 750420j^{-3} + \cdots$ (7.2).
13. Check the paper's claim (Example of Section 3) that $t_3(539\mathrm{c}1, 7) = 1$: compute $v_7(j)$ and the unit part of the Tate parameter of 539c1 at 7, and show that it equals that of 77c1 (7.4, 7.8).
14. For 11a1 at 11 show that the unit part of q_E is $\equiv -1 \pmod{11}$ and that it is a 5th power in $\mathbb{F}_{11}^\times$ (7.8; this is the situation of Lemma lem:tamfactor's extra term).
15. Prove 7.6: compute $E(K)/E_0(K)$ for split and nonsplit multiplicative reduction from the Tate uniformisation, and check $c_2 = 2$, $c_3 = 1$ for 726a1.
16. Check the paper's statement (Definition def:terms, Section sec:wild) that the index of I_m^* at 2 is not the valuation of the Tate parameter, on 304b1 and 832c1 (7.7, 9.4).
17. Check the paper's statement (Remark rem:wild) "a type II at 2 has $P_2 = 1$" against 336f1 and the type II^* curve 1232l1, using PARI's `lfunsympow` and the dictionary of 9.2; explain the answer through the inertia images ($\mathbb{Z}/6$, resp. $\mathbb{Z}/2$) and verify $v_3(\deg \varphi_E) = e_3(E, 2)$ for 1232l1 ($\deg \varphi_E = 288$).

12.6 5. Dependencies and forward pointers

Results from other chapters used here (by chapter label only): * Chapter 1 (`ch:groups`): finite abelian groups ($\mathbb{Z}/n$, $(\mathbb{Z}/2)^2$, $\mathbb{Z}/4$ as component groups), $\mathrm{SL}_2(\mathbb{F}_3)$, Q_8 and cyclic subgroups of $\mathrm{GL}_2(\mathbb{F}_3)$ (for 6.5, 6.6, 9.2). * Chapter 2 (`ch:rings`): v_p on $\mathbb{Q}$, Legendre symbols (2.2), ℓ-th powers in $\mathbb{F}_q^\times$ and the class $c(u)$ (7.4), q^*, χ_{q^*}. * Chapter 3 (`ch:modules`): DVRs, $\mathbb{Z}_\ell$-modules, uniquely divisible groups and the snake lemma (4.5). * Chapter 5 (`ch:galois`): the Kummer cocycle of an element and its ℓ-adic version

(7.5), quadratic characters. * Chapter 6 (`ch:ant`): $\mathbb{Q}(\sqrt{q^*})$ unramified outside q (8.1), decomposition and inertia groups. * Chapter 7 (`ch:local`): $\mathbb{Q}_p$, Hensel (2.4, 4.2), $\mathbb{Q}_p^{\mathrm{ur}}$, unramified extensions (4.3, 8.1), inertia, wild inertia, the tame quotient and the tame relation (6.6, 6.8), the Artin and Swan conductors (6.1), tame quadratic characters at odd q (8.2(d)), the different exponent of quadratic extensions of $\mathbb{Q}_2$ (7.7), the shape table 9.4 and item 9.5 (for the `paperlink`s of 6.8 and 9.2). * Chapter 8 (`ch:curves`): plane curves, singular points (2.1), components and multiplicities of curves on a surface only in words (5.1). * Chapter 9 (`ch:ellcurves`): Weierstrass data and coordinate changes (1.1 to 1.3 there), the group law, $E[m]$, $T_\ell E$, the 3-division polynomial (9.2 here), quadratic twists and their invariants (10.1, 10.2 there), j-invariants (1.4 there), curves over $\mathbb{C}$ (for 7.1's motivation). * Black boxes from later chapters (forward `\Cref`): Neron-Ogg-Shafarevich (Chapter 11; used in 6.1, 6.5(ii)); the Euler factors (Chapter 19; only in `paperlink`s of 6.8, 9.2); PARI file formats (Chapter 22).

Forward pointers (chapters that use this one): * Chapter 9 (`ch:ellcurves`): minimality (1.2, 1.3), $a_p \in \{0, \pm 1\}$ at bad p (2.1), injectivity of reduction on torsion (2.6) for its 8.6 and 9.2, the Neron differential (1.1(iii)) for its period lattice, conductors (6.4), types of the running examples (10.3), Lemma lem:twist on types (8.2), the invariance of t_ℓ under prime-to-ℓ isogenies (7.4(ii)) for its 9.9(iii). * Chapter 11 (`ch:galrep`): 2.6 and 6.5 (Neron-Ogg-Shafarevich and the semistable reduction theorem), 4.5 (inertia invariants and component groups; Chapter 11 cites it instead of proving its own 2.2), 6.6 to 6.8 (Kraus; type IV at 2), 7.3 to 7.5 (the Tate curve and its torsion, from which Chapter 11 proves Lemma lem:tamagawa-exp and the matrix shape), 7.7 (types I_m^*), 8.2 (twists), 9.2, 9.3 (inertia at 2, 3). * Chapter 14 (`ch:moddeg`): Φ_q^{geom}, c_q (Section 1.4 of the paper, (eq:main)), conductors, the minimal model and Neron differential (Manin constant). * Chapter 15 (`ch:modgalrep`): conductor exponents and the twist exchange (8.2) in Theorem thm:qual; the Tate curve at multiplicative primes. * Chapters 16, 17 (`ch:cohomology`, `ch:galcoh`): q_E, u, $c(u)$, c_q (7.4, 7.6, 7.8) in Lemma lem:tamfactor; 11a1's unit part. * Chapter 19 (`ch:lfunctions`): the types at $q \geq 5$ (5.3), Kraus (6.6), G_{ss} at 2 (6.8, 9.2) for Lemma lem:euler and Remark rem:wild, and the answer to point 43. * Chapters 21, 22 (`ch:paper`, `ch:computation`): the Kodaira codes (5.5), PARI outputs (10.2), the running examples (5.6, 7.8, 8.5, 10.3), the corrected readings of Remark rem:wild (Part 8, N1) and of Proposition prop:forms (Part 7).

12.7 6. Notes for the writer

12.7.1 6.1 Pitfalls and conventions

- **Φ^{geom} versus c (R4).** The paper's inequalities are about $\#\Phi_q^{\mathrm{geom}}$; c_q can be smaller (121c1, 121c2, 980i1 at 2: $c = 1$, $\#\Phi^{\mathrm{geom}} = 3$). Say it twice.
- **Two meanings of Φ (R6).** Component group versus semistability group: use G_{ss} for the latter throughout.
- **"Potentially multiplicative" is not "multiplicative".** Type I_n^* curves are additive. The paper's n_q is defined at multiplicative primes ($n_q = v_q(\Delta)$); at additive potentially multiplicative primes use $\nu = -v_q(j)$.
- **Kodaira codes.** "$n+4$" and "$-(n+4)$" need $n \geq 1$; I_0^* is -1 (5.5). PARI's `ellglobalred` lists only bad primes, so code 1 does not occur there.
- **The prime 2 in Tate's algorithm.** The translations of steps 2, 6, 7 differ at $p = 2$ (no division by 2); copy them from Cremona's book, not from memory. A short model $y^2 = x^3 + Ax + B$ is never the right input at 2 or 3.
- **Twisting a general model.** Complete the square first (Chapter 9, 10.1), then recompute the minimal model (1.2); never multiply a_2, a_4, a_6 of a model with a_1 or $a_3 \neq 0$.
- **q versus q^* (point 29) and the sign in the cubic (point 45):** built into 8.2 and 5.4.
- **Tate curve conventions.** The parameter is called q_E, its unit part u, the variable of the Tate curve $\mathfrak{q}$ (or t); never u for the Tate parameter (point 8). The corner of the matrix of a nonsplit or additive curve is $\varepsilon\kappa$ (point 42). The series $\mathfrak{q}(j)$ has third coefficient 750420 (Chapter 7's outline has 356652: Part 8, B-items).
- **Kraus's theorem numbers.** Not checked against Kraus90 (the outline author had no access);

three secondary sources number its theorems inconsistently (6.6). Cite Kraus90 without a theorem number, and cite Dokchitser-Dokchitser Theorem 3.1 (checked in the arXiv version) for 6.7.

- **Do not prove Lemma lem:tamagawa-exp here** (orchestrator decision, important_notes.md B15): 7.5 states $E_{\mathfrak{q}}[\ell^t]$ and the matrices; Chapter 11 proves the lemma. A one-line `paperlink` at 7.5 says where.
- **Bibliography (B14).** Keys needed by several chapters go into `refs.bib`, not into `bib/ch10.bib`: `SerreTate68` (Chapters 7, 9, 10, 11), and `DokDok15`, `KramerTunnell82`, `Lorenzini10`, `Papadopoulos93` (important_notes.md B16; check the other bib files first). Chapter-10-only keys (`Kraus89`, `Saito88`, `BrumerKramer94`) go into `bib/ch10.bib`. Every entry must be checked against the source.
- **Theorem numbers marked "(verify)".** Recalled, not checked: AEC VII.1.1, VII.1.3, VII.2.1, VII.3.1, VII.5.4, VII.5.5, VII.6.1, VIII.8.2; ATAEC IV.9 (Algorithm 9.4, Table 4.1, Corollary 9.2), IV.11.1, V.3.1, V.5.3; BLR 1.2, 1.4, 7.2, 9.5. Print section numbers if in doubt. Checked: Dokchitser-Dokchitser Theorem 3.1 (arXiv:1208.5519v4, page 6), its citation of Kramer-Tunnell Proposition 8.20; Billerey's and Pineda-Martin's statements of Kraus's 2-adic results.

12.7.2 6.2 What to leave to other chapters

- Chapter 7: all local field theory, conductors of Galois representations, Weil-Deligne representations, the shapes table.
- Chapter 9: Weierstrass formulas, twists as curves, $\rho \otimes \chi$, labels, $L(E, s)$.
- Chapter 11: Neron-Ogg-Shafarevich in full; every statement about $\rho(I_q)$, $E[3]^{I_q}$ beyond 4.5; Lemma lem:tamagawa-exp, Lemma lem:IV, Remark rem:types (with points 22, 49); the Serre conductor; Theorem thm:qual's local analysis.
- Chapter 19: $L_q(A_f, s)$, P_q, $e_\ell(E, q)$, Lemma lem:euler, the Euler factors at 2 and 3 (Remark rem:wild's factor statements); Chapter 10 gives them only in `paperlink`s with forward pointers.
- Chapter 22: PARI usage and `ecdata` formats.

12.7.3 6.3 How the relevant important_notes.md points are handled

- **Point 8** (Tate parameter called u; "up to units"): convention R1 and 7.4 name it q_E; 7.2 gives $v(q_E) = -v(j)$; the "up to units" step is Chapter 7's (Hensel) and Chapter 11's (proof of the lemma).
- **Point 15** (e_ℓ not twist-invariant): 8.5's last row and the `paperlink` there; q_E and t_ℓ are twist-invariant (7.4(i)).
- **Point 18** (I_n^* at $q = 2$ in the proof of Theorem thm:main): 7.3 and 7.7 give the local structure (a ramified, possibly wild, quadratic twist of a Tate curve with parameter of valuation $\nu = -v_2(j)$), which is what the fix needs; Chapter 11 does the Galois side.
- **Point 22** (Remark rem:types incomplete): 4.5(iii) gives $\dim E[3]^{I_q}$ for every type; the rest is Chapter 11's.
- **Point 28** ("tame" in Lemma lem:twist): 8.2(d) and 8.3; 9.3 (types $\mathrm{II}, \mathrm{II}^*, \mathrm{IV}, \mathrm{IV}^*$ at 3 are always wild).
- **Point 29** (q versus q^*): 8.2's proof, with Chapter 9, 10.2(iii).
- **Point 43** (type IV at 2): settled; 6.7, 6.8, 4.5 and Part 7 (two independent proofs, both checked on the data).
- **Point 45** (sign in the cubic): 5.4.
- **Point B8** (curve-level twists): the table of 8.5 names the right curves (11a1 to 121d2, 11a3 to 121d1 and 539d1, 121a1 to 121c2, 121c1 to 121a2).
- Also touched: point 23 (c_q versus $c(u)$: convention R4), point 31 (reduction of u: 7.4), point 42 ($\varepsilon\kappa$: 7.5(iii)), point 49 (conductor of E versus Serre conductor: 8.2(d) states the conductor of E only).

12.7.4 6.4 Bibliography additions (complete entries to be checked against the sources)

`Kraus89` (A. Kraus, Acta Arith. 54 (1989), 75 to 80); `Saito88` (T. Saito, "Conductor, discriminant, and the Noether formula of arithmetic surfaces", Duke Math. J. 57 (1988), 151 to 173); `Papadopoulos93` (I. Papadopoulos, J. Number Theory 44 (1993), 119 to 152); `DokDok15` (T. and V. Dokchitser, "Local invariants of isogenous elliptic curves", Trans. Amer. Math. Soc. 367 (2015), pages to be checked; arXiv:1208.5519); `KramerTunnell82` (K. Kramer, J. Tunnell, "Elliptic curves and local epsilon-factors", Compositio Math. 46 (1982), 307 to 352); `Lorenzini10` (D. Lorenzini, "Models of curves and wild ramification", Pure Appl. Math. Q. 6 (2010), pages to be checked); `BrumerKramer94` (A. Brumer, K. Kramer, "The conductor of an abelian variety", Compositio Math. 92 (1994), pages to be checked); `SerreTate68` (B14: `refs.bib`). Optional: Billerey (Kraus's results at 2 over ramified quadratic fields; journal to be checked), Pineda-Martin, arXiv:2609.01485 (Proposition 3 restates Kraus at 2).

12.7.5 6.5 RHB and physics asides (one sentence each)

The Tate curve $\mathbb{C}^\times/\mathtt{q}^{\mathbb{Z}}$ as the exponential picture of a torus (7.1); the Kodaira fibres $\tilde{E}_6, \tilde{E}_7, \tilde{E}_8$ as extended Dynkin diagrams, the same diagrams that classify simple Lie algebras and finite subgroups of SU(2) (5.1; the binary tetrahedral group $\mathrm{SL}_2(\mathbb{F}_3)$ of 6.6 is one of them).

12.7.6 6.6 Verification log

Scripts in `/tmp/ch10/scripts` (PARI/GP 2.17.2 through `cypari2`, Python 3), data in `/tmp/ch10/data` (`ecdata allcurves` files for conductors 0 to 29,999, downloaded from GitHub; arXiv PDFs 1208.5519, 2406.01985, 2607.02678, 2609.01485 and Billerey's paper, read with `pdftotext`). * `running.py`: conductors, Δ, c_4, c_6, j, local data (f, Kodaira code, c, $v(\Delta)$, $v(c_4)$, $v(j)$) and `lfunsympow` Euler factors for 11a1, 11a3, 20a1, 99a1, 121a1, 121c1, 121c2, 121d1, 539c1, 539d1, 77c1, 140b1, 726a1, 980i1 (tables 5.6, 10.3). * `tate_alg.py`: an independent implementation of Tate's algorithm (Cremona's structure); agrees with PARI's `elllocalred` on type, f and c for all 7,778 pairs (curve, bad prime) of the first 3,000 curves of `ecdata` (conductors up to 645), all residue characteristics included; a run on all curves below 10,000 was too slow in pure Python to finish here. Hand traces of 5.6. * `checks.py`: Ogg's formula on 211,455 pairs (no failure); $\max f_2 = 8$, $\max f_3 = 5$; $v_p(\Delta)$ by type for $p \geq 5$ (9,699 pairs); potentially multiplicative pairs are always I_n or I_n^*; the possible c by type; the $(v(c_4), v(c_6), v(\Delta))$ table of 5.3. * `twist.py`: 90,970 (curve number 1, odd bad prime) twists by q^*: exchanges of 8.2 and 8.3, same n, $v(\Delta)$ shift, other primes unchanged. * `misc.py`: $a_p = (\frac{-c_6}{p})$ at 127,592 odd multiplicative pairs; I_n^* at 3 always potentially multiplicative (7,805); (m, ν) for I_m^* at 2; f_3 by type. * c_p at multiplicative primes against the Tate-curve rule (7.6): 86,856 pairs, no mismatch. * `kraus` check of 1.2(iii): 20,000 random integral models and 8,248 minimal models. * `tate.py`: Tate parameters by the series $\mathtt{q}(j)$ (coefficients from `serreverse`) and by `ellinit(E,O(p^k)).tate` (7.8). * `f2.py`, `inertia_fast.py`: types, f_2 and the ramification pattern of the roots of ψ_3 over $\mathbb{Q}_2$ (via `factorpadic` and `nfinit/idealprimedec` on each factor) for all curves with $4 \mid N < 30{,}000$ and $v_2(j) \geq 0$ (table 9.2); `lfunsympow` at 2 on 400 type IV, IV* curves. * `p2dist.py`, `p2cm.py`, `typeII2.py`: PARI's P_2, P_3 by type and f (optimal curves, $N \leq 5{,}000$ and $N < 10{,}000$), CM and non-CM separated, $E[3]$ irreducibility via ψ_3, examples 336f1, 1232l1, modular degrees with `ellmoddegree`. * Not verified: theorem numbers in AEC, ATAEC, BLR, Kraus90 (no access to Kraus90 itself); anything above conductor 30,000; the journal page numbers of the added references.

12.8 7. Open question 43

Question (important_notes.md point 43, before its rewrite): for $E/\mathbb{Q}_2$ of Kodaira type IV or IV*, is the image of inertia on T_3E always cyclic of order 3 or 6, so that the computation of Lemma lem:euler gives $P_2(X) = 1 + 2X$ and $e_3(E, 2) = 1$, and the exception in Proposition prop:forms disappears?

Answer: proved. More precisely, the image is always cyclic of order exactly 3 (order 6 never occurs for these types), the reduction is tame ($f_2 = 2$), and $P_2(X) = 1 + 2X$, $e_\ell(E, 2) = v_\ell(3/2)$ for every odd ℓ. Two independent proofs, which agree:

1. **Via the conductor (this outline, item 6.8).** (a) Type IV or IV* forces $v_2(j) \geq 0$, because an additive potentially multiplicative curve has type I_m^* (item 7.7; Lorenzini 2010, Theorem 2.8, verify; checked on all 3,732 such pairs at 2 below conductor 10,000). (b) By Dokchitser-Dokchitser, "Local invariants of isogenous elliptic curves", Trans. Amer. Math. Soc. 367 (2015), **Theorem 3.1** (checked in arXiv:1208.5519v4, where it cites Kramer-Tunnell 1982, **Proposition 8.20**, for the prime 2): an additive potentially good curve over a finite extension of $\mathbb{Q}_2$ is tame (conductor exponent 2) iff its type is IV or IV*. (c) $f_2 = 2$ and $V^{I_2} = 0$ give Swan conductor 0, so wild inertia acts trivially; the image of I_2 is a finite quotient of the procyclic tame quotient of odd supernatural order, inside a group of order dividing 24 (item 6.5(iii)), hence cyclic of order 3. (d) The tame relation makes Frobenius invert the generator ($2 \equiv -1 \pmod 3$); the proof of Lemma lem:euler then gives the invariant line of $\mathrm{ad}^0 V$ spanned by $2\sigma + 1$ with geometric Frobenius eigenvalue $\chi_{-3}(2) = -1$, so $L_2(A_f, s)^{-1} = 1 + 2^{-s}$, $P_2(X) = 1 + 2X$, $e_3(E, 2) = v_3(1 + 2/4) = 1$. This is also the statement attributed to Kraus90 by Pineda-Martin (arXiv:2609.01485, Proposition 3: at 2, "$\#\Phi_p = 3$ iff the Kodaira symbol is IV or IV*", citing Kraus90 Theorem 2) and is consistent with Billerey's summary of Kraus's 2-adic results (for $v(j) \geq 12$ and $3 \nmid v(j)$: $\#G_{\mathrm{ss}} = 3$ if the type is IV, IV* and 6 otherwise). Kraus90 itself was not accessible, so its theorem number is not confirmed.
2. **Via the component group (Chapter 11 outline, now important_notes.md point 43).** By item 4.5 (which Chapter 11 cites from here), $E[3]^{I_2} \simeq \Phi_2^{\mathrm{geom}}[3] = \mathbb{Z}/3$ for types IV, IV*, so inertia fixes a line of $E[3]$; with determinant 1 its image mod 3 is unipotent of order 3, and reduction mod 3 is injective on finite subgroups of $\mathrm{GL}_2(\mathbb{Z}_3)$, so the image on T_3E is cyclic of order 3. This proof needs (a) above (potentially good, so the image is finite) and nothing about conductors.

What was checked. (i) All 14,431 curves of `ecdata` with conductor below 30,000 and type IV or IV* at 2 (6,421 and 8,010): $v_2(j) \geq 0$, $f_2 = 2$, and the four roots of the 3-division polynomial generate over $\mathbb{Q}_2$ fields of ramification indices $1, 3, 3, 3$ (projective image of inertia cyclic of order 3; together with $f_2 = 2$ this pins the image to $\mathbb{Z}/3$). (ii) Conversely $f_2 = 2$ occurs at 2 for no other additive potentially good type (table 9.2). (iii) PARI's `lfunsympow` gives $P_2(X) = 1 + 2X$ on 400 sampled curves of these types, and on all 1,475 optimal curves of these types with $N \leq 5{,}000$. (iv) The running examples 20a1, 140b1, 980i1 (IV* at 2). The Chapter 11 outline's independent check (4,750 curves below 10,000) agrees.

Consequences. Proposition prop:forms holds with no exception; Remark rem:wild's "a type IV at 2 has $P_2(X) = 1 + 2X$" and Section sec:wild's "a type IV or IV* at 2 has $e_3 = 1$" are theorems; the appeal to the 92,812 curves can be replaced by the argument. The orders 6 seen at 2 (twists of IV, IV* curves by wildly ramified quadratic characters, e.g. 80b1) have other types (II, II*, I_0^*, I_m^*) and also have $P_2 = 1 + 2X$; that is Part 8, N1.

12.9 8. Points for important_notes.md

12.9.1 8.1 New points about the paper (not already recorded; each verified as stated)

- **N1. Remark rem:wild and Section sec:wild: "a type II at 2 has $P_2 = 1$ and $e_3 = 0$" is false in general; so is "a type I_n^* at 2 has potentially good reduction ... with a cubic P_2".** Location: Remark rem:wild (second sentence) and Section sec:wild ("Definition def:terms with PARI's factor explains every one of these minima: a type II at 2 has $P_2 = 1$ and $e_3 = 0$"). Problem: at 2 the Kodaira type does not determine the inertia image (the paper's own point), and type II (and II*) occurs with inertia image Q_8 or $\mathrm{SL}_2(\mathbb{F}_3)$ ($P_2 = 1$), cyclic of order 6 ($P_2 = 1 + 2X$, $e_3 = 1$) or of order 2 (a ramified twist of a curve with good reduction at 2; P_2 a cubic, $e_3 = v_3(9 - a_2(E')^2) \in \{0, 2\}$). Counts (optimal non-CM curves, $N < 10{,}000$, $4 \mid N$): type II at 2: $P_2 = 1$ for 1,092, $1 + 2X$ for 371, cubic for 74; type II*: 514, 74, 191. Examples with

$E[3]$ irreducible (irreducible ψ_3), not CM: 336f1 $[0,1,0,-1,2]$ (type II at 2, $f_2 = 4$, $P_2 = 1+2X$, $e_3(E,2) = 1$, $\deg\varphi_E = 24$); 1232l1 $[0,0,0,32,-16]$ (type II* at 2, $f_2 = 4$, $P_2 = 1+2X-4X^2-8X^3$, $P_2(1/4) = 9/8$, $e_3(E,2) = 2$, $\deg\varphi_E = 288$, so $v_3(\deg\varphi_E) = 2$: here $3 \nmid N = 2^4\cdot 7\cdot 11$, the image is not 3Ns, and Theorem thm:main's bound 2 comes entirely from the type II* prime 2 and is attained). Likewise potentially good I_m^* at 2 (optimal, $N \le 5{,}000$): $P_2 = 1$ for 876, a cubic for 111, $1+2X$ for 8; and they are 61% of all I_m^* at 2 below 10,000 ("about half" is loose). Effect: none on any theorem or count (the rule uses PARI's factor, and the minima in sec:wild are still 0 because most type II primes at 2 have $e_3 = 0$); the explanatory sentences are wrong. Fix: "a type II or II* at 2 has $P_2 = 1$ in most cases, but $1+2X$ (inertia of order 6) or a cubic (inertia of order 2) also occur, with $e_3 = 1$ or 2"; "a potentially good I_n^* at 2 usually has $P_2 = 1$". Confidence: high (PARI 2.17.2 `lfunsympow`, inertia patterns as in 9.2).

- **N2. Proposition prop:forms** (already point 43, rewritten as settled): this outline gives a second, independent proof (Part 7, via Dokchitser-Dokchitser Theorem 3.1) and extends the data check to 14,431 curves below 30,000. No new action beyond point 43's.
- **N3. Section 1.4 and Lemma lem:twist (ii): the exchanges II $\leftrightarrow$ IV*, III $\leftrightarrow$ III*, IV $\leftrightarrow$ II* also hold at $q = 3$.** The paper restricts them to $q \ge 5$ ("the three others for $q \ge 5$"), which is safe; in fact all 5,122 such pairs at 3 below conductor 10,000 exchange as listed, with unchanged (wild) conductor exponent, and a proof sketch via Ogg's formula is item 8.3. Consequence: the max formula for $w_\ell(E,q)$ holds at $q = 3$ too. Observation, optional; confidence medium-high (data complete in the range; the sketch has one cited step to verify).
- **N4. Appendix, Kodaira codes: "$n+4$ for I_n, $-(n+4)$ for I_n^*" holds for $n \ge 1$ only** ($-(0+4) = -4$ is IV*; I_0^* is -1, which the paper does list separately). Fix: add "$(n \ge 1)$". Low importance.
- **N5. Definition def:terms is stated "for a bad prime $q \ne \ell$", but $t_\ell(E,\ell)$ is used**: (eq:rule) sums $t_\ell(E,q)$ over all $q \mid N$, and the appendix's `rule` computes `t` also for `q == l` (e.g. 726a1 at $\ell = q = 3$, type I_3, contributes 1 to the value $[2,2]$ that the paper reports). Fix: define $t_\ell(E,q)$ for every bad q (only e_ℓ needs $q \ne \ell$). Expository; confidence high.

12.9.2 8.2 Book-internal inconsistencies

- **Chapter 7 outline, Exercise 10 and verification log (line "with $q_E = j^{-1} + 744j^{-2} + 356652j^{-3}$").** The third coefficient of $\mathtt{q}(j)$ is 750420 ($\mathtt{q} = j^{-1}+744j^{-2}+750420j^{-3}+872769632j^{-4}+\cdots$, PARI `serreverse`). The conclusions there ($v_{11}(q_E) = 5$, unit part $\equiv -1 \pmod{11}$) are unaffected (they depend only on the first term) and were rechecked with PARI's `E.tate`.
- **Chapter 11 outline, item 2.2 `lem:galrep:component-invariants`**, duplicates this chapter's 4.5. Orchestrator decision: the lemma lives in Chapter 10; Chapter 11 cites it.
- **Chapter 7 outline (9.5(ii), N2), Chapter 21 outline (A1.8, A2.6)** still treat type IV at 2 as open or assumed (B13); Chapter 7's N2 also allows order 6 for these types, which never happens (Part 7). Chapter 21's A2.6 repeats "a type II at 2 has $P_2 = 1$" as a fact (N1 above); Chapter 7's 9.5(iii) is correctly hedged but should cite N1. Fix when written.
- **PLAN.md, Chapter 10 scope: "Kraus's conditions at 2 and 3 as a remark".** These are Kraus 1989 (Acta Arith. 54), not the paper's `Kraus90` (semistability defect); writers must not cite `Kraus90` for them (item 1.2(iii)).
- **PLAN.md Section 7.1 lists the key `TateAlgorithm`**, which is not in `refs.bib`; Tate's algorithm is `Tate75` there.
- No inconsistency found between this chapter's data and the Chapter 9 outline's running-examples table (types at 2, 3 of 20a1, 99a1, 726a1, 980i1 agree), nor with the Chapter 11 outline's statements on I_m^* at 2 (conductor $2a(\chi) \in \{4,6\}$ matches 7.7) and on Ogg-Saito.

Chapter 13

Chapter 11: Galois representations of elliptic curves (ch:galrep, chapters/ch11_galrep.tex)

Stage 1 line-up. Part III (Curves and elliptic curves). Target length 2,775 lines of LaTeX (card excluded; see Part 3), inside the 2,800 lines that PLAN.md Section 7.3 allows for this chapter.

Rigour tags: **P** = proved here, **S** = proof sketched here, **R** = stated; reference. Item format, as in the Chapter 2 to 9 outlines: number, label-ready name, kind, tag; the precise statement with all hypotheses; reference (book and chapter or section; a theorem number only where the outline author is sure, otherwise the section, and numbers marked "(verify)" must be checked before they are printed); the paper's use by its `\label` names, or "background". Every computation quoted below (3-division polynomials, image labels, inertia images per Kodaira type, Euler factors at 2, congruences, Serre conductors, the structure of the 3Ns curves, the Eisenstein data) was checked with PARI/GP 2.15 through `cypari2` in `/tmp/ch11` against Cremona's `ecdata` files `allcurves`, `galrep`, `degphi` for conductors below 10,000 (verification log, Part 6.6).

13.1 0. What this chapter is for

13.1.1 0.1 Purpose

The paper works throughout with $\rho = \rho_{E,\ell}$ on $T = T_\ell E$ and its reduction $\bar{\rho}$ on $E[\ell]$. Every statement it makes about these representations, as opposed to the algebra of the representations (Chapter 4), the field theory (Chapters 5 to 7) or the geometry of the curve (Chapters 9, 10), is collected and proved here. Concretely the chapter must deliver:

1. **The local representation at every prime $p \neq \ell$, by reduction type.** Neron-Ogg-Shafarevich; the Tate-curve shape at a multiplicative prime (DDT Theorem 3.1(e) for elliptic curves, with the corner $\varepsilon\kappa$ of important_notes.md point 42); the paper's **Lemma lem:tamagawa-exp proved in full**; the potentially multiplicative additive case at every q, including $q = 2$ (needed for important_notes.md point 18); the image of inertia at a potentially good additive prime by Kodaira type (the "inertia images and invariants" half of **Lemma lem:euler**); $\delta_p = \dim V^{I_p}$ of Section 1.1.
2. **The mod 3 inertia picture.** The paper's **Lemma lem:IV and Remark rem:types proved in full**, Remark rem:types in its corrected, complete form (important_notes.md point 22), and a new uniform tool: for additive reduction at $q \neq \ell$, $E[\ell^k]^{I_q} \simeq \Phi_q(\overline{\mathbb{F}}_q)[\ell^k]$ (item 2.2). The tool gives a second proof of Lemma lem:IV valid at every $q \neq 3$ and **settles important_notes.md point 43**: at $q = 2$, types IV, IV* have cyclic inertia image of order 3, so $P_2(X) = 1 + 2X$ and $e_3(E, 2) = 1$ are theorems and the exception in Proposition prop:forms disappears (item 4.4).

3. **The representation at ℓ** (good reduction): finite flatness, the Deligne-Fontaine description, the determination of a_ℓ mod ℓ by $\bar\rho|_{D_\ell}$; this is the local analysis at $p = \ell$ in **Lemma lem:old**.
4. **Conductors.** The Serre conductor $N(\bar\rho)$, the formula $a_p(\bar\rho_{E,\ell}) = f_p(E) - \dim_{\mathbb{F}_\ell} \Phi_p(\bar{\mathbb{F}}_p)[\ell]$ (item 6.2), hence $N(\bar\rho) \mid N$; this is what the **optimal-level paragraph of Section 3** and cases (a), (b), (c) of Theorem thm:qual need.
5. **Images.** Serre's open image theorem (stated), the complete list of mod 3 images for every $E/\mathbb{Q}$ proved from the group theory of Chapter 1 (item 7.3), Sutherland's labels checked against `ecdata`, Zywina's lists (stated).
6. **The hypothesis of Theorem thm:main.** Absolute irreducibility on $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{\ell^*}))$; the paper's **Proposition prop:ns proved in full** (corrected as in important_notes.md points 1, 9, 33); the paper's **Lemma lem:h0 proved in full in its corrected form** ($h_\ell(E) = 0$ for $\ell \geq 5$, $h_3(E) \leq 1$, with or without CM; points 5, 6, 10, 14); and a new observation: for $\ell \geq 5$, $\ell \nmid N$, irreducibility already implies absolute irreducibility on $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{\ell^*}))$ (item 8.2), so $\ell \notin S_f$ iff $E[\ell]$ is irreducible and, for $\ell = 3$, the image is not $N_s(\mathbb{F}_3)$ (item 8.6).
7. **Section sec:anomaly.** The structure of the curves with image 3Ns (item 8.4): every such curve has a prime $p \neq 3$ with cyclic inertia of order 4 (type III or III* when $p \geq 5$, with $p \equiv 1$ or $11 \pmod{12}$), at which $e_3(E, p) \geq 1$; the paper reports this as an observation on the tables.
8. **Section sec:eis and Conjecture conj:eis.** Reducible $\bar\rho$: the semisimplification, ordinarity at ℓ, the label dictionary 3B.1.1, 3B.1.2, 3Cs.1.1 (important_notes.md point 44), and the restatement of Chapter 9's proof that $\mathbb{Z}/3 \oplus \mu_3$ and a rational point of order 9 cannot coexist (point 17).
9. **Serre's conjecture for** $\bar\rho_{E,\ell}$ in the optimal-level form the paper uses (statement, with the elliptic-curve inputs proved here; the general theory is Chapter 15).

13.1.2 0.2 Split with the neighbouring chapters (decided here, consistent with PLAN.md Section 4 and the written outlines)

- **Chapter 1** proves the group theory of $\mathrm{GL}_2(\mathbb{F}_3)$: the 55 subgroups, the three irreducible images with surjective determinant and an element of determinant -1, trace 0 (its 6.7), orbits on lines (6.8), Sutherland's labels as intrinsic subgroups (6.10), Dickson (5.10), $1 + \ell M_2(\mathbb{Z}_\ell)$ torsion-free for ℓ odd (7.4). Chapter 11 cites these and adds which of them occur for $E/\mathbb{Q}$ and why.
- **Chapter 4** proves the representation theory: distinct-eigenvalue criterion (2.4), unique stable line of a non-split extension (3.6), Brauer-Nesbitt (5.2, 5.3), Clifford theory for index two (6.6, 6.7), the trace-zero lemma and $H^0(\mathrm{ad}^0 V \otimes \chi)$ (7.2, 7.4), the $\mathrm{GL}_2(\mathbb{F}_3)$ table and the representation-theoretic core of Proposition prop:ns (8.1 to 8.5), unipotent and cyclic shapes and their fixed spaces on V and $\mathrm{ad}^0 V$ (9.1 to 9.4), lattices, homothety and the length lemma behind h_ℓ (10.1 to 10.5). Chapter 11 never reproves these; it supplies the Galois and elliptic-curve inputs (which group, which character, which Kodaira type).
- **Chapter 5** proves continuity, the cyclotomic character, complex conjugation and "irreducible iff absolutely irreducible" for $\det\bar\rho(c) = -1$ (galois:complex-conjugation (d)), restriction to $G_{\mathbb{Q}(\sqrt{\ell^*})}$ (galois:restriction, galois:index-two-restriction), Chebotarev plus Brauer-Nesbitt (galois:chebotarev-brauer-nesbitt, galois:congruent-ap), the determinant obstruction $\chi_\ell^2 \not\equiv 1$ (galois:twist (c)), the quartic-on-lines proposition and the example $y^2 = x^3 - 6x - 8$ (6912d1). Chapter 11 applies them.
- **Chapter 6** proves Frobenius elements, the finite-level tame relation (3.11, 3.12), $\mathbb{Z}_3[\zeta_3]$ as a DVR and $T \simeq \mathbb{Z}_3[\zeta_3]$ (4.7, 4.8), the commutant and the Frobenius action $\chi_{-3}(q)$, $\chi_{-4}(q)$ (4.10), the twist detector by traces at inert primes (6.4), CM and $a_p = 0$ (6.5), Kronecker-Weber (5.5).
- **Chapter 7** proves the local field theory: tame character and Frob t $\mathrm{Frob}^{-1} = t^q$ (5.3, 5.4), finite tame images are cyclic (5.6), wild images (5.7), Kummer classes on inertia (6.2), Weil-Deligne representations, sp(2), the Tate shape (7.6), invariants, L-factors, Swan and Artin conductors (8.1 to 8.4), and the shapes table 9.4, whose Kodaira column it leaves as a black box to Chapters 10 and 11. **Chapter 11 supplies that column as theorems** (items 3.2, 3.4, 4.1, 4.4) and does not repeat the Weil-Deligne or L-factor computations.
- **Chapter 9** proves ρ, $\bar\rho$, $\det\rho = \chi_\ell$, complex conjugation (5.1 to 5.6), division polynomials and ψ_3 (4.4 to 4.6), CM (6.5), rational isogenies, $\mathbb{Z}/\ell \oplus \mu_\ell$ and the no-27-isogeny argument (9.5 to 9.9),

twists $\rho \otimes \chi_d$ (10.3), and the easy half of Neron-Ogg-Shafarevich (8.6).

- **Chapter 10** (outlined in parallel) proves minimal models, reduction types, E_0, E_1 and the formal group, $E(K)/E_0(K) \simeq \Phi_p(k)$, Kodaira types and Tate's algorithm, the component groups $\Phi_q(\bar{\mathbb{F}}_q)$ by type, conductor exponents and Ogg's formula, the semistable reduction theorem, the Tate curve with its Galois module $E_{\mathfrak{q}}[\ell^t]$ (statement), Kraus's semistability defect $12/\gcd(v_q(\Delta), 12)$ for $q \geq 5$, Lemma lem:twist (i), (ii) and Remark rem:wild. **Chapter 11 uses all of these as black boxes with `\Cref{ch:reduction}` and proves the representation-theoretic consequences.** If the Chapter 10 outline proves the Galois-theoretic consequences of the Tate curve itself (the matrix shape, Lemma lem:tamagawa-exp), Chapter 11 keeps them anyway, because PLAN.md puts Lemma lem:tamagawa-exp here; the Chapter 10 writer should then only state them.
- **Chapter 15** proves Galois representations of newforms, local-global compatibility, states the optimal-level theorem in general (DDT Theorem 3.15) and proves only its specialisation to $\bar{\rho}_{E',\ell}$, with the trivial-nebentypus argument (important_notes.md B18), Ribet's level lowering, Lemma lem:old and Theorem thm:qual. Chapter 11 supplies the elliptic-curve inputs (local structure at $p \mid N$ and at ℓ, Serre conductor, finite flatness, Serre weight) and states Serre's conjecture for $\bar{\rho}_{E,\ell}$; the Deligne-Fontaine description is proved (sketched) here for E and quoted for newforms in Chapter 15.
- **Chapters 16, 18, 19, 20.** Chapter 16 has the cohomology of the Tate shape (it imports the shape from here); Chapter 18 proves Lemma lem:tamfactor and restates Lemma lem:h0 (proved here) in Selmer language; Chapter 19 computes $L_q(A_f, s)$ and $e_\ell(E, q)$ (Lemma lem:euler in full) from the inertia images proved here; Chapter 20 uses $\ell \notin S_f$ through item 8.6.
- **Chapters 21, 22** use the card: the tour cites items 3.3, 4.3, 4.5, 8.3, 8.5; the computation chapter uses 7.3, 7.6 (`ns3`, `elldivpol`).

13.2 1. Global conventions fixed in this chapter

- **C1 (objects).** $E/\mathbb{Q}$ is given by a global minimal model, conductor N; ℓ is a prime, odd unless said otherwise; $T = T_\ell E$, $V = T \otimes \mathbb{Q}_\ell$, $\rho = \rho_{E,\ell} \colon G_{\mathbb{Q}} \to \mathrm{Aut}_{\mathbb{Z}_\ell}(T) \simeq \mathrm{GL}_2(\mathbb{Z}_\ell)$, $\rho \bmod \ell^k$ the action on $E[\ell^k] = T/\ell^k T$, $\bar{\rho} = \bar{\rho}_{E,\ell}$ on $E[\ell]$; $A = \mathrm{ad}^0 T$, $A_f = \mathrm{ad}^0 V$ (trace zero, the paper's macro `\ad`). $\delta_p = \dim V^{I_p}$. These are Chapter 9's objects (C1 there) and the paper's (Section 1.1).
- **C2 (local data).** For a prime q: $D_q \simeq G_{\mathbb{Q}_q}$ and I_q via a fixed embedding $\overline{\mathbb{Q}} \to \overline{\mathbb{Q}}_q$ (Chapter 6, 3.10); $f_q = f_q(E)$ the conductor exponent; $n_q = v_q(\Delta_{\min})$ at a multiplicative prime; $q_E \in \mathbb{Q}_q^\times$ the Tate parameter, $v_q(q_E) = -v_q(j)$, and $u = q_E q^{-v_q(q_E)}$ its unit part (important_notes.md point 8: the Tate parameter is never called u); $\Phi_q^{\mathrm{geom}} = \Phi_q(\bar{\mathbb{F}}_q)$ the geometric component group of the Neron model (the paper's macro `\Phig`), $c_q = \#\Phi_q(\mathbb{F}_q)$, which divides $\#\Phi_q^{\mathrm{geom}}$; the group scheme Φ_q is always written with an argument, never as a bare Φ_q (Appendix B, B-C11(b)). Kodaira symbols as in PLAN.md 7.1.
- **C3 (Frobenius).** As in Chapter 7 (L3) and Chapter 9 (C5): Frob_p is an **arithmetic** Frobenius; Frob_p^{-1} is the geometric one, written Frob_p^{-1} and never Φ_p in this chapter, where $\Phi_q(\cdot)$ is the component group (Appendix B, B-C2(b)); group-theoretic statements (conjugation, the Tate matrices) use Frob_p; L-factors use Frob_p^{-1} and are Chapter 19's. On $V_\ell E$ at a good prime the arithmetic Frobenius has characteristic polynomial $X^2 - a_p X + p$, roots α, β; the geometric one has α^{-1}, β^{-1}; on $\mathrm{ad}^0 V$ both give $\alpha/\beta, 1, \beta/\alpha$ (important_notes.md B1). The relation on tame inertia is $\mathrm{Frob}\ t\ \mathrm{Frob}^{-1} = t^q$. The letter Φ is used in this chapter only for the component group, always with an argument ($\Phi_q(\mathbb{F}_q)$, $\Phi_q(\bar{\mathbb{F}}_q)$) or as Φ_q^{geom}; the geometric Frobenius is always Frob^{-1} or Frob_q^{-1}.
- **C4 (characters).** χ_ℓ is the ℓ-adic cyclotomic character, $\bar{\chi}_\ell$ its reduction; χ_d the quadratic character of $\mathbb{Q}(\sqrt{d})$; $q^* = (-1)^{(q-1)/2} q$ and χ_{q^*}, the quadratic character of conductor q (the paper writes χ_q; the book never does, decision D3); ℓ^* likewise. Where χ_3 could mean either, write χ_3^{cyc} or χ_{-3} (important_notes.md point 32); $\bar{\chi}_3 = \chi_{-3}$. $\omega = \bar{\chi}_\ell|_{I_\ell}$ and ω_2 are the fundamental characters (5.2).
- **C5 (standard matrices).** As Chapter 1, 6.2 and Chapter 4, C6: $J = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, $w = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, $u = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$, the subgroups $C_s, N_s, C_{ns}, N_{ns}, \mathrm{B}, \mathrm{U}, Q_8, C_4 = \langle J \rangle$ of $\mathrm{GL}_2(\mathbb{F}_3)$. The paper's σ is an element of order 3 in the image of inertia (Lemmas lem:IV, lem:euler); the matrix $\mathrm{diag}(1, -1)$ is written h here, as in Chapter 4.

- **C6 (labels).** Sutherland's labels (3B, 3B.1.1, 3B.1.2, 3Cs, 3Cs.1.1, 3Ns, 3Nn, ...) as in `ecdata` and the paper, always accompanied in the text by the intrinsic description (7.3, 7.4). "Surjective" has no label. Curve labels are Cremona's (Chapter 9, C6).
- **C7 (typography).** PLAN.md 7.2: no dash ligatures ("Neron-Ogg-Shafarevich", "Deligne-Fontaine", "Serre-Tate", "Khare-Wintenberger", ranges "1 to 8"); ASCII source; labels `...:galrep:...`; the line-up names below are the label stems.

13.3 2. Line-up of the core results, in teaching order

13.3.1 Section 1. The representations $\rho_{E,\ell}$ and $\bar\rho_{E,\ell}$: recap, twists, irreducibility

1.1 `def:galrep:rho` (definition and proposition, recap; P with the proofs cited). For $E/\mathbb{Q}$ and a prime ℓ: T, V, ρ, $\rho \bmod \ell^k$, $\bar\rho$ as in C1. (i) ρ is continuous; the field cut out by $\rho \bmod \ell^k$ is $\mathbb{Q}(E[\ell^k])$ (Chapter 9, 5.4; Chapter 5, galois:continuous-reps). (ii) $\det\rho = \chi_\ell$ (Weil pairing, Chapter 9, 5.2, 5.3); so $\mathbb{Q}(\zeta_\ell) \subset \mathbb{Q}(E[\ell])$, $\det\bar\rho = \bar\chi_\ell$ is surjective onto $\mathbb{F}_\ell^\times$, and $\bar\rho(G_{\mathbb{Q}(\sqrt{\ell^*})}) = \{g \in \bar\rho(G_\mathbb{Q}) : \det g \in (\mathbb{F}_\ell^\times)^2\}$ (Chapter 5, galois:restriction (c)). For $\ell = 3$: $\bar\chi_3 = \chi_{-3}$, the field cut out by $\det\bar\rho_{E,3}$ is $\mathbb{Q}(\sqrt{-3})$ and $\bar\rho_{E,3}(G_{\mathbb{Q}(\sqrt{-3})}) = \bar\rho_{E,3}(G_\mathbb{Q}) \cap \mathrm{SL}_2(\mathbb{F}_3)$. (iii) For ℓ odd and c a complex conjugation, $\rho(c)$ is conjugate in $\mathrm{GL}_2(\mathbb{Z}_\ell)$ to $\mathrm{diag}(1,-1)$ ($T = T^{c=1} \oplus T^{c=-1}$, both of rank one, as $2 \in \mathbb{Z}_\ell^\times$ and $\det\rho(c) = -1$); so $\bar\rho(c)$ has determinant -1 and trace 0 (Chapter 9, 5.6). (iv) ρ is unramified outside $N\ell$ (2.1 below). Reference: SilvermanAEC III.7 and III.8; SerreAbelian I.1. Paper: Section 1.1 (V, T, ρ, $\bar\rho$), Proposition prop:ns ("$\det\bar\rho_{E,3} = \chi_{-3}$ cuts out $\mathbb{Q}(\sqrt{-3})$"; "an element of determinant -1 and trace 0 (complex conjugation)"), Lemma lem:h0, the hypothesis of Theorem thm:main.

1.2 `conv:galrep:notation` (convention). C1 to C6 of Part 1 typeset as `convention` environments, with the dictionary arithmetic/geometric Frobenius (C3) and the warning on χ_3 (C4). Paper: Section 1.1, Lemma lem:euler, Lemma lem:tamfactor (important_notes.md points 8, 11, 32, B1).

1.3 `prop:galrep:twist-isogeny` (proposition; P). Let $E/\mathbb{Q}$, ℓ odd. (i) (Chapter 9, 10.3, cited.) For $d \in \mathbb{Q}^\times$, $\rho_{E^{(d)},\ell} \simeq \rho_{E,\ell} \otimes \chi_d$; hence $\mathrm{ad}^0\,\rho_{E^{(d)},\ell} \simeq \mathrm{ad}^0\,\rho_{E,\ell}$, the projective images of $\bar\rho_{E^{(d)},\ell}$ and $\bar\rho_{E,\ell}$ coincide, and irreducibility of $E[\ell]$ is twist-invariant. (ii) Effect on the mod 3 labels (item 7.3): twisting by χ_{-3} exchanges 3B.1.1 and 3B.1.2 and fixes every other label; twisting by χ_d with $d \notin \{1,-3\} \bmod \mathbb{Q}^{\times 2}$ sends 3B.1.1 and 3B.1.2 to 3B and 3Cs.1.1 to 3Cs, and fixes 3Ns, 3Nn and the surjective image. (Proof: $\mathrm{diag}(1,\chi_{-3}) \otimes \chi_d = \mathrm{diag}(\chi_d, \chi_d\chi_{-3})$; $\left(\begin{smallmatrix}1 & * \\ 0 & \chi_{-3}\end{smallmatrix}\right) \otimes \chi_{-3} = \left(\begin{smallmatrix}\chi_{-3} & * \\ 0 & 1\end{smallmatrix}\right)$; the property $\bar\rho \simeq \bar\rho \otimes \chi_{-3}$ is twist-invariant, which keeps 3Ns and 3Nn apart although $N_s(\mathbb{F}_3) \subset N_{ns}(\mathbb{F}_3)$ (important_notes.md point 2); $\mathrm{GL}_2(\mathbb{F}_3)$ has a unique subgroup of index 2, $\mathrm{SL}_2(\mathbb{F}_3)$, whose projective image is $A_4 \neq S_4$.) (iii) (Chapter 9, 9.9, cited.) If $E[\ell]$ is irreducible, all curves of the isogeny class have isomorphic T_ℓ, hence the same ρ, $\bar\rho$, image label at ℓ, $\mathrm{ad}^0\,T$ and $h_\ell(E)$ (important_notes.md point 27). Reference: elementary. Paper: Lemma lem:twist, Theorem thm:qual (b), (c) ($\bar\rho \otimes \chi$ satisfies the hypotheses since $\bar\rho$ does), Theorem thm:quant ($\mathrm{ad}^0\,T$ unchanged), the proof of Theorem thm:main (I_n^* primes), Section sec:eis.

1.4 `prop:galrep:irreducibility` (proposition; P, from Chapters 4, 5, 9). Let ℓ be odd. (i) $\bar\rho_{E,\ell}$ is irreducible iff it is absolutely irreducible (1.1(iii) and Chapter 4, 2.4: a stable line of $\bar{\mathbb{F}}_\ell^2$ is an eigenline of $\bar\rho(c)$, and both eigenlines are $\mathbb{F}_\ell$-rational). (ii) $\bar\rho_{E,\ell}$ is reducible iff E has a rational ℓ-isogeny (Chapter 9, 9.5); for $\ell = 3$ iff ψ_3 has a rational root. (iii) (Chebotarev and Brauer-Nesbitt, Chapter 5, galois:congruent-ap.) $\bar\rho^{\mathrm{ss}}$ is determined by $a_p(E) \bmod \ell$ for p in any set of density one with $p \nmid N\ell$; if $\bar\rho$ is irreducible, $\bar\rho$ itself is. If h is a newform of weight 2 with coefficients in $\mathcal{O}$, $\lambda \mid \ell$, $k = \mathcal{O}/\lambda$, and $a_p(h) \equiv a_p(E) \pmod\lambda$ for a density-one set of p, then $\bar\rho_{h,\lambda} \simeq \bar\rho_{E,\ell} \otimes_{\mathbb{F}_\ell} k$ when $\bar\rho_{E,\ell}$ is irreducible (absolute irreducibility, (i), is what makes this work: important_notes.md point 10). Reference: as cited. Paper: Lemma lem:old (case $p \nmid \ell N$: "$\bar\rho_h \simeq \bar\rho$ by Chebotarev and Brauer-Nesbitt, $\bar\rho$ being irreducible"), Lemma lem:h0 ("irreducible" means absolutely irreducible), Theorem thm:quant, Conjecture conj:euler ("$E[\ell]$ irreducible").

13.3.2 Section 2. Neron-Ogg-Shafarevich, component groups and inertia invariants

2.1 `thm:galrep:nos` (theorem, the criterion of Neron, Ogg and Shafarevich; P for (a) to (d), S for (e), (f)). Let $K/\mathbb{Q}_p$ be finite, E/K, $\ell \neq p$. The following are equivalent: (a) E has good reduction; (b) I_K acts trivially on $T_\ell E$; (c) I_K acts trivially on $E[m]$ for every m prime to p; (d) I_K acts trivially on $E[m]$ for infinitely many m prime to p. Moreover (e) E has potentially good reduction iff $\rho_{E,\ell}(I_K)$ is finite, for one (equivalently every) $\ell \neq p$; (f) E has potentially multiplicative reduction ($v_K(j) < 0$) iff $\rho_{E,\ell}(I_K)$ is infinite, and then $\rho_{E,\ell}(I_K)$ contains a unipotent element $\neq 1$ (3.2, 3.4). For $E/\mathbb{Q}$: $\rho_{E,\ell}$ is unramified at $p \neq \ell$ iff $p \nmid N$. Proof: (a)⇒(c) is Chapter 9, 8.6 (reduction is injective on prime-to-p torsion); (d)⇒(a): over K^{ur} (through finite unramified K'), if E had bad reduction then $E(K') \supset E_0(K') \supset E_1(K')$ with E_1 pro-p, $E_0/E_1 \simeq \tilde{E}_{\mathrm{ns}}(k')$ equal to k'^{+} or a one-dimensional torus, and $E/E_0 \simeq \Phi_p(k')$ of bounded order (Chapter 10); then $\#E(K^{\mathrm{ur}})[m] \leq m \cdot \#\Phi_p(\bar{k})$ for m prime to p, contradicting $E[m] \subset E(K^{\mathrm{ur}})$ of order m^2 for large m. (e), (f): semistable reduction theorem (Chapter 10) plus (a) to (d) over a finite extension, and the Tate shape 3.2. Reference: Serre-Tate 1968, Section 1, Theorem 1 and Section 2 (`SerreTate68`, in `refs.bib` by decision B14, see Part 6.5); SilvermanAEC VII.7 (Theorem 7.1, verify). Paper: Section 1.1 (δ_p), Lemma lem:tamagawa-exp, Lemma lem:IV ("potentially good", "finite image"), Lemma lem:old ("$\bar{\rho}_h$ unramified"), the optimal-level paragraph ("$N(\bar{\sigma})$ is prime to every prime at which $\bar{\sigma}$ is unramified").

2.2 `lem:galrep:component-invariants` (lemma; R: proved in Chapter 10, its item 4.5 `prop:reduction:component-invariants`, which this item restates and cites with `\Cref{ch:reduction}` instead of proving again: important_notes.md B16). Let $q \neq \ell$ be primes, $E/\mathbb{Q}_q$ with bad reduction, $\Phi_q^{\mathrm{geom}} = \Phi_q(\bar{\mathbb{F}}_q)$, $k \geq 1$. (i) If the reduction is additive, there is a $\mathrm{Gal}(\bar{\mathbb{F}}_q/\mathbb{F}_q)$-equivariant isomorphism

$$E[\ell^k]^{I_q} \;\simeq\; \Phi_q^{\mathrm{geom}}[\ell^k];$$

in particular $E(\mathbb{Q}_q)[\ell^\infty] \simeq \Phi_q(\mathbb{F}_q)[\ell^\infty]$. (ii) If the reduction is multiplicative, there is an exact sequence of $\mathrm{Gal}(\bar{\mathbb{F}}_q/\mathbb{F}_q)$-modules $0 \to \mu_{\ell^k} \otimes \varepsilon \to E[\ell^k]^{I_q} \to \Phi_q^{\mathrm{geom}}[\ell^k] \to 0$, with $\Phi_q^{\mathrm{geom}} \simeq \mathbb{Z}/n_q$ and ε the unramified character of 3.2 (trivial for split reduction). (iii) Consequently, for ℓ odd, $\dim_{\mathbb{F}_\ell} E[\ell]^{I_q}$ is: $1 + [\ell \mid n_q]$ for I_n, $n \geq 1$; 1 for IV, IV* if $\ell = 3$; 0 for every other additive type (the orders of Φ_q^{geom} for additive types are $1, 2, 3, 4$: Chapter 10). Proof (in Chapter 10, 4.5; recalled here in one paragraph only): for each finite unramified $K'/\mathbb{Q}_q$ with residue field k': $E(K')/E_0(K') \simeq \Phi_q(k')$ (Neron model and Hensel, Chapter 10), $E_1(K')$ is a $\mathbb{Z}_q$-module, hence uniquely ℓ-divisible, and $E_0(K')/E_1(K') \simeq \tilde{E}_{\mathrm{ns}}(k')$, which is k'^{+} (additive: uniquely ℓ-divisible) or $k'^{\times}$ (split over large enough k'); the snake lemma for multiplication by ℓ^k, and the union over K', since $E[\ell^k]^{I_q} = E[\ell^k](\mathbb{Q}_q^{\mathrm{ur}})$. Reference: the argument is the one in Serre-Tate 1968, Section 1; SilvermanAEC VII.6 and VII.7 (verify); ATAEC IV.9 for $E(K)/E_0(K) \simeq \Phi_q(k)$. Paper: Lemma lem:IV (second proof, 4.3), Remark rem:types ("$E[3]^{I_q} = 0$", 4.5), Lemma lem:tamagawa-exp (second proof, 3.3), Remark rem:wild and Proposition prop:forms (type IV at 2, 4.4), the final check in the proof of Lemma lem:tamfactor (the ℓ-part of c_q; Chapter 18), the Serre conductor formula 6.2. Verified: at $q = 2$ and $q \geq 5$, for Kodaira types sampled from every class (Part 6.6), $E[3]^{I_q} \neq 0$ exactly for I_n and IV, IV*.

2.3 `thm:galrep:serre-tate` (theorem; R, with (ii) and (iv) P). Let $E/\mathbb{Q}_q$ have potentially good reduction and $\ell \neq q$. (i) (Serre-Tate) The kernel of $\rho_{E,\ell}|_{I_q}$ is $I_L = \mathrm{Gal}(\overline{\mathbb{Q}}_q/L)$ with $L = \mathbb{Q}_q^{\mathrm{ur}}(E[m])$ for any $m \geq 3$ prime to q; L is the smallest extension of $\mathbb{Q}_q^{\mathrm{ur}}$ over which E acquires good reduction. So $\rho_{E,\ell}(I_q) \simeq \mathrm{Gal}(L/\mathbb{Q}_q^{\mathrm{ur}})$ does not depend on ℓ, and the characteristic polynomial of $\rho_{E,\ell}(\sigma)$, $\sigma \in I_q$, lies in $\mathbb{Z}[X]$ and does not depend on ℓ. (ii) $\rho_{E,\ell}(I_q) \subset \mathrm{SL}_2(\mathbb{Z}_\ell)$ (χ_ℓ is trivial on I_q), and for ℓ odd reduction modulo ℓ is injective on it ($1 + \ell M_2(\mathbb{Z}_\ell)$ is torsion-free, Chapter 1, 7.4); a finite subgroup of $\mathrm{SL}_2(\mathbb{Q}_\ell)$ with a nonzero fixed vector is trivial, so $V^{I_q} = 0$ unless the reduction is good. (iii) (Kraus, Serre-Tate) $\rho_{E,\ell}(I_q)$ is cyclic of order $1, 2, 3, 4$ or 6 if $q \geq 5$; one of these or $C_3 \rtimes C_4$ (order 12) if $q = 3$; one of these or Q_8 or $\mathrm{SL}_2(\mathbb{F}_3)$ if $q = 2$. (iv) For $q \geq 5$ it is tame (order prime to q), hence cyclic (Chapter 7, 5.6), and its order is $12/\gcd(v_q(\Delta_{\min}), 12)$ (Kraus's theorem, proved or stated in Chapter 10; cite it, do not reprove). Reference: Serre-Tate 1968, Section 2 (theorem and corollary numbers: verify); Kraus90 (cite the paper; theorem numbers not checked). Paper: Lemma lem:IV ("the semistability defect ... for any $\ell \neq q$"), Lemma lem:euler ("the image of I_q in $\mathrm{Aut}\, V$ is cyclic of order 3 or 6"), Remark rem:types, Remark rem:wild.

13.3.3 Section 3. Good and multiplicative reduction; Lemma lem:tamagawa-exp

3.1 `prop:galrep:good` (proposition; P, citing Chapter 9, 8.6 and 2.1). Let $p \nmid N\ell$. Then ρ is unramified at p, the arithmetic Frobenius has characteristic polynomial $X^2 - a_p X + p$ on V (roots α, β), the geometric one $X^2 - \frac{a_p}{p}X + \frac{1}{p}$; on $\mathrm{ad}^0 V$ both act with eigenvalues $\alpha/\beta, 1, \beta/\alpha$; $\delta_p = 2$; $\bar\rho(\mathrm{Frob}_p)$ has characteristic polynomial $X^2 - a_p X + p \bmod \ell$, and for $\ell = 3$ its orbit pattern on the four lines is the factorisation pattern of $\psi_3 \bmod p$ for $p \nmid 3 \operatorname{disc}\psi_3$ (Chapter 5, galois:frobenius-on-lines). Reference: SilvermanAEC VII.7. Paper: Section 1.1, the table of Section 1 (α, β for E'), Theorem thm:dfg14 (level-raising factor), Lemma lem:old (case $p \,\|\, N$ unramified: "$X^2 - a_p(h)X + p$").

3.2 `thm:galrep:tate-shape` (theorem; P given Chapter 10's Tate curve and Chapter 5's Kummer cocycle). Let q be a prime of multiplicative reduction, $\ell \neq q$, $\varepsilon\colon G_{\mathbb{Q}_q} \to \{\pm 1\}$ the unramified character with $\varepsilon(\mathrm{Frob}_q) = a_q(E)$ ($+1$ split, -1 nonsplit), and κ a Kummer cocycle of q_E (Chapter 5, galois:l-adic-kummer), written in $\mathbb{Z}_\ell$ after fixing a generator of $\mathbb{Z}_\ell(1)$. There is a $\mathbb{Z}_\ell$-basis e_1, e_2 of T in which

$$\rho(g) = \begin{pmatrix} \chi_\ell(g)\varepsilon(g) & \varepsilon(g)\kappa(g) \\ 0 & \varepsilon(g) \end{pmatrix} \qquad (g \in G_{\mathbb{Q}_q}),$$

($T \simeq T_\ell E_{q_E} \otimes \varepsilon$; the corner is $\varepsilon\kappa$, important_notes.md point 42). Consequences: (i) on I_q, $\rho(\sigma) = \left(\begin{smallmatrix} 1 & n_q\, t_\ell(\sigma) \\ 0 & 1 \end{smallmatrix}\right)$ ($\kappa|_{I_q} = v_q(q_E)\, t_\ell$, Chapter 7, 6.2), a unipotent action through the maximal pro-ℓ quotient of I_q, with infinite image; (ii) $V^{I_q} = \mathbb{Q}_\ell e_1$, so $\delta_q = 1$, and Frob_q acts on it by $a_q(E)q$; (iii) $\mathbb{Q}_\ell e_1$ is the unique $G_{\mathbb{Q}_q}$-stable line of V (non-split extension of ε by $\varepsilon\chi_\ell$, Chapter 4, 3.6); (iv) the Weil-Deligne representation is $\mathrm{sp}(2) \otimes \mu_\varepsilon$, conductor exponent 1 (Chapter 7, 7.6, 9.3); (v) $(\mathrm{ad}^0 V)^{I_q} = \mathbb{Q}_\ell e$, $e = e_1 \otimes e_2^\vee$, on which Frob_q acts by q (and the geometric Frobenius by q^{-1}; Chapter 4, 9.1(iii)), so $\dim(\mathrm{ad}^0 V)^{I_q} = 1$. Reference: SilvermanATAEC V.3 and V.5 (Tate curve); SerreAbelian IV, Appendix A.1 (verify); DDT Theorem 3.1(e) as cited by the paper (the elliptic-curve case of a theorem on newforms; letter and number not checked). Paper: Lemma lem:tamagawa-exp, Lemma lem:tamfactor (the matrices, $te_2 = e_2 + n'e_1$, $v_\ell(n') = v_\ell(n)$), Lemma lem:euler (type I_n), Lemma lem:old (cases $p \,\|\, N$), Section 1.1 ($\delta_q = 1$); Chapter 16 imports the shape.

3.3 `lem:galrep:tamagawa-exp` (lemma, the paper's Lemma lem:tamagawa-exp; P, two proofs). Let q be a prime of multiplicative reduction, $n_q = v_q(\Delta_E)$, and $\ell \neq q$ a prime. Then $\rho_{E,\ell} \bmod \ell^t$ is unramified at q iff $\ell^t \mid n_q$; this holds for split and nonsplit reduction. Supplements: (i) the Tamagawa exponent $t_q := \max\{t : \rho \bmod \ell^t \text{ unramified at } q\}$ (Kim and Ota) equals $v_\ell(n_q) = v_\ell(-v_q(j)) = t_\ell(E, q)$; (ii) $\#E[\ell^t]^{I_q} = \ell^{\,t+\min(t, v_\ell(n_q))}$; (iii) if $\ell \nmid n_q$, $\bar\rho|_{D_q}$ is ramified, a non-split extension of ε by $\varepsilon\bar\chi_\ell$ with a unique stable line, and $\bar\rho(I_q)$ has order ℓ; (iv) if $\ell \mid n_q$, $\bar\rho|_{D_q}$ is unramified and $\bar\rho(\mathrm{Frob}_q)$ has eigenvalues $a_q(E)q$ and $a_q(E)$ modulo ℓ. Proof 1 (the paper's, expanded; important_notes.md point 8): over the unramified quadratic extension (or $\mathbb{Q}_q$ if split) $E \simeq E_{q_E}$, which does not change I_q; $E[\ell^t]$ is generated by μ_{ℓ^t} and an ℓ^t-th root of q_E in $\overline{\mathbb{Q}}_q^\times / q_E^{\mathbb{Z}}$; I_q fixes μ_{ℓ^t}, and by Kummer theory over $\mathbb{Q}_q^{\mathrm{ur}}$ (which contains μ_{ℓ^t}, $q \neq \ell$) it fixes q_E^{1/ℓ^t} iff q_E is an ℓ^t-th power in $\mathbb{Q}_q^{\mathrm{ur}}$; units of its valuation ring are ℓ^t-th powers (Hensel), so this holds iff $\ell^t \mid v_q(q_E) = n_q$ (Chapter 7, 6.2(v)). Proof 2: 2.2(ii) with $\Phi_q^{\mathrm{geom}} \simeq \mathbb{Z}/n_q$ gives (ii), and $E[\ell^t]^{I_q} = E[\ell^t]$ iff $\min(t, v_\ell(n_q)) = t$. Reference: SilvermanATAEC V; Chapter 7, 6.2. Paper: Lemma lem:tamagawa-exp, Section 1.5 ("$t_q = v_\ell(n_q) = t_\ell(E, q)$ whether or not the reduction is split"), Remark rem:ko(2), Theorem thm:qual (c), Lemma lem:old (the two cases $p \,\|\, N$), the optimal-level paragraph ($N(\bar\rho)$ prime to q iff $\ell \mid n_q$).

3.4 `prop:galrep:potmult` (proposition; P given Chapter 10). Let $q \neq \ell$ ($q = 2$ allowed) and let E have additive, potentially multiplicative reduction at q ($v_q(j) < 0$). Then there is a ramified quadratic character χ of $G_{\mathbb{Q}_q}$ such that $E \otimes \chi$ has multiplicative reduction over $\mathbb{Q}_q$, and $\rho|_{G_{\mathbb{Q}_q}} \simeq \rho' \otimes \chi$ with ρ' of the shape 3.2 for a Tate parameter of valuation $-v_q(j)$. For q odd one may take $\chi = \chi_{q^*}|_{G_{\mathbb{Q}_q}}$; then $E^{(q)}$ has type I_n, $n = -v_q(j)$, and E has type I_n^* (Chapter 10, Lemma lem:twist (ii)). Consequences: $\rho(I_q) = \{\chi(\sigma)\left(\begin{smallmatrix} 1 & nt_\ell(\sigma) \\ 0 & 1 \end{smallmatrix}\right)\}$ is $\pm$(unipotent), infinite; $V^{I_q} = 0$ ($\delta_q = 0$); $\mathrm{ad}^0 V \simeq \mathrm{ad}^0 V'$ as $G_{\mathbb{Q}_q}$-modules, so $(\mathrm{ad}^0 V)^{I_q}$ is a line on which Frob_q acts by q; $\rho \bmod \ell^t$ is ramified at q for every $t \geq 1$, while $(\rho \otimes \chi) \bmod \ell^t$ is unramified iff $\ell^t \mid v_q(j)$; the conductor exponent is $2a(\chi)$: 2 for q odd, 4 or 6 for $q = 2$ (Chapter 7,

9.3(ii)). At $q = 2$ the index m of the Kodaira symbol I_m^* need not equal $-v_2(j)$, and a symbol I_m^* at 2 can also occur with potentially good reduction (Definition def:terms; Remark rem:wild; Chapter 10). Reference: SilvermanATAEC V.5 (verify); Chapter 7, 7.6, 9.3. Paper: Definition def:terms (t_ℓ via $v_q(j)$, I_n^* at 2), Lemma lem:euler (type I_n^*: "$V = V' \otimes \chi_{q^*}$ with V' Steinberg"), Lemma lem:twist (iii) (conductor exponent 2 of the ramified twist), the proof of Theorem thm:main (the I_n^* primes; this item covers $q = 2$, which the paper's argument via Lemma lem:twist does not: important_notes.md point 18), Theorem thm:qual (c), Remark rem:types.

3.5 `cor:galrep:delta` (corollary; P). $\delta_p = \dim V^{I_p}$ is 2, 1, 0 according as E has good, multiplicative or additive reduction at $p \neq \ell$ (3.1, 3.2; for additive reduction 2.3(ii) and 3.4), and the conductor exponent is $f_p = 2 - \delta_p + \mathrm{sw}_p(V)$ (Ogg-Saito: the conductor of Chapter 10 equals the Artin conductor of $V_\ell E$; stated, Chapter 7, 8.4). Paper: Section 1.1 ("$\delta_p = 2, 1, 0$ for good, multiplicative, additive reduction"), Definition def:terms (naive factor), Theorem thm:dfg14, Section 4.1 (M^Σ).

13.3.4 Section 4. Additive potentially good reduction: the image of inertia by Kodaira type

4.1 `thm:galrep:additive-shapes` (theorem; P given Kraus's theorem (Chapter 10, R) and Chapters 4, 6, 7). Let $q \geq 5$, $\ell \neq q$, and let E have additive potentially good reduction at q. Then $\rho(I_q) = \langle g \rangle$ is cyclic, $g \in \mathrm{SL}_2(\mathbb{Z}_\ell)$, and, with σ denoting an element with $\sigma^2 + \sigma + 1 = 0$ and τ one with $\tau^2 = -1$:

type	$v_q(\Delta_{\min})$	$\#\rho(I_q)$	generator	$(\mathrm{ad}^0 V)^{I_q}$	action of Frob_q on $(\mathrm{ad}^0 V)^{I_q}$
I_0^*	6	2	-1	$\mathrm{ad}^0 V$	that of $\mathrm{ad}^0 V_\ell E^{(q)}$ (unramified; eigenvalues $\alpha'/\beta', 1, \beta'/\alpha'$)
IV, IV*	4, 8	3	σ	$\mathbb{Q}_\ell(2\sigma + 1)$	$\chi_{-3}(q)$
III, III*	3, 9	4	τ	$\mathbb{Q}_\ell \tau$	$\chi_{-4}(q)$
II, II*	2, 10	6	$-\sigma$	$\mathbb{Q}_\ell(2\sigma + 1)$	$\chi_{-3}(q)$

In every case $V^{I_q} = 0$, $f_q = 2$ (tame), $\rho(\mathrm{Frob}_q)\, g\, \rho(\mathrm{Frob}_q)^{-1} = g^q$ for every lift, the action on the line is the same for the arithmetic and the geometric Frobenius, $(2\sigma + 1)^2 = -3$ ("the line spanned by $\sqrt{-3}$"), and $\mathrm{End}_{I_q}(T) = \mathbb{Z}_\ell[g] \simeq \mathbb{Z}_\ell \otimes \mathbb{Z}[\zeta_3]$ (resp. $\mathbb{Z}_\ell \otimes \mathbb{Z}[i]$), $\mathrm{End}_{I_q}(V) = \mathbb{Q}_\ell \otimes \mathbb{Q}(\zeta_3)$ (resp. $\mathbb{Q}_\ell \otimes \mathbb{Q}(i)$), which is not a field when $\ell \equiv 1 \pmod 3$ (resp. $\pmod 4$)) (important_notes.md point 12). For I_0^*: $\rho \otimes \chi_{q^*}$ is unramified at q and equals $\rho_{E^{(q)},\ell}$, $E^{(q)}$ having good reduction. For III, III* the line is spanned by τ itself, which is diagonal only over $\mathbb{Q}_\ell(i)$ (important_notes.md point 34). Proof: order from Kraus (2.3(iv)); cyclic and tame (Chapter 7, 5.6), generated by g with $g^m = 1$, $\det g = 1$, so $g = -1$, σ, τ, $-\sigma$ (Chapter 4, 9.3); the commutant and its trace-zero line (Chapter 4, 9.2; Chapter 6, 4.10); Frob g $\mathrm{Frob}^{-1} = g^q$ (Chapter 7, 5.6; Chapter 6, 3.12), so conjugation acts on $2\sigma + 1$ by $+1$ iff $q \equiv 1 \pmod 3$ ($2\sigma^2 + 1 = -(2\sigma + 1)$) and on τ by $+1$ iff $q \equiv 1 \pmod 4$; the geometric Frobenius gives the same sign because $q^{-1} \equiv q \pmod{12}$; $\rho|_{I_q} = \chi_{q^*}|_{I_q}$ in the I_0^* case because I_q has a unique character of order 2. This is the "inertia images and invariants" part of Lemma lem:euler; the factors $L_q(A_f, s)$ and $e_\ell(E, q)$ are computed from it in Chapter 19 (and the shapes, conductors and L-factors of the Weil-Deligne side are Chapter 7, 9.2, 9.4). Reference: Kraus90; Serre-Tate 1968, Section 2; Chapters 4, 6, 7. Paper: Lemma lem:euler (all additive rows), Lemma lem:IV, Remark rem:types, the table of Section 1. Verified: for each type, 12 curves of conductor below 10,000 with surjective mod 3 image and a prime $q \geq 5$ of that type: the ramification index of q in the 3-division field and the existence of an inertia-fixed 3-torsion point agree with the table and with 4.5 (Part 6.6).

4.2 `rem:galrep:kraus-from-components` (remark; P given 2.2 and 2.3, as an exercise). At $q \geq 5$ the dictionary "type $\mapsto$ order of $\rho(I_q)$" of 4.1 also follows from the component groups alone: for g of order $m \in \{2, 3, 4, 6\}$ in $\mathrm{SL}_2(\mathbb{Z}_\ell)$, the fixed spaces of $\bar{g}$ on $E[2]$ and $E[3]$ are: $m = 2$: $E[2]$ and 0; $m = 3$: 0 and a

line; $m = 4$: a line and 0; $m = 6$: 0 and 0 (reduce g modulo 2 and 3; $\tau \not\equiv 1 \pmod 2$ because $(1+2X)^2 = -1$ has no solution in $M_2(\mathbb{Z}_2)$). Comparing with $\Phi_q^{\mathrm{geom}}[2]$, $\Phi_q^{\mathrm{geom}}[3]$ via 2.2 and the independence of ℓ in 2.3(i) gives $\mathrm{I}_0^* \mapsto 2$ ($\Phi_q^{\mathrm{geom}}[2] = (\mathbb{Z}/2)^2$), $\mathrm{IV}, \mathrm{IV}^* \mapsto 3$, $\mathrm{III}, \mathrm{III}^* \mapsto 4$, $\mathrm{II}, \mathrm{II}^* \mapsto 6$. This is a second route to Kraus's table that uses only the component groups; present it as Exercise 7, not as the proof. Paper: background for Lemma lem:IV and Remark rem:types.

4.3 `lem:galrep:IV` (lemma, the paper's Lemma lem:IV; P, two proofs). Let $q \geq 5$ be a prime at which E has type IV or IV*. Then the image of I_q in $\mathrm{Aut}(T_3E)$ is cyclic of order 3, its image in $\mathrm{Aut}(E[3])$ is a nontrivial unipotent subgroup, $E[3]^{I_q}$ is a line, $\bar\rho_{E,3}$ is tamely ramified at q, and the Serre conductor of $\bar\rho_{E,3}$ has exponent 1 at q. Proof A (the paper's, every step expanded): Kraus gives order 3 (2.3(iv)); tame as $3 \neq q$, hence cyclic, generated by σ with $\det\sigma = \chi_3^{\mathrm{cyc}}(\sigma) = 1$ (I_q fixes μ_3), so by Cayley-Hamilton $\sigma^2 + \sigma + 1 = 0$ on T; T is a module over $\mathbb{Z}_3[X]/(X^2+X+1) \simeq \mathbb{Z}_3[\zeta_3]$ with X acting as σ (important_notes.md point 13), a DVR with uniformiser $\pi = 1 - \zeta_3$, residue field $\mathbb{F}_3$ (Chapter 6, 4.7); torsion-free of $\mathbb{Z}_3$-rank 2, hence free of rank one (Chapter 6, 4.8); $E[3] = T/3T \simeq \mathbb{Z}_3[\zeta_3]/(\pi^2)$ with $\sigma - 1$ acting as $-\pi$, nonzero with square zero: a nontrivial unipotent with fixed line $\pi\mathbb{Z}_3[\zeta_3]/(\pi^2)$; Serre exponent $\dim E[3] - \dim E[3]^{I_q} + \mathrm{sw} = 2 - 1 + 0 = 1$. Proof B (component groups; works for every prime $q \neq 3$): $E[3]^{I_q} \simeq \Phi_q^{\mathrm{geom}}[3] = \mathbb{Z}/3$ (2.2(i)) is a line; I_q acts nontrivially on $E[3]$ (otherwise $\Phi_q^{\mathrm{geom}}[3] \supseteq (\mathbb{Z}/3)^2$); the elements of $\mathrm{SL}_2(\mathbb{F}_3)$ fixing a nonzero vector form a group U of order 3 (Chapter 1, 5.6), so $\bar\rho(I_q) = \mathrm{U}$; $\rho(I_q)$ is finite (potentially good: types IV, IV* have $v_q(j) \geq 0$, Chapter 10) and injects into $\mathrm{GL}_2(\mathbb{F}_3)$ (2.3(ii)), so $\rho(I_q)$ is cyclic of order 3, tame. Reference: Kraus90; Chapter 6, 4.7, 4.8; Chapter 4, 9.4. Paper: Lemma lem:IV; Theorem thm:qual (a), (b) ("By Lemma lem:IV the Serre conductor $M = N(\bar\rho)$ has exponent exactly 1 at q, so $M \mid N/q$"). Example: 121c1 (type IV at 11) has $N(\bar\rho_{121c1,3}) = 11$ and $\bar\rho_{121c1,3} \simeq \bar\rho_{11a1,3}$ (6.3).

4.4 `prop:galrep:IV-at-2` (proposition; P; new, settles important_notes.md point 43). Let $E/\mathbb{Q}$ have Kodaira type IV or IV* at 2. Then: (i) for every odd prime ℓ, $\rho_{E,\ell}(I_2)$ is cyclic of order 3, hence tamely ramified, and $E[3]^{I_2}$ is a line; (ii) $f_2(E) = 2$, i.e. $4 \parallel N$, and $v_2(\Delta_{\min}) = 4$ (type IV) or 8 (type IV*), by Ogg's formula; (iii) the Serre conductor of $\bar\rho_{E,3}$ has exponent 1 at 2; (iv) $(\mathrm{ad}^0 V)^{I_2} = \mathbb{Q}_\ell(2\sigma+1)$ and Frob_2 acts on it by $\chi_{-3}(2) = -1$ (it inverts σ since $2 \equiv -1 \pmod 3$), so $L_2(A_f, s)^{-1} = 1 + 2^{-s}$, $L_2(\mathrm{Sym}^2 f, s)^{-1} = 1 + 2 \cdot 2^{-s}$, i.e. $P_2(X) = 1 + 2X$, and $e_\ell(E,2) = v_\ell(1 + \frac{1}{2}) = v_\ell(3/2)$, which is 1 for $\ell = 3$ and 0 for $\ell \geq 5$. Proof: Proof B of 4.3 at $q = 2$ gives (i) for $\ell = 3$; 2.3(i) transfers the group to every odd ℓ; (ii) $f_2 = \mathrm{codim}\, V^{I_2} + \mathrm{sw} = 2 + 0$ and Ogg $f = v(\Delta) + 1 - m$ with $m = 3$, 7 components; (iii) as in 4.3; (iv) as in 4.1, the order-3 image being a quotient of tame inertia on which Frob_2 acts by squaring (Chapter 7, 5.4, 5.6), and Chapter 19's normalisation $L_2(\mathrm{Sym}^2 f, s) = L_2(A_f, s-1)$. Consequences for the paper: the case excepted in Proposition prop:forms ($\ell = 3$, type IV or IV* at 2) is covered, so (eq:main) and (eq:twist) hold under the hypothesis of Theorem thm:main with no exception; "PARI's factor at 2 is $1 + 2X$ for every one of them" and the value $e_3 = 1$ of Remark rem:wild are theorems; and the argument of Theorem thm:qual (a) applies verbatim at $q = 2$ (with $3 \nmid N$), the Serre conductor having exponent 1 at 2 while $4 \parallel N$. Verified: all 4,750 curves of conductor below 10,000 with type IV (2,094) or IV* (2,656) at 2 have $\bar\rho_{E,3}(I_2)$ of order 3 with a fixed line, $f_2 = 2$, $v_2(\Delta_{\min}) = 4$, resp. 8, and $v_2(j) \geq 0$; 60 of them sampled with `lfunsympow`: $P_2(X) = 1 + 2X$ every time. Reference: Chapter 10 (types, Ogg), 2.2, 2.3. Paper: Proposition prop:forms, Remark rem:wild, Section sec:wild, Theorem thm:qual (a), Definition def:terms.

4.5 `prop:galrep:mod3-inertia` (proposition and table; P; the paper's Remark rem:types in corrected, complete form). Let $q \geq 5$ and $\ell = 3$. The image $\bar\rho_{E,3}(I_q)$, the fixed space, the Serre exponent $a_q(\bar\rho_{E,3})$ and the effect of the twist by χ_{q^*} (which exchanges the types as in Lemma lem:twist (ii), Chapter 10) are:

type at q	$\bar\rho(I_q)$ generated by	order	$\dim E[3]^{I_q}$	$a_q(\bar\rho_{E,3})$	$f_q(E)$	$a_q(\bar\rho_{E^{(q)},3})$
I_0	1	1	2	0	0	2
I_n, $3 \nmid n$	unipotent $\neq 1$	3	1	1	1	2
I_n, $3 \mid n$	1	1	2	0	1	2

type at q	$\bar\rho(I_q)$ generated by	order	$\dim E[3]^{I_q}$	$a_q(\bar\rho_{E,3})$	$f_q(E)$	$a_q(\bar\rho_{E^{(q)},3})$
I_n^*, $3\nmid n$	$-$(unipotent $\neq 1$)	6	0	2	2	1
I_n^*, $3\mid n$	-1	2	0	2	2	0
I_0^*	-1	2	0	2	2	0
II, II*	$-$(unipotent $\neq 1$)	6	0	2	2	1
III, III*	$\bar\tau$, $\bar\tau^2=-1$ (X^2+1 irreducible over $\mathbb{F}_3$)	4	0	2	2	2
IV, IV*	unipotent $\neq 1$	3	1	1	2	2

So: (a) $E[3]^{I_q}=0$ and $a_q(\bar\rho_{E,3})=2=f_q$ for **every** additive type other than IV, IV* (including I_0^*, III, III*), and the vanishing of $E[3]^{I_q}$ holds also at $q=2$ (2.2); (b) the twist by χ_{q^*} lowers the Serre exponent of $\bar\rho_{E,3}$ exactly for II, II*, I_n^* and I_0^*, and lowers the conductor exponent of E itself exactly for I_n^* and I_0^* (II $\leftrightarrow$ IV* keeps $f_q=2$). The paper's remark lists (a) only for "the last two cases" (II, II*, I_n^*) and omits I_0^* from (b); its phrase "lowers the conductor exponent" is right for $\bar\rho_{E,3}$, not for E, in the II case (important_notes.md point 22). For $\ell\geq 5$ ($q\geq 5$, $q\neq\ell$): every additive type has $E[\ell]^{I_q}=0$ and $a_q(\bar\rho_{E,\ell})=2$, and I_n has exponent 0 or 1 according as $\ell\mid n$ or not. Proof: reduce the generators of 4.1, 3.2, 3.4 modulo 3 (Chapter 4, 9.4; reduction is injective on finite subgroups, and a unipotent $\left(\begin{smallmatrix}1&n\\0&1\end{smallmatrix}\right)$ is trivial mod 3 iff $3\mid n$); or read $\dim E[3]^{I_q}$ off 2.2; exponents from 6.2. Verified on samples of every row (Part 6.6). Paper: Remark rem:types, Theorem thm:qual (why only (a), (b), (c)), Lemma lem:IV, the optimal-level paragraph.

4.6 `rem:galrep:wild-inertia` (remark; R/S). At $q=2,3$ the type does not determine $\rho(I_q)$ (2.3(iii)): at $q=2$ the image can be Q_8 or $\mathrm{SL}_2(\mathbb{F}_3)$ (wild, absolutely irreducible, so $(\mathrm{ad}^0 V)^{I_2}=0$ and the factor is 1), cyclic of order 4 or 6 with a wild part, or cyclic of order 3 (only for types IV, IV*, by 4.4; the converse need not hold); at $q=3$ an image of order 3 is wild. What survives at every $q\neq\ell$: 2.2 ($E[\ell]^{I_q}\simeq\Phi_q^{\mathrm{geom}}[\ell]$ for additive reduction) and 6.2. Observed at $q=2$ (conductor below 10,000, samples of 40 per type, surjective mod 3 image): for II, II*, III, III*, I_0^* and potentially good I_m^* the least common multiple of the ramification indices of 2 in the fields of definition of the 3-torsion points is 2, 4, 6 or 8 (image of order 2, 4, 6, or 8 or 24); always $E[3]^{I_2}=0$. The Euler factors at 2 and 3 are PARI's (Chapter 19, 22); Remark rem:wild itself is Chapter 10's. Paper: Remark rem:wild, Section sec:wild, Definition def:terms (I_n^* at 2).

13.3.5 Section 5. The representation at ℓ when E has good reduction at ℓ

5.1 `thm:galrep:good-at-l` (theorem; P for (i) given Chapter 10, R for (ii), (iii)). Let $\ell\nmid N$. (i) For every k, $E[\ell^k]$ is the generic fibre of the finite flat group scheme $\mathcal{E}[\ell^k]$ over $\mathbb{Z}_\ell$ ($\mathcal{E}$ the Neron model, an elliptic curve over $\mathbb{Z}_\ell$); so $\bar\rho|_{D_\ell}$ is "finite" (flat), "good" in the sense of DDT Theorem 3.15. (ii) $V_\ell E|_{G_{\mathbb{Q}_\ell}}$ is crystalline with Hodge-Tate weights 0 and 1, and the crystalline Frobenius on D_{cris} has characteristic polynomial $X^2-a_\ell X+\ell$ (Fontaine; see Berger's survey; Chapter 18 uses this). (iii) The Serre weight of $\bar\rho_{E,\ell}$ (when irreducible) is 2 (Serre87, Section 2, verify; Edixhoven92). If E has multiplicative reduction at ℓ the weight is 2 or $\ell+1$ according as $\ell\mid v_\ell(\Delta)$ or not (stated for orientation only; the paper never needs $\ell\mid N$ in a proof). Reference: SilvermanAEC VII; Serre87; Edixhoven92; Berger. Paper: the optimal-level paragraph ("the restriction to D_ℓ comes from the finite flat group scheme $E'[\ell]$ and is good"), Theorem thm:main (crystalline at ℓ, through DFG04; Chapter 18).

5.2 `def:galrep:fundamental-characters` (definition and proposition; P given Chapter 7, 5.3). The tame inertia I_ℓ^{t} of $\mathbb{Q}_\ell$; $\omega=\omega_1\colon I_\ell\to\mu_{\ell-1}\subset\mathbb{F}_\ell^\times$, $\sigma\mapsto\sigma(\ell^{1/(\ell-1)})/\ell^{1/(\ell-1)}$ mod $\mathfrak{m}$, equal to $\bar\chi_\ell|_{I_\ell}$; $\omega_2\colon I_\ell\to\mathbb{F}_{\ell^2}^\times$, $\sigma\mapsto\sigma(\ell^{1/(\ell^2-1)})/\ell^{1/(\ell^2-1)}$ mod $\mathfrak{m}$, of order ℓ^2-1, with $\omega_2^{\ell+1}=\omega$; conjugation by Frob_ℓ sends ω_2 to ω_2^ℓ, so ω_2 does not extend to $G_{\mathbb{Q}_\ell}$, and $\mathrm{Ind}_{G_{\mathbb{Q}_{\ell^2}}}^{G_{\mathbb{Q}_\ell}}$ of an extension of ω_2 is irreducible. Reference:

Serre72, Section 1 (fundamental characters; verify the numbering). Paper: Lemma lem:old (case $p = \ell$, "the restriction to D_ℓ irreducible").

5.3 `thm:galrep:deligne-fontaine` (theorem; S for E, R in general). Let ℓ be odd and $\ell \nmid N$. (i) If $\ell \nmid a_\ell$ (ordinary):

$$\bar\rho|_{D_\ell} \simeq \begin{pmatrix} \bar\chi_\ell\,\mu(a_\ell^{-1}) & * \\ 0 & \mu(a_\ell) \end{pmatrix}, \qquad \bar\rho|_{I_\ell} \simeq \begin{pmatrix} \omega & * \\ 0 & 1 \end{pmatrix},$$

where $\mu(x)$ is the unramified character with $\mathrm{Frob}_\ell \mapsto x \bmod \ell$ (connected-etale sequence of $\mathcal{E}[\ell]$: the etale quotient is $\tilde E[\ell]$, on which Frob_ℓ acts by the unit root of $X^2 - a_\ell X + \ell$, congruent to a_ℓ). (ii) If $\ell \mid a_\ell$ (supersingular; then $a_\ell = 0$ for $\ell \geq 5$ by Hasse): $\bar\rho|_{I_\ell} \otimes \mathbb{F}_{\ell^2} \simeq \omega_2 \oplus \omega_2^\ell$ and $\bar\rho|_{D_\ell}$ is irreducible (Serre72, Section 1, the formal group of height 2; verify the proposition number). (iii) Hence $a_\ell \bmod \ell$ is determined by $\bar\rho|_{D_\ell}$: it is 0 if $\bar\rho|_{D_\ell}$ is irreducible, and otherwise it is the eigenvalue of Frob_ℓ on the unique unramified one-dimensional quotient (unique because $\omega \neq 1$ for ℓ odd). The same holds for a newform h of weight 2, level prime to ℓ, coefficients in $\mathcal{O}$, with $a_\ell(h) \bmod \lambda$ (Deligne in the ordinary case, Fontaine in the supersingular case; Edixhoven92, Section 2, as cited by the paper; the newform version is Chapter 15's). Reference: Serre72, Section 1; Edixhoven92, Section 2. Paper: Lemma lem:old (case $p = \ell$: "the residue class of a_ℓ is determined by the restriction to D_ℓ"). Example: $a_3(121\mathrm{c}1) = 2 \equiv -1 = a_3(11\mathrm{a}1) \pmod 3$, both ordinary at 3, as $\bar\rho_{121c1,3} \simeq \bar\rho_{11a1,3}$ requires.

5.4 `cor:galrep:ordinary` (corollary; P given 5.3). Let ℓ be odd and $\ell \nmid N$. (i) If $\bar\rho_{E,\ell}$ is reducible, E is ordinary at ℓ and the two characters of $\bar\rho^{\mathrm{ss}}$ restrict to I_ℓ as ω and 1. (ii) If $\bar\rho \simeq \bar\rho \otimes \chi_{\ell^*}$, then $\ell = 3$ and E is ordinary at 3. Proof of (ii): $\chi_{\ell^*}|_{I_\ell} = \omega^{(\ell-1)/2}$ (Chapter 5, galois:sqrt-lstar); in the ordinary case $\{\omega, 1\} = \{\omega^{(\ell+1)/2}, \omega^{(\ell-1)/2}\}$ as multisets of characters of I_ℓ forces $\omega^{(\ell+1)/2} = 1$, i.e. $\ell - 1 \mid (\ell+1)/2$, i.e. $\ell = 3$; in the supersingular case $\omega^{(\ell-1)/2} = \omega_2^{(\ell^2-1)/2} = -1$ and $\{\omega_2, \omega_2^\ell\} = \{-\omega_2, -\omega_2^\ell\}$ would need $\omega_2^{\ell-1} = -1$, i.e. $\ell + 1 \mid (\ell-1)/2$, impossible. Verified: every curve of conductor below 10,000 with a Borel or split Cartan label at $\ell \in \{3, 5, 7, 13, 37\}$, $\ell \nmid N$, has $\ell \nmid a_\ell$ (4,571 pairs); the 62 curves with image 3Ns and $3 \nmid N$ all have $3 \nmid a_3$. Paper: Section sec:eis (background), the hypothesis of Theorem thm:main (8.2, 8.6).

13.3.6 Section 6. Conductors of mod ℓ representations

6.1 `def:galrep:serre-conductor` (definition and proposition; P). For a continuous $\bar\sigma\colon G_{\mathbb{Q}} \to \mathrm{GL}_n(\bar{\mathbb{F}}_\ell)$ and $p \neq \ell$: the Artin exponent $a_p(\bar\sigma) = \dim \bar\sigma - \dim \bar\sigma^{I_p} + \mathrm{sw}_p(\bar\sigma)$ (the definitions of Chapter 7, 8.3(v), applied to the finite image), and the Serre conductor $N(\bar\sigma) = \prod_{p \neq \ell} p^{a_p(\bar\sigma)}$, prime to ℓ by definition. Properties: $a_p = 0$ iff $\bar\sigma$ is unramified at p (so $N(\bar\sigma)$ is prime to every prime at which $\bar\sigma$ is unramified); invariant under extension of scalars and under twisting by characters unramified at p; additive in short exact sequences only for the sw part. Serre weight $k(\bar\sigma)$: definition by reference (Serre87, Section 2; Edixhoven92). Reference: Serre87, Section 1.2 (verify). Paper: the optimal-level paragraph ("$N(\bar\sigma)$ is the Serre conductor"), Lemma lem:IV, Remark rem:types, Theorem thm:qual.

6.2 `prop:galrep:serre-conductor-E` (proposition; P given 2.2, 3.3, 3.5 and Ogg-Saito). Let $p \neq \ell$. Then

$$a_p(\bar\rho_{E,\ell}) = f_p(E) - \dim_{\mathbb{F}_\ell} \Phi_p(\bar{\mathbb{F}}_p)[\ell].$$

Hence $N(\bar\rho_{E,\ell})$ divides the prime-to-ℓ part of N, with $v_p(N(\bar\rho)) = v_p(N)$ unless $\ell \mid \#\Phi_p(\bar{\mathbb{F}}_p)$; explicitly: 0 at good p; 0 or 1 at multiplicative p according as $\ell \mid n_p$ or not; at additive p and ℓ odd, $f_p - 1$ if $\ell = 3$ and the type is IV or IV*, and f_p otherwise; in particular $\bar\rho_{E,\ell}$ (ℓ odd) is ramified at every additive prime $p \neq \ell$. Proof: the wild inertia P_p acts on T through a finite p-group (Chapter 7, 5.7), of order prime to ℓ, so the fixed spaces of its ramification subgroups commute with reduction (Chapter 4, 3.2(ii)) and $\mathrm{sw}_p(\bar\rho) = \mathrm{sw}_p(V)$; then $f_p = 2 - \delta_p + \mathrm{sw}_p(V)$ (3.5) and $\dim E[\ell]^{I_p} = \delta_p + \dim \Phi_p^{\mathrm{geom}}[\ell]$ (2.2, 3.3). Reference: Carayol89 (divisibility, for modular forms); Livne 1989, "On the conductors of mod ℓ Galois representations coming from modular forms", J. Number Theory 31 (add to `bib/ch11.bib` after checking); Serre87, Section 4 (examples for elliptic curves; verify). Paper: the optimal-level paragraph ("$N(\bar\rho_{E',\ell})$ divides the conductor of E' [Carayol89]"), Theorem thm:qual (a), (b), (c) (the level M), Lemma lem:IV, Remark rem:types, Lemma lem:old (hypothesis "$p \nmid M$ for every $p \,\|\, N$ at which $\bar\rho$ is unramified").

6.3 ex:galrep:serre-conductor-examples (worked example; P, all data recomputed). $\ell = 3$: 11a1 (I_5): $N(\bar\rho) = 11$. 121c1 (IV at 11, $f_{11} = 2$): $N(\bar\rho) = 11$, and $a_p(121c1) \equiv a_p(11a1) \pmod 3$ for all $p < 2000$, $p \neq 3, 11$: $\bar\rho_{121c1,3} \simeq \bar\rho_{11a1,3}$. 121a1 (II at 11): $N(\bar\rho) = 121$; its twist by χ_{-11} is 121c2 (IV^*): $N = 11$; $a_p(121a1) \equiv \chi_{-11}(p)a_p(11a1) \pmod 3$ ($p < 2000$). 77c1 (I_3 at 7, I_2 at 11): $N(\bar\rho) = 11$, $\bar\rho_{77c1,3} \simeq \bar\rho_{11a1,3}$ ($p < 2000$). 539c1 (I_3^* at 7): $v_7(N(\bar\rho)) = 2$; twist by χ_{-7} gives 77c1. 726a1 (I_6 at 2, I_3 at $3 = \ell$, III at 11): $N(\bar\rho) = 11^2$. 6912d1 (III at 2 with $f_2 = 8$, II at $3 = \ell$): $N(\bar\rho) = 2^8$. $\ell = 5$: 11a1: $N(\bar\rho_{11a1,5}) = 1$ ($5 \mid n_{11}$; $E[5] \simeq \mathbb{Z}/5 \oplus \mu_5$). 980i1 ($\mathrm{IV}^*$ at 2, I_1 at $5 = \ell$, I_5^* at 7): twist by χ_{-7} is 140b1 (I_5 at 7), $N(\bar\rho_{140b1,5}) = 4$, and $a_p(140b1) \equiv a_p(20a1)$, $a_p(980i1) \equiv \chi_{-7}(p)a_p(20a1) \pmod 5$ ($p < 1224$, $p \neq 2, 5, 7$); but $5 \mid N$ here, so this illustrates the mechanism, not Theorem thm:qual (important_notes.md point 21). Paper: the Example of Section 3, Theorem thm:qual.

13.3.7 Section 7. The images of $\bar\rho_{E,\ell}$

7.1 thm:galrep:open-image (theorem; R). (i) (Serre) If $E/\mathbb{Q}$ has no CM, $\rho_{E,\ell}(G_\mathbb{Q})$ is open in $\mathrm{GL}_2(\mathbb{Z}_\ell)$ for every ℓ, and $\bar\rho_{E,\ell}$ is surjective for all but finitely many ℓ. (ii) If E has CM by an order in K and ℓ is unramified in K, the image of $\bar\rho_{E,\ell}$ lies in the normaliser of a Cartan subgroup, split or nonsplit according as ℓ splits or is inert in K, and is not contained in the Cartan subgroup itself. (iii) (Mazur, Chapter 9, 9.6) A Borel image occurs only for $\ell \in \{2, 3, 5, 7, 11, 13, 17, 19, 37, 43, 67, 163\}$. (iv) Whether $\ell > 37$ forces surjectivity for non-CM E (Serre's uniformity question) is open; say so in one sentence without further claims. Reference: Serre72 (cite sections; the main theorem numbers to be verified); Mazur78. Paper: background for Proposition prop:ns ("without complex multiplication"), Section sec:data ("the images of the mod ℓ Galois representations"), Conjecture conj:eis.

7.2 prop:galrep:image-constraints (proposition; P from Chapter 1, 5.10 and 1.1). Let ℓ be odd and $H = \bar\rho_{E,\ell}(G_\mathbb{Q})$. Then $\det H = \mathbb{F}_\ell^\times$ and H contains an element conjugate to $\mathrm{diag}(1, -1)$. Hence $H \not\subset \mathrm{SL}_2(\mathbb{F}_\ell)$, $H \not\subset C_{ns}(\mathbb{F}_\ell)$ (the only involution of $C_{ns}(\mathbb{F}_\ell) \simeq \mathbb{F}_{\ell^2}^\times$ is -1, of determinant 1), and by Dickson either $H = \mathrm{GL}_2(\mathbb{F}_\ell)$ (if $\ell \mid \#H$ and H is not in a Borel subgroup: then $H \supseteq \mathrm{SL}_2(\mathbb{F}_\ell)$ and $\det H = \mathbb{F}_\ell^\times$), or H lies in a Borel subgroup ($\bar\rho$ reducible), or in the normaliser of a split or nonsplit Cartan subgroup but not in the Cartan subgroup when $\bar\rho$ is irreducible, or H has exceptional projective image A_4, S_4 or A_5. Do not call N_s maximal: $N_s(\mathbb{F}_3) \subset N_{ns}(\mathbb{F}_3)$ and $N_s(\mathbb{F}_5)$ lies in a subgroup with projective image S_4 (important_notes.md point 2). Paper: Proposition prop:ns (first step), background.

7.3 thm:galrep:mod3-images (theorem and table; P for the list, data for occurrence). For **every** $E/\mathbb{Q}$, CM or not, the image $H = \bar\rho_{E,3}(G_\mathbb{Q})$ is conjugate to exactly one of the eight subgroups below; in particular never to $C_{ns}(\mathbb{F}_3)$ (3Cn) or to a subgroup of $\mathrm{SL}_2(\mathbb{F}_3)$. Proof: H has surjective determinant and contains an element of determinant -1 and trace 0 (7.2); by Chapter 1, 6.10 and 6.9 the subgroups of $\mathrm{GL}_2(\mathbb{F}_3)$ with surjective determinant fall into nine conjugacy classes, and exactly one of them, $C_{ns}(\mathbb{F}_3)$, has no such element. The orbit pattern on the four lines is the factorisation pattern of ψ_3 (Chapter 9, 4.5; Chapter 5, galois:quartic-on-lines). Occurrence: the examples. Counts: curves of conductor below 10,000 in `ecdata` (64,687 curves; "CM" are the 462 CM curves, for which `ecdata`'s `galrep` file lists no label when the image is the one forced by CM; those were classified here from ψ_3):

label	order	H (Chapter 1 models)	arithmetic meaning	ψ_3 over $\mathbb{Q}$	$\mathrm{Gal}(\psi_3)$	non-CM	CM	example
(surjective)	48	$\mathrm{GL}_2(\mathbb{F}_3)$	generic	irreducible	S_4	53,760	0	11a1
3Nn	16	$N_{ns}(\mathbb{F}_3)$	irreducible, $\bar\rho \simeq \bar\rho \otimes \chi_K$, $K \neq \mathbb{Q}(\sqrt{-3})$	irreducible	D_4	218	184	245a1

label	order	H (Chapter 1 models)	arithmetic meaning	ψ_3 over $\mathbb{Q}$	$\mathrm{Gal}(\psi_3)$	non-CM	CM	example
3Ns	8	$N_s(\mathbb{F}_3)$	irreducible, $\bar\rho \simeq \bar\rho \otimes \chi_{-3}$	two quadratics	V_4	120	26	726a1, 338d1, 6912d1
3B	12	$\mathrm{B}(\mathbb{F}_3)$	stable line, character $\neq 1, \chi_{-3}$	linear times cubic	S_3	4,072	(154 unlabelled Borel)	50b1, 539a1
3B.1.1	6	$\left(\begin{smallmatrix}1 & *\\ 0 & *\end{smallmatrix}\right)$	rational point of order 3, $E[3]$ non-split	linear times cubic	S_3	2,772	39	20a1, 14a4
3B.1.2	6	$\left(\begin{smallmatrix}* & *\\ 0 & 1\end{smallmatrix}\right)$	$\mu_3 \subset E[3]$, $E[3]/\mu_3 \simeq \mathbb{Z}/3$, non-split	linear times cubic	S_3	2,772	39	20a3, 14a3
3Cs	4	$C_s(\mathbb{F}_3)$	$E[3] \simeq \psi \oplus \psi\chi_{-3}$, ψ quadratic $\neq 1, \chi_{-3}$	two linear times quadratic	C_2	249	18	539a2, 98a3
3Cs.1.1	2	$\{\mathrm{diag}(1, \pm 1)\}$	$E[3] \simeq \mathbb{Z}/3 \oplus \mu_3$	two linear times quadratic	C_2	262	2	14a1, 19a1, 26a1

(The CM column: 26 curves with $j = -32768$ or 8000 give 3Ns, 184 give 3Nn, 154 with $j \in \{0, 54000, -12288000\}$ have an unlabelled Borel image, and the CM curves with labels are counted in their rows.) Remarks to make: 3B.1.1 and 3B.1.2 are exchanged by 3-isogeny and by the twist by χ_{-3} (1.3), and occur in equal numbers; the curves with a rational point of order 3 are exactly those labelled 3B.1.1 or 3Cs.1.1 (2,811 + 264, a data check of the labels); 3Cs.1.1 is **not** the split Cartan subgroup but its subgroup of order 2 (important_notes.md point 44); $N_s(\mathbb{F}_3) \subset N_{ns}(\mathbb{F}_3)$ (point 2); 3Nn is also dihedral (induced from the field cut out by $\bar\rho^{-1}(C_{ns})$), which is why the hypothesis of Theorem thm:main excludes "induced from $\mathbb{Q}(\sqrt{-3})$", not "dihedral" (Chapter 4, 8.3). Reference: Chapter 1; Zywina15 (the list for non-CM curves; the book proves it directly); Sutherland16 (labels). Paper: Proposition prop:ns, Section sec:anomaly, Section sec:eis, Conjecture conj:eis, Section sec:data, the appendix (`ns3`).

7.4 `conv:galrep:labels` (convention; R for the general scheme, data for the check). Sutherland's label of a subgroup of $\mathrm{GL}_2(\mathbb{F}_\ell)$: ℓ, then the type of the smallest standard overgroup (B, Cs, Cn, Ns, Nn, or the exceptional A4, S4, A5), then optional numerical suffixes identifying the subgroup; surjective images carry no label. Give the general rule by reference only (Sutherland16, Section 6; verify), and use the intrinsic descriptions of 7.3 in every statement (important_notes.md point 3: the labels were checked here against `ecdata`, which uses Sutherland's labels: 3B.1.1 is exactly the rational-3-torsion case, 3B.1.2 its isogenous partner, 3Cs.1.1 the case $E[3] \simeq \mathbb{Z}/3 \oplus \mu_3$, 3Ns the $2 + 2$ factorisation, 3Nn the irreducible ψ_3 with group D_4; and 5B.1.1 for 11a3, torsion $\mathbb{Z}/5$, 5B.1.2 for 11a2, 5Cs.1.1 for 11a1, $E[5] \simeq \mathbb{Z}/5 \oplus \mu_5$). Labels occurring at $\ell = 5, 7, 11, 13$ for conductor below 10,000 (from `galrep`; the writer may print this as a table): $\ell = 5$: 5B, 5B.1.1, 5B.1.2, 5B.1.3, 5B.1.4, 5B.4.1, 5B.4.2, 5Cs.1.1, 5Cs.1.3, 5Cs.4.1, 5Nn,

5Ns, 5Ns.2.1, 5S4; $\ell = 7$: 7B, 7B.1.1 to 7B.1.6, 7B.2.1, 7B.2.3, 7B.6.1 to 7B.6.3, 7Ns, 7Ns.2.1, 7Ns.3.1, 7Ns.6.1.2; $\ell = 11$: 11B.1.3 to 11B.1.8, 11B.10.3 to 11B.10.5, 11Nn.1.4; $\ell = 13$: 13B, 13B.3.1, 13B.3.2, 13B.4.1, 13B.4.2, 13B.5.1, 13B.5.2, 13Ns.5.1.4. Paper: Section sec:anomaly (3Ns), Section sec:eis (3Cs.1.1, 3B.1.1), Section sec:data.

7.5 `thm:galrep:zywina` (theorem; R). For non-CM $E/\mathbb{Q}$ and $\ell \in \{2, 3, 5, 7, 11, 13\}$, Zywina15 lists the subgroups of $\mathrm{GL}_2(\mathbb{F}_\ell)$ that occur as $\bar\rho_{E,\ell}(G_\mathbb{Q})$, up to conjugacy, together with the corresponding families of j-invariants (for $\ell = 13$ the list depended on rational points of certain modular curves; say only that later work settled them, without a reference unless one is checked). The book needs only $\ell = 3$, which 7.3 proves. Reference: Zywina15, Section 1 (theorem numbers not checked, important_notes.md point 4). Paper: Proposition prop:ns (the citation of Zywina15), Section sec:data.

7.6 `rem:galrep:computing-images` (remark; S). How images are computed: (a) $\ell = 3$: factor ψ_3 (PARI `elldivpol(E,3)`, `factor`), compute $\mathrm{Gal}(\psi_3)$ (`polgalois`) and the 3-torsion (`elltors`); 7.3 then determines the image (the appendix's `ns3` tests the $2 + 2$ pattern, which detects $N_s(\mathbb{F}_3)$ exactly when $E[3]$ is irreducible); (b) in general, characteristic polynomials of $\bar\rho(\mathrm{Frob}_p)$ modulo ℓ for many p and Chebotarev statistics (Sutherland's algorithm, R: Sutherland16), division polynomials, and Zywina's j-invariant families; (c) the `galrep` files of `ecdata`. Verified here: for all 64,687 curves of conductor below 10,000 the pattern of ψ_3 and $\mathrm{Gal}(\psi_3)$ agree with the `galrep` label as in 7.3. Paper: Section sec:data ("The mod 3 images of the tables were checked against the factorisation of the 3-division polynomial"), the appendix (`ns3`), Section sec:anomaly; Chapter 22.

13.3.8 Section 8. Absolute irreducibility over $\mathbb{Q}(\sqrt{\ell^*})$, Proposition prop:ns and Lemma lem:h0

8.1 `prop:galrep:dihedral` (proposition; P from Chapter 4, 6.6; Chapter 5, galois:index-two-restriction; Chapter 6, 6.4). Let ℓ be odd, $\bar\rho = \bar\rho_{E,\ell}$ irreducible and K a quadratic field. The following are equivalent: (a) $\bar\rho|_{G_K}$ is not absolutely irreducible; (b) $\bar\rho \simeq \bar\rho \otimes \chi_K$ (over $\mathbb{F}_\ell$, equivalently over $\bar{\mathbb{F}}_\ell$); (c) $\bar\rho \otimes \bar{\mathbb{F}}_\ell \simeq \mathrm{Ind}_{G_K}^{G_\mathbb{Q}} \psi$ for a character ψ of G_K; (d) $a_p(E) \equiv 0 \pmod\ell$ for all (equivalently almost all) primes $p \nmid N\ell$ inert in K. When they hold, every isomorphism $\bar\rho \to \bar\rho \otimes \chi_K$ has trace 0 and $H^0(\mathbb{Q}, \mathrm{ad}^0\,\bar\rho \otimes \chi_K)$ is a line. For $K = \mathbb{Q}(\sqrt{\ell^*})$ condition (a) is the failure of the hypothesis of Theorem thm:main. Paper: Theorem thm:main hypothesis, S_f (DFG04 (31)), Theorem thm:qual, Proposition prop:ns, Lemma lem:h0.

8.2 `thm:galrep:lstar-automatic` (theorem; P given 5.3; new in this form, see Part 7). Let ℓ be odd, $\ell \nmid N$, and $\bar\rho_{E,\ell}$ irreducible. If $\ell \geq 5$, then $\bar\rho_{E,\ell}$ is absolutely irreducible on $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{\ell^*}))$. If $\ell = 3$ and $\bar\rho_{E,3}$ is not absolutely irreducible there (image $N_s(\mathbb{F}_3)$), then E is ordinary at 3. Proof: 8.1 and 5.4(ii). Verified: among the curves of conductor below 10,000, those with an ℓNs or ℓNn label, $\ell \geq 5$, for which $a_p \equiv 0 \pmod\ell$ at every $p < 870$, $p \nmid N\ell$, inert in $\mathbb{Q}(\sqrt{\ell^*})$, are 24 curves (5Ns.2.1, 7Ns.2.1, 7Ns.3.1, 7Ns.6.1.2), all with $\ell \mid N$; the 28 curves with 5Ns, 5Nn or 7Ns and $\ell \nmid N$ are dihedral for another quadratic field. Paper: Theorem thm:main (hypothesis), Section sec:forms ("the curves whose mod ℓ representation is induced from $\mathbb{Q}(\sqrt{\ell^*})$ with $\ell \geq 5$, which we did not single out": all of them have $\ell \mid N$), DDT Theorem 3.15 as quoted in Section 3 (its extra hypothesis matters only at $\ell = 3$).

8.3 `prop:galrep:ns` (proposition, the paper's Proposition prop:ns; P, in the corrected form). Let $E/\mathbb{Q}$ (with or without CM: important_notes.md point 9) have $E[3]$ irreducible. Then the image H of $\bar\rho_{E,3}$ is conjugate to $\mathrm{GL}_2(\mathbb{F}_3)$, $N_{ns}(\mathbb{F}_3)$ or $N_s(\mathbb{F}_3)$; $\bar\rho_{E,3}$ is absolutely irreducible on $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{-3}))$ unless $H \sim N_s(\mathbb{F}_3)$; in that case $\bar\rho_{E,3} \otimes \mathbb{F}_9 \simeq \mathrm{Ind}\,\psi$ for a character ψ of $G_{\mathbb{Q}(\sqrt{-3})}$ of order 4 with values in $\mathbb{F}_9^\times$ (the restriction is irreducible but not absolutely irreducible over $\mathbb{F}_3$: important_notes.md point 33), $\bar\rho_{E,3} \simeq \bar\rho_{E,3} \otimes \chi_{-3}$ over $\mathbb{F}_3$, $H^0(\mathbb{Q}, \mathrm{ad}^0\,\bar\rho_{E,3}(1))$ is a line, and $h_3(E) = 1$ (8.5). The 3-division polynomial is a product of two irreducible quadratics in $\mathbb{Q}[x]$ in the N_s case and irreducible in the other two. Proof (the paper's, every step expanded): (1) $\bar\rho(G_{\mathbb{Q}(\sqrt{-3})}) = H \cap \mathrm{SL}_2(\mathbb{F}_3)$ (1.1(ii)); (2) the list (7.3 with $E[3]$ irreducible); (3) determinant-one subgroups $\mathrm{SL}_2(\mathbb{F}_3)$, Q_8, $C_4 = \langle J^{-1}\rangle$ (Chapter 1, 6.7); (4) $\mathrm{SL}_2(\mathbb{F}_3)$ and Q_8 act absolutely irreducibly: "non-abelian and faithful" is not enough (a Borel subgroup is both), the missing step is Chapter 1, 5.7 (commutator subgroups) or Chapter 4, 3.7 (Maschke for Q_8): important_notes.md point 1; (5) C_4 is abelian, $X^2 + 1$ irreducible over $\mathbb{F}_3$, eigenvalues $\pm i \in \mathbb{F}_9$, so $\bar\rho|_{G_{\mathbb{Q}(\sqrt{-3})}} \otimes \mathbb{F}_9 = \psi \oplus \psi^{-1}$ (Chapter 4, 2.5, 8.2); (6) Frobenius reciprocity and Clifford theory (Chapter

4, 6.2, 6.6; Chapter 5, galois:index-two-restriction): $\bar\rho\otimes\mathbb{F}_9\simeq\operatorname{Ind}\psi$ and $\bar\rho\simeq\bar\rho\otimes\chi_{-3}$, the isomorphism descending to $\mathbb{F}_3$ (Chapter 4, 1.4(ii)); (7) H^0 and h_3 by 8.5; (8) lines and roots of ψ_3 (Chapter 9, 4.5; Chapter 1, 6.8). Examples: 726a1: $\psi_3=(x^2-x-74)(3x^2+8x+20)$; 6912d1 ($y^2=x^3-6x-8$, Chapter 5's example): $\psi_3=3(x^2-4x-2)(x^2+4x+6)$; 338d1; 11a1: $\psi_3=3x^4-4x^3-60x^2-237x-21$ irreducible, $\operatorname{disc}=-3^3\cdot11^{10}$, group S_4. Paper: Proposition prop:ns and its proof in Section sec:boundary, Section sec:anomaly, Remark rem:ns, Conjecture conj:euler, the appendix (`ns3`).

8.4 `prop:galrep:ns-structure` (proposition; P; new in this form, see Part 7). Let $E/\mathbb{Q}$ (CM allowed) have $\bar\rho_{E,3}(G_\mathbb{Q})\sim N_s(\mathbb{F}_3)$, and let K_1, K_2 be the splitting fields of the two quadratic factors of ψ_3 (the fixed fields of $\bar\rho^{-1}(C_s)$ and $\bar\rho^{-1}(\{\pm1,\pm w\})$). (i) K_1, K_2, $\mathbb{Q}(\sqrt{-3})$ are the three quadratic subfields of the splitting field of ψ_3, which is biquadratic (Chapter 5, galois:quartic-on-lines (e)). (ii) For every prime $p\neq3$, $\bar\rho(I_p)\subseteq C_4=N_s(\mathbb{F}_3)\cap\mathrm{SL}_2(\mathbb{F}_3)=\langle J\rangle$. Consequently every multiplicative prime $p\neq3$ has $3\mid n_p$ (so $t_3(E,p)\geq1$), every prime $p\geq5$ of type I_n^* has $3\mid n$, no prime $p\geq5$ has type II, II*, and no prime $p\neq3$ has type IV, IV*. (iii) Let R be the set of primes $p\neq3$ ramified in K_1. Then R is also the set of primes $p\neq3$ ramified in K_2, it equals $\{p\neq3:\bar\rho(I_p)=C_4\}$, and it is nonempty (the only quadratic field unramified outside 3 is $\mathbb{Q}(\sqrt{-3})$). For $p\in R$, $\rho_{E,\ell}(I_p)$ is cyclic of order 4 for every odd $\ell\neq p$ (a $\pm$unipotent has order $1,2,3,6$ or ∞, so the reduction is potentially good; then 2.3); if $p\geq5$, p has type III or III*; conversely every prime $p\geq5$ of type III or III* lies in R. (iv) For $p\in R$, Frob_p acts on $(\mathrm{ad}^0V)^{I_p}=\mathbb{Q}_\ell\tau$ by $\chi_{-3}(p)$ ($\bar\rho(\mathrm{Frob}_p)$ commutes with J iff it lies in C_4 iff $\det=p\equiv1\pmod3$), so $L_p(A_f,s)^{-1}=1-\chi_{-3}(p)p^{-s}$ and $e_3(E,p)=v_3(p-\chi_{-3}(p))\geq1$; for $p\geq5$ comparison with 4.1 ($\chi_{-4}(p)$) gives $\chi_{-4}(p)=\chi_{-3}(p)$, i.e. $p\equiv1$ or $11\pmod{12}$; for $p=2$: $P_2(X)=1+2X$, $e_3(E,2)=1$. (v) $a_p(E)\equiv0\pmod3$ for every prime $p\nmid3N$ with $p\equiv2\pmod3$ (8.1(d)). (vi) If $3\nmid N$, E is ordinary at 3 (8.2). (vii) $h_3(E)=1$ (8.5). Verified on all 120 non-CM curves with image 3Ns of conductor below 10,000: for 100 of them R contains primes $p\geq5$, exactly the type III, III* primes ≥5, all $\equiv1$ or $11\pmod{12}$; for the other 20 (classes 3840h, k, o, r, 4864b, c, o, p, 6912d, f, j, l, p, r, v, x) the only prime of R is 2, of type III or III* with $f_2=8$ and $\bar\rho(I_2)$ cyclic of order 4; every multiplicative prime $p\neq3$ has $3\mid n_p$; no type II, II* at $p\geq5$ and no IV, IV* at $p\neq3$; $a_p\equiv0\pmod3$ for $p\equiv2\pmod3$, $p<410$; $3\nmid a_3$ whenever $3\nmid N$; Euler factors $1+11X$ (726a1 at 11), $1-13X$ (338d1 at 13), $1+2X$ (3840h1, 3840h2, 4864b1, 4864c1, 6912d1, 6912j1 at 2). Paper: Section sec:anomaly ("every 3Ns curve of the tables has a prime of type III or III* with a positive term e_3"; "the residue classes … are $q\equiv1$ and $q\equiv11\pmod{12}$ …, with some $q=2$"; "the violation involves a prime $p\equiv-1\pmod3$ with $t_3(E,p)\geq1$"), the appendix ("for 726a1 … the type III term and the correction cancelling"), Remark rem:ns, Conjecture conj:euler.

8.5 `lem:galrep:h0` (lemma, the paper's Lemma lem:h0 in corrected form; P). Let ℓ be odd, $\bar\rho=\bar\rho_{E,\ell}$ irreducible (E with or without CM), $W^D=\mathrm{ad}^0T(1)\otimes\mathbb{Q}_\ell/\mathbb{Z}_\ell$ (Chapter 18's notation; $T^D=\mathrm{Hom}(\mathrm{ad}^0T,\mathbb{Z}_\ell(1))\simeq\mathrm{ad}^0T(1)$ via the trace form for ℓ odd, Chapter 4, 7.6). Then: (i) $H^0(\mathbb{Q},W^D)[\ell^k]=H^0(\mathbb{Q},\mathrm{ad}^0(T/\ell^k)(1))=\mathrm{Hom}_{G_\mathbb{Q}}(T/\ell^k,(T/\ell^k)(1))$, every such map having trace 0 automatically ($\mathrm{End}(T)(1)=\mathrm{ad}^0T(1)\oplus\mathbb{Z}_\ell(1)$ and $H^0(\mathbb{Q},\mu_{\ell^k})=0$); (ii) $H^0(\mathbb{Q},W^D)\simeq\mathbb{Z}/\ell^h$ is finite and cyclic, with $h=h_\ell(E)$ the largest $k\geq0$ such that $\rho\bmod\ell^k\simeq(\rho\otimes\chi_\ell)\bmod\ell^k$ (Chapter 4, 10.5, using absolute irreducibility, 1.4(i)); (iii) $h_\ell(E)=0$ for every $\ell\geq5$, and $h_3(E)\leq1$, with $h_3(E)=1$ iff $\bar\rho_{E,3}\simeq\bar\rho_{E,3}\otimes\chi_{-3}$ iff $\bar\rho_{E,3}$ is not absolutely irreducible on $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{-3}))$ iff the image is $N_s(\mathbb{F}_3)$; (iv) h_ℓ is invariant under quadratic twists and prime-to-ℓ isogenies (1.3). Proof of (iii): an isomorphism modulo ℓ^k gives $\chi_\ell^2\equiv1\pmod{\ell^k}$ on $G_\mathbb{Q}$ by determinants, and $\chi_\ell(G_\mathbb{Q})=\mathbb{Z}_\ell^\times$, so $\ell^k=3$ (Chapter 5, galois:twist (c); Chapter 2, 5.5); for $\ell=3$, $k=1$: $\bar\chi_3=\chi_{-3}$ and 8.1, 8.3. Remarks for the text: (a) the paper's sentence "for E without complex multiplication it is finite" and its reason ("$\rho\simeq\rho\otimes\chi_\ell$ would force $a_p=0$ at all primes inert in $\mathbb{Q}(\sqrt{\ell^*})$") confuse χ_ℓ with χ_{ℓ^*} (important_notes.md point 5); the determinant argument replaces it and needs no CM hypothesis; CM by $\mathbb{Q}(\sqrt{\ell^*})$ forces $\ell\mid N$ anyway (point 14; Chapter 9, 6.5). (b) The criterion for $h_\ell(E)>0$ is $\bar\rho\simeq\bar\rho\otimes\bar\chi_\ell$, not $\bar\rho\simeq\bar\rho\otimes\chi_{\ell^*}$ (point 6); for $\ell\geq5$ the latter can hold only if $\ell\mid N$ (8.2). (c) The numerical check of Section sec:anomaly ("some prime $p<1223$ inert in $\mathbb{Q}(\sqrt{-3})$ has $9\nmid a_p$") is valid but superfluous: $h_3(E)\geq2$ would give $a_p\equiv pa_p\pmod9$ by traces, and $p-1$ is a unit mod 3 for $p\equiv2\pmod3$, so $9\mid a_p$ at every such $p\nmid3N$; the determinant argument of (iii) excludes $h_3(E)\geq2$ for all E at once. (d) "Irreducible" must be read as "absolutely irreducible" in the Clifford step (point 10); for $\bar\rho_{E,\ell}$ they agree (1.4(i)). Paper: Lemma lem:h0, Conjecture conj:euler ($h_\ell(E)$), Section 1.3, Proposition prop:ns, Remark rem:ns, Section sec:anomaly ("$h_3(E)=1$ for every 3Ns curve"), the proof of Theorem

thm:main ($H^0(\mathbb{Q}, W^D) = 0$), the appendix (ns3 as the correction term); Chapter 18 restates it.

8.6 cor:galrep:Sf (corollary; P). Let ℓ be odd and $\ell \nmid N$. The following are equivalent: (a) $\ell \notin S_f$ (DFG04 (31): $\ell \nmid 2N$ and $\bar\rho|_{\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{\ell^*}))}$ absolutely irreducible); (b) $E[\ell]$ is irreducible and $h_\ell(E) = 0$; (c) $E[\ell]$ is irreducible and, if $\ell = 3$, the image of $\bar\rho_{E,3}$ is not $N_s(\mathbb{F}_3)$. So the hypothesis of Theorem thm:main reads "$\ell \nmid 2N$, $E[\ell]$ irreducible, and not ($\ell = 3$ and image 3Ns)", the extra hypothesis of Theorem thm:qual at $\ell = 3$ is "image not 3Ns", and in the range $\ell \nmid N$ the failure of the hypothesis is the vanishing-or-not of $h_\ell(E)$. Proof: 8.1, 8.2, 8.5. Data: 62 curves with image 3Ns and $3 \nmid N$ below conductor 10,000 are the only failures at $\ell = 3$ with $3 \nmid N$ and $E[3]$ irreducible. Paper: Theorem thm:main, Theorem thm:qual, Section 4.2 (S_f), Section sec:forms, Remark rem:ns, Conjecture conj:euler (the two hypotheses "$\ell \notin S_f$" and "$h_\ell = 0$" coincide when $\ell \nmid N$).

13.3.9 Section 9. The Eisenstein case

9.1 prop:galrep:reducible (proposition; P given 2.2, 3.2, 5.3 and Kronecker-Weber, Chapter 6, 5.5). Let ℓ be odd, $\ell \nmid N$, and $E[\ell]$ reducible. (i) $\bar\rho^{\mathrm{ss}} = \psi_1 \oplus \psi_2$ with $\psi_1\psi_2 = \bar\chi_\ell$ (Chapter 9, 5.2(iii)); (ii) E is ordinary at ℓ and $\{\psi_1|_{I_\ell}, \psi_2|_{I_\ell}\} = \{\omega, 1\}$ (5.4(i)); (iii) ψ_1, ψ_2 are unramified at every multiplicative prime ($\bar\rho(I_q)$ is unipotent there, 3.2); (iv) if E is semistable, $\{\psi_1, \psi_2\} = \{1, \bar\chi_\ell\}$, so $a_p \equiv 1 + p \pmod{\ell}$ for all $p \nmid N\ell$, and E or a curve ℓ-isogenous to it has a rational point of order ℓ (the sub is $\mathbb{Z}/\ell$ or μ_ℓ; in the second case E/μ_ℓ contains $E[\ell]/\mu_\ell \simeq \mathbb{Z}/\ell$). Proof of (iv): a character of $G_\mathbb{Q}$ with values in $\bar{\mathbb{F}}_\ell^\times$ unramified outside ℓ factors through $\mathrm{Gal}(\mathbb{Q}(\zeta_{\ell^m})/\mathbb{Q})$ and has order prime to ℓ, hence is a power of $\bar\chi_\ell$ determined by its restriction to I_ℓ. Verified: every curve of conductor below 10,000 with a Borel or split Cartan label at $\ell \in \{3, 5, 7\}$, $\ell \nmid N$, N square-free (1,315, 158, 26 curves) has $a_p \equiv 1 + p \pmod{\ell}$ for all $p < 282$, $p \nmid N\ell$. Reference: Serre72, Section 5 (verify); Mazur77 for the prime-level case. Paper: Section sec:eis, Conjecture conj:eis (torsion as the controlling invariant), Open question 4, background for "Eisenstein".

9.2 prop:galrep:eis-labels (proposition; P, with data). For $\ell = 3$: 3B.1.1 iff E has a rational point of order 3 and $E[3]$ is not split; 3Cs.1.1 iff $E[3] \simeq \mathbb{Z}/3 \oplus \mu_3$; 3B.1.2 iff $\mu_3 \subset E[3]$ with quotient $\mathbb{Z}/3$, non-split; the twist by χ_{-3} exchanges 3B.1.1 and 3B.1.2 (1.3); so $3 \mid \#E(\mathbb{Q})_{\mathrm{tors}}$ iff the label is 3B.1.1 or 3Cs.1.1. Restatement of Chapter 9, 9.8 (important_notes.md point 17): if E has no CM and image 3Cs.1.1, then $E(\mathbb{Q})$ has no point of order 9; so in Conjecture conj:eis $v_3(\#E(\mathbb{Q})_{\mathrm{tors}}) + [E[3] \simeq \mathbb{Z}/3 \oplus \mu_3] \le 2$, and the bound $D \le 2$ follows from the first bound. For $\ell = 5$: 5Cs.1.1 iff $E[5] \simeq \mathbb{Z}/5 \oplus \mu_5$ (11a1), 5B.1.1 iff a rational point of order 5 with $E[5]$ non-split (11a3). Data: 2,811 curves 3B.1.1 and 264 curves 3Cs.1.1 are exactly the curves of conductor below 10,000 with $3 \mid \#E(\mathbb{Q})_{\mathrm{tors}}$; the 264 curves 3Cs.1.1 have torsion of order 3 or 6 (Chapter 9). Paper: Section sec:eis ("the split Cartan image 3Cs.1.1": wrong description, point 44; "We have not proved that the two cannot coexist": point 17), Conjecture conj:eis.

9.3 rem:galrep:eisenstein (remark; R, pointers). Reducible $\bar\rho$ means: T_ℓ of curves in the isogeny class are not homothetic (Chapter 4, 10.4: the lattices of ℓ-isogenous curves), the maximal ideal of the Hecke algebra attached to (f, ℓ) is Eisenstein (Chapter 14), multiplicity one and the theorems of Diamond, Flach and Guo are unavailable, and the modular degree can lose the local divisibility (the deficit D of Conjecture conj:eis). Why the hypothesis "$E[\ell]$ irreducible" appears in Theorem thm:quant, Conjecture conj:euler and Proposition prop:forms. Paper: Section sec:eis, Conjecture conj:eis, Open question 4.

13.3.10 Section 10. Serre's conjecture for $\bar\rho_{E,\ell}$ and the optimal level

10.1 thm:galrep:optimal-level (theorem; R, with the elliptic-curve inputs P). (i) (Serre's conjecture, Khare and Wintenberger; stated) Every odd irreducible $\bar\sigma\colon G_\mathbb{Q} \to \mathrm{GL}_2(\bar{\mathbb{F}}_\ell)$ arises from a newform of weight $k(\bar\sigma)$, level $N(\bar\sigma)$ and character determined by $\det\bar\sigma$. (ii) (The form used by the paper: Ribet, Carayol, Diamond; DDT Theorem 3.15, as quoted in Section 3) Let ℓ be odd and $E'/\mathbb{Q}$ with $\ell \nmid N(E')$ and $\bar\rho = \bar\rho_{E',\ell}$ irreducible, and if $\ell = 3$ with image not $N_s(\mathbb{F}_3)$ (8.6). Then there are a newform g of weight 2 on $\Gamma_0(N(\bar\rho))$ and a prime $\lambda \mid \ell$ of its coefficient field with $\bar\rho_{g,\lambda} \simeq \bar\rho \otimes \bar{\mathbb{F}}_\ell$, i.e. $a_p(g) \equiv a_p(E') \pmod{\lambda}$ for all $p \nmid \ell N(E')$. Elliptic-curve inputs proved here: $\bar\rho$ is modular of level $N(E')$ (modularity, Chapter 14), odd (1.1(iii)), good at ℓ (5.1(i), so $\delta(\bar\rho) = 0$), of weight 2 (5.1(iii)), with $\det\bar\rho = \bar\chi_\ell$ (1.1(ii)) and $N(\bar\rho)$ given by 6.2. That the nebentypus is trivial is proved in Chapter 15; the theorem itself (Ribet's level lowering, Carayol, Diamond) is stated there, not proved (Chapter 15 proves only the specialisation to

$\bar{\rho}_{E',\ell}$; important_notes.md B18). Reference: DDT Theorem 3.15 (as cited by the paper; verify); Ribet90; Carayol89; Diamond95; Serre87; KhareWintenberger. Paper: the optimal-level paragraph of Section 3, Theorem thm:qual cases (a), (b), (c).

10.2 `ex:galrep:level-lowering` (worked example; P for the data). The congruences of the Example of Section 3 read through 4.5 and 6.2: 121c1 (type IV at 11, $\deg\varphi = 6$): $N(\bar{\rho}_{121c1,3}) = 11$, $g = f_{11a}$; 121a1 (type II, $\deg\varphi = 6$): twist by χ_{-11} into class 121c, then 11a, $a_p(121a1) \equiv \chi_{-11}(p)a_p(11a1) \pmod 3$; 77c1 ($\mathrm{I}_3$ at 7, $\deg\varphi = 6$): $N(\bar{\rho}_3) = 11$, 77c1 $\equiv$ 11a1 (mod 3); 539c1 (I_3^* at 7, $\deg\varphi = 288 = 2^5 \cdot 3^2$): twist by χ_{-7} is class 77c, then 11a; 980i1 (I_5^* at 7, $\deg\varphi = 2880$) at $\ell = 5$: twist 140b1, then 20a1, with $5 \mid N$ (illustration only). All congruences checked for $p < 2000$ ($p < 1224$ for $\ell = 5$); modular degrees from `degphi`. Paper: Section 3, Example; Theorem thm:qual.

13.4 3. Section plan with target line counts

Lines of LaTeX, card excluded (PLAN.md 7.3: up to 2,800 for this chapter). Items per section: 4, 3, 5, 6, 4, 3, 6, 6, 3, 2 (42 items).

section	title	contents	target lines
(starred)	Core results at a glance	the card: every item of Part 2, stated as above, with tag and `\Cref`	(excluded)
(starred)	Where the paper uses this chapter	Section 1.1; hypothesis of thm:main and S_f; lem:tamagawa-exp; lem:IV; rem:types; lem:euler (inertia half); lem:old (local analysis); optimal-level paragraph; prop:ns; lem:h0; sec:anomaly; sec:eis; conj:eis; prop:forms and rem:wild (type IV at 2); def:terms, rem:ko	55
1	The representations $\rho_{E,\ell}$ and $\bar{\rho}_{E,\ell}$	conventions C1 to C6 as `convention` environments; 1.1 to 1.4; `paperlink` for Section 1.1 and for "irreducible" (point 10)	180
2	Neron-Ogg-Shafarevich, component groups and inertia invariants	2.1 to 2.3, with the full proof of 2.1 (a) to (d) and of 2.2; `warning` that 2.2 is for additive and multiplicative reduction separately	250

section	title	contents	target lines
3	Good and multiplicative reduction	3.1 to 3.5; the Tate matrices in display; both proofs of lem:tamagawa-exp; `paperlink` for lem:tamagawa-exp (point 8), lem:tamfactor (point 42), thm:main's I_n^* step (point 18)	300
4	Additive potentially good reduction	4.1 to 4.6; the shapes table and the mod 3 table (`booktabs`); both proofs of lem:IV; 4.4 with its consequences for prop:forms; `paperlink` for lem:IV (point 13), rem:types (point 22), lem:euler (points 12, 34), rem:wild (point 43)	380
5	The representation at ℓ	5.1 to 5.4; fundamental characters; the ordinary and supersingular cases; `paperlink` for lem:old ($p = \ell$)	190
6	Conductors of mod ℓ representations	6.1 to 6.3; the formula $a_p = f_p - \dim \Phi_p^{\mathrm{geom}}[\ell]$ with proof; the examples table	200
7	The images of $\bar{\rho}_{E,\ell}$	7.1 to 7.6; the mod 3 image table (may be split in two: group data and arithmetic data); labels at $\ell = 5, 7, 11, 13$ as a small table; `paperlink` for sec:anomaly, sec:eis (point 44), sec:data	320
8	Absolute irreducibility over $\mathbb{Q}(\sqrt{\ell^*})$; prop:ns; lem:h0	8.1 to 8.6; the full proofs of prop:ns (points 1, 9, 33) and lem:h0 (points 5, 6, 10, 14); 8.2 and 8.4 with their data	400
9	The Eisenstein case	9.1 to 9.3; `paperlink` for conj:eis and sec:eis (points 17, 44)	160
10	Serre's conjecture and the optimal level	10.1, 10.2; the Example of Section 3 as a table	140
11	Exercises	the 16 exercises of Part 4.2	150

section	title	contents	target lines
12	Notes and references	Serre-Tate, Serre72, Serre87, Kraus90, Zywina15, Sutherland16, Edixhoven92, Carayol89, Livne, DDT, Silverman; which numbers were cited and which only by section	50
	total		**2,775**

13.5 4. Coverage table and exercises

13.5.1 4.1 Coverage: PLAN.md scope, Section 6 map rows and paper hooks to line-up items

PLAN.md scope item, Section 6 row, or paper hook	line-up items
$\rho_{E,\ell}\colon G_{\mathbb{Q}} \to \mathrm{Aut}(T_\ell E)$, $\bar\rho_{E,\ell}$, continuity	1.1
$\det \rho_{E,\ell} = \chi_\ell$ (Weil pairing); $\det \bar\rho_{E,3} = \chi_{-3}$ cuts out $\mathbb{Q}(\sqrt{-3})$	1.1(ii)
complex conjugation has determinant -1 and trace 0	1.1(iii), 1.4(i)
Neron-Ogg-Shafarevich (unramified iff good; potentially good iff finite inertia image)	2.1, 2.3
local representation at a good prime $(X^2 - a_p X + p)$	3.1
multiplicative prime: Tate curve, extension of ε by $\varepsilon\chi_\ell$, Kummer class in the corner, non-split; DDT Theorem 3.1(e)	3.2
the paper's Lemma lem:tamagawa-exp in full, split and nonsplit	3.3
additive potentially good, $q \geq 5$: inertia cyclic of order $2, 3, 4, 6$ (Kraus), eigenvalues	2.3, 4.1, 4.2
the paper's Lemma lem:IV in full	4.3 (and 4.4 at $q = 2$)
the paper's Remark rem:types in full	4.5
potentially multiplicative additive reduction $(\rho = \rho' \otimes \chi_{q^*})$	3.4
at ℓ: finite flat, crystalline; Deligne-Fontaine description; a_ℓ mod ℓ determined by $\bar\rho|_{D_\ell}$	5.1, 5.2, 5.3, 5.4
Artin conductor and Serre conductor; $N(\bar\rho_{E,\ell}) \mid N$ (Carayol); prime to unramified primes	6.1, 6.2, 6.3
Serre's open image theorem	7.1
classification of images for non-CM $E/\mathbb{Q}$ (Zywina, $\ell = 3, 5, 7, 11, 13$); labels 3B, 3B.1.1, 3Cs.1.1, 3Ns, 3Nn, 3Cs, 3Cn	7.2, 7.3, 7.4, 7.5
reducible representations and rational isogenies; Eisenstein case; $\mathbb{Z}/3 \oplus \mu_3$ as 3Cs.1.1	1.4(ii), 9.1, 9.2, 9.3
irreducible versus absolutely irreducible (complex conjugation, ℓ odd)	1.4(i)

PLAN.md scope item, Section 6 row, or paper hook	line-up items
absolute irreducibility on $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{\ell^*}))$, dihedral and induced case	8.1, 8.2, 8.6
the paper's Proposition prop:ns in full	8.3 (with 7.3, 8.4)
the paper's Lemma lem:h0 in full ($h_\ell(E)$, $\rho \bmod \ell^k \simeq (\rho \otimes \chi_\ell) \bmod \ell^k$, finiteness: corrected)	8.5
semisimplification and Brauer-Nesbitt for $\bar{\rho}_{E,\ell}$ ($a_p \bmod \ell$ on a density-one set)	1.4(iii)
Serre's conjecture for $\bar{\rho}_{E,\ell}$ (optimal level, weight 2, level $N(\bar{\rho})$)	10.1, 10.2
Galois representation of a quadratic twist	1.3
computing mod ℓ images (division polynomials, Sutherland's algorithm)	7.6
Section 6 row: $\rho_{E,\ell}$, $\bar{\rho}$, T, V, $E[\ell]$, absolute irreducibility, restriction to $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{\ell^*}))$	1.1, 1.4, 8.1, 8.2, 8.6
Section 6 row: normaliser of a split Cartan, 3Ns, $\mathrm{GL}_2(\mathbb{F}_3)$ and subgroups, orbits on lines	7.2, 7.3, 8.3, 8.4
Section 6 row: $h_\ell(E)$, $\bar{\rho} \simeq \bar{\rho} \otimes \chi$	8.1, 8.5
Section 6 row: 3-division polynomial and its factorisation pattern	7.3, 7.6, 8.3
Section 6 row: Eisenstein case, rational ℓ-isogeny, $\mathbb{Z}/3 + \mu_3$, torsion	9.1, 9.2, 9.3
Section 6 row: level lowering, Serre conductor, Serre's conjecture	6.1, 6.2, 6.3, 10.1, 10.2
Section 6 row: image of inertia at additive primes, unipotent elements, $\mathbb{Z}_3[\zeta_3]$	2.2, 4.1, 4.3, 4.4, 4.5
Section 6 row: Deligne-Fontaine description of $\bar{\rho}$ at ℓ	5.2, 5.3
Section 6 row: DDT Theorem 3.1(e), local structure at multiplicative primes	3.2, 3.3
Section 6 row: finite flat group schemes, "good at ℓ"	5.1
hook: Section 1.1 (V, T, ρ, $\bar{\rho}$, $A_f = \mathrm{ad}^0 V$, $\delta_p = \dim V^{I_p}$)	1.1, 1.2, 3.5
hook: Theorem thm:main hypothesis ($\ell \notin S_f$)	1.4, 8.1, 8.2, 8.6
hook: Lemma lem:tamagawa-exp	3.3
hook: Lemma lem:IV	4.3, 4.4
hook: Remark rem:types	4.5
hook: Lemma lem:euler (inertia images and invariants)	3.2(v), 3.4, 4.1, 4.4(iv), 8.4(iv)
hook: Lemma lem:old (local analysis at $p = \ell$, at $p \,\|\, N$ ramified or unramified, at $p \nmid \ell N$)	5.3, 3.3, 3.2(iii), 1.4(iii)
hook: the optimal-level paragraph of Section 3	5.1, 6.1, 6.2, 10.1
hook: Proposition prop:ns	7.3, 8.3
hook: Lemma lem:h0	8.5
hook: Section sec:anomaly	7.3, 8.4, 8.5
hook: Section sec:eis	7.3, 9.1, 9.2
hook: Conjecture conj:eis	9.2, 9.3
also: Definition def:terms (t_ℓ from $v_q(j)$; I_n^* at 2), Section 1.5 and Remark rem:ko(2) (Tamagawa exponent)	3.3(i), 3.4

PLAN.md scope item, Section 6 row, or paper hook	line-up items
also: Proposition prop:forms, Remark rem:wild, Section sec:wild (type IV at 2)	4.4, 4.6
also: Theorem thm:qual and the Example of Section 3	4.3, 4.5, 6.2, 6.3, 10.1, 10.2
also: proof of Theorem thm:main (the I_n^* primes, including $q = 2$)	3.4
also: Lemma lem:twist, Theorem thm:quant ($\mathrm{ad}^0 T$ unchanged by twists)	1.3, 3.4
also: Section sec:data, Section sec:forms, the appendix (`ns3`)	7.6, 8.2, 8.6
also: Remark rem:ns, Conjecture conj:euler (h_ℓ, w_3 untouched)	8.4, 8.5, 8.6

13.5.2 4.2 Exercises (Section 11 of the chapter), graded from easy to harder

1. Show that for ℓ odd $\rho_{E,\ell}(c)$ is conjugate in $\mathrm{GL}_2(\mathbb{Z}_\ell)$ to $\mathrm{diag}(1,-1)$, and deduce that $\bar\rho_{E,\ell}$ is irreducible iff absolutely irreducible. Why does the argument fail for $\ell = 2$?
2. Check the paper's claim (Lemma lem:tamagawa-exp) on 11a1 and 11a3: $n_{11} = 5$, resp. 1; show that $\bar\rho_{11a1,5}$ is unramified at 11 ($E[5] \simeq \mathbb{Z}/5 \oplus \mu_5$) while $\bar\rho_{11a3,5}$ is ramified at 11, and compute $\#E[5^t]^{I_{11}}$ for both.
3. Prove the hard direction of the criterion of Neron, Ogg and Shafarevich from $E_0/E_1 \simeq \tilde E_{\mathrm{ns}}$, E_1 pro-p and the finiteness of E/E_0.
4. Prove Lemma 2.2: for additive reduction at $q \neq \ell$, $E[\ell^k]^{I_q} \simeq \Phi_q(\bar{\mathbb{F}}_q)[\ell^k]$. Deduce that the ℓ-part of c_q is $\#E(\mathbb{Q}_q)[\ell^\infty]$ when the reduction is additive (compare the final check in the proof of Lemma lem:tamfactor).
5. Check the paper's claim (Lemma lem:IV) for 121c1: type IV at 11; show $N(\bar\rho_{121c1,3}) = 11$ and verify $a_p(121\mathrm{c}1) \equiv a_p(11\mathrm{a}1) \pmod 3$ for $p < 100$, including $p = 3$ (both ordinary at 3).
6. Check the paper's claim (Remark rem:wild) that a curve of type IV at 2 has $P_2(X) = 1 + 2X$ and $e_3(E, 2) = 1$: prove it with Lemma 2.2 at $q = 2$ and Theorem 2.3, and check it on 20a1 and 980i1 (type IV^* at 2) with `lfunsympow`.
7. Derive Kraus's table for $q \geq 5$ ($\mathrm{I}_0^* \mapsto 2$, $\mathrm{IV}, \mathrm{IV}^* \mapsto 3$, $\mathrm{III}, \mathrm{III}^* \mapsto 4$, $\mathrm{II}, \mathrm{II}^* \mapsto 6$) from the component groups, Lemma 2.2 at $\ell = 2$ and $\ell = 3$, and the independence of ℓ (Remark 4.2).
8. Check the paper's claim (Remark rem:types) row by row (table 4.5): for each additive type at $q \geq 5$ compute $\bar\rho_{E,3}(I_q)$, $E[3]^{I_q}$, the Serre exponent, and the exponent after twisting by χ_{q^*}. Which types does the paper's remark omit?
9. Show that $C_{ns}(\mathbb{F}_\ell)$ is never the image of $\bar\rho_{E,\ell}$ for ℓ odd, and that the image never lies in $\mathrm{SL}_2(\mathbb{F}_\ell)$.
10. Check the paper's claim (Proposition prop:ns) for 726a1: factor ψ_3, identify the three quadratic subfields $\mathbb{Q}(\sqrt{33})$, $\mathbb{Q}(\sqrt{-11})$, $\mathbb{Q}(\sqrt{-3})$, deduce the image $N_s(\mathbb{F}_3)$, and check $a_p \equiv 0 \pmod 3$ for the primes $p \equiv 2 \pmod 3$, $p < 60$, $p \neq 2, 11$.
11. Check the paper's claim (Section sec:anomaly) that $h_3(E) = 1$ for every curve with image 3Ns, using Lemma 8.5 only; then show that the paper's numerical test ("$9 \nmid a_p$ for some inert $p < 1223$") also excludes $h_3(E) \geq 2$, and why it is not needed.
12. Prove Proposition 8.4(iii), (iv): a curve with image 3Ns has a prime $p \neq 3$ with cyclic inertia of order 4, and $e_3(E, p) \geq 1$ there. Check on 726a1 ($p = 11$), 338d1 ($p = 13$) and 6912d1 ($p = 2$).
13. Let $\ell \geq 5$ and $\ell \nmid N$. Using the fundamental characters, show that $\bar\rho_{E,\ell} \not\simeq \bar\rho_{E,\ell} \otimes \chi_{\ell^*}$. Find in `ecdata` a curve with image 5Ns and $5 \nmid N$, and determine the quadratic field from which its mod 5 representation is induced.
14. Let E be semistable, ℓ odd, $\ell \nmid N$, $E[\ell]$ reducible. Show $a_p \equiv 1 + p \pmod \ell$ for $p \nmid N\ell$ and that E or an ℓ-isogenous curve has a rational point of order ℓ. Check on 14a1 and 26a1 at $\ell = 3$.
15. Check the paper's claim (Section sec:eis) that 3Cs.1.1 is "the split Cartan image": identify the images of 14a1 and of 539a2 and show the claim is a misdescription.

16. Compute $N(\bar\rho_{E,3})$ from the formula $a_p = f_p - \dim \Phi_p^{\mathrm{geom}}[3]$ for 539c1, 77c1, 726a1 and 6912d1, and $N(\bar\rho_{E,5})$ for 980i1 and 140b1; say which of the Section 3 congruences of the paper these explain.

13.6 5. Dependencies and forward pointers

Results from earlier chapters used here (by chapter label only, `\Cref{ch:...}`): * Chapter 1 (`ch:groups`): models and subgroups of $\mathrm{GL}_2(\mathbb{F}_3)$ (6.2 to 6.6), the three images (6.7), orbit dichotomy (6.8), elements of determinant -1 and trace 0 (6.9), labels as subgroups (6.10), unipotent elements and their order (5.6), commutator lemma (5.7), Cartan subgroups and normalisers (5.8, 5.9), Dickson (5.10), $1+\ell M_2(\mathbb{Z}_\ell)$ pro-ℓ and torsion-free (7.4). * Chapter 2 (`ch:rings`): $\mathbb{F}_9 = \mathbb{F}_3[i]$, χ_{-3}, χ_{-4} and their values, q^*, $a^2 \equiv 1 \pmod{\ell^k}$ for all units iff $\ell^k = 3$ (ℓ odd) (5.5). * Chapter 3 (`ch:modules`): DVRs, lattices, Nakayama, cyclic modules over a DVR, the snake lemma. * Chapter 4 (`ch:reps`): 2.4, 2.5, 3.2(ii), 3.6, 3.7, 5.2, 5.3, 6.2, 6.6, 6.7, 7.2, 7.4, 7.6, 8.1 to 8.5, 9.1 to 9.4, 10.1 to 10.5, 11.1. * Chapter 5 (`ch:galois`): galois:continuous-reps, galois:cyclotomic-character, galois:complex-conjugation, galois:restriction, galois:index-two-restriction, galois:twist, galois:l-adic-kummer, galois:quartic-on-lines, galois:frobenius-on-lines, galois:chebotarev-brauer-nesbitt, galois:congruent-ap, galois:sqrt-lstar. * Chapter 6 (`ch:ant`): Frobenius and inertia in $G_{\mathbb{Q}}$ (3.7, 3.10), the finite tame relation (3.11, 3.12), $\mathbb{Z}_3[\zeta_3]$ (4.7, 4.8), $\mathbb{Z}_\ell[\zeta_3]$ (4.9), the commutant (4.10), Kronecker-Weber (5.5), the twist detector (6.4), CM (6.5). * Chapter 7 (`ch:local`): $\mathbb{Q}_q^{\mathrm{ur}}$ and unramified base change (4.2, 4.4), tame character and tame relation (5.3, 5.4), Frobenius fixing roots (5.5), finite tame images (5.6), wild images (5.7), Kummer classes on inertia (6.2), quadratic characters of $\mathbb{Q}_q$ (6.3), Weil-Deligne representations and sp(2) (7.2 to 7.6), invariants, Swan and Artin conductors (8.1, 8.3, 8.4), the shapes (9.1 to 9.5). * Chapter 8 (`ch:curves`): only through Chapter 9. * Chapter 9 (`ch:ellcurves`): Weierstrass data (1.1 to 1.6), ψ_3 (4.4 to 4.6), Weil pairing, $T_\ell E$, ρ, det, complex conjugation (5.1 to 5.7), CM (6.3 to 6.5), curves over $\mathbb{F}_q$ (8.3, 8.6), torsion, isogenies, $\mathbb{Z}/\ell \oplus \mu_\ell$, no cyclic 27-isogeny, prime-to-ℓ isogenies (9.3 to 9.10), twists (10.1 to 10.6), labels (12.1). * Chapter 10 (`ch:reduction`, outlined in parallel): minimal models, reduction types, E_0, E_1, formal groups, $E(K)/E_0(K) \simeq \Phi_p(k)$ (Neron model, Hensel), Kodaira types and component groups, Tate's algorithm, conductor exponents and Ogg's formula, the semistable reduction theorem, the Tate curve and $E_{\mathfrak{q}}[\ell^t]$, Kraus's theorem, Lemma lem:twist (i), (ii), Remark rem:wild. Items 2.1 to 2.3, 3.2 to 3.5, 4.1 to 4.6, 6.2 depend on these; they are black boxes here, cited with `\Cref{ch:reduction}`.

Black boxes imported from later chapters (stated with a forward `\Cref`, never used in the proof of a numbered result of this chapter except where said): modularity and the Hecke algebra (Chapters 13, 14; only in 10.1 and 9.3); the Galois representations of newforms and the optimal-level theorem in general (Chapter 15; only in 1.4(iii), 5.3(iii), 10.1); crystalline representations and Bloch-Kato conditions (Chapter 18; only in 5.1(ii)); L-factors in the symmetric-square normalisation and $e_\ell(E,q)$ (Chapter 19; only in 4.4(iv) and 8.4(iv), which use the definition $L_q(A_f,s) = \det(1 - \mathrm{Frob}_q^{-1}\, q^{-s} \mid (\mathrm{ad}^0 V)^{I_q})^{-1}$, with the geometric Frobenius Frob_q^{-1} (never Φ_q here, B-C2(b)) and $L_q(\mathrm{Sym}^2 f, s) = L_q(A_f, s-1)$ as conventions).

Forward pointers (chapters that use this one): * Chapter 14 (`ch:moddeg`): reducible $\bar\rho$ and the Eisenstein ideal (9.1 to 9.3); $\ell \nmid 2N$, $\bar\rho$ irreducible in Remark rem:free (1.4). * Chapter 15 (`ch:modgalrep`): 1.4(iii) (congruences give isomorphic residual representations), 3.2 and 3.3 (the cases $p \,\|\, N$ of Lemma lem:old), 5.3 (the case $p = \ell$), 6.1 and 6.2 (Serre conductor, $N(\bar\rho) \mid N$), 4.3, 4.5 (Serre exponents in Theorem thm:qual), 8.6 (the hypothesis at $\ell = 3$ of DDT Theorem 3.15), 10.1, 10.2. * Chapter 16 (`ch:cohomology`): 3.2 (the Tate shape, unipotent and pro-ℓ action of inertia; motivation for its model). * Chapter 17 (`ch:galcoh`): 3.2 (the Kummer class $\varepsilon\kappa$ in the corner). * Chapter 18 (`ch:blochkato`): 3.2, 3.4 (inputs of Lemma lem:tamfactor, including $q = 2$ for important_notes.md point 18), 5.1 (crystalline at ℓ), 8.5 (Lemma lem:h0, restated there in Selmer language), 8.6 ($\ell \notin S_f$), 2.2 (the ℓ-part of c_q at additive primes). * Chapter 19 (`ch:lfunctions`): 3.1, 3.2(v), 3.4, 4.1, 4.4, 4.6, 8.4(iv) (the inputs of Lemma lem:euler and Remark rem:wild; Chapter 19 computes the factors and e_ℓ). * Chapter 20 (`ch:congruence`): 8.5, 8.6 (the hypothesis of Theorem thm:main and the vanishing of $H^0(\mathbb{Q}, W^D)$), 3.4 (the I_n^* step at $q = 2$), 1.3 (twists in Theorem thm:quant), 4.4 (Proposition prop:forms without exception). * Chapter 21 (`ch:paper`): 3.3, 4.3, 4.4, 4.5, 8.3, 8.4, 8.5, 8.6, 9.2 (the tour must use the corrected statements of 4.5 and 8.5 and may shorten its proofs of lem:tamagawa-exp, lem:IV, rem:types, prop:ns, lem:h0 by pointing here).

* Chapter 22 (ch:computation): 7.3, 7.4, 7.6 (ns3, elldivpol, galrep labels and the CM caveat), 4.4 (the factor $1+2X$ at 2), 6.2.

13.7 6. Notes for the writer

13.7.1 6.1 Conventions to typeset (Section 1, as convention environments)

- C1 to C7 of Part 1. Name the Frobenius every time in a display (Frob_q arithmetic, Frob_q^{-1} geometric); the Tate matrices and the conjugation actions use Frob_q; the "action on $(\mathrm{ad}^0 V)^{I_q}$" in 4.1 is the same for both, except at multiplicative primes (3.2(v): q versus q^{-1}), which is exactly important_notes.md point 11.
- Macros: \ad is ad^0, \Sym is Sym^2, \Ns prints 3Ns, \Phig is the geometric component group (the paper's notation); use \Phig_q for $\Phi_q(\bar{\mathbb{F}}_q)$ throughout (C2; Appendix B, B-C11(b)). There is no \O macro: write \mathcal{O}.
- Labels: def:galrep:..., thm:galrep:..., lem:galrep:..., prop:galrep:..., ex:galrep:..., rem:galrep:..., conv:galrep:..., cor:galrep:...; the paper's lemmas proved here get the paper's name in the optional argument, e.g. \begin{lemma}[the paper's Lemma lem:IV]\label{lem:galrep:IV}.

13.7.2 6.2 Pitfalls

- **The paper's lemmas in corrected form.** State Lemma lem:h0 as 8.5 (not "for E without CM it is finite"), Proposition prop:ns as 8.3 (no CM hypothesis, $\mathrm{Ind}\,\psi$ over $\mathbb{F}_9$, the commutator step), Remark rem:types as 4.5 (complete grouping). In a paperlink quote the paper's sentence and the correction; never silently change the paper's statement.
- **Component group lemma (2.2).** It is for $q \neq \ell$ and distinguishes additive from multiplicative reduction; for good reduction $E[\ell^k]^{I_q} = E[\ell^k]$. It uses $\Phi_q(\bar{\mathbb{F}}_q)$, not $c_q = \#\Phi_q(\mathbb{F}_q)$; the Frobenius-invariant version gives $E(\mathbb{Q}_q)[\ell^\infty]$ only for additive reduction. Prove it through finite unramified extensions (the union over them is $\mathbb{Q}_q^{\mathrm{ur}}$, which is not complete).
- **Type IV at 2 (4.4).** The proof needs only Lemma 2.2, the fact that types IV, IV* are potentially good ($v_2(j) \geq 0$; Chapter 10), and the injectivity of reduction on finite subgroups of $\mathrm{GL}_2(\mathbb{Z}_3)$. Do not invoke Kraus at $q = 2$. The conclusion is about types IV and IV* only: types II, III at 2 have various inertia groups (4.6).
- **Tame versus $q \geq 5$.** "Tame since $q \neq 3$" (Lemma lem:IV) means: the image has order 3, prime to q. At $q = 2$ an image of order 3 is tame; at $q = 3$ it is wild. Say "the order is prime to q", not "$q \geq 5$", where that is what is used.
- **The corner of the Tate matrix is $\varepsilon\kappa$** (important_notes.md point 42); $\varepsilon(\mathrm{Frob}_q) = a_q(E)$; on inertia $\varepsilon = 1$.
- **n_q versus the index of a Kodaira symbol.** $n_q = v_q(\Delta_{\min}) = -v_q(j)$ at a multiplicative prime; at a potentially multiplicative additive prime use $-v_q(j)$; at $q = 2$ the index m of I_m^* is not $-v_2(j)$ in general (3.4).
- **Irreducible and absolutely irreducible.** For $\bar{\rho}_{E,\ell}$, ℓ odd, they agree (1.4(i)); for its restriction to $G_{\mathbb{Q}(\sqrt{\ell^*})}$ they do not (the 3Ns case: C_4 on $\mathbb{F}_3^2$ is irreducible, not absolutely irreducible). Never write "$\bar{\rho}|_{G_{\mathbb{Q}(\sqrt{-3})}}$ is reducible" in the 3Ns case.
- **Dihedral is not the excluded case.** 3Nn images are dihedral (induced from another quadratic field) and satisfy the hypothesis of Theorem thm:main; only "induced from $\mathbb{Q}(\sqrt{\ell^*})$" is excluded. For $\ell \geq 5$, $\ell \nmid N$, that never happens (8.2).
- **χ_ℓ versus χ_{ℓ^*}.** $\bar{\chi}_\ell = \chi_{\ell^*}$ only for $\ell = 3$; h_ℓ concerns $\bar{\chi}_\ell$, the hypothesis of Theorem thm:main concerns χ_{ℓ^*} (important_notes.md points 5, 6).
- **Labels.** Always give the intrinsic description with a label; 3Cs.1.1 is not the split Cartan subgroup (point 44); 3B.1.1 versus 3B.1.2 depends on a convention, fixed here by the data (rational point of

order 3 for 3B.1.1). `ecdata`'s `galrep` files omit the label when the image is the one forced by CM (Part 6.6): say so wherever counts are quoted.

- $N_s(\mathbb{F}_3)$ **is not maximal** (point 2); do not import the large-ℓ picture of maximal subgroups.
- **Serre conductor versus conductor.** $a_p(\bar\rho) \le f_p$ with the defect $\dim \Phi_p^{\mathrm{geom}}[\ell]$ (6.2); a twist can lower one and not the other (4.5(b)).
- **Supersingular at ℓ.** For $\ell \ge 5$, $\ell \mid a_\ell$ means $a_\ell = 0$ (Hasse); for $\ell = 3$, $a_3 \in \{0, \pm 3\}$. The inertia description in 5.3(ii) is over $\mathbb{F}_{\ell^2}$.
- **Theorem numbers.** Only cite numbers marked as sure: none in this outline is fully sure except those the paper itself cites (DDT Theorem 3.15, DDT Theorem 3.1(e), Edixhoven92 Section 2, Zywina15 and Sutherland16 by name); everything marked "(verify)" must be checked or replaced by a section reference (SilvermanAEC VII.7 Theorem 7.1; Serre-Tate 1968 Theorem 1 and Section 2 corollaries; Serre72 propositions on fundamental characters and Section 5; Serre87 Sections 1.2, 2, 4; Sutherland16 Section 6).

13.7.3 6.3 What to leave to other chapters

- Chapter 1: the group theory of $\mathrm{GL}_2(\mathbb{F}_3)$ and Dickson's theorem (cite).
- Chapter 4: all representation theory (Clifford, Brauer-Nesbitt, induced representations over $\mathbb{F}_9$, the length lemma 10.5, the shapes on V and $\mathrm{ad}^0 V$); here only the Galois input.
- Chapter 5: continuity, complex conjugation, restriction and index-two arguments, Chebotarev, Kummer cocycles.
- Chapter 6: $\mathbb{Z}_3[\zeta_3]$, the commutant and $\chi_{-3}(q)$, $\chi_{-4}(q)$ actions, CM and $a_p = 0$.
- Chapter 7: tame inertia and the tame relation, Kummer classes on inertia, Weil-Deligne representations, L-factors, Swan and Artin conductors, the shapes table (its Kodaira column is supplied by 3.2, 3.4, 4.1, 4.4 here).
- Chapter 9: ψ_3, Weil pairing, isogenies, torsion, the no-27-isogeny proof (restated in 9.2 by `\Cref`), twists of models.
- Chapter 10: minimal models, Tate's algorithm, Kodaira types, component groups, Ogg's formula, the Tate curve itself, Kraus's theorem, Lemma lem:twist (i), (ii), Remark rem:wild. Chapter 11 does not reprove Kraus; it proves what follows from it.
- Chapter 15: newforms, local-global compatibility, the optimal-level theorem in general and its proof, Lemma lem:old and Theorem thm:qual (here only the local inputs and the elliptic-curve specialisation).
- Chapter 18: Lemma lem:tamfactor and the Selmer-group reading of Lemma lem:h0; Chapter 19: $L_q(A_f, s)$, P_q, $e_\ell(E, q)$, Lemma lem:euler in full; Chapter 22: PARI usage.

13.7.4 6.4 How each relevant point of important_notes.md is handled

- **Point 1** (prop:ns, "non-abelian and faithful"): 8.3 step (4) supplies the commutator step (Chapter 1, 5.7) or Maschke for Q_8 (Chapter 4, 3.7); `paperlink`.
- **Point 5** (lem:h0 finiteness; $h_\ell \le 1$): 8.5 states and proves the corrected lemma ($h_\ell = 0$ for $\ell \ge 5$, $h_3 \le 1$, with or without CM); remark (a); Exercise 11.
- **Point 6** (criterion for $h_\ell > 0$ uses $\bar\chi_\ell$): 8.5 remark (b), with the sharpening of 8.2 ($\bar\rho \simeq \bar\rho \otimes \chi_{\ell^*}$ for $\ell \ge 5$ needs $\ell \mid N$).
- **Point 8** (lem:tamagawa-exp: notation, "up to units"): 3.3 proof 1 written with q_E and the two separate steps; Chapter 7, 6.2(v).
- **Point 9** (prop:ns: CM hypothesis unused): 8.3 and 7.3 are stated for every $E/\mathbb{Q}$; data include the 26 CM curves with image $N_s(\mathbb{F}_3)$.
- **Point 10** (irreducible versus absolutely irreducible in lem:old and lem:h0): 1.4(i), (iii); 8.5 remark (d).
- **Point 11** (geometric versus arithmetic Frobenius): C3, 3.2(v), 4.1 (same sign for both), 4.4(iv); the L-factors are Chapter 19's.

- **Point 14** (CM by $\mathbb{Q}(\sqrt{\ell^*})$ forces $\ell \mid N$): 8.5 remark (a), citing Chapter 9, 6.5 and Chapter 6, 6.5(vi); not needed for the proof.
- **Point 17** ($\mathbb{Z}/3 \oplus \mu_3$ and a point of order 9 cannot coexist): 9.2 restates Chapter 9, 9.8 in label language.
- **Point 22** (rem:types grouping): 4.5, the complete table, with the addendum on "conductor of E" versus "Serre conductor" (Part 7, N4).
- **Point 33** ($\mathrm{Ind}\,\psi$ over $\mathbb{F}_9$): 8.3 statement and step (6); Chapter 4, 8.2.
- **Point 34** (types III, III*: the line is spanned by τ): 4.1.
- **Point 42** (corner $\varepsilon\kappa$): 3.2.
- **Point 44** (3Cs.1.1 is not the split Cartan image): 7.3, 7.4, 9.2; Exercise 15.
- **B1** (on $V_\ell E$ the arithmetic Frobenius has eigenvalues α, β): C3 and 3.1.
- Also: point 2 (7.2, 7.3), point 3 (7.4: labels checked against `ecdata`), point 4 (7.5: Zywina by section), point 12 (4.1), point 13 (4.3), point 18 (3.4 covers $q = 2$), point 21 (6.3, 10.2: 980i1 illustrates only), point 27 (1.3(iii)), point 32 (C4), point 43 (**settled**: 4.4).

13.7.5 6.5 References and bibliography

`SilvermanAEC`, `SilvermanATAEC`, `SerreAbelian`, `Serre72`, `Serre87`, `Kraus90`, `Zywina15`, `Sutherland16`, `Edixhoven92`, `Carayol89`, `Ribet90`, `Diamond95`, `DDT`, `KhareWintenberger`, `Mazur77`, `Mazur78`, `Berger`, `ecdata`, `PARI` are in `refs.bib`. Add to `refs.bib`, not `bib/ch11.bib`, since both are needed by more than one chapter (decision B14; important_notes.md B16, B18; complete entries, checked against the sources): `SerreTate68` (J.-P. Serre and J. Tate, "Good reduction of abelian varieties", Ann. of Math. 88 (1968), 492-517) and `Livne89` (R. Livne, "On the conductors of mod ℓ Galois representations coming from modular forms", J. Number Theory 31 (1989), 133-141; check the pages). **Coordinate `SerreTate68` (settled by decision B14: it goes into `refs.bib`):** the Chapter 7 outline adds the same key to `bib/ch07.bib` and the Chapter 9 outline plans the same work for `bib/ch09.bib`; `main.tex` reads every chapter bib, so a duplicated key gives a BibTeX "Repeated entry" error in the full build, while `tools/build_chapter.sh 11` reads only `refs.bib` and `bib/ch11.bib` (Part 7, B2).

13.7.6 6.6 Verification log (computations done for this outline)

Scripts `/tmp/ch11/scripts/` (`ec.py`, `inert.py`, `t1.py` to `t10.py`), PARI/GP 2.15 through `cypari2`; data `allcurves`, `galrep`, `degphi` for conductors 0 to 9,999 (64,687 curves) from the `ecdata` GitHub repository. * Mod 3 images (7.3, 7.6): for every curve, the factorisation pattern of `elldivpol(E,3)`, `polgalois` of irreducible ψ_3, and $3 \mid \#E(\mathbb{Q})_{\mathrm{tors}}$, against the `galrep` label. Exact agreement with the table of 7.3; the only unlabelled non-surjective cases are the 462 CM curves ($j \in \{0, 1728, -3375, 8000, -32768, 54000, 287496, -884736, -12288000, 16581375, -884736000, -147197952000\}$), classified from ψ_3 (26 with $2+2$, 184 with D_4, 154 with a rational root and no label). * Inertia mod 3 (2.2, 4.1, 4.5, 4.6): for a prime q of E, the fields $\mathbb{Q}(P)$ of the 3-torsion points (degree up to 8, from the factors of ψ_3 of the short model and $y^2 = f(x)$ via `rnfequation`), `nfinit([pol,[q]])`, `idealprimedec`; the order of $\bar\rho(I_q)$ is the l.c.m. of the ramification indices when tame, and $E[3]^{I_q} \neq 0$ iff some prime above q has $e = 1$. At $q \geq 5$, 12 curves per type with surjective image: I_0^*: 2, no fixed point; II, II*: 6, none; III, III*: 4, none; IV, IV*: 3, a fixed line; I_n: 3 or 1 as $3 \nmid n$ or $3 \mid n$, fixed; I_n^*: 6 or 2, none; conductor exponents $2, 2, 2, 2, 1, 2$. At $q = 2$, 40 per type: as in 4.6. * Type IV, IV* at 2 (4.4): all 4,750 curves; $\bar\rho(I_2)$ of order 3 with a fixed line, $f_2 = 2$, $v_2(\Delta_{\min}) = 4$, resp. 8, $v_2(j) \geq 0$; `lfunsympow(E,2)` and `lfuneuler(S,2)` on 60 of them and on 20a1, 980i1, 140b1: $P_2(X) = 1 + 2X$. * 3Ns curves (8.2, 8.4): all 120; Kodaira types at all bad primes (`ellglobalred`), a_p mod 3 for $p \equiv 2 \pmod 3$ among the first 80 primes, a_3 mod 3, Euler factors at the type III primes for 726a1, 338d1, 3840h1, 3840h2, 4864b1, 4864c1, 6912d1, 6912j1, and the inertia computation at those primes ($\bar\rho(I_p)$ cyclic of order 4). * $\ell \geq 5$ dihedral images (8.2): every curve with an ℓNs or ℓNn label, $\ell \geq 5$: test of $a_p \equiv 0 \pmod \ell$ at the first 150 primes inert in $\mathbb{Q}(\sqrt{\ell^*})$ and not dividing $N\ell$; the label lists of 7.4. * Eisenstein (5.4(i), 9.1): every curve with a Borel or split Cartan label at ℓ, $\ell \nmid N$: $\ell \nmid a_\ell$ (4,571 pairs); square-free N: $a_p \equiv 1 + p$ for the first 60 primes. * Examples (6.3, 10.2): data of 11a1, 11a2, 11a3, 121a1, 121c1, 121c2, 121d1, 121d2, 77c1, 539c1,

980i1, 140b1, 726a1, 6912d1, 338d1, 99a1, 20a1, 14a1 (models, conductors, Kodaira codes, discriminants, j, modular degrees, labels); congruences 121c1 versus 11a1, 121a1 versus $\chi_{-11}\cdot$11a1, 77c1 versus 11a1, 539c1 versus $\chi_{-7}\cdot$77c1 mod 3 for the first 300 primes; 140b1 versus 20a1 and 980i1 versus $\chi_{-7}\cdot$20a1 mod 5 for the first 200 primes; $y^2 = x^3 - 6x - 8$ has conductor 6912 and is 6912d1, $a_5 = 3$. * Not verified: anything beyond conductor 10,000; the theorem numbers of the cited books and papers; the general definition of Sutherland's labels against Sutherland16 itself (only against `ecdata`'s use of them).

13.8 7. Points for important_notes.md

13.8.1 7.1 New points about the paper (not among points 1 to 45 and B1 to B9; each checked as stated)

N1. Proposition prop:forms, Remark rem:wild, Section sec:wild: the type IV, IV* case at 2 is a theorem (settles important_notes.md point 43). Location: Proposition prop:forms ("except possibly when $\ell = 3$ and E has type IV or IV* at 2, where they hold as soon as $e_3(E,2) \geq 1 \ldots$ PARI's factor at 2 is $1 + 2X$ for every one of them"), Remark rem:wild, Section sec:wild. Problem: the paper leaves as an exception, verified only numerically, a case that has a short proof. Fix: for additive reduction at $q \neq \ell$, $E[\ell]^{I_q} \simeq \Phi_q(\bar{\mathbb{F}}_q)[\ell]$ (the prime-to-q torsion of $E_0(\mathbb{Q}_q^{\mathrm{ur}})$ is trivial and E_0 is ℓ-divisible); types IV, IV* have $\Phi_2^{\mathrm{geom}} \simeq \mathbb{Z}/3$ and potentially good reduction, so $E[3]^{I_2}$ is a line, $\bar{\rho}_{E,3}(I_2)$ is the unipotent group of order 3, and $\rho_{E,\ell}(I_2)$ is cyclic of order 3 for every odd ℓ (finite subgroups of $\mathrm{GL}_2(\mathbb{Z}_3)$ inject mod 3; independence of ℓ, Serre-Tate). The image is tame, Frobenius at 2 inverts it ($2 \equiv -1 \bmod 3$), and the proof of Lemma lem:euler for types IV, IV* applies verbatim: $L_2(A_f,s)^{-1} = 1 + 2^{-s}$, $P_2(X) = 1 + 2X$, $e_3(E,2) = v_3(3/2) = 1$ (and $e_\ell(E,2) = 0$ for $\ell \geq 5$). So Proposition prop:forms holds with no exception under the hypothesis of Theorem thm:main (and under Conjecture conj:euler with $h_\ell = 0$). Also $f_2 = 2$, and the Serre conductor of $\bar{\rho}_{E,3}$ has exponent 1 at 2, so the proof of Theorem thm:qual (a) works at $q = 2$ as well (with $3 \nmid N$, $\bar{\rho}$ irreducible and not 3Ns): its hypothesis "$q \geq 5$" can be weakened to "$q \neq 3$" in case (a). Confidence: high (proof above, item 4.4; all 4,750 curves with type IV or IV* at 2 and conductor below 10,000 checked: inertia mod 3 of order 3 with a fixed line, $f_2 = 2$; $P_2 = 1 + 2X$ on more than 60 of them, including 20a1 and 980i1). Importance: medium (removes an exception from a proposition).

N2. Theorem thm:main and Section sec:forms: for $\ell \geq 5$, $\ell \nmid N$, the hypothesis on $\mathbb{Q}(\sqrt{\ell^*})$ is implied by irreducibility. Location: the hypothesis of Theorem thm:main (and the abstract), Section 4.2 (S_f), Section sec:forms ("the hypothesis of Theorem thm:main fails ... for the curves whose mod ℓ representation is induced from $\mathbb{Q}(\sqrt{\ell^*})$ with $\ell \geq 5$, which we did not single out"). Problem: not an error, an unstated simplification. If $\ell \nmid N$ and $\bar{\rho}_{E,\ell} \simeq \bar{\rho}_{E,\ell} \otimes \chi_{\ell^*}$, restrict to inertia at ℓ: $\chi_{\ell^*}|_{I_\ell} = \omega^{(\ell-1)/2}$, while $\bar{\rho}|_{I_\ell}^{\mathrm{ss}}$ is $\omega \oplus 1$ (ordinary) or $\omega_2 \oplus \omega_2^\ell$ (supersingular); comparing the characters forces $\ell = 3$ and ordinary reduction at 3. Hence for $\ell \nmid 2N$: $\ell \notin S_f$ iff $E[\ell]$ is irreducible and, if $\ell = 3$, the image is not 3Ns (equivalently $h_\ell(E) = 0$). The curves "induced from $\mathbb{Q}(\sqrt{\ell^*})$ with $\ell \geq 5$" in Section sec:forms all have $\ell \mid N$, so they are already among the curves with $\ell \mid N$; and the 3Ns curves with $3 \nmid N$ are ordinary at 3. Fix: state the hypothesis of Theorem thm:main as "$\ell \nmid 2N$, $E[\ell]$ irreducible, and the image of $\bar{\rho}_{E,3}$ not the normaliser of a split Cartan subgroup if $\ell = 3$", with this argument (or a reference, if DFG04 or DDT record it), and delete or reword the "did not single out" clause. Confidence: high (the inertia description is Serre72, Section 1; data: the 24 curves below conductor 10,000 with $\bar{\rho}_{E,\ell} \simeq \bar{\rho}_{E,\ell} \otimes \chi_{\ell^*}$, $\ell \geq 5$, all have $\ell \mid N$; all 62 curves with image 3Ns and $3 \nmid N$ have $3 \nmid a_3$). Possibly known in the literature; not stated in the paper.

N3. Section sec:anomaly: the observations on the 3Ns curves are theorems. Location: Section sec:anomaly ("Conversely every 3Ns curve of the tables has a prime of type III or III* with a positive term e_3"; "The residue classes of the type III primes of the violating curves are $q \equiv 1$ and $q \equiv 11 \pmod{12}$ in comparable numbers, with some $q = 2$"), and the appendix (726a1: "the type III term and the correction cancelling"). Problem: presented as data; they follow from the image. If $\bar{\rho}_{E,3}(G_{\mathbb{Q}}) = N_s(\mathbb{F}_3)$, then for $p \neq 3$ the image of I_p lies in $N_s \cap \mathrm{SL}_2(\mathbb{F}_3) = C_4$; the quadratic fields K_1, K_2 of the two factors of ψ_3 differ from $\mathbb{Q}(\sqrt{-3})$, so some prime $p \neq 3$ ramifies in them, and exactly at

such p the image of inertia is cyclic of order 4 (types III, III* when $p \geq 5$; every type III, III* prime $p \geq 5$ is of this kind); $\bar{\rho}(\mathrm{Frob}_p)$ commutes with the image of inertia iff its determinant p is 1 mod 3, which forces $\chi_{-4}(p) = \chi_{-3}(p)$, i.e. $p \equiv 1$ or 11 (mod 12), and $e_3(E,p) = v_3(p - \chi_{-3}(p)) \geq 1$; at $p = 2$ the same argument gives $P_2 = 1 + 2X$ and $e_3(E,2) = 1$. Also every multiplicative prime $p \neq 3$ of a 3Ns curve has $3 \mid n_p$ (so $t_3(E,p) \geq 1$), and no prime $p \geq 5$ has type II, II*, no prime $p \neq 3$ type IV, IV*. Fix: add a sentence or a short proposition in Section sec:boundary (item 8.4 of the Chapter 11 outline). Confidence: high (proof; all 120 3Ns curves below conductor 10,000 agree: 100 have type III or III* primes ≥ 5, all $\equiv 1$ or 11 (mod 12), the other 20 have cyclic inertia of order 4 at 2, of Kodaira type III or III* there, with $P_2 = 1 + 2X$). The Kodaira type at 2 of such a prime is not determined by the argument (observed: III or III* with $f_2 = 8$). Importance: low to medium (explains the anomaly section; strengthens Remark rem:ns).

N4. Remark rem:types, last sentence (addendum to important_notes.md point 22): "these are the types whose twist by χ_q lowers the conductor exponent". Problem: for II, II* the twist (to IV*, IV) does not lower the conductor exponent of E (both are 2); it lowers the Serre exponent of $\bar{\rho}_{E,3}$ (from 2 to 1). For I$_n^*$ and I$_0^*$ both exponents drop. Fix: "whose twist by χ_{q^*} lowers the Serre conductor exponent of $\bar{\rho}_{E,3}$ at q: II, II*, I$_n^*$, and also I$_0^*$" (the last omission is point 22 / Chapter 21's N8). Confidence: high (table 4.5). Importance: low (expository).

No other new problem was found in this chapter's area. Checked and found correct (beyond the points already recorded): Lemma lem:tamagawa-exp (both directions, split and nonsplit; also by the component-group route); Lemma lem:IV (statement and proof, modulo point 13); the inertia descriptions in Lemma lem:euler for $q \geq 5$ (orders, generators, Frobenius signs; modulo points 11, 12, 34); Remark rem:types for the types it lists; the list of images and the orbit argument in Proposition prop:ns (modulo points 1, 9, 33); the trace-zero argument and the length description in Lemma lem:h0 (modulo points 5, 6, 10); the claims of the optimal-level paragraph ("good" at ℓ, $N(\bar{\rho}) \mid N$, prime to unramified primes); the local analysis in Lemma lem:old at $p = \ell$ and at $p \,\|\, N$ ("both non-split because $\bar{\rho}$ is ramified": by 3.3(iii)); the numerical check of Section sec:anomaly (valid, see 8.5 remark (c)); Section sec:eis's labels 3B.1.1 and 3Cs.1.1 as used (modulo point 44); the congruences of the Example of Section 3.

13.8.2 7.2 Book-internal inconsistencies

- **B1. PLAN.md, Chapter 11 entry: "finiteness for non-CM curves via $a_p = 0$ at inert primes".** PLAN.md repeats the paper's argument for Lemma lem:h0, which important_notes.md point 5 shows to be wrong (it confuses χ_ℓ with χ_{ℓ^*}). This outline follows point 5 (item 8.5); PLAN.md need not change, but the Chapter 11 writer must not follow its wording. (Likewise PLAN.md's Chapter 11 entry lists rem:types in the paper's incomplete grouping; item 4.5 gives the complete one.)
- **B2. The bibliography key `SerreTate68`.** The Chapter 7 outline adds it to `bib/ch07.bib`, the Chapter 9 outline plans "Serre-Tate 1968" for `bib/ch09.bib`, and this chapter needs it in `bib/ch11.bib` (Chapter 10 will too). `main.tex` reads all chapter bib files, so one key in several files is a "Repeated entry" error in the full build, while each standalone build (`tools/build_chapter.sh`) reads only `refs.bib` and its own file. Action for the orchestrator: decide once (an exception adding the entry to `refs.bib`, or a rule that only the first chapter defines a shared key and the full build is the reference).
- **B3. Chapter 7 outline, item 9.5(ii) and its point N2; Chapter 21 outline, items A1.8 and A2.6.** They treat the cyclicity of the inertia image for types IV, IV* at 2 as an unverified assumption and keep the exception of Proposition prop:forms. Item 4.4 proves it; the Chapter 7, 19 and 21 writers should cite `\Cref{ch:galrep}` and drop the hedge.
- **B4. important_notes.md point 6 and the Chapter 21 outline, A1.5 commentary.** Both say that for $\ell \geq 5$ "$\bar{\rho} \simeq \bar{\rho} \otimes \chi_{\ell^*}$ can occur (the case excluded by S_f)". True only when $\ell \mid N$ (item 8.2); in the range $\ell \nmid N$ of Theorem thm:main it never occurs. Not a contradiction, but the Chapter 21 writer should add "(only when $\ell \mid N$)".
- **B5. Chapter 10 and Chapter 11 on the Tate curve.** PLAN.md gives Chapter 10 the statement of $E_q[\ell^t]$ and Chapter 11 its representation-theoretic consequences (Lemma lem:tamagawa-exp). If

the Chapter 10 outline (written in parallel) also proves the matrix shape or Lemma lem:tamagawa-exp, one of the two must cite the other; this outline keeps both proofs of lem:tamagawa-exp here, as PLAN.md requires.

- **B6. Chapter 4 outline, table 9.3, and Chapter 7 outline, table 9.4: the "Kodaira types" column.** Both defer it to Chapters 10 and 11; it is supplied by items 3.2, 3.4, 4.1, 4.4 here and agrees with both tables (no conflict). The mod 3 column of Chapter 4's table ("I_n: 1 if $3 \nmid n$, 2 if $3 \mid n$") agrees with 2.2 and 4.5.
- **B7. Counts of mod 3 images.** The Chapter 9 outline quotes "the 264 curves of conductor below 10,000 with image 3Cs.1.1"; this outline splits them as 262 non-CM plus 2 CM (27a1 and 27a3), consistent; Chapter 22 should quote counts with the CM caveat of 7.3 (`galrep` omits the labels forced by CM).

Chapter 14

Chapter 12: Modular forms and modular curves (ch:modforms, chapters/ch12_modforms.tex)

Stage 1 line-up. Part IV (Modular forms and modular curves). Target length 1,970 lines of LaTeX (card excluded; see Part 3), inside the 1,000 to 2,200 range of PLAN.md Section 7.3.

Rigour tags: **P** = proved here, **S** = proof sketched here, **R** = stated; reference. Item format, as in the Chapter 2 to 9 outlines: number, label-ready name, kind, tag; the precise statement with all hypotheses; reference (book and chapter or section; a theorem number only where sure, otherwise marked "(verify)" or given by section); the paper's use by its `\label` names, or "background". Every number quoted below (indices, cusp counts and widths, elliptic points, genera, dimensions, Sturm bounds, q-expansions, Atkin-Lehner signs, Petersson norms, the paper's congruences) was checked by Python scripts in `/tmp/ch12` (`check1.py` to `check7.py`), using PARI/GP 2.17 through `cypari2` and Cremona's `ecdata` files for conductors below 10,000 (verification log, Part 6.6).

14.1 1. Purpose and global conventions

14.1.1 1.1 What this chapter is for

The paper uses modular forms in five places, and the chapter is built around them.

1. **The space $S_2(\Gamma_0(N))$, the newform $f = f_E$ and its q-expansion** (Section 1, "let $f = f_E \in S_2(\Gamma_0(N))$ be the newform attached to it"; Section 3 preamble, "the subring of $\mathrm{End}\, S_2(\Gamma_0(N))$ generated over $\mathbb{Z}$ by all T_n"; Lemma lem:old, "the forms $h(dz)\ \ldots$ span a $\mathbb{T}$-stable subspace of $S_2(\Gamma_0(N))$"). The chapter supplies the upper half plane, $\Gamma_0(N)$, the slash operator, holomorphy at the cusps, the spaces $M_k(\Gamma_0(N))$ and $S_k(\Gamma_0(N))$, q-expansions, their uniqueness (the q-expansion principle in its elementary forms), dimension formulas, and the worked example $S_2(\Gamma_0(11)) = \mathbb{C}\,\eta(z)^2\eta(11z)^2$ whose coefficients are the a_n of 11a1. The Hecke operators, newforms and eigenforms are Chapter 13's; this chapter only points forward to them.
2. **Coefficientwise congruences and integral forms** (Section 1, the definition of r_E: "$f \equiv g \pmod r$ coefficientwise for some integral cusp form g of level N orthogonal to f"; Lemma lem:crit, "$a_n(g) \equiv a_n(f) \pmod\lambda$ for every $n \geq 1$" and "the pairing $S_2(\Gamma_0(N);\mathbb{Z}) \times \mathbb{T} \to \mathbb{Z}$"; Theorem thm:twist, "preserves integrality"; Theorem thm:quant Step 1, "$S_2(\Gamma_0(N);\mathbb{Z}_\ell)$", "the congruence survives"; the Example of Section 3, "checked for $p < 500$"). The chapter supplies $S_k(\Gamma_0(N);A)$ for subrings $A \subset \mathbb{C}$ and $S_2(\Gamma_0(N);B) = S_2(\Gamma_0(N);\mathbb{Z}) \otimes B$ for any ring B (in particular $\mathbb{Z}_\ell$, $\mathbb{F}_\ell$, $\mathcal{O}_\lambda$), Shimura's integral basis theorem (R), Sturm's bound, and the consequences: $S_2(\Gamma_0(N);\mathbb{Z})$ is a $\mathbb{Z}$-structure of $S_2(\Gamma_0(N))$, saturated in $\mathbb{Z}[[q]]$, its truncated q-expansion map is a split injection, and

congruences can be checked on finitely many coefficients. It proves the paper's three congruences of Section 3 for all n (Part 7, point P1). The perfect pairing with $\mathbb{T}$ is Chapter 13's.

3. **The Petersson product and the Petersson norm** (Section 1, "the classical route to the degree is the Petersson norm ... $\deg\varphi_E$ is $\langle f,f\rangle$ divided by the area of the period lattice" (Zagier85, Watkins02); "orthogonal to f" in the definition of r_E; $f^\perp$ in Lemma lem:crit; "orthogonal to the newform f" in Lemma lem:old; $g^\perp$ in Theorem thm:twist). The chapter supplies the definition with an explicit, verified normalisation (unnormalised: $\int_{\Gamma_0(N)\backslash\mathfrak{H}} f\bar{g}\,y^k\,d\mu$), convergence, positivity, behaviour under change of group and under W_N, the identity with $\frac{i}{2}\int\omega_f\wedge\bar{\omega}_g$ in weight 2, and the numerical check $4\pi^2\langle f,f\rangle/\operatorname{vol}(\Lambda_E)=\deg\varphi_E$ on five curves of the paper's world (which fixes the normalisation that Chapter 14 must use). Rankin-Selberg is pointed to Chapters 13 and 19, the Zagier-Watkins formula with the Manin constant to Chapter 14.
4. **The Atkin-Lehner involution** w_N (Section 4.1: "$\hat{\delta}_!$ is the cup product composed with the operator $[UwU]\omega$, $w=\left(\begin{smallmatrix}0&-1\\M^\Sigma&0\end{smallmatrix}\right)$, which for weight 2 and trivial character is the Atkin-Lehner involution w_{M^Σ}"; Lemma lem:hecke and Theorem thm:quant Step 2 through it). The chapter supplies W_d for $d\,\|\,N$, its normalising property, $W_d^2\in d\Gamma_0(N)$, the involutions w_d of $X_0(N)$ and of $M_k(\Gamma_0(N))$, the identity $w_N^*(f\,dz)=(f|_2W_N)\,dz$, isometry and self-adjointness for the Petersson product, the action on cusps (not simple on q-expansions), the moduli description $w_N(E,C)=(E/C,E[N]/C)$, and the sign $f_{11}|W_{11}=-f_{11}$. The identity "the cup-product adjoint of T_n is $w_NT_nw_N$" and the pairing $\hat{\delta}_!$ are Chapter 14's.
5. $X_0(N)$ **as a curve** (Section 1 and the abstract: "$\varphi_E\colon X_0(N)\to E$ its modular parametrisation"; Section 4.1: "$H^1(X_0(M^\Sigma),\mathbb{Z}_\ell)$"). The chapter supplies $Y_0(N)=\Gamma_0(N)\backslash\mathfrak{H}$, its compactification by the cusps, the charts, the structure of a compact Riemann surface, the cusps and their widths, the elliptic points, the genus formula (as Riemann-Hurwitz for $X_0(N)\to X(1)$, Chapter 8), tables of genera, $\dim S_2(\Gamma_0(N))=g(X_0(N))$ (background to multiplicity one), the moduli interpretation by pairs (E,C) over $\mathbb{C}$, and the existence of the model over $\mathbb{Q}$ (stated). Everything about $X_0(N)$ over $\mathbb{Q}$ beyond that statement, $J_0(N)$ and the modular degree is Chapter 14's.

It also discharges two debts of earlier chapters: Chapter 9 (its 7.5(iii) and the "Uniformisation" writer note) defers here the fundamental domain of $\mathrm{SL}_2(\mathbb{Z})$, the q-expansion of j and the bijectivity of $j\colon\mathrm{SL}_2(\mathbb{Z})\backslash\mathfrak{H}\to\mathbb{C}$ (by the valence formula); Chapter 8 (its 5.5(f), 7.3(f), 8.1(f), 9.1, 9.3, 9.4 and Part 6.3) defers here the modular curve as a quotient of the upper half plane, the genus formula as an application of Riemann-Hurwitz, weight-two cusp forms as holomorphic differentials and the Petersson product.

The physics reader's complex analysis pays off here, in one sentence each, at three places: * **q-expansions are Fourier series** (6.3): a function on $\mathfrak{H}$ of period 1 in x is a Fourier series in x, and holomorphy (the Cauchy-Riemann equations) forces the n-th mode to be $a_ne^{2\pi inx}e^{-2\pi ny}$, exactly the separated solutions of Laplace's equation in a half-strip that decay away from the boundary (RHB's Fourier series and separation of variables). * **The valence formula is the residue theorem** (7.3): the number of zeros of a modular form in a fundamental domain is $\frac{1}{2\pi i}\oint f'/f\,dz$ around its boundary (the argument principle of RHB's contour-integration chapter), and the side contributions cancel in pairs by the symmetries T and S, leaving $k/12$. * **The Petersson product is an L^2 inner product** (10.1): $\langle f,g\rangle=\int f\bar{g}\,y^k\,d\mu$ is the weighted L^2 inner product of functions on a finite-area surface, like the weighted inner products of Sturm-Liouville theory in RHB, and Chapter 13's Hecke operators are self-adjoint for it, so their eigenforms are orthogonal (as eigenfunctions of a self-adjoint operator are). PLAN.md's "physics aside: Fourier series on a quotient" is the first of these, placed in 6.3 and recalled in 9.2.

14.1.2 1.2 Global conventions fixed in this chapter

- **C1 (the upper half plane).** $\mathfrak{H}=\{z=x+iy\in\mathbb{C}:y>0\}$ (the paper never names it; Chapter 9 writes $\mathfrak{H}$). $\mathfrak{H}^*=\mathfrak{H}\cup\mathbb{P}^1(\mathbb{Q})$, $\mathbb{P}^1(\mathbb{Q})=\mathbb{Q}\cup\{\infty\}$. $d\mu=dx\,dy/y^2$. $q=e^{2\pi iz}$, and $q_h=e^{2\pi iz/h}$ for a cusp of width h. The letter q is also the paper's name for a bad prime; in this chapter a prime is called p or ℓ and q is always $e^{2\pi iz}$, except in `paperlink` environments, which say so.
- **C2 (matrices).** $S=\left(\begin{smallmatrix}0&-1\\1&0\end{smallmatrix}\right)$, $T=\left(\begin{smallmatrix}1&1\\0&1\end{smallmatrix}\right)$, $\zeta_3=e^{2\pi i/3}$ (not ρ, which is a Galois representation; the matrix T is local to Sections 1 and 2 and their exercises, elsewhere T is $T_\ell E$: Appendix B, B-C15(b),

(j)). $\mathrm{GL}_2^+(\mathbb{R})$ acts on $\mathfrak{H}$ by Moebius transformations; $j(\gamma,z)=cz+d$ for $\gamma=\left(\begin{smallmatrix}a&b\\c&d\end{smallmatrix}\right)$. Indices of subgroups of $\mathrm{SL}_2(\mathbb{Z})$ are written $[\mathrm{SL}_2(\mathbb{Z}):\Gamma]$; the degree of $X(\Gamma)\to X(1)$ is the index of the images in $\mathrm{PSL}_2(\mathbb{Z})$, which equals $[\mathrm{SL}_2(\mathbb{Z}):\Gamma]$ when $-I\in\Gamma$ (always for $\Gamma_0(N)$) and half of it otherwise.

- **C3 (the slash operator).** For $\gamma\in\mathrm{GL}_2^+(\mathbb{R})$, $k\in\mathbb{Z}$ and $f\colon\mathfrak{H}\to\mathbb{C}$:

$$(f|_k\gamma)(z)=(\det\gamma)^{k/2}(cz+d)^{-k}f(\gamma z).$$

  Scalars λI with $\lambda>0$ act trivially, $-I$ acts by $(-1)^k$, and $f|_k\gamma|_k\gamma'=f|_k(\gamma\gamma')$. This is the book's normalisation (Shimura71 and Miyake use it too, verify before saying so in print); Diamond-Shurman use $(\det\gamma)^{k-1}$, which agrees in weight 2 (the only weight the paper uses) and differs by a positive scalar for $\det\gamma\neq1$ otherwise. In weight 2: $(f|_2\gamma)(z)\,dz=\gamma^*(f(z)\,dz)$, because $d(\gamma z)/dz=\det\gamma/(cz+d)^2$. The choice $(\det)^{k/2}$ makes W_N an exact involution and an isometry in every even weight (12.1, 12.2).
- **C4 (Atkin-Lehner).** $W_N=\left(\begin{smallmatrix}0&-1\\N&0\end{smallmatrix}\right)$ is the matrix, exactly the paper's w of Section 4.1 (with M^Σ in place of N); w_N denotes both the holomorphic involution $z\mapsto-1/(Nz)$ of $X_0(N)$ and the operator $f\mapsto f|_kW_N$ on forms. Some books use $\left(\begin{smallmatrix}0&1\\-N&0\end{smallmatrix}\right)=-W_N$; the action on $\mathfrak{H}$ and on forms of even weight is the same. For $d\,\|\,N$ (that is, $d\mid N$ and $\gcd(d,N/d)=1$), W_d is any integer matrix $\left(\begin{smallmatrix}dx&y\\Nz&dw\end{smallmatrix}\right)$ of determinant d.
- **C5 (the Petersson product). Unnormalised**: $\langle f,g\rangle=\langle f,g\rangle_{\Gamma_0(N)}=\int_{\Gamma_0(N)\backslash\mathfrak{H}}f(z)\overline{g(z)}\,y^k\,d\mu$ (linear in the first variable). It is the normalisation in which $4\pi^2\langle f,f\rangle=\deg\varphi_E\cdot\mathrm{vol}(\Lambda_E)$ holds for an optimal curve with Manin constant 1, checked numerically on five curves (10.4); the normalisations of Zagier85 and Watkins02 must be checked against their texts before the chapter says they agree. PARI's `mfpetersson` divides by the index $\mu(N)=[\mathrm{SL}_2(\mathbb{Z}):\Gamma_0(N)]$ (checked on five levels, 10.4); other books divide by $\mathrm{vol}(\Gamma_0(N)\backslash\mathfrak{H})=\pi\mu(N)/3$. The `convention` box says so and never mixes them.
- **C6 (coefficients and integrality).** $a_n(f)$ is the n-th coefficient of the expansion at ∞. For a subring $A\subseteq\mathbb{C}$, $M_k(\Gamma_0(N);A)$ and $S_k(\Gamma_0(N);A)$ are the forms with all $a_n\in A$. For an arbitrary commutative ring B (such as $\mathbb{Z}_\ell$, $\mathbb{F}_\ell$, $\mathcal{O}_\lambda$, which are not subrings of $\mathbb{C}$ in a preferred way), $S_k(\Gamma_0(N);B):=S_k(\Gamma_0(N);\mathbb{Z})\otimes_{\mathbb{Z}}B$, with its q-expansion map to $B[[q]]$ (injective by 11.4). "$f\equiv g\pmod{\mathfrak{a}}$ coefficientwise" means $a_n(f)-a_n(g)\in\mathfrak{a}$ for all $n\geq0$; "at all n" and "at all $p\notin\Sigma$" are distinguished every time (the paper distinguishes them in Lemma lem:crit versus Lemma lem:old).
- **C7 (counts attached to N).** $\mu(N)=[\mathrm{SL}_2(\mathbb{Z}):\Gamma_0(N)]=N\prod_{p\mid N}(1+1/p)$; $\nu_2(N)$, $\nu_3(N)$ the numbers of elliptic points of order 2, 3; $\nu_\infty(N)$ the number of cusps; $g(N)$ the genus of $X_0(N)$; $B_k(N)=\lfloor k\mu(N)/12\rfloor$ the Sturm bound ($B_2(N)=\lfloor\mu(N)/6\rfloor$).
- **C8 (newforms are not defined here).** "Newform", "eigenform", "Hecke operator", "oldform" are Chapter 13's. This chapter uses them only in examples and `paperlinks` with a forward `\Cref{ch:hecke}`; when $\dim S_2(\Gamma_0(N))=1$ the unique normalised form is called "the normalised generator", not "the newform", until Chapter 13.
- **C9 (typography).** PLAN.md Section 7.2: no dash ligatures ("Atkin-Lehner", "Riemann-Hurwitz", "Ramanujan-Petersson", "Diamond-Shurman" with plain hyphens; ranges as "1 to 8"), ASCII source, labels `...:modforms:...`, tables with `booktabs`.

14.2 2. Line-up of the core results, in teaching order

14.2.1 Section 1. The upper half plane and the action of $\mathrm{SL}_2(\mathbb{R})$

1.1 `prop:modforms:upper-half-plane` (definition and proposition; P). $\mathrm{GL}_2^+(\mathbb{R})$ acts on $\mathfrak{H}$ by $\gamma z=\frac{az+b}{cz+d}$, through $\mathrm{GL}_2^+(\mathbb{R})/\mathbb{R}^\times$. For $\gamma\in\mathrm{GL}_2^+(\mathbb{R})$ and $z\in\mathfrak{H}$: (i) $\mathrm{Im}(\gamma z)=\det\gamma\cdot\mathrm{Im}\,z/|cz+d|^2$, so $\gamma z\in\mathfrak{H}$; (ii) $\frac{d}{dz}(\gamma z)=\det\gamma/(cz+d)^2$; (iii) the cocycle identity $j(\gamma\gamma',z)=j(\gamma,\gamma' z)\,j(\gamma',z)$ for $j(\gamma,z)=cz+d$; (iv) $\mathrm{PSL}_2(\mathbb{R})=\mathrm{SL}_2(\mathbb{R})/\{\pm I\}$ acts faithfully and transitively, and the stabiliser of i in $\mathrm{SL}_2(\mathbb{R})$ is $\mathrm{SO}(2)$, so $\mathfrak{H}\simeq\mathrm{SL}_2(\mathbb{R})/\mathrm{SO}(2)$; (v) $d\mu=dx\,dy/y^2$ is $\mathrm{GL}_2^+(\mathbb{R})$-invariant (by (i), (ii) and the Jacobian $|d(\gamma z)/dz|^2$).

Reference: Shimura71 1.1; DiamondShurman 1.1 and 2.1 (cite the sections); SerreCourse VII.1. Paper: background. Physics aside (one sentence): $ds^2 = (dx^2 + dy^2)/y^2$ is the hyperbolic metric and the Moebius maps are its isometries.

1.2 `prop:modforms:element-types` (definition and proposition; P). For $\gamma \in \mathrm{SL}_2(\mathbb{R})$, $\gamma \neq \pm I$: γ is *elliptic* if $|\operatorname{tr}\gamma| < 2$ (exactly one fixed point in $\mathfrak{H}$, and γ has finite order when $\gamma \in \mathrm{SL}_2(\mathbb{Z})$, namely 3, 4 or 6), *parabolic* if $|\operatorname{tr}\gamma| = 2$ (exactly one fixed point, in $\mathbb{P}^1(\mathbb{R})$), *hyperbolic* if $|\operatorname{tr}\gamma| > 2$ (two fixed points in $\mathbb{P}^1(\mathbb{R})$, none in $\mathfrak{H}$). Proof: solve $cz^2 + (d-a)z - b = 0$; discriminant $(\operatorname{tr}\gamma)^2 - 4$. For $\gamma \in \mathrm{SL}_2(\mathbb{Z})$ elliptic, $\operatorname{tr}\gamma \in \{0, \pm 1\}$ and the characteristic polynomial is $X^2 + 1$ or $X^2 \pm X + 1$. The set of fixed points of the parabolic elements of $\mathrm{SL}_2(\mathbb{Z})$ is exactly $\mathbb{P}^1(\mathbb{Q})$ (the *cusps*). Reference: Shimura71 1.1 to 1.3 (cite the sections); DiamondShurman 2.3. Paper: background (elliptic points and cusps of 4.1 to 4.3).

14.2.2 Section 2. The modular group and its fundamental domain

2.1 `thm:modforms:fundamental-domain` (theorem; P). Let $\mathcal{F} = \{z \in \mathfrak{H} : |\operatorname{Re} z| \leq \frac{1}{2},\ |z| \geq 1\}$ and $G = \langle S, T\rangle \subseteq \mathrm{SL}_2(\mathbb{Z})$. (i) Every $z \in \mathfrak{H}$ is G-equivalent to a point of $\mathcal{F}$ (choose $\gamma \in G$ maximising $\operatorname{Im}\gamma z$, which exists by 1.1(i) since $\{(c,d) : |cz+d| \leq 1\}$ is finite, then translate by a power of T; if $|\gamma z| < 1$, S increases the imaginary part). (ii) If $z \neq z'$ in $\mathcal{F}$ are $\mathrm{SL}_2(\mathbb{Z})$-equivalent then $\operatorname{Re} z = \pm\frac{1}{2}$ and $z' = z \mp 1$, or $|z| = 1$ and $z' = -1/z$. (iii) The stabiliser of $z \in \mathcal{F}$ in $\mathrm{PSL}_2(\mathbb{Z})$ is trivial except: $z = i$ (generated by S, order 2), $z = \zeta_3$ (generated by ST, order 3), $z = -\bar{\zeta}_3$ (generated by TS, order 3). (iv) $G = \mathrm{SL}_2(\mathbb{Z})$: S and T generate, with $S^2 = (ST)^3 = -I$. (v) $\operatorname{vol}_\mu(\mathcal{F}) = \pi/3$. Reference: SerreCourse VII.1 (Theorem 1 and Theorem 2; verify the numbering); DiamondShurman 2.3 (cite the section). Paper: background (Chapter 9's deferral: "isomorphism classes of $E/\mathbb{C} = \mathrm{SL}_2(\mathbb{Z})\backslash\mathfrak{H}$").

2.2 `cor:modforms:level-one-quotient` (corollary; P). $\mathrm{SL}_2(\mathbb{Z})\backslash\mathfrak{H}$ is obtained from $\mathcal{F}$ by gluing the two vertical sides by T and the two halves of the arc by S; adding the point ∞ gives $X(1) = \mathrm{SL}_2(\mathbb{Z})\backslash\mathfrak{H}^*$, homeomorphic to a sphere (a closed disc whose boundary is folded and glued; P by an explicit cell decomposition: one vertex class $\{\zeta_3, -\bar{\zeta}_3\}$, the point i, the edges, one face, $\chi = 2$). So $X(1)$ has genus 0 before j is introduced; this feeds the genus formula 5.3 without circularity. The orbits of i and ζ_3 are the only elliptic points of $\mathrm{SL}_2(\mathbb{Z})$, with isotropy orders 2 and 3 in $\mathrm{PSL}_2(\mathbb{Z})$. Reference: DiamondShurman 2.3; SerreCourse VII.1. Paper: background.

14.2.3 Section 3. Congruence subgroups and their indices

3.1 `def:modforms:congruence-subgroups` (definitions and proposition; P). For $N \geq 1$: $\Gamma(N) = \ker(\mathrm{SL}_2(\mathbb{Z}) \to \mathrm{SL}_2(\mathbb{Z}/N))$, $\Gamma_1(N) = \{\gamma \equiv \left(\begin{smallmatrix}1 & *\\ 0 & 1\end{smallmatrix}\right) \bmod N\}$, $\Gamma_0(N) = \{\gamma \equiv \left(\begin{smallmatrix}* & *\\ 0 & *\end{smallmatrix}\right) \bmod N\}$, so $\Gamma(N) \subseteq \Gamma_1(N) \subseteq \Gamma_0(N) \subseteq \mathrm{SL}_2(\mathbb{Z})$; a *congruence subgroup* is any $\Gamma \supseteq \Gamma(N)$ for some N. (i) Reduction $\mathrm{SL}_2(\mathbb{Z}) \to \mathrm{SL}_2(\mathbb{Z}/N)$ is surjective (lift $\bar{\gamma}$ to an integer matrix of determinant $\equiv 1$, adjust a row using gcd and the Chinese remainder theorem; Chapter 2), so $\Gamma(N)$ is normal of index $\#\,\mathrm{SL}_2(\mathbb{Z}/N) = N^3\prod_{p|N}(1-p^{-2})$. (ii) $[\mathrm{SL}_2(\mathbb{Z}) : \Gamma_1(N)] = N^2\prod_{p|N}(1-p^{-2})$ (bottom rows of order N) and $[\Gamma_0(N) : \Gamma_1(N)] = \varphi(N)$ ($\gamma \mapsto d \bmod N$ onto $(\mathbb{Z}/N)^\times$). (iii) $\Gamma_0(N)\backslash\mathrm{SL}_2(\mathbb{Z}) \simeq \mathbb{P}^1(\mathbb{Z}/N)$ via $\Gamma_0(N)\gamma \mapsto (c : d)$, so $\mu(N) = [\mathrm{SL}_2(\mathbb{Z}) : \Gamma_0(N)] = \#\mathbb{P}^1(\mathbb{Z}/N) = N\prod_{p|N}(1+1/p)$. (iv) $-I \in \Gamma_0(N)$ always, and $-I \in \Gamma_1(N)$ iff $N \leq 2$. Examples: $\mu(11) = 12$, $\mu(20) = 36$, $\mu(37) = 38$, $\mu(99) = 144$, $\mu(121) = 132$, $\mu(539) = 672$, $\mu(726) = 1584$, $\mu(980) = 2016$ (verified; the orders of $\mathrm{SL}_2(\mathbb{Z}/N)$ and the $\Gamma_1(N)$ indices by enumeration for $N \leq 12$). Reference: DiamondShurman 1.2 and 3.9 (cite the sections); Shimura71 1.6. Paper: Section 1 ("$S_2(\Gamma_0(N))$"); Remark rem:free ($J_1(N)$, pull-back to $J_0(N)$: Chapter 14).

3.2 `prop:modforms:coset-domains` (proposition; P). If $\mathrm{SL}_2(\mathbb{Z}) = \bigsqcup_{i=1}^{\mu}\Gamma_0(N)\gamma_i$, then $\bigcup_i \gamma_i\mathcal{F}$ is a fundamental domain for $\Gamma_0(N)$ (every point is $\Gamma_0(N)$-equivalent to a point of it; interiors of distinct translates are disjoint and inequivalent), and $\operatorname{vol}_\mu(\Gamma_0(N)\backslash\mathfrak{H}) = \mu(N)\pi/3$. For $N = p$ prime one may take $\gamma_0 = I$ and $\gamma_{j+1} = ST^j$, $0 \leq j \leq p-1$ (bottom rows $(0:1)$ and $(1:j)$). Example: $\Gamma_0(11)$ has 12 translates of $\mathcal{F}$, eleven of them clustered at 0. Reference: DiamondShurman 2.3 (exercises; cite the section). Paper: background (10.1 integrates over this domain; Exercise 3).

14.2.4 Section 4. Cusps, widths and elliptic points of $\Gamma_0(N)$

4.1 `def:modforms:cusp-width` (definitions and proposition; P). The *cusps* of a congruence subgroup Γ are the Γ-orbits on $\mathbb{P}^1(\mathbb{Q})$, finitely many since $\mathrm{SL}_2(\mathbb{Z})$ acts transitively on $\mathbb{P}^1(\mathbb{Q})$ and $[\mathrm{SL}_2(\mathbb{Z}) : \Gamma] < \infty$. The *width* of the cusp $s = \sigma\infty$ ($\sigma \in \mathrm{SL}_2(\mathbb{Z})$) is the least $h \geq 1$ with $\pm\sigma T^h \sigma^{-1} \in \Gamma$; it depends only on the Γ-orbit of s. (i) Cusps of Γ correspond to the orbits of $\langle \pm T \rangle$ acting on the right on $\Gamma\backslash \mathrm{SL}_2(\mathbb{Z})$, and the width is the orbit length; hence $\sum_{\text{cusps}} h_s = [\mathrm{PSL}_2(\mathbb{Z}) : \bar{\Gamma}]$. (ii) When $-I \notin \Gamma$ a cusp is *irregular* if $-\sigma T^h \sigma^{-1} \in \Gamma$ but $\sigma T^h \sigma^{-1} \notin \Gamma$ (example: the cusp $1/2$ of $\Gamma_1(4)$); $\Gamma_0(N)$ has no irregular cusps since $-I \in \Gamma_0(N)$. Reference: DiamondShurman 2.4 and 3.8 (cite the sections); Shimura71 1.4. Paper: background.

4.2 `prop:modforms:cusps-gamma0` (proposition; P for N squarefree and for $N = p^2$, S in general). (i) Every cusp of $\Gamma_0(N)$ is equivalent to some a/c with $c \mid N$, $c \geq 1$, $\gcd(a, c) = 1$; a/c and a'/c' of this form are equivalent iff $c = c'$ and $a \equiv a' \pmod{\gcd(c, N/c)}$. Hence

$$\nu_\infty(N) = \sum_{c \mid N} \varphi(\gcd(c, N/c)),$$

and the cusp a/c has width $N/\gcd(c^2, N)$. (ii) $\infty \sim 1/N$ has width 1 and $0 = 0/1$ has width N. (iii) For N squarefree the cusps are $1/c$, $c \mid N$, $2^{\omega(N)}$ of them, of widths N/c. Examples (each checked by brute force on $\mathbb{P}^1(\mathbb{Z}/N)$ against the orbit lengths of T): $N = 11$: ∞, 0, widths $1, 11$; $N = 20$: six cusps, widths $1, 1, 4, 5, 5, 20$; $N = 99$: eight cusps, widths $1, 1, 1, 9, 11, 11, 11, 99$; $N = 121$: twelve cusps ∞, 0, $a/11$ ($1 \leq a \leq 10$), widths $1, 121$ and ten of width 1; $\nu_\infty(539) = 16$, $\nu_\infty(726) = \nu_\infty(980) = 48$. Reference: DiamondShurman 3.8 (cite the section); Ogg's "Rational points on certain elliptic modular curves" (1973; key `Ogg73`, shared with Chapter 14, so `refs.bib`, decision B14, important_notes.md B19). Paper: background (the cusps of $X_0(N)$; the cusp ∞ is the base point of φ_E, Chapter 14).

4.3 `prop:modforms:elliptic-gamma0` (proposition; P). The elliptic points of order 2 (respectively 3) of $\Gamma_0(N)$ on $Y_0(N)$ are in bijection with the solutions $x \bmod N$ of $x^2 + 1 \equiv 0$ (respectively $x^2 + x + 1 \equiv 0$), hence

$$\nu_2(N) = \begin{cases} 0 & 4 \mid N \\ \prod_{p \mid N} \left(1 + \left(\frac{-4}{p}\right)\right) & \text{otherwise,} \end{cases} \qquad \nu_3(N) = \begin{cases} 0 & 9 \mid N \\ \prod_{p \mid N} \left(1 + \left(\frac{-3}{p}\right)\right) & \text{otherwise,} \end{cases}$$

with $\left(\frac{-4}{2}\right) = \left(\frac{-3}{3}\right) = 0$ (Kronecker symbols, Chapter 2). Proof: an elliptic point of order 2 on $Y_0(N)$ lies over $i \in X(1)$ and corresponds to a coset $\Gamma_0(N)\gamma$ fixed by S, that is, a point $(c : d) \in \mathbb{P}^1(\mathbb{Z}/N)$ with $(d : -c) = (c : d)$; count with the Chinese remainder theorem and Hensel (Chapter 2). Likewise with ST for order 3. Examples: $\nu_2(11) = \nu_3(11) = 0$; $\nu_2(13) = \nu_3(13) = 2$; $\nu_2(37) = \nu_3(37) = 2$; $\nu_2 = \nu_3 = 0$ for $N = 121, 539, 726, 980$. Reference: DiamondShurman 3.7 and 3.8 (cite the sections); Shimura71 Proposition 1.43 (verify). Paper: background.

14.2.5 Section 5. The modular curves as compact Riemann surfaces; genus; moduli

5.1 `thm:modforms:y0-riemann-surface` (theorem; P for (i), S for (ii)). Let Γ be a congruence subgroup. (i) The action of Γ on $\mathfrak{H}$ is properly discontinuous: for compact $K \subset \mathfrak{H}$, $\{\gamma \in \Gamma : \gamma K \cap K \neq \emptyset\}$ is finite (1.1(i) bounds $|cz + d|$, hence c, d, then a, b); every point has a neighbourhood U with $\gamma U \cap U \neq \emptyset$ only for γ in its (finite cyclic) stabiliser modulo $\pm I$. Hence $Y(\Gamma) = \Gamma\backslash\mathfrak{H}$ is Hausdorff and connected. (ii) Charts: near a non-elliptic point, a local inverse of $\mathfrak{H} \to Y(\Gamma)$; near an elliptic point z_0 with stabiliser of order e in $\bar{\Gamma}$, $t = \left(\frac{z - z_0}{z - \bar{z}_0}\right)^e$. Transition maps are holomorphic, so $Y(\Gamma)$ is a Riemann surface. Reference: DiamondShurman 2.1, 2.2; Shimura71 1.5 (cite the sections). Paper: background (Section 1: $X_0(N)$).

5.2 `thm:modforms:x0-compact` (theorem; S). Topologise $\mathfrak{H}^* = \mathfrak{H} \cup \mathbb{P}^1(\mathbb{Q})$ with the horoball neighbourhoods $\{\mathrm{Im}\, z > C\} \cup \{\infty\}$ of ∞ and their $\mathrm{SL}_2(\mathbb{Z})$-translates (discs tangent to $\mathbb{R}$) at the other cusps. Then $X(\Gamma) = \Gamma\backslash\mathfrak{H}^*$ is a compact connected Hausdorff space ($\mathfrak{H}^* = \mathrm{SL}_2(\mathbb{Z})(\mathcal{F} \cup \{\infty\})$ and Γ has finite index), and with the chart $q_h = e^{2\pi i \sigma^{-1}(z)/h}$ at a cusp $s = \sigma\infty$ of width h it is a compact Riemann surface. We

write $X_0(N)$, $X_1(N)$, $X(N)$ for $\Gamma_0(N)$, $\Gamma_1(N)$, $\Gamma(N)$, and $X(1)$ for $\mathrm{SL}_2(\mathbb{Z})$. By Riemann existence and GAGA (Chapter 8, its 8.3) $X_0(N)$ is a smooth projective curve over $\mathbb{C}$, and holomorphic maps between these surfaces are morphisms. Reference: DiamondShurman 2.4; Shimura71 1.5. Paper: Section 1 and the abstract ("$\varphi_E\colon X_0(N)\to E$"), Section 4.1 ($H^1(X_0(M^\Sigma),\mathbb{Z}_\ell)$ is the H^1 of this compact Riemann surface, Chapter 8, its 10.2).

5.3 `thm:modforms:genus-formula` (theorem; P given Chapter 8's Riemann-Hurwitz, its 7.2 and 8.4(c), and 2.2). The map $\pi\colon X_0(N)\to X(1)$, $\Gamma_0(N)z\mapsto \mathrm{SL}_2(\mathbb{Z})z$, is holomorphic of degree $\mu(N)$ (since $-I\in\Gamma_0(N)$), unramified outside the images of i, ζ_3, ∞. Above i there are ν_2 points with $e=1$ and $(\mu-\nu_2)/2$ with $e=2$; above ζ_3, ν_3 points with $e=1$ and $(\mu-\nu_3)/3$ with $e=3$; above ∞ the ν_∞ cusps with e equal to the width. With $g(X(1))=0$ (2.2),

$$g(X_0(N)) = 1 + \frac{\mu(N)}{12} - \frac{\nu_2(N)}{4} - \frac{\nu_3(N)}{3} - \frac{\nu_\infty(N)}{2}.$$

Table (verified: the formula against brute-force counts on $\mathbb{P}^1(\mathbb{Z}/N)$ for every tabulated $N\le 140$, and g against PARI's `mfdim([N,2],1)`, which equals g by 8.2, for every tabulated N):

N	μ	ν_2	ν_3	ν_∞	g	N	μ	ν_2	ν_3	ν_∞	g
1	1	1	1	1	0	37	38	2	2	2	2
11	12	0	0	2	1	43	44	0	2	2	3
13	14	2	2	2	0	77	96	0	0	4	7
14	24	0	0	4	1	99	144	0	0	8	9
20	36	0	0	6	1	121	132	0	0	12	6
23	24	0	0	2	2	140	288	0	0	12	19
25	30	2	0	6	0	539	672	0	0	16	49
27	36	0	0	6	1	726	1584	0	0	48	109
36	72	0	0	12	1	980	2016	0	0	48	145

$g(X_0(N))=0$ exactly for $N\in\{1,\dots,10,12,13,16,18,25\}$ and $g=1$ for $N\in\{11,14,15,17,19,20,21,24,27,32,36,49\}$ (verified for $N\le 2000$ with the formula; that the lists are complete for all N is stated (R), and Exercise 6 gives the elementary lower bound for g that reduces it to a finite check). In particular $X_0(11)$ has genus 1 (PLAN.md). These agree with the Chapter 8 outline's list (its 5.5(f) and 6.6). Reference: DiamondShurman Theorem 3.1.1 and Section 3.9 (verify the number); Shimura71 Proposition 1.40 (verify). Paper: background (Section 1; $\dim S_2(\Gamma_0(N))$ and multiplicity one).

5.4 `thm:modforms:moduli` (theorem; P for (i) to (iii), R for (iv), (v)). (i) Every pair (E,C) with $E/\mathbb{C}$ an elliptic curve and $C\subset E$ a cyclic subgroup of order N is isomorphic to $(\mathbb{C}/\Lambda_\tau, \langle\frac{1}{N}+\Lambda_\tau\rangle)$ for some $\tau\in\mathfrak{H}$, $\Lambda_\tau=\mathbb{Z}+\mathbb{Z}\tau$, and $\Gamma_0(N)\tau\mapsto$ this pair is a bijection from $Y_0(N)$ onto isomorphism classes of such pairs (Chapter 9, its 7.5(i), plus the computation of which $\gamma\in\mathrm{SL}_2(\mathbb{Z})$ preserve $\langle\frac{1}{N}\rangle$). (ii) Likewise $Y_1(N)\leftrightarrow\{(E,P): P \text{ of exact order } N\}$ via $\Gamma_1(N)\tau\mapsto(\mathbb{C}/\Lambda_\tau,\frac{1}{N})$, and $Y(N)\leftrightarrow\{(E,(P,Q)) : \text{basis of } E[N] \text{ with Weil pairing } e^{2\pi i/N}\}$ (with the sign convention of Chapter 9, C7). (iii) The forgetful map $(E,C)\mapsto E$ is π of 5.3, and $(E,C)\mapsto(E/C,E[N]/C)$ is $\Gamma_0(N)\tau\mapsto\Gamma_0(N)(-1/(N\tau))$, that is, w_N (12.1(v)). (iv) (Cusps.) The cusps correspond to degenerate objects: generalised elliptic curves (Neron polygons) with a cyclic subgroup, or the Tate curve $\mathbb{G}_m/q^{\mathbb{Z}}$ with the subgroup μ_N (the cusp ∞) or a subgroup generated by an N-th root of q (the cusp 0); stated with reference to Deligne-Rapoport, no details. (v) ($\mathbb{Q}$-structure.) There is a smooth projective curve $X_0(N)_{\mathbb{Q}}$ over $\mathbb{Q}$ whose complex points are $X_0(N)$, such that the bijection of (i) is $G_{\mathbb{Q}}$-equivariant on $\bar{\mathbb{Q}}$-points ($\sigma\in G_{\mathbb{Q}}$ acting on pairs over $\bar{\mathbb{Q}}$ by conjugating equations); the cusps ∞ and 0 are $\mathbb{Q}$-rational and a cusp of denominator c is defined over $\mathbb{Q}(\zeta_{\gcd(c,N/c)})$. Reference: DiamondShurman 1.5 and 7.7 (cite the sections); Shimura71 6.7 (cite the section); Deligne-Rapoport (Lecture Notes in Math. 349, 1973) and Katz-Mazur (Annals of Math. Studies 108, 1985), keys `DeligneRapoport73`, `KatzMazur85`, shared with Chapter 14, so to be added to `refs.bib` (decision B14, important_notes.md B19); Ogg (1973) for the fields of definition of cusps. Paper: Section 1 ("$X_0(N)$-optimal", "$\varphi_E\colon X_0(N)\to E$"; Chapter 14 builds on (v)), Remark rem:free ($J_1(N)$, via (ii)).

14.2.6 Section 6. Modular forms, q-expansions, nebentypus

6.1 `def:modforms:slash` (definition and proposition; P). The slash operator of C3. (i) $f|_k\gamma|_k\gamma' = f|_k(\gamma\gamma')$ (by 1.1(iii)); (ii) $f|_k(\lambda I) = \mathrm{sign}(\lambda)^k f$; (iii) in weight 2, $(f|_2\gamma)(z)\,dz = \gamma^*(f(z)\,dz)$, the pullback of the differential under $z \mapsto \gamma z$ (by 1.1(ii)); (iv) $\mathrm{Im}(\gamma z)^{k/2}|f(\gamma z)| = \mathrm{Im}(z)^{k/2}|(f|_k\gamma)(z)|$ for $\gamma \in \mathrm{GL}_2^+(\mathbb{R})$ (by 1.1(i)): the function $y^{k/2}|f|$ transforms like a function. Remark: with Diamond-Shurman's $(\det\gamma)^{k-1}$ the same holds in weight 2; (iv) and the isometry 12.2 need $(\det\gamma)^{k/2}$. Reference: Shimura71 2.1; DiamondShurman 1.2 (cite the sections). Paper: Section 4.1 (the operator $[UwU]\omega$ of DFG04 is a slash operator; for weight 2 and trivial character it is $f \mapsto f|_2 W_{M^\Sigma}$).

6.2 `def:modforms:modular-form` (definition; P for the remarks). Let Γ be a congruence subgroup and $k \in \mathbb{Z}$. A *modular form of weight k for* Γ is a holomorphic $f\colon \mathfrak{H} \to \mathbb{C}$ with $f|_k\gamma = f$ for all $\gamma \in \Gamma$ that is *holomorphic at every cusp*: for every $\sigma \in \mathrm{SL}_2(\mathbb{Z})$, $f|_k\sigma$ is bounded on $\{\mathrm{Im}\, z \geq 1\}$ (equivalently, by 6.3, $f|_k\sigma = \sum_{n\geq 0} b_n q_h^n$ with h the width of $\sigma\infty$, for Γ without irregular cusps). It is a *cusp form* if moreover $b_0 = 0$ for every σ (vanishing at every cusp). $M_k(\Gamma) \supseteq S_k(\Gamma)$; $M_k(\Gamma_0(N))$, $S_k(\Gamma_0(N))$ are the spaces of the paper ("$S_2(\Gamma_0(N))$", "cusp form of level N"). Remarks (P): (a) the condition at σ depends only on the cusp $\sigma\infty$ modulo Γ; (b) if $-I \in \Gamma$ and k is odd then $M_k(\Gamma) = 0$, so for $\Gamma_0(N)$ only even k matter; (c) $M_k(\Gamma) \subseteq M_k(\Gamma')$ for $\Gamma' \subseteq \Gamma$; (d) $M_k(\Gamma)M_l(\Gamma) \subseteq M_{k+l}(\Gamma)$; (e) for $f \in M_k(\Gamma_0(M))$ and $d \geq 1$, $f(dz) = d^{-k/2}(f|_k\left(\begin{smallmatrix} d & 0 \\ 0 & 1 \end{smallmatrix}\right))(z)$ lies in $M_k(\Gamma_0(Md))$ (P; the systematic use, oldforms, is Chapter 13's, C8). Reference: DiamondShurman 1.2; Shimura71 2.1; SerreCourse VII.2. Paper: Section 1 ("$f = f_E \in S_2(\Gamma_0(N))$"), Section 3 (the forms $h(dz)$ of Lemma lem:old: (e)), Section 4.1 (f^Σ of level M^Σ, Chapter 13).

6.3 `prop:modforms:q-expansion` (proposition; P). Let $f \in M_k(\Gamma_0(N))$. (i) $f(z+1) = f(z)$, and $f(z) = \sum_{n\geq 0} a_n(f)q^n$, $q = e^{2\pi i z}$, converging absolutely and uniformly on $\{\mathrm{Im}\, z \geq y_0\}$ for every $y_0 > 0$, with $a_n(f) = \int_0^1 f(x+iy)e^{-2\pi i n(x+iy)}\,dx$ for every $y > 0$. (ii) For a holomorphic f of period 1, $f(z) = F(q)$ with F holomorphic on the punctured disc, and "bounded as $\mathrm{Im}\, z \to \infty$" is equivalent to "$F$ has a removable singularity at 0" (Riemann), that is, to $a_n = 0$ for $n < 0$; then $f(\infty) := a_0(f)$. (iii) The expansion at a cusp $s = \sigma\infty$ of width h is that of $f|_k\sigma$ in q_h. (iv) (Analytic q-expansion principle.) $f \mapsto (a_n(f))_{n\geq 0}$ is injective on $M_k(\Gamma_0(N))$ (identity theorem); more precisely, by 8.1, f is determined by $a_0, \ldots, a_{B_k(N)}$. Physics aside: (i) is the Fourier series of a function periodic in x, and holomorphy forces each mode $a_n(y)e^{2\pi i n x}$ to be $a_n e^{-2\pi n y}e^{2\pi i n x}$: Fourier series on the quotient $\mathfrak{H}/\langle T\rangle$, a half-cylinder (Part 1.1). Reference: DiamondShurman 1.1, 1.2; SerreCourse VII.2. Paper: Section 1 ("coefficientwise"), Lemma lem:crit, the Example of Section 3 (all statements about a_n are about this expansion).

6.4 `prop:modforms:hecke-bound` (proposition; P). If $f \in S_k(\Gamma_0(N))$ then $y^{k/2}|f(z)|$ is bounded on $\mathfrak{H}$ (it is $\Gamma_0(N)$-invariant by 6.1(iv), continuous, and tends to 0 at every cusp, since $|f|_k\sigma(z)| \leq Ce^{-2\pi y/h}$ near ∞), and hence $|a_n(f)| \leq C_f n^{k/2}$ for all $n \geq 1$ (integrate over $x \in [0,1]$ at height $y = 1/n$). Consequently $\sum a_n(f)n^{-s}$ converges absolutely for $\mathrm{Re}\, s > k/2 + 1$. For $f \in M_k$ not cuspidal, $a_n = O(n^{k-1+\varepsilon})$ (example E_k: $\sigma_{k-1}(n)$). Reference: SerreCourse VII.4; DiamondShurman 5.9 (cite the section). Paper: background ($L(f,s)$, Chapters 13 and 19; the convergence of the Petersson integral, 10.1).

6.5 `thm:modforms:ramanujan-petersson` (theorem; R). Let $f \in S_k(\Gamma_0(N))$ be a normalised newform (Chapter 13) and $p \nmid N$. Then $|a_p(f)| \leq 2p^{(k-1)/2}$; equivalently the roots α_p, β_p of $X^2 - a_p(f)X + p^{k-1}$ satisfy $|\alpha_p| = |\beta_p| = p^{(k-1)/2}$, so $\beta_p = \bar{\alpha}_p$ and $|\alpha_p/\beta_p| = 1$; and $a_n(f) = O(n^{(k-1)/2+\varepsilon})$. For $k = 2$ this is Eichler-Shimura plus Weil's Riemann hypothesis for curves over $\mathbb{F}_p$ (Chapter 8, its 11.8(d); Chapter 14); for $f = f_E$ it is Hasse's bound (Chapter 9); in general Deligne. Checks: $|a_p(11a1)| \leq 2\sqrt{p}$ for all $p < 10^5$; Ramanujan's $|\tau(p)| \leq 2p^{11/2}$ for $p < 2000$. Reference: DiamondShurman, Chapter 8 (statement; verify the section before citing it); Deligne, La conjecture de Weil I (1974; `Deligne` in `refs.bib`). Paper: background (Lemma lem:euler and Theorem thm:dfg14 use α, β with $\alpha\beta = q$; the level-raising factor $(1 - \frac{\alpha}{\beta}q^{-1})(1 - q^{-1})(1 - \frac{\beta}{\alpha}q^{-1})$ is a nonzero real number because $|\alpha/\beta| = 1$).

6.6 `def:modforms:nebentypus` (definition and proposition; P for (i), (ii), S for (iii)). Let χ be a Dirichlet character modulo N. $M_k(\Gamma_0(N), \chi) = \{f \in M_k(\Gamma_1(N)) : f|_k\gamma = \chi(d)f\ \forall \gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \Gamma_0(N)\}$, and $S_k(\Gamma_0(N), \chi)$ likewise; χ is the *nebentypus* (character). (i) $M_k(\Gamma_0(N), \mathbf{1}) = M_k(\Gamma_0(N))$. (ii) $M_k(\Gamma_0(N), \chi) = 0$ unless $\chi(-1) = (-1)^k$ (apply $\gamma = -I$). (iii) The *diamond operators* $\langle d\rangle f = f|_k\gamma$ ($\gamma \in \Gamma_0(N)$ with lower right entry $\equiv d$) give an action of $(\mathbb{Z}/N)^\times$ on $M_k(\Gamma_1(N))$ and $M_k(\Gamma_1(N)) = \bigoplus_\chi M_k(\Gamma_0(N), \chi)$, S_k likewise (characters of a finite abelian group, Chapter 4). Remark (forward pointer):

the twist of $f \in S_k(\Gamma_0(M))$ by a primitive character χ of conductor r has character χ^2 and level dividing $\mathrm{lcm}(M, r^2)$ (Chapter 13); for the quadratic characters of the paper $\chi^2 = \mathbf{1}$. Reference: DiamondShurman 4.3, 5.2 (cite the sections); Shimura71 3.6 (cite the section). Paper: the optimal-level theorem of Section 3 ("nebentypus of order prime to ℓ"; "$\det\bar\sigma = \bar\chi_\ell$ forces the nebentypus ... to be trivial", Chapter 15), Lemma lem:twist(iii) and Theorem thm:twist (characters of twists, Chapter 13), Remark rem:free ($J_1(N)$).

14.2.7 Section 7. Level one: Eisenstein series, Δ, j and the valence formula

7.1 `prop:modforms:eisenstein-level-one` (proposition; P for $k \geq 4$, S for E_2). For even $k \geq 4$, $G_k(z) = \sum_{(m,n)\neq(0,0)} (mz+n)^{-k} = 2\zeta(k)E_k(z)$ with

$$E_k(z) = 1 - \frac{2k}{B_k}\sum_{n\geq 1}\sigma_{k-1}(n)q^n \in M_k(\mathrm{SL}_2(\mathbb{Z})), \qquad E_4 = 1+240\sum\sigma_3(n)q^n, \quad E_6 = 1-504\sum\sigma_5(n)q^n.$$

Proof: absolute convergence for $k > 2$; $G_k|_k\gamma = G_k$ by reindexing the lattice; the q-expansion from the Lipschitz formula $\sum_{n\in\mathbb{Z}}(z+n)^{-k} = \frac{(-2\pi i)^k}{(k-1)!}\sum_{m\geq 1} m^{k-1}q^m$, obtained by differentiating $\pi\cot\pi z = \frac{1}{z} + \sum_{n\neq 0}(\frac{1}{z+n} - \frac{1}{n})$ (the partial-fraction expansion of RHB, or Poisson summation). $G_k(z) = G_k(\Lambda_z)$ is Chapter 9's lattice sum (its 7.1). $E_2 = 1 - 24\sum\sigma(n)q^n$ converges but is not modular: $E_2(-1/z) = z^2E_2(z) + \frac{6z}{\pi i}$ (S, or R: DiamondShurman 1.2), equivalently $E_2(z) - \frac{3}{\pi y}$ is invariant of weight 2. Reference: SerreCourse VII.2.3 and VII.4.2; DiamondShurman 1.1, 1.2. Paper: background (the Eisenstein case of Section sec:eis and Conjecture conj:eis via 8.4; Chapter 14).

7.2 `prop:modforms:delta-j` (proposition; P for (i), (ii), S for (iii)). (i) $\Delta = (E_4^3 - E_6^2)/1728 \in S_{12}(\mathrm{SL}_2(\mathbb{Z}))$, nonzero, $\Delta = \sum\tau(n)q^n = q - 24q^2 + 252q^3 - 1472q^4 + 4830q^5 - \cdots$ with $\tau(n) \in \mathbb{Z}$; in Chapter 9's notation $\Delta(\Lambda_z) = (2\pi)^{12}\Delta(z)$. (ii) $j = E_4^3/\Delta = 1728E_4^3/(E_4^3 - E_6^2)$ is holomorphic on $\mathfrak{H}$ (Δ has no zero on $\mathfrak{H}$, 7.4(ii), which uses only 7.3), $\mathrm{SL}_2(\mathbb{Z})$-invariant, with $j(z) = q^{-1} + 744 + 196884q + 21493760q^2 + 864299970q^3 + \cdots$, all coefficients integers; $j(z) = j(\Lambda_z)$ of Chapter 9 (its 7.1, 7.5(ii)); $j(i) = 1728$, $j(\zeta_3) = 0$. (iii) (Jacobi.) $\Delta = q\prod_{n\geq 1}(1-q^n)^{24} = \eta^{24}$ (S via 9.1 and $\dim S_{12} = 1$). Reference: SerreCourse VII.3.4 and VII.4.4; DiamondShurman 1.1. Paper: background (Chapter 9's j, Δ; Definition def:terms uses $v_q(j_E)$, which is Chapter 9's algebraic j, equal to this j at the lattice of E).

7.3 `thm:modforms:valence-level-one` (theorem, the valence formula; P). Let $f \neq 0$ be meromorphic on $\mathfrak{H}$, with $f|_k\gamma = f$ for $\gamma \in \mathrm{SL}_2(\mathbb{Z})$, and meromorphic at ∞ ($f = \sum_{n\geq n_0} a_n q^n$). Then

$$v_\infty(f) + \tfrac{1}{2}v_i(f) + \tfrac{1}{3}v_{\zeta_3}(f) + \sum_{p\in\mathrm{SL}_2(\mathbb{Z})\backslash\mathfrak{H},\ p\neq i,\zeta_3} v_p(f) = \frac{k}{12},$$

where v_p is the order at any representative and v_∞ the order in q. Proof: integrate $\frac{1}{2\pi i}f'/f$ around the boundary of $\mathcal{F}$ truncated at height Y and indented by small arcs at i, ζ_3, $-\bar\zeta_3$ (assume no other zeros on the boundary, else indent there too): the top segment gives $-v_\infty$; the vertical sides cancel (T); the two halves of the arc are related by S, and $\frac{f'}{f}(Sz)\,d(Sz) = \frac{f'}{f}(z)\,dz + k\frac{dz}{z}$ gives $\frac{k}{12}$; the indentations give $-\frac{1}{2}v_i$ and $-\frac{1}{3}v_{\zeta_3}$ (angles π and $\pi/3 + \pi/3$). Physics aside: the argument principle counts the "charge" inside a contour (RHB, residue theorem), and the symmetries cancel the sides. Reference: SerreCourse VII.3 (Theorem 3; verify); DiamondShurman 3.1 (exercises). Paper: background (7.4, 7.5; the Sturm bound 11.3).

7.4 `cor:modforms:level-one-structure` (corollary; P). (i) $M_k(\mathrm{SL}_2(\mathbb{Z})) = 0$ for $k < 0$, k odd, and $k = 2$; $M_0 = \mathbb{C}$; $M_k = \mathbb{C}E_k$ for $k = 4, 6, 8, 10, 14$. (ii) Δ has no zero on $\mathfrak{H}$ and a simple zero at ∞; E_4 vanishes only at ζ_3 (simply) and E_6 only at i (simply). (iii) Multiplication by Δ is an isomorphism $M_{k-12} \to S_k$, and $\dim M_k = \lfloor k/12\rfloor$ if $k \equiv 2 \pmod{12}$, $\lfloor k/12\rfloor + 1$ otherwise ($k \geq 0$ even). (iv) $M_* = \bigoplus_k M_k = \mathbb{C}[E_4, E_6]$, E_4, E_6 algebraically independent. (Dimensions verified against PARI for $k \leq 26$.) Reference: SerreCourse VII.3.2 (Theorem 4; verify); DiamondShurman 3.5, 3.6. Paper: background.

7.5 `thm:modforms:j-bijective` (theorem; P). $j\colon \mathrm{SL}_2(\mathbb{Z})\backslash\mathfrak{H} \to \mathbb{C}$ is a bijection, and it extends to an isomorphism of Riemann surfaces $X(1) \to \mathbb{P}^1(\mathbb{C})$ with $j(\infty) = \infty$. Proof: for $\lambda \in \mathbb{C}$, $f_\lambda = E_4^3 - \lambda\Delta \in M_{12}$

has $v_\infty(f_\lambda) = 0$, so by 7.3 $n + \frac{a}{2} + \frac{b}{3} = 1$ with $n, a, b \geq 0$ integers, whose only solutions are $(1,0,0)$, $(0,2,0)$, $(0,0,3)$: exactly one point of $\mathrm{SL}_2(\mathbb{Z})\backslash\mathfrak{H}$ has $j = \lambda$. Holomorphy and injectivity on $X(1)$ make j a degree-one holomorphic map, hence an isomorphism (Chapter 8, 8.2). Consequence (completing Chapter 9, its 7.3(ii) and 7.5(iii), with no circularity: nothing here uses the uniformisation theorem): for every $j_0 \in \mathbb{C}$ there is a lattice Λ with $j(\Lambda) = j_0$, so every elliptic curve over $\mathbb{C}$ is $\mathbb{C}/\Lambda$ (with Chapter 9, its 1.4 and 7.4). Reference: SerreCourse VII.3.3 (Proposition 5; verify); DiamondShurman 1.4 and 2.3. Paper: background (Chapter 9's deferral; the period lattice Λ_E in Section 1).

14.2.8 Section 8. $\Gamma_0(N)$: valence, dimensions, weight-two forms as differentials, weight-two Eisenstein series

8.1 `thm:modforms:valence-gamma0` (theorem; S). Let $f \in M_k(\Gamma_0(N))$, $f \neq 0$. For $P \in X_0(N)$ let $v_P(f)$ be the order of f at a representative $z \in \mathfrak{H}$ (in the variable z) if $P \in Y_0(N)$, and the order in q_h of the expansion at the cusp if P is a cusp, and $e_P \in \{1,2,3\}$ the order of the stabiliser of z in the image $\bar{\Gamma}_0(N)$ of $\Gamma_0(N)$ in $\mathrm{PSL}_2(\mathbb{Z})$ (and $e_P = 1$ at cusps). Then

$$\sum_{P \in X_0(N)} \frac{v_P(f)}{e_P} = \frac{k\,\mu(N)}{12}.$$

In particular $v_\infty(f) \leq k\mu(N)/12$, so f is determined by $a_0(f), \ldots, a_{B_k(N)}(f)$. Sketch: apply 7.3 to the norm $F = \prod_{i=1}^{\mu} f|_k\gamma_i \in M_{k\mu}(\mathrm{SL}_2(\mathbb{Z}))$ (independent of the choice of coset representatives) and compare orders point by point over $X(1)$ using the ramification of 5.3; or, for $k = 2$, use 8.2 and $\deg \mathrm{div}(\omega) = 2g-2$ (Chapter 8, its 5.3(b)) with the corrections at cusps and elliptic points. Example: $f_{11} = \eta(z)^2\eta(11z)^2$ (9.2) has simple zeros at ∞ and at 0 and none elsewhere: $1 + 1 = 2 \cdot 12/12$. Reference: DiamondShurman 3.1 (exercises), Stein (book) Chapter 9 (cite the chapter); Shimura71 2.4. Paper: background (11.3, 6.3(iv)).

8.2 `thm:modforms:weight-two-differentials` (theorem; P given Chapter 8, its 9.1 and 9.3). (i) $f \mapsto \omega_f = f(z)\,dz$ is an isomorphism from $S_2(\Gamma_0(N))$ onto the space $H^0(X_0(N), \Omega^1)$ of holomorphic differentials on $X_0(N)$. (ii) $M_2(\Gamma_0(N))$ maps isomorphically onto the meromorphic differentials that are holomorphic on $Y_0(N)$ and have at most simple poles at the cusps, and $\mathrm{res}_s\,\omega_f = \frac{h_s}{2\pi i} a_0(f|_2\sigma_s)$ at the cusp $s = \sigma_s\infty$ of width h_s. (iii) Hence $\dim S_2(\Gamma_0(N)) = g(X_0(N))$ (Chapter 8, its 9.3), and $\sum_s h_s\, a_0(f|_2\sigma_s) = 0$ for every $f \in M_2(\Gamma_0(N))$ (residue theorem, Chapter 8, its 9.2). Proof: $\Gamma_0(N)$-invariance of $f\,dz$ is weight 2 by 6.1(iii); at a cusp of width h, $dz = \frac{h}{2\pi i}\frac{dq_h}{q_h}$; at an elliptic point of order e with local coordinate w on a disc and chart $t = w^e$, an invariant $F(w)\,dw$ has $F(\zeta w)\zeta = F(w)$ for $\zeta = e^{2\pi i/e}$, so $F(w) = w^{e-1}G(w^e)$ and $F\,dw = \frac{1}{e}G(t)\,dt$ is holomorphic in t: no condition at elliptic points in weight 2. Examples: $S_2(\mathrm{SL}_2(\mathbb{Z})) = 0$ and $g(X(1)) = 0$; $\dim S_2(\Gamma_0(11)) = 1$, $\dim S_2(\Gamma_0(121)) = 6$, $\dim S_2(\Gamma_0(539)) = 49$ (PARI agrees, Part 6.6). Reference: DiamondShurman 3.3 (cite the section); Shimura71 2.4. Paper: background ("$\dim S_2$ and multiplicity one"; Section 1: $\varphi_E^*\omega_E$ is a multiple of ω_f, Chapter 14; Section 4.1: $H^1(X_0(N), \mathbb{C}) = H^0(\Omega^1) \oplus \overline{H^0(\Omega^1)} = \omega_{S_2} \oplus \overline{\omega_{S_2}}$ by Chapter 8, its 10.2(iii)).

8.3 `thm:modforms:dimension-formulas` (theorem; P for $k = 2$, R for $k \geq 4$ with a sketch via Riemann-Roch). For $k \geq 4$ even:

$$\dim S_k(\Gamma_0(N)) = (k-1)(g-1) + \Big(\frac{k}{2} - 1\Big)\nu_\infty + \Big\lfloor \frac{k}{4} \Big\rfloor \nu_2 + \Big\lfloor \frac{k}{3} \Big\rfloor \nu_3, \qquad \dim M_k(\Gamma_0(N)) = \dim S_k(\Gamma_0(N)) + \nu_\infty;$$

for $k = 2$: $\dim S_2 = g$, $\dim M_2 = g + \nu_\infty - 1$; $M_0 = \mathbb{C}$, $S_0 = 0$; $M_k = 0$ for $k < 0$ or k odd. (Verified against PARI for all $N \leq 60$ and $N \in \{77, 99, 121, 539\}$, $k \in \{2, 4, \ldots, 14\}$.) Sketch for $k \geq 4$: $f \mapsto f\,(dz)^{k/2}$ identifies M_k with the sections of a line bundle of degree $\frac{k}{2}(2g-2) + \frac{k}{2}\nu_\infty + \lfloor\frac{k}{4}\rfloor\nu_2 + \lfloor\frac{k}{3}\rfloor\nu_3$ (corrections at cusps and elliptic points), and Riemann-Roch (Chapter 8, its 5.2, 5.3(c)) gives the dimension since the degree exceeds $2g-2$. Reference: DiamondShurman Theorems 3.5.1 and 3.6.1 (verify the numbers; else cite Sections 3.5, 3.6); Shimura71 2.6. Paper: background.

8.4 `prop:modforms:eisenstein-weight-two` (proposition; P given 7.1). For $N > 1$, $E_2^{(N)}(z) := E_2(z) - NE_2(Nz)$ lies in $M_2(\Gamma_0(N))$ (it equals $E_2^*(z) - (E_2^*|_2\left(\begin{smallmatrix} N & 0 \\ 0 & 1 \end{smallmatrix}\right))(z)$ with $E_2^* = E_2 - \frac{3}{\pi y}$ invariant), is not a cusp form ($a_0 = 1 - N$), and $E_2^{(N)} = (1-N) - 24\sum_{n\geq 1}(\sigma(n) - N\sigma(n/N))q^n$ ($\sigma(n/N) = 0$ if $N \nmid n$).

For $N = p$ prime, $M_2(\Gamma_0(p)) = \mathbb{C}E_2^{(p)} \oplus S_2(\Gamma_0(p))$ ($\dim M_2 = g+1$ by 8.3). Worked example (P, given 11.3; verified for $n \le 5000$): for $p = 11$, the form $-\frac{1}{24}E_2^{(11)} = \frac{5}{12} + \sum_{n\ge1}(\sigma(n) - 11\sigma(n/11))q^n$ has 5-integral coefficients and is congruent modulo 5 to f_{11} (9.2), coefficientwise (apply 11.3 to $12(f_{11} + \frac{1}{24}E_2^{(11)}) \in M_2(\Gamma_0(11);\mathbb{Z})$), because the two agree modulo 5 at $n = 0, 1, 2$ ($\frac{5}{12} \equiv 0$; $1 = 1$; $3 \equiv -2$) and the Sturm bound $B_2(11)$ is 2. Hence $a_\ell(11a1) \equiv 1 + \ell \pmod 5$ for every prime $\ell \neq 11$ (and $a_{11} = 1$): the Eisenstein congruence behind the rational 5-torsion of 11a1 (Chapter 9, its 9.7). Reference: DiamondShurman 4.6 (cite the section); Mazur77 (Eisenstein ideal; Chapter 14). Paper: Conjecture conj:eis and Section sec:eis (the Eisenstein case: reducible $\bar\rho$, congruence with an Eisenstein series; Chapter 14 for the Eisenstein ideal), Open question 4.

14.2.9 Section 9. The Dedekind η-function and the level-11 example

9.1 `prop:modforms:eta` (proposition; R for the transformation law, P for the rest). $\eta(z) = q^{1/24}\prod_{n\ge1}(1-q^n)$ (with $q^{1/24} = e^{2\pi iz/24}$) is holomorphic and nonvanishing on $\mathfrak{H}$, $\eta(z+1) = e^{\pi i/12}\eta(z)$ and $\eta(-1/z) = \sqrt{z/i}\,\eta(z)$ (principal branch). Consequences: $\eta^{24} \in S_{12}(\mathrm{SL}_2(\mathbb{Z}))$, hence $\eta^{24} = \Delta$ (7.4(iii)); $\frac{\eta'}{\eta} = \frac{\pi i}{12}E_2$ (logarithmic derivative), which is equivalent to the transformation law of E_2 in 7.1. Reference: Apostol, *Modular functions and Dirichlet series in number theory*, Chapter 3 (add to `bib/ch12.bib`); Knapp, Chapter IX (cite the chapter); Koblitz III.2. Paper: background.

9.2 `thm:modforms:level-eleven` (theorem and worked example; P for (i) to (iii), R for (iv)). Let $f_{11}(z) = \eta(z)^2\eta(11z)^2 = q\prod_{n\ge1}(1-q^n)^2(1-q^{11n})^2$. (i) $f_{11} \in S_2(\Gamma_0(11))$ (from 9.1, checking $f_{11}|_2\gamma = f_{11}$ on generators, or by the criterion of Ligozat for η-quotients, R: $\sum\delta r_\delta = 24 \equiv 0$, $\sum(N/\delta)r_\delta = 24 \equiv 0 \pmod{24}$, $\prod\delta^{r_\delta} = 11^2$ a square); its orders at the cusps ∞ and 0 are 1 and 1. (ii) $S_2(\Gamma_0(11)) = \mathbb{C}f_{11}$ (dimension $g(X_0(11)) = 1$, 8.2), so f_{11} is the normalised generator ($a_1 = 1$). (iii) $f_{11}|_2W_{11} = -f_{11}$: from 9.1, $f_{11}(-1/(11z)) = -11z^2 f_{11}(z)$, so $(f_{11}|_2W_{11})(z) = 11(11z)^{-2}f_{11}(-1/(11z)) = -f_{11}(z)$ (also checked numerically at three points, and $f_{11}(i/\sqrt{11}) \approx 0.1029 > 0$ pins the sign; PARI's `mfatkineigenvalues` gives -1). (iv) Its coefficients are those of 11a1:

$$f_{11} = q-2q^2-q^3+2q^4+q^5+2q^6-2q^7-2q^9-2q^{10}+q^{11}-2q^{12}+4q^{13}+4q^{14}-q^{15}-4q^{16}-2q^{17}+4q^{18}+2q^{20}+\cdots,$$

a_p for $p = 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37$: $-2, -1, 1, -2, 1, 4, -2, 0, -1, 0, 7, 3$, equal to $p+1-\#\tilde E(\mathbb{F}_p)$ for $E =$ 11a1: $y^2 + y = x^3 - x^2 - 10x - 20$ (Chapter 9, its 8.7). Checked for all $n \le 2000$; the same for 11a2 and 11a3 (isogenous). That this holds for all n is Eichler-Shimura for $X_0(11)$, which is itself the elliptic curve 11a1 (Chapter 14; R). Physics aside, recalled from 6.3: f_{11} is a Fourier series on the quotient. Further η-products (verified to $n = 300$ against Cremona's curves): $\eta(z)\eta(2z)\eta(7z)\eta(14z)$ (14a1), $\eta(z)\eta(3z)\eta(5z)\eta(15z)$ (15a1), $\eta(2z)^2\eta(10z)^2$ (20a1), $\eta(2z)\eta(4z)\eta(6z)\eta(12z)$ (24a1), $\eta(3z)^2\eta(9z)^2$ (27a1), $\eta(4z)^2\eta(8z)^2$ (32a1), $\eta(6z)^4$ (36a1). Reference: DiamondShurman 1.2, 3.9 (exercises) and 6.5 (cite the sections); Knapp, Chapter XI (verify); Ligozat (1975; add to `bib/ch12.bib` if cited). Paper: the running example (Section 3 Example: "the form of 11a1"; Theorem thm:qual (a), (b) via the congruences with f_{11}; 121d $= f_{11a} \otimes \chi_{-11}$), PLAN.md's example "$S_2(\Gamma_0(11))$ is one-dimensional spanned by $\eta(z)^2\eta(11z)^2$".

9.3 `ex:modforms:small-levels` (example; P for dimensions, R for the identifications). For each N with $g(X_0(N)) = 1$ ($N = 11, 14, 15, 17, 19, 20, 21, 24, 27, 32, 36, 49$), $S_2(\Gamma_0(N))$ is spanned by the form of the strong Weil curve of conductor N (Chapter 14). $\dim S_2(\Gamma_0(37)) = 2$: the two newforms of 37a1 and 37b1, with W_{37}-eigenvalues $+1$ and -1 (12.4(c)). $\dim S_2(\Gamma_0(121)) = 6$ (four newforms, 121a to 121d, and the two oldforms $f_{11}(z)$, $f_{11}(11z)$; Chapter 13). Paper: the levels of Section 3 (11, 20, 77, 121, 140, 539, 980) and the dimension counts behind Lemma lem:old (Chapter 13).

14.2.10 Section 10. The Petersson inner product

10.1 `def:modforms:petersson` (definition and proposition; P). For $f \in S_k(\Gamma_0(N))$ and $g \in M_k(\Gamma_0(N))$,

$$\langle f, g\rangle = \langle f, g\rangle_{\Gamma_0(N)} = \int_{\Gamma_0(N)\backslash\mathfrak{H}} f(z)\overline{g(z)}\,y^k\,\frac{dx\,dy}{y^2},$$

the integral over any fundamental domain (3.2). (i) The integrand is $\Gamma_0(N)$-invariant (6.1(iv) and 1.1(v)), so the value does not depend on the fundamental domain. (ii) It converges absolutely: on $\sigma_i\mathcal{F}$ near the cusp, $|(f|_k\sigma_i)(z)| \le Ce^{-2\pi y/h}$ and $|g|_k\sigma_i|$ is bounded, against the finite measure $y^{k-2}dx\,dy$ times an exponential. (iii) On $S_k(\Gamma_0(N))$ it is a positive definite Hermitian inner product (sesquilinear; $\langle f,f\rangle = \int |f|^2 y^k\,d\mu > 0$ for $f \neq 0$). (iv) In weight 2: $\langle f,g\rangle = \frac{i}{2}\int_{X_0(N)} \omega_f \wedge \overline{\omega_g} = \iint f\bar{g}\,dx\,dy$ (Chapter 8, its 9.4(b)): the Petersson product is the Hodge inner product of the holomorphic differentials of 8.2. Physics aside: a weighted L^2 inner product on a surface of finite area, as in Sturm-Liouville theory. The `convention` box C5 (unnormalised; PARI divides by $\mu(N)$, others by $\pi\mu(N)/3$). Reference: DiamondShurman 5.4 (cite the section); Shimura71 3.4; SerreCourse VII.5 (remark). Paper: Section 1 ("the Petersson norm", "$\langle f,f\rangle$"), the definition of r_E ("orthogonal to f"), Lemma lem:crit ($f^\perp$), Lemma lem:old ("orthogonal to the newform f"), Theorem thm:twist ($g^\perp$; the paper there uses the $\mathbb{T}$-stable complement, which Chapter 13 shows is the Petersson complement).

10.2 `prop:modforms:petersson-change` (proposition; P). (i) If $\Gamma' \subseteq \Gamma_0(N)$ has finite index then $\langle f,g\rangle_{\Gamma'} = [\bar{\Gamma}_0(N) : \bar{\Gamma}']\,\langle f,g\rangle_{\Gamma_0(N)}$ (bars: images in $\mathrm{PSL}_2(\mathbb{Z})$); in particular $\langle f,g\rangle_{\Gamma_0(NM)} = \frac{\mu(NM)}{\mu(N)}\langle f,g\rangle_{\Gamma_0(N)}$ for f,g of level N viewed at level NM. This is why the normalisation matters when levels change (Chapter 13's oldforms, the paper's M^Σ). (ii) If $\alpha \in \mathrm{GL}_2^+(\mathbb{Q})$ normalises $\Gamma_0(N)$ (for instance $\alpha = W_d$, 12.1), then $\langle f|_k\alpha, g|_k\alpha\rangle = \langle f,g\rangle$ (change of variables $z \mapsto \alpha z$ and 6.1(iv)). (iii) More generally, for $\alpha \in \mathrm{GL}_2^+(\mathbb{Q})$, $\langle f|_k\alpha, g|_k\alpha\rangle_{\alpha^{-1}\Gamma\alpha} = \langle f,g\rangle_\Gamma$. Reference: DiamondShurman 5.4, 5.5 (cite the sections). Paper: background (Section 4.1 changes level from M to M^Σ; Theorem thm:quant Step 1 works at level N with forms of level N_S).

10.3 `prop:modforms:orthogonal-complement` (proposition; P, linear algebra). For $0 \neq f \in S_k(\Gamma_0(N))$, $S_k(\Gamma_0(N)) = \mathbb{C}f \oplus f^\perp$, $f^\perp = \{g : \langle g,f\rangle = 0\}$, and the projection onto $\mathbb{C}f$ along $f^\perp$ is $g \mapsto \frac{\langle g,f\rangle}{\langle f,f\rangle} f$. Forward pointer (Chapter 13): when f is a newform with rational coefficients, $f^\perp$ is stable under all Hecke operators (including U_p, whose Petersson adjoint is $w_N U_p w_N$, with f an eigenvector of w_N), is spanned by forms with rational coefficients, and $L = f^\perp \cap S_2(\Gamma_0(N);\mathbb{Z})$ is a saturated sublattice of rank $\dim S_2 - 1$; none of this is proved here. Paper: Lemma lem:crit ("J the annihilator in $\mathbb{T}$ of the orthogonal complement $f^\perp$", "projection onto $\mathbb{Q}f$ along $f^\perp$").

10.4 `ex:modforms:petersson-numerics` (worked example; P as a computation, verified with PARI). Unnormalised Petersson norms of the normalised forms of five optimal curves, with the area of the period lattice (Chapter 9, its 7.6) and the modular degree (from `ellmoddegree`; Chapter 14):

curve	N	$\mu(N)$	$\langle f,f\rangle$	$\mathrm{vol}(\Lambda_E)$	$4\pi^2\langle f,f\rangle/\mathrm{vol}(\Lambda_E)$	$\deg\varphi_E$
11a1	11	12	0.046900147873495215436234559	1.60000000	1	1
20a1	20	36	0.081349456825338215478284400	1.70000000	1	1
37a1	37	38	0.371754147510796813274078928	2.80000000	2	2
99a1	99	144	0.538519568710863074975105421	5.40000000	4	4
121a1	121	132	0.428239360011957702048195	4.80000000	6	6

(PARI's `mfpetersson` returns $\langle f,f\rangle/\mu(N)$, for example 0.0039083456561246 for 11a1.) So with C5, $\deg\varphi_E = 4\pi^2\langle f,f\rangle/\mathrm{vol}(\Lambda_E)$ for these curves, whose Manin constant is $c_E = 1$. The derivation is one line (Chapter 14 does it): if $\varphi_E^*\omega_E = c_E \cdot 2\pi i\, f(z)\,dz$ then $\deg\varphi_E \cdot \frac{i}{2}\int_E \omega_E \wedge \bar{\omega}_E = \frac{i}{2}\int_{X_0(N)} \varphi_E^*\omega_E \wedge \overline{\varphi_E^*\omega_E}$, that is, $\deg\varphi_E \cdot \mathrm{vol}(\Lambda_E) = 4\pi^2 c_E^2\langle f,f\rangle$ (Chapter 8, its 9.4(b), 11.5(i)). Note where c_E^2 stands (Part 7, B1). Paper: Section 1 ("$\deg\varphi_E$ is $\langle f,f\rangle$ divided by the area of the period lattice of E", Zagier85, Watkins02, Cremona97); the exact formula with the Manin constant is Chapter 14's.

10.5 `rem:modforms:petersson-forward` (remark; R, forward pointers only). (a) (Chapter 13.) For $(n,N) = 1$ the Hecke operator T_n is self-adjoint for $\langle\ ,\ \rangle$; U_p ($p \mid N$) is not, its adjoint being $w_N U_p w_N$; newforms of different eigensystems are orthogonal, and oldforms are orthogonal to newforms. (b) (Chapters 13 and 19.) Rankin-Selberg: for a newform f of weight 2 and level N, $\langle f,f\rangle$ is an explicit elementary factor times $L(\mathrm{Sym}^2 f, 2)$ times correction factors at the primes dividing N (DFG04 Lemma 2.12, Shimura76); Chapter 19 states the exact formula. The Rankin series $\sum a_n(f)^2 n^{-s}$ is **not** the naive adjoint (symmetric square) L-function: at a good prime it carries an extra factor $1 + p^{1-s}$, globally

$\sum a_n^2 n^{-s} = \zeta^{(N)}(s-1)L^{\mathrm{nv}}(\mathrm{Sym}^2 f, s)/\zeta^{(N)}(2s-2)$ up to bad factors (important_notes.md point 30). (c) (Chapter 14.) The Zagier-Watkins formula with the Manin constant (10.4). Paper: Section 1 (Petersson norm, Rankin, Shimura, Flach, "the Euler product of $\sum a_n(f)^2 n^{-s}$", point 30).

14.2.11 Section 11. Integral forms, the Sturm bound and congruences

11.1 `def:modforms:integral-forms` (definition; P for the remarks). For a subring $A \subseteq \mathbb{C}$: $M_k(\Gamma_0(N); A) = \{f \in M_k(\Gamma_0(N)) : a_n(f) \in A\ \forall n \geq 0\}$ and $S_k(\Gamma_0(N); A)$ likewise (an A-module); for $A = \mathbb{Z}$ these are the paper's "integral cusp forms" and $S_2(\Gamma_0(N); \mathbb{Z})$. For any commutative ring B: $S_k(\Gamma_0(N); B) := S_k(\Gamma_0(N); \mathbb{Z}) \otimes_{\mathbb{Z}} B$, with the q-expansion map $S_k(\Gamma_0(N); B) \to B[[q]]$, $f \otimes b \mapsto b \sum a_n(f) q^n$ (C6); for $B = \mathbb{Z}_\ell$ this is the paper's $S_2(\Gamma_0(N); \mathbb{Z}_\ell)$. Remarks: (a) $S_k(\Gamma_0(N); \mathbb{Z})$ is saturated in $\mathbb{Z}[[q]]$ by definition: if $mg \in S_k(\Gamma_0(N); \mathbb{Z})$ with $g \in \mathbb{Z}[[q]]$ then $g \in S_k(\Gamma_0(N); \mathbb{Z})$; (b) the definition uses the expansion at ∞ only (integrality at other cusps is a different and subtler matter, not needed); (c) twisting by a quadratic character preserves integrality (Theorem thm:twist; Chapter 13). Reference: DiamondShurman 6.5 (cite the section); Shimura71 3.5. Paper: Section 1 ("integral cusp form g of level N"), Lemma lem:crit ($S_2(\Gamma_0(N); \mathbb{Z})$; "with $\mathbb{Z}$ replaced by $\mathbb{Z}_\ell$"), Theorem thm:twist ("preserves integrality"), Theorem thm:quant Step 1 ($S_2(\Gamma_0(N); \mathbb{Z}_\ell)$).

11.2 `thm:modforms:integral-basis` (theorem; R). $S_k(\Gamma_0(N))$ has a $\mathbb{C}$-basis consisting of forms with integer coefficients; equivalently the natural map $S_k(\Gamma_0(N); \mathbb{Z}) \otimes_{\mathbb{Z}} \mathbb{C} \to S_k(\Gamma_0(N))$ is an isomorphism. The same holds for $M_k(\Gamma_0(N))$ with $\mathbb{Q}$ (and, after scaling, $\mathbb{Z}$). Proofs: via the rationality of the Hecke eigenvalues and the Galois action on newforms (Shimura), via the algebraic theory of modular forms (the q-expansion principle of Deligne-Rapoport and Katz), or for $k = 2$ via $H^1(X_0(N), \mathbb{Z})$ and Eichler-Shimura (Chapter 14). Reference: Shimura71 Theorem 3.52 (verify the number); DiamondShurman 6.5 (cite the section). Paper: Lemma lem:crit (that $S_2(\Gamma_0(N); \mathbb{Z})$ has rank $\dim S_2$ is used when the perfect pairing with $\mathbb{T}$ is applied, Chapter 13).

11.3 `thm:modforms:sturm` (theorem, Sturm's bound; S). Let $\mathcal{O}$ be the ring of integers of a number field, λ a prime of $\mathcal{O}$, and $f \in M_k(\Gamma_0(N); \mathcal{O})$. If $a_n(f) \in \lambda$ for all $n \leq k\mu(N)/12$, then $a_n(f) \in \lambda$ for all n. For $k = 2$ the condition is on $n \leq B_2(N) = \lfloor \mu(N)/6 \rfloor$. Values (agreeing with PARI's `mfsturm`): $B_2(11) = 2$, $B_2(20) = 6$, $B_2(37) = 6$, $B_2(99) = 24$, $B_2(121) = 22$, $B_2(539) = 112$, $B_2(980) = 336$, $B_2(1089) = 264$, $B_2(4851) = 1344$. Sketch: replace f by the norm $\prod_i f|_k \gamma_i$ suitably normalised, which has coefficients integral at λ (Chapter 6's integrality of algebraic integers, after adjoining μ_N) and lies in $M_{k\mu}(\mathrm{SL}_2(\mathbb{Z}))$; for level one, write it in the $\mathcal{O}$-basis $E_4^a E_6^b \Delta^c$ triangular in q and use the mod λ form of 7.3 (a nonzero mod-λ form of weight K and level one has $v_\infty \leq K/12$). The characteristic-zero statement is 8.1. Reference: Sturm, *On the congruence of modular forms*, Lecture Notes in Math. 1240 (1987), 275-280 (add to `bib/ch12.bib`); Stein (book), Chapter 9 (cite the chapter; the theorem number to be verified). Paper: Section 1 and Lemma lem:crit (coefficientwise congruences "for every n" are decidable), the Example of Section 3 (11.5).

11.4 `cor:modforms:integral-lattice` (corollary; P given 11.2, 11.3). Let $d = \dim S_2(\Gamma_0(N))$ and $B = B_2(N)$. (i) $S_2(\Gamma_0(N); \mathbb{Z})$ is free of rank d and $S_2(\Gamma_0(N); \mathbb{Z}) \otimes \mathbb{C} = S_2(\Gamma_0(N))$. (ii) The truncation $\tau_B \colon S_2(\Gamma_0(N); \mathbb{Z}) \to \mathbb{Z}^B$, $f \mapsto (a_1, \ldots, a_B)$, is injective (8.1) with torsion-free cokernel: if $\tau_B(g) \in m\mathbb{Z}^B$ then, applying 11.3 for each prime $p \mid m$ to g, g/p, ..., $g/m \in S_2(\Gamma_0(N); \mathbb{Z})$. So τ_B is a split injection of free $\mathbb{Z}$-modules (Chapter 3, saturation). (iii) Hence for every commutative ring B' the q-expansion map $S_2(\Gamma_0(N); B') \to B'[[q]]$ is injective, and so is its truncation at B; for $B' = \mathbb{Z}_\ell$ the image is $(S_2(\Gamma_0(N); \mathbb{Q}) \otimes \mathbb{Q}_\ell) \cap \mathbb{Z}_\ell[[q]]$, and for $B' = \mathbb{F}_\ell$ a form is zero iff its first B coefficients are. (iv) For $f, g \in S_2(\Gamma_0(N); \mathcal{O})$ and an ideal $\mathfrak{a} \subseteq \mathcal{O}$, $f \equiv g \pmod{\mathfrak{a}}$ coefficientwise iff $a_n(f) \equiv a_n(g) \pmod{\mathfrak{a}}$ for $n \leq B$ (11.3 applied prime by prime to the powers of the primes of $\mathcal{O}$ dividing $\mathfrak{a}$, as in (ii)). (v) Forms of different levels $M_1, M_2 \mid N$ are compared in $S_2(\Gamma_0(N))$ with $N = \mathrm{lcm}(M_1, M_2)$ (6.2(c)); a congruence "at all $p \notin \Sigma$" must first be turned into one "at all n" (depletion, Chapter 13), which raises the level and the bound. Reference: as for 11.2, 11.3. Paper: Lemma lem:crit ("every step holds verbatim over $\mathbb{Z}_\ell$, the pairing being perfect after tensoring with $\mathbb{Z}_\ell$": (iii) gives the q-expansion side), Theorem thm:quant Step 1 ("projecting h to the $\mathfrak{m}$-component ... the congruence survives": a statement about $S_2(\Gamma_0(N); \mathbb{Z}_\ell)$ and its q-expansions), Lemma lem:old (the passage from a_p to a_n is Chapter 13's recursion, not Sturm).

11.5 `ex:modforms:paper-congruences` (worked example; P given 11.3 and the levels of the twists

from Chapter 13). The three congruences of the Example of Section 3 hold **for every** n and are theorems by 11.4(iv) (Part 7, point P1): (a) $f_{121\mathrm{a}} \equiv f_{11\mathrm{a}} \otimes \chi_{-11} = f_{121\mathrm{d}} \pmod 3$: both in $S_2(\Gamma_0(121);\mathbb{Z})$, $B_2(121) = 22$; the coefficients agree mod 3 for $n \le 22$ (also at $p = 3$: $a_3 = 2$ and -1; at $p = 11$: both 0). (b) $f_{539\mathrm{c}} \equiv f_{11\mathrm{a}} \otimes \chi_{-7} = f_{539\mathrm{d}} \pmod 3$: both of level 539, $B_2(539) = 112$ ($a_3 = -2$ and 1; $a_7 = 0$, 0; $a_{11} = 1$, 1). (c) $f_{980\mathrm{i}} \equiv f_{20\mathrm{a}} \otimes \chi_{-7} \pmod 5$: both of level 980, $B_2(980) = 336$ ($a_2 = 0, 0$; $a_5 = 1, 1$; $a_7 = 0, 0$). Each verified for all $n \le 20000$ (and $n \le 3000$ for (a)). The paper's checks "for $p < 500$" and "for $p < 1000$" therefore already prove these congruences once the excluded a_p are added, since the a_n for $n \le B$ are determined by the a_p, $p \le B$, through the Hecke recursion (Chapter 13). Also (8.4): f_{11} is congruent mod 5 to an Eisenstein series, $B_2(11) = 2$. Paper: the Example of Section 3, Theorem thm:qual (b), (c) (illustrations), Conjecture conj:eis.

14.2.12 Section 12. The Atkin-Lehner involutions

12.1 `def:modforms:atkin-lehner` (definition and proposition; P). Let $d \,\|\, N$ (C4). Integers x, y, z, w with $d^2xw - Nyz = d$ exist (divide by d: solve $dxw - (N/d)yz = 1$, possible since $\gcd(d, N/d) = 1$), and $W_d = \left(\begin{smallmatrix} dx & y \\ Nz & dw \end{smallmatrix}\right)$, $\det W_d = d$. (i) W_d normalises $\Gamma_0(N)$. (ii) $W_d^2 \in d \cdot \Gamma_0(N)$ (explicitly $W_d^2 = d\left(\begin{smallmatrix} dx^2+(N/d)yz & (x+w)y \\ Nz(x+w) & (N/d)yz+dw^2 \end{smallmatrix}\right)$). (iii) Two choices of W_d differ by left multiplication by an element of $\Gamma_0(N)$. (iv) For $d, d' \,\|\, N$, $W_dW_{d'} \in W_{dd'/\gcd(d,d')^2} \cdot \Gamma_0(N)$ up to a positive scalar; the W_d commute modulo $\Gamma_0(N)$ and scalars. Consequently (v) $z \mapsto W_d z$ induces a holomorphic involution w_d of $X_0(N)$, and the w_d generate a group of automorphisms of $X_0(N)$ that is a quotient of $(\mathbb{Z}/2)^{\omega(N)}$ (isomorphic to it when N is squarefree, by 12.3(ii)); $W_N = \left(\begin{smallmatrix} 0 & -1 \\ N & 0 \end{smallmatrix}\right)$ ($x = w = 0$, $y = -1$, $z = 1$) gives $w_N \colon z \mapsto -1/(Nz)$; in moduli terms $w_N(E, C) = (E/C, E[N]/C)$ (5.4(iii); P: scale $\Lambda_{N\tau}$). (vi) $f \mapsto f|_kW_d$ is a well-defined involution of $M_k(\Gamma_0(N))$ and of $S_k(\Gamma_0(N))$ for even k ((ii), (iii), C3), the operators commute, and $w_N = \prod_{p^e \| N} w_{p^e}$. (vii) In weight 2: $w_d^*(\omega_f) = \omega_{f|_2W_d}$ (6.1(iii)). Reference: AtkinLehner70 (Section 2; cite the section); DiamondShurman 5.5 (and its exercises; cite the section); Miyake 4.6 (cite the section). Paper: Section 4.1 ("$w = \left(\begin{smallmatrix} 0 & -1 \\ M^\Sigma & 0 \end{smallmatrix}\right)$ … the Atkin-Lehner involution w_{M^Σ}"), Lemma lem:hecke and Theorem thm:quant Step 2 (through $\hat{\delta}_!$, Chapter 14).

12.2 `prop:modforms:atkin-lehner-petersson` (proposition; P). (i) For even k and $f, g \in S_k(\Gamma_0(N))$: $\langle f|_kW_d, g|_kW_d\rangle = \langle f, g\rangle$ (10.2(ii)) and $\langle f|_kW_d, g\rangle = \langle f, g|_kW_d\rangle$ (apply the first to $g|_kW_d$ and use $W_d^2 \in d\Gamma_0(N)$). So w_d is a self-adjoint isometric involution, $S_k(\Gamma_0(N)) = S_k^+ \oplus S_k^-$ orthogonally into its ±1-eigenspaces, and the w_d are simultaneously diagonalisable. (ii) For $k = 2$: $\dim S_2(\Gamma_0(N))^+$ (for w_N) equals the genus of $X_0(N)/w_N$ (a w_N-invariant holomorphic differential descends holomorphically, by the local computation of 8.2 with $e = 2$; P given Chapter 8). Check (class numbers for the number of fixed points of w_N, R; Riemann-Hurwitz, Chapter 8): $N = 11$: 4 fixed points ($h(-44) + h(-11) = 3 + 1$), $X_0(11)/w_{11}$ of genus 0, $S_2^+ = 0$; $N = 23$: 6 fixed points, genus 0; $N = 37$: 2 fixed points ($h(-148) = 2$), genus 1, $\dim S_2^+ = 1$; $N = 43$: 4, genus 1 (PARI's Atkin-Lehner matrices agree). (iii) (The input to important_notes.md point 39(d).) w_N is an involutive automorphism of the compact Riemann surface $X_0(N)$, so w_N^* is an isometry of the cup product on $H^1(X_0(N), \mathbb{Z})$ with $(w_N^*)^2 = 1$ (Chapter 8, its 12.3(a)); if $f|_2W_N = \varepsilon f$ ($\varepsilon = \pm1$) then w_N^* acts by ε on ω_f, on $\overline{\omega_f}$, and hence on the real 2-plane of $H^1(X_0(N), \mathbb{R})$ spanned by $\mathrm{Re}\,\omega_f$, $\mathrm{Im}\,\omega_f$ (Chapter 8, its 10.2(iii)). Reference: DiamondShurman 5.5; AtkinLehner70. Paper: Section 4.1 ($\hat{\delta}_!$ "perfect and alternating"; Chapter 8, its 12.3(e), Chapter 14), Lemma lem:hecke (important_notes.md points 36 and 39).

12.3 `prop:modforms:atkin-lehner-cusps` (proposition; P). (i) w_N exchanges ∞ and 0, and maps the cusps of denominator c (that is, a/c with $c \mid N$, 4.2) to the cusps of denominator N/c: $W_N(a/c) = -1/((N/c)a)$, and $\gcd((N/c)a, N) = N/c$. (ii) For N squarefree, w_e ($e \mid N$) maps the cusp $1/c$ to the cusp $1/(ce/\gcd(c,e)^2)$; so the group of the w_e acts simply transitively on the $2^{\omega(N)}$ cusps. (Verified for $N = 6, 30, 77$ with explicit W_e.) (iii) In general w_d permutes the cusps; Exercise 17 works out $X_0(20)$ (w_4, w_5, w_{20} on its six cusps). Reference: DiamondShurman 3.8 and 5.5 (exercises; cite the sections); Ogg (1973). Paper: background (the cusps ∞ and 0 of $X_0(N)$; Chapter 14's cuspidal subgroup, Conjecture conj:eis).

12.4 `ex:modforms:atkin-lehner-q-expansions` (examples; P for (a), (b), R for (c)). The action on q-expansions is not simple. (a) $f|_kW_N$ is, up to the factor $N^{k/2}(Nz)^{-k}$, the expansion of f at the

cusp 0; it is not given coefficient by coefficient by the $a_n(f)$ (there is no formula $a_n(f|W_N) = F(a_m(f))$ valid on all of M_k); it is determined globally, through the transformation law. (b) (Illustration with oldforms, which are Chapter 13's.) For $h \in M_k(\Gamma_0(M))$ and $N = Mp$, $p \nmid M$: $W_N = W_M\left(\begin{smallmatrix} p & 0 \\ 0 & 1 \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} p & 0 \\ 0 & 1 \end{smallmatrix}\right)W_N = W_M \cdot pI$, so

$$(h|_k W_N)(z) = p^{k/2}(h|_k W_M)(pz), \qquad \big(h(pz)\big)|_k W_N = p^{-k/2}(h|_k W_M)(z) :$$

w_N exchanges the lines of $h(z)$ and $h(pz)$ (when h is a w_M-eigenform) and is not diagonal in the basis of q-expansions. Example: on $\mathrm{span}\{f_{11}(z), f_{11}(2z)\} \subset S_2(\Gamma_0(22))$, w_{22} has matrix $\left(\begin{smallmatrix} 0 & -1/2 \\ -2 & 0 \end{smallmatrix}\right)$ (from $f_{11}|W_{11} = -f_{11}$, $k = 2$, $p = 2$). (c) (Chapters 13 and 19.) A newform $f \in S_k(\Gamma_0(N))$ is an eigenvector of every w_d with eigenvalue $\varepsilon_d(f) = \pm 1$; for $k = 2$ and $p \,\|\, N$, $a_p(f) = -\varepsilon_p(f)$ (Atkin-Lehner, Theorem 3, verify the part); and the sign of the functional equation of $L(f, s)$ is $-\varepsilon_N(f)$ for $k = 2$. Checks: 11a1: $\varepsilon_{11} = -1$ (9.2(iii)), $a_{11} = 1$, root number $+1$; 37a1: $\varepsilon_{37} = +1$, root number -1 (rank 1). Paper: Section 4.1 (the sign by which w_N acts on $M_{f,\lambda}$, important_notes.md point 39), Lemma lem:crit (f is an eigenvector of w_N, needed for $f^{\perp}$ to be U_p-stable, Chapter 13), Lemma lem:euler and Lemma lem:old ($a_p = \pm 1$ at multiplicative primes, Chapters 13 and 15).

14.3 3. Section-by-section plan with target line counts

Line counts are lines of LaTeX, card excluded (PLAN.md Section 7.3: 1,000 to 2,200).

section	title	contents	target lines
(starred)	Core results at a glance	the card: every item of Part 2 with its tag and `\Cref`; conventions C3, C5 restated in one line each	(excluded)
(starred)	Where the paper uses this chapter	Section 1 ($S_2(\Gamma_0(N))$, f, r_E, Petersson norm, $X_0(N)$); Section 3 preamble and Lemma lem:crit (integral forms, coefficientwise congruences, $f^{\perp}$, over $\mathbb{Z}_\ell$); Lemma lem:old (forms $h(dz)$, orthogonality); the Example of Section 3 (congruences, Sturm); the optimal-level theorem (nebentypus); Section 4.1 (W_{M^Σ}, $H^1(X_0(M^\Sigma))$); Theorems thm:twist and thm:quant Step 1 (integrality, $S_2(\Gamma_0(N); \mathbb{Z}_\ell)$); Conjecture conj:eis (Eisenstein congruence)	40

section	title	contents	target lines
1	The upper half plane	opening paragraph (what the chapter is for, the three physics connections); conventions C1 to C9 as `convention` environments; 1.1, 1.2	90
2	The modular group and its fundamental domain	2.1, 2.2 with a figure of $\mathcal{F}$ (TikZ or described in words) and the gluing; `paperlink` to Chapter 9's moduli statement	130
3	Congruence subgroups	3.1, 3.2; the table of $\mu(N)$ for the paper's levels; the fundamental domain of $\Gamma_0(11)$ described	110
4	Cusps, widths and elliptic points	4.1 to 4.3; worked cusp lists for $N = 11, 20, 99, 121$ with widths; `warning`: irregular cusps for $\Gamma_1(4)$	140
5	Modular curves as compact Riemann surfaces	5.1 to 5.4; the genus table; `paperlink`: Section 1 ($\varphi_E \colon X_0(N) \to E$; $X_0(11)$ has genus 1 and is 11a1, Chapter 14); the moduli dictionary, with $w_N(E, C) = (E/C, E[N]/C)$	200
6	Modular forms, q-expansions, nebentypus	6.1 to 6.6; physics aside (Fourier series on the half-cylinder); `paperlink`: "coefficientwise" means 6.3(i); the optimal-level theorem's nebentypus	170
7	Level one: Eisenstein series, Δ, j, valence	7.1 to 7.5; the contour picture for 7.3 described; physics aside (argument principle); `paperlink`: Chapter 9's deferrals discharged	190
8	$\Gamma_0(N)$: valence, dimensions, differentials	8.1 to 8.4; the Eisenstein congruence for 11a1 mod 5 in full; `paperlink`: Section sec:eis and Conjecture conj:eis	160

section	title	contents	target lines
9	η and the level-11 example	9.1 to 9.3; the first twenty coefficients of f_{11} against a_p(11a1); the table of genus-one η-products	110
10	The Petersson inner product	10.1 to 10.5; physics aside (L^2); the numerical table of 10.4; `warning` on normalisations (C5, PARI); `paperlink`: Section 1 (Zagier, Watkins), point 30	140
11	Integral forms, Sturm, congruences	11.1 to 11.5; `paperlinks`: Lemma lem:crit over $\mathbb{Z}_\ell$, Theorem thm:quant Step 1, the Example of Section 3 proved for all n	160
12	The Atkin-Lehner involutions	12.1 to 12.4; `paperlink`: Section 4.1 (the paper's w is W_{M^Σ}; what Chapter 14 adds: $T^t = w_N T w_N$, $\hat{\delta}_!$; important_notes.md points 36, 39, 41)	160
13	Exercises	the 18 exercises of Part 4.2	130
14	Notes and references	DiamondShurman, Shimura71, SerreCourse VII, Miyake, Koblitz, Knapp, Apostol, Stein (book), Sturm 1987, Ligozat, Ogg, Deligne-Rapoport, Katz-Mazur, AtkinLehner70, Zagier85, Watkins02, Cremona97, PARI; which numbers were cited and which are marked (verify)	40
	total		**1,970**

14.4 4. Coverage table and exercises

14.4.1 4.1 Coverage: PLAN.md scope, Section 6 map rows and paper hooks to line-up items

PLAN.md scope item, Section 6 row, or paper hook	line-up items
the upper half plane and the action of $\mathrm{SL}_2(\mathbb{R})$ and of $\mathrm{SL}_2(\mathbb{Z})$	1.1, 1.2, 2.1
the modular group	2.1(iv), 2.2
fundamental domains	2.1, 2.2, 3.2
the congruence subgroups $\Gamma_0(N)$, $\Gamma_1(N)$, $\Gamma(N)$ and their indices	3.1
cusps and their widths	1.2, 4.1, 4.2, 12.3
elliptic points	1.2, 2.1(iii), 2.2, 4.3
the modular curves $Y_0(N)$, $X_0(N)$ as Riemann surfaces (compactification by cusps)	5.1, 5.2
the genus formula for $X_0(N)$; tables of small genera; $X_0(11)$ has genus 1	5.3 (and 8.1 for the zero count of f_{11})
modular forms and cusp forms of weight k for $\Gamma_0(N)$; holomorphy at the cusps via q-expansions	6.1, 6.2, 6.3
the spaces $M_k(\Gamma_0(N))$ and $S_k(\Gamma_0(N))$	6.2, 8.3
q-expansions and the q-expansion principle	6.3, 8.1, 11.4 (and 11.2 for the algebraic form, R)
Eisenstein series (brief)	7.1, 8.4
the valence formula	7.3, 8.1
dimension formulas; $\dim S_2(\Gamma_0(N))$ = genus of $X_0(N)$	7.4, 8.2(iii), 8.3
weight-2 cusp forms as holomorphic differentials $f(z)\,dz$	6.1(iii), 8.2, 12.1(vii)
Fourier coefficients and the Ramanujan-Petersson bound (statement)	6.4, 6.5
the Petersson inner product (definition, positive definite)	10.1 to 10.5
the Sturm bound and integrality; $S_2(\Gamma_0(N);\mathbb{Z})$, $S_2(\Gamma_0(N);\mathbb{Z}_\ell)$	11.1 to 11.5
modular forms with character (nebentypus), briefly	6.6
the Atkin-Lehner involutions w_d and w_N: definition; action on q-expansions is not simple; action on cusps	12.1, 12.2, 12.3, 12.4
$X_0(N)$ as a moduli space of pairs (E, C); cusps as degenerate pairs; the $\mathbb{Q}$-structure (statement)	5.4
example: $S_2(\Gamma_0(11))$ spanned by $\eta(z)^2\eta(11z)^2$, the form of 11a1; its first coefficients	9.1, 9.2
physics aside: Fourier series on a quotient	6.3, 9.2 (and Part 1.1: valence via residues 7.3, Petersson as L^2 10.1)
Section 6 row: newform f_E, $S_2(\Gamma_0(N))$, q-expansions, coefficientwise congruences (with Chapter 13)	6.2, 6.3, 9.2, 11.4, 11.5 (newforms: Chapter 13)
Section 6 row: Petersson norm, Rankin-Selberg, $L(\mathrm{Sym}^2 f, 2)$, Zagier, Watkins, Flach (with 13, 14, 19)	10.1, 10.4, 10.5
Section 6 row: $H^1(X_0(N), \mathbb{Z}_\ell)$, cup product, Atkin-Lehner, Hecke self-adjointness (Chapters 8, 14; the involution is defined here)	5.2, 8.2, 12.1, 12.2(iii)
hook: $S_2(\Gamma_0(N))$, the newform f (Section 1)	6.2, 8.2, 8.3, 9.2, 9.3
hook: q-expansions and congruences coefficientwise (Section 1, Lemma lem:crit)	6.3, 11.3, 11.4, 11.5

PLAN.md scope item, Section 6 row, or paper hook	line-up items
hook: integral forms $S_2(\Gamma_0(N);\mathbb{Z})$ (Lemma lem:crit)	11.1, 11.2, 11.4
hook: the Petersson norm $\langle f,f\rangle$ (Section 1, Zagier, Watkins)	10.1, 10.4
hook: the Atkin-Lehner involution w_N (Section 4.1, the pairing $\hat{\delta}_!$)	12.1, 12.2, 12.4(c)
hook: $X_0(N)$ as a curve (Section 1)	5.2, 5.3, 5.4
hook: the genus and $\dim S_2$ (background to multiplicity one)	5.3, 8.2, 8.3
also: Section 3 optimal-level theorem (nebentypus trivial)	6.6
also: Theorem thm:quant Step 1 ($S_2(\Gamma_0(N);\mathbb{Z}_\ell)$, congruences survive projection)	11.1, 11.4(iii)
also: Theorem thm:twist ("preserves integrality"; levels of twists: Chapter 13)	6.6 (remark), 11.1(c)
also: the Example of Section 3 (121a1, 539c1, 980i1 congruences)	11.5
also: Conjecture conj:eis, Section sec:eis (Eisenstein congruence; rational 5-torsion of 11a1)	8.4
also: Lemma lem:euler, Theorem thm:dfg14 ($\lvert\alpha/\beta\rvert=1$)	6.5
also: important_notes.md points 30, 36, 39, 41	10.5(b); 12.2(iii); 12.2(iii), 12.4(c); 12.1(v), (vii) and Part 6.4
also: what Chapters 8 and 9 deferred here	2.1, 2.2, 5.2, 5.3, 7.2, 7.5, 8.2, 10.1

14.4.2 4.2 Exercises (Section 13 of the chapter), graded from easy to harder

1. Prove $\mathrm{Im}(\gamma z)=\det\gamma\ \mathrm{Im}\,z/|cz+d|^2$ and the invariance of $dx\,dy/y^2$. Show that the stabiliser of i in $\mathrm{SL}_2(\mathbb{R})$ is $\mathrm{SO}(2)$.
2. Move $z=0.1+0.05i$ into $\mathcal{F}$ by an explicit word in S and T. Show $S^2=(ST)^3=-I$ and find the stabilisers of i, ζ_3, $-\bar{\zeta}_3$.
3. Show $\Gamma_0(N)\backslash\mathrm{SL}_2(\mathbb{Z})\simeq\mathbb{P}^1(\mathbb{Z}/N)$ and $\mu(N)=N\prod_{p\mid N}(1+1/p)$. For $N=11$ write the twelve coset representatives and sketch the fundamental domain $\bigcup\gamma_i\mathcal{F}$; check its hyperbolic area $12\cdot\pi/3$.
4. Find the cusps of $\Gamma_0(N)$ and their widths for $N=p$, $N=p^2$ and $N=20$; check that the widths add up to $\mu(N)$. Show that $X_0(121)$ has twelve cusps, ten of them of the form $a/11$.
5. Show that the elliptic points of order 2 of $\Gamma_0(N)$ correspond to the roots of x^2+1 modulo N, and compute ν_2, ν_3 for $N=5,10,13,37,121$. Deduce $g(X_0(37))=2$.
6. Derive the genus formula from Riemann-Hurwitz (Chapter 8). Compute $g(X_0(N))$ for $N=11,20,37,121,539$ and compare with Chapter 8's list. Show $g(X_0(N))\geq 1+\mu(N)/12-\frac{2^{\omega(N)}}{4}-\frac{2^{\omega(N)}}{3}-\frac{\nu_\infty(N)}{2}$ and use it to explain why only finitely many N have genus 0.
7. Prove the Lipschitz formula from the partial fractions of $\pi\cot\pi z$ and derive the q-expansions of E_4 and E_6 (the constants 240 and -504). Check $\Delta=(E_4^3-E_6^2)/1728=q-24q^2+252q^3-\cdots$.
8. Using the valence formula, show $M_2(\mathrm{SL}_2(\mathbb{Z}))=0$, $\dim M_{12}=2$, and that E_4 vanishes only at ζ_3. Prove that j is a bijection $\mathrm{SL}_2(\mathbb{Z})\backslash\mathfrak{H}\to\mathbb{C}$ and compute $j(i)$, $j(\zeta_3)$.
9. Show that $f\,dz$ is $\Gamma_0(N)$-invariant iff f has weight 2; show that at a cusp $f\,dz$ is holomorphic iff f vanishes there, and that there is no condition at elliptic points. Deduce $\dim S_2(\Gamma_0(N))=g(X_0(N))$ and the residue relation of 8.2(iii).
10. (The Eisenstein congruence.) Show $E_2(z)-11E_2(11z)\in M_2(\Gamma_0(11))$. Using the Sturm bound $B_2(11)=2$, prove that $a_n(f_{11})\equiv\sigma(n)-11\sigma(n/11)\pmod 5$ for all n, hence $a_\ell(11\mathrm{a}1)\equiv 1+\ell\pmod 5$ for all primes $\ell\neq 11$. Relate this to the rational 5-torsion point of 11a1 (Chapter 9) and to

the Eisenstein case of the paper (Section sec:eis).

11. From the transformation law of η, show $f_{11} = \eta(z)^2\eta(11z)^2 \in S_2(\Gamma_0(11))$ and $f_{11}|_2 W_{11} = -f_{11}$. Compute $a_1, \dots, a_{13}$ and compare with $p + 1 - \#E(\mathbb{F}_p)$ for $E = 11\text{a}1$ at $p = 2, 3, 5, 7, 13$.
12. Prove that the Petersson integrand is $\Gamma_0(N)$-invariant, that the integral converges for f cuspidal, and that W_N is an isometry and self-adjoint. Show $\langle f, g\rangle_{\Gamma_0(NM)} = \frac{\mu(NM)}{\mu(N)}\langle f, g\rangle_{\Gamma_0(N)}$.
13. (Normalisations; with PARI.) Compute `mfpetersson` for the form of 11a1 and multiply by $\mu(11) = 12$; compute the area of the period lattice of 11a1 and check $4\pi^2\langle f, f\rangle = \mathrm{vol}(\Lambda_E)$. In the class 11a, check that $5\,\mathrm{vol}(\Lambda_{11\text{a}3})/\,\mathrm{vol}(\Lambda_{11\text{a}1}) = 25$ and deduce from $\deg\varphi \cdot \mathrm{vol}(\Lambda) = 4\pi^2 c_\varphi^2\langle f, f\rangle$ that the parametrisation $\varphi\colon X_0(11) \to 11\text{a}3$ of degree 5 has $c_\varphi = 5$ (Part 7, B1).
14. Prove 11.4(ii): the truncation $S_2(\Gamma_0(N);\mathbb{Z}) \to \mathbb{Z}^{B_2(N)}$ is injective with torsion-free cokernel. Deduce that $S_2(\Gamma_0(N);\mathbb{Z}) \otimes \mathbb{F}_\ell \to \mathbb{F}_\ell[[q]]$ is injective.
15. (Check the paper's claim.) Using Sturm's bound, prove that $a_n(121\text{a}1) \equiv a_n(121\text{d}1) \pmod 3$ for all n, given that both forms lie in $S_2(\Gamma_0(121);\mathbb{Z})$ and that the congruence holds for $n \le 22$. Do the same for 539c1 against the twist of 11a1 by χ_{-7} ($B_2(539) = 112$). Explain why the paper's check "for $p < 500$" suffices once a_3 and a_{11} are compared.
16. Show that W_d exists for $d \,\|\, N$, normalises $\Gamma_0(N)$, and satisfies $W_d^2 \in d\Gamma_0(N)$. Show $w_N(E, C) = (E/C, E[N]/C)$ in the moduli description over $\mathbb{C}$.
17. For squarefree N show that w_e maps the cusp $1/c$ to $1/(ce/\gcd(c, e)^2)$. For $N = 20$ list the six cusps (4.2) and compute the action of w_4, w_5, w_{20} on them.
18. (Atkin-Lehner on oldforms.) For $h \in S_2(\Gamma_0(M))$ and $N = Mp$, $p \nmid M$, prove $(h|_2 W_N)(z) = p\,(h|_2 W_M)(pz)$ and $h(pz)|_2 W_N = p^{-1}(h|_2 W_M)(z)$. Write the matrix of w_{22} on $\mathrm{span}\{f_{11}(z), f_{11}(2z)\}$ and check that it squares to 1. Show that $\dim S_2(\Gamma_0(N))^+$ equals the genus of $X_0(N)/w_N$, and verify it for $N = 11$ and $N = 37$ with the fixed-point counts of 12.2.

14.5 5. Dependencies and forward pointers

Results from other chapters used here: * Background (PLAN.md Section 2): complex analysis (holomorphic and meromorphic functions, Laurent series, Riemann's removable singularity theorem, the identity theorem, the argument principle and the residue theorem, Moebius transformations, $\pi\cot\pi z$ in partial fractions, uniform convergence of series of holomorphic functions); real analysis (double integrals, change of variables, absolute convergence); linear algebra (Hermitian inner products, orthogonal complements, eigenspaces of commuting involutions); RHB (Fourier series, separation of variables, contour integration, Sturm-Liouville inner products). * Chapter 1 (`ch:groups`): group actions, cosets, orbit-stabiliser, $\mathrm{SL}_2(\mathbb{Z}/N)$, $\mathbb{P}^1(\mathbb{Z}/N)$ and lines (its C1), the generators of SL_2 (cited, not reproved: 2.1 proves generation of $\mathrm{SL}_2(\mathbb{Z})$ independently). * Chapter 2 (`ch:rings`): $\mathbb{Z}/N$, the Chinese remainder theorem, $(\mathbb{Z}/N)^\times$, Euler's φ, Kronecker symbols $(\frac{-4}{p})$, $(\frac{-3}{p})$ and the counts of roots of x^2+1, $x^2+x+1 \bmod N$ (4.3), Dirichlet characters and their conductors (6.6), divisor sums. * Chapter 3 (`ch:modules`): free modules, saturated sublattices, split injections, base change $\otimes\mathbb{Z}_\ell$ (11.4). * Chapter 4 (`ch:reps`): characters of finite abelian groups (the decomposition by nebentypus, 6.6(iii)). * Chapter 6 (`ch:ant`): rings of integers $\mathcal{O}$, primes λ (11.3, 11.4(iv)); Chapter 7 (`ch:local`): $\mathbb{Z}_\ell$, $\mathbb{Q}_\ell$ (11.1, 11.4(iii)). * Chapter 8 (`ch:curves`): Riemann surfaces and charts (its 8.1), holomorphic maps, local normal form and degree (its 8.2), Riemann existence and GAGA (its 8.3), topological Riemann-Hurwitz and Euler characteristic (its 8.4(c), 7.2) for 5.3, holomorphic differentials, residues and the residue theorem (its 9.1, 9.2), $\dim H^0(\Omega^1) = g$ (its 9.3) for 8.2, positivity $\frac{i}{2}\int\omega\wedge\bar\omega > 0$ (its 9.4(b)) for 10.1(iv), Riemann-Roch (its 5.2, 5.3) for 8.3, $H^1(X,\mathbb{C}) = H^0(\Omega^1) \oplus \overline{H^0(\Omega^1)}$ (its 10.2(iii)) and automorphisms as isometries of the cup product (its 12.3(a)) for 12.2(iii), $\deg\varphi\cdot\int_Y = \int_X \varphi^*$ (its 11.5(i)) for 10.4. * Chapter 9 (`ch:ellcurves`): lattices, $G_{2k}(\Lambda)$, g_2, g_3, $\Delta(\Lambda)$, $j(\Lambda)$ (its 7.1), homothety classes of lattices $= \mathrm{SL}_2(\mathbb{Z})\backslash\mathfrak{H}$ (its 7.5(i), (ii)) for 5.4 and 7.2, the period lattice and its area (its 7.6) for 10.4, the a_p of the running examples and the rational 5-torsion of 11a1 (its 8.7, 9.7) for 8.4 and 9.2, quadratic twists (its 10.3, 10.4) for 11.5.

Black boxes imported from later chapters (stated with a forward `\Cref`, used only in examples, remarks, exercises and `paperlinks`, never in the proof of a numbered result of this chapter): newforms, Hecke

operators, Petersson adjoints, the recursion for a_n, twisting and the levels of twists, oldforms (Chapter 13; used in 6.5, 9.3, 10.3, 10.5(a), 11.4(v), 11.5, 12.4(b), (c)); modularity, Eichler-Shimura, $X_0(11) \simeq$ 11a1, the modular degree and the Manin constant (Chapter 14; used in 9.2(iv), 10.4); the functional equation and its sign, Rankin-Selberg (Chapter 19; 10.5(b), 12.4(c)).

Forward pointers (chapters that use this one): * Chapter 13 (`ch:hecke`): $M_k(\Gamma_0(N))$, $S_k(\Gamma_0(N))$, the slash operator and C3 (for double cosets), q-expansions (6.3), the maps $f(z) \mapsto f(dz)$ (6.2(e)), the Petersson product and its behaviour under level change (10.1, 10.2), W_N and w_N on forms (12.1, 12.2) for the adjoint of U_p and for Atkin-Lehner theory, $S_2(\Gamma_0(N);\mathbb{Z})$, its rank and saturation and Sturm (11.1 to 11.4) for the perfect pairing with $\mathbb{T}$ and the integral Hecke algebra, nebentypus (6.6) for twists, the dimension formulas (8.3). * Chapter 14 (`ch:moddeg`): $X_0(N)$ as a compact Riemann surface and its $\mathbb{Q}$-model (5.2, 5.4(v)), cusps (4.2, 12.3), $S_2 = H^0(\Omega^1)$ (8.2) for $J_0(N)(\mathbb{C}) = S_2^\vee/H_1$, the involution w_N of $X_0(N)$ and its action on ω_f and on H^1 (12.1(v), (vii), 12.2(iii)) for $T^t = w_N T w_N$ and $\hat{\delta}_!$, the Petersson norm in the C5 normalisation (10.1, 10.4) for the Zagier-Watkins formula, the Eisenstein series of weight 2 (8.4) for the Eisenstein ideal and the cuspidal subgroup. * Chapter 15 (`ch:modgalrep`): nebentypus (6.6) in the optimal-level theorem; congruences and Sturm (11.4, 11.5) for the examples of Theorem thm:qual. * Chapter 19 (`ch:lfunctions`): q-expansions and the Hecke bound (6.3, 6.4) for $L(f,s)$, α_p, β_p and Ramanujan-Petersson (6.5), the Petersson product (10.1) and point 30 (10.5(b)) for Rankin-Selberg, the sign $-\varepsilon_N$ (12.4(c)). * Chapter 20 (`ch:congruence`): $S_2(\Gamma_0(N);\mathbb{Z}_\ell)$ and its q-expansions (11.1, 11.4) for the $\mathfrak{m}$-components; w_N (12.2(iii)). * Chapter 21 (`ch:paper`): A3.1 and A3.2 (integral forms, Petersson orthogonality, coefficientwise congruences), A4.1 (w is W_{M^Σ}), the Example of Section 3 (11.5). * Chapter 22 (`ch:computation`): PARI's `mfinit`, `mfdim`, `mfsturm`, `mfpetersson` (divides by $\mu(N)$, C5), `mfatkineigenvalues`, `mffromell`, `ellmoddegree`, `ellan`, `elltwist` as used in Part 6.6.

14.6 6. Notes for the writer

14.6.1 6.1 Conventions to state (Section 1 of the chapter, as `convention` environments)

C1 to C9 of Part 1.2. In addition one `convention` box in Section 10, "Normalisations of the Petersson product", with exactly these lines: the book's product is unnormalised (C5); PARI's `mfpetersson` is the book's divided by $\mu(N)$; the "normalised" product of some books is the book's divided by $\mathrm{vol}(\Gamma_0(N)\backslash\mathfrak{H}) = \pi\mu(N)/3$ and is independent of the level at which f, g are viewed; the formula $\deg\varphi_E \cdot \mathrm{vol}(\Lambda_E) = 4\pi^2 c_E^2\langle f,f\rangle$ (Chapter 14) holds in the book's normalisation with c_E defined by $\varphi_E^*\omega_E = c_E \cdot 2\pi i\, f(z)\,dz$ (as in Chapter 8, its 4.3).

14.6.2 6.2 Pitfalls

- **Slash normalisation.** $(\det\gamma)^{k/2}$ (C3) versus Diamond-Shurman's $(\det\gamma)^{k-1}$: identical in weight 2, different otherwise. Under $(\det)^{k-1}$, $W_N^2 = -N\cdot I$ acts by N^{k-2} times $(-1)^k$, so w_N is an involution only after rescaling; under C3 it is an involution. Say this once; the paper only needs weight 2.
- **Petersson normalisation (C5).** Never quote a numerical Petersson norm without saying which normalisation; the factor between PARI and the book is $\mu(N)$ (12 for $N = 11$), and between "normalised" and the book $\pi\mu(N)/3$. 10.4 is the check.
- **Index in SL_2 versus PSL_2.** For $\Gamma_0(N)$ they agree ($-I \in \Gamma_0(N)$); for $\Gamma_1(N)$, $N \geq 3$, the degree of $X_1(N) \to X(1)$ is half of $[\mathrm{SL}_2(\mathbb{Z}) : \Gamma_1(N)]$. The genus formula 5.3 is stated for $\Gamma_0(N)$ only.
- **Cusps.** $\infty \sim 1/N$ (width 1), $0 = 0/1$ (width N); the width is relative to $\Gamma_0(N)$, not to $\mathrm{SL}_2(\mathbb{Z})$. Irregular cusps exist for $\Gamma_1(4)$ but not for $\Gamma_0(N)$; do not state "cusp form iff $b_0 = 0$ in q_h" for groups with irregular cusps without care.
- **"q-expansion principle".** Here it means the elementary statements 6.3(iv), 8.1, 11.4; the algebraic q-expansion principle (Deligne-Rapoport, Katz: integrality at ∞ controls integrality of the form over $\mathbb{Z}[1/N]$) is only mentioned in 11.2. Do not claim that integrality at ∞ implies integrality at

the other cusps.

- $S_2(\Gamma_0(N);\mathbb{Z}_\ell)$ is defined as $S_2(\Gamma_0(N);\mathbb{Z})\otimes\mathbb{Z}_\ell$ (C6), because $\mathbb{Z}_\ell\not\subset\mathbb{C}$ canonically; 11.4(iii) shows it is the same as the $\mathbb{Z}_\ell$-integral q-series in $S_2(\Gamma_0(N);\mathbb{Q})\otimes\mathbb{Q}_\ell$. The paper's Lemma lem:crit and Theorem thm:quant use it in that sense.
- **Sturm and the paper's congruences.** Sturm's bound applies to forms in the same space $M_k(\Gamma_0(N);\mathcal{O})$ and to congruences at **all** n. A congruence at all $p\notin\Sigma$ first needs depletion (Chapter 13), which raises the level (for example $121\cdot 9=1089$, $B_2=264$; $539\cdot 9\cdot 11=53361$, $B_2=14784$). For the paper's three examples this is unnecessary because the congruences hold at all n (11.5). For the Eisenstein example multiply by 12 to clear the denominator before applying Sturm.
- $f^\perp$ **is a Petersson complement**, transcendental in definition; that it is defined over $\mathbb{Q}$ and $\mathbb{T}$-stable is Chapter 13's (10.3). Theorem thm:twist uses the $\mathbb{T}$-stable complement; Chapter 13 shows the two agree for a newform.
- **Atkin-Lehner on q-expansions** (12.4): never write $a_n(f|W_N)=\pm a_n(f)$ for a general f; only for eigenforms. Chapter 21's A3.2 draft sentence about adjoints should read: for $(n,N)=1$, T_n is self-adjoint; for $p\mid N$, $U_p^\dagger=w_NU_pw_N$ (Chapter 13); f is a w_N-eigenvector.
- **Signs.** W_N versus $-W_N$ (C4); the Atkin-Lehner sign ε_N versus the root number $-\varepsilon_N$ in weight 2 (12.4(c)); check both on 11a1 ($\varepsilon=-1$, root number $+1$) and 37a1 ($\varepsilon=+1$, root number -1) before printing.
- **Hecke operators.** None here (C8). The card must not contain T_n, U_p, eigenforms or newforms as items; they appear only in forward pointers. Likewise twists of forms (Chapter 13) and $J_0(N)$ (Chapter 14).
- **Modularity.** 9.2(iv) is a numerical fact here; that f_{11} is the form of 11a1 for all n is Eichler-Shimura for $X_0(11)$, Chapter 14. Never present the check to $n=2000$ as a proof.
- j **and uniformisation** (Chapter 9's writer note): 7.5 proves the bijectivity of j by the valence formula without using uniformisation; Chapter 9 states uniformisation (R) and may now cite 7.5 for its first route. No circularity.
- Typography (PLAN.md 7.2): "Atkin-Lehner", "Ramanujan-Petersson", "Riemann-Hurwitz", "Rankin-Selberg", "Eichler-Shimura", "Deligne-Rapoport", "Katz-Mazur", "Diamond-Shurman" with plain hyphens; ranges "1 to 8"; tables with `booktabs`.

14.6.3 6.3 What to leave to Chapters 8, 13, 14 and 19

- **Chapter 8** (`ch:curves`): Riemann surfaces in general, Riemann-Hurwitz, holomorphic differentials and $\dim H^0(\Omega^1)=g$, Riemann-Roch, residues, H^1 and the cup product, correspondences and their adjoints. Chapter 12 applies them to $X_0(N)$ (5.3, 8.2, 8.3, 10.1(iv), 12.2(iii)) and does not reprove them.
- **Chapter 13** (`ch:hecke`): Hecke operators T_n, U_p (double cosets and q-expansion formulas), the Hecke algebra $\mathbb{T}$, its freeness, the perfect pairing $S_2(\Gamma_0(N);\mathbb{Z})\times\mathbb{T}\to\mathbb{Z}$, self-adjointness of T_n and the adjoint of U_p for the Petersson product, eigenforms, newforms and oldforms (Atkin-Lehner theory, multiplicity one, $\varepsilon_d(f)$, $a_p=-\varepsilon_p$, $a_p=0$ for $p^2\mid N$), twisting by characters and Atkin-Li, depleted and stabilised forms, $L(f,s)$, the statement of Rankin-Selberg. Chapter 12 supplies only the spaces, the slash operator, the maps $f(z)\mapsto f(dz)$ as a remark, the Petersson product, w_d, integrality and Sturm.
- **Chapter 14** (`ch:moddeg`): $X_0(N)$ over $\mathbb{Q}$ beyond the statement 5.4(v), its rational points and cusps over $\mathbb{Q}$, $J_0(N)$, the Hecke correspondences, $T^t=w_NTw_N$, the pairing $\hat{\delta}_!=\langle x,w_Ny\rangle$, Eichler-Shimura, modularity, the modular parametrisation, the Manin constant and the Zagier-Watkins formula, the Eisenstein ideal and the cuspidal subgroup.
- **Chapter 19** (`ch:lfunctions`): the functional equation and root numbers, the Rankin-Selberg formula for $\langle f,f\rangle$ with exact local factors, symmetric square and adjoint L-functions, point 30.

14.6.4 6.4 How the relevant important_notes.md points are handled

- **Point 30** (the naive adjoint L-function is not the Euler product of $\sum a_n^2 n^{-s}$). Chapter 12 touches it only where the Petersson norm is related to L-values: remark 10.5(b) states the correct relation $\sum a_n^2 n^{-s} = \zeta^{(N)}(s-1)L^{\mathrm{nv}}(\mathrm{Sym}^2 f, s)/\zeta^{(N)}(2s-2)$ (up to bad factors) and says that the Rankin series is not the naive adjoint L-function; the exact statement is Chapter 19's. The `paperlink` of Section 10 quotes the paper's sentence and the fix.
- **Point 36** (the pairing stays perfect on the $\mathfrak{m}$-component). Chapter 8, its 12.3(e), proves the abstract statement; it needs an involutive automorphism w of the curve acting as an isometry with $w^{*2} = 1$. Chapter 12 supplies exactly that input for $X_0(N)$: w_N is a holomorphic involution of $X_0(N)$ (12.1(v)), hence an isometry of the cup product (12.2(iii)). The identity $T^t = w_N T w_N$ that completes the argument is Chapter 14's.
- **Point 39** (Lemma lem:hecke is always an equality). Its step (d) needs: "w_N preserves the f-part and acts on it by the Atkin-Lehner sign ± 1 (on $f\,dz$ and on its conjugate)". Chapter 12 supplies $w_N^*(\omega_f) = \omega_{f|_2 W_N}$ (12.1(vii)) and the consequence that w_N^* acts by ε on ω_f, $\overline{\omega_f}$ and the real plane they span in $H^1(X_0(N), \mathbb{R})$ (12.2(iii)), given $f|_2 W_N = \varepsilon f$ (Chapter 13 for newforms in general; 9.2(iii) proves $\varepsilon = -1$ for f_{11}). It also fixes that the paper's w is exactly W_{M^Σ} (C4), with no hidden scalar in weight 2 (C3: $(\det)^{k/2} = (\det)^{k-1}$ at $k = 2$). The remaining steps of point 39 are Chapter 14's.
- **Point 41** (Tate module versus H^1 as Hecke modules). Chapter 12 only supplies that w_N is an involution, so that $w_N^* = (w_N^{-1})^* = (w_N)_*$ on H^1 and H_1 and the correction "Poincare duality composed with w_N" is well defined; the rest is Chapter 14's (with Chapter 8, its 12.3(c)).
- Points 15 and B8 (curve-level twists): 11.5 states the congruences at the level of forms (classes), naming 121d, 539d by class only.

14.6.5 6.5 References

DiamondShurman (Chapters 1 to 3, 5, 6, 7; sections cited, theorem numbers 3.1.1, 3.5.1, 3.6.1 marked "verify"), Shimura71 (Chapters 1 to 3, 6.7; Theorem 3.52 and Propositions 1.40, 1.43 marked "verify"), SerreCourse (Chapter VII; theorem numbers marked "verify"), Miyake (4.6), Koblitz (Chapter III), Knapp (Chapters VIII, IX, XI by chapter), AtkinLehner70, Stein (book; Chapter 9 for Sturm), Zagier85, Watkins02, Cremona97, Deligne, Shimura76, DFG04 (Lemma 2.12 as the paper cites it), PARI and PARIusers are in `refs.bib`. Not in `refs.bib`, to be added to `bib/ch12.bib` with complete checked entries if cited: Sturm, *On the congruence of modular forms* (Lecture Notes in Math. 1240, 1987); Apostol, *Modular Functions and Dirichlet Series in Number Theory* (GTM 41); Ligozat, *Courbes modulaires de genre 1* (Bull. Soc. Math. France Memoire 43, 1975); Ogg, *Rational points on certain elliptic modular curves* (Proc. Sympos. Pure Math. 24, 1973); Deligne-Rapoport, *Les schemas de modules de courbes elliptiques* (Lecture Notes in Math. 349, 1973); Katz-Mazur, *Arithmetic Moduli of Elliptic Curves* (Annals of Math. Studies 108, 1985). Every bibliographic detail above was recalled, not re-verified; check before adding. Exception (decision B14, important_notes.md B19): Ogg 1973 (`Ogg73`), Deligne-Rapoport (`DelligneRapoport73`) and Katz-Mazur (`KatzMazur85`) are needed by Chapter 14 too, so they go into `refs.bib`, not `bib/ch12.bib`.

14.6.6 6.6 Verification log (computations done for this outline)

Python 3 with sympy 1.14 and PARI/GP 2.17 through `cypari2`; scripts in `/tmp/ch12`; Cremona `ecdata` files from `/tmp/ch09/data`. * `check1.py`: μ, ν_2, ν_3, ν_∞, g by the formulas versus brute force on $\mathbb{P}^1(\mathbb{Z}/N)$ (fixed points of S and ST, T-orbits) for 44 levels up to 169, all equal; cusp widths by the formula $N/\gcd(c^2, N)$ versus T-orbit lengths for $N \in \{4, 8, 9, 11, 20, 25, 27, 36, 49, 77, 99, 121\}$; $g =$ `mfdim([N,2],1)` for all tested N including 539, 726, 980. Genus 0 and 1 lists checked for $N \le 2000$. * `check2.py`: $\eta(z)^2\eta(11z)^2$ against `ellan` of 11a1 for $n \le 2000$ (and 11a2, 11a3 for $n \le 200$); PARI's newform at level 11; the dimension formulas 8.3 against `mfdim([N,k],1)` and `mfdim([N,k],4)` for $N \le 60$ and $N \in \{77, 99, 121, 539\}$, $k = 2$ to 14, no mismatch; level-one dimensions for $k \le 26$; $B_2(N) =$ `mfsturm([N,2])` for

ten levels. * `check3.py`: $f_{11}|_2 W_{11}/f_{11} = -1$ at three points; $f_{11}(i/\sqrt{11}) \approx 0.1029$; `mfatkineigenvalues` at $N = 11, 20, 37, 99, 121$; root numbers of 11a1, 37a1; `mfpetersson`$\cdot 4\pi^2/\operatorname{vol} = \deg/\mu$ for five curves (so PARI divides by μ). * `check4.py`, `check5.py`: the three congruences of the paper's Section 3 for all $n \le 3000$ (121) and $n \le 20000$ (539, 980), with the twists computed by `elltwist` and their conductors 121, 539, 980; Sturm bounds at the depleted levels; Hasse for 11a1 at $p < 10^5$; Ramanujan's bound for $\tau(p)$, $p < 2000$; the q-expansions of j, E_2, E_4, E_6, Δ. * `check6.py`: $\#\,\mathrm{SL}_2(\mathbb{Z}/N)$ and $[\mathrm{SL}_2(\mathbb{Z}) : \Gamma_1(N)]$ by enumeration for $N \le 12$; the action of w_e on cusps for $N = 6, 30, 77$; fixed points of w_N by class numbers and $\dim S_2^+$ by PARI's Atkin-Lehner matrix for $N = 11, 23, 37, 43$. * `check7.py`: the η-products of 9.2 against 14a1, 15a1, 20a1, 24a1, 27a1, 32a1, 36a1 for $n \le 300$; the Eisenstein congruence mod 5 for $n \le 5000$; the unnormalised Petersson norms and $4\pi^2\langle f, f\rangle/\operatorname{vol}(\Lambda_E) = \deg\varphi_E$ to 30 digits for 11a1, 20a1, 37a1, 99a1, 121a1; the period areas of 11a1, 11a2, 11a3 (ratios $1 : \frac{1}{5} : 5$). * By hand: the W_d^2 identity of 12.1(ii); $W_N = W_M\,\mathrm{diag}(p, 1)$ and the matrix of w_{22} on $\{f_{11}(z), f_{11}(2z)\}$ (also checked numerically); η-transformation for $f_{11}|W_{11}$; $\Delta(\Lambda_z) = (2\pi)^{12}\Delta(z)$; the E_2 transformation from E_2^*; the valence contour angles; the residue of ω_f at a cusp. * Not checked: theorem numbers in DiamondShurman, Shimura71, SerreCourse, Stein, Miyake (marked "verify"); the normalisations used by Zagier85 and Watkins02 in print.

14.7 7. Points for important_notes.md

New points in the area of this chapter ($S_2(\Gamma_0(N))$, q-expansions and coefficientwise congruences, integral forms, the Petersson norm, w_N, $X_0(N)$ as a curve), not among points 1 to 45 and B1 to B9 already recorded. Points 30, 36, 39, 41 are handled in Part 6.4 and not repeated.

Errors in the paper in this area: none found. Checked in particular: the definition of r_E (Section 1 and Section 3 preamble), the use of $S_2(\Gamma_0(N); \mathbb{Z})$ and of its $\mathbb{Z}_\ell$-version in Lemma lem:crit and Theorem thm:quant Step 1, "orthogonal complement" in Lemmas lem:crit and lem:old, the description of w in Section 4.1 (it is exactly W_{M^Σ} with no scalar, C3, C4), the direction of the Zagier-Watkins relation in Section 1 ("$\langle f, f\rangle$ divided by the area": correct, the explicit factor being $4\pi^2 c_E^2$ in the unnormalised normalisation, 10.4), and every congruence of the Example of Section 3 (11.5).

- **P1. Example of Section 3: the three congruences hold at every n, and the paper's finite checks already prove them by Sturm's bound (an optional strengthening, not an error).** Location: the Example after the proof of Theorem thm:qual: "$a_p(121\text{a}1) \equiv \chi_{-11}(p)a_p(11\text{a}1) \pmod 3$ for all $p \neq 3, 11$ (checked for $p < 500$)", "$a_p(539\text{c}1) \equiv \chi_{-7}(p)a_p(11\text{a}1) \pmod 3$ for all $p \neq 3, 7, 11$ (checked for $p < 1000$)", "$a_p(980\text{i}1) \equiv \chi_{-7}(p)a_p(20\text{a}1) \pmod 5$ for all $p \neq 2, 5, 7$ (checked for $p < 1000$)". Observation: in each case both forms are newforms in the same space ($S_2(\Gamma_0(121))$, $S_2(\Gamma_0(539))$, $S_2(\Gamma_0(980))$: the twists have conductors 121, 539, 980) with integer coefficients, and the congruence also holds at the excluded primes (a_3: 2 versus -1; a_3: -2 versus 1, a_7: $0, 0$, a_{11}: $1, 1$; a_2: $0, 0$, a_5: $1, 1$, a_7: $0, 0$). So the forms are congruent at every n, and by Sturm's bound ($B_2 = 22$, 112, 336) this is proved by the coefficients $n \le B_2$, which the stated checks cover once the excluded a_p are included. Fix (optional): "and in fact $f_{121\text{a}} \equiv f_{11\text{a}} \otimes \chi_{-11} \pmod 3$ coefficientwise, which Sturm's bound (22 at level 121) reduces to a finite check", and likewise for the other two. Effect: the illustrations become theorems; nothing else changes. Confidence: high (all $n \le 20000$, $n \le 3000$ for the first; Sturm bounds from PARI agree with $\lfloor \mu/6 \rfloor$).

Book-internal inconsistencies found while preparing this outline:

- **B1. The place of the Manin constant in the Zagier-Watkins formula (fixed in PLAN.md and Chapter 9; notation c_φ, c_E of Appendix B, B-C9).** PLAN.md (Chapter 14 scope: "$\deg\varphi_E = 4\pi^2\langle f, f\rangle/(c_\varphi^2 \operatorname{vol}(\Lambda_E))$ up to the precise normalisation") and the Chapter 9 outline (Part 0, the Chapter 14 bullet: "the formula $\deg\varphi_E = 4\pi^2\langle f, f\rangle/(c_\varphi^2 \operatorname{vol}(\Lambda_E))$") put c_φ^2 in the denominator, while the Chapter 8 outline (its 4.3) defines c_φ by $\varphi_E^*\omega_E = 2\pi i c_\varphi\, f(z)\, dz$. With that definition and the unnormalised Petersson product (C5), pulling back the area form gives $\deg\varphi_E \cdot \operatorname{vol}(\Lambda_E) = 4\pi^2 c_\varphi^2\langle f, f\rangle$, that is, $\deg\varphi_E = 4\pi^2 c_\varphi^2\langle f, f\rangle/\operatorname{vol}(\Lambda_E)$: c_φ^2 in the **numerator**.

Numerical check in the class 11a (where $X_0(11) = 11a1$, deg $= 1$, $c_\varphi = 1$): the parametrisation $X_0(11) \to 11a3$ has degree 5 and $5\,\mathrm{vol}(\Lambda_{11a3})/\,\mathrm{vol}(\Lambda_{11a1}) = 25$, so its constant is $c_\varphi = 5$, an integer as it must be (the Neron differential pulls back integrally); with c_φ^2 in the denominator one would get $c_\varphi = 1/5$. For optimal curves $c_E = 1$ in all computed cases, so no number in the paper or the tables changes. Fix: Chapter 14 must state $\deg \varphi_E \cdot \mathrm{vol}(\Lambda_E) = 4\pi^2 c_\varphi^2 \langle f, f\rangle$ with both conventions (C5 here, c_φ as in Chapter 8) spelled out; PLAN.md and the Chapter 9 outline should be read accordingly. Confidence: high (derivation in 10.4, numerics in Part 6.6).

- **B2. Who proves that $S_2(\Gamma_0(N);\mathbb{Z})$ is a lattice.** PLAN.md puts "integrality (forms with integer coefficients, $S_2(\Gamma_0(N);\mathbb{Z})$, $S_2(\Gamma_0(N);\mathbb{Z}_\ell)$)" in Chapter 12 and "the integral structure: $S_2(\Gamma_0(N);\mathbb{Z})$ is a lattice" in Chapter 13. Resolution adopted here: Chapter 12 states Shimura's integral basis theorem (11.2, R) and proves the consequences (rank, saturation, split truncation, base change to $\mathbb{Z}_\ell$ and $\mathbb{F}_\ell$: 11.4); Chapter 13 cites 11.2 and 11.4 and proves the perfect pairing with $\mathbb{T}$. The Chapter 13 outline should follow this.
- **B3. Chapter 21 outline, A3.2, step (1)**, contains a self-correcting sentence ("since $T_n^* = w_N T_n w_N$ for n prime to $N\ldots$ the writer should use: $\ldots$"). The correct statement for the Petersson product is: T_n is self-adjoint for $(n, N) = 1$, $U_p^\dagger = w_N U_p w_N$ for $p \mid N$, and a newform is a w_N-eigenvector (Chapter 13; 10.5(a) and 12.4(c) here). Not an inconsistency in substance; the sentence should be cleaned when Chapter 21 is written.
- Consistency checks that passed: the Chapter 8 outline's genera ($g(X_0(N)) = 0, 1, 1, 2, 7, 9, 6, 19, 49, 109, 145$ for $N = 1, 11, 20, 37, 77, 99, 121, 140, 539, 726, 980$) and its description of $X_0(N) \to X(1)$ (degree $\mu(N)$, branched over $j = 0, 1728, \infty$) agree with 5.3; the Chapter 9 outline's 11a1 coefficients and the claim that they agree with the η-product (to q^{20}) agree with 9.2 (checked to q^{2000}); Chapter 9's deferral of the bijectivity of j is discharged by 7.5 without circularity.

Chapter 15

Chapter 13: Hecke operators, newforms and twists (`ch:hecke`, `chapters/ch13_hecke.tex`)

Stage 1 line-up. Part IV (Modular forms and modular curves), between Chapter 12 (spaces of modular forms, Petersson product, Atkin-Lehner involutions, integrality and Sturm) and Chapters 14 and 15 (Jacobians and the modular degree; Galois representations and level lowering). Target length 1,930 lines of LaTeX (card excluded; see Part 3), inside the 1,000 to 2,200 range of PLAN.md Section 7.3.

Rigour tags: **P** = proved here, **S** = proof sketched here, **R** = stated; reference. Item format, as in the Chapter 2 to 12 outlines: number, label-ready name, kind, tag; the precise statement with all hypotheses; reference (book and chapter or section; a theorem number only where sure, otherwise marked "(verify)" or given by section); the paper's use by its `\label` names, or "background". Every number quoted below (Hecke matrices, characteristic and minimal polynomials of U_p, dimensions of new and old subspaces, ranks of Hecke algebras, determinants of the pairing, congruence numbers, conductors of twists, Sturm bounds, Petersson Gram matrices, Rankin-Selberg residues) was checked by Python scripts in the session scratchpad (`c1.py` to `c11.py`), using PARI/GP 2.17 through `cypari2` and Cremona's `ecdata` files for conductors below 10,000 (verification log, Part 6.6).

15.1 1. Purpose and global conventions

15.1.1 1.1 What this chapter is for

The paper uses the Hecke theory of $S_2(\Gamma_0(N))$ in six places, and the chapter is built around them.

1. **The Hecke algebra and its perfect pairing with integral forms** (Section 3 preamble: "Let $\mathbb{T}$ be the Hecke algebra of level N, the subring of $\mathrm{End}\, S_2(\Gamma_0(N))$ generated over $\mathbb{Z}$ by all T_n, $n \geq 1$, including the operators U_p for $p \mid N$"; Lemma lem:crit: "$\phi_f \colon \mathbb{T} \to \mathbb{Z}$, $T_n \mapsto a_n(f)$", "The pairing $S_2(\Gamma_0(N); \mathbb{Z}) \times \mathbb{T} \to \mathbb{Z}$, $(g, t) \mapsto a_1(tg)$, is perfect", "the forms killed by I are $\mathbb{Z}f$", "Every step holds verbatim over $\mathbb{Z}_\ell$"; Theorem thm:quant Step 1: "with the pairing $S_2(\Gamma_0(N); \mathbb{Z}_\ell) \times \mathbb{T}_\ell \to \mathbb{Z}_\ell$"). The chapter defines T_n and U_p (double cosets and q-expansions), proves commutativity, multiplicativity and the recursion, proves that $\mathbb{T}$ is free of rank $\dim S_2$ over $\mathbb{Z}$ and that the pairing is perfect (from Chapter 12's integral basis theorem and Sturm bound), and derives the dictionary "normalised eigenforms with values in A = ring homomorphisms $\mathbb{T} \to A$", in particular the eigencharacter ϕ_f and the maximal ideals of $\mathbb{T}$.
2. **Petersson adjoints and the complement $f^\perp$** (Lemma lem:crit: "the f-line is $\mathbb{T}$-stable, and so is $f^\perp$, because f is an eigenvector of the adjoints of the T_n as well"; Lemma lem:old: "orthogonal to the newform f"; Theorem thm:twist: "$g^\perp$, the $\mathbb{T}$-stable complement of the g-line"). The chapter proves

that T_n, $(n,N)=1$, is self-adjoint, that $U_p^\dagger = w_N U_p w_N$, and that $f^\perp$ is $\mathbb{T}$-stable for a newform f; it separates the Petersson complement from the $\mathbb{T}$-stable complement, which differ for non-newforms such as g^S in Theorem thm:quant.

3. **Newforms, oldforms and strong multiplicity one** (Section 1: "the newform $f = f_E$"; Lemma lem:old: "a newform on $\Gamma_0(M)$ with $p^2 \mid M$ has $a_p = 0$ [AtkinLehner70, Theorem 3]"; Theorem thm:twist: "differing from that of g at infinitely many p"; Theorem thm:quant Step 1: "strong multiplicity one identifies the forms with the T_p-eigenvalues of g, $p \nmid N$, with the span of the $g(dz)$, $d \mid N/N_S$"). The chapter states Atkin-Lehner theory (decomposition into old and new parts, newform bases, multiplicity one), strong multiplicity one, and the local facts $a_p = 0$ for $p^2 \mid N$, $a_p = -\varepsilon_p$ for $p \,\|\, N$.
4. U_p **on oldforms; stabilised and depleted forms;** f^Σ**,** M^Σ**,** δ_p (Lemma lem:old: the eigenvalues of U_p on the span of the $h(p^j z)$, the stabilisations, the depleted forms, "U_p is not semisimple on this span when 0 is a repeated eigenvalue"; Section 4.1: "Put $M^\Sigma = M\prod_{p\in\Sigma} p^{\delta_p}$ and let $f^\Sigma = \sum_{(n,\Sigma)=1} a_n(f)q^n$ be the Σ-depleted eigenform, of level M^Σ"; Theorem thm:quant Step 1: "the operators U_q, $q \in S$, have the eigenvalue 0 with multiplicity one", "the others have U_q-eigenvalue $a_q(g) = \pm 1 \ldots$ or a root $\alpha, \beta \ldots$ which are ℓ-adic units"). The chapter proves the full linear algebra (cyclic vector, Jordan blocks), the structure of $\mathbb{T}\otimes\mathbb{Q}$ and of its generalised eigenspaces, and the level of f^Σ, with the example at level 88 where $\mathbb{T}\otimes\mathbb{Q}$ is not reduced (important_notes.md point 19).
5. **Twists** (Lemma lem:twist (iii): "the twist $g\otimes\chi$, with coefficients $a_n(g\otimes\chi) = \chi(n)a_n(g)$, is a newform of weight 2 on $\Gamma_0(M')$, $M' = \mathrm{lcm}(M, q^2)$"; Theorem thm:twist: "$\iota\colon h \mapsto h\otimes\chi_S$ maps $S_2(\Gamma_0(N_S))$ into $S_2(\Gamma_0(\mathrm{lcm}(N_S, D^2)))$, preserves integrality, and sends an eigenform ... to an eigenform with eigenvalues twisted by $\chi_S(n)$"; Theorem thm:quant Step 1: "ι intertwines T_p with $\chi_S(p)T_p$"). The chapter proves the elementary properties of the twist operator (Gauss sums, level, integrality, intertwining, $\iota\circ\iota =$ depletion) and states the theorem of Atkin and Li with its hypothesis $q^2 \nmid M$, with examples where the level drops when $q^2 \mid M$.
6. **Congruences between eigenforms** (Section 1: "$f \equiv g \pmod r$ coefficientwise"; Lemma lem:crit: "$a_n(g) \equiv a_n(f) \pmod\lambda$ for every $n \geq 1$"; Lemma lem:old: "the congruence for all n then follows from the multiplicativity of the coefficients, from $a_{p^k} = a_p^k$ for $p \mid N$, and from the recursion"; Theorem thm:quant Step 1: idempotents of $\mathbb{T}_\ell$). The chapter proves that congruences at all primes give congruences at all n, that congruences away from Σ give congruences of the Σ-depleted forms, and the decomposition $\mathbb{T}_\ell = \prod \mathbb{T}_\mathfrak{m}$ with its effect on forms and on the pairing.

It also states the L-function of a newform (Euler product, functional equation) and the Rankin-Selberg residue formula with the normalisation of Chapter 12 (the exact relation with $L(\mathrm{Sym}^2 f, 2)$ at bad primes is Chapter 19's), and defines the Hecke congruence ideal and the congruence number of a rational newform (the identity between them is the paper's Lemma lem:crit, proved in Chapter 15).

It discharges the debts of earlier chapters: Chapter 12 (its C8, 6.2(e), 6.5, 6.6, 9.3, 10.3, 10.5(a), 11.2 and 11.4 as used for the pairing, 11.4(v), 11.5, 12.4(b), (c), and its B2 = book-internal note B11: Chapter 12 states Shimura's integral-basis theorem and proves its consequences, Chapter 13 proves the perfect pairing); Chapter 3 (its 9.6, 10.4, 10.5, 10.7(v), 10.8, 11.1(iv) and Exercise 14: the modular meaning of the shift operator, the level-88 example, multiplicity one); Chapter 6 (its 7.1: Galois conjugates of newforms are newforms, K_f totally real); Chapter 8 (its 12.1(c): the Hecke algebra is the ring of operators generated by the Hecke correspondences); Chapter 9 (its 8.3 and 10.4: $X^2 - a_pX + p$, the recursion, twists of newforms match twists of curves); Chapter 2 (its forward pointer: twisting by Dirichlet characters, Gauss sums, conductors in Atkin-Li).

The physics reader meets three familiar objects, one sentence each: the Hecke recursion $T_{p^{r+1}} = T_pT_{p^r} - pT_{p^{r-1}}$ is the three-term recursion of the Chebyshev polynomials of the second kind (RHB, special functions), with generating function $\sum_r T_{p^r}X^r = (1 - T_pX + pX^2)^{-1}$ (2.4); the self-adjoint commuting family T_n, $(n,N)=1$, diagonalised by an orthogonal basis of eigenforms is the spectral theorem for commuting Hermitian operators (5.2); the functional equation of $L(f,s)$ comes from the Mellin transform of $f(iy)$ (10.2), RHB's integral transforms.

15.1.2 1.2 Global conventions fixed in this chapter

- **C1 (weight, level, spaces).** $N \geq 1$ is the level and $k \geq 2$ an even weight; everything the paper needs is $k = 2$, and every statement is given for $k = 2$ first, with the general-k form in parentheses where it costs nothing. $S_k(N) = S_k(\Gamma_0(N))$, $M_k(N) = M_k(\Gamma_0(N))$, trivial character throughout (the only characters are the quadratic twists of Section 9, which keep the character trivial because $\chi^2 = 1$). $S_2(N; A) = S_2(\Gamma_0(N); \mathbb{Z}) \otimes A$ for a commutative ring A (Chapter 12, C6).
- **C2 (slash and double cosets).** The slash operator is Chapter 12's C3: $(f|_k\gamma)(z) = (\det\gamma)^{k/2}(cz + d)^{-k} f(\gamma z)$. For $\alpha \in \mathrm{GL}_2^+(\mathbb{Q})$ with $\Gamma\alpha\Gamma = \bigsqcup_i \Gamma\alpha_i$ the double coset operator is

$$[\Gamma\alpha\Gamma]_k f = (\det\alpha)^{k/2-1} \sum_i f|_k\alpha_i.$$

  With this normalisation $[\Gamma\alpha\Gamma]_k$ coincides with Diamond-Shurman's operator (their slash carries $(\det\gamma)^{k-1}$ and no prefactor; the products agree), and in weight 2 the prefactor is 1. Say this once in a `convention` box.
- **C3 (names of Hecke operators; level dependence).** At level N the operators are T_n, $n \geq 1$; for a prime $p \mid N$ the operator T_p is written U_p, as in the paper ("all T_n, $n \geq 1$, including the operators U_p for $p \mid N$"), and $T_{p^r} = U_p^r$. The operator T_p depends on the level: for $h \in S_k(M)$ and $p \nmid M$, the operator T_p of level M and the operator U_p of level Mp differ, $T_p^{(M)}h = U_p^{(Mp)}h + p^{k-1}h(pz)$ (2.3). When two levels are in play the level is written as a superscript. The letter U alone never denotes an operator on q-series; V_d denotes $h(z) \mapsto h(dz)$.
- **C4 (Hecke algebras).** $\mathbb{T} = \mathbb{T}_N = \mathbb{Z}[T_n : n \geq 1] \subset \mathrm{End}_{\mathbb{C}} S_2(N)$ is the full Hecke algebra of level N (the paper's $\mathbb{T}$); $\mathbb{T}^{(N)} = \mathbb{Z}[T_n : (n, N) = 1]$ is the anemic Hecke algebra; $\mathbb{T}_A = \mathbb{T} \otimes_{\mathbb{Z}} A$, $\mathbb{T}_\ell = \mathbb{T} \otimes \mathbb{Z}_\ell$. Unless said otherwise $\mathbb{T}$ acts on cusp forms of weight 2 (not on M_2, where the Eisenstein series would add a factor).
- **C5 (eigenforms).** An *eigenform* is a nonzero simultaneous eigenvector of all of $\mathbb{T}$ (all T_n including the U_p); an *anemic eigenform* is one for $\mathbb{T}^{(N)}$ only; *normalised* means $a_1 = 1$; a *newform* is a normalised eigenform in the new subspace (6.1, 6.2). A *system of eigenvalues* is a ring homomorphism $\mathbb{T} \to \overline{\mathbb{Q}}$ (or $\mathbb{T}^{(N)} \to \overline{\mathbb{Q}}$); its *generalised eigenspace* $V(\phi)$, for a system of eigenvalues ϕ (upright: φ is reserved for maps of curves, Appendix B, B-C13(b)), is as in Chapter 3, its 10.5. The $\mathbb{T}$-*stable complement* of a generalised eigenspace is the sum of the other generalised eigenspaces.
- **C6 (Petersson product and adjoints).** The Petersson product is Chapter 12's C5 (unnormalised, linear in the first variable; PARI's `mfpetersson` divides it by $\mu(N)$). Adjoints are written $t^\dagger$ (book-internal note B4: $*$ is reserved for pullbacks and Pontryagin duals; Chapter 8 uses † too).
- **C7 (characters and twists).** For a Dirichlet character χ of conductor r, $f \otimes \chi = \sum_n \chi(n)a_n(f)q^n$ with $\chi(n) = 0$ when $(n, r) > 1$. For an odd prime q, $q^* = (-1)^{(q-1)/2}q$ and $\chi_{q^*} = \left(\frac{q^*}{\cdot}\right)$ is the quadratic character of conductor q (the paper's χ_q; the chapter writes χ_{q^*}, as Chapter 2 decided, to avoid the clash with the cyclotomic χ_ℓ, important_notes.md point 32). For a finite set S of odd primes, $D = \prod_{q\in S} q$ and $\chi_S = \prod_{q\in S} \chi_{q^*}$, of conductor D.
- **C8 (depletion).** For a finite set Σ of primes and a q-series $h = \sum a_n q^n$, $h^\Sigma = \sum_{(n,\Sigma)=1} a_n q^n$, where $(n, \Sigma) = 1$ means that no prime of Σ divides n (the paper's notation). For a single prime, $h^{\{p\}}$; never $h^{(p)}$, since a parenthesised superscript is reserved for twists $E^{(q)}$ and for removed Euler factors $\zeta^{(N)}$, $\mathbb{T}^{(N)}$ (Appendix B, B-C13(e)).
- **C9 (congruences).** "$f \equiv g \pmod{\mathfrak{a}}$" means coefficientwise at **all** $n \geq 1$ (Chapter 12, C6); a congruence only at the primes outside a set is always written out as such. λ is a prime of the ring of integers $\mathcal{O}$ of a number field, above ℓ.
- **C10 (typography).** PLAN.md Section 7.2: plain hyphens in "Atkin-Lehner", "Atkin-Li", "Rankin-Selberg", "Petersson", "Eichler-Shimura", "Coleman-Edixhoven"; ranges "1 to 8"; labels `...:hecke:...`; tables with `booktabs`.

15.2 2. Line-up of the core results, in teaching order

15.2.1 Section 2. Hecke operators: double cosets and q-expansions

2.1 def:hecke:double-coset (definition and proposition; P). Let $\Gamma = \Gamma_0(N)$ and $\alpha \in \mathrm{GL}_2^+(\mathbb{Q})$. (i) $\Gamma \cap \alpha^{-1}\Gamma\alpha$ has finite index in Γ (both contain $\Gamma(m)$ for a suitable m: scale α to an integer matrix and take $m = N\det\alpha$), so $\Gamma\alpha\Gamma = \bigsqcup_{i=1}^{r} \Gamma\alpha_i$ is a finite disjoint union, $r = [\Gamma : \Gamma \cap \alpha^{-1}\Gamma\alpha]$. (ii) The operator $[\Gamma\alpha\Gamma]_k$ of C2 does not depend on the representatives α_i and maps $M_k(N)$ to $M_k(N)$ and $S_k(N)$ to $S_k(N)$ (right multiplication by $\gamma \in \Gamma$ permutes the cosets $\Gamma\alpha_i$; each $f|_k\alpha_i$ is holomorphic and bounded, respectively vanishing, at every cusp by Chapter 12, its 6.2). Reference: DiamondShurman 5.1 (cite the section); Shimura71 3.1 (cite the section). Paper: background (the definition of $\mathbb{T}$ in the Section 3 preamble).

2.2 prop:hecke:coset-decomposition (proposition and definition; P). Let p be prime and $\Gamma = \Gamma_0(N)$. (i) If $p \nmid N$,

$$\Gamma\left(\begin{smallmatrix}1&0\\0&p\end{smallmatrix}\right)\Gamma = \bigsqcup_{j=0}^{p-1} \Gamma\left(\begin{smallmatrix}1&j\\0&p\end{smallmatrix}\right) \;\sqcup\; \Gamma\left(\begin{smallmatrix}p&0\\0&1\end{smallmatrix}\right) \qquad (p+1 \text{ cosets});$$

(ii) if $p \mid N$, $\Gamma\left(\begin{smallmatrix}1&0\\0&p\end{smallmatrix}\right)\Gamma = \bigsqcup_{j=0}^{p-1} \Gamma\left(\begin{smallmatrix}1&j\\0&p\end{smallmatrix}\right)$ (p cosets). Define $T_p = [\Gamma\,\mathrm{diag}(1,p)\Gamma]_k$ on $M_k(N)$, written U_p when $p \mid N$ (C3). Proof: the coset space is in bijection with the matrices of determinant p with lower left entry divisible by N modulo left multiplication by Γ (row reduction to upper triangular form); for $p \mid N$ every element of the double coset has upper left entry prime to p (it is $\equiv a_1a_2 \pmod p$ with a_1a_2 a unit, since $p \mid c_i$), which excludes $\mathrm{diag}(p,1)$. Reference: DiamondShurman 5.2 (cite the section; their Proposition 5.2.1 is the decomposition for $\Gamma_1(N)$, verify); Shimura71 3.4. Paper: Section 3 preamble ($\mathbb{T}$ contains the U_p).

2.3 prop:hecke:Tp-q-expansion (proposition; P). For $f = \sum_{n\ge0} a_n q^n \in M_k(N)$ and p prime:

$$a_n(T_pf) = a_{pn}(f) + p^{k-1}a_{n/p}(f) \quad (p \nmid N), \qquad a_n(U_pf) = a_{pn}(f) \quad (p \mid N),$$

for all $n \ge 0$, with $a_{n/p} = 0$ when $p \nmid n$. In weight 2: $a_n(T_pf) = a_{pn} + p\,a_{n/p}$. Equivalently, with V_p of C3 and the operator $\sum a_nq^n \mapsto \sum a_{pn}q^n$ on q-series: $T_p^{(M)}h = U_p^{(Mp)}h + p^{k-1}V_ph$ for $h \in M_k(M)$, $p \nmid M$ (the level dependence of C3). Proof: $f|_k\left(\begin{smallmatrix}1&j\\0&p\end{smallmatrix}\right) = p^{-k/2}f(\frac{z+j}{p})$, and $\sum_{j=0}^{p-1} e^{2\pi in(z+j)/p}$ is $p\,q^{n/p}$ or 0; $f|_k\,\mathrm{diag}(p,1) = p^{k/2}f(pz)$; multiply by $p^{k/2-1}$. Reference: DiamondShurman Proposition 5.2.2 (verify) and 5.3; SerreCourse VII.5.3 (level one); Miyake 4.5. Paper: Section 3 preamble, Lemma lem:old ("$U_ph(p^jz) = h(p^{j-1}z)$", "$U_ph = a_p(h)h - p\,h(pz)$"), Theorem thm:twist and Theorem thm:quant Step 1 (the intertwining of 9.1).

2.4 thm:hecke:Tn (definition and theorem; P). Define $T_1 = 1$, $T_{p^{r+1}} = T_pT_{p^r} - p^{k-1}T_{p^{r-1}}$ for $p \nmid N$ and $r \ge 1$, $T_{p^r} = U_p^r$ for $p \mid N$, and $T_n = \prod_{p^r \| n} T_{p^r}$. Then on $M_k(N)$: (i) all the T_n commute; (ii)

$$T_mT_n = \sum_{d\mid(m,n),\ (d,N)=1} d^{k-1}T_{mn/d^2},$$

in particular $T_mT_n = T_{mn}$ for $(m,n) = 1$; (iii) for $f \in M_k(N)$ and $m \ge 1$,

$$a_m(T_nf) = \sum_{d\mid(m,n),\ (d,N)=1} d^{k-1}a_{mn/d^2}(f), \qquad \text{in particular} \qquad a_1(T_nf) = a_n(f);$$

(iv) $\sum_{r\ge0} T_{p^r}X^r = (1 - T_pX + p^{k-1}X^2)^{-1}$ for $p \nmid N$ and $(1 - U_pX)^{-1}$ for $p \mid N$. Proof: (i) for primes, by comparing the coefficient formulas of 2.3 (both $T_pT_{p'}f$ and $T_{p'}T_pf$ have n-th coefficient $a_{pp'n} + p^{k-1}a_{p'n/p} + p'^{k-1}a_{pn/p'} + (pp')^{k-1}a_{n/pp'}$ for $p \ne p'$, with the obvious changes for U); (ii) and (iii) by induction on r from the recursion; (iv) is the recursion. Physics aside: (iv) says $T_{p^r} = p^{r(k-1)/2}U_r\big(T_p/(2p^{(k-1)/2})\big)$ with U_r the Chebyshev polynomial of the second kind. Reference: DiamondShurman Proposition 5.3.1 (verify); Shimura71 Theorem 3.24 (verify); SerreCourse VII.5. Paper: Lemma lem:crit ($a_1(tg)$; $\mathbb{T}$ is the $\mathbb{Z}$-span of the T_n, used in "$\phi_g \equiv \phi_f \pmod\lambda$ on all of $\mathbb{T}$"), Lemma lem:old (multiplicativity, $a_{p^k} = a_p^k$ for $p \mid N$, the recursion), Theorem thm:twist ("both are multiplicative").

2.5 `ex:hecke:eisenstein-and-delta` (examples; P). (i) Level one: $T_pE_k = \sigma_{k-1}(p)E_k$ (from $\sigma_{k-1}(pn) + p^{k-1}\sigma_{k-1}(n/p) = \sigma_{k-1}(p)\sigma_{k-1}(n)$ for all $n \geq 1$, by multiplicativity of σ_{k-1}); $\dim S_{12}(1) = 1$ forces $T_n\Delta = \tau(n)\Delta$, so τ is multiplicative and satisfies the recursion (Mordell). (ii) Level N: the weight-two Eisenstein series $E_2^{(N)}$ of Chapter 12 (its 8.4) satisfies $T_pE_2^{(N)} = (1+p)E_2^{(N)}$ for $p \nmid N$, and for $N = 11$ also $U_{11}E_2^{(11)} = E_2^{(11)}$ (coefficient computation with σ). With Chapter 12's congruence $f_{11} \equiv -\frac{1}{24}E_2^{(11)} \pmod 5$ this says that the eigencharacters of f_{11} and of the Eisenstein series agree modulo 5: the maximal ideal (5) of $\mathbb{T}_{11} = \mathbb{Z}$ contains $T_p - 1 - p$ ($p \neq 11$) and $U_{11} - 1$, so it is Eisenstein (Chapter 14). Paper: Section sec:eis and Conjecture conj:eis (the Eisenstein case), background.

2.6 `rem:hecke:correspondences` (remark; R, forward pointer). In weight 2, $f \mapsto f\,dz$ identifies $S_2(N)$ with the holomorphic differentials on $X_0(N)$ (Chapter 12, its 8.2), and T_p becomes the action on differentials of the Hecke correspondence $X_0(N) \leftarrow X_0(Np) \to X_0(N)$, $z \mapsto z$ and $z \mapsto pz$ (Chapter 8, its 12.1(c)); this is how $\mathbb{T}$ acts on $J_0(N)$ and on $H^1(X_0(N), \mathbb{Z})$ (Chapter 14), and why the same ring $\mathbb{T}$ acts on forms and on cohomology. Nothing in this chapter depends on the remark. Reference: DiamondShurman 6.3 (cite the section); Shimura71 7.2 (cite the section). Paper: Section 4.1 (the Hecke action on $H^1(X_0(M^\Sigma), \mathbb{Z}_\ell)$), Remark rem:free.

15.2.2 Section 3. The Hecke algebra, eigenforms and eigencharacters

3.1 `def:hecke:hecke-algebra` (definition and proposition; P). $\mathbb{T} = \mathbb{T}_N$ and $\mathbb{T}^{(N)}$ as in C4. (i) $\mathbb{T}$ is commutative (2.4(i)). (ii) $\mathbb{T}$ is the $\mathbb{Z}$-span of $\{T_n : n \geq 1\}$ (the span is closed under products by 2.4(ii)), and $\mathbb{T}^{(N)}$ is the $\mathbb{Z}$-span of the T_n with $(n, N) = 1$. (iii) $\mathbb{T}$ preserves $S_2(N; \mathbb{Z})$ (2.4(iii)); hence $\mathbb{T} \subset \mathrm{End}_{\mathbb{Z}} S_2(N; \mathbb{Z})$ is a finitely generated free $\mathbb{Z}$-module (a subgroup of a free module of rank d^2, $d = \dim S_2(N)$), and $\mathbb{T} \otimes A$ acts on $S_2(N; A)$ for every ring A. Reference: DiamondShurman 6.5 (cite the section); Ribet83 Section 2 (verify). Paper: Section 3 preamble (the definition of $\mathbb{T}$), Lemma lem:crit (the $\mathbb{Z}$-span is used in "$\phi_g \equiv \phi_f$ on all of $\mathbb{T}$"), Lemma lem:ddt417 and Lemma lem:hecke ("$\mathbb{T}$ finite and free"; Chapter 20).

3.2 `prop:hecke:eigenforms` (definition and proposition; P). Let $0 \neq f \in S_k(N)$ with $T_nf = \lambda_n f$ for all $n \geq 1$. (i) $a_n(f) = \lambda_n a_1(f)$ (by 2.4(iii)), so $a_1(f) \neq 0$; the *normalised* eigenform $f/a_1(f)$ has $T_nf = a_n(f)f$ and $f = \sum \lambda_n q^n$. (ii) A normalised eigenform satisfies $a_{mn} = a_m a_n$ for $(m, n) = 1$, $a_{p^{r+1}} = a_p a_{p^r} - p^{k-1}a_{p^{r-1}}$ for $p \nmid N$, and $a_{p^r} = a_p^r$ for $p \mid N$. (iii) The *eigencharacter* $\phi_f \colon \mathbb{T} \to \mathbb{C}$, $t \mapsto$ eigenvalue of t on f, is a ring homomorphism with $\phi_f(T_n) = a_n(f)$, and $f \mapsto \phi_f$ is injective on normalised eigenforms; distinct normalised eigenforms are linearly independent (distinct characters of a commutative algebra). (iv) An anemic eigenform need not have $a_1 \neq 0$: $f_{11}(2z) \in S_2(22)$ is an eigenform of $\mathbb{T}^{(22)}$ with $a_1 = 0$. Reference: DiamondShurman 5.8 (cite the section). Paper: Lemma lem:crit ("$\phi_f \colon \mathbb{T} \to \mathbb{Z}$, $T_n \mapsto a_n(f)$"; "normalised eigenform g ... for all of $\mathbb{T}$"), Lemma lem:old ("normalise it by $a_1(g) = 1$"; the relations of (ii)), Theorem thm:twist, Theorem thm:quant Step 1 ("$\pi \colon \mathbb{T} \to \mathbb{Z}$ the eigencharacter of g^S").

3.3 `prop:hecke:coefficient-field` (proposition; P given 4.1 and Chapter 6, its 7.1). Let f be a normalised eigenform in $S_2(N)$. (i) Every $a_n(f)$ is an algebraic integer, $K_f = \mathbb{Q}(a_n(f) : n \geq 1)$ is a number field and $\phi_f(\mathbb{T})$ is an order in $\mathcal{O}_{K_f}$ (Chapter 6, its 7.1, applied to $\mathbb{T}$, free of finite rank by 3.1(iii)). (ii) For $\sigma \in \mathrm{Aut}(\mathbb{C})$, $f^\sigma = \sum \sigma(a_n)q^n$ is again a normalised eigenform in $S_2(N)$ (the T_n have integer matrices in a $\mathbb{Z}$-basis of $S_2(N; \mathbb{Z})$, Chapter 12, its 11.2), and f^σ is a newform if f is (the new subspace is defined over $\mathbb{Q}$, 6.1(iii)). (iii) If f is a newform, K_f is totally real: $a_n \in \mathbb{R}$ for $(n, N) = 1$ by 5.2, and $a_p \in \{0, \pm 1\}$ for $p \mid N$ by 6.4. (iv) Warning: a normalised eigenform of the full $\mathbb{T}$ that is not a newform can have non-real coefficients: the 2-stabilisations of f_{11} in $S_2(22)$ have $a_2 = -1 \pm i$ (7.1). For $f = f_E$: $K_f = \mathbb{Q}$, $\mathcal{O} = \mathbb{Z}$, $\lambda = \ell$ (the paper's "$\lambda = \ell$" in Section 4). Reference: DiamondShurman 6.5 (cite the section); Shimura71 3.5. Paper: Lemma lem:crit and Lemma lem:old ("coefficients in the ring of integers $\mathcal{O}$ of a number field", "after enlarging $\mathcal{O}$ and λ"), Section 4.1.

3.4 `thm:hecke:eigen-dictionary` (theorem; P given 4.1). Let A be a commutative ring. (i) The map $S_2(N; A) \to \mathrm{Hom}_{\mathbb{Z}}(\mathbb{T}, A)$, $g \mapsto (t \mapsto a_1(tg))$, is an isomorphism of $\mathbb{T}$-modules, where $\mathbb{T}$ acts on $\mathrm{Hom}(\mathbb{T}, A)$ by $(t\psi)(u) = \psi(tu)$. (ii) Under it the normalised eigenforms with coefficients in A (elements g with $a_1(g) = 1$ and $tg = \phi(t)g$ for some $\phi(t) \in A$, for all t) correspond bijectively to the ring homomorphisms $\phi \colon \mathbb{T} \to A$, via $g = f_\phi := \sum_n \phi(T_n)q^n$. (iii) Consequences. $A = \mathbb{C}$ (or $\overline{\mathbb{Q}}$): the normalised eigenforms of $S_2(N)$ are the f_ϕ, $\phi \in \mathrm{Hom}(\mathbb{T}, \overline{\mathbb{Q}})$, finitely many. $A = \overline{\mathbb{F}}_\ell$: the mod ℓ eigenforms $\sum \phi(T_n)q^n$ correspond

to $\mathrm{Hom}(\mathbb{T},\overline{\mathbb{F}}_\ell)$, and their kernels are the maximal ideals of $\mathbb{T}$ of residue characteristic ℓ (each maximal ideal $\mathfrak{m}$ corresponding to the $[\mathbb{T}/\mathfrak{m}:\mathbb{F}_\ell]$ Frobenius-conjugate embeddings). $A=\mathcal{O}$: $\phi_f\colon \mathbb{T}\to\mathcal{O}$ and its reduction $\mathbb{T}\to\mathcal{O}/\lambda$, whose kernel $\mathfrak{m}_{f,\lambda}$ is "the maximal ideal attached to (f,λ)". Proof: (i) is 4.1(iii) tensored with A; (ii) ψ is multiplicative with $\psi(1)=1$ iff the corresponding g is a normalised eigenform, because $a_1(tT_ng)=\psi(tT_n)$. Reference: DDT Section 1.6 or 4.1 (cite the section; verify); Ribet83 Section 2 (verify). Paper: Lemma lem:crit ("$\phi_g\colon\mathbb{T}\to\mathcal{O}$", "the kernel of $\mathbb{T}\to\mathcal{O}/\lambda$, a maximal ideal of residue characteristic ℓ"), Lemma lem:hecke ("$\mathfrak{m}$ the maximal ideal of $\mathbb{T}\otimes\mathbb{Z}_\ell$ containing ℓ and $\ker\phi_f$"), Theorem thm:quant Step 1.

15.2.3 Section 4. The integral structure: the perfect pairing

4.1 `thm:hecke:perfect-pairing` (theorem; P given Chapter 12, its 11.2 and 11.4). Let $S=S_2(N;\mathbb{Z})$, $d=\dim S_2(N)$, $B=B_2(N)$ the Sturm bound of Chapter 12 (its C7), $\mathbb{T}=\mathbb{T}_N$. (i) $\mathbb{T}$ acts faithfully on S. (ii) The pairing

$$S\times\mathbb{T}\to\mathbb{Z},\qquad \langle g,t\rangle=a_1(tg),$$

is $\mathbb{Z}$-bilinear, satisfies $\langle ug,t\rangle=\langle g,ut\rangle$, and is perfect. (iii) Hence $\mathbb{T}$ is free of rank d over $\mathbb{Z}$, $S\simeq\mathrm{Hom}_{\mathbb{Z}}(\mathbb{T},\mathbb{Z})$ and $\mathbb{T}\simeq\mathrm{Hom}_{\mathbb{Z}}(S,\mathbb{Z})$ as $\mathbb{T}$-modules, and for every commutative ring A the induced pairing $S_2(N;A)\times\mathbb{T}_A\to A$ is perfect (in particular over $\mathbb{Z}_\ell$). (iv) $\mathbb{T}$ is spanned as a $\mathbb{Z}$-module by $T_1,\dots,T_B$; it is in general not spanned by $T_1,\dots,T_d$. Proof: (i) and injectivity of $\mathbb{T}\to\mathrm{Hom}(S,\mathbb{Z})$: if $a_1(tg)=0$ for all g, then $a_n(tg)=a_1(T_ntg)=a_1(tT_ng)=0$, so $tg=0$; injectivity of $S\to\mathrm{Hom}(\mathbb{T},\mathbb{Z})$: $a_n(g)=\langle g,T_n\rangle$. Surjectivity of $\mathbb{T}\to\mathrm{Hom}(S,\mathbb{Z})$: its image contains the functionals $g\mapsto a_n(g)=\langle g,T_n\rangle$, and these span $\mathrm{Hom}(S,\mathbb{Z})$ because the truncation $S\to\mathbb{Z}^B$, $g\mapsto(a_1,\dots,a_B)$, is a split injection (Chapter 12, its 11.4(ii)), so its dual $\mathbb{Z}^B\to\mathrm{Hom}(S,\mathbb{Z})$ is onto; this also gives (iv). Then both maps are isomorphisms of free modules of the same rank (Chapter 3, its 9.6). Verified (Part 6.6): $\mathrm{rank}_{\mathbb{Z}}\mathbb{T}=d$ and $\det(\langle g_i,t_j\rangle)=\pm1$ in $\mathbb{Z}$-bases at the levels $11,23,33,37,44,77,88,99,121,539$ ($d=1,2,3,2,4,7,9,9,6,49$); $T_1,\dots,T_d$ span a sublattice of index 5 at level 77, 3 at 88, 2 at 99, and are linearly dependent at 121 and 539. Reference: Ribet83 Theorem 2.2 and AU96 Lemma 2.1 (as cited by the paper; verify both numbers); ARS12 Section 3; DDT Section 1 (cite the section). Paper: Lemma lem:crit ("The pairing … is perfect", "Every step holds verbatim over $\mathbb{Z}_\ell$, the pairing being perfect after tensoring with $\mathbb{Z}_\ell$"), Theorem thm:quant Step 1 ("with the pairing $S_2(\Gamma_0(N);\mathbb{Z}_\ell)\times\mathbb{T}_\ell\to\mathbb{Z}_\ell$"), PLAN.md scope.

4.2 `cor:hecke:pairing-consequences` (corollary; P given 4.1 and Chapter 3, its 9.6). In the notation of 4.1: (i) a form $g\in S_2(N;A)$ is determined by the values $a_1(tg)$, $t\in\mathbb{T}$; (ii) for an ideal $\mathfrak{a}\subset\mathbb{T}$ with $\mathbb{T}/\mathfrak{a}$ torsion-free, $S[\mathfrak{a}]\simeq\mathrm{Hom}(\mathbb{T}/\mathfrak{a},\mathbb{Z})$ is saturated in S of rank $\mathrm{rank}\,\mathbb{T}/\mathfrak{a}$; (iii) for a ring homomorphism $\phi\colon\mathbb{T}\to\mathbb{Z}$, $S[\ker\phi]=\mathbb{Z}f_\phi$ (3.4(ii)); in particular for a newform f with rational coefficients, the forms killed by $I=\ker\phi_f$ are $\mathbb{Z}f$; (iv) for ideals I,J with $I\cap J=0$ and torsion-free quotients, $S/(S[I]\oplus S[J])$ has the order of $\mathbb{T}/(I+J)$ (Chapter 3, its 9.6(iii)). Paper: Lemma lem:crit ("under it the forms killed by I are $\mathbb{Z}f$ and the forms killed by J are L"; "Dualising the inclusion … shows that $\mathbb{Z}f\oplus L$ has index r'"), Section 6 map row ("a form is determined by its pairing with $\mathbb{T}$").

4.3 `def:hecke:congruence-ideal` (definition and remark; P for the remark). Let $\mathcal{O}$ be $\mathbb{Z}$, $\mathbb{Z}_{(\ell)}$, $\mathbb{Z}_\ell$ or $\mathcal{O}_\lambda$ and $\phi\colon\mathbb{T}_{\mathcal{O}}\to\mathcal{O}$ a surjective ring homomorphism (an eigencharacter with values in $\mathcal{O}$). Its *Hecke congruence ideal* is $\eta(\phi)=\phi(\mathrm{Ann}_{\mathbb{T}_{\mathcal{O}}}(\ker\phi))\subset\mathcal{O}$ (Chapter 3, its 11.1). For a newform $f\in S_2(N)$ with rational coefficients, the *congruence number* r_f is the largest integer r such that $f\equiv g\pmod r$ for some $g\in S_2(N;\mathbb{Z})\cap f^\perp$. Remark: r_f is finite, because $\mathbb{Z}f\oplus(f^\perp\cap S_2(N;\mathbb{Z}))$ has finite index in $S_2(N;\mathbb{Z})$ (both summands are saturated, of ranks 1 and $d-1$, 8.2) and a congruence modulo r produces the element $(f-g)/r$ of order r in the quotient. The equality $r_f\mathbb{Z}=\eta(\phi_f)$ is the paper's Lemma lem:crit, proved in Chapter 15 (restated with the commutative-algebra viewpoint in Chapter 20); for the $X_0(N)$-optimal curve $r_E=r_{f_E}$ and its relation with $\deg\varphi_E$ (Agashe-Ribet-Stein) is Chapter 14's. The paper also uses $\eta(\pi)$ for the eigencharacter π of a depleted form g^S, which is not a newform ("$\eta^{\mathrm{Hecke}}(g^S)$", Theorem thm:quant Step 1); 8.2 says when the definitions apply to such π. Computed values in 11.2. Paper: Section 1 (the congruence number), Lemma lem:crit, Lemma lem:hecke, Theorem thm:quant Step 1, Section 6 map row (congruence ideal, r_E).

15.2.4 Section 5. Petersson adjoints

5.1 `prop:hecke:double-coset-adjoint` (proposition; P). Let $\Gamma = \Gamma_0(N)$, $\alpha \in \mathrm{GL}_2^+(\mathbb{Q})$ and $\alpha' = (\det\alpha)\alpha^{-1}$. For $f, g \in S_k(N)$: $\langle [\Gamma\alpha\Gamma]_k f, g\rangle = \langle f, [\Gamma\alpha'\Gamma]_k g\rangle$. Proof: change of variables $z \mapsto \alpha z$ in the Petersson integral over $\Gamma \cap \alpha^{-1}\Gamma\alpha$ (Chapter 12, its 10.2(iii)), and the fact that one set of representatives serves for $\Gamma\alpha\Gamma$ on the left and on the right. Reference: DiamondShurman Proposition 5.5.2 (verify); Shimura71 3.4. Paper: background (5.2, 5.3).

5.2 `thm:hecke:self-adjoint` (theorem; P). For $(n, N) = 1$ the operator T_n is self-adjoint on $S_k(N)$. Consequently the anemic algebra $\mathbb{T}^{(N)} \otimes \mathbb{C}$ consists of commuting self-adjoint operators: $S_k(N)$ is the orthogonal direct sum of the eigenspaces of $\mathbb{T}^{(N)}$ (anemic generalised eigenspaces are eigenspaces), and the eigenvalues of T_n, $(n, N) = 1$, are real. Proof: for $p \nmid N$, $\mathrm{diag}(1,p)' = \mathrm{diag}(p,1)$, and $\Gamma\,\mathrm{diag}(1,p)\Gamma = \Gamma\,\mathrm{diag}(p,1)\Gamma$ (both double cosets are the set of integer matrices of determinant p with lower left entry divisible by N, an elementary-divisor argument), so 5.1 applies; then 2.4(ii). Spectral theorem for commuting normal operators (linear algebra). Reference: DiamondShurman Theorem 5.5.3 (verify); SerreCourse VII.5.3 (level one). Paper: Lemma lem:crit (adjoints), Theorem thm:twist and Theorem thm:quant Step 1 (generalised eigenspaces of the T_p, $p \nmid N$), Chapter 12, its 10.5(a).

5.3 `prop:hecke:Up-adjoint` (proposition; P for (i), (ii); S for (iii)). Let $p \mid N$ and $w_N = |_k W_N$ (Chapter 12, its 12.1). (i) $U_p^\dagger = w_N U_p w_N$ on $S_k(N)$. Proof: $W_N\,\mathrm{diag}(1,p)W_N^{-1} = \mathrm{diag}(p,1)$ and W_N normalises Γ, so $[\Gamma\,\mathrm{diag}(p,1)\Gamma]_k = w_N[\Gamma\,\mathrm{diag}(1,p)\Gamma]_k w_N^{-1}$; apply 5.1; $w_N^{-1} = w_N$ (Chapter 12's slash convention C3 makes w_N an exact involution). (ii) U_p is in general neither self-adjoint nor normal: on $\mathrm{span}\{f_{11}(z), f_{11}(2z)\} \subset S_2(22)$ the Gram matrix of the Petersson product is proportional to $\left(\begin{smallmatrix} 1 & -1/3 \\ -1/3 & 1/4 \end{smallmatrix}\right)$, U_2 has matrix $\left(\begin{smallmatrix} -2 & 1 \\ -2 & 0 \end{smallmatrix}\right)$ (columns: images of $f_{11}(z)$, $f_{11}(2z)$), its adjoint has matrix $\left(\begin{smallmatrix} 0 & -1/2 \\ 4 & -2 \end{smallmatrix}\right)$, and w_{22} has matrix $\left(\begin{smallmatrix} 0 & -1/2 \\ -2 & 0 \end{smallmatrix}\right)$ (Chapter 12, its 12.4(b)), with $w_{22}U_2w_{22}$ equal to the adjoint (all verified with PARI's `mfpetersson` and `mfatkininit`). (iii) Gram entries in weight 2: for a newform h of level M and $p \nmid M$, with all products at level Mp: $\langle h(z), h(pz)\rangle = \frac{a_p(h)}{p(p+1)}\langle h, h\rangle$ and $\langle h(pz), h(pz)\rangle = p^{-2}\langle h, h\rangle$ (check: $-2/(2\cdot 3) = -1/3$ and $1/4$ for f_{11}, $p = 2$; the second is a change of variables, the first from the trace $\mathrm{Tr}_M^{Mp}(h(pz)) = p^{-1}T_p^{(M)}h = \frac{a_p}{p}h$ and $\langle h, g\rangle_{\Gamma_0(Mp)} = \langle h, \mathrm{Tr}_M^{Mp} g\rangle_{\Gamma_0(M)}$ for h of level M, with $\langle h, h\rangle_{\Gamma_0(Mp)} = (p+1)\langle h, h\rangle_{\Gamma_0(M)}$ (Chapter 12, its 10.2(i)); the general-weight form, verify in DiamondShurman 5.7). Reference: DiamondShurman 5.5 (cite the section; the identity $T_p^\dagger = w_N T_p w_N^{-1}$ for all p is there, verify); AtkinLehner70. Paper: Lemma lem:crit ("because f is an eigenvector of the adjoints of the T_n as well"), Section 4.1 (the cohomological analogue $w_{M^\Sigma}T_n w_{M^\Sigma}^{-1}$, Chapter 14), Theorem thm:quant Step 1 (why the complement of g^S must be the $\mathbb{T}$-stable one, 8.2).

5.4 `cor:hecke:perp-stable` (corollary; P). Let $V \subset S_k(N)$ be a subspace stable under $\mathbb{T}$ and under w_N. Then $V^\perp$ is stable under $\mathbb{T}$ (and w_N). In particular, if f is an eigenform of all of $\mathbb{T}$ and an eigenvector of w_N (every newform is, 6.4), then $f^\perp$ is $\mathbb{T}$-stable. Proof: $\langle tg, v\rangle = \langle g, t^\dagger v\rangle$ with $t^\dagger = t$ or $w_N t w_N$ (5.2, 5.3), and $t^\dagger V \subset V$. Paper: Lemma lem:crit (the $\mathbb{T}$-stability of $f^\perp$), Lemma lem:old, Theorem thm:twist ($f^\perp$ at level N), Chapter 12, its 10.3.

15.2.5 Section 6. Oldforms and newforms: Atkin-Lehner theory

6.1 `def:hecke:old-new` (definition and proposition; P for (i), (ii); P given 6.2 for (iii)). For $M \mid N$ and $d \mid N/M$ let $V_d\colon S_k(M) \to S_k(N)$, $h \mapsto h(dz) = d^{-k/2}h|_k\,\mathrm{diag}(d,1)$ (Chapter 12, its 6.2(e)). The *old subspace* is $S_k^{\mathrm{old}}(N) = \sum_{p\mid N}\big(S_k(N/p) + V_p S_k(N/p)\big)$ (equivalently the span of all $V_d S_k(M)$, $M \mid N$, $M \neq N$, $d \mid N/M$) and the *new subspace* is its Petersson orthogonal complement $S_k^{\mathrm{new}}(N)$. (i) T_n with $(n, N) = 1$ commutes with every V_d (2.3), so both subspaces are $\mathbb{T}^{(N)}$-stable (with 5.2). (ii) Both are stable under every U_p, $p \mid N$: the old one by the formulas of 7.1 applied to $V_d h$, the new one by 5.4, since w_N maps $V_d S_k(M)$ into $V_{N/(Md)}S_k(M)$ (Chapter 12, its 12.4(b), generalised by the same matrix identity) and so preserves the old subspace. (iii) Both are defined over $\mathbb{Q}$: the new subspace is the sum of the anemic eigenspaces of the newforms of level N (6.2(iii)), and $\mathrm{Aut}(\mathbb{C})$ permutes newforms (3.3(ii)). Reference: AtkinLehner70 (Section 2, cite the section); DiamondShurman 5.6, 5.7 (cite the sections). Paper: Section 1 ("the newform"), Lemma lem:old ("the forms $h(dz)$, $d \mid N/M$"), Section 4.1.

6.2 thm:hecke:atkin-lehner (theorem, Atkin-Lehner, Li; R, with a sketch S). (i) $S_k^{\mathrm{new}}(N)$ has a basis of *newforms*: normalised eigenforms of all of $\mathbb{T}_N$ (all T_n, the U_p included); distinct newforms are orthogonal. (ii) (Multiplicity one.) If f is a newform of level N and L is a multiple of N, the eigenspace in $S_k(L)$ of the system of eigenvalues of f for $\mathbb{T}^{(L)}$ is $\mathrm{span}\{f(dz) : d \mid L/N\}$, of dimension $\sigma_0(L/N)$; in $S_k^{\mathrm{new}}(N)$ it is $\mathbb{C}f$. (iii) (Decomposition.)

$$S_k(N) = \bigoplus_{M \mid N} \bigoplus_{d \mid N/M} V_d\, S_k^{\mathrm{new}}(M),$$

the forms $h(dz)$, h a newform of level $M \mid N$, $d \mid N/M$, form a basis of $S_k(N)$, and the old subspace is the sum of the terms with $M \neq N$. (iv) In particular a newform f of level N is orthogonal to every $h(dz)$ with $h \neq f$ a newform of level $M \mid N$, so $f^{\perp} = \bigoplus_{h \neq f} \mathrm{span}\{h(dz)\}$ (with 5.2: different systems are orthogonal). Sketch (DiamondShurman's route): the Main Lemma (a form whose coefficients a_n vanish for all n prime to N lies in $\sum_{p \mid N} V_p S_k(N/p)$) gives (ii) for the new subspace, since the difference of two new anemic eigenforms with the same system, suitably normalised, has $a_n = 0$ for $(n, N) = 1$; (iii) by induction on N. Reference: AtkinLehner70 (the main theorems; cite by section, the numbering to be verified); Li, *Newforms and functional equations*, Math. Ann. 212 (1975) (proposed key `Li75`, Part 6.5); DiamondShurman Theorem 5.8.3 and the Main Lemma 5.7.1 (verify the numbers); Miyake 4.6. Paper: Section 1 ("$f = f_E \in S_2(\Gamma_0(N))$ the newform"), Lemma lem:old ("span a $\mathbb{T}$-stable subspace V of $S_2(\Gamma_0(N))$ orthogonal to the newform f (whether $M < N$, or $M = N$ and $h \neq f$)"), Lemma lem:crit ($\mathbb{T} \otimes \mathbb{Q} = \mathbb{Q} \times \mathbb{T}'$ needs (ii)), Theorem thm:quant Step 1 ("the forms with the T_p-eigenvalues of g, $p \nmid N$, ... the span of the $g(dz)$, $d \mid N/N_S$").

6.3 thm:hecke:strong-multiplicity-one (theorem; R for (i), S for (ii)). (i) If f and g are newforms of weight k and levels M, M', and $a_p(f) = a_p(g)$ for all but finitely many primes p, then $f = g$ (and $M = M'$). Equivalently: the systems of T_p-eigenvalues of two distinct newforms differ at infinitely many primes. (ii) (Weight 2, density one.) The same conclusion holds if $a_p(f) = a_p(g)$ for all p in a set of primes of Dirichlet density one: the λ-adic Galois representations of f and g (Chapter 15) then have equal traces on a dense set of Frobenius elements (Chebotarev, Chapter 5), hence equal traces everywhere, hence isomorphic semisimplifications (Brauer-Nesbitt, Chapter 4), hence $a_p(f) = a_p(g)$ for all $p \nmid MM'\ell$, and (i) applies. (iii) Consequence (P): for a newform f of level N, a generalised eigenform $h \in S_k(L)$, $N \mid L$, of $\mathbb{T}^{(L)}$ whose system differs from that of f at infinitely many p lies in the sum of the anemic eigenspaces of the newforms $h' \neq f$, and is orthogonal to f and to every $f(dz)$. Reference: Miyake Theorem 4.6.19 (verify); DiamondShurman 5.8 (cite the section); AtkinLehner70. Paper: Theorem thm:twist ("a system of T_p-eigenvalues ... differing from that of g at infinitely many p ... so $\iota(h) \in f^{\perp}$"), Theorem thm:quant Step 1 ("strong multiplicity one"), Lemma lem:old and Theorem thm:qual ($h \neq f$; important_notes.md point 20), PLAN.md scope ("a newform is determined by a_p for p in any set of density one").

6.4 prop:hecke:newform-local (proposition; P for (i) given AtkinLehner70's stability of the old subspace under w_d, P for (ii); R for (iii)). Let f be a newform of weight k, level N, trivial character. (i) For every $d \,\|\, N$, $f|_k W_d = \varepsilon_d(f) f$ with $\varepsilon_d(f) = \pm 1$ (the involution w_d commutes with $\mathbb{T}^{(N)}$ and preserves the new subspace, being a self-adjoint isometry that maps the old subspace into itself (AtkinLehner70), so it preserves $\mathbb{C}f$ by 6.2(ii)). (ii) Hence 5.4 applies: $f^{\perp}$ is $\mathbb{T}$-stable. (iii) (AtkinLehner70, Theorem 3, as cited by the paper; verify which parts) If $p^2 \mid N$ then $a_p(f) = 0$; if $p \,\|\, N$ then $a_p(f) = -p^{k/2-1}\varepsilon_p(f)$, so in weight 2 $a_p(f) = -\varepsilon_p(f) \in \{\pm 1\}$. (iv) For $f = f_E$ (Chapter 14): $a_p = 1, -1, 0$ for split multiplicative, nonsplit multiplicative, additive reduction at p. Checks: f_{11}: $\varepsilon_{11} = -1$, $a_{11} = 1$ (Chapter 12, its 9.2(iii)); $a_{11} = 0$ for 121a to 121d; $a_7 = 0$ for 539c, 539d; $a_7(77\mathrm{c}) = -1$, $a_{11}(77\mathrm{c}) = 1$. Reference: AtkinLehner70 Theorem 3 (as cited); DiamondShurman 5.8 and 6.6 (cite the sections). Paper: Lemma lem:old (case $p^2 \mid N$: "a newform on $\Gamma_0(M)$ with $p^2 \mid M$ has $a_p = 0$"; case $p \,\|\, N$: $a_p(h) = \pm 1$), Lemma lem:crit (the w_N-eigenvector property behind the $\mathbb{T}$-stability of $f^{\perp}$), Lemma lem:euler and Theorem thm:quant Step 1 ($a_q(g) = \pm 1$ at a multiplicative q), Chapter 12, its 12.4(c).

6.5 cor:hecke:dimensions (corollary; P given 6.2). $\dim S_k(N) = \sum_{M \mid N} \sigma_0(N/M) \dim S_k^{\mathrm{new}}(M)$, and by Moebius inversion $\dim S_k^{\mathrm{new}}(N) = \sum_{M \mid N} \beta(N/M) \dim S_k(M)$ with $\beta = \mu * \mu$ multiplicative, $\beta(p) = -2$, $\beta(p^2) = 1$, $\beta(p^r) = 0$ for $r \geq 3$. Table (weight 2; verified with PARI's `mfdim` for the full and the new space and with the formula):

N	11	22	33	44	77	88	99	121	242	539	980	1089
$\dim S_2(N)$	2	3	4	7	9	9	6	22	49	145	109	
$\dim S_2^{\mathrm{new}}(N)$	0	1	1	5	3	4	4	10	34	13	41	

For example $S_2(121)$ is spanned by the four rational newforms 121a, 121b, 121c, 121d and the two oldforms $f_{11}(z)$, $f_{11}(11z)$ (Chapter 12, its 9.3), and $S_2(88)$ by three newforms of level 88, $f_{44}(z)$, $f_{44}(2z)$ and the four $f_{11}(2^j z)$, $0 \le j \le 3$. Reference: as 6.2. Paper: background (the levels of Section 3 and Section 4.1).

15.2.6 Section 7. U_p on oldforms: stabilisation, depletion, non-semisimplicity

7.1 `prop:hecke:Up-oldspan` (proposition; P). Let h be a newform of weight k and level M, p a prime, $e \ge 0$, and $V_{h,p,e} = \mathrm{span}\{h(p^j z) : 0 \le j \le e\}$, of dimension $e+1$ (6.2(iii)), inside $S_k(Mp^e)$; let U_p be the operator of level Mp^e (assume $e \ge 1$ or $p \mid M$). (i) $U_p h(p^j z) = h(p^{j-1}z)$ for $j \ge 1$; $U_p h = a_p(h)h - p^{k-1}h(pz)$ if $p \nmid M$; $U_p h = a_p(h)h$ if $p \mid M$. (ii) $h(p^e z)$ is a cyclic vector, so the characteristic and minimal polynomials of U_p on $V_{h,p,e}$ coincide:

$$X^{e-1}(X^2 - a_p(h)X + p^{k-1}) \quad (p \nmid M,\ e \ge 1), \qquad X^e(X - a_p(h)) \quad (p \mid M).$$

(iii) Every eigenvalue has a one-dimensional eigenspace, spanned by: for $p \nmid M$, writing $X^2 - a_p X + p^{k-1} = (X-\alpha_p)(X-\beta_p)$: the *p-stabilisations* $h_\alpha = h - \beta_p h(pz)$ (eigenvalue α_p) and $h_\beta = h - \alpha_p h(pz)$ (eigenvalue β_p), and, if $e \ge 2$, the *p-depleted form* $h^{\{p\}} = h - a_p h(pz) + p^{k-1}h(p^2 z)$ (eigenvalue 0); for $p \,\|\, M$: h (eigenvalue $a_p = \pm p^{k/2-1}$) and, if $e \ge 1$, $h^{\{p\}} = h - a_p h(pz)$ (eigenvalue 0); for $p^2 \mid M$: $h = h^{\{p\}}$ (eigenvalue $a_p = 0$). (iv) U_p is semisimple on $V_{h,p,e}$ iff its characteristic polynomial is squarefree: iff $e \le 2$ and $\alpha_p \ne \beta_p$ ($p \nmid M$), $e \le 1$ ($p \,\|\, M$), $e = 0$ ($p^2 \mid M$); otherwise the eigenvalue 0 has a single Jordan block of size $e-1$, e or $e+1$. (v) Each eigenvector of (iii) is a normalised eigenform of the full Hecke algebra of level Mp^e, with the eigenvalues of h at the other operators. Verified matrices (columns: images of $f_{11}(p^j z)$): level 22: $\left(\begin{smallmatrix} -2 & 1 \\ -2 & 0 \end{smallmatrix}\right)$, X^2+2X+2; level 44: $X(X^2+2X+2)$; level 88: $X^2(X^2+2X+2)$ with a Jordan block of size 2 at 0; level 121 ($p = 11 \,\|\, M$): $\left(\begin{smallmatrix} 1 & 1 \\ 0 & 0 \end{smallmatrix}\right)$, $X(X-1)$; level 1331: $X^2(X-1)$. Reference: Chapter 3, its 10.7(v) (the linear algebra); AtkinLehner70; DiamondShurman 5.6 and 5.7 exercises (cite the sections). Paper: Lemma lem:old (every formula of its proof: "$U_p h(p^j z) = h(p^{j-1}z)$", "The eigenvalues of U_p there are therefore the roots $\alpha_p, \beta_p \ldots$ together with 0 if $e_p \ge 2$", "the p-stabilisations", "the depleted forms", "U_p is not semisimple on this span when 0 is a repeated eigenvalue"), Theorem thm:quant Step 1 ("on that span the operators $U_q \ldots$ have the eigenvalue 0 with multiplicity one"; "the others have U_q-eigenvalue $a_q(g) = \pm 1 \ldots$ or a root α, β"), Section 4.1 (stabilisations, Chapter 9, its 8.3).

7.2 `cor:hecke:Up-eigenvalues` (corollary; P given 6.2(iii), 6.4(iii), 7.1). Weight 2. Every eigenvalue of U_p on $S_2(N)$, $p \mid N$, is 0, or ± 1, or a root of $X^2 - a_p(h)X + p$ for a newform h of level $M \mid N$ with $p \nmid M$. In particular, for every prime λ of $\overline{\mathbb{Q}}$ not above p, every nonzero eigenvalue of U_p is a λ-adic unit (the roots are algebraic integers with product p), so a generalised eigenform of $\mathbb{T}$ whose U_p-eigenvalue is $\equiv 0 \pmod{\lambda}$ has U_p-eigenvalue 0. Verified characteristic polynomials of U_p (PARI `mfheckemat`): level 88, U_2: $X^7(X^2+2X+2)$; level 99, U_3: $X^6(X+1)(X^2+X+3)$; level 121, U_{11}: $X^5(X-1)$; level 539, U_7: $X^{42}(X-1)^3(X+1)^2(X^2+2X+7)$; level 980, U_7: $X^{126}(X-1)^8(X+1)^9(X^2-2X+7)$; level 1089, U_3: $X^{76}(X-1)^{10}(X+1)^{11}(X^2-2X+3)^2(X^2+X+3)^4$. Paper: Theorem thm:quant Step 1 ("The generalised eigenforms of $\mathbb{T}$ whose eigencharacter is congruent to π modulo $\mathfrak{m}$ have $U_q \equiv 0$ at $q \in S$" and then "systems of eigenvalues with $U_q = 0$": the passage from $\equiv 0$ to $= 0$ is this corollary; "which are ℓ-adic units since $\ell \ne q$").

7.3 `rem:hecke:non-semisimple` (remark and example; P, with (iii) R). (i) At level 88 the minimal polynomial of U_2 on $S_2(88)$ is $X^2(X^2+2X+2)$: U_2 is not semisimple, and $\mathbb{T}_{88} \otimes \mathbb{Q}$ contains a nonzero nilpotent element (Chapter 3, its 10.7(iii)); so the full Hecke algebra need not be reduced (important_notes.md point 19). The source is $V_{f_{11},2,3}$ ($e = 3$) and also $V_{f_{44},2,1}$ ($2^2 \mid 44$, U_2 nilpotent of order 2 there). (ii) A large multiplicity of the eigenvalue 0 does not by itself make U_p non-semisimple: at level 539 the eigenvalue 0 of U_7 has multiplicity 42 but the minimal polynomial $X(X-1)(X+1)(X^2+2X+7)$ is

squarefree (no single $V_{h,7,e}$ has a repeated root). (iii) (Coleman-Edixhoven.) For a newform of weight 2 and level prime to p, $\alpha_p \neq \beta_p$; so U_p is semisimple on the p-old part of level Mp (for rational a_p this is elementary: $a_p^2 \neq 4p$). Reference: ColemanEdixhoven (in `refs.bib`; the theorem number to be verified). Paper: Lemma lem:old ("U_p is not semisimple on this span when 0 is a repeated eigenvalue, but each listed eigenvalue has an eigenvector"), Lemma lem:ddt417 ($\wp \cap I = 0$ is not automatic, point 19).

7.4 `def:hecke:depletion` (definition and proposition; P). Let f be a newform of weight k and level M and Σ a finite set of primes. (i) Put $\delta_p(f) = 2$ if $p \nmid M$, 1 if $p \parallel M$, 0 if $p^2 \mid M$; equivalently δ_p is the degree in p^{-s} of the inverse Euler factor $L_p(f,s)^{-1}$ (10.1), the number of roots of the Hecke polynomial at p; for $f = f_E$, $\delta_p = 2, 1, 0$ for good, multiplicative, additive reduction, which is $\dim V^{I_p}$ for $V = V_\ell E$ (Chapter 11; the paper's definition in Section 1.1). (ii) The Σ-*depleted form* $f^\Sigma = \sum_{(n,\Sigma)=1} a_n(f) q^n$ (C8) equals $\big(\prod_{p\in\Sigma} D_p\big) f$ with $D_p = 1 - a_p V_p + p^{k-1} V_p^2$ ($p \nmid M$), $1 - a_p V_p$ ($p \parallel M$), 1 ($p^2 \mid M$), by multiplicativity and the recursion of 3.2(ii). (iii) $f^\Sigma \in S_k(M^\Sigma)$ with $M^\Sigma = M \prod_{p\in\Sigma} p^{\delta_p}$, and M^Σ is the least level containing f^Σ (by 7.1(iii): the eigenvalue 0 of U_p occurs in $V_{f,p,e}$ only for $e \geq \delta_p$). (iv) f^Σ is a normalised eigenform of the full Hecke algebra of level M^Σ, with $U_p f^\Sigma = 0$ for $p \in \Sigma$ and the eigenvalues of f elsewhere, and its generalised eigenspace in $S_k(M^\Sigma)$ is the line $\mathbb{C} f^\Sigma$: on V_{f,p,δ_p} the characteristic polynomial is $X(X^2 - a_p X + p^{k-1})$ or $X(X - a_p)$ with $a_p \neq 0$, so 0 is a simple root (6.2(iii) and 7.1). (v) At a level N with $M^\Sigma \mid N$, $N \neq M^\Sigma$, the form f^Σ is still a normalised eigenform of $\mathbb{T}_N$, but its generalised eigenspace can be larger (the depleted $f_{11}^{(2)}$ spans its generalised eigenspace at level 44 and lies in a Jordan block of U_2 at level 88). (vi) $L(f^\Sigma, s) = L(f,s) \prod_{p\in\Sigma} L_p(f,s)^{-1}$ (10.1): depletion removes the Euler factors at Σ, and M^Σ raises the level by one factor p per removed root. Examples (verified): $f_{11}^{\{2\}} = f_{11} + 2f_{11}(2z) + 2f_{11}(4z)$, of level 44, has $a_n = 0$ for even n and $a_n = a_n(f_{11})$ for odd n (checked to $n = 4000$); $f_{11}^{\{11\}} = f_{11} - f_{11}(11z)$, level 121; for 121a, $\Sigma = \{11\}$: $f^\Sigma = f$, level 121; for $g = 77$c and $\Sigma = \{7\}$: $g^\Sigma = g - a_7(g) g(7z)$, level 539; for $g = f_{11}$, $\Sigma = \{7\}$: $g^\Sigma = g + 2g(7z) + 7g(49z)$, level 539 ($a_7(f_{11}) = -2$). Reference: DFG04 Section 1.7.3 (for the notation M^Σ, f^Σ; not Section 1.5.3, important_notes.md B26, checked against the published DFG04 by the Chapter 20 outline); DDT Section 4 (cite the section). Paper: Section 4.1 ("$M^\Sigma = M \prod_{p\in\Sigma} p^{\delta_p}$", "the Σ-depleted eigenform, of level M^Σ"), Section 1.1 (δ_p), Theorem thm:dfg14 and the paragraph after it ("adding an additive prime to Σ changes neither the level nor the ideal"), Theorem thm:quant ("$N_S^\Sigma = N$: the S-depleted form g^S lives at level N"), Lemma lem:old (the depleted forms).

15.2.7 Section 8. The structure of $\mathbb{T}$, generalised eigenspaces and congruences

8.1 `thm:hecke:structure` (theorem; P given 6.2, 6.3, 7.1). Weight k, level N. (i) $S_k(N) \otimes \overline{\mathbb{Q}} = \bigoplus_h V_h$, the sum over the newforms h of levels $M_h \mid N$, with $V_h = \mathrm{span}\{h(dz) : d \mid N/M_h\}$; V_h is the eigenspace of $\mathbb{T}^{(N)}$ for the system of h, and as a module over $\overline{\mathbb{Q}}[U_p : p \mid N]$ it is $\bigotimes_{p|N} V_{h,p,e_p}$ with $e_p = v_p(N/M_h)$. (ii) $\mathbb{T} \otimes \overline{\mathbb{Q}} \simeq \prod_h A_h$, where A_h is the image of $\mathbb{T} \otimes \overline{\mathbb{Q}}$ in $\mathrm{End}(V_h)$, and $A_h \simeq \bigotimes_{p|N} \overline{\mathbb{Q}}[X]/(m_{h,p}(X))$ with $m_{h,p}$ the characteristic polynomial of 7.1(ii) (equal to $X - a_p(h)$ when $e_p = 0$); so $\mathrm{rank}\, \mathbb{T} = \sum_h \prod_p (e_p + 1) = \dim S_k(N)$, consistent with 4.1, and $S_k(N) \otimes \overline{\mathbb{Q}}$ is free of rank one over $\mathbb{T} \otimes \overline{\mathbb{Q}}$ (each V_h is cyclic). (iii) The generalised eigenspaces of $\mathbb{T} \otimes \overline{\mathbb{Q}}$ are the joint generalised eigenspaces $V_h(\mu)$, $\mu = (\mu_p)$ a choice of a root of each $m_{h,p}$, of dimension the product of the multiplicities; $\mathbb{T} \otimes \mathbb{Q}$ is reduced iff every $m_{h,p}$ is squarefree (7.1(iv)). (iv) For a newform f of level N with rational coefficients: $V_f = \mathbb{C} f$, and $\mathbb{T} \otimes \mathbb{Q} = \mathbb{Q} \times \mathbb{T}'$ with $\mathbb{T}'$ acting faithfully on $f^\perp = \bigoplus_{h \neq f} V_h$ (this is the paper's "$\mathbb{T} \otimes \mathbb{Q} = \mathbb{Q} \times \mathbb{T}'$"). In general this is "$\mathbb{T} \otimes \mathbb{Q}$ is a product of fields (the new part, K_h for each Galois orbit of newforms of level N) times the old part", the old part possibly non-reduced. Proof: (i) is 6.2(iii) with 6.3; the factors of (ii) are separated by idempotents of $\mathbb{T}^{(N)} \otimes \overline{\mathbb{Q}}$ (distinct anemic systems, 5.2), and on V_h the algebra generated by the U_p is the tensor product of the algebras of 7.1, each acting with a cyclic vector (Chapter 3, its 10.5, 10.8). Verified at level 88: $9 = 3 \cdot 1 + 2 + 4$ (U_2 on $V_{f_{44}}$ is X^2, on $V_{f_{11}}$ is $X^2(X^2 + 2X + 2)$). Reference: DDT Section 1 or 4 (cite the section); Chapter 3, its 10.1, 10.5. Paper: Lemma lem:crit ("Writing $\mathbb{T} \otimes \mathbb{Q} = \mathbb{Q} \times \mathbb{T}'$ with $\mathbb{T}'$ acting faithfully on $f^\perp$"), Lemma lem:old ("V is the tensor product of these local spaces and $\mathbb{T}$ is commutative"), Theorem thm:twist ("generalised eigenform", "$g^\perp$ ... the sum of the generalised eigenspaces of the other systems"), Theorem thm:quant Step 1 ("the generalised eigenspace of π ... is

the line of g^S"), PLAN.md scope.

8.2 `prop:hecke:complements` (proposition; P given 8.1, 5.4). Let g be a normalised eigenform of $\mathbb{T}_N$ with eigencharacter π whose generalised eigenspace is a line $\mathbb{C}g$ (a newform of level N, or f^Σ at level M^Σ by 7.4(iv), or g^S in Theorem thm:quant). Let g^c be the $\mathbb{T}$-stable complement of $\mathbb{C}g$ (C5). (i) $S_k(N) = \mathbb{C}g \oplus g^c$, $\mathbb{T} \otimes \overline{\mathbb{Q}} = \overline{\mathbb{Q}} \times \mathbb{T}''$ with $\mathbb{T}''$ faithful on g^c, and if π has values in $\mathbb{Q}$ then g^c is defined over $\mathbb{Q}$ and $g^c \cap S_2(N;\mathbb{Z})$ is saturated of rank $d-1$; so the arguments of the paper's Lemma lem:crit apply to π and $\eta(\pi)$ (4.3) is the largest modulus of a congruence $g \equiv h$, $h \in g^c \cap S_2(N;\mathbb{Z})$ (proved in Chapter 15 for newforms; the proof uses only (i)). (ii) If g is a newform, $g^c = g^\perp$ (6.2(iv)). (iii) If g is not a newform, $g^c \neq g^\perp$ in general: for $g = f_{11}^{\{2\}}$ at level 44, $g^c \ni f_{11} - \beta f_{11}(2z)$ (a stabilisation), which is not orthogonal to g: with the Petersson Gram matrix of $f_{11}(z), f_{11}(2z), f_{11}(4z)$ at level 44, proportional to $\begin{pmatrix} 1 & -1/3 & 1/24 \\ -1/3 & 1/4 & -1/12 \\ 1/24 & -1/12 & 1/16 \end{pmatrix}$, one finds $\langle g, f_{11} - \beta f_{11}(2z)\rangle = \frac{5}{12}\langle f_{11}, f_{11}\rangle$ (verified with PARI). Paper: Theorem thm:twist ("$g^\perp$, the $\mathbb{T}$-stable complement of the g-line"; at level N_S the newform g makes the two notions agree), Theorem thm:quant Step 1 ("in the $\mathbb{T}$-stable complement of the g^S-line"; there it is not a Petersson complement).

8.3 `prop:hecke:local-components` (proposition; P given Chapter 3, its 10.3, 10.5, 10.6 and 4.1). Let ℓ be a prime and $\mathbb{T}_\ell = \mathbb{T} \otimes \mathbb{Z}_\ell = \prod_{\mathfrak{m}} \mathbb{T}_{\mathfrak{m}}$ over the maximal ideals $\mathfrak{m} \ni \ell$ of $\mathbb{T}$, with idempotents $e_{\mathfrak{m}}$. (i) $S_2(N;\mathbb{Z}_\ell) = \bigoplus_{\mathfrak{m}} e_{\mathfrak{m}} S_2(N;\mathbb{Z}_\ell)$, and $e_{\mathfrak{m}} S_2(N;\mathbb{Z}_\ell) \otimes \overline{\mathbb{Q}}_\ell$ is the sum of the generalised eigenspaces of the $\phi\colon \mathbb{T} \to \overline{\mathbb{Q}}_\ell$ with $\phi^{-1}(\mathfrak{m}_{\overline{\mathbb{Z}}_\ell}) = \mathfrak{m}$, that is, with $\phi \equiv$ the residual character of $\mathfrak{m}$. (ii) A ring homomorphism $\phi\colon \mathbb{T} \to \mathcal{O}_\lambda$ factors through exactly one $\mathbb{T}_{\mathfrak{m}}$, $\mathfrak{m} = \mathfrak{m}_{\phi,\lambda}$, and ϕ kills the other factors: they lie in $\ker\phi$, not in $\mathrm{Ann}(\ker\phi)$, and $\eta(\phi) = \eta(\phi|_{\mathbb{T}_{\mathfrak{m}}})$ (important_notes.md point 35; Chapter 3, its 11.1(v)). (iii) The pairing of 4.1 satisfies $\langle e_{\mathfrak{m}} g, t\rangle = \langle g, e_{\mathfrak{m}} t\rangle$, so it restricts to perfect pairings $e_{\mathfrak{m}} S_2(N;\mathbb{Z}_\ell) \times \mathbb{T}_{\mathfrak{m}} \to \mathbb{Z}_\ell$ and $e_{\mathfrak{m}} S_2(N;\mathbb{Z}_\ell) \simeq \mathrm{Hom}(\mathbb{T}_{\mathfrak{m}}, \mathbb{Z}_\ell)$ (the modular-forms analogue of point 36, which concerns the cohomological pairing of Chapter 14). (iv) $e_{\mathfrak{m}}$ preserves integrality and every $\mathbb{T}$-stable subspace, and fixes the $\mathfrak{m}$-part; so a congruence $g \equiv h \pmod{\ell^t}$ with g in the $\mathfrak{m}$-part and $h \in S_2(N;\mathbb{Z}_\ell) \cap W$, W a $\mathbb{T}$-stable subspace, gives $g \equiv e_{\mathfrak{m}} h \pmod{\ell^t}$ with $e_{\mathfrak{m}} h \in W$ in the $\mathfrak{m}$-part (Chapter 3, its 10.6). Paper: Lemma lem:hecke ("the congruence ideal is computed in $\mathbb{T}_{\mathfrak{m}}$, since the other local factors … lie in the annihilator": the correction of point 35), Theorem thm:quant Step 1 ("$\mathfrak{m}$ the maximal ideal of $\mathbb{T}_\ell$ containing $\ker\pi$ and ℓ", "projecting h to the $\mathfrak{m}$-component (an idempotent of $\mathbb{T}_\ell$, so the projection stays in the complement and the congruence survives)"), Remark rem:free, PLAN.md scope.

8.4 `prop:hecke:congruences` (proposition; P). Let $\mathcal{O}$ be the ring of integers of a number field and λ a prime of $\mathcal{O}$. (i) For normalised eigenforms f, g of $\mathbb{T}_N$ with coefficients in $\mathcal{O}$ the following are equivalent: $a_n(f) \equiv a_n(g) \pmod\lambda$ for all $n \geq 1$; the same for all primes $n = p$ (including $p \mid N$); $\phi_f \equiv \phi_g \pmod\lambda$ on all of $\mathbb{T}$; $\mathfrak{m}_{f,\lambda} = \mathfrak{m}_{g,\lambda}$. (From primes to all n: 3.2(ii), since both satisfy the same multiplicativity, the same recursion with the same p^{k-1} for $p \nmid N$, and $a_{p^r} = a_p^r$ for $p \mid N$; from n to $\mathbb{T}$: 3.1(ii).) (ii) If f, g are normalised eigenforms of levels M_f, M_g, Σ contains every prime dividing $M_f M_g$, and $a_p(f) \equiv a_p(g) \pmod\lambda$ for all $p \notin \Sigma$, then $f^\Sigma \equiv g^\Sigma \pmod\lambda$ coefficientwise, both forms lying in $S_k(L)$, $L = \mathrm{lcm}(M_f^\Sigma, M_g^\Sigma)$. (iii) Without depletion the conclusion can fail at the primes of Σ, and the remedy in the paper is the choice of U_p-eigenvalues in Lemma lem:old (Chapter 15); a finite check of a congruence between two forms of the same space is Sturm's bound (Chapter 12, its 11.3), which is a different tool from the recursion (important_notes.md point 46). Paper: Section 1 ("$f \equiv g \pmod r$ coefficientwise"), Lemma lem:crit ("$a_n(g) \equiv a_n(f) \pmod\lambda$ for every $n \geq 1$", "$\phi_g \equiv \phi_f \pmod\lambda$ on all of $\mathbb{T}$"), Lemma lem:old (its reduction "the congruence for all n then follows from …"), the Example of Section 3, PLAN.md scope ("so that congruences at primes propagate to all n").

15.2.8 Section 9. Twisting

9.1 `prop:hecke:twist-operator` (proposition; P for (i), (iii) to (vi); S for (ii)). Let χ be a primitive Dirichlet character of conductor r, $\mathfrak{g}(\bar\chi) = \sum_{u \bmod r} \bar\chi(u) e^{2\pi i u/r}$ its Gauss sum (Chapter 2), and $f \in S_k(M)$. (i)

$$f \otimes \chi = \sum_n \chi(n) a_n(f) q^n = \mathfrak{g}(\bar\chi)^{-1} \sum_{u \bmod r} \bar\chi(u)\, f\big|_k \left(\begin{smallmatrix} 1 & u/r \\ 0 & 1 \end{smallmatrix}\right)$$

(from $\sum_u \bar\chi(u)e^{2\pi inu/r} = \chi(n)\mathfrak{g}(\bar\chi)$ for primitive χ and all n, Chapter 2). (ii) $f\otimes\chi \in S_k(\Gamma_0(\mathrm{lcm}(M,r^2)),\chi^2)$; for quadratic χ, $f\otimes\chi\in S_k(\mathrm{lcm}(M,r^2))$ (sketch: for $\gamma\in\Gamma_0(\mathrm{lcm}(M,r^2))$ and each u there are u' and $\gamma'\in\Gamma_0(M)$ with $\left(\begin{smallmatrix}1 & u/r\\ 0 & 1\end{smallmatrix}\right)\gamma = \gamma'\left(\begin{smallmatrix}1 & u'/r\\ 0 & 1\end{smallmatrix}\right)$, $u'\equiv d^2u \pmod r$). (iii) If χ is quadratic and $f\in S_k(M;\mathbb{Z})$ then $f\otimes\chi\in S_k(\mathrm{lcm}(M,r^2);\mathbb{Z})$ ($\chi(n)\in\{0,\pm1\}$). (iv) (Intertwining.) Let χ be quadratic and $L=\mathrm{lcm}(M,r^2)$. For a prime $p\nmid r$: $T_p^{(L)}(f\otimes\chi) = \chi(p)\,(T_p^{(M)}f)\otimes\chi$, where on each side T_p means U_p if p divides the level (compare coefficients with 2.3; it uses $\chi(p)^2=1$); for $p\mid r$: $U_p^{(L)}(f\otimes\chi)=0$. So $f\mapsto f\otimes\chi$ maps the eigenspace of $\mathbb{T}^{(Mr)}$ for a system (λ_p) into the eigenspace for $(\chi(p)\lambda_p)$, and a normalised eigenform of $\mathbb{T}_M$ to a normalised eigenform of $\mathbb{T}_L$. (v) (Twisting twice.) $(f\otimes\chi)\otimes\bar\chi = \sum_{(n,r)=1}a_n(f)q^n = f^{\{p|r\}}$; for a normalised eigenform f this is the depleted form of 7.4. In particular $g\otimes\chi\otimes\chi$ is the q-depletion of g, not g (important_notes.md point 20). (vi) (The paper's ι.) For a finite set S of odd primes, D and χ_S as in C7, $\iota=\iota_S\colon S_2(M)\to S_2(\mathrm{lcm}(M,D^2))$, $h\mapsto h\otimes\chi_S$, is $\mathbb{C}$-linear, maps $S_2(M;\mathbb{Z})$ into $S_2(\mathrm{lcm}(M,D^2);\mathbb{Z})$, kills every $h(dz)$ with $(d,D)>1$, satisfies $\iota(h)=\iota(h^S)$, and maps generalised eigenspaces of $\mathbb{T}^{(DM)}$ for (λ_p) into those for $(\chi_S(p)\lambda_p)$. Reference: Shimura71 Proposition 3.64 (verify); Miyake 4.3 (cite the section); AtkinLi78 Section 3 (cite the section); DiamondShurman, the exercises on twists in Chapter 5 (verify, else omit). Paper: Theorem thm:twist ("ι ... maps $S_2(\Gamma_0(N_S))$ into $S_2(\Gamma_0(\mathrm{lcm}(N_S,D^2)))$", "preserves integrality", "sends an eigenform of all T_n, $(n,N_S)=1$, to an eigenform with eigenvalues twisted by $\chi_S(n)$", "kills the coefficients at multiples of D"), Theorem thm:quant Step 1 ("ι intertwines T_p with $\chi_S(p)T_p$ for $p\nmid DN$"; "the level of a twist by χ_S divides $\mathrm{lcm}(N,D^2)=N$"), Theorem thm:qual (b), (c) ($h=g\otimes\chi$).

9.2 `thm:hecke:atkin-li` (theorem, Atkin-Li; R, with a sketch S via conductors). Let g be a newform of weight 2 on $\Gamma_0(M)$, q an odd prime and $\chi=\chi_{q^*}$. (i) If $q\nmid M$, $g\otimes\chi$ is a newform on $\Gamma_0(Mq^2)$. (ii) If $q\parallel M$, $g\otimes\chi$ is a newform on $\Gamma_0(Mq)$. In both cases the level is $\mathrm{lcm}(M,q^2)$; this is the paper's Lemma lem:twist (iii). (iii) If $q^2\mid M$, the q-series $g\otimes\chi$ need not be a newform: it is the q-depletion $g'^{\{q\}}$ of the newform g' with $a_p(g')=\chi(p)a_p(g)$ for $p\neq q$ (6.3), whose level divides M (9.1(ii) and 6.2(ii)) and may be smaller or equal. Examples (conductors of the twisted curves by PARI, labels from `ecdata`): $q\nmid M$: 11a1 by -11 gives 121d2, by -7 gives 539d2; 20a1 by -7 gives 980g2 (not 980i); 37a1 by 5 gives 925e1; $q\parallel M$: 14a1 by -7 gives 98a3, 77c1 by -7 gives 539c1, 15a1 by -3 gives 45a4; $q^2\mid M$: 539c1 by -7 gives 77c1 and 539d1 by -7 gives 11a3 (level drops), 121a1 by -11 gives 121c2, 99a1 by -3 gives 99c1 and 49a1 by -7 gives 49a3 (level kept). Sketch (via Chapter 15): the level of a newform is the conductor of its λ-adic Galois representation ρ_g (Carayol86), and $\rho_{g\otimes\chi}=\rho_g\otimes\chi$; if ρ_g is unramified at q, or unipotent on inertia ($q\parallel M$, Steinberg), then $\rho_g\otimes\chi$ has no inertia invariants and is tame at q, so its conductor exponent is 2. Reference: AtkinLi78 Theorem 3.1 (as cited by the paper; verify the numbering and that the hypothesis $q^2\nmid M$ is the one stated there); Li75 (proposed key; Part 6.5) for the case $q\nmid M$; Shimura71 3.6 (cite the section). Paper: Lemma lem:twist (iii), Theorem thm:qual (b), (c) ("$h=g\otimes\chi$ is a newform on $\Gamma_0(Mq)$", "on $\Gamma_0(Mq^2)$"), Theorem thm:twist and Theorem thm:quant ($\iota(g)=f$ in the cases I_n^*, I_0^*), the Example of Section 3 ($f_{11a}\otimes\chi_{-11}$ is the form of 121d).

9.3 `prop:hecke:twist-transport` (proposition; P given 6.2, 6.3, 9.1). Let S, D, χ_S, ι be as in 9.1(vi). Let g be a newform of weight 2 and level M and f the newform whose system is $(\chi_S(p)a_p(g))_{p\nmid DM}$ (6.3), of level N_f, and let L be a common multiple of $\mathrm{lcm}(M,D^2)$ and N_f. (i) If $h\in S_2(M)$ lies in the sum of the anemic generalised eigenspaces of systems different from that of g (that is, $h\in g^\perp$ at level M, 6.2(iv)), then $\iota(h)\in S_2(L)$ lies in the sum of the anemic eigenspaces of systems different from that of f, hence is orthogonal to every $f(dz)$; if $N_f=L$, $\iota(h)\in f^\perp$. (ii) If $f=\iota(g)$ (as q-series; this holds when $a_n(f)=0$ for all n with $(n,D)>1$, for instance when $q^2\mid N_f$ for all $q\in S$, given that $a_p(f)=\chi_S(p)a_p(g)$ also at the primes $p\mid M$ outside S, where the twist is unramified; for elliptic curves this is $a_p(E^{(d)})=\chi_d(p)a_p(E)$ at a multiplicative $p\nmid d$, Chapter 9), then a congruence $g\equiv h \pmod r$ with $h\in g^\perp\cap S_2(M;\mathbb{Z})$ gives $f\equiv\iota(h)\pmod r$ with $\iota(h)\in f^\perp\cap S_2(L;\mathbb{Z})$. (iii) Conversely $\iota(f)=g^S$, and ι maps $f^\perp$ (at level N_f, with $D^2\mid N_f$) into the $\mathbb{T}$-stable complement of $\mathbb{C}g^S$ at level N_f whenever $g^S\in S_2(N_f)$ and the generalised eigenspace of g^S there is a line (8.2): $f^\perp=\bigoplus_{h\neq f}V_h$, and $\iota(V_h)\subset V_{h'}$ with $h'\neq g$, since $h'=g$ would force the system of h to be that of f. This is the engine of the paper's Theorem thm:twist and of Theorem thm:quant Step 1; the theorems themselves (with congruence numbers) are proved in Chapter 20. Paper: Theorem thm:twist (its whole proof), Theorem thm:quant Step 1, Part 7 point N1.

15.2.9 Section 10. The L-function of a newform; Rankin-Selberg

10.1 `prop:hecke:euler-product` (proposition; P). Let f be a normalised eigenform of $\mathbb{T}_N$ of weight k. Then $L(f,s)=\sum_{n\geq 1}a_n(f)n^{-s}$ converges absolutely for $\operatorname{Re}s>k/2+1$ (Chapter 12, its 6.4) and

$$L(f,s)=\prod_{p\nmid N}\bigl(1-a_pp^{-s}+p^{k-1-2s}\bigr)^{-1}\prod_{p\mid N}\bigl(1-a_pp^{-s}\bigr)^{-1}$$

(3.2(ii) and 2.4(iv)). For a newform of weight 2: the factor at $p\mid N$ is $(1\mp p^{-s})^{-1}$ if $p\,\|\,N$ and 1 if $p^2\mid N$, and $L_p(f,s)^{-1}=(1-\alpha_pp^{-s})(1-\beta_pp^{-s})$ at $p\nmid N$; $L(f^\Sigma,s)=L(f,s)\prod_{p\in\Sigma}L_p(f,s)^{-1}$; $L(f\otimes\chi,s)$ has the factors $(1-\chi(p)a_pp^{-s}+\chi(p)^2p^{1-2s})^{-1}$ at $p\nmid rN$. For $f=f_E$, $L(f_E,s)=L(E,s)$ (modularity, Chapter 14). Reference: DiamondShurman 5.9 (cite the section). Paper: Section 1 (the L-function), Section 1.1 (δ_p is the degree of $L_p(f,s)^{-1}$), background.

10.2 `thm:hecke:functional-equation` (theorem; S). Let $f\in S_k(N)$ with $f|_kW_N=\varepsilon_Nf$ (Chapter 12, its 12.1). Then $\Lambda(f,s)=N^{s/2}(2\pi)^{-s}\Gamma(s)L(f,s)=N^{s/2}\int_0^\infty f(iy)y^{s-1}\,dy$ extends to an entire function, bounded in vertical strips, with $\Lambda(f,s)=i^k\varepsilon_N\Lambda(f,k-s)$; in weight 2 the sign is $-\varepsilon_N$. Sketch: split the Mellin integral at $y=1/\sqrt{N}$ and apply W_N on $(0,1/\sqrt{N})$; each piece converges for all s by the exponential decay of f at the cusps. Physics aside: the Mellin transform, RHB's integral transforms. Checks: 11a1 ($\varepsilon_{11}=-1$, sign +1, rank 0), 37a1 ($\varepsilon_{37}=+1$, sign -1, rank 1) (Chapter 12, its 12.4(c)). Reference: DiamondShurman Theorem 5.10.2 (verify); Miyake 4.3 (cite the section). Paper: background (Section 1; Chapter 19 for the adjoint and symmetric square L-functions).

10.3 `thm:hecke:rankin-selberg` (theorem; S, with numerical checks). Let $f\in S_k(N)$ (any cusp form). (i) $D(f,s)=\sum_{n\geq 1}|a_n(f)|^2n^{-s}$ converges for $\operatorname{Re}s>k$, continues meromorphically to the whole plane (through the continuation of the Eisenstein series below), is holomorphic on $\operatorname{Re}s\geq k$ except for a simple pole at $s=k$, and

$$\operatorname{Res}_{s=k}D(f,s)=\frac{3\,(4\pi)^k}{\pi\,\Gamma(k)\,\mu(N)}\,\langle f,f\rangle,\qquad\text{in weight 2}:\quad \operatorname{Res}_{s=2}D(f,s)=\frac{48\pi}{\mu(N)}\langle f,f\rangle,$$

with the unnormalised Petersson product of Chapter 12 (C5) and $\mu(N)=[\mathrm{SL}_2(\mathbb{Z}):\Gamma_0(N)]$. (ii) For a newform of weight 2 with rational coefficients and squarefree level: $D(f,s)=\zeta^{(N)}(s-1)\,L(\mathrm{Sym}^2f,s)/\zeta^{(N)}(2s-2)$, with $L(\mathrm{Sym}^2f,s)$ in the normalisation of the paper and of PARI's `lfunsympow` ($L_p(\mathrm{Sym}^2f,s)=L_p(A_f,s-1)$), and so $\langle f,f\rangle=N\,L(\mathrm{Sym}^2f,2)/(8\pi^3)$. (iii) In general $D(f,s)$ is **not** the naive symmetric square L-function: at a good prime $\sum_r a_{p^r}^2X^r=(1+pX)/\bigl((1-\alpha^2X)(1-pX)(1-\beta^2X)\bigr)$, an extra factor $1+p^{1-s}$; and at the bad primes the correction factors are Chapter 19's (DFG04 Lemma 2.12, Shimura76). Sketch of (i): unfold $\int_{\Gamma_0(N)\backslash\mathfrak{H}}|f|^2y^kE(z,s)\,d\mu$ against the Eisenstein series $E(z,s)=\sum_{\Gamma_\infty\backslash\Gamma_0(N)}\operatorname{Im}(\gamma z)^s$, whose residue at $s=1$ is $1/\operatorname{vol}(\Gamma_0(N)\backslash\mathfrak{H})=3/(\pi\mu(N))$. Verified numerically (PARI, 30 digits): the residue formula of (i), computed from the Euler factors of `lfunsympow`, for 11a1, 37a1, 20a1, 121a1, 99a1; the formula of (ii) for 11a1 ($\langle f,f\rangle=0.04690014787\ldots$, $L(\mathrm{Sym}^2f,2)=1.05759924459\ldots$) and 37a1. Reference: Shimura76; Rankin, *Contributions to the theory of Ramanujan's function $\tau(n)$ and similar arithmetical functions* I, II, Proc. Cambridge Philos. Soc. 35 (1939) (proposed key `Rankin39`, Part 6.5); Zagier85; DFG04 Lemma 2.12 (as cited by the paper). Paper: Section 1 ("$\langle f,f\rangle$ is the special value $L(\mathrm{Sym}^2f,2)$ up to the local factors at the bad primes (Rankin, Shimura; see DFG04 Lemma 2.12)"; "the Euler product of $\sum a_n(f)^2n^{-s}$", important_notes.md point 30), Section 6 map row (Petersson norm, Rankin-Selberg).

15.2.10 Section 11. Worked examples from the paper's world

11.1 `ex:hecke:level-88` (example; P as a computation). $S_2(88)$, $d=9$: newforms of level 88 (three, with $a_2=0$), f_{44} (one, old twice), f_{11} (old four times). U_2 has characteristic polynomial $X^7(X^2+2X+2)$ and minimal polynomial $X^2(X^2+2X+2)$; on $V_{f_{11},2,3}$ its matrix is $\left(\begin{smallmatrix}-2&1&0&0\\-2&0&1&0\\0&0&0&1\\0&0&0&0\end{smallmatrix}\right)$ with a Jordan block of size 2 at 0, kernel spanned by $f_{11}^{(2)}=f_{11}+2f_{11}(2z)+2f_{11}(4z)$, and the stabilisations $f_{11}-\beta f_{11}(2z)$, $\alpha,\beta=-1\pm i$.

So $\mathbb{T}_{88}\otimes\mathbb{Q}$ is not reduced, $\wp\cap I=0$ is a statement to be proved for the depleted form (important_notes.md point 19), and the paper's "the eigenvalue 0 with multiplicity one" (Theorem thm:quant Step 1) is a statement about the minimal level M^{Σ}, which fails one level higher. Also $\operatorname{rank}\mathbb{T}_{88}=9$, pairing determinant -1, $[\mathbb{T}_{88}:\mathbb{Z}T_1+\cdots+\mathbb{Z}T_9]=3$, congruence number of 88a1 equal to 16 (modular degree 8; $2^3\mid 88$).

11.2 ex:hecke:congruence-numbers (example; P as a computation, by the lattice method of 4.3 and 4.2). Congruence numbers r_f of the rational newforms at small levels, against the modular degrees of the optimal curves (Cremona's `degphi` file):

curve	11a1	33a1	37a1	37b1	44a1	77a1	77b1	77c1	88a1	99a1	99b1	99c1	99d1
r_f	1	3	2	2	2	4	20	6	16	12	12	12	6
$\deg\varphi_E$	1	3	2	2	2	4	20	6	8	4	12	12	6

curve	121a1	121b1	121c1	121d1	539a1	539b1	539c1	539d1
r_f	6	4	6	24	320	192	288	504
$\deg\varphi_E$	6	4	6	24	320	192	288	72

The three differences (88a1 at 2, 99a1 at 3, 539d1 at 7) are at primes p with $p^2\mid N$, as the theorem of Agashe, Ribet and Stein allows (Chapter 14); 99A1 with degree 4 and congruence number 12 is the paper's example (Remark after Theorem thm:qual, from ARS12 Table 1), and 539d1 (degree 72, $r=504=2^3\cdot 3^2\cdot 7$) is a second one. Paper: Section 1, the remark after Theorem thm:qual (99A1).

11.3 ex:hecke:twist-pairs (example; P as a computation). The paper's twisting statements on actual forms. (a) Case (c) of Theorem thm:twist ($N_S=N$): $r(121\text{a})=r(121\text{c})=6$ and $121\text{a}\otimes\chi_{-11}=121\text{c}$ (9.2). (b) Case I_n^*: 539c1 has type I_3^* at 7, $539\text{c}\otimes\chi_{-7}$ corresponds to 77c ($q^2\mid M$, level drops), and $r(77\text{c})=6\mid 288=r(539\text{c})$; with $\ell=3$: $v_3(288)=2=v_3(6)+v_3(7^2-1)$, the equality case of Theorem thm:quant. (c) Case I_0^*: 539d1, $539\text{d}\otimes\chi_{-7}$ corresponds to 11a, $r(11\text{a})=1\mid 504$; $\ell=3$: $v_3(504)=2=0+v_3(6)+v_3(8^2-(-2)^2)$. (d) Theorem thm:quant Step 1 at level 539: the Hecke congruence modules (4.3, 8.2) of the depleted forms $(77\text{c})^{\{7\}}=g-a_7(g)g(7z)$ and $(11\text{a})^{\{7\}}=f_{11}+2f_{11}(7z)+7f_{11}(49z)$ have orders 288 and 504, equal to $r(539\text{c})$ and $r(539\text{d})$ at every prime (Part 7, point N1); and $\iota\big((77\text{c})^{\{7\}}\big)=f_{539\text{c}}$, $\iota\big((11\text{a})^{\{7\}}\big)=f_{539\text{d}}$ (checked to $n=3000$). Paper: Theorem thm:twist (cases (a), (b), (c) and the data paragraph after it; important_notes.md point 16), Theorem thm:quant and its proof, the remark after it.

11.4 ex:hecke:level-121 (example; P). $S_2(121)$: four newforms 121a to 121d (all rational, $a_{11}=0$, $11^2\mid 121$) and $V_{f_{11},11,1}$, on which U_{11} has matrix $\left(\begin{smallmatrix}1&1\\0&0\end{smallmatrix}\right)$ (f_{11} has $a_{11}=1$, $11\,\|\,11$), eigenvectors f_{11} (eigenvalue 1) and $f_{11}-f_{11}(11z)$ (eigenvalue 0, the depleted form of level $M^{\{11\}}=121$); U_{11} on $S_2(121)$ has characteristic and minimal polynomials $X^5(X-1)$ and $X(X-1)$ (semisimple). $121\text{a}\equiv 121\text{d}\pmod 3$ at every n (Chapter 12, its 11.5) and $121\text{d}=f_{11}\otimes\chi_{-11}$ (9.2(i)); $T_1,\dots,T_6$ are linearly dependent although $\operatorname{rank}\mathbb{T}_{121}=6$, and $T_1,\dots,T_{22}$ ($B_2(121)=22$) span $\mathbb{T}_{121}$. Paper: the Example of Section 3 ((b): 121a1, 121c, 121d, 11a1), Theorem thm:qual (b).

15.3 3. Section-by-section plan with target line counts

Line counts are lines of LaTeX, card excluded (PLAN.md Section 7.3: 1,000 to 2,200).

section	title	contents	target lines
(starred)	Core results at a glance	the card: every item of Part 2 with its tag and \Cref; conventions C2, C3, C6, C7 restated in one line each	(excluded)
(starred)	Where the paper uses this chapter	Section 3 preamble ($\mathbb{T}$ with the U_p); Lemma lem:crit (ϕ_f, I, J, $f^\perp$, the perfect pairing, over $\mathbb{Z}_\ell$); Lemma lem:old (U_p on $h(p^j z)$, stabilisations, depleted forms, $a_p = 0$ for $p^2 \mid M$, multiplicativity and recursion); Lemma lem:twist (iii) (Atkin-Li); Theorem thm:twist (ι, generalised eigenforms, $g^\perp \to f^\perp$); Theorem thm:quant Step 1 (strong multiplicity one, multiplicity one of 0 for U_q, units, idempotents, η^{Hecke}); Section 4.1 (f^Σ, M^Σ, δ_p); Section 1 (r_E, Petersson norm and $L(\mathrm{Sym}^2 f, 2)$); Lemma lem:hecke (computed in $\mathbb{T}_{\mathtt{m}}$, point 35)	45
1	Introduction and conventions	opening paragraph (what the chapter is for; the three physics connections); C1 to C10 as convention environments, with the double-coset normalisation box (C2) and the level-dependence warning (C3)	70
2	Hecke operators	2.1 to 2.6; the coset computation written out; the q-expansion formulas; commutativity and the recursion with the Chebyshev aside; the Eisenstein examples; paperlink: Section 3 preamble and Lemma lem:old's formulas	210

section	title	contents	target lines
3	The Hecke algebra and eigenforms	3.1 to 3.4; `warning`: anemic eigenforms may have $a_1 = 0$; `warning`: full eigenforms at non-minimal level may have non-real a_p ($-1 \pm i$ at level 22); `paperlink`: Lemma lem:crit's ϕ_f and maximal ideals	150
4	The integral structure: the perfect pairing	4.1 to 4.3 with the full proof of 4.1 from Chapter 12's 11.2 and 11.4; the table of ranks and pairing determinants; `paperlink`: Lemma lem:crit "perfect", "verbatim over $\mathbb{Z}_\ell$"; the definitions of $\eta(\phi)$ and r_f with the pointer to Chapter 15	160
5	Petersson adjoints	5.1 to 5.4; the level-22 Gram matrix and the adjoint of U_2 as a worked example; `paperlink`: Lemma lem:crit "eigenvector of the adjoints"; cohomological analogue pointer to Chapter 14	100
6	Oldforms and newforms	6.1 to 6.5; the Main Lemma sketched; strong multiplicity one with the density-one corollary via Chapter 15; the dimension table; `paperlink`: Lemma lem:old (orthogonality, $a_p = 0$), Theorem thm:quant Step 1 (strong multiplicity one)	210

section	title	contents	target lines
7	U_p on oldforms; depletion	7.1 to 7.4; the matrices at levels 22, 44, 88, 121; Jordan blocks; f^{Σ}, M^{Σ}, δ_p with the "why δ_p" paragraph; `paperlink`: Lemma lem:old (each case), Section 4.1, Theorem thm:quant Step 1 (units and multiplicity one), point 19	200
8	Structure of $\mathbb{T}$; congruences	8.1 to 8.4; $\mathbb{T} \otimes \overline{\mathbb{Q}} = \prod A_h$; Petersson versus $\mathbb{T}$-stable complement with the level-44 computation; $\mathbb{T}_\ell = \prod \mathbb{T}_{\mathfrak{m}}$ on forms and on the pairing; congruences at primes versus at all n; `paperlink`: Lemma lem:crit's $\mathbb{T} \otimes \mathbb{Q} = \mathbb{Q} \times \mathbb{T}'$, Lemma lem:hecke (point 35), Theorem thm:quant Step 1 (idempotents)	200
9	Twisting	9.1 to 9.3; Gauss-sum formula; level $\mathrm{lcm}(M, r^2)$; intertwining; $\iota \circ \iota =$ depletion (point 20); Atkin-Li with the examples table; the transport lemma; `paperlink`: Lemma lem:twist (iii), Theorem thm:twist, Theorem thm:qual (b), (c)	170
10	L-functions and Rankin-Selberg	10.1 to 10.3; Mellin sketch (physics aside); unfolding sketch; the numerical table of residues; `warning`: the Rankin series is not the naive symmetric square (point 30); pointer to Chapter 19	120

section	title	contents	target lines
11	Worked examples	11.1 to 11.4: level 88; congruence numbers versus modular degrees (99A1, 539d1); the twist pairs at levels 121 and 539 and Step 1 of Theorem thm:quant on actual forms; level 121	110
12	Exercises	the 17 exercises of Part 4.2	140
13	Notes and references	DiamondShurman, Shimura71, Miyake, SerreCourse, Koblitz, AtkinLehner70, Li75, AtkinLi78, Ribet83, AU96, ARS12, DDT, ColemanEdixhoven, Shimura76, Rankin39, Zagier85, DFG04, Stein, PARI; which numbers were cited and which are marked (verify)	45
	total		**1,930**

15.4 4. Coverage table and exercises

15.4.1 4.1 Coverage: PLAN.md scope, Section 6 map rows and paper hooks to line-up items

PLAN.md scope item, Section 6 row, or paper hook	line-up items
Hecke operators T_n on $M_k(\Gamma_0(N))$ via double cosets	2.1, 2.2, 2.4
explicit formula on q-expansions: $a_n(T_p f) = a_{pn} + p^{k-1}a_{n/p}$ $(p \nmid N)$; $a_n(U_p f) = a_{pn}$ $(p \mid N)$	2.3, 2.4(iii)
commutativity; multiplicativity $T_{mn} = T_m T_n$ for coprime m, n; $T_{p^{r+1}} = T_p T_{p^r} - p^{k-1} T_{p^{r-1}}$	2.4
the Hecke algebra $\mathbb{T} = \mathbb{Z}[T_n]$ in $\mathrm{End}(S_2)$, finite free over $\mathbb{Z}$ of rank $\dim S_2$, with the U_p	3.1, 4.1(iii)
self-adjointness of T_n, $(n, N) = 1$; non-self-adjointness of U_p	5.1, 5.2, 5.3
eigenforms and normalised eigenforms ($a_1 = 1$, $T_n f = a_n f$)	3.2
the eigencharacter $\phi_f \colon \mathbb{T} \to \mathbb{Z}$ or $\mathcal{O}$; coefficient field and its ring of integers	3.2(iii), 3.3, 3.4
oldforms and newforms: $f(z) \mapsto f(dz)$, $d \mid N/M$; old subspace; new subspace as orthogonal complement	6.1

PLAN.md scope item, Section 6 row, or paper hook	line-up items
Atkin-Lehner theory: newform basis of the new subspace, eigenforms for all T_n and U_p, multiplicity one	6.2, 6.4
strong multiplicity one (density one; same eigenvalues for almost all p)	6.3
U_p on the span of $h(z)$, $h(pz)$: $X^2 - a_p(h)X + p$, stabilisations $h - \beta h(pz)$, $h - \alpha h(pz)$	7.1(ii), (iii)
on $h(z)$, $h(pz)$, $h(p^2z)$: eigenvalue 0, depleted form $h - a_p h(pz) + ph(p^2z)$	7.1(iii)
$p \parallel M$: U_p acts by $a_p(h)$ on h; $h - a_p h(pz)$ has eigenvalue 0	7.1(iii)
$p^2 \mid M$: $a_p(h) = 0$ (Atkin-Lehner Theorem 3), U_p kills h	6.4(iii), 7.1(iii)
non-semisimplicity of U_p when 0 is a repeated eigenvalue	7.1(iv), 7.3, 11.1
the Σ-depleted form f^Σ, its level $M^\Sigma = M\prod p^{\delta_p}$, $\delta_p = 2, 1, 0$ (explain why)	7.4
$S_2(\Gamma_0(N)) \otimes \overline{\mathbb{Q}} = \bigoplus$ spans of $h(dz)$	6.2(iii), 8.1(i)
generalised eigenspaces of $\mathbb{T}$ and their relation to newforms	8.1(iii), 8.2
$\mathbb{T} \otimes \mathbb{Q}$ as a product of fields times the old part	8.1(ii), (iv)
the orthogonal complement $f^\perp$ of a newform and its $\mathbb{T}$-stability	5.4, 6.2(iv), 6.4(ii), 8.1(iv)
twisting by Dirichlet characters; Atkin-Li ($q^2 \nmid M$: $g \otimes \chi$ newform of level $\mathrm{lcm}(M, q^2)$)	9.1, 9.2
the twist map ι on the whole space; intertwines T_p with $\chi(p)T_p$ for p prime to the conductor	9.1(iv), (vi), 9.3
integral structure: $S_2(\Gamma_0(N); \mathbb{Z})$ is a lattice (Chapter 12, B11)	4.1 (citing Chapter 12, its 11.2, 11.4)
the perfect pairing $S_2(\Gamma_0(N); \mathbb{Z}) \times \mathbb{T} \to \mathbb{Z}$, $(g, t) \mapsto a_1(tg)$ (Ribet 1983, Abbes-Ullmo), proved via the q-expansion principle	4.1
consequences: $\mathbb{T}$ free of rank $\dim S_2$; $\mathrm{Hom}(\mathbb{T}, \mathbb{Z}) = S_2(\mathbb{Z})$; a form is determined by its pairing with $\mathbb{T}$	4.1(iii), 4.2, 3.4
congruences between eigenforms mod λ: all n versus n prime to a set	8.4
the recursion for a_{p^k} and multiplicativity, so congruences at primes propagate to all n	3.2(ii), 8.4(i)
$L(f, s) = \sum a_n n^{-s} = \prod(1 - a_p p^{-s} + p^{1-2s})^{-1} \prod_{p\mid N}(1 - a_p p^{-s})^{-1}$	10.1
analytic continuation and functional equation (statement)	10.2
Rankin-Selberg and $\langle f, f\rangle =$ (explicit factor) $L(\mathrm{Sym}^2 f, 2)$ (statement; precise form in Chapter 19)	10.3
Section 6 row: newform f_E, $S_2(\Gamma_0(N))$, q-expansions, coefficientwise congruences (with Chapter 12)	3.2, 6.2, 8.4
Section 6 row: congruence ideal, congruence number r_E, η^Σ, $M_{f,\lambda}$, $\hat{\delta}_!$ (with 14, 20)	4.3, 8.2, 8.3, 11.2 (η^Σ, $M_{f,\lambda}$, $\hat{\delta}_!$: Chapters 14, 20; f^Σ, M^Σ: 7.4)

PLAN.md scope item, Section 6 row, or paper hook	line-up items
Section 6 row: Petersson norm, Rankin-Selberg, $L(\mathrm{Sym}^2 f, 2)$ (with 12, 14, 19)	10.3
Section 6 row: Atkin-Li, Atkin-Lehner $a_p = 0$, oldforms, U_p eigenvalues, stabilisations, depleted forms	6.4, 7.1, 7.2, 7.4, 9.2
Section 6 row: perfect pairing $S_2(\Gamma_0(N);\mathbb{Z}) \times \mathbb{T} \to \mathbb{Z}$ (Ribet 1983, Abbes-Ullmo)	4.1
hook: Lemma lem:crit ($\mathbb{T}$, ϕ_f, I, J, $f^\perp$, the perfect pairing, $\mathbb{T}/(I+J)$ cyclic of order r)	3.1, 3.2, 3.4, 4.1, 4.2, 4.3, 5.4, 8.1(iv), 8.4 (the proof of the lemma: Chapter 15)
hook: Lemma lem:old (oldforms, U_p eigenvalues, stabilisations, depleted forms, $a_p = 0$ for $p^2 \mid M$, multiplicativity, recursion)	2.3, 3.2(ii), 6.2(iv), 6.4(iii), 7.1, 8.1(i), 8.4 (the proof of the lemma: Chapter 15)
hook: Lemma lem:twist (iii) (Atkin-Li)	9.2
hook: Theorem thm:twist (ι, generalised eigenforms, $g^\perp \to f^\perp$)	6.3, 8.1, 8.2, 9.1, 9.3, 11.3 (the proof: Chapter 20)
hook: Theorem thm:quant Step 1 (strong multiplicity one, stabilisations, U_q eigenvalue 0 with multiplicity one, idempotents)	6.3, 7.1, 7.2, 7.4(iv), 8.2, 8.3, 9.3, 11.3 (the proof: Chapter 20)
hook: Section 4.1 (f^Σ, M^Σ, δ_p)	7.4
hook: Section 1 (congruence number r_E, Petersson norm)	4.3, 10.3, 11.2
also: Lemma lem:hecke ("computed in $\mathbb{T}_{\mathfrak{m}}$"; point 35)	8.3(ii)
also: Theorem thm:qual (b), (c) ($h = g \otimes \chi$ newform; $h \neq f$; point 20)	6.3, 9.1(v), 9.2
also: the Example of Section 3 (121d $= f_{11\mathrm{a}} \otimes \chi_{-11}$; congruences at all n; point 46)	8.4(iii), 9.2, 11.4
also: the remark after Theorem thm:qual (99A1: degree 4, congruence number 12)	11.2
also: Section sec:eis (Eisenstein maximal ideal of $\mathbb{T}_{11}$)	2.5(ii)
also: important_notes.md points 16, 19, 20, 21, 30, 35, 36, 46; B8, B10, B11	11.3; 7.3, 11.1; 9.1(v); 9.2 (980g versus 980i); 10.3(iii); 8.3(ii); 8.3(iii); 8.4(iii); 9.2; 10.3 (C5 only); 4.1
also: what Chapters 2, 3, 6, 8, 9, 12 deferred here	9.1 (Gauss sums), 7.1, 7.3, 8.1 (Chapter 3's 10.7(v), Exercise 14), 3.3 (Chapter 6's 7.1), 2.6 (Chapter 8's 12.1(c)), 3.2(ii), 9.2 (Chapter 9's 8.3, 10.4), 4.1, 5.3, 6.4, 7.4, 8.4 (Chapter 12's C8, 10.3, 10.5(a), 11.4(v), 12.4(c))

15.4.2 4.2 Exercises (Section 12 of the chapter), graded from easy to harder

1. Write down the coset decompositions of $\Gamma_0(N)\,\mathrm{diag}(1,p)\Gamma_0(N)$ for $p \nmid N$ and for $p \mid N$, and derive the q-expansion formulas of 2.3, including the constant term. Show that with Diamond-Shurman's slash normalisation the operator is the same (C2).
2. Prove $T_p T_{p'} = T_{p'} T_p$ and $T_p U_q = U_q T_p$ from the q-expansion formulas. Prove $\sum_r T_{p^r} X^r = (1 - T_p X + pX^2)^{-1}$ and identify T_{p^r} with a rescaled Chebyshev polynomial of the second kind.
3. For 11a1 check with PARI (`ellan`) that $a_{mn} = a_m a_n$ for coprime $m, n \le 200$, that $a_{p^{r+1}} = a_p a_{p^r} - p a_{p^{r-1}}$ for $p \neq 11$, and that $a_{11^r} = 1$. Explain why $a_{11^r} = a_{11}^r$ is a statement about U_{11}.
4. (The perfect pairing at a small level.) At level 33 find a $\mathbb{Z}$-basis of $S_2(33;\mathbb{Z})$ from q-expansions up to the Sturm bound 8, write the matrices of $T_1, \ldots, T_8$, show that they span a lattice of rank 3, and

check that the pairing matrix $(a_1(t_j g_i))$ has determinant ± 1. At level 77 show that $T_1, \dots, T_7$ span a sublattice of index 5 in $\mathbb{T}_{77}$.

5. (Check the paper's claim in Lemma lem:crit that the pairing stays perfect "after tensoring with $\mathbb{Z}_\ell$".) Show that a perfect pairing of free $\mathbb{Z}$-modules stays perfect after $\otimes A$ for every ring A; show that $\mathbb{T}_\ell = \prod_{\mathfrak{m}} \mathbb{T}_{\mathfrak{m}}$ and that the pairing is perfect on each $e_{\mathfrak{m}} S_2(N;\mathbb{Z}_\ell) \times \mathbb{T}_{\mathfrak{m}}$. Do the other local factors lie in $\ker \phi_f$ or in $\mathrm{Ann}(\ker \phi_f)$ (point 35)?
6. At level 22 compute the Petersson Gram matrix of $f_{11}(z)$, $f_{11}(2z)$ (with PARI's `mfpetersson`, remembering that it divides by $\mu(22) = 36$), the matrix of U_2 and of its adjoint, and check $U_2^\dagger = w_{22} U_2 w_{22}$. Show that U_2 is not normal.
7. (Non-semisimple U_p.) Compute the matrix of U_2 on $\mathrm{span}\{f_{11}(2^j z) : 0 \le j \le 3\} \subset S_2(88)$, its characteristic and minimal polynomials, and a basis of eigenvectors and generalised eigenvectors. Deduce that $\mathbb{T}_{88} \otimes \mathbb{Q}$ is not reduced, and explain why the paper's Lemma lem:ddt417 needs $\wp \cap I = 0$ to be proved rather than assumed (point 19).
8. (Check the paper's claims in Lemma lem:old.) For a newform h of level M verify that the stabilisations and depleted forms of 7.1 are U_p-eigenvectors with the stated eigenvalues in each of the cases $p \nmid M$, $p \parallel M$, $p^2 \mid M$, and that every eigenvalue has a one-dimensional eigenspace.
9. (Check the paper's claim in Section 4.1 that f^Σ has level M^Σ.) For $f = f_{11}$ compute f^Σ for $\Sigma = \{2\}$, $\{11\}$, $\{2, 11\}$, $\{3, 5\}$, its level, and show that it is not of smaller level. Show $L(f^\Sigma, s) = L(f,s) \prod_{p \in \Sigma} L_p(f,s)^{-1}$ and explain δ_p as the number of roots removed.
10. (Check the paper's claim in Theorem thm:quant Step 1.) Show that every eigenvalue of U_q on $S_2(N)$ is 0, ± 1 or a root of $X^2 - aX + q$, hence 0 or an ℓ-adic unit for $\ell \neq q$. For $g = 77\mathrm{c}$, $q = 7$, $N = 539$ and for $g = 11\mathrm{a}$, $q = 7$, $N = 539$ show that the eigenvalue 0 of U_7 on $\mathrm{span}\{g(dz)\}$ has multiplicity one, and find the characteristic polynomial of U_7 on $S_2(539)$.
11. Prove the dimension formula of 6.5 and its Moebius inversion; compute $\dim S_2^{\mathrm{new}}(N)$ for $N = 88, 121, 539$ from the dimensions of $S_2(M)$, $M \mid N$, and compare with PARI.
12. (Twists.) Prove the Gauss-sum formula for $f \otimes \chi$ and the intertwining $T_p(f \otimes \chi) = \chi(p)(T_p f) \otimes \chi$ for quadratic χ, $p \nmid r$. Show $(f \otimes \chi) \otimes \chi = f^{\{p|r\}}$. In the paper's case (b) of Theorem thm:qual, show that $h \otimes \chi$ is the q-depletion of g, and use strong multiplicity one to prove $h \neq f$ (point 20).
13. (Check the paper's Lemma lem:twist (iii).) With PARI's `elltwist` and `ellglobalred`, find the conductors of the twists of 11a1 by -11 and -7, of 14a1 and 77c1 by -7, and of 539c1, 539d1, 121a1 by -7, -7, -11. Which satisfy $q^2 \nmid M$, and where does the level drop? Show that the twist of 20a1 by -7 lies in the class 980g, so that the paper's congruence for 980i1 is a congruence between two different newforms of level 980.
14. (Check the paper's claim, from ARS12 Table 1, that 99A1 has modular degree 4 and congruence number 12.) Compute r_f for the newform of 99a by the lattice method of 4.3 at level 99; compare with $\deg \varphi_E$ from `ellmoddegree`, and explain why the discrepancy is at 3. Do the same for 539d1 ($r = 504$, degree 72).
15. (Check case (c) of Theorem thm:twist.) Compute r_f for 121a and 121c at level 121 and verify that they are equal and that $121\mathrm{a} \otimes \chi_{-11} = 121\mathrm{c}$. Check $r(77\mathrm{c}) \mid r(539\mathrm{c})$.
16. (Rankin-Selberg; point 30.) Compute $\langle f, f \rangle$ for 11a1 (unnormalised) and $L(\mathrm{Sym}^2 f, 2)$ with `lfunsympow`, and check $\langle f, f \rangle = 11\, L(\mathrm{Sym}^2 f, 2)/(8\pi^3)$. Show that at a good prime $\sum_r a_{p^r}^2 X^r = (1 + pX)/((1 - \alpha^2 X)(1 - pX)(1 - \beta^2 X))$, and deduce that $\sum a_n^2 n^{-s}$ is not the naive symmetric square L-function.
17. Prove the functional equation of $L(f,s)$ from the Mellin transform and W_N; compute the sign for 11a1 and 37a1 from ε_N and compare with their ranks.

15.5 5. Dependencies and forward pointers

Results from other chapters used here: * Background (PLAN.md Section 2): linear algebra (commuting operators, eigenspaces and generalised eigenspaces, Jordan form as a statement, Hermitian inner products and adjoints, the spectral theorem for commuting normal operators, tensor products of vector spaces); complex analysis (Fourier and Mellin transforms, change of variables in integrals); RHB (Chebyshev

polynomials, Mellin transforms). * Chapter 2 (ch:rings): Dirichlet characters, conductors, primitive characters, Gauss sums and $\sum_u \bar{\chi}(u)\zeta^{nu} = \chi(n)\mathfrak{g}(\bar{\chi})$ (9.1), the Kronecker symbols χ_{q^*} (C7), multiplicative functions and Moebius inversion (6.5), σ_{k-1} (2.5). * Chapter 3 (ch:modules): free modules and saturation (4.1, 4.2), the algebra-module duality 9.6 (4.1, 4.2), finite free algebras and idempotents 10.1 to 10.4 (8.1, 8.3), generalised eigenspaces 10.5 and the idempotent congruence 10.6 (8.1, 8.3), semisimplicity and the shift operator 10.7(v) (7.1, 7.3), common eigenvectors 10.8 (8.1), congruence ideals 11.1 (4.3, 8.3(ii)). * Chapter 4 (ch:reps): Brauer-Nesbitt (only in the sketch of 6.3(ii)). * Chapter 5 (ch:galois): Chebotarev (only in the sketch of 6.3(ii)). * Chapter 6 (ch:ant): algebraic integers, orders and coefficient fields (its 7.1, used in 3.3), primes λ above ℓ (its 7.2). * Chapter 7 (ch:local): $\mathbb{Z}_\ell$, $\mathcal{O}_\lambda$ (8.3). * Chapter 8 (ch:curves): the Hecke correspondences (its 12.1(c), only in 2.6). * Chapter 9 (ch:ellcurves): $X^2 - a_pX + p$ with roots α, β (its 8.3, used in 7.1), quadratic twists of curves and $a_p(E^{(d)}) = \chi_d(p)a_p(E)$ (its 10.4, used in 9.2's examples). * Chapter 12 (ch:modforms): the slash operator C3 (C2 here), M_k, S_k and q-expansions (its 6.2, 6.3), V_d (its 6.2(e)), the Hecke bound (its 6.4), nebentypus (its 6.6, for 9.1(ii)), the Petersson product and its behaviour under change of level and under W_d (its 10.1, 10.2, 12.2), the integral basis theorem and the Sturm bound with their consequences (its 11.2, 11.3, 11.4: the input of 4.1), the Atkin-Lehner involutions (its 12.1 to 12.4), $\mu(N)$ (its 3.1), the level-11 form (its 9.2), the paper's congruences (its 11.5).

Black boxes imported from later chapters (stated with a forward \Cref, used only in sketches, remarks, examples and paperlinks, never in the proof of a numbered result here): the Galois representations of newforms, the equality of level and conductor (Carayol) and Brauer-Nesbitt for them (Chapter 15; used in the sketches of 6.3(ii) and 9.2); modularity $f_E \leftrightarrow E$ and the modular degree (Chapter 14; used in 6.4(iv), 10.1, 11.2); the exact Rankin-Selberg formula with bad factors and $L(\mathrm{Sym}^2 f, s)$ (Chapter 19; used in 10.3(ii), (iii)).

Forward pointers (chapters that use this one): * Chapter 14 (ch:moddeg): $\mathbb{T}$ and its freeness (3.1, 4.1) for the Hecke action on $J_0(N)$ and $H^1(X_0(N), \mathbb{Z})$; the remark 2.6 on correspondences; $U_p^\dagger = w_N U_p w_N$ for the Petersson product (5.3) as the model of the cup-product statement "$w_N T_n w_N$ is the adjoint"; newforms and $a_p = \pm 1, 0$ (6.4); the congruence number r_f and $\eta(\phi_f)$ (4.3, 11.2) for Agashe-Ribet-Stein; the maximal ideals $\mathfrak{m}$ (3.4, 8.3) for multiplicity one and the Gorenstein property; the Eisenstein example 2.5(ii). * Chapter 15 (ch:modgalrep): the perfect pairing and its consequences (4.1, 4.2), $\mathbb{T} \otimes \mathbb{Q} = \mathbb{Q} \times \mathbb{T}'$ (8.1(iv)), the eigencharacter dictionary (3.4) and congruences (8.4) for the proof of Lemma lem:crit; U_p on oldforms, the tensor structure and the depleted forms (7.1, 8.1), $a_p = 0$ for $p^2 \mid M$ (6.4), multiplicativity and the recursion (3.2) for the proof of Lemma lem:old; strong multiplicity one (6.3) and Atkin-Li (9.2) for Theorem thm:qual; the level of a newform as conductor (9.2's sketch). * Chapter 19 (ch:lfunctions): $L(f, s)$, its Euler factors and δ_p (10.1, 7.4), the functional equation (10.2), the Rankin-Selberg residue (10.3) and point 30. * Chapter 20 (ch:congruence): $\mathbb{T}$ finite free (4.1), $\mathbb{T}_\ell = \prod \mathbb{T}_\mathfrak{m}$ on forms and the local pairing (8.3), $\eta(\phi)$ (4.3), f^Σ, M^Σ, δ_p and multiplicity one for f^Σ (7.4), the $\mathbb{T}$-stable complement (8.2), units among U_q-eigenvalues (7.2), the twist operator and the transport lemma (9.1, 9.3) for the proofs of Theorem thm:twist and Theorem thm:quant; point N1 of Part 7. * Chapter 21 (ch:paper): A3.1 to A3.3 (Lemma lem:crit, Lemma lem:old), A2.4 (Lemma lem:twist (iii)), A4.14 to A4.16 (Theorem thm:twist, Theorem thm:quant), the remark after Theorem thm:qual (99A1, 11.2), the Example of Section 3 (9.2, 11.4). * Chapter 22 (ch:computation): PARI's mfinit, mfbasis, mfcoefs, mfheckemat, mfdim (with flags for the full and the new space), mfsturm, mfpetersson, mfatkininit, mffromell, mfbd, elltwist, ellglobalred, lfunsympow, matrixqz as used in Part 6.6; the lattice method for congruence numbers (11.2).

15.6 6. Notes for the writer

15.6.1 6.1 Conventions to state (Section 1 of the chapter, as convention environments)

C1 to C10 of Part 1.2. The three normalisations to box: (a) T_p and U_p (C2, C3, 2.3): $[\Gamma\alpha\Gamma]_k f = (\det\alpha)^{k/2-1} \sum f|_k\alpha_i$ with Chapter 12's slash, giving $a_n(T_p f) = a_{pn} + p^{k-1}a_{n/p}$ and $a_n(U_p f) = a_{pn}$, the same operators as Diamond-Shurman's; $T_{p^r} = U_p^r$ for $p \mid N$; level written as a superscript whenever two

levels occur. (b) The pairing: $\langle g, t\rangle = a_1(tg)$ on $S_2(N;\mathbb{Z}) \times \mathbb{T}_N$, $\mathbb{Z}$-bilinear, with $\langle ug, t\rangle = \langle g, ut\rangle$; the Petersson product is a different, sesquilinear, transcendental pairing and is never called "the pairing". (c) Twists: $f \otimes \chi = \sum \chi(n) a_n q^n$ with $\chi(n) = 0$ for $(n, r) > 1$, so twisting kills the coefficients at multiples of the conductor and twisting twice depletes; χ_{q^*} for the paper's χ_q; $\iota = \iota_S$ for the paper's twist map; $\chi_{-7}(2) = +1$, $\chi_{-11}(2) = -1$.

15.6.2 6.2 Pitfalls

- **Level dependence of T_p** (C3). "T_p" on a form of level M and "U_p" on the same form viewed at level Mp differ by $p^{k-1}V_p$. Lemma lem:old and Theorem thm:quant Step 1 work at level N with forms of lower level; every computation must say at which level the operator acts.
- **Anemic versus full.** Multiplicity one holds for $\mathbb{T}^{(N)}$ on the new subspace and for the full $\mathbb{T}$ on a newform at its own level, not for $\mathbb{T}^{(N)}$ on $S_k(N)$ (the anemic eigenspace of h is V_h, of dimension $\sigma_0(N/M_h)$). An anemic eigenform can have $a_1 = 0$ ($f_{11}(2z)$); only full eigenforms can be normalised.
- **Non-real eigenvalues.** Eigenforms of the full $\mathbb{T}$ that are not newforms can have non-real coefficients ($a_2 = -1 \pm i$ for the 2-stabilisations of f_{11}); "K_f totally real" is for newforms (and for anemic eigenvalues) only. Chapter 6's 7.1 states it for newforms; keep it so.
- **Multiplicity of an eigenvalue versus semisimplicity.** The eigenvalue 0 of U_7 at level 539 has multiplicity 42 and U_7 is semisimple; at level 88 the eigenvalue 0 of U_2 has multiplicity 7 and U_2 is not. The paper's "multiplicity one" in Theorem thm:quant Step 1 is the algebraic multiplicity of 0 on the span of the $g(dz)$; say so.
- **$\mathbb{T}$-stable versus Petersson complement** (8.2). They agree for newforms at their level, not for depleted or stabilised forms (level 44 computation). Theorem thm:twist uses the Petersson complement at level N_S (where g is new) and Theorem thm:quant Step 1 the $\mathbb{T}$-stable complement at level N; the proof of Lemma lem:crit runs for any eigencharacter whose generalised eigenspace is a line, with the $\mathbb{T}$-stable complement.
- **Twist of a q-series versus newform** (9.1(v), 9.2(iii)). $g \otimes \chi$ is a newform only when $q^2 \nmid M$; otherwise it is the q-depletion of a newform of level dividing M. $g \otimes \chi \otimes \chi = g^{\{q\}}$, not g (point 20). Name forms by class, not by curve, when twisting (point 15, B8): 11a $\otimes\, \chi_{-11}$ = 121d, 11a $\otimes\, \chi_{-7}$ = 539d, 20a $\otimes\, \chi_{-7}$ = 980g (not 980i), 77c $\otimes\, \chi_{-7}$ = 539c.
- **$T_1, \ldots, T_d$ do not span $\mathbb{T}$.** The Sturm bound B is the right number (4.1(iv)); at level 121 even linear independence fails.
- **Congruences at primes versus at all n** (8.4). The recursion transfers congruences at all primes (including $p \mid N$) to all n between full eigenforms of the same level; Sturm transfers a finite check to all n between any two forms of the same space; neither turns a congruence at $p \notin \Sigma$ into one at all n without depletion or a choice of U_p-eigenvalues (Lemma lem:old).
- **The Rankin series** (10.3(iii)): $\sum a_n^2 n^{-s}$ is not the naive symmetric square (point 30); the factor $1 + p^{1-s}$ at good primes and the bad-prime corrections must be written out; the squarefree formula $\langle f, f\rangle = NL(\mathrm{Sym}^2 f, 2)/(8\pi^3)$ is with the unnormalised Petersson product and PARI's normalisation of $L(\mathrm{Sym}^2 f, s)$; the general formula is Chapter 19's.
- **Petersson normalisation** (Chapter 12, C5): PARI's `mfpetersson` divides by $\mu(N)$; Gram matrices in 5.3 and 8.2 are given up to the common factor and do not depend on it.
- **Theorem numbers.** Unverified: Ribet83 Theorem 2.2, AU96 Lemma 2.1, AtkinLi78 Theorem 3.1, AtkinLehner70 Theorem 3 (cited as the paper cites them, marked "as cited by the paper"); DiamondShurman 5.2.2, 5.3.1, 5.5.2, 5.5.3, 5.7.1, 5.8.3, 5.10.2; Shimura71 3.24, 3.64; Miyake 4.6.19. Cite sections where unsure.
- **Do not prove here** the paper's Lemma lem:crit or Lemma lem:old (Chapter 15), Theorem thm:twist or Theorem thm:quant (Chapter 20). This chapter supplies every Hecke-theoretic step they quote, and the `paperlink`s say which.
- Typography (PLAN.md 7.2): "Atkin-Lehner", "Atkin-Li", "Rankin-Selberg", "Eichler-Shimura", "Coleman-Edixhoven", "Diamond-Shurman" with plain hyphens; no Unicode; $\mathfrak{g}(\chi)$ for Gauss sums (not g, which names forms).

15.6.3 6.3 What to leave to Chapters 12, 14, 15, 19 and 20

- **Chapter 12** (ch:modforms): the spaces, the slash operator, q-expansions, V_d, the Petersson product and its normalisation, the Atkin-Lehner involutions w_d as operators, Shimura's integral basis theorem, $S_2(N;A)$, Sturm's bound and the split truncation (book-internal note B11). Chapter 13 cites them and does not reprove them.
- **Chapter 14** (ch:moddeg): $X_0(N)$ over $\mathbb{Q}$, $J_0(N)$, the Hecke correspondences and their action on $J_0(N)$ and on $H^1(X_0(N),\mathbb{Z})$, the cup-product adjointness $T_n^\dagger = w_N T_n w_N$ and the pairing $\hat{\delta}_!$, modularity, the modular degree, the theorem of Agashe-Ribet-Stein, multiplicity one $\dim J_0(N)[\mathfrak{m}] = 2$, the Gorenstein property of $\mathbb{T}_\mathfrak{m}$, the Eisenstein ideal. Chapter 13 gives only the Petersson-product analogue (5.3) and the algebraic $\mathfrak{m}$ (3.4, 8.3).
- **Chapter 15** (ch:modgalrep): Galois representations $\rho_{f,\lambda}$, local-global compatibility, the conductor of ρ_f equal to the level, the full proofs of Lemma lem:crit and Lemma lem:old, level lowering and Theorem thm:qual. Chapter 13 gives the Hecke side (4.1, 7.1, 8.1, 8.4, 9.2) and uses Chapter 15 only in sketches.
- **Chapter 19** (ch:lfunctions): $L(\mathrm{Sym}^2 f, s)$ and $L(\mathrm{ad}^0 f, s)$, naive versus true factors, δ_p from the Galois side, the exact Rankin-Selberg formula with bad factors (DFG04 Lemma 2.12, Shimura76), point 30 in full. Chapter 13 states the residue formula and the squarefree case.
- **Chapter 20** (ch:congruence): congruence modules, $\eta^\Sigma_{f,\lambda}$, $M_{f^\Sigma,\lambda}$, Lemma lem:ddt417, Lemma lem:hecke, Theorem thm:dfg14 and the proofs of Theorem thm:twist and Theorem thm:quant (including point N1). Chapter 13 gives f^Σ, M^Σ, δ_p (7.4), the $\mathbb{T}$-stable complement (8.2), $\mathbb{T}_\mathfrak{m}$ (8.3) and the transport lemma (9.3).

15.6.4 6.4 How the relevant important_notes.md points are handled

- **Point 19** (Lemma lem:ddt417: $\wp \cap I = 0$ is not automatic; the full Hecke algebra can contain nilpotents). 7.3(i) and 11.1 prove, with the level-88 matrices, that $\mathbb{T}_{88} \otimes \mathbb{Q}$ is not reduced and locate the Jordan blocks; 7.4(iv) and 8.1(iv) prove the multiplicity-one statements that make $\eta \neq 0$ in the paper's applications (newform at its level; f^Σ at level M^Σ; g^S at level N). The algebraic fix ($\wp \cap I = 0$ iff $\eta \neq 0$) is Chapter 3's 11.1(ii).
- **Point 20** (Theorem thm:qual (b), (c): $g = h \otimes \chi$ is not literal). 9.1(v) proves $(g \otimes \chi) \otimes \chi = g^{\{q\}}$; Exercise 12 derives $h \neq f$ from strong multiplicity one (6.3). Chapter 15 applies it.
- **Point 21** (980i1 is outside the hypotheses of Theorem thm:qual, $5 \mid 980$). Chapter 13 uses 980i1 only as a twisting and level example and says so; 9.2 adds that $20\mathrm{a} \otimes \chi_{-7}$ is 980g, so the paper's congruence for 980i1 is between two different newforms.
- **Point 30** (the naive adjoint L-function is not the Euler product of $\sum a_n^2 n^{-s}$). 10.3(ii), (iii) state the correct relation and the extra factor $1 + p^{1-s}$; Exercise 16 checks it. The precise statement at bad primes is Chapter 19's.
- **Point 35** (Lemma lem:hecke: the other local factors lie in the kernel, not in the annihilator). 8.3(ii) states the correct version for $\mathbb{T}_\ell$ acting on forms; Exercise 5 asks the question.
- **Point 36** (the pairing stays perfect on the $\mathfrak{m}$-component). 8.3(iii) proves the analogue for the pairing $\langle g, t\rangle = a_1(tg)$, which is what Theorem thm:quant Step 1 uses; the cohomological pairing $\hat{\delta}_!$ is Chapter 14's and Chapter 20's, where the same one-line argument (idempotents of a commutative algebra of self-adjoint operators are self-adjoint) applies.
- **Point 46** (the paper's congruences in the Example of Section 3 hold at every n by Sturm's bound). Chapter 12 (its 11.5) proves them; 8.4(iii) explains the division of labour between Sturm and the Hecke recursion; 9.2 and 11.4 identify $f_{11\mathrm{a}} \otimes \chi_{-11}$ with 121d.
- **B10** (the Manin constant in the Zagier-Watkins formula). Not used here; 10.3 uses only the unnormalised Petersson product of Chapter 12 (C5), consistent with B10.
- **B11** ($S_2(\Gamma_0(N);\mathbb{Z})$ is a lattice: Chapter 12 versus Chapter 13). Followed: Chapter 13 cites Chapter 12's 11.2 and 11.4 and proves the perfect pairing (4.1) from them.
- Also touched: **point 16** (Theorem thm:twist data paragraph: strict inequality in case (a) too): 11.3 gives the forms behind the pairs 539c1/77c1 and 121d1/11a1; **points 15 and B8** (twist pairs by

class): 9.2's table names classes and the curves PARI returns; **point 39** is Chapter 14's and is not used.

15.6.5 6.5 References

`refs.bib` already contains: DiamondShurman, Shimura71, Shimura76, Miyake, Koblitz, SerreCourse, Stein, AtkinLehner70, AtkinLi78, Ribet83, AU96, ARS12, DDT, DFG04, Zagier85, ColemanEdixhoven, Carayol86, Sturm87, PARI, PARIusers, Cremona97, ecdata, Deligne. Not in `refs.bib` and not in any `bib/chNN.bib` (all chapter bib files except `bib/chA.bib` are empty; checked): **`Li75`** (W.-C. W. Li, *Newforms and functional equations*, Math. Ann. 212 (1975), 285-315) and **`Rankin39`** (R. A. Rankin, *Contributions to the theory of Ramanujan's function $\tau(n)$ and similar arithmetical functions. I. The zeros of the function $\sum_{n=1}^{\infty}\tau(n)/n^s$ on the line* $\operatorname{Re} s = 13/2$. *II. The order of the Fourier coefficients of integral modular forms*, Proc. Cambridge Philos. Soc. 35 (1939), 351-356 and 357-372). Both are needed by more than one chapter (Li75 by Chapters 13 and 15 for newform theory with character and the levels of twists; Rankin39 by Chapters 13 and 19 for Rankin-Selberg), so by decision B14 they go into `refs.bib`, not into `bib/ch13.bib`. Every bibliographic detail above was recalled, not re-verified; check before adding. Theorem numbers to verify are listed in 6.2.

15.6.6 6.6 Verification log (computations done for this outline)

Python 3 with sympy and PARI/GP 2.17 through `cypari2`; scripts in the session scratchpad (`ch13/c1.py` to `c11.py`); Cremona `ecdata` files from `/tmp/ch09/data` (`allcurves`, `degphi`, `galrep`). * `c1.py`: multiplicativity and the recursion for 11a1 up to $n = 5000$ (and $a_{11^r} = 1$); the formula $a_m(T_n f) = \sum d\, a_{mn/d^2}$ consistent with $T_n f = a_n f$ for $m, n < 60$; $\dim S_2(N)$, $\dim S_2^{\mathrm{new}}(N)$ and the formula of 6.5 for the twelve levels of the table; Sturm bounds $(2, 6, 8, 12, 16, 24, 24, 22, 66, 112, 336, 264)$. * `c2.py`: matrices of U_p on $V_{f_{11},2,e}$, $e = 1, 2, 3$, and $V_{f_{11},11,e}$, $e = 1, 2$, with characteristic polynomials and Jordan forms (sympy); the depleted form $f_{11}^{\{2\}}$ to $n = 4000$; U_2 on all of $S_2(22)$, $S_2(44)$, $S_2(88)$ (characteristic and minimal polynomials, PARI `mfheckemat`). * `c3.py`: integral basis of $S_2(N;\mathbb{Z})$ by saturating the q-expansion lattice (`matrixqz`), the matrices of T_n, $n \le \max(B, 41)$, in that basis (all integral), $\operatorname{rank}_{\mathbb{Z}} \mathbb{T}$, the determinant of the pairing (± 1), the index of $\mathbb{Z}T_1 + \cdots + \mathbb{Z}T_d$, and congruence numbers r_f by the index of $\mathbb{Z}f \oplus (f^{\perp} \cap S_2(N;\mathbb{Z}))$, $f^{\perp}$ computed as $\sum_p \operatorname{im}(T_p - a_p)$ over $p \le 40$, $p \nmid N$ (rank $d-1$ checked), at levels 11, 33, 37, 44, 77, 88, 99, 121, 539; compared with `degphi`. * `c4.py`: Petersson Gram matrix at level 22 (`mfpetersson` on `mfsymbols`), the adjoint of U_2, and w_{22} from `mfatkininit`. * `c5.py`: conductors of the twists of 9.2 (`elltwist`, `ellglobalred`, minimal models matched to `allcurves`); $a_p(11\mathrm{a} \otimes \chi_{-11}) = \chi_{-11}(p) a_p(11\mathrm{a})$ for $p < 3000$. * `c6.py`: Rankin-Selberg residues from the Euler factors of `lfunsympow` against $48\pi\langle f, f\rangle/\mu(N)$ for 11a1, 37a1, 20a1, 121a1, 99a1 (30 digits), and $\langle f, f\rangle = NL(\mathrm{Sym}^2 f, 2)/(8\pi^3)$ for 11a1, 37a1. * `c7.py`: the depleted forms $(77\mathrm{c})^{\{7\}}$, $(11\mathrm{a})^{\{7\}}$ at level 539, ι of them equal to $f_{539\mathrm{c}}$, $f_{539\mathrm{d}}$ (to $n = 3000$), and their Hecke congruence modules (288, 504), computed as the index of $\mathbb{Z}g \oplus (g^c \cap S_2(539;\mathbb{Z}))$ with g^c the image of $t - \pi(t)$ for a random $t \in \mathbb{T}$ for which $\pi(t)$ is a simple root of the characteristic polynomial. * `c8.py`: Theorem thm:quant at $q = 3$ (Part 7, N2): all optimal curves of conductor below 10,000 with type I_0^* at 3, $\ell \in \{5, 7, 11, 13\}$, $\ell \nmid N$, surjective mod ℓ image (`galrep`), $e_\ell > 0$: 388 cases, no case with $v_\ell(\deg \varphi_E) < e_\ell$. * `c11.py`: PARI's $P_3(3^{-2})$ (from `lfunsympow`, `lfuneuler`) against the twist formulas $L_3(A_f, 1)^{-1} = \frac{2}{3} \cdot \frac{16 - a_3(E')^2}{9}$ (type I_0^*) and $\frac{8}{9}$ (type I_n^*) for all 1,100 optimal curves of conductor below 3,000 with one of these types at 3: no mismatch (Part 7, N2). * `c9.py`: $\mathbb{T}_{23} = \mathbb{Z}[T_2]$ with $T_2^2 + T_2 - 1 = 0$ (consistency with Chapter 3, its 10.4(a)). * `c10.py`: the Gram matrix of $f_{11}(z), f_{11}(2z), f_{11}(4z)$ at level 44 and $\langle f_{11}^{\{2\}}, f_{11} - \beta f_{11}(2z)\rangle = \frac{5}{12}\langle f_{11}, f_{11}\rangle$. * By hand: the coset decompositions of 2.2; $W_N \operatorname{diag}(1, p) W_N^{-1} = \operatorname{diag}(p, 1)$; the trace identity of 5.3(iii); the intertwining of 9.1(iv); the conjugation identity of 9.1(ii); $\iota(f) = g^S$ and the inclusion of 9.3(iii). * Not checked: theorem numbers in the references (6.2); the hypotheses of AtkinLi78 Theorem 3.1 in print.

15.7 7. Points for important_notes.md

New points in the area of this chapter (Hecke operators, the Hecke algebra and its pairing, newforms and oldforms, U_p on oldforms, depletion, twists, L-functions of newforms), not among points 1 to 53 and B1 to B16 already recorded. Points 16, 19, 20, 21, 30, 35, 36, 46, B10 and B11 are handled in Part 6.4 and not repeated.

Errors in the paper in this area: none found. Checked in particular: the q-expansion formulas and eigenvalues used in Lemma lem:old (every case, 7.1); "$a_p = 0$ for $p^2 \mid M$" and "$a_p = \pm 1$ for $p \,\|\, M$" (6.4); the $\mathbb{T}$-stability of $f^\perp$ in Lemma lem:crit (5.3, 5.4); the perfect pairing at nine levels (4.1); Lemma lem:twist (iii) on twelve twists, including the failure of newness when $q^2 \mid M$, which the lemma excludes (9.2); Section 4.1's f^Σ and M^Σ (7.4); the claims of Theorem thm:quant Step 1 (multiplicity one of the eigenvalue 0, units, the transport along ι, the idempotent projection: 7.1, 7.2, 8.3, 9.3), with the Hecke congruence modules computed at level 539; the claims of the proof of Theorem thm:twist (9.1, 9.3, 6.3); the 99A1 congruence number 12 (11.2).

- **N1. Theorem thm:quant, Step 1: the inequality is an equality (optional strengthening, not an error).** Location: proof of Theorem thm:quant, Step 1, ending "$v_\ell(r_E) \geq v_\ell(\eta^{\mathrm{Hecke}}(g^S))$", and Remark (2) after the proof. Observation: the transport also runs the other way, with no projection. Twisting f gives $\iota(f) = \sum \chi_S(n)^2 a_n(g) q^n = g^S$; and ι maps $f^\perp = \bigoplus_{h \neq f} V_h$ (newforms h of level dividing N, 6.2(iv)) into $\bigoplus_{h' \neq g} V_{h'}$, because $\iota(V_h)$ lies in the anemic eigenspace of the system $\chi_S(p)a_p(h)$ (9.1(iv), $\mathrm{lcm}(N, D^2) = N$), which is that of g only if h has the system of f, that is, $h = f$ (strong multiplicity one). $\bigoplus_{h' \neq g} V_{h'}$ lies in the $\mathbb{T}$-stable complement of the line $\mathbb{C}g^S$. So a congruence $f \equiv h \pmod r$ with $h \in f^\perp \cap S_2(N;\mathbb{Z})$ gives $g^S \equiv \iota(h) \pmod r$ with $\iota(h)$ integral in the complement, and by the argument of Lemma lem:crit for g^S (its generalised eigenspace is a line), r_E divides the generator of $\eta^{\mathrm{Hecke}}(g^S)$. With Step 1 (which needs $\ell \notin S$ for the units),

  $$v_\ell(r_E) = v_\ell\big(\eta^{\mathrm{Hecke}}(g^S)\big) \quad \text{for every prime } \ell \notin S,$$

  so in the chain of Theorem thm:quant only Step 2 (freeness of $H^1(X_0(N), \mathbb{Z}_\ell)_{\mathfrak{m}}$ at the maximal ideal of g^S) and Step 4 (multiplicity one at level N_S) can be strict. Numerical check: at level 539 the Hecke congruence modules of $(77\mathrm{c})^{\{7\}}$ and $(11\mathrm{a})^{\{7\}}$ have orders 288 and 504, exactly $r(539\mathrm{c})$ and $r(539\mathrm{d})$ (even at $\ell = 2$ and at $\ell = 7 = q$). Fix (optional): add after Step 1 "Conversely $\iota(f) = g^S$ and $\iota(f^\perp)$ lies in the $\mathbb{T}$-stable complement of g^S, so Step 1 is an equality", and sharpen Remark (2) accordingly. Confidence: high for the argument (it uses only 6.2, 6.3, 9.1); the numerics were computed with the lattice method of 11.2 (Part 6.6, `c7.py`).

- **N2. Theorem thm:quant: the restriction to primes $q \geq 5$ is not needed (optional strengthening).** Location: statement of Theorem thm:quant ("let S be the set of primes $q \geq 5$ at which E has type I_0^* or I_n^*") and Step 3 ("the values are those of Lemma lem:euler", a lemma stated for $q \geq 5$). Observation: no step uses $q \geq 5$. Lemma lem:twist (ii) gives the exchanges $\mathrm{I}_n \leftrightarrow \mathrm{I}_n^*$ and $\mathrm{I}_0 \leftrightarrow \mathrm{I}_0^*$ for every odd q; Theorem thm:twist allows any odd primes; and at an I_n^* or I_0^* prime the true factor $L_q(A_f, s)$ is that of the twist E_S, good or multiplicative at q (the adjoint is unchanged by the twist), so Step 3 holds at $q = 3$ with the same formulas: $e_\ell(E,3) = v_\ell(8) = 0$ for I_n^* and $e_\ell(E,3) = v_\ell(2) + v_\ell(16 - a_3(E')^2)$ for I_0^*, which is positive for $\ell = 5$ ($a_3(E') = \pm 1$) and $\ell = 7$ ($a_3(E') = \pm 3$). ($\ell \neq 3$ automatically, as $\ell \nmid N$.) Checks: PARI's $P_3(3^{-2})$ agrees with these formulas for all 1,100 optimal curves of conductor below 3,000 with type I_0^* or I_n^* at 3; and $v_\ell(\deg \varphi_E) \geq e_\ell(E,3)$ holds in all 388 cases with type I_0^* at 3, $\ell \in \{5, 7, 11, 13\}$, $\ell \nmid N$, surjective mod ℓ image and $e_\ell > 0$, below conductor 10,000 (for example 333d1 and 477a1 at $\ell = 7$, 234e1 and 333a1 at $\ell = 5$). Fix (optional): "the set of odd primes q at which ...", and say in Step 3 that the values at $q = 3$ follow from the twist. The same remark applies to Lemma lem:euler's rows I_n^* and I_0^*, which hold for every odd $q \neq \ell$. Confidence: high for the argument, which only uses that the adjoint representation is twist-invariant; this is for the Chapter 20 and Chapter 21 writers to adopt or not (Chapter 21's pitfall W8(d) currently says "S contains only primes $q \geq 5$", which is right as a reading of the paper).

- **N3. Lemma lem:crit and Theorem thm:quant Step 1: "the proof of Lemma lem:crit, carried out" for a non-newform (expository).** Location: Theorem thm:quant Step 1, "Now the proof of Lemma lem:crit, carried out over $\mathbb{Z}_\ell$ with the pairing ..., shows that $\ell^{v_\ell(\eta^{\mathrm{Hecke}}(g^S))}$ is the largest power ... with h ... in the $\mathbb{T}$-stable complement of the g^S-line". Problem: Lemma lem:crit is stated for the newform f with the Petersson complement $f^\perp$; for g^S the Petersson complement is not $\mathbb{T}$-stable (8.2(iii): at level 44, $\langle f_{11}^{\{2\}}, f_{11} - \beta f_{11}(2z)\rangle \neq 0$), and the proof must be run with the $\mathbb{T}$-stable complement, using only that the generalised eigenspace of the eigencharacter is a line. The paper does say "$\mathbb{T}$-stable complement" here, so this is a matter of making the reference precise. Fix (optional): state Lemma lem:crit for any eigencharacter $\pi\colon \mathbb{T} \to \mathbb{Z}$ whose generalised eigenspace is a line, with the $\mathbb{T}$-stable complement, and note that for a newform it is the Petersson complement. Confidence: high; importance low.

Book-internal inconsistencies and coordination notes found while preparing this outline:

- **BI1 (B8, addition).** The twist of 20a1 by χ_{-7} lies in the class 980g (980g2 by PARI's minimal model), not 980i. The paper's congruence $a_p(980\text{i}1) \equiv \chi_{-7}(p)a_p(20\text{a}1) \pmod 5$ is therefore between two different newforms of level 980; Chapter 12's 11.5(c) states it correctly ("$f_{980\text{i}} \equiv f_{20\text{a}} \otimes \chi_{-7}$"), and no outline claims the contrary, but writers should not say "980i is the twist of 20a". The twist of 980i1 by -7 is 140b1 (Chapter 9, its 10.4; Chapter 21).
- **BI2 (PLAN.md, Chapter 13 scope).** "$T_{p^{k+1}} = T_p T_{p^k} - p^{k-1} T_{p^{k-1}}$" uses k both as the exponent and as the weight. The outline writes $T_{p^{r+1}} = T_p T_{p^r} - p^{k-1} T_{p^{r-1}}$ with k the weight ($p^{k-1} = p$ in weight 2). Notation only.
- **BI3 (division of labour for r_E and Lemma lem:crit).** PLAN.md places the proof of Lemma lem:crit in Chapter 15 ("prove it here in full"), mentions it for Chapter 14 ("proved in full here or in Chapter 20") and restates it in Chapter 20. This outline defines $\eta(\phi)$ and r_f (4.3) and supplies every Hecke-theoretic ingredient, but proves neither the identity $r_f\mathbb{Z} = \eta(\phi_f)$ nor the lemma; Chapter 14's outline should cite Chapter 15 for the proof and not prove it a third time.
- Consistency checks that passed: Chapter 3's 10.7(v) (the shift operator: characteristic polynomial $X^{e-1}(X^2 - aX + p)$, Jordan block of size $e-1$) and its Exercise 14 (level 88, $a_2(f_{11}) = -2$) agree with 7.1 and 11.1; Chapter 3's 10.4(a) ($\mathbb{T}_{23} = \mathbb{Z}[(1+\sqrt{5})/2]$) agrees with `c9.py`; Chapter 12's 10.5(a) ($U_p^\dagger = w_N U_p w_N$), 12.4(b) (the matrix of w_{22}) and 11.5 (the congruences) agree with 5.3 and 9.2; Chapter 21's A4.14 and A4.15 (the twist pairs and the checks of Theorem thm:quant with Cremona's degrees) agree with 11.3, where the congruence numbers were computed directly; Chapter 6's 7.1 ("K_f totally real" for newforms) agrees with 3.3.

Chapter 16

Chapter 14: Modular curves over Q, Jacobians and the modular degree (ch:moddeg, chapters/ch14_moddeg.tex)

Stage 1 line-up. Part IV (Modular forms and modular curves). Target length 1,980 lines of LaTeX (the card excluded; see Part 3), inside the 1,000 to 2,200 range of PLAN.md Section 7.3.

Rigour tags: **P** = proved here, **S** = proof sketched here, **R** = stated; reference. Item format, as in the Chapter 2 to 12 outlines: number, label-ready name, kind, tag; the precise statement with all hypotheses; reference (book and chapter or section; a theorem number only where sure, otherwise marked "(verify)" or given by section); the paper's use by its `\label` names, or "background". Every number quoted below (modular degrees, congruence numbers, Petersson norms and period areas, Manin constants of the class 11a, dimensions of $J_0(N)$, cusp counts, images of the cusp 0, the data checks of Watkins's conjecture and of the theorem of Calegari and Emerton) was checked with PARI/GP 2.17 through `cypari2` and Cremona's `ecdata` files for conductors below 10,000 (verification log, Part 6.6).

16.1 1. Purpose and global conventions

16.1.1 1.1 What this chapter is for

The paper uses this chapter in six places, and the chapter is built around them.

1. **The optimal curve, the modular parametrisation and its degree** (abstract; Section 1: "Among the curves in the isogeny class of E there is a distinguished one, the $X_0(N)$-optimal curve, for which the modular parametrisation $\varphi_E\colon X_0(N) \to E$ has minimal degree"; Section 6: "Optimality is determined in the tables for every class of conductor at most 400000"). The chapter supplies $X_0(N)$ over $\mathbb{Q}$, $J_0(N)$, the Eichler-Shimura construction of the quotient J_f, modularity, the factorisation of every modular parametrisation of every curve of the class through the optimal one, the equivalence of "minimal degree", "connected kernel" and "saturated pullback on H^1", and the Cremona labelling convention with its one exception.
2. **The Petersson norm and the Manin constant** (Section 1: "Up to explicit factors $\deg \varphi_E$ is $\langle f, f\rangle$ divided by the area of the period lattice of E (Zagier, Watkins)"). The chapter states and proves the exact formula $\deg\varphi \cdot \mathrm{vol}(\Lambda_E) = 4\pi^2 c_\varphi^2 \langle f, f\rangle$ with $\varphi^*\omega_E = c_\varphi \cdot 2\pi i\, f(z)\, dz$ (important_notes.md B10), defines the Manin constant and states what is known about it.
3. **The congruence number and the theorem of Agashe, Ribet and Stein** (Section 1; Section 3 preamble; Lemma lem:crit; Lemma lem:hecke; the remark after Theorem thm:qual on 99A1). The

chapter defines r_E and the Hecke congruence ideal, states Lemma lem:crit (proved in Chapter 15), states ARS12 Theorem 2.2 and **proves its divisibility half at every prime** from Section 9, and verifies 99A1 ($\deg\varphi_E = 4$, $r_E = 12$) by direct computation.

4. $H^1(X_0(N),\mathbb{Z}_\ell)$**, the Hecke correspondences,** $T^t = w_N T w_N$ **and the pairing** $\hat{\delta}_!$ (Section 4.1: "the pairing $\hat{\delta}_!$ is the cup product composed with ... the Atkin-Lehner involution w_{M^Σ}; it is perfect and alternating ..., and the Hecke operators, U_p included, are self-adjoint for it, because $w_{M^\Sigma} T_n w_{M^\Sigma}^{-1}$ is the adjoint of T_n for the cup product"). The chapter supplies the Hecke correspondences on $X_0(N)$ over $\mathbb{Q}$, their compatibility with the Hecke operators of Chapter 13, the identity $C^t = w_N C w_N$ for every Hecke correspondence (proved by a matrix computation), and, with Chapter 8, its 12.3, the pairing $\hat{\delta}_!(x,y) = \langle x, w_N^* y\rangle$.
5. **Lemma lem:hecke and important_notes.md point 39** (Section 4.1, Lemma lem:hecke, Remark rem:free, Theorem thm:main, Theorem thm:quant Step 4). The chapter proves that for the optimal curve the lattice $M_{f,\lambda}$ is exactly $\varphi_E^* H^1(E,\mathbb{Z}_\ell)$, that $\hat{\delta}_!$ has ideal of values $\deg\varphi_E \cdot \mathbb{Z}_\ell$ on it, and hence that $v_\lambda(\eta_{f,\lambda}^{\emptyset}) = v_\ell(\deg\varphi_E)$ with no multiplicity-one hypothesis, DFG04's η being normalised as the paper describes it (the five checks of 9.6 hold: confirmed against the published DFG04 in the Chapter 20 outline, its Part 7; important_notes.md point 73).
6. **Multiplicity one, Gorenstein, the Eisenstein ideal, Shimura curves, the prime 2** (Remark rem:free; Conjecture conj:eis, Section sec:eis and Open question 4; Section 1.5; Section sec:two and Open question 5). The chapter states the multiplicity one theorem and deduces freeness and the Gorenstein property from Chapter 3's criterion; defines the cuspidal subgroup, the Shimura subgroup and Mazur's Eisenstein ideal, proves Manin-Drinfeld for the cusps 0, ∞ and the formula $\varphi_E(0) = c\,L(E,1)$ modulo Λ_E; states the theorem of Ribet and Takahashi, the abc connection, Watkins's conjecture and the theorem of Calegari and Emerton, with data checks.

The physics reader appears twice, in one sentence each: the Zagier-Watkins formula is "the area of a d-sheeted cover is d times the area of the base" applied to the L^2 norm of a holomorphic one-form (7.3); the projection formula $\langle\varphi^*x,\varphi^*y\rangle = \deg\varphi\,\langle x,y\rangle$ is the same statement for intersection numbers of loops (9.3).

16.1.2 1.2 Global conventions fixed in this chapter

- **C1 (curves and levels).** $N \geq 1$; $X_0(N)$ is the compact Riemann surface of Chapter 12 (its 5.2) and, by 2.1, a smooth projective curve over $\mathbb{Q}$; $Y_0(N)$ its non-cuspidal part; ∞ and 0 the two cusps of Chapter 12, its 4.2; $g = g(X_0(N))$. $\Gamma = \Gamma_0(N)$. $W_N = \left(\begin{smallmatrix} 0 & -1 \\ N & 0 \end{smallmatrix}\right)$ and w_N the involution of $X_0(N)$ (Chapter 12, C4).
- **C2 (Hecke actions; which action where).** $\mathbb{T} = \mathbb{Z}[T_n : n \geq 1] \subset \mathrm{End}_{\mathbb{C}}\, S_2(\Gamma_0(N))$ is Chapter 13's full Hecke algebra, U_p included. A Hecke correspondence $\mathcal{C}$ (3.1) acts on $J_0(N)$, on $H_1(X_0(N),\mathbb{Z})$ and on $T_\ell J_0(N)$ by pushforward $\mathcal{C}_*$ ("Albanese"; on moduli: $T_p(E,C) = \sum_D (E/D, (C+D)/D)$), and on $H^0(\Omega^1)$, $H^1(X_0(N),\mathbb{Z})$, $H^1(X_0(N),\mathbb{Z}_\ell)$ by pullback $\mathcal{C}^*$, the transpose of $\mathcal{C}_*$ (Chapter 8, its 12.2(a)). On $S_2 = H^0(\Omega^1)$, $\mathcal{C}^*$ is Chapter 13's operator (3.2). Because $\mathbb{T}$ is commutative, $t \mapsto t_*$ and $t \mapsto t^*$ are both ring homomorphisms from $\mathbb{T}$, and every statement "$\mathbb{T}$ acts on ..." in this chapter means one of these two, named each time.
- **C3 (notation for adjoints, duals, pullbacks).** As in Chapter 8, C4: $A^\dagger$ is the adjoint of A for a pairing; * is pullback; $M^\vee = \mathrm{Hom}(M,\mathbb{Z}_\ell)$; the paper's Pontryagin duals (*) occur only inside `paperlink` environments, which say so (important_notes.md B4).
- **C4 (Petersson product and the Manin constant).** $\langle f,g\rangle$ is Chapter 12's **unnormalised** product (its C5). For an elliptic curve $E/\mathbb{Q}$, ω_E is the Neron differential of a global minimal model (Chapters 9, 10), Λ_E its period lattice and $\mathrm{vol}(\Lambda_E) = \frac{i}{2}\int_{E(\mathbb{C})} \omega_E \wedge \bar{\omega}_E$ (Chapter 9, its 7.6). For a modular parametrisation φ, the constant c_φ is defined by $\varphi^*\omega_E = c_\varphi \cdot 2\pi i\, f(z)\, dz = c_\varphi f(q)\, dq/q$ (7.1). PARI's `mfpetersson` divides by $\mu(N) = [\mathrm{SL}_2(\mathbb{Z}) : \Gamma_0(N)]$.
- **C5 (Frobenius).** As in Chapter 8, C7: the arithmetic Frobenius acts on $T_\ell J$ as the p-power Frobenius endomorphism π_p; $\hat{\pi}_p = p\pi_p^{-1}$ is the Verschiebung (Appendix B, B-C2(g); not F, V, since V is $V_\ell E$ in the paper); on $H^1 = (T_\ell J)^\vee$ the geometric Frobenius has the same eigenvalues.

- **C6 (optimal; labels).** "Optimal" always means $X_0(N)$-optimal (6.2). Cremona labels as in Chapter 9, its 12.1; ARS12 and Cremona97 print 99A1 for 99a1. $\lambda = \ell$ and $\mathcal{O} = \mathbb{Z}_\ell$ throughout, since the newform of an elliptic curve has rational coefficients.
- **C7 (Eisenstein).** A maximal ideal $\mathfrak{m}$ of $\mathbb{T}$ (or of $\mathbb{T} \otimes \mathbb{Z}_\ell$) is *Eisenstein* if the residual representation $\bar{\rho}_\mathfrak{m}$ (Chapter 15) is reducible; for $\mathfrak{m}$ attached to (f_E, ℓ) this happens iff $E[\ell]$ is reducible (11.3). Mazur's *Eisenstein ideal* I_{Eis} at prime level is a specific ideal (11.2); the two uses are kept apart.
- **C8 (typography).** PLAN.md Section 7.2: no dash ligatures; "Eichler-Shimura", "Agashe-Ribet-Stein", "Zagier-Watkins", "Manin-Drinfeld", "Ribet-Takahashi", "Calegari-Emerton", "Deligne-Rapoport" with plain hyphens; labels `...:moddeg:...`.

16.2 2. Line-up of the core results, in teaching order

16.2.1 Section 1. Overview and standing notation

1.1 `conv:moddeg:setup` (conventions C1 to C8 as `convention` environments, and the standing notation; P). Throughout, $E/\mathbb{Q}$ is an elliptic curve of conductor N, $f = f_E \in S_2(\Gamma_0(N))$ its newform (5.3), $\phi_f \colon \mathbb{T} \to \mathbb{Z}$, $T_n \mapsto a_n(f)$, its eigencharacter, $I_f = \ker \phi_f$ (the paper's I in Lemma lem:crit, the paper's $\wp$ in Lemma lem:ddt417), $f^\perp$ the Petersson orthogonal complement of f (Chapter 12, its 10.3; $\mathbb{T}$-stable by Chapter 13). A roadmap paragraph says which section answers which sentence of the paper (Part 1.1 above). Paper: background.

16.2.2 Section 2. The modular curve $X_0(N)$ over $\mathbb{Q}$

2.1 `thm:moddeg:x0-over-q` (theorem; R). (i) There is a smooth projective geometrically connected curve $X_0(N)_\mathbb{Q}$ over $\mathbb{Q}$ whose complex points form the Riemann surface $X_0(N)$ of Chapter 12, such that for every field $K \subseteq \mathbb{C}$ (indeed every field of characteristic 0) the moduli bijection of Chapter 12, its 5.4(i), is compatible with $\mathrm{Aut}(\mathbb{C}/K)$ on points with $j \neq 0, 1728$. (ii) The involutions w_d ($d \,\|\, N$) and the forgetful map $X_0(N) \to X(1) = \mathbb{P}^1_j$ are defined over $\mathbb{Q}$; $w_N(E, C) = (E/C, E[N]/C)$. (iii) $X_0(N)$ has good reduction at every prime $p \nmid N$ (Igusa; Deligne-Rapoport), and a regular model over $\mathbb{Z}_p$ with semistable reduction for $p \,\|\, N$ (two copies of $X_0(N/p)_{\mathbb{F}_p}$ crossing at the supersingular points). Proof idea (one paragraph): the field $\mathbb{Q}(j, j_N)$ with $j_N(z) = j(Nz)$ is the function field of $X_0(N)$ over $\mathbb{Q}$, cut out by the modular polynomial $\Phi_N(j, j_N) = 0$, which has integer coefficients (Kronecker); the q-expansion principle shows the model is the right one. Reference: Shimura71 6.7 (cite the section); DiamondShurman 7.7 (cite the section); DiamondIm (Sections 8, 9; cite the sections); Deligne-Rapoport (proposed key `DeligneRapoport73`, refs.bib, see Part 6.5) and Katz-Mazur (`KatzMazur85`). Paper: Section 1 ("$\varphi_E \colon X_0(N) \to E$" is a morphism over $\mathbb{Q}$), Section 4.1 (H^1 of a curve over $\mathbb{Q}$).

2.2 `prop:moddeg:cusps-over-q` (proposition; S, with the count P given Chapter 12, its 4.2). (i) The cusps of $X_0(N)$ are the classes a/c with $c \mid N$, $\gcd(a, c) = 1$, a taken modulo $\gcd(c, N/c)$; there are $\varphi(\gcd(c, N/c))$ of denominator c (φ is Euler's function here only). (ii) The cusps of denominator c are defined over $\mathbb{Q}(\zeta_{\gcd(c,N/c)})$ and permuted simply transitively by its Galois group ($\sigma_t \colon \zeta \mapsto \zeta^t$ sends a/c to the cusp a'/c with $a' \equiv t^{\pm 1} a$; the sign is a convention, state one and check it). (iii) In particular ∞ ($c = N$) and 0 ($c = 1$) are $\mathbb{Q}$-rational, and every cusp is rational iff $\gcd(c, N/c) \leq 2$ for all $c \mid N$. Examples (P, checked): $X_0(20)$ has 6 cusps, all rational; $X_0(99)$ has 8 cusps, of which the two of denominator 3 and the two of denominator 33 form two pairs conjugate over $\mathbb{Q}(\sqrt{-3})$; $X_0(121)$ has 12 cusps, ten of them (denominator 11) conjugate over $\mathbb{Q}(\zeta_{11})$. Reference: Ogg 1973 (proposed key `Ogg73`, refs.bib, Part 6.5), DiamondShurman 3.8 (cite the section), Stein (book), Chapter 8 (cite the chapter). Paper: background (the cusp ∞ is the base point of φ_E; the cusp 0 in 11.4; the cuspidal subgroup in 11.1, Conjecture conj:eis, Open question 4).

2.3 `prop:moddeg:rational-points` (proposition; S for (i), R for (ii)). (i) Let K be a field of characteristic 0. If E/K has a cyclic subgroup C of order N stable under G_K, the point (E, C) of $Y_0(N)$ is K-rational; conversely a K-rational point of $Y_0(N)$ with $j \neq 0, 1728$ is represented by such a pair over K, unique

up to quadratic twist. (ii) (Mazur78, Kenku) $Y_0(N)(\mathbb{Q}) \neq \emptyset$ iff $N \in \{1,\dots,19,21,25,27,37,43,67,163\}$; $Y_0(N)(\mathbb{Q})$ is infinite exactly when $g(X_0(N)) = 0$; the rational points of $Y_0(27)$, $Y_0(43)$, $Y_0(67)$, $Y_0(163)$ have CM. Consequence used in the book: a rational cyclic 27-isogeny forces CM (Chapter 9, its 9.8, the proof of important_notes.md point 17). Reference: Mazur78 (cite the section), Kenku (cite in Chapter 9's bib; verify). Paper: Conjecture conj:eis and Section sec:eis (rational ℓ-isogenies, $\mathbb{Z}/3 \oplus \mu_3$, point 17).

2.4 ex:moddeg:small-levels (worked examples; P given 2.1, Chapter 12's 5.3 and 9.2, and 6.2). (a) $X_0(11)$ has genus 1 and the rational point ∞; it is the elliptic curve 11a1 $[0,-1,1,-10,-20]$ with origin ∞ (its j-invariant $-122023936/161051$ is recovered from the period ratio, Chapter 9, its 7.7), $J_0(11) = X_0(11)$ and $\varphi_{11a1} = \mathrm{id}$, of degree 1. (b) $X_0(20)$ is 20a1 and $X_0(14)$ is 14a1 in the same way (degree 1). (c) $X_0(37)$ has genus 2; w_{37} has 2 fixed points and $X_0(37)/w_{37}$ has genus 1 (Chapter 12, its 12.2(ii)); with origin the image of ∞ it is the curve 37a1, the class 37a has one curve, and the quotient map $X_0(37) \to X_0(37)/w_{37}$ is the optimal parametrisation of 37a1, of degree 2 (`ecdata` agrees). Its pullback of differentials is the w_{37}-invariant part $S_2^+ = \mathbb{C} f_{37a}$ ($\varepsilon_{37}(f_{37a}) = +1$, root number -1, rank 1). Paper: background (the smallest cases of φ_E; Section sec:two: 37a1 has rank 1 and $2 \mid \deg$).

16.2.3 Section 3. Hecke correspondences and their adjoints

3.1 def:moddeg:hecke-correspondence (definition and proposition; P for the group theory, S for the moduli description). Let $\alpha \in M_2(\mathbb{Z})$ with $\det\alpha > 0$ lie in $\Delta_0(N) = \{\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) : N \mid c,\ \gcd(a,N) = 1\}$, and $\Gamma_\alpha = \Gamma \cap \alpha^{-1}\Gamma\alpha$. The *Hecke correspondence* of $\Gamma\alpha\Gamma$ is $\mathcal{C}_\alpha = (X(\Gamma_\alpha), \pi_1, \pi_\alpha)$ with $\pi_1(\Gamma_\alpha z) = \Gamma z$ and $\pi_\alpha(\Gamma_\alpha z) = \Gamma\alpha z$ (Chapter 8, its 12.1). For a prime p: $T_p := \mathcal{C}_{\mathrm{diag}(1,p)}$ if $p \nmid N$ and $U_p := \mathcal{C}_{\mathrm{diag}(1,p)}$ if $p \mid N$; in both cases $X(\Gamma_\alpha) \simeq X_0(Np)$ (conjugate by $\mathrm{diag}(p,1)$), $\deg\pi_1 = p+1$ resp. p. (i) $\mathcal{C}_\alpha$ depends only on $\Gamma\alpha\Gamma$, and $\Gamma\alpha\Gamma = \bigsqcup_i \Gamma\alpha_i$ with $\alpha_i = \alpha\gamma_i$, γ_i running over $\Gamma_\alpha\backslash\Gamma$. (ii) On moduli (S): $T_p(E,C) = \sum_D (E/D, (C+D)/D)$ over the $p+1$ subgroups D of order p; $U_p(E,C) = \sum_{D \cap C = 0} (E/D, (C+D)/D)$ over the p subgroups D of order p meeting C trivially. (iii) Hence T_p, U_p and w_d are correspondences defined over $\mathbb{Q}$, and so are all T_n (polynomials in them, Chapter 13). Reference: Shimura71 Chapter 7 (cite the sections); DiamondShurman 5.2, 6.3, 7.9 (cite the sections); DiamondIm (cite the sections). Paper: Section 4.1 ("the Hecke operators, U_p included"), Lemma lem:hecke, Theorem thm:quant Steps 1, 2.

3.2 prop:moddeg:hecke-compatibility (proposition; P for (i), (ii); S for (iii)). (i) For $f \in S_2(\Gamma_0(N))$ and α as in 3.1, $\mathcal{C}_\alpha^*(f(z)\,dz) = \big(\sum_i f|_2\alpha_i\big)(z)\,dz = (f|[\Gamma\alpha\Gamma]_2)(z)\,dz$ (compute $\pi_\alpha^*(f\,dz) = (f|_2\alpha)\,dz$ by Chapter 12, C3, and the trace π_{1*} as the sum over $\Gamma_\alpha\backslash\Gamma$). With $\alpha = \mathrm{diag}(1,p)$ and $\alpha_i \in \{\left(\begin{smallmatrix} 1 & j \\ 0 & p \end{smallmatrix}\right)\} \cup \{\mathrm{diag}(p,1) \text{ if } p \nmid N\}$ this is Chapter 13's T_p (or U_p) in weight 2, where the normalisations $(\det)^{k/2}$ and $(\det)^{k-1}$ agree. (ii) Consequently $t \mapsto t^*$ identifies $\mathbb{T}$ with the subring of $\mathrm{End}\, H^1(X_0(N),\mathbb{Z})$ generated by the $\mathcal{C}^*$ ($H^1(X_0(N),\mathbb{C}) = S_2 \oplus \overline{S_2}$ by Chapter 8, its 10.2(iii), with $\mathcal{C}^*$ acting on $\overline{S_2}$ by the complex conjugate matrix, so faithfulness on S_2 gives faithfulness on H^1); in particular $\mathbb{T}$ is a finite free $\mathbb{Z}$-module (a subring of End of a lattice) and $t \mapsto t_*$ embeds $\mathbb{T}$ in $\mathrm{End}_{\mathbb{Q}}\, J_0(N)$. (iii) (The H^1 route to Shimura's integral basis, Chapter 12, its 11.2, in weight 2.) Since $\mathbb{T} \otimes \mathbb{C} \simeq \mathrm{Hom}_{\mathbb{C}}(S_2,\mathbb{C})$ by the pairing $(t,g) \mapsto a_1(tg)$ (Chapter 13), every $\psi \in \mathrm{Hom}_{\mathbb{Z}}(\mathbb{T},\mathbb{Z})$ is $t \mapsto a_1(tg_\psi)$ for a unique $g_\psi \in S_2$, and $a_n(g_\psi) = \psi(T_n) \in \mathbb{Z}$: so $S_2(\Gamma_0(N);\mathbb{Z})$ has rank g. This item only records the route; Chapters 12 and 13 own the statements (important_notes.md B11). Reference: DiamondShurman 6.3, 6.5 (cite the sections); Shimura71 3.4, 7.3 (cite the sections). Paper: Section 3 preamble ($\mathbb{T}$ "the subring of End $S_2(\Gamma_0(N))$ generated over $\mathbb{Z}$ by all T_n"), Section 4.1 (the Hecke action on H^1).

3.3 thm:moddeg:transpose (theorem; P). (i) For $\alpha \in \Delta_0(N)$ put $\alpha' = (\det\alpha)\alpha^{-1}$. Then $\mathcal{C}_\alpha^t \simeq \mathcal{C}_{\alpha'}$ as correspondences: the map $u\colon X(\Gamma_\alpha) \to X(\Gamma_{\alpha'})$, $\Gamma_\alpha z \mapsto \Gamma_{\alpha'}\alpha z$, is an isomorphism ($\alpha\Gamma_\alpha\alpha^{-1} = \Gamma \cap \alpha\Gamma\alpha^{-1} = \Gamma_{\alpha'}$, scalars acting trivially) with $\pi_1 \circ u = \pi_\alpha$ and $\pi_{\alpha'} \circ u = \pi_1$. (ii) For β normalising Γ (for instance W_N), $\mathcal{C}_{\beta\alpha\beta^{-1}} = w_\beta \circ \mathcal{C}_\alpha \circ w_\beta^{-1}$, so $\mathcal{C}^*_{\beta\alpha\beta^{-1}} = (w_\beta^{-1})^* \mathcal{C}_\alpha^* w_\beta^*$. (iii) $W_N\,\mathrm{diag}(1,p)\,W_N^{-1} = \mathrm{diag}(p,1) = \mathrm{diag}(1,p)'$ (a two-line matrix product). Hence $T_p^t = w_N T_p w_N$ **and** $U_p^t = w_N U_p w_N$ **for every prime** p, and since $t \mapsto t^t$ reverses products and $\mathbb{T}$ is commutative, $t^t = w_N t w_N$ **for every** $t \in \mathbb{T}$ (as correspondences up to equivalence, hence on H^1, H_1, $J_0(N)$). (iv) For $p \nmid N$ also $T_p^t = T_p$ ($\Gamma\,\mathrm{diag}(1,p)\,\Gamma = \Gamma\,\mathrm{diag}(p,1)\,\Gamma$ by elementary divisors in $\Delta_0(N)$, since $\gcd(p,N) = 1$); for $p \mid N$, $U_p^t \neq U_p$ in general (Exercise 2: on the span of $f_{11}(z)$, $f_{11}(2z)$ at level 22). Reference: Shimura71 Chapter 7 (cite the section); DDT Section

4.4 (cite the section, as the paper does); DFG04 Sections 1.6.1, 1.8.2 (as the paper cites them). Paper: Section 4.1 ("$w_{M^\Sigma} T_n w_{M^\Sigma}^{-1}$ is the adjoint of T_n for the cup product (this is what the composition with w is for)"); Chapter 8, its 12.4(d), points here.

3.4 `cor:moddeg:delta-hat` (corollary; P given 3.3 and Chapter 8, its 12.3 and 12.3(e)). Let R be $\mathbb{Z}$, $\mathbb{Z}_\ell$ or $\mathbb{Q}_\ell$ and $\langle\ ,\ \rangle$ the cup product on $H^1(X_0(N), R)$ (Chapter 8, its 10.3). (i) $(t^*)^\dagger = w_N^* t^* w_N^*$ for every $t \in \mathbb{T}$. (ii) The pairing

$$\hat{\delta}_!(x, y) = \langle x, w_N^* y \rangle$$

is perfect and alternating on $H^1(X_0(N), R)$, and every t^*, $t \in \mathbb{T}$ (the U_p^* included), is self-adjoint for it. (iii) Every idempotent of $\mathbb{T} \otimes \mathbb{Z}_\ell$ is self-adjoint, $H^1(X_0(N), \mathbb{Z}_\ell) = \bigoplus_{\mathfrak{m}} H^1(X_0(N), \mathbb{Z}_\ell)_{\mathfrak{m}}$ is an orthogonal decomposition and $\hat{\delta}_!$ is perfect on each summand (important_notes.md point 36). (iv) For weight 2 and trivial character, $\hat{\delta}_!$ is the pairing $\hat{\delta}_!$ of DFG04 (13), (14) **as the paper reads it** in Section 4.1 (the operator $[UwU]\omega$ being w_N, with no scalar, Chapter 12, C3, C4); the comparison of normalisations is item 9.6. Paper: Section 4.1, Lemma lem:hecke ("with the pairing $\hat{\delta}_!$ of Section sec:dfg, perfect and alternating with $\mathbb{T}$ self-adjoint"), Theorem thm:quant Step 2, Lemma lem:ddt417 as applied.

3.5 `rem:moddeg:petersson-analogue` (remark; R, pointer). The same identity on the Petersson side: T_n is self-adjoint for $(n, N) = 1$ and $U_p^\dagger = w_N U_p w_N$ (Chapter 13); a newform is a w_N-eigenvector with eigenvalue $\varepsilon_N(f) = \pm 1$ (Chapter 13; Chapter 12, its 12.4(c)). One sentence on why the two adjoint statements agree: under $S_2 \subset H^1(X_0(N), \mathbb{C})$ the Hodge inner product $\frac{i}{2}\int \omega \wedge \bar{\eta}$ is the Petersson product (Chapter 12, its 10.1(iv)), and complex conjugation exchanges S_2 and $\overline{S_2}$. Paper: Lemma lem:crit ("f is an eigenvector of the adjoints of the T_n as well").

16.2.4 Section 4. The Jacobian $J_0(N)$, its homology and the f-lattice

4.1 `def:moddeg:j0` (definition and theorem; P for (ii), (iv); R for (i), (iii)). (i) $J_0(N) := \mathrm{Jac}(X_0(N)_{\mathbb{Q}})$ is an abelian variety over $\mathbb{Q}$ of dimension $g = \dim S_2(\Gamma_0(N))$ (Chapter 8, its 11.3; Chapter 12, its 8.2), with good reduction at every $p \nmid N$ (from 2.1(iii)). (ii) $J_0(N)(\mathbb{C}) = S_2(\Gamma_0(N))^\vee / H_1(X_0(N), \mathbb{Z})$, with $\gamma \mapsto (f \mapsto \int_\gamma f(z)\,dz)$ (Chapter 8, its 11.2, and $S_2 = H^0(\Omega^1)$). (iii) The Abel-Jacobi map $\iota\colon X_0(N) \to J_0(N)$, $P \mapsto [(P) - (\infty)]$, is defined over $\mathbb{Q}$ because ∞ is rational (2.2), and has the universal property of Chapter 8, its 11.3(iii). (iv) Table (checked: `mfdim([N,2],1)` and the ranks of the integral lattices of Part 6.6): $\dim J_0(N) = 1, 1, 1, 2, 7, 9, 6, 19, 49, 109, 145$ for $N = 11, 14, 20, 37, 77, 99, 121, 140, 539, 726, 980$. Reference: DiamondShurman 6.1, 6.6, 7.9 (cite the sections); Milne, Jacobian varieties (Chapter 8's key). Paper: Section 1, Remark rem:free ("$J_0(N)[\mathfrak{m}]$"), Conjecture conj:eis and Open question 4 (torsion, cuspidal subgroup).

4.2 `prop:moddeg:hecke-on-j0` (proposition; P given 3.1, 3.2 and Chapter 8, its 12.2). (i) $t \mapsto t_*$ is an injective ring homomorphism $\mathbb{T} \to \mathrm{End}_{\mathbb{Q}}(J_0(N))$; on $J_0(N)(\mathbb{C}) = S_2^\vee / H_1$ it is the transpose of t^* on S_2 and t_* on H_1. (ii) On $T_\ell J_0(N) = H_1(X_0(N), \mathbb{Z}_\ell)$ (Chapter 8, its 11.7) the action is t_*, commuting with $G_{\mathbb{Q}}$ (the correspondences are defined over $\mathbb{Q}$). (iii) The involution $t \mapsto t^t = w_N t w_N$ of $\mathbb{T}$ (3.3) fixes ϕ_f for every newform f (since f is a w_N-eigenvector, $\phi_f(U_p^t) = \phi_f(U_p)$), so I_f is stable under it; hence the quotient $J_0(N)/I_f J_0(N)$ and the lattice $M_{f,\lambda}$ of 4.4 are the same whether one uses the Albanese action t_* or the Picard action $(t^t)_*$ (pitfall in Part 6.2). Reference: DiamondShurman 6.3, 6.6 (cite the sections). Paper: Remark rem:free ("the localised Hecke algebra", "the Tate module"), Section 4.1.

4.3 `prop:moddeg:tate-module-vs-h1` (proposition; P for (i), (ii) given Chapter 8, its 11.7, 11.8, 12.3(c); R for the etale input). (i) $H^1(X_0(N), \mathbb{Z}_\ell) = \mathrm{Hom}_{\mathbb{Z}_\ell}(T_\ell J_0(N), \mathbb{Z}_\ell)$, with t^* the transpose of t_*; as $G_{\mathbb{Q}}$-modules (etale cohomology) $H^1_{\mathrm{et}}(X_0(N)_{\bar{\mathbb{Q}}}, \mathbb{Z}_\ell) \simeq T_\ell J_0(N)(-1)$, and the cup product is $G_{\mathbb{Q}}$-equivariant with values in $\mathbb{Z}_\ell(-1)$. (ii) The map $\Psi = w_{N*} \circ \mathrm{PD}\colon H^1(X_0(N), \mathbb{Z}_\ell) \to H_1(X_0(N), \mathbb{Z}_\ell) = T_\ell J_0(N)$ is an isomorphism of $\mathbb{T}$-modules for the actions t^* on the source and t_* on the target (Chapter 8, its 12.3(c): $\mathrm{PD} \circ \mathcal{C}^* = (\mathcal{C}^t)_* \circ \mathrm{PD}$, and 3.3); PD alone intertwines t^* with $(t^t)_*$. Over $\bar{\mathbb{Q}}$, Ψ becomes $G_{\mathbb{Q}}$-equivariant after the twist by $\mathbb{Z}_\ell(-1)$ (w_N is defined over $\mathbb{Q}$). (iii) Hence freeness over a localisation $\mathbb{T}_{\mathfrak{m}}$ transfers between $T_\ell J_0(N)_{\mathfrak{m}}$ and $H^1(X_0(N), \mathbb{Z}_\ell)_{\mathfrak{m}}$ (important_notes.md point 41). Reference: Chapter 8 outline items cited; Milne, Etale cohomology (Chapter 8's citation). Paper: Remark rem:free ("the Tate module is free of rank two"), Section 4.2 (point 26).

4.4 def:moddeg:f-lattice (definition and proposition; P given Chapter 13's multiplicity one). Let f be a newform of level N with rational coefficients. (i) $H^1(X_0(N),\mathbb{Q})\otimes\mathbb{C} = S_2\oplus\overline{S_2}$, and the subspace killed by $I_f\otimes\mathbb{C}$ is $\mathbb{C}\,\omega_f\oplus\mathbb{C}\,\overline{\omega_f}$, of dimension 2 (the eigensystem ϕ_f occurs once in S_2, Chapter 13, and its complex conjugate system, equal to ϕ_f, once in $\overline{S_2}$). So $V_f := H^1(X_0(N),\mathbb{Q})[I_f]$ is a 2-dimensional $\mathbb{Q}$-space with $V_f\otimes\mathbb{R}$ spanned by $\operatorname{Re}\omega_f$, $\operatorname{Im}\omega_f$. (ii) Define

$$M_{f,\lambda} := H^1(X_0(N),\mathbb{Z}_\ell)[I_f] = \{x : tx = \phi_f(t)x\ \forall t\in\mathbb{T}\} = H^1(X_0(N),\mathbb{Z}_\ell)\cap(V_f\otimes\mathbb{Q}_\ell),$$

a saturated $\mathbb{Z}_\ell$-sublattice of rank 2 (saturated as a kernel, Chapter 3). This is the paper's $M_{f,\lambda}$ "the kernel on it of the kernel of the eigencharacter" (Section 4.1 with $\Sigma=\emptyset$). (iii) w_N^* preserves V_f and $M_{f,\lambda}$ and acts on both by $\varepsilon_N(f)$ (Chapter 12, its 12.2(iii); integrality because w_N^* preserves $H^1(X_0(N),\mathbb{Z}_\ell)$). (iv) $M_{f,\lambda}$ is $G_\mathbb{Q}$-stable (the Hecke and Galois actions commute) and is a lattice in the Galois representation of f (Chapter 15). Paper: Section 4.1, Lemma lem:hecke ("$L[\wp] = M_{f,\lambda}$ has rank 2"), Section 4.2 (point 26).

16.2.5 Section 5. Eichler-Shimura and modularity

5.1 thm:moddeg:shimura-quotient (theorem; S for (i), (ii); R for (iii)). Let f be a newform of level N with coefficient field K_f and $I_f = \ker(\mathbb{T}\to K_f)$. (i) $I_fJ_0(N)$ (the subgroup generated by the images $t_*J_0(N)$, $t\in I_f$) is an abelian subvariety over $\mathbb{Q}$ (an image of the abelian variety $J_0(N)^r$ under a homomorphism, hence connected), and $J_f := J_0(N)/I_fJ_0(N)$ (Shimura's A_f; the book writes J_f because A_f is the paper's adjoint $\mathrm{ad}^0 V$, Appendix B, B-C12(c)) is an abelian variety over $\mathbb{Q}$ of dimension $[K_f:\mathbb{Q}]$, on which $\mathbb{T}$ acts through $\mathbb{T}/I_f\hookrightarrow\mathcal{O}_{K_f}$; the quotient map $\pi_f\colon J_0(N)\to J_f$ has connected kernel by construction. (ii) For f with rational coefficients, $J_f =: E_f$ is an elliptic curve over $\mathbb{Q}$ with $\mathbb{T}$ acting through ϕ_f; $\pi_f^*\colon H^1(E_f,\mathbb{Q})\to H^1(J_0(N),\mathbb{Q}) = H^1(X_0(N),\mathbb{Q})$ has image V_f (4.4). (iii) (Shimura) $L(J_f,s) = \prod_{\sigma\colon K_f\to\mathbb{C}} L(f^\sigma,s)$, in particular $L(E_f,s) = L(f,s)$. Reference: Shimura71 Theorem 7.14 (verify the number), Shimura73; DiamondShurman 6.6 and Theorem 6.6.6 (verify the number). Paper: Section 1 (the optimal curve is E_f, 6.2), Lemma lem:hecke (via 9.1), Section 4.2.

5.2 thm:moddeg:eichler-shimura (theorem; R for (i), P for (ii) given (i) and Chapter 8, its 11.8(d)). (i) (Eichler-Shimura congruence relation.) For $p\nmid N$, in $\mathrm{End}((J_0(N))_{\mathbb{F}_p})$: $(T_p)_* = \pi_p + \hat\pi_p$, where π_p is the Frobenius endomorphism and $\hat\pi_p = p\pi_p^{-1}$ the Verschiebung (Kronecker's congruence: modulo p the correspondence T_p is the graph of the Frobenius morphism plus its transpose; proof idea from the moduli description 3.1(ii): for an ordinary curve in characteristic p, the quotient by the connected subgroup of order p is the Frobenius image, and the quotients by the p etale subgroups are the Frobenius preimages). (ii) Consequences: on $T_\ell J_f$ ($\ell\neq p$) the arithmetic Frobenius satisfies $\pi_p^2 - T_p\pi_p + p = 0$; for f rational, $\det(X - \mathrm{Frob}_p \mid T_\ell E_f) = X^2 - a_p(f)X + p$, so $a_p(E_f) = p+1-\#E_f(\mathbb{F}_p) = a_p(f)$ for every $p\nmid N$ (with Chapter 9, its 8.x, and Chapter 8, its 11.8(d)); on H^1 the geometric Frobenius has the same characteristic polynomial (C5). (iii) This is how the Galois representation of a weight-two newform is constructed (Chapter 15). Reference: Shimura71 Section 7.4 (verify the theorem number), DiamondShurman 8.7 (cite the section), DiamondIm Section 10 (cite the section). Paper: Section 1 ("$f = f_E$ the newform attached to it"), Lemma lem:old (a_p as traces of Frobenius), Section 4.2.

5.3 thm:moddeg:modularity (theorem; R). (Wiles, Taylor-Wiles, Breuil-Conrad-Diamond-Taylor.) For every elliptic curve $E/\mathbb{Q}$ of conductor N there is a newform $f_E\in S_2(\Gamma_0(N))$ with rational coefficients and $a_p(f_E) = a_p(E)$ for every prime p (including $p\mid N$; the level equals the conductor by Carayol86); equivalently E is isogenous over $\mathbb{Q}$ to E_{f_E} (5.1), and $L(E,s) = L(f_E,s)$. One paragraph of history: Taniyama, Shimura, Weil; semistable case Wiles95, TaylorWiles; general case BCDT; the role of $R=\mathbb{T}$ (Chapter 20). Reference: Wiles95, TaylorWiles, BCDT, Carayol86. Paper: Section 1 ("let $f = f_E$ be the newform attached to it"), everything.

5.4 cor:moddeg:classes-and-newforms (corollary; S). $E\mapsto f_E$ is a bijection from $\mathbb{Q}$-isogeny classes of elliptic curves of conductor N to newforms of level N with rational coefficients. (Injectivity: isogenous curves have the same a_p; curves with the same a_p for almost all p are isogenous by Faltings's isogeny theorem (R). Surjectivity: E_f has conductor N (Carayol86, R) and $f_{E_f} = f$ by 5.2(ii) and strong multiplicity one, Chapter 13.) Example: 11a $\leftrightarrow f_{11} = \eta(z)^2\eta(11z)^2$; the level 99 has four classes 99a, 99b, 99c, 99d and $\dim S_2(\Gamma_0(99)) = 9 = 4+2\cdot1+3\cdot1$ (newforms of levels 99, 33, 11 counted with the number

of divisors of $99/M$: $4 \cdot 1 + 1 \cdot 2 + 1 \cdot 3$). Paper: Section 1 ("the isogeny class of E"), Section sec:data (classes).

5.5 `ex:moddeg:class-11a` (worked example; P given 2.4, 5.2, 6.2 and the numerics of Part 6.6). The class 11a has three curves, 11a1 $= X_0(11)$, 11a2, 11a3, linked by 5-isogenies (11a1 to 11a2 and to 11a3 of degree 5, 11a2 to 11a3 of degree 25). Since $X_0(11) = J_0(11) =$ 11a1, Eichler-Shimura for $X_0(11)$ proves $a_p(11a1) = a_p(f_{11})$ for every $p \neq 11$ (the identity Chapter 12, its 9.2(iv), checked numerically, is now a theorem). The modular parametrisations of 11a2 and 11a3 are the two 5-isogenies out of 11a1, of degree 5 each (6.2(ii)); 11a3 is $X_1(11)$ (R). Paper: background; Section 3 Example (11a1 as the congruent form).

16.2.6 Section 6. Optimal curves, the modular parametrisation and the modular degree

6.1 `def:moddeg:parametrisation` (definition; P for the equivalence). Let $E/\mathbb{Q}$ have conductor N. A *modular parametrisation* of E is a nonconstant morphism $\varphi\colon X_0(N) \to E$ over $\mathbb{Q}$ with $\varphi(\infty) = O$. By the universal property (4.1(iii)) these correspond bijectively to nonzero homomorphisms $h\colon J_0(N) \to E$ over $\mathbb{Q}$, via $\varphi = h \circ \iota$ and $h = \varphi_*$ (Chapter 8, its 11.4(iv)). Paper: abstract and Section 1 ("$\varphi_E\colon X_0(N) \to E$ its modular parametrisation").

6.2 `thm:moddeg:optimal` (theorem; S for (i), with R inputs; P for (ii) to (iv) given (i) and Chapter 8, its 11.5). Let $\mathcal{I}$ be an isogeny class of conductor N, f its newform (5.4) and $E_f = J_0(N)/I_f J_0(N)$ with $\varphi_{E_f} = \pi_f \circ \iota$ (5.1). (i) Every homomorphism $h\colon J_0(N) \to E$ to a curve $E \in \mathcal{I}$ vanishes on $I_f J_0(N)$. Sketch: the identity component B of $\ker h$ is an abelian subvariety of codimension 1 with $V_\ell B = \ker(h_*\colon V_\ell J_0(N) \to V_\ell E)$; the Galois-stable quotient $V_\ell J_0(N)/V_\ell B \simeq V_\ell E$ has Frobenius traces $a_p(f)$; decomposing $V_\ell J_0(N) \otimes \bar{\mathbb{Q}}_\ell$ into the generalised eigenspaces of $\mathbb{T}$ (Galois-stable, Chapter 3, its 10.5), on which Frobenius satisfies $X^2 - a_p(g)X + p$ by 5.2, strong multiplicity one (Chapter 13) and Chebotarev (Chapter 5) show that only the ϕ_f-eigenspace, of dimension 2 by 4.4, can map onto $V_\ell E$; so $V_\ell B \supseteq V_\ell(I_f J_0(N))$, both have dimension $2g-2$, and an abelian subvariety is determined by its rational Tate module, so $B = I_f J_0(N)$. (ii) **Factorisation.** For every $E \in \mathcal{I}$ and every modular parametrisation φ of E there is a unique isogeny $\psi\colon E_f \to E$ with $\varphi = \psi \circ \varphi_{E_f}$; hence $\deg\varphi = \deg\psi \cdot \deg\varphi_{E_f}$. (iii) **Characterisations.** For $E \in \mathcal{I}$ and a modular parametrisation φ of E the following are equivalent: (a) ψ in (ii) is an isomorphism; (b) $\ker(\varphi_*\colon J_0(N) \to E)$ is connected; (c) $\varphi_*\colon H_1(X_0(N),\mathbb{Z}) \to H_1(E,\mathbb{Z})$ is surjective; (d) $\varphi^* H^1(E,\mathbb{Z})$ is saturated in $H^1(X_0(N),\mathbb{Z})$; (e) $\varphi^*\colon E = E^\vee \to J_0(N)$ is injective; (f) $\deg\varphi$ is the minimum of the degrees of all modular parametrisations of all curves in $\mathcal{I}$. ((b) to (e): Chapter 8, its 11.5(iii); (a)⇔(b) by (i); (a)⇔(f) by (ii).) (iv) **Uniqueness.** The pair (E_f, φ_{E_f}) is the unique pair up to unique isomorphism satisfying these conditions; it is called the $X_0(N)$-*optimal curve* (or strong Weil curve) E of $\mathcal{I}$, and $\deg\varphi_E$ its *modular degree*. Reference: Cremona97 Chapter II (cite the section); ARS12 Section 2 or 3 (cite the section); Stevens, *Stevens89* (proposed key, optional; see Part 6.5). Paper: abstract, Section 1 ("for which the modular parametrisation has minimal degree"), Theorem thm:main and Lemma lem:hecke ("E be $X_0(N)$-optimal"), Section 3 preamble ("let E_0 be the optimal curve in the class").

6.3 `prop:moddeg:degree-descriptions` (proposition; P given 6.2 and Chapter 8, its 11.4, 11.5). Let E be optimal, $d = \deg\varphi_E$, $A = \varphi_E^*(E) \subseteq J_0(N)$ (an elliptic curve isomorphic to E, by 6.2(iii)(e)) and $B = \ker\varphi_{E*} = I_f J_0(N)$. (i) d is the degree of the morphism of curves, in each of the senses of Chapter 8 (field degree, preimage count, pullback of divisors, topological degree). (ii) $\varphi_{E*} \circ \varphi_E^* = [d]$ on E, the composition $E^\vee \to J_0(N) \to E$ of PLAN.md. (iii) $A \cap B = \varphi_E^*(E[d]) \simeq (\mathbb{Z}/d)^2$, so d is the exponent of $A \cap B$ (the "modular exponent" of ARS12). (iv) $\langle \varphi_E^* x, \varphi_E^* y\rangle_{X_0(N)} = d\,\langle x, y\rangle_E$ on H^1 (Chapter 8, its 11.5(i)); the Gram determinant of the cup product on $\varphi_E^* H^1(E,\mathbb{Z})$ is d^2 and its ideal of values is $d\mathbb{Z}$. (v) $\deg\varphi_E$ is computable (Section 12, 12.6). Paper: Section 1 ("the modular degree is computable"), Lemma lem:hecke (via 9.3).

6.4 `prop:moddeg:valuation-in-class` (proposition; P). Let E be optimal in $\mathcal{I}$, ℓ a prime with $E[\ell]$ irreducible, and $E' \in \mathcal{I}$. Then every isogeny $E \to E'$ of minimal degree is cyclic of degree prime to ℓ, so the minimal degree of a modular parametrisation of E' (6.2(ii)) has the same ℓ-adic valuation as $\deg\varphi_E$. (A cyclic isogeny of degree divisible by ℓ has a rational subgroup of order ℓ in its kernel, contradicting irreducibility, Chapter 9.) So under the paper's standing hypothesis "$E[\ell]$ irreducible" the choice of curve

in the class does not affect v_ℓ of the degree (important_notes.md point 27). Without irreducibility it does: 11a1, 11a2, 11a3 at $\ell = 5$ (degrees 1, 5, 5). Paper: Theorem thm:qual ("the $X_0(N)$-optimal curve in the isogeny class of E"), Section 3 preamble (E_0), Theorem thm:main.

6.5 `conv:moddeg:cremona-optimal` (convention and data; R for the data). In Cremona's tables curve number 1 of each class is the optimal curve, with the single exception of 990h, whose optimal curve is 990h3 $= [1, -1, 1, -1568, -4669]$ (modular degree 1728 in `ecdata`'s `degphi` file); optimality is established for every class of conductor at most 400000 and for a large part of the classes up to 500000 (the paper, Section 6: 359,009 further classes); the LMFDB numbering differs (Chapter 9, its 12.1). How optimality is certified (one paragraph, R): compare the period lattices of the curves in the class with the lattice of φ^* computed from modular symbols (Cremona97 Chapter II; Cremona's later papers). Paper: Section 6 ("Optimality is determined in the tables ..."), important_notes.md point 25 (the abstract's "every optimal curve of conductor at most 500000").

6.6 `ex:moddeg:running-degrees` (worked example table; data R from `ecdata`, the consistency checks P). Modular degrees of the optimal curves of the paper's world (all equal to PARI's `ellmoddegree`, Part 6.6):

curve	N	rank	$\#E(\mathbb{Q})_{\mathrm{tors}}$	$\deg \varphi_E$	$\dim J_0(N)$	role
11a1	11	0	5	1	1	$X_0(11)$ itself
20a1	$2^2 \cdot 5$	0	6	1	1	$X_0(20)$ itself
37a1	37	1	1	2	2	$X_0(37)/w_{37}$
99a1 (99A1)	$3^2 \cdot 11$	1	2	$4 = 2^2$	9	$r_E = 12$ (8.5)
121a1	11^2	0	1	$6 = 2 \cdot 3$	6	thm:qual (b), $e_3 = 1$ attained
539c1	$7^2 \cdot 11$	1	2	$288 = 2^5 \cdot 3^2$	49	thm:qual (c), $\ell = 3$
980i1	$2^2 \cdot 5 \cdot 7^2$	0	1	$2880 = 2^6 \cdot 3^2 \cdot 5$	145	Section 3 Example, $\ell = 5$
726a1	$2 \cdot 3 \cdot 11^2$	1	2	$144 = 2^4 \cdot 3^2$	109	3Ns, $r_E = 1584$ (8.5)

Supporting curves (all `ecdata`): 77c1 (6), 140b1 (60), 121c1 (6), 121d1 (24), 539d1 (72), 990h3 (1728). Paper: the Example of Section 3, the remark on 99A1, the appendix (121a1, 726a1), Section sec:anomaly.

16.2.7 Section 7. The Manin constant and the formula of Zagier and Watkins

7.1 `def:moddeg:manin-constant` (definition and proposition; P). Let φ be a modular parametrisation of E (6.1). (i) $\varphi^*\omega_E$ is a nonzero holomorphic differential lying in $V_f \otimes \mathbb{C}$ (by 6.2(ii), φ^* factors through π_f^*, and 5.1(ii)), hence a multiple of $\omega_f = f(z)\,dz$: $\varphi^*\omega_E = c_\varphi \cdot 2\pi i\, f(z)\,dz = c_\varphi\, f(q)\,\frac{dq}{q}$. (ii) $c_\varphi \in \mathbb{Q}^\times$: φ and ω_E are defined over $\mathbb{Q}$ and the cusp ∞ is rational, so the expansion of $\varphi^*\omega_E$ in the local parameter q at ∞ has rational coefficients, and $a_1(f) = 1$. (iii) For $\varphi = \psi \circ \varphi_E$ (6.2(ii)), $c_\varphi = c_{\varphi_E} \cdot u_\psi$ where $\psi^*\omega_{E'} = u_\psi\, \omega_E$, $u_\psi \in \mathbb{Z} \setminus \{0\}$. The *Manin constant* of E is $c_E := c_{\varphi_E}$ for the optimal curve; the sign of ω_E is chosen so that $c_E > 0$ (Appendix B, B-C9; only c_E^2 enters any formula). Paper: background (the "explicit factors" of Section 1).

7.2 `thm:moddeg:manin-known` (theorem; R; every item to be checked against the source before printing). (i) (Edixhoven) $c_E \in \mathbb{Z}$, by the q-expansion principle on the Neron model; more generally $c_\varphi \in \mathbb{Z}$ for every modular parametrisation. (ii) (Manin's conjecture) $c_E = 1$ for the optimal curve (the literature writes $c_E = \pm 1$ when ω_E is not normalised). (iii) Known results: if a prime p divides c_E then $p = 2$ or $p^2 \mid N$ (Mazur78 for $p \nmid 2N$; AU96 for $p^2 \nmid N$); $c_E = 1$ for every optimal curve of conductor

below a large bound (Cremona's computations; ARS's paper on the Manin constant) and for semistable optimal curves (Cesnavicius); the recent bound $c_E \mid \deg\varphi_E$ and related results (Cesnavicius, Neururer, Saha). The writer cites only what is checked; proposed keys in Part 6.5. (iv) For non-optimal curves $c_\varphi \neq 1$ is possible: 7.4. Paper: background; the paper's tables assume $c_E = 1$ implicitly only through the Zagier-Watkins heuristic of Section 1, not in any proof.

7.3 `thm:moddeg:zagier-watkins` (theorem; P given Chapter 8, its 9.4(b), 11.5(i), and Chapter 12, its 10.1(iv)). For every modular parametrisation φ of E,

$$\deg\varphi \cdot \mathrm{vol}(\Lambda_E) = 4\pi^2 c_\varphi^2 \,\langle f, f\rangle,$$

with the unnormalised Petersson product (C4). In particular, for the optimal curve with $c_E = 1$, $\deg\varphi_E = 4\pi^2\langle f,f\rangle/\mathrm{vol}(\Lambda_E)$. Proof (three lines): $\frac{i}{2}\int_{X_0(N)}\varphi^*\omega_E \wedge \overline{\varphi^*\omega_E} = \deg\varphi\cdot\frac{i}{2}\int_E \omega_E\wedge\bar\omega_E$ (a $\deg\varphi$-sheeted orientation-preserving cover away from finitely many points), the right side is $\deg\varphi\cdot\mathrm{vol}(\Lambda_E)$, and the left is $c_\varphi^2\cdot 4\pi^2\cdot\frac{i}{2}\int\omega_f\wedge\bar\omega_f = 4\pi^2c_\varphi^2\langle f,f\rangle$. **The Manin constant stands in the numerator** (important_notes.md B10). Physics aside (one sentence): the L^2 norm of a pulled-back one-form counts every sheet of the cover. Reference: Zagier85; Watkins02; Cremona97 Chapter II (cite the sections; check their normalisations of $\langle\ ,\ \rangle$ before saying they agree). Paper: Section 1 ("Up to explicit factors $\deg\varphi_E$ is $\langle f,f\rangle$ divided by the area of the period lattice of E (Zagier, Watkins)").

7.4 `ex:moddeg:zw-numerics` (worked example; P as a computation, Part 6.6). Unnormalised Petersson norms ($\mu(N)$ times PARI's `mfpetersson`), period areas $\mathrm{vol}(\Lambda_E)$ and $4\pi^2\langle f,f\rangle/\mathrm{vol}(\Lambda_E) = \deg\varphi/c_\varphi^2$ (30 digits):

curve	$\langle f,f\rangle$	$\mathrm{vol}(\Lambda_E)$	$4\pi^2\langle f,f\rangle/\mathrm{vol}$	$\deg\varphi$	c_φ
11a1	0.046900147873495187582436234559591	31771	1	1	
11a2	same	0.370308724691191586354		5	1
11a3	same	9.257718117279796158854		5	5
37a1	0.371754147510696703	3832740789576273907		2	1
99a1	0.538519568710866631	7510542145355032		4	1
121a1	0.428239360011957218	7802048195801690671		6	1

(The 20a1 value 1 is in Chapter 12, its 10.4.) Reading: in the class 11a the parametrisations of 11a2 and 11a3 both have degree 5 (6.2(ii)), and the formula forces $c_\varphi = 1$ for 11a2 and $c_\varphi = 5$ for 11a3 (the 5-isogeny 11a1 $\to$ 11a3 pulls ω_{11a3} back to $5\,\omega_{11a1}$). Paper: Section 1; the Chapter 22 computations.

7.5 `rem:moddeg:rankin-and-computation` (remark; R, pointers). (a) Rankin and Shimura express $\langle f,f\rangle$ through $L(\mathrm{Sym}^2 f, 2)$ with explicit factors at the bad primes (Chapter 19; DFG04 Lemma 2.12 as the paper cites it; Shimura76); combined with 7.3 this is Watkins's method of computing $\deg\varphi_E$ (Watkins02), implemented in PARI's `ellmoddegree`. (b) The Rankin series $\sum a_n^2 n^{-s}$ is not the naive adjoint L-function (important_notes.md point 30; Chapter 12, its 10.5(b); Chapter 19). (c) The "explicit factors" of the paper's Section 1 are $4\pi^2c_E^2$ in the book's normalisation. Paper: Section 1 (Zagier, Watkins, Rankin, Shimura, Flach), Section 6 (`ellmoddegree`).

16.2.8 Section 8. The congruence number and the theorem of Agashe, Ribet and Stein

8.1 `def:moddeg:congruence-number` (definitions recalled from Chapter 13, its 4.3, and proposition; P for (ii), (iii)). (i) (The paper's definition, Section 1 and Section 3 preamble; Chapter 13's r_f for $f = f_E$.) The *congruence number* r_E of E (of $f = f_E$) is the largest integer r such that $f \equiv g \pmod r$ coefficientwise for some $g \in S_2(\Gamma_0(N);\mathbb{Z}) \cap f^\perp$. (ii) The *Hecke congruence ideal* of f is $\eta_f = \phi_f(\mathrm{Ann}_{\mathbb{T}} I_f) \subseteq \mathbb{Z}$, and $\eta_f \otimes \mathbb{Z}_\ell = \phi_f(\mathrm{Ann}_{\mathbb{T}_\ell}(I_f\mathbb{T}_\ell))$ is computed in the localisation $\mathbb{T}_{\mathfrak{m}}$ at the maximal ideal $\mathfrak{m}$ containing ℓ and I_f (Chapter 3, its 11.1(v); important_notes.md point 35: the other local factors lie in $\ker\phi_f$, not in the annihilator). (iii) Since f is a newform its eigensystem has multiplicity one, $\mathbb{T}\otimes\mathbb{Q} = \mathbb{Q}\times\mathbb{T}'$, $I_f \cap \mathrm{Ann}(I_f) = 0$ and $\eta_f \neq 0$ (Chapter 3, its 11.1(ii); important_notes.md point 19). Paper: Section 1 ("the congruence number r_E"), Lemma lem:crit, Lemma lem:hecke.

8.2 `lem:moddeg:crit` (lemma, the paper's Lemma lem:crit; R here, **proved in full in Chapter 15**, restated as a theorem on congruence modules in Chapter 20). $\mathbb{T}/(I_f + J) \simeq \mathbb{Z}/r_E$ with $J = \mathrm{Ann}_{\mathbb{T}}(f^{\perp}) = \mathrm{Ann}_{\mathbb{T}}(I_f)$, so $r_E\mathbb{Z} = \eta_f$; the same over $\mathbb{Z}_\ell$ with r_E replaced by its ℓ-part; and a normalised eigenform $g \in f^{\perp}$ congruent to f at every n modulo a prime $\lambda \mid \ell$ forces $\ell \mid r_E$. Chapter 14 uses only the identity $r_E\mathbb{Z}_\ell = \eta_f\mathbb{Z}_\ell$. Paper: Lemma lem:crit; Lemma lem:hecke ("$r_E\mathbb{Z}_\ell$ is the congruence ideal ... by Lemma lem:crit").

8.3 `thm:moddeg:ars` (theorem; R as a whole; the divisibility is **proved here** in 9.4, the equality for $\ell \nmid 2N$ with $E[\ell]$ irreducible in 10.3). (Agashe, Ribet and Stein, ARS12 Theorem 2.2 as the paper cites it; verify the number and the exact hypotheses.) Let E be optimal. Then $\deg\varphi_E \mid r_E$, and $v_p(\deg\varphi_E) = v_p(r_E)$ for every prime p with $p^2 \nmid N$. Proof idea of ARS (one paragraph): with A, B as in 6.3, the modular degree is the exponent of $H_1(J_0(N),\mathbb{Z})/(H_1(A,\mathbb{Z}) + H_1(B,\mathbb{Z}))$ and the congruence number is the exponent of $S_2(\mathbb{Z})/(S_2(\mathbb{Z})[I_f] + S_2(\mathbb{Z})[J])$; the perfect pairing $S_2(\mathbb{Z}) \times \mathbb{T} \to \mathbb{Z}$ (Chapter 13) and the action of $\mathbb{T}$ on H_1 compare the two, and they agree at p when $H_1(X_0(N),\mathbb{Z}_p)$ is locally free of rank two over $\mathbb{T} \otimes \mathbb{Z}_p$ at the relevant maximal ideal (multiplicity one, which ARS establish for $p^2 \nmid N$, including Eisenstein and $p = 2$ cases, by citing Mazur, Ribet and others). Reference: ARS12 Sections 2 and 3 (cite the sections); Ribet83 (congruence primes); Zagier85 (the first comparison of $\deg\varphi_E$ with congruence primes). Paper: Section 1, Section 3 preamble, Lemma lem:hecke, the remark after Theorem thm:qual.

8.4 `rem:moddeg:abc-size` (remark; R). The size of $\deg\varphi_E$: the conjecture $\deg\varphi_E \ll_\varepsilon N^{2+\varepsilon}$ is equivalent to a form of the abc conjecture (Frey; Mai-Murty; see ARS12 Section 2, as the paper cites it); lower bounds $\deg\varphi_E \gg N^{7/6-\varepsilon}$ are known (Watkins; cite only if checked). Paper: Section 1 ("its size is tied to deep questions").

8.5 `ex:moddeg:99a1-and-726a1` (worked example; P as a computation, Part 6.6; R for the comparison with ARS12 Table 1). Congruence numbers computed directly (integral basis of $S_2(\Gamma_0(N);\mathbb{Z})$ by saturating the q-expansions to beyond the Sturm bound; the Hecke matrices on it; the primitive integral functional λ that is a common left eigenvector with eigenvalues $\phi_f(T_p)$, $p \le 19$, which vanishes exactly on $f^{\perp} \cap S_2(\mathbb{Z})$; then $r_E = |\lambda(f)|$, by the cyclic-quotient argument of Lemma lem:crit):

curve	N	$\deg\varphi_E$	r_E	$r_E/\deg\varphi_E$	primes p with $p^2 \mid N$
11a1	11	1	1	1	none
20a1	20	1	1	1	2
37a1	37	2	2	1	none
77c1	77	6	6	1	none
99a1	99	4	12	3	3
99b1, 99c1	99	12	12	1	3
99d1	99	6	6	1	3
121a1	121	6	6	1	11
539c1	539	288	288	1	7
726a1	726	144	1584	11	11
980i1	980	2880	2880	1	2, 7

So 99A1 has $\deg\varphi_E = 4$ and $r_E = 12$ exactly as ARS12 Table 1 and the paper's remark after Theorem thm:qual say, and the discrepancy is at 3 with $3^2 \mid 99$. The congruence behind $3 \mid r_E$: $f_{99a} \equiv f_{99b} \equiv f_{99d}$ (mod 3) coefficientwise (checked for $n \le 30$, beyond the Sturm bound $B_2(99) = 24$ of Chapter 12, its 11.3; both are newforms of level 99, hence in $f^{\perp}$). The paper's own 726a1 is a second example, with ratio 11 and $11^2 \mid 726$. In every other case of the table the two numbers agree, also at primes p with $p^2 \mid N$: the equality can hold there, it is just not guaranteed. Paper: the remark after Theorem thm:qual ("Table 1 of ARS12 lists the curve 99A1 ... with modular degree 4 and congruence number 12"), Section 1, Lemma lem:hecke ("as $\ell^2 \nmid N$").

16.2.9 Section 9. The modular degree as a congruence ideal (important_notes.md point 39)

The section opens with one paragraph: the paper proves $v_\ell(\deg\varphi_E) \geq v_\lambda(\eta^{\emptyset}_{f,\lambda})$ (Lemma lem:hecke) and needs multiplicity one (Remark rem:free) for equality; for the optimal curve the lattice $M_{f,\lambda}$ is the pullback of $H^1(E)$, the projection formula computes the pairing on it, and the equality holds unconditionally. Everything in 9.1 to 9.5 is proved; 9.6 lists what must still be compared with DFG04.

9.1 `prop:moddeg:mf-is-pullback` (proposition; P given 4.4, 5.1, 6.2 and Chapter 8, its 11.5, 11.7, 11.8, 12.2(a); the Chapter 15 outline's 2.6(ii) states the same result and should cite this item, Part 7, B-c). Let E be optimal with newform f and let ℓ be any prime. (a) **Hecke equivariance.** $\varphi_{E*} = \pi_{f*}\circ\iota_* \colon H_1(X_0(N),\mathbb{Z}) \to H_1(E,\mathbb{Z})$ satisfies $\varphi_{E*}\circ t_* = \phi_f(t)\,\varphi_{E*}$ for $t \in \mathbb{T}$ ($\mathbb{T}$ acts on $E = J_0(N)/I_fJ_0(N)$ through ϕ_f); transposing (C2), $t^*\varphi_E^*x = \phi_f(t)\varphi_E^*x$ for $x \in H^1(E,\mathbb{Z}_\ell)$, so $\varphi_E^*H^1(E,\mathbb{Z}_\ell) \subseteq M_{f,\lambda}$. (b) **Ranks.** Both are free of rank 2 (φ_E^* is injective, since $\varphi_{E*}\varphi_E^* = \deg\varphi_E \neq 0$; $M_{f,\lambda}$ by 4.4). (c) **Saturation.** E is optimal, so φ_{E*} is surjective on H_1 and $\varphi_E^*H^1(E,\mathbb{Z})$ is saturated (6.2(iii)); saturation is kept by $\otimes\mathbb{Z}_\ell$. A saturated sublattice of full rank in the saturated lattice $M_{f,\lambda}$ equals it. Hence

$$M_{f,\lambda} = \varphi_E^*H^1(E,\mathbb{Z}_\ell).$$

(d) **Galois structure** (important_notes.md points 26, 40). φ_E is defined over $\mathbb{Q}$, so in etale cohomology φ_E^* is $G_\mathbb{Q}$-equivariant and $M_{f,\lambda} \simeq H^1_{\text{et}}(E_{\bar{\mathbb{Q}}},\mathbb{Z}_\ell) \simeq \mathrm{Hom}_{\mathbb{Z}_\ell}(T_\ell E,\mathbb{Z}_\ell) \simeq T_\ell E(-1)$ (Weil pairing): an equality of lattices up to this canonical isomorphism, with no irreducibility hypothesis; and $\mathrm{ad}^0\, M_{f,\lambda} \simeq \mathrm{ad}^0\, T_\ell E$ (ad^0 is insensitive to duals and twists, Chapter 4). Paper: Section 4.1 ($M_{f,\lambda}$), Section 4.2 ("the lattice $M_{f,\lambda}$ of DFG04 and $T_\ell E$ are homothetic when $\bar\rho$ is irreducible": correct up to the dual, and exact for the optimal curve), Lemma lem:hecke ("$L[\wp] = M_{f,\lambda}$ has rank 2").

9.2 `lem:moddeg:wn-on-mf` (lemma; P given 4.4(iii)). w_N^* acts on $M_{f,\lambda} = \varphi_E^*H^1(E,\mathbb{Z}_\ell)$ by $\varepsilon_N(f) = \pm1$; so $\hat\delta_!(x,y) = \varepsilon_N(f)\langle x,y\rangle$ on $M_{f,\lambda}$. (Equivalently: w_N preserves $I_fJ_0(N)$ by 4.2(iii), descends to the automorphism $[\varepsilon_N(f)]$ of E, and $\varphi_E\circ w_N = \varepsilon_N(f)\,\varphi_E + \varphi_E(0)$ (evaluate at ∞ to check the constant); Exercise 13.) Paper: Section 4.1 ("composed with … w_{M^Σ}").

9.3 `thm:moddeg:degree-is-eta` (theorem; P). Let E be optimal and ℓ any prime. On $M_{f,\lambda}$ the pairing $\hat\delta_!$ of 3.4 has

$$\hat\delta_!(\varphi_E^*a,\varphi_E^*b) = \varepsilon_N(f)\,\deg\varphi_E\,\langle a,b\rangle_E \qquad (a,b \in H^1(E,\mathbb{Z}_\ell)),$$

with $\langle\ ,\ \rangle_E$ unimodular (Chapter 8, its 10.3). Hence the ideal of values of $\hat\delta_!$ on $M_{f,\lambda}\times M_{f,\lambda}$ is $\deg\varphi_E\cdot\mathbb{Z}_\ell$, and the Gram determinant of $\hat\delta_!$ on a basis of $M_{f,\lambda}$ is $(\deg\varphi_E)^2$ up to a unit (a rank-two alternating form, Chapter 3, its 9.4). **Consequently, if DFG04's ideal $\eta^{\emptyset}_{f,\lambda}$ is the ideal of values of $\hat\delta_!$ on $M_{f,\lambda}$, as the paper's Section 4.1 describes it, then $v_\lambda(\eta^{\emptyset}_{f,\lambda}) = v_\ell(\deg\varphi_E)$ for every prime $\ell \nmid 2N$, with no multiplicity-one, irreducibility or Gorenstein hypothesis.** Proof: 9.1, 9.2, 6.3(iv). Physics aside: the projection formula is the statement that a loop on E lifts to $\deg\varphi$ loops whose intersection numbers add up. Paper: Lemma lem:hecke, Remark rem:free, Theorem thm:main ("with equality when … free"), important_notes.md point 39.

9.4 `cor:moddeg:ars-divisibility` (corollary; P given 9.3, 8.2 and Chapter 3, its 9.3, 11.1, 11.3). For E optimal and every prime ℓ: $v_\ell(r_E) \geq v_\ell(\deg\varphi_E)$; that is, $\deg\varphi_E \mid r_E$ (the divisibility half of 8.3). Proof (the assembly of Chapter 3, its 11.3, in the case at hand; Chapter 20 proves the general statement as Lemma lem:ddt417): take $\mathcal{O} = \mathbb{Z}_\ell$, the algebra $\mathbb{T}_\ell = \mathbb{T}\otimes\mathbb{Z}_\ell$ acting by t^* on $L = H^1(X_0(N),\mathbb{Z}_\ell)$ with the pairing $\hat\delta_!$ (perfect, $\mathbb{T}_\ell$ self-adjoint, 3.4), $\pi = \phi_f$, $\wp = I_f\mathbb{T}_\ell$, $I = \mathrm{Ann}(\wp)$, so that $\pi(I) = r_E\mathbb{Z}_\ell$ by 8.2 and $\wp\cap I = 0$ by 8.1(iii). Then $L[\wp] = M_{f,\lambda}$, $Q = L/(L[\wp]\oplus L[I])$ is killed by r_E and generated by 2 elements, and $Q \simeq \mathrm{coker}(M_{f,\lambda} \to \mathrm{Hom}(M_{f,\lambda},\mathbb{Z}_\ell))$ has order $\#\mathbb{Z}_\ell/(\deg\varphi_E)^2$ by 9.3; so $2v_\ell(\deg\varphi_E) \leq 2v_\ell(r_E)$. Equality holds when $L_\mathfrak{m}$ is free over $\mathbb{T}_\mathfrak{m}$ (Chapter 3, its 11.3(v)), for instance in the situation of 10.3. Paper: Section 1 ("By a theorem of Agashe, Ribet and Stein, $\deg\varphi_E$ divides the congruence number"), Lemma lem:hecke.

9.5 `thm:moddeg:lem-hecke` (theorem, the paper's Lemma lem:hecke in the strengthened form; P given 8.3 (R) and 9.3). Let E be optimal and ℓ an odd prime with $\ell \nmid N$. Then

$$v_\ell(\deg\varphi_E) = v_\ell(r_E) = v_\ell\big(\text{ideal of values of } \hat\delta_! \text{ on } M_{f,\lambda}\big),$$

the first equality by ARS12 Theorem 2.2 ($\ell^2 \nmid N$), the second by 9.3; and, with DFG04's normalisation (as the paper reads it; confirmed in the Chapter 20 outline, Part 7, important_notes.md point 73), both equal $v_\lambda(\eta^{\emptyset}_{f,\lambda})$. The hypotheses "$\bar\rho$ irreducible" and "free over $\mathbb{T}_\mathfrak{m}$" of the paper's Lemma lem:hecke are not needed for this. Consequences (stated, proved in Chapter 20 where the assembly lives): (a) the identity (eq:identity) of Theorem thm:main holds **with equality** whenever $\ell \notin S_f$; (b) Step 4 of the proof of Theorem thm:quant needs no freeness at level N_S (apply 9.3 to the optimal curve in the class of E_S, then ARS at N_S), which partly answers important_notes.md point 24; (c) the first inequality of Theorem thm:quant is an **equality** whenever $\ell \notin S_f$, with no multiplicity-one hypothesis: apply (a) to E and to E_S; the twist does not change $\mathrm{ad}^0 T$, so the Selmer groups agree, the Tamagawa factors agree, and the Euler terms differ by $\sum_{q\in S} e_\ell(E,q)$, so $v_\ell(\deg\varphi_E) = v_\ell(\deg\varphi_{E_S}) + \sum_{q\in S} e_\ell(E,q)$ (important_notes.md point 74, which supersedes the earlier reading that Remark (2) after Theorem thm:quant needs multiplicity one at level N; important_notes.md B26). Paper: Lemma lem:hecke, Remark rem:free, Theorem thm:main, Theorem thm:quant.

9.6 `rem:moddeg:dfg-normalisation` (remark; R: the five checks against DFG04 needed before the paper or the book asserts 9.5 for DFG04's η. **Settled:** the Chapter 20 outline, its Part 7, read the published DFG04 and confirmed all five, item (1) in the form "agrees with $H^1(X_0(N),\mathbb{Z}_\ell)$ at non-Eisenstein maximal ideals", which is all the paper uses; important_notes.md point 73; the writer cites the Chapter 20 outline's Part 7 for the DFG04 numbers). 9.3 is a theorem about the ideal of values of $\langle x, w_N^* y\rangle$ on $H^1(X_0(N),\mathbb{Z}_\ell)[I_f]$. That this is DFG04's $\eta^{\emptyset}_{f,\lambda}$ needs five checks in DFG04 Sections 1.5.3, 1.6.1, 1.6.2, 1.7.3 and (13), (14): (1) **Which curve.** That DFG04's integral structure $M(N,1)_{!,\lambda}$ for weight 2 and trivial character is $H^1(X_0(N),\mathbb{Z}_\ell)$ (as the paper says), and not the trivial-character part of the cohomology of a $\Gamma_1(N)$-type curve: if it were $H^1(X_1(N),\mathbb{Z}_\ell)^{(\mathbb{Z}/N)^\times}$, the pullback along $X_1(N)\to X_0(N)$ would multiply the cup product by the degree $\#(\mathbb{Z}/N)^\times/\{\pm1\}$, which can be divisible by ℓ (for example $\ell=3$, $N=7\cdot m$), and its saturation would involve the Shimura subgroup (11.3). (2) **Interior cohomology.** That "!" means the image of compactly supported cohomology of $Y_0(N)$ in cohomology, which for constant coefficients is $H^1(X_0(N))$ (an exact-sequence argument: $H^1_c(Y)\to H^1(X)$ is onto and $H^1(X)\to H^1(Y)$ is injective), and that group cohomology of $\Gamma_0(N)$ agrees with it for constant coefficients ($\mathrm{Hom}(\Gamma_0(N),\mathbb{Z}_\ell) = \mathrm{Hom}(\pi_1(Y_0(N)),\mathbb{Z}_\ell)$, the elliptic elements being torsion; P). (3) **The operator.** That $[UwU]\omega$ is w_N^* with no scalar factor in weight 2 (Chapter 12, C3, C4 suggest so) and that DFG04's cup product lands in $H^2_c \simeq \mathbb{Z}_\ell$ with the standard normalisation (a factor 2 from ±1 is harmless for odd ℓ; at $\ell=3$ check the treatment of elliptic points of order 3, which exist when every prime factor of N is 1 mod 3 and $9\nmid N$). (4) **Kernel, not projection.** That $M_{f,\lambda}$ is the kernel lattice $L[\wp]$ (saturated), as the paper says, and not the image of L under the projection to the f-component (the dual lattice, which would change the ideal of values by the congruence ideal). (5) **Values versus Gram.** That η is defined as the ideal of values (equivalently the square root of the Gram determinant), as the paper says, and that DFG04 Theorem 3.7 and Proposition 1.4(c) are stated for this ideal. If all five hold, 9.5 applies to DFG04's η verbatim; if (3) or (5) differ by a unit nothing changes; if (1) or (4) differ, the paper's own description of η in Section 4.1 needs correcting, independently of point 39. Confidence of the book: high for 9.3, medium for the identification. Paper: Section 4.1, Lemma lem:hecke, Theorem thm:dfg14, Theorem thm:dfg37.

9.7 `ex:moddeg:99-at-3` (worked example; P given 8.5, 9.3, 9.4 and Chapter 3, its 11.5). For 99a1 and $\ell=3$ (where $3^2 \mid 99$ and the paper's theorems do not apply): the ideal of values of $\hat\delta_!$ on $M_{f,3}$ is $4\mathbb{Z}_3 = \mathbb{Z}_3$ (9.3), while $v_3(r_E)=1$ (8.5). So the inequality of 9.4 is strict, $L_\mathfrak{m} = H^1(X_0(99),\mathbb{Z}_3)_\mathfrak{m}$ is **not** free over $\mathbb{T}_\mathfrak{m}$ at the maximal ideal $\mathfrak{m}$ of 99a1 mod 3, and by Chapter 3, its 11.5 (hypothesis (b) holds, rank $L_\mathfrak{m} = 2\,\mathrm{rank}\,\mathbb{T}_\mathfrak{m}$), $\dim_{\mathbb{T}/\mathfrak{m}} L/\mathfrak{m}L > 2$: multiplicity one fails for $J_0(99)$ at $\mathfrak{m}$, although $\bar\rho_{99\mathrm{a}1,3}$ is irreducible (the class 99a has only a 2-isogeny). This is why Remark rem:free needs $\ell\nmid N$. Paper: the remark after Theorem thm:qual (99A1), Remark rem:free.

16.2.10 Section 10. Multiplicity one, the Gorenstein property and Remark rem:free

10.1 def:moddeg:maximal-ideal (definition; P for (i), R for (ii)). (i) For $f = f_E$ and a prime ℓ, $\mathfrak{m} = \mathfrak{m}_{f,\ell}$ is the maximal ideal of $\mathbb{T}_\ell = \mathbb{T} \otimes \mathbb{Z}_\ell$ containing ℓ and I_f, the kernel of $\mathbb{T}_\ell \to \mathbb{Z}_\ell \to \mathbb{F}_\ell$; $\mathbb{T}_\ell = \prod_{\mathfrak{m}'} \mathbb{T}_{\mathfrak{m}'}$ (Chapter 3, its 10.3). (ii) There is a semisimple representation $\bar\rho_{\mathfrak{m}} \colon G_{\mathbb{Q}} \to \mathrm{GL}_2(\mathbb{T}/\mathfrak{m})$, unramified outside $N\ell$, with $\operatorname{tr} \bar\rho_{\mathfrak{m}}(\mathrm{Frob}_p) = T_p \bmod \mathfrak{m}$ and $\det = \bar\chi_\ell$ (Chapter 15); for $\mathfrak{m}_{f,\ell}$ it is $\bar\rho^{\mathrm{ss}}_{E,\ell}$ (5.2, Chebotarev and Brauer-Nesbitt, Chapters 4, 5). $\mathfrak{m}$ is Eisenstein if $\bar\rho_{\mathfrak{m}}$ is reducible (C7). Paper: Lemma lem:hecke ("$\mathfrak{m}$ the maximal ideal of $\mathbb{T} \otimes \mathbb{Z}_\ell$ containing ℓ and $\ker \phi_f$"), Remark rem:free.

10.2 thm:moddeg:multiplicity-one (theorem; R, with the reduction to $J_1(N)$ sketched as in the paper). Let $\ell \nmid 2N$ and $\mathfrak{m}$ non-Eisenstein. Then $J_0(N)[\mathfrak{m}]$ is 2-dimensional over $\mathbb{T}/\mathfrak{m}$ (hence $\simeq \bar\rho_{\mathfrak{m}}$ by Boston-Lenstra-Ribet). References: Mazur77 (prime level; Chapter II, cite the section); Mazur's argument via the q-expansion principle mod ℓ and the good reduction of $J_0(N)$ at ℓ, as written in Ribet90 Section 5 (verify the theorem number); the paper's route: RibetStein Theorem 3.5 for $J_1(N)$ (as the paper cites it; check the exceptional case "Serre weight ℓ" and that it cannot occur for weight 2 and $\ell \nmid N$: $\det \bar\rho = \bar\chi_\ell$ is ramified at ℓ), and the pullback $J_0(N)[\mathfrak{m}] \to J_1(N)[\mathfrak{m}']$, injective because its kernel lies in the Shimura subgroup (11.3), on which $\mathbb{T}$ acts through Eisenstein ideals. Paper: Remark rem:free.

10.3 thm:moddeg:gorenstein (theorem; P given 10.2, 4.3 and Chapter 3, its 11.4, 11.5). In the situation of 10.2: $\mathbb{T}_{\mathfrak{m}}$ is Gorenstein, and $H^1(X_0(N), \mathbb{Z}_\ell)_{\mathfrak{m}} \simeq \mathbb{T}^2_{\mathfrak{m}} \simeq T_\ell J_0(N)_{\mathfrak{m}}$ as $\mathbb{T}_{\mathfrak{m}}$-modules (Tilouine97; Wiles95 Chapter 2; DDT Section 4, cite the sections). Proof: $L = H^1(X_0(N), \mathbb{Z}_\ell)_{\mathfrak{m}}$ with $\hat\delta_!$ (perfect on the summand, 3.4(iii)) satisfies (a) $\dim L/\mathfrak{m}L = \dim J_0(N)[\mathfrak{m}] = 2$ (the pairing identifies $L/\mathfrak{m}L$ with the dual of $(L/\ell L)[\mathfrak{m}]$, and $L/\ell L$ with $J_0(N)[\ell]_{\mathfrak{m}}$ up to the Hecke-equivariant identification 4.3(ii)) and (b) $\operatorname{rank}_{\mathbb{Z}_\ell} L = 2 \operatorname{rank}_{\mathbb{Z}_\ell} \mathbb{T}_{\mathfrak{m}}$ (4.4(i) for every eigensystem, localised); Chapter 3, its 11.5, applies. Consequence: equality in 9.4 at such ℓ, that is $v_\ell(r_E) = v_\ell(\deg \varphi_E)$ for $\ell \nmid 2N$ with $E[\ell]$ irreducible, an independent proof of this case of 8.3. Paper: Remark rem:free ("the localised Hecke algebra is then Gorenstein and the Tate module is free of rank two \cite{Tilouine97}"), Lemma lem:hecke (equality clause).

10.4 rem:moddeg:rem-free-after-39 (remark; P given 9.5). What Remark rem:free is still needed for, once 9.5 is known: (a) not for the equality in Lemma lem:hecke or in (eq:identity) (9.5); (b) not for the equality in Theorem thm:quant either (9.5(c), important_notes.md point 74); it is still the natural tool for comparing the Hecke congruence ideal and the cohomological one directly for forms not attached to an optimal quotient, such as the depleted form g^S (Step 2 of the paper's proof is an inequality without it), but the equality of Theorem thm:quant no longer goes through that comparison; (c) as the conceptual reason for the equality in ARS12 at $\ell \nmid 2N$ (10.3). It fails for $\ell \mid N$ in general (9.7: $N = 99$, $\ell = 3$). Paper: Remark rem:free, Theorem thm:quant and the remark after it, important_notes.md points 24, 39.

16.2.11 Section 11. The Eisenstein ideal, the cuspidal subgroup and the Shimura subgroup

11.1 def:moddeg:cuspidal-subgroup (definition and theorem; P for (ii), (iii), R for (i) in general). (i) The *cuspidal subgroup* $C_N \subseteq J_0(N)(\bar{\mathbb{Q}})$ is generated by the classes of the degree-zero divisors supported on the cusps; it is finite (Manin-Drinfeld) and $G_{\mathbb{Q}}$-stable (2.2); its $G_{\mathbb{Q}}$-invariants contain the rational cuspidal subgroup generated by the classes of $\mathbb{Q}$-rational cuspidal divisors. (ii) For $p \nmid N$: $T_p(\infty) = (p+1)(\infty)$ and $T_p(0) = (p+1)(0)$ as divisors (the $p+1$ matrices α_i of 3.2(i) fix ∞; they send 0 to 0 and to j/p, which are $\Gamma_0(N)$-equivalent to 0 since $\gcd(p, N) = 1$). (iii) Hence $(T_p - 1 - p)_*$ kills $x = [(0) - (\infty)]$; $T_p - 1 - p$ is an isogeny of $J_0(N)$, its eigenvalues $a_p(g) - 1 - p$ on S_2 being nonzero by the Ramanujan-Petersson bound $|a_p(g)| \le 2\sqrt{p} < p + 1$ (Chapter 12, its 6.5, R); so x is torsion. This is Manin-Drinfeld for the two rational cusps, in full. Reference: Manin (1972) and Drinfeld (1973) for (i), cited through DiamondShurman or Stein (book) by section (verify where it is stated); Ogg73. Paper: Conjecture conj:eis, Section sec:eis, Open question 4 ("presumably through the cuspidal subgroup of $J_0(N)$").

11.2 thm:moddeg:mazur-prime-level (theorem; R, with the example P). (Mazur77.) Let N be prime and $n = \operatorname{num} \frac{N-1}{12}$. (i) The class of $(0) - (\infty)$ has order n and generates C_N. (ii) $J_0(N)(\mathbb{Q})_{\mathrm{tors}} = C_N \simeq \mathbb{Z}/n$. (iii) The *Eisenstein ideal* $I_{\mathrm{Eis}} \subseteq \mathbb{T}$ generated by $T_\ell - 1 - \ell$ ($\ell \neq N$) and $1 + w_N$ satisfies $\mathbb{T}/I_{\mathrm{Eis}} \simeq \mathbb{Z}/n$; it annihilates C_N and the Shimura subgroup $\mathcal{S}_N$ (11.3), which is μ-type of order n and Cartier dual to C_N.

(iv) For ℓ odd, a maximal ideal $\mathfrak{m} \ni \ell$ is Eisenstein in the sense of C7 iff $\mathfrak{m} \supseteq I_{\mathrm{Eis}}$, and such $\mathfrak{m}$ exist iff $\ell \mid n$ (Mazur77, Chapter II; verify the exact statement and the cases $\ell = 2$, $\ell = N$). Example (P, checked): $N = 11$, $n = 5$, $J_0(11)(\mathbb{Q}) = 11\mathrm{a}1(\mathbb{Q}) \simeq \mathbb{Z}/5$, $\ell = 5$ is Eisenstein, and $f_{11} \equiv$ the Eisenstein series $-\frac{1}{24}E_2^{(11)} \pmod 5$ (Chapter 12, its 8.4); $N = 37$, $n = 3$, and 37b1 has torsion $\mathbb{Z}/3$. Reference: Mazur77 (Chapter II; cite the sections; "Theorem 1" of the introduction for (ii), verify). Paper: Conjecture conj:eis, Section sec:eis, Open question 4, Section 6 table of counts (the Eisenstein column).

11.3 `def:moddeg:shimura-subgroup` (definition and proposition; R for (ii), (iii), P for (i)). (i) The *Shimura subgroup* $\mathcal{S}_N = \ker(\pi^*\colon J_0(N) \to J_1(N))$, $\pi\colon X_1(N) \to X_0(N)$ the forgetful map; by Chapter 8, its 11.5(iii) (with $(d) \Leftrightarrow (a)$), $\mathcal{S}_N = 0$ iff $\pi^* H^1(X_0(N), \mathbb{Z})$ is saturated in $H^1(X_1(N), \mathbb{Z})$. (ii) $\mathcal{S}_N$ is finite, of multiplicative type (Cartier dual to a constant group), and T_ℓ acts on it by $1 + \ell$ for $\ell \nmid N$: it is Eisenstein (Mazur77 for prime N; Ling-Oesterle and Ribet88 in general, cite the sections). (iii) So $\mathcal{S}_N[\mathfrak{m}] = 0$ for non-Eisenstein $\mathfrak{m}$, which is the injectivity used in 10.2. (iv) For the optimal curve, $\bar\rho_{E,\ell}$ reducible (a rational ℓ-isogeny, Chapter 11, its 9.1) iff $\mathfrak{m}_{f,\ell}$ is Eisenstein; then 10.2, 10.3, DFG04 and Lemma lem:hecke's hypothesis fail, and the modular degree can lose local divisibility (the deficit D of Conjecture conj:eis; Chapter 11, its 9.3). Paper: Remark rem:free ("the Shimura subgroup, is Eisenstein"), Conjecture conj:eis, Section sec:eis.

11.4 `prop:moddeg:image-of-zero` (proposition; P). Let E be optimal with Manin constant c_E. Then $\varphi_E(0) \in E(\mathbb{Q})_{\mathrm{tors}}$ (the cusp 0 is rational, 2.2, and φ_{E*} maps the torsion class $[(0) - (\infty)]$ of 11.1 to $\varphi_E(0)$), and under $E(\mathbb{C}) = \mathbb{C}/\Lambda_E$,

$$\varphi_E(0) \equiv \pm c_E\, L(E,1) \pmod{\Lambda_E},$$

because $\varphi_E(z) = \int_{i\infty}^{z} c_E \cdot 2\pi i f(\tau)\, d\tau$ and $\int_0^{i\infty} 2\pi i f(\tau)\, d\tau = -2\pi \int_0^\infty f(iy)\, dy = -L(f,1)$ (Chapter 19; the sign depends on the orientation conventions, state one). Since $\Lambda_E \cap \mathbb{R} = \mathbb{Z}\Omega_E$ with Ω_E the least positive real period, the order of $\varphi_E(0)$ is the denominator of $L(E,1)/\Omega_E$ when $c_E = 1$. Checks (P, PARI): $L/\Omega = 1/5$ (11a1, torsion $\mathbb{Z}/5$), $1/6$ (14a1, $\mathbb{Z}/6$), $1/6$ (20a1, $\mathbb{Z}/6$), $1/3$ (19a1, $\mathbb{Z}/3$), $2/3$ (37b1, $\mathbb{Z}/3$), 1 (121a1, trivial torsion), 0 for the rank-one curves 37a1, 99a1, 539c1, 726a1 (so $\varphi_E(0) = O$). Paper: Conjecture conj:eis and Open question 4 (a heuristic link between rational torsion, the cuspidal subgroup and the Eisenstein deficit; the book says it is a heuristic, not an argument).

11.5 `rem:moddeg:eisenstein-and-the-paper` (remark; R, pointers). Why the Eisenstein case is outside every theorem of the paper: Ihara's lemma and Proposition 1.4 of DFG04, multiplicity one (10.2), $\ell \notin S_f$, and the identification of $T_\ell E$ with $M_{f,\lambda}$ up to homothety all use $\bar\rho$ irreducible (9.1 does not, but its use in DFG04 does). The paper's Conjecture conj:eis bounds the deficit by torsion, and Open question 4 asks for an Eisenstein-ideal interpretation; important_notes.md point 17 (Chapter 9, its 9.8) proves the bound $D \le 2$ from the first bound. Paper: Conjecture conj:eis, Section sec:eis, Open question 4.

16.2.12 Section 12. Shimura curves, the prime 2 and the computation of the modular degree

12.1 `def:moddeg:shimura-curve` (definition; R). For D a product of an even number of distinct primes and M prime to D, $X_0^D(M)$ is the Shimura curve attached to an Eichler order of level M in the indefinite quaternion algebra over $\mathbb{Q}$ of discriminant D; it is a smooth projective curve over $\mathbb{Q}$ without cusps. By Jacquet-Langlands (Ribet's form) its Jacobian is isogenous to the D-new quotient of $J_0(DM)$, Hecke-equivariantly. For E of conductor $N = DM$ with $D \,\|\, N$ (so E is multiplicative at every $q \mid D$) there is an optimal parametrisation $\varphi_{E,D}\colon X_0^D(M) \to E$ (defined via the Jacobian, a base point being chosen; its degree does not depend on the choice), of degree δ_D. Reference: Ribet90 (Shimura curves and level lowering; cite the section), RibetTakahashi97, Takahashi01. Paper: Section 1.5 ("compared $\deg \varphi_E$ with the degree of a parametrisation by a Shimura curve").

12.2 `thm:moddeg:ribet-takahashi` (theorem; R; check every hypothesis against the sources before printing). (Ribet and Takahashi; Takahashi.) Let E be optimal and semistable of conductor $N = DM$ as in 12.1. Then for every prime ℓ such that $E[\ell]$ is irreducible (and under the further hypotheses of RibetTakahashi97 and Takahashi01, to be listed exactly)

$$v_\ell(\deg \varphi_E) = v_\ell(\delta_D) + \sum_{q \mid D} v_\ell(n_q), \qquad n_q = v_q(\Delta_E) = \#\Phi_q(\bar{\mathbb{F}}_q),$$

the order of the geometric component group at q (the source must be checked on whether the arithmetic c_q or the geometric n_q appears; they differ at nonsplit q, and the paper's Section 1.5 says n_q). Pasten controls the error term at the reducible primes. Proof idea (one paragraph): Ribet's exact sequence relating the character groups of the tori of $J_0(N)$ at q and of $J_0^D(M)$ at another prime, and the component groups as cokernels of the monodromy pairing. Consequence: the multiplicative part $v_\ell(\deg \varphi_E) \geq \sum_{q|D} t_\ell(E,q)$ of (eq:main) for the primes of D. Paper: Section 1.5 (history), Remark rem:ko (Kim-Ota: the same congruence ideal in another guise; Chapter 15).

12.3 `conj:moddeg:watkins` (conjecture and data; R, with the data P). (Watkins02.) For every optimal $E/\mathbb{Q}$, $2^{\operatorname{rank} E(\mathbb{Q})} \mid \deg \varphi_E$. Data (Part 6.6): all 38,042 optimal curves of conductor below 10,000 satisfy it; examples 37a1 (rank 1, deg = 2), 99a1 (1, 4), 539c1 (1, 288), 726a1 (1, 144). Known results (Dummigan06, CalegariEmerton09, Yazdani11, CaroPasten22, HatleyKundu, BKP as the paper cites them) are listed without statements unless checked. Paper: Section sec:two, Open question 5.

12.4 `thm:moddeg:calegari-emerton` (theorem; R; the statement to be copied from CalegariEmerton09 with its exact hypotheses). The paper's paraphrase: a curve of odd modular degree has at most two odd primes of bad reduction and either a rational point of order 2 or prime conductor. Data (Part 6.6): 103 optimal curves of conductor below 10,000 have odd modular degree; all have at most two odd bad primes; all but two have a rational 2-torsion point or prime conductor; the two exceptions are 27a1 (deg = 1, torsion $\mathbb{Z}/3$) and 243b1 (deg = 9, torsion $\mathbb{Z}/3$), both with $j = 0$ (CM by $\mathbb{Z}[\zeta_3]$). So the paraphrase needs the hypothesis or exception that CalegariEmerton09 state (presumably "without CM", or prime-power conductor allowed); Part 7, point N1. Paper: Section sec:two.

12.5 `rem:moddeg:computing-the-degree` (remark; R). Three methods: (a) Zagier: compute φ_E on H_1 from modular symbols and read the degree off the index of 6.3(iv) (Zagier85; Cremona97 Chapter II); (b) Watkins: $\langle f, f \rangle$ from the symmetric square special value with the bad Euler factors, then 7.3 (Watkins02); (c) Cremona's period-lattice comparison used to certify optimality (6.5). PARI's `ellmoddegree` implements (b) and needs `parisizemax` of a few hundred megabytes in the thousands (the paper's appendix); Chapter 22 has the sessions. Paper: Section 1 ("The modular degree is computable"), Section 6, the appendix.

16.3 3. Section-by-section plan with target line counts

Line counts are lines of LaTeX, card excluded (PLAN.md Section 7.3: 1,000 to 2,200). The card lists the 56 items of Part 2 (numbered 14.1 to 14.56 in the book; conventions C1 to C8 in one line each), about 170 lines, not counted.

section	title	contents	target lines
(starred)	Core results at a glance	the card: every item of Part 2 with its tag and `\Cref`; the formula of 7.3 and the identity of 9.3 displayed	(excluded)

section	title	contents	target lines
(starred)	Where the paper uses this chapter	Section 1 (optimal curve, φ_E, $\deg \varphi_E$, Zagier-Watkins, ARS, r_E, abc); Section 1.5 (Ribet-Takahashi); Section 3 preamble, Lemma lem:crit (stated), the remark on 99A1; Section 4.1 ($H^1(X_0(M^\Sigma), \mathbb{Z}_\ell)$, $\hat{\delta}_!$, $M_{f,\lambda}$); Lemma lem:hecke and Remark rem:free (with the strengthening 9.5 and the checklist 9.6); Section 4.2 ($M_{f,\lambda}$ versus $T_\ell E$); Theorem thm:main (equality), Theorem thm:quant Steps 2, 4; Section 6 (optimality, `ellmoddegree`); Conjecture conj:eis, Section sec:eis, Open question 4; Section sec:two, Open question 5	50
1	Overview and conventions	the opening paragraph (the six uses, Part 1.1); C1 to C8 as `convention` environments; 1.1; a roadmap table "paper sentence $\to$ item"	80
2	The modular curve $X_0(N)$ over $\mathbb{Q}$	2.1 to 2.4; the cusp table for $N = 20, 99, 121$; `paperlink`: Section 1 (φ_E over $\mathbb{Q}$), Conjecture conj:eis (rational isogenies, point 17)	140
3	Hecke correspondences and adjoints	3.1 to 3.5; the matrix computation $W_N \operatorname{diag}(1,p) W_N^{-1} = \operatorname{diag}(p,1)$ displayed; `paperlink`: Section 4.1 (the sentence on $w_{M^\Sigma} T_n w_{M^\Sigma}^{-1}$), point 36	170

section	title	contents	target lines
4	The Jacobian, its homology, the f-lattice	4.1 to 4.4; the dimension table; a `warning` box "Albanese versus Picard, H_1 versus H^1, PD versus $w_N \circ$PD" (points 26, 41)	160
5	Eichler-Shimura and modularity	5.1 to 5.5; the class 11a in full; one page of history of modularity; `paperlink`: Section 1 ("the newform attached to it")	150
6	Optimal curves and the modular degree	6.1 to 6.6; the proof of 6.2(ii) to (iv) in full, (i) sketched; the running table; `paperlink`: Section 1 ("minimal degree"), Section 6 (optimality, point 25), point 27	190
7	The Manin constant and the Zagier-Watkins formula	7.1 to 7.5; the three-line proof of 7.3; the numerical table; `warning`: where c_φ^2 stands (B10) and which Petersson normalisation	120
8	The congruence number and Agashe-Ribet-Stein	8.1 to 8.5; the computation of r_E explained as an algorithm; the table; `paperlink`: the remark after Theorem thm:qual (99A1)	140
9	The modular degree as a congruence ideal	9.1 to 9.7; all proofs in full; a `paperlink` quoting Lemma lem:hecke and stating the strengthened form; the checklist 9.6 as a `remark` with the five items numbered; the 99 at 3 example	200
10	Multiplicity one and the Gorenstein property	10.1 to 10.4; `paperlink`: Remark rem:free, line by line, with what point 39 changes	130

section	title	contents	target lines
11	Eisenstein ideal, cuspidal and Shimura subgroups	11.1 to 11.5; the computation $T_p(0) = (p+1)(0)$; the table of $L(E,1)/\Omega_E$ against torsion; `paperlink`: Conjecture conj:eis, Section sec:eis, Open question 4	140
12	Shimura curves, the prime 2, computing $\deg \varphi_E$	12.1 to 12.5; `paperlink`: Section 1.5, Section sec:two (with point N1 of Part 7), the appendix (`ellmoddegree`)	130
13	Exercises	the 17 exercises of Part 4.2	140
14	Notes and references	Shimura71, DiamondShurman, DiamondIm, Cremona97, Zagier85, Watkins02, ARS12, Mazur77, Mazur78, Ribet90, RibetStein, Tilouine97, Wiles95, TaylorWiles, BCDT, Carayol86, LingOesterle, Ribet88, RibetTakahashi97, Takahashi01, Pasten, CalegariEmerton09, AU96, the proposed keys of Part 6.5; which numbers were cited and which are marked (verify)	45
	total		**1,985**

16.4 4. Coverage table and exercises

16.4.1 4.1 Coverage: PLAN.md scope, Section 6 map rows and paper hooks to line-up items

PLAN.md scope item, Section 6 row, or paper hook	line-up items
$X_0(N)$ as a smooth projective curve over $\mathbb{Q}$ (statement)	2.1
its cusps	2.2, 11.1(ii)
its points over $\mathbb{Q}$	2.3, 2.4

PLAN.md scope item, Section 6 row, or paper hook	line-up items
$J_0(N)$, abelian variety of dimension g, $J_0(N)(\mathbb{C}) = S_2^\vee / H_1(X_0(N), \mathbb{Z})$	4.1
the Hecke action on $J_0(N)$ by correspondences, $T_p(E, C) = \sum(E/D, (C+D)/D)$	3.1, 3.2, 4.2
Eichler-Shimura relation $T_p = \mathrm{Frob}_p + \mathrm{Frob}_p^*$ mod p, $a_p(f)$ = trace of Frobenius	5.2
the Hecke action on $H^1(X_0(N), \mathbb{Z})$ and H_1	3.2, 4.2, 4.3
the cup product on $H^1(X_0(N), \mathbb{Z})$ (perfect, alternating)	3.4 (with Chapter 8, its 10.3)
adjoint of T_n under the cup product is $w_N T_n w_N$; $\langle x, w_N y \rangle$ perfect, alternating, all T_n, U_p	3.3, 3.4, 9.6
self-adjoint; $= \hat{\delta}_!$ of DFG04	
$T_\ell J_0(N) = H_1 \otimes \mathbb{Z}_\ell$ and $H^1(X_0(N), \mathbb{Z}_\ell)$ dual; etale cohomology (statement)	4.3
the modularity theorem (statement, history)	5.3, 5.4
the abelian variety $J_f = J_0(N)/I_f J_0(N)$; for rational f the elliptic curve quotient	5.1, 5.5
the modular parametrisation φ_E ($\infty \mapsto O$)	6.1
the $X_0(N)$-optimal curve: unique, connected kernel, strong Weil curve; Cremona label 1, checked to 400000	6.2, 6.5
the modular degree: degree of the morphism; $E^\vee \to J_0(N) \to E$ is $[\deg \varphi_E]$; computability	6.3, 6.6, 12.5
the Manin constant: definition, the conjecture $c_E = 1$, known results, role in the formulas	7.1, 7.2, 7.4
the Zagier-Watkins formula $\deg \varphi_E = 4\pi^2 c_E^2 \langle f, f \rangle / \mathrm{vol}(\Lambda_E)$ with $\varphi_E^* \omega_E = c_E \, 2\pi i f \, dz$	7.3, 7.4, 7.5
the congruence number r_E and the congruence ideal; Lemma lem:crit (proved in Chapter 15)	8.1, 8.2
Agashe-Ribet-Stein: $\deg \varphi_E \mid r_E$, equality for $p^2 \nmid N$; 99A1 with deg = 4, $r = 12$; proof idea	8.3, 8.5, 9.4, 10.3
multiplicity one, $\mathfrak{m}$, $J_0(N)[\mathfrak{m}]$ two-dimensional (Mazur, Ribet-Stein 3.5, Shimura subgroup)	10.1, 10.2, 11.3
Gorenstein (Tilouine), $H^1(X_0(N), \mathbb{Z}_\ell)_\mathfrak{m}$ free of rank two; Remark rem:free	10.3, 10.4
the Eisenstein ideal, the cuspidal subgroup, the Shimura subgroup; Mazur at prime level; why reducible is Eisenstein	11.1, 11.2, 11.3, 11.5
Shimura curves and Ribet-Takahashi	12.1, 12.2
the abc connection ($\deg \varphi_E \ll N^{2+\varepsilon}$; ARS12 Section 2)	8.4
Watkins's conjecture; Calegari-Emerton	12.3, 12.4
Mazur's theorem on $J_0(N)(\mathbb{Q})_{\mathrm{tors}}$ for prime N	11.2
computing the modular degree (Watkins, Cremona, `ellmoddegree`)	7.5, 12.5
Section 6 row: optimal curve, modular parametrisation, modular degree, Manin constant	6.1 to 6.6, 7.1 to 7.4
Section 6 row: congruence ideal, r_E, η^Σ, $M_{f,\lambda}$, $\hat{\delta}_!$ (with 13, 20)	3.4, 4.4, 8.1, 9.1 to 9.6
Section 6 row: Agashe-Ribet-Stein	8.3, 8.5, 9.4

PLAN.md scope item, Section 6 row, or paper hook	line-up items
Section 6 row: multiplicity one, freeness, Gorenstein, $J_0(N)[\mathfrak{m}]$ (with 3, 20)	10.1 to 10.4, 9.7
Section 6 row: Eisenstein case, torsion, cuspidal subgroup, Eisenstein ideal (with 9, 11)	11.1 to 11.5, 2.3
Section 6 row: Petersson norm, Zagier, Watkins (with 12, 13, 19)	7.3 to 7.5
Section 6 row: $H^1(X_0(N),\mathbb{Z}_\ell)$, cup product, Atkin-Lehner, Hecke self-adjointness (with 8)	3.3, 3.4, 4.3, 4.4
Section 6 row: Ribet-Takahashi, Pollack-Weston, Kim-Ota, Tamagawa exponent (with 15, 20)	12.2 (Pollack-Weston, Kim-Ota: Chapters 15, 20)
Section 6 row: Watkins's conjecture, Calegari-Emerton	12.3, 12.4
Section 6 row: the worked examples 11a1, 20a1, 99A1 (with 9, 15, 21, 22)	2.4, 5.5, 6.6, 7.4, 8.5, 9.7, 11.4
hook: Section 1 (optimal curve, modular degree, Petersson norm, Zagier, Watkins, ARS, congruence number)	6.2, 6.3, 7.3, 8.1, 8.3
hook: Lemma lem:crit (congruence number versus congruence ideal)	8.1, 8.2
hook: Lemma lem:hecke	9.1 to 9.5
hook: Remark rem:free	10.2, 10.3, 10.4
hook: Section 4.1 ($H^1(X_0(N),\mathbb{Z}_\ell)$, $\hat{\delta}_!$, Atkin-Lehner)	3.3, 3.4, 4.4, 9.6
hook: the remark after Theorem thm:qual (99A1)	8.5, 9.7
hook: Section 1.5 (Ribet-Takahashi)	12.1, 12.2
hook: Section sec:two (Watkins, Calegari-Emerton)	12.3, 12.4
hook: Conjecture conj:eis and Section sec:eis (Eisenstein ideal, cuspidal subgroup, Mazur)	11.1 to 11.5
hook: Open question 4	11.2, 11.4, 11.5
also: Section 4.2 ($M_{f,\lambda}$ versus $T_\ell E$; points 26, 40)	9.1(d)
also: Theorem thm:main (equality), Theorem thm:quant Steps 2, 4 and Remark (2) (point 39, point 24)	9.5, 10.4
also: what Chapters 8, 9, 11, 12 defer to Chapter 14 (Part 5)	2.1 to 2.4, 3.1 to 3.4, 4.1 to 4.4, 5.1 to 5.5, 6.1 to 6.5, 7.1, 7.3, 9.1, 11.1 to 11.3

16.4.2 4.2 Exercises (Section 13 of the chapter), graded from easy to harder

1. List the cusps of $X_0(20)$, $X_0(99)$ and $X_0(121)$ with their denominators and fields of definition (2.2); check the counts 6, 8, 12 against Chapter 12, its 4.2, and decide which are $\mathbb{Q}$-rational.
2. Compute $W_N \operatorname{diag}(1,p)\, W_N^{-1}$ and deduce $U_p^t = w_N U_p w_N$ (3.3). On the span of $f_{11}(z)$, $f_{11}(2z)$ in $S_2(\Gamma_0(22))$ write the matrix of U_2 (from $a_2(f_{11}) = -2$) and of w_{22} (Chapter 12, its 12.4(b)); check that U_2 is not self-adjoint for the Petersson product but that $w_{22}U_2w_{22}$ is its adjoint.
3. Prove 3.4(ii), (iii) in detail: $\hat{\delta}_!$ is alternating, every t^* is self-adjoint, idempotents are self-adjoint and the $\mathfrak{m}$-decomposition is orthogonal (important_notes.md point 36).
4. Using Eichler-Shimura (5.2) and $X_0(11) = $ 11a1, prove $\#11a1(\mathbb{F}_p) = p + 1 - a_p(f_{11})$ for $p \neq 11$, and check it for $p = 2, 3, 5, 7, 13$ from the η-product (Chapter 12, its 9.2).
5. Show that $X_0(37) \to X_0(37)/w_{37}$ is a modular parametrisation of 37a1 of degree 2, that 37a1 is optimal, and deduce $\deg \varphi_{37a1} = 2$ without `ecdata` (2.4(c), 6.2).
6. Check the paper's claim that the optimal curve is the curve of minimal modular degree in its class on the class 11a: using 6.2(ii) and the isogeny degrees $5, 5, 25$, list the minimal degrees of the

parametrisations of 11a1, 11a2, 11a3, and confirm them with the volumes of 7.4.

7. (Manin constant.) From $\mathrm{vol}(\Lambda_{11a3}) = 5\,\mathrm{vol}(\Lambda_{11a1})$ and 7.3 show that the parametrisation φ of 11a3 has $c_\varphi = 5$, and that $c_\varphi \in \mathbb{Z}$ in this case directly from the 5-isogeny 11a1 $\to$ 11a3 ($\psi^*\omega = 5\omega$).
8. Check the paper's claim that $\deg \varphi_{121a1} = 6$ with the numbers of 7.4 (or recompute $\langle f, f\rangle$ with PARI's `mfpetersson`, remembering the factor $\mu(121) = 132$).
9. Check the paper's claim (the remark after Theorem thm:qual, quoting ARS12 Table 1) that 99A1 has modular degree 4 and congruence number 12: (a) show $f_{99a} \equiv f_{99b} \pmod 3$ for $n \le 24$ and conclude by Sturm (Chapter 12, its 11.3) and Lemma lem:crit that $3 \mid r_E$; (b) show $4 \mid r_E$ from 9.4; (c) explain why this does not contradict ARS12 Theorem 2.2.
10. In the situation of Exercise 9 prove that $H^1(X_0(99), \mathbb{Z}_3)_{\mathfrak{m}}$ is not free over $\mathbb{T}_{\mathfrak{m}}$ and that $J_0(99)[\mathfrak{m}]$ has dimension at least 3 over $\mathbb{T}/\mathfrak{m}$ (9.7, Chapter 3, its 11.5). Which hypothesis of Remark rem:free fails?
11. Check the paper's 726a1 against ARS12: $\deg \varphi_E = 144 = 2^4 \cdot 3^2$ and $r_E = 1584$. Which prime carries the discrepancy, and why is this allowed? (Describe the algorithm of 8.5; running it is optional.)
12. Prove 6.4: if $E[\ell]$ is irreducible, v_ℓ of the minimal parametrisation degree is the same for every curve of the class (important_notes.md point 27); give the counterexample at $\ell = 5$ in the class 11a.
13. Prove $\varphi_E(0) \equiv \pm c_E L(E,1) \pmod{\Lambda_E}$ and $\varphi_E \circ w_N = \varepsilon_N(f)\varphi_E + \varphi_E(0)$ (11.4, 9.2). Compute the order of $\varphi_E(0)$ for 11a1 and for 37b1 from $L(E,1)/\Omega_E = 1/5,\ 2/3$, and compare with $E(\mathbb{Q})_{\mathrm{tors}}$; explain why $\varphi_E(0) = O$ for 37a1.
14. Prove $T_p(\infty) = (p+1)(\infty)$ and $T_p(0) = (p+1)(0)$ for $p \nmid N$, and deduce Manin-Drinfeld for $[(0)-(\infty)]$ (11.1). For $N = 11$ show that $5[(0)-(\infty)] = 0$ using $T_2 - 3$ and the eigenvalue $a_2(f_{11}) - 3 = -5$.
15. Check the paper's claim (Section sec:two) that Watkins's conjecture holds for 37a1, 99a1, 539c1 and 726a1, and find in `ecdata` the optimal curves of odd modular degree below conductor 100; test the paper's paraphrase of Calegari-Emerton on 27a1 and decide which hypothesis of CalegariEmerton09 excludes it (12.4).
16. Prove 9.1 and 9.3 for $N = 11$ directly ($\varphi_E = \mathrm{id}$, $M_{f,\lambda} = H^1(X_0(11), \mathbb{Z}_\ell)$, $w_{11}^* = -1$), and then for 37a1 using 2.4(c): the pullback of H^1 of $X_0(37)/w_{37}$ has cup-product values $2\mathbb{Z}$.
17. (Harder.) Let $\pi\colon X_1(N) \to X_0(N)$ be the forgetful map. Show $\langle \pi^*x, \pi^*y\rangle = \deg\pi\,\langle x, y\rangle$ and that $\pi^*H^1(X_0(N), \mathbb{Z})$ is saturated iff the Shimura subgroup vanishes (11.3). Explain why this is the first item of the checklist 9.6.

16.5 5. Dependencies and forward pointers

Results from other chapters used here: * Chapter 3 (`ch:modules`): saturated sublattices and kernels (5.5), perfect pairings and the Gram determinant (9.1 to 9.4), adjoints (9.5), finite free algebras and idempotents (10.3, 10.5), congruence ideals and $\wp \cap I = 0$ (11.1, important_notes.md point 19), the module ingredients of Lemma lem:ddt417 (11.3), Gorenstein and the freeness criterion (11.4, 11.5). * Chapter 4 (`ch:reps`): ad^0 is insensitive to duals and twists (9.1(d)); Brauer-Nesbitt (10.1). * Chapter 5 (`ch:galois`): Chebotarev (6.2(i), 10.1). * Chapter 8 (`ch:curves`): curves, morphisms and degree (its 6.4), Riemann existence and GAGA (8.3), positivity and area (9.4), H_1, H^1, cup product, Poincare duality (10.1 to 10.3), the Jacobian and Abel-Jacobi (11.1 to 11.4), **the degree and the cup product, the saturation criterion (11.5)**, torsion and Tate modules (11.6, 11.7), Weil pairing and etale cohomology (11.8), correspondences and their actions (12.1, 12.2), **adjoint = transpose and the self-adjointness lemma (12.3, 12.3(e))**. * Chapter 9 (`ch:ellcurves`): isogenies and dual isogenies, the period lattice and its area (7.6), a_p and $\#E(\mathbb{F}_p)$ (Section 8), rational isogenies and torsion (Section 9, 9.8), labels and the running table (12.1, 12.2). * Chapter 10 (`ch:reduction`): the Neron model and Neron differential (7.1), $n_q = \#\Phi_q(\bar{\mathbb{F}}_q)$ at multiplicative primes (12.2). * Chapter 11 (`ch:galrep`): reducible $\bar\rho$ and rational isogenies (its 9.1), the Eisenstein case (its 9.3). * Chapter 12 (`ch:modforms`): $X_0(N)$ as a compact Riemann surface and its $\mathbb{Q}$-model as a statement (its 5.2, 5.4(v)), cusps (4.2), genus and $\dim S_2$ (5.3, 8.2), the slash operator (C3), W_N and w_N (12.1), w_N^* on ω_f and on H^1 (12.2(iii)), the Petersson product and its normalisation (10.1, C5), Sturm (11.3), the Eisenstein congruence for f_{11} (8.4), the η-product (9.2). *

Chapter 13 (ch:hecke): $\mathbb{T}$, T_p and U_p on q-expansions, newforms, multiplicity one and strong multiplicity one, $\varepsilon_N(f)$, Petersson adjoints, the perfect pairing $S_2(\mathbb{Z}) \times \mathbb{T} \to \mathbb{Z}$, $f^\perp$ $\mathbb{T}$-stable.

Black boxes imported from later chapters (stated with a forward \Cref, used only in examples, remarks and paperlinks, or as named inputs of R items): Lemma lem:crit (Chapter 15; used in 8.2, 9.4, 9.5 as a cited result); the Galois representation $\bar{\rho}_{\mathfrak{m}}$ (Chapter 15; 10.1); the Rankin-Shimura formula and $L(f,1)$ (Chapter 19; 7.5, 11.4); the assembly of Theorem thm:main and Lemma lem:ddt417 in general (Chapter 20; 9.5(a), 9.4 proves only its case $d = 2$ from Chapter 3); DFG04's definitions (Chapter 20; 9.6). Lemma lem:crit is proved in full in Chapter 15, Section 8 (orchestrator decision); Chapter 14 states the congruence number and the congruence ideal (8.1, recalled from Chapter 13, its 4.3), cites lem:crit with \Cref{ch:modgalrep} in 8.2 and uses only $r_E\mathbb{Z}_\ell = \eta_f\mathbb{Z}_\ell$ (in 9.4, 9.5); Chapter 20 restates it.

Forward pointers (chapters that use this one): * Chapter 15 (ch:modgalrep): $J_0(N)$, J_f, Eichler-Shimura (5.1, 5.2) for $\rho_{f,\lambda}$; $M_{f,\lambda} \simeq T_\ell E(-1)$ for the optimal curve (9.1(d)) for Section 4.2 of the paper; the congruence number and ARS (8.1 to 8.3) for Theorem thm:qual ("it suffices to prove $\ell \mid r$"); 12.2 for Section 1.5 and Remark rem:ko. * Chapter 19 (ch:lfunctions): 7.3 (the Petersson norm and the modular degree) for the Bloch-Kato heuristic; 11.4 ($L(E,1)$). * Chapter 20 (ch:congruence): 3.4 ($\hat{\delta}_!$), 4.4 ($M_{f,\lambda}$), 9.1 to 9.6 (Lemma lem:hecke as an equality and the DFG04 checklist), 10.2 to 10.4 (Remark rem:free), 8.1, 8.3, 8.5. Chapter 20 should cite 9.3 and 9.5 rather than re-derive them, and present the equality in (eq:identity) accordingly. * Chapter 21 (ch:paper): A1 (Section 1: optimal curve, Zagier-Watkins, ARS), A1.9 (12.2), A3.7 (99A1: 8.5, 9.7), A4.1 (3.4, 4.4), A4.4 and A4.5 (9.5, 10.3, 10.4), A4.6 (9.1(d)), A6.10 and Open question 4 (11.1 to 11.5), A7.1 (12.3, 12.4). * Chapter 22 (ch:computation): ellmoddegree (12.5, 7.4), mfpetersson and its normalisation (7.4), the congruence-number algorithm (8.5), the optimality convention (6.5), the $L(E,1)/\Omega$ check (11.4).

16.6 6. Notes for the writer

16.6.1 6.1 Conventions to state (Section 1 of the chapter, as convention environments)

C1 to C8 of Part 1.2. In addition a warning box in Section 4: "Four identifications that are easy to confuse: (1) $T_\ell J_0(N) = H_1 \otimes \mathbb{Z}_\ell$ (covariant, action t_*); (2) $H^1(X_0(N), \mathbb{Z}_\ell) = \mathrm{Hom}(T_\ell J_0(N), \mathbb{Z}_\ell)$ (contravariant, action t^*, Galois-equivariantly $T_\ell J_0(N)(-1)$); (3) PD intertwines t^* with $(t^t)_*$; (4) $w_{N*} \circ$ PD intertwines t^* with t_*." And a warning box in Section 7: "c_φ^2 in the numerator; unnormalised Petersson product; PARI divides by $\mu(N)$".

16.6.2 6.2 Pitfalls

- **Albanese versus Picard.** $\mathbb{T}$ acts on $J_0(N)$ by t_* (C2); some books use the Picard action $t \mapsto (t^t)_* = (w_N t w_N)_*$. For T_n with $(n,N) = 1$ the two agree; for U_p they differ by conjugation by w_N. Everything in this chapter (the quotient J_f, the lattice $M_{f,\lambda}$, the congruence ideal) is invariant because ϕ_f is (4.2(iii)), but a writer who mixes the two in one computation will get wrong matrices. State the action every time.
- **"$\hat{\delta}_!$ is DFG04's $\hat{\delta}_!$" is the paper's reading, not a theorem of this chapter.** Say "the pairing of DFG04 as the paper describes it" and point to 9.6.
- **Point 39 is about the optimal curve.** 9.1 fails for a non-optimal curve (Chapter 8, its 11.5(v): the pullback has index $\deg\psi$ in its saturation), and for depleted forms there is no elliptic curve at all (10.4(b)). Do not write "$v(\eta) = v(\deg)$" for g^S.
- **Lemma lem:crit is proved in Chapter 15.** Chapter 14 states it (8.2) and uses only $r_E\mathbb{Z}_\ell = \eta_f\mathbb{Z}_\ell$; do not reprove it here (PLAN.md's Chapter 14 entry allows "here or in Chapter 20"; the split of this book puts the proof in Chapter 15).

- **ARS12 Theorem 2.2.** Quote it exactly as the paper does and mark the number (verify). The book proves the divisibility at every prime (9.4) and the equality only for $\ell \nmid 2N$ with $E[\ell]$ irreducible (10.3); the equality for $p \| N$, for $p = 2$ and in the Eisenstein case is ARS's, stated (R).
- **Equality of** $\deg \varphi_E$ **and** r_E **at** p **with** $p^2 \mid N$ **is possible** (121a1, 539c1, 980i1, 99b1, 99d1 in 8.5); only the guarantee is lost. Do not write "differ at $p^2 \mid N$".
- **The Manin constant.** $c_E = 1$ (with the sign of ω_E fixed by $c_E > 0$, B-C9) is a conjecture in general; the paper never needs it in a proof, only in the Section 1 heuristic. Results about c_E (7.2(iii)) must be checked against the sources; cite nothing from memory with a theorem number.
- **Optimality versus Cremona label** 1: 990h is the exception (990h3); optimality is proved in the tables only up to 400000 and for part of the range above (point 25).
- **Cusps and fields of definition.** The sign in the Galois action on cusps of denominator c is convention-dependent (2.2(ii)); only the field of definition is used.
- **Signs in 11.4.** $\varphi_E(0) = \pm c_E L(E, 1)$: fix the orientation of the path from $i\infty$ to 0 and the sign of ω_E once; the order of the point does not depend on them.
- **Ribet-Takahashi:** n_q **or** c_q**.** The paper says n_q; the book must check the source (12.2) and say which.
- **Calegari-Emerton.** Copy the statement from the paper; the paper's paraphrase fails for 27a1 and 243b1 (12.4; Part 7, N1).
- **Theorem numbers marked (verify)**: Shimura71 7.9 and 7.14, DiamondShurman 6.6.6 and 8.7.x, ARS12 2.2, RibetStein 3.5, Ribet90 Section 5, Mazur77's numbering, Mazur78 on the Manin constant. Cite by section unless checked.
- Typography (PLAN.md 7.2): "Eichler-Shimura", "Manin-Drinfeld", "Agashe-Ribet-Stein", "Ribet-Takahashi", "Calegari-Emerton", "Boston-Lenstra-Ribet", "Jacquet-Langlands" with plain hyphens; tables with `booktabs`.

16.6.3 6.3 What to leave to Chapters 8, 12, 13, 15, 19, 20

- **Chapter 8** (`ch:curves`): general curves, Jacobians, H^1 and the cup product, correspondences, adjoint = transpose, the self-adjointness lemma 12.3(e), the degree-and-cup-product proposition 11.5 with the saturation criterion. Chapter 14 applies them to $X_0(N)$ (3.3, 3.4, 6.2(iii), 9.1, 9.3) and does not reprove them.
- **Chapter 12** (`ch:modforms`): $X_0(N)$ over $\mathbb{C}$, cusps, genus, the moduli bijection over $\mathbb{C}$, W_N and w_N on forms and on $X_0(N)$, the Petersson product and its numerics (its 10.4), Sturm. Chapter 14 adds the $\mathbb{Q}$-structure, rationality of cusps, rational points, and everything about $J_0(N)$.
- **Chapter 13** (`ch:hecke`): Hecke operators on forms and q-expansions, $\mathbb{T}$ as a ring, newforms, multiplicity one, Petersson adjoints, the perfect pairing with $S_2(\mathbb{Z})$, twists (Atkin-Li), oldforms and U_p-eigenvalues. Chapter 14 adds only the correspondences, their compatibility with Chapter 13's operators (3.2) and the cup-product adjoints (3.3, 3.4).
- **Chapter 15** (`ch:modgalrep`): $\rho_{f,\lambda}$, local-global compatibility, Lemma lem:crit and Lemma lem:old in full, level lowering and level raising, Kim-Ota and Pollack-Weston. Chapter 14 states lem:crit (8.2), uses 5.2 to start the construction of ρ_f, and states Ribet-Takahashi (12.2).
- **Chapter 19** (`ch:lfunctions`): $L(f, s)$, $L(\mathrm{Sym}^2 f, s)$, the Rankin-Shimura formula for $\langle f, f \rangle$ with the bad factors, point 30. Chapter 14 uses only $L(f, 1) = 2\pi \int_0^\infty f(iy)\, dy$ (11.4) and points to the rest (7.5).
- **Chapter 20** (`ch:congruence`): congruence modules in general, Wiles's numerical criterion and Lemma lem:ddt417 in full, DFG04's definitions of $M_{f^\Sigma,\lambda}$, η^Σ and their Theorems 3.7, Proposition 1.4, the assembly of Theorem thm:main and the proof of Theorem thm:quant. Chapter 14 supplies Lemma lem:hecke in its strengthened form (9.5), the checklist (9.6) and Remark rem:free (10.3); Chapter 20 should cite them.

16.6.4 6.4 How the relevant important_notes.md points are handled

- **Point 39** (Lemma lem:hecke is always an equality). Proved in full in 9.1 to 9.5: saturation of $\varphi_E^* H^1(E)$ from the connectedness of $\ker(J_0(N) \to E)$ (6.2(iii), Chapter 8, its 11.5), Hecke equivariance (9.1(a)), w_N by the Atkin-Lehner sign (9.2), the projection formula (6.3(iv)); conclusion $v_\lambda(\eta_{f,\lambda}^{\emptyset}) = v_\ell(\deg \varphi_E)$ for every $\ell \nmid 2N$, unconditionally, **DFG04's η being the ideal of values of $\langle x, w_N^* y\rangle$ on $H^1(X_0(N), \mathbb{Z}_\ell)[I_f]$** (confirmed: Chapter 20 outline, Part 7; important_notes.md point 73). The five-item list 9.6, now checked against DFG04, is: the curve ($X_0(N)$ versus a $\Gamma_1(N)$-type curve with trivial-character part), interior cohomology, the operator $[UwU]\omega$ without scalar and the normalisation of H_c^2 (including elliptic points at $\ell = 3$), kernel lattice versus projection lattice, ideal of values versus Gram determinant. 9.5(a) to (c) state the consequences for Theorem thm:main and Theorem thm:quant.
- **Point 40** ($M_{f,\lambda} = \varphi_E^* H^1(E, \mathbb{Z}_\ell) \simeq T_\ell E(-1)$ exactly): 9.1(c), (d).
- **Point 26** ($M_{f,\lambda}$ is homothetic to the dual of $T_\ell E$): 9.1(d), with the `warning` of Section 4.
- **Point 41** (Tate module versus H^1 as Hecke modules): 4.3(ii), (iii): $w_{N*} \circ \mathrm{PD}$ is $\mathbb{T}$-equivariant and, after the twist by $\mathbb{Z}_\ell(-1)$, $G_{\mathbb{Q}}$-equivariant; freeness transfers.
- **Point 36** (the pairing stays perfect on the $\mathfrak{m}$-component): 3.4(iii), from Chapter 8, its 12.3(e), and 3.3.
- **Point 35** (other local factors lie in the kernel): 8.1(ii), citing Chapter 3, its 11.1(v); the `paperlink` for Lemma lem:hecke quotes the corrected wording.
- **Point 19** ($\wp \cap I = 0$): 8.1(iii) and 9.4, citing Chapter 3, its 11.1(ii) (the argument uses $\eta \neq 0$, which multiplicity one of the newform gives; not "$\mathbb{T}$ reduced").
- **Point 27** (optimality plays no role at ℓ when $E[\ell]$ is irreducible): 6.4, proved, with the class 11a at $\ell = 5$ as the counterexample without irreducibility.
- **Point 24** (the abstract calls the Theorem thm:quant argument elementary): 9.5(b) removes multiplicity one from Step 4, but Step 4 still uses ARS12 at level N_S, Step 3 uses DFG04 Proposition 1.4 and Step 2 the commutative algebra; so "elementary" remains an overstatement, now a smaller one. The `paperlink` for Theorem thm:quant says this.
- **Point 25** (optimality established only up to 400000 and for part of the classes beyond): 6.5 states exactly what is known.
- **B3** (Chapter 21 outline, A4.4 and A4.6): the Chapter 21 writer should cite 8.1(ii) for the "kernel, not annihilator" wording, 9.1(d) for $M_{f,\lambda}$ versus $T_\ell E$, and 9.5 for the equality; see also Part 7, B-a below.
- **B10** (the Manin constant in the numerator): 7.3, with the derivation and the class 11a check (7.4), and a `warning` box.
- **B14** (bibliography): every proposed key in 6.5 that another chapter also needs goes into `refs.bib`; the others into `bib/ch14.bib`. Checked: `refs.bib` already has DiamondIm, LingOesterle, Ribet88, Carayol86, Mazur78, AU96, Tilouine97, RibetStein, Shimura73, Stein, Ogg (but Ogg is the 1967 wild ramification paper, not Ogg 1973 on cusps), ecdata; no chapter bib file has any key yet except `bib/chA.bib`.

16.6.5 6.5 References

In `refs.bib` and cited here: Shimura71 (Chapters 6, 7; theorem numbers 7.9, 7.14 marked "verify"), Shimura73, DiamondShurman (Sections 3.8, 5.2, 6.1 to 6.6, 7.7, 7.9, 8.7; theorem numbers marked "verify"), DiamondIm (Sections 7 to 12 by section), Stein, Cremona97 (Chapter II), Zagier85, Watkins02, ARS12 (Theorem 2.2 as the paper cites it; Sections 2, 3; Table 1), Ribet83, AU96, Mazur77 (Chapter II), Mazur78, Ribet90, Ribet88, LingOesterle, RibetStein (Theorem 3.5 as cited), Tilouine97, Wiles95, TaylorWiles, BCDT, Carayol86, DDT (Section 4), DFG04 (Sections 1.5.3 to 1.8.2, (13), (14), Lemma 2.12, Proposition 1.4, Theorem 3.7, as the paper cites them), RibetTakahashi97, Takahashi01, Pasten, CalegariEmerton09, Dummigan06, Yazdani11, CaroPasten22, HatleyKundu, BKP, PARI, PARIusers, ecdata, LMFDB, Conrad (modular curves notes), Loeffler (modular curves notes).

Proposed new keys (complete entries to be checked against the publications before they are added;

none is in any bib file now). By decision B14, a key that another chapter also needs goes in `refs.bib`: * `DeligneRapoport73` (P. Deligne, M. Rapoport, *Les schemas de modules de courbes elliptiques*, Lecture Notes in Math. 349, 1973) and `KatzMazur85` (N. Katz, B. Mazur, *Arithmetic Moduli of Elliptic Curves*, Annals of Math. Studies 108, 1985): needed by Chapters 12 and 14, so **refs.bib**, not `bib/ch12.bib` as the Chapter 12 outline planned (Part 7, B-d). * `Ogg73` (A. Ogg, *Rational points on certain elliptic modular curves*, Proc. Sympos. Pure Math. 24, 1973): Chapters 12 and 14, so **refs.bib**; the existing key `Ogg` is a different paper (Ogg 1967, wild ramification). * `Edixhoven91` (B. Edixhoven, *On the Manin constants of modular elliptic curves*, in *Arithmetic Algebraic Geometry (Texel, 1989)*, Progr. Math. 89, 1991): `bib/ch14.bib`. * `ARS06` (A. Agashe, K. Ribet, W. Stein, *The Manin constant*, Pure Appl. Math. Q. 2, 2006): `bib/ch14.bib`. * `Cesnavicius18` (K. Cesnavicius, *The Manin constant in the semistable case*, Compositio Math., 2018) and `CNS` (K. Cesnavicius, M. Neururer, A. Saha, *The Manin constant and the modular degree*, J. Eur. Math. Soc.; year and volume to be checked): `bib/ch14.bib`, cited only after checking the statements quoted in 7.2. * `Kenku` (M. Kenku's completion of the list of N with $Y_0(N)(\mathbb{Q}) \neq \emptyset$; the Chapter 9 outline also needs it, so **refs.bib**; exact paper to be identified). * `Frey` and `MaiMurty` for the abc connection (8.4): cite only through ARS12 Section 2 unless checked; then `bib/ch14.bib`. * Optional: `Stevens89` (G. Stevens, *Stickelberger elements and modular parametrizations of elliptic curves*, Invent. Math. 98, 1989) for optimal curves and the $X_1(N)$-optimal variant; `bib/ch14.bib`. Every bibliographic detail above was recalled, not re-verified.

16.6.6 6.6 Verification log (computations done for this outline)

Python 3 with PARI/GP 2.17 through `cypari2`; scripts `cong.gp`, `run.py`, `run2.py`, `zw.py` in the session scratchpad (`ch14/`); Cremona's `ecdata` files in `/tmp/ch09/data`. * **Congruence numbers** (`cong.gp`): for N the conductor, take the $\mathbb{Q}$-basis `mfbasis(mfinit([N,2],1))`, its q-expansions to $19(B_2(N)+20)$ terms, the integral lattice $S_2(\Gamma_0(N);\mathbb{Z})$ as the saturation of the span of the truncations at $B_2(N)+20$ (via `matkerint` of the left kernel; Chapter 12, its 11.4(ii) justifies saturating the truncation), the Hecke matrices of T_p (U_p for $p \mid N$) for $p \le 19$ on that lattice from the q-expansion formulas, the common left eigenvector λ with eigenvalues $a_p(f)$ made primitive, and $r_E = |\lambda(f)|$. Results as in 8.5: $1,1,2,6,4 \to 12,12,12,6,6,288,1584,2880$ for 11a1, 20a1, 37a1, 77c1, 99a1, 99b1, 99c1, 99d1, 121a1, 539c1, 726a1, 980i1 (the ranks $1,1,2,7,9,6,49,109,145$ of the lattices equal $\dim S_2$). `ellmoddegree` gave the degrees in the same table; all agree with `ecdata`'s `degphi` file. Sanity check of the method: $r_E = \deg \varphi_E$ whenever $p^2 \nmid N$ for all p, as ARS predicts. * **Level-99 congruences**: $a_n(99a1) \equiv a_n(99b1) \equiv a_n(99d1) \pmod 3$ for all $n \le 30$ ($B_2(99) = 24$); 99c1 and the old forms from 11a1, 33a1 are not congruent to 99a1 mod 3 ($a_2 \equiv 1$ versus 2). * **Zagier-Watkins** (`zw.py`): $\mu(N)\cdot$`mfpetersson`(F) for the newform F of the curve, `E.omega` for the periods, area $|\operatorname{Im}(\bar\omega_1\omega_2)|$; the ratios $4\pi^2\langle f,f\rangle/\operatorname{vol}(\Lambda_E)$ equal $1, 5, 1/5, 2, 4, 6$ to 30 digits for 11a1, 11a2, 11a3, 37a1, 99a1, 121a1, as in 7.4 (and Chapter 12, its 10.4, for 20a1). The level-539 Petersson norm exceeded a 2 GB PARI stack and was not computed. * **Class 11a**: `allisog` gives the isogeny matrix $\left(\begin{smallmatrix} 1 & 5 & 5 \\ 5 & 1 & 25 \\ 5 & 25 & 1 \end{smallmatrix}\right)$ for 11a1, 11a2, 11a3. * **Cusp counts** (by the formula $\sum_{c \mid N} \varphi(\gcd(c, N/c))$): 6, 8, 12 for $N = 20, 99, 121$, agreeing with Chapter 12, its 5.3 table. * **Image of the cusp** 0 (11.4): `bestappr(lfun(E,1)/E.omega[1])` $= 1/5,\ 1/6,\ 1/6,\ 1/3,\ 2/3,\ 1,\ 0,\ 0$ for 11a1, 14a1, 20a1, 19a1, 37b1, 121a1, 539c1 (and 0 for 37a1, 99a1, 726a1 by analytic rank 1); torsion orders $5, 6, 6, 3, 3, 1, 2$. * **Watkins's conjecture and Calegari-Emerton** (12.3, 12.4): for each of the 38,042 lines of `degphi.00000-09999` (one optimal curve per class, conductor below 10,000), the rank and torsion order from `allcurves`: no curve with $2^{\text{rank}} \nmid \deg$; 103 curves of odd degree; all with at most two odd bad primes; the two without rational 2-torsion and with non-prime conductor are 27a1 ($[0,0,1,0,-7]$, $\deg 1$) and 243b1 ($[0,0,1,0,2]$, $\deg 9$), both $j = 0$. * **Optimality and labels**: `degphi` lists 990h with the equation of 990h3 and degree 1728; all other classes up to 10,000 use curve number 1 (as recorded in Chapter 9, its 12.1). * By hand: $W_N \operatorname{diag}(1,p)\, W_N^{-1} = \operatorname{diag}(p,1)$; the transpose computation of 3.3(i); $\mathcal{C}_\alpha^*(f\,dz) = (f|[\Gamma\alpha\Gamma]_2)\,dz$ with the normalisation of Chapter 12, C3; $T_p(0) = (p+1)(0)$; $\int_0^{i\infty} 2\pi i f\,d\tau = -L(f,1)$; $\varphi_E \circ w_N = \varepsilon\varphi_E + \varphi_E(0)$; the projection-formula computation of 9.3; $\dim S_2(\Gamma_0(99)) = 4+2+3$; the interior-cohomology exact sequence in 9.6(2). * Not checked: the theorem numbers marked (verify); the exact statements of ARS12 Theorem 2.2, RibetStein Theorem 3.5, Ribet-Takahashi, Calegari-Emerton and the Manin-constant literature;

DFG04's definitions (9.6).

16.7 7. Points for important_notes.md

New points in this chapter's area ($X_0(N)$ over $\mathbb{Q}$, $J_0(N)$, optimal curves, the modular degree, the Manin constant, r_E and ARS, $\hat{\delta}_!$ and Lemma lem:hecke, multiplicity one, the Eisenstein ideal, Ribet-Takahashi, Section sec:two), not among points 1 to 53 and B1 to B16. Points 19, 24, 25, 26, 27, 35, 36, 39, 40, 41, B3, B10, B14 are handled in Part 6.4 and not repeated.

N1. Section sec:two: the paraphrase of Calegari-Emerton fails for two CM curves. Location: Section sec:two, "in line with the theorem of Calegari and Emerton that a curve of odd modular degree has at most two odd bad primes and a rational 2-torsion point or prime conductor \cite{CalegariEmerton09}". Problem: among the 103 optimal curves of conductor below 10,000 with odd modular degree (`ecdata`, `degphi` and `allcurves`), two have neither a rational point of order 2 nor prime conductor: 27a1 $[0,0,1,0,-7]$ ($N=3^3$, $\deg\varphi_E=1$, $E(\mathbb{Q})_{\mathrm{tors}}\simeq\mathbb{Z}/3$) and 243b1 $[0,0,1,0,2]$ ($N=3^5$, $\deg\varphi_E=9$, torsion $\mathbb{Z}/3$), both with $j=0$ (CM). The other 101 satisfy the paraphrase, and all 103 have at most two odd bad primes. Fix: state the theorem with the hypothesis or exception of CalegariEmerton09 (presumably "E without CM", or a prime-power conductor clause), checked against the paper; the sentence is used only as context for data on non-CM curves, so nothing else changes. Confidence: high for the data (two independent counts from `ecdata`; j computed with PARI); medium on the exact form of the published theorem, which was not consulted.

N2. Lemma lem:hecke and Section 4.1 (refinement of point 39): the precise checklist for DFG04's η, and two corollaries. Location: Section 4.1 ("The integral structure $M(M^\Sigma,1)_{!,\lambda}$ is $H^1(X_0(M^\Sigma),\mathbb{Z}_\ell)\ldots$; the restriction of $\hat{\delta}_!$ to the rank-two lattice $M_{f^\Sigma,\lambda}$ takes its values in an ideal of $\mathbb{Z}_\ell$, which is $\eta^\Sigma_{f,\lambda}$"), Lemma lem:hecke, Remark rem:free. Content: (a) Point 39's "medium confidence on the normalisation" splits into five concrete checks (item 9.6): (1) whether DFG04's weight-2, trivial-character integral structure is $H^1(X_0(N),\mathbb{Z}_\ell)$ or the trivial-character part of the cohomology of a $\Gamma_1(N)$-type curve: in the latter case the pullback multiplies the cup product by $\#(\mathbb{Z}/N)^\times/\{\pm1\}$, which ℓ can divide even when $\ell\nmid N$ (for example $\ell=3$, $7\mid N$), and the Shimura subgroup enters the saturation, so η could differ from $\deg\varphi_E$ by a non-unit; (2) interior cohomology is $H^1(X_0(N))$ for constant coefficients (true, short exact-sequence argument; group cohomology of $\Gamma_0(N)$ agrees because elliptic elements are torsion); (3) $[UwU]\omega=w_N^*$ with no scalar in weight 2, and the normalisation of H^2_c (only 2 and, if elliptic points of order 3 occur, possibly 3 are at risk); (4) $M_{f,\lambda}$ is the kernel lattice, not the projection lattice (the latter would change the ideal of values by the congruence ideal); (5) η is the ideal of values, not the Gram determinant. If (1) or (4) differ, the paper's own description of η in Section 4.1 is wrong, independently of point 39. (b) Corollaries of the point-39 identity that the paper can state: the divisibility $\deg\varphi_E\mid r_E$ follows at **every** prime ℓ from the identity and the inequality half of Lemma lem:ddt417 applied to all of $H^1(X_0(N),\mathbb{Z}_\ell)$ (item 9.4), so Lemma lem:hecke needs ARS12 only for $v_\ell(r_E)\le v_\ell(\deg\varphi_E)$; and the hypothesis "$\bar{\rho}$ irreducible" of Lemma lem:hecke is not used by the identity $v(\text{values of }\hat{\delta}_!\text{ on }M_{f,\lambda})=v_\ell(\deg\varphi_E)$ (it matters only for $\ell\notin S_f$ later). (c) What point 39 does **not** remove: Remark rem:free is still needed for Remark (2) after Theorem thm:quant (the lattice there belongs to the depleted form g^S, which is not attached to an optimal quotient). **Superseded** by important_notes.md point 74: the first inequality of Theorem thm:quant is an equality whenever $\ell\notin S_f$ with no multiplicity-one hypothesis (9.5(c)); B26. Fix: in Section 4.1, after "is $H^1(X_0(M^\Sigma),\mathbb{Z}_\ell)$", cite the exact place in DFG04 where this identification is made (or add the pullback factor if DFG04 work on X_1); state Lemma lem:hecke as an equality with the four-line proof of point 39. Confidence: high for the mathematics of (b), (c) and of the claims in (a)(2); the checks (a)(1), (3) to (5) were open here (DFG04 not consulted) and are now settled: all five hold (Chapter 20 outline, Part 7; important_notes.md point 73).

N3 (confirmation, not an error). The remark after Theorem thm:qual: 99A1 has $r_E=12$, recomputed; 726a1 is a second example from the paper's own list. Location: "Table 1 of \cite{ARS12} lists the curve 99A1, of conductor $9\cdot11$, with modular degree 4 and congruence number 12". Recomputed directly (Part 6.6): $r_E=12$, $\deg\varphi_E=4$; the extra 3 comes from the newforms 99b1 and 99d1, congruent to 99a1 modulo 3 at every n. The paper's 726a1 ($N=2\cdot3\cdot11^2$, the 3Ns example) has

$\deg \varphi_E = 144$ and $r_E = 1584 = 11 \cdot 144$: the paper could cite it as a second instance. Also 121a1, 539c1, 980i1 have $r_E = \deg \varphi_E$ although $p^2 \mid N$. Optional addition; confidence high.

Book-internal inconsistencies found while preparing this outline:

- **B-a. Chapter 21 outline, dependency diagram and hypothesis table** (around its lines 1369 and 1393): "[Tilouine97; RibetStein 3.5; Shimura subgroup] => rem:free => equality in thm:main; thm:quant Step 4" and the row listing multiplicity one as a hypothesis of the equality. After point 39 (item 9.5 here) the equality in (eq:identity) and Step 4 do not go through Remark rem:free; only Remark (2) after Theorem thm:quant does. Extends B3.
- **B-b. Chapter 9 outline, 12.2, and Chapter 21 outline (A3.7, the references table): "$r_E = 12$ ... not recomputed" / "as cited".** It is now recomputed (8.5 here; also by the Chapter 15 outline); both may cite Chapter 14.
- **B-c. Duplicate proof of point 40.** The Chapter 15 outline, its 2.6(ii), proves $M_{f,\lambda} = \varphi^* H^1(E_0, \mathbb{Z}_\ell) \simeq T_\ell E_0(-1)$, which is this chapter's 9.1 (and which PLAN.md and the Chapter 8 outline, its Part 6.3, place in Chapter 14). Proposal: Chapter 14 proves it, Chapter 15 cites `\Cref{ch:moddeg}`.
- **B-d. Bibliography keys planned in `bib/ch12.bib` that Chapter 14 also needs** (`DeligneRapoport73`, `KatzMazur85`, `Ogg73`, and `Kenku` from Chapter 9): by B14 they belong in `refs.bib`. Note also that the existing `refs.bib` key `Ogg` is Ogg's 1967 paper on wild ramification, so the 1973 cusp paper needs its own key.
- **B-e. PLAN.md, Chapter 14 entry.** (i) "Mazur: c | ..." is an unfinished placeholder; 7.2 here lists what to check. (ii) "the product of the c_q or n_q over q | D" (Ribet-Takahashi) leaves the choice open; 12.2 asks the writer to settle it from the source (the paper says n_q; the Chapter 15 outline's 11.2 also says n_q "as the paper's summary"). (iii) "Lemma lem:crit ... proved in full here or in Chapter 20" is superseded by the decision that Chapter 15 proves it (also the Chapter 15 outline's I3).
- **B-f. Ribet-Takahashi is stated in two chapters.** The Chapter 15 outline's 11.2 and this chapter's 12.2 both state it (R). Proposal: Chapter 14 carries the full statement (PLAN.md), Chapter 15 a one-line pointer.
- Consistency checks that passed: Chapter 8's 11.5 and 12.3(e), Chapter 12's 10.4 numerics (20a1, 37a1, 99a1, 121a1, 11a1), genus table and w_{37} data, Chapter 3's 11.1 to 11.5, Chapter 9's 12.1 (990h3) and running table (degrees, torsion, ranks), and Chapter 13's 4.3 definitions agree with the recomputations of Part 6.6.

Chapter 17

Chapter 15: Galois representations of modular forms, level lowering and congruences (ch:modgalrep, chapters/ch15_modgalrep.tex)

Stage 1 line-up. Part IV (Modular forms and modular curves). Target length 2,770 lines of LaTeX (card excluded; see Part 3), inside the 2,800 lines that PLAN.md Section 7.3 allows for this chapter.

Rigour tags: **P** = proved here, **S** = proof sketched here, **R** = stated; reference. Item format, as in the Chapter 2 to 12 outlines: number, label-ready name, kind, tag; the precise statement with all hypotheses; reference (book or paper and chapter or section; a theorem number only where the outline author is sure or where the paper itself cites it, otherwise the section, and numbers marked "(verify)" must be checked before they are printed); the paper's use by its `\label` names, or "background". Every computation quoted below (the congruences of the Example of Section 3 for all n, curve-level twist pairs, levels of twists, Serre conductors, level-raising conditions and the congruent forms they predict, the oldform construction of Lemma lem:old on two examples, the congruence number of 99a1, the new example 52a1 for case (a) at $q = 2$, the 1,051 curves with type IV or IV* at 2) was checked with PARI/GP 2.17 through `cypari2` against Cremona's `ecdata` files `allcurves`, `degphi`, `galrep` for conductors below 10,000 (verification log, Part 6.6).

17.1 1. Purpose and global conventions

17.1.1 1.1 What this chapter is for

Section 3 of the paper proves Theorem thm:qual by attaching Galois representations to newforms, lowering the level of a residual representation, twisting, and converting a congruence between eigenforms into divisibility of the congruence number. Section 4 reads the same mechanism quantitatively: the level-raising Euler factor of Theorem thm:dfg14, the twist map of Theorems thm:twist and thm:quant, and the identification of the lattice $M_{f,\lambda}$ of Diamond, Flach and Guo with $T_\ell E$. This chapter supplies all of it. Concretely it must deliver:

1. **The representation $\rho_{f,\lambda}$ of a newform of weight** 2 (Eichler-Shimura construction through J_f; Deligne for higher weight as a statement): unramified outside $N\ell$, $\operatorname{tr}\rho(\mathrm{Frob}_p) = a_p$, $\det = \chi_\ell$, odd, absolutely irreducible; lattices and the residual representation; congruent newforms have isomorphic residual representations (absolute irreducibility, important_notes.md point 10); twists $\rho_{f^\chi} \simeq \rho_f \otimes \chi$ and the distinction between the *naive twist* $f \otimes \chi$ and the *newform twist* f^χ (the source of important_notes.md point 20 and of the new point N1, Part 7).

2. **The comparison with** $\rho_{E,\ell}$ for $f = f_E$: $T_\ell E$ is a lattice in $\rho_{f,\ell}$; for the optimal curve the lattice $M_{f,\lambda}$ of DFG04 is exactly $\varphi_E^* H^1(E, \mathbb{Z}_\ell) \simeq T_\ell E(-1)$, homothetic to the dual of $T_\ell E$ in general (important_notes.md points 26 and 40, B3), so $\mathrm{ad}^0 M_{f,\lambda} = \mathrm{ad}^0 T_\ell E$ (Section 4.2 of the paper).
3. **Local-global compatibility and local components** (Carayol; Langlands, Deligne): the local Langlands correspondence for $\mathrm{GL}_2(\mathbb{Q}_p)$ as a statement; the local types of a weight-two newform with trivial character (unramified principal series, unramified twist of Steinberg with DDT Theorem 3.1(e), and for $p^2 \mid N$ ramified twist of Steinberg, ramified principal series, supercuspidal); $\delta_p = 2, 1, 0$; $a_p = 0$ and $V^{I_p} = 0$ when $p^2 \mid N$ (a local proof of Atkin-Lehner's Theorem 3); the Kodaira-type dictionary for $f = f_E$ at $q \geq 5$ (the "local components" of Lemma lem:euler); **the conductor of a quadratic twist type by type** (item 3.8), which gives Lemma lem:twist(iii) (Atkin-Li) and the levels in Theorems thm:qual, thm:twist and thm:quant.
4. **The representation at** $\ell \nmid N$ for a newform: crystalline, finite flat residual, Fontaine-Laffaille; the Deligne-Fontaine description and the determination of a_ℓ mod λ by $\bar{\rho}|_{D_\ell}$ (the case $p = \ell$ of Lemma lem:old).
5. **Serre's conjecture** (statement; Khare-Wintenberger), Serre conductor and weight, $N(\bar{\rho}_f) \mid N$ (proved), **Mazur's principle, Ribet's level-lowering theorem, Carayol's lemma, and the optimal-level theorem in the form of DDT Theorem 3.15** with the paper's specialisation to $\bar{\rho}_{E',\ell}$ (trivial nebentypus: proved).
6. **Ribet's level-raising theorem** ($a_q^2 \equiv (q+1)^2 \pmod{\lambda}$; the necessity proved) and the level-raising Euler factor $L_q^{\mathrm{nv}}(A_f, 1)^{-1} = \frac{q-1}{q} \cdot \frac{(q+1)^2 - a_q^2}{q^2}$ of Theorem thm:dfg14 (statement; Chapter 20 has the proof idea).
7. **The paper's Lemma lem:crit proved in full** (PLAN.md: here; Chapter 20 restates it) and **Lemma lem:old proved in full**.
8. **The paper's Theorem thm:qual proved in full**, cases (a), (b), (c), with case (a) extended to $q = 2$ (important_notes.md point 43, settled), the corrections of points 10 and 20 and of the new point N1; the Example (121a1, 539c1, 980i1: congruences for all n by Sturm, point 46; correct curve-level labels, B8; 980i1 outside the hypotheses, point 21); the remark on $\ell \mid N$ and 99A1 (the congruence number 12 recomputed here).
9. **The twist map and levels for Theorems thm:twist and thm:quant**, the remark after Theorem thm:quant read as level raising (points 16, 24, 39), and the statements of Ribet-Takahashi, Pollack-Weston and Kim-Ota (Section 1.5, Remark rem:ko(2)), with a one-page outline of Taylor-Wiles and $R = \mathbb{T}$ saying what DFG04 Theorem 3.7 needs (point 47).

17.1.2 1.2 Split with the neighbouring chapters (decided here, consistent with PLAN.md Section 4 and the written outlines)

- **Chapter 3** proves the commutative algebra: dualising an inclusion of lattices (5.7), the congruence module of a decomposition (5.8), the algebra-module duality for a perfect pairing $S \times \mathbb{T} \to R$ (9.6), Artinian algebras and idempotents (10.1 to 10.6), non-semisimple operators and the shift-operator model of U_p (10.7(v)), common eigenvectors on tensor products (10.8), congruence ideals (11.1). Lemma lem:crit and Lemma lem:old here cite these by `\Cref{ch:modules}` and add only the modular input.
- **Chapter 4** proves the unique stable line of a non-split extension (3.6), Brauer-Nesbitt (5.2, 5.3), lattices, reduction and the homothety lemma (10.1 to 10.3), adjoint invariance under duals, twists and homotheties (7.7). Cited.
- **Chapter 5** proves Chebotarev plus Brauer-Nesbitt in the form galois:congruent-ap, with the Galois representations of newforms as a forward black box; **this chapter discharges that black box** (item 2.1) and restates galois:congruent-ap for newforms (2.5).
- **Chapter 6** gives coefficient fields, orders, primes λ, completions $\mathcal{O}_\lambda$ (its 7.1 to 7.4).
- **Chapter 7** gives the Galois side of local theory: Weil-Deligne representations, sp(2) and its twists, the classification of two-dimensional Weil-Deligne representations (7.5), invariants, L-factors with the geometric Frobenius (8.1, 8.2), Artin and Swan conductors (8.3, 8.4), ϵ-factors (8.5, recalled

here), the shapes table (9.4). **Chapter 15 adds the automorphic side** (local Langlands, local components π_p, Carayol's theorem) and the types of newforms; it does not redo any Weil-Deligne computation.

- **Chapter 11** gives everything specific to $\rho_{E,\ell}$: the Tate shape and Lemma lem:tamagawa-exp (3.2, 3.3), inertia images by Kodaira type and Lemma lem:IV (4.1 to 4.5, with 4.4 at $q = 2$), the representation at ℓ for E (5.1 to 5.4), the Serre conductor formula $a_p(\bar\rho_{E,\ell}) = f_p - \dim \Phi_p^{\mathrm{geom}}[\ell]$ and its examples (6.1 to 6.3), images and Proposition prop:ns, Lemma lem:h0, the equivalence $\ell \notin S_f \iff (E[\ell]$ irreducible and not 3Ns) for $\ell \nmid 2N$ (8.6), and the *statement* of the optimal-level theorem for $\bar\rho_{E,\ell}$ (10.1) with its examples (10.2), leaving "the general theorem, its proof and the trivial-nebentypus argument" to this chapter. **Chapter 15 proves nothing about $\rho_{E,\ell}$ that Chapter 11 proves**; where an elliptic-curve fact is needed it is cited (`\Cref{ch:galrep}`), and Chapter 15's examples quote Chapter 11's Serre conductors instead of recomputing them in the text.
- **Chapter 12** gives $S_2(\Gamma_0(N);\mathbb{Z})$, the integral basis, Sturm's bound and **the proof that the three congruences of the Example hold for every** n (its 11.3 to 11.5, important_notes.md point 46), nebentypus (6.6), Atkin-Lehner involutions (12.1 to 12.4). Cited, not redone.
- **Chapter 13** (outlined in parallel) gives, per PLAN.md: Hecke operators on forms, U_p on oldforms with its eigenvalues, stabilisations and depleted forms, the decomposition of $S_2(\Gamma_0(N)) \otimes \overline{\mathbb{Q}}$ into spans of $h(dz)$, strong multiplicity one, Atkin-Lehner's $a_p = 0$ for $p^2 \mid M$ (their Theorem 3), the Atkin-Li theorem in the paper's form, the twist map ι on the whole space and its compatibility with T_p, the perfect pairing $S_2(\Gamma_0(N);\mathbb{Z}) \times \mathbb{T} \to \mathbb{Z}$, the recursion for a_{p^k}. **Chapter 15 cites all of these** and adds the Galois-theoretic content: why Atkin-Li holds (local components, item 3.8), the conductor of a twist when q^2 divides the level (beyond Atkin-Li), and the reading of ι as $\rho \mapsto \rho \otimes \chi$.
- **Chapter 14** (outlined in parallel) gives $J_0(N)$, Eichler-Shimura $(T_p)_* = \pi_p + \hat\pi_p$ (π_p the Frobenius endomorphism, $\hat\pi_p = p\pi_p^{-1}$, Appendix B, B-C2(g)), J_f as a quotient, modularity, the optimal curve, the modular degree, the congruence number, Agashe-Ribet-Stein (with 99A1 as its example), multiplicity one and Remark rem:free, and important_notes.md point 39. Chapter 15 cites the construction of J_f and Eichler-Shimura for 2.1, and Agashe-Ribet-Stein for the passage $r_E \to \deg \varphi_E$; it recomputes $r(99\mathrm{a}1) = 12$ only as data for the remark of Section 3.
- **Chapter 19** computes $L_q(A_f, s)$, P_q and $e_\ell(E, q)$ (Lemma lem:euler in full); Chapter 15 gives only the local types behind the phrase "local components" and the level-raising factor identity (7.2).
- **Chapter 20** gives congruence ideals in $\mathbb{T}_{\mathfrak{m}}$, Lemma lem:ddt417, Lemma lem:hecke, Theorem thm:dfg14 with its proof idea (Ihara's lemma, degeneracy maps), DFG04 Theorem 3.7, Taylor-Wiles in outline, and **the full proofs of Theorems thm:twist and thm:quant**. Chapter 15 states those two theorems, proves the level and twist facts they rest on (3.8, 10.1, 10.2), and explains the remark after Theorem thm:quant from the Galois side; Lemma lem:crit is proved here and only restated in Chapter 20 (PLAN.md).
- **Chapter 21** (written outline) proves lem:crit, lem:old and thm:qual again in its tour; it may shorten them by pointing here, and must take over the corrections listed in Part 7.2.

17.1.3 1.3 Global conventions fixed in this chapter

- **C1 (newforms).** $f = \sum a_n(f) q^n$ is a normalised newform of weight 2 on $\Gamma_0(N)$ (trivial character) unless said otherwise; $K_f = \mathbb{Q}(a_n(f) : n \geq 1)$, a totally real number field; $\mathcal{O}_f$ its ring of integers; $\lambda \mid \ell$ a prime of $\mathcal{O}_f$ (or of a finite extension $\mathcal{O}$, "after enlarging $\mathcal{O}$"), $K_{f,\lambda}$ and $\mathcal{O}_\lambda$ the completions, k_λ the residue field. $\mathbb{T} = \mathbb{T}(N)$ is the **full** Hecke algebra of level N generated over $\mathbb{Z}$ by all T_n, $n \geq 1$ (so including U_p for $p \mid N$), as in Section 3 of the paper; $\phi_f \colon \mathbb{T} \to \mathcal{O}_f$ its eigencharacter. Chapter 13's notation is used throughout.
- **C2 (Frobenius).** As in Chapters 7 and 11: Frob_p is an **arithmetic** Frobenius, $\Phi_p = \mathrm{Frob}_p^{-1}$ the geometric one; L-factors use Φ_p. The normalisation of $\rho_{f,\lambda}$: $\det(X - \rho_{f,\lambda}(\mathrm{Frob}_p)) = X^2 - a_p(f)X + p$ for $p \nmid N\ell$ and $\det \rho_{f,\lambda} = \chi_\ell$, so that $\rho_{f_E,\ell} \simeq \rho_{E,\ell}$ (Chapter 11). The cohomological realisation $H^1(X_0(N), \cdot)$ gives the **dual** $\rho_{f,\lambda}^\vee \simeq \rho_{f,\lambda}(-1)$, on which the geometric Frobenius has trace a_p (important_notes.md B1, points 26, 40).
- **C3 (characters).** A Dirichlet character χ is also the Galois character with $\chi(\mathrm{Frob}_p) = \chi(p)$ for

$p \nmid \mathrm{cond}(\chi)$; χ_ℓ is the ℓ-adic cyclotomic character; χ_{q^*}, $q^* = (-1)^{(q-1)/2}q$, is the quadratic character of conductor q (q odd), as in Chapter 11 (C4) (the paper's χ_q, never written in the book, decision D3); write χ_{-3}, never χ_3, for the quadratic character of conductor 3 (important_notes.md point 32).

- **C4 (twists: the decisive convention).** For a Dirichlet character χ of conductor r:
  - the **naive twist** $f \otimes \chi := \sum_n \chi(n)a_n(f)q^n$ (the paper's $f \otimes \chi$, the map ι of Theorem thm:twist);
  - the **newform twist** f^χ: the unique newform with $\rho_{f^\chi,\lambda} \simeq \rho_{f,\lambda} \otimes \chi$ (item 2.7);
  - the **q-depletion** $h^{\{q\}} := \sum_{q\nmid n} a_n(h)q^n$ and, for a set S, h^S (Chapter 13's depleted form; the paper's f^Σ); never $h^{(q)}$, since a parenthesised superscript is reserved for twists $E^{(q)}$ (Appendix B, B-C13(e)). $\Sigma(r)$ is the set of primes dividing r. Then $f \otimes \chi = (f^\chi)^{\Sigma(r)}$ (the depletion at the primes dividing r), and $f \otimes \chi = f^\chi$ iff $a_q(f^\chi) = 0$ for every $q \mid r$. The paper writes $f \otimes \chi$ for both; Section 9 below shows where this matters (points 20 and N1).
- **C5 (local components).** For p prime, π_p is the local component at p of the automorphic representation of f; the Galois side is $\mathrm{WD}_p(f) := \mathrm{WD}(\rho_{f,\lambda}|_{G_{\mathbb{Q}_p}})^{\text{F-ss}}$ ($p \neq \ell$), a two-dimensional Frobenius-semisimple Weil-Deligne representation (Chapter 7). Statements are made on the Galois side; the automorphic names (principal series, Steinberg, supercuspidal) are labels via the local Langlands correspondence (3.2), with one fixed normalisation (3.2(iv)).
- **C6 (labels).** Curve labels are Cremona's (Chapter 9, C6); isogeny classes are named without the final digit (11a, 121c), curves with it (121c2). "99A1" in ARS12 is Cremona's 99a1. Mod ℓ image labels as in Chapter 11 (C6).
- **C7 (typography).** PLAN.md 7.2. Names with plain hyphens (Eichler-Shimura, Atkin-Lehner, Atkin-Li, Deligne-Fontaine, Fontaine-Laffaille, Khare-Wintenberger, Ribet-Takahashi, Pollack-Weston, Kim-Ota, Taylor-Wiles); labels `...:modgalrep:...`; the line-up names below are the label stems.

17.2 2. Line-up of the core results, in teaching order

17.2.1 Section 1. Conventions and recollections

1.1 `conv:modgalrep:notation` (convention). C1 to C6 of Part 1.3 typeset as `convention` environments, with the twist dictionary of C4 displayed: for χ of conductor r, $f \otimes \chi = (f^\chi)^{\Sigma(r)}$; $f \otimes \chi = f^\chi$ iff $a_q(f^\chi) = 0$ for all $q \mid r$; $(f \otimes \chi) \otimes \chi = f^{\Sigma(r)}$ for χ quadratic. Example (checked): $f_{11a} \otimes \chi_{-11} = f_{121d}$ exactly ($a_{11}(f_{121d}) = 0$), while $f_{539c} \otimes \chi_{-7} = (f_{77c})^{\{7\}} \neq f_{77c}$ since $a_7(f_{77c}) = -1$. Paper: Section 3 (cases (b), (c)), Lemma lem:twist(iii), Theorem thm:twist (ι), important_notes.md points 20, 32.

1.2 `prop:modgalrep:newforms-recap` (proposition, a recollection; P with the proofs cited from Chapters 12, 13). For f as in C1: (i) $\phi_f(T_n) = a_n(f)$; $a_{mn} = a_m a_n$ for $(m,n) = 1$; $a_{p^{k+1}} = a_p a_{p^k} - p a_{p^{k-1}}$ for $p \nmid N$ and $a_{p^k} = a_p^k$ for $p \mid N$; (ii) $a_p(f) \in \{\pm 1\}$ if $p \| N$ and $a_p(f) = 0$ if $p^2 \mid N$ (Atkin-Lehner, their Theorem 3, as cited by the paper; reproved locally in 3.5); (iii) strong multiplicity one: two newforms (of any levels, trivial character) with a_p equal for all p outside a set of density zero are equal; (iv) $S_2(\Gamma_0(N)) \otimes \overline{\mathbb{Q}} = \bigoplus_{M|N} \bigoplus_h \bigoplus_{d|N/M} \overline{\mathbb{Q}}\, h(dz)$ over newforms h of level M; the span of the $h(dz)$ is $\mathbb{T}$-stable, and these spans are mutually orthogonal for the Petersson product; (v) $L(f,s) = \prod_{p\nmid N}(1-a_p p^{-s}+p^{1-2s})^{-1} \prod_{p|N}(1-a_p p^{-s})^{-1}$. Reference: Chapter 13 (`\Cref{ch:hecke}`); AtkinLehner70; DiamondShurman Chapter 5 (cite the sections). Paper: Lemma lem:old, Lemma lem:crit, Section 3, Theorem thm:quant Step 1.

17.2.2 Section 2. The Galois representation of a newform

2.1 `thm:modgalrep:eichler-shimura` (theorem, the construction of Eichler, Shimura and Deligne-Rapoport for weight 2; S for the construction, R for the inputs). Let f be as in C1, $I_f = \ker(\phi_f \colon \mathbb{T} \to \mathcal{O}_f)$ and $J_f = J_0(N)/I_f J_0(N)$ (Chapter 14; Shimura's A_f, written J_f because A_f is the paper's adjoint $\mathrm{ad}^0 V$, which this chapter also uses, Appendix B, B-C12(c)). Then: (i) J_f is an abelian variety over $\mathbb{Q}$ of dimension

$[K_f:\mathbb{Q}]$, with good reduction at every $p\nmid N$, and $\mathbb{T}/I_f$ (an order in K_f) acts on it by $\mathbb{Q}$-endomorphisms; (ii) for every prime ℓ, $V_\ell(J_f)$ is free of rank 2 over $K_f\otimes\mathbb{Q}_\ell=\prod_{\lambda|\ell}K_{f,\lambda}$, and $V_\lambda(f):=V_\ell(J_f)\otimes_{K_f\otimes\mathbb{Q}_\ell}K_{f,\lambda}$ carries a continuous representation $\rho_{f,\lambda}\colon G_\mathbb{Q}\to\mathrm{GL}_2(K_{f,\lambda})$ with: (a) $\rho_{f,\lambda}$ is unramified at every $p\nmid N\ell$; (b) for such p, $\det(X-\rho_{f,\lambda}(\mathrm{Frob}_p))=X^2-a_p(f)X+p$ (the Eichler-Shimura congruence relation $T_p=\mathrm{Frob}_p+p\,\mathrm{Frob}_p^{-1}$ on J_f mod p, Chapter 14); (c) $\det\rho_{f,\lambda}=\chi_\ell$ (Weil pairing and the Rosati-stable polarisation; or Chebotarev from (b)); (d) $\rho_{f,\lambda}$ is odd: $\det\rho_{f,\lambda}(c)=-1$ for complex conjugation c; (e) $\rho_{f,\lambda}$ is absolutely irreducible (2.2). (iii) The dual $\rho_{f,\lambda}^\vee\simeq\rho_{f,\lambda}(-1)$ is realised on the rank-two $K_{f,\lambda}$-space $H^1_{\mathrm{et}}(X_0(N)_{\overline{\mathbb{Q}}},\mathbb{Q}_\ell)\otimes_{\mathbb{T}\otimes\mathbb{Q}_\ell}K_{f,\lambda}$ (the ϕ_f-eigenspace of the etale cohomology), on which the geometric Frobenius at $p\nmid N\ell$ has trace $a_p(f)$ (Chapter 8, 11.8; Chapter 14). Proof sketch to give: (b) from Eichler-Shimura on $J_0(N)$ and the freeness in (ii) (which follows from $\dim_\mathbb{Q}H_1(X_0(N),\mathbb{Q})[I_f\otimes\mathbb{Q}]=2[K_f:\mathbb{Q}]$, i.e. multiplicity one of ϕ_f in H^1, Chapter 14); (a) from Neron-Ogg-Shafarevich for abelian varieties (Serre-Tate) and good reduction of $J_0(N)$ outside N (Igusa; stated). Reference: Shimura71, Chapter 7 (Theorem 7.14 and the construction in 7.5; verify the numbers); DiamondShurman Chapter 9 (cite the sections); DDT Theorem 3.1 (as cited by the paper); DiamondIm Section 12; Deligne (Bourbaki 355) for the general statement. Paper: Section 1.1 (V, T, ρ), Section 4.2 ($M_{f,\lambda}$), Lemma lem:old ($\bar\rho_h$), the optimal-level paragraph ("arises from a newform").

2.2 `prop:modgalrep:irreducible` (proposition; S). For f as in C1 and every λ, $\rho_{f,\lambda}$ is absolutely irreducible. Sketch to give (weight 2): if $\rho_{f,\lambda}\otimes\overline{K}_{f,\lambda}$ had a stable line, its character ψ_1 and the quotient ψ_2 would be Hodge-Tate (sub-quotients of $V_\ell(J_f)$, Faltings, Tate; stated), hence of the form $\varepsilon_i\chi_\ell^{k_i}$ with ε_i of finite order and $\{k_1,k_2\}=\{0,1\}$ (Serre's theorem on abelian Hodge-Tate characters, stated); then $a_p=\varepsilon_1(p)+\varepsilon_2(p)p$ up to order for a density-one set of p, so $|a_p|\geq p-1>2\sqrt{p}$ for $p\geq 7$, contradicting the Ramanujan-Petersson bound (Chapter 12, 6.5; Weil). Reference: Ribet77 (the general case, add to `bib/ch15.bib`, Part 6.5). Paper: background (the paper uses only residual irreducibility, which is a hypothesis).

2.3 `thm:modgalrep:deligne` (theorem; R). (Deligne; Deligne-Serre for weight 1.) For a normalised newform f of weight $k\geq 2$, level N and character ε, and $\lambda\mid\ell$, there is a continuous irreducible $\rho_{f,\lambda}\colon G_\mathbb{Q}\to\mathrm{GL}_2(K_{f,\lambda})$, unramified outside $N\ell$, with $\det(X-\rho_{f,\lambda}(\mathrm{Frob}_p))=X^2-a_pX+\varepsilon(p)p^{k-1}$ for $p\nmid N\ell$ and $\det\rho_{f,\lambda}=\varepsilon\chi_\ell^{k-1}$; it is odd. Reference: Deligne (Bourbaki 355); DiamondShurman 9.6 (cite the section). Paper: background (only weight 2 is used; the optimal-level theorem allows a nebentypus, 6.4).

2.4 `prop:modgalrep:lattices` (proposition; P from Chapters 4 and 5). (i) $\rho_{f,\lambda}$ stabilises an $\mathcal{O}_\lambda$-lattice L (compactness of $G_\mathbb{Q}$; Chapter 5, galois:continuous-reps); the reduction $\bar\rho_L\colon G_\mathbb{Q}\to\mathrm{GL}_2(k_\lambda)$ has semisimplification independent of L (Chapter 4, 10.2), written $\bar\rho_{f,\lambda}^{\mathrm{ss}}$; (ii) $\bar\rho_{f,\lambda}$ is odd, and **for ℓ odd it is irreducible over k_λ iff absolutely irreducible**: $\bar\rho(c)$ has the distinct eigenvalues $1,-1\in k_\lambda$, and a stable line over $\bar k_\lambda$ is an eigenline of $\bar\rho(c)$, hence defined over k_λ (as Chapter 11, 1.4(i), for E); (iii) if $\bar\rho_L$ is irreducible, every stable lattice is homothetic to L (Chapter 4, 10.3, the lemma PLAN.md calls Ribet's lemma on lattices), so $\bar\rho_{f,\lambda}$ is well defined up to isomorphism; (iv) for $\ell\nmid N$ the "residual representation attached to f and λ" in the paper means $\bar\rho_{f,\lambda}$ for any choice of L. Paper: Section 1.1 ($\bar\rho$), Lemma lem:old, Section 4.2 ("homothetic when $\bar\rho$ is irreducible"), important_notes.md point 10.

2.5 `cor:modgalrep:congruent-newforms` (corollary; P given 2.1, 2.4 and Chapter 5, galois:congruent-ap). Let f, h be newforms of weight 2 with trivial character, of levels N_f, N_h, with coefficients in the ring of integers $\mathcal{O}$ of a number field, and $\lambda\mid\ell$ a prime of $\mathcal{O}$. (i) If $a_p(f)\equiv a_p(h)\pmod\lambda$ for all p in a set of density one, then $\bar\rho_{f,\lambda}^{\mathrm{ss}}\simeq\bar\rho_{h,\lambda}^{\mathrm{ss}}$ over k_λ. (ii) If moreover $\bar\rho_{f,\lambda}$ is irreducible and ℓ is odd (hence $\bar\rho_{f,\lambda}$ is absolutely irreducible, 2.4(ii)), then $\bar\rho_{h,\lambda}\simeq\bar\rho_{f,\lambda}$ (over k_λ; in particular over a common finite extension of $\mathbb{F}_\ell$ when the two are first defined over different fields). (iii) Conversely $\bar\rho_{f,\lambda}^{\mathrm{ss}}\simeq\bar\rho_{h,\lambda}^{\mathrm{ss}}$ gives $a_p(f)\equiv a_p(h)\pmod\lambda$ for every $p\nmid N_fN_h\ell$. Reference: as cited. Paper: Lemma lem:old (case $p\nmid\ell N$: "$\bar\rho_h\simeq\bar\rho$ by Chebotarev and Brauer-Nesbitt, $\bar\rho$ being irreducible"; corrected per important_notes.md point 10), Section 3 ("$\bar\rho_g\simeq\bar\rho$, that is, $a_p(g)\equiv a_p(f)\pmod\lambda$ for all $p\nmid 3N$").

2.6 `thm:modgalrep:comparison-E` (theorem; P given Chapters 8, 11, 14 and modularity). Let $E/\mathbb{Q}$ have conductor N, $f=f_E$ (modularity, Chapter 14), E_0 the $X_0(N)$-optimal curve of the class, $\varphi_{E_0}\colon X_0(N)\to E_0$ the modular parametrisation (curly φ: the upright ϕ is the eigencharacter ϕ_f, Appendix B, B-C13(b)), ℓ any prime. (i) $J_f=E_0$ (the quotient of $J_0(N)$ by $I_fJ_0(N)$ is the optimal quotient; Chapter 14), so $T_\ell E_0$ is a $G_\mathbb{Q}$-stable lattice in $V_\lambda(f)$ with $\lambda=\ell$, and $\rho_{E_0,\ell}=\rho_{f,\ell}$ on it; for every E in the class,

$V_\ell E \simeq V_\ell E_0$ through an isogeny. (ii) **(important_notes.md point 40; R here: proved in Chapter 14, its item 9.1, and cited, important_notes.md B19.)** The lattice $M_{f,\lambda} := H^1(X_0(N), \mathbb{Z}_\ell)[\ker \phi_f]$ of DFG04 (Section 4.1 of the paper, Chapter 20) equals $\varphi_{E_0}^* H^1(E_0, \mathbb{Z}_\ell)$, and $\varphi_{E_0}^*$ gives a $G_\mathbb{Q}$-isomorphism $H^1(E_0, \mathbb{Z}_\ell) = \mathrm{Hom}_{\mathbb{Z}_\ell}(T_\ell E_0, \mathbb{Z}_\ell) \simeq T_\ell E_0(-1)$ onto it; no irreducibility is needed. Proof (Chapter 14, 9.1; recalled in outline only): the kernel of $J_0(N) \to E_0$ is connected, so $\varphi_{E_0 *}\colon H_1(X_0(N), \mathbb{Z}) \to H_1(E_0, \mathbb{Z})$ is surjective and $\varphi_{E_0}^*$ is injective with torsion-free cokernel (Chapter 3, saturation); $\mathbb{T}$ acts on the image through ϕ_f; both sides have rank 2 (multiplicity one of ϕ_f in $H^1(X_0(N), \mathbb{Q})$); a saturated sublattice of full rank in a saturated sublattice is equal to it; Galois-equivariance by the etale comparison (Chapter 8, 11.8(c)). (iii) **(important_notes.md point 26.)** If $\bar\rho_{E,\ell}$ is irreducible, every $G_\mathbb{Q}$-stable lattice in $V_\ell(E)^\vee$, in particular $M_{f,\lambda}$ and $(T_\ell E)^\vee$ for any E in the class, is homothetic to $(T_\ell E)^\vee \simeq T_\ell E(-1)$ (2.4(iii)); the paper's "$M_{f,\lambda}$ and $T_\ell E$ are homothetic" must be read with the dual. (iv) Consequently $\mathrm{ad}^0 M_{f,\lambda} = \mathrm{ad}^0 T_\ell E$ inside $\mathrm{End}(V_\ell E)$ under $x \mapsto -x^t$ and the twist-invariance of ad^0 (Chapter 4, 7.7; Chapter 9, 5.5), which is all the paper uses ("so their adjoints coincide"). Reference: Chapters 8, 9, 14; DFG04 Sections 1.6.2, 1.7.3 (the normalisation of $M_{f,\lambda}$ as the paper describes it: checked against the published DFG04 in the Chapter 20 outline, its Part 7, important_notes.md point 73; M_f is the kernel lattice for the full Hecke algebra). Paper: Section 4.2 (the parenthesis on $M_{f,\lambda}$ and $T_\ell E$), Lemma lem:hecke ($L[\wp] = M_{f,\lambda}$), Theorem thm:main ($A = \mathrm{ad}^0 T$), important_notes.md points 26, 39, 40 and B3.

2.7 `prop:modgalrep:twists` (proposition; P given 2.1, strong multiplicity one and Shimura's twisting lemma). Let g be a newform of weight 2, trivial character, level M, and χ a primitive Dirichlet character of conductor r. (i) $\rho_{g,\lambda} \otimes \chi$ is unramified outside $Mr\ell$ and has $\mathrm{tr}(\rho_{g,\lambda} \otimes \chi)(\mathrm{Frob}_p) = \chi(p) a_p(g)$ for $p \nmid Mr\ell$. (ii) $g \otimes \chi \in S_2(\Gamma_0(\mathrm{lcm}(M, r^2)), \chi^2)$ (Chapter 13; AtkinLi78 for the level $\mathrm{lcm}(M, r^2)$; Shimura71, Proposition 3.64, gives the weaker level Mr^2; verify both numbers), and there is a unique newform g^χ, of character χ^2 and level dividing $\mathrm{lcm}(M, r^2)$, with $\rho_{g^\chi,\lambda} \simeq \rho_{g,\lambda} \otimes \chi$; it satisfies $a_n(g^\chi) = \chi(n) a_n(g)$ for $(n, r) = 1$, and $g \otimes \chi = (g^\chi)^{\Sigma(r)}$ (C4). (iii) For χ quadratic, $(g^\chi)^\chi = g$, and the level of g^χ is $M \prod_{q|r} q^{a(\pi_q \otimes \chi_{q^*}) - a(\pi_q)}$, computed type by type in 3.8. (iv) For $E/\mathbb{Q}$ and d a fundamental discriminant, $f_{E^{(d)}} = (f_E)^{\chi_d}$ (Chapter 9, 10.3, 10.4 and strong multiplicity one). Examples (checked): $f_{11a}^{\chi_{-11}} = f_{121d} = f_{11a} \otimes \chi_{-11}$; $f_{121a}^{\chi_{-11}} = f_{121c} = f_{121a} \otimes \chi_{-11}$; $f_{539c}^{\chi_{-7}} = f_{77c}$, while $f_{539c} \otimes \chi_{-7} = f_{77c}^{\{7\}}$; $f_{11a}^{\chi_{-7}} = f_{539d} = f_{11a} \otimes \chi_{-7}$; $f_{20a}^{\chi_{-7}} = f_{980g}$; $f_{980i}^{\chi_{-7}} = f_{140b}$. Paper: Lemma lem:twist(iii), Theorem thm:qual (b), (c), Theorem thm:twist, Theorem thm:quant, the Example of Section 3.

17.2.3 Section 3. Local-global compatibility and local components

3.1 `rem:modgalrep:weil-deligne-recap` (remark, a recollection; P or R as in Chapter 7). From Chapter 7 (`\Cref{ch:local}`): Weil-Deligne representations (r, N) and $\mathrm{WD}(\rho)$ for $p \neq \ell$ (its 7.3, 7.4), $\mathrm{sp}(2) \otimes \mu$ and the classification of two-dimensional Frobenius-semisimple (r, N) (7.5(ii), (iii)), $V^{I_p} = (\ker N)^{r(I_p)}$ (8.1), $L(V, s) = \det(1 - \Phi_p p^{-s} \mid V^{I_p})^{-1}$ (8.2), the Artin conductor $a(V) = \dim V - \dim V^{I_p} + \mathrm{sw}(V)$ (8.3, 8.4), ϵ-factors (8.5), the shapes table (9.4). Stated in one page, no proofs. Paper: background for Lemma lem:euler, Lemma lem:twist(iii).

3.2 `thm:modgalrep:local-langlands` (theorem; R). (The local Langlands correspondence for $\mathrm{GL}_2(\mathbb{Q}_p)$: Kutzko, Bushnell-Henniart.) Fix an isomorphism $\mathbb{C} \simeq \overline{\mathbb{Q}}_\ell$. There is a unique bijection $\pi \mapsto \mathrm{rec}(\pi)$ from isomorphism classes of irreducible admissible infinite-dimensional representations of $\mathrm{GL}_2(\mathbb{Q}_p)$ to isomorphism classes of two-dimensional Frobenius-semisimple Weil-Deligne representations of $W_{\mathbb{Q}_p}$ such that: (i) for characters μ of $\mathbb{Q}_p^\times$ (identified with characters of $W_{\mathbb{Q}_p}$ by local class field theory, Chapter 7, 6.5), $\mathrm{rec}(\pi \otimes (\mu \circ \det)) = \mathrm{rec}(\pi) \otimes \mu$; (ii) $\det \mathrm{rec}(\pi)$ corresponds to the central character of π; (iii) L- and ϵ-factors of pairs and conductors are preserved, in particular $a(\mathrm{rec}\,\pi) = a(\pi)$, the exponent of the conductor (Casselman's newvector theory); (iv) principal series $\pi(\mu_1, \mu_2) \mapsto \mu_1 \oplus \mu_2$, twisted Steinberg $\mathrm{St} \otimes \mu \mapsto \mathrm{sp}(2) \otimes \mu$, supercuspidals $\mapsto$ irreducible $(r, 0)$; for p odd every supercuspidal corresponds to $\mathrm{Ind}_{W_K}^{W_{\mathbb{Q}_p}} \theta$ for a quadratic $K/\mathbb{Q}_p$ and a character θ not factoring through the norm (dihedral). The book fixes **one** normalisation and states it once: the *arithmetic* (Tate-twisted) normalisation in which a weight-two newform with trivial character has $\mathrm{WD}_p(f) \simeq \mathrm{rec}(\pi_p \otimes |\det|^{-1/2})$, so that $\det \mathrm{WD}_p(f) = \|\cdot\|$, the character of $W_{\mathbb{Q}_p}$ that matches $\det \rho = \chi_\ell$ (Chapter 7, 7.1(ii): $\|w\| = \chi_\ell(w)$); the writer must check the half-integral twist against the chosen reference and never mix two normalisations in one formula. Reference:

BushnellHenniart06 (add to `bib/ch15.bib`; cite chapters, not theorem numbers), Rohrlich94 (Sections 8 to 15 for the elliptic-curve dictionary; add to `refs.bib`, Part 6.5), Casselman73 (conductors; add to `bib/ch15.bib`). Paper: Lemma lem:euler ("local component ... Steinberg"), Lemma lem:twist(iii).

3.3 `def:modgalrep:local-component` (definition and theorem; R). To a newform f of weight 2 and level N with trivial character corresponds a cuspidal automorphic representation $\pi_f = \bigotimes'_v \pi_v$ of $\mathrm{GL}_2(\mathbb{A}_{\mathbb{Q}})$ (Jacquet-Langlands); π_p is its *local component at* p, with trivial central character and conductor exponent $a(\pi_p) = v_p(N)$ (Casselman73). The Hecke eigenvalue $a_p(f)$ is determined by π_p: it is the eigenvalue of the normalised U_p (or T_p) on the new vector. Reference: DiamondIm Section 11 (cite the section); BushnellHenniart06; Loeffler-Weinstein for computing π_p in practice (add `LoefflerWeinstein12` to `bib/ch15.bib`). Paper: Lemma lem:euler, Lemma lem:twist(iii) (the phrase "the local component of g at q").

3.4 `thm:modgalrep:carayol` (theorem, local-global compatibility at $p \neq \ell$; R, with (ii), (iii) P from (i)). (Langlands, Deligne, Carayol.) For f as in C1, every prime p and every $\lambda \mid \ell$ with $\ell \neq p$: (i) $\mathrm{WD}_p(f) \simeq \mathrm{rec}(\pi_p \otimes |\det|^{-1/2})$ (normalisation of 3.2); (ii) the Artin conductor exponent of $\rho_{f,\lambda}|_{G_{\mathbb{Q}_p}}$ is $v_p(N)$, so **the conductor of $\rho_{f,\lambda}$ is** N (for $\ell \nmid N$; for $\ell \mid N$, its prime-to-ℓ part is that of N); (iii) $L_p(f,s) = L(V_\lambda(f)|_{G_{\mathbb{Q}_p}}, s-1) = \det(1 - \Phi_p p^{1-s} \mid V_\lambda(f)^{I_p})^{-1}$, i.e. $a_p(f)$ is the trace of the geometric Frobenius Φ_p on $V_\lambda(f)^{I_p}$ multiplied by p (good p: Φ_p-eigenvalues α^{-1}, β^{-1} and $p\alpha^{-1} + p\beta^{-1} = \beta + \alpha$; multiplicative p: the eigenvalue $\varepsilon(\mathrm{Frob}_p)p^{-1}$ on the line, giving $a_p = \varepsilon(\mathrm{Frob}_p)$); equivalently $a_p(f) = \mathrm{tr}(\Phi_p \mid (V_\lambda(f)^\vee)^{I_p})$, the trace of the geometric Frobenius on the inertia invariants of the cohomological realisation (C2). Reference: Carayol86 (Theoreme A; the paper's Carayol89 is the mod ℓ paper), Langlands73 (add to `bib/ch15.bib`), Deligne73 (add to `refs.bib`), DDT Theorem 3.1 (as cited by the paper for (e)). Paper: Lemma lem:euler, Lemma lem:twist(iii), Lemma lem:old (cases $p \,\|\, N$), Section 1.1 (δ_p), the optimal-level paragraph ("the conductor").

3.5 `thm:modgalrep:local-types` (theorem and table; P on the Galois side given 3.4 and Chapter 7, the automorphic names R). Let f be as in C1, $p \neq \ell$, $V = V_\lambda(f)|_{G_{\mathbb{Q}_p}}$, $\delta_p = \dim V^{I_p}$. Exactly one of the following holds.

case	$v_p(N)$	$\mathrm{WD}_p(f)$	π_p	δ_p	$a_p(f)$
(U)	0	unramified, Frob_p-eigenvalues α, β, $\alpha+\beta = a_p$, $\alpha\beta = p$	unramified principal series	2	a_p
(St)	1	$\mathrm{sp}(2) \otimes \mu$, μ unramified, $\mu^2 = 1$	unramified twist of Steinberg	1	± 1
(RSt)	$2a(\chi)$	$\mathrm{sp}(2) \otimes \mu\chi$, μ unramified, χ ramified quadratic	ramified twist of Steinberg	0	0
(RPS)	$2a(\mu_1) \geq 2$	$\mu_1 \oplus \mu_2$, both ramified, $\mu_1\mu_2$ unramified on I_p	ramified principal series	0	0
(SC)	≥ 2	irreducible, $N = 0$	supercuspidal	0	0

In case (St): $\rho_{f,\lambda}|_{G_{\mathbb{Q}_p}} \simeq \left(\begin{smallmatrix} \varepsilon\chi_\ell & * \\ 0 & \varepsilon \end{smallmatrix}\right)$ with ε unramified, $\varepsilon(\mathrm{Frob}_p) = a_p(f)$, non-split; on an $\mathcal{O}_\lambda$-lattice the same shape holds (DDT Theorem 3.1(e), as cited by the paper; letter and number not re-checked here), and the residual representation $\bar{\rho}|_{D_p}$ is an extension of $\bar{\varepsilon}$ by $\bar{\varepsilon}\bar{\chi}_\ell$, unramified iff the corner vanishes on I_p modulo λ, and non-split whenever it is ramified, with a unique stable line then (Chapter 4, 3.6). In

particular $a_p(f) = 0$ **and** $V^{I_p} = 0$ **whenever** $p^2 \mid N$**, and** $\delta_p = 2, 1, 0$ **according as** $p \nmid N$**,** $p \| N$**,** $p^2 \mid N$**.** Proof (P, to be written in full): by 3.4(ii) $a(V) = v_p(N)$; $\det V = \chi_\ell$ is unramified at p. If $a(V) = 0$, (U). If $a(V) = 1$: V is tame and $\dim V^{I_p} = 1$; $N = 0$ would make $r|_{I_p} = 1 \oplus \psi$ with ψ ramified, contradicting $\det|_{I_p} = 1$; so $N \neq 0$ and $V \simeq \mathrm{sp}(2) \otimes \mu$ (Chapter 7, 7.5(ii)) with μ unramified ($a = 1$), $\mu^2 = 1$ from det, and $a_p(f) = \mu(\mathrm{Frob}_p)$ from 3.4(iii). If $a(V) \geq 2$: $\dim V^{I_p} = 2$ is excluded ($a = 0$); $\dim V^{I_p} = 1$ is excluded as before ($N = 0$ gives $\det|_{I_p} \neq 1$; $N \neq 0$ gives $\mathrm{sp}(2) \otimes \mu$ with μ unramified, $a = 1$); so $V^{I_p} = 0$, $L_p = 1$, and $a_p(f) = 0$ by 3.4(iii); the three subcases are the classification of Chapter 7, 7.5. This is a local proof of Atkin-Lehner's Theorem 3 for weight 2 (the paper cites AtkinLehner70 Theorem 3 in Lemma lem:old). For $f = f_E$ the Galois-side table is also Chapter 11's 3.1, 3.2, 3.4, 4.1. Paper: Section 1.1 ("$\delta_p = 2, 1, 0$"), Lemma lem:euler, Lemma lem:twist(iii), Lemma lem:old (cases $p \| N$, $p^2 \mid N$), Section 4.1 (M^Σ, δ_p), Theorem thm:dfg14 (only $\delta_p \geq 1$ contribute).

3.6 `prop:modgalrep:kodaira-dictionary` (proposition and table; P given Chapter 11, 3.2, 3.4, 4.1 and 3.5, R for the automorphic names; Rohrlich94). Let $f = f_E$, $q \geq 5$ a bad prime, $q \neq \ell$. The local type of f at q by Kodaira type:

Kodaira type at q	$\rho_{E,\ell}(I_q)$	type of π_q
I_n, $n \geq 1$	unipotent, infinite	(St)
I_n^*, $n \geq 1$	$\pm$unipotent	(RSt) with $\chi = \chi_{q^*}$
I_0^*	$\{\pm 1\}$	(RPS) with $\mu_1 = \chi_{q^*}\nu_{\alpha'}$, $\mu_2 = \chi_{q^*}\nu_{\beta'}$ (ν_x unramified, $\mathrm{Frob}_q \mapsto x$; α', β' the Frobenius eigenvalues of $E^{(q)}$)
$\mathrm{II}, \mathrm{II}^*, \mathrm{IV}, \mathrm{IV}^*$	cyclic of order 6, 3	(RPS) if $q \equiv 1 \pmod 3$, (SC) if $q \equiv 2 \pmod 3$
$\mathrm{III}, \mathrm{III}^*$	cyclic of order 4	(RPS) if $q \equiv 1 \pmod 4$, (SC) if $q \equiv 3 \pmod 4$

(On the Galois side the I_0^* row reads $V \simeq V' \otimes \chi_{q^*}$ with $V' = V_\ell E^{(q)}$ unramified; the automorphic labels are stated in the normalisation of 3.2(iv) and are not used in any proof.) Proof of the last two rows: $\rho(I_q) = \langle g \rangle$ cyclic of order m with $\mathrm{Frob}_q\, g\, \mathrm{Frob}_q^{-1} = g^q$ (Chapter 11, 4.1); if $q \equiv 1 \pmod{m'}$ ($m' = 3$ or 4) Frob_q preserves the two eigenlines of g and $V \otimes \overline{K}$ is a sum of two characters (ramified principal series); if $q \equiv -1 \pmod{m'}$ it swaps them and $V \otimes \overline{K}$ is induced from the unramified quadratic extension, irreducible (supercuspidal, "unramified dihedral") (Chapter 7, 9.2(ii)). Examples (checked): 11a1 at 11 (St); 121d2 at 11 (I_5^*, RSt); 539d1 at 7 (I_0^*, RPS); 121c1 at 11 (IV, $11 \equiv 2 \pmod 3$: SC); 121a1 at 11 (II: SC); 726a1 at 11 (III, $11 \equiv 3 \pmod 4$: SC). Paper: Lemma lem:euler (the phrases "local component of f is an unramified twist of the Steinberg representation" and "$V = V' \otimes \chi_{q^*}$ with V' Steinberg"), Lemma lem:twist(iii), Remark rem:types (background).

3.7 `cor:modgalrep:delta-naive` (corollary; P). With δ_p as in 3.5: $a_p(f) \in \{\pm 1\}$ iff $\delta_p = 1$, $a_p(f) = 0$ iff $\delta_p = 0$ (for $p \mid N$); hence the naive local factor of the adjoint, which is built from a_p alone, equals the true one exactly when $\delta_p > 0$ (the comparison is Chapter 19's), and $\mathrm{ad}^0(\rho \otimes \chi) = \mathrm{ad}^0\,\rho$ makes the true adjoint factor twist-invariant while the naive one is not (important_notes.md point 15). Paper: Section 1.1 (definition of L_p^{nv}), Section 4.1 ($M^\Sigma = M \prod p^{\delta_p}$), Theorem thm:dfg14, Theorem thm:quant Step 3 ("$A_g = A_f$").

3.8 `prop:modgalrep:twist-conductor` (proposition and table; P given 3.4, 3.5 and Chapter 7, 8.4). Let q be an odd prime, χ a ramified quadratic character of $G_{\mathbb{Q}_q}$ (tame, $a(\chi) = 1$), and f as in C1. Then $a(\mathrm{WD}_q(f) \otimes \chi)$ is given by:

type of f at q (3.5)	$a(\mathrm{WD}_q(f))$	$a(\mathrm{WD}_q(f) \otimes \chi)$	change of level under $f \mapsto f^{\chi_{q^*}}$
(U)	0	2	$\times q^2$
(St)	1	2	$\times q$

type of f at q (3.5)	$a(\mathrm{WD}_q(f))$	$a(\mathrm{WD}_q(f)\otimes\chi)$	change of level under $f\mapsto f^{\chi_{q^*}}$
(RSt) with the same χ	2	1	$\div q$
(RPS) with $\mu_i=\chi\cdot$(unramified)	2	0	$\div q^2$
(RPS) or (SC) with tame inertia of order 3, 4 or 6	2	2	$\times 1$

Proof: $a(\mathrm{sp}(2)\otimes\nu)=2a(\nu)$ for ramified ν and 1 for unramified ν (Chapter 7, 9.3); for a sum of characters add the conductors; for tame characters ψ of I_q of order 3, 4, 6, $\psi\chi$ has order 6, 4, 3 (so it stays ramified, conductor 1); for (SC) of the tame dihedral kind, WD $\otimes\chi$ is again tame with $V^I=0$, conductor 2. Corollaries: (a) **Lemma lem:twist(iii) (Atkin-Li)**: if $q^2\nmid M$, then $g^{\chi_{q^*}}=g\otimes\chi_{q^*}$ is a newform of level $\mathrm{lcm}(M,q^2)$ (rows (U), (St); $a_q(g^{\chi_{q^*}})=0$ because its type at q is (RSt) or (RPS), so the naive and the newform twist agree, C4); (b) the conductor of $E^{(q)}$ versus E at q by Kodaira type agrees with Lemma lem:twist(ii) (Chapter 10, 8.2): $\mathrm{I}_n\leftrightarrow\mathrm{I}_n^*$ changes $q^1\leftrightarrow q^2$, $\mathrm{I}_0\leftrightarrow\mathrm{I}_0^*$ changes $q^0\leftrightarrow q^2$, and $\mathrm{II}\leftrightarrow\mathrm{IV}^*$, $\mathrm{III}\leftrightarrow\mathrm{III}^*$, $\mathrm{IV}\leftrightarrow\mathrm{II}^*$ keep q^2; (c) at $p\neq q$ nothing changes (χ_{q^*} is unramified at p). Examples (checked by twisting and identifying in `allcurves`): 11a1 (St at 11) to 121d2; 11a3 to 121d1; 11a1 (U at 7) to 539d2; 121a1 (SC) to 121c2; 539c1 (RSt at 7) to 77c1; 539d1 (RPS at 7) to 11a (level $\div 49$); 20a1 (U at 7) to 980g2; 980i1 (RSt at 7) to 140b1. Paper: Lemma lem:twist(iii) and its proof ("whose twist by a ramified quadratic character has conductor exponent 2"), Theorem thm:qual (b) ("$h=g\otimes\chi$ is a newform on $\Gamma_0(Mq)$"), (c) ("on $\Gamma_0(Mq^2)$"), Theorem thm:twist (levels $N_S=N/D$, N/D^2, N), Theorem thm:quant ("$N_S^\Sigma=N$"), Section 1.4.

17.2.4 Section 4. The representation at ℓ when $\ell\nmid N$

4.1 `thm:modgalrep:crystalline` (theorem; R, with (i) S). Let f be as in C1, $\lambda\mid\ell$, $\ell\nmid N$. (i) J_f has good reduction at ℓ, and for every k the $G_{\mathbb{Q}_\ell}$-module $J_f[\ell^k]$ is the generic fibre of a finite flat group scheme over $\mathbb{Z}_\ell$ (the ℓ^k-torsion of the Neron model, an abelian scheme); hence $\bar\rho_{f,\lambda}|_{D_\ell}$ is *finite* ("good" in the sense of DDT Theorem 3.15), as are all its reductions $\rho_{f,\lambda}$ mod λ^k. (ii) $\rho_{f,\lambda}|_{G_{\mathbb{Q}_\ell}}$ is crystalline with Hodge-Tate weights $\{0,1\}$, and the crystalline Frobenius on D_{cris} has characteristic polynomial $X^2-a_\ell(f)X+\ell$ (Fontaine; Faltings; stated); for $\ell\geq 3$ the weights lie in the Fontaine-Laffaille range, so stable lattices correspond to Fontaine-Laffaille modules (FontaineLaffaille; stated, used in Chapter 18 and by DFG04). (iii) The Serre weight of $\bar\rho_{f,\lambda}$ is 2 when it is irreducible (Serre87, Section 2; Edixhoven92; verify the section numbers). Reference: DDT Sections 2.1 and 3.1 (cite the sections); Berger; Chapter 11, 5.1 (the elliptic-curve case). Paper: the optimal-level paragraph ("the restriction to D_ℓ comes from the finite flat group scheme $E'[\ell]$ and is good"), Lemma lem:old (case $p=\ell$), Theorem thm:main (crystalline at ℓ via DFG04, Chapter 18).

4.2 `thm:modgalrep:deligne-fontaine` (theorem; R for (i), (ii), P for (iii)). Let ℓ be odd, f as in C1 with $\ell\nmid N$, $\lambda\mid\ell$, $\bar\rho=\bar\rho_{f,\lambda}$ (any lattice), $\omega=\bar\chi_\ell|_{I_\ell}$, ω_2 a fundamental character of level 2 (Chapter 11, 5.2). (i) (Deligne.) If $a_\ell(f)\not\equiv 0\pmod\lambda$, then $\bar\rho|_{D_\ell}\simeq\begin{pmatrix}\bar\chi_\ell\,\mu(\bar a_\ell^{-1}) & *\\ 0 & \mu(\bar a_\ell)\end{pmatrix}$ with $\mu(x)$ the unramified character sending Frob_ℓ to x. (ii) (Fontaine.) If $a_\ell(f)\equiv 0\pmod\lambda$, then $\bar\rho|_{I_\ell}\otimes\bar k_\lambda\simeq\omega_2\oplus\omega_2^\ell$ and $\bar\rho|_{D_\ell}$ is irreducible. (iii) Consequently $a_\ell(f)$ mod λ is determined by the isomorphism class of $\bar\rho|_{D_\ell}$: it is 0 if $\bar\rho|_{D_\ell}$ is irreducible; otherwise $\bar\rho|_{D_\ell}$ has a unique one-dimensional unramified quotient (unique because $\omega\neq 1$ for ℓ odd), and $a_\ell(f)$ mod λ is the eigenvalue of Frob_ℓ on it. Hence if h is another newform of weight 2 with trivial character and level prime to ℓ and $\bar\rho_{h,\lambda}\simeq\bar\rho_{f,\lambda}$, then $a_\ell(h)\equiv a_\ell(f)\pmod\lambda$. Proof of (iii) from (i), (ii): direct. Reference: Edixhoven92, Section 2 (as cited by the paper; Theorems 2.5 and 2.6 there, verify); Serre87; Wiles88 (ordinary λ-adic representations, for (i)). Paper: Lemma lem:old, case $p=\ell$ ("By the theorems of Deligne and Fontaine describing $\bar\rho_f|_{D_\ell}$ and $\bar\rho_h|_{D_\ell}$ ... Hence $a_\ell(h)\equiv a_\ell(f)$"); the remark after the Example ("$\ell\nmid N$ is used ... in Lemma lem:old at $p=\ell$"). Example (checked): $a_3(121\mathrm{c}1)=2\equiv -1=a_3(11\mathrm{a}1)\pmod 3$ (both ordinary at 3); $a_3(f_{121d})=a_3(11\mathrm{a}1)\chi_{-11}(3)=-1\cdot 1$, and $a_3(121\mathrm{a}1)=2\equiv -1\pmod 3$, as Lemma lem:old requires in the Example.

17.2.5 Section 5. Serre's conjecture: conductor, weight, character

5.1 `def:modgalrep:serre-invariants` (definition; P for the elementary properties). For an odd irreducible continuous $\bar\sigma\colon G_{\mathbb{Q}} \to \mathrm{GL}_2(\bar{\mathbb{F}}_\ell)$: the Serre conductor $N(\bar\sigma) = \prod_{p\neq\ell} p^{a_p(\bar\sigma)}$ (Chapter 11, 6.1: $a_p = \dim\bar\sigma - \dim\bar\sigma^{I_p} + \mathrm{sw}_p$, prime to ℓ by definition, prime to every prime at which $\bar\sigma$ is unramified); the Serre weight $k(\bar\sigma) \geq 2$ (definition by reference: Serre87, Section 2; Edixhoven92), equal to 2 when $\bar\sigma|_{D_\ell}$ is finite (4.1(iii)); the character $\varepsilon(\bar\sigma)$ with $\det\bar\sigma = \varepsilon(\bar\sigma)\bar\chi_\ell^{k-1}$. "$\bar\sigma$ **arises from** a newform g (of level M)" means: $\bar\sigma \simeq \bar\rho_{g,\lambda}\otimes\bar{\mathbb{F}}_\ell$ for some $\lambda \mid \ell$; equivalently (2.5) $\mathrm{tr}\,\bar\sigma(\mathrm{Frob}_p) \equiv a_p(g) \pmod\lambda$ for all $p \nmid M\ell N(\bar\sigma)$ when $\bar\sigma$ is absolutely irreducible. "Modular" means: arises from some newform. Paper: the optimal-level paragraph ("irreducible and modular", "$N(\bar\sigma)$ is the Serre conductor").

5.2 `prop:modgalrep:serre-conductor-divides` (proposition; P). Let f be as in C1, $\lambda \mid \ell$, $p \neq \ell$. Then $a_p(\bar\rho^{\mathrm{ss}}_{f,\lambda}) \leq a_p(\bar\rho_{f,\lambda}) \leq a_p(\rho_{f,\lambda}) = v_p(N)$, with $a_p(\bar\rho) = v_p(N) - (\dim\bar\rho^{I_p} - \dim V^{I_p})$; hence $N(\bar\rho_{f,\lambda}) \mid N$ (the prime-to-ℓ part when $\ell \mid N$). Proof: the wild inertia P_p acts on a stable lattice L through a finite p-group (Chapter 7, 5.7), of order prime to ℓ, so the dimensions of the fixed spaces of the ramification subgroups G_i, $i \geq 1$, are the same on $L\otimes K$ and on $L/\lambda L$ (L is a projective $\mathcal{O}_\lambda[G_i]$-module; Chapter 4, 3.2(ii)) and the Swan conductors agree; L^{I_p} is saturated in L, so $L^{I_p}/\lambda L^{I_p} \hookrightarrow (L/\lambda L)^{I_p}$ and $\dim\bar\rho^{I_p} \geq \dim V^{I_p}$; then 3.4(ii). The first inequality: semisimplification can only enlarge invariants of the tame part (or use the same argument). Reference: Carayol89, Livne89 (the exact list of possible exponents; add `Livne89` to `refs.bib`, Part 6.5). For $f = f_E$ the exact formula $a_p(\bar\rho_{E,\ell}) = f_p(E) - \dim_{\mathbb{F}_\ell}\Phi_p(\bar{\mathbb{F}}_p)[\ell]$ is Chapter 11, 6.2, with its examples in 6.3. Paper: the optimal-level paragraph ("$N(\bar\rho_{E',\ell})$ divides the conductor of E' [Carayol89]"), cases (a), (b), (c) ("$M \mid N/q$", "$M \mid N/q^2$").

5.3 `thm:modgalrep:serre-conjecture` (theorem; R). (Serre's conjecture; Khare-Wintenberger, with Kisin; Khare for level one.) Every odd absolutely irreducible continuous $\bar\sigma\colon G_{\mathbb{Q}} \to \mathrm{GL}_2(\bar{\mathbb{F}}_\ell)$ arises from a newform of weight $k(\bar\sigma)$, level $N(\bar\sigma)$ and character lifting $\varepsilon(\bar\sigma)$. The "strong" form (optimal weight and level) follows from the "weak" form (some weight and level) by the work of Ribet, Carayol, Diamond, Edixhoven, Buzzard and others (Section 6 for the level). Reference: KhareWintenberger; Khare; Serre87; Edixhoven92; Diamond95. Paper: background; the paper uses only the optimal-level direction (6.4) for $\bar\rho_{E,\ell}$, which is modular because E is (modularity, Chapter 14).

17.2.6 Section 6. Lowering the level

6.1 `thm:modgalrep:mazur-principle` (theorem; R, proof idea S). (Mazur's principle.) Let ℓ be odd, f a newform of weight 2 on $\Gamma_0(N)$, $\lambda \mid \ell$ with $\bar\rho = \bar\rho_{f,\lambda}$ irreducible, and $q \,\|\, N$ a prime with $q \neq \ell$, $\bar\rho$ unramified at q and $q \not\equiv 1 \pmod\ell$. Then $\bar\rho$ arises from a newform of weight 2 on $\Gamma_0(N/q)$. Proof idea (one paragraph): $\bar\rho$ occurs in $J_0(N)[\mathfrak{m}]$; at q the Neron model of $J_0(N)$ has toric reduction on the q-new part, with character group X (Grothendieck's monodromy pairing; Raynaud); if $\bar\rho$ were q-new, $J_0(N)[\mathfrak{m}]$ would sit in the toric part, where Frob_q acts as $\pm q$ times something unramified and ± 1 on the quotient, and "unramified at q" with $q \not\equiv 1 \pmod\ell$ forces it to come from the q-old part, i.e. from level N/q (Ihara's lemma, 7.4, separates old from new). Reference: Ribet90, Section 6 (verify); DDT Section 4 (cite the section); Mazur77. Paper: background for "Ribet's theorem" in case (c).

6.2 `thm:modgalrep:ribet` (theorem; R, proof idea S). (Ribet's level-lowering theorem.) Let ℓ be odd, f a newform of weight 2 on $\Gamma_0(N)$ with $\ell \nmid N$, $\lambda \mid \ell$ with $\bar\rho = \bar\rho_{f,\lambda}$ irreducible, and $q \,\|\, N$ with $\bar\rho$ unramified at q. Then $\bar\rho$ arises from a newform of weight 2 on $\Gamma_0(N/q)$. (Ribet90, Theorem 1.1, which also allows $\ell \mid N$ under conditions stated there, in particular $q \not\equiv 1 \pmod\ell$, and "finite at q" when $q = \ell$; quote the hypotheses from the source, verify. The paper's case (c) has $\ell \nmid N$, which is the case stated here.) Proof idea: when $q \equiv 1 \pmod\ell$, Mazur's principle fails; Ribet passes through an auxiliary prime r and the Shimura curve of discriminant qr, comparing the character groups of $J_0(Nr)$ at q and of the Shimura curve's Jacobian at r (Cerednik-Drinfeld uniformisation, "Ribet's exact sequence"), and uses multiplicity one. Reference: Ribet90; DDT Section 4; Ribet91 (without multiplicity one). Paper: case (c) ("this is the case of Ribet's theorem [Ribet90], q exactly dividing the level and $\bar\rho\otimes\chi$ unramified at q").

6.3 `lem:modgalrep:carayol-lemma` (lemma; R). (Carayol's lemma on characters.) Suppose $\bar\sigma$ arises from an eigenform in $S_k(\Gamma_1(M),\varepsilon)$, $k \geq 2$, and let $\varepsilon'\colon (\mathbb{Z}/M)^\times \to \bar{\mathbb{Q}}^\times$ be a character with the same reduction modulo λ as ε. Then $\bar\sigma$ arises from an eigenform in $S_k(\Gamma_1(M),\varepsilon')$ (not necessarily new). In

particular the character may be taken of order prime to ℓ (the Teichmuller lift of $\bar{\varepsilon}$), which is the source of "nebentypus of order prime to ℓ" in 6.4. Reference: Carayol89 (cite the section); DDT Section 4 (the lemma on changing the character; verify the number). Paper: the optimal-level theorem as quoted in Section 3.

6.4 `thm:modgalrep:optimal-level` (theorem; R). (Ribet, Carayol, Diamond; the form of DDT Theorem 3.15, as quoted in the paper.) Let ℓ be odd and $\bar{\sigma}\colon G_{\mathbb{Q}} \to \mathrm{GL}_2(\bar{\mathbb{F}}_\ell)$ irreducible and modular, and if $\ell = 3$ assume $\bar{\sigma}|_{\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{-3}))}$ absolutely irreducible. Then $\bar{\sigma}$ arises from a newform of weight 2 and level $N(\bar{\sigma})\ell^{\delta(\bar{\sigma})}$ with character of order prime to ℓ, where $\delta(\bar{\sigma}) = 0$ when $\bar{\sigma}|_{D_\ell}$ is good (finite flat) and $\delta(\bar{\sigma}) = 1$ otherwise (DDT's precise definition of "good" includes also an ordinary case; quote it). Remarks for the text: (a) the extra hypothesis at $\ell = 3$ is Diamond's; for $\bar{\sigma} = \bar{\rho}_{E,3}$ it excludes exactly the image $N_s(\mathbb{F}_3)$ (Chapter 11, 8.3, 8.6); (b) for $\ell \geq 5$ and $\bar{\sigma} = \bar{\rho}_{E,\ell}$ with $\ell \nmid N$, the analogous condition on $\mathbb{Q}(\sqrt{\ell^*})$ is automatic (important_notes.md point 47; Chapter 11, 8.2), though DDT 3.15 does not need it. Reference: DDT Theorem 3.15 (as cited; verify the statement verbatim); Ribet90; Carayol89; Diamond95. Paper: the optimal-level paragraph of Section 3, cases (a), (b), (c).

6.5 `cor:modgalrep:optimal-level-E` (corollary; P given 6.4, 4.1, 5.2 and Chapter 11). Let ℓ be odd and $E'/\mathbb{Q}$ of conductor N' with $\ell \nmid N'$, $\bar{\sigma} = \bar{\rho}_{E',\ell}$ irreducible, and if $\ell = 3$ image not $N_s(\mathbb{F}_3)$. Then there are a newform g of weight 2 on $\Gamma_0(N(\bar{\sigma}))$ (trivial character) and $\lambda \mid \ell$ with $\bar{\rho}_{g,\lambda} \simeq \bar{\sigma} \otimes k_\lambda$, i.e. $a_p(g) \equiv a_p(E') \pmod{\lambda}$ for every $p \nmid \ell N'$; and $N(\bar{\sigma}) \mid N'$, with $N(\bar{\sigma})$ prime to every prime at which $\bar{\sigma}$ is unramified. Proof (P, every step): $\bar{\sigma}$ is modular (modularity of E'), odd, irreducible, absolutely irreducible (2.4(ii)); $\bar{\sigma}|_{D_\ell}$ comes from $E'[\ell]$, finite flat, so $\delta = 0$ (4.1; Chapter 11, 5.1); 6.4 gives g of weight 2, level $N(\bar{\sigma})$, character ψ of order prime to ℓ; $\det \bar{\rho}_{g,\lambda} = \bar{\psi}\bar{\chi}_\ell$ (2.3) and $\det \bar{\sigma} = \bar{\chi}_\ell$ give $\bar{\psi} = 1$; reduction modulo λ is injective on roots of unity of order prime to ℓ, so $\psi = 1$; finally 5.2 or Chapter 11, 6.2. Remark: if $\ell \mid N'$ and $\bar{\sigma}|_{D_\ell}$ is not finite, the level is $N(\bar{\sigma})\ell$: for 140b1 at $\ell = 5$ (I_1 at 5), $N(\bar{\sigma}) = 4$ and the form is f_{20a} at level 20 (checked: $a_7(f_{20a}) = 2 \equiv (1+7)a_7(140\mathrm{b}1) = -8 \pmod 5$). Paper: the optimal-level paragraph ("So in that case $\bar{\sigma}$ arises from a newform of weight 2 on $\Gamma_0(N(\bar{\sigma}))$"; "$\det \bar{\sigma} = \bar{\chi}_\ell$ forces the nebentypus ... to be trivial").

6.6 `ex:modgalrep:level-lowering` (worked examples; P for the data, all checked). At $\ell = 3$: 121c1 (type IV at 11): $N(\bar{\rho}) = 11$, $g = f_{11a}$; 77c1 (I_3 at 7, I_2 at 11; Ribet's case at $q = 7$): $N(\bar{\rho}) = 11$, $g = f_{11a}$; **52a1** (IV^* at 2, I_2 at 13; the case (a) mechanism at $q = 2$, important_notes.md point 43): $N(\bar{\rho}) = 26$, $g = f_{26b}$ ($a_p(52\mathrm{a}1) \equiv a_p(26\mathrm{b}1) \pmod 3$ for all $p < 80$, $p \neq 2, 3, 13$); 121a1 (II at 11): $N(\bar{\rho}) = 121$, no lowering, but $N(\bar{\rho} \otimes \chi_{-11}) = 11$ (via 121c2). At $\ell = 5$: 140b1 ($\ell \mid N$, $\delta = 1$): level 20; 11a1: $N(\bar{\rho}_{11a1,5}) = 1$, and there is no newform of weight 2 and level 1, consistent with $\bar{\rho}_{11a1,5}$ being reducible ($E[5] \simeq \mathbb{Z}/5 \oplus \mu_5$, Chapter 11), which 6.4 excludes. Serre conductors are quoted from Chapter 11, 6.3 (and recomputed here, Part 6.6). Paper: Section 3, the Example.

6.7 `rem:modgalrep:level-lowering-history` (remark; S). One page: Serre's conjecture (1987) and the epsilon conjecture; Mazur's principle; Ribet's theorem and Fermat's Last Theorem (via the Frey curve, stated in one sentence); Carayol's and Diamond's refinements leading to 6.4; the role of multiplicity one ($J_0(N)[\mathfrak{m}]$ two-dimensional, Chapter 14) and Ribet91 removing it. Reference: Ribet90; Ribet91; Diamond95; DDT Section 4; Serre87. Paper: Section 3 (background).

17.2.7 Section 7. Raising the level

7.1 `thm:modgalrep:level-raising` (theorem; P for necessity, R for sufficiency). (Ribet's level-raising theorem.) Let f be a newform of weight 2 on $\Gamma_0(N)$, $\lambda \mid \ell$ with ℓ odd and $\bar{\rho} = \bar{\rho}_{f,\lambda}$ irreducible, and q a prime with $q \nmid N\ell$. Then $\bar{\rho}$ arises from a newform g of weight 2 on $\Gamma_0(M)$ with $M \mid Nq$ and $q \mid M$ (i.e. new at q) **iff** $a_q(f)^2 \equiv (q+1)^2 \pmod{\lambda}$; and then $a_q(f) \equiv (q+1)a_q(g) \pmod{\lambda'}$ with $a_q(g) = \pm 1$. Necessity (P): g is of type (St) at q (3.5), so $\bar{\rho}_g(\mathrm{Frob}_q)$ has eigenvalues $a_q(g)q$ and $a_q(g)$ (3.5, (St)); $\bar{\rho} \simeq \bar{\rho}_g$ is unramified at q with trace $a_q(f)$, so $a_q(f) \equiv a_q(g)(q+1)$ and $a_q(f)^2 \equiv (q+1)^2$. Sufficiency (R): Ribet84 (the theorem announced at the 1983 ICM; check the hypotheses on ℓ there), DiamondTaylor94 for the general form and for raising the level by q^2 (add to `refs.bib`, shared with Chapter 20, B26); the proof uses Ihara's lemma (7.4) and multiplicity one. Paper: Section 4.1 and Theorem thm:dfg14 ("Wiles's level-raising factor"), the remark after Theorem thm:quant ("the level is raised ... the price is the

level-raising Euler factor"); Lemma lem:old, case $p \,\|\, N$ unramified (the same congruence read backwards: $X^2 - a_p(h)X + p \equiv (X - a_p(f)p)(X - a_p(f))$).

7.2 `prop:modgalrep:level-raising-factor` (proposition; P). Let $p \nmid N$, α, β the roots of $X^2 - a_p(f)X + p$. Then

$$\Big(1 - \frac{\alpha}{\beta p}\Big)\Big(1 - \frac{1}{p}\Big)\Big(1 - \frac{\beta}{\alpha p}\Big) = \frac{p-1}{p} \cdot \frac{(p+1)^2 - a_p(f)^2}{p^2},$$

a nonzero rational (algebraic) number ($|\alpha/\beta| = 1$ by Ramanujan-Petersson, Chapter 12, 6.5). This is $L_p^{\mathrm{nv}}(A_f, 1)^{-1} = L_p(A_f, 1)^{-1}$ at a good p (Chapter 19's normalisation, geometric Frobenius; the eigenvalues $\alpha/\beta, 1, \beta/\alpha$ on $\mathrm{ad}^0 V$ are the same for both Frobenii). At a multiplicative p the factor is $1 - p^{-2}$; at an additive p the naive factor is 1. For $\ell \neq p$ its ℓ-adic valuation is $v_\ell(p-1) + v_\lambda((p+1)^2 - a_p^2)$: the second term is positive iff the level-raising condition of 7.1 holds; the first records congruences at level Np^2 with forms of type (RPS) whose inertial character has ℓ-power order (possible only if $p \equiv 1 \pmod{\ell}$; DiamondTaylor94, stated as a remark, verify). Proof of the identity: $\alpha\beta = p$, $\alpha^2 + \beta^2 = a_p^2 - 2p$, so $(1 - \frac{\alpha}{\beta p})(1 - \frac{\beta}{\alpha p}) = 1 - \frac{a_p^2 - 2p}{p^2} + \frac{1}{p^2} = \frac{(p+1)^2 - a_p^2}{p^2}$ (Chapter 2, 11.4 has the same identity). Paper: Theorem thm:dfg14 and the paragraph after it, the table of Section 1 (I_0^* row, with $a_q(E')$), Lemma lem:euler (I_0^*), Theorem thm:quant Step 3.

7.3 `thm:modgalrep:dfg14` (theorem; R; the paper's Theorem thm:dfg14, stated here, explained in Chapter 20). (DFG04, Proposition 1.4(c), as cited.) For f with rational coefficients, $\ell \nmid 2N$, $\bar\rho_f$ irreducible and Σ a finite set of primes: $\eta_{f,\lambda}^{\Sigma} = \eta_{f,\lambda}^{\emptyset} \prod_{p \in \Sigma} L_p^{\mathrm{nv}}(A_f, 1)^{-1}$; adding an additive prime changes nothing, a good prime contributes 7.2, a multiplicative one $1 - p^{-2}$. The lattices, the pairing and the ideals are defined in Chapter 20 (from the paper's Section 4.1). Reading for this chapter: the Σ-depleted form f^{Σ} of level $M^{\Sigma} = M \prod_{p \in \Sigma} p^{\delta_p}$ (3.5) is "f with its level raised at Σ", and the change of the congruence ideal is the level-raising factor. Paper: Theorem thm:dfg14, proof of Theorem thm:main, Theorem thm:quant Step 3, the remark after Theorem thm:quant.

7.4 `lem:modgalrep:ihara` (lemma; R). (Ihara's lemma, in Ribet's form.) For $q \nmid N$ and $\mathfrak{m}$ a maximal ideal of $\mathbb{T}(N)$ with $\bar\rho_{\mathfrak{m}}$ irreducible, the degeneracy map $J_0(N)^2 \to J_0(Nq)$, $(x, y) \mapsto \alpha_1^* x + \alpha_q^* y$, has kernel contained in the Shimura subgroup embedded antidiagonally, which is Eisenstein; equivalently $H^1(X_0(N), \mathbb{F}_\ell)^2_{\mathfrak{m}} \to H^1(X_0(Nq), \mathbb{F}_\ell)$ is injective. Reference: Ribet84 (verify the theorem number); DDT Section 4 (cite); Ihara's original (not added). Paper: background to Theorem thm:dfg14 (Chapter 20) and to 7.1.

7.5 `ex:modgalrep:level-raising` (worked example; P, checked with PARI's `mfinit`, `mfeigenbasis`, `mffields` and reduction modulo all primes above ℓ). $f = f_{11a}$, $\ell = 3$: $(q+1)^2 - a_q^2$ is $35, 60, 180, 320, 400$ for $q = 5, 7, 13, 17, 19$; it is divisible by 3 exactly for $q = 7, 13$; and among the newforms of level $11q$ exactly these q give a form congruent to f_{11a} modulo a prime above 3: at level 77 the rational form f_{77c} (with $a_7(77\mathrm{c}1) = -1$ and $a_7(11\mathrm{a}1) = -2 \equiv 8 \cdot (-1) \pmod 3$); at level 143 a newform with quartic coefficient field, at one of the two primes above 3 (and not the rational form of 143a1); at levels 55, 187, 209 none. $f = f_{20a}$, $\ell = 5$ ($\bar\rho_{20a,5}$ irreducible): $q = 7$ gives 60 and f_{140b}; $q = 3$ gives 12 and no form at level 60. Paper: background to Theorem thm:dfg14 and to the Example (the congruence $11\mathrm{a} \equiv 77\mathrm{c} \pmod 3$ underlies 539c1, whose twist is 77c1).

17.2.8 Section 8. Congruences between eigenforms: the congruence criterion and the oldform lemma

8.1 `def:modgalrep:congruence-number` (definition and recollection; P for the remarks, R for Agashe-Ribet-Stein). For a newform f of weight 2 on $\Gamma_0(N)$ with rational coefficients (so $f = f_E$): $f^{\perp} \subset S_2(\Gamma_0(N))$ is the orthogonal complement of $\mathbb{C}f$ for the Petersson product; it is $\mathbb{T}$-stable and equals the sum of the generalised eigenspaces of $\mathbb{T} \otimes \mathbb{C}$ for the systems of eigenvalues other than ϕ_f (Chapter 13: f is an eigenvector of the adjoints $T_n^{\dagger}$, since $T_n^{\dagger} = T_n$ for $(n, N) = 1$ and $U_p^{\dagger} = w_N U_p w_N$ with $w_N f = \pm f$; adjoints are †, Appendix B, B-C4(d)); the **congruence number** $r = r_E$ is the largest integer r such that $f \equiv g \pmod r$ coefficientwise for some $g \in S_2(\Gamma_0(N); \mathbb{Z}) \cap f^{\perp}$ (ARS12, Section 2; the paper's definition in Sections 1 and 3). Agashe-Ribet-Stein (ARS12, Theorem 2.2, as cited; Chapter 14): $\deg \varphi_{E_0} \mid r$ and

$v_p(\deg\varphi_{E_0}) = v_p(r)$ for every p with $p^2 \nmid N$. Example (computed here, Part 6.6): for 99a1 (= ARS12's 99A1, $N = 3^2 \cdot 11$), $\deg\varphi = 4$ and $r = 12$, so v_3 differs, as $9 \mid N$ allows. Paper: Section 1, Section 3 (preamble), Lemma lem:crit, the remark after the Example (99A1), Lemma lem:hecke.

8.2 `lem:modgalrep:crit` (lemma, the paper's Lemma lem:crit; P, every step expanded). Let f be as in 8.1, $\mathbb{T}$ the full Hecke algebra of level N, $\phi_f\colon \mathbb{T} \to \mathbb{Z}$, $T_n \mapsto a_n(f)$, $I = \ker\phi_f$, and $J = \mathrm{Ann}_{\mathbb{T}}(f^\perp)$. Then: (i) $I \cap J = 0$, $J = \mathrm{Ann}_{\mathbb{T}}(I)$, and $\mathbb{T}/(I+J)$ is cyclic of order r; equivalently $r\mathbb{Z} = \phi_f(J) = \phi_f(\mathrm{Ann}_{\mathbb{T}} I)$, the congruence ideal of ϕ_f (Chapter 3, 11.1). (ii) If there are a number field with ring of integers $\mathcal{O}$, a prime $\lambda \mid \ell$ of $\mathcal{O}$, and a normalised eigenform $g \in (S_2(\Gamma_0(N)) \otimes \overline{\mathbb{Q}}) \cap f^\perp$ of **all** of $\mathbb{T}$ with coefficients in $\mathcal{O}$ and $a_n(g) \equiv a_n(f) \pmod{\lambda}$ for **every** $n \geq 1$, then $\ell \mid r$. (iii) The same holds after $\otimes\mathbb{Z}_\ell$: $\mathbb{T}_\ell/(I_\ell + J_\ell) \simeq \mathbb{Z}/\ell^{v_\ell(r)}$, $\ell^{v_\ell(r)}$ generates $\phi_f(\mathrm{Ann}_{\mathbb{T}_\ell} I_\ell)$, and it is the largest ℓ^t with $f \equiv h \pmod{\ell^t}$ for some $h \in S_2(\Gamma_0(N);\mathbb{Z}_\ell) \cap (f^\perp \otimes \mathbb{Q}_\ell)$. Proof (the paper's, with every step written out): (1) $\mathbb{T}\otimes\mathbb{Q} = \mathbb{Q}\times\mathbb{T}'$, $\mathbb{T}'$ the image of $\mathbb{T}\otimes\mathbb{Q}$ in $\mathrm{End}(f^\perp)$: the ϕ_f-generalised eigenspace of $\mathbb{T}$ on $S_2(\Gamma_0(N))$ is the line $\mathbb{C}f$ (strong multiplicity one, 1.2(iii), (iv): the system of T_p-eigenvalues, $p \nmid N$, of f occurs on no $h(dz)$ with $h \neq f$), so the idempotent of $\mathbb{T}\otimes\mathbb{Q}$ attached to ϕ_f (Chapter 3, 10.1) projects onto $\mathbb{Q}f$ along $f^\perp$; hence $I = \mathbb{T} \cap (0\times\mathbb{T}')$, $J = \mathbb{T} \cap (\mathbb{Q}\times 0)$, $I \cap J = 0$, $J = \mathrm{Ann}_{\mathbb{T}} I$, and $\mathbb{T}/(I+J) \simeq \phi_f(\mathbb{T})/\phi_f(J) = \mathbb{Z}/\phi_f(J)$ is cyclic, of order r' say. (2) The pairing $S_2(\Gamma_0(N);\mathbb{Z}) \times \mathbb{T} \to \mathbb{Z}$, $(g,t) \mapsto a_1(tg)$, is perfect (Ribet83, Theorem 2.2; AU96, Lemma 2.1; Chapter 13); under it (Chapter 3, 9.6) $S_2[I] = \mathrm{Hom}(\mathbb{T}/I,\mathbb{Z}) = \mathbb{Z}f$ (f is primitive: $a_1(f) = 1$) and $S_2[J] = \mathrm{Hom}(\mathbb{T}/J,\mathbb{Z}) = L := f^\perp \cap S_2(\Gamma_0(N);\mathbb{Z})$. (3) Dualising $\mathbb{T} \hookrightarrow \mathbb{T}/I \times \mathbb{T}/J$, with cokernel $\mathbb{T}/(I+J)$ (Chapter 3, 5.7, 9.6(iii)), $[S_2(\Gamma_0(N);\mathbb{Z}) : \mathbb{Z}f \oplus L] = r'$. (4) The quotient embeds in $\mathbb{Q}f/\mathbb{Z}f$ by projection along $f^\perp$ (Chapter 3, 5.8), so it is cyclic, generated by the class of an h with projection f/r'; then $r'h = f + g_0$ with $g_0 \in L$, so $f \equiv -g_0 \pmod{r'}$ and $r' \leq r$; conversely $f \equiv g_1 \pmod{r}$ with $g_1 \in L$ makes $(f-g_1)/r$ integral with projection f/r, of order exactly r in the quotient, so $r \mid r'$. Hence $r = r'$. (5) For (ii): $\phi_g\colon \mathbb{T} \to \mathcal{O}$ kills J ($g \in f^\perp$) and $\phi_g \equiv \phi_f \pmod{\lambda}$ on all of $\mathbb{T}$ (because $a_n(g) \equiv a_n(f)$ for **every** n, U_p included), so $\phi_g(I) \subset \lambda$; thus $I + J \subset \ker(\mathbb{T} \to \mathcal{O}/\lambda)$, whose image is a subring of a finite field, hence a field: a maximal ideal of residue characteristic ℓ; so $\ell \mid \#\mathbb{T}/(I+J) = r$. (6) Over $\mathbb{Z}_\ell$: $\mathbb{Z}_\ell$ is flat over $\mathbb{Z}$, the pairing stays perfect, and steps (1) to (5) hold verbatim (Chapter 3, 3.4, 11.1(v); Chapter 12, 11.4(iii) for the q-expansion side). Writer's notes: the congruence in (ii) must hold for all n, not only for primes $p \nmid N$: this is why Lemma lem:old exists; the hypothesis "rational coefficients" is the paper's setting ($f = f_E$); the general version (coefficients in $\mathcal{O}_f$, $\phi_f\colon \mathbb{T} \to \mathcal{O}_f$) is Chapter 20's restatement. Paper: Lemma lem:crit, Lemma lem:old (its conclusion), Lemma lem:hecke, Theorem thm:twist, Theorem thm:quant Step 1. Cross-references: Chapter 14 (the same congruence number from the modular-degree side), Chapter 20 (restated as a statement about congruence modules; Chapter 20 does not reprove it).

8.3 `lem:modgalrep:old` (lemma, the paper's Lemma lem:old; P, every case written out). Let f be as in 8.1, ℓ odd with $\ell \nmid N$ and $\bar\rho = \bar\rho_{E,\ell}$ irreducible (hence absolutely irreducible: 2.4(ii), important_notes.md point 10). Let $h \neq f$ be a newform of weight 2 on $\Gamma_0(M)$, $M \mid N$, with coefficients in $\mathcal{O}$, and $\lambda \mid \ell$ with $a_p(h) \equiv a_p(f) \pmod{\lambda}$ for all primes $p \nmid \ell N$. Assume $p \nmid M$ for every prime $p \,\|\, N$ at which $\bar\rho$ is unramified. Then, after enlarging $\mathcal{O}$ and λ, the span V_h of the $h(dz)$, $d \mid N/M$, contains a normalised eigenform g of all of $\mathbb{T}$ with $a_n(g) \equiv a_n(f) \pmod{\lambda}$ for every $n \geq 1$; it lies in $f^\perp$; consequently $\ell \mid r$ (8.2(ii)). Proof: (1) V_h is $\mathbb{T}$-stable and orthogonal to f (1.2(iv); for $M = N$ use $h \neq f$). (2) On V_h: T_n acts by $a_n(h)$ for $(n,N) = 1$; U_p acts by $a_p(h)$ if $p \mid M$ and $v_p(M) = v_p(N)$; for $p \mid N/M$ with $e_p = v_p(N/M)$, on $V_{h,p} = \langle h(p^j z) : 0 \leq j \leq e_p\rangle$: $U_p h(p^j z) = h(p^{j-1}z)$ ($j \geq 1$), $U_p h = a_p(h)h - p\,h(pz)$ if $p \nmid M$, $U_p h = a_p(h)h$ if $p \mid M$ (Chapter 13). (3) Eigenvalues and eigenvectors of U_p on $V_{h,p}$ (Chapter 3, 10.7(v); Chapter 13):

situation at $p \mid N/M$	characteristic polynomial	eigenvalue	eigenvector
$p \nmid M$	$X^{e_p-1}(X^2 - a_pX + p)$	α_p, β_p	$h - \beta_p h(pz)$, $h - \alpha_p h(pz)$
$p \nmid M$, $e_p \geq 2$		0	$h - a_p(h)h(pz) + p\,h(p^2z)$
$p \,\|\, M$	$X^{e_p}(X - a_p)$	$a_p(h) = \pm 1$; 0	h; $h - a_p(h)h(pz)$
$p^2 \mid M$	X^{e_p+1}	0	h ($a_p(h) = 0$)

U_p is not semisimple when 0 is a repeated root, but each listed eigenvalue has an eigenvector. (4) $V_h = \bigotimes_{p|N/M} V_{h,p}$ compatibly with the U_p, and $\mathbb{T}$ is commutative, so for every choice of eigenvalues μ_p ($p \mid N/M$) there is a common eigenform $g \in V_h \otimes \overline{\mathbb{Q}}$ of all of $\mathbb{T}$ with $U_p g = \mu_p g$ ($p \mid N/M$), $U_p g = a_p(h)g$ (other $p \mid N$), $T_n g = a_n(h)g$ ($(n,N)=1$) (Chapter 3, 10.8); normalise $a_1(g)=1$; its coefficients lie in $\mathcal{O}[\alpha_p]$. (5) Choice of μ_p and the congruence at each prime p: * $p \nmid \ell N$: $a_p(g) = a_p(h) \equiv a_p(f)$ by hypothesis; so $\bar\rho_{h,\lambda} \simeq \bar\rho$ (2.5(ii)). * $p = \ell$: weight 2, levels prime to ℓ; $a_\ell(h) \equiv a_\ell(f)$ by 4.2(iii). * $p^2 \mid N$: $a_p(f) = 0$ (3.5). If $v_p(M) = v_p(N)$ then $a_p(h) = 0$ (3.5 for h; AtkinLehner70 Theorem 3). Otherwise $e_p \geq 1$, and $e_p \geq 2$ when $p \nmid M$, so 0 is an eigenvalue: $\mu_p = 0$. * $p \parallel N$, $\bar\rho$ unramified at p: $p \nmid M$ by hypothesis, $e_p = 1$, $\mu_p \in \{\alpha_p, \beta_p\}$. By 3.5 (St) for f, $\bar\rho(\mathrm{Frob}_p)$ has eigenvalues $a_p(f)p$, $a_p(f)$; $\bar\rho_h \simeq \bar\rho$ is unramified at p with characteristic polynomial $X^2 - a_p(h)X + p$ (2.1(ii)(b)), so $X^2 - a_p(h)X + p \equiv (X - a_p(f)p)(X - a_p(f))$ modulo λ, and one of α_p, β_p is $\equiv a_p(f)$ modulo a prime above λ in $\mathcal{O}[\alpha_p]$: take it as μ_p. (This is the level-raising congruence 7.1 read backwards.) * $p \parallel N$, $\bar\rho$ ramified at p: then $p \mid M$ (else $\bar\rho_h$ would be unramified at p, 2.1(ii)(a)), so $v_p(M) = 1 = v_p(N)$ and $a_p(g) = a_p(h) = \pm 1$. Both $\bar\rho|_{D_p}$ and $\bar\rho_h|_{D_p}$ are extensions of an unramified character by its $\bar\chi_\ell$-twist, with Frob_p acting on the unramified quotient by $a_p(f)$, resp. $a_p(h)$ (3.5 (St), on lattices), both non-split because ramified; a non-split extension of two characters has a unique stable line (Chapter 4, 3.6), so the isomorphism $\bar\rho|_{D_p} \simeq \bar\rho_h|_{D_p}$ matches the lines and the quotients: $a_p(h) \equiv a_p(f)$. (Valid also when $p \equiv 1 \pmod{\ell}$, where the two characters coincide.)

(6) From primes to all n: $a_{mn} = a_m a_n$ for coprime m, n (both f and g are normalised eigenforms of all T_n); $a_{p^k} = a_p^k$ for $p \mid N$ ($U_{p^k} = U_p^k$); the recursion $a_{p^{k+1}} = a_p a_{p^k} - p a_{p^{k-1}}$ for $p \nmid N$ (including $p = \ell$). Hence $a_n(g) \equiv a_n(f) \pmod{\lambda}$ for all n. Paper: Lemma lem:old, cases (a), (b), (c). Writer's note: the paper's phrase "$\bar\rho_h \simeq \bar\rho$ by Chebotarev and Brauer-Nesbitt, $\bar\rho$ being irreducible" is corrected by "(hence absolutely irreducible ...) over a common finite extension of $\mathbb{F}_\ell$" in a `paperlink` (important_notes.md point 10); for a weight-two form $\alpha_p \neq \beta_p$ (ColemanEdixhoven), though the argument does not need it.

8.4 `ex:modgalrep:old-examples` (worked examples; P, checked). (a) **Unramified case $p \parallel N$:** $f = f_{77c}$, $\ell = 3$, $h = f_{11a}$ ($M = 11$, $N/M = 7$, $7 \parallel 77$ and $\bar\rho_{77c1,3}$ unramified at 7 since $3 \mid n_7 = 3$): $X^2 + 2X + 7 \equiv (X+1)^2 \pmod 3$, the roots $\alpha, \beta = -1 \pm \sqrt{-6}$ are both $\equiv -1 = a_7(77c1)$ modulo the prime $(3, \sqrt{-6})$ of $\mathbb{Z}[\sqrt{-6}]$; $g = h - \beta h(7z)$ satisfies $g \equiv f_{77c}$ modulo $(3, \sqrt{-6})$ for every $n \leq 3000$ (Sturm bound 16 at level 77, so for all n); hence $3 \mid r(77c1)$, and indeed $r = \deg\varphi = 6$. (b) **Additive case $p^2 \mid N$:** $f = f_{52a}$, $\ell = 3$, $h = f_{26b}$ ($M = 26$, $N/M = 2$, $2 \parallel M$, $e_2 = 1$): $\mu_2 = 0$, $g = h - a_2(h)h(2z) = h - h(2z)$, and $g \equiv f_{52a} \pmod 3$ for all $n \leq 400$ (Sturm bound 14 at level 52, so for all n); $\deg\varphi_{52a1} = 3$. (c) **The Example of Section 3** (8.3 with $M = N$): $V_h = \mathbb{C}h$ and $g = h$; see 9.4. Paper: Lemma lem:old; the Example of Section 3.

17.2.9 Section 9. Proof of Theorem thm:qual; the Example; the remark on $\ell \mid N$

9.1 `thm:modgalrep:qual` (theorem, the paper's Theorem thm:qual, with case (a) extended to $q = 2$; P given the R items 2.1, 3.4, 4.2, 6.4 and Agashe-Ribet-Stein). Let $E/\mathbb{Q}$ have conductor N, ℓ an odd prime with $\ell \nmid N$, $\bar\rho_{E,\ell}$ irreducible and, if $\ell = 3$, absolutely irreducible on $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{-3}))$ (equivalently image not $N_s(\mathbb{F}_3)$, Chapter 11, 8.6). Let $q \neq \ell$ be a prime at which E has additive reduction of one of the kinds: (a) type IV or IV*, $\ell = 3$, and $q \geq 5$ **or** $q = 2$ (the paper: $q \geq 5$; the extension is important_notes.md point 43, proved in Chapter 11, 4.4); (b) type II or II*, $q \geq 5$, $\ell = 3$; (c) type I$_n^*$ with $\ell \mid n$, q odd. Then ℓ divides the modular degree of the $X_0(N)$-optimal curve E_0 of the isogeny class. Remarks: for $\ell \geq 5$ only irreducibility is used; in case (a) at $q = 2$ the hypothesis $3 \nmid N$ excludes $q = 3$; checked on the tables: all 1,051 optimal curves of conductor below 10,000 with type IV or IV* at 2, $3 \nmid N$ and mod 3 image neither reducible nor 3Ns have $3 \mid \deg\varphi$ (Part 6.6). Paper: Theorem thm:qual (statement in Section 1.5), Section sec:proof.

9.2 `sec:modgalrep:proof-qual` (the proof of 9.1, a subsection whose `proof` environment has numbered steps; P). Common start: $f = f_E$, $r = r_{E_0}$; as $\ell \nmid N$, $v_\ell(\deg\varphi_{E_0}) = v_\ell(r)$ (8.1), so it suffices to find h as in Lemma lem:old (8.3). $\bar\rho = \bar\rho_{E,\ell}$. * **Case (a).** $v_q(N) = 2$ (type IV, IV* are additive and tame:

Chapter 11, 4.3 for $q \geq 5$, 4.4 for $q = 2$). By Lemma lem:IV (Chapter 11, 4.3; 4.4 at $q = 2$) the Serre conductor $M = N(\bar{\rho})$ has $v_q(M) = 1$, and $M \mid N$ (5.2), so $M \mid N/q$ and $M < N$. By 6.5 there is a newform g on $\Gamma_0(M)$ with $\bar{\rho}_g \simeq \bar{\rho}$. Put $h = g$: $h \neq f$ (levels), $M \mid N$, and $p \nmid M$ for every p at which $\bar{\rho}$ is unramified (5.1). Lemma lem:old gives $3 \mid r$. * **Case (b).** $v_q(N) = 2$, $\chi = \chi_{q^*}$, $E' = E^{(q)}$. By Lemma lem:twist (i), (ii) (Chapter 10, 8.1, 8.2), E' has type IV* or IV at q, the same types as E elsewhere, conductor N. Its newform is $f_{E'} = f^\chi$ (2.7(iv)), and here $f^\chi = f \otimes \chi$ **because** $a_q(f^\chi) = 0$ ($q^2 \mid N$, 3.5; C4) (the paper's "$f' = f \otimes \chi$, a newform on $\Gamma_0(N)$" is right in this case, but not by Atkin-Li, which needs $q^2 \nmid M$). $\bar{\rho}_{E',3} = \bar{\rho} \otimes \chi$ satisfies the hypotheses of 6.5 (irreducibility and the condition at $\mathbb{Q}(\sqrt{-3})$ are twist-invariant; Chapter 11, 1.3). Lemma lem:IV for E': $M = N(\bar{\rho} \otimes \chi)$ has $v_q(M) = 1$, $M \mid N/q$. 6.5 gives g on $\Gamma_0(M)$ with $a_p(g) \equiv \chi(p) a_p(f) \pmod{\lambda}$ for $p \nmid 3N$. As $q \parallel M$, $h := g \otimes \chi = g^\chi$ is a newform on $\Gamma_0(Mq)$ (3.8(a), Atkin-Li), $Mq \mid N$, and $a_p(h) = \chi(p) a_p(g) \equiv a_p(f)$ for $p \nmid 3N$. $h \neq f$ (important_notes.md point 20): if $h = f$ then $f \otimes \chi = h \otimes \chi = g^{\{q\}}$ (C4: $(g \otimes \chi) \otimes \chi = g^{\{q\}}$), so $f_{E'} = f \otimes \chi$ and g have the same a_p for all $p \neq q$, hence $f_{E'} = g$ by strong multiplicity one; but their levels are N and $M < N$. (The paper's "$g = h \otimes \chi = f \otimes \chi$" is literally false: $h \otimes \chi$ is the q-depletion of g.) Hypothesis of Lemma lem:old: if $p \parallel N$ and $\bar{\rho}$ is unramified at p, then $p \neq q$, $\bar{\rho} \otimes \chi$ is unramified at p, so $p \nmid M$ and $p \nmid Mq$. Lemma lem:old gives $3 \mid r$. * **Case (c).** $v_q(N) = 2$, $\chi = \chi_{q^*}$, $E' = E^{(q)}$ of type I_n at q and conductor N/q (Lemma lem:twist (ii)). Here $f = (f_{E'})^\chi = f_{E'} \otimes \chi$ (Atkin-Li, 3.8(a), as $q \parallel N/q$), while $f \otimes \chi = f_{E'}^{\{q\}}$ **is not a newform on** $\Gamma_0(N/q)$ ($a_q(f_{E'}) = \pm 1$; new point N1, Part 7.1); the paper's "$f' = f \otimes \chi$ is a newform of weight 2 on $\Gamma_0(N/q)$" must be read as "$f' = f_{E'} = f^\chi$". By Lemma lem:tamagawa-exp for E' (Chapter 11, 3.3), $\bar{\rho} \otimes \chi = \bar{\rho}_{E',\ell}$ is unramified at q ($\ell \mid n$), so $M = N(\bar{\rho} \otimes \chi)$ is prime to q and $M \mid N/q^2$. 6.5 (here the step at q is Ribet's theorem 6.2: q exactly divides N/q and $\bar{\rho} \otimes \chi$ is unramified at q) gives g on $\Gamma_0(M)$ with $a_p(g) \equiv \chi(p) a_p(f)$ for $p \nmid \ell N$. As $q \nmid M$, $h := g \otimes \chi = g^\chi$ is a newform on $\Gamma_0(Mq^2)$ (3.8(a)), $Mq^2 \mid N$, congruent to f at all $p \nmid \ell N$. $h \neq f$: if $h = f$ then $g^{\{q\}} = h \otimes \chi = f \otimes \chi = f_{E'}^{\{q\}}$, so $g = f_{E'}$ by strong multiplicity one, of levels M and $N/q \neq M$. The hypothesis of Lemma lem:old holds as in (b). So $\ell \mid r$. * **Conclusion.** $\ell \mid r$ and $v_\ell(\deg \varphi_{E_0}) = v_\ell(r)$. □ Paper: Section sec:proof (all of it).

9.3 `rem:modgalrep:why-these-types` (remark; P). Why exactly these types (Chapter 11, 4.5, table): Theorem thm:qual needs a type whose twist by χ_{q^*} (or itself) has Serre exponent smaller than $v_q(N)$, so that the level can be lowered and raised back to a *different* newform. (i) Types III, III*: Serre exponent 2 for $\bar{\rho}$ and for $\bar{\rho} \otimes \chi_{q^*}$; no lowering (these are the 3Ns-relevant types, Section sec:anomaly). (ii) Type I_0^*: the twist E' has good reduction at q, $M \mid N/q^2$, but 6.5 may return $g = f_{E'}$ itself (when $N(\bar{\rho} \otimes \chi) = N/q^2$), and then $h = g^\chi = f$: no new congruence; the I_0^* term is reached through level raising instead (Theorem thm:quant; the factor $(q+1)^2 - a_q(E')^2$ of 7.2; example 539d1 in 10.4). (iii) Type I_n^* with $\ell \nmid n$: $\bar{\rho} \otimes \chi$ is ramified at q with exponent 1, M has q^1, $h = g^\chi$ has q^2, and 6.5 can return $g = f_{E'}$, giving $h = f$; again Theorem thm:quant handles it. (iv) Types II, IV at $\ell \geq 5$: $\bar{\rho}_{E,\ell}$ has Serre exponent 2 (no ℓ-torsion in Φ_q^{geom}), nothing to lower. Paper: Section 1.5 ("For $\ell \geq 5$ Theorem thm:qual needs only ..."), the remark after Theorem thm:quant ((1): "neither is in Theorem thm:qual").

9.4 `ex:modgalrep:paper-example` (worked example; P, all data checked; the paper's Example after case (c)). Table to typeset:

E	case, ℓ	type at q	twist $E^{(q)}$ (curve)	$N(\bar{\rho} \otimes \chi_{q^*})$, g	$h = g^{\chi_{q^*}}$	$f_E \equiv h$	$\deg \varphi$	bound
121a1	(b), 3	II at 11	121c2 (IV*)	11, f_{11a}	$f_{121d} = f_{11a} \otimes \chi_{-11}$	mod 3, all n (Sturm 22)	6	$e_3 = v_3(12) = 1$
539c1	(c), 3	I_3^* at 7	77c1 (I_3)	11, f_{11a}	$f_{539d} = f_{11a} \otimes \chi_{-7}$	mod 3, all n (Sturm 112)	$2^5 \cdot 3^2$	$e_3 + t_3 = 1 + 1$

E	case, ℓ	type at q	twist $E^{(q)}$ (curve)	$N(\bar\rho\otimes\chi_{q^*})$, g	$h = g^{\chi_{q^*}}$	$f_E \equiv h$	$\deg\varphi$	bound
980i1	(c), 5, $5 \mid N$	I_5^* at 7	140b1 (I_5)	4, level 20 ($\delta=1$), f_{20a}	$f_{980g} = f_{20a}\otimes\chi_{-7}$	mod 5, all n (Sturm 336)	$2^6\cdot 3^2\cdot 5$	$t_5=1$, $e_5=0$
121c1	(a), 3	IV at 11	(none needed)	11, f_{11a}	$h=f_{11a}$, $g=h-h(11z)$	mod 3, all n (Sturm 22)	6	$e_3=1$
52a1	(a) at $q=2,3$	IV^* at 2	(none needed)	26, f_{26b}	$h=f_{26b}$, $g=h-h(2z)$	mod 3, all n (Sturm 14)	3	$e_3(E,2)=1$

Points to make in the text: (i) the congruences hold for **every** n, not only for the primes checked in the paper, by Sturm's bound, since both sides lie in $S_2(\Gamma_0(N))$ (Chapter 12, 11.5; important_notes.md point 46); in the Lemma lem:old language $M=N$ in the first three rows, $V_h=\mathbb{C}h$ and $g=h$. (ii) The curve-level twists are as in the second column (important_notes.md B8): the twist of 121a1 by -11 is 121c2 (type IV^*; 121c1, the optimal curve, has type IV); $f_{11a}\otimes\chi_{-11}$ is the form of the class 121d, whose curve 121d2 is the twist of 11a1 (I_5^*) and 121d1 the twist of 11a3 (I_1^*); $f_{11a}\otimes\chi_{-7}$ is the form of 539d, 539d2 being the twist of 11a1; the twist of 20a1 by -7 is 980g2. (iii) 980i1 has $5\mid 980$, so it lies outside the hypotheses of Theorem thm:qual (and of Theorem thm:main) at $\ell=5$: it illustrates the congruence, not the theorem (important_notes.md point 21); the level of g is $N(\bar\rho)\cdot 5$ because $\bar\rho_{140b1,5}|_{D_5}$ is not finite (6.5, remark). (iv) For 121c1 (case (a) with V_h spanned by $h(z)$, $h(11z)$, $11\,\|\,M=11$, $e_{11}=1$): $\mu_{11}=0$, $g=h-a_{11}(h)h(11z)=f_{11a}-f_{11a}(11z)$, the 11-depletion of f_{11a} ($a_{11}(f_{11a})=1$), congruent to f_{121c} modulo 3 for every n (checked for $n\le 3000$; Sturm bound 22). (v) Modular degrees from `degphi`; e_ℓ, t_ℓ from Definition def:terms (Chapter 19). Paper: the Example of Section 3, Theorem thm:qual, Definition def:terms.

9.5 `rem:modgalrep:l-divides-N` (remark; P for the data). The paper's remark after the Example: $\ell\nmid N$ is used (i) in the optimal-level theorem ($\delta=0$, weight 2: 6.5), (ii) in Lemma lem:old at $p=\ell$ (4.2 needs level prime to ℓ), (iii) to pass from r to $\deg\varphi$, which needs only $\ell^2\nmid N$ and fails without it: 99a1 (ARS12's 99A1, $N=9\cdot 11$) has $\deg\varphi=4$, $r=12$ (both recomputed here, Part 6.6). The data statement "so do they at $q=3$ in cases (a) and (b), where Lemma lem:IV does not apply because the ramification is wild" concerns $q=\ell=3$, i.e. curves with $3\mid N$: the obstruction there is first that $q=\ell$ (inertia at 3 acts on T_3E through an infinite group, so neither Kraus's defect nor Lemma lem:IV makes sense, and Theorem thm:qual requires $q\neq\ell$), and only secondarily wildness (new point N2, Part 7.1). Paper: the remark after the Example.

17.2.10 Section 10. Twisting and the levels in Theorems thm:twist and thm:quant

10.1 `prop:modgalrep:twist-map` (proposition; P given Chapter 13 and 2.7). Let S be a finite set of odd primes, $D=\prod_{q\in S}q$, $\chi_S=\prod_{q\in S}\chi_{q^*}$, and $\iota=\iota_S\colon h\mapsto h\otimes\chi_S=\sum\chi_S(n)a_n(h)q^n$. From Chapter 13 (cited, not reproved): ι maps $S_2(\Gamma_0(M))$ into $S_2(\Gamma_0(\mathrm{lcm}(M,D^2)))$ ($\chi_S^2=1$), preserves integrality, and $\iota\circ T_p=\chi_S(p)\,T_p\circ\iota$ for $p\nmid DM$. Added here: (i) ι sends a generalised eigenform for $(T_p)_{p\nmid DM}$ with system (a_p) to one with system $(\chi_S(p)a_p)$, hence maps the sum of the generalised eigenspaces of systems different from that of g (the $\mathbb{T}$-stable complement $g^\perp$) into the sum of those different from $\chi_S\cdot g$; (ii) on newforms, $\iota(g)=(g^{\chi_S})^S$ (C4), so $\iota(g)=g^{\chi_S}$ iff $a_q(g^{\chi_S})=0$ for all $q\in S$, and ι does not distinguish g from its S-depletion: $\iota(g)=\iota(g^S)$; (iii) Galois meaning: $\rho\mapsto\rho\otimes\chi_S$, so ad^0 and the true local adjoint factors are

unchanged (3.7). Paper: Theorem thm:twist (the map ι and "ι maps $g^{\perp}$ into $f^{\perp}$"), Theorem thm:quant (proof, "$\iota(g^S) = f$", "which kills the coefficients at multiples of D").

10.2 `prop:modgalrep:twist-levels` (proposition; P given 3.8 and Lemma lem:twist(ii)). Let $E/\mathbb{Q}$ have conductor N, S a set of odd primes q with $q^2 \mid N$, $E_S = E \otimes \chi_S$ of conductor N_S, $f = f_E$, $g = f_{E_S}$. (i) If E has type I_n^* at every $q \in S$: $N_S = N/D$; type I_0^* at every $q \in S$: $N_S = N/D^2$; types II, III, IV or a dual at every $q \in S$ with $q \geq 5$: $N_S = N$ (3.8 rows (RSt), (RPS), last row; Chapter 10, 8.2). In each case $\mathrm{lcm}(N_S, D^2) \mid N$, the hypothesis of Theorem thm:twist; in the third case also $\mathrm{lcm}(N, D^2) \mid N_S$. (ii) $\iota(g) = f$ exactly, for all n ($a_n(f) = \chi_S(n)a_n(g)$ for $(n, D) = 1$ by 2.7, and $a_{q^k}(f) = 0$ for $q \in S$ by 3.5). (iii) For $q \in S$ the local type of g at q is (St) in the I_n^* case and (U) in the I_0^* case, so $\delta_q(g) = 1$, resp. 2, and $N_S^S = N_S \prod_{q \in S} q^{\delta_q(g)} = N$: **the S-depleted form g^S has level exactly N**, and $\iota(g^S) = f$. Paper: Theorem thm:twist and the sentence after its proof ("The hypotheses hold when (a) ... (b) ... (c) ..."), Theorem thm:quant (first paragraph of the proof: "$N_S^{\Sigma} = N$: the S-depleted form g^S lives at level N").

10.3 `thm:modgalrep:twist-monotonicity` (theorem; R here, proved in full in Chapter 20). The paper's Theorem thm:twist: with S, D, χ_S, E_S, N_S as in 10.2 and $q^2 \mid N$ for $q \in S$, if $\mathrm{lcm}(N_S, D^2) \mid N$ then $r_{E_S} \mid r_E$; if moreover $\mathrm{lcm}(N, D^2) \mid N_S$ then $r_{E_S} = r_E$. Proof idea in two lines (Chapter 20 writes it out): ι carries a congruence $g \equiv h \pmod r$ with $h \in g^{\perp}$ integral to $f \equiv \iota(h) \pmod r$ with $\iota(h) \in f^{\perp}$ integral (10.1, 10.2), and 8.2. **Data remark (important_notes.md point 16):** the paper's sentence after the proof ("equality in every pair of case (c) and strict inequality only in case (b)") is contradicted by its own Theorem thm:quant in case (a): $v_3(\deg \varphi)$ is 2 versus 1 for 539c1 versus 77c1, 2 versus 1 for 980i1 versus 140b1, 1 versus 0 for 121d1 versus 11a1 (checked from `degphi`), the difference being $e_3(E, q) = v_3(q^2 - 1) = 1$ each time. The book prints the corrected sentence: strict inequality occurs in cases (a) and (b), by the Euler term of Theorem thm:quant. Paper: Theorem thm:twist and the data sentence after it.

10.4 `thm:modgalrep:quant` (theorem; R here, proved in full in Chapter 20; the level-raising reading P). The paper's Theorem thm:quant: for ℓ odd, $\ell \nmid 2N$, $E[\ell]$ irreducible, S the set of primes $q \geq 5$ of type I_0^* or I_n^*, $E_S = E \otimes \chi_S$: $v_\ell(\deg \varphi_E) \geq v_\ell(r_{E_S}) + \sum_{q \in S} e_\ell(E, q) \geq \sum_{q \in S} e_\ell(E, q)$. **Reading as level raising (the remark after Theorem thm:quant, (1)):** Theorem thm:qual lowers the level of $\bar{\rho} \otimes \chi$ and twists back; Theorem thm:quant goes the other way: it raises the level of $g = f_{E_S}$ from N_S to $N_S^S = N$ by depleting at S (10.2(iii)), and by Theorem thm:dfg14 (7.3) the congruence ideal grows by $\prod_{q \in S} L_q^{\mathrm{nv}}(A_g, 1)^{-1}$, which is $\prod(1 - q^{-2})$ over the I_n^* primes and $\prod \frac{q-1}{q} \cdot \frac{(q+1)^2 - a_q(E')^2}{q^2}$ over the I_0^* primes (7.2), i.e. the omitted Euler factors of f (3.7); then ι transports the congruences to f (10.1). Examples (checked): 539d1 (I_0^* at 7, twist 11a, $\ell = 3$): $e_3(E, 7) = v_3(6) + v_3(64 - 4) = 2$, $v_3(r_{11a}) = 0$, $\deg \varphi_{539d1} = 72$, $v_3 = 2$: equality; 539c1 (I_3^* at 7, twist 77c1): $e_3 = v_3(48) = 1$, $v_3(r_{77c}) = v_3(6) = 1$, $v_3(288) = 2$: equality; 121d1 (I_1^* at 11): $e_3 = v_3(120) = 1$, $v_3(r_{11a}) = 0$, $v_3(24) = 1$. Remarks (the paper's remark after the theorem, with the book's corrections): (2) under the hypothesis of Theorem thm:main ($\ell \notin S_f$) the first inequality is an **equality**, with no multiplicity-one hypothesis: by important_notes.md point 39 (Chapter 14) Lemma lem:hecke is an equality for every optimal curve, so (eq:identity) is an equality (point 73), and applying it to E and to E_S (same $\mathrm{ad}^0 T$, same Selmer groups and Tamagawa factors, Euler terms differing by $\sum_{q \in S} e_\ell(E, q)$) gives $v_\ell(\deg \varphi_E) = v_\ell(\deg \varphi_{E_S}) + \sum_{q \in S} e_\ell(E, q)$ (point 74; this supersedes the paper's "when multiplicity one holds at level N", important_notes.md B26); this also removes the freeness hypothesis from Step 4 and partly answers point 24; (3) for the potentially good types II, III, IV and duals, $N_S = N$ (10.2(i)), f is its own depletion, Theorem thm:dfg14 gives no factor, and the omitted factor is reached only through Theorem thm:dfg37 (Chapter 20). The abstract's word "elementary" applies only to Step 1 (important_notes.md point 24). Paper: Theorem thm:quant, the remark after it, Section 1.5 ("The quantitative version of case (c), and of the type I_0^*"), the abstract.

17.2.11 Section 11. The literature on congruence ideals and Tamagawa exponents; Taylor-Wiles in one page

11.1 `def:modgalrep:tamagawa-exponent` (definition and proposition; P from Chapter 11, 3.3). For a prime $q \neq \ell$, the **Tamagawa exponent** $t_q = t_q(\rho_{f,\lambda})$ is the largest $t \geq 0$ such that $\rho_{f,\lambda} \bmod \lambda^t$ is unramified at q (Kim-Ota, Definition 2.12, as cited by the paper: verify). For $f = f_E$ and q multiplicative,

$t_q = v_\ell(n_q) = t_\ell(E, q)$, split or not (Lemma lem:tamagawa-exp; Chapter 11, 3.3(i)). Paper: Section 1.5, Remark rem:ko(2), Definition def:terms.

11.2 `thm:modgalrep:ribet-takahashi` (theorem; R; stated in full in Chapter 14, which this item cites with `\Cref{ch:moddeg}`; here only the one-sentence summary the paper's Section 1.5 needs, important_notes.md B19). (Ribet-Takahashi; Takahashi.) For E optimal of conductor $N = N^+N^-$ with N^- a product of an even number of distinct primes, the degree of the modular parametrisation by $X_0(N)$ and that of the parametrisation by the Shimura curve of discriminant N^- have ratio $\prod_{q|N^-} n_q$ up to primes ℓ at which $E[\ell]$ is reducible (the paper's summary; the writer quotes the exact form, with its hypotheses and the precise local factor, from the sources), and Pasten controls the error term. Reference: RibetTakahashi97, Takahashi01, Pasten; Chapter 14 (where PLAN.md places the full statement). Paper: Section 1.5 (history).

11.3 `thm:modgalrep:pollack-weston` (theorem; R). Pollack and Weston conjectured that moving a prime q into the "new" part of the level changes the congruence ideal by λ^{t_q}, and proved it for weight 2 and square-free level when $\bar\rho$ is ramified at at least two primes (the paper's summary; verify the hypotheses against PollackWeston11 before printing). Paper: Section 1.5.

11.4 `thm:modgalrep:kim-ota` (theorem; R, with the consequence P). (Kim-Ota, Theorem 1.1, as cited.) Let $N = N^+N^-$ with N^- square-free, $\ell \geq 5$, $\ell \nmid N$, $\ell \notin S_f$, and $\bar\rho$ ramified at every $q \mid N^-$ with $q \equiv \pm 1 \pmod{\ell}$. Then $\operatorname{ord}_\lambda \eta_f(N) = \operatorname{ord}_\lambda \eta_f(N^+, N^-) + \sum_{q|N^-} t_f(q)$, where $\eta_f(N)$ is the congruence ideal of f in the full Hecke algebra of level N and $\eta_f(N^+, N^-)$ the congruence ideal in the N^--new quotient. Consequence (P): $\eta_f(N^+, N^-)$ is integral and $\operatorname{ord}_\lambda \eta_f(N) = v_\ell(r_E)$ (8.2(iii)), so $v_\ell(r_E) \geq \sum_{q|N^-} t_\ell(E, q)$; Theorem thm:main contains this inequality for every multiplicative prime, including $q \equiv \pm 1 \pmod{\ell}$ and $\ell = 3$, but not the equality. Paper: Section 1.5, Remark rem:ko(2) (Remark rem:ko(1) is Chapter 18's).

11.5 `rem:modgalrep:taylor-wiles` (remark; S, one page). Taylor-Wiles and $R = \mathbb{T}$ in outline: the universal deformation ring R_Σ of $\bar\rho$ (minimally ramified outside Σ, unrestricted at Σ, flat or Fontaine-Laffaille at ℓ), the Hecke algebra $\mathbb{T}_\Sigma = \mathbb{T}(N^\Sigma)_{\mathfrak{m}}$, the surjection $R_\Sigma \to \mathbb{T}_\Sigma$, the theorem that it is an isomorphism of complete intersections (Wiles95, TaylorWiles; Diamond's refinement), the identification of the dual of the tangent space with the Selmer group $H^1_\Sigma(\mathbb{Q}, W)$, and the numerical criterion (length of the Selmer group equals the valuation of the congruence ideal). What DFG04 Theorem 3.7 needs and why: $\ell \notin S_f$, i.e. $\ell \nmid 2N$ and $\bar\rho|_{\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{\ell^*}))}$ absolutely irreducible (the Taylor-Wiles hypothesis, used to find auxiliary primes); for $\ell \geq 5$, $\ell \nmid N$ and $E[\ell]$ irreducible it is automatic (important_notes.md point 47; Chapter 11, 8.2), so in the range of Theorem thm:main the hypothesis bites only for $\ell = 3$ and image 3Ns; and $\Sigma \not\ni \ell$ with $M_{f,\lambda}$ minimally ramified outside Σ. Details: Chapter 20 (`\Cref{ch:congruence}`); DDT Chapters 2 to 5. Paper: Theorem thm:dfg37, Theorem thm:main hypothesis, Section 1.5 ("by level lowering and twisting rather than by the Taylor-Wiles method").

17.3 3. Section plan with target line counts

Lines of LaTeX, card excluded (PLAN.md 7.3: up to 2,800 for this chapter). Items per section: 2, 7, 8, 2, 3, 7, 5, 4, 5, 4, 5 (52 items).

section	title	contents	target lines
(starred)	Core results at a glance	the card: every item of Part 2, stated as above, with tag and `\Cref`	(excluded)

section	title	contents	target lines
(starred)	Where the paper uses this chapter	Section 1.1 (V, T, δ_p); Section 1.5 (history, thm:qual, Kim-Ota); Section 3 (lem:crit, lem:old, optimal-level paragraph, cases (a), (b), (c), the Example, the remark on $\ell \mid N$); Section 4.1 (M^{Σ}, f^{Σ}); thm:dfg14 and the paragraph after it; Section 4.2 ($M_{f,\lambda}$ versus $T_\ell E$); lem:euler (local components); lem:twist(iii); thm:twist, thm:quant and the remark after it; rem:ko(2); thm:dfg37 (hypothesis)	50
1	Conventions and recollections	1.1 (with the twist dictionary C4 displayed and its two examples), 1.2; `convention` environments C1 to C6	100
2	The Galois representation of a newform	2.1 to 2.7; Eichler-Shimura sketch; the irreducibility sketch; the comparison with $T_\ell E$ and $M_{f,\lambda}$ with the saturation proof; `paperlink` for Section 4.2 (points 26, 40) and for lem:old (point 10)	300
3	Local-global compatibility and local components	3.1 (one-page recall of Chapter 7), 3.2 to 3.8; tables 3.5, 3.6, 3.8 (`booktabs`); the proof of 3.5 in full; `paperlink` for lem:euler ("local components"), lem:twist(iii) (Atkin-Li as a corollary of 3.8), Section 1.1 (δ_p)	380
4	The representation at ℓ	4.1, 4.2; `paperlink` for lem:old ($p = \ell$) and the optimal-level paragraph ("good")	120
5	Serre's conjecture	5.1 to 5.3; the proof of $N(\bar{\rho}_f) \mid N$	150

section	title	contents	target lines
6	Lowering the level	6.1 to 6.7; the trivial-nebentypus proof in full; the examples table (121c1, 77c1, 52a1, 121a1 via 121c2, 140b1, 11a1 mod 5)	270
7	Raising the level	7.1 to 7.5; necessity proved; the factor identity; the level-raising example table for 11a mod 3 and 20a mod 5	200
8	Congruences between eigenforms	8.1 to 8.4; full proofs of lem:crit and lem:old with the U_p table; the two worked oldform examples (77c1 with $\mathbb{Z}[\sqrt{-6}]$, 52a1)	380
9	Proof of Theorem thm:qual	9.1 to 9.5; the proof in numbered steps per case; `paperlink` for points 20, 21, 43, 46, N1, N2; the Example table; 99A1	320
10	Twisting and levels in thm:twist and thm:quant	10.1 to 10.4; `paperlink` for the data sentence after thm:twist (point 16), the remark after thm:quant (points 24, 39)	180
11	Literature; Taylor-Wiles in one page	11.1 to 11.5	130
12	Exercises	the 18 exercises of Part 4.2	150
13	Notes and references	which numbers are cited and which only by section; the new bib entries	50
	total		**2,780**

17.4 4. Coverage table and exercises

17.4.1 4.1 Coverage: PLAN.md scope, Section 6 map rows and paper hooks to line-up items

PLAN.md scope item, Section 6 row, or paper hook	line-up items
$\rho_{f,\lambda}\colon G_{\mathbb{Q}} \to \mathrm{GL}_2(\mathcal{O}_\lambda)$ for a newform of weight 2; Eichler-Shimura construction via $T_\ell J_f$	2.1, 2.4
Deligne for higher weight (statement)	2.3

PLAN.md scope item, Section 6 row, or paper hook	line-up items
unramified outside $N\ell$, $\operatorname{tr}\rho(\mathrm{Frob}_p) = a_p$, $\det = \chi_\ell$, odd, irreducible	2.1, 2.2
residual representation and semisimplification	2.4, 2.5
comparison with $\rho_{E,\ell}$; $T_\ell E$ and $M_{f,\lambda}$ homothetic (to the dual) when $\bar\rho$ irreducible; Ribet's lemma on lattices	2.6 (with Chapter 4, 10.3)
local-global compatibility (Carayol; Langlands-Deligne); $\rho_f\vert_{G_{\mathbb{Q}_p}}$ corresponds to π_p under local Langlands	3.2, 3.3, 3.4
local components of a weight-two newform with trivial character: unramified principal series; unramified twist of Steinberg (DDT 3.1(e)); for $p^2 \mid N$ ramified twist of Steinberg, ramified principal series, supercuspidal	3.5, 3.6
$a_p(f) = 0$ and $V^{I_p} = 0$ when $p^2 \mid N$; $\delta_p = 2, 1, 0$	3.5, 3.7
the twist of a ramified twist of Steinberg has p exactly once in its level	3.8 (row (RSt))
the conductor of ρ_f equals N (Carayol)	3.4(ii)
the Weil-Deligne representation, local L- and ϵ-factors (recalled from Chapter 7)	3.1, 3.4(iii)
the non-split extension at a multiplicative prime has a unique stable line (Lemma lem:old)	3.5 (St), 8.3 case $p \parallel N$ ramified
the local representation at $\ell \nmid N$: crystalline, finite flat, Fontaine-Laffaille	4.1
Deligne-Fontaine description of $\bar\rho\vert_{D_\ell}$; $a_\ell \bmod \lambda$ from it (Edixhoven Section 2)	4.2
Serre conductor and Serre weight; weight 2 for $\bar\rho_{E,\ell}$ with good reduction at ℓ	5.1, 5.2, 4.1(iii)
Serre's conjecture, strong form; Khare-Wintenberger; only the optimal-level direction used	5.3, 6.4
the optimal level theorem of Ribet, Carayol, Diamond (DDT 3.15), hypotheses at $\ell = 3$	6.4
specialisation to $\bar\rho_{E',\ell}$, $\ell \nmid N(E')$: level $N(\bar\rho)$, trivial nebentypus (the paper's argument)	6.5
Ribet's level-lowering theorem ($q \parallel N$, $\bar\rho$ unramified at q), Shimura curves, Ihara/Mazur principle	6.2, 6.1, 7.4, 6.7
Mazur's principle; Carayol's lemma	6.1, 6.3
level raising (Ribet): $a_q^2 \equiv (q+1)^2 \pmod\lambda$	7.1, 7.5
the level-raising factor $(1-\frac{\alpha}{\beta q})(1-\frac{1}{q})(1-\frac{\beta}{\alpha q}) = \frac{q-1}{q}\cdot\frac{(q+1)^2-a_q^2}{q^2}$, the form used in the paper	7.2, 7.3
congruences between newforms: the congruence criterion (Lemma lem:crit, proved in full; cross-references Chapters 14, 20)	8.1, 8.2
the oldform lemma (Lemma lem:old in full, prime by prime: $p \nmid \ell N$, $p = \ell$, $p^2 \mid N$, $p \parallel N$ unramified, $p \parallel N$ ramified)	8.3, 8.4
proof of Theorem thm:qual, cases (a), (b), (c), as a worked application	9.1, 9.2, 9.3

PLAN.md scope item, Section 6 row, or paper hook	line-up items
the examples 121a1, 539c1, 980i1 with the specific congruences	9.4 (with 2.7, 3.8, 6.6)
Ribet-Takahashi, Pollack-Weston, Kim-Ota, Tamagawa exponent t_q (statements with hypotheses)	11.1 to 11.4
Taylor-Wiles and $R = \mathbb{T}$ in one page; what DFG04 Theorem 3.7 needs	11.5
Section 6 row: Steinberg representation, unramified twists, ramified twist has conductor exponent 2	3.5, 3.8 (with Chapter 7)
Section 6 row: level lowering (DDT 3.15), Serre conductor, Serre's conjecture	5.1 to 5.3, 6.1 to 6.6
Section 6 row: level raising (Ribet), Wiles's level-raising factor	7.1 to 7.3
Section 6 row: Deligne-Fontaine description, $a_\ell \bmod \ell$	4.2
Section 6 row: DDT Theorem 3.1(e), local structure at multiplicative primes	3.5 (St)
Section 6 row: Ribet-Takahashi, Pollack-Weston, Kim-Ota, Tamagawa exponent	11.1 to 11.4
Section 6 row: the worked examples 11a1, 20a1, 121a1, 539c1, 980i1, 99A1	6.6, 7.5, 8.1, 9.4, 10.4
hook: Section 3, Lemma lem:crit	8.2
hook: Section 3, Lemma lem:old	8.3, 8.4
hook: Section 3, the optimal-level theorem paragraph	4.1, 5.1, 5.2, 6.4, 6.5
hook: Section 3, cases (a), (b), (c)	9.1, 9.2
hook: Section 3, the Example (121a1, 539c1, 980i1)	9.4
hook: Section 3, the remark on $\ell \mid N$ and 99A1	9.5, 8.1
hook: Section 1.5 (history, Theorem thm:qual, Remark rem:ko)	9.1, 11.1 to 11.4
hook: Section 4.2 ($M_{f,\lambda}$ versus $T_\ell E$)	2.6
hook: Lemma lem:euler (local components: Steinberg and its twists)	3.5, 3.6
hook: Lemma lem:twist(iii) (Atkin-Li)	3.8(a), 2.7
hook: Theorem thm:twist (twist map, eigenforms, levels)	10.1, 10.2, 10.3
hook: Theorem thm:quant (twist map, eigenforms, levels)	10.1, 10.2, 10.4
hook: Theorem thm:dfg14 (the level-raising factor)	7.2, 7.3
hook: the remark after Theorem thm:quant	10.4, 9.3
also: Section 1.1 (δ_p, naive versus true factor at p)	3.5, 3.7
also: Section 4.1 (M^Σ, f^Σ as level raising)	7.3, 10.2(iii)
also: Theorem thm:dfg37 (hypothesis $\ell \notin S_f$)	11.5

17.4.2 4.2 Exercises (Section 12 of the chapter), graded from easy to harder

1. Let ℓ be odd. Show that $\bar{\rho}_{f,\lambda}$ is irreducible over k_λ iff it is absolutely irreducible. Give the step of Lemma lem:old that needs this, and say what goes wrong for $\ell = 2$.
2. Show $(g \otimes \chi) \otimes \chi = g^{\Sigma(r)}$ for χ quadratic of conductor r and $g \otimes \chi = (g^\chi)^{\Sigma(r)}$. Check the paper's claim (Example of Section 3) that $f_{11a} \otimes \chi_{-11}$ is the form of 121d, and show that $f_{539c} \otimes \chi_{-7}$ is **not** the form of 77c.

3. (B8.) Twist 11a1, 11a3, 121a1, 20a1 by the appropriate $-q$, compute minimal models and Kodaira types, and identify the curves in Cremona's tables (121d2, 121d1, 121c2, 980g2). Why does every quantity at $\ell = 3$ in the paper depend only on the class?
4. Prove 3.5: from $a(V) = 1$ and $\det V = \chi_\ell$ deduce $V \simeq \mathrm{sp}(2) \otimes \mu$ with μ unramified, $\mu^2 = 1$; from $a(V) \geq 2$ deduce $V^{I_p} = 0$ and $a_p = 0$. Compare with Atkin-Lehner's Theorem 3.
5. Fill in table 3.8 and check it on 11a1 at 11 and 7, 121a1 at 11, 539c1 and 539d1 at 7, 20a1 at 7 (levels of the twists). Deduce the paper's Lemma lem:twist(iii).
6. Prove $N(\bar\rho_{f,\lambda}) \mid N$ (5.2). Compute $N(\bar\rho)$ for 52a1 at $\ell = 3$ and for 140b1 at $\ell = 5$, and explain why the newform attached to $\bar\rho_{140b1,5}$ has level 20, not 4.
7. Check the paper's claim (optimal-level paragraph) that the nebentypus is trivial: write out the argument for $\bar\rho_{E',\ell}$ with $\ell \nmid N'$. Where is $\ell \nmid N'$ used, and what replaces it for 140b1?
8. (Level raising.) Prove the necessity in 7.1. For f_{11a} and $\ell = 3$ list the $q < 40$, $q \neq 3, 11$, with $3 \mid (q+1)^2 - a_q^2$; for $q = 7$ identify the congruent newform at level 77; for $q = 13$ show the congruent newform at level 143 cannot be rational.
9. Prove the identity of 7.2 and compute the 3-adic valuation of the level-raising factor of f_{11a} at $q = 7$. Relate it to $e_3(539\mathrm{d}1, 7) = 2$ and to $\deg \varphi_{539d1} = 72$.
10. Check the paper's claim (Lemma lem:crit) in the toy algebra $\mathbb{T} = \{(a, b) \in \mathbb{Z}^2 : a \equiv b \pmod r\}$ acting on $\mathbb{Z}^2$ with the standard pairing: compute I, J, $\mathbb{T}/(I + J)$ and the congruence modulus.
11. Check the paper's remark after the Example: compute (by machine) the congruence number of 99a1 as the index of $\mathbb{Z}f \oplus L$ in $S_2(\Gamma_0(99); \mathbb{Z})$ and compare with $\deg \varphi = 4$. Which hypothesis of Agashe-Ribet-Stein fails at 3?
12. Check the paper's claim in Lemma lem:old, case $p \,\|\, N$ unramified, on $f = f_{77c}$, $h = f_{11a}$, $\ell = 3$: compute U_7 on $\langle h, h(7z)\rangle$, its eigenvalues, the prime above 3 in $\mathbb{Z}[\sqrt{-6}]$, and the congruence $h - \beta h(7z) \equiv f_{77c}$ up to the Sturm bound.
13. Check the paper's claim in case (a) on 121c1 and, at $q = 2$, on 52a1: compute $N(\bar\rho_{E,3})$, the lowered newform, the eigenform g of Lemma lem:old, and verify $3 \mid \deg \varphi$. Explain why case (a) holds at $q = 2$ but case (b) is not claimed there.
14. Check the paper's claim in case (b) that $h \neq f$: show that "$g = h \otimes \chi$" is false in general and give the strong multiplicity one argument (important_notes.md point 20).
15. Check the paper's sentence in case (c) "$f' = f \otimes \chi$ is a newform of weight 2 on $\Gamma_0(N/q)$" on 539c1, and correct it.
16. Carry out the construction of Theorem thm:qual for a curve of type I_0^* (539d1, $\ell = 3$) and show that it returns $h = f$. How does Theorem thm:quant nevertheless give $9 \mid \deg \varphi_{539d1}$?
17. Check the paper's data sentence after Theorem thm:twist (important_notes.md point 16) on the pairs 539c1/77c1, 980i1/140b1, 121d1/11a1 at $\ell = 3$, and predict the differences from Theorem thm:quant.
18. Derive $v_\ell(r_E) \geq \sum_{q \| N^-} t_\ell(E, q)$ from the formula of Kim and Ota, and say which of its hypotheses Theorem thm:main removes and which it needs.

17.5 5. Dependencies and forward pointers

Results from earlier chapters used here (by chapter label only, `\Cref{ch:...}`; item numbers refer to their outlines): * Chapter 2 (`ch:rings`): χ_{-3}, χ_{-4}, q^*, Kronecker symbols; the identity $(1 - \frac{\alpha}{\beta q})(1 - \frac{\beta}{\alpha q}) = \frac{(q+1)^2 - a^2}{q^2}$ (11.4). * Chapter 3 (`ch:modules`): saturation (5.5), stacked bases and index (5.6), dualising inclusions (5.7), the congruence module of a decomposition (5.8), algebra-module duality (9.6), Artinian algebras and idempotents (10.1 to 10.4), generalised eigenspaces (10.5, 10.6), non-semisimple operators and the U_p model (10.7), common eigenvectors (10.8), congruence ideals (11.1). * Chapter 4 (`ch:reps`): non-split extensions and the unique stable line (3.6), Brauer-Nesbitt (5.2, 5.3), fixed spaces of groups of order prime to ℓ commute with reduction (3.2(ii)), lattices and homothety (10.1 to 10.3), adjoint invariance (7.7). * Chapter 5 (`ch:galois`): galois:continuous-reps, galois:chebotarev-brauer-nesbitt, galois:congruent-ap (whose newform black box 2.1 discharges). * Chapter 6 (`ch:ant`): coefficient fields, $\mathcal{O}_\lambda$, primes above ℓ (7.1 to 7.4); Frobenius elements; $\mathbb{Z}[\sqrt{-6}]$ for 8.4(a). * Chapter 7 (`ch:local`): Weil group and Weil-Deligne

representations (7.1 to 7.5), invariants and L-factors (8.1, 8.2), Artin, Swan conductors (8.3, 8.4), ϵ-factors (8.5), the shapes (9.1 to 9.4), local class field theory for characters (6.5), quadratic characters of $\mathbb{Q}_q$ (6.3), wild inertia (5.7). * Chapter 8 (`ch:curves`): Jacobians, Tate modules and H^1 as duals (11.7), the etale comparison and the Weil pairing (11.8), pullback and pushforward on H^1 and H_1. * Chapter 9 (`ch:ellcurves`): twists and $\rho_{E^{(d)},\ell} = \rho_{E,\ell} \otimes \chi_d$ (10.3, 10.4), the dual of $T_\ell E$ (5.5), Cremona labels. * Chapter 10 (`ch:reduction`): Lemma lem:twist (i), (ii) (8.1, 8.2), Kodaira types, conductor exponents. * Chapter 11 (`ch:galrep`): $\rho_{E,\ell}$, irreducible iff absolutely irreducible (1.4), twists (1.3), the Tate shape and Lemma lem:tamagawa-exp (3.2, 3.3), potentially multiplicative reduction (3.4), inertia images by type and Lemma lem:IV (4.1, 4.3), type IV at 2 (4.4), the mod 3 table (4.5), the representation at ℓ (5.1 to 5.4), Serre conductors of $\bar{\rho}_{E,\ell}$ (6.1 to 6.3), $\ell \notin S_f$ (8.2, 8.6), the optimal-level statement for E (10.1, 10.2). * Chapter 12 (`ch:modforms`): $S_2(\Gamma_0(N);\mathbb{Z})$, integral basis (11.1, 11.2), Sturm (11.3, 11.4), the three congruences for all n (11.5), nebentypus (6.6), Ramanujan-Petersson (6.5), Atkin-Lehner w_N (12.1). * Chapter 13 (`ch:hecke`, outlined in parallel): Hecke operators, U_p on oldforms, stabilisations and depleted forms, the decomposition of S_2 into spans of $h(dz)$, strong multiplicity one, Atkin-Lehner Theorem 3, Atkin-Li, the twist map ι and its Hecke compatibility, the perfect pairing with $\mathbb{T}$, the recursion for a_{p^k}. **Used in 1.2, 8.2, 8.3, 10.1 as black boxes.** * Chapter 14 (`ch:moddeg`, outlined in parallel): $J_0(N)$, J_f, Eichler-Shimura, modularity, the optimal curve and connected kernel, the modular degree and congruence number, Agashe-Ribet-Stein, multiplicity one, point 39. **Used in 2.1, 2.6, 8.1, 9.2, 10.4.**

Forward pointers (chapters that use this one): * Chapter 16 (`ch:cohomology`): none beyond the shapes (via Chapter 11). * Chapter 17 (`ch:galcoh`): the local types 3.5 for unramified cohomology at $p \mid N$. * Chapter 18 (`ch:blochkato`): crystalline at ℓ (4.1); $A = \mathrm{ad}^0 T$ with $T = T_\ell E$ or $M_{f,\lambda}$ (2.6(iv)); δ_p (3.5). * Chapter 19 (`ch:lfunctions`): local types and the Kodaira dictionary (3.5, 3.6), $L_p(f,s) = L(V_p, s-1)$ (3.4(iii)), the naive factor and δ_p (3.7), the level-raising factor (7.2). * Chapter 20 (`ch:congruence`): Lemma lem:crit (8.2, restated there), the twist map and levels (10.1, 10.2), the level-raising reading of Theorem thm:quant (10.4), Theorem thm:dfg14 preview (7.3), Ihara's lemma (7.4), $\ell \notin S_f$ and Taylor-Wiles (11.5), $M_{f,\lambda}$ (2.6). * Chapter 21 (`ch:paper`): 2.5, 2.6, 3.8, 6.5, 8.2, 8.3, 9.1 to 9.5, 10.3, 10.4, 11.4 (the tour may shorten A3.2 to A3.7 and A4.14 by pointing here; it must adopt the corrections of Part 7.2). * Chapter 22 (`ch:computation`): the verification scripts of Part 6.6 (congruences up to the Sturm bound, `elltwist` and identification, `mfeigenbasis` and reduction modulo λ, the congruence number of 99a1).

17.6 6. Notes for the writer

17.6.1 6.1 Conventions to typeset (Section 1, as `convention` environments)

- C1 to C6 of Part 1.3. The twist dictionary C4 is the single most important convention: write $f \otimes \chi$ only for the naive twist and f^χ for the newform twist, and display $f \otimes \chi = (f^\chi)^{\Sigma(r)}$ once. Every place the paper writes $f \otimes \chi$ or $g \otimes \chi$ gets a `paperlink` saying which is meant.
- Frobenius: as Chapters 7 and 11; $\rho_{f,\lambda}$ normalised by the arithmetic Frobenius trace a_p; the cohomological lattice $M_{f,\lambda}$ is a lattice in the dual. $L_p(f,s) = L(V_\lambda(f)|_{G_{\mathbb{Q}_p}}, s-1)$ with the geometric Frobenius (3.4(iii)); check the formula on 11a1 at 11 ($a_{11} = 1$) and at 2.
- Local Langlands normalisation (3.2): fix one (the book's is the arithmetic one with the $|\det|^{-1/2}$ twist), state it once, and phrase every proof on the Galois side; the automorphic names are labels.
- Macros: `\ad`, `\Sym`, `\nv`, `\Frob`, `\T`, `\degphi`; there is no macro for $\mathcal{O}$; write `\mathcal{O}`. Labels `...:modgalrep:...`; the paper's results proved here get the paper's name in the optional argument (`\begin{lemma}[the paper's Lemma lem:crit]\label{lem:modgalrep:crit}`).

17.6.2 6.2 Pitfalls

- **Naive versus newform twist** (C4): $f \otimes \chi$ is a newform iff $a_q(f^\chi) = 0$ at every $q \mid r$. In case (b) of Theorem thm:qual it is ($q^2 \mid N$); in case (c) the paper's "$f' = f \otimes \chi$ on $\Gamma_0(N/q)$" is false as written (N1) and "$g = h \otimes \chi$" in (b), (c) is false as written (point 20). The conclusions survive by strong multiplicity one; write the arguments as in 9.2.

- **"For all n" versus "for almost all p".** Lemma lem:crit needs congruence of the full eigencharacters (U_p included); a congruence of a_p for $p \nmid \ell N$ is only $\bar{\rho}_h \simeq \bar{\rho}$. Lemma lem:old bridges the gap by choosing U_p-eigenvalues; never shortcut it.
- **Atkin-Li covers only $q^2 \nmid M$.** Case (b) twists a form with $q^2 \mid N$ at the level of f; there the equality $f_{E'} = f \otimes \chi$ holds because both have $a_q = 0$ (3.5), not by Atkin-Li. Use 3.8 for all levels of twists.
- **$\ell \mid N$ examples.** 980i1 and 140b1 have $5 \mid N$: the optimal-level theorem gives level $N(\bar{\rho}) \cdot 5$ there; Lemma lem:old at $p = \ell$ and Agashe-Ribet-Stein are unavailable. Present 980i1 as an illustration only (point 21).
- **Residual irreducibility and absolute irreducibility.** For $\bar{\rho}_{f,\lambda}$ (ℓ odd) they agree (2.4(ii)); for restrictions to $G_{\mathbb{Q}(\sqrt{-3})}$ they do not (Chapter 11). The DDT 3.15 hypothesis at $\ell = 3$ is the latter.
- **Dual versus Tate module.** $M_{f,\lambda} \subset H^1$ is a lattice in $V_\ell E^\vee \simeq V_\ell E(-1)$, not in $V_\ell E$; only ad^0 is insensitive (points 26, 40).
- **Local types at $p = 2, 3$.** Table 3.6 is for $q \geq 5$; at $q = 2$ the dictionary to Kodaira types fails (Chapter 11, 4.6; Remark rem:wild), except types IV, IV* (cyclic inertia of order 3, Chapter 11, 4.4). Do not extend case (b) of Theorem thm:qual to $q = 2$.
- **Level raising hypotheses.** State Ribet's theorem with the hypotheses of the source (7.1 marks them "verify"); the necessity direction is elementary and is what Lemma lem:old uses.
- **Theorem numbers.** Sure only where the paper cites them (DDT Theorems 3.1(e) and 3.15, Ribet83 Theorem 2.2, AU96 Lemma 2.1, AtkinLehner70 Theorem 3, AtkinLi78 Theorem 3.1, ARS12 Theorem 2.2, Edixhoven92 Section 2, DFG04 Proposition 1.4(c), KimOta Theorem 1.1 and Definition 2.12); everything marked "(verify)" must be checked or replaced by a section reference (Shimura71 7.14 and 3.64, Ribet90 Theorem 1.1 and its hypotheses, Carayol86 Theoreme A, Ribet84, DiamondTaylor94, Edixhoven92 Theorems 2.5, 2.6, the Carayol lemma number in DDT).
- **Congruence number versus modular degree.** Lemma lem:crit computes r; the passage to $\deg \varphi$ is Agashe-Ribet-Stein and needs $\ell^2 \nmid N$ (99a1).
- **Optimality.** r depends only on f; $\deg \varphi$ is that of the optimal curve; when $E[\ell]$ is irreducible $v_\ell(\deg \varphi)$ does not depend on which curve of the class is optimal (important_notes.md point 27; Chapter 11, 1.3(iii)).

17.6.3 6.3 What to leave to Chapters 7, 11, 13, 14 and 20

- **Chapter 7:** all Weil-Deligne computations, L- and ϵ-factors, conductors of $\mathrm{sp}(2) \otimes \mu$, the shapes table. Here: a one-page recall (3.1) and the automorphic side.
- **Chapter 11:** every fact about $\rho_{E,\ell}$ (Tate shape, Lemma lem:tamagawa-exp, inertia images, Lemma lem:IV including $q = 2$, Serre conductors of $\bar{\rho}_{E,\ell}$ and their examples, Proposition prop:ns, Lemma lem:h0, $\ell \notin S_f$). Here: the general newform statements and the paper's proofs that combine them (lem:old, thm:qual).
- **Chapter 13:** the q-expansion side of Hecke operators, U_p on oldforms (the formulas and eigenvectors, which 8.3 tabulates by citation), stabilisations and depletions, Atkin-Lehner Theorem 3, Atkin-Li's statement, the twist map ι and its Hecke compatibility, the perfect pairing with $\mathbb{T}$. Here: the Galois reading (3.5, 3.8, 10.1) and the uses.
- **Chapter 14:** $J_0(N)$, J_f, Eichler-Shimura, modularity, modular degree, congruence number, Agashe-Ribet-Stein, multiplicity one, point 39. Here: cited; $r(99a1) = 12$ recomputed only as data.
- **Chapter 20:** congruence ideals over $\mathbb{T}_{\mathfrak{m}}$, Lemmas lem:ddt417, lem:hecke, Theorem thm:dfg14 with its proof idea, DFG04 Theorems 2.7, 3.7, Taylor-Wiles in detail, **the proofs of Theorems thm:twist and thm:quant**. Here: statements (10.3, 10.4), the level and twist facts, the level-raising reading. Lemma lem:crit is proved **here**; Chapter 20 restates it (PLAN.md).

17.6.4 6.4 How each relevant point of important_notes.md is handled

- **Point 10** (irreducible versus absolutely irreducible in Lemma lem:old): 2.4(ii), 2.5(ii), 8.3 (statement and step $p \nmid \ell N$), `paperlink`.

- **Point 16** (data sentence after Theorem thm:twist): 10.3, with the three pairs recomputed; Exercise 17.
- **Point 20** ("$g = h \otimes \chi$" in cases (b), (c)): C4, 9.2 (both cases, with the strong multiplicity one argument); Exercise 14.
- **Point 21** (980i1 outside the hypotheses): 9.4(iii), 6.5 remark ($\delta = 1$), Pitfalls.
- **Point 24** ("elementary" in the abstract): 10.4 remark; with point 39 the middle term needs no freeness.
- **Point 26** ($M_{f,\lambda}$ homothetic to the dual): 2.6(iii), (iv), C2.
- **Point 40** ($M_{f,\lambda} = \varphi_{E_0}^* H^1(E_0, \mathbb{Z}_\ell) \simeq T_\ell E_0(-1)$ exactly for the optimal curve): 2.6(ii), citing Chapter 14, 9.1, which proves it (B19).
- **Point 43** (case (a) of Theorem thm:qual extends to $q = 2$; settled): 9.1 (statement with $q = 2$), 9.2 case (a) (citing Chapter 11, 4.4), the new worked example 52a1 (6.6, 8.4(b), 9.4) and the check on all 1,051 table curves; Exercise 13.
- **Point 46** (the Example's congruences hold for all n by Sturm): 9.4(i), citing Chapter 12, 11.5; the Lemma lem:old reading ($g = h$).
- **Point 47** ($\ell \notin S_f$ automatic for $\ell \geq 5$, $\ell \nmid N$, $E[\ell]$ irreducible): 6.4 remark (b), 11.5.
- **B3** (Chapter 21's A4.4 and A4.6): 2.6 gives the correct statement of the $M_{f,\lambda}$ comparison that A4.6 must use; A4.4 is Chapter 20's and Chapter 3's matter (point 35) and is not touched here.
- **B8** (curve-level twists): 2.7 and 3.8 examples, 9.4 table and (ii), Exercise 3; all pairs recomputed (Part 6.6).
- Also used: point 15 (3.7: the naive factor is not twist-invariant), point 27 (6.2 Pitfalls), point 32 (C3), point 39 (2.6, 10.4), B1 (C2), point 29 (Lemma lem:twist(ii) is Chapter 10's; not repeated).

17.6.5 6.5 References and bibliography

In `refs.bib` already: `DDT`, `Ribet90`, `Ribet91`, `Ribet84`, `Carayol86`, `Carayol89`, `Diamond95`, `DiamondIm`, `Edixhoven92`, `Serre87`, `KhareWintenberger`, `Khare`, `Deligne`, `FontaineLaffaille`, `Wiles88`, `Wiles95`, `TaylorWiles`, `Shimura71`, `AtkinLi78`, `AtkinLehner70`, `Ribet83`, `AU96`, `ARS12`, `DFG04`, `KimOta`, `PollackWeston11`, `RibetTakahashi97`, `Takahashi01`, `Pasten`, `Mazur77`, `ColemanEdixhoven`, `Sturm87`, `DiamondShurman`, `Berger`, `Stein`, `PARI`, `Cremona97`. To add (complete entries, checked against the sources before use; decision B14): * to `refs.bib` (needed by more than one chapter): `Rohrlich94` (D. E. Rohrlich, "Elliptic curves and the Weil-Deligne group", in *Elliptic curves and related topics*, CRM Proc. Lecture Notes 4, Amer. Math. Soc., 1994, 125-157; also planned by Chapter 7), `Deligne73` (P. Deligne, "Les constantes des equations fonctionnelles des fonctions L", in *Modular functions of one variable II*, Lecture Notes in Math. 349, Springer, 1973, 501-597; also planned by Chapter 7), `Livne89` (R. Livne, "On the conductors of mod ℓ Galois representations coming from modular forms", J. Number Theory 31 (1989), 133-141; also planned by Chapter 11); * to `bib/ch15.bib` (this chapter only, so far): `Ribet77` (K. A. Ribet, "Galois representations attached to eigenforms with Nebentypus", in *Modular functions of one variable V*, Lecture Notes in Math. 601, Springer, 1977, 17-51), `Langlands73` (R. P. Langlands, "Modular forms and ℓ-adic representations", in *Modular functions of one variable II*, Lecture Notes in Math. 349, Springer, 1973, 361-500), `BushnellHenniart06` (C. J. Bushnell and G. Henniart, *The local Langlands conjecture for GL(2)*, Grundlehren 335, Springer, 2006), `Casselman73` (W. Casselman, "On some results of Atkin and Lehner", Math. Ann. 201 (1973), 301-314), `LoefflerWeinstein12` (D. Loeffler and J. Weinstein, "On the computation of local components of a newform", Math. Comp. 81 (2012), 1179-1200). Page ranges to be checked. Exception: `DiamondTaylor94` (F. Diamond and R. Taylor, "Non-optimal levels of mod ℓ modular representations", Invent. Math. 115 (1994), 435-462) is also needed by Chapter 20, so it goes into `refs.bib` (decision B14; important_notes.md B26). If Chapter 13 or 19 also cites any of these, move it to `refs.bib` (B14).

17.6.6 6.6 Verification log (computations done for this outline)

Scripts in the session scratchpad (`ch15/v1.py` to `v5.py`, `v2.gp`, `v3.gp`), PARI/GP 2.17 through `cypari2`; data `allcurves`, `degphi`, `galrep` for conductors 0 to 9,999. * **Example of Section 3, for all** n

(point 46): $a_n(121a1) \equiv \chi_{-11}(n)a_n(11a1) \pmod 3$, $a_n(539c1) \equiv \chi_{-7}(n)a_n(11a1) \pmod 3$, $a_n(980i1) \equiv \chi_{-7}(n)a_n(20a1) \pmod 5$ for all $n \le 3000$; `mfsturm` gives 22, 112, 336. Also $f_{121d} = f_{11a} \otimes \chi_{-11}$ and $f_{539d} = f_{11a} \otimes \chi_{-7}$ exactly ($n \le 2000$). * **Newform twists (2.7, 3.8, B8):** a_p equal away from q for 11a $\otimes \chi_{-11}$ = 121d, 11a $\otimes \chi_{-7}$ = 539d, 121a $\otimes \chi_{-11}$ = 121c, 539c $\otimes \chi_{-7}$ = 77c, 980i $\otimes \chi_{-7}$ = 140b ($p < 500$); curve-level: `elltwist` then `ellminimalmodel`, identified in `allcurves`: 11a1 by -11 is 121d2, 11a3 by -11 is 121d1, 11a1 by -7 is 539d2, 121a1 by -11 is 121c2, 539c1 by -7 is 77c1, 980i1 by -7 is 140b1, 20a1 by -7 is 980g2, 121c1 by -11 is 121a2. * **Kodaira types and Serre conductors** (formula of Chapter 11, 6.2): 11a1 (I_5; $\ell = 3$: 11; $\ell = 5$: 1), 11a3 ($\ell = 5$: 11), 121a1 (II; 121), 121c1 (IV; 11), 121c2 (IV^*; 11), 121d1 (I_1^*), 121d2 (I_5^*), 539c1 (I_3^* at 7, I_2 at 11; 539), 539d1 (I_0^* at 7, I_1 at 11), 539d2 (I_0^*, I_5), 77c1 (I_3, I_2; 11), 52a1 (IV^* at 2, I_2 at 13; 26), 26b1 (I_7, I_1; 26), 980i1 (IV^*, I_1 at 5, I_5^* at 7; $\ell = 5$: 196), 140b1 (IV^*, I_1, I_5; 4), 20a1 (IV^*, I_2 at 5; 4), 980g2 (IV^*, I_2, I_0^*; 196), 726a1 (121). * **Level raising (7.5):** for f_{11a}, $\ell = 3$, $q \in \{2, 5, 7, 13, 17, 19\}$, all newforms of level $11q$ (`mfinit([11q,2],0)`, `mfeigenbasis`, `mffields`), reduced modulo every prime above 3 of the coefficient field, compared with $a_p(11a1)$ for $p < 200$, $p \nmid 33q$: congruent forms exactly for $q = 7$ (rational, 77c) and $q = 13$ (quartic field, one of two primes above 3); none for 5, 17, 19 (2: no newforms). For f_{20a}, $\ell = 5$: $q = 7$ yes (rational, 140b), $q = 3$ no. $a_7(77c1) = -1$, $a_7(140b1) = -1$, consistent with $a_q(f) \equiv (q+1)a_q(g)$. 143a1 has $a_2 = 0 \not\equiv -2$. * **Congruence numbers** (index of $\mathbb{Z}f \oplus L$ in the saturated q-expansion lattice of `mfinit([N,2],1)`, with L the image of $T_p - a_p$ for a p separating f): 99a1: $r = 12$, $\deg\varphi = 4$; 11a1: 1; 121a1: $6 = \deg\varphi$; 77c1: $6 = \deg\varphi$; 539c1: $288 = \deg\varphi$. * **Lemma lem:old examples (8.4):** $f_{77c} \equiv f_{11a} - \beta f_{11a}(7z)$ modulo $(3, \sqrt{-6})$ for $n \le 3000$ (Sturm 16); $f_{52a} \equiv f_{26b} - f_{26b}(2z) \pmod 3$ for $n \le 400$ (Sturm 14), while f_{26a} is not congruent; $f_{121c} \equiv f_{11a} - f_{11a}(11z) \pmod 3$ for $n \le 3000$ (Sturm 22). * **Case (a) at $q = 2$ on the tables:** 1,051 curves labelled "1" with type IV or IV^* at 2 (PARI codes ± 4), $3 \nmid N$, mod 3 image surjective or 3Nn (no 3B, 3Cs or 3Ns label): $3 \mid \deg\varphi$ for all of them (`degphi`); first examples 52a1 (deg 3), 76a1 (6), 92b1 (6). * **Theorem thm:quant and point 16 data:** $\deg\varphi$: 539c1 288, 77c1 6, 980i1 2880, 140b1 60, 121d1 24, 11a1 1, 539d1 72; $e_3(539d1, 7) = v_3(6) + v_3(60) = 2$. * Not verified: the theorem numbers marked "(verify)"; the precise hypotheses of Ribet84 and Ribet90 Theorem 1.1 and of Pollack-Weston and Kim-Ota; the normalisation of local Langlands in the chosen reference; DFG04's normalisation of $M_{f,\lambda}$ (2.6(ii) assumes the paper's description); anything beyond conductor 10,000.

17.7 7. Points for important_notes.md

17.7.1 7.1 New points about the paper (not among points 1 to 53 and B1 to B16; each checked as stated)

N1. Theorem thm:qual, proof of case (c): "$f' = f \otimes \chi$ is a newform of weight 2 on $\Gamma_0(N/q)$" is false as written. Location: Section sec:proof, case (c), second sentence ("By Lemma lem:twist, E' has multiplicative reduction of type I_n at q and conductor N/q, so $f' = f \otimes \chi$ is a newform of weight 2 on $\Gamma_0(N/q)$ with $q \,\|\, N/q$"), and its reuse at the end of the case ("$g = h \otimes \chi$ would otherwise equal f', of level $N/q \neq M$"). Problem: with the paper's own definition of $\otimes\chi$ (Lemma lem:twist(iii): $a_n(g \otimes \chi) = \chi(n)a_n(g)$), $f \otimes \chi$ has $a_q = 0$, whereas the newform $f_{E'}$ of the multiplicative curve E' has $a_q(f_{E'}) = \pm 1$; in fact $f = f_{E'} \otimes \chi$ (Atkin-Li, as $q \,\|\, N/q$) and $f \otimes \chi = f_{E'}^{\{q\}}$, the q-depletion of $f_{E'}$, which lives at level N, not N/q. Example: for 539c1 ($q = 7$), $f_{539c} \otimes \chi_{-7}$ has $a_7 = 0$ while f_{77c} has $a_7 = -1$. (In case (b) the analogous sentence is correct, because there $a_q(f_{E'}) = 0$; but the justification is not Atkin-Li, which needs $q^2 \nmid M$, and should be "both have $a_n = 0$ for $q \mid n$".) Fix: in case (c) write "$f' = f_{E'}$, a newform on $\Gamma_0(N/q)$ with $q \,\|\, N/q$ and $f = f' \otimes \chi$ (Lemma lem:twist(iii)), so that $a_p(f') = \chi(p)a_p(f)$ for $p \neq q$", and prove $h \neq f$ as in important_notes.md point 20: if $h = f$, then $g^{\{q\}} = h \otimes \chi = f \otimes \chi = f'^{\{q\}}$, so $g = f'$ by strong multiplicity one, but their levels are M and $N/q \neq M$. Effect: none on the theorem. Confidence: high (definition check; numerical example). Importance: low (expository; companion to point 20).

N2. The remark after the Example of Section 3: the reason given for $q = 3$ in cases (a) and (b) is not the operative one. Location: "… so do they at $q = 3$ in cases (a) and (b), where Lemma lem:IV does not apply because the ramification is wild." Problem: cases (a) and (b) have $\ell = 3$,

so $q = 3$ means $q = \ell$, i.e. $3 \mid N$, which Theorem thm:qual excludes twice ($\ell \nmid N$ and $q \neq \ell$). At $q = \ell$ the inertia group acts on T_3E through an infinite image (it contains the cyclotomic character), so neither Kraus's semistability defect nor the statement of Lemma lem:IV (an image of order 3 in $\mathrm{Aut}\, T_3E$) is meaningful, whatever the wildness; the relevant local theory is the 3-adic one (finite flat or not at 3, the ℓ^δ of the optimal-level theorem), and these are exactly the curves of Conjecture conj:euler's empirical weights w_3. Fix: "... so do they at $q = \ell = 3$ in cases (a) and (b), which lie outside the theorem since then $\ell \mid N$ (this is the situation of the weights w_3 of Conjecture conj:euler)". Confidence: high for the logic. Importance: low (expository).

No other new problem was found in this chapter's area. Checked and found correct (beyond the points already recorded): the statement and proof of Lemma lem:crit (the decomposition $\mathbb{T} \otimes \mathbb{Q} = \mathbb{Q} \times \mathbb{T}'$ needs only that the ϕ_f-generalised eigenspace is $\mathbb{C}f$, which strong multiplicity one gives; the maximal-ideal step; the $\mathbb{Z}_\ell$-version); every case of Lemma lem:old, including the eigenvalue lists and eigenvectors of U_p, the use of DDT Theorem 3.1(e) and the unique-stable-line argument (valid also when $p \equiv 1 \pmod{\ell}$); the optimal-level paragraph (goodness at ℓ, the trivial-nebentypus argument, $N(\bar{\rho}) \mid N$); cases (a), (b), (c) modulo points 20, 43 and N1 (levels $M \mid N/q$, $Mq \mid N$, $Mq^2 \mid N$; the hypothesis of Lemma lem:old); Lemma lem:twist(iii) (also derived from the local conductors, item 3.8); the Example's congruences, curves and degrees (modulo points 21, 46 and B8); the 99A1 data ($\deg \varphi = 4$, $r = 12$); Theorem thm:twist's level hypotheses in its three cases; Step 1 of Theorem thm:quant (every eigenvalue of U_q congruent to 0 modulo $\mathfrak{m}$ is exactly 0, since the other possible eigenvalues are ± 1 or roots of $X^2 - a_q X + q$, units at $\ell \neq q$); the level-raising factor identity of the paragraph after Theorem thm:dfg14.

17.7.2 7.2 Book-internal inconsistencies

- **I1. Chapter 21 outline, A3.5 case (c).** It repeats the paper's "$f' = f \otimes \chi$ on $\Gamma_0(N/q)$" (N1) and then argues "$h \neq f$ (if $h = f$ then $f' = f \otimes \chi$ is the q-depleted form of g; but $a_q(f') = \pm 1 \ldots$ while the depleted form has $a_q = 0 \ldots$)", which is self-contradictory ($f \otimes \chi$ has $a_q = 0$). Action: the Chapter 21 writer uses item 9.2 here (C4 notation, strong multiplicity one), or points to it.
- **I2. Chapter 21 outline, A1.10 (thm-qual) and A3.5.** Case (a) is stated for $q \geq 5$ only; with point 43 settled (Chapter 11, 4.4) the tour should add the extension to $q = 2$ in a commentary (B13 records the stale hedge on type IV at 2 only for Proposition prop:forms and Remark rem:wild).
- **I3. Who proves Lemma lem:crit.** PLAN.md's Chapter 14 entry says "Lemma lem:crit of the paper proved in full here or in Chapter 20", its Chapter 15 entry says "prove it here in full", its Chapter 20 entry says "Chapter 15 proves it, Chapter 20 restates it"; the Chapter 21 outline (A3.2) says "Chapter 15 and 20 also prove it". This outline follows the Chapter 15 and 20 entries (proved here, restated in Chapter 20); Chapters 14, 20 and 21 should cite `\Cref{ch:modgalrep}`.
- **I4. Bibliography keys shared with Chapters 7 and 11 (B14).** The Chapter 7 outline plans `Rohrlich94` and `Deligne73` (with `Tate79`) for `bib/ch07.bib`, and the Chapter 11 outline plans `Livne89` for `bib/ch11.bib`; this chapter needs all three, so by decision B14 they belong in `refs.bib`. Action: whoever writes the first of these chapters adds them to `refs.bib`.
- **I5. Chapter 11 outline, Part 0.2: "Chapter 15 proves ... the optimal-level theorem in general (DDT Theorem 3.15) ... and its proof".** PLAN.md 7.3 classes it as a deep imported theorem; this chapter states it (6.4, R) with a proof idea (6.7) and proves only the specialisation (6.5). The Chapter 11 writer should say "states" (no conflict of content).
- **I6. Chapter 12 outline, 11.5(c).** It writes $f_{20a} \otimes \chi_{-7}$ without naming the class; it is the newform of class 980g (the curve-level twist of 20a1 is 980g2). Not an error; the Chapter 12 writer may add the label for consistency with 9.4 here.
- No conflict found with the Chapter 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 16 outlines on the items cited in Part 5; the numerical data quoted in the Chapter 11 (6.3, 10.2), Chapter 12 (11.3, 11.5) and Chapter 21 (A3.6) outlines agree with the recomputations of Part 6.6.

Chapter 18

Chapter 16. Homological algebra and group cohomology (ch:cohomology, chapters/ch16_cohomology.tex)

Stage 1 line-up (teaching order). Part V, between Chapter 3 (modules; Fitting ideals, lengths, Pontryagin duality, snake lemma) and Chapter 17 (Galois cohomology, Kummer theory, local duality, Selmer groups). Target length: 2000 lines of LaTeX excluding the card (PLAN.md Section 7.3 allows 1000 to 2200).

Conventions fixed for the whole chapter (and for Chapters 17, 18, which inherit them):

- ℓ is a prime; where "ℓ odd" is needed it is said explicitly (it is needed in the model computation, where 2 must be a unit, and in the vanishing of the cohomology of $\mathbb{Z}/2$).
- All cohomology of a profinite group is continuous cochain cohomology. Coefficient modules are of three types: (D) discrete modules (finite, or ℓ-primary torsion such as $W = T \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$); (C) finitely generated $\mathbb{Z}_\ell$-modules with the ℓ-adic topology and a continuous action; (V) finite-dimensional $\mathbb{Q}_\ell$-vector spaces with a continuous action.
- Frobenius is always the arithmetic Frobenius ($x \mapsto x^q$ on $\overline{\mathbb{F}}_q$); $G_{\mathbb{Q}_q}/I_q = \mathrm{Gal}(\overline{\mathbb{F}}_q/\mathbb{F}_q) \simeq \widehat{\mathbb{Z}}$ is topologically generated by it; $\chi_\ell(\mathrm{Frob}_q) = q$ on $\mathbb{Z}_\ell(1) = \varprojlim \mu_{\ell^k}$ for $q \neq \ell$.
- The conjugation action of G/H on $H^n(H, M)$ is $(\sigma \cdot c)(h) = \sigma\, c(\sigma^{-1} h \sigma)$, exactly the paper's formula.
- The relation between Frobenius and a topological generator t of the pro-ℓ tame inertia is $\mathrm{Frob}\, t\, \mathrm{Frob}^{-1} = t^q$, equivalently $\mathrm{Frob}^{-1} t\, \mathrm{Frob} = t^{q^{-1}}$ with $q^{-1} \in \mathbb{Z}_\ell$ (proved in Chapter 7; imported here as a stated fact).
- $\mathrm{ad}^0 T$ (the paper's `\ad T`) is the module of trace-zero endomorphisms of T with $g \cdot \varphi = \rho(g)\varphi\rho(g)^{-1}$.

18.1 1. Core results, in teaching order

Format of each entry: **name** (kind). Statement. *Rigour.* Reference. Paper use.

18.1.1 Section 2: complexes and the long exact sequence

cochain-complex (definition). A cochain complex of R-modules (R commutative, in practice $\mathbb{Z}$ or $\mathbb{Z}_\ell$) is a sequence $C^\bullet = (C^n, d^n \colon C^n \to C^{n+1})_{n \in \mathbb{Z}}$ with $d^{n+1} d^n = 0$. Cocycles $Z^n = \ker d^n$, coboundaries $B^n = \mathrm{im}\, d^{n-1} \subseteq Z^n$, cohomology $H^n(C^\bullet) = Z^n/B^n$. A morphism $f \colon C^\bullet \to D^\bullet$ is a family f^n with $d f^n = f^{n+1} d$; it induces $H^n(f)$, functorially. f, g are homotopic if $f^n - g^n = d s^n + s^{n+1} d$ for maps $s^n \colon C^n \to D^{n-1}$; homotopic morphisms induce the same map on cohomology. A sequence of complexes is

exact if it is exact in each degree. *Proved here.* Weibel, Chapter 1, Sections 1.1 to 1.4. Background (all long exact sequences of Chapters 17 and 18).

snake-lemma (lemma). Given a commutative diagram with exact rows $A' \to A \to A'' \to 0$ and $0 \to B' \to B \to B''$ and vertical maps f', f, f'', the sequence $\ker f' \to \ker f \to \ker f'' \xrightarrow{\delta} \operatorname{coker} f' \to \operatorname{coker} f \to \operatorname{coker} f''$ is exact, where $\delta(a'')$ is defined by lifting $a'' \in \ker f''$ to $a \in A$, observing that $f(a) \mapsto 0$ in B'' so $f(a)$ is the image of a unique $b' \in B'$, and putting $\delta(a'') = [b']$; δ is well defined and R-linear. If $A' \to A$ is injective the sequence begins with $0 \to \ker f'$; if $B \to B''$ is surjective it ends with $\operatorname{coker} f'' \to 0$. *Stated; reference (proved in Chapter 3); the construction of δ is repeated here because it is the connecting map of everything that follows.* Weibel, Lemma 1.3.2. Background.

long-exact-sequence (theorem). Let $0 \to A^\bullet \xrightarrow{i} B^\bullet \xrightarrow{p} C^\bullet \to 0$ be a short exact sequence of cochain complexes. For each n there is a connecting homomorphism $\delta^n \colon H^n(C^\bullet) \to H^{n+1}(A^\bullet)$: represent a class by $c \in Z^n(C)$, lift to $b \in B^n$, then $p(db) = dc = 0$ so $db = i(a)$ for a unique $a \in A^{n+1}$, which is a cocycle, and $\delta^n[c] = [a]$. This is well defined, and $\cdots \to H^n(A) \xrightarrow{i_*} H^n(B) \xrightarrow{p_*} H^n(C) \xrightarrow{\delta^n} H^{n+1}(A) \xrightarrow{i_*} H^{n+1}(B) \to \cdots$ is exact. Naturality: a morphism of short exact sequences of complexes induces a morphism of long exact sequences (commuting ladder). *Proved here.* Weibel, Theorem 1.3.1. Background (Kummer sequences of Chapter 17, the sequence of Lemma lem:dfg21).

18.1.2 Section 3: resolutions, derived functors, Ext and Tor (outline)

projective-injective-resolutions (definition). P is projective if $\operatorname{Hom}_R(P, -)$ is exact (equivalently a direct summand of a free module); I is injective if $\operatorname{Hom}_R(-, I)$ is exact. Every module M has a projective resolution $\cdots \to P_1 \to P_0 \to M \to 0$ (free modules suffice) and an injective resolution $0 \to M \to I^0 \to I^1 \to \cdots$ (R-Mod has enough injectives). Comparison theorem: two projective (resp. injective) resolutions of M are homotopy equivalent, and a map $M \to N$ lifts to a map of resolutions, unique up to homotopy. Over a PID (so over $\mathbb{Z}$ and $\mathbb{Z}_\ell$) every module has a projective resolution of length one, and divisible modules such as $\mathbb{Q}/\mathbb{Z}$, $\mathbb{Q}_\ell/\mathbb{Z}_\ell$ are injective. *Stated; reference.* Weibel, Sections 2.2 and 2.3. Background.

ext-tor (definition). $\operatorname{Ext}^n_R(M,N) = H^n(\operatorname{Hom}_R(P_\bullet, N)) = H^n(\operatorname{Hom}_R(M, I^\bullet))$ for a projective resolution $P_\bullet \to M$ or an injective resolution $N \to I^\bullet$ (the two agree and are independent of choices); $\operatorname{Tor}^R_n(M,N) = H_n(P_\bullet \otimes_R N)$. $\operatorname{Ext}^0 = \operatorname{Hom}$, $\operatorname{Tor}_0 = \otimes$; a short exact sequence in either variable gives a long exact sequence; $\operatorname{Ext}^n(M,N) = 0$ for $n \geq 1$ if M is projective or N injective; over a PID $\operatorname{Ext}^n = \operatorname{Tor}_n = 0$ for $n \geq 2$. Examples: $\operatorname{Ext}^1_{\mathbb{Z}}(\mathbb{Z}/m, N) \simeq N/mN$, $\operatorname{Tor}^{\mathbb{Z}}_1(\mathbb{Z}/m, N) \simeq N[m]$, $\operatorname{Ext}^1_{\mathbb{Z}_\ell}(\mathbb{Z}/\ell^a, \mathbb{Z}/\ell^b) \simeq \mathbb{Z}/\ell^{\min(a,b)}$. *Stated; reference.* Weibel, Sections 2.4 to 2.7 and 3.1 to 3.3. Background.

ext1-extensions (theorem). $\operatorname{Ext}^1_R(M,N)$ is in bijection with the equivalence classes of extensions $0 \to N \to E \to M \to 0$ of R-modules (equivalence: an isomorphism $E \to E'$ inducing the identity on N and on M); 0 corresponds to the split extension $N \oplus M$; the group law is the Baer sum. *Stated; reference.* Weibel, Theorem 3.4.3. Background (the cocycle in the corner of a reducible representation, below).

18.1.3 Section 4: group cohomology by cochains

g-module (definition). For a group G, a G-module is an abelian group M with a left action by automorphisms, i.e. a left $\mathbb{Z}[G]$-module. Invariants M^G; coinvariants $M_G = M/I_G M$ with I_G the augmentation ideal (generated by the $g-1$); for finite G the norm $N_G = \sum_g g$, with $N_G M \subseteq M^G$. Trivial modules. $\operatorname{Hom}_G(M,N) = \operatorname{Hom}_{\mathbb{Z}}(M,N)^G$ for $(g\varphi)(m) = g\varphi(g^{-1}m)$. If M is a $\mathbb{Z}_\ell$-module and G acts $\mathbb{Z}_\ell$-linearly, every cohomology group below is a $\mathbb{Z}_\ell$-module and every map is $\mathbb{Z}_\ell$-linear. *Proved here.* Brown, Chapter I; Serre, Local Fields, Chapter VII, Section 1. Background.

inhomogeneous-cochains (definition). $C^n(G,M) = \operatorname{Map}(G^n, M)$ ($C^0 = M$), with $(d^n\varphi)(g_1, \ldots, g_{n+1}) = g_1\varphi(g_2, \ldots, g_{n+1}) + \sum_{i=1}^{n}(-1)^i \varphi(g_1, \ldots, g_i g_{i+1}, \ldots, g_{n+1}) + (-1)^{n+1}\varphi(g_1, \ldots, g_n)$. Then $d^{n+1}d^n = 0$ and $H^n(G,M) := H^n(C^\bullet(G,M))$. Low degrees: $(d^0 m)(g) = gm - m$; $(d^1\varphi)(g,h) = g\varphi(h) - \varphi(gh) + \varphi(g)$; $(d^2\psi)(g,h,k) = g\psi(h,k) - \psi(gh,k) + \psi(g,hk) - \psi(g,h)$. $H^n(G,-)$ is a covariant functor. *Proved here ($d \circ d = 0$ checked in degrees ≤ 2 in the text; general degree by reference).* Brown,

Chapter III, Section 1; Serre, Local Fields, Chapter VII, Section 3; NSW, Chapter I, Section 2. Every paper hook of the chapter.

cohomology-as-ext (theorem). $H^n(G,M) \simeq \mathrm{Ext}^n_{\mathbb{Z}[G]}(\mathbb{Z},M)$ ($\mathbb{Z}$ with trivial action): the inhomogeneous cochain complex is $\mathrm{Hom}_{\mathbb{Z}[G]}(F_\bullet,M)$ for the bar resolution $F_\bullet$ of $\mathbb{Z}$ by free $\mathbb{Z}[G]$-modules (F_n free on G^n). So $H^n(G,-)$ is the n-th right derived functor of $M \mapsto M^G$ and any projective resolution of $\mathbb{Z}$ over $\mathbb{Z}[G]$ computes it. *Stated; reference.* Brown, Chapter I, Section 5 and Chapter III, Section 1; Weibel, Section 6.5. Background.

h0-h1 (proposition). $H^0(G,M) = M^G$. $H^1(G,M) = Z^1/B^1$ with $Z^1 = \{\varphi\colon G \to M : \varphi(gh) = \varphi(g) + g\varphi(h)\}$ (crossed homomorphisms) and $B^1 = \{g \mapsto gm - m\}$ (principal ones). A 1-cocycle has $\varphi(1) = 0$ and $\varphi(g^{-1}) = -g^{-1}\varphi(g)$. If G acts trivially, $B^1 = 0$ and $H^1(G,M) = \mathrm{Hom}(G,M) = \mathrm{Hom}(G^{\mathrm{ab}},M)$; in particular $H^1(G,M) = 0$ if G is finite and M is torsion-free with trivial action. For $0 \to M' \to M \to M'' \to 0$ exact, $0 \to M'^G \to M^G \to M''^G \xrightarrow{\delta} H^1(G,M') \to H^1(G,M) \to H^1(G,M'')$ is exact with $\delta(m'') = [g \mapsto gm - m]$ for any lift m of m''. *Proved here.* Serre, Local Fields, Chapter VII, Section 3; Brown, Chapter III, Section 1. Lemma lem:tamfactor (κ is a 1-cocycle; the Kummer class in $H^1(G,\mathbb{Z}_\ell(1))$), Lemma lem:dfg21, Lemma lem:h0 (H^0 of W^D).

cocycle-in-the-corner (proposition / worked example). Let R be a commutative ring, $\chi_1,\chi_2\colon G \to R^\times$ characters and $\rho\colon G \to \mathrm{GL}_2(R)$ of the form $\rho(g) = \begin{pmatrix} \chi_1(g) & \kappa(g) \\ 0 & \chi_2(g) \end{pmatrix}$ (so Re_1 is stable, with character χ_1, and the quotient has character χ_2). Then $g \mapsto \kappa(g)\chi_2(g)^{-1}$ is a 1-cocycle with values in $R(\chi_1\chi_2^{-1})$ (R with g acting by $\chi_1(g)\chi_2(g)^{-1}$); its class in $H^1(G,R(\chi_1\chi_2^{-1}))$ depends only on the isomorphism class of (ρ,Re_1) (replacing e_2 by $e_2 + re_1$ changes the cocycle by $d(r)$), and the class is 0 if and only if ρ is split (there is a G-stable complement of Re_1). Every class arises, so $H^1(G,R(\chi_1\chi_2^{-1})) = \mathrm{Ext}^1_{R[G]}(R(\chi_2),R(\chi_1))$ classifies extensions of χ_2 by χ_1. In the paper's shape $\rho(g) = \begin{pmatrix} \chi_\ell(g)\varepsilon(g) & \kappa(g) \\ 0 & \varepsilon(g) \end{pmatrix}$ with $\varepsilon^2 = 1$ the cocycle is $\varepsilon\kappa$ with values in $\mathbb{Z}_\ell(1)$, and on any subgroup where $\varepsilon = 1$ (inertia) it is κ itself; its restriction to inertia is the homomorphism $i \mapsto \kappa(i)$. *Proved here.* Elementary; compare Serre, Local Fields, Chapter VII, Section 3 and DDT, Theorem 3.1(e) for the shape. Lemma lem:tamfactor (the matrix $\left(\begin{smallmatrix} \chi_\ell\varepsilon & \kappa \\ 0 & \varepsilon \end{smallmatrix}\right)$, "$\kappa$ the Kummer class"), Lemma lem:old (a non-split extension of two characters: Chapter 15 uses the uniqueness of the stable line).

h2-extensions (theorem). $H^2(G,M)$ is in bijection with the equivalence classes of group extensions $0 \to M \to E \to G \to 1$ inducing the given action of G on M by conjugation; 0 corresponds to $M \rtimes G$; the cocycle of an extension with a set-theoretic section s, $s(1) = 1$, is $\psi(g,h) = s(g)s(h)s(gh)^{-1}$. *Proof sketched here.* Brown, Chapter IV, Section 3. Background.

functoriality-in-M-and-les (proposition). A G-map $f\colon M \to N$ induces $f_*\colon H^n(G,M) \to H^n(G,N)$. A short exact sequence $0 \to M' \to M \to M'' \to 0$ of G-modules (abstract G) gives an exact sequence of cochain complexes, hence a long exact sequence $0 \to H^0(G,M') \to H^0(G,M) \to H^0(G,M'') \xrightarrow{\delta} H^1(G,M') \to H^1(G,M) \to H^1(G,M'') \xrightarrow{\delta} H^2(G,M') \to \cdots$, natural in the sequence and $\mathbb{Z}_\ell$-linear when the modules are $\mathbb{Z}_\ell[G]$-modules. (Profinite version: les-continuous below.) *Proved here.* Brown, Chapter III, Section 6; NSW, Chapter I, Section 3. Lemma lem:dfg21, Lemma lem:h0, Kummer sequences of Chapter 17.

h0-and-torsion (lemma). (a) For a G-module W and $m \geq 1$: $H^0(G,W[m]) = H^0(G,W)[m]$. (b) For T a $\mathbb{Z}_\ell$-module without ℓ-torsion with $\mathbb{Z}_\ell$-linear G-action and $W = T \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell/\mathbb{Z}_\ell$: $W[\ell^k] \simeq T/\ell^k T$ as G-modules ($t \otimes \ell^{-k} \mapsto t \bmod \ell^k$), hence $H^0(G,W)[\ell^k] = H^0(G,T/\ell^k T)$ and $H^0(G,W) = \varinjlim_k H^0(G,T/\ell^k T)$. (c) A finite ℓ-primary abelian group A with $\#A[\ell] \leq \ell$ is cyclic, $A \simeq \mathbb{Z}/\ell^h$ with $h = \mathrm{length}_{\mathbb{Z}_\ell} A = \max\{k : A[\ell^k]$ has an element of order $\ell^k\}$. (d) Consequently, if $H^0(G,W)$ is finite and $H^0(G,T/\ell T)$ has order at most ℓ, then $\mathrm{length}\, H^0(G,W)$ is the largest k for which $H^0(G,T/\ell^k T)$ contains an element of exact order ℓ^k (an invariant element of $T/\ell^k T$ not divisible by ℓ). *Proved here.* Elementary (lengths from Chapter 3). Lemma lem:h0 ("the ℓ-torsion of $H^0(\mathbb{Q},W^D)$ is $H^0(\mathbb{Q},\mathrm{ad}^0\,\bar\rho \otimes \bar\chi_\ell)$"; "the description of the length follows from the ℓ^k-torsion"), Conjecture conj:euler ($h_\ell(E)$), Remark rem:ns.

18.1.4 Section 5: changing the group

compatible-pairs (definition / proposition). For a homomorphism $f\colon G' \to G$, a G-module M, a G'-module M' and $\varphi\colon M \to M'$ with $\varphi(f(g')m) = g'\varphi(m)$ (a compatible pair), $\varphi \circ (-) \circ f^{\times n}$ maps cocycles to cocycles and coboundaries to coboundaries and induces $(f,\varphi)^*\colon H^n(G,M) \to H^n(G',M')$, compatible with the long exact sequences; composition of pairs gives composition of maps. *Proved here.* Brown, Chapter III, Section 8; NSW, Chapter I, Section 5; Serre, Local Fields, Chapter VII, Section 5. Background (res, inf, conjugation are instances).

res-inf-cor (definition / proposition). (a) Restriction: for $H \le G$ (closed, in the profinite case), $\mathrm{res}\colon H^n(G,M) \to H^n(H,M)$ is induced by $(H \hookrightarrow G, \mathrm{id})$: restrict cochains. (b) Inflation: for $H \trianglelefteq G$, $\mathrm{inf}\colon H^n(G/H, M^H) \to H^n(G,M)$ is induced by $(G \to G/H, M^H \hookrightarrow M)$. (c) Corestriction: for $[G:H] < \infty$ there are maps $\mathrm{cor}\colon H^n(H,M) \to H^n(G,M)$, compatible with long exact sequences, given in degree 0 by the norm $m \mapsto \sum_{g\in G/H} gm$ and in degree 1, for left coset representatives $g_1,\dots,g_r$ and $gg_i = g_{\tau(i)}h_i(g)$ with $h_i(g) \in H$, by $(\mathrm{cor}\,\varphi)(g) = \sum_i g_{\tau(i)}\varphi(h_i(g))$. (d) $\mathrm{cor}\circ\mathrm{res} = [G:H]$ on $H^n(G,M)$ for all n; for H normal, $\mathrm{res}\circ\mathrm{cor} = \sum_{\sigma\in G/H}\sigma_*$ on $H^n(H,M)$ (conjugation action below). *Proved here for (a), (b) and for (c), (d) in degrees* 0 *and* 1*; general degree stated with reference.* Brown, Chapter III, Section 9; Serre, Local Fields, Chapter VII, Sections 5 and 7; NSW, Chapter I, Section 5; Cassels-Frohlich, Chapter IV. Lemma lem:tamfactor (inflation-restriction), Lemma lem:dfg21 (2 kills $H^1(\mathbb{R},-)$).

order-kills (theorem). Let G be finite of order N and M a G-module. Then $N \cdot H^n(G,M) = 0$ for all $n \ge 1$. For $n = 1$ directly: if $\varphi \in Z^1$ and $m = \sum_{g\in G}\varphi(g)$, then summing $\varphi(\sigma g) = \varphi(\sigma) + \sigma\varphi(g)$ over g gives $m = N\varphi(\sigma) + \sigma m$, i.e. $N\varphi = d(-m)$. Consequences: (i) if N is invertible on M then $H^n(G,M) = 0$ for $n \ge 1$; (ii) if M is finitely generated as an abelian group then $H^n(G,M)$ is finite for $n \ge 1$; (iii) if $\ell \nmid N$ and M is an ℓ-primary torsion group or a $\mathbb{Z}_\ell$-module, then $H^n(G,M) = 0$ for $n \ge 1$ and M^G is a direct summand of M (projector $N^{-1}N_G$). *Proved here ($n = 1$ in full; $n \ge 2$ via cor$\circ$res with $H = 1$).* Brown, Chapter III, Section 10; Serre, Local Fields, Chapter VIII, Section 2. Lemma lem:dfg21 and the paragraph after it ($H^1(\mathbb{R}, T^D)$ killed by 2).

conjugation-action (proposition). Let $H \trianglelefteq G$ and M a G-module. For $\sigma \in G$ the pair $(h \mapsto \sigma^{-1}h\sigma,\ m \mapsto \sigma m)$ is compatible and induces $\sigma_*\colon H^n(H,M) \to H^n(H,M)$; on cochains $(\sigma\cdot\varphi)(h_1,\dots,h_n) = \sigma\,\varphi(\sigma^{-1}h_1\sigma,\dots,\sigma^{-1}h_n\sigma)$. This is a left action of G on $H^n(H,M)$, trivial on H (for $n = 1$: $(h_0\cdot\varphi) - \varphi = d(\varphi(h_0))$), hence an action of G/H, and the map induced by a class of G/H does not depend on the representative. It is $\mathbb{Z}_\ell$-linear when M is a $\mathbb{Z}_\ell[G]$-module, so it preserves $H^n(H,M)_{\mathrm{tors}}$, and $H^n(H,M)_{\mathrm{tors}}^{G/H}$ is defined. The restriction $H^n(G,M) \to H^n(H,M)$ lands in $H^n(H,M)^{G/H}$ (for $n = 1$: $\sigma\cdot\mathrm{res}\,\varphi - \mathrm{res}\,\varphi = d(\varphi(\sigma))$). All of this holds verbatim for profinite G, closed normal H, and continuous cochains with coefficients of type (D), (C) or (V). *Proved here ($n = 1$ in full, general n on cochains).* Brown, Chapter III, Section 8; NSW, Chapter I, Section 5; Serre, Local Fields, Chapter VII, Section 5. Lemma lem:tamfactor ("G/I acts on $H^1(I,M)$ by $(\sigma\cdot c)(i) = \sigma c(\sigma^{-1}i\sigma)$"), Section 1.2 of the paper ($\mathrm{Tam}_p(M) = \mathrm{Fitt}(H^1(I_p,M)_{\mathrm{tor}})^{G_{\mathbb{Q}_p}}$), Lemma lem:fpr.

inflation-restriction (theorem). Let $H \trianglelefteq G$ and M a G-module. (a) $0 \to H^1(G/H,M^H) \xrightarrow{\mathrm{inf}} H^1(G,M) \xrightarrow{\mathrm{res}} H^1(H,M)^{G/H}$ is exact. (b) (five-term exact sequence of Hochschild and Serre) There is a transgression tg with $0 \to H^1(G/H,M^H) \to H^1(G,M) \to H^1(H,M)^{G/H} \xrightarrow{\mathrm{tg}} H^2(G/H,M^H) \xrightarrow{\mathrm{inf}} H^2(G,M)$ exact. (c) If $H^i(H,M) = 0$ for $1 \le i \le n-1$ then $0 \to H^n(G/H,M^H) \to H^n(G,M) \to H^n(H,M)$ is exact. (d) Corollary: if $H^1(H,M) = 0$ then $\mathrm{inf}\colon H^1(G/H,M^H) \to H^1(G,M)$ is an isomorphism; in particular if H acts trivially on M and $\mathrm{Hom}(H,M) = 0$ then $H^1(G/H,M) \simeq H^1(G,M)$. Valid for abstract groups; for profinite G, closed normal H and M of type (D); and for continuous cochains with M of type (C) or (V). *(a), (d) proved here; (b), (c) stated with reference, the transgression sketched.* Serre, Local Fields, Chapter VII, Section 6; NSW, Chapter I and Chapter II, Section 4; Weibel, Section 6.8; Milne, Class Field Theory, Chapter II. Lemma lem:tamfactor ("by inflation-restriction $H^1(I,M') = H^1(\langle t\rangle, M')$"), Chapter 17 ($H^1_{\mathrm{ur}} = \ker(H^1(\mathbb{Q}_p,M) \to H^1(I_p,M)) = H^1(G/I, M^I)$).

hochschild-serre (theorem). For $H \trianglelefteq G$ and a G-module M there is a first-quadrant spectral sequence $E_2^{p,q} = H^p(G/H, H^q(H,M)) \Rightarrow H^{p+q}(G,M)$, with G/H acting on $H^q(H,M)$ by conjugation; edge maps are inflation and restriction; its five-term exact sequence is inflation-restriction (b). Same for profinite G, closed normal H, M of type (D). Consequence: if $H^q(H,M) = 0$ for all $q \ge 1$ then

$\inf\colon H^p(G/H, M^H) \to H^p(G, M)$ is an isomorphism for all p. *Stated; reference.* Brown, Chapter VII, Section 6; Weibel, Theorem 6.8.2; NSW, Chapter II, Section 4. Background (only the five-term sequence is used).

18.1.5 Section 6: finite groups

cyclic-group-cohomology (theorem). Let $G = \langle t \rangle$ be cyclic of order m, $N = 1 + t + \cdots + t^{m-1}$, M a G-module. Then $\varphi \mapsto \varphi(t)$ induces $H^1(G, M) \simeq \ker(N\colon M \to M)/(t-1)M$ (a cocycle is determined by $a = \varphi(t)$ via $\varphi(t^k) = (1 + t + \cdots + t^{k-1})a$, and consistency at $t^m = 1$ is $Na = 0$); $H^2(G, M) \simeq M^G/NM$; and $H^{n+2}(G, M) \simeq H^n(G, M)$ for $n \geq 1$. In Tate's notation $\hat{H}^0 = M^G/NM$, $\hat{H}^{-1} = \ker N/(t-1)M$, and $\hat{H}^n \simeq \hat{H}^{n+2}$ for all $n \in \mathbb{Z}$. Examples: trivial action on $\mathbb{Z}$: $H^1 = 0$, $H^2 = \mathbb{Z}/m$; trivial action on $\mathbb{Z}/m$: $H^1 = H^2 = \mathbb{Z}/m$. *H^1 proved here; H^2 and periodicity stated with reference.* Serre, Local Fields, Chapter VIII, Section 4; Cassels-Frohlich, Chapter IV, Section 8; Brown, Chapter III, Section 1. Background for the procyclic computation (PLAN: "$H^1(\langle t \rangle, M) = \ker(\mathrm{Norm})/(t-1)M$").

tate-cohomology (definition). For finite G: $\hat{H}^n(G, M) = H^n(G, M)$ for $n \geq 1$, $\hat{H}^0 = M^G/N_G M$, $\hat{H}^{-1} = \ker N_G/I_G M$, $\hat{H}^{-n} = H_{n-1}(G, M)$ for $n \geq 2$. A short exact sequence of G-modules gives a long exact sequence of $\hat{H}^n$ in all degrees; $\#G$ kills every $\hat{H}^n(G, M)$; induced modules $\mathbb{Z}[G] \otimes A$ have $\hat{H}^n = 0$ for all n; for cyclic G, $\hat{H}^n \simeq \hat{H}^{n+2}$. *Stated; reference.* Serre, Local Fields, Chapter VIII, Section 1; Brown, Chapter VI, Section 4. Background (the real place).

z2-on-l-adic (corollary). Let ℓ be odd, G of order 2 (e.g. $\mathrm{Gal}(\mathbb{C}/\mathbb{R})$ generated by complex conjugation) and M a $\mathbb{Z}_\ell$-module with $\mathbb{Z}_\ell$-linear G-action (type (D) ℓ-primary, (C) or (V)). Then $H^n(G, M) = 0$ for all $n \geq 1$, $M = M^+ \oplus M^-$ with $M^\pm = \{m : \sigma m = \pm m\}$, and $H^0(G, M) = M^+$. For an arbitrary G-module M, $H^n(G, M)$ is killed by 2; so $H^1(\mathbb{R}, M) := H^1(\mathrm{Gal}(\mathbb{C}/\mathbb{R}), M)$ is a 2-torsion group, zero when M is ℓ-adic with ℓ odd. *Proved here.* From order-kills. The paragraph after Lemma lem:dfg21 ("$H^1(\mathbb{R}, T^D)$ is killed by 2, so $H^1_f(\mathbb{R}, T^D) = 0$ for ℓ odd"); PLAN "cohomology of the real place".

18.1.6 Section 7: profinite groups and continuous cohomology

discrete-g-module (definition). Profinite group (recalled): compact Hausdorff totally disconnected, equivalently an inverse limit of finite groups; open normal subgroups form a basis at 1. A discrete G-module is an abelian group with the discrete topology and a continuous action, equivalently every stabiliser is open, equivalently $M = \bigcup_U M^U$ over open normal U. Finite modules and ℓ-primary torsion discrete modules (e.g. $W = T \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$) are the type (D) modules. *Proved here.* Serre, Galois Cohomology, Chapter I, Sections 1 and 2; NSW, Chapter I, Sections 1 and 2. Background.

continuous-cohomology (definition / proposition). For G profinite and M a topological G-module, $C^n_{\mathrm{cont}}(G, M) = \{\varphi\colon G^n \to M \text{ continuous}\}$ is a subcomplex and $H^n(G, M) := H^n(C^\bullet_{\mathrm{cont}}(G, M))$. (D) For discrete M, $H^n(G, M) = \varinjlim_U H^n(G/U, M^U)$ over open normal U with inflation as transition maps (every continuous cochain is locally constant, factors through some $(G/U)^n$ and takes values in some $M^{U'}$). (C) For M a finitely generated $\mathbb{Z}_\ell$-module with the ℓ-adic topology, a continuous action is a continuous homomorphism $G \to \mathrm{Aut}_{\mathbb{Z}_\ell}(M)$ (profinite topology on the target). (V) $M = V$ a finite-dimensional $\mathbb{Q}_\ell$-vector space with continuous action. Convention: all cohomology of profinite groups in this book is continuous cochain cohomology. $H^0(G, M) = M^G$; $H^1(G, M) = Z^1_{\mathrm{cont}}/B^1$; for trivial action $H^1(G, M) = \mathrm{Hom}_{\mathrm{cont}}(G, M)$. For finite G this is the cohomology of inhomogeneous-cochains. *Proved here (the direct limit statement in full).* Serre, Galois Cohomology, Chapter I, Section 2; NSW, Chapter I, Section 2 and Chapter II, Section 7; Tate95, Section 2. Every ℓ-adic cohomology group of the paper: $H^1(G, \mathbb{Z}_\ell(1))$, $H^1(I_q, \mathrm{ad}^0 T)$, $H^1(\mathbb{Q}_p, T^D)$, $H^1_f(\mathbb{Q}, W)$.

les-continuous (proposition). Let G be profinite and $0 \to M' \xrightarrow{i} M \xrightarrow{p} M'' \to 0$ exact with continuous G-maps, i a homeomorphism onto its closed image, and p admitting a continuous section $s\colon M'' \to M$ (continuous map with $ps = \mathrm{id}$, not necessarily additive). Then $0 \to C^\bullet_{\mathrm{cont}}(G, M') \to C^\bullet_{\mathrm{cont}}(G, M) \to C^\bullet_{\mathrm{cont}}(G, M'') \to 0$ is exact and there is a natural long exact sequence $\cdots \to H^n(G, M') \to H^n(G, M) \to H^n(G, M'') \xrightarrow{\delta} H^{n+1}(G, M') \to \cdots$. A continuous section exists in each case used in this book: (i) M'' discrete; (ii) p a surjection of finitely generated $\mathbb{Z}_\ell$-modules (lift generators; on a cyclic factor $\mathbb{Z}/\ell^a$

use the representatives $0,\dots,\ell^a-1\subset\mathbb{Z}_\ell$); (iii) $0\to T\to V\to W\to 0$ with T of type (C) without ℓ-torsion, $V=T\otimes\mathbb{Q}_\ell$, $W=V/T$; (iv) $0\to T\xrightarrow{\ell^k}T\to T/\ell^kT\to 0$. In case (iii) the connecting map $H^0(G,W)\to H^1(G,T)$ has kernel the image of $H^0(G,V)$; its image is the torsion of $H^1(G,T)$ (proved in Chapter 17). *Proved here.* NSW, Chapter II, Section 7; Tate95, Section 2; Rubin, Euler Systems, Appendix B. Lemma lem:tamfactor, Lemma lem:dfg21, Lemma lem:h0 (the sequences $T\to V\to W$ and $T\xrightarrow{\ell^k}T\to T/\ell^k$ of Chapters 17 and 18).

tate-limits (theorem, Tate). Let G be profinite and $T=\varprojlim_k T_k$ an inverse limit of finite discrete G-modules with surjective transition maps, with the inverse-limit topology (e.g. T of type (C), $T_k=T/\ell^kT$). For every $n\geq 0$ there is a natural exact sequence $0\to\varprojlim{}^1{}_k H^{n-1}(G,T_k)\to H^n(G,T)\to\varprojlim_k H^n(G,T_k)\to 0$. In particular (i) $H^1(G,T)=\varprojlim_k H^1(G,T_k)$ always (the $H^0(G,T_k)$ are finite, so Mittag-Leffler holds); (ii) $H^n(G,T)=\varprojlim_k H^n(G,T_k)$ whenever all $H^{n-1}(G,T_k)$ are finite. Moreover $H^n(G,V)=H^n(G,T)\otimes_{\mathbb{Z}_\ell}\mathbb{Q}_\ell$ for $V=T\otimes\mathbb{Q}_\ell$ (a continuous cochain into V has compact image, hence lies in $\ell^{-a}T$). *Stated; reference; the case $n=1$ sketched.* Tate95, Section 2; NSW, Chapter II, Section 7; Rubin, Euler Systems, Appendix B. Lemma lem:tamfactor ("$H^1(G,\mathbb{Z}_\ell(1))=\varprojlim\mathbb{Q}_q^\times/(\mathbb{Q}_q^\times)^{\ell^k}$"), the definition of $H^1(I_q,\mathrm{ad}^0\,T)$.

pitfalls (warning). (i) For M of type (C), $H^n(G,M)$ is not $\varinjlim_U H^n(G/U,M^U)$: $H^1(\mathbb{Z}_\ell,\mathbb{Z}_\ell)=\mathrm{Hom}_{\mathrm{cont}}(\mathbb{Z}_\ell,\mathbb{Z}_\ell)=\mathbb{Z}_\ell$, whereas $\mathrm{Hom}(\mathbb{Z}_\ell,\mathbb{Z}_\ell^{\mathrm{discrete}})=0$ (a continuous homomorphism from a compact group to a discrete torsion-free group is zero). (ii) The long exact sequence needs the continuous section; it always exists in the cases listed. (iii) $\varprojlim^1$ terms can be nonzero in general; they vanish under finiteness. (iv) $H^n(G,V)$ for V of type (V) is a $\mathbb{Q}_\ell$-vector space and sees no torsion; the torsion of $H^1(G,T)$ is governed by $H^0(G,W)$ (Chapter 17). (v) For finite G (discrete topology) continuous and abstract cohomology coincide. (vi) For a procyclic group the formula $H^1=M/(F-1)M$ has precise hypotheses (Section 9); it fails for finite cyclic groups ($\ker N/(F-1)M$) and for non-torsion discrete modules ($H^1(\widehat{\mathbb{Z}},\mathbb{Z})=0$). *Proved here (examples).* Tate95; Serre, Galois Cohomology, Chapter I. Background.

h1-finitely-generated (proposition). Let G be profinite and T of type (C). (i) $H^1(G,T)=\varprojlim_k H^1(G,T/\ell^kT)$ is a compact pro-ℓ $\mathbb{Z}_\ell$-module. (ii) If $H^1(G,T/\ell T)$ is finite then $H^1(G,T)$ is a finitely generated $\mathbb{Z}_\ell$-module, $H^1(G,V)=H^1(G,T)\otimes\mathbb{Q}_\ell$ is finite-dimensional and $H^1(G,T)_{\mathrm{tors}}$ is finite. (iii) The finiteness in (ii) holds for $G=\mathrm{Gal}(\overline{\mathbb{F}}_q/\mathbb{F}_q)$, $G=\mathrm{Gal}(\mathbb{C}/\mathbb{R})$, $G=I_q$ and $G=I_q/K$ with $\ell\neq q$ (proved here via the procyclic and prime-to-ℓ results), and for $G=G_{\mathbb{Q}_p}$ and $G_{\mathbb{Q},S}$ (Chapter 17). *(i) proved here; (ii) sketched (topological Nakayama); (iii) proved here for the listed groups except the last two.* Tate95, Section 2; Rubin, Appendix B. Lemma lem:tamfactor ("$H^1(I,M)_{\mathrm{tor}}$ is $(\mathbb{Z}/\ell^{v_\ell(n)})^2$"), paragraph after Lemma lem:dfg21 ("$H^1_f(\mathbb{Q}_p,T^D)$ is the torsion of $H^1(\mathbb{Q}_p,T^D)$, a finite group").

18.1.7 Section 8: groups of order prime to ℓ and pro-ℓ quotients

supernatural-order-pro-l (definition). The order of a profinite group G is the supernatural number $\#G=\mathrm{lcm}_U\,\#(G/U)$ (U open normal), a formal product $\prod_p p^{n_p}$, $n_p\in\mathbb{N}\cup\{\infty\}$. G is pro-ℓ if $\#G$ is a power of ℓ; G has order prime to ℓ if $\ell\nmid\#G$. The maximal pro-ℓ quotient $G^{(\ell)}=G/K$, K the intersection of the open normal subgroups of ℓ-power index: every continuous homomorphism from G to a pro-ℓ group factors through $G^{(\ell)}$; $G^{(\ell)}=1$ if $\ell\nmid\#G$. The unipotent upper triangular matrices in $\mathrm{GL}_2(\mathbb{Z}_\ell)$ form a pro-ℓ group isomorphic to $\mathbb{Z}_\ell$. Sylow pro-ℓ subgroups exist and are conjugate (stated). *Stated; reference (elementary parts proved).* Serre, Galois Cohomology, Chapter I, Section 1; NSW, Chapter I, Section 1. Lemma lem:tamfactor ("trivial pro-ℓ quotient").

prime-to-l-vanishing (theorem). Let G be profinite of order prime to ℓ. (a) If M is of type (D) and ℓ-primary torsion then $H^n(G,M)=0$ for all $n\geq 1$ (each $H^n(G/U,M^U)$ is killed by $\#(G/U)$ and by a power of ℓ). (b) If M is of type (C) or (V) then $H^n(G,M)=0$ for $n\geq 1$ (from (a) and tate-limits). (c) If G acts trivially on M of type (C) then $H^1(G,M)=\mathrm{Hom}_{\mathrm{cont}}(G,M)=0$, and for this only $G^{(\ell)}=1$ is needed (the image of G in the pro-ℓ group M is a pro-ℓ quotient of G). (d) In all cases M^G is a direct summand of M. *Proved here ((a), (c), (d) in full; (b) via tate-limits).* Serre, Galois Cohomology, Chapter I, Sections 2 and 3; NSW, Chapter I; Brown, Chapter III, Section 10. Lemma lem:tamfactor ("the kernel of $I\to\langle t\rangle$ has trivial pro-ℓ quotient").

cohomology-through-pro-l-quotient (corollary). Let G be profinite, $K \trianglelefteq G$ closed, and M of type (D, ℓ-primary), (C) or (V) with K acting trivially. (a) If $K^{(\ell)} = 1$ then $\inf\colon H^1(G/K, M) \to H^1(G, M)$ is an isomorphism. (b) If $\ell \nmid \#K$ then $\inf\colon H^n(G/K, M) \to H^n(G, M)$ is an isomorphism for all n. (c) If the image of G in $\mathrm{Aut}(M)$ is pro-ℓ (e.g. unipotent upper triangular in a basis of $M \simeq \mathbb{Z}_\ell^2$) then the action factors through $G^{(\ell)}$, and $H^1(G, M) = H^1(G^{(\ell)}, M)$. *(a), (c) proved here; (b) stated (Hochschild-Serre).* As above. Lemma lem:tamfactor.

tame-inertia-example (example). Let q be a prime, $I = I_q$ the inertia subgroup of $G_{\mathbb{Q}_q}$, $P \subset I$ the wild inertia (pro-q), $\ell \neq q$. Facts from Chapter 7: $I/P \simeq \prod_{\ell' \neq q} \mathbb{Z}_{\ell'}(1)$; the maximal pro-$\ell$ quotient of I is $I^{(\ell)} = \mathbb{Z}_\ell(1)$, the quotient map being the tame character t_ℓ; its kernel K has order prime to ℓ. Hence for every M of type (C) on which I acts with pro-ℓ image (e.g. through unipotent matrices), the action factors through $\mathbb{Z}_\ell(1) \simeq \mathbb{Z}_\ell$, topologically generated by an element t, and $H^1(I, M) = H^1(\langle t \rangle, M) = M/(t-1)M$ (procyclic-compact), $H^n(I, M) = 0$ for $n \geq 2$. *Proved here from the imported facts.* Serre, Local Fields, Chapter IV; Chapter 7 of this book. Lemma lem:tamfactor ("the inertia group acts through its maximal pro-ℓ tame quotient, generated by t").

18.1.8 Section 9: procyclic groups

procyclic-groups (definition / lemma). A profinite group is procyclic if topologically generated by one element; these are the quotients of $\widehat{\mathbb{Z}} = \varprojlim \mathbb{Z}/n = \prod_p \mathbb{Z}_p$, i.e. the groups $\prod_p \mathbb{Z}/p^{a_p}$ ($a_p \in \mathbb{N} \cup \{\infty\}$, $\mathbb{Z}/p^\infty := \mathbb{Z}_p$). Examples: $\widehat{\mathbb{Z}} = \mathrm{Gal}(\overline{\mathbb{F}}_q/\mathbb{F}_q)$ generated by Frobenius; $\mathbb{Z}_\ell$; $\mathbb{Z}_\ell(1)$; I_q/P_q. A continuous action of a procyclic Γ with generator F on M of type (C) is determined by $u = F|_M \in \mathrm{Aut}_{\mathbb{Z}_\ell}(M)$: for $\Gamma = \widehat{\mathbb{Z}}$ every u arises; for $\Gamma = \mathbb{Z}_\ell$ exactly the u with $u^{\ell^N} \to 1$ as $N \to \infty$ (the closed subgroup generated by u is pro-ℓ), for instance every unipotent u. *Proved here.* Serre, Galois Cohomology, Chapter I, Section 1; NSW, Chapter I, Section 1. Background; Lemma lem:tamfactor (t).

cocycle-on-powers (lemma). Let Γ be procyclic with generator F, M a Hausdorff topological Γ-module and c a continuous 1-cocycle. Then $c(F^k) = (1 + F + \cdots + F^{k-1})c(F)$ for $k \geq 1$, $c(F^{-k}) = -F^{-k}c(F^k)$, so c is determined by $c(F)$. If M is of type (C), (V) or finite ℓ-primary (so that $(F-1)M$ is closed), then for every $a \in \Gamma$: $c(F^a) \equiv a \cdot c(F) \pmod{(F-1)M}$, where a acts on the $\mathbb{Z}_\ell$-module $M/(F-1)M$ through the canonical map $\Gamma \to \mathbb{Z}_\ell$ (both sides are continuous in a and agree on $\mathbb{Z}$, since $1 + F + \cdots + F^{k-1} \equiv k$ modulo $(F-1)$). *Proved here.* Elementary. Lemma lem:tamfactor ("$c(t^k) \equiv k\,c(t)$ modulo $(t-1)M$", applied with $k = q^{-1} \in \mathbb{Z}_\ell$).

procyclic-discrete (theorem, Serre). Let $\Gamma = \widehat{\mathbb{Z}}$ with generator F and M a discrete torsion Γ-module, or $\Gamma = \mathbb{Z}_\ell$ and M a discrete ℓ-primary torsion Γ-module. Then $H^0(\Gamma, M) = M^{F=1}$, $c \mapsto c(F)$ induces $H^1(\Gamma, M) \simeq M/(F-1)M$, and $H^n(\Gamma, M) = 0$ for $n \geq 2$. If M is finite then $\#H^1(\Gamma, M) = \#H^0(\Gamma, M)$. (Proof of H^1: $H^1(\Gamma, M) = \varinjlim_N \ker(N_{F,N})/(F-1)M^{\Gamma_N}$ over the finite quotients Γ/Γ_N, and every $a \in M$ fixed by Γ_N and of order m satisfies $N_{F,Nm}(a) = m\,N_{F,N}(a) = 0$.) Warning: false for finite cyclic Γ and for non-torsion M. *H^1 and the counting proved here; H^n, $n \geq 2$, stated with reference.* Serre, Local Fields, Chapter XIII, Section 1; Serre, Galois Cohomology, Chapter I, Sections 3 and 4. Chapter 17 (H^1_{ur} with finite coefficients, orders of unramified cohomology).

procyclic-compact (theorem; the identity the paper uses). Let Γ be a procyclic profinite group admitting a continuous surjection onto $\mathbb{Z}_\ell$ (in $\Gamma \simeq \prod_p \mathbb{Z}/p^{a_p}$: $a_\ell = \infty$; examples $\mathbb{Z}_\ell$, $\mathbb{Z}_\ell(1)$, $\widehat{\mathbb{Z}}$, I_q/P_q for $q \neq \ell$), F a topological generator, and M of type (C) or (V). Then: (i) $H^0(\Gamma, M) = \ker(F-1)$; (ii) every $m \in M$ is $c(F)$ for a unique continuous 1-cocycle c (the cocycle $k \mapsto (1 + F + \cdots + F^{k-1})m$ on $\mathbb{Z}$ is uniformly continuous for the topology induced by Γ and extends), and $c \mapsto c(F)$ induces an isomorphism of $\mathbb{Z}_\ell$-modules $H^1(\Gamma, M) \simeq M/(F-1)M$; (iii) $H^n(\Gamma, M) = 0$ for $n \geq 2$. Consequences: $H^1(\Gamma, M)$ is finitely generated over $\mathbb{Z}_\ell$, $H^1(\Gamma, M)_{\mathrm{tors}}$ is finite, and for $M = V$ of type (V), $H^1(\Gamma, V) = V/(F-1)V = 0$ if and only if 1 is not an eigenvalue of F on V, i.e. $\det(1 - F \mid V) \neq 0$. Ingredients of the proof of (ii): the image Γ' of Γ in $\mathrm{Aut}_{\mathbb{Z}_\ell}(M)$ is $C \times \Gamma'_\ell$ with C finite cyclic of order m prime to ℓ and Γ'_ℓ pro-ℓ procyclic generated by $u = F^m|_M$; for such u ($u^{\ell^N} \to 1$) and every n there is N with $(1 + u + \cdots + u^{j-1})M \subseteq \ell^n M$ whenever $\ell^N \mid j$ (write $u^{\ell^{N_0}} = 1 + \ell^n A$). *(i), (ii) proved here in full; (iii) stated with reference.* For (iii): Serre, Galois Cohomology, Chapter I, Sections 3 and 4 ($\mathrm{cd}_\ell(\mathbb{Z}_\ell) = 1$) and Tate95, Section 2; for (ii) also

Rubin, Appendix B. Lemma lem:tamfactor ("$H^1(\langle t\rangle, M') = M'/(t-1)M'$ for every $\mathbb{Z}_\ell$-module M' of finite type with this action"; "$H^1(G/I, \mathbb{Z}_\ell(1)^I) = \mathbb{Z}_\ell(1)/(q-1)$"), the paragraph after Lemma lem:dfg21 ($H^1_{\mathrm{ur}}(\mathbb{Q}_p, V^D) = (V^D)^{I_p}/(\mathrm{Frob}_p - 1) = 0$), Chapter 17 ($H^1_{\mathrm{ur}}(\mathbb{Q}_p, T) = T^{I_p}/(\mathrm{Frob}_p - 1)$).

procyclic-examples (example). (i) Trivial action: $H^1(\mathbb{Z}_\ell, M) = \mathrm{Hom}_{\mathrm{cont}}(\mathbb{Z}_\ell, M) = M$, and $H^1(\widehat{\mathbb{Z}}, M) = M$ via $\widehat{\mathbb{Z}} \to \mathbb{Z}_\ell$. (ii) $\mathbb{Z}_\ell(1)$ with F acting by an integer q prime to ℓ: $H^0 = 0$ if $q \neq 1$ and $H^1(\widehat{\mathbb{Z}}, \mathbb{Z}_\ell(1)) = \mathbb{Z}_\ell/(q-1)\mathbb{Z}_\ell \simeq \mathbb{Z}/\ell^{v_\ell(q-1)}$; with μ_{ℓ^k}: $\#H^1 = \#H^0 = \ell^{\min(k, v_\ell(q-1))}$. (iii) $M = \mathbb{Z}_\ell^2$ with $F = \left(\begin{smallmatrix} 1 & n' \\ 0 & 1 \end{smallmatrix}\right)$: $H^1 = \mathbb{Z}_\ell/n' \oplus \mathbb{Z}_\ell$, torsion $\mathbb{Z}/\ell^{v_\ell(n')}$. (iv) F of finite order prime to ℓ: $M/(F-1)M = M_F$ is a direct summand, isomorphic to M^F. (v) V of type (V): $H^1(\Gamma, V) = 0$ iff $\det(1 - F \mid V) \neq 0$. *Proved here.* Elementary. Lemma lem:tamfactor, the paragraph after Lemma lem:dfg21, Chapter 17 (Kummer: $\mathbb{Q}_q^\times/(\mathbb{Q}_q^\times)^{\ell^k}$).

18.1.9 Section 10: the model computation (the group cohomology of Lemma lem:tamfactor)

model-setting (construction / convention). Data: a profinite group G; a closed normal subgroup $I \trianglelefteq G$; a closed subgroup $K \subseteq I$, normal in G, of order prime to ℓ, with $I/K \simeq \mathbb{Z}_\ell$ topologically generated by t; an element $\mathrm{Frob} \in G$ whose image topologically generates $G/I \simeq \widehat{\mathbb{Z}}$; a positive integer q prime to ℓ with $\mathrm{Frob}\, t\, \mathrm{Frob}^{-1} = t^q$ in G/K, equivalently $\mathrm{Frob}^{-1}\, t\, \mathrm{Frob} = t^{q^{-1}}$ with $q^{-1} \in \mathbb{Z}_\ell$; a finitely generated $\mathbb{Z}_\ell$-module M with a continuous G-action $\rho\colon G \to \mathrm{Aut}_{\mathbb{Z}_\ell}(M)$ trivial on K. In the paper: $G = G_{\mathbb{Q}_q}$, $I = I_q$, $K = \ker(t_\ell\colon I_q \to \mathbb{Z}_\ell(1))$, t a generator of $\mathbb{Z}_\ell(1)$, Frob a lift of the arithmetic Frobenius, and $M = \mathrm{ad}^0\, T_\ell E$ or $T_\ell E$ for E with multiplicative reduction at q (the action of I_q is unipotent, hence factors through $I_q^{(\ell)}$: Chapter 11); the relation is Chapter 7's. *Convention; the imported facts are stated with pointers.* DFG04, p. 708; FPR94, I.4.2.2. Lemma lem:tamfactor, Section 1.2 of the paper.

frobenius-on-h1-inertia (proposition). In the model setting: (a) $\mathrm{inf}\colon H^1(I/K, M) \to H^1(I, M)$ is an isomorphism and $H^1(I, M) \simeq M/(t-1)M$ via $c \mapsto c(t)$. (b) Under this identification the conjugation action of G/I on $H^1(I, M)$ is $\overline{\mathrm{Frob}} \cdot \bar{m} = q^{-1}\rho(\mathrm{Frob})\bar{m}$; this is well defined because $\rho(\mathrm{Frob})(t-1)M = (t^q - 1)\rho(\mathrm{Frob})M \subseteq (t-1)M$, and independent of the lift Frob. (Proof: $(\mathrm{Frob}\cdot c)(t) = \rho(\mathrm{Frob})c(\mathrm{Frob}^{-1}\, t\, \mathrm{Frob}) = \rho(\mathrm{Frob})c(t^{q^{-1}}) \equiv q^{-1}\rho(\mathrm{Frob})c(t)$ by cocycle-on-powers.) (c) $H^1(I, M)_{\mathrm{tors}} = (M/(t-1)M)_{\mathrm{tors}}$ is finite and G/I-stable, and $H^1(I, M)_{\mathrm{tors}}^{G/I} = \{\bar{m} \in (M/(t-1)M)_{\mathrm{tors}} : \rho(\mathrm{Frob})\bar{m} = q\bar{m}\}$. *Proved here.* Combines conjugation-action, cohomology-through-pro-l-quotient and procyclic-compact. Lemma lem:tamfactor ("the arithmetic Frobenius acts on $M/(t-1)M$ as $q^{-1}\rho(\mathrm{Frob})$").

tate-curve-shape-matrices (example). Let $T = \mathbb{Z}_\ell e_1 \oplus \mathbb{Z}_\ell e_2$ with $\rho(g) = \begin{pmatrix} \chi(g)\varepsilon(g) & \kappa(g) \\ 0 & \varepsilon(g) \end{pmatrix}$, where $\varepsilon\colon G \to \{\pm 1\}$ and $\chi\colon G \to \mathbb{Z}_\ell^\times$ are trivial on I with $\chi(\mathrm{Frob}) = q$, and $\kappa|_I\colon I \to \mathbb{Z}_\ell$ is a continuous homomorphism trivial on K with $\kappa(t) = n' \neq 0$, so $te_1 = e_1$, $te_2 = e_2 + n'e_1$; write $\varepsilon = \varepsilon(\mathrm{Frob})$ and $c' = \kappa(\mathrm{Frob})$, so $\rho(\mathrm{Frob}) = \begin{pmatrix} \varepsilon q & c' \\ 0 & \varepsilon \end{pmatrix}$. Let $M = \mathrm{ad}^0\, T$ (trace-zero endomorphisms, action $\varphi \mapsto \rho(g)\varphi\rho(g)^{-1}$) with basis $e = E_{12} = e_1 \otimes e_2^\vee$, $h = E_{11} - E_{22}$, $f = E_{21} = e_2 \otimes e_1^\vee$. Since $\rho(t) = 1 + n'e$, $\rho(t)^{-1} = 1 - n'e$, $eh - he = -2e$, $ehe = 0$, $ef - fe = h$, $efe = e$: $(t-1)e = 0$, $(t-1)h = -2n'e$, $(t-1)f = n'h - n'^2 e$; and $\mathrm{Frob}\cdot e = qe$, $\mathrm{Frob}\cdot h = h - 2\varepsilon c' e$, $\mathrm{Frob}\cdot f = q^{-1}(f + \varepsilon c' h - c'^2 e)$. The data (n', c', ε) of the paper: n' is $n = v_q(\Delta_E)$ times a unit (the restriction of the Kummer class to I is n times the tame character, Chapter 7), $\varepsilon(\mathrm{Frob}) = \pm 1$ according as the reduction is split or nonsplit, and c' is identified in Chapter 17. *Proved here (verified symbolically).* Chapter 11 (Lemma lem:tamagawa-exp), DDT, Theorem 3.1(e), Silverman ATAEC, Chapter V for the origin of the shape. Lemma lem:tamfactor (the basis e, h, f and the three identities).

model-computation (proposition; the group-cohomological content of Lemma lem:tamfactor). Let ℓ be odd, $M = \mathrm{ad}^0\, T$ as in tate-curve-shape-matrices, $v = v_\ell(n')$. (a) $(t-1)M = n'(\mathbb{Z}_\ell e \oplus \mathbb{Z}_\ell h)$ (because $2 \in \mathbb{Z}_\ell^\times$), so $H^1(I, M) = M/(t-1)M = (\mathbb{Z}_\ell/n')\bar{e} \oplus (\mathbb{Z}_\ell/n')\bar{h} \oplus \mathbb{Z}_\ell \bar{f}$, with $H^1(I, M)_{\mathrm{tors}} = (\mathbb{Z}/\ell^v)\bar{e} \oplus (\mathbb{Z}/\ell^v)\bar{h} \simeq (\mathbb{Z}/\ell^v)^2$. (b) $\overline{\mathrm{Frob}}$ acts on the torsion by $\bar{e} \mapsto \bar{e}$, $\bar{h} \mapsto q^{-1}(\bar{h} - 2\varepsilon c'\bar{e})$. (c) $H^1(I, M)_{\mathrm{tors}}^{G/I} = \{x\bar{e} + y\bar{h} : (q-1)y \equiv 0,\ c'y \equiv 0 \pmod{\ell^v}\} \simeq \mathbb{Z}/\ell^v \oplus \mathbb{Z}/\ell^m$ with $m = \min(v, v_\ell(q-1), v_\ell(c'))$; it has order ℓ^{v+m}, length $v + m$ and Fitting ideal $\ell^{v+m}\mathbb{Z}_\ell$. (d) m depends on c' only modulo $\ell^{\min(v, v_\ell(q-1))}\mathbb{Z}_\ell = (n', q-1)\mathbb{Z}_\ell$,

which is exactly the ambiguity of $c' = \kappa(\mathrm{Frob})$ (changing the lift of Frob by t^a changes c' by $qn'a$, changing κ by a coboundary changes it by a multiple of $q-1$); the identification of this class with the unit part of the Tate parameter is Chapter 17's. (e) Check with $M = T$: $H^1(I,T) = (\mathbb{Z}_\ell/n')\bar{e}_1 \oplus \mathbb{Z}_\ell\bar{e}_2$, torsion $(\mathbb{Z}/\ell^v)\bar{e}_1$ on which $\overline{\mathrm{Frob}}$ acts by $q^{-1}\cdot\varepsilon q = \varepsilon$; the invariants are all of it if $\varepsilon = 1$ and 0 if $\varepsilon = -1$ (ℓ odd), with Fitting ideal $\ell^v\mathbb{Z}_\ell$, resp. $\mathbb{Z}_\ell$: the ℓ-part of the Tamagawa number c_q (n for split, 1 or 2 for nonsplit reduction, Chapter 10). *Proved here, every step.* The paper's Lemma lem:tamfactor; FPR94, I.4.2.2; DFG04, p. 708. Lemma lem:tamfactor (all of it except the Kummer identification of c' and the definition of Tam), Remark rem:ko(1), Section sec:tam.

fitting-ideal-of-invariants (corollary / remark). Recalled from Chapter 3: for a finite $\mathbb{Z}_\ell$-module A, $\mathrm{Fitt}_{\mathbb{Z}_\ell}(A) = \ell^{\mathrm{length}\,A}\mathbb{Z}_\ell$, and $\mathrm{length}(\bigoplus \mathbb{Z}/\ell^{a_i}) = \sum a_i$. Hence in the model setting the ideal $\mathrm{Fitt}_{\mathbb{Z}_\ell}\big(H^1(I,M)^{G/I}_{\mathrm{tors}}\big)$ is defined, equals $\ell^{v+m}\mathbb{Z}_\ell$ for $M = \mathrm{ad}^0\,T$ and $\ell^v\mathbb{Z}_\ell$ or $\mathbb{Z}_\ell$ for $M = T$; this ideal is what Chapter 18 (following FPR94, I.4.2.2 and DFG04, p. 708) calls the Tamagawa factor $\mathrm{Tam}_q(M)$, and the formula $v_\ell(\mathrm{Tam}_q(\mathrm{ad}^0\,T)) = v_\ell(n) + \min(v_\ell(n), v_\ell(q-1), v_\ell(c))$ of Lemma lem:tamfactor follows once $v_\ell(n') = v_\ell(n)$ (Chapter 7) and $v_\ell(c') = v_\ell(c)$ in the paper's sense (Chapter 17). *Proved here / recalled.* Chapter 3; FPR94, I.4.2.2; DFG04, p. 708. Section 1.2 of the paper, Lemma lem:tamfactor, Lemma lem:fpr.

18.1.10 Section 11: cup products and Pontryagin duality

cup-product (definition / theorem). For G (abstract; profinite with type (D) modules; or continuous cochains with type (C)) and G-modules A, B: $(a\cup b)(g_1,\dots,g_{p+q}) = a(g_1,\dots,g_p)\otimes g_1\cdots g_p\, b(g_{p+1},\dots,g_{p+q}) \in A\otimes B$ satisfies $d(a\cup b) = da\cup b + (-1)^p a\cup db$ and induces $H^p(G,A)\otimes H^q(G,B)\to H^{p+q}(G,A\otimes B)$; a G-pairing $A\times B\to C$ gives $H^p(G,A)\times H^q(G,B)\to H^{p+q}(G,C)$. Properties: bilinear, associative, graded commutative ($a\cup b = (-1)^{pq}\tau_*(b\cup a)$), compatible with res and inf, with connecting homomorphisms, and $\mathrm{cor}(\mathrm{res}\,a\cup b) = a\cup \mathrm{cor}\,b$. For $p=q=1$: $(a\cup b)(g,h) = a(g)\otimes g\,b(h)$. *Definition and the Leibniz formula in degrees ≤ 2 proved here; the properties stated.* NSW, Chapter I, Section 4; Brown, Chapter V, Section 3; Serre, Local Fields, Chapter VIII, Section 3; Cassels-Frohlich, Chapter IV. Chapter 17 (the pairing of Tate local duality $H^1(\mathbb{Q}_p,M)\times H^1(\mathbb{Q}_p,M^*(1))\to H^2(\mathbb{Q}_p,\mu)$, behind the dual terms of Lemma lem:dfg21).

pontryagin-duality-cohomology (proposition / statement). Recalled from Chapter 3: $A^* = \mathrm{Hom}(A,\mathbb{Q}/\mathbb{Z})$ for finite A ($= \mathrm{Hom}(A,\mathbb{Q}_\ell/\mathbb{Z}_\ell)$ if A is ℓ-primary); $T^* = \mathrm{Hom}_{\mathrm{cont}}(T,\mathbb{Q}_\ell/\mathbb{Z}_\ell)$ for T of type (C) is discrete ℓ-primary (T free of rank r gives $(\mathbb{Q}_\ell/\mathbb{Z}_\ell)^r$), and W^* for W discrete ℓ-primary is of type (C); $(-)^*$ is exact and $A^{**} = A$; lengths of finite modules are preserved; for a G-module A the dual carries $(g\varphi)(a) = \varphi(g^{-1}a)$ and is a G-module of the dual type. Statement: for G profinite and A a discrete torsion G-module, $H^n(G,A)^* \simeq H_n(G,A^*)$ (homology of profinite groups); for G finite, $\hat{H}^n(G,A)^* \simeq \hat{H}^{-n-1}(G,A^*)$ for finite A. The duality the paper actually uses, Tate local duality, is Chapter 17's theorem. *Recalled / stated; reference.* Chapter 3; NSW, Chapter I; Brown, Chapter VI. Lemma lem:dfg21 (the Pontryagin duals $H^1_f(\mathbb{Q}_p,T^D)^*$, $H^0(\mathbb{Q},W^D)^*$).

lengths-in-exact-sequences (lemma). For finite $\mathbb{Z}_\ell$-modules: $\mathrm{length}(A^*) = \mathrm{length}(A)$; dualising an exact sequence gives an exact sequence with reversed arrows; in an exact sequence $0\to A\to B\to C\to D\to 0$ of finite $\mathbb{Z}_\ell$-modules, $\mathrm{length}\,B = \mathrm{length}\,A + \mathrm{length}\,C - \mathrm{length}\,D$. Hence, in the exact sequence of Lemma lem:dfg21 with all terms finite, $\mathrm{length}\,H^1_\Sigma(\mathbb{Q},W) = \mathrm{length}\,H^1_f(\mathbb{Q},W) + \sum_{p\in\Sigma\cup\{\infty\}}\mathrm{length}\,H^1_f(\mathbb{Q}_p,T^D) - \mathrm{length}\,H^0(\mathbb{Q},W^D)$, the term at ∞ being 0 for ℓ odd. *Proved here.* Chapter 3. Proof of Theorem thm:main ("$\mathrm{len}\,H^1_\Sigma = \mathrm{len}\,H^1_f + \sum \mathrm{len}\,H^1_f(\mathbb{Q}_p,T^D)$"), Remark rem:ns (the corrected identity with $-h_\ell(E)$).

18.1.11 Section 12: Galois cohomology, notation

galois-cohomology-convention (convention). For a field K with separable closure $\overline{K}$, $G_K = \mathrm{Gal}(\overline{K}/K)$ is profinite (Chapter 5); a Galois module is a G_K-module of type (D), (C) or (V); $H^n(K,M) := H^n(G_K,M)$ (continuous cochains). For $K = \mathbb{Q}_p$: $I_p \trianglelefteq G_{\mathbb{Q}_p}$, $G_{\mathbb{Q}_p}/I_p = \mathrm{Gal}(\overline{\mathbb{F}}_p/\mathbb{F}_p) \simeq \widehat{\mathbb{Z}}$ generated by the arithmetic Frobenius Frob_p ($x\mapsto x^p$); $H^n(I_p,M)$ carries the conjugation action of $G_{\mathbb{Q}_p}/I_p$; $H^n(\mathbb{R},M) :=$

$H^n(\mathrm{Gal}(\mathbb{C}/\mathbb{R}), M)$; $\mathbb{Z}_\ell(1) = \varprojlim \mu_{\ell^k}$ with $\chi_\ell(\mathrm{Frob}_p) = p$ for $p \neq \ell$; Frob_p never denotes the geometric Frobenius Frob_p^{-1}. The unramified part $H^1_{\mathrm{ur}}(\mathbb{Q}_p, M) := H^1(G_{\mathbb{Q}_p}/I_p, M^{I_p}) = \ker(\mathrm{res}\colon H^1(\mathbb{Q}_p, M) \to H^1(I_p, M))$ (inflation-restriction), studied in Chapter 17. *Convention (the equality of the two descriptions of H^1_{ur} proved here).* Serre, Galois Cohomology, Chapter II, Section 1; NSW, Chapter VII. Sections 1.1 and 1.2 of the paper, Lemma lem:tamfactor, Lemma lem:dfg21 and the paragraph after it.

unramified-h1-preview (example). With $\Gamma = G_{\mathbb{Q}_p}/I_p \simeq \widehat{\mathbb{Z}}$ and M of type (C): $H^1_{\mathrm{ur}}(\mathbb{Q}_p, M) = M^{I_p}/(\mathrm{Frob}_p - 1)M^{I_p}$ (procyclic-compact applied to the closed submodule M^{I_p}); for V of type (V): $H^1_{\mathrm{ur}}(\mathbb{Q}_p, V) = V^{I_p}/(\mathrm{Frob}_p - 1)V^{I_p} = 0$ if and only if $\det(1 - \mathrm{Frob}_p \mid V^{I_p}) \neq 0$. This is the paper's argument that $H^1_f(\mathbb{Q}_p, V^D) = H^1_{\mathrm{ur}}(\mathbb{Q}_p, V^D)$ vanishes for $p \in \Sigma$, because $\det(1 - \mathrm{Frob}_p \mid (V^D)^{I_p}) = L_p(A_f, 1)^{-1} \neq 0$ (the equality $H^1_f = H^1_{\mathrm{ur}}$ at $p \neq \ell$ and the value of the determinant are Chapters 18 and 19). *Proved here.* Serre, Local Fields, Chapter XIII, Section 1; Rubin, Euler Systems, Section 1.3. The paragraph after Lemma lem:dfg21.

18.2 2. Section plan (titles, contents, target line counts; total 2000 lines excluding the card)

0. `\section*{Core results at a glance}` (not counted, about 170 lines): the 48 entries above, numbered 16.1 to 16.48, each with its rigour tag and a `\Cref`.

1. `\section*{Where the paper uses this chapter}` (40 lines): Lemma lem:tamfactor (every step: inflation-restriction, $H^1(\langle t\rangle, M) = M/(t-1)M$, the conjugation action of G/I, the $(t-1)M$ computation, the invariants of the torsion and their Fitting ideal); Section 1.2 of the paper (the object $H^1(I_p, M)^{G_{\mathbb{Q}_p}}_{\mathrm{tor}}$ inside the definition of Tam_p); Lemma lem:fpr (same object); Lemma lem:dfg21 and the paragraph after it (Pontryagin duals, lengths in a four-term exact sequence, $H^1(\mathbb{R}, T^D)$ killed by 2, $H^1_{\mathrm{ur}}(\mathbb{Q}_p, V^D) = (V^D)^{I_p}/(\mathrm{Frob}_p - 1)$); Lemma lem:h0 ($H^0$ and ℓ^k-torsion, the length of a cyclic H^0); Remark rem:ns; the proof of Theorem thm:main (the length bookkeeping).

2. Complexes, cohomology and the long exact sequence (170 lines): cochain-complex, homotopy; snake-lemma recalled with the construction of δ; long-exact-sequence with full proof and naturality; a first example (the complex $0 \to M \xrightarrow{t-1} M \to 0$ and the complex of a cyclic group) to foreshadow Section 9.

3. Resolutions, derived functors, Ext and Tor in outline (100 lines): projective-injective-resolutions, ext-tor, ext1-extensions; one page, statements with references; the examples over $\mathbb{Z}$ and $\mathbb{Z}_\ell$; the remark that over a DVR everything stops at degree one.

4. Group cohomology by cochains: H^0, H^1, H^2 (230 lines): g-module; inhomogeneous-cochains with the degree ≤ 2 formulas and the verification of $d \circ d = 0$ in degree one and two; cohomology-as-ext stated; h0-h1 proved; cocycle-in-the-corner proved (two-dimensional reducible representations and H^1 of a character), with the Tate-curve shape as the running example; h2-extensions sketched; functoriality-in-M-and-les proved; h0-and-torsion proved (this is the H^0 half of Lemma lem:h0). RHB aside: crossed homomorphisms as "cocycles" in the sense of projective representations / phase factors (one sentence).

5. Changing the group (230 lines): compatible-pairs; res-inf-cor (cor in degrees 0, 1 with the explicit formula, cor$\circ$res$= [G : H]$ in degree 1 proved); order-kills with the three consequences; conjugation-action proved in degree 1 with the two coboundary formulas (trivial on H; restriction lands in the invariants) and the independence of the representative; inflation-restriction (a), (d) proved in full with cochains (including the continuous-cochain version: the quotient map $G \to G/H$ is open so the induced cochain on G/H is continuous), (b), (c) stated with the transgression sketched; hochschild-serre stated.

6. Finite groups (150 lines): cyclic-group-cohomology (H^1 proved, the rest stated); tate-cohomology (one page); z2-on-l-adic proved, with the explicit statement about $H^1(\mathbb{R}, -)$; the example $H^1(\mathbb{Z}/2, \mathbb{Z}^-) = \mathbb{Z}/2$ and $H^1(\mathbb{Z}/2, \mathbb{Z}_\ell^-) = 0$.

7. Profinite groups and continuous cohomology (220 lines): discrete-g-module; continuous-cohomology with the three coefficient types and the convention, the direct-limit description for (D) proved; les-continuous proved, with the four cases of continuous sections; tate-limits stated (the $n = 1$ case sketched); pitfalls as a `warning` environment with the examples worked; h1-finitely-generated.

8. Groups of order prime to ℓ and pro-ℓ quotients (110 lines): supernatural-order-pro-l; prime-to-l-vanishing proved; cohomology-through-pro-l-quotient; tame-inertia-example (imports from Chapter 7 clearly marked).

9. Procyclic groups (190 lines): procyclic-groups; cocycle-on-powers; procyclic-discrete (Serre's proof); procyclic-compact proved in full (H^1), the two lemmas on the image $C \times \Gamma'_\ell$ and on topologically unipotent u, the vanishing of H^n ($n \geq 2$) stated; procyclic-examples.

10. The model computation: inertia, Frobenius and the torsion of $H^1(I, \mathrm{ad}^0 T)$ (240 lines): model-setting as a `construction` with the dictionary to the paper; frobenius-on-h1-inertia; tate-curve-shape-matrices with the six identities displayed; model-computation (a) to (e) proved line by line; fitting-ideal-of-invariants; a `paperlink` environment quoting the corresponding sentences of the paper's proof and saying which item justifies each.

11. Cup products and Pontryagin duality (120 lines): cup-product (definition, Leibniz rule in low degree, properties stated); pontryagin-duality-cohomology (recalled from Chapter 3, statements with references, pointer to Tate duality in Chapter 17); lengths-in-exact-sequences proved, with the display of the paper's length identity.

12. Galois cohomology: notation and the road ahead (50 lines): galois-cohomology-convention; unramified-h1-preview; a half-page map of Chapters 17 and 18 (what will be proved about H^1_{ur}, Kummer theory, Tate duality, Selmer groups, Tamagawa factors).

13. Exercises (110 lines; 14 to 16 exercises, graded): (1) $d \circ d = 0$ in degree $2 \to 3$. (2) $H^1(\mathbb{Z}/m, \mathbb{Z}) = 0$, $H^2(\mathbb{Z}/m, \mathbb{Z}) = \mathbb{Z}/m$, $H^1(\mathbb{Z}/m, \mathbb{Z}/m) = \mathbb{Z}/m$.

    (3) $\mathbb{Z}/2$ acting by -1 on $\mathbb{Z}$, on $\mathbb{Z}/2$, on $\mathbb{Z}_\ell$ (ℓ odd): all three cohomologies; check the paper's claim that $H^1(\mathbb{R}, T^D)$ is killed by 2.
    (4) The 2-cocycle of $\mathbb{Z}/4$ as an extension of $\mathbb{Z}/2$ by $\mathbb{Z}/2$; $H^2(\mathbb{Z}/2, \mathbb{Z}/2)$. (5) $H^1_{\mathrm{cont}}(\mathbb{Z}_\ell, \mathbb{Z}_\ell) = \mathbb{Z}_\ell$ and $\mathrm{Hom}_{\mathrm{cont}}(\widehat{\mathbb{Z}}, \mathbb{Z}) = 0$. (6) H^0 and H^1 of $\widehat{\mathbb{Z}}$ on $\mathbb{Z}_\ell(1)$ and on μ_{ℓ^k} with Frobenius acting by q. (7) Verify the six matrix identities of tate-curve-shape-matrices and show $(t-1)^3 = 0$ on $\mathrm{ad}^0 T$; check the paper's claim $H^1(I_q, M)_{\mathrm{tor}} \simeq (\mathbb{Z}/\ell^{v_\ell(n)})^2$.
    (5) Redo model-computation for $\ell = 2$: where does it change? (9) The invariants for $M = T$: split versus nonsplit, and comparison with c_q (Chapter 10).
    (6) (Paper check) For $E = 11\mathrm{a}1$ at $q = 11$, $\ell = 5$: $n = 5$, $q \equiv 1 \pmod 5$, and the unit part of the Tate parameter is $\equiv -1 \pmod{11}$ (Chapter 10 / Chapter 22), a fifth power in $\mathbb{F}_{11}^\times$; deduce that $H^1(I_{11}, \mathrm{ad}^0 T_5 E)^{G/I}_{\mathrm{tors}}$ has order 25; explain why this does not contradict $\deg \varphi_E = 1$ (Theorem thm:main needs $E[5]$ irreducible, and 11a1 has a rational 5-torsion point). (11) Independence of the conjugation action of the lift of Frob.
    (7) $\mathrm{cor} \circ \mathrm{res} = [G : H]$ on H^1 from the explicit formula. (13) Inflation-restriction with continuous cochains for a type (C) module.
    (8) The ℓ^k-torsion of $H^0(G, T \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell)$ and the "largest k" description of the length. (15) Show that for $\Gamma = \mathbb{Z}_p$ ($p \neq \ell$) acting on a type (C) module through an element of order p^a, $H^1(\Gamma, M) = \ker N/(F-1)M$, not $M/(F-1)M$. (16) (harder) Prove inflation-restriction (b) by constructing the transgression.

14. Notes and references (40 lines): Brown, Weibel, Serre (Local Fields; Galois Cohomology), NSW, Cassels-Frohlich, Milne CFT, Tate 1976, Rubin Appendix B; which theorem numbers were cited; the history of the Hochschild-Serre spectral sequence in one sentence; where the paper's conventions come from (DFG04, FPR94).

18.3 3. Results from earlier chapters this chapter relies on

- `ch:modules`: exact sequences and the snake lemma (recalled, not reproved); Hom and $\otimes$; the structure of finitely generated $\mathbb{Z}_\ell$-modules ($\mathbb{Z}_\ell^r \oplus \bigoplus \mathbb{Z}/\ell^{a_i}$); submodules of finitely generated $\mathbb{Z}_\ell$-modules are finitely generated and closed in the ℓ-adic topology; Nakayama's lemma (including the topological form: a compact $\mathbb{Z}_\ell$-module A with $A/\ell A$ finite is finitely generated); length of a finite $\mathbb{Z}_\ell$-module; Fitting ideals ($\mathrm{Fitt}(A) = \ell^{\mathrm{length}\, A}\mathbb{Z}_\ell$ for finite A, Fitting ideal of a direct sum); Pontryagin duality $\mathrm{Hom}(-, \mathbb{Q}_\ell/\mathbb{Z}_\ell)$ between finitely generated $\mathbb{Z}_\ell$-modules and discrete ℓ-primary modules, exactness, $A^{**} = A$, lengths preserved; $T \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$ and its ℓ^k-torsion $T/\ell^k T$; $\mathbb{Q}_\ell/\mathbb{Z}_\ell$ injective (divisible).
- `ch:rings`: $\mathbb{Z}_\ell$, $\mathbb{Z}_\ell^\times$, the valuation v_ℓ, $(\mathbb{Z}/\ell^k)^\times$; an integer q prime to ℓ is a unit of $\mathbb{Z}_\ell$; $v_\ell(q-1)$.
- `ch:groups`: cyclic groups, inverse limits, profinite groups $\mathbb{Z}_\ell$ and $\widehat{\mathbb{Z}} = \prod_p \mathbb{Z}_p$, semidirect products, conjugation and normal subgroups, Sylow theory for finite groups (background to pro-ℓ Sylow subgroups).
- `ch:reps`: $\mathrm{End}(V) = V \otimes V^\vee$, matrix units $E_{ij} = e_i \otimes e_j^\vee$; the trace-zero adjoint ad^0 with the conjugation action $\varphi \mapsto \rho(g)\varphi\rho(g)^{-1}$; unipotent elements.
- `ch:galois`: the Krull topology, G_K profinite, continuous characters and representations, restriction to closed subgroups, complex conjugation as an element of order two.
- `ch:local`: $G_{\mathbb{Q}_p}$, I_p, the wild inertia P_p (pro-p), the tame quotient $I_p/P_p = \prod_{\ell' \neq p} \mathbb{Z}_{\ell'}(1)$, the tame character t_ℓ, the relation $\mathrm{Frob}\; t\; \mathrm{Frob}^{-1} = t^p$ (equivalently $\mathrm{Frob}^{-1}\, t\, \mathrm{Frob} = t^{p^{-1}}$) in $\mathrm{Gal}(\mathbb{Q}_p^{\mathrm{tame}}/\mathbb{Q}_p)$, $G_{\mathbb{Q}_p}/I_p = \mathrm{Gal}(\overline{\mathbb{F}}_p/\mathbb{F}_p) \simeq \widehat{\mathbb{Z}}$ generated by the arithmetic Frobenius, $\mathbb{Z}_\ell(1)$ and $\chi_\ell(\mathrm{Frob}_p) = p$ for $p \neq \ell$, the Kummer class of $u \in \mathbb{Q}_p^\times$ and the fact that its restriction to I_p is $v_p(u)$ times the tame character (used only to explain the datum n' of the model).
- `ch:galrep` (motivation only, not used logically): the shape $\left(\begin{smallmatrix} \chi_\ell \varepsilon & \kappa \\ 0 & \varepsilon \end{smallmatrix}\right)$ of $\rho_{E,\ell}|_{G_{\mathbb{Q}_q}}$ at a prime of multiplicative reduction (DDT, Theorem 3.1(e); Lemma lem:tamagawa-exp), and the unipotence of the inertia action.

18.4 4. Notes for the writer

Conventions to fix and keep (they must agree with Chapters 7, 17, 18):

1. Conjugation action: $(\sigma \cdot c)(h) = \sigma\, c(\sigma^{-1} h \sigma)$, verbatim the paper's formula. With $\mathrm{Frob}^{-1}\, t\, \mathrm{Frob} = t^{q^{-1}}$ this gives the factor q^{-1} in "$\overline{\mathrm{Frob}}$ acts on $M/(t-1)M$ as $q^{-1}\rho(\mathrm{Frob})$". Do not switch to the other common convention $c \mapsto \sigma^{-1} c(\sigma h \sigma^{-1})$ (it gives $q\rho(\mathrm{Frob})^{-1}$; the invariants are the same, the formulas are not).
2. Frobenius is arithmetic throughout; the relation is $\mathrm{Frob}\; t\; \mathrm{Frob}^{-1} = t^q$ ("Frobenius raises tame inertia to the q-th power"); state it as an imported fact from Chapter 7 in exactly the two equivalent forms, and note $q^{-1} \in \mathbb{Z}_\ell$ because $q \neq \ell$.
3. Well-definedness of the Frobenius action on $M/(t-1)M$ needs only $\rho(\mathrm{Frob})(t-1)M \subseteq (t-1)M$, which follows from $(t^q - 1) = (t-1)(1 + \cdots + t^{q-1})$. Do not claim $(t^q - 1)M = (t-1)M$ (true, but it needs $t - 1$ in the radical of $\mathbb{Z}_\ell[t]$; unnecessary).

4. The identity $H^1(\Gamma, M) = M/(F-1)M$ has exact hypotheses: Γ procyclic with a surjection onto $\mathbb{Z}_\ell$ and M of type (C) or (V); or $\Gamma = \widehat{\mathbb{Z}}$ and M discrete torsion; or $\Gamma = \mathbb{Z}_\ell$ and M discrete ℓ-primary. It fails for finite cyclic Γ ($\ker N/(F-1)M$), for $\Gamma = \mathbb{Z}_p$, $p \neq \ell$, on $\mathbb{Z}_\ell$-modules (again $\ker N/(F-1)M$), and for discrete non-torsion M ($H^1(\widehat{\mathbb{Z}}, \mathbb{Z}) = 0$). Say so in a warning; Exercise 15 covers the $\mathbb{Z}_p$ case.
5. The lemma $c(t^a) \equiv a\, c(t) \pmod{(t-1)M}$ for $a \in \mathbb{Z}_\ell$ is the precise form of the paper's "$c(t^k) \equiv k\, c(t)$"; it needs $(t-1)M$ closed, which holds because finitely generated submodules of finitely generated $\mathbb{Z}_\ell$-modules are closed (Chapter 3).
6. "The kernel of $I \to \langle t \rangle$ has trivial pro-ℓ quotient" (paper) is weaker than "has order prime to ℓ"; both hold for tame inertia. For H^1 with trivial action the weaker property suffices ($\mathrm{Hom}_{\mathrm{cont}}(K, M) = 0$), and only the three-term inflation-restriction sequence is needed; no spectral sequence.
7. That the action of I_q on $\mathrm{ad}^0 T_\ell E$ factors through the maximal pro-ℓ quotient is a hypothesis of the model (true because the image of inertia is unipotent, hence pro-ℓ, Chapter 11); state it as such, do not prove the Tate-curve description here.
8. ℓ odd is used in exactly two places: $2 \in \mathbb{Z}_\ell^\times$ (so $(t-1)M = n'(\mathbb{Z}_\ell e \oplus \mathbb{Z}_\ell h)$ rather than $n'(2\mathbb{Z}_\ell e \oplus \mathbb{Z}_\ell(h - n'e))$) and the vanishing of $H^1(\mathbb{Z}/2, -)$ on ℓ-adic modules. Flag both (Exercise 8).
9. In the model, $n' \neq 0$; $v = v_\ell(n')$ may be 0, in which case the torsion is 0 and all Fitting ideals are $\mathbb{Z}_\ell$ (consistent with the paper's formula). The free part $\mathbb{Z}_\ell \bar{f}$ of $H^1(I, \mathrm{ad}^0 T)$ is not Frob-stable ($\mathrm{Frob} \cdot \bar{f} = q^{-1}(\bar{f} + \varepsilon c' \bar{h} - c'^2 \bar{e})$); only the torsion is needed and it is stable.
10. $c' = \kappa(\mathrm{Frob})$ is a cocycle value depending on the lift of Frob and on the cocycle in its class; explain that only its class modulo $(n', q-1)\mathbb{Z}_\ell$ matters and that this is what the paper means by "the image of κ in $H^1(G/I, \mathbb{Z}_\ell(1)^I) = \mathbb{Z}_\ell(1)/(q-1)$". Do not attempt the Kummer-theoretic identification with the unit part of the Tate parameter: that is the end of Lemma lem:tamfactor and belongs to Chapter 17.
11. Continuous cohomology: use continuous cochains throughout, define the three coefficient types once, and never write $H^n(G, T) = \varinjlim H^n(G/U, T^U)$ for compact T. State Tate's $\varprojlim^1$ sequence as a theorem with reference (Tate95, Section 2; NSW, Chapter II, Section 7; Rubin, Appendix B) and use it only through its two corollaries. The long exact sequence for compact or mixed coefficients needs a continuous section: give the four cases.
12. Theorem numbers: cite Weibel 1.3.1, 1.3.2, 3.4.3, 6.8.2 and Serre, Local Fields, Chapter XIII, Section 1, Proposition 1 only if the writer verifies them against the books; otherwise cite chapter and section as in the entries above. Serre's Galois Cohomology is not in refs.bib: add to `refs.bib` (not `bib/ch16.bib`: Chapters 3, 16 and 17 cite it, decision B14, important_notes.md B21) an entry `SerreGaloisCoh` (J.-P. Serre, Galois Cohomology, Springer Monographs in Mathematics, Springer, Berlin, 1997). `Tate95` in refs.bib is Tate's 1976 paper "Relations between K_2 and Galois cohomology" (Invent. Math. 36), the standard reference for continuous cohomology and inverse limits; cite it for tate-limits.
13. Do not duplicate: Chapter 3 proves the snake lemma, Fitting ideals, lengths, Pontryagin duality of modules (recall them in one line each). Chapter 17 proves Hilbert 90, Kummer theory, Tate local duality, H^1_{ur} and its annihilator, the torsion of $H^1(K, T)$ via $H^0(K, W)$, and the Kummer class of the Tate parameter. Chapter 18 defines H^1_f, Tam_p and proves Lemma lem:fpr; Chapter 11 proves Lemma lem:h0 (here only its H^0-and-torsion mechanism). This chapter supplies the group-cohomology engine and the model computation; it must not define Selmer groups, Bloch-Kato conditions or Tamagawa factors (mention them by name with `\Cref{ch:blochkato}` only).
14. Pontryagin duality for cohomology: state only; the real duality used by the paper is Tate's (Chapter 17). The paper's "* is the Pontryagin dual" in Lemma lem:dfg21 is the dual of finite groups, and the only consequences used are that lengths are preserved and that dualising preserves exactness (lengths-in-exact-sequences).
15. Exercise 10 (11a1 at $q = 11$, $\ell = 5$) must carry the caveat that $E[5]$ is reducible for 11a1 ($E(\mathbb{Q}) \simeq \mathbb{Z}/5$), so the paper's theorem does not apply and no contradiction arises with $\deg \varphi_E = 1$; the computation illustrates Lemma lem:tamfactor, which needs no irreducibility. The value $u \equiv -1 \pmod{11}$ comes from $j(11a1) = -2^{12} \cdot 31^3/11^5$ and $q_E = j^{-1}(1 + O(11^5))$; the fifth powers in $\mathbb{F}_{11}^\times$ are $\{\pm 1\}$.
16. Typography: no em or en dashes anywhere (write "Hochschild-Serre", "Fontaine-Perrin-Riou",

"Bloch-Kato" with plain hyphens; "Sections 1.1 to 1.4"); ASCII source only; \ell, not the character. Labels prefixed cohomology: (e.g. \label{thm:cohomology:procyclic-compact}, \label{prop:cohomology:model}). Macros available: \Frob, \Fitt, \len, \ad (which expands to ad^0: write \ad T), \Zl, \Ql, \Qlmod, \Tl, \Hur, \Hf, \Cor, \Inf, \Res, \Ext, \Tor, \Norm, \tors, \sm{a}{b}{c}{d}, \Mtwo, \paperlink, \warning, \construction, \convention. Write $\varprojlim$, $\varinjlim$ with the amsmath commands.

17. Each section opens with one paragraph saying what it is for and where the paper uses it. Every theorem has a proof, a labelled sketch, or "We do not prove this; see ... ".
18. The card must list all 48 entries with rigour tags; the entries of Section 10 are the ones an expert will look up, so their statements must be exactly those above.

18.5 5. Consistency with the later outlines and the Frobenius convention

Added after the Chapters 1, 5 and 6 outlines and important/important_notes.md were written.

1. **Two Frobenius conventions in the paper.** This chapter uses the arithmetic Frobenius throughout, which is the convention of the paper's proof of Lemma lem:tamfactor (rho(Frob) = [[eps q, c'], [0, eps]]) and of the conjugation-action formula above. The L-factors of the paper's Section 1.1, the table of omitted factors and the proof of Lemma lem:euler are correct only with the geometric Frobenius Frobˆ{-1} (important_notes point 11: at a multiplicative prime the inertia-fixed line of ad V has arithmetic eigenvalue q, so the arithmetic convention would give 1 - qˆ{1-s} instead of 1 - qˆ{-1-s}). The writer must keep the arithmetic convention here, say so in a convention box at the start of the chapter, and add a warning that Chapters 18 and 19 compute L-factors with the geometric Frobenius, with a one-line translation: an eigenvalue lambda of the arithmetic Frobenius is lambdaˆ{-1} for the geometric one. Chapters 5 (convention C2) and 6 (item 3.7) fix Frob_P as the arithmetic Frobenius; this agrees.
2. **The class c' in Lemma lem:tamfactor.** Writer note 10 above records that c' = kappa(Frob) depends on the lift of Frobenius and on the cocycle. important_notes point 7 gives the clean fix: write q_E = qˆn u, so kappa = n kappa_q + kappa_u with kappa_u unramified, and choose the lift of Frobenius that fixes a compatible system of l-power roots of q. Then c' = kappa_u(Frob) is the class of u in F_qˆx tensor Z_l, and in general c' is well defined modulo (n', q - 1) Z_l, which does not change min(v_l(n), v_l(q - 1), v_l(c')). The Kummer-theoretic side of this is in Chapters 5 and 17; this chapter proves only the group-cohomological part (Hˆ1 of the procyclic quotient and the invariants).
3. **Absolute irreducibility.** Wherever this chapter's examples use "rhobar irreducible" to conclude that an equivariant map is an isomorphism or that Hˆ0 is a line, say "absolutely irreducible" (important_notes point 10). For rhobar_{E,l} with l odd the two agree because of complex conjugation.

18.6 6. Points for important_notes.md

None beyond those already recorded. The computation of Hˆ1(I_q, ad T) in the paper's Lemma lem:tamfactor, (t - 1)M = n'(Z_l e + Z_l h) and Hˆ1 = (Z_l/n')ˆ2 + Z_l, was rechecked in Section 10 of this outline and is correct, given that l is odd (writer note 8).

Chapter 19

Line-up for Chapter 17: Galois cohomology, local and global, and Selmer groups (ch:galcoh, chapters/ch17_galcoh.tex)

Stage 1 output (core results in teaching order). Part V (Cohomology, Selmer groups and L-functions), between Chapter 16 (group cohomology, continuous cohomology, procyclic groups, the model computation of Lemma lem:tamfactor) and Chapter 18 (Bloch-Kato conditions, Tamagawa factors, the adjoint motive). Target length 2,110 lines of LaTeX excluding the card (Part 11 below), inside the 1,000 to 2,200 range of PLAN.md Section 7.3. Rigour tags: **P** = proved here, **S** = proof sketched here, **R** = stated; reference.

Item format (as in the Chapter 2 to 15 outlines): number, label-ready name, kind, rigour tag; the precise statement with all hypotheses; reference (book and theorem number only where sure, otherwise book and chapter or section; a number followed by "(check)" must be verified by the writer before it is printed, or replaced by the section); the paper's use by its `\label` names, or "background". The numerical examples (Tate parameters and unit parts, full 5-torsion of 11a1 over $\mathbb{Q}_{11}$, orders of local cohomology groups, the toy Selmer computations) were checked with PARI/GP 2.17 through cypari2 and Cremona's ecdata files (verification log, Part 15).

19.1 0. Purpose of the chapter, and what it must deliver

19.1.1 0.1 Purpose

Chapter 16 built the engine (cochains, inflation-restriction, continuous cohomology, procyclic groups) and did the pure group cohomology of Lemma lem:tamfactor. This chapter specialises to Galois groups of fields, local and global, and builds the arithmetic that the paper uses in four places:

1. **Lemma lem:tamfactor (the end of its proof) and Lemma lem:tamagawa-exp.** Hilbert 90 and Kummer theory in cohomological form, $H^1(K,\mu_n) = K^\times/(K^\times)^n$ and $H^1(K,\mathbb{Z}_\ell(1)) = \varprojlim K^\times/(K^\times)^{\ell^k}$; over $\mathbb{Q}_q$ ($q \neq \ell$) the unramified classes are exactly the Kummer classes of units, $H^1(G/I,\mathbb{Z}_\ell(1)) = \mathbb{Z}_\ell(1)/(q-1) \simeq \mathbb{F}_q^\times \otimes \mathbb{Z}_\ell$; the Kummer class of the Tate parameter $q_E = q^n u$ decomposes as $n\kappa_q + \kappa_u$, and the value c' in the corner of $\rho(\mathrm{Frob})$ is, up to the sign $\varepsilon(\mathrm{Frob})$ and modulo the ideal $(n, q-1)\mathbb{Z}_\ell$, the class of the reduction $\bar{u}$ in $\mathbb{F}_q^\times \otimes \mathbb{Z}_\ell$. This proves the sentence "Finally c' is the image of κ in $H^1(G/I,\mathbb{Z}_\ell(1)^I)$ ... the class of the unit part of q_E in $\mathbb{F}_q^\times \otimes \mathbb{Z}_\ell$ by Kummer theory" in a corrected and precise form (important_notes.md points 7, 31, 42; Section 5).

2. **The paragraph after Lemma lem:dfg21.** Unramified cohomology $H^1_{\mathrm{ur}}(\mathbb{Q}_p, M) = M^{I_p}/(\mathrm{Frob}_p - 1)M^{I_p}$; its vanishing for V exactly when $\det(1 - \mathrm{Frob}_p \mid V^{I_p}) \neq 0$ (a condition independent of the Frobenius convention, although the identification of the determinant with $L_p(A_f, 1)^{-1}$ needs the geometric one, point 11); the torsion of $H^1(\mathbb{Q}_p, T)$ as $H^0(\mathbb{Q}_p, W)$ modulo its divisible part; hence "$H^1_f(\mathbb{Q}_p, T^D)$ is the torsion of $H^1(\mathbb{Q}_p, T^D)$, a finite group"; and "$H^1(\mathbb{R}, T^D)$ is killed by 2" (Sections 2, 4, 6).
3. **Lemma lem:dfg21, the proof of Theorem thm:main and Remark rem:ns.** Tate local duality, the exact orthogonality of unramified classes, propagation of local conditions from V to T and W, Selmer structures and their duals, the Greenberg-Wiles formula, the Poitou-Tate sequence, the comparison sequence of two Selmer structures, and an abstract form of DFG04 Lemma 2.1 (relaxing the conditions at a set Σ) with the length identity $\operatorname{length} H^1_{\mathcal{L}^\Sigma}(\mathbb{Q}, W) = \operatorname{length} H^1_{\mathcal{L}}(\mathbb{Q}, W) + \sum_{p\in\Sigma} \operatorname{length} \mathcal{L}_{T^D,p} - \operatorname{length} H^0(\mathbb{Q}, W^D)$, of which the proof of Theorem thm:main uses the case $H^0(\mathbb{Q}, W^D) = 0$ and Remark rem:ns the case of length $h_\ell(E)$ (Sections 8 to 10). Chapter 18 plugs in the Bloch-Kato structure and DFG04 Theorem 2.7.
4. **The Selmer groups of Theorem thm:main.** $H^1_f(\mathbb{Q}, \mathrm{ad}^0 T \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell)$ and $H^1_\Sigma(\mathbb{Q}, W)$ are Selmer groups of Selmer structures in the sense of Section 8; the classical n-Selmer group of an elliptic curve (Section 7) is the model, and it also discharges Chapter 9's deferral of the weak Mordell-Weil theorem.

It also discharges what earlier chapters deferred to it (by `\Cref{ch:galcoh}`): Chapter 5 (Hilbert 90 in profinite form; $H^1(K, \mu_n) = K^\times/K^{\times n}$ without $\mu_n \subset K$; the Kummer class in $H^1(K, \mathbb{Z}_\ell(1))$; $G_{\mathbb{R}} = \{1, c\}$ and $H^1(\mathbb{R}, -)$; the identification of the class of the unit part, end of Lemma lem:tamfactor), Chapter 7 (6.1(ii): the Kummer maps are isomorphisms, via Hilbert 90; the cohomological half of the fix of point 7; the local Euler characteristic count of its 2.7(iv)), Chapter 8 (4.2(iii): Hilbert 90 for $\bar{k}(C)$, stated there), Chapter 9 (9.1: weak Mordell-Weil via the Kummer map and Selmer groups; 5.3, 5.5: $E[n]$, the Weil pairing as $E[n] \simeq E[n]^*(1)$), Chapter 10 (7.4, 7.6, 7.8: q_E, u, $c(u)$, c_q in Lemma lem:tamfactor), Chapter 11 (3.2: the Kummer class $\varepsilon\kappa$ in the corner), Chapter 15 (3.5: the local types used for unramified cohomology at $p \mid N$), Chapter 16 (Hilbert 90, Kummer theory, Tate local duality, H^1_{ur} and its annihilator, the torsion of $H^1(K, T)$ via $H^0(K, W)$, finiteness of H^1 for $G_{\mathbb{Q}_p}$ and $G_{\mathbb{Q},S}$, and the Kummer class of the Tate parameter: its writer notes 10 and 13), Chapter 21 (items A2.7 step (7), A4.9 "attached facts" and proof idea, A4.10 steps (1) to (5) for the inputs, A5.2).

19.1.2 0.2 Global conventions fixed in this chapter (to be typeset as `convention` environments)

- **C1 (fields and groups).** K is a field of characteristic not ℓ (in practice $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{Q}_p$, a finite extension of $\mathbb{Q}_p$, $\mathbb{F}_q$); $\overline{K}$ a separable closure; $G_K = \mathrm{Gal}(\overline{K}/K)$ with the Krull topology. $H^n(K, M) := H^n(G_K, M)$ is continuous cochain cohomology (Chapter 16), with the three coefficient types of Chapter 16: (D) discrete, (C) finitely generated $\mathbb{Z}_\ell$-modules with the ℓ-adic topology, (V) finite-dimensional $\mathbb{Q}_\ell$-vector spaces. For $K = \mathbb{Q}_p$: $I_p \trianglelefteq G_{\mathbb{Q}_p}$, $G_{\mathbb{Q}_p}/I_p \simeq \hat{\mathbb{Z}}$; for $K = \mathbb{R}$: $G_{\mathbb{R}} = \{1, c\}$; $H^n(\mathbb{R}, M) = H^n(G_{\mathbb{R}}, M)$.
- **C2 (Frobenius).** $\mathrm{Frob} = \mathrm{Frob}_p$ is an **arithmetic** Frobenius (acts on $\overline{\mathbb{F}}_p$ by $x \mapsto x^p$; $\chi_\ell(\mathrm{Frob}) = p$ for $\ell \neq p$), as in Chapters 5, 6, 7 (L3), 16 and 21. $\Phi = \mathrm{Frob}^{-1}$ is the geometric Frobenius. Group cohomology (conjugation actions, $H^1_{\mathrm{ur}} = M^I/(\mathrm{Frob} - 1)$, the matrices of Lemma lem:tamfactor) uses Frob. Whenever a local L-factor is named, it is $L_p(V, s) = \det(1 - \Phi p^{-s} \mid V^{I_p})^{-1}$ (Chapter 7, L6). An arithmetic eigenvalue λ is the geometric eigenvalue λ^{-1}. (The Weil-Deligne relation is not used in this chapter; important_notes.md B6.)
- **C3 (twists and duals).** $\mathbb{Z}_\ell(1) = T_\ell\mu = \varprojlim_k \mu_{\ell^k}$, $\mathbb{Q}_\ell/\mathbb{Z}_\ell(1) = \mu_{\ell^\infty}$, $\mathbb{Q}_\ell(1) = \mathbb{Z}_\ell(1) \otimes \mathbb{Q}_\ell$; $M(1) = M \otimes \mathbb{Z}_\ell(1)$. $A^* = \mathrm{Hom}_{\mathrm{cont}}(A, \mathbb{Q}_\ell/\mathbb{Z}_\ell)$ (or $\mathrm{Hom}(A, \mathbb{Q}/\mathbb{Z})$ for finite A) is the **Pontryagin dual**, exactly as in the paper ("* is the Pontryagin dual" in Lemma lem:dfg21) and in Chapter 16. $T^\vee = \mathrm{Hom}_{\mathbb{Z}_\ell}(T, \mathbb{Z}_\ell)$, $V^\vee = \mathrm{Hom}_{\mathbb{Q}_\ell}(V, \mathbb{Q}_\ell)$ are linear duals. The **Kummer dual** (Cartier dual) of a finite or discrete ℓ-primary module M is $M^*(1) = \mathrm{Hom}(M, \mu_{\ell^\infty})$; for a lattice T: $T^D := \mathrm{Hom}_{\mathbb{Z}_\ell}(T, \mathbb{Z}_\ell(1)) = T^\vee(1)$, $V^D = T^D \otimes \mathbb{Q}_\ell = \mathrm{Hom}(V, \mathbb{Q}_\ell(1))$, $W^D := V^D/T^D \simeq \mathrm{Hom}(T, \mathbb{Q}_\ell/\mathbb{Z}_\ell(1)) = T^*(1)$. These are the paper's T^D, V^D, W^D (Section 4.2), there for $T = A = \mathrm{ad}^0 T_\ell E$; the identification $T^D \simeq A(1)$ via

the trace form is Chapter 18's (and Chapter 4's 7.6). The book's symbol $*$ is otherwise overloaded (important_notes.md B4); in this chapter it means Pontryagin dual only.

- **C4 (lattices).** T is a free $\mathbb{Z}_\ell$-module of finite rank with a continuous $\mathbb{Z}_\ell$-linear action, $V = T \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$, $W = T \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell/\mathbb{Z}_\ell = V/T$, $W[\ell^k] = T/\ell^k T$ (Chapter 3, 8.3); $\iota\colon H^1(K,T) \to H^1(K,V)$ and $\pi\colon H^1(K,V) \to H^1(K,W)$ are the maps induced by $T \subset V \to W$. "Length" is $\mathbb{Z}_\ell$-length; $v_\ell(\ell) = 1$.
- **C5 (places, sets).** p runs over primes, v over places of $\mathbb{Q}$ including ∞; $\mathbb{Q}_\infty = \mathbb{R}$. S is a finite set of places containing ∞; $\mathbb{Q}_S$ the maximal extension of $\mathbb{Q}$ in $\overline{\mathbb{Q}}$ unramified outside S; $G_{\mathbb{Q},S} = \mathrm{Gal}(\mathbb{Q}_S/\mathbb{Q})$. A fixed embedding $\overline{\mathbb{Q}} \to \overline{\mathbb{Q}}_v$ for each v identifies $G_{\mathbb{Q}_v}$ with a decomposition group (Chapter 7, 3.5); localisation $\mathrm{res}_v\colon H^1(\mathbb{Q},M) \to H^1(\mathbb{Q}_v,M)$ does not depend on the embedding (conjugate embeddings give the same map on cohomology, Chapter 16 conjugation-action). Σ denotes the paper's finite set of primes (in the paper, the primes dividing N); $\ell \notin \Sigma$.
- **C6 (signs of Kummer maps).** The connecting map of $1 \to \mu_n \to \overline{K}^\times \xrightarrow{n} \overline{K}^\times \to 1$ is $\delta(a)(\sigma) = \sigma(\alpha)/\alpha$ with $\alpha^n = a$, the convention of Chapter 16 (h0-h1: $\delta(m'') = [g \mapsto gm - m]$), of Chapter 5 ($\chi_a$) and of Chapter 7 ($\kappa_x$). No sign is introduced anywhere.
- **C7 (labels and typography).** Labels `...:galcoh:...`; the names below are the label stems. No em or en dashes and no double hyphens anywhere (write "Poitou-Tate", "Greenberg-Wiles", "Bloch-Kato", "Hochschild-Serre" with plain hyphens; ranges "1 to 8"); ASCII source; `\ell`; macros `\Frob`, `\Fitt`, `\len`, `\Tam`, `\ad` and those listed in Chapter 16's writer note 16 (`\Zl`, `\Ql`, `\Qlmod`, `\Hur`, `\Hf`, `\Inf`, `\Res`, `\Cor`, `\tors`); no new macros.

19.1.3 0.3 Scope check (PLAN.md Section 4 entry for Chapter 17)

Each scope item is mapped to line-up items in Part 12 (coverage table). In order of the PLAN entry: Galois cohomology of a field (1.1, 1.2); Hilbert 90 (1.3); Kummer theory $H^1(K,\mu_n) = K^\times/(K^\times)^n$ proved (1.4); Kummer map and class (1.5, 1.6); $H^1(K,\mathbb{Z}_\ell(1)) = \varprojlim K^\times/(K^\times)^{\ell^k}$ and $\mathbb{Z}_\ell(1) = T_\ell\mu$ (2.4, 2.5); Brauer group and $H^2(K,\mu_n)$ (1.7); finiteness of $H^i(\mathbb{Q}_p,M)$ (3.1); local Euler-Poincare characteristic (3.2, 3.5); Tate local duality (3.3, 3.4); H^1_{ur} via inflation-restriction (4.1, 4.3); $H^1_{\mathrm{ur}} = M^{I_p}/(\mathrm{Frob}_p - 1)M^{I_p}$ (4.2); the annihilator of H^1_{ur} (4.4); the real place (6.4); $G_{\mathbb{Q}}$ and $G_{\mathbb{Q},S}$ (6.1, 6.2); finiteness of $H^1(G_{\mathbb{Q},S},M)$ (6.3); the Kummer sequence of an elliptic curve (7.1, 7.2); the n-Selmer group and Sha, finiteness, weak Mordell-Weil (7.3 to 7.5); Selmer structures and dual structures (8.1 to 8.3); Greenberg-Wiles (9.3, 9.4, 9.5); Poitou-Tate (9.1, 9.2); comparison of Selmer groups, the elementary part proved, and its specialisation to Lemma lem:dfg21 (10.1 to 10.6); T, V, W cohomology and $H^1(K,T)_{\mathrm{tors}}$ (2.1 to 2.3); $H^1(\mathbb{Q}_p,T)$ finitely generated with torsion $H^0(\mathbb{Q}_p,W)/\mathrm{div}$, and the paper's "$H^1_f(\mathbb{Q}_p,T^D)$ is the torsion" (2.2, 3.1, 4.6); the Kummer class of the Tate parameter and its image in $H^1(G/I,\mathbb{Z}_\ell(1)^I) = \mathbb{Z}/\ell^{v_\ell(q-1)}$, proved in full (5.1 to 5.6). Additions not in the PLAN entry but needed: propagation of local conditions from V to T and W (4.5; it is what makes the dual terms of Lemma lem:dfg21 the groups $H^1_f(\mathbb{Q}_p,T^D)^*$), the canonical splitting of $H^1(\mathbb{Q}_q,\mathbb{Z}_\ell(1))$ given by the uniformiser q (5.3), and the toy examples 9.4 and 10.5 (class field theory of $\mathbb{Q}$ recovered from Greenberg-Wiles and from the relaxing sequence).

19.2 1. Line-up, Section 1: Galois cohomology of a field; Hilbert 90; Kummer theory

19.2.1 1.1 `conv:galcoh:galois-cohomology` (convention and definition) P

For a field K and a discrete G_K-module M (every element has open stabiliser), $H^n(K,M) = H^n(G_K,M) = \varinjlim_L H^n(\mathrm{Gal}(L/K),M^{G_L})$ over finite Galois L/K inside $\overline{K}$, with inflation as transition maps (Chapter 16, continuous-cohomology (D)); for M of type (C) or (V), continuous cochains. $H^0(K,M) = M^{G_K}$. For $K \subset L \subset \overline{K}$: restriction $H^n(K,M) \to H^n(L,M)$; for L/K Galois, inflation-restriction (Chapter 16) reads $0 \to H^1(\mathrm{Gal}(L/K),M^{G_L}) \to H^1(K,M) \to H^1(L,M)^{\mathrm{Gal}(L/K)}$. Conventions C1 to C6 are fixed

here. Reference: Serre, Galois Cohomology (`SerreGaloisCoh`, see 14.5), Chapter I, Section 2 and Chapter II, Section 1; NSW, Chapter I. Paper: background for every cohomology group of the paper.

19.2.2 1.2 `ex:galcoh:galois-modules` (definitions and examples) P

Discrete G_K-modules: $\overline{K}$ (additive), $\overline{K}^\times$, $\mu_n = \mu_n(\overline{K})$ (always with $\operatorname{char} K \nmid n$), $\mathbb{Z}/n$ (trivial action), $\mathbb{Z}/n(r) = \mu_n^{\otimes r}$, $E[n] = E(\overline{K})[n]$ for an elliptic curve E/K with $\operatorname{char} K \nmid n$ (Chapter 9), $E(\overline{K})$ itself. Type (C): $\mathbb{Z}_\ell(1)$, on which σ acts by $\chi_\ell(\sigma) \in \mathbb{Z}_\ell^\times$ (Chapter 5, cyclotomic character), $T_\ell E$; type (D, ℓ-primary): $\mu_{\ell^\infty} = \mathbb{Q}_\ell/\mathbb{Z}_\ell(1)$, $E[\ell^\infty]$. For K finite, $G_K \simeq \hat{\mathbb{Z}}$ generated by the Frobenius $x \mapsto x^{\#K}$. Reference: Serre, Local Fields, Chapter X, Section 1 (cite the chapter). Paper: background.

19.2.3 1.3 `thm:galcoh:hilbert90` (theorem, Hilbert's Theorem 90 in cohomological form) P

For every field K: (i) $H^1(K, \overline{K}^\times) = 0$; (ii) $H^1(K, \overline{K}) = 0$ (additive; $H^n(K, \overline{K}) = 0$ for all $n \geq 1$, the latter stated); (iii) for every finite Galois L/K, $H^1(\mathrm{Gal}(L/K), L^\times) = 0$ and $H^1(\mathrm{Gal}(L/K), L) = 0$; (iv) the same for every (possibly infinite) Galois L/K, with continuous cochains and $L^\times$ discrete (for instance $L = \bar{k}(C)$ over $K = k(C)$ for a curve C/k, Chapter 8, 4.2(iii)). Proof: (iii) is Chapter 5, galois:hilbert90 (c), recalled with its proof idea (Dedekind independence of characters; normal basis for the additive case); (i), (ii), (iv) follow by the direct limit 1.1 over the finite Galois subextensions (each term vanishes). Reference: Serre, Local Fields, Chapter X, Section 1 (Proposition 2 there, check); `SerreGaloisCoh`, Chapter II, Section 1. Paper: through 1.4 (Lemmas lem:tamagawa-exp and lem:tamfactor).

19.2.4 1.4 `thm:galcoh:kummer` (theorem, Kummer theory in cohomological form) P

Let $n \geq 1$ with $\operatorname{char} K \nmid n$. The sequence $1 \to \mu_n \to \overline{K}^\times \xrightarrow{x \mapsto x^n} \overline{K}^\times \to 1$ is an exact sequence of discrete G_K-modules ($X^n - a$ is separable, so it has a root in $\overline{K}$). Its long exact sequence and 1.3(i) give: (i) the connecting map $\delta\colon K^\times/(K^\times)^n \to H^1(K, \mu_n)$, $\delta(a) = [\sigma \mapsto \sigma(\alpha)/\alpha]$ with $\alpha^n = a$, is an isomorphism of groups, natural in K; (ii) $H^2(K, \mu_n) \simeq \mathrm{Br}(K)[n]$, the n-torsion of $\mathrm{Br}(K) := H^2(K, \overline{K}^\times)$; (iii) if $\mu_n \subset K$, then $H^1(K, \mu_n) = \mathrm{Hom}_{\mathrm{cont}}(G_K, \mu_n)$ and $\delta(a)$ is the Kummer character χ_a of Chapter 5, recovering galois:kummer-classification (c) with the hypothesis $\mu_n \subset K$ removed from (i). (iv) (Restriction and corestriction, stated.) For finite L/K: $\mathrm{res} \circ \delta_K = \delta_L \circ (K^\times/n \to L^\times/n)$ (proved: same cocycle) and $\mathrm{cor} \circ \delta_L = \delta_K \circ N_{L/K}$ (stated; Serre, Local Fields, Chapter XIV). Reference: Serre, Local Fields, Chapter X, Section 3; `SerreGaloisCoh`, Chapter II, Section 1; NSW, Chapter VI, Section 2 (cite chapters). Paper: Lemma lem:tamfactor ("$H^1(G, \mathbb{Z}_\ell(1)) = \varprojlim \mathbb{Q}_q^\times/(\mathbb{Q}_q^\times)^{\ell^k}$", via 2.4), Lemma lem:tamagawa-exp (via 5.6).

19.2.5 1.5 `def:galcoh:kummer-class` (definition and proposition) P

The **Kummer map** of K is δ of 1.4; $\delta(a)$ is the **Kummer class** of a in $H^1(K, \mu_n)$, and a cocycle $\sigma \mapsto \sigma(\alpha)/\alpha$ is a **Kummer cocycle** of a (it depends on the root α: another root $\zeta\alpha$ changes it by the coboundary $\sigma \mapsto \sigma(\zeta)/\zeta$). (i) $\delta(a)|_{G_L} = 0$ iff $a \in (L^\times)^n$, for every separable $L \supseteq K$ (by 1.4(i) for L). (ii) For $\mu_n \subset K$ the kernel of $\delta(a)$ is $G_{K(\alpha)}$, so $K(\alpha)/K$ is cyclic of degree the order of a in $K^\times/K^{\times n}$ (Chapter 5). (iii) Compatibility with Chapter 7, 6.1(i): for a compatible system $(a^{1/\ell^k})_k$ the ℓ-adic Kummer cocycle $\kappa_a(\sigma) = (\sigma(a^{1/\ell^k})/a^{1/\ell^k})_k \in \mathbb{Z}_\ell(1)$ reduces modulo ℓ^k to a Kummer cocycle of a in $H^1(K, \mu_{\ell^k})$. Reference: as 1.4. Paper: Lemma lem:tamfactor ("κ the Kummer class of q_E").

19.2.6 1.6 `ex:galcoh:kummer-finite-field` (worked example) P

$K = \mathbb{F}_q$, $\ell \nmid q$, $v = v_\ell(q-1)$, $G_K = \langle \mathrm{Frob} \rangle \simeq \hat{\mathbb{Z}}$. Two computations of $H^1(\mathbb{F}_q, \mu_{\ell^k})$ agree: Kummer (1.4) gives $\mathbb{F}_q^\times/(\mathbb{F}_q^\times)^{\ell^k}$, cyclic of order $\ell^{\min(k,v)}$ (Chapter 2); the procyclic formula (Chapter 16, procyclic-

discrete) gives $\mu_{\ell^k}/(\mathrm{Frob}-1)\mu_{\ell^k} = \mu_{\ell^k}/\mu_{\ell^k}^{q-1}$, of the same order; the isomorphism between them is $u \mapsto \delta(u)(\mathrm{Frob}) = \alpha^{q-1}$ ($\alpha^{\ell^k} = u$), that is $u \mapsto u^{(q-1)/\ell^k}$ for $k \le v$ (Chapter 5, galois:kummer-finite-field (a)). In the limit: $H^1(\mathbb{F}_q, \mathbb{Z}_\ell(1)) = \mathbb{Z}_\ell(1)/(q-1)\mathbb{Z}_\ell(1) \simeq \mathbb{F}_q^\times \otimes \mathbb{Z}_\ell \simeq \mathbb{Z}/\ell^v$, the second isomorphism canonical, the third after choosing a generator; the class of u has $v_\ell \ge k$ iff $u \in (\mathbb{F}_q^\times)^{\ell^k}$ ($k \le v$; the class 0 has $v_\ell = v$ by convention, important_notes.md point 31). Paper: Lemma lem:tamfactor ("$\mathbb{Z}_\ell(1)/(q-1) \simeq \mathbb{Z}/\ell^{v_\ell(q-1)}$", "$c \in \mathbb{Z}/\ell^{v_\ell(q-1)}$ the class of u"), Section sec:tam.

19.2.7 1.7 `thm:galcoh:brauer` (theorem) R

(i) $\mathrm{Br}(K) = H^2(K, \overline{K}^\times)$; $\mathrm{Br}(\mathbb{F}_q) = 0$ (stated; finite fields are C_1), $\mathrm{Br}(\mathbb{R}) = \mathbb{Z}/2$, $\mathrm{Br}(\mathbb{C}) = 0$. (ii) For K finite over $\mathbb{Q}_p$, the invariant map $\mathrm{inv}_K\colon \mathrm{Br}(K) \xrightarrow{\sim} \mathbb{Q}/\mathbb{Z}$ is an isomorphism (local class field theory; Chapter 7, 6.5), and with 1.4(ii) $H^2(K, \mu_n) \simeq \mathbb{Z}/n$ canonically; inv is compatible with restriction (multiplication by $[L:K]$) and corestriction (identity). (iii) For $\mathbb{Q}$: $0 \to \mathrm{Br}(\mathbb{Q}) \to \bigoplus_v \mathrm{Br}(\mathbb{Q}_v) \xrightarrow{\sum \mathrm{inv}_v} \mathbb{Q}/\mathbb{Z} \to 0$ is exact (Albert-Brauer-Hasse-Noether), with $\mathrm{inv}_\infty\colon \mathrm{Br}(\mathbb{R}) \simeq \frac{1}{2}\mathbb{Z}/\mathbb{Z}$. Reference: Serre, Local Fields, Chapters X, XII, XIII (cite chapters); Cassels-Frohlich, Chapter VI (Serre's "Local class field theory") and Chapter VII (Tate's "Global class field theory"); `SerreGaloisCoh`, Chapter II, Section 5. Paper: background (the target $H^2(\mathbb{Q}_p, \mu) = \mathbb{Q}/\mathbb{Z}$ of the local pairing in 3.3; (iii) is the reciprocity law behind 9.2 and 10.1).

19.3 2. Line-up, Section 2: ℓ-adic coefficients: T, V, W, Kummer duals, and $H^1(K, \mathbb{Z}_\ell(1))$

19.3.1 2.1 `def:galcoh:kummer-duals` (definition and proposition) P

With C3, C4: (i) the evaluation pairings $T \times T^D \to \mathbb{Z}_\ell(1)$, $V \times V^D \to \mathbb{Q}_\ell(1)$, $W \times T^D \to \mathbb{Q}_\ell/\mathbb{Z}_\ell(1)$, $T \times W^D \to \mathbb{Q}_\ell/\mathbb{Z}_\ell(1)$ and, for finite M, $M \times M^*(1) \to \mu_{\ell^\infty}$ are perfect and G-equivariant (G acting on Hom by $(g\varphi)(x) = g\varphi(g^{-1}x)$); (ii) $(T/\ell^k T)^*(1) \simeq T^D/\ell^k T^D$, compatibly with $W[\ell^k] = T/\ell^k T$ and $W^D[\ell^k] = T^D/\ell^k T^D$; (iii) $(T^D)^D = T$, (W^D) is the Kummer dual of T and T^D that of W; (iv) examples: $\mathbb{Z}_\ell^D = \mathbb{Z}_\ell(1)$, $\mathbb{Z}_\ell(1)^D = \mathbb{Z}_\ell$, $(T_\ell E)^D \simeq T_\ell E$ by the Weil pairing (Chapter 9, 5.3, 5.5), $E[n]^*(1) \simeq E[n]$. Reference: Chapter 3, 8.1 to 8.3 (algebra); Rubin, Euler Systems, Section 1.1. Paper: Section 4.2 (T^D, V^D, W^D), Lemma lem:dfg21. (Chapter 18 adds the adjoint-specific $T^D \simeq A(1)$.)

19.3.2 2.2 `prop:galcoh:tvw` (proposition; the cohomology of T, V, W) P

Let G be a profinite group and T as in C4 such that $H^1(G, T/\ell T)$ and $H^2(G, T/\ell T)$ are finite (true for $G = G_{\mathbb{Q}_p}$, 3.1, and $G = G_{\mathbb{Q},S}$, 6.3; for $G = G_{\mathbb{F}_q}$, $G_{\mathbb{R}}$). Then: (i) $H^n(G, T)$ is a finitely generated $\mathbb{Z}_\ell$-module for $n = 1, 2$ and $H^n(G, V) = H^n(G, T) \otimes \mathbb{Q}_\ell$ (Chapter 16, h1-finitely-generated, tate-limits). (ii) **Torsion of $H^1(G, T)$.** The sequence $H^0(G, V) \to H^0(G, W) \xrightarrow{\delta} H^1(G, T) \xrightarrow{\iota} H^1(G, V)$ is exact, $\ker \iota = H^1(G, T)_{\mathrm{tors}}$, and the image of $H^0(G, V)$ in $H^0(G, W)$ is $H^0(G, T) \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$, the maximal divisible subgroup $H^0(G, W)_{\mathrm{div}}$; hence

$$H^1(G, T)_{\mathrm{tors}} \;\simeq\; H^0(G, W)/H^0(G, W)_{\mathrm{div}},$$

and if $H^0(G, V) = 0$ then $H^0(G, W)$ is finite and $\delta\colon H^0(G, W) \xrightarrow{\sim} H^1(G, T)_{\mathrm{tors}}$. (iii) **Divisible part of $H^1(G, W)$.** $0 \to H^1(G, T) \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell \to H^1(G, W) \to H^2(G, T)_{\mathrm{tors}} \to 0$ is exact; the first term is the image of $\pi\colon H^1(G, V) \to H^1(G, W)$ and equals $H^1(G, W)_{\mathrm{div}}$; $H^1(G, W)$ is cofinitely generated of corank $\mathrm{rank}\, H^1(G, T) = \dim H^1(G, V)$. (iv) **Finite levels.** For each k: $0 \to H^{n-1}(G, T)/\ell^k \to H^{n-1}(G, T/\ell^k T) \to H^n(G, T)[\ell^k] \to 0$ and $0 \to H^{n-1}(G, W)/\ell^k \to H^n(G, W[\ell^k]) \to H^n(G, W)[\ell^k] \to 0$ are exact (long exact sequences of $T \xrightarrow{\ell^k} T$ and $W[\ell^k] \to W \xrightarrow{\ell^k} W$, Chapter 16, les-continuous (iii), (iv)). Proof of the key steps: a finite-index divisible subgroup of a cofinitely generated $\mathbb{Z}_\ell$-module is its maximal divisible subgroup; $\ker(H^1(G, T) \to H^1(G, T) \otimes \mathbb{Q}_\ell)$ is the torsion of a finitely generated module.

Reference: Rubin, Euler Systems, Section 1.2 and Appendix B; Tate95, Section 2 (cite sections). Paper: the paragraph after Lemma lem:dfg21 ("$H^1_f(\mathbb{Q}_p, T^D)$ is the torsion of $H^1(\mathbb{Q}_p, T^D)$, a finite group", with 4.6), Lemma lem:h0 and Remark rem:ns ($H^0(\mathbb{Q}, W^D)$ finite when $H^0(\mathbb{Q}, V^D) = 0$, used in 10.3), Chapter 18 (the FPR sequence).

19.3.3 2.3 `ex:galcoh:tvw-examples` (worked examples) P

(a) $G = G_{\mathbb{F}_q}$, $T = \mathbb{Z}_\ell(1)$: $H^0(G, V) = 0$, $H^0(G, W) = \mu_{\ell^\infty}(\mathbb{F}_q)$ of order $\ell^{v_\ell(q-1)}$, and $H^1(G, T) = \mathbb{Z}_\ell(1)/(q-1)$ is all torsion, $\simeq H^0(G, W)$ as 2.2(ii) predicts. (b) $G = G_{\mathbb{R}}$, ℓ odd, any T: all H^n, $n \geq 1$, vanish (6.4) and $H^0(\mathbb{R}, W) = W^{c=1}$ is divisible: $H^1(\mathbb{R}, T)_{\mathrm{tors}} = 0$. (c) $G = G_{\mathbb{Q}_p}$, $p \neq \ell$, $T = \mathbb{Z}_\ell(1)$: by 2.5, $H^1(\mathbb{Q}_p, \mathbb{Z}_\ell(1))_{\mathrm{tors}} = \mathbb{Z}_p^\times \otimes \mathbb{Z}_\ell$, of order $\ell^{v_\ell(p-1)} = \#\mu_{\ell^\infty}(\mathbb{Q}_p) = \#H^0(\mathbb{Q}_p, \mathbb{Q}_\ell/\mathbb{Z}_\ell(1))$, as 2.2(ii) predicts. Paper: background (checks of 2.2).

19.3.4 2.4 `thm:galcoh:kummer-ladic` (theorem) P

For every field K with $\mathrm{char}\, K \neq \ell$: (i) $H^1(K, \mathbb{Z}_\ell(1)) \simeq \widehat{K^\times} := \varprojlim_k K^\times/(K^\times)^{\ell^k}$ (transition maps the projections), the class of $a \in K^\times$ being $\kappa_a = (\delta_k(a))_k$, represented by the ℓ-adic Kummer cocycle of 1.5(iii) (Chapter 5, galois:l-adic-kummer; Chapter 7, 6.1). (ii) $H^1(K, \mathbb{Q}_\ell/\mathbb{Z}_\ell(1)) \simeq K^\times \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$ (direct limit over $K^\times/(K^\times)^{\ell^k}$ with transition maps $a \mapsto a^\ell$). (iii) $\kappa_{ab} = \kappa_a + \kappa_b$; $\kappa_a = 0$ iff $a \in \bigcap_k (K^\times)^{\ell^k}$; the restriction of κ_a to G_L is κ_a for L. Proof: (i) Chapter 16, tate-limits (i): $H^1(K, \mathbb{Z}_\ell(1)) = \varprojlim_k H^1(K, \mu_{\ell^k})$ because the $H^0(K, \mu_{\ell^k})$ are finite, plus 1.4(i) at each level and the check that $\mu_{\ell^{k+1}} \to \mu_{\ell^k}$, $\zeta \mapsto \zeta^\ell$, induces the projection on the Kummer side ($\beta = \alpha^\ell$ is an ℓ^k-th root of a when α is an ℓ^{k+1}-th root); (ii) cohomology of discrete modules commutes with direct limits, and $\mu_{\ell^k} \subset \mu_{\ell^{k+1}}$ induces $a \mapsto a^\ell$. Reference: Rubin, Euler Systems, Section 1.6 (check); NSW, Chapter II, Section 7. Paper: Lemma lem:tamfactor ("κ the Kummer class of q_E in $H^1(G, \mathbb{Z}_\ell(1)) = \varprojlim \mathbb{Q}_q^\times/(\mathbb{Q}_q^\times)^{\ell^k}$").

19.3.5 2.5 `ex:galcoh:kummer-local` (proposition and worked example) P

Let K be finite over $\mathbb{Q}_p$, $\ell \neq p$, residue field $\mathbb{F}_q$, valuation v_K, $\mathcal{O}_K^\times$ the units. (i) (Chapter 7, 2.7(v), recalled.) $0 \to \mathcal{O}_K^\times \otimes \mathbb{Z}_\ell \to \widehat{K^\times} \xrightarrow{v_K} \mathbb{Z}_\ell \to 0$ is exact, $\mathcal{O}_K^\times \otimes \mathbb{Z}_\ell \simeq \mathbb{F}_q^\times \otimes \mathbb{Z}_\ell \simeq \mathbb{Z}/\ell^{v_\ell(q-1)}$; so $H^1(K, \mathbb{Z}_\ell(1)) \simeq \mathbb{Z}_\ell \times \mathbb{Z}/\ell^{v_\ell(q-1)}$ with torsion the unit classes. (ii) $H^1(K, \mu_{\ell^k})$ has order $\ell^k \cdot \#\mu_{\ell^k}(K) = \ell^{k+\min(k, v_\ell(q-1))}$. (iii) $H^1(K, \mathbb{Q}_\ell/\mathbb{Z}_\ell(1)) \simeq \mathbb{Q}_\ell/\mathbb{Z}_\ell$ via v_K ($\mathcal{O}_K^\times \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell = 0$: the units are a finite group times a pro-p group). (iv) $H^1(K, \mathbb{Q}_\ell(1)) \simeq \mathbb{Q}_\ell$, spanned by the class of a uniformiser. Numerical instances (checked): $\mathbb{Q}_7$, $\ell = 3$: $H^1(\mathbb{Q}_7, \mathbb{Z}_3(1)) \simeq \mathbb{Z}_3 \times \mathbb{Z}/3$; $\mathbb{Q}_{19}$, $\ell = 3$: $\mathbb{Z}_3 \times \mathbb{Z}/9$; $\mathbb{Q}_{11}$, $\ell = 5$: $\mathbb{Z}_5 \times \mathbb{Z}/5$; $\mathbb{Q}_5$, $\ell = 3$: $\mathbb{Z}_3$ (every unit of $\mathbb{Z}_5$ is a cube). Paper: Lemma lem:tamfactor (the group $H^1(G, \mathbb{Z}_\ell(1))$ in which κ lives), Section sec:tam.

19.4 3. Line-up, Section 3: local Galois cohomology: finiteness, Euler characteristic, Tate local duality

Throughout this section K is a finite extension of $\mathbb{Q}_p$ (the paper needs only $K = \mathbb{Q}_p$), M a finite discrete G_K-module, $M' = M^*(1) = \mathrm{Hom}(M, \mu)$ its Kummer dual (μ = all roots of unity in $\overline{K}$).

19.4.1 3.1 `thm:galcoh:local-finiteness` (theorem) R (with a sketch for $p \nmid \#M$)

(i) $H^n(K, M)$ is finite for all n and $H^n(K, M) = 0$ for $n \geq 3$ ($\mathrm{cd}\, G_K = 2$). (ii) Consequently, for T of type (C), $H^n(K, T)$ is a finitely generated $\mathbb{Z}_\ell$-module ($n \geq 0$), $H^n(K, V)$ is finite-dimensional, $H^n(K, W)$ is cofinitely generated, and 2.2 applies with $G = G_K$. Sketch for $p \nmid \#M$ and $n \leq 1$ (the only case the paper uses): M is trivial on an open normal subgroup G_L; inflation-restriction reduces H^1 to $\mathrm{Hom}_{\mathrm{cont}}(G_L, M)^{\mathrm{Gal}(L/K)}$, and a continuous homomorphism $G_L \to M$ of order prime to p factors through G_L^{ab}/(wild part), which by Chapter 7 (5.3, 5.4) is topologically generated by

two elements (a Frobenius and a generator of tame inertia), so there are finitely many. Reference: `SerreGaloisCoh`, Chapter II, Section 5 (cite the section); NSW, Chapter VII, Section 2; MilneADT, Chapter I, Section 2. Paper: the paragraph after Lemma lem:dfg21 ("a finite group"), Lemma lem:fpr (Chapter 18), and through 2.2.

19.4.2 3.2 `thm:galcoh:local-euler` (theorem, Tate's local Euler-Poincare characteristic formula) R

$\dfrac{\#H^0(K,M)\,\#H^2(K,M)}{\#H^1(K,M)} = \big|\#M\big|_K = p^{-[K:\mathbb{Q}_p]\,v_p(\#M)}$. In particular the left side is 1 when $p \nmid \#M$. ℓ-adic form: $\dim H^0(K,V) - \dim H^1(K,V) + \dim H^2(K,V) = -[K:\mathbb{Q}_p]\dim V$ if $\ell = p$ and $= 0$ if $\ell \neq p$. Reference: `SerreGaloisCoh`, Chapter II, Section 5.7 (check); MilneADT, Chapter I, Section 2 (Theorem 2.8 there, check); NSW, Chapter VII, Section 3. Paper: through 4.4 (the exact orthogonality of unramified classes), 9.3.

19.4.3 3.3 `thm:galcoh:tate-local-duality` (theorem, Tate local duality) R

(i) For $0 \le i \le 2$ the cup product (Chapter 16) followed by $M \otimes M' \to \mu$ and inv_K (1.7(ii)),

$$H^i(K,M) \times H^{2-i}(K,M') \longrightarrow H^2(K,\mu) \xrightarrow{\mathrm{inv}_K} \mathbb{Q}/\mathbb{Z},$$

is a perfect pairing of finite groups; it is natural in M, compatible with connecting maps (up to the sign $(-1)^i$) and with restriction and corestriction. (ii) (ℓ-adic forms, by passage to the limit with 2.1(ii).) For T as in C4 and $0 \le i \le 2$ there are perfect pairings $H^i(K,W) \times H^{2-i}(K,T^D) \to \mathbb{Q}_\ell/\mathbb{Z}_\ell$ (identifying the discrete group $H^i(K,W)$ with the Pontryagin dual of the compact $H^{2-i}(K,T^D)$), $H^i(K,T) \times H^{2-i}(K,W^D) \to \mathbb{Q}_\ell/\mathbb{Z}_\ell$, and $H^i(K,V) \times H^{2-i}(K,V^D) \to \mathbb{Q}_\ell$ (perfect pairing of finite-dimensional $\mathbb{Q}_\ell$-vector spaces). (iii) (Compatibility.) For $x \in H^1(K,V)$, $y \in H^1(K,T^D)$: $\langle \pi x, y\rangle_{W,T^D} = \big(\langle x, \iota y\rangle_{V,V^D} \bmod \mathbb{Z}_\ell\big)$; and symmetrically with the roles of (T,V,W) and (T^D,V^D,W^D) exchanged. Reference: `SerreGaloisCoh`, Chapter II, Section 5.2 (check); Tate-Duality; MilneADT, Chapter I, Section 2 (Corollary 2.3 there, check); NSW, Chapter VII, Section 2; for (ii), (iii) Rubin, Euler Systems, Section 1.4 (cite the section). Paper: Lemma lem:dfg21 (the Pontryagin duals $H^1_f(\mathbb{Q}_p,T^D)^*$ are the duals of $H^1(\mathbb{Q}_p,W)/H^1_f(\mathbb{Q}_p,W)$, through 4.5 and 10.3).

19.4.4 3.4 `cor:galcoh:local-duality-consequences` (corollary) P (given 3.2, 3.3)

(i) $H^2(K,M) \simeq H^0(K,M')^*$; $H^2(K,\mu_{\ell^k}) \simeq \mathbb{Z}/\ell^k$; $H^2(K,\mathbb{Z}/\ell^k) \simeq \mu_{\ell^k}(K)^*$. (ii) For $\ell \neq p$: $\#H^1(K,M) = \#H^0(K,M)\,\#H^0(K,M')$ and $\dim H^1(K,V) = \dim H^0(K,V) + \dim H^0(K,V^D)$. (iii) For $\ell \neq p$ and T as in C4: $H^2(K,T) \simeq H^0(K,W^D)^*$, finite if $H^0(K,V^D) = 0$. Examples: $\dim H^1(\mathbb{Q}_p,\mathbb{Q}_\ell) = 1$ ($p \neq \ell$; all classes unramified), $\dim H^1(\mathbb{Q}_p,\mathbb{Q}_\ell(1)) = 1$ (2.5(iv)); for $E/\mathbb{Q}_p$ and $\ell \neq p$, $\#H^1(\mathbb{Q}_p,E[\ell]) = \#E(\mathbb{Q}_p)[\ell]^2$ (by 2.1(iv)). Paper: background; 4.4, 4.6.

19.4.5 3.5 `ex:galcoh:local-orders` (worked example) P

(a) $\ell \neq p$, $v = v_\ell(p-1)$, $m = \min(k,v)$: $\#H^i(\mathbb{Q}_p,\mathbb{Z}/\ell^k) = \ell^k,\ \ell^{k+m},\ \ell^m$ and $\#H^i(\mathbb{Q}_p,\mu_{\ell^k}) = \ell^m,\ \ell^{k+m},\ \ell^k$ for $i = 0,1,2$ (Kummer 2.5(ii); $\mathrm{Hom}(G_{\mathbb{Q}_p},\mathbb{Z}/\ell^k) = \mathrm{Hom}(\widehat{\mathbb{Q}_p^\times},\mathbb{Z}/\ell^k)$ by Chapter 7, 6.5; and 3.4(i)): Euler characteristic 1 and duality visible. (b) p odd, $M = \mu_p$ over $\mathbb{Q}_p$: $\#H^0 = 1$, $\#H^1 = \#\mathbb{Q}_p^\times/(\mathbb{Q}_p^\times)^p = p^2$, $\#H^2 = p$, characteristic $p^{-1} = |p|_p$.

(b) 11a1 over $\mathbb{Q}_{11}$, $\ell = 5$: the Tate parameter $q_E = 11^5 u$ with $\bar{u} = -1 \in (\mathbb{F}_{11}^\times)^5$ is a fifth power in $\mathbb{Q}_{11}$ and $\mu_5 \subset \mathbb{Q}_{11}$, so $E[5] \subset E(\mathbb{Q}_{11})$ (all 12 roots of the 5-division polynomial lie in $\mathbb{Q}_{11}$ with rational y; checked with PARI) and $\#H^1(\mathbb{Q}_{11},E[5]) = 5^4$. Paper: background; Chapter 16, Exercise 10 and Section sec:tam (11a1 at $\ell = 5$, an Eisenstein illustration only: $E(\mathbb{Q})_{\mathrm{tors}} = \mathbb{Z}/5$).

19.5 4. Line-up, Section 4: unramified cohomology

In this section p is a prime, $G = G_{\mathbb{Q}_p}$, $I = I_p$, Frob an arithmetic Frobenius, and M a G-module of type (D, torsion), (C) or (V). Statements (iv) of 4.2, 4.4 and 4.6 need $p \neq \ell$.

19.5.1 4.1 `def:galcoh:unramified` (definition and proposition) P

$H^1_{\mathrm{ur}}(\mathbb{Q}_p, M) := \mathrm{im}\big(\inf\colon H^1(G/I, M^I) \to H^1(\mathbb{Q}_p, M)\big) = \ker\big(\mathrm{res}\colon H^1(\mathbb{Q}_p, M) \to H^1(I, M)\big)$; inflation is injective, so $H^1_{\mathrm{ur}}(\mathbb{Q}_p, M) \simeq H^1(G/I, M^I)$ (Chapter 16, inflation-restriction (a), continuous-cochain version). Classes in H^1_{ur} are called unramified; M is unramified if I acts trivially. The definition is made for every p, including $p = \ell$, where it is not the Bloch-Kato condition (Chapter 18). Reference: Rubin, Euler Systems, Section 1.3; `SerreGaloisCoh`, Chapter II, Section 5.5. Paper: the paragraph after Lemma lem:dfg21 ("$H^1_f(\mathbb{Q}_p, V^D) = H^1_{\mathrm{ur}}(\mathbb{Q}_p, V^D)$", the identification $H^1_f = H^1_{\mathrm{ur}}$ at $p \neq \ell$ being Chapter 18's), Lemma lem:tamfactor ($H^1(G/I, \mathbb{Z}_\ell(1)^I)$).

19.5.2 4.2 `prop:galcoh:unramified-computed` (proposition) P

(i) $c \mapsto c(\mathrm{Frob})$ induces $H^1_{\mathrm{ur}}(\mathbb{Q}_p, M) \simeq M^I/(\mathrm{Frob} - 1)M^I$ (Chapter 16: procyclic-compact for (C), (V), procyclic-discrete for torsion (D), applied to $\Gamma = G/I \simeq \hat{\mathbb{Z}}$; M^I is closed, of the same type). (ii) For M finite, $\#H^1_{\mathrm{ur}}(\mathbb{Q}_p, M) = \#H^0(\mathbb{Q}_p, M)$.

(ii) For V: $\dim H^1_{\mathrm{ur}}(\mathbb{Q}_p, V) = \dim H^0(\mathbb{Q}_p, V)$, and the following are equivalent: $H^1_{\mathrm{ur}}(\mathbb{Q}_p, V) = 0$; $H^0(\mathbb{Q}_p, V) = 0$; 1 is not an eigenvalue of Frob on V^I; $\det(1 - \mathrm{Frob} \mid V^I) \neq 0$; $\det(1 - \Phi \mid V^I) \neq 0$ (the last two because $\det(1 - \Phi) = (-1)^d \det(\Phi) \det(1 - \mathrm{Frob})$ on V^I, $d = \dim V^I$). (iv) For T with $V = T \otimes \mathbb{Q}_\ell$: $T^I = T \cap V^I$ is a saturated sublattice of V^I; $H^1_{\mathrm{ur}}(\mathbb{Q}_p, T) = T^I/(\mathrm{Frob} - 1)T^I$; if the conditions of (iii) hold, it is finite of order $\ell^{v_\ell(\det(1-\mathrm{Frob}|V^I))} = \ell^{v_\ell(\det(1-\Phi|V^I))}$ (cokernel of an injective endomorphism of a free $\mathbb{Z}_\ell$-module, Chapter 3; $\det \Phi$ is a unit on the lattice T^I). Reference: Rubin, Euler Systems, Lemma 1.3.2 and Section 1.3 (check); Chapter 16. Paper: the paragraph after Lemma lem:dfg21 ("$(V^D)^{I_p}/(\mathrm{Frob}_p - 1)$ vanishes because $\det(1 - \mathrm{Frob}_p \mid (V^D)^{I_p}) = L_p(A_f, 1)^{-1} \neq 0$": the vanishing is (iii) and does not depend on the Frobenius convention; the identification of the determinant with $L_p(A_f, 1)^{-1}$ needs Φ, important_notes.md point 11, and is Chapter 18/19's), Lemma lem:fpr (Chapter 18 uses (iv): the Euler factor $\ell^{v_\ell(L_p(A_f,1)^{-1})}$ is the order of $H^1_{\mathrm{ur}}(\mathbb{Q}_p, T^D)$).

19.5.3 4.3 `prop:galcoh:ur-sequence` (proposition) P (given Chapter 16)

For M of type (D, torsion), (C) or (V):

$$0 \to H^1_{\mathrm{ur}}(\mathbb{Q}_p, M) \to H^1(\mathbb{Q}_p, M) \xrightarrow{\mathrm{res}} H^1(I, M)^{G/I} \to 0$$

is exact, where G/I acts on $H^1(I, M)$ by the conjugation action of Chapter 16 ($(\sigma \cdot c)(i) = \sigma c(\sigma^{-1} i \sigma)$). Proof: the five-term sequence (Chapter 16, inflation-restriction (b)) ends with $H^2(G/I, M^I)$, which vanishes for $G/I \simeq \hat{\mathbb{Z}}$ and these coefficients (Chapter 16, procyclic-discrete and procyclic-compact (iii), the latter applying because $\hat{\mathbb{Z}}$ surjects onto $\mathbb{Z}_\ell$). For $p \neq \ell$ and M of type (C), $H^1(I, M)$ is computed through the tame quotient (Chapter 16, tame-inertia-example). Remark (forward): taking torsion subgroups in this sequence for $M = T^D$ is the first step of the Fontaine-Perrin-Riou formula, and $\mathrm{Fitt}_{\mathbb{Z}_\ell}\big((H^1(I, M)_{\mathrm{tors}})^{G/I}\big)$ is the Tamagawa factor $\mathrm{Tam}_p(M)$ (Chapter 18; the superscript G applies to the module, not to the ideal: important_notes.md point 37). Reference: Rubin, Euler Systems, Section 1.3; FPR94, I.4. Paper: Section 1.2 (the object $H^1(I_p, M)^{G_{\mathbb{Q}_p}}_{\mathrm{tor}}$), Lemma lem:fpr and Lemma lem:tamfactor (through Chapters 16, 18).

19.5.4 4.4 `thm:galcoh:ur-orthogonal` (theorem, exact orthogonality of unramified classes) P (given 3.2, 3.3)

Let $p \neq \ell$. (i) For M finite of ℓ-power order and $M' = M^*(1)$, $H^1_{\mathrm{ur}}(\mathbb{Q}_p, M)$ and $H^1_{\mathrm{ur}}(\mathbb{Q}_p, M')$ are exact annihilators of each other under the pairing of 3.3(i). (ii) For V and V^D, $H^1_{\mathrm{ur}}(\mathbb{Q}_p, V)^\perp = H^1_{\mathrm{ur}}(\mathbb{Q}_p, V^D)$

under 3.3(ii). Proof (to be written in full): the cup product of two inflated classes is inflated from $H^2(G/I, \mu_{\ell^\infty}) = 0$ (resp. $H^2(G/I, \mathbb{Q}_\ell(1)) = 0$), so the two subgroups are orthogonal; by 4.2(ii), 3.4(i) and 3.2, $\#H^1_{\mathrm{ur}}(M) \cdot \#H^1_{\mathrm{ur}}(M') = \#H^0(M)\,\#H^0(M') = \#H^0(M)\,\#H^2(M) = \#H^1(M)$, so each is the full annihilator of the other; (ii) the same count with dimensions (4.2(iii), 3.4(ii)). Warning (the case $p = \ell$, checked): over $\mathbb{Q}_p$, p odd, $\#H^1_{\mathrm{ur}}(\mathbb{Q}_p, \mathbb{Z}/p) = p$ and $\#H^1_{\mathrm{ur}}(\mathbb{Q}_p, \mu_p) = \#H^0(\mathbb{Q}_p, \mu_p) = 1$, while $\#H^1(\mathbb{Q}_p, \mathbb{Z}/p) = p^2$: the unramified subgroups are not exact annihilators at $p = \ell$ (the annihilator of $H^1_{\mathrm{ur}}(\mathbb{Q}_p, \mathbb{Z}/p)$ is the image of the units in $\mathbb{Q}_p^\times/(\mathbb{Q}_p^\times)^p$, of order p); at ℓ the right local condition is Bloch-Kato's (Chapter 18). Reference: `SerreGaloisCoh`, Chapter II, Section 5.5; MilneADT, Chapter I, Section 2 (Theorem 2.6 there, check); Rubin, Euler Systems, Section 1.4. Paper: Lemma lem:dfg21 (through Chapter 18: $H^1_f(\mathbb{Q}_p, V)^\perp = H^1_f(\mathbb{Q}_p, V^D)$ for $p \neq \ell$ is exactly (ii)).

19.5.5 4.5 `prop:galcoh:propagation` (proposition, propagation of local conditions) P (given 3.3)

Let K be a finite extension of $\mathbb{Q}_p$ (any p), T as in C4. For a $\mathbb{Q}_\ell$-subspace $L_V \subseteq H^1(K, V)$ put $L_T := \iota^{-1}(L_V) \subseteq H^1(K, T)$ and $L_W := \pi(L_V) \subseteq H^1(K, W)$ (the conditions **propagated** from L_V); likewise L_{T^D}, L_{W^D} from $L_{V^D} \subseteq H^1(K, V^D)$. (i) $L_T \supseteq H^1(K,T)_{\mathrm{tors}}$; L_W is divisible and $L_W \subseteq H^1(K,W)_{\mathrm{div}}$. (ii) If L_V and L_{V^D} are exact annihilators under the V-pairing, then L_W and L_{T^D} are exact annihilators under $H^1(K,W) \times H^1(K,T^D) \to \mathbb{Q}_\ell/\mathbb{Z}_\ell$, so $\big(H^1(K,W)/L_W\big)^* \simeq L_{T^D}$; and symmetrically L_T and L_{W^D}. (iii) If $p \neq \ell$, T is unramified and $L_V = H^1_{\mathrm{ur}}(K,V)$, then $L_T = H^1_{\mathrm{ur}}(K,T)$ and $L_W = H^1_{\mathrm{ur}}(K,W)$ (so a propagated structure is unramified at every unramified $p \neq \ell$: $H^1(I,T) = \mathrm{Hom}(I,T)$ is torsion-free, and $H^1_{\mathrm{ur}}(K,W) = W/(\mathrm{Frob}-1)W$ is divisible). Proof of (ii): if $y \in L_{T^D}$ then $\langle \pi x, y\rangle = \langle x, \iota y\rangle \bmod \mathbb{Z}_\ell = 0$ for $x \in L_V$ (3.3(iii)); conversely if $y \perp L_W$, then $\langle \ell^{-k}x, \iota y\rangle \in \mathbb{Z}_\ell$ for all k and all $x \in L_V$, so $\langle x, \iota y\rangle = 0$, $\iota y \in L_V^\perp = L_{V^D}$, $y \in L_{T^D}$; perfectness of the (W, T^D) pairing finishes. Reference: Rubin, Euler Systems, Section 1.3 (propagation, check); MazurRubin04, Section 1.1 (check); Bloch-Kato, Section 3 (for the Bloch-Kato case). Paper: Lemma lem:dfg21 ("$H^1_f(\mathbb{Q}_p, T^D) = \iota^{-1}H^1_f(\mathbb{Q}_p, V^D)$" is a propagated condition, and the dual terms $H^1_f(\mathbb{Q}_p, T^D)^*$ are the quotients $H^1(\mathbb{Q}_p, W)/H^1_f(\mathbb{Q}_p, W)$, by (ii)); Chapter 18 (the Bloch-Kato conditions on T and W are defined this way).

19.5.6 4.6 `cor:galcoh:ur-vanishing` (corollary) P

Let $p \neq \ell$ and suppose $H^1_{\mathrm{ur}}(\mathbb{Q}_p, V) = 0$ (4.2(iii)). Then $H^0(\mathbb{Q}_p, V) = 0$, and for the propagated conditions of $L_V = H^1_{\mathrm{ur}}(\mathbb{Q}_p, V) = 0$: $L_T = \iota^{-1}(0) = H^1(\mathbb{Q}_p, T)_{\mathrm{tors}} \simeq H^0(\mathbb{Q}_p, W)$, a finite group (2.2(ii), 3.1), containing $H^1_{\mathrm{ur}}(\mathbb{Q}_p, T)$ (finite by 4.2(iv)); and $L_W = 0$. In the paper's situation, for $p \in \Sigma$ and $V = V^D = A_f(1)$: "$H^1_f(\mathbb{Q}_p, T^D)$ is the torsion of $H^1(\mathbb{Q}_p, T^D)$, a finite group", and it is $\simeq H^0(\mathbb{Q}_p, W^D)$. Paper: the paragraph after Lemma lem:dfg21 (exactly), Lemma lem:fpr (Chapter 18: length $H^1_f(\mathbb{Q}_p, T^D) =$ length $H^0(\mathbb{Q}_p, W^D)$ is then split into the Euler factor and the Tamagawa factor).

19.5.7 4.7 `ex:galcoh:ur-examples` (worked examples) P

(a) $T = \mathbb{Z}_\ell(1)$, $p \neq \ell$: $H^1_{\mathrm{ur}}(\mathbb{Q}_p, \mathbb{Z}_\ell(1)) = \mathbb{Z}_\ell(1)/(p-1)$, which under 2.4 is the image of $\mathbb{Z}_p^\times$ (5.1). (b) $T = \mathbb{Z}_\ell$, $p \neq \ell$: $H^1(\mathbb{Q}_p, \mathbb{Z}_\ell) = \mathrm{Hom}(G_{\mathbb{Q}_p}, \mathbb{Z}_\ell) = \mathbb{Z}_\ell = H^1_{\mathrm{ur}}$, orthogonal to $H^1_{\mathrm{ur}}(\mathbb{Q}_p, \mathbb{Q}_\ell/\mathbb{Z}_\ell(1)) = 0$, consistent with 4.4. (c) Tate curve (E multiplicative at $p \neq \ell$, Chapter 11, 3.2): $V^I = \mathbb{Q}_\ell e_1$ with Frob-eigenvalue $\varepsilon p \neq 1$, so $H^1_{\mathrm{ur}}(\mathbb{Q}_p, V_\ell E) = 0$ and $H^1(\mathbb{Q}_p, T_\ell E)_{\mathrm{tors}} \simeq E(\mathbb{Q}_p)[\ell^\infty]$. (d) $V = \mathrm{ad}^0 V_\ell E$ at a multiplicative prime: $V^I = \mathbb{Q}_\ell e$ with Frob-eigenvalue p; for $V^D = V(1)$ the eigenvalue is p^2; $\det(1 - \Phi \mid (V^D)^I) = 1 - p^{-2}$ (the paper's $L_p(A_f, 1)^{-1}$ at a multiplicative prime) while $\det(1 - \mathrm{Frob} \mid (V^D)^I) = 1 - p^2$: different numbers, the same ℓ-adic valuation, the same vanishing (point 11). (e) Type III or III* at $q \geq 5$: $(\mathrm{ad}^0 V)^I$ is a line on which Frob acts by $\chi_{-4}(q)$ (Chapter 19; Lemma lem:euler), so $(V^D)^I$ has eigenvalue $q\chi_{-4}(q)$ and $\#H^1_{\mathrm{ur}}(\mathbb{Q}_q, T^D) = \ell^{v_\ell(q-\chi_{-4}(q))} = \ell^{e_\ell(E,q)}$. (f) Good prime: $V_\ell E$ unramified with eigenvalues of absolute value $\sqrt{p}$, so $H^1_{\mathrm{ur}}(\mathbb{Q}_p, V_\ell E) = 0$. Paper: the paragraph after Lemma lem:dfg21, Lemma lem:fpr ("of valuation $e_\ell(E,p)$ at an additive prime and $v_\ell(1-p^{-2})$ at a multiplicative one"), Lemma lem:euler.

19.6 5. Line-up, Section 5: the Kummer class of the Tate parameter (the end of Lemma lem:tamfactor)

Throughout: $q \neq \ell$ primes, ℓ odd where stated, $G = G_{\mathbb{Q}_q}$, $I = I_q$, $v = v_\ell(q-1)$, $t_\ell \colon I \to \mathbb{Z}_\ell(1)$ the tame character (Chapter 7, 5.3). This section proves, in corrected form, the last sentences of the paper's proof of Lemma lem:tamfactor.

19.6.1 5.1 `prop:galcoh:unit-classes` (proposition) P

Let K be finite over $\mathbb{Q}_p$, $p \neq \ell$, with residue field $\mathbb{F}_q$. Under the Kummer isomorphisms 1.4 and 2.4: (i) $H^1_{\mathrm{ur}}(K, \mu_{\ell^k})$ corresponds to the image of $\mathcal{O}_K^\times$ in $K^\times/(K^\times)^{\ell^k}$, and $H^1_{\mathrm{ur}}(K, \mathbb{Z}_\ell(1))$ to $\mathcal{O}_K^\times \otimes \mathbb{Z}_\ell \subset \widehat{K^\times}$; (ii) the sequence 4.3 for $M = \mathbb{Z}_\ell(1)$ is the valuation sequence of 2.5(i):

$$0 \to \mathcal{O}_K^\times \otimes \mathbb{Z}_\ell \to \widehat{K^\times} \xrightarrow{v_K} \mathbb{Z}_\ell \to 0 \quad \text{corresponds to} \quad 0 \to H^1_{\mathrm{ur}}(K, \mathbb{Z}_\ell(1)) \to H^1(K, \mathbb{Z}_\ell(1)) \xrightarrow{\mathrm{res}} \mathrm{Hom}(I_K, \mathbb{Z}_\ell(1))^{G/I} \to 0,$$

via $\mathrm{res}\,\kappa_x = v_K(x)\, t_\ell$ (Chapter 7, 6.2(ii)), $\mathrm{Hom}(I_K, \mathbb{Z}_\ell(1))^{G/I} = \mathbb{Z}_\ell t_\ell$ (t_ℓ is G-equivariant by Chapter 7, 5.4(i); every continuous homomorphism $I_K \to \mathbb{Z}_\ell(1)$ factors through t_ℓ, Chapter 7, 5.3(iv)). Proof: units have all their ℓ-power roots in K^{ur} (Hensel, Chapter 7, 4.2(iv)), so their Kummer classes vanish on I_K; conversely a class with res $= 0$ has $v_K = 0$ by the displayed formula. Reference: `SerreGaloisCoh`, Chapter II, Section 5; Rubin, Euler Systems, Section 1.6 (check). Paper: Lemma lem:tamfactor ("$\kappa \ldots H^1(G/I, \mathbb{Z}_\ell(1)^I)$"), Lemma lem:tamagawa-exp (via 5.6).

19.6.2 5.2 `prop:galcoh:unramified-value` (proposition) P

With $K = \mathbb{Q}_q$: $\mathbb{Z}_\ell(1)^I = \mathbb{Z}_\ell(1)$ (I acts trivially, $q \neq \ell$), and $c \mapsto c(\mathrm{Frob})$ gives $H^1(G/I, \mathbb{Z}_\ell(1)) \simeq \mathbb{Z}_\ell(1)/(q-1)\mathbb{Z}_\ell(1)$ (Chapter 16, procyclic-examples (ii)), a cyclic group of order ℓ^v. For $u \in \mathbb{Z}_q^\times$ with Kummer cocycle κ_u built from ℓ-power roots of u in $\mathbb{Q}_q^{\mathrm{ur}}$: κ_u is inflated from G/I (as a cocycle, not only as a class), $\kappa_u(\mathrm{Frob})$ depends only on the image of Frob in G/I, and its class in $\mathbb{Z}_\ell(1)/(q-1) \simeq \mu_{\ell^v}(\mathbb{F}_q)$ is $\bar{u}^{(q-1)/\ell^v}$, i.e. the class of the reduction $\bar{u}$ in $\mathbb{F}_q^\times \otimes \mathbb{Z}_\ell = \mathbb{F}_q^\times/(\mathbb{F}_q^\times)^{\ell^v}$ (Chapter 5, galois:kummer-finite-field (b), applied to the reductions of the roots; Chapter 7, 6.2(iii)). Hence the composite $\mathbb{Z}_q^\times \otimes \mathbb{Z}_\ell \to H^1_{\mathrm{ur}}(\mathbb{Q}_q, \mathbb{Z}_\ell(1)) \to \mathbb{Z}_\ell(1)/(q-1)$ is the isomorphism "reduce and take the class in $\mathbb{F}_q^\times \otimes \mathbb{Z}_\ell$". After choosing a generator, $\mathbb{Z}_\ell(1)/(q-1) \simeq \mathbb{Z}/\ell^v$; v_ℓ of a class is well defined, with $v_\ell(0) = v$ (important_notes.md point 31). Paper: Lemma lem:tamfactor ("$H^1(G/I, \mathbb{Z}_\ell(1)^I) = \mathbb{Z}_\ell(1)/(q-1) \simeq \mathbb{Z}/\ell^{v_\ell(q-1)}$, the class of the unit part of q_E in $\mathbb{F}_q^\times \otimes \mathbb{Z}_\ell$ by Kummer theory").

19.6.3 5.3 `prop:galcoh:canonical-splitting` (proposition) P

For $K = \mathbb{Q}_q$ the prime q is a canonical uniformiser, and $\mathbb{Q}_q^\times = q^{\mathbb{Z}} \times \mathbb{Z}_q^\times$ canonically. Hence

$$H^1(\mathbb{Q}_q, \mathbb{Z}_\ell(1)) = \mathbb{Z}_\ell\, \kappa_q \;\oplus\; H^1_{\mathrm{ur}}(\mathbb{Q}_q, \mathbb{Z}_\ell(1))$$

canonically, and there is a canonical retraction $\mathrm{pr}_{\mathrm{ur}} \colon H^1(\mathbb{Q}_q, \mathbb{Z}_\ell(1)) \to H^1_{\mathrm{ur}}(\mathbb{Q}_q, \mathbb{Z}_\ell(1)) = H^1(G/I, \mathbb{Z}_\ell(1)^I)$ with $\mathrm{pr}_{\mathrm{ur}}(\kappa_x) = \kappa_{xq^{-v_q(x)}}$: it sends the class of x to the class of its unit part. This is the precise sense in which "the image of κ in $H^1(G/I, \mathbb{Z}_\ell(1)^I)$" exists: not by a functorial map (inflation goes the other way, and over a general K the splitting depends on the uniformiser; important_notes.md point 7), but by the splitting attached to q. The value $\kappa(\mathrm{Frob})$ of a cocycle at a Frobenius lift is a different object (5.4(iii), (iv)). Paper: Lemma lem:tamfactor (the phrase "c' is the image of κ in $H^1(G/I, \mathbb{Z}_\ell(1)^I)$"); Part 16, N1.

19.6.4 5.4 `thm:galcoh:tate-kummer-class` (theorem; the identification of c') P

Let $E/\mathbb{Q}$ have multiplicative reduction at q, $\ell \neq q$ odd, $q_E \in \mathbb{Q}_q^\times$ the Tate parameter, $n = v_q(q_E) = n_q$, $u = q_E q^{-n} \in \mathbb{Z}_q^\times$ its unit part, $\varepsilon \colon G \to \{\pm 1\}$ the unramified character with $\varepsilon(\mathrm{Frob}) = a_q(E)$. Fix compatible systems of ℓ-power roots of q and of u (the latter in $\mathbb{Q}_q^{\mathrm{ur}}$), give q_E the product system, and

let $\kappa = \kappa_{q_E}$, κ_q, κ_u be the Kummer cocycles (Chapter 7, 6.1). By Chapters 10 (7.5(iii)) and 11 (3.2) $T = T_\ell E$ has a basis e_1, e_2 in which $\rho(g) = \left(\begin{smallmatrix} \chi_\ell(g)\varepsilon(g) & \varepsilon(g)\kappa(g) \\ 0 & \varepsilon(g) \end{smallmatrix}\right)$ (the corner is $\varepsilon\kappa$, important_notes.md point 42). Then: (i) $\kappa = n\kappa_q + \kappa_u$ as cocycles; in $H^1(\mathbb{Q}_q, \mathbb{Z}_\ell(1))$ the class of κ is the Kummer class of q_E, and $\mathrm{pr}_{\mathrm{ur}}[\kappa] = [\kappa_u]$ is the class of $\bar{u}$ in $\mathbb{F}_q^\times \otimes \mathbb{Z}_\ell$ (5.2, 5.3). (ii) $\kappa|_I = n\, t_\ell$; for a topological generator t of $I^{(\ell)}$ with $t_\ell(t) = \zeta$ (the generator of $\mathbb{Z}_\ell(1)$ used to write $\mathbb{Z}_\ell(1) = \mathbb{Z}_\ell$), $te_1 = e_1$ and $te_2 = e_2 + ne_1$: the paper's n' can be taken equal to n, and in any case $v_\ell(n') = v_\ell(n)$. (iii) Let Frob_0 be a Frobenius lift fixing the chosen ℓ-power roots of q (Chapter 7, 5.5). Then $\kappa_q(\mathrm{Frob}_0) = 0$ and $\rho(\mathrm{Frob}_0) = \left(\begin{smallmatrix} \varepsilon q & c' \\ 0 & \varepsilon \end{smallmatrix}\right)$ with $\varepsilon = a_q(E)$ and $c' = \varepsilon\, \kappa_u(\mathrm{Frob}_0)$, whose class in $\mathbb{Z}_\ell(1)/(q-1)$ is $\pm$ the class of $\bar{u}$. (iv) For an arbitrary Frobenius lift $\mathrm{Frob} = \mathrm{Frob}_0\, \sigma$ ($\sigma \in I$) and an arbitrary cocycle in the class of κ (another system of roots), the corner c' of $\rho(\mathrm{Frob})$ changes by an element of the ideal $(n, q-1)\mathbb{Z}_\ell$: $\kappa(\mathrm{Frob}_0\, \sigma) = \kappa(\mathrm{Frob}_0) + q\, n\, t_\ell(\sigma)$, and a coboundary changes the value at Frob by $(q-1)z$. (v) Consequently

$$\min\big(v_\ell(n), v_\ell(q-1), v_\ell(c')\big) = v_\ell\big((n, q-1, c')\mathbb{Z}_\ell\big) = \min\big(v_\ell(n), v_\ell(q-1), v_\ell(c)\big),$$

with $c \in \mathbb{Z}/\ell^v$ the class of $\bar{u}$ (any generator; $v_\ell(0) = v$), independently of every choice; with Chapter 16, model-computation, this completes the proof of the formula $v_\ell(\mathrm{Tam}_q(\mathrm{ad}^0\, T)) = v_\ell(n) + \min(v_\ell(n), v_\ell(q-1), v_\ell(c))$ of Lemma lem:tamfactor (the identification of Tam_q with the Fitting ideal of the invariants is Chapter 18's). Reference: SilvermanATAEC, Chapter V (Tate curve, cite the chapter); SerreAbelian, Chapter IV, Appendix A.1 (cite the chapter); DDT, Theorem 3.1(e) as cited by the paper (check). Paper: Lemma lem:tamfactor (end of the proof: "c' is the image of κ ... by Kummer theory, well defined modulo $q-1$; so $v_\ell(c') = v_\ell(c)$ in the sense stated"; the true indeterminacy is the ideal $(n, q-1)$, which is harmless), Remark rem:ko(1), Section sec:tam.

19.6.5 5.5 `ex:galcoh:tate-kummer-examples` (worked examples; checked with PARI's `ellinit(E,O(q^30)).tate` and ecdata) P

At $\ell = 3$, multiplicative primes $q \equiv 1 \pmod 3$ with $3 \mid n$; c = class of $\bar{u}$ in $\mathbb{F}_q^\times/(\mathbb{F}_q^\times)^{3^{v_3(q-1)}}$; "extra" is $\min(v_3(n), v_3(q-1), v_3(c))$; "bound" is the right side of (eq:identity) without the Selmer term, i.e. $\sum_q t_3 +$ extra (no additive primes):

curve, q	a_q	n	$\bar{u} \bmod q$	ℓ-th power?	extra	bound at $\ell = 3$	$v_3(\deg \varphi_E)$	mod 3 image
77c1, 7	-1	3	5	no (cubes mod 7: ± 1)	0	1	1 (deg = 6)	surjective
791a1, 7	-1	3	$6 \equiv -1$	yes	1	2	2 (deg = 144)	surjective
1577a1, 19	$+1$	3	$8 = 2^3$	yes (not a 9th power)	1	2	2 (deg = 720)	surjective
11a1, 11 ($\ell = 5$)	$+1$	5	$10 \equiv -1$	yes (5th powers mod 11: ± 1)	1	(Eisenstein)	0 (deg = 1)	5Cs.1.1

(The other bad primes of 77c1, 791a1, 1577a1 are of type I_2 and contribute 0 at $\ell = 3$; $3 \nmid N$ in each case.) The first three show the extra term of Remark rem:ko(1) and Section sec:tam at work and the bound attained; 11a1 (rational 5-torsion) shows why the paper needs $\bar{\rho}$ irreducible: the local term is 2 but $\deg \varphi_E = 1$. Paper: Section sec:tam, Remark rem:ko(1), Lemma lem:tamfactor; Chapter 22.

19.6.6 5.6 `rem:galcoh:tamagawa-exp-cohomological` (remark) P

Lemma lem:tamagawa-exp in cohomological form: $\rho \bmod \ell^t$ is unramified at q iff the corner $\varepsilon\kappa \bmod \ell^t$ vanishes on I iff $n\,t_\ell \equiv 0 \pmod{\ell^t}$ on I iff $\ell^t \mid n$ (5.4(ii); t_ℓ is surjective), for split and nonsplit reduction alike ($\varepsilon|_I = 1$). Equivalently, by 1.5(i) over $\mathbb{Q}_q^{\mathrm{ur}} \supset \mu_{\ell^t}$: $q_E \in (\mathbb{Q}_q^{\mathrm{ur}\times})^{\ell^t}$ iff $\ell^t \mid n$ (important_notes.md point 8). The lemma itself is proved in Chapter 11 (item 3.3 there); this remark only gives the cohomological reading. Paper: Lemma lem:tamagawa-exp, the history paragraph of Section 1.5 ("$t_q = v_\ell(n_q) = t_\ell(E,q)$").

19.7 6. Line-up, Section 6: global Galois cohomology: $G_{\mathbb{Q}}$, $G_{\mathbb{Q},S}$, the real place

19.7.1 6.1 `def:galcoh:gqs` (definition) P

S, $\mathbb{Q}_S$, $G_{\mathbb{Q},S}$ as in C5; $G_{\mathbb{Q},S}$ is the quotient of $G_{\mathbb{Q}}$ by the closed normal subgroup H_S generated by the inertia groups $I_{\mathfrak{p}}$ of all primes $\mathfrak{p}$ of $\overline{\mathbb{Q}}$ above $p \notin S$. A $G_{\mathbb{Q}}$-module M (type (D) finite, (C) or (V)) is unramified outside S if H_S acts trivially; then it is a $G_{\mathbb{Q},S}$-module. Examples: μ_n with $S \supseteq \{\infty\} \cup \{p \mid n\}$; $E[n]$, $T_\ell E$, $\mathrm{ad}^0 T_\ell E$ with $S \supseteq \{\infty, \ell\} \cup \{p \mid N\}$ (Neron-Ogg-Shafarevich, Chapter 11). Localisation maps res_v (C5). Reference: NSW, Chapter VIII, Section 3 (cite the chapter); Rubin, Euler Systems, Section 1.5. Paper: Section 4.2 (W unramified outside $\Sigma \cup \{\ell\}$), Lemma lem:dfg21.

19.7.2 6.2 `prop:galcoh:unramified-almost-everywhere` (proposition) P

Let M be unramified outside S. (i) Inflation identifies $H^1(G_{\mathbb{Q},S}, M)$ with $\{c \in H^1(\mathbb{Q}, M) : \mathrm{res}_p\, c \in H^1_{\mathrm{ur}}(\mathbb{Q}_p, M)$ for all $p \notin S\}$. (ii) For M finite every $c \in H^1(\mathbb{Q}, M)$ is unramified at almost all p. Proof: (i) inflation-restriction for $H_S \trianglelefteq G_{\mathbb{Q}}$ (Chapter 16): c comes from $G_{\mathbb{Q},S}$ iff $c|_{H_S} \in \mathrm{Hom}(H_S, M)$ vanishes, iff it vanishes on each $I_{\mathfrak{p}}$, which is the unramified condition (conjugation-invariance, C5); (ii) $c|_{G_L}$, L the field cut out by M, is a homomorphism with finite image, so it cuts out a finite extension of L, ramified at finitely many primes. Reference: Rubin, Euler Systems, Section 1.5; MilneADT, Chapter I, Section 4. Paper: Theorem thm:main (the Selmer groups are subgroups of $H^1(G_{\mathbb{Q},S}, W)$), Lemma lem:dfg21 ("unramified elsewhere").

19.7.3 6.3 `thm:galcoh:gqs-finite` (theorem) S

For M finite and unramified outside the finite set S: (i) $H^1(G_{\mathbb{Q},S}, M)$ is finite; (ii) $H^2(G_{\mathbb{Q},S}, M)$ is finite (stated); (iii) for ℓ odd and $S \ni \ell$, $H^n(G_{\mathbb{Q},S}, M) = 0$ for $n \geq 3$ when M is ℓ-primary (stated). Sketch of (i): restrict to the open subgroup $G_{L,S}$ fixing M; $\mathrm{Hom}(G_{L,S}, M)$ is finite because there are only finitely many extensions of L of degree dividing $\#M$ unramified outside S (Hermite-Minkowski, stated); inflation-restriction for $G_{L,S} \trianglelefteq G_{\mathbb{Q},S}$ and the finiteness of H^1 of the finite group $\mathrm{Gal}(L/\mathbb{Q})$ finish. Consequences: for T unramified outside S, 2.2 applies with $G = G_{\mathbb{Q},S}$: $H^1(G_{\mathbb{Q},S}, T)$ is finitely generated, $H^1(G_{\mathbb{Q},S}, W)$ cofinitely generated, $H^1(G_{\mathbb{Q},S}, T)_{\mathrm{tors}} \simeq H^0(\mathbb{Q}, W)/H^0(\mathbb{Q}, W)_{\mathrm{div}}$. Reference: SilvermanAEC, Chapter VIII, Section 1 (Hermite-Minkowski in the proof of weak Mordell-Weil, check); NSW, Chapter VIII, Section 3; MilneADT, Chapter I, Section 4. Paper: Theorem thm:main and Lemma lem:dfg21 (cofinite generation and finiteness statements behind "length"), Lemma lem:h0 (through 2.2: $H^0(\mathbb{Q}, W^D)$ finite when $H^0(\mathbb{Q}, V^D) = 0$).

19.7.4 6.4 `prop:galcoh:real-place` (proposition, recalled from Chapter 16 z2-on-l-adic) P

$G_{\mathbb{R}} = \{1, c\}$. For every $G_{\mathbb{R}}$-module M and $n \geq 1$, $2H^n(\mathbb{R}, M) = 0$; if ℓ is odd and M is of type (D, ℓ-primary), (C) or (V), then $H^n(\mathbb{R}, M) = 0$ for $n \geq 1$, $\hat{H}^0(\mathbb{R}, M) = M^{c=1}/(1+c)M = 0$, and $M = M^+ \oplus M^-$. Consequences: for ℓ odd every local condition at ∞ is $0 = H^1(\mathbb{R}, M)$, the real place contributes nothing to Selmer groups and to the comparison sequences except the factor $1/\#H^0(\mathbb{R}, M)$ in 9.3; and $H^1_f(\mathbb{R}, T^D) = H^1(\mathbb{R}, T^D) = 0$. Example ($\ell = 2$, where it fails): $H^1(\mathbb{R}, \mu_2) = \mathbb{R}^\times/\mathbb{R}^{\times 2} = \mathbb{Z}/2$.

Paper: the paragraph after Lemma lem:dfg21 ("At ∞, $H^1(\mathbb{R}, T^D)$ is killed by 2, so $H^1_f(\mathbb{R}, T^D) = 0$ for ℓ odd"), the summand $p = \infty$ of Lemma lem:dfg21.

19.7.5 6.5 `ex:galcoh:global-kummer` (worked example) P

$H^1(\mathbb{Q}, \mu_n) = \mathbb{Q}^\times/(\mathbb{Q}^\times)^n$; the class of x is unramified at $p \nmid n$ iff $n \mid v_p(x)$ (by 5.1 over $\mathbb{Q}_p$), so for $S \supseteq \{\infty\} \cup \{p \mid n\}$: $H^1(G_{\mathbb{Q},S}, \mu_n) = \{x \in \mathbb{Q}^\times : n \mid v_p(x)\ \forall p \notin S\}/(\mathbb{Q}^\times)^n \simeq \mathbb{Z}_S^\times/(\mathbb{Z}_S^\times)^n$ (class number one), with $\mathbb{Z}_S^\times = \pm\prod_{p\in S} p^{\mathbb{Z}}$: finite, of order $n^{\#S-1} \cdot \#(\pm1)/(\pm1)^n$. The model case of 6.3. Paper: background (the shape of all finiteness arguments).

19.8 7. Line-up, Section 7: elliptic curves: the Kummer map, the n-Selmer group and Sha

19.8.1 7.1 `prop:galcoh:ec-kummer` (proposition) P

Let E/K be an elliptic curve and $\operatorname{char} K \nmid n$. The sequence $0 \to E[n] \to E(\overline{K}) \xrightarrow{[n]} E(\overline{K}) \to 0$ is exact (Chapter 9: $[n]$ is surjective on $\overline{K}$-points, being a nonconstant isogeny); its long exact sequence gives the **Kummer map** $\delta\colon E(K)/nE(K) \hookrightarrow H^1(K, E[n])$, $\delta(P) = [\sigma \mapsto \sigma Q - Q]$ with $nQ = P$, and the exact sequence $0 \to E(K)/nE(K) \to H^1(K, E[n]) \to H^1(K, E)[n] \to 0$, where $H^1(K, E) = H^1(K, E(\overline{K}))$. The analogy with 1.4: replacing E by $\mathbb{G}_m$ gives Kummer theory, and Hilbert 90 says the analogue of $H^1(K, E)$ vanishes. Reference: SilvermanAEC, Chapter VIII, Section 2 and Chapter X, Section 1 (cite the sections). Paper: background (the model for all Selmer groups; Chapter 9, 9.1 deferral).

19.8.2 7.2 `prop:galcoh:ec-local-image` (proposition) P for (i), (ii); R for (iii)

Let $E/\mathbb{Q}_p$ and $p \nmid n$. (i) $\#E(\mathbb{Q}_p)/nE(\mathbb{Q}_p) = \#E(\mathbb{Q}_p)[n] = \#H^0(\mathbb{Q}_p, E[n]) = \#H^1_{\mathrm{ur}}(\mathbb{Q}_p, E[n])$ (snake lemma for $[n]$ on $0 \to E_1(\mathbb{Q}_p) \to E(\mathbb{Q}_p) \to E(\mathbb{Q}_p)/E_1(\mathbb{Q}_p) \to 0$, $E_1(\mathbb{Q}_p)$ uniquely n-divisible by the formal group, Chapter 10; then 4.2(ii)). (ii) If E has good reduction at p, $\delta(E(\mathbb{Q}_p)/n) = H^1_{\mathrm{ur}}(\mathbb{Q}_p, E[n])$: a point Q with $nQ = P$ exists in $E(\mathbb{Q}_p^{\mathrm{ur}})$ (the reduction $\tilde{E}(\overline{\mathbb{F}}_p)$ is n-divisible and $E_1(\mathbb{Q}_p^{\mathrm{ur}})$ is uniquely n-divisible), so $\delta(P)$ vanishes on I_p; equality by (i). (iii) (Tate, stated.) For any reduction type, $\delta(E(\mathbb{Q}_p)/n)$ is its own exact annihilator under 3.3(i) with $E[n] \simeq E[n]^*(1)$ (Weil pairing). (iv) (Example, P, and the link with Tamagawa numbers.) For $p \neq \ell$ and $W = E[\ell^\infty]$: the Kummer image $E(\mathbb{Q}_p) \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$ in $H^1(\mathbb{Q}_p, W)$ is 0, whereas $H^1_{\mathrm{ur}}(\mathbb{Q}_p, W) = W^I/(\mathrm{Frob} - 1)W^I$; for multiplicative reduction of type I_n (ℓ odd) it has order $\ell^{v_\ell(n)}$ if split and 1 if nonsplit (compute W^I as an extension of $\mathbb{Z}/\ell^{v_\ell(n)}$, on which Frob acts by ε, by μ_{ℓ^∞}, which is $(\mathrm{Frob} - 1)$-divisible), i.e. the ℓ-part of the Tamagawa number c_p (Chapter 10, 7.6). This is the classical shadow of the Tamagawa factors of Chapter 18 and agrees with Chapter 16, model-computation (e). Reference: SilvermanAEC, Chapter X, Section 4; MilneADT, Chapter I, Section 3 (for (iii)). Paper: background; Lemma lem:tamfactor (its final check "the ℓ-part of the Tamagawa number $c_q(E)$").

19.8.3 7.3 `def:galcoh:selmer-sha` (definition and proposition) P

For $E/\mathbb{Q}$: $\mathrm{Sel}_n(E/\mathbb{Q}) = \{c \in H^1(\mathbb{Q}, E[n]) : \mathrm{res}_v\, c \in \delta_v(E(\mathbb{Q}_v)/nE(\mathbb{Q}_v))$ for all $v\}$, $\mathrm{Sha}(E/\mathbb{Q}) = \ker\big(H^1(\mathbb{Q}, E) \to \prod_v H^1(\mathbb{Q}_v, E)\big)$, and $0 \to E(\mathbb{Q})/nE(\mathbb{Q}) \to \mathrm{Sel}_n(E/\mathbb{Q}) \to \mathrm{Sha}(E/\mathbb{Q})[n] \to 0$ is exact (diagram chase on 7.1 for $\mathbb{Q}$ and all $\mathbb{Q}_v$). Typeset Sha with the Cyrillic letter only if the preamble provides it; otherwise `\operatorname{Sha}`. Reference: SilvermanAEC, Chapter X, Section 4. Paper: background (Section sec:two mentions ranks, Watkins's conjecture).

19.8.4 7.4 `thm:galcoh:selmer-finite` (theorem and corollary) P (given 6.2, 6.3, 7.2(ii))

With $S = \{\infty\} \cup \{p \mid nN_E\}$: $\mathrm{Sel}_n(E/\mathbb{Q}) \subseteq H^1(G_{\mathbb{Q},S}, E[n])$ (7.2(ii) and 6.2(i)), hence $\mathrm{Sel}_n(E/\mathbb{Q})$ is finite (6.3). Corollary (weak Mordell-Weil): $E(\mathbb{Q})/nE(\mathbb{Q})$ is finite. With the height argument (Chapter 9, 9.1)

this gives Mordell's theorem. Reference: SilvermanAEC, Chapter VIII, Section 1 and Chapter X, Section 4 (Theorem 4.2 there, check). Paper: background (Chapter 9's deferral; "this is the model for all Selmer groups").

19.8.5 7.5 `ex:galcoh:selmer-11a1` (worked example) P (from the data)

11a1: $E(\mathbb{Q}) = \mathbb{Z}/5$, rank 0 and $\#Ш = 1$ (Cremona's allbsd), so $\mathrm{Sel}_5(E/\mathbb{Q}) \simeq E(\mathbb{Q})/5E(\mathbb{Q}) \simeq \mathbb{Z}/5$; and over $\mathbb{Q}_{11}$, $E[5] \subset E(\mathbb{Q}_{11})$ (3.5(c)), so the local condition at 11 is a subgroup of order $\#E(\mathbb{Q}_{11})[5] = 25$ of $H^1(\mathbb{Q}_{11}, E[5])$, of order 625 (7.2(i)). A small model of the difference between "unramified" and "Kummer image" at a prime of bad reduction: here $H^1_{\mathrm{ur}}(\mathbb{Q}_{11}, E[5])$ also has order 25 (4.2(ii)), but the image of $E(\mathbb{Q}_{11})/5$ is a different subgroup: it is $\mathrm{Hom}(G_{\mathbb{Q}_{11}}, \mu_5) \subset \mathrm{Hom}(G_{\mathbb{Q}_{11}}, E[5])$ ($G_{\mathbb{Q}_{11}}$ acts trivially on $E[5] \supset \mu_5$), which contains the ramified class of the uniformiser 11, consistent with $5 \mid c_{11} = 5$ and with 7.2(iv) (Exercise 15). Paper: background.

19.9 8. Line-up, Section 8: Selmer structures and dual Selmer structures

19.9.1 8.1 `def:galcoh:selmer-structure` (definition and examples) P

Let M be a $G_{\mathbb{Q}}$-module unramified outside a finite set S, of one of the types: finite of ℓ-power order; W (discrete, cofinitely generated); T (type (C)). A **Selmer structure** $\mathcal{L}$ on M is a family of subgroups $\mathcal{L}_v \subseteq H^1(\mathbb{Q}_v, M)$, one for each place v, with $\mathcal{L}_v = H^1_{\mathrm{ur}}(\mathbb{Q}_v, M)$ for all $v \notin S_{\mathcal{L}}$, a finite set containing S (for type (C) one also asks $\mathcal{L}_v$ closed). Its **Selmer group** is

$$H^1_{\mathcal{L}}(\mathbb{Q}, M) = \{c \in H^1(\mathbb{Q}, M) : \mathrm{res}_v\, c \in \mathcal{L}_v \ \forall v\} = \ker\Big(H^1(G_{\mathbb{Q},S_{\mathcal{L}}}, M) \to \bigoplus_{v \in S_{\mathcal{L}}} H^1(\mathbb{Q}_v, M)/\mathcal{L}_v\Big)$$

(the equality by 6.2(i)). Examples: (a) the unramified structure; (b) $\mathcal{L}^{\Sigma}$, **relaxed** at a finite set Σ of primes: $\mathcal{L}^{\Sigma}_p = H^1(\mathbb{Q}_p, M)$ for $p \in \Sigma$, $\mathcal{L}^{\Sigma}_v = \mathcal{L}_v$ otherwise; (c) $\mathcal{L}_{\Sigma}$, **strict** at Σ: $\mathcal{L}_{\Sigma,p} = 0$ for $p \in \Sigma$; (d) the classical structure $\delta_v(E(\mathbb{Q}_v)/n)$ on $E[n]$ (a Selmer structure by 7.2(ii)), with Selmer group $\mathrm{Sel}_n(E/\mathbb{Q})$; (e) **propagated structures**: given $\mathbb{Q}_\ell$-subspaces $L_{V,v} \subseteq H^1(\mathbb{Q}_v, V)$ with $L_{V,v} = H^1_{\mathrm{ur}}(\mathbb{Q}_v, V)$ for almost all v, the families $(\iota^{-1}L_{V,v})_v$ on T and $(\pi L_{V,v})_v$ on W (4.5(iii) shows they are Selmer structures). The Bloch-Kato structure is of type (e) (Chapter 18), and the paper's $H^1_f(\mathbb{Q}, W)$ and $H^1_{\Sigma}(\mathbb{Q}, W) = H^1_{f^{\Sigma}}(\mathbb{Q}, W)$ are the Selmer groups of the Bloch-Kato structure and of its relaxation at Σ ("no condition at the primes of Σ, the Bloch-Kato condition at ℓ and the unramified condition elsewhere"). Reference: MazurRubin04, Section 2.1 (check); Rubin, Euler Systems, Section 1.5; DFG04, Section 2.1. Paper: Theorem thm:main ($H^1_f(\mathbb{Q}, \mathrm{ad}^0\, T \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell)$), Section 4.2 ($H^1_{\Sigma}(\mathbb{Q}, W) \supseteq H^1_f(\mathbb{Q}, W)$), Lemma lem:dfg21.

19.9.2 8.2 `def:galcoh:dual-structure` (definition and proposition) P

Let M be finite of ℓ-power order, $M' = M^*(1)$, $\mathcal{L}$ a Selmer structure on M. The **dual structure** $\mathcal{L}^*$ on M' is $\mathcal{L}^*_v = \mathcal{L}^{\perp}_v$ under 3.3(i) (at $v = \infty$ under the analogous pairing on $H^1(\mathbb{R}, -)$; for ℓ odd both sides are 0, 6.4). (i) $\mathcal{L}^*$ is a Selmer structure: at $p \notin S_{\mathcal{L}} \cup \{\ell\}$, $\mathcal{L}^{\perp}_p = H^1_{\mathrm{ur}}(\mathbb{Q}_p, M)^{\perp} = H^1_{\mathrm{ur}}(\mathbb{Q}_p, M')$ by 4.4(i). (ii) $\mathcal{L}^{**} = \mathcal{L}$; $\mathcal{L} \subseteq \mathcal{L}'$ (i.e. $\mathcal{L}_v \subseteq \mathcal{L}'_v$ for all v) implies $\mathcal{L}'^* \subseteq \mathcal{L}^*$; $(\mathcal{L}^{\Sigma})^* = (\mathcal{L}^*)_{\Sigma}$ (relaxed and strict are dual). (iii) (ℓ-adic.) If $\mathcal{L}$ on W is propagated from $(L_{V,v})$ and $L^{\perp}_{V,v} = L_{V^D,v}$ for every v, its dual (under 3.3(ii)) is the structure on T^D propagated from $(L_{V^D,v})$ (4.5(ii)); the relaxation of the first at Σ is dual to the strict version of the second. Reference: MazurRubin04, Section 2.3 (check); Rubin, Euler Systems, Section 1.7. Paper: Lemma lem:dfg21 (the "dual Selmer group" behind $H^0(\mathbb{Q}, W^D)^*$), Chapter 18 (Bloch-Kato structures are self-dual in this sense).

19.9.3 8.3 `prop:galcoh:selmer-l-adic` (proposition) P for (i), (ii); S for (iii)

Let $\mathcal{L}$ be propagated from $(L_{V,v})$ (8.1(e)). (i) $H^1_{\mathcal{L}}(\mathbb{Q},W)$ is cofinitely generated and $H^1_{\mathcal{L}}(\mathbb{Q},T)$ finitely generated (6.3). (ii) If the global $\mathbb{Q}_\ell$-Selmer group $H^1_{L_V}(\mathbb{Q},V)=\{c\in H^1(G_{\mathbb{Q},S},V):\mathrm{res}_v\, c\in L_{V,v}\ \forall v\}$ vanishes, then $H^1_{\mathcal{L}}(\mathbb{Q},T)=H^1(G_{\mathbb{Q},S},T)_{\mathrm{tors}}\simeq H^0(\mathbb{Q},W)/H^0(\mathbb{Q},W)_{\mathrm{div}}$, and $\simeq H^0(\mathbb{Q},W)$ if moreover $H^0(\mathbb{Q},V)=0$ (every class maps to 0 in $H^1(G_{\mathbb{Q},S},V)$, so is torsion by 2.2(ii); conversely torsion classes satisfy every propagated condition, 4.5(i)). (iii) (Stated, with a sketch.) $\mathrm{corank}_{\mathbb{Z}_\ell} H^1_{\mathcal{L}}(\mathbb{Q},W)=\dim H^1_{L_V}(\mathbb{Q},V)$; in particular $H^1_{\mathcal{L}}(\mathbb{Q},W)$ is finite iff $H^1_{L_V}(\mathbb{Q},V)=0$. Reference: Rubin, Euler Systems, Section 1.5 (check); Bloch-Kato, Section 5 (for the Bloch-Kato case, check). Paper: Lemma lem:dfg21 (with T^D in place of T: "$H^1_f(\mathbb{Q},V^D)=0$" forces the dual Selmer group of T^D to be $H^0(\mathbb{Q},W^D)$), Remark rem:ns (finiteness of the Selmer groups in the identity), Theorem thm:dfg27 (through Chapter 18).

19.10 9. Line-up, Section 9: global duality: Poitou-Tate and the Greenberg-Wiles formula

19.10.1 9.1 `thm:galcoh:poitou-tate` (theorem, Poitou-Tate exact sequence) R

Let M be finite, unramified outside S, with $S\ni\infty$ and $S\ni p$ for every $p\mid \#M$; $M'=M^*(1)$; $G_S=G_{\mathbb{Q},S}$. There is an exact sequence

$$0\to H^0(G_S,M)\to\bigoplus_{v\in S}\hat H^0(\mathbb{Q}_v,M)\to H^2(G_S,M')^*\to H^1(G_S,M)\to\bigoplus_{v\in S}H^1(\mathbb{Q}_v,M)\to H^1(G_S,M')^*\to H^2(G_S,M)\to\bigoplus_{v\in}$$

where $\hat H^0(\mathbb{Q}_p,M)=H^0(\mathbb{Q}_p,M)$ for finite p, $\hat H^0(\mathbb{R},M)$ is the Tate group (zero for $\#M$ odd), the maps $\bigoplus_v H^i(\mathbb{Q}_v,M)\to H^{2-i}(G_S,M')^*$ send (c_v) to $y\mapsto\sum_v \mathrm{inv}_v(c_v\cup \mathrm{res}_v\, y)$ (3.3), and the others are restrictions or their duals. In particular the kernel of $H^1(G_S,M)\to\bigoplus_{v\in S}H^1(\mathbb{Q}_v,M)$ (the group $Ш^1_S(M)$) is dual to $Ш^2_S(M')$. Reference: MilneADT, Chapter I, Section 4 (Theorem 4.10 there, check); NSW, Chapter VIII, Section 6; TateDuality; Rubin, Euler Systems, Section 1.7. Paper: Lemma lem:dfg21 (through 10.1, 10.2).

19.10.2 9.2 `prop:galcoh:reciprocity` (proposition) P (given 1.7(iii))

For $x\in H^1(\mathbb{Q},M)$ and $y\in H^1(\mathbb{Q},M')$: $\mathrm{res}_v\, x\cup\mathrm{res}_v\, y$ is unramified, hence has invariant 0, for almost all v, and $\sum_v \mathrm{inv}_v(\mathrm{res}_v\, x\cup\mathrm{res}_v\, y)=0$. Proof: $x\cup y\in H^2(\mathbb{Q},M\otimes M')\to H^2(\mathbb{Q},\mu_{\ell^\infty})\subseteq \mathrm{Br}(\mathbb{Q})$ (via 1.4(ii)), and 1.7(iii); almost all local cups are inflated from $H^2(\hat{\mathbb{Z}},\mu)=0$ (as in 4.4). Paper: through 10.1 (the "complex" part of the comparison sequence, which is all that the inequality direction needs).

19.10.3 9.3 `thm:galcoh:greenberg-wiles` (theorem, the formula of Greenberg and Wiles) R (proof idea S)

Let M be finite of ℓ-power order, $\mathcal{L}$ a Selmer structure on M and $\mathcal{L}^*$ its dual. Then $H^1_{\mathcal{L}}(\mathbb{Q},M)$ and $H^1_{\mathcal{L}^*}(\mathbb{Q},M')$ are finite and

$$\frac{\#H^1_{\mathcal{L}}(\mathbb{Q},M)}{\#H^1_{\mathcal{L}^*}(\mathbb{Q},M')}=\frac{\#H^0(\mathbb{Q},M)}{\#H^0(\mathbb{Q},M')}\prod_v\frac{\#\mathcal{L}_v}{\#H^0(\mathbb{Q}_v,M)},$$

the product over all places including ∞, almost all factors being 1 (4.2(ii)). Proof idea: Poitou-Tate (9.1) and Tate's global Euler characteristic formula (stated: $\#H^0(G_S,M)\#H^2(G_S,M)/\#H^1(G_S,M)=\prod_{v|\infty}\#H^0(\mathbb{Q}_v,M)/|\#M|_\infty$). Reference: Wiles95, Proposition 1.6 (check); DDT, Theorem 2.19 (check); Washington's survey in the Boston FLT volume is an alternative (not in refs.bib; not needed). Paper: background ("the origin of the exact sequence of DFG04 Lemma 2.1", PLAN.md); 9.5.

19.10.4 9.4 `ex:galcoh:gw-check` (worked example) P

$M = \mathbb{Z}/\ell$, ℓ odd, $M' = \mu_\ell$. (a) $\mathcal{L}$ = unramified at every prime (including ℓ): $H^1_{\mathcal{L}}(\mathbb{Q}, \mathbb{Z}/\ell) = \mathrm{Hom}(G_{\mathbb{Q}}, \mathbb{Z}/\ell)^{\mathrm{ur}} = 0$ ($\mathbb{Q}$ has no unramified extension; Minkowski, stated); $\mathcal{L}^*_p = H^1_{\mathrm{ur}}(\mathbb{Q}_p, \mu_\ell)$ for $p \neq \ell$ (4.4) and $\mathcal{L}^*_\ell$ = image of $\mathbb{Z}_\ell^\times$ (4.4, warning), so $H^1_{\mathcal{L}^*}(\mathbb{Q}, \mu_\ell) = \{x : \ell \mid v_p(x)\ \forall p\}/(\mathbb{Q}^\times)^\ell = \{\pm 1\}/\{\pm 1\}^\ell = 0$. Right side: $\frac{\ell}{1} \cdot \frac{1}{\ell}$ (the factor at ∞ is $\#\mathcal{L}_\infty/\#H^0(\mathbb{R}, \mathbb{Z}/\ell) = 1/\ell$; all finite factors are 1 by 4.2(ii)) $= 1$. (b) Relax at one prime $p \neq \ell$: the left side becomes $\#\{$characters of $G_{\mathbb{Q}}$ of order dividing ℓ unramified outside $p\} = \ell^{[p \equiv 1\ (\ell)]}$ over $\#H^1_{(\mathcal{L}^*)_p} = 1$; the right side gains $\#H^1(\mathbb{Q}_p, \mathbb{Z}/\ell)/\#H^1_{\mathrm{ur}} = \#H^0(\mathbb{Q}_p, \mu_\ell) = \ell^{[p \equiv 1\ (\ell)]}$ (3.5(a)). Both sides agree: Greenberg-Wiles recovers the count of cyclic extensions of degree ℓ and conductor p. Paper: background (a check of 9.3 the reader can do by hand).

19.10.5 9.5 `cor:galcoh:gw-comparison` (corollary) P (given 9.3)

For Selmer structures $\mathcal{L} \subseteq \mathcal{L}'$ on finite M: $\dfrac{\#H^1_{\mathcal{L}'}(\mathbb{Q}, M)}{\#H^1_{\mathcal{L}}(\mathbb{Q}, M)} = \prod_v \dfrac{\#\mathcal{L}'_v}{\#\mathcal{L}_v} \cdot \dfrac{\#H^1_{\mathcal{L}'^*}(\mathbb{Q}, M')}{\#H^1_{\mathcal{L}^*}(\mathbb{Q}, M')}$. This is the numerical shadow of the exact sequence 10.1; applied at each finite level to the Bloch-Kato structure and its relaxation at Σ and passed to the limit, it gives the length identity of the proof of Theorem thm:main and of Remark rem:ns (10.4). Paper: Lemma lem:dfg21, Remark rem:ns.

19.11 10. Line-up, Section 10: comparing Selmer structures; the abstract form of DFG04 Lemma 2.1

19.11.1 10.1 `thm:galcoh:comparison` (theorem) P for (i), (ii); S for (iii) (from 9.1)

Let M be finite of ℓ-power order and $\mathcal{L} \subseteq \mathcal{L}'$ Selmer structures on M, so $\mathcal{L}'^* \subseteq \mathcal{L}^*$ on M'. Consider

$$0 \to H^1_{\mathcal{L}}(\mathbb{Q}, M) \to H^1_{\mathcal{L}'}(\mathbb{Q}, M) \xrightarrow{\mathrm{loc}} \bigoplus_v \mathcal{L}'_v/\mathcal{L}_v \xrightarrow{\theta} H^1_{\mathcal{L}^*}(\mathbb{Q}, M')^* \xrightarrow{\mathrm{res}^*} H^1_{\mathcal{L}'^*}(\mathbb{Q}, M')^* \to 0,$$

with $\theta((c_v))(y) = \sum_v \mathrm{inv}_v(c_v \cup \mathrm{res}_v\, y)$ (well defined because $\mathcal{L}^*_v$ annihilates $\mathcal{L}_v$; the sum is finite). (i) (Elementary.) The sequence is exact at $H^1_{\mathcal{L}}$ and at $H^1_{\mathcal{L}'}$ (definitions), res^* is surjective (dual of an inclusion of finite groups), and $\theta \circ \mathrm{loc} = 0$, $\mathrm{res}^* \circ \theta = 0$ (9.2: reciprocity; and $\mathcal{L}'^*_v = \mathcal{L}'^{\perp}_v$). (ii) Hence $\#H^1_{\mathcal{L}'}/\#H^1_{\mathcal{L}} \leq \prod_v \#(\mathcal{L}'_v/\mathcal{L}_v)$, with the defect bounded by $\#(H^1_{\mathcal{L}^*}/H^1_{\mathcal{L}'^*})$. (iii) The sequence is exact (Poitou-Tate 9.1, applied to the middle of the nine-term sequence). Reference: Rubin, Euler Systems, Theorem 1.7.3 (check); MazurRubin04, Section 2.3 (check); Washington's survey (not needed). Paper: Lemma lem:dfg21 (its proof).

19.11.2 10.2 `thm:galcoh:comparison-l-adic` (theorem) S

Let T be as in C4, unramified outside S; $\mathcal{L} \subseteq \mathcal{L}'$ Selmer structures on W propagated from $L_V \subseteq L'_V$ with $L^{\perp}_{V,v} = L_{V^D,v}$, $L'^{\perp}_{V,v} = L'_{V^D,v}$; $\mathcal{L}^*$, $\mathcal{L}'^*$ the dual structures on T^D (8.2(iii)). Then

$$0 \to H^1_{\mathcal{L}}(\mathbb{Q}, W) \to H^1_{\mathcal{L}'}(\mathbb{Q}, W) \to \bigoplus_v \mathcal{L}'_v/\mathcal{L}_v \to H^1_{\mathcal{L}^*}(\mathbb{Q}, T^D)^* \to H^1_{\mathcal{L}'^*}(\mathbb{Q}, T^D)^* \to 0$$

is exact, and $\mathcal{L}'_v/\mathcal{L}_v \simeq (\mathcal{L}^*_v/\mathcal{L}'^*_v)^*$ (4.5(ii)). Sketch: apply 10.1 to $W[\ell^k] = T/\ell^k$ with the induced structures (preimages of $\mathcal{L}_v$, images of $\mathcal{L}^*_v$, which are mutually dual at each level), take the direct limit over k for the W-terms (exact; the error terms $H^0(\mathbb{Q}, W)/\ell^k$ of 2.2(iv) die in the limit because $H^0(\mathbb{Q}, W)$ is torsion) and the inverse limit for the T^D-terms (finite groups, Mittag-Leffler), Pontryagin duality exchanging the two. Reference: Rubin, Euler Systems, Section 1.7 and Appendix B; Greenberg89, Section 4 (check); DFG04, proof of Lemma 2.1. Paper: Lemma lem:dfg21.

19.11.3 10.3 `thm:galcoh:relaxing-sigma` (theorem; the abstract form of DFG04 Lemma 2.1) P (given 10.2, 2.2, 4.5, 6.4, 8.3)

Let ℓ be odd, T unramified outside S, $(L_{V,v})_v$ local conditions on V with $L_{V,v} = H^1_{\mathrm{ur}}(\mathbb{Q}_v, V)$ for $v \notin S$ and $L^\perp_{V,v} = L_{V^D,v}$ for all v; $\mathcal{L}$ the propagated structure on W, $\mathcal{L}^*$ the propagated (dual) structure on T^D. Let Σ be a finite set of primes $p \neq \ell$. Assume (a) $H^1_{L_{V^D}}(\mathbb{Q}, V^D) = 0$, and (b) $H^0(\mathbb{Q}_p, V^D) = 0$ for at least one $p \in \Sigma$ (so $\Sigma \neq \emptyset$; (b) implies $H^0(\mathbb{Q}, V^D) = 0$). Then there is an exact sequence

$$0 \to H^1_{\mathcal{L}}(\mathbb{Q}, W) \to H^1_{\mathcal{L}^\Sigma}(\mathbb{Q}, W) \to \bigoplus_{p \in \Sigma \cup \{\infty\}} \left(\mathcal{L}^*_p\right)^* \to H^0(\mathbb{Q}, W^D)^* \to 0,$$

where $\mathcal{L}^*_p = \iota^{-1}(L_{V^D,p}) \subseteq H^1(\mathbb{Q}_p, T^D)$ and $\mathcal{L}^*_\infty = H^1(\mathbb{R}, T^D) = 0$. Proof (to be written in full): 10.2 with $\mathcal{L}' = \mathcal{L}^\Sigma$, whose dual is $(\mathcal{L}^*)_\Sigma$ (8.2(ii), (iii)); $\mathcal{L}^\Sigma_p/\mathcal{L}_p = H^1(\mathbb{Q}_p, W)/\mathcal{L}_p \simeq (\mathcal{L}^*_p)^*$ (4.5(ii)); by (a) and 8.3(ii), $H^1_{\mathcal{L}^*}(\mathbb{Q}, T^D) = H^1(G_{\mathbb{Q},S}, T^D)_{\mathrm{tors}} = \delta(H^0(\mathbb{Q}, W^D)) \simeq H^0(\mathbb{Q}, W^D)$, finite since $H^0(\mathbb{Q}, V^D) = 0$; the strict group $H^1_{(\mathcal{L}^*)_\Sigma}(\mathbb{Q}, T^D)$ consists of the $\delta(x)$ with $\mathrm{res}_p\, \delta(x) = \delta_p(x) = 0$ for $p \in \Sigma$, and δ_p is injective on $H^0(\mathbb{Q}_p, W^D) \supseteq H^0(\mathbb{Q}, W^D)$ at the prime p of (b) (2.2(ii) locally), so it is 0 and the last map is onto $H^0(\mathbb{Q}, W^D)^*$; the summand at ∞ is 0 (6.4). Paper: Lemma lem:dfg21 is this theorem for the Bloch-Kato structure f (Chapter 18 checks $L^\perp_{V,v} = L_{V^D,v}$: at $p \neq \ell$ it is 4.4(ii), at ℓ Bloch-Kato's Proposition 3.8 (check)), with (a) = DFG04 Theorem 2.7 (Theorem thm:dfg27), and (b) supplied by the paragraph after Lemma lem:dfg21 ($H^1_{\mathrm{ur}}(\mathbb{Q}_p, V^D) = 0$ for $p \in \Sigma$, hence $H^0(\mathbb{Q}_p, V^D) = 0$ by 4.2(iii)); the paper's hypothesis $H^0(\mathbb{Q}, V^D) = 0$ then follows from (b).

19.11.4 10.4 `cor:galcoh:length-identity` (corollary) P

Under 10.3, if $H^1_{\mathcal{L}^\Sigma}(\mathbb{Q}, W)$ is finite then all terms are finite and

$$\operatorname{length} H^1_{\mathcal{L}^\Sigma}(\mathbb{Q}, W) = \operatorname{length} H^1_{\mathcal{L}}(\mathbb{Q}, W) + \sum_{p \in \Sigma} \operatorname{length} \mathcal{L}^*_p - \operatorname{length} H^0(\mathbb{Q}, W^D)$$

(lengths alternate in an exact sequence of finite $\mathbb{Z}_\ell$-modules and are preserved by Pontryagin duality: Chapter 16, lengths-in-exact-sequences); the identity holds in $\mathbb{N} \cup \{\infty\}$ in general. With $\mathcal{L} = f$: $H^0(\mathbb{Q}, W^D) = 0$ gives the identity "length $H^1_\Sigma(\mathbb{Q}, W) =$ length $H^1_f(\mathbb{Q}, W) + \sum_{p \in \Sigma}$ length $H^1_f(\mathbb{Q}_p, T^D)$" of the proof of Theorem thm:main; length $H^0(\mathbb{Q}, W^D) = h_\ell(E)$ gives the displayed identity of Remark rem:ns (with Lemma lem:fpr for the local lengths, Chapter 18). By important_notes.md point 5 (Chapter 11), $h_\ell(E) = 0$ for $\ell \geq 5$ and $h_3(E) \leq 1$, with or without complex multiplication. Paper: proof of Theorem thm:main, Remark rem:ns, Conjecture conj:euler (the term $-h_\ell(E)$).

19.11.5 10.5 `ex:galcoh:relaxing-toy` (worked example) P

$T = \mathbb{Z}_\ell$ (trivial action), ℓ odd, so $V = \mathbb{Q}_\ell$, $W = \mathbb{Q}_\ell/\mathbb{Z}_\ell$, $T^D = \mathbb{Z}_\ell(1)$, $W^D = \mu_{\ell^\infty}$. Conditions on V: $L_{V,p} = H^1_{\mathrm{ur}}(\mathbb{Q}_p, \mathbb{Q}_\ell)$ at every prime including ℓ, 0 at ∞; their annihilators are $L_{V^D,p} = H^1_{\mathrm{ur}}(\mathbb{Q}_p, \mathbb{Q}_\ell(1)) = 0$ for $p \neq \ell$ (4.4(ii)) and $L_{V^D,\ell} = \mathbb{Z}_\ell^\times \otimes \mathbb{Q}_\ell$ at ℓ (unramified characters kill unit classes, local class field theory; dimension count 3.4(ii) fails at ℓ, use 3.2 with $\ell = p$: $\dim H^1(\mathbb{Q}_\ell, \mathbb{Q}_\ell(1)) = 2$). Then: $H^1_{L_{V^D}}(\mathbb{Q}, \mathbb{Q}_\ell(1)) = (\mathbb{Z}^\times \otimes \mathbb{Q}_\ell) = 0$, hypothesis (a); $H^0(\mathbb{Q}_p, \mathbb{Q}_\ell(1)) = 0$ for every p, hypothesis (b); $H^1_{\mathcal{L}}(\mathbb{Q}, W) = \mathrm{Hom}(G_{\mathbb{Q}}, \mathbb{Q}_\ell/\mathbb{Z}_\ell)^{\mathrm{ur}} = 0$; for $\Sigma = \{p\}$, $H^1_{\mathcal{L}^\Sigma}(\mathbb{Q}, W) =$ characters of ℓ-power order unramified outside $p \simeq \mathbb{Z}/\ell^{v_\ell(p-1)}$; $\mathcal{L}^*_p = \iota^{-1}(0) = H^1(\mathbb{Q}_p, \mathbb{Z}_\ell(1))_{\mathrm{tors}} = \mathbb{Z}_p^\times \otimes \mathbb{Z}_\ell \simeq \mathbb{Z}/\ell^{v_\ell(p-1)}$ (2.5); $H^0(\mathbb{Q}, \mu_{\ell^\infty}) = 0$. The sequence of 10.3 reads $0 \to 0 \to \mathbb{Z}/\ell^{v_\ell(p-1)} \xrightarrow{\sim} (\mathbb{Z}/\ell^{v_\ell(p-1)})^* \to 0 \to 0$: the ℓ-part of $(\mathbb{Z}/p)^\times$ classifies the cyclic extensions of ℓ-power degree and conductor p. (Variant with $H^0(\mathbb{Q}, W^D) \neq 0$, the toy model of $h_3(E) = 1$: $\ell = 3$, $T = \mathbb{Z}_3(\chi_{-3})$, $W^D = \mu_{3^\infty} \otimes \chi_{-3}$, for which $H^0(\mathbb{Q}, W^D) = \mathbb{Z}/3$ because $\bar{\chi}_3 = \chi_{-3}$ while $\chi_3^{\mathrm{cyc}} \chi_{-3} \not\equiv 1 \pmod 9$; Exercise 13.) Paper: background for Lemma lem:dfg21 and Remark rem:ns.

19.11.6 10.6 `rem:galcoh:handover-ch18` (remark) P (bookkeeping)

What Chapter 18 adds to obtain the paper's Lemma lem:dfg21 from 10.3: the Bloch-Kato local conditions $H^1_f(\mathbb{Q}_v, V)$ (with $H^1_f = H^1_{\mathrm{ur}}$ at $p \neq \ell$ and $H^1_f(\mathbb{R}, -) = 0$), their exact orthogonality at ℓ (Bloch-Kato), the vanishing $H^1_f(\mathbb{Q}, V^D) = 0$ (DFG04 Theorem 2.7, Theorem thm:dfg27), the identification $\det(1 - \Phi_p \mid (V^D)^{I_p}) = L_p(A_f, 1)^{-1}$ (geometric Frobenius), and the Fontaine-Perrin-Riou splitting of length $H^1_f(\mathbb{Q}_p, T^D) = \text{length } H^0(\mathbb{Q}_p, W^D)$ into $v_\ell(L_p(A_f, 1)^{-1}) + v_\ell(\mathrm{Tam}_p)$ (Lemma lem:fpr). This chapter supplies the duality, the local structure (4.2 to 4.6) and the global sequence (10.1 to 10.4). Paper: Lemma lem:dfg21, Lemma lem:fpr, proof of Theorem thm:main.

19.12 11. Section-by-section plan with target line counts

Line counts are lines of LaTeX, card excluded (PLAN.md Section 7.3: 1,000 to 2,200). The card lists all 54 items above (about 200 lines, not counted). Each section opens with one paragraph saying what it is for and where the paper uses it.

section	title	contents	target lines
(starred)	Core results at a glance	the card: items 1.1 to 10.6, each stated as above, with tag and `\Cref`	(excluded)
(starred)	Where the paper uses this chapter	lem:tamfactor end of proof (5.1 to 5.5); lem:tamagawa-exp (1.4, 5.6); the paragraph after lem:dfg21 (2.2, 4.2, 4.6, 6.4); lem:dfg21 (3.3, 4.4, 4.5, 8.1 to 8.3, 10.1 to 10.3); proof of thm:main and rem:ns (10.4); the Selmer groups of thm:main (8.1); Section 1.2 and lem:fpr through Chapter 18 (4.3); sec:tam, rem:ko(1) (5.4, 5.5)	40
1	Galois cohomology of a field; Hilbert 90; Kummer theory	1.1 to 1.7; conventions C1 to C6 as `convention` environments; `paperlink` for lem:tamfactor ("$H^1(G, \mathbb{Z}_\ell(1)) = \varprojlim \ldots$"); RHB aside: the connecting map as the "phase ambiguity" of choosing an n-th root, one sentence	200

section	title	contents	target lines
2	ℓ-adic coefficients	2.1 to 2.5; C3, C4 restated; the diagram $T \to V \to W$ with its long exact sequence; `warning`: $H^1(G,V)$ sees no torsion, $H^1(G,T)_{\mathrm{tors}}$ comes from $H^0(G,W)$	190
3	Local Galois cohomology: finiteness, Euler characteristic, Tate local duality	3.1 to 3.5; the table of orders 3.5(a); the 11a1 example 3.5(c)	200
4	Unramified cohomology	4.1 to 4.7; `warning` at 4.4 (the case $p = \ell$); `warning` (point 11): the vanishing in 4.2(iii) is convention-free, the value of the determinant is not; `paperlink` for the paragraph after lem:dfg21 (4.2, 4.6, 4.7(d))	230
5	The Kummer class of the Tate parameter	5.1 to 5.6; `paperlink` quoting the end of the proof of lem:tamfactor, with the corrected reading (points 7, 31, 42 and Part 16, N1); the table 5.5	210
6	Global Galois cohomology: $G_{\mathbb{Q}}$, $G_{\mathbb{Q},S}$, the real place	6.1 to 6.5; `paperlink` for "$H^1(\mathbb{R}, T^D)$ is killed by 2" (6.4)	130
7	Elliptic curves: the Kummer map, Selmer and Sha	7.1 to 7.5; the weak Mordell-Weil theorem as the model of all finiteness proofs; 7.2(iv) as the bridge to Tamagawa numbers	170
8	Selmer structures and dual structures	8.1 to 8.3; the dictionary relaxed/strict; `paperlink` for Section 4.2 (H^1_f, H^1_Σ)	150
9	Poitou-Tate and the Greenberg-Wiles formula	9.1 to 9.5; the check 9.4 worked in full	170
10	Comparing Selmer structures; the abstract form of DFG04 Lemma 2.1	10.1 to 10.6; the proof of 10.3 in full; the toy example 10.5; `paperlink` for lem:dfg21, the proof of thm:main and rem:ns	230

section	title	contents	target lines
11	Exercises	18 exercises (Part 12.2)	150
12	Notes and references	Serre (Galois Cohomology; Local Fields), Tate (ICM 1962), Milne ADT, NSW, Cassels-Frohlich, Rubin, Mazur-Rubin, Greenberg, Wiles, DDT, Bloch-Kato, FPR94, Silverman AEC/ATAEC; which numbers were cited (only those verified)	40
	total		**2110**

19.13 12. Coverage table and exercises

19.13.1 12.1 Coverage

PLAN.md scope item, Section 6 map row, or paper hook	line-up items
Galois cohomology of a field $H^i(K, M)$, discrete G_K-modules	1.1, 1.2
Hilbert 90, $H^1(K, \overline{K}^\times) = 0$	1.3
Kummer theory $H^1(K, \mu_n) = K^\times/(K^\times)^n$ (proved)	1.4, 1.6
the Kummer map and the Kummer class of an element	1.5, 2.4
$H^1(K, \mathbb{Z}_\ell(1)) = \varprojlim K^\times/(K^\times)^{\ell^k}$, $\mathbb{Z}_\ell(1) = T_\ell\mu$	2.4, 2.5, C3
the Brauer group and $H^2(K, \mu_n)$ (statement)	1.4(ii), 1.7
finiteness of $H^i(\mathbb{Q}_p, M)$ for finite M	3.1
local Euler-Poincare characteristic formula (statement)	3.2, 3.5
Tate local duality (statement with reference)	3.3, 3.4
$H^1_{\mathrm{ur}}(\mathbb{Q}_p, M) = H^1(G/I, M^I) = \ker(H^1 \to H^1(I, M))$	4.1, 4.3
$H^1_{\mathrm{ur}}(\mathbb{Q}_p, M) = M^{I_p}/(\mathrm{Frob}_p - 1)M^{I_p}$	4.2
the annihilator of H^1_{ur} is H^1_{ur} of the dual ($p \neq \ell$)	4.4
the real place: $H^1(\mathbb{R}, M)$ killed by 2, zero for ℓ odd	6.4
$G_{\mathbb{Q}}$ and $G_{\mathbb{Q},S}$	6.1, 6.2
finiteness of $H^1(G_{\mathbb{Q},S}, M)$	6.3, 6.5
Kummer sequence of an elliptic curve and the Kummer map	7.1, 7.2
n-Selmer group, Sha, $0 \to E(\mathbb{Q})/n \to \mathrm{Sel}_n \to \mathrm{Sha}[n] \to 0$, finiteness, weak Mordell-Weil	7.3, 7.4, 7.5
Selmer structures and $H^1_{\mathcal{L}}(\mathbb{Q}, M)$	8.1, 8.3
the dual Selmer structure via local duality	8.2
the Greenberg-Wiles formula	9.3, 9.4, 9.5

PLAN.md scope item, Section 6 map row, or paper hook	line-up items
the Poitou-Tate exact sequence	9.1, 9.2
comparing Selmer groups for two sets of local conditions, elementary part proved	10.1, 10.2
Lemma lem:dfg21 as the comparison of Bloch-Kato with Σ-relaxed, dual term $H^0(\mathbb{Q}, W^D)^*$	10.3, 10.4, 10.6
$H^1(K,T)$, $H^1(K,V)$, $H^1(K,W)$; $H^1(K,T)_{\mathrm{tors}}$ from $H^0(K,W)$	2.2, 2.3
$H^1(\mathbb{Q}_p,T)$ finitely generated, torsion $H^0(\mathbb{Q}_p,W)/\mathrm{div}$; "$H^1_f(\mathbb{Q}_p,T^D)$ is the torsion"	2.2, 3.1, 4.6
the Tate curve: $H^1(\mathbb{Q}_p,\mathbb{Z}_\ell(1))$, the Kummer class of the Tate parameter, its image in $H^1(G/I,\mathbb{Z}_\ell(1)^I)=\mathbb{Z}/\ell^{v_\ell(q-1)}$ (proved in full)	5.1 to 5.5
Section 6 row: Sigma-imprimitive Selmer group, Bloch-Kato Selmer group, H^1_f, H^1_Σ (17, 18)	8.1, 8.2, 10.3, 10.6
Section 6 row: the computation of $\mathrm{Tam}_q(\mathrm{ad}^0 T)$ at a multiplicative prime (16, 17, 18)	4.3, 5.4, 5.5
Section 6 row: Kummer theory, Kummer classes, $H^1(G,\mathbb{Z}_\ell(1))$, tame character, $\mathrm{Frob}^{-1}\,t\,\mathrm{Frob}=t^{q^{-1}}$ (5, 7, 16, 17)	1.4, 1.5, 2.4, 5.1 to 5.4 (the tame relation is Chapter 7's, used in 5.4(iv))
hook: Lemma lem:tamagawa-exp	1.4, 1.5(i), 5.6
hook: Lemma lem:tamfactor (Kummer theory, c', $H^1(G/I,\mathbb{Z}_\ell(1)^I)$)	1.6, 2.4, 2.5, 5.1 to 5.5
hook: Lemma lem:dfg21 and the paragraph after it	2.2, 3.3, 4.2, 4.4 to 4.7, 6.4, 8.1 to 8.3, 10.1 to 10.3
hook: Remark rem:ns	8.3, 10.4, 10.5 (variant), Exercise 13
hook: the Selmer groups of Theorem thm:main	8.1, 10.4
hook (through Chapter 18): Section 1.2 (Tam_p), Lemma lem:fpr	4.2(iv), 4.3, 4.6, 4.7(d), (e)
hook: Section sec:tam, Remark rem:ko(1)	5.4, 5.5

19.13.2 12.2 Exercises (Section 11 of the chapter), graded from easy to harder

1. (Easy) From Hilbert 90 deduce $H^1(\mathbb{R},\mu_n)=\mathbb{R}^\times/\mathbb{R}^{\times n}$; show it is 0 for n odd and $\mathbb{Z}/2$ for n even; compare with 6.4.
2. Compute $\#H^i(\mathbb{Q}_p,\mathbb{Z}/\ell^k)$ and $\#H^i(\mathbb{Q}_p,\mu_{\ell^k})$, $i=0,1,2$, for $(p,\ell,k)=(7,3,1),(19,3,2),(5,3,2)$; check the Euler characteristic and the duality 3.4(i).
3. Check the paper's claim that $H^1(\mathbb{R},T^D)$ is killed by 2 and that $H^1_f(\mathbb{R},T^D)=0$ for ℓ odd; show that $H^1(\mathbb{R},\mathbb{Z}_2(1))=\mathbb{Z}/2$, so the claim genuinely needs ℓ odd.
4. Check the paper's claim "$H^1(G,\mathbb{Z}_\ell(1))=\varprojlim \mathbb{Q}_q^\times/(\mathbb{Q}_q^\times)^{\ell^k}$" by computing both sides for $q=7$, $\ell=3$ (answer $\mathbb{Z}_3\times\mathbb{Z}/3$) and identify the torsion with $H^0(\mathbb{Q}_7,\mathbb{Q}_3/\mathbb{Z}_3(1))=\mu_3$ (2.2(ii)).
5. Check the paper's claim "$\mathbb{Z}_\ell(1)/(q-1)\simeq\mathbb{Z}/\ell^{v_\ell(q-1)}\ \ldots$ the class of the unit part in $\mathbb{F}_q^\times\otimes\mathbb{Z}_\ell$": for $q=19$, $\ell=3$ compute the classes of $u=2,8,-1$ and their valuations $0,1,2$ (Chapter 7, 2.8(b)); explain why the class 0 must be read as having valuation $v_3(18)=2$ (point 31).
6. Let κ be a Kummer cocycle of $x=q^n u\in\mathbb{Q}_q^\times$. Show that $\kappa(\mathrm{Frob}\,\sigma)-\kappa(\mathrm{Frob})=q\,n\,t_\ell(\sigma)$ for $\sigma\in I_q$ and that a change of cocycle changes $\kappa(\mathrm{Frob})$ by $(q-1)z$. Conclude that the paper's "well defined modulo $q-1$" should read "modulo $(n,q-1)$", and that $\min(v_\ell(n),v_\ell(q-1),v_\ell(c'))$ is nevertheless well defined (point 7).
7. Check the paper's matrix in the proof of Lemma lem:tamfactor (point 42): show that $g\mapsto\left(\begin{smallmatrix}\chi_\ell\varepsilon & \kappa\\ 0 & \varepsilon\end{smallmatrix}\right)$ is not a homomorphism when $\varepsilon\neq1$ and κ is a Kummer cocycle, while $\left(\begin{smallmatrix}\chi_\ell\varepsilon & \varepsilon\kappa\\ 0 & \varepsilon\end{smallmatrix}\right)$ is; what changes in c'?
8. (Paper check, PARI) For 791a1 at $q=7$ and 77c1 at $q=7$ ($\ell=3$) compute the Tate parameter with

`ellinit(E,O(7^30)).tate`, its valuation $n = 3$ and unit part ($\bar{u} = 6$, resp. 5); decide whether $\bar{u}$ is a cube; deduce $v_3(\mathrm{Tam}_7(\mathrm{ad}^0 T)) = 2$, resp. 1, with Chapter 16's model computation; compare with $v_3(\deg \varphi_E) = v_3(144) = 2$, resp. $v_3(6) = 1$ (ecdata). Check the paper's claim in Section sec:tam that the extra term raises the bound exactly where the unit part is an ℓ-th power.

9. Check the paper's claim that $H^1_{\mathrm{ur}}(\mathbb{Q}_p, V^D) = 0$ at a multiplicative prime: compute $(V^D)^{I_p}$ for $V = \mathrm{ad}^0 V_\ell E$, the eigenvalue of Frob (p^2) and of Φ (p^{-2}), and $\det(1 - \Phi) = 1 - p^{-2}$ versus $\det(1 - \mathrm{Frob}) = 1 - p^2$; show that both have the same ℓ-adic valuation and that only the first is $L_p(A_f, 1)^{-1}$ (point 11).
10. Check the paper's claim in Lemma lem:fpr that the Euler factor is an ℓ-adic integer of valuation $e_\ell(E, q)$ at an additive prime, in the case of type III at $q \geq 5$: show $\#H^1_{\mathrm{ur}}(\mathbb{Q}_q, T^D) = \ell^{v_\ell(q - \chi_{-4}(q))}$ using 4.2(iv) and Lemma lem:euler's description of $(\mathrm{ad}^0 V)^{I_q}$.
11. (Greenberg-Wiles by hand) Do 9.4(b) in full: for $M = \mathbb{Z}/\ell$ relaxed at one prime $p \neq \ell$, compute both sides and recover the number of cyclic extensions of $\mathbb{Q}$ of degree ℓ and conductor p.
12. Do 10.5 in full: verify the hypotheses of 10.3 for $T = \mathbb{Z}_\ell$, $\Sigma = \{p\}$, and identify every term of the exact sequence.
13. (The toy model of $h_3(E)$.) Let $T = \mathbb{Z}_3(\chi_{-3})$. Show $H^0(\mathbb{Q}, W^D) = H^0(\mathbb{Q}, \mu_{3^\infty} \otimes \chi_{-3}) \simeq \mathbb{Z}/3$, using $\bar{\chi}_3 = \chi_{-3}$ and an element σ with $\chi_3^{\mathrm{cyc}}(\sigma) = 4$. Show that for $\ell \geq 5$ and any character ψ of order 2, $H^0(\mathbb{Q}, \mu_{\ell^\infty} \otimes \psi) = 0$. Relate this to the paper's claim in Lemma lem:h0 that $H^0(\mathbb{Q}, W^D)$ is finite and vanishes for $\ell \geq 5$, and to the correct bound $h_3(E) \leq 1$ (point 5).
14. Over $\mathbb{Q}_p$, p odd, compute $\#H^i(\mathbb{Q}_p, \mu_p)$ and check the Euler characteristic p^{-1}; show that $H^1_{\mathrm{ur}}(\mathbb{Q}_p, \mathbb{Z}/p)$ and $H^1_{\mathrm{ur}}(\mathbb{Q}_p, \mu_p)$ are not exact annihilators (the warning in 4.4).
15. (PARI) Show that $E[5] \subset E(\mathbb{Q}_{11})$ for $E = 11\mathrm{a}1$ (Tate parameter a fifth power, $\mu_5 \subset \mathbb{Q}_{11}$), deduce $\#H^1(\mathbb{Q}_{11}, E[5]) = 625$, and show that the image of the Kummer map $E(\mathbb{Q}_{11})/5 \to H^1(\mathbb{Q}_{11}, E[5])$ has order 25 but is not $H^1_{\mathrm{ur}}(\mathbb{Q}_{11}, E[5])$; relate this to $c_{11} = 5$ and 7.2(iv).
16. For the Tate curve at $p \neq \ell$ (ℓ odd) compute $H^1_{\mathrm{ur}}(\mathbb{Q}_p, E[\ell^\infty])$ in the split and nonsplit cases and compare with the ℓ-part of c_p (Chapter 10) and with Chapter 16's model computation for $M = T$.
17. (Harder) Prove 4.4(i) for $M = \mathbb{Z}/\ell$, $M' = \mu_\ell$, $p \neq \ell$, directly: the cup product of an unramified character with the Kummer class of a unit is 0 (use local class field theory as stated in Chapter 7, 6.5).
18. (Harder) Show that the hypothesis $H^0(\mathbb{Q}, V^D) = 0$ of Lemma lem:dfg21 follows from $H^0(\mathbb{Q}_p, V^D) = 0$ at one $p \in \Sigma$, and that in the paper's situation $H^0(\mathbb{Q}_p, V^D) = 0$ at every $p \in \Sigma$ (4.2(iii) and the paragraph after the lemma). Where exactly does the proof of 10.3 use a prime of Σ with this property, and what would the last term of the sequence be without it?

19.14 13. Dependencies and forward pointers

Results from earlier chapters used here (by chapter label only): * Chapter 2 (`ch:rings`): ℓ-th powers in $\mathbb{F}_q^\times$, $[\mathbb{F}_q^\times : (\mathbb{F}_q^\times)^{\ell^k}] = \ell^{\min(k, v_\ell(q-1))}$, the class $c(u)$ and v_ℓ of a class (1.6, 5.2, 5.4); q prime to ℓ is a unit of $\mathbb{Z}_\ell$. * Chapter 3 (`ch:modules`): finitely generated $\mathbb{Z}_\ell$-modules, torsion, saturation (4.2(iv)); lengths and their additivity; Pontryagin duality 8.1, 8.2 and T versus W 8.3, cyclic ℓ-primary modules 8.4; cokernel of an injective endomorphism of $\mathbb{Z}_\ell^d$ has order $\ell^{v_\ell(\det)}$; maximal divisible subgroups of cofinitely generated modules. * Chapter 4 (`ch:reps`): 7.6 (duals and the trace form, only as a pointer), unipotent and finite cyclic actions (used in 4.7). * Chapter 5 (`ch:galois`): Hilbert 90 for finite cyclic and finite Galois extensions (galois:hilbert90), Kummer theory with $\mu_n \subset K$ (galois:kummer-character, galois:kummer-classification), galois:kummer-finite-field, galois:l-adic-kummer, the cyclotomic character, profinite Galois groups, complex conjugation and $G_{\mathbb{R}}$. * Chapter 6 (`ch:ant`): decomposition and inertia groups and Frobenius elements (6.1, 6.2), class number one of $\mathbb{Q}$ and the units of $\mathbb{Z}_S$ (6.5). * Chapter 7 (`ch:local`): 2.7 (powers in $K^\times$, $\widehat{K^\times}$), 4.2 ($\mathbb{Q}_q^{\mathrm{ur}}$, Hensel there), 5.3 (tame character, $I^{(\ell)} = \mathbb{Z}_\ell(1)$), 5.4 (tame relation), 5.5 (the Frobenius lift fixing roots of q), 6.1, 6.2 (Kummer cocycles, $\kappa_x|_I = v(x)t_\ell$, κ_u unramified), 6.5 (local class field theory, only in 3.5(a), 10.5, Exercise 17), conventions L3 and L6 (Frobenius and L-factors). * Chapter 9 (`ch:ellcurves`): $E[n]$, surjectivity of $[n]$, the Weil pairing (5.3, 5.5), the descent with heights for Mordell-Weil (9.1). * Chapter 10 (`ch:reduction`): the formal group and E_1 (7.2), the

Tate curve and the Tate parameter q_E, u, $c(u)$ (7.4, 7.5(iii), 7.8), the component groups and c_p (7.6). * Chapter 11 (`ch:galrep`): 3.2 (the matrices $\left(\begin{smallmatrix}\chi_\ell\varepsilon & \varepsilon\kappa\\ 0 & \varepsilon\end{smallmatrix}\right)$ on $T_\ell E$), 3.3 (Lemma lem:tamagawa-exp), Neron-Ogg-Shafarevich (6.1); Lemma lem:h0 in its corrected form (10.4). * Chapter 16 (`ch:cohomology`): everything in its Sections 4 to 11: h0-h1 and connecting maps, inflation-restriction (a) to (d), conjugation-action, order-kills and z2-on-l-adic (6.4), continuous-cohomology, les-continuous, tate-limits, h1-finitely-generated, prime-to-l-vanishing, tame-inertia-example, procyclic-discrete, procyclic-compact, procyclic-examples, model-computation (5.4(v)), cup-product (3.3, 9.2), lengths-in-exact-sequences (10.4), galois-cohomology-convention and unramified-h1-preview (4.1, 4.2 are their full versions).

Black boxes stated here with references, never used to prove an earlier item: inv: $\mathrm{Br}(K) \simeq \mathbb{Q}/\mathbb{Z}$ and Albert-Brauer-Hasse-Noether (1.7); $\operatorname{cd} G_{\mathbb{Q}_p} = 2$ and finiteness (3.1); the local Euler characteristic (3.2); Tate local duality (3.3); Hermite-Minkowski and the finiteness of $H^2(G_{\mathbb{Q},S}, M)$ (6.3); Tate's self-duality of the Kummer image (7.2(iii)); Poitou-Tate (9.1); Greenberg-Wiles (9.3); the corank formula 8.3(iii).

Forward pointers (chapters that use this one): * Chapter 18 (`ch:blochkato`): defines the Bloch-Kato conditions as propagated structures (4.5, 8.1(e)), checks their exact orthogonality (4.4(ii) at $p \neq \ell$; Bloch-Kato at ℓ), proves Lemma lem:fpr from 4.2(iv), 4.3, 4.6, states Lemma lem:dfg21 as 10.3 for f, with DFG04 Theorem 2.7 for hypothesis (a), and assembles Lemma lem:tamfactor from Chapter 16's model computation and 5.4. * Chapter 19 (`ch:lfunctions`): the value $\det(1 - \Phi \mid (V^D)^{I_p}) = L_p(A_f, 1)^{-1}$ and the omitted factors (4.7(d), (e) are the cohomological side). * Chapter 20 (`ch:congruence`): the assembly of Theorem thm:main uses 10.4; the Taylor-Wiles method reads $H^1_\Sigma(\mathbb{Q}, W)$ as a Selmer group of 8.1. * Chapter 21 (`ch:paper`): A2.7 step (7) cites 5.3, 5.4; A4.9 "attached facts" cite 2.2, 4.2, 4.6, 6.4 and its proof idea cites 10.3; A4.10 steps (1), (2) cite 4.3, 4.2(iv); A5.2 cites 10.4. * Chapter 22 (`ch:computation`): Tate parameters and unit parts with PARI (5.5).

19.15 14. Notes for the writer

19.15.1 14.1 Conventions (to be set as `convention` environments at first use)

- **Frobenius.** C2: Frob arithmetic in all cohomological statements; $\Phi = \mathrm{Frob}^{-1}$ only where an L-factor is named. In 4.2(iii) say explicitly that the vanishing of $H^1_{\mathrm{ur}}(\mathbb{Q}_p, V)$ is equivalent to $\det(1 - \mathrm{Frob} \mid V^I) \neq 0$ and to $\det(1 - \Phi \mid V^I) \neq 0$, and that the order of $H^1_{\mathrm{ur}}(\mathbb{Q}_p, T)$ is ℓ^{v_ℓ} of either determinant; then say that only $\det(1 - \Phi \mid (V^D)^{I_p})$ equals $L_p(A_f, 1)^{-1}$ (point 11; Exercise 9).
- **Duals.** C3: * = Pontryagin dual (the paper's), $^\vee$ = linear dual, $M^*(1)$ = Kummer dual of a finite or discrete module, $T^D = T^\vee(1)$, $W^D = V^D/T^D$. Chapter 3's 8.3(vi) says the Cartier dual is Chapter 18's; it is needed here first, for Tate duality. Define it here in general (2.1) and let Chapter 18 add the adjoint-specific identification $T^D \simeq A(1)$ (Part 16, B-a).
- **Twists.** $M(1) = M \otimes \mathbb{Z}_\ell(1)$; Frob acts on $\mathbb{Z}_\ell(1)$ by p ($p \neq \ell$), Φ by p^{-1}. χ_ℓ is the cyclotomic character; the quadratic character of conductor q is χ_{q^*}, never χ_q (the paper's notation, kept only in quotations; important_notes.md point 32, decision D3).
- **Signs.** The Kummer connecting map has no sign (C6). The Tate local pairing is compatible with connecting maps only up to $(-1)^i$ (3.3); no sign matters for any order or length in the chapter, so say this once and never track signs.
- **Kummer cocycle versus class.** Write κ for a cocycle and $[\kappa]$ for its class whenever a value $\kappa(\mathrm{Frob})$ is taken. The paper calls the cocycle in the corner "the Kummer class"; 5.4 must keep the two apart, because $\kappa(\mathrm{Frob})$ depends on the cocycle and the lift.

19.15.2 14.2 Pitfalls

- **c' is a cocycle value, not a class.** 5.3 gives a canonical meaning to "the image of κ in $H^1(G/I, \mathbb{Z}_\ell(1)^I)$" for $\mathbb{Q}_q$ (the retraction attached to the uniformiser q); 5.4(iii) says c' equals $\pm$ that image for the Frobenius lift fixing the roots of q; 5.4(iv) gives the indeterminacy $(n, q-1)\mathbb{Z}_\ell$ in general. Do not write "c' is well defined modulo $q-1$" (the paper's phrase): it is well defined modulo $(n, q-1)$ only, which suffices. State the result as $v_\ell((n, q-1, c')\mathbb{Z}_\ell)$, which is visibly choice-free.
- **The corner is $\varepsilon\kappa$** (point 42); on I the factor ε is 1, at Frob it is $a_q(E) = \pm 1$, harmless for valuations.

- **"u is an ℓ^k-th power in $\mathbb{F}_q^\times$"** means its reduction (point 31); for $\ell \neq q$ this is equivalent to $u \in (\mathbb{Z}_q^\times)^{\ell^k}$ (Chapter 7, 2.7). The class $0 \in \mathbb{Z}/\ell^v$ has valuation v, not ∞, in the formula.
- **Unramified orthogonality needs $p \neq \ell$** (4.4, warning, Exercise 14). At ℓ use Bloch-Kato (Chapter 18). Never write "$H^1_f = H^1_{\mathrm{ur}}$" at $p = \ell$.
- **Unramified for T versus the propagated condition.** For $p \neq \ell$ with T ramified at p, $\iota^{-1}(H^1_{\mathrm{ur}}(\mathbb{Q}_p, V))$ can be strictly larger than $H^1_{\mathrm{ur}}(\mathbb{Q}_p, T)$ (it contains all torsion); the quotient is what Chapter 18 measures by the Tamagawa factor. For T unramified at p they coincide (4.5(iii)). Do not identify them in general.
- **The torsion of $H^1(G,T)$** is $H^0(G,W)$ modulo its divisible part, not $H^0(G,W)$ (2.2(ii)); the two agree when $H^0(G,V) = 0$, which is the case in every application of the paper (4.6, 10.3).
- **Hypothesis (b) of 10.3.** The surjectivity onto $H^0(\mathbb{Q}, W^D)^*$ needs a prime of Σ with $H^0(\mathbb{Q}_p, V^D) = 0$; in DFG04's adjoint setting this is automatic (weights), and in the paper it follows from the paragraph after Lemma lem:dfg21. State it as a hypothesis in the abstract theorem and discharge it for the paper (Exercise 18). Without it, the sequence continues with $\to H^1_{\mathcal{L}^*_\Sigma}(\mathbb{Q}, T^D)^* \to 0$.
- **Continuity.** All ℓ-adic statements use continuous cochains; the limit passages in 2.4 and 10.2 rely on Chapter 16's tate-limits and on finiteness at each level. Do not write $H^1(G,T) = \varinjlim$ of anything.
- **Lengths versus orders.** Lemma lem:dfg21 is stated with Pontryagin duals; only lengths enter the paper; Pontryagin duality preserves them (Chapter 3). Use length throughout Section 10.
- **The real place.** For ℓ odd it contributes nothing to Selmer groups; in Greenberg-Wiles it still contributes the factor $1/\#H^0(\mathbb{R}, M)$ (9.4 shows this factor is essential).
- **Theorem numbers.** Only cite the numbers marked "(check)" after checking them against the books; otherwise cite chapter and section (PLAN.md Section 3). In particular: Serre's Galois Cohomology II.5.2 and II.5.7, Milne ADT I.2.3, I.2.6, I.2.8 and I.4.10, Wiles95 Proposition 1.6, DDT Theorem 2.19, Rubin Theorem 1.7.3 and Lemma 1.3.2, Bloch-Kato Proposition 3.8, AEC X.4.2 were recalled from memory.

19.15.3 14.3 What to leave to neighbouring chapters

- Chapter 5: Hilbert 90 for finite extensions, Kummer theory with $\mu_n \subset K$, the ℓ-adic Kummer cocycle, galois:kummer-finite-field. Recall in one line each; this chapter adds the cohomological form and removes $\mu_n \subset K$.
- Chapter 7: everything field-theoretic about $\mathbb{Q}_q$: Hensel, $\widehat{K^\times}$, the tame character and relation, the Frobenius lift fixing roots (5.5), the decomposition $\kappa_x = a\kappa_\varpi + \kappa_u$ and the formula $\kappa_x|_I = v(x)t_\ell$ (6.2). Cite them; Section 5 here adds only the cohomological statements (5.1 to 5.3) and the identification of c' (5.4).
- Chapters 10, 11: the Tate curve, the matrices of $T_\ell E$ at a multiplicative prime, Lemma lem:tamagawa-exp (proved in Chapter 11, 3.3). Here only the cohomological reading 5.6.
- Chapter 16: the group cohomology of Lemma lem:tamfactor (the $(t-1)M$ computation, the conjugation action, the invariants, their Fitting ideal). Section 5 here uses Chapter 16's model-computation as a black box and supplies only the identification of c'.
- Chapter 18: Bloch-Kato conditions (H^1_f, H^1_g, H^1_e, B_{cris}), the definition of Tam_p, the Fontaine-Perrin-Riou formula (Lemma lem:fpr), the full proof of Lemma lem:tamfactor (assembly), DFG04 Lemma 2.1 in the paper's form and DFG04 Theorem 2.7, $T^D \simeq A(1)$, the adjoint motive. This chapter must not define H^1_f or Tam_p except by forward reference (`\Cref{ch:blochkato}`).
- Chapter 19: local L-factors and their values; $e_\ell(E,q)$; Lemma lem:euler. Here only the cohomological side (4.7(d), (e)).
- Chapter 20: DFG04 Theorem 3.7 and the assembly of Theorem thm:main.

19.15.4 14.4 How each relevant point of important_notes.md is handled

- **Point 5 (Lemma lem:h0: $h_\ell(E) = 0$ for $\ell \geq 5$, $h_3(E) \leq 1$, CM irrelevant).** 10.4 quotes the corrected statement (proved in Chapter 11) when it names length $H^0(\mathbb{Q}, W^D) = h_\ell(E)$; Exercise 13

gives the determinant mechanism in a one-dimensional toy model ($\chi_3^{\mathrm{cyc}}\chi_{-3} \not\equiv 1 \pmod 9$, and $\bar\chi_\ell$ of order ≥ 4 for $\ell \geq 5$). Do not repeat the paper's "finite for E without complex multiplication".

- **Point 7 (Lemma lem:tamfactor, "c' is the image of κ in $H^1(G/I, \mathbb{Z}_\ell(1)^I)$").** Section 5 in full: 5.1, 5.2 (the unramified classes are the unit classes; their value at Frob is the class of $\bar u$), 5.3 (the canonical retraction attached to the uniformiser q, which gives the paper's phrase a precise meaning; Part 16, N1), 5.4 (iii) to (v) (the value c', its indeterminacy $(n, q-1)$, the choice-free minimum). Exercise 6. A `paperlink` quotes the paper's sentence and the corrected reading.
- **Point 8 (Lemma lem:tamagawa-exp phrasing).** 5.6 uses q_E for the Tate parameter and states "q_E is an ℓ^t-th power in $\mathbb{Q}_q^{\mathrm{ur}}$ iff $\ell^t \mid n$".
- **Point 11 (geometric versus arithmetic Frobenius).** C2, 4.2(iii), (iv), 4.7(d), the `warning` in Section 4, Exercise 9: the paragraph after Lemma lem:dfg21 is correct as a vanishing statement in either convention; the equality with $L_p(A_f, 1)^{-1}$ needs Φ.
- **Point 31 (the class of u means the class of its reduction; $v_\ell(0)$).** 1.6, 5.2, 5.4(v), Exercise 5.
- **Point 37 (Tam_p is the Fitting ideal of the invariants).** 4.3, forward remark, written as $\mathrm{Fitt}_{\mathbb{Z}_\ell}\big((H^1(I,M)_{\mathrm{tors}})^{G/I}\big)$.
- **Point 42 (the corner is $\varepsilon\kappa$).** 5.4 setup and (iii); Exercise 7.
- **Point 32 (notation: the paper's χ_q, the book's χ_{q^*}, versus χ_ℓ).** 14.1, Twists.
- **Points 39, 56 (multiplicity one not needed).** Not in this chapter's area; nothing here depends on freeness.
- **B4 (the symbol $*$).** C3 fixes * as Pontryagin dual in this chapter, consistently with the paper and Chapter 16; Appendix B must record it.
- **B6 (PLAN.md's Weil-Deligne relation in geometric form).** Not used here; C2 repeats Chapter 7's convention (arithmetic Frob, geometric $\Phi = \mathrm{Frob}^{-1}$) so that no formula of this chapter depends on PLAN.md's wording.
- **B7 (Chapter 21, A2.7 step (1): κ versus $\varepsilon\kappa$).** 5.4 is the statement Chapter 21 should cite; step (7) there also omits ε in "$c' = \kappa_u(\mathrm{Frob})$" (Part 16, B-c).
- **B14 (bibliography).** See 14.5.

19.15.5 14.5 References and bibliography

In `refs.bib` already: `SerreLocalFields`, `NSW`, `MilneADT`, `TateDuality` (Tate, ICM 1962), `Tate95` (Tate, "Relations between K_2 and Galois cohomology", 1976: the continuous-cohomology reference, despite the key), `CasselsFrohlich`, `Rubin`, `Greenberg89`, `Wiles95`, `DDT`, `BlochKato`, `FPR94`, `DFG04`, `SilvermanAEC`, `SilvermanATAEC`, `SerreAbelian`, `MilneCFT`. To add, following decision B14 (a reference used by more than one chapter goes into `refs.bib`): * `SerreGaloisCoh` (J.-P. Serre, Galois Cohomology, Springer Monographs in Mathematics, Springer-Verlag, Berlin, 1997; translated from Cohomologie galoisienne, Lecture Notes in Math. 5): planned by the Chapter 16 outline for `bib/ch16.bib`, but needed by Chapters 16 and 17 (and mentioned by Chapter 3): **put it in `refs.bib`**, once, and tell the Chapter 16 writer (Part 16, B-b). * `MazurRubin04` (B. Mazur and K. Rubin, Kolyvagin systems, Mem. Amer. Math. Soc. 168 (2004), no. 799): the source of the term "Selmer structure". If Chapter 18 (outlined in parallel) also cites it, it goes in `refs.bib`; otherwise in `bib/ch17.bib`. The writer must check `refs.bib` and `bib/ch18.bib` first. No other new keys are needed; the writer must not add Washington's FLT-volume survey (cited above only as optional).

19.16 15. Verification log (computations done for this outline)

Python with cypari2 (PARI/GP 2.17.2) and Cremona's ecdata files in `/tmp/ch09/data` (allcurves, allbsd, degphi, galrep, conductor below 10,000); scripts in the session scratchpad (`scan.py`). * Tate parameters by `ellinit(E,O(q^30)).tate[3]`, valuation n and $\bar u = q_E q^{-n} \bmod q$, for all curves numbered 1 of conductor at most 2,000 with a multiplicative prime q and $\ell \in \{3,5,7\}$, $\ell \mid n$, $q \equiv 1 \pmod \ell$: 348 pairs, of which 104 have $\bar u$ an ℓ-th power. Rows of 5.5: 11a1 ($q=11$, $n=5$, $\bar u = 10$), 77c1 ($q=7$, $n=3$, $\bar u = 5$, $a_7 = -1$), 791a1 ($q=7$, $n=3$, $\bar u = 6$, $a_7 = -1$), 1577a1 ($q=19$, $n=3$, $\bar u = 8$, $a_{19} = 1$); cubes mod 7: $\{1, 6\}$; cubes

mod 19: $\{1,7,8,11,12,18\}$; ninth powers mod 19: $\{\pm 1\}$. Kodaira codes (`elllocalred`): 791a1 I_3 at 7, I_2 at 113; 77c1 I_3 at 7, I_2 at 11; 1577a1 I_3 at 19, I_2 at 83. Modular degrees (degphi): 77c1 6, 791a1 144, 1577a1 720, 11a1 1. Mod 3 images (galrep): 77c1, 791a1, 1577a1 surjective (no entry); 11a1 has 5Cs.1.1. Hence the "bound" column of 5.5 equals $v_3(\deg\varphi_E)$ in all three irreducible cases. * Observation from the same scan (not used): among split multiplicative pairs with $\bar{u}$ a cube at $\ell = 3$ the mod 3 image is almost always reducible (3Cs.1.1 or 3B.1.1) or 3Nn; nonsplit pairs with $\bar{u}$ a cube are mostly surjective (791a1, 805b1, 874a1, 1066a1, ...). Consistent with the fact that for split reduction, $3 \mid n$, $q \equiv 1 \pmod 3$ and q_E a cube, $E[3]$ is trivial as a $G_{\mathbb{Q}_q}$-module. * 11a1 over $\mathbb{Q}_{11}$: all 12 roots of the 5-division polynomial lie in $\mathbb{Q}_{11}$ (`polrootspadic`), each with $4x^3 + b_2x^2 + 2b_4x + b_6$ a square in $\mathbb{Q}_{11}$; so $\#E(\mathbb{Q}_{11})[5] = 25$. allbsd: 11a1 rank 0, torsion 5, Tamagawa product 5, Sha 1. * Orders of local cohomology (3.5(a)) by hand from the Kummer description: e.g. $(p,\ell,k) = (19,3,2)$ gives $3^2, 3^4, 3^2$ for both $\mathbb{Z}/9$ and μ_9; $(5,3,2)$ ($v_3(4) = 0$, $m = 0$): $\#H^i(\mathbb{Q}_5, \mathbb{Z}/9) = 9,9,1$ and $\#H^i(\mathbb{Q}_5, \mu_9) = 1,9,9$. Euler characteristic 1 in each case. * $\mathbb{Q}_p$, μ_p (p odd): $\#\mathbb{Q}_p^\times/(\mathbb{Q}_p^\times)^p = p^2$ (valuation and $(1+p\mathbb{Z}_p)/(1+p\mathbb{Z}_p)^p \simeq \mathbb{Z}/p$); $H^2 = \mathbb{Z}/p$; characteristic p^{-1}. * Toy example 10.5 and 9.4 by hand (class field theory of $\mathbb{Q}$). Toy model of h_3: $\chi_3^{\mathrm{cyc}}(\sigma) = 4$ forces $\chi_{-3}(\sigma) = 1$ and $4 \not\equiv 1 \pmod 9$. * Not verified (recalled): every theorem number marked "(check)" in Parts 1 to 10.

19.17 16. Points for important_notes.md

19.17.1 16.1 New points about the paper

N1 (addendum to point 7). Lemma lem:tamfactor, end of the proof: the phrase "the image of κ in $H^1(G/I, \mathbb{Z}_\ell(1)^I)$" has a precise meaning after all; what is imprecise is "c' is" that image and "well defined modulo $q-1$". Location: proof of Lemma lem:tamfactor, the sentence "Finally c' is the image of κ in $H^1(G/I, \mathbb{Z}_\ell(1)^I) = \mathbb{Z}_\ell(1)/(q-1) \simeq \mathbb{Z}/\ell^{v_\ell(q-1)}$, the class of the unit part of q_E in $\mathbb{F}_q^\times \otimes \mathbb{Z}_\ell$ by Kummer theory, well defined modulo $q-1$". Problem: point 7 says that no such map exists. That is true functorially, but for $K = \mathbb{Q}_q$ the prime q is a canonical uniformiser, $\mathbb{Q}_q^\times = q^{\mathbb{Z}} \times \mathbb{Z}_q^\times$ canonically, and so $H^1(\mathbb{Q}_q, \mathbb{Z}_\ell(1)) = \mathbb{Z}_\ell\kappa_q \oplus H^1_{\mathrm{ur}}(\mathbb{Q}_q, \mathbb{Z}_\ell(1))$ canonically, with $H^1_{\mathrm{ur}} = H^1(G/I, \mathbb{Z}_\ell(1)^I)$ (inflation is injective); the projection sends the Kummer class of q_E to the class of its unit part, which is the paper's "image" (5.3). The genuine imprecision is elsewhere: c' is the value at a chosen Frobenius lift of the cocycle $\varepsilon\kappa$ (point 42), equal to $\varepsilon(\mathrm{Frob})$ times a representative of that image only for a lift fixing a compatible system of ℓ-power roots of q, and congruent to it modulo the ideal $(n, q-1)\mathbb{Z}_\ell$ (not modulo $q-1$) for an arbitrary lift (5.4(iii), (iv)). Fix: "Write $H^1(G, \mathbb{Z}_\ell(1)) = \mathbb{Z}_\ell\kappa_q \oplus H^1(G/I, \mathbb{Z}_\ell(1))$ using the uniformiser q; the second component of κ is the class of the unit part u, i.e. of $\bar{u}$ in $\mathbb{F}_q^\times \otimes \mathbb{Z}_\ell = \mathbb{Z}_\ell(1)/(q-1)$, and $c' \equiv \varepsilon(\mathrm{Frob})\cdot$(that class) modulo $(n, q-1)$; so $v_\ell((n, q-1, c')) = v_\ell((n, q-1, c))$." Effect: none on any result; the formula of Lemma lem:tamfactor stands. Confidence: high (the splitting is elementary; the indeterminacy is the two-line computation of Exercise 6, the same as in Chapter 16's model-computation (d) and Chapter 7's 6.2(iv)). Action: if point 7 is acted on, use this wording, which keeps the paper's phrase and corrects the modulus.

N2 (observation, not an error). Lemma lem:dfg21: the hypothesis $H^0(\mathbb{Q}, V^D) = 0$ is implied by what the paragraph after the lemma proves, and the surjectivity onto $H^0(\mathbb{Q}, W^D)^*$ uses exactly that. Location: the statement of Lemma lem:dfg21 and the paragraph after it. In the derivation of the four-term sequence from global duality (10.3), the last map is onto $H^0(\mathbb{Q}, W^D)^*$ because $\delta_p\colon H^0(\mathbb{Q}_p, W^D) \to H^1(\mathbb{Q}_p, T^D)$ is injective for a prime $p \in \Sigma$ with $H^0(\mathbb{Q}_p, V^D) = 0$; for $p \in \Sigma$ the paragraph after the lemma proves $H^1_{\mathrm{ur}}(\mathbb{Q}_p, V^D) = 0$, which is equivalent to $H^0(\mathbb{Q}_p, V^D) = 0$ (4.2(iii)), and this implies $H^0(\mathbb{Q}, V^D) = 0$. In DFG04's adjoint setting this is automatic by weights; as the paper quotes the lemma for an abstract V^D it is worth one clause. Fix (optional): after "vanishes, because ... $\neq 0$", add "in particular $H^0(\mathbb{Q}_p, V^D) = 0$, and hence $H^0(\mathbb{Q}, V^D) = 0$". Confidence: high. Low importance.

No other new problem was found in this chapter's area. The following statements of the paper were rechecked here and are correct as stated (with the conventions of points 11 and 42): "$H^1(G, \mathbb{Z}_\ell(1)) = \varprojlim \mathbb{Q}_q^\times/(\mathbb{Q}_q^\times)^{\ell^k}$"; "$H^1(G/I, \mathbb{Z}_\ell(1)^I) = \mathbb{Z}_\ell(1)/(q-1) \simeq \mathbb{Z}/\ell^{v_\ell(q-1)}$"; "$H^1_f(\mathbb{Q}_p, V^D) = H^1_{\mathrm{ur}}(\mathbb{Q}_p, V^D) = (V^D)^{I_p}/(\mathrm{Frob}_p - 1)$ vanishes" (as a vanishing statement, in either Frobenius convention); "$H^1_f(\mathbb{Q}_p, T^D)$ is

the torsion of $H^1(\mathbb{Q}_p, T^D)$, a finite group" (it is moreover $\simeq H^0(\mathbb{Q}_p, W^D)$); "$H^1(\mathbb{R}, T^D)$ is killed by 2, so $H^1_f(\mathbb{R}, T^D) = 0$ for ℓ odd"; the four-term sequence of Lemma lem:dfg21 under its hypotheses (10.3); the length identities in the proof of Theorem thm:main and in Remark rem:ns (10.4). The data of Section sec:tam were spot-checked on 77c1, 791a1 and 1577a1 (5.5): the extra term and the bound behave as the paper says. The known issues in this area are points 5, 7, 8, 11, 31, 37, 42.

19.17.2 16.2 Book-internal inconsistencies

- **B-a. Who defines the Kummer (Cartier) dual.** The Chapter 3 outline, item 8.3(vi), says "Twists by $\mathbb{Z}_\ell(1)$, the Cartier dual $T^D = \mathrm{Hom}(T, \mathbb{Z}_\ell(1))$ and $W^D = V^D/T^D$ are Chapter 18". Tate local duality (this chapter, 3.3) needs them first. Resolution: this chapter defines $M^*(1)$, T^D, V^D, W^D in general (2.1); Chapter 18 adds the adjoint-specific facts ($T^D \simeq A(1)$ via the trace form, $(\mathrm{ad}^0 V)^\vee \simeq \mathrm{ad}^0 V$) and cites 2.1. The Chapter 3 writer should change "Chapter 18" to "Chapters 17 and 18".
- **B-b. `SerreGaloisCoh` must go into `refs.bib`, not `bib/ch16.bib`.** The Chapter 16 outline (writer note 12) and the Chapter 3 outline (item 6.6) plan the key for `bib/ch16.bib`; Chapter 17 cites it too, so by decision B14 it belongs in `refs.bib`. Likewise `MazurRubin04` if Chapter 18 cites it.
- **B-c. Chapter 21 outline, item A2.7.** Step (7) writes "$c' = \kappa_u(\mathrm{Frob})$", omitting the factor $\varepsilon(\mathrm{Frob})$ (important_notes.md B7 records the same omission in step (1) only); and its commentary "There is no such map" should be softened by N1: there is a canonical retraction for $\mathbb{Q}_q$, and the imprecision is the cocycle value and the modulus $(n, q-1)$. Chapter 21 should cite this chapter's 5.3 and 5.4.
- **B-d. Chapter 16 outline, Section 12 ("unramified-h1-preview") and this chapter's 4.2.** No conflict: Chapter 16 previews the formula $H^1_{\mathrm{ur}} = M^I/(\mathrm{Frob}-1)M^I$ and the vanishing criterion; 4.2 here is the full statement with the order formula (iv) and the convention remark. The Chapter 16 writer should keep the preview to the two displayed formulas and point here.
- **B-e. The deferrals to this chapter are all honoured:** Chapter 5 (Hilbert 90 profinite, Kummer without $\mu_n \subset K$, $H^1(K, \mathbb{Z}_\ell(1))$, $G_{\mathbb{R}}$): 1.3, 1.4, 2.4, 6.4; Chapter 7 (6.1(ii), the Euler-characteristic count of 2.7(iv), the cohomological half of point 7): 1.4, 2.4, 3.5, 5.1 to 5.4; Chapter 8 (Hilbert 90 for $\bar{k}(C)$): 1.3(iv); Chapter 9 (weak Mordell-Weil; Kummer sequence; $E[n] \simeq E[n]^*(1)$): 7.1 to 7.4, 2.1(iv); Chapter 10 (q_E, u, $c(u)$, c_q): 5.4, 5.5, 7.2(iv); Chapter 11 ($\varepsilon\kappa$): 5.4; Chapter 15 (local types for unramified cohomology): 4.7; Chapter 16 (all items of its writer note 13 assigned here): 1.3, 1.4, 3.3, 4.2, 4.4, 2.2, 3.1, 6.3, 5.4; Chapter 21 (A2.7 step (7), A4.9, A4.10 inputs, A5.2): 5.3, 5.4, 2.2, 4.2, 4.3, 4.6, 6.4, 10.3, 10.4.

Chapter 20

Chapter 18: Bloch-Kato Selmer groups, Tamagawa factors and the adjoint motive (ch:blochkato, chapters/ch18_blochkato.tex)

Stage 1 line-up. Part V (Cohomology, Selmer groups and L-functions). Target length 2,630 lines of LaTeX (the card excluded; see Part 3), inside the 2,800 lines that PLAN.md Section 7.3 allows for this chapter.

Rigour tags: **P** = proved here, **S** = proof sketched here, **R** = stated; reference. Item format, as in the Chapter 2 to 16 outlines: number, label-ready name, kind, tag; the precise statement with all hypotheses; reference (book and chapter or section; a theorem number only where sure, otherwise the section, and numbers marked "(verify)" must be checked before they are printed); the paper's use by its `\label` names, or "background". Every number quoted below (Tate parameters and unit parts, Tamagawa factors, modular degrees, the counts for Section sec:tam, the freeness of $\mathrm{ad}^0 T$ over $\mathbb{Z}_3[C_3]$, the brute-force check of the Tamagawa formula and of its twist) was checked with PARI/GP 2.17 through `cypari2` and Cremona's `ecdata` files for conductors below 10,000 in `/tmp/ch09/data` (verification log, Part 6.6).

20.1 1. Purpose and global conventions

20.1.1 1.1 What this chapter is for

The paper's Theorem thm:main is an identity between the modular degree and three kinds of terms: the length of a Bloch-Kato Selmer group, Tamagawa factors, and omitted Euler factors. This chapter owns the first two and the bookkeeping that connects them. Concretely it must deliver:

1. **The objects of Section 1.2 of the paper.** ℓ-adic representations, lattices T, divisible modules W, Tate twists, the three duals (linear, Kummer-Cartier $V^D = V^\vee(1)$, Pontryagin), the trace form and $T^D \simeq A(1)$ for $A = \mathrm{ad}^0 T$ and ℓ odd; the Bloch-Kato local conditions at $p \neq \ell$, at ℓ (crystalline, with p-adic Hodge theory as a black box) and at ∞; their propagation to T and W; the global Selmer groups H^1_f and the Σ-imprimitive H^1_Σ of DFG04 Section 2.1; the Tamagawa factor $\mathrm{Tam}_p(M) = \mathrm{Fitt}_{\mathbb{Z}_\ell}\big((H^1(I_p, M)_{\mathrm{tor}})^{G_{\mathbb{Q}_p}}\big)$ with the parenthesis in the right place (important_notes.md point 37).
2. **Lemma lem:fpr in full for** $p \neq \ell$ (Fontaine and Perrin-Riou): $\mathrm{Fitt}\, H^1_f(\mathbb{Q}_p, T^D) = L_p(A_f, 1)^{-1}\, \mathrm{Tam}_p(T^D)$, with the index description $\#\big(H^1_f(\mathbb{Q}_p, M)/H^1_{\mathrm{ur}}(\mathbb{Q}_p, M)\big) = \ell^{v_\ell(\mathrm{Tam}_p(M))}$, and $\mathrm{Tam}_p(T^D) = \mathrm{Tam}_p(A)$ proved in every case that occurs for an elliptic curve (DFG04 (57) in general).

3. **Lemma lem:tamfactor in full, every step**, assembled from Chapter 16 (the group cohomology), Chapter 7 and Chapter 17 (the Kummer class and the identification of c', important_notes.md point 7 and its addendum), Chapter 11 (the Tate shape with the corner $\varepsilon\kappa$, point 42), with the arithmetic Frobenius named (point 11), the reading of $v_\ell(c)$ fixed (point 31) and the class renamed $c(u)$ (point 23); its extension to additive potentially multiplicative primes, including $q = 2$ (point 18).
4. **New: the Tamagawa factor of $\mathrm{ad}^0 T$ at every additive prime.** It is trivial at every prime $q \neq \ell$ of potentially good reduction, for every odd ℓ (item 9.3: the image of inertia has order dividing 24, and for $\ell = 3$ the lattice $\mathrm{ad}^0 T$ is free over $\mathbb{Z}_3[C_3]$). This answers the first half of the paper's Open question 3 and makes (eq:identity) fully explicit (item 9.4).
5. **The Selmer side of the proof of Theorem thm:main**: the exceptional set S_f, DFG04 Theorem 2.7 (stated), DFG04 Lemma 2.1 (proved from Chapter 17's comparison of Selmer structures, with every term identified), the restatement of Lemma lem:h0 in its corrected form (proved in Chapter 11; points 5, 6, 10, 14) as the identification $H^1_f(\mathbb{Q}, T^D) \simeq H^0(\mathbb{Q}, W^D)$, and the length identity $\operatorname{length} H^1_\Sigma = \operatorname{length} H^1_f + \sum(\text{Euler} + \text{Tamagawa}) - h_\ell(E)$, which is the proof step of Theorem thm:main when $h_\ell(E) = 0$ and Remark rem:ns when $h_3(E) = 1$.
6. **Remark rem:ko(1), Section sec:tam, Open questions 2 and 3, the term $h_\ell(E)$ of Conjecture conj:euler**, with a reproduction of the Tamagawa test of Section sec:tam on conductors below 10,000.
7. **The Tamagawa number conjecture and the adjoint motive**: the shape of the conjecture, BSD as the model case, the adjoint motive of f, $L(\mathrm{ad}^0 f, s) = L(\mathrm{Sym}^2 f, s + 1)$ in the paper's normalisation, DFG04 Theorems 0.1 and 0.2 (stated), Flach's theorem; why (eq:identity) is the Tamagawa number conjecture for the adjoint read through the modular degree.

The physics reader appears twice, in one sentence each: a Tate twist is a change of "weight" exactly like multiplying a field by a character of the scaling group (2.2); the Tamagawa factor is the size of a local "defect" counted by the torsion of a first cohomology group, as an index of a sublattice is a volume ratio (7.2).

20.1.2 1.2 Split with the neighbouring chapters (decided here, consistent with PLAN.md Section 4 and the written outlines)

- **Chapter 3** proves Fitting ideals over $\mathbb{Z}_\ell$ ($\mathrm{Fitt}(A) = \ell^{\operatorname{length} A}\mathbb{Z}_\ell$ for finite A: its 7.4 to 7.6), lengths, Pontryagin duality and the duality between T and $T \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$ (8.1 to 8.4). Cited, not reproved.
- **Chapter 4** proves the trace form and $\mathrm{Hom}(A, \mathbb{Z}_\ell(\chi)) \simeq A(\chi)$ for ℓ odd (its 7.6), the invariance of ad^0 under twists, duals and homotheties (7.7), the cyclic and unipotent shapes (9.1 to 9.4), and the twist-length lemma behind h_ℓ (10.5).
- **Chapter 7** fixes the Frobenius conventions (Frob arithmetic, $\Phi = \mathrm{Frob}^{-1}$ geometric, $L(V, s) = \det(1 - \Phi q^{-s} \mid V^I)^{-1}$: its L3, L6, 8.2), proves the Kummer class on inertia and the field-theoretic half of the identification of c' (6.2), and tabulates the local shapes and adjoint L-factors (9.1 to 9.4). Cited.
- **Chapter 10** defines the Tate parameter $q_E = \mathfrak{q}(j_E)$, its unit part u and the class $c(u)$ (its 7.2, 7.4), the component groups and c_q (7.6), additive potentially multiplicative reduction (7.7) and Kraus's defect.
- **Chapter 11** proves the Tate shape with corner $\varepsilon\kappa$ (its 3.2), Lemma lem:tamagawa-exp (3.3), the potentially multiplicative case at every q including 2 (3.4), the inertia images by Kodaira type (Section 4), crystallinity at ℓ (5.1(ii), stated), and **Lemma lem:h0 in corrected form** (8.5) with the description of S_f (8.6). Chapter 18 restates lem:h0 in Selmer language and does not reprove it.
- **Chapter 14** proves important_notes.md point 39 (its 9.1 to 9.5: $v_\lambda(\eta^{\emptyset}_{f,\lambda}) = v_\ell(\deg \varphi_E)$ for the optimal curve) and lists the DFG04 checklist (9.6, point 56), now settled against the published DFG04 (Chapter 20 outline, Part 7; important_notes.md point 73). Chapter 18 cites it for "equality in (eq:identity)".
- **Chapter 16** proves the group cohomology of the Tate shape (its model-computation, frobenius-on-h1-inertia, procyclic-compact, inflation-restriction, lengths-in-exact-sequences). Chapter 18 cites those

items for Steps 2 and 3 of Lemma lem:tamfactor and adds the definition of Tam, the identification of c' and the conclusion.

- **Chapter 17** (its outline `outlines/ch17_galcoh.md` is written) owns Galois cohomology proper: the Kummer duals T^D, V^D, W^D in general (its 2.1), Kummer theory, Tate local duality for finite, compact and discrete coefficients, local Euler characteristics, H^1_{ur} and its annihilator, the cohomology of the real place, $G_{\mathbb{Q},S}$, classical Selmer groups, Selmer structures and their duals, Greenberg-Wiles and Poitou-Tate, **the comparison exact sequence for two Selmer structures** (the abstract form of DFG04 Lemma 2.1), the sequences relating $H^1(K,T)$, $H^1(K,V)$, $H^1(K,W)$, and the Kummer class of the Tate parameter with $H^1(G/I,\mathbb{Z}_\ell(1)) = \mathbb{Z}_\ell(1)/(q-1)$. Chapter 18 never reproves these; it instantiates them with the Bloch-Kato conditions. Chapter 18 needs from Chapter 17 exactly the items listed in Part 5 (inputs K1 to K8, with Chapter 17's item numbers): in particular the identification of c' (its 5.3, 5.4), the propagation and orthogonality of local conditions (4.4, 4.5), $H^1_f(\mathbb{Q}_p,T^D) = H^1(\mathbb{Q}_p,T^D)_{\mathrm{tor}} \simeq H^0(\mathbb{Q}_p,W^D)$ for $p \in \Sigma$ (4.6), and the abstract form of DFG04 Lemma 2.1 (10.3, 10.4). Chapter 18 adds the Bloch-Kato conditions, Tam_p and Lemma lem:fpr, the assembly of Lemma lem:tamfactor, Lemma lem:dfg21 in the paper's form (Chapter 17, 10.3 applied to the Bloch-Kato structure with DFG04 Theorem 2.7 as hypothesis), and $T^D \simeq A(1)$.
- **Chapter 19** (outlined in parallel) owns L-functions: the table of omitted Euler factors, Lemma lem:euler, $e_\ell(E,q)$, P_q, the naive factor, $L(\mathrm{Sym}^2 f, s)$ and the Rankin-Shimura formula. Chapter 18 uses only the values $v_\ell(L_q(A_f,1)^{-1})$ (stated with a `\Cref{ch:lfunctions}`) and their integrality (proved here, 3.5).
- **Chapter 20** owns congruence ideals and the DFG04 assembly: $\eta^{\Sigma}_{f,\lambda}$, Proposition 1.4 (thm:dfg14), Theorem 3.7 (thm:dfg37), the Taylor-Wiles method, Lemma lem:ddt417, Lemma lem:hecke, and the proof of Theorem thm:main as an assembly. Chapter 18 supplies the Selmer half (Section 11 here: item 11.1) and Chapter 20 cites it. Theorems 0.1 and 0.2 of DFG04 are stated here (14.4, 14.5) because they are statements about Selmer groups and Tamagawa factors; Theorem 3.7 is stated here only as a pointer (14.5) and lives in Chapter 20.
- **Chapters 21, 22** use the card: the tour cites 7.1, 7.4, 8.2, 9.3, 10.4, 11.1, 12.1; the computation chapter uses 8.5 and 13.1.

20.1.3 1.3 Global conventions fixed in this chapter

- **C1 (objects).** As Chapter 11, C1: $E/\mathbb{Q}$ with global minimal model and conductor N; ℓ an odd prime unless said otherwise; $T = T_\ell E$, $V = T \otimes \mathbb{Q}_\ell$, $\rho = \rho_{E,\ell}$, $\bar\rho$; $A = \mathrm{ad}^0 T$ (trace zero, the paper's `\ad`), $A_f = \mathrm{ad}^0 V = A \otimes \mathbb{Q}_\ell$, $W = A \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$. Σ is the set of primes dividing N (so $\ell \notin \Sigma$ when $\ell \nmid N$). $\delta_p = \dim V^{I_p}$.
- **C2 (duals; important_notes.md B4).** For a $\mathbb{Z}_\ell$-module M: $M^\vee = \mathrm{Hom}_{\mathbb{Z}_\ell}(M,\mathbb{Z}_\ell)$ (linear dual; $V^\vee = \mathrm{Hom}_{\mathbb{Q}_\ell}(V,\mathbb{Q}_\ell)$; Bloch-Kato and PLAN.md write V^*); $M(n) = M \otimes \mathbb{Z}_\ell(1)^{\otimes n}$; the **Kummer (Cartier) dual** $M^D = \mathrm{Hom}_{\mathbb{Z}_\ell}(M,\mathbb{Z}_\ell(1)) = M^\vee(1)$, and for the paper's objects $T^D = \mathrm{Hom}(A,\mathbb{Z}_\ell(1))$, $V^D = T^D \otimes \mathbb{Q}_\ell = A_f^\vee(1)$, $W^D = V^D/T^D = \mathrm{Hom}(A,\mathbb{Q}_\ell/\mathbb{Z}_\ell(1))$; the **Pontryagin dual** of a compact or discrete ℓ-primary module is $X^* = \mathrm{Hom}_{\mathrm{cont}}(X,\mathbb{Q}_\ell/\mathbb{Z}_\ell)$, the paper's * (as in Chapter 16). The symbol * never means a linear dual in this chapter.
- **C3 (Frobenius; points 11, B1, B6, B7).** As Chapter 7, L3 and L6: Frob_p is an arithmetic Frobenius, $\Phi_p = \mathrm{Frob}_p^{-1}$ the geometric one. Group-theoretic statements (conjugation on $H^1(I_p,-)$, the Tate matrices, Lemma lem:tamfactor) use Frob_p; L-factors use Φ_p: $L_p(M,s) = \det(1 - \Phi_p p^{-s} \mid M^{I_p})^{-1}$. Consequences: $L_p(M(1),s) = L_p(M,s+1)$, so $L_p(A_f(1),0) = L_p(A_f,1)$ (Lemma lem:fpr); $L_p(V_\ell E,s) = L_p(E,s+1)$ while $L_p(E,s) = L_p(V_\ell E^\vee,s)$ (Chapter 7, 8.2(v); B7, B9). On $V_\ell E$ at a good prime the arithmetic Frobenius has eigenvalues α,β (roots of $X^2 - a_p X + p$), the geometric one α^{-1},β^{-1}; on A_f both give $\alpha/\beta, 1, \beta/\alpha$ (B1). In cohomology the choice is invisible: $(\mathrm{Frob} - 1)M = (\Phi - 1)M$, and $\det(1 - \mathrm{Frob} \mid X)$, $\det(1 - \Phi \mid X)$ have the same ℓ-adic valuation for a lattice X (they differ by the unit $\det(-\Phi \mid X)$); but as rational numbers only $\det(1 - \Phi \mid (V^D)^{I_p})$ equals $L_p(A_f,1)^{-1}$.
- **C4 (Hodge-Tate weights).** $\mathbb{Q}_\ell(1)$ has Hodge-Tate weight 1, so $V_\ell E$ has weights $\{0,1\}$ (as Chapter 15, 4.1), A_f has $\{-1,0,1\}$, $V^D = A_f(1)$ has $\{0,1,2\}$. Some sources (including, possibly, DFG04

and BlochKato) use the opposite sign; every formula below that depends on the sign (4.5) says so.

- **C5 (cohomology).** Continuous cochain cohomology throughout (Chapter 16, continuous-cohomology), with coefficients of type (D), (C), (V) as there; $H^i(K, M) = H^i(G_K, M)$; $H^i(\mathbb{R}, M) = H^i(\mathrm{Gal}(\mathbb{C}/\mathbb{R}), M)$; $G_{\mathbb{Q},S}$ for a finite set S of places containing ℓ, ∞ and the primes where M ramifies. "Length" is $\mathbb{Z}_\ell$-length; v_ℓ of a nonzero ideal $\ell^a\mathbb{Z}_\ell$ is a.
- **C6 (Tate data at a prime with $v_q(j) < 0$; points 8, 23, 31, 42).** $q_E = \mathfrak{q}(j_E) \in \mathbb{Q}_q^\times$ the Tate parameter (Chapter 10, 7.4), never called u; $\nu = v_q(q_E) = -v_q(j_E)$ ($= n = n_q$ at a multiplicative prime); $u = q_E q^{-\nu} \in \mathbb{Z}_q^\times$ its unit part and $\bar{u} \in \mathbb{F}_q^\times$ its reduction; $c(u)$ the class of $\bar{u}$ in $\mathbb{F}_q^\times/(\mathbb{F}_q^\times)^{\ell^{v_\ell(q-1)}}$, and $v_\ell(c(u))$ is the largest $k \le v_\ell(q-1)$ with $\bar{u} \in (\mathbb{F}_q^\times)^{\ell^k}$ (so "$v_\ell(0)$" reads $v_\ell(q-1)$). The paper writes c in Lemma lem:tamfactor and c_q in Section sec:tam; the book writes $c(u)$ (or $c_q(u)$) and keeps c_q for the Tamagawa number. ε is the unramified character with $\varepsilon(\mathrm{Frob}_q) = a_q(E) = \pm 1$ (multiplicative case) and the corner of the Tate matrices is $\varepsilon\kappa$.
- **C7 (characters).** χ_ℓ the cyclotomic character, $\bar{\chi}_\ell$ its reduction, χ_{-3}, χ_{-4}, χ_{q^*} quadratic (Chapter 11, C4; point 32). $\bar{\chi}_3 = \chi_{-3}$.
- **C8 (typography and labels).** PLAN.md 7.2: no dash ligatures ("Bloch-Kato", "Fontaine-Perrin-Riou", "Diamond-Flach-Guo", "Poitou-Tate", "Greenberg-Wiles", "Fontaine-Laffaille", "Taylor-Wiles" with plain hyphens; ranges "1 to 8"); ASCII source; labels `...:blochkato:...`; the names in the line-up are the label stems.

20.2 2. Line-up of the core results, in teaching order

20.2.1 Section 1. Overview and conventions

1.1 `conv:blochkato:setup` (conventions C1 to C8 as `convention` environments, with a roadmap; P). A table "sentence of the paper → item" for Section 1.2, the proof of Theorem thm:main, Lemma lem:tamfactor, Lemma lem:fpr, Remark rem:ko(1), Remark rem:ns, Section sec:tam, Open questions 2 and 3. Paper: background.

20.2.2 Section 2. ℓ-adic representations, lattices, twists and duals

2.1 `def:blochkato:l-adic-rep` (definition and proposition; P). Let K be a field and G_K its absolute Galois group. An ℓ-adic representation is a finite-dimensional $\mathbb{Q}_\ell$-vector space V with a continuous homomorphism $G_K \to \mathrm{GL}(V)$. (i) V has a G_K-stable $\mathbb{Z}_\ell$-lattice T (the image of G_K is compact, so the sum of the translates of any lattice is a lattice). (ii) Put $W = V/T = T \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$, a discrete ℓ-primary G_K-module; $0 \to T \to V \to W \to 0$ is exact with a continuous section (Chapter 16, les-continuous (iii)), and $W[\ell^k] \simeq T/\ell^k T$ (Chapter 3, 8.3). (iii) Any two stable lattices are commensurable; if $T/\ell T$ is absolutely irreducible all stable lattices are homothetic (Chapter 4, 10.3). Reference: SerreAbelian, Chapter I; Rubin, Section 1.1. Paper: Section 1.1 (V, T), Section 4.2 (A, W).

2.2 `def:blochkato:twists` (definition and proposition; P). $\mathbb{Z}_\ell(1) = \varprojlim \mu_{\ell^k}$ with G_K acting by χ_ℓ (Chapter 5), $\mathbb{Q}_\ell(1) = \mathbb{Z}_\ell(1) \otimes \mathbb{Q}_\ell$, $\mathbb{Z}_\ell(n) = \mathbb{Z}_\ell(1)^{\otimes n}$ ($n \in \mathbb{Z}$, negative powers via the dual), $M(n) = M \otimes \mathbb{Z}_\ell(n)$. (i) $M(n)(m) = M(n+m)$, $\mathrm{Hom}(M(n), N) = \mathrm{Hom}(M, N)(-n)$. (ii) For $p \neq \ell$, I_p acts trivially on $\mathbb{Z}_\ell(1)$ and Frob_p acts by p; so $M(n)^{I_p} = M^{I_p}(n)$ and the Frobenius eigenvalues on $M(n)^{I_p}$ are those on M^{I_p} times p^n (arithmetic) or p^{-n} (geometric). (iii) $\wedge^2 T_\ell E \simeq \mathbb{Z}_\ell(1)$ by the Weil pairing, hence $T_\ell E^\vee \simeq T_\ell E(-1)$ (Chapter 9). Physics aside (one sentence): a twist rescales Frobenius eigenvalues by a power of p as a character of the scaling group rescales a field of given weight. Reference: Rubin 1.1; Washington. Paper: Section 4.2 ($\mathbb{Z}_\ell(1)$, $A(1)$), Lemma lem:h0 ($\mathrm{ad}^0 T(1)$), Lemma lem:fpr ($A_f(1)$).

2.3 `def:blochkato:duals` (definition and proposition; recalled from Chapter 17, 2.1, which proves (i), (ii); P for (iii), (iv)). Conventions C2. (i) $V^D = V^\vee(1)$, $T^D = T^\vee(1)$, $W^D = V^D/T^D$; then $W^D \simeq \mathrm{Hom}_{\mathbb{Z}_\ell}(T, \mathbb{Q}_\ell/\mathbb{Z}_\ell(1)) = \mathrm{Hom}(T, \mu_{\ell^\infty})$ and $T^D \simeq \mathrm{Hom}(W, \mu_{\ell^\infty})$, so the evaluation pairings $T \times W^D \to \mathbb{Q}_\ell/\mathbb{Z}_\ell(1)$ and $W \times T^D \to \mathbb{Q}_\ell/\mathbb{Z}_\ell(1)$ are perfect and G_K-equivariant (Chapter 3, 8.2, 8.3). (ii)

$(T^D)^D = T$. (iii) X^* (Pontryagin) of a finite module has the same length; the only Pontryagin duals in the paper are of finite groups (Lemma lem:dfg21). (iv) Dictionary with the literature: Bloch-Kato and FPR94 write $V^*(1)$ for V^D; DFG04 write T^D, V^D, W^D exactly as the paper. Reference: BlochKato Section 3; Rubin 1.1; DFG04 Section 2.1. Paper: Section 4.2 (T^D, V^D, W^D), Lemma lem:dfg21 (*).

2.4 `prop:blochkato:adjoint-self-dual` (proposition; P given Chapter 4, 7.6 and 7.7). Let ℓ be odd and $A = \mathrm{ad}^0 T$. (i) The trace form $(x, y) \mapsto \mathrm{tr}(xy)$ is a perfect G_K-invariant pairing on A (Gram determinant -2 in the basis e, h, f), so $A \simeq A^\vee$ and

$$T^D = \mathrm{Hom}(A, \mathbb{Z}_\ell(1)) \simeq A(1), \qquad V^D \simeq A_f(1), \qquad W^D \simeq \mathrm{ad}^0 T(1) \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell.$$

For $\ell = 2$ the image of A in $A^\vee$ has index 2. (ii) ad^0 is insensitive to twisting T by a character, to replacing T by a homothetic lattice or by $T^\vee$: hence $\mathrm{ad}^0 M_{f,\lambda} = \mathrm{ad}^0 T_\ell E$ for the lattice $M_{f,\lambda} = \varphi_E^* H^1(E, \mathbb{Z}_\ell) \simeq T_\ell E(-1)$ of the optimal curve (Chapter 14, 9.1(d); important_notes.md points 26, 40), and $\mathrm{ad}^0 T$ is the same $G_{\mathbb{Q}_q}$-module for E and for any quadratic twist of E. Paper: Section 4.2 ("$T^D = \mathrm{Hom}(A, \mathbb{Z}_\ell(1)) \simeq A(1)$ (via the trace form)", "$M_{f,\lambda}$ and $T_\ell E$ are homothetic ... so their adjoints coincide"), Lemma lem:h0, proof of Theorem thm:main (I_n^* primes: "$\mathrm{ad}^0 T$ is unchanged by the twist").

2.5 `def:blochkato:geometric` (definition and theorem; R, with (iii) P given Chapter 9). (i) An ℓ-adic representation V of $G_\mathbb{Q}$ is *geometric* (Fontaine-Mazur) if it is unramified outside a finite set of primes and de Rham at ℓ (4.1). (ii) $V_\ell E$, $\mathbb{Q}_\ell(n)$, and every tensor construction on them (A_f, V^D, $\mathrm{Sym}^2 V$) are geometric (4.2, 4.3; Grothendieck for $p \neq \ell$). (iii) (purity at good primes) For $p \nmid N\ell$ the arithmetic Frobenius eigenvalues on $V_\ell E$ have absolute value $p^{1/2}$ (Hasse; Chapter 9), so those on A_f have absolute value 1 and those on $V^D = A_f(1)$ absolute value p. Reference: Fontaine-Mazur (cite through Berger, Section 1; verify); Chapter 9, Hasse bound. Paper: background (the shape of the Bloch-Kato conjecture; 3.3).

2.6 `ex:blochkato:adjoint-of-E` (worked example; P). For E and odd ℓ with $\bar\rho$ absolutely irreducible: (i) $H^0(\mathbb{Q}, A_f) = 0$ and $H^0(\mathbb{Q}, W) = 0$ ($H^0(\mathbb{Q}, W)[\ell] = H^0(\mathbb{Q}, \mathrm{ad}^0 \bar\rho)$ consists of trace-zero scalars by Schur, which are 0 as 2 is invertible; Chapter 16, h0-and-torsion). (ii) $H^0(\mathbb{Q}, V^D) = 0$ for every E (by 2.5(iii) no Frobenius eigenvalue on V^D at a good prime equals 1). (iii) $H^0(\mathbb{Q}, W^D)$ is finite, cyclic, of length $h_\ell(E)$ (Chapter 11, 8.5; restated in 10.3). (iv) The three modules A, $A(1) = T^D$ and $A(-1)$ are pairwise non-isomorphic, and only $A(1)$ carries the Tamagawa and Euler data that enter Lemma lem:fpr. Paper: Section 4.2, Lemma lem:h0, Theorem thm:dfg27 ("$H^0(\mathbb{Q}, V^D) = 0$").

20.2.3 Section 3. Local behaviour at $p \neq \ell$: invariants, Euler factors, weights

3.1 `conv:blochkato:frobenius-dictionary` (convention with proofs; P). Convention C3 with its four consequences proved: (a) $L_p(M(1), s) = L_p(M, s+1)$ with Φ_p, and $L_p(M(1), s) = L_p(M, s-1)$ would follow with Frob_p (so the paper's "$L_p(A_f, 1)^{-1} = L_p(A_f(1), 0)^{-1}$" forces Φ_p; point 11); (b) $(\mathrm{Frob} - 1)X = (\Phi - 1)X$ for any Frob-stable X; (c) for a lattice X with $\det(1 - \Phi \mid X) \neq 0$, $\#X/(\mathrm{Frob} - 1)X = \ell^{v_\ell(\det(1-\Phi|X))}$ (the Smith form of $\mathrm{Frob} - 1$, and $\det \mathrm{Frob} \in \mathbb{Z}_\ell^\times$); (d) on $V_\ell E$ the geometric Frobenius gives $L_p(E, s+1)$ and the dual gives $L_p(E, s)$ (Chapter 7, 8.2(v); important_notes.md B7, B9). Reference: Chapter 7, L3, L6, 8.2; DFG04 Section 1.7.2 (checked: DFG04 use the geometric Frobenius; Chapter 20 outline, Part 7; important_notes.md point 73). Paper: Section 1.1 ($L_p(A_f, s)$), the paragraph after Lemma lem:dfg21, Lemma lem:fpr, Lemma lem:tamfactor (arithmetic Frob).

3.2 `prop:blochkato:local-adjoint` (proposition and table; P given Chapter 7, 9.4 and Chapter 11, 3.2, 3.4, Section 4). Let $p \neq \ell$. The inertia invariants and the Frobenius action on A_f and on $V^D = A_f(1)$, by reduction type at p (eigenvalues of Φ_p; those of Frob_p are the inverses):

reduction at p	δ_p	$\dim A_f^{I_p}$	Φ_p on $A_f^{I_p}$	Φ_p on $(V^D)^{I_p}$	$L_p(A_f, 1)^{-1} = \det(1 - \Phi_p \mid (V^D)^{I_p})$
good	2	3	$\alpha/\beta, 1, \beta/\alpha$	$p^{-1}\alpha/\beta, p^{-1}, p^{-1}\beta/\alpha$	$(1 - \frac{\alpha}{\beta p})(1 - \frac{1}{p})(1 - \frac{\beta}{\alpha p})$

reduction at p	δ_p	$\dim A_f^{I_p}$	Φ_p on $A_f^{I_p}$	Φ_p on $(V^D)^{I_p}$	$L_p(A_f,1)^{-1} = \det(1-\Phi_p \mid (V^D)^{I_p})$
multiplicative I_n	1	1	p^{-1}	p^{-2}	$1-p^{-2}$
I_n^* (potentially multiplicative)	0	1	p^{-1}	p^{-2}	$1-p^{-2}$
I_0^* (twist of good E')	0	3	as for E'	as for E'	$\frac{p-1}{p}\cdot\frac{(p+1)^2-a_p(E')^2}{p^2}$
inertia cyclic of order 3 or 6	0	1	$\chi_{-3}(p)$	$\chi_{-3}(p)p^{-1}$	$1-\chi_{-3}(p)p^{-1}$
inertia cyclic of order 4	0	1	$\chi_{-4}(p)$	$\chi_{-4}(p)p^{-1}$	$1-\chi_{-4}(p)p^{-1}$
inertia non-abelian ($p=2$: Q_8, $\mathrm{SL}_2(\mathbb{F}_3)$)	0	0	none	none	1

For $p \geq 5$ the last three rows are the Kodaira types $\mathrm{II}, \mathrm{II}^*, \mathrm{IV}, \mathrm{IV}^*$; $\mathrm{III}, \mathrm{III}^*$; (none); at $p=2$ type $\mathrm{IV}, \mathrm{IV}^*$ is the order-3 row (point 43); at $p=2,3$ the type does not determine the row (Remark rem:wild; Chapter 19 computes P_p with PARI). The table is a citation: the values of $e_\ell(E,p)$ and Lemma lem:euler are Chapter 19's. Paper: Section 1.1, the table of Section 1, Lemma lem:fpr ("valuation $e_\ell(E,p)$ at an additive prime and $v_\ell(1-p^{-2})$ at a multiplicative one"), the paragraph after Lemma lem:dfg21.

3.3 `lem:blochkato:no-local-invariants` (lemma; P). Let $p \neq \ell$. (i) $H^0(\mathbb{Q}_p, V^D) = 0$, and $\det(1-\Phi_p \mid (V^D)^{I_p}) = L_p(A_f,1)^{-1}$ is a nonzero rational number. (ii) $H^0(\mathbb{Q}_p, A_f)$ is the Φ_p-fixed part of $A_f^{I_p}$: it is 0 at multiplicative primes, at I_n^* primes, at the cyclic primes of 3.2 with $\chi_{-3}(p) = -1$ (resp. $\chi_{-4}(p) = -1$) and at the non-abelian primes; it is a line at good primes ($\alpha \neq \beta$ because $a_p^2 \neq 4p$) and at the cyclic primes with $\chi(p) = +1$; at an I_0^* prime it contains the line of the eigenvalue 1. Proof: read off 3.2; for (i) every eigenvalue of Φ_p on $(V^D)^{I_p}$ has complex absolute value p^{-1} (2.5(iii) and the table), so none is 1. **Warning for the writer:** $H^0(\mathbb{Q}_p, A_f) \neq 0$ does occur at bad primes (for example type III at $p \equiv 1 \pmod 4$, and every I_0^* prime), so $H^1(\mathbb{Q}_p, T^D)$ need not be finite there (3.4); only (i) enters the paper's argument. Paper: the paragraph after Lemma lem:dfg21 ("$\det(1-\mathrm{Frob}_p \mid (V^D)^{I_p}) = L_p(A_f,1)^{-1} \neq 0$", true with Frob_p read as geometric), Lemma lem:fpr ("a nonzero rational number").

3.4 `cor:blochkato:local-finiteness` (corollary; P given Chapter 17's local Euler characteristic and Tate local duality, inputs K1, K2). Let $p \neq \ell$. (i) $H^1(\mathbb{Q}_p, T^D)$ is a finitely generated $\mathbb{Z}_\ell$-module of rank $\dim H^0(\mathbb{Q}_p, V^D) + \dim H^0(\mathbb{Q}_p, A_f) = \dim H^0(\mathbb{Q}_p, A_f)$ (by 3.3(i)). (ii) In particular, at every $p \in \Sigma$ where $H^0(\mathbb{Q}_p, A_f) = 0$ (multiplicative and I_n^* primes, cyclic primes with $\chi(p) = -1$, non-abelian primes), $H^1(\mathbb{Q}_p, T^D)$ is **finite**, equal to its torsion, and $H^1(\mathbb{Q}_p, W) \simeq H^1(\mathbb{Q}_p, T^D)^*$ is finite. (iii) In all cases $H^1(\mathbb{Q}_p, T^D)_{\mathrm{tor}}$ is finite (Chapter 16, h1-finitely-generated). Paper: the paragraph after Lemma lem:dfg21 ("$H^1_f(\mathbb{Q}_p, T^D)$ is the torsion of $H^1(\mathbb{Q}_p, T^D)$, a finite group").

3.5 `lem:blochkato:euler-integral` (lemma; P). For $p \neq \ell$ let $X = T^D$. Then X^{I_p} is a Φ_p-stable lattice in $(V^D)^{I_p}$, so $L_p(A_f,1)^{-1} = \det(1-\Phi_p \mid X^{I_p}) \in \mathbb{Z}_{(\ell)} \cap \mathbb{Q}^\times$ and $\#X^{I_p}/(\mathrm{Frob}_p - 1)X^{I_p} = \ell^{v_\ell(L_p(A_f,1)^{-1})}$ (3.1(c), 3.3(iii)). Its valuation is $v_\ell(1-p^{-2}) = v_\ell(p^2-1)$ at multiplicative and I_n^* primes, and $e_\ell(E,p)$ at additive primes (Definition def:terms: the naive factor is 1 there; values in Chapter 19). Paper: Lemma lem:fpr ("a nonzero rational number which is an ℓ-adic integer"), Theorem thm:dfg14 (the factor $1-p^{-2}$).

20.2.4 Section 4. The representation at ℓ: p-adic Hodge theory as a black box

The section opens by saying that nothing in the paper's proofs uses p-adic Hodge theory directly: it enters only through the definition of $H^1_f(\mathbb{Q}_\ell, -)$ and through DFG04. Everything here is R, with precise

statements.

4.1 `def:blochkato:period-rings` (definition; R). Fontaine's rings $B_{\mathrm{dR}} \supset B_{\mathrm{st}} \supset B_{\mathrm{cris}}$ for $\mathbb{Q}_\ell$, with $G_{\mathbb{Q}_\ell}$-actions, the filtration on B_{dR} ($\mathrm{gr}^i B_{\mathrm{dR}} \simeq \mathbb{C}_\ell(i)$), the Frobenius φ on B_{cris} and B_{st}, the monodromy N on B_{st}, $B_{\mathrm{dR}}^{G_{\mathbb{Q}_\ell}} = B_{\mathrm{cris}}^{G_{\mathbb{Q}_\ell}} = \mathbb{Q}_\ell$. For V an ℓ-adic representation of $G_{\mathbb{Q}_\ell}$: $D_\bullet(V) = (B_\bullet \otimes V)^{G_{\mathbb{Q}_\ell}}$, $\dim D_\bullet(V) \leq \dim V$, and V is de Rham, semistable, crystalline if equality holds for $\bullet = \mathrm{dR}, \mathrm{st}, \mathrm{cris}$; crystalline $\Rightarrow$ semistable $\Rightarrow$ de Rham. Hodge-Tate weights with convention C4. Reference: Fontaine (Asterisque 223); Berger (Sections 2 to 4). Paper: background (the definition of H^1_f at ℓ, Section 1.2).

4.2 `thm:blochkato:crystalline-good-reduction` (theorem; R). Let $A/\mathbb{Q}_\ell$ be an abelian variety. $V_\ell A$ is crystalline iff A has good reduction (Fontaine; Coleman-Iovita for the converse), semistable iff A has semistable reduction; for E with good reduction at ℓ, $D_{\mathrm{cris}}(V_\ell E)$ is two-dimensional with φ of characteristic polynomial $X^2 - a_\ell X + \ell$ (in the normalisation where $V_\ell E$ has Hodge-Tate weights $0, 1$; verify the inverse-polynomial convention in Berger) (Chapter 11, 5.1(ii); Chapter 15, 4.1(ii)). Reference: Berger, Section 4 (verify); Fontaine. Paper: Theorem thm:main ($\ell \nmid N$: crystalline at ℓ, through DFG04).

4.3 `prop:blochkato:crystalline-closure` (proposition; R for the formal properties, P for the consequences). Crystalline representations are stable under $\oplus$, $\otimes$, duals, subquotients and Tate twists, and D_{cris} is compatible with these. Hence for $\ell \nmid N$: A_f, $A_f(1) = V^D$ and $V_\ell E^\vee$ are crystalline, with Hodge-Tate weights $\{-1, 0, 1\}$, $\{0, 1, 2\}$, $\{-1, 0\}$. Paper: Theorem thm:main (the Bloch-Kato condition at ℓ for W and W^D).

4.4 `thm:blochkato:fontaine-laffaille` (theorem; R). For ℓ odd, the functor of Fontaine and Laffaille is an equivalence between filtered Dieudonne modules over $\mathbb{Z}_\ell$ with weights in an interval of length at most $\ell - 2$ and $G_{\mathbb{Q}_\ell}$-stable lattices in crystalline representations with Hodge-Tate weights in that interval; it is exact and compatible with subquotients. For $V_\ell E$ ($\ell \nmid N$, weights $\{0, 1\}$) this applies for $\ell \geq 3$, and it identifies $T_\ell E$ with the lattice of the finite flat group schemes $\mathcal{E}[\ell^k]$ (Chapter 11, 5.1(i)). DFG04 (Sections 1 and 2) use this theory to define integral structures and the local condition at ℓ for the adjoint. **Settled (Part 7, Q1; Chapter 20 outline, Part 7; important_notes.md point 73):** although the adjoint has weights spanning an interval of length 2, which exceeds $\ell - 2$ when $\ell = 3$, DFG04 (31) excludes only the primes dividing $Nk!$ ($= 2N$ in weight 2) and the residually induced case, and the local condition at ℓ is defined through extensions of short objects (DFG04 Proposition 2.2, Corollary 2.3, needing only $\ell > 2$); so DFG04 Theorem 3.7 covers $\ell = 3$ in weight 2 and the paper's reading of S_f is the complete list of excluded primes. Reference: FontaineLaffaille; DFG04 Sections 1.1 to 1.3 and 2.1 (verify the section numbers); BlochKato Section 4. Paper: background (DFG04's hypotheses), the optimal-level paragraph ("good" at ℓ).

4.5 `prop:blochkato:dimension-formula` (proposition; R). For V de Rham, with convention C4:

$$\dim_{\mathbb{Q}_\ell} H^1_f(\mathbb{Q}_\ell, V) = \dim H^0(\mathbb{Q}_\ell, V) + \dim D_{\mathrm{dR}}(V)/\mathrm{Fil}^0 D_{\mathrm{dR}}(V),$$

and $\dim D_{\mathrm{dR}}/\mathrm{Fil}^0$ is the number of positive Hodge-Tate weights (counted with multiplicity). For $\ell \nmid N$: $\dim H^1_f(\mathbb{Q}_\ell, A_f) = 1$, $\dim H^1_f(\mathbb{Q}_\ell, V^D) = 2$, and $1 + 2 = 3 = \dim H^1(\mathbb{Q}_\ell, A_f)$, as the duality 5.4 requires. Examples: $H^1_f(\mathbb{Q}_\ell, \mathbb{Q}_\ell(1)) = \mathbb{Z}_\ell^\times \hat{\otimes} \mathbb{Q}_\ell$ (dimension 1), $H^1_f(\mathbb{Q}_\ell, \mathbb{Q}_\ell) = H^1_{\mathrm{ur}}(\mathbb{Q}_\ell, \mathbb{Q}_\ell)$ (dimension 1). Reference: BlochKato Corollary 3.8.4 (verify the number); Berger; Rubin 1.3. Paper: background (6.7, the Euler-characteristic check).

20.2.5 Section 5. Bloch-Kato local conditions

5.1 `def:blochkato:local-conditions` (definition; P for the equalities stated). Let V be an ℓ-adic representation of $G_{\mathbb{Q}_p}$. (i) For $p \neq \ell$: $H^1_f(\mathbb{Q}_p, V) = H^1_{\mathrm{ur}}(\mathbb{Q}_p, V) = \ker\big(H^1(\mathbb{Q}_p, V) \to H^1(I_p, V)\big) = H^1(G_{\mathbb{Q}_p}/I_p, V^{I_p})$ (inflation-restriction, Chapter 16). (ii) For $p = \ell$: $H^1_e = \ker(H^1(\mathbb{Q}_\ell, V) \to H^1(\mathbb{Q}_\ell, B_{\mathrm{cris}}^{\varphi=1} \otimes V))$, $H^1_f = \ker(H^1(\mathbb{Q}_\ell, V) \to H^1(\mathbb{Q}_\ell, B_{\mathrm{cris}} \otimes V))$, $H^1_g = \ker(H^1(\mathbb{Q}_\ell, V) \to H^1(\mathbb{Q}_\ell, B_{\mathrm{dR}} \otimes V))$, with $H^1_e \subseteq H^1_f \subseteq H^1_g$; $H^1_f(\mathbb{Q}_\ell, V)$ classifies the crystalline extensions of $\mathbb{Q}_\ell$ by V. (iii) At ∞: $H^1_f(\mathbb{R}, V) = 0$ ($= H^1(\mathbb{R}, V)$, which is 0 anyway, Chapter 16, z2-on-l-adic). Only H^1_f is used in the paper; H^1_e, H^1_g are defined for orientation. Reference: BlochKato Section 3 (Definition 3.7, verify); FPR94 I.3 (verify); Rubin 1.3. Paper: Section 1.2 ("the Bloch-Kato Selmer group"), Section 4.2 ($H^1_f(\mathbb{Q}_p, V^D)$).

5.2 `def:blochkato:propagation` (definition; P). For a stable lattice $T \subset V$, $W = V/T$, $\iota\colon H^1(\mathbb{Q}_p,T) \to H^1(\mathbb{Q}_p,V)$, $\pi\colon H^1(\mathbb{Q}_p,V) \to H^1(\mathbb{Q}_p,W)$: $H^1_f(\mathbb{Q}_p,T) = \iota^{-1}H^1_f(\mathbb{Q}_p,V)$ and $H^1_f(\mathbb{Q}_p,W) = \pi H^1_f(\mathbb{Q}_p,V)$; likewise $H^1_f(\mathbb{Q}_p,T^D)$, $H^1_f(\mathbb{Q}_p,W^D)$ from V^D. Properties: (a) $H^1_f(\mathbb{Q}_p,T) \supseteq H^1(\mathbb{Q}_p,T)_{\mathrm{tor}} = \ker\iota$; (b) $H^1_f(\mathbb{Q}_p,W) \simeq (\mathbb{Q}_\ell/\mathbb{Z}_\ell)^{\dim H^1_f(\mathbb{Q}_p,V)}$ is divisible (the image of $H^1_f(\mathbb{Q}_p,T)$ in $H^1_f(\mathbb{Q}_p,V)$ is a lattice), so it is 0 exactly when $H^1_f(\mathbb{Q}_p,V)=0$; (c) $H^1_f(\mathbb{Q}_p,T)/H^1(\mathbb{Q}_p,T)_{\mathrm{tor}}$ is a lattice in $H^1_f(\mathbb{Q}_p,V)$. Reference: BlochKato Section 3; Rubin 1.3 (verify). Paper: Lemma lem:dfg21 ("$H^1_f(\mathbb{Q}_p,T^D) = \iota^{-1}H^1_f(\mathbb{Q}_p,V^D)$").

5.3 `prop:blochkato:local-unramified` (proposition; P given Chapter 16, procyclic-compact and inflation-restriction, and Chapter 17, H^1_{ur}, input K3). Let $p \neq \ell$ and T a $G_{\mathbb{Q}_p}$-stable lattice. (i) $H^1_f(\mathbb{Q}_p,V) = V^{I_p}/(\mathrm{Frob}_p - 1)V^{I_p}$, of dimension $\dim H^0(\mathbb{Q}_p,V)$. (ii) There is an exact sequence $0 \to H^1_{\mathrm{ur}}(\mathbb{Q}_p,T) \to H^1(\mathbb{Q}_p,T) \to H^1(I_p,T)^{G_{\mathbb{Q}_p}} \to 0$ (the H^2 of $\widehat{\mathbb{Z}}$ vanishes on type (C) modules), with $H^1_{\mathrm{ur}}(\mathbb{Q}_p,T) = T^{I_p}/(\mathrm{Frob}_p - 1)T^{I_p}$. (iii) $H^1_f(\mathbb{Q}_p,T)$ is the preimage of $H^1(I_p,T)_{\mathrm{tor}}$, so

$$H^1_f(\mathbb{Q}_p,T)/H^1_{\mathrm{ur}}(\mathbb{Q}_p,T) \;\simeq\; \left(H^1(I_p,T)_{\mathrm{tor}}\right)^{G_{\mathbb{Q}_p}}.$$

(iv) $H^1_f(\mathbb{Q}_p,W)$ is the maximal divisible subgroup of $H^1_{\mathrm{ur}}(\mathbb{Q}_p,W)$. (v) If $H^0(\mathbb{Q}_p,V)=0$ then $H^1_f(\mathbb{Q}_p,V) = 0$, $H^1_f(\mathbb{Q}_p,W)=0$ and $H^1_f(\mathbb{Q}_p,T) = H^1(\mathbb{Q}_p,T)_{\mathrm{tor}}$; if moreover $H^0(\mathbb{Q}_p,V^D)=0$ then $H^1(\mathbb{Q}_p,T)$ is finite and $H^1_f(\mathbb{Q}_p,T) = H^1(\mathbb{Q}_p,T)$ (3.4). Paper: the paragraph after Lemma lem:dfg21 ("$H^1_f(\mathbb{Q}_p,V^D) = H^1_{\mathrm{ur}}(\mathbb{Q}_p,V^D) = (V^D)^{I_p}/(\mathrm{Frob}_p - 1)$ vanishes ... so $H^1_f(\mathbb{Q}_p,T^D)$ is the torsion of $H^1(\mathbb{Q}_p,T^D)$"), Lemma lem:fpr.

5.4 `prop:blochkato:local-duality-f` (proposition; P for $p \neq \ell$ given Chapter 17's Tate local duality and the annihilator of H^1_{ur}, inputs K1, K4; R for $p=\ell$). Under the local Tate pairing $H^1(\mathbb{Q}_p,V) \times H^1(\mathbb{Q}_p,V^D) \to H^2(\mathbb{Q}_p,\mathbb{Q}_\ell(1)) = \mathbb{Q}_\ell$ the subspaces $H^1_f(\mathbb{Q}_p,V)$ and $H^1_f(\mathbb{Q}_p,V^D)$ are exact annihilators of each other, for every p (for $p=\ell$: BlochKato Proposition 3.8, verify). Integrally, under the perfect pairing $H^1(\mathbb{Q}_p,W) \times H^1(\mathbb{Q}_p,T^D) \to \mathbb{Q}_\ell/\mathbb{Z}_\ell$, the exact annihilator of $H^1_f(\mathbb{Q}_p,W)$ is $H^1_f(\mathbb{Q}_p,T^D)$, hence

$$H^1(\mathbb{Q}_p,W)/H^1_f(\mathbb{Q}_p,W) \simeq H^1_f(\mathbb{Q}_p,T^D)^*.$$

Proof for $p \neq \ell$: the $\mathbb{Q}_\ell$ statement from the unramified annihilator and dimensions (5.3(i), 3.3); the integral statement by taking preimages and images (Rubin, Proposition 1.4.3, or Flach, verify). Paper: Lemma lem:dfg21 (the local terms $H^1_f(\mathbb{Q}_p,T^D)^*$).

5.5 `ex:blochkato:ur-versus-f` (worked example and warning; P). "Unramified" and "f" differ for W. Let E have split multiplicative reduction at $p \neq \ell$ with $\ell \mid n_p$, and $W = E[\ell^\infty]$. Then $H^1_f(\mathbb{Q}_p,V_\ell E) = 0$ (the Frob_p-eigenvalue on $V^{I_p} = \mathbb{Q}_\ell e_1$ is $p \neq 1$), so $H^1_f(\mathbb{Q}_p,W)=0$, while $W^{I_p} = (\mathbb{Q}_\ell/\mathbb{Z}_\ell)e_1 \oplus (\mathbb{Z}/\ell^{v_\ell(n_p)})e_2$ and $H^1_{\mathrm{ur}}(\mathbb{Q}_p,W) = W^{I_p}/(\mathrm{Frob}_p - 1)W^{I_p} \simeq \mathbb{Z}/\ell^{v_\ell(n_p)} \neq 0$: the ℓ-part of the component group. This is the classical fact that the Kummer condition at $p \neq \ell$ is $E(\mathbb{Q}_p) \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell = 0$, not the unramified one, and the reason Tamagawa factors appear. For the adjoint at a multiplicative prime $H^1_f(\mathbb{Q}_q,W)=0$ always (5.3(v), 3.3), and $H^1_{\mathrm{ur}}(\mathbb{Q}_q,W)=0$ when $\ell \nmid n_q$ (Exercise 4). Paper: background to Lemma lem:dfg21 and Lemma lem:fpr (the difference between the "unramified" and "Bloch-Kato" conditions in DFG04's H^1_Σ).

20.2.6 Section 6. Global Bloch-Kato Selmer groups and the Σ-imprimitive Selmer group

6.1 `def:blochkato:selmer` (definition and proposition; P). Let M be one of V, T, W, V^D, T^D, W^D for a geometric V of $G_{\mathbb{Q}}$, and S a finite set of places containing ℓ, ∞ and the primes where V ramifies. $H^1_f(\mathbb{Q},M) = \{c \in H^1(G_{\mathbb{Q},S},M) : \mathrm{loc}_p(c) \in H^1_f(\mathbb{Q}_p,M) \text{ for all } p \in S\}$. It does not depend on S: for $p \notin S$, $H^1_f(\mathbb{Q}_p,M) = H^1_{\mathrm{ur}}(\mathbb{Q}_p,M)$ (5.3, M unramified), and a class of $H^1(\mathbb{Q},M)$ unramified outside S comes from $G_{\mathbb{Q},S}$ (inflation-restriction). Reference: BlochKato Section 5 (verify); Rubin 1.5; DFG04 Section 2.1. Paper: Section 1.2 ("Let $H^1_f(\mathbb{Q},\mathrm{ad}^0\,T \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell)$ be the Bloch-Kato Selmer group"), Section 4.2.

6.2 `prop:blochkato:selmer-relations` (proposition; P given Chapter 17's T, V, W sequences, input K6, R for (iii)). (i) $H^1_f(\mathbb{Q},T)$ is the preimage of $H^1_f(\mathbb{Q},V)$ under $H^1(G_{\mathbb{Q},S},T) \to H^1(G_{\mathbb{Q},S},V)$, and $H^1_f(\mathbb{Q},T)_{\mathrm{tor}} = H^1(G_{\mathbb{Q},S},T)_{\mathrm{tor}}$ is the image of $H^0(\mathbb{Q},W)$. (ii) $H^1_f(\mathbb{Q},W)$ is a cofinitely generated $\mathbb{Z}_\ell$-module.

(iii) Its maximal divisible subgroup is the image of $H^1_f(\mathbb{Q},V)$, so $\operatorname{corank} H^1_f(\mathbb{Q},W) = \dim H^1_f(\mathbb{Q},V)$; in particular $H^1_f(\mathbb{Q},W)$ is finite iff $H^1_f(\mathbb{Q},V)=0$ (Rubin, Section 1.5, verify the statement number; Flach92 for the adjoint). Paper: Theorem thm:main ("$\operatorname{length}_{\mathbb{Z}_\ell} H^1_f(\mathbb{Q},\operatorname{ad}^0 T\otimes\mathbb{Q}_\ell/\mathbb{Z}_\ell)$" is finite by 10.2 and (iii)).

6.3 `def:blochkato:sigma-selmer` (definition; P). For a finite set Σ of primes not containing ℓ: $H^1_\Sigma(\mathbb{Q},W) = \{c\in H^1(\mathbb{Q},W) : \operatorname{loc}_\ell c \in H^1_f(\mathbb{Q}_\ell,W),\ \operatorname{loc}_p c\in H^1_{\mathrm{ur}}(\mathbb{Q}_p,W) \text{ for } p\notin\Sigma\cup\{\ell\}\}$, no condition at $p\in\Sigma$ (none is needed at ∞, ℓ odd). For $p\notin\Sigma\cup\{\ell\}$ with W unramified at p, $H^1_{\mathrm{ur}}(\mathbb{Q}_p,W) = W/(\mathrm{Frob}_p-1)W$ is divisible and equals $H^1_f(\mathbb{Q}_p,W)$ (5.3(iv)); so $H^1_\Sigma(\mathbb{Q},W)\supseteq H^1_f(\mathbb{Q},W)$ is "Bloch-Kato outside Σ, nothing on Σ" as soon as Σ contains the primes $p\neq\ell$ where W ramifies. For the paper Σ is the set of primes dividing N; it contains the I_0^* primes, where A_f is unramified (3.2), which is harmless. Reference: DFG04 Section 2.1 (verify the notation H^1_Σ). Paper: Section 4.2 ("with no condition at the primes of Σ, the Bloch-Kato condition at ℓ and the unramified condition elsewhere").

6.4 `prop:blochkato:sigma-finite` (proposition; P). If $H^1_f(\mathbb{Q},V)=0$ then $H^1_\Sigma(\mathbb{Q},W)$ is finite: the kernel of $H^1_\Sigma(\mathbb{Q},W)\to\bigoplus_{p\in\Sigma} H^1(\mathbb{Q}_p,W)/H^1_f(\mathbb{Q}_p,W)$ is $H^1_f(\mathbb{Q},W)$ (finite by 6.2(iii)), and each target is cofinitely generated with $H^1(\mathbb{Q}_p,W)/H^1_f(\mathbb{Q}_p,W)\simeq H^1_f(\mathbb{Q}_p,T^D)^*$ finite (5.4, 3.4(iii), 5.3(v) for T^D). Paper: Theorem thm:dfg37 ("$H^1_\Sigma(\mathbb{Q},W)$ is finite").

6.5 `ex:blochkato:classical-selmer` (example; S given Chapter 17's classical Selmer group, R for the comparison at ℓ). For $T=T_\ell E$, $W=E[\ell^\infty]$: $H^1_f(\mathbb{Q},E[\ell^\infty]) = \mathrm{Sel}_{\ell^\infty}(E/\mathbb{Q})$. At $p\neq\ell$ both local conditions are 0: $E(\mathbb{Q}_p)\otimes\mathbb{Q}_\ell/\mathbb{Z}_\ell=0$, and $H^1_f(\mathbb{Q}_p,V_\ell E)=0$ since $H^0(\mathbb{Q}_p,V_\ell E)=0$ (5.3(i); compare 5.5); at ℓ the Kummer image $E(\mathbb{Q}_\ell)\otimes\mathbb{Q}_\ell/\mathbb{Z}_\ell$ is $H^1_f(\mathbb{Q}_\ell,W)$ (BlochKato Example 3.11, verify). So the Bloch-Kato Selmer group is the model of Chapter 17 rewritten, and Tamagawa factors are invisible in the Selmer group itself: they live in the comparison with relaxed conditions (Section 10). Paper: background (orientation for Section 1.2).

6.6 `def:blochkato:adjoint-selmer` (definition; P). The paper's objects: $W=\operatorname{ad}^0 T\otimes\mathbb{Q}_\ell/\mathbb{Z}_\ell$, $H^1_f(\mathbb{Q},W)$ (6.1), $H^1_\Sigma(\mathbb{Q},W)$ with Σ the primes dividing N (6.3), $W^D\simeq \operatorname{ad}^0 T(1)\otimes\mathbb{Q}_\ell/\mathbb{Z}_\ell$ (2.4), $H^1_f(\mathbb{Q},V^D)$, $H^0(\mathbb{Q},W^D)$. The Selmer group of Theorem thm:main is $H^1_f(\mathbb{Q},W)$ with discrete coefficients; the Selmer group of A_f with $\mathbb{Q}_\ell$ coefficients is 0 under the hypothesis (10.2), so "the length of the Bloch-Kato Selmer group of $\operatorname{ad}^0 V_\ell E$" in the paper's abstract means $\operatorname{length} H^1_f(\mathbb{Q},W)$ (Part 7, N3). Paper: abstract, Section 1.2, Theorem thm:main, Section 4.2.

6.7 `prop:blochkato:euler-characteristic` (proposition; S given the Greenberg-Wiles formula with $\mathbb{Q}_\ell$ coefficients, input K7). For $V=A_f$ with $\ell\nmid N$: $\dim H^1_f(\mathbb{Q},A_f)-\dim H^1_f(\mathbb{Q},V^D) = \dim H^0(\mathbb{Q},A_f)-\dim H^0(\mathbb{Q},V^D)+\sum_v\big(\dim H^1_f(\mathbb{Q}_v,A_f)-\dim H^0(\mathbb{Q}_v,A_f)\big)=0$: the terms at $p\neq\ell$ vanish by 5.3(i), the term at ℓ is $+1$ (4.5), the term at ∞ is $-\dim A_f^{c=1}=-1$ (complex conjugation acts on A_f with eigenvalues $1,-1,-1$), and $H^0(\mathbb{Q},A_f)=H^0(\mathbb{Q},V^D)=0$ (2.6). Hence $H^1_f(\mathbb{Q},A_f)=0$ iff $H^1_f(\mathbb{Q},V^D)=0$. Reference: Wiles95 Proposition 1.6 for finite modules and FPR94 for the $\mathbb{Q}_\ell$ version (verify both); Chapter 17. Paper: Theorem thm:dfg27 ("and the same for $A\otimes\mathbb{Q}_\ell$"), Remark rem:ns (the hypothesis $H^1_f(\mathbb{Q},V^D)=0$ for the 3Ns curves).

20.2.7 Section 7. Tamagawa factors (Fontaine and Perrin-Riou)

7.1 `def:blochkato:tamagawa` (definition; P for well-definedness). For a prime $p\neq\ell$ and a $G_{\mathbb{Q}_p}$-stable $\mathbb{Z}_\ell$-lattice M (free of finite rank, continuous action):

$$\mathrm{Tam}_p(M) = \mathrm{Fitt}_{\mathbb{Z}_\ell}\Big(\big(H^1(I_p,M)_{\mathrm{tor}}\big)^{G_{\mathbb{Q}_p}}\Big),$$

the Fitting ideal of the $G_{\mathbb{Q}_p}$-invariants (for the conjugation action of $G_{\mathbb{Q}_p}/I_p$, Chapter 16, conjugation-action) of the torsion subgroup of $H^1(I_p,M)$; it is $\ell^a\mathbb{Z}_\ell$ with a the length of that finite module (Chapter 16, h1-finitely-generated; Chapter 3, 7.6), and $v_\ell(\mathrm{Tam}_p(M)):=a$. The paper's display $\mathrm{Fitt}_{\mathbb{Z}_\ell}(H^1(I_p,M)_{\mathrm{tor}})^{G_{\mathbb{Q}_p}}$ must be read this way (important_notes.md point 37). Reference: FPR94 I.4.2.2 (as cited by the paper); DFG04 p. 708. Paper: Section 1.2, Theorem thm:main, Lemma lem:tamfactor, Lemma lem:fpr.

7.2 `prop:blochkato:tam-index` (proposition; P from 5.3(iii)). For $p \neq \ell$ and M a lattice:

$$\ell^{v_\ell(\mathrm{Tam}_p(M))} = \#\big(H^1(I_p, M)_{\mathrm{tor}}\big)^{G_{\mathbb{Q}_p}} = [H^1_f(\mathbb{Q}_p, M) : H^1_{\mathrm{ur}}(\mathbb{Q}_p, M)].$$

This is the "index" description of the Tamagawa factor (PLAN.md): it measures how much bigger the Bloch-Kato condition for M is than the naive unramified one. Physics aside: an index of lattices as a volume ratio. Paper: Section 1.2, Lemma lem:fpr.

7.3 `prop:blochkato:tam-basic` (proposition; P). Let $p \neq \ell$. (i) If M is unramified at p, $\mathrm{Tam}_p(M) = \mathbb{Z}_\ell$ ($H^1(I_p, M) = \mathrm{Hom}(I_p, M) = M(-1)$ is torsion-free). (ii) If I_p acts on M through a finite quotient of order prime to ℓ, $\mathrm{Tam}_p(M) = \mathbb{Z}_\ell$ (9.1 with $H^1(\Delta, M) = 0$). (iii) $\mathrm{Tam}_p(M)$ depends only on the $G_{\mathbb{Q}_p}$-module M; $\mathrm{Tam}_p(cM) = \mathrm{Tam}_p(M)$ for $c \in \mathbb{Q}_\ell^\times$. (iv) $\mathrm{Tam}_p(\mathrm{ad}^0 T) = \mathrm{Tam}_p(\mathrm{ad}^0(T \otimes \chi))$ for every character χ of $G_{\mathbb{Q}_p}$ (2.4(ii)). In particular $\mathrm{Tam}_p(\mathrm{ad}^0 T) = \mathbb{Z}_\ell$ at every prime of good reduction and at every I_0^* prime ($\mathrm{ad}^0 T$ unramified there). Paper: Section 1.2 ("at a prime of good reduction it is 0"), proof of Theorem thm:main (I_n^* primes).

7.4 `thm:blochkato:fpr-local` (theorem, the first identity of the paper's Lemma lem:fpr, in general for $p \neq \ell$; P given 5.3 and 3.1(c)). Let $p \neq \ell$ and X a $G_{\mathbb{Q}_p}$-stable lattice with $H^0(\mathbb{Q}_p, X \otimes \mathbb{Q}_\ell) = 0$. Then $H^1_f(\mathbb{Q}_p, X) = H^1(\mathbb{Q}_p, X)_{\mathrm{tor}}$ is finite and there is an exact sequence

$$0 \to X^{I_p}/(\mathrm{Frob}_p - 1)X^{I_p} \to H^1(\mathbb{Q}_p, X)_{\mathrm{tor}} \to \big(H^1(I_p, X)_{\mathrm{tor}}\big)^{G_{\mathbb{Q}_p}} \to 0,$$

so $\mathrm{Fitt}_{\mathbb{Z}_\ell} H^1_f(\mathbb{Q}_p, X) = \det(1 - \Phi_p \mid X^{I_p}) \cdot \mathrm{Tam}_p(X)$. (Cross-check: $H^1(\mathbb{Q}_p, X)_{\mathrm{tor}} \simeq H^0(\mathbb{Q}_p, X \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell)$ by Chapter 17, 4.6, so $\#H^0(\mathbb{Q}_p, W^D) = \ell^{v_\ell(L_p(A_f,1)^{-1}) + v_\ell(\mathrm{Tam}_p(T^D))}$ for $X = T^D$.) For $X = T^D$ and $p \in \Sigma$ this is $\mathrm{Fitt}\, H^1_f(\mathbb{Q}_p, T^D) = L_p(A_f, 1)^{-1}\, \mathrm{Tam}_p(T^D)$ (3.3(i), 3.5). Proof: (1) inflation-restriction with $H^2(\widehat{\mathbb{Z}}, X^{I_p}) = 0$ (Chapter 16, procyclic-compact (iii)) gives $0 \to H^1(G/I, X^I) \to H^1(G, X) \to H^1(I, X)^{G/I} \to 0$; (2) $H^1(G/I, X^I) = X^I/(\mathrm{Frob} - 1)X^I$ is finite of order $\ell^{v_\ell \det(1-\Phi|X^I)}$ (3.1(c)); (3) taking torsion keeps exactness (a torsion class on the right lifts to a class some multiple of which lies in the finite left term, hence to a torsion class); (4) Fitting ideals of finite $\mathbb{Z}_\ell$-modules multiply in exact sequences (lengths add; Chapter 3, 7.5, 7.6); (5) $H^1_f(\mathbb{Q}_p, X) = \iota^{-1}(0) = H^1(\mathbb{Q}_p, X)_{\mathrm{tor}}$ by 5.3(v). Reference: FPR94 I.4.2.2; the proof of DFG04 Theorem 2.15 (as cited by the paper). Paper: Lemma lem:fpr, proof of Theorem thm:main ("each local term is $v_\ell(L_p(A_f, 1)^{-1}) + v_\ell(\mathrm{Tam}_p(A))$"), Remark rem:ns.

7.5 `prop:blochkato:tam-dual` (proposition, the second identity of Lemma lem:fpr; P for the elliptic-curve adjoint in every case, R in general). For $p \neq \ell$, ℓ odd and $A = \mathrm{ad}^0 T_\ell E$: $\mathrm{Tam}_p(T^D) = \mathrm{Tam}_p(A(1)) = \mathrm{Tam}_p(A)$. Proof by cases: good, I_0^* and potentially good primes: both sides trivial (7.3, 9.3; I_p acts on $\mathbb{Z}_\ell(1)$ trivially, so $H^1(I_p, A(1)) = H^1(I_p, A)(1)$ and the finite-image argument is unchanged); multiplicative and potentially multiplicative primes: $H^1(I, A(1))_{\mathrm{tor}} = H^1(I, A)_{\mathrm{tor}}(1)$, on which Frob acts as $q \cdot q^{-1}\, \mathrm{Ad}\,\rho(\mathrm{Frob}) = \mathrm{Ad}\,\rho(\mathrm{Frob})$, and the invariants are $\{x\bar{e} + y\bar{h} : (q-1)x \equiv 2\varepsilon c' y \pmod{\ell^v}\}$, the kernel of a linear form on $(\mathbb{Z}/\ell^v)^2$ whose image has order $\ell^{v - \min(v, v_\ell(q-1), v_\ell(c'))}$, so the order is $\ell^{v + \min(v, v_\ell(q-1), v_\ell(c'))}$, the same as for A (8.2); in fact both invariant modules are $\simeq \mathbb{Z}/\ell^v \oplus \mathbb{Z}/\ell^m$, $m = \min(v, v_\ell(q-1), v_\ell(c'))$ (Smith form of the linear form; Exercise 8). Reference: DFG04 (57) (as cited by the paper) for the general statement. Paper: Lemma lem:fpr ("$\mathrm{Tam}_p(T^D) = \mathrm{Tam}_p(A)$").

7.6 `prop:blochkato:tam-of-E` (proposition; P at multiplicative primes, R in general). For $p \neq \ell$, $v_\ell(\mathrm{Tam}_p(T_\ell E)) = v_\ell(c_p(E))$, the ℓ-part of the Tamagawa number. At a multiplicative prime this is the check at the end of Lemma lem:tamfactor (8.2, Step 6: $v_\ell(n)$ if split, 0 if nonsplit, ℓ odd); at a type IV or IV* prime with $\ell = 3$, $H^1(I_p, T)_{\mathrm{tor}} = H^1(C_3, T) = T/\pi T \simeq \mathbb{F}_3$ (9.2), and its Frob-invariants are $\mathbb{F}_3$ or 0 according as $c_p = 3$ or 1. General case: via $H^1_{\mathrm{ur}}(\mathbb{Q}_p, E[\ell^\infty]) \simeq$ the dual of $\Phi_p(\mathbb{F}_p)[\ell^\infty]$ (Grothendieck, SGA 7, Expose IX, verify; or Rubin, Section 1.6, verify). Paper: Lemma lem:tamfactor ("As a check ... the ℓ-part of the Tamagawa number $c_q(E)$, as it should").

20.2.8 Section 8. The Tamagawa factor of $\mathrm{ad}^0 T$ at a multiplicative prime: Lemma lem:tamfactor

8.1 `conv:blochkato:tate-data` (convention and recollection; P by citation). Convention C6, and the inputs: (a) the Tate shape: a $\mathbb{Z}_\ell$-basis e_1, e_2 of T with $\rho(g) = \left(\begin{smallmatrix} \chi_\ell \varepsilon & \varepsilon\kappa \\ 0 & \varepsilon \end{smallmatrix}\right)(g)$ on $G = G_{\mathbb{Q}_q}$, κ a

Kummer cocycle of q_E in $H^1(G,\mathbb{Z}_\ell(1)) = \varprojlim \mathbb{Q}_q^\times/(\mathbb{Q}_q^\times)^{\ell^k}$ (Chapter 11, 3.2; Chapter 10, 7.5(iii); point 42); (b) $\kappa|_I = n\,t_\ell$ (Chapter 7, 6.2(ii)), so a generator t of the pro-ℓ tame quotient acts by $te_1 = e_1$, $te_2 = e_2 + n'e_1$ with $v_\ell(n') = v_\ell(n)$; (c) $\mathrm{Frob}\; t\; \mathrm{Frob}^{-1} = t^q$ with Frob arithmetic (Chapter 7, 5.4; point 11); (d) the decomposition $\kappa = n\kappa_q + \kappa_u$ for $q_E = q^n u$ (Chapter 7, 6.2(i)); (e) the class of κ_u in $H^1(G/I,\mathbb{Z}_\ell(1)) = \mathbb{Z}_\ell(1)/(q-1) \simeq \mathbb{F}_q^\times \otimes \mathbb{Z}_\ell$ is the class of $\bar u$ (Chapter 7, 6.2(iii); Chapter 17, input K8). Paper: Lemma lem:tamfactor (statement and first sentences of the proof).

8.2 `lem:blochkato:tamfactor` (lemma, the paper's Lemma lem:tamfactor, in corrected form; P, every step). Let q be a prime of multiplicative reduction, $n = n_q = v_q(\Delta_E)$, ℓ an odd prime, $\ell \neq q$, $M = \mathrm{ad}^0 T$, and q_E, u, $c(u)$ as in C6. Then $H^1(I_q,M)_{\mathrm{tor}} \simeq (\mathbb{Z}/\ell^{v_\ell(n)})^2$ and

$$v_\ell\big(\mathrm{Tam}_q(\mathrm{ad}^0 T)\big) = v_\ell(n) + \min\big(v_\ell(n),\, v_\ell(q-1),\, v_\ell(c(u))\big).$$

In particular $v_\ell(\mathrm{Tam}_q(\mathrm{ad}^0 T)) \geq t_\ell(E,q)$, with equality unless $q \equiv 1 \pmod\ell$, $\ell \mid n$ and $\bar u$ is an ℓ-th power in $\mathbb{F}_q^\times$. Proof, in six steps, each citing its source: **Step 1** (shape) 8.1(a), (b). **Step 2** (reduction to a procyclic group) the kernel of $I \to \langle t\rangle$ has trivial pro-ℓ quotient, so $H^1(I,M') = M'/(t-1)M'$ for every $\mathbb{Z}_\ell$-module M' of finite type with this action (Chapter 16, tame-inertia-example, procyclic-compact). **Step 3** (the computation) in the basis $e = e_1\otimes e_2^\vee$, $h = e_1\otimes e_1^\vee - e_2\otimes e_2^\vee$, $f = e_2\otimes e_1^\vee$: $(t-1)e = 0$, $(t-1)h = -2n'e$, $(t-1)f = n'h - n'^2 e$; $2 \in \mathbb{Z}_\ell^\times$ gives $(t-1)M = n'(\mathbb{Z}_\ell e \oplus \mathbb{Z}_\ell h)$ and $H^1(I,M) = (\mathbb{Z}_\ell/n')\bar e \oplus (\mathbb{Z}_\ell/n')\bar h \oplus \mathbb{Z}_\ell \bar f$ (Chapter 16, model-computation (a)). **Step 4** (Frobenius) the conjugation action $(\sigma\cdot c)(i) = \sigma c(\sigma^{-1}i\sigma)$ with $\mathrm{Frob}^{-1}\, t\, \mathrm{Frob} = t^{q^{-1}}$ and $c(t^a) \equiv a\,c(t)$ modulo $(t-1)M$ makes the **arithmetic** Frobenius act on $M/(t-1)M$ as $q^{-1}\,\mathrm{Ad}\,\rho(\mathrm{Frob})$; with $\rho(\mathrm{Frob}) = \left(\begin{smallmatrix} \varepsilon q & c' \\ 0 & \varepsilon \end{smallmatrix}\right)$, $c' = \varepsilon\kappa(\mathrm{Frob})$, one has $\mathrm{Ad}\,\rho(\mathrm{Frob})e = qe$, $\mathrm{Ad}\,\rho(\mathrm{Frob})h = h - 2\varepsilon c' e$, so on the torsion $\bar e \mapsto \bar e$, $\bar h \mapsto q^{-1}(\bar h - 2\varepsilon c'\bar e)$ and the invariants are $\{x\bar e + y\bar h : (q-1)y \equiv 0,\ c'y \equiv 0 \pmod{\ell^{v_\ell(n)}}\} \simeq \mathbb{Z}/\ell^{v_\ell(n)} \oplus \mathbb{Z}/\ell^{\min(v_\ell(n),v_\ell(q-1),v_\ell(c'))}$ (Chapter 16, frobenius-on-h1-inertia, model-computation (b), (c)). **Step 5** (the class c'; important_notes.md point 7 with its addendum; Chapter 17, 5.3, 5.4) $c' = \varepsilon\kappa(\mathrm{Frob})$ is a cocycle value: changing the lift by t^a changes it by $\varepsilon q n' a$, changing κ by a coboundary changes it by a multiple of $q-1$, so c' is well defined only modulo $(n, q-1)\mathbb{Z}_\ell$ (not modulo $q-1$ as the paper says), which leaves $\min(v_\ell(n), v_\ell(q-1), v_\ell(c'))$ unchanged (Chapter 16, model-computation (d)). The paper's phrase "the image of κ in $H^1(G/I,\mathbb{Z}_\ell(1)^I)$" is kept, with its precise meaning: q is a canonical uniformiser of $\mathbb{Q}_q$, so $H^1(\mathbb{Q}_q,\mathbb{Z}_\ell(1)) = \mathbb{Z}_\ell\kappa_q \oplus H^1_{\mathrm{ur}}(\mathbb{Q}_q,\mathbb{Z}_\ell(1))$ canonically, and the projection $\mathrm{pr}_{\mathrm{ur}}$ along κ_q sends $[\kappa]$ to $[\kappa_u]$, the class of $\bar u$ in $H^1(G/I,\mathbb{Z}_\ell(1)) = \mathbb{Z}_\ell(1)/(q-1) \simeq \mathbb{F}_q^\times\otimes\mathbb{Z}_\ell$ (Chapter 17, 5.3; it is a splitting, not a functorial map, since inflation goes the other way). For the lift Frob_0 fixing a compatible system of ℓ-power roots of q (Chapter 7, 5.5), $\kappa_q(\mathrm{Frob}_0) = 0$ and $c' = \varepsilon\,\kappa_u(\mathrm{Frob}_0)$, i.e. ε times that image (Chapter 17, 5.4(iii)); so $v_\ell(c') = v_\ell(c(u))$ in the sense of C6 for Frob_0, and the minimum is the same for every lift (Chapter 17, 5.4(v)). Chapter 18 cites Chapter 17, 5.4 for this step and does not reprove it. **Step 6** (definition and check) by 7.1 the Fitting ideal of the invariants has valuation $v_\ell(n) + \min(\dots)$; for $M = T$ the same steps give $H^1(I,T)_{\mathrm{tor}} = (\mathbb{Z}/\ell^{v_\ell(n)})\bar e_1$ with Frob acting by ε, invariants of length $v_\ell(n)$ (split) or 0 (nonsplit), the ℓ-part of c_q (7.6; data check in Part 6.6). Paper: Lemma lem:tamfactor (all of it), Section 1.2, Theorem thm:main ((eq:thm)), Remark rem:ko(1), Section sec:tam, important_notes.md points 7, 11, 23, 31, 42.

8.3 `cor:blochkato:tamfactor-consequences` (corollary; P). (i) $v_\ell(\mathrm{Tam}_q(\mathrm{ad}^0 T)) - t_\ell(E,q) = \min(v_\ell(n), v_\ell(q-1), v_\ell(c(u))) \in \{0,\dots,v_\ell(n)\}$. (ii) The excess is positive iff $\ell \mid n$, $q \equiv 1 \pmod\ell$ and $\bar u \in (\mathbb{F}_q^\times)^\ell$; for $q = 2$ it is 0. (iii) (Remark rem:ko(1)) At such a prime, $v_\ell(\mathrm{Tam}_q) \geq t_\ell(E,q) + 1$, and (eq:identity) raises the bound of (eq:thm) by the excess, which can exceed 1 only if $\ell^2 \mid n$ and $\ell^2 \mid q-1$. (iv) The excess does not depend on whether the reduction is split. Paper: Remark rem:ko(1), Section sec:tam, the abstract (Part 7, N2).

8.4 `prop:blochkato:tam-potmult` (proposition; P given Chapter 11, 3.4 and Chapter 10, 7.4(i)). Let $q \neq \ell$ ($q = 2$ allowed) be a prime of additive, potentially multiplicative reduction, $\nu = -v_q(j_E)$, $q_E = \mathfrak{q}(j_E)$ and $u = q_E q^{-\nu}$. Then $v_\ell(\mathrm{Tam}_q(\mathrm{ad}^0 T)) = v_\ell(\nu) + \min(v_\ell(\nu), v_\ell(q-1), v_\ell(c(u))) \geq t_\ell(E,q)$. Proof: there is a ramified quadratic character χ of $G_{\mathbb{Q}_q}$ with $T\otimes\chi$ of the Tate shape for the Tate parameter q_E (which depends only on j); $\mathrm{ad}^0 T = \mathrm{ad}^0(T\otimes\chi)$ as $G_{\mathbb{Q}_q}$-modules (2.4(ii)); apply 8.2 to the shape (8.2 uses only the shape, not a global curve). For q odd one may use the global twist $E^{(q)}$ of Lemma lem:twist, as the paper does; for $q = 2$ the local argument is needed (important_notes.md point 18). At $q = 2$, $v_\ell(q-1) = 0$ and the value is $t_\ell(E,2) = v_\ell(-v_2(j))$, which is not v_ℓ of the index of the Kodaira symbol (Chapter 10,

7.7). Paper: proof of Theorem thm:main ("at such a prime p the curve $E_{\{p\}}$ ... is multiplicative with the same Tate parameter valuation"), Definition def:terms (t_ℓ via $v_q(j)$).

8.5 `ex:blochkato:tamfactor-examples` (worked examples; P, every number computed; Part 6.6). Tate data and Tamagawa factors ($v = v_\ell(q-1)$, $k = v_\ell(c(u))$):

curve, prime	type	ν	$\bar{u}$ mod q	ℓ	v	k	$v_\ell(\mathrm{Tam}_q(\mathrm{ad}^0_\ell T))$	$v_\ell(\mathrm{Tam}_q(\mathrm{ad}^0 T))$	remark
11a1, 11	I_5	5	$10 = -1$	5	1	1	2	1	$E[5]$ reducible: the lemma, not the theorem (Chapter 16, Exercise 10)
14a1, 7	I_3	3	$6 = -1$	3	1	1	2	1	$E[3]$ reducible (torsion $\mathbb{Z}/6$), $\deg \varphi_E = 1$
539c1, 7	I_3^*	3	5	3	1	0	1	1	5 is not a cube mod 7; same q_E as 77c1
791a1, 7	I_3	3	$6 = -1$	3	1	1	2	1	$\deg \varphi_E = 144$, $v_3 = 2$: bound (eq:thm) 1, refined bound 2
791a1, 113	I_2	2	106	3	0	0	0	0	$113 \equiv 2 \pmod 3$
1184f1, 37	I_3	3	$36 = -1$	3	2	2	2	1	the cap $v_\ell(\nu) = 1$ binds
1748e1, 19	I_6	6	12	3	2	1	2	1	12 a cube, not a ninth power, mod 19

(791a1 and 1184f1, 1748e1 have $E[3]$ irreducible, $3 \nmid N$, non-CM, image not 3Ns.) Paper: Lemma

lem:tamfactor, Remark rem:ko(1), Section sec:tam.

20.2.9 Section 9. Tamagawa factors at additive primes, and Open question 3

9.1 `lem:blochkato:finite-inertia-image` (lemma; P). Let $q \neq \ell$ and M a lattice on which I_q acts through a finite quotient $\Delta = I_q/K$. Then $H^1(I_q, M)_{\mathrm{tor}} = H^1(\Delta, M)$ (inflated), and $H^1(I_q, M)/H^1(\Delta, M)$ embeds in $\mathrm{Hom}(K, M)^{\Delta} \subseteq \mathrm{Hom}(\mathbb{Z}_\ell(1), M) = M(-1)$, which is torsion-free. Proof: inflation-restriction (Chapter 16), $H^1(K, M) = \mathrm{Hom}_{\mathrm{cont}}(K, M)$ factors through the maximal pro-ℓ abelian quotient of K, which is $\mathbb{Z}_\ell(1)$ (K is the inertia group of a finite extension of $\mathbb{Q}_q^{\mathrm{ur}}$, Chapter 7, 5.2); $H^1(\Delta, M)$ is finite (killed by $\#\Delta$). Paper: Open question 3.

9.2 `prop:blochkato:ad-free-c3` (proposition; P; verified by computer). Let $\sigma \in \mathrm{GL}_2(\mathbb{Z}_3)$ have order 3 and $T = \mathbb{Z}_3^2$. Then $\mathrm{ad}^0 T$ is a free $\mathbb{Z}_3[\langle\sigma\rangle]$-module of rank one; hence $\hat{H}^i(\langle\sigma\rangle, \mathrm{ad}^0 T) = 0$ for all i, in particular $H^1(\langle\sigma\rangle, \mathrm{ad}^0 T) = 0$. By contrast $H^1(\langle\sigma\rangle, T) = T/\pi T \simeq \mathbb{F}_3$. Proof: σ has characteristic polynomial $X^2 + X + 1$, so T is free of rank one over $\mathbb{Z}_3[X]/(X^2 + X + 1) \simeq \mathbb{Z}_3[\zeta_3]$ (a DVR with uniformiser $\pi = 1 - \zeta_3$; Chapter 6, 4.7, 4.8; important_notes.md points 12, 13), and $\bar{\sigma}$ is a nontrivial unipotent on $T/3T$ (Lemma lem:IV; Chapter 4, 9.4); $\mathrm{ad}^0(T/3) \simeq \mathrm{Sym}^2(T/3) \otimes \det^{-1}$ (Chapter 4) and Sym^2 of a 2×2 Jordan block is a single 3×3 Jordan block in characteristic 3, i.e. the regular module $\mathbb{F}_3[C_3] = \mathbb{F}_3[X]/(X-1)^3$; by Nakayama a lift of a generator gives a surjection $\mathbb{Z}_3[C_3] \to \mathrm{ad}^0 T$ of free $\mathbb{Z}_3$-modules of rank 3, hence an isomorphism. (Check: in the basis e, h, f and $\sigma = \left(\begin{smallmatrix} 0 & -1 \\ 1 & -1 \end{smallmatrix}\right)$, $e, \sigma e, \sigma^2 e$ have determinant ± 1.) For T: the norm $1 + \sigma + \sigma^2$ is 0 on T and $(\sigma - 1)T = \pi T$. Paper: Open question 3; Lemma lem:euler and Lemma lem:IV (the same lattice).

9.3 `thm:blochkato:tam-additive` (theorem; P given 9.1, 9.2 and the classification of inertia images, R; new, see Part 7, N1). Let ℓ be odd, $q \neq \ell$ a prime of additive reduction of E, $A = \mathrm{ad}^0 T_\ell E$. (i) If the reduction at q is potentially good, $\mathrm{Tam}_q(A) = \mathrm{Tam}_q(A(1)) = \mathbb{Z}_\ell$. (ii) If it is potentially multiplicative, $\mathrm{Tam}_q(A)$ is given by 8.4. Proof of (i): the image Δ of I_q is finite, of order dividing 24: cyclic of order $2, 3, 4, 6$ for $q \geq 5$ (Kraus90), and one of $C_2, C_3, C_4, C_6, Q_8, \mathrm{SL}_2(\mathbb{F}_3)$ at $q = 2$, of order dividing 12 at $q = 3$ (Serre-Tate, Kraus90; cite by section, verify); by 9.1, $H^1(I_q, A)_{\mathrm{tor}} = H^1(\Delta, A)$. If $\ell \nmid \#\Delta$ this is 0 (Chapter 16, order-kills). If $\ell \mid \#\Delta$ then $\ell = 3$ and $q \neq 3$; the 3-primary part of $H^1(\Delta, A)$ injects into $H^1(P, A)$ for a Sylow 3-subgroup P (Chapter 16, res-inf-cor (d)), which is cyclic of order 3 because $9 \nmid 24$, and $H^1(P, A) = 0$ by 9.2 ($P \subset \mathrm{GL}_2(\mathbb{Z}_3)$ is generated by an element of order 3). Twisting by $\mathbb{Z}_\ell(1)$ changes nothing on I_q. Consequences: $\mathrm{Tam}_q(\mathrm{ad}^0 T)$ is trivial at every potentially good prime, of every Kodaira type, at every $q \neq \ell$ including $q = 2$, even for $\ell = 3$ and types $\mathrm{II}, \mathrm{II}^*, \mathrm{IV}, \mathrm{IV}^*$, where $\mathrm{Tam}_q(T)$ can be 3 (7.6). Paper: Open question 3 ("Compute the Tamagawa factors of $\mathrm{ad}^0 T$ at the additive primes"), proof of Theorem thm:main ("$v_\ell(\mathrm{Tam}_p(\mathrm{ad}^0 T)) \geq 0$ at an additive prime": it is 0 there unless the reduction is potentially multiplicative), Section 1.2.

9.4 `cor:blochkato:explicit-identity` (corollary; P given 9.3, 8.2, 8.4, 7.3). For ℓ odd, $\ell \nmid N$:

$$\sum_{q \mid N} v_\ell(\mathrm{Tam}_q(\mathrm{ad}^0 T)) = \sum_{q:\, v_q(j_E) < 0} \Big(t_\ell(E,q) + \min\big(t_\ell(E,q),\, v_\ell(q-1),\, v_\ell(c_q(u))\big)\Big).$$

Hence, under the hypothesis of Theorem thm:main, (eq:identity) reads

$$v_\ell(\deg \varphi_E) \;\geq\; \mathrm{length}\, H^1_f(\mathbb{Q}, W) + \sum_{q \mid N} e_\ell(E,q) + \sum_{q:\, v_q(j) < 0} \big(t_\ell(E,q) + \min(t_\ell(E,q), v_\ell(q-1), v_\ell(c_q(u)))\big),$$

with equality by Chapter 14, 9.5 (important_notes.md points 39, 56; the DFG04 checklist there is settled, Chapter 20 outline, Part 7; important_notes.md point 73). So the excess of $v_\ell(\deg \varphi_E)$ over the explicit local bound is the length of the adjoint Bloch-Kato Selmer group. Paper: Theorem thm:main, Remark rem:ko(1), Section sec:tam, Open question 3.

9.5 `rem:blochkato:open-question-3` (remark; P for the data). What remains of Open question 3 is the Selmer length, which 9.4 computes from $\deg \varphi_E$ (the DFG04 checklist being settled: Chapter 20 outline, Part 7; important_notes.md point 73). On the optimal non-CM curves of conductor below 10,000 with $\ell \nmid N$, $E[\ell]$ irreducible and (for $\ell = 3$) image not 3Ns, the excess $v_\ell(\deg \varphi_E)$ − (explicit bound) is $0, 1, 2, 3, \geq 4$ for $8688, 3288, 942, 248, 37$ of 13,203 curves at $\ell = 3$; $19019, 3993, 328, 19, 2$ of 23,361 at $\ell = 5$;

25133, 2888, 111, 2, 0 of 28,134 at $\ell = 7$, and is never negative. Reading: $H^1_f(\mathbb{Q}, W) = 0$ for about two thirds of the curves at $\ell = 3$ and 81%, 89% at $\ell = 5, 7$. The Tamagawa factor at $q = \ell$ is not defined by 7.1 (it needs p-adic Hodge theory) and belongs to Open question 1 (Chapter 20). Paper: Open questions 1 and 3, Section sec:sharp (the mean excess).

20.2.10 Section 10. The comparison of Selmer groups (DFG04 Lemma 2.1) and $H^0(\mathbb{Q}, W^D)$

10.1 `def:blochkato:exceptional-set` (definition and proposition; P given Chapter 11, 8.6). DFG04's exceptional set, for $f = f_E$ (weight 2, trivial character, rational coefficients): $S_f = \{\ell : \ell \mid 2N$, or $\bar\rho|_{\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{\ell^*}))}$ is not absolutely irreducible$\}$ (DFG04 (31), as the paper reads it for $k = 2$; see 4.4 and Part 7, Q1). For $\ell \nmid 2N$ the following are equivalent: $\ell \notin S_f$; $E[\ell]$ is irreducible and $h_\ell(E) = 0$; $E[\ell]$ is irreducible and not ($\ell = 3$ and image 3Ns) (important_notes.md point 47). Paper: Section 4.2 ("the hypothesis of Theorem thm:main is $\ell \notin S_f$"), Theorem thm:main, Remark rem:ns.

10.2 `thm:blochkato:dfg27` (theorem, the paper's Theorem thm:dfg27 = DFG04 Theorem 2.7; R, with a sketch). If $\ell \notin S_f$ then $H^0(\mathbb{Q}, V^D) = H^1_f(\mathbb{Q}, V^D) = 0$, and $H^0(\mathbb{Q}, A_f) = H^1_f(\mathbb{Q}, A_f) = 0$. The H^0 statements hold for every E with $\bar\rho$ absolutely irreducible (2.6); the content is the vanishing of the adjoint Selmer group with $\mathbb{Q}_\ell$ coefficients, which comes from the Taylor-Wiles method ($R_\emptyset \otimes \mathbb{Q}_\ell = \mathbb{T}_\emptyset \otimes \mathbb{Q}_\ell$, so the tangent space of the deformation ring, $H^1_f(\mathbb{Q}, A_f)$, vanishes; DFG04 Section 2.3, verify), and the two Selmer statements are equivalent by 6.7. Consequence: $H^1_f(\mathbb{Q}, W)$ is finite (6.2(iii)) and $H^1_\Sigma(\mathbb{Q}, W)$ is finite (6.4). Reference: DFG04 Theorem 2.7 (as cited); Flach92 for the finiteness of the integral Selmer group of $\mathrm{Sym}^2 T_\ell E$ under his hypotheses (14.6). Paper: Theorem thm:dfg27, proof of Theorem thm:main.

10.3 `prop:blochkato:h0-selmer` (proposition, Lemma lem:h0 in Selmer language; P given Chapter 11, 8.5 for (ii)). Let ℓ be odd with $\bar\rho$ absolutely irreducible (for $\bar\rho_{E,\ell}$, equivalently irreducible: Chapter 11, 1.4). (i) If $H^0(\mathbb{Q}, V^D) = H^1_f(\mathbb{Q}, V^D) = 0$ then $H^1_f(\mathbb{Q}, T^D) = H^1(\mathbb{Q}, T^D)_{\mathrm{tor}}$, and the connecting map gives $H^0(\mathbb{Q}, W^D) \xrightarrow{\sim} H^1_f(\mathbb{Q}, T^D)$ (6.2(i) for T^D; torsion classes satisfy every local condition by 5.2(a)). (ii) (the paper's Lemma lem:h0, corrected; Chapter 11, 8.5, important_notes.md points 5, 6, 10, 14) $H^0(\mathbb{Q}, W^D) \simeq \mathbb{Z}/\ell^{h_\ell(E)}$ is finite and cyclic, $h_\ell(E)$ is the largest k with $\rho \bmod \ell^k \simeq (\rho \otimes \chi_\ell) \bmod \ell^k$, $h_\ell(E) = 0$ for every $\ell \geq 5$, and $h_3(E) \leq 1$ with equality iff $\bar\rho_{E,3} \simeq \bar\rho_{E,3} \otimes \chi_{-3}$ iff the image of $\bar\rho_{E,3}$ is $N_s(\mathbb{F}_3)$; this holds for every E, with or without CM and whether or not $\ell \mid N$. The criterion is $\bar\rho \simeq \bar\rho \otimes \bar\chi_\ell$, not $\bar\rho \simeq \bar\rho \otimes \chi_{\ell^*}$ (point 6); the proof is the determinant argument $\chi_\ell^2 \equiv 1 \pmod{\ell^k}$ on $G_\mathbb{Q}$, which forces $\ell^k = 3$, and not the paper's appeal to CM, which confuses χ_ℓ with χ_{ℓ^*} (point 5). Paper: Lemma lem:h0, Conjecture conj:euler ($h_\ell(E)$), Section 1.3, proof of Theorem thm:main ("$H^0(\mathbb{Q}, W^D) = 0$ by Lemma lem:h0"), Remark rem:ns.

10.4 `lem:blochkato:dfg21` (lemma, the paper's Lemma lem:dfg21 = DFG04 Lemma 2.1; P given Chapter 17, 10.3, the abstract form). Suppose $H^0(\mathbb{Q}, V^D) = H^1_f(\mathbb{Q}, V^D) = 0$, and let Σ be a nonempty finite set of primes, $\ell \notin \Sigma$, containing every prime $p \neq \ell$ at which W is ramified. Then there is an exact sequence

$$0 \to H^1_f(\mathbb{Q}, W) \to H^1_\Sigma(\mathbb{Q}, W) \to \bigoplus_{p \in \Sigma \cup \{\infty\}} H^1_f(\mathbb{Q}_p, T^D)^* \to H^0(\mathbb{Q}, W^D)^* \to 0,$$

in which the term at ∞ is 0 for ℓ odd and, for $p \in \Sigma$, $H^1_f(\mathbb{Q}_p, T^D) = H^1(\mathbb{Q}_p, T^D)_{\mathrm{tor}} \simeq H^0(\mathbb{Q}_p, W^D)$ is finite (5.3(v); Chapter 17, 4.6). Proof: apply Chapter 17, 10.3 to the Bloch-Kato conditions $L_{V,v} = H^1_f(\mathbb{Q}_v, V)$. Its hypotheses: $L_{V,v} = H^1_{\mathrm{ur}}$ outside S (5.1(i)); $L^\perp_{V,v} = L_{V^D,v}$ for all v (5.4: Chapter 17, 4.4(ii) at $p \neq \ell$, BlochKato at ℓ); (a) $H^1_f(\mathbb{Q}, V^D) = 0$ (here a hypothesis; for $\ell \notin S_f$ it is thm:dfg27, 10.2); (b) $H^0(\mathbb{Q}_p, V^D) = 0$ for some $p \in \Sigma$, which holds for every $p \neq \ell$ (3.3(i)) as soon as $\Sigma \neq \emptyset$. Chapter 17's propagated conditions are Chapter 18's H^1_f on T, W (5.2; Chapter 17, 4.5), and its relaxed structure $\mathcal{L}^\Sigma$ is DFG04's H^1_Σ (6.3). What each hypothesis does (to be said in the text): $H^1_f(\mathbb{Q}, V^D) = 0$ makes $H^1_f(\mathbb{Q}, T^D)$ torsion, $\simeq H^0(\mathbb{Q}, W^D)$ (10.3(i)); $\Sigma \neq \emptyset$ with $H^0(\mathbb{Q}_p, V^D) = 0$ kills the strict dual Selmer group and gives the surjection onto $H^0(\mathbb{Q}, W^D)^*$; ℓ odd kills ∞. The paper's hypothesis $H^0(\mathbb{Q}, V^D) = 0$ is implied by (b). Reference: DFG04 Lemma 2.1 (as cited); Rubin, Theorem 1.7.3 (verify). Paper: Lemma lem:dfg21 and the paragraph after it, proof of Theorem thm:main, Remark rem:ns.

10.5 `cor:blochkato:length-identity` (corollary; P given 10.4, 7.4, 7.5, 6.2, 6.4). Under the hypotheses of 10.4 and $H^1_f(\mathbb{Q}, A_f) = 0$ (automatic by 6.7), all terms are finite and

$$\operatorname{length} H^1_\Sigma(\mathbb{Q}, W) = \operatorname{length} H^1_f(\mathbb{Q}, W) + \sum_{p\in\Sigma}\Big(v_\ell\big(L_p(A_f,1)^{-1}\big) + v_\ell\big(\operatorname{Tam}_p(\mathrm{ad}^0\, T)\big)\Big) - h_\ell(E).$$

Proof: lengths in a four-term exact sequence of finite modules (Chapter 16, lengths-in-exact-sequences), Pontryagin duality preserves length, $\operatorname{length} H^1_f(\mathbb{Q}_p, T^D) = v_\ell(L_p(A_f,1)^{-1}) + v_\ell(\operatorname{Tam}_p(T^D))$ (7.4) and $\operatorname{Tam}_p(T^D) = \operatorname{Tam}_p(\mathrm{ad}^0\, T)$ (7.5), $\operatorname{length} H^0(\mathbb{Q}, W^D) = h_\ell(E)$ (10.3). Paper: proof of Theorem thm:main (the display "length H^1_Σ = length H^1_f + $\sum$ length $H^1_f(\mathbb{Q}_p, T^D)$"), Remark rem:ns (the display with $-h_\ell(E)$), Conjecture conj:euler.

20.2.11 Section 11. The Selmer half of the proof of Theorem thm:main

11.1 `thm:blochkato:selmer-identity` (theorem; P given 10.2 (R), 10.5, 9.4). Let E be an elliptic curve over $\mathbb{Q}$, $\ell \notin S_f$, Σ the set of primes dividing N (nonempty: $N > 1$). Then $H^1_f(\mathbb{Q}, W)$ and $H^1_\Sigma(\mathbb{Q}, W)$ are finite and

$$\operatorname{length} H^1_\Sigma(\mathbb{Q}, W) = \operatorname{length} H^1_f(\mathbb{Q}, W) + \sum_{q\mid N} v_\ell\big(L_q(A_f,1)^{-1}\big) + \sum_{q:\, v_q(j)<0}\Big(t_\ell(E,q) + \min\big(t_\ell(E,q), v_\ell(q-1), v_\ell(c_q(u))\big)\Big).$$

($h_\ell(E) = 0$ by 10.1; Tamagawa terms by 9.4.) Optimality of E is not used here. Paper: proof of Theorem thm:main (from "By Theorem thm:dfg27 the hypotheses of Lemma lem:dfg21 hold" to "each local term is ..."), Theorem thm:main.

11.2 `rem:blochkato:assembly` (remark; forward pointer, nothing proved). The assembly lives in Chapter 20: $v_\ell(\deg \varphi_E) = v_\lambda(\eta^{\emptyset}_{f,\lambda})$ (Chapter 14, 9.5: an equality for the optimal curve, important_notes.md points 39, 56; the DFG04 checklist of Chapter 14, 9.6 is settled, Chapter 20 outline, Part 7; important_notes.md point 73), $v_\lambda(\eta^{\Sigma}) = v_\lambda(\eta^{\emptyset}) + \sum_{p\in\Sigma} v_\ell(L_p^{\mathrm{nv}}(A_f,1)^{-1})$ (thm:dfg14, Chapter 20), $v_\lambda(\eta^{\Sigma}) = \operatorname{length} H^1_\Sigma(\mathbb{Q}, W)$ (thm:dfg37, Chapter 20), and 11.1. The naive factor equals the true one at multiplicative primes ($\delta_p = 1$) and is 1 at additive ones ($\delta_p = 0$), so the Euler terms that survive are $\sum_q e_\ell(E,q)$ (Definition def:terms). Result: (eq:identity) **with equality** for $\ell \notin S_f$ (no freeness hypothesis; Chapter 14, 9.5), and (eq:thm) from $\operatorname{length} H^1_f \geq 0$ and 8.3, 8.4 (including I^*_n primes at $q = 2$, point 18). The chapter gives this as a one-paragraph `remark` and a `paperlink`; it must not reprove Chapter 20's steps. Paper: Theorem thm:main and its proof.

11.3 `ex:blochkato:selmer-examples` (worked examples; P computed; the DFG04 checklist they depended on is settled, Chapter 20 outline, Part 7; important_notes.md point 73). At $\ell = 3$: 121a1 (type II at 11, $e_3 = v_3(12) = 1$, no potentially multiplicative prime, $\deg \varphi_E = 6$): excess 0, so $H^1_f(\mathbb{Q}, \mathrm{ad}^0\, T_3 \otimes \mathbb{Q}_3/\mathbb{Z}_3) = 0$. 791a1 (8.5): bound (eq:thm) 1, explicit bound 2, $v_3(144) = 2$: excess 0, the Tamagawa excess is needed for equality. 1184f1: explicit bound 2, $v_3(1728) = 3$: Selmer length 1. 1748e1: explicit bound 3, $v_3(3240) = 4$: Selmer length 1. (Each curve: $3 \nmid N$, $E[3]$ irreducible, image not 3Ns, non-CM.) Paper: Theorem thm:main, Section sec:sharp ("the identity (eq:identity) says what the excess is").

20.2.12 Section 12. Beyond the hypothesis: Remark rem:ns, $h_\ell(E)$, Open question 2

12.1 `rem:blochkato:rem-ns` (remark with proof of the conditional statement; P given 10.4, 10.5). Let $\ell = 3$ and the image of $\bar\rho_{E,3}$ be $N_s(\mathbb{F}_3)$, $3 \nmid N$. Then $H^0(\mathbb{Q}, V^D) = 0$ (2.6), $\Sigma \neq \emptyset$, $h_3(E) = 1$ (10.3), and **if** $H^1_f(\mathbb{Q}, V^D) = 0$ (equivalently $H^1_f(\mathbb{Q}, A_f) = 0$, 6.7), 10.5 gives Remark rem:ns's identity $\operatorname{length} H^1_\Sigma = \operatorname{length} H^1_f + \sum_p (v_\ell(L_p^{-1}) + v_\ell(\operatorname{Tam}_p)) - h_3(E)$. DFG04 Theorem 2.7 does not supply the vanishing of $H^1_f(\mathbb{Q}, V^D)$ here ($3 \in S_f$); the book states it as a hypothesis, as the paper's "whenever" does, and records that it is predicted by the Bloch-Kato conjecture (14.2) and that later modularity-lifting results may prove it (the writer cites one only after checking it; Part 7, N4). Paper: Remark rem:ns, Section sec:anomaly, Conjecture conj:euler.

12.2 `rem:blochkato:conj-euler-h` (remark; P). In Conjecture conj:euler the correction $h_\ell(E)$ is 0 for $\ell \geq 5$ and is 1 exactly for the curves with mod-3 image $N_s(\mathbb{F}_3)$ (10.3; any N); it is the term $H^0(\mathbb{Q}, W^D)^*$ of Lemma lem:dfg21; the appendix's `ns3(E)` (two quadratic factors of the 3-division polynomial) computes it (Proposition prop:ns; Chapter 11, 8.3). Worked example: 726a1 at $\ell = 3$ (outside Theorem thm:main as $3 \mid 726$): type III at 11 with $e_3 = v_3(12) = 1$, image 3Ns, `rule(E,3)` and `rule(E,3,1)` both return $[2, 2]$ (recomputed). Paper: Conjecture conj:euler, Section 1.3, Section sec:anomaly, the appendix.

12.3 `rem:blochkato:open-question-2` (remark; nothing proved). Open question 2 asks for $v_\lambda(\eta^{\Sigma}_{f,\lambda}) =$ length $H^1_\Sigma(\mathbb{Q}, W)$ when the image is $N_s(\mathbb{F}_3)$. On the Selmer side, 12.1 shows what the identity would give and which input is missing ($H^1_f(\mathbb{Q}, A_f) = 0$); on the congruence side the missing input is Taylor-Wiles for a residual representation induced from $\mathbb{Q}(\sqrt{-3})$ (Chapter 20). The data of Section sec:anomaly (failures by exactly 1, never more) are consistent with the identity with $-h_3(E)$. Paper: Open question 2, Remark rem:ns.

20.2.13 Section 13. Section sec:tam: testing the Tamagawa factors on the tables

13.1 `ex:blochkato:sec-tam` (worked example with data; P for the recomputation). The paper's test: for curves satisfying the hypothesis of Theorem thm:main with a multiplicative prime $q \equiv 1 \pmod{\ell}$, $\ell \mid n_q$, compare $v_\ell(\deg \varphi_E)$ with the bound raised by the Tamagawa excess (8.3). The paper reports, on the working set, no exception, and equality rising from 45.0% to 60.6% ($\ell = 3$, 46,222 curves) and from 67.0% to 78.7% ($\ell = 5$, 10,940 curves). Recomputation for conductors below 10,000 (optimal non-CM curves, $\ell \nmid N$, $E[\ell]$ irreducible, image not 3Ns, with the Tamagawa terms of 9.4 at all potentially multiplicative primes): at $\ell = 3$, 1,050 curves have a potentially multiplicative prime $q \equiv 1 \pmod 3$ with $3 \mid \nu_q$; equality rises from 596 (56.8%) to 714 (68.0%); 186 curves have a positive excess; at $\ell = 5$, 252 curves, $203 \to 220$; at $\ell = 7$, 18 curves, $17 \to 18$; no violation of the refined inequality at $\ell = 3, 5, 7$ among 13,203, 23,361 and 28,134 curves. The rates are not comparable with the paper's (smaller conductors, and I_n^* primes included here). Paper: Section sec:tam.

13.2 `rem:blochkato:what-sec-tam-tests` (remark; P). Given 11.1, 11.2 and the equality of Chapter 14, 9.5, a violation of the refined inequality is impossible, and the proportion of equality is the proportion of curves with $H^1_f(\mathbb{Q}, W) = 0$ (9.5). So Section sec:tam tests the Tamagawa term of the identity, as the paper says, but its phrase "the extra divisibility of the modular degree is exactly where the Tamagawa factor of the adjoint exceeds ℓ^{t_ℓ}" means only that the excess is accounted for in those cases; extra divisibility also arises from the Selmer group. Paper: Section sec:tam.

20.2.14 Section 14. The Tamagawa number conjecture and the adjoint motive

14.1 `def:blochkato:adjoint-motive` (definition and proposition; S). The adjoint motive $\mathrm{ad}^0 M_f$ of a weight-2 newform: its ℓ-adic realisations A_f (with $A_f \simeq \mathrm{Sym}^2 V \otimes \det^{-1}$, Chapter 4), Betti and de Rham realisations; pure of weight 0 with Hodge numbers $h^{1,-1} = h^{0,0} = h^{-1,1} = 1$; $L(\mathrm{ad}^0 f, s) = \prod_p L_p(A_f, s)$ (true Euler factors; at $p = \ell$ via D_{cris}, independent of ℓ), and $L(\mathrm{ad}^0 f, s) = L(\mathrm{Sym}^2 f, s+1)$ in the paper's normalisation $L_p(\mathrm{Sym}^2 f, s) = L_p(A_f, s-1)$ (Chapter 19). The point $s = 1$ is critical (Deligne79; DFG04 Section 1, verify), $L(\mathrm{ad}^0 f, 1) \neq 0$ (Shimura76; GelbartJacquet), and it is the value related to $\langle f, f\rangle$ (Chapter 19). Reference: DFG04 Section 1; Deligne79; Flach92. Paper: Section 1 ("the Selmer group of $\mathrm{Sym}^2 f$", "Rankin, Shimura"), Section 1.1.

14.2 `conj:blochkato:tnc` (conjecture; R). The Tamagawa number conjecture of Bloch and Kato, in the shape used here (the form of FPR94, verify): for a motive M over $\mathbb{Q}$ with $L(M, 0)$ critical and nonzero, a prime ℓ, a lattice T in the ℓ-adic realisation, $W = T \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$ and $W^D = T^D \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$,

$$\frac{L(M,0)}{\Omega(M)} \sim_\ell \frac{\prod_{p \neq \ell} \#\big(\mathbb{Z}_\ell/\operatorname{Tam}_p(T)\big) \cdot \#H^1_f(\mathbb{Q}, W^D)}{\#H^0(\mathbb{Q}, W) \cdot \#H^0(\mathbb{Q}, W^D)},$$

where $\sim_\ell$ means equality up to an ℓ-adic unit and the contributions at ℓ and ∞ are absorbed in the choice of the period $\Omega(M)$ (Bloch-Kato's Tamagawa measure). For $M = A_f(1)$ ($L(M, 0) = L(\mathrm{ad}^0 f, 1)$, $T = T^D$,

so the roles of W and W^D swap) the right side is $\prod \mathrm{Tam}_p(T^D) \cdot \#H^1_f(\mathbb{Q}, W)/(\#H^0(\mathbb{Q}, W^D)\#H^0(\mathbb{Q}, W))$: the adjoint Selmer group of Theorem thm:main, the Tamagawa factors of Lemma lem:fpr, and $h_\ell(E)$ in the denominator, the same sign as in Remark rem:ns. The book never uses the conjecture in a proof and says which normalisations differ between BlochKato Section 5, FPR94 and DFG04 Section 2 (verify each). Paper: Section 1 ("expresses them through the adjoint Selmer group and the adjoint L-value"), Theorem thm:main (context), Remark rem:ns.

14.3 `ex:blochkato:bsd` (example; R). For $M = h^1(E)(1)$ the conjecture is the Birch and Swinnerton-Dyer formula: $L(E,1)/\Omega_E$ against $\#\mathrm{Sha} \cdot \prod_p c_p/\#E(\mathbb{Q})^2_{\mathrm{tors}}$ (rank 0); the local terms $\mathrm{Tam}_p(T_\ell E)$ are the ℓ-parts of the c_p (7.6), the Selmer term is $\#\mathrm{Sha}[\ell^\infty]$ (6.5), and the H^0 terms are the torsion. Orientation only. Reference: BlochKato Section 6 (verify); Chapter 9. Paper: background.

14.4 `thm:blochkato:dfg-01` (theorem, DFG04 Theorem 0.1; R). For a newform f of weight $k \geq 2$ and $\lambda \notin S_f$, the λ-part of the Bloch-Kato conjecture holds for $L(\mathrm{ad}^0 M_f, 1)$ (and $L(\mathrm{ad}^0 M_f, 0)$). For $f = f_E$ the identity reads, up to the exact normalisation of the period in DFG04 Section 1.7 (verify): v_ℓ of the period-normalised $L(\mathrm{ad}^0 f, 1)$ (equivalently, by Hida's theorem and the Rankin-Shimura method, of the congruence ideal $\eta^{\emptyset}_{f,\lambda}$ up to the factors at Σ) equals length $H^1_f(\mathbb{Q}, W) + \sum_{p|N} v_\ell(\mathrm{Tam}_p(A))$ plus the Euler-factor terms. The book quotes the statement with the reference and explains that (eq:identity) is its reading through the modular degree (14.7). Reference: DFG04 Theorem 0.1 (as cited by the paper). Paper: Theorem thm:main ("the hypothesis is the one under which Diamond, Flach and Guo prove the Tamagawa number conjecture for the adjoint motive").

14.5 `thm:blochkato:dfg-02` (theorem, DFG04 Theorem 0.2 = Theorem 3.7 in the paper's reading; R; home: Chapter 20). If $\lambda \notin S_f$ and Σ is a finite set of primes, $\ell \notin \Sigma$, outside which $M_{f,\lambda}$ is minimally ramified, then $H^1_\Sigma(\mathbb{Q}, W)$ is finite of length $v_\lambda(\eta^{\Sigma}_{f,\lambda})$. Stated here only so that 11.1 can be read against it; the statement, the meaning of "minimally ramified" and the Taylor-Wiles sketch are Chapter 20's (thm:dfg37). Paper: Theorem thm:dfg37.

14.6 `thm:blochkato:flach` (theorem; R). (Flach 1992.) For $E/\mathbb{Q}$ modular and ℓ a prime outside an explicit finite set (including the primes of bad reduction and 2, 3 in the original statement; verify the exact hypotheses), the Bloch-Kato Selmer group of $\mathrm{Sym}^2 T_\ell E \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$ is finite, and is annihilated by an explicit integer related to $\deg \varphi_E$ (verify). The first place where the modular degree meets the symmetric-square Selmer group. Reference: Flach92. Paper: Section 1 ("Flach's finiteness theorem for the symmetric square ... is the first place where the modular degree meets the Selmer group of $\mathrm{Sym}^2 f$").

14.7 `rem:blochkato:degree-as-l-value` (remark; heuristic, clearly labelled). The chain: $\deg \varphi_E \cdot \mathrm{vol}(\Lambda_E) = 4\pi^2 c_E^2 \langle f, f\rangle$ (Chapter 14, 7.3); $\langle f, f\rangle$ is $L(\mathrm{Sym}^2 f, 2) = L(\mathrm{ad}^0 f, 1)$ times explicit factors and local factors at the bad primes (Chapter 19); 14.4 expresses $L(\mathrm{ad}^0 f, 1)$ over a period by Selmer and Tamagawa terms; the ratio of naive to true local factors produces $\sum e_\ell(E, q)$. Theorem thm:main is the theorem behind this heuristic, proved through congruence ideals instead of L-values. One paragraph; details Chapters 19 and 20. Paper: Section 1 (the classical route via the Petersson norm), Theorem thm:main.

20.3 3. Section-by-section plan with target line counts

Line counts are lines of LaTeX, card excluded (PLAN.md Section 7.3 allows up to 2,800 for this chapter). The card lists the 65 items of Part 2 (numbered 18.1 to 18.65 in the book; conventions C1 to C8 in one line each), about 210 lines, not counted. Total of the targets: **2,630**.

section	title	contents	target lines
(starred)	Core results at a glance	the card: every item of Part 2 with its tag and `\Cref`; the formulas of 7.1, 7.4, 8.2, 9.4, 10.4, 10.5, 11.1 displayed	(excluded)
(starred)	Where the paper uses this chapter	Section 1.2 (Tam_p, H^1_f); Theorem thm:main and its proof (thm:dfg27, lem:dfg21 and the paragraph after it, lem:fpr, lem:h0, the length bookkeeping, the I^*_n step); Lemma lem:tamfactor; Remark rem:ko(1); Remark rem:ns; Section sec:tam; Open questions 2, 3; Conjecture conj:euler (h_ℓ); the abstract's sentences on the Selmer group and the Tamagawa formula	45
1	Overview and conventions	the opening paragraph (Part 1.1, seven uses); C1 to C8 as `convention` environments; 1.1 with the roadmap table "paper sentence $\to$ item"; a `warning` box on the three duals (C2) and one on the two Frobenii (C3)	80
2	ℓ-adic representations, lattices, twists and duals	2.1 to 2.6; the 3×3 trace-form Gram matrix displayed; `paperlink`: Section 4.2 (trace form, $T^D \simeq A(1)$; homothety of $M_{f,\lambda}$, points 26, 40)	190
3	Local behaviour at $p \neq \ell$	3.1 to 3.5; the table of 3.2 (cited from Chapters 7, 11, 19); `paperlink`: the paragraph after Lemma lem:dfg21 (geometric Frobenius, point 11); a `warning` that $H^0(\mathbb{Q}_p, A_f) \neq 0$ at some bad primes (3.3)	170

section	title	contents	target lines
4	The representation at ℓ: p-adic Hodge theory as a black box	4.1 to 4.5, all R, one paragraph each; the dimension check $1+2=3$; the writer's check on Fontaine-Laffaille at $\ell=3$ (Part 7, Q1) as a footnote-sized remark, not as a claim	140
5	Bloch-Kato local conditions	5.1 to 5.5; the three exact sequences of 5.3 displayed; the worked example 5.5 in full ($E[\ell^\infty]$ at a split multiplicative prime)	240
6	Global Selmer groups and H^1_Σ	6.1 to 6.7; the classical Selmer group as 6.5; the Euler-characteristic count of 6.7 as a four-row table ($p\neq\ell$, ℓ, ∞, H^0)	190
7	Tamagawa factors	7.1 to 7.6; the definition with the parenthesis placed (point 37) and a `paperlink` quoting the paper's display; 7.4 proved in five numbered steps; 7.5 by cases	230
8	Lemma lem:tamfactor	8.1 to 8.5; the proof in six numbered steps, each with its source chapter in the margin of the text; the matrices displayed; a `paperlink` per corrected sentence (points 7, 11, 23, 31, 42, 18); the table of 8.5	280
9	Additive primes and Open question 3	9.1 to 9.5; the Jordan-block computation of 9.2 displayed; the explicit identity 9.4 displayed; the excess distribution table of 9.5	160
10	DFG04 Lemma 2.1 and $H^0(\mathbb{Q}, W^D)$	10.1 to 10.5; the five-term comparison sequence and its specialisation, with a "where each hypothesis enters" list; the length identity	240

section	title	contents	target lines
11	The Selmer half of Theorem thm:main	11.1 to 11.3; the assembly paragraph 11.2 with forward `\Cref`s only; the four worked curves	130
12	Beyond the hypothesis	12.1 to 12.3; 726a1	100
13	Section sec:tam	13.1, 13.2; the comparison table (paper versus recomputation)	80
14	The Tamagawa number conjecture and the adjoint motive	14.1 to 14.7; the BSD dictionary as a two-column table; the heuristic 14.7 in a clearly labelled `remark`	170
15	Exercises	the 16 exercises of Part 4.2	140
16	Notes and references	BlochKato, FPR94, Fontaine, Berger, Rubin, NSW, DFG04, Flach92, Wiles95, TateDuality, MilneADT, Greenberg89; which theorem numbers were cited and which are marked (verify); the history (Bloch-Kato 1990, Fontaine-Perrin-Riou 1994, Flach 1992, DFG 2004) in one paragraph	45

20.4 4. Coverage table and exercises

20.4.1 4.1 Coverage: PLAN.md scope, Section 6 map rows and paper hooks to line-up items

PLAN.md scope item, Section 6 row, or paper hook	line-up items
ℓ-adic representations of $G_{\mathbb{Q}}$, lattices T, $W = V/T$	2.1
Tate twists $V(n)$	2.2
duals V^* (here $V^\vee$), $V^D = V^*(1)$, T^D, W^D	2.3 (C2)
trace form on $\mathrm{ad}^0 V$, self-duality, $A(1) = T^D$ for ℓ odd	2.4
geometric representations; local behaviour at $p \neq \ell$, Weil-Deligne, inertia invariants, $L_p(V, s)$	2.5, 3.1, 3.2 (Weil-Deligne by citation of Chapter 7)
Frobenius convention; good prime eigenvalues $\alpha/\beta, 1, \beta/\alpha$; $L_p(A_f, 1)$ at a good prime	3.1, 3.2, C3

PLAN.md scope item, Section 6 row, or paper hook	line-up items
de Rham, crystalline, semistable; B_{dR}, B_{cris}, B_{st}; D_{cris}; $T_\ell E$ crystalline; adjoint crystalline	4.1, 4.2, 4.3 (and 4.4, 4.5)
Bloch-Kato local conditions at $p \neq \ell$, ℓ, ∞; H^1_g, H^1_e	5.1
conditions on T and W	5.2, 5.3, 5.5
global $H^1_f(\mathbb{Q}, V)$, $H^1_f(\mathbb{Q}, T)$, $H^1_f(\mathbb{Q}, W)$ and their relations	6.1, 6.2
the Σ-imprimitive Selmer group of DFG04 Section 2.1	6.3, 6.4, 6.6
duality of the Bloch-Kato conditions for V and V^D	5.4
FPR Tamagawa factor $\mathrm{Tam}_p(T)$ (definition as in the paper; FPR94 I.4.2.2; DFG04 p. 708)	7.1
the index description and Lemma lem:fpr; $\mathrm{Tam}_p(T^D) = \mathrm{Tam}_p(T)$	7.2, 7.4, 7.5
the Tamagawa factor of an elliptic curve is the ℓ-part of c_p	7.6, 8.2 (Step 6)
the full computation of Lemma lem:tamfactor; $v_\ell(\mathrm{Tam}) \geq t_\ell$ with the equality criterion	8.1 to 8.5
the Tamagawa factor at an additive prime (Open question 3)	9.1 to 9.5, 8.4
the Tamagawa factor at a good prime is trivial	7.3
the Tamagawa number conjecture; the shape for the adjoint; DFG04 Theorems 0.1, 0.2 with $\ell \notin S_f$	14.2, 14.4, 14.5, 10.1
the exact sequence of DFG04 Lemma 2.1 (lem:dfg21)	10.4
DFG04 Theorem 2.7 (thm:dfg27)	10.2, 6.7
$H^0(\mathbb{Q}, W^D)$ and Lemma lem:h0 (restated; proved in Chapter 11)	10.3, 2.6
the corrected sequence of Remark rem:ns ($-h_\ell(E)$)	10.5, 12.1
the adjoint motive; $L(\mathrm{ad}^0 f, s) = L(\mathrm{Sym}^2 f, s+1)$; Hodge numbers; critical values; Flach's theorem	14.1, 14.6
the symmetric square and the Petersson norm (connection)	14.7
Section 6 row: adjoint, Sym^2, trace form, $A_f(1) = \mathrm{Hom}(A_f, \mathbb{Z}_\ell(1))$ (with 4)	2.4, 2.6, 14.1
Section 6 row: Σ-imprimitive and Bloch-Kato Selmer groups, H^1_f, H^1_Σ (with 17)	5.1 to 5.5, 6.1 to 6.7
Section 6 row: FPR Tamagawa factor (with 3, 16)	7.1 to 7.6
Section 6 row: computation of $\mathrm{Tam}_q(\mathrm{ad}^0 T)$ at a multiplicative prime (with 16, 17)	8.1 to 8.5
Section 6 row: $h_\ell(E)$, $\bar{\rho} \simeq \bar{\rho} \otimes \chi$ (with 4, 11)	10.3, 12.2
Section 6 row: Taylor-Wiles, DFG04 Theorems 0.1, 0.2, 2.7, 3.7, Proposition 1.4, Lemma 2.1, Theorem 2.15 (with 20)	10.2, 10.4, 7.4 (Theorem 2.15), 14.4, 14.5 (3.7 and 1.4: Chapter 20)
Section 6 row: finite flat group schemes, "good at ℓ" (with 11)	4.2, 4.4
Section 6 row: Kummer classes, $H^1(G, \mathbb{Z}_\ell(1))$, tame character (with 5, 7, 16, 17)	8.1, 8.2 (Step 5)
hook: Section 1.2 (definition of $\mathrm{Tam}_p(M)$; the Bloch-Kato Selmer group)	7.1, 7.3, 6.1, 6.6
hook: Theorem thm:main and its proof in Section 4.3	11.1, 11.2, 9.4, 10.5

PLAN.md scope item, Section 6 row, or paper hook	line-up items
hook: Theorem thm:dfg37	14.5 (home Chapter 20), 6.4
hook: Theorem thm:dfg27	10.2
hook: Lemma lem:dfg21 and the paragraph after it	10.4, 5.3, 3.3, 3.4
hook: Lemma lem:fpr	7.4, 7.5, 3.5
hook: Lemma lem:h0	10.3
hook: Lemma lem:tamfactor (in full)	8.2 (with 8.1, 8.3, 8.5)
hook: Remark rem:ko(1)	8.3(iii), 8.5
hook: Remark rem:ns	12.1, 10.5
hook: Section sec:tam	13.1, 13.2
hook: Open question 2	12.3
hook: Open question 3	9.3, 9.4, 9.5
hook: Conjecture conj:euler (the term $h_\ell(E)$)	12.2, 10.3
also: the abstract ("the length of the Bloch-Kato Selmer group of $\mathrm{ad}^0 V_\ell E$"; "raises the bound by one whenever ...")	6.6, 8.3 (Part 7, N2, N3)
also: Section sec:sharp ("the identity (eq:identity) says what the excess is")	9.5, 11.3
also: what Chapters 3, 4, 11, 15, 16, 21 defer to Chapter 18 (Part 5)	2.4, 4.2, 4.4, 7.1, 7.4, 8.2, 10.3, 10.4

20.4.2 4.2 Exercises (Section 15 of the chapter), graded from easy to harder

1. (Frobenius conventions.) Compute $L_p(\mathbb{Q}_\ell(1), s)$ with Φ_p and with Frob_p. Show $L_p(M(1), s) = L_p(M, s+1)$ with Φ_p. Check the paper's claim $L_p(A_f, 1)^{-1} = L_p(A_f(1), 0)^{-1} = 1 - p^{-2}$ at a multiplicative prime, and show that the arithmetic Frobenius would give $1 - p^2$ (same ℓ-adic valuation, different number).
2. Compute the Gram matrix of the trace form on $\mathrm{ad}^0 T$ in the basis e, h, f, show $T^D \simeq A(1)$ for ℓ odd, and show the image of A in $A^\vee$ has index 2 when $\ell = 2$ (2.4).
3. For each row of the table 3.2 find $H^0(\mathbb{Q}_p, V^D)$ and $H^0(\mathbb{Q}_p, A_f)$, and (granting the local Euler characteristic) the $\mathbb{Z}_\ell$-rank of $H^1(\mathbb{Q}_p, T^D)$. Which bad primes have $H^1(\mathbb{Q}_p, T^D)$ infinite? Check the paper's claim that $H^1_f(\mathbb{Q}_p, T^D)$ is the torsion of $H^1(\mathbb{Q}_p, T^D)$ in each case.
4. Let q be a multiplicative prime. Show $H^1_f(\mathbb{Q}_q, W) = 0$ for $W = \mathrm{ad}^0 T \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$, and that $H^1_{\mathrm{ur}}(\mathbb{Q}_q, W) = 0$ when $\ell \nmid n_q$ ($W^{I_q} = (\mathbb{Q}_\ell/\mathbb{Z}_\ell)e$). For $W = E[\ell^\infty]$, split reduction and $\ell \mid n_q$, show $H^1_{\mathrm{ur}}(\mathbb{Q}_q, W) \simeq \mathbb{Z}/\ell^{v_\ell(n_q)}$ while $H^1_f(\mathbb{Q}_q, W) = 0$ (5.5).
5. Check the paper's claim $H^1(I_q, \mathrm{ad}^0 T)_{\mathrm{tor}} \simeq (\mathbb{Z}/\ell^{v_\ell(n)})^2$ for 14a1 at $q = 7$, $\ell = 3$ ($n = 3$, $\bar{u} \equiv -1$), compute the Frobenius invariants and $v_3(\mathrm{Tam}_7(\mathrm{ad}^0 T)) = 2$, and explain why $\deg \varphi_{14a1} = 1$ is no contradiction ($E[3]$ is reducible; Lemma lem:tamfactor needs no irreducibility).
6. Check the paper's claim that c' is "well defined modulo $q-1$": show that changing the lift of Frob by t^a and the cocycle by a coboundary changes c' by an element of $(n, q-1)\mathbb{Z}_\ell$ (not only of $(q-1)\mathbb{Z}_\ell$), that $\min(v_\ell(n), v_\ell(q-1), v_\ell(c'))$ is unchanged, and that the "image of κ" exists through the splitting given by the uniformiser q, but not through a map defined for every uniformiser (point 7 and its addendum).
7. Check the paper's claim that "the same computation for $M = T$ gives ... the ℓ-part of the Tamagawa number $c_q(E)$": for 11a1 at $q = 11$, $\ell = 5$ (split, $c_{11} = 5$) and for 14a1 at $q = 2$, $\ell = 3$ (nonsplit, $n = 6$, $c_2 = 2$).
8. Prove $\mathrm{Tam}_q(A(1)) = \mathrm{Tam}_q(A)$ at a multiplicative prime by computing the invariants of $H^1(I_q, A(1))_{\mathrm{tor}}$ as the kernel of the linear form $(x, y) \mapsto (q-1)x - 2\varepsilon c' y$ on $(\mathbb{Z}/\ell^v)^2$, and show both invariant modules are $\mathbb{Z}/\ell^v \oplus \mathbb{Z}/\ell^m$.
9. Prove 9.2: for $\sigma \in \mathrm{GL}_2(\mathbb{Z}_3)$ of order 3, $\mathrm{ad}^0 T$ is free over $\mathbb{Z}_3[\langle\sigma\rangle]$ while $H^1(\langle\sigma\rangle, T) \simeq \mathbb{F}_3$. Deduce that $\mathrm{Tam}_q(\mathrm{ad}^0 T_3 E)$ is trivial at a prime $q \geq 5$ of type IV, while $\mathrm{Tam}_q(T_3 E)$ is 3 when $c_q = 3$ (for

example 50a1 at $q = 5$, type IV, $c_5 = 3$).

10. Check the paper's claim in Remark rem:ko(1) on 791a1 ($[1, 0, 1, -31, 117]$, $N = 7 \cdot 113$): $n_7 = 3$, $\bar{u} \equiv -1 \pmod 7$ is a cube, so $v_3(\mathrm{Tam}_7(\mathrm{ad}^0 T)) = 2$; the bound (eq:thm) is 1, the refined bound 2, and $v_3(\deg \varphi_E) = v_3(144) = 2$.
11. Check the paper's claim in Lemma lem:fpr that $L_p(A_f, 1)^{-1}$ is an ℓ-adic integer of valuation $e_\ell(E, p)$ at an additive prime, for 121a1 at $p = 11$ (type II, $\ell = 3$): $1 - \chi_{-3}(11)/11 = 12/11$, $v_3 = 1$.
12. Prove the surjectivity at the right end of the sequence of Lemma lem:dfg21 (10.4, step 4), and show by an example of the general shape what fails when $\Sigma = \emptyset$.
13. Check the paper's claim (proof of Theorem thm:main) that $H^0(\mathbb{Q}, W^D) = 0$ under its hypothesis, by the determinant argument; compute $h_3(726a1) = 1$ and explain why the paper's finiteness argument via $a_p = 0$ at inert primes is not the right one (points 5, 6).
14. Verify the four local terms of 6.7 and deduce $H^1_f(\mathbb{Q}, A_f) = 0 \iff H^1_f(\mathbb{Q}, V^D) = 0$. Where is the sign convention C4 used?
15. Check the paper's claims of Section sec:tam on a small range: with the series $\mathtt{q}(j)$ of Chapter 10, 7.2, compute Tate parameters and unit parts, and reproduce, for conductors at most 2000 and $\ell = 3$, the counts 128 curves in the population, equality $87 \to 102$, and 19 curves with a positive Tamagawa excess (13.1; Part 6.6).
16. (Harder.) Show that the shape 14.2 for $M = V_\ell E$ reproduces the local and torsion terms of the Birch and Swinnerton-Dyer formula, and for $M = A_f(1)$ identifies each term with a term of (eq:identity); where does $h_\ell(E)$ appear?

20.5 5. Dependencies and forward pointers

Results from earlier chapters used here (cited with `\Cref{ch:...}`, never reproved): * Chapter 2 (`ch:rings`): v_ℓ of rationals $(1 - p^{-2}$, $(q + 1)^2 - a^2)$, the ℓ^k-th powers in $\mathbb{F}_q^\times$ and the index $\ell^{\min(k, v_\ell(q-1))}$ (for $c(u)$). * Chapter 3 (`ch:modules`): Fitting ideals and lengths over $\mathbb{Z}_\ell$ (7.4 to 7.6), Pontryagin duality and T versus $T \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$ (8.1 to 8.4), lengths in exact sequences. * Chapter 4 (`ch:reps`): the trace form and $\mathrm{Hom}(A, \mathbb{Z}_\ell(\chi)) \simeq A(\chi)$ (7.6); invariance of ad^0 (7.7); $\mathrm{ad}^0 \simeq \mathrm{Sym}^2 \otimes \det^{-1}$; unipotent and cyclic shapes (9.1 to 9.4); the twist-length lemma (10.5). * Chapter 5 (`ch:galois`): $\mathbb{Z}_\ell(1)$, χ_ℓ, Kummer cocycles (galois:l-adic-kummer), $\mathbb{F}_q$-Kummer theory (galois:kummer-finite-field), the determinant obstruction (galois:twist (c)). * Chapter 6 (`ch:ant`): $\mathbb{Z}_3[\zeta_3]$ as a DVR and $T \simeq \mathbb{Z}_3[\zeta_3]$ (4.7, 4.8); the commutant (4.10). * Chapter 7 (`ch:local`): L3, L6, 4.3 (Frobenius dictionary), 5.2 to 5.5 (inertia, tame character, Frob t $\mathrm{Frob}^{-1} = t^q$, Frobenius fixing roots), 6.2 (the Kummer class on inertia and $\kappa = a\kappa_\varpi + \kappa_u$), 8.1, 8.2 ($V^I$, L-factors), 9.1 to 9.4 (local shapes and adjoint L-factors). * Chapter 9 (`ch:ellcurves`): Weil pairing, Hasse bound, c_p and the BSD orientation. * Chapter 10 (`ch:reduction`): Tate parameter $\mathtt{q}(j)$ and unit part (7.2, 7.4, 7.8), c_q at multiplicative primes (7.6), potentially multiplicative additive reduction (7.7), Kodaira types, Kraus's defect. * Chapter 11 (`ch:galrep`): the Tate shape with corner $\varepsilon\kappa$ (3.2), Lemma lem:tamagawa-exp (3.3), potentially multiplicative primes at every q (3.4), inertia images by type (Section 4), crystalline at ℓ (5.1), Lemma lem:h0 (8.5), S_f (8.6). * Chapter 14 (`ch:moddeg`): 9.5 (Lemma lem:hecke as an equality: points 39, 56) and the checklist 9.6; 9.1(d) ($M_{f,\lambda} \simeq T_\ell E(-1)$); 7.3 (Zagier-Watkins, for 14.7). * Chapter 15 (`ch:modgalrep`): 4.1 (crystalline at ℓ, Fontaine-Laffaille range for M_f). * Chapter 16 (`ch:cohomology`): continuous-cohomology, les-continuous, h0-and-torsion, inflation-restriction, res-inf-cor (d), order-kills, z2-on-l-adic, conjugation-action, cohomology-through-pro-l-quotient, tame-inertia-example, procyclic-compact, cocycle-on-powers, frobenius-on-h1-inertia, tate-curve-shape-matrices, model-computation, fitting-ideal-of-invariants, lengths-in-exact-sequences, h1-finitely-generated, galois-cohomology-convention, unramified-h1-preview. * **Chapter 17** (`ch:galcoh`; its outline is now written, and the inputs below are its items, given in brackets): **K1** [3.3, 3.4] Tate local duality $H^i(\mathbb{Q}_p, M) \times H^{2-i}(\mathbb{Q}_p, M^D) \to \mathbb{Q}_\ell/\mathbb{Z}_\ell$ for finite M, and its versions for (W, T^D) and (V, V^D); **K2** [3.2] the local Euler characteristic formula (for $p \neq \ell$: $\#H^0 \#H^2/\#H^1 = 1$ for finite M; ranks for lattices); **K3** [4.1 to 4.3] $H^1_{\mathrm{ur}}(\mathbb{Q}_p, M) = M^{I_p}/(\mathrm{Frob}_p - 1)M^{I_p}$ and the exact sequence $0 \to H^1_{\mathrm{ur}} \to H^1 \to H^1(I_p, M)^G \to 0$; **K4** [4.4, 4.5] for $p \neq \ell$, H^1_{ur} of M and of M^D are exact annihilators (finite M of order prime to p; the

divisible and compact versions); **K5** [10.1 to 10.4] the comparison exact sequence for two Selmer structures $\mathcal{F} \subseteq \mathcal{G}$ (Poitou-Tate), in the five-term form used in 10.4, with discrete coefficients W and compact dual T^D; **K6** [2.2] the sequences relating $H^1(K,T)$, $H^1(K,V)$, $H^1(K,W)$ and $H^1(K,T)_{\mathrm{tor}} = \delta H^0(K,W)$; **K7** [9.3, 9.5] the Greenberg-Wiles formula (finite coefficients) and, if Chapter 17 states it, its $\mathbb{Q}_\ell$ version (otherwise 6.7 states the $\mathbb{Q}_\ell$ version as R); **K8** [5.1 to 5.4] the Kummer class of the Tate parameter and the identification of $H^1(G/I, \mathbb{Z}_\ell(1)) = \mathbb{Z}_\ell(1)/(q-1)$ with $\mathbb{F}_q^\times \otimes \mathbb{Z}_\ell$ (the cohomological half of point 7). Chapter 17 also proves $H^1_f(\mathbb{Q}_p, T^D) = H^1(\mathbb{Q}_p, T^D)_{\mathrm{tor}} \simeq H^0(\mathbb{Q}_p, W^D)$ for $p \in \Sigma$ (its 4.6) and the abstract length identity (its 10.4); Chapter 18 cites them. If any input turns out to be missing in the written Chapter 17, Chapter 18 states it as R with reference (NSW, MilneADT, Rubin) rather than proving it.

Black boxes imported from later chapters (stated with a forward `\Cref`, used only in remarks, examples and `paperlinks`): the values $e_\ell(E,q)$ and Lemma lem:euler (Chapter 19; used in 3.2, 3.5, 11.3); $L(\mathrm{Sym}^2 f, s)$, Rankin-Shimura and $L(\mathrm{ad}^0 f, 1) \neq 0$ (Chapter 19; used in 14.1, 14.7); thm:dfg14, thm:dfg37, Lemma lem:hecke and the assembly (Chapter 20; used in 11.2, 14.5).

Forward pointers (chapters that use this one): * Chapter 19 (`ch:lfunctions`): 3.1 (Frobenius dictionary for L-factors), 3.2, 3.5 (integrality of $L_q(A_f,1)^{-1}$), 14.1 (the adjoint motive and $L(\mathrm{ad}^0 f, s) = L(\mathrm{Sym}^2 f, s+1)$), 14.2 (the shape of the Bloch-Kato conjecture for the adjoint). * Chapter 20 (`ch:congruence`): 10.1 (S_f), 10.2 (thm:dfg27), 10.4 (lem:dfg21), 7.4, 7.5 (lem:fpr), 10.3 (lem:h0 in Selmer form), **11.1 (the Selmer half of the proof of Theorem thm:main)**, 9.4 (the explicit Tamagawa terms), 12.1 (Remark rem:ns, Open question 2), 14.4 (DFG04 Theorem 0.1). Chapter 20 should cite 11.1 rather than repeat the Selmer bookkeeping. * Chapter 21 (`ch:paper`): A1.3 (cite 7.1, 7.3), A2.7 (cite 8.2 and 8.4; the step list there matches 8.2), A4.6 to A4.11 (cite 6.3, 10.1 to 10.5, 7.4, 7.5), the proof of Theorem thm:main (11.1, 11.2), Remark rem:ns (12.1), Section sec:tam (13.1), Open questions 2 and 3 (12.3, 9.3 to 9.5). * Chapter 22 (`ch:computation`): 8.5 and the script of Part 6.6 (Tate parameters via `q(`j`)` or `ellinit(E,O(q^k)).tate`; unit parts; the Tamagawa terms), 13.1 (reproducing Section sec:tam), 9.5 (the excess distribution).

20.6 6. Notes for the writer

20.6.1 6.1 Conventions to typeset (Section 1, as `convention` environments)

C1 to C8 of Part 1.3. In addition two `warning` boxes in Section 1: "**Three duals.** $M^\vee = \mathrm{Hom}(M, \mathbb{Z}_\ell)$ (linear; Bloch-Kato's V^*), $M^D = \mathrm{Hom}(M, \mathbb{Z}_\ell(1))$ (Kummer-Cartier; the paper's ^{D}), $X^* = \mathrm{Hom}(X, \mathbb{Q}_\ell/\mathbb{Z}_\ell)$ (Pontryagin; the paper's *). Tate local duality pairs $H^1(\mathbb{Q}_p, W)$ with $H^1(\mathbb{Q}_p, T^D)$, and the Pontryagin dual of a finite group is the only * in Lemma lem:dfg21." and "**Two Frobenii.** Frob_p arithmetic in group theory (Lemma lem:tamfactor), $\Phi_p = \mathrm{Frob}_p^{-1}$ in L-factors (Section 1.1, Lemma lem:fpr, the paragraph after Lemma lem:dfg21); the paper names neither (important_notes.md point 11). In cohomology the choice is invisible up to ℓ-adic units; in L-factors it is not."

20.6.2 6.2 Pitfalls

- **Tate twists and Frobenius.** On $\mathbb{Z}_\ell(1)$ the arithmetic Frobenius at $p \neq \ell$ acts by p and the geometric one by p^{-1}. So on $(V^D)^{I_p} = A_f^{I_p}(1)$ at a multiplicative prime Frob_p acts by $p \cdot p = p^2$ and Φ_p by p^{-2}; $\det(1 - \Phi_p) = 1 - p^{-2} = L_p(A_f,1)^{-1}$. Never write $1 - q^{1-s}$ for the multiplicative adjoint factor (point 11).
- **The dual V^D is $V^\vee(1)$, not $V(1)$.** For A_f they agree only through the trace form, which needs ℓ odd (2.4). Write the isomorphism once, name it, and use $A(1)$ thereafter as the paper does.
- **"Unramified" is not "f" for W.** $H^1_f(\mathbb{Q}_p, W)$ is the divisible part of $H^1_{\mathrm{ur}}(\mathbb{Q}_p, W)$ (5.3(iv)); the difference is exactly the Tamagawa phenomenon (5.5). For T the inclusion goes the other way: $H^1_{\mathrm{ur}}(\mathbb{Q}_p, T) \subseteq H^1_f(\mathbb{Q}_p, T)$ with index $\ell^{v_\ell(\mathrm{Tam}_p(T))}$ (7.2). DFG04's H^1_Σ imposes the unramified condition outside $\Sigma \cup \{\ell\}$, which equals the f condition there only because W is unramified there (6.3).

- $H^1(\mathbb{Q}_p, T^D)$ **need not be finite at** $p \in \Sigma$ (3.3, 3.4: type III with $p \equiv 1 \pmod 4$, every I_0^* prime); the paper's "$H^1_f(\mathbb{Q}_p, T^D)$ is the torsion of $H^1(\mathbb{Q}_p, T^D)$, a finite group" is right as stated; do not "simplify" it to "$H^1(\mathbb{Q}_p, T^D)$ is finite".
- **The parenthesis in** Tam_p (point 37): $\mathrm{Fitt}\big((H^1(I_p, M)_{\mathrm{tor}})^{G_{\mathbb{Q}_p}}\big)$, the Fitting ideal of the invariants.
- **The class** c' (point 7 and its addendum): "the image of κ in $H^1(G/I, \mathbb{Z}_\ell(1)^I)$" is meaningful only via the canonical splitting given by the uniformiser q (Chapter 17, 5.3); c' is a cocycle value, equal to ε times that image only for the lift Frob_0 fixing the ℓ-power roots of q, and in general well defined modulo $(n, q-1)\mathbb{Z}_\ell$, not $q-1$; only the minimum matters.
- **The corner** $\varepsilon\kappa$ (point 42) and **arithmetic** Frob **in Lemma lem:tamfactor** (point 11): $\rho(\mathrm{Frob}) = \left(\begin{smallmatrix} \varepsilon q & \varepsilon\kappa(\mathrm{Frob}) \\ 0 & \varepsilon \end{smallmatrix}\right)$.
- $v_\ell(c(u))$ **when** $\bar{u}$ **is an** $\ell^{v_\ell(q-1)}$**-th power** (point 31): read as $v_\ell(q-1)$; when $\ell \nmid q-1$ the minimum is 0 whatever $\bar{u}$ is.
- c_q **means the Tamagawa number** (point 23); the class of the unit part is $c(u)$ or $c_q(u)$.
- I_n^* **at** $q = 2$ (point 18): use 8.4 (local twist over $\mathbb{Q}_2$), not Lemma lem:twist, and $\nu = -v_2(j)$, not the index of the symbol.
- $h_\ell(E)$: the criterion is $\bar{\rho} \simeq \bar{\rho} \otimes \bar{\chi}_\ell$; $h_\ell = 0$ for $\ell \geq 5$ and $h_3 \leq 1$ for every E (points 5, 6, 14). The paper's sentence on finiteness for non-CM curves is replaced by the determinant argument; do not repeat "for E without complex multiplication it is finite".
- **Hodge-Tate sign** (C4): fix it once; 4.5 depends on it; when quoting DFG04 or BlochKato check their sign.
- **The equality in (eq:identity)** is Chapter 14's theorem (point 39); the DFG04 checklist (point 56) is settled (Chapter 20 outline, Part 7; important_notes.md point 73): say "with equality whenever $\ell \notin S_f$, by `\Cref{ch:moddeg}` and `\Cref{ch:congruence}`", never "with equality when multiplicity one holds".
- **Remark rem:ns is conditional** on $H^1_f(\mathbb{Q}, V^D) = 0$ for the 3Ns curves (12.1); state the condition every time.
- **Section sec:tam's population** in the paper is type I_n only; 9.4 shows that I_n^* primes carry the same Tamagawa excess; say so, do not change the paper's numbers.
- **Theorem numbers marked (verify)**: BlochKato Definition 3.7, Proposition 3.8, Corollary 3.8.4, Example 3.11, Section 5, Section 6; FPR94 I.3, I.4.2.2, III; Rubin 1.3, 1.4.3, 1.5, 1.6, 1.7.3; DDT 2.19; Wiles95 Proposition 1.6; DFG04 Sections 1.1 to 1.7, 2.1 to 2.3, (31), (57), Theorems 0.1, 0.2, 2.7, 2.15, 3.7, p. 708 (cite as the paper cites them); Kraus90 for the inertia groups at 2 and 3; Flach92's hypotheses; SGA 7 Expose IX for 7.6. Cite by section unless checked.

20.6.3 6.3 What to leave to Chapters 7, 11, 16, 17, 19, 20

- **Chapter 7**: Frobenius conventions, Weil-Deligne representations, L-factors and the local shapes table; Kummer classes on inertia and the field-theoretic identification of c'. Chapter 18 cites them (3.1, 3.2, 8.1).
- **Chapter 11**: the Tate shape, Lemma lem:tamagawa-exp, the potentially multiplicative case, inertia images by Kodaira type, crystallinity at ℓ (as a statement), **Lemma lem:h0 in corrected form (its 8.5)** and S_f (8.6). Chapter 18 restates lem:h0 in Selmer form (10.3) and does not reprove the determinant argument beyond one sentence.
- **Chapter 16**: all group cohomology, in particular the computation $(t-1)M = n'(\mathbb{Z}_\ell e \oplus \mathbb{Z}_\ell h)$, $H^1 = (\mathbb{Z}_\ell/n')^2 \oplus \mathbb{Z}_\ell$, the Frobenius action and the invariants (its Section 10). Chapter 18's 8.2 cites Steps 2 to 4 from there and adds Steps 1, 5, 6 and the definition.
- **Chapter 17**: Tate local duality, local Euler characteristic, H^1_{ur} and its annihilator, Poitou-Tate and Greenberg-Wiles, the comparison sequence for Selmer structures, the T, V, W sequences, the Kummer class of the Tate parameter (inputs K1 to K8). Chapter 18 instantiates them with the Bloch-Kato conditions; it does not reprove them.
- **Chapter 19**: the table of omitted Euler factors, Lemma lem:euler, $e_\ell(E, q)$, P_q, the naive factor (point 30), $L(\mathrm{Sym}^2 f, s)$, Rankin-Shimura. Chapter 18 uses only the values and their integrality.
- **Chapter 20**: congruence ideals, thm:dfg14, thm:dfg37 (Taylor-Wiles), Lemma lem:ddt417 (point 19

lives there), Lemma lem:hecke, the assembly of Theorem thm:main, Theorem thm:quant, Remark rem:ko(2). Chapter 18 supplies 11.1 and states 14.5 only as a pointer.

20.6.4 6.4 How each relevant point of important_notes.md is handled

- **Point 5** ($h_\ell = 0$ for $\ell \geq 5$, $h_3 \leq 1$, CM irrelevant; the paper's finiteness argument confuses χ_ℓ with χ_{ℓ^*}): 10.3(ii), citing Chapter 11, 8.5; Exercise 13; pitfall in 6.2.
- **Point 6** (the criterion for $h_\ell > 0$ is $\bar{\rho} \simeq \bar{\rho} \otimes \bar{\chi}_\ell$): 10.3(ii), 12.2; the `paperlink` for Section 1.3 quotes the corrected wording.
- **Point 7** (with the addendum from the Chapter 17 outline): the paper's "image of κ in $H^1(G/I, \mathbb{Z}_\ell(1)^I)$" is read via the canonical splitting $H^1(\mathbb{Q}_q, \mathbb{Z}_\ell(1)) = \mathbb{Z}_\ell \kappa_q \oplus H^1_{\mathrm{ur}}$ given by q; $c' = \varepsilon \cdot$(that image) for Frob_0; indeterminacy $(n, q-1)$, not $q-1$: 8.2, Step 5, citing Chapter 17, 5.3 and 5.4; Exercise 6.
- **Point 11** (geometric Frobenius in L-factors, arithmetic in Lemma lem:tamfactor): C3, 3.1, 3.3, 8.2 Step 4; warning box; Exercise 1.
- **Point 12** ($\mathrm{End}_{I_q}(T) = \mathbb{Z}_\ell[\sigma]$, not $\mathrm{End}_{I_q}(V)$ a field): 9.2 works with $T \simeq \mathbb{Z}_3[\zeta_3]$ and says "the commutant of σ in $\mathrm{End}(T)$ is $\mathbb{Z}_3[\sigma]$"; the V-statement is never used.
- **Point 19** ($\wp \cap I = 0$ in Lemma lem:ddt417): not used in Chapter 18; it belongs to Chapters 3 and 20. Mentioned only in 6.3.
- **Point 26** ($M_{f,\lambda}$ is homothetic to the dual of $T_\ell E$): 2.4(ii), citing Chapter 14, 9.1(d).
- **Point 31** (the reduction of u; the reading of $v_\ell(0)$): C6, 8.2.
- **Point 37** (the parenthesis in Tam_p): 7.1, with a `paperlink` quoting the display.
- **Point 39** (Lemma lem:hecke is always an equality, so (eq:identity) is an equality for $\ell \notin S_f$): 9.4, 11.2, citing Chapter 14, 9.5; consequence in 9.5 and 11.3 (the excess is the Selmer length).
- **Point 40** ($M_{f,\lambda} = \varphi_E^* H^1(E, \mathbb{Z}_\ell) \simeq T_\ell E(-1)$ exactly): 2.4(ii).
- **Point 42** (corner $\varepsilon\kappa$): C6, 8.1(a), 8.2 Step 4.
- **Point 47** ($\ell \notin S_f$ iff $E[\ell]$ irreducible and $h_\ell(E) = 0$, for $\ell \nmid 2N$): 10.1.
- **Point 56** (the DFG04 checklist): settled by point 73 (Chapter 20 outline, Part 7); 11.2 and 9.4 cite Chapter 14, 9.6 and the Chapter 20 outline's Part 7.
- **B1** (on $V_\ell E$ the arithmetic Frobenius has eigenvalues α, β): C3, 3.2.
- **B6** (PLAN.md's Weil-Deligne relation is the geometric form): not needed here; C3 follows Chapter 7's L3, L4.
- **B7** (Chapter 21's C2(b): $\det(1 - \Phi p^{-s} \mid V_\ell E)^{-1} = L_p(E, s+1)$, and point 42 in its A2.7): C3, 3.1(d); 8.2 is the version Chapter 21's A2.7 should cite.
- **B14** (bibliography): checked `refs.bib` and every `bib/*.bib`. All keys this chapter needs are already in `refs.bib`: BlochKato, FPR94, DFG04, Fontaine, Berger, FontaineLaffaille, Flach92, Rubin, NSW, MilneADT, TateDuality, Greenberg89, Wiles95, Washington, Deligne79, Shimura76, GelbartJacquet, Kraus90, SilvermanATAEC, DDT, Hida81, Tate95, PARI, ecdata. The chapter bib files are empty except `bib/chA.bib`. Proposed new keys: none required. Optional, only if the writer cites them after checking: `SerreTate68` (already proposed for refs.bib by B16; needed here for the inertia groups in 9.3), `SGA7` (Grothendieck, *Modeles de Neron et monodromie*, SGA 7 I, Expose IX, Lecture Notes in Math. 288, 1972; for 7.6; Chapter 10 may also want it, so refs.bib), `ColemanIovita99` (R. Coleman, A. Iovita, *The Frobenius and monodromy operators for curves and abelian varieties*, Duke Math. J. 97 (1999); for the converse in 4.2; bib/ch18.bib). Bibliographic details recalled, not re-verified.

20.6.5 6.5 References

Main sources, in the order of the chapter: Rubin (Euler Systems, Chapter 1, for local and global conditions, propagation, the comparison sequence; cite sections), BlochKato (Sections 3 to 6), FPR94 (I.3, I.4, III), Fontaine and Berger (for B_{dR}, B_{cris}), FontaineLaffaille, NSW and MilneADT (local duality, Poitou-Tate), Wiles95 (Greenberg-Wiles), DFG04 (Sections 1, 2, 3; (31), (57), p. 708; Theorems 0.1, 0.2, 2.7, 2.15; Lemma 2.1), Flach92, Deligne79 (critical values), Shimura76 and GelbartJacquet (the symmetric square),

Kraus90 (inertia images), the paper's own references for the data (Cremona97, PARI, ecdata).

20.6.6 6.6 Verification log (computations done for this outline)

Python 3 with PARI/GP 2.17 through `cypari2`; scripts `h1c3.py`, `tate2.py`, `tamcheck.py`, `tamcheck2.py`, `brute.py`, `rule.gp` in the session scratchpad (`ch18/`); Cremona's `ecdata` files (`allcurves`, `degphi`, `galrep`) for conductors below 10,000 in `/tmp/ch09/data`. * **Tate parameters.** $\mathtt{q}(j)$ as `serreverse(1/ellj(x+O(x^60)))` (Chapter 10, 7.2: $j^{-1}+744j^{-2}+750420j^{-3}+\cdots$), evaluated at $1/j_E$ in $\mathbb{Q}_q$; agrees with `ellinit(E,O(q^40)).tate[3]` for 11a1 at 11, 14a1 at 7 and at 2. Unit parts as in 8.5 (11a1: 10; 14a1 at 7: 6, at 2: 1; 539c1: 5 at 7, 3 at 11, agreeing with Chapter 10, 7.8; 791a1: 6 at 7, 106 at 113; 874b1: 18 at 19; 1184f1: 36 at 37; 1748e1: 12 at 19). * **Brute force of Lemma lem:tamfactor and of 7.5** (`brute.py`): for $\ell\in\{3,5\}$, $v\in\{0,1,2\}$, all primes $q<120$, $q\neq\ell$, $\varepsilon=\pm1$, $c'\in\{0,\dots,\ell^3-1\}$, the number of Frob-invariants of $(\mathbb{Z}/\ell^v)\bar e\oplus(\mathbb{Z}/\ell^v)\bar h$ for the action $q^{-1}\operatorname{Ad}\rho(\mathrm{Frob})$ (module A) and $\operatorname{Ad}\rho(\mathrm{Frob})$ (module $A(1)$) equals $\ell^{v+\min(v,v_\ell(q-1),v_\ell(c'))}$ in all 52,896 cases. * $\mathrm{Tam}(T)$ **versus** c_q at multiplicative primes: for all curves of `allcurves` below conductor 3000 and $\ell\in\{3,5,7\}$, $\ell\neq q$, $v_\ell(c_q)=v_\ell(n_q)$ if $a_q=1$ and 0 if $a_q=-1$: 101,336 pairs, no mismatch. * **9.2** (`h1c3.py`): with $\sigma=\left(\begin{smallmatrix}0&-1\\1&-1\end{smallmatrix}\right)$ acting on ad^0T by conjugation (basis e,h,f), $\ker N=(\sigma-1)\,\mathrm{ad}^0T$ (Smith form of the inclusion: $[1,1]$), so $H^1(C_3,\mathrm{ad}^0T)=0$; $\{e,\sigma e,\sigma^2e\}$ is a $\mathbb{Z}$-basis (determinant ±1), so $\mathrm{ad}^0T\simeq\mathbb{Z}[C_3]$; by hand $H^1(C_3,T)=T/\pi T\simeq\mathbb{F}_3$. * **The refined inequality and Section sec:tam** (`tamcheck2.py`): optimal curves of `degphi` below 10,000, non-CM (j not among the 13 CM values; CM curves carry no mod-3 label in `galrep`, and the CM curves 121b1, 256a1, 256d1, 1936f1, with mod-3 image of split Cartan type, are the only "violations" if CM curves are not removed), $\ell\in\{3,5,7\}$, $\ell\nmid N$, $E[\ell]$ irreducible (no label ℓB or ℓCs), and for $\ell=3$ no label 3Ns. $e_\ell(E,q)$ from `lfunsympow`/`lfuneuler` exactly as the paper's `rule`; Tamagawa terms by 9.4 at all potentially multiplicative primes. Results as in 9.5 and 13.1; no violation. For conductor at most 2000: at $\ell=3$, 2,158 curves, population 128, equality $87\to102$, positive excess 19 (Exercise 15). * **The paper's `rule`** (appendix, copied verbatim into `rule.gp`): 121a1 gives $[1,1]$, 726a1 gives $[2,2]$ with and without `iii`, as the paper says; 791a1 gives $[2,1]$ (the Tamagawa excess is not in `rule`). * **Inertia types**: 50a1 has type IV at 5 with $c_5=3$; 196b1 (IV* at 7), 225b1, 225e1 likewise (Exercise 9). * By hand: the matrix identities of 8.2 Step 4 ($FhF^{-1}=h-2\varepsilon c'e$, $FeF^{-1}=qe$); the change of c' under $\mathrm{Frob}\mapsto\mathrm{Frob}\,t^a$; the local terms of 6.7 (complex conjugation on A_f: eigenvalues $1,-1,-1$); Sym^2 of a 2×2 Jordan block in characteristic 3 is a 3×3 Jordan block; the exactness argument of 10.4 step 4; $W^I=(\mathbb{Q}_\ell/\mathbb{Z}_\ell)e$ at a multiplicative prime with $\ell\nmid n$; $H^1_{\mathrm{ur}}(\mathbb{Q}_p,E[\ell^\infty])\simeq\mathbb{Z}/\ell^{v_\ell(n)}$ in 5.5. * Not checked: the theorem numbers marked (verify); DFG04's normalisations ((31), (57), Section 1.7) and its treatment of $\ell=3$ (Part 7, Q1); the exact list of inertia groups at $q=2,3$ (Kraus90) beyond the facts that their orders divide 24 and that a Sylow 3-subgroup has order 3.

20.7 7. Points for important_notes.md

New points in this chapter's area (Bloch-Kato Selmer groups, Tamagawa factors, Lemma lem:fpr, Lemma lem:tamfactor, Lemma lem:dfg21, Remark rem:ns, Section sec:tam, Open questions 2 and 3), not among points 1 to 62 and B1 to B20 and not among the Chapter 17 outline's N1, N2. Points 5, 6, 7 (with its addendum), 11, 12, 18, 19, 23, 26, 31, 37, 39, 40, 42, 47, 56, B1, B6, B7, B14 are handled in Part 6.4 and not repeated.

N1. Open question 3, first half, has an answer: $\mathrm{Tam}_q(\mathrm{ad}^0T)$ **is trivial at every additive prime of potentially good reduction, and is given by Lemma lem:tamfactor (with the Tate parameter of j_E) at every potentially multiplicative prime.** Location: Open question 3 ("Compute the Tamagawa factors of ad^0T at the additive primes"); the proof of Theorem thm:main ("$v_\ell(\mathrm{Tam}_p(\mathrm{ad}^0T))\geq0$ at an additive prime"). Content: let ℓ be odd and $q\neq\ell$ additive. (a) Potentially good: inertia acts on T through a finite group Δ of order dividing 24, and $H^1(I_q,\mathrm{ad}^0T)_{\mathrm{tor}}=H^1(\Delta,\mathrm{ad}^0T)$ (inflation-restriction; the rest of $H^1(I_q,\mathrm{ad}^0T)$ embeds in $\mathrm{ad}^0T(-1)$, torsion-free). If $\ell\geq5$ this is 0 ($\#\Delta$ prime to ℓ). If $\ell=3$ its 3-part injects into $H^1(P,\mathrm{ad}^0T)$ for a Sylow 3-subgroup P of order 3 ($9\nmid24$), and

$\mathrm{ad}^0 T$ is a free $\mathbb{Z}_3[P]$-module (the reduction is the adjoint of a nontrivial 2×2 unipotent in characteristic 3, a single 3×3 Jordan block, i.e. the regular module; Nakayama), so $H^1(P, \mathrm{ad}^0 T) = 0$. Hence $\mathrm{Tam}_q(\mathrm{ad}^0 T) = \mathbb{Z}_\ell$ (and $\mathrm{Tam}_q(\mathrm{ad}^0 T(1)) = \mathbb{Z}_\ell$), even at types IV, IV* with $\ell = 3$, where $\mathrm{Tam}_q(T)$ can be 3 (for example 50a1 at 5). (b) Potentially multiplicative: $\mathrm{ad}^0 T \simeq \mathrm{ad}^0(T\otimes\chi)$ for a ramified quadratic χ of $G_{\mathbb{Q}_q}$ making $T\otimes\chi$ of Tate shape with parameter $\mathsf{q}(j_E)$, so $v_\ell(\mathrm{Tam}_q(\mathrm{ad}^0 T)) = t_\ell(E,q)+\min(t_\ell(E,q), v_\ell(q-1), v_\ell(c_q(u)))$, also at $q=2$ (this also settles point 18). Consequently (eq:identity) becomes fully explicit: $v_\ell(\deg\varphi_E) - \big[\sum_q e_\ell(E,q) + \sum_{v_q(j)<0}\big(t_\ell(E,q) + \min(t_\ell(E,q), v_\ell(q-1), v_\ell(c_q(u)))\big)\big] = \mathrm{length}\, H^1_f(\mathbb{Q}, W)$ (with equality in (eq:identity) by point 39; the checklist of point 56 is settled by point 73), which also answers the second half of Open question 3 as far as it can be answered from $\deg\varphi_E$. Data (optimal non-CM curves, conductor below 10,000, $\ell \nmid N$, $E[\ell]$ irreducible, not 3Ns): the excess is never negative; it is 0 for 8,688 of 13,203 curves at $\ell = 3$, 19,019 of 23,361 at $\ell = 5$, 25,133 of 28,134 at $\ell = 7$. Fix: replace "$v_\ell(\mathrm{Tam}_p(\mathrm{ad}^0 T)) \geq 0$ at an additive prime" by the exact values, and restate Open question 3 as the computation of the Selmer length. Confidence: high (the group theory is elementary; the freeness was checked by computer with an explicit basis; the only imported fact is that potentially good inertia images over $\mathbb{Q}_q$, $q \neq \ell$, have order dividing 24, Serre-Tate and Kraus90, which should be cited with the exact statement).

N2. Abstract: the Tamagawa sentence omits the condition $\ell \mid n_q$. Location: abstract, "the data confirm the Tamagawa factor formula, which raises the bound by one whenever a Steinberg prime $q \equiv 1 \pmod{\ell}$ has a Tate parameter whose unit part is an ℓ-th power". Problem: by Lemma lem:tamfactor the excess is $\min(v_\ell(n_q), v_\ell(q-1), v_\ell(c))$, which is 0 when $\ell \nmid n_q$, whatever the unit part; Remark rem:ko(1) and Section sec:tam state the condition $\ell \mid n$ correctly. Also "by one" is "by at least one" (the excess can be 2 when ℓ^2 divides n_q and $q-1$). Fix: "whenever a multiplicative prime $q \equiv 1 \pmod{\ell}$ with $\ell \mid n_q$ has a Tate parameter whose unit part is an ℓ-th power (by $\min(v_\ell(n_q), v_\ell(q-1), v_\ell(c))$, so by at least one)". Confidence: high.

N3. Abstract: "the length of the Bloch-Kato Selmer group of $\mathrm{ad}^0 V_\ell E$". Location: abstract ("the length of the Bloch-Kato (written with a double hyphen in the source) Selmer group of $\mathrm{ad}^0 V_\ell E$ plus the Tamagawa factors plus the omitted Euler factors"). Problem: the Selmer group of the $\mathbb{Q}_\ell$-representation $\mathrm{ad}^0 V_\ell E$ is a $\mathbb{Q}_\ell$-vector space, which is 0 under the hypothesis (DFG04 Theorem 2.7, "and the same for $A\otimes\mathbb{Q}_\ell$"); the length meant is that of $H^1_f(\mathbb{Q}, \mathrm{ad}^0 T\otimes\mathbb{Q}_\ell/\mathbb{Z}_\ell)$, as in Theorem thm:main. Fix: "of $\mathrm{ad}^0 T_\ell E\otimes\mathbb{Q}_\ell/\mathbb{Z}_\ell$". Also "when multiplicity one holds at level N equals" can become "equals" by point 39 (the DFG04 checklist of point 56 being settled by point 73). Confidence: high; low importance.

N4. Remark rem:ns: the vanishing of $H^1_f(\mathbb{Q}, V^D)$ for the 3Ns curves is an unverified hypothesis, and it is the only one. Location: Remark rem:ns ("the exact sequence of Lemma lem:dfg21 still does [apply] whenever $H^0(\mathbb{Q}, V^D) = H^1_f(\mathbb{Q}, V^D) = 0$"), Section 1.3 ("the correction is the term that Lemma 2.1 of DFG04 produces"), Open question 2. Problem: $H^0(\mathbb{Q}, V^D) = 0$ always holds (weights), but for image $N_s(\mathbb{F}_3)$ one has $3 \in S_f$, so DFG04 Theorem 2.7 does not give $H^1_f(\mathbb{Q}, V^D) = 0$; the paper leaves open whether the condition holds, and Section 1.3 states the conclusion without it. By the Greenberg-Wiles count with $\mathbb{Q}_\ell$ coefficients the condition is equivalent to $H^1_f(\mathbb{Q}, \mathrm{ad}^0 V_3 E) = 0$. Fix: in Section 1.3 and Open question 2, say "provided $H^1_f(\mathbb{Q}, \mathrm{ad}^0 V_3 E) = 0$ (not covered by DFG04 Theorem 2.7 in this case)", or cite a result proving it for residually dihedral representations after checking it. Confidence: medium-high (the logic is certain; whether later literature already proves the vanishing for these curves was not checked).

N5 (optional). Section sec:tam: the test can include the I^*_n primes, and one phrase overstates. Location: Section sec:tam ("at a prime q of type I_n with $\ell \mid n$ and $q \equiv 1 \pmod{\ell}$"; "the extra divisibility of the modular degree is exactly where the Tamagawa factor of the adjoint exceeds ℓ^{t_ℓ}"). Content: by N1(b) a prime of type I^*_n (any $q \neq \ell$) carries the same Tamagawa excess, computed from $\mathsf{q}(j_E)$; including such primes, our recomputation below conductor 10,000 ($\ell = 3$: 1,050 curves, equality 56.8% $\to$ 68.0%; $\ell = 5$: 252 curves, 203 $\to$ 220) shows no violation. The phrase "exactly where" should read "is accounted for, in these cases, by"; extra divisibility also comes from the Selmer group (N1). Confidence: high for the data and the mathematics; optional.

Q1 (open check, not claimed as an error; now SETTLED: DFG04 covers $\ell = 3$ in weight 2, Chapter 20 outline, Part 7; important_notes.md point 73). DFG04 at $\ell = 3$ for weight 2. The

paper reads DFG04 (31) as $S_f = \{\lambda \mid 2N\} \cup \{\text{not absolutely irreducible over } \mathbb{Q}(\sqrt{\ell^*})\}$ for $k = 2$. DFG04 build integral structures and the local condition at ℓ with Fontaine-Laffaille theory; for M_f (weights $0, 1$) this needs $\ell \geq 3$, but the adjoint has weights spanning an interval of length $2 > \ell - 2$ when $\ell = 3$. Before the book asserts Theorem thm:main at $\ell = 3$, the writer must check in DFG04 that $\ell = 3$ is not excluded for $k = 2$ (for instance by a condition such as "$\ell > k + 1$" or "$\ell \nmid (k + 1)!$" in (31) or in Sections 1 and 2) and how the adjoint's local condition at ℓ is defined there. If $\ell = 3$ were excluded, Theorem thm:main at $\ell = 3$ would lose its Taylor-Wiles input (the data would be unaffected). Confidence: none either way here (DFG04 not consulted); settled since by the Chapter 20 outline (Part 7): DFG04 covers $\ell = 3$ (point 73).

Book-internal inconsistencies found while preparing this outline:

- **B-a. PLAN.md, Chapter 18 entry.** (i) "$\mathrm{Tam}_p(T) = \mathrm{Fitt}_{\mathbb{Z}_\ell}(H^1(I_p, T)_{\mathrm{tors}})^{G_{\mathbb{Q}_p}}$" repeats the misplaced parenthesis of point 37. (ii) "the Tamagawa factor at an additive prime (what is known: it is at least 0; the paper does not compute it; Open question 3)" is superseded by 9.3 (N1).

  (iii) PLAN.md writes $V^* = \mathrm{Hom}(V, \mathbb{Q}_\ell)$ for the linear dual while the paper and Chapter 16 use * for the Pontryagin dual (extends B4); this outline uses $V^\vee$ for the linear dual and keeps * for Pontryagin (C2), consistent with Chapters 14 and 16.

- **B-b. Chapter 21 outline.** A4.10 says $\mathrm{Tam}_p(T^D) = \mathrm{Tam}_p(A)$ is checked "in the multiplicative case" only; 7.5 here proves it in every case that occurs (with 9.3), and A4.10 can cite 7.5. A1.3 says nothing about additive primes; it can cite 9.3. A4.9's proof idea should cite Chapter 17, 10.3 and 10.4 here. Its A2.7 should cite 8.2 (with the ε factor and the modulus $(n, q - 1)$, as the Chapter 17 outline's B-c says).
- **B-c. Chapter 3 outline, line 106** ("T^D ... the Cartier dual, Chapter 18"): the general definition is Chapter 17's 2.1, the adjoint-specific $T^D \simeq A(1)$ is 2.4 here (same as the Chapter 17 outline's B-a).
- **B-d. Chapter 16 outline, Exercise 10 (11a1 at $q = 11$, $\ell = 5$)** and 8.5 here use the same example; 8.5 cites it rather than repeating the computation. No conflict.
- **B-e. The paper's "$H^1_f(\mathbb{Q}_p, T^D)$ is the torsion of $H^1(\mathbb{Q}_p, T^D)$"** must not be simplified to "$H^1(\mathbb{Q}_p, T^D)$ is finite": the latter fails at I_0^* primes and at cyclic primes with $\chi(p) = 1$ (3.3, 3.4). The Chapter 17 outline's 4.6 states it correctly; any later chapter should follow it.
- Consistency checks that passed: Chapter 4's 7.6 (Gram determinant -2), Chapter 7's 6.2 and 8.2, Chapter 10's 7.4 and 7.8 (unit parts of 11a1, 539c1), Chapter 11's 3.2, 3.4, 8.5, 8.6, Chapter 14's 9.5, Chapter 16's model-computation, and Chapter 17's 4.4 to 4.6, 5.3, 5.4, 10.3, 10.4 agree with the line-up and with the recomputations of Part 6.6.

Chapter 21

Chapter 19: L-functions: Euler products, local factors, the adjoint and the symmetric square (ch:lfunctions, chapters/ch19_lfunctions.tex)

Stage 1 line-up. Part V (Cohomology, Selmer groups and L-functions), after Chapter 18 (Bloch-Kato Selmer groups, Tamagawa factors, the Tamagawa number conjecture) and before Chapter 20 (congruence ideals and the theorem of Diamond, Flach and Guo). Target length 1,975 lines of LaTeX (card excluded; see Part 3), inside the 1,000 to 2,200 range of PLAN.md Section 7.3.

Rigour tags: **P** = proved here, **S** = proof sketched here, **R** = stated; reference. Item format, as in the Chapter 2 to 15 outlines: number, label-ready name, kind, tag; the precise statement with all hypotheses; reference (book and chapter or section; a theorem number only where sure, otherwise marked "(verify)" or given by section); the paper's use by its `\label` names, or "background". Every number quoted below (Euler factors, $P_q(X)$, valuations, Petersson norms, L-values, modular degrees, conductors of $L(\mathrm{Sym}^2 f, s)$, the outputs of `rule(E, l)`) was checked by Python scripts `t0.py` to `t8.py` in the session scratchpad (`.../scratchpad/ch19/`), using PARI/GP 2.17.2 through `cypari2` and Cremona's `ecdata` files for conductors below 10,000 in `/tmp/ch09/data` (verification log, Part 6.6).

21.1 1. Purpose and global conventions

21.1.1 1.1 What this chapter is for

The paper's title names the object of this chapter: the local factors of the adjoint L-function. The chapter is built around seven uses.

1. **Naive versus true adjoint L-function** (Section 1: "The L-function that appears there is the *naive* one: the Euler product of $\sum a_n(f)^2 n^{-s}$, which has the true local factor at the primes of good or multiplicative reduction and the trivial factor at the additive primes"). The chapter defines both, proves the exact relation of the naive one with the Rankin series $\sum a_n^2 n^{-s}$ and with Shimura's series $\sum a_{n^2} n^{-s}$ (important_notes.md point 30), and explains why the naive factor is 1 exactly where $\delta_p = 0$.
2. **The local terms of Section 1.1** ("$\delta_p = \dim V^{I_p}$ ...; $L_p(A_f, s) = \det(1 - \mathrm{Frob}_p\, p^{-s} \mid A_f^{I_p})^{-1}$ is the true local factor, and the naive factor is $L_p^{\mathrm{nv}}(A_f, s) = L_p(A_f, s)$ if $\delta_p > 0$ and 1 if $\delta_p = 0$. In the

normalisation of the symmetric square, $L_p(\mathrm{Sym}^2 f, s) = L_p(A_f, s-1)$, and PARI's `lfunsympow(E,2)` returns these factors at every prime, including 2 and 3"). The chapter fixes the Frobenius (geometric, point 11), proves the shift by one, and states what PARI returns.

3. **Definition def:terms, the table of Section 1 and Lemma lem:euler in full** (the omitted factor by Kodaira type at a tame prime $q \geq 5$). The chapter assembles the inertia images of Chapter 11 and the local factors of Chapter 7 into the table, adds the column $P_q(X)$, and proves the valuations.
4. **Remark rem:wild and Section sec:wild** (the primes 2 and 3, where "the Kodaira type does not determine $L_q(A_f, s)$, but PARI does"). The chapter proves a classification valid at every prime $q \neq \ell$ (new item 10.1): P_q is one of 1, $1 - X$, $1 \mp qX$, or the type I_0^* cubic of a twist of good reduction, and it records which sentences of the remark are theorems (point 43), false (point 50) or approximate.
5. **Theorem thm:dfg14 and the paragraph after it** ("adding a good prime p multiplies the ideal by $(1 - \frac{\alpha}{\beta}p^{-1})(1 - p^{-1})(1 - \frac{\beta}{\alpha}p^{-1})$, Wiles's level-raising factor; adding a multiplicative prime multiplies it by $1 - p^{-2}$"), the proof of Theorem thm:main (the cancellation of the naive factors) and Theorem thm:quant Step 3. The chapter computes every naive factor at $s = 1$; the theorems themselves are Chapter 20's.
6. **Lemma lem:fpr, its L-part** ("$L_p(A_f, 1)^{-1} = L_p(A_f(1), 0)^{-1}$ is a nonzero rational number which is an ℓ-adic integer, of valuation $e_\ell(E, p)$ at an additive prime and $v_\ell(1 - p^{-2})$ at a multiplicative one") and the paragraph after Lemma lem:dfg21 ("$\det(1 - \mathrm{Frob}_p \mid (V^D)^{I_p}) = L_p(A_f, 1)^{-1} \neq 0$"). The Fitting-ideal half is Chapter 18's.
7. **The Petersson norm** (Section 1: "Up to explicit factors $\deg \varphi_E$ is $\langle f, f \rangle$ divided by the area of the period lattice of E (Zagier, Watkins), and $\langle f, f \rangle$ is the special value $L(\mathrm{Sym}^2 f, 2)$ up to the local factors at the bad primes (Rankin, Shimura; see DFG04 Lemma 2.12); Flach's finiteness theorem for the symmetric square is the first place where the modular degree meets the Selmer group of $\mathrm{Sym}^2 f$"). The chapter proves the exact formula $\langle f, f \rangle = \frac{N}{8\pi^3} L^{\mathrm{nv}}(\mathrm{Sym}^2 f, 2)$ for every level (new in this form for the book, item 8.3), deduces $\deg \varphi_E = N\, L^{\mathrm{nv}}(\mathrm{ad}^0 f, 1)/(2\pi \operatorname{vol} \Lambda_E)$ and the unconditional identity $v_\ell(\deg \varphi_E) = v_\ell(B_E) + \sum_{q \mid N} e_\ell(E, q)$ (item 8.4), which is the conceptual reason why the omitted factors appear in the paper's theorem.

It also covers the general theory that PLAN.md places here and that nothing earlier supplies: Dirichlet series and Euler products, the Riemann zeta function, Dirichlet L-functions and Dedekind zeta functions of abelian fields, L-functions of ℓ-adic representations and of compatible systems, $L(E, s)$ with its functional equation, the analytic theory of $L(\mathrm{Sym}^2 f, s)$ (Shimura, Gelbart-Jacquet), critical values and Deligne's conjecture (one paragraph), Flach's theorem (statement), the shape of the Bloch-Kato formula for the adjoint (one page, pointer to Chapter 18), and the mathematical content of the appendix function `rule(E, 1)` (the line-by-line reading is Chapter 22's).

It discharges the debts of earlier chapters: Chapter 2 (its 11.3 to 11.6 are "given the table of Section 1 as stated in Chapters 19 and 21"; its 11.6(f) asks for the label of the I_0^* example: it is 539d1, item 12.1 here); Chapter 4 (its 7.5(iii): $L_p(\mathrm{Sym}^2 f, s) = L_p(A_f, s-1)$, "the L-factor bookkeeping is Chapter 19's"); Chapter 6 (its 3.7: Euler factors with the geometric Frobenius); Chapter 7 (its forward list: convention 8.2, the shapes table 9.4, the adjoint computations 9.1 to 9.3, the wild primes 9.5, "the symmetric square normalisation and PARI's P_q"); Chapter 9 ("Chapter 19 supplies analytic continuation and the functional equation of $L(E, s)$ and the local factors via Weil-Deligne representations"); Chapter 10 ("the Euler factors $L_q(A_f, s)$, $P_q(X)$, $e_\ell(E, q)$, Lemma lem:euler in full and the Euler factors at 2 and 3"); Chapter 11 (its 4.1, 4.4, 4.6, 8.4(iv): "the factors $L_q(A_f, s)$ and $e_\ell(E, q)$ are computed from it in Chapter 19"); Chapter 12 (its 10.5(b): "Chapter 19 states the exact formula"); Chapter 13 (its 10.1 to 10.3: "the exact relation with $L(\mathrm{Sym}^2 f, 2)$ at bad primes is Chapter 19's"); Chapter 14 (its 7.5: the Rankin-Shimura formula with the bad factors); Chapter 15 (its 3.7: "the comparison is Chapter 19's").

The physics reader meets three familiar objects, one sentence each: the Mellin transform that turns a theta series or a q-expansion into a Dirichlet series (RHB, integral transforms; item 2.3); Poisson summation for the Gaussian, which gives the functional equation of ζ (RHB, Fourier transforms; item 3.1); and the Euler product as the partition function of a gas of non-interacting "primons", one bosonic mode per prime (one sentence in Section 2, as an aside only).

21.1.2 1.2 Global conventions fixed in this chapter

- **C1 (objects).** $E/\mathbb{Q}$ is given by a global minimal model, conductor N; $f = f_E \in S_2(\Gamma_0(N))$ its newform (Chapter 14); ℓ a prime; $T = T_\ell E$, $V = T \otimes \mathbb{Q}_\ell$, $\rho = \rho_{E,\ell}$; $A = \mathrm{ad}^0 T$, $A_f = \mathrm{ad}^0 V$ (trace zero; the paper's macro `\ad`); $\delta_p = \dim V^{I_p}$ for $p \neq \ell$ (for $p = \ell$ use $V_{\ell'}E$, $\ell' \neq p$; Part 7, N4). Wherever a statement holds for every newform of weight 2 with trivial character and rational coefficients, say so; the paper needs only $f = f_E$.
- **C2 (Frobenius).** As Chapter 7 (L3) and Chapter 11 (C3): Frob_p is an **arithmetic** Frobenius, $\Phi_p = \mathrm{Frob}_p^{-1}$ the **geometric** one. Group theory uses Frob_p (Frob g Frob$^{-1} = g^p$ on tame inertia); L-factors use Φ_p. On $V_\ell E$ at a good prime the arithmetic Frobenius has characteristic polynomial $X^2 - a_p X + p$, roots α, β; the geometric one has α^{-1}, β^{-1}; on $V_\ell E^\vee = H^1$ the geometric one has α, β; on $\mathrm{ad}^0 V$ both give $\alpha/\beta, 1, \beta/\alpha$ (important_notes.md B1, B9). In the paper the Frobenius of Section 1.1, of Lemma lem:euler, of Lemma lem:fpr and of the paragraph after Lemma lem:dfg21 is the geometric one; that of Lemma lem:tamfactor is the arithmetic one (point 11).
- **C3 (local factor).** For $p \neq \ell$ and a continuous representation M of $G_{\mathbb{Q}_p}$ on a finite-dimensional $\mathbb{Q}_\ell$-space: $L_p(M, s) = \det(1 - \Phi_p\, p^{-s} \mid M^{I_p})^{-1}$ (Chapter 7, 8.2). Consequences used throughout: $L_p(M(n), s) = L_p(M, s+n)$, $L_p(\mathbb{Q}_\ell(1), s) = (1 - p^{-1-s})^{-1}$, $L_p(V_\ell E^\vee, s) = L_p(E, s)$, $L_p(V_\ell E, s) = L_p(E, s+1)$.
- **C4 (the four normalisations; display as one table, item 7.4).** $L(\mathrm{ad}^0 f, s) = L(A_f, s) = \prod_p L_p(A_f, s)$: the paper's and DFG04's adjoint, value used at $s = 1$, functional equation $s \leftrightarrow 1 - s$. $L(\mathrm{Sym}^2 f, s) = \prod_p L_p(\mathrm{Sym}^2 V^\vee, s) = L(A_f, s-1)$: the paper's symmetric square and PARI's `lfunsympow(E,2)`, value used at $s = 2$, functional equation $s \leftrightarrow 3 - s$. L^{nv}: the naive versions (C6). $D(f, s) = \sum a_n(f)^2 n^{-s}$: the Rankin series. Never write "$L(\mathrm{Sym}^2 f, s)$" for the Rankin series.
- **C5 (P_q).** For every prime q (good or bad), $P_q(X) \in \mathbb{Z}[X]$ is the polynomial with $L_q(\mathrm{Sym}^2 f, s) = P_q(q^{-s})^{-1}$; so $P_q(q^{-2}) = L_q(A_f, 1)^{-1}$. PARI's `lfuneuler(lfunsympow(E,2),q)` returns the rational function $1/P_q(x)$, $x = q^{-s}$.
- **C6 (naive factor).** For $p \nmid N$: $L_p^{\mathrm{nv}} = L_p$. For $p \mid N$: $L_p^{\mathrm{nv}}(A_f, s) = L_p(A_f, s)$ if $\delta_p > 0$, $= 1$ if $\delta_p = 0$ (the paper, Section 1.1, after DFG04 1.7.2); equivalently $L_p^{\mathrm{nv}}(A_f, s)^{-1} = 1 - a_p(f)^2 p^{-1-s}$ for $p \mid N$ (item 8.1). $L^{\mathrm{nv}}(\mathrm{ad}^0 f, s) = \prod_p L_p^{\mathrm{nv}}(A_f, s)$, $L^{\mathrm{nv}}(\mathrm{Sym}^2 f, s) = L^{\mathrm{nv}}(\mathrm{ad}^0 f, s-1)$.
- **C7 (characters).** χ_{-3}, χ_{-4} are the quadratic characters of $\mathbb{Q}(\sqrt{-3})$, $\mathbb{Q}(i)$; for an odd prime q, $q^* = (-1)^{(q-1)/2} q$ and χ_{q^*} is the quadratic character of conductor q (the paper's χ_q; Chapter 2 and Chapter 13 write χ_{q^*} to avoid the clash with the cyclotomic χ_ℓ, point 32). Twists: $E^{(d)} = E \otimes \chi_d$.
- **C8 (Kodaira data).** Kodaira symbols as in PLAN.md 7.1; PARI's codes (Chapter 10): $n+4$ for I_n and $-(n+4)$ for I_n^* **for** $n \geq 1$ (I_0 is 1, I_0^* is -1; point 52), $\pm 2, \pm 3, \pm 4$ for II/II*, III/III*, IV/IV*.
- **C9 (Petersson product).** Unnormalised, as Chapter 12 (C5): $\langle f, g \rangle = \int_{\Gamma_0(N)\backslash \mathfrak{H}} f \bar{g}\, y^2\, d\mu = \int f \bar{g}\, dx\, dy$ for weight 2; PARI's `mfpetersson` is this divided by $\mu(N) = [\mathrm{SL}_2(\mathbb{Z}) : \Gamma_0(N)] = N \prod_{p \mid N}(1 + p^{-1})$. Period lattice Λ_E of the Neron differential of the minimal model, $\mathrm{vol}(\Lambda_E)$ its covolume (PARI: `E.area`).
- **C10 (typography).** PLAN.md 7.2: plain hyphens in "Rankin-Selberg", "Gelbart-Jacquet", "Bloch-Kato", "Weil-Deligne", "Coates-Schmidt"; ranges "1 to 8"; labels `...:lfunctions:...`; no new macros (write `\operatorname{Sym}^2` where the paper's `\Sym` would be ambiguous, `\operatorname{vol}`, `\operatorname{Res}`).

21.2 2. Line-up of the core results, in teaching order

21.2.1 Section 2. Dirichlet series and Euler products

2.1 `def:lfunctions:dirichlet-series` (definition and proposition; P). A Dirichlet series is $D(s) = \sum_{n \geq 1} a_n n^{-s}$ with $a_n \in \mathbb{C}$, $s = \sigma + it$. (i) If $\sum |a_n| n^{-\sigma_0} < \infty$ then D converges absolutely and uniformly on $\mathrm{Re}\, s \geq \sigma_0$ and is holomorphic on $\mathrm{Re}\, s > \sigma_0$ (Weierstrass M-test; uniform limits of holomorphic functions,

Appendix A). (ii) The abscissa of absolute convergence is $\sigma_a = \inf\{\sigma_0\}$; if $a_n = O(n^c)$ then $\sigma_a \le c+1$; if $a_n = O(n^{c+\varepsilon})$ for every $\varepsilon > 0$, likewise. (iii) Uniqueness: if $D(s) = 0$ for all real s large enough then every $a_n = 0$ (multiply by m^s inductively). (iv) Products of Dirichlet series are Dirichlet series with the convolution $\sum_{de=n} a_d b_e$. Reference: Davenport, Chapter 1 and Chapter 8 (cite the chapters; proposed key `Davenport00`, Part 6.5); SerreCourse VI.2 (cite the section). Paper: background.

2.2 `prop:lfunctions:euler-product` (proposition; P). Let $a\colon \mathbb{N} \to \mathbb{C}$ be multiplicative ($a_{mn} = a_m a_n$ for coprime m, n, $a_1 = 1$) with $\sum |a_n| n^{-\sigma} < \infty$. Then for $\operatorname{Re} s \ge \sigma$: $\sum_n a_n n^{-s} = \prod_p \bigl(\sum_{k\ge 0} a_{p^k} p^{-ks}\bigr)$, the product converging absolutely; it is nonzero at s if every factor is nonzero there. If moreover $\sum_k a_{p^k} X^k = P_p(X)^{-1}$ with $P_p \in \mathbb{C}[X]$, $P_p(0) = 1$, $\deg P_p \le d$ for all p, the series has an Euler product of degree d. Examples: ζ ($d = 1$), $L(s,\chi)$ ($d = 1$), $L(f,s)$ ($d = 2$, Chapter 13, its 10.1), $L(\mathrm{ad}^0 f, s)$ and $L(\mathrm{Sym}^2 f, s)$ ($d = 3$, Section 6), $\zeta(s)\zeta(s-1) = \sum \sigma(n) n^{-s}$ (exercise). Physics aside: $\zeta(s) = \prod_p (1-p^{-s})^{-1}$ is the partition function of free bosonic "primons" at inverse temperature s (one sentence). Reference: Davenport, Chapter 1 (cite the chapter). Paper: Section 1 ("the Euler product of $\sum a_n(f)^2 n^{-s}$"), Definition def:terms (P_q).

2.3 `lem:lfunctions:mellin` (lemma; P for (i), (iii), S for (ii)). (i) $\Gamma(s) = \int_0^\infty e^{-y} y^{s-1} dy$ ($\operatorname{Re} s > 0$) satisfies $\Gamma(s+1) = s\Gamma(s)$, $\Gamma(1/2) = \sqrt{\pi}$, and extends meromorphically to $\mathbb{C}$ with simple poles at $0, -1, -2, \ldots$ and no zeros. Put $\Gamma_{\mathbb{R}}(s) = \pi^{-s/2}\Gamma(s/2)$, $\Gamma_{\mathbb{C}}(s) = 2(2\pi)^{-s}\Gamma(s)$; (ii) duplication: $\Gamma_{\mathbb{R}}(s)\Gamma_{\mathbb{R}}(s+1) = \Gamma_{\mathbb{C}}(s)$ (Legendre; S). (iii) Mellin: if $a_n = O(n^c)$ and $F(y) = \sum_{n\ge 1} a_n e^{-2\pi n y}$ then $\int_0^\infty F(y) y^{s-1} dy = (2\pi)^{-s}\Gamma(s) \sum a_n n^{-s}$ for $\operatorname{Re} s > c+1$ (Fubini). Physics aside: the Mellin transform is the Fourier transform on the multiplicative group $\mathbb{R}_{>0}$ (RHB, integral transforms). Reference: SerreCourse VII.4 (cite the section); Davenport, Chapter 9. Paper: background (Chapter 13, its 10.2; the gamma factors of 7.2).

21.2.2 Section 3. The Riemann zeta function, Dirichlet L-functions and Dedekind zeta functions

3.1 `thm:lfunctions:zeta` (theorem; P for (i) to (iii), S for (iv), R for (v)). (i) For $\operatorname{Re} s > 1$, $\zeta(s) = \sum n^{-s} = \prod_p (1-p^{-s})^{-1} \ne 0$. (ii) $\zeta(s) - \frac{1}{s-1}$ extends holomorphically to $\operatorname{Re} s > 0$ (compare with $\int_1^\infty x^{-s} dx$ by partial summation); so ζ has a simple pole at $s = 1$ with residue 1. (iii) $\zeta(2) = \pi^2/6$ (Parseval for the Fourier series of x on $(-\pi, \pi)$, RHB). (iv) (Riemann) $\xi(s) = \Gamma_{\mathbb{R}}(s)\zeta(s)$ extends meromorphically to $\mathbb{C}$ with simple poles at $s = 0, 1$ only, and $\xi(s) = \xi(1-s)$. Sketch: $\theta(y) = \sum_{n\in\mathbb{Z}} e^{-\pi n^2 y}$ satisfies $\theta(1/y) = \sqrt{y}\,\theta(y)$ (Poisson summation; the Gaussian is its own Fourier transform, RHB); $\xi(s) = \frac{1}{2}\int_0^\infty (\theta(y) - 1) y^{s/2-1} dy$; split at $y = 1$ and substitute $y \mapsto 1/y$. (v) $\zeta(1+it) \ne 0$ for real $t \ne 0$ (Hadamard, de la Vallee Poussin; R). Reference: SerreCourse VI.3 and VII (cite sections); Davenport, Chapters 8, 13; Washington, Chapter 4. Paper: background; $\zeta(2)$ and the residue at 1 are used in 8.3.

3.2 `def:lfunctions:partial-zeta` (definition and proposition; P). For an integer $N \ge 1$, $\zeta^{(N)}(s) = \zeta(s)\prod_{p\mid N}(1-p^{-s})$ (Euler factors at $p \mid N$ removed). Then $\zeta^{(N)}(s-1)$ has a simple pole at $s = 2$ with residue $\prod_{p\mid N}(1-p^{-1})$, and $\zeta^{(N)}(2) = \frac{\pi^2}{6}\prod_{p\mid N}(1-p^{-2})$; hence $\operatorname{Res}_{s=2}\zeta^{(N)}(s-1)/\zeta^{(N)}(2s-2) = \frac{6}{\pi^2}\prod_{p\mid N}(1+p^{-1})^{-1} = \frac{6N}{\pi^2\mu(N)}$ with $\mu(N) = [\mathrm{SL}_2(\mathbb{Z}) : \Gamma_0(N)]$ (Chapter 12). Paper: Section 1 (through point 30 and 8.2, 8.3).

3.3 `thm:lfunctions:dirichlet-L` (definition and theorem; P for (i), (ii), (iv)(a); S for (iii); R for (iv)(b), (v)). Let χ be a Dirichlet character modulo m (Chapter 2). (i) $L(s,\chi) = \sum \chi(n) n^{-s} = \prod_p (1-\chi(p)p^{-s})^{-1}$ for $\operatorname{Re} s > 1$. (ii) If $\chi \ne 1$ the partial sums of $\chi(n)$ are bounded, so $L(s,\chi)$ converges (conditionally) and is holomorphic on $\operatorname{Re} s > 0$. (iii) If χ is primitive of conductor $m > 1$ and $\chi(-1) = (-1)^a$, $a \in \{0,1\}$, then $\Lambda(s,\chi) = (m/\pi)^{(s+a)/2}\Gamma(\frac{s+a}{2})L(s,\chi)$ is entire and $\Lambda(s,\chi) = W(\chi)\Lambda(1-s,\bar\chi)$ with $W(\chi) = \tau(\chi)/(i^a\sqrt{m})$, $|W(\chi)| = 1$, $\tau(\chi)$ the Gauss sum (Chapter 2); for real χ, $W(\chi) = 1$ (Gauss's determination of the sign of the quadratic Gauss sum; stated). Sketch: the theta series $\sum \chi(n) n^a e^{-\pi n^2 y/m}$ and Poisson summation twisted by χ. (iv) (a) $L(1,\chi_{-4}) = 1 - \frac{1}{3} + \frac{1}{5} - \cdots = \frac{\pi}{4}$ (Leibniz) and $L(1,\chi_{-3}) = \frac{\pi}{3\sqrt{3}}$ (P, by integrating a geometric series); (b) $L(1,\chi) \ne 0$ for every $\chi \ne 1$ (R; for quadratic χ via 3.4 and the class number formula). (v) Dirichlet's theorem on primes in arithmetic progressions (R). Reference: Davenport, Chapters 1, 4, 9; Washington, Chapter 4; SerreCourse VI. Paper: background; χ_{-3}, χ_{-4}, χ_{q^*} of the table, of Lemma lem:twist and of Section 1.4 are these characters (Chapter 2).

3.4 `thm:lfunctions:dedekind` (theorem; P for (i), R for (ii), (iii)). For a number field K, $\zeta_K(s) = \sum_{\mathfrak{a}} N\mathfrak{a}^{-s} = \prod_{\mathfrak{P}}(1 - N\mathfrak{P}^{-s})^{-1}$ ($\mathrm{Re}\, s > 1$; Chapter 6). (i) For $K = \mathbb{Q}(\sqrt{d})$ with discriminant D_K: $\zeta_K(s) = \zeta(s)L(s, \chi_{D_K})$; the Euler factor at p is $(1-p^{-s})^{-1}(1-\chi_{D_K}(p)p^{-s})^{-1}$ by the splitting law (Chapter 6). Examples: $\zeta_{\mathbb{Q}(i)} = \zeta\, L(s, \chi_{-4})$, $\zeta_{\mathbb{Q}(\sqrt{-3})} = \zeta\, L(s, \chi_{-3})$. (ii) For $K/\mathbb{Q}$ abelian, $\zeta_K(s) = \prod_{\chi} L(s, \chi)$ over the primitive characters attached to the characters of $\mathrm{Gal}(K/\mathbb{Q})$ (Washington, Theorem 4.3 (verify)). (iii) Artin formalism (R): Artin L-functions are additive and invariant under induction; $\zeta_K = L(\mathrm{Ind}_{G_K}^{G_{\mathbb{Q}}} \mathbf{1}, s)$. Remark (one sentence): the factor $(1 - \chi_{-3}(q)q^{-s})^{-1}$ of the paper's table is the Euler factor at q of $L(s, \chi_{-3})$, because the inertia-invariant line of $\mathrm{ad}^0 V$ at a prime of type II to IV* is, as a $G_{\mathbb{Q}_q}$-module, the restriction of the global character χ_{-3} (item 9.3; likewise χ_{-4}). Reference: Neukirch VII.5, VII.10 (cite sections); Washington, Chapter 4. Paper: background (the characters of the table).

21.2.3 Section 4. L-functions of ℓ-adic representations

4.1 `def:lfunctions:local-factor` (definition and proposition; recalled from Chapter 7 for $p \neq \ell$, P for (ii), (iii); R for (iv)). (i) For $p \neq \ell$: $L_p(M, s) = \det(1 - \Phi_p p^{-s} \mid M^{I_p})^{-1}$, which depends only on the Weil-Deligne representation of M (Chapter 7, 8.1, 8.2). (ii) $L_p(M \oplus M', s) = L_p(M, s)L_p(M', s)$; $L_p(M(n), s) = L_p(M, s + n)$. (iii) Writing the factor with the arithmetic Frob_p inverts the eigenvalues; the two agree iff the eigenvalue multiset on M^{I_p} is stable under $\lambda \mapsto \lambda^{-1}$ (true for $\mathrm{ad}^0 V$ at a good prime, false for $\mathrm{ad}^0 V$ at a multiplicative prime and for $V_\ell E$; point 11). (iv) At $p = \ell$, for M crystalline: $L_\ell(M, s) = \det(1 - \varphi\, \ell^{-s} \mid D_{\mathrm{cris}}(M))^{-1}$, in the normalisation in which $D_{\mathrm{cris}}(\mathbb{Q}_\ell(1))$ has $\varphi = \ell^{-1}$ (so that (ii) holds at ℓ as well); for E with good reduction at ℓ, φ on $D_{\mathrm{cris}}(V_\ell E^\vee)$ has characteristic polynomial $X^2 - a_\ell X + \ell$ (Katz-Messing; R), so $L_\ell(V_\ell E^\vee, s) = L_\ell(E, s)$ and $L_\ell(A_f, s)$ is given by the good-prime formula 6.2. Reference: Chapter 7; Fontaine (refs.bib `Fontaine`), Berger (cite the section); Tate79 Section 4 (proposed refs.bib key, Part 6.5). Paper: Section 1.1 (the definition of $L_p(A_f, s)$; the paper uses it only at $p \neq \ell$).

4.2 `def:lfunctions:compatible-system` (definition and theorem; R). A *weakly compatible system* is a family $(M_\ell)_\ell$ of representations of $G_{\mathbb{Q}}$ on $\mathbb{Q}_\ell$-spaces (or over completions of a number field), unramified outside a finite set S, such that for $p \notin S \cup \{\ell\}$ the characteristic polynomial of Φ_p on M_ℓ has rational coefficients independent of ℓ. Its L-function is $L(M, s) = \prod_p L_p(M_{\ell(p)}, s)$ with $\ell(p) \neq p$, which is well defined when the (Frobenius-semisimplified) Weil-Deligne representation at each bad p is independent of $\ell \neq p$. Known in the cases needed: $V_\ell E$, $V_\ell E^\vee$ and all tensor constructions of them (Sym^2, ad^0), by the Tate curve, Grothendieck's monodromy theorem and the Neron-Ogg-Shafarevich criterion (Chapters 7, 10, 11; SerreTate68, R); $V_\lambda(f)$ for newforms (Carayol86; Chapter 15, its 3.4, R). In general it is a conjecture. Paper: Section 1.1 ("the true local factor"; "PARI's `lfunsympow(E,2)` returns these factors at every prime").

4.3 `prop:lfunctions:convergence` (proposition; P given the Hasse bound, Chapter 9). Let (M_ℓ) be a weakly compatible system, pure of weight w: for $p \notin S$ the eigenvalues of Φ_p have absolute value $p^{w/2}$ under every complex embedding, and for $p \in S$ the eigenvalues on M^{I_p} have absolute value at most $p^{w/2}$. Then $L(M, s)$ converges absolutely and is nonzero for $\mathrm{Re}\, s > 1 + w/2$ (compare with $\zeta(\sigma - w/2)^{\dim M}$). Examples: $V_\ell E^\vee$ ($w = 1$): $L(E, s)$ on $\mathrm{Re}\, s > 3/2$; A_f ($w = 0$; at bad primes the eigenvalues are q^{-1}, ± 1 or α'/β' of modulus ≤ 1, Section 9): $L(\mathrm{ad}^0 f, s)$ on $\mathrm{Re}\, s > 1$, hence $L(\mathrm{Sym}^2 f, s)$ on $\mathrm{Re}\, s > 2$; $\mathbb{Q}_\ell(1)$ ($w = -2$): $\zeta(s + 1)$. So $s = 1$ for $L(\mathrm{ad}^0 f, s)$ and $s = 2$ for $L(\mathrm{Sym}^2 f, s)$ lie on the edge of absolute convergence: the values the paper uses are values of the analytic continuation (7.1), or of the Rankin integral (8.3). Paper: background.

4.4 `ex:lfunctions:finite-inertia` (worked example; P given Chapter 7, 9.2, 9.3). For V two-dimensional with $\det V|_{I_q} = 1$ and $\rho(I_q)$ finite cyclic of order $m \in \{2, 3, 4, 6\}$, $q \nmid m$: $L_q(V, s) = 1$ in every case ($V^{I_q} = 0$), while $L_q(\mathrm{ad}^0 V, s)$ is the cubic factor of the unramified twist ($m = 2$) or $(1 - \chi(q)q^{-s})^{-1}$ with $\chi = \chi_{-3}$ ($m = 3, 6$) or χ_{-4} ($m = 4$). This is the PLAN item "the L-function of a representation with finite image of inertia (the three Kodaira cases)"; the proof is Chapter 7's, the dictionary to Kodaira types is item 9.3. Paper: Lemma lem:euler (background to it).

21.2.4 Section 5. The L-function of an elliptic curve and of its newform

5.1 `def:lfunctions:LE` (definition and proposition; P given Chapters 9, 11). $L(E,s)=\prod_p L_p(E,s)$ with $L_p(E,s)^{-1}=1-a_pp^{-s}+p^{1-2s}$ (good p), $1-a_pp^{-s}$ with $a_p=+1$ (split) or -1 (nonsplit) (multiplicative), 1 (additive); uniformly $a_p=p+1-\#\tilde{E}(\mathbb{F}_p)$ for every p, with $\tilde{E}$ the reduction of the minimal model, singular point included (equivalently $a_p=p-\#\tilde{E}_{\mathrm{ns}}(\mathbb{F}_p)$ at bad p: $\tilde{E}_{\mathrm{ns}}$ is $\mathbb{G}_m$, its nonsplit form or $\mathbb{G}_a$; Chapter 10). (i) For every $p\neq\ell$, $L_p(E,s)=L_p(V_\ell E^\vee,s)=L_p(V_\ell E,s-1)$, and $\delta_p=\deg L_p(E,s)^{-1}$ as a polynomial in p^{-s} (Chapter 11, its 3.1, 3.2, 3.4, 4.1, 3.5). (ii) Absolute convergence on $\mathrm{Re}\,s>3/2$ (4.3). (iii) Isogenous curves have the same L-function. Reference: SilvermanAEC C.16 (verify); Cremona97 Chapter II. Paper: Section 1.1 (δ_p).

5.2 `thm:lfunctions:LE-feq` (theorem; R for modularity, S for the rest given Chapter 13, its 10.2). Let $E/\mathbb{Q}$ have conductor N. (i) $L(E,s)=L(f_E,s)$ (modularity, Chapter 14). (ii) $\Lambda(E,s)=N^{s/2}(2\pi)^{-s}\Gamma(s)L(E,s)=N^{s/2}\int_0^\infty f_E(iy)y^{s-1}dy$ extends to an entire function and $\Lambda(E,s)=w_E\Lambda(E,2-s)$ with $w_E=-\varepsilon_N(f_E)\in\{\pm1\}$, where $f_E|W_N=\varepsilon_N(f_E)f_E$ (Chapter 12, Chapter 13, its 10.2). (iii) (R) $w_E=-\prod_{p|N}w_p$ with $w_p=-a_p$ at multiplicative p; the local root numbers at additive p are given by Rohrlich's formulas (Rohrlich94; not needed here). Examples: 11a1 ($w=+1$), 37a1 ($w=-1$) (Chapter 13). Reference: DiamondShurman 5.10 (cite the section); Rohrlich94 (proposed refs.bib key, B18). Paper: background (Section 1; Chapter 14, its 11.4 for $L(f,1)$).

21.2.5 Section 6. The adjoint and the symmetric square: definitions and the good primes

6.1 `def:lfunctions:ad-sym` (definition and proposition; P given Chapter 4, its 7.5, and 4.1, 4.2). Let f be a newform of weight 2, trivial character, rational coefficients (the paper: $f=f_E$), and $V=V_\ell E$ (or $V_\lambda(f)$). Define

$$L(\mathrm{ad}^0 f,s)=L(A_f,s)=\prod_p L_p(A_f,s),\qquad L(\mathrm{Sym}^2 f,s)=\prod_p L_p(\mathrm{Sym}^2 V^\vee,s),$$

each local factor computed with any $\ell\neq p$ (4.2). Then: (i) $\mathrm{Sym}^2 V^\vee\simeq A_f(-1)$ as $G_{\mathbb{Q}}$-representations, because $\mathrm{Sym}^2 W\simeq\mathrm{ad}^0 W\otimes\det W$ for $\dim W=2$ (Chapter 4, 7.5) and $\mathrm{ad}^0 V^\vee\simeq\mathrm{ad}^0 V$, $\det V^\vee=\chi_\ell^{-1}$; hence $L_p(\mathrm{Sym}^2 f,s)=L_p(A_f,s-1)$ for every p and $L(\mathrm{Sym}^2 f,s)=L(\mathrm{ad}^0 f,s-1)$ (C3). (ii) For every p, $P_p(X):=L_p(\mathrm{Sym}^2 f,s)^{-1}|_{p^{-s}=X}$ lies in $\mathbb{Z}[X]$, $P_p(0)=1$, $\deg P_p=\dim A_f^{I_p}\in\{0,1,3\}$ (integrality from the explicit list 6.2, 9.2, 9.3, 10.1), and $P_p(p^{-2})=L_p(A_f,1)^{-1}\in\mathbb{Z}[1/p]$. (iii) The eigenvalues of Φ_p on $\mathrm{Sym}^2 V^\vee$ at a good p are $\alpha^2,\alpha\beta=p,\beta^2$, so $P_p(X)=(1-\alpha^2X)(1-pX)(1-\beta^2X)$. Reference: elementary; DFG04 1.7.2 (as quoted by the paper; the citation is correct and DFG04 use the geometric Frobenius: Chapter 20 outline, Part 7; important_notes.md point 73). Paper: Section 1.1 ("In the normalisation of the symmetric square, $L_p(\mathrm{Sym}^2 f,s)=L_p(A_f,s-1)$"), Definition def:terms ("$P_q(X)$ is the polynomial with $L_q(\mathrm{Sym}^2 f,s)=P_q(q^{-s})^{-1}$").

6.2 `prop:lfunctions:good-factor` (proposition; P). Let $p\nmid N$, $p\neq\ell$, $a=a_p(f)$, $\alpha+\beta=a$, $\alpha\beta=p$. (i) Φ_p and Frob_p both have eigenvalues $\alpha/\beta,1,\beta/\alpha$ on A_f, and

$$L_p(A_f,s)^{-1}=\Big(1-\frac{\alpha}{\beta}p^{-s}\Big)(1-p^{-s})\Big(1-\frac{\beta}{\alpha}p^{-s}\Big)=(1-p^{-s})\big(1-(a^2-2p)p^{-1-s}+p^{-2s}\big),$$

$P_p(X)=(1-pX)(1-(a^2-2p)X+p^2X^2)$. (ii) $L_p(A_f,1)^{-1}=\frac{p-1}{p}\cdot\frac{(p+1)^2-a^2}{p^2}$ (Chapter 2, 11.4), which is Wiles's level-raising factor; it is nonzero by the Hasse bound $|a|\le2\sqrt{p}$ ($(p+1)^2-a^2\ge(p-1)^2>0$). (iii) For $\ell\neq p$ odd: $v_\ell(L_p(A_f,1)^{-1})>0$ iff $p\equiv1\pmod\ell$ or $a\equiv\pm(p+1)\pmod\ell$; the second condition is Ribet's level-raising condition (Chapter 15). Reference: elementary; DDT (Wiles's factor; cite the section). Paper: Theorem thm:dfg14 and the paragraph after it ("Wiles's level-raising factor"), the I_0^* row of the table (the same formula for E'), Theorem thm:quant Step 3.

21.2.6 Section 7. Analytic theory, normalisations and critical values

7.1 `thm:lfunctions:sym2-analytic` (theorem; R, with (iii) P given 8.3). Let f be as in 6.1. (i) (Shimura 1975) $L(\mathrm{Sym}^2 f, s)$ extends meromorphically to $\mathbb{C}$ (Shimura proved this, and holomorphy under conditions that the writer must copy from the paper, via the series $\sum a_{n^2} n^{-s}$ of 8.2(B), a theta series and an Eisenstein series of half-integral weight; proposed key `Shimura75`, Part 6.5). (ii) (Gelbart and Jacquet 1978) there is an automorphic representation $\mathrm{Ad}(\pi_f)$ of $\mathrm{GL}_3(\mathbb{A}_{\mathbb{Q}})$, cuspidal unless f has complex multiplication, whose standard L-function is $L(s, \pi_f, \mathrm{Ad}) = L(\mathrm{ad}^0 f, s)$ (unitary normalisation, centre $1/2$; agreement of the local factors at the bad primes with the Galois side by local-global compatibility, Chapter 15, its 3.4, R); consequently $L(\mathrm{ad}^0 f, s)$ is entire (also in the CM case, where it is $L(s, \chi_K)$ times a Hecke L-function of a nontrivial character; verify) and satisfies $\Lambda(\mathrm{ad}^0 f, s) = \varepsilon\, \Lambda(\mathrm{ad}^0 f, 1-s)$ with $\Lambda(\mathrm{ad}^0 f, s) = N_{\mathrm{ad}^0}^{s/2} \Gamma_{\mathbb{R}}(s+1)\Gamma_{\mathbb{C}}(s+1) L(\mathrm{ad}^0 f, s)$, N_{ad^0} the conductor of A_f, $\varepsilon = \pm 1$ (the root number; PARI's data give $\varepsilon = +1$ for every curve examined; state "$\varepsilon = +1$" only with a reference, verify). (iii) $L(\mathrm{ad}^0 f, 1) > 0$ (P: by 8.3, $L^{\mathrm{nv}}(\mathrm{ad}^0 f, 1) = 8\pi^3 \langle f, f\rangle / N > 0$, and every omitted factor $L_q(A_f, 1)^{-1}$ is positive, by 9.3 and 10.1). Reference: Shimura75; GelbartJacquet; Shimura76 (cite sections; verify). Paper: Section 1 ("Rankin, Shimura"); the special value $L(\mathrm{Sym}^2 f, 2) = L(\mathrm{ad}^0 f, 1)$ of Section 1.

7.2 `prop:lfunctions:ad-conductor` (proposition; P given Chapter 7, 9.1 to 9.4; the gamma factor R). (i) The conductor exponent of A_f at $q \geq 5$ ($q \neq \ell$) is 2 for the types I_n, I_n^* ($n \geq 1$), II, II*, III, III*, IV, IV*, and 0 for I_0^* (where A_f is unramified); so $N_{\mathrm{ad}^0} = \prod_{q \mid N,\ q \text{ not } \mathrm{I}_0^*} q^2$ (the square of the radical of N with the I_0^* primes removed, not N^2) when $(N, 6) = 1$. (ii) The gamma factor of $L(\mathrm{Sym}^2 f, s)$ is $\Gamma_{\mathbb{R}}(s)^2 \Gamma_{\mathbb{R}}(s+1) = \Gamma_{\mathbb{R}}(s)\Gamma_{\mathbb{C}}(s)$, that of $L(\mathrm{ad}^0 f, s)$ is $\Gamma_{\mathbb{R}}(s+1)\Gamma_{\mathbb{C}}(s+1)$ (2.3(ii)). Checked: PARI's `lfunparams(lfunsympow(E,2))` is `[N_sym, 3, [0, 0, 1]]` (motivic weight 3, so $s \leftrightarrow 3-s$; gamma shifts $0, 0, 1$), with $N_{\mathrm{sym}} = 121$ for 11a1, 121a1, 539d1, 275b1 (the I_0^* primes 7, 5 contribute nothing), 100 for 20a1 (type IV* at 2: exponent 2), 81 for 27a1 (wild at 3), 64 for 64a1, $1764 = 42^2$ for 336f1, $5929 = 77^2$ for 1232l1 (no contribution at 2: case 10.1(b)), and sign $+1$ in all these cases. Paper: background (Section 1.1, `lfunsympow`).

7.3 `rem:lfunctions:critical` (remark; R, with the rationality P given 8.3). Deligne's conjecture (Deligne79): for a motive M over $\mathbb{Q}$, an integer n is critical if neither $L_\infty(M, s)$ nor $L_\infty(M^\vee, 1-s)$ has a pole at $s = n$, and then $L(M, n)/c^{\pm}(M(n)) \in \mathbb{Q}^\times$ for Deligne's period $c^{\pm}$. For ad^0 of a weight-2 form ($L_\infty = \Gamma_{\mathbb{R}}(s+1)\Gamma_{\mathbb{C}}(s+1)$, self-dual) the critical integers are $s = 0, 1$; for $L(\mathrm{Sym}^2 f, s)$ they are $s = 1, 2$. By 8.3, $L(\mathrm{ad}^0 f, 1)/(\pi^3 \langle f, f\rangle) = 8/\big(N \prod_{q^2 \mid N} L_q(A_f, 1)^{-1}\big) \in \mathbb{Q}^\times$, which is Deligne's conjecture at $s = 1$ once his period is identified with $\pi^3 \langle f, f\rangle$ up to $\mathbb{Q}^\times$ (Sturm proved the analogous statement for all weights; proposed key `Sturm80`, verify; Zagier's work on $L(\mathrm{Sym}^2 f, s)$ in level one; cite sections only). One paragraph in the chapter. Paper: background (Section 1, "the special value $L(\mathrm{Sym}^2 f, 2)$").

7.4 `rem:lfunctions:normalisations` (remark with table; P for every identity, R for what each source writes). The objects that occur, in one table (the writer checks each source's own notation before attributing it; marked "verify" where not checked):

name	source	relation to $L(A_f, s) = L(\mathrm{ad}^0 f, s)$	functional equation	value the paper uses	factor at additive p
$L(\mathrm{ad}^0 f, s)$	paper Sections 1, 1.1; DFG04 (1.7.2, $L(\mathrm{ad}^0 M_f, s)$ at $s = 1$; checked, Chapter 20 outline, Part 7; important_notes.md point 73)	itself	$s \leftrightarrow 1 - s$	$s = 1$	true

name	source	relation to $L(A_f, s) = L(\mathrm{ad}^0 f, s)$	functional equation	value the paper uses	factor at additive p
$L(s, \pi_f, \mathrm{Ad})$	Gelbart-Jacquet (unitary)	$= L(A_f, s)$ (weight k: $L(A_f, s) = L(\mathrm{Sym}^2, s + k - 1)$)	$s \leftrightarrow 1 - s$	$s = 1$	true
$L(\mathrm{Sym}^2 f, s)$	paper; PARI `lfunsympow(E,2)`; Watkins02; CoatesSchmidt87 (verify)	$L(A_f, s - 1)$	$s \leftrightarrow 3 - s$	$s = 2$	true
Shimura's series	Shimura75: $\zeta^{(N)}(2s - 2) \sum a_{n^2} n^{-s}$	$L^{\mathrm{nv}}(A_f, s - 1)$ (8.2(B))	(via 7.1)		naive (1)
$L^{\mathrm{nv}}(\mathrm{ad}^0 f, s)$	paper Section 1.1; DFG04 (the naive factors of Proposition 1.4)	$L(A_f, s) \prod_{p^2 \mid N} L_p(A_f, s)^{-1}$	(Atkin–Lehner)	$s = 1$	naive (1)
Rankin series $D(f, s)$	Rankin, Selberg, Shimura76; Chapter 13, 10.3	$\zeta^{(N)}(s - 1) L^{\mathrm{nv}}(A_f, s - 1) / \zeta^{(N)}(2s - 2)$ (8.2(A))	via $E(z, s)$	residue at $s = 2$	naive (1)
$L(f \times f, s)$	Rankin-Selberg convolution	$\zeta^{(N)}(2s - 2) D(f, s) = \zeta^{(N)}(s - 1) L^{\mathrm{nv}}(A_f, s - 1)$		pole at $s = 2$	naive (1)

Pitfall stated in the chapter: Hida's and Sturm's papers use weight k and the classical variable; for weight k, $L(\mathrm{ad}^0 f, s) = L(\mathrm{Sym}^2 f, s + k - 1)$, so "$L(\mathrm{Sym}^2 f, k)$" is $L(\mathrm{ad}^0 f, 1)$. Paper: Section 1.1 (normalisation sentence); Appendix B (conventions).

21.2.7 Section 8. Naive versus true; Rankin-Selberg; the Petersson norm and the modular degree

8.1 `def:lfunctions:naive` (definition and proposition; P given Chapter 15, its 3.5, 3.7, and Chapter 11, its 3.5). For $p \mid N$ let $L_p^{\mathrm{nv}}(A_f, s) = L_p(A_f, s)$ if $\delta_p > 0$ and 1 if $\delta_p = 0$; for $p \nmid N$, $L_p^{\mathrm{nv}} = L_p$ (C6). (i) For $p \mid N$, $L_p^{\mathrm{nv}}(A_f, s)^{-1} = 1 - a_p(f)^2 p^{-1-s}$, because $a_p(f) = \pm 1$ iff $\delta_p = 1$ (and then $L_p(A_f, s)^{-1} = 1 - p^{-1-s}$, item 9.2) and $a_p(f) = 0$ iff $\delta_p = 0$. So $L^{\mathrm{nv}}(\mathrm{ad}^0 f, s)$ is determined by the Fourier coefficients $a_p(f)$ alone, through 6.2(i) at good p and this formula at bad p; this is what "naive" means, and why the constructions of DFG04 built from q-expansions and depleted forms see it. (ii) $L^{\mathrm{nv}}(\mathrm{ad}^0 f, s) = L(\mathrm{ad}^0 f, s) \prod_{p^2 \mid N} L_p(A_f, s)^{-1}$: the two differ exactly at the additive primes, by the omitted factors, which are invisible in $a_p(f) = 0$ (the local representation at an additive prime is not determined by a_p; Chapter 15, 3.5). Paper: Section 1 ("the naive one ... the trivial factor at the additive primes"), Section 1.1, Definition def:terms, Theorem thm:dfg14, Lemma lem:fpr.

8.2 `prop:lfunctions:rankin-series` (proposition; P; important_notes.md point 30, sharpened). Let f be as in 6.1. For $\operatorname{Re} s > 2$:

$$\text{(A)}\quad \sum_{n\ge1} a_n(f)^2 n^{-s} = \frac{\zeta^{(N)}(s-1)}{\zeta^{(N)}(2s-2)}\, L^{\mathrm{nv}}(\mathrm{Sym}^2 f, s), \qquad \text{(B)}\quad \sum_{n\ge1} a_{n^2}(f)\, n^{-s} = \frac{L^{\mathrm{nv}}(\mathrm{Sym}^2 f, s)}{\zeta^{(N)}(2s-2)},$$

exactly, with no correction at the bad primes. Proof: both sides are Euler products (2.2); at a good p, with $X = p^{-s}$, $\sum_k a_{p^k}^2 X^k = \frac{1+pX}{(1-\alpha^2X)(1-pX)(1-\beta^2X)}$ and $\sum_k a_{p^{2k}} X^k = \frac{1-p^2X^2}{(1-\alpha^2X)(1-pX)(1-\beta^2X)}$ (write $a_{p^k} = \frac{\alpha^{k+1}-\beta^{k+1}}{\alpha-\beta}$ and sum geometric series; or check with the recursion of Chapter 13); at $p \mid N$, $a_{p^k} = a_p^k$ with $a_p^2 \in \{0,1\}$, so both local series equal $(1 - a_p^2 p^{-s})^{-1} = L_p^{\mathrm{nv}}(\mathrm{Sym}^2 f, s)$ (8.1(i)), and $\zeta^{(N)}$ has no factor at p. Consequences: (a) the Rankin series is **not** the naive symmetric square: at a good p it carries the extra factor $1 + p^{1-s}$, e.g. for 11a1 the coefficient of 2^{-s} is $a_2^2 = 4$ in the Rankin series and $a_2^2 - 2 = 2$ in $L^{\mathrm{nv}}(\mathrm{Sym}^2 f, s)$; (b) Shimura's series (B), completed by $\zeta^{(N)}(2s-2)$, **is** the naive symmetric square. `warning` environment quoting the paper's sentence and the corrected reading: "the naive one, $L^{\mathrm{nv}}(\mathrm{Sym}^2 f, s) = \zeta^{(N)}(2s-2)\sum a_{n^2} n^{-s}$, which differs from $\sum a_n(f)^2 n^{-s}$ by the factor $\zeta^{(N)}(s-1)/\zeta^{(N)}(2s-2)$". Verified: the local identities as power series to order 20 for $(p, a_p) = (5,2), (7,-3), (11,0), (13,5)$; the global identities at $s = 6$ for 11a1, 20a1, 121a1 (agreement to 10^{-12}, the truncation error). Reference: Shimura75 (for (B)); Rankin39, Shimura76 (for (A)); elementary. Paper: Section 1 (point 30).

8.3 `thm:lfunctions:petersson-sym2` (theorem; S for the unfolding, which is Chapter 13's 10.3(i); P for the rest). Let f be as in 6.1 and $\langle\ ,\ \rangle$ the unnormalised Petersson product (C9). Then

$$\langle f, f\rangle = \frac{N}{8\pi^3}\, L^{\mathrm{nv}}(\mathrm{Sym}^2 f, 2) = \frac{N}{8\pi^3}\, L(\mathrm{Sym}^2 f, 2) \prod_{p^2\mid N} P_p(p^{-2}) = \frac{N}{8\pi^3}\, L(\mathrm{ad}^0 f, 1) \prod_{p^2\mid N} L_p(A_f, 1)^{-1}.$$

Proof: Chapter 13, 10.3(i) gives $\operatorname{Res}_{s=2} D(f,s) = \frac{48\pi}{\mu(N)}\langle f, f\rangle$; by 8.2(A) and 3.2 the residue is $\frac{6N}{\pi^2\mu(N)} L^{\mathrm{nv}}(\mathrm{Sym}^2 f, 2)$; compare. (The point $s = 2$ is on the edge of absolute convergence, 4.3: write $L^{\mathrm{nv}}(\mathrm{Sym}^2 f, s) = D(f,s)\zeta^{(N)}(2s-2)/\zeta^{(N)}(s-1)$, which is holomorphic at $s = 2$ because D has at most a simple pole there (Chapter 13) and $\zeta^{(N)}(s-1)$ has one; its value is the value at $s = 2$ of the continuation 7.1, which is what PARI computes.) For square-free N this is Chapter 13's 10.3(ii). So "$\langle f, f\rangle$ is $L(\mathrm{Sym}^2 f, 2)$ up to the local factors at the bad primes" (paper, Section 1) holds in the precise form: up to $N/(8\pi^3)$ and the omitted factors at the **additive** primes only; the multiplicative primes contribute nothing beyond N. Verified (PARI, 18 digits; $\langle f, f\rangle$ as $\mu(N)$ times `mfpetersson`, $L(\mathrm{Sym}^2 f, 2)$ by `lfun(lfunsympow(E,2),2)`): 11a1, 14a1, 20a1, 27a1, 36a1, 49a1, 50a1, 64a1, 99a1, 121a1, 121c1, 121d1, 162a1, 52a1, 80a1, with correction factors $1, 1, 3/2, 1, 2, 8/7, 6/5, 1, 4/3, 12/11, 12/11, 120/121, 4/3, 3/2, 1$ (table 12.2). Reference: Rankin39; Selberg40 (proposed `bib/ch19.bib` key); Shimura76 (verify which statement); Zagier85; Watkins02 (the method of `ellmoddegree`; cite the section); DFG04 Lemma 2.12 (as cited by the paper; checked: it gives the Petersson norm with the naive L-value, Chapter 20 outline, Part 7; important_notes.md point 73; the writer compares its constant with this exact formula before attributing the formula to it). Paper: Section 1 ("Rankin, Shimura; see DFG04 Lemma 2.12").

8.4 `cor:lfunctions:degree-formula` (corollary; P given 8.3 and Chapter 14, its 7.3). (i) For every modular parametrisation $\varphi\colon X_0(N) \to E$ with Manin constant c_φ: $\deg\varphi \cdot \mathrm{vol}(\Lambda_E) = c_\varphi^2\, N\, L^{\mathrm{nv}}(\mathrm{ad}^0 f, 1)/(2\pi)$. (ii) For the optimal curve ($c_E = 1$ for the curves of the tables, with the sign of ω_E fixed by $c_E > 0$, Chapter 14; Appendix B, B-C9): $\deg\varphi_E = N\, L^{\mathrm{nv}}(\mathrm{ad}^0 f, 1)/(2\pi\,\mathrm{vol}(\Lambda_E))$. (iii) Put $B_E = N\, L(\mathrm{ad}^0 f, 1)/(2\pi\,\mathrm{vol}(\Lambda_E))$. Then $B_E = \deg\varphi_E \prod_{q^2\mid N} P_q(q^{-2})^{-1} \in \mathbb{Q}^\times$ (for $c_E = 1$) and, for every prime $\ell \nmid N$,

$$v_\ell(\deg\varphi_E) = v_\ell(B_E) + \sum_{q\mid N} e_\ell(E, q)$$

(at a multiplicative q, $e_\ell = 0$; at an additive q, $e_\ell(E,q) = v_\ell(L_q(A_f,1)^{-1}) = v_\ell(P_q(q^{-2}))$, 9.1). This identity is unconditional: the paper's inequality (eq:thm) is equivalent to $v_\ell(B_E) \ge \sum_{q\mid N} t_\ell(E,q)$. (iv) $v_\ell(B_E)$ can be negative when the hypotheses of Theorem thm:main fail: 539d1 ($B = 343/5$, $E[5]$ reducible),

121d1 ($B = 121/5$, $E[5]$ reducible), 98a1 ($B = 49/3$, $E[3]$ reducible), 121b1 ($B = 11/3$, CM by $\mathbb{Q}(\sqrt{-11})$, mod 3 image induced) (table 12.3). Verified: B_E computed both as $\deg \varphi_E / \prod P_q(q^{-2})$ (Cremona's `degphi`) and numerically from `lfun` and `E.area` agree for the 19 curves of 12.1 (e.g. 121a1: 11/2; 539c1: 294; 980i1: 1960; 1232l1: 256; 592c1: 128). Paper: Section 1 ("Up to explicit factors $\deg \varphi_E$ is $\langle f, f\rangle$ divided by the area of the period lattice"), Theorem thm:main (the role of the omitted factors), Section sec:sharp ("the size of the Petersson norm"). See Part 7, N2.

8.5 `rem:lfunctions:bloch-kato-reading` (remark; S, a reading, not a proof; forward pointer to Chapters 18, 20). For $\ell \notin S_f$, the paper's identity (eq:identity), which holds with equality (important_notes.md points 39, 56; Chapters 14, 20), reads $v_\ell(\deg \varphi_E) = \operatorname{length} H^1_f(\mathbb{Q}, W) + \sum_q v_\ell(\mathrm{Tam}_q(\mathrm{ad}^0 T)) + \sum_q e_\ell(E, q)$. Comparing with 8.4(iii) gives $v_\ell(B_E) = \operatorname{length} H^1_{\tilde f}(\mathbb{Q}, W) + \sum_q v_\ell(\mathrm{Tam}_q(\mathrm{ad}^0 T))$: the ℓ-part of the Bloch-Kato formula for $L(\mathrm{ad}^0 f, 1)$ with the period $2\pi \operatorname{vol}(\Lambda_E)/N$, which is the shape of DFG04 Theorem 0.1 (Chapter 18; the period normalisation of DFG04 is not checked here, and the chapter must say so). This is the conceptual content of the paper's sentence "the primes at which the naive and the true adjoint L-functions differ are exactly the primes at which the modular degree acquires extra local divisibility": the Petersson norm, hence $\deg \varphi_E$, sees L^{nv}; Bloch-Kato sees L; the ratio is the product of the omitted factors. Paper: Section 1 (the paragraph before Section 1.1), Theorem thm:main.

8.6 `thm:lfunctions:flach` (theorem; R). (Flach 1992.) Let $E/\mathbb{Q}$ be modular (now always) and p an odd prime satisfying the hypotheses of Flach92 (good reduction at p and a condition on the image of $\rho_{E,p}$; the writer copies them from the paper). Then the Bloch-Kato Selmer group of $\mathrm{Sym}^2 T_p E$ (for p odd $\mathrm{Sym}^2 T_p E \simeq \mathrm{ad}^0 T(1)$, the module T^D of Chapter 18) with $\mathbb{Q}_p/\mathbb{Z}_p$ coefficients is finite, and it is annihilated by an explicit integer built from the modular degree (verify the exact annihilator and which Selmer group, of $\mathrm{Sym}^2 T_p E$ or of its Kummer dual, against Flach92 before writing anything finer). The proof uses an Euler-system-type construction of classes from K_2 of $X_0(N) \times X_0(N)$ (Beilinson-Flach elements; one sentence). Reference: Flach92 (cite the main theorem by section). Paper: Section 1 ("Flach's finiteness theorem for the symmetric square is the first place where the modular degree meets the Selmer group of $\mathrm{Sym}^2 f$"). Chapter 18 may restate it in Selmer language; the statement lives here.

21.2.8 Section 9. The local terms and Lemma lem:euler

Standing hypotheses: $q \mid N$ a bad prime, $q \neq \ell$; the inertia images are Chapter 11's (Kraus's theorem via Chapter 10), the Weil-Deligne shapes and local factors Chapter 7's; this section assembles them, adds P_q, the naive factor and the valuations.

9.1 `def:lfunctions:terms` (definition, the paper's Definition def:terms, with the corrections of points 15 and 53; P for the properties). For a bad prime $q \neq \ell$:

$$e_\ell(E, q) = v_\ell\Big(\frac{L_q(A_f, 1)^{-1}}{L_q^{\mathrm{nv}}(A_f, 1)^{-1}}\Big) = v_\ell\Big(\frac{P_q(q^{-2})}{1 - a_q(E)^2 q^{-2}}\Big);$$

for every bad prime q (including $q = \ell$, as `rule(E, l)` and (eq:rule) use it; point 53): $t_\ell(E, q) = v_\ell(-v_q(j_E))$ if $v_q(j_E) < 0$, else 0. Properties: (i) $e_\ell(E, q)$ is a finite nonnegative integer ($P_q(q^{-2}) \in \mathbb{Z}[1/q] \setminus \{0\}$, 9.6); it is 0 at a multiplicative q (naive = true, 9.2) and $v_\ell(P_q(q^{-2}))$ at an additive q ($a_q = 0$). (ii) $t_\ell(E, q) = v_\ell(n)$ for type I_n at any q and for I_n^* at odd q (Chapter 10); at $q = 2$ a symbol I_m^* may have $-v_2(j) \neq m$, or $v_2(j) \geq 0$ (potentially good), and then $t_\ell = 0$. (iii) Twists: 9.5 (t_ℓ invariant; e_ℓ not, point 15). Paper: Definition def:terms, Theorem thm:main, Conjecture conj:euler, the appendix.

9.2 `prop:lfunctions:multiplicative` (proposition; P given Chapter 11, its 3.2(v), and Chapter 7, its 9.3). Let $q \neq \ell$ be a prime of multiplicative reduction (any q, split or nonsplit). Then $A_f^{I_q}$ is the line spanned by $e = e_1 \otimes e_2^\vee$ (the nilpotent N of the Weil-Deligne representation), Φ_q acts on it by q^{-1} (Frob_q by q; the sign ε cancels in the adjoint), so $L_q(A_f, s)^{-1} = 1 - q^{-1-s}$, $P_q(X) = 1 - X$, $L_q(A_f, 1)^{-1} = 1 - q^{-2}$, the naive factor equals the true one ($a_q^2 = 1$) and $e_\ell(E, q) = 0$. With Frob_q in place of Φ_q one would get $1 - q^{1-s}$ (point 11). Paper: the I_n row of the table; Lemma lem:euler (type I_n); Definition def:terms ("it is 0 at a multiplicative prime"); Theorem thm:dfg14 ("adding a multiplicative prime multiplies it by $1 - p^{-2}$"); Lemma lem:fpr; Section sec:iso ("Multiplicative primes show no Euler term").

9.3 `lem:lfunctions:euler` (lemma, the paper's Lemma lem:euler; P given Kraus's theorem (R, via Chapters 10 and 11)). Let $q \geq 5$ be a bad prime, $q \neq \ell$. Then $L_q(A_f, s)$, P_q and $e_\ell(E, q)$ are as follows, where σ denotes an element of order 3 and τ one of order 4 in $\mathrm{SL}_2(\mathbb{Z}_\ell)$, $E' = E^{(q^*)} = E \otimes \chi_{q^*}$ (good reduction at q for type I_0^*) and $a' = a_q(E')$:

type at q	$\rho(I_q)$	$A_f^{I_q}$ spanned by	Φ_q on it	$L_q(A_f, s)^{-1}$	$P_q(X)$	$P_q(q^{-2})$	$e_\ell(E, q)$
I_n	unipotent, infinite	e	q^{-1}	$1 - q^{-1-s}$	$1 - X$	$1 - q^{-2}$	0
I_n^*	$\chi_{q^*}\cdot$unipotent	e	q^{-1}	$1 - q^{-1-s}$	$1 - X$	$1 - q^{-2}$	$v_\ell(q^2 - 1)$
$\mathrm{IV}, \mathrm{IV}^*$ / $\mathrm{II}, \mathrm{II}^*$	$\langle\sigma\rangle$ / $\langle-\sigma\rangle$	$2\sigma + 1$ ($(2\sigma + 1)^2 = -3$)	$\chi_{-3}(q)$	$1 - \chi_{-3}(q)q^{-s}$	$1 - \chi_{-3}(q)qX$	$1 - \chi_{-3}(q)/q$	$v_\ell(q - \chi_{-3}(q))$
$\mathrm{III}, \mathrm{III}^*$	$\langle\tau\rangle$	τ	$\chi_{-4}(q)$	$1 - \chi_{-4}(q)q^{-s}$	$1 - \chi_{-4}(q)qX$	$1 - \chi_{-4}(q)/q$	$v_\ell(q - \chi_{-4}(q))$
I_0^*	$\{\pm 1\}$	all of $A_f = \mathrm{ad}^0 V_\ell E'$	$\alpha'/\beta', 1, \beta'/\alpha'$	$(1 - \frac{\alpha'}{\beta'}q^{-s})(1 - q^{-s})(1 - \frac{\beta'}{\alpha'}q^{-s})$	$(1 - qX)(1 - (a'^2 - 2q)X + q^2X^2)$	$\frac{(q-1)((q+1)^2 - a'^2)}{q^3}$	$v_\ell(q-1) + v_\ell((q+1)^2 - a'^2)$

Proof (to be written in five steps, each citing its source): (1) inertia images: Chapter 11, its 3.2, 3.4, 4.1 (Kraus for the orders; tame since $q \geq 5$, hence cyclic). (2) Invariants: $A_f^{I_q}$ is the trace-zero part of the commutant of $\rho(I_q)$ (Chapter 4); for $\langle\pm\sigma\rangle$ the commutant is $\mathbb{Q}_\ell[\sigma] \simeq \mathbb{Q}_\ell \otimes \mathbb{Q}(\zeta_3)$, whose trace-zero line is spanned by $2\sigma + 1$ (the paper's "$\sqrt{-3}$"); the sign of $-\sigma$ disappears in ad^0; for $\langle\tau\rangle$ the trace-zero line is spanned by τ itself (point 34: τ is diagonalisable over $\mathbb{Q}_\ell$ only if $\ell \equiv 1 \pmod 4$, so "the diagonal element" must not be written); for the unipotent shapes it is $\ker[N, \cdot\,] = \mathbb{Q}_\ell N$ (Chapter 7, 9.3); for $\{\pm 1\}$ everything. The commutant statements are for $\mathrm{End}_{I_q}(T) = \mathbb{Z}_\ell[g]$ and $\mathrm{End}_{I_q}(V) = \mathbb{Q}_\ell[g]$, the latter $\mathbb{Q}_\ell \otimes \mathbb{Q}(\zeta_3)$ or $\mathbb{Q}_\ell \otimes \mathbb{Q}(i)$, which is not a field when $\ell \equiv 1 \pmod 3$, resp. $\pmod 4$ (point 12). (3) Frobenius: $\rho(\mathrm{Frob}_q)\, g\, \rho(\mathrm{Frob}_q)^{-1} = g^q$ (Chapter 7, 5.6); $\sigma^q = \sigma$ or σ^{-1} as $q \equiv 1$ or $2 \pmod 3$, and $2\sigma^{-1} + 1 = -(2\sigma + 1)$; $\tau^q = \tau$ or $-\tau$ as $q \equiv 1$ or $3 \pmod 4$; Φ_q gives the same sign since $q^{-1} \equiv q$ modulo 3 and modulo 4; for I_0^*, $V \simeq V_\ell E' \otimes \chi_{q^*}$ (Chapter 10, Lemma lem:twist (ii); Chapter 11, 4.1) and $\mathrm{ad}^0 V = \mathrm{ad}^0 V_\ell E'$ is unramified with eigenvalues $\alpha'/\beta', 1, \beta'/\alpha'$ (6.2); for I_n^*, $V \simeq V' \otimes \chi_{q^*}$ with V' of multiplicative shape and $\mathrm{ad}^0 V = \mathrm{ad}^0 V'$ (9.2). (4) P_q by 6.1(i). (5) Valuations by Chapter 2, 11.3 to 11.5: the naive factor is 1 for the additive rows ($a_q = 0$) and $1 - q^{-2}$ for I_n. Well-definedness for I_0^*: the other ramified quadratic character of $\mathbb{Q}_q$ gives the twist with $a_q = -a'$, and only a'^2 enters. Verified: on **all** 64,687 curves of conductor below 10,000, every one of the 117,232 pairs (curve, bad prime $q \geq 5$) has PARI's P_q equal to the table (I_n: 102,309; I_n^*: 6,224; I_0^*: 2,241, with a' computed from `elltwist`; II: 1,575; II^*: 760; III, III^*: 1,394 each; IV: 760; IV^*: 1,575); no exception. Reference: Kraus90; SerreTate68; Chapters 4, 7, 10, 11. Paper: Lemma lem:euler, the table of Section 1, the abstract.

9.4 `rem:lfunctions:euler-values` (remark; P, citing Chapter 2, its 11.5). When the omitted factor is an ℓ-adic non-unit: for $\ell = 3$, always for II to IV^* ($q \equiv \chi_{-3}(q) \pmod 3$) and for I_n^*; for III, III^* iff $q \equiv \pm 1 \pmod{12}$; for I_0^* iff $q \equiv 1 \pmod 3$ or $3 \mid a'$. For $\ell \geq 5$: iff $q \equiv \chi(q) \pmod \ell$ (rows 3, 4), $q \equiv \pm 1 \pmod \ell$ (I_n^*), $q \equiv 1 \pmod \ell$ or $a' \equiv \pm(q+1) \pmod \ell$ (I_0^*). Examples (12.1): 121a1 (II at 11: $e_3 = 1$), 147b1 (IV^* at 7: $e_3 = v_3(6) = 1$), 726a1 (III at 11: $e_3 = v_3(12) = 1$), 539c1 (I_3^* at 7: $e_3 = v_3(48) = 1$), 539d1 (I_0^* at 7, $E' =$ 11a3, $a' = -2$: $e_3 = v_3(6) + v_3(60) = 2$), 275b1 ($\mathrm{I}_0^*$ at 5, $a' = \pm 1$: $e_7 = v_7(35) = 1$, $\deg \varphi_E = 28$). Paper: Proposition prop:forms (proof), the Remark after Theorem thm:quant, (1), Sections sec:iso, sec:anomaly.

9.5 `prop:lfunctions:twists` (proposition; P). Let χ be a quadratic character and $E^\chi = E \otimes \chi$. (i) A_f and $A_{f\otimes\chi}$ are isomorphic ($\mathrm{ad}^0(\rho \otimes \chi) = \mathrm{ad}^0\, \rho$), so $L_q(A_f, s)$, P_q and $L(\mathrm{ad}^0\, f, s)$ are invariant under

every quadratic twist. (ii) If χ is unramified at q, then δ_q, a_q^2, the naive factor and $e_\ell(E,q)$ are unchanged. (iii) For q odd and $\chi=\chi_{q^*}$: e_ℓ is unchanged for the exchanges II $\leftrightarrow$ IV*, III $\leftrightarrow$ III*, IV $\leftrightarrow$ II* ($q\geq 5$; $\delta_q=0$ on both sides), but changes for $\mathrm{I}_n \leftrightarrow \mathrm{I}_n^*$ ($0\leftrightarrow v_\ell(q^2-1)$) and $\mathrm{I}_0\leftrightarrow \mathrm{I}_0^*$ (e_ℓ undefined at a good prime; $v_\ell(q-1)+v_\ell((q+1)^2-a_q^2)$ on the I_0^* side); in every case $e_\ell(E,q)+v_\ell(L_q^{\mathrm{nv}}(A_f,1)^{-1})=v_\ell(L_q(A_f,1)^{-1})$ is invariant. (iv) $t_\ell(E,q)$ is invariant under every quadratic twist (j is). Example: 11a1 (type I_5 at 11, $e_3=0$) and its twist 121d2 = 11a1 $\otimes\chi_{-11}$ (type I_5^*, $e_3=v_3(120)=1$) (important_notes.md point 15, B8). Paper: Definition def:terms ("Both terms are invariant under quadratic twist": true for t_ℓ only), Theorem thm:quant Step 3 ("$A_g=A_f$ … its naive factor is the true one, which is the factor that f omits").

9.6 `prop:lfunctions:fpr-L-part` (proposition; P; the L-part of the paper's Lemma lem:fpr). Let ℓ be odd, $p\mid N$, $p\neq\ell$, $V^D=\mathrm{Hom}(A_f,\mathbb{Q}_\ell(1))\simeq A_f(1)$ (trace form, Chapter 4, 7.6). Then

$$\det\big(1-\Phi_p \;\big|\; (V^D)^{I_p}\big)=L_p(A_f(1),0)^{-1}=L_p(A_f,1)^{-1}=P_p(p^{-2}),$$

a nonzero element of $\mathbb{Z}[1/p]\subset\mathbb{Z}_{(\ell)}$, of valuation $e_\ell(E,p)$ at an additive p and $v_\ell(1-p^{-2})=v_\ell(p^2-1)$ at a multiplicative p. Nonvanishing: $1-p^{-2}$, $1\mp p^{-1}$ and $(p-1)((p+1)^2-a'^2)/p^3$ (Hasse) are nonzero, and 10.1 lists the only other shapes. With Frob_p in place of Φ_p the determinant changes by $\det(-\mathrm{Frob}_p\mid (V^D)^{I_p})$, a unit of $\mathbb{Z}_{(\ell)}$ (a power of p up to sign): the vanishing of $H^1_{\mathrm{ur}}(\mathbb{Q}_p,V^D)=(V^D)^{I_p}/(\mathrm{Frob}_p-1)$ (Chapter 17) holds in either reading, but the equality with $L_p(A_f,1)^{-1}$ needs Φ_p (point 11). Paper: Lemma lem:fpr ("$L_p(A_f,1)^{-1}=L_p(A_f(1),0)^{-1}$ is a nonzero rational number which is an ℓ-adic integer, of valuation $e_\ell(E,p)$ at an additive prime and $v_\ell(1-p^{-2})$ at a multiplicative one"), the paragraph after Lemma lem:dfg21. The Fitting-ideal identity $\mathrm{Fitt}\,H^1_f(\mathbb{Q}_p,T^D)=L_p(A_f,1)^{-1}\,\mathrm{Tam}_p(T^D)$ is Chapter 18's.

9.7 `prop:lfunctions:naive-factors` (proposition; P). At $s=1$: $L_p^{\mathrm{nv}}(A_f,1)^{-1}=\frac{(p-1)((p+1)^2-a_p^2)}{p^3}$ ($p\nmid N$), $1-p^{-2}$ ($p\,\|\,N$), 1 ($p^2\mid N$). Consequently, for a finite set Σ of primes dividing N, $p\neq\ell$: $\sum_{p\in\Sigma}\big(v_\ell(L_p(A_f,1)^{-1})-v_\ell(L_p^{\mathrm{nv}}(A_f,1)^{-1})\big)=\sum_{p\in\Sigma}e_\ell(E,p)$, the cancellation in the proof of Theorem thm:main; and for $g=f_{E_S}$ as in Theorem thm:quant ($\delta_q(g)\geq 1$ at $q\in S$): $v_\ell(L_q^{\mathrm{nv}}(A_g,1)^{-1})=v_\ell(L_q(A_f,1)^{-1})=e_\ell(E,q)$ (9.5(i), 9.3). Paper: Theorem thm:dfg14 and the paragraph after it ("Only the primes with $\delta_p\geq 1$ contribute"), the proof of Theorem thm:main, Theorem thm:quant Step 3, the Remark after Theorem thm:quant, (3) ("there the quadratic twist does not lower the level, f is its own depletion, and Theorem thm:dfg14 gives no factor"). The theorem of DFG04 itself is Chapter 20's.

21.2.9 Section 10. Every prime, including 2 and 3

10.1 `prop:lfunctions:all-shapes` (proposition; P given Grothendieck/Neron-Ogg-Shafarevich (Chapter 11) and local class field theory only for the existence of quadratic characters (Chapter 7, 6.3); new in this form). Let $q\neq\ell$ be a prime of bad reduction, any q (including 2 and 3). Exactly one of the following holds.

(a) *Potentially multiplicative* ($v_q(j)<0$): $P_q(X)=1-X$, $L_q(A_f,1)^{-1}=1-q^{-2}$; $e_\ell(E,q)=0$ if the reduction is multiplicative and $v_\ell(q^2-1)$ if it is additive.

(b) *Potentially good with* $\rho(I_q)=\{\pm1\}$; equivalently, some quadratic twist $E^{(d)}$ has good reduction at q (for q odd, $d=q^*$ works; for $q=2$, one of $d=-4,8,-8$ works). Then, with $a'=a_q(E^{(d)})$ (its square does not depend on the choice), $P_q(X)=(1-qX)(1-(a'^2-2q)X+q^2X^2)$ and $e_\ell(E,q)=v_\ell(q-1)+v_\ell((q+1)^2-a'^2)$: the I_0^* formula, at every q.

(c) *Potentially good with* $\rho(I_q)$ *cyclic of order* 3, 4 *or* 6: $A_f^{I_q}$ is a line spanned by $x=2g-\mathrm{tr}\,g$ (g a generator), Φ_q acts on it by a sign $c\in\{\pm1\}$, $P_q(X)=1-cqX$ and $e_\ell(E,q)=v_\ell(q-c)$. For $q\geq 5$, $c=\chi_{-3}(q)$ or $\chi_{-4}(q)$ (9.3); for $q=2,3$ the type does not determine c.

(d) *Potentially good with* $\rho(I_q)$ *non-abelian* (Q_8 or $\mathrm{SL}_2(\mathbb{F}_3)$ at $q=2$, $\mathbb{Z}/3\rtimes\mathbb{Z}/4$ at $q=3$; Chapter 11, its 4.6; Kraus90, verify the list): $A_f^{I_q}=0$, $P_q=1$, $e_\ell(E,q)=0$.

Proof: $\rho(I_q)$ is finite in the potentially good case and lies in SL_2 (Chapter 11). A finite subgroup of the upper triangular group in characteristic 0 meets the unipotent radical trivially, so it is abelian;

hence a non-abelian finite subgroup of $\mathrm{SL}_2(\overline{\mathbb{Q}}_\ell)$ is absolutely irreducible and its commutant is the scalars (Schur): (d). A finite abelian subgroup of $\mathrm{SL}_2(\overline{\mathbb{Q}}_\ell)$ is diagonalisable, hence cyclic; if it is not in $\{\pm 1\}$ its commutant is $\overline{\mathbb{Q}}_\ell[g]$, of dimension 2, with trace-zero line spanned by x, and $x^2 = (\operatorname{tr} g)^2 - 4$ is a nonzero scalar; Φ_q normalises $\rho(I_q)$, hence preserves the line, and $\Phi_q x \Phi_q^{-1} = cx$ with $c^2 = 1$ because conjugation preserves x^2: (c). If $\rho(I_q) = \{\pm 1\}$, the character $\theta = \rho|_{I_q}$ is invariant under conjugation by $G_{\mathbb{Q}_q}$ (-1 is central), so $\ker\theta$ is normal in $G_{\mathbb{Q}_q}$ and cuts out a quadratic extension M of $\mathbb{Q}_q^{\mathrm{ur}}$ with $\mathrm{Gal}(M/\mathbb{Q}_q)$ a central extension of $\hat{\mathbb{Z}}$ by $\mathbb{Z}/2$, which splits; so θ extends to a quadratic character χ of $G_{\mathbb{Q}_q}$, $\rho \otimes \chi$ is unramified, $E \otimes \chi$ has good reduction (Neron-Ogg-Shafarevich), χ is the restriction of some χ_d with d as stated (for $q = 2$, the ramified square classes of $\mathbb{Q}_2^\times$ are represented by $-1, \pm 2$ up to the unramified class 5), and $\mathrm{ad}^0 V = \mathrm{ad}^0 V_\ell E^{(d)}$: (b) by 6.2. (a) is 9.2 and its twist (Chapter 11, its 3.4, for $q = 2$ as well). Consequences for the wild primes: at $q = 2$, $e_3(E,2) \in \{0,1,2\}$ (1 in cases (a)-additive and (c) with $c = -1$; 2 in case (b) with $a' = 0$), $e_5(E,2) \in \{0,1\}$ (1 exactly in case (b) with $a' = \pm 2$), $e_\ell(E,2) = 0$ for $\ell \geq 7$; at $q = 3$, $e_5(E,3) = 1$ exactly in case (b) with $a' = \pm 1$, $e_7(E,3) = 1$ exactly in case (b) with $a' = \pm 3$, and $e_\ell(E,3) = 0$ otherwise (since $v_\ell(8) = v_\ell(4/3) = v_\ell(2/3) = 0$ for $\ell \geq 5$; $|a'| \leq 2\sqrt{q}$). Verified on all curves of conductor below 10,000 (all 64,687, not only optimal ones), counting pairs (curve, q):

q	type	(a) $1 - X$	(b) cubic	(c) $1 + qX$	(c) $1 - qX$	(d) 1
2	I_n	25,751				
2	I_m^*	3,732	716	146		5,073
2	I_0^*			792		1,712
2	II / II*		136 / 291	469 / 108		1,109 / 1,267
2	III / III*			42 / 42	56 / 56	2,012 / 2,202
2	IV / IV*			2,094 / 2,656		
3	I_n / I_n^*	25,283 / 7,805				
3	I_0^*		2,283			
3	II / II*			145 / 91	135 / 86	986 / 699
3	III / III*			1,553 / 1,553		
3	IV / IV*			91 / 145	86 / 135	699 / 986

Every cubic factor (1,143 at 2, 2,283 at 3) equals the formula of (b) for a twist $E^{(d)}$ with good reduction found by `elltwist`, and no curve without such a twist has a cubic factor; at 2, I_m^* has $P_2 = 1 - X$ exactly when $v_2(j) < 0$. No factor of degree 2 occurs. Examples: 1232l1 = 77a1 $\otimes\chi_{-4}$, $a_2(77a1) = 0$, $P_2 = (1-2X)(1+2X)^2$, $e_3(E,2) = v_3(9/8) = 2$, $\deg\varphi_E = 288 = 2^5 \cdot 3^2$ (point 50's cubic, explained); 592c1 = 37a1 $\otimes\chi_{-4}$, $a_2(37a1) = -2$, $P_2 = (1-2X)(1+4X^2)$, $e_5(E,2) = v_5(5/8) = 1$, $\deg\varphi_E = 80$ (the prime 2 contributes at $\ell = 5$; all 93 optimal curves below 10,000 with this factor, $5 \nmid N$ and $E[5]$ irreducible have $5 \mid \deg\varphi_E$); 336f1 (type II at 2, case (c), $P_2 = 1 + 2X$, $e_3 = 1$, $\deg\varphi_E = 24$; point 50); every type IV, IV* at 2 is case (c) with $c = \chi_{-3}(2) = -1$ (Chapter 11, its 4.4; point 43). Reference: elementary given Chapters 7, 11; SerreTate68. Paper: Remark rem:wild, Section sec:wild ("Definition def:terms with PARI's factor explains every one of these minima"), Proposition prop:forms, Conjecture conj:euler (the terms at 2, 3), the Remark after Theorem thm:quant and point 60 (the I_0^* formula at $q = 3$: case (b)).

10.2 `rem:lfunctions:rem-wild-status` (remark; P for the statuses, by 10.1 and the data). The sentences of Remark rem:wild, one by one: (i) "At $q = 2, 3$ the Kodaira type does not determine $L_q(A_f, s)$, but PARI does": true (10.1 table); 10.1 says what does determine it (the shape of $\rho(I_q)$ and c or a'). (ii) "a type IV at 2 has $P_2(X) = 1 + 2X$ and $e_3 = 1$": a theorem (Chapter 11, its 4.4; point 43; case (c) of 10.1). (iii) "a type II at 2 has $P_2 = 1$ and $e_3 = 0$": false in general (point 50; 336f1, 1232l1; all four shapes occur for type II at 2). (iv) "a type I_n^* at 2 has potentially good reduction in about half of the cases, with a cubic P_2 and $t_\ell = 0$": below conductor 10,000, 5,935 of the 9,667 pairs (61%) are potentially good, and

among them the cubic occurs for 716 only ($P_2 = 1$ for 5,073, $1 + 2X$ for 146); the cubic there always has $a' = \pm 1$, so it contributes nothing for odd ℓ; $t_\ell = 0$ is right (point 50). (v) "At $q = \ell$ the Euler factor is not the relevant quantity": consistent with Definition def:terms ($q \neq \ell$); t_ℓ is used at $q = \ell$ (point 53). Paper: Remark rem:wild; Section sec:wild.

21.2.10 Section 11. Computing the factors: PARI and `rule(E, l)`

11.1 `rem:lfunctions:pari` (remark; P as a computation). `S = lfunsympow(E, 2)` is the L-function $L(\mathrm{Sym}^2 f, s)$ of C4 (motivic weight 3; `lfunparams(S)` = `[N_sym, 3, [0, 0, 1]]`, 7.2), with the true factors at every prime; `lfuneuler(S, q)` is $1/P_q(x)$, $x = q^{-s}$ (C5); `lfun(S, 2)` is $L(\mathrm{Sym}^2 f, 2) = L(\mathrm{ad}^0 f, 1)$. Hence $e_\ell(E, q)$ = `valuation(subst(1/lfuneuler(S,q), x, 1/q^2) / (1 - a_q^2/q^2), l)` (9.1). The bad factors are computed by Watkins's method (Watkins02; cite the section, verify; Coates-Schmidt for the potentially good types at $q \geq 5$, verify). How to install and call PARI, `lfuncreate`, precision and memory are Chapter 22's. Paper: Section 1.1 ("PARI's `lfunsympow(E,2)` returns these factors at every prime, including 2 and 3"), Section sec:data, the appendix.

11.2 `ex:lfunctions:rule` (worked example; P as a computation). The mathematical content of the appendix function: for $q \neq \ell$, `r` $= P_q(q^{-2}) = L_q(A_f, 1)^{-1}$ (9.1), divided by $1 - q^{-2}$ when the code `k` ≥ 5 (type I_n, $n \geq 1$: the naive factor is the true one, 9.2), and `e` $= v_\ell$(`r`) $= e_\ell(E, q)$, except for the codes ± 3 ($\mathrm{III}, \mathrm{III}^*$) unless `iii` $= 1$; for $q = \ell = 3$ the weights w_3 of Conjecture conj:euler; `t` $= t_\ell(E, q)$ from `valuation(j, q)` at every q, including $q = \ell$ (point 53); finally `ns3` $= h_3(E)$ subtracted when `iii` $= 1$ (Chapter 11, its 8.4, 8.5). At an additive q no division is made because the naive factor is 1 ($a_q = 0$), exactly as in Definition def:terms. Values, recomputed with PARI 2.17.2: 121a1 returns `[1, 1]` ($q = 11$, type II: `r` $= 12/11$, `e` $= 1$, `t` $= 0$; $\deg \varphi_E = 6$); 726a1 $= [1, 1, 0, -35, -51]$ returns `[2, 2]` both with `iii` $= 0$ and `iii` $= 1$ ($q = 2$, I_6: `r` $= 1$, `t` $= v_3(6) = 1$; $q = 3 = \ell$, I_3: $w_3 = 0$, `t` $= 1$; $q = 11$, III: `r` $= 12/11$, `e` $= 1$ only with `iii`, cancelled by `ns3` $= 1$ (image 3Ns); $\deg \varphi_E = 144 = 2^4 \cdot 3^2$). The line-by-line reading, the Kodaira codes and the data protocol are Chapter 22's. Paper: the appendix; Conjecture conj:euler; Section sec:anomaly.

21.2.11 Section 12. Worked examples

12.1 `ex:lfunctions:examples` (worked example; P as a computation; PARI and Cremona's `degphi`). Bad primes with type, P_q, $P_q(q^{-2})$; $\deg \varphi_E$; $B_E = \deg \varphi_E / \prod_{q^2 \mid N} P_q(q^{-2})$ (8.4), which agrees with $N\, L(\mathrm{ad}^0 f, 1)/(2\pi \operatorname{vol} \Lambda_E)$ numerically in every row.

curve	bad primes: type, $P_q(X)$, $P_q(q^{-2})$	nonzero e_ℓ	$\deg \varphi_E$	B_E	remark
11a1	11: I_5, $1 - X$	none	1	1	$t_5(E, 11) = 1$
121a1	11: II, $1 + 11X$, $12/11$	$e_3 = 1$	6	$11/2$	paper's Example (b); `rule` $[1, 1]$
121c1	11: IV, $1 + 11X$, $12/11$	$e_3 = 1$	6	$11/2$	twist of 121a1 by χ_{-11}
147b1	3: I_1; 7: IV^*, $1 - 7X$, $6/7$	$e_3(7) = 1$	42	49	$\chi_{-3}(7) = +1$
200a1	2: II^*, 1; 5: IV^*, $1 + 5X$, $6/5$	$e_3(5) = 1$	120	100	surjective mod ℓ images
350e1	2: I_{11}; 5: II^*, $1 + 5X$, $6/5$; 7: I_2	$e_3(5) = 1$	1320	1100	$t_{11}(E, 2) = 1$

curve	bad primes: type, $P_q(X)$, $P_q(q^{-2})$	nonzero e_ℓ	$\deg \varphi_E$	B_E	remark
726a1	2: I_6; 3: I_3; 11: III, $1+11X$, 12/11	$e_3(11)=1$	144	132	3Ns, $h_3=1$; `rule` $[2,2]$
539c1	7: I_3^*, $1-X$, 48/49; 11: I_2	$e_3(7)=1$	288	294	paper's Example (c), $\ell=3$
539d1	7: I_0^*, $(1-7X)(1+10X+49X^2)$, 360/343; 11: I_1	$e_3=2$, $e_5=1$	72	343/5	$E'=$ 11a3, $a'=-2$; $E[5]$ reducible
980i1	2: IV^*, $1+2X$, 3/2; 5: I_1; 7: I_5^*, 48/49	$e_3(2)=e_3(7)=1$	2880	1960	paper's Example (c), $\ell=5$: $t_5=1$
275b1	5: I_0^*, $(1-5X)(1+9X+25X^2)$, 28/25; 11: I_1	$e_7(5)=1$	28	25	$a'=\pm1$
1232l1	2: II^*, $(1-2X)(1+2X)^2$, 9/8; 7: I_2; 11: I_1	$e_3(2)=2$	288	256	77a1 $\otimes\chi_{-4}$
592c1	2: II^*, $(1-2X)(1+4X^2)$, 5/8; 37: I_1	$e_5(2)=1$	80	128	37a1 $\otimes\chi_{-4}$
336f1	2: II, $1+2X$, 3/2; 3: I_1; 7: I_2	$e_3(2)=1$	24	16	point 50
52a1	2: IV^*, $1+2X$, 3/2; 13: I_2	$e_3(2)=1$	3	2	Chapter 15's case (a) at $q=2$
98a1	2: I_2; 7: I_1^*, 48/49	$e_3(7)=1$	16	49/3	$E[3]$ reducible: bound fails
121b1	11: III, $1+11X$, 12/11	$e_3=1$	4	11/3	CM by $\mathbb{Q}(\sqrt{-11})$: bound fails
121d1	11: I_1^*, $1-X$, 120/121	$e_3=e_5=1$	24	121/5	$E[5]$ reducible

Reading: whenever ℓ is odd, $\ell \nmid N$ and the hypothesis of Theorem thm:main holds, $v_\ell(B_E)\geq 0$ in this table; the four rows with $v_\ell(B_E)<0$ are exactly where the hypothesis fails (reducible $E[\ell]$, or CM with induced mod 3 image). Paper: the Example of Section 3, the appendix, Sections sec:iso, sec:wild, sec:eis.

12.2 `ex:lfunctions:petersson-table` (worked example; P as a computation). For 11a1, 14a1, 20a1, 27a1, 36a1, 49a1, 50a1, 64a1, 99a1, 121a1, 121c1, 121d1, 162a1, 52a1, 80a1: $\langle f,f\rangle$ (unnormalised; e.g. 11a1: 0.04690014787349518782, 20a1: 0.08134945682538172856, 27a1: 0.06846465111921223464, 49a1: 0.12524576465640295885, 121a1: 0.42823936001195717857, 121d1: 0.51163797680176568534), the correction $\prod_{p^2\mid N} P_p(p^{-2})$ $(1,1,3/2,1,2,8/7,6/5,1,4/3,12/11,12/11,120/121,4/3,3/2,1)$, the ratio of $\langle f,f\rangle$ to the right side of 8.3 (1 to 10^{-15} in every case), and $NL^{\mathrm{nv}}(\mathrm{ad}^0 f,1)/(2\pi \operatorname{vol}\Lambda_E) = \deg\varphi_E$ $(1,1,1,1,1,1,2,2,4,6,6,24,12,3,4)$. $L(\mathrm{Sym}^2 f,2)$ for 11a1 is 1.05759924459096… (as in Chapter 13, its 10.3). Paper: Section 1 (Zagier, Watkins, Rankin, Shimura).

21.3 3. Section-by-section plan with target line counts

Line counts are lines of LaTeX, card excluded (PLAN.md Section 7.3: 1,000 to 2,200).

section	title	contents	target lines
(starred)	Core results at a glance	the card: every item of Part 2 with its tag and `\Cref`; C2, C3, C4, C6 restated in one line each; the table of 9.3 in compact form	(excluded)
(starred)	Where the paper uses this chapter	Section 1 (naive versus true, Petersson norm, Zagier, Watkins, Rankin, Shimura, Flach, DFG04 Lemma 2.12; point 30); Section 1.1 (δ_p, L_p, L_p^{nv}, `lfunsympow`, the shift); Definition def:terms and the table; Lemma lem:euler; Remark rem:wild and Section sec:wild; Theorem thm:dfg14 and the paragraph after it; Lemma lem:fpr (L-part) and the paragraph after Lemma lem:dfg21; the proof of Theorem thm:main (cancellation); Theorem thm:quant Step 3 and the Remark after it; Proposition prop:forms; Conjecture conj:euler; the appendix (`rule`)	50
1	Introduction and conventions	what the chapter is for; the physics asides; C1 to C10 as `convention` environments; a `warning` on the two Frobenii (point 11)	110
2	Dirichlet series and Euler products	2.1 to 2.3; the primon aside; Mellin transform	100
3	ζ, Dirichlet L-functions, Dedekind zeta	3.1 to 3.4; theta function and Poisson summation; $\zeta(2)$; $L(1,\chi_{-4})$, $L(1,\chi_{-3})$; $\zeta_{\mathbb{Q}(i)}$, $\zeta_{\mathbb{Q}(\sqrt{-3})}$	190

section	title	contents	target lines
4	L-functions of ℓ-adic representations	4.1 to 4.4; local factors recalled from Chapter 7; compatible systems; convergence and weights; the finite-inertia examples	140
5	The L-function of E	5.1, 5.2; functional equation and root number; examples 11a1, 37a1	90
6	Adjoint and symmetric square	6.1, 6.2; the shift by one; P_p; Wiles's factor and Ribet's condition; `paperlink`: Section 1.1, Theorem thm:dfg14	120
7	Analytic theory, normalisations, critical values	7.1 to 7.4; the normalisation table; conductor of ad^0 by type; Deligne's conjecture in one paragraph	140
8	Naive versus true; Rankin-Selberg; Petersson norm and modular degree	8.1 to 8.6; the `warning` on "the Euler product of $\sum a_n^2 n^{-s}$" (point 30); the proof of 8.3 from Chapter 13; the identity 8.4(iii) and its reading 8.5; Flach	230
9	The local terms and Lemma lem:euler	9.1 to 9.7; the full proof of Lemma lem:euler in five steps; the table with P_q; twists (point 15); Lemma lem:fpr's L-part; the naive factors of Theorem thm:dfg14; `paperlink` for each paper item	300
10	Every prime, including 2 and 3	10.1 with proof and the data table; 10.2 (Remark rem:wild sentence by sentence, points 43, 50)	150
11	Computing the factors	11.1, 11.2 (`lfunsympow`, `lfuneuler`, the mathematics of `rule`); pointer to Chapter 22	70
12	Worked examples	12.1, 12.2 (the two tables, with a short reading)	90
13	Exercises	the 18 exercises of Part 4.2	150

section	title	contents	target lines
14	Notes and references	Davenport, Washington, SerreCourse, Neukirch, Rankin39, Selberg40, Shimura75, Shimura76, GelbartJacquet, Deligne79, Sturm80, Flach92, Zagier85, Watkins02, CoatesSchmidt87, DFG04, Kraus90, SerreTate68, Tate79, Rohrlich94, PARI; which numbers were cited and which are marked (verify)	45
	total		**1,975**

21.4 4. Coverage table and exercises

21.4.1 4.1 Coverage: PLAN.md scope, Section 6 map rows and paper hooks to line-up items

PLAN.md scope item, Section 6 row, or paper hook	line-up items
Dirichlet series and their convergence	2.1
Euler products	2.2
Riemann zeta: Euler product, continuation and functional equation via theta functions, Mellin transforms (physics)	2.3, 3.1
Dirichlet L-functions: definition, Euler product, functional equation, $L(1,\chi)\neq 0$, Dirichlet's theorem	3.3
Dedekind zeta function and its factorisation for abelian fields	3.4
$L(E,s)$: local factors at good, multiplicative ($a_p=\pm 1$), additive primes; the conductor	5.1 (conductor: Chapter 10)
completed $\Lambda(E,s)$, functional equation with root number, as a consequence of modularity; $L(E,s)=L(f_E,s)$	5.2
L-functions of ℓ-adic representations: $L_p(V,s)=\det(1-\mathrm{Frob}_p\, p^{-s}\mid V^{I_p})^{-1}$ for $p\neq\ell$; at $p=\ell$ via D_{cris}	4.1
the Weil-Deligne definition, independence of ℓ; conjecture and theorem (modular forms)	4.1, 4.2
the L-function of a representation with finite image of inertia (the three Kodaira cases)	4.4, 9.3
adjoint and symmetric square L-functions: definitions; $\mathrm{ad}^0\,\rho=\mathrm{Sym}^2\,\rho\otimes\det^{-1}$; $L(\mathrm{Sym}^2\, f,s)=L(\mathrm{ad}^0\, f,s-1)$	6.1

PLAN.md scope item, Section 6 row, or paper hook	line-up items
normalisations in Shimura, Gelbart-Jacquet, DFG04, PARI	7.4 (with 7.1, 8.2(B), 11.1)
the naive adjoint L-function and the Rankin-Selberg convolution (PLAN's formula corrected: Part 7, B-a)	8.1, 8.2, 7.4
$L_p^{\mathrm{nv}} = L_p$ when $\delta_p > 0$, $= 1$ when $\delta_p = 0$, and why (built from a_p alone)	8.1
the Rankin-Selberg method (unfolding) and the theorem of Rankin and Shimura with the exact factors at bad primes	8.3 (unfolding: Chapter 13, 10.3)
the true local factors at bad primes: the paper's table derived in full (Lemma lem:euler), each Kodaira type at $q \geq 5$	9.2, 9.3, 9.4
the invariant line spanned by $\sqrt{-3}$ in $\mathbb{Z}_\ell[\zeta_3]$ (corrected, point 12), by τ for III (point 34)	9.3
I_0^*: $V = V' \otimes \chi_{q^*}$, $L_q(A_f,1)^{-1} = \frac{q-1}{q}\frac{(q+1)^2 - a_q(E')^2}{q^2}$	6.2, 9.3
the wild primes 2 and 3 (type does not determine the factor; the examples of Remark rem:wild)	10.1, 10.2
how PARI computes these (`lfunsympow(E,2)`, `lfuneuler`, P_q, $e_\ell = v_\ell(P_q(q^{-2})/(1-a_q^2q^{-2}))$)	11.1, 11.2
the definition of $e_\ell(E,q)$ (Definition def:terms) "and its invariance under quadratic twist" (false for e_ℓ: point 15)	9.1, 9.5
Flach's theorem on the finiteness of the Selmer group of Sym^2	8.6
special values and the Bloch-Kato conjecture for the adjoint (one page); connection to the modular degree via the Petersson norm	8.4, 8.5 (precise statements: Chapter 18)
Deligne's conjecture on critical values (one paragraph)	7.3
functional equation and analytic continuation of $L(\mathrm{Sym}^2 f, s)$ (Shimura; Gelbart-Jacquet)	7.1, 7.2
Section 6 row: Euler factors of the adjoint, naive versus true, δ_p, $P_q(X)$, $e_\ell(E,q)$	6.1, 8.1, 9.1 to 9.7, 10.1
Section 6 row: Petersson norm, Rankin-Selberg, $L(\mathrm{Sym}^2 f, 2)$, Zagier, Watkins, Flach (with 12, 13, 14)	8.2 to 8.6, 12.2
Section 6 row: level raising, Wiles's level-raising factor (with 15, 20)	6.2, 9.7
hook: Section 1 (Petersson norm, Zagier, Watkins, Rankin, Shimura, Flach, naive versus true, DFG04 Lemma 2.12)	8.1 to 8.6
hook: Section 1.1 (δ_p, $L_p(A_f,s)$, the naive factor, `lfunsympow`, Definition def:terms, the table)	C1 to C6, 6.1, 8.1, 9.1, 9.3, 11.1
hook: Lemma lem:euler (in full)	9.3 (with 9.2, 6.2, 10.1)
hook: Remark rem:wild	10.1, 10.2
hook: Theorem thm:dfg14 and the paragraph after it (the naive factors)	6.2, 9.2, 9.7
hook: Lemma lem:fpr ($L_p(A_f,1)^{-1} = L_p(A_f(1),0)^{-1}$, an ℓ-adic integer of valuation e_ℓ or $v_\ell(1-p^{-2})$)	9.6

PLAN.md scope item, Section 6 row, or paper hook	line-up items
hook: the appendix (`rule(E, l)`)	11.2, 12.1
also: proof of Theorem thm:main (cancellation of the naive factors); Theorem thm:quant Step 3; the Remark after thm:quant, (1), (3)	9.7, 9.4, 9.5
also: Proposition prop:forms (proof: $e_3 = v_3(q - \chi_{-3}(q)) \geq 1$; the exception at 2, point 43)	9.4, 10.1(c)
also: Conjecture conj:euler and Section sec:wild (terms at 2, 3; t_ℓ at $q = \ell$, point 53)	9.1, 10.1, 11.2
also: Section sec:iso ("Multiplicative primes show no Euler term"), Section sec:sharp ("the size of the Petersson norm")	9.2, 8.4
also: important_notes.md points 11, 12, 15, 30, 34, 43, 50, 52, 53, 60; B1, B6, B7, B9, B10	C2, 4.1(iii), 9.2, 9.6; 9.3; 9.5; 8.2; 9.3; 10.1, 10.2; 10.1, 10.2; C8, 11.2; 9.1, 11.2; 10.1(b); C2; C2 (Chapter 7's convention); C3; C3, 5.1; 8.4 (Chapter 14's formula)
also: what Chapters 2, 4, 6, 7, 9 to 15 deferred here	9.4, 12.1 (Chapter 2); 6.1 (Chapter 4); C2, C3 (Chapter 6); 4.1, 4.4, 9.2, 9.3, 10.1 (Chapter 7); 5.1, 5.2 (Chapter 9); 9.3, 10.1 (Chapters 10, 11); 8.3 (Chapters 12, 13); 8.4 (Chapter 14); 8.1 (Chapter 15)

21.4.2 4.2 Exercises (Section 13 of the chapter), graded from easy to harder

1. Prove $\zeta(s)\zeta(s-1) = \sum \sigma(n)n^{-s}$ and $\zeta(s)^2 = \sum d(n)n^{-s}$ from the Euler products, and deduce $\zeta^{(N)}(s) = \zeta(s)\prod_{p|N}(1-p^{-s})$.
2. Show $L(1,\chi_{-4}) = \pi/4$ and write down the Euler factors of $\zeta_{\mathbb{Q}(i)}$ at 2, 3, 5; check that they are those of $\zeta(s)L(s,\chi_{-4})$.
3. (Mellin.) Show $\int_0^\infty (\theta(y)-1)y^{s/2-1}dy = 2\pi^{-s/2}\Gamma(s/2)\zeta(s)$ and complete the proof of 3.1(iv) from $\theta(1/y) = \sqrt{y}\theta(y)$.
4. Prove $L_p(M(1),s) = L_p(M,s+1)$ with the geometric Frobenius, and that the arithmetic one would give $L_p(M,s-1)$. Check the paper's claim that at a multiplicative prime "$(\mathrm{ad}^0 V)^{I_q}$ is a line with eigenvalue q^{-1}", and say for which Frobenius it holds.
5. Prove $\mathrm{Sym}^2 V^\vee \simeq \mathrm{ad}^0 V(-1)$ for $V = V_\ell E$, and check the paper's claim that $L_p(\mathrm{Sym}^2 f, s) = L_p(A_f, s-1)$ at a good prime by computing both sides from α, β.
6. Prove the identities $\sum_k a_{p^k}^2 X^k = \frac{1+pX}{(1-\alpha^2X)(1-pX)(1-\beta^2X)}$ and $\sum_k a_{p^{2k}}X^k = \frac{1-p^2X^2}{(1-\alpha^2X)(1-pX)(1-\beta^2X)}$. Check the paper's claim that the naive adjoint L-function is "the Euler product of $\sum a_n(f)^2 n^{-s}$" on 11a1 at $p = 2$ (coefficients 4 and 2), and state the correct relation.
7. Check the paper's claim that $\langle f, f\rangle$ is $L(\mathrm{Sym}^2 f, 2)$ "up to the local factors at the bad primes": derive 8.3 from the residue formula of Chapter 13 and verify it numerically for 20a1 and 27a1 with PARI (`mfpetersson` times $\mu(N)$, `lfun(lfunsympow(E,2),2)`).
8. Using 8.4, compute B_E for 121a1 and 539c1 and check that $v_3(\deg\varphi_E) = v_3(B_E) + \sum e_3(E,q)$.
9. Check the paper's claim that for type I_0^*, $L_q(A_f,1)^{-1} = \frac{q-1}{q}\cdot\frac{(q+1)^2 - a_q(E')^2}{q^2}$, and show that it does not depend on which ramified quadratic twist E' of good reduction is chosen. Compute it for 539d1 at 7 ($E' = 11$a3).
10. Prove that the trace-zero commutant of a cyclic subgroup $\langle g\rangle \subset \mathrm{SL}_2(\overline{\mathbb{Q}}_\ell)$, $g \neq \pm 1$, of finite order is the line spanned by $2g - \mathrm{tr}\, g$, and that any element normalising $\langle g\rangle$ acts on it by ± 1. Deduce the III row of the table without diagonalising τ (point 34).
11. Check the paper's claim that "a type IV at 2 has $P_2(X) = 1 + 2X$ and $e_3 = 1$" from 10.1(c) and Chapter 11, and that "a type II at 2 has $P_2 = 1$" fails for 336f1 and 1232l1. Identify 1232l1 as a

twist of 77a1 and compute $e_3(E,2)=2$ from the I_0^* formula.

12. Check the paper's claim that "Both terms are invariant under quadratic twist" (Definition def:terms): prove it for t_ℓ and disprove it for e_ℓ with 11a1 and 121d2. Show that $v_\ell(L_q(A_f,1)^{-1})$ is invariant.
13. Check the paper's claim in Lemma lem:fpr that $L_p(A_f,1)^{-1}$ is a nonzero ℓ-adic integer of valuation $e_\ell(E,p)$ at an additive prime and $v_\ell(1-p^{-2})$ at a multiplicative one; where is the Hasse bound used?
14. Check the paper's claim after Theorem thm:dfg14 that adding a good prime multiplies the ideal by Wiles's level-raising factor: compute $L_p^{\mathrm{nv}}(A_f,1)^{-1}$ for 11a1 at $p=7$ and find the primes ℓ dividing it; relate $a_7\equiv\pm 8 \pmod{\ell}$ to Ribet's condition.
15. For $\ell=3$, list the residue classes of $q\geq 5$ modulo 12 for which each additive type gives $e_3(E,q)\geq 1$, and check the paper's claim (Remark after Theorem thm:quant, (1)) that a prime of type I_0^* with $q\equiv 1 \pmod 3$ forces $3 \mid \deg\varphi_E$ at the level of the Euler factor.
16. Show that the critical integers of $L(\mathrm{ad}^0 f,s)$ are 0 and 1 from $\Gamma_{\mathbb{R}}(s+1)\Gamma_{\mathbb{C}}(s+1)$, and deduce from 8.3 that $L(\mathrm{ad}^0 f,1)/(\pi^3\langle f,f\rangle)\in\mathbb{Q}^\times$.
17. Compute B_E for 98a1, 121b1, 121d1 and 539d1, find the primes ℓ with $v_\ell(B_E)<0$, and explain each by the failure of a hypothesis of Theorem thm:main.
18. Run `rule` on 121a1 and 726a1 (with and without `iii`) and account for each term with Sections 9 to 11; explain why the type III term and the correction cancel for 726a1.

21.5 5. Dependencies and forward pointers

Results from earlier chapters used here (by chapter label only): * Chapter 2 (`ch:rings`): Dirichlet characters, Gauss sums, χ_{-3}, χ_{-4}, χ_{q^*}; the valuations 11.3 to 11.5 and the identity 11.4 (used in 6.2, 9.3, 9.4, 10.1). * Chapter 4 (`ch:reps`): $\mathrm{ad}^0 V$, commutants of cyclic and non-abelian groups, Schur, $\mathrm{Sym}^2 W\simeq \mathrm{ad}^0 W\otimes\det W$ (7.5), the trace form (7.6); used in 6.1, 9.3, 9.6, 10.1. * Chapter 6 (`ch:ant`): Dedekind domains, splitting of primes in quadratic fields (3.4), Frobenius elements and the geometric convention (3.7). * Chapter 7 (`ch:local`): conventions L3, L6; 5.6 (Frob g $\mathrm{Frob}^{-1}=g^q$ on tame inertia); 6.3 (quadratic characters of $\mathbb{Q}_q$); 8.1, 8.2 (L-factors); 8.4 (conductors); 9.1 to 9.5 (the shapes and their adjoints). * Chapter 9 (`ch:ellcurves`): a_p, α, β, the Hasse bound, $L(E,s)$ as an Euler product, quadratic twists. * Chapter 10 (`ch:reduction`): Kodaira types, Tate's algorithm, Kraus's theorem (R), Lemma lem:twist (i), (ii), PARI's codes, the conductor. * Chapter 11 (`ch:galrep`): 3.2 (Tate shape), 3.4 (potentially multiplicative), 3.5 (δ_p), 4.1 (inertia images at $q\geq 5$), 4.4 (types IV, IV* at 2), 4.6 (wild inertia), 8.4, 8.5 (3Ns, h_3). * Chapter 12 (`ch:modforms`): the Petersson product (C5), $\mu(N)$, W_N. * Chapter 13 (`ch:hecke`): $L(f,s)$ and its functional equation (10.1, 10.2), the Rankin-Selberg residue (10.3(i)). * Chapter 14 (`ch:moddeg`): modularity; the Zagier-Watkins formula with c^2 in the numerator (7.3; B10); $c_E=1$ on the tables. * Chapter 15 (`ch:modgalrep`): local-global compatibility (3.4), the local types and δ_p (3.5), the naive-factor corollary (3.7). * Chapters 17, 18 (`ch:galcoh`, `ch:blochkato`, being outlined in parallel): only as forward black boxes in 8.5, 8.6 and 9.6 (H^1_{ur}, Bloch-Kato Selmer groups, Tamagawa factors, DFG04 Theorem 0.1); never used in a proof here.

Forward pointers (chapters that use this one): * Chapter 18 (`ch:blochkato`): 9.6 (the L-part of Lemma lem:fpr, $\det(1-\Phi_p \mid (V^D)^{I_p}) = L_p(A_f,1)^{-1}\neq 0$), 8.5 (the Bloch-Kato reading), 8.6 (Flach), C3 ($L_p(M(1),s)=L_p(M,s+1)$). Chapter 18 owns Bloch-Kato, the Tamagawa factors, the Tamagawa number conjecture, DFG04 Theorems 0.1, 0.2, 2.7, Lemma 2.1 and the Fitting-ideal half of Lemma lem:fpr. * Chapter 20 (`ch:congruence`): 9.7 (the naive factors at $s=1$ and the cancellation), 6.2 (Wiles's factor), 9.5 ($A_g=A_f$), 9.3 (the values in Theorem thm:quant Step 3). Chapter 20 owns DFG04 Proposition 1.4 (Theorem thm:dfg14) and the congruence ideals. * Chapter 21 (`ch:paper`): 9.1 (Definition def:terms, corrected), 9.3 (Lemma lem:euler), 10.2 (Remark rem:wild, sentence by sentence), 8.2 (point 30), 8.4 (the identity behind Theorem thm:main), 12.1 (the paper's numerical examples). * Chapter 22 (`ch:computation`): 11.1 (the normalisation of `lfunsympow` and `lfuneuler`), 11.2 (what `rule` computes), 12.1. Chapter 22 owns PARI usage, the line-by-line reading of `rule`, the data protocol and `ellmoddegree`. * Appendix B (`ch:notation`): C2, C4, the normalisation table 7.4.

21.6 6. Notes for the writer

21.6.1 6.1 Conventions to state (Section 1 of the chapter, as `convention` environments)

- C2 and C3 in one `convention` box, with the dictionary "arithmetic eigenvalue λ = geometric eigenvalue λ^{-1}" and the four displays $L_p(\mathbb{Q}_\ell(1), s) = (1 - p^{-1-s})^{-1}$, $L_p(V_\ell E^\vee, s) = L_p(E, s)$, $L_p(V_\ell E, s) = L_p(E, s+1)$, $L_p(A_f, s)^{-1} = 1 - p^{-1-s}$ at a multiplicative p. State, as in Chapters 7, 9, 11: on $V_\ell E$ the **arithmetic** Frobenius has eigenvalues α, β (B1, B9; the Chapter 21 outline's C2(b) still has the opposite, which its writer must fix).
- C4 and C6 together, then the normalisation table 7.4 as soon as all its objects are defined (end of Section 7 or Section 8).
- The variable: $L(\mathrm{ad}^0 f, s)$ at $s = 1$ and $L(\mathrm{Sym}^2 f, s)$ at $s = 2$ are the same number; write both every time a value appears, as the paper does ($L_q(A_f, 1)^{-1} = P_q(q^{-2})$).
- Petersson product unnormalised (C9); say once that PARI's `mfpetersson` divides by $\mu(N)$.
- Characters by discriminant (χ_{-3}, χ_{-4}, χ_{q^*}); say once that the paper's χ_q is χ_{q^*}.

21.6.2 6.2 Pitfalls

- **Frobenius.** The table needs Φ (row I_n: $1 - q^{-1-s}$, not $1 - q^{1-s}$); Lemma lem:tamfactor uses Frob (point 11). Rows II to III* and I_0^* are the same for both (inversion-stable eigenvalues); say so, so that the reader sees exactly where the choice matters.
- **Sym versus ad.** $L(\mathrm{Sym}^2 f, s) = L(\mathrm{ad}^0 f, s-1)$ in weight 2; for weight k the shift is $k-1$. Never call the Rankin series "$L(\mathrm{Sym}^2 f, s)$"; never call $\zeta^{(N)}(2s-2)\sum a_{n^2} n^{-s}$ the true symmetric square (it is the naive one, 8.2(B)).
- **Point 30.** The paper's "the Euler product of $\sum a_n(f)^2 n^{-s}$" must be quoted in a `warning` with the exact relation 8.2(A). PLAN.md's "$L(f \times f, s)/\zeta_N(2s-2)$" is also wrong (Part 7, B-a): with $L(f \times f, s) = \zeta^{(N)}(2s-2)\sum a_n^2 n^{-s}$ one has $L^{\mathrm{nv}}(\mathrm{Sym}^2 f, s) = L(f \times f, s)/\zeta^{(N)}(s-1)$. Do not reproduce PLAN's formula.
- **Naive factor only at bad primes.** The formula $1 - a_q^2 q^{-2}$ of Definition def:terms is the naive factor at $s = 1$ only for $q \mid N$; at a good prime the naive factor is the cubic of 6.2.
- **"The diagonal element"** (type III): write τ (point 34). **"**$\mathbb{Z}_\ell[\zeta_3] \simeq \mathrm{End}_{I_q}(V)$**"**: write $\mathrm{End}_{I_q}(T) = \mathbb{Z}_\ell \otimes \mathbb{Z}[\zeta_3]$ and $\mathrm{End}_{I_q}(V) = \mathbb{Q}_\ell \otimes \mathbb{Q}(\zeta_3)$, not a field for $\ell \equiv 1 \pmod 3$ (point 12).
- a' **for** I_0^***.** $a' = a_q(E^{(q^*)})$ is defined up to sign by the choice of ramified quadratic character; only a'^2 enters.
- $q \geq 5$ **is used only for the shape of** $\rho(I_q)$ (tame, cyclic, determined by the type). At $q = 2, 3$ the type does not determine the shape (10.1); do not write "type II at 2 gives $P_2 = 1$" (point 50). Type IV, IV* at 2 is a theorem (point 43). The observation "type III at 3 always gives $1 + 3X$" below conductor 10,000 is data, not a theorem: do not state it as one.
- e_ℓ **is not twist-invariant** (point 15); t_ℓ and $L_q(A_f, s)$ are. PLAN.md's scope sentence repeats the paper's error (Part 7, B-a).
- δ_p **at** $p = \ell$**.** $V_\ell E^{I_\ell}$ is not 2-dimensional at a good ℓ; define δ_p for $p \neq \ell$ or with $V_{\ell'} E$ (Part 7, N4).
- **The Rankin-Shimura formula.** Do not attribute the exact form 8.3 to DFG04 Lemma 2.12 without checking it; attribute the unfolding to Rankin and Selberg, the bad-prime bookkeeping to this chapter, and cite Shimura76 and Watkins02 by section.
- B_E**.** 8.4(iii) is an identity; 8.5 is a reading through Chapters 18 and 20. Never present $v_\ell(B_E) = \mathrm{length}\, H^1_f + \sum v_\ell\,\mathrm{Tam}$ as proved here, and do not claim that $2\pi\,\mathrm{vol}(\Lambda_E)/N$ is DFG04's period.
- **Flach.** The hypotheses, the module ($\mathrm{Sym}^2 T_p E$ or its dual) and the annihilator must be copied from Flach92; the outline does not certify them.
- **Analytic statements.** 7.1 is R; the root number of $L(\mathrm{ad}^0 f, s)$ is not claimed without a reference; "entire in the CM case" needs a sentence of justification or a reference.
- **Petersson normalisation.** PARI's `mfpetersson` is normalised by $1/\mu(N)$; the formulas of 8.3, 8.4 use the unnormalised product (C9), as Chapters 12 to 14 do.

- **Shimura75 versus Shimura76.** Shimura75 (holomorphy of the symmetric square, Proc. London Math. Soc.) is not the refs.bib key `Shimura76` (special values of the zeta functions associated with cusp forms, Comm. Pure Appl. Math.).

21.6.3 6.3 What to leave to Chapters 7, 11, 13, 14, 18, 20 and 22

- Chapter 7: the definition of Weil-Deligne representations, L-factors and conductors, and the local shapes with their adjoints (9.1 to 9.5 there). Cite; Section 9 here only assembles them with the Kodaira dictionary and the valuations.
- Chapter 11: the inertia images by Kodaira type (4.1, 3.2, 3.4), types IV, IV* at 2 (4.4), Lemma lem:IV, Remark rem:types, the 3Ns structure (8.4). Cite.
- Chapter 13: the Euler product and functional equation of $L(f,s)$ and the Rankin-Selberg unfolding (10.1 to 10.3 there). Cite; 8.3 here adds only the exact bad factors.
- Chapter 14: the Zagier-Watkins formula and the Manin constant (7.3, 7.4 there); point 39 (Lemma lem:hecke is an equality). 8.4 here only combines.
- Chapter 18: Bloch-Kato Selmer groups, Tamagawa factors, the Tamagawa number conjecture, DFG04 Theorem 0.1, the Fitting-ideal half of Lemma lem:fpr.
- Chapter 20: DFG04 Proposition 1.4 (Theorem thm:dfg14), the congruence ideals η^{Σ}, Theorems thm:main and thm:quant.
- Chapter 22: PARI usage, `ellmoddegree`, the line-by-line reading of `rule`, the data protocol and the tables of Section 6 of the paper.

21.6.4 6.4 How the relevant important_notes.md points are handled

- **Point 11 (geometric Frobenius).** C2, C3, 4.1(iii), 9.2 ("with Frob_q one would get $1-q^{1-s}$"), 9.6 (the determinant changes by a unit, so vanishing is unaffected but the equality needs Φ); Exercise 4.
- **Point 12 (End_{I_q}).** 9.3, Step (2); pitfall.
- **Point 15 (e_ℓ not twist-invariant).** 9.1(iii), 9.5 with the 11a1/121d2 example; Exercise 12.
- **Point 30 (the naive L-function and $\sum a_n^2 n^{-s}$).** 8.2, with the exact global identities (A) and (B) (no correction at bad primes), the `warning`, Exercise 6; Part 7, N1 (addendum).
- **Point 34 (τ, not "the diagonal element").** 9.3, Step (2); Exercise 10.
- **Point 43 (types IV, IV* at 2).** 10.1(c), 10.2(ii), 12.1 (52a1, 980i1); cite Chapter 11, 4.4.
- **Point 50 (type II at 2).** 10.1 explains every observed factor: the cubic of 1232l1 is the I_0^* formula for 77a1 $\otimes\chi_{-4}$; 10.2(iii), (iv); Exercise 11.
- **Point 52 (Kodaira codes for $n \geq 1$).** C8; 11.2.
- **Point 53 (t_ℓ at $q=\ell$).** 9.1 defines $t_\ell(E,q)$ at every bad q; 11.2 (726a1 at $q=3$).
- **Point 60 (I_0^* formula at $q=3$).** 10.1(b) proves it for every q (including 2 and 3) whenever the inertia image is $\{\pm 1\}$; all 2,283 cases at 3 checked.
- **B1, B7, B9 (which Frobenius has α,β; the Euler factor of E).** C2, C3, 5.1(i) state them correctly.
- **B6 (Weil-Deligne relation).** Not restated; C2 refers to Chapter 7's L4.
- **B10 (Manin constant in the numerator).** 8.4(i) has c_φ^2 on the right side with $\deg\varphi$ on the left, consistent with Chapter 14's 7.3.
- Adjacent points: 18 (the I_n^* prime at $q=2$ in the proof of Theorem thm:main): 10.1(a) gives its L-factor and $e_\ell(E,2)$; the Tamagawa argument is Chapter 18's. 39, 56 (Lemma lem:hecke an equality): used only in the reading 8.5. 47, 48 (3Ns): 11.2, 12.1.

21.6.5 6.5 References

In `refs.bib`: `DFG04`, `Flach92`, `Zagier85`, `Watkins02`, `Shimura76`, `Shimura71`, `GelbartJacquet`, `Deligne79` (Deligne's conjecture), `BlochKato`, `Fontaine`, `Berger`, `Kraus90`, `SilvermanAEC`, `SilvermanATAEC`, `DiamondShurman`, `SerreCourse`, `Neukirch`, `Washington`, `Carayol86`, `Cremona97`, `ecdata`, `PARI`, `PARIusers`, `DDT`, `Miyake`. Proposed additions, following decision B14 (checked: none

of these keys is in `refs.bib` or in any `bib/chNN.bib`, all of which are empty except `chA.bib`): * to `refs.bib` (needed by more than one chapter): `Rankin39` (R. A. Rankin, "Contributions to the theory of Ramanujan's function $\tau(n)$ and similar arithmetical functions. I, II", Proc. Cambridge Philos. Soc. 35 (1939), 351-372; Chapters 13, 19; B20), `Deligne73` (Chapters 7, 15, 19; B18), `Tate79` (J. Tate, "Number theoretic background", Proc. Sympos. Pure Math. 33, Part 2 (1979), 3-26; Chapters 7, 19), `Rohrlich94` (Chapters 7, 19; B18), `SerreTate68` (B14, B16), `Shimura75` (G. Shimura, "On the holomorphy of certain Dirichlet series", Proc. London Math. Soc. (3) 31 (1975), 79-98; Chapters 18 and 19 both need the analytic theory of $L(\mathrm{Sym}^2 f, s)$; if Chapter 18 does not cite it, `bib/ch19.bib`); * to `bib/ch19.bib`: `Selberg40` (A. Selberg, "Bemerkungen uber eine Dirichletsche Reihe, die mit der Theorie der Modulformen nahe verbunden ist", Arch. Math. Naturvid. 43 (1940), 47-50), `CoatesSchmidt87` (J. Coates and C.-G. Schmidt, "Iwasawa theory for the symmetric square of an elliptic curve", J. Reine Angew. Math. 375/376 (1987), 104-156), `Sturm80` (J. Sturm, "Special values of zeta functions, and Eisenstein series of half integral weight", Amer. J. Math. 102 (1980), 219-240), `Davenport00` (H. Davenport, *Multiplicative number theory*, 3rd ed., revised by H. L. Montgomery, Grad. Texts in Math. 74, Springer, 2000). If Chapter 22 cites Dokchitser's algorithm for `lfun`, `Dokchitser04` belongs in `refs.bib`. The writer checks every bibliographic detail above against the source before adding it (they were recalled, not looked up). No theorem number from these works is cited in this outline except where the paper itself cites one (DFG04 Lemma 2.12, Proposition 1.4, Theorem 0.1) and Washington Theorem 4.3 (marked verify).

21.6.6 6.6 Verification log (computations done for this outline)

Scripts in the session scratchpad, `.../scratchpad/ch19/`, PARI/GP 2.17.2 via `cypari2`, `ecdata` files below conductor 10,000. * `t1.py`: for all 64,687 curves, every bad prime: PARI's P_q against the table of 9.3 at $q \geq 5$ (117,232 pairs, no mismatch; I_0^* via `elltwist` and `ellap`); the distribution of P_q at $q = 2, 3$ by type (table of 10.1); the I_0^* formula at $q = 3$ (2,283 of 2,283). * `t6.py`: every cubic P_2 (1,143) and P_3 (2,283) equals the formula of 10.1(b) for a twist by $d \in \{-4, 8, -8\}$, resp. -3, with good reduction; no curve with P_q of degree 0 or 1 at $q = 2, 3$ has such a twist. * `t4.py`, `t5.py`: I_m^* at 2: $P_2 = 1 - X$ iff $v_2(j) < 0$ (3,732 of 9,667); type II, II* at 2 with $e_3(E, 2) > 0$ or $e_5(E, 2) > 0$ on optimal curves: with $E[\ell]$ irreducible and $\ell \nmid N$, $v_\ell(\deg \varphi_E) \geq e_\ell(E, 2)$ always (162 curves at $\ell = 3$, 93 at $\ell = 5$); the only exception with $\ell \mid N$ is the CM curve 144a1. * `t2.py`: the Petersson formula 8.3 and the degree formula 8.4(ii) for the 15 curves of 12.2 (ratio 1 to 10^{-15}; degrees from `ellmoddegree`). * Power series and global checks of 8.2 (A), (B) (see 8.2). * `t8.py`: the rows of 12.1 (B_E exactly and numerically); `t7.py`: small examples by type; `lfunparams` and the sign of `lfunsympow` for 11a1, 121a1, 336f1, 1232l1, 539d1, 275b1, 20a1, 27a1, 64a1, 121d2 (7.2). * `t3.py` with `rule.gp` (the appendix code verbatim): 121a1 gives `[1, 1]`; 726a1 gives `[2, 2]` with and without `iii`; also P_2 and $\deg \varphi_E$ of 336f1 ($1 + 2X$, 24) and 1232l1 (cubic, 288); $e_3(121\mathrm{d}2, 11) = 1$ against $e_3(11\mathrm{a}1, 11) = 0$. * Not verified (recalled): the contents of DFG04 Lemma 2.12 and 1.7.2; the hypotheses and annihilator of Flach92; the exact statements of Shimura75, Sturm80, Coates-Schmidt and Gelbart-Jacquet (in particular the root number of $L(\mathrm{ad}^0 f, s)$ and the CM case); section numbers of Davenport, Washington, SerreCourse, Neukirch, Watkins02.

21.7 7. Points for important_notes.md

21.7.1 7.1 New points about the paper

No new error that changes a result was found in this chapter's area. The table of Section 1, Lemma lem:euler, the values in Remark rem:wild for type IV at 2, the naive factors after Theorem thm:dfg14, the L-part of Lemma lem:fpr and the outputs of `rule(E, 1)` for 121a1 and 726a1 were all confirmed (Part 6.6). The known issues in the area are points 11, 12, 15, 30, 34, 43, 50, 52, 53, 60. Four additions, none of which changes a theorem:

N1 (addendum to point 30; Introduction, "the naive one: the Euler product of $\sum a_n(f)^2 n^{-s}$"). The relation is exact, with no correction at the bad primes: $\sum a_n(f)^2 n^{-s} = \zeta^{(N)}(s-1) L^{\mathrm{nv}}(\mathrm{Sym}^2 f, s) / \zeta^{(N)}(2s-2)$ (at $p \mid N$ both local series are $(1 - a_p^2 p^{-s})^{-1}$). Moreover Shimura's

series gives the naive function on the nose: $L^{\mathrm{nv}}(\mathrm{Sym}^2 f, s) = \zeta^{(N)}(2s-2)\sum a_{n^2}(f)n^{-s}$. So the paper's sentence can be fixed with no loss of brevity: "the naive one, $\zeta^{(N)}(2s-2)\sum a_{n^2}(f)n^{-s}$ (Shimura), which has the true local factor ...". Point 30's "up to the factors at the bad primes" can be dropped. Confidence: high (local identities as power series; global identities checked numerically at $s = 6$ for 11a1, 20a1, 121a1).

N2 (optional strengthening; Section 1, "Up to explicit factors $\deg\varphi_E$ is $\langle f, f\rangle$ ... and $\langle f, f\rangle$ is the special value $L(\mathrm{Sym}^2 f, 2)$ up to the local factors at the bad primes"). Combining the Rankin-Selberg residue with the Zagier-Watkins formula gives, for every level and every optimal curve with Manin constant 1,

$$\langle f, f\rangle = \frac{N}{8\pi^3} L^{\mathrm{nv}}(\mathrm{Sym}^2 f, 2), \qquad \deg\varphi_E = \frac{N\, L^{\mathrm{nv}}(\mathrm{ad}^0 f, 1)}{2\pi\,\mathrm{vol}(\Lambda_E)} = B_E \prod_{q^2\mid N} L_q(A_f, 1)^{-1}, \quad B_E = \frac{N\, L(\mathrm{ad}^0 f, 1)}{2\pi\,\mathrm{vol}(\Lambda_E)} \in \mathbb{Q}^\times,$$

so $v_\ell(\deg\varphi_E) = v_\ell(B_E) + \sum_{q\mid N} e_\ell(E, q)$ for every $\ell \nmid N$, unconditionally. "Up to the local factors at the bad primes" is exactly "up to $N/(8\pi^3)$ and the omitted factors at the additive primes"; the multiplicative primes contribute nothing. This makes the paper's guiding sentence ("the primes at which the naive and the true adjoint L-functions differ are exactly the primes at which the modular degree acquires extra local divisibility, and the amount is the valuation of the omitted factor") an identity, and reduces (eq:thm) to $v_\ell(B_E) \geq \sum_q t_\ell(E, q)$. Suggested action: one displayed formula in Section 1, citing Rankin, Shimura and Watkins. Confidence: high for the formula (proved from the residue formula; checked to 15 digits on 15 curves with additive primes at 2, 3, 7, 11, and B_E exactly on 19 curves); the attribution to DFG04 Lemma 2.12 not checked here (DFG04 Lemma 2.12 gives the Petersson norm with the naive L-value: Chapter 20 outline, Part 7; important_notes.md point 73).

N3 (optional strengthening; Remark rem:wild and Section sec:wild: the factor at $q = 2, 3$ is not mysterious). At every bad prime $q \neq \ell$, including 2 and 3, P_q is one of: $1 - X$ (potentially multiplicative); $(1 - qX)(1 - (a'^2 - 2q)X + q^2X^2)$ with $a' = a_q(E^{(d)})$ when some quadratic twist $E^{(d)}$ has good reduction at q (inertia image $\{\pm 1\}$; the I_0^* formula, at every q); $1 - cqX$ with $c = \pm 1$ (cyclic inertia image of order 3, 4 or 6); 1 (non-abelian inertia image). Proof: commutants in SL_2 plus Schur (item 10.1). Checked on all curves below conductor 10,000: all 1,143 cubic P_2 and 2,283 cubic P_3 are the twist formula, and no other shapes occur. Consequences worth a sentence in the paper: (a) the cubic of point 50's example 1232l1 is $(1 - 2X)(1 + 2X)^2$, because 1232l1 = 77a1 $\otimes\chi_{-4}$ with $a_2(77\mathrm{a}1) = 0$, so $e_3(E, 2) = v_3(9/8) = 2$; (b) the prime 2 can contribute at $\ell = 5$: 592c1 = 37a1 $\otimes\chi_{-4}$, $a_2(37\mathrm{a}1) = -2$, $e_5(E, 2) = v_5(5/8) = 1$, and $\deg\varphi_E = 80$ (all 93 optimal curves below 10,000 with this factor, $5 \nmid N$ and $E[5]$ irreducible have $5 \mid \deg\varphi_E$); the paper's "For $\ell = 5, 7$ no term is needed at $q = \ell$ or at $q = 2, 3$ beyond Definition def:terms" is consistent with this, but the reader may assume the terms at 2 vanish for $\ell \geq 5$, which is false; (c) the possible values are $e_3(E, 2) \leq 2$, $e_5(E, 2) \leq 1$, $e_\ell(E, 2) = 0$ for $\ell \geq 7$, $e_5(E, 3), e_7(E, 3) \leq 1$, $e_\ell(E, 3) = 0$ for $\ell \geq 11$; (d) Remark rem:wild's "a type I_n^* at 2 has potentially good reduction in about half of the cases" is 61% below conductor 10,000 (5,935 of 9,667), and the cubic there has $a' = \pm 1$, contributing nothing at odd ℓ (complements point 50). Confidence: high (proof plus complete check below 10,000); importance low to medium.

N4 (Section 1.1, "for a prime p let $\delta_p = \dim V^{I_p}$, so $\delta_p = 2, 1, 0$ for good, multiplicative, additive reduction"). At $p = \ell$ this is false ($V_\ell E$ is ramified at ℓ; $V^{I_\ell} = 0$ at a good ℓ), and $L_p(A_f, s) = \det(1 - \mathrm{Frob}_p\, p^{-s} \mid A_f^{I_p})^{-1}$ is not the right factor at $p = \ell$ (it needs D_{cris}). The paper only uses δ_p and L_p at $p \neq \ell$ (Σ never contains ℓ), so nothing breaks. Fix: "for a prime $p \neq \ell$", or define δ_p with $V_{\ell'}E$ for any $\ell' \neq p$. Confidence: high; importance low (expository).

21.7.2 7.2 Book-internal inconsistencies

- **B-a. PLAN.md, Chapter 19 entry.** (1) "the naive adjoint L-function (the Euler product of $\sum a_n^2 n^{-s}$ after removing the zeta factor, i.e. the Rankin-Selberg convolution $L(f\times f, s)/\zeta_N(2s-2)$" is wrong twice: with $L(f\times f, s) = \zeta^{(N)}(2s-2)\sum a_n^2 n^{-s}$ one has $L^{\mathrm{nv}}(\mathrm{Sym}^2 f, s) = L(f\times f, s)/\zeta^{(N)}(s-1)$, and the factor to remove from $\sum a_n^2 n^{-s}$ is $\zeta^{(N)}(s-1)/\zeta^{(N)}(2s-2)$ (item 8.2). (2) "and its invariance under quadratic twist" repeats the paper's error (point 15). (3) "type II at 2 has $P_2 = 1$; a type

I_n^* at 2 with potentially good reduction has a cubic P_2" repeats point 50. The outline follows the corrected statements; PLAN.md should be annotated.

- **B-b. Chapter 2 outline, 11.6(e), (f).** (f) asks to check that the I_0^* example with $a_7(E') = -2$ is 539d1: confirmed (539d1 has type I_0^* at 7, $E' = 539\mathrm{d}1 \otimes \chi_{-7} = 11\mathrm{a}3$, $a_7 = -2$, $e_3(E, 7) = 2$, $\deg \varphi_E = 72$). (e) still quotes "type II at 2 with $P_2 = 1$: $e_3 = 0$" from Remark rem:wild; it is right as a conditional, but should cite 336f1 ($P_2 = 1 + 2X$) as a counterexample to the unconditional sentence.
- **B-c. Chapter 13 outline, 10.3(ii); Chapter 12 outline, 10.5(b); Chapter 14 outline, 7.5(b).** Chapter 13 states $\langle f, f\rangle = N\, L(\mathrm{Sym}^2 f, 2)/(8\pi^3)$ for square-free level only, and Chapters 12 and 14 state point 30's identity "up to bad factors". Not contradictions: item 8.3 here gives the general level (only additive primes contribute, by $P_p(p^{-2})$) and 8.2 the exact identity. When those chapters are written they can cite `\Cref{ch:lfunctions}` for the general form.
- **B-d. Chapter 21 outline, convention C2(b).** Still says the geometric Frobenius on $V_\ell E$ has eigenvalues α, β and that $\det(1 - \Phi p^{-s} \mid V_\ell E)^{-1}$ is $L_p(E, s)$ (already recorded as B1, B7, B9; not new). Its part (c) ($L_p(\mathrm{Sym}^2 f, s) = L_p(A_f, s - 1)$, P_q, `lfuneuler`) agrees with this outline.
- **B-e. Chapter 7 outline, 9.5(ii), (iii).** Still hedges on type IV at 2 and quotes "type II at 2 has $P_2 = 1$" as consistent with a non-abelian image (B13, B16; not new). Item 10.1 here supersedes both: case (c) for IV, IV*, and all four shapes for II.
- No other conflict found: Chapters 4 (7.5(iii)), 6 (3.7), 7 (8.2, 9.1 to 9.4), 9 (C5), 11 (C3, 4.1, 4.4), 15 (3.4(iii), 3.7) use the same normalisations as this outline.

Chapter 22

Chapter 20: Hecke algebras, congruence ideals and the theorem of Diamond, Flach and Guo (`ch:congruence`, `chapters/ch20_congruence.tex`)

Stage 1 line-up. Part VI (Congruences, the modular degree and the paper). Target length 2,670 lines of LaTeX (the card excluded; see Part 3), inside the 2,800 lines that PLAN.md Section 7.3 allows this chapter.

Rigour tags: **P** = proved here, **S** = proof sketched here, **R** = stated; reference. Item format, as in the Chapter 2 to 19 outlines: number, label-ready name, kind, tag; the precise statement with all hypotheses; reference (book and chapter or section; a theorem number only where sure, otherwise marked "(verify)" or given by section); the paper's use by its `\label` names, or "background"; and, for an item owned by another chapter, the item there (as "Chapter NN, its x.y").

DFG04 was read for this outline. The published text (Ann. Sci. Ec. Norm. Sup. (4) 37 (2004), 663-727, from Numdam, 65 pages) and the longer arXiv version (arXiv:2512.02348, "August 2001, with minor update December 2025", 109 pages, with different section numbers) were downloaded and read in the parts the paper cites. All DFG04 numbers below are those of the **published** version, which is the one the paper cites (`DFG04` in `refs.bib`). Part 7 ("DFG04 checklist") records what DFG04 says, item by item, with quotations. Every number quoted below (modular degrees, Euler terms, twist pairs) was checked with PARI/GP 2.17 through `cypari2` and Cremona's `ecdata` files for conductors below 10,000 (verification log, Part 6.6).

22.1 1. Purpose and global conventions

22.1.1 1.1 What this chapter is for

Chapter 20 is the place where the paper's Section 4 is proved, and where the commutative algebra of congruences meets the theorem of Diamond, Flach and Guo. The chapter **assembles**; it does not re-derive what other chapters prove. Its own content:

1. **Congruence modules and congruence ideals** (Section 1: "The congruence number is the congruence ideal of f in the full Hecke algebra of level N (Lemma lem:crit), and the theory of such ideals, from Hida to Diamond, Flach and Guo, expresses them through the adjoint Selmer group

and the adjoint L-value"). The abstract congruence module $\mathbb{T}/(\wp + I) \simeq \mathcal{O}/\eta$ (from Chapter 3, its 11.1), its reading in $\mathbb{T}_{\mathfrak{m}}$, the congruence ideal of a newform and of a depleted eigenform, Lemma lem:crit restated in this language (proved in Chapter 15, its 8.2, decision B17), and Hida's theorem as context.

2. **Wiles's numerical criterion and Lemma lem:ddt417 in full** (Section 4.1, Lemma lem:ddt417). Statement of the criterion (Wiles, Lenstra) for orientation, and the inequality half that the paper uses, proved in the paper's generality with the corrections of important_notes.md points 19 and 38, every ingredient cited from Chapter 3 (its 9.3, 9.4, 9.5, 11.1, 11.3).
3. **The lattices and ideals of DFG04** (Section 4.1). f^{Σ}, M^{Σ}, the integral structure, the kernel lattice $M_{f^{\Sigma},\lambda}$, the pairing $\hat{\delta}_!$, the naive Σ-finite congruence ideal $\eta^{\Sigma}_{f,\lambda}$, exactly as DFG04 Sections 1.5.3, 1.6.1, 1.6.2, 1.7.3 define them, with the comparison to $H^1(X_0(M^{\Sigma}), \mathbb{Z}_\ell)$ that the paper asserts (Part 7, items 1 to 5). Lemma lem:hecke and Remark rem:free are **cited** from Chapter 14 (its 9.3, 9.5, 9.6, 10.3), not re-derived; this chapter adds that the DFG04 checklist of important_notes.md point 56 is now confirmed, so Lemma lem:hecke is an equality for DFG04's own η.
4. **DFG04 Proposition 1.4** (Theorem thm:dfg14): the statement, the three cases (additive: no change; good: Wiles's level-raising factor; multiplicative: $1 - p^{-2}$), and the proof idea ($\gamma = \sum_m \epsilon_m \phi_m$, $\gamma^t \gamma = \phi \cdot \prod L_p^{\mathrm{nv}}(A_f, 1)^{-1}$, Ihara).
5. **The exceptional set S_f, Theorem 3.7 and the Taylor-Wiles method in outline** (Theorem thm:dfg37, Theorem thm:dfg27, Section 4.2). DFG04 (31), "minimally ramified", Theorems 0.1, 0.2, 2.7, 3.7, the axiomatic Theorem 3.1 of DFG04, and a two-page account of why the length of a Selmer group equals the valuation of a congruence ideal. Open check Q1 is settled: DFG04 covers $\ell = 3$ in weight 2.
6. **The assembly of Theorem thm:main**, step by step, each input cited to the chapter that proves it (Chapter 14 for Lemma lem:hecke, Chapter 18 for the Selmer half, its 11.1, Chapter 19 for the naive factors, its 9.7), with the corrections of important_notes.md points 18, 39, 56, 63, 64, and the reading "(eq:identity) holds with equality for $\ell \notin S_f$".
7. **Theorem thm:twist and Theorem thm:quant in full**, with the remark after Theorem thm:quant, the corrections of points 16, 24, 59, 60, 61, and a new observation (Part 8, N2): the first inequality of Theorem thm:quant is an **equality** whenever $\ell \notin S_f$, with no multiplicity-one hypothesis (verified on all 25,477 test pairs below conductor 10,000).
8. **Remark rem:ko, Remark rem:ns, Open questions 1 and 2**, with point 67.

The physics reader meets two familiar ideas, one sentence each: a congruence module measures how far an orthogonal decomposition of a lattice fails to be a direct sum over $\mathbb{Z}$ (the "overlap" of two eigenspaces, as for normal modes of a crystal lattice whose unit cells do not split); and the numerical criterion is an index theorem ("the dimension of the tangent space bounds the defect, with equality exactly in the complete-intersection case").

22.1.2 1.2 Split with the neighbouring chapters (decided here, consistent with PLAN.md Section 4 and the written outlines)

- **Chapter 3** proves all module theory: saturation (5.5), perfect pairings and the Gram cokernel (9.1 to 9.3), alternating rank-two pairings (9.4: $\Delta =$ (ideal of values)^2), adjoints and idempotents (9.5), algebra-module duality (9.6), finite free algebras, idempotent lifting, generalised eigenspaces (10.1 to 10.6), congruence ideals with $\wp \cap I = 0 \iff \eta \neq 0$ and the computation in $\mathbb{T}_{\mathfrak{m}}$ (11.1), toy congruence algebras (11.2), **every ingredient of Lemma lem:ddt417** (11.3), Gorenstein (11.4), Mazur's freeness criterion (11.5). Chapter 20 cites these and writes only the assembly of Lemma lem:ddt417 (as Chapter 3, its 11.3, says).
- **Chapter 13** proves the Hecke algebra $\mathbb{T}$ and its perfect pairing with $S_2(N; \mathbb{Z})$ (4.1, 4.2), the Hecke congruence ideal $\eta(\phi)$ and r_f (4.3), Petersson adjoints (5.1 to 5.4), newforms, strong multiplicity one, U_p on oldforms, depletion f^{Σ}, M^{Σ}, δ_p (6.2 to 7.4), the $\mathbb{T}$-stable complement (8.2), $\mathbb{T}_\ell = \prod \mathbb{T}_{\mathfrak{m}}$ on forms (8.3), congruences (8.4), the twist operator ι and the transport lemma (9.1, 9.3), Atkin-Li (9.2), the computed congruence modules at level 539 (11.3). Chapter 20 cites them; Chapter 13

explicitly leaves the proofs of Theorems thm:twist and thm:quant to Chapter 20.

- **Chapter 14** proves the cup-product adjoint $t^t = w_N t w_N$ and the pairing $\hat{\delta}_!$ (3.3, 3.4, with the orthogonal $\mathfrak{m}$-decomposition, point 36), the lattice $M_{f,\lambda}$ (4.4), the optimal curve (6.2), the congruence number and Agashe-Ribet-Stein (8.1 to 8.5), **point 39 in full** (9.1 to 9.5: $M_{f,\lambda} = \varphi_E^* H^1(E, \mathbb{Z}_\ell)$, $\hat{\delta}_!$ has ideal of values $\deg \varphi_E \mathbb{Z}_\ell$, Lemma lem:hecke as an equality), the DFG04 checklist (9.6), the 99 at 3 example (9.7), multiplicity one, Gorenstein and Remark rem:free (10.1 to 10.4). Chapter 20 cites 9.3, 9.5, 9.6, 10.3 (important_notes.md B19), and resolves 9.6 (Part 7 here).
- **Chapter 15** proves Lemma lem:crit (8.2) and Lemma lem:old (8.3), states Theorem thm:dfg14 (7.3), proves the level-raising factor identity (7.2), states Ihara's lemma (7.4), proves the levels of twists (10.1, 10.2), states thm:twist and thm:quant (10.3, 10.4) for Chapter 20 to prove, states Pollack-Weston and Kim-Ota (11.3, 11.4) and gives a one-page Taylor-Wiles remark (11.5). Chapter 20 does not reprove lem:crit; it restates it as a statement about congruence modules (3.4 here).
- **Chapter 17** proves the abstract form of DFG04 Lemma 2.1 (its 10.3) and the length identity (10.4).
- **Chapter 18** proves Lemma lem:fpr (7.4, 7.5), Lemma lem:tamfactor (8.2), the Tamagawa factors at every additive prime (8.4, 9.3: point 64), Lemma lem:dfg21 in the paper's form (10.4), Lemma lem:h0 in Selmer language (10.3, from Chapter 11, its 8.5), the length identity (10.5), **the Selmer half of Theorem thm:main** (11.1), Remark rem:ns (12.1), and states DFG04 Theorems 0.1, 0.2, 2.7 (14.4, 14.5, 10.2) and S_f (10.1). Chapter 20 cites 11.1 rather than repeating the Selmer bookkeeping (as Chapter 18 asks).
- **Chapter 19** proves Lemma lem:euler (9.3), the naive factors at $s = 1$ and the cancellation (9.7), $A_g = A_f$ (9.5), the L-part of Lemma lem:fpr (9.6), the Petersson formula with L^{nv} (8.3).
- **Chapter 21** is the guided tour; its A4 items walk through the same proofs at the level of the paper's text and should cite this chapter's card for the full arguments (Part 5).

22.1.3 1.3 Global conventions fixed in this chapter

- **C1 (the newform and the coefficients).** f is a newform of weight 2 on $\Gamma_0(M)$ (trivial character). In Sections 2 to 8 the coefficients may lie in a number field K with a prime $\lambda \mid \ell$ and $\mathcal{O} = \mathcal{O}_{K,\lambda}$; from Section 9 on, $f = f_E$ has rational coefficients, $K = \mathbb{Q}$, $\lambda = \ell$, $\mathcal{O} = \mathbb{Z}_\ell$ (the paper's "$\lambda = \ell$"). ℓ is odd and $\ell \nmid M$ unless said otherwise.
- **C2 (Hecke algebras).** $\mathbb{T} = \mathbb{T}_M = \mathbb{Z}[T_n : n \geq 1] \subset \mathrm{End}\, S_2(\Gamma_0(M))$, the **full** Hecke algebra, U_p included (Chapter 13, C4); DFG04's $\mathbb{T}$ of Section 1.6.1 is the same ring (generated by the T_p for all p, the double coset of $\mathrm{diag}(p, 1)$ at $p \mid M$ being U_p). $\mathbb{T}_\ell = \mathbb{T} \otimes \mathbb{Z}_\ell$, $\mathfrak{m}$ a maximal ideal of $\mathbb{T}_\ell$, $e_\mathfrak{m}$ its idempotent. $\mathbb{T}$ acts on $H^1(X_0(M), \mathbb{Z}_\ell)$ by t^* (Chapter 14, C2).
- **C3 (congruence data).** For a surjective $\mathcal{O}$-algebra map $\pi\colon \mathbb{T}_\mathcal{O} \to \mathcal{O}$: $\wp = \ker \pi$, $I = \mathrm{Ann}(\wp)$, $\eta(\pi) = \pi(I)$, congruence module $\Omega(\pi) = \mathbb{T}_\mathcal{O}/(\wp + I) \simeq \mathcal{O}/\eta(\pi)$ (Chapter 3, its 11.1; Chapter 13, its 4.3). The paper's letters: I in Lemma lem:crit is the book's $\wp$, the paper's J is the book's I; the card says so once (a `warning`).
- **C4 (two congruence ideals).** $\eta^{\mathrm{Hecke}}(\pi)$ is $\eta(\pi)$ of C3 (algebraic, computed in $\mathbb{T}_\mathfrak{m}$); $\eta^{\Sigma}_{f,\lambda}$ is DFG04's naive Σ-finite congruence ideal (cohomological, 5.4). Lemma lem:ddt417 compares them: $v(\eta^{\mathrm{Hecke}}) \geq v(\eta^{\mathrm{coh}})$.
- **C5 (lengths).** All counts are $\mathcal{O}$-lengths, not orders (important_notes.md point 38); for $\mathcal{O} = \mathbb{Z}_\ell$ the two agree.
- **C6 (Frobenius, duals, twists).** As in Chapters 7 and 18: Frob_p arithmetic, $\Phi_p = \mathrm{Frob}_p^{-1}$ geometric, $L_p(V, s) = \det(1 - \Phi_p p^{-s} \mid V^{I_p})^{-1}$ (DFG04 Section 1.1.1 writes Frob_p for the geometric Frobenius: confirmed, Part 7); $V^D = \mathrm{Hom}(V, \mathbb{Q}_\ell(1))$, T^D, W^D; * is the Pontryagin dual only (B4), $A^\dagger$ the adjoint, $M^\vee = \mathrm{Hom}(M, \mathcal{O})$.
- **C7 (characters and twists).** χ_{q^*} for the paper's χ_q (point 32), $\chi_S = \prod_{q \in S} \chi_{q^*}$, $D = \prod_{q \in S} q$, $\iota = \iota_S\colon h \mapsto h \otimes \chi_S$ (Chapter 13, its 9.1(vi)); $E_S = E \otimes \chi_S$, N_S its conductor, $g = f_{E_S}$; $E^{(q)}$ for the paper's $E_{\{q\}}$.
- **C8 (typography).** PLAN.md Section 7.2; "Diamond-Flach-Guo", "Taylor-Wiles", "Agashe-Ribet-Stein", "Fontaine-Perrin-Riou", "Fontaine-Laffaille", "Kim-Ota", "Pollack-Weston" with plain

hyphens; labels `...:congruence:...`.

22.2 2. Line-up of the core results, in teaching order

22.2.1 Section 1. Overview and conventions

1.1 `conv:congruence:setup` (conventions C1 to C8 as `convention` environments, with a roadmap; P). A roadmap table "sentence of the paper's Section 4 → item of this chapter → chapter that proves the input". A `warning` box for C3 (the paper's I, J versus $\wp$, I) and for C4 (two congruence ideals). Paper: Section 4 (all of it).

22.2.2 Section 2. Hecke algebras over $\mathbb{Z}_\ell$ and their local components (recollection)

2.1 `prop:congruence:hecke-local` (proposition, a recollection; P by citation). (i) $\mathbb{T}$ is free of rank $\dim S_2(\Gamma_0(M))$ over $\mathbb{Z}$ (Chapter 13, its 4.1), so $\mathbb{T}_\ell$ is a finite free $\mathbb{Z}_\ell$-algebra and $\mathbb{T}_\ell = \prod_{\mathfrak{m}} \mathbb{T}_{\mathfrak{m}}$ over its finitely many maximal ideals (Chapter 3, its 10.2, 10.3). (ii) For a newform f and $\lambda \mid \ell$, $\mathfrak{m} = \mathfrak{m}_{f,\lambda}$ is the kernel of $\mathbb{T}_\ell \to \mathcal{O} \to \mathcal{O}/\lambda$, generated by λ and the $T_n - a_n(f)$ (Chapter 13, its 3.4). (iii) $e_{\mathfrak{m}} S_2(M;\mathbb{Z}_\ell) \otimes \overline{\mathbb{Q}}_\ell$ is the sum of the generalised eigenspaces of the eigensystems $\phi\colon \mathbb{T} \to \overline{\mathbb{Q}}_\ell$ with $\phi \equiv \phi_f$ modulo the maximal ideal of $\overline{\mathbb{Z}}_\ell$; for newforms g this means $\bar\rho_g \simeq \bar\rho_f$ (semisimplifications; Chapter 13, its 8.3; Chapter 15, its 2.5). (iv) The pairing $S_2(M;\mathbb{Z}_\ell) \times \mathbb{T}_\ell \to \mathbb{Z}_\ell$ is perfect on each $\mathfrak{m}$-component (Chapter 13, its 8.3(iii)); $\hat\delta_!$ on $H^1(X_0(M),\mathbb{Z}_\ell)$ is perfect on each $\mathfrak{m}$-component, the decomposition being orthogonal (Chapter 14, its 3.4(iii); important_notes.md point 36). Paper: Lemma lem:hecke ("$\mathfrak{m}$ the maximal ideal of $\mathbb{T} \otimes \mathbb{Z}_\ell$ containing ℓ and $\ker \phi_f$"), Remark rem:free, Theorem thm:quant Step 1, PLAN.md scope.

2.2 `ex:congruence:level-539` (worked example; P as a computation, from Chapter 13, its 11.3). At level $539 = 7^2 \cdot 11$ and $\ell = 3$: the newforms of 539c and 539d, the depleted forms $(77\text{c})^{\{7\}}$ and $(11\text{a})^{\{7\}}$, and which of them share a maximal ideal of $\mathbb{T}_{539} \otimes \mathbb{Z}_3$; the Hecke congruence modules of the depleted forms have orders 288 and 504, equal to $r(539\text{c})$ and $r(539\text{d})$. Used as the running example of Sections 3, 12. Paper: Theorem thm:quant (background).

22.2.3 Section 3. Congruence modules and congruence ideals

3.1 `def:congruence:congruence-module` (definition and proposition; P by citation of Chapter 3, its 11.1). Let $\mathcal{O}$ be a complete DVR, $\mathbb{T}$ a commutative $\mathcal{O}$-algebra, finite and free over $\mathcal{O}$, $\pi\colon \mathbb{T} \to \mathcal{O}$ surjective. With C3: (i) $\Omega(\pi) = \mathbb{T}/(\wp + I) \simeq \mathcal{O}/\eta(\pi)$; (ii) $\eta(\pi) \neq 0 \iff \wp \cap I = 0 \iff$ the factor of $\mathbb{T} \otimes K$ through which π factors is K (multiplicity one); (iii) if $\eta \neq 0$, $I \simeq \eta$ is free of rank one, $\mathrm{Ann}(I) = \wp$, and $\eta = \{a \in \mathcal{O} : ae_\pi \in \mathbb{T}\}$ ("the denominator of the idempotent"); (iv) $\eta(\mathbb{T},\pi) = \eta(\mathbb{T}_{\mathfrak{m}}, \pi|_{\mathbb{T}_{\mathfrak{m}}})$: the other local factors lie in $\ker\pi$, **not** in $\mathrm{Ann}(\ker\pi)$ (important_notes.md point 35); (v) the second module $\Phi(\pi) = \wp/\wp^2$, the cotangent space, finite iff $\mathbb{T} \otimes K$ is reduced at π (P; one line: $\wp \otimes K/\wp^2 \otimes K$ is the cotangent space of $\mathbb{T}_K$ at π, zero iff the factor is K). Reference: DDT Section 4 and Section 5.1 (cite the sections); Lenstra95 (proposed key, Part 6.5); Hida81 Section 1 (cite the section). Paper: Lemma lem:crit, Lemma lem:ddt417, Lemma lem:hecke, Theorem thm:quant Step 1, Section 6 map row (Hida's congruence modules).

3.2 `ex:congruence:toy` (examples; P, from Chapter 3, its 11.2). $\mathbb{T}_r = \{(a,b) \in \mathcal{O}^2 : a \equiv b \pmod{\varpi^r}\}$: $\Omega = \mathcal{O}/\varpi^r$, $\Phi = \mathcal{O}/\varpi^r$ ("two eigenforms congruent modulo ϖ^r"); $\mathbb{T}_3^\flat$ (three pairwise congruent forms): $\eta = \varpi\mathcal{O}$ but $\Phi \simeq (\mathcal{O}/\varpi)^2$, so $\#\Phi > \#\Omega$: not a complete intersection; $\mathcal{O}[\varepsilon]/(\varepsilon^2)$: $\eta = 0$, $\Phi = \mathcal{O}$. These three examples are the whole content of the numerical criterion (4.1) in miniature. Paper: background for Lemma lem:ddt417.

3.3 `def:congruence:newform-ideal` (definition and proposition; P). For a newform f of level M with coefficients in $\mathcal{O}_K$ and $\lambda \mid \ell$, $\phi_f\colon \mathbb{T} \otimes \mathcal{O} \to \mathcal{O}$ (Chapter 13, its 3.4) is surjective after enlarging $\mathcal{O}$ to contain the $a_n(f)$ (its image is an order; replace $\mathbb{T} \otimes \mathcal{O}$ by the $\mathcal{O}$-algebra it generates). The *Hecke congruence*

ideal $\eta_f = \eta(\phi_f)$ is nonzero (strong multiplicity one: the ϕ_f-generalised eigenspace is $\mathbb{C}f$, Chapter 13, its 8.1(iv)). For $f = f_E$: $\eta_f\mathbb{Z}_\ell = r_E\mathbb{Z}_\ell$ (Lemma lem:crit, 3.4). Paper: Section 1 ("the congruence ideal of f in the full Hecke algebra"), Lemma lem:crit, Lemma lem:hecke.

3.4 `thm:congruence:crit` (theorem, the paper's Lemma lem:crit restated; R here, **proved in full in Chapter 15, its 8.2**; the module-theoretic reading P). Let $f = f_E$, $\mathbb{T}$ the full Hecke algebra of level N, ϕ_f, $\wp = \ker\phi_f$ (the paper's I), $I = \mathrm{Ann}_{\mathbb{T}}(\wp) = \mathrm{Ann}_{\mathbb{T}}(f^\perp)$ (the paper's J). Then $\mathbb{T}/(\wp + I) \simeq \mathbb{Z}/r_E$, i.e. the congruence module of ϕ_f is cyclic of order the congruence number, and the same over $\mathbb{Z}_\ell$ with r_E replaced by its ℓ-part; a normalised eigenform $g \in f^\perp$ of all of $\mathbb{T}$ congruent to f at **every** n modulo $\lambda \mid \ell$ forces $\ell \mid r_E$. Module-theoretic reading (P, from Chapter 3, its 5.8, 9.6): under the perfect pairing $S_2(N;\mathbb{Z}) \times \mathbb{T} \to \mathbb{Z}$, $\Omega(\phi_f)^* \simeq S_2(N;\mathbb{Z})/(\mathbb{Z}f \oplus L)$ with $L = f^\perp \cap S_2(N;\mathbb{Z})$: **the congruence module is dual to the failure of $S_2(N;\mathbb{Z})$ to split along the eigenspace decomposition.** Paper: Lemma lem:crit, Section 1, Lemma lem:hecke, Theorem thm:twist, Theorem thm:quant Step 1.

3.5 `prop:congruence:crit-general` (proposition; P given Chapter 13, its 8.2, and Chapter 15, its 8.2). Let $\pi\colon \mathbb{T}_N \to \mathbb{Z}$ be an eigencharacter whose generalised eigenspace in $S_2(\Gamma_0(N)) \otimes \overline{\mathbb{Q}}$ is a line $\mathbb{C}g$ (a newform at its level; a depleted form f^Σ at level M^Σ, Chapter 13, its 7.4(iv); g^S at level N in Theorem thm:quant). Let g^c be the $\mathbb{T}$**-stable** complement of $\mathbb{C}g$ (the sum of the other generalised eigenspaces). Then the proof of 3.4 runs verbatim with g^c in place of $f^\perp$: $\mathbb{T}/(\ker\pi + \mathrm{Ann}\ker\pi) \simeq \mathbb{Z}/r(\pi)$ with $r(\pi)$ the largest modulus of a congruence $g \equiv h$, $h \in g^c \cap S_2(N;\mathbb{Z})$; the same over $\mathbb{Z}_\ell$. **Warning** (important_notes.md point 61): for g not a newform the Petersson complement is not $\mathbb{T}$-stable (level 44: Chapter 13, its 8.2(iii)); the paper's text uses the $\mathbb{T}$-stable complement, which is right. Paper: Theorem thm:quant Step 1 ("the proof of Lemma lem:crit, carried out over $\mathbb{Z}_\ell$ ... in the $\mathbb{T}$-stable complement of the g^S-line").

3.6 `thm:congruence:hida` (theorem; R). (Hida 1981.) For a newform f of weight $k \geq 2$ and level N and a prime λ outside an explicit finite set (dividing N, $(k-2)!$ and 6 in Hida's statement; check), the congruence primes of f (the primes dividing η_f) are exactly the primes dividing the algebraic part $L^{\mathrm{nv}}(A_f,1)/\Omega_f^+\Omega_f^-$ of the adjoint L-value, with $\langle f, f\rangle$ related to $L^{\mathrm{nv}}(A_f,1)$ by the Rankin-Shimura formula (Hida81 Theorem 5.1, as DFG04 Lemma 2.12 cites it: verified there, Part 7). The book states this as context only: the paper never uses Hida's theorem in a proof, and DFG04's $\eta_{f,\lambda}^{\emptyset}$ is the refinement used instead. Reference: Hida81 (cite by section except Theorem 5.1, verified as cited by DFG04); HidaBook (cite the chapter); Ribet83 (congruence primes); Zagier85. Paper: Section 1 ("from Hida to Diamond, Flach and Guo"), Section 6 map row (Hida's congruence modules).

3.7 `rem:congruence:two-ideals` (remark; P). The three ideals of this chapter and their relations, displayed once: $r_E\mathbb{Z}_\ell = \eta_f^{\mathrm{Hecke}}$ (3.4), $v(\eta_f^{\mathrm{Hecke}}) \geq v(\eta_{f,\lambda}^{\emptyset})$ (Lemma lem:ddt417, 4.3 and 5.6), $v(\eta_{f,\lambda}^{\emptyset}) = v_\ell(\deg\varphi_E)$ for the optimal curve (Chapter 14, its 9.3, with Part 7 here), $v_\ell(\deg\varphi_E) = v_\ell(r_E)$ for $\ell^2 \nmid N$ (ARS12). Hence all three agree at $\ell \nmid N$, and the inequality of Lemma lem:ddt417 is strict exactly where freeness fails (99a1 at 3: Chapter 14, its 9.7). Paper: Lemma lem:hecke, Remark rem:free.

22.2.4 Section 4. Wiles's numerical criterion and Lemma lem:ddt417

4.1 `thm:congruence:numerical-criterion` (theorem; R, with the proof idea S). (Wiles; Lenstra.) Let $\mathcal{O}$ be a complete DVR with finite residue field, $\mathbb{T}$ a local $\mathcal{O}$-algebra, finite and free over $\mathcal{O}$, $\pi\colon \mathbb{T} \to \mathcal{O}$ surjective, $\Phi = \wp/\wp^2$, $\eta = \eta(\pi) \neq 0$. Then (i) $\mathrm{length}\,\Phi \geq v(\eta)$; (ii) equality holds (with Φ finite) iff $\mathbb{T}$ is a complete intersection over $\mathcal{O}$; (iii) (the form used in Taylor-Wiles) if $R \twoheadrightarrow \mathbb{T}$ is a surjection of complete local Noetherian $\mathcal{O}$-algebras compatible with π and $\mathrm{length}\,\Phi_R \leq v(\eta_{\mathbb{T}}) < \infty$, then $R \to \mathbb{T}$ is an isomorphism of complete intersections. Proof idea (one paragraph): for a complete intersection $\mathbb{T} = \mathcal{O}[[x_1,\ldots,x_n]]/(f_1,\ldots,f_n)$ both sides equal $v(\det(\partial f_i/\partial x_j)(\pi))$ (Tate's computation of $\mathrm{Ann}(\wp)$ through the Jacobian); in general reduce to it by a flat-deformation argument (Lenstra). Examples 3.2 show each case. Reference: Wiles95 Chapter 2 (Appendix); Lenstra95; DDT Section 5.1 (Wiles's numerical criterion; the theorem number to be verified: the paper's "DDT Theorem 5.3" in PLAN.md is not confirmed); Mazur's survey in the Cornell-Silverman-Stevens volume (cite by section). Paper: background (Section 6 map row "Wiles's numerical criterion"); the inequality half in the presence of a module is 4.3.

4.2 `rem:congruence:why-a-module` (remark; P). Why the paper needs a version **with a module**:

the ideal that DFG04 control is not $\eta(\pi)$ (an invariant of the ring) but the ideal of values of a pairing on the π-part of a module L (5.4). Lemma lem:ddt417 compares the two; it is an equality exactly when L is free over $\mathbb{T}$, which is what multiplicity one gives (Chapter 3, its 11.5; Chapter 14, its 10.3). Paper: Section 4.1 ("We also need the following piece of commutative algebra, which is the inequality half of DDT Lemma 4.17").

4.3 `lem:congruence:ddt417` (lemma, the paper's Lemma lem:ddt417; **P**, every step, in corrected form). Let $\mathcal{O}$ be a DVR, $\mathbb{T}$ a commutative $\mathcal{O}$-algebra, finite and free over $\mathcal{O}$, L a $\mathbb{T}$-module free of finite rank over $\mathcal{O}$ with a perfect $\mathcal{O}$-bilinear pairing $\langle\ ,\ \rangle$ for which every $t \in \mathbb{T}$ is self-adjoint, $\pi\colon \mathbb{T} \to \mathcal{O}$ surjective, $\wp = \ker\pi$, $I = \mathrm{Ann}_{\mathbb{T}}(\wp)$, $\eta = \pi(I)$. Suppose $L[\wp]$ is free of rank d over $\mathcal{O}$ with Gram determinant $\Delta \neq 0$. Then $\eta^d \subseteq \Delta\mathcal{O}$, i.e. $d\,v(\eta) \geq v(\Delta)$; equality holds if L is free over $\mathbb{T}$. Proof (the paper's, with every step expanded and lengths in place of orders, point 38): (0) if $\eta = 0$ there is nothing to prove; assume $\eta \neq 0$, so $\wp \cap I = 0$ (3.1(ii), Chapter 3, its 11.1(ii): **not** "because $\mathbb{T}$ is reduced", point 19; $\mathbb{T}_{88} \otimes \mathbb{Q}$ is not reduced, Chapter 13, its 7.3). (1) Fix $t \in I$ with $\pi(t)$ a generator of η; for $x \in L$, $tx \in L[\wp]$ ($\wp I = 0$) and $(\pi(t) - t)x \in L[I]$ ($I(\pi(t) - t) \subseteq I\wp = 0$); so $Q = L/(L[\wp] + L[I])$ is killed by η. (2) $L[\wp] \cap L[I] = 0$ (an element killed by $\wp + I \ni \pi(t)$ is killed by $\pi(t) \neq 0$; Chapter 3, its 11.3(ii)). (3) $L[\wp]$ is saturated and $L[\wp]^{\perp} = L[I]$ (Chapter 3, its 11.3(i), (iii)). (4) The map $L \to \mathrm{Hom}(L[\wp], \mathcal{O})$, $y \mapsto \langle -, y\rangle$, is onto with kernel $L[I]$ (Chapter 3, its 9.3(i)), so $Q \simeq \mathrm{coker}(L[\wp] \to \mathrm{Hom}(L[\wp], \mathcal{O}))$, of length $v(\Delta)$ (Chapter 3, its 9.3(ii)). (5) Q is generated by d elements and killed by η, hence a quotient of $(\mathcal{O}/\eta)^d$: $v(\Delta) = \mathrm{length}\, Q \leq d\,v(\eta)$. (6) If $L \simeq \mathbb{T}^r$ then $L[\wp] = I^r$, so $r = d$ (as I is free of rank one), $L[I] = \mathrm{Ann}(I)^r = \wp^r$ and $Q \simeq (\mathbb{T}/(\wp + I))^d \simeq (\mathcal{O}/\eta)^d$, of length $d\,v(\eta)$ (Chapter 3, its 11.3(v)). Remarks to print: (a) in the lemma's generality $\wp \cap I = 0$ can fail ($\mathcal{O}[\varepsilon]/(\varepsilon^2)$ acting on $\mathcal{O}$), but then $\eta = 0$; (b) if $\eta = 0$ and L is free, then $\Delta = 0$, so the hypothesis $\Delta \neq 0$ already forces $\eta \neq 0$ in the free case (Chapter 3, its 11.6); (c) self-adjointness is used only in (3). Reference: DDT Lemma 4.17 (as cited by the paper; the book proves only the inequality half and the equality clause, which is what the paper states). Paper: Lemma lem:ddt417, Lemma lem:hecke, Theorem thm:quant Step 2.

4.4 `ex:congruence:ddt417-checks` (examples; P, from Chapter 3, its 11.6, and Chapter 14, its 9.7). (a) $L = \mathbb{T}_r$ with the trace-type pairing: $d = 1$, equality. (b) $L = \mathcal{O}^2$ with $\mathbb{T}_r$ acting diagonally and the standard pairing: $\Delta = 1$, $v(\eta) = r$, strict; L is not free. (c) $L = \mathbb{T}_r^2$ with an alternating pairing: $d = 2$, $\Delta = \eta^2$: the shape of Lemma lem:hecke. (d) The modular instance: $L = H^1(X_0(99), \mathbb{Z}_3)_{\mathfrak{m}}$ for 99a1: $v_3(\eta^{\mathrm{Hecke}}) = 1$, $v_3(\Delta) = 0$ (the ideal of values is $4\mathbb{Z}_3 = \mathbb{Z}_3$), strict, so L is not free (Chapter 14, its 9.7). Paper: Lemma lem:ddt417, the remark after Theorem thm:qual (99A1).

22.2.5 Section 5. The lattices and ideals of Diamond, Flach and Guo; Lemma lem:hecke and Remark rem:free

5.1 `def:congruence:depletion-recall` (definition, recalled from Chapter 13, its 7.4; P by citation). For a newform f of level M and a finite set Σ of primes: $\delta_p = 2, 1, 0$ according as $p \nmid M$, $p \,\|\, M$, $p^2 \mid M$ (for $f = f_E$: good, multiplicative, additive; $= \dim V^{I_p}$, and DFG04 Section 1.7.2 defines $\delta_p = \dim M_{f,\lambda}^{I_p}$, the same number since $M_{f,\lambda}$ is the dual of V); $M^{\Sigma} = M\prod_{p\in\Sigma} p^{\delta_p}$; $f^{\Sigma} = \sum_{(n,\Sigma)=1} a_n q^n$, a normalised eigenform of the full $\mathbb{T}_{M^{\Sigma}}$ with $U_p f^{\Sigma} = 0$ ($p \in \Sigma$), whose generalised eigenspace at level M^{Σ} is the line $\mathbb{C}f^{\Sigma}$ (Chapter 13, its 7.4(iv)). The naive factor: $L_p^{\mathrm{nv}}(A_f, s) = L_p(A_f, s)$ if $\delta_p > 0$, 1 if $\delta_p = 0$ (DFG04 Section 1.7.2; Chapter 19, its 8.1). Paper: Section 4.1 ("Put $M^{\Sigma} = \ldots$"), Section 1.1 ("We follow the conventions of DFG04 Section 1.7.2").

5.2 `def:congruence:integral-structure` (definition and proposition; R for DFG04's construction, P for the comparison (ii) given Chapters 8, 14 and the argument of Part 7, item 1). (i) (DFG04 Sections 1.2, 1.4.1, 1.4.2.) DFG04's integral structure at level M and trivial character is $M(M,1)_{!,\lambda}$: the image of $\left(H_c^1(Y_{N'}, \mathbb{Z}_\ell)\right)^{U_0(M)} \to \left(H^1(Y_{N'}, \mathbb{Z}_\ell)^{U_0(M)}\right)_{\mathrm{tf}}$, where $Y_{N'}$ is the (adelic, geometrically disconnected) full level-N' modular curve for an auxiliary $N' \geq 3$ with $U_{N'} \subset U_0(M)$, and $U_0(M)$ acts through $U_0(M)/U_{N'}$ (independent of N', DFG04 Section 1.4.1). (ii) (Comparison, the paper's "$M(M^{\Sigma},1)_{!,\lambda}$ is $H^1(X_0(M^{\Sigma}), \mathbb{Z}_\ell)$".) Pullback identifies $H^1(X_0(M), \mathbb{Z}_\ell)$ with a $\mathbb{T}$-stable sublattice of finite index of $M(M,1)_{!,\lambda}$; the index is 1 when ℓ does not divide the order of the relevant covering group (transfer), and in general the quotient is a subquotient of $\mathrm{Hom}(U_0(M)/U', \mathbb{Q}_\ell/\mathbb{Z}_\ell)$ for a torsion-free congruence subgroup

U', i.e. it comes from abelian unramified subcovers of congruence covers of $X_0(M)$, of Shimura-subgroup type, on which $\mathbb{T}$ acts through Eisenstein ideals. So for every **non-Eisenstein** $\mathfrak{m}$, $M(M,1)_{!,\lambda,\mathfrak{m}} = H^1(X_0(M),\mathbb{Z}_\ell)_\mathfrak{m}$ (S; the Eisenstein claim cited to Ling-Oesterle and Ribet88 for the Shimura covering and stated as a sketch in general, Part 7 item 1). DFG04's own Lemma 1.8 and Corollary 1.9 use the same "kernel and cokernel are Eisenstein" device. (iii) The paper only ever uses the $\mathfrak{m}$-part for $\mathfrak{m} = \mathfrak{m}_{f,\lambda}$ with $\bar\rho_f$ irreducible, where (ii) applies. Reference: DFG04 Sections 1.2.2, 1.2.3, 1.4.1, 1.4.2; Chapter 14, its 4.3, 9.6, 11.3. Paper: Section 4.1 ("The integral structure $M(M^\Sigma,1)_{!,\lambda}$ is $H^1(X_0(M^\Sigma),\mathbb{Z}_\ell)$, with its Hecke action").

5.3 `def:congruence:dfg-pairing` (definition and proposition; P for (ii) to (iv) given Chapter 14, its 3.3, 3.4; R for DFG04's normalisation (i)). (i) (DFG04 Sections 1.5.1 to 1.5.3, (13), (14).) DFG04's $\hat\delta_!$ is the cup product into H^2_c, restricted to U-invariants and divided by $[U:U_{N'}]$ (their $\bar\delta$, (11), (12)), composed with the operator $[UwU]_\omega$, $w = \left(\begin{smallmatrix} 0 & -1 \\ M & 0\end{smallmatrix}\right) \in \prod_{p|M} \mathrm{GL}_2(\mathbb{Z}_p)$, which for weight 2 and trivial character is the Atkin-Lehner involution (DFG04 Section 1.5.3: $[UwU]_\omega$ "coincides with $\psi(-1_\infty)N^{k-2}[Uw^{-1}U]_\omega$", so for $k=2$, $\psi=1$ it is an involution, with no scalar); "since $M(N,\psi)=0$ unless $\psi(-1_\infty)=(-1)^{k-2}$, ... the corresponding perfect pairing is alternating". (ii) On pulled-back classes $\hat\delta_!(\pi^*x,\pi^*y) = c\cdot\langle x, w_M^* y\rangle_{X_0(M)}$ with $c = \deg(Y_{N'} \to Y_0(M))/[U:U_{N'}] \in \{1,\frac12\}$ ($-1 \in U$ acts trivially): a unit for odd ℓ. So on $H^1(X_0(M),\mathbb{Z}_\ell)_\mathfrak{m}$, DFG04's $\hat\delta_!$ is, up to a unit, the pairing $\langle x, w_M^* y\rangle$ of Chapter 14, its 3.4 (which the book also calls $\hat\delta_!$, Appendix B, B-C13(f)). (iii) It is perfect and alternating, and every $t \in \mathbb{T}$ (U_p included) is self-adjoint for it (Chapter 14, its 3.3, 3.4(ii); DFG04 Section 1.5.2 last paragraph and 1.8.2). (iv) Its restriction to every $\mathfrak{m}$-component is perfect (Chapter 14, its 3.4(iii); point 36). Paper: Section 4.1 ("the pairing $\hat\delta_!$ is the cup product composed with the operator $[UwU]\omega$ ... perfect and alternating ... self-adjoint"), Lemma lem:hecke, Theorem thm:quant Step 2.

5.4 `def:congruence:eta-sigma` (definition and proposition; P for (iii), (iv), R for (i), (ii) as DFG04's definitions). Let $\lambda \nmid 2M$ (C1). (i) (DFG04 Section 1.6.2.) For a normalised eigenform h of $\mathbb{T}_{M'}$ with eigencharacter θ_h, $M_{h,\lambda} := M(M',1)_{!,\lambda}[\ker\theta_h]$, the **kernel** lattice ("Let $I_f = \ker\theta_f$ and $M_f = M(N,\psi)_![I_f]$", with $M[I]$ the kernel of $M \to M^r$, $x \mapsto (x_1x,\dots,x_rx)$, DFG04 Section 1.1.2). (ii) (DFG04 Section 1.7.3.) "The pairing $\hat\delta_!$ restricts to an alternating pairing on M_{f^Σ}, which Proposition 1.4 below shows is non-degenerate ... The image of $\wedge^2 M_{f^\Sigma}$ therefore defines an integral structure ... necessarily of the form $\eta_f^\Sigma M_\psi(1-k)$ ... We call η_f^Σ the (naive, Σ-finite) congruence ideal of f." So $\eta_{f,\lambda}^\Sigma$ is the **ideal of values** of $\hat\delta_!$ on $M_{f^\Sigma,\lambda}$ (the image of $\wedge^2$). (iii) $M_{f^\Sigma,\lambda}$ is saturated of rank 2, $G_\mathbb{Q}$-stable, and lies in the $\mathfrak{m}_{f,\lambda}$-component (θ factors through $\mathbb{T}_\mathfrak{m}$, so $\theta(e_\mathfrak{m})=1$, $1-e_\mathfrak{m} \in \ker\theta$ and $x = e_\mathfrak{m}x$ for $x \in M_{h,\lambda}$; Chapter 3, its 10.6); its Gram determinant is $(\eta_{f,\lambda}^\Sigma)^2$ up to a unit (Chapter 3, its 9.4(ii): a rank-two alternating form has Gram matrix $\left(\begin{smallmatrix} 0 & a \\ -a & 0\end{smallmatrix}\right)$). (iv) For $\Sigma=\emptyset$ and $f=f_E$, $M_{f,\lambda}$ is homothetic to the **dual** $\mathrm{Hom}(T_\ell E,\mathbb{Z}_\ell) = T_\ell E(-1)$, and equal to $\varphi_E^* H^1(E,\mathbb{Z}_\ell)$ for the optimal curve (Chapter 14, its 9.1; important_notes.md points 26, 40; confirmed by DFG04 (18): geometric Frobenius on $M_{f,\lambda}$ has characteristic polynomial $X^2 - a_pX + p$, which is that of H^1); $\mathrm{ad}^0 M_{f,\lambda} = \mathrm{ad}^0 T_\ell E$ (Chapter 4). Paper: Section 4.1, Section 4.2 ("$M_{f,\lambda}$ ... and $T_\ell E$ are homothetic": read "the dual of $T_\ell E$"), Lemma lem:hecke, Theorem thm:quant Step 2.

5.5 `ex:congruence:eta-values` (worked example; P given Chapter 14, its 9.3, 8.5, and 5.6 here). $v_\ell(\eta_{f,\lambda}^\emptyset) = v_\ell(\deg\varphi_E)$ for $\ell \nmid 2N$ with $E[\ell]$ irreducible: 121a1 ($\deg 6$; $v_3=1$), 539c1 (288; $v_3=2$), 77c1 (6), 11a1 (1); and the depleted forms of 2.2 with $\Sigma=\{7\}$: by Theorem thm:dfg14, $v_3(\eta_{77c,3}^{\{7\}}) = v_3(\eta_{77c,3}^\emptyset) + v_3(1-7^{-2}) = 1+1 = 2$, matching $v_3(288)$ (this is Theorem thm:quant read backwards, 12.4). Paper: Lemma lem:hecke, Theorem thm:dfg14.

5.6 `thm:congruence:lem-hecke` (theorem, the paper's Lemma lem:hecke in the strengthened form; R here, **proved in Chapter 14, its 9.3 and 9.5**; the identification with DFG04's η P given Part 7). Let E be $X_0(N)$-optimal, $\ell \nmid 2N$, $\bar\rho$ irreducible. Then

$$v_\ell(\deg\varphi_E) = v_\ell(r_E) = v_\lambda(\eta_{f,\lambda}^\emptyset),$$

with no freeness hypothesis: the first equality is ARS12 Theorem 2.2 ($\ell^2 \nmid N$), the second is Chapter 14, its 9.3 (the ideal of values of $\hat\delta_!$ on $M_{f,\lambda} = \varphi_E^* H^1(E,\mathbb{Z}_\ell)$ is $\deg\varphi_E\mathbb{Z}_\ell$), combined with 5.2(ii), 5.3(ii), 5.4(ii), which identify that ideal with DFG04's $\eta_{f,\lambda}^\emptyset$ (Part 7, items 1 to 5, all confirmed). The paper's own

proof (via Lemma lem:ddt417 applied to $L = H^1(X_0(N), \mathbb{Z}_\ell)_{\mathfrak{m}}$, $d = 2$) is printed as well, with the two corrections: the other local factors of $\mathbb{T}_\ell$ lie in $\ker \phi_f$, not in its annihilator (point 35; 3.1(iv)); and $\hat{\delta}_!$ is perfect on the $\mathfrak{m}$-component (point 36; 5.3(iv)). It gives the inequality $\geq$ unconditionally and equality when L is free (Remark rem:free); point 39 shows the equality holds anyway. Paper: Lemma lem:hecke, proof of Theorem thm:main, Theorem thm:quant Step 4, Remark rem:free.

5.7 `rem:congruence:rem-free` (remark, the paper's Remark rem:free; R, **proved in Chapter 14, its 10.2, 10.3**). For $\ell \nmid 2N$ and $\mathfrak{m}$ non-Eisenstein, $J_0(N)[\mathfrak{m}]$ is two-dimensional (Mazur; RibetStein Theorem 3.5 for $J_1(N)$ and the Shimura subgroup), $\mathbb{T}_{\mathfrak{m}}$ is Gorenstein and $H^1(X_0(N), \mathbb{Z}_\ell)_{\mathfrak{m}} \simeq \mathbb{T}_{\mathfrak{m}}^2$ (Tilouine97; Chapter 3, its 11.5). What it is still needed for after point 39 (Chapter 14, its 10.4, corrected by 12.6 here): **not** for the equality in Lemma lem:hecke or in (eq:identity); **not** for Step 4 of Theorem thm:quant; **not** for Remark (2) after Theorem thm:quant (12.6, new); it remains the conceptual reason why the Hecke and the cohomological congruence ideals agree at $\ell \nmid 2N$, and it fails for $\ell \mid N$ in general (99a1 at 3). Paper: Remark rem:free, Theorem thm:main ("with equality when ... free").

22.2.6 Section 6. The variation of the congruence ideal with Σ: DFG04 Proposition 1.4

6.1 `def:congruence:gamma` (definition; P). (DFG04 Section 1.7.3.) For $m \mid M^\Sigma/M = \prod_{p\in\Sigma} p^{\delta_p}$ let $\epsilon_m \colon M(M,1)_! \to M(M^\Sigma,1)_!$ be the degeneracy map $m^{-1}[U_0(M^\Sigma)\left(\begin{smallmatrix} m & 0 \\ 0 & 1 \end{smallmatrix}\right)U_0(M)]$ (on forms: $h \mapsto h(mz)$ up to normalisation; in weight 2 the factor m^{1-k} is m^{-1}), ϕ_m the endomorphism of $M(M,1)_!$ with $\phi_1 = 1$, $\phi_p = -T_p$, $\phi_{p^2} = p$ (weight 2, trivial character), multiplicative in coprime m, and

$$\gamma = \sum_{m \mid M^\Sigma/M} \epsilon_m \phi_m \colon M(M,1)_! \to M(M^\Sigma,1)_!,$$

with adjoint γ^t for the pairings $\hat{\delta}_!$ at the two levels. On q-expansions $\gamma(f) = \prod_{p\in\Sigma}(1 - a_p V_p + pV_p^2)f$ or $(1 - a_p V_p)f$, which is f^Σ (Chapter 13, its 7.4(ii)). Paper: background to Theorem thm:dfg14.

6.2 `thm:congruence:dfg14` (theorem, DFG04 Proposition 1.4, the paper's Theorem thm:dfg14; R for (b), S for (a), (c), P for the reading). Let f be a newform of weight 2, level M, coefficients in K, $\lambda \nmid 2M$, Σ a finite set of primes not containing ℓ. (a) γ restricts to an isomorphism $M_f \to M_{f^\Sigma}$ (over K) with $\gamma_{\mathrm{dR}}(f) = f^\Sigma$. (b) $\gamma^t \circ \gamma = \phi_{M^\Sigma/M} \prod_{p\in\Sigma} L_p^{\mathrm{nv}}(A_f, 1)^{-1}$ on M_f, so $\hat{\delta}_!$ is non-degenerate on M_{f^Σ}. (c) If $\bar{M}_{f,\lambda}$ is irreducible, γ induces an isomorphism $M_{f,\lambda} \to M_{f^\Sigma,\lambda}$ and

$$\eta_{f,\lambda}^\Sigma = \eta_{f,\lambda}^\emptyset \prod_{p\in\Sigma} L_p^{\mathrm{nv}}(A_f, 1)^{-1}.$$

(The quotations are from the published text, Part 7.) Here $\phi_{M^\Sigma/M}$ acts on M_f by $\prod_{\delta_p=1}(-a_p)\prod_{\delta_p=2} p$, a λ-adic unit ($a_p = \pm 1$ at $\delta_p = 1$; $p \neq \ell$). Proof of (c) from (b) (P): an irreducible $G_{\mathbb{Q}}$-lattice has only homothetic sublattices (Chapter 4), so $\gamma(M_{f,\lambda}) = \lambda^n M_{f^\Sigma,\lambda}$, and $n = 0$ because $\gamma_{\mathrm{dR}}(f) = f^\Sigma$ is primitive; then the ideal of values on $M_{f^\Sigma,\lambda}$ is $\hat{\delta}_!(\gamma x, \gamma y) = \hat{\delta}_!(x, \gamma^t\gamma y)$, i.e. $\eta^\emptyset$ times the factor. Proof idea for (b) (S): double-coset computations as in Wiles95 Chapter 2 and DDT p. 121 (DFG04's references); in weight 2 at a good p it is the 3×3 Gram matrix of $\epsilon_1, \epsilon_p, \epsilon_{p^2}$ under $\hat{\delta}_!$, whose determinant, divided by that at level M, is the level-raising factor (Exercise 6). Reference: DFG04 Proposition 1.4 (verified, Part 7); Wiles95 Chapter 2; DDT Section 4 (cite); Ribet84 (Ihara's lemma, for the integral statement behind (c) in Wiles's approach). Paper: Theorem thm:dfg14, proof of Theorem thm:main, Theorem thm:quant Step 3.

6.3 `prop:congruence:three-cases` (proposition; P given Chapter 19, its 9.7, and Chapter 15, its 7.2). The factor $L_p^{\mathrm{nv}}(A_f, 1)^{-1}$ at $p \in \Sigma$: (i) additive ($\delta_p = 0$): 1; level and ideal unchanged; (ii) good ($\delta_p = 2$): $(1 - \frac{\alpha}{\beta}p^{-1})(1 - p^{-1})(1 - \frac{\beta}{\alpha}p^{-1}) = \frac{p-1}{p} \cdot \frac{(p+1)^2 - a_p^2}{p^2}$, Wiles's level-raising factor, of ℓ-adic valuation $v_\ell(p-1) + v_\ell((p+1)^2 - a_p^2)$ (identity checked symbolically, Part 6.6); (iii) multiplicative ($\delta_p = 1$): $1 - p^{-2}$ (the factor $(1 - p^{-1-s})^{-1}$ of DFG04 (23) at $s = 1$, $\pi_p(f)$ special). Hence "only the primes with $\delta_p \geq 1$ contribute". Interpretation: v_ℓ of the factor counts the congruences between f^Σ and forms new at p (level raising: Chapter 15, its 7.1, 7.5). Paper: the paragraph after Theorem thm:dfg14.

6.4 `rem:congruence:ihara` (remark; R). Where irreducibility enters: Ihara's lemma (Chapter 15, its 7.4; DFG04 Section 1.8.3, Lemma 1.11 in the refined setting) says that the degeneracy maps are injective on $\mathfrak{m}$-parts mod λ, i.e. that $\gamma(M_{f,\lambda})$ is saturated; DFG04's proof of (c) uses instead the homothety argument of 6.2. In the Eisenstein case the factor can change by a non-unit (Section sec:eis, Conjecture conj:eis: outside every theorem of the paper). Paper: Theorem thm:dfg14 ("If $\bar\rho_f$ is irreducible").

6.5 `rem:congruence:refined-structures` (remark; R). DFG04 Section 1.8 ("Refined integral structures") modifies the level at the exceptional primes $\Sigma_1 = \{p \in \Sigma_e : p \equiv -1 \bmod \lambda\}$ (for twist-minimal f, Σ_e is the set of p with $\delta_p = 0$ and $L_p(A_f, s) \neq 1$, i.e. $M_{f,\lambda}|_{G_p}$ induced from the unramified quadratic extension), adds an auxiliary prime $r > 3$ with $L_r(A_f, 1)^{-1}$ a unit (which makes the level sufficiently small with index prime to ℓ, so that $\hat\delta^{\Sigma}_{!,\lambda}$ is perfect integrally: Corollary 1.6), and proves the generalisation Proposition 1.12 ($\eta^{\Sigma}_{f,1} \subset \eta^{\emptyset}_{f,1} \prod_{p\in\Sigma} L_p(A_f,1)^{-1}$ with the **true** factors); "if $\Sigma_1 \cup \{r\} \subset \Sigma$, then $\eta^{\Sigma}_{f,1} = \eta^{\Sigma}_{f,\lambda}$". This is the mechanism by which the true Euler factors at exceptional primes enter DFG04's proof of Theorem 2.15. The paper never uses the refined ideals; one paragraph. Paper: background (why Theorem thm:dfg37 is about the naive ideal).

22.2.7 Section 7. The exceptional set and the theorems of Diamond, Flach and Guo

7.1 `def:congruence:exceptional-set` (definition and proposition; R for the definition as DFG04 (31), P for (ii) given Chapter 11, its 8.6, and Chapter 18, its 10.1). (i) (DFG04 (31), verified.) S_f is the set of primes λ of K such that "$\lambda \mid Nk!$, or the $(\mathcal{O}_K/\lambda)[G_F]$-module $\bar M_{f,\lambda}$ is not absolutely irreducible, where $F = \mathbb{Q}(\sqrt{(-1)^{(\ell-1)/2}\ell})$ and $\lambda \mid \ell$". For $k = 2$: $\lambda \mid 2N$ or $\bar\rho|_{\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{\ell^*}))}$ not absolutely irreducible. There is **no** further condition on ℓ relative to the weight beyond $\lambda \nmid k!$ (Part 7, Q1). (ii) For $f = f_E$ and $\ell \nmid 2N$: $\ell \notin S_f$ iff $E[\ell]$ is irreducible and $h_\ell(E) = 0$ iff $E[\ell]$ is irreducible and not ($\ell = 3$ and image 3Ns) (important_notes.md point 47). (iii) S_f is finite (DFG04 Lemma 2.4, with Lemma 2.5: for $\lambda \nmid N(2k-1)(2k-3)k!$, irreducible implies absolutely irreducible over F; for $k = 2$ the extra primes are 3; point 47 is the sharper elliptic-curve statement). Paper: Section 4.2 ("the hypothesis of Theorem thm:main is $\ell \notin S_f$"), Theorem thm:main, Section 1.2.

7.2 `def:congruence:selmer-sigma` (definition, recalled from Chapter 18, its 6.3, with DFG04's wording; P for the comparison). (DFG04 Section 2.1.) For a finite set Σ of primes not containing ℓ: $H^1_\Sigma(\mathbb{Q}, W) = \ker\big(H^1(\mathbb{Q}, W) \to \bigoplus_{p\notin\Sigma\cup\{\ell,\infty\}} H^1(\mathbb{Q}_p, W)/H^1_{\mathrm{ur}}(\mathbb{Q}_p, W) \oplus H^1(\mathbb{Q}_\ell, W)/H^1_f(\mathbb{Q}_\ell, W)\big)$: no condition at $\Sigma\cup\{\infty\}$, Bloch-Kato at ℓ, **unramified** (not H^1_f) elsewhere ("slightly relaxed local conditions at primes p where $H^1_{\mathrm{ur}}(\mathbb{Q}_p, W)$ is not divisible"). For $\Sigma \supseteq \{p \mid N\}$ the two agree outside Σ (W unramified there, $W^{I_p} = W$ divisible). DFG04 define $H^1_f(\mathbb{Q}_\ell, W)$ for the adjoint as the extensions in the essential image of the Fontaine-Laffaille functor ((32); their Proposition 2.2 compares it with Bloch-Kato's), which is why only $\ell \nmid k!$ is needed (Part 7, Q1). Paper: Section 4.2 ("the group defined in DFG04 Section 2.1").

7.3 `def:congruence:minimally-ramified` (definition; R, DFG04 Section 3.1; P for (ii)). (i) A lift ρ of $\rho_0 = \bar\rho$ is *minimally ramified at* $p \neq \ell$ if: when $\ell \nmid \#\rho_0(I_p)$, $\rho(I_p) \simeq \rho_0(I_p)$; when $\ell \mid \#\rho_0(I_p)$, $\dim V_\rho^{I_p} = 1$ (DFG04 Section 3.1; for deformations to A: M/M^{I_p} free of rank one). (ii) $M_{f,\lambda}$ is minimally ramified outside Σ whenever Σ contains every prime $p \neq \ell$ at which ρ_f ramifies; in particular for $\Sigma = \{p \mid N\}$, the paper's choice (P: unramified at p means $\rho(I_p) = 1 = \rho_0(I_p)$). (iii) Remark: the paper never needs a smaller Σ; DFG04 use smaller Σ internally (the minimal deformation problem, $\Sigma = \emptyset$, at the twist-minimal level). Paper: the paragraph after Theorem thm:dfg37.

7.4 `thm:congruence:dfg37` (theorem, DFG04 Theorem 3.7, the paper's Theorem thm:dfg37; R, proof idea in Section 8). (Verified statement.) "Let f be a newform of weight $k \geq 2$ and level N with coefficients in the number field K. Suppose that λ is a prime of K not in the set S_f defined in (31) ... Suppose that Σ is a finite set of primes not containing ℓ such that $M_{f,\lambda}$ is minimally ramified outside Σ. Then the $\mathcal{O}_\lambda$-module $H^1_\Sigma(\mathbb{Q}, A_{f,\lambda} \otimes K_\lambda/\mathcal{O}_\lambda)$ has length $v_\lambda(\eta^\Sigma_f)$." Here $A_{f,\lambda} = \mathrm{ad}^0 M_{f,\lambda}$ ($= \mathrm{ad}^0 T_\ell E$ for $f = f_E$, 5.4(iv)) and η^Σ_f is the naive ideal of 5.4. Its proof in DFG04 applies the axiomatic Theorem 3.1 to the twist of minimal conductor and transfers back ("the theorem holds for a twist of f, hence for f itself"). Paper: Theorem thm:dfg37, proof of Theorem thm:main, Remark rem:ns, Open question 2.

7.5 `thm:congruence:dfg02` (theorem, DFG04 Theorem 0.2 "(= Theorem 3.7)"; R). "Suppose Σ is a finite set of rational primes containing those dividing N. Suppose that λ is a prime of K which is not in S_f and does not divide any prime in Σ. Then the $\mathcal{O}_{K,\lambda}$-module $H^1_\Sigma(G_\mathbb{Q}, A_{f,\lambda}/\mathcal{A}_{f,\lambda})$ has length $v_\lambda(\eta_f^\Sigma)$." With $\Sigma = \{p \mid N\}$ and $\ell \nmid N$ this is exactly the paper's use (verified, Part 7). DFG04's gloss: "η_f^Σ ... is a generalization of the congruence ideal of Hida and Wiles; it can also be viewed as measuring the failure of the pairing on M_f to be perfect on $\mathcal{M}_f$." Chapter 18, its 14.5, states this as a pointer; this is its home. Paper: the sentence "this is the form given as Theorem 0.2 there".

7.6 `thm:congruence:dfg27` (theorem, DFG04 Theorem 2.7, the paper's Theorem thm:dfg27; R; **stated in Chapter 18, its 10.2**, cited here). For $\lambda \notin S_f$: $H^0(\mathbb{Q}, A_{f,\lambda}) = H^1_f(\mathbb{Q}, A_{f,\lambda}) = H^1_f(\mathbb{Q}, B_{f,\lambda}) = H^0(\mathbb{Q}, B_{f,\lambda}) = 0$, with $B_f = A_f(1)$, so $V^D = B_{f,\lambda}$ (verified wording: DFG04 state it as "Conjecture 2.6 holds for A_f and B_f", with these vanishings as the "more precisely"). Its proof: $L(A_f, 1) \neq 0$ via Lemma 2.12, and $H^1_f(\mathbb{Q}, A_{f,\lambda}) = 0$ because H^1_f of the divisible module sits in the finite group of Theorem 3.7. Paper: Theorem thm:dfg27, proof of Theorem thm:main; point 63 (only $H^1_f(\mathbb{Q}, V^D) = 0$ is needed from it).

7.7 `thm:congruence:dfg01` (theorem, DFG04 Theorem 0.1 "(= Theorem 2.15)"; R; **stated in Chapter 18, its 14.4**, cited here). For $\lambda \notin S_f$ the λ-part of the Bloch-Kato conjecture holds for A_f and $A_f(1)$. One paragraph on how DFG04 prove it: the proof of Theorem 2.15 takes $\Sigma = \{p \mid N\}$, uses Proposition 1.4 for γ^t, Lemma 2.1 for H^1_Σ versus H^1_f, the Fontaine-Perrin-Riou local formula $\operatorname{Fitt} H^1_f(\mathbb{Q}_p, M_\lambda) = L_p(M, 0)^{-1} \operatorname{Tam}^0_p(M_\lambda)$ ("By [41, Proof of I.4.2.2]"), and Theorem 3.7: **the same assembly as the paper's proof of Theorem thm:main** (Section 9), with the period computation (Theorem 2.10, Lemma 2.12) in place of the modular degree. Paper: Section 1.2 ("the hypothesis is the one under which Diamond, Flach and Guo prove the Tamagawa number conjecture for the adjoint motive of f \cite[Theorem 0.1]{DFG04}"), Theorem thm:main.

7.8 `lem:congruence:dfg212` (lemma, DFG04 Lemma 2.12; R; verified). For a newform of conductor N, weight k: $\langle \tau(f), \tau(f) \rangle_{\Gamma_1(N)} = (k-1)!\, \delta(N)\, N\, \varphi(N)\, L^{\mathrm{nv}}(A_f, \tau, 1)/(4^k \pi^{k+1})$, with $\delta(N)$ as in DFG04 (53); proved from Hida81 Theorem 5.1 (Rankin-Shimura) by a local comparison (54). So the Petersson norm sees the **naive** adjoint value, as Chapter 19, its 8.3, proves in the book's normalisation (unnormalised $\Gamma_0(N)$ product: $\langle f, f \rangle = N\, L^{\mathrm{nv}}(\mathrm{ad}^0 f, 1)/(8\pi^3)$; consistent with DFG04 since $[\bar{\Gamma}_0(N) : \bar{\Gamma}_1(N)] = \varphi(N)/2$ for $N \geq 3$). Paper: Section 1 ("see DFG04 Lemma 2.12"). Chapter 19 states the formula; this item records the verification of the citation.

22.2.8 Section 8. The Taylor-Wiles method in outline

8.1 `def:congruence:deformation-ring` (definition and theorem; R). (Mazur; DDT Chapter 2.) For $\rho_0 = \bar{\rho}_{f,\lambda}$ absolutely irreducible and a finite set Σ, the deformation problem "of type Σ" (DFG04 Section 3.1: minimally ramified outside $\Sigma \cup \{\ell\}$, determinant δ, short crystalline at ℓ) is representable by a complete local Noetherian $\mathcal{O}$-algebra R_Σ; its reduced cotangent space at a point $\pi\colon R_\Sigma \to \mathcal{O}$ satisfies $\mathrm{Hom}_\mathcal{O}(\wp_R/\wp_R^2, K/\mathcal{O}) \simeq H^1_\Sigma(\mathbb{Q}, \mathrm{ad}^0\, M_\pi \otimes K/\mathcal{O})$ (Selmer group with the type-Σ local conditions). Reference: DDT Sections 2.2 to 2.8 (cite the sections); Mazur's "deformation theory" survey (cite by section). Paper: background to Theorem thm:dfg37 (PLAN.md: "the Selmer group as the tangent space / dual of the cotangent space").

8.2 `def:congruence:hecke-side` (definition and proposition; R for the existence of the modular point, S for the rest). $\mathbb{T}_\Sigma = \mathbb{T}(N^\Sigma)_\mathfrak{m}$ (the Hecke algebra acting on $L_\Sigma = M(\sigma^\Sigma)_{!,\lambda,\mathfrak{m}}$, DFG04 Section 3.2), with the modular deformation $R_\Sigma \to \mathbb{T}_\Sigma$ (surjective: $\mathbb{T}_\Sigma$ is generated by traces of Frobenius); π the point of f; $\eta(\mathbb{T}_\Sigma, \pi)$ compared with η_f^Σ by Lemma lem:ddt417 and multiplicity one (L_Σ free over $\mathbb{T}_\Sigma$, DFG04 Lemma 3.2 to 3.4 and Lemma 1.10). Paper: background.

8.3 `thm:congruence:tw-axiomatic` (theorem, DFG04 Theorem 3.1; R, with the idea S). For ρ_0 odd, absolutely irreducible over G_F (F the quadratic subfield of $\mathbb{Q}(\mu_\ell)$), of minimal conductor among its twists, with Serre weight $2 \leq k \leq \ell - 1$; given a set $\mathcal{N}$ of allowable (short crystalline) lifts, a trellis L and a system of perfect pairings φ satisfying the level-raising compatibility $\varphi_\rho^\Sigma = \varphi_\rho^\emptyset \beta_\rho^\Sigma \prod_{p \in \Sigma} L_p(\mathrm{ad}^0, 1)^{-1}$ and the Taylor-Wiles conditions at horizontal primes ($q \equiv 1 \bmod \ell$, ρ_0 unramified at q with distinct Frobenius eigenvalues): every allowable lift is modular, and length $H^1_\Sigma(\mathbb{Q}, A_\rho) = \mathrm{length}\, \mathcal{O}(\delta)/\varphi_\rho^\Sigma(\wedge^2 L_\rho)$ for

$\Sigma \supseteq \Sigma_\rho$. Idea: patching over sets Q_n of horizontal primes gives $R_\emptyset \simeq \mathbb{T}_\emptyset$, a complete intersection (4.1(iii)); then the numerical criterion and Wiles's comparison of Φ_{R_Σ} and η^Σ under change of Σ (both change by the same local factors: the Selmer side by Chapter 17's comparison, the Hecke side by Proposition 1.4) give the equality at every Σ. Reference: DFG04 Theorem 3.1 and Section 3 (verified statement); Wiles95; TaylorWiles; DDT Chapters 3 to 5; Diamond96 (proposed key). Paper: background to Theorem thm:dfg37.

8.4 `rem:congruence:where-hypothesis` (remark; P). Where each hypothesis of $\ell \notin S_f$ is used: $\ell \nmid N$ (the representation is crystalline at ℓ, Fontaine-Laffaille applies, the level structure is prime to ℓ); $\ell \nmid k! = 2$ (Fontaine-Laffaille needs the Hodge-Tate weights $0, 1$ in $[0, \ell-2]$); absolute irreducibility over $\mathbb{Q}(\sqrt{\ell^*})$ (existence of horizontal primes making the dual Selmer group vanish: the Chebotarev argument needs an element g of $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\mu_{\ell^n}))$ with $\bar\rho(g)$ having distinct eigenvalues, which fails exactly when $\bar\rho$ is induced from $\mathbb{Q}(\sqrt{\ell^*})$). For elliptic curves with $\ell \nmid 2N$ the last condition bites only for $\ell = 3$, image 3Ns (point 47). Paper: Section 1.5 ("by level lowering and twisting rather than by the Taylor-Wiles method"), Section sec:anomaly, Open question 2.

8.5 `rem:congruence:why-length-equals-eta` (remark; S). The slogan "length of the Selmer group equals the valuation of the congruence ideal" in three lines: $H^1_\Sigma(\mathbb{Q}, W)^* \simeq \Phi_{R_\Sigma} = \wp_R/\wp_R^2$ (8.1); $R_\Sigma = \mathbb{T}_\Sigma$ is a complete intersection (8.3), so $\mathrm{length}\,\Phi_{\mathbb{T}_\Sigma} = v(\eta(\mathbb{T}_\Sigma))$ (4.1(ii)); and L_Σ is free over $\mathbb{T}_\Sigma$, so $\eta(\mathbb{T}_\Sigma) = \eta_f^\Sigma$ (4.3, equality case). Paper: Theorem thm:dfg37 (the reader's question "why should a Selmer group know about congruences?").

22.2.9 Section 9. The assembly: proof of Theorem thm:main

9.1 `thm:congruence:main` (theorem, the paper's Theorem thm:main in the strengthened form; P as an assembly of cited items). Let E be $X_0(N)$-optimal, $f = f_E$, ℓ an odd prime with $\ell \nmid 2N$ and $\ell \notin S_f$ (equivalently, by 7.1(ii), $E[\ell]$ irreducible and not ($\ell = 3$ and image 3Ns)), $T = T_\ell E$, $W = \mathrm{ad}^0 T \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$. Then

$$v_\ell(\deg \varphi_E) = \mathrm{length}_{\mathbb{Z}_\ell} H^1_f(\mathbb{Q}, W) + \sum_{q|N} v_\ell\big(\mathrm{Tam}_q(\mathrm{ad}^0 T)\big) + \sum_{q|N} e_\ell(E,q) \qquad \text{(eq:identity), with equality,}$$

and consequently $v_\ell(\deg \varphi_E) \geq \sum_q e_\ell(E,q) + \sum_q t_\ell(E,q)$ (eq:thm). The Tamagawa terms are explicit (Chapter 18, its 8.2, 8.4, 9.3: 0 at good and potentially good primes, $t_\ell + \min(t_\ell, v_\ell(q-1), v_\ell(c(u)))$ at potentially multiplicative primes, $q = 2$ included), so (eq:identity) reads $\mathrm{length}\, H^1_f(\mathbb{Q}, W) = v_\ell(\deg \varphi_E) - B_\ell^{\mathrm{expl}}(E)$ (Chapter 18, its 9.4). The equality (rather than the paper's "$\geq$, with equality when ... free") is important_notes.md points 39 and 56, with the DFG04 checklist confirmed (Part 7). Paper: Theorem thm:main, (eq:identity), (eq:thm).

9.2 `sec:congruence:assembly` (the proof of 9.1, a subsection whose `proof` environment has numbered steps, each naming its input and the hypothesis it uses; P). $\Sigma = \{q \mid N\}$ (nonempty as $N > 1$). Step 1 (degree to congruence ideal; uses optimality and $\ell \nmid 2N$, $\bar\rho$ irreducible): $v_\ell(\deg \varphi_E) = v_\lambda(\eta_{f,\lambda}^\emptyset)$ (5.6 = Chapter 14, its 9.3, 9.5, with Part 7). The paper's version ($\geq$ via Lemma lem:ddt417, equality under freeness) is printed in a `paperlink`. Step 2 (vary Σ; uses $\bar\rho$ irreducible, $\ell \nmid 2N$): $v(\eta^\emptyset) = v(\eta^\Sigma) - \sum_{q\in\Sigma} v_\ell(L_q^{\mathrm{nv}}(A_f,1)^{-1})$ (6.2(c)); only the multiplicative primes contribute, each $v_\ell(1-q^{-2}) = v_\ell(q^2-1)$ (6.3; Chapter 19, its 9.7). Step 3 (Taylor-Wiles; uses $\ell \notin S_f$): $v(\eta^\Sigma) = \mathrm{length}\, H^1_\Sigma(\mathbb{Q}, W)$ (7.4; $M_{f,\lambda}$ is minimally ramified outside Σ, 7.3(ii); $\mathrm{ad}^0 M_{f,\lambda} = \mathrm{ad}^0 T$, 5.4(iv)). Step 4 (the Selmer half; uses $\ell \notin S_f$ through thm:dfg27, ℓ odd, Lemma lem:h0): $\mathrm{length}\, H^1_\Sigma(\mathbb{Q}, W) = \mathrm{length}\, H^1_f(\mathbb{Q}, W) + \sum_{q\in\Sigma}\big(v_\ell(L_q(A_f,1)^{-1}) + v_\ell(\mathrm{Tam}_q(\mathrm{ad}^0 T))\big)$ (**Chapter 18, its 11.1**, which packages Lemma lem:dfg21 (its 10.4, from Chapter 17, its 10.3), Lemma lem:fpr (its 7.4, 7.5), Lemma lem:h0 (its 10.3: $h_\ell(E) = 0$), thm:dfg27 (its 10.2)). Only $H^1_f(\mathbb{Q}, V^D) = 0$ is needed from thm:dfg27; $H^0(\mathbb{Q}, V^D) = 0$ is automatic (point 63; Chapter 17, its 10.3(b)). Step 5 (cancellation; Definition def:terms): Steps 1 to 4 give (eq:identity) with the Euler sum $\sum_q \big(v_\ell(L_q^{-1}) - v_\ell((L_q^{\mathrm{nv}})^{-1})\big) = \sum_q e_\ell(E,q)$: at a multiplicative q true and naive factors coincide, at an additive q the naive factor is 1 (Chapter 19, its 9.7). Good primes do not divide N. Step 6 ((eq:thm)): drop $\mathrm{length}\, H^1_f \geq 0$; $v_\ell(\mathrm{Tam}_q(\mathrm{ad}^0 T)) \geq t_\ell(E,q)$ at every potentially multiplicative q, **including an I_n^* prime at $q = 2$** (important_notes.md point 18: the paper's appeal to "$E_{\{p\}}$ of Lemma lem:twist" covers

only odd p; the local argument of Chapter 18, its 8.4, covers every $q \neq \ell$, using $n = -v_q(j)$, not the index of the Kodaira symbol); $v_\ell(\mathrm{Tam}_q(\mathrm{ad}^0 T)) = 0$ at good and potentially good primes (point 64, Chapter 18, its 9.3). `paperlink`: the paper's proof of Theorem thm:main, sentence by sentence, with the step that justifies it. Paper: Section 4.3.

9.3 `rem:congruence:hypothesis-table` (remark; P). Table "input → hypothesis used → what fails without it": optimality (Step 1: for a non-optimal curve the pullback lattice has index $\deg\psi$ in its saturation; harmless for v_ℓ when $E[\ell]$ is irreducible, point 27); $\ell \nmid N$ (Steps 1, 3, 5: ARS, Fontaine-Laffaille, $q \neq \ell$ in Lemma lem:fpr); ℓ odd (Step 4 at ∞; ad^0 self-dual via the trace form); $\bar\rho$ irreducible (Steps 1, 2: non-Eisenstein $\mathfrak{m}$, homothety); absolute irreducibility over $\mathbb{Q}(\sqrt{\ell^*})$ (Steps 3, 4: Taylor-Wiles and $h_\ell = 0$); multiplicity one (no longer needed, 5.7). Paper: Theorem thm:main; Chapter 21's hypothesis table (B2 there) should be aligned with this one.

9.4 `ex:congruence:main-examples` (worked examples; P, data from Chapter 18, its 11.3, and Part 6.6). At $\ell = 3$: 121a1 (II at 11, $e_3 = 1$, $\deg 6$): $H^1_f(\mathbb{Q}, W) = 0$; 791a1: Tamagawa excess needed for equality; 1184f1 and 1748e1: Selmer length 1. Each line: $\ell \nmid N$, $E[3]$ irreducible, image not 3Ns, so $3 \notin S_f$ and the identity is a theorem; the Selmer lengths are therefore **computed**, not conjectured. Paper: Theorem thm:main, Section sec:sharp ("the identity says what the excess is").

22.2.10 Section 10. Remark rem:ko: Tamagawa excess and the formula of Kim and Ota

10.1 `rem:congruence:rem-ko-1` (remark, Remark rem:ko(1); P given 9.1 and Chapter 18, its 8.3). At a multiplicative prime $q \equiv 1 \pmod\ell$ with $\ell \mid n_q$ and unit part u of the Tate parameter an ℓ-th power in $\mathbb{F}_q^\times$, $v_\ell(\mathrm{Tam}_q(\mathrm{ad}^0 T)) \geq t_\ell(E,q) + 1$; precisely the excess is $\min(v_\ell(n), v_\ell(q-1), v_\ell(c(u)))$ (points 23, 65: "by at least one", and the class is $c(u)$, not c_q). By 9.1 this raises the bound exactly by that amount for $\ell \notin S_f$, also at potentially multiplicative additive primes (point 68). Paper: Remark rem:ko(1), Section sec:tam, the abstract.

10.2 `thm:congruence:kim-ota` (theorem, Remark rem:ko(2); R, **stated in Chapter 15, its 11.4**; the consequence P). (KimOta Theorem 1.1, as cited; their Definition 2.12 for the Tamagawa exponent.) For $N = N^+N^-$ with N^- square-free, $\ell \geq 5$, $\ell \nmid N$, $\ell \notin S_f$, and $\bar\rho$ ramified at every $q \mid N^-$ with $q \equiv \pm 1 \pmod\ell$: $\mathrm{ord}_\lambda \eta_f(N) = \mathrm{ord}_\lambda \eta_f(N^+, N^-) + \sum_{q\mid N^-} t_f(q)$. Consequence: $v_\ell(r_E) \geq \sum_{q\mid N^-} t_\ell(E,q)$, contained in (eq:thm) without $\ell \geq 5$ and $q \not\equiv \pm 1$. What this chapter adds: (i) Kim-Ota's $\eta_f(N)$ must be checked to be the Hecke congruence ideal of 3.3 (then $= r_E\mathbb{Z}_\ell$ by 3.4) and not a cohomological one (if cohomological, 5.6 gives the same value at $\ell \nmid N$ anyway); (ii) with 9.1 the comparison reads: their equality plus (eq:identity) for E give $\mathrm{ord}_\lambda \eta_f(N^+, N^-) = \mathrm{length}\, H^1_f(\mathbb{Q}, W) + \sum_{q\mid N^-} \min(\dots) + \sum_{q\mid N^+}(\dots)$, a Selmer-theoretic reading of the N^--new congruence ideal (S, one paragraph, clearly marked as a consequence of both theorems). Reference: KimOta; PollackWeston11; RibetTakahashi97 (Chapter 14, its 12.2). Paper: Remark rem:ko(2), Section 1.5.

22.2.11 Section 11. Twist monotonicity: Theorem thm:twist

11.1 `prop:congruence:twist-inputs` (proposition, recollection; P by citation). From Chapter 13, its 9.1, 9.3, and Chapter 15, its 10.1, 10.2: $\iota = \iota_S$ maps $S_2(\Gamma_0(M))$ into $S_2(\Gamma_0(\mathrm{lcm}(M, D^2)))$, preserves integrality, intertwines T_p with $\chi_S(p)T_p$ for $p \nmid DM$, kills the q-coefficients at multiples of D ($\iota(h) = \iota(h^S)$), and maps the sum of the generalised eigenspaces of systems different from that of g into the sum of those different from the twisted system; $(h \otimes \chi) \otimes \chi = h^{\{q\}}$ (point 20). Levels (Chapter 15, its 10.2): type I_n^* at all $q \in S$: $N_S = N/D$; type I_0^*: $N_S = N/D^2$; types II, III, IV or duals at $q \geq 5$: $N_S = N$. Paper: Theorem thm:twist (its proof), the sentence after it.

11.2 `thm:congruence:twist` (theorem, the paper's Theorem thm:twist; **P**, every step). Let $E/\mathbb{Q}$ have conductor N, S a set of odd primes, D, χ_S, E_S, N_S as in C7. If $q^2 \mid N$ for every $q \in S$ and $\mathrm{lcm}(N_S, D^2) \mid N$, then $r_{E_S} \mid r_E$; if moreover $\mathrm{lcm}(N, D^2) \mid N_S$, then $r_{E_S} = r_E$. Proof (the paper's, expanded): (1) $\iota(g) = f$ for $g = f_{E_S}$: both multiplicative; $a_p(f) = \chi_S(p)a_p(g)$ for $p \nmid D$ (at $p \mid N$, $p \notin S$, the twist is unramified at p, Chapter 9, its 10.4); $a_{q^k}(f) = 0$ for $q \in S$ since $q^2 \mid N$ (Chapter 13, its 6.4(iii)). (2) $g^\perp$ (Petersson complement at level N_S, where g is new, so it equals the $\mathbb{T}$-stable complement, Chapter

13, its 8.2(ii)) is the sum of the generalised eigenspaces of the other systems, each differing from g's at infinitely many p (strong multiplicity one, Chapter 13, its 6.3); ι sends such a space into the generalised eigenspace of the twisted system, which differs from f's at infinitely many p, hence lies in $f^{\perp}$ (Chapter 13, its 6.3(iii)). (3) A congruence $g \equiv h \pmod r$ with $h \in g^{\perp}$ integral gives $f \equiv \iota(h) \pmod r$ with $\iota(h) \in f^{\perp}$ integral (Chapter 13, its 9.1(iii)), so $r \mid r_E$ by the definition of r_E (no need for Lemma lem:crit here). (4) Under the second hypothesis $E = (E_S)_S$ and the roles exchange. Paper: Theorem thm:twist.

11.3 `ex:congruence:twist-cases` (worked examples and data; P, Part 6.6). The paper's cases: (a) I_n^*: 539c1/77c1, 980i1/140b1, 121d1/11a1; (b) I_0^*: 539d1/11a1; (c) potentially good: 121a1/121c1 ($r = 6 = 6$). Data check of case (c) (Part 6.6): for all 14,876 pairs (E optimal non-CM with at least one prime $q \geq 5$ of type II, III, IV or a dual, S the set of such primes, $\ell \in \{3, 5, 7, 11, 13\}$, $\ell \nmid N$, $E[\ell]$ irreducible) below conductor 10,000, $v_\ell(\deg \varphi_E) = v_\ell(\deg \varphi_{E_S})$, as the theorem predicts. **Correction of the paper's data sentence** (important_notes.md point 16): strict inequality occurs in case (a) as well as case (b), the difference being exactly the Euler term of Theorem thm:quant (12.6: 25,477 pairs, equality in every one). Paper: the paragraph after the proof of Theorem thm:twist.

22.2.12 Section 12. Theorem thm:quant

12.1 `thm:congruence:quant` (theorem, the paper's Theorem thm:quant, with the extension to $q = 3$; **P** given 6.2 (R) and Chapter 14 (R inputs)). Let ℓ be odd, $\ell \nmid 2N$, $E[\ell]$ irreducible; let S be the set of **odd** primes q at which E has type I_0^* or I_n^* (the paper: $q \geq 5$; the extension to $q = 3$ is important_notes.md point 60, P here, as $\ell \neq 3$ is then automatic and the twist lemma for these exchanges holds at every odd q); $E_S = E \otimes \chi_S$. Then $v_\ell(\deg \varphi_E) \geq v_\ell(r_{E_S}) + \sum_{q \in S} e_\ell(E, q) \geq \sum_{q \in S} e_\ell(E, q)$. Paper: Theorem thm:quant.

12.2 `sec:congruence:quant-setup` (the set-up of the proof; P). By Lemma lem:twist (q odd; Chapter 10), E_S is good at the I_0^* primes ($\delta_q(g) = 2$) and multiplicative at the I_n^* primes ($\delta_q(g) = 1$), so $N_S^S = N_S \prod_{q \in S} q^{\delta_q(g)} = N$ (Chapter 15, its 10.2(iii)): the depleted form g^S lives at level N and $\iota(g^S) = \iota(g) = f$. The eigencharacter π of g^S on $\mathbb{T}_N$ has generalised eigenspace $\mathbb{C}g^S$ (Chapter 13, its 7.4(iv)), so 3.5 applies to it. Paper: proof of Theorem thm:quant, first paragraph.

12.3 `lem:congruence:quant-step1` (lemma, Step 1, in the sharpened form; P). $v_\ell(r_E) = v_\ell(\eta^{\mathrm{Hecke}}(g^S))$ (the paper proves $\geq$). Proof of $\geq$ (the paper's): $\mathfrak{m} \ni U_q$ ($q \in S$); every U_q-eigenvalue on $S_2(\Gamma_0(N))$ is 0, ± 1 or a root of $X^2 - aX + q$, so 0 or an ℓ-adic unit (Chapter 13, its 7.2); hence the $\mathfrak{m}$-component is $\mathbb{C}g^S$ plus generalised eigenspaces of systems with $U_q = 0$ from newforms $h \neq g$ (Chapter 13, its 7.1, 8.1); by 3.5 over $\mathbb{Z}_\ell$, $\ell^{v(\eta^{\mathrm{Hecke}}(g^S))}$ is the largest ℓ^t with $g^S \equiv h \pmod{\ell^t}$, h integral in the $\mathbb{T}$-stable complement (point 61); project by $e_{\mathfrak{m}}$ (Chapter 13, its 8.3(iv)); apply ι, which lands in $f^{\perp}$ (11.1). Proof of $\leq$ (point 59, Chapter 13, its 9.3(iii)): $\iota(f) = g^S$ and $\iota(f^{\perp})$ lies in the $\mathbb{T}$-stable complement of $\mathbb{C}g^S$, so a congruence for f gives one for g^S of the same modulus. Check: level 539, orders 288 and 504 (2.2). Paper: Step 1, Remark (2) after the proof.

12.4 `lem:congruence:quant-steps234` (lemma, Steps 2 to 4; P given 4.3, 6.2 (R), 5.6). Step 2: Lemma lem:ddt417 with $L = H^1(X_0(N), \mathbb{Z}_\ell)_{\mathfrak{m}}$, $\hat{\delta}_!$ perfect on the $\mathfrak{m}$-part (5.3(iv), point 36), $L[\wp] = M_{g^S,\lambda}$ of rank 2, Gram determinant $(\eta_{g,\lambda}^S)^2$: $v(\eta^{\mathrm{Hecke}}(g^S)) \geq v(\eta_{g,\lambda}^S)$ (no freeness). Step 3: 6.2(c) for g ($\bar{\rho}_g = \bar{\rho} \otimes \chi_S$ irreducible, $\ell \nmid 2N_S$): $v(\eta_g^S) = v(\eta_g^{\emptyset}) + \sum_{q \in S} v_\ell(L_q^{\mathrm{nv}}(A_g, 1)^{-1}) = v(\eta_g^{\emptyset}) + \sum_{q \in S} e_\ell(E, q)$, because $A_g = A_f$ (Chapter 19, its 9.5(i)), $\delta_q(g) \geq 1$ (so its naive factor is the true one), and the true factor of A_f at q is the one f omits (Chapter 19, its 9.7). At $q = 3$ the values are the twist formulas $L_3(A_f, 1)^{-1} = \frac{2}{3} \cdot \frac{16 - a_3(E')^2}{9}$ (I_0^*) and $\frac{8}{9}$ (I_n^*), checked against PARI (point 60, Part 6.6). Step 4: $v(\eta_g^{\emptyset}) = v_\ell(\deg \varphi_{E_S^{\mathrm{opt}}}) = v_\ell(r_{E_S})$ by 5.6 for the optimal curve of the class of E_S (point 39: **no freeness**, which removes point 24's worry about Step 4; v_ℓ is a class invariant as $E_S[\ell]$ is irreducible, Chapter 14, its 6.4). Paper: Steps 2 to 4.

12.5 `rem:congruence:outer-inequality` (remark; P). The outer inequality $v_\ell(\deg \varphi_E) \geq \sum_{q \in S} e_\ell(E, q)$ needs only Steps 1 to 3 and the integrality of $\eta_g^{\emptyset}$; neither multiplicity one nor ARS at level N_S. Still not "elementary" (point 24): Step 3 is DFG04 Proposition 1.4 and Step 2 is Lemma lem:ddt417. The abstract's wording should be "by a twisting argument combined with the formula of Diamond, Flach and Guo". Paper: the abstract, Section 1.5.

12.6 `thm:congruence:quant-equality` (theorem, the remark after Theorem thm:quant, (2), in the strengthened form; P given 9.1; **new**, Part 8, N2). If $\ell \notin S_f$, then

$$v_\ell(\deg \varphi_E) = v_\ell(r_{E_S}) + \sum_{q \in S} e_\ell(E, q),$$

and every inequality in the chain of 12.3 and 12.4 is an equality (in particular $v(\eta^{\mathrm{Hecke}}(g^S)) = v(\eta^S_{g,\lambda})$ at level N, a freeness-type statement for the depleted form obtained without multiplicity one). Proof: apply 9.1 (equality) to E and to the optimal curve in the class of E_S ($\ell \notin S_{f_{E_S}}$: twisting does not change absolute irreducibility over $\mathbb{Q}(\sqrt{\ell^*})$, and $N_S \mid N$); the Selmer groups coincide ($\mathrm{ad}^0 T$ is twist-invariant), so do the Tamagawa factors (local, $\mathrm{ad}^0 T$ unchanged); the Euler sums differ exactly by $\sum_{q \in S} e_\ell(E, q)$ (at $q \notin S$ the twist is unramified, same type and e_ℓ; at $q \in S$, E_S is good or multiplicative, contributing 0). Hence $v_\ell(\deg \varphi_E) - v_\ell(\deg \varphi_{E_S}) = \sum_{q \in S} e_\ell(E, q)$, and with $v_\ell(\deg \varphi_{E_S}) = v_\ell(r_{E_S})$ ($\ell^2 \nmid N_S$) the chain $v_\ell(r_E) \geq v(\eta^{\mathrm{Hecke}}(g^S)) \geq v(\eta^S_g) = v_\ell(r_{E_S}) + \sum e_\ell = v_\ell(r_E)$ closes. The paper's Remark (2) asks for multiplicity one at level N; point 56 (and Chapter 14, its 9.5(c), 10.4(b)) say it is still needed; **it is not**. Data (Part 6.6): equality in all 25,477 pairs (optimal non-CM curves of conductor below 10,000 with $S \neq \emptyset$, $q = 3$ included, $\ell \in \{3, 5, 7, 11, 13\}$, $\ell \nmid N$, $E[\ell]$ irreducible), including the 8 pairs at $\ell = 3$ with image 3Ns, which lie outside the theorem. Examples: 539c1 ($2 = 1 + 1$), 980i1 ($2 = 1 + 1$), 121d1 ($1 = 0 + 1$), 539d1 ($2 = 0 + 2$). Paper: the remark after Theorem thm:quant, (2).

12.7 `rem:congruence:after-quant` (remark, the remark after Theorem thm:quant, (1) and (3); P). (1) Theorem thm:quant is level raising: from N_S to $N_S^S = N$, at the price of the level-raising Euler factors (Chapter 15, its 10.4). For $\ell = 3$ a single I_n^* prime $q \geq 5$ forces $3 \mid \deg \varphi_E$ ($3 \mid q^2 - 1$), as does an I_0^* prime with $q \equiv 1 \pmod 3$; neither needs absolute irreducibility over $\mathbb{Q}(\sqrt{-3})$. (3) For the potentially good types $N_S = N$, f is its own depletion, and Theorem thm:dfg14 gives no factor; the omitted factor is reached only through Theorem thm:dfg37, i.e. with $\ell \notin S_f$, and Section sec:anomaly shows the hypothesis cannot be dropped (3Ns curves fail through a type III term). Paper: the remark after Theorem thm:quant.

22.2.13 Section 13. Beyond the hypothesis: Remark rem:ns and Open questions 1 and 2

13.1 `rem:congruence:rem-ns` (remark, Remark rem:ns; P for the conditional statement given Chapter 18, its 12.1). For $\ell = 3$ and image $N_s(\mathbb{F}_3)$, $3 \nmid N$: Theorem thm:dfg37 does not apply ($3 \in S_f$); Lemma lem:dfg21 applies **provided** $H^1_f(\mathbb{Q}, V^D) = 0$ (equivalently $H^1_f(\mathbb{Q}, \mathrm{ad}^0 V_3 E) = 0$), which DFG04 Theorem 2.7 does not supply here (point 67); then $\operatorname{length} H^1_\Sigma = \operatorname{length} H^1_f + \sum_p (\dots) - h_3(E)$ with $h_3(E) = 1$ (Chapter 11, its 8.5; point 5). If, in addition, $v(\eta^\Sigma) = \operatorname{length} H^1_\Sigma$ held, the assembly 9.2 would give (eq:identity) with $-h_3(E)$, the shape of Conjecture conj:euler. Paper: Remark rem:ns, Section 1.3, Section sec:anomaly.

13.2 `rem:congruence:open-question-2` (remark, Open question 2; P for the analysis). What is missing for 3Ns curves: (a) on the Hecke side, the identity $v(\eta^\Sigma_{f,\lambda}) = \operatorname{length} H^1_\Sigma(\mathbb{Q}, W)$ needs Taylor-Wiles for $\bar\rho$ induced from $\mathbb{Q}(\sqrt{-3})$, where horizontal primes fail to kill the dual Selmer group (8.4); (b) on the Selmer side, $H^1_f(\mathbb{Q}, \mathrm{ad}^0 V_3 E) = 0$. The data (failures of (eq:thm) by exactly 1, never more) and the equality of 12.6 at the 8 3Ns pairs are consistent with the corrected identity. The book records that later work on residually dihedral representations may settle (b) and (a); it cites nothing it has not checked. Paper: Open question 2.

13.3 `rem:congruence:open-question-1` (remark, Open question 1; P for the analysis). At $\ell \mid N$: Step 1 fails ($r_E \neq \deg \varphi_E$ possible at $\ell^2 \mid N$: 99a1, 726a1, 539d1; Chapter 14, its 8.5), Step 3 fails ($\lambda \mid N$ puts ℓ in S_f; DFG04's integral structures and Fontaine-Laffaille at ℓ need $\ell \nmid N$), and the Euler factor at $q = \ell$ is not the relevant local quantity (Remark rem:wild). What would replace it: a local condition at ℓ (finite flat, ordinary, or semistable deformation conditions), with the empirical weights w_3 as the target. One paragraph. Paper: Open question 1, Conjecture conj:euler.

22.2.14 Section 14. The imported statements, verified against DFG04

14.1 `rem:congruence:imported-table` (remark with table; P for the verification record). A `booktabs` table: each DFG04 statement the paper cites (Theorems 0.1, 0.2, 2.7, 2.15, 3.7; Proposition 1.4; Lemmas 2.1, 2.12; (13), (14), (31), (57); p. 708; Sections 1.5.3, 1.6.1, 1.6.2, 1.7.2, 1.7.3, 1.8.2, 2.1), what it says (quoted in the book's notation), where the paper uses it, and the status "verified against the published text" (Part 7 here gives the record). Also the two differences of wording the reader should know: DFG04 Lemma 2.1 asks that Σ contain every prime where $H^1_f(\mathbb{Q}_p, W) \neq H^1_{\mathrm{ur}}(\mathbb{Q}_p, W)$ (the paper's "every $p \neq \ell$ at which W is ramified" implies it); DFG04's H^1_Σ uses H^1_{ur}, not H^1_f, outside $\Sigma \cup \{\ell\}$ (the same for $\Sigma \supseteq \{p \mid N\}$). And the warning that the arXiv version (2025) is a longer text with **different numbering** (its Sections 1.1 to 3.5 do not match the published 1.1 to 3.2); the book cites the published version. Paper: every citation of DFG04.

22.3 3. Section-by-section plan with target line counts

Line counts are lines of LaTeX, card excluded (PLAN.md Section 7.3: this chapter may run to 2,800). The card lists the 59 items of Part 2 (numbered 20.1 to 20.59 in the book; conventions in one line each), about 230 lines, not counted. Every section opens with one paragraph saying what it is for and where in the paper it is used.

section	title	contents	target lines
(starred)	Core results at a glance	the card: every item of Part 2 with its tag and `\Cref`; the identity of 9.1 and the chain of 12.6 displayed	(excluded)
(starred)	Where the paper uses this chapter	Section 1 (Hida, DFG04, congruence ideal, Lemma 2.12); Lemma lem:crit (restated, 3.4); Section 4.1 (5.1 to 5.4, 6.2); Lemma lem:ddt417 (4.3); Lemma lem:hecke (5.6); Remark rem:free (5.7); Section 4.2 (7.1 to 7.7); thm:dfg37, thm:dfg27, lem:dfg21, lem:fpr, lem:h0 (cited items); proof of thm:main (9.2); Remark rem:ko (10.1, 10.2); thm:twist (11.2); thm:quant and the remark after it (12.1 to 12.7); Remark rem:ns (13.1); Open questions 1, 2 (13.2, 13.3)	50

section	title	contents	target lines
1	Overview and conventions	1.1; C1 to C8 as `convention` environments; the roadmap table; `warning` boxes for C3 and C4	90
2	Hecke algebras over $\mathbb{Z}_\ell$	2.1, 2.2 (recollection with citations; the level-539 example)	110
3	Congruence modules and congruence ideals	3.1 to 3.7; the module-theoretic reading of Lemma lem:crit; the $\mathbb{T}$-stable complement; Hida; physics aside on the "overlap"	210
4	Wiles's numerical criterion and Lemma lem:ddt417	4.1 to 4.4; the criterion with the toy examples; the full proof of Lemma lem:ddt417 with steps (0) to (6); `paperlink`: the paper's proof sentence by sentence (points 19, 38)	260
5	The DFG04 lattices; Lemma lem:hecke; Remark rem:free	5.1 to 5.7; quotations from DFG04 Sections 1.4 to 1.7; the comparison with $H^1(X_0(M),\mathbb{Z}_\ell)$; `paperlink`: Section 4.1 and Lemma lem:hecke with points 35, 36, 39	240
6	DFG04 Proposition 1.4	6.1 to 6.5; γ, $\gamma^t\gamma$; the three cases; the level-raising factor; Ihara; refined structures in one paragraph	190
7	S_f and the theorems of DFG04	7.1 to 7.8; quotations of (31), Theorems 0.1, 0.2, 2.7, 3.7, Lemma 2.12; minimally ramified	180
8	Taylor-Wiles in outline	8.1 to 8.5; deformation rings, tangent spaces, horizontal primes, patching in words, DFG04 Theorem 3.1; where each hypothesis is used	200

section	title	contents	target lines
9	The assembly of Theorem thm:main	9.1 to 9.4; the six-step proof; the hypothesis table; the examples; `paperlink`: Section 4.3 sentence by sentence (point 18)	230
10	Remark rem:ko	10.1, 10.2	80
11	Theorem thm:twist	11.1 to 11.3; full proof; the data paragraph corrected (point 16)	140
12	Theorem thm:quant	12.1 to 12.7; Steps 1 to 4; the sharpened Step 1 (point 59); $q = 3$ (point 60); the equality theorem 12.6 with the data; the remark after the theorem	290
13	Beyond the hypothesis	13.1 to 13.3; Remark rem:ns (point 67); Open questions 1, 2	110
14	The imported statements	14.1, the verification table	70
15	Exercises	the 16 exercises of Part 4.2	160
16	Notes and references	DFG04 (published versus arXiv numbering), Hida81, Wiles95, TaylorWiles, Lenstra95, Diamond96, DDT, FPR94, KimOta, PollackWeston11, ARS12, Tilouine97, RibetStein, Ribet84, DiamondTaylor94; which numbers were verified	60
	total		**2,670**

22.4 4. Coverage table and exercises

22.4.1 4.1 Coverage: PLAN.md scope, Section 6 map rows and paper hooks to line-up items

PLAN.md scope item, Section 6 row, or paper hook	line-up items
Hecke algebras as finite flat $\mathbb{Z}$-algebras; $\mathbb{T}_{\mathfrak{m}}$; $\mathfrak{m}$ attached to (f, ℓ); $\mathbb{T} \otimes \mathbb{Z}_\ell = \prod \mathbb{T}_{\mathfrak{m}}$	2.1 (citing Chapters 3, 13)

PLAN.md scope item, Section 6 row, or paper hook	line-up items
the $\mathfrak{m}$-component of $S_2(\Gamma_0(N);\mathbb{Z}_\ell)$ via idempotents; which newforms belong to $\mathfrak{m}$ ($\bar\rho_g \simeq \bar\rho_f$)	2.1(iii), 2.2
congruence modules and ideals: $\eta = \pi(\mathrm{Ann}\,\wp)$, $\mathcal{O}/\eta = \mathbb{T}/(\wp + I)$	3.1, 3.2, 3.3
Hida's definition and "largest modulus of a congruence with $f^\perp$"; Lemma lem:crit restated (proof in Chapter 15)	3.4, 3.5, 3.6
Hida's theorem on congruence modules and the adjoint L-value	3.6, 7.8
Wiles's numerical criterion (statement)	4.1, 3.2
Lemma lem:ddt417 in full, in the paper's generality, every ingredient from Chapter 3	4.3, 4.4
Lemma lem:hecke (proof in full) and Remark rem:free	5.6 (proved in Chapter 14, its 9.3, 9.5, cited per B19), 5.7
DFG04 Sections 1.5.3, 1.6.2, 1.7.3: integral structure, $M_{f^\Sigma,\lambda}$, $\hat\delta_!$, η^Σ; Gram determinant a square	5.1 to 5.5
DFG04 Proposition 1.4(c), the three cases, proof idea	6.1 to 6.5
the exceptional set S_f, DFG04 (31)	7.1
DFG04 Theorem 3.7, "minimally ramified", why the primes dividing N qualify	7.3, 7.4, 7.5
DFG04 Theorem 2.7; Theorems 0.1, 0.2	7.6, 7.7, 7.5
Taylor-Wiles in outline (R_Σ, $R_\Sigma \to \mathbb{T}_\Sigma$, numerical criterion, patching, Selmer as tangent space)	8.1 to 8.5
the assembly of Theorem thm:main step by step (lem:hecke, thm:dfg14, thm:dfg37, thm:dfg27, lem:dfg21, lem:h0, lem:fpr, def:terms, lem:tamfactor; cancellation; I_n^* primes)	9.1, 9.2, 9.3, 9.4
Theorem thm:twist in full	11.1, 11.2, 11.3
Theorem thm:quant in full, Steps 1 to 4, $\eta^{\mathrm{Hecke}}(g^S)$ versus the cohomological ideal	12.1 to 12.6
the remark after Theorem thm:quant (why the argument fails for II, III, IV)	12.7
Kim-Ota and the Tamagawa exponent (Remark rem:ko(2))	10.2
Open questions 1 and 2	13.3, 13.2
Section 6 row: congruence ideal, r_E, η^Σ, $M_{f,\lambda}$, $\hat\delta_!$ (with 13, 14)	3.3, 3.4, 5.2 to 5.5
Section 6 row: multiplicity one, freeness, Gorenstein, $J_0(N)[\mathfrak{m}]$ (with 3, 14)	5.7, 4.2, 4.4(d)
Section 6 row: Hida's congruence modules, Wiles's numerical criterion, DDT Lemma 4.17 (with 3)	3.1, 3.6, 4.1, 4.3
Section 6 row: Taylor-Wiles, $R = \mathbb{T}$, DFG04 Theorems 0.1, 0.2, 2.7, 3.7, Proposition 1.4, Lemma 2.1, Theorem 2.15 (with 18)	6.2, 7.4 to 7.7, 8.1 to 8.5, 14.1 (Lemma 2.1 and Theorem 2.15: Chapter 18, its 10.4, 7.4)
Section 6 row: level raising, Wiles's level-raising factor (with 15, 19)	6.3, 12.7
Section 6 row: Ribet-Takahashi, Pollack-Weston, Kim-Ota, Tamagawa exponent (with 14, 15)	10.2
hook: Section 4 in full (4.1 to 4.4)	Sections 4 to 12

PLAN.md scope item, Section 6 row, or paper hook	line-up items
hook: Theorem thm:dfg14	6.2, 6.3
hook: Lemma lem:ddt417	4.3
hook: Lemma lem:hecke	5.6
hook: Remark rem:free	5.7
hook: Theorem thm:dfg37	7.3, 7.4, 8.5
hook: Theorem thm:dfg27	7.6 (stated in Chapter 18, its 10.2)
hook: Lemma lem:dfg21	9.2 Step 4 (proved in Chapter 18, its 10.4), 14.1
hook: Lemma lem:fpr	9.2 Step 4 (proved in Chapter 18, its 7.4, 7.5), 7.7, 14.1
hook: Lemma lem:h0	9.2 Step 4 (proved in Chapter 11, its 8.5; Chapter 18, its 10.3)
hook: the proof of Theorem thm:main	9.2
hook: Remark rem:ko	10.1, 10.2
hook: Theorem thm:twist	11.2
hook: the proof of Theorem thm:quant and the remark after it	12.2 to 12.7
hook: Section 1 (Hida, DFG04, congruence ideal)	3.3, 3.6, 7.7, 7.8
hook: Lemma lem:crit	3.4, 3.5
hook: Remark rem:ns	13.1
also: Open question 3 (answered in Chapter 18, its 9.3; the identity 9.1 makes the Selmer length computable)	9.1, 9.4
also: the abstract ("elementary twisting argument"; "Bloch-Kato Selmer group of $\mathrm{ad}^0 V_\ell E$"; "with equality when multiplicity one holds")	12.5, 9.1
also: important_notes.md points 16, 18, 19, 24, 35, 36, 38, 39, 47, 56, 59, 60, 61, 63, 64, 67, Q1, B3, B17, B19	Part 6.4

22.4.2 4.2 Exercises (Section 15 of the chapter), graded from easy to harder

1. For $\mathbb{T}_r = \{(a,b) \in \mathbb{Z}_\ell^2 : a \equiv b \pmod{\ell^r}\}$ and $\pi(a,b) = a$ compute $\wp$, I, η, Ω and $\Phi = \wp/\wp^2$; do the same for $\mathbb{T}_3^\flat$ and show $\operatorname{length}\Phi > v(\eta)$ there. Which of the two is a complete intersection?
2. Prove that $\wp \cap I = 0$ iff $\eta \neq 0$ (3.1(ii)), and give an example with $\wp \cap I \neq 0$. Explain why the paper's Lemma lem:ddt417 survives this (important_notes.md point 19).
3. Run the proof of Lemma lem:ddt417 (4.3) on $L = \mathbb{T}_r$, $L = \mathbb{Z}_\ell^2$ with diagonal action, and $L = \mathbb{T}_r^2$ with an alternating pairing; identify Q in each case.
4. (Check the paper's claim that "the Gram determinant of the pairing on $M_{f^\Sigma,\lambda}$ is $(\eta_{f,\lambda}^\Sigma)^2$ up to a unit".) Prove that for a rank-two alternating form the Gram determinant is the square of a generator of the ideal of values, and that the image of $\wedge^2$ is that ideal (DFG04's definition, 5.4(ii)).
5. Prove that the idempotents of $\mathbb{T}_\ell$ are self-adjoint for $\hat{\delta}_!$ and that the restriction of $\hat{\delta}_!$ to $H^1(X_0(N), \mathbb{Z}_\ell)_{\mathfrak{m}}$ is perfect (point 36). Show that $M_{f,\lambda}$ lies in the $\mathfrak{m}_{f,\lambda}$-component.
6. (Check the paper's paragraph after Theorem thm:dfg14.) For a good prime p and roots α, β of $X^2 - a_p X + p$, prove $(1 - \frac{\alpha}{\beta p})(1 - \frac{1}{p})(1 - \frac{\beta}{\alpha p}) = \frac{p-1}{p} \cdot \frac{(p+1)^2 - a_p^2}{p^2}$. For $f = f_{11\mathrm{a}}$ and $\ell = 3$ list the $p < 40$ with v_3 of this factor positive, and compare with the levels $11p$ at which a newform congruent to f modulo 3 exists (Chapter 15, its 7.5).
7. (Check DFG04 Proposition 1.4(b) in the smallest case.) For $f = f_{11}$ and $\Sigma = \{2\}$, compute $f^\Sigma = f + 2f(2z) + 2f(4z)$ and verify that the factor $L_2^{\mathrm{nv}}(A_f, 1)^{-1} = \frac{1}{2} \cdot \frac{9-4}{4} = \frac{5}{8}$ is a unit at $\ell = 3$, so $\eta_{f,3}^{\{2\}} = \eta_{f,3}^{\emptyset}$; and for $\Sigma = \{7\}$ that the factor $\frac{6}{7} \cdot \frac{64-4}{49}$ has $v_3 = 2$.
8. Show that for $f = f_E$ and $\ell \nmid 2N$, $\ell \notin S_f$ iff $E[\ell]$ is irreducible and not ($\ell = 3$ with image 3Ns)

(7.1(ii); point 47), using the description of $\bar{\rho}|_{I_\ell}$ by fundamental characters.

9. (Check the paper's description of H^1_Σ.) Show that for $\Sigma \supseteq \{p \mid N\}$ DFG04's local condition H^1_{ur} at $p \notin \Sigma \cup \{\ell\}$ equals H^1_f. Give a W and a prime at which $H^1_{\mathrm{ur}}(\mathbb{Q}_p, W) \neq H^1_f(\mathbb{Q}_p, W)$.
10. Write out the assembly 9.2 for 121a1 at $\ell = 3$: compute each term ($\Sigma = \{11\}$, type II, $L^{\mathrm{nv}}_{11} = 1$, $e_3 = v_3(12) = 1$, $\mathrm{Tam}_{11}(\mathrm{ad}^0 T) = \mathbb{Z}_3$), and deduce $H^1_f(\mathbb{Q}, \mathrm{ad}^0 T_3 \otimes \mathbb{Q}_3/\mathbb{Z}_3) = 0$ from $\deg \varphi_E = 6$.
11. (Check the paper's Remark rem:ko(1).) For a multiplicative prime with $n_q = 9$, $q \equiv 1 \pmod 9$ and unit part a cube in $\mathbb{F}_q^\times$, compute the excess of $v_3(\mathrm{Tam}_q(\mathrm{ad}^0 T))$ over $t_3(E, q)$; find such a curve in the tables, if one exists below conductor 10,000.
12. Prove Theorem thm:twist (11.2) for 121a1 and 121c1 directly: show $\iota(f_{121\mathrm{c}}) = f_{121\mathrm{a}}$ and conclude $r(121\mathrm{a}) = r(121\mathrm{c})$.
13. (Check the paper's sentence after Theorem thm:twist; point 16.) For the pairs 539c1/77c1, 980i1/140b1, 121d1/11a1 compute v_3 of the two modular degrees and the Euler term $v_3(q^2 - 1)$; which case of the theorem are they, and is the inequality strict?
14. Prove the sharpened Step 1 (12.3): $\iota(f) = g^S$ and ι maps $f^\perp$ into the $\mathbb{T}$-stable complement of $\mathbb{C}g^S$. Why must the complement be the $\mathbb{T}$-stable one (point 61)?
15. (Theorem thm:quant at $q = 3$; point 60.) Verify with PARI that $L_3(A_f, 1)^{-1} = \frac{2}{3} \cdot \frac{16 - a_3(E')^2}{9}$ for a curve of type I_0^* at 3 (for example from Chapter 13, its 6.6, `c11.py`: 333d1 or 477a1 at $\ell = 7$) and that $v_7(\deg \varphi_E) \geq e_7(E, 3)$.
16. (Harder.) Prove 12.6: for $\ell \notin S_f$, $v_\ell(\deg \varphi_E) = v_\ell(\deg \varphi_{E_S}) + \sum_{q \in S} e_\ell(E, q)$, from (eq:identity) with equality for E and E_S. Deduce that every inequality in the proof of Theorem thm:quant is an equality, and verify it on 539d1 ($\ell = 3$: $2 = 0 + 2$).

22.5 5. Dependencies and forward pointers

Results from other chapters used here (by item): * Chapter 3 (`ch:modules`): saturation (5.5), the congruence module of a decomposition (5.8), perfect pairings, orthogonals and the Gram cokernel (9.1 to 9.3), rank-two alternating pairings (9.4), adjoints and idempotents (9.5), algebra-module duality (9.6), finite free algebras and idempotent lifting (10.2, 10.3), idempotent congruences (10.6), congruence ideals (11.1), toy algebras (11.2), the ingredients of Lemma lem:ddt417 (11.3), Gorenstein and Mazur's freeness criterion (11.4, 11.5), the toy checks (11.6). * Chapter 4 (`ch:reps`): homothety of lattices for irreducible residual representations (used in 6.2(c)); ad^0 invariant under duals and twists (5.4(iv), 12.6). * Chapter 9 (`ch:ellcurves`): quadratic twists and $a_p(E^{(d)})$ (11.2). * Chapter 10 (`ch:reduction`): Lemma lem:twist (i), (ii) (12.2), Kodaira types. * Chapter 11 (`ch:galrep`): $\ell \notin S_f$ versus 3Ns (its 8.6; 7.1), Lemma lem:h0 corrected (its 8.5; 13.1). * Chapter 13 (`ch:hecke`): $\mathbb{T}$ and the perfect pairing (4.1, 4.2), $\eta(\phi)$ and r_f (4.3), newforms and strong multiplicity one (6.2 to 6.4), U_p on oldforms and units (7.1, 7.2), depletion (7.4), structure of $\mathbb{T}$ (8.1), the $\mathbb{T}$-stable complement (8.2), local components (8.3), congruences (8.4), twist operator and transport (9.1, 9.3), Atkin-Li (9.2), the level-539 computations (11.3). * Chapter 14 (`ch:moddeg`): $\hat{\delta}_!$ (3.3, 3.4), $M_{f,\lambda}$ (4.4), optimal curves (6.2, 6.4), r_E and ARS (8.1 to 8.5), **9.1 to 9.6** (point 39 and the checklist), 9.7, 10.1 to 10.4, the Shimura subgroup (11.3). * Chapter 15 (`ch:modgalrep`): Lemma lem:crit (8.2), level raising and the factor identity (7.1, 7.2), Ihara (7.4), twist levels (10.1, 10.2), Kim-Ota (11.4), the Taylor-Wiles remark (11.5). * Chapter 17 (`ch:galcoh`): the abstract DFG04 Lemma 2.1 (10.3) and the length identity (10.4). * Chapter 18 (`ch:blochkato`): H^1_Σ (6.3), Tamagawa factors (7.1 to 7.5, 8.2 to 8.4, 9.3, 9.4), S_f (10.1), thm:dfg27 (10.2), lem:h0 (10.3), lem:dfg21 (10.4), the length identity (10.5), **the Selmer half 11.1**, examples (11.3), Remark rem:ns (12.1), DFG04 Theorems 0.1 and 0.2 (14.4, 14.5). * Chapter 19 (`ch:lfunctions`): naive factors (8.1, 9.7), the Petersson formula with L^{nv} (8.3), $A_g = A_f$ (9.5), Lemma lem:euler (9.3).

Forward pointers (chapters that use this one): * Chapter 21 (`ch:paper`): A4.1 to A4.16 should cite this card for full proofs: A4.1 (5.2 to 5.4, with the corrected "dual of $T_\ell E$"), A4.2 (6.2), A4.3 (4.3), A4.4 and A4.5 (5.6, 5.7: equality without freeness; points 35, 39), A4.6 to A4.8 (7.1 to 7.6, with the verified DFG04 wording), A4.12 (9.2), A4.13 (10.1, 10.2), A4.14 (11.2, 11.3), A4.15, A4.16 (12.3 to 12.7, including 12.6),

A8.1 (13.2, 13.3), B3 (14.1: every DFG04 status becomes "verified"). * Chapter 22 (`ch:computation`): the twist test of Part 6.6 (`quant.py`, `twistc.py`) as a worked session; e_ℓ at $q = 3$ from `lfunsympow`. * Appendix B (`ch:notation`): $\eta^{\Sigma}_{f,\lambda}$, η^{Hecke}, $\Omega(\pi)$, $\Phi(\pi)$, $M_{f^\Sigma,\lambda}$, $\hat{\delta}_!$, S_f, f^Σ, M^Σ, δ_p, ι_S, E_S, g^S.

22.6 6. Notes for the writer

22.6.1 6.1 Conventions to state (Section 1 of the chapter, as `convention` environments)

C1 to C8 of Part 1.3. Two `warning` boxes: (a) the paper's I and J in Lemma lem:crit are this chapter's $\wp$ and I (C3); (b) the Hecke congruence ideal and DFG04's cohomological ideal are different objects, compared by Lemma lem:ddt417 and equal at $\ell \nmid 2N$ for the optimal curve's newform (C4, 3.7).

22.6.2 6.2 Pitfalls

- **Assemble, do not re-derive.** Lemma lem:crit is Chapter 15's (B17); Lemma lem:hecke as an equality is Chapter 14's (9.3, 9.5; B19); the Selmer half is Chapter 18's (11.1); the naive factors are Chapter 19's (9.7). Each step of 9.2 names the chapter; the proofs there are not repeated. The only full proofs written here are Lemma lem:ddt417 (4.3), Proposition 1.4(c) from (b) (6.2), the assembly (9.2), Theorems thm:twist and thm:quant (11.2, 12.3, 12.4), and 12.6.
- **Published versus arXiv DFG04.** The arXiv text arXiv:2512.02348 has different section and theorem numbers (e.g. its Section 2 is "Premotivic structures for modular forms"). Cite only the published numbering, which the paper uses and which Part 7 verified.
- $H^1(X_0(M), \mathbb{Z}_\ell)$ **versus DFG04's integral structure.** They agree on non-Eisenstein $\mathfrak{m}$-components (5.2(ii)), which is all the paper uses. Do not claim equality of the whole lattices.
- **The pairing normalisation.** DFG04 divide the cup product by $[U : U_{N'}]$; on $X_0(M)$ this is the cup product times 1 or $\frac{1}{2}$ (5.3(ii)). Say "up to a unit at odd ℓ".
- **Ideal of values, not Gram determinant.** DFG04 define η^Σ via the image of $\wedge^2$; the Gram determinant is its square (5.4).
- **Naive versus true factors.** Proposition 1.4 has the **naive** factors; DFG04's Proposition 1.12 (refined structures) has the true ones. The paper and this chapter use only Proposition 1.4.
- **Twist-minimality.** DFG04 assume twist-minimal f in Sections 1.8 and 2, and prove Theorem 3.7 for a twist and transfer it. The paper applies Theorem 3.7 to f_E directly, which is fine because 3.7 is stated for every newform; but η^Σ_f itself depends on f, not only on A_f (that is the content of Theorem thm:quant), so the transfer is about H^1_Σ and η^Σ for the same Σ, not about $\eta^\emptyset$.
- I^*_n **at** 2 (point 18): use $n = -v_2(j)$ and the local twist argument of Chapter 18, its 8.4; the Kodaira index is not the Tate valuation at 2.
- **The $\mathbb{T}$-stable complement** for g^S (point 61); the Petersson complement is right only for newforms at their level.
- $q = 3$ **in Theorem thm:quant** (point 60): allowed; $\ell \neq 3$ then, since $\ell \nmid N$. State the extension as the book's, with the proof.
- **Equality statements.** 9.1 and 12.6 are equalities for $\ell \notin S_f$; they depend on point 39 (Chapter 14) and on the checklist of Part 7, now confirmed. Present them as theorems of the book that strengthen the paper, with the paper's weaker statements in `paperlink`s.
- **Theorem numbers.** Verified against the published DFG04 (Part 7): Theorems 0.1, 0.2, 2.7, 2.15, 3.1, 3.6, 3.7; Propositions 1.3, 1.4, 1.12, 2.2; Lemmas 1.2, 1.5, 1.8, 1.10, 1.11, 2.1, 2.4, 2.5, 2.12; Corollaries 1.6, 1.9, 2.3; (13), (14), (18), (22), (23), (25), (31), (53), (57). Not verified: DDT Lemma 4.17 and Theorem 5.3 (cite "DDT Section 4" and "Section 5.1" unless checked), Hida81 Theorem 5.1 (as DFG04 cite it), KimOta Theorem 1.1 and Definition 2.12, RibetStein Theorem 3.5, ARS12 Theorem 2.2 (as the paper cites them).

- Typography (PLAN.md 7.2): "Diamond-Flach-Guo", "Taylor-Wiles", "Fontaine-Laffaille", "Fontaine-Perrin-Riou", "Kim-Ota" with plain hyphens; no Unicode; η^{Hecke} in roman.

22.6.3 6.3 What to leave to Chapters 3, 13, 14, 15, 17, 18, 19 and 21

As in Part 1.2: the module theory (Chapter 3), Hecke theory and twists (Chapter 13), $\hat{\delta}_!$, point 39 and multiplicity one (Chapter 14), Lemma lem:crit, level raising and Kim-Ota (Chapter 15), the abstract Selmer comparison (Chapter 17), Bloch-Kato, Tamagawa factors and the Selmer half (Chapter 18), Euler factors (Chapter 19), and the line-by-line tour of the paper's text (Chapter 21).

22.6.4 6.4 How the relevant important_notes.md points are handled

- **Point 16** (strict inequality not only in case (b)): 11.3, with the corrected sentence; 12.6 shows the difference is exactly the Euler term.
- **Point 18** (I_n^* at $q = 2$): 9.2 Step 6, citing Chapter 18, its 8.4.
- **Point 19** ($\wp \cap I = 0$): 4.3 step (0), 3.1(ii), Exercise 2.
- **Point 24** ("elementary"): 12.5 (outer inequality needs Steps 1 to 3; still not elementary), and 12.4 Step 4 (no freeness, by point 39).
- **Point 35** (kernel, not annihilator): 3.1(iv), 5.6.
- **Point 36** (pairing perfect on the $\mathfrak{m}$-part): 5.3(iv), 5.6, 12.4, Exercise 5.
- **Point 38** (lengths): C5, 4.3.
- **Point 39** (Lemma lem:hecke always an equality): 5.6 cites Chapter 14, its 9.3, 9.5; Part 7 confirms the identification with DFG04's η; 9.1 and 12.4 use it.
- **Point 47** ($\ell \notin S_f$ automatic for $\ell \geq 5$): 7.1(ii), 8.4, Exercise 8.
- **Point 56** (the DFG04 checklist): Part 7, all five items confirmed (item 1 in the form needed, on non-Eisenstein components); the second bullet ("Remark (2) after thm:quant still needs multiplicity one") is superseded by 12.6 (Part 8, N2).
- **Point 59** (Step 1 is an equality): 12.3.
- **Point 60** ($q = 3$ allowed in thm:quant): 12.1, 12.4 (with the PARI check), Exercise 15.
- **Point 61** ($\mathbb{T}$-stable complement): 3.5, 12.3, Exercise 14.
- **Point 63** ($H^0(\mathbb{Q}, V^D) = 0$ automatic): 9.2 Step 4, 7.6.
- **Point 64** (Tamagawa factors at additive potentially good primes trivial): 9.1, 9.2 Step 6.
- **Point 67** (Remark rem:ns needs $H^1_f(\mathbb{Q}, V^D) = 0$, unknown for 3Ns): 13.1, 13.2.
- **Open check Q1** (DFG04 at $\ell = 3$, weight 2): settled, Part 7 (Q1): covered.
- Also: points 5, 6, 10, 14 (Lemma lem:h0) through Chapter 18, its 10.3 (9.2 Step 4); points 26, 40 ($M_{f,\lambda}$ versus the dual of $T_\ell E$) in 5.4(iv), now confirmed by DFG04 (18); point 11 (geometric Frobenius is DFG04's convention) confirmed (C6); points 23, 65, 68 in 10.1; point 27 in 9.3.
- **B3** (Chapter 21's A4.4 and A4.6): Part 5 forward pointers tell Chapter 21 which items to cite (kernel not annihilator; dual of $T_\ell E$).
- **B17** (lem:crit proved in Chapter 15): 3.4 restates and cites; the module-theoretic reading is the only addition.
- **B19** (cite Chapter 14's 9.3, 9.5, 9.6, 10.3): 5.6, 5.7, Part 7.
- **B14** (bibliography): see 6.5.

22.6.5 6.5 References

In `refs.bib` and cited here: DFG04 (published numbering), DDT (Sections 1 to 5, Lemma 4.17 as the paper cites it), Hida81, HidaBook, Wiles95, TaylorWiles, Ribet83, Ribet84, ARS12, Tilouine97, RibetStein, Mazur77, FPR94, BlochKato, FontaineLaffaille, KimOta, PollackWeston11, RibetTakahashi97, LingOesterle, Ribet88, Zagier85, Flach92, PARI, ecdata.

Proposed new keys (none is in `refs.bib` or any `bib/chNN.bib` now; all chapter bib files except `bib/chA.bib` are empty, checked): * `DiamondTaylor94` (F. Diamond, R. Taylor, *Non-optimal levels of mod ℓ modular representations*, Invent. Math. 115 (1994), 435-462; the data as printed in DFG04's

reference [26]): planned by Chapter 15 for `bib/ch15.bib` and needed here (level raising at $p \equiv -1$, Ihara), so by decision B14 it goes in `refs.bib`. * `Diamond96` (F. Diamond, *On deformation rings and Hecke rings*, Ann. of Math. 144 (1996), 137-166; DFG04 [22]): `bib/ch20.bib` unless Chapter 15 also cites it (its 11.5 mentions "Diamond's refinement"; then `refs.bib`). * `Lenstra95` (H. W. Lenstra, *Complete intersections and Gorenstein rings*, in *Elliptic curves, modular forms and Fermat's last theorem* (Hong Kong, 1993), International Press, 1995, 99-109; pages to be checked): `bib/ch20.bib` (Chapter 3 mentions complete intersections only as a pointer here). * `DFG04long` (F. Diamond, M. Flach, L. Guo, *Adjoint motives of modular forms and the Tamagawa number conjecture*, arXiv:2512.02348, 2001, updated 2025): `bib/ch20.bib`, cited once in the Notes for the foundational details omitted from the published version. The data of DiamondTaylor94 and Diamond96 were read from DFG04's bibliography; Lenstra95's pages were recalled, not re-verified.

22.6.6 6.6 Verification log (computations and readings done for this outline)

Downloads to `/tmp/ch20`: the published DFG04 from Numdam (`dfg04.pdf`, 65 pages, text by `pdftotext -layout`) and arXiv:2512.02348v2 (109 pages). Scripts in the session scratchpad (`ch20/quant.py`, `ch20/twistc.py`, `ch20/ecross.py`), Python 3 with PARI/GP 2.17 through `cypari2`, Cremona's `ecdata` files in `/tmp/ch09/data` (`degphi`, `galrep`). * **Reading of DFG04** (Part 7): Sections 0.2, 0.3, 1.1.1, 1.1.2, 1.2.2, 1.2.3, 1.4.1, 1.4.2, 1.5.1 to 1.5.3, 1.6.1 to 1.6.3, 1.7.1 to 1.7.3, 1.8.1, 1.8.2, Proposition 1.12, Section 2.1, Lemmas 2.1, 2.4, 2.5, Theorem 2.7, Lemma 2.12, Section 2.4 with (55) to (58), Theorem 2.15 (beginning of the proof), Section 3.1 (Theorem 3.1 and its set-up), Section 3.2 (Theorems 3.6, 3.7). * **Theorem thm:quant, equality** (`quant.py`): for every optimal curve of conductor below 10,000 (from `degphi`), not CM, with at least one odd prime of type I_0^* or I_n^* (Kodaira code -1 or ≤ -5 from `elllocalred`), S the set of these primes ($q = 3$ included), $E_S =$ `elltwist` by $\prod q^*$, minimal model, its class located in `degphi` by conductor and a_p for $p < 80$ (always found); for $\ell \in \{3, 5, 7, 11, 13\}$, $\ell \nmid N$, $E[\ell]$ irreducible (no `galrep` label with B or Cs at ℓ): $e_\ell(E, q) = v_\ell(q^2 - 1)$ (I_n^*) or $v_\ell(q-1) + v_\ell((q+1)^2 - a_q(E')^2)$ (I_0^*). Result: 25,477 pairs (15,607 with $3 \in S$, 9,862 with S of primes $q \ge 5$, 8 with image 3Ns at $\ell = 3$), **no violation of Theorem thm:quant and equality** $v_\ell(\deg \varphi_E) = v_\ell(\deg \varphi_{E_S}) + \sum_{q \in S} e_\ell(E, q)$ **in every one**. * **Euler terms against PARI** (`ecross.py`): $P_q(q^{-2})$ from `lfunsympow(E,2)` and `lfuneuler` against the twist formulas $\frac{(q-1)((q+1)^2 - a_q(E')^2)}{q^3}$ (I_0^*) and $1 - q^{-2}$ (I_n^*): 400 random (curve, odd prime) pairs below conductor 10,000, $q = 3$ included, no mismatch. * **Theorem thm:twist, case (c)** (`twistc.py`): S the set of primes $q \ge 5$ of types $\mathrm{II}, \mathrm{III}, \mathrm{IV}$ or duals; 14,876 pairs (42 of them 3Ns at $\ell = 3$): $v_\ell(\deg \varphi_E) = v_\ell(\deg \varphi_{E_S})$ in every one. * **Level-raising identity** (6.3, Exercise 6): $(1 - \frac{\alpha}{\beta q})(1 - \frac{1}{q})(1 - \frac{\beta}{\alpha q})q^3 = (q-1)((q+1)^2 - (\alpha + \beta)^2)$ with $\beta = q/\alpha$, checked symbolically in PARI. * By hand: the proof of Lemma lem:ddt417 on the three toy modules (4.4); the factor $c \in \{1, \frac{1}{2}\}$ in 5.3(ii) (-1 acts trivially on the upper half plane); the inflation-restriction description of the comparison in 5.2(ii); the chain in 12.6; the values in Exercises 7 and 10. * Not checked: DDT's and Hida81's theorem numbers; KimOta's definition of $\eta_f(N)$ (10.2(i)); the general Eisenstein claim of 5.2(ii) beyond the Shimura covering (sketched; Part 7, item 1).

22.7 7. DFG04 checklist

Source read: F. Diamond, M. Flach, L. Guo, *The Tamagawa number conjecture of adjoint motives of modular forms*, Ann. Sci. Ecole Norm. Sup. (4) 37 (2004), no. 5, 663-727 (Numdam, item ASENS_2004_4_37_5_663_0; DOI 10.1016/j.ansens.2004.09.001), the version the paper cites. A longer version is now on arXiv (arXiv:2512.02348, dated "August 2001, with minor update December 2025"), with **different** numbering; nothing below refers to it. Quotations are from the published text (mathematical symbols transcribed; page numbers of the journal).

22.7.1 7.1 The five items of important_notes.md point 56

Item 1. Which curve: is DFG04's integral structure $H^1(X_0(N),\mathbb{Z}_\ell)$? Confirmed in the form the paper needs (non-Eisenstein components); not literally for the whole lattice. What DFG04 say. Section 1.2.2 (p. 674): $M_B = H^1(X^{an},\mathcal{F}_B^k)$, $M_{c,B} = H^1_c(X^{an},\mathcal{F}_B^k)$ for the full level-N' curve $X = X_{N'}$ (open, adelic, with $\varphi(N')$ geometric components); 1.2.3: the ℓ-adic realisations are $H^1(X_{\bar{\mathbb{Q}}},\mathcal{F}_\ell^k)$ and H^1_c, "$\mathbb{Z}_\ell \otimes M_B \simeq M_\ell$". Section 1.4.1 (p. 678-679): for a representation σ of an open compact U, "$M(\sigma) = (M\otimes V)^U$ … $M(\sigma)_! = \mathrm{im}(M(\sigma)_c \to M(\sigma)_{\mathrm{tf}})$. We remark that $M(\sigma)_!$ may lie properly in $(M_!\otimes V)^U$ … $M(\sigma)_c$ and $M(\sigma)_!$ are independent of N." Section 1.4.2 (p. 679): $U = U_0(N)$, $\sigma = \sigma(N,\psi)$ the character $\left(\begin{smallmatrix} a & b \\ c & d\end{smallmatrix}\right) \mapsto \psi^{-1}(a_N)$, "We let $M(N,\psi)_\bullet$ denote the premotivic structure $M(\sigma)_\bullet$". So for $k=2$, $\psi=1$: $M(N,1)_{!,\lambda}$ is the image of the $U_0(N)$-invariants of H^1_c of the full-level open curve in the torsion-free quotient of the invariants of H^1. It is **not** defined as $H^1(X_0(N),\mathbb{Z}_\ell)$, and it is not a $\Gamma_1(N)$-type trivial-character part either. Comparison (this outline, 5.2(ii)). Pullback along $Y_{N'} \to Y_0(N)$ maps $H^1(X_0(N),\mathbb{Z}_\ell)$ (interior cohomology of $Y_0(N)$, item 2) into $M(N,1)_{!,\lambda}$ with finite cokernel; by inflation-restriction the cokernel is a subquotient of $\mathrm{Hom}(U_0(N)/U',\mathbb{Q}_\ell/\mathbb{Z}_\ell)$ for a sufficiently small U', i.e. it comes from cyclic unramified subcovers of congruence covers of $X_0(N)$ (generalised Shimura coverings), which are Eisenstein; it vanishes when ℓ does not divide the order of the covering group (transfer). DFG04 themselves arrange this in Section 1.8.1 with an auxiliary prime $r>3$, $L_r(A_f,1)^{-1}$ a unit, and $U' = U^\Sigma \cap U_1(r)$ of index $r-1$ prime to ℓ ("Lemma 1.2 with $U' = U^\Sigma\cap U_1(r)$ yields … Corollary 1.6"), and their Lemma 1.8 and Corollary 1.9 use the same "kernel and cokernel are Eisenstein" device after localising at $\mathfrak{m}$. Since $M_{f,\lambda} = M(N,1)_{!,\lambda}[I_f]$ lies in the $\mathfrak{m}_{f,\lambda}$-component, and $\mathfrak{m}_{f,\lambda}$ is non-Eisenstein when $\bar\rho_f$ is irreducible, $M_{f,\lambda}^{\mathrm{DFG}} = H^1(X_0(N),\mathbb{Z}_\ell)[I_f]$, which is what Lemma lem:hecke and point 39 need. Verdict: confirmed for the use made of it; the paper's sentence "The integral structure $M(M^\Sigma,1)_{!,\lambda}$ is $H^1(X_0(M^\Sigma),\mathbb{Z}_\ell)$" should read "agrees with $H^1(X_0(M^\Sigma),\mathbb{Z}_\ell)$ after localisation at the non-Eisenstein maximal ideals". Confidence: high for the definitions; medium-high for the Eisenstein claim in general (certain for the Shimura covering $X_1(N)\to X_0(N)$; Ling-Oesterle, Ribet88), which is the only part not read off DFG04.

Item 2. Interior cohomology. Confirmed. DFG04 Section 1.4.1 define $M(\sigma)_!$ as the image of compactly supported in ordinary cohomology (quoted above). For constant coefficients on the open curve $Y_0(N)$ this image is $H^1(X_0(N))$ (the exact-sequence argument of Chapter 14, its 9.6(2)), and group cohomology of $\Gamma_0(N)$ with coefficients $\mathbb{Z}_\ell$ agrees with $H^1(Y_0(N),\mathbb{Z}_\ell)$ since elliptic elements are torsion. Confidence: high.

Item 3. The operator $[UwU]_\omega$ and the normalisation of the pairing. Confirmed, up to a unit at odd ℓ. DFG04 Section 1.5.3 (p. 682): "Let w denote $\left(\begin{smallmatrix} 0 & -1 \\ N & 0\end{smallmatrix}\right)$ in $\prod_{p|N}\mathrm{GL}_2(\mathbb{Z}_p)$. The operator $[UwU]_\omega$ defines an isomorphism (13) $M_\bullet(N,\psi)\to M_\bullet(\sigma')$ … One finds that $[UwU]_\omega^{-1} = [Uw^{-1}U]_{\omega^t\otimes\psi(\det w)}$ and $[UwU]_\omega$ is adjoint to $N^{k-2}[Uw^{-1}U]_{\omega^t\otimes\psi(\det w),c}$ and coincides with $\psi(-1_\infty)N^{k-2}[Uw^{-1}U]_\omega$. Composing the operator $[UwU]_\omega$ with the duality morphism $\bar\delta$, we obtain a duality isomorphism (14) $\hat\delta\colon M(N,\psi)\to \mathrm{Hom}_K(M(N,\psi)_c, M_\psi(1-k))$. Similar assertions hold for $M(N,\psi)_!$ yielding an isomorphism $\hat\delta_!$. Since $M(N,\psi)=0$ unless $\psi(-1_\infty)=(-1)^{k-2}$, we find that the corresponding perfect pairing is alternating." For $k=2$, $\psi=1$: $N^{k-2}=1$, $[UwU]_\omega$ is an involution (it equals $[Uw^{-1}U]_\omega$, its own inverse), the Atkin-Lehner involution, **with no scalar**. The pairing $\bar\delta$ is defined in Section 1.5.2, (11), (12): the cup product into H^2 of the full-level curve, restricted to invariants, "normalizing by dividing by $[U:U_N]$, we get an isomorphism … independent of N". On classes pulled back from $X_0(N)$ this is $\deg(Y_{N'}\to Y_0(N))/[U:U_{N'}]$ times the cup product on $X_0(N)$, and the degree is $[U:U_{N'}]$ or $[U:U_{N'}]/2$ according as -1 acts trivially (it does: $-1\in\mathrm{GL}_2(\mathbb{Q})$ acts trivially on $\mathfrak{H}^\pm$); so the factor is $\frac12$, a unit for odd ℓ (point 56 anticipated "a factor 2 … harmless for odd ℓ"). Integrality and perfectness: Section 1.5.2, "Poincare duality on X_U then shows that the isomorphism $\bar\delta$ arises from an injective morphism … whose cokernel C satisfies $C_\ell = 0$ for $\ell\nmid N(k-2)!$" (U sufficiently small), Lemma 1.2, Corollary 1.6. **Elliptic points of order 3 at $\ell=3$**: DFG04 avoid them by working at a sufficiently small level of index prime to ℓ (the auxiliary prime r, item 1); on the non-Eisenstein components the result is the cup product on $X_0(N)$ (a smooth curve, unimodular Poincare duality). Confidence: high.

Item 4. Kernel lattice, not projection lattice. Confirmed. DFG04 Section 1.6.2 (p. 683): "Let

$I_f = \ker \theta_f$ and $\mathcal{M}_f = M(N,\psi)_![I_f]$ in the notation of Section 1.1.2"; Section 1.1.2 (p. 671): "If I is an $\mathcal{O}_K$-submodule of $\mathrm{End}\,\mathcal{M}$, then we define an object $\mathcal{M}[I]$ ... as the kernel of $(x_1,\dots,x_r)\colon \mathcal{M} \to \mathcal{M}^r$ where $x_1,\dots,x_r$ generate I." And Section 1.7.3: "The construction of Section 1.6.2 thus yields a premotivic structure $\mathcal{M}_{f^\Sigma}$ ... contained in $M(N^\Sigma,\psi)_!$." The Hecke algebra $\mathbb{T}$ of Section 1.6.1 is generated by the T_p for **all** primes p (the double coset of $\left(\begin{smallmatrix} p & 0 \\ 0 & 1 \end{smallmatrix}\right)$ at $p \mid N$ is U_p), so I_f is the kernel on the full Hecke algebra, as in the paper. Confidence: high.

Item 5. Ideal of values versus Gram determinant. Confirmed (ideal of values). DFG04 Section 1.7.3 (p. 687): "The pairing $\hat{\delta}_!$ defined in Section 1.5.3 restricts to an alternating pairing on $\mathcal{M}_{f^\Sigma}$, which Proposition 1.4 below shows is non-degenerate, hence induces an isomorphism $\wedge^2_K M_{f^\Sigma} \to M_\psi(1-k)$. The image of $\wedge^2_{\mathcal{O}_K}\mathcal{M}_{f^\Sigma}$ therefore defines an integral structure for $M_\psi(1-k)$, necessarily of the form $\eta_f^\Sigma \mathcal{M}_\psi(1-k)$ for some fractional ideal $\eta_f^\Sigma \subset K$. We call η_f^Σ the (naive, Σ-finite) congruence $\mathcal{O}_K$-ideal of f." The image of $\wedge^2$ of a rank-two lattice under an alternating pairing is the ideal of values; its square is the Gram determinant. Theorem 3.7 and Proposition 1.4(c) are stated for this η_f^Σ ("where η_f^Σ was defined before Proposition 1.4"). Confidence: high.

Consequence. All five items hold in the form the paper uses. Hence **important_notes.md point 39 applies to DFG04's own ideal**: $v_\lambda(\eta^\emptyset_{f,\lambda}) = v_\ell(\deg \varphi_E)$ for the optimal curve and every $\ell \nmid 2N$ with $\bar{\rho}$ irreducible; Lemma lem:hecke is an equality, (eq:identity) holds with equality for $\ell \notin S_f$, and Remark rem:free is not needed for any statement of Theorem thm:main.

22.7.2 7.2 Open check Q1: does DFG04 Theorem 3.7 cover $\ell = 3$ in weight 2? Yes (confirmed).

- (31) (p. 695): $S_f = \{\lambda : \lambda \mid Nk!$, or $\bar{M}_{f,\lambda}$ is not absolutely irreducible on G_F, $F = \mathbb{Q}(\sqrt{(-1)^{(\ell-1)/2}\ell})\}$. For $k=2$, $k!=2$: the only conditions on ℓ are $\ell \nmid 2N$ and the irreducibility. The abstract says the same: "λ is any prime of K not dividing $Nk!$".

- Fontaine-Laffaille: a crystalline representation is *short* (Section 1.1.2, p. 670) if $\mathrm{Fil}^0 D = D$, $\mathrm{Fil}^{\ell-1} D = 0$ and no nonzero quotient V' has $V' \otimes \mathbb{Q}_\ell(\ell-1)$ unramified; for $M_{f,\lambda}$ (Hodge-Tate weights $0,1$) at $\ell = 3$ this holds. The adjoint is **not** short at $\ell = 3$ (Section 1.7.1: one can twist A_f into $\mathcal{MF}^0$ only "if $\ell - 1 > 2(k-1)$"), but DFG04 never need that: in Section 2.1 they define $H^1_f(\mathbb{Q}_\ell, W_n)$ for $W = \mathrm{Hom}(L_1, L_2) \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$ with L_1, L_2 short, as the extensions (32) in the essential image of the Fontaine-Laffaille functor, and Proposition 2.2 and Corollary 2.3 (which needs only $\ell > 2$) compare it with Bloch-Kato's condition.

- Taylor-Wiles: Section 3.1 requires ρ_0 odd, absolutely irreducible on G_F, twist-minimal, with Serre weight $2 \le k \le \ell - 1$ (for $k=2$, $\ell=3$: $2 \le 2$); Section 1.8 requires, "if $3 \in \lambda$, ... its restriction to $G_{\mathbb{Q}(\mu_3)}$ to be absolutely irreducible", which for $\ell = 3$ is the same as the S_f condition ($\mathbb{Q}(\sqrt{-3}) = \mathbb{Q}(\mu_3)$). Verdict: Theorem thm:main at $\ell = 3$ (with $3 \nmid N$, image not 3Ns) has its Taylor-Wiles input. Confidence: high (the conditions were read in all three places where ℓ enters).

22.7.3 7.3 The theorem numbers and references the paper cites from DFG04

paper's citation	what DFG04 contain there (published numbering)	verdict
Theorem 0.1 (Section 1.2, after thm:main)	"Theorem 0.1 (= Theorem 2.15). Let f be a newform of weight $k \ge 2$ and level N with coefficients in K. If λ is not in S_f, then the λ-part of the Bloch-Kato conjecture holds for A_f and $A_f(1)$."	confirmed

paper's citation	what DFG04 contain there (published numbering)	verdict
Theorem 0.2 ("the form given as Theorem 0.2 there")	"Theorem 0.2 (= Theorem 3.7) ... Suppose Σ is a finite set of rational primes containing those dividing N. Suppose that λ is a prime of K which is not in S_f and does not divide any prime in Σ. Then $H^1_\Sigma(G_\mathbb{Q}, A_{f,\lambda}/\mathcal{A}_{f,\lambda})$ has length $v_\lambda(\eta_f^\Sigma)$."	confirmed
Theorem 3.7 (thm:dfg37)	stated for Σ "a finite set of primes not containing ℓ such that $M_{f,\lambda}$ is minimally ramified outside Σ"; η_f^Σ "defined before Proposition 1.4"; proof: for a twist of minimal conductor, then for f	confirmed (statement as the paper quotes it)
Theorem 2.7 (thm:dfg27)	"Conjecture 2.6 holds for both $M = A_f$ and $M = B_f$ if λ is not in S_f. More precisely ... $H^0(\mathbb{Q}, A_{f,\lambda}) \simeq H^1_f(\mathbb{Q}, A_{f,\lambda}) \simeq H^1_f(\mathbb{Q}, B_{f,\lambda}) \simeq H^0(\mathbb{Q}, B_{f,\lambda}) \simeq 0$", $B_f = A_f(1)$	confirmed ($V^D = B_{f,\lambda}$)
Proposition 1.4(c) (thm:dfg14)	"If $\bar{M}_{f,\lambda}$ is an irreducible $(\mathcal{O}_K/\lambda)[G_\mathbb{Q}]$-module, then γ induces an isomorphism $\mathcal{M}_{f,\lambda} \to \mathcal{M}^\Sigma_{f,\lambda}$ ... and $\eta^\Sigma_{f,\lambda} = \eta^\emptyset_{f,\lambda} \prod_{p\in\Sigma} L_p^{\mathrm{nv}}(A_f, 1)^{-1}$"; standing hypothesis of Section 1.7.3: $\lambda \notin S_N^K$ (for $k = 2$: $\ell \nmid 2N$)	confirmed
Lemma 2.1 (lem:dfg21)	the four-term sequence, under "Σ is nonempty and contains all primes where $H^1_f(\mathbb{Q}_p, W) \neq H^1_{\mathrm{ur}}(\mathbb{Q}_p, W)$, and ... $H^0(\mathbb{Q}, V^D) = H^1_f(\mathbb{Q}, V^D) = 0$"; sum over $p \in \Sigma \cup \{\infty\}$; proof via Flach's Proposition 1.4 ([32])	confirmed; the paper's hypothesis ("every $p \neq \ell$ at which W is ramified") implies DFG04's
Theorem 2.15 (proof) and FPR94 I.4.2.2 (lem:fpr)	"By [41, Proof of I.4.2.2] we have for $p \in \Sigma$, $\mathrm{Fitt}_{\mathcal{O}_\lambda} H^1_f(\mathbb{Q}_p, M_\lambda) = L_p(M, 0)^{-1} \mathrm{Tam}_p^0(M_\lambda)$ since $L_p(M, 0)^{-1} \neq 0$" (p. 709), applied with $M = A_f(1)$	confirmed ([41] is Fontaine-Perrin-Riou, `FPR94`)
p. 708 (definition of Tam_p)	"The argument of [41, I.4.2.2] shows that if $p \neq \ell$, then $\mathrm{Tam}_p^0(M_\lambda) = \mathrm{Fitt}_{\mathcal{O}_\lambda}\big(H^1(I_p, M_\lambda)_{\mathrm{tor}}\big)^{G_{\mathbb{Q}_p}}$" (the factors defined "as in I.4.1 (and II.5.3.3) of [41]")	confirmed; the same misplaced superscript as the paper (point 37: the invariants are of the torsion module)

paper's citation	what DFG04 contain there (published numbering)	verdict
(57)	"$\mathrm{Tam}_p^0(M_\lambda) = \mathrm{Tam}_p^0(M_\lambda^D)$" for $p \neq \ell$, deduced from the formula above	confirmed (general, not only for the adjoint)
(31)	as in 7.2	confirmed
Lemma 2.12 (Section 1, Petersson norm)	"$\langle \tau(f), \tau(f) \rangle_{\Gamma_1(N)} = \frac{(k-1)!\delta(N)N\varphi(N)L^{\mathrm{nv}}(A_f,\tau,1)}{4^k\pi^{k+1}}$", from Hida81 Theorem 5.1	confirmed: the **naive** value, as Chapter 19, its 8.3, states
Section 1.7.2 (conventions: δ_p, L_p, L_p^{nv})	$\delta_p = \dim M_{f,\lambda}^{I_p}$ ($= 2, 1, 0$ for $p \nmid N$; $p \mid N$, $a_p \neq 0$; $p \mid N$, $a_p = 0$); $L_p^{\mathrm{nv}} = L_p$ if $\delta_p > 0$, 1 if $\delta_p = 0$; at $\delta_p = 1$, $L_p(A_f, s) = (1 - p^{-1-s})^{-1}$ if $\pi_p(f)$ is special; Frob_p is the **geometric** Frobenius (Section 1.1.1)	confirmed; also confirms point 11
Section 2.1 (Bloch-Kato and H^1_Σ)	as in 7.2 here; "H^1_Σ ... without local conditions at $p \in \Sigma \cup \{\infty\}$ and slightly relaxed local conditions at primes where $H^1_{\mathrm{ur}}(\mathbb{Q}_p, W)$ is not divisible"	confirmed
Sections 1.5.3, 1.6.2, 1.7.3; (13), (14)	as in items 1 to 5	confirmed (item 1 in the form needed)
Sections 1.6.1, 1.8.2 (self-adjointness)	1.6.1, Proposition 1.3: the Hecke action is compatible with "the duality morphisms $\hat{\delta}$ and $\hat{\delta}_!$ of (14)"; 1.5.2: $[U'gU]^T$ is the adjoint of $[U'gU]$; 1.8.2: "the Hecke operators are self-adjoint with respect to the pairing of (25)"	confirmed

Also confirmed: (18) the geometric Frobenius on $M_{f,\lambda}$ has characteristic polynomial $X^2 - \psi(p)^{-1}a_pX + \psi(p)^{-1}p^{k-1}$, that of H^1, so $M_{f,\lambda}$ is the dual of the Tate module (points 26, 40); (22) $A_{f\otimes\psi'} \simeq A_f$; DFG04's proof of Theorem 2.15 takes $\Sigma = \{p \mid N\}$ and runs the same steps (Proposition 1.4, Lemma 2.1, the FPR formula, Theorem 3.7) as the paper's proof of Theorem thm:main.

22.8 8. Points for important_notes.md

New points in this chapter's area (Section 4 of the paper, DFG04, congruence ideals), not among points 1 to 72, Q1 and B1 to B23. Points 16, 18, 19, 24, 35, 36, 38, 39, 47, 56, 59, 60, 61, 63, 64, 67, Q1, B3, B17, B19 are handled in Part 6.4 and not repeated, except where N1 and N2 settle or supersede them.

N1. Point 56 and open check Q1 settled: DFG04 confirms the paper's reading, with one rewording (DFG04 read). Location: Section 4.1 ("The integral structure $M(M^\Sigma, 1)_{!,\lambda}$ is $H^1(X_0(M^\Sigma), \mathbb{Z}_\ell)$, with its Hecke action; ... the pairing $\hat{\delta}_!$ is the cup product composed with the operator $[UwU]\omega$ ... which for weight 2 and trivial character is the Atkin-Lehner involution"); Lemma lem:hecke; Remark rem:free; Theorem thm:main ("with equality when ... free"); Section 4.2 (S_f). Content: the published DFG04 was read (Part 7). Items 2 to 5 of point 56 hold verbatim: $M(\sigma)_!$ is the image of compactly

supported cohomology (interior cohomology); $[UwU]_\omega$ is the Atkin-Lehner involution with no scalar for $k = 2$ ($N^{k-2} = 1$), the duality being the cup product divided by $[U : U_{N'}]$, i.e. the cup product on $X_0(N)$ up to the unit $\frac{1}{2}$; $\mathcal{M}_f = M(N, \psi)_![I_f]$ is the kernel lattice for the full Hecke algebra; η_f^Σ is defined by the image of $\wedge^2 \mathcal{M}_{f^\Sigma}$, i.e. the ideal of values. Item 1 holds in the form needed: DFG04's integral structure is not defined as $H^1(X_0(N), \mathbb{Z}_\ell)$ but as the image of the $U_0(N)$-invariants of the compactly supported cohomology of the full-level curve; it agrees with $H^1(X_0(N), \mathbb{Z}_\ell)$ on non-Eisenstein components (the discrepancy, if any, comes from abelian unramified subcovers of congruence covers and is Eisenstein; DFG04 avoid it with an auxiliary prime r, Section 1.8.1), and the paper only uses $\mathfrak{m}_{f,\lambda}$ with $\bar\rho$ irreducible. Q1: S_f in (31) excludes only $\lambda \mid Nk!$ ($= 2N$ for $k = 2$) and residually induced-from-$\mathbb{Q}(\sqrt{\ell^*})$ forms; Fontaine-Laffaille is applied to M_f (weights $0, 1$, short at $\ell = 3$), and the adjoint's local condition is defined through extensions of short objects ((32), Proposition 2.2, Corollary 2.3 for $\ell > 2$); the Taylor-Wiles set-up (Section 3.1) requires Serre weight $2 \le k \le \ell - 1$; so $\ell = 3$ is covered in weight 2. The paper's theorem numbers from DFG04 (Theorems 0.1, 0.2, 2.7, 2.15, 3.7; Proposition 1.4(c); Lemmas 2.1, 2.12; (31), (57); p. 708; Sections 1.5.3, 1.6.1, 1.6.2, 1.7.2, 1.7.3, 1.8.2, 2.1; (13), (14)) are all correct for the published version. Consequences: point 39 holds for DFG04's η: Lemma lem:hecke is an equality, (eq:identity) is an equality for $\ell \notin S_f$, the phrase "with equality when $H^1(X_0(N), \mathbb{Z}_\ell)_\mathfrak{m}$ is free" and the dependence of the equality on Remark rem:free can be removed; Theorem thm:main at $\ell = 3$ stands. Fix: (a) in Section 4.1 write "The integral structure $M(M^\Sigma, 1)_{!,\lambda}$ agrees, after localisation at any non-Eisenstein maximal ideal of the Hecke algebra, with $H^1(X_0(M^\Sigma), \mathbb{Z}_\ell)$" and "the pairing ... is, up to the unit $\frac{1}{2}$, the cup product composed with w_{M^Σ}"; (b) state Lemma lem:hecke as an equality with the proof of point 39; (c) a footnote that the arXiv version of DFG04 (2025) has different numbering. Confidence: high for the readings (quoted in Part 7); medium-high for the Eisenstein nature of the comparison cokernel beyond the Shimura covering (sketched, not read off DFG04).

N2. Remark (2) after Theorem thm:quant: the first inequality is an equality whenever $\ell \notin S_f$, with no multiplicity-one hypothesis; the data show equality in every tested case (supersedes the second bullet of point 56 and Chapter 14, its 9.5(c), 10.4(b)). Location: the remark after the proof of Theorem thm:quant, (2): "so the first inequality of Theorem thm:quant is an equality when $H^1(X_0(N), \mathbb{Z}_\ell)_\mathfrak{m}$ is free, that is, under multiplicity one at level N"; Section 1.5 ("the identity (eq:identity) for E and for E_S shows that the first inequality is then an equality when multiplicity one holds at both levels"). Problem: the hypothesis is not needed. With point 39 confirmed for DFG04's ideal (N1), (eq:identity) holds with equality for E and for the optimal curve of the class of E_S whenever $\ell \notin S_f$ (then also $\ell \notin S_{f_{E_S}}$: $N_S \mid N$, and twisting does not change absolute irreducibility over $\mathbb{Q}(\sqrt{\ell^*})$). The Selmer groups and Tamagawa factors of E and E_S coincide ($\mathrm{ad}^0 T$ is twist-invariant), and the Euler sums differ by exactly $\sum_{q\in S} e_\ell(E, q)$, so $v_\ell(\deg \varphi_E) = v_\ell(\deg \varphi_{E_S}) + \sum_{q\in S} e_\ell(E, q)$; with ARS at level N_S ($\ell \nmid N_S$) and the chain of Steps 1 to 3, every inequality in the proof of Theorem thm:quant is an equality (in particular $v(\eta^{\mathrm{Hecke}}(g^S)) = v(\eta^S_{g,\lambda})$ at level N, a freeness-type statement for the depleted form). Data: for all optimal non-CM curves of conductor below 10,000 with at least one odd prime of type I_0^* or I_n^* ($q = 3$ included), $\ell \in \{3, 5, 7, 11, 13\}$, $\ell \nmid N$, $E[\ell]$ irreducible: 25,477 pairs, equality $v_\ell(\deg \varphi_E) = v_\ell(\deg \varphi_{E_S}) + \sum_{q\in S} e_\ell(E, q)$ in **every** one (including 8 pairs at $\ell = 3$ with image 3Ns, outside the argument). Examples: 539c1 ($2 = 1 + 1$), 980i1 ($2 = 1 + 1$), 121d1 ($1 = 0 + 1$), 539d1 ($2 = 0 + 2$). Fix: in Remark (2) and Section 1.5, replace "when ... free, that is, under multiplicity one at level N" by "whenever $\ell \notin S_f$ (by (eq:identity) for E and E_S, each an equality by Lemma lem:hecke in its strengthened form)", and add the data line; the data paragraph after Theorem thm:twist can then say that the difference in cases (a) and (b) is **exactly** the Euler term (point 16). Confidence: high (the argument uses only statements proved in the book, given N1; the data are a complete count from `ecdata`).

N3. Lemma lem:dfg21: the paper's hypothesis is a sufficient form of DFG04's (confirmation, expository). Location: Lemma lem:dfg21 ("let Σ be a nonempty finite set of primes containing every prime $p \neq \ell$ at which W is ramified"). Content: DFG04 Lemma 2.1 asks that Σ be nonempty and contain "all primes where $H^1_f(\mathbb{Q}_p, W) \neq H^1_{\mathrm{ur}}(\mathbb{Q}_p, W)$" (with $\ell \notin \Sigma$ from Section 2.1); at an unramified $p \neq \ell$ the two groups agree, so the paper's condition implies DFG04's. Similarly the paper's description of H^1_Σ ("unramified elsewhere") is DFG04's (they use H^1_{ur}, not H^1_f, outside $\Sigma \cup \{\ell\}$). No change needed; optionally say "(a sufficient form of the hypothesis of DFG04)". Confidence: high.

N4. Section 1 and Lemma lem:tamfactor: "the conventions of DFG04 Section 1.7.2" include the geometric Frobenius (confirmation of point 11). DFG04 Section 1.1.1 (p. 668): "Frob_p for the geometric Frobenius element in G_p/I_p", and Section 1.7.2 defines $\delta_p = \dim M_{f,\lambda}^{I_p}$ with $M_{f,\lambda}$ the H^1-lattice; so the paper's $L_p(A_f, s)$ "following DFG04 Section 1.7.2" is with the geometric Frobenius, as point 11 says, and the paper's $\delta_p = \dim V^{I_p}$ ($V = V_\ell E$, the dual) is the same number. Point 11's fix for Lemma lem:tamfactor (say "arithmetic") stands. Confidence: high.

None other found. Checked and found correct (beyond the points above): Lemma lem:ddt417's statement and its use with $d = 2$ (modulo points 19, 38); Theorem thm:dfg14's three cases and the factor $1 - p^{-2}$ (DFG04 (23) at $s = 1$); Theorem thm:dfg37 and the minimal-ramification remark; Theorem thm:dfg27 (as DFG04's "more precisely"); the assembly of Section 4.3 (it is DFG04's own proof of Theorem 2.15 with the modular degree in place of the period); Theorem thm:twist (proof, and case (c) equality on 14,876 pairs); Theorem thm:quant (proof, no violation on 25,477 pairs); the claim $\iota(g^S) = f$; Step 3's identification of the naive factor of g with the omitted factor of f (400 PARI checks, $q = 3$ included).

Book-internal inconsistencies found while preparing this outline:

- **B-a. Point 56, second bullet; Chapter 14 outline, 9.5(c) and 10.4(b); Chapter 15 outline, 10.4 remark (2); Chapter 21 outline, A4.16(2).** All say the equality in Theorem thm:quant (Remark (2)) still needs multiplicity one at level N. Superseded by N2 (item 12.6 here): it holds whenever $\ell \notin S_f$. The Chapter 14 writer should change 9.5(c) and 10.4(b) to cite `\Cref{ch:congruence}`.
- **B-b. "Not checked against DFG04" statuses.** Chapter 14's 9.6 ("medium for the identification"), Chapter 18's Q1 and 10.1 ("see Q1"), Chapter 19's 8.3 ("DFG04 Lemma 2.12 ... content not re-checked") and 8.5 ("the period normalisation of DFG04 is not checked"), and Chapter 21's B3 table ("as cited") can now cite Part 7 here: all DFG04 citations are verified (the period normalisation of DFG04 Theorem 2.10 was **not** checked here either).
- **B-c. Chapter 13 outline, 7.4 reference line**: "DFG04 Sections 1.5.3, 1.7.3 (as cited by the paper, for the notation M^Σ, f^Σ)": N^Σ and f^Σ are in DFG04 Section 1.7.3 only; 1.5.3 is the duality. Minor.
- **B-d. Chapter 21 outline, A4.1** says M_{f^Σ} for $\Sigma = \emptyset$ is "isomorphic to $\mathrm{Hom}(T_\ell E, \mathbb{Z}_\ell) = T_\ell E(-1)$ up to homothety", A4.6 still says "homothetic to $T_\ell E$" (already B3), and A4.5 routes the equality through Remark rem:free (already B19). Not new; Chapter 21 should cite 5.4(iv), 5.6, 5.7 here.
- **B-e. Chapter 18 outline, 14.5**: "DFG04 Theorem 0.2 = Theorem 3.7 in the paper's reading": DFG04 print "Theorem 0.2 (= Theorem 3.7)" themselves, with the hypothesis "Σ containing those dividing N" in 0.2 and "minimally ramified outside Σ" in 3.7. Both readings are now quoted in 7.4, 7.5 here.
- **B-f. PLAN.md, Chapter 20 entry.** "DDT Theorem 5.3 / Lemma 4.17" for the numerical criterion: Theorem 5.3 is not verified (cite DDT Section 5.1); "the Gram determinant of the pairing on $M_{f^\Sigma,\lambda}$ is $(\eta^\Sigma)^2$" is right, but DFG04 define η^Σ via $\wedge^2$ (ideal of values), not via the Gram determinant; "Remark rem:free on when freeness holds" no longer bears on Theorem thm:main (N1).
- **B-g. Bibliography (B14).** `DiamondTaylor94` is planned by Chapter 15 for `bib/ch15.bib` and is needed here: it belongs in `refs.bib`. `Diamond96`, `Lenstra95`, `DFG04long` go in `bib/ch20.bib` unless another chapter cites them.
- Consistency checks that passed: Chapter 3's 11.1 to 11.6 (the ingredients and toy checks of Lemma lem:ddt417), Chapter 13's 7.2, 7.4, 8.2, 8.3, 9.1, 9.3, 11.3 (Step 1 of thm:quant, the level-539 congruence modules), Chapter 14's 9.1 to 9.7 (now unconditional), Chapter 15's 7.2, 10.1 to 10.4, Chapter 17's 10.3, 10.4, Chapter 18's 10.1 to 11.1 and 14.4 (the Selmer half and the DFG04 statements, now verified), Chapter 19's 8.3, 9.5 to 9.7 (Lemma 2.12 verified; the naive factors) agree with this line-up and with Part 7.

Chapter 23

Chapter 21. Reading the paper: a guided tour (label ch:paper, file chapters/ch21_paper.tex)

Stage 1 line-up, third revision (14 October 2026). Target length: at most 2,800 lines of LaTeX (PLAN.md 7.3 allows this for Chapter 21), card excluded; the section plan in Section 4 totals 2,795.

The chapter is a commentary on the paper in the paper's order. Every item below is either (A) a statement of the paper restated precisely in the book's shared notation, (B) a convention the reader must fix to read the paper, or (C) a device of the tour (dependency diagram, hypothesis table, list of imported statements, worked verifications). Rigour tags follow PLAN.md 3: the paper's own lemmas are *proved here* with every compressed step expanded; the imported theorems are *stated; reference* with the exact numbering; the long arguments reproduced from the paper (the assembly of Theorem thm:main, Steps 1 to 4 of Theorem thm:quant) are *proved here* because every ingredient is a card entry of an earlier chapter. Where an earlier chapter proves a paper lemma in full, the tour still walks through the paper's proof but cites that chapter's item for routine verifications (writer note W3).

Frobenius convention (fixed in item C2, binding Appendix B conventions B-C2 and B-C3). The book's `Frob_p` is ALWAYS the arithmetic Frobenius: it acts on `mu_{l^k}` by `zeta -> zeta^p`, so `chi_l(Frob_p) = p`. The geometric Frobenius is `Phi_p = Frob_p^{-1}`; because this chapter also uses the component groups `Phig_q` of the paper (B-C11), it writes the geometric Frobenius at a named prime as `Frob_q^{-1}` (B-C2(b)) and never as `Phi_q`. (a) Group theory and cohomology use `Frob`: the relation `Frob^{-1} t Frob = t^{q^{-1}}`, the conjugation action of `G/I` on `H^1(I, M)`, unramified cohomology `H^1_ur = M^I/(Frob - 1)M^I`, the Tate matrices of Lemma lem:tamfactor. (b) L-factors use the geometric Frobenius: `L_p(M, s) = det(1 - Frob_p^{-1} p^{-s} | M^{I_p})^{-1}` for `p != l`, the convention of DFG04 Section 1.7.2 (verified against the published text: Chapter 20 outline, Part 7.3). The two are forced by the paper itself: at a Steinberg prime the inertia-invariant line of `ad V_l E` carries the arithmetic eigenvalue `q` (computed in Lemma lem:tamfactor: `Ad(rho(Frob)) e = q e`) and the paper's factor is `1 - q^{-1-s}`, which needs the geometric eigenvalue `q^{-1}`; and `L_p(A_f(1), 0) = L_p(A_f, 1)` (Lemma lem:fpr) holds because `Frob_p^{-1}` acts on `Z_l(1)` by `p^{-1}`. On `V_l E` the arithmetic Frobenius has the eigenvalues `alpha, beta` (roots of `X^2 - a_p X + p`) and the geometric one has `alpha^{-1}, beta^{-1}`, a different set; on `ad V` both have `alpha/beta, 1, beta/alpha` (B-C3). So the convention is invisible at good primes and on the lines where Frobenius acts by `chi_{-3}(q), chi_{-4}(q) = +-1`, and visible exactly at multiplicative primes. (This is important_notes.md point 11 with B1, B7, B9; DFG04 use the geometric Frobenius, Chapter 20 Part 7.3, confirming point 11. The book states the convention once, in C2, and repeats it in a one-line Commentary at Lemma lem:euler, Lemma lem:tamfactor and the paragraph after Lemma lem:dfg21.)

How the commentary is marked. Wherever the paper's text is wrong, imprecise or has a gap, the item carries a paragraph **Commentary** (typeset as a `warning` environment titled "Commentary") with

three parts: what the paper says; the correct statement; how the book presents it. Every point 1 to 80 of important/important_notes.md about the paper is placed, in its final state (with all addenda, corrections and settlements applied), at the item it concerns; Part C is the index. A point that is a theorem the book proves elsewhere is cited by that chapter's item number. Points 15 to 28 were found by the first version of this outline (its N1 to N14); the old labels are not used any more.

Status of this file. First written in two sessions (A0 to A4.10, then the rest); revised in this third session to apply Appendix B (conventions B-C1 to B-C16 and the Chapter 21 entries F1, F2, T1, T2, D5, D6, X5, L6 of its Part 3), the book-internal notes B1, B3, B7, B8, B9, B12, B13, B16, B18, B19, B21, B25, B26, B27, and points 29 to 80 with the settlements of points 39 (confirmed via 56 and 73), 43 and 74. The list of changes is writer note W1.

23.1 Part A. Core results, in teaching order

23.1.1 A0. Conventions and standing notation (Section 21.1 of the chapter)

C1 conv-standing (convention; *stated*; reference: the paper, Section 1 and Section 1.1; DFG04 1.7.2; Appendix B, B-C1, B-C10, B-C12, B-C13; paper use: everywhere). Statement. Throughout, E/Q is an elliptic curve of conductor N, X_0(N)-optimal in its isogeny class, varphi_E : X_0(N) -> E its modular parametrisation (curly varphi for maps of curves) and deg varphi_E its modular degree; f = f_E in S_2(Gamma_0(N)) is the newform with a_p(f) = a_p(E) for all p; phi_f : T -> Z, T_n -> a_n(f), its eigencharacter (upright phi); l is an odd prime; V = V_l E, T = T_l E, rho = rho_{E,l} : G_Q -> Aut(T) = GL_2(Z_l), rho bar = rho mod l on E[l]; A = ad T and A_f = ad V are the trace-zero adjoints (the macro \ad already contains the superscript 0; A_f is never an abelian variety: Shimura's quotient attached to f is J_f, B-C12); delta_p = dim V^{I_p} equals 2, 1, 0 for good, multiplicative, additive reduction at p != l (at p = l the paper's formulas are not used, point 72); v_l is the l-adic valuation with v_l(l) = 1, extended to Q^x and to ideals of Z_l, and for a class c in Z/l^m, v_l(c) := min(v_l(c~), m) (B-C10); length means Z_l-length. q denotes a bad prime (q | N), p any prime. The Hecke algebra T is the full one: the subring of End S_2(Gamma_0(N)) generated over Z by all T_n, n >= 1, including U_p for p | N. E^{(q)} is the quadratic twist of E by Q(sqrt(q*)), q* = (-1)^{(q-1)/2} q; E_S = E tensor chi_S. Depletion is f^Sigma, f^{p}, f^S (never a parenthesised superscript, which is reserved for twists, B-C13(e)). The Manin constant is c_E; c_q = #Phi_q(F_q) is the Tamagawa number; the class of the unit part of the Tate parameter is c(u) (B-C14); a bare c is not used in the book's own text. Characters (B-C6): chi_l with a letter subscript is the l-adic cyclotomic character and chi bar_l its reduction; a numerical subscript always means the quadratic character chi_d of Q(sqrt(d)) (chi_{-3}, chi_{-4}, chi_{-7}, chi_{-11}); the cyclotomic character at 3 is chi_3^cyc; the quadratic character of conductor q is chi_{q*}, never chi_q (PLAN.md 7.5, decision D3). **Commentary (important_notes.md point 32).** The paper uses chi_q for the quadratic character of conductor q and chi_l for the cyclotomic character, so chi_3 is ambiguous, and it names one character both chi_11 and chi_{-11}. Notation only; the book follows the rule just stated and writes chi_{q*} and chi_{-11}.

C2 conv-frobenius (convention; *stated*; reference: Appendix B, B-C2, B-C3, B-C7; DFG04 1.1.1 and 1.7.2 (verified, Chapter 20 outline 7.3); Serre, Abelian l-adic representations, I.2; \Cref{ch:local} 4.3 and 5.4, \Cref{ch:ant} 3.7, \Cref{ch:lfunctions} 7.4; paper use: Section 1.1, Definition def:terms, Lemma lem:euler, Lemma lem:tamfactor, the paragraph after Lemma lem:dfg21, Lemma lem:fpr). Statement. (a) Frob_p (arithmetic) is an element of a decomposition group D_p = G_{Q_p} inducing x -> x^p on the residue field, well defined modulo I_p and up to conjugation; chi_l(Frob_p) = p; with t a topological generator of the pro-l part of tame inertia, Frob t Frob^{-1} = t^p, equivalently Frob^{-1} t Frob = t^{p^{-1}} (exponent in Z_l; \Cref{ch:local} 5.4). (b) The geometric Frobenius is Phi_p = Frob_p^{-1}; in this chapter it is written Frob_p^{-1} (B-C2(b), because Phig_q denotes component groups). For a G_{Q_p}-representation M on a finite-dimensional Q_l-space, p != l: L_p(M, s) = det(1 - Frob_p^{-1} p^{-s} | M^{I_p})^{-1}. On V = V_l E at a good prime p, the ARITHMETIC Frobenius has characteristic polynomial X^2 - a_p X + p, eigenvalues alpha, beta

(alpha + beta = a_p, alpha beta = p), and the geometric one has alpha^{-1} = beta/p, beta^{-1} = alpha/p: not the same set. The geometric Frobenius has the eigenvalues alpha, beta on the dual V^vee = H^1(E_Qbar, Q_l) = V(-1). Hence L_p(E, s) = (1 - a_p p^{-s} + p^{1-2s})^{-1} = L_p(V^vee, s), while L_p(V, s) = L_p(E, s + 1) (B-C3(c)). On A_f = ad V at a good prime both Frobenii have the eigenvalues alpha/beta, 1, beta/alpha, an inversion-stable set, so L_p(A_f, s)^{-1} = (1 - (alpha/beta) p^{-s})(1 - p^{-s})(1 - (beta/alpha) p^{-s}) in either convention; at a multiplicative prime the arithmetic Frobenius acts on (ad V)^{I_q} by q and the geometric one by q^{-1}, and only the latter gives the paper's 1 - q^{-1-s}. (c) L(Sym f, s) = L(ad f, s - 1), so L_p(Sym f, s) = L_p(A_f, s - 1); L_q(Sym f, s) = P_q(q^{-s})^{-1} defines P_q in Z[X], and L_q(A_f, 1)^{-1} = P_q(q^{-2}); PARI's lfunsympow(E, 2) returns L(Sym f, s) in this normalisation at every prime, and lfuneuler(S, q) is 1/P_q(x) as a rational function of x = q^{-s} (verified in PARI 2.17.2: lfuneuler(S, 11) = 1/(11*x + 1) for 121a1, \Cref{ch:computation} R2). (d) L_p(M(1), s) = L_p(M, s + 1), so L_p(A_f(1), 0) = L_p(A_f, 1).

C3 conv-duals (convention; *stated*; reference: Appendix B, B-C4, B-C5, B-C12(b); DFG04 Section 2.1 and (57); \Cref{ch:modules} 8.1 to 8.3, \Cref{ch:reps} 7.6, 7.7, \Cref{ch:blochkato} 2.3, 2.4; paper use: Section 4.2, Lemma lem:dfg21, Lemma lem:fpr, Lemma lem:h0). Statement. One symbol for each kind of dual: M^vee = Hom(M, Z_l) (linear dual; for representations the contragredient), X^* = Hom_cont(X, Q_l/Z_l) (Pontryagin dual; the paper's * in Lemma lem:dfg21), M^D = Hom(M, Z_l(1)) = M^vee(1) (Kummer dual), A^dagger (adjoint of an operator for a pairing); a star after the name of a MAP is a pullback (varphi_E^*). Z_l(1) = T_l mu = lim mu_{l^k}, with Frob_p acting by p; M(1) = M tensor Z_l(1). For A = ad T: W = A tensor Q_l/Z_l, T^D = Hom(A, Z_l(1)), V^D = T^D tensor Q_l, W^D = V^D/T^D. For l odd the trace form (x, y) -> tr(xy) on ad T is perfect and G_Q-equivariant (\Cref{ch:reps} 7.6), so T^D = A(1) and W^D = ad T(1) tensor Q_l/Z_l, the module whose H^0 defines h_l(E); H^0(Q, W^D)[l] = H^0(Q, ad rho bar tensor chi bar_l). ad is insensitive to duals and twists: ad(T^vee) = ad T = ad(T(n)) = ad(T tensor chi) (\Cref{ch:reps} 7.7), so ad M_{f,lambda} = ad T_l E although M_{f,lambda} = T_l E(-1) (A4.1).

23.1.2 A1. The paper's abstract and Section 1: the local terms and the statements (Sections 21.2 to 21.4)

A1.0 abstract-and-introduction (reading guide (the abstract and the first two paragraphs of Section 1); *stated*, with the corrected formulas *proved* in the chapters cited; reference: Zagier85, Watkins02, ARS12 Theorem 2.2, Hida81, DFG04 Lemma 2.12 (verified, Chapter 20 outline 7.3), Flach92; \Cref{ch:moddeg} 7.3, 8.3, \Cref{ch:lfunctions} 8.2 to 8.4; paper use: abstract, Section 1). Content. The abstract states (eq:thm), describes e_l and t_l, names the inputs (DFG04, FPR94, ARS12, the Tamagawa factor at a multiplicative prime), the identity (eq:identity), the twisting statements, the 3Ns boundary and the data. The introduction recalls the Petersson-norm route to the degree (Zagier, Watkins: deg varphi_E vol(Lambda_E) = 4 pi^2 c_E^2 <f, f> with c_E^2 in the NUMERATOR and the unnormalised Petersson product, B-C8, B-C9, important_notes.md B10; \Cref{ch:moddeg} 7.3), the special value L(Sym f, 2) (DFG04 Lemma 2.12 gives <f, f> with the NAIVE value, \Cref{ch:lfunctions} 8.3), ARS (deg varphi_E | r_E, equal p-valuations for p^2 not dividing N), and the naive versus true adjoint L-function. **Commentary (important_notes.md points 30 and 70).** The paper says the naive adjoint L-function is "the Euler product of sum a_n(f)^2 n^{-s}". That is the Rankin series, which differs by a factor at every good prime: sum a_n^2 n^{-s} = zeta^{(N)}(s - 1) L^nv(Sym f, s)/zeta^{(N)}(2s - 2) exactly, with no correction at the bad primes. Correct statement: the naive function is L^nv(Sym f, s) = zeta^{(N)}(2s - 2) sum a_{n^2} n^{-s} (Shimura's series), or is defined by its local factors as Section 1.1 does. The book quotes the paper's sentence, prints the identity, and cites \Cref{ch:lfunctions} 8.2 for the proof. **Commentary (important_notes.md point 71, optional strengthening).** Combining <f, f> = N L^nv(Sym f, 2)/(8 pi^3) with Zagier-Watkins gives deg varphi_E = N L^nv(ad f, 1)/(2 pi vol(Lambda_E)) for optimal E (c_E = 1 in Cremona's range); writing B_E for the same expression with the true L-value, v_l(deg varphi_E) = v_l(B_E) + sum_q e_l(E, q) unconditionally, and (eq:thm) is equivalent to v_l(B_E) >= sum_q t_l(E, q). This makes the role of the omitted factors visible in one line. The book states it as a remark with a pointer

to \Cref{ch:lfunctions} 8.4 and its table 12.1 (the attribution of the Petersson formula to DFG04 Lemma 2.12 is verified, Chapter 20 outline 7.3). **Commentary (important_notes.md points 25, 65, 66; and 24, 39, 77 for the abstract, discussed at their items).** The abstract (i) says the bound is an equality "when multiplicity one holds at level N": the book proves it is an equality whenever l is not in S_f (A1.4); (ii) calls the Euler terms of types I_0^*, I_n^* "elementary": only Step 1 of the twisting argument is (A1.11); (iii) says the Tamagawa factor formula "raises the bound by one whenever a Steinberg prime q = 1 mod l has a Tate parameter whose unit part is an l-th power": the condition l | n_q is missing, and the raise is min(v_l(n), v_l(q - 1), v_l(c(u))), "by at least one" (point 65; A2.7, A4.13); (iv) speaks of "the Bloch-Kato Selmer group of ad V_l E": it is that of ad T tensor Q_l/Z_l, the Q_l-space version being 0 under the hypothesis (point 66; DFG04 Theorem 2.7); (v) says the corrected inequality holds "for every optimal curve of conductor at most 500000" and that 3Ns curves fail "in a third of the cases": the statement concerns non-CM curves with E[l] irreducible, optimality is known for every class only up to 400000 (and for 359,009 classes beyond), and the failure rates are 31%, 23%, 18% on the three ranges (point 25). The book quotes the abstract once in Section 21.2 with these five corrections in a list.

A1.1 def-terms (definition; *stated* (it is the paper's Definition def:terms, with the corrections of points 15 and 53); reference: paper Definition def:terms; \Cref{ch:lfunctions} 9.1, 9.5, \Cref{ch:ellcurves} 10.7; paper use: Definition def:terms, Theorem thm:main, Conjecture conj:euler, appendix). Statement. For a bad prime q != l: e_l(E, q) = v_l(L_q(A_f, 1)^{-1} / L_q^nv(A_f, 1)^{-1}) = v_l(P_q(q^{-2}) / (1 - a_q(E)^2 q^{-2})), where L_q^nv(A_f, s) = L_q(A_f, s) if delta_q > 0 and = 1 if delta_q = 0; and, for EVERY bad prime q (including q = l, as (eq:rule) and rule(E, l) use it), t_l(E, q) = v_l(-v_q(j_E)) if v_q(j_E) < 0, = 0 otherwise. Facts attached: e_l(E, q) = 0 at a multiplicative prime (naive = true); at an additive prime a_q(E) = 0, so e_l = v_l(P_q(q^{-2})); e_l(E, q) is a finite nonnegative integer (P_q(q^{-2}) is a nonzero element of Z[1/q], \Cref{ch:lfunctions} 9.6); t_l(E, q) = v_l(n) for type I_n at any q and for I_n^* at odd q (v_q(j) = -n); at q = 2 a symbol I_m^* may have -v_2(j) != m, or v_2(j) >= 0 (potentially good), and then t_l = 0 (\Cref{ch:reduction} 7.7, 9.4). Twist behaviour: t_l(E, q) and the true factor L_q(A_f, s) are invariant under every quadratic twist (j and ad V are unchanged); e_l(E, q) is invariant under twists unramified at q, and under twists ramified at q that keep the reduction potentially good of type II, III, IV or a dual (a_q = 0 on both sides); it is NOT invariant when the twist moves q between multiplicative and I_n^*, or between good and I_0^*. **Commentary (important_notes.md point 15, as corrected; B8).** The paper ends the definition "Both terms are invariant under quadratic twist". False for e_l at the twisting prime: 11a1 has type I_5 at 11 and e_3(11a1, 11) = 0; its twist by -11 is 121d2 (class 121d), of type I_5^* at 11, with e_3 = v_3(11^2 - 1) = 1 (the curve 121d1 is the twist of 11a3, type I_1^*, same e_3); and 11a1 twisted by -7 is 539d2, in the class 539d whose optimal curve 539d1 has type I_0^* at 7 and e_3(539d1, 7) = v_3(6) + v_3(64 - 4) = 2, where 11a1 has no term at 7. The change of e_l under such twists is exactly what Theorem thm:quant exploits. Correct statement: as in the Facts above. The book states the corrected invariance in the definition and uses these pairs (curve-level identifications from \Cref{ch:reduction} 8.5). **Commentary (important_notes.md point 53).** Definition def:terms is stated for q != l, but (eq:rule) sums t_l(E, q) over all q | N and rule(E, l) adds t at q = l (726a1 at q = l = 3: t_3(E, 3) = v_3(3) = 1). The book defines t_l(E, q) for every bad q by the same formula, as above.

A1.2 prop-table (proposition; *proved here* (it is Lemma lem:euler, proved in Section 21.6); reference: paper table of Section 1; \Cref{ch:lfunctions} 9.3, 10.1, \Cref{ch:rings} 11.4, 11.5; paper use: the table, Lemma lem:euler, Theorem thm:qual, Theorem thm:quant Step 3). Statement. Let q >= 5 be a bad prime, q != l, chi_{-3} and chi_{-4} the quadratic characters of Q(sqrt(-3)) and Q(i) (chi_{-3}(q) = 1 iff q = 1 mod 3; chi_{-4}(q) = 1 iff q = 1 mod 4), E' the twist of E by Q(sqrt(q*)), and for type I_0^* let alpha, beta be the roots of X^2 - a_q(E') X + q. Then: I_n: L_q(A_f, s)^{-1} = 1 - q^{-1-s}, e_l = 0; I_n^*: L_q(A_f, s)^{-1} = 1 - q^{-1-s}, e_l = v_l(q^2 - 1); II, II^*, IV, IV^*: L_q(A_f, s)^{-1} = 1 - chi_{-3}(q) q^{-s}, e_l = v_l(q - chi_{-3}(q)); III, III^*: L_q(A_f, s)^{-1} = 1 - chi_{-4}(q) q^{-s}, e_l = v_l(q - chi_{-4}(q)); I_0^*: L_q(A_f, s)^{-1} = (1 - (alpha/beta) q^{-s})(1 - q^{-s})(1 - (beta/alpha) q^{-s}), e_l = v_l(q - 1) + v_l((q + 1)^2 - a_q(E')^2). Remark for the book

(important_notes.md point 69): at EVERY bad q, including 2 and 3, P_q(X) has one of four shapes: 1 - X (potentially multiplicative), the I_0^* cubic (1 - qX)(1 - (a'^2 - 2q)X + q^2 X^2) with a' the trace of a quadratic twist with good reduction at q, 1 - c q X with c = +-1 (cyclic inertia of order 3, 4 or 6 with an invariant line), or 1 (\Cref{ch:lfunctions} 10.1). The table is the case q >= 5 of this classification. PARI agrees with the table on all 67,745 pairs (optimal curve, bad q >= 5) below conductor 10,000 (\Cref{ch:computation} R4).

A1.3 def-tamagawa-factor (definition; *stated; reference*; reference: FPR94 I.4.2.2, DFG04 p. 708 (verified, Chapter 20 outline 7.3); \Cref{ch:blochkato} 7.1, 7.3, 9.3; paper use: Section 1.2, Theorem thm:main, Lemma lem:tamfactor, Lemma lem:fpr). Statement. For a prime p != l and a G_{Q_p}-stable Z_l-lattice M, Tam_p(M) = Fitt_{Z_l}((H^1(I_p, M)_tor)^{G_{Q_p}}), the Fitting ideal of the G_{Q_p}-invariants of the torsion of H^1(I_p, M) (G_{Q_p}/I_p acts by conjugation); it is an ideal l^v Z_l and v_l(Tam_p(M)) means v. For M = T_l E it is the l-part of c_p(E); for M = ad T it is Z_l at a good prime and, by the theorem of \Cref{ch:blochkato} 9.3 (important_notes.md point 64), at every potentially good additive prime too; at a potentially multiplicative prime its valuation is given by Lemma lem:tamfactor (A2.7, \Cref{ch:blochkato} 8.4). The Bloch-Kato Selmer group H^1_f(Q, ad T tensor Q_l/Z_l) is the one of DFG04 Section 2.1 (\Cref{ch:blochkato} 6.1, 6.6). **Commentary (important_notes.md point 37).** The paper's display Fitt_{Z_l}(H^1(I_p, M)_tor)^{G_{Q_p}} reads as the invariants of an ideal; what is meant (and what FPR94 and DFG04 mean, though DFG04 p. 708 prints the same superscript) is the Fitting ideal of the invariants. Notation only; the book moves the parenthesis (B-C10).

A1.4 thm-main (theorem; *proved here* in Section 21.11 in the strengthened form (equality), as an assembly of A4.* (the assembly is proved; its inputs DFG04 3.7, 2.7, Proposition 1.4(c), Lemma 2.1, FPR94, ARS12 are stated with references; the equality uses the theorem of \Cref{ch:moddeg} 9.3); reference: the paper, Theorem thm:main; \Cref{ch:congruence} 9.1; paper use: Theorem thm:main). Statement as in the paper. Let E be X_0(N)-optimal and l an odd prime with l not dividing 2N such that rho bar_{E,l} restricted to Gal(Qbar/Q(sqrt(l*))), l* = (-1)^{(l-1)/2} l, is absolutely irreducible. Then (eq:identity) v_l(deg varphi_E) >= length_{Z_l} H^1_f(Q, ad T tensor Q_l/Z_l) + sum_{q | N} v_l(Tam_q(ad T)) + sum_{q | N} e_l(E, q), with equality when H^1(X_0(N), Z_l)_m is free over T_m; in particular (eq:thm) v_l(deg varphi_E) >= sum_{q | N} e_l(E, q) + sum_{q | N} t_l(E, q). Statement as the book gives it (important_notes.md points 39, 56, 73; \Cref{ch:congruence} 9.1). Same hypotheses. Then (eq:identity) holds WITH EQUALITY, with no multiplicity-one hypothesis, and the Tamagawa terms are explicit: 0 at good and at potentially good primes, t_l(E, q) + min(t_l(E, q), v_l(q - 1), v_l(c(u_q))) at potentially multiplicative primes, q = 2 included (\Cref{ch:blochkato} 9.4). So length H^1_f(Q, W) = v_l(deg varphi_E) - B^expl_l(E), the excess of the degree over an explicit local bound; and (eq:thm) follows. Reading of the hypotheses (the hypothesis table B2): for l not dividing 2N, "rho bar absolutely irreducible on Gal(Qbar/Q(sqrt(l*)))" is l not in S_f (DFG04 (31), verified), and it is equivalent to "E[l] irreducible and not (l = 3 and image N_s(F_3))" (important_notes.md point 47; \Cref{ch:galrep} 8.2, 8.6): for l >= 5 not dividing N it is automatic once E[l] is irreducible. **Commentary (important_notes.md points 39, 56, 73).** What the paper says: "with equality when H^1(X_0(N), Z_l), localised ..., is free over the localised Hecke algebra" (also in the abstract, and Remark rem:free supplies freeness). Correct statement: for the optimal curve, M_{f,lambda} = varphi_E^* H^1(E, Z_l) and the projection formula gives the pairing hat delta_! on it the ideal of values deg varphi_E Z_l (\Cref{ch:moddeg} 9.1 to 9.3); DFG04's integral structure, operator, kernel lattice and eta are exactly the ones this computation needs (the five items of point 56, verified against the published DFG04: \Cref{ch:congruence} Part 7.1, point 73); so v_lambda(eta^empty_{f,lambda}) = v_l(deg varphi_E) for every l not dividing 2N with rho bar irreducible, and (eq:identity) is an equality whenever l is not in S_f. How the book presents it: Theorem 21.x is stated with equality; the paper's freeness clause is quoted in a Commentary; Remark rem:free is explained (A4.5) but used for nothing in the proof.

A1.5 conj-euler (conjecture; *stated* (a conjecture of the paper, with its evidence in Section 6); reference: paper Conjecture conj:euler; \Cref{ch:blochkato} 12.2, \Cref{ch:computation} 9.2; paper use: Conjecture conj:euler, Section sec:data, appendix). Statement. Let E be X_0(N)-optimal without CM, l an odd prime with E[l] irreducible, and h_l(E) = length_{Z_l} H^0(Q, ad T(1) tensor

Q_l/Z_l). Then (eq:rule) v_l(deg varphi_E) >= sum_{q | N, q != l} e_l(E, q) + sum_{q | N} t_l(E, q) + [l = 3] w_3(E) - h_l(E), where w_3(E) = 1 for the types II, IV, III^* at 3, 2 for II^*, IV^*, 0 for III, I_0^*, and w_3(E) = 0 when 3 does not divide N or the reduction at 3 is potentially multiplicative. Here e_l(E, q) at q = 2, 3 is Definition def:terms with PARI's P_q, and t_l(E, q) at q = l is as extended in A1.1. Facts (theorems, A4.11): h_l(E) = 0 for every l >= 5; h_3(E) <= 1, with h_3(E) = 1 exactly when the image of rho bar_{E,3} is N_s(F_3) (label 3Ns), for every E with rho bar irreducible, CM or not. So the correction is 1 exactly on the 3Ns curves and 0 otherwise. **Commentary (important_notes.md point 6 and B13).** The paper (paragraph after the conjecture, and the first sentence of Section 1.3) says h_l(E) "is positive exactly when rho bar = rho bar tensor chi_{l*}" and speaks of "a correction term when the representation is induced from Q(sqrt(l*))". Correct statement: the criterion is rho bar = rho bar tensor chi bar_l (residual cyclotomic character), which forces l = 3, where chi bar_3 = chi_{-3}; for l >= 5 the condition rho bar = rho bar tensor chi_{l*} can hold only when l | N (point 47, \Cref{ch:galrep} 8.2), and h_l(E) = 0 regardless. The book writes: "h_l(E) is positive exactly when l = 3 and rho bar_{E,3} is induced from Q(sqrt(-3)), that is, when the image is 3Ns". **Commentary (important_notes.md point 67).** The paper presents the correction term as "the term that Lemma 2.1 of DFG04 produces". That identification (Remark rem:ns) needs H^1_f(Q, V^D) = 0 for 3Ns curves, which DFG04 Theorem 2.7 proves only for l not in S_f, and 3Ns curves at l = 3 are exactly in S_f. The book says that the derivation is conditional on this vanishing (A5.2), which is predicted by the Bloch-Kato conjecture and not proved by the paper.

A1.6 def-component-groups-and-weights (definition; *stated* with the orders proved in \Cref{ch:reduction} 5.1, 8.4; reference: Silverman ATAEC IV.9; paper use: Section 1.4, (eq:main), (eq:twist), Proposition prop:forms, Conjecture conj:eis). Statement. For a bad prime q, Phig_q = Phi_q(Fbar_q) (B-C11(b)); #Phig_q = n for I_n; 1 for II, II^*; 2 for III, III^*; 3 for IV, IV^*; 4 for I_0^* and I_n^*. The Tamagawa number c_q = #Phi_q(F_q) divides #Phig_q; the paper's statements are about Phig_q. For odd q, E^{(q)} has the same Kodaira type as E at every p != q, and at q the types are exchanged I_n <-> I_n^*, I_0 <-> I_0^* (every odd q), II <-> IV^*, IV <-> II^*, III <-> III^* (q >= 5; the latter three also hold at q = 3 with the conductor exponent unchanged, important_notes.md point 51, an optional strengthening checked on 5,122 pairs, \Cref{ch:reduction} 8.3). Twist-symmetrised weight: w_l(E, q) = v_l(n) if the type is I_n, or I_n^* with q odd; = 1 if l = 3 and the type is IV or IV^*, or II or II^* with q odd; = 0 otherwise. Then w_l(E, q) = max(v_l(#Phig_q(E)), v_l(#Phig_q(E^{(q)}))) for q >= 5 (\Cref{ch:reduction} 8.4), and w_l(E, 2) = v_l(#Phig_2(E)).

A1.7 ineq-main-twist (statement; *stated* (inequalities found and tested in Section sec:forms, proved under the hypothesis of Theorem thm:main by Proposition prop:forms); paper use: Section 1.4, Section sec:forms, Conjecture conj:eis). Statement. (eq:main) v_l(deg varphi_E) >= sum_{q | N} v_l(#Phig_q); (eq:twist) v_l(deg varphi_E) >= sum_{q | N} w_l(E, q). Since w_l(E, q) >= v_l(#Phig_q) termwise, (eq:twist) implies (eq:main).

A1.8 prop-forms (proposition; *proved here*, with no exception; reference: paper Proposition prop:forms; \Cref{ch:galrep} 4.4, \Cref{ch:reduction} 6.8; paper use: Proposition prop:forms). Statement as in the paper: under the hypothesis of Theorem thm:main, (eq:main) and (eq:twist) hold, except possibly when l = 3 and E has type IV or IV^* at 2, where they hold as soon as e_3(E, 2) >= 1; the same under Conjecture conj:euler with h_l(E) = 0; for all 92,812 working-set curves of type IV or IV^* at 2, P_2(X) = 1 + 2X. Statement as the book gives it (important_notes.md point 43, SETTLED): under the hypothesis of Theorem thm:main, and under Conjecture conj:euler with h_l(E) = 0, (eq:main) and (eq:twist) hold with no exception. Proof plan (termwise comparison of (eq:twist) with (eq:thm), resp. (eq:rule)): type I_n: t_l = v_l(n) = w_l; I_n^*, q odd: t_l = v_l(n) = w_l; I_n^* at q = 2: w_l(E, 2) = v_l(4) = 0; q >= 5 of type II, II^*, IV, IV^*, l = 3: e_3 = v_3(q - chi_{-3}(q)) >= 1 = w_3 because q = chi_{-3}(q) mod 3; l >= 5: both 0; q = 3, these types, l = 3: w_3(E) >= 1 = w_3(E, 3) (only under (eq:rule): in Theorem thm:main 3 does not divide N); q = 2, type II, II^*: w_l(E, 2) = 0; q = 2, type IV, IV^*, l = 3: w_3(E, 2) = 1 <= e_3(E, 2) = 1 by the theorem of \Cref{ch:galrep} 4.4; types III, III^*, I_0^*: w_l = 0. All other terms are >= 0. **Commentary (important_notes.md point 43).** The paper leaves the case l = 3, type IV or IV^* at 2, as an exception covered by data

(92,812 curves with P_2 = 1 + 2X). It is a theorem: for additive reduction at q != l the l-power torsion fixed by inertia is the l-power torsion of Phig_q (\Cref{ch:reduction} 4.5), which for types IV, IV^* is Z/3, so inertia fixes a line of E[3] and acts through a unipotent of order 3; reduction mod 3 is injective on finite subgroups of GL_2(Z_3), so the image of I_2 on T_3 E is cyclic of order 3, tame, and the same for every odd l; Lemma lem:euler's computation then gives the invariant line of ad V with Frobenius eigenvalue chi_{-3}(2) = -1, P_2(X) = 1 + 2X, e_3(E, 2) = v_3(3/2) = 1 (\Cref{ch:galrep} 4.4; a second proof via the conductor in \Cref{ch:reduction} 6.8, Dokchitser-Dokchitser). The book replaces the exception and the count by this argument and records the data (3,226 optimal curves below 10,000 with type IV or IV^* at 2, all P_2 = 1 + 2X, \Cref{ch:computation} R4) as a check.

A1.9 history-tamagawa-exponent (theorem (statements of Ribet-Takahashi, Pollack-Weston, Kim-Ota); *stated; reference*; references: RibetTakahashi97, Takahashi01, Pasten, PollackWeston11, KimOta Theorem 1.1 and Definition 2.12 (as cited); \Cref{ch:moddeg} 12.2, \Cref{ch:modgalrep} 11.1 to 11.4, \Cref{ch:congruence} 10.2; paper use: Section 1.5, Remark rem:ko(2)). Statement. (i) Ribet-Takahashi: for N = N^+ N^- with N^- a product of an even number of distinct primes and E optimal, the ratio of deg varphi_E to the degree of the parametrisation by the Shimura curve of discriminant N^- is prod_{q | N^-} n_q up to primes l at which E[l] is reducible (precise hypotheses in the cited papers; Pasten controls the error). (ii) The Tamagawa exponent at q: the largest t with rho_{E,l} mod l^t unramified at q; for multiplicative q it is v_l(n_q) = t_l(E, q) (Lemma lem:tamagawa-exp). (iii) Kim-Ota Theorem 1.1 (as in A4.13). Theorem thm:main contains the inequality v_l(deg varphi_E) >= sum_{q || N} t_l(E, q) for every multiplicative prime, including q = +-1 mod l and l = 3, but not Kim-Ota's equality.

A1.10 thm-qual (theorem; *proved here* in Section 21.9, with case (a) extended to q = 2; reference: paper Theorem thm:qual; \Cref{ch:modgalrep} 9.1, 9.2; paper use: Theorem thm:qual, Section sec:proof). Statement as the book gives it. Let E/Q have conductor N, l an odd prime not dividing N with rho bar_{E,l} irreducible and, if l = 3, absolutely irreducible on Gal(Qbar/Q(sqrt(-3))) (equivalently image not 3Ns, \Cref{ch:galrep} 8.6). Let q != l be a prime at which E has additive reduction of one of the kinds (a) type IV or IV^*, l = 3, and q >= 5 OR q = 2; (b) type II or II^*, q >= 5, l = 3; (c) type I_n^* with l | n, q odd. Then l divides the modular degree of the X_0(N)-optimal curve in the isogeny class of E. (For l >= 5 only irreducibility of rho bar is used.) **Commentary (important_notes.md point 43, B18).** The paper states case (a) for q >= 5 only. Lemma lem:IV holds at q = 2 for types IV, IV^* (A1.8 Commentary; \Cref{ch:galrep} 4.4(iii): the Serre conductor of rho bar_{E,3} has exponent 1 at 2 while 4 || N), so the proof of case (a) applies verbatim at q = 2 (with 3 not dividing N). Worked example: 52a1 (type IV^* at 2, Serre conductor 26, g = f_{26b} - f_{26b}(2z) congruent to f_{52a} mod 3, deg varphi = 3; \Cref{ch:modgalrep} 6.6, 9.4); all 1,051 such optimal curves below conductor 10,000 with mod 3 image neither reducible nor 3Ns have 3 | deg varphi. The book states the extended theorem and marks the extension as the book's.

A1.11 thm-quant (theorem; *proved here* in Section 21.13, with q odd in place of q >= 5 and with the equality clause; reference: paper Theorem thm:quant; \Cref{ch:congruence} 12.1 to 12.6; paper use: Theorem thm:quant, Section sec:twisting). Statement as in the paper. Let l be an odd prime with l not dividing 2N and E[l] irreducible, S the set of primes q >= 5 at which E has type I_0^* or I_n^*, E_S the twist of E by chi_S = prod_{q in S} chi_{q*}. Then v_l(deg varphi_E) >= v_l(r_{E_S}) + sum_{q in S} e_l(E, q) >= sum_{q in S} e_l(E, q). Statement as the book gives it. (i) The same with S the set of ODD primes of type I_0^* or I_n^* (important_notes.md point 60: the twist lemma for I_n <-> I_n^* and I_0 <-> I_0^* holds at every odd q, l != 3 is automatic when 3 in S since 3 then divides N, and PARI's P_3 agrees with the twist formula on all 1,100 curves checked; \Cref{ch:congruence} 12.1, 12.4). (ii) (important_notes.md point 74) If moreover l is not in S_f, the first inequality is an EQUALITY: v_l(deg varphi_E) = v_l(r_{E_S}) + sum_{q in S} e_l(E, q), with no multiplicity-one hypothesis (\Cref{ch:congruence} 12.6). Reading: the outer inequality uses Steps 1 to 3 and the integrality of eta^empty_g; the middle term uses Step 4, which needs no freeness at level N_S (point 39: Lemma lem:hecke is an equality for the optimal curve of the class of E_S). **Commentary (important_notes.md point 24, revised by 39).** The abstract calls this "an elementary twisting argument". Only Step 1 (transport of congruences along the twist) is elementary; Step 2 is Lemma lem:ddt417 and Step 3 is DFG04 Proposition 1.4(c) (Ihara's lemma). The old worry that the middle term needs multiplicity one at level N_S is removed by point 39 (Step 4). The book's wording:

"by a twisting argument combined with the formula of Diamond, Flach and Guo for the variation of the congruence ideal". **Commentary (important_notes.md point 74).** The paper's sentence after the theorem ("the first inequality is then an equality when multiplicity one holds at both levels") and Remark (2) after the proof ask for multiplicity one. Correct statement: apply the identity (eq:identity), now an equality (A1.4), to `E` and to the optimal curve of the class of `E_S`: `ad T` is unchanged by the twist, so the Selmer groups agree; the Tamagawa factors agree (trivial at `I_0^*` primes by point 64, equal at `I_n^*` primes since `E_S` is multiplicative there with the same Tate parameter); the Euler sums differ by exactly `sum_{q in S} e_l(E, q)`. Data: equality in all `25,477` pairs (curve, `l`) below conductor `10,000` with `S` nonempty, including `15,607` pairs with `3 in S` and the `8` pairs with image 3Ns, which lie outside the theorem (`\Cref{ch:congruence}` 12.6). The book states the equality as part of the theorem.

A1.12 `prop-ns` (proposition; *proved here* in Section 21.14, in the corrected form; reference: paper Proposition prop:ns; `\Cref{ch:galrep}` 8.3, 8.4, `\Cref{ch:groups}` 5.7, 6.3, 6.7, 6.8, `\Cref{ch:reps}` 6.2, 6.6, 8.2, `\Cref{ch:galois}` galois:quartic-on-lines; paper use: Proposition prop:ns, Section sec:anomaly, Conjecture conj:euler). Statement (book's form). Let `E/Q` (with or without CM) have `E[3]` irreducible. Then the image `H` of `rho bar_{E,3}` is conjugate to `GL_2(F_3)`, `N_ns(F_3)` or `N_s(F_3)`; `rho bar_{E,3}` is absolutely irreducible on `Gal(Qbar/Q(sqrt(-3)))` unless `H ~ N_s(F_3)`; in that case `rho bar_{E,3} tensor F_9 = Ind psi` for a character `psi` of order 4 of `G_{Q(sqrt(-3))}` with values in `F_9^x`, `rho bar_{E,3} = rho bar_{E,3} tensor chi_{-3}` over `F_3`, `H^0(Q, ad rho bar_{E,3}(1))` is a line, and `h_3(E) = 1`. The 3-division polynomial is a product of two irreducible quadratics in the `N_s` case and irreducible in the other two. Commentary (points 1, 2, 4, 9, 33, 48): see A5.1, where the proof is walked through. In short: the "non-abelian and faithful" step needs the commutator argument (point 1); `N_s(F_3)` is not maximal (point 2); Zywina15 and Dickson are cited by section (point 4) and the list of images is proved directly; "without CM" is not used (point 9); "induced" holds over `F_9` only (point 33); and the main observations of Section sec:anomaly about 3Ns curves are theorems (point 48, `\Cref{ch:galrep}` 8.4). Data check: among the `38,042` optimal curves below conductor `10,000`, `ns3` (two quadratic factors) agrees with Cremona's 3Ns label on all `37,853` non-CM curves (`90` each), and is `1` on `13` CM curves (121b1, 256a1, ...; `j = -32768` or `8000`) (`\Cref{ch:computation}` R5).

A1.13 `conj-eis` (conjecture; *stated*, with the bound `D <= 2` *proved here* from the first bound; reference: paper Conjecture conj:eis; `\Cref{ch:ellcurves}` 9.8, `\Cref{ch:galrep}` 9.2, `\Cref{ch:moddeg}` 11; paper use: Conjecture conj:eis, Section sec:eis, Open question 4). Statement. Let `E` be optimal, not CM, with a rational `l`-isogeny, `l` odd, and `D = sum_{q | N} v_l(#Phig_q) - v_l(deg varphi_E)` the deficit of (eq:main). Then `D <= v_l(#E(Q)_tors) + [E[3] = Z/3 + mu_3]`, and `D <=` the number of curves in the isogeny class of `E` with a rational point of order `l`. In particular `D <= 0` without rational `l`-torsion, `D <= 1` for `l >= 5`, `D <= 2` for `l = 3`. **Commentary (important_notes.md point 17).** The paper says that a point of order `9` and `E[3] = Z/3 + mu_3` "do not occur together in the tables" and, not having proved it, that `D <= 2` "is part of the conjecture". They cannot occur together for non-CM `E` (proved in `\Cref{ch:ellcurves}` 9.8): with `phi: E -> E' = E/mu_3` and its dual `phi^`, the subgroup `(phi^)^{-1}(<P>)` of `E'` has order 27, is rational, and does not contain `E'[3]` (since `phi^(E'[3]) = mu_3` is not in `<P>`), so it is cyclic; a rational cyclic 27-isogeny forces CM (`\Cref{ch:ellcurves}` 9.6(iii)). So `D <= 2` follows from the first bound and Mazur's theorem; likewise `D <= 1` for `l = 5, 7` and `D <= 0` for `l >= 11`. The book proves this and states only the first two bounds as conjectural. **Commentary (important_notes.md points 3 and 44).** Section sec:eis calls 3Cs.1.1 "the split Cartan image". 3Cs.1.1 is the order-2 subgroup `{diag(1, +-1)}` of the split Cartan subgroup, the image exactly when `E[3] = Z/3 + mu_3`; the full split Cartan image is 3Cs (`\Cref{ch:groups}` 6.10, `\Cref{ch:galrep}` 9.2). The labels 3Ns, 3Cs.1.1, 3B.1.1 were checked against Cremona's `galrep` file (consistent with the intrinsic descriptions: 3B.1.1 and 3Cs.1.1 exactly the curves with rational 3-torsion), not yet against the tables of Sutherland16 (`\Cref{ch:galrep}` 7.4). The book uses the intrinsic description everywhere.

23.1.3 A2. The paper's Section 2: local invariants (Sections 21.5 to 21.7)

A2.1 `lem-tamagawa-exp` (lemma; *proved here*; reference: Silverman ATAEC V.3 and V.5, DDT Theorem 3.1(e); `\Cref{ch:reduction}` 7.4, 7.5, `\Cref{ch:galrep}` 3.2, 3.3 (two proofs); paper use: Lemma lem:tamagawa-exp, Theorem thm:qual case (c), Section 1.5, Remark rem:ko). Statement. Let `q` be

a prime of multiplicative reduction, n_q = v_q(Delta_E) = -v_q(j_E), and l != q a prime. Then rho_{E,l} mod l^t is unramified at q iff l^t | n_q; this holds for split and for nonsplit reduction. Expanded proof: over the unramified quadratic extension K of Q_q (or Q_q itself if split) E is the Tate curve E_{q_E}, v_q(q_E) = n_q; I_q = I_K. E_{q_E}[l^t] is generated by mu_{l^t} and an l^t-th root of q_E; I_q fixes mu_{l^t} (l != q) and moves the root by the Kummer cocycle. By Kummer theory over Q_q^ur (which contains mu_{l^t}), inertia acts trivially iff q_E is an l^t-th power in Q_q^ur; units of its valuation ring are l^t-th powers (Hensel, l != q), so this holds iff l^t | v_q(q_E) = n_q. **Commentary (important_notes.md point 8).** The paper's proof calls the Tate parameter u (Lemma lem:tamfactor uses u for its unit part) and says "an l^t-th power in Q_q^ur up to units", which mixes two statements. No mathematical error. The book names the Tate parameter q_E (B-C14) and writes the two steps separately as above.

A2.2 lem-IV (lemma; *proved here*; reference: Kraus90; \Cref{ch:ant} 4.7, 4.8 (Z_3[zeta_3] is a DVR with uniformiser pi = 1 - zeta_3), \Cref{ch:reduction} 6.6, \Cref{ch:galrep} 4.3, 4.4; paper use: Lemma lem:IV, Theorem thm:qual (a), (b), Remark rem:types). Statement. Let q >= 5 be a prime at which E has type IV or IV^*. Then the image of I_q in Aut(T_3 E) is cyclic of order 3, its image in Aut(E[3]) is a nontrivial unipotent subgroup, E[3]^{I_q} is a line, rho bar_{E,3} is tamely ramified at q, and the Serre conductor of rho bar_{E,3} has exponent 1 at q. (The same holds at q = 2: \Cref{ch:galrep} 4.4, A1.8.) Expanded proof: v_q(Delta_min) in {4, 8}; Kraus: the image of I_q on T_l E (any l != q) has order 12/gcd(v_q(Delta), 12) = 3; tame since q != 3, so cyclic, generated by sigma of order 3 with det sigma = chi_3^cyc(sigma) = 1 (the cyclotomic character at 3 is trivial on inertia at q != 3; B-C6(a)); eigenvalues primitive cube roots of unity; sigma^2 + sigma + 1 = 0 on T (Cayley- Hamilton); T is a torsion-free module over Z_3[X]/(X^2 + X + 1) = Z_3[zeta_3] with X acting as sigma, a DVR with uniformiser pi = 1 - zeta_3, pi^2 = -3 zeta_3, residue field F_3; rank 2 over Z_3 forces T free of rank one; E[3] = T/3T = Z_3[zeta_3]/(pi^2) with sigma acting as zeta_3 = 1 - pi; so sigma - 1 acts as -pi, nonzero with square 0 there: a nontrivial unipotent; fixed line pi Z_3[zeta_3]/(pi^2); conductor exponent 2 - 1 + 0 = 1. **Commentary (important_notes.md point 13).** The paper writes "Z_3[sigma]/(sigma^2 + sigma + 1) = Z_3[zeta_3]"; Z_3[sigma] already satisfies the relation. The intended statement is Z_3[X]/(X^2 + X + 1) with sigma acting as X. Notation only.

A2.3 rem-types (proposition (the paper's Remark rem:types, in corrected and complete form); *proved here* (three lines of linear algebra mod 3, the table of \Cref{ch:galrep} 4.5 cited); reference: Kraus90; paper use: Remark rem:types, Theorem thm:qual (why only (a), (b), (c))). Statement (book's form, q >= 5, l = 3): (a) E[3]^{I_q} = 0 and the Serre conductor exponent of rho bar_{E,3} at q is 2 = f_q(E) for EVERY additive type other than IV, IV^*: the generator of inertia acts mod 3 as -1 (I_0^*, and I_n^* with 3 | n), as tau with tau^2 = -1 (III, III^*), or as -u with u a nontrivial unipotent (II, II^*, and I_n^* with 3 not dividing n); none fixes a vector. (b) The twist by chi_{q*} lowers the Serre exponent of rho bar_{E,3} exactly for II, II^* (to 1), I_n^* (to 1, or 0 if 3 | n) and I_0^* (to 0); it lowers the conductor exponent of E itself exactly for I_n^* and I_0^* (II <-> IV^* keeps f_q = 2). **Commentary (important_notes.md points 22 and 49).** The paper states (a) only "in the last two cases" (II, II^*, I_n^*) and omits I_0^* from (b) (point 22); and its phrase "these are the types whose twist by chi_q lowers the conductor exponent" is right for the Serre conductor of rho bar_{E,3}, not for the conductor of E, in the type II case (point 49). Nothing downstream depends on this. The book states the complete version above and explains separately why I_0^* is not in Theorem thm:qual (A3.5).

A2.4 lem-twist (lemma; (i), (ii) *proved here*; (iii) *stated; reference* Atkin-Li Theorem 3.1; reference: \Cref{ch:reduction} 8.1, 8.2, 8.3, 5.4, \Cref{ch:hecke} 9.2, \Cref{ch:modgalrep} 3.8, \Cref{ch:local} 9.3; paper use: Lemma lem:twist, Section 1.4, Lemma lem:euler, Theorem thm:qual (b), (c), Theorem thm:main (I_n^* primes), Theorem thm:twist, Theorem thm:quant). Statement. Let q be odd, chi = chi_{q*} (the paper's chi_q; B-C6(c)), E^{(q)} the twist of E by chi. (i) At every p != q, E and E^{(q)} have the same Kodaira type, the same Phig_p and the same conductor exponent. (ii) If E has type I_n at q (n >= 0) then E^{(q)} has type I_n^* at q and conversely, with the same n. If q >= 5 and E has type II, III, IV at q then E^{(q)} has type IV^*, III^*, II^* respectively, and conversely. The conductor exponent of E^{(q)} at q is 2 when additive and 1 when of type I_n, n >= 1.

(iii) (Atkin-Li) Let g be a newform of weight 2 on Gamma_0(M) with q^2 not dividing M. Then g tensor chi = sum chi(n) a_n(g) q^n is a newform on Gamma_0(lcm(M, q^2)). Expanded proof of (i): chi is unramified at p, so E and E^{(q)} become isomorphic over an unramified extension of Q_p, and type, Phig_p, conductor exponent are invariant under unramified base change (\Cref{ch:reduction} 4.3). (ii): the twist by q* of a minimal model has a model with c_4' = (q*)^2 c_4, c_6' = (q*)^3 c_6, Delta' = (q*)^6 Delta, j' = j (model y^2 = x^3 + q* a_2 x^2 + (q*)^2 a_4 x + (q*)^3 a_6). Potentially multiplicative: v_q(j) < 0, type I_n or I_n^* with n = -v_q(j) according as the reduction is multiplicative or additive, and a ramified quadratic twist exchanges them. I_0 <-> I_0^*: for the twisted model the cubic produced by Tate's algorithm after removing powers of q is T^3 + e a_2 T^2 + a_4 T + e a_6, e = q*/q = +-1, whose roots are e times those of T^3 + a_2 T^2 + a_4 T + a_6, so "distinct roots mod q" (the criterion for I_0^*, \Cref{ch:reduction} 5.4) is unaffected; conversely a type I_0^* curve at odd q has such a model, and its twist is good. q >= 5, potentially good: the type is read off v_q(Delta_min) in {0, 2, 3, 4, 6, 8, 9, 10}; twisting adds 6, and the model is minimal iff the result is < 12, else subtract 12. Conductor exponents: 2 for the additive side of each listed exchange, since the reduction becomes semistable over a tamely ramified quadratic extension (also at q = 3). (iii) reference; for q || M the local component is an unramified twist of Steinberg, whose twist by a ramified quadratic character has conductor exponent 2 (\Cref{ch:local} 9.3(ii)). **Commentary (important_notes.md points 29 and 45).** The paper writes the invariants of the twist as q^2 c_4, q^3 c_6, q^6 Delta (those of the twist by q, which differs from the twist by q* in the sign of c_6 when q = 3 mod 4) and the twisted cubic as T^3 + a_2 T^2 + a_4 T + a_6 (the correct cubic carries the sign e). Only q-adic valuations and the distinctness of roots are used, so the argument is unaffected. The book writes q* and the cubic with e. **Commentary (important_notes.md point 28).** "The conductor exponents are the standard ones for q odd (tame)": at q = 3 a general additive curve can be wild (exponent up to 5). For the exchanges the lemma lists the additive side has exponent 2, as above. The book restricts the sentence. **Commentary (important_notes.md point 51, optional).** The exchanges II <-> IV^*, III <-> III^*, IV <-> II^* also hold at q = 3, with the conductor exponent unchanged (all 5,122 pairs below conductor 10,000, \Cref{ch:reduction} 8.3, \Cref{ch:computation} R9); the paper states them for q >= 5 only and needs no more. The book mentions it in one sentence, without proof.

A2.5 lem-euler (lemma; *proved here* (the computation of (ad V)^{I_q} and its Frobenius eigenvalue in each case; Chapter 19 proves it in its own language, \Cref{ch:lfunctions} 9.3); reference: Kraus90; \Cref{ch:local} 9.2 to 9.4, \Cref{ch:galrep} 3.2, 4.1, \Cref{ch:ant} 4.9, \Cref{ch:reps} 7.3, 7.7; paper use: Lemma lem:euler, the table, Theorem thm:quant Step 3, Lemma lem:fpr). Statement: as A1.2. Expanded proof plan. General principle: L_q(A_f, s)^{-1} = prod (1 - lambda q^{-s}) over the eigenvalues lambda of the GEOMETRIC Frobenius Frob_q^{-1} on (ad V)^{I_q} (C2(b)); a good prime has eigenvalues alpha/beta, 1, beta/alpha in either convention. I_n: in a basis e_1, e_2 of T, rho|_{G_{Q_q}} = (chi_l eps, eps kappa; 0, eps) (arithmetic Frobenius, corner eps kappa, B-C14); inertia acts through t with t e_1 = e_1, t e_2 = e_2 + n' e_1, n' != 0; on ad V (basis e = e_1 tensor e_2^vee, h = e_1 tensor e_1^vee - e_2 tensor e_2^vee, f = e_2 tensor e_1^vee) (t - 1)e = 0, (t - 1)h = -2n' e, (t - 1)f = n' h - n'^2 e, so (ad V)^{I_q} = Q_l e. The arithmetic Frobenius sends e to q e, so the geometric one acts by q^{-1} and L_q(A_f, s)^{-1} = 1 - q^{-1-s}; the naive factor at s = 1 is 1 - a_q^2 q^{-2} = 1 - q^{-2} (a_q^2 = 1), so e_l = 0. (The same line is ker N of the Weil-Deligne representation sp(2) twisted by an unramified character, \Cref{ch:local} 9.3(i).) **Commentary (important_notes.md point 11).** The paper says "a line with eigenvalue q^{-1}" without naming the Frobenius, and computes with the arithmetic one in Lemma lem:tamfactor. The eigenvalue is q for the arithmetic and q^{-1} for the geometric Frobenius; the table is right with the geometric one, which is DFG04's convention (verified). The book says "geometric" here and "arithmetic" in A2.7. I_n^*: V = V' tensor chi_{q*}, V' Steinberg, ad V = ad V', same true factor, but a_q = 0 so the naive factor is 1 and e_l = v_l(1 - q^{-2}) = v_l(q^2 - 1). II, II^*, IV, IV^*: image of I_q cyclic of order 3 or 6 generated by sigma or -sigma, eigenvalues zeta_3^{+-1}; on ad V the sign disappears; End_{I_q}(T) = Z_l[sigma] = Z_l tensor Z[zeta_3] and End_{I_q}(V) = Q_l tensor Q(zeta_3) (a field if l = 2 mod 3, Q_l x Q_l if l = 1 mod 3; \Cref{ch:ant} 4.9); (ad V)^{I_q} is its trace-zero part, the line spanned by 2 sigma + 1 (square -3; it plays the role of sqrt(-3)). The

arithmetic Frobenius acts on <sigma> by sigma -> sigma^q (\Cref{ch:local} 5.4), so it centralises sigma if q = 1 mod 3 and inverts it if q = 2 mod 3, acting on 2 sigma + 1 by chi_{-3}(q) = +-1 (as 2 sigma^2 + 1 = -(2 sigma + 1)); a sign is its own inverse, so the geometric Frobenius acts by the same chi_{-3}(q); L_q(A_f, s)^{-1} = 1 - chi_{-3}(q) q^{-s} and e_l = v_l(1 - chi_{-3}(q) q^{-1}) = v_l(q - chi_{-3}(q)). III, III^*: image of order 4 generated by tau, tau^2 = -1, eigenvalues +-i; End_{I_q}(V) = Q_l tensor Q(i); the invariant line of ad V is spanned by tau itself (it commutes with tau and has trace 0); Frobenius acts on it by tau -> tau^{+-1}, i.e. by chi_{-4}(q); e_l = v_l(q - chi_{-4}(q)). **Commentary (important_notes.md point 12).** The paper writes "sqrt(-3) in Z_l[zeta_3] = End_{I_q}(V)". Correct: End_{I_q}(T) = Z_l tensor Z[zeta_3] and End_{I_q}(V) = Q_l tensor Q(zeta_3), which is not a field when l = 1 mod 3. The argument only needs the element 2 sigma + 1 and the action of Frobenius through Gal(Q(zeta_3)/Q), so it is unaffected. **Commentary (important_notes.md point 34).** For types III, III^* the paper speaks of "the diagonal element"; when l = 3 mod 4, i is not in Q_l, tau is not diagonalisable over Q_l, and there is no diagonal element. The invariant line is spanned by tau, on which Frobenius acts by chi_{-4}(q) as the paper says. The book writes tau. I_0^*: V = V' tensor chi_{q*}, V' unramified with arithmetic Frobenius eigenvalues alpha, beta (those of E', alpha + beta = a_q(E'), alpha beta = q); ad V = ad V' unramified with eigenvalues alpha/beta, 1, beta/alpha; L_q(A_f, 1)^{-1} = (1 - alpha/(beta q))(1 - 1/q)(1 - beta/(alpha q)) = ((q - 1)/q) ((q + 1)^2 - a^2)/q^2 with a = a_q(E') (\Cref{ch:rings} 11.4); e_l = v_l(q - 1) + v_l((q + 1)^2 - a^2). Pitfalls for the writer: alpha/beta lies in Q_l(alpha), the product is rational; chi_{-3}(q) = (q/3) = (-3/q). The paper's closing sentence ("confirmed against PARI's lfunsympow on every curve of the tables") is reproduced on all 67,745 pairs below conductor 10,000 (\Cref{ch:computation} R4).

A2.6 rem-wild (remark (the paper's Remark rem:wild), in corrected form; *stated* with PARI-verified values; reference: \Cref{ch:lfunctions} 10.1, 10.2, \Cref{ch:galrep} 4.4, 4.6, \Cref{ch:reduction} 9.2 to 9.4, \Cref{ch:computation} 6.6; paper use: Remark rem:wild, Conjecture conj:euler, Section sec:wild, Proposition prop:forms). Statement (book's form). At q = 2, 3 the Kodaira type does not determine L_q(A_f, s); what does is the shape of rho(I_q) (\Cref{ch:lfunctions} 10.1): P_q is 1 - X, the I_0^* cubic of a quadratic twist with good reduction, 1 - c q X (c = +-1), or 1. Definition def:terms applies verbatim with PARI's factor. Type IV or IV^* at 2: P_2(X) = 1 + 2X and e_3(E, 2) = v_3(3/2) = 1, a THEOREM (A1.8). Type II at 2: all four shapes occur; for instance 336f1 has P_2 = 1 + 2X, e_3 = 1, deg 24; type II^* at 2: 1232l1 (= 77a1 tensor chi_{-4}) has the cubic (1 - 2X)(1 + 2X)^2, e_3(E, 2) = v_3(9/8) = 2, deg 288 = 2^5 3^2, the bound at l = 3 attained by the prime 2 alone; 592c1 (= 37a1 tensor chi_{-4}) has (1 - 2X)(1 + 4X^2), e_5(E, 2) = v_5(5/8) = 1, deg 80: the prime 2 can contribute at l = 5. Type I_m^* at 2: potentially good in 61% of the pairs below conductor 10,000 (5,935 of 9,667), and then P_2 = 1 for 5,073, 1 + 2X for 146, a cubic (with a' = +-1, contributing nothing at odd l) for 716; t_l(E, 2) = 0 in that case. At q = l the Euler factor is not the relevant quantity; the weights w_3 of Conjecture conj:euler are empirical. **Commentary (important_notes.md points 50 and 69; B16).** The paper says "a type II at 2 has P_2 = 1 and e_3 = 0" and "a type I_n^* at 2 has potentially good reduction in about half of the cases, with a cubic P_2". Both are false as general statements: below conductor 10,000 type II at 2 has P_2 = 1 for 1,092 curves, 1 + 2X for 371 and a cubic for 74 (type II^*: 514, 74, 191), with 336f1 and 1232l1 as counterexamples; and the cubic is a minority shape for potentially good I_m^* at 2. Correct statement: the type at 2 does not determine P_2; the four shapes and their meaning are as above. No theorem or count of the paper changes, since the rule and tables use PARI's factor. The book replaces the two sentences by "the Kodaira type at 2 does not determine P_2; for instance ..." and gives the classification with a pointer to \Cref{ch:lfunctions} 10.1. "A type IV at 2 has P_2 = 1 + 2X" is kept, as a theorem.

A2.7 lem-tamfactor (lemma; *proved here*, every step, in corrected form; reference: FPR94 I.4.2.2, DFG04 p. 708 (verified), DDT Theorem 3.1(e); \Cref{ch:blochkato} 8.1 to 8.5 (the lemma proved there), \Cref{ch:galcoh} 5.1 to 5.5 (the identification of c'), \Cref{ch:cohomology} model-computation, frobenius-on-h1-inertia, \Cref{ch:modules} 7.6; paper use: Lemma lem:tamfactor, Theorem thm:main (eq:thm), Remark rem:ko(1), Section sec:tam, Open question 3). Statement (book's notation, B-C14). Let q be a prime of multiplicative reduction, n = n_q = v_q(Delta_E), l an odd prime, l != q, T = T_l E, M = ad T. Let q_E in Q_q^x be the Tate parameter (v_q(q_E) = n), u

= q_E q^{-n} in Z_q^x its unit part, u bar in F_q^x its reduction, and c(u) the class of u bar in F_q^x/(F_q^x)^{l^{v_l(q-1)}}, a cyclic group of order l^{v_l(q-1)} identified with Z/l^{v_l(q-1)} by a choice of generator; v_l(c(u)) does not depend on the choice and is read as in B-C10 (so v_l(0) = v_l(q - 1)), and v_l(c(u)) >= k iff u bar is an l^k-th power in F_q^x, for k <= v_l(q - 1). Then H^1(I_q, M)_tor = (Z/l^{v_l(n)})^2 and v_l(Tam_q(ad T)) = v_l(n) + min(v_l(n), v_l(q - 1), v_l(c(u))). In particular v_l(Tam_q(ad T)) >= t_l(E, q), with equality unless q = 1 mod l, l | n and u bar is an l-th power in F_q^x. Expanded proof, the exact facts invoked: (1) [\Cref{ch:galrep} 3.2, \Cref{ch:reduction} 7.5; DDT 3.1(e)] Over the unramified quadratic extension E is a Tate curve; T has a basis e_1, e_2 in which G = G_{Q_q} acts by rho(g) = (chi_l(g) eps(g), eps(g) kappa(g); 0, eps(g)), eps unramified of order <= 2 with eps(Frob) = a_q(E), kappa the Kummer cocycle of q_E in H^1(G, Z_l(1)) = lim Q_q^x/(Q_q^x)^{l^k}. The corner is eps kappa: the nonsplit curve is the Tate curve twisted by eps, and (chi_l, kappa; 0, 1) tensor eps = (chi_l eps, eps kappa; 0, eps). (2) [\Cref{ch:local} 5.3, 6.2] I = I_q acts through its maximal pro-l tame quotient Z_l(1), generated by t; kappa|_I = n times the tame character and eps|_I = 1, so t e_1 = e_1, t e_2 = e_2 + n' e_1 with v_l(n') = v_l(n) (one may take n' = n, \Cref{ch:galcoh} 5.4(ii)). (3) [\Cref{ch:cohomology} cohomology-through-pro-l-quotient, procyclic-compact] The kernel of I -> <t> has order prime to l, so inflation-restriction gives H^1(I, M') = H^1(<t>, M') = M'/(t - 1)M' via c -> c(t), for every finitely generated Z_l-module M'. (4) [computation] In the basis e, h, f of M = ad T: (t - 1)e = 0, (t - 1)h = -2n' e, (t - 1)f = n' h - n'^2 e; 2 is a unit, so (t - 1)M = n'(Z_l e + Z_l h), H^1(I, M) = (Z_l/n') e bar + (Z_l/n') h bar + Z_l f bar, torsion (Z/l^{v_l(n)})^2. (5) [\Cref{ch:cohomology} conjugation-action, frobenius-on-h1-inertia] G/I acts by (sigma . c)(i) = sigma c(sigma^{-1} i sigma); with Frob^{-1} t Frob = t^{q^{-1}} (C2(a), ARITHMETIC Frobenius) and c(t^k) = k c(t) modulo (t - 1)M, the arithmetic Frobenius acts on M/(t - 1)M as q^{-1} Ad(rho(Frob)). (6) [computation] rho(Frob) = (eps q, c'; 0, eps) with eps = eps(Frob) = +-1 and c' = eps(Frob) kappa(Frob); Ad sends e -> q e, h -> h - 2 c' eps e; so on the torsion e bar -> e bar and h bar -> q^{-1}(h bar - 2 c' eps e bar). Fixed points x e bar + y h bar: (q - 1) y = 0 and c' y = 0 in Z/l^{v_l(n)}; l^{v_l(n)} choices of x, l^{min(v_l(n), v_l(q-1), v_l(c'))} of y; the Fitting ideal of this finite group has valuation v_l(n) + min(v_l(n), v_l(q - 1), v_l(c')) (\Cref{ch:modules} 7.6). (7) [\Cref{ch:galcoh} 5.1 to 5.4] Identify c'. Write q_E = q^n u, so kappa = n kappa_q + kappa_u as cocycles (product system of roots). Since q != l, inertia acts trivially on Z_l(1), so H^1(G/I, Z_l(1)^I) = Z_l(1)/(q - 1) = F_q^x tensor Z_l; and since q is a canonical uniformiser, H^1(Q_q, Z_l(1)) = Z_l kappa_q + H^1_ur canonically (\Cref{ch:galcoh} 5.3). Projection onto the unramified summand along kappa_q sends [kappa] to [kappa_u], the class of u bar in F_q^x tensor Z_l: this is the precise meaning of the paper's "image of kappa in H^1(G/I, Z_l(1)^I)". For the Frobenius lift that fixes a compatible system of l-power roots of q (possible since Q_q(q^{1/l^infty}) is totally ramified), kappa_q(Frob) = 0 and c' = eps(Frob) kappa_u(Frob), which is c(u) up to a unit (the sign eps(Frob) does not change the valuation). For another lift Frob t^k the corner becomes c' + eps q k n' (compute rho(Frob t^k)), and another choice of roots changes kappa by a coboundary; so in general c' is well defined only modulo (n', q - 1) Z_l, which does not change min(v_l(n), v_l(q - 1), v_l(c')). Hence the minimum equals min(v_l(n), v_l(q - 1), v_l(c(u))). (8) [\Cref{ch:blochkato} 7.1; FPR94 I.4.2.2 as recalled in DFG04 p. 708] Tam_q(M) = Fitt((H^1(I, M)_tor)^G). (9) Check for M = T: H^1(I, T)_tor = (Z/l^{v_l(n)}) e_1 bar with Frob acting by q^{-1} eps q = eps; invariants all of it if split, 0 if nonsplit (l odd): v_l(Tam_q(T)) = v_l(c_q(E)), as the paper says (\Cref{ch:blochkato} 7.6; checked on 101,336 pairs). (10) The "in particular": t_l(E, q) = v_l(n); the extra term is >= 1 iff v_l(n) >= 1, v_l(q - 1) >= 1 and v_l(c(u)) >= 1; it is 0 at q = 2 and does not depend on whether the reduction is split (\Cref{ch:blochkato} 8.3). **Commentary (important_notes.md point 42; B7, T1).** The paper writes the corner of the Tate matrix as kappa. For a homomorphism the corner c must satisfy c(gh) = chi_l eps(g) c(h) + c(g) eps(h), which kappa does only when eps is trivial; eps kappa always works. Correct statement: corner eps kappa, as in (1). Nothing changes: eps is trivial on inertia, and at Frobenius c' changes by the sign eps(Frob). The book writes eps kappa and c' = eps(Frob) kappa(Frob). **Commentary (important_notes.md point 7 with its addendum; B21).** The paper says "c' is the image of kappa in H^1(G/I, Z_l(1)^I) ... well defined modulo q - 1". The phrase has a precise meaning via

the splitting given by the uniformiser q (step (7)); what is imprecise is that c' is a cocycle VALUE, equal to eps(Frob) times that image only for the Frobenius lift fixing a compatible system of l-power roots of q, and in general well defined only modulo (n, q - 1), not modulo q - 1. The conclusion is unaffected. The book keeps the paper's phrase, adds "via the splitting given by the uniformiser q", and states the indeterminacy. **Commentary (important_notes.md point 11).** The matrix rho(Frob) = (eps q, c'; 0, eps) and the relation Frob^{-1} t Frob = t^{q^{-1}} are for the ARITHMETIC Frobenius (B-C2(c)); the book says so in this proof, while the L-factors of A2.5 use the geometric one. **Commentary (important_notes.md points 23 and 31; T2).** The paper's c (and the c_q of Section sec:tam) collides with the Tamagawa number c_q, and "the class of u in F_q^x" should be "of the reduction of u", with v_l(0) read as v_l(q - 1). The book writes c(u) (or c(u_q)) in its own text and keeps c only inside quotations of the paper. Worked values (PARI-checked, \Cref{ch:blochkato} 8.5, \Cref{ch:computation} R3, 6.8): 791a1 at 7 (I_3, u bar = 6 = -1, a cube: v_3(Tam) = 2, t_3 = 1); 539c1 at 7 (I_3^*, same q_E as 77c1, u bar = 5, not a cube: v_3(Tam) = 1); 1577a1 at 19 (u bar = 8, a cube, not a ninth power: extra 1); 11a1 at 11, l = 5 (u bar = 10 = -1, a fifth power: v_5(Tam) = 2, but E[5] reducible: the lemma applies, the theorem does not). The formula was also checked exhaustively for l = 3, 5 (52,896 brute-force checks, \Cref{ch:blochkato} Part 6.6).

23.1.4 A3. The paper's Section 3: proof of Theorem thm:qual (Sections 21.8 to 21.9)

A3.1 def-congruence-number (definition; *stated*; reference: ARS12 Section 2 and Theorem 2.2 (as cited); \Cref{ch:hecke} 4.3, 11.2, \Cref{ch:moddeg} 8.1, 8.3, 8.5, 9.4, 10.3; paper use: Section 1, Section 3 preamble, Lemma lem:crit, Lemma lem:hecke, Theorem thm:twist). Statement. r_E (for E optimal of level N, newform f) is the largest integer r such that f = g mod r coefficientwise for some g in S_2(Gamma_0(N); Z) orthogonal to f for the Petersson product. Agashe-Ribet-Stein Theorem 2.2: deg varphi_E | r_E, and v_p(deg varphi_E) = v_p(r_E) for every prime p with p^2 not dividing N. (The divisibility is reproved in \Cref{ch:moddeg} 9.4 from the projection formula, and the equality for l not dividing 2N with E[l] irreducible in \Cref{ch:moddeg} 10.3.) Examples (all recomputed, \Cref{ch:moddeg} 8.5, \Cref{ch:hecke} 11.2; important_notes.md B19, points 58, 62): 99a1 (ARS12's 99A1, N = 3^2 11): deg = 4, r = 12, the gap at 3 with 3^2 | N, the congruence f_{99a} = f_{99b} = f_{99d} mod 3 beyond the Sturm bound; 726a1: deg = 144, r = 1584 = 11 * 144, the gap at 11 with 11^2 | 726; 539d1: deg = 72, r = 504, the gap at 7 with 7^2 | 539; 88a1: deg 8, r = 16. In every other computed case the two agree, also at primes p with p^2 | N (121a1, 539c1, 980i1).

A3.2 lem-crit (lemma; *proved here* (the tour follows the paper's proof line by line; the full proof is \Cref{ch:modgalrep} 8.2, decision B17; \Cref{ch:moddeg} 8.2 and \Cref{ch:congruence} 3.4 restate it); reference: Ribet83 Theorem 2.2, AU96 Lemma 2.1, ARS12 Section 3 (as cited); \Cref{ch:hecke} 4.1, 5.2 to 5.4, \Cref{ch:modules} 5.8, 9.6; paper use: Lemma lem:crit, Lemma lem:old, Lemma lem:hecke, Theorem thm:twist, Theorem thm:quant Step 1). Statement. Let phi_f : T -> Z, I = ker phi_f, J = Ann_T(f^perp). Then T/(I + J) is cyclic of order r_E; equivalently r_E generates the congruence ideal phi_f(J) = phi_f(Ann_T I). Consequently, if g in S_2(Gamma_0(N)) tensor Qbar is a normalised eigenform for all of T, orthogonal to f, with coefficients in the ring of integers O of a number field, and lambda | l with a_n(g) = a_n(f) mod lambda for every n, then l | r_E. The same over Z_l with the l-part of r_E. (In this lemma I and J are the paper's letters; in Lemma lem:ddt417 the paper's I is the annihilator, B-C13(c).) Expanded proof steps: (1) T tensor Q = Q x T' with T' faithful on f^perp. The f-line and f^perp are T-stable: for (n, N) = 1, T_n is self-adjoint for the Petersson product (\Cref{ch:hecke} 5.2); for p | N, U_p^dagger = w_N U_p w_N^{-1} (\Cref{ch:hecke} 5.3), and f is an eigenvector of w_N; so f is an eigenvector of every T_n^dagger, and the orthogonal complement of a line stable under all T_n and all T_n^dagger is T-stable (\Cref{ch:hecke} 5.4). Hence I = T cap (0 x T'), J = T cap (Q x 0), I cap J = 0, T/(I + J) = Z/phi_f(J), cyclic of order r'. (2) The pairing S_2(Gamma_0(N); Z) x T -> Z, (g, t) -> a_1(tg), is perfect (\Cref{ch:hecke} 4.1); the forms killed by I are Z f, those killed by J are L = f^perp cap S_2(Gamma_0(N); Z). (3) Dualising T -> T/I x T/J (cokernel T/(I + J)) gives [S_2(Z) : Z f + L] = r'. (4) S_2(Z)/(Z f + L) embeds in Q f/Z f, so it is cyclic, generated by h with projection f/r'; r' h = f + g_0 gives r' <= r_E, and a congruence

modulo r_E gives an element of order r_E, so r_E | r'. (5) phi_g kills J and phi_g = phi_f mod lambda, so I + J lies in the maximal ideal ker(T -> O/lambda) and l | r_E. (6) Over Z_l verbatim. **Commentary (B12, D5; book-internal, not about the paper).** The first version of this item wrote adjoints with a star and contained a sentence that corrected itself midway. The paper's own sentence ("f is an eigenvector of the adjoints of the T_n as well") is right; the book justifies it by the clean statement of step (1), with adjoints written T_n^dagger, U_p^dagger (B-C4(d)).

A3.3 lem-old (lemma; *proved here* (the five prime-by-prime cases; full proof \Cref{ch:modgalrep} 8.3); reference: Atkin-Lehner Theorem 3, Edixhoven92 Section 2, DDT Theorem 3.1(e) (as cited); \Cref{ch:hecke} 6.3, 6.4, 7.1, 7.2, 7.4, \Cref{ch:modgalrep} 4.2, \Cref{ch:galois} galois:chebotarev-brauer-nesbitt, \Cref{ch:reps} 3.7, 5.2; paper use: Lemma lem:old, cases (a), (b), (c)). Statement. Let h be a newform of weight 2 on Gamma_0(M), M | N, h != f, coefficients in O, and lambda | l with a_p(h) = a_p(f) mod lambda for all p not dividing l N. Assume p does not divide M for every p || N at which rho bar is unramified. Then, after enlarging O and lambda, the span of the h(dz), d | N/M, contains an eigenform g as in Lemma lem:crit; so l | r_E. Expanded proof: as in the paper, with the U_p-structure of the old span from \Cref{ch:hecke} 7.1, 7.2 (eigenvalues alpha_p, beta_p and 0, the eigenvectors h - beta_p h(pz), h - alpha_p h(pz) and the depleted forms; U_p is not semisimple when 0 is repeated, \Cref{ch:hecke} 7.3), and the five cases: p not dividing l N; p = l (Deligne-Fontaine via Edixhoven: a_l mod lambda is determined by rho bar|_{D_l}, \Cref{ch:modgalrep} 4.2); p^2 | N (a_p(f) = 0; Atkin-Lehner Theorem 3; mu_p = 0); p || N unramified (one of alpha_p, beta_p is congruent to a_p(f), via DDT 3.1(e): eigenvalues a_p(f) p, a_p(f) of rho bar(Frob_p)); p || N ramified (non-split extensions have a unique stable line). **Commentary (important_notes.md point 10).** "rho bar_h = rho bar by Chebotarev and Brauer-Nesbitt, rho bar being irreducible": irreducibility gives only equal semisimplifications over a common finite field. Insert "hence absolutely irreducible, complex conjugation having distinct eigenvalues +1 and -1" (\Cref{ch:reps} 3.7, \Cref{ch:galrep} 1.4) and "over a common finite extension of F_l".

A3.4 thm-optimal-level (theorem; *stated; reference*; reference: DDT Theorem 3.15 (Ribet90, Carayol89, Diamond95, as cited); \Cref{ch:modgalrep} 6.2, 6.4, 6.5, \Cref{ch:galrep} 6.1, 6.2, 10.1; paper use: the optimal-level paragraph of Section 3, cases (a), (b), (c), the remark after the example). Statement. Let l be odd and sigma bar : G_Q -> GL_2(Fbar_l) irreducible and modular, absolutely irreducible on Gal(Qbar/Q(sqrt(-3))) if l = 3. Then sigma bar arises from a newform of weight 2 and level N(sigma bar) l^{delta(sigma bar)} with nebentypus of order prime to l, N(sigma bar) the Serre conductor and delta(sigma bar) = 0 when sigma bar|_{D_l} is good. Corollary (\Cref{ch:modgalrep} 6.5): for sigma bar = rho bar_{E',l} with E' of conductor prime to l, sigma bar arises from a newform on Gamma_0(N(sigma bar)). Also N(rho bar_{E',l}) | N(E') (Carayol), and N(sigma bar) is prime to the primes where sigma bar is unramified. Case q || N, unramified at q: Ribet's theorem (\Cref{ch:modgalrep} 6.2).

A3.5 proof-thm-qual (theorem (proof of Theorem thm:qual, cases (a), (b), (c)); *proved here*; reference: paper Section sec:proof; \Cref{ch:modgalrep} 9.2, 9.3, 2.7, 3.8, \Cref{ch:hecke} 6.3, 9.2; paper use: Theorem thm:qual). Notation (B-C13(d)): f^chi is the newform twist (rho_{f^chi} = rho_f tensor chi), f tensor chi = sum chi(n) a_n(f) q^n the naive twist, and f tensor chi is the q-depletion of f^chi; (g tensor chi) tensor chi = g^{q}, the q-depletion of g. Common start: r = r_{E_0}; as l does not divide N, v_l(deg varphi_{E_0}) = v_l(r), so it suffices to produce h as in Lemma lem:old. Case (a): q^2 || N, l = 3, q >= 5 or q = 2; Lemma lem:IV (A2.2; at q = 2 A1.8) gives v_q(N(rho bar)) = 1, so M = N(rho bar) | N/q; the optimal-level theorem gives g on Gamma_0(M) with rho bar_g = rho bar; take h = g (h != f by levels); p does not divide M whenever rho bar is unramified at p; Lemma lem:old gives 3 | r. Case (b): q^2 || N, l = 3, type II/II^* at q >= 5; chi = chi_{q*}, E' = E^{(q)} of type IV^*/IV at q, the same types elsewhere, conductor N; its newform is f_{E'} = f^chi, and here f^chi = f tensor chi because a_q(f^chi) = 0 (q^2 | N); this is NOT Atkin-Li, which needs q^2 not dividing the level. rho bar tensor chi satisfies the hypotheses (irreducibility and the condition over Q(sqrt(-3)) are twist-invariant). Lemma lem:IV for E': M = N(rho bar tensor chi) has v_q(M) = 1, M | N/q; the optimal-level theorem gives g on Gamma_0(M) with a_p(g) = chi(p) a_p(f) mod lambda, p not dividing 3N; as q || M, h = g^chi is a newform on Gamma_0(Mq) (Atkin-Li), Mq | N, a_p(h) = chi(p)^2 a_p(f) = a_p(f) mod lambda. h != f: if h = f then f tensor chi =

h tensor chi = g^{q}, so the newforms f_{E'} and g have the same a_p for all p != q, hence f_{E'} = g by strong multiplicity one (\Cref{ch:hecke} 6.3), contradicting the levels N != M. Hypothesis of Lemma lem:old: as in the paper. 3 | r. Case (c): type I_n^* at odd q != l, l | n, q^2 || N; E' = E^{(q)} has type I_n at q, conductor N/q; f = (f_{E'})^chi = f_{E'} tensor chi (Atkin-Li, q || N/q), while f tensor chi = f_{E'}^{q} is the q-depletion of f_{E'}, not a newform (a_q(f_{E'}) = +-1); so the paper's f' must be read as f_{E'} = f^chi. Lemma lem:tamagawa-exp for E': rho bar tensor chi is unramified at q (l | n), so M = N(rho bar tensor chi) is prime to q, M | N/q^2; Ribet's theorem gives g on Gamma_0(M) with a_p(g) = chi(p) a_p(f) mod lambda; h = g^chi is a newform on Gamma_0(M q^2), M q^2 | N, congruent to f at p not dividing l N. h != f: if h = f then g^{q} = h tensor chi = f tensor chi = f_{E'}^{q}, so g = f_{E'} by strong multiplicity one, of levels M != N/q. l | r. **Commentary (important_notes.md points 20 and 54; B18).** In cases (b) and (c) the paper argues "h != f: otherwise g = h tensor chi = f tensor chi = f'". But h tensor chi is the q-depletion of g, not g (point 20); and in case (c) the paper's "f' = f tensor chi is a newform of weight 2 on Gamma_0(N/q)" is false, since E' is multiplicative at q and its newform has a_q = +-1 while f tensor chi has a_q = 0 (point 54; check: 539c1 twisted by -7 is 77c1, type I_3 at 7, a_7 = -1). The conclusion h != f is right, by strong multiplicity one as above. The book writes f' = f_{E'} = f^chi and gives the corrected one-line argument; the first version of this outline repeated the slip and argued case (c) inconsistently (B18), which is now removed. Why I_0^* (and I_n^* with l not dividing n) is not in Theorem thm:qual (one sentence in the book; \Cref{ch:modgalrep} 9.3): the lowered form may be f_{E'} itself, and then h = f; Theorem thm:quant reaches these terms by level raising instead.

A3.6 example-abc (example; worked verification with PARI and Sturm's bound; reference: paper Example after case (c); \Cref{ch:modgalrep} 9.4, \Cref{ch:modforms} 11.3, 11.5, \Cref{ch:reduction} 8.5, \Cref{ch:computation} 6.1, 6.3, 6.4; paper use: the example of Section 3). Statement (the paper's three congruences, with the curve-level identifications corrected per B8 and B20). (b) 121a1 (type II at 11): its twist by -11 is 121c2 (type IV^*; the optimal curve 121c1 of that class has type IV), whose mod 3 representation has Serre conductor 11 and is that of 11a1; f_{11a} tensor chi_{-11} is the newform of the class 121d (121d2 is the twist of 11a1, type I_5^*; 121d1 the twist of 11a3); f_{121a} = f_{121d} mod 3 for EVERY n (both in S_2(Gamma_0(121); Z), Sturm bound 22); deg varphi_{121a1} = 6, e_3(E, 11) = v_3(12) = 1, bound attained. (c) at l = 3: 539c1, type I_3^* at 7; its twist by -7 is 77c1 (conductor 77, I_3 at 7); f_{11a} tensor chi_{-7} is the newform of the class 539d (539d2 is the twist of 11a1); f_{539c} = f_{539d} mod 3 for every n (Sturm bound 112); deg = 288 = 2^5 3^2; e_3(E, 7) = v_3(48) = 1, t_3(E, 7) = 1, bound 2, attained. (c) at l = 5: 980i1, type I_5^* at 7; its twist by -7 is 140b1 (I_5 at 7); the twist of 20a1 by -7 is 980g2 and f_{20a} tensor chi_{-7} is the newform of the class 980g, NOT 980i; the congruence is f_{980i} = f_{980g} mod 5 for every n (Sturm bound 336); deg = 2880 = 2^6 3^2 5. PARI values (\Cref{ch:computation} R1): rule(121a1, 3) = [1, 1]; rule(539c1, 3) = [2, 2]; rule(980i1, 3) = [2, 2] (with e_3(E, 2) = 1 from type IV^* at 2, a theorem, and e_3(E, 7) = 1); rule(980i1, 5) = [1, 1] (t_5(E, 7) = 1). Level-raising conditions visible: a_7(11a1) = -2, (7 + 1)^2 - 4 = 60, divisible by 3; a_7(20a1) = 2, 60 divisible by 5. **Commentary (important_notes.md point 46, optional strengthening).** The paper checks the congruences for p < 500 or p < 1000, excluding a few primes. Both sides lie in the same S_2(Gamma_0(N); Z), so Sturm's bound (22, 112, 336) turns the finite check into a proof for every n, the excluded primes included (\Cref{ch:modforms} 11.5). The book says "for all n, by Sturm's bound". **Commentary (important_notes.md point 21).** 5 | 980, so the 980i1 example is outside the hypothesis l not dividing N of Theorem thm:qual (and of Theorem thm:main): it illustrates the congruence mechanism of case (c), not the theorem; the level of g is 20 = 4 * 5 because rho bar_{140b1,5} is not finite at 5 (\Cref{ch:modgalrep} 9.4(iii)). The book says so and points to the remark after the example. **Commentary (B8, B20; book-internal).** The first version of this outline paired curves within classes loosely (121d1 for the twist of 11a1; "twist of 11a by -7" for 539d1). The class-level statements are right, and every quantity at l = 3 depends only on the class; the book names the curves as above, from \Cref{ch:reduction} 8.5.

A3.7 rem-l-divides-N (example (the paper's remark on l | N and 99A1); *stated*, with recomputed data; reference: ARS12 Table 1; \Cref{ch:modgalrep} 9.5, \Cref{ch:moddeg} 8.5, 9.7; paper use: the remark after the example). Statement. l not dividing N is used (i) in the optimal-level theorem (delta

= 0, weight 2), (ii) in Lemma lem:old at p = l, (iii) to pass from r to deg varphi, where only l^2 not dividing N is needed and cannot be dropped: 99A1 (N = 9 * 11) has deg = 4, r = 12 (both recomputed, \Cref{ch:moddeg} 8.5 and \Cref{ch:hecke} 11.2, important_notes.md B19); further instances 726a1 (r = 1584, deg 144, gap at 11, point 58) and 539d1 (r = 504, deg 72, gap at 7, point 62). At 99a1 and l = 3, H^1(X_0(99), Z_3)_m is not free over T_m (\Cref{ch:moddeg} 9.7). The data show the conclusion of Theorem thm:qual for curves with l || N or l^2 | N. **Commentary (important_notes.md point 55).** The paper adds that the data show the conclusion "at q = 3 in cases (a) and (b), where Lemma lem:IV does not apply because the ramification is wild". In those cases q = l = 3, so 3 | N and the curves are outside Theorem thm:qual in any case; the obstruction is first that q = l (inertia at 3 acts on T_3 E through an infinite group), and only secondarily wildness. The book says "where q = l".

23.1.5 A4. The paper's Section 4: the congruence ideal and the adjoint Selmer group (Sections 21.10 to 21.13)

A4.1 constr-dfg-lattice (construction; *stated; reference* (the objects of DFG04 read for weight 2, trivial character; DFG04 Sections 1.5.3, 1.6.1, 1.6.2, 1.7.3, 1.8.2, (13), (14) verified against the published text, \Cref{ch:congruence} 7.1, 7.3); reference: DDT Section 4.4; \Cref{ch:curves} 10.3, \Cref{ch:moddeg} 3.4, 4.4, 9.1, \Cref{ch:congruence} 5.1 to 5.4, \Cref{ch:modules} 9.4; paper use: Section 4.1, Lemma lem:hecke, Theorem thm:quant Step 2). Statement. Let f be a newform of weight 2 on Gamma_0(M) with rational coefficients, lambda = l not dividing 2M, Sigma a finite set of primes. M^Sigma = M prod_{p in Sigma} p^{delta_p}, f^Sigma = sum_{(n, Sigma) = 1} a_n(f) q^n the Sigma-depleted form (DFG04 Section 1.7.3; \Cref{ch:hecke} 7.4), an eigenform of the full Hecke algebra of level M^Sigma with U_p f^Sigma = 0 for p in Sigma. The integral structure M(M^Sigma, 1)_{!,lambda} agrees with H^1(X_0(M^Sigma), Z_l) after localisation at the non-Eisenstein maximal ideals (A4.1 Commentary). M_{f^Sigma,lambda} = H^1(X_0(M^Sigma), Z_l)[ker phi_{f^Sigma}], the kernel lattice (saturated, free of rank 2). For Sigma empty and f = f_E, E optimal: M_{f,lambda} = varphi_E^* H^1(E, Z_l) = Hom(T_l E, Z_l) = T_l E(-1) exactly, as G_Q-lattices, with no irreducibility hypothesis (\Cref{ch:moddeg} 9.1; important_notes.md point 40), and ad M_{f,lambda} = ad T_l E (C3). The pairing hat delta_!(x, y) = <x, w y> is the cup product composed with the Atkin-Lehner involution w = w_{M^Sigma} (DFG04's [UwU]_omega in weight 2, with no scalar, N^{k-2} = 1); it is perfect and alternating, and every T_n and U_p is self-adjoint for it, because w T_n w^{-1} is the cup-product adjoint T_n^dagger (\Cref{ch:moddeg} 3.3, 3.4). Alternating: w preserves the cup product, so <y, w x> = <w y, x> = -<x, w y>. The naive Sigma-finite congruence ideal eta^Sigma_{f,lambda} is the ideal of values of hat delta_! on M_{f^Sigma,lambda} (DFG04: the image of wedge^2, verified); equivalently the Gram determinant on a basis is (eta^Sigma_{f,lambda})^2 up to a unit (a rank-two alternating form has Gram matrix (0, a; -a, 0), \Cref{ch:modules} 9.4). **Commentary (important_notes.md points 26, 40; B3, D6).** Section 4.2 of the paper says "the lattice M_{f,lambda} of DFG04 and T_l E are homothetic when rho bar is irreducible". They live in dual representations (V_l E^vee versus V_l E, determinants chi_l^{-1} and chi_l; DFG04 (18): the geometric Frobenius on M_{f,lambda} has the characteristic polynomial of H^1, verified). Correct statement: for the optimal curve M_{f,lambda} = varphi_E^* H^1(E, Z_l) = T_l E(-1) = (T_l E)^vee, an equality, and for any curve of the class the dual of T_l E up to homothety when rho bar is irreducible; the conclusion "their adjoints coincide" holds because ad(T^vee) = ad T = ad(T(-1)). The book states it this way. **Commentary (important_notes.md point 73, item 1 and item 3).** The paper says "The integral structure M(M^Sigma, 1)_{!,lambda} is H^1(X_0(M^Sigma), Z_l)". DFG04 define it as the image of the U_0(N)-invariants of compactly supported cohomology of the full-level curve; it agrees with H^1(X_0(M^Sigma), Z_l) after localisation at non-Eisenstein maximal ideals, which is all the paper uses (definitions certain; the Eisenstein nature of the comparison cokernel in general is the Chapter 20 outline's sketch, medium-high). DFG04's pairing is the cup product on X_0(N) times 1/2, a unit for l odd. The book rewords the sentence accordingly and mentions the factor 1/2 (\Cref{ch:congruence} 5.2, 5.3).

A4.2 thm-dfg14 (theorem; *stated; reference*; reference: DFG04 Proposition 1.4(c) (verified, \Cref{ch:congruence} 7.3); \Cref{ch:congruence} 6.1 to 6.4, \Cref{ch:modgalrep} 7.2 to 7.4, \Cref{ch:lfunctions} 9.7; paper use: Theorem thm:dfg14, proof of Theorem thm:main, Theorem

thm:quant Step 3, the remark after Theorem thm:quant). Statement. If rho bar_f is irreducible (and l does not divide 2M, DFG04's standing hypothesis of Section 1.7.3), then eta^Sigma_{f,lambda} = eta^empty_{f,lambda} prod_{p in Sigma} L_p^nv(A_f, 1)^{-1}. Reading: additive p in Sigma: factor 1; good p: Wiles's level-raising factor (1 - (alpha/beta) p^{-1})(1 - p^{-1})(1 - (beta/alpha) p^{-1}) = ((p - 1)/p)((p + 1)^2 - a_p^2)/p^2; multiplicative p: 1 - p^{-2}. Proof idea: Ihara's lemma (\Cref{ch:modgalrep} 7.4) and the degeneracy maps; one paragraph, with reference.

A4.3 lem-ddt417 (lemma; *proved here* in corrected form (full proof \Cref{ch:congruence} 4.3); reference: DDT Lemma 4.17 (inequality half, as cited); \Cref{ch:modules} 5.5, 9.2, 9.3, 11.1, 11.3, 11.6; paper use: Lemma lem:ddt417, Lemma lem:hecke, Theorem thm:quant Step 2). Statement. Let O be a DVR with finite residue field (or state it with lengths), T a commutative O-algebra, finite and free as an O-module, acting on an O-module L free of finite rank with a perfect O-bilinear pairing for which every element of T is self-adjoint; pi : T -> O a surjective homomorphism, wp = ker pi, I = Ann_T(wp), eta = pi(I); suppose L[wp] is free of rank d with Gram determinant Delta != 0. Then eta^d subset Delta O, i.e. d v(eta) >= v(Delta), with equality if L is free over T. Expanded proof: (0) If eta = 0 there is nothing to prove. (1) wp cap I = 0: take t in I with pi(t) a generator of eta; for x in wp cap I, t x = 0 (t kills wp) and (pi(t) - t) x = 0 (pi(t) - t in wp, x kills wp), so eta x = 0 and x = 0, T being torsion-free. Hence L[wp] cap L[I] = 0. (2) For x in L, t x in L[wp] and (pi(t) - t) x in L[I], so Q = L/(L[wp] + L[I]) is killed by eta. (3) L[wp] is saturated, so L -> Hom(L[wp], O) is onto with kernel L[wp]^perp (\Cref{ch:modules} 9.2). (4) L[wp]^perp = L[I] (self-adjointness, as in the paper). (5) Q = coker(L[wp] -> Hom(L[wp], O)), of length v(Delta) (\Cref{ch:modules} 9.3), generated by d elements and killed by eta: v(Delta) <= d v(eta). (6) If L = T^d then Q = (T/(wp + I))^d = (O/eta)^d (Ann(I) = wp, by the same decomposition). **Commentary (important_notes.md point 19, as corrected).** The paper's proof begins "Since L is torsion-free and wp cap I = 0" without proving wp cap I = 0. The reason is NOT that T is reduced (the full Hecke algebra can have nilpotents: U_2 at level 88 on the span of f_11(z), f_11(2z), f_11(4z), f_11(8z), \Cref{ch:hecke} 7.3, 11.1); it is the two-line argument of step (1), valid when eta != 0, the case eta = 0 being trivial. The book adds those two lines. **Commentary (important_notes.md point 38).** The lemma is stated for an arbitrary DVR but counts #O/Delta and #(O/eta)^d, finite only for a finite residue field; for O = Z_l, the only case used, this is fine. The book states it with lengths (B-C10).

A4.4 lem-hecke (lemma; *proved here* in the strengthened form (equality); reference: ARS12 Theorem 2.2 (as cited); \Cref{ch:moddeg} 9.3 to 9.5, \Cref{ch:congruence} 5.6, \Cref{ch:modules} 10.3; paper use: Lemma lem:hecke, proof of Theorem thm:main, Theorem thm:quant Step 4). Statement as in the paper. Let E be X_0(N)-optimal, l odd with l not dividing 2N and rho bar irreducible, m the maximal ideal of T_l = T tensor Z_l containing l and ker phi_f. Then v_l(deg varphi_E) = v_l(r_E) >= v_lambda(eta^empty_{f,lambda}), with equality if H^1(X_0(N), Z_l)_m is free over T_m. Statement as the book gives it (important_notes.md points 39, 56, 73; \Cref{ch:moddeg} 9.5). For E optimal and l odd with l not dividing N: v_l(deg varphi_E) = v_l(r_E) = v_l(ideal of values of hat delta_! on H^1(X_0(N), Z_l)[ker phi_f]), with no irreducibility and no freeness hypothesis; when moreover l does not divide 2N and rho bar is irreducible, this ideal is DFG04's eta^empty_{f,lambda} (their integral structure agrees with H^1(X_0(N), Z_l) at the non-Eisenstein m, A4.1), so v_l(deg varphi_E) = v_lambda(eta^empty_{f,lambda}). Expanded proof: (1) v_l(deg) = v_l(r_E): ARS12 Theorem 2.2 (l^2 does not divide N). (2) The ideal of values of hat delta_! on M_{f,lambda} = varphi_E^* H^1(E, Z_l) is deg varphi_E Z_l: by the projection formula <varphi^* a, varphi^* b> = <a, varphi_* varphi^* b> = deg varphi_E <a, b>_E, the cup product on E is unimodular, and w_N acts on M_{f,lambda} by the Atkin-Lehner sign +-1 (\Cref{ch:moddeg} 9.2, 9.3). (3) This is DFG04's eta^empty_{f,lambda} (A1.4 Commentary). The paper's route, kept as a remark: r_E Z_l = phi_f(Ann_T ker phi_f) Z_l (Lemma lem:crit); T_l = prod T_{m'}; the congruence ideal can be computed in T_m; Lemma lem:ddt417 with L = H^1(X_0(N), Z_l)_m, which gives >= and needs freeness for =. **Commentary (important_notes.md point 35; B3).** The paper says "the other local factors of T tensor Z_l are killed by phi_f and lie in the annihilator". They lie in the KERNEL ker phi_f = wp, not in Ann(wp) in general (if e' is the idempotent of another local factor, e' wp = wp cap T_{m'}, usually nonzero). The conclusion is right: phi_f factors through the projection to T_m, and the annihilator

of wp decomposes along the idempotents, Ann(wp) being the annihilator of wp_m inside T_m. The book writes this (\Cref{ch:modules} 10.3). **Commentary (important_notes.md point 36).** Applying Lemma lem:ddt417 to L = H^1(X_0(N), Z_l)_m needs the pairing to be perfect on the m-component: the idempotent e_m is self-adjoint (all Hecke operators are), so H^1 splits orthogonally into its local components and the pairing is perfect on each. One sentence in the book. **Commentary (important_notes.md point 39).** The inequality and the freeness clause are superseded by the equality above; the book prints the paper's argument as a remark and proves the equality.

A4.5 rem-free (proposition (the paper's Remark rem:free); *proof sketched here*; reference: Tilouine97, Ribet-Stein Theorem 3.5, Mazur77 (as cited); \Cref{ch:moddeg} 10.1 to 10.4, 11.3, \Cref{ch:congruence} 5.7, \Cref{ch:modules} 11.4, 11.5; paper use: Remark rem:free). Statement. For l not dividing 2N and m non-Eisenstein, J_0(N)[m] is two-dimensional over T/m (Ribet-Stein Theorem 3.5 for J_1(N), the exception "Serre weight l" not occurring in weight 2 with l not dividing N, and J_0(N)[m] -> J_1(N)[m'] injective because its kernel lies in the Shimura subgroup, which is Eisenstein); hence T_m is Gorenstein and H^1(X_0(N), Z_l)_m = T_m^2 (Tilouine; \Cref{ch:moddeg} 10.3). Consequence: equality in Lemma lem:ddt417 for L = H^1(X_0(N), Z_l)_m, an independent proof of v_l(r_E) = v_l(deg varphi_E) at such l. Role after the corrections: none in the proof of Theorem thm:main (A4.4) and none in the equality of Theorem thm:quant (A4.16, point 74). It explains WHY Hecke and cohomological congruence ideals agree in the paper's range, and it fails for l | N (99a1 at l = 3: H^1(X_0(99), Z_3)_m is not free, \Cref{ch:moddeg} 9.7). **Commentary (important_notes.md point 41).** Identifying T_l J_0(N) with H^1(X_0(N), Z_l) as Hecke modules uses Poincare duality composed with w_N; this identification is not Galois-equivariant, but freeness over T_m transfers. One clarifying clause.

A4.6 setup-dfg-selmer (definition; *stated; reference*; reference: DFG04 Section 2.1 and (31) (verified, \Cref{ch:congruence} 7.1 to 7.3); \Cref{ch:blochkato} 6.3, 6.6, 10.1, \Cref{ch:galrep} 8.6; paper use: Section 4.2, Theorem thm:dfg37, Theorem thm:dfg27, Lemma lem:dfg21, Lemma lem:h0, Remark rem:ns). Statement. f = f_E, Sigma = {q : q | N}, lambda = l not dividing 2N, A = A_{f,lambda} = ad T (equal to ad M_{f,lambda}, M_{f,lambda} being T_l E(-1), the dual of T_l E, A4.1), W = A tensor Q_l/Z_l, T^D = Hom(A, Z_l(1)) = A(1) (trace form), V^D = T^D tensor Q_l, W^D = V^D/T^D. H^1_f(Q, W) is the Bloch-Kato Selmer group; H^1_Sigma(Q, W) has no condition at p in Sigma, the Bloch-Kato condition at l, and the unramified condition elsewhere (DFG04 use H^1_ur outside Sigma cup {l}, the same for Sigma containing the primes dividing N). S_f = {lambda : lambda | Nk! (that is 2N for k = 2), or M bar_{f,lambda} is not absolutely irreducible on G_F, F = Q(sqrt(l*))} (DFG04 (31), verified); the hypothesis of Theorem thm:main is l not in S_f. Fact (important_notes.md point 47; \Cref{ch:galrep} 8.2, 8.6): for l not dividing 2N, l not in S_f iff E[l] is irreducible and h_l(E) = 0, iff E[l] is irreducible and not (l = 3 and image 3Ns). For l >= 5 not dividing N, a dihedral rho bar induced from Q(sqrt(l*)) is impossible (restrict to inertia at l: the characters {omega, 1} (ordinary) or {omega_2, omega_2^l} (supersingular) cannot be permuted by the twist by chi_{l*}|_{I_l} = omega^{(l-1)/2} unless l = 3); for l = 3 the failure is exactly image 3Ns, and those curves are ordinary at 3. **Commentary (important_notes.md point 26; B3).** The first version of this item repeated "homothetic to T_l E"; corrected as in A4.1.

A4.7 thm-dfg37 (theorem; *stated; reference*; reference: DFG04 Theorem 3.7 = Theorem 0.2 (verified, including l = 3, \Cref{ch:congruence} 7.2, 7.4, 7.5); \Cref{ch:congruence} 7.3, 8.1 to 8.5, \Cref{ch:blochkato} 14.5; paper use: Theorem thm:dfg37, proof of Theorem thm:main, Remark rem:ns, Open question 2). Statement. If l is not in S_f, then H^1_Sigma(Q, W) is finite, of length v_lambda(eta^Sigma_{f,lambda}). (DFG04 Theorem 3.7: for a finite set Sigma not containing l such that M_{f,lambda} is minimally ramified outside Sigma; Theorem 0.2: for Sigma containing the primes dividing N, lambda not in S_f and not dividing any prime of Sigma; the set of primes dividing N qualifies.) Idea: Taylor-Wiles; H^1_Sigma is dual to the cotangent space of R_Sigma, R_Sigma = T_Sigma complete intersection, and the numerical criterion identifies the length with the congruence ideal (\Cref{ch:congruence} 8.5). **Commentary (important_notes.md point 73, open check Q1; B26).** The first version marked "covers l = 3?" as unchecked. Settled: S_f excludes only lambda | 2N and the residually induced case; the local condition at l is defined through extensions of short objects (DFG04 Proposition 2.2, Corollary 2.3, needing only l > 2), the Taylor-Wiles conditions 2 <= k

<= l - 1 hold at l = 3 for k = 2, and the extra condition at 3 (absolute irreducibility on G_{Q(mu_3)}) is the S_f condition. So Theorem thm:main at l = 3 (with 3 not dividing N, image not 3Ns) has its Taylor-Wiles input.

A4.8 thm-dfg27 (theorem; *stated; reference*; reference: DFG04 Theorem 2.7 (verified: "Conjecture 2.6 holds for A_f and B_f = A_f(1) if lambda is not in S_f"); \Cref{ch:blochkato} 10.2, 3.3; paper use: Theorem thm:dfg27, proof of Theorem thm:main). Statement. If l is not in S_f, then H^0(Q, V^D) = H^1_f(Q, V^D) = 0 (and the same for A tensor Q_l). **Commentary (important_notes.md point 63).** The vanishing of H^0(Q, V^D) is automatic here: H^0(Q_p, V^D) = 0 at every p in Sigma, since the local Euler factor L_p(A_f, 1)^{-1} is nonzero (\Cref{ch:blochkato} 3.3, \Cref{ch:galcoh} 10.3(b)). So only H^1_f(Q, V^D) = 0 is needed from DFG04 Theorem 2.7. Observation, one sentence.

A4.9 lem-dfg21 (lemma; *proved here* given Chapter 17's abstract comparison (\Cref{ch:galcoh} 10.3, 10.4; \Cref{ch:blochkato} 10.4); reference: DFG04 Lemma 2.1 (verified); paper use: Lemma lem:dfg21, proof of Theorem thm:main, Remark rem:ns, Conjecture conj:euler). Statement. Suppose H^0(Q, V^D) = H^1_f(Q, V^D) = 0, and let Sigma be a nonempty finite set of primes containing every p != l at which W is ramified. Put H^1_f(Q_p, T^D) = iota^{-1} H^1_f(Q_p, V^D), iota : H^1(Q_p, T^D) -> H^1(Q_p, V^D). Then there is an exact sequence 0 -> H^1_f(Q, W) -> H^1_Sigma(Q, W) -> sum_{p in Sigma cup {infty}} H^1_f(Q_p, T^D)^* -> H^0(Q, W^D)^* -> 0, * the Pontryagin dual. (Here iota is the map T -> V on cohomology; the twist map of Theorem thm:twist is written iota_S in this chapter, B-C15(c).) Attached facts (the paragraph after the lemma): for p in Sigma, H^1_f(Q_p, V^D) = H^1_ur(Q_p, V^D) = (V^D)^{I_p}/(Frob_p - 1) vanishes because det(1 - Frob_p^{-1} | (V^D)^{I_p}) = L_p(A_f(1), 0)^{-1} = L_p(A_f, 1)^{-1} != 0; so H^1_f(Q_p, T^D) is the TORSION of H^1(Q_p, T^D), a finite group (the paper's wording is right and must not be simplified: H^1(Q_p, T^D) itself is infinite at I_0^* primes and at primes with cyclic inertia where chi(p) = 1, important_notes.md B22). At infty, H^1(R, T^D) is killed by 2, so H^1_f(R, T^D) = 0 for l odd (\Cref{ch:galcoh} 6.4). Proof idea (Greenberg-Wiles / Poitou-Tate): compare "Bloch-Kato everywhere" with "no condition on Sigma"; the cokernel maps to sum_{p in Sigma} H^1(Q_p, W)/H^1_f(Q_p, W), dual to sum H^1_f(Q_p, T^D) by local duality; the defect is the dual Selmer group of T^D with strict conditions at Sigma, whose divisible part vanishes by H^1_f(Q, V^D) = 0 and whose finite part is H^0(Q, W^D). **Commentary (important_notes.md point 11).** "det(1 - Frob_p | (V^D)^{I_p}) = L_p(A_f, 1)^{-1}" holds with the GEOMETRIC Frobenius; with the arithmetic one the determinant at a multiplicative prime is 1 - q^2, a different number with the same valuation. The book writes Frob_p^{-1}. **Commentary (important_notes.md point 75).** DFG04's hypothesis is that Sigma contain every prime where H^1_f(Q_p, W) != H^1_ur(Q_p, W); the paper's "every p != l at which W is ramified" implies it. The paper's statement is a correct special case.

A4.10 lem-fpr (lemma; first identity *proved here*; second identity *proved here* for the elliptic-curve adjoint and *stated; reference* in general; reference: FPR94 I.4.2.2, DFG04 proof of Theorem 2.15 and (57) (verified: (57) is Tam^0_p(M_lambda) = Tam^0_p(M^D_lambda) for p != l, general); \Cref{ch:blochkato} 7.2 to 7.5, 3.5, \Cref{ch:lfunctions} 9.6, \Cref{ch:cohomology} inflation-restriction, procyclic-compact, lengths-in-exact-sequences; paper use: Lemma lem:fpr, proof of Theorem thm:main, Remark rem:ns). Statement. For p in Sigma, Fitt_{Z_l} H^1_f(Q_p, T^D) = L_p(A_f, 1)^{-1} Tam_p(T^D) and Tam_p(T^D) = Tam_p(A); L_p(A_f, 1)^{-1} = L_p(A_f(1), 0)^{-1} is a nonzero l-adic integer, of valuation e_l(E, p) at an additive prime and v_l(1 - p^{-2}) at a multiplicative one. Expanded proof of the first identity (\Cref{ch:blochkato} 7.4): inflation-restriction with G/I = Zhat gives 0 -> X^I/(Frob - 1)X^I -> H^1(G, X) -> H^1(I, X)^{G/I} -> 0 for X = T^D; Frob - 1 is injective on X^I (the determinant is the nonzero Euler factor; geometric and arithmetic give the same valuation) and the quotient is finite of length v_l(L_p(A_f, 1)^{-1}); taking torsion and adding lengths gives the identity (Fitt = l^{length} for finite modules). Integrality: P_p has integral coefficients and p^{-2} is an l-adic unit. The second identity: both sides are Fitting ideals of the Frobenius invariants of the same finite module H^1(I, A)_tor, with the action twisted by chi_l(Frob); DFG04 (57) in general, and for the adjoint of E directly: at potentially good primes both are Z_l (\Cref{ch:blochkato} 9.3), at potentially multiplicative primes the count of A2.7 gives the same number for both (\Cref{ch:blochkato} 7.5).

A4.11 lem-h0 (lemma; *proved here* in the corrected form; reference: \Cref{ch:galrep} 8.5 (proved

there), 8.1, 8.3, \Cref{ch:blochkato} 10.3, \Cref{ch:reps} 6.6, 10.5; paper use: Lemma lem:h0, proof of Theorem thm:main, Conjecture conj:euler, Proposition prop:ns, Section sec:anomaly). Statement as in the paper. Let l be odd with rho bar irreducible. Then H^0(Q, W^D) is finite, and it vanishes iff rho bar is not isomorphic to rho bar tensor chi bar_l. For l >= 5 it always vanishes; for l = 3 it vanishes iff rho bar_{E,3} is absolutely irreducible on Gal(Qbar/Q(sqrt(-3))). Its length h_l(E) is the largest k such that rho mod l^k = (rho tensor chi_l) mod l^k, and for E without CM it is finite. Statement as the book gives it (points 5, 14). Same hypotheses, E with or without CM. Then h_l(E) = 0 for every l >= 5, and h_3(E) <= 1 with equality iff rho bar_{E,3} = rho bar_{E,3} tensor chi_{-3} iff the image is N_s(F_3). Expanded proof. (1) rho bar is absolutely irreducible (irreducible, l odd, complex conjugation has distinct eigenvalues +1, -1). (2) H^0(Q, W^D)[l^k] = Hom_{G_Q}(T/l^k, (T/l^k)(1)): End(T)(1) = ad T(1) + Z_l(1) and H^0(Q, mu_{l^k}) = 0, so no trace condition is needed. (3) k = 1: a nonzero map is an isomorphism (Schur) and the space has dimension at most one; so H^0(Q, W^D) is cyclic, Z/l^h. (4) A generator of H^0[l^k] is an isomorphism rho mod l^k -> (rho tensor chi_l) mod l^k (Nakayama), and conversely. (5) Determinants: such an isomorphism gives chi_l^2 = 1 mod l^k on G_Q; chi_l(G_Q) = Z_l^x, so l = 3, k = 1 (4 is not 1 mod 9). (6) l = 3: chi bar_3 = chi_{-3}; an isomorphism rho bar -> rho bar tensor chi_{-3} exists iff rho bar is induced from G_{Q(sqrt(-3))} (over F_9) iff the image is N_s(F_3) (Clifford theory for an index-two subgroup, \Cref{ch:reps} 6.6; Proposition prop:ns). **Commentary (important_notes.md point 5).** The paper's finiteness sentence ("rho = rho tensor chi_l would force a_p = 0 at all primes inert in Q(sqrt(l*))") confuses the cyclotomic character chi_l with the quadratic character chi_{l*}: rho = rho tensor chi_l is impossible for every E (it would give p a_p = a_p). The correct statement is (5), stronger and elementary, with no CM hypothesis. Consequence for Section sec:anomaly: "h_3(E) <= 1 because some prime p < 1223 inert in Q(sqrt(-3)) has 9 not dividing a_p" is replaced by "h_3(E) <= 1 by Lemma lem:h0" (the numerical check is valid but unnecessary). (B13: PLAN.md's Chapter 11 entry repeats the flawed argument; this chapter must not.) **Commentary (important_notes.md point 14).** Even read with chi_{l*}, the CM alternative cannot occur when l does not divide N: CM by Q(sqrt(l*)) needs class number one, so l in {3, 7, 11, 19, 43, 67, 163}, and the prime ramified in the CM field divides N. Moot by point 5; one sentence. **Commentary (important_notes.md point 10).** The Clifford step needs ABSOLUTE irreducibility; step (1) supplies it.

A4.12 proof-thm-main (theorem (the proof of Theorem thm:main, Section 4.3 of the paper), in the strengthened form; *proved here* as an assembly; every input is a card entry: Lemma lem:hecke (A4.4), Theorem thm:dfg14 (A4.2), Theorem thm:dfg37 (A4.7), Theorem thm:dfg27 (A4.8), Lemma lem:dfg21 (A4.9), Lemma lem:h0 (A4.11), Lemma lem:fpr (A4.10), Definition def:terms (A1.1), Lemma lem:tamfactor (A2.7), Lemma lem:twist (A2.4); reference: \Cref{ch:congruence} 9.1, 9.2, \Cref{ch:blochkato} 8.4, 9.3, 9.4, 11.1; paper use: Theorem thm:main, (eq:identity), (eq:thm)). The steps, each with the hypothesis it uses. Step 1 (degree to congruence ideal; uses optimality and l not dividing 2N): v_l(deg varphi_E) = v_lambda(eta^empty_{f,lambda}) (A4.4, an EQUALITY; no freeness). Step 2 (vary Sigma; uses rho bar irreducible): v(eta^empty) = v(eta^Sigma) - sum_{p in Sigma} v_l(L_p^nv(A_f, 1)^{-1}); only multiplicative primes contribute, each v_l(1 - p^{-2}) = v_l(p^2 - 1). Step 3 (Taylor-Wiles; uses l not in S_f): v(eta^Sigma) = length H^1_Sigma(Q, W). Step 4 (compare Selmer groups; uses l not in S_f through Theorem thm:dfg27 (only H^1_f(Q, V^D) = 0, point 63), l odd, Lemma lem:h0): length H^1_Sigma = length H^1_f(Q, W) + sum_{p in Sigma} length H^1_f(Q_p, T^D). Step 5 (local terms; uses p != l): length H^1_f(Q_p, T^D) = v_l(L_p(A_f, 1)^{-1}) + v_l(Tam_p(A)). Step 6 (assemble; cancellation): (eq:identity) with EQUALITY; the Euler sum is sum_{p | N} e_l(E, p) (naive = true at multiplicative p, naive = 1 at additive p). Step 7 (explicit Tamagawa terms, and (eq:thm)): v_l(Tam_p(ad T)) = 0 at good and at potentially good primes (point 64, \Cref{ch:blochkato} 9.3); = t_l(E, p) + min(t_l(E, p), v_l(p - 1), v_l(c(u_p))) at every potentially multiplicative p, INCLUDING an I_n^* prime at p = 2: over Q_p, E is a quadratic twist of a Tate curve with parameter of valuation -v_p(j), ad T|_{G_{Q_p}} is unchanged, and Lemma lem:tamfactor applies (it needs only p != l, l odd), with n = -v_p(j), not the index of the Kodaira symbol (\Cref{ch:blochkato} 8.4). Dropping length H^1_f >= 0 and the min gives (eq:thm). **Commentary (important_notes.md point 18).** For the I_n^* primes the paper uses "the curve E_{\{p\}} of Lemma lem:twist", which is defined for odd p only, while Theorem thm:main allows 2 | N;

at p = 2 with v_2(j) < 0 the term t_l(E, 2) can be positive. The argument is local and goes through as in Step 7. The book states Step 7 for every potentially multiplicative p, p = 2 included; the paper's E_{\{p\}} is the book's E^{(p)}. **Commentary (important_notes.md points 39, 56, 73, 64).** The paper's proof gives >= in Step 1 and "equality in the free case"; the book's Step 1 is an equality (A4.4), so (eq:identity) is an equality for l not in S_f. With point 64 every Tamagawa term is explicit, so the identity reads length H^1_f(Q, W) = v_l(deg varphi_E) - B^expl_l(E): the Selmer length is COMPUTED from the degree. Examples (\Cref{ch:blochkato} 11.3, \Cref{ch:congruence} 9.4; l = 3, 3 not dividing N, E[3] irreducible, image not 3Ns): 121a1 (e_3 = 1, deg 6): Selmer length 0; 791a1 (explicit bound 2 thanks to the Tamagawa excess at 7, deg 144): 0; 1184f1 (bound 2, deg 1728): 1; 1748e1 (bound 3, deg 3240): 1. Rigour: the assembly is *proved here*; DFG04 Theorems 3.7, 2.7, Proposition 1.4(c), Lemma 2.1, (57), FPR94, ARS12 are *stated; reference.*

A4.13 rem-ko (remark (the paper's Remark rem:ko); (1) *proved here* from A2.7 and A4.12; (2) *stated; reference* KimOta Theorem 1.1 and Definition 2.12 (as cited); \Cref{ch:congruence} 10.1, 10.2, \Cref{ch:modgalrep} 11.4, \Cref{ch:blochkato} 8.3; paper use: Remark rem:ko, Section sec:tam, Section 1.5). Statement. (1) At a multiplicative prime q = 1 mod l with l | n and u bar an l-th power in F_q^x, v_l(Tam_q(ad T)) >= t_l(E, q) + 1; precisely the excess is min(v_l(n), v_l(q - 1), v_l(c(u))), and by A4.12 it raises the bound by exactly that amount for l not in S_f, also at potentially multiplicative additive primes. (2) Kim-Ota: for N = N^+ N^-, N^- square-free, l >= 5, l not dividing N, l not in S_f, rho bar ramified at every q | N^- with q = +-1 mod l: ord_lambda eta_f(N) = ord_lambda eta_f(N^+, N^-) + sum_{q | N^-} t_f(q); so v_l(r_E) >= sum_{q | N^-} t_l(E, q), which (eq:thm) contains without l >= 5 and q != +-1 mod l; the equality is theirs. Expansion: check that Kim-Ota's eta_f(N) is the Hecke congruence ideal (then = r_E Z_l); if cohomological, A4.4 gives the same value at l not dividing N. **Commentary (important_notes.md points 65 and 23).** As in the abstract, the paper's "(1) ... raises the bound accordingly" is right with the condition l | n, and the raise can exceed one (only if l^2 | n and l^2 | q - 1); the class is c(u), not c_q.

A4.14 thm-twist (theorem (the paper's Theorem thm:twist); *proved here* (full proof \Cref{ch:congruence} 11.2); reference: Atkin-Li, Shimura71 by section (level of a twist); \Cref{ch:hecke} 6.3, 8.2, 9.1, 9.3, \Cref{ch:modgalrep} 10.1, 10.2, \Cref{ch:ellcurves} 10.4, 10.6; paper use: Theorem thm:twist, Section sec:twisting, Theorem thm:quant Step 1). Statement. Let E/Q have conductor N, S a set of odd primes, D = prod_{q in S} q, chi_S = prod_{q in S} chi_{q*}, E_S = E tensor chi_S of conductor N_S. If q^2 | N for every q in S and lcm(N_S, D^2) | N, then r_{E_S} | r_E; if moreover lcm(N, D^2) | N_S, then r_{E_S} = r_E. Expanded proof (the twist map is written iota_S in this chapter, B-C15(c), L6). (1) iota_S : h -> h tensor chi_S maps S_2(Gamma_0(N_S)) into S_2(Gamma_0(lcm(N_S, D^2))) subset S_2(Gamma_0(N)), preserves integrality, intertwines T_p with chi_S(p) T_p for p not dividing D N_S, and kills the coefficients at multiples of D (\Cref{ch:hecke} 9.1). (2) iota_S(g) = f for g = f_{E_S}: both multiplicative, a_p(f) = chi_S(p) a_p(g) for p not dividing D (the twist is unramified there), a_{q^k}(f) = 0 for q in S (q^2 | N). (3) g^perp, the T-stable complement of the g-line (the sum of the generalised eigenspaces of the other systems; at level N_S, where g is new, it equals the Petersson complement, \Cref{ch:hecke} 8.2), is mapped into f^perp (strong multiplicity one, \Cref{ch:hecke} 6.3). (4) A congruence g = h mod r, h in g^perp integral, gives f = iota_S(h) mod r, so r | r_E. (5) Under the second hypothesis the roles exchange. Hypothesis check: (a) I_n^* at each q in S: N_S = N/D; (b) I_0^*: N_S = N/D^2; (c) II, III, IV or a dual at q >= 5: N_S = N, equality (\Cref{ch:modgalrep} 10.2). Worked instances (degrees from Cremona's degphi file, congruence numbers from \Cref{ch:hecke} 11.2, 11.3): (c) 121a1 and 121c1: r = 6 = 6; (a) 539c1 (deg 288) and 77c1 (deg 6), r(77c) = 6 | 288 = r(539c); 980i1 (deg 2880) and 140b1 (deg 60); 121d1 (deg 24) and 11a3 (class 11a, deg 1); (b) 539d1 (deg 72, r = 504) and 11a3 (class 11a, r = 1). The paper's data paragraph: "On the working set, for the 157,938 pairs (curve, prime of type I_n^*) and the 292,580 pairs (curve, prime of potentially good additive type), v_l(deg varphi_E) >= v_l(deg varphi_{E_q}) holds for l in {3, 5, 7, 11, 13}, l^2 not dividing N, with no exception, with equality in every pair of case (c) and strict inequality only in case (b), where the difference is the Euler term of Theorem thm:quant." **Commentary (important_notes.md point 16).** "Strict inequality only in case (b)" contradicts Theorem thm:quant, which predicts a gap of e_l(E, q) = v_l(q^2 - 1) in case (a) too. Instances at l = 3 (3 not dividing N, E[3] irreducible): 539c1 versus 77c1 (v_3: 2

versus 1), 980i1 versus 140b1 (2 versus 1), 121d1 versus 11a3 (1 versus 0). Reproduced below conductor 10,000 (\Cref{ch:computation} 11.7): case (a) 24,373 triples, strict in 2,779; case (b) 8,348, strict in 1,308; case (c) 19,554, all equal (and 14,876 pairs with equality, \Cref{ch:congruence} 11.3). Correct sentence: "strict inequality in cases (a) and (b), the difference being exactly the Euler term of Theorem thm:quant when l is not in S_f" (A4.16). The book quotes the paper and prints the corrected sentence.

A4.15 proof-thm-quant (theorem (proof of Theorem thm:quant, Steps 1 to 4), with q odd; *proved here*, with DFG04 Proposition 1.4(c) as a stated input (full proof \Cref{ch:congruence} 12.2 to 12.5); reference: paper Section sec:twisting; \Cref{ch:hecke} 7.1, 7.2, 7.4, 8.1, 8.3, 9.3, \Cref{ch:modules} 10.3, \Cref{ch:moddeg} 6.4, 9.3, 9.5; paper use: Theorem thm:quant). Set-up. f = f_E, g = f_{E_S} of level N_S, S the odd primes of type I_0^* or I_n^*, D = prod_{q in S} q, O = Z_l. By Lemma lem:twist, E_S is good at the I_0^* primes (delta_q(g) = 2) and multiplicative at the I_n^* primes (delta_q(g) = 1), so N_S^S = N: the depleted form g^S lives at level N, its generalised eigenspace is the line C g^S (\Cref{ch:hecke} 7.4(iv)), and iota_S(g^S) = iota_S(g) = f. Step 1 (transport of congruences; elementary). T the full Hecke algebra of level N, pi : T -> Z the eigencharacter of g^S, m the maximal ideal of T_l containing ker pi and l; U_q in m for q in S. Every U_q-eigenvalue on S_2(Gamma_0(N)) is 0, +-1 or a root of X^2 - aX + q, hence 0 or an l-adic unit (\Cref{ch:hecke} 7.2); so the m-component is C g^S plus generalised eigenspaces of systems with U_q = 0 from newforms h != g. The proof of Lemma lem:crit over Z_l, run for g^S with the T-STABLE complement of the g^S-line, shows that l^{v(eta^Hecke(g^S))} is the largest l^t with g^S = h mod l^t, h integral in that complement; project by e_m (\Cref{ch:hecke} 8.3); apply iota_S, which lands in f^perp. So v_l(r_E) >= v_l(eta^Hecke(g^S)). Step 2 (Hecke versus cohomological ideal): Lemma lem:ddt417 with L = H^1(X_0(N), Z_l)_m (pairing perfect on the m-part, point 36), L[wp] = M_{g^S,lambda} of rank 2 with Gram determinant (eta^S_{g,lambda})^2: v(eta^Hecke(g^S)) >= v(eta^S_{g,lambda}), no freeness needed. Step 3 (DFG04 Proposition 1.4(c) for g; rho bar_g = rho bar tensor chi_S irreducible, l not dividing 2N_S): v(eta^S_g) = v(eta^empty_g) + sum_{q in S} e_l(E, q), because A_g = A_f, delta_q(g) >= 1 (naive factor = true factor), and the true factor of A_f at q is the one f omits. At q = 3 the values are the twist formulas L_3(A_f, 1)^{-1} = (2/3)(16 - a_3(E')^2)/9 (I_0^*) and 8/9 (I_n^*), checked against PARI (point 60). Step 4 (level N_S): v(eta^empty_g) = v_l(deg varphi_{E_S^opt}) = v_l(r_{E_S}) by A4.4 for the optimal curve of the class of E_S (point 39: NO freeness; v_l(deg) is a class invariant as E_S[l] is irreducible, \Cref{ch:moddeg} 6.4, \Cref{ch:ellcurves} 9.9). With v_l(deg varphi_E) = v_l(r_E) this proves the first inequality; the second because r_{E_S} is an integer. **Commentary (important_notes.md point 61).** "The proof of Lemma lem:crit carried out for g^S" must use the complement of the g^S-line stable under the full Hecke algebra; the Petersson orthogonal complement is not T-stable when U_p is involved (level 44). The paper's text says "the T-stable complement", which is right; the book must not substitute the Petersson one. **Commentary (important_notes.md point 59, optional).** Step 1 is an equality: iota_S sends f to g^S and f^perp into the T-stable complement of g^S, and conversely, so v_l(r_E) = v_l(eta^Hecke(g^S)) (\Cref{ch:congruence} 12.3; at level 539 the Hecke congruence modules of the two depleted forms have orders 288 and 504, equal to r(539c) and r(539d), \Cref{ch:hecke} 11.3(d)). **Commentary (important_notes.md point 60).** The restriction q >= 5 is not needed: the argument works for every odd q in S (the twist lemma for I_n <-> I_n^*, I_0 <-> I_0^* holds at every odd q); the bound held in all 388 cases with type I_0^* at 3. The book states the theorem with q odd and marks the extension as the book's. **Commentary (important_notes.md points 24 and 39).** Without point 39 the middle term needed multiplicity one at level N_S (Lemma lem:hecke gives only v_l(r_{E_S}) >= v(eta^empty_g), the wrong direction); with point 39 it does not. The outer inequality never needed it. Steps 2 and 3 are not elementary (A1.11). Worked checks (\Cref{ch:congruence} 12.6, \Cref{ch:hecke} 11.3; Cremona degrees): at l = 3: 539c1: 2 = v_3(r_{77c}) + v_3(48) = 1 + 1; 980i1: 2 = v_3(r_{140b}) + v_3(48) = 1 + 1; 121d1: 1 = 0 + v_3(120); 539d1: v_3(72) = 2 = 0 + v_3(6) + v_3(60). All equalities, as A4.16 predicts. "E[l] irreducible" cannot be dropped: 539d1 at l = 5 has e_5 = v_5(6) + v_5(60) = 1 but v_5(72) = 0 (image 5B.4.1); 539a1 (I_2^* at 7, image 3B) has e_3 = 1 and v_3(320) = 0; 98a1 (I_1^* at 7, E[3] reducible) has e_3 = 1, deg 16.

A4.16 rem-after-quant (remark (the remark after the proof of Theorem thm:quant), with (2) in the strengthened form; (1), (3) *proved here*, (2) *proved here* from A4.12; reference: \Cref{ch:congruence}

12.6, 12.7; paper use: the remark after Theorem thm:quant). Statement. (1) Theorem thm:quant is the twist-and-lower argument of Section sec:proof read the other way: the level is raised from N_S to N, at the price of the level-raising Euler factors. For l = 3 a single I_n^* prime q >= 5 forces 3 | deg varphi_E (3 | q^2 - 1), and so does an I_0^* prime with q = 1 mod 3; neither needs the hypothesis on Q(sqrt(-3)), neither is in Theorem thm:qual. (2) If l is not in S_f, then v_l(deg varphi_E) = v_l(r_{E_S}) + sum_{q in S} e_l(E, q), and every inequality in the chain of Steps 1 to 4 is an equality (in particular v(eta^Hecke(g^S)) = v(eta^S_{g,lambda}), a freeness-type statement for the depleted form obtained without multiplicity one). Proof: (eq:identity), an equality (A4.12), for E and for the optimal curve of the class of E_S (l not in S_{f_{E_S}}: twisting preserves absolute irreducibility over Q(sqrt(l*))); the Selmer groups coincide (ad T twist-invariant); the Tamagawa factors coincide (trivial at I_0^* primes by point 64; at I_n^* primes E_S is multiplicative with the same Tate parameter); the Euler sums differ by exactly sum_{q in S} e_l(E, q) (outside S the twist is unramified; at q in S, E_S is good or multiplicative and contributes 0). (3) Step 1 fails for II, IV, III and duals: the twist does not lower the level, f is its own depletion, Theorem thm:dfg14 gives no factor; these terms come only through Theorem thm:dfg37, i.e. with l not in S_f, and Section sec:anomaly shows the hypothesis cannot be dropped (3Ns curves fail through a type III term). **Commentary (important_notes.md point 74; B26).** The paper's (2) concludes "the first inequality of Theorem thm:quant is an equality when H^1(X_0(N), Z_l)_m is free, that is, under multiplicity one at level N". Correct statement: equality whenever l is not in S_f, with no multiplicity-one hypothesis (proof above). Data: equality in all 25,477 pairs below conductor 10,000 (15,607 with 3 in S), and also in the 8 pairs with image 3Ns, which lie outside the theorem. The first version of this item (and \Cref{ch:moddeg} 9.5(c), 10.4(b), \Cref{ch:modgalrep} 10.4) said the equality still needs multiplicity one; that claim is withdrawn.

23.1.6 A5. The paper's Section 5: the boundary of the hypothesis (Section 21.14)

A5.1 proof-prop-ns (proposition (proof of Proposition prop:ns); *proved here* (full proof \Cref{ch:galrep} 8.3); reference: Zywina15 (by section, point 4); \Cref{ch:groups} 5.7, 6.3, 6.5 to 6.8, \Cref{ch:reps} 1.4, 2.5, 6.2, 6.6, 8.2, \Cref{ch:galois} galois:quartic-on-lines, galois:index-two-restriction, \Cref{ch:ellcurves} 4.5; paper use: Proposition prop:ns, Section sec:anomaly, Conjecture conj:euler). Statement: A1.12. Expanded proof, step by step. (1) det rho bar_{E,3} = chi bar_3 = chi_{-3} (Weil pairing) cuts out Q(sqrt(-3)), so rho bar(G_{Q(sqrt(-3))}) = H cap SL_2(F_3). (2) Possible H: E[3] irreducible, det surjective, and H contains rho bar(c) of determinant -1 and trace 0; pure group theory (enumeration of the 55 subgroups of GL_2(F_3), \Cref{ch:groups} 6.3, 6.7) leaves GL_2(F_3) (48), N_ns(F_3) (16, semidihedral), N_s(F_3) (8, dihedral); the nonsplit Cartan C_ns(F_3) = F_9^x is excluded (its elements of determinant -1 have order 8, hence nonzero trace). (3) Determinant-one subgroups: SL_2(F_3), Q_8, C_4 = <J>, J = (0, 1; -1, 0). (4) SL_2(F_3) and Q_8 act absolutely irreducibly. (5) C_4 is abelian; X^2 + 1 is irreducible over F_3, so the restriction to G_{Q(sqrt(-3))} is irreducible over F_3 but not absolutely irreducible: over F_9 it is psi + psi^{-1} with psi of order 4, values in F_9^x (\Cref{ch:reps} 2.5). (6) Frobenius reciprocity: rho bar tensor F_9 = Ind psi; a representation induced from an index-two subgroup is invariant under the twist by that subgroup's character, so rho bar tensor F_9 = (rho bar tensor chi_{-3}) tensor F_9, and the isomorphism descends to F_3 (\Cref{ch:reps} 1.4(ii)); hence H^0(Q, ad rho bar(1)) != 0 and h_3(E) = 1 (A4.11). (7) Lines: the four roots of psi_3 are the x-coordinates of the four lines of E[3]; N_s(F_3) has two orbits of size two, N_ns(F_3) and GL_2(F_3) are transitive; psi_3 has no rational root (E[3] irreducible); so psi_3 is (leading coefficient 3) times two irreducible quadratics in the first case and irreducible otherwise. Examples: 726a1, psi_3 = (x^2 - x - 74)(3x^2 + 8x + 20) (PARI); 6912d1 (y^2 = x^3 - 6x - 8), 3(x^2 - 4x - 2)(x^2 + 4x + 6); 11a1, irreducible with group S_4. Where the hypotheses enter: "E[3] irreducible" in (2) and (7); "without CM" nowhere. **Commentary (important_notes.md point 1).** The paper justifies (4) by "they are non-abelian and act faithfully". Not enough (a Borel subgroup is non-abelian, faithful and reducible). Missing step: a subgroup fixing a line over Fbar_3 lies in a Borel subgroup, so its commutator subgroup lies in the unipotent radical, abelian of exponent 3 (\Cref{ch:groups} 5.7); Q_8 has commutator subgroup {+-1} (not unipotent) and SL_2(F_3) has commutator subgroup

Q_8 (not abelian). The book inserts this sentence. **Commentary (important_notes.md point 33).** "rho bar = Ind psi" and "induced from Gal(Qbar/Q(sqrt(-3)))" are statements over F_9: the restriction is irreducible over F_3, and psi takes values in F_9. Write rho bar tensor F_9 = Ind psi; the isomorphism rho bar = rho bar tensor chi_{-3} then descends to F_3, as in (6). **Commentary (important_notes.md point 2).** With the standard models (\Cref{ch:groups} 6.2), N_s(F_3) lies in N_ns(F_3) with index two, and N_s(F_5) in a subgroup of order 96 with image S_4 in PGL_2(F_5). The paper does not claim maximality; the book warns readers importing the large-l picture not to call N_s maximal at l = 3, 5. **Commentary (important_notes.md point 4).** The theorem numbers in Zywina15 (list of mod 3 images) and Serre72 (Dickson) are cited by section; step (2) is proved directly, so nothing depends on them. **Commentary (important_notes.md point 9).** "Without complex multiplication" is not used: the list in (2) follows from the surjective determinant, irreducibility and complex conjugation. CM curves with 3 split in the CM field have image in N_s(F_3) and 2 + 2 factorisation (121b1, CM by Q(sqrt(-11)); 13 CM optimal curves below conductor 10,000 have ns3 = 1). The book drops the hypothesis.

A5.2 rem-ns (remark (the paper's Remark rem:ns); the displayed identity *proved here* CONDITIONALLY from Lemma lem:dfg21, Lemma lem:fpr, Lemma lem:h0; the 3Ns structure *proved here* from \Cref{ch:galrep} 8.4; data *stated*; reference: \Cref{ch:blochkato} 12.1 to 12.3, \Cref{ch:congruence} 13.1, 13.2; paper use: Remark rem:ns, Conjecture conj:euler, Open question 2). Statement. When l = 3 and the image is N_s(F_3), Theorem thm:dfg37 does not apply, but IF H^1_f(Q, V^D) = 0 the sequence of Lemma lem:dfg21 gives length H^1_Sigma = length H^1_f(Q, W) + sum_{p in Sigma}(v_l(L_p(A_f, 1)^{-1}) + v_l(Tam_p(A))) - h_3(E), with h_3(E) = 1 (H^0(Q, V^D) = 0 is automatic, point 63). If v(eta^Sigma) = length H^1_Sigma still held, (eq:identity) would hold with h_3(E) subtracted: the shape of Conjecture conj:euler. Data: h_3(E) = 1 for every 3Ns curve; (eq:thm) fails for 31%, 23%, 18% of them on the three ranges, always by exactly one, never for another curve. Structure of 3Ns curves (important_notes.md point 48; \Cref{ch:galrep} 8.4, theorems): for every prime p != 3, rho bar(I_p) subset C_4; so every multiplicative p != 3 has 3 | n_p (t_3(E, p) >= 1), no p >= 5 has type II, II^*, no p != 3 has type IV, IV^*; the set R of primes p != 3 with rho bar(I_p) = C_4 is nonempty, and for p in R the image of inertia on T_l E is cyclic of order 4, p >= 5 has type III or III^*, e_3(E, p) = v_3(p - chi_{-3}(p)) >= 1 with chi_{-4}(p) = chi_{-3}(p), i.e. p = 1 or 11 mod 12; for p = 2 in R, P_2 = 1 + 2X; a_p = 0 mod 3 for p = 2 mod 3 not dividing 3N; if 3 does not divide N, E is ordinary at 3. **Commentary (important_notes.md point 67).** The paper's remark is phrased conditionally ("whenever"), but Section 1.3 presents the correction term as what Lemma 2.1 of DFG04 produces. The needed vanishing H^1_f(Q, V^D) = 0 (equivalently H^1_f(Q, ad V_3 E) = 0) is DFG04 Theorem 2.7, proved only for l not in S_f; 3Ns curves are exactly l = 3 in S_f. The book states the identity as conditional on this vanishing (predicted by the Bloch-Kato conjecture), and cites no later result unless checked. **Commentary (important_notes.md point 48).** Section sec:anomaly reports as observations that every 3Ns curve has a type III or III^* prime with positive e_3, that the residue classes are 1 and 11 mod 12 (with some q = 2), and that each violation involves a prime p = -1 mod 3 with t_3(E, p) >= 1. The first two are theorems (structure above), checked on all 120 non-CM 3Ns curves below conductor 10,000 (100 with such a prime p >= 5, 20 with p = 2); the third is explained in part (every multiplicative prime p != 3 of a 3Ns curve has t_3(E, p) >= 1) and observed for all 36 violators below 10,000 (\Cref{ch:computation} 11.3), but that every violator has such a prime p = -1 mod 3 is not proved. The book states the first two as a proposition and the third as an observation.

23.1.7 A6. The paper's Section 6: the computations (Sections 21.15 to 21.17)

General note for all of A6. The paper's tables are \input from files tables/*.tex that are NOT in this repository. The book describes each table from its caption and the numbers quoted in the text, and never invents table entries. Every count of the paper is *stated* (attributed to the paper). The book's own reproduction on conductor below 10,000 (38,042 optimal curves, PARI/GP 2.17.2, Cremona's ecdata files; \Cref{ch:computation} Section 11 and Part 7) is quoted beside the paper's numbers where it helps, always labelled as the book's, with the warning that rates fall with N and are not comparable across

ranges (\Cref{ch:computation} 11.8). Chapter 22 covers PARI, ecdata and the protocol; here the tour explains what each statement means and what it tests.

A6.1 data-protocol (data description; *stated*; reference: Cremona97, ecdata commit 25cec5e, PARI 2.15, Sutherland16; \Cref{ch:computation} 4.1, 4.3, 9.1, 9.2; paper use: Section sec:data, Section 6.1). Content. Tables: every curve of conductor below 500000 with modular degrees, isogeny classes, torsion, mod l images (Sutherland's labels); the per-prime Kodaira symbols and Tamagawa numbers are recomputed with PARI from the a-invariants (Cremona's files contain none; the modular-degree file is degphi, not alldegphi, important_notes.md B27). Optimality determined for every class with N <= 400000 and for 359,009 further classes up to 500000 (curve number 1 is not always optimal: 990h3, not 990h1, is the optimal curve of 990h, \Cref{ch:computation} R0). Ranges: working set (N <= 300000), hold-out, extension, out-of-table set. Order of discovery as in the paper. Test status, read off the paper (the book prints \Cref{ch:computation} 9.2 as a table): (eq:main) and Conjecture conj:eis: found on the working set, blind-tested once on the hold-out, then on the extension. (eq:twist): replaced a weaker form on the working set and was "tested once" on hold-out and extension; it is blind only if it was fixed before the hold-out was opened for (eq:main), which the text does not say, and the paper does not state the blind status of the out-of-table curves for it. Conjecture conj:euler (the Euler terms and the weights w_3): hold-out and extension are a confirmation only; the out-of-table curves, generated before the rule was formulated, are its blind test. The correction term h_l: never blind-tested, since the out-of-table set contains no 3Ns curve. The III-free form: a reformulation chosen after the 3Ns exceptions. Lemmas lem:euler, lem:twist, Proposition prop:forms, h_3 = 1: theorems; the data are checks of computations. **Commentary (important_notes.md point 77; B27).** The paper does not say that the correction term h_l was fitted on, and only confirmed by, data already seen, nor the blind status of the out-of-table curves for (eq:twist). The first version of this item (and Appendix A) called (eq:twist) blind-tested without this caveat. Correct statement: as above. The book adds both caveats. **Commentary (important_notes.md point 25).** "Every optimal curve of conductor at most 500000" (abstract) means the classes where optimality is known; the extension counts are over the 359,009 determined classes.

A6.2 table-uniform (how to read Table tab:uniform; *stated*; \Cref{ch:computation} 11.1; paper use: Section 6.2). Columns (from the caption): for each range and l in {3, 5, 7, 11, 13}: curves with E[l] irreducible, not CM, positive bound in (eq:rule); violations (all zero); equality rate; violations when h_l is dropped (all 3Ns curves at l = 3); the "III-free" rule (type III, III^* terms omitted, no correction): positive bounds, violations, equality rate. Reading: the III-free rule has no exception because the 3Ns failures always involve a type III term (A5.2: a theorem). Book's small range (N < 10,000, l = 3): 33,851 curves, 20,317 with a positive bound, 0 violations, equality 71.0%, 36 violations without h, III-free 20,127 positive, 0 violations. **Commentary (important_notes.md point 80, second item).** The number of 3Ns violations "when the correction term is dropped" depends on whether the terms at q = l (the t_l(E, l) and w_3) are included: below conductor 10,000 it is 36 with them and 24 with (eq:thm) read literally (terms at q != l only; 726a1 then has bound 2 and no violation). The paper does not say which reading its counts use. The book states the reading it uses (with the q = l terms, as rule(E, 3, 1) without ns3) and gives both numbers.

A6.3 calibration (data argument (Section sec:sharp); *stated*, arithmetic *checked here*; \Cref{ch:computation} 10.1 to 10.6; paper use: Section sec:sharp). Content. Among 366,151 working-set curves with E[3] irreducible, not CM and bound 0, 53.2% have 3 not dividing deg varphi_E; under independence the 867,578 curves with a positive bound would give about 605,000 violations (69.7% of them); there are none. Equality rates 57.5%, 75.6%, 84.1%, 91.6%, 92.4% at l = 3, 5, 7, 11, 13; mean excess 0.63 at l = 3. Odd part of the degree: equal to prod l^{b_l} for 1.85% of 1,226,216 curves. Expansion: (1) the null model sum_E P(v < b(E)) from the bound-0 population; the v = 0 part alone gives at least 0.532 x 867,578 = 461,551. (2) Book's range: 14,275 expected (70.3%), 0 observed; a stratified permutation null gives about 7,600; the reference population changes the number by a factor of two, the conclusion not at all. (3) Under Theorem thm:main (now an equality) the excess over the explicit bound IS the Selmer length (A4.12); the paper's "plus the excess of the Tamagawa factors over t_l" is the same statement with the Tamagawa excess counted separately. **Commentary (important_notes.md point 79).** The calibration shows that the statement is strong (far from vacuous),

not that it is valid: on the working set the rule was fitted, so validity rests on the blind tests (A6.1). The expected count depends on the reference population, and the odd-part statistic is strongly range-dependent (1.85% on the working set, 17.1% below 10,000). The book says so.

A6.4 tamagawa-test (data (Section sec:tam); *stated*, percentages *checked*; \Cref{ch:blochkato} 13.1, 13.2, \Cref{ch:computation} 11.5; paper use: Section sec:tam, Remark rem:ko(1), Lemma lem:tamfactor). Content. At a prime of type I_n with l | n and q = 1 mod l the bound is raised by min(v_l(n), v_l(q - 1), v_l(c(u))). l = 3: 46,222 curves, 12,095 with a positive extra term (total 12,125), no exception, equality from 45.0% to 60.6%; l = 5: 10,940, 1,646, equality 67.0% to 78.7%. Book's range (only I_n primes, as the paper): l = 3: 989 curves, equality 55.6% to 66.9%; with I_n^* primes included: 1,050 curves, 56.8% to 68.0%; no violation. Reading: given the equality (A4.12), a violation is impossible, and the proportion of equality is the proportion of curves with H^1_f(Q, W) = 0; the test checks the Tamagawa term of the identity. **Commentary (important_notes.md points 23, 68).** c_q here is the class c(u) of Lemma lem:tamfactor, not the Tamagawa number (point 23). The test can include potentially multiplicative additive (I_n^*) primes, where the formula also applies (point 68, A4.12 Step 7); and "the extra divisibility of the modular degree is exactly where the Tamagawa factor of the adjoint exceeds l^{t_l}" overstates: the Selmer length also contributes extra divisibility. The book rewords both.

A6.5 isolated-terms (data (Section sec:iso, Table tab:iso); *stated*; \Cref{ch:computation} 11.4; paper use: Section sec:iso). Content. Working-set curves with exactly one additive prime q >= 5, q != l, and no multiplicative prime with l | n: in every cell (type, e_l(E, q)), for l = 3, 5, 7, the minimum of v_l(deg) equals e_l, and for I_n^* the excess over e_l has minimum v_l(n); cells with e = 0 (III: 5,534; I_0^*: 5,394) have minimum 0; I_0^* up to e = 6. Multiplicative primes alone: min(v_l(deg) - v_l(n)) = 0 for every v_l(q^2 - 1) <= 6. Two primes of type II/II^*: 9 | deg for all 2,502 curves, and never via a congruence with the own twist. Book's range: the same pattern in every cell. Reading Table tab:iso: rows Kodaira type, columns e_3(E, q), entries minimum (number of curves).

A6.6 corollary-forms (data (Section sec:forms, Table tab:counts); *stated*; \Cref{ch:computation} 11.2; paper use: Section sec:forms, Proposition prop:forms, Conjecture conj:eis). Content. Positive bounds and violations (none) of (eq:main) and (eq:twist) per range and l; Eisenstein column. Equality rates (eq:twist) 39.4%, 71.1%, 81.2%, 91.5% at l = 3, 5, 7, 13; (eq:main) 36.9%, 69.7%, 80.4%, 91.2%; the rule bounds 867,578 curves instead of 705,563 at l = 3. Where the hypothesis of Theorem thm:main fails on the working set: 1,170 3Ns curves; curves with l | N; "the curves whose mod l representation is induced from Q(sqrt(l*)) with l >= 5, which we did not single out". **Commentary (important_notes.md point 47).** That last class consists only of curves with l | N: for l >= 5 not dividing N and E[l] irreducible, l is automatically not in S_f (A4.6). The book says so. **Commentary (important_notes.md point 43).** With A1.8, Proposition prop:forms has no exception, so the zero violations of (eq:main) and (eq:twist) under the hypothesis are predicted by a theorem; the data test them outside it.

A6.7 ns3-boundary (data (Section sec:anomaly, Table tab:ns); *stated*, with the theoretical parts *proved here* (A4.11, A5.1, A5.2); \Cref{ch:computation} 11.3, \Cref{ch:galrep} 8.4; paper use: Section sec:anomaly, Remark rem:ns, Conjecture conj:euler). Content. (eq:thm) with all types and no correction: 364, 68, 43, 0 violations on the four ranges, all at l = 3, all 3Ns curves (each with psi_3 = two quadratics); none by GL_2(F_3) or N_ns(F_3) curves. Every 3Ns curve has a type III/III^* prime with positive e_3; the inequality fails for 31%, 23%, 18% of them, always by one; the violation involves a prime p = -1 mod 3 with t_3(E, p) >= 1. h_3(E) = 1 for every 3Ns curve. Type III primes of violators: q = 1 and q = 11 mod 12 in comparable numbers, some q = 2. Book's range: 90 optimal non-CM 3Ns curves, all with a positive type III term (74 at a prime q >= 5, 16 at q = 2: conductors 3840, 4864, 6912), 36 violators with the q = l terms (24 without), residues 1 (12), 11 (20), q = 2 (4), shortfall always 1; every violator has a prime p = -1 mod 3 with t_3(E, p) >= 1. Example: 726a1 (B4). **Commentary (important_notes.md point 5, second half).** "h_3(E) <= 1 because some prime p < 1223 inert in Q(sqrt(-3)) has 9 not dividing a_p" is replaced by "h_3(E) <= 1 by Lemma lem:h0" (A4.11 step (5)). **Commentary (important_notes.md point 48).** The observations on the type III primes and their residues mod 12 are theorems (A5.2); the book states them as such and keeps the counts as illustration. **Commentary (important_notes.md point 80).** Which reading of

(eq:thm) the counts use: see A6.2. **Commentary (important_notes.md point 3).** 3Ns is checked against Cremona's `galrep` labels (`ns3` agrees on all non-CM curves below `10,000`); the definition against Sutherland16 is still to be checked; the book always gives the intrinsic description.

A6.8 `wild-primes` (data (Section sec:wild, Table tab:isolated); *stated*, with the explanations *proved* (A1.8, A2.6); reference: `\Cref{ch:lfunctions}` 10.1, 10.2, `\Cref{ch:computation}` 6.6, R5; paper use: Section sec:wild, Remark rem:wild, Conjecture conj:euler, Proposition prop:forms). Content. Table tab:isolated: minima of `v_3(deg)` by additive type and number of primes (untwisted rows), and by total weight of (eq:twist) (twisted rows). Findings: `2` contributes nothing through `II, II^*` for `14,182` of the `44,672` curves whose only candidate is such a prime, and nothing through `I_n^*` with `3 || n` (with `9 | n` the minimum is `1`); explained by Definition def:terms with PARI's factor; with the weights of (eq:twist) at `2` counted as if `2` were odd, (eq:twist) fails thousands of times per range; with Definition def:terms never. At `q = 3`, potentially good, `l = 3`: minima `1` for `II, IV, III^*`, `2` for `II^*, IV^*`, `0` for `III, I_0^*`: the weights `w_3` (reproduced exactly in the book's range, `\Cref{ch:computation}` R5). For `l = 5, 7` nothing beyond Definition def:terms is needed. **Commentary (important_notes.md points 50 and 69).** The paper explains the minima by "a type `II` at `2` has `P_2 = 1` and `e_3 = 0`", which is false in general (A2.6); the correct explanation is that `P_2` depends on the shape of the inertia image, not on the type, and that `P_2 = 1` is the most frequent shape for type `II` at `2`. The sentence "For `l = 5, 7` no term is needed at `q = l` or at `q = 2, 3` beyond Definition def:terms" is right (the term is in def:terms), but the reader must not infer that `2` never contributes at `l >= 5`: 592c1 has `e_5(E, 2) = 1` (A2.6). "A type `IV` or `IV^*` at `2` has `e_3 = 1`" is a theorem (A1.8). **Commentary (important_notes.md point 53).** "A type `I_n^*` at `2` has `t_3 = v_3(-v_2(j))` when potentially multiplicative" uses `-v_2(j)`, not the index of the symbol, as A1.1 defines it.

A6.9 `out-of-table` (data (Section sec:oot, Table tab:oot); *stated*; `\Cref{ch:ellcurves}` 9.10, `\Cref{ch:computation}` 9.1; paper use: Section sec:oot). Content. Curves with prescribed local types and `500000 < N <= 4 x 10^6` (families with a type `IV/IV^*` prime, with two or three type `II` primes, mixed types, random), kept if the isogeny class is trivial (so optimal and `E[l]` irreducible for every `l`, `\Cref{ch:ellcurves}` 9.10) and not CM; degrees by `ellmoddegree`. None has image 3Ns; Conjecture conj:euler and both corollary forms hold for all `893`; equality in (eq:rule) for `52.5%` of the `828` with a positive bound at `l = 3` (`435` curves). **Commentary (important_notes.md point 77).** This is the blind test of the Euler terms of the rule, and only of them: with no 3Ns curve among the `893`, the correction term `h_l` was never blind-tested (A6.1).

A6.10 `eisenstein` (data and conjecture (Section sec:eis); *stated*, with `D <= 2` *proved* (A1.13); reference: `\Cref{ch:galrep}` 9.1 to 9.3, `\Cref{ch:moddeg}` 11.1 to 11.5, `\Cref{ch:computation}` 11.6; paper use: Section sec:eis, Conjecture conj:eis, Open question 4). Content. Working set, `l = 3`: `75,430` optimal non-CM curves with a rational `3`-isogeny, `67,845` with a positive bound in (eq:main), `2,777` with `D > 0`, `D = 2` for `22` (`17` with `E[3] = Z/3 + mu_3`, `5` with a point of order `9`). No curve without a rational `3`-torsion point has a deficit. None of the `2,065` curves with `N <= 400000` and image 3Cs.1.1 has a point of order `9`; the `18` optimal curves with a point of order `9` have image 3B.1.1. Torsion bound attained with `D > 0` in `2,773` cases. `l = 5, 7`: `D <= 1`, only with rational `l`-torsion (`192` and `18` cases). Against (eq:twist) the deficit is not controlled by torsion, so the conjecture is stated for (eq:main) only. Book's range: `4,002` Eisenstein curves, `431` with `D > 0`, `D = 2` for `14` (twelve with image 3Cs.1.1, two with a point of order `9`: 714i1, 6942n1), no violation. **Commentary (important_notes.md point 76; B27).** Two descriptive sentences are false as written: (i) the curves with `E[3] = Z/3 + mu_3` and `D = 2` are said to have torsion `Z/3`, but 14a1 (image 3Cs.1.1, torsion `Z/6`, types `I_6` at `2` and `I_3` at `7`, `deg 1`) has `D = 2`; (ii) "with a rational point of order `3` and mod `3` image of Borel type, `D <= 1`" is false for curves with a point of order `9`: 714i1 (torsion `Z/9`, image 3B.1.1, `deg 1080 = 2^3 3^3 5`, `D = 2`) and 6942n1 (torsion `Z/9`, `deg 65664 = 2^7 3^3 19`, `D = 2`). The paper's next clause names the order-9 case as the other source of `D = 2`, so (ii) is a wording slip: "without a point of order `9`". Neither affects Conjecture conj:eis. The first version of this item repeated both slips; the book corrects them. **Commentary (important_notes.md point 17).** "We have not proved that the two cannot coexist": they cannot (A1.13); `D <= 2` is a theorem given the first bound. **Commentary (important_notes.md point 44).** "The split Cartan image 3Cs.1.1": 3Cs.1.1 is the subgroup `{diag(1, +-1)}`, i.e. `E[3] = Z/3 + mu_3`; the split Cartan image is 3Cs (A1.13). Expansion for the book: what "Eisenstein" means (reducible `rho bar`, maximal ideal containing `T_p -`

1 - p), why Section 4 fails there (Ihara's lemma, Lemma lem:hecke's non-Eisenstein m, DFG04), the role of rational torsion and the cuspidal subgroup (Mazur77), and the small examples 539a1 (image 3B, I_2^* at 7, e_3 = 1, v_3(320) = 0, while (eq:main) holds) and 11a1 at l = 5 (rule(E, 5) = [0, 1], a fifth-power unit part, D = 1 = v_5(#E(Q)_tors); \Cref{ch:computation} 6.7): in the Eisenstein case the additive Euler terms do not contribute reliably.

23.1.8 A7. The paper's Section 7: the prime 2 (Section 21.18)

A7.1 prime-two (data and literature (Section sec:two); *stated; reference*: Watkins02, Dummigan06, CalegariEmerton09, Yazdani11, CaroPasten22, HatleyKundu, BKP; \Cref{ch:moddeg} 12.3, 12.4, \Cref{ch:computation} R9; paper use: Section sec:two, Open question 5). Content. For working-set optimal curves without a rational 2-isogeny and not CM: v_2(deg) >= sum_{q || N} v_2(n_q) + #{q : type III, III^*, I_0^* or I_n^*} with no exception (964,432, 325,510, 313,741 curves); counting I_0^* with weight 2 fails (123, 26, 20). Isolated minima: III, III^*, I_n^* contribute 3, II, IV, I_0^* contribute 1, "in line with" Calegari-Emerton. Not reconciled with the literature or with Euler factors at l = 2 (outside DFG04). Book's range: the inequality holds on all 25,086 curves; with I_0^* counted twice it fails 6 times. What the book says: l = 2 is outside every theorem of the paper (l odd is used in Lemma lem:tamfactor, the trace form, the real place, S_f); this section is data plus pointers; Watkins's conjecture stated precisely (\Cref{ch:moddeg} 12.3). **Commentary (important_notes.md point 57).** The paper paraphrases Calegari-Emerton as "a curve of odd modular degree has at most two odd bad primes and a rational 2-torsion point or prime conductor". 27a1 and 243b1 have odd modular degree, torsion Z/3 and non-prime conductor; both have CM (j = 0). So the published theorem excludes CM curves or has extra cases; the paraphrase must say so. The book copies the statement from CalegariEmerton09 with its exact hypotheses (\Cref{ch:moddeg} 12.4, still to be checked against the source).

23.1.9 A8. Open questions (Section 21.18)

A8.1 open-questions (list; *stated*, with the book's partial answers; \Cref{ch:congruence} 13.2, 13.3, \Cref{ch:blochkato} 9.3 to 9.5, 12.3; paper use: Open questions 1 to 5). (1) Extend Theorem thm:main to l | N, identifying w_3 with a local term at l. Book: at l | N Step 1 fails (r_E != deg possible: 99a1, 726a1, 539d1), Step 3 fails (l in S_f; Fontaine-Laffaille needs l not dividing N), and the Euler factor at q = l is not the relevant local quantity; what would replace it is a local deformation condition at l (\Cref{ch:congruence} 13.3). (2) Prove the corrected identity of Remark rem:ns for 3Ns curves. Book: two inputs are missing, H^1_f(Q, ad V_3 E) = 0 (point 67) and Taylor-Wiles for a residual representation induced from Q(sqrt(-3)); the data (failures by exactly one; equality of Theorem thm:quant in the 8 3Ns pairs) are consistent with it. (3) Compute Tam_q(ad T) at additive primes and the Selmer length. Book (important_notes.md point 64): the first half is ANSWERED: Tam_q(ad T) is trivial at every potentially good prime (\Cref{ch:blochkato} 9.3), and at potentially multiplicative primes it is given by Lemma lem:tamfactor (A4.12 Step 7); with the equality of A4.12 the Selmer length is then v_l(deg varphi_E) - B^expl_l(E), computed curve by curve (below 10,000, at l = 3: excess 0, 1, 2, 3, >= 4 for 8,688, 3,288, 942, 248, 37 of 13,203 curves, never negative); what remains is to explain that length. (4) The Eisenstein deficit and the cuspidal subgroup (Mazur77; \Cref{ch:moddeg} 11). (5) Section sec:two and Watkins's conjecture. **Commentary (important_notes.md point 64).** Open question 3 is half answered; the book says which half and reduces the question to the Selmer group.

23.1.10 A9. The appendix: reproducibility and the function rule(E, l) (Section 21.19)

A9.1 pari-rule (construction (the paper's scripts/euler_rule.gp); *explained line by line*, outputs verified in PARI/GP 2.17.2; reference: PARI manual (ellglobalred, elllocalred, lfunsympow, lfuneuler, ellmoddegree, elldivpol, factor); \Cref{ch:reduction} 5.5, \Cref{ch:lfunctions} 11.1, 11.2, \Cref{ch:computation} 5.1 to 5.4, 6.1 to 6.7, R1, R2; paper use: the appendix). Line by line (rule(E, l,

{iii = 0})): (1) R = ellglobalred(E): [N, [u,r,s,t], c, faN, L]; P = R[4][,1] the bad primes; K = R[5] the elllocalred data [f_p, kod, [u,r,s,t], c_p], so K[i][2] is the Kodaira code at P[i]. (2) S = lfunsympow(E, 2): L(Sym f, s), with L_p(Sym f, s) = L_p(A_f, s - 1); b the bound, j = E.j. (3) Loop over bad q: k the code, vj = valuation(j, q). (4) If q != l: F = 1/lfuneuler(S, q) is P_q(x) (verified: lfuneuler(S, 11) = 1/(11*x + 1) for 121a1); r = P_q(q^{-2}) = L_q(A_f, 1)^{-1}; if k >= 5 (type I_n, n >= 1) divide by the naive factor 1 - q^{-2}, so r = 1, e = 0; at additive q the naive factor is 1; e = v_l(r) unless the type is III or III^* (abs(k) == 3) and iii = 0: Definition def:terms verbatim. (5) If q == l = 3: e = w_3(E) (codes 4, 2, -3 give 1; -4, -2 give 2; others 0). (6) t = v_l(-vj) if vj < 0, at every bad q, q = l included (A1.1). (7) b += e + t. (8) If iii and l = 3, subtract ns3(E) (the correction h_3(E), which is 1 exactly for image 3Ns, A4.11, A5.1). (9) Return [v_l(ellmoddegree(E)), b]. ns3(E): factor psi_3 = elldivpol(E, 3) over Q (PARI drops the content 3); true iff two irreducible factors, the first of degree 2 (pattern 2 + 2). For non-CM curves with E[3] irreducible this is image 3Ns (A1.12 data check). Modes: rule(E, 1) is the "III-free" bound without correction; rule(E, l, 1) is (eq:rule). Verified outputs (PARI/GP 2.17.2, \Cref{ch:computation} R1): 121a1 [1, 1] (both modes); 726a1 [2, 2] in both modes, the type III term and the correction cancelling; 539c1 [2, 2] at l = 3; 980i1 [2, 2] at l = 3 and [1, 1] at l = 5; 99a1 [0, 0]; 336f1 [1, 1]; 1232l1 [2, 2]; 592c1 [1, 1] at l = 5; 11a1 [0, 1] at l = 5 (Eisenstein). The paper ran PARI 2.15; the book's runs are 2.17.2 (W6). Points the book must make. (a) ellmoddegree(E) returns the degree of the modular parametrisation of the GIVEN model (for a non-optimal curve, not the paper's quantity): 990h1 gives [4, 3] while the optimal 990h3 gives [3, 3]; 11a3 at l = 5 gives [-1, 0]. (b) valuation(j, q) of j = 0 is +oo, so t = 0. (c) The test abs(k) != 3 excludes III and III^* at q != l, while at q = l = 3 the III^* weight w_3 = 1 is counted even when iii = 0 (it is not an Euler term). (d) The PARI codes n + 4, -(n + 4) are valid for n >= 1 only (A9.2). **Commentary (important_notes.md point 78).** rule gives wrong-looking answers when misused: on a non-optimal curve (990h1: [4, 3]; 990h3, the optimal curve of the class, [3, 3]; 11a3 at l = 5: [-1, 0]) and at an Eisenstein prime (11a1 at l = 5: [0, 1]). The appendix should say that E must be the optimal curve and E[l] irreducible (and, for the statement of Conjecture conj:euler, E without CM). The book says so. **Commentary (important_notes.md point 80, first item).** "ellmoddegree needs parisizemax of a few hundred megabytes for conductors in the thousands": in PARI 2.17.2 at most about 32 MB suffices below conductor 10,000 (true for 2.15 at larger conductors). Low importance; one clause.

A9.2 kodaira-codes (convention; *stated*; reference: PARI manual (elllocalred), \Cref{ch:reduction} 5.5, Appendix B B-C11(c); paper use: the appendix). Table (PARI code: Kodaira symbol, #Phig, conductor exponent at q >= 5, v_q(Delta_min) at q >= 5): 1: I_0, 1, 0, 0; n + 4 (n >= 1): I_n, n, 1, n; 2: II, 1, 2, 2; 3: III, 2, 2, 3; 4: IV, 3, 2, 4; -1: I_0^*, 4, 2, 6; -(n + 4) (n >= 1): I_n^*, 4, 2, n + 6; -2: II^*, 1, 2, 10; -3: III^*, 2, 2, 9; -4: IV^*, 3, 2, 8. (Hence k >= 5 detects multiplicative reduction.) **Commentary (important_notes.md point 52).** The paper's list "n + 4 for I_n, -(n + 4) for I_n^*" is valid for n >= 1 only; I_0 is 1 and I_0^* is -1, not 4 and -4 (which are IV and IV^*). Low importance; the table above is the correction.

A9.3 reproducibility (description; *stated*; \Cref{ch:computation} 12.1, 12.2; paper use: the appendix's first and last paragraphs, the statement on the use of an AI system). Content: the repository and directory named in the paper (scripts for Euler factors, the rule, the boundary, the Tamagawa test, the corollary forms; pinned data commit; rerun from a clean clone reproduced Table tab:counts; Lemma lem:twist(ii) checked on 4,577 random pairs; Lemma lem:euler against lfunsympow on every curve). The book lists these as the paper's claims and does not reproduce the scripts (the contents of the repository were not inspected); the book's own reproduction is \Cref{ch:computation} Part 7 (Lemma lem:twist on 85,848 pairs, Lemma lem:euler on 67,745, ellmoddegree against Cremona's degphi file on all 38,042 optimal curves below 10,000, no mismatch). The statement on the use of an AI system is part of the paper; the tour mentions it in one sentence in Section 21.1 (how the paper was produced, and that every count can be rerun), with no further commentary.

23.2 Part B. Devices of the tour (Sections 21.20 and 21.21)

23.2.1 B1. Dependency diagram of the paper's results, as corrected (Section 21.20)

To be drawn as a `tikzcd` or plain `tabular` diagram (no colour needed). Arrows mean "is used in the proof of". Imported statements in square brackets; the book's additions (results the paper does not have, which change the structure) are marked `(book)`.

```
[Kraus90]                                              => lem:IV (q >= 5); rem:types
[Neron component groups; Ch. 10, 4.5] (book)           => lem:IV at q = 2 (Ch. 11, 4.4) => prop:forms (no e
lem:IV                                                 => thm:qual (a), (b)
[Tate curve; DDT 3.1(e)]                               => lem:tamagawa-exp
lem:tamagawa-exp                                       => thm:qual (c); rem:ko(2); Section 1.5
[Tate's algorithm, Silverman IV.9]                     => lem:twist (i), (ii)
[Atkin-Li 3.1]                                         => lem:twist (iii)
lem:twist                                              => lem:euler; thm:qual (b), (c); thm:twist; thm:quan
[Kraus90] + lem:twist                                  => lem:euler => table of Section 1 => def:terms (val
lem:euler                                              => thm:quant Step 3; prop:forms
[FPR94 I.4.2.2] + [Ch. 16, 17 cohomology]              => lem:tamfactor (all potentially multiplicative q,
[Ch. 18, 9.3: Tam trivial at pot. good q] (book)       => explicit Tamagawa terms in eq:identity; Open ques
lem:tamfactor                                          => thm:main Step 7 (eq:thm); rem:ko(1); sec:tam
[Ribet83 2.2; AU96 2.1]                                => lem:crit
lem:crit                                               => lem:old; lem:hecke (paper's route); thm:twist; th
[Edixhoven92; Atkin-Lehner 3; DDT 3.1(e); Chebotarev; Brauer-Nesbitt] => lem:old
lem:old + [DDT 3.15] + [ARS12 2.2]                     => thm:qual
[DDT 4.17] = lem:ddt417                                => thm:quant Step 2 (and the paper's route to lem:he
[projection formula; Ch. 14, 9.1 to 9.3] (book) + [DFG04 Part 7 checklist] => lem:hecke WITH EQUALITY
[ARS12 2.2] + lem:hecke                                => thm:main Step 1; thm:quant Step 4
[Tilouine97; RibetStein 3.5] => rem:free               => (no longer used by any proof; explains Hecke = co
[DFG04 Prop 1.4(c)] = thm:dfg14                        => thm:main Step 2; thm:quant Step 3
[DFG04 Thm 3.7 = 0.2] = thm:dfg37                      => thm:main Step 3
[DFG04 Thm 2.7] = thm:dfg27                            => H^1_f(Q, V^D) = 0 in lem:dfg21 (H^0(Q, V^D) = 0 i
[DFG04 Lemma 2.1] = lem:dfg21                          => thm:main Step 4; rem:ns (conditionally)
[FPR94; DFG04 2.15, (57)] = lem:fpr                    => thm:main Step 5; rem:ns
[Clifford theory; group theory of GL_2(F_3)]           => prop:ns <=> lem:h0 => thm:main Step 4; conj:euler
[inertia at l: Ch. 11, 5.3, 8.2] (book)                => "l not in S_f" automatic for l >= 5, l not dividi
thm:main (equality)                                    => prop:forms => (eq:main), (eq:twist) under the hyp
thm:twist                                              => thm:quant (inequality)
thm:main (equality) for E and for E_S (book)           => thm:quant with EQUALITY for l not in S_f (remark
lem:dfg21 + lem:fpr + lem:h0 + [H^1_f(Q, V^D) = 0, assumed] => rem:ns => conj:euler (the correction te
[Ch. 9, 9.8: no rational cyclic 27-isogeny] (book) => conj:eis: D <= 2 is a theorem given the first bo
data (Section 6)                                       => conj:euler (w_3, h_l); conj:eis (first bound); se
```

The three spines: (i) local: lem:tamagawa-exp, lem:IV, lem:twist, lem:euler, lem:tamfactor (with the two additions on type `IV` at `2` and Tamagawa factors at potentially good primes); (ii) congruences: lem:crit, lem:old, optimal level, thm:qual; (iii) Selmer: lem:hecke (now an equality, through the projection formula, not through rem:free), thm:dfg14, thm:dfg37, thm:dfg27, lem:dfg21, lem:fpr, lem:h0, thm:main. Theorem thm:quant uses (i) and (ii), and from (iii) only thm:dfg14, lem:ddt417 and lem:hecke; it never uses thm:dfg37, which is why its inequality needs no hypothesis on `Q(sqrt(l*))`; its EQUALITY uses thm:main for `E` and `E_S`, hence `l` not in `S_f`. Multiplicity one (rem:free) feeds nothing.

23.2.2 B2. Hypothesis table, as corrected (Section 21.20)

Aligned with \Cref{ch:congruence} 9.3.

hypothesis	where it is used	what fails without it
l odd	Lemma lem:tamfactor (2 a unit in (t - 1)M); trace form T^D = A(1); real place (H^1(R, T^D) = 0); S_f contains 2; Lemma lem:h0 (complex conjugation gives absolute irreducibility; the determinant argument); optimal-level theorem (DDT 3.15 is for odd l); Lemma lem:old at p = l	everything in Sections 2 to 4 at l = 2; Section sec:two shows a different 2-adic pattern (types III, III^*, I_n^* weigh 3, I_0^* weighs 1), and Watkins's conjecture brings in the rank
l not dividing N (and l^2 not dividing N)	ARS12 Theorem 2.2 (only l^2 not dividing N); DFG04 (l not in S_f; Fontaine-Laffaille at l); Lemma lem:old at p = l; optimal-level theorem (delta = 0, weight 2, trivial nebentypus); Lemmas lem:euler, lem:fpr, lem:tamfactor (q != l)	99a1 (N = 9 * 11): deg = 4, r = 12 (recomputed), so r_E and the degree differ at l = 3, and H^1(X_0(99), Z_3)_m is not free; 726a1 (r = 1584, deg 144, at 11); 539d1 (r = 504, deg 72, at 7); at q = l = 3 the Euler factor is replaced by the empirical weights w_3; no theorem covers l | N, though the data show (eq:rule) there for l >= 5, l not dividing N, nothing beyond E[l] irreducible: the condition is AUTOMATIC (point 47); at l = 3 it fails exactly for image 3Ns, where (eq:thm) fails for 31%, 23%, 18% of the curves, by exactly one (726a1: bound 3 with the q = l terms, v_3(deg) = 2); the correction h_3(E) = 1 repairs it conditionally (Remark rem:ns). Theorem thm:quant's inequality does not need it
rho bar absolutely irreducible on Gal(Qbar/Q(sqrt(l*))) (l not in S_f)	DFG04 Theorem 3.7 (Taylor-Wiles) and Theorem 2.7 (H^1_f(Q, V^D) = 0); Lemma lem:h0 (h_l = 0); optimal-level theorem at l = 3 (Theorem thm:qual); the equality in Theorem thm:quant	
rho bar irreducible (weaker; implied by the previous)	Theorem thm:dfg14 (Ihara's lemma); the non-Eisenstein m (DFG04's integral structure agrees with H^1(X_0(N), Z_l) there); Lemma lem:old (Brauer-Nesbitt); Theorem thm:quant; Theorem thm:qual	Eisenstein case: 539a1 (image 3B, I_2^* at 7): e_3 = 1, v_3(deg) = 0; 539d1 at l = 5 (image 5B.4.1, I_0^* at 7): e_5 = 1, v_5(deg) = 0; 98a1 (I_1^* at 7, E[3] reducible); only the weaker (eq:main) survives, with the torsion-controlled deficit of Conjecture conj:eis (20a1: D = 1 = v_3(#E(Q)_tors); 14a1: D = 2)

hypothesis	where it is used	what fails without it
X_0(N)-optimality	ARS12; the definition of deg varphi_E; Lemma lem:hecke's equality (M_{f,lambda} = varphi_E^* H^1(E) needs the optimal map); rule (ellmoddegree of the given model)	nothing at l when E[l] is irreducible: the curves of the class are linked by isogenies of degree prime to l, so v_l(deg), v_l(n_q), e_l, t_l, ad T and h_l are class invariants (point 27; \Cref{ch:ellcurves} 9.9, \Cref{ch:moddeg} 6.4); it matters for Conjecture conj:eis and for rule (990h1 gives [4, 3], the optimal 990h3 [3, 3])
multiplicity one (H^1(X_0(N), Z_l)_m free over T_m)	in the paper: the equality in (eq:identity) (via Lemma lem:ddt417 in Lemma lem:hecke) and the equality in Theorem thm:quant (remark (2)); in the book: NOWHERE	nothing: (eq:identity) is an equality for l not in S_f by the projection formula (points 39, 56, 73), Theorem thm:quant's middle term needs no freeness (point 39) and its equality follows from (eq:identity) (point 74). Freeness does hold in the paper's range (Remark rem:free) and fails for l | N (99a1 at 3)

The book adds one sentence under the table: for l not dividing 2N, the hypotheses of Theorem thm:main are those of DFG04's Theorem 0.1 for weight 2 and trivial character, i.e. "E[l] irreducible and not (l = 3 and image 3Ns)", plus optimality, which is harmless at l.

23.2.3 B3. Statements imported from the literature, with the paper's numbering (Section 21.20)

Rule for the book (PLAN.md 3): numbers are given as the paper cites them. "Verified" means checked against the published source by an outline agent (for DFG04: the Chapter 20 outline, Part 7, read from the published text, Ann. Sci. Ec. Norm. Sup. (4) 37 (2004), 663-727); "as cited" means taken from the paper and not checked; the writer verifies these before printing or replaces them by section references listed in the Notes. **Warning:** a 2025 arXiv version of DFG04 (arXiv:2512.02348, a longer text dated "August 2001, with minor update December 2025") uses DIFFERENT numbering (its Sections 1.1 to 3.5 do not match the published 1.1 to 3.2); every DFG04 number in the paper and the book refers to the published version, and the book's bibliography entry must say so.

citation (as in the paper)	exact content used	where in the paper	book item	status
DFG04 Theorem 0.1 (= Theorem 2.15)	Bloch-Kato conjecture for A_f, A_f(1) at lambda not in S_f	Section 1.2	A1.4, B2	verified

citation (as in the paper)	exact content used	where in the paper	book item	status
DFG04 Theorem 0.2 (= Theorem 3.7)	`H^1_Sigma` has length `v_lambda(eta^Sigma)` for `Sigma` containing the primes dividing `N`, `lambda` not in `S_f`	Section 4.2	A4.7	verified
DFG04 Theorem 2.7	`H^0 = H^1_f = 0` for `A_f` and `B_f = A_f(1)`, `lambda` not in `S_f`	Theorem thm:dfg27	A4.8	verified
DFG04 Theorem 3.7	length of `H^1_Sigma(Q, W)` is `v_lambda(eta^Sigma)`, `Sigma` finite, `l` not in `Sigma`, minimally ramified outside `Sigma`; covers `l = 3` in weight `2`	Theorem thm:dfg37	A4.7	verified
DFG04 Proposition 1.4(c)	`eta^Sigma = eta^empty prod L_p^nv(A_f, 1)^{-1}` for `M bar_{f,lambda}` irreducible	Theorem thm:dfg14	A4.2	verified
DFG04 Lemma 2.1	the four-term sequence (DFG04's hypothesis on `Sigma` is implied by the paper's)	Lemma lem:dfg21	A4.9	verified
DFG04 Theorem 2.15 (proof)	FPR local formula `Fitt H^1_f(Q_p, M) = L_p(M, 0)^{-1} Tam^0_p(M)`	Lemma lem:fpr	A4.10	verified
DFG04 (31)	`S_f = {lambda : lambda | Nk!, or M bar_{f,lambda} not abs. irreducible on G_{Q(sqrt(l*))}}`	Section 4.2	A4.6	verified
DFG04 (57)	`Tam^0_p(M) = Tam^0_p(M^D)`, `p != l`, general	Lemma lem:fpr	A4.10	verified; multiplicative case also by hand (Exercise 8)

citation (as in the paper)	exact content used	where in the paper	book item	status
DFG04 p. 708	`Tam^0_p(M) = Fitt(H^1(I_p, M)_tor)^{G_{Q_p}}` (same misplaced superscript as the paper, point 37)	Section 1.2, Lemma lem:tamfactor	A1.3	verified
DFG04 Lemma 2.12	`<f, f>` with the NAIVE value `L^nv(A_f, 1)` (from Hida81 Theorem 5.1)	Section 1	A1.0	verified
DFG04 Sections 1.1.1, 1.5.3, 1.6.1, 1.6.2, 1.7.2, 1.7.3, 1.8.2, 2.1; (13), (14), (18), (22)	geometric Frobenius; the operator `[UwU]_omega = w_N`; kernel lattice; ideal of values; self-adjointness; `H^1_Sigma`; `M_{f,lambda}` dual of the Tate module	Sections 1.1, 4.1, 4.2	C2, A4.1, A4.6	verified (the integral structure agrees with `H^1(X_0(N), Z_l)` at non-Eisenstein `m`)
DDT Theorem 3.1(e)	local structure at a multiplicative prime	Lemma lem:old, Lemma lem:tamfactor	A2.7, A3.3	as cited
DDT Theorem 3.15	optimal level for odd `l`, with the `Q(sqrt(-3))` condition at `l = 3`	Section 3	A3.4	as cited
DDT Lemma 4.17	numerical criterion; inequality half used	Lemma lem:ddt417	A4.3	as cited
DDT Section 4.4	Hecke adjoints and the twisted pairing	Section 4.1	A4.1	as cited
FPR94 I.4.2.2	`Tam_p(M) = Fitt((H^1(I_p, M)_tor)^{G_{Q_p}})` for `p != l`	Section 1.2, Lemmas lem:tamfactor, lem:fpr	A1.3, A2.7, A4.10	as cited (DFG04's recollection verified)
ARS12 Theorem 2.2	`deg varphi_E | r_E`; equal `p`-valuations when `p^2` does not divide `N`	Sections 1, 3, Lemma lem:hecke	A3.1, A4.4	as cited; the divisibility reproved in `\Cref{ch:moddeg}` 9.4

citation (as in the paper)	exact content used	where in the paper	book item	status
ARS12 Section 2; Section 3; Table 1	abc connection; perfect pairing; 99A1 with `deg 4`, `r = 12`	Sections 1, 3	A3.1, A3.7, B4	`deg(99a1) = 4` and `r(99a1) = 12` both recomputed (`\Cref{ch:moddeg}` 8.5, `\Cref{ch:hecke}` 11.2)
Atkin-Li Theorem 3.1	twist of a newform of level `M`, `q^2` not dividing `M`, is new of level `lcm(M, q^2)`	Lemma lem:twist(iii)	A2.4	as cited
Atkin-Lehner Theorem 3	`a_p = 0` for a newform with `p^2 | M`	Lemma lem:old	A3.3	as cited
Ribet83 Theorem 2.2; AU96 Lemma 2.1	perfect pairing `S_2(Gamma_0(N); Z) x T -> Z`	Lemma lem:crit	A3.2	as cited (the pairing is proved in `\Cref{ch:hecke}` 4.1)
Ribet-Stein Theorem 3.5	multiplicity one for `J_1(N)[m]`, exception Serre weight `1`	Remark rem:free	A4.5	as cited
Kim-Ota Theorem 1.1; Definition 2.12	`ord eta_f(N) = ord eta_f(N^+, N^-) + sum t_f(q)`; the Tamagawa exponent	Remark rem:ko(2), Section 1.5	A1.9, A4.13	as cited
Kraus90	order of the image of inertia `12/gcd(v_q(Delta), 12)` for `q >= 5`	Lemmas lem:IV, lem:euler	A2.2, A2.5	no number cited
Zywina15	possible mod 3 images	Proposition prop:ns	A5.1	by section (point 4); the list is proved directly
Sutherland16	labels 3Ns, 3Cs.1.1, 3B.1.1	Sections 1.3, sec:anomaly, sec:eis	A6.7, A6.10	consistent with Cremona's `galrep` (checked); definitions against the source still to check (point 3)
Silverman ATAEC IV.9; V.3, V.5	Tate's algorithm; Tate curves and their twists	Sections 1.4, 2	A1.6, A2.1, A2.4	standard
Edixhoven92 Section 2	`rho bar|_{D_l}` determines `a_l mod lambda`	Lemma lem:old	A3.3	as cited

citation (as in the paper)	exact content used	where in the paper	book item	status
Carayol89; Ribet90; Diamond95	Serre conductor divides N; level lowering	Section 3	A3.4	as cited
Tilouine97	J_0(N)[m] two-dimensional implies Gorenstein and freeness	Remark rem:free	A4.5	as cited
CalegariEmerton09	odd modular degree	Section sec:two	A7.1	as cited; the paraphrase fails for CM curves (point 57)
Mazur77; Watkins02; Zagier85; Cremona97; Flach92; Hida81; the l = 2 literature	context	Sections 1, 6, 7, Open questions	A1.0, A1.9, A7.1, A8.1	as cited

23.2.4 B4. Worked verifications of the numerical examples (Section 21.21)

All values below are PARI-verified (PARI/GP 2.17.2 through cypari2) or recomputed exactly, as recorded in important_notes.md and in the outlines cited: Kodaira types and local data from ellglobalred; P_q from lfuneuler(lfunsympow(E, 2), q); modular degrees from ellmoddegree, agreeing with Cremona's degphi file on all 38,042 optimal curves below 10,000; congruence numbers by the lattice method (\Cref{ch:moddeg} 8.5, \Cref{ch:hecke} 11.2); Tate data from ellinit(E, O(q^30)) (\Cref{ch:computation} 3.5; ellinit(E, q) reduces mod q and is not the q-adic curve, important_notes.md B27); images from Cremona's galrep file and elldivpol. The writer prints each verification as a short table with the source item cited.

1. **121a1** [1,1,1,-30,-76]. Type II at 11 (local data [2, 2, 1]); P_11(X) = 1 + 11X, P_11(11^{-2}) = 12/11, e_3(E, 11) = v_3(12/11) = 1 (chi_{-3}(11) = -1); t_3 = 0; bound 1; deg = 6, r = 6: attained; rule(E, 3) = [1, 1] in both modes; B^main_3 = 0, B^tw_3 = 1 (the curve on which (eq:twist) is strictly stronger than (eq:main)). Twist by -11: 121c2 (type IV^*); the optimal curve 121c1 of that class has type IV, deg 6, r = 6 (case (c) of Theorem thm:twist with equality). f_{121a} = f_{121d} = f_{11a} tensor chi_{-11} mod 3 for every n (Sturm bound 22). Mod 3 image surjective. Theorem thm:main applies (3 does not divide 121): H^1_f(Q, ad T_3 tensor Q_3/Z_3) = 0 (A4.12).
2. **539c1** [1,0,1,170,-3237]. Types I_3^* at 7 (code -7), I_2 at 11 (code 6); P_7 = 1 - X, P_7(7^{-2}) = 48/49, e_3(E, 7) = 1, t_3(E, 7) = 1, t_3(E, 11) = 0; bound 2; deg = 288 = 2^5 3^2, r = 288: attained; rule(E, 3) = [2, 2], rule(E, 5) = [0, 0]. Tate data at 7: same q_E as 77c1, u bar = 5, not a cube mod 7, so no Tamagawa excess. Twist by -7: 77c1 (I_3 at 7, a_7 = -1, deg 6 = r); Theorem thm:quant: v_3(288) = 2 = v_3(6) + v_3(48), equality (A4.16). f_{539c} = f_{539d} = f_{11a} tensor chi_{-7} mod 3 for every n (Sturm bound 112); 539d2 is the twist of 11a1. Level raising 11 -> 77 at 7: a_7(11a1) = -2, (7 + 1)^2 - 4 = 60 = 0 mod 3. Mod 3 image surjective.
3. **980i1** [0,0,0,1568,-72716]. Types IV^* at 2 (code -4), I_1 at 5, I_5^* at 7 (code -9); deg = 2880 = 2^6 3^2 5, r = 2880. At l = 3 (3 does not divide N, image surjective): P_2 = 1 + 2X, e_3(E, 2) = v_3(3/2) = 1 (a theorem, A1.8), e_3(E, 7) = v_3(48/49) = 1, bound 2 = v_3(2880); rule(E, 3) = [2, 2]; Theorem thm:quant with S = {7}: E_S = 140b1 (also IV^* at

2: -7 = 1 mod 8, so chi_{-7} is trivial on G_{Q_2}), deg 60, 2 = v_3(60) + v_3(48), equality. At l = 5 (5 | N: outside the theorems): t_5(E, 7) = 1, e_5 = 0 everywhere, rule(E, 5) = [1, 1]; the congruence is f_{980i} = f_{980g} = f_{20a} tensor chi_{-7} mod 5 for every n (Sturm bound 336); the twist of 20a1 by -7 is 980g2, not a curve of class 980i (B20); a_7(20a1) = 2, 64 - 4 = 60 = 0 mod 5.

4. **726a1** [1,1,0,-35,-51]. Local data 2: [1, 10, 2] (I_6), 3: [1, 7, 1] (I_3), 11: [2, 3, 2] (III); image 3Ns, psi_3 = (x^2 - x - 74)(3x^2 + 8x + 20), ns3 = 1; deg = 144 = 2^4 3^2, r = 1584 = 11 * 144 (gap at 11, 11^2 | 726). Breakdown at l = 3 (rule_terms): q = 2: ‘P_2 = 1
   - X,e = 0,t = v_3(6) = 1;q = 3 = l:w_3 = 0,t = v_3(3) = 1;q = 11:P_11 = 1 + 11X,e_3 = v_3(12/11) = 1(counted only withiii = 1).rule(E, 3) = [2, 2];rule(E, 3, 1) = 2 + 1 - 1 = 2:[2, 2], the cancellation the paper describes. Without the correction the bound is3 > 2: a violation by one, involvingp = 2 = -1 mod 3witht_3(E, 2) = 1. With (eq:thm) read literally (terms atq != 3only) the bound is2and there is no violation (A6.2).3 | N, so Theorem thm:main does not apply atl = 3‘ in any case.
5. **99A1** = Cremona 99a1 [1,-1,1,-2,0]. N = 3^2 11; type III at 3 (code 3, v_3(j) = 9), I_1 at 11; deg = 4, r = 12 (recomputed: the congruence f_{99a} = f_{99b} = f_{99d} mod 3 beyond the Sturm bound 24); rule(E, 3) = [0, 0] (w_3 = 0 for type III at 3; PARI's P_3 = 1 + 3X is not used at q = l). v_3(deg) = 0 < 1 = v_3(r), allowed by ARS12 since 9 | N; H^1(X_0(99), Z_3)_m is not free over T_m (\Cref{ch:moddeg} 9.7). No violation of Conjecture conj:euler at l = 3 (mod 3 image surjective).
6. Further examples used in the book (same sources): **121d1** [0,-1,1,-40,-221] (I_1^* at 11, twist of 11a3; deg 24 = r): e_3 = v_3(120) = 1 = v_3(24); with 11a1 and 121d2 shows that e_l is not twist invariant (A1.1); Theorem thm:quant: 1 = 0 + 1. **539d1** [0,1,1,-16,-66] (twist of 11a3 by -7, I_0^* at 7, deg 72, r = 504): P_7 = (1 - 7X)(1 + 10X + 49X^2), e_3 = v_3(6) + v_3(60) = 2 = v_3(72); e_5 = 1 but v_5(72) = 0, image 5B.4.1 (Eisenstein at 5); r and deg differ at 7 (7^2 | 539). **539a1** (I_2^* at 7, image 3B, deg 320): e_3 = 1, v_3(deg) = 0. **52a1** (IV^* at 2, I_2 at 13, deg 3): e_3(E, 2) = 1; case (a) of Theorem thm:qual at q = 2 (Serre conductor 26, g = f_{26b} - f_{26b}(2z)). **336f1** (II at 2, P_2 = 1 + 2X, deg 24, rule(E, 3) = [1, 1]), **1232l1** (II^* at 2, P_2 = (1 - 2X)(1 + 2X)^2, e_3(E, 2) = 2, deg 288, rule(E, 3) = [2, 2]), **592c1** (II^* at 2, P_2 = (1 - 2X)(1 + 4X^2), e_5(E, 2) = 1, deg 80, rule(E, 5) = [1, 1]): the type at 2 does not determine P_2. **791a1** (I_3 at 7, u bar = 6 a cube: Tamagawa excess 1; explicit bound 2 = v_3(144), Selmer length 0); **1184f1** (explicit bound 2, deg 1728, Selmer length 1); **1748e1** (explicit bound 3, deg 3240, Selmer length 1). **20a1** (IV^* at 2, I_2 at 5, torsion Z/6, image 3B.1.1, deg 1): D = v_3(3) - 0 = 1 = v_3(#E(Q)_tors). **14a1** (image 3Cs.1.1, torsion Z/6, I_6 at 2, I_3 at 7, deg 1): D = 2. **714i1** (torsion Z/9, image 3B.1.1, deg 1080, D = 2). **11a1** at l = 5 (image 5Cs.1.1, rule(E, 5) = [0, 1], u bar = 10 a fifth power, v_5(Tam_11(ad T)) = 2, D = 1): outside every theorem. **990h1/990h3** (rule [4, 3] versus [3, 3]: run rule on the optimal curve).

Arithmetic checks (all consistent): the I_0^* identity (1 - alpha/(beta q))(1 - 1/q)(1 - beta/(alpha q)) = (q - 1)((q + 1)^2 - a^2)/q^3; e_3 for P_2 = 1 + 2X is v_3(3/2) = 1; the sec:tam percentages (45.05%, 60.59%, 67.03%, 78.70%); 364/1,170 = 31.1%; 17 + 5 = 22; 435/828 = 52.5%; 14,182/44,672 = 31.7%; 0.532 x 867,578 = 461,551.

23.3 Part C. Where each point of important_notes.md is handled (Section 21.22)

The chapter's Section 21.22 “Corrections to the paper used in this book” is a longtable with one line per point about the paper: paper location, the problem in one sentence, the book's statement (final state), the item. Status: E = error in a statement or proof (the result survives unless said), G = gap, X = expository

or notation, S = strengthening or new theorem, D = data or wording about data, C = confirmation (no change). Every point is in its final state: addenda, corrections and settlements are applied.

point	paper location	final state	status	handled at
1	proof of prop:ns, "non-abelian and faithful"	commutator argument added	G	A5.1, A1.12
2	context of prop:ns	`N_s(F_3)` not maximal (index 2 in `N_ns(F_3)`)	X	A5.1
3	Sutherland labels	consistent with `galrep`; check against Sutherland16	X	A1.13, A6.7, A6.10, B3
4	Zywina15, Dickson	cited by section; list proved directly	X	A5.1, B3
5	lem:h0 finiteness; sec:anomaly `p < 1223`	determinant argument: `h_l = 0` (`l >= 5`), `h_3 <= 1`, CM allowed	E	A4.11, A6.7, A1.5
6	Section 1.3: `chi_{l*}`	criterion is `rho bar = rho bar tensor chi bar_l`, forcing `l = 3`	E	A1.5
7 (+ addendum)	lem:tamfactor, "image of `kappa`"	meaningful via the splitting by `q`; `c'` defined mod `(n, q - 1)`	X	A2.7 step (7)
8	lem:tamagawa-exp, `u`	Tate parameter `q_E`; two steps separated	X	A2.1
9	prop:ns "without CM"	not used; dropped	X	A5.1
10	lem:old, lem:h0: "irreducible"	"absolutely irreducible" (complex conjugation)	G	A3.3, A4.11
11	Frobenius conventions	`L`-factors geometric, lem:tamfactor arithmetic (DFG04 geometric, verified)	X	C2, A2.5, A2.7, A4.9, A4.10
12	lem:euler, `End_{I_q}(V)`	`Q_l tensor Q(zeta_3)`, `T`-version `Z_l tensor Z[zeta_3]`	X	A2.5
13	lem:IV, `Z_3[sigma]/(...)`	`Z_3[X]/(X^2 + X + 1)`	X	A2.2

point	paper location	final state	status	handled at
14	lem:h0, CM alternative	impossible when `l` does not divide `N`; moot	X	A4.11
15 (corrected)	def:terms, twist invariance	true for `t_l`, false for `e_l` at the twisting prime (11a1/121d2)	E	A1.1
16	after thm:twist, "strict only in case (b)"	strict in (a) and (b)	D	A4.14
17	conj:eis, sec:eis, `D <= 2`	a theorem (no rational cyclic 27-isogeny)	S	A1.13, A6.10
18	proof of thm:main, `I_n^*` at `2`	local argument covers `q = 2`	G	A4.12 Step 7
19 (corrected)	lem:ddt417, `wp cap I = 0`	two-line proof (not "T reduced")	G	A4.3
20	thm:qual (b), (c), `h != f`	`h tensor chi` is the depletion of `g`; strong multiplicity one	E	A3.5
21	example 980i1	`5 | 980`: illustrates the congruence only	X	A3.6
22	rem:types	complete grouping (all additive types but `IV, IV^*`; `I_0^*` lowers)	X	A2.3
23	sec:tam, `c_q`	class `c(u)`, not the Tamagawa number	X	A2.7, A6.4
24 (revised by 39)	abstract, "elementary"	only Step 1 elementary; no multiplicity one needed	X	A1.0, A1.11, A4.15
25	abstract, ranges and "a third"	precise ranges; `31%, 23%, 18%`	D	A1.0, A6.1
26	Section 4.2, homothety	`M_{f,lambda}` is the dual of `T_l E`	X	A4.1, A4.6
27	optimality	irrelevant at `l` when `E[l]` irreducible	C	B2
28	lem:twist proof, "tame"	right only for the listed exchanges	X	A2.4
29	lem:twist (ii), invariants by `q`	write `q*`	X	A2.4
30	intro, naive `L` as Rankin series	Rankin series differs; use Shimura's series	E	A1.0
31	lem:tamfactor, class of `u`	of the reduction of `u`; `v_l(0) = v_l(q - 1)`	X	A2.7

point	paper location	final state	status	handled at
32	chi_q versus chi_l (paper)	chi_{q*}, never chi_q (D3); numerical chi_d; chi_3^cyc	X	C1, A2.2 (B-C6)
33	prop:ns, lem:h0, "Ind psi"	over F_9; descends to F_3	X	A5.1, A1.12
34	lem:euler, "diagonal element"	the invariant line is spanned by tau	X	A2.5
35	lem:hecke, "annihilator"	the other factors lie in the kernel	E	A4.4
36	lem:hecke, thm:quant Step 2	pairing perfect on the m-component (self-adjoint e_m)	G	A4.4, A4.15
37	Section 1.2, Tam_p parenthesis	Fitting ideal of the invariants	X	A1.3
38	lem:ddt417, orders	lengths, or finite residue field	X	A4.3
39 (confirmed by 56, 73)	thm:main, lem:hecke, rem:free	v_l(deg) = v(eta^empty); (eq:identity) an equality for l not in S_f	S	A1.4, A4.4, A4.12, B1, B2
40	Section 4.2	M_{f,lambda} = varphi_E^* H^1(E) = T_l E(-1) exactly	S	A4.1
41	rem:free	T_l J_0(N) versus H^1: not Galois-equivariant; freeness transfers	X	A4.5
42	lem:tamfactor, corner kappa	corner eps kappa	E	A2.7, A2.5
43 (settled)	prop:forms exception; thm:qual (a)	a theorem: P_2 = 1 + 2X for IV, IV^* at 2; (a) extends to q = 2	S	A1.8, A1.10, A2.2, A2.6, A3.5
44	sec:eis, "split Cartan image 3Cs.1.1"	3Cs.1.1 is E[3] = Z/3 + mu_3; split Cartan is 3Cs	X	A1.13, A6.10
45	lem:twist (ii), the cubic	cubic with the sign e = q*/q	X	A2.4
46	example of Section 3	congruences for all n by Sturm's bound	S	A3.6
47	Theorem thm:main hypothesis; sec:forms	l not in S_f automatic for l >= 5, l not dividing N, E[l] irreducible	S	A1.4, A4.6, A6.6, B2
48	sec:anomaly observations	theorems about 3Ns curves	S	A5.2, A6.7

point	paper location	final state	status	handled at
49	rem:types, "conductor exponent"	of `rho bar_{E,3}`, not of `E`, for type `II`	X	A2.3
50	rem:wild, sec:wild, type `II` at `2`	false in general (336f1, 1232l1)	E	A2.6, A6.8
51	lem:twist (ii) at `q = 3`	the exchanges hold at `q = 3` too (optional)	S	A2.4, A1.6
52	appendix, codes	`n + 4`, `-(n + 4)` only for `n >= 1`	X	A9.2
53	def:terms versus (eq:rule), `rule`	`t_l` defined at `q = l` too	X	A1.1, A6.8
54	thm:qual (c), `f' = f tensor chi`	`f' = f_{E'} = f^chi`; `f tensor chi` is its depletion	E	A3.5
55	remark after the example, "wild"	the obstruction is `q = l`	X	A3.7
56	refinement of 39; checklist	all five items verified (73)	C	A1.4, A4.1, A4.4
57	sec:two, Calegari-Emerton	paraphrase fails for CM curves (27a1, 243b1)	E	A7.1
58	remark on `l | N`	726a1: `r = 1584`, `deg 144`	S	A3.1, A3.7
59	thm:quant Step 1	an equality (optional)	S	A4.15
60	thm:quant, `q >= 5`	`q` odd suffices	S	A1.11, A4.15
61	thm:quant Step 1	the `T`-stable complement, not the Petersson one (paper right)	C	A4.15
62	remark on `l | N`	539d1: `r = 504`, `deg 72`	S	A3.1, A3.7
63	lem:dfg21	`H^0(Q, V^D) = 0` automatic	X	A4.8, A4.12
64	Open question 3	`Tam_q(ad T)` trivial at potentially good primes: first half answered	S	A1.3, A4.12, A8.1
65	abstract, rem:ko(1)	needs `l | n_q`; raises by at least one	E	A1.0, A4.13
66	abstract, `ad V_l E`	Selmer group of `ad T tensor Q_l/Z_l`	X	A1.0
67	rem:ns, Section 1.3	conditional on `H^1_f(Q, V^D) = 0` for 3Ns curves	G	A1.5, A5.2, A8.1

point	paper location	final state	status	handled at
68	sec:tam	`I_n^*` primes can be included; "exactly where" overstates	X	A6.4
69	rem:wild, sec:wild	four shapes of `P_q` at every `q`; `2` can contribute at `l = 5`	S	A1.2, A2.6, A6.8
70	intro (addendum to 30)	exact Rankin relation; Shimura's series	X	A1.0
71	intro (optional)	`v_l(deg) = v_l(B_E) + sum e_l`	S	A1.0
72	Section 1.1, `delta_p`, `L_p` at `p = l`	definitions for `p != l`; harmless	X	C1
73	point 56 and Q1	settled: DFG04 confirms; covers `l = 3`; arXiv numbering differs	C	A1.4, A4.1, A4.7, B3
74	remark (2) after thm:quant	equality for `l` not in `S_f`, no multiplicity one	S	A1.11, A4.16, B1, B2
75	lem:dfg21 hypothesis	a sufficient form of DFG04's	C	A4.9
76	sec:eis, two sentences	14a1 (torsion `Z/6`); order-9 curves have `D = 2`	D	A6.10
77	sec:data, protocol	`h_l` never blind-tested; (eq:twist) status unstated	D	A6.1, A6.9
78	appendix, `rule`	run on the optimal curve, `E[l]` irreducible	X	A9.1
79	sec:sharp	shows strength, not validity; range-dependent	D	A6.3
80	appendix memory; 3Ns counts	32 MB in 2.17.2; `36` versus `24`	D	A9.1, A6.2, A6.7

Book-internal notes (not about the paper; they do not go into Section 21.22 but are applied in this outline): B1, B7, B9, B25 (Frobenius in C2 and the opening: C2, opening paragraph, A2.5, A2.7, A4.9); B3 (A4.4 kernel, A4.6 dual: A4.4, A4.6); B8 and B20 (twist pairs at curve level: A1.1, A3.6, A4.14, B4); B12 (A3.2 sentence: A3.2); B13 (type `IV` at `2`, `chi_{l*}` for `l >= 5`: A1.5, A1.8, A2.6, A4.6); B16 (A2.6 false claim: A2.6); B18 (A3.5 case (c), A1.10: A3.5, A1.10); B19 (dependency diagram and hypothesis table through rem:free; `r(99A1)`: B1, B2, A3.1, A3.7, B4); B21 (A2.7 step (7) `eps`, "no such map": A2.7); B22 (cite Chapter 18 items; the "torsion of `H^1(Q_p, T^D)`" wording: A1.3, A2.7, A4.9, A4.10); B24 (conventions: C1 to C3); B25 (duals `^vee`, adjoints `^dagger`, `chi_3` numerical: C3, A3.2, A2.2); B26 (A4.16 multiplicity one; "not checked against DFG04": A4.16, B3, A4.7); B27 (`alldegphi`; A6.1 blind

caveat; A6.10 slips: A6.1, A6.10, B4); B17 (lem:crit proved in Chapter 15: A3.2). Appendix B Part 3 entries: F1 (opening), F2 (C2(b)), T1 (A2.5, A2.7 corner), T2 (A2.7 `c(u)`), D5 (A3.2), D6 (A4.1), X5 (A2.2 `chi_3^cyc`), L6 (A4.14 `iota_S`).

23.4 4. Section-by-section plan with target line counts

Lines of LaTeX. PLAN.md 7.3 allows Chapter 21 up to 2,800 lines; the total below (card excluded) is 2,795. The card ("Core results at a glance") lists every A-item with its rigour tag (about 65 entries) and is excluded from the count as for every chapter. Compared with the first plan (2,770): 21.2 grows by 30 (A1.0, the abstract corrections), 21.20 by 10 (the corrected diagram and table), 21.21 by 20 (PARI-verified data), 21.22 by 60 (80 points instead of 26), 21.1, 21.18 and 21.24 by 5 each (`+135`); to compensate, 21.3, 21.5, 21.6, 21.13 (the equality shortens Step 4 and the remark), 21.15, 21.16 and 21.23 lose 10 each, 21.8 loses 20 (lem:crit and lem:old are proved in Chapter 15 and only walked through here), and the opening paragraph, 21.12, 21.17 and 21.19 lose 5 each (`-110`; the small-range reproductions are cited from Chapter 22, not reprinted). Net `+25`.

section	title	contents (items)	target lines
(starred)	Core results at a glance	every A-item, stated as above, tagged, with `\Cref`	(excluded)
(starred)	Where the paper uses this chapter	one paragraph: "all of it"; how to read the tour alongside the paper	25
21.1	Conventions and how to read the tour	C1, C2 (Frobenius, B-C2, B-C3), C3 (duals, B-C4); how Commentaries are marked; one sentence on the AI statement (A9.3)	95
21.2	The abstract, the introduction and the local terms	A1.0 (with the five corrections of the abstract), A1.1, A1.2, A1.3	140
21.3	The theorem, the conjecture, the component-group forms	A1.4 (equality; reading of the hypotheses), A1.5, A1.6, A1.7, A1.8 (no exception)	140
21.4	History and the statements without the hypothesis	A1.9, A1.10 (with `q = 2`), A1.11 (with `q` odd and equality), A1.12, A1.13 (with `D <= 2` proved)	110
21.5	Tate curves and inertia	A2.1, A2.2, A2.3	100
21.6	Twists and the omitted Euler factors	A2.4, A2.5, A2.6 (corrected Remark rem:wild)	150
21.7	The Tamagawa factor at a multiplicative prime	A2.7, every step, with the corner `eps kappa`, the identification of `c'` and the check for `M = T`	130
21.8	The congruence criterion and oldforms	A3.1, A3.2, A3.3	110

section	title	contents (items)	target lines
21.9	Level lowering and the proof of Theorem thm:qual	A3.4, A3.5, A3.6 (Sturm, curve-level twists), A3.7	150
21.10	Congruence ideals; the modular degree as a congruence ideal	A4.1, A4.2, A4.3, A4.4 (equality), A4.5	150
21.11	The adjoint Selmer group and the proof of Theorem thm:main	A4.6 to A4.13 (Step 7 with explicit Tamagawa terms)	190
21.12	Twist monotonicity	A4.14 (with the data paragraph and its correction)	75
21.13	Theorem thm:quant and its equality	A4.15, A4.16	140
21.14	The boundary of the hypothesis	A5.1, A5.2 (conditional identity; structure of 3Ns curves)	90
21.15	The computations I: data, protocol, the rule, calibration	A6.1 (test-status table), A6.2, A6.3	100
21.16	The computations II: Tamagawa factors, isolated terms, corollary forms	A6.4, A6.5, A6.6	80
21.17	The computations III: 3Ns, wild primes, out of table, Eisenstein	A6.7, A6.8, A6.9, A6.10	115
21.18	The prime 2 and the open questions	A7.1, A8.1 (Open question 3 half answered)	65
21.19	The appendix: the function `rule(E, l)`	A9.1 (listing, line-by-line table, verified outputs), A9.2, A9.3	95
21.20	The map of the paper	B1 (corrected diagram), B2 (corrected hypothesis table), B3 (imported statements with statuses and the arXiv warning)	160
21.21	Worked verifications	B4	120
21.22	Corrections to the paper used in this book	Part C table (80 lines) with a short introduction	120
21.23	Exercises	Section 6 below (18 exercises)	100

section	title	contents (items)	target lines
21.24	Notes and references	which numbers were verified (DFG04 against the published text), which are as cited (B3), the arXiv warning, sources and software for the data	45
	total		**2795**

23.5 5. Coverage table: every label and section of the paper

paper label or section	items
abstract	A1.0 (corrections: points 24, 25, 39, 65, 66), A1.4, A1.11, A6.1, A6.9
Section 1 (introduction, Petersson norm, ARS, naive versus true)	A1.0 (points 30, 70, 71), C1, A1.1, A3.1, A1.9
Section 1.1 “The local terms” (`V, T, rho, A_f, delta_p, L_p, L_pˆnv`, PARI)	C1, C2, C3, A1.1
def:terms	A1.1
the table of Section 1	A1.2, A2.5
Section 1.2 “The theorem” (`Tam_p`, Bloch-Kato Selmer group)	A1.3, A1.4
thm:main, eq:identity, eq:thm	A1.4, A4.12
Section 1.3 “Beyond the hypotheses”	A1.5
conj:euler, eq:rule	A1.5, A5.2, A6.2, A9.1
Section 1.4 “Local component groups”, eq:main, eq:twist	A1.6, A1.7
prop:forms	A1.8, A6.6
Section 1.5 “History . . . ”	A1.9
thm:qual	A1.10, A3.5
thm:quant	A1.11, A4.15, A4.16
prop:ns	A1.12, A5.1
conj:eis	A1.13, A6.10
sec:local	A2.1 to A2.7
lem:tamagawa-exp	A2.1
lem:IV	A2.2
rem:types	A2.3
lem:twist	A2.4
lem:euler	A2.5
rem:wild	A2.6, A6.8
lem:tamfactor	A2.7
sec:proof	A3.1 to A3.7
lem:crit	A3.2
lem:old	A3.3
optimal-level paragraph of Section 3	A3.4
cases (a), (b), (c)	A3.5
the example (121a1, 539c1, 980i1)	A3.6, B4
the remark after the example (`1 | N`, 99A1)	A3.7, B4

paper label or section	items
sec:quant	A4.1 to A4.16
sec:dfg	A4.1
thm:dfg14	A4.2
lem:ddt417	A4.3
lem:hecke	A4.4
rem:free	A4.5
Section 4.2 “The theorem of Diamond, Flach and Guo” (set-up, `S_f`)	A4.6
thm:dfg37	A4.7
thm:dfg27	A4.8
lem:dfg21 and the paragraph after it	A4.9
lem:fpr	A4.10
lem:h0	A4.11
Section 4.3 “Proof of Theorem thm:main”	A4.12
rem:ko	A4.13
sec:twisting, thm:twist and the data paragraph after it	A4.14
proof of thm:quant (Steps 1 to 4)	A4.15
the remark after the proof of thm:quant	A4.16
sec:boundary (proof of prop:ns)	A5.1
rem:ns	A5.2
sec:data, Section 6.1 “Data and protocol”	A6.1
Section 6.2 “The rule on whole ranges”, tab:uniform	A6.2
sec:sharp	A6.3
sec:tam	A6.4
sec:iso, tab:iso	A6.5
sec:forms, tab:counts	A6.6
sec:anomaly, tab:ns	A6.7
sec:wild, tab:isolated	A6.8
sec:oot, tab:oot	A6.9
sec:eis	A6.10
sec:two	A7.1
Open questions 1 to 5	A8.1
appendix “Reproducibility”, `rule`, `ns3`, Kodaira codes	A9.1, A9.2, A9.3
statement on the use of an AI system	A9.3 (one sentence in 21.1)
PLAN.md extras: dependency diagram; hypothesis table; imported statements; worked verifications	B1; B2; B3; B4
important_notes.md points 1 to 80	Part C

23.6 6. Exercises (Section 21.23)

Graded from routine to harder; most are “check the paper’s claim that . . . ”. Data for the curves are in B4; point counting by hand or with a short program, or PARI, is allowed (the book says which).

1. Check that 121a1 `= [1,1,1,-30,-76]` has type `II` at `11` and `e_3(E, 11) = 1`: compute `Delta`, `c_4`, `j`, and `11 - chi_{-3}(11)`; compare with `deg varphi_E = 6`.
2. Check that the twist of 121a1 by `-11` has the same `j` and `v_11(Delta) = 8`, hence type `IV^*` (Lemma lem:twist(ii)); explain why the optimal curve 121c1 of the twisted class has type `IV` instead.

3. Check a_p(121a1) = chi_{-11}(p) a_p(11a1) mod 3 for p < 30, p != 3, 11, by counting points; then use Sturm's bound to prove f_{121a} = f_{121d} mod 3 for every n.
4. For 539c1 check the types I_3^* at 7 and I_2 at 11, the bound e_3 + t_3 = 2 and its attainment (deg = 288). Identify the twist by -7 (77c1, deg 6) and check Theorem thm:quant with equality.
5. Check the paper's sentence that strict inequality v_l(deg varphi_E) > v_l(deg varphi_{E_q}) occurs "only in case (b)" against 539c1/77c1, 980i1/140b1 and 121d1/11a3 at l = 3. Which case are they in? Deduce the corrected statement from Theorem thm:quant and its equality.
6. Show that e_l(E, q) is not invariant under quadratic twist by comparing 11a1 and 121d2 at q = 11, and 11a1 and 539d2 at q = 7. Prove the corrected invariance statement of A1.1.
7. For 539d1 (type I_0^* at 7) compute e_3(E, 7) and e_5(E, 7) from a_7(11a1) = -2, compare with deg = 72, and explain the behaviour at l = 5. Do the same for 539a1 at l = 3.
8. Lemma lem:tamfactor in coordinates: for l = 3, n = 9, q = 19 and each class c in Z/9, count the Frobenius-fixed points of (Z/9) e + (Z/9) h under e -> e, h -> q^{-1}(h - 2 c eps e) and check the formula. Then twist the action by chi_l(Frob) = q and show the count is unchanged (Tam_q(A(1)) = Tam_q(A)).
9. Check that (chi_l eps, kappa; 0, eps) is a homomorphism only if eps is trivial, and that (chi_l eps, eps kappa; 0, eps) always is (important_notes.md point 42). Check the paper's "as a check" for M = T.
10. Check the identity (1 - alpha/(beta q))(1 - 1/q)(1 - beta/(alpha q)) = (q - 1)((q + 1)^2 - a^2)/q^3, and prove that for type IV at 2, P_2(X) = 1 + 2X and e_3(E, 2) = 1 (A1.8).
11. Prove the sharp form of Lemma lem:h0: if rho mod l^k = (rho tensor chi_l) mod l^k then chi_l^2 = 1 mod l^k on G_Q; deduce h_l(E) = 0 for l >= 5 and h_3(E) <= 1. Where does the paper's own argument go wrong?
12. Check the table of w_l(E, q) = max(v_l(#Phig_q(E)), v_l(#Phig_q(E^{(q)}))) for all Kodaira types at q >= 5, and the proof of Proposition prop:forms case by case, with no exception.
13. Prove that a non-CM elliptic curve over Q cannot have both E[3] = Z/3 + mu_3 and a rational point of order 9. Deduce D <= 2 in Conjecture conj:eis from its first bound. Check D = 2 for 14a1 and for 714i1.
14. For 726a1 determine the types at 2, 3, 11, factor psi_3, and simulate rule(E, 3) and rule(E, 3, 1) by hand, recovering [2, 2]. Count the bound of (eq:thm) without correction in both readings (with and without the terms at q = l).
15. Lemma lem:ddt417: prove wp cap I = 0 when eta != 0 by the two-line argument of A4.3, and show that U_2 at level 88 has a nonzero nilpotent part, so "T reduced" is not the reason.
16. The degree as a congruence ideal: for E optimal, use the projection formula to show that the ideal of values of <x, w_N y> on varphi_E^* H^1(E, Z_l) is deg varphi_E Z_l; check that the Gram determinant of a rank-two alternating form is the square of a generator of its ideal of values.
17. Check the calibration arithmetic of Section sec:sharp: the bound-zero proportion 53.2% alone predicts at least 461,000 violations among 867,578 bounded curves; explain what further information gives 605,000, and why the figure measures strength, not validity.
18. 99A1: check N = 9 * 11 and type III at 3; explain why deg = 4, r = 12 does not contradict ARS12, why H^1(X_0(99), Z_3)_m is not free, and why Conjecture conj:euler at l = 3 is not violated (w_3 = 0).

23.7 7. Notes for the writer

W1. Changes made in the third session (this revision), by part. Opening and C2: Frobenius convention rewritten per B-C2, B-C3 (F1, F2; B1, B7, B9, B25): Frob always arithmetic, the geometric one written Frob_p^{-1} (Phi_p in Appendix B), the eigenvalues alpha, beta belong to the arithmetic Frobenius on V_l E, and L_p(V_l E, s) = L_p(E, s + 1). C1, C3: notation per B-C1, B-C4, B-C6, B-C10, B-C12 to B-C14 (duals ^vee, ^*, ^D, adjoints ^dagger, c(u), c_E, J_f, depletion f^Sigma). A1.0 new (abstract and introduction: points 24, 25, 30, 65, 66, 70, 71, B10). A1.1: point 15 in corrected form (121d2), point 53. A1.3: points 37, 64. A1.4: equality (points 39, 56, 73), the reading of the hypothesis (point 47). A1.5:

points 6, 67, B13. A1.6: point 51. A1.8: no exception (point 43, B13). A1.10: case (a) at `q = 2` (B18). A1.11: `q` odd (point 60), equality (point 74), point 24 revised. A1.12: points 33, 48 added. A1.13: points 17, 44, 3. A2.2: `chi_3^cyc` (X5). A2.3: points 22, 49. A2.4: points 28, 29, 45, 51. A2.5: corner `eps kappa`, points 11, 12, 34, 69. A2.6 rewritten (points 50, 69, 43; B16). A2.7: corner `eps kappa` (point 42, T1, B7), step (7) with `eps` and the softened Commentary (point 7 addendum, B21), `c(u)` (T2, points 23, 31). A3.1: `r(99A1)` recomputed (B19), points 58, 62. A3.2: lem:crit proved in Chapter 15 (B17), adjoints `^dagger`, the self-correcting sentence replaced (D5, B12). A3.5: points 20, 54, case (c) argument repaired (B18), case (a) at `q = 2`. A3.6: Sturm (point 46), curve-level twists (B8, B20), PARI outputs. A3.7: point 55, data recomputed. A4.1: `M_{f,lambda} = T_l E(-1)` (points 26, 40; D6, B3), point 73 items 1 and 3. A4.3: point 19 as corrected, point 38. A4.4: equality; kernel not annihilator (point 35, B3); point 36. A4.5: role reduced to none; point 41. A4.6: dual of `T_l E` (B3), point 47. A4.7: covers `l = 3` (Q1, point 73). A4.8: point 63. A4.9: `iota` versus `iota_S` (L6), points 11, 75, B22 wording. A4.10: verified (57). A4.12: equality, explicit Tamagawa terms (point 64), point 18. A4.13: points 23, 65. A4.14: `iota_S`, point 16 with data. A4.15: points 24, 39, 59, 60, 61. A4.16: equality (point 74, B26). A5.1: point 33. A5.2: points 48, 67. A6.1: test-status table, point 77 (B27), `degphi` (B27), point 25. A6.2: point 80. A6.3: point 79. A6.4: points 23, 68. A6.6: points 43, 47. A6.7: points 48, 80. A6.8: points 50, 53, 69. A6.9: point 77. A6.10: point 76 (B27), points 17, 44. A7.1: point 57. A8.1: point 64 (Open question 3 half answered). A9.1: verified outputs, points 78, 80. A9.2: point 52. B1, B2 rebuilt (B19; points 39, 47, 74). B3: DFG04 statuses "verified" with the arXiv warning (B26; point 73). B4: PARI-verified data. Part C: points 1 to 80. Section 4: plan adjusted. Section 9: the first version's N1 to N14 are important_notes.md points 15 to 28. W2. Chapter numbers are written `\Cref{ch:slug}`: 1 `ch:groups`, 2 `ch:rings`, 3 `ch:modules`, 4 `ch:reps`, 5 `ch:galois`, 6 `ch:ant`, 7 `ch:local`, 8 `ch:curves`, 9 `ch:ellcurves`, 10 `ch:reduction`, 11 `ch:galrep`, 12 `ch:modforms`, 13 `ch:hecke`, 14 `ch:moddeg`, 15 `ch:modgalrep`, 16 `ch:cohomology`, 17 `ch:galcoh`, 18 `ch:blochkato`, 19 `ch:lfunctions`, 20 `ch:congruence`, 22 `ch:computation`. The item numbers cited in this outline ("`\Cref{ch:moddeg}` 9.3") are those of the outlines as of 14 October 2026; in the LaTeX they become `\Cref{ch:moddeg}` only (PLAN.md 7.1) until the cross-reference pass of PLAN.md 8.4 upgrades them to theorem labels. Environments: `theorem`, `lemma`, `proposition` for the paper's results with the paper's names in the optional argument (`\begin{lemma}[the paper's Lemma lem:tamfactor]`); labels with the prefix `paper` (`lem:paper:tamfactor`, `thm:paper:main`). Commentaries are `warning` environments titled "Commentary"; `paperlink` is not needed here. The paper's labels are quoted as text, never `\ref`'d. W3. Restating versus re-proving. Where another chapter proves a paper result in full (Chapter 10: lem:twist (i), (ii); Chapter 11: lem:tamagawa-exp, lem:IV and its extension to `q = 2`, rem:types, prop:ns, lem:h0; Chapter 14: lem:hecke in the strengthened form (9.3, 9.5); Chapter 15: lem:crit (decision B17), lem:old, thm:qual; Chapter 18: lem:tamfactor, lem:fpr, the additive Tamagawa factors; Chapter 19: lem:euler; Chapter 20: lem:ddt417, thm:main, thm:twist, thm:quant and its equality), this chapter still walks through the paper's proof line by line (PLAN.md requires it) but shortens routine verifications by citing the chapter; it must not contradict those chapters. Where those outlines are older than the settlements (`\Cref{ch:moddeg}` 9.5(c), 10.4(b), 9.6 and `\Cref{ch:modgalrep}` 10.4 still say the equality in Theorem thm:quant needs multiplicity one, or that the DFG04 normalisation is unchecked), this chapter follows points 73 and 74 and `\Cref{ch:congruence}` 12.6, Part 7 (important_notes.md B26). W4. Frobenius. State C2 once; in every proof say which Frobenius is meant. Never write `rho(Frob) = (eps q, c'; 0, eps)` and `L_q = det(1 - Frob_q^{-1} q^{-s})` in the same paragraph without the adjectives. Never write `Phi_q` for a Frobenius in this chapter (it would clash with `Phig_q`). W5. Data. The paper's tables are not in the repository: describe them from their captions and the numbers in the text only, and attribute every number to the paper. The book's own small-range numbers come from `\Cref{ch:computation}` (Section 11, Part 7), `\Cref{ch:blochkato}` 13.1, `\Cref{ch:galrep}` 8.4 and are always labelled as such, with the range. W6. PARI. Every output printed in this chapter was obtained with PARI/GP 2.17.2 (`\Cref{ch:computation}` R1 to R9); the paper used 2.15. State the version once. `lfuneuler(S, q)` returns `1/P_q(x)`; `ellinit(E, q)` reduces mod `q` (use `ellinit(E, O(q^k))` for the `q`-adic curve); `ellmoddegree` returns the degree for the given model; `ellsearch` needs the `elldata` package. Cremona's modular-degree file is `degphi`; the files contain no per-prime Kodaira symbols. W7. References. B3 lists every imported statement with its status. The DFG04 numbers are verified against the published version; cite that version, and add a sentence in the Notes that arXiv:2512.02348 numbers differently.

Verify before printing: ARS12 Theorem 2.2, Kim-Ota Theorem 1.1 and Definition 2.12 (and whether their `eta_f(N)` is the Hecke congruence ideal), Ribet-Stein Theorem 3.5, FPR94 I.4.2.2, Edixhoven92 Section 2, Atkin-Li Theorem 3.1, Atkin-Lehner Theorem 3, DDT 3.1(e), 3.15, 4.17, CalegariEmerton09 (point 57), Sutherland16 (point 3). Bibliography keys follow decision B14 (a key needed by more than one chapter goes in `refs.bib`). W8. Pitfalls. (a) `chi_{q*}`, never `chi_q`, for the quadratic character of conductor `q` (D3); numerical subscripts are quadratic characters `chi_d` (`chi_{-7}(2) = +1`, `chi_{-11}(2) = -1`); the cyclotomic character at `3` is `chi_3^cyc` (B-C6). (b) `t_l` at `q = 2` uses `-v_2(j)`, not the index of the symbol `I_n^*`. (c) Theorem thm:qual needs `l` not dividing `N`; 980i1 is outside it at `l = 5`. (d) In Theorem thm:quant the book allows every odd `q in S`; say that the paper has `q >= 5`. (e) Lemma lem:twist(ii)'s conductor statement at `q = 3` holds for the listed exchanges only. (f) Do not call `N_s(F_3)` maximal. (g) "Optimal" is Cremona's curve number `1` except 990h (990h3) below `10,000`, and is known for every class only up to `400000`. (h) The twist map is `iota_S`; `iota` is `H^1(T) -> H^1(V)`. (i) `f tensor chi` (naive twist) versus `f^chi` (newform twist). (j) Never identify the class 980g (the twist of 20a) with 980i. W9. Style. Short sentences; each section opens with a paragraph saying which part of the paper it covers. No dashes (write "Diamond-Flach-Guo" with a plain hyphen, ranges "1 to 4"). ASCII only in the source. W10. Length control. If the chapter runs long, shorten 21.15 to 21.17 (data) first, then the restatements of statements proved in Chapters 11, 14, 15, 18 to 20; never shorten 21.11, 21.13, 21.20 or 21.22.

23.8 8. Dependencies and verification log

23.8.1 8.1 Dependencies on Chapters 1 to 20 and 22 (by label; item numbers as cited in Part A)

- `ch:groups`: 5.7 (Borel commutator), 6.2, 6.3, 6.5 to 6.8, 6.10 (subgroups of `GL_2(F_3)`, orbits on lines, labels) (A5.1, A1.12, A1.13).
- `ch:rings`: 9.5, 9.6, 10.5 (Kronecker symbols, `q*`, the paper's characters), 11.4, 11.5 (the `I_0^*` identity, local terms) (A1.1, A1.2, A2.5).
- `ch:modules`: 5.5, 5.8, 7.6, 9.2 to 9.6, 10.3, 11.1, 11.3 to 11.6 (saturation, Fitting ideals, Gram determinants, alternating forms, idempotents, congruence ideals, Gorenstein) (A2.7, A4.1, A4.3, A4.4, A4.5).
- `ch:reps`: 1.4, 2.5, 3.7, 5.2, 6.2, 6.6, 7.3, 7.6, 7.7, 8.2, 10.5 (absolute irreducibility, Brauer-Nesbitt, Clifford, induction, adjoint, trace form) (A3.3, A4.11, A5.1, C3).
- `ch:galois`: galois:quartic-on-lines, galois:index-two-restriction, galois:chebotarev-brauer-nesbitt, galois:sqrt-lstar, galois:kummer-finite-field (A3.3, A5.1, A2.7).
- `ch:ant`: 3.7 (Frobenius convention), 4.7 to 4.9 (`Z_3[zeta_3]`, `Z_l tensor Z[zeta_3]`) (A2.2, A2.5, C2).
- `ch:local`: 4.3, 5.3, 5.4, 6.2, 9.2 to 9.4 (Frobenius convention, tame character and relation, Kummer class on inertia, local shapes) (A2.1, A2.5, A2.7, C2).
- `ch:curves`: 10.3, 11.5 (cup product, projection formula) (A4.1, A4.4).
- `ch:ellcurves`: 4.5, 9.6, 9.8 to 9.10, 10.4, 10.6, 10.7 (division polynomial, no cyclic 27-isogeny, prime-to-`l` isogenies, trivial classes, twists) (A1.1, A1.13, A4.14, A5.1, A6.9, B2).
- `ch:reduction`: 4.3, 4.5, 5.1, 5.4, 5.5, 6.6, 6.8, 7.4, 7.5, 7.7, 8.1 to 8.5, 9.2 to 9.4 (component groups, Tate's algorithm, codes, Kraus, type `IV` at `2`, Tate curves, twists, wild primes) (A1.1, A1.6, A1.8, A2.1 to A2.6, A9.2).
- `ch:galrep`: 1.4, 3.2, 3.3, 4.3 to 4.6, 6.1, 6.2, 7.4, 8.1 to 8.6, 9.1 to 9.3 (A1.4, A1.8, A1.12, A2.2, A2.3, A4.6, A4.11, A5.1, A5.2).
- `ch:modforms`: 11.3, 11.5 (Sturm, the paper's congruences for all `n`) (A3.6).
- `ch:hecke`: 4.1, 5.2 to 5.4, 6.3, 7.1 to 7.4, 8.1 to 8.3, 9.1 to 9.3, 11.1 to 11.3 (perfect pairing, adjoints, strong multiplicity one, old spaces, depletion, twists, congruence numbers) (A3.1 to A3.5, A4.14, A4.15).
- `ch:moddeg`: 3.3, 3.4, 6.4, 8.1 to 8.5, 9.1 to 9.7, 10.1 to 10.4, 11, 12.3, 12.4 (A3.1, A4.1, A4.4, A4.5, A6.10, A7.1).

- ch:modgalrep: 2.7, 3.8, 4.2, 6.2, 6.4 to 6.6, 7.2 to 7.4, 8.2, 8.3, 9.1 to 9.5, 10.1, 10.2, 11.1 to 11.4 (A1.9, A1.10, A3.2 to A3.7).
- ch:cohomology: inflation-restriction, conjugation-action, cohomology-through-pro-l-quotient, procyclic-compact, model-computation, frobenius-on-h1-inertia, lengths-in-exact-sequences (A2.7, A4.10).
- ch:galcoh: 5.1 to 5.5, 6.4, 10.3, 10.4 (A2.7, A4.8, A4.9).
- ch:blochkato: 3.3, 6.1, 6.3, 6.6, 7.1 to 7.6, 8.1 to 8.5, 9.1 to 9.5, 10.1 to 10.5, 11.1, 11.3, 12.1 to 12.3, 13.1, 13.2, 14.5 (A1.3, A2.7, A4.6 to A4.12, A5.2, A6.4, A8.1).
- ch:lfunctions: 7.4, 8.2 to 8.4, 9.1, 9.3, 9.5 to 9.7, 10.1, 10.2, 11.1, 12.1 (A1.0 to A1.2, A2.5, A2.6, A6.8).
- ch:congruence: 3.4, 4.3, 5.2 to 5.7, 6.2, 7.1 to 7.5, 8.5, 9.1 to 9.4, 10.1, 10.2, 11.2, 11.3, 12.1 to 12.7, 13.1 to 13.3, Part 7 (A1.4, A1.11, A4.1 to A4.16, A5.2, A8.1, B2, B3).
- ch:computation (comes after this chapter; forward references by \Cref{ch:computation} only): 3.5, 4.1, 4.3, 5.1 to 5.4, 6.1 to 6.8, 9.1, 9.2, 10.1 to 10.6, 11.1 to 11.8, 12.1, R0 to R9 (A6, A9, B4). Chapter 21 explains what each statement means and tests; Chapter 22 explains how to run it.

23.8.2 8.2 Verification log

First sessions (kept): Python 3 with sympy and Cremona's ecdata files for N < 10,000 (allcurves, degphi, galrep): Weierstrass invariants and Kodaira types of the example curves; a_p by point counting for p < 1000; the congruences of the paper's example; the 2 + 2 factorisation of psi_3 exactly for the 90 non-CM optimal 3Ns curves; label consistency of 3B.1.1, 3Cs.1.1; exhaustive checks of the Tamagawa formula for l = 3, 5. This revision: no new computation was run; every number added was taken from the PARI-verified records cited at its item (important_notes.md and the outlines of Chapters 10, 11, 13, 14, 15, 18, 19, 20, 22), and the arithmetic of each quoted example was rechecked by hand (v_3(2880) = 2, v_3(1728) = 3, v_3(3240) = 4, v_3(9/8) = 2, v_5(5/8) = 1, 1584 = 11 * 144, 504/72 = 7, the line counts of Section 4). Not verified here: the paper's large-range counts; the theorem numbers marked "as cited" in B3; the definitions of Sutherland's labels against Sutherland16; the exact statement of CalegariEmerton09.

23.9 9. Points for important_notes.md

The first version's points N1 to N14 are important_notes.md points 15 to 28 (N1 = 15, ..., N14 = 28) and are no longer listed here. This revision found no new point about the paper. Book-internal items found while revising (for the orchestrator):

R1. Appendix B: B-C2(b) says that in Chapter 21 the geometric Frobenius at a named prime is written Frob_q^{-1}, never Phi_q, while the Part 3 entry F1 for Chapter 21 says "write Phi_p" in the opening paragraph. This outline follows B-C2(b) (Frob_p^{-1}) and names Phi_p once as Appendix B's symbol for the same element; Appendix B Part 3, F1, should be reworded accordingly. R2. \Cref{ch:moddeg} 9.5(c), 9.6, 10.4(b) and \Cref{ch:modgalrep} 10.4 still carry the pre-settlement claims (multiplicity one needed for the equality in Theorem thm:quant; the DFG04 normalisation "to be checked"); \Cref{ch:blochkato} 9.4, 11.2, 11.3 say "subject to the DFG04 checklist". All are superseded by points 73 and 74 (already recorded as B26); listed here so the Chapter 14, 15, 18 writers apply them. R3. Item numbers of the other outlines are cited throughout Part A; any renumbering of those outlines must be propagated here.

Chapter 24

Chapter 22: Computation, data and the statistical protocol (ch:computation, chapters/ch22_computation.tex)

Stage 1 line-up. Part VI, last chapter, after Chapter 21 (the guided tour). Chapter 21 explains what each computational statement of the paper means and what it tests; this chapter explains how to run every computation, how to read Cremona's data files, and how to judge the paper's protocol and statistics. Target length 1,940 lines of LaTeX (card excluded; Part 3), inside the 1,000 to 2,200 range of PLAN.md Section 7.3.

Rigour tags, as in the Chapter 13 to 19 outlines: **P** = proved here, **S** = proof sketched here, **R** = stated; reference. Two further tags are used for computational items, and each maps to one of the three tags PLAN.md allows on the card: **C** = computed here (a session or count whose exact output is printed and was reproduced for this outline, Part 7; on the card it is tagged *(proved here)*, the proof being the printed computation) and **D** = data reported by the paper and not reproducible in this book beyond conductor 10,000 (on the card: *(stated; reference)*, the reference being the paper's Section 6). Item format: number, label-ready name, kind, tag; the precise statement or description; reference; the paper's use by its `\label` names. Every number below marked C was produced by the scripts of Part 7 with PARI/GP 2.17.2 (through `cypari2` 2.2.4) and Cremona's `ecdata` files for conductor below 10,000 in `/tmp/ch09/data`.

24.1 1. Purpose and global conventions

24.1.1 1.1 What this chapter is for

The paper's Section 6 (sec:data and its eight subsections, five tables), its appendix (the GP functions `rule(E, l, iii)` and `ns3(E)`) and a dozen sentences elsewhere ("PARI's `lfunsympow(E,2)` returns these factors at every prime"; "In each case the values were confirmed against PARI's `lfunsympow` on every curve of the tables"; "PARI's factor at 2 is $1 + 2X$ for every one of them"; "checked for $p < 500$") rest on software and data that the reader has never used, and on a statistical protocol that the reader's basic statistics course equips them to judge. The chapter has four jobs.

1. **Tools.** PARI/GP from zero, with exactly the functions the paper and the book use (`ellinit`, `ellglobalred`, `elllocalred`, `ellap`, `ellan`, `elltors`, `ellisomat`, `elldivpol`, `factor`, `valuation`, `elltwist`, `ellminimalmodel`, `lfunsympow`, `lfuneuler`, `lfunparams`, `ellinit` over $\mathbb{Q}_q$ and `.tate`, `ellmoddegree`, `ellweilcurve`), and Cremona's `ecdata` files (formats as found, Part 7). This discharges the deferrals of Chapters 2, 3, 9, 10, 11, 12, 13, 14, 15, 17, 18 and 19 (Part 5).

2. **The appendix, line by line.** `rule(E, l, iii)` and `ns3(E)` are printed verbatim, read line by line as code (the mathematics of each line is Chapter 19's 11.2 and Chapter 21's A9.1, cited, not repeated), proved to compute the right side of (eq:rule) and of its III-free form under stated conditions, and run on the running examples 121a1, 726a1, 539c1, 980i1, 99a1, 336f1, 1232l1, 592c1 and 11a1, with every term accounted for. The pitfalls (non-optimal models, CM curves, Eisenstein curves, the class 990h) are shown by running the function where it gives the wrong answer.
3. **The protocol and the statistics.** Working set, hold-out, extension and out-of-table curves; pre-registration; which statement was blind-tested, which only confirmed, which is a theorem merely checked; base rates, equality rates, the mean excess; the calibration argument of Section sec:sharp made exact (null model, expectation by linearity, variance, the choice of reference population, a permutation version); multiple comparisons; and an honest list of what the data do and do not show. Appendix A (`ch:prereq`, its Section on probability and statistics, already written) supplies the vocabulary (base rate, hypothesis test, hold-out, pre-registration, linearity of expectation, Markov and Chebyshev); this chapter applies it and does not restate it.
4. **Reproduction.** A small version of the paper's tables (tab:uniform, tab:counts, tab:ns, tab:iso, the Tamagawa test of sec:tam, the Eisenstein counts of sec:eis, the twist data after Theorem thm:twist, the calibration statistic and the odd-part statistic of sec:sharp) on all 38,042 optimal curves of conductor below 10,000, with the scripts that produce them, a comparison with the paper's numbers, and a reproducibility checklist. Every count of the paper on the large ranges is quoted as data (tag D); the book's own counts (tag C) are what the reader can rerun in minutes.

The physics reader knows the situation from experimental practice: a blinded analysis with a signal region kept closed until the selection is frozen is the hold-out; "unblinding" is the one-time test; a look-elsewhere correction is the multiple-comparisons caveat (one sentence each in Sections 9 and 10, no more).

24.1.2 1.2 Global conventions fixed in this chapter

- **K1 (software and versions).** The paper used PARI/GP 2.15 and `ecdata` at commit `25cec5e` (Section sec:data, the appendix). The book's outputs were produced with PARI/GP 2.17.2 through `cypari2` 2.2.4 (Python 3.13), and with a local copy of six `ecdata` files for conductor below 10,000 whose commit is not recorded (Part 6.5: the writer must pin it). Every printed output says which version produced it. Where 2.15 and 2.17 may differ (the return type of `ellmoddegree`, the availability of `readstr` and `strsplit`, the memory needed by `ellmoddegree`), the chapter says so and the writer checks the 2.15 manual.
- **K2 (models and labels).** Curves are given by Cremona's reduced global minimal models $[a_1, a_2, a_3, a_4, a_6]$ (Chapter 9, C3; Chapter 10). Labels are Cremona's (`\curve{121a1}`, macro `\curve` in `preamble.tex`); the LMFDB's labels appear only in Section 4. "Optimal" means $X_0(N)$-optimal (Chapter 14).
- **K3 (the quantities).** v_ℓ is the ℓ-adic valuation; $e_\ell(E,q)$, $t_\ell(E,q)$ (also at $q=\ell$, point 53), $w_3(E)$, $h_\ell(E)$ as in Definition def:terms and Conjecture conj:euler (Chapter 19, 9.1; Chapter 21, A1); $w_\ell(E,q)$ and Φ_q^{geom} as in Section 1.4 of the paper (Chapter 10). For a curve E and an odd prime ℓ the chapter names five bounds: $B_\ell^{\mathrm{rule}}(E)$ = right side of (eq:rule) = the second entry of `rule(E,l,1)`; $B_\ell^{\mathrm{nc}}(E) = B_\ell^{\mathrm{rule}}(E) + h_\ell(E)$ (no correction; "all types, no correction" of Section sec:anomaly); $B_\ell^{\mathrm{III}}(E)$ = second entry of `rule(E,l)` (types III, III* left out at $q \neq \ell$, no correction); $B_\ell^{\mathrm{main}}(E) = \sum_q v_\ell(\#\Phi_q^{\mathrm{geom}})$ (eq:main); $B_\ell^{\mathrm{tw}}(E) = \sum_q w_\ell(E,q)$ (eq:twist). The value $v = v_\ell(\deg \varphi_E)$ is the *observed valuation*.
- **K4 (violation, equality, excess).** For a curve with bound b and observed valuation v: a *violation* is $v < b$; an *equality* is $v = b$ with $b > 0$; the *excess* is $v - b$. Rates are always among curves with $b > 0$, as in the paper.
- **K5 (populations).** The *base population at ℓ* is the set of optimal curves (one per isogeny class) without CM whose mod ℓ representation is irreducible; CM is decided by j (the thirteen CM j-invariants, Chapter 9), irreducibility by the `galrep` labels (reducible iff a label starts with "ℓB"

or "ℓCs"; Section 4). The *Eisenstein population at ℓ* is the optimal non-CM curves with reducible mod ℓ representation. The *hypothesis population* adds $\ell \nmid N$ and, at $\ell = 3$, image not 3Ns (Theorem thm:main; important_notes.md point 47).

- **K6 (ranges).** The paper's ranges: working set $N \le 300000$, hold-out $300000 < N \le 400000$, extension $400000 < N \le 500000$, out-of-table $500000 < N \le 4 \cdot 10^6$. The book's: $N < 5000$ ("small working set"), $5000 \le N < 10000$ ("small hold-out"), and their union. The book's split is for teaching only: nothing was discovered on it.
- **K7 (Kodaira codes).** PARI's codes as in Chapter 10, item 5.5 (cited, not reprinted in full): 1 for I_0, $n+4$ for I_n and $-(n+4)$ for I_n^* when $n \ge 1$, -1 for I_0^*, $\pm 2, \pm 3, \pm 4$ for II/II*, III/III*, IV/IV* (point 52). The idioms used in code: `k >= 5` iff multiplicative; `k <= -5` iff I_n^*, $n \ge 1$; `abs(k) == 3` iff III or III*.
- **K8 (P_q and the symmetric square).** As Chapter 19, C4 and C5: `lfuneuler(lfunsympow(E,2), q)` is $1/P_q(x)$ with $x = q^{-s}$, and $P_q(q^{-2}) = L_q(A_f, 1)^{-1}$. This chapter never re-derives a factor; it computes and cites.
- **K9 (code typography).** GP code in `lstlisting` with `language=GP` (defined in `preamble.tex`), Python in `lstlisting` without language or in `verbatim`; GP comments start with `\\`. No code line contains two consecutive hyphens (decrement is written `k -= 1`, never with the double-hyphen operator), so that even code blocks pass a strict dash grep. Outputs are printed exactly as PARI prints them.
- **K10 (statistics vocabulary).** As Appendix A (`ch:prereq`): definitions def:prereq:prob, def:prereq:test, def:prereq:holdout, theorem thm:prereq:expectation, example ex:prereq:calibration, paperlink pl:prereq:protocol. Cite them by `\Cref{ch:prereq}` only (cross-chapter rule of PLAN.md 7.1); never restate the definitions.
- **K11 (typography).** PLAN.md 7.2: plain hyphens ("Agashe-Ribet-Stein", "Fontaine-Perrin-Riou"); ranges "1 to 8"; labels `...:computation:...`; ASCII source; numbers with thousands separators as `{,}` in math or plain commas in text, as the paper does.

24.2 2. Line-up of the core items, in teaching order

24.2.1 Section 1. Introduction and conventions

1.1 `conv:computation:versions` (convention; P as a record). K1 and K2 as one `convention` environment: the versions (PARI/GP 2.15 for the paper, 2.17.2 for the book), the data (`ecdata` commit `25cec5e` for the paper; the book's pinned commit, to be filled in by the writer, with the SHA-256 sums of the six files used, Part 6.5), and the rule that every printed output names its version. Paper: Section sec:data ("PARI/GP 2.15", "commit 25cec5e"), the appendix.

1.2 `conv:computation:bounds` (convention; P as a definition). K3 and K4: the five bounds B^{rule}, B^{nc}, B^{III}, B^{main}, B^{tw}, the observed valuation, violation, equality, excess, and the rule that rates are among positive bounds. One sentence: (eq:thm) is B^{nc} restricted to $q \ne \ell$ with no w_3 term, which agrees with B^{nc} whenever $\ell \nmid N$. Paper: Section sec:data, all tables.

1.3 `conv:computation:populations` (convention; P as a definition). K5 and K6, with the counts of the book's base populations below conductor 10,000 (C): 38,042 optimal curves, of which 189 CM; base population at $\ell = 3$: 33,851; Eisenstein population at $\ell = 3$: 4,002; at $\ell = 5, 7, 11, 13$ the base populations have 37,319, 37,691, 37,839, 37,823 curves. Paper: Section sec:data ("curves with $E[\ell]$ irreducible, not CM").

24.2.2 Section 2. PARI/GP from zero

2.1 `rem:computation:install` (remark; R, with the book's setup C). (i) Two ways to run PARI: the `gp` interpreter (Debian/Ubuntu package `pari-gp`; the PARI web site for other systems; the writer gives no version-specific install command it has not run) and the Python package `cypari2` (`pip install`

cypari2; then `import cypari2; pari = cypari2.Pari(); pari('...')` evaluates any GP expression; `pari('read("file.gp")')` loads a GP file). The book's outputs come from the second route (Part 7). (ii) Optional data packages: `elldata` (Cremona's tables inside PARI, needed by `ellsearch` and `ellidentify`) and `galdata`. Without `elldata`, `ellsearch(11)` fails with "error opening elldata file" (C: this is the state of the book's installation), which is why the chapter reads `ecdata` files directly (Section 4) instead of using `ellsearch` as PLAN.md suggests. (iii) Memory: `parisize` (current stack) and `parisizemax` (the ceiling to which PARI may grow the stack); defaults in the book's installation: 8,000,000 bytes and 8,003,584 bytes; `default(parisizemax, 10^9)` in GP or `pari.allocatemem(10**9)` in Python. Reference: PARIusers (cite the chapters on defaults and on elliptic curves; no section numbers unless checked). Paper: the appendix ("`ellmoddegree` needs `parisizemax` of a few hundred megabytes").

2.2 `rem:computation:gp-language` (remark; P as a description). The part of the GP language the chapter needs, in one page: integers and rationals are exact; `x` is a reserved polynomial variable (never assign to it: `lfuneuler` returns rational functions in `x`); vectors `[a, b]` are 1-indexed, `#v` is the length, `v[2..#v]` a slice; matrices `[1, 2; 3, 4]`, `M[i, j]`, a column `M[, 1]`; functions `f(a, {b = 0}) = ...` with optional arguments; local variables `my(...)`; braces `{ ... }` for a definition spanning several lines in a file; comments `\\ ...` to the end of the line; `if(c, a, b)`, `for(i = 1, n, ...)`, `foreach(v, s, ...)`, `next`; `Map()` with `mapput`, `mapisdefined`; strings with `Str`, `strsplit`, and `readstr(file)` (a vector of the lines of a file; available in 2.17.2, verify in 2.15); `valuation(r, l)` for rationals ($v_\ell(0)$ is `+oo`, so tests like `vj < 0` are false at $j = 0$); `factor`; `kronecker`; `O(q^k)` for q-adic precision. Paper: the appendix (every construct of `rule` is in this list).

2.3 `ex:computation:first-session` (worked example; C). On 121a1 $= [1, 1, 1, -30, -76]$: `E = ellinit([1,1,1,-30,-76])`; the member functions `E.a1`, ..., `E.a6`, `E.b2`, `E.c4`, `E.c6`, `E.disc`, `E.j` give $1, 1, 1, -30, -76$, $b_2 = 5$, $c_4 = 1441$, $c_6 = 54703$, $\Delta = -121$, $j = -24729001$; `ellan(E, 12)` gives `[1, -1, 2, -1, 1, -2, 2, 3, 1, -1, 0, -2]` ($a_{11} = 0$: additive at 11); `ellap(E, 2)` $= -1$. On 726a1: `elltors(E)` $=$ `[2, [2], [[6, -3]]]` (torsion $\mathbb{Z}/2$ generated by $(6, -3)$); `ellisomat(E)[2]` $=$ `[1, 2; 2, 1]` (class 726a has two curves joined by a 2-isogeny); on 11a1 the matrix is `[1, 5, 5; 5, 1, 25; 5, 25, 1]`, matching `ecdata`'s `allisog` (item 4.1). Paper: the appendix (`ellinit`, `E.j`).

2.4 `warn:computation:pitfalls` (warning; P). (a) `ellinit` accepts any model; `ellglobalred` reports the change to a minimal model, and `ellminimalmodel` returns it; local data are always those of the minimal model. (b) Do not assign to `x`. (c) q-adic objects carry precision `O(q^k)`; results are correct only to that precision. (d) Stack overflow is an error, not a wrong answer: raise `parisizemax`. (e) GP is 1-indexed, Python 0-indexed: `K[i][2]` in GP is `R[4][i-1][1]` in `cypari2`, and the entry `R[4][i, 1]` of the factorisation matrix is `R[3][i-1, 0]`; `cypari2` does not accept Python slices of a PARI matrix (C: `R[3][:,0]` raises a `TypeError`), so whole columns are taken in GP, `pari('(g) -> g[4][,1]~')(R)`, as S4 does. (f) `factor` of a polynomial with integer coefficients returns the irreducible factors over $\mathbb{Q}$ without the content: `factor(3*x^4 + 6)` is `Mat([x^4 + 2, 1])` (C); this is what makes `ns3` independent of the leading coefficient 3 of ψ_3. Paper: the appendix.

24.2.3 Section 3. Invariants of a curve in PARI

3.1 `def:computation:ellglobalred` (definition of the output; C). `ellglobalred(E)` returns `[N, [u, r, s, t], c, faN, L]`: the conductor, the change of variables to a global minimal model, the product $c = \prod_p c_p$ of the Tamagawa numbers, the factorisation matrix of N and the vector L of local data, $L[i] =$ `elllocalred(E, p_i)` $= [f_p, k_p, [u, r, s, t], c_p]$ (conductor exponent, Kodaira code, local change of variables, Tamagawa number). So the paper's `R[4][,1]` is the column of bad primes and `R[5][i][2]` (written `K[i][2]`) the Kodaira code at the i-th one. Exact output for 726a1 (C, 2.17.2): `[726, [1, 0, 0, 0], 4, [2, 1; 3, 1; 11, 2], [[1, 10, 0, 2], [1, 7, 0, 1], [2, 3, 0, 2]]]` (types I_6, I_3, III; $c_2 = 2$, $c_3 = 1$, $c_{11} = 2$; PARI prints a trivial change of variables as `0`); `elllocalred(ellinit([0,1,0,4,4]), 2)` $=$ `[2, -4, [1, 0, 0, 0], 3]` (20a1: IV^*, $f_2 = 2$, $c_2 = 3$). The mathematics (Tate's algorithm, Φ^{geom} versus c_q) is Chapter 10's (its 5.2, 5.5, 10.2); this item is the data structure only. Paper: the appendix (`R = ellglobalred(E)`, `P = R[4][,1]`, `K = R[5]`, `K[i][2]`), Section sec:data ("The Kodaira types and discriminant exponents at the bad primes were recomputed with PARI/GP 2.15 from the a-invariants").

3.2 `conv:computation:code-idioms` (convention; P). K7 with a five-line table of the tests used in `rule` and in the book's scripts (code test, Kodaira types selected, paper term affected): `k >= 5` (I_n, $n \geq 1$: divide by the naive factor; $\#\Phi^{\mathrm{geom}} = n = k-4$); `k <= -5` ($\mathrm{I}_n^*$, $n \geq 1$: $n = -k-4$); `k == -1` (I_0^*); `abs(k) == 3` (III, III*: skipped unless `iii`); `k == 4 || k == 2` / `k == -4 || k == -2` / `k == -3` (the weights $w_3 = 1, 2, 1$ at $q = \ell = 3$). Point 52 ($n+4$ and $-(n+4)$ only for $n \geq 1$) in one sentence. Full dictionary: Chapter 10, 5.5. Paper: the appendix ("Here `K[i][2]` is PARI's Kodaira code ... ").

3.3 `rem:computation:lfunsympow` (remark; C, mathematics cited from Chapter 19). `S = lfunsympow(E, 2)` is an L-function object for $L(\mathrm{Sym}^2 f, s)$ in the normalisation of Chapter 19, C4: `lfunparams(S)` is `[121, 3, [0, 0, 1]]` for 121a1 (conductor, motivic weight 3, gamma shifts). `lfuneuler(S, q)` is the Euler factor as a rational function of $x = q^{-s}$: for 121a1, `lfuneuler(S, 11)` = `1/(11*x + 1)` and `lfuneuler(S, 2)` = `1/(-8*x^3 - 2*x^2 + x + 1)` (a good prime, $a_2 = -1$: $P_2(X) = (1-2X)(1+3X+4X^2)$). Hence $P_q(q^{-2})$ is `subst(1/lfuneuler(S, q), x, 1/q^2)` (exact rational: 12/11 at $q = 11$), and $e_\ell(E, q)$ = `valuation(` that `/ (1 - a_q^2/q^2), l)` (Definition def:terms; Chapter 19, 11.1). Cost: the factors at the bad primes are computed when `S` is created; creating `S` and reading all bad factors took about 0.2 milliseconds per curve below conductor 10,000 (C). `lfun(S, 2)` is $L(\mathrm{Sym}^2 f, 2)$, used by `ellmoddegree` (3.8). Paper: Section 1.1 ("PARI's `lfunsympow(E,2)` returns these factors at every prime, including 2 and 3"), Section sec:data, the appendix (`S = lfunsympow(E, 2)`, `F = 1/lfuneuler(S, q)`, `r = subst(F, x, 1/q^2)`).

3.4 `ex:computation:euler-by-hand` (worked example and check; C). For $q \geq 5$ the factor is read off the Kodaira type (Lemma lem:euler; Chapter 19, 9.3): a GP function `euler_by_type(E, q)` returning $L_q(A_f, 1)^{-1}$ from the code, from $\chi_{-3}(q)$, $\chi_{-4}(q)$ (`kronecker(-3, q)`, `kronecker(-4, q)`) and, for I_0^*, from $a' =$ `ellap(ellinit(elltwist(E, q*)), q)` with q^* = `if(q % 4 == 1, q, -q)`; compared with 3.3 on every pair (optimal curve of conductor below 10,000, bad prime $q \geq 5$): 67,745 pairs (I_n 58,860, I_n^* 2,709, II to IV* 3,588, III, III* 1,676, I_0^* 912), no mismatch (C). Chapter 19's 9.3 did the same on all 64,687 curves (117,232 pairs); this item is the reader's version of that check and of the paper's sentence. Paper: Lemma lem:euler ("In each case the values were confirmed against PARI's `lfunsympow` on every curve of the tables"), the appendix (last sentence).

3.5 `ex:computation:tate-parameter` (worked example; C). `Eq = ellinit(E, O(q^30))` is E over $\mathbb{Q}_q$ to precision q^{30}; for multiplicative reduction (split or nonsplit) `Eq.tate[3]` is the Tate parameter $q_E \in \mathbb{Q}_q$, with $v_q(q_E) = n = -v_q(j)$ (Chapter 10, 7.3, 7.4; the series $\mathsf{q}(j)$ of Chapter 10, 7.2 gives the same q_E, as Chapter 18's verification log checked). The GP function `tatedata(E, q, l)` (S2) returns $[n, q_E, \bar{u}, k]$ with $\bar{u}$ the reduction of the unit part $q_E q^{-n}$ and k the largest $k \leq v_\ell(q-1)$ such that $\bar{u}$ is an ℓ^k-th power in $\mathbb{F}_q^\times$ (test: $\bar{u}^{(q-1)/\ell^k} = 1$). Outputs (C): 11a1 at 11, $\ell = 5$: $n = 5$, $q_E = 10 \cdot 11^5 + 2 \cdot 11^6 + \cdots$, $\bar{u} = 10$, $k = 1$; 77c1 at 7 ($a_7 = -1$, nonsplit): $[3, \ldots, 5, 0]$; 539c1 at 7: the same q_E as 77c1 (its twist), $\bar{u} = 5$, $k = 0$; 791a1 $= [1, 0, 1, -31, 117]$ at 7: $\bar{u} = 6$, $k = 1$; 1577a1 $= [0, -1, 1, 30, -335]$ at 19: $\bar{u} = 8$, $k = 1$. These agree with Chapter 17, 5.5 and Chapter 18, 8.5. Two warnings: `elltatepairing` is the Tate pairing on torsion points and has nothing to do with the Tate parameter; and `ellinit(E, q)` with an integer prime q (PLAN.md's former suggestion, now corrected there) reduces E modulo q: it returns `[]` at a bad prime (C: 77c1 at 7) and the curve over $\mathbb{F}_q$ at a good one, never the curve over $\mathbb{Q}_q$; the q-adic curve needs `O(q^k)` (Part 8, B-c). Notation: the class of $\bar{u}$ is written $c(u)$, never c_q (points 23, N9 of Chapter 21). Paper: Section sec:data ("the Tate parameters with `ellinit` over $\mathbb{Q}_q$"), Section sec:tam, Lemma lem:tamfactor, Remark rem:ko(1).

3.6 `ex:computation:mod3-image` (worked example and check; C). `elldivpol(E, 3)` is ψ_3 (degree 4, leading coefficient 3); `factor` gives the pattern. 726a1: $\psi_3 = 3x^4 + 5x^3 - 210x^2 - 612x - 1480 = (x^2 - x - 74)(3x^2 + 8x + 20)$, pattern $2+2$, so `ns3` $= 1$ (image 3Ns, Proposition prop:ns). `polgalois` needs an irreducible polynomial and raises an error on 726a1's ψ_3 (C): use it only on irreducible ψ_3 (Chapter 11, 7.3 for the pattern-to-image table, cited). Check (C): among the 37,853 non-CM optimal curves below 10,000, `ns3(E) = 1` exactly for the 90 curves whose `galrep` entry contains 3Ns (no mismatch either way); among the 189 CM optimal curves `ns3` is 1 for 13 (all with $j = -32768$ or 8000, e.g. 121b1, 256a1), which `galrep` does not label (CM caveat, Chapter 11's B7). Paper: the appendix (`ns3`), Section sec:data ("The mod 3 images of the tables were checked against the factorisation of the 3-division polynomial"), Proposition prop:ns, Section sec:anomaly.

3.7 `ex:computation:twists` (worked example and check; C). `elltwist(E, d)` is the quadratic twist

by $\mathbb{Q}(\sqrt{d})$ (a model, not necessarily minimal; `ellminimalmodel` normalises it; `ellglobalred` of the twist gives its conductor). The GP function `twistcheck(E)` (S2) lists, for each odd bad prime q, the codes of E and of $E \otimes \chi_{q^*}$ at q and the conductor of the twist: 121a1 gives `[[11, 2, -4, 121]]` (II $\to$ IV*, conductor unchanged); 539c1 gives `[[7, -7, 7, 77], [11, 6, -6, 5929]]`; 980i1 gives `[[5, 5, -5, 4900], [7, -9, 9, 140]]`. Check of Lemma lem:twist (i), (ii) on all optimal curves below 10,000 (C): 85,848 pairs (curve, odd bad prime q, with $q \geq 5$ for the potentially good exchanges) with the predicted code and conductor exponent at q, no mismatch; no change of code at any odd $p \neq q$; and at $q = 3$ the exchanges II $\leftrightarrow$ IV*, III $\leftrightarrow$ III*, IV $\leftrightarrow$ II* hold in all 5,122 pairs (important_notes.md point 51). This is the book's version of the paper's `twist_lemma_check.gp`, which the paper ran on 4,577 random pairs. Paper: Lemma lem:twist, the appendix (last paragraph).

3.8 `rem:computation:ellmoddegree` (remark; C for the facts about the function, S for the method). (i) `ellmoddegree(E)` returns the degree of the modular parametrisation of the given curve divided by the square of its Manin constant (PARI manual; verify the exact wording in 2.15): on 11a1, 11a2, 11a3 it returns 1, 5, 1/5 (C). So it is $\deg \varphi_E$ only on the optimal curve. (ii) `ellweilcurve(E)` returns the curves of the isogeny class with lattice data; the optimal curve is the one whose entry is `[1, 1]` (C: for class 11a the entries are `[1, 1/5]`, `[1, 1]`, `[5, 1]`, marking 11a1; for class 990h they are `[3, 1]`, `[1, 1]`, `[6, 1]`, `[2, 1]`, marking $[1, -1, 1, -1568, -4669] = 990\text{h}3$; the writer checks the meaning of the entries in the manual before printing more than "the entry `[1, 1]` marks the optimal curve"). (iii) The method: Watkins's, through $L(\mathrm{Sym}^2 f, 2)$ and the Petersson norm (Chapter 14, 7.3 and 12.5; Chapter 19, 8.4); the `lfun` machinery prints `_lfuninit_worker` warnings when it grows the stack (C). (iv) Cost and memory (C, 2.17.2): about 0.065 seconds per curve on a random sample below 10,000; the default 8 MB stack suffices for most curves (e.g. 9999m1, degree 5632) but not for the largest degrees (9510e1, degree 100,254,000; 9690m1, 68,217,600; 7910b1, 31,272,384), which need 16 to 32 MB. (v) Full check: `ellmoddegree` agrees with `ecdata`'s `degphi` on all 38,042 optimal curves below 10,000 (Part 7, R8; no mismatch). Paper: Section 1 ("The modular degree is computable"), Section sec:data ("fifteen modular degrees spanning the configurations were recomputed with `ellmoddegree`"), Section sec:oot, the appendix.

24.2.4 Section 4. Cremona's tables and the `ecdata` files

4.1 `def:computation:ecdata-files` (definition by description; C for the formats as found). Cremona's tables (Cremona97 for $N \leq 1000$ in print; the `ecdata` repository, refs.bib key `ecdata`, for everything since) are plain text files, one per range of 10,000 conductors, one line per curve or per isogeny class, fields separated by single spaces, a-invariants written `[a1,a2,a3,a4,a6]` without spaces. The six files used in this book, for conductor 0 to 9,999, with their formats (C, read off the files):

file	lines	one line per	fields
`allcurves.00000-09999`	64,687	curve	N, class, number, $[a_1, \ldots, a_6]$, rank r, $\#E(\mathbb{Q})_{\mathrm{tors}}$
`allbsd.00000-09999`	64,687	curve	as `allcurves`, then $\prod_p c_p$, real period Ω, $L^{(r)}(E,1)/r!$, regulator, analytic order of Sha
`allisog.00000-09999`	38,042	class	N, class, 1, $[a_i]$ of curve 1, the list of $[a_i]$ of the class, the isogeny-degree matrix
`alllabels.00000-09999`	64,687	curve	Cremona's N, class, number, then the LMFDB's N, class, number

file	lines	one line per	fields
`degphi.00000-09999`	38,042	class	N, class, number of the optimal curve, $\deg \varphi_E$, the set of primes dividing $\deg \varphi_E$, $[a_i]$ of the optimal curve
`galrep.00000-09999`	64,687	curve	label, then the Sutherland labels of the non-surjective mod ℓ images (nothing if all are surjective)

Sample lines (C): `11 a 1 [0,-1,1,-10,-20] 0 5` (allcurves); `11 a 1 [0,-1,1,-10,-20] 0 5 5 1.26920930427955 0.253841860855911 1.00000000000000 1` (allbsd: $\prod c_p = 5$, $L(E,1)/\Omega = 1/5$); `11 a 1 1 {} [0,-1,1,-10,-20]` (degphi: degree 1, no prime divisors); `726 a 1 144 {2,3} [1,1,0,-35,-51]`; `11a1 5Cs.1.1` and `726a1 2B 3Ns` (galrep); `11 a 2 11 a 1` (alllabels: 11a2 is 11.a1). Facts the reader needs: the Kodaira symbols and the local Tamagawa numbers are **not** in these six files (only $\prod_p c_p$ is, in `allbsd`), so they are recomputed with PARI (3.1), as the paper did; `degphi` has one line per class and gives the optimal curve's equation; `galrep` gives images only for ℓ where the image is not $\mathrm{GL}_2(\mathbb{F}_\ell)$. The repository has further directories (generators, a_p lists, integral points, optimality and Manin constant data, and others); the writer names only those checked in the repository at the pinned commit (Part 6.5), and must not write "`alldegphi`" (PLAN.md's former name, now corrected there; the file is `degphi`, Part 8). Paper: Section sec:data ("Cremona's tables in the form of the `ecdata` repository at commit `25cec5e`, which contain every elliptic curve over $\mathbb{Q}$ of conductor below 500000 with ...").

4.2 `ex:computation:reading-ecdata` (worked example; C). Reading the files in GP: `readstr(file)` gives the vector of lines, `strsplit(s, " ")` the fields, `eval` turns `"[0,-1,1,-10,-20]"` into a vector (C, 2.17.2). In Python: `line.split()` and `ast.literal_eval` (or `eval` on trusted data). The GP functions `readgal` (a `Map` from labels to image labels) and the loop of S3 are printed in full. One caution: the label of the optimal curve is assembled from the `degphi` line, whose "number" field is 1 even for 990h, whose equation is that of 990h3 (4.3); the scripts look the equation up in `allcurves` instead of trusting the number. Paper: background (the appendix's scripts are not in the repository of the book).

4.3 `rem:computation:optimality-data` (remark; R, with C checks). (i) In Cremona's labelling, curve number 1 of each class is the optimal curve, with exactly one exception below 10,000: class 990h, whose optimal curve is 990h3 $= [1,-1,1,-1568,-4669]$, the equation listed in `degphi` (C: of the 38,042 `degphi` equations, exactly one is not curve number 1 in `allcurves`). Chapter 9, 12.1 states this; here it is checked and its computational consequence shown: `ellmoddegree` gives 1728 on 990h3 and 5184 on 990h1, and `rule` gives `[3, 3]` and `[4, 3]` at $\ell = 3$ (C). (ii) Optimality is proved for all classes with $N \le 400000$ and for 359,009 classes above (the paper, Section sec:data; Chapter 14, its 6.5 for what "proved" means); the paper's extension range uses only those classes (point 25). (iii) `ellweilcurve` (3.8(ii)) identifies the optimal curve inside PARI for moderate conductors. (iv) Point 27 / Chapter 21's N13: when $E[\ell]$ is irreducible, v_ℓ of the degree does not depend on which curve of the class is optimal; when it is reducible, it does (11a1, 11a2, 11a3 at $\ell = 5$: `rule` gives `[0, 1]`, `[1, 0]`, `[-1, 0]`, C). Paper: Section 1 ("Among the curves in the isogeny class of E there is a distinguished one"), Section sec:data ("Optimality is determined in the tables for every class of conductor at most 400000 and for 359,009 further classes up to 500000"), the appendix.

4.4 `conv:computation:labels` (convention; R). (i) Cremona labels N class number, class codes a, b, ..., z, ba, ...; the capitalised form 99A1 of Cremona97 and ARS12 is 99a1. (ii) LMFDB labels N.class number, ordered differently; the correspondence is the `alllabels` file (examples only: 11a1 = 11.a2, 726a1 = 726.b2; the full list of the running examples is Chapter 9's 12.1, cited). (iii) Sutherland's labels of mod ℓ images as they occur in `galrep` (the general scheme and the $\ell = 3$ table are Chapter 11's 7.3, 7.4, cited, not reprinted); what the chapter needs: an image is reducible iff its label starts with "ℓB" or "ℓCs" (Borel or split Cartan subgroups and their subgroups), so `galrep` decides membership in the base population

of K5; 3Ns (image $N_s(\mathbb{F}_3)$), 3Cs.1.1 ($E[3] \simeq \mathbb{Z}/3 \oplus \mu_3$, the order-2 subgroup of the split Cartan, point 44) and 3B.1.1 (a rational point of order 3, $E[3]$ non-split) are the three labels the paper uses. Counts among the 37,853 non-CM optimal curves below 10,000 (C): 3B.1.1 1,858, 3B 1,567, 3B.1.2 503, 3Nn 170, 3Ns 90, 3Cs.1.1 49, 3Cs 25. CM caveat: `galrep` omits the labels forced by CM (Chapter 11, 7.3), so CM curves are excluded by j, not by labels. (iv) Point 3 (labels to be checked against Sutherland16): the intrinsic descriptions of Chapter 11 are used in every statement; the data checks of 3.6 and of Chapter 11 confirm the `ecdata` usage. Paper: Section 1.3 (3Ns), Section sec:anomaly, Section sec:eis (3Cs.1.1, 3B.1.1), Section sec:data.

24.2.5 Section 5. The paper's function `rule(E, l, iii)`, line by line

5.1 `ex:computation:rule-listing` (construction; C). The appendix code, verbatim (it is the paper's `scripts/euler_rule.gp`), in an `lstlisting` with line numbers: the 14 lines of `rule` and the one line of `ns3`, exactly as in the paper (they contain no double hyphen; K9). Loaded with `read("euler_rule.gp")` in GP or `pari('read("euler_rule.gp")')` in Python (the code cannot be pasted into `pari(...)` as one string, because a GP file joins the lines of a braced block, while a single string does not: C, the first attempt failed with a syntax error). Paper: the appendix.

5.2 `rem:computation:rule-reading` (remark; P, a reading of code). One paragraph per line, saying what the line computes as a program (types, indices, branches) and which term of (eq:rule) it is; the mathematics of each term is cited (Chapter 19, 11.2; Chapter 21, A9.1) and not re-proved. The points that are specific to the code: (a) `P = R[4][,1]` is a column vector of the bad primes in increasing order, and `K[i]` refers to the same order; (b) `F = 1/lfuneuler(S, q)` is a polynomial in `x`, and `subst(F, x, 1/q^2)` an exact rational; (c) the test `k >= 5` is "multiplicative", and dividing by `1 - 1/q^2` makes `r` equal to 1 there (the naive factor is the true one, $e_\ell = 0$); (d) at an additive prime no division is made, because the naive factor is 1 ($a_q = 0$); (e) `abs(k) != 3 || iii` skips the types III, III* at $q \neq \ell$ unless `iii` is set; (f) the `else` branch of `if(q != l, ...)` handles $q = \ell$: no Euler term, and for $\ell = 3$ the weight w_3 by code (1 for IV, II, III*; 2 for IV*, II*; 0 otherwise), so `rule(E, 3)` counts the III* weight at $q = 3$ even with `iii = 0` (Chapter 21, A9.1(c)); (g) `t` is computed at every bad prime, including $q = \ell$ (point 53), and `valuation(j, q)` of $j = 0$ is `+oo`, so `t = 0` there; (h) the correction `ns3(E)` is subtracted only when `iii` is set and $\ell = 3$ (for $\ell \geq 5$, $h_\ell = 0$ always: Lemma lem:h0 with point 5); (i) the returned pair is $[v_\ell(\text{ellmoddegree}(E)),\ b]$, and `ellmoddegree` is by far the slowest step (3.8). Paper: the appendix.

5.3 `prop:computation:rule-correct` (proposition; P). Let E be an optimal curve given by any model, ℓ an odd prime. (i) For every bad prime $q \neq \ell$ the number `e` computed by `rule(E, l, 1)` is $e_\ell(E, q)$ of Definition def:terms, and `t` is $t_\ell(E, q)$. (ii) If E has no CM and $E[3]$ is irreducible, `ns3(E)` $= h_3(E)$. (iii) Hence, under these conditions, `rule(E, l, 1)` returns $[v_\ell(\deg \varphi_E), B_\ell^{\text{rule}}(E)]$ and `rule(E, l)` returns $[v_\ell(\deg \varphi_E), B_\ell^{\text{III}}(E)]$; and $B_\ell^{\text{nc}} = B_\ell^{\text{rule}} +$ `ns3`. Proof: (i) the code is 1 exactly at I_0 (never a bad prime), ≥ 5 exactly at multiplicative primes, where $a_q^2 = 1$ and the naive factor at $s = 1$ is $1 - q^{-2}$; at every other bad prime $a_q = 0$ and the naive factor is 1 (Chapter 19, 8.1); so `r` is the ratio of Definition def:terms; `t` is the definition of t_ℓ verbatim. (ii) `ns3` is 1 iff ψ_3 factors as $2 + 2$ over $\mathbb{Q}$ (item 3.6, `factor` ignores content and lists factors by increasing degree, so patterns 4, $1 + 3$, $1 + 1 + 2$, $1 + 1 + 1 + 1$ give 0); for $E[3]$ irreducible and no CM this is image $N_s(\mathbb{F}_3)$ (Proposition prop:ns, Chapter 11, 7.3), and then $h_3(E) = 1$, otherwise 0 (Lemma lem:h0 corrected by point 5: $h_3 \leq 1$, with equality iff $\bar{\rho} \simeq \bar{\rho} \otimes \chi_{-3}$). (iii) follows, $\deg \varphi_E$ being `ellmoddegree(E)` for the optimal curve (3.8; Manin constant 1 on the tables, Chapter 14). Check (C): the book's independent Python implementation of the five bounds (S5) agrees with `rule` on the running examples and with the GP loop S3 on every optimal curve below conductor 2,000 at $\ell = 3, 5, 7$. Paper: the appendix ("`rule(E,l)` leaves the type III terms out and `rule(E,l,1)` is (eq:rule)").

5.4 `warn:computation:rule-pitfalls` (warning; C for each example). Where `rule` gives an answer that is not the paper's quantity: (a) a non-optimal model: `ellmoddegree` returns $\deg / c_\varphi^2$ of that model (c_φ the Manin constant of its parametrisation, Appendix B, B-C9), so `rule` on 11a3 at $\ell = 5$ returns `[-1, 0]` (a negative "valuation of a degree") and on 990h1 at $\ell = 3$ returns `[4, 3]` instead of 990h3's `[3, 3]`; (b) CM curves: `ns3` is 1 on 13 CM optimal curves below 10,000, where h_3 has no meaning in Conjecture conj:euler (which excludes CM); (c) Eisenstein curves: 11a1 at $\ell = 5$ returns `[0, 1]`, an apparent violation,

but $E[5]$ is reducible (5Cs.1.1) and the curve is outside every statement of the paper; the population filter K5 is part of the statement, not an afterthought; (d) $\ell \mid N$: `rule` runs and the data satisfy (eq:rule) there, but no theorem applies (Theorem thm:main needs $\ell \nmid N$); (e) the codes $n+4$, $-(n+4)$ are for $n \geq 1$ (point 52); (f) versions: in 2.17.2 `ellmoddegree` returns an integer (or a rational on a non-optimal model); the writer checks the return type in 2.15 before saying the code runs unchanged there; (g) speed: for a table, take $\deg \varphi_E$ from `degphi` and compute only the bound (the function `rulebound` of S3). Paper: the appendix ("A single curve can be checked against (eq:rule) in PARI/GP"; important_notes.md points 27, 52, 53; Part 8, N3).

24.2.6 Section 6. Worked sessions

Each item: the input, the exact output (2.17.2), the per-prime breakdown printed by `rule_terms(E, 1)` of S2 (rows $[q, k, P_q(x), P_q(q^{-2}), e, w, t]$), and the reading. Mathematics of each term: Chapters 19 and 21 (cited).

6.1 `ex:computation:121a1` (worked example; C). `rule(ellinit([1,1,1,-30,-76]), 3) = [1, 1]` (with or without `iii`). Breakdown: `[[11, 2, 11*x + 1, 12/11, 1, 0, 0]]`: type II at 11, $P_{11} = 1 + 11X$, $e_3 = v_3(12/11) = 1$, $t_3 = 0$; $\deg \varphi_E = 6$ (`ellmoddegree` and `degphi`). Also $B_3^{\mathrm{main}} = 0$ (type II has $\#\Phi^{\mathrm{geom}} = 1$) and $B_3^{\mathrm{tw}} = 1$: the curve on which (eq:twist) is strictly stronger than (eq:main). Paper: the appendix ("returns `[1, 1]`"), the Example of Section 3 (b).

6.2 `ex:computation:726a1` (worked example; C). `rule(E, 3)` and `rule(E, 3, 1)` both return `[2, 2]` for 726a1 $= [1, 1, 0, -35, -51]$. Breakdown with `iii = 1`: `[[2, 10, -x + 1, 3/4, 0, 0, 1], [3, 7, -x + 1, 8/9, 0, 0, 1], [11, 3, 11*x + 1, 12/11, 1, 0, 0]]`; with `iii = 0` the last row has `e = 0`. So $B_3^{\mathrm{III}} = 1+1 = 2$ (t_3 at 2 from I_6, at 3 $= \ell$ from I_3), $B_3^{\mathrm{nc}} = 3$, `ns3` $= 1$ ($\psi_3 = (x^2-x-74)(3x^2+8x+20)$), $B_3^{\mathrm{rule}} = 2$; $\deg \varphi_E = 144 = 2^4 3^2$. Reading: without the correction the bound fails by exactly one, the violation involves the prime $2 \equiv -1 \pmod 3$ with $t_3(E, 2) = 1$ (as Section sec:anomaly says of every violation), and $3 \mid N$ puts the curve outside Theorem thm:main anyway (with (eq:thm) read literally, terms at $q \neq 3$ only, the bound is 2 and there is no violation: Part 8, N6). The congruence number $r_E = 1584 = 11 \cdot 144$ (Chapter 14; point 58) is not computable by `rule`. Paper: the appendix ("for 726a1 ... both return `[2, 2]`, the type III term and the correction cancelling"), Section sec:anomaly.

6.3 `ex:computation:539c1` (worked example; C). `rule(ellinit([1,0,1,170,-3237]), 3) = [2, 2]`: $q = 7$, code -7 (I_3^*), $P_7 = 1 - X$, $P_7(7^{-2}) = 48/49$, $e_3 = v_3(48/49) = 1$ (no division: additive), $t_3 = v_3(3) = 1$; $q = 11$, code 6 (I_2), $e_3 = 0$, $t_3 = 0$; $\deg \varphi_E = 288$. At $\ell = 5$: `[0, 0]`. `tatedata(E, 7, 3)` $= [3, q_E, 5, 0]$: the Tate parameter of the twist 77c1, and 5 is not a cube mod 7, so the Tamagawa factor adds nothing beyond t_3 (Lemma lem:tamfactor). Paper: the Example of Section 3, (c) at $\ell = 3$ ("the two terms $e_3(E, 7) = v_3(48) = 1$ and $t_3(E, 7) = v_3(3) = 1$ are both attained").

6.4 `ex:computation:980i1` (worked example; C). 980i1 $= [0, 0, 0, 1568, -72716]$: `rule(E, 3) = [2, 2]` ($q = 2$, code -4, IV^*, $P_2 = 1 + 2X$, $e_3 = v_3(3/2) = 1$; $q = 7$, I_5^*, $e_3 = v_3(48/49) = 1$); `rule(E, 5) = [1, 1]` ($t_5(E, 7) = v_5(5) = 1$; $e_5 = 0$ everywhere). $\deg \varphi_E = 2880 = 2^6 3^2 5$. Reading: at $\ell = 3$ the prime 2 contributes through PARI's factor (point 43: a theorem); at $\ell = 5$, $5 \mid N$, so the example illustrates the congruence, not Theorem thm:qual (point 21). Paper: the Example of Section 3, (c) at $\ell = 5$.

6.5 `ex:computation:99a1` (worked example; C). 99a1 $= [1, -1, 1, -2, 0]$, $N = 3^2 \cdot 11$: `rule(E, 3) = [0, 0]`: $q = 3 = \ell$, code 3 (III), $w_3 = 0$, $t_3 = 0$ ($v_3(j) = 9$); $q = 11$, I_1. $\deg \varphi_E = 4$; the congruence number $r_E = 12$ (ARS12, Table 1; recomputed in Chapter 14) is outside PARI's built-in functions. PARI's $P_3 = 1 + 3X$ is not used by `rule` at $q = \ell$. Reading: the bound is 0 and there is no violation; the gap between r_E and $\deg \varphi_E$ at 3 is allowed because $9 \mid N$. Paper: the Remark after the Example of Section 3 (99A1).

6.6 `ex:computation:wild-two` (worked example; C). Three curves where the prime 2 contributes through PARI's factor alone: 336f1 $= [0, 1, 0, -1, 2]$ (type II at 2, $P_2 = 1 + 2X$, $e_3 = 1$; `rule(E, 3) = [1, 1]`, $\deg \varphi_E = 24$); 1232l1 $= [0, 0, 0, 32, -16]$ (type II^* at 2, `lfuneuler` gives $P_2 = -8x^3 - 4x^2 + 2x + 1 = (1 - 2x)(1 + 2x)^2$, $e_3 = v_3(9/8) = 2$; `rule(E, 3) = [2, 2]`, $\deg \varphi_E = 288$); 592c1 $= [0, 0, 0, -16, -16]$ (type II^* at 2, $P_2 = -8x^3 + 4x^2 - 2x + 1 = (1 - 2x)(1 + 4x^2)$, $e_5 = v_5(5/8) = 1$; `rule(E, 5) = [1, 1]`, $\deg \varphi_E = 80$). Reading: the type at 2 does not determine P_2 (point 50; the classification of Chapter 19, 10.1, cited), and the prime 2 can contribute at $\ell = 5$ (point 69). Paper: Remark rem:wild, Section sec:wild.

6.7 `ex:computation:11a1-eisenstein` (worked example; C). 11a1 at $\ell = 5$: `rule(E, 5) = [0, 1]`; `galrep` gives 5Cs.1.1 ($E[5] \simeq \mathbb{Z}/5 \oplus \mu_5$), `tatedata(E, 11, 5) = [5, q_E, 10, 1]`: $\bar{u} = 10 \equiv -1$ is a fifth power in $\mathbb{F}_{11}^{\times}$, so the Tamagawa factor of $\mathrm{ad}^0 T$ at 11 has valuation 2 (Lemma lem:tamfactor), while $\deg \varphi_E = 1$. Reading: an Eisenstein curve, outside every statement of the paper; the example shows why the population filter (K5) is part of each statement (Chapter 16's Exercise 10, Chapter 18's 8.5 make the same point in cohomology). Paper: Section sec:eis, Conjecture conj:eis (for 11a1 at $\ell = 5$: $D = 1 = v_5(\#E(\mathbb{Q})_{\mathrm{tors}})$).

6.8 `ex:computation:tamagawa-session` (worked example; C). At $\ell = 3$, multiplicative primes $q \equiv 1 \pmod 3$ with $3 \mid n$: 791a1 at 7 ($\bar{u} = 6$, a cube: extra term 1; $B_3^{\mathrm{rule}} = 1$, refined bound 2, $\deg \varphi_E = 144$, $v_3 = 2$: the refined bound is attained); 77c1 at 7 ($\bar{u} = 5$, not a cube: extra 0; bound 1, $\deg \varphi_E = 6$); 1577a1 at 19 ($\bar{u} = 8 = 2^3$, a cube and not a ninth power: extra 1; bound 2, $\deg \varphi_E = 720$). Agrees with Chapter 17, 5.5 (cited for the cohomology). Paper: Section sec:tam, Remark rem:ko(1).

24.2.7 Section 7. Computing modular degrees and congruence numbers

7.1 `rem:computation:moddeg-methods` (remark; S). Three methods, each in one paragraph, with pointers: Zagier's and Cremona's through modular symbols and the period lattice (Zagier85; Cremona97, Chapter II), and Watkins's through the symmetric square L-value and the Petersson norm (Watkins02), which is what `ellmoddegree` implements (3.8). Why it is expensive (S, heuristic; the writer cites PARI's documentation of `lfun` or Dokchitser's method, verify): the number of coefficients needed to evaluate $L(\mathrm{Sym}^2 f, 2)$ grows roughly like the square root of the conductor of $L(\mathrm{Sym}^2 f, s)$, which is at most N^2 (Chapter 19, 7.2), so roughly like N; Cremona's tables computed degrees up to 500000, the paper only fifteen with `ellmoddegree` inside the tables and 893 outside. The mathematics is Chapter 14's 7.3, 12.5 and Chapter 19's 8.3, 8.4 (cited). Paper: Section 1 ("The modular degree is computable \cite{Cremona97,Watkins02}"), Section sec:data, Section sec:oot.

7.2 `ex:computation:moddeg-check` (worked example; C). `ellmoddegree` against `degphi` on all 38,042 optimal curves of conductor below 10,000 (script S8, four processes, Part 7, R8): no mismatch. The paper's sentence "fifteen modular degrees spanning the configurations were recomputed with `ellmoddegree`, all agreeing with the tables" is thereby extended, in the book's range, to every curve. Paper: Section sec:data.

7.3 `rem:computation:congruence-number` (remark; R, values C in Chapter 14). PARI has no function for the congruence number r_E. Chapter 14 (its 8.5 and verification log) computes it from `mfinit`, `mfbasis` and the Hecke action on the integral lattice; quote its values $r(99\mathrm{a}1) = 12$ against $\deg \varphi_E = 4$, $r(726\mathrm{a}1) = 1584$ against 144 (point 58), $r(539\mathrm{d}1) = 504$ against 72 (point 62), and say only where the algorithm lives. Paper: the Remark after the Example of Section 3 (99A1), Lemma lem:crit; PLAN.md's scope item "99A1 (where `ellmoddegree` gives 4 and the congruence number 12 is not given by PARI directly)".

24.2.8 Section 8. Populations, violations and descriptive statistics

8.1 `def:computation:units` (definition; P). What is counted. (i) The unit is a curve, one per isogeny class (the optimal one), at a fixed ℓ; "pairs (E, ℓ)" (Section sec:forms) and "pairs (curve, prime q)" (after Theorem thm:twist; Lemma lem:twist checks) are different units and their counts are not comparable with curve counts. (ii) The populations of K5, with the reason for each filter: CM curves are excluded by the statements (Conjecture conj:euler, Conjecture conj:eis); reducible $E[\ell]$ moves a curve to the Eisenstein population, where a different statement is made (Conjecture conj:eis); the hypothesis population is where Theorem thm:main applies, so that data there are checks of a theorem, not tests. (iii) Why only one curve per class: $v_\ell(\deg \varphi_E)$ is attached to the optimal curve, and counting all curves of a class would count the same newform several times (point 27: for $E[\ell]$ irreducible the local terms and v_ℓ are class invariants). Paper: Section sec:data, Tables tab:uniform, tab:counts, tab:ns, Section sec:forms ("pairs (E, ℓ)").

8.2 `def:computation:rates` (definition and remark; P). For a population and a bound B: the number with $B > 0$; the number of violations; the *equality rate* (equalities over positive bounds); the *mean excess* (average of $v - B$ over positive bounds); the *base rate* of an attribute (Appendix A, def:prereq:prob). Remarks: (a) curves with $B = 0$ cannot violate, so including them would inflate every rate; (b) an equality

rate is a property of the population and the range, not of "elliptic curves": in the book's data at $\ell = 3$ it is 75.1% for $N < 5000$ and 68.0% for $5000 \le N < 10000$, against the paper's 57.5% for $N \le 300000$ (C, D); degrees grow with N, and with them the room for extra divisibility; (c) the mean excess is 0.31 and 0.42 on the two small ranges, 0.37 on their union (C), against the paper's 0.63 (D); under the hypothesis of Theorem thm:main the excess over B^{rule} is the Selmer length plus $\sum_q (v_\ell(\mathrm{Tam}_q(\mathrm{ad}^0 T)) - t_\ell(E,q))$ (eq:identity, an equality by points 39 and 56); the Tamagawa factor is trivial at the potentially good additive primes (point 64), so the second sum runs over the potentially multiplicative primes only and is given by Lemma lem:tamfactor (points 18, 68); the data measure the excess but do not explain it (Open question 3). Paper: Section sec:sharp ("the mean excess at $\ell = 3$ is 0.63"; "the identity says what the excess is"), Section sec:forms.

8.3 `ex:computation:base-rates` (worked example; C). Base rates at $\ell = 3$ in the base population below 10,000 (33,851 curves): $3 \mid \deg \varphi_E$ for 76.1%. Conditional on having at least one prime $q \ge 5$ of a given type (a curve with several types is counted in several rows):

type at some $q \ge 5$	curves	share with $3 \mid \deg \varphi_E$
II / II*	946 / 418	100% / 100%
IV / IV*	468 / 896	100% / 100%
I_n^*	2,233	100%
I_n with $3 \mid n$	5,134	100%
III / III*	813 / 807	75.4% / 75.5%
I_0^*	692	73.1%
I_n with $3 \nmid n$	31,185	74.6%

Reading: the rows at 100% are exactly the types with a term that is always positive at $\ell = 3$ ($e_3 \ge 1$ for II to IV* and I_n^* at $q \ge 5$; $t_3 \ge 1$ for $3 \mid n$); the rows near the base rate are the types whose term is positive only for some q (III: $q \equiv \pm 1 \pmod{12}$; I_0^*: $q \equiv 1 \pmod 3$ or $3 \mid a'$). A base-rate table is the first thing to compute and the weakest evidence: it is the shadow of the inequality on a coarse partition. Paper: Section sec:iso (the same idea, cell by cell), PLAN.md ("the base rates by Kodaira type").

8.4 `rem:computation:simpson` (remark; P, with a C illustration). Rates pooled over ranges or over types can mislead (aggregation, Simpson's paradox in its mildest form): the pooled equality rate on $N < 10000$ (71.0%) is a weighted average of the two ranges' rates, with weights the numbers of positive bounds (8,510 and 11,807), and a comparison of the paper's 57.5% with the book's 71.0% says nothing about a difference in the curves' arithmetic; it says the ranges differ. The fair comparison is like with like: the same statement, the same population definition, the same range. Paper: Section sec:sharp, Section sec:oot ("equality in (eq:rule) for 52.5%").

24.2.9 Section 9. The protocol

9.1 `rem:computation:protocol` (remark; R for the definitions, which are Appendix A's; D for the paper's description). The four ranges (K6) with what is known about each: the working set ($N \le 300000$, every class optimal-determined), the hold-out ($300000 < N \le 400000$, the same), the extension ($400000 < N \le 500000$, only 359,009 classes optimal-determined, so the extension counts are over those classes: point 25), the out-of-table curves (893 curves with $500000 < N \le 4 \cdot 10^6$, generated in families with prescribed local types and kept when the isogeny class is trivial, so that the curve is optimal and every $E[\ell]$ is irreducible; their degrees by `ellmoddegree`). Appendix A's def:prereq:holdout and pl:prereq:protocol are cited for the vocabulary (working set, hold-out, pre-registration, blind test, confirmation). Paper: Section sec:data, Section sec:oot.

9.2 `prop:computation:test-status` (proposition, read off the paper's text; P as a reading, D for the facts). The status of each data claim, from the paper's own account (Section sec:data):

statement	formulated on	blind test	confirmation only	remark
(eq:main)	working set	hold-out (once), then extension		nothing adjusted after the hold-out test; the out-of-table curves are a further check whose status the paper does not state
Conjecture conj:eis	working set	hold-out (once), then extension		stated for (eq:main) only; $D \leq 2$ is a theorem (point 17)
(eq:twist)	working set (replacing a weaker form)	hold-out and extension, "tested once"		blind only if fixed before the hold-out was opened for (eq:main); the text does not say (Part 8, N2); out-of-table as for (eq:main)
Conjecture conj:euler: the Euler terms	working set (minima by type, Section sec:iso)	out-of-table	hold-out, extension	the terms at 2, 3 "fixed afterwards by PARI's Euler factors"
the weights w_3	working set (minima at $q = 3$)	out-of-table, only on curves with $3 \mid N$ and potentially good reduction at 3, if any	hold-out, extension	empirical by construction (Section sec:wild)
the correction h_ℓ	from the exceptions	none: the out-of-table set has no 3Ns curve	all ranges	$h_3 = 1$ on 3Ns is a theorem (Lemma lem:h0, point 5); the inequality with it is not
the III-free form	after the 3Ns exceptions	none	all ranges	a reformulation chosen to have no exceptions
Lemmas lem:euler, lem:twist; Proposition prop:forms; $h_3 = 1$				theorems: the data are checks of computations, not tests
the Tamagawa refinement (sec:tam)	from Lemma lem:tamfactor and (eq:identity)	working set, as a prediction of a theorem		a test of the identity's Tamagawa term (point 68 on its wording)

The proposition is the table plus the sentence: a statement is blind-tested on a range only if it was written down before any quantity of that range entering the statement was looked at; generating data before a statement is formulated is necessary but not sufficient (it is sufficient if the data were not

examined). Paper: Section sec:data ("Since the hold-out and the extension had already been used for (eq:main) and (eq:twist), they are a confirmation and not a blind test for Conjecture conj:euler; the out-of-table curves, generated before the rule was formulated, are").

9.3 `rem:computation:what-is-tested` (remark; P). What a blind test of a universal inequality establishes: a single violation refutes; zero violations on n fresh curves is evidence whose weight depends on how likely a violation would have been if the statement were false (Section 10: calibration), and on whether the fresh curves exercise the statement (the out-of-table families were built with types IV, IV*, two or three II, mixed and random; they cannot test the 3Ns correction, which needs 3Ns curves, or the III terms unless such primes occur). "Nothing was adjusted after the hold-out test" means: no term, weight or population filter of (eq:main) and Conjecture conj:eis was changed after their hold-out counts were seen; it does not cover later statements. Paper: Section sec:data, Section sec:oot, the AI statement ("a list of tests fixed before any was run").

9.4 `rem:computation:multiple-comparisons` (remark; P for the arithmetic, R for the general theory: Wasserman04, chapter on multiple testing, cite by chapter). (i) The paper's statements came out of a search: "the pre-registered search, which produced the candidate relation" (AI statement), then successive refinements (Tamagawa number to Φ^{geom}, then twist-symmetrised weights, then Euler factors, then w_3, then h_ℓ). (ii) If m false candidate statements each survive a working-set check with probability π, about $m\pi$ false statements survive; a hold-out check with survival probability π' for false statements leaves $m\pi\pi'$; this is the whole case for holding data back (exercise). (iii) For a universal inequality the relevant π' is tiny when the calibration of Section 10 says hundreds of thousands of violations were expected under independence: so the multiple-comparisons risk sits mostly in the *choice of form* (which weights, which correction, which population filter), where the working set was used repeatedly, not in the bare existence of a relation. (iv) Post-hoc features: the w_3 were defined as observed minima, so the working set cannot violate them; the III-free form was chosen to have no exceptions; such features are validated only by data not used to choose them. The book's toy version (9.5) shows the procedure. Physics aside (one sentence): this is the look-elsewhere effect. Paper: Section sec:data, Section sec:wild ("the weights w_3 ... are empirical"), Section sec:iso.

9.5 `ex:computation:toy-protocol` (worked example; C). On the book's data: freeze the small working set $N < 5000$; read off the minima of $v_3(\deg \varphi_E)$ over curves whose only possible contribution at 3 is a potentially good prime $q = 3$ (the paper's recipe for w_3): II 1 (239 curves), II* 2 (156), III 0 (221), III* 1 (221), IV 1 (156), IV* 2 (239), I$_0^*$ 0 (299): exactly the paper's w_3. Then test once on the small hold-out $5000 \le N < 10000$: the same minima (255, 186, 251, 251, 186, 255, 347 curves), and the full rule with these frozen weights has no violation among the 11,699 base-population curves with $3 \mid N$ there. Reading: a replication of the recipe, not a discovery; and a reminder that the small working set was not blind to the paper (the weights were known in advance), so this is a demonstration of the procedure only. Paper: Section sec:wild, Conjecture conj:euler.

24.2.10 Section 10. The calibration argument of Section sec:sharp

10.1 `prop:computation:null-model` (proposition; P). Let a population be split by a bound $b(E) \in \{0, 1, 2, \dots\}$, with observed valuations $v(E)$. Null hypothesis H_0: v is independent of b, all curves having the same distribution F of v. Let $\hat{F}$ be the empirical distribution of v on the curves with $b = 0$, and X the number of violations among the curves with $b > 0$. Then (i) under H_0, $E[X] = \sum_{b(E)>0} F(v < b(E))$ by linearity (Appendix A, thm:prereq:expectation(a)), with no independence between curves needed, and $\hat{E} = \sum_{b(E)>0} \hat{F}(v < b(E))$ estimates it; (ii) if moreover the $v(E)$ are independent across curves, X is a sum of independent indicators, $\mathrm{Var}\, X = \sum p_E(1 - p_E) \le E[X]$, $P(X = 0) = \prod(1 - p_E) \le e^{-E[X]}$, and by Chebyshev $P(X = 0) \le \mathrm{Var}\, X / E[X]^2 \le 1/E[X]$ (pairwise independence suffices for the last bound); (iii) under H_0 the pooled distribution of v over all curves (bound 0 and positive) estimates the same F, so a large difference between the pooled and the bound-0 distributions is itself evidence against H_0; a permutation test makes this exact: shuffle v among the curves (within strata of conductor and number of bad primes, to respect the dependence of v on N) and recount. Proofs: linearity; the product formula and $1 - p \le e^{-p}$; Chebyshev. Reference: Appendix A; Feller68 (chapters on expectation and on Chebyshev's inequality; cite chapters). Paper: Section sec:sharp ("If the valuation of the degree were independent of

the local data, the 867,578 working-set curves with a positive bound at $\ell = 3$ would produce roughly 605,000 violations (summing over them the proportion of bound-0 curves whose valuation is smaller than the bound); there are none").

10.2 `ex:computation:calibration-paper` (worked example; D, with arithmetic P). The paper's numbers: 366,151 bound-0 curves, of which 53.2% have $3 \nmid \deg \varphi_E$; 867,578 curves with a positive bound; expected violations about 605,000, that is 69.7% of them; observed 0. Checks: the $v = 0$ part alone gives at least $0.532 \cdot 867{,}578 \approx 461{,}551$ (Chapter 21, A6.3), and 605,000 requires an average $\hat{F}(v < b)$ of 0.697, consistent with the book's own ratio 70.3% (10.3); under independence the standard deviation of X is at most $\sqrt{605000} \approx 778$, so 0 is about 780 standard deviations below the mean, and the "p-value" is below $e^{-605000}$: meaningless as a number, which is why the paper (and Appendix A's ex:prereq:calibration) states the comparison of 605,000 with 0 and no p-value. Paper: Section sec:sharp.

10.3 `ex:computation:calibration-small` (worked example; C). The same statistic on the base population at $\ell = 3$ below 10,000: bound-0 curves 13,534, with $\hat{F}$: $v = 0$: 8,084 (59.7%), 1: 4,079, 2: 1,081, 3: 251, 4: 33, 5: 5, 6: 1; positive bounds 20,317 (bound 1: 13,553; 2: 5,534; 3: 1,109; 4: 112; 5: 9); expected violations $\hat{E} = 14{,}275$ (70.3% of the bounded curves), standard deviation under independence 61.5; observed 0. By range: $N < 5000$: $\hat{E} = 6{,}091$ ($\hat{F}(0) = 0.630$); $5000 \le N < 10000$: 8,119 ($\hat{F}(0) = 0.566$). At $\ell = 5, 7, 11, 13$ (union): $\hat{E} = 5{,}901$, 3,476, 1,187, 698; observed 0 each. Robustness of the reference population (C): stratified by conductor bins of width 2500 and by $\omega(N)$ (capped at 4), $\hat{E} = 14{,}177$; a permutation of v within the same strata over all curves (200 shuffles, seed 0) gives a mean of 7,638 violations (standard deviation 49, minimum 7,517): the reference population changes the expected count by a factor of two, and the conclusion not at all. The pooled share of $v = 0$ is 23.9% against 59.7% among bound-0 curves: under H_0 these would agree (10.1(iii)). Paper: Section sec:sharp.

10.4 `rem:computation:calibration-reading` (remark; P). What the calibration shows: the zero count is not an artefact of 3 dividing most modular degrees (base rate 76% below 10,000): the bound picks out, among all curves, those whose degree is divisible, and a bound unrelated to the degree would fail tens of thousands of times in the book's range and hundreds of thousands in the paper's. What it does not show: (a) that the inequality is true beyond the data (that is the theorem's job under its hypotheses, and the conjecture's outside them); (b) anything on the working set about validity, because the rule was fitted there (weights read off minima, terms at 2, 3 chosen after seeing minima): on the working set the calibration measures the *strength* of the statement; its *validity* is supported by the blind tests of Section 9; (c) independence across curves is an idealisation (twists and congruent forms tie curves together), so the standard deviations are indicative only; (d) the null "independent of the local data" is crude; a sharper null would condition on N and on the number of bad primes, as the stratified version does. The equality rates and the mean excess are descriptive (8.2). Paper: Section sec:sharp ("The zero counts are not tautological"; "The bounds are attained").

10.5 `ex:computation:sharpness` (worked example; C). How sharp the bound is, against chance: under H_0 the expected number of equalities among the 20,317 bounded curves at $\ell = 3$ is $\sum \hat{F}(v = b) \approx 4{,}548$; observed 14,420 (71.0%). Equality rates by ℓ (union of the small ranges): 71.0%, 84.5%, 90.7%, 96.5%, 96.0% at $\ell = 3, 5, 7, 11, 13$ (paper, working set: 57.5%, 75.6%, 84.1%, 91.6%, 92.4%). Mean excess at $\ell = 3, 5, 7, 11, 13$: 0.37, 0.17, 0.10, 0.04, 0.04. Paper: Section sec:sharp.

10.6 `ex:computation:odd-part` (worked example; C and D). The paper's last statistic: among optimal non-CM curves with no isogeny of odd degree (so $E[\ell]$ irreducible for every odd ℓ; decided from the `allisog` matrix), the share whose odd part of $\deg \varphi_E$ equals $\prod_\ell \ell^{B_\ell^{\mathrm{tw}}}$. Book (C): 33,137 curves; equality for 5,675 (17.13%); the quotient is always an integer ((eq:twist) holds for every odd ℓ); most common quotients 3 (5,926), 1, 5 (2,978), 9 (2,532), 15, 7, 21, 11; the smallest prime factor of a nontrivial quotient is 3 in 59% of cases and divides N in 48%. Paper (D, working set): 1,226,216 curves, equality 1.85%, quotients most often 9, 3, 15, 45, 27, 21, smallest prime factor 3 in 63%, dividing N in 52%. Reading: the local terms explain a sharp part of $\deg \varphi_E$ and less and less of the rest as N grows (the rest is the Selmer group and the size of the Petersson norm, Chapter 19, 8.4, 8.5); the paper's phrase "its smallest prime factor is 3 in 63% of the cases and divides N in 52%" is read as "the smallest prime factor of the quotient divides N" (the writer says so). Paper: Section sec:sharp ("The whole odd part of $\deg \varphi_E$ is another matter").

24.2.11 Section 11. Reproducing the paper's tables on conductor below 10,000

Each item prints a table produced by the scripts S4, S5 (Part 7), with the paper's corresponding numbers (D) beside it where the paper gives them in the text (the paper's table files `tables/*.tex` are not available; Chapter 21, A6: never invent their entries). All counts are over the base population at ℓ (K5) unless stated.

11.1 `ex:computation:repro-uniform` (worked example; C). Table tab:uniform in small. Columns: range, ℓ, base population, $B^{\mathrm{rule}} > 0$, violations, equality rate, violations of B^{nc}, $B^{\mathrm{III}} > 0$, violations, equality rate.

range	ℓ	curves	rule > 0	viol.	eq.	viol. without h	III-free > 0	viol.	eq.
$N <$ 5000	3	15,102	8,510	0	75.1%	16	8,430	0	75.0%
	5	16,863	3,078	0	88.4%	0	3,068	0	88.3%
	7	17,107	1,644	0	91.5%	0	1,642	0	91.5%
	11	17,190	515	0	97.5%	0	515	0	97.5%
	13	17,180	296	0	97.3%	0	296	0	97.3%
$5000 \le N <$ 10000	3	18,749	11,807	0	68.0%	20	11,697	0	67.6%
	5	20,456	4,398	0	81.7%	0	4,368	0	81.7%
	7	20,584	2,291	0	90.2%	0	2,287	0	90.2%
	11	20,649	741	0	95.8%	0	741	0	95.8%
	13	20,643	432	0	95.1%	0	428	0	95.1%
$N <$ 10000	3	33,851	20,317	0	71.0%	36	20,127	0	70.7%
	5	37,319	7,476	0	84.5%	0	7,436	0	84.5%
	7	37,691	3,935	0	90.7%	0	3,929	0	90.8%
	11	37,839	1,256	0	96.5%	0	1,256	0	96.5%
	13	37,823	728	0	96.0%	0	724	0	96.0%

The paper (D): no violation of (eq:rule) at any ℓ on any range; the violations without h are all 3Ns curves (364, 68, 43, 0 on the four ranges); equality 57.5% at $\ell = 3$ on the working set. Agreement in kind: no violation, the uncorrected failures are exactly the 3Ns curves (11.3), and the III-free form never fails. Paper: Section 6.2, Table tab:uniform, Section sec:sharp.

11.2 `ex:computation:repro-counts` (worked example; C). Table tab:counts in small: for each range and ℓ, $B^{\mathrm{main}} > 0$, violations, equality rate; $B^{\mathrm{tw}} > 0$, violations, equality rate; Eisenstein population, positive B^{main}, positive deficit D, violations of the torsion bound, violations of the class bound, $D = 2$.

range	ℓ	main > 0	viol.	eq.	twist > 0	viol.	eq.	Eis.	main > 0	$D >$ 0	viol.	viol. (class)	$D =$ 2
$N <$ 5000	3	5,908	0	59.0%	6,996	0	61.3%	2,096	1,725	284	0	0	11
	5	2,478	0	86.5%	2,681	0	87.2%	335	199	45	0	0	0
	7	1,351	0	89.8%	1,452	0	90.6%	91	35	9	0	0	0
	11	450	0	96.4%	472	0	97.2%	8	0	0	0	0	0
	13	272	0	96.3%	286	0	97.2%	18	2	0	0	0	0
$5000 \le N <$ 10000	3	8,069	0	49.1%	9,545	0	51.6%	1,906	1,624	147	0	0	3

range	ℓ	main > 0	viol.	eq.	twist > 0	viol.	eq.	Eis.	main > 0	$D > 0$	viol.	viol. (class)	$D = 2$
	5	3,307	0	76.6%	3,636	0	78.6%	199	139	8	0	0	0
	7	1,719	0	86.9%	1,894	0	87.8%	71	32	3	0	0	0
	11	579	0	94.3%	631	0	94.9%	6	0	0	0	0	0
	13	369	0	93.8%	389	0	94.1%	12	2	0	0	0	0
$N <$ 10000	3	13,977	0	53.3%	16,541	0	55.7%	4,002	3,349	431	0	0	14
	5	5,785	0	80.8%	6,317	0	82.2%	534	338	53	0	0	0
	7	3,070	0	88.1%	3,346	0	89.0%	162	67	12	0	0	0
	11	1,029	0	95.2%	1,103	0	95.9%	14	0	0	0	0	0
	13	641	0	94.9%	675	0	95.4%	30	4	0	0	0	0

The paper (D, working set): equality in (eq:twist) 39.4%, 71.1%, 81.2%, 91.5% at $\ell = 3, 5, 7, 13$, in (eq:main) 36.9%, 69.7%, 80.4%, 91.2%; the rule bounds 867,578 curves instead of 705,563 at $\ell = 3$ (book: 20,317 instead of 16,541). Same ordering (main below twist below rule) in both. Paper: Section sec:forms, Table tab:counts, Conjecture conj:eis.

11.3 `ex:computation:repro-ns` (worked example; C). Table tab:ns in small (optimal non-CM curves with image 3Ns):

range	3Ns curves	positive III term	violate B^{nc}	other curves violating	`ns3` $= 1$	residues mod 12 of the III primes of violators	shortfall
$N < 5000$	30	30	16 (53%)	0	30	1: 4, 11: 8, $q = 2$: 4	always 1
$5000 \le N <$ 10000	60	60	20 (33%)	0	60	1: 8, 11: 12	always 1
$N <$ 10000	90	90	36 (40%)	0	90	1: 12, 11: 20, $q = 2$: 4	always 1

Every violator has a prime $p \equiv -1 \pmod 3$ with $t_3(E, p) \ge 1$ (36 of 36); 12 of the 36 have $3 \mid N$. Reading of "violate (eq:thm)": the book uses B^{nc} (the rule with every term and no correction, which is what "violations when the correction term is dropped" in Table tab:uniform means); if (eq:thm) is read literally, with terms at $q \ne \ell$ only and no w_3, the count is 24 instead of 36 (726a1 then has bound 2 and no violation), and still no non-3Ns curve violates (C). The writer states which reading is used (Part 8, N6). Of the 90 curves, 74 have a type III or III* prime $q \ge 5$ with $e_3 \ge 1$ and the other 16 (conductors 3840, 4864, 6912) get their positive III term at $q = 2$ from PARI's factor (consistent with Chapter 21, A6.7 and with point 48). The paper (D): 364, 68, 43, 0 violations, 31%, 23%, 18% of the 3Ns curves. Reading: the same phenomenon at higher rates in the small range; the falling rate with N is visible already inside the book's range. Paper: Section sec:anomaly, Table tab:ns, Remark rem:ns.

11.4 `ex:computation:repro-iso` (worked example; C). Section sec:iso in small: curves of the base population with exactly one additive prime, $q \ge 5$, $q \ne \ell$, and no multiplicative prime with $\ell \mid n$; minimum of $v_\ell(\deg \varphi_E)$ by type and $e_\ell(E, q)$. At $\ell = 3$: II $e = 1$: min 1 (264 curves), $e = 2$: 2 (27); II* 1: 1 (135), 2: 2 (11); III 0: 0 (193), 1: 1 (44), 2: 2 (2); III* the same counts; IV 1: 1 (150), 2: 2 (11); IV* 1: 1 (249), 2: 2 (27); I_0^* 0: 0 (146), 1: 1 (42), 2: 2 (88), 3: 3 (50); I_n^* 1: 1 (813), 2: 2 (57), 3: 3 (3), with the minimum of $v_3(\deg \varphi_E) - e_3 - t_3$ equal to 0 in each I_n^* cell. At $\ell = 5$ and $\ell = 7$: every cell's minimum equals e (cells listed in Part 7, R5). Multiplicative primes alone: for curves with a single multiplicative prime q with

$\ell \mid n$ and no additive prime, $\min(v_\ell(\deg \varphi_E) - v_\ell(n)) = 0$ for every value of $v_\ell(q^2-1)$ ($\ell = 3$: values 0 to 4, 4,367 curves; $\ell = 5$: 0 to 2, 2,588 curves). Two odd primes of type II/II* and nothing else at $\ell = 3$: 8 curves, $9 \mid \deg \varphi_E$ for all 8. The paper (D): the same pattern in every cell; III: 5,534, I_0^*: 5,394 curves with $e = 0$; I_0^* up to $e = 6$; 2,502 curves with two II-type primes. Paper: Section sec:iso, Table tab:iso.

11.5 `ex:computation:repro-tam` (worked example; C). Section sec:tam in small: hypothesis population ($\ell \nmid N$, $E[\ell]$ irreducible, not CM, at $\ell = 3$ not 3Ns) with at least one prime of type I_n, $\ell \mid n$, $q \equiv 1 \pmod{\ell}$; extra term $\min(v_\ell(n), v_\ell(q-1), v_\ell(c(u)))$ summed over such primes, from `tatedata`. $\ell = 3$: 989 curves, 179 with a positive extra term (total 179), no violation of the refined bound, equality from 550 (55.6%) to 662 (66.9%). $\ell = 5$: 246 curves, 24 positive, no violation, equality 197 (80.1%) to 214 (87.0%). $\ell = 7$: 18 curves, 1 positive (4171a1 at 43), equality 17 to 18. The paper (D, working set): $\ell = 3$: 46,222, 12,095, 45.0% to 60.6%; $\ell = 5$: 10,940, 1,646, 67.0% to 78.7%. Note: following the paper, only type I_n primes are used; the refinement also applies at I_n^* primes (point 68; Chapter 18 reports 1,050 curves at $\ell = 3$ when they are included). Paper: Section sec:tam, Remark rem:ko(1), Lemma lem:tamfactor.

11.6 `ex:computation:repro-eis` (worked example; C). Section sec:eis in small, $\ell = 3$: 4,002 Eisenstein curves, 3,349 with $B^{\mathrm{main}} > 0$, 431 with $D > 0$, $D = 2$ for 14; no curve without a rational 3-torsion point has $D > 0$; the torsion bound is attained with $D > 0$ in 427 cases; $D = 2$ occurs for twelve curves with image 3Cs.1.1 (14a1, 26a1, 35a1, 38a1, 77b1, 158d1, 278b1, 326c1, 485a1, 4385b1, 7094c1, 8027a1), all of rank 0, with torsion $\mathbb{Z}/3$ except 14a1 (torsion $\mathbb{Z}/6$), and for two curves with a point of order 9 (714i1, 6942n1, image 3B.1.1). Of the 49 optimal non-CM curves with image 3Cs.1.1, none has a point of order 9 (torsion $\mathbb{Z}/3$ for 48, $\mathbb{Z}/6$ for 14a1); the four optimal curves with a point of order 9 (714i1, 1482l1, 1554n1, 6942n1) have image 3B.1.1. At $\ell = 5$, 7: $D \le 1$, and $D > 0$ only with a rational ℓ-torsion point (53 and 12 curves). Against (eq:twist) the deficit exceeds the conjectured bound for 180 curves at $\ell = 3$ and 35 at $\ell = 5$. The paper (D): 75,430, 67,845, 2,777, $D = 2$ for 22 (17 with 3Cs.1.1 "all of rank 0 and torsion $\mathbb{Z}/3$", 5 with a point of order 9); for the 14a1 discrepancy see Part 8, N1. Paper: Section sec:eis, Conjecture conj:eis (with point 17: $D \le 2$ is provable).

11.7 `ex:computation:repro-twist` (worked example; C). The data paragraph after Theorem thm:twist in small: for each optimal non-CM curve, each odd q with $q^2 \| N$ in case (a) (I_n^*), (b) (I_0^*) or (c) (potentially good, $q \ge 5$), the twist E_q by q^* (found in `allcurves` through `ellminimalmodel`), and each $\ell \in \{3, 5, 7, 11, 13\}$ with $\ell \ne q$, $\ell^2 \nmid N$: compare v_ℓ of the two optimal degrees. Pairs (curve, q): (a) 6,348, (b) 2,183, (c) 5,150. Triples (curve, q, ℓ): (a) 24,373, all with $v_\ell(\deg \varphi_E) \ge v_\ell(\deg \varphi_{E_q})$, strict in 2,779 (e.g. 121d1 versus 11a1 at $\ell = 3$: 24 versus 1; 75b1 versus 15a1; 150c1 versus 30a1); (b) 8,348, strict in 1,308 (e.g. 275b1 versus 11a1 at $\ell = 7$: 28 versus 1); (c) 19,554, all equal. So strict inequality occurs in cases (a) and (b), not "only in case (b)" (important_notes.md point 16, confirmed). Paper: the paragraph after Theorem thm:twist, Theorem thm:quant.

11.8 `rem:computation:comparison` (remark; P). Why the book's numbers differ from the paper's and what does agree: the ranges differ by a factor of fifty in conductor; rates of equality, of 3Ns failures and of the odd-part statistic all fall with N, inside the book's range already; the qualitative claims (no violations of (eq:rule), (eq:main), (eq:twist) and of Conjecture conj:eis; uncorrected failures exactly on 3Ns curves and by exactly one; minima equal to e cell by cell; the weights w_3; the Tamagawa refinement without exception; strict twist inequalities in cases (a) and (b)) all reproduce. One sentence of the paper does not reproduce verbatim (Part 8, N1: 14a1). Paper: Section 6.

24.2.12 Section 12. Reproducibility practice

12.1 `rem:computation:paper-repository` (remark; R, a description only). The paper's appendix names a repository and directory with `scripts/step10_sym2_euler.py` (Euler factors), `scripts/step10_rule.py` and `scripts/step11_tw_boundary.py` (the rule, the correction term, image statistics), `scripts/step12_tamagawa.py` (Tamagawa factors), `scripts/conj_check.py` (the two corollary forms), `scripts/euler_rule.gp` (the appendix code) and `twist_lemma_check.gp`, a pinned data commit and the result tables, and states that the pipeline was rerun from a clean clone and reproduced every count of Table tab:counts. The book lists these as the paper's claims; it does not reproduce or link the scripts beyond the URL the paper gives, and the writer must not describe their contents beyond the appendix's own words. Paper: the appendix, the AI statement.

12.2 `conv:computation:checklist` (convention; P). A reproducibility checklist, applied to the book's own runs: (1) pin the data (repository commit hash, and SHA-256 sums of the files read: Part 6.5 lists the sums of the book's six files); (2) record software versions (PARI, `cypari2`, Python) and non-default settings (`parisizemax`; random seeds, e.g. seed 0 for the permutations of 10.3); (3) separate data, code and results; write results as plain files with the counts, not only as summaries; (4) rerun from a clean copy and compare counts; (5) cross-check independently (here: GP and Python implementations of the bound agree; `ellmoddegree` against `degphi`; `ns3` against `galrep`; Lemma lem:euler against `lfunsympow`); (6) state what was not checked. Paper: the appendix ("The pipeline was rerun from a clean clone"), the AI statement ("The scripts, the result files and the pinned data commit are in the repository ... so that every count can be rerun").

12.3 `rem:computation:book-scripts` (remark; P). The scripts the chapter prints, with their role (the code itself is in Part 7 of this outline, tested; the writer prints S1 to S3 in full and S4 to S8 in shortened form, about 40 lines each at most): S1 `euler_rule.gp` (the paper's appendix, verbatim); S2 `sessions.gp` (`rule_terms`, `tatedata`, `twistcheck`); S3 `table_small.gp` (`rulebound`, `readgal`, `reducible`, `tablerow`: one row of Table 11.1 in pure GP); S4 `compute.py` (one JSON record per optimal curve: invariants, local data, P_q, $P_q(q^{-2})$, a_q, $v_q(j)$, `ns3`, torsion counts in the class, isogeny degrees); S5 `analyse.py` and `tables*.py` (the five bounds and Tables 11.1 to 11.4, 11.6, 11.7, and 8.3); S6 `calib2.py` (10.3, 10.5, the stratified and permutation versions); S7 `tam.py` (11.5); S8 `moddeg_check.py` (7.2). Paper: the appendix.

24.3 3. Section-by-section plan with target line counts

Lines of LaTeX, card excluded (PLAN.md Section 7.3: 1,000 to 2,200). The card lists all items of Part 2 (about 150 lines, not counted); computational items appear on the card with the exact output in one line (for example "`rule` on 121a1 returns `[1, 1]`") and the tag *(proved here)* for C, *(stated; reference)* for D.

section	title	contents	target lines
(starred)	Core results at a glance	the card: every item of Part 2 with its tag and `\Cref`; K3 to K5 restated in two lines; the five bounds	(excluded)

section	title	contents	target lines
(starred)	Where the paper uses this chapter	Section 1.1 (lfunsympow); Section 1.4 (component groups from the tables); Lemma lem:euler and Lemma lem:twist (the PARI checks); Remark rem:wild; Proposition prop:ns (ns3); Proposition prop:forms (the factor at 2); the Example of Section 3 and the 99A1 remark; Section sec:data and all its subsections and tables; the paragraph after Theorem thm:twist; the abstract's counts; the appendix	45
1	Introduction and conventions	purpose; 1.1 to 1.3 as convention environments (K1 to K11 condensed); the physics aside; the debts of Chapters 2, 3, 9 to 15, 17 to 19 in one paragraph	80
2	PARI/GP from zero	2.1 to 2.4; a first session in lstlisting; the pitfalls warning	150
3	Invariants of a curve in PARI	3.1 to 3.8; the outputs of ellglobalred, lfunsympow, lfuneuler, .tate, elldivpol, elltwist, ellmoddegree, ellweilcurve; the code-idiom table; S2 printed	230
4	Cremona's tables and ecdata	4.1 to 4.4; the file table with sample lines; reading files in GP and Python; optimality and 990h3; labels; paperlink for Section sec:data	140

section	title	contents	target lines
5	The function `rule(E, l, iii)`	5.1 (S1 printed with line numbers), 5.2 (line by line), 5.3 (proposition and proof), 5.4 (pitfalls `warning`); `paperlink` for the appendix	170
6	Worked sessions	6.1 to 6.8, each with input, exact output, breakdown and a two-line reading	210
7	Modular degrees and congruence numbers	7.1 to 7.3; pointers to Chapters 14 and 19	70
8	Populations, violations and descriptive statistics	8.1 to 8.4; the base-rate table; Appendix A cited for the vocabulary	110
9	The protocol	9.1 to 9.5; the status table of 9.2; multiple comparisons; the toy protocol; `paperlink` for Section sec:data	150
10	The calibration argument	10.1 (with proof), 10.2 to 10.6; the reading of 10.4 as a `remark` with two lists (“shows”, “does not show”)	150
11	Reproducing the paper’s tables	11.1 to 11.8; four `booktabs` tables and short readings; S3 printed, S4 to S7 shortened	190
12	Reproducibility practice	12.1 to 12.3; the checklist; the book’s data pin	60
13	Exercises	the 18 exercises of Part 4.2	140
14	Notes and references	PARI, PARIusers, Cremona97, CremonaBook, ecdata, LMFDB, Sutherland16, Zywina15, Watkins02, Zagier85, Feller68, Wasserman04 (after B-b of Part 8), RHB06; the paper’s appendix; what was verified and with which versions	45
	total		**1,940**

24.4 4. Coverage table and exercises

24.4.1 4.1 Coverage: PLAN.md scope, Section 6 map rows and paper hooks to line-up items

PLAN.md scope item, Section 6 map row, or paper hook	line-up items
PARI/GP for the reader who has never used it (installation)	2.1, 2.2, 2.3, 2.4
`ellinit`, `ellglobalred` and its output (conductor, minimal model change, Kodaira symbols and Tamagawa numbers per prime, the local data structure with the Kodaira codes)	3.1, 3.2 (codes: Chapter 10, 5.5)
`ellap`, `ellan`, `elltors`, `ellisomat`	2.3
`ellmoddegree` (and its memory needs)	3.8, 7.1, 7.2
`elldivpol`	3.6
`lfuncreate` / `lfunsympow` / `lfuneuler` and the conventions for the symmetric square Euler factor	3.3, 3.4 (`lfuncreate` mentioned in 3.3 as the general constructor, not needed)
`ellinit` over $\mathbb{Q}_q$ and the Tate parameter (not `elltatepairing`), its valuation and unit part	3.5, 6.8
`valuation`, `factor`	2.2, 2.4(f), 3.6
Cremona's tables and `ecdata`: allcurves, allbsd, `degphi` (PLAN.md formerly wrote alldegphi), allgens, alllabels, allisog; Kodaira symbols and Tamagawa numbers; mod ℓ images with Sutherland's labels; optimality and the label "1"; LMFDB labels	4.1 to 4.4 (`allgens` not in the local copy: 4.1 says so)
reading a Cremona label (11a1, 121a1, 121c, 121d, 539c1, 980i1, 726a1, 99A1, capitalisation)	4.4 (full list: Chapter 9, 12.1)
the data protocol: four ranges; pre-registration; "nothing was adjusted"; blind-tested versus confirmed; the out-of-table set for Conjecture conj:euler	9.1, 9.2, 9.3, 9.5
elementary statistics: what a violation is; the calibration argument with the numbers; what it does and does not show; equality rates and mean excess as descriptive statistics; base rates by Kodaira type; multiple comparisons; why a hold-out	8.2, 8.3, 8.4, 9.4, 10.1 to 10.6
`rule(E, l, iii)` reproduced and explained line by line (`R[4]`, `R[5]`, the code `k`, $r = P_q(q^{-2})$ with the multiplicative correction, the III exclusion, w_3, t_ℓ from $v_q(j)$, `ns3`)	5.1 to 5.4
worked sessions: 121a1 `[1, 1]`; 726a1 `[2, 2]` with and without `iii`; 539c1; 980i1; 99A1 (degree 4, congruence number 12 not in PARI)	6.1 to 6.5, 7.3
computing $e_\ell(E, q)$ by hand from the type and checking against `lfunsympow`	3.4
computing $t_\ell(E, q)$	3.5, 6.1 to 6.8
the mod 3 image 3Ns via `elldivpol`	3.6, 6.2
the twist lemma check (`twist_lemma_check.gp`: how to write it)	3.7 (S2's `twistcheck`), Exercise 14

PLAN.md scope item, Section 6 map row, or paper hook	line-up items
reproducing a row of the paper's tables on a small range (a GP loop over `ecdata`)	11.1 (S3), Exercise 7
how the modular degree is computed (Watkins, Cremona; why expensive)	7.1
the repository named in the appendix (descriptive list)	12.1
reproducibility practice (pinned commits, clean clone)	12.2, 1.1
Section 6 row: PARI/GP, Cremona's tables, ecdata, LMFDB, Sutherland's labels	2, 3, 4
Section 6 row: the statistical protocol, calibration, hold-out sets	8, 9, 10
Section 6 row: the function `rule(E, l)` and the Kodaira codes (with Chapter 10)	3.2, 5
Section 6 row: the worked examples 11a1, 20a1, 121a1, 539c1, 980i1, 726a1, 99A1 (with 9, 14, 15, 21)	2.3, 3.1, 3.5, 6.1 to 6.7
hook: Section 1 ("PARI's `lfunsympow(E,2)` returns these factors at every prime, including 2 and 3")	3.3, 6.6
hook: Remark rem:wild	6.6 (mathematics: Chapter 19, 10.1, 10.2)
hook: Lemma lem:euler ("confirmed against PARI's `lfunsympow` on every curve of the tables")	3.4
hook: Section 1.4 (component groups; Cremona's tables and optimality)	3.1, 3.2, 4.3, 11.2
hook: Proposition prop:ns (3-division polynomial)	3.6, 5.3(ii)
hook: Proposition prop:forms (92,812 curves, factor $1+2X$ at 2)	Part 7, R4 (3,226 optimal curves below 10,000, all $1+2X$); point 43
hook: Section sec:data, protocol paragraph	1.1, 9.1, 9.2, 9.3
hook: Section 6.2 and Table tab:uniform	11.1
hook: Section sec:sharp	8.2, 10.1 to 10.6
hook: Section sec:tam	3.5, 6.8, 11.5
hook: Section sec:iso and Table tab:iso	8.3, 11.4
hook: Section sec:forms and Table tab:counts	11.2
hook: Section sec:anomaly and Table tab:ns	3.6, 6.2, 11.3
hook: Section sec:wild and Table tab:isolated	6.6, 9.5 (the w_3 minima)
hook: Section sec:oot and Table tab:oot	9.1, 9.2, 9.3 (no reproduction possible: the curves are not published in the paper)
hook: Section sec:eis (image labels 3Cs.1.1, 3B.1.1)	4.4, 11.6
hook: the paragraph after Theorem thm:twist (157,938 and 292,580 pairs)	11.7
hook: the Example of Section 3 (congruences "checked for $p < 500$", $p < 1000$)	2.3 (`ellan`, `ellap`); Exercise 2; Sturm's bound: Chapter 12, point 46
hook: the appendix (all paragraphs)	1.1, 5, 6, 12.1
hook: the abstract's counts ("893 curves", "57%", "a third of the cases")	9.1, 11.3 (point 25)
also: what Chapters 2, 3, 9, 10, 11, 12, 13, 14, 15, 17, 18, 19 deferred here	Part 5

24.4.2 4.2 Exercises (Section 13 of the chapter), graded from easy to harder

1. Run `ellglobalred` on 20a1 and 726a1, read off the conductor, the bad primes, the Kodaira codes and the c_p, and say which entries the paper's `R[4][,1]` and `K[i][2]` are. Why does code 1 never occur in the list?
2. With `ellan` compute a_n for 121a1 and 11a1 for $n \le 500$ and check the paper's claim $a_p(121a1) \equiv \chi_{-11}(p)a_p(11a1) \pmod 3$ for $p \neq 3, 11$; then say why a check up to Sturm's bound (Chapter 12) proves it for all n.
3. Using `lfunsympow` and `lfuneuler`, compute P_q and $e_3(E, q)$ for 121a1 at 11, 147b1 at 7, 539d1 at 7 and 726a1 at 11, and check them against the table of Section 1 of the paper (Lemma lem:euler) by the function `euler_by_type` of 3.4.
4. Check the paper's claim that `rule(ellinit([1,1,1,-30,-76]), 3)` returns `[1, 1]` and that for 726a1 `rule(E,3)` and `rule(E,3,1)` both return `[2, 2]`; print the breakdown with `rule_terms` and identify each term with a term of (eq:rule).
5. Run `rule` on 11a1, 11a2, 11a3 at $\ell = 5$ and on 990h1, 990h3 at $\ell = 3$. Explain each output, in particular the negative entry for 11a3, using `ellmoddegree` and `ellweilcurve`. Which of these runs does the paper's statement concern?
6. Show that `ns3(E)` is 1 iff ψ_3 is a product of two irreducible quadratics over $\mathbb{Q}$, whatever the content of ψ_3 and whatever order `factor` lists the factors in; find a CM curve with `ns3(E) = 1` and explain why Conjecture conj:euler excludes it.
7. Using S3, reproduce the row $\ell = 5$, $N < 2000$ of Table 11.1 and compare it with the Python result $[5622, 915, 0, 845]$ (curves, positive bounds, violations, equalities).
8. List the 3Ns optimal curves of conductor below 2000 that violate the uncorrected bound, and check the paper's claims that the shortfall is always 1 and that each violation involves a prime $p \equiv -1 \pmod 3$ with $t_3(E, p) \ge 1$.
9. Compute the Tate parameter of 791a1 at 7 with `ellinit(E, O(7^30)).tate`, decide whether its unit part is a cube in $\mathbb{F}_7^\times$, and check the paper's claim (Section sec:tam) that the extra Tamagawa term then raises the bound to $v_3(\deg \varphi_E)$.
10. (Calibration.) Let X be a sum of independent indicators with $E[X] = \mu$. Prove $P(X = 0) \le e^{-\mu}$ and, using only pairwise independence, $P(X = 0) \le \operatorname{Var} X/\mu^2 \le 1/\mu$. Apply both to the paper's $\mu \approx 605{,}000$ and to the book's $\mu = 14{,}275$.
11. Show that under the null hypothesis of 10.1 the bound-0 distribution and the pooled distribution of v estimate the same thing, and explain why the book's values 59.7% and 23.9% for $P(v = 0)$ are evidence against the null.
12. Check the paper's claim that among bound-0 curves 53.2% have $3 \nmid \deg \varphi_E$ against the book's 59.7% below 10,000, and compute the same share on $N < 5000$ and on $5000 \le N < 10000$. What does the trend suggest about the paper's ranges?
13. (Multiple comparisons.) Suppose $m = 200$ false candidate inequalities each survive a working-set check with probability $\pi = 0.05$ and a hold-out check with probability $\pi' = 0.05$, independently. How many false survivors are expected after the working set, and after the hold-out? What changes if the hold-out is reused for a second round of candidates?
14. Write `twist_lemma_check.gp`: for every optimal curve below 2000 and every odd bad prime q, compare the Kodaira code of the twist by q^* at q with Lemma lem:twist (ii), and check (i) at the other odd primes. Extend to $q = 3$ and test important_notes.md point 51.
15. Classify each data statement of Section 6 of the paper as blind-tested, confirmed only, or a theorem checked, quoting the sentence of Section sec:data that decides it; say what a fresh hold-out would have to contain to test the correction term h_3.
16. Reproduce the list of Eisenstein curves at $\ell = 3$ below 10,000 with deficit $D = 2$ and check the paper's claim that the 3Cs.1.1 curves among them have "rank 0 and torsion $\mathbb{Z}/3$". Then prove that no non-CM curve has both $E[3] \simeq \mathbb{Z}/3 \oplus \mu_3$ and a point of order 9 (point 17).
17. (Toy protocol.) Read off the minima of $v_3(\deg \varphi_E)$ at $q = 3$ on $N < 5000$ as in 9.5, freeze them, and test them on $5000 \le N < 10000$. Then do the same for a statement you choose yourself on the working set, and report honestly whether it survived.

18. Compute the odd-part statistic of 10.6 on $N < 5000$ and on $5000 \le N < 10000$, and explain why the paper's 1.85% is so much smaller than the book's 17.1%.

24.5 5. Dependencies and forward pointers

Results and conventions from earlier chapters used here (by chapter label only; PLAN.md Section 5 lists 9, 10, 11, 14, 19, 21 as the prerequisites): * Chapter 2 (`ch:rings`): valuations of rationals, χ_{-3}, χ_{-4}, Kronecker symbols, ℓ-th powers in $\mathbb{F}_q^\times$ (3.4, 3.5); its deferral "PARI's `valuation`, `kronecker`, `znprimroot`, `ffgen`, `polisirreducible` as the computational counterparts" is met in 2.2 for `valuation` and `kronecker` (the other three are not used by the paper; one sentence). * Chapter 3 (`ch:modules`): its deferral "`matsnf`, `mathnf` (one sentence)" is met by one sentence in Section 7 (the congruence module of Chapter 14 is a Smith normal form computation). * Chapter 9 (`ch:ellcurves`): Weierstrass data and PARI's member functions (C3 there), division polynomials, isogeny classes, labels (12.1, 12.2), the 990h exception, the running-examples table. * Chapter 10 (`ch:reduction`): Tate's algorithm, Kodaira codes (5.5), `ellglobalred` and `elllocalred` (10.2), the Tate parameter (7.3, 7.4, 7.8), Lemma lem:twist (8.1, 8.2). * Chapter 11 (`ch:galrep`): mod 3 images and the ψ_3 patterns (7.3), Sutherland's labels and the `galrep` check (7.4, 7.6), the CM caveat (B7), 3Ns and h_3 (8.4, 8.5), types IV, IV* at 2 (4.4). * Chapter 12 (`ch:modforms`): `mfinit`, `mfdim`, `mfsturm`, `mfpetersson` (its normalisation by $\mu(N)$); Sturm's bound for the congruences of the Example of Section 3 (point 46). Its deferral list is met by one paragraph in Section 7 naming these functions and pointing back. * Chapter 13 (`ch:hecke`): `mfbasis`, `mfcoefs`, `mfheckemat`, `mfatkininit`, `mffromell`, `mfbd`, `matrixqz` (one sentence each in Section 7, with the pointer to Chapter 13's lattice method). * Chapter 14 (`ch:moddeg`): optimal curves (6.5), the Zagier-Watkins formula and Watkins's method (7.3, 12.5), `ellmoddegree`, the congruence-number algorithm and its values (8.5, verification log), Watkins's conjecture and Calegari-Emerton (12.3, 12.4). * Chapter 15 (`ch:modgalrep`): the verification scripts of its Part 6.6 (congruences up to the Sturm bound, `elltwist`, `mfeigenbasis` and reduction modulo λ): cited as further examples of the same technique in Section 7, not reprinted. * Chapters 17, 18 (`ch:galcoh`, `ch:blochkato`): Lemma lem:tamfactor and its examples (17's 5.5, 18's 8.5); the computation of the Tate parameter (18's verification log). * Chapter 19 (`ch:lfunctions`): the normalisation of `lfunsympow` and `lfuneuler` (C4, C5, 11.1), the mathematics of `rule` (11.2), the table of P_q by shape (10.1), the examples (12.1). * Chapter 21 (`ch:paper`): A6 (what each data statement means and tests), A9.1 (the mathematics of `rule`, line by line), A9.2 (codes), B4 (the hand verifications, now run with PARI here). Chapter 22 does not repeat A6's readings; it adds the computations, the statistics and the status table 9.2. * Appendix A (`ch:prereq`): its section on elementary probability and statistics (base rates, expectation, Markov and Chebyshev, hypothesis tests, hold-out sets and pre-registration, the calibration example, the protocol `paperlink`). Chapter 22 cites it and never restates it.

Forward pointers (chapters that use this one): none in the main text (this is the last chapter). Appendix B (`ch:notation`): the PARI Kodaira codes (with Chapter 10) and the names B^{rule}, B^{nc}, B^{III}, B^{main}, B^{tw} if the writer of Appendix B wants them (they are chapter-local otherwise). The cross-reference pass of PLAN.md Section 8, step 4, may upgrade Chapter 21's `\Cref{ch:computation}` to labels of items 5.1 to 5.4, 9.2, 10.1 to 10.3, 11.1 to 11.7.

24.6 6. Notes for the writer

24.6.1 6.1 Conventions to state (Section 1 of the chapter)

- K1 (versions, data pin) and K9 (code typography) in one `convention` box at the start; every `lstlisting` with output says "(PARI/GP 2.17.2)" in its caption or the sentence before it.
- K3 to K5 (bounds, violation, equality, excess, populations) in a second box; these names are used in every table.
- K10: one sentence sending the reader to Appendix A for base rates, hypothesis tests, hold-out sets, pre-registration, linearity of expectation, Markov, Chebyshev; no definition from Appendix A is repeated.

24.6.2 6.2 Pitfalls

- **Do not invent table entries.** The paper's `tables/*.tex` are not available (Chapter 21, A6). Quote only the numbers in the paper's text, tag them D, and put the book's own counts (C) beside them, never in their place.
- **Do not present the book's small-range counts as tests.** The book's ranges were not blind (the statements were known); they are reproductions. Say so in Section 11 and in 9.5.
- **Optimal curve, not curve number 1.** Every script looks the optimal equation up in `degphi`; 990h3 is the exception below 10,000. Never run `rule` on a non-optimal model and call the output a modular degree (5.4(a)).
- **Population filters belong to the statements.** Do not count Eisenstein or CM curves as counterexamples (6.7, 5.4(b), (c)).
- **CM detection.** By j (thirteen values), not by `galrep` labels (which are absent for CM-forced images).
- **Irreducibility from `galrep`.** "Reducible iff a label begins with ℓB or ℓCs"; compare the prime part exactly (a label like 37B.8.1 must not be read as a 3-adic label; the book's scripts compare the leading digits, Part 7).
- **Units of counting.** Curves versus pairs (E, ℓ) versus pairs (curve, prime); rates only among positive bounds.
- **Statistics.** No p-values for the calibration (10.2 says why); say "expected number under the null" and give the null; say that the reference population matters by a factor of two (10.3) and does not change the conclusion; say that on the working set the calibration measures strength, not validity (10.4(b)).
- **Versions.** Do not assert that the paper's code runs unchanged in 2.15 without checking `ellmoddegree`'s return type and `readstr`/`strsplit` there (the book's runs are 2.17.2).
- **Names of files and functions.** `degphi`, not `alldegphi`; `ellweilcurve`'s output described only as far as checked (3.8(ii)); `elltatepairing` is not the Tate parameter.
- **Code blocks.** No double hyphen even inside code (K9); GP comments `\\`; `lstlisting` with `language=GP`.
- **Typography of numbers.** The paper writes 867,578; in running text the writer may write 867,578. Percentages with one decimal as in the paper.

24.6.3 6.3 What to leave to other chapters

- Chapter 10: the mathematics of Tate's algorithm, the full Kodaira code dictionary, the Tate curve. Cite 5.5, 7.3, 7.4, 7.8, 10.2.
- Chapter 11: the classification of mod 3 images, Sutherland's labelling scheme, the ψ_3 patterns. Cite 7.3, 7.4.
- Chapter 14: optimality, Manin constants, the congruence-number algorithm and its values. Cite 6.5, 8.5, 12.5.
- Chapter 19: the normalisations of the symmetric square, the classification of P_q, the mathematics of `rule`. Cite C4, C5, 10.1, 11.1, 11.2.
- Chapter 21: what each data statement means (A6), the line-by-line mathematics of `rule` (A9.1), the corrections table (Part C).
- Appendix A: the definitions of elementary probability and statistics.

24.6.4 6.4 How the relevant important_notes.md points are handled

- **Point 25 (abstract wording: "every optimal curve", "a third of the cases").** 9.1 (optimality above 400000 is partial), 11.3 (rates 31%, 23%, 18% on the paper's ranges; 53%, 33% on the book's), Exercise 15.
- **Point 43 (types IV, IV* at 2: a theorem).** 6.4 (980i1), Part 7 R4: all 3,226 optimal curves below 10,000 with type IV or IV* at 2 have $P_2 = 1 + 2X$ (C); the paper's appeal to the 92,812

curves of the working set is then a check of a theorem, and 9.2 lists Proposition prop:forms among the theorems checked.

- **Point 48 (the 3Ns observations are theorems).** 11.3 reports the observations (every 3Ns curve has a positive Ш term; all violators have a prime $p \equiv -1 \pmod 3$ with $t_3 \geq 1$) as checks, with a sentence that Chapter 11 (8.4, 8.5) proves them.
- **Point 50 (type II at 2 does not give $P_2 = 1$ in general).** 6.6 (336f1, 1232l1 with PARI outputs).
- **Point 52 (codes $n+4$ and $-(n+4)$ only for $n \geq 1$).** K7, 3.2, 5.4(e).
- **Point 53 (t_ℓ at $q = \ell$).** 5.2(g), 5.3(i), 6.2 (726a1 at $q = 3$).
- **Point 57 (Calegari-Emerton and CM curves).** One sentence in Section 7 or 11.8 with the data (C): the CM optimal curves of odd modular degree below 10,000 are 27a1, 32a1, 36a1, 49a1, 243b1, of which 27a1 and 243b1 have torsion $\mathbb{Z}/3$ and non-prime conductor; the writer quotes the published hypothesis only after checking it (Chapter 14 owns the statement). Also (C, R9): the paper's 2-adic inequality of Section sec:two holds for all 25,086 non-CM optimal curves below 10,000 without a rational 2-isogeny, and the variant counting I_0^* with weight 2 fails 6 times (e.g. 99d1, 931b1, 2299d1), as the paper's 123, 26, 20 exceptions lead one to expect.
- **Point 58 (726a1: $r = 1584$, $\deg = 144$) and point 62 (539d1: $r = 504$, $\deg = 72$).** 7.3, 6.2.
- **Point 64 ($\mathrm{Tam}_q(\mathrm{ad}^0 T)$ trivial at potentially good additive primes).** 8.2(c): the excess is the Selmer length plus the Tamagawa excess at potentially multiplicative primes; the mean excess is therefore an average Selmer length plus a computable term.
- **Point 68 (sec:tam can include I_n^* primes; "exactly where" overstates).** 11.5 (the book follows the paper's choice of I_n only, says that I_n^* could be included, and quotes Chapter 18's count); 9.2 (the test is of the identity's Tamagawa term, with the Selmer term as the remaining excess).
- **Point 69 (P_q has four shapes; 592c1 at $\ell = 5$).** 6.6.
- Further points used: 3 and 44 (labels; 4.4), 5 and 6 ($h_\ell = 0$ for $\ell \geq 5$, $h_3 \leq 1$; 5.2(h), 5.3(ii)), 15 (e_ℓ not twist invariant; 11.7 compares degrees, not e_ℓ), 16 (strict twist inequalities in case (a); 11.7 confirms with counts), 17 ($D \leq 2$ provable; 11.6, Exercise 16), 21 (980i1 outside thm:qual; 6.4), 23 and N9 of Chapter 21 ($c(u)$, not c_q; 3.5), 27 (optimality irrelevant at ℓ for irreducible $E[\ell]$; 4.3(iv)), 39 and 56 (equality in (eq:identity); 8.2(c)), 46 (Sturm; Exercise 2), 47 ($\ell \notin S_f$ automatic for $\ell \geq 5$, $\ell \nmid N$: K5's hypothesis population), 51 (exchanges at $q = 3$; 3.7), 60 (I_0^* formula at $q = 3$; Chapter 19 checked it, 3.4 cites), 65 (abstract's Tamagawa sentence; 11.5 states "$\ell \mid n$" and "at least one").
- Book-internal: B7 of Chapter 11 (counts with the CM caveat; 3.6, 4.4), B14 (bibliography keys; Part 8, B-b), B16 (key `Tate75`, not `TateAlgorithm`, if Tate's algorithm is cited).

24.6.5 6.5 References and the reproducibility note

In `refs.bib`: `PARI` (version 2.15, as the paper), `PARIusers`, `Cremona97`, `ecdata`, `LMFDB`, `Sutherland16`, `Zywina15`, `Watkins02`, `Zagier85`, `ARS12`, `SilvermanATAEC`. In `bib/chA.bib` (Appendix A): `Feller68`, `Wasserman04`, `RHB06`; Chapter 22 needs them too, so by decision B14 they must move to `refs.bib` before Chapter 22 cites them (otherwise the stand-alone build of Chapter 22, which reads only `refs.bib` and `bib/ch22.bib`, fails; Part 8, B-b). Proposed additions to `bib/ch22.bib`: a `misc` entry `PARI217` for PARI/GP 2.17.2 (the version of the book's runs; the writer checks the release year), and `cypari2` (the SageMath developers' Python interface, version 2.2.4; the writer checks the authorship line). If the writer cites Dokchitser's algorithm for `lfun`, `Dokchitser04` goes to `refs.bib` (Chapter 19's note). No theorem number from these works is cited.

Reproducibility note (for the chapter's Notes and for the writer). Software: PARI/GP 2.17.2 (`version()` returns `[2, 17, 2]`), `cypari2` 2.2.4, Python 3.13, on Linux with 4 cores and 15 GB of memory; `parisize` and `parisizemax` set by `pari.allocatemem(10**9)` (or $2 \cdot 10^9$) unless stated. Data: six `ecdata` files for conductors 0 to 9,999, copied to `/tmp/ch09/data` by an earlier agent; the commit was not recorded. SHA-256 sums (C): `allbsd.00000-09999` b078b5226d276fbc64289833abfd8430e3f5969c657bc32f94db546e8f96e8fd; `allcurves.00000-09999` 259f3846329395b371e8079c77a6f1097adaebc98a054974573e241416efa968; `allisog.00000-09999` 9e033f1d092509f280d23e2f2666577c96af9593911568ca9e9678c2e35b1d82; `alllabels.00000-09999` bc06d9bd10dbba36ddb84ad2fc407d2a71c8c89f22361befae8eda08aa18a377; `degphi.00000-09999`

65d96cbab0de966409810b666e55b731515728c4d19904ea95b3750d48f6dd68; galrep.00000-09999 7faf8a6a92d85eba62dda9d025474a543bcd3ae61b8c489f581124f2ec34742d. Before the chapter is printed the writer downloads the same six files from the ecdata repository at the paper's commit 25cec5e (or at a recorded later commit), compares the sums, reruns S4 to S8 (about 15 seconds for S4, seconds for S5 to S7, about 25 minutes on four cores for S8), and prints the commit hash and the sums in the chapter's Notes. If the sums differ, the counts of Section 11 must be recomputed, not edited by hand.

24.7 7. Reproduction log

Everything below was run for this outline in the session scratchpad (/tmp/claude-0/-home-user-modular-math-book/404f75d6-d3ce-544f-a9b9-1e1ed5d22ed3/scratchpad/ch22/), with PARI/GP 2.17.2 through cypari2 2.2.4 (no gp binary is installed; every GP file was loaded with pari('read("file.gp")')), Python 3.13, and the six ecdata files of 4.1 in /tmp/ch09/data (SHA-256 sums in 6.5). Population: the 38,042 optimal curves of degphi.00000-09999.

R0. Data files (formats as found). Six files, conductor 0 to 9,999: allcurves (64,687 lines: N class number [a1,...,a6] rank torsion), allbsd (64,687: the same, then $\prod c_p$, Ω, $L^{(r)}(1)/r!$, regulator, Sha; checked on 11a1: 5, 1.2692..., $0.2538\cdots = \Omega/5$, 1, 1), allisog (38,042: one line per class with the curves and the isogeny matrix), alllabels (64,687: Cremona's then the LMFDB's N, class, number), degphi (38,042: N class number degree {primes} [a1,...,a6] of the optimal curve), galrep (64,687: label then non-surjective image labels; 20,930 lines with no label). No file contains per-prime Kodaira symbols or Tamagawa numbers. Exactly one degphi equation is not curve number 1: 990h3. There is no gp and no elldata package: ellsearch(11) and ellidentify fail with "error opening elldata file".

R1. The appendix code. euler_rule.gp extracted verbatim from the paper (sed between rule(E, l, {iii = 0}) = and the ns3 line); it contains no double hyphen. Pasting it into pari(...) as one string fails (syntax error); read works. Outputs [rule(E,1), rule(E,l,1)] and ns3, ellmoddegree, ellglobalred local data [f, code, c]:

curve	$[a_i]$	local data	$\deg \varphi_E$	ns3	$\ell = 3$	$\ell = 5$	$\ell = 7$
121a1	$[1,1,1,-30,-76]$	11: [2, 2, 1]	6	0	[1, 1], [1, 1]	[0, 0]	[0, 0]
726a1	$[1,1,0,-35,-251]$	2: [1, 10, 2]; 3: [1, 7, 1]; 11: [2, 3, 2]	144	1	[2, 2], [2, 2]	[0, 0]	[0, 0]
539c1	$[1,0,1,170,-3237]$	7: [2, −7, 4]; 11: [1, 6, 2]	288	0	[2, 2]	[0, 0]	[0, 0]
980i1	$[0,0,0,1568,-27216]$	2: [2, −4, 1]; 5: [1, 5, 1]; 7: [2, −9, 2]	2880	0	[2, 2]	[1, 1]	[0, 0]
99a1	$[1,-1,1,-2,0]$	3: [2, 3, 2]; 11: [1, 5, 1]	4	0	[0, 0]	[0, 0]	[0, 0]
336f1	$[0,1,0,-1,2]$	2: [4, 2, 1]; 3: [1, 5, 1]; 7: [1, 6, 2]	24	0	[1, 1]	[0, 0]	[0, 0]
1232l1	$[0,0,0,32,-16]$	2: [4, −2, 1]; 7: [1, 6, 2]; 11: [1, 5, 1]	288	0	[2, 2]	[0, 0]	[0, 0]

curve	$[a_i]$	local data	$\deg \varphi_E$	ns3	$\ell = 3$	$\ell = 5$	$\ell = 7$
592c1	$[0, 0, 0, -16, -216]$	2: [4, −2, 1]; 37: [1, 5, 1]	80	0	[0, 0]	[1, 1]	[0, 0]
11a1	$[0, -1, 1, -10, -20]$	11: [1, 9, 5]	1	0	[0, 0]	[0, 1]	[0, 0]

(Where one pair is shown, rule(E,l) and rule(E,l,1) agree.) The paper's two claims (121a1 [1, 1]; 726a1 [2, 2] in both modes) are confirmed. Per-prime breakdowns (rule_terms of S2) as printed in 6.1 to 6.6. Non-optimal models: 11a2 and 11a3 at $\ell = 5$ give [1, 0] and [-1, 0] (ellmoddegree 5 and 1/5); 990h3 at $\ell = 3$ gives [3, 3], 990h1 gives [4, 3] (ellmoddegree 1728 and 5184, each within a 100 MB stack). ellweilcurve marks 11a1 and 990h3 with [1, 1].

R2. PARI behaviour used in the text. factor(3*x^4+6) = Mat([x^4 + 2, 1]) (content dropped); factor(elldivpol(E,3)) for 726a1 = [x^2 - x - 74, 1; 3*x^2 + 8*x + 20, 1]; polgalois of that ψ_3 raises "not an irreducible polynomial". lfunparams(lfunsympow(E,2)) for 121a1 = [121, 3, [0, 0, 1]]; lfuneuler(S,11) = 1/(11*x + 1); lfuneuler(S,2) = 1/(-8*x^3 - 2*x^2 + x + 1). elltors (726a1) = [2, [2], [[6, -3]]]; ellisomat(E)[2] = [1, 2; 2, 1] (726a1), [1, 5, 5; 5, 1, 25; 5, 25, 1] (11a1); ellan (121a1, 12 terms) = [1, -1, 2, -1, 1, -2, 2, 3, 1, -1, 0, -2]; ellglobalred (726a1) = [726, [1, 0, 0, 0], 4, [2, 1; 3, 1; 11, 2], [[1, 10, 0, 2], [1, 7, 0, 1], [2, 3, 0, 2]]]; elllocalred(ellinit([0,1,0,4,4]), 2) = [2, -4, [1, 0, 0, 0], 3]; ellinit(ellinit([1,1,0,4,11]), 7) = [] and ellinit(..., 5) is a curve over $\mathbb{F}_5$ (so ellinit(E, q) is reduction, not $\mathbb{Q}_q$). readstr and strsplit exist in 2.17.2. Timing: lfunsympow plus all bad lfuneuler about 0.2 ms per curve; S4 on all 38,042 curves in 14 s.

R3. Tate parameters (S2 tatedata, ellinit(E, O(q^30)).tate[3]). 11a1 at 11: $n = 5$, $q_E = 10 \cdot 11^5 + 2 \cdot 11^6 + 6 \cdot 11^7 + \cdots$, $\bar{u} = 10$, $k = 1$ ($\ell = 5$); 77c1 at 7: $n = 3$, $\bar{u} = 5$, $k = 0$; 539c1 at 7: identical q_E to 77c1, $\bar{u} = 5$, $k = 0$; 791a1 at 7: $\bar{u} = 6$, $k = 1$; 1577a1 at 19: $\bar{u} = 8$, $k = 1$. valuation(qE, q) $= n$ asserted for every site used in R7.

R4. Lemma lem:euler and the factor at 2 (S5). For the 67,745 pairs (optimal curve, bad $q \geq 5$): PARI's $P_q(q^{-2})$ equals the table of Section 1 in every case (I_n 58,860; I_n^* 2,709; II, II*, IV, IV* 3,588; III, III* 1,676; I_0^* 912, with a' from elltwist). All 3,226 optimal curves with type IV or IV* at 2 have $P_2 = 1 + 2X$.

R5. The five bounds on the whole range (S4, S5; Python, and S3 in GP for $N < 2000$). Tables 11.1 to 11.4 and 11.6 above are the output. GP (S3) and Python agree on $N < 2000$: $\ell = 3$ [4951, 2517, 0, 2046], $\ell = 5$ [5622, 915, 0, 845], $\ell = 7$ [5735, 468, 0, 446] (curves, positive bounds, violations, equalities). ns3 versus galrep's 3Ns on the 37,853 non-CM optimal curves: 0 mismatches (90 each); ns3 = 1 on 13 CM optimal curves (121b1, 256a1, 256d1, 1089e1, 1936f1, 2304c1, 2304k1, 3025a1, 5929a1, 6400f1, 6400r1, 7744a1, 7744s1; $j = -32768$ or 8000). Section sec:iso cells at $\ell = 5$: I_0^* $e = 0, 1, 2$: minima $0, 1, 2$ (94, 75, 10 curves); II 0: 0 (210), 1: 1 (12); II* 0: 0 (94), 1: 1 (5); III, III* 0: 0 (152, 150), 1: 1 (11 each); IV 0: 0 (95), 1: 1 (5); IV* 0: 0 (209), 1: 1 (12); I_n^* 0: 0 (427), 1: 1 (141). At $\ell = 7$: every cell's minimum equals e ($e \leq 1$; the largest cell I_n^* with $e = 0$, 806 curves). The weights w_3 (curves whose only possible contribution at 3 is a potentially good prime $q = 3$, whole range): II 1 (494), II* 2 (342), III 0 (472), III* 1 (472), IV 1 (342), IV* 2 (494), I_0^* 0 (646): the paper's weights exactly. 3,239 base-population curves at $\ell = 3$ have a positive w_3.

R6. Calibration (S6). As in 10.3 and 10.5. The permutation test: 200 shuffles of v_3 within strata (conductor bin of width 2500 times $\min(\omega(N), 4)$), Python's random.seed(0); mean 7,638, standard deviation 49.2, minimum 7,517.

R7. Tamagawa test (S7). As in 11.5; the first examples with a positive extra term at $\ell = 3$: 791a1 (7), 805b1 (7, $\bar{u} = 1$), 805c1, 805d1, 874a1 (19, $\bar{u} = 12$), 874b1 (19, $\bar{u} = 18$); at $\ell = 5$: 4136e1, 4136f1 (11, $\bar{u} = 10$), 5673a1 (61), 6507a1, 6507b1 (241), 6588e1; at $\ell = 7$: 4171a1 (43, $\bar{u} = 1$).

R8. ellmoddegree against degphi (S8). All 38,042 optimal curves, four processes, parisizemax $3 \cdot 10^9$: 38,042 curves checked (9,511 + 9,511 + 9,510 + 9,510), 0 mismatches, 0 errors, about 16 minutes per process.

R9. Further checks. Lemma lem:twist: 85,848 pairs, 0 mismatches; 5,122 pairs at $q = 3$ with the potentially good exchanges, 0 mismatches; no code change at odd $p \neq q$. Twist pairs after Theorem thm:twist: as 11.7. Odd-part statistic: as 10.6 (33,137 curves, no non-integral quotient). Section sec:two at $\ell = 2$: on the 25,086 non-CM optimal curves without a label 2B or 2Cs, $v_2(\deg \varphi_E) \geq \sum_{q \| N} v_2(n_q) + \#\{q : \text{III}, \text{III}^*, \text{I}_0^*, \text{I}_n^*\}$ holds always; with I_0^* counted twice it fails 6 times (99d1, 931b1, 2299d1, ...). CM optimal curves of odd degree: 27a1, 32a1, 36a1, 49a1, 243b1. Eisenstein $D = 2$ list as in 11.6 (with ranks 0 and torsion orders $6, 3, 3, 3, 3, 3, 3, 3, 3, 9, 3, 9, 3, 3$ in the order 14a1, 26a1, 35a1, 38a1, 77b1, 158d1, 278b1, 326c1, 485a1, 714i1, 4385b1, 6942n1, 7094c1, 8027a1). `ellmoddegree` memory: the default 8 MB stack suffices for 9999m1 but overflows for 9510e1, 9690m1, 7910b1, which succeed with stacks of 16 to 32 MB (`_lfuninit_worker` warnings); average time about 0.065 s per curve on a random sample of 10.

Not verified. Anything above conductor 10,000 (all D items); the paper's out-of-table curves (not published); the behaviour of PARI 2.15 (all runs are 2.17.2); the contents of the paper's repository; the meaning of every entry of `ellweilcurve`'s output beyond the `[1, 1]` mark; the commit of the local `ecdata` copy.

24.7.1 7.1 The scripts (tested as printed here)

S1 is the paper's appendix code, extracted verbatim (the chapter prints it from the paper, with line numbers):

```
rule(E, l, {iii = 0}) =
{ my(R = ellglobalred(E), P = R[4][,1], K = R[5], S = lfunsympow(E, 2), b = 0, j = E.j);
  for(i = 1, #P,
    my(q = P[i], k = K[i][2], e = 0, t = 0, vj = valuation(j, q));
    if(q != l,
      my(F = 1/lfuneuler(S, q), r = subst(F, x, 1/q^2)); \\ P_q(q^{-2}): omitted factor
      if(k >= 5, r /= (1 - 1/q^2));                      \\ multiplicative: naive one out
      if(abs(k) != 3 || iii, e = valuation(r, l)),       \\ III, III* (codes 3,-3) if iii
      if(l == 3, e = if(k == 4 || k == 2, 1,
                     if(k == -4 || k == -2, 2, if(k == -3, 1, 0))))));   \\ w_3 at q = 3
    if(vj < 0, t = valuation(-vj, l));                    \\ Tamagawa exponent
    b += e + t);
  if(iii && l == 3, b -= ns3(E));                         \\ the correction term h_3
  [valuation(ellmoddegree(E), l), b]; }
ns3(E) = my(F = factor(elldivpol(E, 3))); (#F[,1] == 2 && poldegree(F[1,1]) == 2);
```

S2 `sessions.gp` (worked sessions; the outputs of 6.1 to 6.8 and of 3.4, 3.5, 3.7 come from it; `eulercheck` returned `[]` on all 9,515 optimal curves below 3,000):

```
\\ sessions.gp: per-prime breakdown of the rule, Tate data, and the twist check.
\\ rule_terms(E, l, iii): one row [q, code, P_q(x), P_q(q^-2), e, w, t] per bad prime q.
rule_terms(E, l, {iii = 1}) =
{ my(R = ellglobalred(E), P = R[4][,1], K = R[5], S = lfunsympow(E, 2), M = []);
  for(i = 1, #P,
    my(q = P[i], k = K[i][2], F = 1/lfuneuler(S, q), r = subst(F, x, 1/q^2),
       e = 0, w = 0, t = 0, vj = valuation(E.j, q));
    if(q != l,
      my(rr = if(k >= 5, r / (1 - 1/q^2), r));
      if(abs(k) != 3 || iii, e = valuation(rr, l)),
      if(l == 3, w = if(k == 4 || k == 2, 1, if(k == -4 || k == -2, 2, if(k == -3, 1, 0))))));
    if(vj < 0, t = valuation(-vj, l));
    M = concat(M, [[q, k, F, r, e, w, t]]));
  M; }
\\ tatedata(E, q, l): [n, Tate parameter q_E, unit part mod q, largest k <= v_l(q-1) with ubar an l^k-
```

```
tatedata(E, q, l) =
{ my(Eq = ellinit(E, O(q^30)), qE = Eq.tate[3], n = valuation(qE, q),
     u = lift(qE / q^n + O(q)), v = valuation(q - 1, l), k = 0);
  while(k < v && Mod(u, q)^((q - 1) / l^(k + 1)) == 1, k++);
  [n, qE, u, k]; }
\\ twistcheck(E): for each odd bad prime q, the Kodaira code of E and of its twist by q* at q
twistcheck(E) =
{ my(P = ellglobalred(E)[4][,1], V = []);
  for(i = 1, #P, my(q = P[i]); if(q == 2, next);
    my(qs = if(q % 4 == 1, q, -q), Et = ellinit(elltwist(E, qs)));
    V = concat(V, [[q, elllocalred(E, q)[2], elllocalred(Et, q)[2], ellglobalred(Et)[1]]]));
  V; }
\\ euler_by_type(E, q): L_q(A_f,1)^(-1) from the Kodaira type, for a bad prime q >= 5 (Lemma lem:euler
euler_by_type(E, q) =
{ my(k = elllocalred(E, q)[2]);
  if(k >= 5 || k <= -5, return(1 - 1/q^2));
  if(abs(k) == 2 || abs(k) == 4, return(1 - kronecker(-3, q)/q));
  if(abs(k) == 3, return(1 - kronecker(-4, q)/q));
  if(k == -1, my(a = ellap(ellinit(elltwist(E, if(q % 4 == 1, q, -q))), q));
    return((q - 1)*((q + 1)^2 - a^2)/q^3));
  error("not a bad prime"); }
\\ compare with PARI's factor: returns the list of q >= 5 where they differ (expected: [])
eulercheck(E) =
{ my(S = lfunsympow(E, 2), P = ellglobalred(E)[4][,1], bad = []);
  for(i = 1, #P, my(q = P[i]); if(q < 5, next);
    if(subst(1/lfuneuler(S, q), x, 1/q^2) != euler_by_type(E, q), bad = concat(bad, q)));
  bad; }
```

S3 `table_small.gp` (one row of Table 11.1 in pure GP; needs S1 for `ns3`; `tablerow(2000, 5, 1, dir)` returns `[5622, 915, 0, 845]`; each call reads the files once and takes about 0.3 s below 2000):

```
\\ table_small.gp: the paper's rule on Cremona's optimal curves of conductor < Nmax.
\\ Needs euler_rule.gp (the paper's appendix) for ns3; reads ecdata files degphi.* and galrep.*.
rulebound(E, l, iii) =
{ my(R = ellglobalred(E), P = R[4][,1], K = R[5], S = lfunsympow(E, 2), b = 0, j = E.j);
  for(i = 1, #P,
    my(q = P[i], k = K[i][2], e = 0, t = 0, vj = valuation(j, q));
    if(q != l,
      my(r = subst(1/lfuneuler(S, q), x, 1/q^2));
      if(k >= 5, r /= (1 - 1/q^2));
      if(abs(k) != 3 || iii, e = valuation(r, l)),
      if(l == 3, e = if(k == 4 || k == 2, 1, if(k == -4 || k == -2, 2, if(k == -3, 1, 0)))));
    if(vj < 0, t = valuation(-vj, l));
    b += e + t);
  if(iii && l == 3, b -= ns3(E));
  b; }
{ CMJ = [0, 1728, -3375, 8000, -32768, 54000, 287496, -884736, -12288000, 16581375,
       -884736000, -147197952000, -262537412640768000]; }
\\ galrep: label -> list of non-surjective image labels
readgal(file) =
{ my(M = Map()); foreach(readstr(file), s, my(w = strsplit(s, " "));
    mapput(M, w[1], w[2..#w])); M; }
reducible(G, lab, l) =
{ my(L); if(!mapisdefined(G, lab, &L), return(0));
```

```
  for(i = 1, #L, my(s = L[i]);
    if(strprefix(s, Str(l, "B")) || strprefix(s, Str(l, "Cs")), return(1)));
  0; }
strprefix(s, p) = #s >= #p && Vecsmall(s)[1..#p] == Vecsmall(p);
\\ one row of the table: [curves, positive bound, violations, equalities]
tablerow(Nmax, l, iii, dir) =
{ my(G = readgal(Str(dir, "/galrep.00000-09999")), c = [0, 0, 0, 0]);
  foreach(readstr(Str(dir, "/degphi.00000-09999")), s,
    my(w = strsplit(s, " "), N = eval(w[1]));
    if(N >= Nmax, next);
    my(E = ellinit(eval(w[6])), deg = eval(w[4]), lab);
    if(setsearch(Set(CMJ), E.j), next);
    \\ the label of the optimal curve is w[1] w[2] w[3], except 990h (990h3)
    lab = Str(w[1], w[2], if(N == 990 && w[2] == "h", 3, w[3]));
    if(reducible(G, lab, l), next);
    my(b = rulebound(E, l, iii), v = valuation(deg, l));
    c[1]++; if(b > 0, c[2]++; if(v < b, c[3]++); if(v == b, c[4]++)));
  c; }
```

S4 `compute.py`, shortened to its core (the full version also stores a_q, c_q, $P_q(x)$, rank, torsion, the number of curves in the class with a rational point of order ℓ for $\ell = 3, 5, 7$, and the isogeny degrees; 14 s for all 38,042 curves):

```
# S4 (shortened): one JSON record per optimal curve, N < 10000 (PARI/GP 2.17.2 via cypari2)
import cypari2, json
from fractions import Fraction
pari = cypari2.Pari(); pari.allocatemem(2*10**9)
D = '/path/to/ecdata/'                          # the pinned copy (Part 6.5)
bya = {}                                         # [a1,...,a6] -> Cremona label
for line in open(D + 'allcurves.00000-09999'):
    w = line.split(); bya[tuple(eval(w[3]))] = w[0] + w[1] + w[2]
gal = {w[0]: w[1:] for w in (l.split() for l in open(D + 'galrep.00000-09999'))}
out = open('records.jsonl', 'w')
for line in open(D + 'degphi.00000-09999'):
    w = line.split(); a = eval(w[5]); lab = bya[tuple(a)]   # 990h -> 990h3
    E = pari.ellinit(a); gr = pari.ellglobalred(E); S = pari.lfunsympow(E, 2)
    P = [int(p) for p in pari('(g) -> g[4][,1]~')(gr)]
    primes = []
    for i, q in enumerate(P):
        k = int(pari('(g, i) -> g[5][i][2]')(gr, i + 1))
        r = Fraction(str(pari.subst(1/pari.lfuneuler(S, q), 'x', pari(f'1/{q}^2'))))
        vj = int(pari.valuation(E.j(), q)) if E.j() != 0 else 99
        primes.append(dict(q=q, k=k, r=[r.numerator, r.denominator], vj=vj))
    ns3 = int(pari('(E) -> my(F = factor(elldivpol(E, 3))); (#F[,1] == 2 && poldegree(F[1,1]) == 2)')(
    out.write(json.dumps(dict(lab=lab, N=int(w[0]), deg=int(w[3]), j=str(E.j()), gal=gal.get(lab, []),
                              ns3=ns3, primes=primes, ainvs=a)) + '\n')
```

S5 to S8 are not printed in full in the chapter. S5 (`analyse.py`, `tables.py` to `tables4.py`) reads `records.jsonl` and computes, for each curve and ℓ, the per-prime terms exactly as `rule` does (`e` from the stored $P_q(q^{-2})$ with the division at codes ≥ 5; `w` at $q = \ell = 3$; `t` from $v_q(j)$), the five bounds of K3, $\#\Phi_q^{\mathrm{geom}}$ and $w_\ell(E, q)$ from the codes, and the counts of Section 11; the population filter reads `galrep` labels, comparing the prime part of each label exactly; CM by the thirteen j-invariants. S6 (`calib2.py`) computes $\hat{F}$, $\hat{E}$, the variance, the stratified version and the permutation test. S7 (`tam.py`) calls `ellinit(a, O(q^30))` and `.tate[3]` at every qualifying multiplicative prime. S8 (`moddeg_check.py`)

compares `ellmoddegree` with `degphi`, line by line, in four processes with `parisizemax` $3 \cdot 10^9$, catching errors. The writer prints the core loop of S5 (about 30 lines) and of S6 (about 25 lines).

24.8 8. Points for important_notes.md

24.8.1 8.1 New points about the paper

Checked against the paper's text and, where numerical, by the runs of Part 7. None changes a theorem.

N1. Section sec:eis, two descriptive sentences contradict the data (14a1; the points of order 9). Location: Section sec:eis, "With a rational point of order 3 and mod 3 image of Borel type, $D \le 1$; $D = 2$ occurs only when $E[3] \simeq \mathbb{Z}/3 \oplus \mu_3$ (the split Cartan image `3Cs.1.1` ..., seventeen curves, all of rank 0 and torsion $\mathbb{Z}/3$) or when E has a point of order 9 (five curves)." Problem: (a) 14a1 $= [1, 0, 1, 4, -6]$ is optimal, not CM, of conductor 14 (so in the working set), with image 3Cs.1.1, rank 0, $\deg \varphi_E = 1$ and types I_6 at 2, I_3 at 7, so $D = v_3(6) + v_3(3) - 0 = 2$; its torsion is $\mathbb{Z}/6$, not $\mathbb{Z}/3$. (b) A curve with a point of order 9 has Borel image (3B.1.1 for all four such optimal curves below 10,000), so the first clause, read literally, contradicts the second: 714i1 and 6942n1 (torsion $\mathbb{Z}/9$, image 3B.1.1) have $D = 2$. Fix: "With a rational point of order 3, no point of order 9 and Borel image, $D \le 1$"; "all of rank 0, with torsion $\mathbb{Z}/3$ or $\mathbb{Z}/6$" (the paper already says "their torsion is $\mathbb{Z}/3$ or $\mathbb{Z}/6$" of the 2,065 curves with image 3Cs.1.1). Confidence: high for (a) and (b) as statements about the curves (`degphi`, `galrep`, `allcurves` and PARI's local data agree); the count "seventeen" itself could not be checked (above 10,000).

N2. Section sec:data, the protocol paragraph: what was blind-tested needs two qualifications. Location: "the correction term $h_\ell(E)$ was identified last, from the exceptions. Since the hold-out and the extension had already been used for (eq:main) and (eq:twist), they are a confirmation and not a blind test for Conjecture conj:euler; the out-of-table curves, generated before the rule was formulated, are." Problem: (a) the out-of-table set contains no curve with image 3Ns (Section sec:oot: "none has image 3Ns"), so the correction term, and with it the claim that the 3Ns failures are repaired exactly, has had no blind test on any range; only $h_3 = 1$ itself is a theorem (Lemma lem:h0 with point 5); likewise the III-free form was chosen after the exceptions were seen. (b) "Generated before the rule was formulated" makes a blind test only if the modular degrees of those curves were not examined while the rule (the Euler terms, the weights w_3, the terms at 2 and 3) was being formulated; the paper does not say, and Section sec:oot suggests the families were built to probe the additive types. (c) For (eq:twist), "then tested once on the hold-out" is a blind test only if (eq:twist) was fixed before the hold-out counts for (eq:main) were seen; the order is not stated. Fix: one sentence: "The out-of-table curves contain no 3Ns curve, so the correction term has been confirmed but not blind-tested; their modular degrees were [were not] examined before the rule was fixed", and say when (eq:twist) was fixed relative to the hold-out test of (eq:main). Confidence: high for (a) (from the paper's own words); (b) and (c) are requests for information, not errors.

N3. The appendix: `rule(E, 1)` must be run on the optimal model, and curve number 1 is not always optimal. Location: "A single curve can be checked against (eq:rule) in PARI/GP ... with the following functions, which return $[v_\ell(\deg \varphi_E), \text{bound}]$". Problem: `ellmoddegree(E)` returns the degree of the given model's parametrisation divided by the square of its Manin constant, so on a non-optimal curve the first entry is not $v_\ell(\deg \varphi_E)$: `rule` on 11a3 at $\ell = 5$ returns `[-1, 0]`, on 990h1 at $\ell = 3$ returns `[4, 3]` against `[3, 3]` for the optimal 990h3 (Cremona's curve number 1 is optimal in every class below 10,000 except 990h). Fix: "for the $X_0(N)$-optimal curve E (Cremona's curve number 1 except in a few classes such as 990h, where it is 990h3; `ellweilcurve` identifies it)". Confidence: high (PARI 2.17.2; the 2.15 manual should be checked for the same normalisation of `ellmoddegree`).

N4 (optional; Section sec:sharp, the calibration). Location: "The zero counts are not tautological. ... would produce roughly 605,000 violations ...; there are none." Observation: (a) on the working set the rule was fitted (the weights w_3 are observed minima, the terms at 2 and 3 were fixed after the minima were seen), so the working-set zero count shows that the statement is strong, not that it is valid; validity rests on the blind tests (N2). (b) The expected count depends on the reference population: below conductor 10,000 the paper's recipe (bound-0 curves) gives 14,275 expected violations at $\ell = 3$, a stratified version 14,177, and a permutation of v_3 over all curves within conductor strata about 7,600; all against 0 observed.

The paper's conclusion stands under every version; a sentence saying that the bound-0 population is the reference, and that its v_3-distribution differs sharply from the pooled one (59.7% against 23.9% with $v_3 = 0$ below 10,000), would make the argument complete. Confidence: high for the numbers; importance low.

N5 (low; the appendix, memory). "`ellmoddegree` needs `parisizemax` of a few hundred megabytes for conductors in the thousands": with PARI 2.17.2 the default 8 MB stack suffices for most curves below 10,000 and 16 to 32 MB for the largest degrees (9510e1, $\deg \varphi_E = 100{,}254{,}000$); the statement may be right for 2.15. Fix: "up to a few hundred megabytes, depending on the version". Confidence: high for 2.17.2; not checked for 2.15.

N6 (low; Sections sec:anomaly and 6.2, Remark rem:ns: which inequality the 3Ns counts refer to). Location: "Table tab:ns records what happens to the inequality (eq:thm), with all types included and no correction"; Table tab:uniform's "violations when the correction term is dropped". Problem: for curves with $\ell \mid N$, (eq:thm) as stated has no term at $q = \ell$, while (eq:rule) without h_ℓ has $t_\ell(E, \ell)$ and $w_3(E)$; about half of the 3Ns curves have $3 \mid N$ (44 of 90 below 10,000), and the two readings give different counts (36 against 24 violators below 10,000; 726a1 violates the first and not the second). Fix: say "(eq:rule) without the correction term" in Section sec:anomaly and in Remark rem:ns, if that is what was counted. Confidence: high that the two readings differ; which one the paper used cannot be decided from the text.

For the record (not new): the runs confirm the paper's claims on `rule` for 121a1 and 726a1, Lemma lem:euler on every optimal curve below 10,000, Lemma lem:twist (i), (ii) and the exchanges at $q = 3$ (point 51), the factor $1 + 2X$ at type IV, IV* at 2 (point 43), the 3Ns observations (point 48), strict twist inequalities in case (a) (point 16), the w_3 minima, the absence of any violation of (eq:rule), (eq:main), (eq:twist) and Conjecture conj:eis, and the Tamagawa refinement, all below conductor 10,000. Unverified sentence: "Cremona's tables ... contain ... its Kodaira symbols, Tamagawa numbers": the six standard files checked contain only $\prod_p c_p$; per-prime data may be in other files of the repository (the writer checks; not reported as an error).

24.8.2 8.2 Book-internal inconsistencies

- **B-a. File name `alldegphi`.** PLAN.md (Chapter 22 scope), the Chapter 11 outline (lines 13, 957) and the Chapter 21 outline (A6, 8.2) write `alldegphi`; the `ecdata` file is `degphi.00000-09999` (Part 7, R0). Writers must use `degphi`. (Fixed in PLAN.md and in the Chapter 11 outline; the Chapter 21 outline is left to its reviser.)
- **B-b. Statistics references live in `bib/chA.bib`.** `Feller68`, `Wasserman04`, `RHB06` are defined in `bib/chA.bib`, but Chapter 22 must cite them; by decision B14 they move to `refs.bib` (the stand-alone build of Chapter 22 reads only `refs.bib` and `bib/ch22.bib` and would fail).
- **B-c. PLAN.md, Chapter 22 scope, on PARI.** (1) "extract the Tate parameter ... via `ellinit(E, q)`": `ellinit(E, q)` with an integer prime reduces modulo q (returns `[]` at a bad prime); the q-adic curve is `ellinit(E, O(q^k))` and the parameter `.tate[3]`. (2) "a GP script that loops over curves ... from `ellsearch`": `ellsearch` needs the optional `elldata` package, absent here; the chapter reads `ecdata` files instead. (Both are now corrected in PLAN.md's Chapter 22 entry.)
- **B-d. Appendix A (written) and the Chapter 21 outline on blind tests.** Appendix A's paperlink `pl:prereq:protocol` and the Chapter 21 outline (A6.1: "blind test for (eq:main), (eq:twist), Conjecture conj:eis") present the (eq:twist) test as blind without the qualification of N2(c), and do not say that the correction term was never blind-tested (N2(a)). If N2 is accepted, both should add one clause.
- **B-e. Chapter 21 outline, A6.10.** Repeats "all rank 0, torsion $\mathbb{Z}/3$" for the seventeen 3Cs.1.1 curves; adopt N1.
- **B-f. Units of counts.** The Chapter 11 outline counts all curves below 10,000 (120 non-CM 3Ns curves, 262 non-CM 3Cs.1.1), the Chapter 21 outline and this one count optimal curves (90 3Ns, 49 3Cs.1.1). Both are right; every writer must say which.
- **B-g. Chapter 14 outline, 12.5,** repeats the paper's memory claim for `ellmoddegree`; see N5.

- No conflict found with Chapters 9, 10, 17, 18 and 19 on any number: the Tate parameters of 77c1, 539c1, 791a1, 1577a1, 11a1, the outputs of `rule` for 121a1 and 726a1, the P_q of 336f1, 1232l1, 592c1, the modular degrees and the Kodaira data of the running examples all agree. The Chapter 18 outline's Tamagawa test (1,050 curves at $\ell = 3$) includes I_n^* primes (point 68), this one (989) follows the paper's restriction to I_n: not a conflict.

Chapter 25

Appendix B: Notation, conventions and index of symbols (ch:notation, chapters/chB_notation.tex)

Stage 1 output for Appendix B. Target length of the LaTeX appendix: 400 to 800 lines (PLAN.md Section 7.3); the plan in Part 4 aims at about 750.

This file is the **specification of the book-wide conventions**. Part 1 fixes every convention that more than one chapter needs, each as a boxed statement (to be typeset as a `convention` environment) with its reason and the chapters it binds. Part 2 is the index of symbols. Part 3 lists every place where an outline (Chapters 1 to 19 and 21, as they stand on 14 October 2026) disagrees with Part 1, with the change needed. Part 4 is the section plan of the appendix and the notes for its writer. Part 5 records what could not be settled here.

Sources read: PLAN.md (in full, in particular the Appendix B entry and Section 7), the paper `note_Oct10_7pm.tex` (in full), `important/important_notes.md` (in full; the book-internal notes B1 to B23 and points 5, 6, 7, 11, 23, 26, 29, 30, 32, 37, 40, 42, 52, 70), and the outlines `outlines/ch01_groups.md` to `outlines/ch21_paper.md`: their global convention sections and writer notes in full, their item lists, and every line found by a systematic search for `Frob`, `\Phi`, `^*`, `^\vee`, `dagger`, `chi_`, `c^2`, `Petersson`, `\omega`, `A_f`, `\varphi`, `\phi`, `\pi`, `h^{(p)}`, `n+4` (the search scripts and their raw output were kept in the session scratchpad; the counts are in Part 3). Chapters 20 (`ch:congruence`) and 22 (`ch:computation`) are being outlined at the same time; their symbols are taken from PLAN.md and the paper and cited by chapter label only.

How items are cited. "ch:local 5.4" is item 5.4 of the Chapter 7 outline; Chapters 5 and 16 name items by label stems ("ch:galois galois:cyclotomic-character", "ch:cohomology model-setting"); Chapter 21 by its A-numbers and C1 to C3; a convention of an outline by its own name ("ch:rings C5", "ch:local L3", "ch:reduction R4"). "B-C4" is convention 4 of Part 1 below. In the LaTeX appendix the item numbers are replaced by `\Cref{ch:slug}` only (PLAN.md 7.1).

Typography of this file: ASCII; LaTeX in `$...$` as in the other outlines; no dash ligatures of any kind; table rules with single hyphens.

25.1 Part 1. Book-wide conventions

Each convention has: the **boxed statement** (to be typeset verbatim, up to LaTeX markup, as a `convention` environment in the appendix, and quoted or cited by the chapters), the **reason**, the **chapters bound**, and the **changes** (pointers to Part 3). Where an outline already fixed the convention, the source is named; nothing below contradicts a decision recorded in important_notes.md.

25.1.1 B-C1. Letters for primes, and the collisions of q and ℓ

Convention B-C1. p, q, ℓ denote primes. ℓ is the coefficient prime (odd in every use of the paper); q is a bad prime of E ($q \mid N$); p is any prime. $\ell^* = (-1)^{(\ell-1)/2}\ell$ and $q^* = (-1)^{(q-1)/2}q$ for odd ℓ, q. The letter q also denotes: the size of a finite field $\mathbb{F}_q$ (Chapters 1, 2, 7, 9: when not prime, Chapter 9 writes $\mathfrak{q}$); the variable $q = e^{2\pi i z}$ of q-expansions (Chapters 12, 13, 14); the variable of the Tate curve, written $\mathfrak{q}$ (Chapter 10). The Tate parameter of E is always q_E, never u and never $\mathfrak{q}$ alone. A chapter that uses two of these meanings says so in its first convention box and never puts two of them in one display.

Reason: the paper uses q for bad primes and for q-expansions; Chapters 1, 2, 7, 10, 12 each found the collision independently (ch:groups C4, ch:rings C2, ch:local L1, ch:reduction R1, ch:modforms C1). Chapters bound: all. Changes: none (all outlines comply).

25.1.2 B-C2. Frobenius: arithmetic Frob, geometric Φ

Convention B-C2 (Frobenius). (a) Frob_p (or Frob when the prime is clear) is an **arithmetic** Frobenius at p: an element of a decomposition group $D_p \simeq G_{\mathbb{Q}_p}$ inducing $x \mapsto x^p$ on the residue field, well defined modulo I_p and up to conjugation. It acts on μ_n ($p \nmid n$) by $\zeta \mapsto \zeta^p$, so $\chi_\ell(\mathrm{Frob}_p) = p$ for $\ell \neq p$. For a finite field $\mathbb{F}_q$, $\mathrm{Frob}_q \in \mathrm{Gal}(\overline{\mathbb{F}}_q/\mathbb{F}_q)$ is $x \mapsto x^q$. (b) $\Phi_p = \mathrm{Frob}_p^{-1}$ (or Φ) is the **geometric** Frobenius. An eigenvalue λ of Frob is the eigenvalue λ^{-1} of Φ. Where component groups $\Phi_q(\cdot)$ (B-C11) occur in the same chapter (Chapters 10, 11, 21, 22), the geometric Frobenius at a named prime is written Frob_q^{-1}, never Φ_q; Φ without subscript is always the geometric Frobenius of a local field. $\mathrm{Frob}^{\mathrm{geom}}$, $\mathrm{Frob}^{\mathrm{arith}}$ and "Frob_p, the geometric one" are not used. (c) **Group theory and cohomology use** Frob: the tame relation, conjugation actions on $H^1(I_p, -)$, unramified cohomology $H^1_{\mathrm{ur}} = M^I/(\mathrm{Frob}-1)M^I$, the Tate matrices of Lemma lem:tamfactor, Frobenius on Kummer classes. This binds Chapters 4, 5, 6, 7, 11, 16, 17, 18 and the commentary of Chapter 21. (d) **L-factors use** Φ: for a representation M of $G_{\mathbb{Q}_p}$ on a finite-dimensional $\mathbb{Q}_\ell$-space ($p \neq \ell$),

$$L_p(M, s) = \det\bigl(1 - \Phi_p\, p^{-s} \;\big|\; M^{I_p}\bigr)^{-1},$$

so $L_p(\mathbb{Q}_\ell(1), s) = (1 - p^{-1-s})^{-1}$ and $L_p(M(n), s) = L_p(M, s+n)$. The paper's $L_p(A_f, s)$ (Section 1.1), the table of Section 1, the proof of Lemma lem:euler, Lemma lem:fpr and the sentence "$\det(1 - \mathrm{Frob}_p \mid (V^D)^{I_p}) = L_p(A_f, 1)^{-1}$" after Lemma lem:dfg21 are read with Φ; the proof of Lemma lem:tamfactor is read with Frob (important_notes.md point 11). DFG04 1.7.2 uses the geometric convention for L-factors. (e) **The tame relation.** For a topological generator t of the tame (or pro-ℓ tame) inertia quotient of $G_{\mathbb{Q}_q}$:

$$\mathrm{Frob}\; t\; \mathrm{Frob}^{-1} = t^q, \qquad \text{equivalently} \qquad \mathrm{Frob}^{-1}\, t\, \mathrm{Frob} = t^{q^{-1}}\ (q^{-1} \in \mathbb{Z}_\ell), \qquad \Phi\, t\, \Phi^{-1} = t^{q^{-1}}.$$

The paper's "$\mathrm{Frob}^{-1}\, t\, \mathrm{Frob} = t^{q^{-1}}$" is the second form, with the arithmetic Frobenius. (f) **The Weil-Deligne relation.** With $\|w\| = q^{d(w)}$, $w|_{K^{\mathrm{ur}}} = \mathrm{Frob}^{d(w)}$ (so $\|\mathrm{Frob}\| = q$, $\|\Phi\| = q^{-1}$, $\|w\| = \chi_\ell(w)$ for $\ell \neq p$), a Weil-Deligne representation (r, N) satisfies

$$r(w)\, N\, r(w)^{-1} = \|w\|\, N, \quad \text{that is} \quad r(\mathrm{Frob})\, N\, r(\mathrm{Frob})^{-1} = q\, N, \quad r(\Phi)\, N\, r(\Phi)^{-1} = q^{-1} N.$$

(PLAN.md's "$\mathrm{Frob}\, N\, \mathrm{Frob}^{-1} = p^{-1}N$" is the geometric form with the wrong name, important_notes.md B6.) (g) **Frobenius endomorphisms and the Artin map.** The q-power Frobenius morphism of a curve or abelian variety over $\mathbb{F}_q$ is π_q (or π); it acts on $\overline{\mathbb{F}}_q$-points as Frob_q and on T_ℓ as Frob_q; its dual isogeny (Verschiebung) is $\hat{\pi}_q = q\pi_q^{-1}$. The Eichler-Shimura relation reads $(T_p)_* = \pi_p + \hat{\pi}_p$ on $J_0(N)_{\mathbb{F}_p}$. The local Artin map sends a uniformiser to an arithmetic Frobenius (ch:ant 5.6, ch:local 6.5); sources that send it to the geometric one (Deligne) are flagged where cited.

Reason: (a) to (e) are the decisions of ch:galois C2, ch:ant 3.7, ch:local L3, L4, L6, 4.3, 5.4, 8.2, ch:galrep C3, ch:cohomology (opening conventions and writer note 2), ch:galcoh C2, ch:blochkato C3, ch:lfunctions C2, C3, and important_notes.md points 11, B6. The only new decisions are the clash rule in (b) (the paper's Φ_q is the component group, B-C11) and the notation π_q, $\hat{\pi}_q$ in (g), which removes $V = pF^{-1}$ (clash with $V = V_\ell E$) from Chapter 14 and ϕ_q (clash with ϕ_f, φ_E) from Chapters 8 and 9. Chapters bound: 4 to 11, 14 to 19, 21, 22. Changes: Part 3, entries F1 to F9, and P1, P2, P5 (PLAN.md).

25.1.3 B-C3. Which Frobenius has the eigenvalues α, β

Convention B-C3. Let $E/\mathbb{Q}$ have good reduction at $p \neq \ell$, $V = V_\ell E$, $a_p = p + 1 - \#\tilde{E}(\mathbb{F}_p)$, and let α, β be the roots of $X^2 - a_p X + p$ (so $\alpha + \beta = a_p$, $\alpha\beta = p$, $|\alpha| = |\beta| = \sqrt{p}$). (a) On $V = V_\ell E$ the **arithmetic** Frobenius has characteristic polynomial $X^2 - a_p X + p$, eigenvalues α, β; the geometric one has $\alpha^{-1} = \beta/p$, $\beta^{-1} = \alpha/p$ (a different set). (b) On $V^\vee = H^1(E_{\overline{\mathbb{Q}}}, \mathbb{Q}_\ell) \simeq V(-1)$ the **geometric** Frobenius has eigenvalues α, β (and trace a_p). (c) Hence $L_p(E, s) = (1 - a_p p^{-s} + p^{1-2s})^{-1} = L_p(V^\vee, s)$, while $L_p(V, s) = L_p(E, s+1)$. The same holds for $\rho_{f,\lambda}$ normalised by $\det(X - \rho_{f,\lambda}(\mathrm{Frob}_p)) = X^2 - a_p(f)X + p$: $L_p(f, s) = L_p(V_\lambda(f), s-1)$. (d) On $\mathrm{ad}^0 V$ both Frobenii have the eigenvalues $\alpha/\beta, 1, \beta/\alpha$ (an inversion-stable set), so the good factor $L_p(A_f, s)^{-1} = (1 - \frac{\alpha}{\beta}p^{-s})(1 - p^{-s})(1 - \frac{\beta}{\alpha}p^{-s})$ does not depend on the convention; neither do the lines on which Frobenius acts by $\chi_{-3}(q)$ or $\chi_{-4}(q) = \pm 1$. It does at a multiplicative prime: on $(\mathrm{ad}^0 V)^{I_q}$ the arithmetic Frobenius acts by q, the geometric one by q^{-1}, and only the latter gives the paper's $1 - q^{-1-s}$. (e) In Lemma lem:old the paper's α_p, β_p are the roots of $X^2 - a_p(h)X + p$ for the newform h; they are the U_p eigenvalues of the p-stabilisations $h - \beta_p h(pz)$, $h - \alpha_p h(pz)$ (ch:hecke 7.1).

Reason: important_notes.md B1, B7, B9; ch:ellcurves C5, 5.7; ch:local 8.2(v); ch:galrep C3; ch:modgalrep C2; ch:blochkato C3; ch:lfunctions C2; ch:curves C7. Chapters bound: 6, 7, 8, 9, 11, 13, 14, 15, 18, 19, 21. Changes: Part 3, F1, F2 (Chapter 21, convention C2(b)), F5.

25.1.4 B-C4. Duals, adjoints, pullbacks

Convention B-C4 (one symbol for each kind of dual). (a) **Linear dual** $M^\vee = \mathrm{Hom}_R(M, R)$ ($R = \mathbb{Z}$, $\mathbb{Z}_\ell$, $\mathcal{O}$, $\mathbb{Q}_\ell$, a field); for a representation, the contragredient $V^\vee$ with $(g\lambda)(v) = \lambda(g^{-1}v)$. Never M^*, never M'. (b) **Pontryagin dual** $X^* = \mathrm{Hom}_{\mathrm{cont}}(X, \mathbb{Q}_\ell/\mathbb{Z}_\ell)$ (or $\mathrm{Hom}(X, \mathbb{Q}/\mathbb{Z})$ for a finite group), for X finite, compact (finitely generated over $\mathbb{Z}_\ell$) or discrete ℓ-primary. This is the paper's * in Lemma lem:dfg21. (c) **Kummer (Cartier) dual** M^D: for a lattice or vector space, $M^D = \mathrm{Hom}(M, \mathbb{Z}_\ell(1)) = M^\vee(1)$ (resp. $\mathrm{Hom}(V, \mathbb{Q}_\ell(1))$); for a finite or discrete ℓ-primary module, $M^D = \mathrm{Hom}(M, \mu_{\ell^\infty}) = M^*(1)$. Hence $V^D = V^\vee(1)$ and $W^D = T^*(1)$ for $W = T \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$. In the paper (Section 4.2) and in Chapters 18, 20, 21 the symbols T^D, V^D, W^D are built from the **adjoint** lattice $A = \mathrm{ad}^0 T$: $T^D = \mathrm{Hom}(A, \mathbb{Z}_\ell(1)) \simeq A(1)$, not $\mathrm{Hom}(T_\ell E, \mathbb{Z}_\ell(1))$ (index, entry T^D). (d) **Adjoint of an operator** A for a pairing $\langle\ ,\ \rangle$: $A^\dagger$, defined by $\langle Ax, y\rangle = \langle x, A^\dagger y\rangle$ (Petersson product, cup product, abstract perfect pairings alike). Never A^*. (e) **Pullback and pushforward** along a morphism or correspondence: φ^*, φ_*, $\mathcal{C}^*$, $\mathcal{C}_*$, and for $t \in \mathbb{T}$ the two actions t^* (on H^1, S_2, $H^0(\Omega^1)$) and t_* (on H_1, $J_0(N)$, $T_\ell J_0(N)$) of ch:moddeg C2. A star after the name of a **map** is a pullback; a star after the name of a **module** is a Pontryagin dual. Every other star is one of the fixed exceptions listed in the index (Kodaira symbols II* etc., q^*, ℓ^*, $\mathfrak{H}^*$, E_2^*, the dual Selmer structure $\mathcal{L}^*$ of ch:galcoh 8.2). (f) **Transpose** of a matrix: A^T (ch:modules C8) or A^t; of a correspondence: $\mathcal{C}^t$ (ch:curves C4). **Dual isogeny**: $\hat{\varphi}$; dual abelian variety: $A^\vee$ (so $E^\vee = E$ via the principal polarisation).

Reason (important_notes.md B4, B22). The current usage:

outline	linear dual	Pontryagin dual	Kummer dual	adjoint of an operator	pullback
ch:modules (3)	$M^\vee$ (C3)	M^* (8.1, 8.2)	T^D (C3)	A^* (9.5)	none
ch:reps (4)	$V^\vee$ (C2)	reserved *	none	none	none
ch:curves (8)	$M^\vee$ (C5)	none	none	$A^\dagger$ (C4)	φ^* (C4)
ch:ellcurves (9)	$V_\ell E^\vee$	none	none	none	φ^*
ch:modforms (12)	none	none	none	$U_p^\dagger$	γ^*
ch:hecke (13)	none	none	none	$t^\dagger$ (C6)	none
ch:moddeg (14)	$M^\vee$ (C3)	in `paperlink` only	none	$A^\dagger$ (C3)	* (C3)
ch:modgalrep (15)	$\rho^\vee$ (C2)	none	none	T_n^* (8.1, line 495)	ϕ^*
ch:cohomology (16)	none	A^*	none	none	$(f,\varphi)^*$
ch:galcoh (17)	$T^\vee$ (C3)	A^* (C3)	$M^*(1)$, $T^D = T^\vee(1)$ (C3)	none	none
ch:blochkato (18)	$M^\vee$ (C2)	X^* (C2)	$M^D = M^\vee(1)$ (C2)	none	φ_E^*
ch:lfunctions (19)	$V_\ell E^\vee$	none	none	none	none
ch:paper (21)	`V_l E^*`, `T^*` (lines 678, 680, 1894)	`^*`	`T^D`	`T_n^*` (line 518)	none
PLAN.md	V^* (Chapters 4, 14, 18 entries)	*	$V^D = V^*(1)$	none	φ^*, Frob_p^*

Every outline but three already writes $^\vee$ for linear duals and * for Pontryagin duals; four use † for adjoints against three that use *; pullbacks by * are universal and standard (the Manin constant, the modular parametrisation, Hecke correspondences) and cannot be renamed without rewriting Chapters 8, 9, 12, 14. So the choice above (linear $^\vee$, Pontryagin *, Kummer ^{D}, adjoint †, pullback * on maps) costs: Chapter 3 (adjoints in 9.5 and its writer note; $V^* \otimes W$ in 5. Dependencies), Chapter 15 (one line), Chapter 21 (four lines), PLAN.md (three entries). Any other choice costs more: adjoints by * would change Chapters 8, 12, 13, 14; linear duals by * would change eleven outlines. Changes: Part 3, D1 to D9.

25.1.5 B-C5. Tate twists, Hodge-Tate weights and other signs

Convention B-C5. (a) $\mathbb{Z}_\ell(1) = T_\ell\mu = \varprojlim_k \mu_{\ell^k}$ (transition maps $\zeta \mapsto \zeta^\ell$), on which G_K acts through χ_ℓ; $\mathbb{Z}_\ell(n) = \mathbb{Z}_\ell(1)^{\otimes n}$, $\mathbb{Z}_\ell(-1) = \mathrm{Hom}(\mathbb{Z}_\ell(1), \mathbb{Z}_\ell)$, and $M(n) = M \otimes_{\mathbb{Z}_\ell} \mathbb{Z}_\ell(n)$ for every $n \in \mathbb{Z}$; $\mathbb{Q}_\ell/\mathbb{Z}_\ell(1) = \mu_{\ell^\infty}$. So Frob_p acts on $\mathbb{Z}_\ell(1)$ by p, Φ_p by p^{-1}, and $L_p(M(n), s) = L_p(M, s+n)$. (b) The Weil pairing (Silverman's e_m, ch:ellcurves C7) gives $\wedge^2 T_\ell E \simeq \mathbb{Z}_\ell(1)$, $\det \rho_{E,\ell} = \chi_\ell$, $T_\ell E \simeq (T_\ell E)^\vee(1)$, and $H^1(E, \mathbb{Z}_\ell) = (T_\ell E)^\vee \simeq T_\ell E(-1)$; for a curve X, $H^1_{\mathrm{et}}(X, \mathbb{Z}_\ell) \simeq T_\ell J(X)(-1)$ (ch:curves 11.8). (c) **Hodge-Tate weights**: $\mathbb{Q}_\ell(1)$ has Hodge-Tate weight $+1$; so $V_\ell E$ has weights $\{0,1\}$, $\mathrm{ad}^0 V$ has $\{-1,0,1\}$, $V^D = \mathrm{ad}^0 V(1)$ has $\{0,1,2\}$, $V_\ell E^\vee$ has $\{-1,0\}$ (ch:blochkato C4). Sources with the opposite sign are flagged where cited. (d) Other fixed signs: the cup product $\langle \alpha, \beta \rangle = \int_X \alpha \wedge \beta$ with the complex orientation (ch:curves C6; the paper needs only sign-invariant properties); the Kummer map $\delta(a)(\sigma) = \sigma(\alpha)/\alpha$, $\alpha^n = a$ (ch:galcoh C6); the conjugation action $(\sigma \cdot c)(h) = \sigma\, c(\sigma^{-1} h \sigma)$ of G/H on $H^n(H, M)$, verbatim the paper's (ch:cohomology writer note 1); the slash operator

$(f|_k\gamma)(z) = (\det\gamma)^{k/2}(cz+d)^{-k}f(\gamma z)$ (ch:modforms C3) and $W_N = \left(\begin{smallmatrix} 0 & -1 \\ N & 0 \end{smallmatrix}\right)$ (ch:modforms C4).

Reason: ch:galois C6, ch:galcoh C3, ch:blochkato C2, C4, ch:curves 11.8, ch:ellcurves 5.5, ch:local 8.2. Chapters bound: 5, 7, 8, 9, 11, 12, 14, 15, 16, 17, 18, 19, 21. Changes: none in the outlines (all comply); the writers of Chapters 18 and 20 must check DFG04's and Bloch-Kato's sign of the Hodge-Tate weight before quoting a dimension formula.

25.1.6 B-C6. Characters: χ_ℓ, χ_d, χ_{q^*}

Convention B-C6 (characters). (a) $\chi_\ell\colon G_{\mathbb{Q}} \to \mathbb{Z}_\ell^\times$ is the ℓ-adic **cyclotomic** character, $\sigma(\zeta) = \zeta^{\chi_\ell(\sigma)}$ on μ_{ℓ^∞}; $\bar\chi_\ell\colon G_{\mathbb{Q}} \to \mathbb{F}_\ell^\times$ its reduction; $\omega_\ell\colon a \mapsto a \bmod \ell$ the Dirichlet character that corresponds to $\bar\chi_\ell$ (ch:rings C5). A letter subscript (χ_ℓ, χ_p) always means the cyclotomic character. At a numerical prime the cyclotomic character is written χ_3^{cyc}, χ_5^{cyc}; its reduction $\bar\chi_3$ (with the bar) is unambiguous, quadratic characters never carrying a bar. (b) For a non-square $d \in \mathbb{Q}^\times$, χ_d is the **quadratic** character of $\mathbb{Q}(\sqrt{d})$, $\sigma(\sqrt{d}) = \chi_d(\sigma)\sqrt{d}$; as a Dirichlet character it is the Kronecker symbol $\left(\frac{D}{\cdot}\right)$, D the discriminant of $\mathbb{Q}(\sqrt{d})$, of conductor $|D|$. A **numerical** subscript always means χ_d with d written with its sign: χ_{-3}, χ_{-4}, χ_{-7}, χ_{-11}, χ_5, χ_{13}. The paper's χ_{11} is written χ_{-11}; the symbols χ_3, χ_7, χ_{11} are not used. (c) For an odd prime q, the quadratic character of conductor q is χ_{q^*}, the character of $\mathbb{Q}(\sqrt{q^*})$ (χ_d with $d = q^*$ the discriminant), **never** χ_q. The paper's χ_q appears only when the paper is quoted (B.5 and Part 4, item 4), followed by the book's χ_{q^*}. A letter subscript χ_ℓ is reserved for the ℓ-adic cyclotomic character, and χ_3^{cyc} is written where a number is needed (PLAN.md 7.5, decision D3, which replaces the earlier allowance for the paper's χ_q). Never write χ_ℓ for a quadratic character: $\chi_{\ell^*} = \bar\chi_\ell^{(\ell-1)/2}$ is the quadratic character of $\mathbb{Q}(\sqrt{\ell^*})$, and $\chi_{\ell^*} = \bar\chi_\ell$ iff $\ell = 3$, where $\bar\chi_3 = \chi_{-3}$ (important_notes.md points 5, 6, 32). (d) $\chi_S = \prod_{q\in S}\chi_{q^*}$ for a finite set S of odd primes, of conductor $D = \prod_{q\in S} q$. ε is an unramified character of $G_{\mathbb{Q}_q}$ of order at most 2 (the Tate data, B-C14), with $\varepsilon(\mathrm{Frob}_q) = a_q(E)$ at a multiplicative prime. ψ is the character of order 4 of Proposition prop:ns, with values in $\mathbb{F}_9^\times$. (e) A Dirichlet character χ is also the Galois character with $\chi(\mathrm{Frob}_p) = \chi(p)$ for $p \nmid \operatorname{cond}\chi$ (arithmetic Frobenius; ch:ant 5.7, ch:modgalrep C3).

Reason: the paper names one character χ_{11} and χ_{-11} and uses χ_3 both ways (point 32). Rule (b) and (c) together name every quadratic character by its discriminant (decision D3, PLAN.md 7.5): Chapters 2, 13, 18, 19 already wrote χ_{q^*}, and the paper's χ_q in Chapters 5, 7, 9, 10, 11, 15, 21 (and elsewhere) has been converted to χ_{q^*}. Only seven lines use a numerical χ_3 in the book's own text. Chapters bound: 2, 4, 5, 6, 7, 9, 10, 11, 13, 15, 17, 18, 19, 21, 22. Changes: Part 3, X1 to X5.

25.1.7 B-C7. Normalisations of the adjoint and symmetric square L-functions

Convention B-C7. Let f be a newform of weight 2 on $\Gamma_0(N)$, $V = V_\lambda(f)$ (for $f = f_E$, $V = V_\ell E$), $A_f = \mathrm{ad}^0 V$.

$$L(\mathrm{ad}^0 f, s) = L(A_f, s) = \prod_p L_p(A_f, s), \qquad L(\mathrm{Sym}^2 f, s) = \prod_p L_p(\mathrm{Sym}^2 V^\vee, s) = L(\mathrm{ad}^0 f, s-1),$$

with L_p as in B-C2(d) ($p \neq \ell$; at $p = \ell$ through D_{cris}, or with $V_{\ell'}$ for $\ell' \neq p$). So $L_p(\mathrm{Sym}^2 f, s) = L_p(A_f, s-1)$, and the special values are $L(\mathrm{Sym}^2 f, 2) = L(\mathrm{ad}^0 f, 1)$. $P_q(X) \in \mathbb{Z}[X]$ is defined by $L_q(\mathrm{Sym}^2 f, s) = P_q(q^{-s})^{-1}$; hence $P_q(q^{-2}) = L_q(A_f, 1)^{-1}$. The **naive** factor is $L_p^{\mathrm{nv}}(A_f, s) = L_p(A_f, s)$ if $\delta_p = \dim V^{I_p} > 0$ and 1 if $\delta_p = 0$ (equivalently $L_p^{\mathrm{nv}}(A_f, s)^{-1} = 1 - a_p(f)^2 p^{-1-s}$ for $p \mid N$); L^{nv} is their product. The normalisations of the sources:

object	source	equals	functional equation	point used
$L(\mathrm{ad}^0 f, s)$, $L(\mathrm{ad}^0 M_f, s)$	the paper; DFG04 1.7.2	$L(A_f, s)$	$s \leftrightarrow 1-s$	$s=1$
$L(s, \pi_f, \mathrm{Ad})$	Gelbart-Jacquet (unitary)	$L(A_f, s)$ (weight k: $L(\mathrm{Sym}^2, s+k-1)$)	$s \leftrightarrow 1-s$	$s=1$
$L(\mathrm{Sym}^2 f, s)$	the paper; PARI `lfunsympow(E,2)`; Watkins02	$L(A_f, s-1)$	$s \leftrightarrow 3-s$	$s=2$
Shimura's series $\zeta^{(N)}(2s-2)\sum a_{n^2} n^{-s}$	Shimura75	$L^{\mathrm{nv}}(A_f, s-1)$	via the above	
Rankin series $D(f,s) = \sum a_n^2 n^{-s}$	Rankin39	$\zeta^{(N)}(s-1)L^{\mathrm{nv}}(\mathrm{Sym}^2 f, s)/\zeta^{(N)}(2s-2)$	none	

The Rankin series is never called $L(\mathrm{Sym}^2 f, s)$ (important_notes.md points 30, 70). $e_\ell(E,q) = v_\ell\big(L_q(A_f,1)^{-1}/L_q^{\mathrm{nv}}(A_f,1)^{-1}\big) = v_\ell\big(P_q(q^{-2})/(1-a_q^2 q^{-2})\big)$ (Definition def:terms).

Reason: ch:lfunctions C4, C5, C6, 7.4 (whose table this is, with the rows marked "verify" there still to be checked against DFG04 and Watkins02); ch:reps 7.5(iii); ch:local 9.5; ch:blochkato 14.1; ch:paper C2(c). All outlines agree. Chapters bound: 4, 7, 13, 15, 18, 19, 20, 21, 22. Changes: none in the outlines; PLAN.md Chapter 19 entry already fixed (B23).

25.1.8 B-C8. The Petersson product

Convention B-C8. For $f, g \in S_k(\Gamma_0(N))$,

$$\langle f, g\rangle = \int_{\Gamma_0(N)\backslash\mathfrak{H}} f(z)\overline{g(z)}\, y^k\, \frac{dx\,dy}{y^2},$$

unnormalised (not divided by the index or the volume), linear in the first variable; in weight 2, $\langle f,g\rangle = \frac{i}{2}\int_{X_0(N)} \omega_f \wedge \overline{\omega_g}$ with $\omega_f = f(z)\,dz$. PARI's `mfpetersson` returns $\langle f,g\rangle/\mu(N)$, $\mu(N) = [\mathrm{SL}_2(\mathbb{Z}) : \Gamma_0(N)] = N\prod_{p\mid N}(1+p^{-1})$; other books divide by $\mathrm{vol}(\Gamma_0(N)\backslash\mathfrak{H}) = \pi\mu(N)/3$. Never quote a number without its normalisation.

Reason: ch:modforms C5 (checked numerically on five curves, 10.4), ch:hecke C6, ch:moddeg C4, ch:lfunctions C9; important_notes.md B10. Chapters bound: 12, 13, 14, 19, 21, 22. Changes: none (all comply).

25.1.9 B-C9. The Manin constant and the formula of Zagier and Watkins

Convention B-C9. For a modular parametrisation $\varphi\colon X_0(N) \to E$ (sending ∞ to O) and the Neron differential ω_E of a global minimal model, $\varphi^*\omega_E = c_\varphi \cdot 2\pi i\, f(z)\,dz = c_\varphi\, f(q)\,dq/q$; the **Manin constant** of the optimal curve is $c_E = c_{\varphi_E}$. The sign of ω_E is chosen so that $c_E > 0$; Manin's conjecture reads $c_E = 1$. Then, with the product of B-C8,

$$\deg\varphi \cdot \mathrm{vol}(\Lambda_E) = 4\pi^2 c_\varphi^2\,\langle f, f\rangle, \qquad \mathrm{vol}(\Lambda_E) = \tfrac{i}{2}\int_{E(\mathbb{C})} \omega_E \wedge \bar{\omega}_E,$$

c^2 **in the numerator** of $\deg\varphi_E = 4\pi^2 c_E^2\langle f,f\rangle/\mathrm{vol}(\Lambda_E)$ (important_notes.md B10; check: the degree-5 parametrisation of 11a3 has $c_\varphi = 5$). The Manin constant is never written as a bare c in the book's own text (c alone is taken by the paper's class in Lemma lem:tamfactor, B-C14, and c_q by Tamagawa numbers).

Reason: ch:moddeg C4, 7.1, 7.3; ch:modforms 10.1, B1; ch:lfunctions 8.4. The sign normalisation $c_E > 0$ is new: Chapter 14 writes "c_E well defined up to sign" and Chapters 14 and 19 write "$c_E = \pm 1$"; only c_E^2 enters any formula, so nothing changes but the wording. Chapters bound: 9, 12, 14, 18, 19, 21, 22. Changes: Part 3, M1 to M5.

25.1.10 B-C10. Valuations, lengths, orders, Fitting ideals

Convention B-C10. v_ℓ is the ℓ-adic valuation on $\mathbb{Q}$ (and on $\mathbb{Q}_\ell$), normalised by $v_\ell(\ell) = 1$, with $v_\ell(0) = +\infty$; for a nonzero ideal $\ell^a\mathbb{Z}_\ell$, $v_\ell(\ell^a\mathbb{Z}_\ell) = a$; for a DVR $\mathcal{O}$ with uniformiser ϖ, $v(\varpi) = 1$, and the paper's v_λ for $\lambda = \ell$ is v_ℓ. For a class $c \in \mathbb{Z}/\ell^m$, $v_\ell(c) := \min(v_\ell(\tilde{c}), m)$ (so the paper's "$v_\ell(0)$" in Lemma lem:tamfactor reads $v_\ell(q-1)$). "**Length**" without subscript means $\mathbb{Z}_\ell$-length, length $=$ length$_{\mathbb{Z}_\ell}$, and length$(\mathbb{Z}_\ell/\ell^a) = a$; for finite $\mathbb{Z}_\ell$-modules $\#X = \ell^{\text{length}\,X}$. Orders $\#\mathcal{O}/\Delta$ are used only for $\mathcal{O}$ with finite residue field (important_notes.md point 38); otherwise lengths. $\text{Fitt}_{\mathbb{Z}_\ell}(X) = \ell^{\text{length}\,X}\mathbb{Z}_\ell$ for finite X. $\text{Tam}_p(M) = \text{Fitt}_{\mathbb{Z}_\ell}\big((H^1(I_p, M)_{\text{tor}})^{G_{\mathbb{Q}_p}}\big)$: the Fitting ideal **of the invariants** (important_notes.md point 37).

Reason: PLAN.md Appendix B entry; ch:rings C3; ch:modules C2, C6; ch:galcoh C4; ch:blochkato C5, C6. Chapters bound: all from 2 on. Changes: none in the outlines; PLAN.md Chapter 18 entry (the misplaced parenthesis, B22).

25.1.11 B-C11. Kodaira symbols, component groups, PARI codes

Convention B-C11. (a) Kodaira symbols are typeset $\text{I}_0, \text{I}_n, \text{II}, \text{III}, \text{IV}, \text{I}_0^*, \text{I}_n^*, \text{IV}^*, \text{III}^*, \text{II}^*$ (PLAN.md 7.1); "type I_n" assumes $n \geq 1$ unless I_0 is meant; at $q = 2$ the index m of a symbol I_m^* need not be $-v_2(j)$ and the reduction may be potentially good (ch:reduction 9.4). (b) Φ_q is the component group of the special fibre of the Neron model of E at q, a finite etale group scheme over $\mathbb{F}_q$; it is always written **with an argument**: $\Phi_q(\mathbb{F}_q)$, $\Phi_q(\overline{\mathbb{F}}_q)$. The **geometric component group** $\Phi_q(\overline{\mathbb{F}}_q)$ is Φ_q^{geom} (the paper's macro `\Phig`), and $\Phi_q^{\text{geom}}[\ell^k]$, $\#\Phi_q^{\text{geom}}$ are written with it; the **Tamagawa number** is $c_q = \#\Phi_q(\mathbb{F}_q) = [E(\mathbb{Q}_q) : E_0(\mathbb{Q}_q)]$, which divides $\#\Phi_q^{\text{geom}}$. A bare "Φ_q" for a group is not used. The group $\text{Gal}(L/K^{\text{ur}})$ measuring the semistability defect is G_{ss} (ch:reduction R6), never Φ. (c) PARI's Kodaira codes (`elllocalred(E,p)[2]`, the paper's `K[i][2]`): 1 for I_0; $n+4$ for I_n and $-(n+4)$ for I_n^* **for** $n \geq 1$ **only**; -1 for I_0^*; $2, 3, 4$ for $\text{II}, \text{III}, \text{IV}$ and $-2, -3, -4$ for $\text{II}^*, \text{III}^*, \text{IV}^*$ (important_notes.md point 52; the paper's appendix states "$n+4$" without $n \geq 1$). Codes appear only in tables, `lstlisting` and `paperlink` environments. (d) Conductor exponent $f_p = f_p(E)$, Swan conductor $\text{sw}_p(E)$; the letter δ_p is reserved for $\dim V^{I_p}$.

Reason: ch:reduction R3, R4, R5, R6, 4.2, 4.4, 5.5; ch:lfunctions C8; ch:paper A9.2; the clash of Φ_q with the geometric Frobenius (B-C2(b)) and with the semistability group. Chapters bound: 10, 11, 14, 19, 21, 22. Changes: Part 3, K1 (Chapter 11), K2 (Chapter 15), P4 (PLAN.md, the codes).

25.1.12 B-C12. The adjoint, the symmetric square, the paper's A, A_f, and the abelian variety J_f

Convention B-C12. (a) For a representation V (or lattice T), $\text{End}(V) = V \otimes V^\vee$ with $g \cdot x = \rho(g)x\rho(g)^{-1}$ is written $\text{End}(V)$ or $\text{Ad}\,V$; the **trace-zero adjoint** is $\text{ad}^0\,V$ (the paper's macro `\ad`, which already contains the superscript 0: never `\ad`); $\text{Sym}^2\,V$ (macro `\Sym`, exponent included) is the symmetric square. For $\dim V = 2$ the basis of $\text{ad}^0\,V$ is $e = e_1 \otimes e_2^\vee$, $h = e_1 \otimes e_1^\vee - e_2 \otimes e_2^\vee$, $f = e_2 \otimes e_1^\vee$ (ch:reps C3; the paper's Lemma lem:tamfactor). $\text{ad}^0\,V \simeq \text{Sym}^2\,V \otimes \det^{-1}$ and, for ℓ odd, the trace form makes $\text{ad}^0\,T$ self-dual, $\text{ad}^0\,T \simeq (\text{ad}^0\,T)^\vee$. (b) The paper's objects: $V = V_\ell E$, $T = T_\ell E$, $\rho = \rho_{E,\ell}$, $\bar{\rho}$; $A = \text{ad}^0\,T$, $A_f = \text{ad}^0\,V = A \otimes \mathbb{Q}_\ell$, $W = A \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$, $T^D = \text{Hom}(A, \mathbb{Z}_\ell(1)) \simeq A(1)$, $V^D = T^D \otimes \mathbb{Q}_\ell$, $W^D = V^D/T^D = \text{ad}^0\,T(1) \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$. ad^0 is insensitive to duals and twists: $\text{ad}^0(T^\vee) \simeq \text{ad}^0\,T = \text{ad}^0(T(n)) = \text{ad}^0(T \otimes \chi)$, so $\text{ad}^0\,M_{f,\lambda} = \text{ad}^0\,T_\ell E$ although $M_{f,\lambda} \simeq T_\ell E(-1)$ (points 26, 40). (c) A_f **is the adjoint, never**

an abelian variety. The quotient $J_0(N)/I_f J_0(N)$ attached to a newform f (Shimura's A_f) is written J_f, with the quotient map $\pi_f\colon J_0(N) \to J_f$; for rational f, $J_f = E_f$ is the optimal curve of the class.

Reason: the paper's A_f (Section 1.1) is the adjoint; PLAN.md's Chapter 14 entry and the Chapter 14 and 15 outlines use Shimura's A_f for the abelian variety, and Chapter 15 uses both meanings (its 2.1 and 3.7). Renaming the abelian variety touches two outlines; renaming the paper's adjoint is not allowed (PLAN.md 7.1: the paper's notation is the book's). Chapters bound: 4, 11, 14, 15, 18, 19, 20, 21. Changes: Part 3, A1 to A3, P5.

25.1.13 B-C13. Hecke algebras, eigencharacters, twists and depletion

Convention B-C13. (a) $\mathbb{T} = \mathbb{T}_N = \mathbb{Z}[T_n : n \geq 1] \subset \mathrm{End}_{\mathbb{C}} S_2(\Gamma_0(N))$ is the **full** Hecke algebra (the operators $U_p = T_p$ for $p \mid N$ included); $\mathbb{T}^{(N)} = \mathbb{Z}[T_n : (n, N) = 1]$ the anemic one; $\mathbb{T}_\ell = \mathbb{T} \otimes \mathbb{Z}_\ell$; for a maximal ideal $\mathfrak{m} \ni \ell$, $\mathbb{T}_\mathfrak{m} = (\mathbb{T} \otimes \mathbb{Z}_\ell)_\mathfrak{m}$, a direct factor of $\mathbb{T}_\ell$ cut out by an idempotent $e_\mathfrak{m}$ (ch:modules C7). (b) $\phi_f\colon \mathbb{T} \to \mathcal{O}_f$ (upright ϕ) is the **eigencharacter** of f, $T_n \mapsto a_n(f)$; a general system of eigenvalues is also written ϕ. φ (curly) is reserved for **maps of curves**: $\varphi_E\colon X_0(N) \to E$, isogenies φ. (c) The paper's two uses of I: in Lemma lem:crit $I = \ker \phi_f$ and $J = \mathrm{Ann}_\mathbb{T}(f^\perp)$; in Lemma lem:ddt417 $\wp = \ker \pi$ and $I = \mathrm{Ann}_\mathbb{T}(\wp)$. So the J of lem:crit is the I of lem:ddt417 (for $\pi = \phi_f$). The book keeps each lemma's letters inside that lemma and writes $\ker \phi_f$ (or $\wp$, or I_f in Chapter 14) and $\mathrm{Ann}_\mathbb{T}(\wp)$ in its own text. (d) Twists: for a Dirichlet character χ of conductor r, $f \otimes \chi = \sum \chi(n) a_n(f) q^n$ is the **naive twist** (the paper's $f \otimes \chi$ and the map ι of Theorem thm:twist); f^χ is the **newform twist** ($\rho_{f^\chi} \simeq \rho_f \otimes \chi$); then $f \otimes \chi$ is the r-depletion of f^χ (ch:modgalrep C4). $E^{(d)} = E \otimes \chi_d$ is the quadratic twist of a curve by $\mathbb{Q}(\sqrt{d})$; $E^{(q)} := E^{(q^*)}$ for odd q (the paper's); $E_S = E \otimes \chi_S$. (e) **Depletion**: $f^\Sigma = \sum_{(n,\Sigma)=1} a_n q^n$ for a set Σ of primes (the paper's notation), $f^{\{p\}}$ for one prime, f^S for the set S. A parenthesised superscript (p), (q), (d) is reserved for **twists** of curves ($E^{(q)}$) and for removing Euler factors ($\zeta^{(N)}$, $\mathbb{T}^{(N)}$); it never denotes depletion. (f) $W_N = \left(\begin{smallmatrix} 0 & -1 \\ N & 0 \end{smallmatrix}\right)$, w_N the involution of $X_0(N)$ and the operator $f \mapsto f|_k W_N$; $\varepsilon_N(f) = \pm 1$ its eigenvalue on a newform. $\hat{\delta}_!$ (always with the subscript) is the pairing $\langle x, w_N y\rangle$ on $H^1(X_0(N), \mathbb{Z}_\ell)$ (DFG04's $\hat{\delta}_!$ in weight 2, trivial character); $M_{f,\lambda} = H^1(X_0(N), \mathbb{Z}_\ell)[\ker \phi_f]$ (with $\lambda = \ell$ for rational f); $\eta^\Sigma_{f,\lambda}$ its ideal of values on $M_{f^\Sigma,\lambda}$.

Reason: ch:hecke C3 to C8, ch:modgalrep C1, C4, ch:moddeg C2, C3, ch:modules C7; the clash of $h^{(p)}$ with $E^{(q)}$; the clash of a curly φ for eigenvalue systems (ch:hecke C5) and of an upright ϕ for the modular parametrisation (ch:modgalrep 2.6) with the paper's ϕ_f and φ_E. Chapters bound: 3, 12, 13, 14, 15, 20, 21, 22. Changes: Part 3, H1 to H5.

25.1.14 B-C14. The Tate data at a prime with $v_q(j) < 0$

Convention B-C14. $q \neq \ell$; $q_E \in \mathbb{Q}_q^\times$ the Tate parameter (never u); $\nu = v_q(q_E) = -v_q(j_E)$, equal to $n = n_q = v_q(\Delta_E)$ at a multiplicative prime; $u = q_E q^{-\nu} \in \mathbb{Z}_q^\times$ its unit part, $\bar{u} \in \mathbb{F}_q^\times$ its reduction; $c(u)$ the class of $\bar{u}$ in $\mathbb{F}_q^\times/(\mathbb{F}_q^\times)^{\ell^{v_\ell(q-1)}}$ (the paper's c in Lemma lem:tamfactor and c_q in Section sec:tam), with $v_\ell(c(u))$ read as in B-C10. $\kappa = \kappa_{q_E}$ the Kummer cocycle of q_E, $\kappa = \nu\kappa_q + \kappa_u$ (ch:galcoh 5.3); t a topological generator of the pro-ℓ tame inertia quotient. In a basis e_1, e_2 of $T_\ell E$:

$$\rho(g) = \begin{pmatrix} \chi_\ell(g)\varepsilon(g) & \varepsilon(g)\kappa(g) \\ 0 & \varepsilon(g) \end{pmatrix},$$

with corner $\varepsilon\kappa$, not κ (important_notes.md point 42); $c' = \varepsilon(\mathrm{Frob})\kappa(\mathrm{Frob})$ for the arithmetic Frobenius, well defined modulo $(n, q-1)\mathbb{Z}_\ell$ (point 7 and its addendum).

Reason: ch:blochkato C6, ch:reduction R1, R4, 7.4, ch:galcoh 5.3, 5.4, ch:local 7.6, ch:cohomology cocycle-in-the-corner; important_notes.md points 7, 8, 23, 31, 42. Chapters bound: 2, 7, 10, 11, 16, 17, 18, 21, 22. Changes: Part 3, T1, T2 (Chapter 21).

25.1.15 B-C15. Other letters with several meanings (the rules)

Convention B-C15. (a) Uniformisers of local fields and DVRs are ϖ (ch:modules C2, ch:local L1); the paper's $\pi = 1 - \zeta_3$ (Lemma lem:IV) and $\pi\colon \mathbb{T} \to \mathcal{O}$ (Lemma lem:ddt417) keep their names inside those lemmas; π_f is the quotient map to J_f; π_q the Frobenius endomorphism (B-C2(g)). (b) $\zeta_n = e^{2\pi i/n}$; a primitive cube root of unity is always ζ_3, never ω or ρ. ω is a differential: ω_E (Neron or invariant differential), $\omega_f = f(z)\,dz$; ω_1, ω_2 are periods (Chapters 8, 9); the fundamental characters of Chapters 11 and 15 are ω (level 1, $= \bar{\chi}_\ell|_{I_\ell}$) and ω_2 (level 2), used only there and never in a chapter with periods; ω_ℓ is the Dirichlet character of B-C6(a). (c) ι is the twist map of Theorem thm:twist in Chapters 13, 15, 20, 21 and the map $H^1(K,T) \to H^1(K,V)$ in Chapters 17, 18, 21 (both are the paper's); a chapter using both writes ι_S for the twist map. (d) D: the conductor $\prod_{q\in S} q$ of χ_S (Theorem thm:twist); the deficit of Conjecture conj:eis; the discriminant of a quadratic field (Chapters 2, 5, 6, 9); the quaternion discriminant of a Shimura curve (Chapter 14). D_p is a decomposition group, D_{cris} Fontaine's functor, ^{D} the Kummer dual. Each chapter uses at most two of these and says which. (e) Σ is the paper's finite set of primes; the Shimura subgroup of $J_0(N)$ is $\mathcal{S}_N$ (not Σ_N); S is the set of primes of Theorem thm:quant, S_f DFG04's exceptional set, S a set of places in $G_{\mathbb{Q},S}$ (Chapter 17). (f) N is the conductor or level; the monodromy operator of a Weil-Deligne representation is N only in Chapters 7, 15 (local statements); N_s, N_{ns} are normalisers of Cartan subgroups; Norm is a norm map. (g) μ_n roots of unity; $\mu(N) = [\mathrm{SL}_2(\mathbb{Z}) : \Gamma_0(N)]$; μ an unramified character (Chapter 7) and the U_p-eigenvalue choice μ_p of Lemma lem:old are local names. (h) $\eta^{\Sigma}_{f,\lambda}$, η congruence ideals; $\eta(z)$ the Dedekind eta function (Chapter 12 only). Δ the discriminant, $\Delta(z)$ the cusp form of level one, Δ the Gram determinant of Lemma lem:ddt417 (local). Λ_E period lattice, $\Lambda(\cdot, s)$ completed L-functions. (i) $t_\ell(E,q)$ (the paper's term) always carries its arguments; the tame character $t_\ell\colon I \to \mathbb{Z}_\ell(1)$ of Chapter 7 is written with its domain or an argument $t_\ell(\sigma)$; t alone is a generator of tame inertia. (j) V is $V_\ell E$ or a representation; the Verschiebung is $\hat{\pi}_p$; $V_d\colon h(z) \mapsto h(dz)$ (Chapter 13) and V_f (Chapter 14, the f-part of H^1) are local names. T is $T_\ell E$; $\mathbb{T}$ the Hecke algebra; T_n Hecke operators; the matrix $\left(\begin{smallmatrix}1 & 1\\ 0 & 1\end{smallmatrix}\right)$ is T only in Chapter 12, Sections 1 and 2. (k) ε: unramified character (B-C14); $\varepsilon_N(f)$ Atkin-Lehner sign; a root number is w or $\varepsilon(\cdot)$ with its argument; the non-square of Chapter 1's standard models is ε only in Chapter 1, Section 5. (l) $\delta_p = \dim V^{I_p}$; $\hat{\delta}_!$ the pairing; δ a connecting map; $\delta(\bar{\sigma})$ the DDT exponent of the optimal level theorem (Chapter 15). (m) ψ: the order-4 character (prop:ns); ψ_m division polynomials (Chapter 9); an isogeny ψ (Chapter 14, 7.1).

Reason: the collisions recorded in the index (Part 2). Chapters bound: as named. Changes: Part 3, L1 to L6.

25.1.16 B-C16. Labels, names and typography

Convention B-C16. Curve labels are Cremona's, as printed by the paper: 11a1 (a curve), 121c (an isogeny class); ARS12's 99A1 is Cremona's 99a1; LMFDB labels are written 11.a2. Mod ℓ image labels are Sutherland's (3Ns, 3Cs.1.1, 3B.1.1, ...; the macro `\Ns` prints 3Ns), always with the intrinsic description; the groups are $N_s(\mathbb{F}_3)$, $N_{ns}(\mathbb{F}_3)$ (standard models of ch:groups 6.2). "Optimal" means $X_0(N)$-optimal. Double names take plain hyphens (Bloch-Kato, Diamond-Flach-Guo, Fontaine-Perrin-Riou, Atkin-Lehner); no dash ligatures anywhere (PLAN.md 7.2).

Changes: none.

25.2 Part 2. Index of symbols

One row per symbol. "Defined in" gives the chapter label and the outline item that defines the symbol (or the convention box that fixes it); where a symbol is defined in one chapter and specialised in another, both are given. Chapters 20 and 22 are cited by label only. "Paper" names the place in the paper where the symbol first occurs. The last column flags every other meaning the symbol has in the book or in the paper (**clash**) and the rule that resolves it (Part 1). In the LaTeX appendix the rows are grouped exactly as here, alphabetically inside each group (Latin, then Greek, then decorations).

25.2.1 2.1 Numbers, valuations, rings and modules

symbol	meaning	defined in	conflicts and notes
p, q, ℓ	primes; q a bad prime, ℓ the coefficient prime	ch:rings C2; B-C1	q also field size, $e^{2\pi i z}$, Tate variable $\mathfrak{q}$ (B-C1)
q^*, ℓ^*	$(-1)^{(q-1)/2}q$, $(-1)^{(\ell-1)/2}\ell$	ch:rings 9.6; ch:galois galois:sqrt-lstar	star exception (B-C4(e))
$\mathfrak{q}$	size of a finite field when not prime	ch:ellcurves C2	-
$\mathbb{F}_q$, $\mathbb{F}_q^\times$, $(\mathbb{F}_q^\times)^{\ell^k}$	finite field, its units, ℓ^k-th powers	ch:rings 6.1, 6.2, 7.1	index $[\mathbb{F}_q^\times : (\mathbb{F}_q^\times)^{\ell^k}] = \ell^{\min(k, v_\ell(q-1))}$ (7.1)
$\mathbb{Z}/n$, $(\mathbb{Z}/n)^\times$	integers mod n, units	ch:rings 5.1	-
μ_n, $\mu_n(K)$, μ_{ℓ^∞}	roots of unity	ch:rings 8.1, C6	$\mu(N)$ index, μ characters, μ_p of lem:old (B-C15(g))
ζ_n	$e^{2\pi i/n}$	ch:rings C6	never ω or ρ for ζ_3 (B-C15(b))
$\Phi_n(x)$	n-th cyclotomic polynomial	ch:rings 8.2	**clash**: geometric Frobenius Φ, component group $\Phi_q(\cdot)$; integer subscript and a polynomial argument identify it
$\sqrt{-3}$	$1 + 2\zeta_3 = i\sqrt{3}$	ch:rings C6; ch:ant 4.6	spans $(\mathrm{ad}^0 V)^{I_q}$ for types II, IV (lem:euler)
$\mathbb{Z}[i]$, $\mathbb{Z}[\zeta_3]$, $\mathbb{Z}_3[\zeta_3]$	Gaussian and Eisenstein integers; a DVR	ch:rings 3.4; ch:ant 4.5, 4.7	-
π (in lem:IV)	uniformiser $1-\zeta_3$ of $\mathbb{Z}_3[\zeta_3]$	ch:ant 4.7	**clash**: $\pi\colon \mathbb{T} \to \mathcal{O}$ (lem:ddt417), π_f, π_q, 3.14159 (B-C15(a))
v_p, v_ℓ	p-adic valuation, $v_\ell(\ell) = 1$	ch:rings 11.1, C3; B-C10	$v_\ell(c)$ for $c \in \mathbb{Z}/\ell^m$ (B-C10)
$\lvert x\rvert_p$	$p^{-v_p(x)}$	ch:rings 11.2	-
$\left(\frac{a}{p}\right)$, $\left(\frac{a}{n}\right)$	Legendre, Jacobi, Kronecker symbols	ch:rings 9.1, 9.4, 9.5, C4	-
$R^\times$	unit group	ch:rings 1.1	-
$\mathcal{O}$, ϖ, k, K	DVR, uniformiser, residue field, fraction field	ch:modules C2, 6.1	paper's $\mathcal{O}$ also the integers of a number field (ch:ant 1.2); ch:reps C7 writes λ for the uniformiser

symbol	meaning	defined in	conflicts and notes
$\mathcal{O}_K$, $\mathcal{O}_f$, K_f	integers of K; coefficient field of f and its integers	ch:ant 1.2, 7.1; ch:modgalrep C1	-
λ	prime of $\mathcal{O}_f$ above ℓ; $\lambda = \ell$ for rational f	ch:ant 7.2; ch:modgalrep C1	eigenvalues λ (local)
v_λ	valuation of λ	ch:ant 7.2	$= v_\ell$ when $\lambda = \ell$
$\mathbb{Z}_\ell$, $\mathbb{Q}_\ell$, $\mathbb{Q}_\ell/\mathbb{Z}_\ell$	ℓ-adic integers, numbers, the divisible group	ch:modules 6.4, 6.5; ch:groups 7.2	-
length, length_R	length; length = $\mathrm{length}_{\mathbb{Z}_\ell}$	ch:modules 7.1, 7.2, C6; B-C10	-
$\mathrm{Fitt}_R(M)$	Fitting ideal	ch:modules 7.4 to 7.6	-
$\mathrm{Ann}_R(M)$, $M[\mathfrak{a}]$, $M[\ell^k]$, M_{tor}	annihilator, $\mathfrak{a}$-torsion, torsion	ch:modules 7.3, C3, 5.4	-
$\#X$	order of a finite set	B-C10	$\#X = \ell^{\mathrm{length}\, X}$
L, $L[\wp]$, $L[I]$	lattice with perfect pairing; its $\wp$- and I-torsion	ch:modules 9.1, 11.3	L-functions $L(\cdot, s)$
saturated	M/L torsion-free	ch:modules 5.5	-
$S^\perp$	orthogonal of S for a pairing	ch:modules C5, 9.1	Pontryagin $B^\perp$ (ch:modules 8.1)
Gram determinant Δ	$\det(\langle e_i, e_j\rangle)$; in lem:ddt417 of $L[\wp]$	ch:modules 9.3; paper lem:ddt417	**clash**: discriminant Δ, $\Delta(z)$ (B-C15(h))
$\wp$	$\ker(\pi\colon \mathbb{T} \to \mathcal{O})$	ch:modules 11.1; paper lem:ddt417	Weierstrass $\wp(z)$ (ch:ellcurves 7.2); Artin-Schreier $\wp$ (ch:galois)
η	congruence ideal $\pi(\mathrm{Ann}_{\mathbb{T}}\, \wp)$	ch:modules 11.1	Dedekind $\eta(z)$ (B-C15(h))
$\mathcal{O}/\eta$	congruence module $\mathbb{T}/(\wp + \mathrm{Ann}\, \wp)$	ch:modules 11.1	-
$\mathbb{T}_\mathfrak{m}$, $e_\mathfrak{m}$, $L_\mathfrak{m}$	factor of $\mathbb{T} \otimes \mathbb{Z}_\ell$ at $\mathfrak{m}$, its idempotent, $e_\mathfrak{m}(L \otimes \mathbb{Z}_\ell)$	ch:modules C7, 10.3, 10.4	$\mathbb{T}_{(\mathfrak{m})}$ algebraic localisation
$V(\phi)$	generalised eigenspace of a system ϕ	ch:modules 10.5; ch:hecke C5	ch:hecke writes φ (H1)
Gorenstein	$\mathrm{Hom}_\mathcal{O}(\mathbb{T}, \mathcal{O})$ free of rank one	ch:modules 11.4	-
Hom, $\otimes$, End	as usual	ch:modules 3.1, 3.2	-

25.2.2 2.2 Groups and representations

symbol	meaning	defined in	conflicts and notes
C_n, D_n, Q_8, SD_{16}, V_4	cyclic, dihedral (order $2n$), quaternion, semidihedral, Klein	ch:groups C3, 2.4 to 2.6	D_4 has order 8
GL_2, SL_2, PGL_2, PSL_2	matrix groups	ch:groups 5.1	-
B, U	Borel subgroup, unipotent subgroup	ch:groups 5.5, 5.6	-

symbol	meaning	defined in	conflicts and notes
C_s, N_s, C_{ns}, N_{ns}	split and nonsplit Cartan subgroups and their normalisers (standard models)	ch:groups 5.8, 6.2	$N_s(\mathbb{F}_3) \subset N_{ns}(\mathbb{F}_3)$, not maximal (point 2)
J, u, w, a, h	standard matrices of $\mathrm{GL}_2(\mathbb{F}_3)$	ch:groups 6.2; ch:reps C6	$h = \mathrm{diag}(1,-1)$ (ch:groups writes σ); w also w_N, $w_\ell(E,q)$
ε (Chapter 1)	fixed non-square of $\mathbb{F}_q$ (-1 at $q=3$)	ch:groups 5.8	**clash**: characters ε (B-C15(k))
$\mathbb{P}^1(F)$, $[v]$	lines of F^2	ch:groups C1, 5.3	-
3Ns, 3Nn, 3Cs, 3Cs.1.1, 3B, 3B.1.1, 3B.1.2	Sutherland's labels of mod 3 images	ch:groups 6.10; ch:galrep 7.4	`\Ns` prints 3Ns (B-C16)
ρ, $\rho(g)$	a representation	ch:reps 1.1	$\rho_{E,\ell}$; ch:modforms C2 uses $\rho = e^{2\pi i/3}$ (L3)
$V^\vee$	contragredient	ch:reps C2; B-C4(a)	-
$\mathrm{Hom}_G(V,W)$	G-maps, $= (V^\vee \otimes W)^G$	ch:reps 1.3	PLAN.md writes $V^* \otimes W$ (D7)
tr_V	trace function (character)	ch:reps C4, 4.1	not χ_V
ρ^{ss}, $\bar\rho^{ss}$	semisimplification	ch:reps 3.5	-
Ind_H^G, Res	induction, restriction	ch:reps 6.1; ch:galois galois:restriction	Res also cohomological restriction
ψ	character of order 4 of C_4 with values in $\mathbb{F}_9^\times$, $\psi(J) = i$	ch:reps C6, 8.2	division polynomials ψ_m; isogeny ψ (B-C15(m))
$\mathrm{End}\, V$, $\mathrm{Ad}\, V$	full adjoint	ch:reps C3, 7.3	-
$\mathrm{ad}^0 V$, $\mathrm{ad}^0 T$	trace-zero adjoint (macro `\ad`)	ch:reps 7.3; B-C12	never `\ad`
$\mathrm{Sym}^2 V$	symmetric square (macro `\Sym`)	ch:reps 7.5	ch:reps names the map $\Phi\colon \mathrm{Sym}^2 V \to \mathrm{ad}^0 V$ (F3)
e, h, f	basis $e_1 \otimes e_2^\vee$, $e_1 \otimes e_1^\vee - e_2 \otimes e_2^\vee$, $e_2 \otimes e_1^\vee$ of ad^0	ch:reps C3; ch:cohomology model-setting	e idempotent, e_ℓ; h newform; f newform (local to lem:tamfactor)
trace form	$(x,y) \mapsto \mathrm{tr}(xy)$ on $\mathrm{ad}^0 T$	ch:reps 7.6	perfect for ℓ odd
σ	generator of order 3 of the inertia image (lem:IV, lem:euler)	ch:reps C6, 9.3	Chapter 1's $\sigma = \mathrm{diag}(1,-1)$ is h elsewhere
τ	generator of order 4 of the inertia image	ch:reps 9.3	-
homothetic	$L' = cL$	ch:reps C7, 10.1, 10.3	-

25.2.3 2.3 Fields, Galois groups, Frobenius, characters

symbol	meaning	defined in	conflicts and notes
$\overline{\mathbb{Q}}$, $G_{\mathbb{Q}}$, G_K	algebraic closure inside $\mathbb{C}$; absolute Galois groups	ch:galois C1, galois:absolute-galois-group	-

symbol	meaning	defined in	conflicts and notes
$G_{\mathbb{Q},S}$, $\mathbb{Q}_S$	Galois group of the maximal extension unramified outside S	ch:galcoh 6.1	S (B-C15(e))
c (complex conjugation)	an element of order 2 of $G_{\mathbb{Q}}$	ch:galois galois:complex-conjugation	**clash**: $c(u)$, c_q, c_E (always decorated)
D_p, I_p, P_p	decomposition, inertia, wild inertia groups at p	ch:ant 3.10; ch:local L2, 5.2	D (B-C15(d))
Frob_p, Frob	arithmetic Frobenius	ch:ant 3.7, 3.8; ch:local L3, 4.3; B-C2	$\mathrm{Frob}^{\mathrm{geom}}$ not used
Φ_p, Φ	geometric Frobenius Frob_p^{-1}	ch:local L3, 4.3; B-C2(b)	**clash**: component group $\Phi_q(\cdot)$, cyclotomic Φ_n, ch:reps's arithmetic Φ (F3)
π_q, $\hat{\pi}_q$	q-power Frobenius endomorphism, Verschiebung	B-C2(g); ch:ellcurves 8.1	ch:curves ϕ_q, ch:moddeg F, V (F6, F7)
χ_ℓ, $\bar{\chi}_\ell$	ℓ-adic cyclotomic character, its reduction	ch:galois galois:cyclotomic-character, C4; B-C6(a)	the paper's χ_q, the book's χ_{q^*} (B-C6(c), D3)
χ_3^{cyc}	cyclotomic character at a numerical prime	B-C6(a); ch:galrep C4	-
ω_ℓ	Dirichlet character $a \mapsto a \bmod \ell$	ch:rings C5, 10.5	fundamental character ω (B-C15(b))
χ_d	quadratic character of $\mathbb{Q}(\sqrt{d})$	ch:galois C3, galois:quadratic-characters; ch:rings 9.5	numerical subscripts always χ_d (B-C6(b))
χ_{-3}, χ_{-4}	characters of $\mathbb{Q}(\sqrt{-3})$, $\mathbb{Q}(i)$	ch:rings 10.5; ch:ant 5.8	$\chi_{-3} = \bar{\chi}_3$
χ_{q^*}	quadratic character of conductor q (the paper writes χ_q; never in the book, D3)	ch:rings C5; ch:galois C3; B-C6(c)	the paper's χ_q **clashes** with χ_ℓ when $q = \ell$ (point 32)
χ_{ℓ^*}	character of $\mathbb{Q}(\sqrt{\ell^*})$, $= \bar{\chi}_\ell^{(\ell-1)/2}$	ch:galois C4	$= \bar{\chi}_\ell$ iff $\ell = 3$
χ_S, D	$\prod_{q\in S} \chi_{q^*}$, its conductor $\prod_{q\in S} q$	ch:hecke C7; ch:ellcurves 10.6	D (B-C15(d))
ε	unramified character of order ≤ 2 of $G_{\mathbb{Q}_q}$	ch:local 7.6; ch:galrep 3.2; B-C14	$\varepsilon_N(f)$, Chapter 1's non-square
$\chi(\mathrm{Frob}_p) = \chi(p)$	Dirichlet characters as Galois characters	ch:ant 5.7; ch:modgalrep C3	-
$\mathbb{Z}_\ell(1)$, $M(n)$	Tate module of μ, Tate twists	ch:galois C6; ch:blochkato 2.2; B-C5	-
$\mathbb{Q}(\sqrt{\ell^*})$	quadratic subfield of $\mathbb{Q}(\zeta_\ell)$	ch:galois galois:sqrt-lstar; ch:ant 5.3	-
$\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{\ell^*}))$	the group on which $\bar{\rho}$ must be absolutely irreducible	ch:galois galois:restriction	-

symbol	meaning	defined in	conflicts and notes
χ_a, κ_x	Kummer character of a; Kummer cocycle of $x \in K^\times$	ch:galois galois:kummer-character, galois:l-adic-kummer; ch:local 6.2	-
θ_m	tame character $I_K \to \mu_m$	ch:local 5.3	$\theta = \rho\|_{I_q}$ (ch:lfunctions 10.1, local)
Artin map	uniformiser to arithmetic Frobenius	ch:ant 5.6; ch:local 6.5	Deligne's convention opposite (B-C2(g))
Art	local Artin map	ch:local 6.5	-
density	Dirichlet density	ch:galois galois:density	-

25.2.4 2.4 Local fields and Weil-Deligne representations

symbol	meaning	defined in	conflicts and notes
K, $\mathcal{O}_K$, $\mathfrak{m}_K$, ϖ_K, k_K, v_K	a p-adic field and its data	ch:local L1	ch:reduction R1 writes π for the uniformiser (L1)
$\mathbb{Q}_p^{\mathrm{ur}}$, K^{tame}	maximal unramified, tame extensions	ch:local L2, 4.2	-
I_K, P_K, $I_K^{(\ell)}$	inertia, wild inertia, maximal pro-ℓ quotient of I_K	ch:local L2, 5.2, 5.3	-
$t_\ell \colon I_K \to \mathbb{Z}_\ell(1)$, t_ζ	tame character; its $\mathbb{Z}_\ell$-valued version	ch:local 5.3, L5	**clash**: $t_\ell(E,q)$ (B-C15(i))
t	topological generator of tame (pro-ℓ) inertia	ch:local 5.3(iii); ch:cohomology tame-inertia-example	-
W_K, $d(w)$, $\|w\|$	Weil group, its degree map, $q^{d(w)}$	ch:local 7.1, L4	W, W_N
(r, N), $\mathrm{WD}(\rho)$	Weil-Deligne representation	ch:local 7.3, 7.4	N (B-C15(f))
$\mathrm{sp}(n)$	special (Steinberg) representation	ch:local 7.5	-
$a(V)$, $\mathrm{sw}(V)$	Artin conductor exponent, Swan conductor	ch:local 8.3, 8.4	a_p coefficients
$L(V,s)$, $L_p(V,s)$	local L-factor, with Φ	ch:local 8.2; B-C2(d)	-
$\epsilon(r,\psi,dx)$	local epsilon factor	ch:local 8.5	-
$\mathbb{Q}_p^\times = p^{\mathbb{Z}} \times \mu_{p-1} \times (1+p\mathbb{Z}_p)$	structure of $\mathbb{Q}_p^\times$	ch:local 2.5	-
ω, ω_2	fundamental characters of levels 1, 2 at ℓ	ch:galrep 5.2	**clash**: periods ω_1, ω_2, differentials (B-C15(b))
π_p, $\mathrm{WD}_p(f)$	local component of f; its Weil-Deligne representation	ch:modgalrep C5, 3.3	**clash**: Frobenius endomorphism π_p (Chapters 8, 9, 14 only; B-C15(a))
D_{cris}, B_{cris}, B_{dR}, B_{st}	Fontaine's rings and functor	ch:blochkato 4.1	-

25.2.5 2.5 Curves, elliptic curves, reduction, Tate curves

symbol	meaning	defined in	conflicts and notes
g	genus	ch:curves C2, 5.3	newform g; generator g (local)
φ^*, φ_*	pullback, pushforward	ch:curves C4; B-C4(e)	Pontryagin * on modules
$\mathcal{C}$, $\mathcal{C}^t$, $\mathcal{C}^*$, $\mathcal{C}_*$	correspondence, transpose, its actions	ch:curves 12.1, 12.2	-
$A^\dagger$	adjoint for a pairing	ch:curves C4, 12.3; B-C4(d)	ch:modules writes A^* (D1)
$H_1(X,\mathbb{Z})$, $H^1(X,\mathbb{Z})$, $H^1(X,\mathbb{Z}_\ell)$	homology, cohomology of a curve	ch:curves 10.1, 10.2	-
$\langle\ ,\ \rangle$ (cup product)	$\int_X \alpha\wedge\beta$	ch:curves C6, 10.3	Petersson product (B-C8)
PD	Poincare duality $H^1 \to H_1$	ch:curves C6, 10.3	-
Pic^0, $J(C)$, $T_\ell J$	Picard group, Jacobian, Tate module	ch:curves 4.2, 11.1, 11.7	-
$\deg\varphi$	degree of a morphism of curves	ch:curves 6.2	-
$a_1,\dots,a_6$; $b_2,\dots,b_8$; c_4,c_6; Δ; j	Weierstrass data	ch:ellcurves 1.1, C3	Δ, c (B-C15(h))
ω_E	invariant (Neron) differential $dx/(2y+a_1x+a_3)$	ch:ellcurves C3; ch:moddeg C4	ω (B-C15(b))
$E[m]$, $E(K)$, $E(\mathbb{Q})_{\mathrm{tors}}$	torsion, points, torsion subgroup	ch:ellcurves 3.5, 9.1, 9.3	-
ψ_m	division polynomials, $\psi_2 = 2y+a_1x+a_3$	ch:ellcurves 4.1	ψ (B-C15(m))
e_m	Weil pairing (Silverman's)	ch:ellcurves 5.1, C7	-
$\hat\varphi$, $[m]$	dual isogeny, multiplication by m	ch:ellcurves 3.4, 3.5	-
$T_\ell E$, $V_\ell E$	Tate module, $T_\ell E\otimes\mathbb{Q}_\ell$	ch:ellcurves 5.3	paper's T, V (Section 1.1)
Λ_E, $\mathrm{vol}(\Lambda_E)$, ω_1,ω_2	period lattice of ω_E, its covolume, periods	ch:ellcurves 7.1, 7.6	ω_2 (B-C15(b)); PARI `E.area`
$\mathfrak{H}$, $\mathfrak{H}^*$	upper half plane, with cusps	ch:modforms C1; ch:ellcurves 7.5	star exception
$a_p(E)$, $a_{\mathfrak{q}}$	$p+1-\#\tilde E(\mathbb{F}_p)$	ch:ellcurves 8.1, 8.6	$a(V)$ conductor
α, β	roots of X^2-a_pX+p	ch:ellcurves 8.3; B-C3	arithmetic eigenvalues on $V_\ell E$
$E^{(d)}$, $E^{(q)}$	quadratic twist by $\mathbb{Q}(\sqrt{d})$; by $\mathbb{Q}(\sqrt{q^*})$	ch:ellcurves 10.1, 10.5; ch:reduction 8.2	depletion superscripts (H3, H4)
E'	twist $E^{(q)}$ of good reduction at an I_0^* prime (paper's table); in thm:qual, $E^{(q)}$	ch:ellcurves 10.5	-

symbol	meaning	defined in	conflicts and notes
E_S, N_S	$E \otimes \chi_S$, its conductor; the paper's $E_{\{p\}}$ (proof of thm:main) and E_q (after thm:twist) are $E^{(p)}$, $E^{(q)}$	ch:ellcurves 10.6	the Tate curve is $E_{\mathtt{q}}$, not E_q (L5)
$L(E,s)$, $L_p(E,s)$	Hasse-Weil L-function	ch:ellcurves 11.1; ch:lfunctions 5.1	$L_p(E,s) = L_p(V_\ell E^\vee, s)$ (B-C3)
CM, j-invariants	complex multiplication	ch:ellcurves 6.3, 6.4	-
11a1, 121c, 99A1, 11.a2	Cremona, ARS12, LMFDB labels	ch:ellcurves 12.1, C6; B-C16	-
minimal model, $\Delta_{\min}$	(global) minimal Weierstrass model	ch:reduction 1.1, 1.3	-
$E_0(K)$, $E_1(K)$, $\tilde{E}_{\mathrm{ns}}$	filtration of $E(K)$; nonsingular points of the reduction	ch:reduction 2.3, 2.4	E_0 also the optimal curve of the class (paper Section 3)
$\mathcal{E}$, $\Phi_q(\cdot)$	Neron model; its component group scheme	ch:reduction 4.1, 4.2; B-C11(b)	**clash**: Φ_p geometric Frobenius
Φ_q^{geom}	geometric component group $\Phi_q(\overline{\mathbb{F}}_q)$	ch:reduction R4, 4.2; paper Section 1.4	ch:galrep writes Φ_q (K1)
c_q, c_p	Tamagawa number $\#\Phi_q(\mathbb{F}_q)$	ch:reduction 4.4	never for $\#\Phi_q^{\mathrm{geom}}$ or $c(u)$
$\mathrm{I}_n, \ldots, \mathrm{II}^*$	Kodaira symbols	ch:reduction 5.1; B-C11(a)	-
Kodaira code k	PARI's code, `K[i][2]`	ch:reduction 5.5; B-C11(c)	valid formula $\pm(n+4)$ only for $n \geq 1$
m_p	number of components of the special fibre	ch:reduction 6.2	-
f_p, N	conductor exponent, conductor $\prod p^{f_p}$	ch:reduction 6.1, 6.4	f newform; N monodromy
$\mathrm{sw}_p(E)$	wild part of f_p	ch:reduction R5	-
G_{ss}, e	semistability group, defect $12/\gcd(v_p(\Delta),12)$	ch:reduction R6, 6.5, 6.6	never Φ_K
$E_{\mathtt{q}}$, $\mathtt{q}$	Tate curve, its parameter variable	ch:reduction 7.1	q (B-C1)
q_E	Tate parameter of E	ch:reduction 7.4; B-C14	never u
n, n_q	$v_q(\Delta_E) = v_q(q_E)$ at a multiplicative prime	ch:galrep C2; paper lem:tamagawa-exp	-
ν	$v_q(q_E) = -v_q(j_E)$	ch:blochkato C6; B-C14	-
u, $\bar{u}$	unit part $q_E q^{-\nu}$, its reduction	ch:reduction 7.4; B-C14	the paper's lem:tamagawa-exp calls q_E "u" (point 8)
$c(u)$	class of $\bar{u}$ in $\mathbb{F}_q^\times/(\mathbb{F}_q^\times)^{\ell^{v_\ell(q-1)}}$	ch:rings 7.2; ch:reduction 7.4; ch:blochkato C6	paper's c, c_q (point 23)
$w_\ell(E,q)$	twist-symmetrised weight	ch:reduction 8.4; ch:paper A1.6	w_N, w

25.2.6 2.6 Galois representations of elliptic curves and of newforms

symbol	meaning	defined in	conflicts and notes
$\rho = \rho_{E,\ell}$, $\bar{\rho} = \bar{\rho}_{E,\ell}$	action of $G_{\mathbb{Q}}$ on $T_\ell E$, on $E[\ell]$	ch:ellcurves 5.4; ch:galrep 1.1, C1	-
$\rho \bmod \ell^k$	action on $E[\ell^k] = T/\ell^k T$	ch:galois C5; ch:galrep C1	-
V, T	$V_\ell E$, $T_\ell E$ (paper Section 1.1)	ch:galrep C1	V_d, V_f, Verschiebung (B-C15(j)); T matrix, $\mathbb{T}$
A, A_f	$\mathrm{ad}^0 T$, $\mathrm{ad}^0 V$	ch:galrep C1; B-C12(b)	**clash**: Shimura's abelian variety, renamed J_f (A1, A2)
δ_p	$\dim V^{I_p}$: $2, 1, 0$ for good, multiplicative, additive $p \neq \ell$	ch:galrep 3.5; ch:local 8.1	defined for $p \neq \ell$ only (point 72)
absolutely irreducible	irreducible over $\bar{k}$	ch:reps 2.3	-
$N(\bar{\rho})$, $k(\bar{\rho})$	Serre conductor, Serre weight	ch:galrep 6.1; ch:modgalrep 5.1	-
$\delta(\bar{\sigma})$	DDT's exponent at ℓ in the optimal level theorem	ch:modgalrep 6.4	δ_p, $\hat{\delta}_!$ (B-C15(l))
$\bar{\sigma}$	a mod ℓ representation in DDT 3.15	ch:modgalrep 6.4	-
$\rho_{f,\lambda}$, $\bar{\rho}_{f,\lambda}$, $V_\lambda(f)$	Galois representation of a newform	ch:modgalrep 2.1	normalised by $\det(X - \rho(\mathrm{Frob}_p)) = X^2 - a_p X + p$
$h_\ell(E)$	length of $H^0(\mathbb{Q}, \mathrm{ad}^0 T(1) \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell)$	ch:galrep 8.5; ch:blochkato 10.3	$h_\ell(E) = 0$ for $\ell \geq 5$, $h_3 \leq 1$ (point 5)
S_f	DFG04's exceptional set: $\lambda \mid 2N$ or $\bar{\rho}$ not absolutely irreducible on $G_{\mathbb{Q}(\sqrt{\ell^*})}$	ch:galrep 8.6; ch:blochkato 10.1	S, S_2
D_ℓ	decomposition group at ℓ	ch:ant 3.10	-
ι $(T \to V)$	$H^1(K,T) \to H^1(K,V)$	ch:galcoh C4	**clash**: twist map ι (B-C15(c))

25.2.7 2.7 Modular forms, Hecke algebras, modular curves, the modular degree

symbol	meaning	defined in	conflicts and notes
$\Gamma_0(N)$, $\Gamma_1(N)$, $\Gamma(N)$	congruence subgroups	ch:modforms 3.1	-
$\mu(N)$	$[\mathrm{SL}_2(\mathbb{Z}) : \Gamma_0(N)] = N \prod_{p \mid N}(1 + p^{-1})$	ch:modforms C7	μ_n (B-C15(g))
$\nu_2(N)$, $\nu_3(N)$, $\nu_\infty(N)$	elliptic points, cusps	ch:modforms C7, 4.3	ν (B-C14)
$Y_0(N)$, $X_0(N)$, $X_1(N)$	modular curves	ch:modforms 5.1, 5.2; ch:moddeg 2.1	-
$q = e^{2\pi i z}$	q-expansion variable	ch:modforms C1	q prime (B-C1)
$f\|_k \gamma$	slash operator with $(\det \gamma)^{k/2}$	ch:modforms C3, 6.1; B-C5(d)	-

symbol	meaning	defined in	conflicts and notes
M_k, S_k, $S_2(\Gamma_0(N))$, $S_2(N;A)$	modular and cusp forms; integral forms	ch:modforms 6.2, 11.1, C6; ch:hecke C1	-
$a_n(f)$	n-th coefficient at ∞	ch:modforms C6	-
$\langle f,g\rangle$	Petersson product, unnormalised	ch:modforms C5, 10.1; B-C8	cup product
$f^\perp$	Petersson orthogonal complement	ch:modforms 10.3; ch:hecke 5.4	not the $\mathbb{T}$-stable complement when U_p is involved (point 61)
E_k, $\Delta(z)$, $j(z)$, $\eta(z)$	Eisenstein series, discriminant form, j, Dedekind eta	ch:modforms 7.1, 7.2, 9.1	Δ, η (B-C15(h))
$B_k(N)$	Sturm bound	ch:modforms C7, 11.3	B_E
W_N, w_N, W_d	Atkin-Lehner matrix and involution	ch:modforms C4, 12.1	w
$\varepsilon_N(f)$	eigenvalue of w_N on a newform	ch:hecke 10.2; ch:moddeg 9.2	-
T_n, T_p, U_p	Hecke operators; $U_p = T_p$ for $p \mid N$	ch:hecke C3, 2.4	level-dependence $T_p^{(M)}$
$[\Gamma\alpha\Gamma]_k$	double coset operator	ch:hecke C2, 2.1	-
$\mathbb{T}$, $\mathbb{T}_N$, $\mathbb{T}^{(N)}$, $\mathbb{T}_\ell$, $\mathbb{T}_A$	full Hecke algebra, anemic algebra, scalar extensions	ch:hecke C4, 3.1; B-C13(a)	-
ϕ_f	eigencharacter $\mathbb{T} \to \mathcal{O}_f$	ch:hecke 3.2, 3.4; ch:modgalrep C1	ch:hecke C5 writes φ (H1)
I, J (lem:crit)	$\ker \phi_f$, $\mathrm{Ann}_{\mathbb{T}}(f^\perp)$	ch:modgalrep 8.2; B-C13(c)	**clash**: $I = \mathrm{Ann}\,\wp$ in lem:ddt417; I_p inertia
I_f	$\ker \phi_f$ (Chapter 14)	ch:moddeg 5.1	-
r_E, r	congruence number	ch:moddeg 8.1; ch:modgalrep 8.1	-
f^Σ, $f^{\{p\}}$, f^S	depletion	ch:hecke C8, 7.4; B-C13(e)	not $h^{(p)}$ (H3, H4)
M^Σ	$M\prod_{p\in\Sigma} p^{\delta_p}$, level of f^Σ	ch:hecke 7.4; paper Section 4.1	-
V_d	$h(z) \mapsto h(dz)$	ch:hecke C3	V
$f\otimes\chi$, f^χ	naive twist, newform twist	ch:hecke 9.1; ch:modgalrep C4, 2.7	-
ι	twist map $h \mapsto h\otimes\chi_S$	ch:hecke 9.1; ch:modgalrep 10.1	ι of Chapter 17 (B-C15(c))
$L(f,s)$	L-function of a newform	ch:hecke 10.1	-
α_p, β_p, μ_p	roots of $X^2 - a_p(h)X + p$; chosen U_p-eigenvalue (lem:old)	ch:hecke 7.1; ch:modgalrep 8.3	B-C3(e)
$J_0(N)$, $J_1(N)$	Jacobians	ch:moddeg 4.1	-
t^*, t_*	the two Hecke actions (on H^1, on H_1)	ch:moddeg C2, 4.2	pullback star (B-C4(e))
J_f, π_f, E_f	$J_0(N)/I_fJ_0(N)$, the quotient map; the elliptic curve for rational f	ch:moddeg 5.1 (written A_f there); B-C12(c)	renamed (A1)

symbol	meaning	defined in	conflicts and notes
φ_E, $\deg \varphi_E$	modular parametrisation, modular degree (macro `\degphi`)	ch:moddeg 6.1, 6.2	ch:modgalrep writes ϕ (H2)
E_0	the optimal curve of the class (paper Section 3)	ch:modgalrep 2.6	$E_0(K)$
c_φ, c_E	Manin constants	ch:moddeg 7.1; B-C9	never bare c
$\hat{\delta}_!$	$\langle x, w_N y\rangle$ on $H^1(X_0(N), \mathbb{Z}_\ell)$	ch:moddeg 3.4; B-C13(f)	ch:moddeg writes $\hat{\delta}$ (H5)
$M_{f,\lambda}$, $M_{f^\Sigma,\lambda}$	$H^1(X_0(M^\Sigma), \mathbb{Z}_\ell)[\ker \phi_{f^\Sigma}]$, rank two	ch:moddeg 4.4, 9.1; paper Section 4.1	ch:moddeg writes $M_{f,\ell}$ (H5)
$\mathfrak{m}$	maximal ideal of $\mathbb{T} \otimes \mathbb{Z}_\ell$ attached to (f, ℓ)	ch:moddeg 10.1	$\mathfrak{m}_K$
I_{Eis}, C_N, $\mathcal{S}_N$	Eisenstein ideal, cuspidal subgroup, Shimura subgroup	ch:moddeg 11.1 to 11.3	ch:moddeg writes Σ_N (L4)
δ_D	degree of the Shimura curve parametrisation (Ribet-Takahashi)	ch:moddeg 12.2	-

25.2.8 2.8 Cohomology, Selmer groups, Tamagawa factors

symbol	meaning	defined in	conflicts and notes
$C^n(G, M)$, Z^1, B^1	inhomogeneous cochains, cocycles, coboundaries	ch:cohomology inhomogeneous-cochains, h0-h1	C_n cyclic group
$H^n(G, M)$, $H^n(K, M)$	(continuous) group and Galois cohomology	ch:cohomology continuous-cohomology, galois-cohomology-convention; ch:galcoh 1.1	types (D), (C), (V) of coefficients
$\hat{H}^n$	Tate cohomology	ch:cohomology tate-cohomology	-
Res, Inf, Cor	restriction, inflation, corestriction	ch:cohomology res-inf-cor	-
$(\sigma \cdot c)(h) = \sigma c(\sigma^{-1} h \sigma)$	conjugation action of G/H on $H^n(H, M)$	ch:cohomology conjugation-action; B-C5(d)	-
n'	$v_\ell(n') = v_\ell(n)$, the unipotent entry of $\rho(t)$	ch:cohomology model-setting	-
κ, κ_q, κ_u	Kummer cocycle of q_E, of q, of u	ch:galcoh 1.5, 5.3; B-C14	κ bijection of ch:ellcurves 2.3 (local)
c'	$\varepsilon(\mathrm{Frob})\kappa(\mathrm{Frob})$, well defined modulo $(n, q-1)$	ch:galcoh 5.4; ch:cohomology frobenius-on-h1-inertia	-
$H^1_{\mathrm{ur}}(\mathbb{Q}_p, M)$	$\ker(H^1(\mathbb{Q}_p, M) \to H^1(I_p, M)) = M^{I_p}/(\mathrm{Frob} - 1)M^{I_p}$	ch:galcoh 4.1, 4.2	macro `\Hur`

symbol	meaning	defined in	conflicts and notes
T, V, W	lattice, $T \otimes \mathbb{Q}_\ell$, $T \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$	ch:galcoh C4	in the paper $W = A \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$ (B-C12(b))
T^D, V^D, W^D	Kummer duals: $\mathrm{Hom}(T, \mathbb{Z}_\ell(1))$ etc.; **in the paper and Chapters 18, 20, 21 built from** $A = \mathrm{ad}^0 T$	ch:galcoh C3, 2.1; ch:blochkato C2, 2.3, 6.6	**clash**: paper's T^D is A^D, not $(T_\ell E)^D$
$M^*(1)$	$\mathrm{Hom}(M, \mu_{\ell^\infty})$ for finite or discrete M, $= M^D$	ch:galcoh C3	-
$\mathrm{Sel}_n(E/\mathbb{Q})$, Sha	n-Selmer group, Shafarevich-Tate group	ch:galcoh 7.3	-
$\mathcal{L}$, $\mathcal{L}^*$, $H^1_{\mathcal{L}}$	Selmer structure, dual structure, Selmer group	ch:galcoh 8.1, 8.2	star exception (B-C4(e))
$\mathcal{L}^\Sigma$, $\mathcal{L}_\Sigma$	relaxed and strict at Σ	ch:galcoh 8.2	-
$H^1_f(\mathbb{Q}_p, M)$, $H^1_f(\mathbb{Q}, M)$	Bloch-Kato local conditions and Selmer groups	ch:blochkato 5.1, 5.2, 6.1	macro `\Hf`
H^1_g, H^1_e	other Bloch-Kato conditions	ch:blochkato 5.1	-
$H^1_\Sigma(\mathbb{Q}, W)$	Σ-imprimitive Selmer group of DFG04 2.1	ch:blochkato 6.3, 6.6	-
$\mathrm{Tam}_p(M)$	$\mathrm{Fitt}_{\mathbb{Z}_\ell}((H^1(I_p, M)_{\mathrm{tor}})^{G_{\mathbb{Q}_p}})$	ch:blochkato 7.1; B-C10	point 37 (placement of the invariants)
Σ	the paper's finite set of primes (in thm:main, the primes dividing N)	ch:galcoh C5; ch:blochkato C1	$\mathcal{S}_N$; in thm:quant $\Sigma = S$
$G_{\mathbb{R}}$, $H^1(\mathbb{R}, M)$	the real place	ch:galcoh C1, 6.4	killed by 2
X^*	Pontryagin dual	ch:modules 8.1, 8.2; B-C4(b)	pullback stars on maps
$\mathrm{ad}^0 M_f$	the adjoint motive	ch:blochkato 14.1	-

25.2.9 2.9 L-functions and the paper's local terms

symbol	meaning	defined in	conflicts and notes
$\zeta(s)$, $\zeta^{(N)}(s)$	Riemann zeta; with the factors at $p \mid N$ removed	ch:lfunctions 3.1, 3.2	ζ_n roots of unity
$L(s, \chi)$, $\zeta_K(s)$	Dirichlet and Dedekind L-functions	ch:lfunctions 3.3, 3.4	-
$\Gamma_{\mathbb{R}}$, $\Gamma_{\mathbb{C}}$	$\pi^{-s/2}\Gamma(s/2)$, $2(2\pi)^{-s}\Gamma(s)$	ch:lfunctions 2.3, 7.2	$\Gamma_0(N)$
$L_p(M, s)$	$\det(1 - \Phi_p p^{-s} \mid M^{I_p})^{-1}$	ch:lfunctions 4.1, C3; ch:local 8.2; B-C2(d)	-
$L(\mathrm{ad}^0 f, s)$, $L(A_f, s)$	adjoint L-function	ch:lfunctions 6.1, C4; B-C7	-
$L(\mathrm{Sym}^2 f, s)$	symmetric square, $= L(\mathrm{ad}^0 f, s-1)$	ch:lfunctions 6.1, C4; B-C7	never the Rankin series
L_p^{nv}, L^{nv}	naive factors and naive L-function (macro `\nv`)	ch:lfunctions 8.1, C6; B-C7	-

symbol	meaning	defined in	conflicts and notes
$D(f,s)$	Rankin series $\sum a_n(f)^2 n^{-s}$	ch:lfunctions C4, 8.2	D (B-C15(d))
$P_q(X)$	$L_q(\mathrm{Sym}^2 f, s) = P_q(q^{-s})^{-1}$	ch:lfunctions C5	-
$e_\ell(E,q)$	$v_\ell(P_q(q^{-2})/(1-a_q^2 q^{-2}))$, the omitted Euler factor	ch:lfunctions 9.1; ch:paper A1.1	not twist-invariant (point 15)
$t_\ell(E,q)$	$v_\ell(-v_q(j_E))$ if $v_q(j_E)<0$, else 0	ch:lfunctions 9.1; ch:paper A1.1	tame character t_ℓ (B-C15(i)); extended to $q=\ell$ (point 53)
t_q, $t_f(q)$	Tamagawa exponent (Kim-Ota): largest t with $\rho \bmod \ell^t$ unramified at q	ch:modgalrep 11.1	$= t_\ell(E,q)$ at multiplicative q
N_{ad^0}	conductor of A_f	ch:lfunctions 7.2	-
$\Lambda(\mathrm{ad}^0 f, s)$	completed adjoint L-function	ch:lfunctions 7.1	Λ_E
B_E	$N\,L(\mathrm{ad}^0 f,1)/(2\pi\,\mathrm{vol}(\Lambda_E))$	ch:lfunctions 8.4	$B_k(N)$
ε (root number)	sign of a functional equation	ch:lfunctions 7.1; ch:hecke 10.2	ε characters (B-C15(k))

25.2.10 2.10 Congruence ideals and the theorems of Diamond, Flach and Guo (Chapter 20, outlined in parallel)

symbol	meaning	defined in	conflicts and notes
$\pi\colon \mathbb{T}\to\mathcal{O}$, $\wp$, $I=\mathrm{Ann}_{\mathbb{T}}\wp$, $\eta=\pi(I)$	the data of lem:ddt417	ch:modules 11.1, 11.3; ch:congruence	**clash**: I of lem:crit (B-C13(c)); π (B-C15(a))
d, Δ, Q	rank of $L[\wp]$, its Gram determinant, $L/(L[\wp]\oplus L[I])$	ch:modules 11.3; ch:congruence	Δ discriminant
$M(M^\Sigma,1)_{!,\lambda}$	DFG04's integral structure, $= H^1(X_0(M^\Sigma),\mathbb{Z}_\ell)$	ch:congruence; paper Section 4.1	-
$\eta^\Sigma_{f,\lambda}$, $\eta^\emptyset_{f,\lambda}$	naive Σ-finite congruence ideal; level N	ch:congruence; paper Section 4.1	$\eta(z)$
$\eta^{\mathrm{Hecke}}(g^S)$	congruence ideal of g^S in $\mathbb{T}$	ch:hecke 4.3; ch:congruence	-
R_Σ, $\mathbb{T}_\Sigma$	deformation ring, Hecke algebra of Taylor-Wiles	ch:congruence; ch:modgalrep 11.5	-
g, g^S	f_{E_S}, its S-depletion (thm:quant)	ch:modgalrep 10.1; ch:congruence	genus g
N_S^Σ	$N_S\prod_{q\in S} q^{\delta_q(g)} = N$	ch:modgalrep 10.1	-

25.2.11 2.11 The paper's named quantities, statements and data

symbol	meaning	defined in	conflicts and notes
$w_3(E)$	weight of the prime 3 in Conjecture conj:euler	ch:paper A1.5	$w_\ell(E,q)$
D (conj:eis)	deficit $\sum_q v_\ell(\#\Phi_q^{\mathrm{geom}}) - v_\ell(\deg \varphi_E)$	ch:paper A1.13	D (B-C15(d))
$[\,\cdot\,]$	Iverson bracket (1 if true, 0 if false)	ch:paper A1.5, A1.13	-
(eq:main), (eq:twist), (eq:rule), (eq:identity), (eq:thm)	the paper's displayed inequalities	ch:paper A1.4, A1.5, A1.7	cited by these names
$E[3] \simeq \mathbb{Z}/3 \oplus \mu_3$	the image 3Cs.1.1	ch:galrep 9.2; ch:ellcurves 9.7	not "the split Cartan image" (point 44)
working set, hold-out, extension, out-of-table	the four data ranges	ch:computation; ch:paper A6.1	-

25.2.12 2.12 Computation (Chapter 22, outlined in parallel)

symbol	meaning	defined in	conflicts and notes
`rule(E, l, iii)`	the paper's function returning $[v_\ell(\deg \varphi_E), \mathrm{bound}]$	ch:computation; ch:lfunctions 11.2; ch:paper A9.1	-
`ns3(E)`	1 iff ψ_3 factors as two quadratics	ch:computation; ch:paper A9.1	the correction h_3
`R[4]`, `R[5]`, `K[i][2]`	`ellglobalred` output: bad primes, local data, Kodaira code	ch:computation; ch:reduction 5.5	B-C11(c)
`lfunsympow(E,2)`, `lfuneuler`	$L(\mathrm{Sym}^2 f, s)$ and $1/P_q(x)$, $x = q^{-s}$	ch:lfunctions C5, 11.1	normalisation B-C7
`mfpetersson`	Petersson product divided by $\mu(N)$	ch:modforms C5	B-C8
`E.area`	$\mathrm{vol}(\Lambda_E)$	ch:lfunctions C9	-
`ellmoddegree`, `elldivpol`, `ellinit`	modular degree, division polynomial, curve	ch:computation	-
`ecdata` files	Cremona's tables (allcurves, alldegphi, ...)	ch:computation	-

25.2.13 2.13 Decorations and operators

decoration	meaning	defined in	conflicts and notes
$M^\vee$	linear dual	B-C4(a)	-
X^*	Pontryagin dual (modules)	B-C4(b)	pullback φ^* (maps); exceptions in B-C4(e)
M^D	Kummer dual $M^\vee(1)$ or $M^*(1)$	B-C4(c)	D
$A^\dagger$	adjoint of an operator	B-C4(d)	-
φ^*, φ_*	pullback, pushforward	B-C4(e)	-

decoration	meaning	defined in	conflicts and notes
A^{T}, $\mathcal{C}^t$	transpose	B-C4(f)	-
$\hat{\varphi}$, $\hat{\pi}_q$	dual isogeny	B-C4(f)	$\hat{\delta}_!$, $\hat{H}$
$M(n)$	Tate twist	B-C5(a)	-
ρ^{ss}	semisimplification	ch:reps 3.5	-
$\bar{\rho}$, $\bar{\chi}_\ell$, $\bar{u}$	reduction mod ℓ (or mod q)	ch:galois C5	complex conjugate $\bar{z}$
M^G, M^{I_p}	invariants	ch:cohomology h0-h1	-
f^Σ	depletion	B-C13(e)	-
$(\cdot)^{(N)}$	with the factors at N removed; anemic	B-C13(e)	-
$E^{(d)}$	twist of a curve	B-C13(d)	-
f^χ, $f \otimes \chi$	newform twist, naive twist	B-C13(d)	-
$S^\perp$, $f^\perp$	orthogonal complement	ch:modules C5; ch:modforms 10.3	-
Ш* etc., q^*, ℓ^*, $\mathfrak{H}^*$, E_2^*, $\mathcal{L}^*$	the fixed exceptions to B-C4(e)	B-C4(e)	-
$(\cdot)_{\mathrm{tor}}$	torsion submodule	ch:modules C3	-
$[\ell \mid n]$	Iverson bracket	ch:paper A1.13	-

Number of index rows: 2.1: 35, 2.2: 23, 2.3: 25, 2.4: 15, 2.5: 44, 2.6: 14, 2.7: 39, 2.8: 23, 2.9: 16, 2.10: 8, 2.11: 6, 2.12: 8, 2.13: 19; in all 275 rows, carrying 490 symbol entries (the math or code items of the first column; several rows carry two to six related symbols).

25.3 Part 3. Where the outlines disagree with Part 1

Found by searching every outline for `Frob`, `\Phi`, `^*` (with the Kodaira symbols, q^*, ℓ^* filtered out), `^\vee`, `dagger`, `chi_`, `c^2`, `Petersson`, and then for the clashes these searches revealed (`\omega`, `A_f`, `\phi`, `\varphi`, `\pi`, `h^{(p)}`, `\Sigma_N`, `E_q`, `iota`, `n+4`). Line numbers refer to the outline files as of 14 October 2026. "Required" means the outline as written would put a wrong or ambiguous formula into the book; "cosmetic" means the outline is correct but uses a symbol that Part 1 reserves for something else. The searches for `c^2` (the Zagier-Watkins formula) and `Petersson` (the normalisation) found **no disagreement**: every outline that states the formula puts c^2 in the numerator (ch:ellcurves line 82, ch:modforms 10.x and B1, ch:moddeg C4 and 7.3, ch:blochkato 14.7, ch:lfunctions 8.4) and every outline that fixes the Petersson product uses the unnormalised one with PARI's division by $\mu(N)$ (ch:modforms C5, ch:hecke C6, ch:moddeg C4, ch:lfunctions C9). The searches for the $L(\mathrm{Sym}^2 f, s)$ normalisation, the Tate twist, v_ℓ, "length" and the tame relation likewise found no disagreement.

25.3.1 3.1 By outline

ch02_rings.md (`ch:rings`) * F9 (cosmetic). 6.3 `prop:rings:frobenius`, line 340: "$\phi\colon x \mapsto x^p$" for the Frobenius automorphism of $\mathbb{F}_q$. Write Frob_p (B-C2(a)); ϕ is the eigencharacter ϕ_f elsewhere. * D8 (cosmetic). 10.3 and 10.4, lines 522 and 537 to 538: "χ^*" for the primitive character inducing χ. Write χ^{prim} (B-C4(e): a star on a character is not one of the fixed exceptions). * L2 (cosmetic). C6, line 90, and 3.4 (and about forty further uses): "$\omega = \zeta_3$". Write ζ_3 (B-C15(b)); if the writer keeps ω for the Eisenstein integers, the chapter's convention box must say that ω is local to Chapter 2.

ch03_modules.md (`ch:modules`) * D1 (required). 9.5 `prop:modules:adjoint`, lines 592 to 599: the adjoint is written A^* ("$A^* = G^{-1}A^{\mathsf{T}}G$", "$(AB)^* = B^*A^*$", "$A^* = wAw^{-1}$", "$w^* = w$"). Write $A^\dagger$

throughout (B-C4(d)). * D2 (required). Section 6, notes for the writer ("Three duals"), lines 972 to 974: "The adjoint of an operator is A^* (9.5)". Replace by $A^\dagger$ and add "(* is the Pontryagin dual only)". * D3 (cosmetic). Section 5, dependencies, line 944: "$\mathrm{Hom}_G(V,W) = (V^* \otimes W)^G$". Write $V^\vee$.

ch04_reps.md (ch:reps) * F3 (required). C8, lines 121 to 122; 9.1(iii), lines 602 to 604; table 9.3, header of the last-but-one column, line 629; Exercise 14, line 851; Section 6, line 962; Section 8, line 1053: the **arithmetic** Frobenius is called Φ ("an abstract $\Phi \in \mathrm{GL}(V)$ with $\Phi g \Phi^{-1} = g^q$"; "the geometric Frobenius Φ^{-1}"). This inverts the book's convention. Write F for the abstract arithmetic Frobenius (as ch:ant 3.12 does) and F^{-1} for the geometric one. * F3, second half (cosmetic). 7.5 prop:reps:sym2, lines 498 to 500; Exercise 10, line 843; verification log, lines 1018 and 1025: the isomorphism $\mathrm{Sym}^2 V \otimes \det^{-1} \to \mathrm{ad}^0 V$ is called Φ. Rename it Ξ (any letter not taken; Ψ is used by ch:moddeg 4.3).

ch05_galois.md (ch:galois) * X1 (required). galois:quadratic-characters (b), line 532: "$\chi_{-3} = \chi_{-1}\chi_3$" with χ_3 the character of $\mathbb{Q}(\sqrt{3})$, while the paper's χ_3 is χ_{-3}. Use an example without a numerical χ_3, for instance $\chi_{-15} = \chi_{-3}\chi_5$, or write $\chi_{-3} = \chi_{-4}\chi_{12}$. * X2 (required). Exercise 14 (in the section plan), line 789: "$(\rho \otimes \chi_3)$ mod 9" with the cyclotomic character. Write χ_3^{cyc} (B-C6(a)). * Lines 212, 539, 542, 753 used the paper's χ_q with the letter q: converted to χ_{q^*} under B-C6(c) (decision D3), as is C3.

ch07_local.md (ch:local) * X3 (cosmetic). 13.2 (pitfalls), line 974: "the tame $\chi_3 = \chi_{-3}$". Write χ_{-3}. * Lines 359 ("$\chi_p(I_p) = \mathbb{Z}_p^\times$", cyclotomic at p with a letter subscript) and 520 to 535 (now χ_{q^*}, decision D3) comply. * L4 states the Weil-Deligne relation in both forms; it is the reference for B-C2(f). No change.

ch08_curves.md (ch:curves) * F6 (cosmetic). 6.5 prop:curves:separability-frobenius, lines 347 to 350: the q-power Frobenius morphism is ϕ_q. Write π_q (B-C2(g)). * A3 (required). 11.8, line 604: "$\rho_{f,\lambda}$ via T_ℓ of A_f". Write J_f (B-C12(c)). * D9 (cosmetic). C4, lines 88 to 89 and Part 7, lines 1018 to 1019: the remark "Chapter 3 writes A^* for the adjoint" becomes stale after D1; replace by "as in Appendix B".

ch09_ellcurves.md (ch:ellcurves) * F4 (cosmetic). C5, line 116, and 8.1 def:ellcurves:frobendo, line 502: the arithmetic Frobenius of $G_{\mathbb{F}_\mathfrak{q}}$ is "$\phi_\mathfrak{q}$". Write $\mathrm{Frob}_\mathfrak{q}$; keep π for the Frobenius endomorphism ($\pi_\mathfrak{q}$ when the field must be shown). * F5 (cosmetic). Section 6, line 974, and Part 8 (B1 extension), line 1074: "$\mathrm{Frob}^{\mathrm{arith}}$", "$\mathrm{Frob}^{\mathrm{geom}}$". Write Frob_p and Φ_p. (The mathematics there is right and is the source of B-C3.) * M4 (cosmetic). Section 0, line 82: "$\deg \varphi_E = 4\pi^2 c^2 \langle f, f\rangle / \mathrm{vol}(\Lambda_E)$" with a bare c. Write c_E. * C5's "$\det(1 - \mathrm{Frob}_p\, p^{-s} \mid V_\ell E)^{-1}$ with the arithmetic Frobenius" is a correct identity (B-C3(c)); keep it, and add the book's form $L_p(E,s) = L_p(V_\ell E^\vee, s)$ with Φ_p next to it.

ch10_reduction.md (ch:reduction) * L1 (cosmetic). R1, line 93: the uniformiser is π ("$\mathfrak{m} = \pi\mathcal{O}$", "$v(\pi) = 1$"). Write ϖ (B-C15(a)), as Chapters 3 and 7. * L5 (cosmetic). Lines 43, 61, 747, 750, 856: "E_q", "$E_q[\ell^t]$" for the Tate curve, while R1 and 7.1 fix $E_\mathfrak{q}$. Write $E_\mathfrak{q}$ (the paper's E_q is a twist, index entry E_S). The same cross-reference phrase occurs in ch:galois line 826, ch:local line 993, ch:ellcurves line 938, ch:galrep lines 89 and 831: change there too. * R3 to R6 and 5.5 are the source of B-C11. No other change.

ch11_galrep.md (ch:galrep) * K1 (required). C2, line 115: "$\Phi_q = \Phi_q(\bar{\mathbb{F}}_q)$ the geometric component group of the Neron model (the paper's Φ_q^{geom}), $c_q = \#\Phi_q(\mathbb{F}_q)$" defines Φ_q as a group and then evaluates it at $\mathbb{F}_q$. Write "$\Phi_q^{\mathrm{geom}} = \Phi_q(\bar{\mathbb{F}}_q)$". Then in 2.2 (lines 203 to 210: "$\Phi_q[\ell^k]$", "$\Phi_q \simeq \mathbb{Z}/n_q$"), 4.6 (line 393), 6.2 (lines 452 to 458: "$a_p(\bar{\rho}_{E,\ell}) = f_p - \dim \Phi_p[\ell]$"), the section plan (line 705), the exercises (line 805) and the pitfalls (line 896) write $\Phi_q^{\mathrm{geom}}[\ell^k]$, $\Phi_p^{\mathrm{geom}}[\ell]$, $\#\Phi_q^{\mathrm{geom}}$; keep $\Phi_q(\mathbb{F}_q)$ and $\Phi_q(\bar{\mathbb{F}}_q)$ where they occur (lines 876, 992). * F8 (required). Section 5, line 838: "$L_q(A_f, s) = \det(1 - \Phi_q q^{-s} \mid (\mathrm{ad}^0 V)^{I_q})^{-1}$" in the chapter where Φ_q is the component group. Write Frob_q^{-1} (B-C2(b); C3 of this very outline says so). * X5 (cosmetic). 4.3 or its proof, line 334: "$\det \sigma = \chi_3(\sigma) = 1$" (cyclotomic). Write χ_3^{cyc} or χ_ℓ with $\ell = 3$. * L5: lines 89, 831 (above).

ch12_modforms.md (ch:modforms) * L3 (cosmetic). C2, line 85: "$\rho = e^{2\pi i/3}$". Write ζ_3 (ρ is the Galois representation). The matrix $T = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ of the same line may stay (B-C15(j)) if the chapter says it is local to Sections 1 and 2. * M3 (cosmetic). 10.x, line 507; 4.2 of the exercises, line 734; Section 6.1,

lines 805 to 806; Part 7 B1, line 951: the Manin constant is a bare c. Write c_φ (general parametrisation) or c_E (optimal curve), B-C9.

ch13_hecke.md (ch:hecke) * H1 (cosmetic). C5, line 102: a system of eigenvalues and its generalised eigenspace are φ, $V(\varphi)$; 8.3, lines 460 to 461 mix φ and ϕ. Write ϕ, $V(\phi)$ (B-C13(b)); φ is the modular parametrisation. * H3 (required). C8, line 112 ("$h^{(p)} = h^{\{p\}}$") and 7.1, lines 374 to 375 ("$h^{(p)} = h - a_p h(pz) + p^{k-1}h(p^2z)$"): depletion by a parenthesised superscript, which the book reserves for twists $E^{(q)}$. Write $h^{\{p\}}$ only (B-C13(e)).

ch14_moddeg.md (ch:moddeg) * A1 (required). 1.1 (line 24), 5.1 (lines 240 to 242: "$A_f := J_0(N)/I_f J_0(N)$", "$\pi_f\colon J_0(N) \to A_f$", "$A_f =: E_f$"), 5.2 (line 252: "$T_\ell A_f$"), the coverage table (line 717), Section 5 (line 831), Section 6.2 (line 857): the abelian variety is A_f, the paper's adjoint. Write J_f (B-C12(c)). * F7 (required). C5, line 77, and 5.2, lines 249 to 253: "F the Frobenius endomorphism, $V = pF^{-1}$ the Verschiebung", "$(T_p)_* = F + V$", "$F^2 - T_p F + p = 0$". V is $V_\ell E$ in the paper. Write π_p, $\hat\pi_p = p\pi_p^{-1}$, $(T_p)_* = \pi_p + \hat\pi_p$. * M1 (cosmetic). 7.1 (lines 356 to 358, "c_E well defined up to sign"), 7.2 (ii), (iii) and 7.3 (line 372, "$c_E = \pm 1$"), 11.4: fix the sign of ω_E by $c_E > 0$ and write $c_E = 1$ (B-C9). * L4 (cosmetic). 11.3 def:moddeg:shimura-subgroup and line 604 (four occurrences): the Shimura subgroup is Σ_N, while Σ is the paper's set of primes. Write $\mathcal{S}_N$. * H5 (cosmetic). $\hat\delta$ without the subscript (17 occurrences, from 3.4 on) and $M_{f,\ell}$ (22 occurrences, C6 and Section 9): the paper writes $\hat\delta_!$ and $M_{f,\lambda}$. Write the paper's symbols, with "$\lambda = \ell$" said once (C6 already says it).

ch15_modgalrep.md (ch:modgalrep) * A2 (required). 1.1 (line 26), 1.2 (lines 87, 89), 2.1 (lines 150 to 155: "$A_f = J_0(N)/I_f J_0(N)$", "A_f is an abelian variety"), 2.6 (line 196, "$A_f = E_0$"), 4.1 (lines 338 to 339, "A_f has good reduction at ℓ", "$A_f[\ell^k]$"), the coverage table (line 755), Section 5 (line 873), Section 6.3 (line 935): the abelian variety is A_f, while 3.7 (line 311, "$A_g = A_f$") uses the paper's adjoint. Write J_f for the abelian variety. * D4 (required). 8.1 def:modgalrep:congruence-number, line 495: "f is an eigenvector of the adjoints T_n^*, since $T_n^* = T_n$ for $(n, N) = 1$ and $U_p^* = w_N U_p w_N$". Write $T_n^\dagger$, $U_p^\dagger$. * H2 (required). 1.1 (line 32), 2.6 (lines 196 to 201: "$\phi = \varphi_{E_0}$", "$\phi^* H^1(E_0, \mathbb{Z}_\ell)$", "$\phi_*$"), 6.4 (line 948): the modular parametrisation is an upright ϕ, the letter of the eigencharacter ϕ_f (C1 of the same outline). Write φ_{E_0}. * H4 (required). C4, lines 117 to 118 ("$h^{(q)}$", "$h^{(S)}$", "$f \otimes \chi = (f^\chi)^{(r)}$"), 1.1 (line 135), 10.1 (lines 653 to 654, 663 to 664: "$(g^{\chi_S})^{(S)}$", "$\iota(g^{(S)})$"), Exercise 2 (line 812), writer note (line 893): depletion by a parenthesised superscript. Write $h^{\{q\}}$, h^S, and $(f^\chi)^{\Sigma(r)}$ with $\Sigma(r)$ the set of primes dividing r (B-C13(e)). * K2 (cosmetic). Section 1.2, line 73 ("$\dim \Phi_p[\ell]$") and 6.x, line 615 ("ℓ-torsion in Φ_q"): write $\Phi_p^{\mathrm{geom}}[\ell]$, Φ_q^{geom}. Line 379 ("$\Phi_p(\bar{\mathbb{F}}_p)[\ell]$") complies.

ch17_galcoh.md (ch:galcoh) * X4 (cosmetic). 14.4, line 908 ("$\chi_3\chi_{-3} \not\equiv 1 \pmod 9$") and Part 15, line 963 ("$\chi_3(\sigma) = 4$"): cyclotomic character at 3 written χ_3. Write χ_3^{cyc}. * C3 ("the Kummer dual $M^*(1)$") agrees with B-C4(c); the dual Selmer structure $\mathcal{L}^*$ (8.2) is a listed exception. No change.

ch18_blochkato.md (ch:blochkato) * M5 (cosmetic). 14.7, line 676: "$4\pi^2 c^2 \langle f, f\rangle$". Write c_E^2. * C2, C3, C4, C6 are the sources of B-C4(c), B-C2, B-C5(c), B-C14. No other change.

ch19_lfunctions.md (ch:lfunctions) * M2 (cosmetic). 8.4, lines 349 to 351, and Section 5, line 724: "$c_E = \pm 1$". Write $c_E = 1$ (B-C9). * C2 to C9 are the sources of B-C2, B-C3, B-C7, B-C8, B-C11(c). No other change.

ch21_paper.md (ch:paper) * F1 (required). Opening paragraph, lines 17 to 18: "In every L-FACTOR, `L_p(M, s) = det(1 - Frob_p p^{-s} | M^{I_p})^{-1}` with `Frob_p` the GEOMETRIC Frobenius `= Frob_arith^{-1}`". The book's Frob_p is always arithmetic. Write `Phi_p` there. * F2 (required). C2(b), lines 62 to 66: "`Frob_p^{geom}`", "for `M = V_l E` and `p` good this is `(1 - a_p p^{-s} + p^{1-2s})^{-1}`", "`det(1 - Frob^{geom} X | V) = (1 - alpha X)(1 - beta X)`", "on `V_l E` the geometric Frobenius has eigenvalues `alpha, beta`; the arithmetic one has `p/alpha = beta` ... the same set". All wrong (important_notes.md B1, B7, B9). Replace by B-C3: on $V_\ell E$ the arithmetic Frobenius has α, β, the geometric α^{-1}, β^{-1} (not the same set); $(1 - a_p p^{-s} + p^{1-2s})^{-1} = L_p(V_\ell E^\vee, s) = L_p(E, s)$, while $L_p(V_\ell E, s) = L_p(E, s+1)$; write Φ_p for the geometric Frobenius. * T1 (required). A2.5, line 385, and A2.7, lines 450 and 466: the Tate matrices with corner `kappa`. Write `eps kappa` (B-C14; point 42, B7), and in step (6) `c' = eps(Frob) kappa(Frob)`. * T2 (cosmetic). A2.7, line 477 (and the other book-text uses of a bare `c`, `v_l(c)`): write `c(u)`; keep `c` inside quotations of the paper (point 23, B-C14). * D5 (required). A3.2, lines 518 to 519: "`T_n^* = w_N T_n w_N`", "`U_p^* = w_N U_p w_N`" for adjoints.

Write `T_n^dagger`, `U_p^dagger`; and replace the self-correcting sentence (B12) by the clean statement of ch:hecke 5.2, 5.3. * D6 (required). A4.1 commentary, lines 678 and 680, and Section 9 N12, line 1894: "`V_l E^*`", "`ad(T^*)`" for **linear** duals. Write `V_l E^vee`, `ad(T^vee)`. * X5 (cosmetic). A2.2 (or its proof), line 309: "`chi_3(sigma) = 1`" with the cyclotomic character. Write `chi_3^{cyc}`. * L6 (cosmetic). A4.9 (lines 795 to 803) uses `iota` for `H^1(Q_p, T^D) -> H^1(Q_p, V^D)` and A4.14 (lines 954 to 965) for the twist map. Write `iota_S` for the twist map in this chapter (B-C15(c)). * The writer note W4 (line 1713) already asks for the adjectives; with F1, F2 the symbols carry them.

ch01_groups.md, ch06_ant.md, ch16_cohomology.md: no disagreement. (ch:ant writes the geometric Frobenius as Frob_p^{-1}, which B-C2(b) allows; ch:groups's ε and σ are local and already renamed by Chapters 4 and 11.)

25.3.2 3.2 PLAN.md (not an outline; for the orchestrator)

- P1. Chapter 7 entry: "$\mathrm{Frob}\, N\, \mathrm{Frob}^{-1} = p^{-1}N$". Write "$\Phi N \Phi^{-1} = p^{-1}N$, equivalently $\mathrm{Frob}\, N\, \mathrm{Frob}^{-1} = pN$" (B6, B-C2(f)).
- P2. Chapter 18 and Chapter 19 entries: "$L_p(V,s) = \det(1 - \mathrm{Frob}_p\, p^{-s} \mid V^{I_p})^{-1}$ (... the Frobenius here is geometric or arithmetic: fix the convention ...)". Write Φ_p (B-C2(d)).
- P3. Chapter 4 entry "$\mathrm{Hom}_G(V,W) = (V^* \otimes W)^G$", Chapter 14 entry "$J_0(N)(\mathbb{C}) = S_2(\Gamma_0(N))^*/H_1(X_0(N),\mathbb{Z})$", Chapter 18 entry "duals $V^* = \mathrm{Hom}(V, \mathbb{Q}_\ell)$ and ... $V^D = V^*(1)$": write $^\vee$ (B-C4(a)). Also D7.
- P4. Chapter 10 and Chapter 22 entries: "$n+4$ for I_n, $-(n+4)$ for I_n^*": add "for $n \geq 1$; I_0^* is -1" (B-C11(c)).
- P5. Chapter 14 entry: "the abelian variety A_f attached to a newform" (write J_f) and "the Eichler-Shimura relation $T_p = \mathrm{Frob}_p + \mathrm{Frob}_p^*$" (write $(T_p)_* = \pi_p + \hat{\pi}_p$, B-C2(g)).
- P6. Chapter 18 entry: "$\mathrm{Tam}_p(T) = \mathrm{Fitt}_{\mathbb{Z}_\ell}(H^1(I_p,T)_{\mathrm{tors}})^{G_{\mathbb{Q}_p}}$": the invariants go inside (point 37, B22).
- P7. Appendix B entry: its list names "c"; the book has c_q (Tamagawa number), $c(u)$ (class of the unit part), c' and c_E (Manin constant), never a bare c.

25.3.3 3.3 Counts

outline	required	cosmetic	entries
ch01_groups	0	0	none
ch02_rings	0	3	F9, D8, L2
ch03_modules	2	1	D1, D2, D3
ch04_reps	1	1	F3 (both halves)
ch05_galois	2	0	X1, X2
ch06_ant	0	0	none
ch07_local	0	1	X3
ch08_curves	1	2	A3, F6, D9
ch09_ellcurves	0	3	F4, F5, M4
ch10_reduction	0	2	L1, L5
ch11_galrep	2	2	K1, F8, X5, L5
ch12_modforms	0	2	L3, M3
ch13_hecke	1	1	H3, H1
ch14_moddeg	2	3	A1, F7, M1, L4, H5
ch15_modgalrep	4	1	A2, D4, H2, H4, K2
ch16_cohomology	0	0	none
ch17_galcoh	0	1	X4
ch18_blochkato	0	1	M5
ch19_lfunctions	0	1	M2

outline	required	cosmetic	entries
ch21_paper	5	3	F1, F2, T1, D5, D6, T2, X5, L6
total	20	28	48 entries (plus 7 in PLAN.md)

By theme: Frobenius 10 entries (F1 to F9, with F3 in two halves), duals and adjoints 9 (D1 to D9), characters 6 (X1 to X5, X5 in two outlines), Manin constant 5 (M1 to M5), component groups 2 (K1, K2), the abelian variety A_f 3 (A1 to A3), Hecke notation 5 (H1 to H5), Tate data 2 (T1, T2), other letters 7 (L1 to L6, L5 in two outlines). Per searched pattern: `Frob` 6 (F1, F2, F4, F5, F8, F9 and P1, P2), `\Phi` 4 (F3, F8, K1, K2), `^*` 7 (D1 to D6, D8), `^\vee` 1 (D6), `dagger` 2 (D4, D5, both "write dagger"), `chi_` 5 (X1 to X5), `c^2` 0 (M3 to M5 concern the letter, not the formula), `Petersson` 0.

25.4 Part 4. Section plan for `chapters/chB_notation.tex` and notes for the writer

25.4.1 4.1 Section plan (target 750 lines of LaTeX; PLAN.md allows 400 to 800)

The appendix has no card of core results (it states no theorems), no exercises, and no "Where the paper uses this chapter" section in the usual form; instead its first section says how to use it. It does have a short "Notes and references" section. Labels: `\label{ch:notation}` on the chapter, `sec:notation:...`, `conv:notation:...`, `tab:notation:...`.

section	content	source here	lines
B.1 How to use this appendix	one paragraph: the conventions bind every chapter; the index gives the chapter where each symbol is defined (`\Cref{ch:slug}` only, no theorem numbers, PLAN.md 7.1); the paper's notation is the book's; where the paper is ambiguous the book's reading is given	opening of this file	20
B.2 Letters, primes, valuations, lengths	`convention` boxes B-C1, B-C10; a five-line table of the meanings of q	B-C1, B-C10	35

section	content	source here	lines
B.3 Frobenius	B-C2 and B-C3 as two `convention` boxes; one `warning` box "Where the paper uses which Frobenius" (Section 1.1, table, lem:euler, lem:fpr, after lem:dfg21: geometric; lem:tamfactor: arithmetic); the dictionary table (arithmetic eigenvalue λ = geometric λ^{-1}; $V_\ell E$, $V_\ell E^\vee$, $\mathrm{ad}^0 V$, $\mathbb{Z}_\ell(1)$, the Steinberg line)	B-C2, B-C3	80
B.4 Duals, adjoints, pullbacks, twists	B-C4 and B-C5 as `convention` boxes; a table "symbol / meaning / example in the paper" for $^\vee$, *, ^{D}, †, φ^*, (n); the list of fixed exceptions to the star rule	B-C4, B-C5	60
B.5 Characters	B-C6 as a box, with the table of the paper's characters (χ_{-3}, χ_{-4}, χ_{q^*} (the paper's χ_q), χ_{ℓ^*}, χ_ℓ, $\bar{\chi}_\ell$, ω_ℓ, χ_S, ε, ψ: field cut out, conductor, value at p)	B-C6; ch:rings 10.5; ch:ant 5.8	40
B.6 L-functions, Petersson product, Manin constant	B-C7 (with its table of normalisations), B-C8, B-C9; a `warning` on the Rankin series and on PARI's `mfpetersson`	B-C7 to B-C9	60
B.7 Local data at a bad prime	B-C11 (Kodaira symbols, Φ_q, Φ_q^{geom}, c_q, PARI codes as a ten-row table), B-C14 (Tate data and the Tate matrix with corner $\varepsilon\kappa$)	B-C11, B-C14	50
B.8 Adjoint, Hecke algebras, twists, depletion	B-C12, B-C13; the dictionary of the two uses of I (lem:crit, lem:ddt417) as a three-line table	B-C12, B-C13	45

section	content	source here	lines
B.9 Index of symbols	Part 2 as `longtable`s (four columns: symbol, meaning, chapter, see also), grouped as 2.1 to 2.13, every row kept; the "conflicts" column shortened to "see also" pointers to B.2 to B.8 and B.10; item numbers of the outlines dropped	Part 2	300
B.10 Letters with several meanings	B-C15 as one table (letter, meanings, where, rule)	B-C15	35
B.11 Notes and references	which sources use which conventions (Deligne73 and Tate79 for $\|\cdot\|$; DFG04 1.7.2; Bloch-Kato and FPR94 write $V^*(1)$ for V^D; Silverman's Weil pairing; Diamond-Shurman's slash; Shimura75, Gelbart-Jacquet; PARI manual); B-C16	B-C16 and the references named in Part 1	25
total			750

The index dominates: 275 rows of Part 2 at one LaTeX line each, plus thirteen group headings and the `longtable` preambles, give about 300 lines; the boxes are therefore kept to their statements, with the reasons reduced to one sentence and a pointer to the chapter that proves the identity. If the total exceeds 800 lines, merge index rows that differ only in decoration (as $\Gamma_0(N)$, $\Gamma_1(N)$, $\Gamma(N)$ already are) rather than dropping symbols.

25.4.2 4.2 Notes for the writer

1. **Wording of the boxes.** The boxed statements of Part 1 are final; the writer only converts them to LaTeX (macros of `preamble.tex`: `\Frob`, `\ad`, `\Sym`, `\nv`, `\Tam`, `\Fitt`, `\len`, `\Phig`, `\Ns`, `\degphi`, `\Zl`, `\Ql`, `\Qlmod`, `\Hom`, `\End`, `\Ann`, `\Hf`, `\Hur`). There is no macro for the geometric Frobenius: write `\Phi_p`. Do not define new macros (PLAN.md 7.1).
2. **Cross-references.** Only `\Cref{ch:slug}`. The item numbers of Part 2 are for the orchestrator and the chapter writers; once the chapters exist, the cross-reference pass of PLAN.md Section 8, step 4, may replace them by precise labels.
3. **Chapters 20 and 22.** Their outlines are being written in parallel. When they land, check that they use π, $\wp$, $I = \operatorname{Ann}\wp$, $\eta^{\Sigma}_{f,\lambda}$, $\hat{\delta}_!$, $M_{f,\lambda}$, J_f (not A_f), `rule(E, l, iii)`, the Kodaira codes with $n \geq 1$, $c(u)$ and Φ_p as here, and add their item numbers to Part 2.
4. **Paper quotations.** Inside `paperlink` environments and verbatim quotations the paper's own symbols stand (χ_q for the book's χ_{q^*}, decision D3; c, c_q in sec:tam, χ_{11}, $E_{\{p\}}$, "Frob$_p$" in Section 1.1), followed by the book's reading in brackets.
5. **Do not "fix" the paper's numbers.** No number in the paper changes under any convention of Part 1 (points 11, B1, B10). The appendix says so once, in B.3.

6. **Typography.** No dash ligatures, ASCII source, tables with `booktabs` and `longtable` (check that `longtable` is loaded by `preamble.tex`; if not, split B.9 into several `tabular` environments, since chapters may not load packages).
7. **What not to put in the appendix.** Proofs (the boxes cite the chapters that prove the identities: $L_p(M(1), s) = L_p(M, s+1)$ is ch:local 8.2, $\mathrm{ad}^0 V \simeq \mathrm{Sym}^2 V \otimes \det^{-1}$ is ch:reps 7.5, the Zagier-Watkins identity is ch:moddeg 7.3).

25.5 Part 5. What could not be settled here

1. **DFG04's own conventions.** B-C2(d) and B-C7 assume DFG04 1.7.2 uses the geometric Frobenius and $L(\mathrm{ad}^0 M_f, s)$ with the value at $s = 1$, as the Chapter 7, 19 and 21 outlines report; nobody has checked the text (Chapter 19 marks its table row "verify"). Likewise the Hodge-Tate sign of DFG04 and Bloch-Kato (B-C5(c)) and the checklist of important_notes.md point 56.
2. **The sign of the Manin constant.** B-C9 fixes $c_E > 0$ by choosing the sign of ω_E. This is a book decision, harmless (only c_E^2 enters), but the literature states Manin's conjecture as $c_E = \pm 1$ or $c_E = 1$ depending on whether ω_E is normalised; the Chapter 14 writer must phrase the cited results accordingly.
3. **Fundamental characters ω, ω_2 versus periods ω_1, ω_2.** Kept both (B-C15(b)) because they never meet in one chapter as the outlines stand; if Chapter 11 or 15 ends up mentioning periods, rename the fundamental characters $\theta_{\ell-1}$, θ_{ℓ^2-1} (Serre's notation, which is Chapter 7's θ_m).
4. **The paper's χ_q.** Resolved by decision D3 (PLAN.md 7.5): the stricter rule is adopted, χ_{q^*} everywhere in the book's own text (as Chapters 2, 13, 18, 19 already did), and the outlines of Chapters 5, 6, 7, 9, 10, 11, 15, 17, 21 have been converted. The paper's χ_q survives only inside quotations of the paper (Part 4, item 4).
5. **Two uses of I and of ι in the paper.** B-C13(c) and B-C15(c) keep the paper's letters locally rather than renaming them, because Chapter 21 must quote the paper verbatim. If the paper itself is revised, the cleanest fix is J for $\mathrm{Ann}_{\mathbb{T}}(\wp)$ in lem:ddt417 and ι_S for the twist map.
6. **Line numbers.** Part 3 cites lines of the outlines as they stand on 14 October 2026; the Chapter 20 and 22 outlines and any revision of the others will shift them. The entry identifiers (F1, D1, ...) are stable.
7. **Reading.** The outlines were read through their convention sections, item lists, writer notes and every line matched by the searches listed at the head of Part 3; prose paragraphs that matched none of the patterns were not all read line by line, so a disagreement phrased without any of the searched symbols could have been missed.